Light Metals 2016

Edited by
EDWARD WILLIAMS

TMS

Springer

Light Metals 2016

TMS2016

145th Annual Meeting & Exhibition

FEBRUARY 14-18 DOWNTOWN NASHVILLE, TENNESSEE **MUSIC CITY CENTER**

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Light Metals 2016

Proceedings of the symposia sponsored
by the Aluminum Committee of the Light Metals Division
of The Minerals, Metals & Materials Society (TMS)

held during

TMS2016
145th Annual Meeting & Exhibition

FEBRUARY 14-18 DOWNTOWN NASHVILLE,
TENNESSEE MUSIC CITY CENTER

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Editor
Edward Williams

ISBN 978-3-319-48615-4 ISBN 978-3-319-48251-4 (eBook)
DOI 10.1007/978-3-319-48251-4

Chemistry and Materials Science: Professional

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Published by Springer International Publishers, Switzerland, 2016
Reprint of the original edition published by John Wiley & Sons, Inc., 2016, 978-1-119-22579-9

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Printed on acid-free paper

This Springer imprint is published by Springer Nature
The registered company is Springer International Publishing AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

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PREFACE

It is my honor to present the *Light Metals 2016* proceedings, which represents an enormous investment in time and effort by all of the authors and their supporting organizations in conducting the research and in preparing the manuscripts published in this volume. I greatly appreciate all of their work, and I would like to thank them for their efforts. For the past four decades, *Light Metals* has been the pre-eminent annual publication for work in aluminum production processes, and this is due to the contributions of the authors.

The global aluminum industry has gone through dramatic shifts over the past several years, with China now accounting for more than half of the world's primary aluminum production. The Gulf States (GCC) have also seen significant increases in total smelting capacity and they are now the second largest region in terms of aluminum production. These geographic changes in smelting are reflected in the papers represented in this volume, with roughly a third each coming from North America and Europe, and the remaining third from China and the Middle East. This global diversity in the source of the research work strengthens the quality of these proceedings, and helps to make sure that is staying relevant in today's truly global production environment.

As in past years, this volume contains collected research and development work for aluminum processes organized into symposia: Aluminum and Bauxite, Electrode Technology, Aluminum Reduction Technology, and Cast Shop for Aluminum Production. This year, there has been a change in the structure of the programming, combining the former Aluminum Alloys and Aluminum Processing into one new symposia: Aluminum Alloys, Processing and Characterization. This new grouping covers work in the areas of aluminum alloy development, microstructure, deformation processes, and characterization.

Additionally, a specialized symposia of Strip Casting of Light Metals co-sponsored by the Aluminum Committee and Magnesium Committee will be held in parallel to the traditional aluminum sessions. Included in this volume are six manuscripts from the Strip Casting of Light Metals Symposium, which is organized by Kai F. Karhausen (Hydro Aluminium Rolled Products GmbH), Dietmar Letzig (MagIC – Helmholtz Zentrum Geesthacht), Jan Bohlen (MagIC – Helmholtz Zentrum Geesthacht), and Murat Dundar (Assan Aluminium).

I would like to express my appreciation to the Subject Chairs: Angelique Adams, Stephan Broek, Mohamed Ibrahim Hassan Ali, Paul McGlade, and Steven Long as well as their Session Chairs who helped put this volume together by soliciting and reviewing papers as well as organizing the sessions. I also greatly appreciate the guidance from my predecessors in this role, Margaret Hyland and John Grandfield. Lastly, I would also like to express my great appreciation to all of the great TMS staff for their support in the preparation of this volume.

Edward Williams
Editor, *Light Metals 2016*

EDITOR'S BIOGRAPHY

EDWARD WILLIAMS
LIGHT METALS 2016 EDITOR

Edward Williams is currently the Division Manager of Casting Technology at the Alcoa Technical Center. He began his career at Reynolds Aluminum in the recycling and reclamation group in 1989 as an engineering co-op. After graduating, he worked as a research engineer in the Reynolds Casthouse R&D organization until 2000, when he transferred to Ingot Technology at the Alcoa Technical Center with the merger of the two companies. With Alcoa, Eddie has worked at both ATC as a research engineer and as production plant engineer at Alcoa's Warrick Operations. Over his career, he has worked in a large variety of roles in casthouse research and plant support, including molten metal treatment, aluminum melting furnace efficiency, scrap recycling, CFD modeling of molten aluminum processes, continuous casting, Al-Li ingot casting, and ingot plant capital engineering as well as R&D management. He has served TMS as a Cast Shop session chair twice and as the Cast Shop subject chair in 2014. He earned the Light Metals Award from TMS in 2001. Eddie has his B.S. and M.S. in mechanical engineering from Virginia Tech.

PROGRAM ORGANIZERS

ALUMINA AND BAUXITE

Paul McGlade is a metallurgical engineer, who after completing his master's degree, commenced work as a research engineer at the Comalco Research Centre in Melbourne Australia. Paul worked on the molten metal side of the business developing semi-continuous casting technology and providing technical assistance to Comalco's (now Rio Tinto's) cast houses in Australia, New Zealand and the US. Paul left Comalco and spent a considerable time as a key technical resource for the Australian Magnesium Corporation project to develop what would have been the world's largest magnesium production facility, before becoming an independent consultant to the aluminum and magnesium industries. Paul subsequently joined GHD and is now Manager – Aluminium. Most of Paul's work at GHD centers on the bauxite and alumina industry, especially the development of residue management areas, managing teams that plan and design both greenfield and brownfield facilities across the world.

ALUMINUM ALLOYS, PROCESSING AND CHARACTERIZATION

Zhengdong Long is Kaiser Technical Expert and Senior Alloy Development Engineer at Kaiser Aluminum in Spokane, Washington, United States. Dr. Long has a Ph.D. in Material Science and Engineering from China Central Iron and Steel Research and Institute in Beijing, and MBA from University of Kentucky in Lexington, Kentucky, USA. Dr. Long has been active for over a decade in the areas of physical and mechanical metallurgies of aluminum alloys and superalloys. Dr. Long's diverse experience includes new alloys and products development and applications, process development and optimization, product integration and quality improvement, and project management. His expertise and interests are casting and thermo-mechanical processing, static and dynamic material properties, corrosion behaviors, advanced material characterization as well as material and process modeling. He has numerous publications and patents.

ALUMINUM REDUCTION TECHNOLOGY

Stephan Broek graduated in Amsterdam in 1991 with a bachelor degree in Chemical Engineering. He also obtained a post-graduate diploma in Process Technology at Twente University. His forming years in his career were at the Hoogovens Steel Company where he worked as a process engineer at Hoogovens Technical Services. It was during this time that he was exposed to large scale flue gas desulphurization technologies. He learned this from some of the pioneers in this field. In 1997 he relocated to Canada to work in the aluminum product group. He also was a junior member of the North American management team. In 2003 he joined Hatch. The first major position was as director of technology commercialization. His keen interest in research and development exposed him to many different industries while with Hatch. In this role he led efforts in technology development, intellectual property management, fund raising, and also served as the representative from Hatch on the board of the Steel Research Center at McMaster University. His largest contribution was his leadership in the development of the N-Solv technology, a clean process for recovery of bitumen from tar sands. In 2009 he joined the Aluminium Center of Excellence in Hatch Light Metals. He started building a small team of technical specialists that dedicate their work to environmental projects in primary aluminum smelters worldwide. This has become a leading group in this field working for many aluminum companies. Today, Stephan operates from the Abu Dhabi offices of Hatch to serve the GCC smelters and to build strong relationships across the plants. Sharing his knowledge is something he is passionate about. He is a regular speaker and author at international conferences and is very much involved in TMS. He is a member-at-large of the Aluminum Committee.

CAST SHOP TECHNOLOGY: AN LMD SYMPOSIUM IN HONOR OF WOLFGANG SCHNEIDER

Mohamed Ibrahim Hassan Ali is assistant professor in the Mechanical and Material Engineering Department at Masdar Institute of Science and Technology, Abu Dhabi, UAE. Dr. Ali has an M.Sc. and Ph.D. in Mechanical Engineering. His Ph.D. was on the channel program between the University of Helwan, Cairo, Egypt and the University of Michigan, Ann Arbor, Michigan, USA. Dr. Ali has a Professional Engineer license in the state of Kentucky and served as a postdoctoral and senior research scientist before he joined Masdar Institute. Dr. Ali has multidisciplinary experimental and computational experience in a broad area of thermal and material science. He developed several CFD and FE models for several industrial and energy sustainability applications covering reverberatory furnaces design, DC casting, thermal-fluid science, fluid flow-structure interaction, thermo-electrical-mechanical modeling for anode assembly, industrial furnaces burner design, combustion and flames, fire safety and heat engines. His research and consulting work have been sponsored by several grants from the National Science Foundation (NSF), U.S. Department

of Energy, University of Kentucky (UK) Center of Aluminum Technology, UK Energy Research Center, Masdar Institute internal projects, Global foundation, Emiratis Global Aluminum (EGA) and Masdar/MIT flagship. Actively involved in many professional organizations including ASME, TMS, SAE, ICAE, Dr. Ali has over 100 journal and peer-reviewed conference publications and over 80 technical presentations in international conferences.

ELECTRODE TECHNOLOGY

Angelique N. Adams is the Director of Global Technology Development in the Smelting Center of Excellence at Alcoa. Dr. Adams holds a bachelor's degree in chemical engineering and doctorate in Fuel Science from Penn State. She has over 16 years of experience in aluminum manufacturing and has held roles in operations, technical support, and research and development. Since 2012, she has lead the development and execution of the innovation strategy for Alcoa's smelting operations, managing a multimillion dollar technology portfolio and a team of scientists and engineers located across the globe. In 2014, Dr. Adams was named one of six Rising Stars by STEMConnector.org, a consortium of over 110 companies, academic institutions, and government entities concerned with education and workforce development in Science, Technology, Engineering and Math (STEM). Dr. Adams is actively involved in her community and is passionate about children and STEM education. She currently serves on the Board of Directors for the Boys and Girls Club of the Tennessee Valley.

STRIP CASTING

Kai F. Karhausen is head of the research department Rolling Technology at the R&D Centre of Hydro Aluminium Rolled Products GmbH in Bonn, Germany. He graduated in Materials Science and received his Ph.D. in 1994 at the University of Technology RWTH Aachen. He joined the Hydro Aluminium R&D Centre in Karmøy, Norway and conducted research in the field of aluminum extrusion before moving to the R&D Centre in Bonn to work on rolling technology of aluminum. He is author of a large number of papers with a focus on metal forming technology, numerical simulation, and integration of material models into process simulation tools. He served as chair of the Aluminum Processing Committee of TMS for five years.

Dietmar Letzig is head of the department Wrought Magnesium Alloys at the Magnesium Innovation Centre of the Helmholtz Zentrum Geesthacht in Germany. He is the author of many papers in peer-reviewed scientific journals and proceedings of international conferences. His area of expertise is microstructure and mechanical properties of wrought magnesium alloys processed by severe production processes. He is leading fundamental as well as application-oriented research projects in metal forming processes and is an expert in the processing of magnesium sheet including twin roll casting and the optimization of the magnesium sheet performance.

Jan Bohlen is research scientist with the Magnesium Innovation Centre (MagIC) at the Helmholtz-Zentrum Geesthacht in Germany. He graduated in Material Physics in 1996 and obtained his Ph.D. from the University of Göttingen (Germany) in 2000. His work in the wrought magnesium alloys group at MagIC is focused on the development of new wrought magnesium alloys and understanding the effect of processing on microstructural development.

Murat Dündar holds the position of Director of Technology in Assan Aluminum. He joined Assan Aluminum in May 1999 as a research specialist. He has primarily focused on performance improvement of aluminum foil and sheet products produced out of Twin Roll Casting Technology (TRC), managing research and development projects on developing new alloys compatible with TRC technology, characterization of as-cast structures and related casting defects, interface between liquid metal and caster shell surface, solidification in TRC and related microstructures, tailoring microstructural features starting from casting and in further downstream operations, improvement in productivity of casting process, and finally casting of high Mg-bearing and 6000 series alloys with TRC. He holds a B.Sc. degree in Metallurgical Engineering from Middle East Technical University, Turkey, a M.Sc. degree in Materials Science from State University of New York at Buffalo, USA, and a Ph.D. degree in Materials Engineering from New Mexico Institute of Mining and Technology, USA.

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AN IMPROVED LIME SINTER PROCESS TO PRODUCE Al_2O_3 FROM LOW-GRADE Al-CONTAINING RESOURCES

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Keywords: Lime sinter process, Leaching, Sintering, $Na_4Ca_3(AlO_2)_{10}$, Al-containing resource

Abstract

Low-grade bauxites and non-bauxite resources are widely distributed, which are suitable to extract alumina by the lime sinter process. The application of the lime sinter process is restricted by the high consumption of calcium, the large amount of slag and the poor alumina leaching property of clinkers. An improved lime sinter process based on the dry-sintering is proposed by adding Na_2O-containing compounds to decrease CaO addition and increase alumina leaching property. The molar ratios of CaO to Al_2O_3 and Na_2O to Al_2O_3 are 0.6 and 0.4, respectively. $Na_4Ca_3(AlO_2)_{10}$ as well as trace amounts of $CaO•Al_2O_3$ and $12CaO•7Al_2O_3$ are formed after sintering at 1200°C for 30min in $CaO-Al_2O_3-Na_2O$ system, the alumina leaching efficiency of which is above 90%. The minerals after sintering in $CaO-Al_2O_3-SiO_2-Na_2O$ system include $Na_4Ca_3(AlO_2)_{10}$, $12CaO•7Al_2O_3$, $2CaO•SiO_2$ and $2CaO•Al_2O_3•SiO_2$. The leaching efficiency of clinkers increases as the sintering temperature increases to 1300°C.

Introduction

Low-grade bauxites and non-bauxite resources such as nepheline, fly ash and red mud are widely distributed in the world [1~3]. The methods that are widely used in the alumina production industry mainly include the Bayer process, the sinter process and the comprehensive application of the above two processes. The alumina extraction efficiency is very low when the Bayer process is used to deal with the low-grade resources [4~7]. It is suitable to extract alumina by the sinter process since the compounds containing SiO_2 transform to $2CaO•SiO_2$ (C_2S) [8~11]. The sinter processes mainly include the lime sinter process and the soda-lime sinter process. The economic benefit of the soda-lime sinter process is limited by the high energy consumption. The application of the lime sinter process is restricted by several defects, which include the high consumption of calcium, the large amount of slag and the poor leaching property of clinkers. The compounds in clinkers of the sinter processes which can decompose in the sodium carbonate and sodium hydroxide solution mainly include $CaO•Al_2O_3$ (CA), $12CaO•7Al_2O_3$ ($C_{12}A_7$) and $Na_2O•Al_2O_3$ (NA). Yu Hai-yan [12] et al. found that the doping of Na_2O in clinkers can improve the leaching efficiency of Al_2O_3. C. Ostrowski [13] et al. observed that CA can be transformed into $Na_4Ca_3(AlO_2)_{10}$ and $C_{12}A_7$ in the Na_2O-doped $CaO-Al_2O_3$ system. In addition, Tian Yong-pan [14] pointed out that the leaching efficiency of $Na_4Ca_3(AlO_2)_{10}$ is better than that of CA and $C_{12}A_7$.

An improved lime sinter process based on the formation of $Na_4Ca_3(AlO_2)_{10}$ with dry-sintering process is proposed by adding Na_2O-containing compounds to decrease CaO addition and increase alumina leaching property in this paper. The phase constitutions and microstructures of $Na_2O-CaO-Al_2O_3$ system and $Na_2O-CaO-Al_2O_3-SiO_2$ system clinkers were investigated. The leaching properties of these clinkers were also investigated.

Experimental

Materials

The leaching liquid was prepared with the analytically pure sodium carbonate, sodium hydroxide. The clinkers containing $Na_4Ca_3(AlO_2)_{10}$ were synthesized by analytically pure reagents $CaCO_3$, Al_2O_3, SiO_2, and Na_2CO_3.

Equipments

The mixed materials were sintered in a MoSi2 resistance furnace (KSL-1700-A2) at a heating rate of 8°C /min. The XRD data were collected on a Philips PW3040/60 with CuKα1 (λCu=1.54056 Å) radiation operated at 40KV and 40mA. The data was analyzed by MDI Jade. The microstructural analyses were carried out by SEM (SHIMADZU SSX-550) and EDS (DX-4). The chemical compositions of clinkers and residues were analyzed by XRF (XRF-1800).

Synthesis and analysis

The molar ratio of CaO to Al_2O_3 (the remaining CaO subtracted the composition to form C_2S in $Na_2O-CaO-Al_2O_3-SiO_2$ system) was 0.6 and the molar ratio of Na_2O to Al_2O_3 was 0.4. The materials were sintered at the range of 1100°C ~ 1300°C. The sintered clinkers were polished flat in water-free ethylene glycol using SiC papers and cleaned by ultra-sound in water-free ethylene glycol for SEM analysis.

The leaching experiments were carried out in a thermostatic water bath. The sintered clinkers were leached by the sodium carbonate and sodium hydroxide solution at 80°C. The liquid-to-solid ratio for leaching was 10. The concentrations of sodium carbonate and sodium hydroxide were 80 g/L and 15g/L (all in the form of Na_2O), respectively. The leaching residue was washed to neutrality with hot water and dried at 90°C. The alumina leaching efficiency of $Na_2O-CaO-Al_2O_3$ system clinkers and $Na_2O-CaO-Al_2O_3-SiO_2$ system clinkers were calculated according to formula (1) or formula (2), respectively:

$$\eta_{A0} = \left(1 - \frac{(A/C)_{residue}}{(A/C)_{clinker}}\right) \times 100\% \quad (1)$$

$$\eta_{A0} = \left(1 - \frac{(A/S)_{residue}}{(A/S)_{clinker}}\right) \times 100\% \quad (2)$$

where $(A/C)_{residue}$ and $(A/C)_{clinker}$ are the mass ratio of Al_2O_3 to CaO of leached residues and sintered clinkers, respectively.

$(A/S)_{residue}$ and $(A/S)_{clinker}$ are the mass ratio of Al_2O_3 to SiO_2 of leached residues and sintered clinkers, respectively. η_{AO} is the leaching efficiency of alumina.

Results and discussions

Phase formation

Part of the clinkers was taken out the furnace during the process of the increasing sinter temperature from 1100°C to 1300°C. The XRD patterns of the Na_2O-CaO-Al_2O_3 system clinkers are shown in Fig. 1. The clinkers mainly contain $Na_4Ca_3(AlO_2)_{10}$. Only trace amount of CA and $C_{12}A_7$ exist in the clinkers. Small amount of CaO exists in the clinker sintered at 1100°C and then reacts absolutely with the temperature increases to 1200°C. According the intensity of the characteristic peak of $C_{12}A_7$ at 33.2°, it can be seen that the content of $C_{12}A_7$ is the least when the sinter temperature is 1200°C. But another characteristic peak appears at 25°. It cannot be confirmed which phase it belongs to. Hence, it may be explained that the crystal structures of phases in clinkers are not stable. For that reason, the clinkers are sintered at 1200°C for 10min and 30min, the XRD patterns of which are shown in Fig. 2. The content of $C_{12}A_7$ increases apparently when the clinker is sintered for 10min and then decreases when the sinter time extends to 30min. The characteristic peaks of $Na_4Ca_3(AlO_2)_{10}$ in this paper are almost the same with the results reported by C. Ostrowski [13], while they are different from the information on the card of PDF#02-1003. The compositions of the clinkers sintered at 1200°C for 30min are shown in Table 1. The calculated C/A ratio and N/A ratio are 0.62 and 0.42, respectively. The values are almost the same with the C/A ratio and N/A ratio of $Na_4Ca_3(AlO_2)_{10}$, which confirms the formation of $Na_4Ca_3(AlO_2)_{10}$.

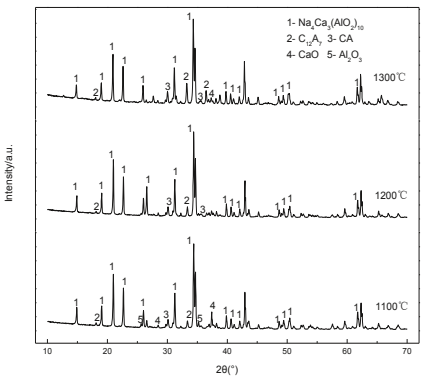


Fig.1 XRD patterns of Na_2O-CaO-Al_2O_3 system clinkers when heated to different temperatures

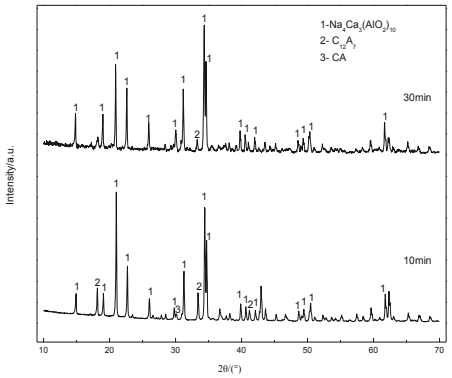


Fig.2 XRD patterns of Na_2O-CaO-Al_2O_3 system clinkers sintered at 1200°C for different times

Table 1 Compositions of clinkers sintered at 1200°C for 30min/%

Na_2O	Al_2O_3	CaO
16.04	62.62	21.34

According to the analysis results of Na_2O-CaO-Al_2O_3 system clinkers, the Na_2O-CaO-Al_2O_3-SiO_2 system clinker is sintered at 1200°C for 30min. The result is shown in Fig. 3. $C_{12}A_7$, CA, and $Na_4Ca_3(AlO_2)_{10}$ are also found in the clinkers. The compounds containing SiO_2 include $2CaO{\cdot}Al_2O_3{\cdot}SiO_2$ (C_2AS), β-C_2S and α-C_2S. Comparing with the Na_2O-CaO-Al_2O_3 system clinkers, the new compound containing Al_2O_3 is C_2AS and part of CaO and Al_2O_3 still remain in the clinkers. Hence, the clinkers are sintered at 1200°C, 1300°C and 1350°C for 60min. The compounds are shown in Fig. 4. The compositions of clinkers are shown in Table 2. Al_2O_3 reacted absolutely with the increasing of the sinter temperature and time. But small amount of CaO still exists in the clinkers sintered at 1200°C and 1300°C. According to the change of the intensity of the characteristic peak of CaO, The content of CaO decreases obviously when the temperature increases to 1350°C. Trace amount of $C_{12}A_7$ exists in clinkers sintered at 1200°C and 1300°C, while the content of $C_{12}A_7$ increases apparently when the sinter temperature is 1350°C. The reason may be the decreasing of Na_2O content in the clinkers. The intensity of the strongest peak decreases apparently with the decreasing of Na_2O content, which suggest the decline of the content of $Na_4Ca_3(AlO_2)_{10}$ in clinkers sintered at 1350°C.

Table 2 Compositions of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers after sintering for different temperatures/%

Temperature/°C	Na_2O	Al_2O_3	SiO_2	CaO
1150	7.86	31.94	16.33	41.00
1200	7.21	31.06	16.68	40.88
1250	6.74	32.66	16.76	42.04
1300	6.08	33.34	16.78	42.31
1350	2.45	34.33	18.11	45.11

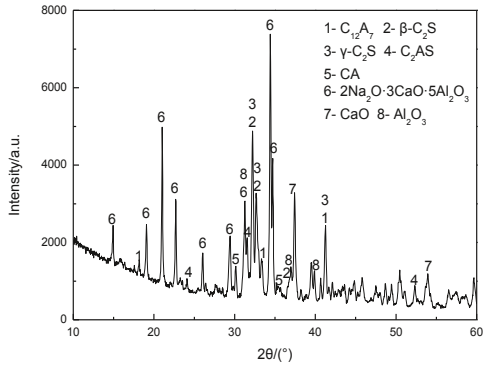


Fig.3 XRD patterns of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers sintered at 1200℃ for 30min

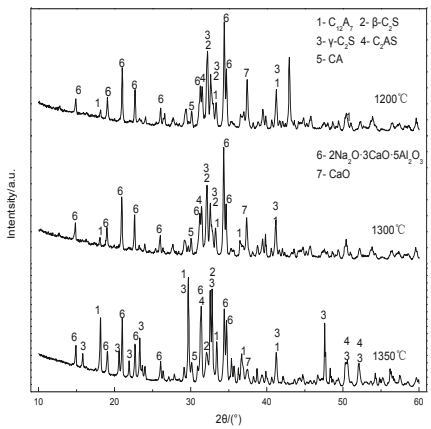
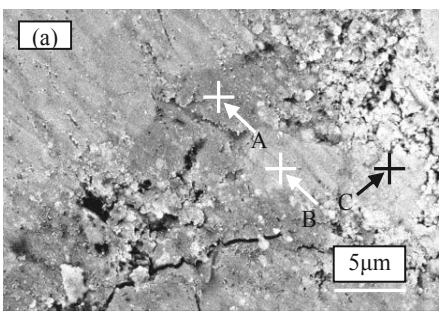


Fig.4 XRD patterns of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers sintered at different temperatures for 60min

Microstructure analysis

The materials were pressed into cylindrical clinkers. Part of the clinkers were taken out the furnace when the temperature increased to 900°C. The rest are sintered at 1350°C for 60min. The surfaces of the clinkers were polished for the SEM analysis to see the distribution characteristics of phases. Microstructures of clinkers are shown in Fig. 5 and Fig. 6. It can be seen that the gray scales of different regions are different from each other. The analysis results of EDS are given in Table 3.



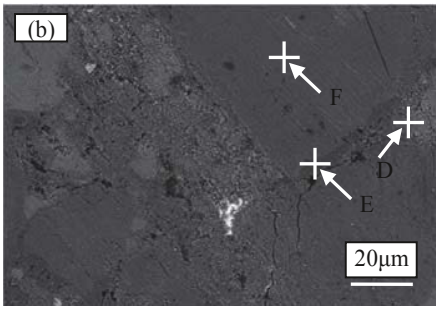
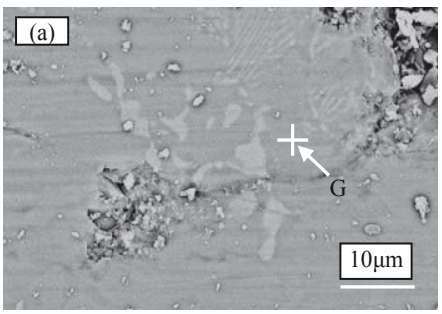


Fig.5 BSE image of sintered Na_2O-CaO-Al_2O_3-SiO_2 system clinkers heat-treated to 900°C



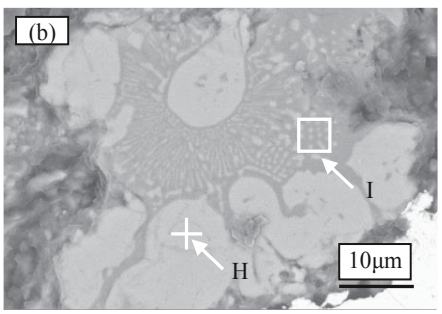


Fig.6 BSE image of sintered Na_2O-CaO-Al_2O_3-SiO_2 system clinkers heat-treated at 1350°C for 60min

Table 3 Atomic percentages of clinkers corresponding to Fig. 5 and Fig. 6/%

Point	O	Al	Ca	Na	Si
A	57.22	21.54	6.60	7.32	7.32
B	53.76	5.78	20.33	4.82	15.30
C	52.27	1.78	24.61	1.17	12.17
D	56.77	2.11	18.07	7.05	16.00
E	58.82	36.01	3.80	1.16	0.21
F	53.74	29.63	1.96	14.67	~0
G	51.71	25.58	18.76	3.51	0.45
H	55.92	0.46	27.77	1.57	14.28
I	52.20	19.43	13.74	9.58	5.04

As shown in Table 3, the content of Al_2O_3 in point A is the highest while the contents of Na_2O, CaO and SiO_2 do not have much difference. The regions represented by B, C and D are the reaction regions of CaO and SiO_2. The region represented by point F mainly contains Na_2O and Al_2O_3. Moreover, the molar ratio of Na_2O to Al_2O_3 is 0.5, while the average molar ratio of Na_2O to Al_2O_3 of the clinker is 0.4. The region represented by E, which distributes at the edge of region F, mainly contains Al_2O_3.

Hence, it can be concluded that Na_2CO_3 mainly diffuses into the region of Al_2O_3. There two kinds of reaction zones, which have distinctive characteristics. The first kind is the reaction zones between Na_2O, CaO and Al_2O_3 while small amount of SiO_2 exist in the regions, which is represented by E and F. The second kind is the reaction between CaO and SiO_2, which is represented by B, C, and D.

The phase distribution features of clinkers sintered at 1350°C are shown in Fig. 6. According to the EDS results, the region of point G is constituted by $C_{12}A_7$ and $Na_4Ca_3(AlO_2)_{10}$, which corresponds to the first kind of the reaction zones. The $C_{12}A_7$ and $Na_4Ca_3(AlO_2)_{10}$ coexist with each other in the Na_2O-CaO-Al_2O_3-SiO_2 system and Na_2O-CaO-Al_2O_3 system. Comparing the microstructures of clinkers of the different reaction stage, it can be concluded that $Na_4Ca_3(AlO_2)_{10}$ is formed though the direct reactions of Na_2O, CaO and Al_2O_3. The region represented by point H is the distribution of C_2S corresponding to the second reaction zones. The phases distributed in the region of I are NA, C_2S and $Na_4Ca_3(AlO_2)_{10}$, which presents a grid-like distribution feature.

Phase contents and leaching efficiency

The leaching time of pure $Na_4Ca_3(AlO_2)_{10}$ is 5min. The contents of CaO and Al_2O_3 are shown in Table 4. The leaching efficiency, which is calculated according to formula (1), is 95.85%. Therefore, the leaching property of $Na_4Ca_3(AlO_2)_{10}$ is perfect.

Table 4 Compositions of leaching residues of $Na_4Ca_3(AlO_2)_{10}$/%

Al_2O_3	CaO	η_{AO}
5.99	50.45	95.85

The Na_2O-CaO-Al_2O_3-SiO_2 system clinkers were leached for 120min. The compositions of leached residues are shown in Table 5. The leaching efficiency, which is calculated according to formula (2) and the contents of Al_2O_3 and SiO_2, are shown in Fig. 7. It can be seen that the leaching efficiency increases with increasing sinter temperature when the temperature is under 1300°C. The maximum is about 80% when sinter temperature is 1300°C. Due to the declines of the $Na_4Ca_3(AlO_2)_{10}$ content, the leaching efficiency of clinkers sintered at 1350°C decreases to 70%. As shown in Fig. 8, almost all the $C_{12}A_7$ and $Na_4Ca_3(AlO_2)_{10}$ can be leached absolutely when the leaching time of sintered clinkers is 120min and the compound containing Al in residues is C_2AS. Therefore, the formation of C_2AS is the reason why the leaching efficiency of the Na_2O-CaO-Al_2O_3-SiO_2 system clinker is lower than the pure $Na_4Ca_3(AlO_2)_{10}$. More experiments are needed to be done to decrease the content of C_2AS.

Table 5 Compositions of leaching residues of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers/%

Temperature/°C	Na_2O	Al_2O_3	SiO_2
1150	2.30	10.79	19.86
1200	2.02	12.29	23.78
1250	1.20	10.92	22.04
1300	0.65	8.61	20.77
1350	0.30	15.14	18.29

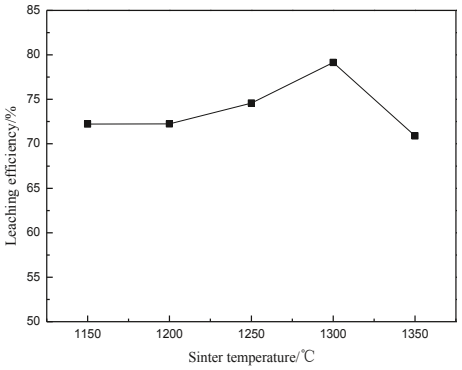


Fig.7 Leaching efficiency of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers

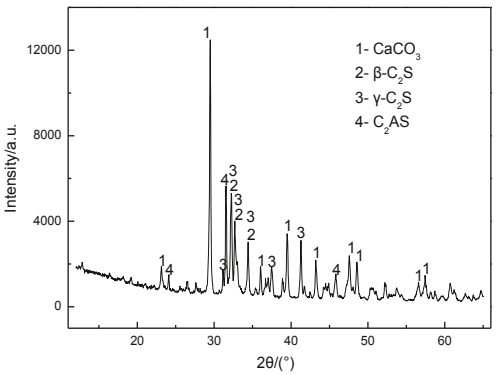


Fig.8 XRD patterns of leached residues of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers sintered at 1300°C

Conclusions

1. $Na_4Ca_3(AlO_2)_{10}$ with trace amount of CA and $C_{12}A_7$ can be formed when the mixtures of Na_2CO_3, $CaCO_3$ and Al_2O_3 are sintered at 1200°C for 30min. The formation process of $Na_4Ca_3(AlO_2)_{10}$ mainly involves the transformation reactions with $C_{12}A_7$. The leaching efficiency is above 95%.

2. There are two distinctive reaction zones in the microstructures of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers, which are the reactions forming $Na_4Ca_3(AlO_2)_{10}$ between Na_2O, CaO and Al_2O_3 and the reactions forming C_2S between CaO and SiO_2.

3. The leaching efficiency of Na_2O-CaO-Al_2O_3-SiO_2 system clinkers can be about 80% when the sinter temperature is 1300°C while the C/A ratio and N/A ratio are 0.6 and 0.4, respectively. The formation of C_2AS is the reason why the leaching efficiency is lower than the pure $Na_4Ca_3(AlO_2)_{10}$.

Acknowledgment

The authors greatly acknowledge the financial supports from the National Nature Science Foundation of China (No. 51174054, 51374065).

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INVESTIGATION OF FLOTATION BEHAVIORS OF REFRACTORY HIGH-SILICA BAUXITE

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Keywords: High-silica bauxite; Flotation; Al_2O_3 recovery; A/S

Abstract

High content of silica results in high soda consumption, which limits the use of high-silica bauxite in the production of Bayer alumina. In this work,the main flotation parameters of high silica bauxite (A/S 5.0) including particle size, pulp pH, collector dosage and dispersant dosage were studied.With an optimal condition of 33% solid density, particle size of -74 μm above 93%, sodium oleate dosage 2200 $g \cdot t^{-1}$, Na_2CO_3 dosage 2000 $g \cdot t^{-1}$, sodium hexametaphosphate dosage 30 $g \cdot t^{-1}$, slurry pH 9.4 and agitation speed 2280 rpm, the recovery of aluminum minerals of 56%, the A/S of concentrate of 12.2, and the separation index of 0.4 can be obtained.

Introduction

Bauxite resource in China is mainly diasporic type with a fatal disadvantage of low Al_2O_3 to SiO_2 (A/S) mass ratio ranging from 4 to 6[1, 2], which cannot meet the minimum requirement for the Bayer process. Flotation desilication is a low cost method to increase the A/S compared to other measures, such as biological processing and chemical processing[3]. Therefore, it is particularly useful for separating finely ground ores from their associated gangue minerals, so as to apply the Bayer process[4].

Bauxite beneficiation belongs to the typical fine sludge system, with fine disseminated grain size and easy argillization[5].The study of micro-scale bauxite efficient sorting technology to further optimize production processes and reduce production costs has become a hot bauxite beneficiation research technology. As diaspore dissemination size is relatively small, most between 0.01-0.06mm, grinding fineness is required to ensure the effective dissociation[6,7]

In this work, a kind of diaspore was used as raw materials and its microstructure was investigated by the Polarization Microscope. Flotation experiments were carried out to separate silicate minerals from diaspore.

Experiments

Materials

Bauxite The ores used in this paper were collected from Henan province in China. And the main chemical compositions of ore were listed in Table I, while the XRD pattern of bauxite was shown in Figure 1.

Table I. Chemical Analysis of Actual Mineral (wt.%).

Al_2O_3	SiO_2	Fe_2O_3	Na_2O	K_2O	MgO	CaO	S	TiO_2	LOI
69.26	13.85	1.61	0.17	2.15	0.14	0.25	0.1	2.08	8.22

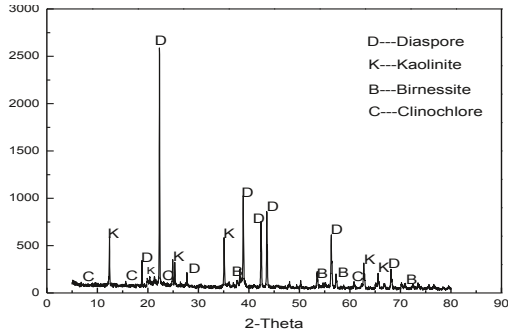
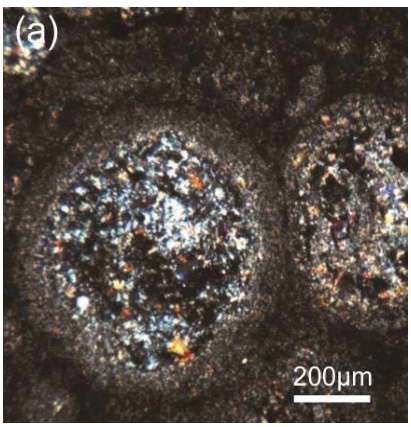


Figure 1. XRD pattern of bauxite

Seen from Table I, the content of Al_2O_3 and SiO_2 were 69.26%, 13.85%, respectively. A/S of this sample was 5.0. At the same time, it can be concluded that the major gangue minerals in bauxite were kaolinite, clinochlore and birnessite based on Figure 1.

Besides, the microstructure of minerals in bauxite was investigated by Axio Scope A1+J&M200 and results were given in Figure 2. Seen from Figure 2, it can be known that the diaspore grains were mixed with kaolinite of dispersion state, which constituted the main base of the ore. Better crystallization of diaspore can be seen in Figure 2 (a) compared with Figure 2 (b). Diaspore was of small crystal size, most between 0.01-0.06 mm. Aggregate were often shaped with beans granular, oolitic and spindle, the diameter of beans oolitic aggregate was generally between 0.06-2 mm.



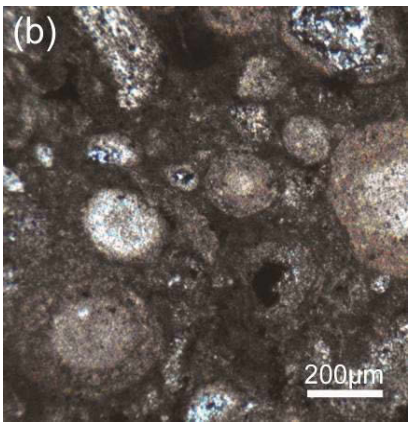


Figure 2. Diaspore aggregates of oolitic, flakes (+)

Other Materials Sodium oleate used as collector was provided by Aladdin Biotech Shares in Shanghai Province, China. The sodium hexametaphosphate used as dispersant was brought by Kermel Chemical Reagent Co. Ltd in Tianjin, China. Solutions of HCl and NaOH were used to adjust the pH of the system.

Flotation Tests

The particles below 1 mm were put into RK/ZQM（BM）ball mill after crushing, screening, and grinding concentration was 60% with 300 g ore and 200 g water. Flotation experiments were performed in flotation machine (RK/FD III type single flotation cell, China) with the effective cell volume of 0.5L and the agitation speed of 2280 rpm.

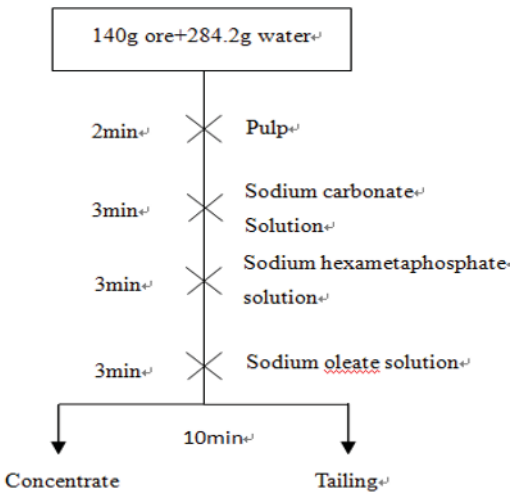


Figure 3. Flotation flow chart for this work

Flow chart of flotation was illustrated in Figure 3. The floated and unfloated fraction were collected, dried, and weighed to calculate the recovery and the yield for the flotation of the actual minerals. Besides, the floated and unfloated actual minerals were separately sampled to analyze the contents of Al_2O_3 and SiO_2 for calculating the A/S and the Al_2O_3 recovery. The value of Al_2O_3 recovery is calculated from this equation:

$$\varepsilon=\gamma\cdot\beta/\alpha \quad (1)$$

Where ε is Al_2O_3 recovery, γ is the yield of concentrate, α and β represent the Al_2O_3 grade of raw ore and concentrate, respectively.

Results and Discussions

Effect of Particle Size

Generally, the content of particles below 74 μm is an index to express the particle size in floatation experiment. In this paper, effect of particle size on flotation was researched and results were shown in Figure 4. Experiment was carried out with the condition of Na_2CO_3 dosage 2000 $g\cdot t^{-1}$ (pH=9.4), sodium oleate dosage 1600 $g\cdot t^{-1}$, sodium hexametaphosphate dosage 40 g·t−1, solid density 33%, agitation speed 2280 rpm, temperature 30°C.

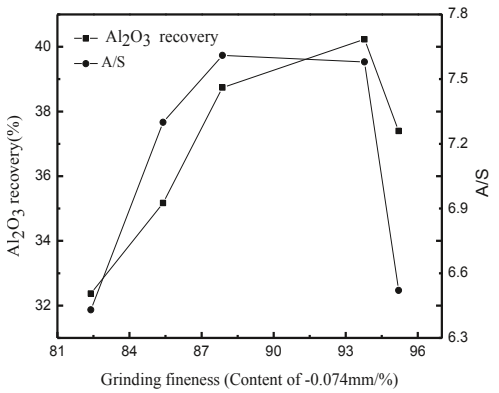


Figure 4. Effect of particle size on floatability

Al_2O_3 recovery and A/S increased first and then had a reduction with increasing content of -74 μm. A/S rose up to maximum, corresponding Al_2O_3 recovery was 38.74% when content of -74 μm reached 87.87%. While Al_2O_3 recovery reached the highest value, and corresponding A/S was 7.58 when content of -74μm reached 93.78%. Considering these two factors, it was appropriate to keep the content of the particle size (-74μm) as 93.78%.

Effect of Pulp pH

Effect of pulp pH on Al_2O_3 recovery and A/S was shown in Figure 5. And experimental conditions were fixed as sodium oleate dosage 1600 $g\cdot t^{-1}$, sodium hexametaphosphate dosage 40 $g\cdot t^{-1}$, solid density 33%, agitation speed 2280 rpm, temperature 30°C, content of the particle size (-74μm) 93.78%.

When pH value was increased from 8.5 to 11.0, Al_2O_3 recovery rose first and then fell, Al_2O_3 recovery went up to the maximum at pH 9.4. A/S dropped first and then increased, there is a later decrease when pH>9.4. When pH was 8.5, maximum of A/S can be obtained, while Al_2O_3 recovery was much lower compared to the Al_2O_3 recovery at pH 9.4. Therefore, the optimal pH was 9.4.

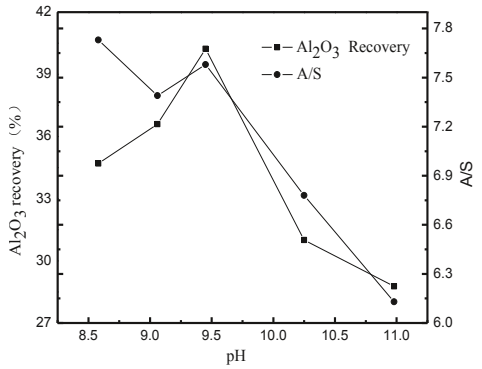


Figure 5. Effect of pH on floatability

Effect of Dispersant Dosage

When sodium hexametaphosphate dosage was changed from 20 $g \cdot t^{-1}$ to 60 $g \cdot t^{-1}$, effect of sodium hexametaphosphate dosage on Al_2O_3 recovery and A/S was investigated. Experimental results were shown in Figure 6. Flotation experiment was performed with the condition of pH 9.4, sodium oleate dosage 1600 $g \cdot t^{-1}$, solid density 33%, agitation speed 2280 rpm, temperature 30°C, content of the particle size below 74μm 93.78%.

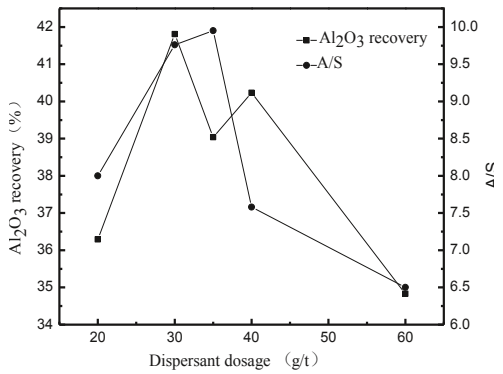


Figure 6. Effect of dispersant dosage on floatability

With the increase of the amount of sodium hexametaphosphate, A/S increased and then decreased. Al_2O_3 recovery changed intricately, the recovery improved significantly with dosage of 30 $g \cdot t^{-1}$, and corresponding A/S was 9.8. Though A/S rose up to maximum when hexametaphosphate dosage is 35 $g \cdot t^{-1}$, Al_2O_3 recovery was low. Therefore, the optimal hexametaphosphate dosage was 30 $g \cdot t^{-1}$.

Effect of Collector Dosage

When sodium oleate dosage was changed from 1400 $g \cdot t^{-1}$ to 2400 $g \cdot t^{-1}$, effect of collector on Al_2O_3 recovery and A/S was investigated. Experimental results were shown in Figure 7. Flotation experiment was performed with the condition of pH 9.4, sodium hexametaphosphate dosage 30 $g \cdot t^{-1}$, solid density 33%, agitation speed 2280 rpm, temperature 30°C, content of the particle size(-74μm) 93.78%.

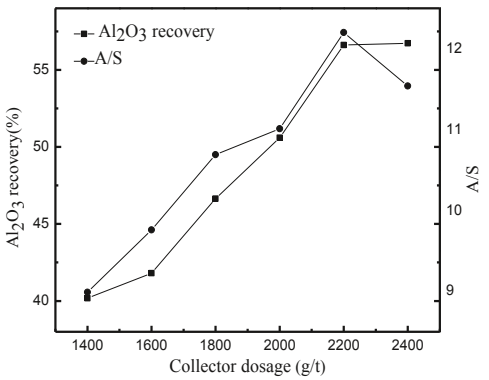


Figure 7. Effect of collector dosage on floatability

With the increase of sodium oleate dosage, both the Al_2O_3 recovery and A/S gradually increased when sodium oleate dosage was lower than 2200 $g \cdot t^{-1}$. A/S had a slight decrease and the recovery stayed almost constant when the amount of sodium oleate was more than 2200 $g \cdot t^{-1}$. So, the optimal sodium oleate dosage was 2200 $g \cdot t^{-1}$.

Conclusions

The main flotation parameters of high silica bauxite (A/S =5.0) including particle size, pulp pH, dispersant dosage and collector dosage were studied. The flotation optimum conditions were concluded as follows: the content of particles -74μm above 93%, sodium oleate dosage 2200 $g \cdot t^{-1}$, Na_2CO_3 dosage 2000 $g \cdot t^{-1}$, sodium hexametaphosphate dosage 30 $g \cdot t^{-1}$. The bauxite concentrate with A/S of 12.2 and the recovery of Al_2O_3 of 56% were obtained from the low grade bauxite ore under the conditions. Laboratory test showed that it was technically feasible to separate diaspore from bauxite ore, foam concentrate with A/S above 7.0 can be obtained for low grade bauxite using one stage flotation.

Acknowledgments

The financial supports from the National Science Fund for Young Scholars of China (No.51404213 and No.51404214) and the Development Fund for Outstanding Young Teachers of Zhengzhou University (No.1421324065) are gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Study on effective extraction of Al and Fe from high-iron bauxite through "calcification-carbonization" method

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Keywords: High-iron bauxite, Calcification-carbonization, Direct reduction, Novel red mud

Abstract

With the rapid development of China alumina production, efficient utilization of plentiful low grade high-iron bauxite has received widespread attention in the aluminum industry. Based on this, "calcification-carbonization-direct reduction" method is proposed for extracting effectively valuable metals Al and Fe from high-iron bauxite. The research results indicate that: after the calcification-carbonization process for high-iron bauxite, leaching rate of Al_2O_3 can reach 83.67%, higher than extraction rate of Al_2O_3 through Bayer method. Then, direct reduction of calcification-carbonization residue is investigated. Metallization ratio of Fe is 80.5% when reduction temperature and time is 1050℃ and 2h respectively. Moreover, metallization ratio of Fe can reach 93.6% with the addition of additive A. After the "calcification-carbonization-direct reduction" process, valuable metals Al and Fe can be extracted effectively from high-iron bauxite. The main component of final residue is $CaCO_3$ and $CaSiO_3$, which can be directly applicable to the cement industry.

Introduction

With the rapid development of alumina industry in China, high quality bauxite ore in China has been exhausting [1-2]. Unfortunately, most foreign country started to restrict their export of bauxite ores to China since 2013, so China's alumina industry is facing to a serious situation. However, there are a lot of low grade bauxite ores in China, especially the high-iron bauxite, which is abundant in the province of Guangxi, Yun nan, Shan xi, and He nan. Therefore, efficient utilization of the plentiful low grade high-iron bauxite has received widespread attention in China [3-4].

The high-iron bauxite in China, containing high iron, high silica, low aluminum, and low A/S, is more than 1.5 million tons. However, it has not been widely used in China's aluminum industry, because the iron or alumina grade cannot reach their industrial requirement respectively [5]. Therefore, it is not economical to use them as bauxite ore or iron ore in industry respectively.

Owing to the isomorphism of iron and aluminum, iron and aluminum are disseminated each other in fine granularity, cemented together [6-7]. Therefore, it is difficult to separate and recover the iron and alumina respectively. At present, many methods are studied on the high-iron bauxite research. However, these methods usually suffer low dissociate rate, low comprehensive recover efficiency, high energy cost, serious equipment corrosion and environment pollution [8-11]. To make good use of the high-iron bauxite and extract effectively valuable metals Al and Fe from high-iron bauxite, a novel method of "calcification-carbonization-direct reduction" is proposed in this paper [12-13]. In the new method, all the silica and most of alumina are concentrated in the calcium-alumina-silica compound (hydrogarnet) in the calcification process, then the calcification slag is carbonized with CO_2 and finally the carbonation slag is treated in low alkali solution at low temperature. After the above processes, the alumina is extracted in the alkali solution, the silica in the form of calcium silicate and iron is concentrated in the slag, which will be reduced directly to recover iron. In the end, a novel red mud with low iron, low aluminum which can be used as ceramic materials directly is received. All the valuable elements can be used through the whole processes.

Experimental

Materials

The high-iron bauxite ores used in these experiments originates from the Yunnan Wenshan deposits in China. The mineral composition and phase analysis of the bauxite ore were analyzed, and the results were shown in Figure 1 and Table 1. The analysis shows that the bauxite has a low alumina content, high iron content and low A/S. The contents of Al_2O_3 and SiO_2 are 46.25% and 17.43%, respectively, resulting in a bauxite A/S of 2.65. The content of Fe_2O_3 is 17.86%. Therefore, Wenshan bauxite is a kind of high-iron bauxite. The main minerals of the Wenshan bauxite include diaspore (AlOOH), Fe_2O_3, aluminum silicon compound and titanium dioxide (TiO_2).

Table 1. Chemical composition of Wenshan high-iron bauxite (wt, %)

Composition	Al_2O_3	SiO_2	Fe_2O_3	TiO_2	CaO	Na_2O	L.O.I
Content, %	46.25	17.43	17.86	3.42	0.19	0.70	14.27

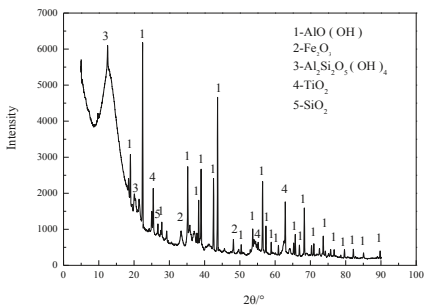

Figure 1. XRD phase analysis of Wenshan high-iron bauxite

Table 2 shows the composition of the reducing agent coal used in the reduction process of carbonization slag.

Table 2. Composition of reduction coal (wt, %)

Composition	C	volatile	Fe_2O_3	ash	Water	Al_2O_3	SiO_2
Content, %	29.05	56.28	0.53	9.04	5.63	1.78	5.26

Experimental Procedure

In this paper, the whole process of treating the high-iron bauxite is shown in Figure 2.
The experiments include three parts: calcification, carbonization, and direct reduction. The calcification and carbonization processes are taken in an autoclave, and the chemical reactions are as follows:

Calcification process:

$$Al_2O_3\cdot 2SiO_2\cdot 2H_2O+Al_2O_3\cdot H_2O+3CaO+H_2O\rightarrow 3CaO\cdot Al_2O_3\cdot xSiO_2\cdot(6-2x)H_2O \quad (1)$$

Carbonization process:

$$3CaO\cdot Al_2O_3\cdot xSiO_2\cdot(6-2x)H_2O+(3-2x)CO_2\rightarrow xCa_2SiO_4+(3-2x)CaCO_3+2Al(OH)_3+(3-2x)H_2O \quad (2)$$

The reactions show that main compositions of the carbonization slag are Ca_2SiO_4, $CaCO_3$, and $Al(OH)_3$. Then the $Al(OH)_3$ can be extracted by low alkali solution and the Ca_2SiO_4 and $CaCO_3$ are remained in the slag. In order to recover iron from the carbonization slag, a reduction process by the reducing coal was taken in a baking furnace.
After drying, all the slag samples produced in different processes were examined by XRD to identify the various compounds and by XRF to calculate the chemical compositions.

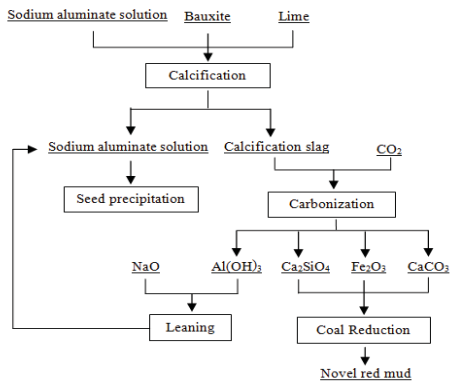

Figure 2. Process chart of calcification carbonization method for high-iron bauxite

Results and discussion

Calcification

The raw materials were calcified in an autoclave, and the experimental conditions were: calcification temperature was 260 ℃, original N_k (concentration of caustic sodium) was 240 g/L, molecular ratio was 1.5, Calcification time was 2h, mass ratio (Ca/Si) was 3: 0.64, agitation speed was 300 r/min. After calcification, the raw materials were analyzed, and the results were shown in Table 3 and Figure 3.
As shown in Figure 3 and Table 3, andradite-grossular hydrogarnet is formed after calcification. Most of the alumina is in the andradite-grossular hydrogarnet phase; Meanwhile a part of iron exist in the andradite-grossular hydrogarnet phase, others is still remained in the form of hematite; And the excessive CaO is in the form of $Ca(OH)_2$ by the reason of hydrolysis during the process. In addition, the content of Na_2O in the calcification slag is only 0.35%, which is far lower than the red mud produced by Bayer method.

Table 3. Chemical composition of calcification slag (wt, %)

Composition	Al_2O_3	SiO_2	TFe	TiO_2	CaO	Na_2O	L.O.I
Content, %	19.87	10.27	8.40	2.01	38.95	0.35	19.56

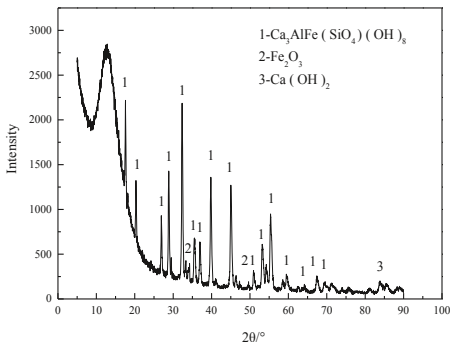

Figure 3. XRD phase analysis of calcification slag

Carbonization and alumina extraction

The slag produced in the calcification was then carbonized in an autoclave, under the experimental conditions: carbonization temperature was 120 ℃, carbonization time was 2h, CO_2 pressure was 1 Mpa, and agitation speed was 300 r/min. To extract alumina, the slag was treated in a water bath kettle after the carbonization process with the conditions: water temperature was 60 ℃, reaction time was 1 h, the concentration of NaOH was 100 g/L, L/S was 12:1, and the agitation speed was 300r/min. Table 4 and Figure 4 showed the chemical composition and XRD patterns of the alumina extraction slag respectively.
As shown in the XRD patterns, after carbonization and alumina extraction, the main mineral phases in the slag are $CaCO_3$ and andradite-grossular hydrogarnet. And the relative extraction rate of Al_2O_3 can be calculated as follow.

$$\eta=\frac{(A/S)_{ore}-(A/S)_{mud}}{(A/S)_{ore}-1}\times 100\%=83.67\%$$

Table 4. Chemical composition of carbonization slag

(wt, %)

Composition	Al_2O_3	SiO_2	TFe	TiO_2	CaO	Na_2O	L.O.I
Content, %	12.27	9.62	8.36	1.99	39.73	0.07	27.50

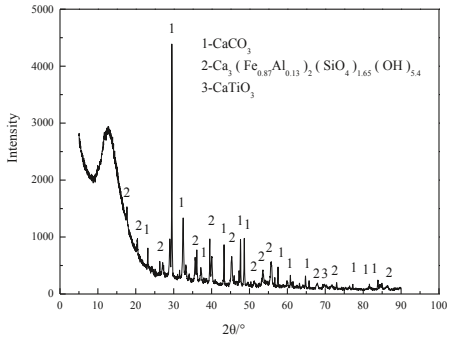

Figure 4. XRD phase analysis of carbonization slag

Comparing with the Bayer method, the relative extraction rate of alumina by "calcification carbonization" method is higher, which can be reached to 83.67%. This is because the alumina of hydrogarnet in "calcification-carbonization" method can be extracted efficiently. Therefore, the total extraction ratio of Al_2O_3 from raw materials is higher. From Table 3, the content of Na_2O can be reduced to 0.07%, but the content of iron is still high, it is necessary to recover iron from the slag. Due to the isomorphism phenomenon in the high-iron bauxite, the iron and alumina are always nested together for their similar chemical properties. Therefore, after calcinations and carbonation, the effective extraction of alumina can destroy the coexistence relationship of iron and alumina, which make it easy to extract iron from the slag. In the following, extraction of iron from the above slag by reduction process was investigated.

Iron reduction

Effect of reduction time. When reduction temperature was 1050 ℃, effect of reduction time on reduction behavior of Fe in carbonization slag was investigated. The experimental results were shown in Figure 5 and Figure 6.

While reduction temperature and time was 1050 ℃ and 2h, XRD phase analysis of reduction product was shown in Figure 5. The results indicated that main phase was CaO, metal Fe and Ca_2SiO_4. Obviously, Fe_2O_3 and andradite-grossular hydrogarnet in carbonization slag has been reduced to metal Fe by coal reduction.

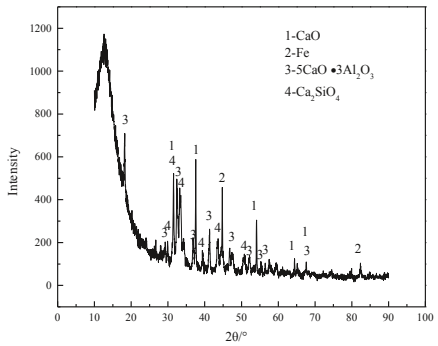

Figure 5. XRD phase analysis of reduction product

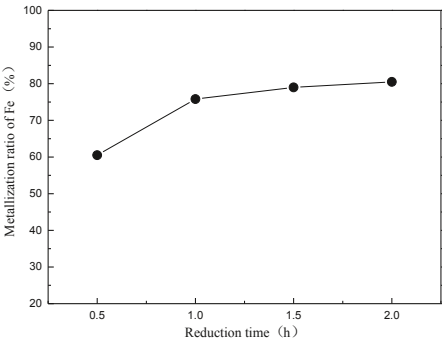

Figure 6. Effect of reduction time on reduction behavior of Fe

With the increase of reduction time, metallization ratio gradually raised from 60.5% at 0.5h to 80.5% at 2h, as shown in Figure 6. Therefore, reduction time has an obvious influence on the reduction behavior of Fe, and extension of reduction time contributes to the reduction of Fe in carbonization slag.

Effect of additive A on reduction behavior of Fe

When reduction temperature and time were 1050 ℃, 1 h respectively. The effect of additive A dosage on reduction behavior of Fe in carbonization slag was investigated. The experimental results were shown in Figure 7.

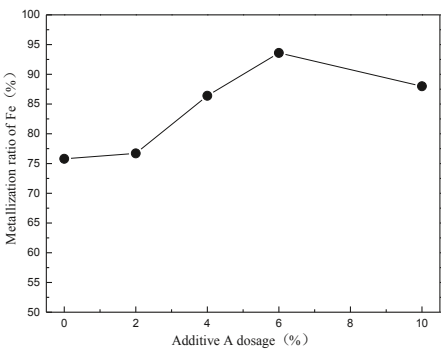

Figure 7. Effect of additive A dosage on reduction behavior of Fe

The results in Figure 7 showed that, Fe metallization ratio of reduction product without additive A was 75.8%. With the increase of additive A dosage, metallization ratio has a obvious increase. When additive A dosage increased to 6%, metallization ratio of reduction product reached the maximum of 93.6%. Meanwhile, the corresponding XRD phase analysis was studied in Figure 8. The results indicated that the main phase was CaO, metal Fe and complex calcium compound.

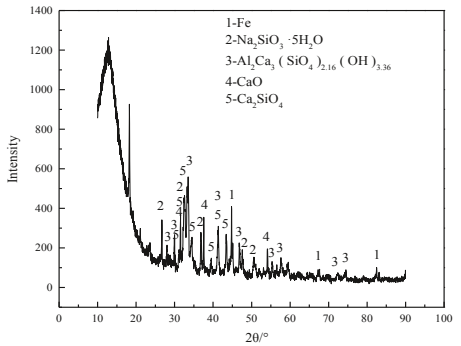

Figure 8. XRD phase analysis of reduction product with additive

Conclusion

1) After the calcification-carbonization process for high-iron bauxite, leaching rate of Al_2O_3 can reach 83.67%, which is higher than extraction rate of Al_2O_3 through Bayer method.

2) Then, direct reduction of "calcification carbonization" residue is investigated. Metallization ratio of Fe is 80.5% when reduction temperature and time are 1050℃ and 2h respectively. Moreover, metallization ratio of Fe can reach 93.6% with the addition of additive A.

3) After the "calcification-carbonization-direct reduction" process, valuable metals of Al and Fe can be extracted effectively from high-iron bauxite. The main component of final residue is $CaCO_3$ and $CaSiO_3$, which can be directly applicable to the cement industry.

Acknowledgement

This research was supported by the National Natural Science Foundation of China (U1202274, 51504059), the Research Fund for the Doctoral Program of Higher Education of China (20120042 110011) and the Fundamental Research Funds for the Central Universities of China (N140203005, N140204015).

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

WAYS TO IMPROVE OF ALUMINUM CONTENT RAW MATERIAL TREATMENT BY SINTERING METHOD

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Keywords: vaporization, desilication, aluminate liquor, alumina

Abstract

The main problem is facing the wide use sintering process of the aluminum refining raw material (nepheline, ash from coal, etc.) are a high power consumption and an expensiveness of alumina. Improvements are aimed at reducing the consumption of heat and more efficient use of secondary energy resources. Reducing the cost of alumina is expected to be reached by inclusion of technogenic raw materials, enhancing the capacity of production equipment and reducing the use of expensive chemical products. It is shown that the production of high demand products and the creation of new processes for their preparation for alumina refineries can not only increase the profitability of processing of raw materials, but also eliminate or reduce process steps, which produces the main amount of waste.

Introduction

The basis of processing aluminum raw material by sintering is the chemical reaction at temperatures above 1000 °C between aluminum-containing components of feedstock and calcium oxide to form water soluble alkali metal aluminates and belite mud in which the main component is dicalcium silicate.

The aluminate liquor received after leaching sintered material and subjected to desilication and then decomposition with precipitation of aluminum hydroxide, alumina, soda and potash. Belite mud can be produced to cement [1-3].

Sintering method for processing an alkali alum silicate raw material for alumina is highly energy-intensive. High power consumption greatly affects the cost of the finished product [3].

The largest expenditure of energy takes place in the sintering operation, desilication and evaporation. Reduced energy consumption during sintering can be achieved by changing the construction of the furnace unit and by reducing the moisture content of the charge. Reducing energy consumption in the desilication and evaporation can be achieved both by improving the equipment, and using new technological scheme of processing of raw materials.

Lowering the cost of production can be achieved not only by improving the technology and equipment, but also by expanding the range of products, the organization of production of new and expensive species. An integrated approach to the processing of mineral raw materials and aluminum-containing raw materials in particular, is the determining factor in modern conditions.

Development and implementation of technologies for more liquid products and the creation of new processes for their preparation in alumina companies can not only improve the profitability of processing of raw materials, but also eliminate or reduce the processing steps or process steps, which form the principal amount of waste.

Heat Technology Of Alumina Production From Nepheline Or Other Alumina-Containing Raw Materials

Alumina production by the sintering method consumes a large amount of low pressure steam. The supply of steam is mainly determined by two technological processes: autoclave desilication of aluminate liquor and evaporation of soda liquor.

Desilication of aluminate liquors is an energy intensive process which tends to coat the metal surfaces and not allow for better heat recovery for preheating the feed liquor to the surface of the heater.

Improving the energy efficiency of heat technology requites two innovations [4,5]:

1. To put in the high silica content and low A/C ratio aluminate liquor (liquor with molar ratio of $(Na_2O+K_2O)/Al_2O_3$ less than 1,6) evaporation process step with preliminary aging before entering the first evaporator battery. This eliminates the process of soda liquor concentrating evaporation. Soda liquor evaporation has low capacity due to the intense incrustation of evaporator heat exchange surface; especially in the first evaporator battery. Also it is provides high performance of evaporator battery if it is stable and works with high efficiency.

2. To implement the desilication autoclave process with regeneration of desilicated pulp heat; it is based on the inclusion of special mixing to the scheme of heat exchangers, which eliminates precipitation overgrowing.

The heat technology of alumina production from nepheline raw materials can be improved by heat recovery of the compressed end gas of calcination ovens, for example this can be used for liquor evaporation. Similar engineering solution can be used for the sintering ovens end gases if these gases well cleaned from impurities or can be possibly used for a blower or compressor.

This upgrade technology of heat in the aluminum raw material processing can significantly reduce steam consumption in the evaporation and desilication aluminate and soda liquors.

Complex Processing Of Aluminum-Containing Raw Materials

Reducing the payback period and the cost of producing alumina from alkali aluminosilicate raw materials supposes using cheaper raw materials and energy resources, as well as qualitative changes in the structure of production towards producing expensive and highly liquid products based on the creation of high tonnage production [6-11].

Expanding the range of products in the processing of aluminum-containing materials is achieved in various ways. Especially by:

1. Processing intermediates alkali alumosilicate raw starting components for catalysts and sorbents;
2. Produce oxides and hydroxides of aluminum for non-metallurgical purposes;
3. Produce coagulants of different quality and range.

Production Of Initial Materials For Catalysts And Adsorbents

Catalysts and sorbents may be produced from silica white, zeolites, active alumina oxide or hydroxide, calcium aluminates or rubidium-cesium concentrates. In comparison with existing catalytic production the manufacture of catalysts and source materials for their use at alumina plants allows to make the process more economic and ecologically safe. It is defined by use of the technological expedients which are typical for the basic manufacture, by the opportunity waste recovery of filtrates and washings in alumina and soda-potassium cycles, by the decrease cost of transportation of initial materials [7].

Silica White Production. Silica white is used for manufacture of a wide spectrum of chemical products including zeolites, glasses, synthetic detergents, silicates and as extenders, sorbents and desiccants. In parallel with natural diatomite silica white is applied as the basis of vanadium and other catalysts.

The leaching by sodium liquors of belite mud allows partial (at 30-35%) recovery of the silicon from mud in liquor:

$$2CaO \cdot SiO_2 + 2R_2CO_3 + aq = 2CaCO_3 + R_2SiO_3 + 2ROH \quad (1)$$

Where: R - Na, K.

The precipitation of silica gel from the liquor is by caustic alkali neutralization. The residue of silica gel is washed by water from impurity, depending on assignment, subject to hydrochemical activation, and then dried.

Zeolites Production. Zeolites are alum silicates of alkali and alkaline earth metals with special properties that determine their practical use. If the natural zeolites are used primarily as a pozzolanic additive in cements, the main areas of application for synthetic zeolites are synthetic detergents, catalysts for petroleum refining industry as well as dehumidifiers and sorbents for various purposes.

Most commercially important zeolites are synthetic zeolites NaA, NaX, NaY and ZSM-5. The first type is mainly used for the production of detergents, and the others - in cracking and hydrocracking catalysts in petroleum refining, synthetic fuel or other purposes [8].

Synthetic zeolites are prepared by using aluminum- and silicon-containing ore materials. With respect to the alumina plant, they can be obtained directly from the aluminate liquor and sodium silicate liquor under the scheme:

$$2NaAl(OH)_4 + Na_2SiO_3 = 2Na_2O \cdot xAl_2O_3 \cdot ySiO_2 + 4H_2O$$

Rubidium-Cesium Concentrate Production. Rubidium and cesium compounds can be used as promoters of vanadium and other catalysts. The rubidium and cesium extraction from alkaline alum silicate raw materials processing by sintering can be implemented from sodium and potassium production circulating liquors or from dusts of sintering furnaces in which has higher portion of potassium from the alkalis sum. The essence of technology is alternation of evaporation and cooling operations with use of quantitative laws of distribution of alkaline metals. Concentrates of rare alkaline metals for catalysts can be produced with the equipment of alumina plants that processing of alkaline alumosilicate raw materials without additional expenses for reconstruction [7].

Preparation Of Aluminum Hydroxide Non-Metallurgical Destination

An advantage this method is the possibility of obtaining sintered aluminum hydroxides with a low content of coloring impurities. Furthermore, the feature of the technology allows it to obtain hydroxides (oxides) aluminum which can be sold at a price much higher than that of alumina and recycle waste produced.

Production of aluminum hydroxides of a small-capacity for non-metallurgical purpose because of its high profitability, will reduce the payback period of investment in the complex as a whole.

Thus, calculations show that the proportion of release of aluminum hydroxide as pseudoboehmite 10% of the total alumina in the final product payback period of investment in the facility capacity of 250 thousand. T. of alumina per year can be reduced by a factor of 1.2 [9].

Coagulants Production

Coagulants, the receipt of which is expedient in the complexes of the aluminum raw material processing are the basic aluminum chlorides and sulfates and alkali metal aluminates [10,11].

Basic chlorides and sulfates are prepared from aluminum hydroxide and acid. Subject to the estimated amounts of the reactants eliminates the ingress of chloride and sulfate ions in a closed water cycle of alumina production.

Basic chlorides (oxychlorides) aluminum: $Al_2(OH)nClz$ where n = 3-5, l, z = 3÷0,9 and n + z = 6, the most preferred form which is A12(OH)5C1 can be prepared by scheme:

$$2A1(OH)_3 + HC1 = A1_2(OH)_5C1 + H_2O$$

Oxychlorides in liquor (13-15% $A1_2O_3$) or dry product: $A1_2(OH)_5C1 \cdot 2{,}5H_2O$ (42-43% $A1_2O_3$).
Aluminum sulphate is obtained by reacting aluminum hydroxide with a liquor of sulfuric acid by the reaction:

$$2Al(OH)_3 + 3H_2SO_4 = Al_2(SO_4)_3 + 6H_2O).$$

The end product is a melt or a liquor of aluminum sulphate with a concentration of 7,2% Al_2O_3.
Hydroaluminate sodium: $Na_2O \cdot Al_2O_3 \cdot nH_2O$, where n - 2,5÷6 may be obtained by evaporation of aluminate liquor alumina production under the scheme:

$$2Na + 2A1(OH)_4\text{-} = Na_2O \cdot Al_2O_3 \cdot nH2O,$$

are widely used mainly as coagulating agents for acidic water.

The most promising technology for producing sodium aluminate based on crystallization of low A/C ratio aluminate liquors with $\alpha_{caustic} < 1,8$ (in this case it is possible to carry out evaporation to lower concentrations, and thereby to reduce by several orders of magnitude compared to the evaporation of a relatively high A/C ratio liquors (liquor with molar ratio of $(Na_2O+K_2O)/Al_2O_3$ more than 3,5 unit), the corrosion performance of the evaporators.

Hydroaluminate may be implemented as a product, and used to adjust the caustic aluminate liquor module if necessary productivity redistribution decomposition.

New Components in Charge

Using cheaper raw materials for the preparation of batches of sintering instead of the traditional and the transition to multi-batch suppose reduce production costs by increasing productivity and reducing the basic equipment costs.
The result made of chemical-technological research was the specification of the conditions of use as a component of the calcium carbonate batches sintering waste ammonia production, as well as the aluminum component - bauxite, reducing the cost of obtaining and broaden the base of technologically suitable raw materials.

The Use Of Bauxite as Component of Charge

An indication of completeness of passage of physicochemical interactions between the components of charge during sintering is the degree of transition of alumina and alkali in the liquor of the cake with the subsequent leaching. One of the factors contributing to the increase in the extraction of alumina in liquor is an increase in the Al_2O_3 content in the sinter. Increasing the share of alumina processed batch kiln increases.

One way of increasing the alumina content of the sinter is the addition to the charge of materials with a high level of aluminium, including bauxite [12,13].

With increasing proportion of the bauxite in the mixture in sinter increased content of aluminum and iron oxide and decreased content of silicon and calcium oxides.

The research shown that addition of 5-30% of bauxite to the charge based on aluminum silicate ore and sintering the this composition increases extraction of alumina and alkali for an average of 1-2%, the sintering temperature is reduces for approximately 25 °C and parameters of the sludge settling after leaching retains.

The addition of bauxite to nepheline charge cause to a small increase of the alumina and alkali extraction to a liquor; dependence is linear. For example, addition of 10% of Severoonezhsk bauxite to the charge with the same sintering temperature, increased extraction of alumina by 1.5-2.5%, sodium oxide - at 0.7-1.5% potassium oxide - by 0.8-2.0%. Furthermore, with the addition bauxite temperature of sinter formation shifts to lower temperatures at 25 °C, this is caused by a changing in the phase composition of sinters.

Acceptability appraisal of leach slimes as raw cement component is realized by the major components content (Al_2O_3, CaO, SiO_2, Fe_2O_3), and damaging impurities (Na_2O, K_2O, MgO, P_2O_5, Cr_2O_3). Characteristics of produced sludge should be in the range: saturation coefficient $CaO\cdot(1,65Al_2O_3 + 0,35Fe_2O_3)/2,8SiO_2 = 0,63\div0,64$; silicate ratio $SiO_2/(Al_2O_3 + Fe_2O_3) = 3,81 \div 8,10$; alumina ratio $Al_2O_3/Fe_2O_3 = 0,63\div1,13$. It is provides the possibility of using the sludge as a raw batch component for cement production.

The bauxite using as the aluminum component of the charge based on alkaline aluminosilicate ore increases equipment capacity and, accordingly, the release of the final product - alumina (aluminum hydroxide), also the specific consumption of energy reduces.

The bauxite using in the charge based on alkaline aluminosilicate ore justified at high (as now) prices of alumina, because it is reduces release of soda products proportionally to the bauxite addition.

The main advantage of bauxite and nepheline raw materials joint processing is increasing of alumina production capacity, ie reduction of specific material streams for charge preparation, sintering and leaching sinters. However, it reduces the release of connected products. The dependence of the release of alumina and soda from the bauxite share component of the raw material mixture is inversely proportional. It is due to the need to increase turnover soda-alkaline liquor to provide the desired alkali charge module.

The Use of Waste as Technological Component of Charge

An important component of the charge for sintering is calcium carbonate. It is used as a natural mineral - limestone. Limestone is limited for making the charge to content of impurities of silicon, magnesium and iron. Limestone has to be crush and grind and these processes need grinding equipment and energy costs. At the same time, there is man-made chemical production waste, the basis of which calcium carbonate. One such waste is carbonate production of ammonium nitrate, which is obtained by the reaction:

$$Ca(NO_3)_2 + 2NH_3 + CO_2 + H_2O = 2NH_4NO_3 + CaCO_3$$

Such carbonate waste, consisting of 88-95% calcite, has been used as a component of industrial batch sintering [14].

In comparison with a natural limestone, waste does not contain regulated impurities of silicon, magnesium and iron, and does not require grinding. Besides, there was an increased reactivity of the carbonate waste in comparison with limestone.

Conclusion

Improved processing technologies alkali alumosilicate materials should be aimed at reducing the cost of the final products, the optimization of process parameters, reducing energy consumption, increasing of rational utilization of feedstock and thereby reducing harmful emissions into the environment. This may be achieved by:
-improving hardware and technological solutions;
-using cheaper raw materials;

-correcting charge composition, the sintering temperature and the desilication depending on their composition;
-reducing irretrievable losses;
-using the new technological properties of intermediate products processing.

At the same time the negative impacts of products of mining and mineral processing on the environment can be reduced by engaging in waste management and waste dumps, reducing material flows and liquid waste, the use of intermediate products of processing of mineral raw materials for a new purpose.

It is found that during sintering physicochemical interaction in charges obtained using these bauxites and industrial waste calcium occur at lower temperatures than conventional raw batch components. On the basis of certain dependencies it proved that ratio control components of the charge, the sintering temperature and concentration modes of leaching can be avoided decrease extraction of useful components by leaching cakes and phase separation difficulties associated with the change of the charge.

New features are able to reduce the payback period of investments in the creation of new industries based on sintering technology also arise due to the release of aluminum hydroxides, fine low-alkali spheroidal hydroxides, pseudoboehmite, coagulants, zeolites, sodium aluminate, as well as compounds of rare metals.

Fabrication of additional economical feasible products and creation of essentially new processes of their production at the refinery allows not only to increase profitability of raw material processing, but also to exclude or to reduce steps of processing or technological stages on which a main amount wastes is formed.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EXTRACTION OF ALUMINIUM BY AUTOCLAVE HYDROCHLORIC ACID LEACHING OF BOEHMITE-KAOLINITE BAUXITE

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Keywords: Pressure leaching, Bauxite, Boehmite, Kaolinite, Hydrochloric acid.

Abstract

In this paper we investigate the process of autoclave extraction of boehmite-kaolinite bauxite for Severoonezhsk deposits by hydrochloric acid. The effect of temperature and duration of the process on transition of aluminum into solution has been studied. The apparent activation energy was defined for the process. The influence of hydrochloric acid concentration on alumina transition in solution was shown. Determined the apparent degree of the process on reagents. The obtained results allow to allocate 3 limiting stages of the process: 150 – 160°C – a chemical reaction (E_a = 70.88 kJ/mol), 160 – 170°C – intermediate region (E_a = 40.22 kJ/mol), 170 – 180°C – diffusion area (E_a = 22.66 kJ/mol). Thermodynamic modeling of physicochemical processes underlying the preparation of aluminum containing solutions has been performed. It has been demonstrated that the major aluminum species in solution are aluminum chlorides and hydroxychlorides.

Introduction

Russia is the world leader in aluminum production, but the bulk of the feedstock (~ 70% of bauxite and alumina) is imported from abroad [1]. In terms of the economy in the direction of "import substitution" the reorientation of the industry on the domestic ore base is very relevant. In this regard, the development of technology for the processing of low-quality acid bauxite is a promising task and will result in the construction of new alumina refineries in the European part of Russia, provide raw material security.

Currently, hydrochloric acid method for high-silicon aluminum-containing raw materials is intensively developing, which allows to delete an oxide of silicon in the head of the process and significantly reduce material flows.

Compared to other mineral acids, it has several significant advantages acids:

- low solubility of silica in HCl [2];
- possibility of selective crystallization of $AlCl_3 \cdot 6H_2O$ [3, 4];
- industrial implemented of HCl capture technology with production of hydrochloric acid for reuse in the process of dissolution of raw materials [5, 6];

In addition to the main product – alumina – the new complex technology allows to obtain a number of co-products: mixed coagulants [7, 8, 9], gallium, scandium, REM, concentrate for iron metallurgy, raw materials for metallurgical silicon [10] and building materials. This approach provides a comprehensive usage of Russian ore.

Since that time, new corrosion resistant materials (enamel, the alloys of Ta, Zr, Nb) and durable equipment designed based on them that really allows you to create hardware scheme acidic alumina refinery [11, 12].

The developed technology does not require significant expenses for designing of new metallurgical apparatus, because the individual process stages have already been introduced in the chemical (autoclaves for high pressure decomposition of aluminum hydroxide in hydrochloric acid at JSC «AURAT» (Moscow, Russia)) and titanium industry (oxy-hydration of ferric chloride in a fluidized bed reactor at the plant of QIT-Fer et Titane (Quebec, Canada) [13]).

Developments in hydrochloric direction are now being actively developed in Canada [14, 15], and especially in China [16–20], i.e. in advanced aluminum industry and lack of its traditional raw materials base.

In this work as a raw material have been used domestic high-silica bauxite of Severoonezhsk deposits (Arkhangelsk region). To date, the main direction of processing of such bauxite is the production of refractories at JSC «Ulyanovsk refractory ceramic Mine» and JSC «Novomoskovskogneupor». Increasing of complexity of the processing of such raw materials will meet the needs of basic aluminum refineries in European Russia (Nadvoitsk, Volkhov, Volgograd, Kandalaksha) in metallurgical alumina. At present, these plants are stopped or converted to the production of metal aluminum for the manufacture of automotive components. The introduction of acid method will lead to a significant increase in the competitiveness of domestic factories, improve the quality of life in the remote regions of Russia, the creation of new jobs, training of qualified specialists in the field of mining and mineral processing.

Experimental

Decomposition of bauxite in hydrochloric acid was performed in 2 ways. The first method involves leaching was carried out in a glass round-bottom flask mounted in a heating mantle. Dissolution was performed bauxite hydrochloric acid concentration of 20% with mechanical stirring and a temperature of 110 °C for 1 – 5 hours.

In the second method, the leaching was carried out in titanium ampoule autoclaves of two types: a volume of 36 cm^3 and 250 cm^3, installed in the thermostat. Mixing of the pulp was provided by rotation of the autoclave "over the head". Inside the ampoules were inserted fluoroplastic vessels to prevent the interaction of hydrochloric acid with the walls of the autoclave. Dissolution of bauxite was carried out with hydrochloric acid with a concentration of 10, 15, 20, 25, 30% at temperatures of 150 – 200 ° C for 1 – 3 hours. The time of heating from 25 °C to the desired temperature was 1 hour. The set temperature is maintained automatically with an accuracy of ±1 °C. Experiments in the autoclave increased volume was conducted to clarify the parameters of leaching and obtain more reliable results. At the end

of the experience ampoule autoclave was cooled for 10 minutes under running cold water

Then, the solution was separated from the solid residue (Si-stoff) on a vacuum filter. The Si-stoff was washed with distilled water, then dried at 110 °C in a drying oven for two hours. The resulting solutions, washes and Si-stoff were analyzed for the content of Al, Fe, Si and impurity elements. Chemical analysis was performed by the photometric method using a spectrophotometer PE 5400VI («EKOHIM», Russia) and by atomic emission spectrometry on atomic emission spectrometer with inductively coupled plasma Optima-4300 DV («Perkin-Elmer», USA). XRD analysis of bauxite was carried out on data obtained by diffractometer Ultima IV («Rigaku», Japan).

Experimental

Severoonezhsk bauxite relates to a boehmite-kaolinite type with silica module of 2.5. According to X-ray diffraction (see Figure 1), the main minerals of bauxite are boehmite (~ 41.5%), kaolinite (~ 25%), gibbsite (~ 9%), hematite (~ 4%), goethite (~ 5%), anatase (~ 3%). The chemical composition, %: Al_2O_3 – 50.0; SiO_2 – 20.0; Cr_2O_3 – 0.86; Fe_2O_3 – 7.94; TiO_2 – 2.94.

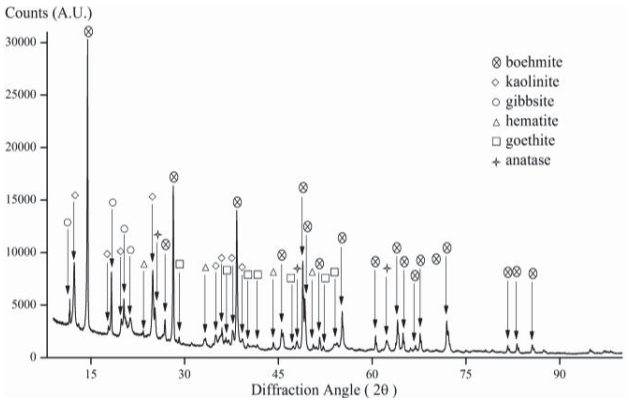

Figure 1. The XRD pattern of the bauxite from Severoonezhsk deposits.

Decomposition of bauxite was carried out with hydrochloric acid at a temperature of 110 °C and S:L = 1:6. As result of leaching the solution is transferred into Al and Fe, but the recovery rate of aluminum is very low and does not exceed 23%. Obtained solution had the following chemical composition, g/L: Al, 10.6; Fe, 7.69; Si, 0,02.

Dissolution bauxite in time is uneven, so in the first two hours of dissolution goes most intensively due to complete dissolution of easy-extracted in hydrochloric acid minerals – gibbsite, goethite and hematite, kaolinite and boehmite thus dissolved slightly [2].

Autoclave decomposition

To speed up the process of ore leaching were investigated the process of autoclave digestion on bauxite by 20% hydrochloric acid in the temperature range of 150 – 200 °C at duration of 1 – 3 hours and ratio S:L = 1:6 (see Figure 5). Dosage of acid was taken in view of stoichiometric quantity to dissolve all aluminum and iron present in the bauxite.

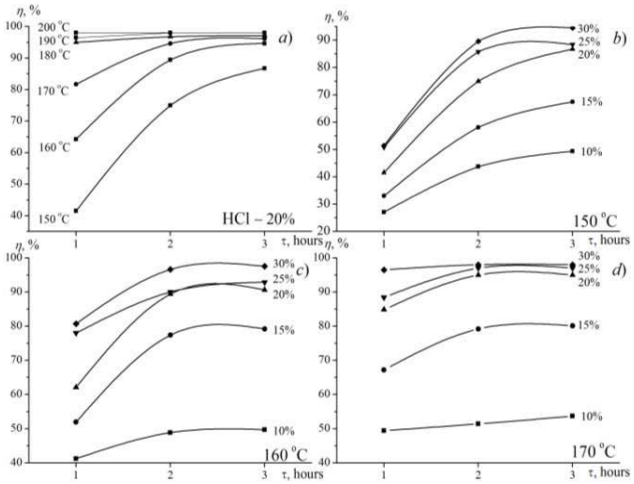

Figure 5. The dependence of the degree of aluminum extraction (η) into solution on duration of leaching (τ) and temperature.

As can be seen from Figure 5*a*, at increasing the process temperature above 170 °C, the extraction rate of aluminum into solution exceeds 95% and the duration of the process has almost no effect on the extraction [7, 9].

To determine the kinetic parameters of the leaching process of boehmite-kaolinite bauxite by hydrochloric acid was used the experimental data of dependence of transition degree of alumina in a solution on the temperature, duration and concentration of hydrochloric acid. Based on these data, using the known equations kinetics and Arrhenius equation, the values were calculated for the apparent activation energy.

$$K = A \times e^{-E/R} \quad (5)$$

$$V = K \times C^n. \quad (6)$$

Where:

V – the rate of reaction;

K – reaction rate constant;

C – concentration of the reactant;

n – reaction order;

A – pre-exponential factor or frequency factor – constant characteristic of the reaction;

E – activation energy;

R – universal gas constant.

Determination of E_a conducted under standard reactant concentration (in this case 20% HCl) for four temperatures (150, 160, 170 and 180 °C). After the transformation of the equation (5) were obtained rate constants and E_a [21, 22]:

$$\lg K = \lg A + E / 4.575 \times T; \quad (7)$$

$$E_a = 4.575 \times \operatorname{tg} \alpha \quad (8)$$

Figure 6 shows the dependence of lg*K* reaction rate on reverse temperature. This chart can be divided into 3 areas, which indicates to a change in the nature of the interaction between the reacting components. For each plot at a slope (8) has been determined the value of the apparent activation energy: at 150 °C E_a = 70.22 kJ/mol at 160 °C – 40.88 kJ/mol at 170 °C – 22.66 kJ/mol. Data on Fig. 7 shows that the increase in the degree of aluminium transition into a solution the values E_a of leaching process varies considerably, due to the staggered nature of the dissolution of a boehmite-kaolinite bauxite because of different solubility of its constituent minerals.

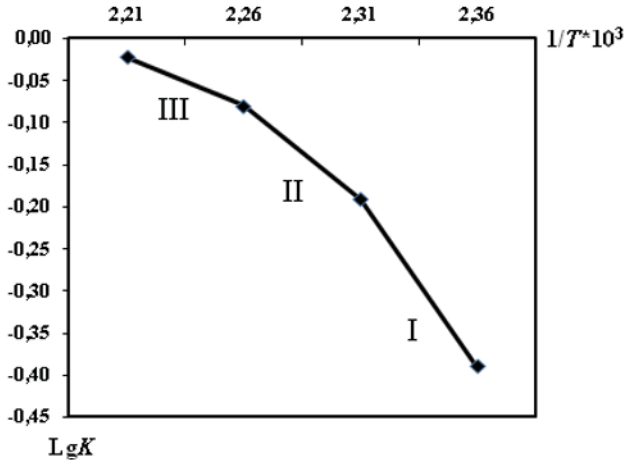

Figure 6. The dependence of lg K reaction rate on reverse temperature.

To determine the apparent order of the process according to reagent (HCl) were taken experimentally obtained dependences degree of aluminum leaching (Figure 5*b, c, d*) on its duration at various hydrochloric acid concentrations and constant temperature and concentration of other reagents. Logarithmica equation (6), we obtain:

$$\lg V = \lg K + n \lg C. \quad (9)$$

Where two points can be obtained the following expression:

$$n = \frac{\lg V_1 \times \lg V_2}{\lg C_1 \times \lg C_2}. \quad (10)$$

Using equations (6) and (10), determine the values of reaction order and rate constants. The values of reaction order and rate constants for different temperatures and concentrations of hydrochloric acid are presented in Table I.

Table I. Kinetic characteristics of the decomposition reaction of bauxite with hydrochloric acid

T, °C	Consentration HCl, %	Rate constans, L/mol×sec×10^{-6}	Reaction order
150	10 - 15	7,322	0,52
	15 - 20	3,058	0,61
	20 - 25	0,425	0,9
	25 - 30	0,092	1,11
160	10 - 15	4,725	0,42
	15 - 20	2,019	0,61
	20 - 25	0,286	0,81
	25 - 30	0,061	0,88
170	10 - 15	2,153	0,73
	15 - 20	0,811	0,92
	20 - 25	0,117	1,09
	25 - 30	0,028	1,33

At the initial stage of leaching bauxite hydrochloric acid takes place in the kinetic region at which the rate of the reaction depends on the temperature, as evidenced by the value of activation energy (E_a = 70.22 kJ/mol) and fractional order of reaction (0.54 – 0.61). The reaction proceeds on the surface of bauxite and between its layers, where access is still free HCl. Thus, as the reaction product forms gel-like silicic acid, which partially covers the surface of the particles and also fills the space between the layers in the structure of bauxite. As a result, the supply of hydrochloric acid to alumina octahedra and complicated reaction occurs in the transition region. This explains the decrease in the activation energy (E_a = 40.88 kJ/mol). Further the reaction leads to the formation of significant amounts of silica and kaolinite dissolution reaction becomes diffusion region (E_a = 22.66 kJ/mol, the order of reagent equal to 1.09), as the supply of the reactant becomes more difficult and eventually stops completely. This is evidenced by a decrease of the rate constant (see Table 2). Especially greatly reduced internal diffusion (between layers of bauxite).

To confirm the kinetic characteristics was studied the structure of the particles of the Si-stoff of the two experiments corresponding to the kinetic region (150 °C, τ = 1 hour) and diffusion mode (180 °C, τ = 1 hour). The gradual dissolution of bauxite begins at a temperature of 150 °C. In Figure 7 shows the presence of individual particles of aluminum particle size of 50-100 μm that have not interacted with hydrochloric acid.

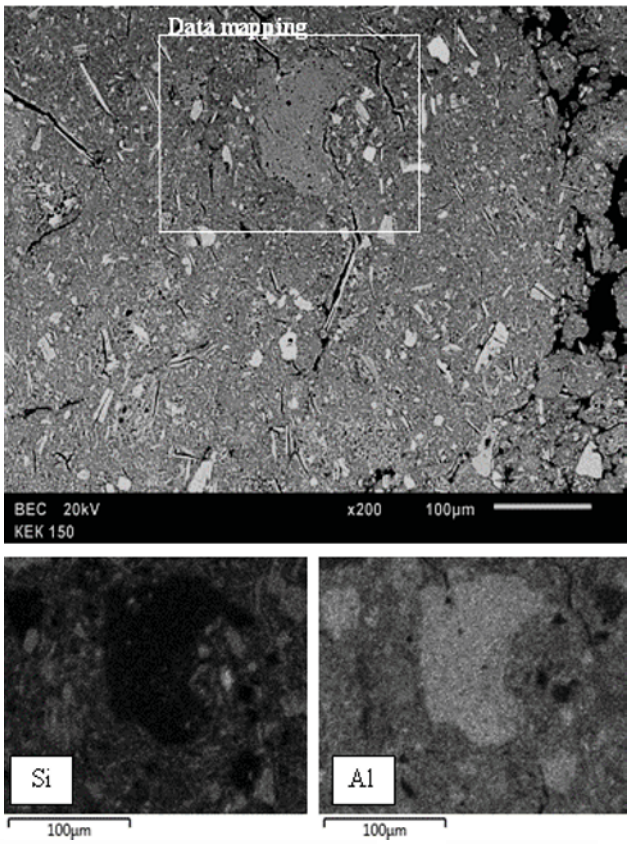

Figure 7. Micrograph of particles of the Si-stoff (SEM method).

However, at this temperature, begins the destruction of the particles of kaolinite to separate minor particles with fineness of 5-10 μm (see Figure 8).

The structure of the particles in the diffusive decomposition of bauxite shows the presence of a large amount of silicon oxide, which prevents the dissolution of aluminum residues (see Figure 9–11). You can allocate as large particles of aluminum oxide (~ 100 μm) coated silicon (see Figure 9), and a small fraction in the form of scales or plates with a length of ~10 μm (see Figure 10).

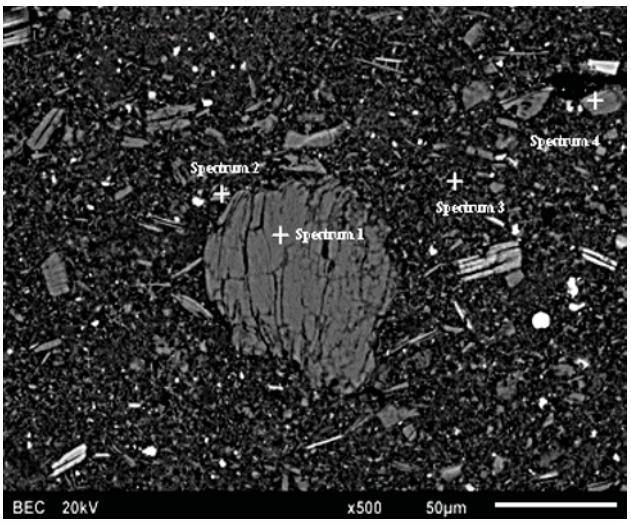

Figure 8. Micrograph of silicon particles made by the method of SEM (marks allocated point x-ray microanalysis, the results of which are shown in the table).

Spectrum	Content of element, %			
	Al	O	Si	Ti
1	23,0	52,7	24,3	-
2	35,6	48,9	11,0	3,8
3	22,8	52,6	24,5	-
4	22,7	52,4	24,1	0,5

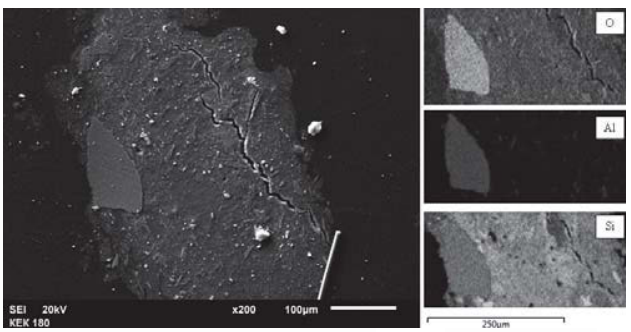

Figure 9. Micrograph of particles of the Si-stoff (SEM method).

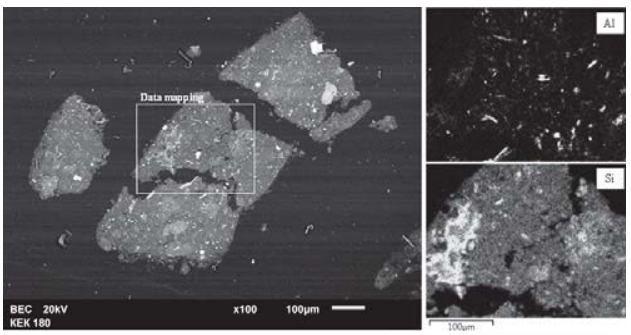

Figure 10. Micrograph of particles of the Si-stoff (SEM method).

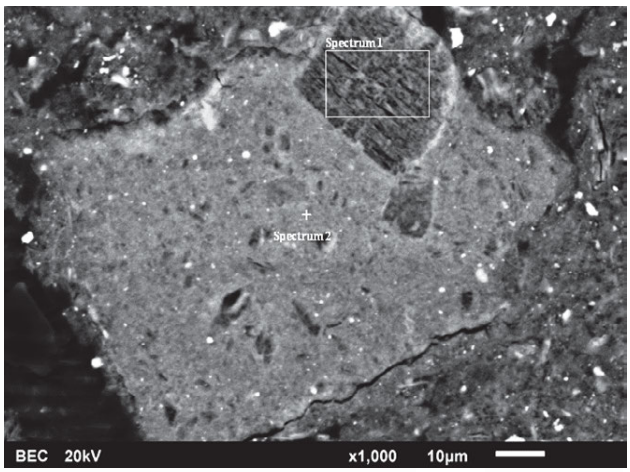

Figure 11. Micrograph of particles of the Si-stoff, made by the method of SEM (marks allocated point x-ray microanalysis, the results of which are shown in the table).

Spectrum	Content of element, %		
	Al	Si	O
1	3,4	47,3	48,9
2	–	47,9	49,8

If the silicon component is not destroyed to micron size it can be large particles up to 50 µm (see Figure 11). In the inner part of such particles may contain aluminum, which for hydrochloric acid is severely limited. Thus, in conditions of high temperature silicon oxide substantially prevents full opening of bauxite and transfer of aluminum in the solution. For complete decomposition requires excess acid from 130 to 300% of the required number for the dissolution of aluminum and iron in the ore, excluding HCl to bind Ca, Cr, Mg, K [23].

Physico-chemical modeling

In the present work, the processes that occur in the course of the hydrochloric acid treatment of bauxite in the Al–Si–Fe–Cl–C–N–H–O–e system were studied with the use of the Selector–S program package to determine the forms of aluminum in solution.
The component composition of the multicomponent system under consideration, including all possible phases and chemical compounds involved in acidic decomposition, is presented in the Table II .
Calculations of the equilibrium composition of the Al–Si–Fe–Cl–C–N–H–O–e multicomponent system were performed by the Gibbs energy (G(x), isobaric isothermal potential of the system) minimization method [24, 25]. All thermodynamic parameters of dependent components, as well as parameters of the HKF equation for the acidic solution components, were taken from the Selektor, a_Sprons98, s_Sprons98, g_Reid, and s_Yokokawa software data–bases. These are six condensed phases, 51 component of an aqueous solution, and 12 components of the gas phase [26–28].
Thermodynamic calculations have shown that the increase in solubility of aluminum minerals is due to the change in the ionic composition of a solution. In particular, the fraction of cationic and hydroxide aluminum complexes decreases, while the content of aluminum chloride and hydroxychloride complexes increases [29].

Table II. Dependent components of the Al–Si–Fe–Cl–C–N–H–O–*e* system

Aqueous solution	
$Al(OH)^{2+}$, Al^{3+}, AlO^{+}, AlO^{2-}, CO_2, CO_3^{2-}, Cl^{-}, ClO^{-}, Fe^{2+}, Fe^{3+}, $FeCl^{+}$, $FeCl^{2+}$, $FeCl_2$, FeO, FeO^{+}, FeO_2^{-}, $FeOH^{+}$, $FeOH^{2+}$, $HAlO_2$, HCO_3, HCl, HClO, $HFeO_2$, HNO_2, HNO_3, $HSiO_3$, N_2, NO_2^{-}, NO_3^{-}, O_2, SiO_2, $Al(OH)_3$, $Al(OH)_4^{-}$, $AlCl(OH)_2$, $AlCl_2(OH)$, $AlCl_2^{+}$, $AlCl_3$, $AlCl_4^{-}$, $AlCl_5^{2-}$, $Fe(OH)_2^{+}$, $Fe(OH)_3$, $Fe(OH)_4^{-}$, $FeCl_2^{+}$, $FeCl_3$, $FeCl_4^{-}$, $H_3SiO_3^{+}$,$H_3SiO_4^{-}$, H_4SiO_4, OH^{-}, H^{+}, H_2O	
Gas phase	
CO_2, CO, H_2, H_2O, O_2, O_3, HCl, Cl_2, NO, N_2, NO_2, N_2O	
Condensed phases	
Boehmite AlO(OH)	Kaolinite $Al_2Si_2O_5(OH)_4$
Hematite Fe_2O_3	Gibbsite $Al(OH)_3$
Quartz SiO_2	Goethit FeOOH

The equilibrium component composition of hydrochloric acid solutions has been determined. At 110 °C, aluminum mainly exists as Al_3^{+} and $AlCl_4^{-}$ species, and the content of $AlCl_3$ is estimated at the level of $10^{-3} – 10^{-4}$ mol/kg H_2O (see Figure 12*a*). The other species have no noticeable effect on aluminum solubility since their content is less than 10^{-6} mol/kg H_2O.

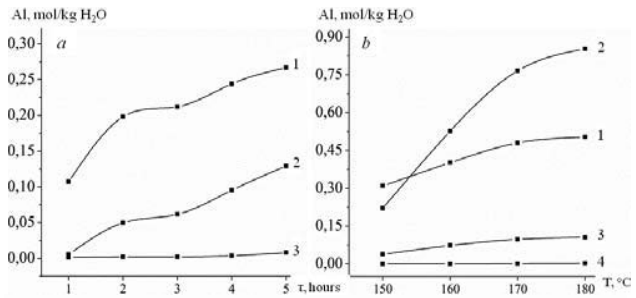

Figure 12. The distribution of aluminum in chemical forms in the hydrochloric acid solution during the leaching of bauxite in ambient (*a*) and autoclaved (*b*) conditions: 1 – Al^{3+}; 2 – $AlCl_4^{-}$; 3 – $AlCl_3$; 4 – $AlCl_2(OH)$.

In solutions obtained by autoclave decomposition of bauxite is the main form of aluminum becomes $AlCl_4^{-}$, there is a negligible amount $AlCl_2(OH)$ (see Figure 12*b*).

The existence of aluminum in the form of $AlCl_4^{-}$ is consistent well with the Lewis acid–base theory; in the framework of this theory, aluminum chloride is an aprotic acid; i.e., the compound can accept an electron pair into a vacant orbital. Aluminum chloride with six electrons in the outer level has an incomplete electronic configuration and can accept two electrons into the vacant orbital [30].

Conclusion

1. It is found that at autoclave leaching of bauxite with hydrochloric acid recovery rate of aluminum in the solution reaches 94 – 98%.
2. It is shown that the process takes place in 3 steps at temperatures of 150 – 160 °C in the kinetic region (E_a = 70.22 kJ/mole) at 160 – 170 °C – in the intermediate region (E_a = 40.88 kJ/mole) at 170 – 180 °C – in the diffusion region (E_a = 22.66 kJ/mol).
3. The increase in aluminum concentration in an acidic solution is evidence of the formation of aluminum chloride species.

Acknowledgments

The reported study was partially supported by RFBR, research project No. 15-38-50545 mol_nr.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

FT-IR, XPS AND DENSITY FUNCTIONAL THEORY STUDY OF ADSORPTION MECHANISM OF SODIUM FORMATE ONTO GOETHITE OR HEMATITE

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Keywords: goethite, hematite, sodium formate, adsorption, FT-IR, XPS

[*]Project (Grant No.51134007) and Project (Grant No. 51174231) supported by National Natural Science Foundation of China.
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Abstract

The adsorption of sodium formate on goethite or hematite surface was investigated by FT-IR and XPS, respectively. Periodic plane-wave density functional theory (DFT) calculations were performed on models of carboxylate surface complexes on the goethite (101), (100) surfaces or hematite (001) surface. The photoemission core-level shifts (CLS) of the adsorbed surface iron sites calculated with periodic interfacial structures were compared with the experimental XPS. FT-IR results reveal that the interfacial structures may be assigned to bidentate mononuclear (BM) complexes formed via bonding of one surface iron atom of goethite or hematite to both oxygen atoms of carboxylate group. In agreement with the adsorption energies determined by DFT calculations, the calculated results show that only the calculated Fe2p CLS of the adsorbed surface iron sites on goethite (100) and hematite (001) surfaces are correspondingly in accordance with the experimental observed Fe2p CLS, which confirms that the BM structure occurs upon adsorption.

Introduction

Goethite and hematite, the most common iron-containing minerals in red mud, play an important role in the entrainment, adsorption and precipitation of the dissolved organic species in the Bayer liquor due to their high surface areas and high densities of reactive surface sites, which consequently makes the balancing of the inputs and outputs of the organic carbon to alumina refineries notoriously difficult to achieve with any degree of accuracy [1, 2]. On the other hand, several procedures[3, 4] have been proposed to remove the dissolved organic compounds from the Bayer liquor due to the deleterious effects of dissolved organic compounds on the alumina production. However, these procedures are technically too complicated or require higher operation costs. For example, Schepers et al[3] investigated a method for the removal of dissolved organic compounds from the Bayer liquor, i.e., magnesium salt was added to the Bayer liquor to form a hydroxide mixture containing magnesium hydroxide and aluminum hydroxide, followed by the adsorption of dissolved organic compounds in the Bayer liquor. Several studies[5-8] have also reported that goethite and hematite in red mud or activated red mud can be utilized for the adsorption of organic pollutants (such as carboxylate and its derivatives, dye stuff and so on) from wastewater in paper, textiles, printing and leather industries. Therefore, it is expected to be possible for goethite and hematite as potential adsorbents to remove dissolved organic compounds from Bayer liquors. In the meanwhile, aliphatic carboxylate is one of the typical dissolved organic compounds in the Bayer liquor[2]. Owing in part to the possible utilization of the adsorption of dissolved organic compounds (such as sodium formate) on the goethite or hematite surface under the condition of the Bayer process, an effort of molecular-level understanding of interfacial structure-property relationships between sodium formate and the goethite or hematite surface was made in this paper.

The interfacial structure of carboxylate adsorbed on the goethite or hematite surface was commonly investigated via Fourier Transform Infrared Spectroscopy (FT-IR)[9-12] . Persson[10] assumed that oxalate and malonate both form outer-sphere and inner-sphere surface complex on the goethite surface with ATR-FTIR, and that these complexes coexisted over a broad pH interval. Persson[11] also proposed that two surface complexes, ie one solvent-surface hydration-separated ion pair and one surface hydration-shared ion pair, may form upon adsorption of acetate or benzoate at the water/goethite interface. Therefore, accurate interfacial structures of carboxylate adsorbed on iron (hydr)oxide surfaces could not be identified just by FT-IR spectra.

X-ray photoelectron spectroscopy (XPS) is one of the most extensively used surface analytical techniques due to its high sensitivity of 0.1 at.%[13] and the ability to probe the electronic and geometric structures of solid surface with adsorbed molecules[14].

The periodic DFT calculations are valuable for estimating interfacial behaviors because they explicitly include details of the solid surface[15]. Theoretical study of the adsorption/dissociation reactions of formic acid on the α-Al_2O_3 (0001) surface was carried out by Ruan et al[16]. Kundu et al[17] studied the interfacial structure and adsorption behavior of formic acid on the magnetite surface. Furthermore, the adsorption and reaction of SO_2 on clean and oxygen-precovered Pd (100) was investigated with XPS and DFT calculations, and the adsorbed SO_2 species were identified by comparing the calculated core level shifts (CLS) with experimental observed CLS[18]. This (incomplete, but demonstrative) summary represents the emerging role of DFT modeling applied to chemical questions about various interface reactivity.

In this paper, sodium formate was chosen as a model of aliphatic carboxylate in the Bayer liquor. Goethite (101), (100) or hematite (001) surface, which represents the high reactivity properties of goethite or hematite, is taken into account. A combination of FT-IR, XPS and DFT calculations is performed to study the interfacial structure of sodium formate bound to the goethite or hematite surface. Modeling of the structures, energetics, electronic structures and core level shifts of sodium formate on goethite (100) and (101) or hematite (001) surface is performed in a self-consistent manner. Possible interfacial structures of sodium formate on (100) and (101) surfaces of goethite or (001) surface of hematite and the relative stabilities of the interfacial structures are obtained. And CLS of Fe2p and charge transfers of adsorbed surface iron site calculated by density functional theory (DFT) with periodic interfacial structures are confronted to the X-ray photoemission experiments.

Methods

Experimental Details.

Materials and characterization. Goethite and hematite were prepared and characterized by the method described in our previous work[19]. The XRD patterns are presented in Figure 1. The results of surface area and particle size distribution are given in Table 1. Sodium formate (Figure 2) was of analytical grade. The samples of goethite or hematite after the treatment of sodium formate (treated goethite or hematite) were obtained by a method described in our previous work[19]. Goethite or hematite before the treatment of sodium formate (untreated goethite or hematite) was prepared in the same manner except that sodium formate was not added to the goethite or hematite suspension.

The infrared spectra of the samples were measured by a Nicolet-6700 FT-IR spectrometer. The samples were analyzed on X-ray photoelectron spectrometer (ESCALAB 250XI, Thermo Scientific Co., USA) utilizing a monochromatic Al Kα X-ray at1486.6 eV. All measurements were carried out at the pressure below 10^{-8} Pa and with a flood gun for charge neutralization. All spectra were charge-referenced so that the unfunctionalized aliphatic C1s component occurs at 284.8 eV. Curve fittings of Fe2p spectra were performed using the Avantage software.

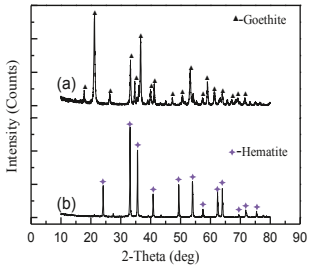

Figure 1. XRD patterns of (a) goethite, (b) hematite

Table 1 Properties of goethite and hematite samples

Properties	Goethite	Hematite
Surface area (m^2/g)	1.4	5
Particle size D_{10} (μm)	2.224	0.530
Particle size D_{50} (μm)	8.367	1.685
Particle size D_{90} (μm)	36.095	9.828
XRD	Goethite	Hematite

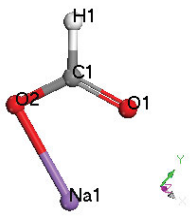

Figure 2. The optimized structure of sodium formate

Calculational details. Periodic slab models of the goethite surfaces were created by cleaving the surfaces (101)[20] and (100)[21] (Figure 3a and b) from the crystal structure of goethite (space group Pnma[22]), which are modeled in a supercell geometry with a vacuum of 10 Å and extended to (1×3). The thickness of the slabs amounted to eight iron layers for goethite (101) and four for goethite (100), respectively. The Periodic slab model of hematite surface (Figure 3c) was created by cleaving the surface (001) from bulk hematite (space group $R\bar{3}c$[23]). The surface (001) of hematite was extended to (1×2)[24] with an excess of 15 Å of vacuum. Initial magnetic moments of Fe atoms were assigned so as to obtain an antiferromagnetic goethite or hematite slab in each case. The goethite (101), (100) or hematite (001) surfaces was passivated by water dissociation products to surface dangling Fe and O atoms (Figure 3). The free sodium formate was added to goethite (101), (100) surfaces or hematite (001) surface in bidentate mononuclear (BM) structures as a model of sodium formate-goethite (101) system, sodium formate-goethite (100) system or sodium formate-hematite (001) system, respectively (Figure 4).

DFT calculations were carried out with the Castep Code[25] in Materials Studio 6.0. Plane wave basis sets were used to solve the Kohn-Sham equations. The GGA electron exchange and correlation effects were described using the Perdew Burke Ernzerhof (PBE)[26]. The on-site Coulomb interaction of 3d electrons, the GGA+U method was applied to the Fe atoms to improve the description of the electronic properties[27]. For the electronic integration in reciprocal space, the Brillouin zone was sampled according to the Monkhorst–Pack scheme[28]. A single k point was used for sodium formate, k points set was fine in the calculation for surfaces with or without sodium formate.

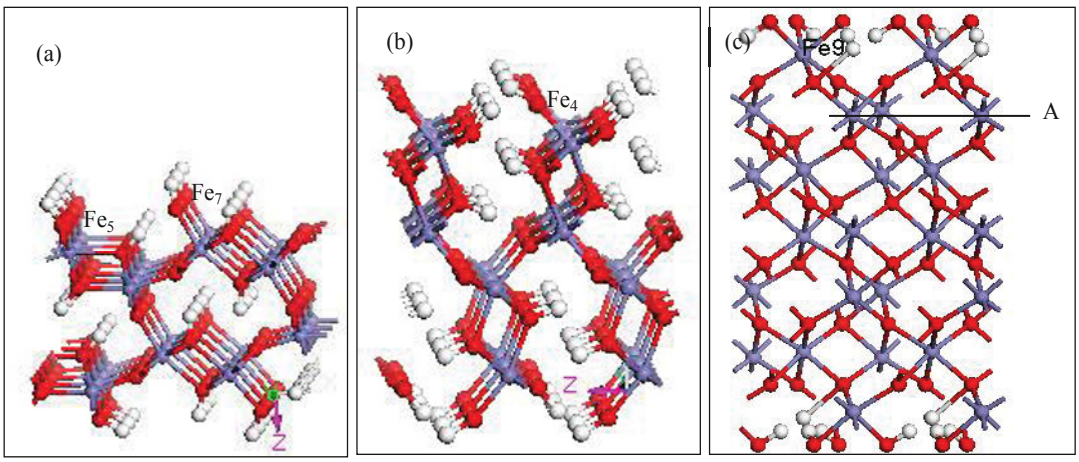

Figure 3. Truncated side view of the models (a) goethite (101), (b) goethite (100) and (c) hematite (001) surfaces. Oxygen atoms are shown in red, iron atoms in grayish blue and hydrogen atoms in white. All the color set in this paper was same.

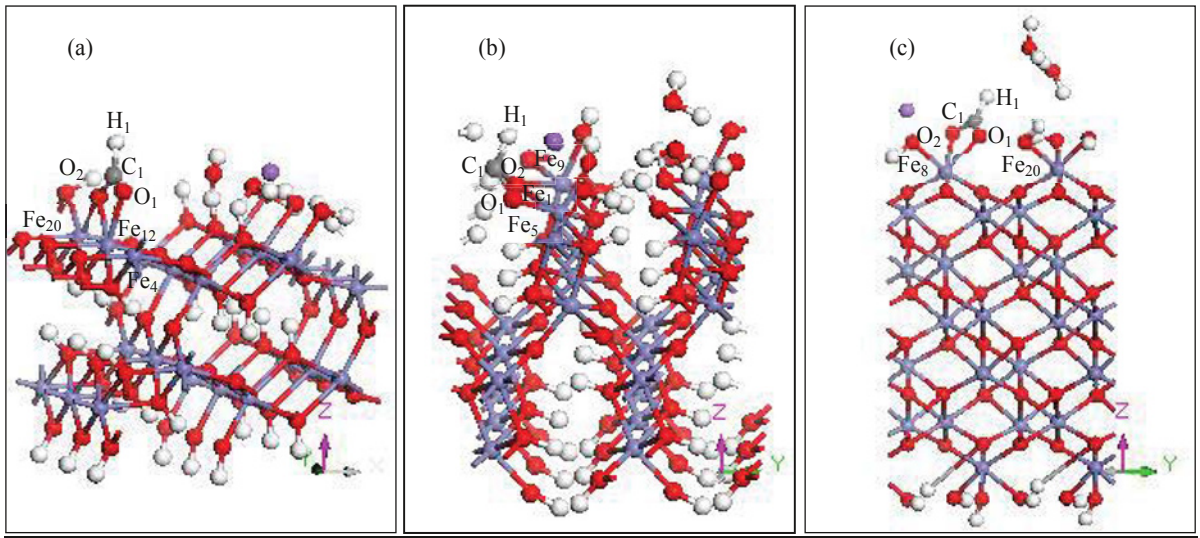

Figure 4. Illustration of the optimized interfacial structures of (a) sodium formate-goethite (101) system, (b) sodium formate-goethite (100) system and (c) sodium formate-hematite (001) system.

For sodium formate, the ionic cores were described by the norm conserving pseudopotentials[29]. The wave functions were expanded with an energy cutoff of 750 eV for geometry optimization and vibrational analysis. All positions were fully relaxed and optimized up to a force convergence of 0.03 eV/Å. The final optimized structure was subjected to a single-point energy calculation with the ultrasoft pseudopotentials[30] and an energy cutoff of 340 eV for the plane-wave basis set to achieve an accuracy of total energy differences of 2×10^{-5} eV /atom. As to the models of surface slabs and BM structures, the structures of half outer iron layers of goethite (101), (100) surfaces or hematite (001) surface was relaxed and fully optimized with the same optimized level as the single-point energy calculation for sodium formate up to a force convergence of 0.05 eV/Å.

The adsorption energy, $E_{ads,}$ of the adsorbate at goethite or hematite surface in (eV) was described as Eq. (1)

$$E_{\text{ads}} = E_{\text{adsorbate}-surface} - E_{surface} - E_{adsorbate} \qquad Eq.\ (1)$$

For the calculation of CLS of Fe2p, geometry-optimized BM interfacial structures were used as models. We modeled CLS of Fe2p as total energy differences between the system with a core hole on the excited Fe atom and the unperturbed system. To describe the excited Fe atom, we generated a special Fe pseudopotential on the basis of an iron atom which included a core hole in the 2p level. In this work, the reference was taken to be the surface iron atoms located at the same position on surface slabs.

Results and discussion

FT-IR spectroscopy analysis.

The FT-IR spectra of (a) free sodium formate, (b) untreated goethite and (c) treated goethite are depicted in Figure 5. The FT-IR spectra of (a) sodium formate, (b) untreated hematite and (c) treated hematite are depicted in Figure 6.

Several studies[19] have demonstrated that it is possible to identify the structures on the basis of the carboxylate asymmetric (ν_{asym}) and symmetric (ν_{sym}) stretches and their separation ($\Delta\nu=\nu_{asym}-\nu_{sym}$) when the carboxyl group of carboxylic acids or their salts is directly involved in the adsorption on solid surface. When $\Delta\nu$(adsorbed) is smaller than $\Delta\nu$(salt), a bidentate mononuclear structure (BM) is observed (two oxygen atoms of the carboxylate group form two bonds with one metal atom on the solid surface). However, when $\Delta\nu$(adsorbed) is greater than $\Delta\nu$(salt), a monodentate mononuclear structure (MM) is seen (only one oxygen atom of the carboxylate group binds with one metal atom on the solid surface). In bidentate binuclear structure (BB) (two oxygen atoms of the carboxylate group bind with two metal atoms on the solid surface), $\Delta\nu$(adsorbed) is almost equal to $\Delta\nu$(salt).

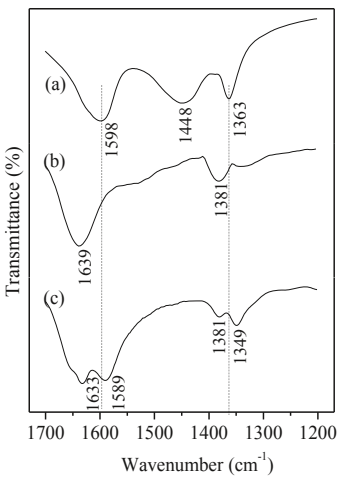

Figure 5. FT-IR spectra (left) of (a) sodium formate, (b) goethite, (c) treated goethite

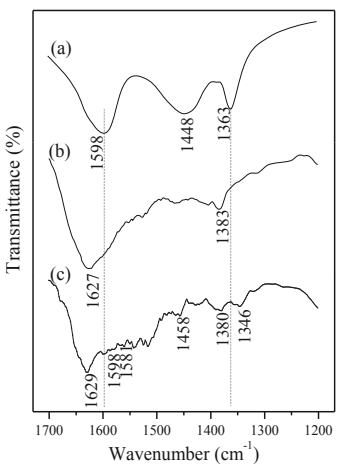

Figure 6. FT-IR spectra (right) of (a) sodium formate, (b) hematite, (c) treated hematite

In the FT-IR spectrum of free sodium formate (Figure 5a), the asymmetric (ν_{asym}) and symmetric (ν_{sym}) stretches of the carboxylate group are located at 1598 cm^{-1} and 1363 cm^{-1}, respectively. Furthermore, the weak band at 1448 cm^{-1} is assigned to the C-H bending vibration of free sodium formate.

For goethite after the treatment of sodium formate, spectral bands at 1589 cm^{-1} and 1349 cm^{-1} are indicative of the asymmetric (ν_{asym}) and symmetric (ν_{sym}) stretches of carboxylate group (Figure 5c). $\Delta\nu$(salt) for free sodium formate is 235 cm^{-1}, while

Δν(adsorbed) of adsorbed sodium formate is 240 cm^{-1}. This suggests that a bidentate mononuclear structure (BM) may be formed between two oxygen atoms of carboxylate group in sodium formate and one surface iron atom of goethite.

For hematite after the treatment of sodium formate, spectral bands at 1581 cm^{-1} and 1346 cm^{-1} are indicative of the asymmetric (ν_{asym}) and symmetric (ν_{sym}) stretches of carboxylate group Figure 6c). Δν(salt) for free sodium formate is 235 cm^{-1}, while Δν(adsorbed) of adsorbed sodium formate is also 235 cm^{-1}. This suggests that a bidentate mononuclear structure (BM) may be formed between two oxygen atoms of carboxylate group in sodium formate and one surface iron atom of hematite.

X-ray photoelectron spectroscopy analysis.

For the analysis of Fe2p spectra, a Shirley background is used for the Fe2p1/2 and Fe2p3/2 envelopes. The Fe2p1/2 and Fe2p3/2 envelopes are fitted using peaks corresponding to the Gupta and Sen (GS) multiplets, surface structures and shake-up satellites[31]. The fitting of each compound follows the GS predictions well. After the Fe2p spectrum was fitted, the main peak centre of gravity (CG) was determined using the GS multiplets only, which is the Peak2 in the fitting curve. The Fe2p spectra and fitting curves for goethite or hematite before and after treatment of sodium formate are depicted in Figure 7. The fitting peaks were listed in Table 2

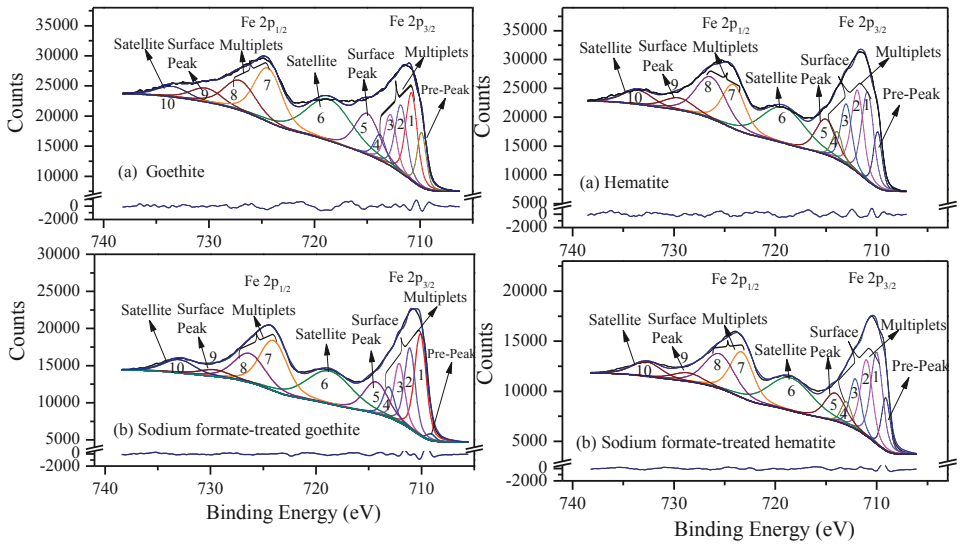

Figure 7. The Fe2p spectra and the fitting curves (Left panel) of (a) goethite, (b) treated goethite; the Fe2p spectra and the fitting curves (right panel) of (a) hematite, (b) treated hematite

Table 2 The fitting peaks of Fe2p spectra for goethite and hematite before and after the treatment of sodium formate

Fitting peaks	Untreated goethite	treated goethite	Untreated hematite	Treated hematite
Pre-Peak	709.8[1.0][a]	709.1[1.0]	709.9[1.0]	709.1[1.0]
Peak1	710.8[1.3]	710.1[1.3]	710.9[1.4]	710.0[1.4]
Peak2	711.8[1.3]	711.1[1.2]	711.9[1.4]	711.0[1.4]
Peak3	712.8[1.3]	712.1[1.2]	713.0[1.3]	712.1[1.4]
Peak4	713.8[1.3]	713.1[1.3]	713.9[1.2]	713.0[1.2]
Peak5	715.0[2.7]	714.3[2.7]	715.0[2.4]	714.1[2.4]
Peak6	718.5[5.0]	718.7[5.0]	719.2[5.0]	718.5[5.0]
Peak7	724.4[3.1]	724.0[3.1]	724.1[2.7]	723.3[2.7]
Peak8	727.0[3.5]	726.3[3.5]	726.4[3.5]	725.5[3.5]
Peak9	730.3[3.5]	729.6[3.5]	729.5[3.5]	728.6[3.5]
Peak10	733.5[3.5]	733.0[3.5]	733.6[3.5]	732.7[3.5]

[a]The value in the square brackets is the FWHM of each peaks.

As presented in Figure 7 and Table 2, it is observed that the binding energies of all fitting peaks of Fe2p for goethite or hematite before the treatment of sodium formate were decreased compared with that after the treatment of sodium formate. Compared with untreated goethite, the peaks in Fe2p spectrum for treated goethite are correspondingly downshifted by 0.7 eV relative to untreated goethite. In comparison with untreated hematite, the peaks in the Fe2p spectrum for treated hematite are both downshifted by 0.9 eV. In this work, the downshift of binding energies of Fe2p spectra for treated goethite or hematite will be explained by the atomic potential model[32]. According to the atomic potential model, the ion atoms may be partially reduced to a lower valence state. The coordination between oxygen atoms of carboxylate group from sodium formate and surface iron atoms of goethite or hematite may explain the partial reduction of iron atoms on the surfaces, because these iron atoms accepted electron clouds from oxygen atoms of carboxylate group from sodium formate. However, the binding energy shift of the peaks of Fe 2p spectra for treated goethite was a little smaller than that of treated hematite, which may be due to the differences of the interfacial structures between treated goethite and hematite.

Quantum chemical calculations.

Interfacial structures. The O^H-O^H distance (2.925Å) at Fe site of goethite (101) or (100) surface and the O^H-O^H distance (2.622 Å) at Fe site of hematite (001) surface matches well with the O-O distance (2.359 Å) of sodium formate, which allows sodium formate to form a BM structure on the goethite (101) and (100) surfaces. Here BM structure, which specifies the linkage between two oxygen atoms of sodium formate and surface iron site, was verified in detail (Figure 4).

The interatomic distances and the lattice parameters for bulk goethite or hematite are listed in Table 3. The Fe-O distances in optimized bulk goethite are in the range from 1.864 Å to 2.008 Å, which owns a difference within 0.128 Å relative to the previously calculated data in the range from 1.901 to 2.136 Å for bulk goethite[33]. The Fe-O distances in optimized bulk hematite are 2.000 Å and 2.024 Å, which is in agreement with the observed data of 1.9511 Å and 2.1028 Å from the single crystal of hematite[34].

The results indicate that the calculation method used in this work is reasonable.

Table 3

The interatomic distances (Å) and the lattice parameters (Å) for bulk goethite or hematite in this study, in Ref 33 or in Ref 34

Sample	Fe-O		Lattice parameter	
Bulk goethite	1.864	1.901[33]	a=4.395	a=4.598 [33]
	1.879 (×2)	1.979 (×2)	b=9.510	b=9.951 [33]
	1.993 (×2)	2.121 (×2)	c=2.925	c=3.018 [33]
	2.008	2.136	α=β=γ=90	α=β=γ=90
			a=b=5.035	a=b=5.037
Bulk hematite	2.000 (×3)	1.951 (×3)[34]	c=13.720	c=13.771[34]
	2.024 (×3)	2.103 (×3)[34]	α=β=90	α=β=90
			γ=120	γ=120

Values in square brackets give the number of bond.

The interatomic distances and the bond angles for the BM structures of sodium formate on goethite (101), (100) and hematite (001) surfaces are listed in Table 4.

Table 4 The calculated interatomic distances (Å) and bond angles (°) for the interfacial structures of sodium formate adsorbed on goethite (101), (100) or hematite (001) surfaces

d and ∠	Sodium formate-goethite (101)		Sodium formate-goethite (100)		Sodium formate-hematite (001)	
d_{Fe-O}	Fe_{12}-O_2	2.011	Fe_1-O_2	1.928	Fe_8-O_2	2.076
	Fe_{12}-O_1	2.071	Fe_1-O_1	1.959	Fe_8-O_1	2.041
	Fe_{20}-O_2	2.128	Fe_9-O_2	2.135	-	-
	Fe_4-O_1	2.927	Fe_5-O_1	2.515	Fe_{11}- O_1	3.845
d_{C-O}	C_1-O_1	1.272	C_1-O_1	1.466	C_1-O_1	1.284
d_{C-O}	C_1-O_2	1.286	C $_1$-O_2	1.482	C_1-O_2	1.270
$\angle_{O-Fe-O}$	O_1-Fe_{12}-O_2	65.2	O_1-Fe_1-O_2	72	O_1-Fe_8-O_2	64.1
$\angle_{Fe-O-C}$	Fe_{12}-O_1-C_1	87.0	Fe_1-O_1-C_1	90.9	Fe_8-O_1-C_1	89.7
$\angle_{Fe-O-C}$	Fe_{12}-O_2-C_1	89.3	Fe_1-O_2-C_1	91.0	Fe_8-O_2-C_1	88.6
$\angle_{O-C-O}$	O_1-C_1-O_2	118.5	O_1-C_1-O_2	99.9	O_1-C_1-O_2	117.6

As listed in Table 4, after goemetry optimization of the interfacial structures of sodium formate adsorbed on the goethite (101), (100) or hematite (001) surface, the average Fe-O distance for the BM structures is approximately 2 Å. The interatomic distances of Fe_{12}-O_2 and Fe_{12}-O_1 are correspondingly 2.011 Å and 2.071 Å in the interfacial structures of sodium formate adsorbed on goethite (101) surface, and the interatomic distances of Fe_1-O_2 and Fe_1-O_1 are correspondingly 1.928 Å and 1.959 Å in the interfacial structures of sodium formate adsorbed on goethite (100) surface.

In addition, it can be observed that the interatomic distance between one of the oxygen atoms of sodium formate and the neighboring iron atom almost equals to the average Fe-O distance of the BM structure. Thus, the adsorbed sodium formate on goethite (101) or (100) surface forms a complex as a BM structure, and one of the oxygen atoms of sodium formate simultaneously attach to the neighboring surface iron atom (Figure 4a and b). The interatomic distances of Fe_8-O_2 and Fe_8-O_1 are correspondingly 2.076 Å and 2.041 Å in the interfacial structures of sodium formate adsorbed on hematite (001) surface, whereas, the interatomic distance of Fe_{11}-O_1 in the interfacial structure of sodium formate adsorbed on hematite (001) surface is 3.845 Å, which is much larger than the average Fe-O distance of 2 Å in the BM structures. Therefore, the formation of a five-membered ring complex as a BM structure occurs on hematite (001) surface without any extra attachments (Figure 4c).

Adsorption energies. On the basis of adsorption energies, goethite (101), (100) or hematite (001) surfaces are predicted to be capable of adsorbing sodium formate. The adsorption energies of the BM complex of sodium formate adsorbed on goethite (101) and (100) surface are correspondingly -1.45 eV and -7.58 eV. The favorable interfacial structure of sodium formate adsorbed on hematite (001) is the BM complex with Eads of -6.78 eV.

Electronic properties and core level shift. We choose to probe the electronic structures of sodium formate and the BM complexes on goethite (101), (100) or hematite (001) surfaces. The calculated Mulliken charge values are listed in Table 5.

Table5 Calculated Mulliken charges (e), Charge transfer (e) and photoemmission CLS (eV) of atoms

Structure	Atom	Mulliken charge	Charge transfer	Calculated CLS
Sodium formate		O_1:-0.71, O_2:-0.71	-	-
Goethite (101)	Fe_5	1.01	-	-
Goethite (100)	Fe_9	1.05	-	-
Hematite (001)	Fe_9	1.17	-	-
Sodium formate adsorbed on the goethite (101)		O_1:-0.56, O_2:-0.54	-	-
	Fe_4	0.96	-0.05	-
	Fe_{12}	0.97	-0.04	-0.08
	Fe_{20}	1.00	-0.01	-
Sodium formate adsorbed on the goethite (100)		O_1:-0.65, O_2:-0.59	-	-
	Fe_1	1.04	-0.01	-0.46
	Fe_5	0.91	-0.14	-0.55
	Fe_9	1.05	-0.00	-
Sodium formate adsorbed on the hematite (001)		O_1:-0.55, O_2:-0.55	-	-
	Fe_8	1.03	-0.14	-0.72
	Fe_{20}	1.08	-0.09	-0.39

Table 5 indicates that the negative charges of sodium formate are mainly clustered on O_1 and O_2 atoms of carboxylate group. Hence O_1 and O_2 atoms in carboxylate group of sodium formate are the electron-donating centers and the chemically reactive centers. After the adsorption of sodium formate on the goethite (101), (100) or hematite (001) surfaces occurred, the charges of two oxygen atoms of the carboxylate group increased, and the charges of the adsorbed surface Fe sites decreased. This means that electron clouds are transferred from O atoms of carboxylate group to adsorbed surface Fe sites or their nearest neighboring iron atoms on the goethite (101), (100) or hematite (001) surfaces. Therefore, the Fe 2p peaks of the adsorbed surface iron sites may appear at lower binding energy than that of surface iron atoms at clean mineral surfaces. Here we compare the calculated CLS with the experimental Fe2p core level photoemission measurements.

The fitting peaks for treated goethite were downshifted by 0.7 eV compared with untreated goethite (Table 2). The calculated CLS of Fe2p (-0.08 eV) on goethite (101) surface is not consistent with the experimentally observed CLS of Fe2p (-0.7 eV). The calculated CLS of Fe2p (-0.55 eV) on goethite (100) surface is in good agreement with the experimentally observed CLS of Fe2p (-0.7 eV). Thus, the goethite (100) surfaces may be predicted to be capable of adsorbing sodium formate as a BM structure. On the other hand, the fitting peaks for treated hematite are downshifted by 0.9 eV relative to untreated hematite. Our calculated CLS of Fe2p (-0.72 eV) on hematite (001) surface is in good agreement with the experimentally observed CLS of Fe2p (-0.9 eV). This consistency suggests that our optimized interfacial structure for sodium formate-hematite (001) system can be regarded as a reasonable and realistic structure. Furthermore, due to the extra bond to the neighboring iron atoms in the sodium formate-goethite system, the calculated or the observed CLS of the fitting peaks of Fe 2p spectra for treated goethite was a little smaller than that of treated hematite.

Conclusion

The adsorption of sodium formate on goethite or hematite surface was investigated by FT-IR, XPS and density functional theory (DFT) calculations, respectively. The goemetry optimization by DFT indicates that the adsorbed sodium formate on goethite (101) and (100) forms a BM structure, and one oxygen atom of sodium formate simultaneously attach to one neighboring surface iron atom. Whereas, the formation of a BM structure occurs on hematite (001) surface without any extra attachments between two oxygen atoms of sodium formate and the neighboring surface iron atoms.

The calculated CLS of Fe2p for the interfacial structure on goethite (101) surface was not consistent with the experimental observed CLS of Fe2p, and the calculated CLS of Fe2p for the interfacial structure on goethite (100) surface and hematite (001) surface are in good agreement with the experimentally observed CLS of Fe2p. Thus, the goethite (100) surface and hematite (001) surface may be predicted to be capable of adsorbing sodium formate as BM structures.

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Light Metals 2016

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Effect of Different Silica Mineral Compositions on the Digestion Results in Bayer Process

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Abstract: The bauxite in the northern region of China belongs to diasporic type ore with high silica. The main silica minerals in the bauxite is kaolinite，illite and chamosite. Technological investigations were carried out based on the Bayer process. The presence of high silica causes high bound-soda losses in the red mud and the decrease of leaching rate. The dissolution behavior of different silicate minerals has obvious effect on the digestion of bauxite. Predesilication characteristic of the bauxite, the effects of dissolution temperature and retention time on Al2O3 digestion efficiency were determined. Experiment results showed that of chamosite and illite mineral characteristics and dissolution characteristics of chamosite helped to lower alkali consumption to improve the digestion rate of bauxite.

Keywords: different silica minerals, settling separation, Bayer process, diasporic bauxite

1. Chemical composition of the bauxite samples

After crushing and mixing of each sample, their chemical composition was analyzed. The results are listed in Table I. It is obvious that only a small number of bauxite samples contained low SiO_2 content less than 7%, so it is not likely to produce alumina with it through Bayer process. But fortunately, part of silica in this bauxite exists as chamosite, and which does not react with caustic to form sodium-aluminum hydrosilicates causing soda losses in Bayer digestion process. Therefore, the possibility of using this type of bauxite as raw material to produce alumina by Bayer process still exists. We divided the bauxite samples into 2 groups (high and low silica content groups), mixed the samples into 6 mixed samples and ground them to pass 100 mesh. The mixed samples were used for predesilication, digestion and red mud settling tests by Bayer process.

Table I Chemical Composition of the bauxite Samples.

Sample Number	Composition(Wt %)				
	SiO_2	Al_2O_3	Fe_2O_3	TiO_2	Ignition Loss
c-1	10.2	52.9	21.4	3.5	10.9
c-2	15.1	48.9	21.0	3.0	11.3
c-3	9.5	51.8	20.9	3.9	12.0
c-4	13.9	44.9	21.9	3.6	11.8
c-5	6.4	49.2	26.4	5.4	11.6
c-6	16.3	49.0	14.9	5.7	11.9

2. Mineral composition and structural characteristics of the bauxite samples

The mineral composition and structural characteristics of the bauxite samples were determined by X-ray diffractometry, Infrared Spectrophotometry, Polarization Microscopy and Scanning Electron Microscopy analyses.

Figures 1 and 2. illustrate the x-ray diffraction patterns and IR absorption spectra for the six bauxites, respectively. Alumina is present primarily as diaspore. Silica is present mainly as kaolinite and chamosite(as Figures 3 and 4) with trace amounts as illite. Hematite is the main iron mineral (wt % 19.0 consistent), but limonite, siderite exist in small quantities. Rutile and anatase are the main titanium minerals. There are some other minerals such as calcite and zircon in the bauxite.

The distribution of alumina and silica in different minerals was calculated on the basis of quantitative analysis of main minerals in various bauxite samples and the results are listed in Table II.

Further investigation on structural characteristics of the bauxite was carried out. It was found that the bauxite did not belong to bedded ore deposit and the texture is rather complicated. The main textural characteristics of the bauxite are crystal, microlitc and aphanitic texture.

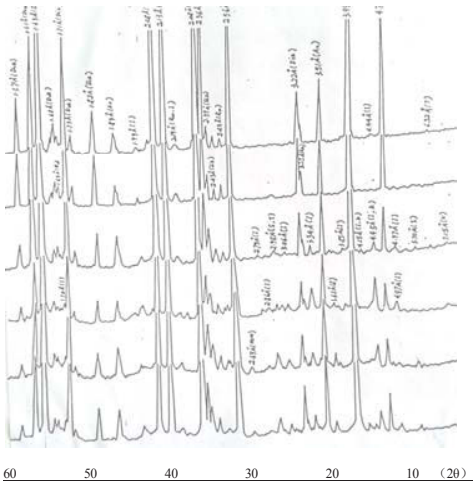


Figure 1 X-ray diffraction pattern of the bauxite samples

Note: a to f from top to bottom

a. c-1, b. c-2, c. c-3, d. c-4, e. c-5, f. c-6

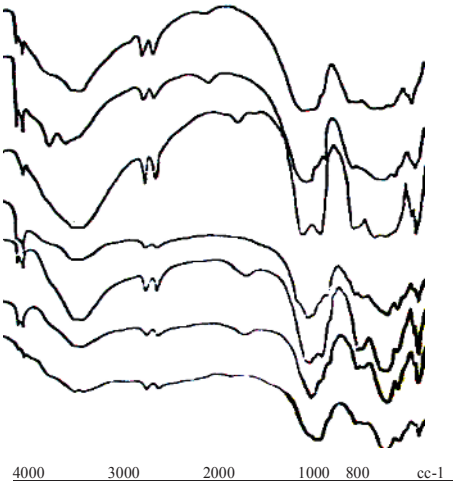


Figure 2 IR absorption spectrum of the bauxite samples

Note: a to f from top to bottom second spectral line

a. c-1, b. c-2, c. c-3, d. c-4, e. c-5, f. c-6

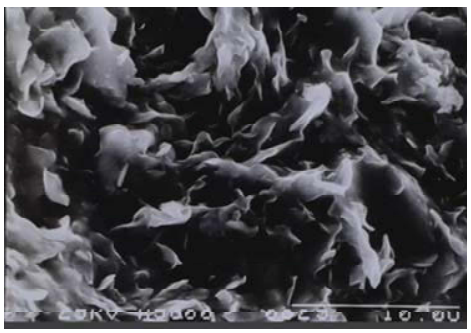


Figure 3 chamosite crystal in bauxiteSEM

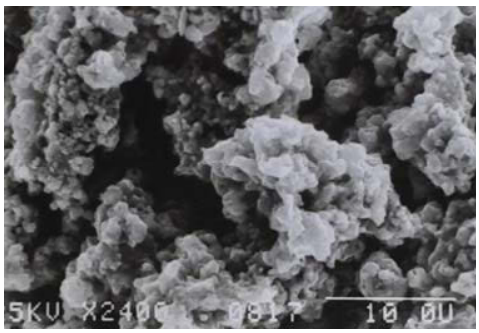


Figure 4 kaolinite crystal in bauxiteSEM

Table II. The distribution of the main components of bauxite in individual mineral phases

Components	Composition(Wt %)					
	c-5	c-6	c-3	c-4	c-1	c-2
Al_2O_3 in diaspore	44.2	33.7	43.9	35.3	42.2	27.7
Al_2O_3 in kaolinite	4.0	6.4	1.7	10.0	4.8	10.4
Al_2O_3in chamosite	2.1	2.8	1.8	2.2	3.4	1.6
SiO_2 in kaolinite	3.3	4.8	2.6	2.2	3.7	1.5
SiO_2 in chamosite	4.6	7.6	2.0	11.9	5.7	12.3
Fe_2O_3 in hematite	19.5	18.9	18.7	20.5	24.7	13.5

3. Predesilication tests

The purpose of predesilication tests was to determine the extent of reaction between SiO_2 of the bauxite and alkaline liquor to form sodium-aluminum-hydrosilicate in slurry tanks. Raw materials of the tests is mixed bauxite samples，Calcium oxide content in raw lime is 90.1%.

The Digestion liquor was prepared from chemically pure NaOH, Na_2CO_3, $Al(OH)_3$ and de-ion water.

The results of predesilication efficiency are listed in Table III.

The results of predesilication tests showed: 1) when the concentration of Na_2O_T in liquor is in the range from 150 to 190 g/L, the rate of sodium-aluminum-hydrosilicate formation was not significantly influenced by the concentration of

Table III. The results of predesilication tests of the bauxite samples

Sample Number	Amount of Bauxite (g/L)	Amount of Lime (g/L)	Composition of Digestion liquor (g/L)			Temperature of Reaction (°C)	Retention Time (hr.)	Predesilication Efficiency (%)
			Na_2O_T	Al_2O_3	Na_2O_K			
c-1	181.7	12.9	162.1	67.5	139.8	98	2	53.3
c-2	205.1	19.0	192.3	73.3	159.9	98	8	57.3
c-3	149.3	10.2	162.0	67.4	139.4	98	2	25.5
c-4	149.1	10.5	162.2	67.3	139.3	98	8	43.6
c-5	192.5	12.4	162.3	67.5	139.6	98	2	25.3
c-6	192.7	12.5	161.9	67.2	139.5	98	8	30.5

Note: Na_2O_T=Na_2O_K+Na_2O_C, the same as in following table

Na_2O_k; 2) Predesilication efficiencies of these six bauxite samples were different due to their percentage differences in chamosite contents.

4. Effect of chamosite on Bayer process

The purpose of the tests was to determine the digestion efficiency of the bauxite in different digestion conditions (temperature, retention time, concentration of Na_2O_k and amount of lime addition). The raw materials for the tests were the same as those used in predesilication tests.

Four kinds of digestion liquor were used in the tests. They are all made up of chemically pure NaOH, Na_2CO_3, $Al(OH)_3$ and de-ion water. Their chemical compositions are as below:

Table IV Composition of Digestion Liquor(g/L)

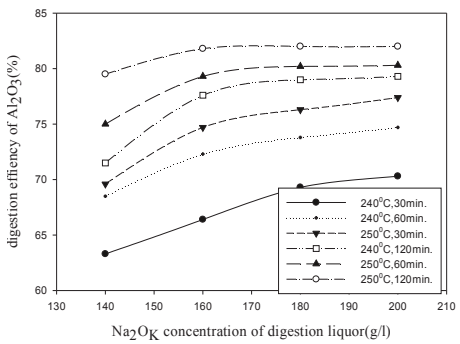
Na_2O_T	Al_2O_3	Na_2O_k	SiO_2	$(Na_2O_k)/(Al_2O_3)$
221.2	95.0	201.0	0.2	3.45
201.0	85.2	180.0	0.2	3.48
182.3	74.8	160.0	0.1	3.52
163.4	66.4	140.0	0.1	3.47

Note: molecular formula plus ( ) represents the mole ratio of sodium oxide and alumina

Technological investigations were carried out based on the Bayer process. Steel Bombs in melt salts were used for the test. The results are shown in Figure 5.

The presence of high silica causes high bound-soda losses in the red mud in the Bayer process, so most of the high silica digestion were carried out on the samples by Soda-lime sintering, but a part of silica content in the bauxite was in the form of chamosite, the bound-soda losses can be greatly reduced with chamosite mineral by using Bayer process.

Fig 5 The influence of temperature, retention time and concentration of Na_2O_k on the digestion efficiency.



Because the bauxites are the diasporic type ore, digestion temperature of 240-250ºC and lime addition were used in digestion test. Besides that, several digestion tests were conducted at 260ºC(the highest digestion temperature used in autoclave for alumina production in the world) and 280-300ºC (to investigate the possibility of using tube digester). The highest digestion temperature，the better digestion efficiency.

It shows that digestion temperature has the great influence on Al_2O_3 digestion efficiency. The results of Bayer digestion tests can be seen from the curves in Figure 3 that higher digestion efficiency can be achieved at 250ºC and 120 minutes retention time. At the moment, the influence of Na_2O concentration (in

140-200 g/L Na_2O) on digestion efficiency is very small. If the digestion temperature is 250 ºC and the retention time is only 60 minutes, in order to obtain better digestion efficiency, a digestion liquor containing 160-200 g/L Na_2O is needed.

The Silica content in sample c-1 was 10.2%. If all the silica in the bauxite was transformed into Na-Al-hydrosilicate, bound-soda losses in red mud would be more than 100kg Na_2O /T Al_2O_3. However a portion of the silica content in the bauxite was in the form of chamosite, so only part of it reacted with alkaline liquor, and the bound-soda losses in red mud was just 82.3 kg Na_2O /T Al_2O_3. Because of low silica content in c-5 sample, so bound-soda losses in red mud was about 73.5 kg Na_2O /T Al_2O_3.

The silica content in the c-3 sample was less than that in c-1, however the chamosite content was also less than that in c-1, therefore, its bound-soda losses in red mud reached 110.5 kg Na_2O /T Al_2O_3.

Different content and mineralogical form of silica in these three bauxites samples had different influence on Al_2O_3 digestion character, and led to great difference in digestion efficiency of Al_2O_3.

The influence results of lime addition on Al_2O_3 digestion efficiency show that the optimum amount of lime addition varies from bauxite to bauxite. It is in about 0.03 to 0.06 CaO g/g bauxite range.

The Al_2O_3 digestion data shows that digestion rate will be increased as digestion temperature increases. At 280-300 ºC, digestion reaction is finished in 5 minutes. That shows there is a possibility of using tube digester to process this kind of bauxite.

5. Settling tests of Bayer red mud

The purpose of the tests was to determine the settling rate of red mud in slurry after digestion, heights and compressed liquid/solid ratios (weight) of settled red mud. Methods of the tests: Red mud slurry produced from digestion in the Steel Bombs was added to the settling tubes, and the settling rate, heights and compressed liquid/solid ratios of settled red mud were measured.

The results are listed in Table V.

The results show that the settling properties are improved after adding flocculants. The settling rate is about 1.3-2.2 cm/min. at first 5 minutes while the height of settled red mud is about 30% after 60 minutes. The liquid/solid ratios of compressed red mud are about 4.

However, the compressibility of red mud is not so good, the height of compressed red mud reaches approximately 60% (without flocculants), while water content in red mud is about 80-85%. Therefore, it is not suitable to use settling separation for the red mud separation, while fast liquid/solid separation equipment should be used instead.

6. Conclusion

1 The results of digestion tests show that a part of silica content in this kind of bauxite exists in chamosite form, and low silica content samples are chosen for digestion by Bayer process, bound soda losses in red mud come to 73.5-82.3kg Na_2O /T Al_2O

Table V. Settling properties of Bayer Red Mud

Clear liquid zone			Settling rate (cm/min)				Height of settled red mud (%)	Compressed liquid/solid of red mud (w/w)
Concentration of Al_2O_3(g/L)	(Na_2O_K) /(Al_2O_3)	Specific gravity(ml/g)	0-5 min	0-10 min	0-15 min	0-60 min		
129.5	1.54	1.22	1.49	0.87	0.62	0.20	33.3	3.78
153.6	1.55	1.25	1.25	0.86	0.62	0.20	33.7	4.12
123.6	1.54	1.20	1.95	1.08	0.75	0.21	26.6	3.11
145.9	1.55	1.24	1.26	0.84	0.61	0.20	32.1	3.57
148.3	1.55	1.23	2.17	1.17	0.81	0.23	20.6	4.08
145.1	1.56	1.23	1.88	1.07	0.74	0.21	24.8	3.76
145.5	1.56	1.24	1.45	0.88	0.63	0.20	32.5	3.43

Note: Specific gravity(ml/g) which refers to the degree of polymerization of the flocculants

2 Digestion properties of the bauxite in Bayer process are similar to other region diasporic bauxite, but their silica content is higher, so it should be considered to find low silica content bauxite resource (less than 6% SiO_2) or silica content in the bauxite was in chamosite form, in order to decrease bound soda losses.

3 It is not suitable to use settling separation for the red mud separation, while fast liquid/solid separation equipment and high polymer should be used instead.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF LIME ADDITION DURING DIGESTION ON STABILITY OF DIGESTED LIQUOR OF DIASPORIC BAUXITE

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Keywords: Bayer process, Sodium aluminate solution, Digestion, Hydrolysis, Desilication product

Abstract

The hydrolysis loss of digested alumina from bauxite reentering into the bauxite residue has always been the research focus during the Bayer process. The stability of sodium aluminate solution after digestion with different lime additions during the dilution process from a diasporic bauxite was studied, and the mechanism by which the lime addition affects was also discussed. The molar ratio, silica index and hydrolysis degree of diluted liquor after digestion are highly affected by the lime addition, and the corresponding alumina extraction efficiency of bauxite and the Na_2O to SiO_2 ratio of bauxite residue are changed. The stability of diluted liquor is firstly increased by the lime addition below 15% of the bauxite, and then decreased as the lime addition increases. The mineral content and proportion of desilication products precipitating during digestion are the main factors affecting the stability of sodium aluminate solution after digestion.

Introduction

As the world leader in alumina production, China produced over 51.25 million tonnes of alumina, comprising nearly half of the global alumina output of 112.27 million tonnes in 2014 [1]. Approximately 98% of China's bauxite ore is diasporic bauxite with low alumina to silicate ratio, which has the characteristic of high alumina and silicate content and low iron content. Comparing with gibbsite and boehmite, extraction from diasporic-bauxite by the Bayer process requires higher temperatures (240～260 °C) and sodium concentrations (220～260 g/L in the form of Na_2O) [2]. Meanwhile, lime must be added during the Bayer digestion process to produce insoluble titanates instead of sodium titanate to minimize or even eliminate the unfavorable influences of TiO_2 on the alumina recovery, otherwise an impervious nanometer thick layer of sodium titanate will cover the bauxite particles to seriously hinder further reaction of alumina hydrates in the bauxite [3, 4].

The Bayer process for alumina production is mainly comprised of bauxite digestion, solid-liquid separation, gibbsite precipitation and calcination. In order to meet the needs of further desilication and the following settling of bauxite residue after the bauxite digestion, the digested slurry usually has to be diluted. However, the diluted slurry is not stable due to the low molar ratio of Na_2O to Al_2O_3 in sodium aluminate solution, and correspondingly the gibbsite will subsequently reprecipitate known as the hydrolysis reaction. The hydrolysis degree of the diluted slurry during the settling process is usually over 2 %, and sometimes even over 5 %, which leads to a large number of alumina losing into the bauxite residue [5].

The hydrolysis loss of digested alumina reentering into the bauxite residue during the Bayer process has always been the research focus of the enterprises and researchers. The stability of sodium aluminate solution after digestion with different lime additions during the dilution process from a diasporic bauxite was studied in this paper, and the mechanism by which the lime addition affects was also discussed.

Experimental

Materials

The main chemical composition of gibbsitic bauxite is shown in Table 1. As presented in Table 1, the contents of Al_2O_3 and SiO_2 in the bauxite are 51.80 wt% and 8.11 wt%, and the calculated weight ratio of Al_2O_3 to SiO_2 is 6.39. The XRD pattern of the bauxite is shown in Fig. 2. According to Fig. 2, the bauxite is comprised of diaspore (AlOOH), kaolinite ($Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$), hematite ($Fe_2O_3$), goethite (FeOOH), quartz (SiO_2), and anatase (TiO_2).

The bauxite was ground to less than 0.28 mm in particle size, and lime with 93 wt.% reactive CaO was produced by calcination of industrial $CaCO_3$ at 1050 °C for 3 h. Sodium aluminate solution for the digestion experiments was prepared by dissolving industrial sodium hydroxide and aluminum hydroxide into the industrial Bayer liquor. The concentrations of caustic Na_2O (N_K) and Na_2CO_3 (N_C, in the form of Na_2O) were 240 g/L and 15 g/L respectively, and the molar ratio of caustic Na_2O to Al_2O_3 (AO) is 2.90.

Table 1 Chemical composition of the diasporic bauxite (wt%)

Al_2O_3	Fe_2O_3	SiO_2	TiO_2	MgO	CaO	Na_2O	LOI
51.80	24.69	8.11	2.66	0.23	0.10	0.02	12.03

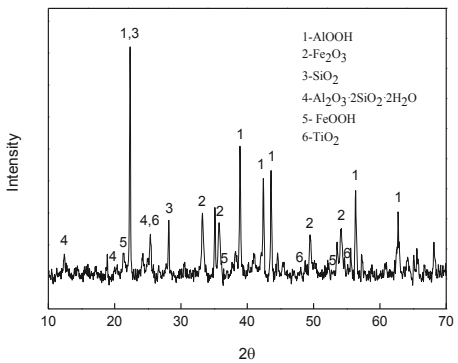

Fig. 1 XRD pattern of the diasporic bauxite

Experimental procedures

The digestion experiments were carried out in a 150 mL bomb reactor. The reactor was heated by molten salt with a temperature control accuracy of ±1 °C, and the bomb was rotated at 48 rpm fixedly. The slurry mixed by the bauxite and Bayer liquor (28.57 g bauxite in 100 ml liquor) with different lime dosages was put into the bombs, sealed and rotated. The digestion experiments were performed at 280 °C for 1 h, cooled and separated by centrifuging. The solids were then washed and dried.

The diluted desilication experiments to simulating the settling process of digested slurry were carried out in a stainless steel vessel with a nominal capacity of 500 mL. The digested slurry was diluted by hot deionized water and added into the vessel immediately. The N_K of diluted slurry were between 170-180 g/L. The sealed vessel was then put into the pre-heated oil bath followed by agitation at 100 rpm. The diluted slurry was held at 95 °C for different durations, cooled and separated by centrifuging.

The concentrations of Al_2O_3 and Na_2O in sodium aluminate solution were analyzed by the volumetric method, and the concentration of SiO_2 was analyzed by the spectrophotometry using a 722S spectrophotometer. The solids were washed and dried for chemical analysis. The compositions of Al_2O_3 and SiO_2 were analyzed by X fluorescence spectrometer (XRF, ZSX100e), while the composition of Na_2O was analyzed by atomic absorption spectrometry. X-ray diffraction measurements were performed to identify the minerals of samples by an X-ray powder diffractometer (XRD, PANalytical PW3040/60) using Cu-Kα X-radiation with a scan rate of 3° 2θ/min.

The hydrolysis degree of sodium aluminate solution was calculated by the following formula:

$$\eta = \frac{\alpha_{Kt} - \alpha_{K0}}{\alpha_{Kt}} \times 100\% \quad (1)$$

where α_{K0} and α_{Kt} denote the molar ratio of caustic Na_2O to Al_2O_3 in sodium aluminate solution at the time 0 and t.

The alumina digestion efficiency of bauxite was calculated by the following formula:

$$\eta_{Al_2O_3} = \frac{(A/S)_{ore} - (A/S)_{residue}}{(A/S)_{ore}} \times 100\% \quad (2)$$

where $(A/S)_{ore}$ and $(A/S)_{residue}$ denote the weight ratio of Al_2O_3 to SiO_2 in bauxite ore and bauxite residue respectively.

Results and discussion

The digestion and dilution results after digestion with different lime additions are listed in Table 2. The XRD patterns of the bauxite residues with different lime additions after digestion are presented in Fig. 2. As for the digestion of bauxite without lime addition, most gibbsite remains in the bauxite residue. The main minerals in the bauxite residues with different lime additions are hematite (Fe_2O_3), zeolite ($Na_2O{\cdot}Al_2O_3{\cdot}1.68SiO_2{\cdot}1.73H_2O$), grossular hydrogarnet ($Ca_{2.93}Al_{1.97}Si_{0.64}O_{2.56}(OH)_{9.44}$), andradite-grossular ($Ca_3AlFeSiO_4(OH)_8$) and perovskite ($CaTiO_3$).

Table 2 Digestion and diluted desilication results with different lime additions

Lime addition (%)	Time (min)	Liquor concentration ($g{\cdot}L^{-1}$)			Solid composition (wt.%)		
		N_K	AO	SiO_2	Al_2O_3	SiO_2	Na_2O
0	0	170.3	136.4	0.42	41.96	9.68	6.52
	60	177.2	138.7	0.35	42.36	9.56	5.57
	120	177.2	134.6	0.25	44.58	9.84	6.39
7.5	0	171.5	198.2	1.11	15.26	13.53	5.96
	60	175.2	200.4	1.02	15.46	13.36	5.28
	120	176.1	199.6	0.90	16.32	13.76	5.77
15	0	172.6	199.3	1.03	14.86	11.80	3.53
	60	180.7	207.8	0.94	15.12	11.94	2.78
	120	178.4	204.7	0.84	15.30	12.05	3.53
22.5	0	176.0	202.5	0.93	14.29	10.87	0.65
	60	180.2	206.3	0.82	14.41	10.87	0.63
	120	178.5	203.6	0.73	15.11	11.27	0.82
30	0	178.4	196.8	0.53	15.38	10.35	0.32
	60	182.1	199.7	0.41	15.36	10.29	0.43
	120	181.8	198.5	0.31	15.95	10.57	0.28

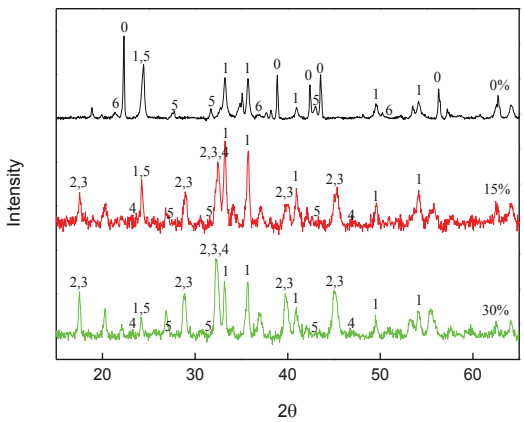

Fig. 2 XRD patterns of bauxite residues digested with different lime additions
0-gibbsite; 1-hematite; 2-grossular hydrogarnet; 3-andradite-grossular and; 4-perovskite; 5-zeolite; 6-goethite

The possible dissolution reactions of minerals in diasporic bauxite during digestion are listed as Eq. (3) to (9).

$$AlOOH(s) + NaOH(aq) + H_2O \rightarrow NaAl(OH)_4(aq) \quad (3)$$

$$Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O(s) + 6NaOH(aq) \rightarrow 2NaAl(OH)_4(aq) + 2Na_2SiO_3(aq) + H_2O \quad (4)$$

$$SiO_2(s) + 2NaOH(aq) \rightarrow Na_2SiO_3(aq) + H_2O \quad (5)$$

$$TiO_2(s) + 2NaOH(aq) \rightarrow Na_2TiO_3(aq) + H_2O \quad (6)$$

$$CaO(s)+H_2O \rightarrow Ca(OH)_2(aq) \quad (7)$$

$$Na_2TiO_3(aq)+Ca(OH)_2(aq) \rightarrow CaTiO_3(s)+2NaOH(aq) \quad (8)$$

$$2FeOOH(s) \rightarrow Fe_2O_3(s)+ H_2O \quad (9)$$

The desilication reactions during the digestion process according to the XRD results may occur as presented from Eq. (10) to (12).

$$2NaAl(OH)_4(aq)+1.68Na_2SiO_3(aq) \rightarrow Na_2O \cdot Al_2O_3 \cdot 1.68SiO_2 \cdot 1.73H_2O(s)+3.36NaOH(aq)+0.59H_2O \quad (10)$$

$$1.97NaAl(OH)_4(aq)+0.64Na_2SiO_3(aq)+2.93Ca(OH)_2(aq) \rightarrow 2.93CaO \cdot 0.985Al_2O_3 \cdot 0.64SiO_2 \cdot 4.72H_2O(s)+3.25NaOH(aq)+0.525H_2O \quad (11)$$

$$NaAl(OH)_4(aq)+Na_2SiO_3(aq)+3Ca(OH)_2(aq)+0.5Fe_2O_3(s) \rightarrow 3CaO \cdot 0.5Fe_2O_3 \cdot 0.5Al_2O_3 \cdot SiO_2 \cdot 4H_2O(s)+NaOH(aq)+0.5H_2O \quad (12)$$

The molar ratio (α_K) of diluted liquor after digestion with different lime additions are shown in Fig. 3. The α_K of digested liquor without lime addition is much larger than those with lime additions. The α_K of diluted liquor with different lime additions between 7.5% and 22.5% are almost the same. As the dilution duration increases, the α_K increases, and the increased degree without lime addition is much larger than those with lime additions.

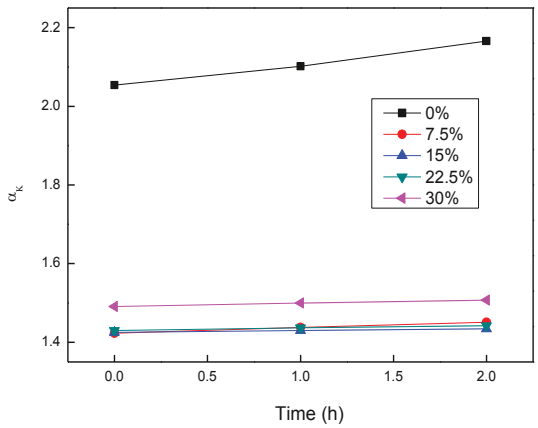

Fig. 3 Molar ratio of diluted liquor after digestion with different lime additions

The silica index (defined as the concentration ratio of alumina to silica in sodium aluminate solution) of diluted liquors after digestion was calculated as shown in Fig. 4, which is introduced to represent the desilication performance during the dilution process. All the silica indexes increase with the increasing dilution duration. Due to the lime addition, the silica index firstly decreases when the lime addition is 7.5%, and then gradually increases as the increase of lime addition, demonstrating that the addition of lime can promote the desilication reaction during the digestion process.

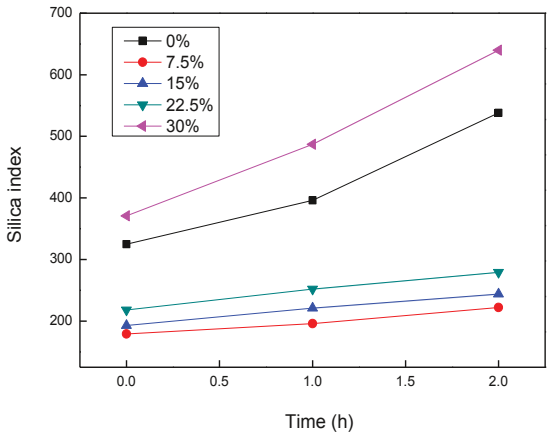

Fig. 4 Silica index of diluted liquor after digestion with different lime additions

The weight ratios of Na_2O to SiO_2 (N/S) in the bauxite residues are shown in Fig. 5. The N/S values do not change obviously during the dilution process. However, the lime additions can greatly affect the N/S values after the digestion process, which decrease largely as the lime addition increases.

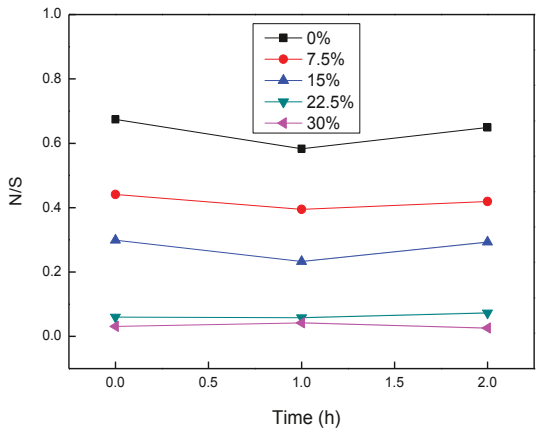

Fig. 5 Na_2O to SiO_2 ratio of bauxite residues digested with different lime additions

The hydrolysis degrees of diluted liquors after digestion were calculated according to Formula (1) as shown in Fig. 6. The hydrolysis degrees of diluted liquors digested with different lime additions increase as the duration increases. At the same duration, the hydrolysis degree of diluted liquor without lime addition during the digestion is much larger than those with lime additions, which is over 5% when the diluted duration is 2 h, demonstrating that the diluted liquor without lime addition is much unstable. When added by 7.5% lime during the digestion, the hydrolysis degree of diluted liquor greatly decreases, indicating that the lime addition can greatly promote the stability of sodium aluminate solution. As the lime addition increases to 15%, the hydrolysis degree of diluted liquor decreases to 0.63% after dilution for 2 h, which hardly hydrolyzes during the dilution. Then the hydrolysis

degrees of diluted liquor increase slightly when the lime addition is over 15%, though they are still below 1%.

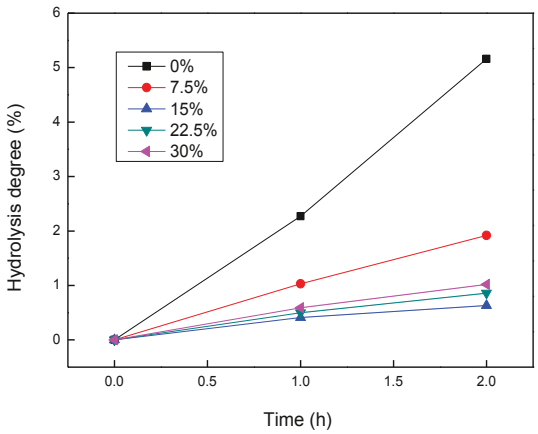

Fig. 6 Hydrolysis degrees of diluted liquors after digestion with different lime additions

The alumina digestion efficiencies of the bauxite under different dilution durations are in good accordance with the hydrolysis results, as shown in Fig. 7. As the dilution duration increases, all the alumina digestion efficiencies of the bauxite decrease, and the decreased degree without lime addition is obviously larger than those with lime additions. Meanwhile, at the same duration, the alumina extraction efficiency without lime addition is very low, which is in accordance with the liquor results. The alumina extraction efficiency reaches the largest when the lime addition is 7.5%, and then decreases gradually as the lime addition increases.

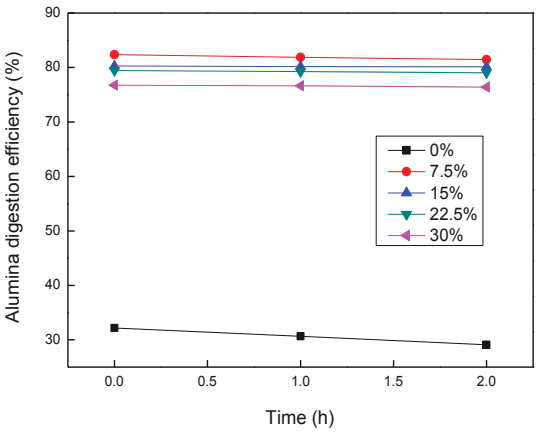

Fig. 7 Alumina extraction efficiencies of the bauxite with different lime additions

The main chemical reactions during the dilution process after the bauxite digestion include the hydrolysis and the desilication of sodium aluminate solution, as listed in Eqs. (13) and (14).

$$NaAl(OH)_4(aq) \rightarrow Al(OH)_3(s) + NaOH(aq) \quad (13)$$

$$Na_2SiO_3(aq) + NaAl(OH)_4(aq) \rightarrow Na_2O \cdot Al_2O_3 \cdot xSiO_2 \cdot yH_2O(s) + NaOH(aq) \quad (14)$$

According to the previous research [6], the crystal composition, crystallinity and microstructure of DSPs in the bauxite residue can affect the stability of sodium aluminate solution. The actual weights of DSPs in the bauxite residues were calculated by the method proposed by the authors [7] as shown in Fig. 8. The zeolite content in the bauxite residue decreases gradually with the increase of lime addition, which corresponds to the decrease of Na_2O content. The contents of grossular hydrogarnet and andradite-grossular hydrogarnet increase gradually with the increasing lime addition. Correspondingly, the sum of DSPs in the bauxite residues increase as the lime addition increases.

Comparing the mineral contents of DSPs in the bauxite residue after digestion with the hydrolysis results after dilution, the following conclusions can be obtained: both the content and the proportion of DSPs precipitating during digestion can affect the stability of sodium aluminate solution after digestion; the decrease of zeolite and the increase of hydrogarnet are beneficial to the stability of digested liquor.

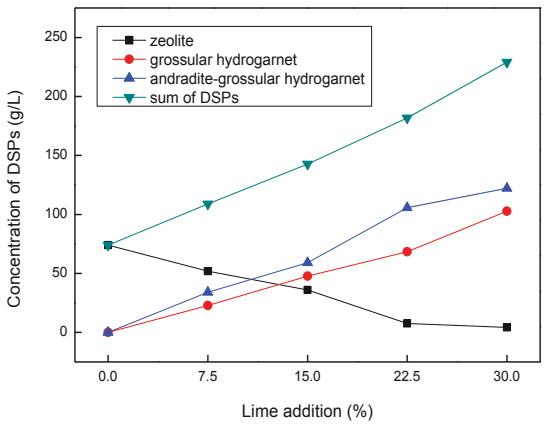

Fig. 8 Mineralogical weights of bauxite residues with different lime additions

Conclusions

The molar ratio, silica index and hydrolysis degree of diluted liquor after digestion in diasporic bauxite are highly affected by the lime addition, and the corresponding alumina extraction efficiency of bauxite and the Na_2O to SiO_2 ratio of bauxite residue are changed. The stability of diluted liquor is greatly increased by the lime addition below 15% of the bauxite, and then slightly decreased as the lime addition increases. Both the content and the proportion of DSPs precipitating during digestion can affect the stability of sodium aluminate solution after digestion, and the decrease of zeolite and the increase of hydrogarnet are beneficial to decrease the hydrolysis degree of digested liquor.

Acknowledgment

We acknowledge the financial supports from the National Natural Science Foundation of China (No. 51104041) and the Fundamental Research Funds for the Central Universities (No. N130402010).

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INFLUENCE FACTORS OF STIRRING SPEED OF SELF-STIRRING TUBULAR REACTOR USED IN BAUXITE DIGESTION PROCESS

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Keywords: self-stirring, stirring speed , dimensional analysis method, reactor

Abstract

Aim at improvement leaching reaction and heat transfer efficiency, A novel self-stirring tubular reactor driven by pressure energy used in bauxite digestion process was presented originally by Northeastern University, which combined the advantages of autoclaves and conventional tubular digestion equipment. Quasi-logarithmic equation between the stirring speed and the various factors was established by dimensional analysis method. The results demonstrated that higher pressure promoted the appearance of more obvious turbulence and improved the mixing effect of media in the reactor. The quasi-logarithmic equation between the stirring speed and the various factors was:

$$N = 356.7053\rho^{-0.43227}\mu^{-0.13546}P^{0.56773}H^{-0.86454}$$

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Introduction

In hydrometallurgy, solid-liquid reaction is one kind of important reactions, which refers to leaching of ores and so on. The leaching reaction process is always completed in pressure hydrometallurgy reactors (equipment), such as tubular leaching equipment used for alumina production, large mechanical agitation tank used for direct leaching of zinc sulfide concentrate, and mechanical agitation equipment in nickel-cobalt hydrometallurgy[1-5]. Therefore, the research of efficient solid-liquid pressure leaching reactors is always a hot spot focused on by academia and industry[6-9].

A kind of self-stirring tubular reactor driven by pressure energy is presented originally by the Northeastern University[10]. The advantages of autoclaves and traditional tubular reactors are combined in this kind of reactor, the own energy of high pressure fluid is utilized to drive agitator to rotate, which realizes the stirring action of reaction medium to promote the reaction process. It structurally solves the sealed problem of traditional mechanical agitation reactors.

Dimensional analysis is adopted to research the effects of various conditions (fluid density, viscosity, liquid level and pressure) on stirring speed of tubular self-stirring reactor.

Research equipment and method

Research equipment

The specific sizes of tubular self-stirring reactor are shown as following: length of the reactor cylinder 1000mm, diameter of reactor cylinder φ 198mm. The structure of the reactor is shown in Figure1.

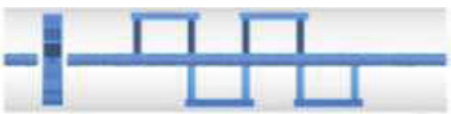

Figure 1. Structure of self-stirring reactor driven

by pressure energy

The instrumentation plan of liquid level is divided into six equal pieces according to the reactor diameter.

Research content and method

Dimensional analysis is a kind of mathematical analysis method, through which the relation among different variables can be analysed accurately and experiment and result can be simplified. As same as method of the similarity theory to guide the experiment of chemical engineering research, dimensional analysis has been widely used in the study of engineering disciplines[11]. With dimensional analysis, dimensionless variables which control physical phenomenon would be derived if the control parameters is determined, thus the basis of model design will be obtained. Quasi-logarithmic equation between the stirring speed and the various factors (fluid density ρ, viscosity μ, pressure, liquid level, etc.) of the reactor is established and the transformation rule from pressure energy to mechanical energy is confirmed by dimensional analysis method in this paper.

Dimensional analysis of stirring speed

The main parameters affecting stirring speed

The stirring speed of self-stirring reactor is mainly affected by the following factors:

fluid density ρ

fluid viscosity μ

inlet pressure P

liquid level H

diameter of reactor cylinder D

length of reactor cylinder L

Through the analysis above, dimensional analysis is adopted to obtain the general function form:

$$N = f\left(\rho, \mu, P, H, D, L\right) \tag{1}$$

or

$$f\left(N, \rho, \mu, P, H, D, L\right) = 0 \tag{2}$$

All dimensional expressions of physical quantities can be summarized as the general form of $[a]=L^{l}T^{t}M^{m}$ to represent it, in this form [a] represents dimension of physical quantities, L represents length dimension, T represents time dimension and M represents mass dimension. In hydrodynamics basic three dimensions generally involved are M, L, T. So all these variables dimensions can be listed in the table below.

table 1. variable dimension

	N	ρ	μ	P	H	D	L
M	0	1	1	1	0	0	0
L	0	-3	-1	-1	1	1	1
T	-1	0	-1	-2	0	0	0

Establishment of dimensionless equation of undetermined parameters

It can be known from the analysis principle of PI theorem that sum of variables is 7, independent variable k=3, so sum of dimensionless combination n-k=4 can be established. ρ, μ, P are chosen to be independent variables. Each Π is respectively:

$$\Pi_0 = \rho^{\alpha_0}\mu^{\beta_0}P^{\gamma_0}N \tag{3}$$

$$\Pi_4 = \rho^{\alpha_4}\mu^{\beta_4}P^{\gamma}H \tag{4}$$

$$\Pi_5 = \rho^{\alpha_5}\mu^{\beta_5}P^{\gamma_5}D \tag{5}$$

$$\Pi_6 = \rho^{\alpha_6}\mu^{\beta_6}P^{\gamma_6}L \tag{6}$$

To Π_0, through substitution of dimension of variables dimension relation can be obtained:

$$M^0L^0T^0=\left[ML^{-3}\right]^{\alpha_0}\left[ML^{-1}T^{-1}\right]^{\beta_0}\left[ML^{-1}T^{-2}\right]^{\gamma_0}T^{-1} \quad (7)$$

Thereout, exponential equations can be obtained:

$$\begin{aligned} &M:0=\alpha_0+\beta_0+\gamma_0\\ &L:0=-3\alpha_0-\beta_0-\gamma_0\\ &T:0=-\beta_0-2\gamma_0\ -1 \end{aligned} \quad (8)$$

Solution is $\alpha_0=0$, $\beta_0=1$, $\gamma_0=-1$, so

$$\Pi_0=\mu\ P^{-1}N$$

Similarly，to Π_4, through substitution of dimension of variables dimension relation can be obtained:

$$M^0L^0T^0=\left[ML^{-3}\right]^{\alpha_4}\left[ML^{-1}T^{-1}\right]^{\beta_4}\left[ML^{-1}T^{-2}\right]^{\gamma_4}L^1 \quad (9)$$

Thereout, exponential equations can be obtained:

$$\begin{aligned} &M:0=\alpha_4+\beta_4+\gamma_4\\ &L:0=-3\alpha_4-\beta_4-\gamma_4+1\\ &T:0=-\beta_4-2\gamma_4 \end{aligned} \quad (10)$$

Solution is $\alpha_4=\frac{1}{2}$, $\beta_4=-1$, $\gamma_4=\frac{1}{2}$, so

$$\Pi_4=\rho^{1/2}\mu^{-1}P^{1/2}H$$

Similarly, to Π_5, through substitution of dimension of variables dimension relation can be obtained:

$$M^0L^0T^0=\left[ML^{-3}\right]^{\alpha_5}\left[ML^{-1}T^{-1}\right]^{\beta_5}\left[ML^{-1}T^{-2}\right]^{\gamma_5}L^1 \quad (10)$$

Thereout, exponential equations can be obtained:

$$\begin{aligned} &M:0=\alpha_5+\beta_5+\gamma_5\\ &L:0=-3\alpha_5-\beta_5-\gamma_5+1\\ &T:0=-\beta_5-2\gamma_5 \end{aligned} \quad (11)$$

Solution is $\alpha_5=\frac{1}{2}$, $\beta_5=-1$, $\gamma_5=\frac{1}{2}$, so

$$\Pi_5=\rho^{1/2}\mu^{-1}P^{1/2}D$$

Similarly, to Π_6, through substitution of dimension of variables dimension relation can be obtained:

$$M^0L^0T^0=\left[ML^{-3}\right]^{\alpha_6}\left[ML^{-1}T^{-1}\right]^{\beta_6}\left[ML^{-1}T^{-2}\right]^{\gamma_6}L^1 \quad (11)$$

Thereout, exponential equations can be obtained:

$$\begin{aligned} &M:0=\alpha_6+\beta_6+\gamma_6\\ &L:0=-3\alpha_6-\beta_6-\gamma_6+1\\ &T:0=-\beta_6-2\gamma_6 \end{aligned} \quad (12)$$

Solution is $\alpha_6=\frac{1}{2}$, $\beta_6=-1$, $\gamma_6=\frac{1}{2}$, so

$$\Pi_6=\rho^{1/2}\mu^{-1}P^{1/2}L$$

Thus it is obtained:

$$f\left(\mu P^{-1}N,\rho^{1/2}\mu^{-1}P^{1/2}H,\rho^{1/2}\mu^{-1}P^{1/2}D,\rho^{1/2}\mu^{-1}P^{1/2}L\right)=0 \quad (13)$$

The form above can be expressed as

$$f_1=\left(\mu P^{-1}N,\rho^{1/2}\mu^{-1}P^{1/2}H,\rho^{1/2}\mu^{-1}P^{1/2}D,\rho^{1/2}\mu^{-1}P^{1/2}L\right) \quad (14)$$

To obtain the expression of stirring speed *N*, it could be expressed as explicit function:

$$\mu P^{-1}N=f_2\left(\rho^{1/2}\mu^{-1}P^{1/2}H,\rho^{1/2}\mu^{-1}P^{1/2}D,\rho^{1/2}\mu^{-1}P^{1/2}L\right) \quad (15)$$

According to the specific experimental condition in this paper, agitator diameter *D* and length *L* are fixed value, so it can be obtained:

$$\mu P^{-1}N=f_3\left(\rho^{1/2}\mu^{-1}P^{1/2}H\right) \quad (16)$$

Confirmation of relation of dimensionless equation

According to the parameter relation (15), plus specific

experimental condition, corresponding specific empirical formula can be deduced and established.

Criterion relation of different phenomena within a certain range of independent variable can be expressed in the form of power function. By the analysis above, empirical criterion formula can be fitted as:

$$\mu P^{-1} N = \mathrm{A}\ (\rho^{1/2}\mu^{-1}P^{1/2}H)^{\mathrm{B}} \tag{16}$$

of which A, B are fitting coefficient.

Log of both sides of the form above is solved:

$$\ln(\mu P^{-1}N) = \ln \mathrm{A} + \mathrm{B}\ln\ (\rho^{1/2}\mu^{-1}P^{1/2}H) \tag{17}$$

According to the linear relation of form above, experimental data is processed to gain fitting coefficient A and B.

Experimental data is substituted into to obtain: A=356.7053, B=-0.86454

Fitting coefficient A and B are substituted into equation (17) to obtain empirical formula:

$$N = 356.7053\rho^{-0.43227}\mu^{-0.13546}P^{0.56773}H^{-0.86454} \tag{18}$$

of which *N*-stirring speed, r/min;

ρ-fluid density, $\times 10^3$ kg·m^3;

μ-fluid viscosity, $\times 10^{-3}$ Pa·s;

P-inlet pressure, MPa;

H-liquid level, cm.

With this method the relation is obtained on the basis of experiment in this paper. Therefore, for the purpose of stirring speed of this experiment study, as long as the main physical parameters , structural parameters and operational parameters are given, it can be predicted by using the above relation. But it should be pointed out, applying dimensional analysis formula is restricted under the experimental condition in this paper.

Verification of dimensional formula

All the experimental data are substituted into form （16）to process dimensional analysis, so the data obtained distribute with limits, which has representative significance. Fitted value of stirring speed is obtained under different conditions by means of dimensional formula (18), and is compared with experimental value under same conditions.

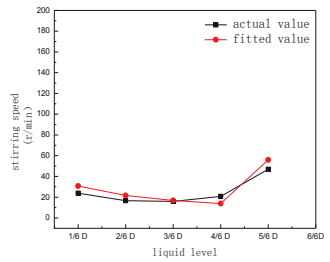

Figure 2. Comparison of fitted value and actual value of stirring speed under different liquid levels （ρ=1.0×10^3 kg/m^3，μ=21.8×10^{-3} Pa·s，*P*=0.5 MPa）

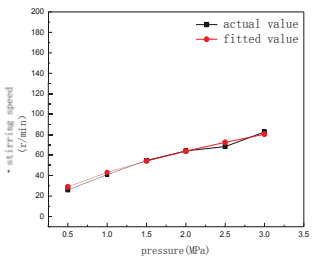

Figure 3. Comparison of fitted value and actual value of stirring speed under different inlet pressure （ρ=1.0×10^3 kg/m^3，μ=33.0×10^{-3} Pa·s，*H* =2/6D）

Correlation curves of fitted and actual value under different conditions are shown in Figure 2 and 3. It can be found from figures that dimensional formula is deduced ideally, difference with the actual value is minor, error of individual point is obvious. The main reason of appearance of error of empirical formula is considered analytically that random error exists in the measurement process of physical parameters of main experiment including two phase density, viscosity, etc and major error exists in the measurement process of liquid level and stirring speed.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

LEACHING KINETICS FOR RECOVERING ALUMINA FROM WASTE TRICALCIUM ALUMINATE GENERATED AFTER FILTRATION OF BAYER'S LIQUOR

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Abstract

A study was performed on the leaching kinetics to recover the alumina from Tricalcium aluminate (TCA) generated as waste after removal of suspended red mud particles in the Bayer process by different concentrations of Na_2CO_3 solution at different temperatures and solid:liquid ratios. The influences of initial sodium carbonate and stirring speed on reaction rate constant were examined and the corresponding apparent activation energy was calculated. The results indicate that the leaching process is a first order reaction, and the reaction rate constant 8.64×10^{-6} mol/(L.m). Optimization of recovery and minimization of Na_2CO_3 in liquor were studied, and the dissolution rate of TCA and conversion of $CaCO_3$ also increases. The apparent activation energy is 52.86 kJ/mol, and temperature has great influence on the leaching rate of TCA. The leaching rate is controlled by diffusion of reacting reagents in the solid product layer. A higher leaching rate of TCA is obtained by higher temperature and/or lower solid g/l during leaching process..

Keywords: TCA, Katoite, leaching kinetics and calcite.

1. Introduction

Alumina is produced industrially via extraction from bauxite ore, using the Bayer process(1). In the Bayer process, tricalcium aluminate (TCA) is typically produced on site through the addition of slaked lime to saturated aluminate pregnant liquor, depending on operating conditions(2). Tricalcium aluminate hexahydrate ($3CaO{\cdot}Al_2O_3{\cdot}6H_2O$, TCA) is used as a filter aid when separating pregnant liquor from fine solids and colloidal material, prior to gibbsite crystallization. The main role of TCA is to reduce the tendency for the filtered solids to blind the cloth and shorten the cloth life by forming a porous solid coating on the cloth industrially via Kelly or Diastar filter under pressure. This depends on particle size distribution of TCA to ensure adequate filter throughput and effective solid separation from solution. After separation of solids from aluminate liquor the TCA is contaminated with the fine goethite/hematite particles disposed through the washer circuit to Residue Storage Areas as waste.

The solubility of calcium hydroxide in highly concentrated sodium hydroxide solution at 298 K also has been studied(3). The solubility of calcium hydroxide was found to steadily decrease with the increasing sodium hydroxide concentration. The solubility of TCA in the $KOH–K_2CO_3–H_2O$ system in the range of 30–80 °C has been studied. Solubility determination can be obtained by either the dissolution method or the precipitation method.(4) The experimental technique employed in this research is the most common method of determining solubility, which is by mixing an excess amount of solute into a solvent and gently mixing until equilibrium is reached then establishing the amount of solute that has dissolved(5). For the thermodynamic predictions, relevant data was reviewed using critically assessed databases and specialist research papers where required. Extraction of alumina from Calcium aluminate slag obtained from blast furnace during smelting of Iron was studied (6).

No study of the solubility of TCA ($Na_2O–CaO–Al_2O_3–CO_3–H_2O$) for extraction of residual alumina has been reported. This study is mainly to recover Alumina from waste filter aid (TCA mud). At present Alumina refinery waste filter aid mud is normally disposed off through washer circuit. Therefore the present work is to determine the solubility of well-characterized TCA in sodium carbonate solution over a range of temperatures (343–373K), stirring condition and solid: liquid ratio by Na_2CO_3 solution compositions. Comparing the outcomes of the solubility experiments in these well-defined systems with thermodynamic predictions provides insight into the nature of the chemical equilibrium. Based on the investigation into the influences of various factors on the reaction rate of leaching mechanism, the dominant procedure could be determined and kinetic developed equation.

2. Experimental

2.1 Synthesis of Tricalcium Aluminate

Saturated Sodium aluminate solution of bayers process having required mass of Na_2O:Al_2O_3 ratio (1.08) and calcium oxide

slurry (as 100 g CaO/L H_2O) were added to the reaction tank having aluminate liquor and the mixture was heated to 368 K. After 2 hours, the slurry with the TCA particles was quickly mixed with saturated aluminate liquor of settler overflow and passed through filter cloth at 2.5Kg/cm^3 pressure to separate the fine iron oxide particles. Then the solid was washed to a storage tank for further disposal to washers. The contaminated solids were collected and washed with hot water then dried in a hot air oven at 373 K for 10 h prior to leaching experiments.

2.2 EXPERIMENTAL

The tricalcium aluminate is contaminated with the fine goethite/hematite particles generated after filtration of saturated aluminate liquor from Alumina Plant (waste filter aid) was added into Na_2CO_3 solution at different solid-liquid ratio (g/L) and then it was put into a constant oil bath reactor to react at different temperatures such as 70, 80, 90 and 100^{o}C, different Na_2CO_3 concentrations ranging from 15.5 to 19.5 g/L and different solid-liquid ratio. Solutions specimens were taken out to measure the mass concentration of Alumina in a certain time interval. The leaching rate of Al_2O_3 (f,%) is calculated by the following formula : f =(1- m/m$_o$×100)% where m is the mass of Al_2O_3 in leached mud and m$_o$ is the mass of Al_2O_3 in un-leached mud established to find out the key factors and their influence degree on the leaching process.

The extraction of alumina from the dump mud is determined kinetically. Therefore, the leaching process could be optimized accordingly. The muds generated during leaching process were analyzed by Energy Dispersive X-Ray Fluorescence spectrometer (EDXRF). The surface morphology and crystalline phases were analyzed by SEM and XRD techniques. The conversion of tricalcium aluminate to calcium carbonate is characterised by DTA method.

3. RESULTS AND DISCUSSION

3.1 Characterization of Mud

It is elucidated that the diminishing of Al *Kα* and Al *Kβ* in Fig-2 in comparison to Fig-1 shows that the leaching 90%of alumina from dump mud achieved successfully.

The XRD pattern of pre-filter aid (Calcium aluminate) and leached samples are shown in Fig.3 In case of calcium aluminate in spectra (a) the 2θ value at 32.6 shows the presence of alumina in the un-leached condition as calcium aluminate (Katoite phase **A**) where as in (b) the intensity of katoite phase diminishing due to formation of $NaAlO_2$ at 16.5 g/l of Na_2CO_3 solution at optimum condition of leaching for 360 min at 100 ^{o}C. The presence of Fe_2O_3, P_2O_5, SiO_2 Na_2O and CO_3 dominated in leached residue as Calcium Carbonate (portandlite) calcium

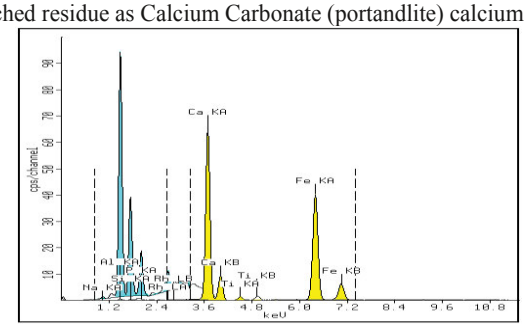

Figure-1 EDXRF Spectra of Dump mud collected from Alumina Plant

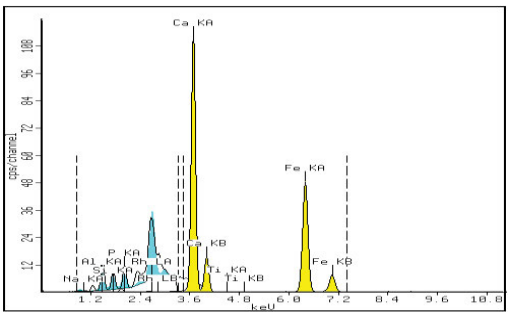

Figure-2 EDXRF Spectra of leached 50 g/l Dump mud with 16.5 g/l Na_2CO_3 solution

Table-1 Chemical analysis of Mud

Constituents (%)	Pre-filter aid solid	Post-Filter aid Dump Mud	Leached Dump Mud
Al_2O_3	20.89	21.11	2.12
Fe_2O_3	6.24	15.37	17.85
TiO_2	1.45	2.10	2.23
SiO_2	1.87	5.62	4.41
Na_2O	0.87	1.73	2.40
P_2O_5	3.23	2.57	5.70
V_2O_5	0.042	0.045	0.047
CaO	36.87	27.73	33.65
CO_2	7.58	7.98	7.83
H_2O	20.67	15.56	23.12

apatite and hematite phase along traces of alumina in sodalite phase. The presence of carbonate along with water molecule in the crystal lattice of leached residue is characterized by DTA method. (Fig.4)

From the DTA/DSC analysis it is presumed that surface moisture is eliminated at 97.94 ^{o}C amounting 4.55%. when the temperature raised from 165.93 ^{o}C to 285.48 ^{o}C the water molecule released from the crystal lattice and then the caobonate released from leached mud at temperature range of

451.29 to 548.42 °C amounting 7.83%. When the samples was heated further the collapse of crystal structure takes place and a small change take place at 954.44 °C showing a well crystalline phase of Calcium ferrite/Calcium apatite.

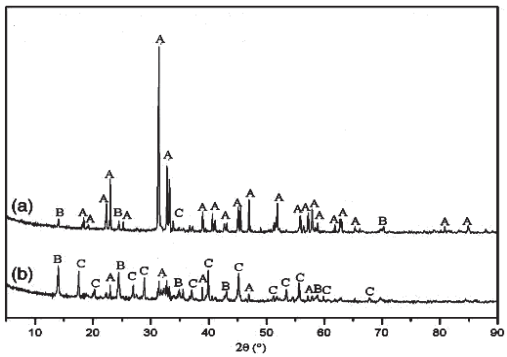

Figure-3 XRD Spectra of un-leached (a) and leached (b) Dump mud 50 g/l with 16.5 g/l Na_2CO_3 solution at 100 °C for 360 min

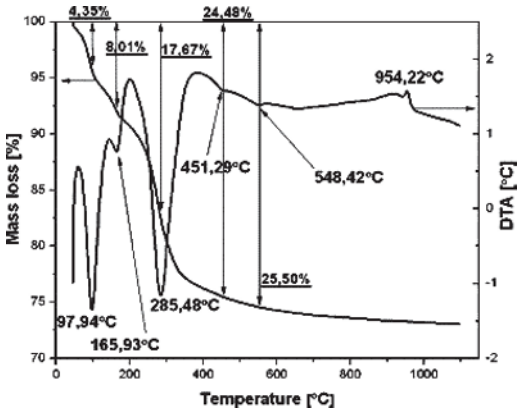

Figure-4 DTA/DSC Spectra of leached Dump mud 50 g/l with 16.5 g/l Na_2CO_3 solution at 100 °C for 360 min

Figs.5. shows the surface morphology of pre filter aid (**A**) (Calcium aluminate) shows well defined crystal morphology of trigonal pyramidal geometry. The surface morphology of the post filter aid mud (Dump mud) (**B**) shows agglomerated with hematite after filtering of saturated sodium aluminate liquor for separating the fine iron hydroxide particles. When the dump mud is leached with Na_2CO_3 solution the crystal morphology has been changed from trigonal to aglomorated flaky structure (**C**) at 16.5 g/l and well defined platy of oval shapes produced after extraction of alumina from dump mud (17.5 g/l for 360min at 100 °C) having portandlite in nature shown in (**D**).

Determination of Kinetic Equation

The leaching reaction of Dump mud (Calcium Aluminate) with Na_2CO_3 solution is expressed as follows:

$$xCaO.yAl_2O_3\text{-}xCaO\text{-}zSiO_2\text{-}xCaO\text{-}P_2O_5\text{-}Na_2O\text{-}V_2O_5\text{-}CO_3\text{-}H_2O + Na_2CO_3 \rightarrow (NaAl(OH)_4)_{y-1} + (Na_2O\text{–}(Al_2O_3)_1\text{–}SiO_2\text{-}P_2O_5\text{–}V_2O_5\text{-}H_2O) + 3CaCO_3 + NaOH \quad (1)$$

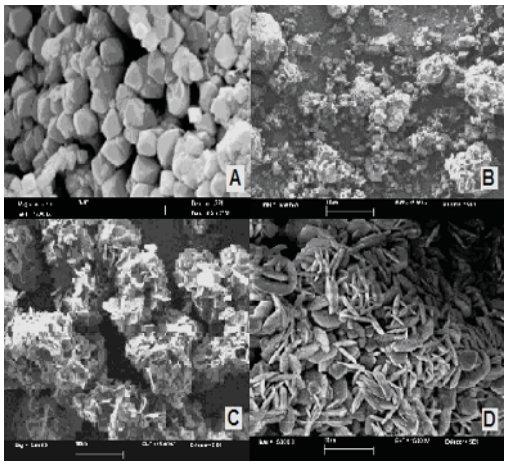

Figure-5 SEM micrograph of unleached and leached Dump mud 50 g/l with 16.5 g/l Na_2CO_3 solution at 100 °C for 360 min

The reaction rate equation of formula (1) can be expressed as:

$$v = -\frac{dC}{dt} = kC_{Na_2CO_3}^{n1} C_{CaAlO_2^-}^{n2} \quad (2)$$

Calcium aluminate is slightly soluble at very solubility and its solubility is very slow. Therefore its activity can be approximately taken as a constant. Suppose $k = k'C_{CaAlO_2^-}^{n1}$ then Eq.(2) can be transferred into:

$$v = -\frac{dC_{Na_2CO_3}}{dt} = kC_{Na_2CO_3}^{n} \quad (3)$$

In addition, from Eq.(1) we can deduce that:

$$v = -\frac{dC_{Na_2CO_3}}{dt} = \frac{1}{2}\frac{dC_{Na_2CO_3}}{dt} = \frac{1}{2}\frac{dC_{AlO_2^-}}{dt} = kC_{Na_2CO_3}^{n} \quad (4)$$

In Eqs.(2)~(4), C is the reactants concentration (mol/L), t reaction time (min), k' and k reaction rate constants [mol/(L.min)], $C_{Na_2CO_3}$, Na_2CO_3 concentration (mol/L), $C_{CaAlO_2^-}$ $CaAlO_2$ concentration(mol/L), n_1,n_2 and n reaction order. $CAlO_2$ is Al_2O_3 concentration in mud respectively.

According to Eq.(4)

$$\ln v = \ln\frac{1}{2}[\frac{dC_{AlO_2^-}}{dt}]t_0 = \ln k + n\ln C_{Na_2CO_3}^{n} \quad (5)$$

Based on Table-2 and Eq.(5) is obtained which shows the relationship between ln$v$ vs ln$f$..

According to Figure.-7, the reaction rate constant k and reaction order n were calculated as: k= 0.0569433 g/L.min) = $8.64x10^{-6}$ mol/L.m) and n=0.565319 ≈ 0.57.

Therefore, the rate equation of the leaching reaction is expressed as: $$\nu = (\frac{1}{2})\frac{dC_{AlO_2^-}}{dt} = 8.64\times10^{-6} C_{Na2CO3}^{n}$$

From this rate equation, the influential factors on the reaction rate of leaching process can be shown clearly, which is absent in other related researchers.

3.2 Influential Factors of Reaction Rate Constant

3.2.1 Effect of initial Na_2CO_3 on reaction rate constant

The effect of initial Na_2CO_3 on reaction rate constant was studied by performing experiments at different initial 15.5, 16.5, 17.5, 18.5 and 19.5g/L different initial Na_2CO_3 mass concentrations under 100 ^oC of temperature and 300 r/min of stirring speed. By considering the Al_2O_3 concentration in leached Waste filter aid and leaching rate at 17.5 g/l of Na_2CO_3 concentration with time from Fig-6, the rate constant *k* was calculated by differential method.

Fig. 6 shows the variation of AlO_2 concentration and leaching rate with time when 17.5g/l. Initial concentration differential method was also used to measure the rate constant *k*.

Figure 6 was obtained by drawing a graph with ln*v* against ln*f*. The rate constant was calculated from Fig.6 when Na_2CO_3 is 17.5 g/l the rate constant *k*=1.24×10−5 mol/(L/s). Simultaneously, the rate constants when 15.5, 16.5, 17.5, 18.5 and 19.5 g/l were also calculated and the results were shown in Table 2. Table 2 indicates that the reaction rate constant *k* increases with the increase of initial Na_2CO_3 g/l, which means that the calcium aluminate fits in leaching under alkaline condition. This is because $CaCO_3$ in reaction (1) more easily forms under alkaline condition to make the reaction move to right, resulting in the increase of the dissolution rate of Calcium Aluminate. The principle of this method is that some curves were made according to the variation of resultants concentration *f* with time t at different initial reactants based on the experimental data, then the reaction rate ν_o when t-0 at different initial concentrations could be calculated, finally the kinetic parameters could be obtained by plotting of ln*f* vs ln ν_o. on the basis of Eq.(2) and the data in Fig.-6, the reaction rates under different initial Na_2CO_3 concentrations when t=0 , $\nu_o=[dC_{AlO2}-/dt]_{to}/_2$, were obtained.

3.2.2 Effect of Temperature at 17.5 g/L Na_2CO_3 concentration on reaction rate constant

The leaching of AlO_2^- from calcium aluminate at 17.5 g/l Na_2CO_3 under conditions of different leaching period at

Table.-2: Reaction rate at the initial reaction *v*o under different Na_2CO_3 concentration.

Na_2CO_3 concentration (g/L)	*v*o [g/(L min)]
15.5	0.47
16.5	1.45
17.5	2.85
18.5	4.33
19.5	6.21

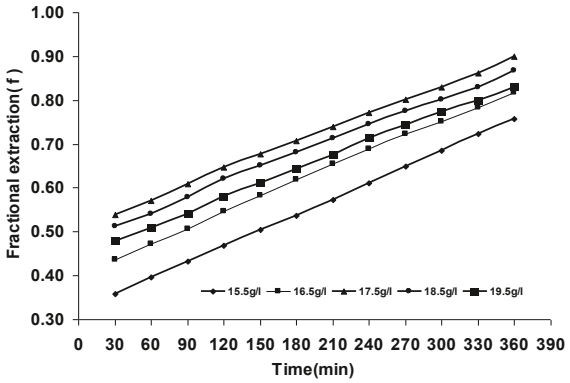

Figure-6 Leaching efficiency of Alumina from Dump Mud with variation of Na_2CO_3 concentration.

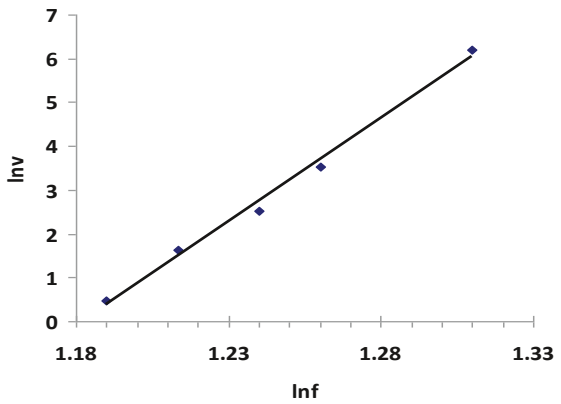

Figure-7 Relationship between ln*v* and ln*f* at different Na_2CO_3 concentration.

different temperature varying from 70-100 ^{o}C and 300 r/min shown in Fig.8 With variation of temperature the AlO_2^- in mud the differential method was adopted to measure the kinetics properties (7). The results are shown in Table-3.

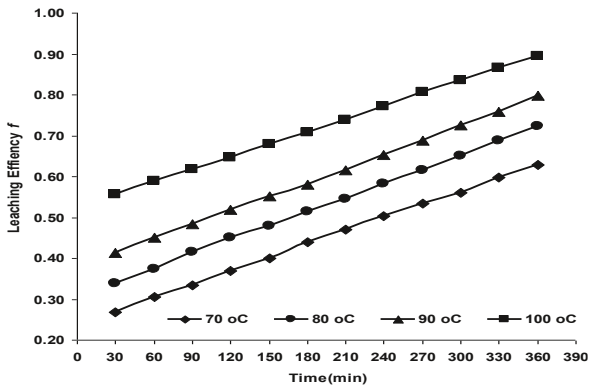

Figure-8 Leaching efficiency of Alumina from Dump Mud with variation of temperature at 17.5 g/l Na_2CO_3 concentration.

Table-3 Effect of Temperature on rate constant at 17.5 g/L of Na_2CO_3 and 1:20 solid-liquid ratio

T ^{o}C	kv_o [10^{-3} mol/(l.min)
70	0.55
80	0.88
90	1.21
100	1.69

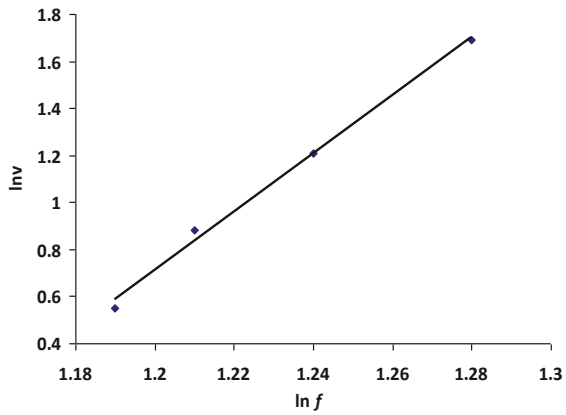

Figure-9 Relationship between ln$v$ and ln f at different leaching temperature at 17.5 g/L of Na_2CO_3 and 1:20 solid-liquid ratio.

3.2.3 Effect of ratio of solid-liquid on reaction rate constant

The variation graphs of AlO_2 concentration and leaching rate with time at different ratios of liquid to solid were obtained by investigating the leaching process at the ratios of 50, 100,150 and 200g/l and different initial Na_2CO_3 concentrations under 100 ^{o}C of temperature and Na_2CO_3 concentration of 17.5 g/l value. Figs.10 and 11 show the f–t and ln v − ln f curves, respectively.

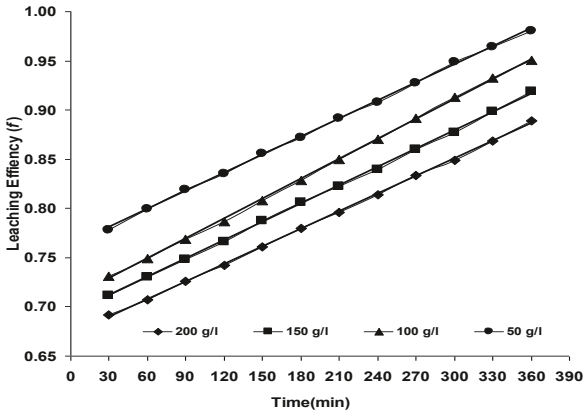

Figure-10 Leaching efficiency of Alumina from Dump Mud with variation solid in system

Table-4 Effect of Ratio of liquid to solid on rate constant

Ratio of Liquid to Solid mL/g	kv_o [10^{-3} mol/(l.min)
0.05	0.58
0.10	0.80
0.15	1.13
0.20	1.54

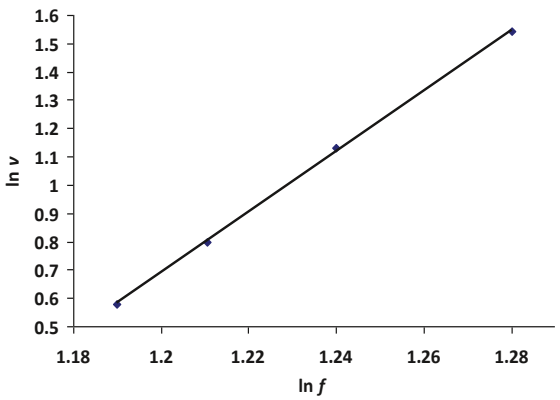

Figure-11 Relationship between ln$v$ and ln$f$ at different liquid to solid at 100 ^{o}C with 17.5 g/L Na_2CO_3 concentration.

The reaction rate constants at each ratio were calculated using the same method. The results are shown in Table 4

Table 4 indicates that the reaction rate constant k increases with the increase of solid-liquid ratio, furthermore, the increase trend becomes slow as the solid-liquid ratio is more than 5. This is because when the ratio is less than 2 the contact between

calcium aluminate and leaching solution is not sufficient, resulting in a low reaction rate; when the ratio increases, the contact between calcium aluminate and leaching solution is more and more sufficient, resulting in the increase of reaction rate. When the ratio is more than 20, the contact between calcium aluminate and leaching solution is enough sufficient, the reaction rate increases slowly with the increase of solid-liquid ratio. Therefore the suitable solid-liquid ratio is 1:20.

3.2.4 Determination of apparent activation energy

Table 3, obtained under the experimental conditions of the stirring speed of 300 r/min, initial Na_2CO_3 mass concentrations of 15.5, 16.5, 17.5, 18.5 and 19.5 respectively and the temperatures of 60, 70, 80, 90 and 100°C respectively, shows the reaction rate constants under different temperatures which were obtained by also using initial concentration differential method Based on the data in Table 3, Fig.12 can be obtained by drawing a graph with ln$k$ against T^{-1}. It indicates that the leaching of calcium aluminate by Na_2CO_3 is an endothermic process. The following equation was deduced by fitting the data:

$$\ln k = 1.166 - 5441.0T^{-1}$$

According to the Arrhenius equation (7):

$$\ln k = -Ea/(RT) + A \qquad (6)$$

where Ea is apparent activation energy (kJ/mol), R gas constant [J/(mol.k)], T thermodynamic temperature (K), and A Arrhenius constant, respectively. Ea of the leaching process was calculated as 52.86kJ/mol. This is relatively high, indicating the great influence of temperature on leaching rate [7].

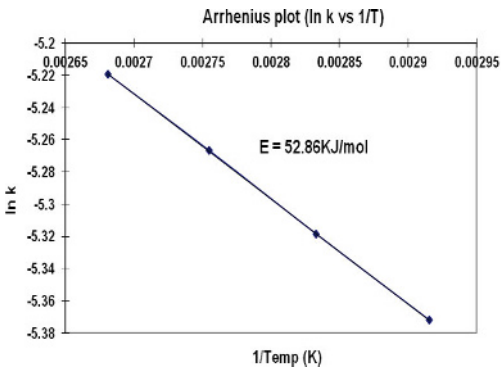

Figure- 12 Relationship between ln$k$ and T^{-1}.

4 CONCLUSIONS

The waste filter aid mud contains ~20-22% Al_2O_3, which can be utilizing for recover by leaching with Na_2CO_3 solution. Approximately 10,000 MT of alumina / year can be recovered by this method at Alumina Refinery, Damanjodi, NALCO by utilizing the waste filter aid mud. This study may give an insight to other Alumina refineries for utilization of this waste to recover alumina.

1. The leaching kinetic equation of calcium aluminate by Na_2CO_3 is expressed as $\mathbf{v} = 8.64 \times 10^{-6} C_{Na2CO_3}^{0.57}$ (g/L.m)
2. The amount of solid depends on the leaching efficiency, with decrease of solid in the leached solution suggests more extraction of Al_2O_3 from waste filter aid mud.
3. The apparent activation energy Ea of the leaching reaction of Calcium aluminate by Na_2CO_3 is 52.56kJ/mol. The temperature has great influence on the leaching rate of Calcium aluminate and the relationship between rate constant k and temperature can be expressed as $\ln k = 1.166 - 5441.0T^{-1}$
4. The increase of carbonate concentration in the leaching process facilitates the calcite formation.

Acknowledgement

The author is grateful NALCO management for encouragement for publishing this study as future waste utilization project.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INDUSTRIAL IMPLEMENTATION CHARACTERISTICS OF ALUMINATES LIQUOR LOW-TEMPERATURE DESILICATION TECHNOLOGY

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Keywords: aluminum-containing raw materials, nepheline, alumina, sintering, desilication aluminate liquor, hydrolysis

Abstract

Low-temperature desilication process was investigated during alkali aluminosilicate raw materials treatment by mode of production of alumina by sintering method at industrial conditions. Nepheline concentrate used as seed for desilication. During the research process, temperature changed from 85 to 100 °C and seeding from 9.6 to 31.0 kg/m^3. Accomplishment of desilication at the first stage with use of nepheline concentrate as seed reduces steam consumption for the desilication process and allows to simplify the equipment decision in comparison with autoclave conditions. However, when the non autoclave desilication is applied, the amount of solid phase in a state of circulating slurry substantially increase and contains up to 30 % of the liquid phase in the form of desilicated liquor. The best results achieved at temperatures close to the maximum and amount of seed more than 20 kg/m^3. It was established that during low-temperature desilication aluminate liquors dilute with nepheline pulp liquid phase while entering desilication in contrast to autoclave one. This dilution results in aluminate liquor hydrolysis, and the material flows gain which needs increasing volume of settling tanks and equipment; the latter leads to decrease of feeding sinter leaching liquor concentration. As the result of hydrolysis the solid phase after the desilication has in its composition an average amount of mass Al_2O_3 equal to 3,0 % in the type of $Al(OH)_3$. As well, liquor dilution results in increase of heat consumption for the liquor evaporation in the soda production area.

Introduction

Despite the almost universal use of bauxite as raw material for the production of alumina, alternative sources of raw materials, in particular alkali alumosilicate raw materials (nepheline, products of chemical enrichment of aluminum silicates, etc.), continue to be used. For countries with limited supply of high-quality bauxite, these sources of raw materials are of commercial interest. Process of sintering is a way to alumina production and related products from alumosilicate raw material [1-3]. The technology of the sintering process are the physico-chemical interaction between the charge sintering alumosilicates and calcium oxide to form water soluble alkali metal aluminates and controlling the slurry of dicalcium silicate. Aluminate liquor prior to separating the aluminum hydroxide is subjected to two-stage desilication, wherein the first step involves the removal of silicon compounds from the liquid phase in the form of alkali metal hydroalumosilicates. The first stage is carried out in the desilication autoclave conditions, which allows you to quickly reach a liquor in the desired weight ratio aluminum and silicon oxides - 300-400 units. [4].

Liquors desilication without autoclave is an alternative to autoclave solution desilication method which requires using a large amount of seed material and prolonged aging of the pulp to achieve acceptable results for the first stage. As advantages of non-autoclave desilication indicate low consumption of heating steam, ease of equipment design and low operating costs.

The technology of low-temperature desilication known for some time, recent interest has arisen again, and it is regarded as having prospects. As a primer for the proposed use of the desilication slurry nepheline.

The purpose of this study is to assess the comparative results of the industrial use of technology and a low-temperature desilication autoclave cycle nepheline raw materials by sintering method.

Experimental

As initial feedstock nepheline concentrate containing an average of % wt.: 27,7 - Al_2O_3, 17,3 - R_2O (where R_2O the sum Na_2O - 13.87 - and K_2O - 8.17 in terms of Na_2O), 41, 6 - SiO_2, 2,45 - CaO, 1,65 – LOI used.

The flowsheet low temperature desilication aluminate liquor obtained after leaching cake containing on average, g/dm^3: 76.6 - Al_2O_3; 74.2 - R_2O_{total}; 63.2 - R_2O_{ku} (in ROH and $RAlO_2$); 2,09 - SiO_2. The ratio of Na_2O and K_2O in liquor varies little compared with nepheline. The liquor after leaching cake was mixed with an aqueous slurry of nepheline concentrate having a moisture content of 45 %, based on 9,6÷31 kg dry nepheline and 1 m^3 aluminate liquor. Nepheline preliminarily pulverized to a particle size sieve residue of 0.08 mm, not more than 10 %, i.e. the proportion of particles larger than 0.08 mm from the sum of all of the particles should not exceed 10 %. The mixed slurry was heated in a tubular heater to a predetermined temperature, which might be changed in the range of 85 to 100 °C, and subjected desilication continuously with a mean residence time of the pulp in the reaction vessels with mixing devices of 6 hours. The solid phase after the desilication phase is separated from the liquid by filtration, followed by condensation and returned to the batch preparation sintering. The resulting liquor after the desilication contained on average in g/dm^3: 66.2 - Al_2O_3; 70.8 - R_2O_{total}; 63.3 - R_2O_{ku} (in ROH and $RAlO_2$).

Results and discussion

The dependence of the weight ratio of Al_2O_3/SiO_2 (silica modules) obtained after low temperature desilication of the liquor temperature in the reaction chamber is shown

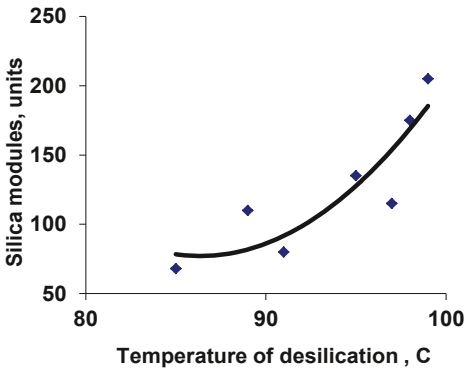

Fig.1 Dependence of silica module of aluminate liquor after the desilication temperature at seeding dosage of 25 g/dm^3

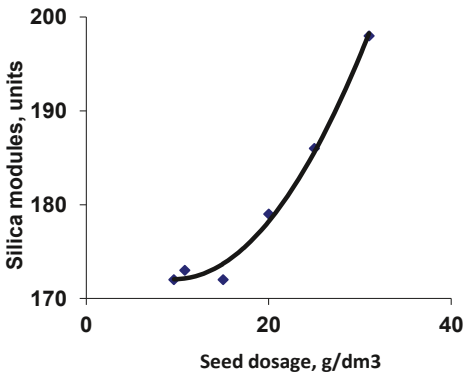

Fig. 2. Dependence of silica module of aluminate liquor desilication after the dosage of seed at 98 °C

From Figure 1, it follows that under the conditions of the low-temperature desilication even at temperatures close to 100 °C silica module liquor does not exceed 200 units. I. e. without autoclave desilication does not provide a liquor with a high silica modulus. The degree of desilication increases sharply only at temperatures above 97 °C.

Dependence silica module obtained after low-temperature liquor desilication from seed dosage is shown in Figure 2.

From Figure 2 it follows that with increasing dosage degree of desilication increases, but it has a minimal number of priming effect compared to temperature, because of the circulation of the liquor in the reaction vessel. From the data obtained, the dosage of seed should be at least 20 g/dm^3 on dry nepheline concentrate.

X-ray analysis results of thoroughly washed solids obtained after low temperature desilication showed that in comparison with the original seed material they exhibit hydroalumosilicates presence of alkali metals and aluminum trihydroxide as gibbsite (Figure 3).

Thus it can be concluded that at the non-autoclave desilication occurs in addition to the main reaction with the SiO_2 isolation from liquor in solid phase according to the scheme:

$$2NaAl(OH)_4 + 2Na_2SiO_2(OH)_2 = Na_2O \cdot Al_2O_3 \cdot 2SiO_2 \cdot H_2O + 4NaOH + 3H_2O,$$

hydrolysis of the aluminate liquor scheme

$$NaAlO_2 + 2H_2O = Al(OH)_3 + NaOH.$$

Hydrolysis returns to the solid phase after the desilication an average of 3.0 % by weight of Al_2O_3 in the form of $Al(OH)_3$, which is confirmed by the analysis of both liquid and solid phases.

Basic hydrolysis most likely cause is the dilution of the aluminate liquor after leaching liquid phase of the seed material.

Averaged over the first seven months of performance of alumina production in the autoclave (145÷150 °C) and non-autoclave (97÷99 °C) desilication at otherwise unchanged conditions are presented in Table I.

From the Table I it follows that the transition to a low temperature desilication has almost the same composition of nepheline concentrate but accompanied by increased alumina and alkalis in the charge and sinter because of the increase in the amount of white sludge used in preparation of the charge, and also due to lower consumption of cake for 1 ton of alumina. The difference between the amount of Al_2O_3 in the sinter for 1 ton of alumina in the two circuits will characterize Al_2O_3 loss from changing to non-autoclave desilication (an average of 60 kg per 1 ton of alumina). When non-autoclave desilication number seed to a first stage desilication increases by 3-5 times, and therefore increases the amount of liquid phase - aluminate liquor supplied to the batch preparation. This eventually causes a concentration increase in Al_2O_3 and R_2O in wash water. Increased extraction of Al_2O_3 and R_2O at leaching of cake due to return to the "head" of virtually commercial products - sodium aluminate and aluminum hydroxide. Out of cake for 1 ton of alumina has decreased, but the content of Al_2O_3 and R_2O has increased due to a greater return in the "head" of the process.

Implementation of the first stage of the desilication using as seed nepheline concentrate, lets do some reduce steam consumption and simplify hardware design, compared with the implementation

of redistribution under autoclave conditions. However, when non-autoclave desilication significantly increases the amount of solids in a circulating slurry containing up to 30 % of the liquid phase in the form desilicated liquor.

Concentration of Al_2O_3 in liquor decreased during the desilication test 5,6÷12.8 g/dm^3, including by hydrolysis, diluting the pulp and seed nepheline dilution steaming.

Comparative results of industrial applications and technologies low temperature desilication autoclave cycle nepheline sintering raw material by the process have shown that the major drawback is the low temperature desilication hydrolysis low modulus liquors at low temperatures in the reaction equipment and related operations in the phase separation.

In addition, the transition from the autoclave to non-autoclave desilication accompanied by increased material flow leads to the need to increase the volume of the tank and a decanting apparatus that further reduces the concentration of the liquor fed to the leaching of the cake, and an increase in expenditure ratios. Dilution liquor also leads to an increase in consumption of heat for the evaporation of liquors in soda production.

Conclusion

1. As a result of industrial exploitation scheme provides low-temperature (100 °C) desilication liquors in the first stage marked by substantial hydrolysis of aluminate liquors, leading to the eventual reduction of commodity output due to irretrievable loss sintering even with some increase extraction of useful components of the cake by leaching. Compared with the processing technologies, providing desilication autoclave, additional deadweight loss per 1 ton of alumina amounted to 12.7 kg of Al_2O_3, and 1.1 kg - R_2O.

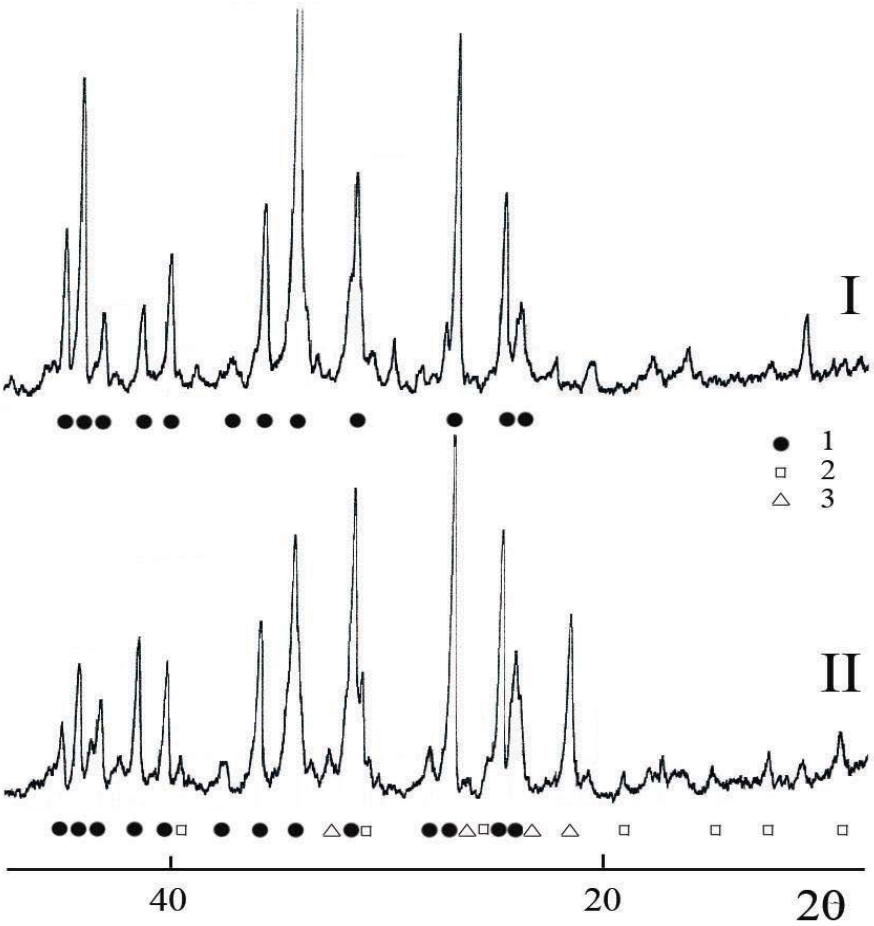

Fig.3 Diffractograms of initial alkali alumosilicate raw material used as a seed (I), and the solid phase after the desilication at 98 °C for 6 hours and seeded 20 g/dm^3 on dry nepheline (II):
1 - nepheline and feldspar;
2- hydroalumosilicates of sodium;
3 – gibbsite

Table I

Comparative performance of alumina production from nepheline concentrate by sintering method at autoclave and non-autoclave desilication

№	Denomination	Units	Autoclave desilication	Non-autoclave desilication
1	Content in nepheline concentrate:	%, wt.		
	Al_2O_3		28,4	28,3
	R_2O		17,3	17,4
2	Content in wash water for charge preparation:	g/dm^3		
	Al_2O_3		16,6	28,6
	R_2O_o		22,6	29,6
3	Content and ratio in charge:	%, wt.		
	Al_2O_3		11,6	12,4
	R_2O		7,2	7,3
	CaO/SiO_2	%, mol.	1,95÷1,96	1,95÷1,96
	R_2O/Al_2O_3	%, mol.	0,97÷1,02	0,97÷1,02
4	Content and ratio in sinter:	%, wt.		
	Al_2O_3		16,0	16,8
	R_2O		9,6	9,7
	CaO/SiO_2	%, mol.	1,94÷1,95	1,94÷1,95
	R_2O/Al_2O	%, mol.	0,97÷0,99	0,97÷0,99
5	Recovery at leaching of sinter:	%		
	Al_2O_3		82,0	82,4
	R_2O		84,0	85,1
6	Content in belite mud:	%, wt.		
	Al_2O_3		3,26	3,74
	R_2O		1,91	1,84
7	Molar ratio of R_2O/Al_2O_3 in liquor after leaching	units	1,35	1,36
8	Content of seed for first stage of desilication	g/dm^3	5÷8	20÷25
9	Weight ratio of Al_2O_3/SiO_2 in liquor after desilication	units	287	166
10	Amount used sinter per ton of alumina among other:	t		
	Al_2O_3		1,31	1,37
	R_2O		0,91	0,91
11	Specific heat consumption for alumina production	GJ/t alumina	10,58	9,69
	Specific heat consumption for soda production	GJ/t soda production	21,25	25,68
	Alumina production	units	1	0,845

2. The most likely cause of the hydrolysis is a low concentration of alumina in liquor due to a constant dilution of steaming and waste water, when introduced seeded desilication. Hydrolysis can be avoided by increasing the concentration of aluminate liquors.

3. Implementation of desilication in non-autoclave conditions requires the following conditions:

- strict observance of temperature (the temperature of the liquor entering the desilication should not be below 98÷100 °C);
- liquors which arrive at desilication should have a caustic module which eliminates aluminate liquor hydrolysis from the desilication process;
- implementation of hardware and technical measures to reduce the input of water in or take it out of the cycle of processing of raw materials.

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Light Metals 2016
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TMS (The Minerals, Metals & Materials Society), 2016

Study on the Influence of Chemical Additives during the Digestion of Bauxite

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Abstract: The synthetic polymer flocculant has been applied to Bayer process for thirty years. In recent years, it is pointed out that the flocculation is related to the presence of other ions. The experimental results show that the effect of mud sedimentation and mud compaction will affect the production capacity and purity of alumina product. The potential of using water absorbing polymer or synthetic dispersant to change the rheological property of the slurry is also summarized in the end. The combination of these effects, with the new bio polymer to improve the primary pour into the washing bath of the overflow liquid clarity and recovery of sodium hydroxide also made a narrative.

Keywords : Bayer process, carbonate, causticizing, flocculation

1 Outline

Synthetic macro-molecular polymer as the substitute of starch and wheat flour has been used in red-mud slurry flocculation of alumina industry for almost 30 years, but some alumina plants still holds the traditional starch flocculation technology. People believed that synthetic macro-molecular polymer could get over the interference of organic impurity and enhance the efficiency of settlement separation at the same time, but in the red-mud slurry which contains a great deal of carbonate ions, starch's use was helpful to the clarification of the slurry. It was found by research that the interference of carbonate ions could be eliminated by slaked lime causticizing and synthetic macro-molecular polymer could reduce the disturbance of humic salt and organic impurity. Factory test shows that small amount of synthetic macro-molecular polymer can replace large use of starch. More adsorption place could be set aside on the red-mud ore grain that promote the deeper adsorption on the surface of ore grain for more humic salt and organic impurity' s removal.

2 Experiment

2.1 Experiment Reagent

Polymer made by Nanchang University: HPAM, PAAS; Ordinary starch; Soda ash(AR), Hydroxide sodium(AR), Hydroxide calcium(white lime CP).

2.2 Experiment methods

After crushed and comminuted, bauxite was confected to a mixture by proportional correction liquor in a blend tank. Put it into an autoclave and lixiviated under high pressure for 90min. The digested ore slurry was refrigerated and decompressed to normal pressure. Then add in solution and dilute to a certain concentration before settlement separation experiment.

Lab settlement experiments were carried out under aqueous thermostat. The size of settlement tube is Φ30*220mm. The turbidity of overflowing liquid or supernatant was determined by eyes' observation, filtration and weighing , and light-electricity turbidimetric method. As for the low turbidity samples, their turbidity was determined by checking calibration curve which was drawn according to the absorbence of turbidity standard solution determined by spectrophotometer with 1cm tube at 660nm.

2.3 Experiment materals

Chemical composition of diasporic bauxite in Chinese are presented in Table 1. Lime chemical composition (lime mixed with

incompletely burnt limestone): CaO 71.08%; MgO 6.85%. Chemical composition of bauxite digestion liquor is shown in Table 2.

Table 1 Chemical composition of Bauxite

composition	SiO_2	Fe_2O_3	Al_2O_3	CaO	TiO_2
Content（%）	16.66	14.40	63.82	0.86	2.97

Table 2 Bayer diluted digested slurry composition

composition	Al_2O_3	Na_2O_k	Na_2O_T	Solids	α_k
Content (g/l)	133.60	137.40	155.84	51.7	1.69

（Na_2O_k:Content of caustic alkali，Na_2O_T:Content of all alkali, α_k: the mole ratio of sodium oxide and alumina）

3 Experimental results and Discussion

3.1 Effects of carbonate ions on flocculation settlement

The diluted digested ore slurry contained caustic alkali and the content of carbonate sodium amounted to 31.63g/L. It was proved by experiments that if such digested ore slurry components was carried through settlement separation only depending on synthetic macro-molecular polymer, the velocity of settlement separation and the packing characteristics of red-mud were both subjected to disadvantage affects.

Table 3 Effects of adding hydroxide calcium on flocculation settlement

Test Number	Hydroxide Calcium Capacity（g/l）	Settlement Velocity（cm/min）				Turbidity（g/l）
		Polymer Capacity（mg/l）				
		1.0	2.0	3.0	4.0	
1	0	3.8	8.1	9.6	11.3	0.24
2	5	24.5	55.7	85.3	119.2	0.13
3	10	31.4	62.5	100.8	136.7	0.11

Table 4 Effects of adding hydroxide calcium on under-set solid content

Test Number	Hydroxide Calcium Capacity（g/l）	Mass ratio of Liquid-solids of Compressed Red-mud（L/S）				Turbidity（g/l）
		Polymer Capacity（mg/l）				
		1.0	2.0	3.0	4.0	
A	0	5.88	5.71	5.40	5.12	0.24
B	5	4.08	4.0	3.84	3.77	0.13
C	10	3.39	3.35	3.33	3.30	0.11

Further research showed that the affects were due to the ore grain in the mixture which could adsorb carbonate ions firstly and interfere synthetic macro-molecular flocculant's

3.2 Effects of Humic Salt Impurity

Diasporic bauxite generally contains a little humic salt impurity, which is a disadvantage to alumina production technology. In digested ore slurry, such organic substance concentrate and to a certain extent adsorbed on the surface of ore grain, which often cause foam spillover in the top of setting bath, and bring bad effects to normal production.

The existence of humic salt and other organic carbon impurity, either alone or mixed together, will affect the efficiency of red-mud's settlement separation and the clarification of supernatant badly. Such bad affects can be removed by using synthetic macro-molecular polymer replacing starch and more adsorption place could be set aside on the red-mud ore grain, which also help the organic impurity's adsorption and removal. Synthetic polymer flocculants degree of different ions on the upper "clear liquid" in the light transmission rate is significant (shown in Figure 4).

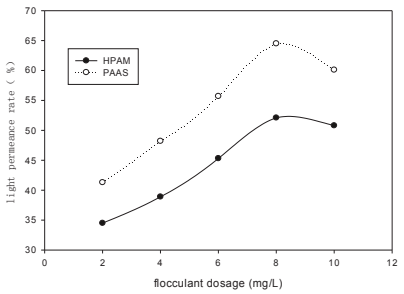

Figure 4 Effects of flocculation on light permeance rate

Figure 4 shows that with the increased use of synthetic polymer, the upper "clear liquid" in the transmittance tended to increase. When the amount of the flocculant reaches a certain volume, the upper "clear liquid" and through Light rates decline with the volume of the flocculant increasing. High degree of synthetic ion polymer to the red mud surface electrical properties have strong surface charge neutralization, small degree of synthetic ion flocculant on the mineral surface charge have weak neutralization

4 Conclusions

4.1 The disadvantageous effect of carbonate ion in red-mud slurry on flocculation settlement can be reduced by hydroxide calcium causticizing.

4.2 The carbonate ion, humic salt and organic impurity in alumina red-mud slurry will be firstly adsorbed on the surface of slurry ore grains, which will affect flocculation settlement. The effects of carbonate ion on settlement efficiency of adding starch should not be ignored.

4.3 The technology of using synthetic macro-molecular polymer instead of starch to clarify slurry can make more place available to adsorb humic salt and organic impurity, which will be removed from slurry with red-mud, and the effect of clarification can be further promoted.

4.4 When the sodium carbonate in the digested ore slurry was causticized by adding white lime the packing characteristics of red-mud could also be improved.

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flocculation. When the carbonate sodium in the digested ore slurry were causticized again by adding white lime, synthetic macro-molecular flocculant's flocculation; especially the velocity of settlement separation and the packing characteristics of red-mud, could be improved. The test dates are showed in table3,4 and Figure 1,2,3.

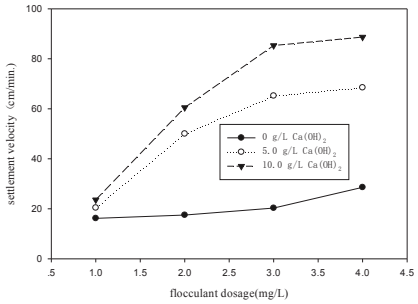

Figure **1 Influence of adding hydroxide calcium on settlement velocity**

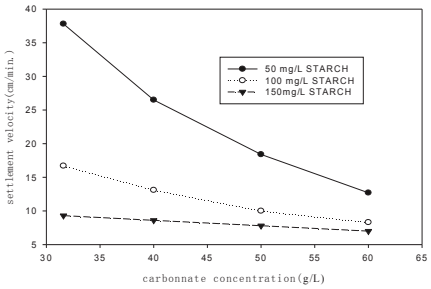

Figure **3 Effects of Carbonate on Settlement Efficiency of Adding Starch**

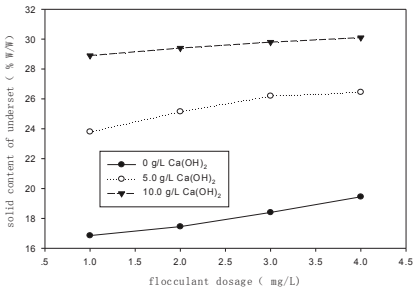

Figure **2 Influence of adding hydroxide calcium on Red-mud compressed**

Because of the particularity of alumina production technology, the digested ore slurry tends to contain a great deal of carbonate during production process and make the red-mud's settlement separation more complex. Until today, some alumina plants still adopt the traditional most effective starch as digested ore slurry flocculation. But, experiments indicated that adding starch for red-mud's settlement separation, carbonate ions still largely affected the settlement separation's velocity and clarification of red-mud slurry, which could be seen in table5. So many carbonate's remove by adding white lime and causticizing, and use of synthetic macro-molecular polymer replacing starch, do good not only to settlement separation's velocity of red-mud slurry, but also to improve the red-mud's packing.

Table5 Effects of Carbonate Sodium on Settlement Efficiency of Adding Starch

Test Number	Starch Capacity（mg/l）	Carbonate Sodium Capacity（g/l）	Settlement Velocity（cm/min）	Turbidity（g/l）
a	100	31.6	9.3	0.24
b	100	40.0	8.6	0.22
c	100	50.0	7.5	0.41

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Light Metals 2016

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TMS (The Minerals, Metals & Materials Society), 2016

GOING FAR (FLOATING ALUMINA REFINERY)

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Keywords: Floating, Compact, Innovation

Abstract

The Shell "Prelude" project is embarking on a revolutionary step change in LNG (liquefied natural gas) processing. Prelude will be a 488m long floating facility that will process, liquefy, store and transfer LNG at sea directly above the natural gas field. Could the same floating principle be applied to an alumina refinery moored at a shipping berth close to the bauxite supply?
"Going FAR" is, for now, an abstract concept for a Floating Alumina Refinery. The FAR concept potentially reduces the sovereign risk from politically unstable or socially fragile countries by simply moving to another location, protecting the majority of the refinery asset and investment. Going FAR may also provide the opportunity to process small isolated bauxite deposits where it is uneconomical to build a conventional land-based refinery.
Bringing this concept to commercial reality will require outside the square thinking, and many critical questions need to be answered. What are the benefits of FAR? How do the economics compare to a traditional onshore refinery? What is the maximum refinery capacity that would be possible as a floating facility? What changes are required in technology and equipment? What are the major changes to safety and environmental risks and how will they be controlled and managed? Will the FAR design inspire innovations which have application for land-based refineries? This paper will explore the edges of these key questions in search of the future direction for this concept.

Introduction

Greenfield alumina refinery design has traditionally centred on locating the refinery in the most economic location either at or near the bauxite mine, alumina smelter or cheap energy sources. Depletion of the bauxite resource or an increase in energy price can impact on refinery operating costs and viability, decreasing the value of the refinery asset. The refinery asset value can also decrease in politically unstable or socially fragile regions where the refinery can no longer operate as intended, due to outside influences. A portable refinery asset can be moved to another location and, with some infrastructure capital, continue to operate as a refinery. A portable asset can potentially hold a higher value to its owner than a land based asset in the same region.
Following similar principles to recent floating liquefied natural gas projects, Going FAR explores the design considerations for a portable floating alumina refinery.
While operating, Going FAR will be located at a purpose built shallow water mooring facility. Storage of raw materials and major commodities consume large footprint areas and have a relatively low capital cost compared to refinery processes. The hull of Going FAR will be used to store liquid raw materials including fuel (possibly LNG), caustic, desalinated water and other liquid chemicals. In-process liquors will be stored in separate compartments in the hull to provide liquor volume surge for the refinery.
Alumina is assumed to be direct loaded from Going FAR onto alumina ships for transport to the smelter or customer. Infrastructure will be required on shore to support Going FAR. In particular, the on-shore facilities will cater for storage and inventories of bauxite, lime and rotating spare equipment and parts.
Although not explored, storage of materials and services could also be supported on separate barges or ships as opposed to land based infrastructure.
All employees would live at facilities nearby the on shore infrastructure, along with associated services. Major maintenance of refinery components would be carried out in on shore facilities although the logistics of moving equipment between ship and shore will be a challenge requiring resolution during the design process. Bauxite residue would be neutralised and back loaded to the mine, returning the residue to its original location.
When the bauxite reserve is depleted or other factors make the operation no longer viable, tug boats would move Going FAR from its moored location to a new location and bauxite supply.

Refinery Capacity

Typical land based refineries produce between 100 and 150 ktpa per hectare for the refinery process areas, excluding stockpiles, utilities and services and residue storage.
Compared to land based refineries Going FAR requires new processes and technologies to provide more compact facilities with a layout design that maximises usage of available space. Vertical integration of facilities in the layout design, utilising multiple floor levels will be required to reduce the footprint. Based on the technology and layout discussed in the paper, it is assumed a threefold increase in production can be achieved for the same process area footprint.
The area for Going FAR would be around 36,000 m2 (3.6 hectares) based on Prelude. The estimated refinery capacity for Going FAR is in the range of 1.1 to 1.6 Mtpa.

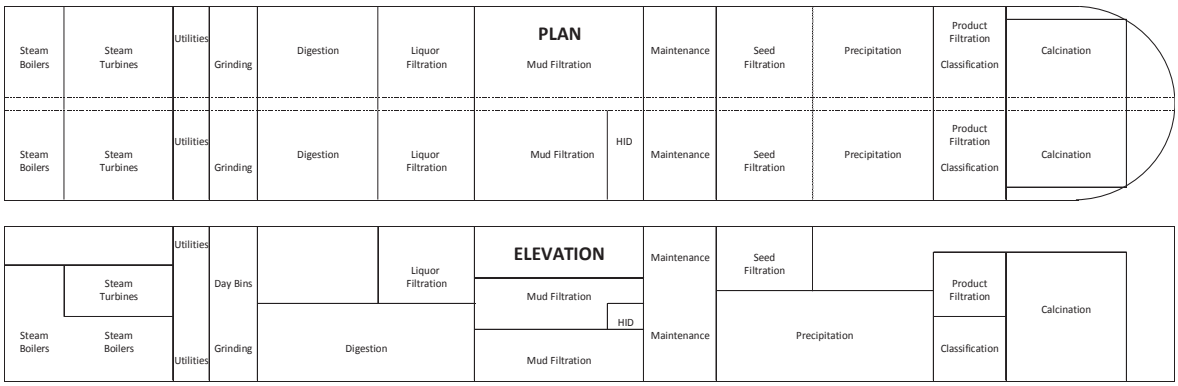

Figure 1. Conceptual Layout

Technology

A key criterion in the technology selection for Going FAR is compact high efficiency units that consume the small footprint volume (length x breadth x height).
Some key areas of the refinery that require significant changes in technology have been examined. Gravity separation is one technology commonly used for separating liquor and solids in several areas of the refinery. Gravity separation is very space intensive and needs to be replaced with alternate technology like pressure driven separation (hydrocyclones, filters or centrifuges) to enable a significant reduction in the footprint for Going FAR.

Water Balance

During the design and operation of a refinery the internal and external water balances must be understood.
A basic refinery external water balance can be completed with bauxite as the input and hydrate and residue as outputs as per Figure 2.

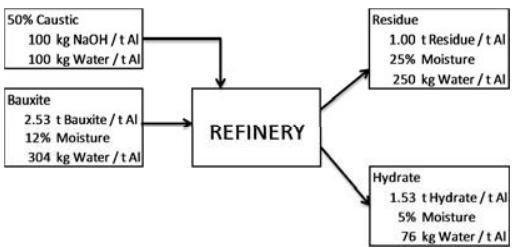

Figure 2. External Water Balance

For filtration of Bayer residue solids, there is a net water input to the circuit of approximately 70 - 80 kg water per tonne of alumina. For a 1.5 Mtpa refinery, there is a net water input of 13 t/h which must be removed from the circuit as water liquid or vapour. For the same 1.5 Mtpa refinery achieving 50% residue solids with gravity thickening, there is a net water loss of 30 t/h which must be supplemented with fresh water.
The refinery will require a cooling source. Cooling towers can lose 1.5 t/h of water per MW of cooling. For a 1.5 Mtpa refinery with approximately 60 MW of cooling in Precipitation (20 MW for Heat of Reaction, 40 MW for a 10°C drop in slurry temperature), water losses of 90 t/h would result.

Single pass seawater cooling does not have evaporation losses and thus no additional make-up water is required. Seawater cooling may be a suitable technology for Going FAR providing the environmental impacts of elevated temperature are managed, and any risk of caustic contamination can be ruled out.
The internal water balance considers the water input and output internal to the Bayer circuit. Management of the refinery internal water balance will also be a key element for Going FAR. To minimise the total evaporation requirement, water input to the process needs to be minimised. The key water inputs are from washing product hydrate and mud which are discussed further in following sections. Management of other parasitic water inputs will be extremely important. Going FAR will have limited online liquor volume surge capacity and evaporation. Parasitic water will quickly swell plant volume and impact on residue washing performance. The grace of poor water management occurring in some refineries with large evaporation rates will not be possible on Going FAR. The design of the refinery to minimise potential sources of parasitic water, including purge water and hose water, will also be key to world class parasitic water management.

Grinding

For the purposes of this paper Going FAR is based on processing low impurity gibbsitic bauxite using low temperature digestion. The bauxite for Going FAR would be crushed and or beneficiated near the bauxite mine. Bauxite ore ready for processing would be stored on stockpiles near the mine and or port providing surge between the refinery and mining operations.
Wet grinding of the bauxite using caustic is assumed. This enables all Bayer unit operations using caustic liquors to be contained on the ship and significantly reduces the transfer of caustic between land and the ship. This strategy minimises the potential for loss of caustic containment to the environment by elimination of sources. Elimination or substitution of hazards / risks should always be preferred over procedural controls.
Bauxite transport from the mine to the ship will be by whatever is the most viable method for the given location, whether by truck, conveyor, rail or pipeline capsule. The latter offers an interesting fit with the Going FAR concept as it is relatively adaptable to a range of logistic and topological conditions. Pipeline slurry transport could also be considered but would require some modifications to the on-ship processes.

Digestion

Digestion of the bauxite slurry using multiple flash stages with heat recovery using shell and tube heat exchangers provides a compact digestion design. Single stream digestion using vertical slurry shell and tube heat exchangers provides efficient heat exchange with feed slurry at 70°C being heated by approximately 50°C and digestion slurry with a similar mass flow also being cooled by 50°C.

In a typical low temperature digestion refinery significant evaporation occurs in Digestion. With space limitations on Going FAR, a separate evaporation facility only adds to the total footprint of the facilities. The design temperature of the digestion facility can be set to cater for all evaporation to occur in Digestion.

Standard low temperature digestion at 150°C provides approximately 6.2 t/h of evaporation per 100 t/h of Digestion feed liquor flow. For a set refinery production, the design Digestion flow depends on the Precipitation yield and refinery availability. To meet the minimum water balance requirements of the Bayer circuit for Going FAR, a digestion temperature of 160°C would be required. This may require an additional flash heating stage in Digestion increasing the Digestion footprint by 15 to 20%. However providing the same evaporation rate in a separate facility would use a significantly larger footprint.

Mud Separation

Separation of the residue from the digestion slurry has conventionally used gravity separation thickeners. However due to the space constraints an alternative technology is required.

Direct filtration of the undiluted blow off slurry would significantly reduce the footprint. Recent advances in vertical pressure filter technology with fully automated cleaning cycles enables the filters to have a very short cycle time (minutes) providing a much higher filtration rate compared to older style filters with cycle times of several hours. Based on an assumed filtration rate of 0.5 m3/h/m2, then 12 to 13 online filters would be required. In addition dilution liquor filters may be required plus spare filters for cleaning and maintenance, requiring an installation of approximately 20 vertical pressure filters.

Direct filtration of hot digestion slurries could also be achieved using hyperbaric filtration (Reference 3) in a compact footprint. Separation of residue solids prior to cooling helps to reduce hydrate re-precipitation in the residue. Hyperbaric filtration can incorporate multi-stage cake washing in the same process step, significantly reducing the number of residue wash filters downstream providing a further reduction in footprint.

Residue Washing & Disposal

The development of synthetic flocculants allowed in a significant step change in the diameter of dense bed mud separation thickeners / washers. Going FAR requires another significant step change or technology change to reduce the footprint, as 4 to 6 gravity separation vessels, each with a diameter and height of 16 to 20m, occupy a significant footprint.

Pressure filtration of the residue mud is now well proven as the last washing step to provide a mud residue cake with a moisture content of 22 to 30wt%. The filter cake generated has a dry, crumbly or sandy texture, significantly different to the pasty sticky material produced using vacuum filtration or high density gravity thickening and compaction.

Residue from a pressure filter can be more easily transported to the residue storage area. One stage of pressure filters provides a similar soda loss to four high rate washing stages. Table I shows a comparison of the assumptions and inputs and result.

Table I. Soda loss comparison with constant wash water ratio of 2.1 t/t

Equipment Quantity and Type	Discharge Density (wt %)	Total Sodium Concentration (g/L Na_2CO_3)	Soda Loss (kg NaOH / t mud)
4 Washers	36 to 46	26	23
5 Washers	36 to 46	16	14
6 Washers	36 to 46	9.3	8.1
1 Filter Stage	70	72	22
2 Filter Stages	70 to 75	7.1	1.8

The impact of washing performance at different wash water ratios is much less sensitive for filtration compared to dense bed washers as shown in Figure 3 due to the high discharge density significantly reducing the ratio of discharge liquor to wash water.

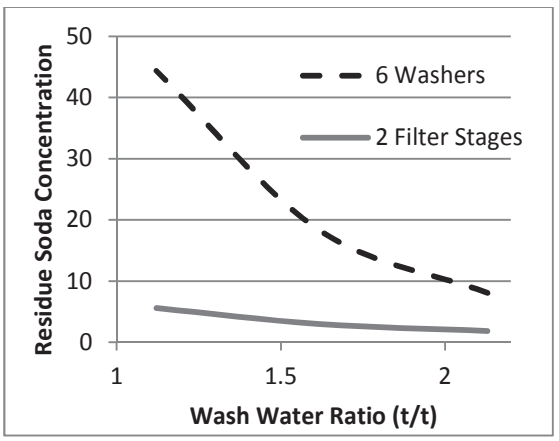

Figure 3. Impact of wash water ratio

However filtration rate of mud residues with higher caustic concentration is impacted by the carrier liquid viscosity. As per Darcy's Law, the filtration rate is inversely proportional to the viscosity.

Darcy's Law

$$Q = \frac{-kA}{\mu}\frac{(p_b - p_a)}{L} \quad (1)$$

Where Q = Filtrate Discharge Rate
k = permeability of the medium
A = filtration area
μ = filtrate viscosity
pb - pa = total pressure drop
L = filtration path length

With residue washing using a two stage filtration process, the viscosity will be higher for the first filtration stage. Thus a larger number of filters will be required in the first filtration stage

compared to the second filtration stage. Due to the exponential relationship of viscosity and caustic concentration, reductions in wash water ratio will increase the viscosity at both filtration steps and thus impact on the number of filters required.
For the same caustic losses the wash water rate can be reduced in a filter washing circuit. This reduces the evaporation capacity required. However the trade-off is in an increase in pressure filters required for residue washing due to the increase in viscosity. For a reduction in wash water ratio from 2 to 1, the number of operating filters increases by 20%.
The cleaning requirements of the filters processing high scaling slurries will influence the number of spare filters. Residue pressure filtration has been proven as the last washing stage. However, using pressure filters for residue with high caustic and A/C is not proven and would require pilot testing and trials to prove the operability of the concept. Automated chemical wash facilities may be required to remove hydrate scale to maintain filtration rate across different cycles.
The discharge from the final pressure filter stage is expected to be a cake of 22 to 30% moisture suitable for transportation. The residue cake can then be back loaded via the bauxite transport system (be it conveyors, trucks, pipeline capsules etc) to the mine area. Suitable neutralisation and/or encapsulation procedures would be employed based on site requirements.

Precipitation

For Precipitation, the footprint is dictated by the volume of slurry to be retained which is a result of the required holding time. Precipitation tanks consume a large proportion on the land-based refineries layout. Novel ideas are required to reduce the Precipitation footprint on Going FAR.
Large growth precipitation holding times are required to give high yields which may not be the optimum for Going FAR. However for a constant refinery production, a reduction in holding time and yield requires higher plant flow. Higher plant flow requires holding a larger volume of slurry to maintain holding time.
Just focussing on the Precipitation area, the trade-off of yield versus flow gives an optimum holding time where the holding volume is minimised for a constant production. In the first precipitators, the yield gained per unit volume is greater than the extra volume required due to increased flow. For the back end precipitators, the opposite relationship applies. By using holding volume as an indicator of footprint size in both plan and elevation, there is a minimum footprint for Going FAR. The exact location of the optimum depends on the target precipitation temperature, seed charge and seed density as well as the location and magnitude of row cooling. A typical optimum shown in Figure 4 would mean around 12 to 15 hours holding time would minimise the size of the precipitation area.

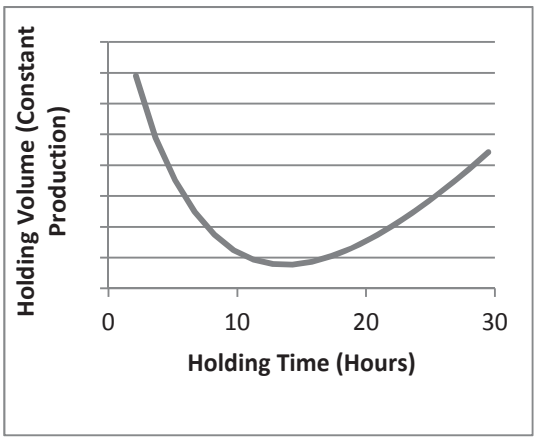

Figure 4. Optimum holding time

However, after considering the impact on other plant areas, and also on product quality issues, it is likely that the optimum yield for Going FAR will be closer to 20 hours.
The tank volume required for 20 hours holding time represents the biggest single challenge to achieving a sufficiently compact footprint to make the concept viable. A technology breakthrough in this area will have a significant impact on the economics of Going FAR. A move away from conventional cylindrical tanks may allow all or part of the precipitation volume to be stored in the hull providing improvements in agitation technology are found to drive this. Tube precipitation with the tubes cooled by seawater is another concept. However over 220 tube passes the length of the ship would be required for 12 hours holding time. Therefore, large tube technology does not provide a compact footprint.

Classification and Filtration

As per the modern day standard refinery design, product and seed will be classified using hydrocyclones. Hydrocyclones provide a significantly more compact layout compared to gravity classification. The cyclones can be located on top of the precipitators utilising gravity flow to seed filtration, or direct to precipitation. Alternatively the cyclones can be located on the ship floor level utilising the head in the precipitators to drive the flow at the required cyclone pressure.
For Going FAR, all hydrate seed will be filtered to minimise spent liquor recycle and thereby reduce required precipitation volume.
Locating seed disc filters on top of the precipitation provides a compact footprint, as additional filtration capacity does not consume valuable ship floor space. The filtrate is spent liquor that returns through HID. Sealing within the filter elements and drum as well as cloth selection and management is important for recovery of solids, as there is no gravity polishing step. The height of the seed filter can be used to advantage. Leaving the required height to provide a barometric leg for the filters, the filtrate can then be driven through HID by gravity to spent liquor storage in the hull.

Calcination

Calcination of hydrate to alumina is energy intensive, consuming approximately 3GJ of energy per tonne of alumina. While Calcination will be possible on Going FAR, do the economics support it? Going FAR will be located close to the bauxite source which does not necessarily mean it is also located near a cheap

energy supply. Smelters, due to the high energy consumption, are typically located close to low cost energy. Should Going FAR ship hydrate or alumina? Shipping hydrate allows for the calcination energy to be sourced from a cheaper energy supplies, but the free and bound moisture are now transported with the hydrate, increasing transport costs.
Preliminary analysis indicates a break-even price of about US$60 - US$80 per barrel of oil depending on freight distance and gas prices at the calcination location. There are many factors that can influence the world oil price including increases in production of low cost stranded gas.
The economic decision on the location of Calcination needs will be based on the predicted long term energy price. Despite a recent drop in oil price it is assumed that the long term oil price will continue to rise.
Thus it has been assumed Going FAR will calcine hydrate to alumina prior to transportation.

Steam Power Station and Utilities

The focus of this paper is on the technology around the key refinery areas with little focus on the key technology changes around utilities. Some considerations for the utility areas are as follows.
To be self-sufficient, the steam power station on Going FAR must provide full cogeneration to meet process steam and electrical power demands. The power demands for Going FAR will be slightly higher than for land based refineries. The refinery flow will be high due to the reduced holding time, increasing power consumption. Also, the multiple level layout increases the height of some facilities and thus increases pumping power. While clever layout and design can utilise the gravitational energy to advantage, there will be inefficiencies that increase the pump power costs above conventional refineries.
A supply of clean fresh water will be essential to replace water losses in the refinery. Reverse osmosis desalination can provide Going FAR with a compact secure fresh water source using only 1/3 of the footprint of multi-flash desalination and at lower cost. Desalinated water would be stored in the hull for use in the power station and throughout the refinery.

OTHER CONSIDERATIONS

Operations and Maintenance

The layout design for Going FAR will be critical to the success. Automation of several maintenance and operations activities will be required. Investment in specialised cranage, lifting devices and tools will be necessary to remove equipment for maintenance through tight access envelopes. The layout design will have two competing forces.
1. A force to minimise the footprint and operations and maintenance envelopes to eliminate wasted space.
2. A force to provide sufficient maintenance and operations envelopes to complete the required tasks ensuring acceptable health and safety risks.
Clear definition and description of all operating and maintenance tasks, logistics and scheduling will be required early in the design process. Risk assessments of all operations and maintenance activities will be required during each step of the design to ensure the health and safety requirements are satisfied. Successful change management during design and construction is also a key. Experience from recent floating LNG facilities will provide valuable learnings.
At the shallow water mooring, Going FAR will use stability legs to reduce the impacts of wave motion on the ship. Wave motion will impact the operation of the refinery, including the effects of tank sloshing and level swings, plus unloading and loading of materials to barges or ships. The stability legs will not be permanent structures, allowing the ship to raise the stability legs when needed to move to a different location. Possibly similar technology to off shore platform legs could be used for the stability legs. Each leg may be a column that can be filled with seawater when in position to provide stability. The legs can then be emptied of water and raised by filling with air prior to ship relocation.

Environmental Considerations

Going FAR will require several measures to protect the neighbouring seawater environment. The design basis for Going FAR limits the transfer of caustic liquors to and from the ship reducing the risk of environmental contamination.
All caustic liquors on Going FAR will need to be contained on the ship to prevent environmental harm. The design of Going FAR will need to ensure that liquors from unplanned failures and events like power failures, tank overflows, gland failure, flange failures, pressure relief events are fully contained on the ship. Compartments in the hull will provide emergency storage of plant liquors, tank overflows and precipitation rundown. Separate compartments will be used for storage of sump liquors from drainage and flushing operations.
It is important to maintain high environmental standards and not become complacent about environment issues although the impact of a small caustic release into seawater poses lower environmental consequence compared to release of crude oil from the hydrocarbon industry.

Constructability

Going FAR does not require any earthworks, foundations or civil infrastructure for the refinery areas reducing cost and schedule. Some civils may be required for the land based areas, but these areas containing stockpiles or maintenance sheds require much simpler civil works and foundations.
Going FAR is suited to using lower cost construction labour in South East Asia or a similar location. Construction of modularised facilities can occur close to the construction port significantly reducing construction hours and work force compared to stick built facilities. Modularisation and the lack of civil works will provide a significant schedule advantage of several months enabling production from Going FAR to be produced earlier than a conventional refinery improving the economics.
All the pre-commissioning, wet and dry testing can be completed at the construction port, again reducing man-hours and schedule. Modularised construction for land based refineries still requires significant pre-commissioning. Going FAR can pre-commission all refinery areas together at the completion of construction.
An estimated cost for Going FAR is in the range of US$2000 – US$2500 per tonne of alumina using low construction labour costs in Asia. Materials costs assume two times the quantity of steel per tonne of production for typical land based refineries due to the multiple layer compact concepts that are required for Going FAR.

CONCLUSION

Going FAR is currently an abstract concept that, at first, would not seem likely to become reality in the near or distant future.

While the capital cost is higher than would be considered for a land based greenfield refinery, Going FAR brings other advantages such as reduced political and sovereign risk in certain regions and the ability to unlock small isolated bauxite deposits. Going FAR may also have a shorter construction schedule to produce first alumina ahead of conventional land based refineries. Once these advantages are factored in, it seems that a favourable business case for Going FAR is much closer to reality than might have been originally thought.

However, the technical risk associated with construction of the first Going FAR facility is a significant barrier to be overcome. The risks associated with operating complex and compact plants on large floating facilities will be better understood thanks to the Shell Prelude project. The lessons learned from the design and operation of Prelude and future floating facilities will be invaluable to Going FAR.

The Bayer technical risk for Going FAR can be reduced with elements of the technology installed in existing refineries as part of efficiency upgrades. Many of the technical improvements needed for Going FAR provide significant opportunity to reduce operating costs in existing refineries.

With continuing improvement and innovation in technology, it may be years and not decades before the first “floating” alumina is produced from Going FAR.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SUSTAINING CAPITAL OF ALUMINA REFINERY PROJECTS – IMPORTANT BUT UNLOVED

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Keywords: Alumina Refinery Opex, Alumina Operating Cost, Sustaining Capital Cost, Sustaining Capex, Susex

Abstract

Sustaining capital cost / expenditure, abbreviated as sustaining capex or Susex, is a project cost item which is sometimes forgotten and often disliked. To many it is unknown, and it may be (partly) hidden in other operating cost items. The relation – and the difference – with repair and maintenance costs is not always clear and may inadvertently lead to incomplete cost estimates.

This paper explores several facets of sustaining capital and aims at clarifying some of these issues for alumina refinery projects.

Overview – Section 2

Sustaining Capital – General & Ala Refinery Project (2.1)

Sustaining capital expenditure ("Susex") in general refers to expenditure related to capital asset additions, replacements or improvements required to maintain/sustain existing assets. Required investments covered under Susex contribute to ensuring that the operation's overall reliability is maintained at the existing production level and that all (existing and new) reliability, legislative, regulatory, environmental and safety requirements are met. Considered from an economic perspective: Susex does not generate additional revenues. Drivers of Susex include:

- Reliability and risk mitigation which includes but is not limited to life extensions, end of life replacements, and system reinforcements (designed and implemented to meet existing and new requirements)
- Regulatory compliance
- Health, Safety & Environment compliance and improvements
- Social requirements
- Product market requirements.

Summarizing: for an alumina refinery project Susex is the ongoing capital expenditure required to avoid the eventual discontinuation of the operation, to maintain its production level, its product at target quality and to maintain Health Safety and Environment (HSE) on target / in compliance. It covers items such as:

- Replacing worn out equipment, and major equipment repair[1]
- Developing bauxite residue ("red mud") disposal areas
- Repair of facility structures
- Investments to meet (existing and new) HSE requirements
- Investments to meet (existing and new) alumina product quality standards
- Investments to meet social obligations.

Often an alumina refinery project includes other project elements such as a Bauxite mine and Infrastructure (port, railway, roads, personnel housing, etc). In that case most of the above items apply also to the additional project elements, and supplementary Susex items may include (but are not necessarily limited to):

- Mine mobile equipment replacement
- Mine fleet expansion to maintain plant throughput capacity (if applicable)
- Resettlement costs of local communities (if applicable)
- Port dredging (if applicable)
- Railway rolling stock and locomotive replacement (if applicable).

Operation Closure Costs (2.2)

In the development of greenfield alumina refinery projects, ordinarily an allowance is included in the financial model, covering the costs of closure of the operation, and the costs of rehabilitation of the affected mine lease and refinery proper areas at the end of mine life. Mine area rehabilitation is normally a progressive process, but final closure of the operational residue area and the refinery site itself will require expenditure at the end of life.

After operation closure, port and housing facilities and any other usable infrastructure are sometimes handed over to the country's government without further expense. Operation closure costs are not included in this paper's Susex cost.

Difference with Repair & Maintenance Costs (2.3)

The principle difference between Susex and Repair and Maintenance (R&M) costs (including materials and contractors) as included in the alumina refinery cash operating cost is that Susex is depreciated (spread) over a period of time while R&M costs are expensed, i.e. form part of a plant's cash operating costs (Opex).

However when comparing Susex between different projects, elements of Repair and Maintenance costs should be considered as well because plants do not always use the same basis for these two cost items (refer section 5).

Sustaining Capital Cost Values – Section 3

The actual Susex of a project depends on aspects such as its location, age, technologies used, operational and maintenance methods, infrastructure, production capacity, etc. It should also be remembered that due to its nature actual Susex values may vary (significantly) from year to year, oscillating around a long term average.

Table I – Typical Susex Range and Value

	Sustaining Capex, $/tA	
Project Element	**Indicative Range**	**Typical Value**
Bauxite Mine	2.5-5.5	~3.5
Infrastructure	1.5-3.5	~2.5
Alumina Refinery	6-12	~8
Total	10-21	~14

The Susex numbers in Table I are based on available global historical data and include typical Bauxite Mine and Alumina Refinery Susex values and ranges (for both low and high

[1] Mobile equipment, communication, control and IT systems, mills, tanks, filters, boilers, etc

temperature digestion plants) for existing bauxite and alumina projects, and Infrastructure values based on greenfield data.

The approach used in this paper to reconcile an alumina refinery's Susex is based on the underlying capex build-up and the lifetime of its components as shown in Table II.

The table illustrates that, based on typical values for refinery capex and lifetime of equipment and structures, a typical annual long-term alumina refinery Susex value may be derived (8.7 $/tA) which is reasonably in line with the actual typical value as shown in Table I (~8 $/tA).

Table II – Alumina Refinery Sustaining Capex Build-up

Alumina Refinery	Typical Greenfield Capex Values, $/Annual tA		Lifetime, Years		Calculated Annual Long Term Susex				
					Greenfield Project Basis				Mature Portfolio Basis④
Capex Element	Indicative Range	Typical Value	Indicative Range⑤	Typical	Indicative Range, $/tA	Typical, $/tA	% of Direct, Total Capex (Range)	% of Direct, Total Capex (Typical)	% of Direct, Total Capex (Range)
Equipment		150	40-60	50	2.5-4	3			
Commodities① / Structures		370	60-70	65	5-6.5	5.7			
Direct Capex		520			7.5-10.5	8.7	1.4-2	1.7	2.8-4
Indir. Capex②		480							
Total Capex	850-1,400③	1,000					0.8-1.1	0.9	1.6-2.2

① Structural Steel, Concrete, piping, wire and cable, raceway (cable trays), Instrumentation, Electrical Equipment, Earthworks, etc
② Freight, EPCM, temporary construction support, commissioning, insurance, owner's engineering, etc.
③ Dependent on production capacity, location, bauxite quality, technologies, etc
④ A portfolio with (Bauxite and) Alumina projects which on average are midway their lifetime
⑤ Refer for instance [1]

The indicative range derived this way (7.5-10.5 $/tA), although being narrower than the actual range, refer Table I (6-12 $/tA), is not inconsistent as it falls completely within the range of Table I .

Sustaining Capital – Industry Majors – Section 4

To provide a context for the numbers in section 3, Susex values in relation to Properties, Plant and Equipment (P, P & E) of Alcoa [1], Hydro [2], and Rio Tinto [3] have been compared for the years 2013 and 2014 based on their respective annual reports. The result is shown in Table IV.

Land and land rights (Alcoa, Hydro) and Mining properties and leases (Rio Tinto) have been excluded from P, P & E in this comparison under the assumption that no sustaining capital has been spent on these items. Construction Work-in-Progress, respectively Capital Works in Progress (WIP, Alcoa, respectively Rio Tinto) and Plant under Construction (PUC, Hydro) are separately indicated. These are excluded in the calculation of Susex as percentage of P, P & E under the assumption that they have not (yet) required sustaining capital expenditure.

When interpreting the numbers of Table IV it should be realized that Alcoa, Hydro and Rio Tinto differ significantly in many respects such as scale (2014 Revenue: Alcoa ~24BUS$; Hydro ~12.5 BUS$; Rio Tinto ~48 BUS$) and business focus (Alcoa 2014, an integrated Aluminium company with significant downstream activities: Alumina + Primary Metal ~43% of Revenue; Hydro 2014, an integrated aluminium company with limited downstream and other activities: Bauxite & Alumina + Primary Metal + Metal Markets ~69% of Revenue; Rio Tinto 2014, a major mining company (iron ore, copper, diamonds) with integrated aluminium activities: Rio Tinto Alcan ~25% of Revenue).

The differences between these industry majors are reflected in the range of Susex as percentage of P, P & E on "at cost" basis (2.2-3.8%) and to a lesser extent on "net of depreciation" basis (5.0-6.5%). Table III provides a comparison of the results.

Table III – Susex Comparison

Alumina Refinery Sustaining Capex					
	Indic. Range $/tA	Typ. Value $/tA	% of Total Capex (typ.)	% of Total Capex greenf. / "at cost" (range)	% of Total Capex mature / net of deprec. (range)
Actual (Table I)	6-12	8			
Calculated (Table II)	7.5-10.5	8.7	0.9	0.8-1.1	1.6-2.2
Industry Majors 2013-2014 (Table IV)				2.2-3.8	5.0-6.5

Noticeable from this table is that the range of Susex as percentage of P, P & E both on "at cost" as well as on "net of depreciation" basis is a factor 2-4 times larger as reported in the annual reports of Alcoa, Hydro, and Rio Tinto than it is as derived in section 3. However due to the availability of only limited detail information on Susex provided in annual reports and presentations in general, and on alumina refinery operations in particular it is difficult to reconcile this difference.

Table IV – Years 2013 and 2014 Sustaining Capex of Alcoa, Hydro and Rio Tinto in relation to their Properties, Plant & Equipment

(Numbers based on 2014 Annual Reports)	ALCOA				HYDRO				RIO TINTO			
	2014		2013		2014		2013		2014		2013	
	MUS$	As % of Cost	MUS$	As % of Cost	MNOK	As % of Cost	MNOK	As % of Cost	MUS$	As % of Cost	MUS$	As % of Cost
Properties, Plant, and Equipment (excl. Land and Mining Properties & Leases), at cost	33,503	100%	34,660	100%	102,856	100%	96,199	100%	80,680	100%	79,757	100%
Accumulated Depreciation, depletion, and amortization	-19,091	-57%	-19,227	-55%	-50,757	-49%	-47,274	-49%	-33,785	-42%	-33,881	-42%
Sub Total	14,412	43%	15,433	45%	52,099	51%	48,925	51%	46,895	58%	45,876	58%
Construction Work in Progress / Plant under Construction (WIP/PUC)	1,466	4%	1,567	5%	2,687	3%	3,020	3%	9,885	12%	14,071	18%
Properties, Plant, and Equipment, Net	15,878	47%	17,000	49%	54,786	53%	51,945	54%	56,780	70%	59,947	75%
Sustaining Capital	735		770		3,300		2,700		2,700		3,000	
Sustaining Capex as % of "P P & E, at Cost" Value, excl. WIP/PUC	2.2%		2.2%		3.2%		2.8%		3.3%		3.8%	
Sustaining Capex as % of "P P & E, Net" Value, excl. WIP/PUC	5.1%		5.0%		6.3%		5.5%		5.8%		6.5%	

Relationship between Susex and Maintenance Materials & Contract Services – Section 5

The following aspect may explain at least part of the discrepancy found in Table III above. To properly compare Susex values between projects, costs of "Maintenance Supplies/Materials" and "Contract Services" in an operation's operating cost should also be taken into account, because some operations account for maintenance project costs under these headings rather than under Susex. The same could well apply to the collection of operations reported on in an annual report.

Available information of existing operating alumina refineries (both low and high temperature digestion) indicates a range of 18-30 US$/tA for the total of [Susex + Maintenance Materials + Contract Services].

Conclusions – Section 6

The actual typical Susex value (~8 US$/tA) and range (6-12 US$/tA) for an alumina refinery project can be reconciled reasonably well by using an approach based on the underlying capex build-up and lifetimes of major capex elements.

The discrepancy between the range of Susex as percentage of Property, Plant & Equipment both on "at cost" as well as on "net of depreciation" basis in the annual reports of some of the industry majors (2.2-3.8%, respectively 5.0-6.5%) and the equivalent ranges arrived at by using the earlier mentioned approach (0.8-1.1%, respectively 1.6-2.2%) cannot easily be resolved.

When comparing Susex values between operations, costs of "Maintenance Supplies/Materials" and "Contract Services" should be taken into account: for alumina refinery projects the sum of Susex plus these two items ranges between 18-30 US$/tA.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Reaction behavior and conversion of anatase in alumina production process with calcification-carbonization method

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Keywords: anatase; reaction behavior; alumina; calcification-carbonization

Abstract

The reaction behavior and conversion of anatase in the calcification-carbonization process were investigated experimentally in the pure substance systems of Na_2O-Al_2O_3-CaO-TiO_2-H_2O and Na_2O-Al_2O_3-SiO_2-CaO-TiO_2-H_2O. The calcification experiments were conducted respectively in the digestion conditions of diaspore and gibbsite. In calcification process, with temperature increasing, anatase converts to firstly $Ca_3TiSi_2(Al_2Si_{0.5}Ti_{0.5})O_{14}$ and then $CaTiO_3$. The increase of sodium aluminate solution concentration and additive TiO_2-SiO_2 ratio can promote this conversion. In carbonization process, CO_2 can't make $CaTiO_3$ decompose. $CaTiO_3$ is the most stable titanate phase in calcification-carbonization process.

Introduction

With the rapid development of the aluminum industry in China, domestic high grade bauxite has been unable to meet the needs of the industry. The processing methods of low grade bauxite and other sources of aluminum while difficult to process have been given more and more attention[1-3]. The NEU research group has proposed a calcification–carbonization method to break the traditional combination of Na, Al and Si in red mud, reduce the content of aluminum and alkali in red mud and realize clean and efficient process of low grade bauxite and red mud[4]. We plan to process a kind of low grade and high titanium bauxite from Yunnan Wenshan area with calcification–carbonization method. In the composition of the bauxite, the content of TiO_2 is more than 6.5% and TiO_2 in form of anatase accounts for more than 70%.

Research shows that in digestion conditions anatase reacts with alkali ahead of diaspore to produce colloidal membrane sodium titanate surrounded on the mineral surfaces, thereby preventing the digestion of alumina. Addition of lime can eliminate the harmful effects, but will produce substantial titanium scale which is difficult to clean up, reduce the production capacity of the equipment, and increase energy consumption and costs[5-6]. At present, the formation mechanism of titanium scale in Bayer digestion process is not clear. Also further research on the reaction behavior of titanium minerals in processing low grade bauxite with calcification–carbonization method is needed[7].

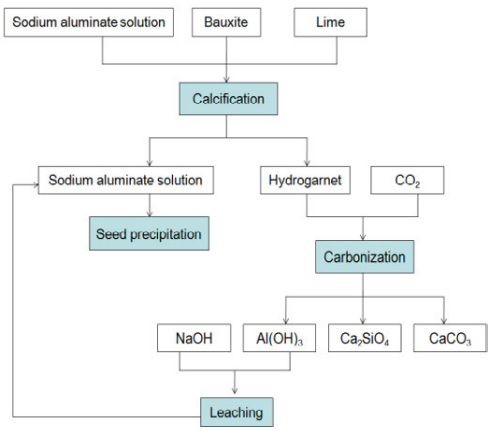

Fig 1. Process chart of calcification–carbonization method

In order to avoid the effect of impurity elements, the reaction behavior of anatase in the process of calcification-carbonization was studied in pure material systems. In calcification process, the reaction behaviors of anatase in the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O and

Na_2O-Al_2O_3-SiO_2-CaO-TiO_2-H_2O were separately investigated under the digestion conditions of diaspore and gibbsite. In carbonization process, the reaction behavior of titaniferrous calcified product was studied under different temperatures and pressures.

Experimental

Starting materials

The reagents (AR): TiO_2 (anatase) powder, $Al(OH)_3$ powder, NaOH flake, CaO powder and $Na_2SiO_3 \cdot 9H_2O$ crystal. The sodium aluminate solution was prepared with a certain proportion of $Al(OH)_3$ powder and NaOH flake in deionized water.

Experimental method

All the calcification and carbonization experiments were conducted in a KTFOZ-6.0 autoclave. After calcification, the obtained calcified products were cooled by flowing cold water and subsequently filtered with a vaccum filter. Then the filter cakes were washed for 3 times and put in a drying box at 95±5℃ for 12h. The prepared calcified precipitate subsequently was used for the carbonization. As the same as calcified products, the carbonized products also need cooling, filtering, washing and drying. The resultant calcified and carbonized precipitate can be used for XRD and chemical analysis. A Bruker D8 Advance diffractometer was used for the phase analysis of solid products and the XRD patterns were analyzed by the software of X' Pert High score Plus.

Results and discussion

Calcification conversion of anatase in the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O

Diaspore is the most difficult to dissolve among the minerals in diasporic bauxite. It needs to add lime and perform under the conditions of elevated temperature (240～260℃) and strong alkaline solution (180～250g/L Na_2O)[8-9]. Relatively speaking, gibbsite can dissolve more easily in the alkaline solution (100～140g/L Na_2O). Therefore, the lower temperature (140～160℃) can satisfy the reaction conditions. In this work, sodium aluminate solution with different alkaline concentrations (Na_2O 240g/L and 140 g/L) were used to clarify the conversion and formation of Ti-containing minerals. $n(Na_2O)$ / $n(Al_2O_3)$ =3.1, $n(CaO)$ / $n(TiO_2)$ =1. The calcification time and the stirring speed are 1h and 300r/min respectively. The digestion conditions of diaspore and gibbsite are separately 200～260℃ (Na_2O 240g/L) and 140～200℃ (Na_2O 140g/L).

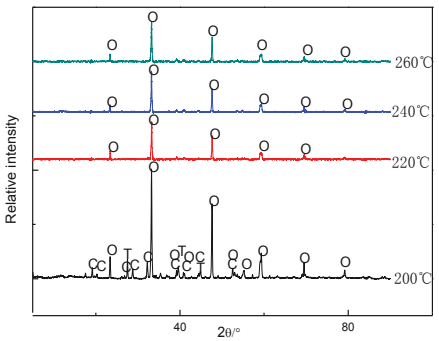

Figure 2. XRD patterns of solid products produced under the digestion condition of diasporic bauxite O-$CaTiO_3$;T-TiO_2;C-$(CaO)_3Al_2O_3(H_2O)_6$;

Fig. 2 shows the solid products produced under the digestion condition of diasporic bauxite in the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O. From 200 ℃ to 260℃，$CaTiO_3$ is the main stable phase. At 200℃, there are some TiO_2 and $(CaO)_3Al_2O_3(H_2O)_6$ unreacted because of the low temperature. The results of XRD indicate that TiO_2 and $(CaO)_3Al_2O_3(H_2O)_6$ in the sodium aluminate solution react completely to form stable $CaTiO_3$ when the temperature is above 220℃.

Table 1. Compositions of calcified precipitates at different temperatures

Temperature	TiO_2	CaO	Al_2O_3	Else	$n(TiO_2)/n(CaO)$
200℃	46.21%	33.24%	5.87%	14.56%	0.97
220℃	50.71%	35.08%	2.97%	11.24%	1.01
240℃	52.75%	36.96%	1.97%	8.32%	1.00
260℃	52.59%	37.50%	2.29%	7.62%	0.98

In the calcified precipitates, the mole number of TiO_2 and CaO is virtually identical which corresponds with the XRD results. In the calcification process, Al_2O_3 reacts with $Ca(OH)_2$ to form $(CaO)_3Al_2O_3(H_2O)_6$ in the sodium aluminate solution. When the temperature increases from 200℃ to 240℃, the formation of $CaTiO_3$ is promoted, while the formation of $(CaO)_3Al_2O_3(H_2O)_6$ is inhibited which leads to the decrease of Al_2O_3 concent in the precipitate gradually. When the temperature is above 240℃, the composition varies little and tends to a balance.

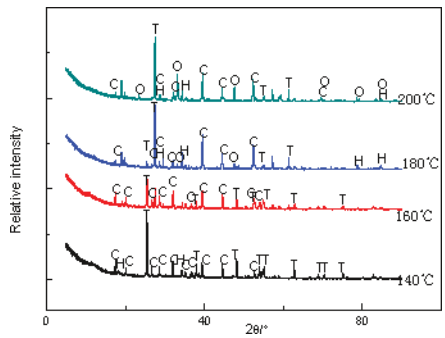

Figure 3. XRD patterns of solid products produced under the digestion condition of gibbsite
O-$CaTiO_3$; C-$(CaO)_3Al_2O_3(H_2O)_6$; H-$Ca(OH)_2$; T-TiO_2(01-073-1764 or 03-065-1119)

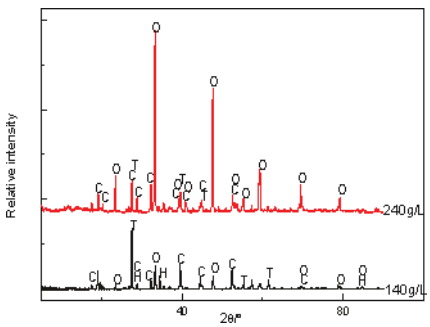

Figure 4. XRD patterns of solid products in different Na_2O concentrations at 200℃
O-$CaTiO_3$;T-TiO_2;C-$(CaO)_3Al_2O_3(H_2O)_6$; H-$Ca(OH)_2$;

Fig. 3 shows the solid products produced under the digestion condition of gibbsite in the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O. At 140℃, $(CaO)_3Al_2O_3(H_2O)_6$ have been formed although there are some TiO_2 and $Ca(OH)_2$ unreact. When the temperature increase to 200℃, $CaTiO_3$ is formed stably. Fig. 4 shows that the formation of $CaTiO_3$ can be promoted by higher concentration in the sodium aluminate solution. Above all, whether in the digestion condition of diaspore or gibbsite, TiO_2 gradually converts to $CaTiO_3$ with the increase of temperature in the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O.

Calcification conversion of anatase in the system of Na_2O-Al_2O_3-SiO_2-CaO-TiO_2-H_2O

Silicon is the most harmful impurity in the alkaline process of alumina production. It not only has an effect on the digestion process, but also become the main cause for alumina and alkali chemical loss. Moreover, silicon has a significant impact on product quality, device scaling and red mud processing[10]. Therefore, silicon should not be ignored in the designate system. In this work, a corresponding amount of Na_2SiO_3 $9H_2O$ was added in the solution instead of SiO_2 to realize that the weight of SiO_2 is 2.04 times that of TiO_2. Any other parameter is the same as the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O.

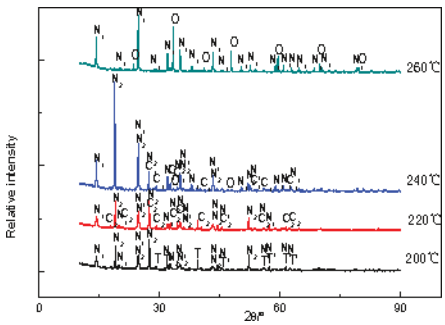

Figure 5. XRD patterns of solid products produced under the digestion condition of diasporic bauxite
N_1-$Na_4(Si_3Al_3O_{12})(OH)$ (00-011-0401) or $Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$ (01-088-1190); N_2-$3NaAlSiO_4{\cdot}NaOH$(00-015-0734); O- $CaTiO_3$; T-$Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$;$C_1$-$Ca_3Al_2(SiO_4)(OH)_8$; C_2-$Ca_3Al_2(SiO_4)_{1.25}(OH)_7$

The XRD results show that silicon complicates the reaction products because of the formation of varied

sodium aluminosilicate and calcium aluminosilicate. With the increase of calcification temperature, the reactant TiO_2 firstly convert to $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ at 200℃. Then $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ decompose in the solution to generate new equilibrium phase such as $Ca_3Al_2(SiO_4)_{1.25}(OH)_7$ and $Ca_3Al_2(SiO_4)(OH)_8$. When the temperature reaches 260℃, the final product $CaTiO_3$ is formed.

Table 2. Solid products at different calcification temperatures in Fig. 5

Temperature	Solid products
200℃	$Na_4(Si_3Al_3O_{12})(OH)$, $3NaAlSiO_4{\cdot}NaOH$, $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$
220℃	$Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$, $3NaAlSiO_4{\cdot}NaOH$, $Ca_3Al_2(SiO_4)_{1.25}(OH)_7$
240℃	$Na_4(Si_3Al_3O_{12})(OH)$, $Ca_3Al_2(SiO_4)(OH)_8$, $3NaAlSiO_4{\cdot}NaOH$, $CaTiO_3$
260℃	$Na_4(Si_3Al_3O_{12})(OH)$, $CaTiO_3$

From Fig. 6 can be seen, the formation of $CaTiO_3$ can be promoted when TiO_2-SiO_2 mass ratio increases to a certain degree. When the TiO_2-SiO_2 mass ratio is above 0.5, the diffraction peaks of $CaTiO_3$ almost have no changes. Just like the findings in the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O, $CaTiO_3$ is still the most stable Ti-containing mineral in the system of Na_2O-Al_2O_3-SiO_2-CaO-TiO_2-H_2O.

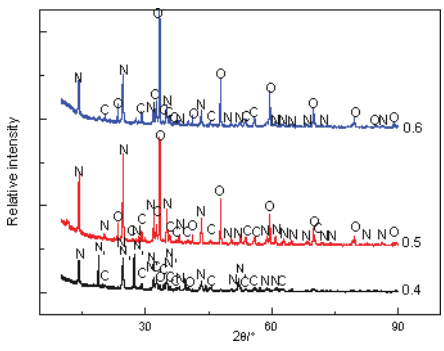

Figure 6. XRD patterns of solid products produced in different TiO_2-SiO_2 mass ratios at 240℃
N-$Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$;$N_1$-$3NaAlSiO_4{\cdot}NaOH$;
C-$Ca_3Al_2(SiO_4)(OH)_8$; O-$CaTiO_3$;

Because of the low temperature and concentration, there are still some TiO_2 unreacted in the digestion process of gibbsite from 140℃ to 160℃. $Ca_{2.93}Al_{1.97}Si_{0.64}O_{2.56}(OH)_{9.44}$ is stable from 140℃ to 180℃. When the temperature increase to 200℃, $Ca_{2.93}Al_{1.97}Si_{0.64}O_{2.56}(OH)_{9.44}$ and TiO_2 decompose and convert to $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ gradually.

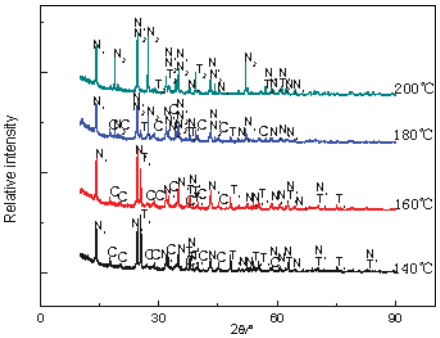

Figure 7. XRD patterns of solid products produced under the digestion condition of gibbsite
N_2-$3NaAlSiO_4{\cdot}NaOH$; N_1-$Na_4(Si_3Al_3O_{12})(OH)$ or $Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$;
T_1-TiO_2; T_2-$Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$;
C-$Ca_{2.93}Al_{1.97}Si_{0.64}O_{2.56}(OH)_{9.44}$

Table 3. Solid products at different calcification temperatures in Fig. 7

Temperature	Solid products
140℃	$Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$、$TiO_2$、$Ca_{2.93}Al_{1.97}Si_{0.64}O_{2.56}(OH)_{9.44}$
160℃	$Na_4(Si_3Al_3O_{12})(OH)$、TiO_2、$Ca_{2.93}Al_{1.97}Si_{0.64}O_{2.56}(OH)_{9.44}$
180℃	$Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$、$Ca_{2.93}Al_{1.97}Si_{0.64}O_{2.56}(OH)_{9.44}$、$3NaAlSiO_4{\cdot}NaOH$、$TiO_2$
200℃	$Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$、$Na_4(Si_3Al_3O_{12})(OH)$、$Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$

Ti exits in the form of $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ at 200℃ (Fig. 8). The sodium aluminate concentration in the solution has little effect on the Ti-containing phase. In terms of composition, the percentage concent of TiO_2 and CaO are 12.85% and 8.39% separately when Na_2O concentration of sodium aluminate solution is 140 g/L. The percentage concent of TiO_2 and CaO increase to 15.02% and 9.62% accordingly when Na_2O is 240 g/L. The percentage of TiO_2 and CaO becames raised with the increase of Na_2O concentration. It indicates that the formation of $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ can be promoted by raising the concentration of sodium aluminate solution.

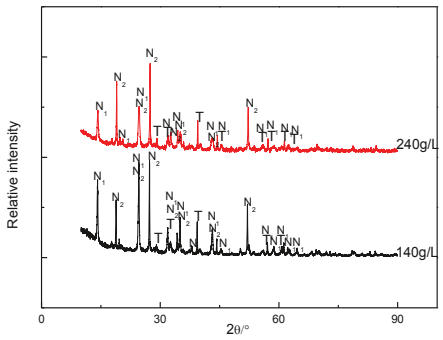

Figure 8. XRD patterns of solid products in different Na_2O concentrations at 200℃

N_2-$3NaAlSiO_4 \cdot NaOH$; N_1-$Na_4(Si_3Al_3O_{12})(OH)$ or $Na_8(Al_6Si_6O_{24})(OH)_2(H_2O)_2$;

T-$Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$;

To sum up, in the system of Na_2O-Al_2O_3-SiO_2-CaO-TiO_2-H_2O, TiO_2 converts to $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ at lower temperature and in lower concentration. When the two factors increase, $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ can convert to $CaTiO_3$ as the final product which is the most stable titanate in the process of calcification.

$CaTiO_3$ conversion in carbonization process

As the final product of Ti-containing minerals in calcification process, whether $CaTiO_3$ can be decomposed by CO_2 in carbonization process or not were confirmed experimentally. In general, the carbonization parameters depend on the the process itself. The carbonization temperature usually varies from 100℃ to 120℃ and the pressure of CO_2 ranges from 1.0MPa to 1.2MPa. The calcified precipitate produced at 240℃ (Table 1) was used for carbonization as starting materials. The carbonized precipitate prepared at different temperatures and pressures were compared with the calcified precipitate separately in Fig. 9.

The XRD results show that CO_2 can't make $CaTiO_3$ decompose when the carbonization temperature increases to 140℃ and the CO_2 pressure reached 1.8MPa. It indicates that $CaTiO_3$ is the most stable titanate phase in calcification-carbonization process.

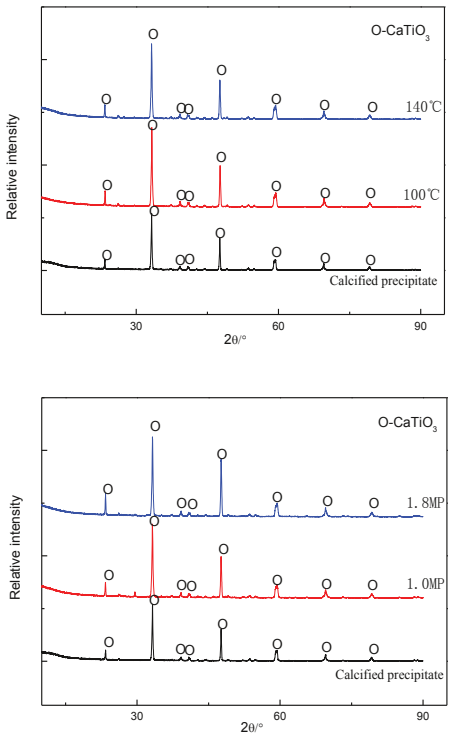

Figure 9. XRD patterns contrast between calcified precipitate and carbonized precipitate

Carbonization conditions: liquid-solid ratio 10:1, time 1.5h. CO_2 pressure 1.0MPa (above), temperature 120℃ (below)

As far as $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ is concerned, higher purity sample is needed to be prepared for carbonization research. As a result, some additional work is required to be done to explore the hydrothermal synthesis scheme of $Ca_3TiSi_2(Al_2Si_{.5}Ti_{.5})O_{14}$ specifically.

Conclusion

The reaction behavior and conversion of anatase in the calcification-carbonization process was investigated based on the experiments in the pure substance systems of Na_2O-Al_2O_3-CaO-TiO_2-H_2O and Na_2O-Al_2O_3-SiO_2-CaO-TiO_2-H_2O. The main conclusions are as follows:

1) In the system of Na_2O-Al_2O_3-CaO-TiO_2-H_2O, anatase converts to $CaTiO_3$ gradually with the increase of temperature and sodium aluminate

concentration in the calcification process of diaspore and gibbsite.

2) In the system of Na_2O-Al_2O_3-SiO_2-CaO-TiO_2-H_2O, with temperature increasing, anatase converts to firstly $Ca_3TiSi_2(Al_2Si_{0.5}Ti_{0.5})O_{14}$ and then $CaTiO_3$. The increase of sodium aluminate solution concentration and additive TiO_2-SiO_2 ratio can promote this conversion.

3) In carbonization process, CO_2 can't make $CaTiO_3$ decompose when the carbonization temperature increases to 140℃ and the CO_2 pressure reached 1.8MPa.. $CaTiO_3$ is the most stable titanate phase in calcification-carbonization process.

Acknowledgements

This research was supported by the National Natural Science Foundation of China (Nos. U1202274, 51204040), the Research Fund for the Doctoral Program of Higher Education of China (20120042 110011) and the Fundamental Research Funds for the Central Universities of China (N140203005, N140204015).

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ALKALINITY PRECIPITATION MEASUREMENT ON CARBONATION OF BAUXITE RESIDUE

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Keywords: Bauxite residue, carbonation, carbon dioxide, alkalinity precipitation.

Abstract

The alumina production Bayer process generates 0.7 to 2.0 ton of Bauxite residue (BR) and an average of 1.0 ton of CO_2 per ton of alumina produced. The direct use of exhaust gases to react and reduce the alkalinity of BR may allow a triple gain: improving the storage conditions, opening a range of new applications for BR and sequester from 16 to 102 kg of CO_2 per ton of alumina. This paper shows the first results of a lab scale program to measure the effects of adding different percentages of CaO and MgO before carbonation in order to precipitate the alkalinity on stable compounds. . The inlet and exhaust gases were analyzed. The precipitate after carbonation was evaluated to monitor the amount of stable compounds like dawsonite. The pH of the suspension was monitored during and after the reaction to evaluate the buffer effect.

Introduction

Aluminum is used extensively in transportation, electrical transmission, packaging, construction and many other industries. It is produced by electrolytic reduction of aluminum oxide known as alumina. The alumina is mainly produced from the refining of bauxite ore using the Bayer process. The extraction of aluminum oxide from the ore is obtained by dissolution in caustic soda at high temperature followed by separation of insoluble residues (BR). The last step is the seed precipitation and calcination of the alumina. The BR poses risks to the environment due to its large volume and high alkalinity. According to the latest technology BR is stacked in specially constructed sealed deposit areas. To date the immense majority of the BR that have been produced are stored. With exception to some efforts in Russia and China [1] very little is reused or incorporated into other production processes. However, a strategic goal was set by the International Aluminium Institute (formed by most industries) in 2010 to reuse 20% of BR by 2025 [2].

This paper summarizes the lab scale experiments and the first results of the reaction of CO_2 diluted on air with a water suspension of BR with addition of CaO. The alkalinity of the BR was substantially reduced after reaction with the CO_2. The stabilization pH showed a tendency downwards after 200 days of the carbonation.

These lab scale experiments were designed to enable the continuation of the pilot scale experiments with flue gas and BR presented at 2013[3]. As can be observed at Figure 1 on that experiments it was verified the possibility of carbonating the BR with flue gas containing around 8% CO_2. The pH after carbonation reached 8.5. Although the pH rebuff after 150 days to an average of 10.5. Since we are dealing with a low Ca BR, it was decided to run a lab scale experiments to measure the impact of adding different amounts of CaO and or MgO.

This work is part of a long term effort to design and test an efficient new process of reducing the alkalinity of BR, capture CO_2 and SO_2 minimizing power consumption and capital costs.

The pH of the BR slurry was measured during the reaction and for up to 400 days after reaction to access the extent of pH stabilization over time. Lower the alkalinity of BR reduces the environmental risk with its storage [4] and potentially facilitates the reuse of this material in different new applications.

At previous experiments, Figure 2, the stabilization pH having a standard deviation (0.3) much smaller than the value at the end of experiment (0.5) clearly indicated that the stabilization pH is being determined by slow release of ions from the solid portion of the BR. Unlike similar processes available at the literature, in this case the tricalcium aluminate could not be the main factor due the low Ca content shown in X-ray fluorescence. This was the main reason to test the addition of CaO on a lab experiment in order to control the stabilization pH.

Bayer Process for Alumina Production

The Bayer process to obtain alumina from bauxite can be divided in four steps:

Digestion

The minerals containing aluminum at the bauxite are selectively extracted from the non soluble components through the dissolution in sodium hydroxide.

$$Al(OH)_3 + Na+ + OH^- \rightarrow NaAlO_2 + 2H_2O \quad (1)$$

Clarification

After the extraction, the insoluble residue must be separated through a settling and filtering process. The aluminum containing liquor is transferred to the precipitators.

Precipitation

Crystalline aluminum trihydroxide “hydrate” is precipitated from the liquor.

$$NaAlO_2 + 4H_2O \rightarrow Al(OH)_3 + Na^+ + OH^- \quad (2)$$

The hydrate is fed into the calcination kiln.

Calcination

The hydrate is calcined to form alumina trough the separation of the water.

$$2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O \quad (3)$$

Bauxite Residue

Alumina production through Bayer process generates from 0.7 to 2.0 tons of residue per ton of alumina produced. It is usually called Bauxite Residue (BR) also called red mud. The BR composition include hematite and e other iron oxides, quartz and titanium oxides.

On the BR treatment only part of the sodium hydroxide is removed. The BR has a considerable amount of sodium hydroxide and several concentrated ores. On the other hand, the bulk of its alkalinity is in the solids and reacts slowly after the pH of the liquid is lowered. Contributes to this the addition of lime and other additives during the Bayer process with the objective of avoiding silica dissolution, helping the settling and improving the production. Not integrating BR in other processes is a waste of mineral resources, but it must be done in a energetic and economically viable way. Table 1 shows the chemical analysis of the BR from Alumina do Norte do Brasil S/A (ALUNORTE – Belém-PA-Brazil).

Table 1 - Dry BR Composition [6].

CONTENT	%	CONTENT	%
SiO_2	18.3	Na_2O	9.31
Al_2O_3	22.6	V_2O_5	0.28
Fe_2O_3	34.9	K_2O	0.13
TiO2	5.56	MgO	0.04
CaO	1.32	FL	7.56

BR Complex Alkalinity and the Reaction with Carbon Dioxide

The chemical reactions that control the alkalinity of BR and its neutralization behavior are complex. The time and the pH are the two most important variables. The pH is governed by the presence of alkaline solids. BR solutions pH average is 12.3, with standard deviation of 1.0 and ranges between 9.7 and 12.8. Therefore BR is hazardous due its high alkalinity [8], and high volume. The pH neutralization process is affected by the presence of hydroxides, carbonates, aluminates and polyacrylates [7-8]. These solids that are formed on the desilication process, on the Bayer process and also by addition of lime. The buffering action of this multiple alkaline solids causes the neutralization behavior of BRs to be complex with several steps and time dependent. Washing with water is not a practical way to remove the alkalinity from BR [8]. This chemistry and alkalinity impact on different aspects of BR. Storage requirements are affected by the reactivity with sealants. Bulk density, sedimentation rates, compaction, hydraulic conductivity, drying rates, dusting behavior, and physical strength are also alkalinity impacted [11]. The reuse as a raw material other process requires at least reduced reactivity to facilitate handling, interaction with other raw materials and transport.

The carbonation neutralizes the alkalinity according to the following equations:

$$CO_2\,(g) + OH^- \leftrightarrow HCO_3^- \quad (5)$$

$$HCO_3^-\,(aq) \leftrightarrow H^+ + CO_3^{2-}\,(aq) \quad (6)$$

As pH decreases the following reactions occur:

$$Al\,(OH)_4^-\,(aq) \leftrightarrow Al(OH)_3\,(s) + OH^-\,(aq) \quad (7)$$

$$NaAl(OH)_4\,(aq) + CO_2 \leftrightarrow NaAl(OH)_2CO_3\,(s) + H_2O\,(l) \quad (8)$$

Previous Experiments

These lab scale experiments were designed to enable the continuation of the pilot scale experiments with flue gas and BR presented at 2013[5]. As we can observe at Figures 1 and 2, that experiments verified the possibility of carbonating the BR with flue gas containing around 8% CO_2. Althow the pH rebuffed after 150 days to an average of 10.5. Since were dealing with low Ca BR, it was decided to run a lab scale experiments to measure the impact of adding different amounts of CaO and or MgO.

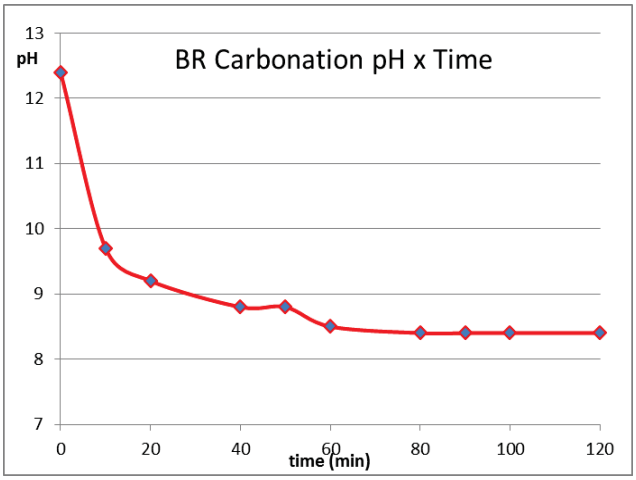

Figure 1 - pH evolution during carbonation at pilot plant

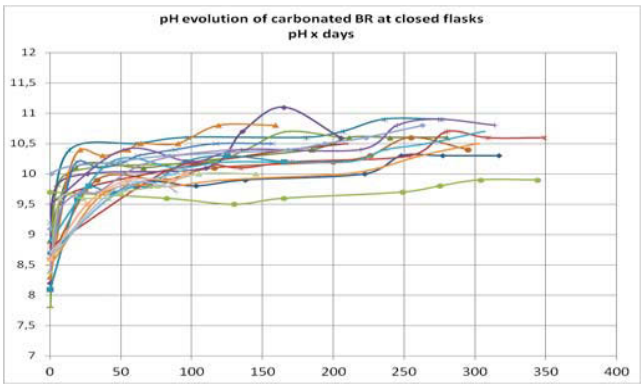

Figure 2 – pH evolution of several experimentsafter carbonation at pilot plant.

As can be seen at the comparison of the XRD patterns of carbonated and uncarbonated BR, shown in Figure 3 : the decrease of the peaks of tricalcium aluminate identified by peaks at 21; 33.2º; 41 and 47.6º. This fact indicates the dissolution of tricalcium aluminate during the carbonation process. We also note the increase of the peaks relating to dawsonite peaks ~ 18 and 34,4º and calcium carbonate (calcite) 29,3777º 2θ [3.04034 Å] showing the formation of these crystals. These changes are consistent with that described by Khaitan et al. (2009)[13].

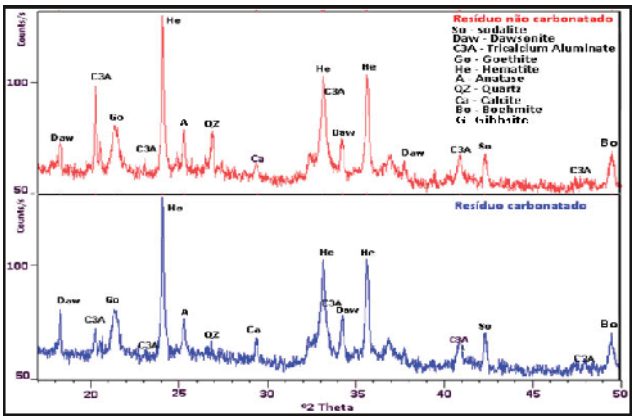

Figure 3 – XRD of BR before and after carbonation.

Lab Scale Experiment Equipment

The main objective of this experiment is to evaluate the possibility of influence the precipitation of the alkalinity of the carbonated BR on stable compounds and therefore obtain a lower long term stabilization pH.
The laboratory experiment consists of following steps:

1. Mixture of dried BR, with CaO. The amount of CaO vary from 2.5% to 50% ww.
2. Adition of water in order to obtain a suspension with 27% BR/CaO and 73% water ww.
3. Carbonation on a bubbling tower with a mixture of CO_2 and air 50% vv. The carbonation process duration is 120 minutes.
4. Monitoring the pH of the reacted material.

The preliminary results presentes here are an average of six experiments that were monitored for 400 days.
At Figure 4 , can be observed the flowchart for the laboratory process.

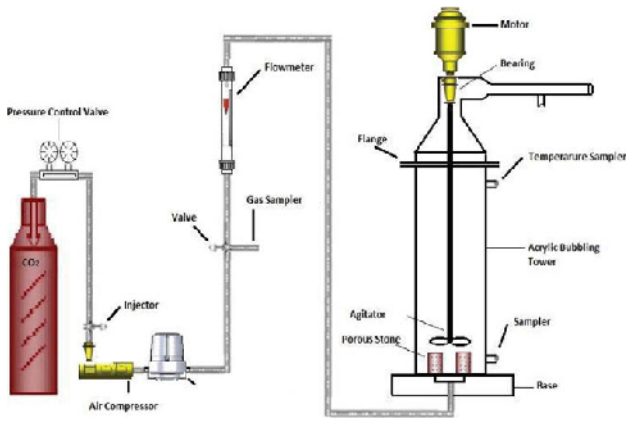

Figure 4 - Process flowchart for the lab experiment.

At Figures 5,6,7 8 and 9 details of the equipment used at the experiment can be observed.

Figure 5 – Porous stones and agitator

Figure 6 – Assemble of the bubbling tower and equipment.

Figure 7 – Carbonation process ongoing.

Figure 8 – Infrared gas analyzer measuring CO_2 at reactor exit.

Figure 9 – Complete assemble of carbonation.

Results to Date

As can observed at the Figure 10 the average pH after 120 minutes of carbonation is 7.5.

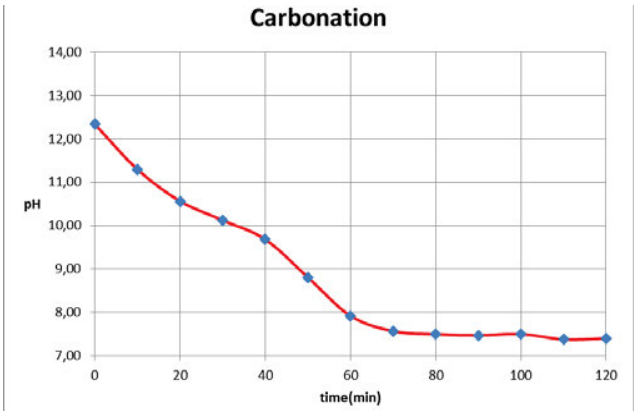

Figure 10 - Evolution of pH during carbonation process

As can be seen at Figure11, the behavior of the pH on the 400 days after after carbonation process is substantially different with CaO added (red) and pure BR (blue) after carbonation.

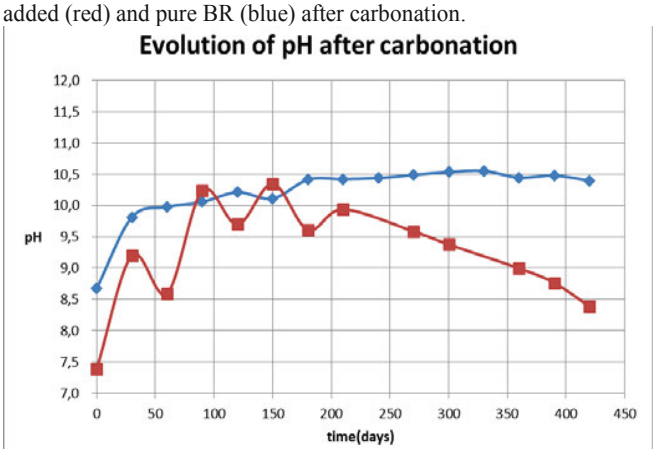

Figure 11 - BR pH evolution after carbonation with ■ and without ● CaO.

As can be observed at Figure 12 the evolution of BR with % of CaO from 5 to 50 have a very similar behavior.

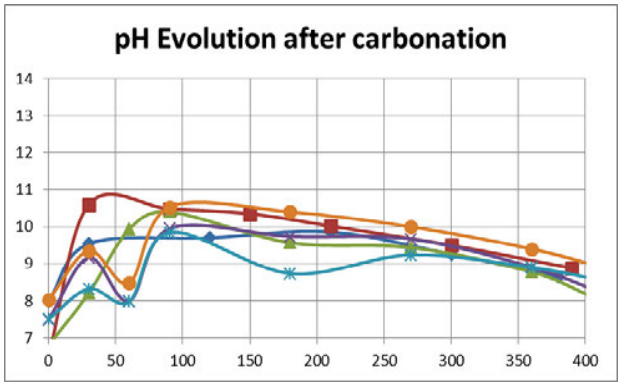

Figure 12 – pH evolution after carbonation of six experiments with amounts of CaO from 5 to 50% ww.

As can be seen at Figure 13, there is a substantial grow of the Dawsonite, compares with Figure 3.

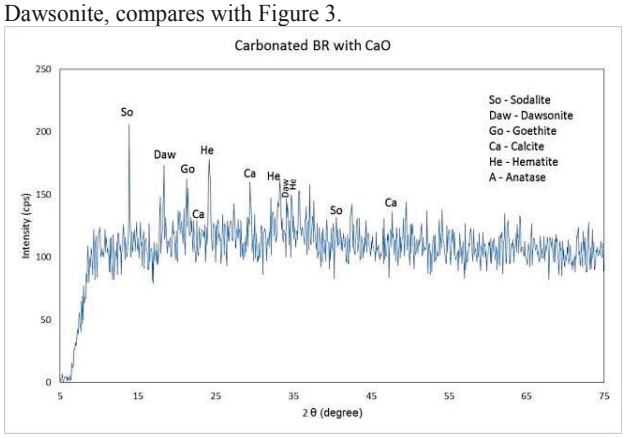

Figure 13 – XRD of 400 days carbonated BR with CaO

Conclusions

The first results of the addition of CaO before carbonation shows a strong influence on the behavior of the stabilization pH getting a very desirable downtrend.
The pH downtrend does not show relation with increase of the percentage of CaO.
There is a substantial grow on the amount of Dawsonite on the material carbonated with CaO.
The lower percentage of addition should be explored to understand the limits.

Next Steps

Continue the experiments with addition of CaO, especially with the lower percentages to understand the sensitivity.
Make experiments with the addition of MgO.
Make experiments with the addition of CaO and MgO.
Design the next pilot carbonation experiment in order to identify the process flow, the best equipment and measure the consequences on the physical properties of the BR.
Evaluate the capital and energy consumption for the next steps.
Evaluate mineral containing CaO and MgO easily available at the refinery sites.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Extraction of alumina from the magnetic separation tailings derived from reductive roasting of bauxite residue

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Keywords: iron, reductive roasting, magnetic separation, alumina, alkali leaching

Abstract

Much work has been done to recycle iron from bauxite residue via reductive roasting-magnetic separation process, of which more than 85% iron could be recycled as powdered metallic iron, but insufficient efforts have been made to utilize the magnetic separation tailings which were rich in valuable components like aluminum, titanium etc. This study was therefore focused on the alumina recovery from the above-mentioned tailings based on alkali leaching. The effects of the parameters like concentration of leaching agent (NaOH), temperature and time to the extraction of alumina have been investigated. Besides, the influence of the up-stream reductive roasting process and sodium additives on the extraction of alumina have also been studied. The results showed the reductive roasting process was favorable to the extraction of alumina while the sodium additives demonstrated an adverse effect. 82.90% alumina was leached under the optimal conditions from a bauxite residue containing 19.59 wt. % Al_2O_3

Introduction

Bauxite residue is a waste generated during the production of alumina from bauxite ores by Bayer process. Approximately 0.8-1.5 tons [1] of bauxite residue are discharged for producing 1 ton of alumina depending on the difference of the bauxite ores properties and the extraction conditions. It's estimated that more than 90Mt of bauxite residue are generated annually in the world [2], putting a great pressure on the surrounding environment like the underground water, soil and air due to its high alkalinity (pH=10-13) and high content of powdered dust, therefore plenty of capital and large land areas are needed to manage the bauxite residue. Therefore, the utilization of bauxite residue has been become a big challenge and necessary issue confronting the whole aluminum industry.

Many efforts have been made to treat and utilize bauxite residue in various areas, such as to produce building materials like cement [3], bricks [1], glass-ceramics [4] in construction industry or apply it to neutralize and absorb poisonous compounds in waste water, gas and soil in environmental protection taking advantage of its high alkalinity and large specific area [5, 6]. Bauxite residue can also be used as catalyst in hydrogenation, hydro-dechlorination, exhaust gas clean-up and other areas after modified [7]. All these applications have contributes a lot to the comprehensive utilization of bauxite residue, but in fact large-scale applications of bauxite residue have been limited on account of new problems occurring and low production efficiency.

Bauxite residue has also become a significant secondary resource and attracted a wide range of attentions because of various valuable components existed in bauxite residue such as Fe_2O_3, Al_2O_3, TiO_2, Na_2O etc., and other scattered metals, especially rare earth elements. The contents of Fe_2O_3 and Al_2O_3 in bauxite residue, in general, are 14-45% and 5-14%, respectively [2], which generally accounts for almost half of the total bauxite residue. Thus, the recovery of the iron and alumina from bauxite residue is not only beneficial to partly relief the tension of resources scarcity, but also greatly reduce the bauxite residue output. As a result, the cost to dispose bauxite residue as well as the damage to the environment has been profitably cut down, which is very favorable to the overall utilization of bauxite residue.

There are already many studies to recover metals from bauxite residue. Sintering process [8], hydrothermal process [9] and hydro-chemical process [10] are the basic methods for the recovery of alumina from bauxite residue. In the sintering process, bauxite residue is always mixed with soda and lime to transform the insoluble alumina-bearing compounds to soluble sodium aluminate, which can be recycled in the next step by alkali leaching with low concentration. Generally, the alkali leaching is also used in hydrothermal and hydro-chemical process to extract alumina, but they both require much higher concentration of alkaline solution and leaching temperature than Bayer process.

Reductive roasting is preferred to availably recovery of iron [11], while sodium additives are proved extremely advantageous to promote the reduction process [12] for contributing to the gathering and growing of the metallic iron particles. Through magnetic separation, over 85% iron could be recycled in iron concentrate, which can be used as the raw material of steel making. But the magnetic tailings rich in lots of valuable components like alumina, titanium, calcium, etc. are abandoned without further utilization, resulting in a heavy resource waste.

In this paper, to make a continuous use of the above-mentioned magnetic tailings, a method of alkali leaching was applied to extract the alumina from the magnetic tailings. The effects of the parameters like concentration of leaching agent (NaOH), leaching temperature and time to the extraction of alumina have been investigated. In addition, the influence of the up-stream reductive roasting process and sodium additives on the extraction of alumina have also been studied.

Experimental

Materials

Magnetic separation tailings derived from reductive roasting of bauxite residue The process of reductive roasting and magnetic separation of bauxite residue have been intensively explored in previous work. The iron could be effectively recycled under the optimum conditions as followed: adding additives with a certain proportion of 15wt. % Na_2SO_4 and 3wt. % $Na_2B_4O_7$, roasting time of 60min at temperature of 1100℃, magnetic separation of the reduced briquettes after grinding up to 97.61% passing 0.074mm at a magnetic field intensity of 0.1T. The magnetic concentrate contained 91.41% of iron and the metallization degree reached 91.66% as well as the overall recovery of iron was about 85%.

The main chemical compositions of the above-mentioned magnetic tailings (SW), and the magnetic tailings derived from

non-additives reductive roasting process (NW) were shown in Table Ⅰ. The original bauxite residue(RM) was taken from Guangxi province of China, and its content was also shown in Table Ⅰ. It was exhibited that alumina and other elements have concentrated in the magnetic tailings.

Table Ⅰ. Main chemical composition of magnetic tailings derived from reductive roasting of bauxite residue with the function of sodium additives (SW) or without the function of sodium additives (NW) and original bauxite residue(RM) (wt. %).

Material	Al_2O_3	SiO_2	Fe_2O_3	TiO_2	CaO	Na_2O
SW	23.5	17.8	6.3	7.6	18.7	11.2
NW	26.5	16.7	17.1	7.3	17.5	6.3
RM	19.6	12.7	31.1	5.2	12.2	5.6

XRD patterns of the original bauxite residue and the magnetic tailings (SW) were shown in Fig.1 and Fig.2, respectively. It was obvious that the primary iron-bearing and titanium-bearing minerals such as hematite (Fe_2O_3), and sphene ($CaTiSiO_4O$) in the original bauxite residue have disappeared, instead perovskite ($CaTiO_3$) and some magnetite (Fe_3O_4) and andradite ($Ca_3Fe_2(Si_{1.58}Ti_{1.42}O_{12})$) have emerged. The main alumina-bearing minerals in the magnetic tailings were sodium aluminum silicate and calcium aluminum silicate.

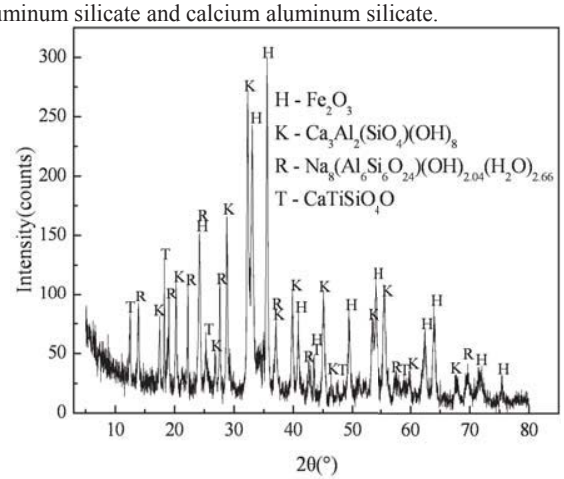

Fig. 1 X-ray diffraction pattern of bauxite residue (RM)

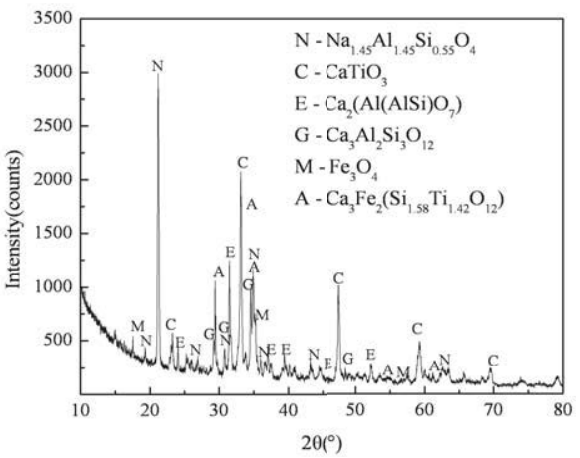

Fig. 2 XRD pattern of magnetic tailings derived from the reductive roasting process with the function of sodium additives (SW)

Experimental methods

The method to obtain the magnetic tailings The main steps to obtain the magnetic tailings were depicted in Fig.3. Dried bauxite residue or mixed thoroughly with a certain proportion of 15wt. % Na_2SO_4 and 3wt. % $Na_2B_2O_7$, was put into a cylinder with a diameter of 10 mm and height of 10mm to be briquetted. Add 40g above-mentioned briquettes into the cylindrical stainless steel pot with 50g soft coal underplayed on the bottom in advance. Then the pot was transferred into the vertical resistance furnace with a diameter of 80mm, which has already been preheated to the given temperature. The roasting process was conducted in the furnace at 1100℃ for 60 min, the roasting temperature, roasting time and the amount of sodium additives were determined by previous probe trials. At the end of the reduction roasting, the steel pot was taken out with coal covered on the top to avoid oxidization of the briquettes.

Before magnetic separation, the briquettes were ground to 97% under 0.074mm in an XMQ Φ 240 × 90 ball mill. The magnetic separation was conducted in the XCGS-73 Davis Tube with a magnetic field intensity of 0.1T. The magnetic concentrate, can be used in iron industry, and the magnetic tailings were subjected to alkali leaching subsequently.

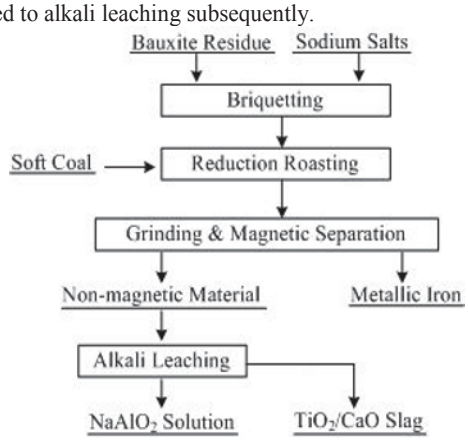

Fig. 3 Experimental procedure

Alkali leaching The magnetic tailings obtained were used to extract alumina by alkali leaching under high pressure in the autoclave with molten-salt as its internal heating medium and an automatic control system to manage the heating rate, stirring speed and temperature.

After dried and calculated, a certain mass of the above-mentioned magnetic tailings and well-prepared leaching agent (NaOH solution) were put into the stainless reaction tank with a volume of 100mL, subsequently the tank was sealed up tight. As the temperature of the autoclave reached the set value, the reaction tank was then fixed on the autoclave according to the requirements. After the lid of the autoclave was put on, the stirring of the reaction tank was started and the leaching process was also started. At the end of the leaching process, the reaction tank was taken out to be cooled in flowing water. The leachate was filtrated with hot deionized water, and the residue was dried in an oven at 110 ºC for further analysis.

Results and discussion

Extraction of alumina by alkali leaching

Effects of NaOH concentration and leaching temperature. The effects of NaOH concentration and leaching temperature on the extraction ratio of alumina and A/S of leached residue were investigated in this part. It was shown in Fig. 4 and Fig. 5 that at relatively low NaOH concentration (35%, 40%), the extraction ratio of alumina increased greatly and the corresponding A/S of leached residue decreased substantially as the temperature changed from 170℃ to 260℃, while at high NaOH concentration (45%, 50%), the extraction ratio of alumina increased obviously until 230℃ but less afterwards, and the A/S of leached residue reduced fast before 230℃ but slowly from 230 ℃ to 260℃. The maximum extraction ratio reached 82.90% with 50% NaOH at 260℃. So it could be deduced that high NaOH concentration and leaching temperature were relatively preferred to extract alumina from the magnetic tailings.

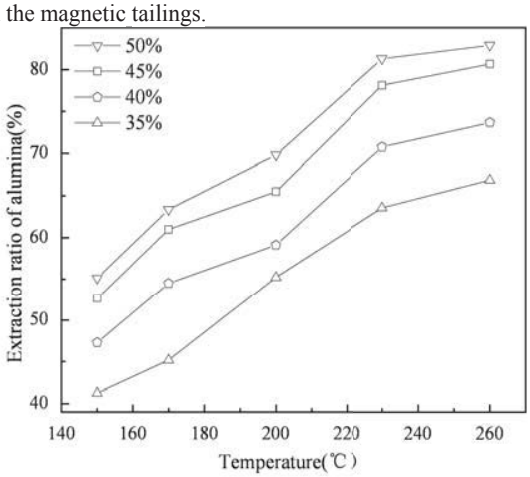

Fig. 4 The effect of NaOH concentration and temperature on the extraction ratio of alumina (liquid-solid ratio of 10, leaching time of 120min)

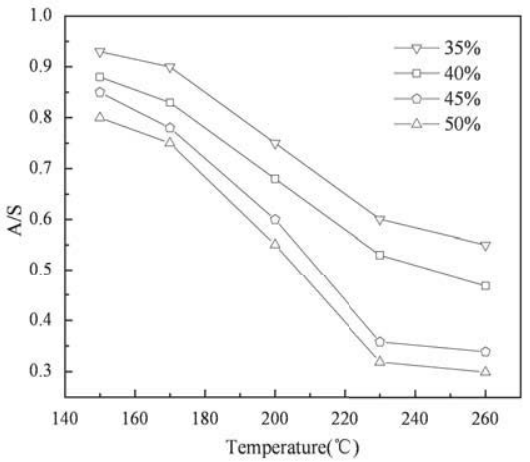

Fig. 5 The effect of NaOH concentration and temperature on the A/S of leached residue (liquid-solid ratio of 10, leaching time of 120min)

The XRD patterns of the material before and after leaching were shown in Fig 6. It was obviously that the main Al-bearing minerals $Na_{1.45}Al_{1.45}Si_{0.55}O_4$ and $Ca_3(Al(AlSi)O_7)$ in the magnetic tailings have disappeared after alkali leaching, instead $NaAlSi_2O_8$ has emerged and the peak of $Ca_3Al_2(SiO_4)_3$ has been strengthened, which indicated that in the duration of alkali leaching, the main alumina-bearing minerals $Na_{1.45}Al_{1.45}Si_{0.55}O_4$ and $Ca_3(Al(AlSi)O_7)$ have transformed into $NaAlSi_2O_8$ and $Ca_3Al_2(SiO_4)_3$, resulting in the reduction of A/S and achieving the extraction of alumina.

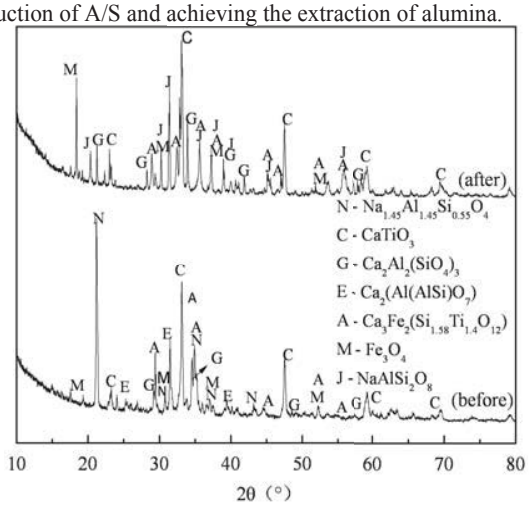

Fig. 6 XRD patterns of the materials before and after leaching (liquid-solid ratio of 10, 50% NaOH solution, 260℃120min,)

Effect of leaching time The effect of leaching time on extraction of alumina and A/S of leached residue was primarily studied. The leaching time was taken a range from 90 min to 180 min. As shown in Fig.7, the extraction ratio of alumina got a sharp increase when time changed from 90 min to 120 min, but gradually decreased as time lasted, while the corresponding A/S of leached residue declined quickly and reached the minimum of 0.3 when time was 120 min, then went up when time continued increasing. This phenomenon could be explained by their XRD patterns in Fig. 8

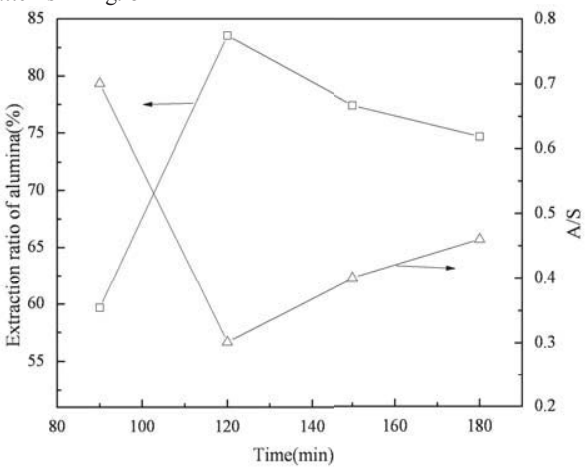

Fig. 7 Effect of leaching time on extraction ratio of alumina and A/S of leached residue (liquid-solid ratio of 10, 50% NaOH solution, 260℃)

The XRD patterns in different time were shown in Fig.8. In 90min, the main Al-bearing mineral was $(Na_2O)_{1.31}Al_2O_3(SiO_2)_{2.01}(H_2O)_{1.65}$, as reaction proceeded, it has transformed into $NaAl(SiO_3)_2$, resulting in the reduction of the A/S of leached residue from 90min to 120 min, but when time lasted, the peak of $NaAl(SiO_3)_2$ got greatly weaken and the grossular ($Ca_3Al_2(SiO_4)_3$) peak got obviously strengthened, and the A/S of leached residue thus increased. It indicated that when time was

too long, the soluble $NaAlO_2$ in alkaline solution could react with SiO_2 and CaO in the residue to form insoluble $Ca_3Al_2(SiO_4)_3$ in turn, resulting in the increase of A/S of the leached residue and the reduction of extraction ratio of alumina.

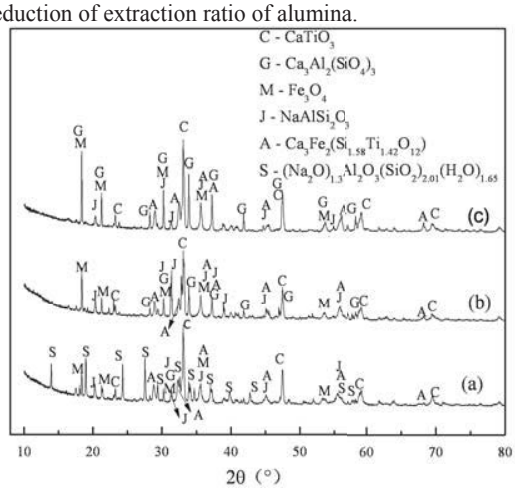

Fig .8 XRD patterns of leached residue after leaching at different leaching time (liquid-solid ratio of 10, 260℃, 50%NaOH solution): (a) 90min, (b) 120min, (c) 180min

The effects of the roasting process and the sodium additives on the extraction of alumina

In order to examine the effects of the up-stream roasting process and sodium additives on the extraction of alumina, the same treatment to the original bauxite residue (RM) and the magnetic tailings derived from the reductive roasting process with (SW) or without sodium additives (NW) were conducted in this segment.

It was shown in Fig. 9 and Fig. 11 that the extraction ratio of alumina from the above three materials at each leaching temperature and NaOH concentration was in the following order: NW>SW>RM. At the same time, as exhibited in Fig. 10 and Fig. 12 that the A/S of leached residue at each temperature and NaOH concentration was in the sequence: RW>SW>NW. So it can be concluded that the reductive roasting process was favorable to the extraction of alumina, but the sodium additives adding in the roasting process had a negative effect on the subsequent extraction of alumina. This phenomenon can be explained by their XRD patterns.

The main chemical compositions of the leached residue from the three materials were shown in Table Ⅱ, the content of alumina in the leached residue reduced greatly, while other elements have been concentrated, especially titanium dioxide and calcium dioxide.

Table Ⅱ. Main chemical compositions of the leached residue leached under the optimum conditions (wt. %)

Leached residue	Al_2O_3	SiO_2	Fe_2O_3	TiO_2	CaO	Na_2O
From RM	4.6	13.2	45.8	7.6	17.9	9.0
From NW	3.0	17.8	29.8	9.2	23.3	8.6
From SW	5.8	19.2	9.3	10.9	26.9	9.2

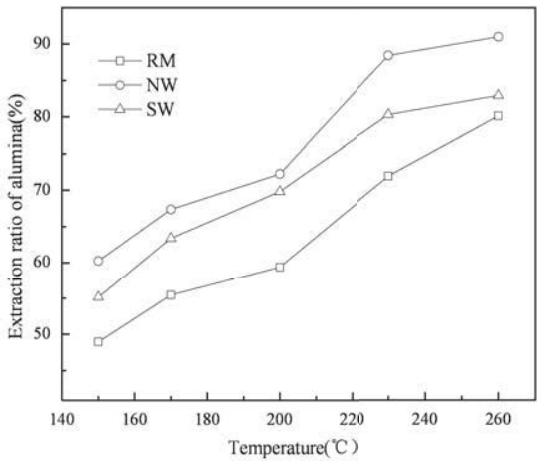

Fig. 9 The extraction ratio of alumina from the three different materials in the function of temperature (liquid-solid ratio of 10, 50% NaOH solution, 120min)

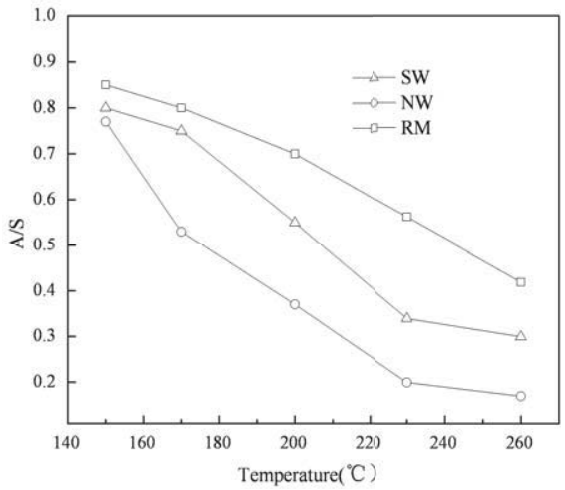

Fig. 10 The A/S of leached residue from the three different materials in the function of temperature (liquid-solid ratio of 10, 50% NaOH solution, 120min)

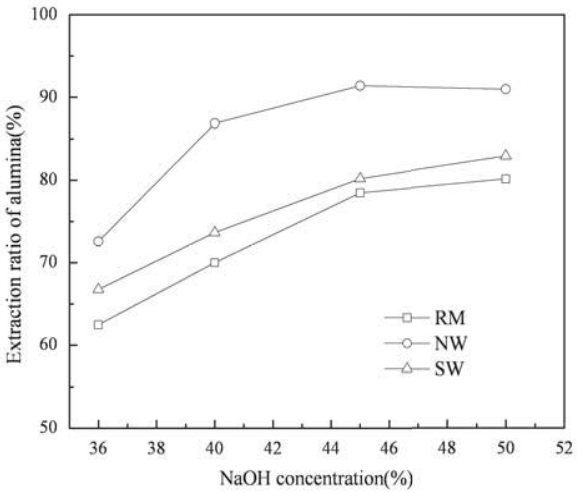

Fig. 11 The extraction ratio of alumina from the three different materials in the function of NaOH concentration (liquid-solid ratio of 10, 260℃, 120min)

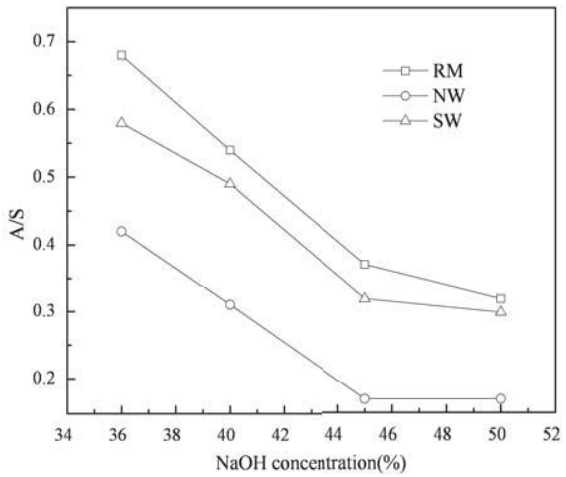

Fig. 12 The A/S of leached residue from the three different materials in the function of NaOH concentration (liquid-solid ratio of 10, 260℃, 120min)

. The XRD patterns of NW were shown in Fig. 13. Compared with the XRD patterns of bauxite residue (RM) in Fig.1 and SW in Fig. 2, the main difference among the three materials was the iron-bearing mineral, which in RM was hematite (Fe_2O_3), in NW was mainly fayalite (Fe_2SiO_4), and in SW was andradite ($CaFe_2(Si_{1.58}Ti_{1.42}O_{12})$), while the content of fayalite (Fe_2SiO_4) in NW was much higher than andradite ($CaFe_2(Si_{1.58}Ti_{1.42}O_{12})$) in SW according to their chemical analysis in Table Ⅰ. The minerals that fixed with silica would have a favorable effect on the extraction of alumina during the alkali leaching process, because that would prevent the reaction between alumina and silica to form insoluble sodium alumina silicate, and thus prevents the reduction of extraction ratio of alumina. So the presence of fayalite (Fe_2SiO_4) in NW and andradite ($CaFe_2(Si_{1.58}Ti_{1.42}O_{12})$) in SW were pretty beneficial to the extraction of alumina from NW or SW, but since the content of fayalite (Fe_2SiO_4) in NW was much higher than that of andradite ($CaFe_2(Si_{1.58}Ti_{1.42}O_{12})$) in SW, so the extraction ratio of alumina from NW was better than that from SW, and both of which were better than that from RM in the meantime.

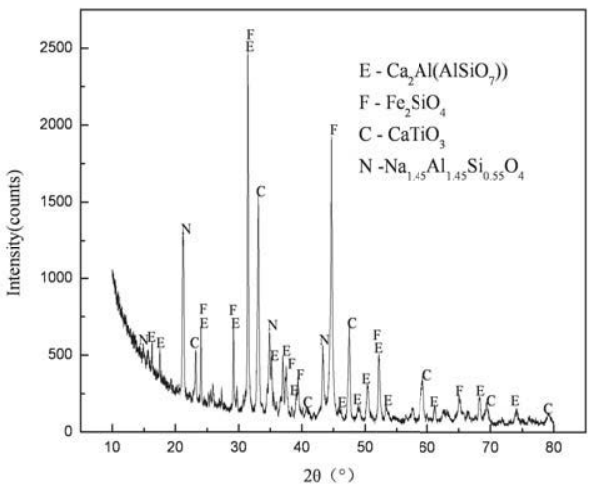

Fig. 13 XRD pattern of the magnetic tailings derived from reductive roasting of bauxite residue without the function of sodium additives (NW)

Conclusions

The extraction of alumina from the magnetic tailings derived from the reductive roasting process with the function of sodium additives has been successfully conducted, furthermore, the influence of the roasting process and the sodium additives on the extraction of alumina was also investigated, the conclusions could be obtained as followed:

(1) The reductive roasting–magnetic separation process of recycling iron from bauxite residue have been applied, and the magnetic tailings concentrated in valuable components such as Al_2O_3, TiO_2, CaO etc. are necessary to manage. Under the optimum conditions of leaching at 260 ℃ for 120min with 50% NaOH and a liquid to solid ratio of 10, 82.90% alumina from the magnetic tailings could be extracted.

(2) Through the above-mentioned process, the output of bauxite residue can be reduced significantly, thus it's very favorable to relief the pressure of disposing bauxite residue. At the same time, the product of iron and $NaAlO_2$ were obtained, which can be used in steel-making industry and aluminum industry, making compensations to the cost of the above-mentioned process.

(3) After alkali leaching, TiO_2, CaO, SiO_2 are the dominant components in the leached residue, which should be recycled in the subsequent work to further cut down the output of bauxite residue and create more extra value.

(4) At each temperature and NaOH concentration, the extraction ratio of alumina from the above three materials are in this order: NW>SW>RM, which indicates that the upstream reductive roasting process has a beneficial influence on the subsequent extraction of alumina, while the sodium additives prove to be adverse, but overall the reductive roasting process with the function of sodium additives still has a positive effect on the subsequent extraction of alumina.

Acknowledgements

The authors would like to gratefully acknowledge the financial supports from National Science foundation of China (No 51174230, 51234008) and the Program for New Century Excellent Talents in University (NCET-11-0515). This work was also financially supported by Co-Innovation Center for Clean and Efficient Utilization of Strategic Metal Mineral Resources.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Research on activated alumina obtained by spray pyrolysis method

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Keywords: Aluminium chloride,Activated alumina, Spray pyrolysis,Crystallization process

Abstract

Activated Alumina was prepared by directly spray pyrolysis $AlCl_3$ solution. Batch experiments was performed to choose better reaction device among muffle furnace, propane graphite furnace and tubular resistance furnace. The alumina was characterized by XRD and SEM, and the specific surface area of the prepared alumina was also examined. The experiment results showed that using the tubular resistance furnace by spray pyrolysis $AlCl_3$ solution could effectively improve the efficiency of the pyrolysis, obviously change the morphology of product and effectively increase the specific surface area of product granule. Compared with the muffle furnace roasting, the product index of spray pyrolysis by using tubular resistance furnace was better, and the products conformed to the requirements of the index of activated alumina. The present paper provided a new way of preparing activated alumina.

Introduction

Alumina production includeS three ways as alkaline process, acid process and acid-alkali-combination process[1~3]. The bayer process and sintering process do not suit for processing high iron bauxite, fly ash and other non-traditional aluminum resources, at the same time they may produce a large amount of solid wastes;While traditional acid process has long process flow and high energy consumption.

Spray pyrolysis technology is a method that using the metal salt solution as precursor after atomized to prepare metal oxide powder at high temperature. In recent years, researchers apply spray pyrolysis technology in preparation of thin film materials [4~8], ultrafine metal and oxide powder preparation [9~17]. Aimed at the characteristics of bauxite resources in China and the existing problems in the alumina production, it was a breakthrough for the paper that adopted $AlCl_3$ solution by spray pyrolysis for alumina. This method simplifies the process in traditional acid process for alumina, reduces energy consumption, HCl can process bauxite by cycle in double-cycle method. This paper studies on different furnace type in different ways of pyrolysis in different temperature on the effect of pyrolysis efficiency, surface structure, the phase composition, systematically studies the changes of mechanism of pyrolysis reaction.

Experimental

Materials

Reagents for the experiments are $AlCl_3 \cdot 6H_2O$ (Analytically pure), distilled water (Homemade).

Experimental process

Activated alumina were prepared by muffle furnace(SX2-5-12), graphite furnace and tube furnace(Homemade) at different conditions, such as different solution concentrations, pyrolysis temperatures, carrier gas pressures. The physical characterization of activated alumina were studied by using XRD (Model PW3040/60) and surface structure of the products was measured by using SEM (Model Hitachi SU8010). The open-framework structure of the products with different conditions were carried out by using ASAP (Model ASAP2020).

Results and discussion

Performance of AlCl3 saturated solution by pyrolysis in muffle furnace

Fig.1 shows the chlorine content of pyrolysis products. From Fig.1(a), with the pyrolysis time extends, the chlorine content of pyrolysis products reduces. The decline speed of chlorine content slows down. This is because that there were amount of crystal water and chlorine in raw material at the beginning of reaction which was easy to response to positive, but as the crystal water and chloride reduced, reaction speed slowed down obviously. But with the reaction temperature rose, the chemical reaction rate and water diffusion speeded up, the chlorine content in pyrolysis products decreased in a short period of time obviously. But after 800℃, the chlorine content changed little with pyrolysis temperature rised. Compared with the chlorine content of pyrolysis product in 700℃, The chlorine content in the products had no obvious decline in 800℃.

Performance of AlCl3 solution by spray pyrolysis in graphite furnace

From Fig.1(b), as the pyrolysis temperature increased, the chlorine content of the pyrolysis products reduced. The chlorine content in the product is 3.41% in 600℃ which is much better than that in 800℃in muffle furnace. Meanwhile reaction time by spray pyrolysis was much shorter than that in muffle furnace. That two points both explain that the performance by spray pyrolysis in graphite furnace is much better than that in muffle furnace.

Performance of $AlCl_3$ solution by spray pyrolysis in tube furnace

From Fig.1(c), the chlorine contents were gradually reduced with temperature rised. But the pyrolysis efficiency of the process isulated in 700℃ for 30 minutes was higher than tthat occured directly under 900℃, but it was lower than the efficiency under 1000℃. So the thermal insulation for a certain time was beneficial to the pyrolysis process under low temperature, but with pyrolysis temperature rised, the impact of temperature to the efficiency was more obvious than the thermal insulation. The performance by spray pyrolysisin tube furnace is much better than that both in muffle furnace and graphite furnace.

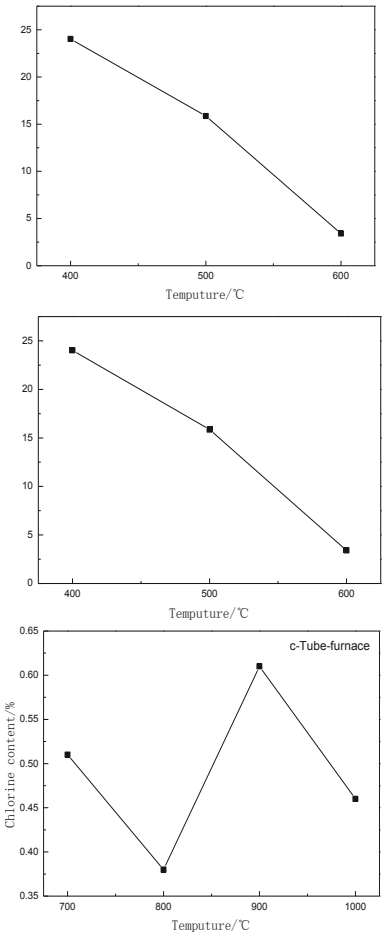
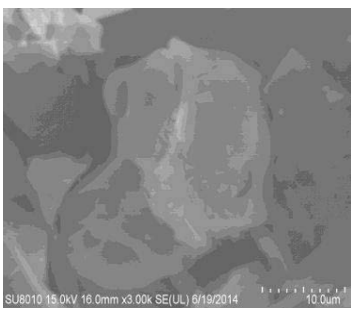


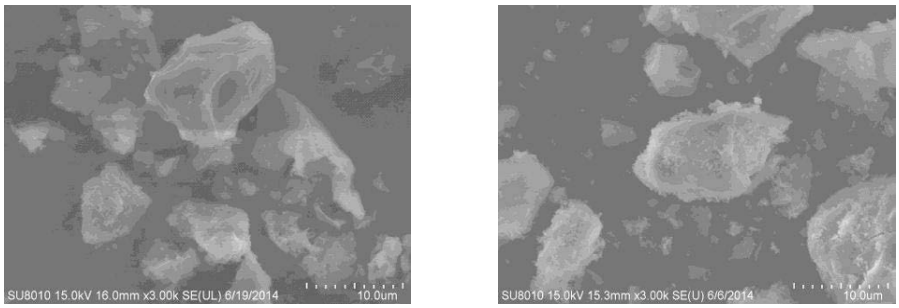
Fig.1 Chlorine content of pyrolysis products

Analysis of apparent morphology of the pyrolysis products

Fig. 2 shows the SEM images of the pyrolysis products in muffle furnace, graphite furnace and tube furnace. From the figure the products coexist as granular and layered in 800℃ by muffle furnace, however, the products exist as porous and blocks in graphite furnace, that exists as granulate in tube-resistance furnace. In muffle furnace by pyrolysis process, the products sintered for 1h, and with increasing of temperature, product surface began to be irregular sunken. This is because of that products may be generated solid particles in the whole droplets or precipitation may be generated a hollow shell particles if the precipitate could not reach the droplets inside. As temperature increased, pores in product collapsed, forming irregular concave surface morphology. And for the hollow shell particles in sintering process, if the pressure inside the shell was too high, shell would be broken, and formed debris, and eventually formed lamellar shape particles structure.



a- Muffle furnace

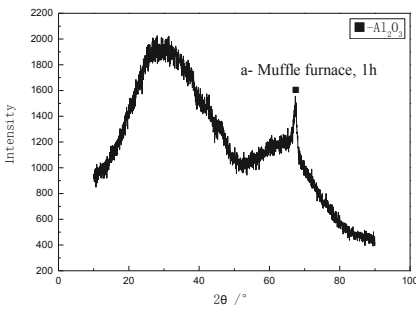


b-Graphite furnace c-Tube-furnace

Fig. 2 SEM analysis of the pyrolysis products in different furnaces

Analysis of pyrolysis mechanism

From Fig. 3, pyrolysis products only appear one diffraction peak in (440)crystal plane in muffle furnace, moreover, products in graphite furnace do not exist diffraction peak, but in tube furnace the products have multiple diffraction peaks, generate γ-Al2O3 and α-Al2O3. Compared with Fig.3(a) and Fig.3(b), in muffle furnace by static pyrolysis, the products needed a long time to ensure complete alumina transformation process, before the process alumina products mostly existed as the amorphous state. In graphite furnace in 600℃, products mostly existed in this station. Compared with static pyrolysis, spray pyrolysis is a better way to response. But graphite furnace adopts flame spraying, intense gas flow, instable gas flow and uneven temperature field in the furnace, at the same time droplets state interfere with gas. So when spray pyrolysis response in tube furnace in 900℃, pyrolysis products appeared two kinds of alumina morphology in Fig. 3(d). Compared with the results before, the pyrolysis products were in crystal phase transition. From the above three kinds of furnace, the crystallization of pyrolysis products in tube furnace more completed, pyrolysis effect was better.



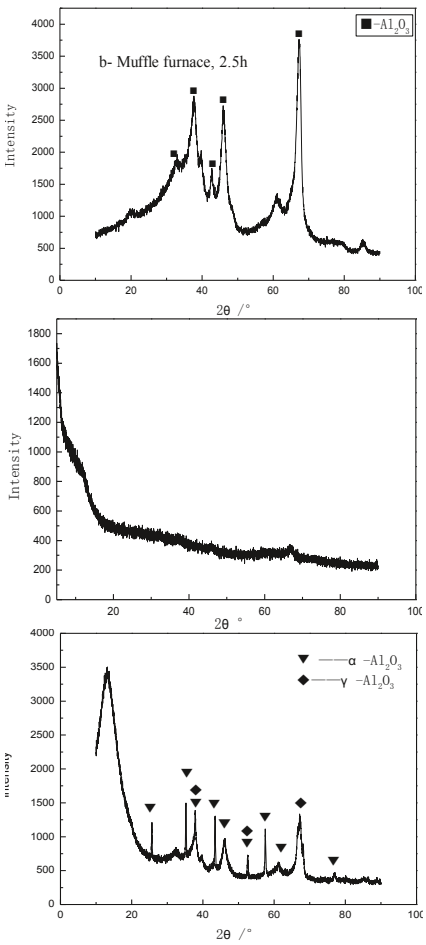


Fig.3 XRD analysis of pyrolysis products

Table 1 shows that the specific surface area of pyrolysis products in graphite furnace and tube furnace was measured by nitrogen adsorption method. From Table 1, the pyrolysis products have more the BET and pore volume and pore width. This is mainly because that in tube furnace pyrolysis temperature could reach 1000℃, compared with 600℃ in graphite furnace, the increase of pyrolysis temperature promoted the microporous generated for pyrolysis products, and part of the micro hole collapsed, fused. A large number of new microporous generation made the parameters of the BET and pore volume and pore width increased. The products conformed to the requirements of the index of activated alumina and

provided a new way of preparing activated alumina.

Table 1 The BET and Pore volume and Pore width of spray products under different furnaces

Furnace	BET /m^2/g	average pore size/nm	pore volume/cm^3/g
Graphite furnace	28.1	0.0330	4.70
Tube furnace	132.3	0.2655	9.54

Conclusions

1) By means of different furnace type and different reaction conditions the pyrolysis experiments obtained the optimum conditions for each reaction: 700℃, 1h with 4.81% of chlorine content in muffle furnace; 600℃, 10%, 0.3MPa with 3.41% of chlorine content in graphite furnace; 1000℃, 10%, 0.3MPa with 0.42% of chlorine content in tube furnace; In tube furnace by spray pyrolysis, in 800℃, 20%, 0.2MPa, chlorine residue can be below 0.5%, so adopting tube furnace by spray pyrolysis can effectively improve the efficiency of the pyrolysis reaction.

2) Products coexist as granular and layered in muffle furnace, meanwhile, the products exist as porous and blocks in graphite furnace, that exists as granulate in tube furnace. As temperature increased, pores in product collapsed, forming irregular concave surface morphology. And for the hollow shell particles in sintering process, if the pressure inside the shell was too high, shell would be broken, and formed debris, and eventually formed lamellar shape particles structure.

3) Pyrolysis products in muffle furnace only appear one diffraction peak in the (440) crystal plane, moreover, products in graphite furnace do not exist diffraction peak, but in tube furnace the products have multiple diffraction peaks, generate γ-Al2O3 and α-Al2O3. In muffle furnace by static pyrolysis, the products needed a long time to ensure complete alumina transformation process, before the process alumina products mostly existed as the amorphous state. From the above three kinds of furnace, the crystallization of pyrolysis products in tube furnace more completed, effect of pyrolysis was better. The products in tube furnace conformed to the requirements of the index of activated alumina and provided a new way of preparing activated alumina.

Acknowledgement

This work was financially supported by the Joint Funds of the National Natural Science Foundation of China(No.U1202274), the National Natural Science Foundation of China (No.51004033), the National Science and Technology Support Program (No.N100302005), doctor start up foundation in Taiyuan University of Science and Technology(20142001), science and technology research projects of Liaoning education department(L2014096), specialized Research Fund for the Doctoral Program of Higher Education(20120042110011).

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ALUMINUM ALLOYS, PROCESSING AND CHARACTERIZATION

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ALUMINUM ALLOYS, PROCESSING AND CHARACTERIZATION

Alloy Development and Applications

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

CHARACTERIZATION OF NEAR NET-SHAPE CASTABLE RARE EARTH MODIFIED ALUMINUM ALLOYS FOR HIGH TEMPERATURE APPLICATION

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Keywords: Aluminum, Rare Earth, Alloy, Cerium, High Temperature Alloy

Abstract

The modern family of aluminum alloys has a broad range of applications. However, there are currently few aluminum alloys capable of operating at elevated temperatures while maintaining mechanical properties necessary for high performance automotive and aerospace applications. The high temperature phase equilibrium of Al-REE-X (REE-Rare earth elements and X-traditional alloying elements) is not well understood yet this class of materials shows promise as a lightweight material for stiffness-driven high temperature applications. To date, we have determined RE alloys of Al to be castable across a broad range of temperatures and compositions. In addition, room temperature physical properties of Al-RE alloys appear to be in line with existing aluminum alloys with superior microstructural stability and associated properties at temperatures exceeding 150 °C. The CALPHAD work, microstructure, and compositional data are presented for representative example of Al-RE alloys at room and elevated temperatures.

Introduction

Aluminum alloys are valued for their high specific strength. However, at high temperatures, mechanical properties for aluminum alloys rapidly degrade. There do exist aluminum alloys which can function well enough at high temperature, but these alloys are too cost prohibitive for wide-spread application due to their use of elements such as scandium and zirconium [1]. An economic high-temperature aluminum alloy can be realized through the use of cerium, as an alloying addition. Cerium has been shown in powder metallurgy to greatly increase the high temperature mechanical properties of aluminum alloys beyond that of existing high-temperature aluminum alloys [2]. Investigations into the castability and mechanical properties of an Al-Ce alloy have been conducted for the purpose and resulted in the realization of an economical high-temperature aluminum alloy for wide spread application at temperatures up to 250 °C and possible higher.

CALPHAD WORK

Prior to casting any alloys CALPHAD methods were employed to determine the compositions of interest to the project. All possible binary diagrams of the Al-Fe-Ce-Si family were calculated using a proprietary database; additionally, Al-Cu and Al-Mg were calculated used the light metal alloys v3 (TCAL3) Thermo-Calc database. It was determined that initial castings in the Al-Ce binary system should be made surrounding the eutectic. By casting around and on the eutectic effects of both primary aluminum and $Al_{11}Ce_3$ crystallization on alloy mechanical properties could be determined. Looking at the phase diagram in Figure 1, the eutectic point falls at ~10 wt% Ce. As a result one casting was selected on eutectic and two were selected above and below the eutectic point. Ternary and quaternary additions were determined via similar methods. Figure 2 shows the Al-Ce-Si calculated for comparisons between calphad and experimental

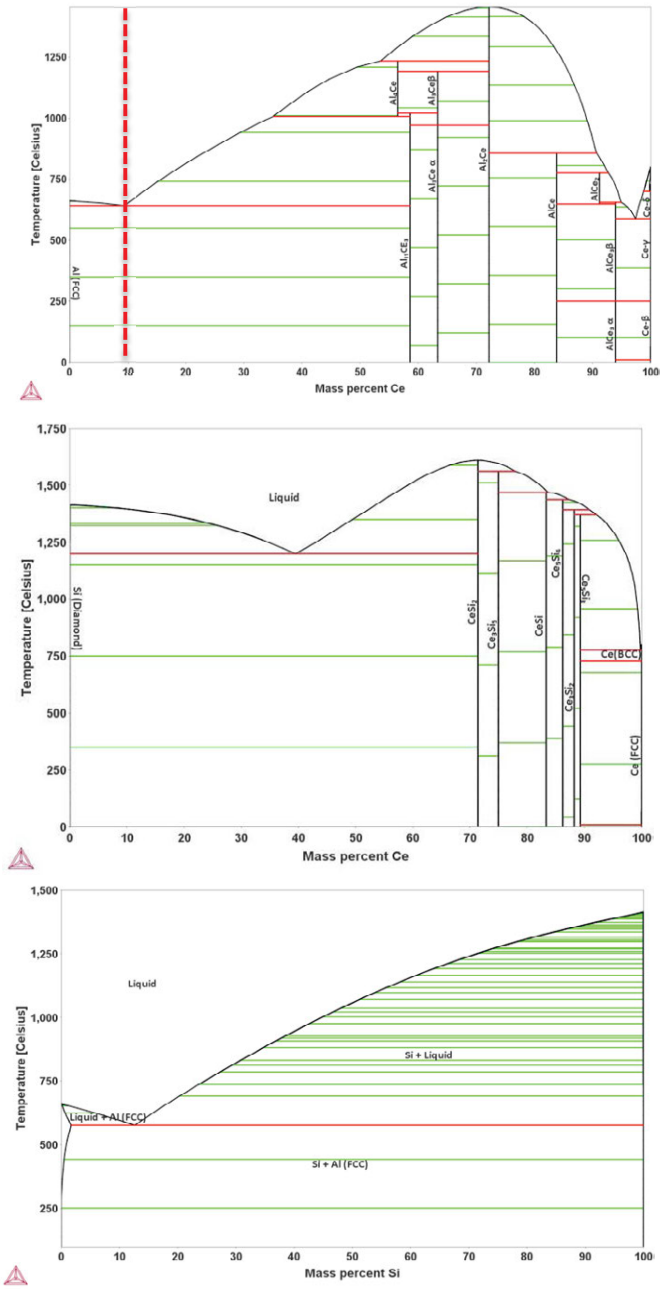

Figure 1. Top) Al-Ce diagram used to determine casting compositions [red line denotes eutectic] Middle) Ce-Si binary phase diagram Lower) Al-Si binary phase diagram

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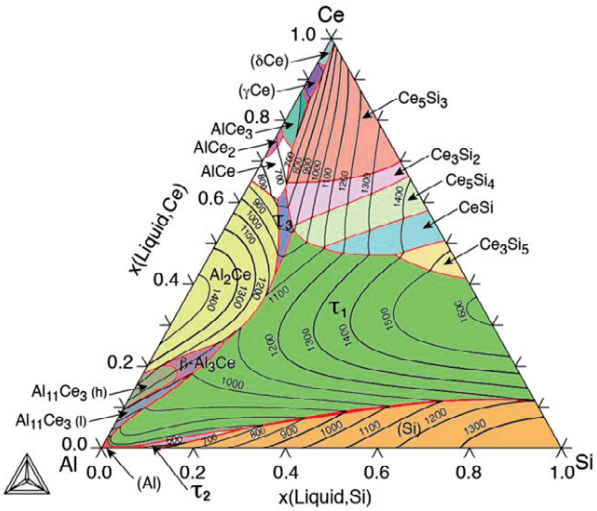


Figure 2. Al-Ce-Si ternary liquidus diagram calculated using proprietary database.

Experimental

Castability

In order to determine the castability of the Al-Ce alloy system it was important to begin at a scale that was comparable to foundry practices. Thus melts of 20-40 kg were made and poured into test bars for mechanical property measurements. In the binary Al-Ce system melts were prepared at 6, 8, 10, 12, 16 wt% Ce (16 wt% was excluded from the hot-tear and step-plate castings as the primary intermetallic left it far too brittle to be a reliable casting alloy). Once test bars were poured, hot-tear molds and step-plates were cast from sand molds with a polymer binder. Step-plates were poured in two different molds. The first step-plate having a small iron chill located at the front of the step-plate under the finger like risers. The second set of step-plates used a full coverage Cu chill.

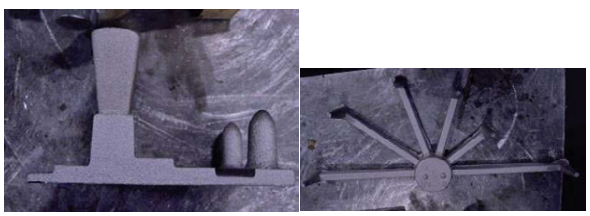


Figure 3. Left) Step-plate mold of Al-12wt% Ce. Right) Hot-tear mold of Al-12wt% Ce

Initial observation of the binary Al-Ce alloys showed no issue with mold filling, very little shrinkage, and almost no hot-tearing in all samples, with one exception, Al-6wt% Ce exhibited some hot tearing at the base of the hot-tear constraints. These conclusions were further supported by X-Ray imaging of the molds which showed few internal defects in the bulk of the samples.

Computed tomography was used to further confirm uniformity of the casting internal structure. Scans were made along the arms of hot-tear molds and at all compositions binary Al-Ce alloys proved castable.

Following the success of binary alloys, ternary and quaternary additions were made using: Fe, Mg, Cu, Mn, and Si. Herein we will focus on the Mg, Cu, and Si additions. A melt of Al-12Ce-0.4Mg (all further values will be in weight percent) was prepared and poured into identical molds as the binary alloys. Castability of the Al-Ce-Mg alloy was measured to be greater than those of the Al-Ce binary alloy system. Overall fewer micro-voids and less shrinkage occurred in the molds. In the case of Si, it was added as quaternary addition to Al-8Ce-0.4Mg and Al-12Ce-0.4Mg in two different mounts 4wt% Si and 1 wt% Si. At 4 wt% Si the castings were very hard to pour as melt fluidity had dropped drastically. Once separated from the sand molds the hot-tear mold of the 4wt% Si was fractured into pieces and the step-plates upon X-ray imaging had multiple large voids. However, when Si was reduced to only 1wt% it did not inhibit castability and was later determined to increase the mechanical properties through a reaction with the binary Al-Ce intermetallic compound. Cu additions, 2.25wt% and 4.5wt%, proved very interesting. In the permanent mold used for the test-bars the alloy poured very well filling the mold with very little shrinkage or cracking, but when the step-plates were removed from the molds it was clear a reaction had occurred between the Al-Ce-Cu alloy and the iron chill used to aid in alloy solidification. It is not yet known what reaction occurred, but due to the widespread use of iron chills in the aluminum casting industry it is worth determining if this alloy can be cast.

Overall, the Al-Ce-X alloy system has proven very castable over a wide range of compositions.

Solidification Characteristics

DSC was utilized to determine the solidus and liquidus temperature of each alloy. Scans were made from room temperature to 750 °C at a scan rate of 20 °C/min. Upon investigation it was determined binary Al-Ce alloys solidify in a nearly isothermal state. Isothermal solidification is desirable for casting alloys as a mechanism to maintain mechanical property uniformity throughout cast pieces.

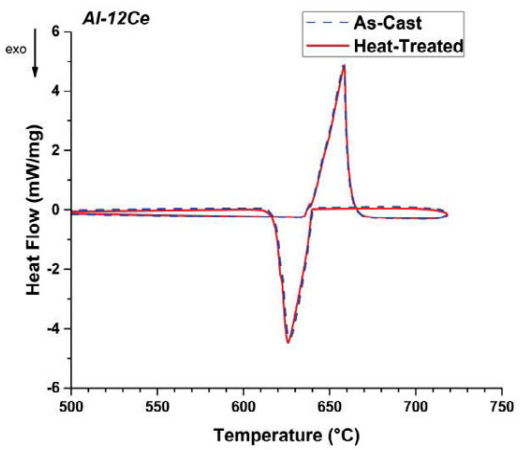


Figure 4. DSC curves for as-cast and T6 heat-treated Al-Ce binary alloys showing near isothermal melting and solidification important to casting practices.

In the case of binary alloys (Fig 4), heat-treatment does not affect the liquidus and solidus temperatures. Nor does heat-treatment affect the heat-flow peaks. It can then be concluded from the DSC curves that phases and phase fractions are unchanged during heat-treatment.

Ternary and quaternary additions often create very complex melting and solidification characteristics in the Al-Ce-X alloys. In most cases however onset of phase transformations does not begin until well above the proposed operating temperature of the Al-Ce alloys. The solidification window increases for the Al-Ce-Mg-Si alloys. This increase again does not create an alloy whose melting onset approaches the upper limit for the alloy operating temperature.

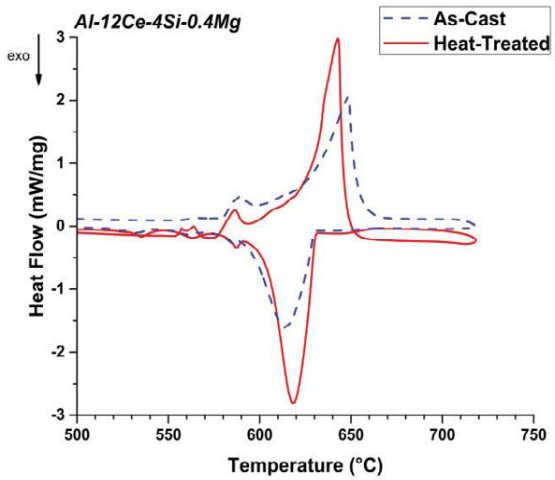


Figure 5. DSC curves for Al-12Ce-4Si-1Si as-cast and T6 heat treated showing complex behavior when compared to the binary alloy DSC curves.

The DSC curve in figure 5 shows different behavior before and after heat-treatment for the Al-Ce-Si-Mg alloy. This change in behavior results from the presence of new ternary intermetallic phase, AlSiCe, which is precipitated during the T6 heat-treatment. The new ternary AlCeSi phase is important for strengthening the binary alloys.

Microstructure

The microstructure has been analyzed for morphology via SEM and optical microscopy. In the case of the binary alloys at and below the Al-Ce eutectic point (~12wt%), intermetallic solidification shows eutectic morphology. In the primary solidification region depicted by the Al-16Ce sample, large primary crystals have formed which account for the rapid decrease in ductility following the move from 12 to 16wt% Ce. Heat-treatment affects the microstructure by decreasing the interconnectedness of the eutectic microstructure. Given the insolubility of cerium in FCC aluminum though, the phase fractions are not affected. Large primary crystals also remain largely unchanged by the heat-treatment.

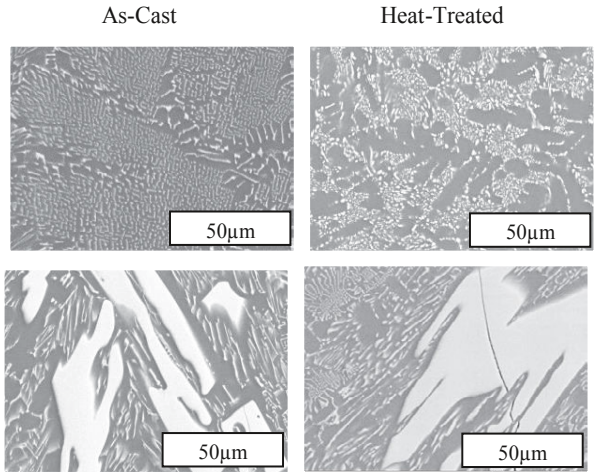


Figure 6. Upper) SEM images of 12wt% Ce before and after heat-treatment. Lower) SEM images of 16wt% Ce before and after heat-treatment.

The eutectic microstructure is very fine in some locations it can have rod diameter as small as 50nm. This size and distribution may be one of the causes for the high-ductility present in the Al-Ce alloy.

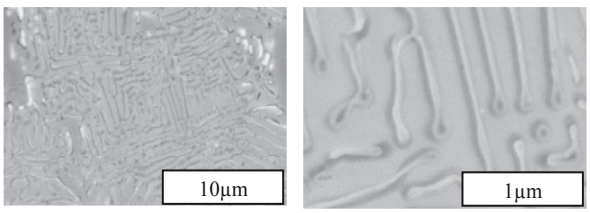
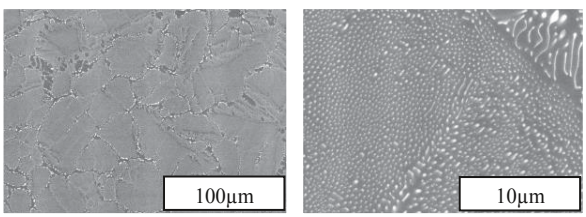


Figure 7. SEM images of Al-10Ce showcasing the very small diameter of the eutectic microstructure.

Cooling Rate Analysis

Al 12 wt% step plates were cross-sectioned then pieces cut from each step. These pieces were polished and then imaged in a Hitachi S-4700 SEM. The cooling rate has a large effect on the microstructure of the system. For steps located directly over the iron chill solidification was driven toward eutectic solidification with very find rod structure, often as small as 100nm. For the sections taken for the larger middle step it was determined that solidification drifted toward the primary $Al_{11}Ce_3$ intermetallic with small crystals forming upon solidification.



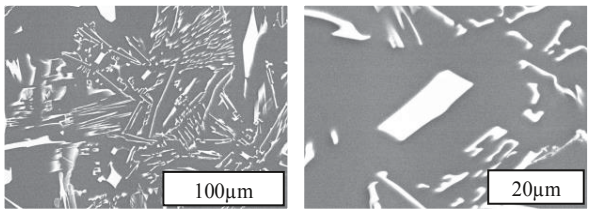


Figure 8. Upper) SEM images taken from step-plate sections positioned over chills. Lower) Step-plate sections taken from large middle step away from chill.

It remains to measure the effects of the cooling rate dependent microstructure on the mechanical properties of the alloys.

Conclusions

Initial characterization of Al-Ce alloys show promise for providing a cost–effective high-temperature aluminum alloy. Fine microstructures and nearly isothermal solidification lead to a highly castable and ductile alloy. Mechanical properties can be increased through ternary and quaternary additions making Al-Ce-X alloys useful in elevated temperature applications, such as automotive engines.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ON THE EFFECT OF ALLOYING ELEMENT RANGE ON THE MECHANICAL PROPERTIES OF RECYCLED ALUMINIUM ALLOY EN AB-46000

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Keywords: Aluminium, recycling, alloying elements, casting, solidification, mechanical properties, microstructure.

Abstract

Aluminium can be produced from the raw material bauxite or by recycling aluminium scrap. When aluminium is being recycled, the material strength is then depending on the alloying and trace elements in the aluminium scrap. This paper aims to investigate the sole effect of the alloying element range of Si, Cu, Mg, Mn and Fe on the mechanical properties of the recycled aluminium alloy EN AB-46000 by producing directional solidified samples with low defect levels.
The tensile test results and microscopic investigations reveal clearly that the variations within the alloying element range lead to remarkable differences in the mechanical properties that can be quantified to 73 MPa in tensile strength and 49 MPa in yield strength. The elongation to failure was also affected as a function of the alloying elements which is due to increased levels of intermetallics.

Introduction

The aluminium consumption in the world is increasing and continues to find new applications. High demands are put on the material strength of cast products and the market is continuously asking for a material with high and reliable quality. The quality of aluminium components mainly depends on casting process and parameters, such as solidification rate, but also chemical composition, obviously, significantly affects the mechanical properties [1].

The cast alloy EN AB-46000 is mainly used for high pressure die-casting processes. It is characterized by high fluidity, high strength at room temperature and a lower price than its competitors [2, 3]. The primary and most important alloying element in this cast alloy is Si, followed by Cu and Mg which is characterized by high fluidity and reduction in shrinkage due to the high Si content that is nearly 9-10% [2, 4, 5].

Secondary alloys, particularly those destined to HPDC process can contain iron levels up to 1.5 %. For most foundry alloys, the presence of iron is detrimental due to its tendency to form very harmful Fe-based intermetallic compounds, such as the acicular β-Al_5FeSi phase. The sharp tips of these brittle phases can act as stress raiser, decreasing ductility and overall properties of the castings. In order to enhance the strength of these alloys, the morphology of the β-Al_5FeSi phases can be altered, for example by adding Mn, to a more compact, less harmful shape, such as the α-phase $Al_{15}(Fe,Mn)_3Si_2$, with Chinese script like morphology [1]. However, even if Mn is present in the alloy, Al_5FeSi-needles and $Al_{15}(Fe,Mn)_3Si_2$ are likely to coexist and the fraction of each of them, depends on the cooling rate and the ratio Fe:Mn, which is usually recommended to 2:1 [5, 11, 12, 13].

Cu is a commonly used as an alloying element to increase the material strength in cast alloys, especially when heat treatment is applied. In Al-Si alloys, Cu is usually added in levels between 1.5 – 3.5 % and forms the intermetallic phase Al_2Cu [11, 12]. If the addition of Cu is fully soluble in the Al-matrix an increase in the material strength as well as a relatively unchanged ductility can be expected. On the other hand that Cu creates a continuous network at the grain boundaries the consequence will be a serious reduction in ductility [5, 12].

The addition of Mg has the same effect as Cu and is added to increase the strength. Depending on the cooling conditions and alloying elements, Mg may be soluble in the Al-matrix, forming Mg_2Si and/or forming Mg-rich intermetallics together with the alloying elements. If the size of Mg_2Si is controlled and formed in solid solution, it helps to increase the strength [13, 14, 15].

A commonly used alloying element to modify the morphology of Si in these alloys is Sr. A small addition, up to 200 ppm to molten alloy, results in reduction of size and morphology changes of the eutectic Si-particles. As a result of adding Sr, the material will experience slight increase in the strength and ductility [13].

Defects in the cast Al products can seriously lower the mechanical properties and cause problems with the quality of the material. Oxides, pores and sludge are defects that commonly found in these alloys. Oxides form due to the great affinity of Al to react with the environment. Molten Al tends to form thin layer of oxides on the surface of the melt that both, can be protective form continuous oxidation but also dangerous if these oxides enters the bulk material due to stirring and filling. The oxides in the melt may form cracks causing leakage related problems. Pores will always be formed during solidification, both in the form of gas or shrinkage, due to hydrogen in the melt and due to shrinkage during the solidification. Oxides will also assist in the formation of porosity as it tend to unfurl during solidification [16].

The tendency of Al-Si-Cu alloys to form primary Fe-rich particles can be predicted by the sludge factor (SF), equation 1, which represents an attempt to gather the combined effects of Fe, Mn and Cr. The accepted empirical equation of SF is expressed as [17, 18, 19]:

$$SF = (1 \times wt.\%Fe) + (2 \times wt.\%Mn + (3 \times wt.\%Cr) \quad (1)$$

Due to the influence of various parameters on the quality and mechanical properties of Al castings, the aim of the current work is to highlight the influence of alloying range of the main alloying elements on the tensile properties of Al-Si-Cu-Mg alloys.

Experimental

Alloys and materials

The Company XY produces at least eighth different variations of the AB 46000 for pressure die casting [20]. The purpose of this work is to quantify the effect of the ranges of the alloying elements on the mechanical properties of two of these alloys; EN AB-46000 (in this worked called alloy A) which is the standard alloy, according to EN 1676:2010, and a modified version of EN AB-46000 called "Modified 46000" (in this worked called alloy B). The compositions of the both alloys are given in Table 1.

Table 1. Chemical composition of the standard alloys.

Alloy	Elements (wt. %)							
	Si	Fe	Cu	Mn	Mg	Zn	Cr	Sr
A	8.0-11.0	0.6-1.3	2.0-4.0	-0.55	0.15-0.55	-1.2	-0.15	-0.02
B	8.7-9.4	0.5-0.6	2.7-3.3	0.3-0.47	0.35-0.45	-1.2	-	0.03-0.05

The first task was to select two alloy ranges for the A and B that will have the theoretical best (called alloy H) and worst (called alloy L) mechanical properties according earlier publications [2-17]. The recommended theoretical chemical compositions for the alloys are presented in Table 2

Table 2. Recommended theoretical chemical composition of the standard alloys.

Alloy	Elements (wt. %)							
	Si	Fe	Cu	Mn	Mg	Zn	Cr	Sr
AH	9.0	0.60	4.00	0.40	0.55	0.70	0.05	0.02
AL	9.0	1.10	2.00	0.20	0.15	0.70	0.05	0.03
BH	9.0	0.50	3.30	0.35	0.45	0.70	0.05	0.03
BL	9.0	0.60	2.70	0.30	0.35	0.70	0.05	0.03

Casting Procedures

The alloys that have been studied in order to understand and assess the relationship between alloying element range and the mechanical properties were melted using a resistance furnace set to 730 °C. The alloying elements were preheated in a furnace to 300 °C before addition to melting furnace. The melt was always skimmed before any addition was made. 15 min after Sr additions, the chemical composition of the melt was examined using an atomic emission spectrometer.

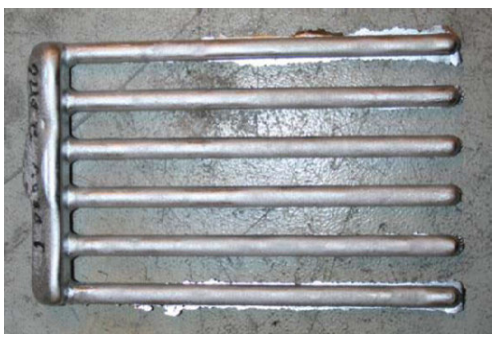


Figure 1. The cylindrical cast rods.

After that the alloy was fully molten and met the requirements, cylindrical metal rods with length 18 cm and 1 cm in diameter, see Figure 1, were casted in a permanent cooper mould that was preheated to 250 °C.

Directional Solidification

To achieve specimens with low levels of oxide films and with low content of porosity as well as providing a well-fed a homogenous microstructure, the specimens were remelted and solidified using the directional solidification technique. In this work a resistance, heated furnace with an electrically driven elevator was used, see Figures 2 and 3. A cylindrical turned metal bar, which is machined from the casting specimens shown in figure 1, enclosed within a graphite coated steel tube, was inserted into furnace, see figure 3. The temperature in the furnace was set to 710°C and the sample was left at the temperature for 30 minutes for remelting under a protecting atmosphere (Ar). After remelting, the furnace is moved with a constant velocity of 3 mm/s, and the bar is continuously cooled as it passes a water cooled zone to produce a cast bar with a Secondary Dendrite Arm Spacing (SDAS) ~ 10µm that is comparable to HPDC microstructures.

Figure 2. Directional solidification furnace.

For alloy A, 27 specimens (15 specimens for the AH and 12 specimens for the AL alloys) and for alloy B 26 specimens (14 specimens for the BH and 12 specimens for the BL alloys) were produced, investigated and compared.

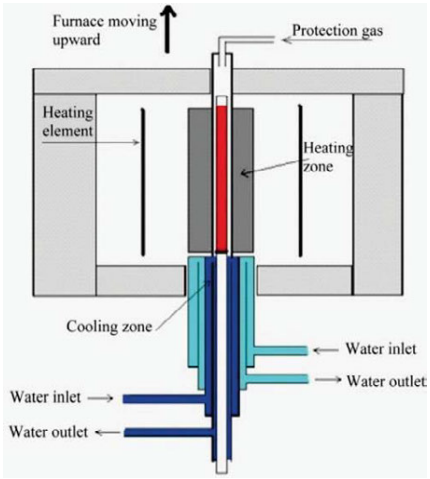
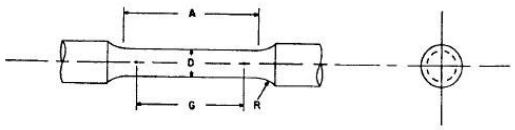


Figure 3. A schematic of the Directional solidification furnace.

Mechanical Strength Specimens

The room temperature tensile properties have been evaluated for all the samples. Specimens for tensile testing were machined from the remelted bars according standard ASTM B 557M-07 (Standard Test Methods for Tension Testing Wrought and Cast Aluminum- and Magnesium-Alloy Products), with gauge length 30±0.06 mm and diameter 6.00±0.10 mm, see Figure 4 [1].



Nominal Diameter	Dimensions, mm	
	Standard Specimen	Small-Size Specimens Proportional to Standard
	12.5	6
G—Gage length	62.50 ± 0.10	30.00 ± 0.06
D—Diameter (Note 1)	12.50 ± 0.25	6.00 ± 0.10
R—Radius of fillet, min	9	6
A—Length of reduced section, min (Note 2)	75	36

Figure 4. Specimen used for tensile testing [1].

The tensile tests were conducted using a tensile test device called MTS 318.10. The samples were loaded in tension with increasing load until fracture; the elongation rate was 1.0 mm/minute. Strain was measured using an axial extensometer (MTS 634.31F-24) according standard ASTM B 557M-07 (E83 Method of Verification and Classification of Extensometers). All the data from experiments in form of load and displacement was monitored, stored and analyzing used Matlab to obtain ultimate tensile strength, yield strength and elongation.

Microstructure Analysis

The specimens from the tensile testing were used for both the microstructure and macrostructure examinations. The macroscopy investigations were carried out on the fracture surface and then the samples were grinded and polished and the microstructure was investigated. The microstructure was etched with 0.5 % HF for ten seconds at room temperature. To determine the different features of the microstructure, a light microscope and image processing software were used.

Results

Alloys

The mean values of the chemical compositions for the investigated alloys along with SF and Fe:Mn are presented in table 3 that presents the mean values of the achieved contents upon which the calculation of SF and Fe:Mn was performed. The Cr in the investigated alloys A and B was 0.01% and 0.02% respectively.

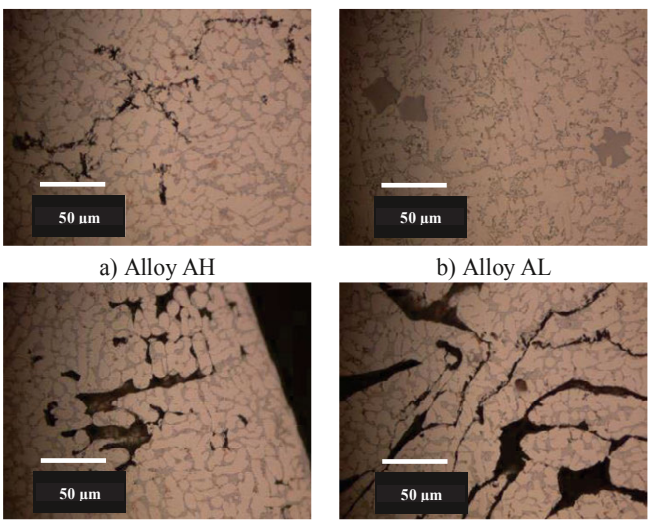
Microstructure

The directional solidified microstructure of the alloys, as illustrated in figure 5, is composed of Al matrix, Al-Si eutectic and Fe-rich particles where the latter were in the form of both β-Al_5FeSi phases, $Al_{15}(Fe,Mn)_3Si_2$ and notably the sludge particles that are clearly seen in figure 5b.

Table 3. The composition of the investigated alloys, SF and Fe:Mn

Alloy	Elements (wt. %)							
	Si	Fe	Cu	Mn	Mg	Sr	SF	Fe:Mn
AH	9.50	0.80	3.90	0.42	0.66	0.020	1.67	1.90
AL	9.50	1.30	2.20	0.20	0.13	0.001	1.73	6.50
BH	9.53	0.45	3.24	0.30	0.69	0.028	1.11	1.50
BL	9.36	0.52	2.59	0.68	0.34	0.042	1.91	0.76

The SDAS was approximately 10µm (±2) and the microstructure was found homogenous. The Cu was found in the microstructure but the stoichiometry of the phases was not analysed. No detectable Mg_2Si was identified and the major part of Mg-rich phases is assumed to be tied to intermetallics and in solid solution. One can also observe the typical defects present in these alloys that are oxides and shrinkage related porosity, see figure a, c and d. Even though the aim was to produce sound and high quality samples for investigation of the sole role of alloying elements, some samples revealed some defects as outlined.



a) Alloy AH b) Alloy AL

c) Alloy BH d) Alloy BL

Figure 5. Micrographs of typical defects in some samples from alloys A and B.

Mechanical properties

During the tensile testing, a number of samples failed at low strain and stress values due to large defect area fractions; see figure 5 for defect types. These samples were considered not representative and were further excluded from tensile data presented in this section. A closer look at this data can be found in Appendix 1.

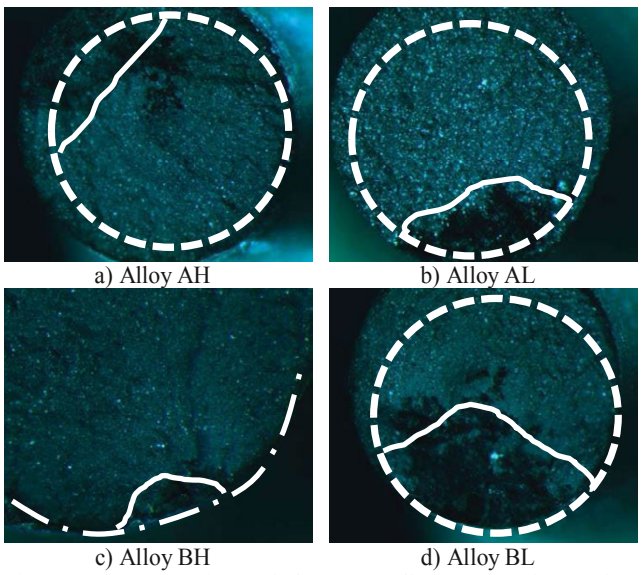
The average tensile test data of the alloys in terms of tensile strength (Rm), proof stress ($R_{p0,2}$) and elongation (A_{50}) for both alloys A and B with the corresponding H and L are presented in table 4. The results reveal that the expected variations between A and B are clearly observed especially when B is optimized and considered a "better" performer. In terms of strength, alloys AH and BH are performing according to expectations being the stronger ones. Concerning the A_{50} the BL was found to have the best elongations to failure that may be due to lower levels of the major alloying elements mainly Cu and approximately 50% lower Mg content.

Table 4. Tensile test data for alloys A and B, both H and L.

Alloy	Tensile strength (R_m) MPa	Proof stress ($R_{p0,2}$) MPa	Elongation (A_{50}) %
AH	380 (10)	217 (3)	3.8
AL	307 (16)	168 (8)	3.1
BH	384 (8)	205 (6)	4.2
BL	366 (6)	186 (4)	5.5

Metallographic Analysis

A number of samples exhibited premature failures during the tensile testing which are derived to defects available on fracture surfaces, as reported in figure 6. Besides shrinkage porosity, large areas covered by oxides, see marked regions on the fracture surface in figure 6 a-d were detected. It is worth mentioning that the investigated alloys were not exposed to melt treatments to minimize the oxide level nor did have the possibility to measure the level of oxides and inclusions.



a) Alloy AH b) Alloy AL

c) Alloy BH d) Alloy BL

Figure 6. Cross-sections of the 6 mm-diameter specimens for alloys A and B with corresponding H and L.

Discussion

Comparing alloy AH with AL, AH exhibited larger values of strength; the difference in Rm is 73 MPa, in $R_{p0,2}$ is 49 MPa and 0.7% in A_{50}. The difference in strength is explained by the variations of Cu and Mg in these two alloys, see table 3. It is assumed that the level of Cu and Mg in the matrix as well as the fraction of precipitates are higher in AH compared to AL. According to research from Yildirim et al. [9] and Seifeddine [10] who have studied the influence of alloying elements on the properties of Al-Si cast alloys, the higher levels of Mg and Cu contribute to increased material strength. In research of Pucher et al. [58] the influence of alloying elements is studied with the conclusion that an increased level of Mg (0.1-0.5 %) and Cu (2-3.5 %) has positive effect on the yield strength. This corresponds with the result of the tensile testing where the alloys (AH, BH, BL) with higher amounts of Mg and Cu, obtained increased values for strength. The Rm and A_{50}values generated in this study for alloys A and B is also larger than the specified standard values of alloy, see table 5, according to EN1676-2010 [20]. Unfortunately, all $R_{p0,2}$ values were lower than the recommended standard values, see table 5. Any reasons for that behavior could not be explained but it is worth mentioning that the data upon which alloy B is based in table 5 are data generated by samples cast and analyzed at the company XY. Variations in production and experimental set up could be one of the reasons for this variation.

Table 5. Mechanical properties of the standard alloys [20].

Alloy	Tensile strength (R_m) MPa, min.	Proof stress ($R_{p0,2}$) MPa, min.	Elongation (A_{50}) %, min.
A	240	140	<1
B	352	226	2.6

As far as the elongation A_{50} is concerned, all investigated alloys performed better than the specified values in the standard, see table 5. This can be explained by the lower level of defects and avoidance of turbulent filling in the specimens produced by directional solidification compared to HPDC processes that additionally leads to laminations and air entrapments. The higher levels in Fe and SF in AL resulted in lower ductility compared to AH which also is supported by research from Taylor [6] and Seifeddine et al. [1]. Alloys BH and BL obtained the highest value for elongation which depends on their lower Fe level in comparison to alloys AH and AL [1, 5, 6, 7, 8] and better values for BL compared to BH due to lower Cu and Mg contents.

Conclusions

It can be concluded that a wider alloying element range cause increased differences in the material strength. The influence of alloying element range have been quantified for the alloy A to at least 73 MPa in tensile strength and 49 MPa in yield strength as well as 0.7% in elongation. Corresponding values for the B alloy are 18 MPa in tensile strength, 19 MPa in yield strength and 1.3 % in elongation. These variations are directly linked to the levels of Cu and Mg. Increased Cu and Mg (in both AH and BH) leads to increased levels of solutes and precipitates and hence increased strength. Besides, higher levels of Cu and Mg results in increased levels of intermetallics that cause reduction and/or variations in

elongation which also can be accelerated by the increased levels of Fe in in the alloys.

Acknowledgements

The authors would like to thank Linnaeus University, Faculty of Technology, Department of Mechanical Engineering, Växjö, Sweden and special thanks to Bertil Enquist. Thanks to the staff at the School of Engineering, Department of Material and Manufacturing – Jönköping University, Sweden and special thanks to Anton Bjurenstedt.

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Appendix 1

Table X1. Results from tensile testing for alloy AH.

Alloy AH	Tensile strength (R_m) MPa	Proof stress ($R_{p0,2}$) MPa	Elongation (A_{50}) %	Remarks
AH1	302	220	2.0	Oxides
AH2	392	217	4.4	
AH3	391	222	4.3	
AH4	286	218	1.6	Oxides
AH5	372	214	3.4	
AH6	380	216	3.8	
AH7	375	223	3.8	
AH8	288	220	1.5	Oxides
AH9	250	223	1.1	Oxides
AH10	237	215	1.1	Oxides
AH11	357	212	3.1	
AH12	382	213	4.1	
AH13	387	218	3.9	
AH14	176	176	0.6	Oxides
AH15	382	214	4.0	

Table X2. Results from tensile testing for alloy AL.

Alloy AL	Tensile strength (R_m) MPa	Proof stress ($R_{p0,2}$) MPa	Elongation (A_{50}) %	Remarks
AL1	340	174	3.9	
AL2	87	87	0.4	Oxides
AL3	307	175	2.8	
AL4	253	170	1.7	Oxides, pores
AL5	304	165	3.2	
AL6	322	167	3.3	
AL7	318	169	3.1	
AL8	295	152	3.4	
AL9	161	161	0.4	Oxides, pores
AL10	297	162	2.9	
AL11	314	173	3.0	
AL12	280	178	1.8	

Table X3. Results from tensile testing for alloy BH.

Alloy BH	Tensile strength (R_m) MPa	Proof stress ($R_{p0,2}$) MPa	Elongation (A_{50}) %	Remarks
BH1	384	203	4.6	
BH2	390	204	4.6	
BH3	394	201	5.3	
BH4	387	204	4.8	
BH5	391	204	4.5	
BH6	385	202	4.4	
BH7	360	205	2.7	Oxides
BH8	373	198	3.9	
BH9	388	208	4.4	
BH10	383	197	4.3	
BH11	385	205	4.1	
BH12	392	220	4.5	
BH13	385	215	3.7	
AH14	374	208	3,3	

Table X4. Results from tensile testing for alloy BL.

Alloy AL	Tensile strength (R_m) MPa	Proof stress ($R_{p0,2}$) MPa	Elongation (A_{50}) %	Remarks
BL1	201	175	1.1	Oxides
BL2	275	191	1.8	Oxides
BL3	376	186	6.8	
BL4	268	187	1.7	Oxides
BL5	368	190	5.0	
BL6	207	175	1.1	Oxides
BL7	366	186	5.7	
BL8	199	182	1.1	Oxides
BL9	356	185	3.9	
BL10	377	191	5.5	
BL11	362	180	5.9	
BL12	362	182	5.7	

Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Phase and Thermal Stability Analysis of Al-Fe-V-Si-Y Alloys After Solidification at Intermediate Cooling Rates

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Keywords: High Temperature, Aluminum, Dispersion, Icosahedral, Approximant

Abstract

Samples of RS-8009 (Al-Fe-V-Si alloy) prepared by both copper mold chill casting and rapid solidification processing were analyzed using SEM and TEM methods including CBED, selected area electron diffraction (SAD), and electron backscatter diffraction (EBSD). Diffraction analysis revealed a hexagonal phase identified as h-AlFeSi (P6/mmm a=2.45 nm c=1.25 nm) with evidence of a structural relationship to the icosahedral quasicrystalline (QC) phase and a further relationship with the α-Al-Fe-Si phase ($Im\bar{3}$ a=1.26 nm), which is the desirable intermetallic in this system and also a quasicrystal approximant. The analysis confirmed the findings of earlier studies on the system, which suggested the same structural relationships using different methods. As will be shown, both phases form across a wide range of cooling rates and appear to have good thermal stabilities.

Introduction

The Al-Fe-V-Si system was developed from research in the Al-Fe-Si system, which has been active since the early 1980s [1]. Iron additions to Al alloys are known to form fine icosahedral dispersoids during Rapid Solidification Processing (RSP) that are advantageous from the standpoint of cost and slow coarsening rates, but other additions are required to form more stable dispersoids. One such alloy, RS8009 (Al-8.5Fe-1.3V-1.7Si), has excellent room temperature strength as well as strength at high temperature and good corrosion resistance [2]. During rapid solidification, the cubic α-$Al_{12}(Fe/V)_3Si$ phase ($Im\bar{3}$ with a~1.26 nm) forms spherical dispersoids of about 20-60 nm in diameter in a microcellular or cellular matrix of Al, which is typically somewhat supersaturated with the alloying additions.

The α-$Al_{12}(Fe/V)_3Si$ dispersoids are more stable than the QC dispersoids that are observed in Al-Fe and Al-Fe-X alloys. This phase is known as a QC approximant; such phases are defined as a rational approximation of a corresponding QC structure, with a few specific commonalities. First, an approximant will share the same icosahedral structural unit with the QC phase. A direct consequence of the shared structural unit is a close relationship between the electron diffraction patterns of the two phases. Typically, a specific orientation relationship (OR) between the two phases also exists. Finally, the two phases will have approximately the same valence electron concentration, and will have similar composition [3]. The result is a crystalline phase that exhibits pseudo-symmetric diffraction patterns when viewed from angles associated with the quasicrystal, including pseudo 5-fold patterns.

A competing phase in this system during rapid cooling is the hexagonal h-AlFeSi phase reported by Koh et al. [4], who identified the phase in strip and spray-cast Al-Fe-V-Si alloys. They used convergent beam electron diffraction (CBED) to characterize this phase. In their analysis, they determined that the phase has a space group of P6/mmm, with lattice parameters of a = 2.514 and c = 1.257. In addition to the h-AlFeSi particles, coarse icosahedral particles and the cubic α-Al-Fe-Si phase were also observed. Composite electron diffraction patterns across particle boundaries were used to propose ORs and, by extension, structural relationships between the three phases. The OR between the icosahedral phase and the bcc α-$Al_{12}(Fe/V)_3Si$ phase was determined to be $\langle i3\rangle_{icosa}//[111]_{bcc}$, $\langle i2\rangle_{icosa}//[001]_{bcc}$, $(110000)_{icosa}//(100)_{bcc}$, and $(110000)_{icosa}//(\bar{3}5\bar{2})_{bcc}$. The OR between the icosahedral phase and the hexagonal phase was determined to be $\langle i2\rangle_{icosa}//[11\bar{2}0]_{hexa}$ and $(110000)_{icosa}//(1\bar{1}00)_{hexa}$. This OR was previously unreported, but has been corroborated subsequently by other literature [3]. Finally, the OR between the hexagonal phase and α-$Al_{12}(Fe/V)_3Si$ was determined to be $[0001]_{hexa}//[001]_{bcc}$, $[1\bar{1}00]_{hexa}//[010]_{bcc}$, $[11\bar{2}0]_{hexa}//[100]bcc$, $(11\bar{2}0)_{hexa}//(100)_{bcc}$, $(0001)_{hexa}//(001)_{bcc}$, and $(1\bar{1}00)_{hexa}//(010)_{bcc}$. Although ORs were found between all three intermetallic phases in the alloy, no relationship was found between any of the phases and the Al matrix. This was confirmed by Ezersky et al. [3] who extended the work of Koh et al. [4] in considering the h-AlFeSi phase to be an approximant. Experimental and simulated HRTEM along the $[11\bar{2}0]$ zone axis revealed the same structure proposed by Koh, composed of double Mackay icosahedra. In addition, the composition of the phase was measured and found to be 77.4Al-14.8Fe/V-6.9Si in atomic %, which is very close to that of the α-$Al_{12}(Fe/V)_3Si$ phase, as expected.

Experimental Procedure

In this study, the base material, for both experimental and baseline purposes, was an extrusion of RS8009 provided by Honeywell. The extrusion was 96mm in diameter, and was produced with an extrusion ratio of 8.8:1. The composition of RS8009 is Al-8.5Fe-1.7Si-1.3V, in wt% [2]. Three Y-doped 8009 alloys were prepared by remelting pieces of the extrusion with pure (99.9%) Y chips (Table 1).

Table 1. Compositions of the experimental alloys, listed in atomic %. Note that the compositions are based on the nominal composition for RS8009 with a measured mass of Y added.

Alloy	Al (at%)	Fe (at%)	V (at%)	Si (at%)	Y (at%)
RS8009	93.2	4.3	0.7	1.7	N/A
0.25 Y	93	4.3	0.7	1.7	0.25
0.5 Y	92.8	4.3	0.7	1.7	0.5
1 Y	92.3	4.3	0.7	1.7	1

These alloys were induction melted and cast in a copper wedge mold, producing cooling rates on the order of 10^3 to 10^4 °C/s.

Sections of the extrusion were also sectioned and prepared in the as-received state. Standard SEM and TEM methods were used to analyze the microstructures. Thermal stability was tested by comparing microhardness before and after exposure to 400°C for 2h in air.

Results and Discussion

The cast microstructures (Figures 1 and 2) consisted of regions of microeutectic near the prior chill surfaces, corresponding to the highest cooling rates, and primary intermetallics in the sample interiors. The addition of Y did not significantly alter the structure of the cast alloys, but rather introduced a uniform dispersion of YV_2Al_{20}. particles throughout the castings, presumably due to formation in the melt.

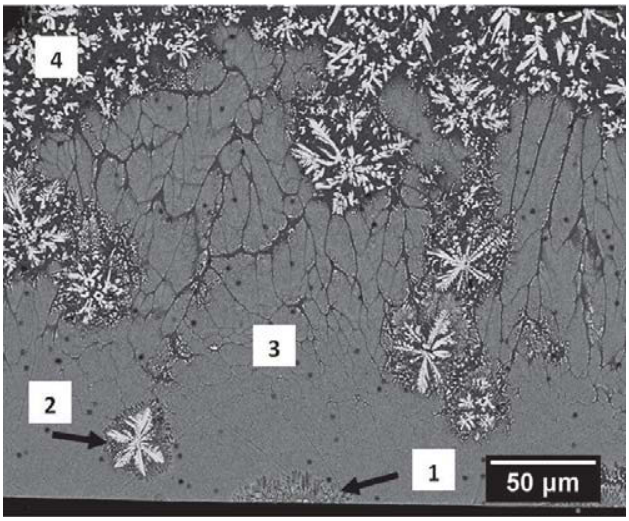

Figure 1. SEM BSE micrograph showing the structure of the 8009 casting, with the chill wall surface shown at the bottom. The labeled structures are 1: Primary α-Al dendrites 2: Primary hexagonal intermetallics 3: α-Al + cubic $Al_{12}(Fe/V)_3Si$ microeutectic, and 4: Primary hexagonal/other intermetallics in Al matrix.

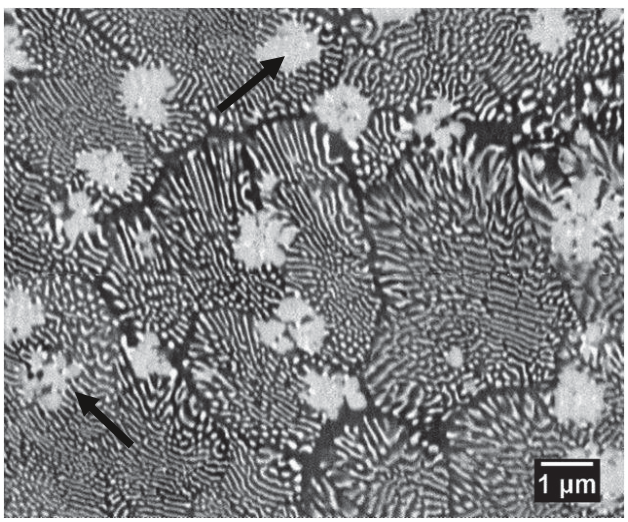

Figure 2. SEM BSE micrograph showing a typical structure from the 8009+0.25Y sample. Arrows highlight the YV_2Al_{20} particles distributed within a microeutectic matrix. The microeutectic was observed in all samples, regardless of Y content.

As previously stated, a heat treatment of 400°C for 2h was chosen. The treatment simulated the thermal history experienced by rapidly solidified ribbons during consolidation and extrusion, based on the work of Park et al. [5]. The sample sections were taken near the tip, where the cooling rates were the highest; these contained the highest volume fraction of the microeutectic. The microstructure of the eutectic in the 8009 casting before and after the anneal along with hardness data from all of the alloys (Figure 4) indicate that all the castings experienced coarsening of the microeutectic and a corresponding hardness drop, on the order of 20%. However, even after annealing, all cast samples still maintained hardnesses higher than those of the RS8009 extrusion. Sections of extrusion were also included in the annealing study, but as expected, showed no hardness loss due to the softening that would have occurred during thermomechanical processing. Furthermore, no evidence of phase transformations was detected, which is consistent with the results of Koh et al. [4] who conducted in-situ hot-stage TEM analysis of the h-AlFeSi phase and found no phase transformations below 500°C. Similarly, no transformations or significant changes in the morphology of the larger YV_2Al_{20} particles were observed.

Convergent beam analysis of the hexagonal phase revealed an icosahedral pseudo-symmetry when considering the strongest Kikuchi bands. Not all Kikuchi bands visible in each pattern in Figures 5 and 6 are typical of hexagonal materials. In addition to strong bands parallel to the $\{11\bar{2}0\}$, $\{10\bar{1}0\}$, and (0001), the strongest bands (and most intense reflections) were found to be associated with high-index planes with spacings of approximately 0.21 nm, consistent with {220022} spacings in the icosahedral structure. All told, the strongest Kikuchi bands observed were: $\{10\ 0\ \bar{1}0\ 0\}$, $\{6\ 6\ \bar{1}2\ 0\}$, $\{\bar{4}\ \bar{6}\ 10\ 3\}$, $\{6\ 10\ \bar{6}\ 5\}$, $\{10\ 0\ \bar{1}0\ 2\}$, and (0 0 0 6). When plotted on a stereographic projection (Figure 7), a relationship between the structures of the icosahedral and h-AlFeSi phases becomes evident. When the full hexagonal projection is reduced to the form shown in Figure 8b, the planes shown (those that correspond to the strongest Kikuchi lines in Figures 5 and 6) closely match the 2-fold, 3-fold and 5-fold planes of the icosahedral structure. It was determined that three icosahedral stereographic projections, sharing the common 2-fold projection direction shown in Figure 8b but rotated by 0° 60° and 120°, form the complete h-AlFeSi projection shown in Figure 7. From this relationship, the structural relationship can be inferred, and an OR derived. The common 2-fold icosahedral direction is parallel to the [0001] in the hexagonal phase, and the orthogonal 2-fold icosahedral directions are parallel to two of the 6 $<11\bar{2}0>$ directions in the hexagonal phase. This relationship is identical to that found by both Ezersky et al. [3] and Koh et al. [4]. Also, a previous analysis of the cubic $Im\bar{3}$ phase that is part of the microeutectic as well as the primary dispersoid phase in RS8009 by Hwang et al. [6] revealed a close relationship between the cubic phase and the icosahedral phase, from which structural and ORs were determined. However, as shown by Hwang et al., only a single icosahedral stereographic projection was needed to correlate the two structures. Finally, no OR with the FCC Al matrix was found, consistent with the work of Ezersky et al. [3] and Koh et al. [4]

Another interesting observation is that the strongest Kikuchi bands from the hexagonal phase actually consist of double bands – while this is not uncommon, in this case the parallel bands, e.g., $\{10\ 0\ \overline{10}\ 0\}$ and $\{16\ 0\ \overline{16}\ 0\}$ are approximately related to each other by the Golden Mean (τ) value (1.618). This phenomenon is characteristic of approximant phases [7].

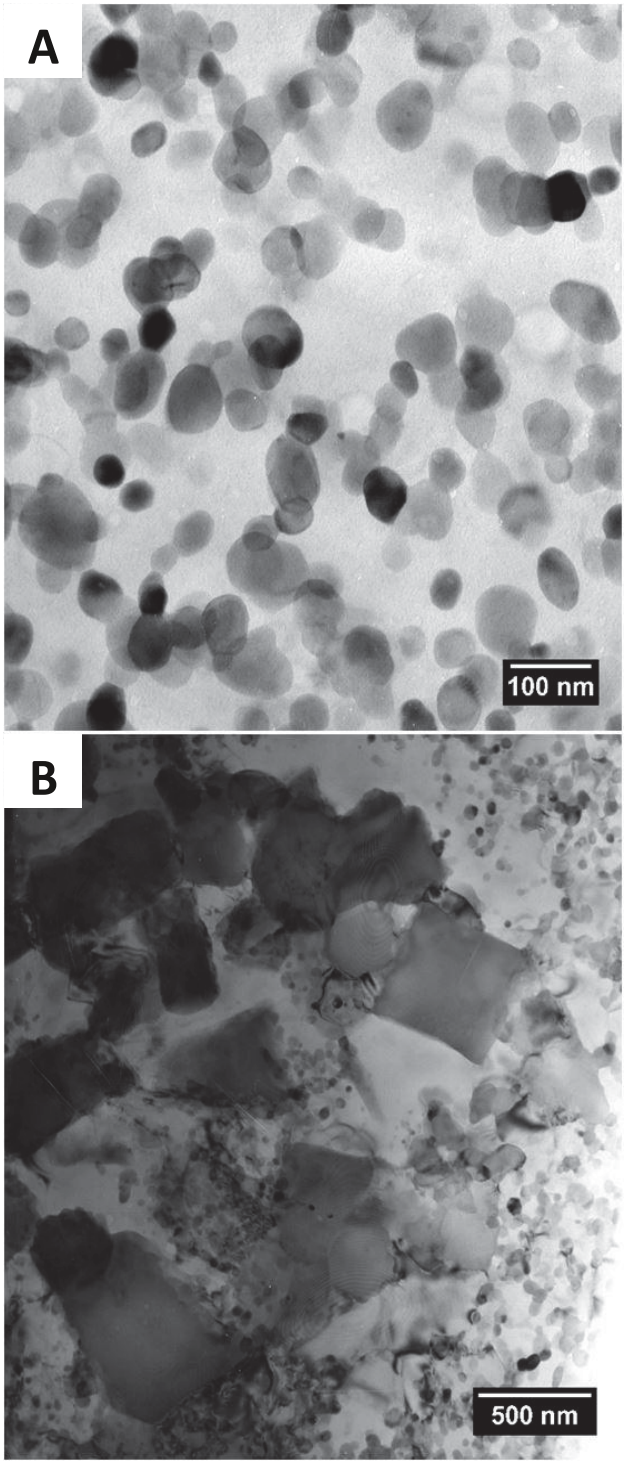

Figure 3. BF TEM images taken from the as-received RS8009 extrusion. (a) typical desirable microstructure containing fine, spherical α-$Al_{12}(Fe/V)_3Si$ particles in an Al matrix; and (b) coarse, faceted h-AlFeSi phase as observed in the as-received condition .

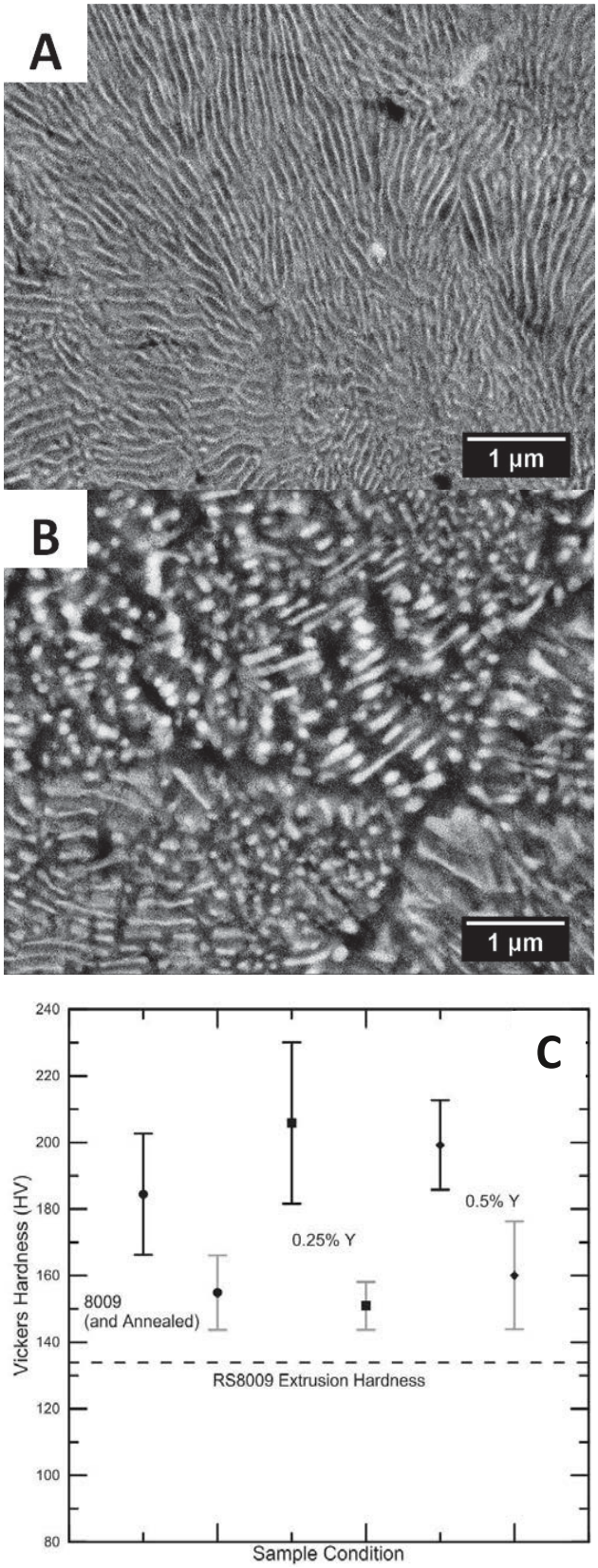

Figure 4. BSE images of as-cast 8009 (a), annealed (400°C, 2h) 8009 (b), and microhardness of 8009 with and without Y additions (c).

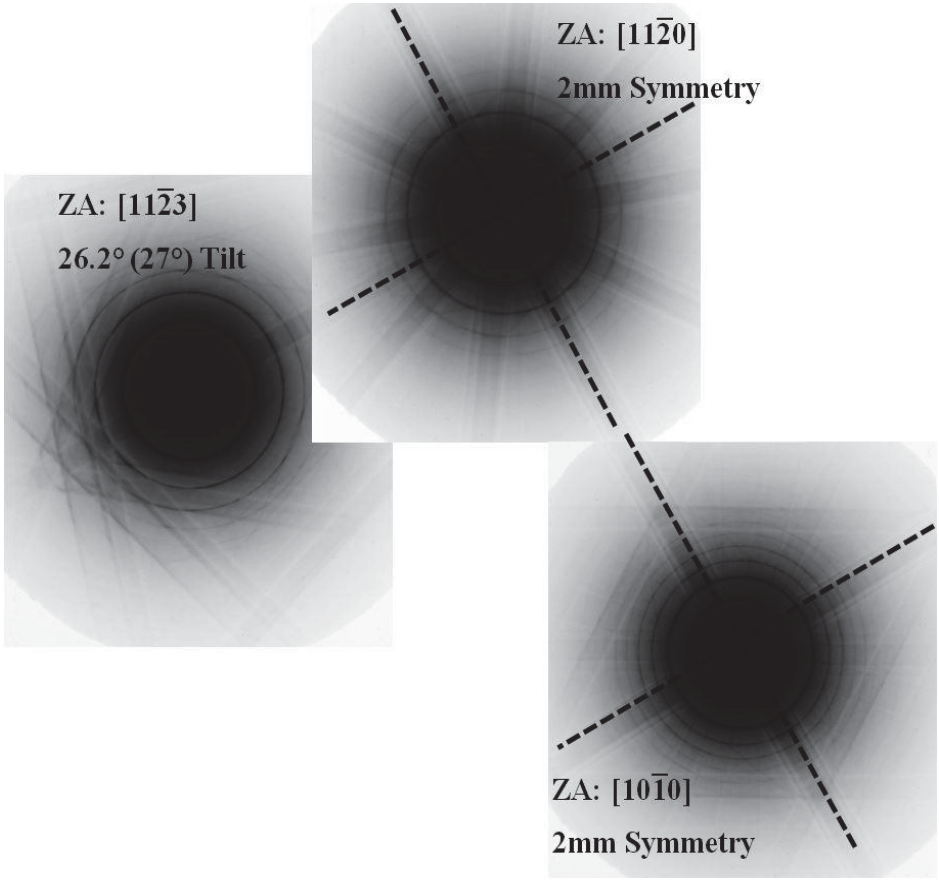

Figure 5. CBED patterns from three zone axes of the hexagonal phase. The mirror planes required for point group determination are marked with dotted lines. Each strong Kikuchi band has an associated wider Kikuchi band that can be seen, with a width ratio of approximately 1.6 (the Golden Mean).

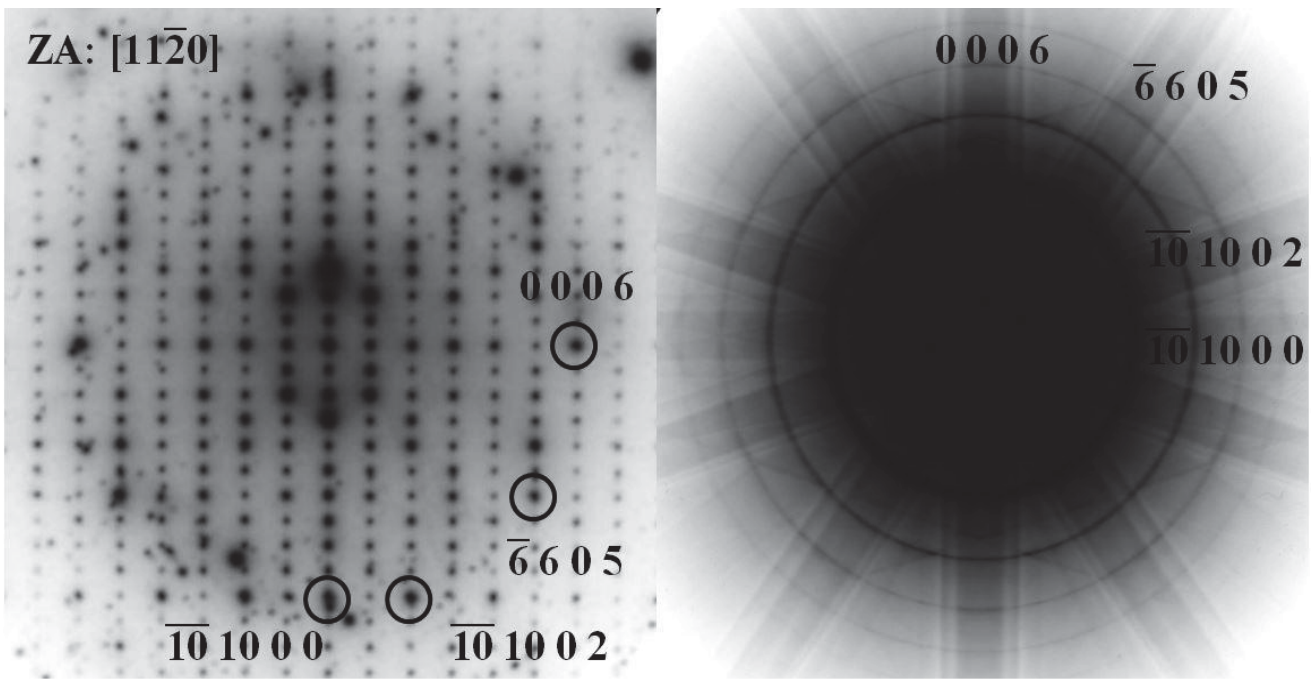

Figure 6. SAD and CBED patterns from the [11$\bar{2}$0] zone axis of the hexagonal phase. The intense reflections in the SAD pattern correspond to the strong Kikuchi bands observed in the CBED pattern. Note that all intense reflections were associated with spacings of approximately 0.21 nm.

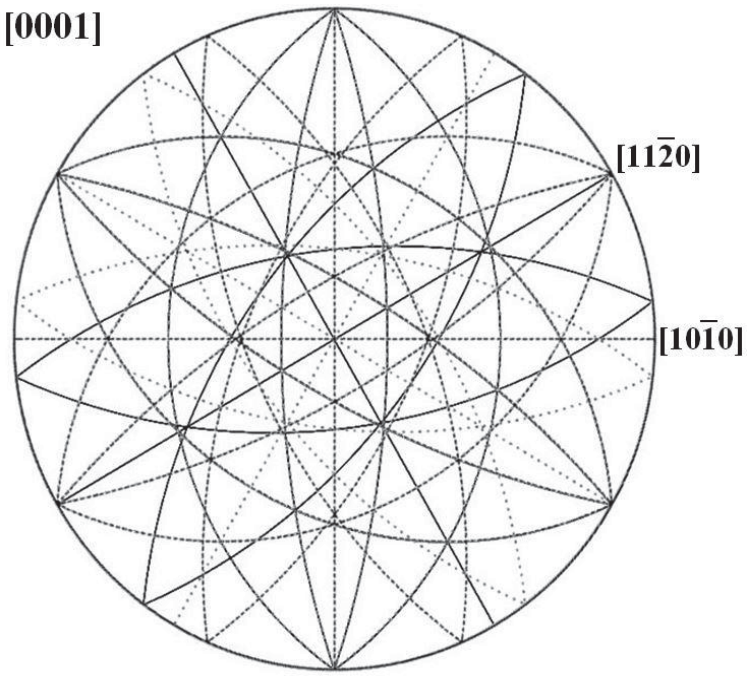

Figure 7: Full hexagonal [0001] stereographic projection created by summation of 3 orientations of the icosahedral projection, separated by 60°. Each orientation is drawn with a different line style (light dash, medium dash, or solid). Note the overlapping lines along all {10 0 $\overline{10}$ 2} and {6 0 $\bar{6}$ 5} plane traces.

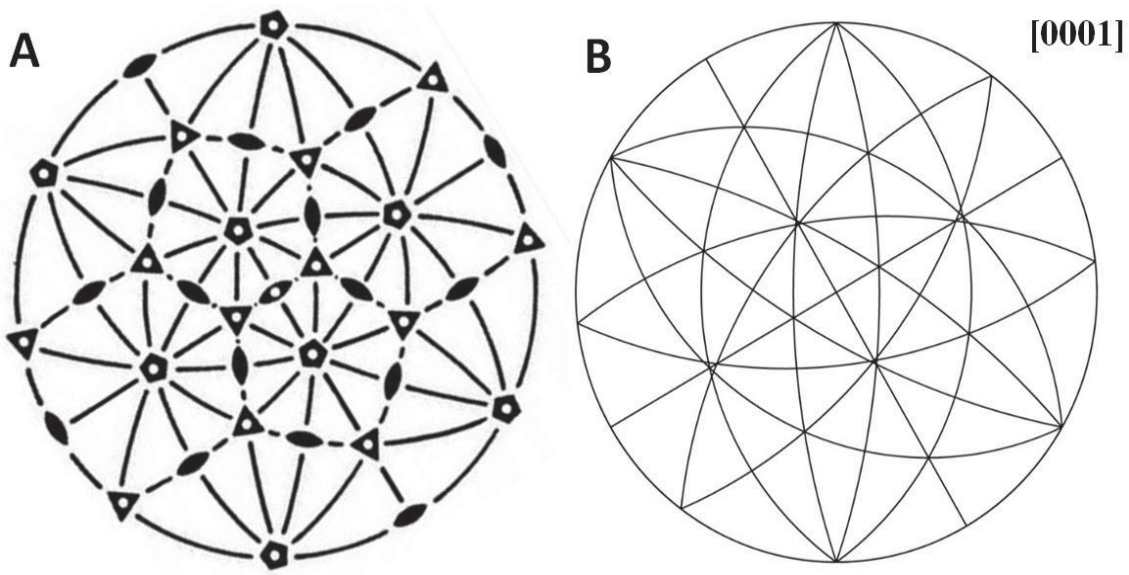

Figure 8. Stereographic projections. (a) is from the icosahedral m$\bar{3}$5 phase rotated to align a 5-fold axis vertically [6] and (b) is a reduced [0001] projection from the h-AlFeSi phase, displaying only the plane traces necessary to approximate the icosahedral projection. Three such projections are required to complete the projection in Figure 7. In total, 15 planes are required by the icosahedral projection: 1 {6 6 $\overline{12}$ 0}, 1 {10 0 $\overline{10}$ 0}, 4 {$\bar{4}$ $\bar{6}$ 10 3}, 4 {6 0 $\bar{6}$ 5}, 4 {10 0 $\overline{10}$ 2} planes, and the (0 0 0 6) plane.

Conclusions

As-received samples of RS8009 extrusion and cast samples of remelted 8009 with and without Y additions were analyzed using TEM and SEM.

- The cast structures showed favorable hardness and thermal stability, consistent with potential applications at high temperature.
- The same phases were observed across all samples, which were created with very different cooling rates.
- A structural relationship between the α-$Al_{12}(Fe/V)_3Si$, h-AlFeSi, and icosahedral phases was identified using convergent beam electron diffraction.
- The strong Kikuchi bands closely mimic the icosahedral Kikuchi pattern, and it was shown that a combination of three icosahedral stereographic projections overlays a single h-AlFeSi projection.
- The proposed orientation relationship was consistent with previous findings in the literature obtained using different methods.

Acknowledgements

The authors would like to thank the National Science Foundation and the members of the Center for Advanced Non-Ferrous Structural Alloys (CANFSA) for their generous financial and technical support of this research.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MICROSTRUCTURE AND PHASE EVOLUTION IN A201 ALLOYS WITH ADDITIONS OF SILICON

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Keywords: A201 alloys, Silicon, Precipitation, Microstructure, Mechanical properties

Abstract

A201 alloys and A201 alloys with the addition of 1 wt. % silicon were investigated in the as cast, solution treated and aged conditions. They were investigated by a combination of HRSEM, TEM and EDS. The addition of Si was used to improve the castability of the basic alloy, but it was found that 1 wt. % Si wasn't enough to improve fluidity and prevent hot tearing. It was found that the addition of Si alloy increased the average grain size after solution treatment at 550°C for 19 hours, and large particles of secondary phases were precipitated at the GBs, also increased the precipitation of Ω phase during aging and, by these means, increased the microhardness of the A201 alloy, despite of coarser grain structure of this alloy. There are two kinds of phases were precipitated during aging at 170°C in both investigated alloys, tetragonal lath-shape θ' phase and orthorhombic polygonal Ω phases with different of the amount, size and shape of precipitates.

Introduction

The most widely studied age-hardening alloy system is Al–Cu. Alloy A201 is an important commercial alloy due to its high mechanical properties, excellent machinability and good formability. It is commonly used in aerospace and automobile parts [1-6].

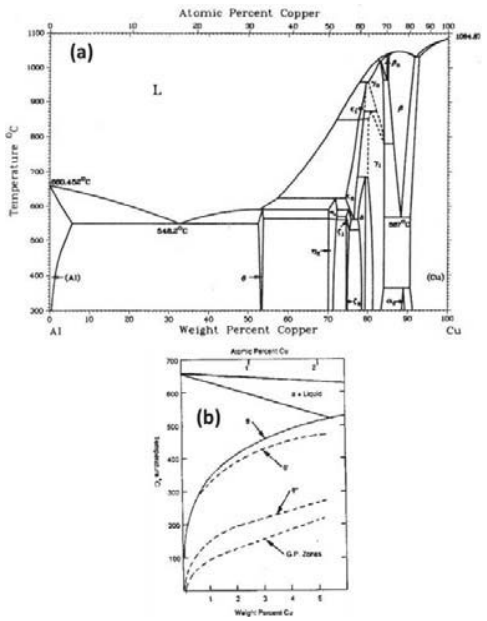

Fig. 1. (a) Al–Cu phase diagram; (b) Al–Cu phase diagram showing the metastable solvus boundaries for GP zones, θ" and θ', together with the equilibrium solvus line for the θ phase.

Fig. 1a shows the Al-rich corner of the equilibrium Al–Cu phase diagram, and includes the metastable solvus boundaries for Guinier–Preston (GP) zones, θ" and θ'.

Over many studies [1-3, 7-13], it has been proposed that the precipitation sequence varies depending on the degree of super-saturation and the aging temperature while the decomposition sequence in this system contains one or more of the following processes:

Supersaturated Solid Solution (SSSS) → GP zones → θ" → θ' → θ. The four structures of the precipitates can be recognized as shown in Table. 1.

Table 1: summary of properties of the Al-Cu precipitation

	Notes
GP zones	Thin plates of Cu atoms on {100} planes in α series
θ"	Coherent phase, 2 layers of Cu atoms separated by 3 of Al
θ'	Semi-coherent plates of tetragonal $CuAl_2$ on {100} planes in a phase
θ	Incoherent equilibrium phase, body centered tetragonal $CuAl_2$

The precipitation processes from which this hardening is derived are well known to be sensitive to the presence of selected trace element additions or micro-alloying, which can change the process and/or kinetics of precipitation in many age hardenable alloys [1-2, 9].

The effect of different alloying elements on the mechanical properties of Al alloys has been investigated, but the influence of each alloying element on the other is not fully understood [14-17]. The outstanding effect of silicon in aluminum alloys is the improvement of casting characteristics. The level of Si primarily controls the amount of Al-Si eutectic that forms during solidification and also influenced the precipitation sequence.

In the present work two A201 alloys were investigated: A201 and A201-1 wt. % Si alloys with additions of Mn, Fe and Mg. Their specific compositions were selected in order to achieve maximum age hardening response, improved castability, grain refinement and strength. They were investigated by a combination of high resolution scanning electron microscopy (HRSEM), transmission electron microscopy (TEM) and energy dispersive X-ray spectrometry (EDS). Microhardness measurements were performed to reveal the effect of precipitation on the mechanical properties of the alloy.

Experimental

The chemical compositions of the alloys analyzed in this study are shown in Table 2.

Table 2: Composition of A201alloys (mass fraction, %)

Al	Cu	Ag	Mn	Mg	Fe	Si
Bal.	4.97	0.56	0.33	0.24	0.05	0.068
Bal.	4.8	0.46	0.292	0.28	0.033	0.72

Casting the addressed alloys to three plates that having different thicknesses in a metal die as shown in Fig. 2. Casting and dies temperatures will be the same for all alloys. Specimens were cut from the cast plates. The as cast specimens were first followed two stages of solution treatment at 515°C for 5hr; 550°C for 14hr in a heat treatment furnace under Ar atmosphere in order to prevent oxidation during solution treatment, then quenched in water. Then the ST samples were aged at 170°C in an oil bath for various periods of time of up to 16 days, followed by water quenching.

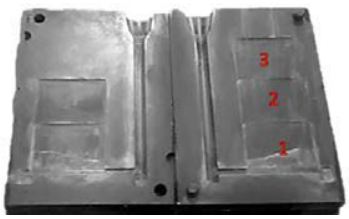

Fig. 2. Three plates having different thicknesses in a metal die

The microhardness of aged specimens was measured by a Vickers hardness tester under a load of 100gram and holding time of 15sec. 10-15 indentations were conducted on each specimen.

A comprehensive characterization utilized HRSEM, EDS and TEM was carried out in order to elucidate the homogenized and aged micro-structure of the alloys. Metallographic specimens mounted in epoxy resin for polishing preparation. The mounts were polished using four steps (plane grinding, fine grinding, diamond polishing and oxide polishing). Thin foils for TEM characterization were prepared by ion beam thinning, and were examined at the acceleration voltage of 200 kV. The compositions of the matrix and of secondary were analyzed by EDS instrument attached to the HRSEM, operating at 5kV.

Results

As cast microstructure

Figs.3-4 show the as-cast alloys (a) for A201 alloy and (b) with the addition of ~1 wt. % Si. From Fig. 3, it can be concluded that 1 wt. % Si wasn't enough to improve castability due to the occurrence of crack.

The microstructures of as cast alloys consist mainly of the α-Al phase as a matrix, eutectic-like structures in the near-grain boundary regions (Fig. 4).

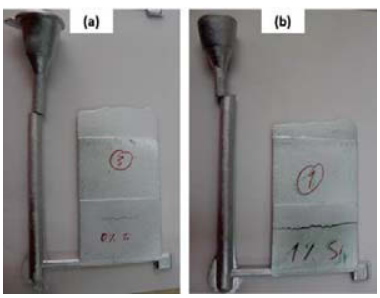

Fig. 3. The casts of (a) A201 alloy and (b) A201 alloy with addition of 1 wt. % Si.

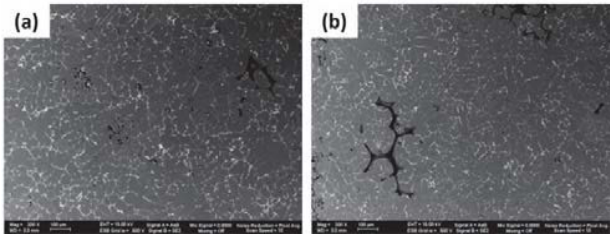

Fig. 4. HRSEM images of the as-cast sample (a) for A201 alloy and (b) for A201-1 wt. % Si alloy.

The grain size (d=L/n), was measured using the intercept line method, when n is the number of grain boundaries and L is intercepting line. The average grains size of the as-cast is about 37±3 [μm] for A201 alloys and 32±3 [μm] for A201-1 wt. % Si alloys.

Fig. 4 show the phases at the grain boundaries for the two as-cast alloys. The compositions of the grain boundaries were analyzed by EDS HRSEM shown in Tables. 3.

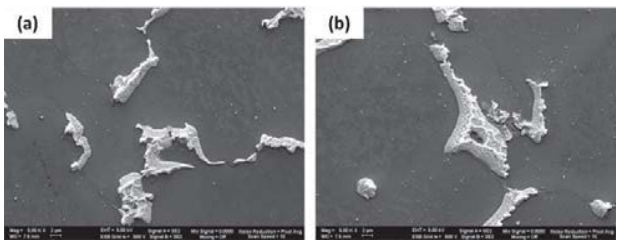

Fig. 5. HRSEM images of the GBs in the as-cast sample (a) for A201 alloy and (b) for A201-1 wt. % Si alloy.

Table 3: Compositions of phases at the GB in A201 and A201-1wt. % Si alloys (Fig.4), measured by EDS HRSEM.

	GB (Fig.4)	Al (wt.%)	Cu (wt.%)	Mn (wt.%)	Fe (wt.%)	Ag (wt.%)	Si (wt.%)
A201	gray	32.41	64.45	1.25	0.93	0.75	0.21
	white	1.36	94.95	--	--	4.11	0.09
A201-1wt.% Si	gray	32.88	64.13	0.85	0.45	1	0.69
	white	0.5	94.41	0.69	0.02	4.38	--

ST microstructure

The purpose of homogenization for precipitate hardened A201 alloys is to dissolve soluble as cast phases into α-Al matrix and obtain supersaturated solid solution by quenching. Fig. 6 shows that the GB is almost dissolved in A201 alloy after ST, however partially dissolved in A201-1 wt. % Si alloy and the microstructure became to be with very coarse grains. The average grains size of the ST is about 108 [μm] for A201 alloys and ~250 [μm] for A201-1 wt. % Si alloys

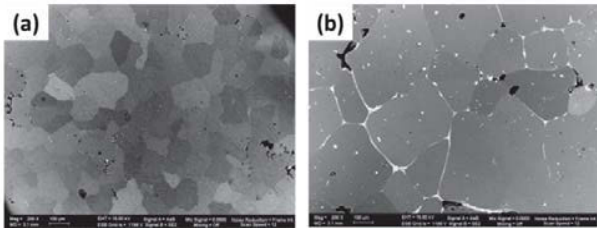

Fig. 6. HRSEM images of the ST sample (a) for A201 alloy and (b) for A201-1 wt. % Si alloy.

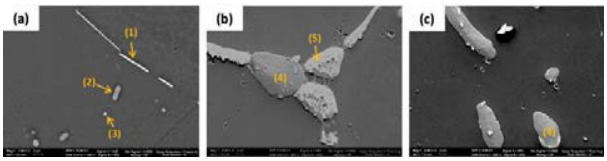

Fig. 7. Microstructure of the ST sample (a) for A201 alloy and (b, c) for A201-1 wt. % Si alloy. HRSEM images show the secondary phases (1-5).

It was observed that the homogenized alloy contained primary α-Al solid solution, secondary phases in the near-grain boundary regions and inside the grains (Fig. 7). The compositions of the secondary phases were analyzed by EDS HRSEM shown in Tables.4

Table 4: Compositions of different secondary phases in A201 and A201-1wt. % Si alloys (Fig.5), measured by EDS HRSEM.

	Phases (Fig.5)	Al (wt.%)	Cu (wt.%)	Mn (wt.%)	Fe (wt.%)	Ag (wt.%)	Si (wt.%)
A201	1	36.4	41.6	1.6	20.4	0.2	--
	2	45.5	16.9	19.6	15.4	2.6	--
	3	66.7	29.7	1	--	3	--
A201-1 wt.% Si	4	40.4	3.2	39.1	8.7	0.8	7.9
	5	1.56	96.74	--	--	2.16	--

Age-hardening behavior

The age hardening curves of the two investigated alloys, A201 and A201-1wt. % Si alloys during isothermal aging at 170°C up to 16 days with error bars of 5 HV are presented in Fig. 8.

Microhardness of the A201 alloy reached a maximum value of ~ 138 HV after 6 h of aging. In the A201- 1 wt. % Si alloy, initially the hardness decreases during the 40 min and reaches a maximum value of ~ 155 HV also after 6 h of aging. Further aging led to decrease in the microhardness.

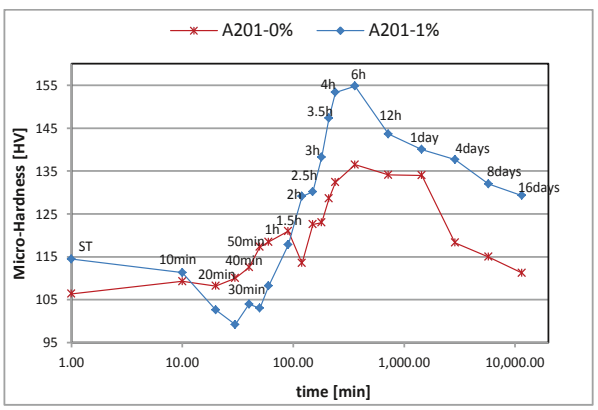

Fig. 8. Age hardening curve of the two alloys aged at 170°C

Microstructure of aged samples

Fig.9 show the big different in the microstructure between the two samples: A201 and A201-1 wt. %Si alloys at the maximum hardness (after 6h of aging).

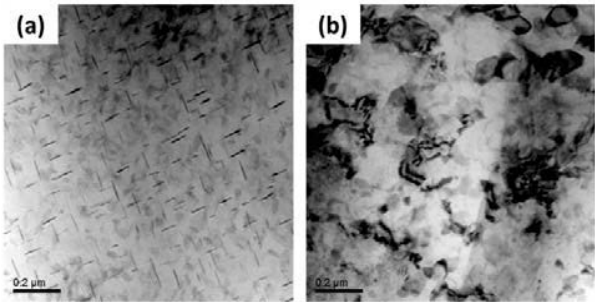

Fig.9. Bright-field (BF) TEM micrographs (a) for A201 alloy and (b) for A201-1wt. % Si alloy, after 6h of aging.

BF TEM micrographs and there corresponding selected area electron diffraction (SAED) patterns for A201 alloy after 6 h of aging are shown in Fig. 10. It can be seen that the precipitates are aligned in two directions of <220> for Z.A $[001]_{\alpha\text{-matrix}}$, and <111> directions for Z.A $[112]_{\alpha\text{-matrix}}$.

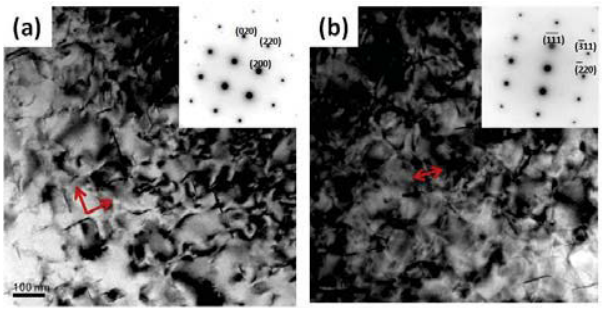

Fig.10. BF-TEM images and corresponding SAED patterns of A201 alloy aged for 6 h at 170˚C: Z.A $[001]_{\alpha}$ in (a) and Z.A $[112]_{\alpha}$ in (b).

Discussion

The addition of Si to the as-cast A201 alloy didn't improve the castability of the alloy (Fig. 3) as expected. This addition of 1 wt. % Si results in large hot tearing (Fig.3b), which means that the addition of 1 wt. % Si wasn't enough to improve castability.

The addition of Si to the A201 alloy influenced the solution treatment microstructure: it increased at least twice the average grain size, and large particles (several tens of microns) of secondary phases were precipitated at grain boundaries. This phenomenon can be connected with the influence of Si on the Al-Cu phase diagram (Fig. 11). The ST for Al-4wt. %Cu with 1wt. % of Si conducted at 550˚C apparently provided appearance of liquid phase (Fig. 7 b-c) resulted in the increased grain growth and fast GB precipitation.

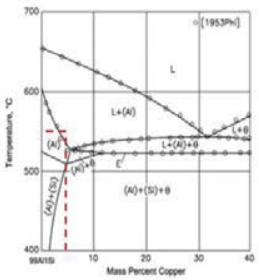

Fig.11. Al-Cu-Si computed vertical section at 1 mass % Si [18]

Precipitation during aging was observed in both A201 and A201-1 wt. % Si alloys. However, the presence of Si changed substantially the amount, size and shape of precipitates (Fig. 9). The major amount of precipitates in the A201 alloy were lath-shape θ' phase with tetragonal structure, while in the A201+Si they were mostly polygonal Ω precipitates with orthorhombic structure. The effect of the 1% Si addition on the precipitation of Ω phase looks to be similar to the effect of Ag addition to the Al-Cu-Mg alloy system reported by Palmerin at al.[10].

Age hardening curves for both alloys (Fig. 8) demonstrated the maximum hardness after ~ 6 h aging. Despite of coarser grain structure of the A201+Si alloy, the maximum hardness of this alloy is substantially higher (~ 15%) than that of A201 alloy. It can be connected with substantial precipitation of Ω phase in the Si containing alloy.

Conclusion

The results can be summarized as follows:

1) It should be added more than 1 wt. % of Si to improve the castability of the A201 alloy.
2) The addition of Si to the A201 alloy increased the average grain size after ST, and large particles of secondary phases were precipitated at grain boundaries.
3) The tetragonal lath-shape θ' phase and orthorhombic polygonal Ω phases were precipitated during aging in both investigated alloys.
4) The Si addition increased the precipitation of Ω phase during aging and, by these means, increased the microhardness of the A201 alloy.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

HIGH TEMPERATURE CREEP EVOLUTION IN Al-Si ALLOYS DEVELOPED FOR AUTOMOTIVE POWERTRAIN APPLICATIONS: A NEUTRON *IN-SITU* STUDY ON *HKL*-PLANE CREEP RESPONSE

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Keywords: Aluminum alloy, automotive powertrain, high temperature creep, neutron diffraction

Abstract

Recent trend in the automotive industry towards lightweighting and downsizing the powertrain components, without compromising the power output, have led to increased engine power density. Increased power density frequently requires these lighter components to operate in conditions of increased temperature and pressure, which is challenging for many aluminum alloys in use today in the powertrain manufacturing. Meeting the challenge requires not only improving high-temperature performance of known alloys or developing new ones, but also developing new advanced techniques to understand the long-term behaviour of the alloys.

This *in-situ* neutron diffraction study evaluated creep strain for individual *hkl* planes in three automotive alloys under tensile load. The conditions of temperature and pressure were typical of or similar to those experienced by engine heads in service. High temperature plastic and elastic creep properties of three cast aluminum alloys were characterized. The properties determined for these alloys can be used as a benchmark for further development of advanced alloy systems suitable for the engine head application.

The *in-situ* neutron diffraction measurements provided data on d-spacing evolution for various *hkl* crystallographic planes as a function of temperature, tensile load, and time; thereby revealing *hkl*-specific evolution of elastic strains. This information contributes to in-depth understanding of why a particular alloy exhibits particular service properties (eg. creep).

Introduction

Characteristics of the primary creep regime, including the initial strain required before the onset of creep, the strain hardening characteristic that retards the rate of creep up to the steady state secondary creep regime can all be used to determine the quality of an alloy and help predict the long term behavior. This kind of information becomes extremely valuable in selecting/developing alloys for the most critical applications in automotive powertrain, for example, cylinder heads.

Significant improvements in fuel economy and powertrain weight can be achieved by enhancing engine efficiency and through replacement of large engines by smaller ones in the same vehicle class. Weight reduction potential by reduced engine size is also increased by the ability to reduce the engine support structure, allowing further weight reductions.

In a simulation of a diesel engine, a 60% increase in power and torque requires increase in maximum cylinder head pressure from 18 MPa to 30 MPa, increase in peak head temperature from 215 to 275^{o}C and from 150 to 190^{o}C on the head combustion face and water jacket face respectively. Authors [1] studied a 6-cylinder turbo medium duty diesel engine and reported that temperature distribution across the cylinder head deck face is ranging from 200 to 350^{o}C, while typical temperatures for the valves and injector regions are 250~260^{o}C. These service parameters (high temperatures and increased pressures) are very challenging for aluminum alloys currently used in engine applications. To achieve the future 2017-2025 industry-mandatory Corporate Average Fuel Economy (CAFE) targets, significant technological changes of engine materials are required.

The objective of this study is to develop a basis for identification of a cast aluminum alloy with improved high temperature properties, including creep and high cycle fatigue strength over the potential service life of the engine. The expected running life of an engine is between 2000 and 4000 hours. Such long term testing is both very expensive and time prohibitive to enable comprehensive evaluation of an alloy. A study of the creep behavior, including the primary and secondary creep regions, and identifying the transition point, will help identify the alloy chemistry and processing conditions that are capable of sustaining the thermal mechanical loads over the expected life cycle of an engine.

Experimental

Sample material

Three different aluminum alloys were cast in ASTM B108 steel mold and then used to machine the 80-mm-long creep specimens, as shown in Figure 1. The specimens were machined from the “gauge” section of the cast test bar.

Table 1 represents the typical chemical composition of the three alloys investigated. The well-known commercial A380 material was produced as a reference point for the “base” and “new” material performance evaluation. The principal difference between the two Al-7wt%Si alloys is the addition Zr, Ti, and V into the alloy with the designation “new”.

Table I
Chemical Composition of the alloys under investigation.

Alloy	Alloying Elements (Wt%)									
	Si	Cu	Mg	Fe	Sr	Mn	Zr	Ti	V	Al
A380	9.5	3.4	0.071	0.90	<0.001	0.19	0.007	<0.001	0.009	Bal.
"Base"	7.1	1.02	0.50	0.090	0.0086	0.005	-	-	-	Bal.
"New"	7.02	0.95	0.48	0.090	0.012	0.005	0.56	0.20	0.32	Bal.

Figure 1. The aluminum test bar (*a*.) cast using ASTM B108 steel mold and machined creep specimen (*b*.). Total length of the specimen is 80 mm, with the length of the gauge section of 32 mm.

Neutron Diffraction: In-Situ Creep Strain Measurements

The creep strain measurements were performed at L3 neutron diffractometer of the Canadian Neutron Beam Center (CNBC), Canadian Nuclear Laboratories, in Chalk River, ON, Canada. A detailed description of the testing procedure was published earlier in [2-5]. Elastic strain (ε) of the atomic lattice was measured using the peak shift method. This method calculates strain using the change in lattice spacing:

$$\varepsilon = {(d_0 - d)}/{d_0} \quad (1)$$

Where d_0 and d are the lattice spacing before and after creep, respectively. The lattice spacing was calculated using Bragg's law:

$$n\lambda = 2dsin(\theta) \quad (2)$$

Where λ is the neutron wavelength (1.55 Å), n=1 for the first order diffraction, and θ is the neutron scattering angle measured during the neutron diffraction experiment.

The elastic strains were measured in the direction parallel to the applied load for several crystallographic planes: (111), (311), (200), and (220). The neutron-diffraction experimental setup, shown in Figure 2, includes a high-temperature extensometer that measures the total value of strain (both plastic and elastic components).

Experiment Procedure and Selection of Experimental Parameters

One of the objectives of this study was evaluation of the elastic/plastic tensile primary-and-secondary creep properties near the yield strength of the material because critical areas of the automotive engine are often operated near these limits. As mentioned earlier, the temperature requirements for aluminum alloys in the automotive industry have increased and will increase further, therefore the test data are now frequently collected at higher temperatures. In this study the analysis was focused on the primary and secondary creep behavior at 200°C and 250°C, as oppose to the typical for earlier studies range of 150-175°C as aluminum was rarely used for the higher-temperature applications.

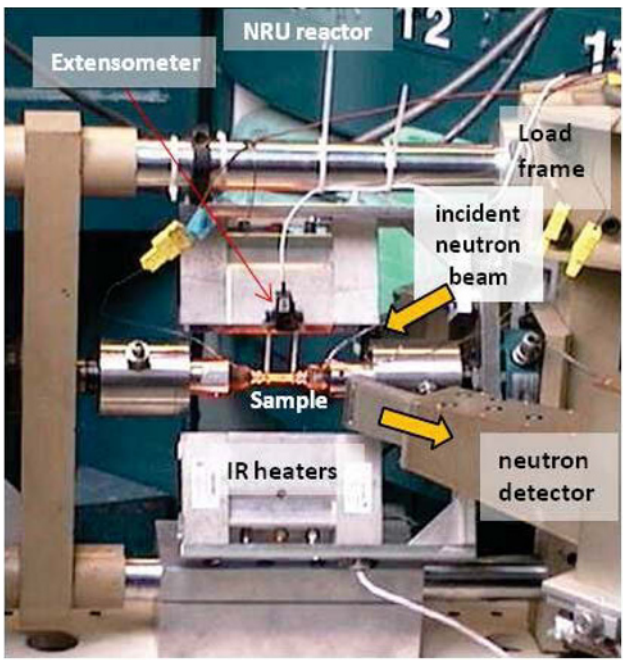


Figure 2. Experimental setup: In-situ neutron diffraction measurement of strain evolution in high-temperature tensile creep.

The procedure for the neutron experiment included the following:

- Sample "conditioning" at the experiment temperature for 200 hours prior to the experiment;
- Upon installation of the sample at the neutron diffractometer (Fig. 2), heating the sample to the experiment temperature (either 200 or 250°C) and measuring the diffraction angles for the selected *hkl* planes (for calculating the corresponding values of *d*-spacing, equation /2/), prior to applying the tensile load. These no-load values for the *d*-spacing are later used as a reference in calculation of the induced strains (equation /1/).
- Applying the tensile load, initially at ~90% of the yield strength, and maintaining it for as long as it takes for the

creep to enter the steady-state region (the secondary creep phase) and exhibit the constant creep rate for at least 14~16 hours, while continuously measuring the angular peak position for the selected neutron diffraction planes.

- Releasing the applied load and measuring the new diffraction peaks' positions, therefore determining the new reference value of *d*-spacing for the strain calculation (if the value is different from the original).
- Adding ~10% to the original tensile load value and applying this load to the sample. Again, maintaining the new load value, allowing the sample exhibit the steady-state creep for at least 14-16 hours. Continuously measuring the angular position for the selected neutron diffraction peaks.
- Releasing the applied load and measuring the new no-load diffraction peaks' positions.
- Continuing the above steps while adding 10% to the value of the applied tensile load until the eventual sample fracture.
- Continuously recording the total over-all strain, thermal+plastic+elastic, throughout the experiment.

In the technical literature there are several available sets of tensile properties of the A380 alloy tested at various temperatures. For example, tensile properties of die cast alloy A380 have been measured [6] and are listed in Table II. The listed values for yield strength at 200 and 250^{0}C were used in determining the initial load values, as per the experiment procedure outlined above. This starting load values are presented in Table III.

Table II
Tensile properties of die cast alloy A380 at various temperatures [6]

Temperature (C)	Tensile Strength (MPa)	Yield Strength (MPa)	Elongation (%)
150	235	152	5
205	165	110	8
260	90	55	20

Table III
"Starting" experimental parameters

Temperature (C)	Initial Operating Load (MPa)
200	90
250	72

Results and Discussion

Experimental results

The experimental results are presented in Figures 3 and 4. The data show that the elastic strains developed at different crystallographic planes vary with respect to load, temperature, and alloy compositions. These variations can be linked to the creep properties of the alloys under investigation enabling a better understanding of the service performance of these alloys in the automotive powertrain applications.

As presented in Figure 3, at the lower test temperature of 200^{o}C, the (111) crystallographic planes, typically followed by the (311) planes, have the highest level of elastic strains compared to the (200) and (220) planes. An exception to this trend is found in A380 alloy at the second loading stage, where the (220) planes show an increased level of strains similar to those obtained for the (111) and (311) planes.

However, as the testing temperature increased to 250^{o}C (Figure 4), the strains developed at (200) and (220) planes increase greatly, in some cases being at par or even exceeding those obtained for the (111), and (311) planes.

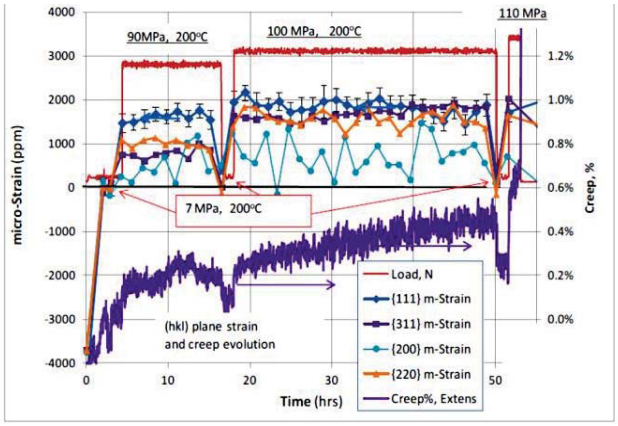
The "base" alloy performance at 200^{o}C was comparable to the other two alloys, but as the temperature increased to 250^{o}C the initial load setting had to be reduced to 45 MPa for this alloys, while maintaining the initial load setting of 72 MPa for the other two alloys. This was due to significantly reduced material strength that led to pronounced elongation of the sample; this sample failed shortly after the load reached 54.5 MPa, while the A380 alloy was still tested at 80 MPa, and the "new" alloy sample reached 88 MPa before failing.

Analysis of the total (thermal+plastic+elastic) creep strains and strain rates exhibited by the three alloys at different experiment conditions allowed for calculations of the power-law stress exponent, further used in the analysis of the creep mechanisms.

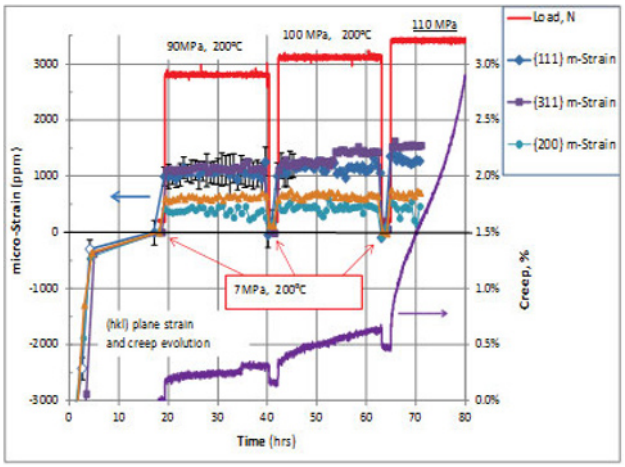
Creep Mechanism

According to the calculations of homologous temperatures, T_m/T (T_m: melting temperature of alloy; T: test temperature of 200-250^{o}C), and normalized stresses, σ/G (σ: applied stress of 45-130MPa; G: shear modulus of 27GPa). The normalized stresses for the alloys under investigation are located in the range of 2.2~2.9 $\times 10^{-3}$ for the applied stress of 45 MPa to 5×10^{-3} for the applied stress of 130 MPa. Thus the main suggested creep mechanism is dislocation creep, which is dominant at normalized stress range of $10^{-4} < \sigma/G < 10^{-2}$ [2].

Deformation mechanism map developed in an earlier study [3] for pure aluminum is shown in Figure 5. The normalized stress here, σ/G, is plotted versus homologous temperature, T_m/T, in order to represent the different creep mechanisms with respect to temperature and applied stress level. The creep mechanisms are presented for the range of temperatures from a high of $T=T_m$ (T_m/T=1) to a low of less than $0.2T_m$ (T_m/T=5). The region applicable to our test conditions is highlighted in this figure. It follows that for pure aluminum the imposed conditions would cause the power-law breakdown (PLB) creep mode, bordering with dislocation creep on one side and pure plasticity on the other.



(a)



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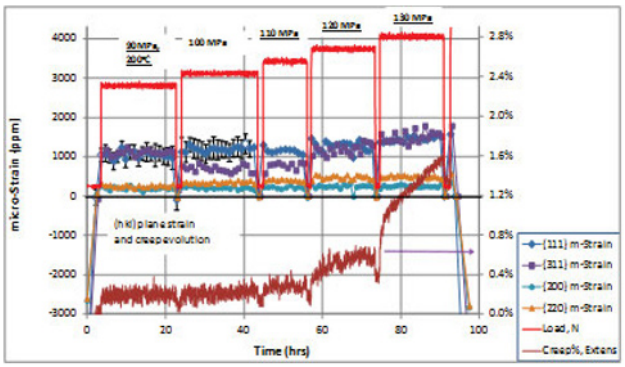
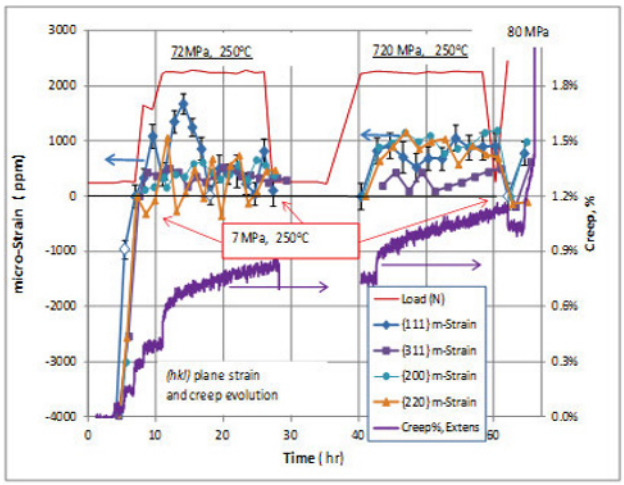
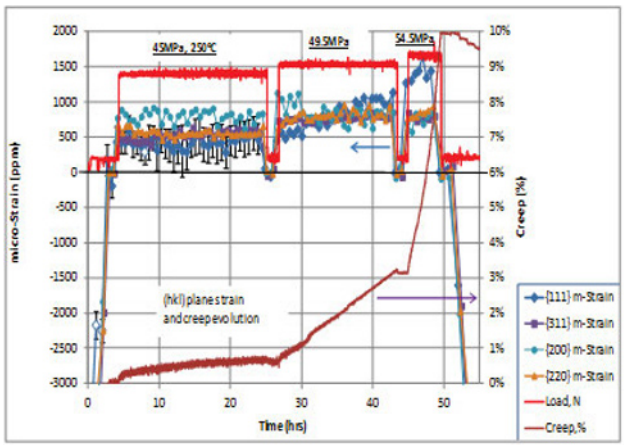


(c)

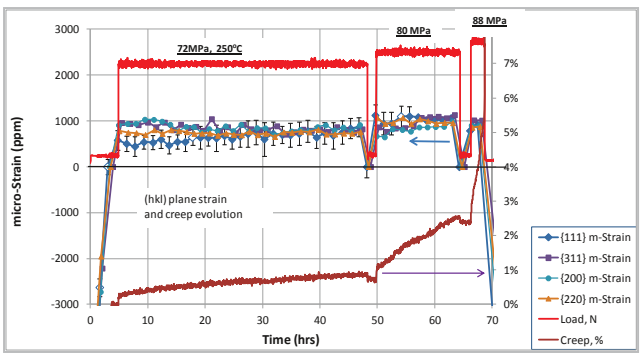
Figure 3. Creep strain (%) and elastic microstrain (ppm) evolution at elected *hkl* planes at 200°C during ascending cyclic loading for A380 alloy (a), "base" alloy (b), and "new" alloy (c)



(a)



(b)



(c)

Figure 4. Creep strain (%) and elastic microstrain (ppm) evolution at selected *hkl* planes at 250°C under ascending cyclic loading for A380 alloy (a), "base" alloy (b), and "new" alloy (c)

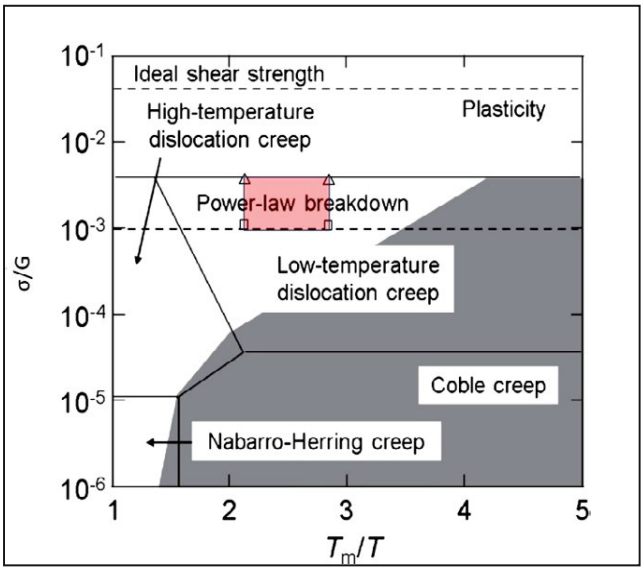


Figure 5. Deformation mechanism map for pure aluminum [3]

Most likely, however, that in the cycling-increasing-load testing conditions of the present study the creep in the tested samples begins as power-law governed dislocation creep and later transitions to PLB creep and finally to plasticity. This hypothesis is supported by the calculated power-law stress exponents (n) for the A380, "base", and "new" alloys, as presented in Figure 6. The governing power-law equation is:

$$\dot{\varepsilon} = A_0\, \sigma^n \exp\left(-\frac{Q}{R\,T}\right), \qquad (3)$$

where $\dot{\varepsilon}$ is the minimum creep rate, 1/s; A_o is dimensionless constant, σ is the applied stress, MPa, Q is the creep activation energy, kJ/mol; R is the universal gas constant, J/(mol K); and T is the absolute temperature, K.

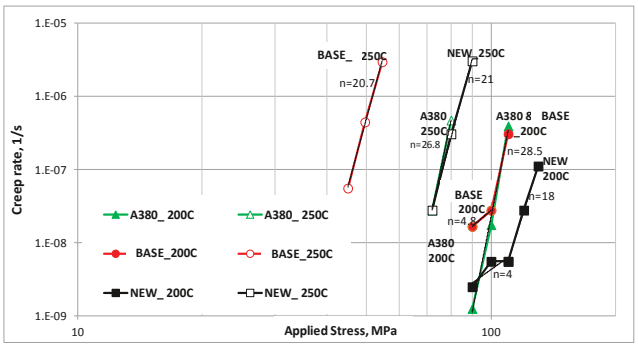


Figure 6. Calculated stress exponents of the tested alloys at the temperatures of 200°C and 250°C

The experiment-based calculated results presented in Figure 6 mostly display large values of stress exponent, typically exceeding 18. According to reference [3], these high values of n represent the PLB creep region, which starts when the value of stress exponent exceeds 5. For $1<n<5$ the creep is characterized by dislocation creep for which the power law (Equation 3) is applicable.

Figure 6 also indicates that there are two exceptions from the typical for this experiment PLB creep ($n>18$): specifically, at the lower temperature of 200°C the "base" alloy exhibited the dislocation power-law creep at applied stresses of 90 and 100 MPa with n =4.8; the "new" alloy performs similarly at the same temperature at applied stresses of 90 and 110 MPa with n=4.

Creep Deformation with Respect to Crystallographic Planes

The experimental results presented in Figure 1 indicate that for the lower temperature of 200°C the load is mostly supported by the (111) and (311) planes, because the measured elastic μ-strains are the largest in the [111] and [311] directions. However, at the higher temperature of 250°C (Figure 2), there is a greater contribution by the (220) and (200) planes to supporting the load.

As reported previously [3], the dislocations can move easier on certain crystallographic planes than on others and in certain preferred crystallographic directions of atoms than in others. The close-packed planes and directions (i.e. the planes and directions having the highest planar and linear atomic packing densities) are the most preferred dislocation slipping planes and directions allowing the dislocations to move easier. In aluminum with its face-centered cubic (FCC) crystal structure, the close-packed planes are (111), which have the highest planar density of 14 atoms/nm^2, and the direction is [110]. The higher planar density of the (111) plane leads to its higher load carrying capacity than the less-packed (220) and (200) planes.

Although the (311) planes have a lower planar density, 8 atoms/nm^2, it also provides a high capacity in supporting the applied load, as shown in Figure 1. This is attributed to the fact that the [311] direction is oriented with difference of just 30° in relation to the [111] direction. The relatively small difference in direction makes the (311) planes able to carry almost the same portion of the load as (111) planes, which is indicated by the similar *hkl* microstrain levels.

During the second and subsequent re-loading steps, the less-packed planes can be activated to carry more of the load, in some cases these planes are strained almost as much as (111) planes. This activation can be attributed to accumulated plastic deformation resulting in grain deformation and re-orientation that may lead to cross-slip from the (111) plane to the (311) and (220) planes. Similar deformation behavior was reported in a previous study [4], in which the cross-slip is activated from the (111) plane onto the (001) planes as the load increases.

Temperature is also a significant factor in the strain. Although FCC metals deform easily by glide on the octahedral planes, such as (111), it is believed that the glide can also take place on less-packed planes such as (220) and (200) at high temperatures. This believe is in agreement with other studies [5-7], which report that above 200°C aluminum deforms easily by glide on the non-

compact family planes of (100), (110), (112) and (113) types and conclude that those planes are thermally activated. The activation enthalpies for less-packed slip planes in aluminum are in the range of 1.2 to 1.9eV depending on which planes are activated [8, 9]. This range agrees with the creep activation enthalpies above 200°C, which range between 1.3 and 1.6eV [8].

It is worth mentioning that the critical resolved shear stress of less-packed planes can be reduced significantly as the temperature increases. For example, in hexagonal close-packed (HCP) metal such as zirconium, the critical resolved shear stress for slip on the $(10\bar{1}0)[\bar{1}2\bar{1}0]$ system is decreased from 10MPa to 0.2MPa, as the temperature increases from 77K to 575K [3]. Accordingly, in the present study the increase of temperature from 200°C to 250°C can result in a significant activation of less-packed crystallographic planes of (220) and (200) enabling them to better support the applied load, as evidenced in Figure 2.

Conclusions

1. Three aluminum alloys have been subjected to *in-situ* creep testing under the high-temperature high-load conditions, typical for automotive powertrain applications. Both total (plastic and elastic) and elastic-only strains have been measured and analyzed. The creep mechanism typical for each stage of incremental loading has been discussed.
2. The main creep mechanism is dislocation creep that starts at the lower load setting that was close to the yield strength and is associated with the power-law-governed creep. As the applied load increased, the creep mode shifted to the power-law breakdown (PLB) that was dominant for most of the testing conditions.
3. For the lower temperature of 200°C the load is mostly supported by the (111) and (311) planes, as the measured elastic μ-strains are the largest in the [111] and [311] directions. These measured strains increase throughout the experiment correlating to the increase in the appplied load. The (220) planes may also become activated, as shown for A380 alloy at 100 MPa.
4. The close-packed planes and directions, eg. {111}, are the most preferred dislocation slipping planes and directions where the dislocations movement is easier. The higher packing density of the {111} planes enable them to better support the applied stress than the less-packed (220) and (200) planes.
5. The [311] direction is only 30° from the [111] direction, and this small angular difference makes the (311) planes able to carry almost the same portion of the load as (111) planes even though the (311) planes have a relatively low planar density.
6. Under certain conditions the less-packed planes can be activated to carry a significant part of the load:
   - accumulated plastic deformation resulting in grains deformation and re-orientation may lead to the cross-slip from (111) to (311) and (220) planes.
   - high temperature may lead to thermal activation of the (200) and (220) glide.
7. The "new" alloy has shown superior performance in conditions simulating an in-service automotive powertrain component, such as an engine head. Its performance was superior in both ultimate strength and total (plastic and elastic) creep, which can be explained in part by activation all the observed *hkl* planes to support the load.

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Light Metals 2016

ALUMINUM ALLOYS, PROCESSING AND CHARACTERIZATION

Corrosion Resistance

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INFLUENCING INTERGRANULAR CORROSION VIA SURFACE TREATMENT

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Keywords: Aluminum, Corrosion, Surface Treatment, 6xxx, chassis parts

Abstract

The amount of aluminum used in cars by the automotive industry has doubled during the last 10 years. Further growth is expected for the following years. Especially 6xxx aluminum alloys, which have the potential to make cars lighter and more economical. The consequential growing demand for high quality aluminum extrusions could be constrained by the problem of intergranular corrosion. The demand for intergranular corrosion resistant aluminum parts is an important issue for the aluminum industry.
TRIMET has evaluated a method, which is used for increasing the corrosion resistance of steel. Shot peening is in use for corrosion prevention of steel parts. The prevention mechanism is the introduction of near surface compressive stresses. Additionally some papers also report positive effects of shot peening on the corrosion resistance of aluminum parts. The paper describes the corrosion testing of surface and heat treated aluminum profiles and presents the results concerning corrosion and applicability.

Introduction

The reduction of intergranular corrosion, especially of 6xxx extrusion profiles is an important topic for the automotive industry. Due to the requirements and the specifications of the automotive industry the increase of the corrosion resistance of 6xxx alloys is essential. Special rapid tests of the automotive industry make the business difficult, too. For example the intergranular corrosion of chassis parts for VW, according to their rapid corrosion test (VW PV 1113) [1], is not allowed to be more than 50µm. Corrosion protection by coating is much too expensive, while intergranular corrosion can be a serious problem for 6xxx-chassis parts. These facts are the trigger for many projects concerning intergranular corrosion reduction of 6xxx alloys.
The target of the project presented here is the reduction of the intergranular corrosion by a surface treatment, which means in this case, shot-blasting. Such a surface treatment could be a possibility to improve the corrosion properties, because the corrosion starts at the surface of the component. Shot-blasting modifies the geometry and the residual stresses of the surface and the near surface areas.
Many chassis parts are produced using 6xxx alloys. One usual industrial process is:

- Aluminum production (primary or secondary production)
- Billet casting
- Billet heat treatment
- Profile extrusion
- Profile heat treatment
- Forming process
- Heat treatment during and after the forming process
- Machining

All these production steps influence the corrosion properties of the final product. Beside the usual solutions of alloy development and heat treatments, one option is the surface treatment of the final chassis part. Shot -blasting and glass bead blasting are mentioned in many publications to reduce intergranular corrosion or stress corrosion cracking of steel and aluminum [2, 3, 4].

Experimental

Experience shows that the intergranular corrosion behavior of the profile after the extrusion process and the heat treatment is comparable to the corrosion behavior of the chassis part profiles that were used in this investigation. The data for the specimen production were:

- Alloy types: EN AW-6060, EN AW-6061, EN AW-6082
- Billet diameter: 290mm
- Casting technology: float/spout
- Billet homogenizing: 580°
- Extrusion profiles: Figure 1, Figure 2, Figure 3
- Profile homogenizing: T4, T5, T6

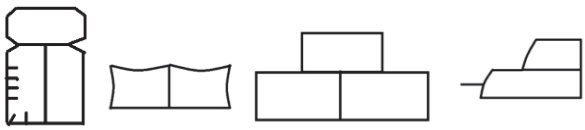

Figure 1: Schematic of the testes profiles

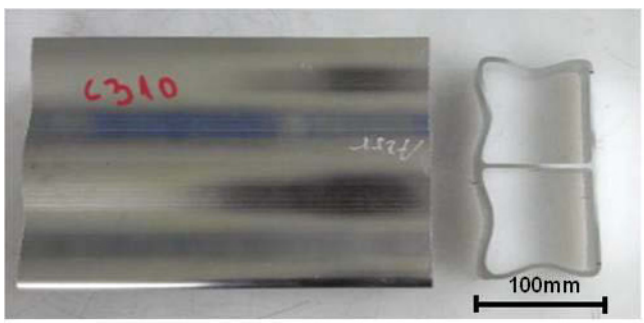

Figure 2: Example for the used profile types

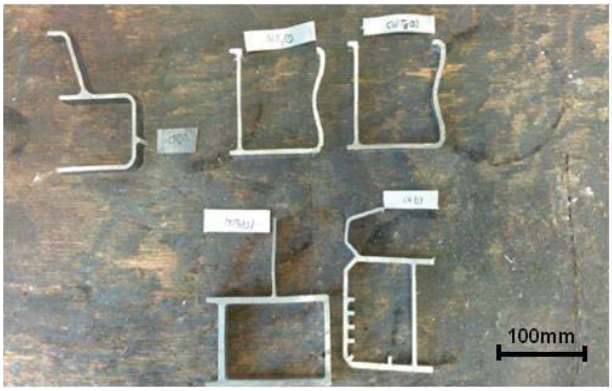

Figure 3: Profile slices prepared for the corrosion testing.

Glass beads were used as the material for the jet blasting to avoid iron accumulation on the aluminum surface. Such iron particles could influence the corrosion properties negatively.
Two different glass bead granulations were tested:

- Coarse: 300 - 400µm
- Fine 40 - 70µm

Both glass bead granulations were used with a pressure of 5.3bar.
The distance between specimen and jet nozzle was 15cm.
The glass bead blasting time was

- 30s
- 60s
- 90s
- 120s.

All surfaces were evaluated with a "Keyence Digital Microscope VHX 600" digital microscope with the additional possibility to measure the surface profile, Figure 4.
Before and after the intergranular corrosion testing the microstructure was investigated. Therefore grain size, phase structure and after the corrosion testing the formation and the depth of the corrosion were examined.
The intergranular corrosion testing was carried out according to PV 1113 [1]:

- The intersections has to be prepared with an iron-free sandpaper (120) cooled with water.
- Surface cleaned with acetone and in an ultrasonic bath with alcohol.
- Test medium
  - 1000 ml H_2O
  - 20 g NaCl
  - 100 ml HCl (25 %)
  - Unique utilization
- The minimum volume of the test medium is 8ml per cm^2 of the specimen surface (specimen completely under test medium).
- Test duration is 2h (+0,5h) at room temperature.
- Cleaning only with flowing water immediately after test duration.

The depth of the corrosion was measured like visible in Figure 5. It is the distance between the surface of the specimen and the deepest corrosion attack. For all types of alloys and surface treatments 10 specimens were tested. 3 micrographs from each specimen were taken for measuring the depth of the corrosion.

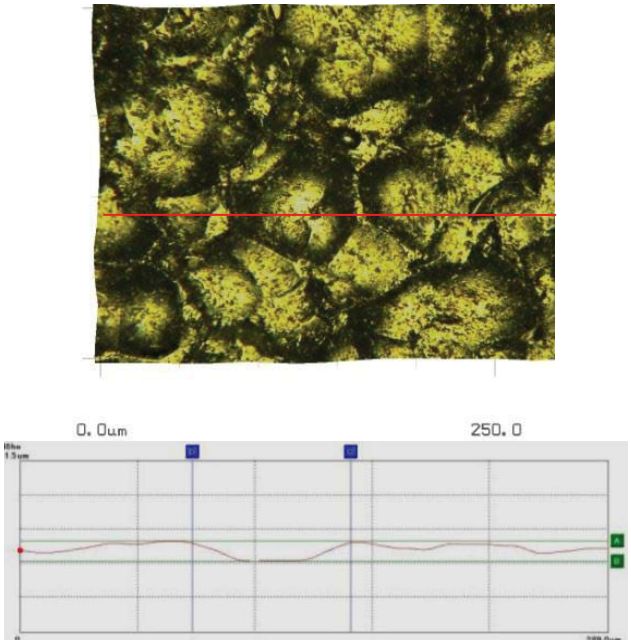

Figure 4: Example for the measurement of a surface profile of a glass bead blasted specimen (EN AW-6060; T6; glass bead granulation 40-70µm; blasting time 90s)

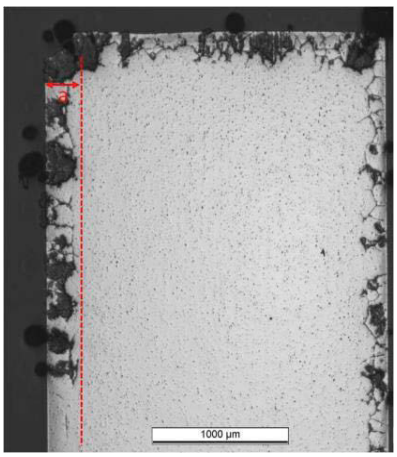

Figure 5: Example for the measuring method for the depth of the corrosion.

Results

The differences between surfaces produced by the two different glass bead granulations and the original unblasted surface are pictured below. Figure 6 shows an overall impression of the two surfaces produced by the two different glass bead granulations. Some examples for more details of the surface can be seen in the pictures below (Figure 7 to Figure 9).
The surface roughness is measured and is exemplified here by the depth of the dents, Table 1 (glass bead blasting time 90s). The surface roughnesses for other glass bead blasting times are nearly identical with the data in Table 1.

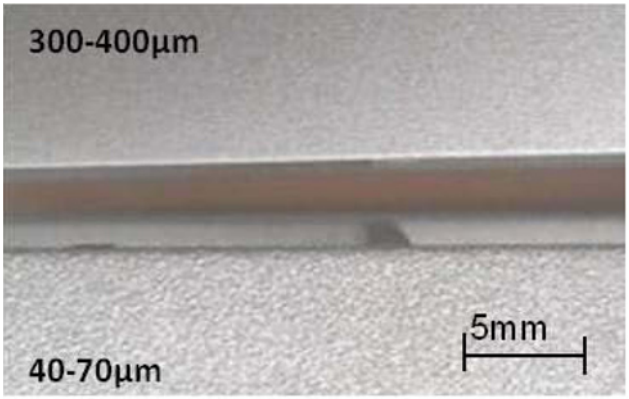

Figure 6: Specimen blasted with fine and coarse glass beads respectively. (Pressure 5.3bar, glass bead blasting time 90s)

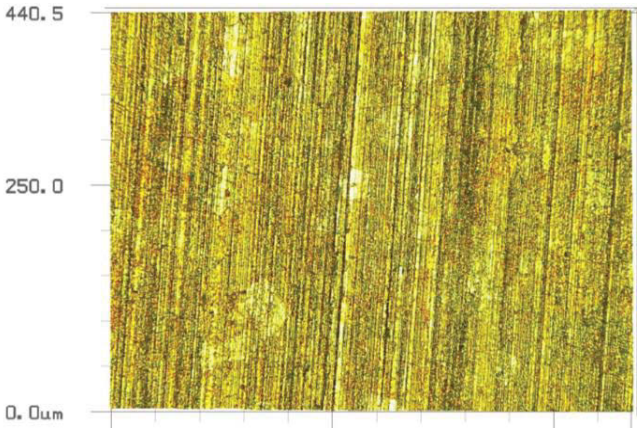

Figure 7: Surface profile of an untreated profile surface.

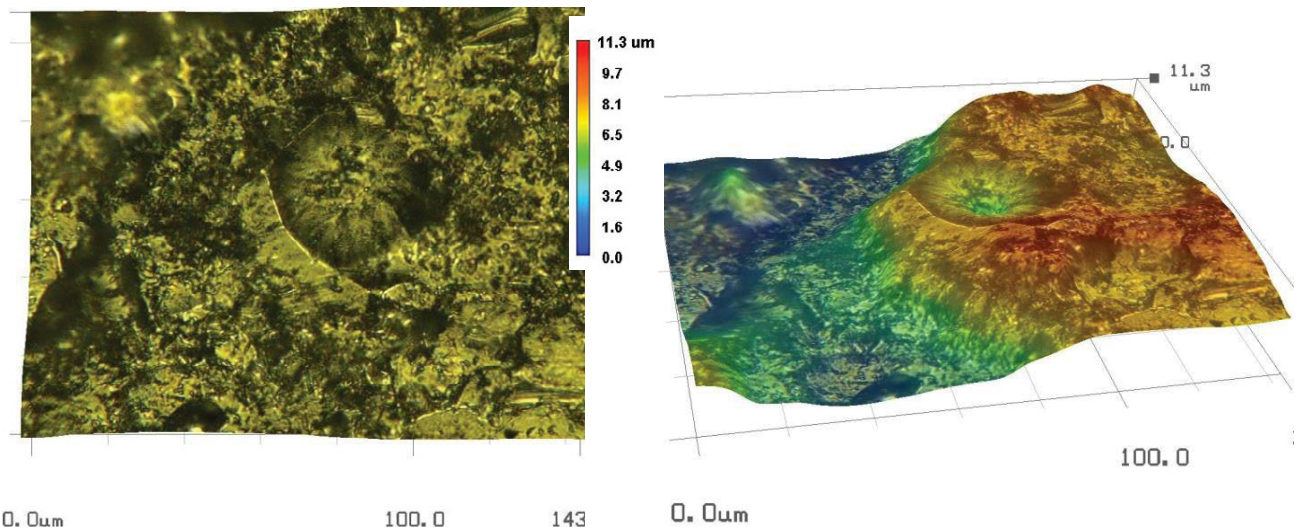

Figure 8: Left: Picture of a glass bead blasted profile surface (Fine glass beads, 5.3bar, 90s glass bead blasting time); Right: Surface profile of a glass bead blasted profile surface (Fine glass beads, 5.3bar, 90s glass bead blasting time)

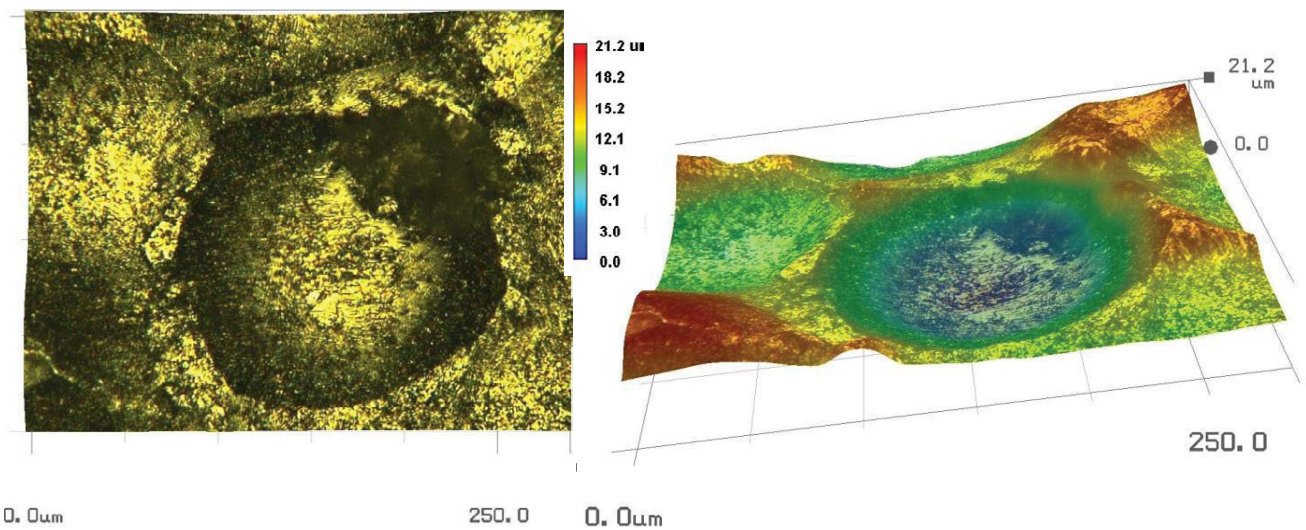

Figure 9: Left: Picture of a glass bead blasted profile surface (Coarse glass beads, 5.3bar, 90s glass bead blasting time); Right: Surface profile of a glass bead blasted profile surface (Coarse glass beads, 5.3bar, 90s glass bead blasting time)

Table 1: Examples for the mean depth of the glass bead blasting dents (all data from trials with 90s glass bead blasting time)

Alloy	Homogenization	Granulation	Mean depth of dents
6060-1	T6	Coarse	18µm
		Fine	10µm
6060-2	T6	Coarse	15µm
		Fine	8µm
6061	T4	Coarse	16µm
		Fine	7µm
	T6	Coarse	30µm
		Fine	9µm

After the corrosion testing, micrographs of slice are taken. Some samples can be seen in Figure 13 to Figure 16. The corrosion depth was measured for all specimen variations. All corrosions depths were compared to untreated reference specimens.
The two most important criteria are:

- Overall impression by visual examination
- Corrosion depth measured in µm

Some examples for the visual impression can be seen in the pictures below (Figure 13 to Figure 16). It is clearly visible that:

- The corrosion is getting worse with longer glass bead blasting times.
- The corrosion is significant.
- The original surface is completely destroyed after the corrosion testing for all specimens with more than 30s glass bead blasting time.

When analyzing the corrosion depth according to PV1113 attention should be paid to one important measurement error. The problem is visible in Figure 10. In the case of a strong corrosion attack, there is the possibility that the original surface is completely removed. Consequently the measurement of the corrosion depth is too small, because the measuring points of the surface are arbitrary.

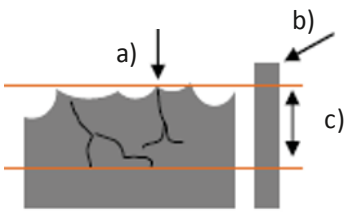

Figure 10: Principle of the measurement error in case of strong corrosion attack (a: surface taken for measurement; b: original surface; c: measured value)

The analysis of the corrosion depths, visible in Figure 11 and Figure 12 shows:

- According to the visual impression the corrosion depth increases with the increase of the glass bead blasting time
- One exception from this behavior due to the destruction of the original surface
- The corrosion results are better for the fine glass bead granulation
- All treated specimen corroded significantly more than the untreated reference specimens

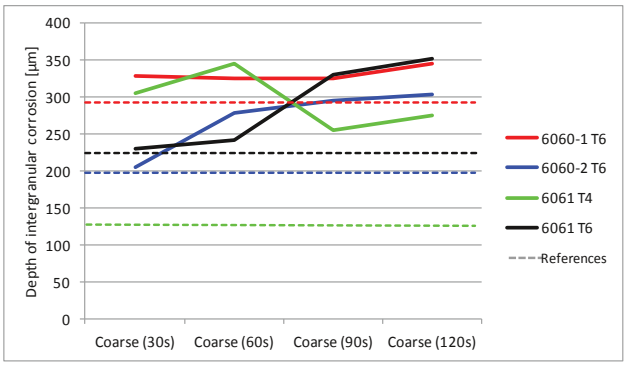

Figure 11: Depths of the intergranular corrosion for different alloys and heat treatment variations (Coarse glass bead granulation)

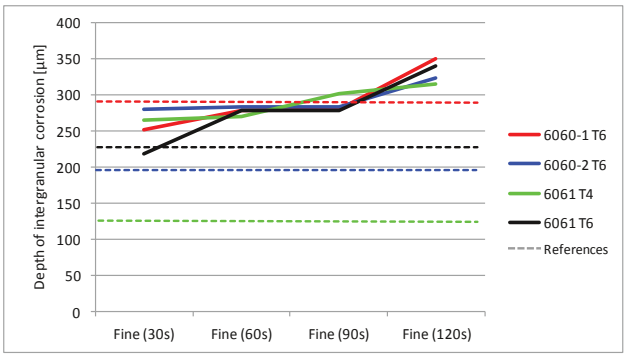

Figure 12: Mean depths of the intergranular corrosion for different alloys and heat treatment variations (Fine glass bead granulation)

Conclusions

In contrast to the results of the dissertation of Christian Dindorf the corrosion attack is getting worse after a treatment with jet blasting technologies, [2]. The glass bead blasting deteriorates the intergranular corrosion properties of all investigated specimens.
This corrosion behavior could be due to:

- The different alloys (6xxx instead of 7xxx)
- The slightly different focus of the projects. The project presented here was focused specially on intergranular corrosion, while the focus of Ch.Dindorf is the stress corrosion cracking. The reduction of the cathodic current density and the residual compressive stresses generated by Ch.Dindorf via bead blasting reduced the stress corrosion cracking and the surface corrosion. Possibly the effect of the residual compressive stresses

is not effective against intergranular corrosion at 6xxx-alloys.This could be one reason for differences between the two studies.

Finally it is safe to say that glass bead blasting is no option for increase of the intergranular corrosion properties of 6xxx alloys used for automotive chassis parts.
Additional trials with other jet blasting parameters could achieve better results, because other parameters yield different compressive stresses. According to [2], [3] and [4] the compressive stresses should yield to the better corrosion properties.
However, if such trials give better results, the outcome of this project shows that the surface treatment via jet blasting is not a reliable method for corrosion prevention. There is a significant risk to influence the corrosion properties negatively.

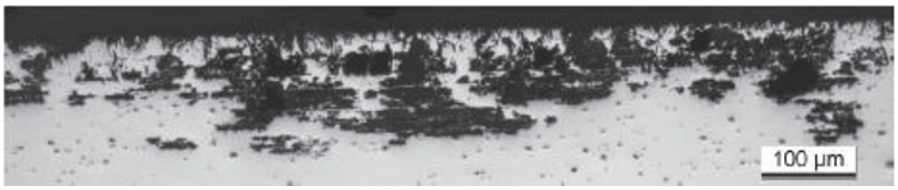

Figure 13: Specimen: C28 T4; corrosion depth: 170µm; glass bead blasting time: 30 sec

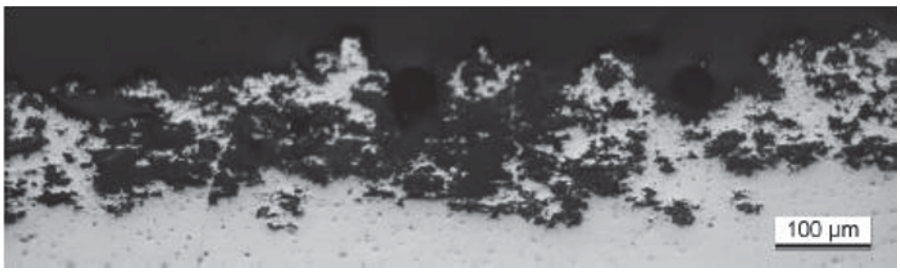

Figure 14 Specimen: C28 T4; corrosion depth: 230µm; glass bead blasting time 60 sec.

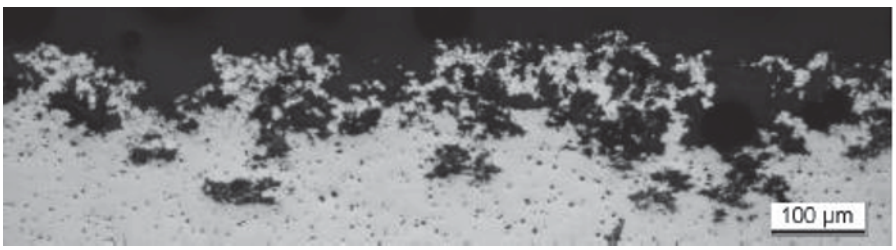

Figure 15: Specimen: C28 T4; corrosion depth: 235µm; glass bead blasting time 90 sec.

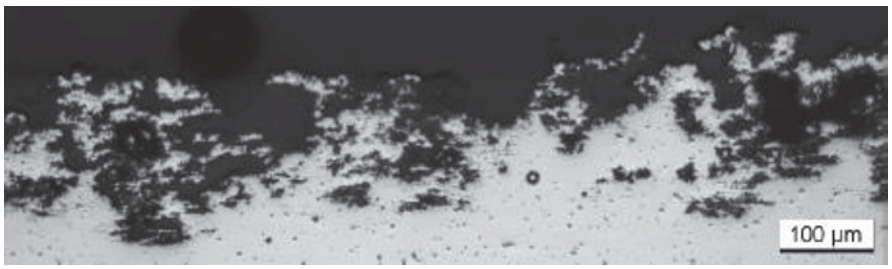

Figure 16: Specimen: C28 T4; corrosion depth: 277µm; 100x; glass bead blasting time 120 sec

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MECHANICAL CHARACTERIZATION AND CORROSION TESTING OF X608 Al ALLOY

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Keywords: Aluminum, Al-6xxx, class-8 truck, A-pillar cover, light-weighting, formability, paint-bake, corrosion

Abstract

This paper describes the mechanical characterization and corrosion testing of X608 Al alloy that is being considered for A-pillar covers for heavy-duty truck applications. Recently, PNNL developed a thermo-mechanical process to stamp A-pillar covers at room temperature using this alloy, and the full-size prototype was successfully stamped by a tier-1 supplier. This study was conducted to obtain additional important information related to the newly developed forming process, and to further improve its mechanical properties. The solutionization temperature, pre-strain and paint-bake heat-treatment were found to influence the alloy's fabricability and mechanical properties. Natural aging effect on the formability was investigated by limiting dome height (LDH) tests. Preliminary corrosion experiments showed that the employed thermo-mechanical treatments did not significantly affect the corrosion behavior of Al X608.

Introduction

In August 2011, the U.S. government announced the nation's first-ever fuel economy and greenhouse gas emissions regulations for the medium- and heavy-duty trucks built after 2014. These regulations require improvements to cut greenhouse gas emissions by about 270 million metric tons and reduce the nation's oil consumption by 530 million barrels over the life of vehicles sold during 2014 through 2018 [1].

Among the prospective light-weighting alternative materials (aluminum, high-strength steel and composites), aluminum has a potential to reduce mass by 40-60 percent [1]. Aluminum (Al) is predominantly used to make engines and wheels, due to its superior performance, durability and environmental advantages. Automakers are now aggressively applying today's advanced aluminum alloys and designs to fabricate other components, as a part of a holistic approach to reduce the vehicle weight to boost the fuel economy, reduce emissions and maintenance costs, and to increase payload and durability.

The average Class 8 truck today uses over 1,000 pounds of Al (about 4% of total tractor weight). With emerging applications, it is possible to reduce the vehicle weight by up to 3,300 pounds [1-3]. Recent research shows that a one ton weight reduction can save up to 3,400 gallons of fuel over the lifetime of a vehicle [4].

The current 6xxx series Al sheet alloys do not have the required formability to allow their use in the manufacture of many aerodynamic cab components and structures. The current automotive grade 6xxx-series Al alloys are limited to the equivalent of 18-20% tensile elongation at room temperature, whereas, an equivalent elongation of over 40% is needed for some of the applications/components being addressed in this work. Another limitation in the widespread use of Al for truck components is the poor post-formed tensile properties (typically < 100 MPa yield strength) after the hot forming processes. Hence, hot-formed Al components are not suitable for applications that require higher post-formed strength for long-term fatigue and dent resistance. The post-formed parts are required to retain sufficient room temperature ductility so that they can be assembled with the rest of the structure via riveting and hemming operations. Therefore, there is a need to develop a manufacturing process that can achieve the required formability and the post-formed mechanical properties in Al alloys.

The objective of this project is to demonstrate lightweight materials manufacturing methods that will increase the efficiency of Class 8 trucks by enabling more widespread use of mass-saving Al, and enabling aerodynamic styling through the use of a new approach to Al sheet forming. Efforts are ongoing to develop forming technology that will enable Al sheet to replace steel sheet and molded fiberglass reinforced composite components, providing individual panel weight savings of approximately 40%.

Determining the right alloy for the body structure and hang-on panels has been the subject of considerable R&D effort [5]. For skin sheet materials, such as A-pillar covers, the focus is on achieving a good balance between formability, strength after the paint-bake, and a high surface quality after the painting operation. Hence, the age-hardenable 6xxx Al alloys with good paint-bake response are the primary choice for these applications [6].

A 6xxx series Al alloy, specifically developed for PACCAR cab structure applications, was provided by Novelis with the designation X608. Recently, PNNL developed a two-step forming process to stamp A-pillar cover at room temperature using Novelis X608 Al alloy. In the first step, the blank was heated to the solutionizing temperature (525°C), water-quenched, and partially formed at room temperature. In the second step, the pre-form was re-heated and water-quenched, as in the first step with the final forming being completed at room temperature. The formed A-pillar covers were subjected to the paint-bake treatment (180°C for 20 minutes). The full-size prototype was successfully stamped by a tier-1 supplier, as shown in Figure 1.

(a) (b)

Figure 1(a): A schematic of A-pillar cover: Length: 650 mm (b) A black-painted A-pillar cover (between the door and windshield)

Mechanical testing was performed on specimens obtained from as-fabricated and paint-baked A-pillar covers. The paint-bake yield strength was ~132 MPa (required ~180 MPa) and failure

strain (engineering) was >23%. Additional information regarding the process development for stamping A-Pillar covers with Al X608, and the resulting mechanical properties can be found here [7].

The objective of the current work is to obtain the following additional information related to the two-step forming process to stamp A-pillar cover at room temperature using Novelis X608 Al alloy: (1) optimum solutionizing temperature and time, and their limits; (2) strength improvement by varying the forming process parameters; (3) natural aging effect on the formability in a commercial production scenario; and (4) the effect of thermo-mechanical process on the corrosion behavior. It is anticipated that this information can serve as a guide to the manufacturers to tailor the forming process in order to achieve the desired strength and corrosion resistance in the finished components.

Al-Mg-Si (6xxx series) alloys are generally considered to have good corrosion resistance compared to Cu or Zn rich high-strength Al-alloys [8, 9]. 6xxx series alloys represent the medium strength class of Al-alloys, because these respond to age-hardening via Mg_2Si precipitation, making them stronger than 5xxx Al-Mg solid solution alloys, but not as strong as 2xxx or 7xxx alloys. In order to obtain an optimal combination of mechanical properties, a small amount of Cu or a large excess of Si compared to the stoichiometric Mg/Si ratio corresponding to the Mg_2Si phase is often used in 6xxx alloys [8,10,11]. This can result in an increased susceptibility to localized corrosion such as pitting and intergranular (IG) corrosion [8,12,13].

Experimental

The alloy selected for this work is Novelis X608 Al alloy (6xxx series) of thickness 1.27 mm, specifically developed for PACCAR cab structure applications. In our previous work [7], the average tensile yield strength of A-pillar covers following a paint-bake (180°C-20 minutes) heat-treatment was ~132 MPa (below the 180 MPa target). Subsequent analysis suggested that the below-target yield strength was likely due to: (1) sheet heat-treatment temperature being below the solutionization temperature, and/or (2) insufficient plastic strain during room temperature forming.

It is very important to determine the right solutionizing temperature and time, besides the limit within which the mechanical properties wouldn't be affected significantly, due to possible variations in the furnace temperature and heating time that can occur in a commercial production scenario. Hence, three temperatures (500, 525 and 545°C) and heating durations (15, 30, and 50 min) were selected for this study. X608 sheet samples were solutionized, water-quenched, and then tensile tested (constant cross-head velocity) using an MTS load frame at a nominal strain rate of 6E-4/s at ambient temperature, according to the ASTM E8 standard [14].

Plastic strain during forming can influence the mechanical properties of the finished component (after paint-bake). Hence, an experiment was conducted to simulate different levels of plastic strain during forming. In the first step, an X608 Al (T4) sheet was machined into ASTM E8 sub-size tensile specimens that were heated to the solutionizing temperature (525°C for 30 min) and then water-quenched. The samples were then pre-strained at room temperature to 2%, 10% and 15% strain (to simulate different levels of plastic strain during forming; nominal strain rate of 5E-4/s) [14]. Finally, the pre-strained samples were paint-baked at 180°C for 20 minutes. The paint-baked samples (i.e. representing a formed component that has been e-coated and paint-baked) were re-tested at room temperature to determine their yield strength.

It is important to study the effect of natural aging in Novelis X608 Al alloy after solutionizing. In a commercial production setting, the sheets could be solutionized and forming operations may be delayed due to gaps in production runs; hence the material invariably undergoes natural aging (days/weeks).

Limited dome height (LDH) tests were performed on 1.27 mm thick alloy sheet to assess its formability attributes after natural aging. For the LDH tests, specimens were prepared from the X608 sheet material in 178x102 mm and 178x178 mm configurations, designated as 7x4 and 7x7, respectively. The specimens were solutionized at 525°C for 30 minutes, and then immediately quenched with UCON quenchant media consisting of 20:80 ratio of oil to water. Three populations (consisting of each specimen configuration) were created, where one population was tested immediately following solutionizing and quenching (specimens designated with an "SQ"); the second was naturally aged for 7 days ("SQ7NA") prior to testing; and the third population was naturally aged for 30 days ("SQ30NA") prior to testing. All specimens were etched with a square grid pattern (2.54x2.54 mm) for the post-deformation strain analysis.

Room temperature LDH tests were performed utilizing a 10 cm round punch along with a 25,000 lbs. clamp test load and a punch speed of 1.27 mm per second. A Teflon sheet along with 3-4 drops of A9 oil was used as a lubricant for all specimen configurations. Post-deformation strains were quantified by analyzing the deformed grids via the ASAME (v.4.1, ASAME Technology LLC) image analysis software.

The effect of thermo-mechanical process on the corrosion behavior of Al X608 was investigated by using accelerated cyclic corrosion tests along with the immersion and electrochemical testing techniques in various environments of interest.

For corrosion evaluations, ASTM E-8 tensile specimens were prepared from Al X608 alloy (T4). The tensile specimens were solutionized (525°C for 30 minutes) and water quenched. Three populations were created from these specimens, where varying subsequent plastic strains were employed (0%, 5%, and 15%) utilizing an MTS load frame at a nominal strain rate of 5E-4/s.[14] Following plastic deformation (or no strain condition), all specimens were then heat treated to simulate a paint bake (PB) and top coat (TC) treatment, where specimens were exposed to 180°C for 20 minutes, and 104°C for 12 minutes, respectively.

Cyclic corrosion tests on Al X608 were performed according to the SAE J2334 standard [15]. Four specimens from each of the three populations were weighed then exposed to cyclic conditions consisting of a humid stage (50°C & 100% RH for 6 h), a salt spray application stage with 5 weight% NaCl (15 min duration at ambient temperature), and a dry stage (60°C & 50% RH for 17 h 45 min) with the test cycle repeated daily. Two specimens from each population were evaluated at the end of 15 and 30 days of exposure. After exposure, specimens were cleaned with mild soap and water followed by alcohol rinse. Final weight measurements were performed after cleaning. Visual analysis was then performed using a stereoscope to evaluate the observed corrosion.

The corrosion potential and localized corrosion behavior of Al X608 was evaluated at room temperature in 3.5 weight% NaCl, by using the standard three-electrode electrochemical corrosion testing technique. Graphite rod was used as the counter electrode, test specimen was the working electrode and the saturated calomel electrode (SCE) inside a Luggin probe was used as a reference electrode. A Potentiostat (Gamry Instruments, Warminster, PA) and electrochemical corrosion cell (Ametek/Princeton Applied Research, Oak Ridge, TN) that holds the test specimen (1 cm^2 exposed surface area) in a 900 ml of test solution were utilized for this study. The accuracy of the electrochemical test equipment was evaluated according to the ASTM G5 standard using Type 430 SS (UNS S43000; obtained from the Metal Samples Company, Munford, AL) in 1.0 N H_2SO_4 [16].

The pre-strained tensile specimens prepared for the corrosion study as mentioned earlier, along with the as-received (AR) shim, and an AR shim subjected to solutionizing and quenching only (SQ) were also employed for the electrochemical testing. Thus, electrochemical testing was performed on the following four samples employing 2 specimens per condition: (1) AR (2) SQ (3) SQ + 0% strain + PB + TC and (4) SQ + 15% strain + PB+ TC. A hydraulic press was used to punch discs of diameter 15.8 mm. The test discs were prepared for electrochemical testing using 320 and 600 grit SiC paper (wet grinding and polishing using water), followed by cleaning using DI water, degreasing using ethanol, rinsing using DI water and air drying. The test specimen was then quickly inserted into the corrosion cell and the open-circuit specimen potential (corrosion potential) was recorded after 55 min of immersion. After 1 h of immersion, potentiodynamic polarization was performed with a potential sweep rate of 0.6 V/h, and the corresponding changes in the current density were recorded in the form of a potential (E) versus current density (I) plot, commonly known as the polarization diagram, as per the ASTM G5 standard [16]. The corrosion behavior of tested specimens was evaluated using a stereo microscope.

Results and Discussion

PNNL recently developed a two-step forming process to stamp A-pillar covers at room temperature using Novelis X608 Al alloy. Mechanical testing was performed on specimens obtained from the as-fabricated and paint-baked A-pillar covers. The paint-bake yield strength (YS) was ~132 MPa (required about 180 MPa) and failure strain (engineering) was >23% [7].

Subsequent analysis suggested that the below-target YS was likely due to: (1) sheet heat treatment temperature being below the solutionization temperature, and/or (2) insufficient plastic strain during room temperature forming. Incomplete solutionization can reduce the precipitation strengthening potential, while insufficient plastic strain can reduce the kinetics of the precipitation during the paint-bake heat-treatment. Consequently, the full strengthening potential of the precipitation hardening phase could not be utilized in the prototype A-pillar covers.

Determination of the Solutionization Temperature and Time:
Identifying appropriate solutionizing temperature and time is important for pre-heating during forming to ensure the maximum potential for subsequent paint-bake heat treatment. A "fully" solutionized material would have the highest yield strength (due to solid-solution strengthening) relative to a partially solutionized material. In addition, efforts were made to determine the parameter range within which the mechanical properties wouldn't be significantly affected, due to the possible variations in the furnace temperature and heating time that can occur in a commercial production scenario. Figure 2 below show the results of the study that involved three temperatures (500, 525 and 545°C) and three heating durations (15, 30, and 50 minutes). The room temperature quasi-static yield strength of the samples solutionized at 500°C ranged between 110-115 MPa, while that for the samples heated at 525°C and 545°C were found to lie between 120-125 MPa. The failure strain exceeded the physical limits of the extensometer and hence, extensometer strain >23% could not be recorded. As a result, the specimens "appear" to fail abruptly at 23% strain. Thus, heating at 525°C for 30 minutes was selected as the "fully solutionized" condition.

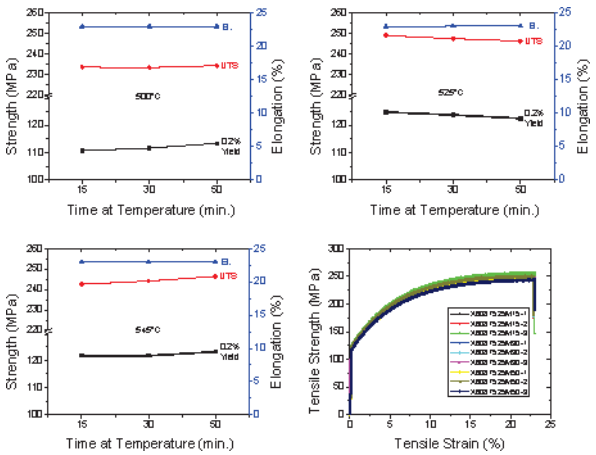

Figure 2: Mechanical Properties of Al X608, as a function of solutionization temperature and time.

Effect of Pre-strain on the Paint-bake Response:
Efforts were made to identify an apt plastic strain during room temperature forming to enhance the precipitation kinetics during the paint-bake heat-treatment, and eventually improve upon the paint-bake yield strength obtained in the previous study from about 132 MPa to 180 MPa [7]. As shown in Figure 3 below, a pre-strain of about 10%, prior to the paint-bake would be sufficient to produce yield strength greater than 180 MPa. These results highlight the combined effects of proper solutionization heat treatment and room temperature pre-strain on the final yield strength of X608 Al sheet alloy. These results can serve as a guide to a manufacturer to tailor the forming process to achieve the desired strength levels in the finished components.

Effect of Natural Aging on the Formability:
LDH tests were performed to study the effect of natural aging in X608 Al alloy's formability. In the LDH testing, the 7x4 and 7x7 specimen configurations were designed to undergo plane strain and biaxial stress conditions, respectively. In Figures 4 and 5, the measured safe and unsafe strains from representative specimens are plotted in the forming limit diagram. Figure 4 compares the SQ population (no aging) and SQ7NA population (naturally aged for seven days). In both specimen configurations, a slight increase in the formability was observed in the solutionized and quenched sheet that has been naturally aged for 7 days.

While comparing the SQ population with the SQ30NA population (naturally aged for 30 days), a slight decrease in the formability was observed with 30-day aging, as shown in Figure 5. No

significant difference in the formability was observed between the 7 and 30-day naturally aged specimens. These observations indicate that once the X608 sheet has undergone solutionizing and quenching, any delay in the stamping process within a 30-day window should not affect its formability, and the material will not have to undergo another solutionizing treatment.

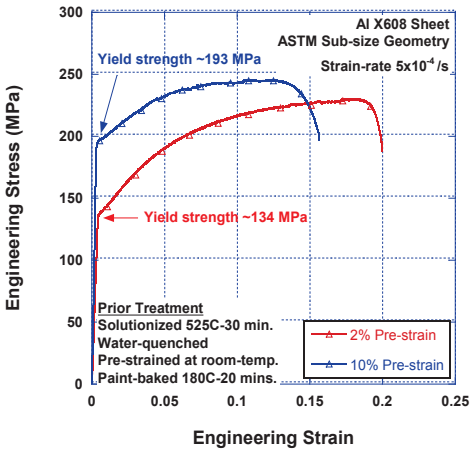

Figure 3: Stress-strain behavior of Al X608, as a function of plastic strain during room temperature forming.

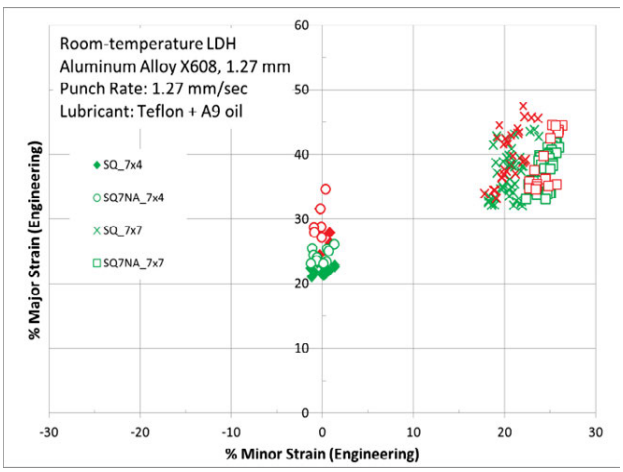

Figure 4: Forming limit diagram comparing SQ and SQ7NA specimens for both configurations. Safe (colored green) and unsafe (colored red) strains are shown.

Cyclic Corrosion Testing:

At the end of exposure in a cyclic corrosive environment, some degree of pitting and intergranular (IG) corrosion appear to be occuring in all specimens. Figure 6 below is a representative image of the corrosion observed. Although 6xxx series aluminum alloys typically have good corrosion resistance, intergranular corrosion resistance can vary depending on the chemical constituents present, such as silicon and copper. In these precipitation hardened alloys, the primary precipitate is Mg_2Si. Alloys with excess silicon (i.e. more than the stoichiometric amount needed to form Mg_2Si) tend to be more susceptible to IG corrosion due to silicon segragation to the grain-boundaries, thus producing a silicon-depleted region along the boundaries [17]. This region becomes anodic to the boundary and the matrix. In addition, depending on the copper content, IG susceptibility can also increase, if copper-depleted regions develop along the grain boundaries.

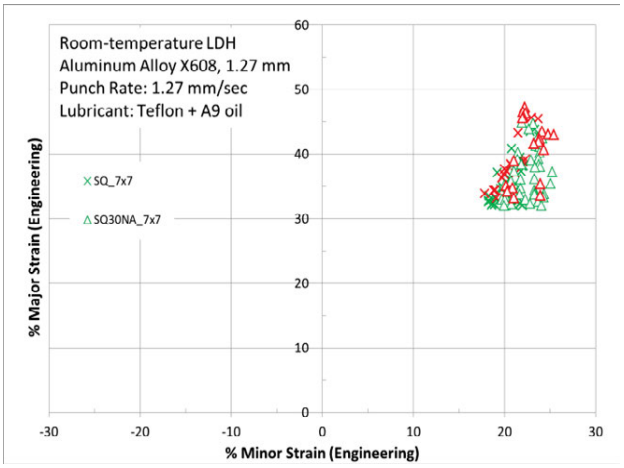

Figure 5: Forming limit diagram comparing SQ and SQ30NA specimens for 7x7 configuration. Safe (colored green) and unsafe (colored red) strains are shown.

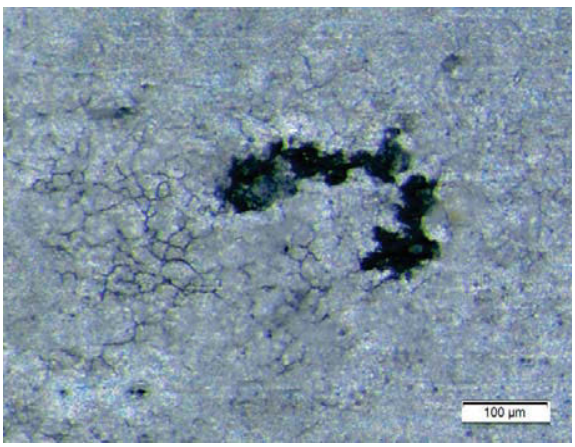

Figure 6: A representative image of the corrosion observed on all specimens under cyclic corrosion conditions. Image is of an SQ specimen (no strain) after 30 days of cyclic exposure.

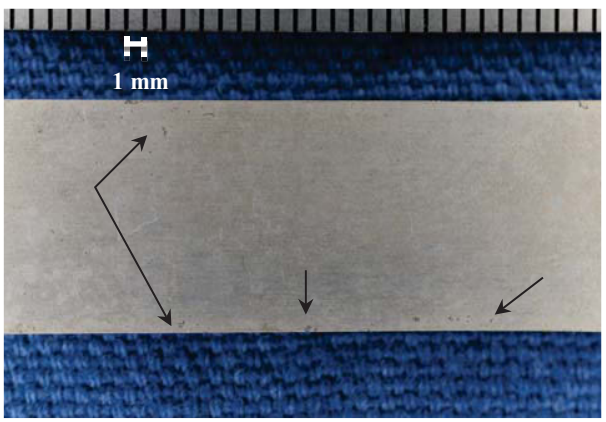

Figure 7: A representative image of the gage region of an SQ specimen (15% strain) after 30 days of cyclic exposure with arrows pointing to areas of corrosion present on the surface.

The specimens exposed to the cyclic corrosion for 30 days showed that there was no measureable weight loss, and the degree of corrosion varied slightly. No significant differences were observed between specimens (prestrain = 0% and 15%) after 15 and 30 days of exposure. Pits and IG corrosion were observed sporadically along the specimen gage region. The most corrosion was observed among specimens that were strained at 15%, and exposed to 30 days of cyclic corrosive conditions (see Figure 7). To fully understand the severity and extent of corrosion present, metallographic examination is needed to identify and characterize sites that are preferentially attacked in the X608 alloy specimens.

Electrochemical Testing:

Before performing electrochemical (EC) testing, the accuracy of the equipment was evaluated using Type 430 SS in 1N H_2SO_4 (calibration). The EC testing was performed on the following four samples (Al X608) employing 2 specimens per condition in 3.5 weight% NaCl solution: (1) AR (2) SQ (3) SQ + 0% strain + PB + TC and (4) SQ + 15% strain + PB + TC.

When a metal is immersed in a test solution, EC reactions characteristic of the metal-solution interface occur at the surface of the metal, causing the metal to corrode. These reactions create an electrochemical potential, called the corrosion potential or the open circuit potential (measured in mV), at the metal-solution interface. Since the corrosion potential is determined by the specific chemistry of the system, it is a characteristic of the specific metal-solution system. The corrosion potential of test specimens was recorded after 55 min of immersion. There was no significant difference in corrosion potential between the four tested samples. For all four tested conditions, the corrosion potential (E_{corr}) changed during the first few hundred seconds, and then stabilized to about -730 mV (V_{SCE}), as shown in Figure 8.

After 1 hour of immersion, potentiodynamic polarization was performed on these samples with a potential sweep rate of 0.6 V/h, and the corresponding changes in the current density were recorded in the form of a potential (E) versus current density (I) plot, as shown in Figure 9. The alloy did not show any passive region in the tested conditions. There was no significant difference in the corrosion current (I_{corr}) between the four tested samples, and the values were between 1E-6 to 9E-7 A/cm^2.

Based upon the E_{corr} and I_{corr} values, it appears that prior thermo-mechanical treatments employed here did not change the corrosion behavior of Al X608. However, visual observation of the tested specimens revealed pitting and IG corrosion in all four samples, as shown in Figures 10-13, and the extent of attack appears to be least severe in “AR”, followed by “SQ”, “SQ + 0% strain + PB + TC” and “S + WQ + 15% strain + PB + TC” samples. Additional R&D work will be beneficial to comprehensively understand the corrosion behavior of Al X608 in terms of alloy composition (versus other 6xxx alloys) and various thermo-mechanical treatments.

Conclusion

Recently, PNNL developed a two-step thermo-mechanical process to stamp A-pillar covers at room temperature using Al X608, and the full-size prototype was successfully stamped by a tier-1 supplier. The present study was conducted to obtain additional important information related to the influence of thermo-mechanical processing on subsequent mechanical properties,

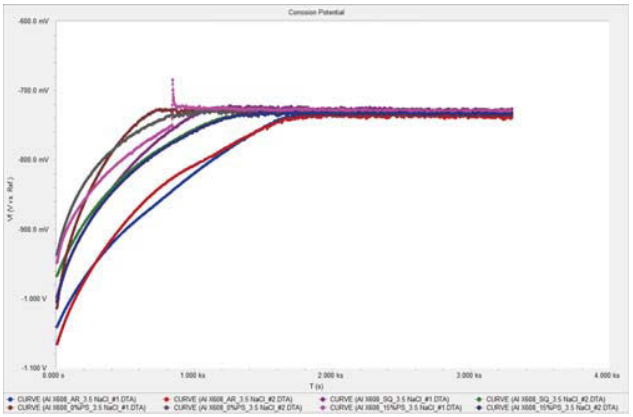

Figure 8: Corrosion Potential of Al X608 in four different conditions using 3.5 weight% NaCl: (1) AR (2) SQ (3) SQ + 0% strain + PB + TC and (4) SQ + 15% strain + PB + TC

Figure 9: Potentiodynamic Polarization Curves of Al X608 in four different conditions using 3.5 weight% NaCl: (1) AR (2) SQ (3) SQ + 0% strain + PB + TC and (4) SQ + 15% strain + PB + TC

Figure 10: Al X608 (AR) - Pitting and IG corrosion observed

Figure 11: Al X608 (SQ) - Pitting and IG corrosion observed

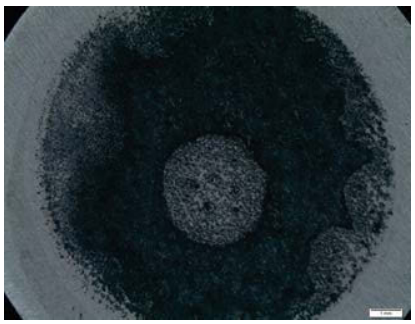

Figure 12: Al X608 (SQ + 0% strain + PB + TC) - Pitting and IG corrosion observed

Figure 13: Al X608 (SQ + 15% strain + PB + TC) - Pitting and IG corrosion observed

formability and corrosion behavior of Al X608. The primary conclusions of this work are as follows:

- Pre-heating at 525°C for 30 min was identified as the appropriate solutionizing schedule for X608 Al to maximize its precipitation strengthening potential during subsequent processing.
- A certain level of plastic strain was found to be necessary to enhance the paint bake response of X608 Al. A pre-strain of about 10% was found to be sufficient to produce a post-paint-bake yield strength greater than 180 MPa.
- Effect of natural aging (up to 30 days) on formability was studied and the results indicate that starting with a solutionized and water quenched sheet, natural aging within a 30-day window has minimal effect on its room-temperature formability.
- Preliminary corrosion experiments showed that the employed thermo-mechanical treatments did not significantly affect the corrosion behavior of Al X608.
- The results of this study can serve as a guide to manufacturers to tailor their thermo-mechanical process to achieve the desired formability and strength in the finished components.

Acknowledgments

The PNNL is operated by Battelle Memorial Institute for the U.S. Department of Energy under contract DE-AC05-76RL01830. This work was sponsored by W. Joost in association with the U.S. Department of Energy, Office of Vehicle Technologies, as part of the Lightweight Materials program. This work was done in collaboration with PACCAR, Magna International-Stronach Center for Innovation and Novelis, and the authors are grateful for the technical discussions and assistance provided by K. Dunn and N. Larson (PACCAR); B. Kokosza & B. Shulkin (Magna International-SCFI); and A. Brooks (Novelis). The authors also acknowledge the technical support provided by the PNNL staff: M. Smith, K. Mattlin, M. Dahl, T. Roosendaal and A. Guzman.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SIMULTANEOUS IMPROVEMENT OF MECHANICAL AND CORROSION PROPERTIES OF ALUMINUM ALLOYS

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Keywords: Aluminum, Nanocrystalline, Corrosion, Mechanical properties

Abstract

The use of Al-alloys is limited in many applications due to a dramatic deterioration of the corrosion properties with any effort made to increase the strength. The hardening phases, uniformly distributed fine intermetallics, act as electrochemical heterogeneities and promote galvanic interactions, thus causing localized corrosion. The chemical composition, size, number, and distribution of the intermetallics in the matrix govern the corrosion of Al alloys. Refining the size and doping of intermetallics via microalloying have been found effective in improving corrosion resistance of Al alloys. This paper presents an overview of recent research on: 1) understanding the influence of intermetallics on corrosion of Al alloys and methods to optimize corrosion and mechanical properties by controlling intermetallic formation via advanced production and post-production processing techniques, and 2) development of ultra-high strength and corrosion resistant Al alloys by combining suitable alloying additions and alloy production techniques.

Introduction

Aluminum alloys are among the most versatile metallic materials for engineering parts in different sectors such as the aircraft and construction industry. The wide range of application comes from the ability of aluminum alloys to vary their mechanical properties by a number of production and post-production processes. However, one of the most viable methods to strengthen these alloys, age hardening, involves the appearance of second phase particles which increases the susceptibility to localized corrosion [1]. With the exception of a few elements such as magnesium, the solubility of most alloying additions in aluminum at room temperature is low (less than 1%) [2, 3]. Therefore, in an attempt to generate the most optimal mechanical properties, most commercial alloys exceed the solubility limit (at room temperature) of their alloying elements which generate uniformly distributed fine secondary phase particles during an age hardening processes that provide precipitation strengthening. Depending on their chemical composition, these secondary phase particles are likely to exhibit significantly distinct electrochemical properties than that of the aluminum matrix thus causing galvanic interactions and initiation of localized corrosion [4]. The influence of secondary phases on corrosion of Al alloys also depends upon spatial distribution and size of these particles [5, 6].

In precipitation strengthened Al alloys, an increase in strength deteriorates the corrosion performance significantly, as the precipitation strengthening process causes significant increase in electrochemical heterogeneities. Figure 1 shows a graph of pitting potential (E_{pit}) vs compressive yield strength for different aluminum alloys, a clear tendency of commercial alloys to be more susceptible to pitting as the strength of the alloy increases can be observed [7–9]. Most recently, it has been shown that the simultaneous improvement in corrosion and mechanical properties of Al alloys is possible by combining novel alloying addition and processing technology, i.e., high-energy ball milling [8, 10, 11]. Figure 1, shows that Al and Al-20Cr alloys as prepared by high-energy ball milling followed by spark plasma sintering (SPS) exhibited ultra-high strength and superior corrosion resistance. Various strategies that could increase the strength and corrosion of aluminum alloys simultaneously will be discussed in this paper.

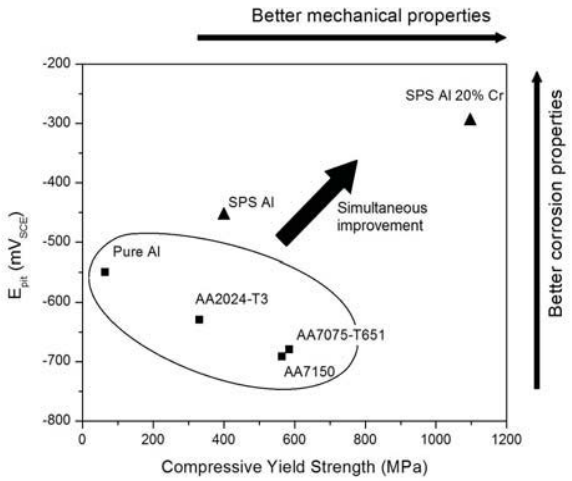

Figure 1. Pitting potential (E_{pit}) vs compressive yield strength in several aluminum alloys [7–9]. Most of the commercial alloys show a decrease in E_{pit} with increase in strength. Simultaneous improvement of E_{pit} and yield strength is exhibited by novel alloys, SPS Al and SPS Al20%Cr.

The effect of several metallurgical factors such as grain size, second phase particles size, chemical composition and spatial distribution is also reviewed in this paper. Special attention is given to the role of size, distribution and chemical composition of secondary phases as these factors have great effect on both mechanical and corrosion properties. A strict correlation may not be possible due to the fact that individual processes intrinsically affect many factors simultaneously, however, a general notion may be inferred from a numerous studies performed at different conditions for various Al alloys. In addition, advanced technologies in aluminum alloy production for simultaneously improving mechanical and corrosion properties are presented.

Effect of grain size

The main purpose of grain refinement is to improve strength and wear resistance [12–14]. Grain refinement leads to a significant increase in grain boundary volume fraction. For example, volume

fraction of grain boundaries in a material with a grain size of 10 nm was estimated to be 0.3 [15–18]. Grain boundaries are high-energy sites with higher reactivity and therefore grain refinement is expected to accelerate the corrosion, which has in fact been reported for several material/environment systems [5, 19–23]. On contrary, increasing the reactivity by grain refinement is also expected to facilitate the passivation and therefore corrosion performance of material showing passive behavior is expected to increase [10, 24–26]. The Al and Al alloys are passive in most environments, therefore, an increased corrosion resistance due to the grain refinement is postulated and also reported in literature [8, 11, 19, 27–29]. However, researchers have studied the influence of grain size on corrosion properties of several materials (including Al) in various environment and no clear relationship has been established [12, 19, 24], which could be attributed to the fact that the process of grain refinement also influences several other metallurgical parameters (i.e., size and distribution of secondary phases, subsequent precipitation and solid solubility, residual stresses) and therefore a detailed discussion in the context of Al alloy and production technique used for grain refinement is imperative.

Grain refinement in Al alloys could be achieved by numerous methods. For instance, addition of refinement agents, i.e., Al-Ti master alloy in casting process is one of most common methods of grain refinement. Added Al-Ti master alloy forms an insoluble Al_3Ti intermetallic which act as a nucleant for induced heterogeneous solidification [2, 30] and leads to significant grain refinement. However, evidence has been found that the presence of refining Ti and micro-segregation in nearby regions can have a detrimental effect by causing localized galvanic cells resulting in higher susceptibility to pitting corrosion [22].

One additional process where grain refinement can be achieved is by mechanical working [3, 12, 30]; cold working, rolling and extrusion are basic methods to produce directional grained materials which will behave differently from other grain morphologies such as those obtained by electrodeposition or severe plastic deformation [24]. Several studies have been performed on the alloys produced by equal channel angular pressing (ECAP). The obtainable grain size depends upon certain parameters such as temperature and number of passes, for a commercial pure aluminum the grain size decreases to 500 nm at room temperature after 16 passes. Song et al. [29] investigated the corrosion behavior of this alloy and reported that both pitting potential and corrosion current density decreased significantly. This behavior was attributed to the formation of a more stable and protective passive film due to the enhanced reactivity caused by grain boundaries. Chung et al. investigated the effect of the number of ECAP passes on corrosion properties of AA 1050 and reported increase in pitting potential and improved passive layer stability as measured by EIS at the passivation and breakdown states. Improved corrosion performance was attributed to the reduction in size of silicon rich impurities caused by ECAP, rather than the grain size.

ECAP results in Al alloys with grain size of ~ 100 nm, usually termed as ultrafine grain materials. Al alloys with smaller grain can be produced by the consolidation of high-energy ball milled powders, such as back-pressure [31] and forward extrusion [32]. Various Al alloys, i.e., AA 1050 [33], pure industrial Al [29] and Al-Mg [34] as produced by high-energy ball milling have been reported to exhibit improved corrosion properties.

Nanocrystalline materials generally have lower ductility compared to their coarser-grained counterparts, the decrease in ductility has been attributed to the limited dislocation activity [35, 36]. Some of nanocrystalline Al alloys are reported to possess excellent ductility, which indicates that the lack of ductility is not an inherent property of nanocrystalline materials albeit it could be introduced due to the defects introduced during processing [8, 37, 38]. Moreover, significant ductility while maintaining high yield strength could also be achieved by polydispersed grain size distribution. For instance, Witkin et al. reported an increase in ductility in nanocrystalline Al-7.5Mg alloy by incorporating unmilled alloy powder of the same composition [39].

Effect of chemical composition of intermetallics

Aluminum can produce a vast amount of different second phase particles due to a relatively low solubility of most alloying elements in it. Some intermetallics are purposely generated in order to increase mechanical properties[1, 12, 40] by means of precipitation strengthening. Most commonly found intermetallics in commercial alloys include Al_2CuMg, Al_3Fe, $MgZn_2$, Al_2Cu, Mg_2Si [41], the appearance of these species vary depending on the chemical composition and heat treatment of the alloy. The influence of intermetallics on corrosion of an Al alloy depends upon their electrochemical characteristics, governed by composition and size. Intermetallics with the more noble potentials are not necessarily those who can sustain largest cathodic currents and similarly for the most anodic particles are not necessarily those who yield the higher anodic currents. Moreover, individual studies focused directly on synthesized intermetallic particles of interest have been performed in order to characterize their electrochemical behavior and evaluate their effect when interacting with a bulk aluminum matrix [42–46].

Intermetallics with similar corrosion potentials to the matrix are preferred and usually yield the highest corrosion properties [1, 40]. The corrosion potential of Al-Mn intermetallics has been reported be close to that of Al and therefore Al-Mn was reported to be one of least deleterious intermetallic [47–49]. Zamin [47] reported that increasing the Mn/Fe ratio would result in increased corrosion resistance in various cast alloys containing Mn, Fe, Si and minor additions of Mg.

Some of the intermetallics, such as Al_4Sr and $Al_{11}Nd_3$ were reported to cause no significant influence on pitting corrosion [50–52]. Although electrochemical characteristic of such intermetallics are not well investigated, their presence is expected to not cause localized corrosion. Incorporation of certain elements in secondary phases was reported to improve corrosion performance by moderating the reactivity. For instance, incorporation of Cu to $MgZn_2$ and Mg_2Al_3 phases was reported to improve the corrosion performance of Al-Cu-Mg and Al-Mg alloys [53–55]. Similarly, incorporation of Mn, Cr and Si to Al-Fe intermetallics was reported to decrease their deleterious influence on corrosion performance of Al alloys [8, 11].

Effect of second phase particle size and distribution

Second phases present in Al alloys act as electrochemical heterogeneities, rendering alloys prone to various forms of

localized corrosion. The secondary phases could be in the form of 1) uniformly distributed fine precipitates which provide strength to age hardenable alloys, 2) dispersoids, and 3) constituent particles which come from impurities of the alloy that are often combined with alloying additions [2]. These heterogeneities are also the cause of electrochemical potential differences leading to localized corrosion. The nature of secondary phases such as constituent particles, intermetallic and dispersoids is critical for the assessment of the impact to corrosion properties.

Intermetallics, i.e. Al_3Fe [56], Al_7Cu_2Fe [57], $MgZn_2$ [58], Mg_2Si [46], Mg_2Al_3 [59] are found detrimental to corrosion properties in commercial alloys such as 7075-T6 and 2024-T3 [60]. A decrease in size and number of constituent particles is reported to cause improved corrosion performance. For instance, Dorin et al. [61] investigated the influence of size and shape of Fe intermetallics in Al-0.1 wt %Fe and Al-2wt. % Fe alloys via potentiodynamic polarization and immersion tests in 0.1 M NaCl. A significant improvement in corrosion properties was reported due to the decrease in size and number of intermetallics. The influence of the surface area fraction of the of iron intermetallics on the pitting potential and corrosion current density for an Al-Fe is presented in Figure 2, which clearly demonstrate that as the area fraction increases, the overall corrosion properties decrease having a critical point at about 10 percent. Iron is known to be especially detrimental to structural-sensitive properties of an aluminum alloy [62], but it can also improve wear resistance and reduce friction coefficient in nanocrystalline structure [63]. The detrimental behavior of surface area of intermetallics to corrosion properties can be extrapolated to most alloying elements, as heterogeneous structures are the origin to surface potential differences and micro-galvanic effects.

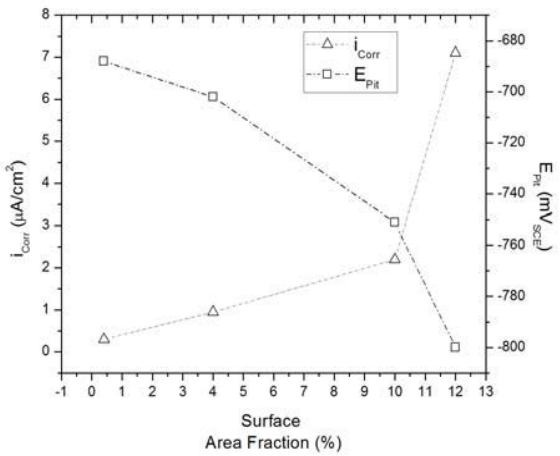

Figure 2. Corrosion current density (i_{corr}) and pitting potential (E_{pit}) for Al-2Fe alloys as a function of surface area fraction of intermetallics [61].

More recently, the influence of chemical composition, size, and distribution of the intermetallics on corrosion and mechanical properties have been investigated for a range of alloy compositions. The impact of intermetallics upon ductility is as expected from age hardening mechanisms, where it is often possible to increase strength at the expense of ductility. This effect becomes evident when comparing a material undergone different heat treatments; for example, artificially aged 2024-T8 has higher yield strength with lower ductility and fracture toughness than that of the naturally aged 2024-T3 [64]. Similarly, results have been reported for conventionally cast Al-Si-Cu and Al-Si-Mg alloys that the increase in size of β-iron intermetallics (Al_5FeSi) deteriorated the ductility and ultimate tensile strength of the alloy [65]. ECAP has been reported to break rod-like particles, in conventional Al alloys, into small spherical particles dispersed uniformly throughout the matrix restricting grain growth and effectively increasing tensile elongation before failure at high temperatures. Xu et al. [66] found that the presence of fine $MgZn_2$ and Al_3Zr particles in ultrafine AA7034 processed by ECAP at 473K would result in high strain rate superplasticity.

Moreover, Ralston et al. [54, 55] investigated the corrosion resistance and mechanical properties of an Al-Cu-Mg alloys as a function of aging time (and therefore precipitate size and density) and reported an existence of the critical size of precipitates below which pitting does not initiate. It was hypothesized that below the critical precipitate size, precipitates would stop interfering with the formation of a continuous protective passive layer and it would have similar protective capabilities of that of a pure aluminum alloy.

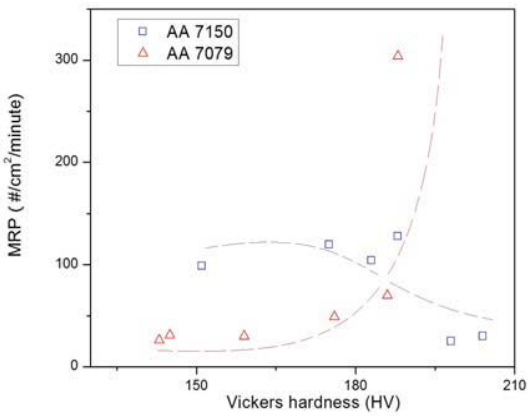

Figure 3. Metastable pitting rate (MPR) vs Vickers hardness for two commercial Al alloys [7] .

Gupta et al. [7] attempted to establish a relationship among the corrosion resistance, hardness, and aging time for AA7079 and AA7150 alloy. The corrosion performance of the alloys after several aging times was investigated using potentiodynamic polarization, metastable pitting rate measurements during potentiostatic polarization, and surface analysis after constant immersion tests. The size, number, and chemical composition of hardening phases were reported to be a strong function of the alloy composition and aging time. In AA7079 alloy, where the chemical composition of the hardening phases was not altered significantly with aging time, size of the intermetallics was the most critical parameters. It was reported that below a critical precipitate size, corrosion was not initiated. A sudden increase in pitting corrosion, as measured via metastable pitting rate, was reported. In AA7150, where Cu content was higher, the change in chemical composition of the hardening precipitates was reported to dominate the corrosion performance. For instance, after one hour of aging time, in spite of a significant increase in precipitate size, a significant decrease in pitting corrosion was observed. This

behavior was attributed to Cu enrichment of hardening precipitates. These investigations suggest that importance of the role of alloy composition, and characteristics of precipitates on corrosion performance of Al alloys. Via selection of suitable heat treatment and alloy composition, mechanical and corrosion properties can be optimized. Figure 3 demonstrates an example where metastable pitting rate (MPR), a representative of the extent of pitting corrosion, was plotted against Vickers hardness. It is interesting to note the behavior of the two alloys, AA7150 showed significant decreased in MPR while increasing strength. Alloys showing such a behavior are desired, whereas for AA7079 a sudden decrease in corrosion resistance beyond a hardness of 160 HV was noticed. Such studies are useful in designing novel alloys and heat treatment technologies to optimize properties.

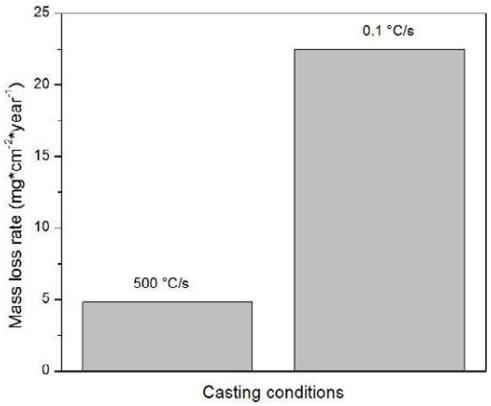

Figure 4. Mass loss rate from constant immersion test in 0.1 M NaCl for two Al-2Fe alloys produced using two different cooling rates during casting [61].

Formation of precipitates during service condition in Al-Mg alloys was reported to cause intergranular corrosion (IGC) due to the formation of Mg_2Al_3 on the grain boundaries. It was reported that IGC was dependent upon size, shape, and distribution of the Mg_2Al_3 [59, 67]. A microstructure with lower number of Mg_2Al_3 and higher distance between Mg_2Al_3 particles was reported exhibit superior corrosion performance [59, 68].

Figure 4 shows a graph of mass loss rate for two Al-2Fe alloys as produced by cooling rates of 500 °C/s and 0.1 °C/s. A significant improvement in corrosion resistance for the alloy produced by higher cooling rate was attributed to the combined inference of 1) significant decrease in volume fraction of Al_3Fe, 2) decrease in size of the intermetallics, and 3) increase in solid solubility of Fe in Al. These examples confirm that the corrosion resistance of Al alloys is a strong function of size, distribution and chemical composition of intermetallics, which could be manipulated to improve corrosion resistance.

Strategies to simultaneously improve corrosion and mechanical properties

Controlling the characteristics of the intermetallics, grain size, and composition of the matrix can be used to optimize the corrosion performance and mechanical properties of Al alloys. Any processing technology leading to a homogenous microstructure and removal of electrochemical heterogeneities is expected to improve corrosion performance. Al alloys relying on grain refinement and solid solution strengthening as main strengthening mechanism are expected to possess superior corrosion resistance. For example nanocrystalline Al as produced by high-energy ball milling of Al exhibited excellent corrosion resistance and compressive yield strength of 394 MPa [8]. Such an excellent combination of properties was attributed to nanocrystalline structure and possible incorporation of O and N from the high-energy ball milling media.

Sputter deposited thin films Al-M (M: Cr, Ti, W, Mo, Ta) were reported to exhibit significantly improved corrosion resistance, which was attributed to the extended solid solubility of M. These studies suggested that any process causing extended solid solubility of certain alloying elements should results in improved corrosion resistance. More recently, it has been shown that high-energy ball milling also results in extended solid solubility. For instance, solubility of Cr in a high-energy ball milled Al-20Cr alloy was reported to be as much as 13 wt.% [8, 11]. An Al-20Cr as produced by high-energy ball milling followed consolidation using spark plasma sintered exhibited compressive yield strength of 1094 MPa and an excellent corrosion resistance [11]. Simultaneous improvement in corrosion resistance and mechanical properties of Al-20Cr alloy were attributed to 1) extended solid solubility of Cr, 2) nanocrystalline structure, and 3) uniformly distributed fine Al-Cr intermetallics. Subsequently, ECAP can also produce ultra-fine grains and refine the size of constituent particles [20, 29, 69]. Similarly electrodeposition [15, 49, 70, 71] and sputtering [63, 72, 73] can also be effective ways to create a nanocrystalline structure, although their application is limited to the surface of a base metal.

Conclusions

Mechanical and corrosion behavior of an aluminum alloy depend upon the chemical composition, shape, size and distribution of intermetallics, grain size, and matrix composition which is also a function of alloy composition, production and post production preprocessing techniques. Controlling these factors is key to obtaining the best combination of mechanical and corrosion properties. Commercial alloys that are produced by casting rely on their chemical composition and subsequent thermomechanical processes to achieve optimum corrosion and mechanical properties. The mechanical and corrosion properties of the Al alloys could be improved by combining modern processing/production techniques, i.e., high-energy ball milling, ECAP, and novel alloying additions. For instance, Al-Cr alloys as produced by the consolidation of high-energy ball milled alloy powders exhibited excellent combination of corrosion resistance and strength. The improved properties were attributed to the nanocrystalline structure and extended solid solubility of alloying element.

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Light Metals 2016

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Plasticity Behavior

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NEW METHODOLOGY TO DETERMINE STABLE TEXTURE COMPONENTS UNDER COLD ROLLING IN FCC METALS

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Keywords: Rolling, Crystal plasticity Texture

Abstract

In this work evolution of texture components during deformation of AA5754 aluminum alloy sheet under cold rolling is studied by analyzing the evolution of element-rotation-distribution calculated using a rate-dependent crystal plasticity finite element model (CPFEM). The proposed criteria can successfully predict the stability of a given textural component for cold rolling deformation in FCC materials with high stacking fault energy that deform predominantly by slip. Comparison of simulation results with experimental data shows that this approach successfully captures the stable textures reported for cold rolled AA5754 sheets. With the initial texture and the strain path, it is believed that, the method described in this work can be used to predict the final stable textures without any need for expensive crystal plasticity based numerical simulations and this could be of immense help for simulating large strain deformation of macroscale sheet samples exhibiting texture evolution.

Introduction

It is known that polycrystalline material textures evolve during deformation. This in turn affects their forming behavior. Individual grains inside the material undergo rotation and shape change as the local strains can be very different from the global strain path applied to the material [1]. Local strains result in texture evolution inside the material and could even result in non-uniform textures across the material through-thickness.

Crystal plasticity finite element (CPFEM) simulations are used to predict the material response under any strain path [2–4]and unlike phenomenological approaches [5, 6], can also predict the texture evolution. Industrial processes, like cold rolling can also be simulated using CPFEM to study the element-rotation-evolution by studying the misorientation between the initial and final Euler angles of each element.

Crystal plasticity framework used in this paper [7] is used in conjunction with statistically representative 3D microstructures of AA5754 sheet [8] to simulate cold rolling. It is noted that the analysis presented in this work can be performed on results from other crystal models available in literature.

In this paper, analysis of element-rotation distribution leads to a new stability criteria used to find stable texture components under cold rolling of AA5754 sheets. This new approach can be used to predict stable textures under any strain path for FCC materials.

Constitutive Model

Crystal plasticity formulation used in this paper is based on the model by Asasro et al.[9]. The constitutive model was implemented as a user-defined material subroutine (UMAT) in commercial software LS-DYNA [10]. The crystal plasticity material model used in this paper is based on the works by [2–4] and works primarily on slip system mechanism to model plastic deformation in polycrystal FCC materials. Plastic deformation in FCC materials is on twelve {1 1 1} <1 1 0> slip systems. The constitutive equation for this rate-dependent model can be written as:

$$\hat{\sigma} = ED - \dot{\sigma}^0 - \sigma\, trS \tag{1}$$

Where $\hat{\sigma}$ is the Jaumann rate of Cauchy stress, E is the elastic stress tensor, S is the strain-rate tensor and $\dot{\sigma}^0$is the viscoplastic type stress state based on slip rates. Equation (2) is used to calculate the slip rates for each slip system (α).

$$\dot{\gamma}_\alpha = \dot{\gamma}_0 sgn\, \tau_\alpha \left|\frac{\tau_\alpha}{g_\alpha}\right|^{1/m} \tag{2}$$

Where $\dot{\gamma}_0$ is the reference shear rate, $\tau_\alpha$is the resolved shear stress on each slip system α, $g_\alpha$is the hardness of each slip system and incorporates single slip hardening and m is the strain-rate sensitivity index.

Single slip hardening law used in this paper is defined as:

$$h_\beta = h_0 \left(\frac{h_0 \gamma_\alpha}{\tau_\alpha n} + 1\right)^{n-1} \tag{3}$$

Where h_0 is the initial hardening rate, γ_α is the accumulated slip on a slip system and n is the hardening exponent.

Microstructure and FE Model

In this paper, symmetric cold rolling is performed similar to the work by Ali et al. [11]. Half symmetry is applied to model a 2.5 mm AA5754-O sheet and the sample is 175 x 175 x 1250 µm with solid cube elements. Statically equivalent 3D microstructure is generated to model the AA5754 sheet. The method employed is similar to the work by Brahme et al. [8]. 3D microstructure is created from 2D Electron Backscatter Diffraction (EBSD) data. Synthetic microstructure matches the orientation distribution function (ODF) and misorientation distribution function (MDF). The resulting microstructure is then converted to FE mesh with multiple elements one grain. The model contained ~800 grains with an average grain size of 22 µm.

Figure 1 shows the initial microstructure of the AA5754 sheet used to model cold rolling. The starting microstructure is matched to the works from Jin and Lloyd [12] to provide a model that can be used for validation as presented in [11]. The starting

microstructure has high volume fraction of Cube (10%) with minute quantities of Goss (0.8%) and other texture components.

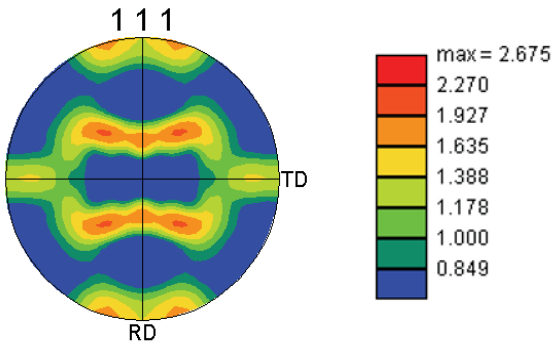

Figure 1: Initial <1 1 1> pole figure of the AA5754 sheet

Vertical (Z-Direction - ND) and horizontal (Y-Direction - RD) velocities on the top surface were used to model the cold rolling procedure described in this paper (Figure 2). Bottom of the model was fixed to apply the half-symmetry conditions. The model was compressed to 60% thickness reduction to model the conditions imposed by Jin and Lloyd [12] and is shown in Figure 3.

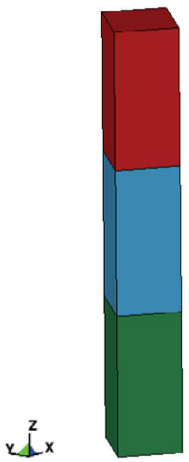

Figure 2: Finite element model

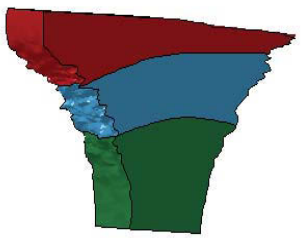

Figure 3: Deformed mesh at 60% thickness reduction

Proposed Methodology

Crystal plasticity results from the cold rolling simulations are analyzed using an in-house (gCode) that calculates the misorientation between the original and final orientations of an element [11, 13, 14]. Initial and final orientations of each element are extracted as Euler angles $(\varphi_1, \emptyset, \varphi_2)$ and are then converted to quartonians to find the misorientation between them [15]. The misorientation between the initial and final orientations of each element is called element rotation.

Euler angles at 0%, 10%, 20%...60% are extracted and element rotations at each deformation step are calculated. Element rotations are then parsed into 6 bins with elements that rotate 0-10°, 10-20°…50-60°. Initial texture for each bin is extracted as Euler angles and plotted as pole figures. This processes shows the evolution of textures during cold rolling and is used to derive the proposed criteria for stable textures.

Results and Discussion

Results from cold rolling simulations parsed into 6 bins are plotted on <1 1 1> pole figures shown in Figure 4. All figures are plotted on the same scale and each column shows the element-rotation-distribution for a given thickness reduction (20% and 60%) and each row shows the orientation of elements belonging to different element-rotation-bins (0-10°, 10-20°, 20-30°, 30-40°). Results from 40° to 60° are not shown as they represent a very small volume fraction of elements with respect to the overall sample. First two rows show typical rolling textures (0-20°) with high Cube. However, subsequent element-rotation-bins show less Cube as shown in Figure 4.

Pole figures shown in Figure 4 provide insight into the preferential element rotation during the deformation. Preferential element rotation leads to the analysis of stable textures using the element-rotation-distribution. Different methods are used to find stable textures under different loading conditions i.e. [1, 16]. However these methods do not look at stability that occurs throughout the deformation but only at a certain time step. This paper describes a new, element rotation based, stability criteria (SR) that looks at stable textures throughout the deformation and independent of time. The new definition states that stable textures are the ones that undergo the least change in R_s during deformation; where R_s is defined as the rate of change of a texture component over time. Texture components are calculated as volume fractions for quantification.

In-line with the definition, perfect stability would result in $R_s \sim 0$. This does not imply that the volume fraction of a certain component remains same throughout the deformation but that the elements rotation in and out of a orientation remain the same.

Based on the results from Figure 4, maximum change in textures can be seen between the 10-20° and 20-30° bins. Therefore elements that rotate less than 20° during deformation are considered as stable textures in this paper.

Volume fraction calculations at several locations in the deformation are then used to calculate the percentage change in R_s during the deformation. Results show the least change in R_s is found in Copper (5%) followed by Brass, Goss (8%), S (17%) and Cube (33%). This means that the most stable texture during rolling is Copper followed by Brass, Goss, S and Cube. These results are in-line with the results from Zhou et al. [16] that shows Copper, Goss and Brass to be stable textures during cold rolling.

Conclusions

A rate-dependent crystal plasticity framework is used in this paper to simulate cold rolling. Results from the cold rolling simulations are used to analyze the element-rotation-distribution evolution in the rolled sheet. Analysis of element-rotation-distribution leads to a new stability criteria ($R_s \sim 0$). It is shown the new stability criteria is able to successfully predict stable textures under cold rolling. The current model can be extended to predicting stable textures under any loading conditions for FCC materials.

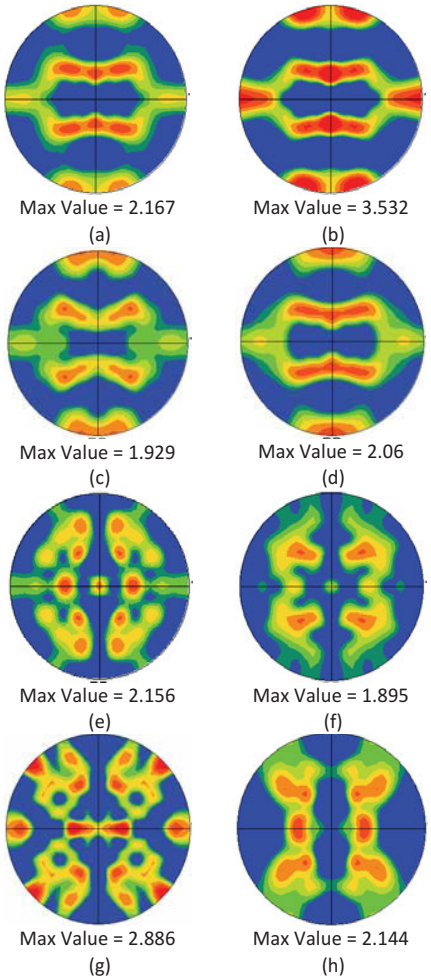

Figure 4: Evolution for element-rotation-distribution represented as <1 1 1> pole figures for (a) 20% thickness reduction for 0-10° (b) 60% thickness reduction for 0-10° (c) 20% thickness reduction for 10-20° (d) 60% thickness reduction for 10-20° (e) 20% thickness reduction for 20-30° (f) 60% thickness reduction for 20-30° (g) 20% thickness reduction for 30-40° (h) 60% thickness reduction for 30-40°

Acknowledgements

This work was supported by the Natural Sciences and Engineering Research Council Automotive Partnership Collaboration (NSERC-APC) Program and General Motors of Canada. The authors would also like to acknowledge the high performance computing center at the University of Sherbrooke. The authors would also like to acknowledge the help of Jaspreet Nagra and Waqas Muhammad for useful discussions.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Recrystallization in Al-Mg Alloys after Hot Compression

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Keywords: Aluminum, Al-Mg Alloys, Hot Deformation, Recrystallization

Abstract

Because the 5000-series aluminum alloys are not heat treatable, controlling their microstructures during deformation processing to achieve fine grain sizes is particularly important. Recrystallization during or immediately following hot deformation provides an opportunity for controlling grain size prior to cold deformation. The present study investigates the effects of magnesium concentration, temperature, and deformation rate on recrystallization during and immediately after hot upsetting. Three aluminum alloys with magnesium concentrations of 0.0, 0.5, and 4.5 wt.% were upset at 400 and 500 °C at fixed strain rates of 0.1, 1.0, and 10.0 s^{-1}. The specimens were rapidly quenched after upsetting to preserve their microstructures. Portions of the specimens were then annealed for 10 min. to allow static recrystallization. Cross-polarized optical microscopy was used to examine the as quenched and annealed specimens. These results are interpreted to explain microstructural evolution during and after hot deformation.

Introduction

Deformation processing controls the microstructure developed in commercial wrought alloy products, and that microstructure controls properties and performance. By manipulating material composition, deformation conditions, and heat treatments, microstructure and the resulting material properties can be tailored for specific applications. Rolling is particularly important in the processing of aluminum plate and sheet. Hot rolling is defined as occurring above the recrystallization temperature and is typically applied directly to both chill-cast and continuously-cast materials. It therefore provides the first opportunity to evolve the cast microstructure into a more desirable form. Improved microstructural evolution during hot rolling can reduce the need for subsequent processing and heat treatment. Hot rolling is most often applied as a series of rolling passes. This makes it important to understand the microstructural evolution during both the deformation of rolling and the annealing that occurs before and after each rolling pass [1]. Microstructural evolution is affected by recovery, recrystallization, and grain growth. These can occur during deformation, dynamically, or after the deformation is complete, statically. Distinguishing between recrystallization that occurs dynamically and that which occurs statically is often difficult but is important to understanding the mechanisms of recrystallization.

The 5xxx series aluminum alloys are alloyed primarily with magnesium and are not generally heat treatable. Thus, the 5000-series alloys rely upon a fine grain size and cold work for strength, in addition to moderate solid-solution strengthening from magnesium. Because of their acceptable room-temperature formabilities and excellent formabilities at elevated temperatures, the 5000-series alloys have found use in automobiles to lower mass and thus improve vehicle performance parameters such as fuel economy [1]. Wrought AA5083 sheet and closely related alloy sheet materials are among the most common materials for superplastic forming in the automotive and aerospace industries. Excellent superplastic performance is possible in these materials by achieving a grain size of approximately 10 μm or less [2,3], which is typically accomplished through recrystallization after very large cold rolling strains. Approaches that produce grain refinement without such large cold rolling strains, such as using hot or warm rolling would be attractive. Al-Mg alloys with 2–5 wt.% Mg can produce tensile elongations of 200–300% independent of grain size, which provides an alternative to superplasticity for some hot forming applications [4].

This study investigates the effect of magnesium content, deformation temperature, and deformation strain rate on the evolution of microstructure during hot deformation and subsequent annealing. The effects of recovery and recrystallization both during deformation, dynamic, and immediately following deformation, static, are probed. Hot compression is used to mimic hot rolling. Microstructures are studied as quenched immediately after hot compression and following additional post-deformation annealing.

Table I: Chemical compositions of the materials, in wt.%

Alloy	Cu	Si	Fe	Mn	Mg	Zn	Cr	Ti
Al	0.004	0.09	0.11	0.05	0.00	0.016	0.05	0.01
Al-0.5Mg	0.002	0.10	0.11	0.05	0.50	0.006	0.05	0.01
Al-4.5Mg	0.004	0.11	0.11	0.05	4.39	0.004	0.05	0.01

Experimental Procedure

Three aluminum alloys with magnesium contents ranging from 0 to 4.5 wt.% are studied. The chemical compositions of these materials are provided in Table I. These alloys were direct-chill cast, scalped and then homogenized. Homogenization was at 520 °C for 5 hrs. and was followed by air cooling. This produced an average grain size of approximately 100 μm. Cylindrical test specimens were then machined to a diameter of 8 mm and a height of 12 mm; the cylinder axis was oriented along the short transverse direction of the casting. Specimens were tested in compression using a customized instrument that provides elevated temperatures and control of compression rate, as well as direct quenching of specimens with He gas following testing. The specimens were hot compressed at 400 or 500 °C at fixed engineering strain rates of 0.1, 1, or 10 s^{-1} to a final height of 6 mm, which resulted in a total compressive engineering strain of 0.5. Upon test completion, rapid quenching by He gas was used to preserve the deformed microstructures. The specimens were characterized in the as quenched condition and were then annealed for 10 minutes in a salt bath at their respective test temperatures for further characterization.

Microstructures were observed using cross-polarized light optical microscopy. This required sectioning, mounting, and metallographic preparation of the test specimens. A schematic of a sectioned Al-0.5Mg specimen deformed at 400 °C at a strain rate of 0.1 s^{-1} in the as quenched condition is shown in Figure 1 to reveal a typical microstructure, identify important microstructural regions and define reference directions. Metallographic preparation involved standard grinding and polishing followed by electrolytic etching in Barker's reagent (1.8 vol.% HBF_4 in deionized water) at a voltage of 25 V or greater for approximately 3 to 4 min. Horizontal (radial direction) and vertical (axial direction) grain sizes were measured in the deformed centers of the specimens using the directional lineal-intercept method in accordance with ASTM standard E112 [5].

Results

Cross-polarized light optical photomicrographs from the deformed regions of specimens as quenched immediately after upsetting are presented in Figure 2. In the pure

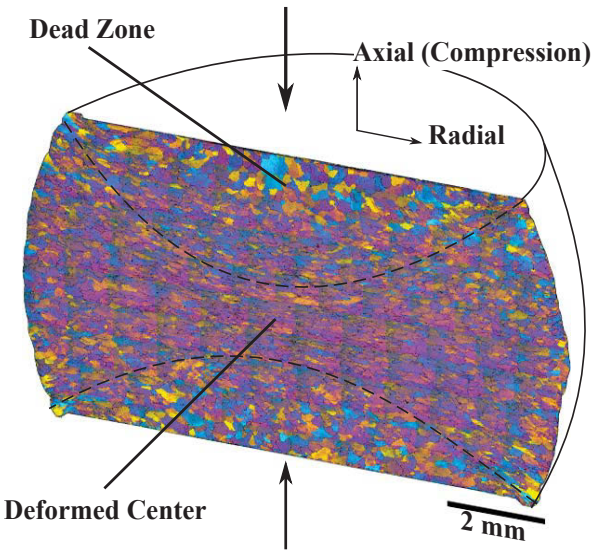

Figure 1: This schematic shows a specimen sectioned and mounted for metallographic observation. The microstructure shown is that of the Al-0.5Mg alloy tested at 400 °C and a strain rate of 0.1 s^{-1} in the as-deformed as-quenched condition. Important microstructural regions and reference directions are identified.

Al and the Al-0.5Mg alloy, grains are elongated along the radial direction to form the classic "pancake" shape expected from upsetting. For the Al-4.5Mg alloy, the microstructure developed after deformation at 400 °C at a strain rate of 0.1 s^{-1} shows a similar pancake morphology, but with small equiaxed grains appearing along grain boundaries for the four following deformation conditions: 400 °C at 1 s^{-1}, 400 °C at 10 s^{-1}, 500 °C at 0.1 s^{-1}, and 500 °C at 1 s^{-1}. These small equiaxed grains have the appearance of recrystallized grains that originated at the deformed grain boundaries. Full recrystallization is evident in the Al-4.5Mg microstructure produced by deformation at 500 °C and 10 s^{-1}, as all the grains are equiaxed and approximately equal in size.

Photomicrographs of the deformed regions of specimens after annealing for 10 min. at the same temperature as upsetting are displayed in Figure 3. The microstructures of the Al specimens deformed and annealed at both 400 and 500 °C evidence grains elongated

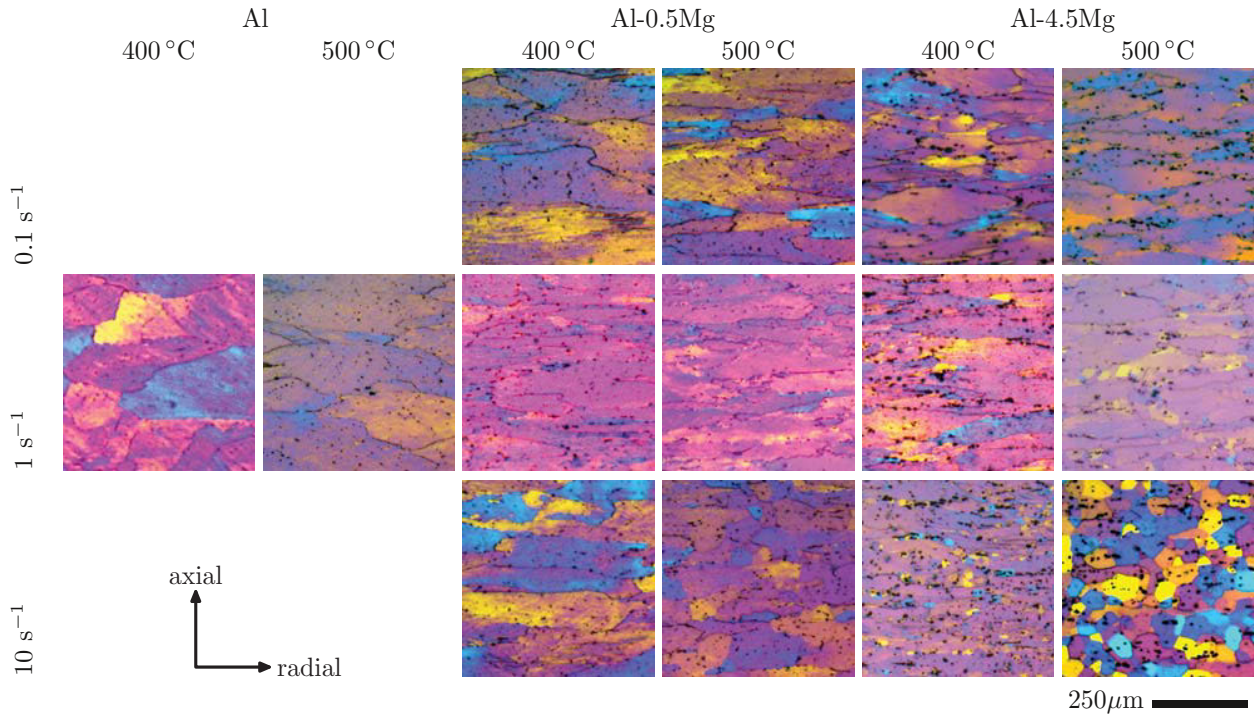

Figure 2: Cross-polarized light optical photomicrographs from the deformed zone of specimens are shown in the as-deformed, as-quenched condition.

along the radial direction and suggest some grain growth during annealing. The microstructures of the Al-0.5Mg specimens deformed at 400 °C for strain rates of both 0.1 and 1 s^{-1} and deformed at 500 °C and 0.1 s^{-1} present a few large equiaxed grains within microstructures of large grains elongated along the radial direction. These features suggest that recovery and recrystallization occurred concurrently in these Al-0.5Mg specimens. Fully recrystallized microstructures, as indicated by equiaxed grains, are observed after annealing for the following specimens: Al-0.5Mg deformed at 400 °C and 10 s^{-1}; Al-0.5Mg deformed at 500 °C and both 1 and 10 s^{-1}; and all specimens of Al-4.5Mg.

Serrated grain boundaries are observed in several of the specimens as quenched after upsetting. Examples are presented in Figure 4, which displays photomicrographs of the Al, Al-0.5Mg, and Al-4.5Mg specimens deformed at 500 °C and a strain rate of 1 s^{-1}. Grains in the two alloys containing magnesium are serrated, but those in Al generally do not display such serrations. The Al-4.5Mg alloy is also partially recrystallized. Serrations imply the occurrence of dynamic recovery as the grain boundary migrates to absorb subgrain boundaries [6,7].

Discussion

The addition of magnesium increases grain refinement during hot deformation. This is evident from the data presented for the Al, Al-0.5Mg, and Al-4.5Mg materials in Table II, which include average lineal intercept grain size, calculated through the geometric mean (as recommended by ASTM E112 [5]), and the calculated density of recrystallization nuclei from observations of the annealed specimens. As the weight percent of magnesium increases, the grain size decreases. An exception to this trend is noted for the specimens deformed at 500 °C and 0.1 s^{-1}. The recrystallized grain size increases with an increase in temperature. This is likely a result of increased grain growth. No trend is observed for the effect of strain rate on recrystallized grain size, possibly because of the limited amount of data and their inherent scatter. However, strain rate likely has a strong effect on the incubation time for recrystallization. As the strain rate increases, the times required for deformation and dynamic recovery are expected to decrease.

The literature presents multiple hypotheses for the mechanism of recrystallization in hot-deformed Al-Mg alloys [7,8]. The strongest hypothesis is that recrystallization is a primarily static phenomenon in which nuclei are initiated by a process similar to geometric dynamic recrystallization, GDRX. During deformation, dynamic recovery occurs and causes the deformed grains to become serrated at their boundaries. Once deformation is complete, the serrations pinch off during subsequent heating to form new grains with high-

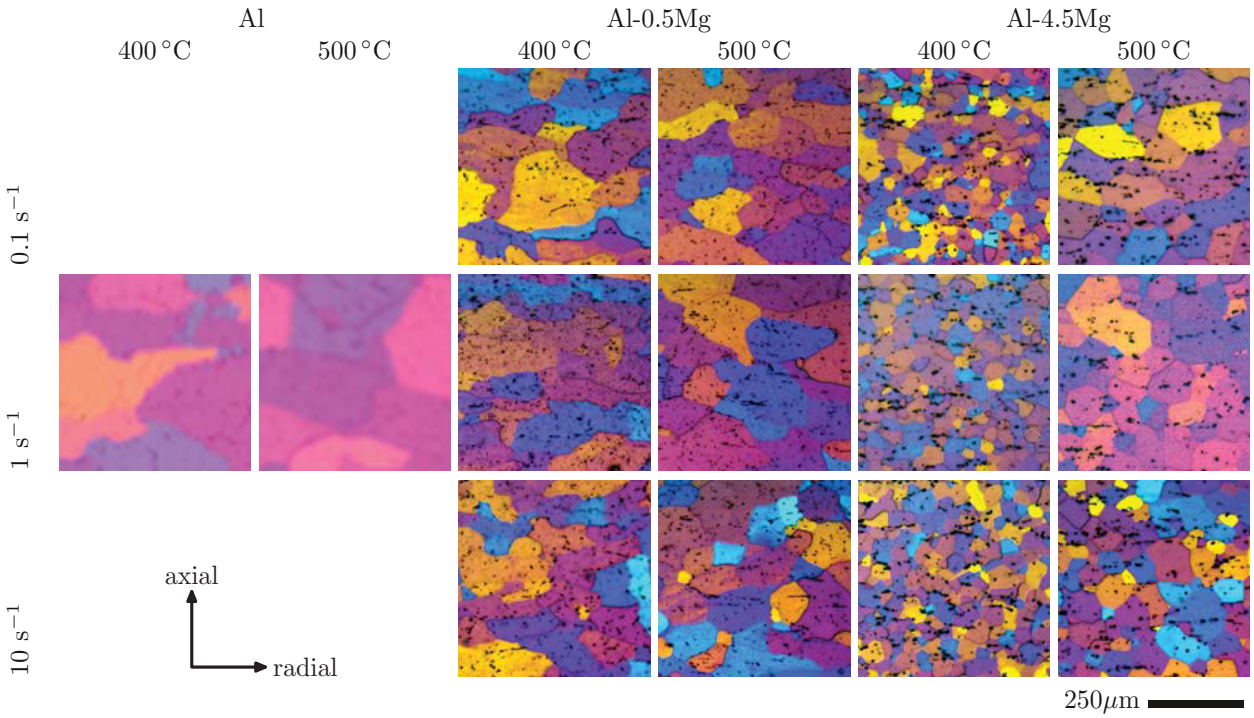

Figure 3: Cross-polarized light optical photomicrographs from the deformed zone of specimens are shown in the annealed condition.

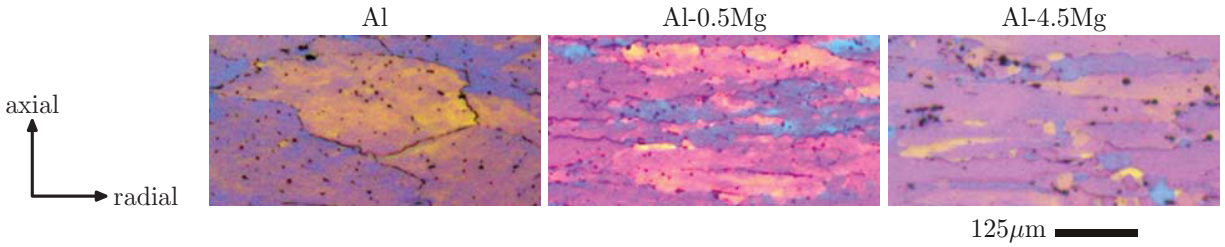

Figure 4: Specimens deformed at 500 °C and a strain rate of 1 s^{-1} are shown in the as-deformed, as-quenched condition. The Al-4.5Mg specimen is approximately 0.34% recrystallized. No recrystallization is present in the Al or Al-0.5Mg specimens.

angle boundaries [9]. The mobility of high-angle grain boundaries allows these pinched off grains to consume the surrounding deformed microstructure [9,10]. The partially recrystallized microstructures of Figure 2 are in accordance with this argument. The nuclei, indicated by the small equiaxed grains, are primarily within serrations along the deformed grain boundaries. These are evidenced by the small, equiaxed grains colored predominantly as yellow and light-blue, and these distinct colors suggest preferred orientations distinct from the surrounding deformed microstructure [11]. These characteristics imply that there are specific subgrain orientations that are particularly suited to migration of high-angle grain boundaries and, therefore, to pinching off from the serrated boundaries. These small, equiaxed grains are evident even in the as-quenched Al-4.5Mg material, for example, because the incubation time for these nuclei to pinch off is less than the time to quench after testing. This explanation is consistent with the experimental results from Chang *et al.* [9], who demonstrated similar behaviors in Al-Mg material tested in tension at elevated temperatures. Reports of shortened incubation times for static recrystallization in Al-Mg alloys with magnesium content over 1% are described by McQueen *et al.* [12]. The completion of recrystallization after annealing supports that recrystallization of the Al-Mg alloys is predominantly a static phenomenon, not dynamic.

Table II: The nuclei per unit area and grain size in the annealed condition for the Al, Al-0.5Mg, and Al-4.5Mg alloys.

Testing Conditions		$N_{A,Annealed}$ $\times 10^7$ [nuclei/m^2]		$\bar{d}$ [μm]		
T[°C]	$\dot{\epsilon}$ [s^{-1}]	Al-0.5Mg	Al-4.5Mg	Al	Al-0.5Mg	Al-4.5Mg
500	0.1	-	10.2±5.12	-	90±8*	112±28
	1	6.94±3.17	25.8±5.31	303±127	135±31	70±7
	10	17.0±5.63	34.7±9.02	-	87±14	61±8
400	0.1	-	83.8±17.5	-	110±25*	39±4
	1	-	45.0±11.2	183±55	87±15*	53±7
	10	22.5±7.70	79.2±18.0	-	75±13	40±5

* Grain sizes marked with an asterisk were specimens that experienced partial recovery and recrystallization.

Conclusions

The effects of magnesium content and deformation conditions on microstructural evolution during hot deformation and subsequent annealing were investigated. Focus was placed on identifying the dynamic and static contributions to microstructural evolution. In the as-deformed as-quenched condition, the Al-4.5Mg alloy presents four partially recrystallized microstructures and one fully recrystallized microstructure, that tested at the highest temperature and fastest rate. After annealing, all the Al-4.5Mg microstructures fully recrystallized, as did three of the six Al-0.5Mg microstructures. This microstructural evolution is explained as static recrystallization initiated in the serrations along the boundaries of deformed grains; the serrations are associated with dynamic recovery. Subgrains with particular orientations are more susceptible to pinching off from these serrations. Increasing magnesium content and decreasing temperature decrease the recrystallized grain size, until temperature becomes too low to induce recrystallization. From the limited data available through this study, no particular relationship between strain rate and recrystallized grain size is observed.

Acknowledgments

The authors are grateful for technical contributions provided by Drs. Koji Ichitani and Shinya Yasuda of UACJ Corp., 1351 Uwanodai, Fukaya-city, Saitama, 366-8511, Japan. This study received support and assistance from Nippon Steel and Sumitomo Metal Corp., UACJ Corp., and the U.S. National Science Foundation (DMR-1105468).

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Large Strain Cyclic Simple Shear Behavior of Aluminum Extrusions: An Experimental and Numerical Study

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Keywords: Simple shear, Backstress, Crystal plasticity, Texture, Extrusion

Abstract

It has been shown that dislocation cells induce intragranular backstresses due to blockage of dislocation passage. The present work focuses on the effects of dislocation induced backstresses on the strain hardening behavior of AA 6063-T6 extruded aluminum alloy during cyclic planar simple shear deformation. Based on the previous work of Brahme et al. [1], the internal stresses are predicted using a close form solution based on the Eshelby's inclusion problem. Using this approach the inherent dislocation microstructure is modeled as a two-phase material consisting of highly dense harder cell walls and less dense soft cell interiors. The new formulation also takes into account the evolution of dislocation cell structure with plastic deformation. Simulations are performed for simple shear and the predictions are compared against experimentally obtained flow stresses for forward, reverse and subsequent forward in-plane simple shearing.

Introduction

Deformation induced dislocation microstructures have been observed in metals with medium to high stacking fault energies such as aluminum and copper. Several transmission electron microscopy (TEM) studies have shown the existence of these microstructures in face centered cubic (FCC) metals and alloys [2-4]. These microstructures tend to form dislocation cells or nets which are characterized by three-dimensional tangles of dislocations that serve as "cell walls – hard phase" and less dense internal regions (cell interiors – soft phase) [4-6]. It has been shown that the presence of such dislocation microstructure induces intragranular backstresses due to blockage of dislocation passage leading to the experimentally observed Bauschinger effect at the macroscopic scale [7-9]. Furthermore, these backstresses are of opposite signs in the soft and hard phase and may result due to incompatibilities in dislocation motion as dislocations tend to glide much easily in the soft phase as compared to the hard phase [10]. The internal backstress has been widely used to explain the mechanical Bauschinger effect induced by dislocations. Using TEM analysis, Hasegawa et al. [11] studied dislocation mechanisms responsible for the Bauschinger effect during stress reversal in polycrystalline aluminum. The authors correlated the observed Bauschinger effect and the decrease in hardening rate after loading reversal to the evolution of the dislocation microstructure in pure aluminum. In particular, the authors linked the decrease in hardening rate to the partial dissolution of dislocation cells after stress reversal.

Several methodologies have been proposed to incorporate the effects of backstresses in computational models. Among these models, the Armstrong–Frederick type, nonlinear kinematic hardening models have been employed by Voyiadjis and Kattan [12], Khan and Cheng [13], McGinty and McDowell [14], Goh et al. [15] and Yoshida and Uemori [16]. Such phenomenological modeling approaches are frequently applied to model the effects of backstresses including the Bauschinger effect. Other approaches such as those proposed by Bayley et al. [17], Mughrabi [5] and Peeters et al. [18] make use of the geometrically necessary dislocation models to model backstress at slip system level. More recently, discrete dislocation dynamics (DDD) has been used for computing the long-range stress field induced by the dislocation microstructures [19]. However, DDD simulations are comparatively time consuming and the results cannot be coupled easily with crystal plasticity finite element (CPFE) framework afterwards.

There has been several efforts made-to-date to incorporate the backstress within CPFE framework. Yapici et al. [20] incorporated backstress into a viscoplastic self-consistent (VPSC) polycrystal model to study the stress–strain asymmetry under different strain paths. Kim et al. [21] used a simplified dislocation density model along with a representative volume element (RVE) approach to study Bauschinger effect in dual phase steels. Brahme et al. [1] made use of electron backscatter diffraction (EBSD) maps in conjunction with crystal plasticity finite element formulation to study the effects of dislocation cell boundaries and strain path on the overall texture evolution of direct chill cast AA5754. More recently, Li et al., [22] used a texture-based representative volume element (TBRVE) crystal plasticity model to successfully predict the Bauschinger effect in rolled polycrystalline aluminum alloy 7075 under tension-compression cyclic loading.

The work hardening behavior of aluminum alloys at large strain is of special interest in the design and modeling of sheet metal forming processes. Shear tests are therefore important since they avoid the early localization that is observed in uniaxial tensile tests. Furthermore, characterization of large strain shear behavior is also important as it is usually observed in sheet metal forming applications, when sheet element moves through the tool radii from the flange area by the action of punch [23].

The purpose of the current work is twofold. Firstly, a comprehensive experimental investigation of the quasi-static, cyclic in-plane simple shear response of extruded AA6063-T6 at room temperature is presented. Secondly, a hardening law is proposed to incorporate the effects of intragranular backstresses due to the formation of dislocation cells structure with plastic deformation. The new hardening model is incorporated into an existing crystal plasticity finite element framework. Cyclic simple shear simulations are performed and results are compared against the experimental data.

Material Characterization

Experimental Procedures

An extruded profile from a commercially available Aluminum alloy 6063 in a T6 tempered condition was used in the present study. The chemical composition of the alloy is given in Table I.

The as-received extruded alloy has an extrusion wall thickness of 1.8 mm.

Table I: Chemical composition of as-received AA6063-T6 extruded material.

Material	Chemical composition (Max. wt%)						
	Mg	Si	Cu	Fe	Mn	Ti	Al
AA6063	0.49	0.4	0.01	0.16	0.029	0.01	Bal

Displacement controlled forward-reverse-forward (FRF) simple shear experiments were performed along the extrusion direction (ED) to characterize the in-plane cyclic simple shear behavior of the alloy. The term "FRF" stands for forward shearing followed by reverse shear in the opposite direction and then subsequent forward shearing in the initial direction. The shear sample geometry used in this study was originally proposed by Kang et al. [24]. However, the sample dimensions were modified to allow for cyclic planar simple shear testing and to prevent buckling of the sample. The modified sample geometry is shown in Figure 1. The length and width of the shear zone are 6.35 mm and 1.2 mm respectively. As proposed by Kang and Gong [25], a pair of notches of 0.25 mm in the thickness direction was machined on both sides of the specimen to ensure simple shear state in the shear zone. All shear tests were performed at room temperature using an MTS Landmark 370 Servo-hydraulic tensile machine with a load cell capacity of 100 KN and a constant cross-head speed of 1.0 mm/min.

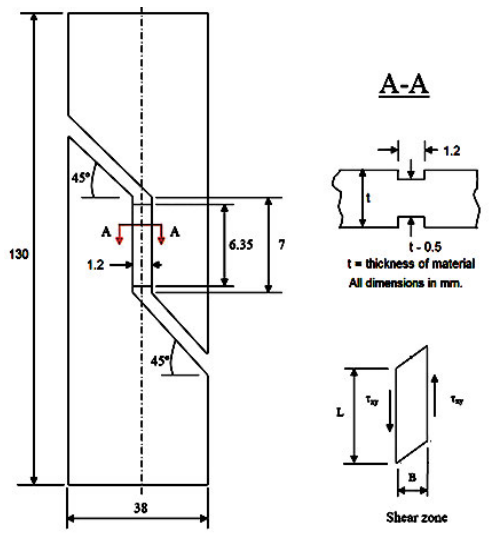

Figure 1: The proposed sample geometry for cyclic planar simple shear tests.

Throughout the deformation process, a commercially available digital image correlation (DIC) system "ARAMIS^{TM}" was used to capture surface strain evolution within the shear zone. For this purpose, a stochastic black-and-white speckle pattern was spray painted on the specimen surface prior to testing. The DIC measurements were made from a set of snapshots collected automatically at a preset frame rate of 1 frame per second during the shear tests. The recorded data was further analyzed to obtain the full-field strain maps using the DIC software. It is noted that the shear strains were calculated from the measured shear angles in the ARAMIS system [26]. For this purpose, the measured shear angles θ were averaged over the entire shear zone and then converted to shear strain γ, using $\gamma = \tan(\theta)$ for each recorded load step.

EBSD texture measurements were performed for characterizing the initial texture of the extruded AA6063-T6 material. The measurements were made using a field-emission Nova NanoSEM^{TM} equipped with a TSL EBSD camera. A step size of 0.5µm was used for all EBSD measurements and the data was further processed using the TSL OIM^{TM} software.

Experimental Results and Discussion

The initial through-thickness texture corresponding to ED EBSD plane (see Figure 2 for reference) is shown in Figure 3.

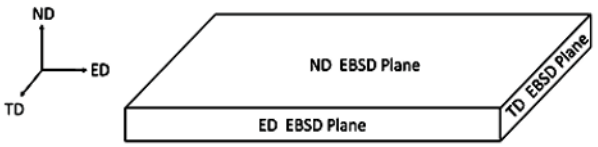

Figure 2: The terminology used for the different EBSD planes. ED represents the extrusion direction and ND represents the through thickness direction.

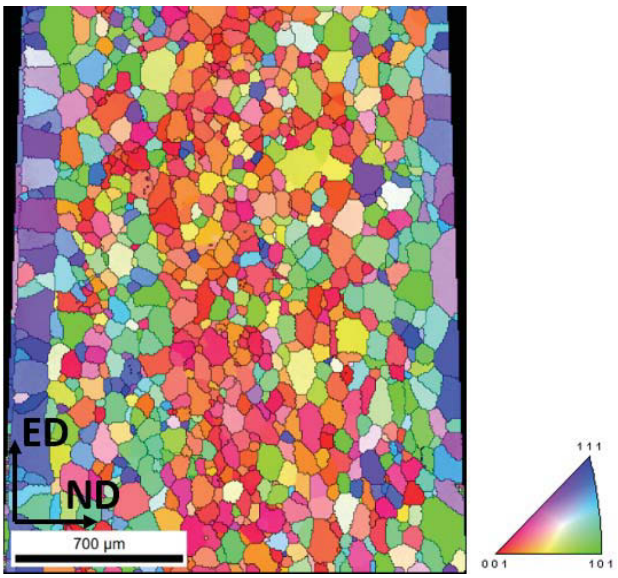

Figure 3: Through thickness texture of the extruded AA6063-T6 as received material.

The inverse pole figure (IPF) map in Figure 3 shows that the through thickness texture and grain size of the extruded material are different in the center as compared to in vicinity of the extrusion walls. Figure 4 shows the initial <111> pole figure of the extruded material. The <111> pole figure shows texture from within the gage section of the shear sample which is also later used as the starting texture for the numerical modeling work. It can be observed that the pole figure shows a predominant Cube texture followed by a comparatively weaker Goss texture.

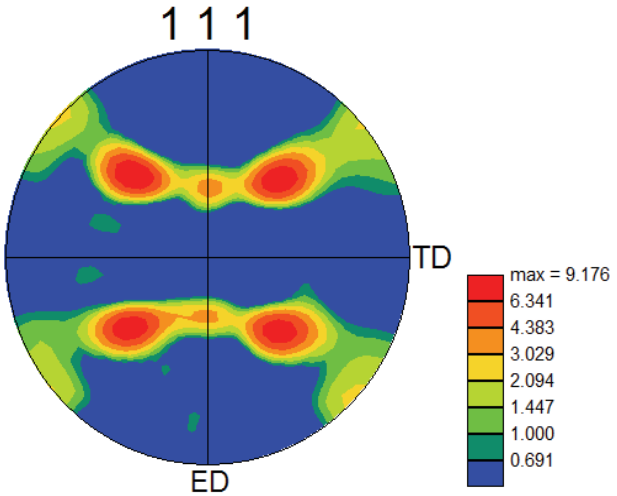

Figure 4: Pole figure of the initial microstructure within gage section of the sample

Figure 5 shows the strain mapping results for shear strains corresponding to initially fully forward, subsequently fully reverse and successively fully forward deformation stages. For convenience, these deformation stages are also marked as "(1)", "(2)" and "(3)" on the cyclic FRF shear stress-strain curve in Figure 6. It can be observed from the strain mapping results that the deformation within the shear zone is macroscopically homogeneous during the entire deformation process with no sign of shear bands formation.

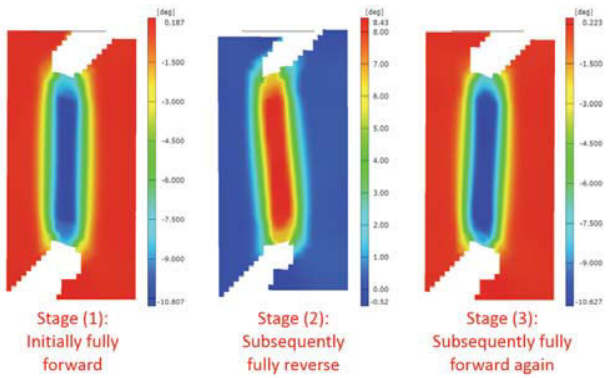

Figure 5: Shear strain distribution at different stages during cyclic FRF deformation using digital image correlation. Note that shear strains are defined as shear angles in the figure.

Figure 6 shows the experimentally obtained shear stress – shear strain curve for cyclic FRF planar simple shear loading. The test was repeated three times and the repeatability of the experiment was reasonably good, with an average absolute deviation from the mean of approximately 3 MPa. It can be observed from Figure 6 that the flow behavior of the material is fairly symmetric during cyclic planar simple shear deformation. However, for a more clear comparison of flow behavior between the initial forward, reverse and subsequent forward loadings, data in Figure 6 is replotted as shown in Figure 7.

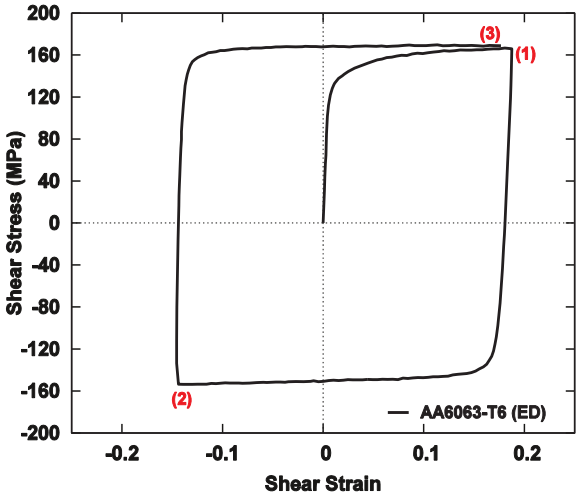

Figure 6: Shear stress vs. shear strain response under cyclic (FRF) planar simple shear loading. Stage (1) on graph corresponds to fully forward in initial shearing, Stage (2) corresponds to subsequently fully reverse loading and Stage (3) corresponds to subsequent fully forward shear loading.

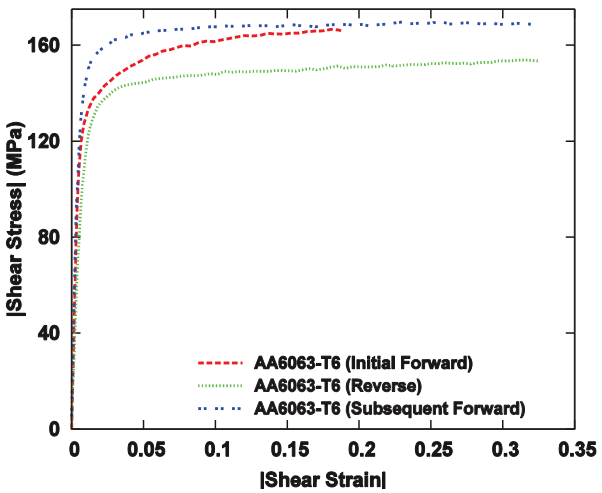

Figure 7: Absolute shear stress vs. absolute shear strain curve for initial forward, reverse and subsequent forward planar simple shear loadings in extrusion direction.

Figure 7 shows that the hardening rate during the initial forward simple shear loading is significantly higher than the subsequent reverse and forward loadings. However, the hardening rate during reverse loading is comparatively similar to that during subsequent forward loading. These differences in hardening rates may well be related with the development of the underlying dislocation microstructure as well as texture evolution with continuous plastic deformation. Furthermore, the occurrence of a moderate Bauschinger effect can also be observed as the yield stress during reverse loading is relatively lower than that during the initial forward loading.

The next section of the manuscript will present the proposed numerical modeling approach which takes into account the effects of intragranular backstresses as they can eventually lead to the occurrence of the observed Bauschinger effect.

Constitutive Modelling

Modelling Framework

The polycrystal modeling framework proposed by Asaro and Needleman [27] will be used in this analysis. The details of the model can be found in Wu et al. [28]. However, for completeness, the main ideas of the model are summarized next:

The total deformation of a crystallite is based on two distinct physical mechanisms: crystallographic slip due to dislocation motion on the active slip systems and elastic lattice distortion. Following this, the deformation gradient is divided into an elastic and a plastic part: elastic part accounting for rigid body rotation and plastic part due to slip on specific slip systems. The rate sensitive constitutive behavior of each crystal is specified by the following equation:

$$\overset{\nabla}{\boldsymbol{\sigma}} = LD - \dot{\sigma}^0 - \sigma \mathbf{tr} D \tag{1}$$

where $\overset{\nabla}{\boldsymbol{\sigma}}$ is the Jaumann rate of Cauchy stress, L is the elastic moduli tensor, D is the strain rate tensor and $\dot{\sigma}^0$ is the visco-plastic stress rate which depends on the slip rates of the slip systems. In Eq. 1, the term **tr***D* is the trace (i.e. the first invariant) of the 2nd order strain rate tensor. The slip rates are calculated by adapting the Saimoto and VanHoutte [29] relationship, previously implemented in polycrystal plasticity by Brahme et al. [30], given below:

$$\tau = \left[(2+\beta)(\alpha\mu b)^2 \frac{1}{4b} \right]^{\frac{1}{(2+\beta)}} \dot{\gamma}^{\frac{1}{(2+\beta)}} C_{(P/A)} \dot{\gamma}^{m} \tag{2}$$

where the term $C_{(P/A)}$ is given by:

$$C_{(P/A)} = C_f \left(\frac{P}{A} \frac{1}{C_1} \right)^{\frac{1}{(2+\beta)}} \tag{3}$$

In Eq. 2 and Eq. 3, the parameter α is related to the strength of obstacles, μ is the shear modulus, b is the burgers vector and m is the strain rate sensitivity index. The terms P, A, C_1, C_f and β are material dependent parameters and are obtained from stress strain curve using the procedure given in Ref. [29]. The relationship in Eq. 2 can be re-written to have a similar form as originally proposed by Asaro and Needleman [27] for implementation within finite element code.

Backstress Model

Based on the previous works of M. Sauzay [10] and Brahme et al. [1], the dislocation microstructure is modeled as a two-phase material consisting of highly dense hard cell walls and less dense soft cell interiors as shown schematically in Figure 8. The formation of such cell structures is a source of internal stresses. Assuming equiaxed dislocation cells and by using Eshelby's results for an embedded spherical inclusion [31], one can compute the backstress τ_b due to dislocation cells as follows [10]:

$$\tau_b = \left(\frac{f_w}{1-f_w} \right) \mu (1-\chi_E) \gamma \tag{4}$$

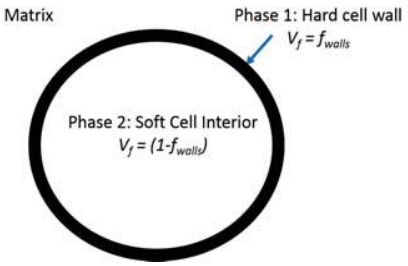

Figure 8: Schematic showing the two-phase approach used for computing intragranular stresses

In Eq. 4, f_w is the total volume fraction of cell walls and χ_E is the factor from Eshelby's analysis of spherical inclusion given by:

$$\chi_E = \left(\frac{2}{15} \right) \left(\frac{4-5\nu}{1-\nu} \right) \tag{5}$$

where ν is the Poisson's ratio of the material. By combining Eq. 2 and Eq. 4, one can compute the overall flow stress contribution including backstress effects.

Problem Formulation and Results

Cyclic FRF planar simple shear simulations are performed using the preceding polycrystal modeling approach embedded within a finite element framework [1,32,33]. The initial texture used for the simulation is shown in Figure 4. An EBSD map of 286 grains has been incorporated to a finite element mesh using 60,000 elements. Each pixel in the EBSD map is represented as an element in the FEM mesh. Each element in the mesh has a dimension of 5 μm x 5 μm. The finite element framework includes certain parallel computing algorithms so that simulations can be performed with models containing a sufficiently large numbers of elements. Details regarding the parallelization method can be found in Inal et al. [33-34].

Figure 9 compares the simulated results of cyclic simple shear with the experimental data. The "without backstress" results in the figure are computed by solely using the relationship given in Eq. 2. It is observed that without backstress, the polycrystal model is able to predict the shear stress – shear strain response during initial forward shearing. However, once the loading is reversed, the model is no longer able to take in to account the early yielding (i.e. Bauschinger) phenomenon during reverse loading and the simulation results start to differ from thereafter. On the other hand, simulation results with the backstress model show an excellent conformity to the experimental data.

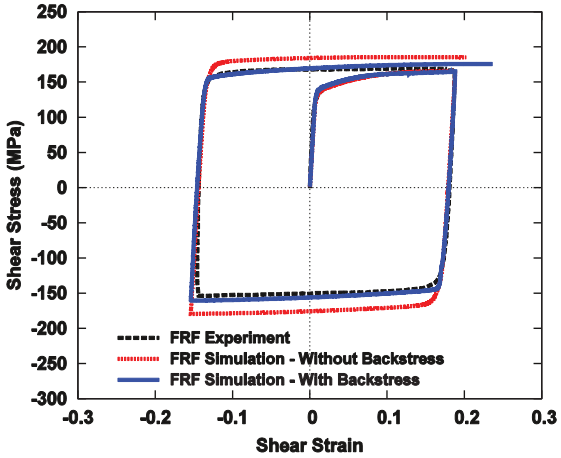

Figure 9: Comparison between simulated FRF simple shear response and experimental data for extruded AA6063-T6

Summary and Conclusions

Deformation induced dislocation microstructures tend to form in FCC metals and alloys during deformation. These microstructures are composed of domains or cells which are bounded by boundaries made up of dense dislocation walls and a less dense cell interior. The formation of these dislocation microstructures induce backstresses which can give rise to the Bauschinger effect at the macroscopic scale. A backstress model is proposed to compute the effects of these intragranular backstresses and is implemented in to a rate-dependent crystal plasticity constitutive model. Cyclic simple shear simulations are performed and the results are compared against the experimental data. The simulated results using the new backstress model show an excellent agreement to the experimental data. In particular, the model is able to reproduce the experimentally observed Bauschinger effect.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Quasi and Dynamic Compression of ECAP Processed AA 6082

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Keywords: Equal channel angular pressing, 6082 AA, Aging, Dynamic compression, Modeling.

Abstract

AA 6082 in as received T651 condition was solution treated and immediately processed via equal channel angular pressing (ECAP) to one and two passes and then naturally aged to 100 days. The processed samples were subjected to quasi and dynamic compression testing using regular universal testing machine and split Hopkinson pressure bar (SHPB), respectively. The testing was conducted at a wide range of strain rates and temperatures and the constitutive Johnson model constants were identified. The effect of cryo temperature versus room temperature during dynamic compression was, also, evaluated in terms of the evolution of microstructure and stress – strain response.

Introduction

Aluminum alloy (AA6082) belongs to a class of heat-treatable Al-Mg-Si alloys. These alloys offer better corrosion resistance and strength levels among all the 6xxx series aluminum alloys. These features make them a good candidate to be used in the aerospace and the automotive applications. Numerous studies have been performed in the last two decade on ways or methods to increase the strength of aluminum alloys [1-4]. Of the many factors that can contribute to enhance the mechanical properties of these alloys, reducing their grain-size and manipulating the morphology and chemical compositions of the precipitated second phase particles were observed to be the most significant factor. By using various thermo mechanical heat treatments, the initial grain size of the polycrystalline material can be reduced to several micrometers. However, to reduce the grain size below the micrometer range to nanometer range, alternative techniques needs to be implemented.

In order to synthesize Ultrafine grain (UFG) materials from a coarse-grained material, it is imperative to subject the coarse grained material to severe plastic deformation that induces remarkably high strains. Recently, "top-down approach" with equal-channel angular pressing (ECAP) technique being the most prominent, has been widely used for producing UFG metals and alloys. ECAP is a processing technique whereby an intense plastic strain is imposed by pressing a sample in a special die [5-9].

The objective of the current work is to investigate the synergetic effect of sever plastic deformation, post deformation natural aging and strain rate of deformation in enhancing the strength of light weight 6082 aluminum alloy.

Experimental Procedures

The material used in this study is the commercial 6082 aluminum alloy, which belongs to the heat-treatable Al–Mg–Si alloys. The chemical composition of the alloy in wt% is 0.75 Mg, 1.2 Si, 0.79 Mn, 0.25 Fe and rest is Al. The material was obtained in plate form whose initial dimensions were 2m x 2m x 15 mm. Round billets of 75 mm length and 10.9 mm in diameter was machined from the plate material. Machining was done in the presence of coolant so as to avoid any heating of the material. Initially, all the samples in this study were solution treated (ST) to 570 ^{o}C for 2 h and quenched in water at room temperature before ECAP processing.

The die used to ECAP process the samples is designed and manufactured from three assembled tool steel blocks. The die angles were machined to be: Ψ=0 and ϕ=90^{o}. The combination of these angles would impose an equivalent strain of 1.2 on the ECAP processed material for each pass. The die material was H13 tool steel heat treated to HRC of about 50. The billet samples were ECAP processed using Route C (where the sample is rotated by 180° between each subsequent passes) to one and two passes. For successful processing of samples ECAP processing was performed within 10 minutes after quenching the samples in water. This was done to avoid any natural ageing that might otherwise occur if there is any significant delay in processing.

The ECAP processed samples were then machined to obtain compression samples of 14.2 mm length, 10.9 mm in diameter for quasi-static experiments (1s^{-1} to 10^{-4} s^{-1}) and 2.4 mm length, 8 mm in diameter for dynamic experiments. Molycote grease in combination of teflon sheets was used as a lubricant. All the quasi-static experiments were performed using Instron Material Testing System and the dynamic experiments were performed using Split Hopkinson Pressure Bar (SHPB). The conditions investigated are as follows:

a. Solution Treated (570 ^{o}C for 2 hrs followed by quenching) → **compressed at different strain rates and temperatures**.

b. Solution Treated (570 ^{o}C for 2 hrs followed by quenching) → ECAP (to 1 pass) → Natural Aged (for 100 Days at Room temperature) → **compressed at different strain rates and temperatures**.

c. Solution Treated (570 ^{o}C for 2 hrs followed by quenching) → ECAP (to 2 passes) → Natural Aged (for 100 Days at Room temperature). → **compressed at different strain rates and temperatures**.

d. Solution Treated (570 ^{o}C for 2 hrs followed byquenching) → ECAP (to 2 passes) → **compressed at high strain rates at room as well as cryogenic temperatures**.

Condition (a) is taken as the reference condition to which the strength improvement, achieved in the other conditions (b, c and d), is compared to. The Vickers micro hardness measurements were recorded before performing compression

Results and Discussions

The original Johnson-Cook (JC) constitutive model is given as,

$$\sigma = \left[A + B\varepsilon_p^n\right] \times \left[1 - \left(\frac{T - T_{ref}}{T_m - T_{ref}}\right)^m\right] \times \left[1 + CLn\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)\right] \quad (1)$$

Two important observations were made based on the experimental results. They were as follows:

1. The yield stress of the material was non-linearly dependent on temperature and
2. The work hardening of the material decreases with increase in temperature.

Hence, to model the experimental responses of the ECAP processed material, it was deemed imperative to have two different material constants (m_1 and m_2) for the temperature term (in the modified model) as oppose to one material constant (m) in the original JC model.

The flow stress, σ of the modified Johnson-Cook (JC) constitutive model used in the current study is expressed as [10-12],

$$\sigma = \left[A \times \left(\frac{T_m - T}{T_m - T_{ref}}\right)^{m_1} + B \times \varepsilon_p^n \left(\frac{T_m - T}{T_m - T_{ref}}\right)^{m_2}\right] \times \left[1 + C \ln\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)\right] \quad (2)$$

Where σ is the flow stress, A is the yield stress (in MPa) of the material obtained from experimental result performed at room temperature at a strain-rate of 10^0 s^{-1}, B is the strain hardening coefficient (in MPa), ε_p is the plastic strain, n is the strain hardening exponent, $\dot{\varepsilon}$ is the strain-rate, $\dot{\varepsilon}_0$ is the reference strain rate (at which the material constant A, B and n are determined) taken here as 1 s^{-1}, C is the strain-rate sensitivity exponent, m_1 and m_2 are the temperature sensitivity exponents, T_m, T, T_{ref} is the melting, test and the reference temperature respectively.

The strain rate sensitivity (m) was determined using the relationship:

$$m = \frac{\ln\left(\sigma_2 / \sigma_1\right)}{\ln\left(\dot{\varepsilon}_2 / \dot{\varepsilon}_1\right)} \quad (3)$$

where σ_1 and σ_2 are the flow stress value at the estimated strain, from the experiments performed with strain rates of $\dot{\varepsilon}_1$ and $\dot{\varepsilon}_2$, respectively. An average m value was then calculated at the three quasi-static strain rates (exclusive of the dynamic test conducted) at room temperature by calculating separate m values at four different strain values of 5%, 10%, 15%, and 30%.

Fig. 1 shows the room temperature true stress-strain responses of AA6082 at different strain-rates. The as-received alloy was subjected to condition (a). Three room temperature quasi-static and one dynamic experiment were performed on this alloy. These experiments were performed to study the responses of the as-received alloy subjected to solution treatment solely. The results indicate that the alloy showed very slight positive strain-rate sensitivity with significant work hardening at room temperature. The room temperature strain-rate sensitivity (m, from Eqn. 3) was found to be 0.0065. The yield strength, from the experimental response at a strain rate of $10^0 s^{-1}$ and 4000 s^{-1}, of the alloy was found to be 175 and 210 MPa, respectively. These values are taken as the reference yield strength value, for quasi and dynamic compression, to which the improvement in strength values achieved in the other conditions of testing is compared to. Vickers micro hardness of solution treated samples is about 59.14 HRV. Fig. 2 shows the true stress-true plastic strain responses of the as-received AA6082 (Solution treated- treatment without ECAP processing; condition (a)) along with correlations from Johnson-Cook (JC) Model. The experimental results are shown by symbols and the model correlations are shown with lines. The initial material constants were further refined using the MATLAB least square optimization procedure. The refined material constants are shown in Table 1. From the figure, it is very clear that the correlations using the modified JC model is in close proximity to the experimental observation. Also, the yield stress values predicted from the model (170 MPa) is close to the observed experimental values (175 Mpa).

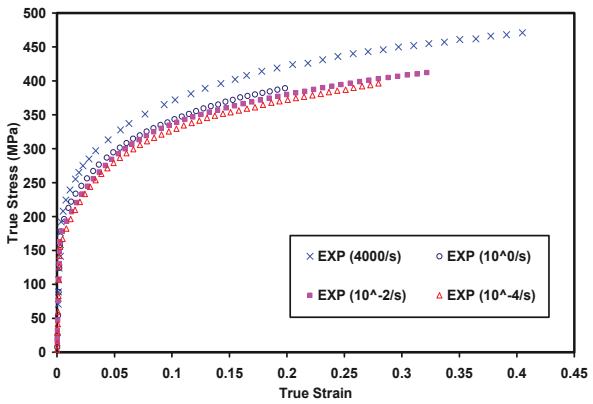
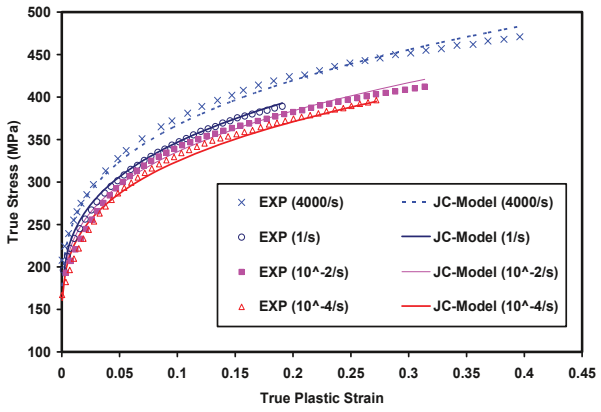
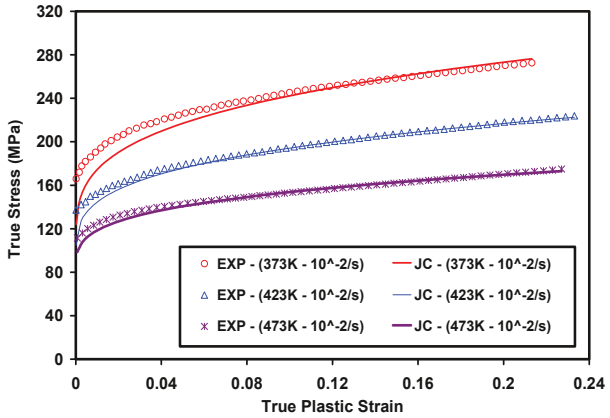


Fig.1 True stress-strain responses of AA6082, subjected to solution treatment (condition (a)), at different strain-rates at room temperature.



(a)



(b)

Fig. 2. (a) Room Temperature True Stress-Strain Responses (symbols) of solution treated AA6082 (condition (a)), along with correlations from the modified JC Model (Lines). (b) High temperature stress – strain response for strain rate of 10^{-2} s^{-1} along with JC Model (Lines).

Figs. 3 and 4 shows the room and high temperature true stress-strain responses of the AA6082 at different strain rates, for conditions (b) and (c), respectively. After ECAP processing to 1 and 2 passes, the samples were allowed to age at room temperature for 100 days (conditions (b and c)). Due to the solution treatment, conducted prior to ECAP, the solute atoms in the alloy (Si, Mg) are completely dissolved in the aluminum matrix. Quenching results in trapping the solute atoms in the aluminum matrix creating a super saturated solid solution. There by clearing the grain boundaries. Aging the material, results in the solute atoms to precipitate at the grain boundaries hindering the dislocation motion. Thus, allowing the material to age harden. Three room temperature quasi-static and one dynamic compression experiments, besides three different temperatures at a strain rate of 10^{-2} s^{-1}, were performed on each ECAP and naturally aged samples. It is observed that, the alloy exhibited very slight positive strain-rate sensitivity and reduced work hardening behavior at room temperature. The yield strength (from the experimental response at strain rate of $10^{0}s^{-1}$) of the alloy was found to be 400 and 470 MPa, after ECAP processing to 1 and 2 passes followed by natural aging, respectively. The relative increase in the material strength in condition (c), compared to the reference condition (condition (a)) is about 170%. The strain-rate sensitivity parameter (m) was found to be 0.0059. The yield strength of the sample ECAP processed to 2 passes prior to natural ageing exhibited a value of 450 MPa [13]. This means that natural aging has resulted in a 30 MPa increase in the yield strength for the case of strain rate of $10^{0}s^{-1}$.

Figs. 5 and 6 show the true stress-true plastic strain responses of AA6082 (condition (b)) along with correlations from the modified Johnson-Cook (JC) Model, for room and high temperature, respectively. Figs. 7 and 8 shows the true stress-true plastic strain responses of AA6082 (condition (c)) along with correlations from the modified Johnson-Cook (JC) Model, for room and high temperature, respectively. The experimental results are shown by symbols and the model correlations are shown with lines. The initial material constants were further refined using the MATLAB least square optimization procedure. The optimized material constants are shown in Table 1.

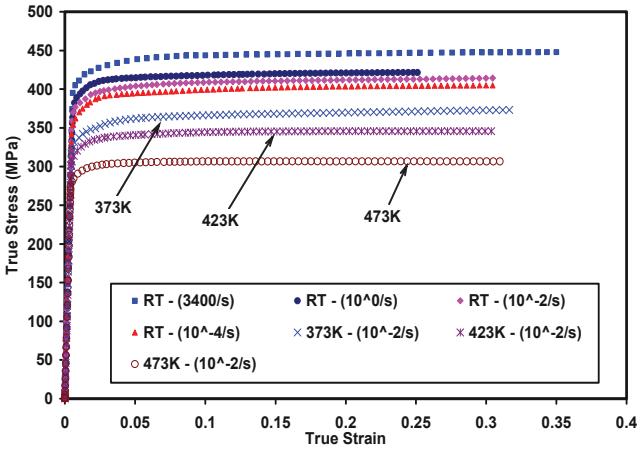


Fig. 3. True stress-strain responses of AA6082 (subjected to solution treatment + ECAP to 1Pass followed by Natural Aging for 100 Days), at different strain-rates at room temperature and three different temperatures at strain- rate of $10^{-2}s^{-1}$.

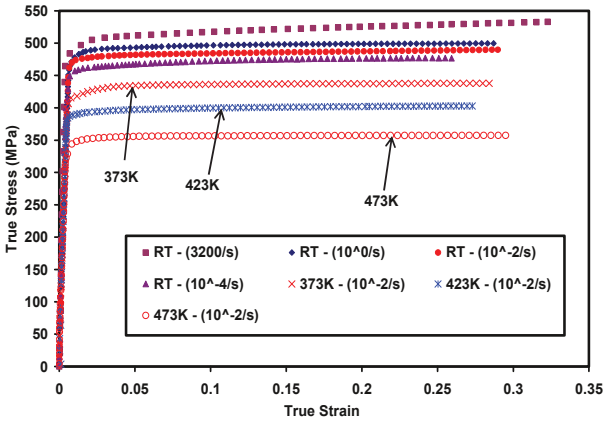


Fig. 4. True stress-strain responses of AA6082 (subjected to solution treatment + ECAP to 2Passes followed by Natural Aging for 100 Days), at different strain-rates at room temperature and three different temperatures at strain- rate of $10^{-2}s^{-1}$.

Fig.9 represent the true stress – true strain for samples ECAP processed to 2 passes and dynamically compressed at almost same strain rate at cryogenic and room temperature (no post ECAP natural aging). The idea of cryogenically compressing the ECAP processed material is to suppress the possibility of dynamic recovery that can accompany the high speed deformation of high stacking fault energy alloys such as the investigated 6082 aluminum alloy. The yield strength of cryogenic dynamic compression was around 620 MPa and room temperature dynamic was around 475 MPa. Almost same strain rate is employed, thus the difference in strength (155 MPa) is mainly due to the cryogenic temperature and suppression of dynamic recovery. Comparing the cryogenic dynamic to the reference condition (a) in the dynamic test (210 MPa), the relative increase in yield strength is about 197%. Comparing the room temperature dynamic yield strength of 475 MPa (Fig. 9) to the room temperature dynamic yield strength of 490 MPa achieved when natural aging was involved (Fig. 4), gives the effect of natural aging, which is about 25 MPa increase.

ECAP and cryogenic dynamic compressing the processed billets has proven to give better strength improvement over when natural aging for 100 days was involved (620 MPa compared to 500 MPa, which is the dynamic yield strength for condition (c)).

Other themo-mechanical processing, presented elsewhere [13], for this alloy involved dynamic aging by warm ECAP processing to 1 pass (samples were solution treated, quenched and immediately ECAP processed at a temperature of 150 ^{o}C) which gave a dynamic yield strength of 500 MPa, and another treatment that involved post ECAP artificial aging (samples were solution treated, ECAP processed top 2 passes and then aged at 140^{o}C for 7.5 hrs), which gave a dynamic yield strength of 520 MPa. These values are comparable to what is reported here for ECAP and naturally aged samples (condition (b)).

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Conclusions

6082 aluminum alloy was subjected to different thermo-mechanical treatments with the objective of improving the achieved yield strength. The room temperature quasi and dynamic yield strength of the solution treated alloy are 175 and 210 MPa respectively. These values were increased to 470 and 490 MPa when ECAP to 2 passes and natural aging were employed. Cryogenic dynamic compression has proven to be an efficient tool to bring the yield strength, of the ECAP processed samples (2 passes), to about 620 MPa. Comparing this value to the 210 MPa presents the synergetic effect of ECAP and cryogenic dynamic compression that is capable of imparting about 196% relative increase in strength.

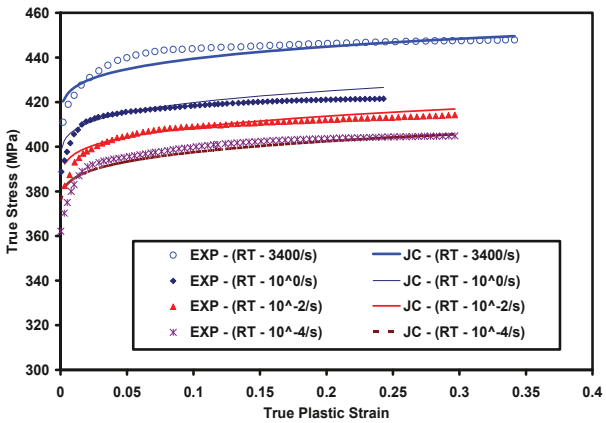

Fig. 5. Room Temperature True Stress-Strain Responses (symbols) of AA6082 along with correlations from the modified JC Model (Lines) for condition (b).

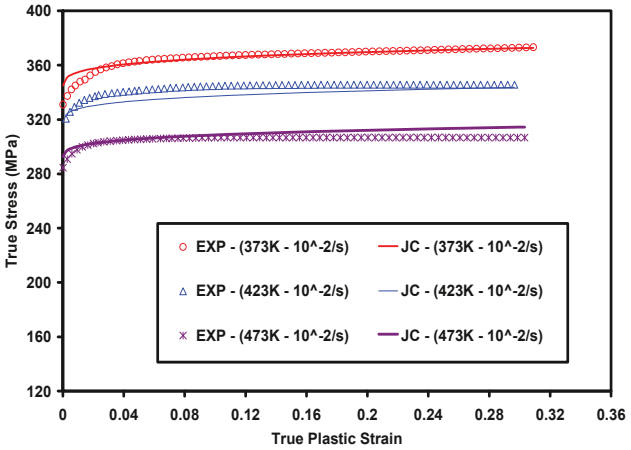

Fig. 6. High temperature True Stress-Strain responses (symbols) of AA6082 along with correlations from the modified JC Model (Lines) for condition (b).

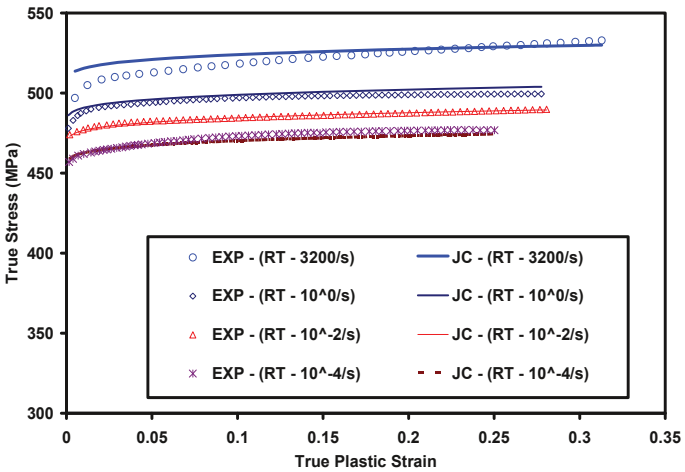

Fig. 7. Room Temperature True Stress-Strain Responses (symbols) of AA6082 along with correlations from the modified JC Model (Lines) for condition (c).

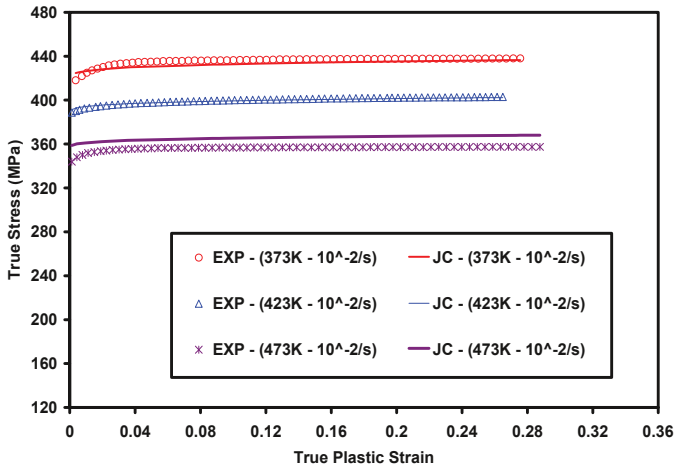

Fig. 8. True Stress-Strain Responses (symbols) of AA6082 along with correlations from the modified JC Model (Lines) for condition (c).

Table 1 Modified Johnson-Cook (JC) materials constants.

Material Type	A (MPa)	B (MPa)	*n*	*C*	m_1	m_2
Solution Treated Condition **(a)**	123.2	434	0.2876	0.007	1.1	3.12
Solution Treated Condition **(b)**	388	52.4	0.217	0.00575	0.78	1.073
Solution Treated Condition **(c)**	478	34.1	0.21	0.0062	0.77	1.78

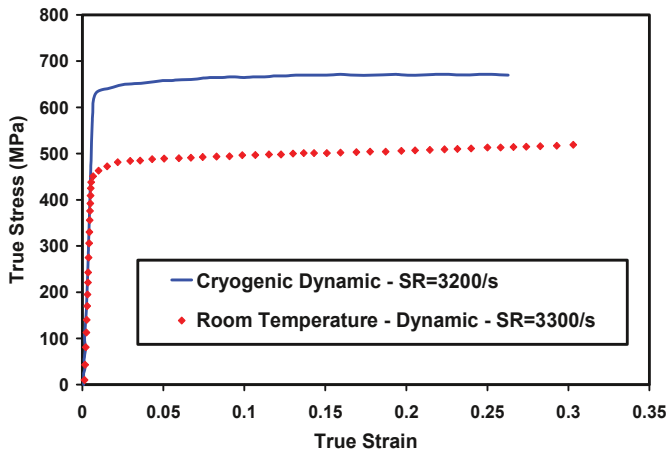

Fig. 9. Dynamic true stress-strain responses of AA6082 ECAP processed to 2 passes and consequently subjected to dynamic loading at cryogenic and room temperature.

Acknowledgment

This project was funded by the National Plan for Science, Technology and Innovation (MARRIFAH), King Abdulaziz City for Science and Technology, Kingdom of Saudi Arabia, Award Number (**10-ADV1032-02**)

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

STUDY ON HOT SIZING AND CREEP-AGEING BEHAVIOR OF Al-Cu-Mn CAST ALLOY

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Keywords: Al-Cu-Mn alloy, Hot sizing process, Creep-ageing behavior, Constitutive model, Genetic algorithm

Abstract

The deformation of large structural cast Al components during quenching is difficult to control and easily out of tolerance. Hot sizing based on creep-forming principle is a new technology to correct inaccuracies of parts in shape and dimensions. Thus, creep-ageing curves of Al-Cu-Mn alloy were conducted under constant tensile stresses ranging 10MPa to 150MPa at 175℃。The curves at high stresses were sparse, while at low stresses were dense. Then, a hyperbolic sine type constitutive model was achieved with genetic algorithm. The creep-ageing model was derived mathematically, but in good consistence with experimental curves with average relative error of 3.85%.

Introduction

Deformation of aluminum alloy structural components during heat treatment largely increases the production cost and lowers down the productive efficiency. Heat treatment, typically referred as T6, includes three processes: solid-solution treatment, quenching and ageing treatment. Structural components mainly are distorted during quenching due to rapid cooling. Then, precipitates, in a finely dispersed way, are introduced in the alloy to increase the yield strength. If heat-treatment distortion, particularly during quenching, is beyond the allowed value, some correcting processes, such as reheat-treatment and manual sizing, have to be used[1]. The more processes mean longer productive time and lower efficiency.

Hot sizing process can decrease quenching distortion of structural components without more remedial methods. Hot sizing is combination of age hardening and creep forming[2, 3]. After quenching, the solid solution with low strength is easy to correct the deformation. If stresses were applied on the components during ageing process, creep and age hardening occur together and simultaneously[4]. So, hot sizing process not only makes materials hardening, but corrects or reduces deformation of components.

To make the new process available, the creep-ageing behavior of materials has to be studied firstly. With the help of creep-ageing model of a material, the crucial applied stress can be deduced to decrease the deformation. In this paper, the relationship between applied stress and creep-ageing behavior was studied in detail.

Materials and Experiments

Al-Cu-Mn alloy, one of high strength cast aluminum alloys, is widely used in aeronautics and astronautics to produce large structural components. Quenching distortion is a very serious manufacturing problem for Al-Cu-Mn alloy, which is studied in this paper. The chemical composition of Al-Cu-Mn alloy is showed in Table 1.

Table 1. Material composition in weight percent of Al-Cu-Mn alloy

Element	Cu	Mn	Ti	Cd	V	Zr	B	Al
Wt (pct)	4.6~5.3	0.3~0.5	0.15~0.35	0.15~0.25	0.05~0.3	0.05~0.5	0.005~0.06	balance

The creep behavior can be derived from creep tensile curves, which are obtained by constant temperature and stress tests. Tensile samples were designed and machined into rod shape with two screw thread ends according to GB/T 2039-1997, as Figure 1. Two raised earring-like structures near the gage were designed and used to set up displacement sensors. The samples were stretched on RD-50TM constant temperature and stress test machine, as Figure 2, which has maximum loading of 5 tons, maximum elongation of 10mm with 2um displacement precision, and maximum temperature of 600℃ with 2℃ temperature precision.

Figure 1. Creep tensile samples

Figure 2. RD-50TM test machine

All samples were heated in Muffle furnace at 540℃ for 10 hours in order to obtain fully solution treatment. After that, samples were quenched in 11% UCON quenchant. And then, samples were tested on RD-50TM test machine at 175℃ ageing temperature and different constant tensile stress from 10MPa to 130MPa for 15 hours.

Results and Discussion

Experiment curves

Creeping tensile curves of Al-Cu-Mn alloy at 7 tensile stresses (10MPa, 30MPa, 50MPa, 70MPa, 90MPa, 110MPa and 130MPa) were obtained according to the above conditions, as Figure 3.

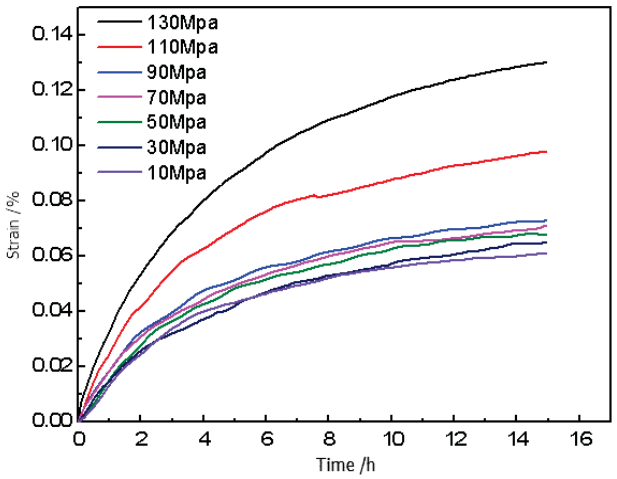

Figure 3. Experimental curves of Al-Cu-Mn alloy at 175℃

At constant tensile stress, creep-plastic strain of Al-Cu-Mn alloy becomes larger over time. At the initial stage, the curves are steep with greater slope, which means higher strain rate. As creep-ageing time goes on, the curves become flat, which means lower strain rate. Strain rate curves at 90MPa, 110MPa and 130MPa are given in Figure 4. At the first 2 hours, strain rates are high and almost remain unchanged. After that, the strain rates decrease rapidly and deformation of samples slows down.

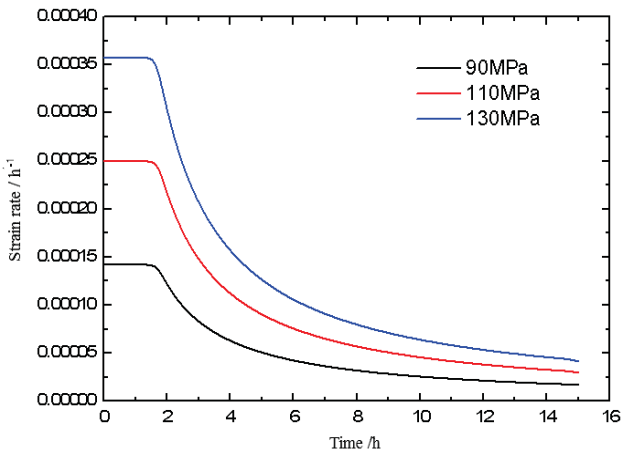

Figure 4. the strain rate curves under applied stresses of 90MPa, 110MPa and 130MPa

At the same time, the higher the stresses, the bigger plastic strains of Al-Cu-Mn due to creep tension. The amount of plastic strain at 8 hours as a function of stress is shown in Figure 5. Firstly, the higher stress level can promote the movement of dislocation more in the creeping process. Furthermore, the movement of atoms and holes in the grains and boundaries of Al-Cu-Mn alloy will be easier to arouse

variation of the microstructure and deformation of the alloy under more stress. Besides, much stronger stresses help the moving of grain boundary, resulting in greater plastic strain.

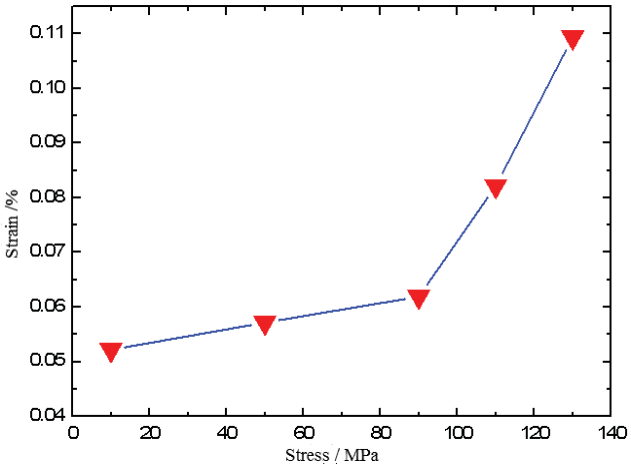

Figure5. The strain vs stress curve at 8 hours

The experimental results showed that the plastic strain due to creeping in Al-Cu-Mn alloy was very sensitive to the changes of the applied stresses. As shown in Figure 3 and Figure 6, with the increment of applied stresses, the creep curves of Al-Cu-Mn alloy are dispersed gradually from denseness to sparseness, and the distance between curves increases obviously. When the applied stress is more than 150MPa, the elongation of tensile samples will greater than 0.6% after 15-hour creep. The elongation after 15-hour creep increased by nearly 0.5% with the applied stress increasing from 130MPa to 150MPa. However, the elongation only increases by nearly 0.02% with the applied stress increment of 20MPa in the range of 90~130MPa, while nearly 0.005% in the range of 10~90MPa. As a result, the applied stress has a nonlinearly huge impact on the movement of dislocation, motion of atoms and holes and variation of grain boundary, changing the microstructure of creep Al-Cu-Mn alloy, which means that the sensitivity of plastic strain and deformation to applied stress is different and nonlinear in the macroscopic.

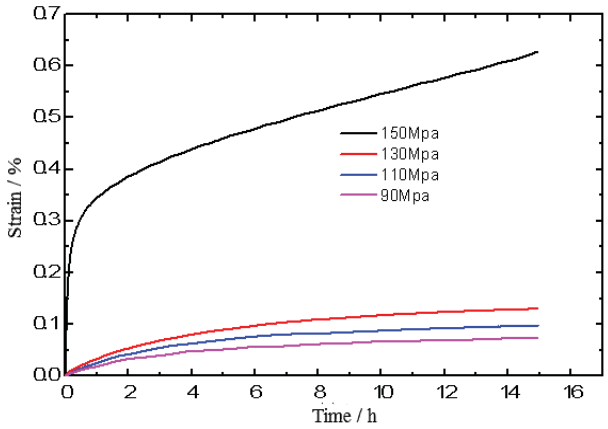

Figure 6. The compared curves under applied stresses of 150MPa

Creep-ageing model

Hyperbolic sine type constitutive model, comprising three state variables to characterize creep ageing mechanism, is able to express the primary, secondary and tertiary creep stage[5]. In hot sizing process, only primary and secondary creeps make a difference. Thus, creep damage, which is related only to tertiary creep, does not appear in hot sizing[6]. So, the simplified form of hyperbolic sine model is derived.

$$\dot{\varepsilon}_c = A\sinh[B\sigma(\sigma-\sigma_0)(1-H)^{m_0}] \quad (1)$$

$$\dot{H} = \left(\frac{h_c}{\sigma^{m_1}}\right)\dot{\varepsilon}_c\left(1-\frac{H}{H^*}\right) \quad (2)$$

The material constants A and B are related to secondary (steady state) creep. Primary creep is described by using H, which represents the dislocation hardening and varies from 0 at the beginning of creep process to H*. The model is combination of creep deformation and age-hardening, the interaction between them is not considered in the above equations, although the equation can be used to predict spring-back.

In the creep-ageing model of Al-Cu-Mn alloy, the expression of $\dot{\varepsilon}_c$ contains the state variable H which depends on time. And the value of H in turn varies with the magnitude of $\dot{\varepsilon}_c$. Besides, variables H and $\dot{H}$ are coupled in a differential form, as shown in Eq 2. Thus, it is very difficult to get the expression of strain, that is $\varepsilon = f(\sigma, t)$. Meanwhile, the constitutive model contains 7 undetermined material parameters, which also make an explicit solution

unachievable. Therefore, a modern optimization algorithm, genetic algorithm (GA), was developed to get an optimal solution for the constitutive model.

Genetic algorithm, as a good optimization method that mimics the process of natural selection, has features of robustness and global convergence. Because the creep behavior of Al-Cu-Mn alloy presented huge difference in different scopes, the hyperbolic sine model in the low stress range (10MPa~90MPa) was fitted based on the experimental curves. The fitted model is shown in Eq 3 and Eq 4. The mean square error between calculated values based on the fitted model and experimental curves is only 7.96×10^{-5} with average relative error of 3.85%. And the compared results between calculated values and experimental results is shown in Figure 7. Consequently, the creep-ageing model fitted by genetic algorithm is suitable for the creep ageing behavior of Al-Cu-Mn alloy in low stress range.

$$\dot{\varepsilon} = 9.79\sinh\left[1.19\times10^{-5}(\sigma+209.1)(1-H)^{-2.15\times10^{3}}\right] \tag{3}$$

$$\dot{H} = \frac{-2.40\times10^{-2}}{\sigma^{0.047}}\dot{\varepsilon}\left(1-\frac{H}{3.328\times10^{13}}\right) \tag{4}$$

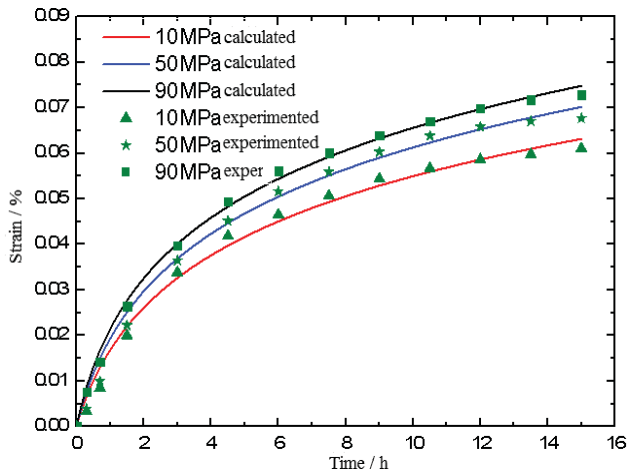

Figure 7. Comparison between calculated values and experimental results in the low stress range

Conclusions

Hot sizing process based on creep ageing behavior of Al-Cu-Mn alloy was proposed and studied in this paper to reduce the distortion caused by rapid cooling in quenching. Many groups of constant temperature and stress tensile test were designed and conducted to get the creep ageing behavior of Al-Cu-Mn alloy at aging condition (175℃, 15 hours). Three experimental rules can be summarized and concluded based on the experimental curves:

(1) The creep strain of Al-Cu-Mn alloy under the same applied stress increases gradually as creep-ageing time goes on;

(2) When the applied stress, unable to make the alloy yield, increases, the plastic strain caused by creep becomes greater;

(3) The sensitivity of plastic strain curves to applied stress is complex and as a function of applied stress. When the applied stress is stronger, the increment of plastic strain is larger.

The hyperbolic sine constitutive model was fitted by genetic algorithm, which simplified the tertiary (damage stage) creep. The constitutive model is in good accordance to the experimental curves of Al-Cu-Mn alloy in the low stress range (10MPa ~ 90MPa). The calculated values are very consistent with experimental results with average relative error of 3.85%.

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Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

GRAIN REFINEMENT MECHANISM OF ALUMINUM BY Al-Ti-B MASTER ALLOYS

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Keywords: Aluminum, grain refinement, Al-Ti-B, Al-5Ti-1B, grain refiner

Abstract

Al-Ti-B type master alloys are routinely added into aluminum for a grain refinement purpose. However, a consensus about the mechanism of grain refinement is still yet to reach. This work is aiming at further understanding the grain refinement mechanism through studying the stability of TiB_2 in pure aluminum and TiB_2/Al_3Ti interfaces in aluminum by transmission electron microscopic (TEM) techniques. It is found that TiB_2 reacts with aluminum forming $(Ti,Al)B_2$ solid solution and releasing titanium into aluminum. A crystallographic orientation relationship exists between TiB_2 and Al_3Ti in Al-Ti-B alloys. A dynamic nucleation theory for the grain refinement is therefore proposed as a dynamic Al_3Ti layer in the thickness of a couple of atoms existing on the surface of $(Ti,Al)B_2$ and being responsible for nucleating aluminum grains.

Introduction

Grain refinement by adding master alloys is a routing industrial process for the casting of aluminum, particularly DC casting [1, 2]. To achieve better mechanical properties and easy processing, rolling and extrusion etc., fine grains are desired. By adding Al-Ti-B type master alloys, the casting grain size can be reduced down to a 200 μm level. However, there are still issues unsolved, such as poisoning and fading in cases [3]. These are mainly due to the lack of understanding of the grain refinement mechanisms though several mechanisms have been proposed since 1945 when Al-Ti-B grain refiners were discovered [4-6]. In general, the proposed mechanisms are based on either to promote nucleation of α-Al grains or to restrict the growth of the α-Al grains after nucleation. For the latter, it is the constitutional undercooling effects of alloying elements, titanium in this case because the concentration of boron in solution is very low in the presence of titanium [7, 8]. Meanwhile the former is based on heterogeneous nucleation of α-Al grains on TiB_2 directly or indirectly through Al_3Ti on its surface [6, 9, 10]. However, due to the low concentration of titanium and boron in the final castings, no direct observation has been reported. This paper is therefore focusing on the stability of TiB_2 in aluminum and the phase relationships between borides and Al_3Ti for the heterogeneous nucleation of α-Al grains.

A typical master alloy for aluminum grain refinement is Al-5wt%Ti-1wt%B, which gives 0.01wt%Ti and 0.002wt%B in the final castings [4]. Two kinds of particles are present in the master alloys, borides and aluminides [1, 2, 4, 11, 12]. There is not a dispute about the aluminide of being Al_3Ti, while the borides have been identified as a solid solution of $(Ti,Al)B_2$ with a variable Ti/Al ratio [12, 13]. The borides are generally treated as TiB_2 due mainly to its high stability in aluminum. The effects of the formation of $(Ti,Al)B_2$ solid solution on the heterogeneous nucleation of α-Al grains have been ignored despite solid solution of $(Ti,Al)B_2$ has long been reported [4, 11, 12]. Al_3Ti is not as stable as boride in aluminum and dissolves into aluminum in a few seconds as reported. The Al-rich side of an Al-Ti phase diagram is shown in Figure 1, where a peritectic reaction exists at 665ºC and α-Al contains 1.15wt% titanium while liquid aluminum only contain 0.15wt% of titanium [5]. Both concentrations are much higher than the concentration of titanium in solution, 0.01wt%. The nucleation of α-Al grain is therefore promoted by the presence of boron in the form of boride particles. It is therefore important to understand the stability of TiB_2 and the formation of $(Ti,Al)B_2$ solid solution and the crystallographic relationship between the boride phases and Al_3Ti in aluminum melt.

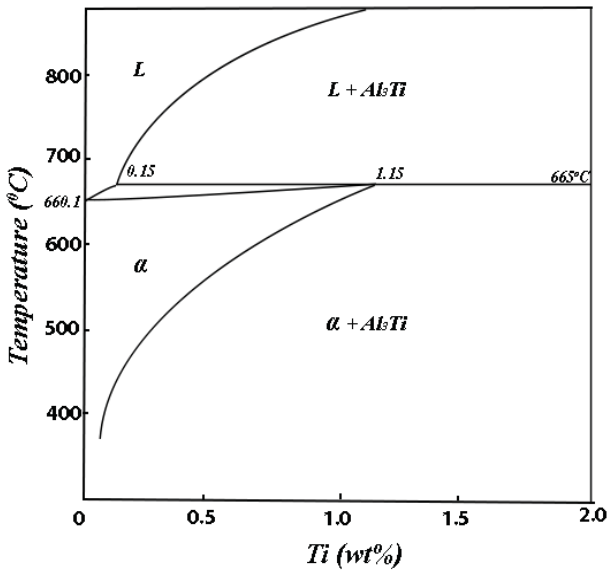

Figure 1, Al-Ti phase diagram showing aperitectic reaction of liquid aluminum and Al_3Ti to form α-Al [5]

Recent research on the interfaces of TiB_2 and Al_3Ti reported the existence of an Al_3Ti monoatomic layer on the surface of TiB_2 particles [14, 15]. A single layer of titanium atoms on the surface of TiB_2 was observed by Z-contrast technique under a TEM and it is claimed Al_3Ti, which is responsible for the nucleation of α-Al grains [14]. *Ab initio* calculations also suggested that a layer of Al_3Ti can be stable on the surface of TiB_2 particles [15]. These claims overlooked one important microstructure of Al-Ti-B master alloys, where large size compounded particles of TiB_2-Al_3Ti are in the α-Al grain centers and small TiB_2 particles are pushed into the α-Al grain boundaries. The unique microstructure of Al-Ti-B master alloys demonstrates that a stable Al_3Ti phase on the surface of TiB_2 particles leads to the agglomeration of Al_3Ti and a small amount of TiB_2 particles while a large population of the TiB_2 particles are pushed into the α-Al grain boundaries. The coarsening of Al_3Ti phase in aluminum can be found in the open literature [4, 16]. It is therefore, for an effective grain refinement of aluminum, stable Al_3Ti on the surface of TiB_2 should be avoided. Instead, only a thin dynamic layer of Al_3Ti on the surface of TiB_2 is effective in grain refinement of aluminum, i.e., promoting the nucleation of

large number of α-Al grains in solidification while avoiding coarsening of the Al_3Ti on boride particles, $(Ti,Al)B_2$ or TiB_2 [6, 13].

Experiments

The investigation consists of two parts: 1) the stability of TiB_2 in aluminum and 2) Al_3Ti/TiB_2 phase relationships in Al-Ti-B master alloys.

To analyze the stability of TiB_2 in aluminum, a paste of TiB_2 particles with coal tar pitch was applied to the surface of cathodes in an electrolysis cell in a Hall–Héroult process. Aluminum was syphoned out of the electrolysis cell after a month of time and analyzed for the concentrations of titanium and boron by optical emission microscope (OEM).

Commercial Al-5wt%-1wt%B and Al-3wt%Ti-1wt%B alloys were used to investigate the phase relationships of TiB_2 (in fact $(Ti,Al)B_2$ solid solution) and Al_3Ti to understand the heterogeneous nucleation of Al_3Ti on TiB_2. The master alloys were produced by chemical reactions of fluoride salts and aluminum, which produce different products depending on the Ti/B ratio and temperature. Alloy samples were taken and prepared following a standard metallographic sample preparation procedure, mounting, grinding, polishing, followed by deep etching with a solution of tartaric acid and iodine in methanol [4]. The solution dissolves the aluminum matrix away effectively and leaves the boride and intermetallic particles untouched. In some cases, the aluminum matrix was totally dissolved away, residual particles were rinsed by methanol and dried up. Scanning electron microscopy (SEM) was done on particles that were stuck on a double sticky electrical conductive tape that has been stuck onto a small piece of steel block on the other side. TEM samples were taken in a focused ion beam (FIB) microscope in a particular way to analyze crystal orientation relationships of the particles. The equipment that have been used in this study are Hitachi S-4800 Field Emission SEM operated at 30keV, a Helios 600i dual column Focused Ion Beam/field emission Scanning Electron Microscope (FESEM) and a Titan 80-300 kV Environmental Electron Microscope operated at 300kV.

Results and discussions

Figure 2 shows the solubility of TiB_2 in aluminum. Five data points were chosen in up to 60 minutes of holding time after aluminum being taken out of the electrolysis cell to minimize the influence of solid TiB_2 particles. A stable concentration of 68.2 ppm of titanium and 12.4 ppm of boron were reached after one month. By considering of 10 ppm titanium in aluminum as impurity in a standard electrolysis cell in the absence of TiB_2 particles, there is about 58.2 ppm of titanium from the dissolution of TiB_2. Interesting enough, the Ti/B ratio is about 5/1, which is the same Ti/B ratio in the most effective grain refiner, Al-5Ti-1B, and final aluminum castings. PoDFA tests were also performed and the concentration of titanium and boron showed no differences, meaning the measured titanium and boron are in solution.

Boride particles in Al-Ti-B type master alloys are called TiB_2, which is, however, a solid solution of $(Ti,Al)B_2$ with a variable Ti/B ratio in fact [4, 11-13]. The opinions about the phase relationships of TiB_2 and AlB_2 are highly divided. Some believe that $(Ti,Al)B_2$ transform into TiB_2 with holding time, while others believe TiB_2 in an aluminum melt transform into a $(Ti,Al)B_2$ solid solution [4, 12, 17-21]. The effects of the stability of TiB_2 on grain refinement of aluminum has been ignored though it is affecting the chemistry of the boride particles, particularly the surface chemistry of the boride particles. And the heterogeneous nucleation theories are based on the nucleation of Al_3Ti on the surface of the boride particles [1, 2, 6, 9, 10]. It is therefore important to understand the stability of TiB_2 in aluminum melt.

This study shows that TiB_2 is not as stable as reported, i.e. insoluble in aluminum. The stoichiometry of TiB_2 gives a Ti/B weight ratio of 2.2, which is lower than the measured number of 4.7. It is an evidence of the replacement of titanium by aluminum and the formation of a $(Ti,Al)B_2$ solid solution [13].

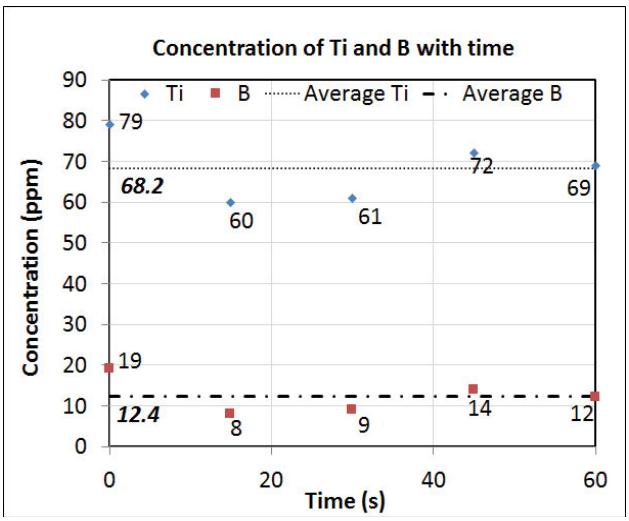

Figure 2, Concentration of titanium and boron in aluminum from the dissolution of TiB_2 particles

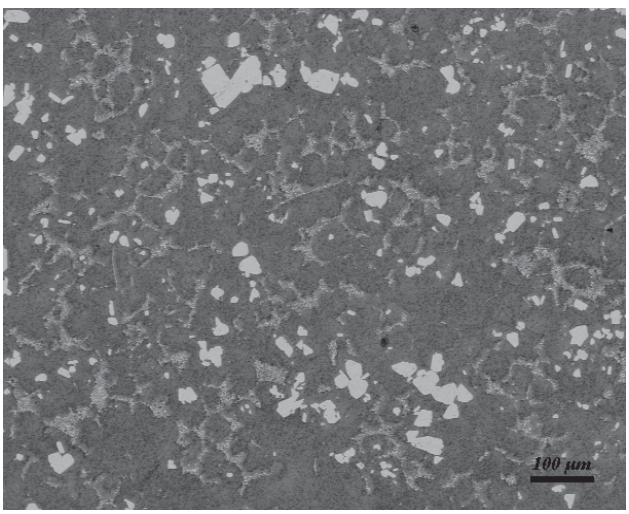

Figure 3 A typical backscattered SEM image of Al-5Ti-1B master alloys showing large Al_3Ti particles in the centers of α-Al grains while smaller boride particles are pushed into the α-Al grain boundaries.

Figure 3 is a typical SEM image of Al-Ti-B master alloys, big Al_3Ti particles in α-Al grain centers and smaller TiB_2 particles at the α-Al grain boundaries. Deep etched sample revealed that small boride

particles at the α-Al grain boundaries form agglomeration with impurities and big particles in the α-Al grain centers are compounded particles of Al_3Ti contain a large number of boride particles, as shown in figure 4. It can be elucidated from figures 3 and 4 that the affinity of boride particles with Al_3Ti is stronger than that of boride with aluminum, i.e. there is a priority for Al_3Ti to nucleate on boride particles in comparison with the nucleation of α-Al on boride particles. However, a stable layer of Al_3Ti on the surface of boride particles leads to the formation of compounded particles, i.e. the growth of Al_3Ti consumes a large amount of titanium in solution and leaves a majority of boride particles uncoated by an Al_3Ti layer. The details of the compounded particles are shown in Figure 5, where individual Al_3Ti nucleated on the surface of boride particles that form the compounded particles in the α-Al grain centers. The growth of Al_3Ti is largely affected by the temperature and the chemistry of the aluminum melt leading to different morphologies, blocky, flaky and petal-like [4]. These microstructures confirm that stable Al_3Ti in an aluminum melt in the presence of boride particles grow into compounded particles and deactivate the majority of boride particles in nucleating α-Al grains.

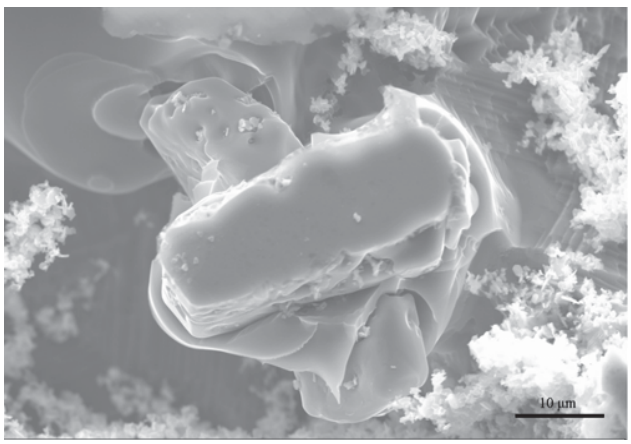

Figure 4 A deep etched sample revealing compounded particles in the α-Al grain centers and boride clusters at the α-Al grain boundaries.

It is widely accepted that Al_3Ti on the surface of TiB_2 is responsible for nucleating α-Al in grain refinement [4, 6, 7, 9, 10, 13, 14, 19, 22]. Recent efforts have been made in identifying an Al_3Ti layer on the surface of TiB_2 through either experimental observation or theoretical calculations despite the stability of Al_3Ti on the surface of TiB_2 particles is simply demonstrated in the microstructure of widely available Al-Ti-B master alloys [14, 15]. To understand the grain refinement mechanism, the question is how a layer of Al_3Ti nucleates α-Al instead of agglomerates into large compounded particles in industrial practice.

Figure 6, a high resolution TEM image, demonstrates the existence of an orientation relationship between Al_3Ti and boride particles. The individual Al_3Ti particles as shown in figure 5 inside the compounded particles nucleate on boride particles. The formation of the compounded particles are through the agglomeration of the individual Al_3Ti particles. To achieve a better grain refinement, the agglomeration of Al_3Ti particles that nucleated on the surface of boride particles must be minimized. By thinking of the effectiveness of adding Al-Ti-B master alloys into aluminum melt for grain refinement and the agglomeration of Al_3Ti in the Al-Ti-B master alloys, the agglomeration of Al_3Ti is avoided during dilution of the Al-Ti-B master alloys in an industrial practice. Further investigation of the dilution of Al-Ti-B master alloys is under way in our lab to understand the agglomeration of the Al_3Ti particles and therefore their effects on grain refinement of aluminum.

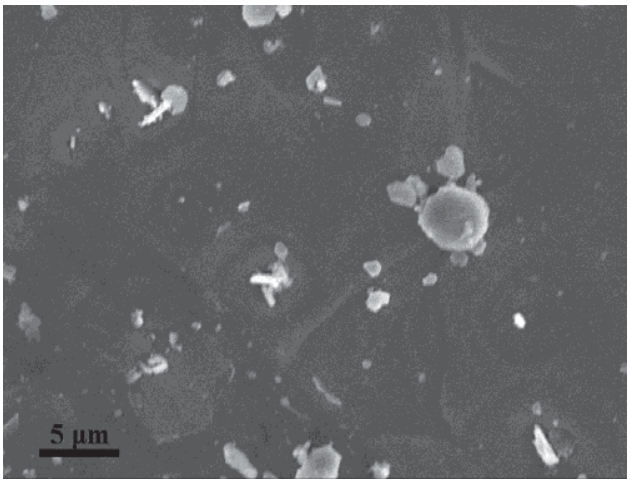

Figure 5 Details of a compounded particle showing agglomeration of individual Al_3Ti particles that nucleated on boride particles.

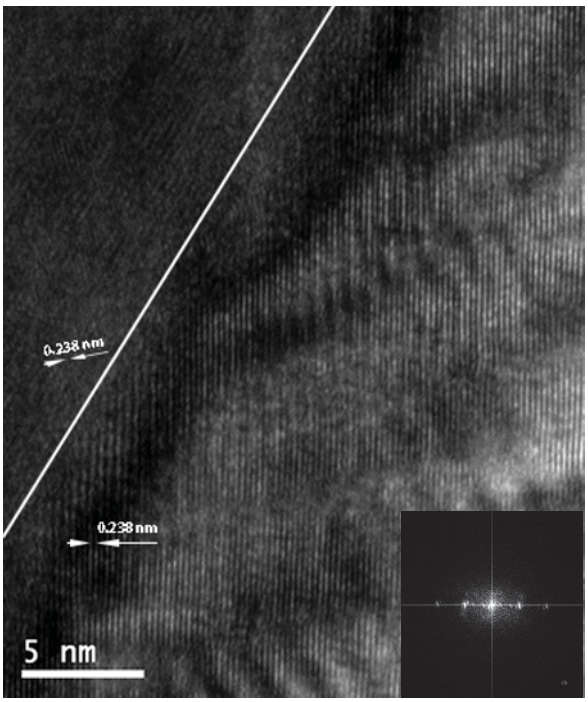

Figure 6 A high resolution TEM image of an Al_3Ti/boride interface showing a coherent interface between these two as indicated by a white line and the growth of Al_3Ti on the boride particles. Insert is the Fourier transformation of the image showing a few degree of mismatching of the two phases in the observed lattice planes.

$$TiB_2 + Al = (Ti,Al)B_2 + (Ti) \quad (1)$$
$$TiB_2 + Al = (Ti,Al)B_2 + Al_3Ti \quad (2)$$

Based on the dissolution of TiB_2 in aluminum and the formation of compounded Al_3Ti/TiB_2 particles in α-Al grain centers and clusters of boride particles at the α-Al grain boundaries, a dynamic layer of Al_3Ti on the surface of boride particles is proposed. After added into an aluminum melt, large Al_3Ti particles dissolve into aluminum and boride particles start to react with aluminum. The reaction of boride particles with aluminum is a slow process may take up to more than a month. The reaction is shown in reaction 1, where titanium is replaced by aluminum to form a solid solution of $(Ti,Al)B_2$ with a variable Ti/Al ratio. By thinking of 0.01wt%Ti and 0.002wt%B in aluminum when an Al-5Ti-1B master alloy is added into aluminum, both concentrations of titanium and boron are above the solubility of TiB_2 in aluminum. The titanium in reaction 1 may not in solution exactly, especially when temperature drops down to the solidification temperatures, 665°C as shown in Figure 1 for pure aluminum. The titanium will form Al_3Ti and reaction 1 can be rewritten as reaction 2. However, the concentration of Al_3Ti is low enough in grain refinement practice to avoid growing into a larger size and leading to agglomeration, which would otherwise effectively deplete the surface Al_3Ti layer on most boride particles. The Al_3Ti is also not stable by thinking of the slow formation of $(Ti,Al)B_2$ solid solution and finally the peritectic reaction of aluminum and Al3Ti in solidification.

Conclusion remarks

The grain refinement of aluminum by adding Al-Ti-B master alloys is realized through the formation of a dynamic layer of Al_3Ti on the surface of boride particles. The concentration of titanium must be maintained to a level that is high enough to form a dynamic layer of Al_3Ti in solidification and low enough to avoid the agglomeration or growth of Al_3Ti. The dynamic layer by nature will be only a couple of atoms thick, with a possibility of being a single atomic layer as reported by others. TiB_2 is not as stable as reported before. It will react with aluminum to form $(Ti,Al)B_2$ solid solution leading to the constant release of titanium into aluminum to maintain a dynamic layer of Al_3Ti in the presence of excess titanium and boron than the stoichiometry of TiB_2 when an Al-Ti-B master alloy is added. TiB_2 promotes the formation of an Al_3Ti layer through a coherency interface and the constant release of titanium through the formation of $(Ti,Al)B_2$ solid solution.

Acknowledgement

X. Wang would like to acknowledge Purdue Polytechnic Institute for the financial support through a startup fund for his tenure track position.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE MICROSTRUCTURES AND MECHANICAL PROPERTIES OF Al-5Mg-1.4Mn ALLOY CAST UNDER NEAR-RAPID COOLING

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Keywords: Aluminum alloy, Al-Mg-Mn alloy, Microstructure, Mechanical properties, Near-rapid solidification

Abstract

A high Mn content alloy (Al-5Mg-1.4Mn) has been cast under the condition of near-rapid cooling. The microstructures and mechanical properties of the alloy have been evaluated. The results indicated that under the condition of near-rapid cooling, the as-cast microstructures of the alloy consisted of α-Al matrix plus two types of intermetallics: $Al_6(Fe,Mn)$ and Mg_2Si. The intermetallics were small and uniformly distributed. During hot and cold rolling, they broke into particles. The particle structures were dense, but small. Some cavities were observed among the $Al_6(Fe,Mn)$ fragments. Both the hot band and cold rolled sheet demonstrated excellent mechanical properties. The significant improvements were achieved by increasing Mn to a higher level, while being cast under the condition of near-rapid cooling.

Introduction

The AA5083 alloy is a typical Al-Mg-Mn alloy with 4-5wt%Mg and 0.4-1.0wt%Mn[1]. Mn is added to neutralize the negative effects of Fe by modifying the Fe-rich intermetallics from the platelet-like morphology into Chinese script morphology which is expected to have less deteriorating effect on the mechanical properties of the alloy[2,3]. Mn also contributes to the strength of the alloy through solid solution strengthening. With the increase of the Mn content, the strength of the alloy increases accordingly. Alloys AA5182 and AA5083 have roughly the same contents of Mg. The tensile strength of AA5083 alloy is 15 MPa higher than that of AA5182 alloy at O temper[4]. The reason for the better strength of the AA 5083 alloy is the higher Mn content in the alloy. However, higher contents of Mn can also promote the formation of coarse intermetallic $Al_6(Fe,Mn)$ during casting[5]. This intermetallic is undissolvable during heat treatment. During deformation, they break into fragments and create microcracks or cavities among the fragments[5]. The coarse fragment and cavities can become nucleation sites of cracks and lead to premature fracture of the alloy[6,7]. Therefore, the Mn content must be limited to a certain low level in order to avoid the formation of those intermetallics, while maintaining tensile strength. In traditional production practice, Mn has been limited to 1.0wt% in most Al-Mg based alloys[1].

A recent research[8] demonstrated that the coarse Mn-rich intermetallics could be significantly refined during solidification by increasing the cooling rate to 20 $^{o}C\ s^{-1}$, which was a near-rapid cooling. It was assumed that the fine intermetallics would have less deteriorating effect on the mechanical properties and therefore the content of Mn in Al-Mg-Mn alloy could be further increased. Another research[9] proved that under the condition of near-rapid cooling, the content of Mn could be increased to 1.6wt% and the mechanical properties of the Al-5.0Mg-1.6Mn alloy increased significantly.

The cooling rate in the continuous strip casting process of aluminum alloys is quite high. In the twin-belts continuous strip casting process, the molten metal is continuously poured into a moving mold formed by two rolling steel belts to produce a cast slab. The cast slab is then immediately fed into hot rolling mills forming hot band products. The thickness of the cast slab is about 20 mm. In order to achieve high cooling rate during casting, the two rolling steel belts which form the casting mold are continuously cooled by cold water. The cooling rate in the continuous strip casting is in the range of 10^1-10^2 $^{o}C\ s^{-1}$.

The continuous strip casting process of aluminum alloy sheet has recently received more and more attention from manufacturers and end users[11] due to its advantage of high productivity and low production cost. Al-Mg-Mn aluminum alloy sheet materials are often produced using this process[12-16]. Currently, attentions have been paid to the production process and product quality, little works has been done on new material development through this process. It is assumed that the continuous strip cast process is not just a production method with high productivity and low cost; it can also be a good process for producing high performance materials through its characteristic of near-rapid cooling.

A research on the solidification structures of continuous strip cast AA3104 alloy indicated that the intermetallic $Al_6(Fe,Mn)$ was significantly refined due to fast cooling[10]. It was assumed that the process helped more alloying elements to retain solid solution in the cast slab. Therefore, continuous strip casting is most likely a good process to produce high Mn containing Al-Mg-Mn alloys with improved mechanical properties.

To this end, this work evaluated the microstructures and mechanical properties of an Al-5Mg-1.4Mn alloy cast under near-rapid cooling. The results could be comparable as if the materials were produced by the continuous strip cast process. The ultimate aim was to develop and produce high Mn containing Al-Mg alloys with higher mechanical properties using the high productivity and low cost continuous strip casting process.

Experiment procedures

A double-side water cooled casting apparatus was designed, as shown in Figure 1. The solidification process in the apparatus was similar to the one in the twin-belt continuous strip caster. The casting mold with 150 mm X 150 mm X 15 mm in size was continuously cooled from two sides by cold water. The average cooling rates before the start of solidification were about 20 $^{o}C\ s^{-1}$ which was measured by recording the temperature evolution with time.

The studied alloy: Al-5Mg-1.4Mn-0.12Cr-0.25Fe-0.12Si (in wt%) was designed by copying the composition of AA5083 alloy. Commercial pure aluminum ingots, magnesium ingot (99.9wt%), Al-10Mn, Al-10Fe, Al-10Cr and Al-30Si mater alloys were used

to prepare the studied samples. 4 kg of raw material was melted in a graphite-clay crucible in an electric resistance furnace. The molten aluminum was degassed by injecting Ar into the pool and grain refined by adding Al-5Ti-B grain refiner, then casted with the apparatus shown in Figure 1 at around 725 ^{o}C. The chemical compositions of the cast slabs were analyzed by emission spectroscopy; the results were listed in Table 1.

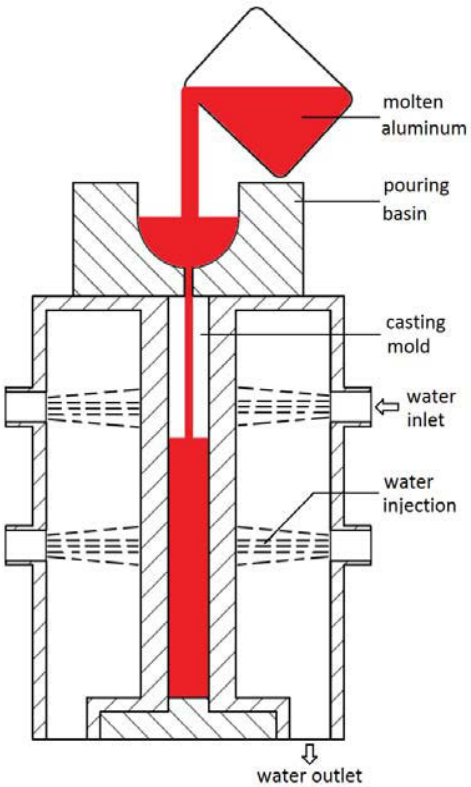

Fig 1 Schematic sketch of the double-side water cold casting apparatus

Table 1 Chemical composition of the alloy, wt%

Mg	Mn	Fe	Si	Cr	Ti	Al
5.21	1.39	0.24	0.14	0.17	0.011	Bal.

The casted slabs were heated and hot rolled at 420 ^{o}C to form 3.5 mm thick hot bands. The total reduction rate was about 75%. Some of the hot bands were annealed at 420 ^{o}C for 2 hours. Then, the annealed hot bands were cold rolled to a thickness of 1.4 mm with a reduction rate of 60%. The cold rolled sheets were then subjected to stabilization treatment at 150 ^{o}C for 2 hours. All the annealed hot bands and the stabilization treated cold rolled sheets were subjected to tensile tests with tensile specimens shown in Figure 2. The tensile tests were performed at a tensile speed of 2 mm/min.

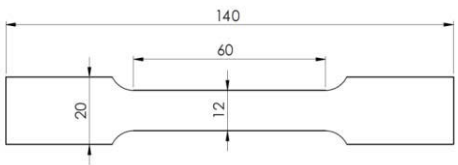

Fig 2 Dimensions of the tensile specimens (unit in millimeter)

The recrystallization behavior of the hot band and the cold rolled sheet was investigated by heating the materials at different temperature. The microstructures of the materials were examined with optical microscopy (OM) and scanning electronic microscopy (SEM). The metallographic samples were prepared using standard metallographic procedures. The samples were electropolished at a voltage of 27 V D.C. for 15 s using 60 mL $HClO_4$ + 140 mL H_2O +800 mL C_2H_5OH. The samples for grain structure observation were anodized at a voltage of 18 V D.C. for 60 s using Barker's reagent (50 mL HBF_4 + 950 mL H_2O). The anodized samples were examined using an Olympus GX71 optical microscope with polarized light. A Zeiss scanning electron microscopy with energy dispersive X-ray (EDX) analyzer was employed to identify intermetallic compounds and observe the fracture surface of the tensile specimens.

Results

Microstructures

Figure 3a shows the as-cast microstructures of the alloy. They consisted of α-Al matrix plus two types of intermetallics, marked as I and II on the figure. According to the previous research[8], the type I intermetallic was $Al_6(Fe,Mn)$, the type II intermetallic was Mg_2Si. They were small and uniformly distributed. Figure 3b shows the grain structures of the alloy. The average grain size was about 75 microns. Both the grains and intermetallics were significantly refined by the near-rapid cooling process.

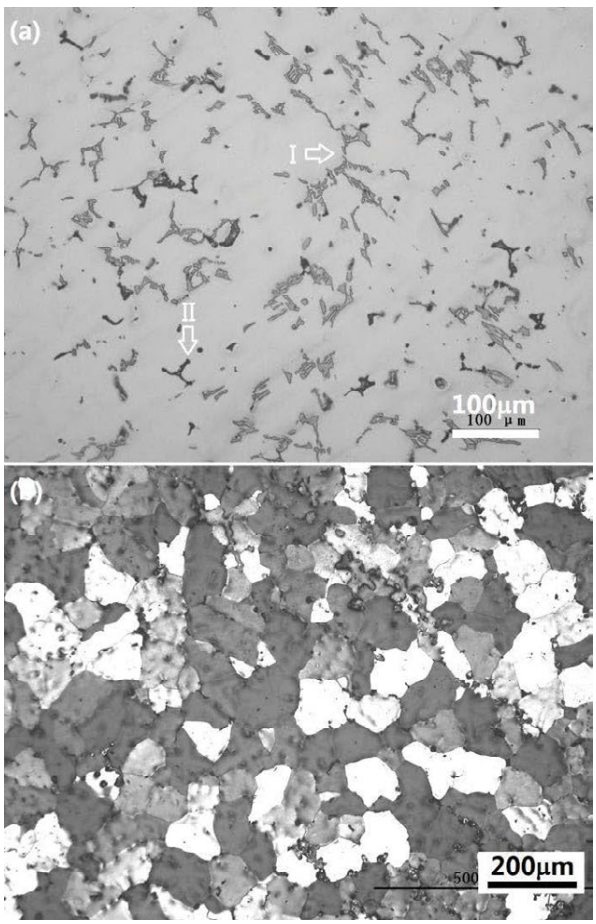

Fig 3 (a) The as-cast microstructures and (b) grain structures of the cast slab

Figure 4 shows the particle structures of the hot band. They were dense, but small. Under high magnification, two types of particles were observed, as seen in Figure 4b, and distinguished by color, marked as A and B. Some cavities were observed among the type A particles. EDX analysis indicated that the type A particles in light gray mainly contained Al, Fe and Mn and roughly met the formula of $Al_6(Fe,Mn)$; the type B particles mainly contained Mg and Si, see Table 2. These particles were the product of hot rolling. During hot rolling, the intermetallics formed during solidification broke into particles. Due to the blocky structure and the brittle feature of the intermetallic $Al_6(Fe,Mn)$, movement of the broken fragments are difficult. Therefore, some cavities were left among the $Al_6(Fe,Mn)$ fragments.

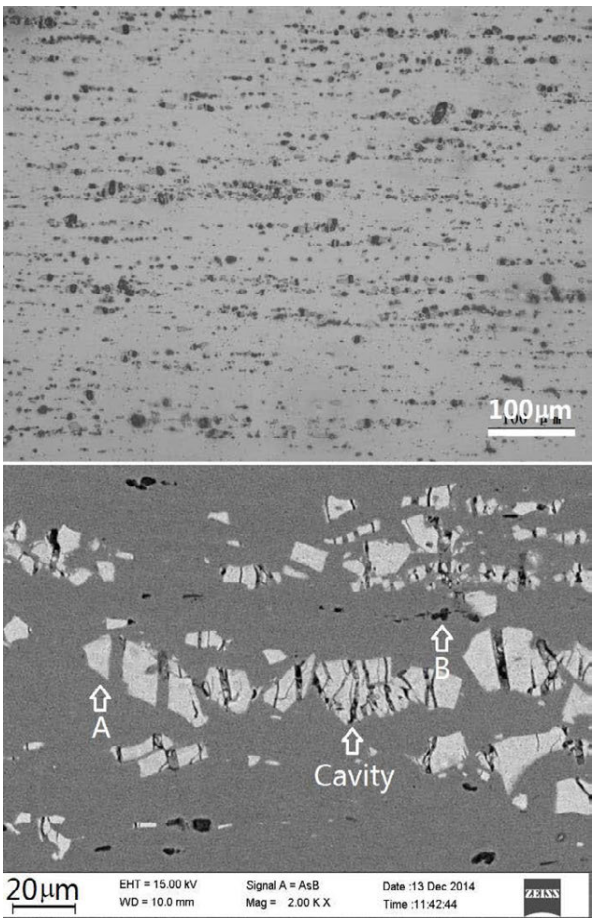

Fig 4 The particle structures (a) OM image; (b) SEM image

Table 2 The EDX analysis results of the intermetallics, at%

Intermetallic type	Al	Mn	Fe	Mg	Si
A	81.20	13.18	5.62		
B	6.07			67.06	26.87

Recrystallization Behavior

Figure 5 shows the softening curves of the hot band and the cold rolled sheet when annealed at different temperature for 20 min. The initial hardness of the hot band was relatively low. It could be inferred that dynamic recrystallization occurred during hot rolling. The cold rolled sheet was obtained by cold rolling the fully annealed hot band with a reduction rate of 60%.

It could be seen from the curves that the hardness of the materials dropped rapidly. After being soaked for 20 min at 325 ^{o}C, the hardness of the cold rolled sheet almost reached its lowest level. Most likely, the recrystallization temperature of the cold rolled sheet was about 325 ^{o}C. However, the recrystallization temperature of the hot band was higher. The hardness of the hot band did not reach its lowest level until annealed at 350 ^{o}C for 20 min. As the annealing temperature increased further, the hardness of the materials did not change remarkably. It was assumed that the recrystallization temperature for the hot band and the cold rolled sheet were 350 ^{o}C and 325 ^{o}C, respectively.

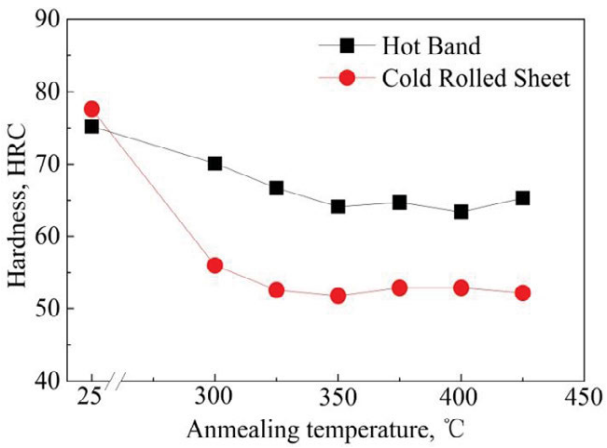

Fig 5 The softening curves of the cold rolled sheet and hot band

Figure 6 shows the grain structures of the hot band before and after annealing. Before annealing, the hot band displayed a partially recrystallized grain structure (Figure 6a). This observation confirmed the above theory that dynamic recrystallization occurred during hot rolling. After being annealed for 20 min at 350 ^{o}C, the deformed grain structures developed into recrystallized grain structures with uniform equiaxed shape (Figure 6b), However, some elongated grains were still observed (Figure 6b). In the sample annealed at 425 ^{o}C for 20 min, all the grains showed equiaxed morphology. The cold rolled sheet displayed heavily deformed grain structures (Figure 7a). They corresponded to the high deformation rate of the cold rolling. After being annealed at 325 ^{o}C for 20 min, the heavily deformed grains were fully recrystallized with uniform equiaxed shape (Figure 7b).

Mechanical Properties

Table 3 provides the tensile properties of the hot band and the cold rolled sheet. The hot band was fully recrystallized; the cold rolled sheet was only subjected to stabilization treatment at 150 ^{o}C

for 2 h, no annealing was conducted. The materials demonstrated excellent mechanical properties. The ultimate tensile strength and yield strength of the fully recrystallized hot band reached 360 MPa and 210 MPa, respectively. The elongation was 24%. They were much higher than the mechanical property specification of the AA5083 alloy. The mechanical properties of the cold rolled sheets were also listed in Table 3. It could be seen that the stabilization treatment greatly increased the ductility of the cold

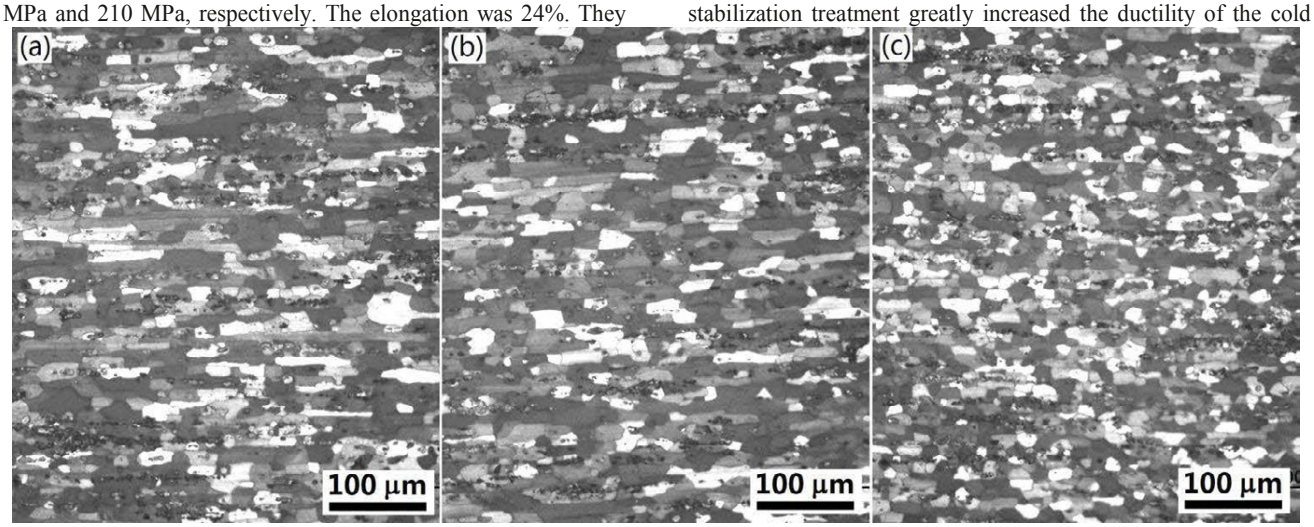

Fig 6 The grain structures of the hot band after annealed at different temperature for 20 min.
(a) Unannealed ; (b) 350 ^{o}C; (c) 425 ^{o}C

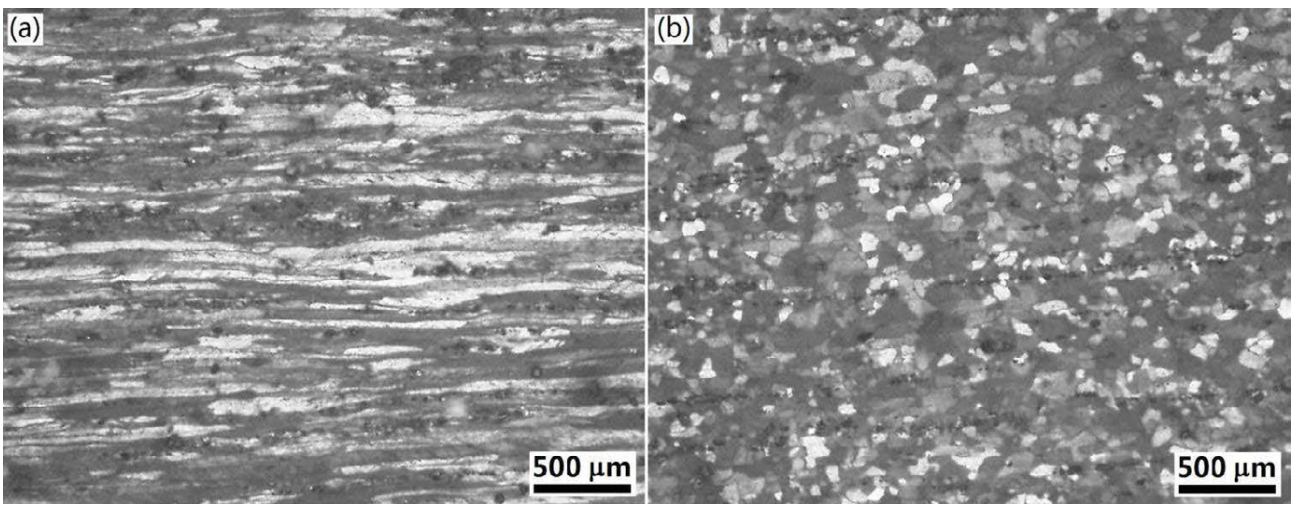

Fig 7 The grain structures of the cold rolled sheet.
(a) Unannealed ; (b) annealed at 325 ^{o}C for 20min.

rolled sheet, while the tensile strength was maintained at a high level. Fracture surfaces of the hot band were observed using SEM. It was found that the material displayed ductile fracture mode. The fracture surface showed a mixture of fine dimples and holes, see Figure 8. Some particles were observed in the holes. It was assumed that the holes were created by the intermetallic particles which fractured or left from the matrix. Some fractured particles remained in the holes. These characteristics were consistent with the microstructures of the material.

Discussion

Mg and Mn were the main alloying elements; Fe and Si were impurities in the studied alloy. During solidification, Mn combined with Al and Fe to form intermetallic $Al_6(Fe,Mn)$; Mg combined with Si to form intermetallic Mg_2Si. According to the Al-Mn phase diagram[17], the eutectic composition of α-Al + Al_6Mn was 1.7 wt%Mn. For the hypoeutectic Al-Mn alloy, the solidification started with the precipitation of α-Al matrix.

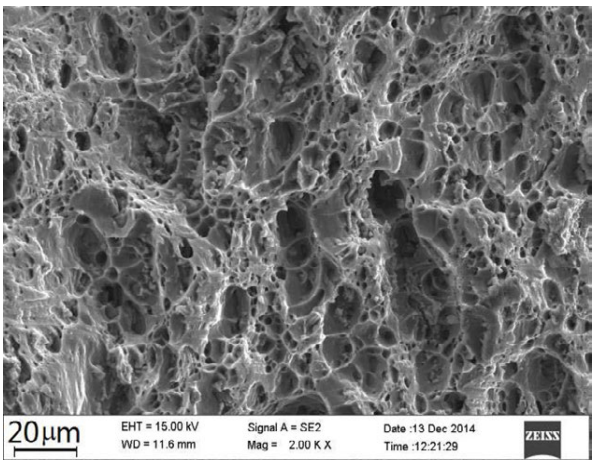

Fig 8 The fracture surface of the hot band

Table 3 The mechanical properties of the materials

Material	UTS, MPa	YS, MPa	El, %
Hot band	342.3	199.4	21.7
Cold rolled sheet*	397.6	364.8	9.6

* subjected to stabilization treatment at 150 ^{o}C for 2 h

A portion of Mn atoms would be solid solutioned in the α-Al matrix. The remaining Mn atoms would form intermetallic Al_6Mn. The higher the content of Mn in the alloy, the more Mn atoms solid solutioned in the α-Al matrix, and the more intermetallic Al_6Mn formed in the solidification structures. Once the content of Mn exceeded the eutectic point, almost all of the Mn atoms would form Al_6Mn as primary phase. The amount of intermetallic Al_6Mn would increase significantly. This was confirmed by the research on Al-5Mg-xMn alloys[9]. The coarse intermetallic would break into fragments during deformation and deteriorate the mechanical properties. Therefore, the content of Mn should be kept at a low level in order to minimize the amount of the intermetallic Al_6Mn.

Fe had a strong tendency of segregating into the interface front and forming Fe-rich intermetallics. In the Al-Mg-Mn alloy, the appearance of Fe caused the intermetallic Al_6Mn to transform into $Al_6(Fe,Mn)$. The content of Fe was also of great influence on the amount of the intermetallic $Al_6(Fe,Mn)$. Therefore, the content of Fe also had to be limited to a low level in order to make sure that the sum of Fe and Mn was less than the composition of eutectic reaction. In this study, the sum of the Fe and Mn contents was less than 1.7 wt%, therefore, the composition of the alloy was still hypoeutectic.

On the other hand, the cooling rate during casting was of great influence on the size, morphology and distribution of the intermetallics. Although the contents of Mn, Fe and Si were high, especially the sum of Mn and Fe, which was close to the eutectic point, the resulting size of the intermetallics was small. Under the condition of near-rapid cooling, the intermetallics were significantly refined[8]; the morphology of the intermetallics was modified from mass blocks to Chinese script.

During hot and cold rolling, the intermetallics broke into particles. The size and morphology of the particles were determined by the features of the intermetallics formed during casting. In this study, the as-cast intermetallics $Al_6(Fe,Mn)$ and Mg_2Si were refined by the near-rapid cooling. Therefore, the particles in the hot band and cold rolled sheet were small and uniform.

In the Al-Mg-Mn alloy, the content of Mn was of great influence on the mechanical properties. Increasing Mn would increase the tensile strength. For example, increasing Mn from 0.35wt% in AA5182 alloy to 0.7wt% in AA5083 alloy, the tensile strength increased by 15MPa at O temper[4]. The improvement of the tensile strength was essentially attributed to the higher amount of manganese atoms being solid solutioned in the matrix. Solute strengthening played an important role in the strengthening of non-heat treatable aluminum alloys, such as AA5083 alloy. This was confirmed by Huskins et al.[18] and Ryen et al.[19]. Therefore, increasing the content of Mn in the alloy would increase the amount of manganese atoms being solid solutioned in the matrix, and as a result, the tensile strength of the alloy would increase accordingly.

However, on the other hand, increasing Mn gave rise to the increase of the coarse intermetallics being formed during solidification and the amount of particles remained in the hot bands and cold rolled sheets. The remaining particles in the materials, especially the cavities between the fragments formed during deformation, acted as the nucleation sites of cracks and result in the premature failure of the materials. Therefore, increasing Mn would deteriorate the mechanical properties of the materials. In the traditional production of Al-Mg-Mn alloys via direct chill casting process, the content of manganese had to be limited to a low level (<1.0 wt%) in order to avoid the formation of coarse manganese-rich intermetallic compounds[7].

Under the condition of near-rapid cooling, the Mn- and Fe-rich intermetallics were refined to a great extent. Therefore, a higher level of Mn could be allowed. The content of Mn could be increased to a relatively higher level without deteriorating the mechanical properties of the alloy. More manganese atoms were solid solutioned in the matrix to increase mechanical properties. In this study, under the condition of near-rapid cooling, the content of Mn was increased to 1.4wt% and the mechanical properties of the alloy were greatly improved.

Conclusions

1. Under the condition of near-rapid cooling, the as-cast microstructures of the Al-5Mg-1.4Mn alloy consisted of α-Al matrix plus two types of intermetallics: $Al_6(Fe,Mn)$ and Mg_2Si. They were small and uniformly distributed.
2. The intermetallics $Al_6(Fe,Mn)$ and Mg_2Si broke into particles during hot and cold rolling. The particle structures of the hot band were dense, but small. Some cavities were observed among the $Al_6(Fe,Mn)$ fragments.
3. Both the hot band and the cold rolled sheet demonstrated excellent mechanical properties. They were significantly improved by increasing Mn to a higher level, while being cooled by near rapid cooling.

Acknowledgement

The authors are grateful to the Research Foundation of Shenyang Aerospace University for its financial support.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF THE SHAPE OF SOLID PARTICLES ON THE DISTRIBUTION OF PARTICLES IN JIS AC4CH (A356) ALUMINUM ALLOY SEMI-SOLID HIGH-PRESSURE DIE CASTING

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Keywords: Semi-solid Process, Aluminum Alloy, Density, Shear Rate

Abstract

In the semi-solid high-pressure die casting process, the slurry flows as a solid–liquid two-phase flow. In this study, the effect of the shape of solid particles on their distribution in the slurry was investigated. The solid particles were concentrated in the center of the flow direction in the case of globular-shaped solid particles and high-flow-velocity conditions. Moreover, the concentration ratio of the solid particles increased with an increasing Reynolds number. This phenomenon was explained as follows. The shape of the solid particles affected the viscosity of the slurry; the viscosity was decreased by the globular particles. The solid particles applied Saffman force, which is generated by the velocity gradient, and moved in the direction away from the solid–liquid interface. The Reynolds number increased with decreasing viscosity or increasing flow velocity; therefore, the Saffman force increased with an increasing Reynolds number.

Introduction

The high-pressure die casting (HPDC) process is commonly used to form aluminum alloy products with complex configurations. The HPDC process features high productivity. However, HPDC products generally exhibit low engineering performance because of the existence of inherent defects such as porosity, hot cracks, and oxide inclusions; these defects stem from the high-velocity injection of liquid metal, which has a relatively low viscosity, into the mold. The semi-solid HPDC process is expected to enable the casting of high-quality aluminum products because the viscosity of semi-solid aluminum is greater than that of liquid aluminum, which reduces air entrainment and facilitates the casting of aluminum with fewer gas defects than that cast via general HPDC. Additionally, solidification shrinkage would be reduced in the semi-solid state compared to that in the liquid state, which is expected to reduce shrinkage porosity and improve dimensional accuracy. The semi-solid process has been studied since the 1970s [1,2] and has recently been developed to fabricate high-quality aluminum alloy products [3,4]. However, semi-solid HPDC has a lower formability than general HPDC because the fluidity of semi-solid metals is lower than that of liquid metals. To solve this problem, researchers developed a method to increase the fluidity of the semi-solid slurry by applying shear stress during the forming processes for semi-solid magnesium injection [5,6] and semi-solid aluminum HPDC [7-9].

The microstructures of the products cast by semi-solid HPDC consist of primary solid α-phase particles and liquid matrix phase, which are solid particles and liquid phase during the injection forming process, respectively. The primary α-phase particles tend to aggregate at the center of the flow direction [10]. Moreover, their distribution changes depending on the flow velocity of the semi-solid slurry [11]. The primary solid α-phase particles and the liquid matrix phase have different properties. Therefore, the properties of the semi-solid HPDC products could be improved by controlling the distribution of the primary α-phase. In this study, the effect of the morphology of solid particles on their distribution was investigated. Slurries with different initial forms of solid particles were injected with different shear stresses at the gate, thus altering the shapes of the solid particles. The distribution of the primary α-phase in each specimen was observed.

Experimental procedure

The alloy used for the experiments was an Al–7wt%Si–0.3wt%Mg alloy (AC4CH in the JIS, equivalent to A356 in the ASTM standard). The semi-solid slurry was prepared by electromagnetic stirring method (EMS) [12] or mechanical vibration method (MV) [13]. In both of these methods, the molten AC4CH alloy was poured into a stainless steel container at room temperature. Therefore, heat was drawn from the molten AC4CH alloy by the container. With this process, the temperature of the molten AC4CH alloy was decreased to the semi-solid temperature.

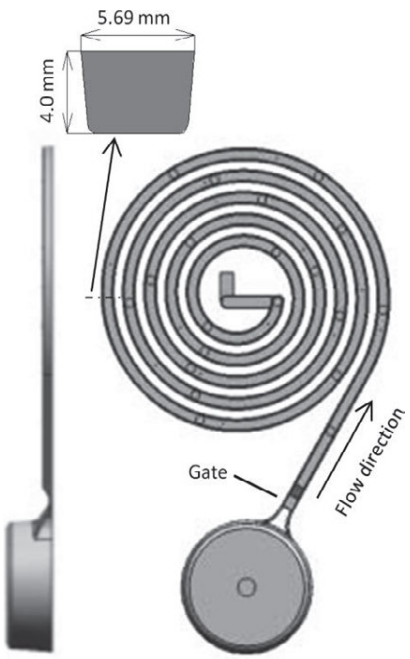

Fig. 1 Appearance of the spiral cavity of the permanent mold

Moreover, electromagnetic stirring (in the EMS method) or mechanical vibration (in the MV method) was applied during cooling, which resulted in slurries with dispersed small and fine solid particles. In this study, the poured temperature of the AC4CH alloy was 700°C, and the pouring weight was approximately 220 g. The stainless steel container was 90 mm high, with an inner diameter and wall thickness of 38.7 mm and 2.0 mm, respectively. In the EMS method, EMS was applied in the vertical direction for 5 s, then in the rotational direction for 10 s. In the MV method, mechanical vibration at a frequency of 50 Hz and an acceleration amplitude of 166.6 m/s^2 (17 G) was applied for 20 s.

In this study, a 125 t HPDC machine (TOYO BD-125V5) was used. The specimens were prepared by injection into a permanent mold with spiral cavity, as shown in Fig. 1. The spiral cavities of the permanent mold were 5.7 mm wide, 1350 mm long, and 4.0 mm thick; the mold's gate was 5 mm wide, and the gate velocity was controlled by varying the thickness from 1.0 to 2.2 to 4.0 mm. The injection velocity (piston speed) was set to 0.1 or 0.35 m/s. The microstructure of the surface perpendicular to the flow direction was observed by optical microscopy at the points representing 50% of their total fluidity length. The specimens were polished by grinding with emery paper, followed by polishing with diamond paste; the specimens were subsequently etched with 0.5% HF solution.

Results

Microstructure before injection

Fig. **2** shows the microstructures of the slurry before injection. The specimens were prepared by rapid cooling with water after standing for 15 s. These micrographs show that the AC4CH aluminum alloy specimens exhibited microstructures with a dispersion of primary α-Al particles in their matrix (eutectic α-Al and Si). These observations suggest that the primary α-Al particles were solid phase and that the matrix was liquid phase before the samples were rapidly cooled. The shape of the primary α-Al particles appears similar between the EMS slurry and the MV slurry at the center. At the surface region, the primary α-Al particles of the MV slurry exhibit a more complex shape than those of the EMS slurry.

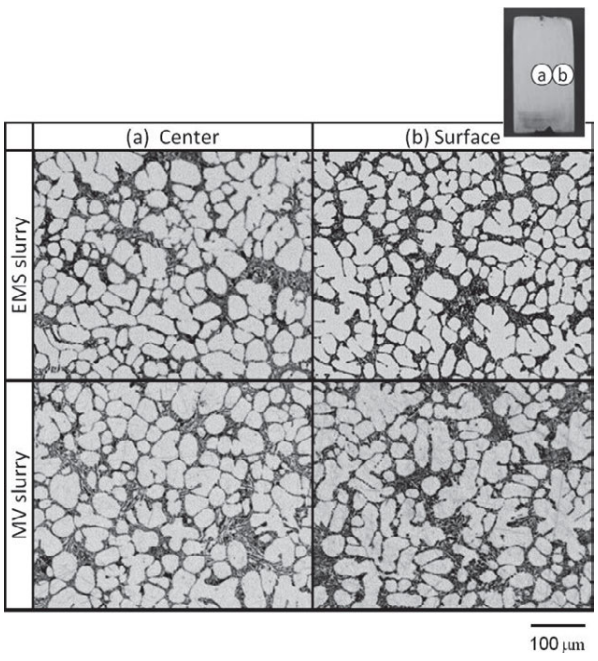

Fig. 2 Microstructures of the slurry before injection

To evaluate the size and shape of the primary α-Al particles quantitatively, the particle roundness, R, and the particle diameter, d [μm], were measured by image analysis. The roundness of a particle was calculated by the following equation [5]:

$$R = L^2/(4\pi A),$$

where L [μm] and A [μm^2] are the boundary length and area of the particle, respectively. When $R = 1$, the particle is a perfect sphere; the value of R increases as the particle shape deviates from being round. We then calculated the particle diameter (equivalent circle diameter) by the following equation:

$$d = 2\sqrt{A/\pi}$$

The area-weighted mean roundness, R_s, and area-weighted mean diameter, d_s [μm], were calculated by

$$R_s = \Sigma R_i A_i / \Sigma A_i,$$
$$d_s = \Sigma d_i A_i / \Sigma A_i,$$

where d_i [μm], A_i [μm^2], and R_i are the diameter, area, and roundness of the particle, respectively. Fig. **3** shows the mean particle diameter and roundness of the primary α-Al particles in each slurry, as calculated using the aforementioned equations. The results indicate that the solid particles in the MV slurry exhibit a larger and more complex shape than those in the EMS slurry before injection.

Microstructure after injection

The microstructures of the central section of the fluidity test specimen at the middle part of the total flow length are shown in Fig. 4. These photographs were taken at the points representing 50% of their total fluidity length. A comparison of shear rate at the gate, as determined by calculation of the mean particle diameter and roundness from these micrographs, is shown in Fig.

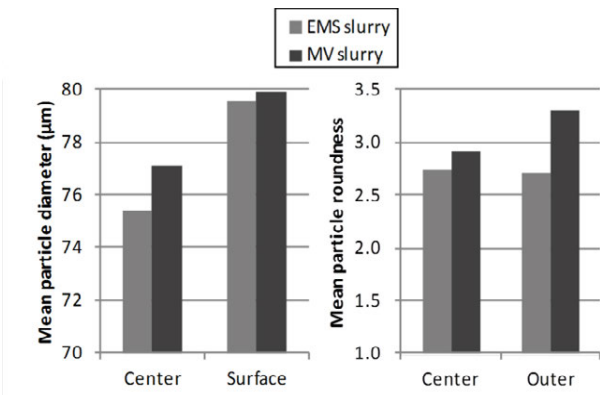

Fig. 3 Mean particle diameter and roundness of primary α-Al particles in the slurry before injection

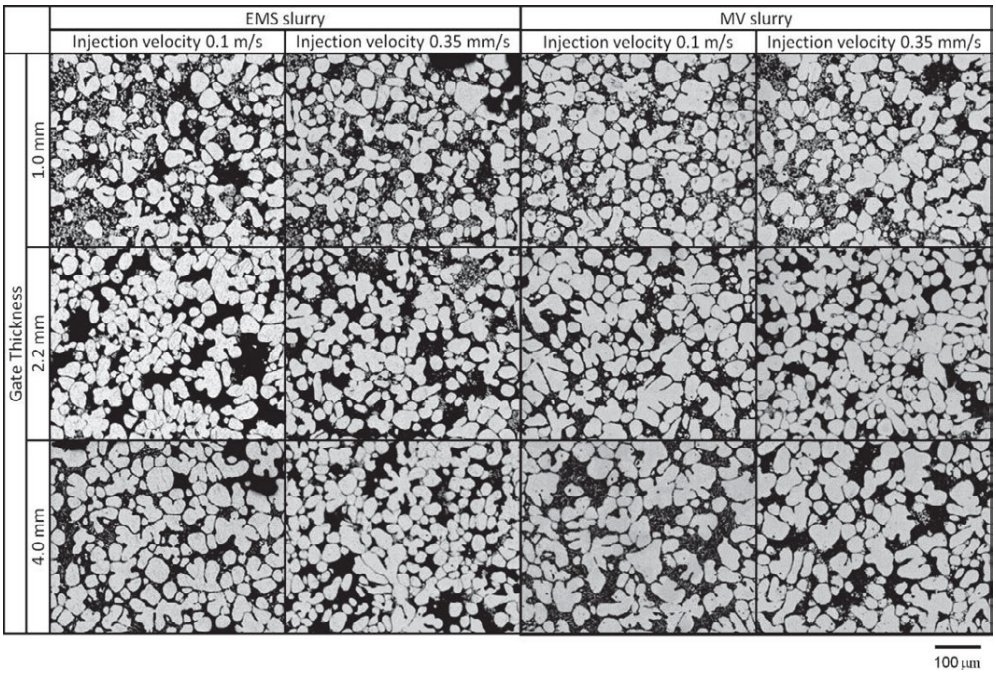

Fig. 4 Microstructures of the central section of the specimen at the middle part of the total flow length. These photographs were taken at the points representing 50% of their total fluidity length.

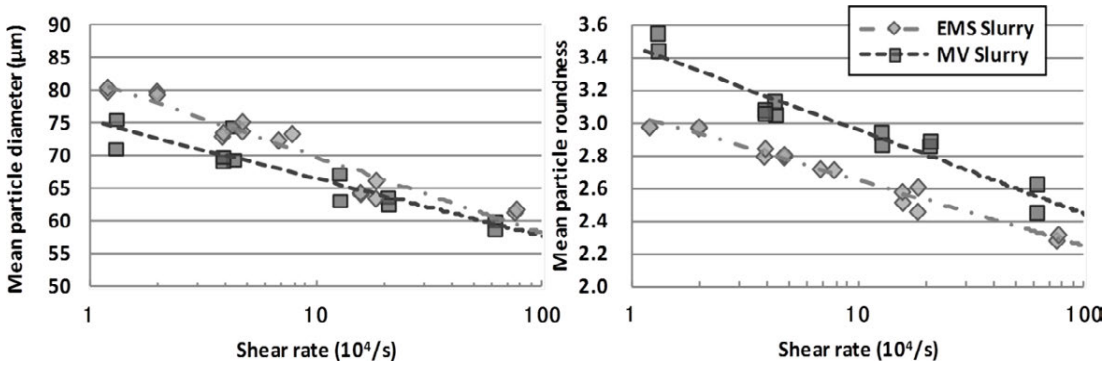

Fig. 5 Comparison of the effects of the shear rate at the gate on both the mean roundness and the diameter of primary α-Al particles between the slurry subjected to mechanical vibration and that subjected to electromagnetic stirring.

5. The mean particle diameter in the EMS slurry is slightly greater than that in the MV slurry. The mean particle roundness in the MV slurry is even larger than that in the EMS slurry. However, the difference of the mean particle diameter and roundness between the MV slurry and the EMS slurry decreased with an increasing shear rate. These results indicate that the solid particles in the MV slurry have finer and more complex shapes than those in the EMS slurry under the same injection conditions.

Distribution of solid particles

The distribution of the primary α-Al particles in the cross sections of specimens is shown in Fig. 6. The primary α-Al particles were concentrated in the center under high-injection-velocity conditions. However, under low-injection-velocity conditions, the primary α-Al particles were distributed uniformly in the cross section. The same tendencies were observed, irrespective of the slurry preparation method.

Next, to quantitatively evaluate the distribution of particles, the axial concentration degree, C, was calculated as follows [11]. First, the center of gravity of all particles was calculated by image analysis and was used as a dispersion center. Next, the area of each particle, A_{si}, and the distance from the dispersion center, r_i, were calculated. From these values, the mean particle distance from the dispersion center, r_a, was calculated by the following area-weighted formula:

$$r_a = \sum(A_{si} r_i) / \sum A_{si}$$

Moreover, to remove the effect of the difference of the cross-sectional area (A) of the specimen, the shape of the cross section was approximated by a circle and r_a was divided by the equivalent circle radius, $\sqrt{A/\pi}$. The dimensionless value r_a' was then calculated as

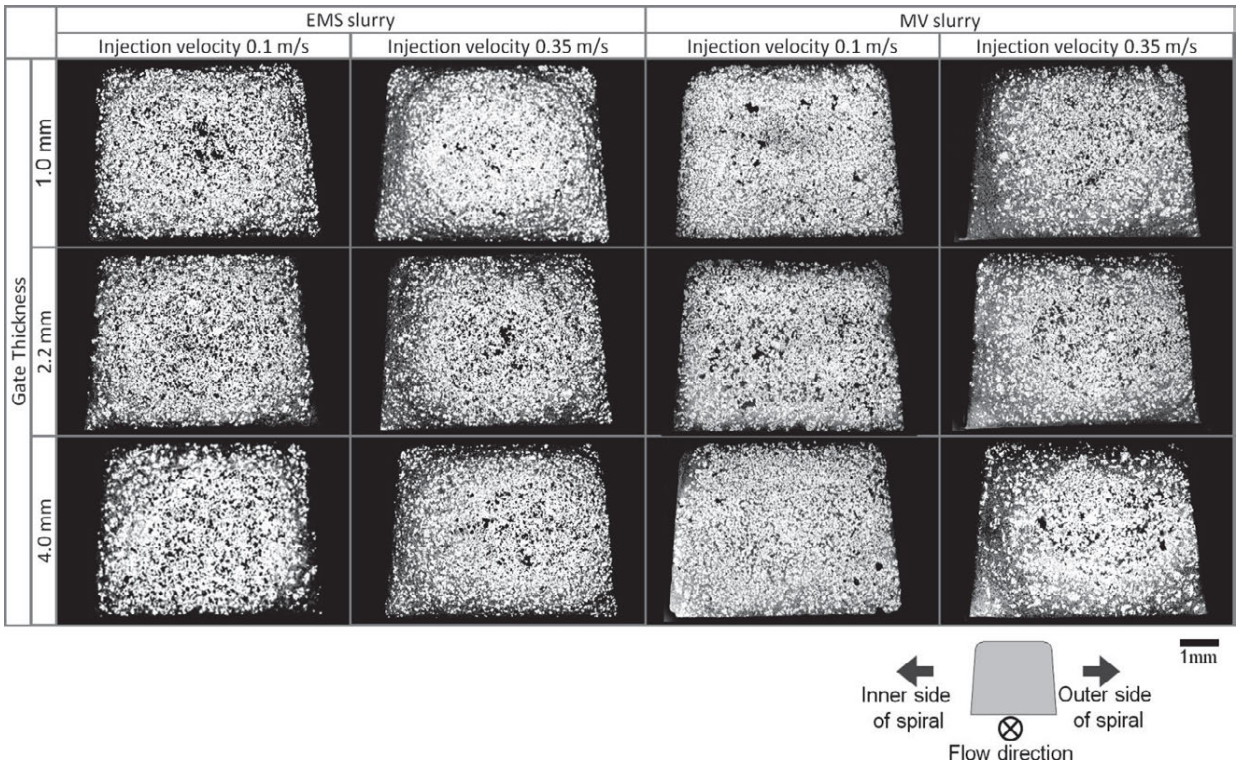

Fig. 6 Distribution of the primary α-Al particles in the vertical section of the specimens at the middle part of the total flow length. These photographs were taken at the points corresponding to 50% of their total fluidity length.

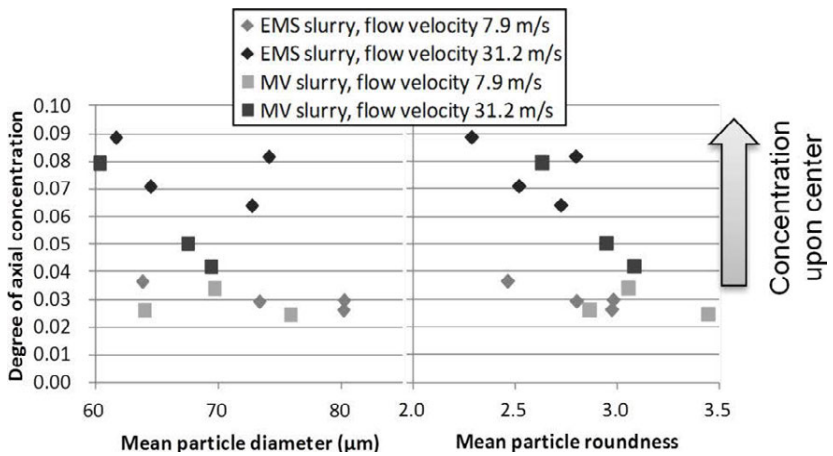

Fig. 7 Effect of the mean diameter and roundness of primary α-Al particles on the degree of axial concentration

$$r_a' = \sum(A_{si}r_i)/\left(\sqrt{\frac{A}{\pi}}\sum A_{si}\right)$$

Therefore, if a particle is located at the dispersion center, then $r_a' = 0$, and if a particle is located at the wall, then $r_a' = 1$.

Here, if particles sufficiently smaller than the cross section are uniformly dispersed, the value of r_a' is considered as follows. When particles with an infinitesimal area are uniformly close packed in the circular cross section, r_a' in the range D ($0 < r < a$, $0 < \theta < 2\pi$) is calculated as

$$r_a' = \iint_D r\,dS/\left(a\iint_D dS\right)$$
$$= \int_0^a\int_0^{2\pi}(r\cdot r)\mathrm{d}\theta\mathrm{d}r/\left(a\cdot\int_0^a\int_0^{2\pi}r\mathrm{d}\theta\mathrm{d}r\right) = \frac{2}{3}$$

That is, if the particles are dispersed in the circular cross section with a radius of 1, then $r_a' = 2/3$. Hence, $r_a' < 2/3$ indicates that the particle distribution is near the central axis, and $r_a' > 2/3$ indicates that the particle distribution is near the wall. Therefore, the axial concentration degree, C, is defined as 2/3 minus r_a'.

$$C = \frac{2}{3} - \sum(A_{si} r_i) / \left(\sqrt{\frac{A}{\pi}} \sum A_{si} \right)$$

Thus, if the particles are uniformly distributed, then $C = 0$; if the particles are concentrated near the central axis, then $C > 0$; and if the particles are concentrated near the wall, then $C < 0$. Next, the maximum value of C is considered if particles sufficiently smaller than the cross-sectional area are dispersed near the central axis with uniform close packing and with a primary α-phase fraction of 0.5. Under these conditions, the density of the close packing of a true circle in two dimensions is $\sqrt{3}\pi/6$; therefore, in a circular cross section of radius 1, the particles are distributed with a radius of $\sqrt{3/\sqrt{3}\pi}$. Hence, the mean distance from the dispersion center is calculated as $a' = \sqrt{3/\sqrt{3}\pi} \times \frac{2}{3} \approx 0.495$. Thus, the maximum value of C can be calculated as $C = \frac{2}{3} - r_a' \approx 0.172$.

Fig. 7 shows the effect of the mean diameter and roundness on the degree of axial concentration. The degree of axial concentration was higher under high-flow-velocity conditions, which means that the solid particles were concentrated toward the center axis. Moreover, the degree of axial concentration shows a strong correlation with the mean particle roundness; thus, the degree of axial concentration increased with decreasing mean particle roundness. The viscosity of the slurry is decreased by the globular solid particles. This result suggests that viscosity affects the distribution of solid particles, i.e., the solid particles were concentrated toward the center axis with decreasing viscosity.

Discussion

To evaluate the effect of flow morphology on the distribution of particles, here, we approximate the Reynolds number of the slurry. Hirai et al. modified the Mori–Ototake viscosity formula [14] as follows and reported that this equation is consistent with the observed apparent viscosity of Al–10%Cu semi-solid slurry [15]:

$$\eta_a = \eta_{La} \left\{ 1 + \frac{d \cdot S_r}{2\left(\frac{1}{f_S} - \frac{1}{f_{SCr}}\right)} \right\},$$

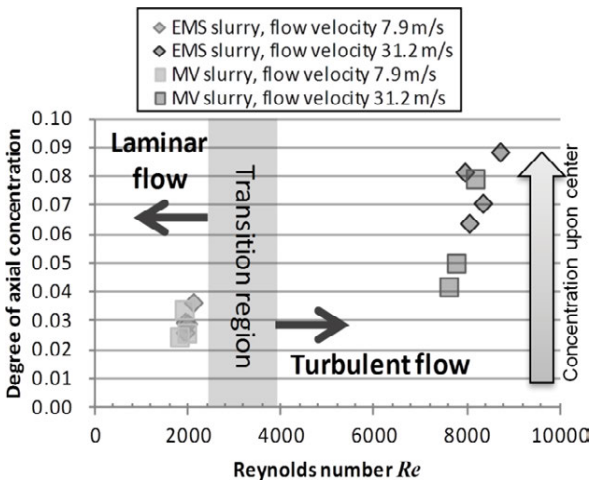

Fig. 9 Effect of Reynolds number on the degree of axial concentration

where η_a is the viscosity of the semi-solid alloy (Pa·s), η_{La} is the apparent viscosity of the liquid alloy (Pa·s), d is the effective average diameter of the particles in suspension (m), S_r is the volume base specific surface (surface area/volume) (m^{-1}), f_S is the mass fraction of solid, and f_{SCr} is the critical mass fraction of solid. Next, the Reynolds number in the pipe was calculated as follows:

$$Re = \frac{\rho v D_H}{\eta},$$

where ρ is the density of fluid (kg/m^3), v is the flow velocity (m/s), η is the viscosity of the fluid (Pa·s), and D_H is the hydraulic diameter of the pipe (m). D_H is also defined as

$$D_H = \frac{4A}{P},$$

where A is the cross-sectional area (m^2) and P is the wetted perimeter (m).

The relation between the degree of axial concentration and the Reynolds number is shown in **Error! Reference source not found.**. The degree of axial concentration is correlated with the Reynolds number; i.e., the degree of axial concentration increased with an increasing Reynolds number. The velocity-reduced particle near the wall surface receives Saffman lift force in the direction of the center of the pipe, which is generated by the velocity difference between the particle and the fluid [16]. When the Reynolds number is <2300, the fluid exhibits a laminar flow and a parabolic velocity distribution (**Error! Reference source not found.**). In contrast, when the Reynolds number is >4000, the fluid exhibits a turbulent flow and the velocity boundary layer appears**Error! Reference source not found.**. This layer becomes narrow with an increasing Reynolds number; therefore, the velocity gradient increases, in turn, increasing the Saffman lift force. Consequently, the solid particles were considered to move toward the center, resulting in the solid particles being concentrated at the center of the specimen under high-Reynolds-number conditions.

Conclusions

An AC4CH (A356) aluminum alloy slurry with solid particles of different shapes was injected into a spiral-shaped cavity. The distributions of the primary α-Al particles were observed, and the effect of the size and shape of the solid particles was evaluated. The distribution of the primary α-Al particles was uniform under low-flow-velocity conditions. In contrast, under high-flow-velocity conditions, the primary α-Al particles became concentrated around the flow direction axis in the cross section perpendicular to the flow direction. Moreover, under high-flow-

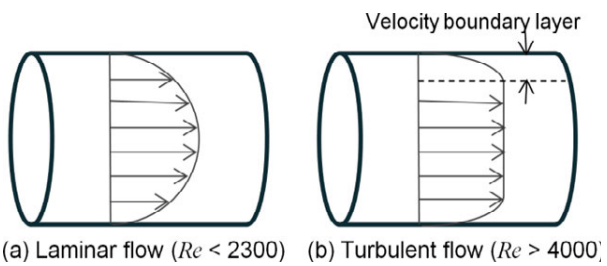

Fig. 8 Schematic of the difference in velocity distribution between the laminar and turbulent flows

velocity conditions, the concentration degree of the primary α-Al particles increased with decreasing mean particle diameter.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

A HIGH STRENGTH ALUMINIUM ALLOY FOR HIGH PRESSURE DIE CASTING

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Keywords: Aluminium alloys; High pressure die casting; Mechanical property; Microstructure

Abstract

In order to achieve the mechanical properties of thin-wall components made by high pressure die-casting at a level of 300MPa of yield strength, 420MPa of UTS and 3% of elongation, the composition of the new aluminium alloy and casting parameters have been optimized to reduce casting defects in final components. In the present paper, we describe the alloy development in an industrial scale HPDC system.

Introduction

Aluminium alloys are being increasingly used as lightweight materials in transport since weight reduction has been proved as one of the most effective approaches to improve fuel economy and reduce CO_2 emissions [1,2]. High pressure die casting (HPDC) is a widely used process to make thin-wall aluminium components for structural applications. The solution and ageing heat treatment have been used to enhance the mechanical properties of castings [3]. However, the existing aluminium alloys for HPDC process are mainly based on Al-Si, Al-Si-Cu, and Al-Mg-Si systems. Most of these alloys are not particularly developed for property enhancement by heat treatment, especially not suitable for solution heat treatment because of the unsatisfied mechanical properties for industrial application [4,5]. Therefore, the development of high strength aluminium alloys becomes significant to promote the application of aluminium alloys where an improved strength and lightweight are essential.

In the present paper, we describe the alloy development including the effect of Mg, Mn and Zn in the Al-Mg_2Si system on the mechanical properties and microstructure of the cast alloys. The effect of solution and ageing heat treatment on the mechanical properties and microstructure was also studied

Experimental

Commercial purity aluminium, magnesium, zinc, Al-20wt.%Mn, and Al-50wt.%Si master alloys were used as base materials to make the aluminium alloys. 10kg melt were made each time using a clay-graphite crucible in an electrical resistance furnace. During manufacturing, Al, Si and Mn were melted first and 15 ppm Be was added to the melt 10 minutes prior to the addition of Mg ingots. The melt was degassed at 750ºC using a rotatory degassing equipment with a rotating speed of 500 rpm for 5 minutes. After degassing, the melt was maintained at a temperature of 700ºC. A mushroom casting with a bottom flat of ϕ60×10mm was made by a steel mould for composition analysis. The casting was cut 5mm off from the bottom and ground down to 800 grid sand SiC paper. The composition was obtained from a well-calibrated optical mass spectroscope, in which at least five spark analyses were tested and the average value was taken as the chemical composition of the alloy.

The castings were produced using a 4500 kN cold chamber high pressure die casting machine, in which most of the operation parameters including injection speed and intensification pressure are monitored. One set of die was used to cast standard tensile samples, during which three 6.35mm round samples and three flat tensile samples at 2mm, 3mm, and 5mm were made in one shot. The mechanical properties were obtained from 6.35mm round tensile samples. All parameters had been optimized for the HPDC machine during casting.

The tensile tests were conducted following ASTM standard B557, using an Instron 5500 Universal Electromechanical Testing System equipped with Bluehill software. All tensile tests were performed at ambient temperature. The gauge length of the extensometer was 25 mm and the ramp rate for extension was 2 mm/min. The microstructure of each alloy was examined using a Zeiss optical microscope with quantitative metallography, and a Zeiss SUPRA 35VP SEM.

Results

1. Effect of extra Mg in Al-Mg_2Si alloys

Fig. 1 shows the equilibrium phase diagrams at Al-rich side for Al-Mg_2Si system. The pseudo-binary Al-Mg_2Si system behaves like a normal binary eutectic system with a eutectic point at 13.9 wt.% Mg_2Si. It shows that a narrow three-phase region for L+α-Al+Mg_2Si co-existence between 591 °C and 578 °C. With extra Mg in the Al-Mg_2Si system, the shape of equilibrium phase diagram exhibits some changes. The eutectic point is shifted towards a lower Mg_2Si concentration and the area of co-existence for the three phases is enlarged. This means that, for a given Mg_2Si content, an extra Mg addition results in a decrease of the volume fraction of the primary α-Al phase and an increase of the volume fraction of Al-Mg_2Si eutectic phase.

To confirm the calculated results, a series of experiments were performed with different amounts of extra Mg in the Al-Mg_2Si alloys. The representative microstructures are shown in Fig. 2. The primary phase presented in the as-cast microstructure was different. With 1.9wt.% extra Mg, the alloy showed a typical hypoeutectic microstructure, consisting of dendrites or fragmented dendrites of primary α-Al phase surrounded by Al-Mg_2Si eutectic phase. However, the microstructure was much different when the extra Mg was 5.8 wt.%. The coarse primary α-Al phase almost disappeared and the microstructure was dominated by fine primary α-Al phase and Al-Mg_2Si eutectics. In general, the experimental results were consistent with the calculation results, although the Al-Mg_2Si alloys without extra Mg cannot be obtained in high pressure die casting because of the severe die soldering problem.

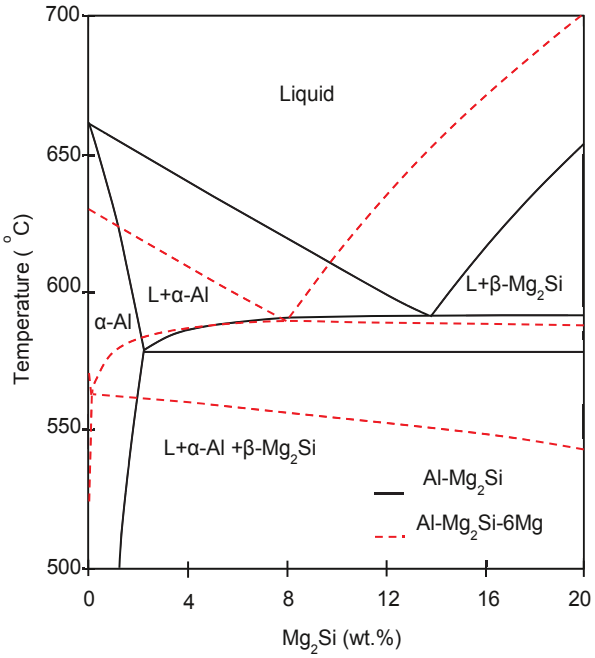


Fig. 1 Equilibrium phase diagrams of Al-Mg_2Si pseudo-binary system and Al-Mg_2Si-Mg ternary system with 6wt.%Mg.

The yield strength, UTS and elongation of the as-cast Al-Mg_2Si-Mg alloys can be varied by the content of extra Mg. The yield strength was increased with increasing Mg content, but showed no significant difference with increasing the Mg_2Si phase. Conversely, the elongation was decreased with the increase of Mg and Mg_2Si in the alloys. The UTS were different for the alloys with different extra Mg contents. The decrease of the UTS was significant for the Al-9.9wt.%Mg_2Si and Al-12.7wt.% Mg_2Si alloys with the increase of extra Mg in the alloys. However, it is noted that the yield strength is much lower than the target of 300MPa.

2. Effect of Mn in Al-Mg_2Si-Mg alloys

Fig. 3 shows the typical microstructure of the intermetallics in the Al-7.8wt.%Mg_2Si- 5.8wt.%Mg-0.59wt.%Mn alloy with 0.12 wt.% Fe. Intermetallics with two different sizes were observed and the volume fraction of the intermetallics increased with the increase of Mn addition. The quantitative analysis showed that the average size of the fine intermetallics were about 0.8μm. An obvious increase of the volume fraction of the Mn-rich intermetallics could be seen for the increasing Mn content. The composition of the Mn-rich intermetallic phase was measured with different amounts of Mn additions and the results confirmed that the Mn-rich intermetallics were α-AlFeMnSi phase. The measured constituent of the intermetallics could be described by $Al_{42-46}(Fe,Mn)_{3.5}$-Si_7. The Mn-rich intermetallics are essentially associated with Fe in the die cast alloys.

The increase of Mn content in the alloys resulted in a slight increase of the yield strength and UTS, but a slight decrease of the elongation for the Al-7.8wt.%Mg_2Si-5.8wt.%Mg alloy under as-cast condition. The elongation was decreased from 6.9% to 6.5% when the Mn content was increased from 0.19wt.% to 0.59wt.%, which is the level of the commonly used in the die-cast Al-Mg-Si alloy.

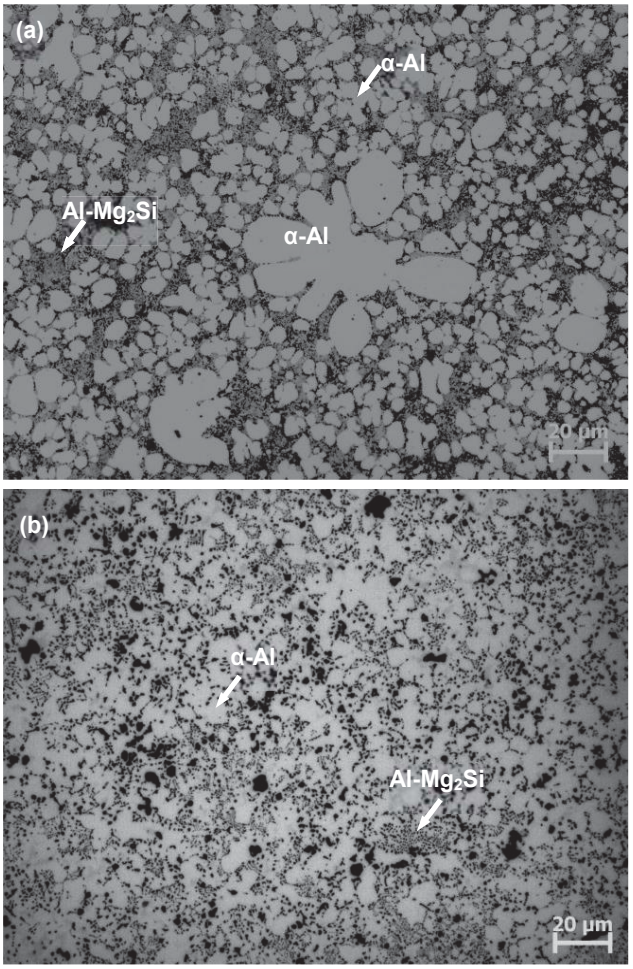


Fig. 2 Optical micrographs showing the effect of excess Mg on the as-cast microstructure of Al-7.8 wt.%Mg_2Si alloy with (a) 1.9wt.%Mg, (b) 5.8wt.%Mg.

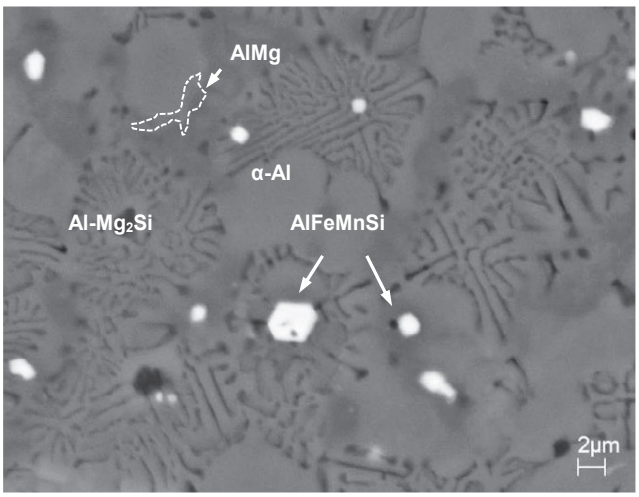


Fig. 3 The morphology of Mn-rich intermetallic phase and AlMg intermetallic phase in the Al-7.8 wt.%Mg2Si-5.8 wt.%Mg-0.59wt.%Mn alloy with 0.12wt.%Fe.

3. Effect of Zn in Al-Mg_2Si-Mg-Mn alloys

The typical microstructure of Al-Mg_2Si-Mg-Zn-Mn alloy is shown in Fig. 4. An AlMgZn phase was observed, which were characterised by irregular shapes. The Zn-containing intermetallic compounds were located at the boundaries of the primary α-Al phase or between the eutectic cells and the primary α-Al phase. The measured volume fractions of the AlMgZn phase were increased significantly with increasing the Zn addition. The quantitative measurement confirmed that the intermetallic compounds contained 25±1 at.% Mg, 10±0.7 at.% Zn, which corresponded to a formula of $Al_{13}Mg_5Zn_2$. Therefore, the Zn-rich intermetallic phase was essentially the same compound in the experimental alloys.

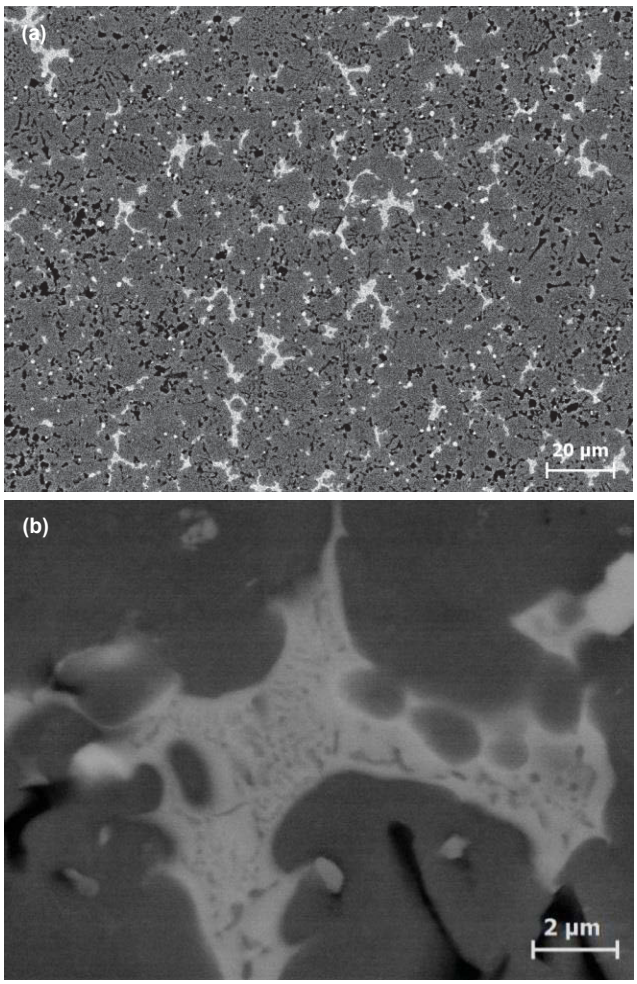
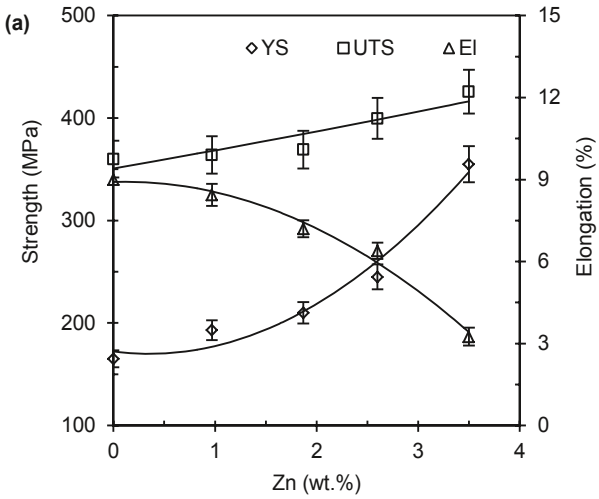


Fig. 4 Backscattered SEM images showing (a) the microstructure and (b) the morphology of Zn- containing intermetallics in the Al-7.8wt.%Mg_2Si-5.8wt.%Mg-0.59wt.%Mn alloy with 3.51wt.% Zn.

The yield strength was increased but the elongation was decreased significantly with increasing the Zn content in the Al-7.8wt.% Mg_2Si-5.8wt.%Mg-0.59wt.%Mn alloy under as-cast condition. When Zn content was 3.51wt.%, the elongation was still higher than 2% and the yield strength was at a level of 250MPa and the UTS was at a level of 350MPa.

4. Effect of solution and ageing in Al-Mg_2Si-Mg-Zn-Mn alloys



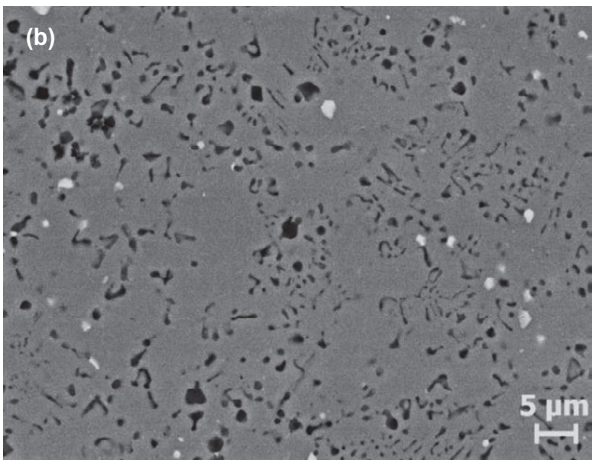


Fig. 5 (a) Mechanical properties with different Zn contents in the Al-7.8wt.%Mg_2Si-5.8wt.%Mg-0.59wt.%Mn and (b) backscattered SEM microstructure of the alloy with 3.51wt.%Zn alloy under a condition of solution treated at 490°C for 30 minutes and subsequent aged at 180°C for 90 minutes.

Solution and ageing treatments were performed at 490°C and 180°C, respectively. Fig. 5(a) shows the effect of Zn on the yield strength, UTS and elongation of the alloys after solution and ageing heat treatment. Significant increases in yield strength and UTS and a decrease in elongation were observed in the alloy with increasing the Zn content. The tensile properties can be easily achieved at a level of 340MPa of yield strength, 430MPa of UTS and 3.5% of elongation when Zn is 3.51wt.%.

The typical microstructures after solution at 490°C for 30 minutes and subsequent aged at 180°C for 90 minutes are shown in Fig. 5b. During solutionising, the AlMgZn intermetallics were gradually dissolved into the matrix. However, no obvious change was observed for the AlFeMnSi intermetallics and eutectic Mg_2Si phase after the solution treatment. The TEM examination revealed that a large numbers of fine MgZn precipitates were formed in the α-Al phase after solutioning and ageing, which provide effective precipitates strengthening.

Discussion

In the Al–Mg_2Si–Mg–Zn alloy, the as-cast microstructure contains several phases formed during solidification, including primary α-Al, eutectic Al–Mg_2Si, β-AlMgZn and α-AlFeMnSi phases.

The microstructural characteristics of the alloy can be understood from the solidification and phase formation process. According to the equilibrium phase diagram of the Al–Mg_2Si–Mg–Zn–Mn system on the cross section of Al–8 wt%Mg_2Si–6 wt%Mg–0.6 wt%Mn calculated by Pandat software [5]. It is seen that the solidification sequence of the alloy is L-α-Al-Mg_2Si- AlMgZn-AlFeMnSi-AlMg when Zn is with in the calculation range. The prior phase is α-AlFeMnSi, followed by the formation of other phases with temperatures being reduced. However, the calculation results are slightly different to the experimental results where no AlMg phase was actually found in the experimental results. The disappearance of the AlMg intermetallics is mainly because of the existence of Zn in the alloy. Although the exact mechanism is still not clear, the formation temperatures of both AlMg and AlMgZn intermetallics is close to each other according to the equilibrium shown in given non-equilibrium solidification, the AlMgZn phase is formed before AlMg phase, resulting in suppression of the AlMg phase.

As the microstructure in the diecastings compromises the primary α-Al phase, Al–Mg2Si eutectics and intermetallic phases. Mg,Si, Zn and Mn can provide solution strengthening to the α-Al phase. The precipitates of intermetallics under as-cast condition can strengthen the alloy by acting as pins to prevent dislocations from sliding under stress. Therefore, the fine and compact α-AlFeMnSi phase can reduce the detrimental effect of needle-shaped intermetallics. However, the intermetallics themselves are brittle and can initialize cracks, resulting in the reduction in ductility. The further strengthening by solution and ageing heat treatment results from the precipitation of MgZn phase in the α-Al phase. The AlMgZn compounds decompose during solution and subsequently precipitate during ageing. The fine precipitates provide much more effective strengthening effect in the alloy, during which the detrimental effect from irregular AlMgZn phase is minimized and the elongation is thus improved. On the other hand, during heat treatment, no significant change is observed for the α-AlFeMnSi andMg_2Si phase because of their high thermal stability, indicating that the enhancement in mechanical properties from the α-AlFeMnSi andMg_2Si intermetallics is limited after the heat treatment.

Summary

A high strength aluminium alloy for high pressure die casting has been developed based on Al-Mg_2Si-Mg-Zn system. The typical alloying elements comprises of 8.0wt.%Mg_2Si, 6.0wt.%Mg, 3.5wt.%Zn, 0.6wt.%Mn. The alloy consists of primary α-Al, Al-Mg_2Si eutectics, β-AlMgZn, and α-AlFeMnSi phases in the as-cast microstructure. The typical tensile properties of the die-cast tensile samples are 300MPa of yield strength, 420MPa of ultimate tensile strength, and 3% of elongation under solution and aged condition. A quick solution and ageing is an effective way to enhance the mechanical properties, by which the AlMgZn intermetallics are decomposed into Al matrix and fine Zn-rich intermetallics are precipitated during the subsequent ageing treatment.

Acknowledgement

The authors acknowledge the Engineering and Physical Sciences Research Council (EPSRC), Technology Strategy Board (TSB) and Jaguar Land Rover (JLR) in United Kingdom for financial support.

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Light Metals 2016

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PRECIPITATION OF Al_3Zr DISPERSOIDS DURING HOMOGENIZATION OF Al-Zn-Cu-Mg-Zr ALLOYS

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Keywords: homogenization, precipitation, nucleation

Abstract

Dispersoids of Al_3Zr in Al-Zn-Cu-Mg-Zr alloys are important as they pin grain boundaries and inhibit recrystallization during extrusion and solution heat treatment. A high volume fraction, V_{Al_3Zr}, and low dispersoid radius, r_{Al_3Zr}, give V_{Al_3Zr}/r_{Al_3Zr} values above a critical value which are essential to prevent recrystallization. Precipitation of Al_3Zr during homogenization is studied for different homogenization schedules. A 1D finite difference model based on classical nucleation and growth theory for a multicomponent system is developed to simulate precipitation of Al_3Zr dispersoids. The 1D domain represents one half of a secondary dendrite arm spacing with initial concentration gradients based on Scheil type solidification. Single-step homogenization, slow heating to homogenization temperatures, and two-step homogenization are evaluated for maximum dispersoid number density, maximum volume fraction and minimum dispersoid radius. V/r ratios along the length of the SDAS are compared. Two step homogenization (10hrs at 420°C and more than 10hrs at 475°C) provides better V/r ratios across the grain and provides the maximum recrystallization resistance.

Introduction

Al-Zn-Cu-Mg alloys (7xxx series) are widely used in aerospace structure applications. A trace amount of Zr in Al-Zn-Cu-Mg alloys is essential to form nanoscale coherent Al_3Zr dispersoids during homogenization. These inhibit grain boundary motion during high temperature thermal (solution heat treatment) and thermo-mechanical treatment (hot rolling, extrusion) and so prevent recrystallization [1], which causes undesirable degradation in strength and fracture toughness [2,3].

Recrystallization can be prevented by a sufficient volume fraction of coherent dispersoids, small enough in size to exert a Zener drag pressure, P_Z which can be expressed mathematically as [4]

$$P_Z = \frac{1.5V\gamma}{r} \quad (1)$$

where V is the dispersoid volume fraction, r is the dispersoid radius and γ is the interface energy of the dispersoid with the matrix. However, the number density and volume fraction of the dispersoids across the grains is not uniform due to microsegregation during casting. Al-Zr system is a peritectic system with partition coefficient for Zr >1 (k^{Zr}~1.4), causing lower concentrations of Zr in regions near the grain boundaries. The Zr concentration can be low enough to cause a dispersoid-free zone in that region. It is difficult to prevent recrystallization in these dispersoid-free zones [5].

The as-cast microstructure in these alloys also includes an interdendritic η phase ($MgZn_2$), which transforms to the S phase (Al_2CuMg) during homogenization [6]. Homogenization of these alloys serves three main purposes: (i) reduction of microsegregation; (ii) dissolution of the low melting interdendritic phases (η and S) and (iii) precipitation of high density, tiny dispersoids throughout the grain. All these coupled processes occur simultaneously. The microstructure after homogenization should have no microsegregation, no η and S phases and a fine, numerous distribution of dispersoids.

There have been both experimental [1, 5, 7, 8] and numerical [9-11] studies on precipitation of Al_3Zr in 7xxx alloys. The numerical studies have considered the classical Kampmann and Wagner (KWN) [12] model to simulate nucleation for a binary Al-Zr system. Also, precipitation of these dispersoids has been modeled with the assumption of either no change to the interdendritic particles or their complete dissolution. Both these assumptions do not hold in reality as these processes are simultaneous. All these studies emphasize the importance of multistage homogenization. During the first lower temperature stage, the supersaturation is higher producing a finer dispersoid radii. The second stage is at a higher temperature, sufficient enough to dissolve the interdendritic particles. However, the literature does not provide one optimum homogenization schedule.

The aim of this study is to evaluate homogenization at different temperatures for different times to determine an optimized homogenization schedule for AA7050 alloy. We will specifically compare the number densities, mean radius and volume fraction to mean radius ratios along the length of the SDAS for three schedules: (i) single step homogenization; (ii) multistep homogenization and (iii) slow heating to the homogenization. The model is different from previous studies as it includes a multicomponent model of dispersoid nucleation and growth and considers simultaneous transformation of the η to S, redistribution of alloying elements, and the nucleation and growth of the Al_3Zr dispersoids leading to realistic nucleation and growth rates and microstructures.

Numerical Model

Model Description

Two models have been developed, the first to simulate the η to S phase transformation in the interdendritic regions and the second the precipitation of Al_3Zr dispersoids across the SDAS. These models are coupled to prediction of alloying element redistribution across the grain. The homogenization model simulating the η to S phase transformation has been validated against experiments. The present work describes the precipitation model in detail.

A 1D precipitation model has been developed to predict nucleation and diffusion-based growth of phases depending on temperature for a 5-component alloy. Thermo-Calc™ and DICTRA™ softwares were used to predict thermodynamic (equilibrium concentrations at different temperatures) and kinetic (diffusion coefficients) data respectively. An aluminum-based mobility database is used for the calculation of diffusion coefficients.

A finite volume based numerical scheme is used to predict the multicomponent diffusion. The 1D domain is divided into finite cells each of which contains all α (fcc Al matrix) phase, all η/S (precipitate) phase or an interface between α and the precipitate. An interface cell has a volume fraction of phase η/S between 0 and 1, the remainder volume being α. Each cell in the SDAS domain consists of another, smaller scale 1D domain consisting of cells in which dispersoid size classes are tracked. The time evolution of dispersoid number density in these size classes are calculated separately for different positions in the larger SDAS domain using

$$\frac{\partial N}{\partial t} = -\frac{\partial (Nv)}{\partial r} + S, \tag{2}$$

where N is the number of dispersoids in the r size class, v is the velocity of the dispersoid-matrix interface and S is the source term representing nucleation [13].

The velocity of the dispersoid-matrix interface representing the growth of the dispersoids can be expressed as [13]

$$v = \frac{dr}{dt} = \frac{\bar{C} - C_i}{C_p - C_i} \frac{D_{Zr}}{r}, \tag{3}$$

where C_p is the concentration of Zr in the dispersoid phase, C_i is the interface concentration at a given radius, r is the position of the particle/matrix interface. That interface concentration, C_i is calculated from a composition given by the phase diagram, C_e, using the Thompson-Freundlich equation to take interfacial curvature into account. D_{Zr} is the diffusion coefficient of Zr in Al. The source term, S, in eqn. (2) is modeled by classical nucleation theory for heterogeneous nucleation [14]

$$\mathrm{S} = (\mathrm{N_{max}}\text{-}\mathrm{N_{tot}})\mathrm{Z}\beta \exp\left[-\frac{\Delta \mathrm{G_{hom}} f(\theta)}{\mathrm{k_B T}}\right] \exp\left(\frac{\tau}{t}\right), \tag{4}$$

where N_{max} is the number of heterogeneous nucleation sites present in the alloy estimated here as the total number of dislocation intersections [15], as 8×10^{19}/m^3 and N_{tot} is the total number of existing dispersoids. Z is the Zeldovitch's factor accounting for fluctuations in nucleus size due to movement of atoms from the nucleus to the matrix [16]

$$Z = \frac{\Delta G_N^2}{2\pi V \gamma^{3/2} \sqrt{N_{Av} RT}}, \tag{5}$$

Here ΔG_N is the driving force for nucleation, V is the molar volume of the dispersoid phase (1×10^{-5}m^3/mole), γ is the interface energy of the dispersoid with the matrix (0.1 J/m^2), N_{Av} is the Avagadro number and T is the temperature. The values considered in this work have been taken from previous numerical studies [9-10].
The coefficient β in equation (4) is the rate of joining of solute atoms from matrix to the nucleus expressed for a multicomponent alloy as [17]

$$\beta = \frac{4\pi r_c^2}{a^4}\, Min(D_i x_i), \tag{6}$$

where r_c is the critical radius of the dispersoids given by:

$$r_c = \frac{2\gamma V}{\Delta G_N}. \tag{7}$$

The lattice parameter of fcc Al (a) is estimated at 4.08×10^{-10} m and x_i is the concentration of the ith alloying element at a particular position on the SDAS. In eqn (4), ΔG_{hom} is given by

$$\Delta G_{hom} = \frac{16\pi\gamma^3 V^2}{3\Delta G_N^2}, \tag{8}$$

Present work assumes heterogeneous nucleation as opposed to homogeneous nucleation used by previous studies. The wetting angle function, $f(\theta)$ is introduced into eqn. (4).

$$f(\theta) = 1/4(2 + cos\theta)(1 - cos\theta)^2 \tag{9}$$

The wetting angle θ is taken as 45° [15]. The incubation time (τ) in eqn. (4) is expressed as [14]:

$$\tau = \frac{2}{\pi\beta x_{Zr}^2}. \tag{10}$$

The driving force for nucleation in a multicomponent is calculated as [16]:

$$\Delta G_N = RT \sum_{i=1}^{n} x_i^p \ln\left(\frac{x_i^\infty}{x_i^m}\right), \tag{11}$$

where x_i^m and x_i^p are the equilibrium concentrations in the precipitate and the matrix phases and x_i^∞ is the average composition of the matrix.
The driving force for nucleation and the critical nuclei are calculated at each time step for a given concentration of the alloying elements. Taking into account size fluctuations, S nuclei are added to size class corresponding to [14]:

$$r_{eff} = r_c + 1/2\sqrt{\frac{k_B T}{\pi\gamma}}, \tag{12}$$

The size evolution is calculated using equation (2) for the desired heat treatment schedules. The governing equation (2) is discretized using an implicit time scheme and solved using the Tridiagonal Matrix Algorithm.

Model Validation

The precipitation model has been validated against experimental work by Robson et al.[18]. An initial model microstructure consisting of interdendritic η phase at the grain boundaries and microsegregation of alloying elements across the grain is homogenized at 500°C. The transient evolution of number density and mean radius of the Al_3Zr dispersoids is predicted. Table 1 shows the comparison of the numerical predictions with the experimental observations.

Table 1: Comparison of the experimental and predicted number densities and mean radii for homogenization at 500^{0}C [18].

Time	Experimental[18]		Numerical	
	Number density	Mean radius	Number density	Mean radius
2 hrs	26.3±8	10±4	32.0	13.3
5 hrs	43.2±11	12.5±3	36.5	15.6
10 hrs	40.5±14	18.5±5	39.7	20.3

The numerical predictions match the experimental results well. The number density and mean radius values fall within the uncertainty of the experimental results. It must be noted that the number of heterogeneous nucleation sites in the alloy has been estimated to match the experimental number densities at 500°C.

Results and Discussion

Three different types of homogenization schedules are evaluated for maximum number density and minimum dispersoid radius to get V/r ratios above the critical value to avoid recrystallization. The different test cases that were run are provided in Table 2. For the SSH and TSH schedules, the homogenization temperatures are reached instantaneously involving no ramp from the room temperature. Same is true for SHH to reach 300°C after which there is a slow ramp.

Table 2: Homogenization schedule details for the different test cases run.

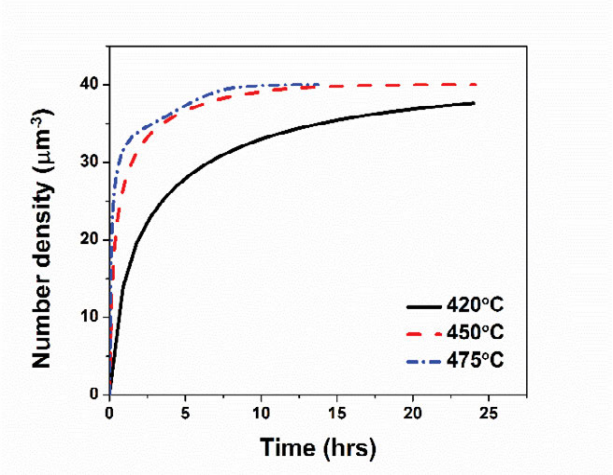
	Homogenization schedule
SSH1	Single step homogenization at 420°C
SSH2	Single step homogenization at 450°C
SSH3	Single step homogenization at 475°C
SHH1	Slow heating from 300°C to 475°C at 5°/hr followed by homogenization at 475°C
SHH2	Slow heating from 300°C to 475°C at 10°/hr followed by homogenization at 475°C
SHH3	Slow heating from 300°C to 475°C at 20°/hr followed by homogenization at 475°C
TSH1	Two step homogenization: 380°C and 475°C
TSH2	Two step homogenization: 400°C and 475°C
TSH3	Two step homogenization: 420°C and 475°C

Single step homogenization

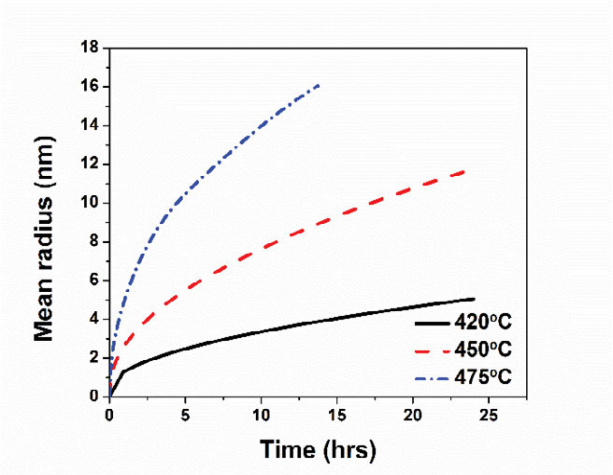
The most straightforward schedule for 7xxx series alloys is to homogenize at a single temperature at which the η to S phase transformation and their dissolution is fast. With these phases melting at temperatures above 475°C [19], it seems that the alloy should be heat treated below that temperature for fast elimination of the η and S phases without forming liquid. Such a homogenization schedule would save both time and cost for the heat treatment. However, the dispersoids formed near this incipient melting temperatures coarsen very fast, which could be minimized by lowering the homogenization temperature at the cost of time needed to eliminate the interdendritic phases.

Homogenization at the maximum temperature of 475°C and the lower temperatures (420°C and 450°C) was simulated. Figure 1(a) and (b) show the evolution of number density and dispersoid radius with time at these 3 temperatures, respectively. There is a steep increase in the number density initially which saturates later on. Because of faster diffusion of Zr in Al at higher temperatures, the saturation is faster. There is little difference in the saturated number densities at these three temperatures but the mean dispersoid radii are different at these temperatures. While the growth of dispersoids is very slow at 420°C, it is relatively rapid at 475°C due to faster diffusion of Zr in Al.

It might seem that lower temperatures are ideal for homogenization as the dispersoids are smaller in size. However, η to S phase transformation is very sluggish at lower temperatures. The equilibrium volume fraction of the η and S phases are also higher at lower temperatures so not all of those are eliminated. To give an idea, after 24 hrs of homogenization at 420°C, the predicted amount of η and S remaining in the alloy is 3.57% and 1.64% respectively, while after 14 hrs at 475°C, all the η is eliminated (transforms to S) and 1.53% S phase remains. The segregation existing in the as-cast microstructure also is removed more slowly at lower temperatures.



(a)



(b)

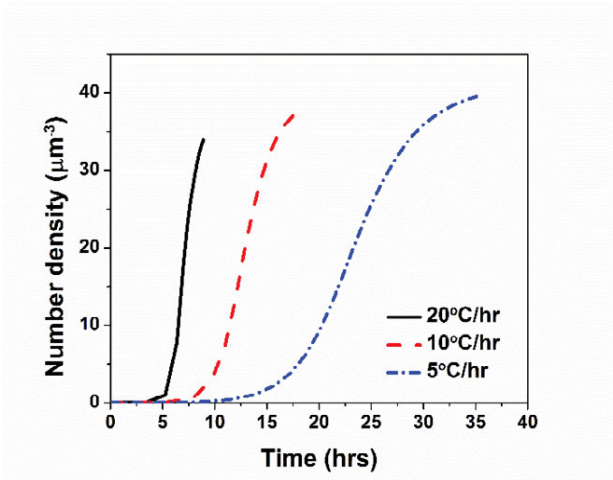
Figure 1: Evolution of (a) number density and (b) mean radius of dispersoids during single step homogenization at different temperatures.

Slow heating and homogenization

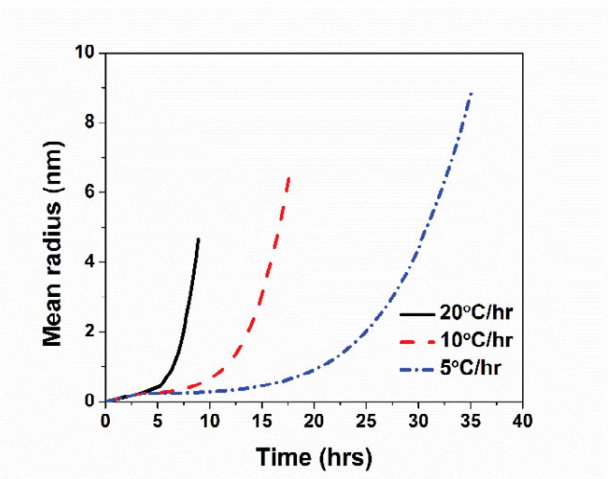
Because number density evolution is faster and mean radius coarsening is slower at lower temperatures while elimination of microsegregation and η/S phases is faster at higher temperatures,

slow heating to homogenization temperatures of 475°C seems to be a viable homogenization schedule which is also more realistically attainable in practice. In this work three, different ramp rates of 20°/hr, 10°C/hr and 5°/hr are evaluated to form an industrially acceptable microstructure in a reasonable amount of time.

Figure 2(a) and (b) show the evolution of number density and mean radius of dispersoids with time respectively, at different heating rates between 5-20°C/hr. The number density increases very slowly initially after which it accelerates until it gradually saturates. The initial number density curve is steeper for faster heating rates because of the rapid increase in temperature. As more time is spent at lower temperatures (400°C-425°C) where increase in number density is considerable, the number densities at lower heating rates is higher. It can be seen that the saturated number density at 475°C is higher at lower heating rates. The mean radius evolution shows a trend different from the number density. It increases very slowly at low temperatures gradually increases parabolically in the temperature range 400°C to 450°C and then increases linearly above 450°C to 475°C. The growth rate is higher at high temperatures owing to higher diffusion leading to very rapid coarsening of the dispersoids.



(a)



(b)

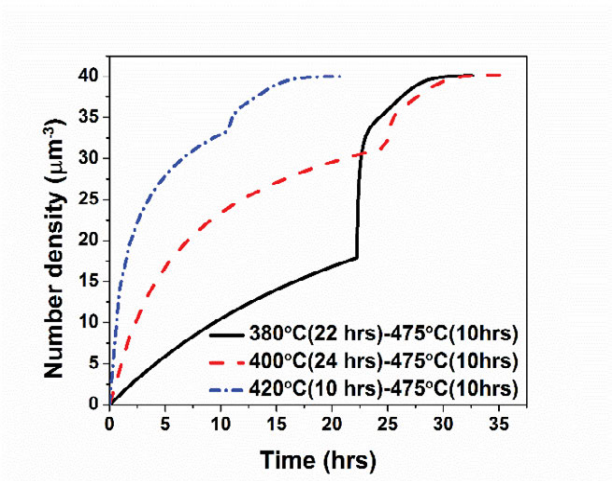
Figure 2: Evolution of (a) number density and (b) mean radius of dispersoids with time during slow heating from 300°C to 475°C at different heating rates.

While faster heating rate may seem to be an attractive homogenization schedule option as far as mean radius of dispersoids is concerned, the number densities for the three cases are comparable. However, we should also keep in mind the time taken to eliminate the η and S phases. The predicted volume fractions of η and S phases remaining at 475°C in the alloy are, 4.12% and 0.83%; 2.53% and 1.15%, and 0% and 2.47% at heating rates of 20°C/hr, 10°C/hr and 5°C/hr respectively. It can be seen that slower heating rates lead to more transformation of η to S phase, which can be eliminated in a shorter time when compared to faster heating rates, during holding at 475°C. So an intermediate heating rate seems to be an answer to having a distribution of fine dispersoids with a high number density as well as more η to S phase transformation.

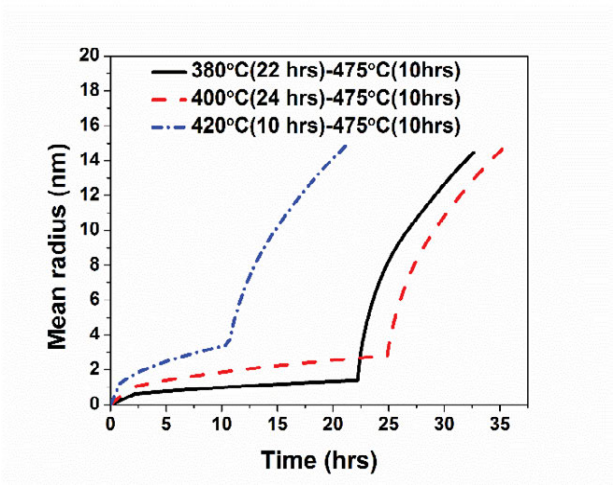
Two step homogenization

To take advantage of the faster nucleation rates at lower temperatures (400-450°C) and faster η to S phase transformation and dissolution at higher temperatures (450-475°C), it would be useful to evaluate a two-step homogenization treatment schedule, consisting of a lower temperature stage for dispersoid nucleation followed by a higher temperature stage for interdendritic phase dissolution. In the present work, we have evaluated 3 homogenization schedules: (i) holding at 380°C for 22 hours followed by 475°C for 10 hrs; (ii) holding at 400°C for 24 hours followed by 475°C for 10 hrs; and (iii) holding at 420°C for 10 hours followed by 475°C for 10 hrs.

Figure 3 (a) and (b) show the time evolution of number density and mean radius of the dispersoids for the above-mentioned homogenization schedules. The number density of the dispersoids increases steadily, approaching saturation during the first step. There is a steep increase in dispersoid production rate at the beginning of the second step. The number density then proceeds towards saturation. The mean radius of the dispersoids show a similar trend during the first step, however at higher temperatures the mean radius increases steeply and continues to do so after 10 hrs. Both the number density and mean radius of the dispersoids during the first step is higher at higher temperatures. However, there is little difference in the final number density and mean radius of the dispersoids after the second step at 475°C for 10 hrs.



(a)



(b)

Figure 3: Evolution of (a) number density and (b) mean radius of dispersoids with time during two step homogenization.

As there is no difference in the final number density and mean radius after 10 hrs of holding at 475°C, the factor that differentiates the schedules is the time taken to eliminate the η and S phases. Because the S phase melts at ~473°C, most of the S phase dissolves at 475°C, leaving 0.95%, 0.92% and 0.89% η for first step at 380°C, 400°C and 420°C, respectively. So, holding at 420°C for the first step seems to be a time and energy saving step due to more η to S phase transformation.

Comparison of the V/r for different homogenization schedules

To inhibit grain boundary movement and minimize recrystallization, the dispersoids must exert enough Zener pressure, which is proportional to the volume fraction and depends inversely on the mean radius of the dispersoids (eqn. 1). An optimized homogenization schedule can be determined after evaluating the Zener pressure by the precipitated dispersoids. Table 3 shows the comparison of the volume fractions, mean radius and V_{Al_3Zr}/r_{Al_3Zr} ratios for the nine test cases.

Table 3: Comparison of volume fraction, mean radius and V_{Al_3Zr}/r_{Al_3Zr} ratio for the different test cases run. The time for various stages are mentioned along with the test cases.

Test case	Volume fraction	Mean radius (nm)	V_{Al_3Zr}/r_{Al_3Zr} (μm^{-1})
SSH1 (24hrs)	0.000027	5.04	0.0054
SSH2 (24hrs)	0.000348	11.79	0.0295
SSH3 (14hrs)	0.000879	16.05	0.0547
SHH1 (5°C/hr,7hrs)	0.000606	14.25	0.0425
SHH2 (10°C/hr,7hrs)	0.000489	13.22	0.0370
SHH3 (20°C/hr,7hrs)	0.000423	12.41	0.0341
TSH1 (24hrs,10hrs)	0.000654	14.45	0.0453
TSH2 (24hrs,10hrs)	0.000687	14.77	0.0465
TSH3 (24hrs,10hrs)	0.000694	14.85	0.0467

To inhibit recrystallization, the V_{Al_3Zr}/r_{Al_3Zr} ratio should be above a critical value. For the single step homogenization, the volume fraction is highest for holding at 475°C for 14 hrs, however the dispersoid radius is higher than other cases in this category. Similarly, for slow heating, the volume fraction is highest heating at a rate of 5°C/hr but the dispersoid radius is also highest for this case. For two step homogenization, the volume fraction and mean radii are comparable for all the three schedules, V_{Al_3Zr}/r_{Al_3Zr} being the best at 420°C.

It should however, be kept it mind that there is a variation of Zr concentration from the center to the edge of the grain leading to variation in volume fractions and mean radius from the center to the edge of the grain. To be able to evaluate the best homogenization schedule we should have a look at the V_{Al_3Zr}/r_{Al_3Zr} ratio variation from the center to the edge of the grain. Figure 4 shows the variation of V_{Al_3Zr}/r_{Al_3Zr} ratios along the SDAS for one test case from each homogenization schedule.

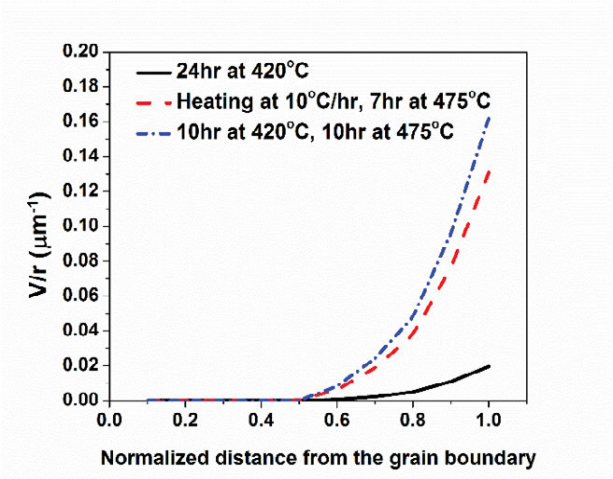


Figure 4: Comparison of V_{Al_3Zr}/r_{Al_3Zr} ratios from the center to the edge of the grain for different homogenization schedules.

Comparison of V_{Al_3Zr}/r_{Al_3Zr} ratios for the three homogenization schedules in Figure 4 shows that V_{Al_3Zr}/r_{Al_3Zr} ratios for slow heating and two step homogenization are comparable and both are better than the single step homogenization. Two step homogenization is slightly better than slow heating. Two step homogenization provides the best recrystallization resistance across the grain. It is also the most cost and energy efficient schedule as it takes less time to remove microsegregation and dissolve the interdendritic phases.

Conclusions

A 1D finite difference precipitation model has been developed based on classical theory of heterogeneous nucleation and growth which works in tandem with a homogenization model which predicts simultaneous transformation interdendritic η to S phase and their dissolution. The precipitation model is able to predict number densities and mean radius of dispersoids which match published experiments. Three different homogenization schedules: (i) single step homogenization, (ii) slow heating and homogenization and (iii) two step homogenization, are evaluated for maximum recrystallization resistance, which is proportional to V_{Al_3Zr}/r_{Al_3Zr}. To study the effect of the different schedules, the number density and mean radius evolution with time are compared. The V_{Al_3Zr}/r_{Al_3Zr} ratio variations along the length of the SDAS are also compared. The two step homogenization with first step of 10 hrs at 420°C and second step of more than 10 hrs at 475°C provides

the best V_{Al_3Zr}/r_{Al_3Zr} ratio in the grain providing the highest recrystallization resistance.

Acknowledgments

The authors acknowledge financial support for this research from Shandong Nanshan Aluminum Co., Beijing Nanshan Institute of Aeronautical Materials.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

APPLICATION OF SECONDARY SHEAR EFFECTS IN THE EXTRUSION MACHINING PROCESS TO EXPLORE RECRYSTALLIZATION MECHANICS DURING CONVENTIONAL EXTRUSION OF 7050 ALUMINUM

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Keywords: Aluminum, extrusion, machining, recrystallization, 7050 alloy

Abstract

Extrusion machining combines cutting with simultaneous extrusion using an additional constraining tool to induce very large and well-characterized strains. Unlike conventional extrusion, accurate analytical models exist for extrusion machining to calculate values of effective strain, strain rate, and temperature rise as a function of processing parameters. Additionally, the size, shape, and microstructural effects of a secondary shear zone that forms due to friction during extrusion machining have been characterized. This secondary shear zone can be manipulated by varying processing parameters, and can produce a severely inhomogeneous microstructure. This microstructure is also seen during conventional extrusion, as the material on the outside of the extrudate is strained more than the inside material, potentially causing an incomplete recrystallization across the face of the product. The secondary shear effect in the extrusion machining process is utilized to reproduce the inhomogeneous microstructure seen in conventional extrusion in order to better understand the recrystallization behavior during AA7050 extrusion.

Introduction

Extrusion is among the most commonly applied industrial processing techniques for aluminum alloys. Thanks to their high formability at elevated temperatures, it is possible to extrude aluminum alloys into long products with complex cross-sectional profiles having mechanical properties that would be difficult to achieve using other techniques.

For certain high-strength aluminum alloys, such as AA7050, surface quality is of particular importance, due to its impact on corrosion susceptibility and fatigue performance. Unfortunately, extrusion is not a homogeneous process, and inherent inhomogeneities in strain, strain rate, and temperature during the process can lead to a defect known as the peripheral coarse grain (PCG) defect [1].

PCG results when the material at the periphery of the extrudate experiences a higher effective strain than that in the center, causing the outer material to recrystallize while the inner material retains a very elongated microstructure. During subsequent heat treatments, recrystallized grains coarsen, resulting in a PCG layer at the outside of the extrudate. The presence of a PCG layer is associated with inhomogeneous mechanical properties, insufficient corrosion resistance, and, ultimately, scrapped products [2].

It is currently impossible to measure local deformation variables (such as temperature, stress, strain, and strain rate) in an industrial or experimental extrusion setting. Analytical models exist to predict specific values, but fail to account for the interactions between changing variables (e.g., change in temperature and flow stress).

To account for these interactions, finite element modeling (FEM) is useful to predict local deformation variables. Commercial FEM packages are increasingly common, and several have been designed specifically to handle the large-scale plastic strains that occur in deformation processes such as extrusion. These packages are quite powerful, and have been the subject of considerable research [3–8].

Recent breakthroughs in FEM of extrusion include microstructural models that can predict recrystallized grain size and PCG layer thickness [3,4]. These breakthroughs are promising and remain the subject of considerable research efforts, but still have several practical shortcomings. In particular, FE models can be highly sensitive to difficult-to-obtain coefficients. Values for these coefficients can be obtained empirically, but must be obtained and verified individually for each set-up, severely limiting the practicality of such models.

Past attempts to observe the microstructural evolution in an experimental setting have employed hot torsion as an analog for the conventional extrusion process. Hot torsion involves a simple shear deformation state at elevated temperature. With sufficient knowledge of the deformation state and measurement of the temperature change, it is possible to correlate the resultant hot torsion microstructure with those seen after conventional extrusion [1,2,9].

Van Geertruyden has used this technique to show that the critical effective strain, regardless of temperature, required to achieve complete dynamic recrystallization seen in extrusion of AA6061 is 3.5 – implying that PCG results when the outside of the extrudate experiences an effective strain above 3.5, while the inside of the extrudate does not [2].

This work employs a constrained extrusion machining process known as large strain extrusion machining (LSEM) [10–12] as an experimental analog for the conventional extrusion process. LSEM differs from hot torsion in that the material is translated through a *deformation zone*, similar to material flow in extrusion, rather than simply flowing around an axis.

Adiabatic heating occurs exclusively in this deformation zone, and the extruded material quickly carries the majority of the generated heat away from the deformation zone. The increased similarity of the LSEM heat schedule to that of the conventional extrusion process is complemented by the introduction of

secondary shear effects due to friction from translational flow against the cutting force. These differences from hot torsion and the analytical models to predict processing parameters make LSEM an attractive alternative analog to study microstructural evolution during aluminum extrusion.

Background

LSEM Mechanics

Strain Figure 1 shows a schematic of the LSEM process. The sharp cutting tool and constraining edge are located to achieve a predetermined cut depth (t_o) and extrudate thickness (t_c). At constant velocity (V_o), these tools can be configured to achieve relatively high strains in a single-step process by taking advantage of large hydrostatic pressure and high temperatures due to adiabatic heating.

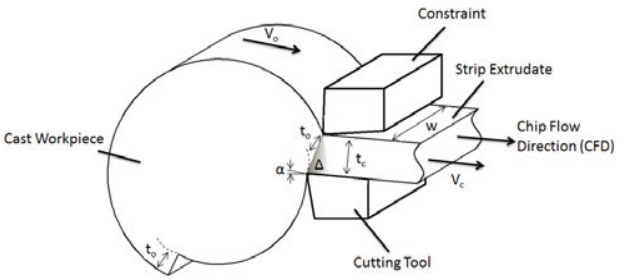

Figure 1. Schematic of LSEM in a rotary setup.

The tooling configuration determines variables such as the tool rake angle (α) and chip thickness ratio ($\lambda=t_c/t_o$), which in turn control the effective strain (ε) and hydrostatic pressure (P) during deformation. Previous work shows that the deformation zone (Δ) is narrow, ~50-150 μm thick, so the analytical solution of an upper-bound analysis, which idealizes the deformation zone as a single plane, is valid. The upper-bound analysis for the shear strain and von Mises effective strain during deformation yields the following solutions [10,13]:

$$\gamma = \frac{\lambda}{\cos\alpha} + \frac{1}{\lambda\cos\alpha} - 2\tan\alpha \quad (1)$$

$$\varepsilon = \frac{1}{\sqrt{3}}\gamma \quad (2)$$

Temperature The large plastic strains that occur in the small deformation zone mean that adiabatic heating is concentrated to a very small area. The temperature in the deformation zone is given by $T = T_o + \Delta T$, where T_o is the initial workpiece temperature and ΔT is the increase in temperature due to adiabatic heating, given by:

$$\Delta T = \frac{(1-\Gamma)u_s}{\rho c}, \quad (3)$$

where ρ is the density, c is the specific heat, and Γ is the fraction of shear plane heat flowing into the workpiece [11].

The specific shear energy of strip formation, u_s, or the energy per unit volume dissipated at the shear plane, is a function of the cutting force, F_c, and the strip dimensions, t and w. The energy is approximated as

$$u_s = \frac{2}{3}\frac{F_c}{tw}, \quad (4)$$

based on the fact that 2/3 of the total energy of strip formation is dissipated at the shear plane. For AA7050, $\rho = 2.83$ g/cm^3 and c = 0.86 J/gK.

ΔT tends to increase with V_o and γ, and T_o can be controlled by preheating the workpiece, so that T can effectively be varied over a wide range.

Secondary Shear Secondary shear effects due to friction are confined to a small portion of the strip, and are therefore disregarded in the analytical models that predict γ and T. There have been, however, recent attempts to characterize the size of the secondary shear zone as a function of process parameters, as well as its role on deformation microstructure and properties. The thickness of the secondary shear zone as a function of strip thickness is given by:

$$\frac{t_{ss}}{t_c} = 1 - \frac{\lambda}{\cos\eta}\sin(\theta+\eta), \quad (5)$$

where η is the friction angle, or the angle friction lines make with the tool, and θ is the fan angle, both of which are functions of both the strip-tool friction and λ. Secondary shear zones extend deeper into the strip as λ decreases below 1, as the LSEM process more closely resembles conventional extrusion [12].

The extent of increased effective strain in the secondary shear zone has not been characterized.

The Peripheral Coarse Grain Defect

AA7050 is typically DC cast and homogenized prior to indirect (or backward) extrusion. Despite the reduced tendency for PCG in indirect extrusion, as compared to direct extrusion, PCG is a limiting factor in the speed and economy of the AA7050 extrusion process [9,14,15].

During aluminum extrusion, large equiaxed grains are typically elongated in the extrusion direction. These elongated grains are separated by high-angle grain boundaries, and are made up of finer, equiaxed subgrains with low-angle grain boundaries. This is considered to be a desirable microstructure prior to aging, as subgrain boundaries are evenly dispersed and act as nucleation points for fine second phase particles that provide strengthening [9].

It has been shown that at high effective strains, like those seen at the periphery of the extrudate, insoluble second phase particles act as nucleation points for grains to form serrations. At some critical strain, ε_c, the magnitude of these serrations becomes too large and the grains impinge on each other, forming new, smaller grains. This mechanism is known as geometric dynamic recrystallization (gDRX). When gDRX grains grow during subsequent heat treatments, they have inferior mechanical properties to the rest of the extrudate. This is the basis for the PCG defect [2,9].

Experimental

Unconstrained free machining (FM) and LSEM were performed on an AA7050-T6 workpiece purchased from McMaster-Carr, Inc. Figure 2 shows the microstructure for this material, as-received.

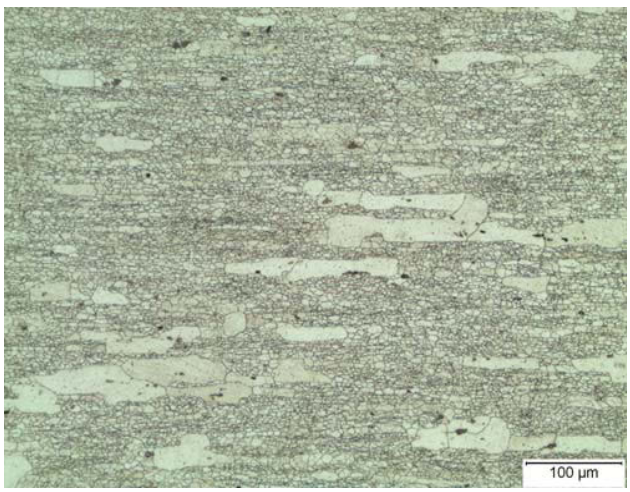

Figure 2. Optical micrograph of as-received AA7050-T6 microstructure seen in the rolling direction.

Elongated grains indicate that this material was previously cold-rolled. The presence of both large, highly-elongated grains and finer, more equiaxed grains may indicate that the material was solution treated and recrystallized after cold-rolling. The material was then aged for strengthening to the T6 condition and a Vickers hardness of 169.

The workpiece was machined into a ~1.24 cm thick, ~10 cm diameter disk for FM and subsequently machined into a 0.3 cm thick, ~7.6 cm diameter disk for LSEM in order to reduce the necessary cutting force so as not to overload cutting force measurements. Cutting force measurements were taken during LSEM using a piezoelectric dynamometer (Kistler Model No. 9257B) to use with equations 3 and 4 to calculate ΔT. For both FM and LSEM experiments, hardened M2 steel cutting tools with rake angle $\alpha = 5°$ were used.

FM experiments were conducted as a baseline for comparison with LSEM, both of which were able to produce continuous strips at varied V_o and λ. For both FM and LSEM, V_o was varied from 0.25-5.03 m/s (50-1000 ft/min). For FM, λ was not determined a priori, but was a function of depth of cut and cutting speed. λ for LSEM was pre-determined and varied from 0.75-1.5. Table I summarizes the 16 different FM and LSEM cuts attempted.

Vickers hardness values were obtained using a Leco LM247AT microhardness tester. A load of 100 g was applied for 10 indents per sample at a distance at least 2.5d from each other and the edges of the sample to avoid interactions.

The machined strips were sectioned along their centerlines and mounted in epoxy. Samples were then polished with 320 through 2000 grit SiC paper, and 0.05 µm alumina and 0.02 µm colloidal silica on napped cloths. Samples were etched with Graff and Sargent's etchant [16] to reveal grain boundaries, which were observed using optical microscopy.

Table I. Grid of machining operations. All experiments were performed at room temperature (T_o = 25 C) without preheating.

FM			
$\lambda = 2.05$ $V_o = 0.25$ m/s	$\lambda = 1.8$ $V_o = 2.03$ m/s	$\lambda = 1.45$ $V_o = 4.06$ m/s	$\lambda = 1.2$ $V_o = 5.08$ m/s
LSEM			
$\lambda = 0.75$ $V_o = 0.25$ m/s	$\lambda = 0.75$ $V_o = 2.03$ m/s	$\lambda = 0.75$ $V_o = 4.06$ m/s	$\lambda = 0.75$ $V_o = 5.08$ m/s
$\lambda = 1$ $V_o = 0.25$ m/s	$\lambda = 1$ $V_o = 2.03$ m/s	$\lambda = 1$ $V_o = 4.06$ m/s	$\lambda = 1$ $V_o = 5.08$ m/s
$\lambda = 1.5$ $V_o = 0.25$ m/s	$\lambda = 1.5$ $V_o = 2.03$ m/s	$\lambda = 1.5$ $V_o = 4.06$ m/s	$\lambda = 1.5$ $V_o = 5.08$ m/s

Results and Discussion

Secondary Shear Effects

Sticking friction increases at slower cutting speeds, to the extent that at 0.25 m/s and $\lambda = 0.75$, the workpiece did not bite when machining was attempted. This greater sticking friction indicates that the secondary shear zone represents a larger fraction of the strip thickness.

This is confirmed by optical microscopy, as the secondary shear zone decreases in size from ~$0.20t_c$ when $\lambda = 1$ to ~$0.12t_c$ when $\lambda = 1.5$ (both at $V_o = 0.25$ m/s), as illustrated in Figures 3 and 4.

In both figures the flow is right-to-left and the secondary shear zone is located at the top of the image. Flow in the secondary shear zone is dominated by workpiece-cutting tool friction and 'drags' behind material that is not as influenced by friction towards the center of the strip. The secondary shear zone is defined as the region between the edge of the strip and the point at which material flow is no longer dominated by friction, which appears as the left-most point on a flow line.

For most LSEM strips, like the condition shown in Figure 4, the microstructure consisted of both fine equiaxed and large elongated grains. This is similar to the microstructure seen in the as-received condition and indicates that some, but not all material has recrystallized.

In the slow speed, low strain condition seen in Figure 3, there are no large grains present. This condition had the highest measured hardness of all of the LSEM samples, which indicates that the material was completely recrystallized, but the grains did not grow, and the second-phase particles that provide strengthening retained a sufficiently fine distribution.

This trend indicates that for deformations with large primary shear strains, a faster cutting speed is desirable to minimize friction and the thickness of the secondary shear zone. For conventional extrusion, an increase in speed may decrease the thickness of the PCG layer.

As cutting speed increases and λ remains constant, the secondary shear zone shrinks to ~$0.05t_c$, as shown in Figure 5.

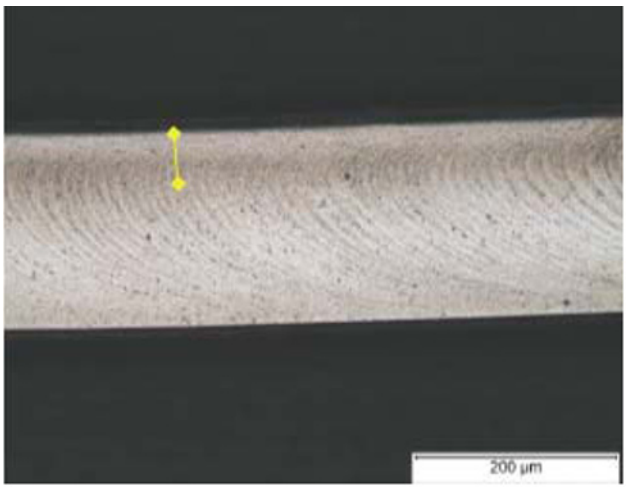

Figure 3. Optical micrograph of an AA7050 strip created by LSEM with V_o = 0.25 m/s and λ = 1. Thickness of the secondary shear zone is highlighted by the line between two points.

Figure 3. Optical micrograph of an AA7050 strip created by LSEM with V_o = 0.25 m/s and λ = 1.5. Thickness of the secondary shear zone is highlighted by the line between two points.

Figure 5. Optical micrograph of an AA7050 strip created by LSEM with V_o = 4.03 m/s and λ = 1. Thickness of the secondary shear zone is highlighted by the line between two points.

An increase in cutting speed is typically associated with an increase in adiabatic heat generation, but these results deviate somewhat from that trend. Figure 6 shows LSEM cutting temperature as a function of λ for different cutting speeds. The general trend is that cutting temperature decreases as λ increases and friction becomes less significant. This effect is more pronounced at slower cutting speeds, but cutting speed itself does not have a consistent correlation with cutting temperature.

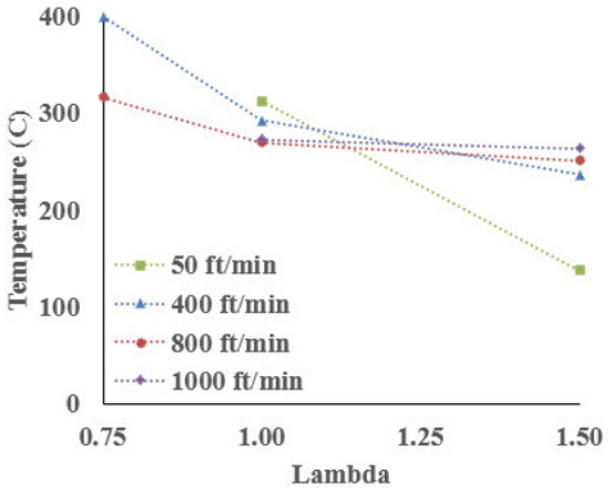

Figure 6. LSEM cutting temperature as a function of λ for different cutting speeds.

This deviation from the models may be due to the large effect of friction and secondary shear for strips this thin, which the upper-bound analysis used to develop these models ignores. The effect of friction at slow cutting speeds is illustrated again in Figure 7. The majority of machined hardness values do not deviate significantly from those of the as-received workpiece, softening slightly regardless of cutting speed. At V_o = 0.25 m/s, however, the samples hardened considerably during LSEM. This may indicate that the secondary shear zone is large enough to affect the magnitude of effective shear in the entire strip.

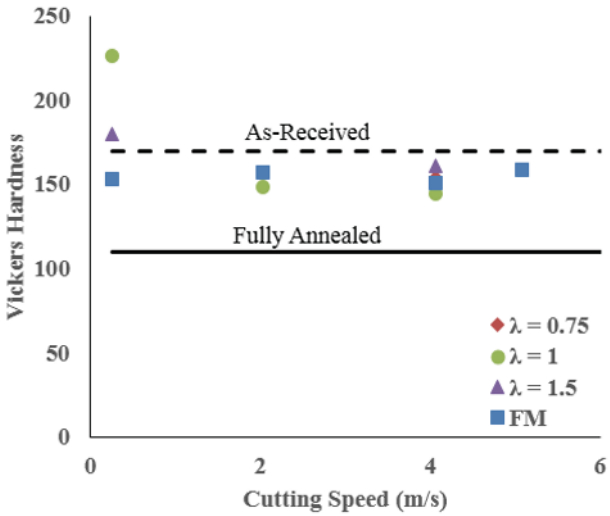

Figure 7. Vickers Hardness as a function of cutting speed for different chip thickness ratios. Dashed and solid lines at 169 and 110 represent H_v values for the as-received and annealed material, respectively.

Edge Plastic Instability

As λ decreased, machining became increasingly unstable. Plastic instability at the sides of the strips caused a side-curling defect, illustrated in Figure 8. If this defect occurred in an industrial setting, any affected product would require an additional machining step to remove the curled regions.

Figure 8. Side curls observed on an AA7050 strip produced by LSEM with V_o = 2.03 m/s and λ = 0.75.

This defect is due to plastic instability at the sides of the strips, as this material is no longer in the plane-strain condition. The material in the center of the strip was still in the plane-strain condition. This defect made the strip extremely unstable at V_o = 4.06 m/s and λ = 0.75, to the extent that LSEM was not attempted at V_o = 5.08 m/s and λ = 0.75.

Future Work

Future work will involve machining annealed material that more closely resembles material that would be used in conventional extrusion than as-received material. In addition, preheated workpieces will be used to explore effects at higher temperatures, closer to those seen during conventional extrusion.

The magnitude of strain in the secondary shear zone will also be investigated, with the intended goal of eventually incorporating it into the total effective strain. This will be useful, especially under conditions that produce different microstructures in the secondary shear zone than in the rest of the strip, which is a desired microstructure for this work in order to study PCG in conventional extrusion.

Conclusions

1. Deformation temperature decreases with increasing lambda. This effect is more pronounced at slower cutting speeds, as friction and secondary shear dominates.
2. As cutting speed decreases, secondary shear zone thickness increases. The size of this zone can become large enough to increase the hardness in the middle of the strip.
3. As chip thickness ratio decreases, cutting becomes unstable, loses its plane strain condition at the sides of the strip, and a side-curling defect has been observed.
4. Future experiments will relate experiments more closely with the conventional extrusion process in an attempt to draw more general conclusions.

Acknowledgements

Support for this research from Shandong Nanshan Aluminum Co., Beijing Nanshan Institute of Aeronautical Materials and NSF CMMI-MEP-1363524 are gratefully acknowledged. Andrew Kustas is also acknowledged for his extensive advice and help with experiments.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

LARGE STRAIN EXTRUSION MACHINING ON 6013 ALUMINUM ALLOY

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Keywords: Deformation processing, aluminum alloys, microstructure development

Abstract

Heat-treatable 6xxx series aluminum alloys are attractive to the automotive industry for car body sheet applications, particularly for their low densities. Traditionally, aluminum alloy sheets are manufactured by multi-step rolling of cast slabs and annealing. To reduce the processing steps required to make sheet, large strain extrusion machining (LSEM), a shear-based single-step deformation process, was used to create continuous strip from a 6013 aluminum alloy as a model material system. Microstructure and mechanical property of the strips, subjected to a variety of deformation parameters (strains, strain rates and temperatures), were characterized using optical microscopy and Vickers hardness test.

Introduction

Aluminum alloys have been employed in aircraft construction since the 1930s, especially age-hardenable 2xxx and 7xxx high-strength aluminum alloy. Now, in order to decrease energy consumption, the automotive industry is showing increasing interest in aluminum alloys. The solid-solution and work-hardenable 5xxx aluminum alloys have traditionally been used for automotive applications due to their relatively low cost and high formability. A problem with these alloys, however, is the appearance of the Lüders bands in forming, which tend to still show after the paint finish, so they are not generally used for exterior parts which are very critical for the surface finish [1].

Heat-treatable 6xxx aluminum alloys are particularly attractive to the automotive industry for body panels and bumpers because of their good combination of formability, surface finishing, and mechanical properties. Common automotive aluminum alloys include 6016, 6061 and 6063 alloy [2]. Alloy 6013, the subject of this paper, is an automotive aluminum alloy with higher Cu and Mg contents for higher strength due to the stronger precipitation-hardening effect.

Commercial production of aluminum alloy sheets requires multiple hot- and cold-rolling reduction steps from thick cast slabs before reaching the final gauge around 1 mm in thickness [3]. Rolling has many advantages for large-scale production, but requires a large infrastructure and is mechanically inefficient due to the incremental reductions steps. To reduce the processing steps required to produce sheet metals, large strain extrusion machining (LSEM), a highly confined shear deformation process, is under development at Purdue [4, 5]. LSEM is a single-step manufacturing technique in which machining and extrusion happen simultaneously. This paper is a first investigation of microstructure development in the production of continuous strip from 6013 aluminum alloy by LSEM. The results are discussed in comparison to the traditional multi-step rolling processes and the associated microstructure development.

Backgrounds

Commercial production of aluminum alloy sheets by rolling

The commercial production of 6xxx aluminum sheets comprises of several thermomechanical processes. The alloys are usually direct-chill cast into large ingots (slabs) with dimensions up to 0.6 m thick, 2 m wide and 6 m long. The ingots must be scalped on the rolling surfaces in order to remove the surfaces defects. In order to prepare for the following hot-rolling, the ingots are preheated to a temperature between 480-580 °C for about 48 h to reduce the short-range intercellular segregation (coring) and dissolve the soluble phases in the material [3]. This homogenization is necessary for the subsequent deformation processing [6]. The hot ingots are then transferred to the hot-rolling line, which typically consists of a reversible breakdown mill and followed by a high-speed, multi-stand tandem mill. In the breakdown mill, the ingots are reversibly hot-rolled to a transfer gauge of around 30 mm in thickness, called transfer slab. The microstructure of the transfer slab typically comprises slightly elongated recrystallized grains because of the high strain and deformation temperature. In the following multi-stand hot-rolling process, the transfer slab undergoes 3 to 4 steps with about 50% reduction in each step to a thickness around 5 mm, called hot band. The microstructure of the hot band is comprised of a highly elongated grain structure due to the large reduction.

The hot band is then coiled and cooled before it is cold-rolled to the final gauge of around 1 mm. After the cold rolling, the sheets have a highly elongated, deformed microstructure. In order to get the maximum age-hardening response and good formability, the sheet is then solution-treated through a continuous annealing line. In this line, the materials is rapidly heated to about 560 °C to dissolve the hardening phases and then quenched to obtain the supersaturated solid solution. Correspondingly, recrystallization occurs in this process, leading to a typical recrystallized microstructure consisting of fine, slightly elongated grains. After the final anneal, the sheets are leveled, possibly pre-aged for stabilization and to improve the age-hardening response [7] in the paint bake cycle, and generally pre-lubricated or pre-coated before the blanking and stamping process [3]. After the forming process, the deformed sheet will undergo the paint bake cycle, which is usually at temperatures between 160 and 200 °C for about 1 h. The final strength of the manufactured part is achieved by age-hardening through the paint bake cycle.

Large strain extrusion machining (LSEM)

In contrast to rolling, the LSEM process [8-10] is a single-step severe plastic deformation (SPD) technique for making sheet that involves simultaneously cutting and extrusion, as shown in Fig. 1. LSEM employs a sharp, wedge tool set at a preset cut depth (t_0) into the workpiece, removing material from the bulk in the form of a chip-extrudate at a constant surface velocity V_0. Apart from the cutting tool, a constraining edge is used to set the final chip

thickness (t_c). The constraining edge can be configured to achieve different degrees of constraint with different chip thickness ratio $\lambda = t_c/t_0$, resulting in high shear strains from ~1 to 5 in a single step, with large hydrostatic pressures and high temperatures from adiabatic heating in the highly localized deformation zone. Effective strains (ε) and hydrostatic pressures are controlled by the tool rake angle (α) and chip thickness ratio. The deformation during the sheet formation is confined to a very narrow zone, and can be approximated as simple shear along plane PQ (Fig. 1).

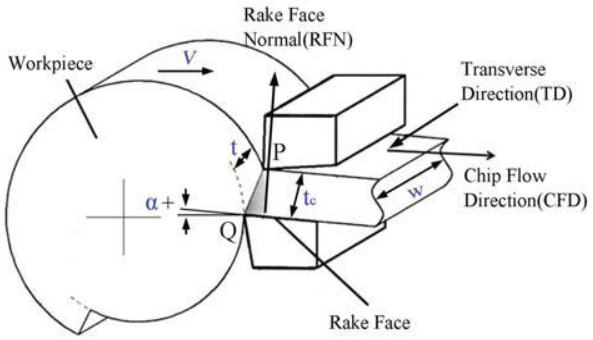

Fig. 1 Schematic of plane-strain LSEM showing sheet formation by simultaneously cutting and extrusion.

When idealized as a single shear plane, upper-bound analysis yields the following analytical expression for the imposed shear strain during deformation:

$$\gamma = \frac{\lambda}{\cos(\alpha)} + \frac{1}{\lambda \cdot \cos(\alpha)} - 2 \cdot \tan(\alpha)$$

where α and λ are the tool rake angle and chip thickness ratio, respectively [11].

Besides the fundamentally different strain path compared to rolling and the wide range of control of deformation parameters, the single-step LSEM process is less energy intensive compared to the commercial rolling process for producing sheet. Thus it is a potential alternative to traditional multi-step rolling process.

Experimental procedure

The material studied was commercial 6013-T6 aluminum alloy plate of 14 mm in thickness obtained from McMaster-Carr, Elmhurst, Illinois. The chemical composition limits according to ASME B211 are: (wt.%) 0.6-1.0 Si, 0.5 Fe, 0.6-1.1 Cu, 0.2-0.8 Mn, 0.8-1.2 Mg, 0.1 Cr, 0.25 Zn, 0.1 Ti, 0.15 others. A disk with 150 mm in diameter and 7 mm in thickness (width, w in Fig. 1) was cut from the as-received (-T6) plate for the workpiece for both free machining (FM), i.e., conventional machining without the constraint, and LSEM, in the rotary configuration shown in Fig. 1. Various strips were obtained under different machining FM and LSEM conditions. Free machining at $V_0 = 0.5$ and 4 m/s and LSEM at $V_0 = 0.5$, 2, and 3 m/s with $\lambda = 1.5$ were performed on the disk to obtain strips. The through thickness microstructure of FM and LSEM strips were investigated by optical microscopy. Microhardness of the bulk and strip specimens were measured by Vickers indentation with loads ranging from 50 g to 100 g to ensure the similar indentation size in the bulk and strip samples. And at least 10 indentations were measured to obtain an average of the hardness.

Results and Discussion

Characterization of as-received 6013-T6 aluminum alloy

The microstructure of the as-received 6013-T6 plate was not uniform through the thickness. The grains near the surface are equiaxed while they are elongated in the rolling direction in the middle (Fig. 2). The grain size near the surface is much finer than that in the middle. This structure results from the inhomogeneous deformation during the hot-rolling process. The large thickness of the starting slab makes $l/h < 0.5$, where l is the depth of deformation and h the slab half-thickness, so that the deformation and recrystallization only happens on the surface [12]. Thus, the workpiece was cut from one side of the plate thickness, and then the disk was obtained as shown in Fig. 2. This was done intentionally so that the response of both the fine-grain and coarse-grain starting structures could be studied simultaneously in the same specimens.

The precipitation states were observed to be similar between the surface and middle part based on Fig. 3c and d. Apart from the large amount of fine Mg_2Si precipitations dispersed in the grains, there are some large Mg_2Si particles (green arrows in Fig. 3c) distributed on the grain boundaries or within the grain. Large primary Al(FeMn)Si intermetallic particles (purple arrows in Fig. 3c) align parallel to the rolling direction because of the rolling processing. There is no big difference in the hardness between the surface, which is 137 HV and the middle, which is 135 HV. By optical microscopy (Fig. 2), the thickness of the coarse grain zone is around three times as large as that of the fine grain zone. It is very interesting to observe the large amount of sub-grain structure in the middle part, that is not on the surface as shown in Fig. 3a and b. The sub-grains actually align in specific orientations (Fig. 3b and d). This sub-grain structure has a significant effect on the microstructure of the strips obtained by FM and LSEM.

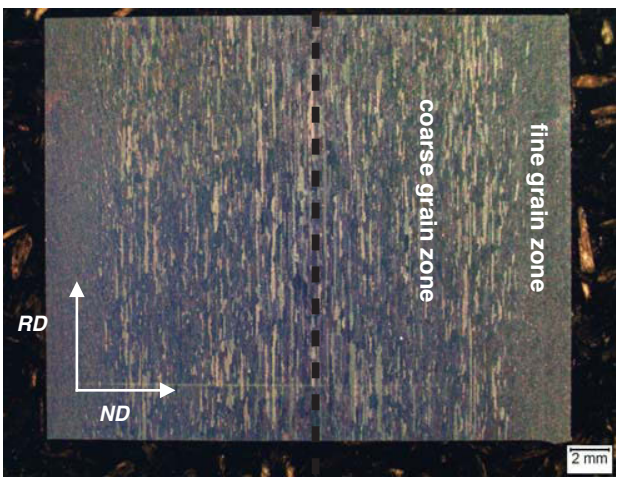

Fig. 2 Macrograph of etched cross-section of as-received 6013-T6 plate showing coarse grain core and fine grain near surfaces.

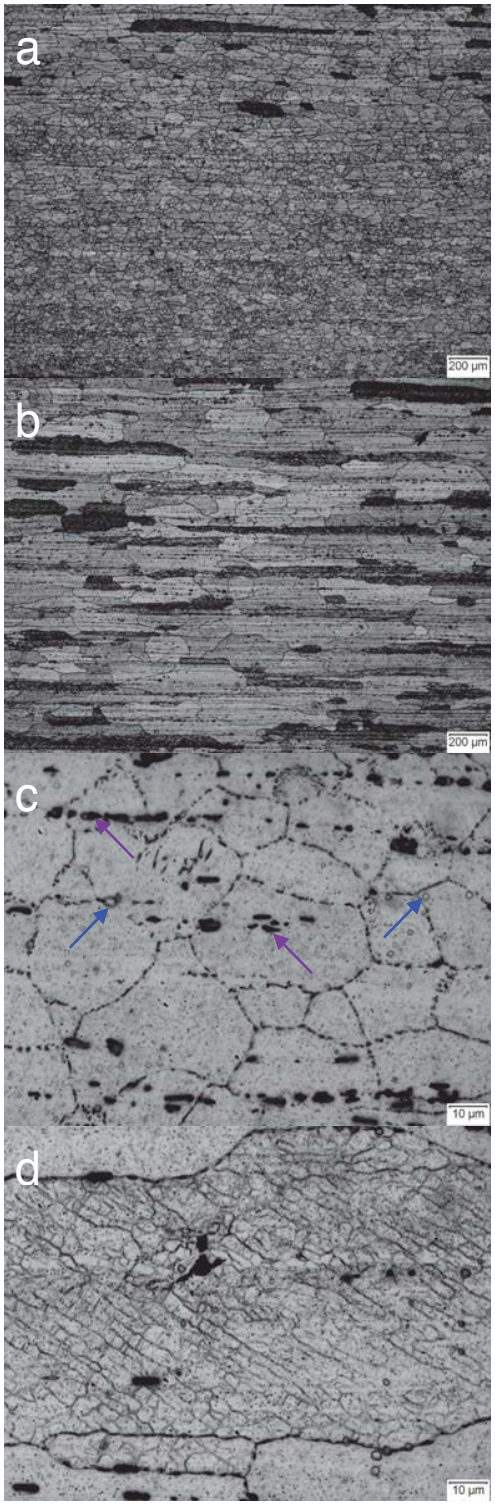

Fig. 3 Microstructure of as-received 6013-T6 plate: (a) and (c) are near the surface; (b) and (d) are from the middle.

Free machining of 6013-T6 strips

The FM experiments were performed on the 7mm thick 6013-T6 disk workpiece. The free machining parameters and hardness values are shown in Table 1. Here the hardness is the average of the coarse- and fine-grain zone.

Table 1. Free machining parameters and hardness of the strip

α	t_0 mm	t_c mm	V_0 m/s	λ	γ	ε	w mm	Hv
5º	0.25	0.89	0.5	3.5	3.62	2.1	7	176
5º	0.25	0.56	4	2.2	2.49	1.44	7	156

The free surfaces of the FM strips have a roughness characteristic of the grain size, whereas the tool (rake face) side of the strip is relatively smooth. The thickness of the chip with lower V_0 is larger than that with higher V_0 as shown in Fig. 4a and c. Built-up edge phenomenon occurs when machining at 0.5 m/s as shown in Fig. 4a and b, while at 4 m/s it does not. So it is preferable to machine at higher speed [13].

There are many second phase particles in the strip. The primary phase (black particles) preferably distribute along the grain boundaries, while the small precipitations distribute dispersively in the grains, which exerts a strong hardening effect on the strip. The average hardness of the chip at V_0=0.5 and 4 m/s are 176 and 156 HV, respectively, which are much higher than that of the as-received plate, 136 HV. The difference in hardness between the fine- and coarse-grain zone is small. At speed of 0.5 m/s, the hardnesses in the fine- (Fig. 4a) and coarse-grain zone (Fig. 4b) are 178 and 174 HV. Whereas, at 4 m/s, the hardnesses in the fine- (Fig. 4c) and coarse-grain zone (Fig. 4d) are 158 and 154 HV. So it seems that the initial grain size does not have much effect on the hardness of the strips.

The particles provide clear markers for the direction of plastic flow in the process. The primary shear zone spans across most of the chip thickness, with secondary shear against the tool rake face adding additional shear (strain) in the direction of flow near the rake face side. The primary shear angle between the (grain) flow direction and cutting tool rake face at 0.5 m/s are larger than that at 4 m/s (Fig. 4a and c), indicating a higher lever of shear strain. This is part of the reason why the hardness with low cutting speed is higher. Also, When cutting at speed of 4 m/s, the local temperature is much higher than the room temperature, which facilitate the recovery process of the strip, so the strip becomes softer, but still the temperature is not high enough to initiate the dynamic recrystallization.

The primary shear angles between the grain flow directions and the rake face at 0.5 m/s and 4 m/s are about 65º and 55º according to Fig. 4a and c. In this case, tan(65º) = 2.14 and tan(55º) = 1.42, which are very similar to the effective shear strains from free machining. So for free machining, shear angle is a reflection of the effective strain.

The microstructure in the coarse-grain zone is much more interesting, because there are so many elongated sub-grain structure within the grain. It is likely that these sub-structures originate from the sub-structures in the as-received 6013-T6 plate, as shown in Fig. 3b and d. During machining, the sub-structure with certain orientation tends to rearrange itself toward specific direction, which is parallel to the alignment of the grain flow direction. In the fine-grain zone, however, this phenomenon is barely observable.

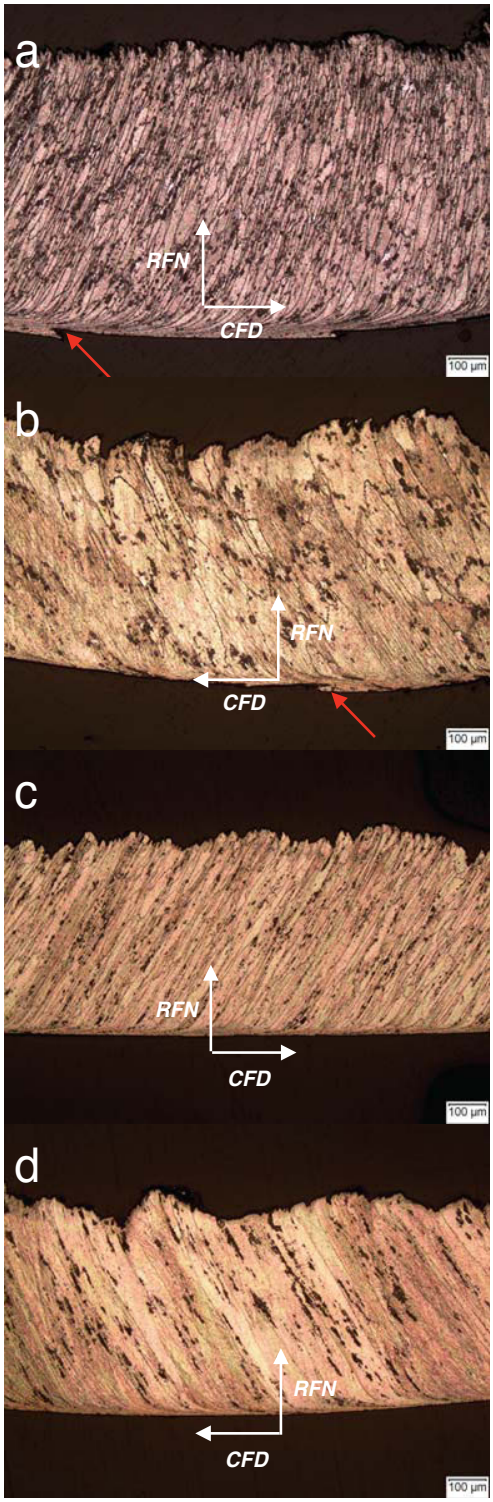

Fig. 4 Microstructure of FM 6013-T6 strip at different V_0. (a) fine-grain zone (b) coarse-grain zone at V_0 = 0.5 m/s; (c) fine-grain zone (d) coarse-grain zone at V_0 = 4 m/s.

LSEM of 6013-T6 strips

Table 2 is the machining parameters of LSEM and the hardness of the strip. The hardness is the average of the coarse and fine grain zone.

Table 2. LSEM parameters and hardness of the strip

α	t_0 mm	t_c mm	V_0 m/s	λ	ɣ	ε	w mm	Hv
5°	0.25	0.38	0.5	1.5	2	1.15	7	138
			2					115
			3					105

Fig. 5 are microstructures of LSEM strip machined at different speeds. Note that unlike the FM strips, both surfaces of the LSEM strip are smooth. Shear from the constraining tool, in the opposite sense of that from the cutting tool (rake face) side, is also a distinct difference. For FM, the flow of the material is free. Whereas for LSEM, the constraint confines the flow of the material and act as an extrusion die. Thus a smooth surface is obtained. The microstructure of each side of the strip is different due to the inhomogeneity of the through thickness microstructure (Fig. 2) of the workpiece disk, as shown in Fig. 5a, b, and 5c, d. With increasing V_0, the area of the secondary shear zone and constrained zone become smaller, while the primary shear zone becomes larger. The inclusions in the as-received plate align themselves parallel to the rolling direction and distribute alternately in the normal direction. After subsequent LSEM, however, the inclusions always arrange along the shear flow direction, whatever the machining parameters are.

Microhardnesses of the strips under different machining parameters were investigated and the results showed that with increasing of the cutting speed, the microhardness of the chip drops from 138 HV at 0.5 m/s to 115 HV at 2 m/s, and to 105 HV at 3 m/s. So recovery must happen during this process and make the chip softer and softer. With increasing cutting speed, the temperature increases correspondently (estimated from previous experiments should be ~150 °C [4]), which facilitates recovery, but still not enough for the dynamic recrystallization as can seen from the microstructure. Actually, some nucleation around large particles at 3 m/s is observed, as shown in white arrows in Fig. 5c and d. Whereas, at 0.5 m/s such phenomenon does not occur. The grain size in the coarse-grain zone after LSEM is larger than that in the fine-grain zone and the difference in the sub-grain structure is similar to the FM condition.

Summary

Free machining (FM) and Large strain extrusion machining (LSEM) experiments were performed on 6013-T6 aluminum alloy. Continuous strip can be obtained through FM and LSEM. For FM, the machining surface is smooth while the opposite is not. It is favorable to machine at high speed to eliminate the built-up edge phenomenon. For LSEM, both sides of the strip are smooth because of the constraint. For both FM and LSEM experiment, the second phase precipitations align along the grain flow direction, whatever the machining parameters are. Also the hardness of the FM and LSEM strip decreases with increasing V_0 due to the dynamic recovery process during machining.

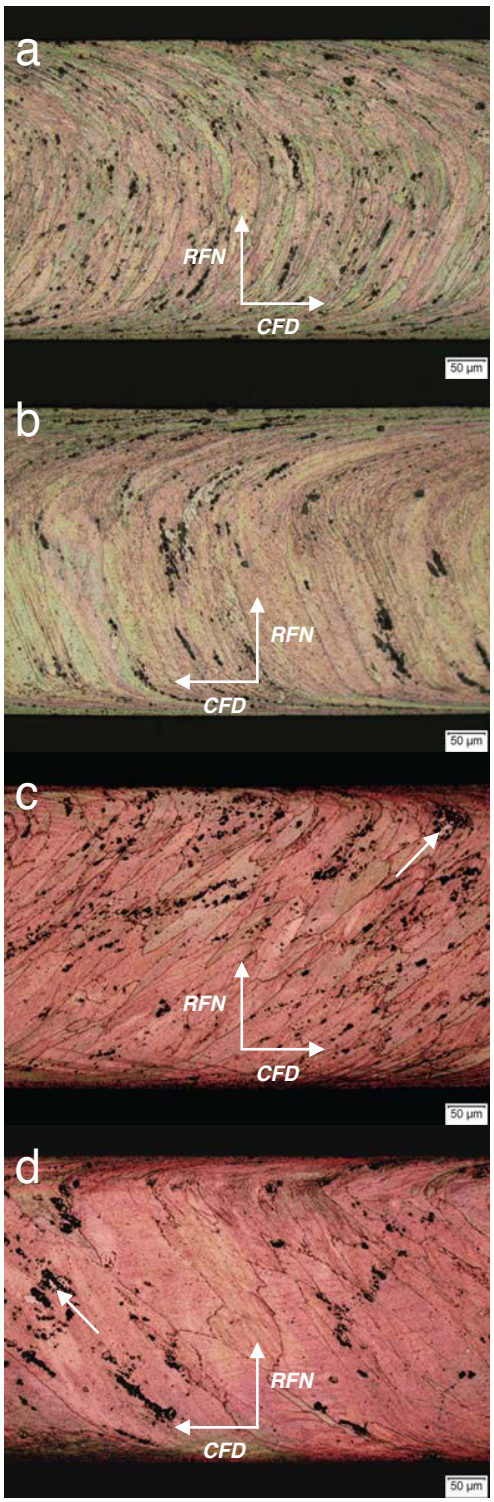

Fig. 5 Microstructure of LSEM 6013-T6 strip at different V_0 at λ = 1.5. (a) fine-grain zone (b) coarse-grain zone at V_0 = 0.5 m/s; (c) fine-grain zone (d) coarse-grain zone at V_0 = 3 m/s

The microstructure in the strip from the fine- and coarse-grain sides of the workpiece is different, especially the sub-structure. The sub-structure in the coarse-grain zone is obvious and aligns along the flow direction; while the sub-structure in fine-grain zone is barely observable. There seems to be no big difference in hardness between the coarse-and fine-grain zone of the strip, however, according to the hardness test.

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Acknowledgment

Support for this research from the National Science Foundation through Grant CMMI-1363524 is gratefully acknowledged. Also thank Ms Yiwei Sun and Dr. Dinakar Sagapuram for assistance with specimen preparation and helpful discussion.

Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF EXTRUSION RATIO ON MECHANICAL PROPERTIES OF Al-20Si PRODUCED BY SPRAY DEPOSITION AND CONTINUOUS EXTRUSION TECHNIQUE

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Keywords: Al-Si alloy, extrusion ratio, Spray deposition and continuous extrusion, mechanical Properties

Abstract

Spray deposition and continuous extrusion (SD-CE) forming technique is a novel technology that combines spray forming and continuous extrusion. Al-20Si alloy rods with a diameter of 8 mm have been fabricated using SD-CE with different extrusion ratios. The microstructure and mechanical properties of these alloy rods were investigated. Al-20Si alloy can be refined effectively by SD-CE, resulting in a great improvement of mechanical properties. The strength and hardness of these alloys gradually increase with increasing the extrusion ratio. However, the elongation rapidly rises with the extrusion ratio, and then gradually drops as the extrusion ratio reach critical level.

Introduction

Hypereutectic aluminum silicon alloys are attractive candidate materials for automobile[1], aerospace and electronics applications due to their excellent properties, such as low weight, high stiffness, good wear resistance and low coefficient of thermal expansion. These properties can be improved effectively by further increasing the silicon content of the alloys[2]. However, it is difficult to produce such alloys with high silicon content by conventional casting process since coarser primary silicon will result in reduction of mechanical properties and deterioration of workability[3]. Effective primary silicon refinement can be obtained after improving the solidification rate[4] and with different extrusion ratios[8]. Many investigations show that the spray formed products are characterized by fine, uniform and macro-segregation free microstructures due to rapid solidification of gas atomized molten droplets during flight and deposition[5, 6]. The ability of the process to produce near-net-shape components employing a minimum number of processing steps is highly expected. In the present study, a novel technology named as spray deposition and continuous extrusion (SD-CE) forming technique[7] was used to prepare Al-20Si alloy. The microstructure and mechanical properties of the alloys with different extrusion ratio were investigated.

The influence of extrusion ratio on the mechanical property of aluminum alloy in the literature [9-10] has been clearly discussed. In [11, 12] study, experimental extrusion ratio is 40, the results show that the material during the extrusion process occurs plastic deformation about 70-80%. Aluminum alloy under the lower extrusion ratio, its organization is similar to the casting aluminum alloy (coarse grains), he organization has poor mechanical properties. Therefore, when aluminum alloy extruded under the condition of the extrusion ratio less than 10, compared with the literature[13], its mechanical performance and physical performance difference is bigger. In addition, some studies have shown that high extrusion ratio (⩾39) compared to low extrusion ratio (⩽24) ,aluminum alloy refining is better[13]. Milenin et al.[14] test using extrusion ratio of 5, for the extruded sample analysis showed that increased pressure on the extrusion ratio is generated was effective, this is reported in the study used the smallest extrusion ratio.

Experimental details

Materials preparation

Rods with a diameter of 8mm with a nominal composition (wt. %) of Al-20Si were fabricated by SD-CE, which is a relatively novel technology that combines elements of spray forming and continuous extrusion. In SD-CE, as shown in Fig.1, molten metal is atomized with a high velocity inert gas, and then the resultant droplets are quenched in flight, and the spray deposits onto the groove of extrusion wheel. In-flight convection heat transfer and conduction heat transfer at the groove of extrusion wheel rapidly removes the latent heat. The deposited material on the groove is drawn by the frictional loads between the material and the groove. The groove is closed at the other end by an abutment so that the material can emerge only from the die. Processing parameters are listed in Table I.

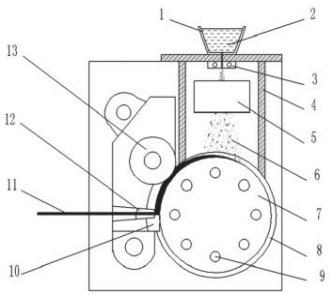

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Fig.1 Schematic diagram of SD-CE
1--- Hopper 2--- Molten metal 3--- Atomizer 4--- Spray chamber 5--- Flow controller 6--- Droplets 7--- Extrusion wheel 8--- Groove of extrusion wheel 9---The cooling channels 10--- Abutment 11---Extrusion 12---- Die 13--- Backup roller

Different extrusion ratio test

Experimental parameters of different extrusion ratio have shown in Table I.

Table I Parameters of SD-CE Al-20Si alloy with different extrusion ratio

Item	Parameter Description	Value
material	Alloy and quality	Al-20Si about 1400g
Smelting	Melting furnace type	Box-type resistance furnace
	Heating power	(10～20)A×220V
	Smelting time	120min
	Smelting temperature	900℃
	Holding time	50min
Spray deposition	Diversion pipe	cross-section of ports 3×7mm^2
	Diversion pipe heating temperature	900℃
	Atomizing gas pressure	0.25MPa
	Flow controller spindle speed	1100 r · min^{-1}
	Motor rotation	clockwise
Continuous Extrusion	Spray deposition distance	425mm
	Speed	10 r · min^{-1}
	Rotation	clockwise
	Extrusion ratio	10、13、17、20
	Radial / tangential extrusion	Radial
	Compression cylinder pressing force	40MPa

Results and discussion

Al-20Si Hypereutectic alloy organization contrast

Figure 2 shows the microstructure characteristic of SD-CE Al-20Si alloys with different extrusion ratio. the SD-CE Al-20Si alloys are characterized by fine and equiaxed primary silicon crystals, and the silicon phase is well distributed in the matrix of Al phase. The microstructural examination indicates that the silicon phase is at the size 3.2~5.6 μm. Moreover, with the increasing extrusion ratio, the size and the shape of primary silicon become finer and more spherical.

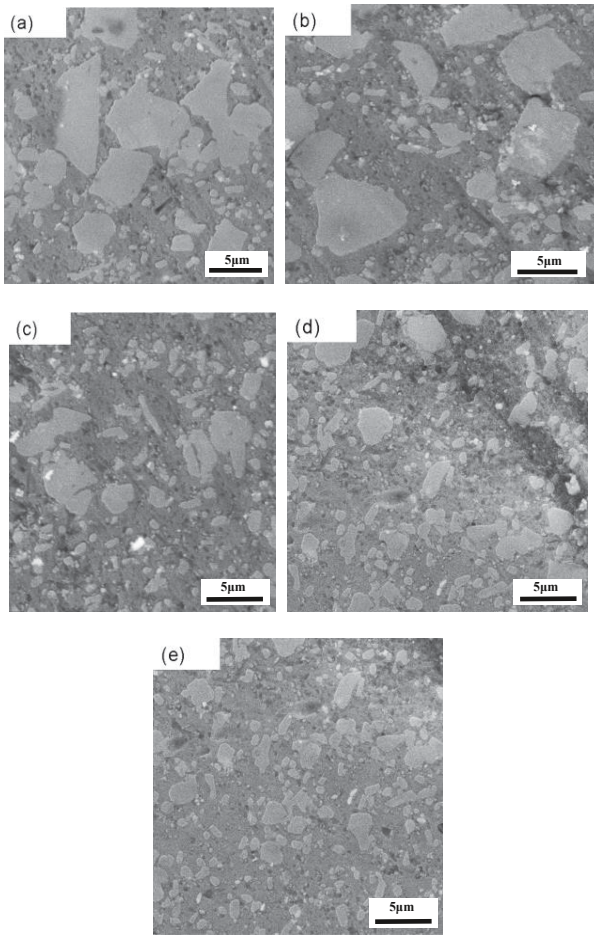

Fig.2 The microstructure of Al-20Si alloys with different extrusion ratio (a) extrusion ratio is 7:1 (b) extrusion ratio is 10:1 (c) extrusion ratio is 13:1 (d) extrusion ratio is 17:1 (e) extrusion ratio is 20:1

As mentioned above, SD-CE benefits of spray forming and continuous extrusion. Using the Image Pro - Plus image analysis system to analyze microstructure of figure 2, the relevant results are shown in table II: the equivalent diameter of primary silicon decreases with extrusion ratio increase, and the equivalent diameter of primary silicon can be up to 3.2μm when the extrusion ratio reaches 20:1, at the same time the roundness is better and the mean free path is greater between the primary silicon with the increase of extrusion ratios, which indicates distribution of crystalline silicon is more uniform.

Table II The characterization parameters of primary silicon in Al-20Si alloy with different extrusion Ratio

Extrusion ratio	7:1	10:1	13:1	17:1	20:1
Equivalent diameter D_{eq}	5.6	5.2	4.8	3.6	3.2
Roundness R	1.3	1.4	1.3	1.2	1.2
Mean free path d_{MFP}	8.5	9.2	9.6	10.3	10.6

Mechanical properties analysis

Al-20Si alloy test of different extrusion ratios are completed under tensile strain rate of 0.5mm / min condition, As can be seen from Figure 3, Al-20Si alloy of SD-CE with the increase of extrusion ratio, the ultimate tensile strength gradually tend to rise, when the extrusion ratio is 20: 1, the ultimate tensile strength of product reached a maximum value of 213.7 MPa, compared with as-cast of 85.6 MPa, Al-20Si alloy of SD-CE increased 149.6%. the elongation is first increases and then decreases, elongation reaches maximum (15.8%) when extrusion ratio is 10:1, compared with the as-cast (2.9%)increased above 4 times. At the initial stage, with the increasing extrusion ratio, the size of the primary silicon reduces, and the density of alloys gradually increases, both are beneficial to the improvement of plastic. While, the decreasing of elongation at the final stage should be ascribed to the large deformation and increase of dislocation density.

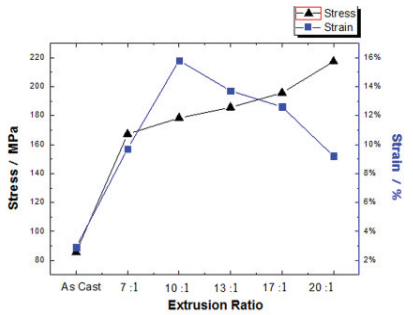

Fig.3 The stress -strain comparisons of Al-20Si alloys with different extrusion ratio

As can be seen from Figure 4, the fracture appearance of as-cast Al-20Si alloy is characterized by brittle cleavage, there are a lot of irregular primary silicon phase, tensile stress acts on the brittleness of Si phase boundary is easy to produce the intergranular fracture, or across the grain produced transgranular fracture, which will cause serious influence on the strength and plastic of the alloy. However, the samples produced through SD-CE, the fracture surface has a lot of fine equiaxed flexible nest, which can effectively hinder crack propagation and help to improve the strength and plastic of the alloy.

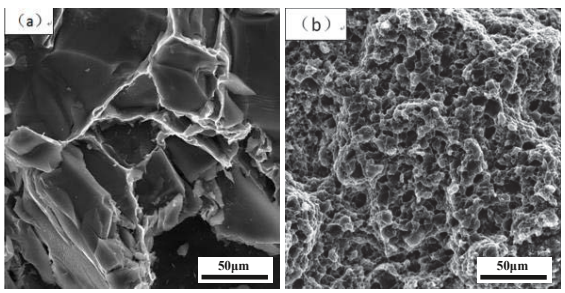

Fig.4 The fracture morphology of Al-20Si alloy (a) Brittle cleavage profile of as-cast Al-20Si alloy (b) Toughness fracture of SD-CE Al-20Si

Conclusions

Spray deposition with following continous extrusion (SD-CE) is a new tube/rod/section/wire casting technology that promise for processing a wide variety of Al alloys. Al-20Si alloy rods with a diameter of 8 mm have been fabricated by SD-CE at different extrusion ratios. Microstructure and wear resistance of Al-20 Si alloy have been investigated. The results show that the Al-20Si alloy can be effectively refined by SD-CE. Compared with as-cast ingot, the primary silicon size of aluminum-silicon alloy is about 3.6 ~ 5.6μm, structure significantly refined. With the increase of extrusion radio, the limit tensile strength of Al-20Si alloy increases significantly due to increase of density of alloy, but the elongation reached maximum value when the extrusion ratio is 10, and then continue to decrease with the extrusion ratio increases due to large deformation and increase of dislocation density.

Acknowledgements

The Authors acknowledge the financial support from National Natural Science Foundation of China (project no.l50874055), Natural Science Foundation of Province Yunnan(project no. 2007E0013Z) and Education Department Foundation of Province Yunnan(project no. 2013Y316).

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Light Metals 2016

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Precipitation Behavior

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Ramasis Goswami
Naval Research Laboratory
Washington, DC, USA

Light Metals 2016
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THE INFLUENCE OF LOW TEMPERATURE CLUSTERING ON STRENGTHENING PRECIPITATION IN Al-Mg-Si ALLOYS

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Keywords: Aluminum, 6000 series, natural aging, artificial aging, clustering, precipitation, strengthening, hardening, composition

Abstract

Heat-treatable 6000 series aluminum alloys are the most commonly extruded materials in the world. The precipitation process in these alloys is both complex and well characterized. The earliest clustering stage has been shown to have a large effect on subsequent strengthening precipitation, however little is known about the influence of clustering as a function of composition and processing parameters. The current work examines this influence considering the factors of relative and absolute magnesium and silicon content, and the extent of natural aging. Billets were cast and extruded prior to heat-treatment, and the hardening response was evaluated with hardness, conductivity, and transmission electron microscopy (TEM). This work advances the current understanding of Al-Mg-Si precipitation by correlating the kinetics of age hardening to composition and processing, and may lead to further optimization of 6000 series alloy strength and toughness.

Introduction

Aluminum alloys of the 6000 series are of great technological interest, especially in recent years due to their growing utilization in the automotive industry. These heat-treatable extrusion alloys have been well studied over the years, and their precipitation sequence has been found to be quite complex. The phase progression, in the simplest case, is shown below (where supersaturated solid solution is abbreviated SSSS).

$$SSSS \rightarrow Mg\ \&\ Si\ clusters \rightarrow MgSi\ co\text{-}clusters \rightarrow GP\ zones \rightarrow \beta'' \rightarrow \beta' \rightarrow \beta$$

Numerous additional metastable phases have been identified, occurring both before and after the most potent strengthener, β'' [1]–[6]. The precipitation sequence begins immediately during and after quenching from a solution treatment with the formation Mg and Si clusters, followed by Mg-Si co-clusters [2], [7], [8]. Due to non-equilibrium distributions of quenched-in vacancies, this process is active even at low (e.g. ambient) temperatures [9], [10]. The process of clustering involves the grouping of solute atoms, forming finely distributed inhomogeneities in the solid solution.

The extent to which clusters form during natural aging (NA) has a large impact on precipitation of strengthening phases, particularly β'', during subsequent artificial aging (AA). Cluster size and its influence on strengthening precipitation has been shown to be a function of temperature and time [11], [12]. For the current study, it is assumed that cluster composition is affected by bulk composition. For most high strength 6000 series alloys, room temperature storage has a detrimental effect on AA response, usually by inducing a decrease in density and/or an increase in size of strengthening precipitates [1], [13], [14]. In some alloys, this negative influence has been shown to be reduced with a prior pre-aging treatment at an intermediate temperature, in which a different "type" of clusters are thought to form [11], [15]. For low strength alloys, natural aging can have the opposite effect, although there is debate regarding the key compositional variable responsible for this difference.

The current work investigates the influence of the duration of natural aging (or the extent of cluster formation) on strengthening behavior as a function of composition variables. The results and discussion will qualitatively show that this influence is affected by both silicon and magnesium content, however their effects are not identical due to differences in solute properties.

Experimental

Six experimental billets 89 mm (3.5'') in diameter were cast in a permanent mold. The experimental compositions span 6000 series limits in terms of Mg/Si ratio and Mg+Si content (Table I). The alloy naming convection used throughout this text consists of two numbers, the intended weight ratio of magnesium to silicon, and the intended total weight percent of magnesium and silicon. Actual values are also given in Table I. The terms Mg-rich and Si-rich will also be used, and they indicate an excess of solute with respect to the metastable strengthening phase β'' (Mg/Si = 0.73 by weight). A moderate amount of iron was included and held relatively constant to maintain consistency with production alloys and to control cast and extruded (recrystallized) grain structure. Trace amounts of manganese were included to stabilize α Al-(Fe,Mn)-Si during homogenization, as it's morphology is preferred for extrusion. No inoculant was used during casting to avoid possible interactions with the clustering process.

Based on cast grain size and Thermocalc/DICTRA simulations, a homogenization treatment of 848K (575°C) for 144 hours was performed. Homogenized billets were then extruded into 22.2 mm (7/8'') rod using a laboratory scale 550 ton direct extrusion press. Extruded rod was then cut into 12.7 mm (½'') cylinders to be heat treated and tested. Prior to the aging heat treatment, a solutionization treatment based on DICTRA simulations was performed to dissolve any post-extrusion secondary precipitation and to equalize grain structure throughout the samples; extruded cylinders were put into a box furnace at 838K (565°C) for 90 minutes, reaching a final temperature of 831K (558°C) prior to quenching in water.

Table I. Experimental alloy nomenclature and compositions.

Alloy	Mg	Si	Fe	Mn	Mg/Si	Mg+Si	remarks
2.5 0.7	0.47	0.20	0.20	0.05	2.35	0.67	soft, Mg-rich
2.5 1.4	1.01	0.41	0.21	0.05	2.46	1.42	hard, Mg-rich
0.73 0.7	0.31	0.39	0.21	0.05	0.79	0.70	soft, balanced for β'' (Mg_5Si_6)
0.73 1.4	0.57	0.84	0.28	0.07	0.68	1.41	hard, balanced for β'' (Mg_5Si_6)
0.27 0.7	0.15	0.58	0.19	0.19	0.26	0.73	soft, Si-rich
0.27 1.4	0.30	1.18	0.23	0.23	0.25	1.48	hard, Si-rich
All data in wt%, tested during casting with an Optical Emission Spectrometer							

For each composition, two aging treatments were performed with different amounts of prior natural aging. Twenty samples for each composition were naturally aged for 30 minutes prior to artificial aging at 448±2K (175°C) in a convection furnace (which roughly simulates the traditional T6 treatment); An additional set of twenty samples were naturally aged for 4 hours prior to an identical and simultaneous artificial aging treatment. A thick aluminum baseplate was pre-heated in the convection furnace to ensure temperature uniformity, and due to this a relatively fast heating rate (shown in Appendix 1) was achieved.

To establish hardening behavior samples were quenched in water with equal spacing in the logarithm of time (beginning at one hour to allow temperatures to fully equilibrate). Samples were then stored in an ultra-cold freezer with a maximum temperature of 193K (-80°C) to severely limit, or eliminate further precipitation prior to hardness and conductivity testing. After polishing to 6 µm, Vickers macro hardness testing was done with a 20 kg load and 15 s dwell time. Electrical conductivity measurements were made using an Eddy current conductivity probe. For each sample, a minimum of 6 hardness and 10 conductivity measurements were made. For the first five aging time points (in which remaining supersaturation was likely), samples were stored in a dry ice-isopropyl alcohol solution between during sample preparation. Based on the hardness curves, select samples were imaged in a TEM at 200keV to qualitatively assess strengthening precipitate type and distribution.

Results – Hardness

The general shape and magnitude of the hardness curves for each composition are similar for the two aging conditions (Figs. 1 & 2), however closer inspection reveals several key differences (Figs. 3-5, 7-9).

Alloy 0.27 1.4

The high strength, Si-rich alloy shows a clear delay in initial hardening when samples undergo 4 hours of initial natural aging vs. 0.5 hr of natural aging. Note all AA hardness curves begin one hour after samples were initially put into the furnace. Although the process of natural aging is known to increase hardness, it is assumed that all samples (for each composition) began artificial aging at relatively similar hardness values (±5Hv), as the process of clustering does not significantly harden the material in these time scales [16].

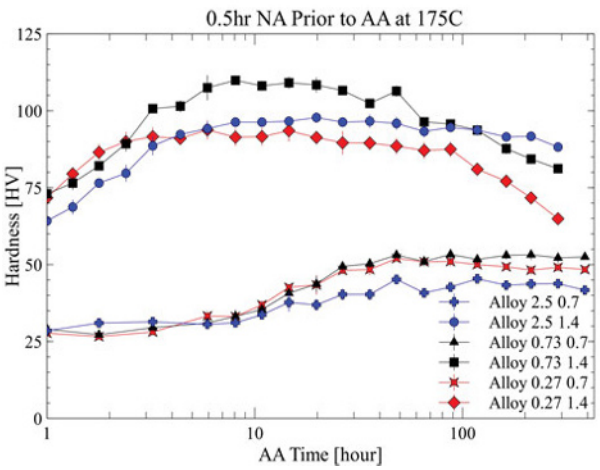


Figure 1. AA hardness curves for all alloys with 0.5 hr prior NA.

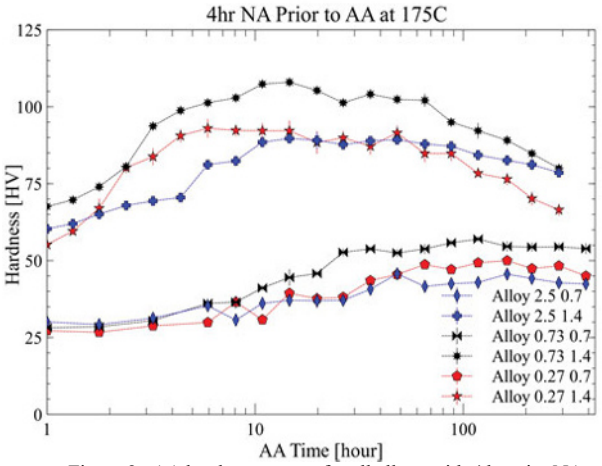


Figure 2. AA hardness curves for all alloys with 4 hr prior NA.

This delay in hardening kinetics is well known for Si-rich alloys [14]. However as can be seen in Figure 3, the hardness is fully recovered at around 4.4 hrs AA, and the two curves remain nearly constant for the remainder of AA. TEM imaging of select samples was performed and can be seen in Figure 6. At approximately the peak aged condition of 11 hrs AA, both specimens had a dense network of needle-like precipitates (tens of nanometers long in 2D projection), consistent with the expected presence of β''. The composition of this alloy contains silicon in excess of that needed for β'', and accordingly, nearly pure silicon plates of similar size were also identified. Qualitatively, β'' needles appear to be longer in the specimen that was naturally aged for 4 hrs. This effect of prior NA, as well as a decrease in number density has been previously reported [13], [17]; however due to inherent difficulties (differences in foil thickness, precipitate overlap, etc.), no density measurements were attempted in the current work.

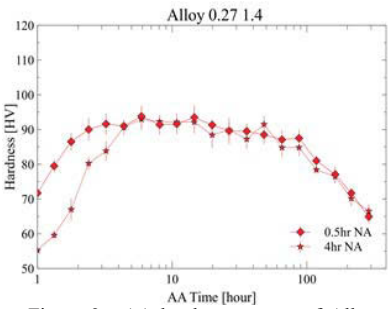


Figure 3. AA hardness curves of Alloy 0.27 1.4 for both NA conditions.

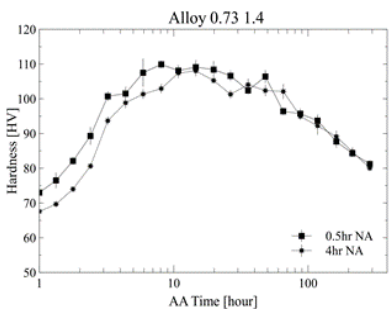


Figure 4. AA hardness curves of Alloy 0.73 1.4 for both NA conditions.

Unlike the alloys artificially peak aged, the nanostructure after 1.8 hrs of AA shows a remarkable difference between specimens of different prior natural aging. The sample with 0.5 hr NA contained a dense network of fine β'' needles, however very few needles were present in the 4hr NA condition. Instead, several-nanometer sized spherical phases were found. These phases were silicon rich and believed to be an earlier stage in the precipitation sequence, such as GP-zones or another pre-β'' phase. This observation agrees with the delayed hardening response seen in this

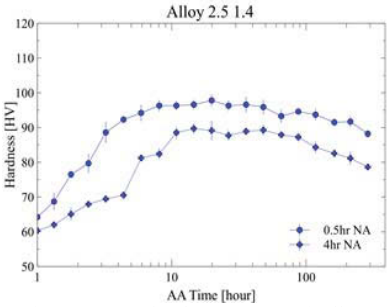


Figure 5. AA hardness curves of Alloy 2.5 1.4 for both NA conditions.

alloy. It should also be mentioned that an assumption is being made, namely that the precipitation sequence in alloys of the same composition follows the same general route, regardless of the NA condition. Under this assumption, the alloy with 0.5 hr NA made the β'' transition before 1.8 hr AA, and the increased initial (1hr AA) hardness is presumably due to presence of GP-zones or pre-β'' phases.

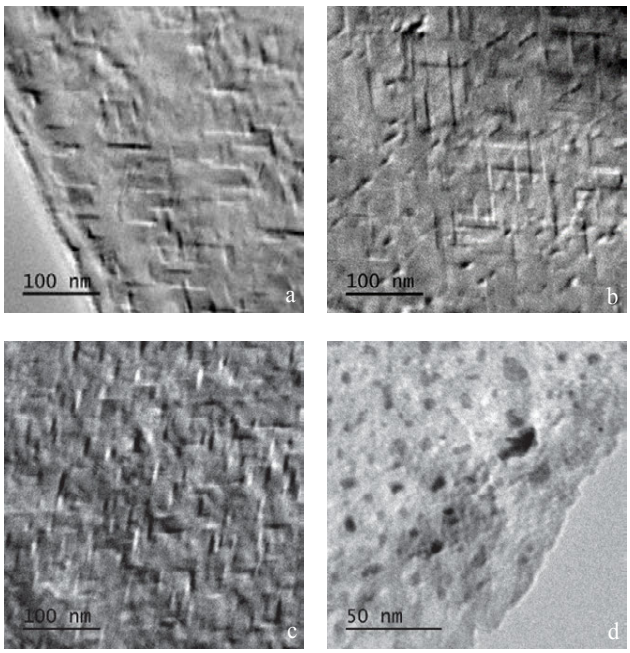


Figure 6. TEM images of Alloy 0.27 1.4 with 0.5 hr NA and 10.8 hrs AA (a); 4 hrs NA and 10.8 hrs AA (b); 0.5hr NA and 1.8 hrs AA (c); 4 hrs NA and 1.8hr AA (d).

Alloy 2.5 1.4

The difference in hardness curves (due to natural aging and increased cluster formation) of the strong, Mg-rich alloy is distinctly different from that of the strong, Si-rich alloy. After the first hour of AA, the sample with 4 hrs NA is softer as expected, and hardening kinetics are once again slower than its 0.5 hr NA counterpart. Somewhat similar to Alloy 0.27 1.4, this alloy displays a delay in significant hardening response, which begins around 4.38hrs AA. However, unlike either of the other high strength alloys investigated, the hardness is never fully recovered. After peak hardness is achieved, the difference in hardness curves (around 5-10 HV) remains relatively constant for the duration of artificial aging.

Alloy 0.73 1.4

Not surprisingly, the high strength alloy that is compositionally balanced for β'' exhibits superior hardening, as the driving force and (metastable) equilibrium volume fraction of the most potent strengthening precipitate is optimized. Interestingly, however, the influence of extended NA appears to have characteristics similar to both high strength Mg-rich and Si-rich alloys. In other words, the negative effect of NA, although not as severe, is present for longer AA times than the Si-rich alloy, but it is eventually fully recovered (around 10.8 hrs AA or 35.8 hrs AA).

X 0.7 Alloys

The hardening response of the soft alloys (with Mg + Si = 0.7 wt%) is much less significant, however the influence of natural aging, as a function of Mg/Si, appears to be substantially different than alloys with higher solute content. After one hour of AA, alloys of each composition had nearly identical hardness values, regardless of the natural aging condition. Alloys of similar compositions have been shown to have decreased natural age hardening [1], [16], however the clustering process is thought to still be active.

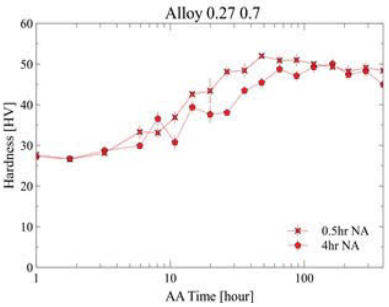


Figure 7. AA hardness curves of Alloy 0.27 0.7 for both NA conditions.

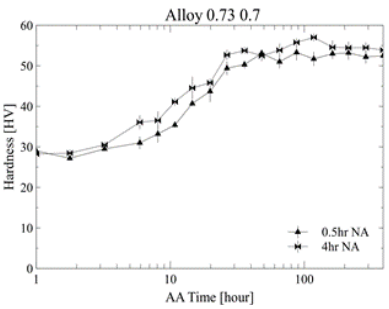


Figure 8. AA hardness curves of Alloy 0.73 0.7 for both NA conditions.

Natural aging had a slight positive effect on only the compositionally balanced Alloy 0.73 0.7. The influence of prolonged NA had minimal effect for the soft, Mg-rich Alloy 2.5 0.7, whereas the largest decrease in hardness (due to NA) was seen for the soft, Si-rich Alloy 0.27 0.7. This alloy was the only one to behave similarly to its high-strength counterpart (i.e. Alloy 0.27 1.4), as its two hardness curves converge with extended time at the artificial aging temperature. However in this case, the convergence does not occur until after the onset of over-aging (around 118 hrs AA).

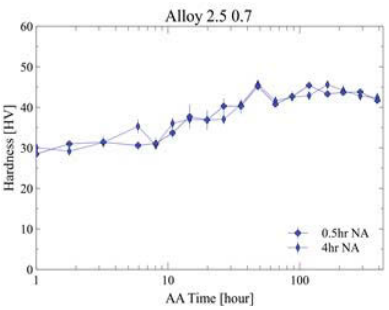


Figure 9. AA hardness curves of Alloy 2.5 0.7 for both NA conditions.

Results – Conductivity

Classic precipitation sequences (like that of β'') generally increase conductivity by depleting the matrix of solute, allowing easier flow of electrons through the material. In contrast, since the spacing of solute clusters is on the order of the distance between electronic scattering centers, cluster formation is known to decrease conductivity [10]. Figures 10 and 11 show the full conductivity

curves for each aging treatment. Like hardness, the trends in conductivity are similar for both NA conditions.

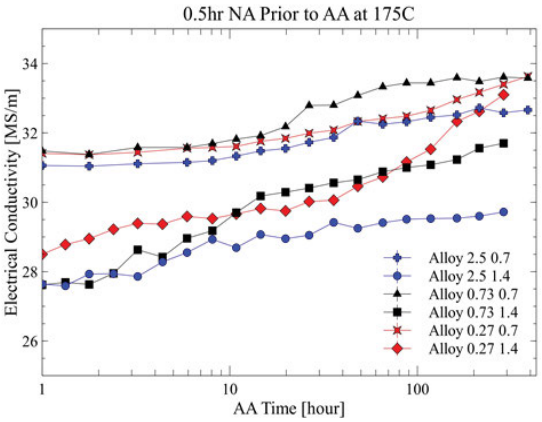


Figure 10. AA conductivity curves for all alloys with 0.5 hr prior NA.

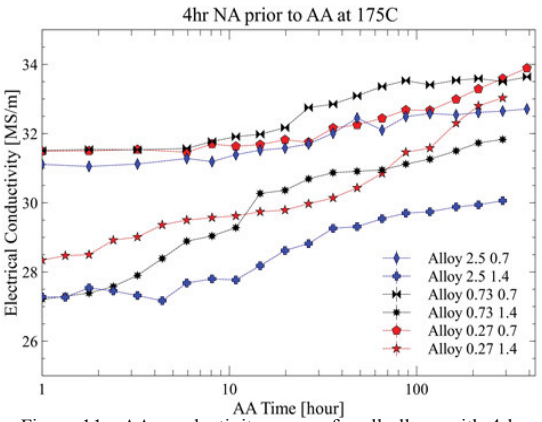


Figure 11. AA conductivity curves for all alloys with 4 hrs prior NA.

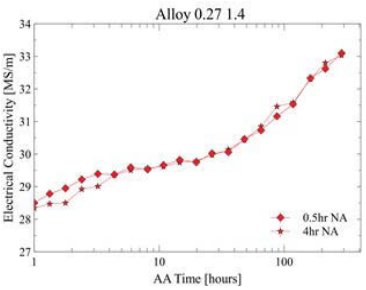


Figure 12. AA conductivity curves of Alloy 0.27 1.4 for both NA conditions.

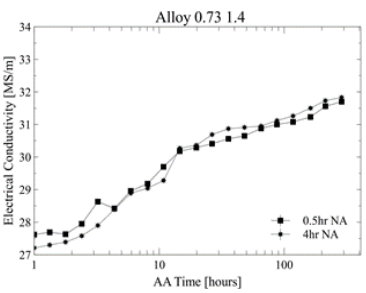


Figure 13. AA conductivity curves of Alloy 0.73 1.4 for both NA conditions.

Compositions with low solute content (X 0.7 alloys) show very little difference in conductivity with varying NA condition, so individual plots were not included in this text. For X 1.4 alloys, extended NA produced significantly different hardness curves throughout the course of AA. The relative impact of NA for each X 1.4 chemistry was consistent with the trends in hardness, with the Mg-rich alloy showing the largest difference (Figs. 12-14). For these alloys the conductivity of samples with 4 hrs prior NA was lower in the early stages of AA, followed by a recovery. For all compositions investigated, the conductivity of samples with 4 hrs NA was as high or higher in the over aged condition.

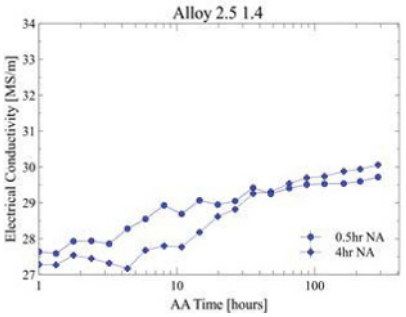


Figure 14. AA conductivity curves of Alloy 2.5 1.4 for both NA conditions.

Discussion

Little is conclusively known regarding cluster formation, although several theories have been developed. Zurob et.al. theorized that clusters form solely due to solute interatomic energies, and that their formation mechanism is based on a version of the vacancy pump model [18], [19]. This version incorporates the probability that the interaction energy between a cluster of solutes and a vacancy is likely larger than that of a single solute and a vacancy. Using first principles calculations, it has been shown that both magnesium and silicon have attractive solute-vacancy binding energies in aluminum (Si being slightly more attractive than Mg) [20]. These attractive interaction energies lead to the rate limiting step in cluster growth being proportional to the probability that a vacancy can escape a growing cluster, in order to aid in transport of the next solute [18]. As a consequence, more extensive cluster formation would reduce the number of vacancies in the matrix available to assist diffusion during precipitation strengthening (i.e. during artificial aging).

One of the key features of Zurob's clustering model is the sole driving force of interatomic energies. Under this model, solute supersaturation is not needed for cluster formation [18]; this is an idea that is not often discussed in related literature. Because of this, cluster dissolution may not occur in the classical sense, instead solute clusters may dissipate as solutes diffuse to growing precipitates.

Alloy 0.27 1.4

The high strength silicon rich alloy clearly displays slower initial hardening kinetics with prolonged NA, and with prolonged AA, an eventual recovery of strength. This may be explained using the concepts above. With four hours of natural aging, silicon rich cluster formation is enhanced. The presence of large, Si-rich clusters then increases the probability of free vacancy depletion in the matrix due to the strong attractive Si-vacancy binding energy (and presumably stronger Si cluster-vacancy binding energy). In the early stages of artificial aging, this vacancy depletion severely limits the solute diffusion needed for precipitation of β''. Additionally, the presence of Si-rich clusters may reduce the effective matrix supersaturation and thus the driving force for precipitate nucleation. The combination of these effects would decrease the initial hardening kinetics (Figure 3).

The solubility of silicon in aluminum at 175°C is limited, and as such, the silicon rich clusters formed during natural aging are

supersaturated with respect to the precipitate phases. As the duration of artificial aging is increased, the silicon rich clusters may dissipate as this solute diffuses to the growing strengthening phases. The cluster dissipation would then liberate trapped vacancies, allowing them back into the matrix to aid in diffusion and growth of β''. This concept is consistent with both the recovery of hardness (Figure 3), and the qualitatively longer β'' needles (Figure 6b).

Alloy 2.5 1.4

In contrast to Alloy 0.27 1.4, further natural aging in Alloy 2.5 1.4 produces both a decrease in initial artificial hardening response and a decrease in hardness throughout the course of AA. Magnesium has also been calculated to have an attractive interaction energy with vacancies in aluminum, although to a slightly lesser extent than that of silicon [20]. The key difference, however, is believed to be the high solubility of magnesium in aluminum. Figure 15 shows a Thermocalc simulation of the equilibrium weight fraction of solute (Mg+Si) in the matrix as a function of Mg/Si ratio in the "X 1.4" alloy system. It should be noted that the phases used for the calculation (indicated on the plot) are common precipitates in the Al-Mg-Si system, and were the phases identified in a previous work where experimental alloys of similar chemistry were analyzed [5]. In this system, with a Mg/Si ratio beyond 1.6, the total amount of solute remaining in the matrix after full decomposition, begins to sharply increase. It is worth mentioning that the same phases may not be present in the alloys used for the current work; further suspension or addition of simulated phases alters the plot, but the positive slope is consistent. For example, if only β'' and pure Si is allowed to precipitate, the simulation is plotted in Figure 16. During the course of aging, these systems slowly transition towards these (metastable) equilibrium states. The high strength Mg-rich alloy has a Mg/Si ratio of around 2.5, and as a consequence, if prior solute clustering is allowed to occur (i.e. during natural aging), much of this Mg-rich equilibrium solute content will have no driving force to dissipate. These clusters may then be considered as stable during the process of artificial aging and don't contribute to precipitation hardening.

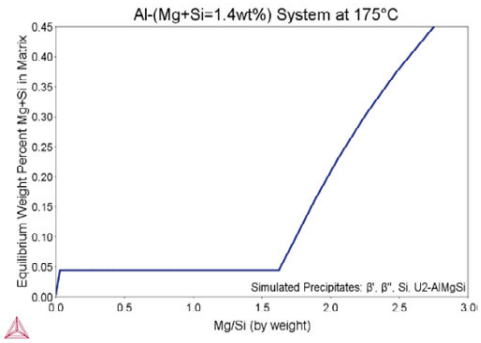


Figure 15. Plotted Thermocalc simulation in X 1.4 alloy system at the AA temperature (calculated with phases FCC matrix, β', β'', Si, U2-AlMgSi).

The implications of this concept can be assessed with respect to differences in observed hardness curves as a function of natural aging (Figure 5). In the early stages of natural aging, larger Mg-rich clusters deplete the matrix of supersaturation and trap excess vacancies (although likely to a lesser extent than Si-rich clusters). This produces a slower and delayed initial hardening response, similar to Alloy 0.27 1.4. Some of these solute clusters are supersaturated with respect to the strengthening phase, causing them to eventually disperse. This replenishes the matrix of excess vacancies and additional solute available to grow the precipitate and increase the hardness. However, some clusters may remain stable, continue to trap vacancies and deplete the matrix of readily available precipitation driving force. For this reason, the hardness is never fully recovered compared to the 0.5hr NA condition. It should be noted that in relation to ambient temperatures, the increased thermal energy of AA conditions increases the probability of vacancy escape. This probability is dependent on the cluster-vacancy binding energy, and the rate at which vacancies escape due to thermal energy is likely much lower than during the dissipation of supersaturated clusters.

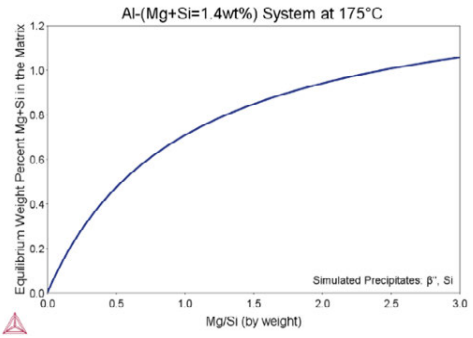


Figure 16. Plotted Thermocalc simulation in X 1.4 alloy system at the AA temperature (calculated with phases FCC matrix, β'', Si).

The idea that cluster dissipation is affected by supersaturation with respect to the strengthening phase may explain the differences in cluster stability that is often seen in the literature. In one case, a slightly silicon rich alloy (with respect to β'') with prior NA displayed a decrease in hardness during the early stages of AA, which was thought to be due to the dissipation of many of the clusters [14]. In another case, a naturally aged Mg-rich alloy showed no such decrease [13]. The discussion here would suggest that many of the clusters in the former work were silicon rich, and dissipated due to super saturation. In contrast, the clusters in the latter case may have been more magnesium rich and a larger proportion of them stable. This would lead to less cluster dissipation, and the possible absence of a hardness trough. These results roughly agree with the concepts presented here; there are, however, several additional works that do not show the same trends.

Alloy 0.73 1.4

The ideas presented to explain the influence of prolonged NA on high strength Si-rich and Mg-rich alloys can be applied to the compositionally balanced alloy. This alloy has an intermediate Mg/Si ratio of 0.73. Using Figs. 15 and 16 as rough guides, the metastable equilibrium solute content of this alloy will be similar to that of Alloy 0.27 1.4. However, the clusters formed during natural aging would be more magnesium rich. As discussed, the clustered Mg solutes have less driving force to dissipate, which may explain the slower recovery of strength that is apparent, when compared to a more silicon rich clustered solid solution.

X 0.7 Alloys

The influence of natural aging on three soft alloys that were investigated obviously show stark contrast to their highly alloyed counterparts. Of all the compositions tested, only Alloy 0.73 0.7 exhibited an increased hardening response with extended natural

aging. The magnesium rich alloy (2.5 0.7) showed very little impact of prior NA throughout the course of natural aging, while the silicon rich alloy (0.27 0.7) again showed delayed hardening, but ultimately, NA had very little impact on peak-over aged hardness.

Due to their low alloying content, all alloys in this group have significantly less driving force for precipitation, in comparison to the X 1.4 alloy group. As a consequence, nucleation rate and overall transformation kinetics will be significantly slower, even in the absence of any natural aging. Under the conditions of the proposed concept, this would allow any clusters formed during natural aging additional time at the artificial aging temperature.
For unstable clusters, like in the case of the silicon rich alloy, this would equate to Si-rich clusters dissipating over a longer period of time. An increase in the duration of the negative hardness effect would then be expected, as is seen in Figure 7.

On the other hand, for stable clusters in solid solution, the elevated (aging) temperature may allow enhanced diffusion and further cluster growth. As stated earlier, prior works have shown that "pre-aging" at an intermediate temperature (between room temperature and the AA temperature) allows larger clusters to form, which are thought to be of a different "type" than naturally aged clusters. These "pre-aged" clusters may minimize the deleterious effects on hardening response, because they are thought to be large enough to act as nucleation sites for β'' (it is also believed that some clusters may transform directly in to β'') [11], [15]. It is possible, that without the driving force for dissipation, the stable clusters in solid solution continue to grow, eventually having sufficient surface energy to act as nucleation sites. If this transition occurs before significant growth of β'', the nucleation advantage may then offset the negative effects of natural aging that are seen in stronger alloys. This may be the case for Alloys 0.73 0.7 and 2.5 0.7. As can be seen, however, the balanced Alloy 0.73 0.7 displayed improved hardening in the 4hr NA condition. Again considering the implications of the addition of nucleation sites, this could be explained with the following: Of the three soft alloys, the balanced alloy has the largest driving force for β''. With a larger driving force, transformation kinetics may then have the ability to take advantage of less energy-rich nucleation sites (i.e. smaller clusters). This could create a denser precipitate network, leading to the increased hardness that is seen. As an aid for the discussion above, Figure 17 shows calculated (using Thermocalc) driving forces for precipitation of β''. Alloys with Mg/Si ratio around 0.73 contain the largest driving force, and the investigated X 0.7 compositions have around 60-65% of the driving force compared to their X 1.4 counterparts.

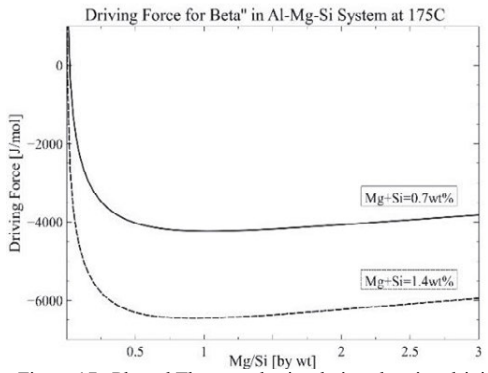


Figure 17. Plotted Thermocalc simulation showing driving force for β'' precipitation in all experimental alloys.

For ease of explanation, the analysis of conductivity has not been included in the discussion thus far, and will be addressed now. Relative magnitudes are roughly as expected based on the metastable equilibrium concentration of solute (Figs. 15 and 16). These trends are also consistent with the proposed influence of natural aging based on solubility and degree of solute supersaturation.

The decreased conductivity (in the 4hr NA condition) during the early stages of AA is likely due to both the direct impact of the presence solute clusters (decreased conductivity), as well as the negative influence on precipitation kinetics (slower increase in conductivity). The relative recovery times are consistent with both the hardness curves and the discussion presented above. At very long aging times, the slightly enhanced conductivity may simply be explained by coarser precipitate structure (caused by prior solute clustering). The current theory would also suggest that the presence of stable solute clusters, as in Alloy 2.5 1.4, may continue to grow and possibly coarsen during the course of AA. At some point this may lead to cluster spacing larger than that of electronic scattering centers, and increased conductivity.

Conclusions

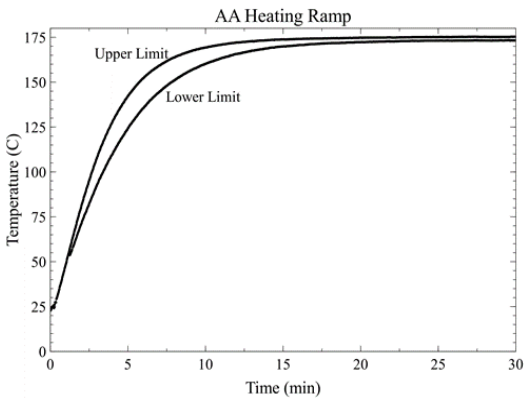
- In Al-Mg-Si alloys, the influence of the duration of natural aging (0.5 hr or 4 hrs) on artificial aging response varies with both absolute and relative solute content.
- High strength alloys (Mg+Si=1.4wt%):
    - For alloys that are rich in silicon (with respect to β''), natural aging has a damaging effect on hardening response for short AA times. For longer durations of AA, the negative effect of NA diminishes.
    - For alloys that are rich in magnesium, longer natural aging produces a softer alloy throughout the course of artificial aging.
    - The highest peak hardness is achieved for alloys compositionally balanced for β''. The influence of natural aging appears to be intermediate, between Si-rich and Mg-rich alloys. The negative effect of NA is removed with prolonged artificial aging.
    - The effect of prolonged natural aging on electrical conductivity (as a function of Mg/Si ratio) during the course of AA shows similar trends to hardness. Extended NA produces decreased conductivity in the early stages of AA, and slightly improved conductivity for long AA times.
- For low strength alloys (Mg+Si=0.7wt%):
    - Extended natural aging in silicon rich alloys results in slower hardening kinetics. The difference in hardness diminishes with time, but does not disappear until very long artificial aging times.
    - Four hours of prior natural aging has minimal impact on the hardening response of magnesium rich alloys.
    - The influence of prolonged natural aging on hardening response is slightly positive for low strength alloys, compositionally balanced for β''.
    - Natural aging has minimal impact on electrical conductivity for alloys with low solute content.
- The observed results are believed to be due to differences in solute clusters (formed during natural aging), and their actions during the strengthening precipitation sequence. The

explanations presented are based on the current understanding of the clustering process, but further work is required to understand the mechanisms involved.

Acknowledgments

This work was funded by SAPA AS - Extrusions North America and performed as part of a doctoral research project at Michigan Technological University.

Appendix 1



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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SYNTHESIS OF Al-TiC NANOCOMPOSITES BY AN IN-SITU GAS-LIQUID METHOD

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Keywords: aluminum, nanocomposites, titanium carbide, gas-liquid in-situ method

Abstract

Next generation of aluminum automotive engines will have to operate at temperatures approaching 300^{0}C. Traditional aluminum alloys cannot perform at these temperatures, but aluminum alloys reinforced with nanoparticles can. The synthesis of aluminum-titanium carbide nanocomposites by an in-situ gas-liquid reaction implies methane to be injected into molten aluminum that has been pre-alloyed with titanium. The gas is introduced by means of a rotating sparger-impeller unit into the hot alloy, and under the correct conditions of temperature, gas flow and rotation speed, it reacts preferentially with titanium to form titanium carbide nanoparticles that are well dispersed in the metal matrix. The apparatus design, the multi-physics phenomena and a mechanism proposal for nanoparticles formation is first given. The operation window in which to allocate the parametric analysis is next calculated. Finally, characterization of initial obtained material, its relationship to the processing parameters and guidelines to obtain the nanoparticles is done.

Introduction

Metal ceramic nanocomposites, i.e. ones with ceramic particles that are under 100nm in average diameter, are believed to provide both high strength and ductility at low and high temperature. Among the various methods for making titanium carbide composites, in-situ methods have gained relevancy over ex-situ methods due to their more stable particle matrix interface, lower agglomeration tendency, and more competitive cost. Three in-situ methods have been developed; the solid-liquid phase sintering method, the solid-molten metal method, and the gas-liquid in-situ method [1]. Limitations of the first two methods include particle size over 0.2µm and very wide particle size distribution up to d90=10µm. This is due to the initial carbon particle size distribution, normally graphite powder that acts as precursor for the TiC particles that form by titanium diffusion into the particle's graphite core. The newly-formed TiC particle eventually breaks into polygonal particles due to strain incompatibility as indicated in Figure 1 [2, 3]

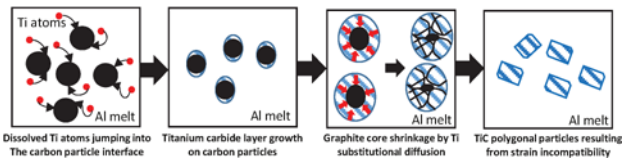


Figure 1. Formation of TiC micro particles by the solid-liquid process.

The gas-liquid in-situ method was developed in the early 1990's [4-6]. The method utilizes a hollow tube submerged in molten aluminum. The tube may have a single nozzle or a static bubbler with several nozzles in its bottom. A reactive gas is bubbled through the tube into the melt. The gas and liquid react at the bubble-liquid boundary layer producing the ceramic particles that are released into the melt bulk. The outcome of this process is not ideal because of the lack of turbulence in the melt and the low gas dispersion intensity [7]. The novelty of the present work is in the use of a hollow tube that has attached to its bottom a gas sparger-impeller unit that can rotate at high speed. In this way, the gas is very efficiently dispersed in the form of very fine bubbles that allow production of nanoparticles.

Apparatus design and procedures

Apparatus design

The apparatus is schematically shown in Figure 2. It is divided in four main subsystems: the gas abatement subsystem, the furnace and protective atmosphere subsystem, the rotation subsystem, and the impeller-crucible subsystem. The gas abatement subsystem includes moisture and oxygen traps and flowmeters. The furnace contains a ceramic retort that is sealed from the outside and operated with a slight argon overpressure. A thermocouple measures the temperature in the melt and the oxygen inside the furnace is continuously monitored to reveal operational anomalies and to protect the furnace against explosion.

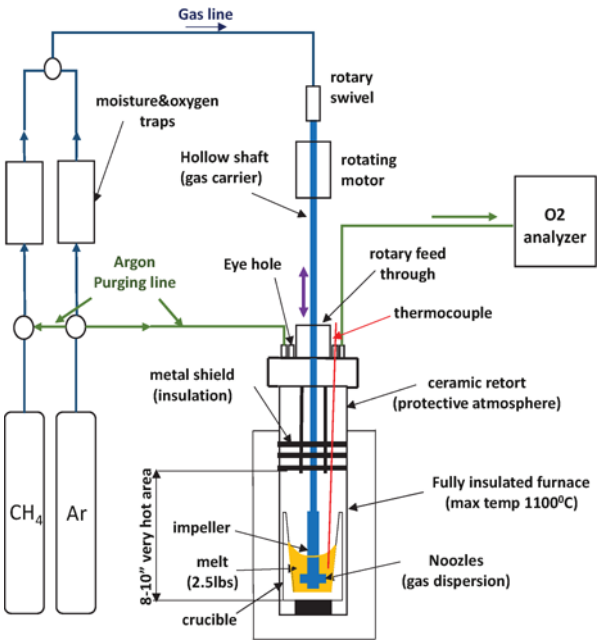


Figure 2. Schematic representation of the apparatus.

The rotational subsystem (shaft-impeller-rotary feed through) has been designed to go up to 750rpm with acceptable vibration level. The crucible capacity is about 2.5lbs of aluminum. The impeller is 3D printed from mullite. Whereas the gas abatement subsystem and

the furnace and protective atmosphere subsystems are built based on standard engineering calculations with components available from industry suppliers, the impeller-crucible subsystem has been designed to meet key specifications including gas-dispersion efficiency and ability to handle viscous fluids.

Procedures

Figure 3 describes the operational procedure. A careful heating up step is first required to avoid thermal fatigue cracking of the ceramic retort. An extra hour allows the melt to reach the maximum temperature of 1050^0C and provides sufficient time for the Al_3Ti intermetallics that are present in the master alloy to completely dissolve in the melt. The experiment is then carried out for a set time (e.g., 1 hour in Figure 3). Argon is pumped through the impeller, which is immersed in the melt and the impeller speeds up to a set rotation speed. CH_4 is added to the argon flow and the experiment begins. During the experiment time, TiC is produced. Once the trial is finished, the CH_4 flow and the impeller are consecutively stopped. The impeller is then raised, and the argon flow is stopped. Temperature is allowed to drop and the ingot solidifies.

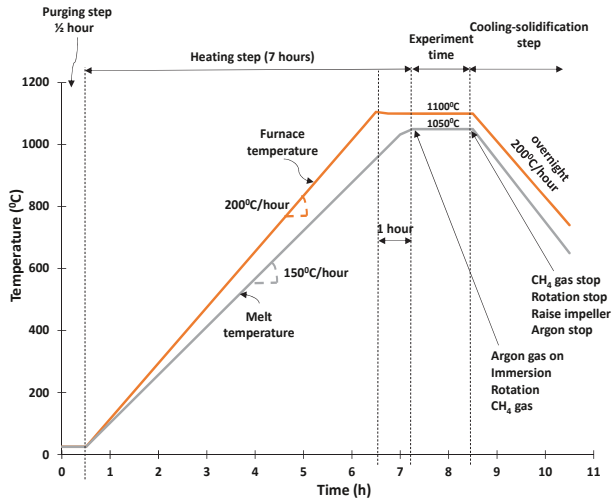
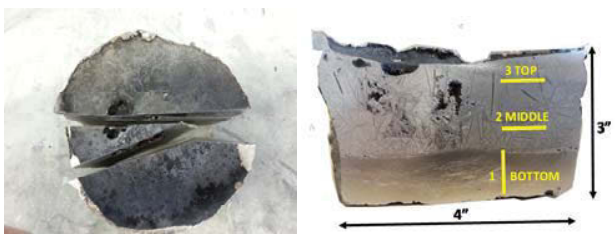


Figure 3. Schematic representation of the temperature vs.time profile.

The ingot is removed from the crucible and is cut as shown in Figure 4(a). Next, three samples are sectioned from 3 different locations as shown in Figure 4(b).



(a) (b)

Figure 4. Ingot sectioning and location of specimens.

The specimens are prepared for microscopy and X-ray diffraction analyses. Diallyl Phthalate mounting powder is used. SiC paper with grit sizes 240, 320, and 400 is used for grinding the samples, and an automatic polisher (Struers Tergramin 20) is used to polish the samples in 3 steps: (1) polishing with 9µm diamonds for 10min, (2) polishing with 3µm diamond for 5 minutes, and (3) polishing with 1µm diamond for 5 minutes. A 20N force is applied in all 3 polishing steps. The prepared samples are analyzed in their as-polished condition with an optical microscope (EPHIPOT 200) and also with a scanning electron microscope (SEM) (JEOL-7000F). For XRD measurements, a Panalytical diffractometer is used. Chemical analysis is performed by selective digestion of the aluminum matrix and the Al_3Ti particles in a 1:1 water:hydrochloric acid solution at 120^0C for one hour under atmospheric pressure. Other constituents, e.g., TiC and aluminum and titanium oxides remains as insoluble powder.

Mechanism of nanoparticle formation

At the macroscale (Figure 5), the variables to consider are: the melt and temperature profile, the impeller and tank design, the rotation speed, the bubble size, the gas residence time, and the bubble-melt interfacial area[8,9]. Thermodynamics allow us to calculate the melt and temperature profile, whereas the remaining variables are initially dependent on the prevailing gas-liquid fluid-dynamics regime. The temperature field is considered constant throughout the melt.

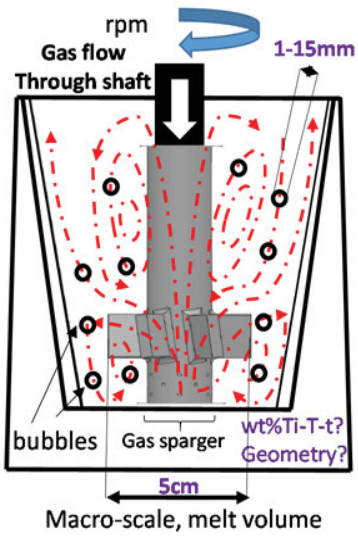


Figure 5. System macroscale and main variables

The proposed mechanism suggests that at the microscale (Figure 6), the gas diffuses across the gas-boundary layer where it is chemisorbed into the bubble-melt interface [10, 11] and finally it is dissolved in form of carbon atoms in the melt. Once in the melt, the carbon atoms diffuse across the liquid boundary layer where they instantly form TiC nuclei. These nuclei continue to grow in size until they eventually exit the boundary layer controlled by the surface renewal rate of the nanoparticles that is based on the theory of Higbie and Danckwert [12, 13]. The theory proposes that continuous interaction between the turbulent fluid eddies and the laminar stagnant packages of nanoparticles that arrive at the interface causes a next flux of nanoparticles to exit the liquid boundary layer. Once they leave the boundary layer, the particles stop growing as the liquid bulk concentration of dissolved carbon is lower than its equilibrium value (Figure 6). We call this process the *growth interruption mechanism* (GIM).

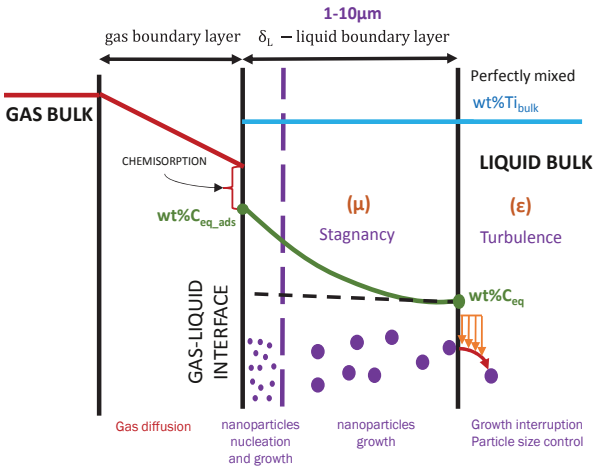


Figure 6. Microscale representation of the gas-liquid interface

It is also necessary to analyze the particle nucleation and growth at the particle-melt interface (Figure 7). The particle growth rate can be either interface limited or diffusion limited.

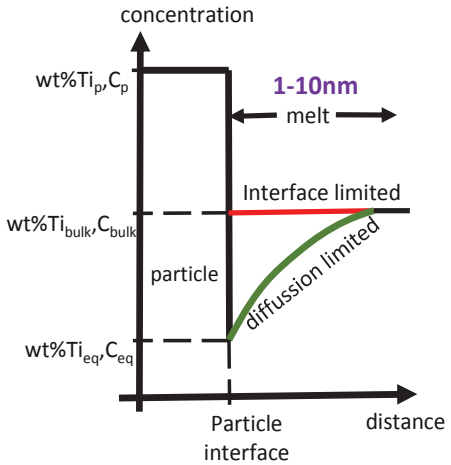


Figure 7. Nanoscale representation at the nanoparticle-melt interface

The particle size produced by this mechanism in the liquid boundary layer is given by

$$\int_{Rp_{nuclei}}^{Rp_F} dR_p = \int_0^{t_F} v_{growth} dt$$

If the growth velocity is considered constant, then

$$R_{pF} = R_{pnuclei} + v_{growth} \cdot t_F \qquad (1)$$

Based on the above, it is concluded that the particle size can be controlled by monitoring its nucleus size, its growth rate, and its residence time in the boundary layer. The residence time depends on the thickness of the liquid boundary layer (δ_L in Figure 6) and on the nanoparticles surface renewal rate. Both magnitudes depend to a large extent on the extent of turbulence and gas dispersion efficiency, which are controlled by macro-scale variables; namely, rotation speed and gas flow. Simultaneously, mass transfer and chemical kinetics at the micro and nanoscale control particle nucleation and growth.

Impeller-crucible design, processing window, and selection of the reactive gas

Thermodynamic considerations

The composition vs. temperature window in which titanium carbide is stable is determined. The carbide displacement reaction to be considered when assessing the thermodynamic feasibility of carbide formation in molten aluminum results from the subtraction of the two reactions shown below:

$$\begin{array}{ll} Ti(d) + C(d) \leftrightarrow TiC(s) & \Delta G_{R1} \\ - & \\ \frac{4}{3}Al(l) + C(d) \leftrightarrow \frac{1}{3}Al_4C_3(s) & \frac{1}{3}\Delta G_{R2} \\ \hline Ti(d) + \frac{1}{3}Al_4C_3(s) \leftrightarrow TiC(s) + \frac{4}{3}Al(l) & \Delta G_{R_disp} = \Delta G_{R1} - \frac{1}{3}\Delta G_{R2} \end{array}$$

Note that the dissolved carbon activity does not play a role in the displacement reaction. If the reaction shifts to the right, metal carbide is stable and no Al_4C_3 forms. Al_4C_3 is a saline carbide. Saline carbides are soft and water reactive and its formation must be avoided. The Van't Hoff equation for the displacement reaction dictates that at equilibrium

$$\Delta G_{R_disp} = \Delta G^0_{R_disp} + RTlnK_a = 0$$

$$\Delta G^0_{R_disp} = -RTlnK_a = -RTln\left(\frac{a_{TiC} \cdot a_{Al}^{4/3}}{a_{Ti} \cdot a_{Al4C3}^{1/3}}\right)$$

$$as\ a_{TiC} = a_{Al4C3} = 1$$

$$\Delta G^0_{R_disp} = -RTln\left(\frac{a_{Al}^{4/3}}{a_M^{a/b}}\right)\ where\ a_i = \gamma_i(T, X_i) \cdot X_i$$

$$\text{being } -\frac{\Delta G^0_{R_disp}}{RT} = lnK_a = ln\frac{[\gamma_{Al}(X_{Al},T)\cdot X_{Al}]^{4/3}}{[\gamma_{Ti}(X_{Ti},T)\cdot X_{Ti}]}$$

and solving

$$lnX_{Ti} = \frac{b}{a}\left(\frac{\Delta G^0_{R_disp}}{RT} + \frac{4}{3}ln\gamma_{Al} + \frac{4}{3}lnX_{Al} - \frac{a}{b}ln\gamma_{Ti}\right)$$

For a two-component mixture $X_{Al} = 1 - X_{Ti}$, and

$$ln\frac{X_{Ti}}{(1-X_{Ti})^{4/3}} = \left(\frac{\Delta G^0_R}{RT} + \frac{4}{3}ln\gamma_{Al} - ln\gamma_{Ti}\right)$$

Finally, assuming the Henryan state for titanium and the Raoultian state for aluminum, we have

$$wt\%Ti = \frac{100MW_{Ti}}{MW_{Al}} exp\left(\frac{\Delta G^0_{R_disp}}{RT} - ln\gamma^H_{Ti}\right) \qquad (2)$$

Expressions for $\Delta G^0_{R_disp}$ and $ln\gamma^H_{Ti}$ are taken from [14] and (2) is plotted as Figure 8. The horizontal line confirms the stability of TiC for titanium contents in liquid aluminum above 0.5wt%, independent of melt temperature. The liquidus line is taken directly from the aluminum rich corner of the Al-Ti phase diagram. Over the liquidus line both TiC and Al_3Ti coexist as solid phases in a liquid slurry.

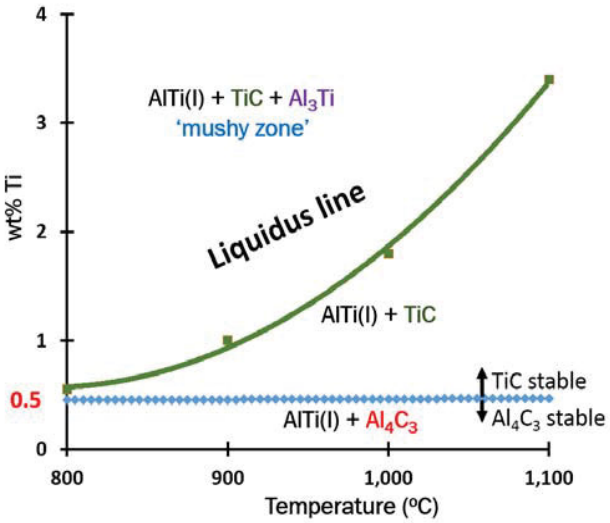


Figure 8. Operation window for TiC formation.

Fluid-dynamic considerations

The investigation of the gas-liquid fluid dynamics is initially limited to a dilute melt containing nanoparticles that is fully mixed and fully turbulent. Therefore, there are no rheology effects derived from the presence of the nanoparticles in the melt. Nanoparticles trajectories are initially random (no buoyancy-sedimentation, see [15]) and they are not in suspension. The gas dispersion takes place in the impeller boundary area and it is mechanistically explained in 3 steps [9]. Low-pressure areas, called trailing vortexes, appear behind the blade tips (see Figure 11). In these areas the shear rate and the energy dissipation rate are maximum. First (step 1 in Figure 11), the gas bubbles attending to the ratio of buoyancy forces/liquid centrifugal forces are sucked by the trailing vortex. Second (step 2 in Figure 11), the bubbles break up by bubble-eddy interactions. Third (step 3 in Figure 11) the bubbles are evacuated from the vortex with the pumped flow. The liquid trailing vortex is called a cavity if it is mixed with gas.

Next, we introduce the **π**-set of the system for a given impeller design. See [16] for a better understanding of dimensional analysis. Following Zlokarnik [8]

$$\prod \equiv \left\{ Ne, Q, Fr, Re, \frac{c}{D}, \frac{D}{TT}, \frac{H}{TT} \right\} \qquad (3)$$

The last three numbers in the π-set are geometry numbers. Geometry will be first approached.

Impeller-crucible geometry. In general, the crucible-impeller geometry is fully defined by the dimensions shown in Figure 9. Efficient gas dispersion is achieved only under turbulence. The Newton number can be correlated to the Reynolds and geometry numbers resulting in the so called characteristic curve of the mixing system that establishes the mixing regime intervals. The characteristic curve is analogous to the friction coefficient vs. Re characteristic curve in a pipeline and it is independent of Fr and also Q for small Q values [8].

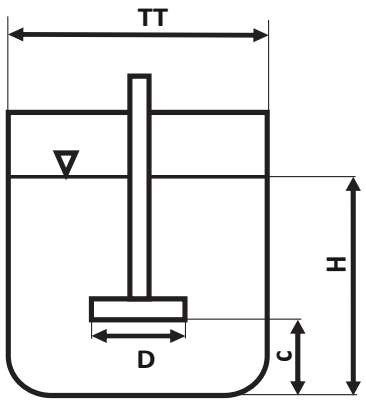


Figure 9. Main dimensions of crucible-impeller geometry

The geometry numbers strongly affect the characteristic curve of the mixing system and the Reynolds interval at which the turbulent regime occurs. Adding baffles to the tank and decreasing the clearance between the impeller and the tank enhances the turbulence as shown in Table I.

Table I. Reynolds threshold values [8]

Design	**Laminar**	**Transient**	**Turbulent**
Baffled , clearance	$Re<10^2$	$Re=10^2$-10^3	$Re>10^3$
Unbaffled ,large clearance	$Re<10^3$	$Re=10^3$-10^4	$Re>10^4$

Table II shows the typical geometrical relations between the impeller and the tank.

Table II. Geometrical relationships for our design and typical values for low and high viscous fluids [17]

Geometry numbers	D/TT	H/TT	c/D	w/D	L/D
for high viscous liquids	0.5-0.75	0.8-1	<0.6	0.35	0.29
for low viscous liquids	0.2-0.6	0.8-1	>0.6		
Our design	0.6	0.85	0.5		

Figure 10 and Figure 11 show the impeller-tank design for the current project. The unbaffled graphite crucible has an 8^0 draft angle with the baffled crucible has 8^0 wall angles (see Figure 11). Off bottom distance (c) is 1 inch and c/D = 0.5 to equally spread the pumped flow upwards and downwards. The impeller is a flat blade disk turbine with 6 blades. This is the traditional Rushton turbine that is customarily used for gas dispersion [8]. The blade's angle to the vertical line is 8^o to guarantee perpendicular flow into the wall and equal flow upwards and downwards. The blade's surface (width 'w' and length 'L') is large enough to enable good pumping capacity. The design of the gas sparger is based on [20, 21]. The sparger is located under the impeller and it has 12 nozzles with a diameter of 2.54mm.

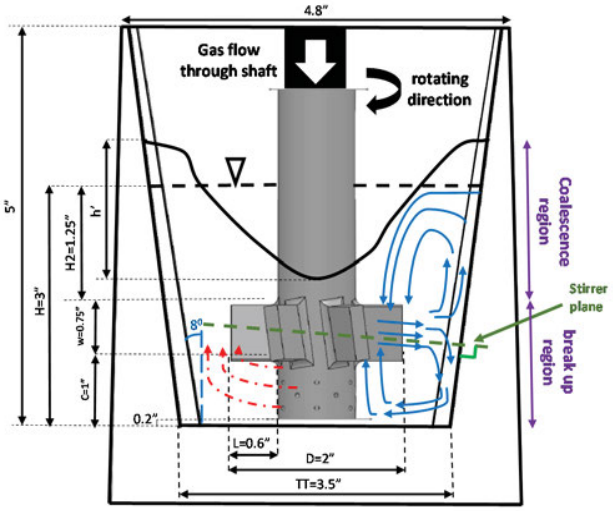


Figure 10. Detailed geometry of the crucible and impeller.

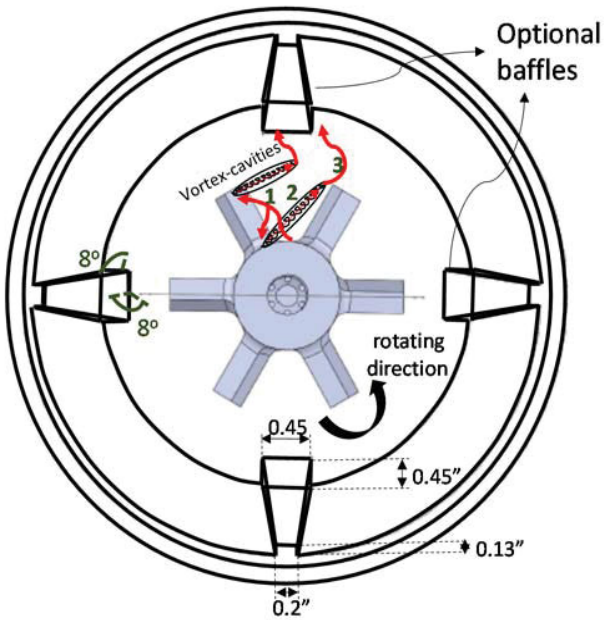


Figure 11. Top-view of the crucible and impeller geometry.

Gas flow and rotation speed. Depending on the gas flow rate and the impeller speed there are four main flow regimes associated with four types of cavity systems behind the blades in the impeller region [9]. Figure 12 depicts the situation for constant rotation speed and increasing gas flow from left to right. With very small gas flow rate, the melt is sucked by the impeller vortex cavity and we have what is commonly known as a complete recirculation regime. An increase in gas flow increases the gas fraction in the cavities creating clinging cavities and we have what is known as a complete dispersion regime. These two regimes are characterized by a high shear rate in the cavities and high gas dispersion efficiency. Higher gas flow rates loads the cavities and three of the cavities become partially flooded whereas the other three remain active. Gas is not efficiently dispersed in the large flooded cavities and the impeller is not able to spread the bubbles along the full tank volume. This is called the loaded regime.

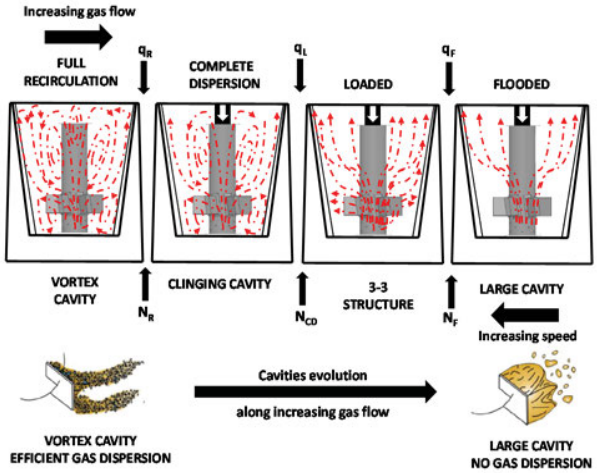


Figure 12.Gas-liquid flow regimes.

With further increase in gas flow rate, the flooded gas flow regime is reached and the cavities are all large, the impeller dispersion ability is overcome, and gas is not dispersed. After the impeller and tank geometry have been selected, the geometry numbers can be removed from the π-set of equation (3). Newton number is removed as well as the required external power is not considered relevant to the dispersion regime calculation [9]. The remaining π-set is

$$\prod \equiv \{Re, Fr, Q\} \tag{4}$$

After Smith et al. [22], the correlations shown as equations (5-8) for a six blades radial turbine allocate the different gas-liquid flow regimes (Figure 12) for the **ππ**-set $\prod \equiv \{Re, Fr, Q\}$ with Re > 10^3 (turbulent conditions).

Gas Flow number intervals	Gas-liquid regime	
$Q < 0.025\left(\frac{D}{TT}\right)^{-0.5}$	Full recirculation	(5)
$0.025\left(\frac{D}{TT}\right)^{-0.5} < Q < 0.2 \cdot \left(\frac{D}{TT}\right)^{0.5} \cdot Fr^{0.5}$	Full dispersion	(6)
$0.2 \cdot \left(\frac{D}{TT}\right)^{0.5} \cdot Fr^{0.5} < Q < 30Fr\left(\frac{D}{TT}\right)^{3.5}$	Loading	(7)
$Q > 30Fr\left(\frac{D}{TT}\right)^{3.5}$	Flooding	(8)

Solving the correlations for $Q=\frac{q}{N \cdot D^3}$ and $Fr=\frac{N^2 \cdot D}{g}$ allows the construction of Figure 13, which is plotted for a turbine with 2 inch diameter. Note that Figure 13 is scale independent if q and N are substituted by Q and Fr. Based on the *growth interruption mechanism* (GIM) in Figure 6, it is suggested that production of TiC nanoparticles requires high turbulence and dispersion intensity in the cavities for a high surface renewal rate and a thin boundary layer, i.e. short residence time (1). Hence, the proposed operation window is allocated in the full recirculation regime in Figure 13; a rotation speed from 450 rpms (see [7]) to 750 rpm and a gas flow rate in the impeller nozzles of 1-20 L/min. The gas flow rate in Figure 13 is the gas flow rate at the impeller nozzles, which are at 1000°C, but the gas flow rate can only be monitored in the flowmeter at room temperature. Assumed ideal gas, the gas flow at room temperature is

$$q_{flowmeter}=q_{noozle} \cdot \frac{T_{flowmeter}}{T_{noozle}} = q_{noozle} \cdot \frac{298K}{1273K} = q_{noozle} \cdot 0.23$$

The previous calculation limits the operation window to the ranges 450-750 rpm and 1-5 L/min.

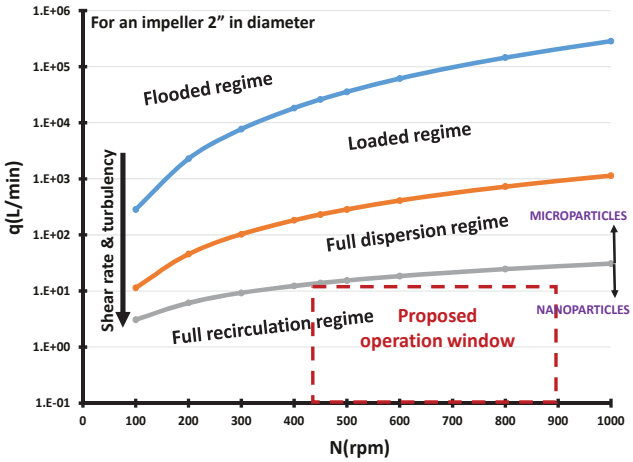


Figure 13. Gas-liquid flow rate map and proposed operation window.

Results and discussion

Four experiments were performed at 1050^{0}C, 450 rpm and gas flow rate = 1 L/min. The titanium content, methane partial pressure and time have been varied inside the calculated operation window. The crucible is unbaffled. Table III shows the design of experiments

Table III. Performed experiments.

Exp. #	Ti (wt. %)	CH_4 (%)	Ti_F (wt. %)	Time (hours)
1	5.65	15	4	0.5
2	5.65	15	3	2
3	2.65	5	2	0.5
4	2.65	5	1.5	2

The Ti content in Table III has been measured at the beginning and at the end of the experiment. It refers to the available titanium, i.e. Ti present in the form of Al_3Ti and as dissolved Ti, but not TiC. All the resulting ingots exhibit the structure schematically illustrated in Figure 14. The top area is populated by Al_3Ti needles inside the α-aluminum grains whereas the bottom shows a dark region full of Al_3Ti polygonal particles and/or small needles in the grain bulk surrounded by TiC microparticles segregated within the interdendritic space (Figure 15 and Figure 16). The TiC microparticles are especially visible in Exp. #2 and Exp. #4, so the analysis will focus on these experiments. Exp. #2 has the bottom populated by polygonal particles whereas Exp. #4 has an intermediate number of needles and almost no polygonal particle.

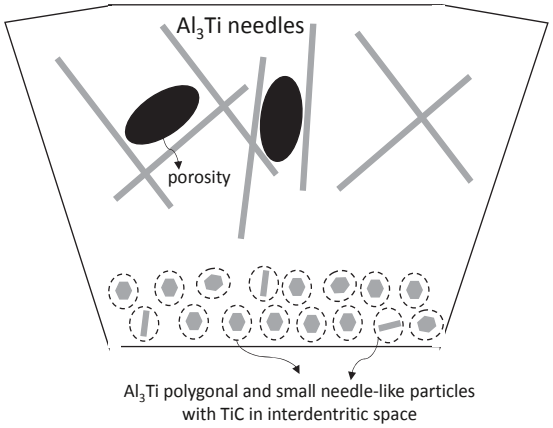


Figure 14. Typical ingot structure.

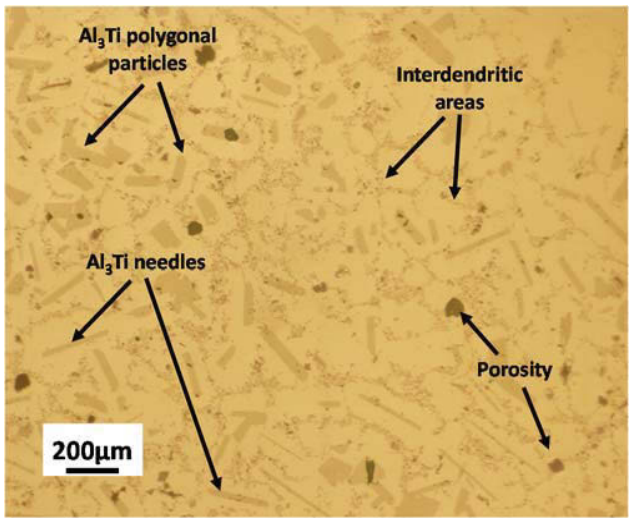


Figure 15. LOM picture at the ingot bottom from Exp. #2.

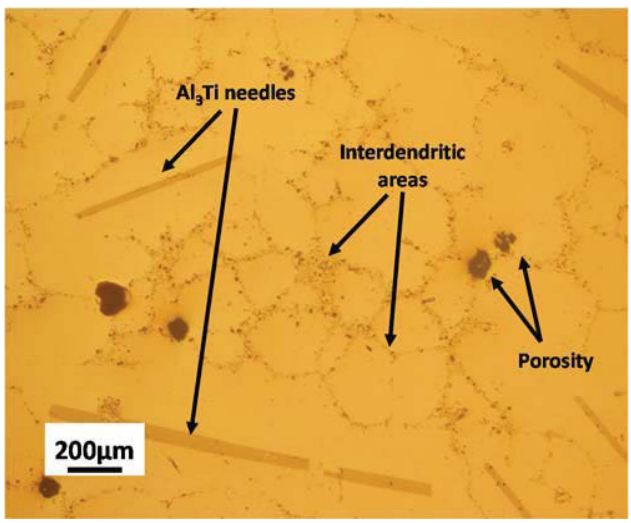


Figure 16. LOM picture at the ingot bottom from Exp. #4.

Figure 17 shows a BSE-EDS map of the bottom of an ingot produced according to Exp. #2. A significant amount of oxygen-containing particles is detected along the interdendritic space next to the TiC particles.

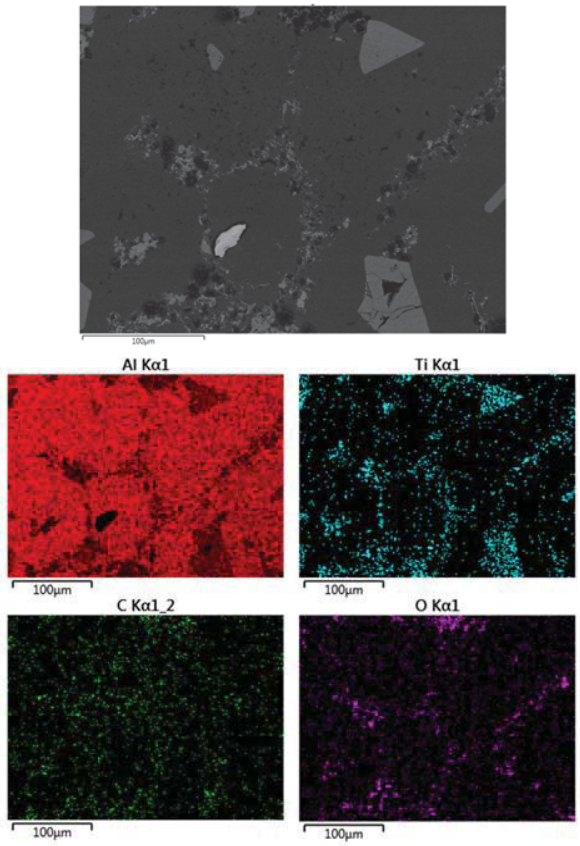


Figure 17. BSE-EDS map of the bottom of the ingot produced by Exp. #2.

Figure 18 shows XRD spectra taken from the different locations indicated in Figure 4.TiC peaks are higher and more numerous at the bottom of the ingot. γ-TiO_2 and $AlTi_2C$ are the main impurities. No Al_4C_3 is produced.

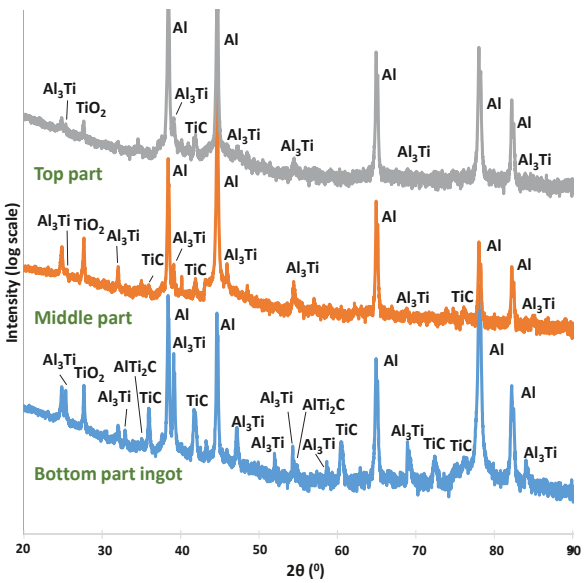


Figure 18. XRD spectrum of samples from exp#2

A 5 steps cooling-solidification mechanism that results in the observed ingot macrostructure is proposed (Figure 19). Note that step 1 in Figure 19 begins after 8½ hours in Figure 3, i.e. just after the impeller is raised at the end of the experiment. In step 1, TiC particles are in suspension in the melt. They sink (step 2) due to their higher density and then solidification begins. From Figure 2 it is assumed that heat is extracted from the top of the apparatus. The Al-Ti phase diagram at the aluminum rich corner is shown in Figure 20.

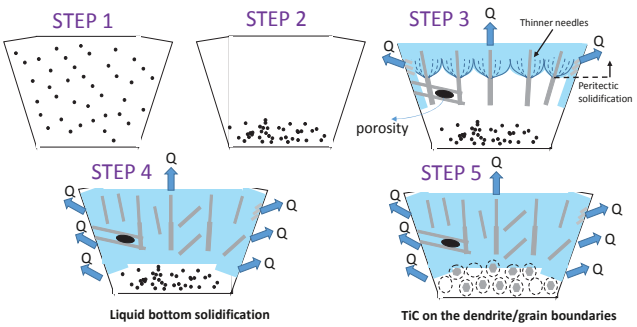


Figure 19. Proposed ingot solidification mechanism

Step 3 implies a peritectic solidification reaction in which the Al_3Ti nuclei grow in a faceted fashion in the [110] and by diffusional growth in the [001] direction resulting in elongated needles growing parallel to the direction of heat extraction [23]. When the temperature reaches the peritectic point, α-Al nucleates along the Al_3Ti needles that become thinner (see Figure 21) due to the peritectic reaction.

$$Al_3Ti(s) + Al(l) \leftrightarrow \alpha\text{-}Al(s) \quad \text{peritectic reaction}$$

As solidification progresses, the temperature gradient (becomes two-dimensional and a complex network of needles appears and traps liquid in its interior.

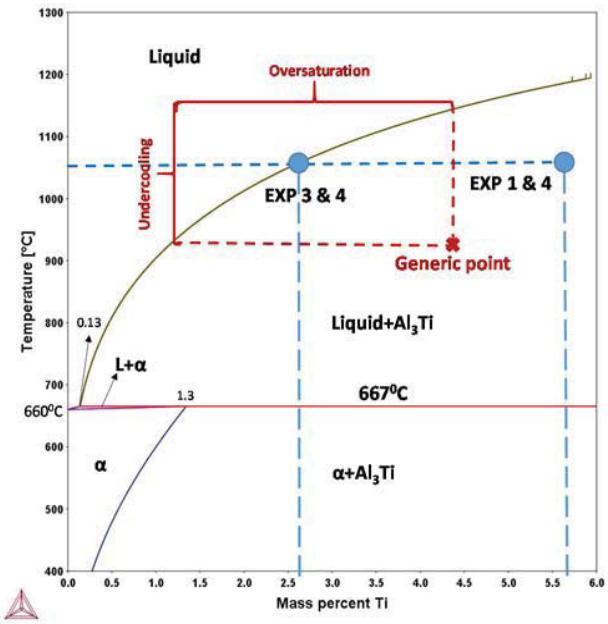


Figure 20. Ti-rich corner of the Al-Ti phase diagram.

Solidification progresses into the bottom part of the ingot, (step 4), and it is characterized by two facts: (1) The Ti content increases due to the vertical compositional segregation, which suggests higher oversaturation (or undercooling), and (2) the cooling rate may have increased due to the larger volume of solid aluminum coupled with heat extraction through the crucible walls. Both of these occurrences cause the Al_3Ti nuclei to grow homogeneously in several directions resulting in a high fraction of polygonal particles (step 5) onto which the α-Al nucleates pushing the TiC particles into the interdendritic space. Several researchers, e.g., [24], confirm the stronger refining effect of Al_3Ti versus TiC particles.

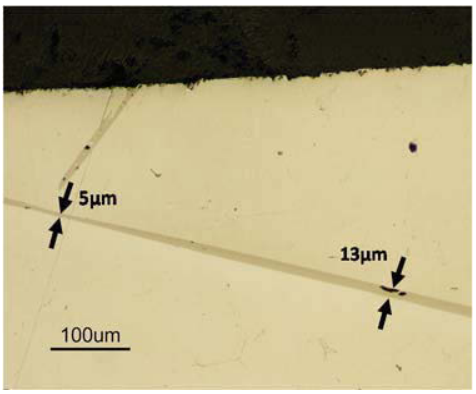


Figure 21. LOM image showing needle thinning Exp. #4.

The Al_3Ti intermetallics

The Al_3Ti particles are deleterious. They can affect the mechanical properties of the A356 reinforced alloy, and they deactivate the nucleation ability of the TiC particles that get segregated into the interdendritic space. Figure 22 shows the variation in average characteristic length of the Al3Ti particles with the Ti content of the melt.

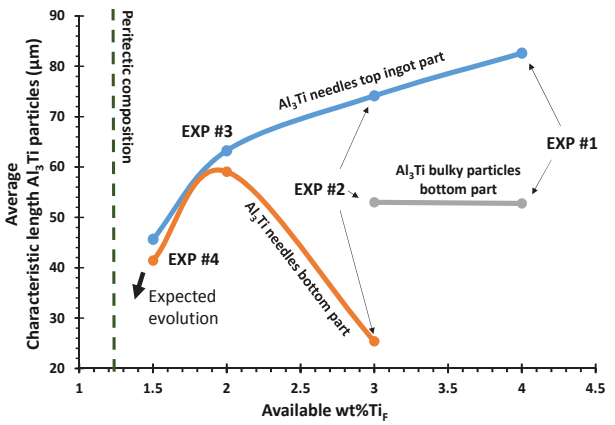


Figure 22. Particle characteristic length with wt.% Ti evolution

It can be inferred from this Figure that a closer Ti content to the peritectic composition reduces the thickness of the Al_3Ti needles as the Ti oversaturation decreases and the ratio of radial diffusional growth to faceted front growth increases. The needles might eventually disappear if the Ti content gets close enough to the peritectic composition or if the composition moves into the hypo-peritectic side where only α-Al exists in equilibrium. It can also be inferred from Figure 22 that for a Ti_F content < 3 wt. %, the bulky particles disappear in favor of the needle-like particles.

The TiC particles

Figure 23 and Figure 24 show the TiC particles that are segregated into the interdendritic regions. In Figure 23 the complexity in morphologies and size distribution is obvious. A size distribution that ranges from less than 100nm up to 5µm and morphologies with very irregular shapes and different aspect ratios is observed. Large oxide particles are also observed.

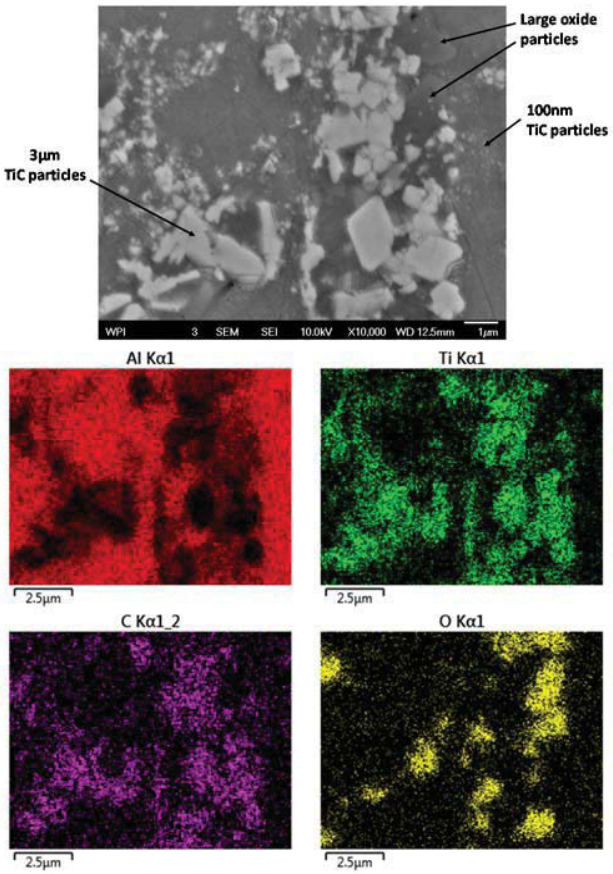


Figure 23. SEI-EDS detail of interdendritic area in exp#2

Figure 24 shows an interdendritic area with a fine dispersion of submicron size TiC particles and also coarse TiC particles that are several micrometers in size. Oxides and $Al_xNi_yFe_z$ intermetallic particles are also present. Nickel (<0.05wt.%) and iron (<0.15wt.%) are impurities in the AlTi initial alloys.
References [4][5] help to explain the TiC microparticles synthesis in the present work. Sahoo et al. [4] report the release of amorphous graphite nanoparticles that are larger than 50nm as a result of methane pyrolysis (also called homogeneous decomposition). These amorphous graphite nanoparticles are dragged by the liquid and results in polygonal TiC microparticles. The soot flakes are not observed in the XRD analysis shown in Figure 18 because they are amorphous. The route described in detail in Figure 6, which involves transfer of non-decomposed methane into the gas-liquid interface where it chemisorbs, is preferred for producing nanoparticles with a narrow particle size distribution. Figure 25 summarizes both TiC production routes.

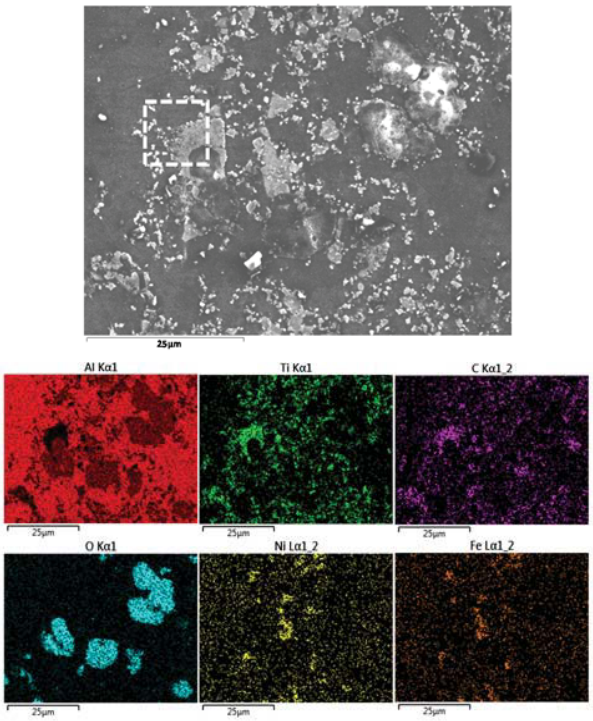


Figure 24. BSE-EDS of the interdendritic area Exp. #4.

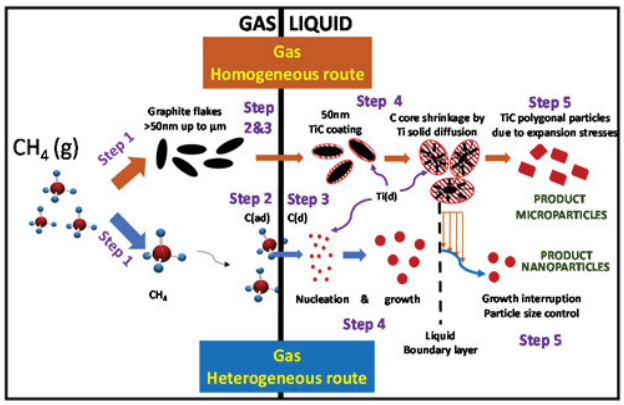


Figure 25. Summary of possible TiC formation routes.

It is unclear whether the oxygen chemisorption is canceling the methane chemisorption. Previous authors [25] dealing with a similar problem in the production of aluminum nitrides propose the addition of lithium up to 2.5wt% in the melt to remove the oxygen in form of $Al_xLi_yO_z$ compounds and allow the main nitride reaction to happen. We will install a tube furnace containing titanium chips in the gas line right before the melt furnace. Titanium is the lowest metal in the Ellingham diagram for metals oxidation with a high melting point and it is expected to fully remove the oxygen. The big number of existing submicron particles (Figure 23 and Figure 24) suggests a combined route (Figure 25) of soot formation and methane chemisorption for TiC formation. However, the dotted area in Figure 24 is zoomed in Figure 26 and it confirms that submicron particles are also formed by TiC microparticles crumbling due to strain incompatibility.

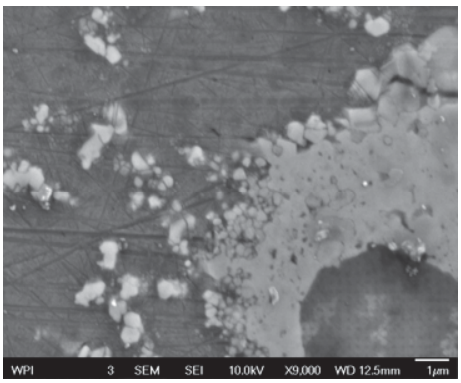


Figure 26. SEM-SEI showing TiC crumbling.

With this in mind, our strategy for producing nanoparticles involves encouraging heterogeneous methane decomposition instead of homogeneous methane decomposition along the hollow shaft. The overall reaction along the gas tube is:

$$CH_4(g) \leftrightarrow C(s) + 2H_2(g)$$

And from Van't Hoff equilibrium equation

$$\Delta G_R = 91863 - 112T = RTlnK = RTln\frac{\%volH_2^2}{\%volCH_4}$$

Reducing the furnace temperature and introducing hydrogen in the bubbling gas mixture shift the reaction to left side. In terms of gas reaction kinetics, Shah et al. [26] allocated the homogeneous methane decomposition at temperatures above 900°C and the heterogeneous decomposition at temperatures under 800°C for a space velocity (SV) of 4 ml/(cm^3·min). The SV is a measure of how fast the gas goes through the reactor tube, i.e. the inverse of the residence time. High SV values correspond to short residence times and vice versa as shown in the following equation.

$$SV = \frac{q[L/min]}{V_{reactor}[cm^3]}$$

In our system, the internal diameter of the hollow shaft is 7mm and so it is active for soot formation in its last 10inches below the metal shield where the temperature is high (see Figure 2). In this case, for a gas flow rate of 1L/min and vol. %CH_4=15, the space velocity is 150 ml/(cm^3·min), an order of magnitude larger than the SV of Shah et al. [26], which is initially advantageous. A better understanding of the soot formation mechanism and kinetics is crucial in order to design new experiments with temperature decrease and hydrogen introduction.

Summary

- An apparatus for the production of Al-TiC nanocomposites has been designed and constructed.
- A mechanism that describes the formation of the nanoparticles is proposed.
- The multi-physics involved in the in-situ synthesis of TiC nanoparticles in aluminum are described.
- An operation window for the process is calculated.
- A first set of experiments have been performed and initial results are reported.

- Oxide particles are the main impurity present in the resulting composite material.
- The resulting ingots typically contain thick Al_3Ti needles at their top and polygonal Al_3Ti particles at their bottom.
- TiC microparticles with an irregular shape and a wide size distribution form and concentrate near the bottom part of the ingots segregated in the interdendritic space.

Nomenclature

δ_L	Liquid boundary layer thickness	[L]
μ	Melt viscosity	$[M\cdot L^{-1}\cdot T^{-1}]$
ε	Melt turbulence	$[L^2\cdot T^{-3}]$
R_{pF}	TiC Particle final size	[L]
$R_{pnuclei}$	Initial nuclei size	[L]
t_F	Residence time in liquid boundary layer	[T]
v_{growth}	Particle growth velocity	$[L\cdot T^{-1}]$
ΔG_{Ri}	Free energy of reaction 'i'	$[M\cdot L^3\cdot T^{-2}]$
(d),(l),(s)	Liquid, dissolved or solid state	none
a_i	activity	dimensionless
γ_i^H	Henry coefficient substance 'i'	dimensionless
T	Temperature	[θ]
X_i	Liquid or gaseous mol fraction	$[^0/_1]$
MW_i	Molecular weight substance 'I'	[M]
K_a	Activities equilibrium constant	dimensionless
R	Gas constant	$[M\cdot L^3\cdot T^{-2}\cdot \theta^{-1}]$
TT	Crucible diameter	[L]
D	Impeller diameter	[L]
c	Impeller off-bottom distance	[L]
H	Melt height	[L]
w	Blade width	[L]
N	Rotation speed	$[T^{-1}]$
q	Gas flow	$[L^3\cdot T^{-1}]$
g	Gravity	$[L\cdot T^{-2}]$
Ne	Newton number	dimensionless
Q	Flow number	dimensionless
Fr	Froude number	dimensionless
Re	Reynolds number	dimensionless
ρg	Gas density	$[L^3\cdot M^{-1}]$
d_h	Nozzle diameter	[L]
v_g	Gas velocity	$[L\cdot T^{-1}]$

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Acknowledgment

The authors gratefully acknowledge the member companies of the Advanced Casting Research Center (ACRC) for their support of this work, and for their continued support of research focused on the science and technology of metal casting at Worcester Polytechnic Institute.

Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ORIENTATION RELATIONSHIPS OF PRECIPITATES WITH THE MATRIX IN AN ALUMINUM QUASICRYSTALLINE ALLOY

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Keywords: Aluminum, Quasicrystal, Precipitate, Transmission Electron Microscopy

Abstract

Aluminum quasicrystalline alloys present a novel class of high-strength alloys. They possess a high potential for practical applications in many fields. In this study, we used a quasicrystalline Al-alloy with a copper addition to achieving strengthening by heat treatment. The alloys were prepared by melt-spinning and by casting into a copper mold. In both cases, quasicrystals formed during solidification either as a primary phase or as a part of a quasicrystal-containing eutectic. Afterward, the samples were subjected to T5 heat treatment. The hardness of the melt-spun ribbons increased considerably while the hardness of the gravitationally cast samples did not change noticeably. A detailed investigation using transmission electron microscopy (TEM) showed that different types of precipitates can form. At lower temperatures, binary Al-Cu precipitates formed, in the intermediate region prevailed the quasicrystalline precipitates, while at higher temperatures formed T-precipitates ($Al_{20}Mn_3Cu_2$). Particular attention was given to determination of their orientation relationships with the Al-rich matrix.

Introduction

It has passed more than thirty years since the discovery of quasicrystals [1]. During the following years, several hundreds of quasicrystals were discovered. In respect to mechanical properties, the quasicrystals resemble intermetallic phases in metallic alloys. They have a ductile-to-brittle transition, so they are brittle at lower temperatures. The quasicrystals are not periodic crystals, as opposed to intermetallic phases. For example, icosahedral quasicrystals (IQC's) have a long-range order but lack periodicity. IQC's have a large degree of symmetry. They possess six fivefold axes, ten threefold axes, and 15 twofold axes, which is much more than in any periodic intermetallic phase. Some consequences of their high symmetry are good matching with many metallic matrices [2], and resistance to breakage or decohesion during severe plastic deformation [3]. Thus, the presence of quasicrystals in metallic matrices can increase both the strength and ductility of the alloys; and this was also found for the quasicrystal-strengthened Al-alloys [4]. However, an icosahedral quasicrystalline phase (IQC) in these alloys is metastable [5]. Thus, the alloys have to be fabricated by rapid solidification and then consolidated. However, the alloying of Al-Mn alloys with, for instance, Be, Si and Fe, can significantly stabilize the IQC-phase [6-9], allowing formation of IQC at much lower cooling rates. The strengths of these alloys are high even in the as-cast state [4, 10]. However, in many cases there is still room for enhancing the strength by an appropriate heat treatment.

In this work, we present the effect of heat treatment of the alloy Al-Mn-Be-Cu, in which IQC can form during melt spinning [11] and casting into a copper mold [12]. Upon heat treatment, different types of precipitates appeared [13]. The main purpose of this work was to determine the crystallographic orientations between precipitates and aluminum matrix, and discern possible differences in the precipitation behavior of melt-spun and gravitationally cast samples.

Experimental

The investigated alloy contained 4.24 wt% Mn, 0.68 wt% Be, 4.44 wt% Cu and 90.64 wt% Al, as determined using AES-ICP (Atomic Emission Spectroscopy – Inductively Coupled Plasma). The alloy prepared using vacuum induction melting in alumina crucible and cast. Afterwards, the alloy was melt-spun using a laboratory melt spinner to achieve cooling rates in the order of 10^5–10^6 K/s, or cast into a copper mold to achieve cooling rates around 1000 K/s. The samples were heat-treated by direct artificial aging at 200°C, 300°C, 400°C and 500°C for 24 hours in the air (T5 treatment). XRD (X-Ray Diffraction) was carried out using synchrotron X-rays with a wavelength of 0.1 nm (Sincrotrone Elettra, Trieste, Italy). Lamellae for transmission electron microscopy (TEM) were prepared using focused ion beam (FIB by FEI, Helios) for the as-cast and heat-treated specimens. High-resolution TEM (HRTEM), energy-filtered TEM (EFTEM), scanning TEM (STEM), electron energy-loss spectroscopy (EELS) and energy-dispersive X-ray spectroscopy (EDXS) were carried out in an FEI Titan 60-300 image corrected electron microscope.

Results and discussion

Fig. 1 shows the microstructures after melt spinning and casting into the copper mold. The melt-ribbons contained primary quasicrystal particles with sizes up to 0.5 μm, and smaller Al_2Cu particles (Fig. 1a), whereas the particles in the mold-cast sample were larger (Fig. 1b). The primary IQC-particles were few micrometers in size. Some had a polygonal shape while others were branched. The microstructure also contained a binary eutectic (Al_{ss} + IQC), and dendrites of Al_{ss}. At grain boundaries, the binary ((Al_{ss} + Al_2Cu) was also present.

The hardness of the melt-spun ribbons and castings were rather similar in the as-cast state (Fig. 2). It remained the same even after heat treatment at 200 °C, whereas the hardness of melt-spun ribbons was considerably higher than that of the mold cast-samples at higher temperatures. The hardness of the Al-Mn-Be-Cu melt-spun ribbons increased by 50% in comparison with the as-cast state by heat treatment at 300°C and 400°C. Such a strong effect may arise from a high supersaturation of the solid solution matrix with the alloying elements obtained by high cooling rates. Consequently, a high precipitate density can be formed in the supersaturated matrix during heat treatment.

During the heat treatment, several types of precipitates can form in Al_{ss} [13]. Fig. 3 shows the precipitates in the samples heat-treated for 24 h at 400 °C. The comparison of TEM-micrographs of melt-spun and mold-cast samples reveals much smaller

precipitates in melt-spun ribbons, which is the main reason for their higher hardness.

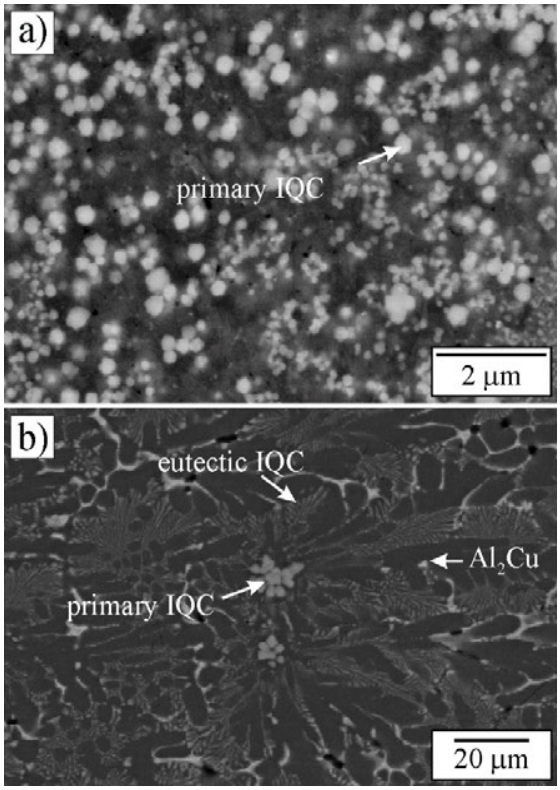

Fig. 1: Backscattered electron micrographs of the alloy Al-Mn-Be-Cu after (a) melt spinning and (b) after casting into a copper mold.

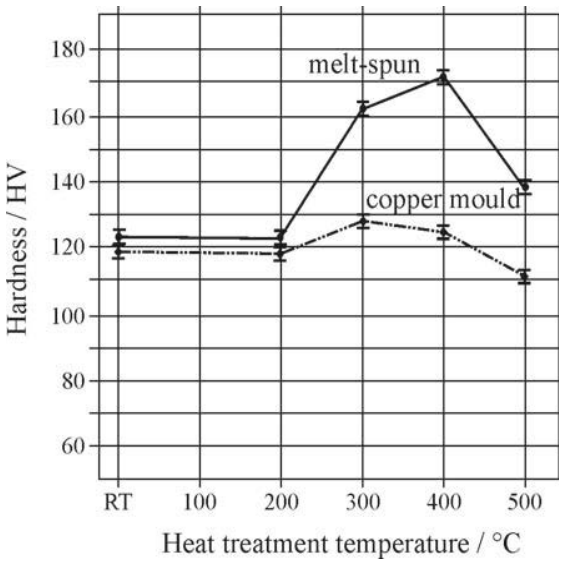

Fig. 2: The room temperature hardness of the melt-spun and mold-cast samples in the as-cast condition and after heat treatment at different temperatures for 24 h.

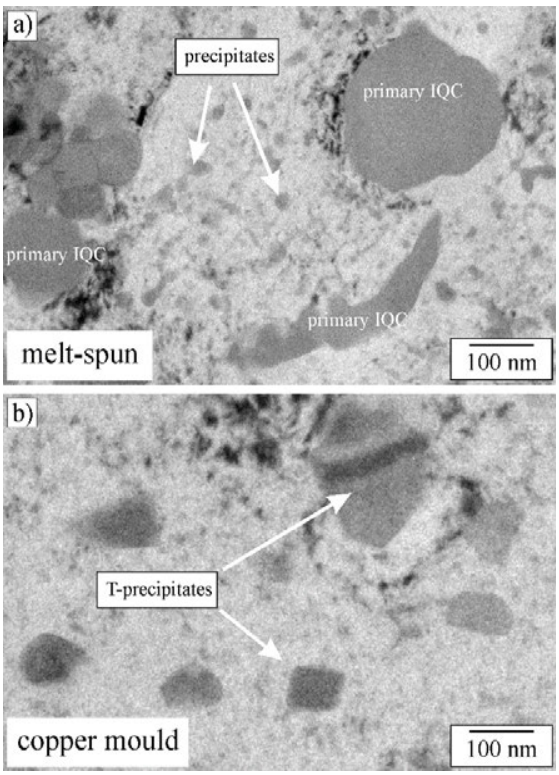

Fig. 3: Bright-field TEM micrographs of heat treated samples at 400°C for 24 h a) melt-spun ribbon and b) mold-cast sample

Θ'-precipitates

Θ'-precipitates are the same precipitates as formed in the binary Al-Cu alloys. In the mold-cast sample formed at 200 °C, while just a few of them were observed in the melt-spun ribbons at 300°C only. The EDS showed that they composed only of Al and Cu, having a composition of Al_2Cu, with the following crystallographic characteristics: $I\bar{4}m2$; a = 0.404 nm, c = 0.58 nm [14].

Fig. 4a shows an HR-STEM micrograph of a Θ'- precipitate within Al_{ss}-matrix. The Fast Fourier Transform (FFT) revealed that the zone axis of the matrix was [110] (Fig. 4b). The inspection of the remaining spots clearly showed that the precipitate was oriented within a [001] zone axis. The planes (001) of Al_{ss} and (110) of θ' were parallel to each other, and the θ'-precipitates formed on (001)-planes of Al_{ss}. The interplanar distances of planes $\{220\}$ from both phases were equal, making possible the formation of a (semi)coherent interface between both phases on (001)-plane of Al_{ss}, which is consistent with previous works [14, 15].

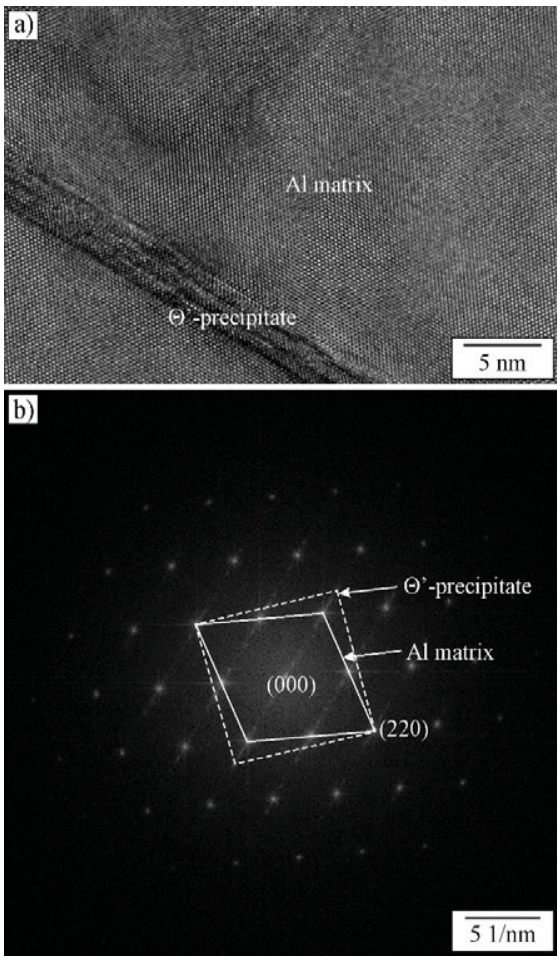

Fig. 4: Θ'-precipitate in a melt-spun ribbon after 24 h heat treatment at 200 °C. (a) The HR-STEM image, b) the corresponding FFT.

ICQ-precipitates

IQC-precipitates were identified in the melt-spun ribbons heat-treated at 300°C and 400°C, and only at 300°C in the mold-cast samples. The HR-STEM-image of one precipitate is shown in Fig. 5a, and its corresponding FFT in Fig. 5b. The FFT of the precipitate is not periodic. The distances between spots increased with the golden mean $\tau\,((1+\sqrt{5})/2 \approx 1.62)$, which is typical for icosahedral quasicrystals. The quasicrystal is oriented along one of its threefold axes. A detailed investigation of the mutual orientation with the Al_{ss}-matrix showed that a threefold [111] axis of Al_{ss} was apparently parallel with the threefold axis of IQC. In addition, it was further revealed that the orientations and interplanar distances of IQC $\{3\,2\,2\,111\}$ planes were the same as for planes $\{2\,2\,0\}$ of Al_{ss}. The threefold axis of Al_{ss}: [111] was parallel to the threefold axis of IQC with indices $[\,0\,1\,1\,0\,\bar{1}\,0\,]$ [16]. This orientation relationship allows nice matching between the matrix and the precipitate in the HRTEM-image. Good matching allowed the creation of a low energy interface between IQC and Al_{ss},. IQC-precipitates have been found in several alloys [17-20].

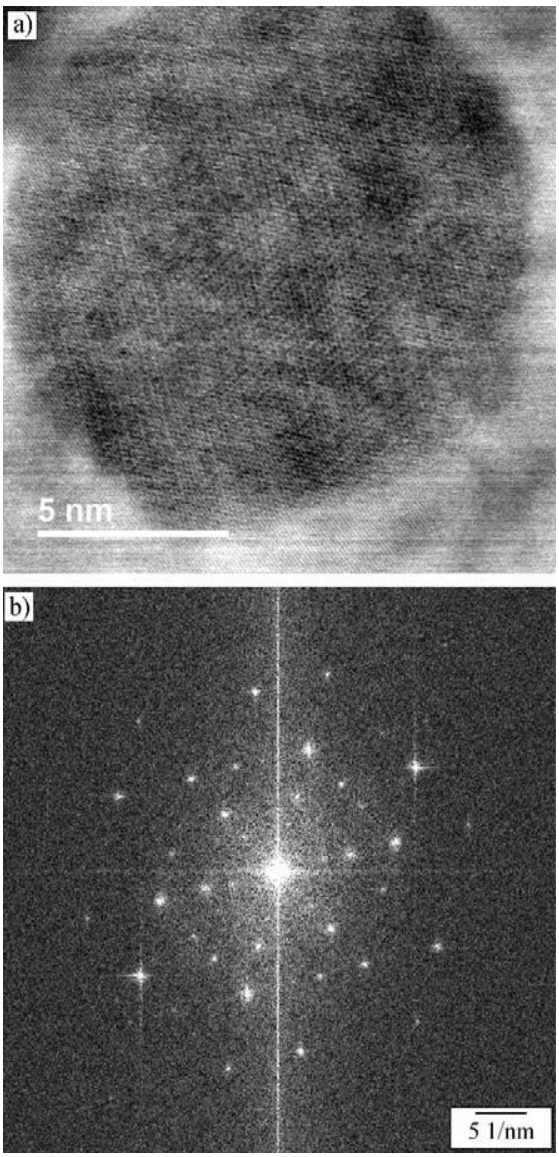

Fig. 5: IQC-precipitate in a melt-spun ribbon after 24 h heat treatment at 400 °C. (a) HR-STEM image of a precipitate in the Al-matrix, and (b) its corresponding threefold diffraction pattern.

T-precipitates ($Al_{20}Mn_3Cu_2$)

Fig. 7a shows a precipitate that was about 50 nm thick. Its FFT (Fig. 7b) exhibits a rectangular lattice, which corresponds to the [100] zone axis of T- $Al_{20}Mn_3Cu_2$. The longitudinal axis of the particle lied parallel to its [0 1 0] direction, the same as was found by Chen et al. [21]. Figs. 7c and 7d show the HR-STEM image and its FFT along the [011] zone axis. In this orientation, the atomic resolution was unobtainable probably because of the tiny distances between atomic columns (Fig 7c). However, it revealed a complex structure as the unit cell possesses a parallel tessellation of similar hexagon subunits, encompassing 156 atoms [22].

In all cases, the T-phase grew preferentially along its *b*-axis. The lattice mismatch between (060)-T and (020)-Al_{ss} was only 3%. Chen et al. [21] found three orientation relationships between T-phase and Al_{ss} that may provide many possibilities for good matching of the side interfaces.

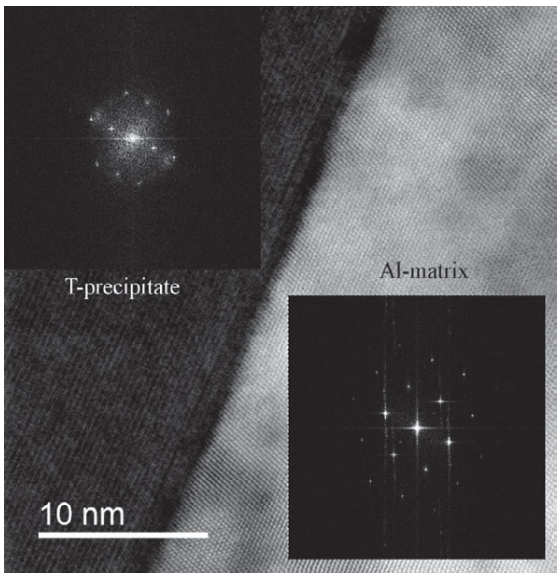

Fig. 6. The HR-STEM micrograph of the interface between T-precipitate and Al-matrix, with corresponding FFT's.

Conclusions

In an quasicrystalline aluminum alloy, Al-Mn-Be-Cu different precipitates formed during heat treatment, mainly dependent on the heat treatment temperature. Three types of precipitates were identified: Θ'-Al_2Cu, icosahedral quasicrystalline (IQC) and T-precipitates ($Al_{20}Mn_3Cu_2$).

The sizes of precipitates were increasing with the rising temperature.

All types of precipitates had specific orientation relationships with the Al-matrix, which allows them to form low-energy interfaces. The optimization of heat treatment is necessary to obtain enhanced mechanical properties and increase the possibility for their practical applications.

Acknowledgement

This work was partly financed by the research program P2—0120 (Slovenian Research Agency – ARRS). The TEM investigations at TU Graz have received funding from the European Union Seventh Framework Program under Grant Agreement 312483 - ESTEEM2 (Integrated Infrastructure Initiative–I3).

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Light Metals 2016

ALUMINUM ALLOYS, PROCESSING AND CHARACTERIZATION

Joining Technologies

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Important considerations for laser marking an identifier on aluminum

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Keywords: Aluminum, laser, marking, identifier

Abstract

Industrial laser marking of aluminium is becoming a key technology for both metal producers and transformers. Its low maintenance and consumables costs due to the fiber-based laser sources, combined with the speed and precision of the marking process makes it a very attractive tool for such industries. This paper aims to explain how to obtain and maintain good contrast laser identifiers on aluminium at various temperatures. The most important laser parameters (spot size, marking speed and line spacing) are analyzed with regards to their effect on the achievable contrast. The results indicate a significant variation of the optimal parameters for the studied temperature range, i.e. between 25°C and 400°C. The marking system tolerance to the piece positioning is also analysed for two different optical configurations.

Introduction

In 2015, parts traceability is now known as a necessary process in many manufacturing industries. Among others, industrial sectors such as automotive, aeronautics and medical have very strict rules on parts traceability. In order to meet these requirements, manufacturers have different choices of technology that can be used. Direct inkjet marking, micro-percussion drilling, part labelling and laser marking are the main technologies available. Compared to the other tools, laser marking is attractive because of its low maintenance, the absence of consumables and the high precision and robustness of the marks.

Laser marking an identifier on a piece of metal such as aluminum is usually made with a laser emitting at a wavelength of 1.06 µm. This wavelength is the preferred choice because of the relatively high absorption of the light by the metal. Generally, there are two choices of industrial laser sources that can produce this wavelength: Nd-YAG and fiber laser. Nd-YAG lasers are solid state lasers that use flash lamp to pump a crystal and typically generate a few ns pulsed train. Fiber lasers use an optical fiber as the gain medium which is pumped by laser diodes and emits light in different pulsed duration as well as in continuous wave (CW). Fiber laser based marking systems are now the best choice for industrial laser marking on metal since they require no maintenance and have a very high expected lifetime of approximately 100,000 hours. The two most common fiber laser sources type used for industrial metal marking applications are Q-switched and Master Oscillator Power Amplifier (MOPA).

Choosing the right laser marking process to produce a high contrast identifier on aluminum requires considering many aspects of the application. Besides the fundamental characteristics that the laser source should exhibit, such as short pulses, high pulse energy and highly absorptive wavelength, important considerations on the particular characteristics of the application must be taken into account in order to obtain and maintain excellent marking and robust detection.

In this article, we studied the performance of a 100 W average power Q-switched pulsed fiber laser operating at 1.06 µm wavelength in regards of the achievable contrast when marking on aluminum at various temperatures. We also pointed out some of the most important parameters and their impact on contrast. The results show important considerations for anyone who wants to laser mark and identifier on aluminum.

Experimental setup

Laser marked identifiers are realized with a combination of two different laser marking processes. The first process consists of a whitening of the background of the piece and the second is the realization of dark marks on the top of this white background. Figure 1 shows an example of a laser marked identifier on aluminum.

Figure 1: Laser marked identifier on aluminum alloy

The contrast of the laser marked identifier is defined by the grey level of the white (perfect white has a grey level of 255) minus the grey level of the dark (perfect black has a grey level of 0). Although other parameters are necessary for a good detection of the identifier, having a good contrast is a very important aspect of a good marking.

The experimental setup used to realize the laser marking operation is shown on Figure 2.

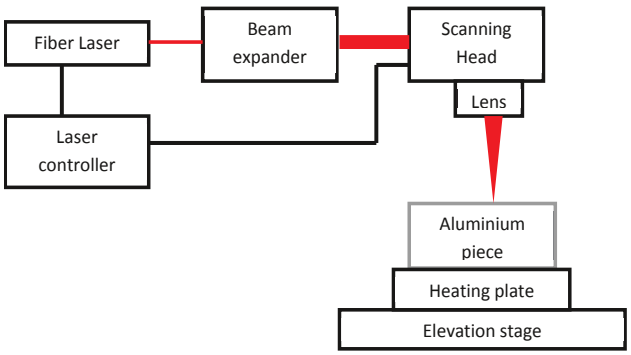


Figure 2: Schematic of the experimental setup

For each temperature studied, i.e. 25°C, 100°C, 300°C and 400°C, a matrix of squares is marked by varying the marking speed and line spacing between each square. Each square within this matrix is the result of the partial superposition of several lines separated by a specific line spacing. They are all produced with a different combination of marking speed and line spacing values. Figure 3 shows an example of such matrix realized for both the background (left) and the identifier (right).

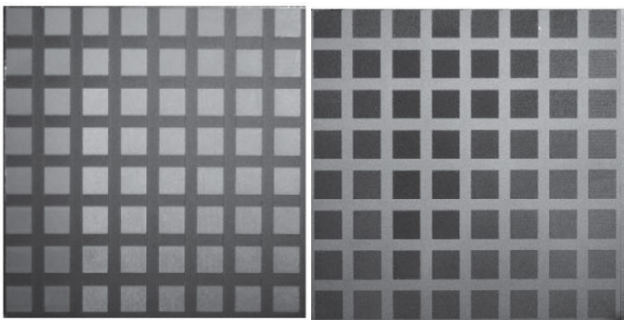


Figure 3: Matrix of squares used for finding the optimal marking parameters for the background (left) and for the identifier (right)

The white coloration background between the black squares of the identifier optimization matrix (Fig. 3, right) is made with the optimal parameters found in the background optimization matrix (Fig. 3, left) for each temperature. In order to measure and compare the contrast obtained, we took a photograph of each matrix and analyzed the images with an algorithm developed under Matlab. The photographs were taken under a white light illumination at a 20 degree angle from the vertical. This angle was found to give the highest level of contrast. The time required to mark the squares was recorded in order to have a time base to compare each contrast obtained.

The study was realized using a 100 W average power Q-switched laser for two different lens focal length, 160 mm and 420 mm, that lead to theoretical spot size values of 43 µm and 112 µm, respectively. Table 1 indicates the characteristics of the laser source.

Laser source type	Q-Switched
Manufacturer	IPG Photonics
Model	YLP series
Max average power	100 W
Cooling	Air
Pulse duration	100 ns
Pulse repetition rate	100 kHz
Pulse energy	1 mJ
M^2	1.6

Table 1: Fiber laser characteristics

Results

The results shown in this section were obtained with the use of alloy 6061 as the target material. The tests were also realized with alloy 5052, without observing a significant difference of the results. The study was also made with two different lenses, one with a focal length (FL) of 420 mm and the other with a FL of 160 mm. The results show higher contrast and much lower marking time with the 420 mm FL, so the results are shown only for this configuration, except in the section about the positioning precision impact, where we graphically highlight the clear advantage of the 420 mm FL.

Marking speed and line spacing

We first studied the effect of the marking speed and the line spacing with a 420 mm FL at 25 °C. The results are shown of Figure 4. We plotted the contrast obtained for various marking speed, each point of the curves representing a different line spacing. The *x*-axis is a time base showing the time required to mark one square centimeter of black identifier.

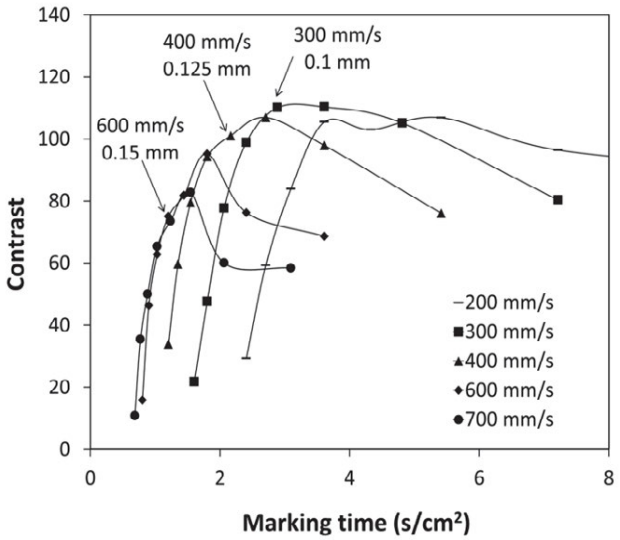


Figure 4: Contrast *vs* marking time for 420 mm FL and 25 °C material temperature

In this picture, we see that the optimal parameters to achieve the highest contrast depend on the time allowed for marking the identifier. The labels in Fig. 4 shows examples of three different combinations of marking speed and line spacing that are the best set of parameters for some time base requirements. It is also interesting to note that more time allowed does not necessarily lead to higher contrast. In fact, there is an optimal time to achieve

the best contrast, which is around 2.9 s/cm^2, for this set of temperature and FL. It is achieved with a 300 mm/s marking speed and a 0.1 mm line spacing.

Temperature

In order to study the influence of the material temperature on the optimal marking parameters, we repeated the experiment of the previous section for three other temperatures. The same behavior was observed for the higher temperatures, i.e. the best parameter set (marking speed and line spacing) varies depending on the time allowed for marking. However, we found out that the optimal parameters depend on the material temperature and that the optimal marking time in second per square centimeter decreases when the material temperature increases. This could be simply explained as follows: When the material temperature is higher, less laser energy is required to reach the sweet marking spot. If we use the same laser power, this phenomenon leads to a reduction of the time required to apply the required energy on the material. Figure 5 shows the curves obtained and Table 2 enumerates the optimal marking parameters along with the resulting contrast for each temperature studied. When the material temperature increases from 25 °C to 400 °C, the optimal time decreases from 2.9 s/cm^2 to 1.7 s/cm^2.

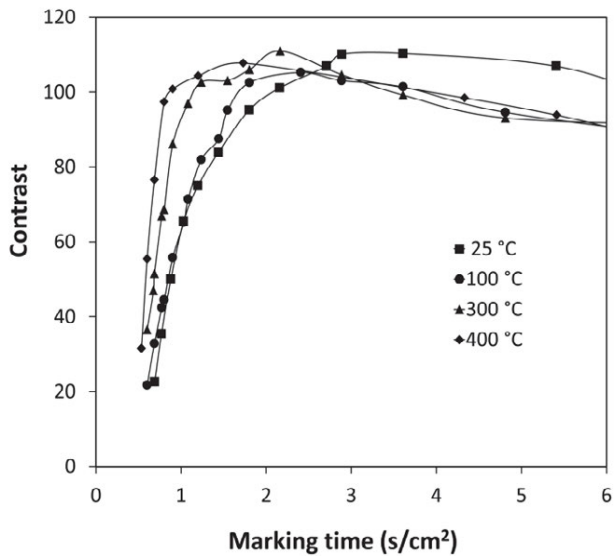


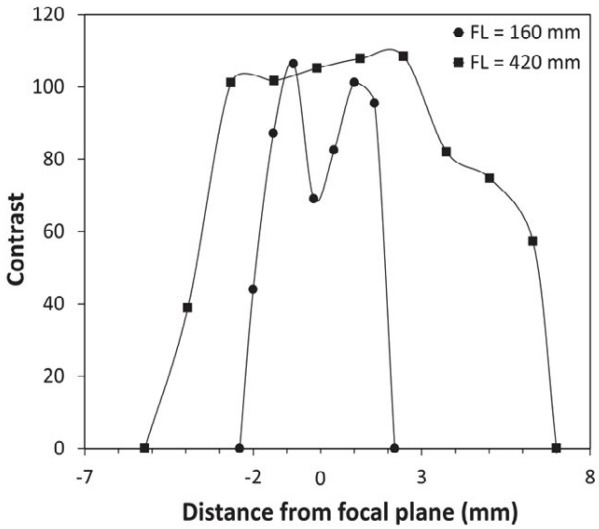
Figure 5: Contrast vs marking time with 420 mm FL for various temperatures

	Material temperature			
	25 °C	100 °C	300 °C	400 °C
Marking speed	300 mm/s	300 mm/s	400 mm/s	500 mm/s
Line spacing	0.125 mm	0.15 mm	0.125 mm	0.125 mm
Time per unit area	2.9 s/cm^2	2.4 s/cm^2	2.2 s/cm^2	1.7 s/cm^2
Contrast	110.3	105.2	111.0	107.7

Table 2: Optimal contrasts and marking parameters for various temperatures

Positioning precision impact

The piece positioning process is inherent to any laser marking application and it is important to understand the effect of its precision to the achievable contrast. In order to do so, the effect of an off-focus positioning of the piece was studied by realizing the optimization matrices at various locations over and under the focal plane. The results are shown on Figure 6 for the two FL studied, i.e. 160 mm and 420 mm.



It is clear from this graph that the 420 mm FL offers a much larger effective positioning range that the 160 mm. Actually, the effective marking range is approximately 6 mm (± 3 mm) with the 420 mm FL compare to 3 mm (± 1.5 mm) for the 160 mm FL. Another interesting information shown by this figure is that the optimal contrast obtained at the focal plane for the 160 mm FL is much lower than the one obtained with the 420 mm FL. We can also observed that offsetting the target surface from the focal plane either in the plus or minus direction increase the optimal contrast for the 160 mm FL when it is not the case for the 420 mm FL. These observations points out that the highest level of contrast achievable is dependent on the spot size of the laser beam, which is controlled by the optical configuration, including the FL of the scanning lens used in the marking system. The contrast increase observed for the 160 mm FL when the surface is off set from the focal plane is caused by a bigger spot size that becomes close to the spot size obtained with the 420 mm FL, so that is why the contrast becomes similar to the one obtained with the 420 mm FL.

Discussion

Generally, the time allowed for marking an identifier on a piece of aluminum is constant from piece to piece on a same production line. This means that the optimal parameters can be found during the process development and then store in the laser marking system memory. If the temperature of the material differs from piece to piece, different laser parameters should be selected in order to keep optimal contrast identifiers. These parameters sets should be determined and then stored as various jobs within the laser marking system controller. The piece temperature can then be read in-line with an appropriate sensor and the appropriate job is selected according to the sensor reading.

Various optical configurations could be used to obtain a 112 µm spot size. In fact, the spot size diameter (Θ) is dependent of 4 parameters: The quality of the laser beam defined by its M^2 value, the wavelength (λ), the input beam diameter on the focusing lens (D) and the focusing lens FL (f). It is given by the following formula [1]:

$$\Theta = \frac{\lambda f M^2}{\pi D}$$

We found out that for a 1.06 µm fiber laser having an M^2 value around 1.6, using a 10 mm input beam size combined with a 420 mm FL scanning lens is a very good choice for an industrial laser metal marking application. The 420 mm FL is long enough to allow an efficient and relatively easy removal of the metal dust produced by the marking process. This helps to keep the lens or the protective window away from contamination that will cause a deterioration of the quality of the identifier and may also damage the laser marking system due to back reflections. The 10 mm input beam diameter is a good choice for two reasons: It is large enough to generate a laser intensity on the optical components that is well below the damage threshold of standard coatings, even for high energy pulses, and it is small enough to allow using relatively small scanning mirrors, which is important to preserve very high speed scanning capability.

Another important advantage of using a relatively large spot size such as 112 µm is that this increases the effective marking range below and above the focal plan. This is very important to ease piece positioning within the effective marking range.

Finally, the laser maximum average power is a crucial parameter because it will drive the price of the solution. It is important to realize proper testing to determine how much laser power is necessary for your application, remembering that the faster you want to mark, the more power you will need.

Conclusion

Laser marking an identifier on aluminum can be efficiently done with a Q-switched fiber emitting a 1.06 µm wavelength. The optimal marking parameters such as marking speed and line spacing are dependent on the time allowed to realize the marking process: Less time available points out higher marking speed and bigger line spacing values. The piece temperature also has an impact on the optimal laser marking parameters: The warmer the piece is, the lower the amount of energy required for the optimal contrast result.

In addition, the optical configuration is a crucial characteristic to be optimized to obtain high contrast identifiers: A 10 mm entrance beam combined with a 420 mm FL lens, which lead to a theoretical spot size of 112 µm, is a very good choice for laser marking an identifier on aluminum.

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Light Metals 2016

ALUMINUM ALLOYS, PROCESSING AND CHARACTERIZATION

Poster Session

Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

CORROSION RESISTANCE OF DIFFERENT ALUMINUM ALLOYS IN ETHANOL

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Keywords: Aluminum-Silicon alloys, Aluminum-Magnesium alloys, Ethanol, Corrosion, Supported electrolyte.

Abstract

The aim of this work is to study the corrosion behavior of Aluminum alloys with silicon and magnesium, by potentiodynamic polarization tests and Electrochemical Impedance Spectroscopy in bioethanol solution (92º - 96º), room temperature and pH 6.5, evaluating the effects of the addition of CH_3COOK and ClO_4Li as supported electrolytes. The results show that adding ClO_4Li to the alcoholic solution induces pitting corrosion, possibly reducing ClO_4^- to Cl^-. Whereas that, CH_3COOK causes inhibition of the corrosion, and increasing of passivation of the metal, a phenomenon which is attributable to the generation of corrosion products.

Introduction

The current global community has a marked dependence on petroleum, which directly generates social, political and economic problems. The increased usage results in the shortages and environmental problems by. [1,2].

There are different alternatives that are projected as potential solutions to these negative impacts, highlighting biofuels, renewable energy produced from biological sources. Bioethanol is a fuel obtained by fermentation of sugar-rich vegetable juices, and is one of the main substitutes for fossil fuels, especially gasoline [3]. For handling, production and handling, the implementation of materials with suitable mechanical and physicochemical properties is necessary, appearing as an option the Aluminum base alloys [4, 5].

Corrosion of materials exposed to organic solvents such as ethanol has been studied for several years already, and has demonstrated the complexity of these analyzes, mainly in the electrochemical tests given the low conductivity of the medium [5-7].

There are different methods to achieve correction of the ohmic drop generated, and one of them is the addition of soluble salts capable of minimizing the resistance of the medium remaining inert [8-10]. It is important to note that the electrochemical behavior of metal / solution system is directly dependent on the contaminants or foreign substances, such as water, acetic acid, chlorides and metals, among others, that can interact with the supporting electrolyte [11].
The aim of this work is to study the electrochemical behavior of different Aluminum alloys when these are exposed to ethanol and add to it as supporting electrolyte salts, by conducting potentiodynamic curves and contrast results with tests Electrochemical Impedance Spectroscopy.

Experimental

The materials tested were Aluminum based alloys with silicon (Al-10wt.% Si and Al-12wt.% Si) and magnesium (Al-2wt.% Mg and Al-4wt.%Mg).

The surfaces of the specimens were prepared by grinding using papers of SiC with different particle size (# 600 to # 1500), this process is performed 15 minutes before each corrosion test to unify the surface preparation of all samples.

The electrolyte used was bioethanol (92º-96º, pH = 6), obtained from sugar cane of sugar refinery in San Javier, located in the province of Misiones, Argentina. From the same two additional solutions were prepared: the first, ClO_4Li dissolving analytical grade up to a concentration of 0.1 M (EtOH + ClO_4Li); and the second dissolving CH_3COOK up to the same concentration, pH corrected by addition of CH_3COOH (about 1 ml per 300 ml solution) (EtOH + CH_3COOK).

Potentiodynamic measurements were carried out using an electrochemical cell of three electrodes and Gamry Reference 600Y potentiostat, a potential sweep rate of 0.16 mV/s, + 500mV between about -300 mV and the open circuit potential (ECA) at 25 °C, using Ag / AgCl / KCl (sat) reference electrode.

Prior to testing, the specimens were exposed to the solutions for one hour, and then held the EIE experience using a Gamry potentiostat Reference 600, the system applying a sinusoidal voltage ± 10 mV in a frequency range of 100 kHz to 0.001 Hz and 25 °C.

Results

Figures 1 and 2 give the results of polarization tests for different Aluminum alloys and solutions.

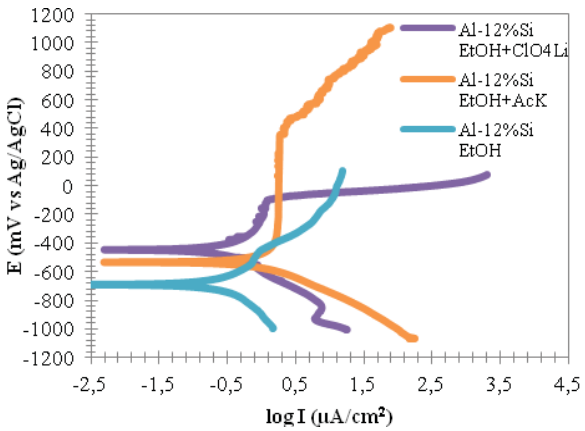

Figure1. Potentiodynamic polarization curves for Al-12wt.%Si alloy in ethanol (EtOH), and ClO_4Li + ethanol, and ethanol + CH_3COOK.

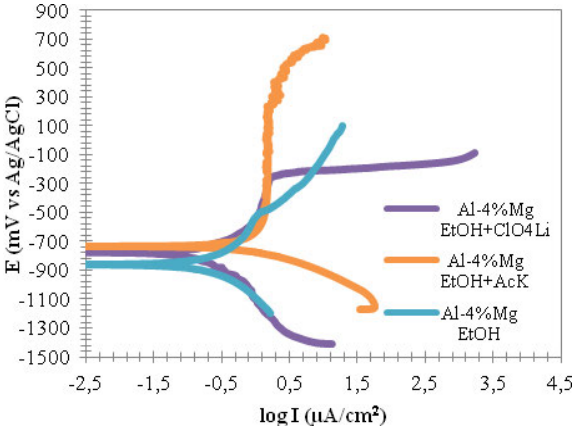

Figure 3. Potentiodynamic polarization curves for Al-4wt.%Mg alloy in ethanol (EtOH), and ClO_4Li + ethanol, and ethanol + CH_3COOK.

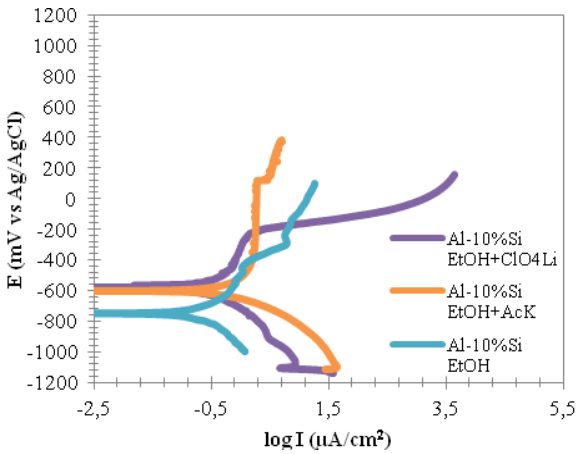

Figure2. Potentiodynamic polarization curves for Al-10wt.%Si alloy in ethanol (EtOH), and ClO_4Li + ethanol, and ethanol + CH_3COOK.

In Figures 3 and 4 potentiodynamic curves for Aluminum base alloys with magnesium in all solutions are represented.

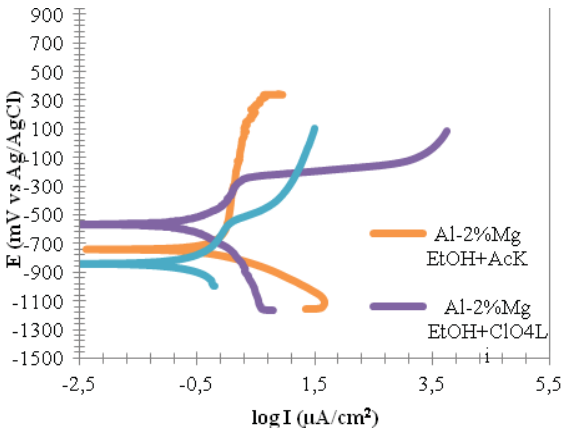

Figure 4. Potentiodynamic polarization curves for Al-2wt.% Mg alloy in ethanol (EtOH), and ClO_4Li + ethanol, and ethanol + CH_3COOK.

The corrosion potentials of potentiodynamic curves obtained are presented in Table 1.

Table 1. Corrosion potential (Ecorr, in mV) for all alloys, exposed to different solutions.

Alloys	Solutions		
	Ethanol	**Ethanol+ClO_4Li**	**Ethanol+CH_3COOK**
Al-12wt.%Si	-693.2	-446.4	-534.4
Al-10wt.%Si	-750.5	-581.7	-602.3
Al-4wt.%Mg	-861.6	-773.9	-735.3
Al-2wt.%Mg	-844.3	-556.8	-742.5

From the curves obtained (Figure 1, 2, 3 and 4) different behaviors as alloys and electrolytes supports used were observed.

The polarization curves were obtained using ClO_4Li as supporting electrolyte generally showed an increase in the corrosion potential (Ecorr) compared to studies with ethanol only. This may be due to the fact that Li^+ ions can react with both the dissolved oxygen, as with corrosion products and impurities of the medium, forming a layer that can adhere to the surface [11], and elevating the corrosion potential to noblest values.

Furthermore, since the solution was prepared with a considerable concentration of supporting electrolyte, and also, that the potential for trial reduction of ClO_4^- to Cl^- ion (Eq. 1) [12] becomes possible. In all cases were observed a lower pitting potential and a rapid increase in current, in comparison with the curves without supporting electrolyte solution.

$$ClO_4^- + 8H^+ + 8e^- \rightleftharpoons Cl^- + 4H_2O \quad (1)$$

For the particular case of the Al-4wt.%Mg alloy with ClO_4Li solution, the increased in Ecorr was slightly lower compared to the change in other alloys (see Table 1), because it has a higher concentration of magnesium in the alloy, the reaction thereof with the dissolved oxygen competes with the formation of oxides of lithium phenomenon.

Analyzing the tests in ethanol with CH_3COOK, all curves described both an increase of corrosion potential, Ecorr, respect to obtained only in ethanol (see Table 1), as well as an increase in the passivation of material on the results of analysis without addition of support electrolyte. This variation is caused by the adsorption of composites which are formed from corrosion products combined with acetate ion from the electrolyte support [10].

Comparatively was denoted that Aluminum alloys with silicon basis, the increase in the passivation range was less so than in the case of Aluminum based alloys with magnesium. In addition, alloys with higher concentration of alloying (Al-12wt.%Si and Al-4wt.%Mg), recorded a further increase in the range of passivation.

For all tests, it should be noted that impurities from ethanol distillery can also generate undesirable interactions with the electrolyte and so consequently modify the measurements.

Responses to EIE assays are described in Figures 5 and 6.

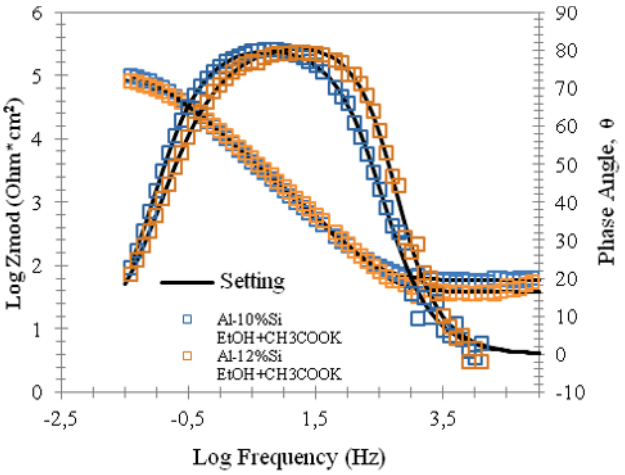

Figure 5. Bode diagram for aluminum base alloys with silicon in ethanol + CH_3COOK.

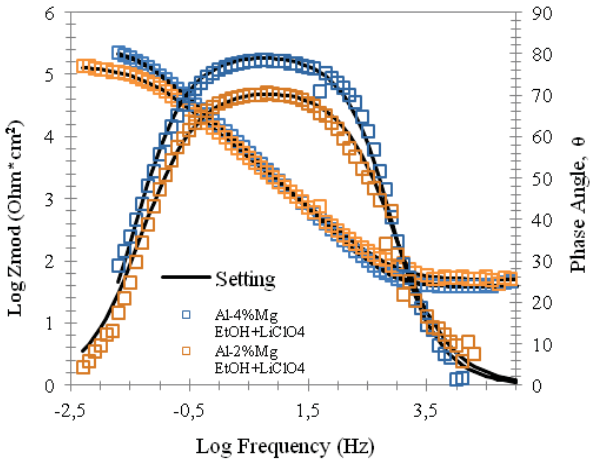

Figure 6. Bode diagram for Aluminum base alloys with magnesium in ethanol + $LiClO_4$.

By adjusting the results, the electrical circuit that represents the alloy / solution interface for all combinations of basic Aluminum and alloys in bioethanol supporting electrolyte was obtained (see Figure 7).

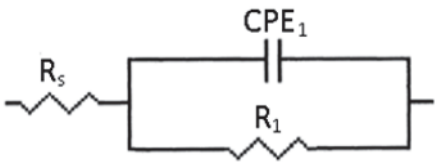

Figure 7. Equivalent electrical circuit for system Alloys / Bioethanol + electrolytes support.

The values of the parameters for each component of the circuit system can be observed in Table 2.

Table 2. Parameters for alloys from the adjusted circuits.

		Rs ($\Omega.cm^2$)	R_1 ($\Omega.cm^2$)	CPE_1 ($\Omega^{-1}cm^2s^n$)	n_1
Ethanol + CH_3COOK	Al-2%Mg	8.30E+01	1.01E+05	1.65E-05	7.82E-01
	Al-4%Mg	1.08E+02	1.54E+05	1.24E-05	8.61E-01
	Al-10%Si	5.83E+01	1.13E+05	1.19E-05	9.13E-01
	Al-12%Si	4.71E+01	8.35E+04	1.08E-05	9.18E-01
Ethanol + $LiClO_4$	Al-2%Mg	4.91E+01	1.34E+05	1.53E-05	8.03E-01
	Al-4%Mg	3.88E+01	2.52E+05	1.03E-05	8.92E-01
	Al-10%Si	5.37E+01	1.36E+05	1.55E-05	8.48E-01
	Al-12%Si	2.17E+01	5.86E+04	1.32E-05	8.99E-01

Comparing the results of the above table indicates that the resistance of the surface layer (R_1) increases as the test is performed using $LiClO_4$ as supporting electrolyte for cases of Al-2wt.%Mg, Al-4wt.%Mg alloys, and Al-10wt.%Si. The opposite effect happening in the case of Al-12wt.%Si alloy.

The aluminum base alloys with silicon have a coefficient of identity (n) close to unity when the test is done in CH_3COOK + ethanol, showing a resistive layer formed with less porous.

Summary and Conclusions

From the analysis of results the following conclusions can be extracted:

- by potentiodynamic polarization tests was demonstrated that a resistive layers were formed by using support electrolyte to increase the conductivity of ethanol, showing a different behavior of the original system comprised solely of alloys in bioethanol.
- The application of electrochemical impedance spectroscopy technique allowed validates the formation of different layers with different characteristics when the material is analyzed in CH_3COOK and in $LiClO_4$.
- For the test conditions, it is not recommended for use CH_3COOK and ClO_4Li as electrolytes support, since they do not meet the condition of being inert towards the metal.

Acknowledgements

The authors thank: 1) San Javier sugar refinery, for their assistance in donating samples of regional bioethanol. 2) The FONCyT-ANPCyT by the subsidy received to fund this research (PICT-2012-2952). 3) G.R. Kramer thanked the CONICET through a scholarship for this work.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Refinement of Primary Silicon Crystals by Novel Al-ZnS Master Alloy in Solidification of Hypereutectic Al-Si Alloys

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Keywords: Hypereutectic Al-Si alloys, refinement, primary silicon, master alloy

Abstract

A novel Al-ZnS master alloy was developed by in situ reaction of Zn and Na_2S in the Al melt. The results from this study left little doubt that this novel Al-ZnS master alloy was a promising refiner in solidification of hypereutectic Al-Si alloys. It refined the primary silicon to the same extent as that achieved by adding P via Cu-P following the same refinement mechanism. The SEM work confirmed that there were many pre-formed ZnS particles contained in the master alloy. The optical metallography showed that the morphologies of the primary silicon crystals in the solidified Al-22Si alloy were drastically changed from irregular coarse morphology to fine regular particles. The primary silicon crystal was refined from 74 µm to 26, 22, 24 µm by adding 0.05, 0.1, 0.15 wt% ZnS respectively with holding time of 20 min, whereas the average particle size of the primary silicon refined by 200 ppm P reduced to 20 µm. With the increase of the holding time up to 1.5 hr, the primary silicon size will increase up to 33 µm in adding 0.1 wt% ZnS.

Key words Hypereutectic Al-Si alloys, refinement, primary silicon, master alloy.

Introduction

Hypereutectic Al-Si alloys are of increasing interest for applications that require a combination of light weight and high wear resistance, such as pistons, liner-less engine blocks and pumps. The wear resistance of this class of alloys is due to the presence of hard primary Si particles formed during casting [1, 2]. For hypereutectic Al-Si alloys, the final properties in the cast component are controlled by the size and distribution of the primary silicon phase in addition to the degree of modification of the eutectic silicon matrix [3]. To maximise machine tool life while taking advantage of the high wear resistance of hypereutectic Al-Si alloys, the primary Si crystals formed during solidification should be as fine and uniformly dispersed as possible. The current commercial practice is to add phosphorous in the form Cu-P, Al-Cu-P and Al-Fe-P master alloys, such that primary Si nucleates heterogeneously on AlP particles. The crystal structure and lattice parameter of AlP are close to that of silicon with the minimal mismatch between them, less than1% [4].

Refining agents like Ga, Ge, Se, Be, Te, Li, Cd, Zn, Mn, V, Cb, Bi, Mo, Hf and S either individually or in combination have also been studied. These agents along with Na would give simultaneous silicon refinement and eutectic modification [5]. A significant refinement of primary silicon in 390 alloy has been reported with the use of Germanium, Gallium, Selenium, Tellurium, Lithium, Cadmium, and Lithium Chloride [6].

Due to the structural similarity of ZnS in its zincblende form, and its similar lattice parameter to those of Si and AlP, its use as an inoculant to refine the primary Si in solidification of hypereutectic Al-Si alloys was examined in this study. ZnS occurs in two common polytypes, zincblende (also called sphalerite: cubic with lattice parameter ao =5.41 Å) and wurtzite (hexagonal with lattice parameters a_o = 3.2495 Å, c_o = 5.2069 Å). ZnS crystallizes with the cubic zincblende structure below 1020 °C and with the hexagonal wurtzite structure above this phase transition temperature [7]. The refinement efficiency of ZnS in solidification of the hypereutectic Al-Si alloys has been investigated. Experiments were conducted to refine the primary silicon particles in Al-Si alloy by adding the new Al-ZnS master alloy.

Experimental procedures

The initial batches of Al-Si alloys were prepared in an electric resistance furnace by melting and diluting an Al-50Si master alloy with commercial purity aluminium LM0 at 1100 °C for 3 h in a clay crucible. The compositions of raw materials are listed in Table 1. For homogeneity, the molten alloy was manually stirred for a few seconds and then cast.

Table 1 Aluminium alloys composition wt%

Alloy	Cu	Mg	Si	Fe	Mn	Ni	Zn	Pb	Sn	Ti	Cr	Al
LM0	0.03	0.03	0.30	0.40	0.03	0.03	0.07	0.03	0.03	-	-	Bal.
Al-50Si	0.08	0.28	51.0	0.32	0.02	0.01	0.02	0.02	0.01	0.09	0.03	Bal.

One synthesis method for zincblendeZnS nanoparticles is by chemical reaction using Zn and Na_2S [8]. Experiments were carried on to prepare the Al-ZnS master alloy by adding the stoichiometric amount of Zn metal followed by adding 50% excess of Na_2S powder to high purity aluminium melt with gentle mixing at 800 ºC for 20 min. The produced master alloy cast in cylindrical mould pre-heated to 200 ºC, and cooling in air at a cooling rate of approximately 16 K/s.

Another set of experiments was conducted to refine Al-22Si alloy by adding different amount of the Al-ZnS master alloy for different holding time. In order to compare with P refinement, the Al-Si alloy was remelted to 800 ºC and then P was added in the form of CuP shot (Supplied by Aura Metals Ltd). The molten alloy was manually stirred for a few seconds and sample was taken after 20 min. Samples for all above experiments were taken by a Boron Nitride coated steel mould (35mm in diameter and 40mm in height with long handle) preheated at 200 ºC which was then water cooled at approximately 15 K/s. Longitudinal sections of resulting samples were prepared by the standard technique of grinding with SiC abrasive papers and polishing with 1 µm

diamond suspension followed by silica suspension. The chemical composition of all alloys was analysed using a "Worldwide Analysis System (WAS) AG, Foundry master". Microstructure characterization was accomplished using an optical microscope (Carl Zeiss Axioskop 2 MAT) equipped with image analysis software. The scanning electron microscopy (SEM) examination was carried out using a Zeiss Supera 35 FEG and JEOL 840 microscopes. The microscope equipped with an energy dispersive spectroscopy (EDS) facility, operated at an accelerating voltage of 5-20 KV.

Results and discussion

Preparation of Al-ZnS Master Alloy

Al-ZnS master alloy was prepared by adding the stoichiometric amount of Zn followed by adding excess amount of Na_2S to high purity aluminium at 800 ºC. The SEM micrographs and EDX analysis confirm the formation of ZnS particle in the master alloy as shown in Figure 1. The higher Al content can be attributed to the small size of the ZnS particle and large beam size used for EDX measurement. It is clear that ZnS particle can be precipitated in the melt and its formation is according to the following chemical equation;

$$Na_2S + Zn^{+} \longrightarrow ZnS + 2Na^{+}$$

The same principle is used to remove Zn from the wastewater or etching solution as ZnS by adding Na_2S [9]. ZnS crystallizes with the cubic zincblende structure below 1020 °C and with the hexagonal wurtzite structure above this phase transition temperature [10]. Since, the master alloy prepared at 800 ºC so the precipitated particles are zincblendeZnS. From the chemical analysis of the master alloy and according to the chemical equation, the composition of the master alloy is Al-1.5wt%ZnS. Further work is in progress to separate these ZnS particles for full characterisation and optimize the master alloy composition and results will be reported elsewhere.

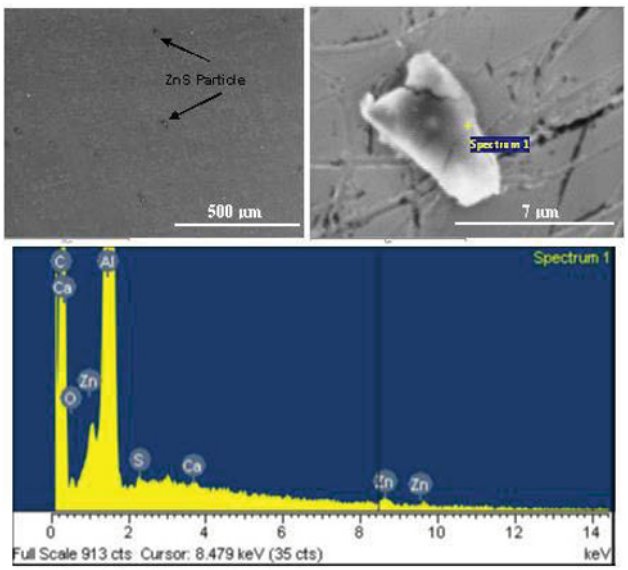

Figure 1 SEM micrographs and EDX analysis graphs showing the formation and structure of precipitated ZnS particles.

Refinement of Primary Silicon by Al-ZnS Master Alloy

Primary results encouraged us to use this master alloy to examine its refinement performance in solidification of the higher Si content of hypereutectic Al-Si alloys. The optical micrographs in Figure 2 show the refinement performance of Al-1.5ZnS master alloy in solidification of Al-22Si alloy. The untreated Al-22Si, Figure 2 (a&b) contained irregular coarse primary Si with average particle size of approximately 74 µm and because of high Ca content in commercial purity alloy (> 200 ppm) the eutectic Si had a mostly fibrous morphology.

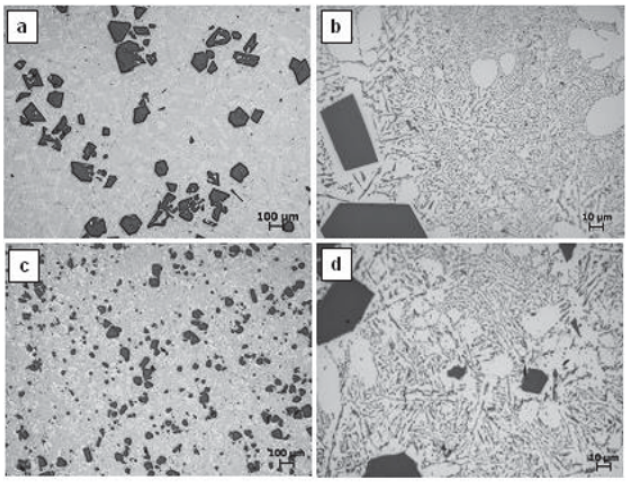

Figure 2 The optical micrographs of; (a, b) Al-22Si, (c, d) Al-22Si with the addition of 0.05wt% ZnS cast after 20 min. (a, c) low magnification and (b, d) high magnification.

Refinement with Al-1.5ZnS master alloy show that the particle size of primary silicon reduced to 26, 22 and 26 µm in adding 0.05, 0.1 and 0.16 wt% of ZnS respectively with 20 min holding time as shown in Figure 2 (c&d) and 3. The morphologies of primary silicon crystals in solidification of Al-22Si alloy are drastically changed from irregular morphologies to regular morphologies with some lose in the modification of the eutectic Si matrix.

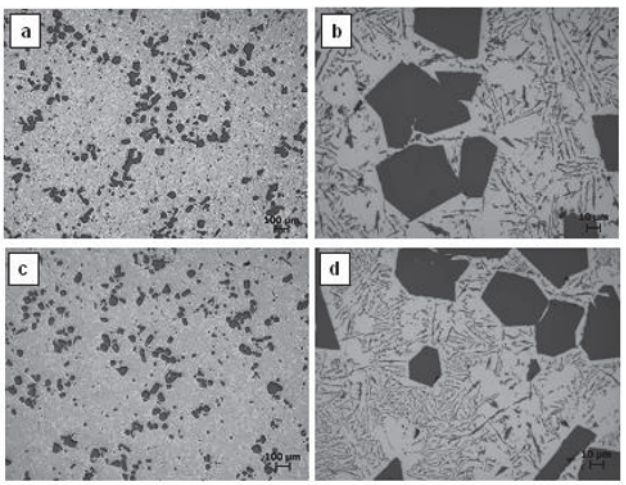

Figure 3 The optical micrographs of Al-22Si cast after 20 min; (a, b) Si with the addition of 0.1% ZnS, (c, d) with the addition of 0.16% ZnS (a, c) low magnification and (b, d) high magnification.

It was clear that with the increase of ZnS content the average particle size of the primary silicon decrease with up to 0.1wt% ZnS. Above 0.1wt% ZnS the size of primary silicon increases again, this could be due to the agglomeration of ZnS particles in the melt. With the increase of the holding time up to 90 min, the average particle size of primary silicon reduced from 74 μm to 33 μm by adding 0.1 wt% of ZnS in solidification of Al-22Si as shown in Figure 4.

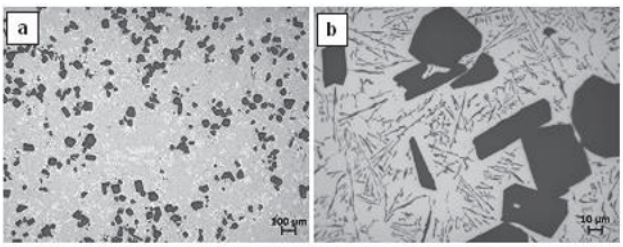

Figure 4 The optical micrographs of Al-22Si with the addition of 0.1% ZnS cast after 1.5 hr. (a) low magnification and (b) high magnification.

This inefficiency in the refinement of primary Si at high holding time could be due to the agglomeration of ZnS particles which then produce larger size if compared with that produced in 20 min holding time. Also, the high holding time leads to continuous removal of the Ca by oxidation, and then the eutectic Si matrix loses its modification.

The chemical composition of Al-22Si alloy after adding the master alloy (0.1%ZnS) analysed using a "Worldwide Analysis System (WAS) AG, Foundry master" showed that the Si composition reduced to around 20%Si because of dilution. Hence, to compare with P refinement, an experiment was conducted by adding 200 ppm of P to Al-20Si alloy. The average particle size of primary silicon was reduced to 20 μm, as shown in Figure 5.

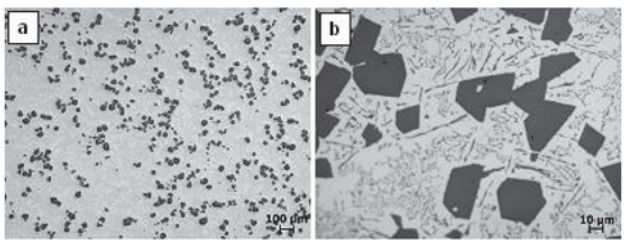

Figure 5 The optical micrographs of Al-20Si alloy with the addition of 200 ppm P. (a) low magnification and (b) high magnification.

In using P for refinement of primary silicon in hypereutectic Al-Si alloy, AlP particles will form in-situ which has a zincblende crystal structure with a lattice constant of 5.431 Å. These particles are suspended in the melt and act as potent sites for epitaxial nucleation of primary silicon [11].

It is very clear that the in-situ prepared zincblendeZnS have the same efficiency of P refinement of primary silicon and follow the same refinement mechanism.

Summary

The conclusions from the results of this study are:

1. According to the heterogeneous nucleation theory, ZnS particles act as potent substrate for nucleation of primary silicon. The lattice parameter of zincblende cube ZnS is 5.41 Å which is similar to that of silicon with a lattice mismatch of less than 1 %. The primary Si nucleates and then grows by wrapping around the ZnSnucleant to develop as a compact particle.
2. The results from this study leave no doubt that the novel Al-ZnS master alloy is a promising refiner in solidification of hypereutectic Al-Si alloys. It has the same efficiency of CuP in the refinement of primary silicon and follows the same refinement mechanism. It seems that there is no interaction between ZnS and the Ca exists in the melt and hence there was no drop in modification of the eutectic Si in commercial purity Al-Si alloy.

Acknowledgement

The authors wish to thank Dr. Kumar Sundaram (Department of Materials, University of Oxford) for his helpful discussions on the contents of this article.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECTS OF ALLOYING ELEMENTS ON MICROSTRUCTURE, MECHANICAL PROPERTIES AND FORMABILITY OF Al-Si-Fe-Cu-Mn BASED ALLOYS FOR MICRO-CHANNEL TUBE OF HEAT EXCHANGER

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Aluminum alloy, Mn addition, Microstructure, Mechanical properties, Micro-channel tube

Abstract

Effects of Mn addition on microstructure and mechanical properties of as-cast and as-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys were investigated. After the as-cast billets were kept at 400 ºC for 1 h, hot extrusion was carried out with a reduction ratio of 38:1. Mn addition to Al-0.15Si-0.2Fe-0.3Cu-0.15Zn based alloys resulted in the formation of $Al_6(Fe,Mn)$ and $Al_{15}(Fe,Mn)_3Si_2$ intermetallic compounds and grain refinement. With increasing of addition of Mn, yield strength and ultimate tensile strength was increased and maximum value of yield strength and ultimate strength was 29 and 96 MPa at 2.1 Mn containing alloy, respectively. In the tensile properties at 420 ºC, Mn containing intermetallic compounds plays an important role for the improvement of strength and elongation due to grain refinement and suppression of grain growth by the formation of thermally stable Mn containing intermetallic compounds.

Introduction

Demand for more fuel-efficient vehicles and heat exchanger to reduce energy consumption and air pollution is a challenge for the automotive and electronic industry. The characteristic properties of aluminum alloys, high strength stiffness to weight ratio, good formabillity, good corrosion resistance, and recycling potential make it the ideal candidate to replace heavier materials. One of the increasing applications of Al alloys is heat exchanger. The Al1000 and 3000 series have generally been used for condenser tubes because of their excellent heat efficient and corrosion resistance [1-4]. In recent, Al alloys for heat exchanger required in order to improve the corrosion resistance, mechanical properties and formability to meet the challenge of the reduction in thickness of tube. In this study, effects of trace elements in Al-Si-Fe-Cu-Mn-Zn alloy on microstructure, mechanical properties, formability and corrosion were investigated. In first, alloys were prepared by gravity mold cast process. Al alloys were hot-extruded into a rod that was 12mm in thickness with a reduction ratio of 38:1. The microstructure of specimens was examined by OM, XRD, SEM and EBSD. The mechanical properties of the Al alloys were maintained by a tensile test at room temperature and 420℃.

Experimental procedure

The nominal compositions of the four studied alloys are Al-0.15wt%Si-0.2wt%Fe-0.3wt%Cu-0.15wt%Zn-xMn alloys (x = 0.9, 1.5, and 2.1 wt%). Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys were fabricated under an induction melting furnace in a graphite crucible. The alloy melts were cast into a pre-heat steel mold (D 75 mm × H 250 mm) at 200 ºC. After the as-cast billets were kept at 400 ºC for 1 h, hot extrusion was carried out with a reduction ratio of 38:1. The diameter of the extruded bar was 12 mm. Then, as-extruded specimens were annealed at 570 ºC for 16 hr. We examined the microstructures of the as-cast and as-extruded alloys using an optical microscope (OM), a scanning electron microscope with an energy-dispersive X-ray spectrometer (EDS) and electron backscatter diffraction (EBSD) system. Phase analyses were performed using an X-ray diffractometer with Cu Kα radiation. Samples were cut and ground mechanically into a mirror-like surface using abrasive papers and diamond pastes. Tensile tests were performed using a universal material test machine at room temperature and elevated temperature. The mechanical tests were carried out at an initial strain rate of 1.0×10^{-3} s^{-1}.

Results and discussion

Fig. 1 shows optical micrographs and SEM-BEI images of the as-cast Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys. The microstructure of the as-cast Al-0.15Si-0.2Fe-0.3Cu-xMn alloys exhibited dendritic morphology grains, as shown in Fig. 1 (a), (b) and (c). As addition of Mn was increased from 0.9wt% to 2.1wt%, grain size refined. In the SEM-BEI images, volume fraction of intermetallic compounds was increased with increasing Mn addition as shown in Fig. (d), (e) and (f).

To confirm intermetallic compounds, SEM-EDS analyses conducted. Fig. 2 shows SEM images and an EDS analysis of the as-cast Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys. Microstructure of the as-cast Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys consists of α-Al matrix and Mn containing intermetallic compounds. In this microstructure of as-cast alloys, these two kind of Mn containing phases were observed: $Al_6(Fe,Mn)$ intermetallic compound with lamellar morphology and $Al_{15}(Fe,Mn)_3Si_2$ intermetallic compound with polygon shape. Mn

containing intermetallic compounds was located mainly in the grain boundary. Therefore, it is implied that grain refinement resulted from the formation of the Mn containing intermetallic compounds during solidification.

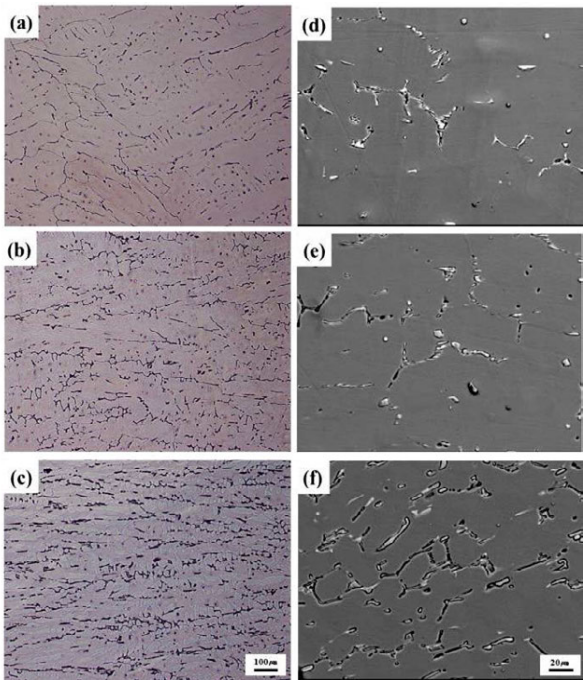

Fig. 1 Optical micrographs and SEM-BEI images of the as-cast Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys. (x = 0.9 (a), 1.5 (b) and 2.1 wt% (c))

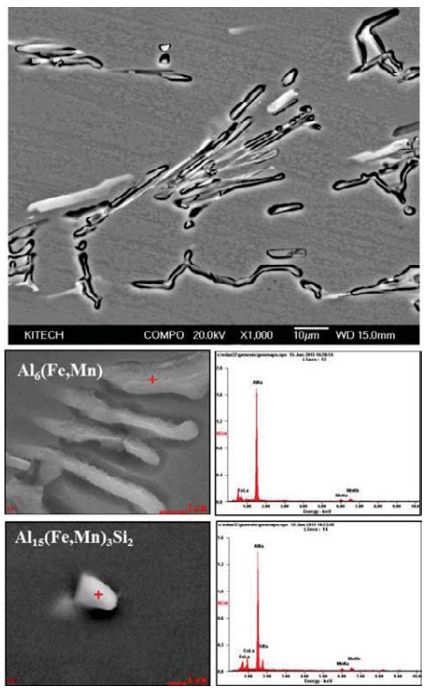

Fig. 2 SEM images and an EDS analysis of the as-cast Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys. (x = 0.9 (a), 1.5 (b) and 2.1 wt% (c))

Fig. 3 shows photo of the extruded bar and extrusion pressure with time during hot extrusion processes for Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys. All Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys extruded without extrusion cracks and defects as shown in Fig. 3 (a), (b) and (c). Maximum extrusion pressure was increased from 217 to 251 and 265 Kgf/cm2 as Mn content was increased from 0.9 to 1.5 and 2.1wt%.

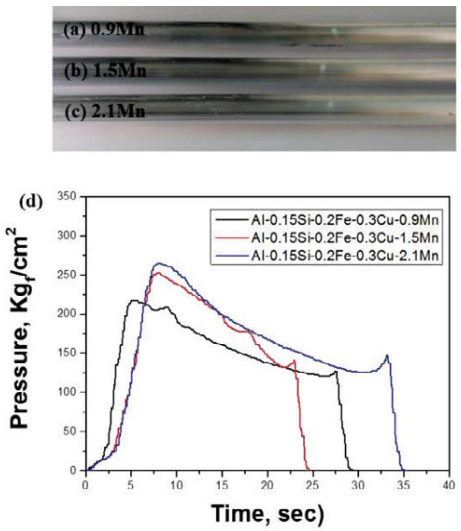

Fig. 3 Photo of the extruded bar ((a), (b) and (c)) and variation of extrusion pressure (d) with time during hot extrusion processes for Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys. (x = 0.9 (a), 1.5 (b) and 2.1 wt% (c))

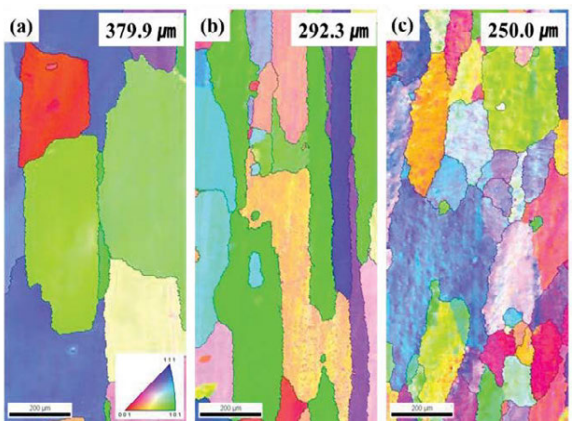

Fig. 4 The inverse pole figure images of the as-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn (x = 0.9 (a), 1.5 (b) and 2.1 wt% (c)) alloys after annealing heattreatment at 570 ºC for 16 hr.

The inverse pole figure images of Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys with different Mn contents after annealing heattreatment at 570 ºC and 16 hr are shown in Fig. 4. The inverse pole figure images were taken parallel to the extrusion direction. Grains were elongated to extrusion direction due to severe deformation during hot extrusion. In all alloys, large grains with above 250 was observed. As Mn addition was from 0.9 to 1.5 and

2.1 wt%, average grain size of as-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys was refine from 380 to 292 and 250 ㎛. In the microstructure of 0.9Mn containing alloy, grains were exhibited as coarse rectangle morphology. On the other hand, grains of 2.1 Mn contiaing alloy was observed as equiaxed morphology. Therefore, Mn addition resulted in the grain refinement and the formation of the equiaxed grain morhlogy due to the suppression of the grain growth during hot extrusion and heattreatment process by the formation of the thermally stable Mn containing intermetallic compounds.

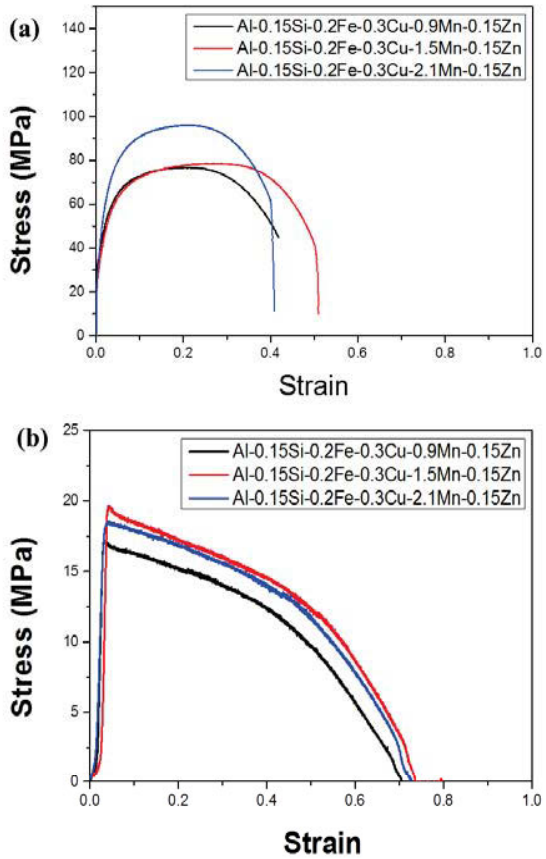

Fig. 5 Tensile properties of the as-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn after annealing heattreatment at room temperature (a) and elevated temperature (420 ºC and 20min) (b).

Fig. 5 shows tensile properties of the as-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn after annealing heattreatment at room temperature and elevated temperature (420 ºC and 20min). In stress-strain curves (Fig. 5 (a)), With increasing of addition of Mn, yield strength and ultimate tensile strength was increased and maximum value of yield strength and ultimate strength was 29 and 96 MPa at 2.1 Mn containing alloy, respectively. The tensile strength of the as-extruded alloys was improved by addition of Mn due to the formation of Mn-containing intermetallic compounds and grain refinement. And elongation was improved from 42 to 51% as Mn content was increased from 0.9 to 1.5wt%. In the case of 2.1 Mn containing alloy, elongation was lowered than that of 1.5wt%Mn containing alloy because of the increase volume fraction of intermetallic compounds. In the stress-strain curves of the as-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn at 420 ºC, ultimate tensile strength was much lower than that of the specimens at room temperature. Elongation was improved to around 74%. As Mn content was from 0.9 to 1.5wt%, tensile strength and elongation was increased. Mn containing intermetallic compounds by Mn addition plays an important role for the enhancement of mechanical properties (strength and elongation) at elevated temperature of 420 ºC. This may be due to grain refinement and suppression of grain growth by the formation of thermally stable Mn containing intermetallic compounds.

Conclusions

In this research, the effects of Mn addition on the microstructure and mechanical properties of as-cast and hot-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys have been investigated. In the microstructure of as-cast alloys, these two kind of Mn containing phases were observed: $Al_6(Fe,Mn)$ and $Al_{15}(Fe,Mn)_3Si_2$ intermetallic compounds. Volume fraction of intermetallic compounds was increased with increasing Mn addition. As Mn addition was from 0.9 to 1.5 and 2.1 wt%, average grain size of as-extruded Al-0.15Si-0.2Fe-0.3Cu-0.15Zn-xMn alloys was refine from 380 to 292 and 250 μm. Mn containing intermetallic compounds plays an important role for the improvement of strength and elongation at elevated temperature of 420 ºC due to grain refinement and suppression of grain growth by the formation of thermally stable Mn containing intermetallic compounds.

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CONCEPTION OF "DREAM CELL" IN ALUMINIUM ELECTROLYSIS

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Keywords: Aluminium electrolysis, slurry, non-Newtonian fluid physics, mathematical model of cell, bipolar cell, inert anodes, low-temperature electrolysis.

Abstract

The behavior of bubbles was studied in a three-phase mixture of alumina, gas and bath (an investigation in a water model).
Thermal and electrical balances were calculated for a low-temperature aluminium cell with bipolar electrodes. The decomposition of alumina dissolved in the AlF_3–KF–Al_2O_3 bath is considered as the basis of cell operation.
The cell is designed for the use of vertically oriented bipolar inert anodes. The specific productivity (kg per m^3) of such a cell is several times higher than the productivity of a traditional cell. The specific power consumption is about 13 kWh/kg Al.
The anode-cathode space (ACS) is filled with a slurry containing more than 30% of alumina.

Introduction

A few years ago, a rather unconventional aluminium production process was presented. According to this process, aluminium is produced by electrolysis of cryolite-alumina melts [1,2]. One of the characteristics of the above process is that the anode-cathode space (ACS) is filled with a slurry, which contains more than 30% of alumina and the rest is a fluoride melt. According to the above idea, undissolved alumina, which separates the anode from the cathode, is no longer the pot operator's "enemy" that generates problems like sediments, muck, increased MHD convection, extra anode effects, etc. Undissolved alumina becomes both an "ally" and an integral part of the bath sub-system. The main benefits of the above technical solution are as follows:

1. Effective separation of aluminium and anode gases by means of the above slurry. In turn, it helps suppress back reactions under low convection conditions, increase current efficiency and reduce the anode-cathode distance (ACD) by 2 to 4 times. Low ACDs fully compensate for the growth of resistance which grows due to the presence of non-conductive alumina.
2. Ability to use electrolytes with a low solubility of alumina, which helps produce aluminium at temperatures less than 973 K.
3. Ability to use vertically oriented or inclined electrodes, bipolar or monopolar. Such orientation increases cell specific productivity per 1 m^2 or 1 m^3. Apparatus performance vs. volume according to Beck [3] is presented in Figure 1. Moreover, there are our data in this figure. The data are based on the assumption that each electrochemical cell including anode, cathode and ACS in the multi-electrode reduction cell has a thickness of 4 cm and a height of 0.5 m. So, the apparatus volume as well as its effective area, under such conditions, is reduced by several times compared to the dimensions of the current industrial cell. An increased electrode surface area provides for a decrease in current density and the electrical work of dissipation, and, hence, a decrease in specific power consumption.

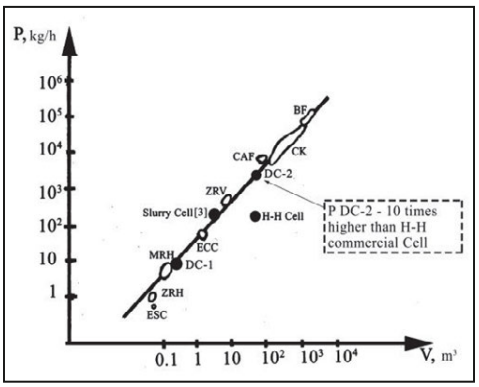

Figure 1.Apparatus productivity vs. volume, according to [3], including the points representing the dream cell (DC) parameters (versions 1 and 2)

Blast Furnace	BF	Electrolytic chlorine cell	ECC	Zink Retort (Horizontal)	ZRH
Carbide arc furnace	CAF	Electrolytic Sodium Cell	ESC	Zink Retort (Vertical)	ZRV
Cement kiln	CK	Magnesium retort (Horizontal)	MRH	Dream Cell version 1	DC-1
Slurry Cell	[3]	Hall-Heroult cell	H-H Cell	Dream Cell version 2	DC-2

4. Ability to use new electrolytes based on potassium fluoride, for instance, which allows for the use of inert anodes. Suppressed convection reduces the transport of anode corrosion products to the cathode and improves metal purity.
5. The use of new designs of the vertical electrode makes it possible to design brand new cells with both monopolar and bipolar electrodes. Bipolar electrodes will allow for the use of a busbar of low weight. Besides, the MHD convection and instability problem will disappear.
6. Low-temperature electrolytes with an alumina slurry allow for new pit designs, which exclude metal inserts and other materials, except for lining materials made of high-alumina concrete. Such cells should be much cheaper and have a longer life compared to the current ones.
7. A cell with vertically oriented electrodes and a low operation temperature will allow engineering a continuous metal tapping system, i.e. a system that doesn't require operators.

The idea of developing new designs and reduction technologies for a cell with vertically oriented electrodes and an alumina slurry in the anode-cathode space seems to be feasible, but requires solving a number of problems, such as:

1. What should be the parameters of a healthy cell which is considered to be a dissipative system, and its sub-systems and parts?

2. How should sub-systems interact (flows of mass, heat, and electricity, corresponding forces)? What should be the vital operating conditions for such a dissipative system?
3. What are the electric and energy balances of the cell?
4. How to provide for a uniform temperature field excluding local over-heating?
5. How does the three-phase 'gas–bath–solid phase' flow look like?
6. How to provide for trouble-free metal tapping through a wettable cathode? What should be the material and design of the cathode?
7. What should be the material and design of the anode and the gas collecting system?
8. How to provide for uniform alumina feeding?
9. What cell design is preferable: bipolar or monopolar?
10. Does high-alumina concrete really provide for durability? What should be the composition?

The following part of the paper tries to answer some of the above questions.

Fluid physics of the three-phase flow (TPF)

It is required to answer the following question: how freely can anode gases leave the space "constrained" by the slurry. The water model [4], which is used to investigate the behavior of bubbles, simulates the electric decomposition of the alumina slurry in the high temperature cell. For the model, an aqueous solution of sulfuric acid was used. The water model provides for an approximate equality between the criteria of similarity between high-temperature and low-temperature electrolysis, as shown in Table 1.

Table 1. Analysis of Similarity of Model and Nature

Parameter	Nature	Model
Electrolyte	50 mol. %KF-50%AlF_3	20% H_2SO_4
Temperature T, K	973	323
Electrolyte density ρ_l, kg/m^3	2114	1120
Kinematic viscosity of the electrolyte ν, m^2/s	$1.28*10^{-6}$	$7.41*10^{-7}$
Dynamic viscosity of the electrolyte, Pa*s	0.00271	0.00083
Surface tension of the electrolyte, N/m	0.138	0.0746
Gas generation rate υ, m^3/m^{2}s	0.0001	0.0002
Anode current density kA/m^2	0.73	2.5
Electrode height, m	0.1	0.1
ACD, m	0.02	0.02
Similarity criteria		
Modified Reynolds number=$\upsilon \cdot h/(\nu \cdot \rho_l)$	16.24	16.24
Galileo's number=$g \cdot h^3/(\nu^2)$	$2*10^{10}$	$1.2*10^{10}$
Weber's number=$\sigma \cdot h/(\nu \cdot \rho_l)$	$5.96*10^6$	$9.8*10^6$
Geometrical simplex= h/l	0.2	0.2
Non-Newtonian simplex=V_s/V	0.3	0.3
Sedimentation simplex= ρ_l/ρ_s	0.47	0.83

In Table 1, h – electrode height, ρ_s – solid's density, σ – electrolyte's surface tension, l – anode-cathode distance, V_s – volume of solids in the slurry, V – volume of the slurry.

The cell vessel consists of two plates made of organic glass, including the rubber plate (2 mm in thickness) clamped between the two plates. The alumina content in the slurry was set to be 30%. The electrodes made of stainless steel were connected to the DC supplier (Autolab PGSTAT302). All the observations were made with the help of a 5x magnifier. The evolution of bubbles was recorded by means of the Nikon D3100 camera (30 fps.) The main cell dimensions are shown in Figure 2.

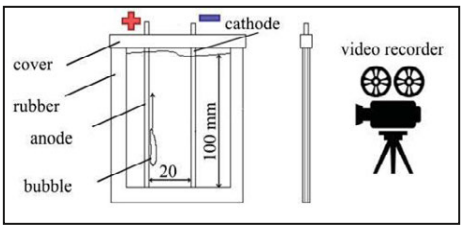

Figure 2. Diagram of Cell to be used to Simulate Electrolysis of "Cryolite Melt–Alumina" Slurry

The task of the experiment was to determine bubble motion behavior vs. current density (gas generation rate). The measurements were taken within the current density range of 0.05 to 0.25 A/cm^2.

The results show the following:

1. The TPF appears between the oxygen evolving electrode and the motionless slurry layer after the electrical current has passed through the electrolyte (Figure 3). A foam layer of small bubbles is formed on the surface of the ACS due to a high interface resistance (atmosphere–to–electrolyte).
2. The TPF consists of approx. 50% gas, 40% liquid electrolyte and 10% alumina.
3. The alumina particles velocity was less than 10 cm/s in the TPF, and it reduced almost down to zero at a distance of 5 mm from the anode.
4. There is a slug-type flow in the TPF layer [5]. The rate of the slug's growth is controlled by coalescence. "Slugs" are of oval or Fortin shape [6], but when they approach the electrolyte-atmosphere interface, they are likely to turn into a sphere. Due to the bubbles ("slugs") movement TPF is characterized by low hydrodynamic resistance (compared to the initial slurry).

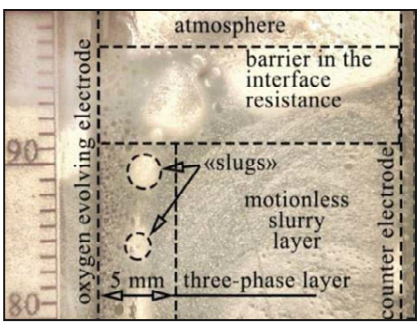

Figure 3. ACS near the electrolyte–atmosphere interface@ 0.25 A/cm^2

5. The three-phase layer has its own structure, which is not to be considered in this paper.
6. A small amount of gas is removed by the bubbles with a size of less than 0.1 mm.
7. The vertical velocity depends on the gas generation rate. Such dependence (125 measurements) is shown in Figure 4.

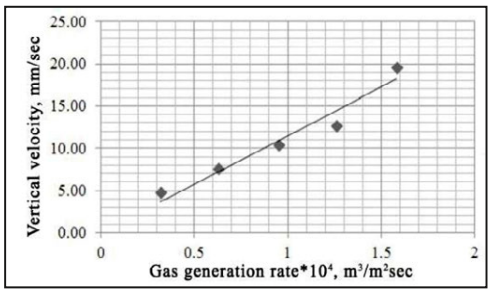

Figure 4. Dependence of the average "slug" vertical velocity on the gas generation rate

8. The average bubble thickness increases when the bubbles approach the electrolyte–atmosphere interface (Figure 5).
9. The "slug" dimensions depend on the gas generation rate (Figure 6).
10. The number of "slugs" crossing the interface goes up with an increase in the gas generation rate (Figure 7).

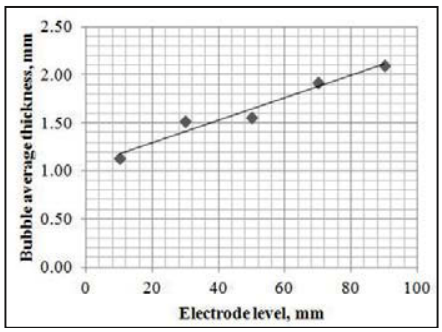

Figure 5. Average "slug" thickness @ various heights at a current density of 0.25 A/cm^2

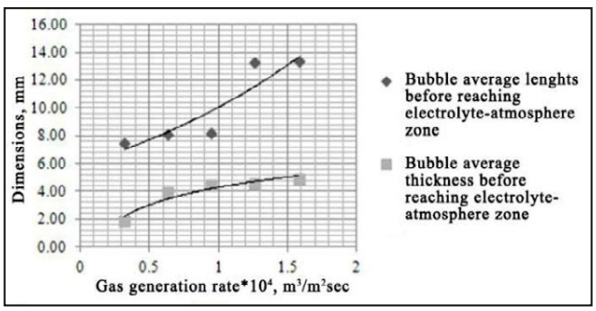

Figure 6. Dependence of the "slug" dimensions on the gas generation rate

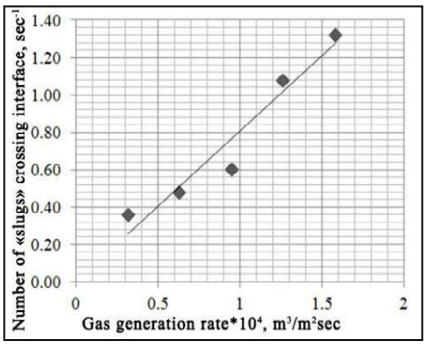

Figure 7. Dependence of the number of "slugs" crossing the interface on the gas generation rate

The presented data evidence that removal of the bubbles from such a dense slurry under free convection conditions is quite effective. The presence of the secondary slurry at the TPF border evidences that convection in the cathode area of the reduction cell will be suppressed, and back reactions between anode and cathode products will be inhibited. As for the ACS, we suggest that the anode-cathode distance be even less than 2 cm, especially if gas removal is properly arranged.

Bipolar Cell Math Model

From a great variety of electro-chemical cells, which have been used for the process of electrolysis of molten salts, we chose the one that uses vertical bipolar electrodes. The main advantages of such a cell are as follows:

1. Very high specific productivity P.
2. Small weight of a busbar (per 1 kA of current), no MHD convection.
3. Ability to design an automated cell with inert anodes, which could operate without any human involvement.

The two-dimensional cell diagram is presented in Figure 8.

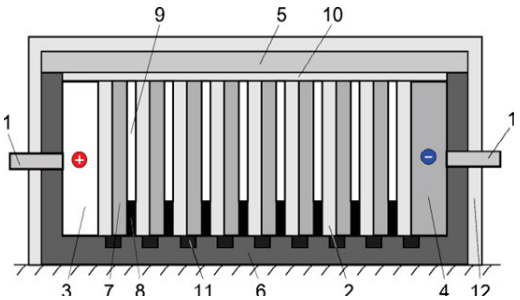

Figure 8. Cell with bipolar vertical electrodes: 1—current lead; 2—slurry bath; 3—anode; 4—cathode; 5—cover; 6—lining; 7—cathode part of the bipolar electrode; 8—bipolar electrode's insulation insert; 9—anode part of the bipolar electrode; 10—atmosphere; 11—aluminium; 12–thermal insulation.

It is necessary to know the temperature and potential fields to get an apprehension of the cell's energy balance.
The model with boundary conditions determines the temperature and potential fields that satisfy the law of energy conservation:

$$W = T\Delta S + Q \tag{1}$$

where W —dissipative work done upon the system, J; TΔS— changes in the system's bound energy, J; Q — heat dissipating into the atmosphere (AS), J.

Differential equations

The thermal-electric problem can be solved by means of the temperature and potential fields that satisfy the laws of energy and charge conservation. Equation system (2) can be solved by the finite elements method with the help of ANSYS [7].
For the steady state, we have:

$$\begin{cases} \frac{\partial}{\partial x}(-\lambda(x,y,z,T)\frac{\partial T}{\partial x})+\frac{\partial}{\partial y}(-\lambda(x,y,z,T)\frac{\partial T}{\partial y})+\frac{\partial}{\partial z}(-\lambda(x,y,z,T)\frac{\partial T}{\partial z})=q_j(x,y,z)+q_s(x,y,z) \\ \frac{\partial}{\partial x}(-\chi(x,y,z,T)\frac{\partial \varphi}{\partial x})+\frac{\partial}{\partial y}(-\chi(x,y,z,T)\frac{\partial \varphi}{\partial y})+\frac{\partial}{\partial z}(-\chi(x,y,z,T)\frac{\partial \varphi}{\partial z})=0 \end{cases}$$

(2)

where λ—thermal conductivity, $W*m^{-1}*K^{-1}$; q_j —specific heat liberated when the current passes through (Joule heat), J/m^3; q_s.— specific heat liberated from the outside sources (Peltier effect, etc.), J/m^3; χ —specific electrical conductivity, $Ohm^{-1}*m^{-1}$; — electric potential, V; x, y, z — coordinates, m.
The Joule heat in the right side of the first equation in system (2) can be found by using equation (8):

$$q_j(x,y,z) = \frac{i^2(x,y,z)}{\chi(x,y,z,T)} \quad (3)$$

where i^2 —square of the current density's vector modulus.
The current density's vector field can be found by using the potential field:

$$i = -\chi(T)\cdot\nabla\varphi \quad (4)$$

The specific heat from the outside sources q_s in this particular area can be calculated as a sum of heat from many outside sources $\sum q_g$ (Table 3).

$$q_s(x,y,z) = \sum q_g(x,y,z) \quad (5)$$

Boundary conditions

To solve system (2), we need to define the model's boundary conditions. The borders are shown in Figure 9 as dashed lines.

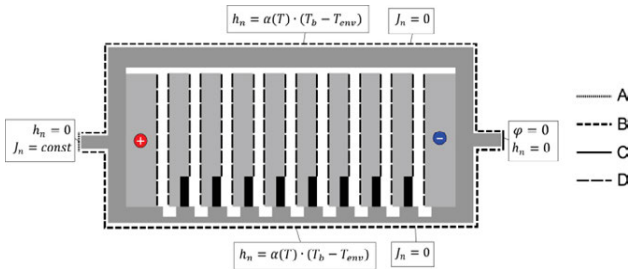

Figure 9. Model's borders (Lines A, B and C), and the layers where Peltier heat liberation, and anode and cathode overvoltage take place (Line D)

Normal current component J_n defines the amount of current entering the model (Fig. 9, Line A):

$$J_n = \text{const} \quad (6)$$

Zero potential defines the place where the current leaves the model (Fig. 9, Line C):

$$\varphi = 0 \quad (7)$$

Electrical insulation condition (Fig. 9, Line B):

$$J_n = 0 \quad (8)$$

Third type boundary condition (Fig. 9, Line B):

$$h_n = \alpha(T)\cdot(T_b - T_{env}) \quad (9)$$

Where h_n — heat flow surface normal, W/m^2; α—heat transfer coefficient, taking into account the radiation and convection components of the heat flow, $W*m^2*K^{-1}$; T_b—border temperature, K; T_{env}—environmental temperature, K.
The heat flow h_f is the vector field, which we can get from the temperature field:

$$h_f = -\lambda(T)\cdot\nabla T \quad (10)$$

The model doesn't allow for side ledge formation, therefore, the lining should be insulated from all the sides. To reduce the number of elements and the time required for calculation, we introduce heat transfer coefficient α_{ef} (W/m^2K):

$$\alpha_{ef} = \frac{1}{\frac{L}{\lambda_t} + \frac{1}{\alpha}} \quad (11)$$

where L – thermal insulation thickness, m; $\lambda_{t.i.}$ – thermal conductivity of the insulation, W/(m*K).
Heat flow boundary condition:

$$h_n = \alpha_{ef}\cdot(T_g - T_{env}) \quad (12)$$

Due to a small section and for the sake of simplifying our calculations, let us assume that the following condition is fulfilled at the A and C boundaries (fig. 9):

$$h_n = 0 \quad (13)$$

Other conditions:

- The cell has two parallel rows of electrodes, and it is symmetric about the plane passing the center of the cell between the rows. It means that we can simulate just a half of the cell up to the plane of symmetry. The model's sectional view at the center of one electrode row is shown in Figure 10. The cathode current density is 0.4 A/cm^2. The cathode height is h=50 cm; the width is b=20 cm. The distance between the bipolar electrodes is 2 cm. The current, which passes through one electrode row, is I=400 A. It is considered that 400 amperes pass through each electrode row.
- The bath composition is 50 mol % KF – 50% AlF_3 [11].
- There is 30% of undissolved alumina in the bath.
- The thermal insulation layer's thickness is 10–17 cm at side walls, 10 cm at cover, 2.9–13.2 cm at end walls, 36 cm at bottom. The concrete floor thickness is 50 cm.
- There is no convection in the slurry.
- Metal and gases are removed continuously.
- Both the anode and the cathode do not react with the atmosphere, bath, and reaction products.

The whole task area is divided into elements. Each element consists of 8 nodes. There are 252,212 elements and 268,970 nodes in the model [7].
The material properties are set to be close to those that can be found in the literature (Table 2).
Below are the data used to calculate the outside heat sources q_s: alumina heating, dissolution, phase transition's thermal energy, Peltier effect's heat, and anode and cathode overvoltage.

Those places, where q_s represents the sum of the Peltier heat and the heat generated by electrode overvoltage, are thin layers on the anode and cathode surfaces (Fig.9, Line D).
The other outside heat sources are located in the ACS.

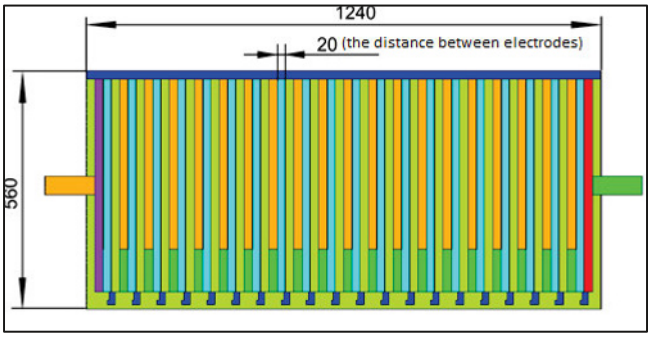

Figure 10. Finite Element Model of Half a Cell – one row of electrodes (sectional view of the bipolar electrode's center). Different parts are marked with different colors. The elements are not shown.

Table 2. Material properties as a function of temperature

The cell's subsystem	Electrical conductivity, $Ohm^{-1}*m^{-1}$	Thermal conductivity, W/(m*K)
	973K	
Anode, and the anode part of the bipole	$7.042*10^6$	82.2
Cathode, and the cathode part of the bipole	$6.3*10^6$	68
Electrolyte	$0.65*10^2$	0.42
Current leads	$12.1*10^6$	382.5
Insulation insert, lining and cover	2	0.92
Aluminium	$6.9*10^6$	90.7
Thermal insulation	-	0.15
Concrete floor	-	1.7

Table 3. Data used to calculate the outside heat sources q_s [10]

Parameter	Value
Aluminium deposited in one anode-cathode pair, kg/day	3.04
Alumina consumption in one anode-cathode pair, kg/day	5.875
Thermal energy for alumina heating, kJ/kg	–735.0
Alumina dissolution heat, kJ/kg	235.1
Alumina moisture evaporation, kJ/kg	–65.0
Alumina phase transition (γ–α) heat, kJ/kg	–27.9
Peltier effect's heat for one anode-cathode pair, near the anode, J/s	–640
Peltier effect's heat for one anode-cathode pair, near the cathode, J/s	64
Anode overvoltage in one anode-cathode pair, V [12]	0.25
Cathode overvoltage in one anode-cathode pair, V	0.045

Figure 11 shows Heat Transfer Coefficient for vertical and horizontal walls, as well as for the boundary condition at the bottom, vs. wall temperature, taking into account the convection and radiation components of the heat flux. For calculating $\alpha(T)$, two models were developed and used to solve the Navier-Stokes equation for the air flow near the wall border at a constant temperature.

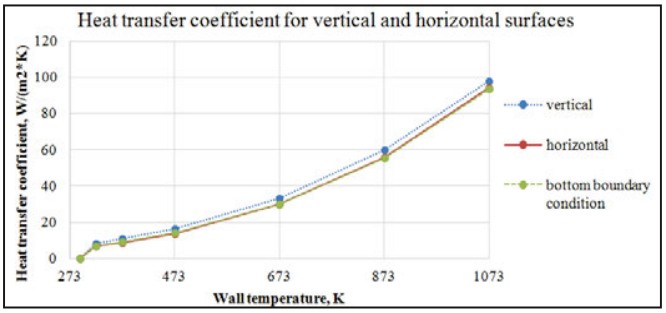

Figure 11. Heat transfer coefficient for vertical and horizontal walls, and the boundary condition at the bottom vs. wall temperature

Calculation results

The model's temperature field is shown in Figure 12.

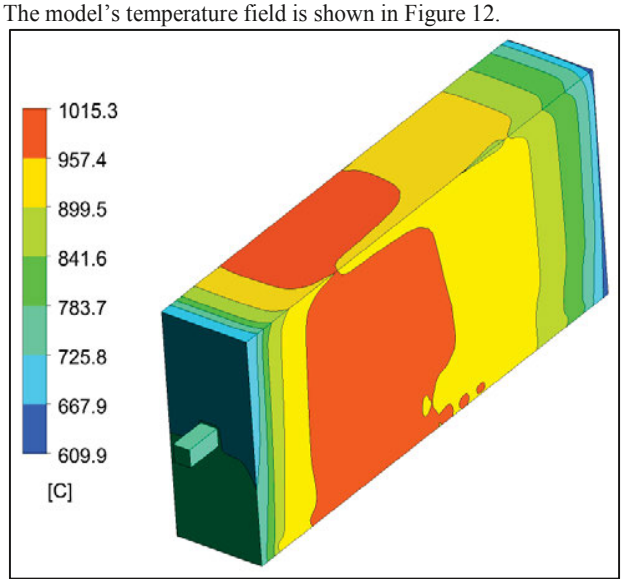

Figure 12. Temperature field on the cell's surfaces, °C

The voltage drop in various parts of the model is shown in Table 4. The EMF in one pair of electrodes was set to be 2.367 V [8]. The rates of heat generation and dissipation from the cell's surfaces by convection are presented in Table 5 and Table 6.

Table 4. Voltage drop, V

Parameter	Value	Percentage of total voltage, %
Ohmic voltage drop in the cell, V	30.84	36.7%
Total (combined) anode overvoltage, V	5	5.9%
Total (combined) cathode overvoltage, V	0.9	1.0%
EMF, V	2.367	–
Sum of decomposition voltages for each pair of electrodes, V	47.34	56.4%
Total voltage, V	**84.08**	**100%**

Table 5. Heat evolution in the model, J/s

Heat dissipation and absorption sources	Heat, J/s
Joule heat for the anodes and anode parts of the bipolar electrodes	0.92
Joule heat for the cathodes and cathode parts of the bipolar electrodes	1.59
Joule heat in lining	133.84
Joule heat in insulation inserts	145.52
Joule heat in current leads	3.29
Joule heat in bath	24392.73
Joule heat in aluminium	0.00
Peltier effect's heat for the anodes	-25600
Peltier effect's heat for the cathodes	2560
Alumina heating	-2002.02
Alumina dissolution	-640.49
Alumina moisture evaporation	-177.05
Alumina phase transition	-75.88
Anode overvoltage	6400
Cathode overvoltage	720
Total	**3462.45**

Table 6. Heat dissipation by convection, J/s

Heat dissipation source	Heat, J/s	Percentage of total heat, %
Cover	659.77	18.9 %
Longitudinal wall	1,496.35	42.9 %
End wall (upstream)	231.42	6.7 %
End wall (downstream)	857.96	24.6 %
Bottom	240.92	6.9 %
Total	**3,486.42**	**100 %**

The discrepancy (dissipated heat minus generated heat) was 23.97J/s, which is 0.69 % of the total heat generated in the model.

Final remarks

Cell productivity (kg/h):

$$p = \sum_{j=1}^{n+1} (k \cdot i_c \cdot h \cdot b \cdot \eta \cdot (1 - \mu_j)) \quad (14)$$

where k – electro-chemical equivalent, 0.336 g/(A*h); i_c–0.4 A/cm^2; n – number of bipolar electrodes (n=19); h–50 cm; b–40 cm (total width of the two parallel rows of electrodes); η – current efficiency, 0.95 [9]; μ_j – part of the total current, which by-passes one pair of electrodes j.
The cell productivity is p=5.107 kg/h.
If the cell dimensions are 1.24x0.46x0.56 m, the specific productivity P per m^3 of the cell will be:

$$P = \frac{p}{1.24 \cdot 0.46 \cdot 0.56} = 16 \frac{kg}{m^3 \cdot h} \quad (15)$$

In our case, P will be several times higher than it is for high-amperage cells with a current intensity of 600 kA (Fig. 1).
Energy consumption W (kWh/t):

$$W = \frac{U \cdot i \cdot h \cdot b}{p} = \frac{84.08 \cdot 0.4 \cdot 50 \cdot 40}{5.072} = 13263 \frac{kW \cdot h}{t} \quad (16)$$

where U –cell's voltage, V.

The technical & economic indicators of the cell can be found below:
The cell's dimensions: 1.24x0.46x0.56 m;
Current intensity: 800 A;
The distance between the bipolar electrodes: 20 mm;
Productivity: 5.107 kg/h Al;
Specific productivity: 16 (kg/h*m^3) Al;
Productivity: 4.505kg/h O_2;
Energy consumption: 13263kWh/t; and
Current efficiency: 94.4% [9].

Conclusion

The cell, which uses an alumina slurry to separate the cathode from the anode, is feasible. It could provide for acceptable fluid dynamics and performance, and high technical and economic indicators. Using our approach, a cells which corresponds to those with 1,000 kA (today's version) could be designed.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE IMPACT OF THE CAVITY ON THE TOP HEAT LOSSES IN ALUMINUM ELECTROLYSIS CELLS

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Keywords: Aluminum, Electrolysis, Top heat losses, Anode cover material, Crust, Cavity, Heat transfer, Thermal radiation

Abstract

The primary aluminum producers are continuously raising the line amperage to increase the productivity. Hence, the top heat losses of aluminum electrolysis cells are amplified which leads to higher thermal constraints on the anode cover. A finite element model is developed in order to predict the heat fluxes and temperatures in the top parts of the cell. The model is validated against measurements from the top surfaces of real industrial cells. It considers the cavity above the liquid bath, which is formed during the melting and falling of anode crust during the life of the anode. The heat flux escaping the anode cover increases with the height of the cavity according to the results. Considering the cavity inside the model is mandatory in order to obtain a good correspondence between the measurements and the model predictions.

Introduction

The efficiency of the Hall-Héroult process is continuously improving in order to achieve better economic and environmental performances. During the last decades, the primary aluminum producers increased the productivity by raising the current delivered to the electrolysis cells. Accordingly, the thermal constraints on the anode cover and anode assembly has increased, which creates additional risk of operating problems. The deterioration of the anode cover induces higher heat losses, fluoride fumes emissions, and air burn of carbon anode. Furthermore, the melting and falling of anode crust due to higher thermal constraints cause the formation of sludge or deposit on the surface of the cathode block. The process of melting and falling of anode crust into the bath also creates a cavity under the remaining anode crust. Figure 1 depicts the location of the cavity within the top parts of an industrial electrolysis cell.

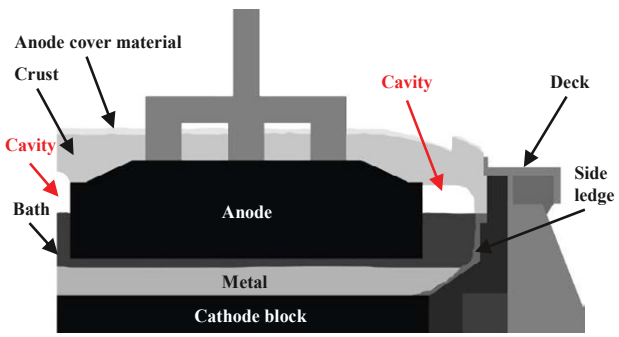

Figure 1. Schematic of an aluminum electrolysis cell.

During the operation of the electrolysis cell, the carbon anode is consumed and it must be replaced after a few weeks. At each anode change, the anode cover material (ACM) is deposited over the newly set anode and on the layer of frozen bath surrounding the anode. The typical ACM is mainly composed of recycled crushed bath and alumina. A mix of fine to coarse particles characterizes the granulometry of this material. The thermal conductivity of various mixtures of ACM has been measured experimentally [1, 2]. The thermal conductivity of ACM increases linearly with temperature between 160 °C and 600 °C [1]. The granulometry or particle size distribution of the ACM has a major impact on its thermal conductivity [2]. The effect of chemical composition on thermal conductivity is less important than granulometry, according to Wijayaratne et al [2]. A larger amount of coarse bath in the mixture of ACM increases the thermal conductivity while higher proportions of fine particles decrease the thermal conductivity.

Once a new anode is set and the ACM is deposited, the layer of frozen bath begins to melt due to the thermal insulation made by the ACM. Afterward, liquid bath begins to penetrate into the particles of ACM by capillary force. The rise of the liquid bath in the ACM generates the formation of a crust consisting of embedded alumina particles into solidified bath [3]. The consolidation of ACM into solid crust is largely controlled by heat transfer [4, 5]. The heat losses from the bulk liquid bath increase the temperature of the ACM, causing the melting of the bath surrounding the alumina particles. The boundary of the liquid front in the ACM progresses with the temperature profile and depends on melting temperature of the mixture, set at 685 °C by Zhang et al. to model the evolution of the anode crust [6]. The acidic and calcium phases in the ACM (mainly $Na_5Al_3F_{14}$, $Na_2Ca_3Al_2F_{14}$, and $NaCaAlF_6$) melt at a lower temperature than the cryolite [7-9]. In the Na_3AlF_6-AlF_3-Al_2O_3 system, the eutectic crystallization of a highly acidic liquid bath is approximately 685 °C [7, 10].

The hardness and thermal conductivity of crust have been investigated by a few researchers [11, 12]. The hot wire and hot strip methods were used to determine the thermal conductivity, which increases with the apparent density of the crust by an almost linear relationship [12]. The thermal conductivity and emissivity of the crust were determined by Rye et al. with an experimental setup [13]. The thermal conductivity of the crust increases strongly with the content of crushed bath in this material. The melting and falling of anode crust create a gas-filled gap between the bulk bath and the crust. The emissivity of the crust in this cavity was estimated to 0.3 [13]. The risk of crust failure may be enhanced by the dissolution and melting of the crust in contact with the liquid bath, by mechanical disturbances due to anode movements and by partial collapse of a weak crust [14].

The heat transfer in the top parts of the electrolysis cell was recently investigated by CFD modeling [15]. A correlation for the convection coefficients was developed in function of the

volumetric flow rate of gas collected from the pot. The heat dissipation and temperature distribution have been measured on the ACM and along the anode rod in industrial cells [16, 17]. The thickness of the ACM layer has a strong impact on the heat flux released by the anode cover [17].

The effect of the cavity on the heat dissipation of industrial electrolysis cell was not covered in any of the previous studies. Moreover, most previous modeling studies are limited to 1D or 2D analysis and are rarely based on measurements from industrial cells. In this study, a 3D thermoelectric model of an industrial electrolysis cell is presented. The thickness of the ACM, crust and cavity were measured and used to predict the heat losses. Furthermore, the results are compared with measurements of the heat fluxes and temperatures at specific locations on the top surfaces of the cells.

Numerical model

The impact of the cavity on the heat losses is quantified by modeling four geometries, with and without cavity, of P155 prebake cells. The cavity is located in the side channel (SC), between the anodes (BA) and in the center channel (CC). The general geometry is a slice of an industrial cell containing two cathode blocks and an anode and a half. Various slices of mid-life anodes were measured and predicted in order to ensure the validity of the prediction of the numerical model. Each of the geometries was based on measurements taken at Rio Tinto's Grande-Baie plant located in Québec. The selected mid-life anodes were far enough from the feeder holes or the corners of the cell to neglect their effect. Figure 2 shows an example of the geometry used to predict the heat losses.

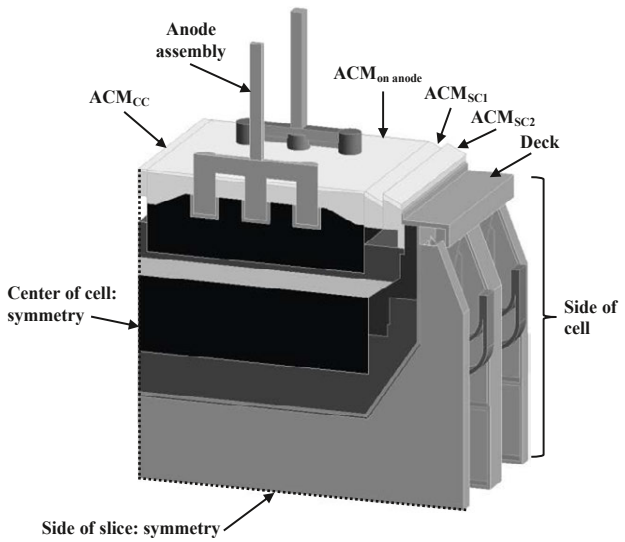

Figure 2. Example of geometry used for the simulation

The ACM is divided into four zones ("CC", "on anode", "SC1" and "SC2"). The ACM and crust thicknesses and the boundary conditions above the anode cover vary based on measurements. In general, the ACM is roughly dispersed on the top of the anode in operating cells. Moreover, the consolidation of the ACM is non-uniform and depends on the local mass transfer and thermal constraint.

Assumptions

In order to simulate the top heat losses of electrolysis cells, the following assumptions were taken into account: (1) the heat transfer from a mid-life anode is close to steady-state; (2) the mass transfer effects on heat losses are neglected; (3) the ACM granulometry and thermal conductivity are similar to the data presented in [1]; (4) the properties of all materials are known; (5) the effect of the flow circulation of the liquid bath is accounted by an "augmented effective thermal conductivity"; (6) the heat transfer in the cavity is radiation dominated and (7) the surfaces in the cavity are gray and diffuse.

Governing equations

The top heat losses and heat generation were predicted based on a coupled thermal-electrical finite element analysis (FEA). The equations were solved using ANSYS Mechanical software (release 16.0). The domain for the FEA simulation is a slice of an electrolysis cell (Figure 2). Equation 1 describes the governing equation for the steady state heat conduction with the volumetric heat generation rate [18]:

$$\frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_y \frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k_z \frac{\partial T}{\partial z}\right) = \dot{q} \quad (1)$$

The volumetric heat source term $\dot{q}$ is calculated by the Joule effect:

$$\dot{q} = \sigma E^2 \quad (2)$$

The electrical field can be derived from:

$$\mathbf{E} = -\nabla V \quad (3)$$

The potential distribution is computed from the governing equation for steady-state electric conduction [18]:

$$\nabla \cdot ([\sigma]\nabla V) = 0 \quad (4)$$

The general equations of heat and electric conduction were described. Further information on the thermoelectric coupling is available in the ANSYS Mechanical documentation [18]. The radiosity method was used in order to calculate the heat transfer in the cavity. The radiosity solution was obtained using the Gauss-Seidel iterative solver. A set of linear algebraic equations is used to obtain the outgoing radiative fluxes (radiosity) for each surface [18].

$$[A]\{q''^{,0}\} = \{D\} \quad (5)$$

Where:

$$A_{ij} = \delta_{ij} - (1 - \varepsilon_i)F_{ij}$$

$q_j''^{,0}$ = radiosity flux for surface i

$$D_i = \varepsilon_i \sigma T_i^4$$

At the surface facing the cavity, the boundary conditions for the heat conduction are provided by the net surface radiosity fluxes. The radiation view factors, F_{ij}, are required to solve the radiosity

module. The overall view factors for each surface are calculated using the hemicube method [18].

Boundary conditions and properties

The description of the boundary conditions and properties will be detailed for the top parts of the cell and given globally for the side and bottom. Insulating conditions have been applied to the symmetric faces of the numerical model, the center of the cell and the two sides of the slice (Figure 2). A potential of 0 V has been applied to the top of each cathode flex. The current delivered to the electrolysis cell was divided by the number of anodes and the corresponding value was applied to the top of each anode rod. An "effective convection coefficient" considering both the convection and radiation has been applied to the side and bottom of the electrolysis cell. Table I depicts the boundary conditions of the top parts of the numerical model.

Table I. Boundary conditions of the top parts

Boundary	Condition
Anode assembly	Convection (h=70 W/m^2·K, T_∞= measured)
$ACM_{on\ anode}$	Convection (h= calculated, T_∞= measured)
ACM_{SC}	Convection (h= calculated, T_∞= measured)
Deck	Convection (h= calculated, T_∞= measured)
Top of anode rod	Temperature (T = measured)

The measured and calculated values will be provided in the results and discussion section. In the numerical model, an "effective convection coefficient" was applied to the top of the ACM on anode and on the side channel. The convection coefficients were determined from the heat flux sensors placed on the mid-life anodes and the measurements of the gas temperature (T_∞). The convection coefficient applied on the anode assembly (70 W/m^2·K) came from the measurements made by Shen et al. [16]. The thermophysical properties of the materials are described in Table II.

Table II. Thermophysical properties [1,11,12,13,19,20,21]

Boundary	Value
ε_{ACM}	0.3
$\varepsilon_{liquid\ bath}$	0.97
$\varepsilon_{carbon\ anode}$	0.9
k_{ACM} (W/m·K)	$0.155 + 0.616 \times 10^{-3}$ T (160 – 600 °C)
k_{crust} (W/m·K)	1.6
$k_{anode\ rod}$ (W/m·K)	Confidential
$k_{stubs\text{-}yoke}$ (W/m·K)	Confidential
k_{anode} (W/m·K)	$-0.000003 \times T^2 + 0.0056 \times T + 3.4$
$k_{bath\ solid}$ (W/m·K)	1.2
$k_{bath\ liquid}$ (W/m·K)	3000

The ACM of the mid-life anodes investigated in this study had a similar chemical composition than the 60 % Al_2O_3 and coarse crushed bath (CCB) mixture described in [1]. The granulometry of the ACM in this study was also similar to the CCB. The thermal conductivity of the ACM in the model was adjusted to consider the transformation of ACM into crust. At temperatures of 700 °C and above, k_{ACM} was increased to 1.6 W/m·K. The thermal conductivity of the liquid bath was augmented to reflect the movements generated from the magnetohydrodynamics effect and induced by CO_2 bubbles.

Measurement method

The thicknesses of ACM and crust were measured on the anode and on the side channel, at a location close to the measurement of the heat flux (using a Hukseflux HF01 sensor). The estimated accuracy of these sensors is ± 10 % according to the product manual. Figure 3 depicts the locations of the measurements.

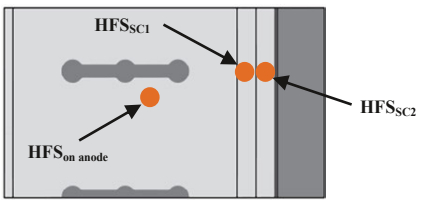

Figure 3. Locations of the heat flux sensors (HFS) on the ACM

In each electrolysis cell, the heat flux escaping the anode cover was measured on the anode, close to the stubs. On mid-life anode, the top surface of the anode cover located at the SC is non-uniform. One part of the crust remains fixed to the anode assembly (SC1) while the other part sticks to the sidewall (SC2). The lowering of the anode assembly during the life of the anode causes the formation of a "step", the higher part of the crust which remains attached to the sidewall. The side ledge might create an obstruction to the movement of the crust during the lowering of the anode panel. In two industrial cells, the heat fluxes were measured on the "step" (SC2). In the two others, the heat fluxes were measured close to the anode (SC1). The ACM and crust located on the center channel were hardly accessible. Thus, they were not measured in this study. All measurements were done during the summer and the ventilation of each pot was kept at a "normal" rate. The draft rate exiting the pots was 1.3 Nm^3/s (normal cubic meters per second). The temperatures of the potroom and the gas exiting the pot were approximately 45 °C and 140 °C respectively.

Results and Discussion

The numerical model was run 8 times to simulate 4 different slices with mid-life anodes (4 with and 4 without cavity). These slices are representing 4 sets of measurements conducted on mid-life anodes. Measurements of the height of the cavity were performed on various cells to find a typical size to consider in the model. The anode cover was broken in order to roughly measure the height of the cavity with a ruler. The height was determined from the top of the liquid bath to the bottom of the anode crust. The measurements of the cavity in the side channel were done on 8 different mid-life anodes. The heights of the cavity located between anodes were measured on 6 different mid-life anodes. Figure 4 shows the average heights of the cavity in the side channel and between anodes along with the standard deviation.

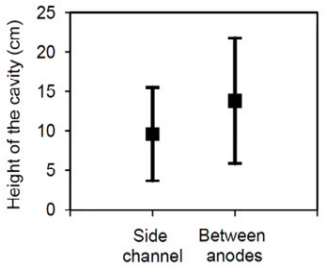

Figure 4. Height of the cavity of mid-life anodes

The average heights of the side channel and between anodes cavities were 10 cm and 14 cm respectively. In the numerical model, the heights of the cavities (in the SC and BA) were set to 10 cm and 15 cm respectively. The cavities are approximated by rectangular prisms in the numerical model. The measurement of the cavity in the center channel is difficult during the operation of the industrial cell. Thus, the height of the cavity in the center channel was also set to 15 cm (equal to the cavity BA).

Predictions and measurements of top heat losses

The top heat losses of 4 slices of mid-life anodes are predicted using the numerical model. The results are compared with the measured values of temperatures and heat fluxes. The following measured values were imposed in each simulated slice: heights of ACM and crust on anode, height of ACM on the side channel, temperatures of air and convection coefficients above the anode and above the side channel, and temperature on the anode rod. The "effective convection coefficients" were calculated from the heat flux measurements (q'') with equation 6, knowing the surface (T_s) and gas (T_∞) temperatures.

$$q'' = h\,(T_s - T_\infty) \qquad (6)$$

The calculated convection coefficients based on measurements were imposed in each slice to consider the real operating conditions and to reflect the local effects. The $T_{air,on\ anode}$ came from the average of two measurements taken on each side of the anode rod, one side close to the center and the other side close to the hood. A negative pressure is imposed inside the pot due to the ventilation system. Thus, the air of the potroom enters by the gaps between the hoods and is heated due to the heat dissipation from the top parts of the cell. The $T_{air,SC}$ was measured close to the gap between the hoods to consider the cooling effect of the air above the ACM on the side channel. The detailed boundary conditions are presented in Table III.

The results of the heat flux measurements and the predictions from the model (without and with cavities) are presented in Table IV. The locations where the sensors were positioned are described in Figure 3.

According to previous study, the heat dissipation from the anode cover increases quickly during the first hours and stabilizes after 24 to 48 h [22]. Thus, the slices with mid-life anodes investigated in this study were simulated at steady-state. However, even if the crust thermal behavior can be considered as stabilized, the heat dissipation will continue to vary slowly, because of the carbon anode consumption and anode panel movements (anode changes, metal tapping, etc.). For all simulated slices, the transformation of ACM into crust and the cavity formation were considered to be completed and stabilized. In the simulations without cavity, the anode crust was in contact with the liquid bath. In the case of slices with cavities, "empty" spaces were added instead of the anode crust. Accordingly, the heat fluxes were computed by the radiosity module. The slices with cavities offer a better representation of the electrolysis cell as they provide more realistic anode crust geometries.

The heat fluxes from the ACM on anode were well predicted in both cases of simulation (with and without cavity). However, the prediction of the heat fluxes from the ACM_{SC} presents a large deviation in the simulations without cavity compared to the measurements. Table V describes the prediction errors of models without and with cavities.

Table V. Average prediction errors of the simulations compared with the measurements

Average prediction errors	$q''_{on\ anode}$ (%)	q''_{SC1} (%)	q''_{SC2} (%)
Without cavity	3.8	-21.8	-39.5
With cavities	9.9	-7.3	-20.8

The average prediction errors of the models with and without cavity were calculated with the average results of heat fluxes in each slice. As expected, the heat fluxes simulated in the models with cavities were higher than without cavity. The predictions of the heat fluxes escaping the ACM on the side channel close to the anode assembly (SC1) were closer to the measurement in the simulations with cavities (-21.8 % vs -7.3 %). The average prediction errors of the q''_{SC2} (closer to the side wall) were larger than the others.

Table III. Boundary conditions for the 4 slices of electrolysis cells based on measurements

Slice	**Heights**					**Boundary conditions**			
	$ACM_{on\ anode}$ (cm)	$Crust_{on\ anode}$ (cm)	*$Cavity_{BA}$ (cm)	ACM_{SC} (cm)	*$Cavity_{SC}$ (cm)	$T_{air,on\ anode}$ (°C)	$h_{on\ anode}$ (W/m²·K)	$T_{air,SC}$ (°C)	h_{SC} (W/m²·K)
A	2.0	10.0	15	0.5	10	120	19	74	27
B	1.5	10.0		3.0		103	25	70	32
C	2.5	8.0		2.0		134	29	92	42
D	3.5	5.0		3.0		162	25	83	25

*The height of the cavity was imposed based on the average height, as measured in various cells

Table IV. Results of the measurements and the predictions

Slice	**Measurements**			**Without cavity**			**With cavities**		
	$q''_{on\ anode}$ (W/m²)	q''_{SC1} (W/m²)	q''_{SC2} (W/m²)	$q''_{on\ anode}$ (W/m²)	q''_{SC1} (W/m²)	q''_{SC2} (W/m²)	$q''_{on\ anode}$ (W/m²)	q''_{SC1} (W/m²)	q''_{SC2} (W/m²)
A	3478	5649	-	3490	4240	-	3685	5006	-
B	3874	3887	-	4023	3221	-	4262	3835	-
C	3566	-	1920	3790	-	1200	4008	-	1557
D	3450	-	1742	3612	-	1017	3842	-	1345

The "step" made of crust located on the side channel is characterized by an irregular shape. Thus, it is difficult to model precisely the "step" in the simulations. Figure 5 depicts the "step" observed in the industrial electrolysis cells.

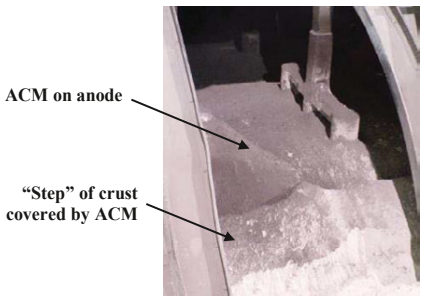

Figure 5. Photo of the "step" of crust in the slice D

Only a general shape of crust covered by ACM was used in the simulations to represent the "step". Except for q''_{SC2}, the predictions of the simulations with cavities were very close to the experimental error of the heat flux sensors (± 10 %). Accordingly, the combination of the chosen thermophysical properties with the measurements of the dimensions of each material enabled a prediction close to the measured values. A part of the error can also be attributed to the general rectangular shape of constant height used to represent the cavity.

General model and boundary conditions

A general model is built to provide the typical heat losses distribution for the studied mid-life anodes, with the average height for each material and boundary conditions previously described. Table VI and Table VII describe the conditions of the mid-life anodes in this general model.

Table VI. Average height of each material and cavity

$ACM_{on\ anode}$ (cm)	$Crust_{on\ anode}$ (cm)	$Cavity_{BA}$ (cm)	ACM_{SC} (cm)	$Cavity_{SC}$ (cm)
2	8	15	2	10

Table VII. Average boundary conditions above the ACM

$T_{air,on\ anode}$ (°C)	$h_{on\ anode}$ (W/m^2·K)	$T_{air,SC}$ (°C)	h_{SC} (W/m^2·K)
130	25	80	32

Figure 6 gives the temperature distribution in the top parts of the electrolysis cell, based on the simulation of the averaged model with cavities.

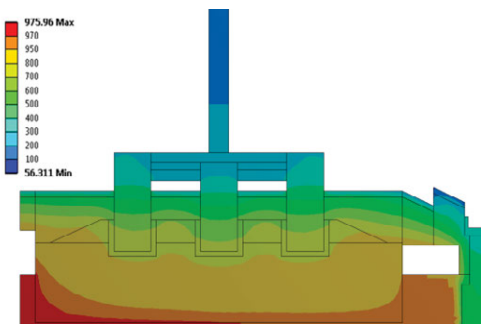

Figure 6. Temperature distribution in a mid-life anode

The temperatures predicted with this model on the surface of the ACM are very close to the measured temperatures. The average measured temperature on the surface of the ACM on anode was 282 °C while the predicted was 298 °C. The average measured temperatures on the surface of ACM_{SC1} and ACM_{SC2} were 240 °C and 145 °C respectively, and the predicted was 219 °C and 128 °C respectively. The error on the prediction of temperature follows the same trend as the heat flux (Table V). A part of the error can be attributed to the non-uniform leveling of the ACM and crust in reality. The heat losses distribution without cavity and with cavities is presented in Table VIII.

Table VIII. Heat losses distribution

Parts	Without cavity (%)	With cavities (%)
Anode rod	3.9	3.8
Stubs and yoke	16.0	15.9
Anode cover (on anode)	17.4	18.2
Anode cover (SC)	3.2	3.9
Top of deck	6.4	6.9
Side shell and cradles	36.1	34.7
Bar and flex	10.4	10.2
Bottom	6.6	6.4

The general model with cavities predicted higher top heat losses (48.7 % vs 46.9 %). The presence of cavities increased principally the heat losses from the anode cover and from the top of the deck. The heat losses from the side shell and the cradles decreased in the model with cavities. Hence, the top heat losses increased by roughly 2 % when cavities were included in the model, mainly reducing the side heat losses.

The impact of the height of the cavity on the heat flux was evaluated with the general model (conditions in Table VI and VII). The heights of the cavities in the center channel and between anodes were fixed at 15 cm. The height of the cavity in the SC was varied from 0 cm to 15 cm. Figure 7 demonstrates the impact of the cavity in the side channel on the heat flux escaping the ACM.

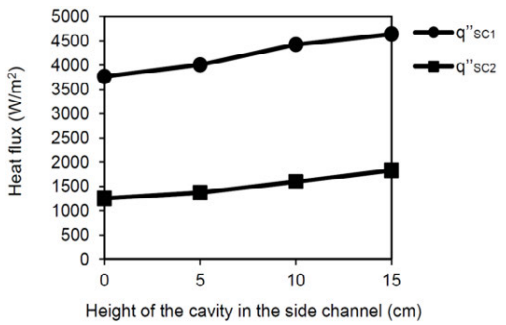

Figure 7. Impact of the cavity in the SC on the heat flux

The q''_{SC1} in the model without cavity in the SC was 3765 W/m^2 and was 4639 W/m^2 with 15 cm of cavity (increase of 23 %). With the same condition, the heat fluxes exiting the ACM_{SC2} was 1255 W/m^2 and 1833 W/m^2 respectively (increase of 46 %). The thickness of the crust is reduced by the process of melting and falling. Accordingly, the thermal resistance of the anode crust decreases with increasing height of cavity, thus explaining the growth of the q''_{SC1} and q''_{SC2} values in Figure 7. In steady-state, the thermal balance of an electrolysis cell with a larger cavity will

be characterized by larger top heat losses and lower heat dissipation on the side.

Conclusions

The heat balance of electrolysis cells have to be rigorously controlled to ensure the optimal performance of aluminum production. The consideration of the cavity for the modeling of heat losses is mandatory in order to obtain a good prediction of the heat fluxes from the anode cover. The average prediction error of the heat flux from the ACM_{SC} close to the anode assembly is reduced from 21.8 % to 7.3 % when the cavities are included in the model. The average heights of cavities surrounding the mid-life anodes investigated in this study are approximately 15 cm between the anodes and 10 cm in the side channel. The heat flux from the anode cover in the side channel increases with the height of the cavity, implying the decrease of the side heat losses.

Acknowledgements

This work is financed and supported by Rio Tinto and "Natural Sciences and Engineering Research Council of Canada" (NSERC). Finally, a part of the research presented in this paper was financed by the "Fonds de Recherche du Québec - Nature et Technologies" (FRQNT) by the intermediary of the Aluminium Research Centre-REGAL.

Nomenclature

A	area (m^2)
E	electric field (V/m)
h	convection heat transfer coefficient ($W/m^2 \cdot K$)
k	thermal conductivity ($W/m \cdot K$)
q	heat transfer rate (W)
q''	heat flux (W/m^2)
$\dot{q}$	rate of energy generation per unit volume (W/m^3)
T	temperature (K or °C)
V	electric potential (V)
δ_{ij}	Kronecker delta
ε	effective emissivity
σ	electrical conductivity (S/m); Stefan-Boltzmann constant ($W/m^2 \cdot K^4$)

bold	a vector
[brackets]	a matrix

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

RIO TINTO AP44 CELL TECHNOLOGY DEVELOPMENT AT ALMA SMELTER

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Keywords: AP4X, AP40, long anodes, shell, energy consumption, thermal balance

Abstract

The AP30 platform reached an important milestone at the Alma Smelter. This latter is the first to operate above 400 kA. Following this success, Rio Tinto Aluminium group launched the AP44 cell development to offer a technology capable of operating above 440 kA. This represents fifty percent more than the original AP30 cell. The technology is expected to deliver world-class performance such as metal production of 3274 kg/day/cell and an energy consumption of approximately 13.23 kWh/kg.

Advanced modeling tools and process control system were developed and used to cope with the challenge in designing and operating high amperage cells. Consequently, the AP44 cells integrate the most recent AP technology developments.

The industrial piloting is underway at the Alma Smelter on eight dedicated cells to demonstrate the technology at both 405 kA and 440 kA to validate the transition parameters. At the end of September 2016, the new AP44 technology would be available for implementation in existing or new aluminium smelters.

AP4X Technology Evolution

The AP30 technology has already demonstrated robustness and spectacular capacity to increase metal production. In 1986, the first AP30 platform at 280 kA was installed in one of our plants, in France, and continuously improved in current density and productivity. Rio Tinto has gradually and successfully developed the AP30 platform towards higher amperage technologies. Over the years, advanced pot lining materials and control logics combined with low anode-cathode distance (ACD) operation have largely contributed to maximizing AP30 platform performance. In 2015, Alma Works was the first smelter to operate above 400 kA by using the available technological breakthrough.

Considering the relatively recent construction of the last four smelters equipped with the AP4X technology, there is now more than 25 AP4X based potlines aiming at increasing the overall performance. Accordingly, the cell development considers market needs by working on both energy consumption and high productivity package.

To exceed 400 kA with the AP30 platform

In recent years, Rio Tinto has actively launched several technology development initiatives in an effort to continue improving the AP4X technology, and to achieve operating amperage above 440 kA. Tests conducted at various plant sites, dedicated booster circuits (Alma, Dunkirk, Alouette, Grande-Baie) and research centres have contributed to develop and industrialise technological bricks for specific AP30 applications, while ensuring compatibility with existing series and lower capital expense.

The technological bricks are also developed to be combined in various ways to optimise performance and address different energy and market requirements associated with retrofit or greenfield applications.

At present, many technological bricks have been industrialised and fully integrated into the AP4X technology:

- Alma – 432 cells at 405 kA
- Sohar – 360 cells at 388 kA
- Alouette – 594 cells at 380 kA
- Kitimat – starting at 405 kA

The AP4X development program, in collaboration with various plants (e.g., Alma, Dunkirk, St-Jean, and Alouette), includes among others the following newly developed technological bricks:

- low-ACD operation for minimal power consumption or increased amperage;
- low-electrical-resistance anode assemblies and slot design;
- low-cost lining components to lower cell production costs;
- optimized shell design;
- advanced ALPSYS control logic.

The combined advantages of these technological bricks justified the development of a new AP4X platform. In order to maximise the overall project value, two strategies were proposed and studied:

- high production with the AP44 technology targeting operation above 440 kA,
- low energy cell with the AP42 technology targeting low specific energy consumption at 12.300 kWh/kg and at an amperage up to 420 kA.

AP4X Platform Trials

The new AP4X platform operates with longer anodes to support high amperage increase. Engineering studies were launched to identify the maximum anode size according to various constraints. The anode design needed to take into account a sufficient central channel space to dissolve alumina from the feeders, and a sufficient lateral channel space to create a sidewall ledge protection.

The new platform was first tested at Aluminium Dunkirk Works, in 2012. The objective of the trial was to evaluate the potential of

the new AP4X platform for low energy cell development (12.5 - 12.7 kWh/kg at 380 – 390 kA). The cell performance targets focussed on low energy consumption, robustness and low capital investment (CAPEX) including retrofitting for existing AP30 plants. One cell was built and operated in a plant environment to evaluate, understand and mitigate possible risks related to a new platform.

During cell operation, the thermo-mechanical behaviour of the new shell design was measured. Aluminium Dunkirk is equipped with a forced convection cooling network (FCN) aiming at cooling the shell wall. Upon blowing air along the side of the shell, the FCN increases the heat extraction to better support high amperage technologies and cools the sidewall of the shell. This technology allowed the development team to modulate the shell temperature and measure the resulting shell deformation. The shell deformation increases or decreases according to the change in temperature. The magnitude of the measured deflection was similar to that of the thermo-mechanical model predictions.

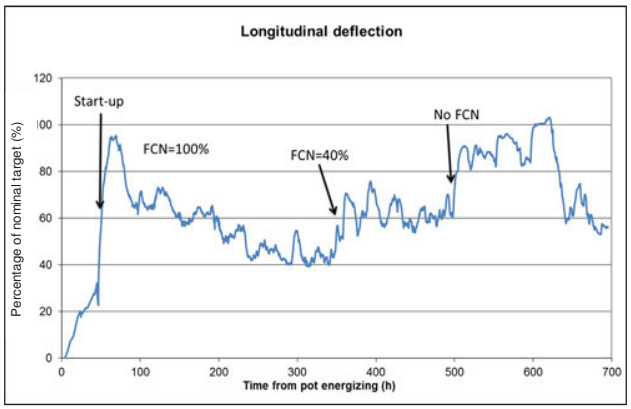

Figure 1: Shell deflection obtained at different FCN levels

Testing also allowed confirming the various issues or technological risk associated with long anode assemblies. The first risk associated with very long anode operation is the alumina feeding. In fact, the smaller liquid surface available could affect the alumina shot dissolution.

The second risk, especially for brownfield plant (retrofit), is associated with the asymmetric anodes assemblies that are required to compensate for the fixed anode beam position. Theoretical non-uniform current density could also lead to a non-horizontal butt surface wear. Another consequence is the importance of the angle between the stem and the vertical axis during anode change. This point could particularly introduce difficulties on anode change when removing and positioning the anode near the positive risers.

Figure 2: Long anode that is evenly consumed

Consequently, the Aluminium Dunkirk trials were used to demonstrate and validate various operational concerns with bigger anodes such as; i) anode change, ii) tapping, iii) anode plan movement , iv) process control, and, v) superstructure behaviour. In addition, specific measurements were taken to assess the shell structure behaviour and process related environmental aspects including; i) cover state, ii) pot hooding, and iii) general operating condition. The Dunkirk trial was also used to validate and implement retrofitting solutions to enable upgrade from the AP30 series to the AP4X platform.

The successful demonstration of the AP4X at Dunkirk provided valuable information to begin and accelerate a new cell development program to reach 440 kA. The AP44 cell potential should improve the current Rio Tinto production with more than 200 kt/yr, and the capital investment for implementation is estimated at lower than 2000 US/mt.

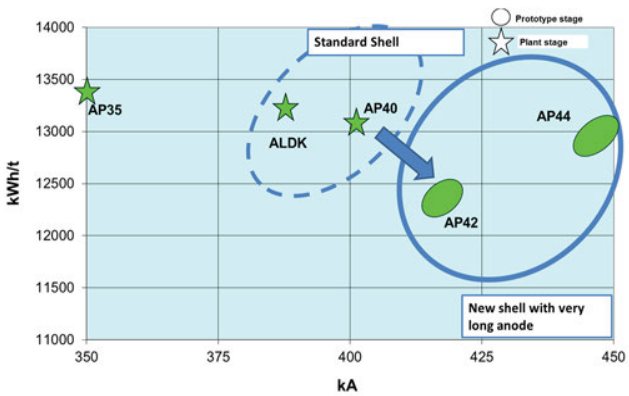

Figure 3: AP4X development roadmap

The operating windows shown in Figure 4 represent the set of operating points for which the thermal performance of the cell meets defined criteria within a given ACD range. The upper boundary is the high thermal limit while the lower boundary is the low thermal limit. Advanced modeling is used to set these

thermal limits. Depending on the cell design, the operating window moves towards higher amperage or low energy consumption. Figure 4 shows how combining different technological bricks can improve AP4X cell performance for the same range of anode-cathode distance. The brick combinations are customised to client's constraints. Further R&D is still required to achieve much higher amperages while reducing the negative effects on the magnetic stability of the cells.

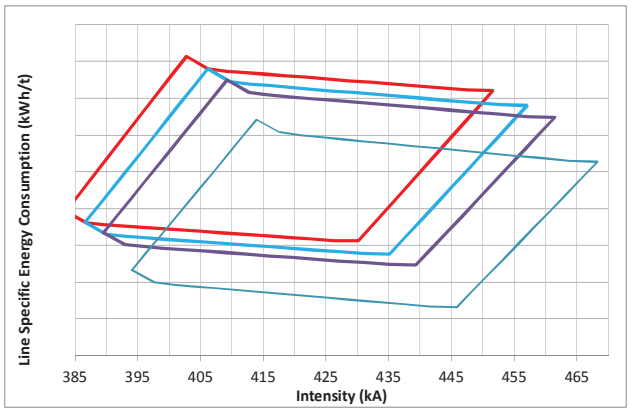

Figure 4: Impact of various technological bricks on performance operating window

AP44 target performance at Alma Works

In 2015, Alma Works achieved record amperages above 400 kA for the AP30 platform upgraded with AP40 technology. Consequently, the Alma plant was selected for the industrial piloting of the AP44 technology including the following reasons:

- two dedicated boosters capable of increasing the amperage by 50 kA, for high-amperage cell development;
- ALPSYS control system and potential R&D improvements;
- short distance from the Arvida R&D Centre and access to expert resources;
- opportunity from the pot relining campaign in 2016 to accelerate development.

Based on this opportunity, the plant's specific arrangements and existing technologies were considered during the early design stage. Indeed, Alma's existing conductors and standard anode assemblies were incorporated into the cell design. The expected performance during the industrial piloting is shown in Table 1.

Moreover, during the conversion, the two technologies will need to coexist at the highest possible amperage in the former technology. When converting from AP40 to AP4X, the latter must be able to operate at 400 kA and, preferably, with the same smaller anodes. This requires various technological component combinations and settings. The industrial pilot includes a design to operate at low amperage with the existing smaller anodes, and another design to operate at the targeted amperage of 440 kA with longer anodes.

Performance Indicator	Target	
Current intensity [kA]	405	440
Metal production [kg/cell/d]	2989	3274
Energy consumption [kWh/kg]	13.37	13.23
Anode effect frequency [EA/cell/d]	<0.1	<0.1

Table 1: AP44 performance targets for the Alma industrial pilot

Figure 5 shows the expected operating window for the AP44 technology designed for industrial piloting.

The low-amperage and high-amperage configurations will confirm the conversion strategy from the existing 400 kA cell line to the new AP44 platform.

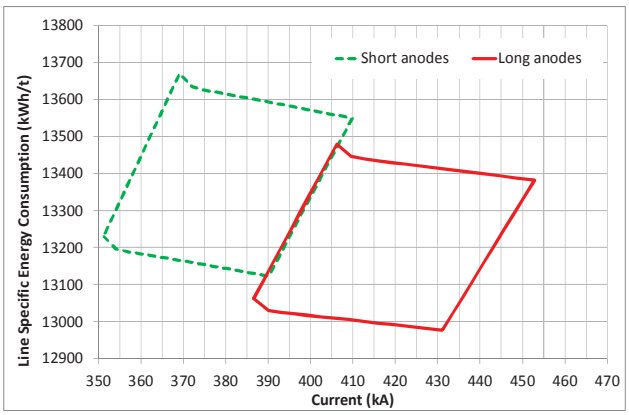

Figure 5: Operating windows of the industrial pilot cells

Dedicated Booster Circuit

To test the new high-amperage platform, a dedicated booster circuit had to be designed and implemented. It is known that higher amperages will amplify adverse MHD effects. Moreover, the presence of a booster circuit, in a cell line, can negatively affect cell stability. To lessen MHD disturbances and ensure the magnetic stability of the test cells, Rio Tinto MHD experts modelled and designed a busbar setup with a magnetic field suitable for both 440 kA pilot cells and 405 kA line cells. In line with the modelling, the Alma Works booster circuit was entirely reconfigured to allow starting and operating five cells at 440 kA. This was done in the spring 2015 to be ready for the 440 kA cell start-ups in July 2015. Figure 6 shows the five-cell booster arrangement.

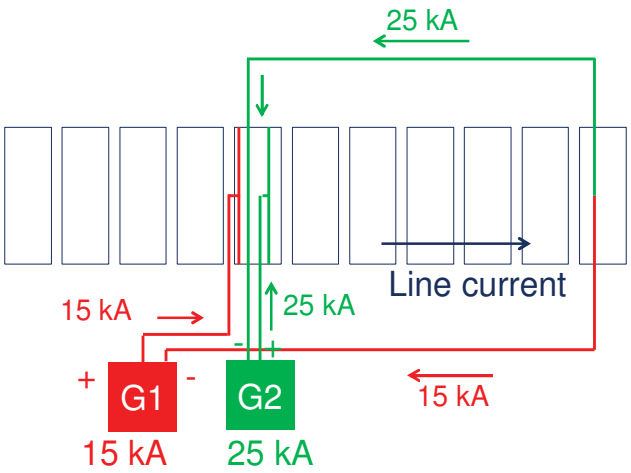

Figure 6: Diagram of the booster circuit arrangement

Experimental results

The industrial pilot includes a total of eight cells. Five cells were started at 440 kA, in the booster circuit, and three other cells were started at 405 kA line amperage. This setup was designed to confirm that the new AP4X platform could be started and operated in different current conditions. It was intended also to confirm that the technology could withstand the transition from the existing platform to 440 kA.

A total of eight test cells were successfully started in July 2015, at Alma Works. A new start-up method, helping to keep the cell voltage constant during the preheating stages, was also used. The size of the preheating bed was adjusted according to amperage levels. The heating rate of the cathode blocks was within acceptable limits and below the maximum rate of 40°C/h.

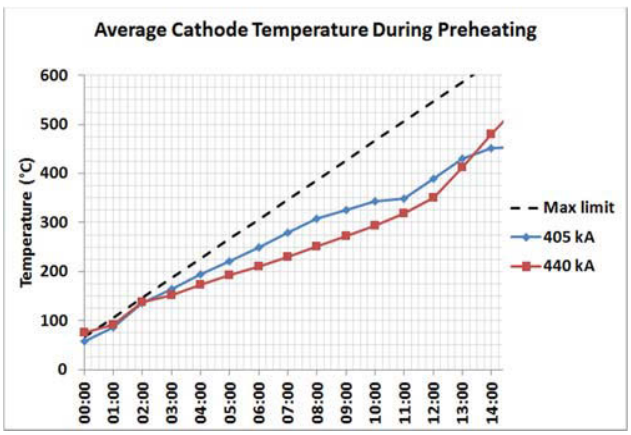

Figure 7: Average cathode temperature during preheating

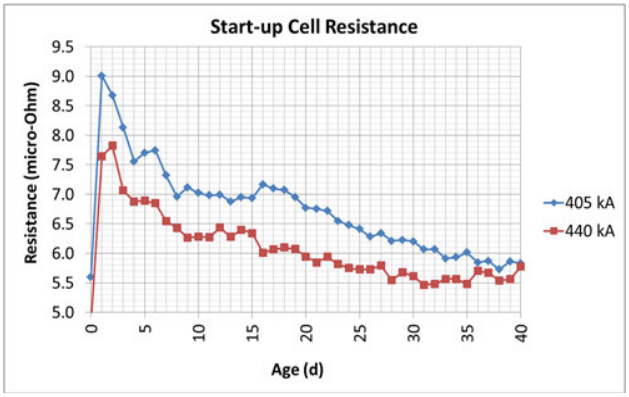

Figure 8: Cell resistance at start-up

Thermocouples were installed in the cell linings to compare readings with thermo-electric model predictions and also to assess start-up robustness by looking for signs of liquid infiltration in the linings. Initial measurements showed that the isotherms were properly positioned, in line with the prediction.

No sign of infiltration was detected during start-up.

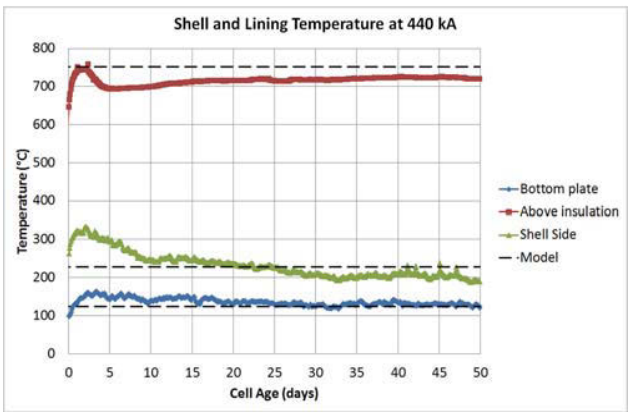

Figure 9: Lining temperature

Conductors

Apart from validating the cell technology, other technical aspects had to be checked experimentally to confirm their capacity to withstand the new current levels.

The higher amperage generates extra power in the cell-to-cell electrical conductors under the walkways and crossovers. First-generation conductors, initially designed for 300 kA, are the most limited in terms of thermal expansion and were modelled. The most aggressive condition for conductors is the electrical bypassing of more than three consecutive cells. Temperatures up to 260 °C, in some areas, can unavoidably generate extensive thermal expansion. To start up the cells at 440 kA, five consecutive cells had to be shut down and bypassed, meaning that heat expansion occurred and the thermo-mechanical behaviour of the conductors was assessed to confirm that the conductor network was able to withstand 440 kA.

The conductors deformation was observed, and temperature measurements were taken at various stages of the tests. Reference marks were made to assess conductor displacements according to line amperage and the number of bypassed cells.

Prior to starting up the cells, wedging zones were prepared to limit the electrical resistance between the wedges and busbars. Voltage drop and temperature measurements of the bypass wedges were taken at 440 kA with satisfactory performance. Figure 10 shows that voltage drop in the bypass wedges, was below 44 mV, and temperatures were below 150 °C with maximum tolerance being 200 °C. With properly prepared wedging zones, the AP30 wedges can withstand 440 kA.

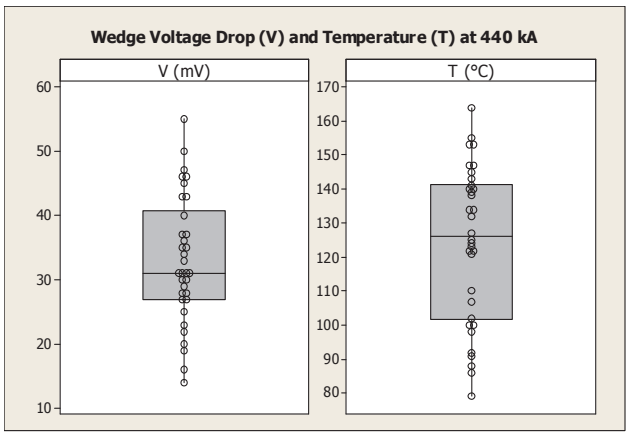

Figure 10: Voltage drop and temperature of short-circuiting wedge

Only the switching fuse was resized to allow for sufficient explosion delays at 440 kA, to efficiently protect the wedging zones.

Environment

For industrialisation purposes, the impacts of significant amperage increase on workplace conditions and occupational hygiene must be determined. The airflow inside the buildings and shop-floor temperatures were modelled. A temperature increase in the 3 °C to 7 °C range was expected, compared with AP40 technology operating at 400 kA. These results will be physically measured during testing.

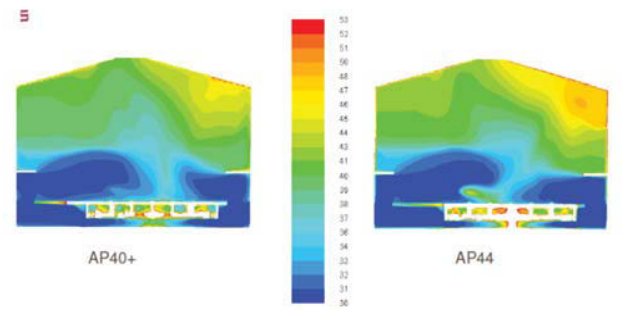

Figure 11: Temperature prediction in the building

Timeframe and next steps

The performance of the AP44 technology will be validated in the spring of 2016.

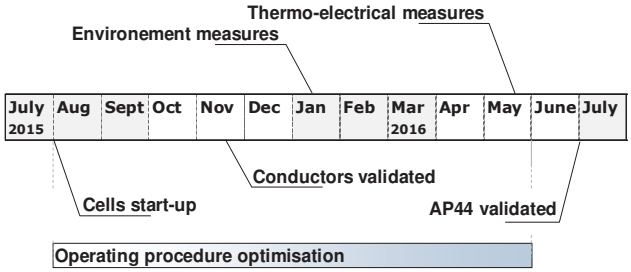

Figure 12: AP44 validation timeline

Test cell operation is currently ongoing and kept stable to ensure thermal balance. The upcoming milestones are:

- performance validation of various cell components;
- exhaustive thermo-electrical measurement campaigns and performance analysis to confirm AP44 performance at targeted and transition amperages;
- completion of the financial analyses for the purposes of industrialisation at Alma Works.

Conclusions

Since its first industrialization, the AP30 technology had been continuously developed following the economical worldwide context evolution.

The technological brick development approach allowed fast evolution of the different variants of Rio Tinto's AP4X technology.

The Aluminium Dunkirk trial has allowed validation of the use of bigger anodes and optimized shell design. Consequently, this pot technology is considered as the best pot upgrade to be implemented in AP30 based smelters given its easy implementation concept with changes mostly at low technological risks.

AP44 is currently being developed at Alma Works. Testing will allow validating the technology at two separate amperage levels and confirming the conversion strategy for an existing potline.

The withstanding capacity of various components at current levels in the 440 kA range, as well as long anode loads, will be confirmed.

The technology will be validated in the spring of 2016, in line with growth plans for Alma Works. The AP44 technology will be made available for industrialisation purposes.

Acknowledgements

The authors wish to thank Rio Tinto for the permission to publish the present work. The authors would like to express their gratitude to Messrs. S. Guérard, B. Bardet, D. Laroche, and M. Forte for their support in the cell development and to Messrs. C. Dreyer, R. Gariepy and J. Langlais for their support and guidance during the industrial piloting of the technology. Finally, special thanks to the Alma Work personal, R&D organisation and engineering group for their support and contribution during the industrial piloting.

Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

HYDRO'S CELL TECHNOLOGY PATH TOWARDS SPECIFIC ENERGY CONSUMPTION BELOW 12 KWH/KG

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Keywords: Aluminium Reduction, Cell Technology, Low Energy Consumption, Hydro Aluminium

Abstract

Hydro's HAL Ultra cell technology development has achieved a major milestone to demonstrate greener aluminium production by operating two different electrolysis cell platforms at energy consumption levels below 12 kWh/kg. Two cells based on HAL300 and HAL4e technology have been rebuild with newly developed solutions to reduce cell resistance, enhance heat conservation and further improve standard operational procedures. Applying Hydro's portfolio of modelling tools, materials knowledge, process know-how and operational competence enabled the development team to design high performance cell components and methods to reduce process variability. These elements were pre-tested on several cells before merging into full demonstrators. The HAL4e and HAL4e Ultra cell technologies will be installed as part of the Karmøy technology pilot plant.

Introduction

The continuous focus on environmental footprint, cost position and productivity in Hydro's aluminium reduction assets has been further intensified in 2010 by starting a major R&D initiative called HAL Ultra. Hydro's long-term vision to be able to produce aluminium at a specific DC energy consumption of only 10 kWh/kg without net CO_2 emissions has been defined [1] and is one of the cornerstones of the cell technology agenda.

The challenge of such an undertaking is to increase in parallel the fundamental knowledge of design bottlenecks and critical process steps, preferably together with academic partners, enhance the capabilities of relevant modelling tools, and organize work streams to be able to select an appropriate portfolio of new technology elements from the in-house technology innovation engine or trigger focused innovations. An intense phase of early testing such solutions on different cells allows the separation of effects and identification of potentials and side effects before bundling them together in demonstrators. The deep staggering of Primary Metals Technology (PMT) organization into competences within fundamental research, cell design, and process control down to engineering and operational teams has demonstrated the creation of fast and efficient development scenarios.

The announcement of Hydro's plan to build a technology pilot facility at Karmøy with both HAL4e and HAL4e Ultra cell technologies has given the development of HAL Ultra cells an early implementation perspective and a tight time-line as well. The plan is to begin operation at the Karmøy Technology Pilot (KTP) in 2017 at the earliest. An additional 75 000 tonnes of aluminium will be produced from 60 new cells, all based on hydropower.

New technology elements can of course be installed in existing smelters as tailor-made upgrade packages, subject to individual fit, strategy and business case.

Roadmap to 12 kWh/kg

The main technical and operational parameters influencing the operational window and performance of a cell can be structured into a) those that establish the built-in voltage drops and effects of process variability on cell voltage, b) those that influence dynamic anode-cathode distance (ACD) and bath voltage as well as magneto-hydrodynamic (MHD) stability, and lastly c) those design and operational parameters that determine the thermal balance. Schematically this is presented in Figure 1. The optimizing of the first set can be grouped by the term 'milli-volt chasing', as the main target is to reduce steady state or temporal electric resistances consequently. It was identified that there are typically significant improvement potentials in the area of:

- design of anode yokes
- design of cathode assemblies
- busbar system and contacts
- bubble effects and over-voltages
- process variability causing additional voltage drop or voltage compensation needs

The reduction of electric resistances in the anode yoke and cathode assembly by design changes typically has effects on the thermal characteristics, which can be favorable with respect to the selected operational window, specifically when amperage shall increase. However, geometric design and material selection can often be combined to get the desired result.

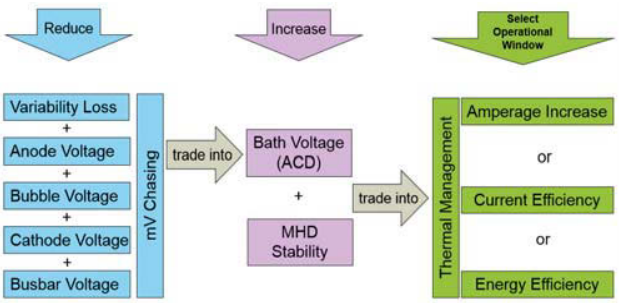

Figure 1: Technology elements and operational strategy

Those voltage gains that are related to joule heating participating in controlling the bath temperature, can be – at least abstractly - traded into a higher bath voltage by increasing ACD, thus staying in the same thermal operational domain. The cell design has at this point realized additional MHD stability just from ACD increase, but potentially this can be further increased by improvements to the busbar system and its magnetic pattern, or by a modified cathode current distribution.

The thermal design will allow using this stability potential to either increase amperage for a volume strategy, or reduce specific energy consumption, or realize higher current efficiency, or a combination of these. Besides the cathode lining design a large effect on the thermal characteristics can also be attributed to anode cover material properties and height, external insulation to the cathode shell [2], and the way the pot gas suction system is designed and operated.

A numerical model has been developed to evaluate the relationship between key design parameters, resulting operational window, and economic consequences within the frames of various financial parameters. Specifically the model consists of the following inputs and elements:

- key technical design parameters
- voltage balance break-down
- current efficiency model
- heat loss estimates and limits
- full OPEX cost model
- metal price assumptions

The current efficiency model is based on empirical data and functional relationships derived from MHD instability models [3]. The heat loss model is based on the parameters of the voltage break down and heat transfer relationships derived from 3D thermo-electric models.

The model can specifically help to analyze the economic consequences of a low-energy versus high volume strategy depending on key parameters like energy price and metal price, and the ability of an implemented technology to adapt to market conditions. Those technology elements on the left side of Figure 1 are typically always beneficial independent of strategy; however, built-in cathode lining characteristics are limiting the ability to adjust cell amperage and ACD.

Figure 2 presents an example of the calculated DC energy consumption of a HAL Ultra cell on the vertical axis based on a grid in the operational regime in terms of current efficiency and cavity heat loss, mapped to the axis parameters cell current and ACD. The data are scaled with respect to a reference set-point at 100%. The colored surface grid has lines of same cavity heat loss increasing from front to back, and constant current efficiency lines starting from 90% on the right to 96% far left.

The operational domain is confined by the limit of heat conservation and maximum heat loss. This is determined by cell robustness and additional operational measures, like anode cover material (ACM) height and flexible external insulation.

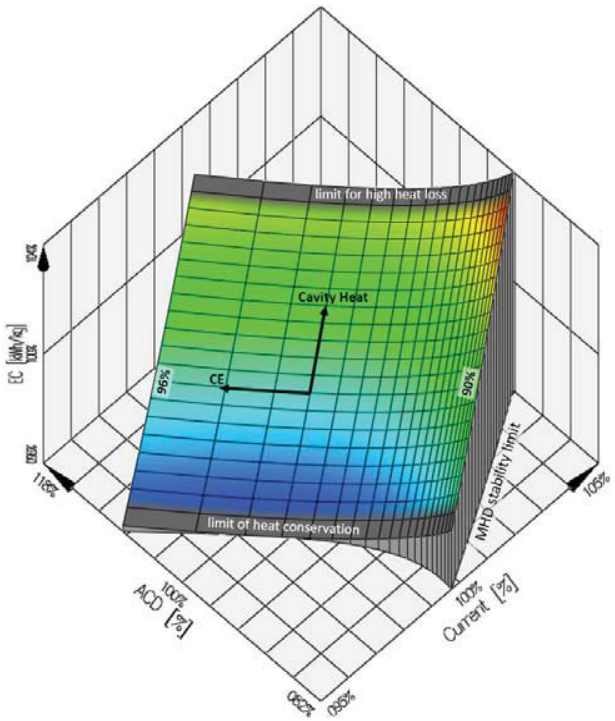

Figure 2: Specific energy consumption in the operational window of an HAL Ultra cell

The right side with increasing energy consumption is affected by decreasing MHD stability and thus dropping current efficiency, a consequence of the modelled non-linearly relationship to high amperage load and small ACD. The left side with high current efficiency is cut-off at 96% because of increasing unattractiveness of high cell voltage for marginal current efficiency gains.

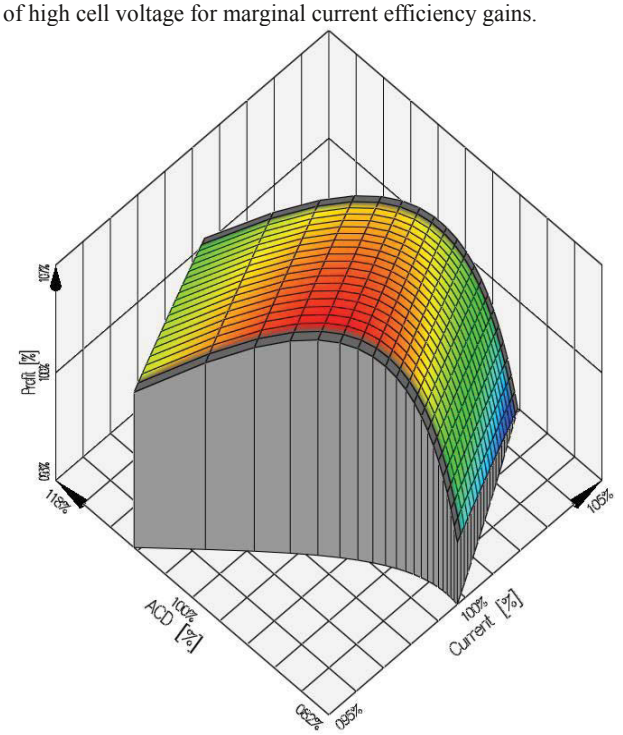

Figure 3: Economic result as function of the operational set-point

A detail can be noticed: the back reaction at low current efficiency actually deforms and extends the operational regime slightly at the right side due to the additional heating effect, noticeable as curved boundary close to lowest ACD.

External insulation of the shell has a major effect on the width of the operational window as it stretches the distance between low and high heat loss limit due to its flexibility when attached, adjusted, or removed.

As expected, the regime with lowest specific energy consumption is located on the front side of the diagram towards smaller ACD and low amperage, where the operational window is limited by feasible heat conservation. The minimum is located in this example at an intermediate location of the low heat loss side and can be used for operational parameter setting.

Applying the operational parameters in the same domain to the economic sub-model gives a direct prediction of financial margins and expected profit distribution. Figure 3 presents such an example where cost and price scenarios are favoring an operation close to the minimum of energy consumption. However, for high margins the optimum will shift to high heat loss and higher energy consumption. The model allows in this way to evaluate the options to maneuver the cell to a favorable set-point, but as well to evaluate and compare with alternative designs. Such evaluation will show the sensitivity and potential of a chosen strategy.
The expected wide applicability of low-energy technology components to different production strategies, as well as the specific advantage of low energy consumption components in a challenging market or at limited power availability, resulted in Hydro's development strategy towards low energy cells with flexible medium-to-high productivity ratio.

HALsee and HAL4e Ultra cells

Based on component tests, which had started already in 2009, the first low energy cell called HALsee was started in 2012. A photo is provided in Figure 4. HALsee stands for the abbreviation of Hydro Aluminium and super energy efficient. The main performance target was to operate this cell below 12 kWh/kg. The cell is using the footprint of the HAL300 cell that was originally designed for 300 kA, which is implemented at the smelters in Sunndalsøra and Qatalum.

Figure 4: HALsee cell in Hydro's cell technology Reference Centre

Although the original HAL300 cell has been the basis, the adaptation to a low-energy cell required significant modifications to most of the existing structures, including busbar, shell, superstructure and wall interfaces. The cell was equipped with a distributed pot suction system (DPS), which allows to reduce the off-gas suction rate without increasing the potroom emission level [4,5].

To monitor cell performance aspects such as lining temperature development, shell bending, or gas suction performance, more than 100 additional sensors were installed and allowed further calibration of our in-house modelling tools.
The design and performance target required us to focus on appropriate process adaptations and precise standard operational procedures (SOPs). In total more than 50 SOPs where either newly written or adjusted to the new requirements. A low-energy start-up free of anode-effects is most environment-friendly, cell voltage has been during start-up only slightly above 5 Volt for a very short time. By avoiding a high thermal shock the insulation in the cathode lining can be protected against early deterioration. Early operation is targeting an early stabilization of a stable thick ledge and further reduction of power input by balancing the heat with increasing conservation measures.

The recorded specific energy consumption of HALsee – presented as 6-month rolling average in Figure 5 – shows that in the first 180 days the energy mean energy consumption was already at 12.1 kWh/kg in average. Further process optimizations, including the tuning of the DPS system, allowed running the cell consistently below 12 kWh/kg thereafter.

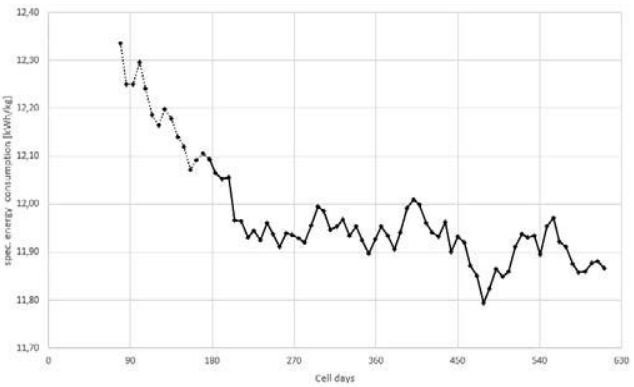

Figure 5: Specific DC energy consumption of HALsee cell. The solid line gives the 6-month rolling average.

The HAL4e Ultra cell is an evolution of the high amperage HAL4e technology [6]. Like the HALsee cell it has a similar set of design elements to reduce voltage drop and increase the cell's thermal insulation. The first HAL4e Ultra cell was started in 2014 and demonstrated energy consumption levels of 11.9 kWh/kg and less, based on metal inventory corrected 6-month averages (See Figure 6). As for the HALsee cell it must be noted that further tuning efforts on this operating cell will likely cause only smaller improvements as the temporal evolution becomes likely more asymptotic.

Cell voltage has been typically at 3.75 Volt when operating at 11.9kWh/kg.

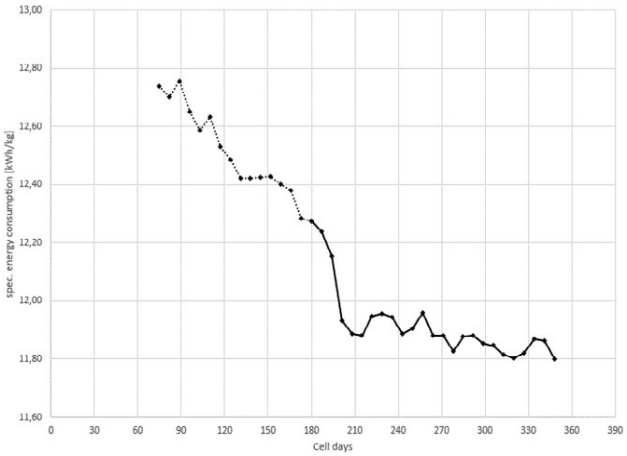

Figure 6: Development of 6-month rolling average energy consumption of HAL4e Ultra

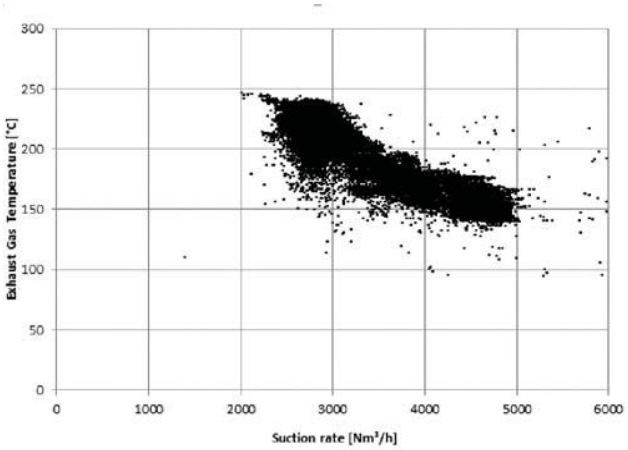

Figure 7: Exhaust gas temperature measurement points as function of suction rate for HALsee

Effect of distributed pot suction (DPS)

The distributed pot suction system in combination with an overall improved gas capture efficiency of the cell and specific thermal design of the superstructure, allows to vary the suction rate within a wide area. The DPS system can deliver the exhaust gas at higher temperatures because of vicinity to the point-feeder holes and higher exhaust gas concentration. Heat losses to the superstructure are significantly reduced by a special thermal design, including the cell covers. At the same time the equipment mounted on the superstructure is less exposed to heat stress thereby potentially prolonging the lifetime of this equipment. The suction rate thus becomes part of the thermal management of the cell. The thermal design of the covers has a positive effect on the heat stress level for operators as there is less heat radiation from cell superstructure.

While operating in DPS mode the suction rate per cell was varied in a range from 5000 Nm³/h down to 2500 Nm³/h. The gas temperature at the cell exhaust gas outlet changed from typically 150°C to levels close to a maximum of 240°C (See Figure 7). During pot tending operations the suction rate is temporarily increased. The measurements points from this increased suction are outside the diagram and not considered here because of the short-term effect.

The heat removed by the suction gases, calculated from temperature increase, under-pressure, and concentration levels, is decreasing at the same time (See Figure 8). At lowest suction rate about 50kW of thermal power are conserved in the cell, or more than 0.4 kWh/kg if converted to thermal contribution. However, the net effect is likely smaller due increased heat loss from other parts of the superstructure.

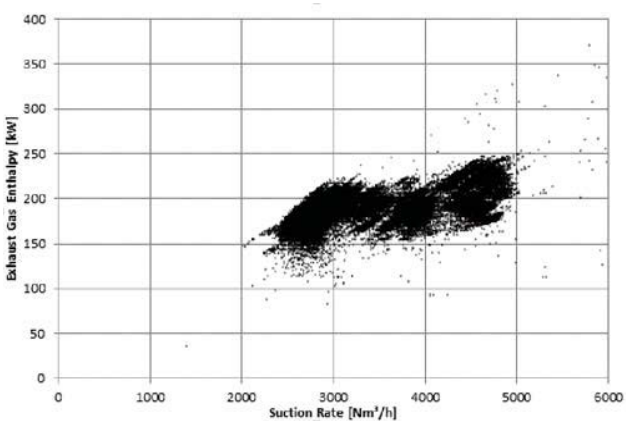

Figure 8: Thermal energy removed in the exhaust gas from HALsee

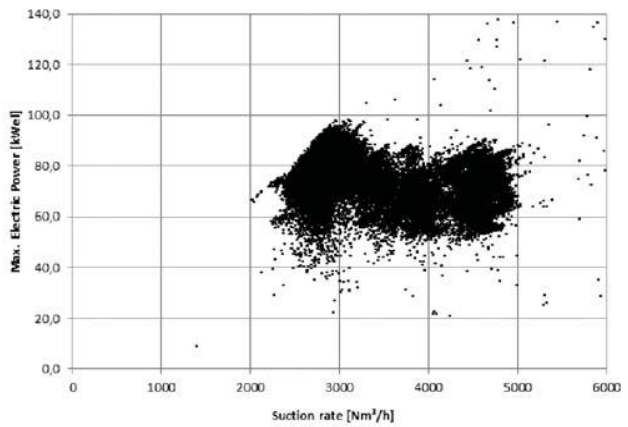

Figure 9: Electric equivalent of the thermal power in the exhaust gas when converted to electricity by the idealized Carnot cycle.

To indicate the value of elevated temperature level to a potential heat recovery system to convert heat to electricity, the idealized Carnot cycle is applied as shown in Figure 9. It has been shown that elevated temperatures from a DPS system are very attractive when considering a heat recovery system [7]. Although it is

obvious that the calculated electric power figures are far too optimistic when applying this theoretical maximum efficiency, it can be observed that the higher temperatures mostly compensate for the smaller amount of enthalpy at low suction rate. The low suction rate does therefore not necessarily result in less recovered energy.

Although a heat recovery system has not been installed with the HAL Ultra cells so far, the elevated temperatures in the off-gas are a critical ingredient and can help to further reduce the energy consumption substantially. The lower suction rate also has consequences for the design base of a new gas treatment plant when planning a full potline. In addition, the more than doubled concentration of CO_2 in the exhaust gas can be essential in unlocking the potential to a future CO_2 treatment.

Summary

Hydro's initiative to develop cell technologies for very low energy consumption has resulted in several technological design elements and operational procedures allowing the demonstration of full cell operation at and below 12 kWh/kg. These are applied to two different cell platforms. This is a very essential step to validate that this level of energy consumption can be achieved in a controlled way. Importantly, it encourages the development of even more energy efficient technologies. Thus far the HAL4e Ultra technology has achieved the planned performance figures and is to be further tuned and evaluated in the remaining time before the technology will be installed at the Karmøy Technology Pilot.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE SUCCESSFUL IMPLEMENTATION OF DUBAL DX+ TECHNOLOGY AT EMAL

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Keywords: High amperage, DX+ technology, performance test.

Abstract

DUBAL (also known as EGA Jebel Ali Operations) DX+ Technology was selected for the construction of EMAL (also known as EGA Al Taweelah Operations) Potline 3. The 444 DX+ cells installed in Potline 3 successfully started-up over 9 months from September 2013. Amperage was increased from 440 kA to 444 kA during the start-up and reached 450 kA two months after completing start-up. As required in the Technology Licence Agreement, a Performance Test of the DX+ Technology was carried out by the DUBAL Technology team on a group of 30 adjacent cells over a period of 28 days in November 2014. The key performance indicators achieved by the test cells and by the rest of the potline exceeded the guaranteed performance criteria. In this paper, detailed analysis of the results of DX+ cell operation at 450 kA and a description of the Technology Performance Test are given.

Introduction

DUBAL developed the proprietary DX+ Technology and proved its performance in five DX+ demonstration cells in DUBAL's Eagle Section from July 2010 to January 2014 [1]. The demonstration cells were started up at 420 kA and cell amperage was increased in stages with long time operation at 440 kA and 450 kA before the cells were shut-down in January 2014. An industrial scale potline, comprising 444 DX+ cells was started up in Potline 3 at EMAL (also known as EGA Al Taweelah Operations) between September 2013 and June 2014 at 440 kA to 444 kA [2]. The amperage reached 450 kA in August 2014 and 455 kA in March 2015, where it currently operates. As required in the Technology Licence Agreement, a Performance Test of the DX+ Technology was carried out on a group of 30 adjacent reduction cells over a period of 28 days in November 2014.

In this paper, detailed analysis of the results of DX+ cell operation at 450 kA and a description of the Technology Performance Test are given. Contractual and non-contractual data were monitored and are reported here.

Milestones in the development of DX+ Technology

DX+ Eagle Demonstration Cells: The goal of designing and building DX+ demonstration cells was to test a high productivity upgrade of DX cells for the planned Phase II expansion of EMAL The cells were started up in July and August 2010 at 420 kA and were all in stable operation by December 2010, with good performance parameters facilitating EMAL's decision to select DX+ Technology for Phase II. With subsequent amperage increases, the DX+ Eagle demonstration cells had been operating at 450 kA for four-and-a-half months by the time EMAL Potline 3 began the start-up of its 444 DX+ cells. Excellent performance of the DX+ demonstration cells had been proven at all amperage levels from 420 kA to 450 kA [1]. Table I shows the average performance of the five DX+ Eagle cells, together with EMAL Potline 3 cells prior to the Performance Test, both at nominal amperage of 450 kA.

DX+ EMAL Potline 3 Cells (Figure 1): EMAL began constructing Potline 3 in October 2011 and started-up the first cell on 11 September 2013, three months ahead of the original start-up plan. The 444th cell was started up on 13 June 2014. EMAL Phase II was built and started-up in shorter time than planned and below the projected budget.

Potline 3 was designed for the amperage range of 420 kA to 460 kA. The start-up began at 440 kA and ended at 444 kA. Two months later, on 26 August 2014, the amperage was already at 450 kA at which point it was decided to carry out the Performance Test. Table 1 gives key performance indicators (KPIs) at 450 kA in order to show that the operation was well established before the Performance Test.

Figure 1. DX+ Potline 3 in EMAL.

Amperage Increase to 455 kA in DX+ EMAL Potline 3 Cells: Amperage increase from 450 kA started on 30 November 2014 and 455 kA was achieved on 11 March 2015. Table 2 gives the performance for the full period at 450 kA from September 2014 to February 2015 and the period at 455 kA from March to August 2015. The former includes the transition period from 450 kA to 455 kA.

Table I. DX+ Eagle and DX+ EMAL Potline 3 key performance indicators at 450 kA.

KPI	Units	DX+ Eagle 29.4.2013 to 12.1.2014	DX+ EMAL 1.9.2014 to 31.10.2014
Amperage	kA	449.049	450.05
Current efficiency	%	94.92	95.21
Metal production	kg/pot-day	3 433	3 450
Net cell voltage*	V	4.26	4.24
DC net specific energy consumption*	kWh/kg Al	13.38	13.28
Gross carbon consumption	kg C/t Al	526	547
Net carbon consumption	kg C/t Al	409	412
Bath temperature	°C	961	956
Excess AlF_3	%	11.4	10.2
Fe	%	0.048	0.039
Si	%	0.026	0.023
AE frequency	AE/pot- day	0.064	0.120
AE duration	s	14.5	8.5
PFC emissions CO_2 Equivalent**	kg/t Al	17	19

*Net voltage and net specific energy exclude contribution of busbar linkages.
**Calculated using the method described in [3].

Table II. DX+ EMAL Potline 3 key performance indicators at 450 kA and 455 kA.

KPI	Units	September 2014 to Januaryr.2015	February to August 2015
Amperage	kA	449.8	454.7
Current efficiency	%	94.63	94.99
Metal production	kg/pot-day	3 427	3 476
Net cell voltage	V	4.24	4.26
DC net specific energy consumption	kWh/kg Al	13.35	13.39
Gross carbon consumption	kg C/t Al	545	556
Net carbon consumption	kg C/t Al	412	416
Bath temperature	°C	958	960
Excess AlF_3	%	9.9	10.0
Fe	%	0.041	0.042
Si	%	0.024	0.025
AE frequency	AE/pot-day	0.19	0.11
AE duration	s	56.7	26.0
PFC emissions CO_2 Equivalent	kg/t Al	195	52

EMAL Phase II Performance Test

Performance Test Criteria: The two performance criteria to be met by the test pots as per the Technology Licence Agreement are:

- Specific DC energy consumption lower or equal to 13.45 kWh/kg Al and,
- Average primary aluminum production per pot and per day greater or equal to 3 200 kg.

Performance Test Procedure: As specified in the Technology Licence Agreement, a Performance Test of the reduction technology provided by DUBAL was carried out on a group of 30 adjacent typical reduction cells (3B076 to 3B105) over a period of 28 days starting, on 1 November 2014. During the test period, the line amperage was stable at 450 kA.

The performance of the test cells was monitored throughout the 28-day test period by individual pots and as a group. DUBAL advisors were present (shift coverage) to monitor the Performance Test cells, evidencing excellent collaboration with and support of the EMAL operation team.

Metal tapping operations took place every 32 hours for each of the test cells, without any delay or backlog. During the test period, metal was sampled in the cells every eight days as well as at the beginning and the end of the test period. The metal reserve was measured for each of the test cells before and after the test period, using the copper dilution method. Corrections were made for solid metal recovered from crucible cleaning and metal skimming, as well as from bath tapping sows.

Bath samples for the analysis of excess AlF_3 and CaF_2 concentration were taken in every pot every four days. Therefore seven series of bath analyses were carried out during the test period. Bath temperatures of the test pots were measured every two days, therefore 14 series of bath temperatures were reported during the test period. At the beginning of the test, 72 anodes were weighed and during the test 72 anode butts were evaluated for net carbon consumption.

Total fluoride emissions through the roof were measured according to the US EPA 14A cassettes method in the section adjacent to the performance test group in Potroom B. It was also measured in one section in Potroom A with the same cassettes method. Figure 2 shows the location of the Performance Test group and fluoride monitoring sections. All sections are also equipped with Boreal Laser.

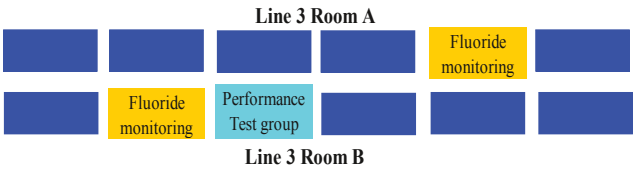

Figure 2. Location of Performance Test group (pale blue) and fluoride monitoring sections (orange) in Potline 3.

Performance Test Results: Table III gives KPIs for the Performance Test and full Potline 3 for the month of November 2014.

Table III. DX+ EMAL Potline 3 key performance indicators during the Performance Test in November 2014.

KPI	Units	Test group, 1 – 29 November	Full Potline 3, 1 – 30 November
Amperage	kA	450.1	450.1
Current efficiency	%	95.0 94.9*	94.6
Metal production	kg/pot-day	3 443	3 428
Net cell voltage	V	4.23	4.24
DC net specific energy consumption	kWh/kg Al	13.27 13.28*	13.35
Gross carbon consumption	kg C/t Al	546	550
Net carbon consumption	kg C/t Al	411	411
Bath temperature	°C	958	957
Excess AlF_3	%	10.4	10.2
Fe	%	0.044	0.040
Si	%	0.027	0.026
AE frequency	AE/pot-day	0.075	0.089
AE duration	s	6.7	7.8
PFC emissions CO_2 equivalent	kg/t Al	9	13

*After metal inventory and solid metal corrections.

Standard deviations of some pot parameters were also calculated. These were: bath temperature 6.1 °C and 6.3 °C, excess AlF_3 1.0 and 1.1, metal height 0.7 cm and 0.8 cm, respectively for the test group and full potline. The standard deviation for bath height in the test group was 1.8 cm.

The net specific DC energy consumption does not include the share of linkage busbar voltage drops. The inclusion of these gives gross voltage and gross specific energy consumption, which were 4.242 V and 13.31 kWh/kg Al, respectively for the test group.

Total fluoride emissions through the potroom roof, measured with cassettes according to EPA 14A method, were 0.17 kg/t Al; and total particulate emissions were 0.25 kg/t Al during the test period.

Graphs of EMAL Potline 3 Performance Test

The evolution of cell parameters in the test group of pots during the Performance Test is shown in Figures 3 to 15.

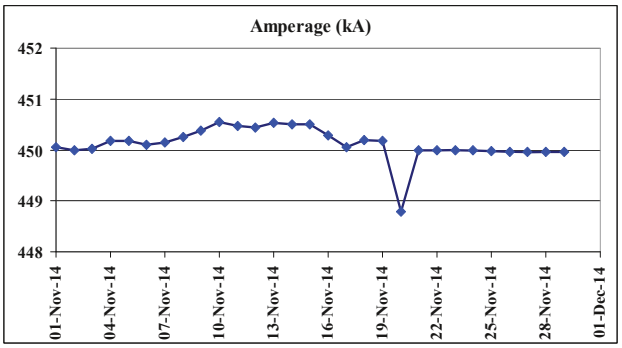

Figure 3. Amperage.

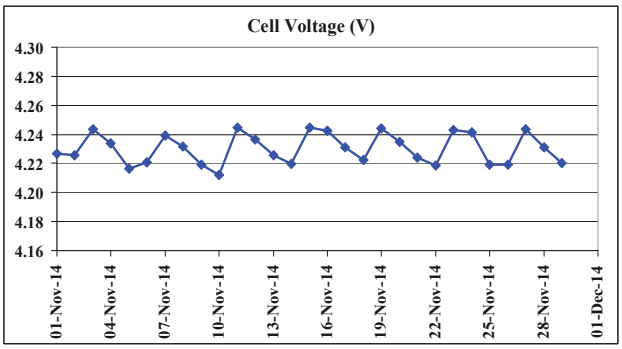

Figure 4. Cell voltage.

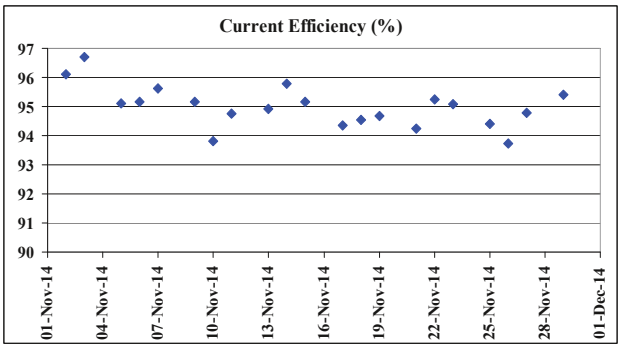

Figure 5. Current efficiency.

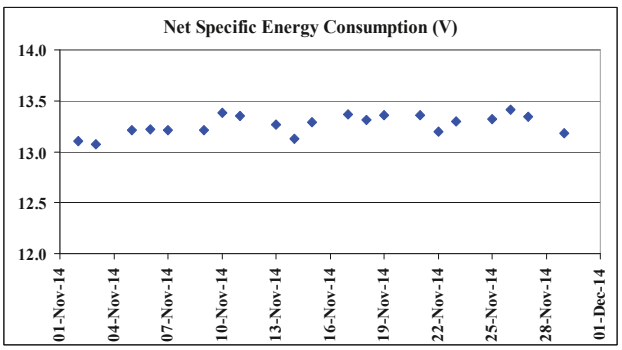

Figure 6. Net specific energy consumption.

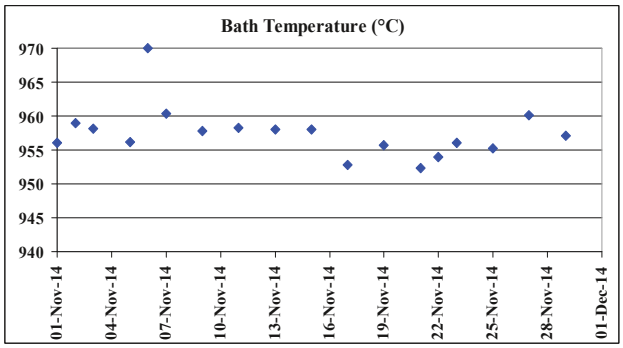

Figure 7. Bath temperature.

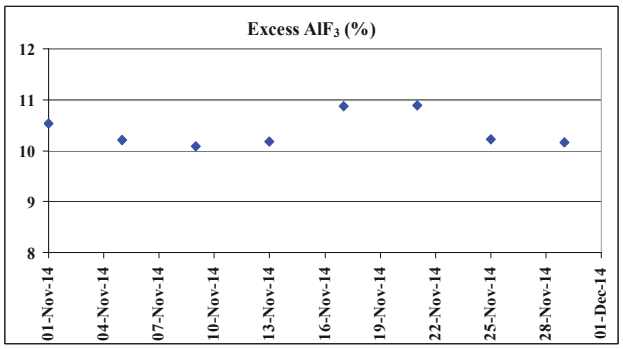

Figure 8. Excess AlF_3.

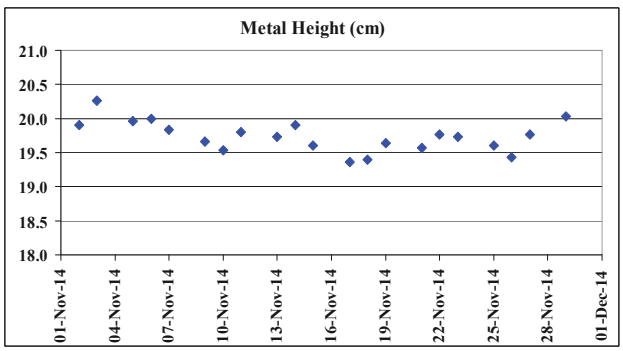

Figure 9. Metal height.

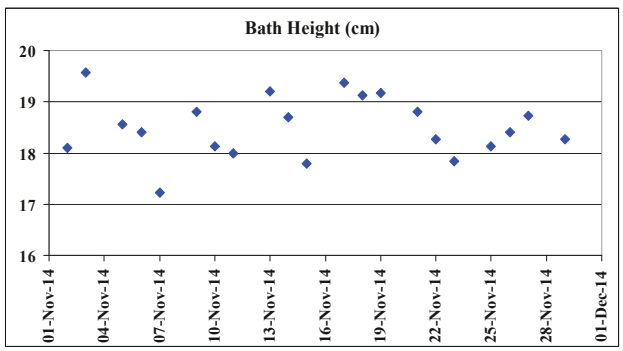

Figure 10. Bath height.

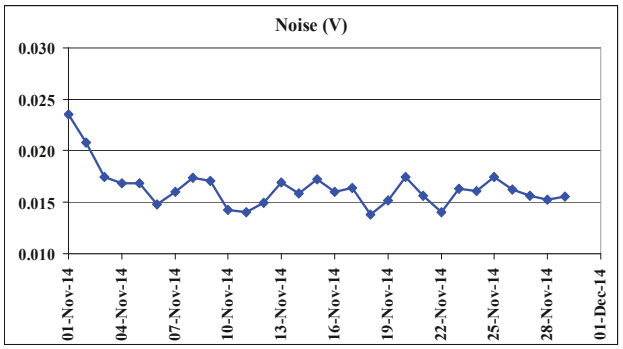

Figure 11. Noise.

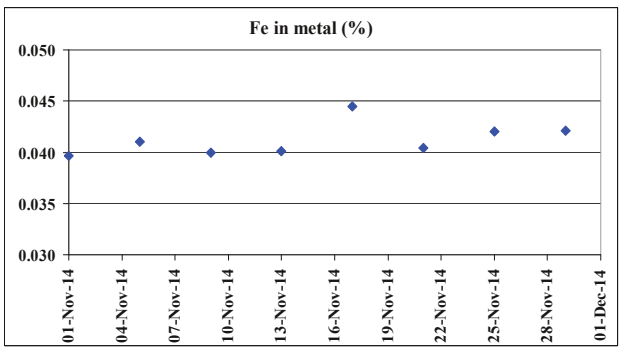

Figure 12. Fe in metal.

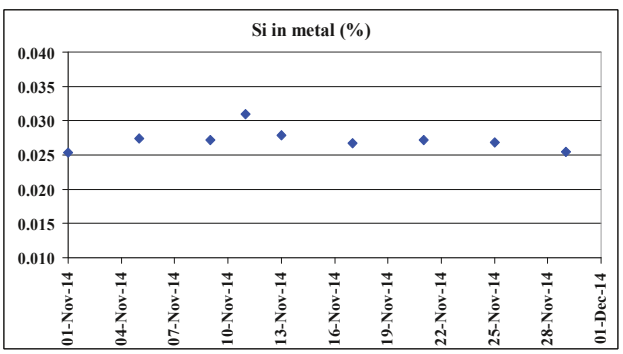

Figure 13. Si in metal

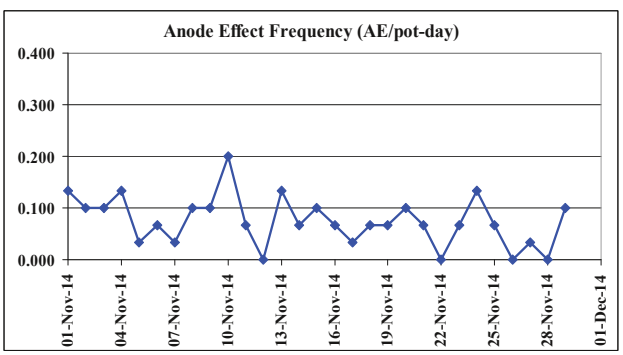

Figure 14. Anode effect frequency.

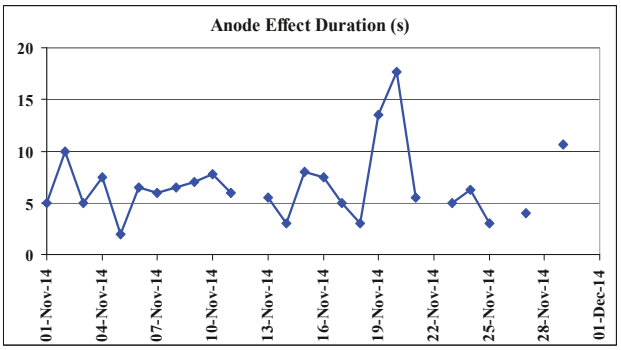

Figure 15. Anode effect duration.

Conclusions

DUBAL DX+ Technology has progressed from the conception and modeling phases through to industrial implementation in a very short period of time. In EMAL Potline 3, the Performance Test confirmed that the DX+ pot performance exceeded by a large margin the Performance Criteria set in the Technology Licence Agreement. Other performance parameters not specified in the Licence Agreement were also world-class.

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ALUMINA DISSOLUTION MODELING IN ALUMINIUM ELECTROLYSIS CELL CONSIDERING MHD DRIVEN CONVECTION AND THERMAL IMPACT

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Keywords: Alumina, dissolution, concentration

Abstract

In the last decade, the cell productivity has been dramatically improved as demonstrated on AP30 technology with amperage increase from 350 kA to 440 kA. At the same time, available liquid bath volume in the pots has been drastically reduced by the anode surface enlargement. In order to achieve such impressive step change, management of alumina dissolution is a key point to optimize both cell design and process control algorithms.
Rio Tinto Aluminium and EPFL have developed a 3D model taking into account the cell geometry, metal and bath velocities, alumina granulometry and thermal balance. It enables to optimize number, location and feeding rate of point feeding systems.
In this paper, we present the model development, the latest progress and the model validation with measurements performed on AP technology pots.

Introduction

About two tons of alumina are needed for one ton of produced aluminium. In recent pot technology, the alumina is introduced into the bath through point feeding systems. The typical rate of those feeders is to introduce at once 1 kg of alumina every minute. This shot of alumina falls on the bath air interface. A part stays on the interface and a part sinks. The whole mechanism is described by Dassylva [1]. The shots locally decrease the bath temperature. The surrounding bath freezes on the shot and agglomerates the alumina. This agglomerate is convected. and is heated up by surrounding bath and melts depending on local temperature and local bath chemistry. The location of the dissolution is strongly dependent of the fluid flow, thermal balance and local chemistry.

In the same time, with the evolution of modern technology the alumina concentration distribution has become a major constraint to continue to improve the productivity while lowering the energy consumption.

The productivity of aluminium electrolysis cells has dramatically increased in the past decade. On AP30 technology, amperage was increased from 350 kA to 440 kA without number of feeder modification. This means that the shot that was introduced every minute is now introduced every 45 seconds. This leads to an increase of temperature variation in the bath.

In the same time, the development of low energy cells (see Martin [2]) has led to a decrease of the available power to melt and dissolve the shots of alumina. Typically the available power by Joule effect has decreased by 15 - 20 % in the anode-to-cathode distance (ACD).

The reduction of the ACD has lead to a lower bath volume typically by 10 – 15 % and the increase of the anodic surface has lead to another bath volume decrease by an extra 15 - 20 %. The bath is the dissolved alumina tank. The lower the bath volume the more reactive the pot is.

Because of the this trends in modern aluminium production, it is necessary to control the way we introduce alumina in pots to avoid local anode effects (see Tarcy [3]) and sludge production because of a very poor dissolution.

To answer these major challenges, Rio Tinto Aluminium has developed a suite of chained models to provide good electrical balance in busbars configuration, satisfactory MHD stability in the cells, thermal balance ensuring a ledge protection of the lining and good alumina distribution. Those coupled models were described by Langlois [4].

Model description

This paper is focusing on the alumina dissolution software.

Two fluids Navier-Stokes

The first step in the alumina dissolution model is to calculate the stationary fluid flow for the two fluids (bath and aluminium), taking into account Lorentz Forces. The stationary calculation was described by Flueck [5].

The incompressible Navier-Stokes equations are solved:

$$\begin{cases} \rho \frac{\partial \vec{u}}{\partial t} + \mathrm{div}\left(\tau(\vec{u}, p)\right) + \rho(\vec{u}.\vec{\nabla})\vec{u} = \vec{j} \times \vec{B} + \rho\vec{g} \\ \mathrm{div}(\vec{u}) = 0 \end{cases} \quad (1)$$

With $\vec{u}$ the velocity vector of the fluid, ρ the fluid density, $\vec{j}$ the current density vector, $\vec{B}$ the magnetic field vector, $\vec{g}$ the gravity vector and with $\tau(\vec{u}, p)$ the stress tensor defined as :

$$\tau(\vec{u}, p) = 2\mu\varepsilon(\vec{u}) - p\mathbb{I} \quad (2)$$

With μ the local viscosity of the fluid, $\mathbb{I}$ is the identity matrix, p the pressure and $\varepsilon(\vec{u})$ the strain rate tensor defined as:

$$\varepsilon(\vec{u}) = \frac{1}{2}\left(\vec{\nabla}\vec{u} + (\vec{\nabla}\vec{u})^T\right) \quad (3)$$

^{T} denotes the transposed vector.

The free aluminium-bath interface is solved with a level set method using an arbitrary Lagrangian-Eulerian formulation. The φ function is used to track the interface and is defined as follow:

$$\varphi(t,\vec{x}) = \begin{cases} > 0, for\ \vec{x}\ in\ bath \\ < 0, for\ \vec{x}\ in\ aluminium \\ = 0, for\ \vec{x}\ on\ the\ interface \end{cases} \quad (4)$$

The φ transport equation to track the interface is solved:

$$\frac{\partial \varphi}{\partial t} + \vec{u}.\vec{\nabla}\varphi = 0 \quad (5)$$

At each iteration of mesh deformation, the ACD is kept constant.

Turbulence

The turbulence model is described by Steiner [6]. It is a mixing-length turbulence model:

$$\mu(x) = \mu_L + \mu_T = \mu_L + \rho\alpha_t h^2\sqrt{2\epsilon(\vec{u}):\epsilon(\vec{u})} \quad (6)$$

With $\mu_L$the laminar viscosity, μ_T the turbulent viscosity, α_t a coefficient and *h* the typical mesh element size.

Maxwell equations

To obtain the magnetic field and current density vectors, the Maxwell equations are solved (see Flueck [7]):

$$\begin{cases} rot\ \vec{H} = \vec{j} \\ div\ \vec{B} = 0 \\ rot\ \vec{B} = \mu o\ \vec{j} \\ div\ \vec{j} = div\left(-\sigma\vec{\nabla}\emptyset + \sigma\vec{u} \times \vec{B}\right) = 0 \end{cases} \quad (7)$$

With $\vec{H}$ the magnetization vector, σ the electrical conductivity and ∅ the electric potential.

Alumina dissolution model

The alumina dissolution model was introduced by Hofer [8]. A particule population n_p is introduced in the bath and is driven by the following equation:

$$\frac{\partial n_p}{\partial t} + \vec{u}.\vec{\nabla}n_p + \frac{\partial}{\partial R}(f(T,c,R)n_p) = S \quad (8)$$

With *R* the radius of the particle, *c* the dissolved Al_2O_3 concentration, *S* is the source term and f the dissolution function defined as:

$$f(T,c,R) = -\alpha\frac{c_{sat}-c}{c_{sat}}f_{R(R)}\,H(T-T_{liq}) \quad (9)$$

With c_{sat} the dissolved alumina saturation concentration of the alumina in the bath, *T* the local temperature, $T_{liq}$the liquidus temperature of the bath, $f_{R(R)}$ is the dissolution speed of grain of radius R and *H* the Heaviside function.

When the particles melt, it increases the local dissolved alumina concentration:

$$\frac{\partial c}{\partial t} + \vec{u}.\vec{\nabla}c - D\Delta c = Q_1 + Q_2 \quad (10)$$

With D the diffusion coefficient of the dissolved alumina in the bath, Q_1 the source term related to the melt particles and Q_2 the concentration consumption due to the current.
The concentration diffusion coefficient *D* is strongly dependent of the turbulence:

$$D(\vec{x}) = \alpha_D\frac{\mu_T}{\mu_L} \quad (11)$$

In order to use the equation (9), we need to know the local temperature:

$$\frac{\partial T}{\partial t} + \vec{u}.\vec{\nabla}T - D\Delta T = P_1 + P_2 + P_3 \quad (12)$$

With P_1 the negative source term related to the introduction of the particle population n_p, P_2 is the negative source term relative to the dissolution, and P_3 is the Joules heating : $P_3 = \alpha_j\vec{j}^2/\sigma$. $\sigma$is the electrical conductivity of the bath. α_j ensures the energy conservation as follow :

$$\alpha_j = \frac{\int(P_1+P_2)\,dV}{\int^{j^2}/_{\sigma}dV} \quad (13)$$

The whole temperature dependency was introduced by Foetisch [9].

Time resolution

It is very important to model the good alumina shot weight. The shot size has a strong impact on the local bath temperature and time needed to melt the shot. This can be only achieved through an unstationary resolution and a precise description of the feeding strategy.

Thermal impact of dissolution

Two simulations in the same configuration were performed to evaluate the impact of the local temperature in the dissolution process, through Equations (9) and (12). It was decided to make one simulation with thermal equation (named Simulation 1) and one simulation without it (named Simulation 2). In simulation 2, a constant delayed time with no dissolution is added to take into account the freezing and melting of the alumina shot. The equation (9) is then written as follows:

$$f(c,R) = -\alpha\frac{c_{sat}-c}{c_{sat}}f_{R(R)}H(t-to) \quad (14)$$

The Heaviside function of time allows treating the delayed time with no dissolution. It is not necessary to solve equation (12) in Simulation 2.

A 510 kA APXe with 5 alumina feeders was modelled. Figures 1 and 2 are the horizontal concentration maps in the middle of the Anode Cathode Distance (ACD) respectively for Simulations 1 and 2. In the simulation 1 the concentration range is 2.1 - 3.5 %. In simulation 2 the range is 2.1 - 6.2 %.

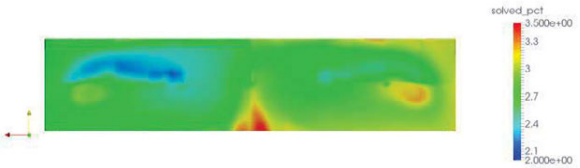

Figure 1. Mid-ACD dissolved alumina concentration map for simulation 1.

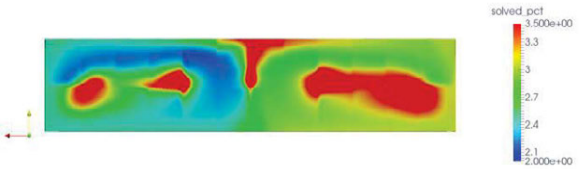

Figure 2. Mid-ACD dissolved alumina concentration map for simulation 2.

Main comparison criteria are given in Table 1. Simulation 2 results are very different than Simulation 1 ones. Thermals impacts of dissolution and melting of alumina has to be taken into account in order to model accurately alumina concentration distribution.

Table 1. Main comparison criteria for Simulations 1 and 2.

	Simulation 1	Simulation 2
Mean C (%)	**2.78**	**2.98**
Min C (%)	**2.1**	**2.1**
Max C (%)	**3.5**	**6.2**
Std Dev C (%)	**0.05**	**0.37**

One can be surprised by the fact that the mean dissolved concentrations are different in Simulation 1 and 2. This is due to the way the software is built. The alumina injection is made to ensure a constant mass ratio of alumina at 3 % (dissolved and undissolved). This means that in the APXe bath there is around 0.22 % of undissolved alumina.

The comparison between Simulation 1 and 2 demonstrates the major impact of heat on alumina dissolution. The local superheat drives the dissolution speed and cannot be neglected. The coupling with temperature is used in following simulations.

Model validation

Some measurement campaigns were done on 510 kA APXe LRF prototype to validate the alumina dissolution model. The measurements were described by Caratini [10].
We decided to make some bath samples when modifying the pot feeding strategy.
In a first step, the feeding strategy uses 6 feeders. The measurement point locations are given on Figure 3.

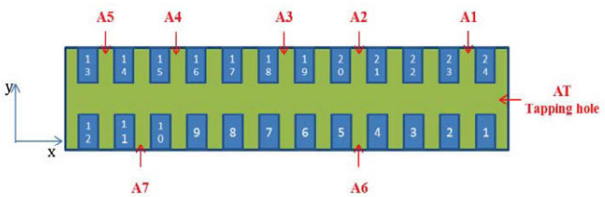

Figure 3. Measurement points (A) locations.

In the Figure 4 is shown the alumina distribution in the middle of the ACD, obtained with the alumina dissolution model. We can see that the dissolved alumina distribution vary from 2.11 to 5.96 %. This range is huge and is mainly due to low energy in the pots pots as discussed previously.

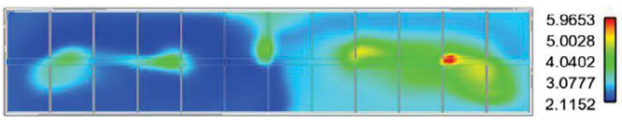

Figure 4. APXe dissolved alumina concentration map with 6 feeders.

This result looks surprising. The right side of the pot is over-fed and the left one is under-fed, with a concentration gap of 0.3 - 0.4 % between the both sides of the pot. The design of the feeder location was done for pot amperage about 570 kA in a different magnetic configuration. The same superstructure was kept to operate the 510 kA APXe configuration.

The results of the measurement campaign are given on Figure 5. The comparison between measurements and model confirms the gap between the two sides of the pot.

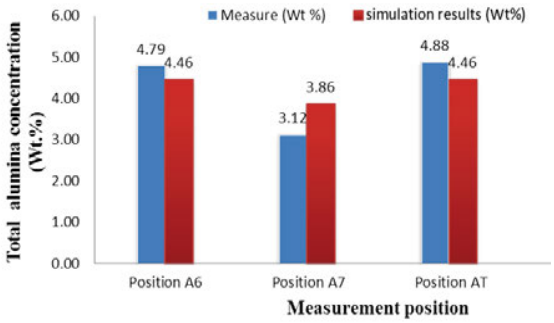

Figure 5. Model and measurements comparison.

In order to confirm further deeply this trend, we decided to increase the gap and the number of measurement points. To increase the gap the feeder F6 was stopped. This feeder is on the left side of the pot.

The comparison between measurements and model result is given on Figure 6. We validated an increase of the gap to 1.1 - 1.7 % when the Feeder F6 is stopped. With the increase of the number of measurement points, the concentration pattern given by the model was validated too.

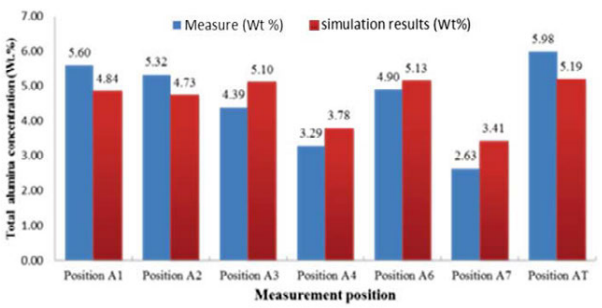

Figure 6. Model and measurements comparison, when Feeder F6 is stopped.

Design improvement by feeding strategy

A work is currently underway to optimize the dissolved alumina concentration distribution in the LRF APXe prototype.
It was demonstrated that the feeder locations are not optimal in the 510 kA magnetic configuration of the LRF prototype potroom. In Figure 7, the map of velocity is given at metal mid-height. In Figure 8, the streamlines in the bath are sketched. This Figure shows the importance of channels in alumina shot dispersion.

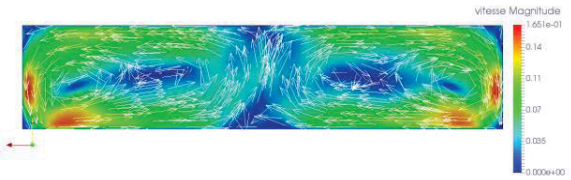

Figure 7. 510 kA APXe aluminium horizontal velocity map.

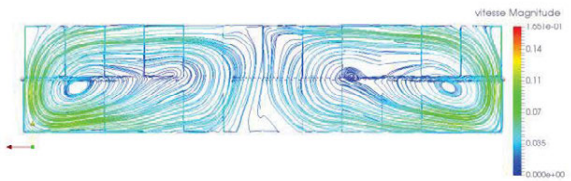

Figure 8. 510 kA APXe velocity streamlines in the bath.

In Figure 9, the dissolved concentration map is given for a 6 feeders configuration. The concentration is too low on the left side of the pot. The Figure 9 is the same as the Figure 4 with a different color range.

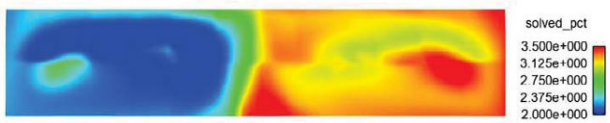

Figure 9. APXe dissolved alumina concentration map with 6 feeders.

A quick solution could be to stop right side Feeder F1 to improve the mass balance between both vortexes. In the Figure 10, the map of dissolved alumina concentration is sketched. We can see that this result is the opposite to the 6 feeders solution on Figure 9.

It may be surprising to see the difference between the 5 and 6 feeder results. This shows the extreme sensitivity of the alumina concentration distribution to the location where the two vortexes separates (see Figure 8). The left vortex is just a little smaller than the right one. This lead to this particular alumina distribution and the optimum is in between both configurations.

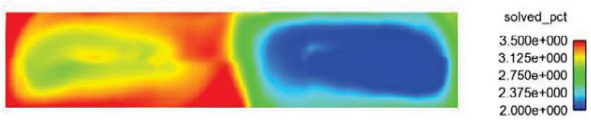

Figure10. APXe dissolved alumina concentration map with 5 feeders.

Fortunately, ALPSYS has a facility that allows applying different feeding period feeder by feeder. It was decided to increase the feeding rate in the under-fed vortex.

An optimization process is setup to find an optimal feeding rate for each feeder. Linear combinations of the dissolved alumina concentration map of different simulations are done. An optimum is found related to a new feeding rate. This new rate vector is simulated and the new map is re-injected in the optimization process.

In Figure 11, the optimized dissolved alumina concentration map is shown. The dissolved alumina concentration is closer to the mean alumina concentration in the whole ACD.

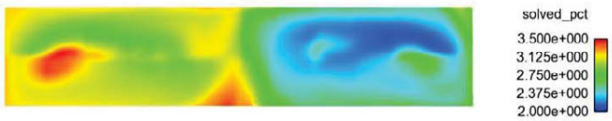

Figure 11. APXe dissolved alumina concentration map with 6 feeders and optimized feeding strategy.

The main comparison criteria are given in Table 2. We can see that the local minimum concentration is increased by 0.3 % and the standard deviation divided by 3.

Table 2. Main comparison criteria for 5, 6 and optimized 6 feeders for APXe CNG1.

	6 feeders	5 feeders	optimized 6 feeders
Mean C (%)	**2.79**	**2.78**	**2.81**
Min C (%)	**1.93**	**1.8**	**2.1**
Std Dev C (%)	**0.40**	**0.31**	**0.12**

The trials will be performed in coming months.

Conclusions

Alumina dissolution has become a major design criteria for low energy aluminium reduction pots. Bad alumina dissolution can strongly deteriorate production results with local or global anode effects, mud, electrical noise and low current efficiency. Low ACD, increased anodic plane surface and amperage increase are in opposition with a good alumina concentration distribution. This is why modelling software is required to improve existing and future designs (feeder locations and feeding strategy). In combination with a flexible pot control system, these tools can really improve existing plants at low cost.

It is now demonstrated that to model precisely the alumina concentration distribution in the bath, local temperature has to be taken into account. The feeding strategy also has to be defined precisely in order to optimize the process control.

We validated with success our alumina dissolution model through a set of measurements campaigns. The model is able to predict trends and we can be confident in obtained concentration patterns.

The software is used to optimize feeder locations and feeding strategy and tests are already in try. The simulations performed on the first pot have shown that the dissolved alumina distribution is very sensitive to small changes in the magnetic configurations.

In order to increase the software predictability the 3D ledge profile from MHD-TE model (see Langlois [4]) will be taken into account. The ledge strongly reduces side channel sizes and pot corners.

Acknowledgement

We would like to thank:

- Ycoor System SA, which supports us in model implementation.
- Moustapha Diallo (Ecole Centrale Paris), who supported us in model validation.
- Véronique Dassylva-Raymond, Laszlo Kiss and Sandor Poncsak (UQAC) for their expertise on alumina dissolution.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

NUMERICAL INVESTIGATION ON THE IMPACT OF ANODE CHANGE ON HEAT TRANSFER AND FLUID FLOW IN ALUMINUM SMELTING CELLS

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Keywords: aluminum smelting cell; anode change; magnetohydrodynamic; two-phase flow; phase change.

Abstract

In order to understand the impact of anode change on heat transfer and magnetohydrodynamic (MHD) flow in aluminum smelting cells, a transient three-dimensional (3D) coupled mathematical model has been developed. The solutions of the mass, momentum and energy conservation equations were simultaneously implemented by the finite volume method with full coupling of the Joule heating and Lorentz force through solving the electrical potential equation. The volume of fluid (VOF) approach was employed to describe the two-phase flow. The phase change of molten electrolyte (bath) as well as molten aluminum (metal) was modeled by an enthalpy – based technique, where the mushy zone is treated as a porous medium with porosity equal to the liquid fraction. The effect of the new anode temperature on recovery time was also analyzed. A reasonable agreement between the test data and simulated results is obtained. The results indicate that the temperature of the bath under cold anodes first decreases reaching the minimal value and rises under the effect of increasing Joule heating, and finally returns to steady state. The colder bath decays the velocity, and the around ledge becomes thicker.

Introduction

Aluminum is used in various products and industries such as transportation, construction, medicine, etc. More than 40 million tons of aluminum are produced per year, mostly in China, Russia and Canada [1].

The Hall-Héroult process is used to transform alumina (Al_2O_3) into aluminum. A direct electric current passes through a bath layer between anodes and cathodes. The overall electrochemical reaction taking place in the aluminum smelting cell can be expressed as: $2Al_2O_3 + 3C + \text{electricity} \rightarrow 4Al + 3CO_2$.

The carbon entering into this reaction comes from the anodes. Therefore, anodes are constantly consumed by the process and need to be periodically replaced. Typically, an anode can last approximately 20 days in a cell. Considering a cell with 40 anodes, this means that almost every day one pair of anodes needs to be replaced in the cell.

Anode change can significantly affect the thermal balance within the cell. New anodes are usually in room temperature much colder than old anodes. The bath in the cell is kept only 5 K to 15 K above its primary temperature of crystallization, which would be frozen if cold anodes are added. Moreover, the influence can prolong more than 24 hours. However, a well-maintained thermal balance of the cell provides the foundation for better operation and higher energy efficiency. Therefore, it is necessary to understand the effect of new anode on heat transfer and MHD flow in the cell.

Considering the complex transient and coupled phenomena taking place in the replacement of anodes, the understanding of heat transfer and fluid flow is challengeable. It is difficult to observe the varying of the temperature distribution in the entire cell by measurement. With the continuous improvement of computational resources, numerical simulation has become an attractive method to achieve this objective. Cheung et al. [2] have developed a thermal model to estimate temperature field and bath ledge thickness in a cell under different anode current distributions. The flow field however was ignored. Dupuis et al. [3] and Safa et al. [4] investigated thermal behavior coupled with MHD flow. They used finite element method to calculate electromagnetic and temperature fields, and then the fluid flow and phase change were studied. They employed a one-way coupled instead of two-way coupled approach.

As mentioned above, there is currently a lack of works concerning the impact of anode change on the heat transfer and MHD flow in the cell. A transient 3D coupled mathematical model has been developed to study the effect of the anode replacement in aluminum smelting cells. The normal conditions of the heat transfer and MHD flow in the cell were investigated, and then the transient conditions were demonstrated when new anodes were substituted.

Mathematical Model

Physical domain

An actual smelting cell contains many components, and in order to keep the computational time reasonable, it was necessary to simplify the geometry. The physical domain included in the present model is shown in Fig. 1. It represents a part of a typical aluminum smelting cell with a 300 kA line current and a total of 40 prebaked anodes. The anodes are aligned by pairs in the cell and are represented by blocks.

The bottom of the anodes is in contact with the bath. The bath layer is also part of the domain considered. In practice, the top portion of the anodes is protected from the air by an anode cover. However, the anode cover was not included in the present model. As will be described below, effective heat transfer conditions were imposed at the surfaces of the upper portion of the anode blocks. Similarly, the details of the shell of the smelting cell were considered through effective heat transfer conditions at the sidewalls of the bath layer.

The fluctuated bath-metal interface in the cell causes the deformation of the bath layer, and as a result the Joule heating distribution changes. Therefore, it is necessary to take the metal into account.

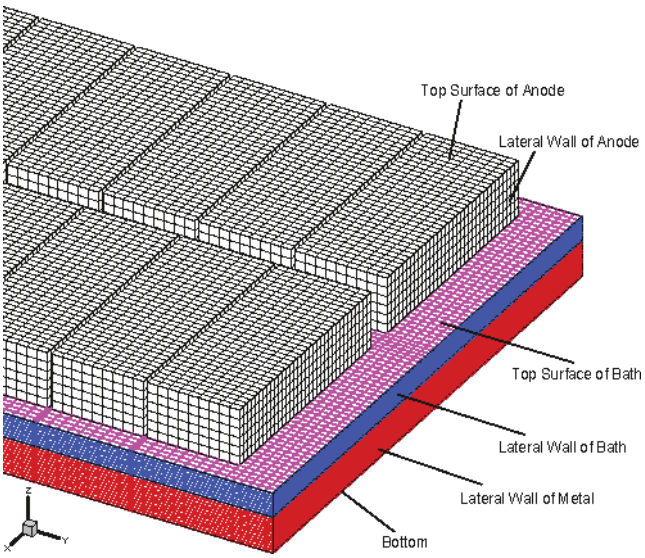

Fig. 1. Mesh model and boundaries.

In order to limit the computational burden, the gas formed at the bottom of the anodes was ignored, and all other components such as the anode assemblies, cathodes, bus bars, etc., were disregarded in the present model.

Two-phase flow

The conservation equations for mass and momentum were solved in the bath and metal [5]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \tag{1}$$

$$\left[\frac{\partial (\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \times \vec{v}) \right] = -\nabla p + \mu_{eff} \nabla^2 \vec{v} + \vec{F}_e + \vec{F}_b + \vec{F}_d \tag{2}$$

where $\vec{F}_e = \vec{J} \times \vec{B}$ (3)

The bath and metal were assumed to be incompressible. The densities change with temperature by Boussinesq approximation. The flow was assumed to be Newtonian. In the momentum equation, the term $\vec{F}_e$ was included to account for the Lorentz force which strongly influences the flow pattern as well as the bath-metal interface fluctuation. The buoyancy force was embodied by $\vec{F}_b$. The term $\vec{F}_d$ was employed to gradually block the velocity to zero in the mushy zone as will be described below in next section.

Turbulence was modeled with RNG *k*-ε model. An enhanced wall function was used to cooperate with the turbulence model, because the bath near the cell lateral wall had a low Prandtl number. The bath-metal interface was tracked with the VOF approach [6]. The continuum surface force model was implemented to consider the surface tension between the bath and the metal.

Heat transfer and phase change

The energy conservation equation was also considered [7]:

$$\frac{\partial}{\partial t}(\rho H) + \nabla \cdot (\rho \vec{v} H) = \nabla \cdot (k_{eff} \nabla T) + Q_J \tag{4}$$

where $H = h + f_\ell L$ and $h = h_{ref} + \int_{T_{ref}}^{T} c_p dT$ (5)

$$Q_J = \frac{\vec{J} \cdot \vec{J}}{\sigma} \tag{6}$$

where the source term Q_J represented the Joule heating. In Eq. (5), melting or solidification of the bath and metal was modeled with the enthalpy method in which the liquid fraction was calculated from:

$$f_\ell = \begin{cases} 0 & T < T_s \\ \dfrac{T - T_\ell}{T_\ell - T_s} & T_s < T < T_\ell \\ 1 & T > T_\ell \end{cases} \tag{7}$$

As mentioned above in Eq. (2), the velocity field must be blocked in the solid phase with the term $\vec{F}_d$. An enthalpy-porosity formulation was used. It treats the mushy zone in the momentum equation as a "pseudo" porous medium in which the porosity gradually decreased from 1 to 0 as the bath and metal solidifies:

$$\vec{F}_d = \frac{(1 - f_\ell)^2}{f_\ell^3} A_{mush} \vec{v} \tag{8}$$

Current density field

The current density distribution required in Eqs. (3) and (6) was calculated by the electrical potential approach. It consists of simultaneously solving for the electrical potential φ, as well as the magnetic potential vector $\vec{A}$. The electrical potential equation was extracted from the conservation of the electric current [8]:

$$\nabla \cdot \vec{J} = 0 \text{ with } \vec{J} = -\sigma \frac{\partial \vec{A}}{\partial t} - \sigma \nabla \varphi \tag{9}$$

At the same time, the magnetic potential vector was related to the magnetic field by:

$$\vec{B} = \nabla \times \vec{A} \tag{10}$$

In the present study, it was decided not to calculate explicitly the magnetic field. This field is dominated by the current distribution in the busbar of the cell. Instead, a typical steady state magnetic field obtained from our previous work was used [5]. The term $\partial \vec{A} / \partial t$ in Eq. (9) was also ignored.

Boundary conditions

A no-slip wall was imposed at all boundaries except at the top surface of bath, where a zero shear stress was employed.

The heat transfer at the top surface of bath, an effective heat coefficient was employed for convection and radiation. The temperature of ambient air was set to 393 K. Natural convection occurred at the top surface of anode, anode-air interface, lateral

walls and bottom. Besides, a thermal contact resistance was added at the anode-bath interface because of the existence of anode gas.

The line current of the cell was 300 kA which was distributed uniformly at the anode top surface. A zero potential was imposed at the bottom. The anode-air interface, bath top surface and lateral walls were assumed to be insulated where there was no electric current going through the boundaries. Due to the anode gas, an electrical contact resistance was employed at the anode-bath interface.

Results and Discussion

Description of steady-state conditions

The model was first exploited to simulate a situation in which a steady-state is achieved. This scenario would represent the situation prevailing before an anode change is performed, i.e. when a long time has gone by since the last anode change.

Fig. 2 illustrates the electrical streamlines and Joule heating distributions. The passage of the electric current from the anodes to the cathodes creates the Joule heating in the highly resistive bath. Little electric current flows to the bath through the anode lateral wall-bath interface because of a higher electrical resistance. The bath between anodes therefore creates less Joule effect. Besides, more current moves to the both ends of the cell resulting in greater Joule heating.

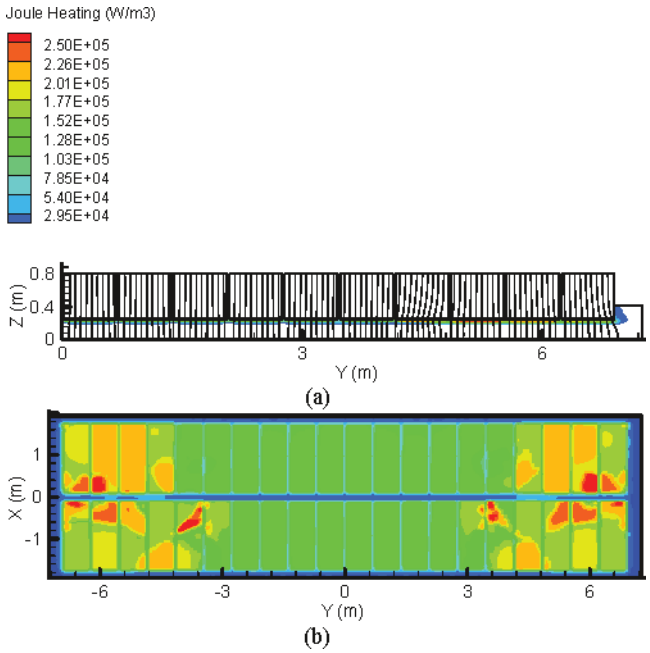

Fig. 2. (a) Electrical streamlines and Joule heating distribution of the steady-state condition (X = 0.965 m). (b) Joule heating distribution (Z = 0.225 m, i.e., the half-height of the bath layer).

Fig. 3 demonstrates the temperature distributions in the cell. Due to less heat and more heat dissipation, the temperature of the anode upper part is lowest. As mentioned above, much heat is generated in the bath, which is sufficient to heat the anodes. The anodes bottom therefore becomes warmer. A hotter zone is observed at the center of the bath layer. The temperature of the outermost bath sharply decreases giving rise to a steep thermal gradient around the cell. A similar temperature distribution is found in the metal. Due to a better heat conduction property, the temperature of the metal is more uniform, and the highest temperature and the temperature difference drop. The thermal gradient around the cell happened in the bath disappears.

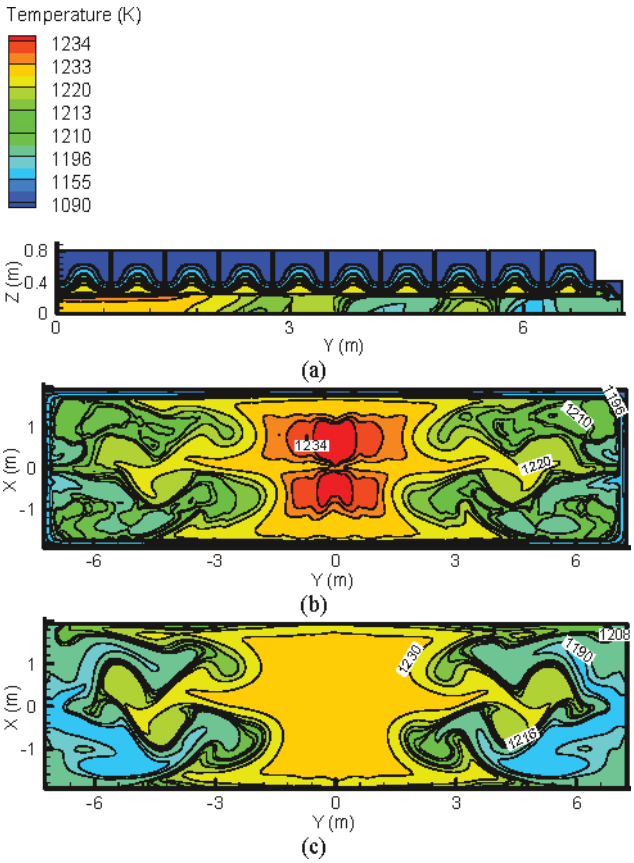

Fig. 3. Temperature distributions of the steady-state condition. (a) X = 0.965 m. (b) Z = 0.225 m (c) Z = 0.15 m, i.e., three fourths height of metal layer from bottom.

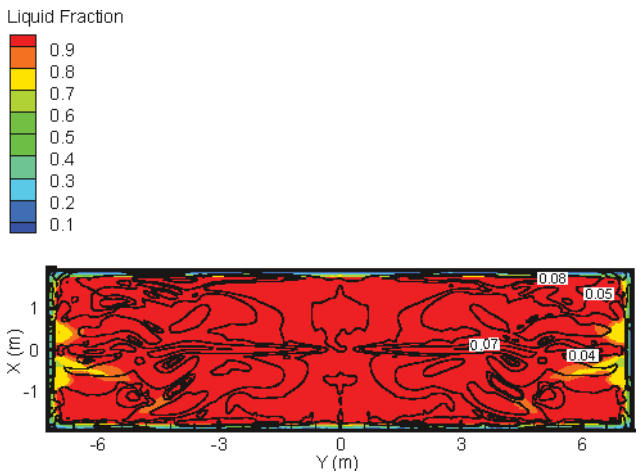

Fig. 4. Velocity streamlines and liquid fraction distribution of the steady-state condition (Z = 0.225 m).

Fig. 4 shows the velocity streamlines and solidification in the bath. Two large vortices as well as many broken eddies appear at

both ends of the cell as a result of the Lorentz force and thermal buoyancy. It can be seen that the temperature distribution aligns well with the flow pattern. The stronger flow promotes the heat transfer around the cell and consequently we found a ledge formed by the frozen bath. Besides, the bath around the cell freezes forming a ledge. Moreover, the temperature of the bath near the middle of both sides also decreases and mushy regions are created, and the mid-mush diminishes under the effect of washing. In particular, the metal is always in liquid phase due to the excellent thermal conductivity and lower solidification temperature.

Transient conditions after anode change

In order to study the effect of the anode change, we simultaneously replace the anodes 20 and 21 by new anodes with room temperature (298 K). For simplicity we still keep a uniform current density at top surfaces of anodes.

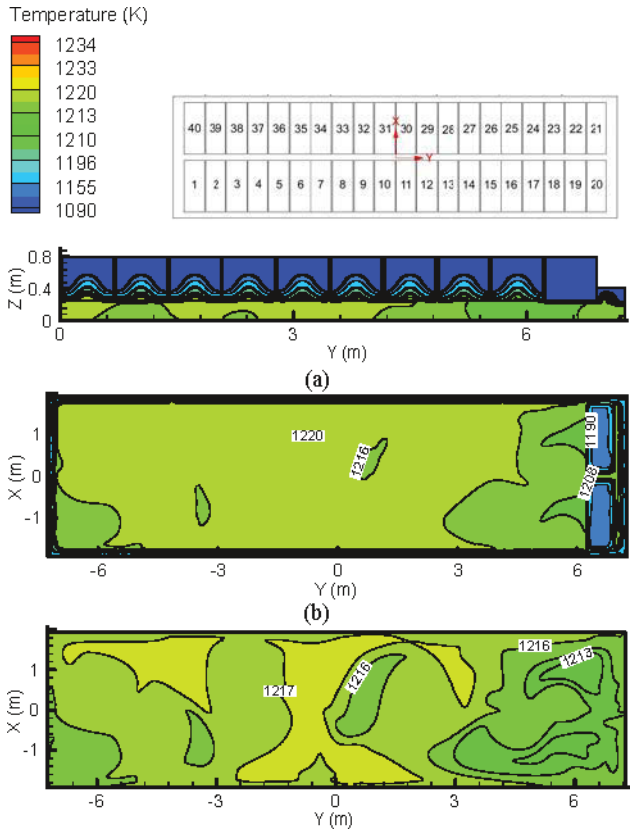

Fig. 5. Temperature distributions after one hour of the anode change, and the new anodes are in room temperature (298 K). (a) X = 0.965 m. (b) Z = 0.225 m. (c) z = 0.15 m.

Fig. 5 displays the temperature profiles after one hour. The new anodes are still cold. The temperature of the bath under anodes 20 and 21 decreases from 1210 K to 1190 K, and the influence area is able to extend the bath below anodes 23 and 18. The temperature of the lower metal also drops as expected, and the influence region is larger than that in the bath because of a higher thermal conductivity.

Fig. 6 represents the flow pattern and solidification. The bath under new cold anodes as well as the surrounding bath gradually solidifies as time progresses. As a result, the thickness of the ledge around the corner increases. Besides, the viscosity of the colder bath reduces and hence decays the flow. The velocity decreases from 0.05 m/s to 0.01 m/s. In spite of the addition of cold anodes, all metal is still far away from freeze.

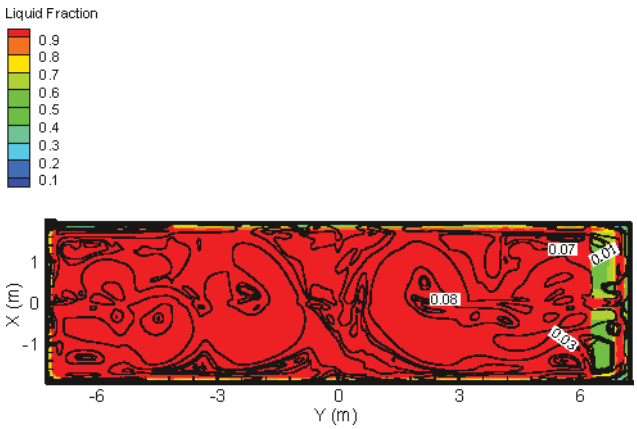

Fig. 6. Velocity streamlines and liquid fraction distribution after one hour of the anode change (Z = 0.225 m).

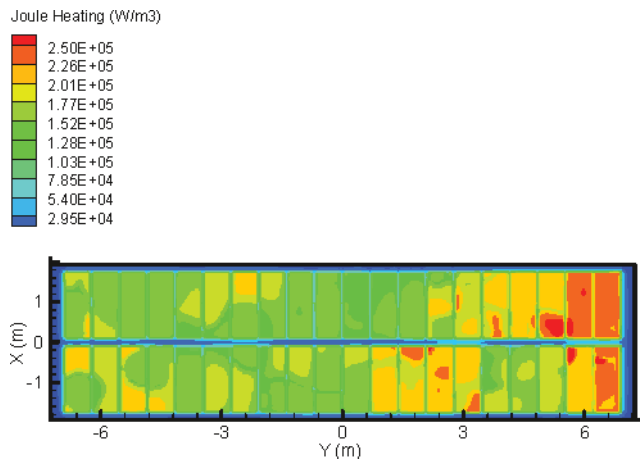

Fig. 7. Joule heating distribution after one hour of the anode change (Z = 0.225 m).

Fig. 7 indicates the Joule heating distribution. We can find that the Joule effect under new anodes becomes higher, because the colder bath electrical conductivity reduces. It can be inferred the bath temperature would continue to decrease while the increasing Joule heating warms the bath as well as the new anodes. After reaching the lowest temperature, the temperature of the bath and the new anodes turns to rise.

Conclusions

In order to study the effect of the anode change on the heat transfer and MHD flow in aluminum smelting cells, a transient 3D coupled mathematical model has been developed. The solutions of the mass, momentum and energy conservation equations were simultaneously implemented by the finite volume method with full coupling of the Joule heating and Lorentz force through solving the electrical potential equation. The VOF approach was employed to solve the two-phase flow. The phase change of bath and metal was modeled by an enthalpy – based technique, where the mushy zone is treated as a porous medium with porosity equal

to the liquid fraction. The impact of the new anode temperature was also analyzed. The simulation closely agrees with the experiment. Under normal conditions, lower temperature regions are located at the both sides of the cell where two pairs of large vortices are found. A bath ledge is formed around the cell. After anode change, the temperature of the bath under cold anodes first decreases reaching the minimal value and rises under the effect of the increasing Joule heating, and finally achieves the steady state. The colder bath decays the velocity, and the around ledge becomes thicker. The influence area of the new anodes can extend to the surrounding anodes, and the influence region in the metal is larger than that in the bath.

Acknowledgement

The authors wish to thank National Natural Science Foundation of China (51434005).

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ON THE IMPORTANCE OF FIELD VALIDATION IN THE USE OF CELL THERMAL BALANCE MODELING TOOLS

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Keywords: Modeling, thermal balance, model validation, cell autopsy

Abstract

Mathematical models have become essential for the design of modern, efficient high-amperage reduction cells, but the models are only one part of the cell design process. Since many of the inputs to a thermal balance model are difficult to evaluate, a process for validation of the predictions of the model is essential.

The validation process typically includes two sources of feedback: from operational pots and from post-mortem examination. Measurements of temperatures, heat fluxes and ledge shape must be made on the operating pots. In addition, prototype pots are shut down for post-mortem examination (cell autopsy) which is the only way to evaluate transformations of materials.

The transformation of materials during operation is one of the major reasons why model predictions do not match real operation. The risk of using unvalidated models to carry on design work is highlighted through the presentation of a real example from the past.

Introduction

As described in the introduction to the modeling section of Essential Reading in Light Metals Vol. 2 [1], modeling of Hall-Héroult cells went from essentially non-existing, to being very expensive and mostly fruitless, to finally becoming a big success story, finally becoming indispensable in the process of designing modern, efficient high-amperage cells [2]:

"Hall-Héroult cells are very challenging to model. This is true now and it was especially true half a century ago. The aluminum industry has invested huge resources in the development of mathematical models especially to support its cell design activities. Today the use of mathematical models is considered indispensable to the successful design of efficient high current cells".

In order to be successful, the design of high amperage cells must be conducted using the "cell development cycle" method [3]. The method comprises several steps, from modeling to measurement campaigns that must be used in repetitive cell design improvement cycles:

1) Cell design through modeling
2) Cell engineering and prototype cells construction
3) Prototype cells operation
4) Prototype cells measurement campaigns
5) Prototype cells postmortem autopsy
6) Model calibration and validation

This cycle is required for two reasons, one is to measure the behavior of cell prototypes that are testing innovative design ideas, for example the replacement of carbon side blocks with SiC or collector bars with copper inserts.

The second reason is the collection of data to be used for model validation. This is very important because models are only as good as the quality of the inputs used to set them up, hence the famous expression: "garbage in, garbage out".

The temperature dependent material properties used in models are such critical model inputs. In models, those properties are set to be temperature dependent but this image is misleading as it is not the complete picture.

For example, refractory brick manufacturers do provide temperature dependent thermal conductivity data obtained by testing their bricks at different operating temperatures.

However, in a Hall-Héroult cell, the cathode lining is exposed to chemical degradation that will significantly affect its properties. Those affected properties are the ones that need to be used as inputs in models and those properties are especially difficult to assess [4, 5].

Only data obtained from cell prototypes in steps 4 and 5 of the cell development cycle can be used in order to get reliable predictions from cell thermal balance models.

Prototype cells heat balance measurement campaigns

The first way to verify the accuracy of cell heat balance model predictions is to directly compare them with data obtained from prototype cell heat balance measurement campaigns [6, 7].

This technique consists of measuring the heat flux on enough locations of the external surface of the cell to be able to calculate the global heat loss of the cell such as the one presented in Table 1.

Cell heat balance models will generate equivalent cell heat balance predictions that can be compared at posteriori to the cell heat balance measurements. Considering the accuracy of individual cell measurements of +/- 5% in the best cases, several such cell heat balance measurements are required to ensure that the model is calibrated using reliable data.

Comparison of the global heat loss and the heat loss partition will tell if the model predictions were right or not but will not directly tell what need to be adjusted if the predictions were not perfect.

Figure 1: Measurement of the heat flux on the crust and the anode yoke using heat flux probes from Japan in the 70's

Table I: Measured cell heat loss from Hirakud HSS 55 kA cell, table II in [7]

Heat Balance Results

date:	5-Nov-03	Cell:	Cell 265	
Cathode Heat Losses		W / m2	kW	%
Shell side coverplate		821	2.26	2.00
Shell side spacer between boxes		1796	3.59	3.18
Shell side bottom box		3105	6.41	5.69
Collector bars to air		1337	7.96	7.06
Collector bars to flexible			2.40	2.13
Shell side wall collector bar level		625	0.57	0.50
Shell side wall insulation level		981	1.80	1.60
Shell side vertical boxes		918	1.97	1.75
Shell side floor perimeter section		1172	9.56	8.48
Shell side floor center section		1884	8.14	7.22
Shell end coverplate		809	2.21	1.96
Shell end spacer between boxes		1789	3.55	3.15
Shell end bottom box		3065	6.29	5.58
Shell end wall collector bar level		623	0.81	0.72
Shell end wall insulation level		969	1.82	1.61
Shell end vertical boxes		1150	2.16	1.92
Shell end floor perimeter section		1165	6.34	5.62
Total for the cathode part			67.82	60.17
Anode Heat Losses				
Crust		306	2.85	2.53
First side channel		4002	12.11	10.74
Second side channel		2037	6.16	5.47
Third side channel		1206	3.65	3.24
Forth side channel		739	2.24	1.98
Above forth side channel		554	1.70	1.51
Anode top		552	4.13	3.67
Studs			12.06	10.70
Total for the anode part			44.9	39.83
Total for the cell			112.7	100.00
Cell internal heat			117.8	
Blits closing			95.71%	

Prototype cells postmortem autopsy

The only way to directly obtain the required used material properties required as inputs in models is to stop the prototype cells, dig out samples and measure the properties of those samples at a range of operating temperatures.

This is typically done in parallel with postmortem cell autopsies which more typically are carried out to determine the mechanism of failure of cells that tapped out or had to be stopped just before it happened [8]. Yet, it is not uncommon to stop some prototype cells for the purpose of obtaining the type of data presented in Figure 2 for example. This is very costly, of course, illustrating well the importance of using a properly validated model in order to come up with an optimum cell design using a minimum of cell development cycles.

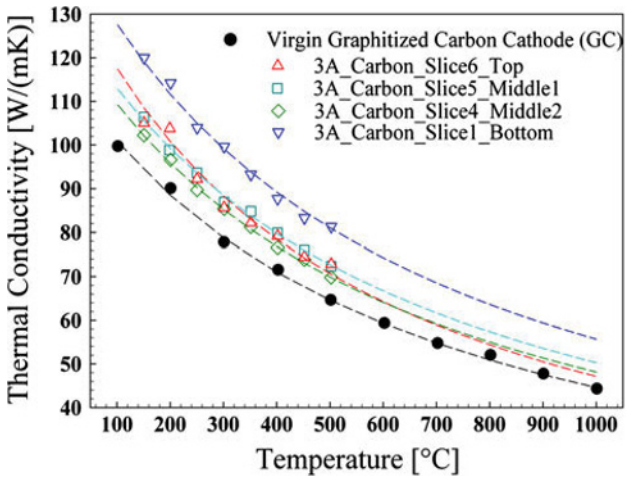

Figure 2: Example of thermal conductivity of spent cathode blocks obtained from pot autopsy, Figure 12 in [5]

The over insulated cathode example

Very often, technical papers and training courses discuss the importance of respecting certain guidelines having in mind some past mistakes and failures without actually discussing any of those mistakes and failures.

In this paper, such a practical example from the recent past will be presented in detail in order to clearly highlight the consequences of using an unvalidated cell heat balance model to come up with a cathode lining design.

This practical example is quite well covered in the literature [8, 9, 10]. It is the case of the over insulated Alcoa P155 cathode lining as explicitly specified in [10]. Notice that the original Alcoa A697 lining design is essentially the same. The original lining design is presented in Figure 3 taken from Figure 1 in [10]. There are 2 very thick layers of calcium silicate slabs just above the potshell floor. Calcium silicate is a fairly good thermal insulator; typical temperature dependent thermal conductivity of that material is presented in Figure 4.

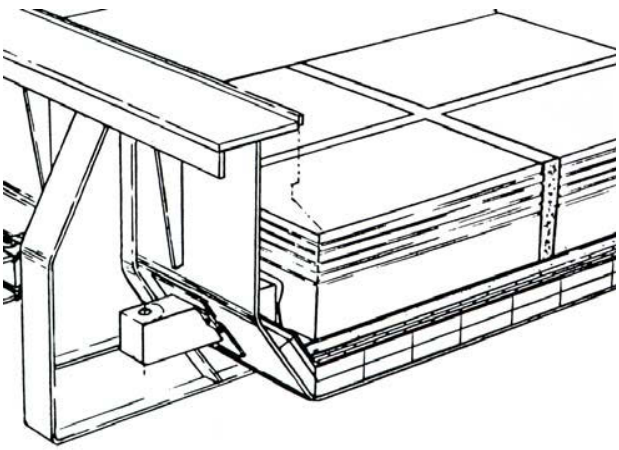

Figure 3: Original lining design of the Alcoa P155 cell, Figure 1 in [10]

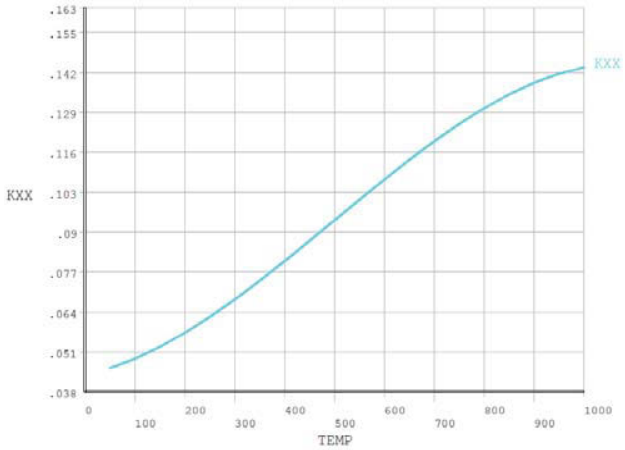

Figure 4: Temperature dependent thermal conductivity of new calcium silicate material

In [10] several types of cathode models were presented; Figure 5 shows the cathode side slice model. The navy blue material above the potshell floor is the calcium silicate.

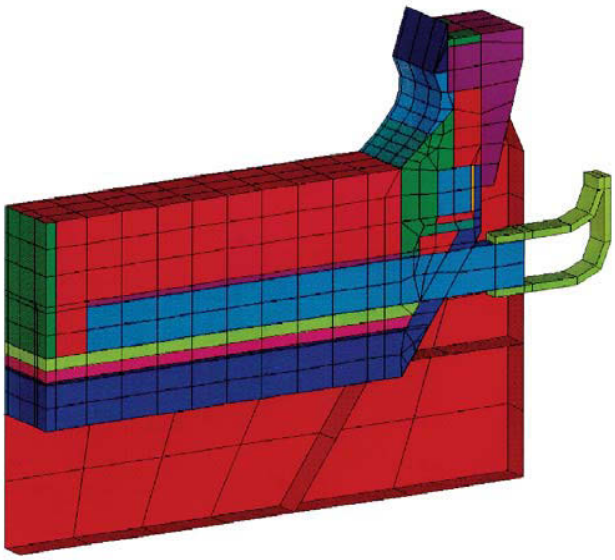

Figure 5: P155 cathode side slice model mesh as developed in 1993, Figure 5 of [10]

That model is no longer available to the authors but a full cell slice of the A697 developed by GeniSim and presented in Figure 6 is available instead. In that model, the calcium silicate is gold.

That A697 full cell slice model was run first using the new calcium silicate temperature dependent thermal conductivity presented in Figure 4. The obtained temperature solution is presented in Figure 7. Notice that contrary to the cathode slice model presented in [10], the full cell slice model presented in here calculates first the cell internal heat from that anode-to-cathode distance (ACD) specified by the user as input and converged in parallel the cell superheat and the ledge thickness so that the presented converged solution is in thermal balance i.e., the cell dissipates exactly the calculated cell internal heat.

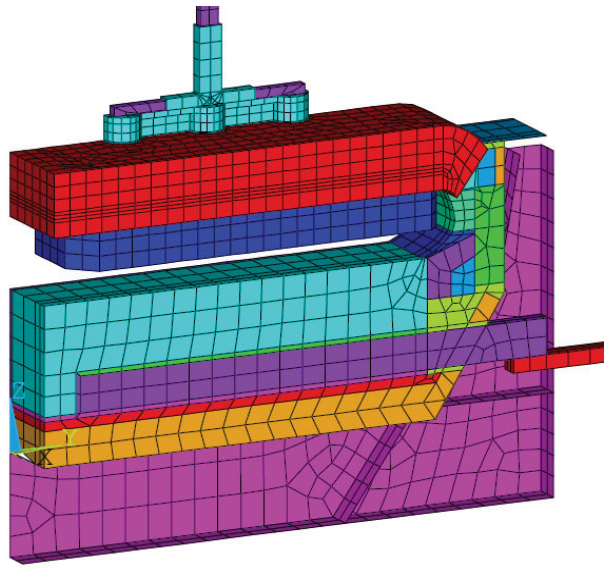

Figure 6: A697 full cell slice model mesh used in this study

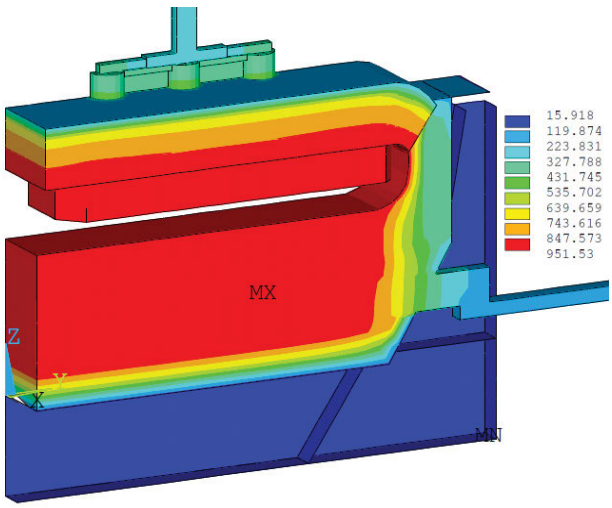

Figure 7: A697 full cell slice model temperature solution using new calcium silicate property

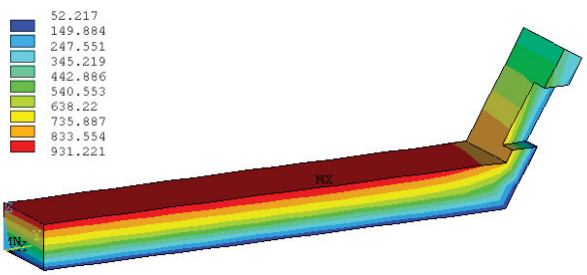

Figure 8: Calcium silicate temperature solution using new calcium silicate property

The temperature solution of only the calcium silicate material is presented in Figure 8, we can see that the top section is reaching above 900 °C.

In the operating cell, the light insulating materials are exposed to degradation both by high temperatures and by reaction with bath chemicals. It has been shown that above about 700 °C, these materials will lose much of their insulating property as first presented in [11] and reproduced in Figure 9. In addition, cathodic bath components, including sodium, permeate the lining materials. If these components reach the insulation in liquid form, they will penetrate the porous structure of the insulation, destroying the insulating value.

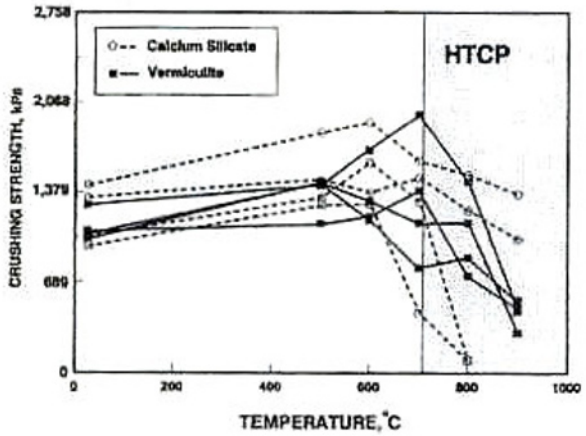

Figure 9: Observed degradation of insulating material properties above about 700 °C from [11]

As noted in [10] the sign of that degradation was seen in a cell autopsy but those autopsy results for the P155 cell were not presented in [10]. Fortunately, very similar autopsy results were presented in [8] for the A697 cell, those autopsy results confirm the chemical degradation of very large portion of the thick insulation layer as presented in Figure 10.

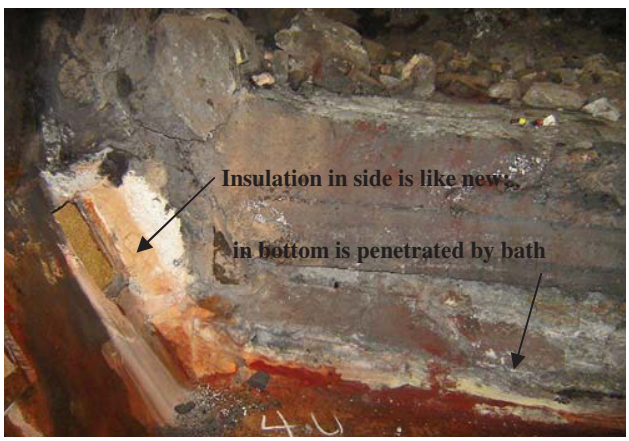

Figure 10: Picture from a A697 cell autopsy showing the chemical degradation of the insulation layer from [8]

Since it is not that easy to observe from that picture the chemical degradation of the insulation layer, the sketch describing the state of the cell lining prepared during another autopsy is presented in Figure 11.

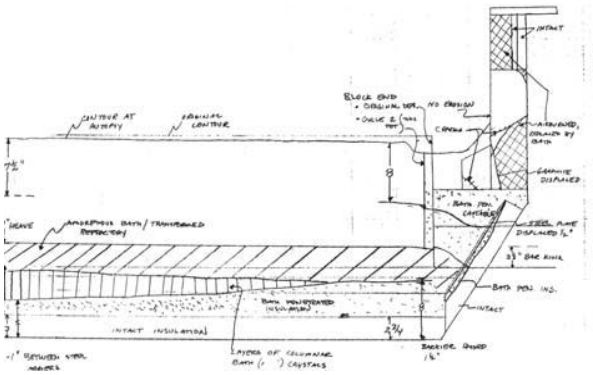

Figure 11: Sketch from a A697 cell autopsy describing the chemical degradation of the insulation layer from [8]

As discussed in [10], in order for the model to take into account that chemical degradation that occurs in the calcium silicate exposed at high temperature, the temperature dependent thermal conductivity needs to be adjusted as presented in Figure 12.

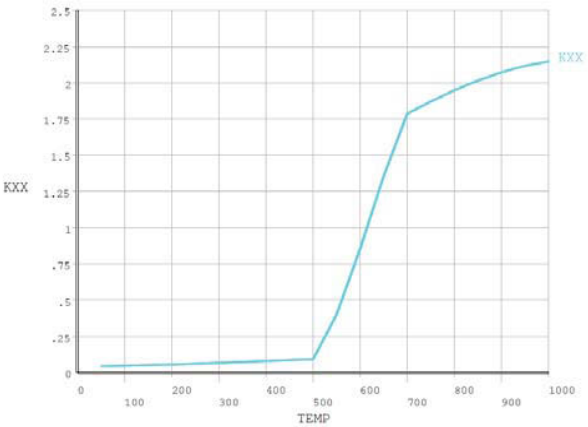

Figure 12: Temperature dependent thermal conductivity of calcium silicate material used in cell lining

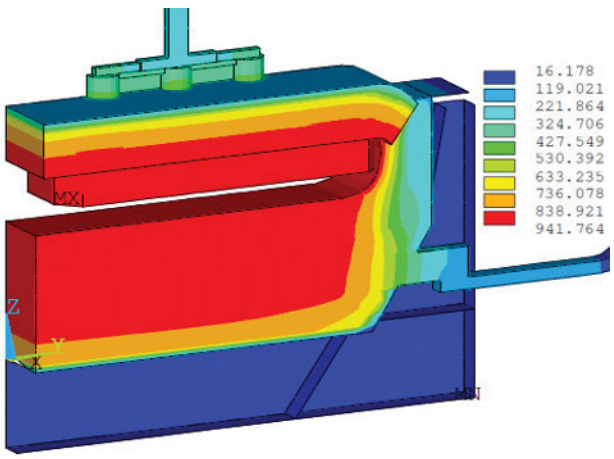

Figure 13: A697 full cell slice model temperature solution using the modified calcium silicate property

The thermal conductivity curve presented in Figure 12 essentially assumed that below a temperature threshold, 500 °C in this case, the new calcium silicate properties remain intact and that above a second temperature threshold, 700 °C in this case, we are dealing with a completely different new material. The ramp between the 2 temperature threshold transition zones is numerically required to ensure that the finite element (FE) solver will be able to converge that very non-linear problem.

Since the cell internal heat was kept the same, the converged cell heat loss remained the same but the heat loss partition is now very different: the cell bottom floor dissipates more heat while cell side wall now dissipates less heat due to the predicted decrease of the cell superheat and the corresponding predicted increase of the ledge thickness. This change in ledging results in changes in the operational behavior of the cell, and may require additional energy input to maintain stability. In any case the cell behaves quite differently than predicted by the model prior to the corrections made as a result of the validation process. The obtained temperature solution is presented in Figure 13, while the temperature solution of only the calcium silicate material is presented in Figure 14.

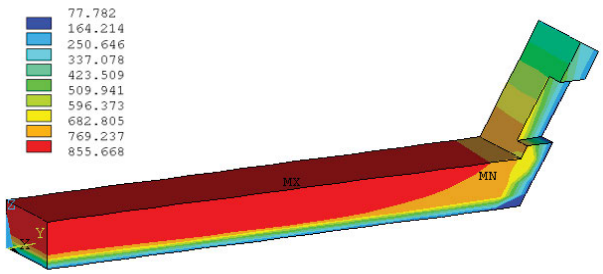

Figure 14: Calcium silicate temperature solution using the modified calcium silicate property

That change of repartition of the cathode heat loss would be quite easy to observe as it significantly affects the potshell temperature. Figure 15 presents the potshell temperature solution of the first case when using the new calcium silicate property while Figure 16 presents the potshell temperature solution of the second case when using modified calcium silicate property.

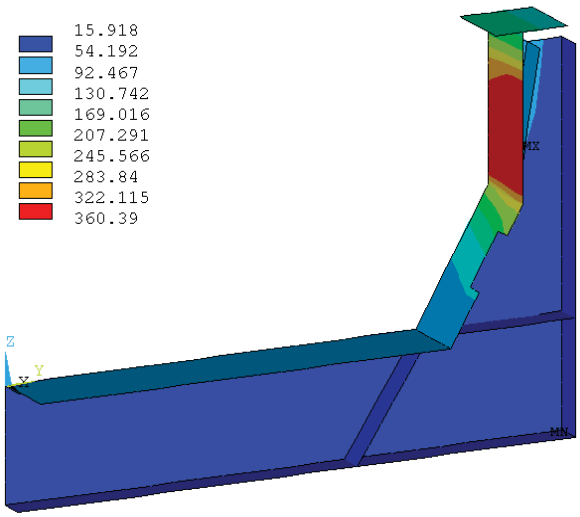

Figure 15: Potshell temperature solution using new calcium silicate property

Of course this change would not occur overnight, the first case would be representative of the condition a few weeks after startup when cell operations have stabilized while the second case would be representative of the condition after full penetration of the bath constituents into the cathode lining.

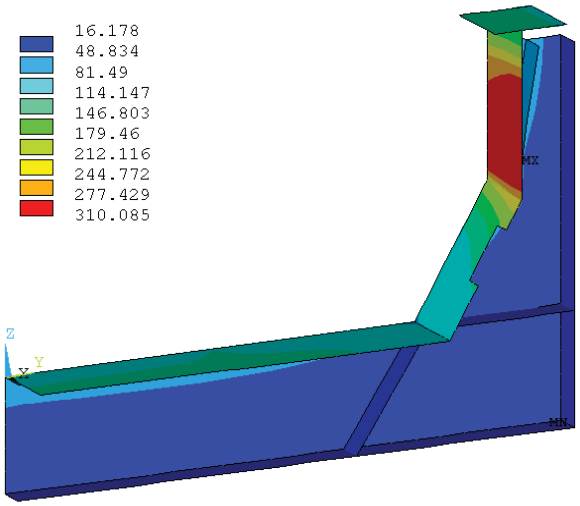

Figure 16: Potshell temperature solution using the modified calcium silicate property

In fact, that type change of the potshell floor temperature have been measured and reported in [4] on a 155 kA cell, those measurements are presented in Figure 17.

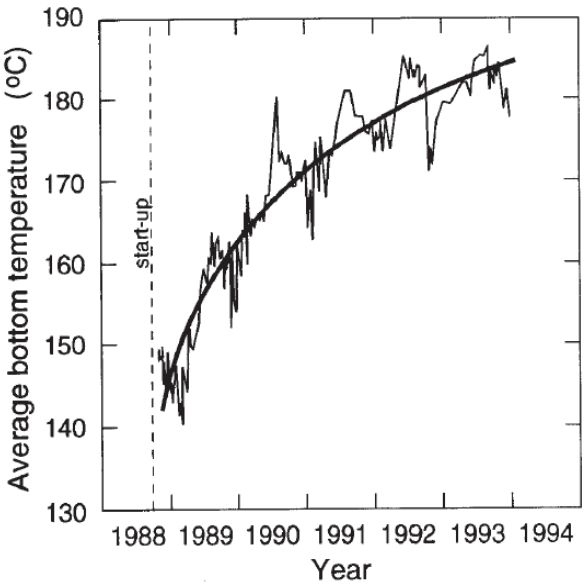

Figure 17: Averaged measured change of the potshell floor temperature of 16 155 kA prebaked cells, Figure 17 of [4]

The need for more field validation work

In retrospective the story of the over insulated cathode lining with a too thick layer thickness of calcium silicate that could not possibly avoid chemical degradation regardless of the type of physical barrier put in place to try to protect it is quite instructive as it is not unique.

Calcium silicate is not the only material that is susceptible to chemical attack and significant properties changes when put in cell operating conditions. Anode cover material is another example of equal importance that also has been the object to significant studies over the years including quite recently: [12, 13].

On the other hand some material would need more characterization work, for example the dry barrier material extensively use in Chinese lining designs. Despite the recent work done [14], there is certainly a need for more work in order to come up with a validated temperature dependent thermal conductivity property to be used in cell thermal balance modeling.

Conclusions

In order to be successful, the cell design of high amperage cells must be conducted using the "cell development cycle" method [3]. The method comprises several steps, from modeling to measurement campaigns that must be used in repetitive cell design improvement cycles:

1) Cell design through modeling
2) Cell engineering and prototype cells construction
3) Prototype cells operation
4) Prototype cells measurement campaigns
5) Prototype cells postmortem autopsy
6) Model calibration and validation

Only data obtained from cell prototypes in steps 4 and 5 of the cell development cycle can be used in order to get reliable predictions from cell thermal balance models.

The first way to verify the accuracy of a cell heat balance model predictions is to directly compare them with data obtained from prototype cell heat balance measurement campaigns [6, 7].

The only way to directly obtain the used material properties required as inputs in models is to stop the prototype cells, dig out samples and measure the properties of those samples at a range of operating temperature [8, 14].

For example, it is well known that above about 700 °C, light insulating materials such as calcium silicate will get exposed to chemical degradation and will lose much of its insulating property.

In order for the model to take into account the chemical degradation that occurs in the insulation exposed at high temperature, the temperature dependent thermal conductivity needs to be adjusted.

The thermal conductivity of other materials such as the dry barrier material used in Chinese lining designs also must be characterized after exposure to high temperatures and penetration by cathodic bath materials.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SIDELEDGE FACING METAL IN ALUMINIUM ELECTROLYSIS CELLS: FREEZING AND MELTING IN THE PRESENCE OF A BATH FILM

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Keywords: Sideledge, Freezing, Melting, Model

Abstract

Proper understanding of the sideledge in aluminium electrolysis cells is important for making the right decisions in cell design and operation. A conceptual model for the lower ledge, based on the existence of an alumina-saturated bath film stemming from sludge, was published earlier. In the current study, the rates of freezing and melting were studied in a simplified model, without taking diffusion across the film into account. By considering the heat- and mass balances during freezing or melting of sideledge, it turned out that the system will approach thermal balance as the film creeps upwards along the ledge. As a result, melting and freezing are rapid close to the bottom of the film, while ledge near the top of the metal changes only slowly depending on the thickness of the film. Melting or freezing may also be initiated by variation in the concentration of aluminium fluoride in the sludge.

Introduction

Historically, the heat flow from metal to sideledge has been treated in the same manner as the heat flow between bath and sideledge; *i.e.*, the heat flow is regarded as being the product between the heat transfer coefficient and the superheat. The heat transfer coefficient for the lower ledge is usually set to about the same value as for the upper ledge.

For given flow conditions, the heat transfer coefficient between a liquid and a wall is roughly proportional with the thermal conductivity of the liquid raised to the power of 2/3. The thermal conductivity of bath (λ_b) is about 0.8 $Wm^{-1}K^{-1}$ [1] while the metal has a thermal conductivity of more than 100 $Wm^{-1}K^{-1}$ [2]. This entails that the heat transfer coefficient at the lower ledge (h_m) will be more than one order of magnitude higher than at the upper ledge; *e.g.*; h_m = 20 000 $Wm^{-2}K^{-1}$.

The existence of a bath film between metal and lower ledge was postulated by Haupin [3] in order to explain that the apparent heat transfer coefficients are approximately equal at bath and metal. The heat flux between metal and ledge in the presence of a bath film can then be expressed by replacing the heat transfer coefficient by the heat transmittance coefficient (U),

$$U = \left(\frac{1}{h_m} + \frac{\delta}{\lambda_b} \right)^{-1} \qquad (1)$$

where δ is the thickness of the film. Taking h_m = 20 000 $Wm^{-2}K^{-1}$ (assuming that the film behaves as a solid surface) and λ = 0.75 $Wm^{-2}K^{-1}$, the bath film needs to be 0.9 mm thick to obtain a heat transmission coefficient of 800 $Wm^{-2}K^{-1}$.

It appears that Haupin and many others envisioned a stationary bath film formed by wetting phenomena alone, in analogy with the spreading of an oil film on water. However, gravity will try to push the bath out. The liquid in the film will, therefore, be set in motion upwards, and some kind of external force must act on the film to prevent it from drying out. Mechanisms for sustaining the bath film were discussed earlier [4]. It was concluded that a film can probably only exist if there is a source of liquid at the bottom of the cell, *i.e.*, alumina-saturated bath stemming from sludge.

The idea of having a bath film between metal and ledge is very fruitful, since it explains the very existence of the lower sideledge. A conceptual model for the lower ledge based on an alumina-saturated bath film was published earlier [4, 5]. The current study represents the first attempt at quantification of that conceptual model. The model is based on a simplified treatment of the bath film as such and phase diagram data for the system Na_3AlF_6-AlF_3-CaF_2-Al_2O_3 [6, 7].

Bath Film between Metal and Ledge

Velocity Profile and Mass Flow

The behaviour of the bath film was described earlier [4, 5]. The flow in the film can be derived from the Navier-Stoke equation. Provided that the drag between the bath and the metal is negligible, the velocity profile is given by

$$u = \frac{\Delta\rho\, g_x}{\mu} \cdot \left(\tfrac{1}{2} y^2 - \delta y\right) \qquad (2)$$

where u is the velocity in the x-direction (see Figure 1), Δρ is the density difference between metal and bath, g_x is acceleration of gravity in the x-direction, μ is dynamic viscosity, y is the distance across the film, and δ is the thickness of the film.

The average velocity in the film can be found by integration,

$$u_{av} = -\frac{\Delta\rho\, g_x \delta^2}{3\mu} \qquad (3)$$

and the mass flow rate per unit length of ledge (j) becomes

$$j = -\frac{\rho \Delta\rho\, g_x \delta^3}{3\mu} \ [kgs^{-1}m^{-1}] \qquad (4)$$

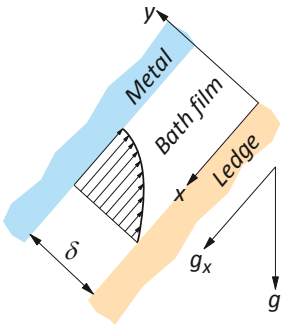

Figure 1. Bath film between metal and sideledge. The velocity profile in the film is shown, provided that the drag between metal and bath film is negligible.

The Need for a Detailed Film Flow Model

A key point in the development of a model for the lower ledge is the need for a detailed flow model. During melting or freezing of ledge there will be diffusion in the y-direction. However; if there is enough time and small dimensions, any concentration gradients will be levelled out due to diffusion. Unsteady state diffusion problems can be described as functions the dimensionless time (or Fourier number, Fo), defined by

$$Fo = Dt/L^2 \qquad (5)$$

where D is the diffusion coefficient [m^2s^{-1}] , t is time [s], and L is a characteristic length [m], in this case, the thickness of the bath film. As a rule of thumb, the concentration differences will be levelled out if Fo > 1. Taking the time scale to be the height of the lower sideledge (H, measured along the ledge) divided by the average flow velocity, give

$$Fo = \frac{DH}{u_{av}\delta^2} = \frac{3\mu DH}{\Delta\rho g_x \delta^4} \qquad (6)$$

Using H = 0.283 m (0.2 m metal height, ledge at 45° angle), D = $1.5 \cdot 10^{-9}$ m^2s^{-1} (diffusion coefficient for alumina suggested by Thonstad [8]), $\Delta\rho$ = 190 kgm^{-3} [9], and μ = 3.38 $kgm^{-1}s^{-1}$ [10] it can easily be calculated that Fo > 1 holds up to $\delta \approx 0.24$ mm.

On the above background, it was decided that the first version of the lower ledge model should be based on the assumption that the bath in the film is well mixed. This means that neither diffusion nor the exact velocity profile is taken into consideration.

Thickness of the Bath Film

An order-of-magnitude estimate of the thickness of the film can be made from the rate of sludge removal (self-feeding). If we consider a 400 kA cell with 10 percent self-feeding taking place entirely through the bath film, the alumina flow in the film is 570 kg/day and the total flow in the film is approximately 7000 kg/day (8.1 wt% alumina). If this flow is spread uniformly along the ledge (assuming 37 m periphery), Equation 4 gives the average film thickness $\delta \approx 0.2$ mm. It should be noted, however, that sludge (or alumina-saturated bath) can find its way back to the bath also outside the bath film. Moreover, there is no reason to assume that the bath film is evenly distributed along the ledge. Therefore, the local thickness of the film may vary from zero to much more than 0.2 mm.

Composition of Bath Film and Sideledge

Sludge and Bath Film

The main concept in the model is that formation of ledge facing metal takes place from an alumina-saturated bath film. The source of the film is sludge at the bottom of the cell, which is "thinned out" either by removal of alumina particles [11] or by addition of solid cryolite formed at the cathode [12]. It can be assumed that the composition of the bath film is the same as the liquid part of the alumina sludge.

The composition of alumina sludge was studied by Thonstad *et al*. [13], Taylor *et al*. [14], and Geay *et al*. [15]. Typical "soft" sludge contains about 40 wt% alumina, most of this as solid particles [13]. The liquid part of the sludge contains less excess aluminium fluoride than the bath. As shown in Figure 2; typical sludge contains about 3 wt% excess AlF_3, and apparently, there is no correlation between the concentrations in the bath and in the sludge. The content of calcium fluoride is close to 4 wt%, which is only slightly lower than the concentration in the bath.

The eutectic temperature for the liquid part of the sludge samples in Figure 1 was in the range 947.4-958 °C [14, 15], with average value 954.0 °C. There is no information available that gives reason to assume that the sludge composition is much dependent of the electrolysis technology.

The liquidus diagram for the system Na_3AlF_6-AlF_3-Al_2O_3-5 wt% CaF_2 is shown in Figure 3.

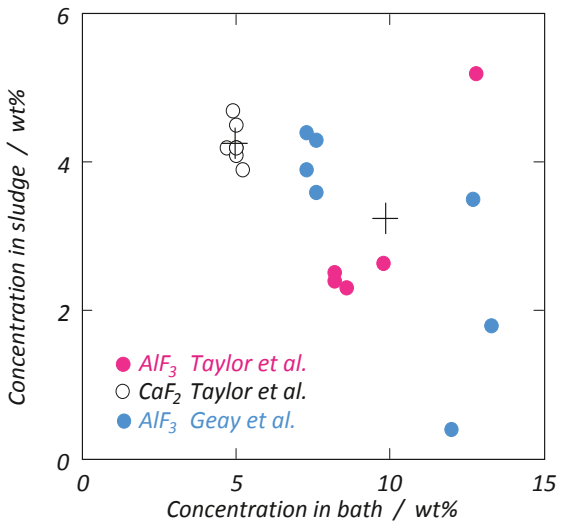

Figure 2. Concentrations of aluminium fluoride and calcium fluoride in the liquid part of sludge samples from industrial cells as a function of the concentrations in the bath. Data taken from Taylor *et al*. [14] and Geay *et al*. [15]. The crosses represent average values.

Ledge

Calcium Fluoride. According to Chrenkova *et al*. [16], four out of nine phase diagram studies in the system Na_3AlF_6-CaF_2 (including Chrenkova *et al*.) concluded that CaF_2 has solid solubility in

cryolite, while five studies concluded that there is no solid solubility. Thonstad *et al*. [17] analysed samples taken from ledge formed in a large laboratory cell, while Ødegård *et al*. [18] sampled the freeze formed on a newly set anode. By plotting the ratio between the CaF_2 concentration in the freeze and in the bath as a function of the same ratio for Al_2O_3 or AlF_3, it appears that the data do not extrapolate through the origin, as shown in Figure 4. This means that CaF_2 to a less degree than AlF_3 and Al_2O_3 is retained in the bath during freezing, which is a strong indication of solid solubility.

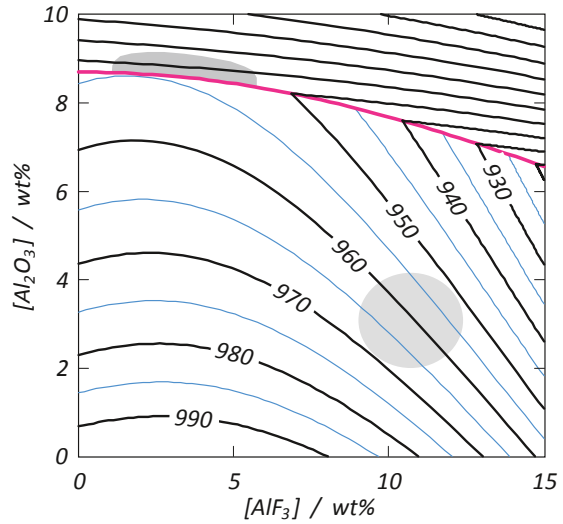

Figure 3. Phase diagram for the system Na_3AlF_6-AlF_3-Al_2O_3-5 wt% CaF_2, based on Solheim *et al*. [7] and Skybakmoen *et al*. [8]. Typical compositions of bath and sludge are indicated.

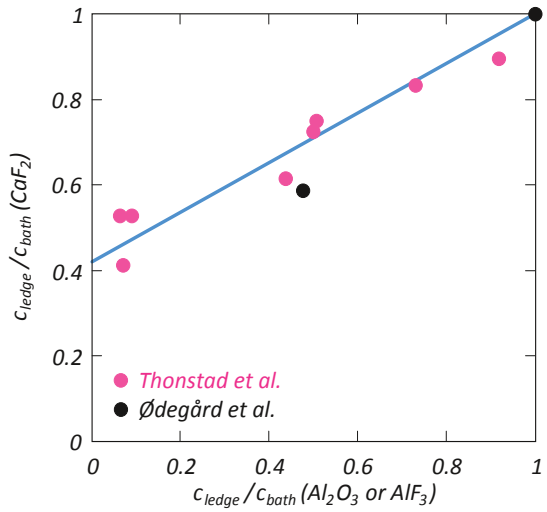

Figure 4. Plots of ratio between concentrations [wt%] in the ledge (freeze) and in the bath (see text). The data by Thonstad *et al*. [17] are based on Al_2O_3 and CaF_2, while the data by Ødegård *et al*. [18] are based on AlF_3 and CaF_2.

Alumina. When sideledge is frozen out formed from an alumina-saturated solution, the alumina concentration will follow the invariant line separating the primary crystallization fields of cryolite and alumina at about 8 wt% Al_2O_3 (Figure 3).

It is known that crystallization of alumina is very sluggish, as is illustrated in Figure 5. The dotted line was obtained by carrying out thermal analysis at high cooling rate (10 °C/min) which gave a supercooling of more than 100 °C in the field for primary crystallization of alumina. Thonstad *et al*. [19] made bottom ledge in laboratory cells, and it was noted that the nucleation of corundum from sludge required 15-20 °C supercooling, while 5-10 °C supercooling was sufficient to obtain sustained crystal growth after nucleation. Since the crystallization is partly dependent on kinetics, it is difficult to predict the alumina concentration. In the present version of the model it was assumed that the alumina concentration in the film as well as in the lower ledge follows the eutectic.

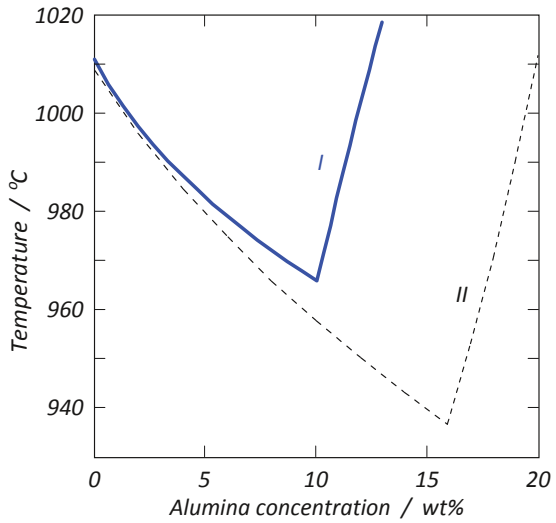

Figure 5. The cryolite-alumina phase diagram. Full line – data obtained by thermal analysis at 0.7 °C/min [6] (cryolite side) and saturation measurements [7] (alumina side), broken line – data obtained by thermal analysis at 10 °C/min cooling rate (Abramov, cited in "Aluminium Electrolysis" [20]).

Description of the Model

The bath film between metal and lower sideledge was divided into a number of control volumes with length Δx along the film (in the current version 200 elements; see Figure 6). Alumina saturated bath is flowing into volume n from volume (n-1). The average velocity is $u_{av(n-1)}$ and the film thickness is $\delta_{(n-1)}$. A heat balance is performed in element n. The heat flux from metal to ledge (q_f, see Figure 6) is calculated by

$$q_f = (T_m - T_s) \cdot \left(\frac{\delta_{(n-1)}}{\lambda_{(n-1)}} + \frac{1}{h_m} \right)^{-1} \qquad (6)$$

where $\lambda_{(n-1)}$ is the thermal conductivity of the bath film in control volume (n-1), h_m is fixed at 20 000 Wm^{-2}, T_m is fixed, and T_s was calculated from the eutectic temperature of the local bath film composition. The rate of freezing is given by

$$r_f = \frac{q_f - q_l}{\Delta H_m} \ [kgm^{-2}s^{-1}] \qquad (7)$$

where the heat flow from the ledge and out (q_l) is fixed at each point and ΔH_m is the heat of melting, assumed to be equal to the heat of melting for cryolite (approx. 510 kJ/kg [21]). The retention time in element n is known from Δx and $u_{av(n-1)}$, so the changes in volume flow, velocity, and film thickness due to melting of freezing can be calculated. These data are used in control volume no. (n+1) along with the new bath composition, calculated by

$$[AlF_3]_n = \frac{[AlF_3]_{n-1}}{1 - m_{f(n)}/m_n} \quad \text{and} \quad [CaF_2]_n = \frac{[CaF_2]_{n-1}}{1 - 0.58\, m_{f(n)}/m_n} \qquad (8)$$

The number 0.58 reflects the assumption that the concentration of calcium fluoride incorporated in the solid phase will be 42 percent of the calcium concentration in the bath film, see Figure 4. The new alumina content and the new eutectic temperature are calculated for each control volume.

It follows from the above that the bath film volume flow, thickness, and linear velocity – as well as the local bath composition, eutectic temperature, and freezing rate – are tracked along the film.

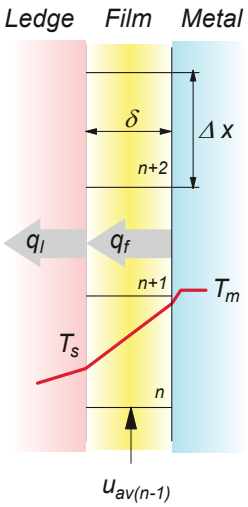

Figure 6. Schematic representation of the model (see text).

Results

Effect of Temperature and Bath Film Thickness

In the results shown below, the bath film was assumed to contain 3 weight percent aluminium fluoride and 4 weight percent calcium fluoride, in accordance with the data in Figure 2. The metal height was set to 0.20 m and the ledge was assumed to be slanting at 45°, giving a total ledge length in the x-direction of 0.282 m. The heat flow from ledge and out was fixed at q_l = 6000 Wm^{-2}, and the rate of freezing (given as mm/h) was calculated using the density ρ_l = 2900 kgm^{-3} for the ledge.

When considering Figure 7, some striking features can be observed. Firstly, it appears that the rate of freezing can be very high at the bottom of the film, while the upper half of the lower sideledge melts and freezes very slowly. There is also a tendency that for a given initial film thickness, the rate of melting is more evenly distributed along the ledge than the rate of freezing. This means that the average thickness of the lower ledge will become higher near the cell bottom with time, which will, eventually, be balanced by lower heat flow through the ledge near the bottom (q_l in Figure 6). The variation in thickness is also larger near the cell bottom.

The above effect can probably partly be related to the fact that the film thickness increases upwards during melting, while it decreases upwards during freezing. The variations in the film thickness and the transport rate are shown in Figure 8 for the case with initial film thickness δ_0 = 0.20 mm. The composition of the bath film for the same case is shown in Figure 9.

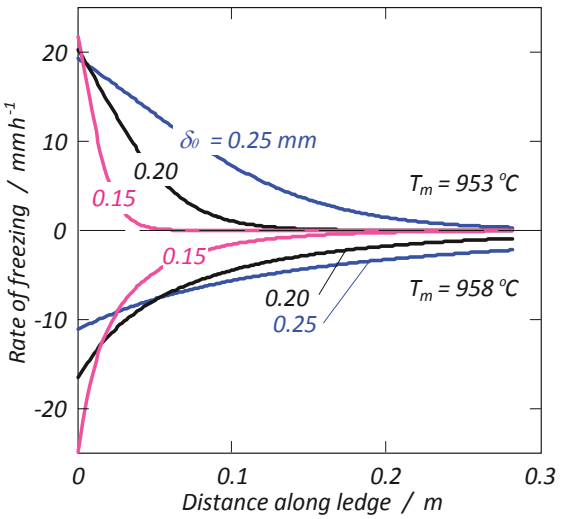

Figure 7. Rate of freezing or melting (negative numbers) as a function of the distance along the ledge, calculated with three different bath film thicknesses at the bottom of the film (δ_0) and two metal temperatures (T_m).

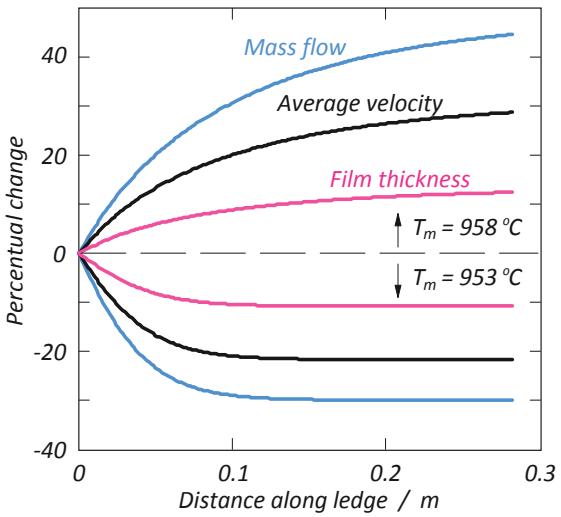

Figure 8. Changes in film mass flow, average flow velocity, and film thickness as a function of the distance along the ledge at initial film thickness 0.20 mm and two metal temperatures (same calculation as in Figure 7).

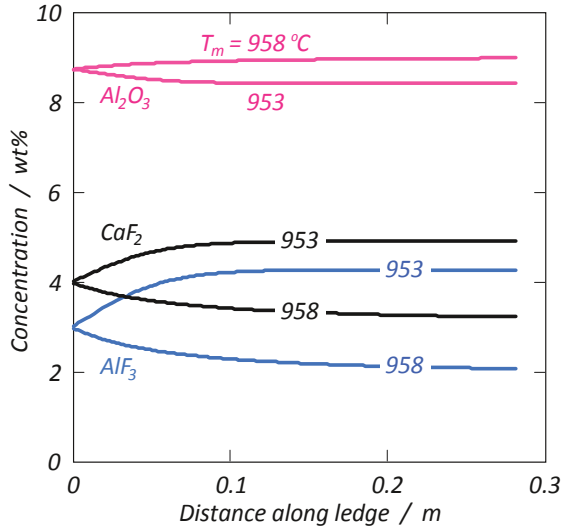

Figure 9. Bath film composition as a function of the distance along the ledge at initial film thickness 0.20 mm and two metal temperatures (same calculation as in Figure 7).

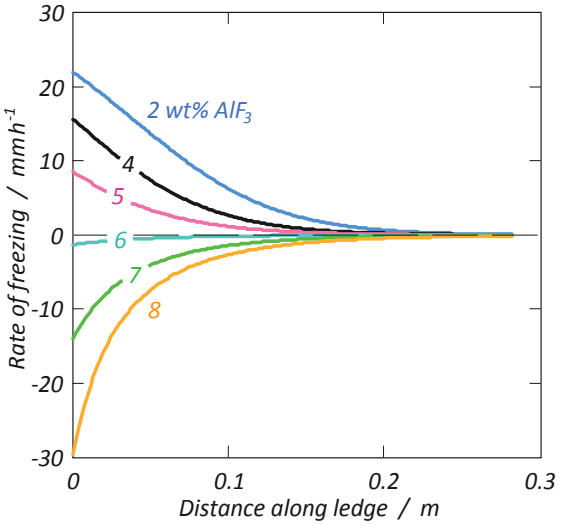

Figure 10. Rate of freezing or melting (negative numbers) as a function of the distance along the ledge, calculated with different amounts of excess aluminium fluoride in the bath film. 4 wt% calcium fluoride, film thickness at the bottom of the film δ_0 =0.23 mm, metal temperature T_m = 953 °C.

Effect of Sludge and Bath Composition

The current study implies that the freezing or melting of the lower sideledge depends on the sludge composition (eutectic temperature) as well as the metal temperature. This is illustrated in Figure 10, where the concentration of aluminium fluoride in the film was varied, keeping the metal temperature constant.

The metal temperature is mainly a function of the bath temperature (*e.g.*, a couple of degrees lower), so the metal temperature is more or less a direct function of the bulk bath composition and superheat. Increasing superheat may induce melting of the lower ledge as well as the upper ledge, provided that the sludge composition stays constant. If the sludge composition varies, the upper and the lower ledge may apparently behave independent of each other.

Regions of Freezing and Melting

Based on phase diagram data and assumptions concerning the temperature, bath and bath film composition, heat flux, and bath film thickness; a generalized diagram describing the fields of melting or freezing of the lower ledge could be constructed. The following data were used in Figure 11: $T_{bath} - T_m = 2$ °C, $q_l = 6000$ Wm^{-2}, $\delta = 0.2$ mm, 3.5 wt% Al_2O_3 and 5 wt% CaF_2 in the bath, 4 wt% CaF_2 in the bath film. If the ledge initially is at thermal balance, melting can occur if the aluminium fluoride concentration in the bulk of the bath decreases (represented by Arrow i in Figure 11), if the superheat increases (Arrow ii) or if the amount of excess aluminium fluoride in the bath film increases (Arrow iii).

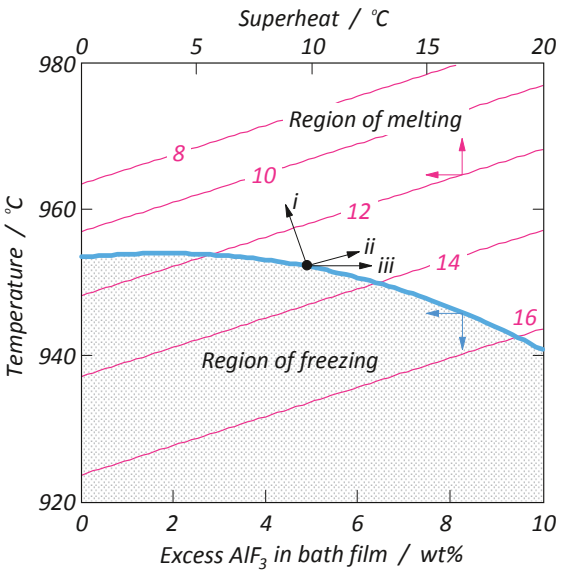

Figure 11. Regions of melting and freezing of the lower ledge. The thin lines represent the ledge surface temperature at different aluminium fluoride concen-trations in the bath (numbers on the lines) and superheat, while the thick curve represents the eutectic temperature of the bath film.

Discussion

Contrary to what was predicted in earlier work [4, 5], the current model seems to imply that the lower part of the lower sideledge participates in the general energy balance for the cell, in a similar way as for the ledge facing bath. For the region close to the bottom, variations in ledge thickness will take place at similar rates as for the ledge facing bath. It should be emphasized, however, that the model is at its initial stage, and it has not been verified.

Future development of the model will probably include:

- Better description of the dynamics of the bath film. A CFD model is already in development [22]. Besides the formation of the film, the aim is to describe the behaviour of the film as a function of the metal flow pattern.

- Taking into account the effect of diffusion across the bath film. It is likely that concentration gradients to a certain degree affect the local liquidus temperature, in such a way that the rates of melting and freezing are reduced [23].

There is a lack of fundamental knowledge when it comes to the alumina concentration in the ledge and film; *i.e.*; the role of kinetics. While bottom ledge may be the result of sludge being supercooled [19] in such a way that the temperature and composition is between the alumina liquidus line and the extrapolated cryolite liquidus line (Figure 5), what are the exact conditions for formation of bottom ledge or a eutectic mixture? What is the correct or "best" alumina concentration to use in the bath film and the ledge? If the alumina concentration in the ledge depends on the conditions at the time the ledge was formed, a rigorous treatment of the melting process may, in fact, require that the history of the ledge is known.

There is still a lot to learn about this topic.

Acknowledgement

The current study was supported by the project "Superior Technology for Energy Efficient Aluminium Production" (STEP), financed by Hydro Aluminium and the Research Council of Norway. Permission to publish the results is gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
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MODELLING OF METAL FLOW AND METAL PAD HEAVING IN A REALISTIC REFERENCE ALUMINIUM REDUCTION CELL

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Keywords: MHD, metal heaving, metal flow, computational fluid dynamics

Abstract

A CFD based numerical model for a realistic generic aluminium reduction cell is published. The model takes into account the complex physics of the magnetohydrodynamic problem in the reduction cell by coupling of two immiscible liquid flows, Lorenz force, flow turbulence and large length-scale with complex cell geometry. The current model extends the traditional box model with a detailed cell geometry including side and bottom ledge profiles and all channels (side, central and cross channels). The simulation tests of this model focus on quantifying the sensitivity of different side and bottom ledge profiles, and the cross channel by comparing the predicted metal flow and metal pad heaving. In addition, the model dependencies upon model assumptions of open/closed channel top are discussed. Finally a physical mechanism for the metal pad heaving that seems to explain at least parts of this complex phenomenon are highlighted.

Introduction

Primary aluminium is produced mostly using the Hall-Héroult cell process of electrolytic reduction of aluminium oxide. A large electric current is supplied via busbars to the carbon anodes, and flows through the shallow layers of molten salt electrolyte and of molten aluminium (also known as bath and metal layers) into a carbon cathode lining, where they are collected by iron collector bars. The dissolved aluminium oxide is reduced at the interface between the two layers to form aluminium droplets which sink to the metal pool at the base of the reduction cell. Due to the low electrical conductivity in the bath layer practically all Joule heating is generated here. Besides maintaining the favourable cell operation temperature for the chemical reactions in the electrolysis process, a large amount of Joule heat is wasted through heat lost to the ambient. In order to improve the cell energy efficiency, it is very important to keep the anode cathode distance (ACD) as small as technologically possible. The technical barrier on reducing the ACD is that the interface between the two layers becomes unstable with respect to its sloshing motion resulted by the coupling effects of electromagnetic forces, interface stability, cell geometry and operation conditions. Hence, understanding the magnetohydrodynamic (MHD) flow physics in reduction cell and developing a proper tool to predict metal pad heaving and metal flow are of great importance to improve the current and energy efficiency as well as to optimize the cell operation process.

The investigation of the metal flow and metal pad deformation in the aluminium reduction cells is hindered by the harsh cell operation conditions for the available measuring technique to tolerate the high temperature and highly erosive media. Accurate direct measurement of the velocity in bath and metal layers and the interface height is still difficult. Numerical simulations [1, 2] are therefore the most feasible way to understand the metal flow pattern and metal pad heaving and the reasons for interface stability. Due to the complexity of cell geometry and MHD problem, some simplifications are applied in the previous numerical studies. In the studies of [3, 4], the shallow-water model was used to characterize the bath and metal layers separately. The model averages the vertical variation in each layer and the effects of transversal channels between anodes are neglected. Computational Fluid Dynamics (CFD) based 3D two-phase MHD model [5] was applied to simulate the metal heaving in a rectangular box model. The effect of channels is ignored as well as the effect of side ledge profile in realistic cells. A CFD based fully coupled two-phase MHD model [6] was developed for the same purpose. This model coupled effects of the electric potential/current distribution, the melt flows in the bath and metal layers, the interface deformation, and the anode consumption with a constant ACD. The model is still based on a rectangular box. Although the details of all channels (side channels, central channel and cross channels) were taken into account, the effect of side ledge profile was ignored. By using this simplification the model deviates from the realistic situation for aluminium reduction cells.

To overcome this deficiency, the model in [6] is extended further in this paper with the capability to account for the effect of side ledge profiles so that the model can be used for studying realistic aluminium reduction cells. The detailed information of the cell geometry and cell operation conditions for a generic cell is provided in this paper so that the simulations cases can be tried by other researchers. In order to understand the effects of side ledge, simulations with different side ledge profiles were conducted and the predictions on metal pad heaving and flows are compared. In addition, the effects of cross channels and open channel top are also modelled and discussed.

Fundamentals and model implementation

The numerical model is developed and implemented on the platform of a commercial CFD tool package ANSYS FLUENT. By using so called User Defined Functions (UDF) the coupling among two-phase liquid flow, interface deformation, electrical potential, current density distribution and the Lorentz force is realized. The FLUENT two-phase VOF model is used for solving the governing equations for flow fields and tracking the interface deformation. The governing equations of the continuity and the momentum conservation of the two-phase flow system with incompressible fluids read,

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{\partial}{\partial t}(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla P + \nabla \cdot \left[\mu\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)\right] + \mathbf{F}_E + \rho \mathbf{g} \quad (2)$$

where $\mathbf{u}$ represents the flow field, and P is the pressure. $\mathbf{g}$ refers to the gravitational acceleration and $\mathbf{F}_E$ represents the electromagnetic forces. ρ and μ are the fluid density and viscosity, respectively. For a two-fluid system, the fluid properties are calculated with the fluid volume fraction weighted averaging,

$$\rho = \rho_1 \alpha_1 + \rho_2 \alpha_2 \quad (3)$$

$$\mu = \mu_1 \alpha_1 + \mu_2 \alpha_2 \quad (4)$$

where the subscripts 1 and 2 denote the primary phase and the secondary phase, respectively, and α the fluid volume fraction.

The phase-interface tracking is accomplished by solving the continuity equation for the volume fraction of the secondary phase (α_2)

$$\frac{\partial \alpha_2}{\partial t} + \mathbf{u} \cdot \nabla \alpha_2 = 0 \quad (5)$$

The primary-phase volume fraction (α_1) will be determined by the phase continuity constraint: $\alpha_1 = 1 - \alpha_2$.

An user defined scalar equation is set up to solve the governing equation for electric potential distribution inside the reduction cell

$$\nabla \cdot \sigma \nabla \varphi = 0. \quad (6)$$

where σ is the electric conductivity. A volume fraction weighted harmonic average method is mandatory to calculate the distribution of electrical conductivity,

$$\frac{1}{\sigma} = \frac{\alpha_1}{\sigma_1} + \frac{\alpha_2}{\sigma_2} \quad (7)$$

From the electric potential field, the electric current distribution can be derived as

$$\mathbf{J} = -\sigma \nabla \varphi \quad (8)$$

And, the electromagnetic force (Lorentz force) density is given as

$$\mathbf{F}_E = \mathbf{J} \times \mathbf{B}, \quad (9)$$

where **B** is a *predefined* magnetic flux density field which mimics the magnetic field produced from the bus bars of a realistic cell.

To mimic the anode burn-off in real cell operations, special efforts are put on developing the model capability to ensure a constant ACD. A UDF is developed to calculate the vertical distance between the bath-metal interface and the anode bottom at each time step, and the mesh grids at anode bottom are adjusted with respect to the interface by using the ANSYS FLUENT functions in handling sliding and dynamic mesh.

To simulate the melts flow in a realistic reduction cell with dimensions about 16 m in length and 4 m in width, a proper turbulence model is necessary. In this study, the standard k-ε turbulence model with standard wall functions is applied to calculate the turbulent viscosity.

Realistic reference model

CFD model

The revised benchmark problem [6] is supplemented by a side ledge profile for the long side as well as for the short side. The generic model shown in Figure 1 has a length of 16 m and width of 4 m. The thickness of the metal layer is 0.24 m, and that of bath layer is 0.21 m. The ACD is set to 0.04 m. The reduction cell consists of thirty anodes in total. Each anode has the dimensions of 0.704 m × 1.61 m. The width of central channel and cross channels is 0.2 m and 0.04 m, respectively. The width of the end- and side-channels is now defined by the side-ledge profile.

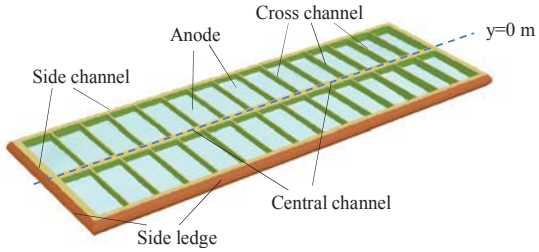

Figure 1: Model for a realistic reference alumina reduction cell.

Figure 2 shows the normalized side ledge profile introduced here which turns the model [6] in a realistic reference model for a reduction cell. The side ledge is normalized with respect to the cell height, the total thickness of bath and metal layers. An extended side ledge profile is also shown in Figure 2. It is used to investigate the effect of side ledge profile on the metal pad heaving and flow field.

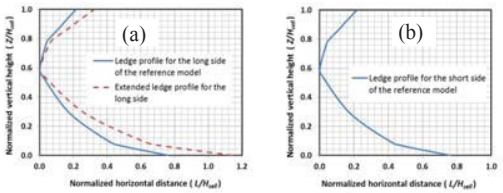

Figure 2: (a) Ledge profiles for the long side of the reference model and for a test model to investigate its significance; (b) Ledge profile for the short side of the reference model.

A structured mesh of hexahedral cells is used for the CFD model. In the horizontal directions, the central channel and side channel are meshed with four mesh cells, the cross channel is meshed with two mesh cells, and each anode is meshed with 20×12 cells. In the vertical direction, the model is divided into three zones. The top zone covering the bath layer above the anode bottom is meshed with twelve cells. The middle zone is the interface deformation zone. It has a thickness of 0.26 m which covers the ACD zone and part of metal layer. It is meshed with eighteen cells with fine meshes to capture the interface deformation. The bottom zone, which has a height of 0.086 m, is meshed with ten mesh cells. The CFD model contains 305216 hexahedral cells in total.

Table 1: Material Properties

Property	Unit	Electrolyte	Liquid Aluminium
Density	kg/m^3	2070	2270
Viscosity	mPa s	1.25	2.5
El. conductivity	S m^{-1}	250	3.0E6

Table 1 summarizes the CFD model settings of material properties (density, viscosity and electric conductivity) for the fluids: electrolyte and liquid aluminium.

Boundary conditions

For fluid flow, no slip boundary conditions are applied on all solid wall surfaces. The free surface on the channel top is simplified as slip boundary with zero shear stress. Standard wall functions are assumed on all solid walls for solving the k-ε turbulence model.

As for the boundary conditions for the electric potential equation, zero electric potential is set on the bottom and side of the anodes. Electric insulation conditions are applied on the side walls of the reduction cell and the channel top, where the normal current density is set zero. The normal current density (A/m^2) on the cathode surface is specified explicitly as

$$J_z = J_k(-2181 - 3013Y^2). \quad (10)$$

where J_k is a scalar factor with unit A/m^2, which ensures that the total current on cathode matches the cell amperage 300 kA for the simulation tests with different ledge profiles. Y is the normalized coordinate $Y = y/L_0$, where L_0 is the length scale unit L_0=1 m. The origin point of the coordinate system locates at the centre of cell bottom.

The external magnetic field (mT) imposed upon both bath and metal layers inside the reduction cell is assumed as,

$$\begin{cases} B_x = B_0(-1.5 - 0.2X + 8.0Y - 0.05XY) \\ B_y = B_0(-0.7 - 1.0X + 0.2Y - 0.01XY) \\ B_z = B_0(-0.02 - 0.1X - 0.5Y - 0.7XY) \end{cases} \quad (11)$$

where B_0 is the magnetic flux density scale in unit mT. The normalized coordinate $X = x/L_0$.

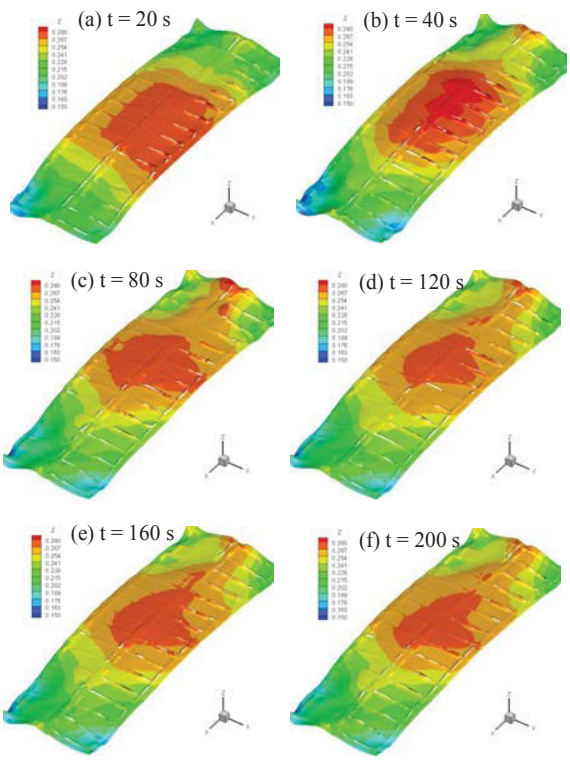

Figure 3: Predicted temporal variation of metal pad height inside the reference reduction cell.

Both the prescribed normal current density J_z and the imposed magnetic field **B** should show the main characteristics of those in realistic reduction cells.

The initial fluid flow field inside the reduction cell is assumed to be stationary. The bath-metal interface is initialized as a flat horizontal surface with an ACD of 0.04 m under the anode bottom. The initial electric potential is set to zero everywhere.

Solution method

General numerical schemes provided by ANSYS FLUENT were applied: "SIMPLE" for pressure-velocity coupling, the spatial discretization scheme "PRESTO!" for pressure, the "Geo-Reconstruct" scheme for volume fraction, and "First Order Upwind" for other equations. Transient simulation is adopted. "First Order Implicit" scheme is applied for the transient formulation. The time step size is set to constant as 0.04 s. The steadiness of the transient simulation results is estimated by averaging the transient data over a certain period of 4 s. It is found that the simulations reach quasi-steady state after 200 s simulation time (5000 time steps).

Results and discussion

Results of realistic reference model

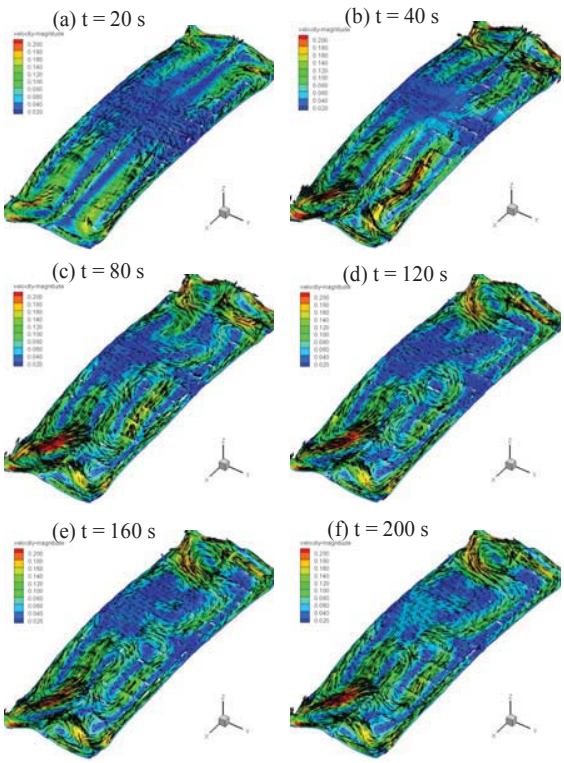

Figure 4: Predicted temporal variation of flow pattern on the bath-metal interface inside the reference reduction cell.

The temporal variations of the metal pad height and the flow pattern on the metal pad for the reference reduction cell are shown Figures 3 and 4, respectively. Just 20 s after the start, the bath-

metal interface, as shown in Figure 3(a), significantly heaves at the cell centre, and sinks at the four corners of the reduction cell. The heaving amplitude of metal pad is almost the same along the short axis of the cell, but the metal pad height varies significantly along the long axis of the cell. The corresponding flow pattern on the bath-metal interface is shown in Figure 4(a). Under the effect of the electromagnetic forces, circulating flows are started at the four corners of the reduction cell. It is believed that the irrotational part of the electromagnetic force is the dominating cause of the metal pad heaving before the flow pattern is built up inside the reduction cell. Certainly, the circulating flows also contribute to lower the bath-metal interface at the corners of the reduction cell.

As the melt flow develops further inside the reduction cell as shown in Figures 4(b), (c) and (d), under the effect of the circulating flows at the cell corners, several vortexes are induced at the cell centre. As a result, the bath-metal interface is also adjusted accordingly. As shown in Figures 3(b), (c) and (d), a dome-shaped metal pad is formed inside the reduction cell. Finally, both the melts flow pattern on the bath-metal interface as shown in Figures 4(e) and (f), and the metal pad heaving as shown in Figures 3(e) and (f) reach a quasi-steady state. There are almost no changes with time.

The simulation results for the reference cell model indicate that the maximum difference of metal pad height is about 0.13 m, and maximum velocity on the bath-metal interface is about 0.22 m/s. These are typical values for realistic reduction cells under similar operating conditions.

Effect of side ledge profile

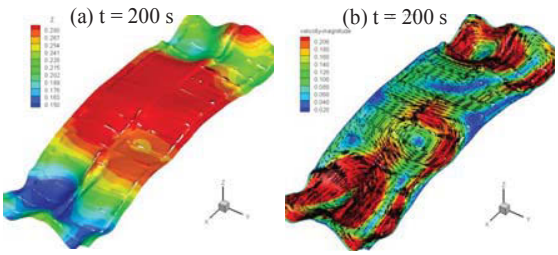

Figure 5: Predicted (a) metal pad height and (b) flow pattern on the bath-metal interface with a box cell model.

In order to understand the effects of side ledge profile, two simulations with artificial side ledge profiles are studied, using a box cell model (cf. Figure 1 of [6]) and a modified reference model with the extended side ledge profile as shown in Figure 2. The concept of box cell model was used in many early studies to simplify the complex cell geometry. Figure 5 shows (a) the predicted metal pad height and (b) the flow pattern on the bath-metal interface. When they are compared to the results for the reference model shown in Figures 3(f) and 4(f), it is clear that the box cell model over predicts the metal heaving amplitude and produces much stronger circulating flows at cell corners, which deviates from the realistic scenarios.

Figure 6 shows the simulation results for the modified reference cell model with the extended side ledge profile for the long side only. Compared to those for the reference model, the metal pad heaving amplitude becomes smaller, the flow pattern on the bath-metal interface is changed and the flow becomes weaker.

A comparison of the metal pad deformation (averaged from t=140 s to t=200 s) along the central channel under the different side ledge profiles is shown in Figure 7. The side ledge profile has significant effects on the metal pad heaving. In other word, the metal pad heaving and flow pattern can be adjusted by controlling the side ledge profile.

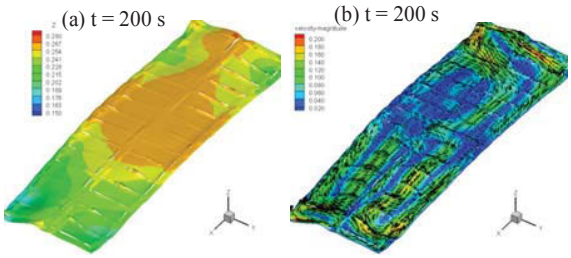

Figure 6: Predicted (a) metal pad height and (b) flow pattern on the bath-metal interface with extended side ledge profile.

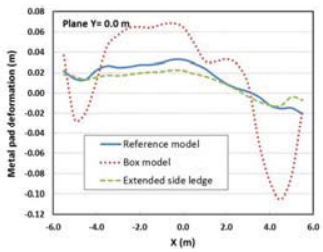

Figure 7: Comparison of the metal pad deformation along central channel under different side ledge profiles.

Effect of cross channel

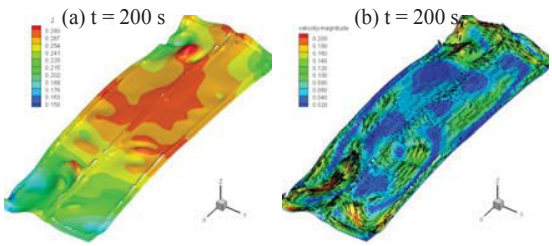

Figure 8: Predicted (a) metal pad height and (b) flow pattern on the bath-metal interface without cross channel.

In the previous study [6], the effect of cross channels, the small channels between adjacent anodes along the long side of the cell, was discussed for the box cell model. After introducing the side ledge profile, the effect of cross channel has been reanalysed. Figure 8 shows the predictions of an artificial cell model modified from the reference model by ignoring the cross channels. When they are compared to the results of the reference model, it is found the steady dome-shaped metal pad is no longer re-produced. Instead, a wavy metal pad is found as shown in Figure 8(a). The transient results of the variation of metal pad height distribution with time are shown in Figure 9. The bath-metal interface becomes unstable when ignoring the cross channels. Waves are

formed at several spots which are associated with the strong flow vortexes.

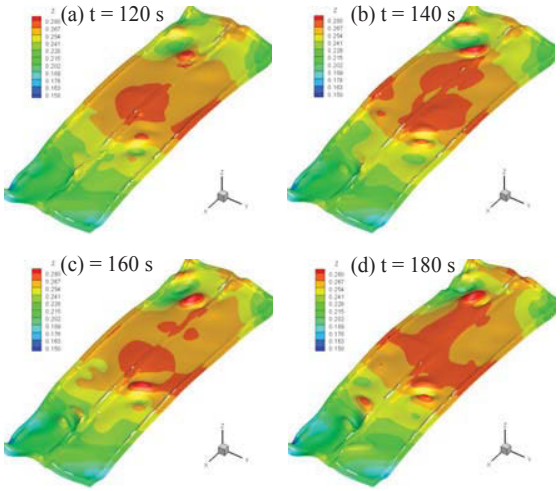

Figure 9: Predicted temporal variation of metal pad height distribution for a modified cell model without cross channels.

On the other hand, the effect of increased cross channel width is studied in another simulation test by doubling the cross channel width of the reference model. The simulation results shown in Figure 10 indicate that quasi-steady metal pad and flow pattern are obtained, and are very similar to those of the reference model.

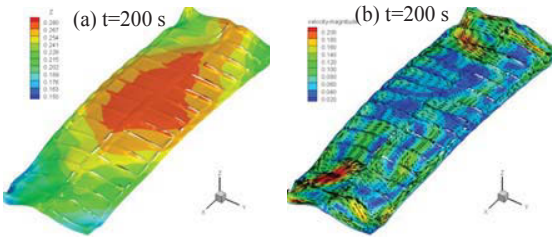

Figure 10: Predicted (a) metal pad height and (b) flow pattern on the bath-metal interface for a modified reference cell model with doubled cross channel width.

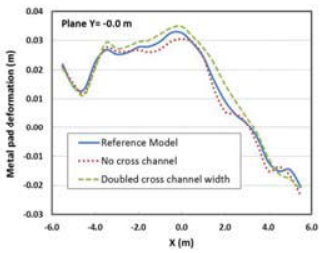

Figure 11: Comparison of the metal pad deformation along central channel under different cross channel width.

A comparison of metal pad deformation (averaged from t=140 s to t=200 s) along the central channel under the effect of cross channel width is shown in Figure 11. The overall distribution of metal pad deformation over the long axis of the reduction cell (along the central channel) has less dependence upon the cross channel width. Even though that the transient metal pad becomes wavy and unstable when the cross channel is neglected in the model, the averaged metal pad profile still follows the overall metal pad deformation pattern. In addition, a close study of Figure 11 indicates that the larger cross channel width can induce slightly larger metal pad deformation.

Effect of open channel top

In practical cell operation, the top of bath layer is open to the ambient environment at the channels. To take this into consideration, the boundary condition at the channel top is modified as the pressure outlet condition so that it is open to the ambient with a constant pressure. With the updated boundary condition at channel top, the calculation for the reference model is continued for another 180 s. The results for metal pad and flow pattern from this simulation are shown in Figure 12. A quasi-steady dome shaped metal heaving is predicted. Compared to the results of the reference model, the metal pad heaving is further enhanced at the cell enter, while the metal pad is lower at the sides. The circulating flow at the cell corners becomes weaker.

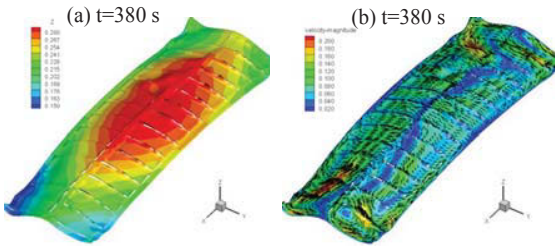

Figure 12: Comparison of the metal pad deformation along central channel under different cross channel width.

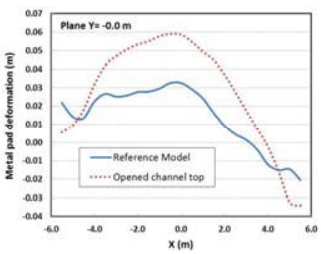

Figure 13: Comparison of the metal pad deformation along central channel under the effect of channel top openness.

A comparison of the metal pad deformation along the central channel under the effect channel top is shown in Figure 13. As reported in [5], the open channel top has significant effect on metal pad heaving. The open channel causes larger metal pad deformation in the reduction cell.

Metal heaving mechanism and open channel

Metal-heaving in the electrolysis cell may have several causes. Before the melts flow is built up, the dynamics of the cell is determined by the presence of the Lorentz force. By applying Ampere's law the Lorentz force can be decomposed according to

$$\mathbf{F} = \mathbf{J} \times \mathbf{B} = \left(\frac{1}{\mu_0} \nabla \times \mathbf{B} \right) \times \mathbf{B} = \frac{1}{\mu_0} (\mathbf{B} \cdot \nabla) \mathbf{B} - \frac{1}{2\mu_0} \nabla (\mathbf{B}^2) \qquad (12)$$

The last term in this decomposition is purely irrotational, the first term can be partly rotational and irrotational. For instance, in areas where the current is zero the first term is purely irrotational and identical to the second term. Assuming that all the rotational force components are small compared to the irrotational force components, the static force balance simplifies to

$$0 = -\nabla p + \rho \mathbf{g} - \frac{1}{2\mu_0}\nabla\left(\mathbf{B}^2\right) \qquad (13)$$

Integrating equation (13) we obtain the static pressure

$$p = -\rho g z - \frac{1}{2\mu_0}\mathbf{B}^2 + p_0 \qquad (14)$$

As we see the static pressure is modified by the magnetic field. The effect of the magnetic flux density is to reduce the pressure locally.

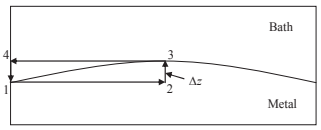

Figure 14: A closed cell with an assumed metal heaving at the bath/metal interface. The integration path used for the derivation of the metal-heaving amplitude Δz is indicated.

Now we consider a cell with no channels (also known as closed cells), and integrate ∇p , where p is given by (14), along the path shown in Figure 14. The metal pad heaving is given as,

$$\left(\rho_m - \rho_b\right) g \Delta z = 0 \qquad (15)$$

where ρ_m is the bath density, and ρ_b is metal density. From equation (15) we obtain $\Delta z = 0$. This means that no metal pad heaving occurs in a cell without channels under the above mentioned assumption.

An open cell with channels in contact with atmospheric pressure is shown in Figure 15. In the channels basically no electrical current will flow and there will be hydrostatic conditions. This means that the pressure at point 3 and 4 in Figure 15 will be identical.

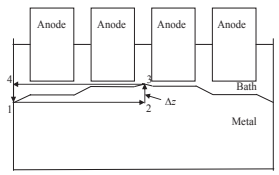

Figure 15: An open cell with four anodes and an assumed metal heaving at the bath/metal interface. The integration path used in the derivation of the metal-heaving amplitude Δz is indicated.

Integrating the pressure gradient along the path shown in Figure 15, we now obtain a non-zero metal-heaving given by

$$\left(\rho_m - \rho_b\right) g \Delta z = \frac{1}{2\mu_0}\left(\mathbf{B}_1^2 - \mathbf{B}_3^2\right) \qquad (15)$$

We denote this mechanism for *irrotational metal pad heaving*. Under the anodes there are currents and a magnetic pressure contribution. The conditions are similar to the closed cell. Thus, under the anodes this mechanism predicts a flat interface as indicated in Figure 15. Knowing the general shape of $\|\mathbf{B}\|$ in a real cell, equation (15) defines a dome shaped metal heaving. For the reference model, $\|\mathbf{B}_3\| = 0$ and $\|\mathbf{B}_1\| = 0.0205\ T$. We get a maximum interface heaving amplitude about 0.085 m according to (15).

In a real cell this simplified irrotational metal-heaving will be smoothed and perturbed by the rotational contributions to the flow. We denote this effect as rotational metal-heaving. For instance the presence of vortices will contribute to enhance the level of irrotational metal-heaving. This simplified model highlights the importance of open channels to metal pad heaving that was already stated in [7]. It agrees well with the findings from the CFD simulations. When cross channels and side ledge are introduced the flow is weakened and the dome shaped irrotational metal pad heaving seems to dominate. When the channels are removed the flow becomes stronger and the mode of metal pad heaving is different and more unstable.

Conclusion

A realistic reference aluminium reduction cell model for predicting metal flow and metal pad heaving is published. The CFD based models describes the complex physical couplings in the MHD problem for the electrolysis process: two phase liquid flow, interface deformation, electric current field and electromagnetic force. In addition, it is necessary for the realistic model to include the geometry details of the reduction cells such as side ledge profile, all channels, open channel top to the ambient and constant ACD. The simulation results show that the side ledge and open channel top affect the metal pad heaving amplitude, the cross channel can affect the interface stability of the metal pad. The metal pad heaving mechanism proposed seems to agree well with the numerical results presented in this study.

Acknowledgement

The present work was supported by the project “Gas and Alumina Distribution and Transport” (GADT), financed by the Research Council of Norway, Institute for Energy Technology and Hydro Primary Metal Technology.

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Light Metals 2016

ALUMINUM REDUCTION TECHNOLOGY

Fundamentals in Chemistry I

SESSION CHAIR
Arne P. Ratvik
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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

CHARACTERIZATION OF BUBBLE BEHAVIOR IN ALUMINUM REDUCTION CELLS

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Keywords: Bubble behavior, cell voltage, pot control, aluminum electrolysis

Abstract

The better understanding of anode bubble behavior during aluminum electrolysis can help not only improve the "noise control", but also optimize the operational conditions and parameters. The effects of current density, electrolysis temperature and alumina content on bubble behavior were investigated in this study. The voltage signals were analyzed using Fast Fourier transform to study variations in the frequency responses. This study demonstrates that the size of bubbles become smaller with the increase of current density, electrolysis temperature and alumina content. Bubble release frequency increases with increase of current density, electrolysis temperature and alumina content. The amplitude of cell voltage fluctuations decrease from 80 mV to 15 mV as current density increases from 0.4 to 1.0 A/cm^2. The domain frequency increases from 0.1 Hz to 0.2 Hz as temperature increases from 950 °C to 970 °C.

Introduction

In the aluminum electrolysis process, anodic gas bubbles are generated on the anode bottom [1]. The gas bubbles accumulate at anode bottom before they are released. The phenomenon of bubble formation and sliding underneath the horizontal surface are complex due to the bubble shape, surface tension, and the anode surface characteristics. The gas bubbles induce flow in the cell which plays an important role in homogenization of the alumina distribution and the temperature field in the electrolytic bath. Conversely, the gas bubbles increase the ohmic voltage drop due to their negligible conductivity compared to the electrolyte, which in turn results in higher energy consumption for the smelting process. The presence of bubbles on anode will effectively decrease the effective area of the anode and increase the current density. Then the activation polarization will increases according to the Tafel equation. Estimates of the contribution of gas bubbles are various, but are in the order of 0.25 V [2]. The bubbles are also believed to decrease the current efficiency, because it can promote remixing of metal. In addition, the behavior of bubbles on the surface of the anode will create "bubble noise" [3], and interfere with noise analysis, then affecting the cell control. Therefore, the detailed analysis of bubble noise not only can improve the "noise control", but also can optimize the operational conditions and parameters. However, the analysis of bubble noise must be based on reliable study and characterization of the bubble behavior.

From the current literature, there are two main research methods to study the bubble behavior, i.e. laboratory electrolysis experiments and physical models. For the physical models, their modeling results have many uncertainties because of the differences in the bubble-generation mechanism and physical and chemical parameters compared to the operational cell [4]. However, the advantage of electrolysis experimental study on bubbles is having the same bubble-generation mechanism and melts and interface properties with the operational cell [4]. Although the size of anode is different between laboratory and industry, the characterization of bubble behavior is the same at equal conditions. And the characterization of bubble behavior is important for the pot control. Laboratory studies have demonstrated that the bubble behavior is affected by the anode current density, alumina content, anode geometry and inclination [4-10]. But these results are not consistent, and detailed understanding of bubble behavior is not fully understood. Some studies have reported that increasing current density is helpful in the formation of smaller bubbles that detach and release more frequently and reduces the thickness of the bubble layer [4,5]. For example, Cassayre [5] observed the bubble diameter on graphite decreased from 18 mm to 7 mm as the current density (CD) increased from 0.2 A/cm^2 to 1.6 A/cm^2 by see-through cell and radiography techniques. The bubble layer thickness decreased from 5 mm to 4.2 mm as the current density was increased from 0.2 A/cm^2 to 1.6 A/cm^2. Other studies have found an inverse result. Wang [6] observed that the bubble size increased rapidly with increasing current density for a 40 mm anode. In addition, the influence of alumina concentration on bubble behavior is more uncertain. Haupin and McGrew [7] have reported that bubble size decreases with increasing alumina content by see-through cell. However, Cassayre [8] found that the alumina content in the electrolyte have no effect on the bubble size. Therefore, it is meaningful to study the behavior of bubbles.

In this work, the bubble behavior is investigated by analyzing voltage signals measured under various conditions, in both time and frequency domains. The impacts of anode current density, alumina concentration, and operation temperature on the bubble behavior are explored. On the basis of these experimental results, the information of bubble noise have been analyzed, which may contribute to a better understanding of cell operation conditions and improving the pot control.

Experimental

Experimental Principle

The bubble behavior is investigated by analyzing the voltage signals and frequency spectrum. It is well known that cell voltage is composed of three different types: decomposition voltage, excess voltage due to polarization of the electrode interface and concentration (anode reaction and concentration overvoltage and cathode-concentration overvoltage), ohmic voltage drops due to the resistance of various zones of the overall cell (electrolyte, anode gas bubbles, anode, cathode, and external voltage drop) [11]. The decomposition voltage is determined by the reaction. Cathode concentration overvoltage and anode concentration overvoltage can be considered as constant within a short time. There are several ohmic voltage drops within the cell voltage and some of which are relatively constant within a limited time, such as, the anode resistance, the cathode resistance and the external voltage drop. Then, oscillation of voltage represents the change in bubble resistance.

Experiment

A 40 by 60 mm (diameter×height) anode was made from graphite and placed in cylindrical graphite crucible lined with a corundum crucible. The diameter of the corundum crucible was 100 mm. The bottom of the corundum crucible with a 15 mm diameter hole was fitted into the graphite crucible, which was placed on a stainless steel pallet. The anode was fixed to a steel rod. The anode was connected to a direct current power supple allowing for currents up to 100 A. In total, 500 g of bath was kept in the graphite crucible covered with a corundum lid. The sketch of the experimental setup is shown in Figure 1. The bath composition was NaF/AlF_3 with molar ratio 2.3, Al_2O_3 2.5 wt%, CaF_2 5 wt%, MgF_2 0.5 wt%.

To understand the behavior of bubble formation and release, acquisition of voltage signals was collected at a rate of 2 Hz with an accuracy of 0.001 V. The voltage signals were continuously recorded under the electrolysis process. The voltage signals were analyzed using Fast Fourier transform to study variations in the frequency response.

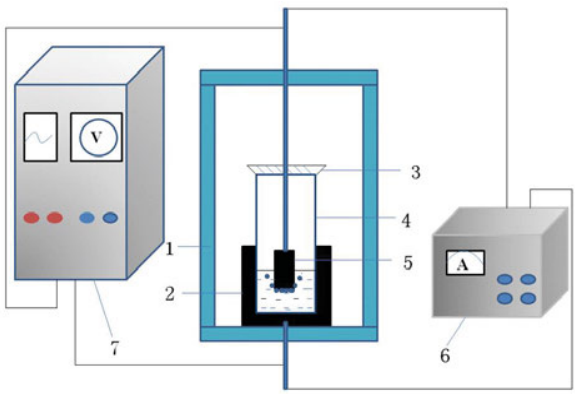

Figure 1. A schematic view of the experimental system
1-Furnace; 2-Graphite Crucible; 3- Corundum lid; 4-Corundum Crucible; 5-Anode; 6-Direct current supplier; 7-Cell controller.

Results and Discussion

Figure 2 shows the variation of cell voltage with time during electrolysis at 960 °C. Two distinct regions are identified from the profile shown in Figure 2, when a new anode is set in the bath, it immediately forms a coating of frozen bath around it due to a large amount of heat is extracted from the bath and transferred into heating the new cold anode [12]. Therefore, voltage is high, 3.8 V, at the start of electrolysis. The voltage decreases from 3.8 to 3.263 V with the melting of the freeze. A slowdown of the voltage decline rate observed at 2000 s suggests that the process of electrolysis have reached stable conditions, and the freeze has melted completely. Also, it can be seen from Figure 2 that the amplitude of cell voltage fluctuations is different at the two regions. This may be due to different bubble behavior. The bubble generation takes mostly place at the outer part of the anode bottom in the process of freeze melting. As a result, bubbles have less distance to travel and less chance for coalescence and growth before they escape from the anode. Therefore, the amplitude of the cell voltage fluctuations is smaller. Bubbles have long distance to travel and more chance for coalescence and growth before they escape at the stage of stable electrolysis. Then, the amplitude of cell voltage fluctuation is increasing.

The cell voltage profiles obtained at various conditions show a similar overall trend. Although there are small differences in the time to reach stable conditions, the average time at which the electrolysis process reaches a stable voltage is approximately 0.5 h. In order to ensure the constant conditions of electrolysis, a short time (200 s) was selected to study the bubble behavior. The corresponding frequency spectrum is shown.

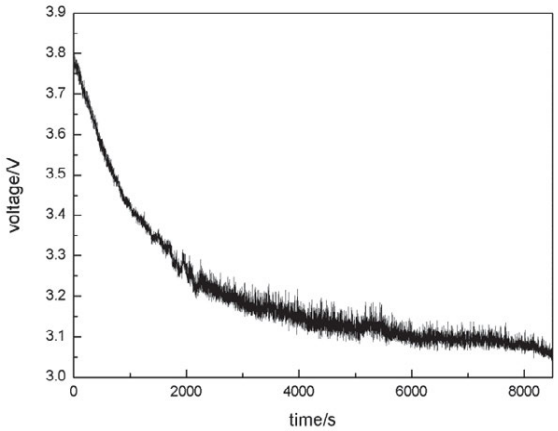

Figure 2. Variation of cell voltage with time in the process of electrolysis (960 °C).

The Influence of Current Density on Bubble Behavior

The impact of CD on bubble behavior is reported in several papers [6-10]. Those studies compared fluctuation of cell voltage in distinct time frames, without investigating the frequency responses of cell voltage signals under different CDs. To find the behavior of bubble variations for different cell voltages, the frequency responses of cell voltage signals are studied for different CDs. The results are shown in Figure 3 and Figure 4, respectively.

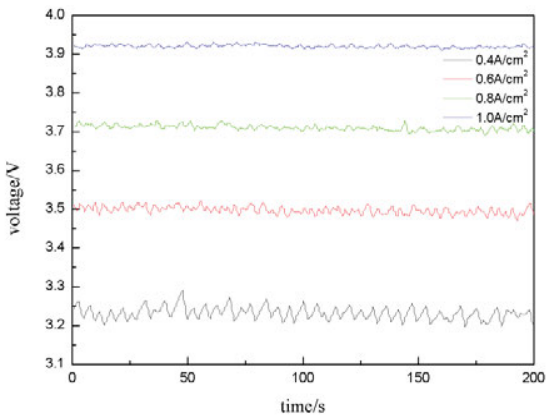

Figure 3. Fluctuations of cell voltage at different CDs in 200 s periods.

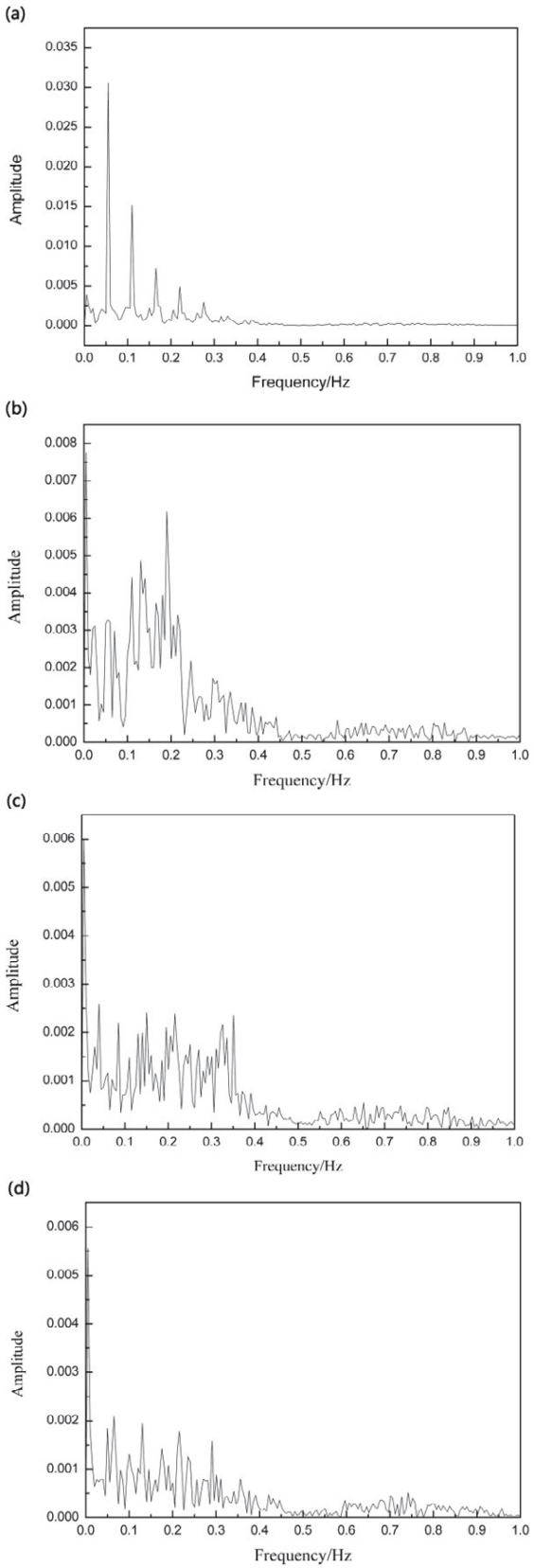

Figure 4. Frequency response of cell voltage at different CD: (a) 0.4; (b) 0.6; (c) 0.8; (d) 1.0 A/cm^2.

The voltage fluctuations at different CDs, as illustrated in Figure 3, show that the cell voltage increases with the increase of CD, which is consistent with what can be expected according to Ohm's law. It can also be seen from Figure 3 that the amplitude of cell voltage fluctuations decreases with increasing CD. The amplitude of cell voltage fluctuation is around 80 mV at CD=0.4 A/cm^2, and decrease to 40 mV at CD=0.6 A/cm^2. When the CD reach to 0.8 A/cm^2, the amplitude of cell voltage fluctuation is around 30 mV. The amplitude of cell voltage fluctuation is around 15 mV at CD=1.0 A/cm^2.

As seen in Figure 4, the spectrum shows the presence of five peaks in the range between 0.05 and 0.3 Hz at CD=0.4 A/cm^2. The prominent peak is located around the frequency of 0.05 Hz. The spectrum shows the presence of prominent peak at the frequency around 0.2 Hz when CD=0.6 A/cm^2. When CD=0.8 A/cm^2 and 1.0 A/cm^2, it shows that the reduced amplitude is evenly distributed over a range of frequencies from 0.05-0.35 Hz.

This may be because the number of nucleation sites is small when CD=0.4 A/cm^2. Fewer bubbles form on the bottom surface of the anode, then grow and coalesce and form bigger bubbles and finally release. The bubbles are stuck on the anode and, without much agitation in bath, detach with difficulty. As a result, bubble coverage varies significantly because the size of the releasing bubbles is large, leading to greater fluctuations in cell voltage. When CD=0.6 A/cm^2, the number of nucleation sites increases and bubbles have little chance to coalescence as much as CD=0.4 A/cm^2. Then, the bubbles become smaller and more easily form at a higher frequency. Therefore, bubble coverage varies less for the released bubbles. When CD is increased to 0.8 and 1.0 A/cm^2, the rate of bubble generation is faster. The amount of gas formed increases, creating more agitation in the melt. Therefore, the process of bubble release becomes more even.

The Influence of Electrolysis Temperature on Bubble Behavior

The cell voltage fluctuations at different electrolysis temperature, as illustrated in Figure 5, show that the amplitude of the cell voltage fluctuations decreases with increasing temperature. The amplitude of cell voltage fluctuation is around 40 mV at 950 °C and decrease to 30 mV at 960 °C. When the temperature reaches 970 °C, the amplitude of the cell voltage fluctuations are around 20 mV.

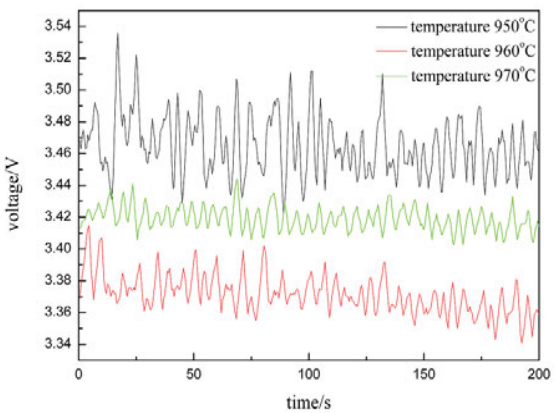

Figure 5. Fluctuation of cell voltage at different electrolysis temperature

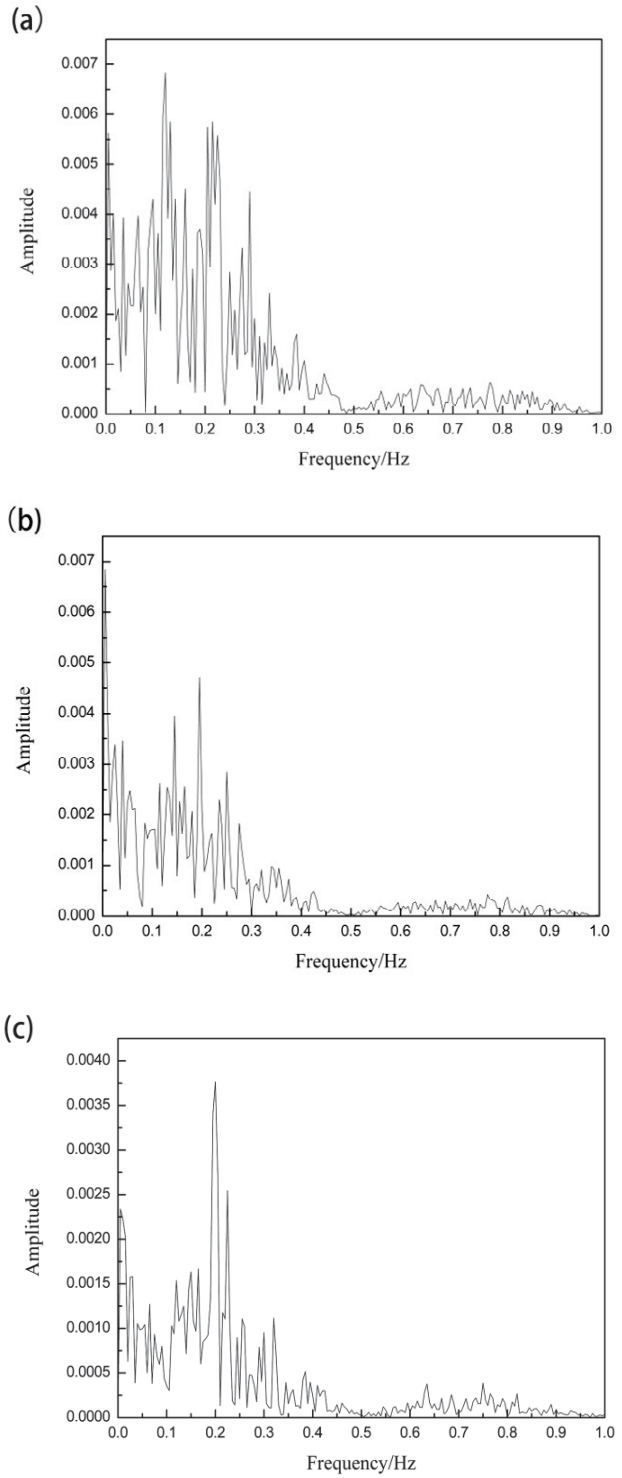

Figure 6. Frequency response of cell voltage at different electrolysis temperature: (a) 950 °C; (b) 960 °C; (c) 970 °C

As shown in Figure 6, the amplitude of peak decreases as the temperature increases. The frequency of the prominent peak is located at around the frequency of 0.1, 0.2, 0.2 Hz at 950, 960, 970 °C, respectively. Bubble release frequency increased with temperature.

This may be due to the viscosity of melt decreases with increase of temperature. The viscosity of melt will affect transport and release of bubble. It is well known that the behavior of bubble is a complex dynamic balance of surface tension, buoyancy force, viscosity friction between molten and bubbles, and contact angle. In the release stage, the buoyancy force must be strong enough to overcome the friction between the molten salt and the bubbles to detachment under anode. The release of bubble is inhibited with the increase of melt viscosity. Then, the size of bubbles released will increase before overcoming viscosity friction. The bubbles grow and coalesce and form bigger bubbles before being released at the lower temperatures. And angle of wetting between anode and cryolitic melts decreases with increase of temperature [1]. The melts have a better wetting to the anode. This is beneficial for the bubble's release.

The Influence of Alumina Concentration on Bubble Behavior

The cell voltage fluctuation at different alumina content, as illustrated in Figure 7, show that the amplitude of cell voltage fluctuations decrease with the increase of alumina concentration. The amplitude of voltage fluctuations are around 40 mV when the content of alumina is 1.5 %, and decrease to 30 mV when the content of alumina is 2.5 %. When the content of alumina reaches 3.5 %, the amplitude of the cell voltage fluctuation is around 15 mV. This may be the reason for the anode effect. It is well known that the reason for anode effect is a low alumina content [13]. The bubble coverage area increases at low content of alumina. Then, the thickness of CO_2 layer increases until it becomes insulating, causing the anode effect onset.

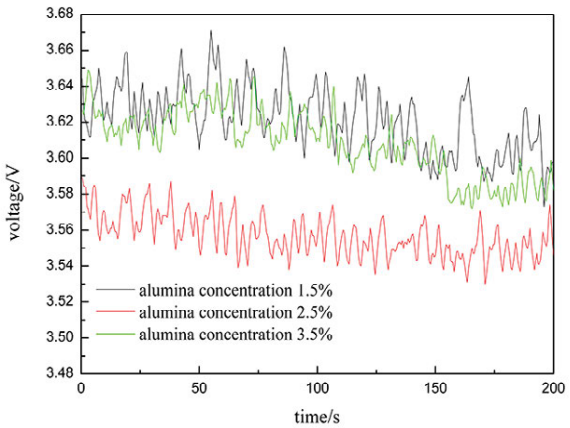

Figure 7. Fluctuation of cell voltage at different alumina content.

It can be seen from Figure 8 that the peak amplitude decreases when the alumina increases. The prominent peak is located around the frequency of 0.05 when the content of alumina is 1.5 %. When the content of alumina is 2.5 %, the reduced amplitude is evenly distributed over a range of frequencies from 0.05-0.3 Hz. When the content of alumina is 3.5 %, the reduced amplitude is evenly distributed over a range of frequencies from 0.1-0.3 Hz. This may be because a change in the anode/bath interfacial tension and wetting angle occurs for different contents of alumina. The wetting angle decreases with increasing alumina. Then, electrolyte has a better wetting of the anode and bubbles have less chance to coalescence. Therefore, the bubbles become smaller and release more easily. This is agreement with the results of Haupin [7].

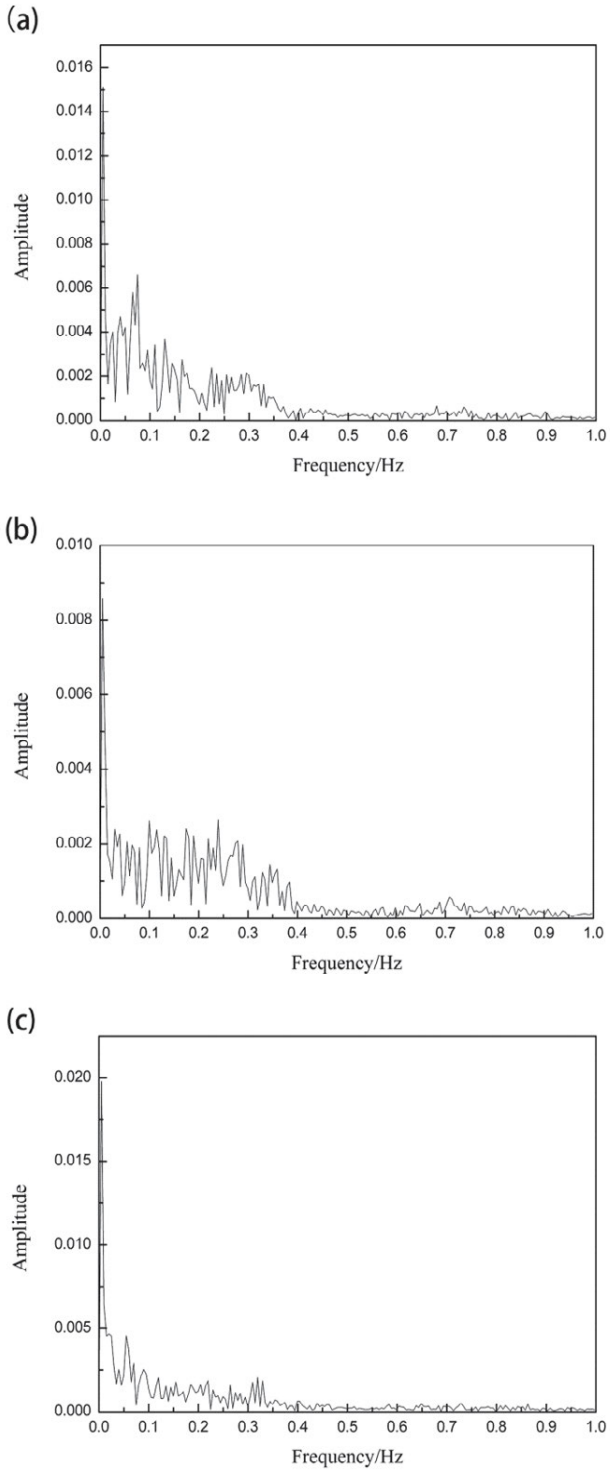

Figure 8. Frequency response of cell voltage at different alumina: (a) 1.5 %; (b) 2.5 %; (c) 3.5 %

Conclusions

This work presents a method to study the bubble behavior based on time and frequency domain analysis of cell voltage signals. The effects of current density, electrolysis temperature, alumina content on bubble behavior were investigated. The results demonstrated that the amplitude of cell voltage fluctuation decreases with increasing CD, electrolysis temperature and content of alumina. The amplitude of cell voltage fluctuations decrease from 80 mV to 15 mV as current density increases from 0.4 A/cm^2 to 1.0 A/cm^2. The amplitude of cell voltage fluctuation decreases from 40 mV to 20 mV when electrolysis temperature increases from 950 °C to 970 °C. And the dominating frequency increases from 0.1 Hz to 0.2 Hz. The amplitude of cell voltage fluctuations decrease from 40 mV to 15 mV as the content of alumina increases from 1.5 % to 3.5 %. The dominating frequencies are evenly distributed over a range of frequencies from 0.1-0.3 Hz when the content of alumina is 3.5 %. Bubble release frequency increased with electrolysis temperature and content of alumina.

Acknowledgement

The authors thank the financial support of the National Science & Technology Pillar Program of China (2012BAE08B02).

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ELIMINATION OF LITHIUM FROM ALUMINIUM ELECTROLYTE BY ACID LEACHING METHOD

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Keywords: Lithium elimination; Aluminium electrolyte; Acid leaching; $LiNO_3$

Abstract

This paper report on an acid leaching method to eliminate lithium from aluminium electrolyte system (CR=3). Both deionized water and HNO_3 were used. Also, the effects of the influences of acid concentration, leaching time, liquid-to-solid (L/S) ratio and temperature on the leaching efficiency were investigated. The results show that the effect of HNO_3 on leaching efficiency is much better than deionized water. The leaching efficiency increases with the increase of HNO_3 concentration, leaching time, L/S ratio and temperature. The optimized conditions for removing lithium for aluminium bath are presented as following: the concentration of HNO_3 is 14 mol/L; the leaching time is 120 min; the L/S ratio is 12.5 $cm^3 \cdot g^{-1}$ and the leaching temperature is 80 °C, the leaching efficiency is up to 93.21 %. This new process generates Na_3AlF_6 and $LiNO_3$ as products from aluminium electrolyte containing lithium salts.

Introduction

With the rapid development of electrolytic aluminium industry and the increasing need of aluminium, massive middle to low grade bauxite ores are exploited to produce metallurgical grade alumina [1-3]. Alumina that is produced from low-grade bauxite ores contains a minor amount of lithium salt. Adding this alumina into the electrolyte in aluminium reduction cells causes serious problems due to the continuous accumulation of lithium salts in the electrolyte [4-5]. It has been reported that the liquidus temperature of aluminium electrolytes can be decreased 7.8 °C with 1 % increase in LiF [6]. The lower liquidus temperature results in higher superheat in the cell, leading to a thinner ledge, and eventually decreases the cell life [7]. In addition, the increase of LiF decreases the alumina solubility in the electrolyte [8]. Thus, the working life of the cell and the current efficiency will decrease with the reduced alumina dissolution. In the cathode part, the cathode may be damaged by LiF due to the produced Li_2C_2 at the cathode surface [9].

Also, the high content of lithium in the electrolyte reduces the quality of aluminium produced. High content of LiF may affect the surface and the texture of aluminium, making it rough and crisp [10].

Thus, eliminating lithium from the electrolyte is very important both to the cell performance and the quality of produced aluminium. However, little progress has been made in the elimination of lithium. J. S. Zhang [11] investigated the existing phases of lithium in cryolite. With the cryolite ratio (CR) <3, the lithium salt was present as Li_2NaAlF_6 while LiF phase dominated for the CR ≥3.

This paper used an acid leaching method as LiF reacts easier with acid than Li_2NaAlF_6. A one- step acid leaching method to remove LiF from an electrolyte with CR of 3.0 was used. The effects of the acid concentration, leaching time, liquid-to-solid (L/S) ratio and temperature on the leaching efficiency were investigated.

Experimental

The compositions of the electrolyte used in this experiment were based on a mixture anhydrous LiF, anhydrous NaF and anhydrous AlF_3 shown in Table I. Anhydrous AlF_3 was prepared by thermal decomposition of $(NH_4)_3AlF_6$ [12]. All compositions were dried at 400 °C for more than 3 hours and stored in a dry container.

Table I. The composition of electrolyte.

Material	wt%	Manufacturer	Specifications
NaF	54	SCRC, China	Impurity：＜1%
LiF	10	SCRC, China	Impurity：＜1%
AlF_3	36	SCRC, China	Impurity：＜1%

All the leaching experiments were conducted using a mechanical stirring system. It operated at 200 rpm and under room atmosphere. The experimental parameters for the leaching experiments are shown in Table II, where T is the leaching temperature, t is the leaching time and L/S is the liquid to solid ratio.

Table II. Leaching experimental parameters.

Step	Deionized water	HNO_3
Concentration/(mol/L)	----	6-14
T/°C	20	20–100
t/min	30	30–150
(L/S)/($cm^3 \cdot g^{-1}$)	280	5–15

SEM and XRD were applied in the analysis of the samples. The surface micro-topography of the electrolyte containing LiF before and after acid leaching was obtained by a Shimadzu ssx-550 scanning electron microscope (SEM). The electrolyte containing LiF, the filterable matter from the leaching experiments and the evaporated crystal substances of the leaching solution were characterised by MPDDY2094 X-ray diffraction (XRD) with a scanning angle from 10° to 80°. The content of lithium was determined by Z-2300 atomic absorption spectrometer in the wave length range from 190 to 900 nm.

Results and Discussion

Characterisation of the Electrolyte Containing LiF

The electrolyte containing LiF was analysed by XRD, as shown in Figure 1. From Figure 1, the characteristic peaks 38.469°, 44.716° and 65.090° in the XRD pattern belong to LiF and the other

characteristic peaks belong to Na_3AlF_6. Thus the compositions of the electrolyte were Na_3AlF_6 and LiF.

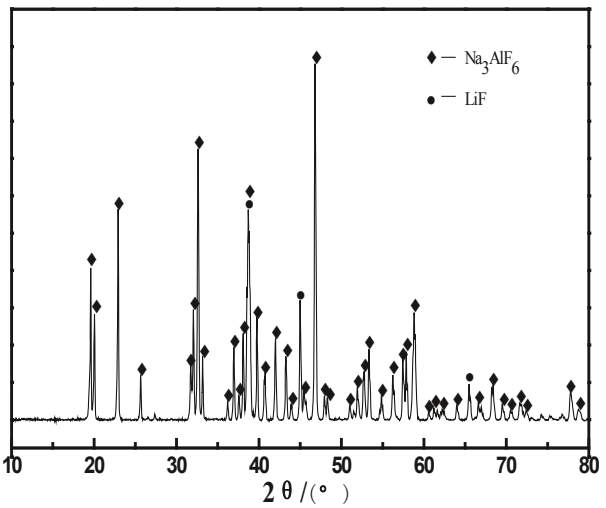


Figure 1. XRD pattern of electrolyte containing LiF.

Effect of Acid Strength

With a solubility of LiF in water equal 2.7 g/L, LiF could possibly be removed from the electrolyte if sufficient water was used. Hence, the effect of leaching with deionized water was investigated. The electrolyte containing LiF was leached at 20 °C for 30 min using a L/S ratio of 280 $cm^3 \cdot g^{-1}$ with deionized water, which was compared with the leaching experiment operated with 6 mol/L HNO_3. The experiments were conducted at 20 °C for 30 min using L/S ratio of 5 $cm^3 \cdot g^{-1}$ with the different concentrations of HNO_3 in the range from 6 mol/L to 14 mol/L.

The results are shown in Table III, showing that the effect of HNO_3 is much better than deionized water. Possibly, LiF exists not only on the surface of the electrolyte, but also inside the electrolyte. The inside part of LiF cannot be dissolved in deionized water.

Table III also shows that the leaching efficiency of LiF from the electrolyte increases with an increase of HNO_3 concentration. Because the maximum leaching efficiency was obtained with 14 mol/L HNO_3 under ambient atmosphere, 14 mol/L HNO_3 was chosen for the following acidic leaching.

Table III. Effects of deionized water and different concentrations of HNO_3 on leaching.

Step	Deionized water	HNO_3				
Concentration /(mol/L)	----	6	8	10	12	14
Leaching efficiency of LiF / %	2.19	8.61	12.33	18.25	22.51	24.19

Effect of Leaching Time

The electrolyte containing LiF was leached at 20 °C with a HNO_3 concentration of 14 mol/L and a L/S ratio of 5 $cm^3 \cdot g^{-1}$, which is used as the baseline to quantify the leaching effect. The reaction time was varied from 30 to 150 min.

The results of acid leaching are shown in Figure. 2, which indicate that the reaction time substantially affect the dissolution of LiF. With an increase in the reaction time from 30 to 120 min, the leaching efficiency rapidly increased from 24.19 % to 41.16 %. The rate of leaching levels out with increasing time. There is nearly no change in the level of leaching from 120 to 150 min. Hence, a reaction time of 120 min was chosen for the consecutive experiments.

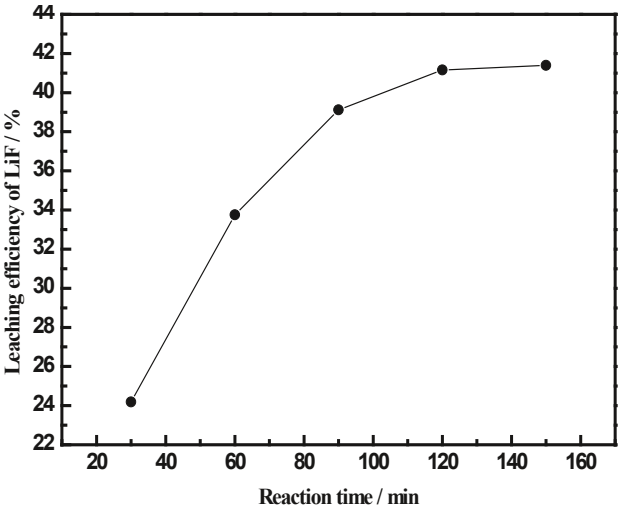


Figure 2. Effect of leaching time.

Effect of L/S ratio

The electrolyte containing LiF was leached at 20 °C for 120 min in an acidic solution with a HNO_3 concentration of 14 mol/L and a L/S ratio of 5 $cm^3 \cdot g^{-1}$, which is used as the baseline to quantify the leaching effect. The L/S ratio was varied in the range of 5 to15 $cm^3 \cdot g^{-1}$ in the experiments.

The results are shown in Figure 3. It was observed that an increasing L/S ratio plays a positive role in the leaching efficiency of LiF. The L/S ratio did not strongly affect the leaching efficiency being increased from 12.5 to 15 $cm^3 \cdot g^{-1}$. The increasing L/S ratio could provide more H^+ to react with the soluble LiF. A L/S ratio of 12.5 $cm^3 \cdot g^{-1}$ was found to be sufficient for subsequent experiments.

Effect of Temperature

In these experiments, the electrolyte containing LiF was leached in a solution with 14 mol/L HNO_3 for 120 min and the L/S ratio is 12.5 $cm^3 \cdot g^{-1}$. This is the base line experiment. In the following experiments, the temperature was varied from 20 °C to 100 °C.

The results of the acidic leaching experiments are presented in Figure 4. The results indicate that the reaction temperature substantially affect the dissolution of LiF and the leaching efficiency had a marked increase in the temperature range of 40 °C to 80 °C. The increase in the reaction velocity constant *K* with increasing temperature may contribute to this. Over the 80 °C, the leaching efficiency dropped rapidly from 93.39 % to

85.05 % at 100 °C. This trend is believed to be due to chemical decomposition of HNO_3 at 86 °C, causing a rapid decrease in the concentration of HNO_3 when the temperature is above 100 °C. The results indicate that the concentration of HNO_3 dominates the leaching efficiency at 100 °C. From the curve, it is found that the leaching efficiency was not much affected between 80 °C and 85 °C. Consequently, the reaction temperature of 80 °C was chosen for the experiments.

From the above experimental results, the optimum parameters for the lithium leaching were explored. With the following parameters: concentration of HNO_3 14mol/L; leaching time 120 min; L/S ratio 12.5 $cm^3 \cdot g^{-1}$ and leaching temperature 80 °C, the leaching efficiency was up to 93.21 %.

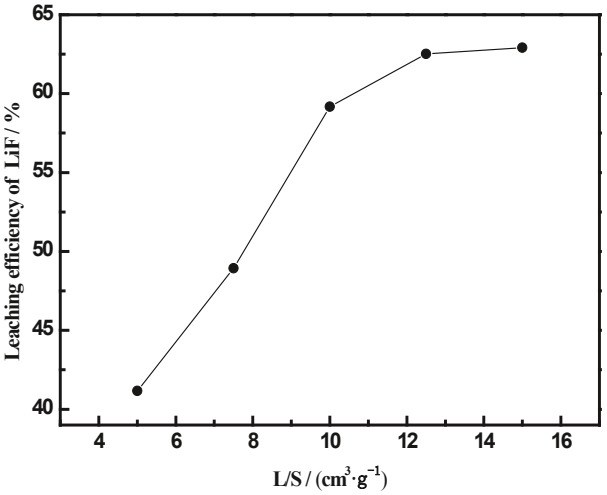


Figure 3. Effect of L/S ratio.

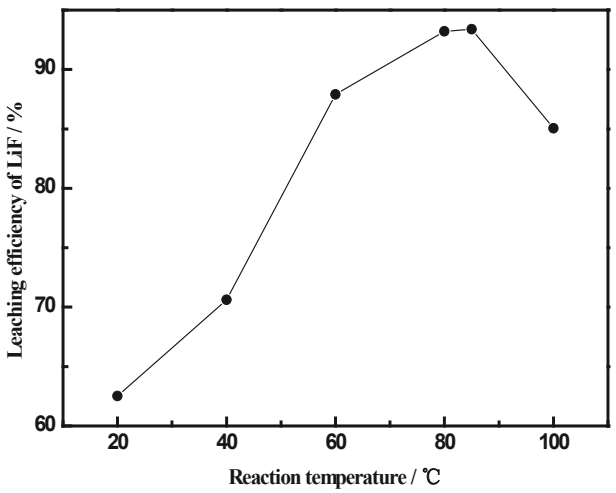
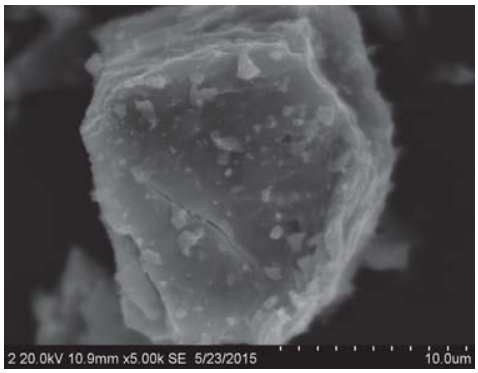


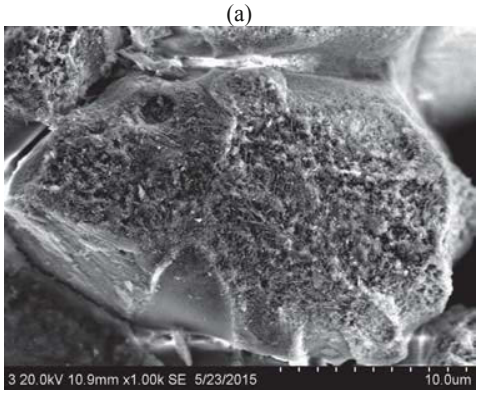
Figure 4. Effect of temperature

SEM Analysis of the Electrolyte before and after Leaching

SEM pictures of the electrolyte before and after the leaching experiment are shown in Figure 5. Before the leaching (Figure 5a), the surface of the electrolyte was smooth and dense with a particle size of 10 μm. After the leaching process (Figure 5b), the particle surface was corroded by acid leaching and many eroded holes appeared on the surface of the particles. The electrolyte was solidified from a high temperature melt that contained NaF, AlF_3 and LiF. LiF could not only exist at the surface of the electrolyte, but also inside it. HNO_3 reacted first with LiF on the surface of the electrolyte before reacting with LiF inside the electrolyte via the eroded holes. However, deionized water has no such effect. This could explain why the effect of HNO_3 is much better than the deionized water.



(a)



(b)

Figure 5. SEM pictures of electrolyte before (a) and after (b) leaching.

Also, the two photos can explain the observations in Figure 2, Figure 3 and Figure 4: with the increase in HNO_3 concentration, leaching time, L/S ratio and temperature, more holes appear on the particle surface, through which more LiF will be leached, and eventually the leaching efficiency increases.

XRD Analysis after Acidic Leaching

The XRD patterns of the electrolyte containing LiF after being leached with a leaching efficiency of 93.21 % are shown in Figure 6.

The XRD patterns show that the characteristic peaks that belong to LiF disappear in the XRD. The results indicate that LiF from the electrolyte dissolves in the nitric acid solution and the component of the electrolyte after acidic leaching is Na_3AlF_6.

With effective elimination of lithium from the aluminium electrolyte by HNO_3 leaching, the remaining electrolyte can be reused for aluminium smelting.

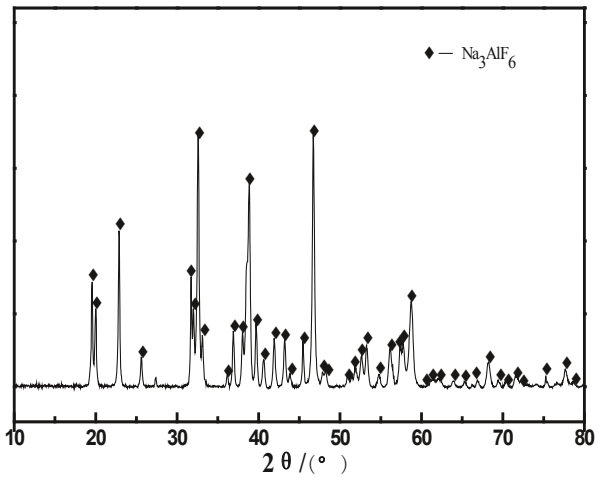


Figure 6. XRD pattern of electrolyte containing LiF after being leached.

XRD of the Precipitated Material

The XRD pattern of the precipitate obtained after evaporation of the filtrate is shown in Figure 7. The optimum experimental parameters were applied.

The XRD pattern shows that the precipitate is $LiNO_3$. The result indicates that LiF from leaching the electrolyte existed in the filtrate as Li^+ and precipitate as $LiNO_3$ in the dried concentrate. This also provides a possibility for Li salt production from the filtrate.

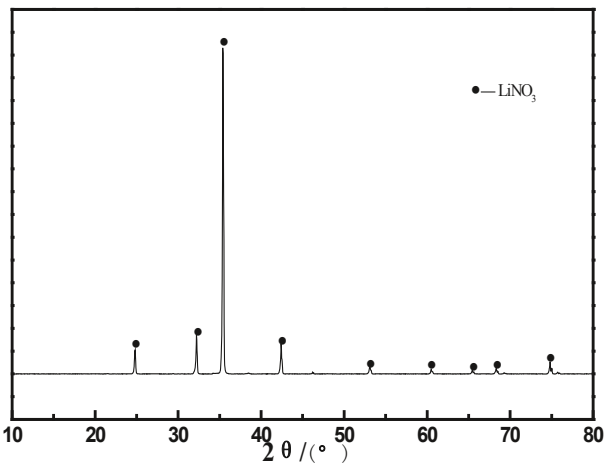


Figure 7. XRD pattern of precipitate after evaporation of filtrate.

Conclusion

This paper investigated an acid leaching method for the elimination of lithium from aluminium electrolyte. From this study, some key conclusions are obtained:

1. The effect of HNO_3 on leaching is much better than deionized water. Acid leaching effectively reduces lithium from the aluminium electrolyte.

2. The leaching efficiency increases with increasing HNO_3 concentration, leaching time, L/S ratio and temperature. This trend can be confirmed by the SEM analysis: lots of corroded holes appeared after the leaching experiment and the reaction area increased.

3. The optimum parameters for the lithium leaching are: HNO_3 concentration 14 mol/L; leaching time 120 min; L/S ratio 12.5 $cm^3 \cdot g^{-1}$ and leaching temperature 80 °C, resulted in a leaching efficiency of up to 93.21 %.

4. The electrolyte containing LiF can be reused for aluminium production because purified Na_3AlF_6 is the main product after acid leaching.

5. This study also provides a possibility for obtaining a $LiNO_3$ rich salt from purification of an aluminium electrolyte with high content of lithium salts.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

IMPACT OF THE HEAT FLUX ON SOLIDIFICATION OF CRYOLITE BASED BATH

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Keywords: solidification, cold finger, electrolytic bath, cooling rate, structure

Abstract

The solidification of the electrolytic bath in Hall-Héroult cells includes the formation of a sideledge, crust as well as temporary freeze layers around alumina agglomerates, new anodes or any cold object inserted into the bath. The structure and the chemical composition of solidified phases depend on the cooling heat flux (or cooling rate), which has a huge impact on the formation of crystals, amorphous phases and pores and on the extent of ion migration. The structure and chemical composition determine the heat loss through the side ledge, and the duration of the existence of a frozen bath around alumina agglomerates or new anodes. The authors of this paper created frozen bath samples under different heat flux conditions in well controlled laboratory environment in order to study the variation of the chemical composition and structure with the cooling rate. The results of this study are presented in this article.

Introduction

The solidification of the electrolytic bath in Hall-Héroult cells includes the formation of a sideledge, crust as well as temporary frozen layers around alumina agglomerates, new anodes or any cold object inserted into the bath. The structure of the solidified phase, as well as the difference between the chemical compositions of molten and frozen bath depend on the cooling heat flux or in other words, on the cooling velocity. In fact, the diffusion of ions is promoted during the solidification by the high melting point of pure cryolite [1-3], however, an elevated rate of solidification can block this movement. Consequently, the slow formation of a big volume of frozen bath (low cooling) can alter the bath composition and thus its physical properties such as the liquidus temperature [1-4]. The cooling rate has a huge impact also on the number of the nucleation sites and pores, on the size, the orientation and the structure of the different crystals, and on the proportion of the amorphous phase. The structure and chemical composition determine the heat loss through the side ledge [5], and the duration of the existence of a frozen bath around alumina agglomerates or new anodes. The first layer of the frozen electrolyte around any cold object inserted into the bath is formed by an elevated heat flux. On the other hand, the additional layers or the side ledge are created at a much lower cooling velocity.

Some information can be found about the structure and the chemical composition of frozen bath in sideledge [2-7] and in the crust [8-9], but in the most cases, samples are taken from industrial cells with unknown and varying conditions. The authors of this paper created frozen bath samples under different heat flux conditions in well controlled laboratory environment in order to study the variation of the chemical composition and structure with the cooling rate. Structure was correlated with heat flux and solidification rate, computed with an in-house developed mathematical model using measured temperature histories as input. The results of this study are presented in this article.

Methodology

Production of Solidified Bath Samples using Cold Finger Technique

Solidified bath samples were produced using the so-called cold finger technique that includes the insertion of a cold object into the molten bath. Samples were produced in a laboratory cell which contained an unstirred molten synthetic cryolite based bath with known composition. Both the composition and the superheat of the bath corresponded to those used in many RTA plants. The quantity of the bath was high enough to avoid any significant perturbation of bulk liquid temperature by the probe itself (semi-infinite liquid condition around the solid block). As no electrolysis was carried out during the tests, the absorption of heat during the insertion of the probe and the loss toward the environment were compensated by the same heating furnace that melted the bath.

The solidified bath samples were produced under both transient and near steady-state conditions, denoting strongly varying and relatively constant, low heat fluxes respectively.

The first scenario was realized by the insertion of a cold, solid stainless steel probe (12.5 x 11 x 2.54 cm) into the bath. Stainless steel was chosen as it can assure good mechanical and chemical resistance to high temperature bath as well as limited oxidation rate while it still has an acceptable thermal conductivity compared to the frozen bath. The time of insertion to obtain the thickest possible frozen crust (sample) around the probe was estimated using our mathematical model, developed earlier [10].

The near steady-state solidification was carried out using a hollow steel probe (same outer dimensions as the transient probe) cooled from inside by circulating air. The air conducting channels inside the probe were designed in a way that could assure a relatively homogenous temperature distribution along the outer surface of the probe. Compressed air was first dried and then cooled to -3 °C using a vortex tube. As the probe was relatively cold at the moment of insertion, a transient crust formed fast around it. The circulation of the air was started only after the re-melting of this initial crust. The re-melting was assumed to be complete when the temperature inside the probe stabilized at a value close to the temperature of the bulk. This assumption was confirmed by visual verification. Even with constant flow rate and input temperature of the cooling air, there was some temporal variation of the heat flux through the walls of the probe due to the initial thickening of the solidified crust around it. However, this variation is very small compared to that can be obtained with the transient probe. The probe was removed when the temperature inside the probe reached an almost constant value.

In all cases, the temperature histories were measured by N type thermocouples (pre-selected by calibration), inserted both into the center of the probes and into the bulk liquid. The probes were preheated to 150 °C before insertion into the liquid in order to remove superficial humidity. Several wall thicknesses were tested for the steady-state (hollow) probe, because a thinner wall results in a higher heat flux and thus thicker sample. On the other hand, below a certain thickness, the mechanical resistance of the wall cannot be sufficient at elevated temperatures to prevent deformations.

Mathematical Model to Estimate the Heat Flux and Solidification Rate during Bath Solidification around the Transient and Steady-State Probes Inserted into the Liquid

Our mathematical model of transient freeze formation [10] was used to estimate the heat flux and solidification rate during the freezing of the bath around the transient and steady-state probes as well as to calculate the time necessary to produce the maximal thickness of the crust around the transient probe. As the probes had planar shape form, a 1D model with planar (Cartesian) coordinate could be used. Due to the high height-to-thickness ratio of the probes, the end effects were neglected for estimation purpose.

The mathematical model was solved using the finite difference method on a fixed grid. Simple explicit method was preferred to the Crank-Nicolson method because of the non-linear relations. The software MatLab® was selected for solving the heat and mass balance equations. More details can be found in the above mentioned paper [10].

Image Analysis

Scanning Electron Microscope (SEM JEOL JSM 6480 LV with EDS Oxford SiLi detector and Inca software) was used to obtain information about surface topography. When secondary electron imaging mode (SEI) is applied, the SEM permits to produce a very high-resolution image about the structure of the sample surface with a relatively large depth of field.

Measurement of Thermal Conductivity of Frozen Bath Samples

The thermal diffusivity and conductivity were measured by laser flash method using a FlashLineTH 5000 (Anter Corp.) apparatus with infrared laser source and radiometric detector between 100 and 500 °C. Measurement at higher temperatures could not be carried out because of the possible appearance of liquid phase [11]. The laser flash method is presented in details in an earlier paper [5]. The variation of density in the same temperature range, necessary to obtain the thermal conductivity, was determined with UNITHERM^{TM} 1101 (Anter Corp.) dilatometer.

Results and Discussion

Cold Finger Test

Figure 1 shows an example of the frozen bath crust, formed on one of the cold finger probes. The crust had almost constant thickness around the middle of the large lateral sides of the probes. Some easily removable, round shaped, dark colored impurity particles were adhered to it at a few places. The average thickness of the continuous crust was about 6 mm for the transient probe and 5.1 mm for the steady state one. The thickness values were measured at room temperature, after the crust shrank during cooling down. The mathematical model underestimated the thickness of the crust around the transient probe by about 20 %. The difference is caused mainly by the fixed grid used in the model. In fact, once the bath in an element of the grid is completely frozen, the grid itself does not shrink due to the cooling. Other minor error sources are related to the liquid adhered to the crust and solidified after the removal of the probe from the bath as well as the end effect and the impact of impurities which were neglected by the model. However, this still can be considered as a good agreement for such a complex heat and mass transfer problem with phase change and the model gives a good approximation of the solidification rate and heat flux.

Samples for image analysis were cut from the crust next to the middle of the large lateral sides of the probes without excluding the impurity particles.

Figure 1. Frozen bath crust formed on the transient cold finger probe.

Steady-state (cooled) probes were tested with wall thickness of 6.35 mm (1/4 inch) and 1.59 mm (1/16 inch), respectively, as mentioned above. The 1.59 mm was too thin to keep the probe shape when exposed for long time to the high temperature of the molten bath and under the pressure of the air circulating inside. Consequently, the crust formed on the inflated, curved side walls could not be considered as one formed by perfectly unidirectional heat flux. The probe with a wall thickness of 6.35 mm had no any visible deformation. The crust obtained with the thinner probe wall was not used for analysis.

Figure 2 shows the temperature histories, measured in the middle of the transient (above) and steady-state (below) probes during their submersion into the molten bath. After a short delay, the temperature inside the transient probe rose with a gradually slowing rate until the probe was removed after 200 seconds. The slow-down of the temperature rise can be explained by the simultaneous effects of the decreasing temperature gradient between the probe and the bulk in time and the gradual thickening of the poor heat conducting solid formed around the probe.

The variation of the temperature inside the steady-state probe during the first 15 minutes corresponds to the formation and re-

melting of the initial transient crust. The curve has similar nature as the one recorded with the transient probe. This relatively long period compared to the 200 seconds can be explained by the observation (supported by the simulations) that the re-melting of the crust takes much longer time than the initial solidification. The small overshoot at 15 minutes was caused by the shaking of the probe in order to accelerate the disintegration of the last layers of the crust. After 15 minutes, the air circulation in the probe was started, resulting in the slow decrease of temperature toward a new equilibrium value. Once again, the variation was slowing in time as the crust became more and more thick. The probe was removed after 32 minutes (when temperature became almost constant) with a 5.1 mm thick crust, measured at room temperature after thermal shrinkage.

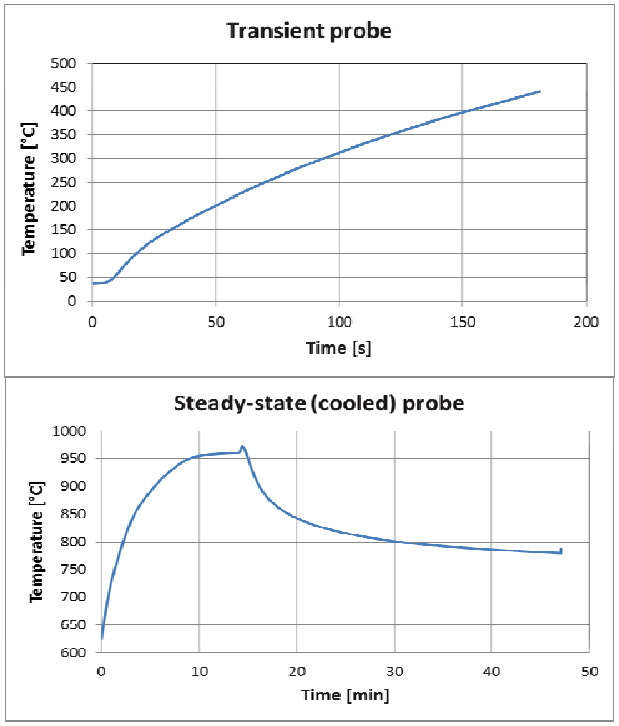


Figure 2. Temperature history measured at the center of the of the two cold finger probes.

Even if the slightly variating temperature in the hollow (air-cooled probe) suggests that the formation of the crust did not take place at perfectly steady-state condition, the total variation of temperature at the center of the probe is about 2.5 times smaller while the average solidification rate is even 11.25 times lower when compared to the case of the transient probe.

Appearance and Image Analysis

Small planar samples were cut from the solid crusts. Only inclusion free zones, close to the middle of the large sides were used for this purpose. The cut faces, normal to the probe surface, were visualized with SEM in order to study the evolution of the structure and the elementary chemical composition in the successively formed frozen bath layers. The details of this study will be discussed in a future paper.

Figure 3 shows the variation of the structure in the normal direction to the probe surface for both transient and steady-state cases using 17x and 25x magnifications respectively. The layer in contact with the probe can be found on the left side of the pictures. The thin, light colored layers along the contact interface are the sides of the samples, which were not perfectly perpendicular to the studied face.

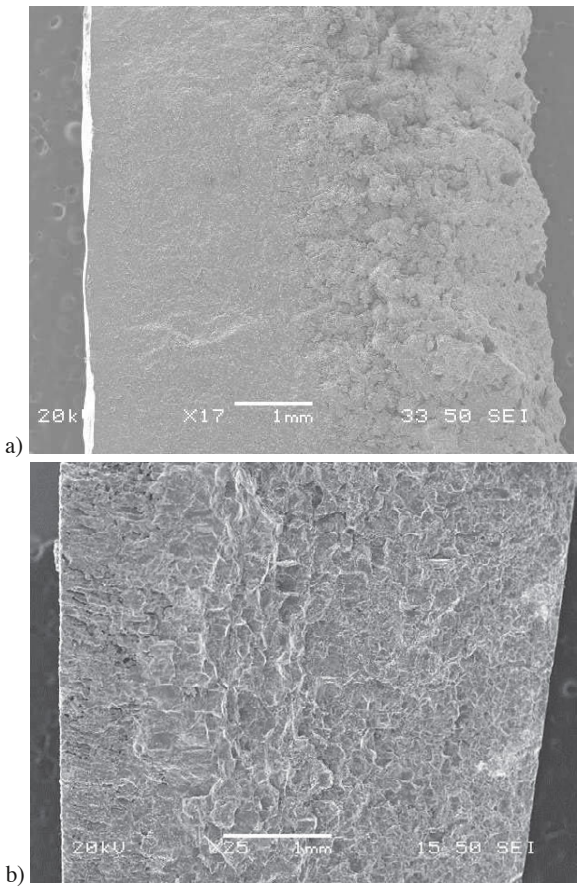
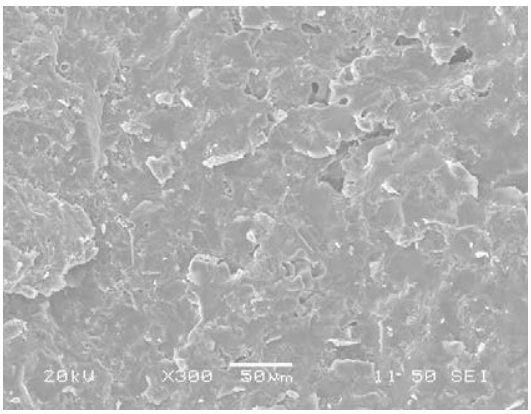


Figure 3. SEM images of the face of the frozen bath in the normal direction to the probe surface; a) samples obtained with transient probe and b) steady-state probe.

The morphology seems to be very different for the two samples. According to the expectations, the structure varies more significantly with distance from the probe face in the transient case as the cooling rate itself change more strongly during the whole solidification process. The first solidified layers of the transient sample (about the first 2.8mm on the left side of Figure 3a) – formed with very high cooling rate (and overcooling) – are relatively compact and they are harder for machining. This condition promotes the formation of the so-called micro-aphanitic structure [8] where a big amount of very small crystals are formed due to the high nucleation and low crystal growing rate. Higher magnification reveals more details about the same zone on Figure 4a.

On the other hand, the last layers formed at much lower cooling rate are strongly porous and quite brittle (Figures 3a and 4c). The transition between the two zones (2.8-4.7 mm) is even more porous (Figures 3a and 4b).

a)


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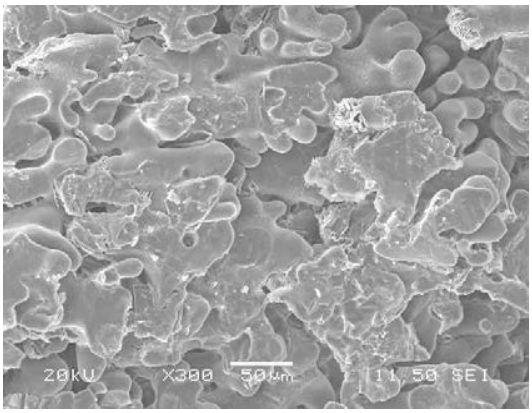

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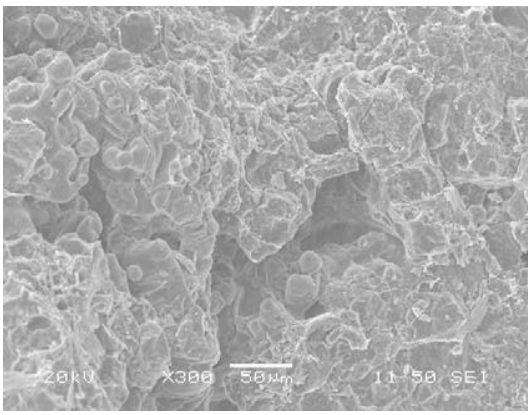


Figure 4. SEM images of the face of the frozen bath in the normal direction to the transient probe surface using 300x magnification, taken from the three main zones.

In those zones, situated farther from the probe (especially in the transition region), rounded shaped particles can be clearly observed, especially around the big pores. Such forms cannot be found in the first layers. Such shape is the indicator of ″crystal corrosion″ caused by a liquid phase. In fact, the cryolite crystal is formed first due to its highest melting point in the system. The remaining bath can be entrapped between the growing crystals if the rate of freezing is relatively fast, but not high enough to form micro-aphanitic structure [6,8]. The liquid entrapped in the pockets becomes more and more acidic with increasing AlF_3 excess and thus decreasing liquidus temperature. For this reason, the liquid can last for relatively long time between the growing crystals and etching them into a round shape.

Using even higher magnification (3000x), exolution (indicated by small acne-like dots on the rounded particule surfaces) can be observed, mainly in the transition zone (Figure 5). This indicates probably the recrystalisation of cryolite to chiolite in the presence of excess AlF_3 from the liquid around 750 °C [12].

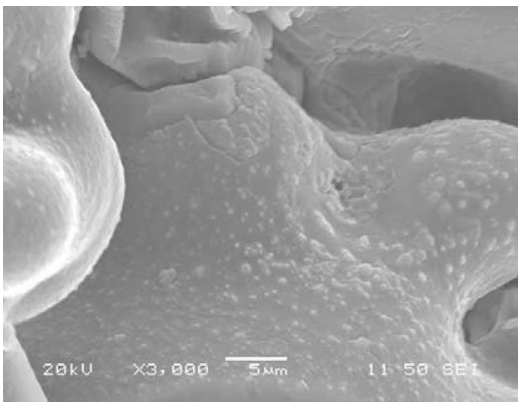
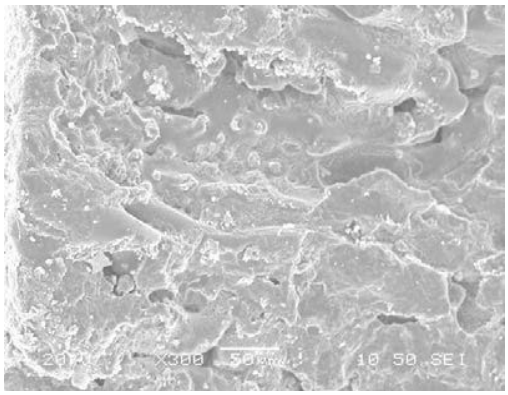


Figure 5, SEM image showing exolution on the surface of froozen bath particles, situated in the transition zone of a transient sample.

The samples obtained with steady-state probe seem to be much more homogenous (Figure 3b and 6a-b). Difference between the first (6a) and last (6b) layers is less significant. However, some orientation of the crystals can be observed in the first layers next to the solid surface of the probe (first 0.87 mm at the left side of Figure 3b). Contrary to the transient case where the initially very high cooling rate promotes homogenous nucleation, this indicates heterogeneous nucleation where the stainless steel wall served as nuclei for crystal growing. Exsolution can be observed in the same zone, close to the probe surface (Figure 6 and 7). The porosity is relatively low and crystals are easier identifiable with straighter boundaries, especially farther from the probe surface (Figure 3b and 8). A thin layer with similar structure could be observed on the transient sample, close to the surface exposed to the bath continuously. Such macro-crystalline texture can be formed when the solid is exposed to the bath for long time and the cooling rate is low [8]. This condition reduces nucleation rate and promotes crystal growing.

The knowledge of the order of magnitude of the heat fluxes and solidification rate during the formation of the different zones can help to explain the diversity of the observed structures. The variation of the heat flux was estimated with the above mentioned mathematical model using temperature histories and thermo-physical properties of the stainless steel and frozen bath as input data. Variation of the computed heat fluxes and solidification rates with crust thickness are shown by Figure 9. Both the heat flux and the solidification rate follow very similar patterns for the two probes. Their initial values in the transient case are very high but decrease fast in time, following a hyperbolic tendency. In the steady-state case, the initial values are almost zero in the absence of significant temperature gradients (there is only a very small gradient due to the heat loss through the support of the probe). After the cooling of the probe by air is launched, the heat flux is increasing gradually until a maximum, followed by a light decrease. This pattern is the result of the competing tendencies of the increasing thermal gradient versus thickening crust which is thermally isolating.

a)



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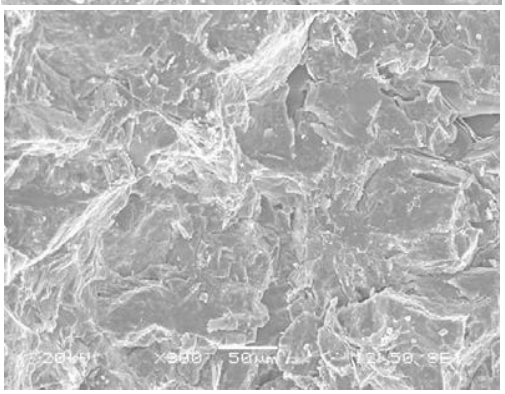


Figure 6. SEM images of the face of the frozen bath in the normal direction to the steady-state probe surface using 300x magnification.

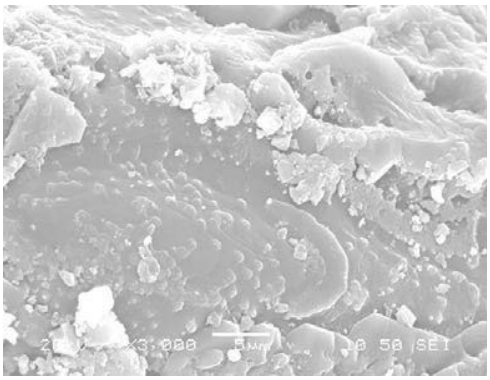


Figure 7. SEM images of the face of the frozen bath in the normal direction to the steady-state probe surface using 3000x magnification; zone close to the probe.

Red ellipsoids in Figure 9 are showing the thickness interval where significant "crystal corrosion" and exsolution were observed. They can be seen in both the transition layer of the transient samples and in some layers close to the probe in the steady state samples. At those places, the heat flux and the solidification rates are quite similar. One can conclude from the results that around 50-95 kW/m^2 with a solidification rate of 35-150 g/m^{2}s represent the favoring conditions for those phenomena.

Contrary to the liquid that can be renewed continuously when in contact with the bulk, the entrapped liquid can become very rich in AlF_3 as more pure cryolite is solidifying from it. This can permit the presence of liquid with very low liquidus temperature at the re-crystallisation of cryolite to chiolite at 750 °C, as explained above.

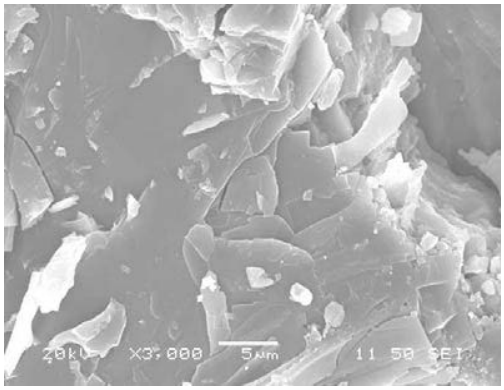
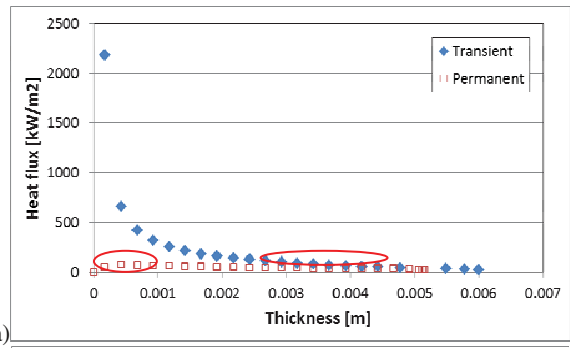


Figure 8. SEM images of the face of the frozen bath in the normal direction to the steady-state probe surface using 3000x magnification; zone farther from the probe.

a)



b)

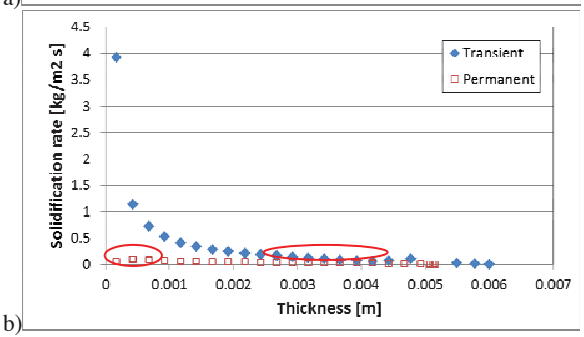


Figure 9. Variation of heat flux (a) and solidification rate (b) with crust thickness, computed with mathematical model [10].

The heat fluxes at the maximum crust thickness are identical for the two probes. This can be surprising at first view; however that can be easily explained by the fact that at this moment, the solidification rates are zero and the heat flux depends only on the heat transfer between the crust and the neighbouring warmer liquid (with same bulk temperature and liquid flow condition). The boundaries of the crystals at this position are the most regular and straight lined for both samples, indicating a slow formation of the crystals and the absence of liquid corrosion, but also a similar liquidus.

The typical values of heat flux passing through the side ledge in industrial cells lay around 5-10 kW/m^2. This range is lower than the values found in this study. However, they are closer to the conditions of formation of the crust around the steady-state probe.

On the other hand, the transient test corresponds more to the formation of the first layers of frozen bath around a new anode or any other cold objects inserted into the liquid during different operations.

The thermal diffusivity and conductivity of the samples were measured using flash method between 100 and 500 °C. The results correspond to the less conductive sideledge samples, recovered from post mortem industrial cells [5]. The thermal conductivity of the steady-state samples were about 20 % higher than that of the more porous transient samples.

Conclusions

Frozen cryolite based bath samples were produced with cold finger technique. Two different probes design were used to assure transient (strongly varying heat fluxes) and almost steady-state (slightly varying, low heat fluxes) conditions. Samples were cut from the solidified bath (crust) formed around the probes for image analysis and thermal conductivity measurement. A mathematical model was used to estimate the variation of the heat flux and solidification rate through the crust in the two cases in order to correlate them with the observed structures. The following conclusions can be drawn from the analysis of the results:

- The crust formed under close to steady-state condition became more homogenous than that produced with the transient probe. The latest was composed of three main zones: a more compact and hard layer close to the probe covered by a more porous and fragile outer layer. There was a particularly porous transition layer between the two main zones. A thin layer of well-formed big crystals with straight boundaries covered the crust.
- Liquid corrosion and exsolution can be observed in the porous layers, indicating the formation of chiolite around 750°C from cryolite in contact with an AlF_3 rich liquid
- In the crust formed around the steady-state probe, orientation of crystals can be observed in the first layer next to the probe, indicating heterogeneous nucleation. Some liquid corrosion and exolution can be identified here. The farthest layers of the crust from the same probe contain relatively big, well-formed crystals with straight boundaries.
- The heat flux through the crust, during its solidification around the transient probe, is estimated to decrease from 2200 to 28 kW/m^2, following an exponentially diminishing tendency. The heat flux through the crust formed under steady-state condition, increases very fast from 0 to 95 kW/m^2 and then decreases slowly to 28 kW/m^2. The variation of the solidification rate followed a very similar pattern.
- The heat fluxes measured at the outer shell of industrial cells are significantly lower than those met in this study. This fact suggests that in all cases, side-ledge is formed relatively slowly, even after an anode effect or anode frame lifting. That explains why the composition of the side-ledge differs from that of the bath.
- The thermal conductivity of the samples produced with cold finger technique in this study corresponds to the less conductive side ledge, recovered from post mortem industrial cells. The steady-state samples conducted about 20 % better the heat when compared to the transient ones. The main reason for this is probably the higher porosity at certain places.

Acknowledgment

This work could not be realized without the valuable financial support of *Rio Tinto Alcan* and the *National Sciences and Engineering Research Council of Canada* (NSERC). The authors would like to thank *Adam Ugron* for his contribution to the design of the steady-state (cooled) cold finger probe. The technical support of Mr. Zhan Zhang with the SEM is thanked as well for performing image analysis.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INVESTIGATION OF SODIUM SULFATE ADDITIONS INTO CRYOLITE-ALUMINA MELTS

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Keywords: sulfur, sodium sulfate, cryolite-alumina melt, aluminium electrolysis

Abstract

In the Hall-Héroult process, sulfur impurities may not only emit harmful gases but also reduce current efficiency. To better understand this process, the behaviour of sulfur compounds in a cryolite-alumina melt at 1253 K (980 °C) was investigated in a laboratory cell. Sodium sulfate (Na_2SO_4) was added into the molten bath as a sulfur source. Furnace off-gases were passed through a mass spectrometer for qualitative assessment. The stability of sodium sulfate in the cryolite melt was found to depend on the presence of carbon ($t_{1/2}$ = 116 min) and carbon-aluminium ($t_{1/2}$ = 29 min). It changed dramatically during electrolysis ($t_{1/2}$ = 5-8 min). Detected sulfurous gases included SO_2, COS, CS_2, and H_2S.

Introduction

Sulfur enters the Hall-Héroult cell from various sources, and it is therefore important to study the behaviour of sulfur compounds in alumina-cryolite melts. Even small amounts of sulfur impurities in the bath can impact the course of electrolysis [1]. The work presented in this paper includes experimental data obtained from a laboratory-scale cell. It provides insight into the fundamental behaviour of sulfur in the cryolite bath during electrolysis and has direct practical implications.

In aluminium electrolysis, the majority (80%) of sulfur enters the cell from carbon anodes. The remainder comes from alumina (20 %) and aluminium fluoride (1 %) [2]. Sulfur content of the anode ranges from 1-4 wt% depending on the type of coke utilized [3]. Anodes are usually prepared from a mix of calcined petroleum coke (or pitch coke) and recycled butts. These are bound together with coal tar or petroleum pitch [4]. Petroleum coke is the main source of sulfur in aluminium electrolysis cells [5]. Sulfur is an inevitable impurity in aluminium anodes. Petroleum coke inherits sulfur from crude oil, and minor amounts of sulfur come from the pitch, where sulfur is a part of the structure. Sulfur that originates from alumina and fluoride additions is typically in the form of sulfate [2].

Analysis of industrial bath samples revealed that the bath contains both sulfate and sulfide [6]. Burnakin et al. [7] noted that sulfate was moderately stable in cryolite melts without aluminium. Several other researchers [8-12] have also studied the stability and phase diagrams of sulfates in cryolite melts.

Sulfur does not accumulate in the bath. Instead, it is readily released as sulfurous gas emissions. Sulfur emissions from an aluminium smelter with an annual production of 200000 tonnes can be as high as 1000 tonnes of sulfur per year [13]. The most important detected sulfurous gases from aluminium reduction cells are SO_2, COS, H_2S, and CS_2 [5, 13-15]. Sulfurous gases are known for their detrimental effects on local air quality (particulate matter), acid rain, human health, and global climate. It has been reported that particulate matter is a key ingredient of polluted air and it has been estimated to kill more than 500000 people each year [16]. However, emission of sodium sulfide and conversion into sulfate is also argued to be an important factor in reducing global warming by backscattering solar radiation into space [17].

Earlier studies on current efficiency are contradictory. Bullough and Marshall [18] investigated the effect of sulfur on current efficiency using prebaked anodes (0.8-1.6 wt% S) in a 10 kA reduction cell and in small scale test equipment. The plant correlation of sulfur data with current efficiency recordings over a long period of time was also given. The reported current efficiency reduction was 5±2 % per each percent in sulfur increase. However, later Barillon and Pinon [19] concluded from their study that overall even a substantial increase in anode sulfur (1-3.5 wt%) has no effect on potline current efficiencies. Gilmore and Bullough [20] also confirmed that there was no sulfur effect, either on current efficiency or metal quality, from industrial and laboratory tests. Recent work that motivated our investigation was performed by Pietrzyk and Thonstad [21], who performed laboratory studies on anodes containing sulfur in the range of 0 to 3.8 wt% and claimed that sulfur has negative effects on current efficiency.

Our work investigates the stability of sodium sulfate in cryolite-alumina melts under various conditions. The study suggests that sodium sulfate is not a stable compound in the melts even though many phase diagrams exist. Our results can be used to study the effect of sulfur on current efficiency, an important process parameter in aluminium electrolysis. They also suggest a possible mechanism that can contribute to current efficiency loss.

Experimental

The bath mixture of the cryolite-alumina melt was composed of AlF_3 (25.2 g), Na_3AlF_6 (319 g), CaF_2 (18.92 g), and Al_2O_3 (15.13 g). This corresponds to a cryolite ratio of 2.5 with excess amounts of AlF_3, Al_2O_3, and CaF_2 of 6.75 wt%, 4 wt% and 5 wt%, respectively. The crucible containing all the above components was kept overnight inside an oven at 473 K (200 °C). Samples were melted in a gas-tight furnace at 1253 K (980 °C). Argon was passed through the oven to maintain inert conditions. Reaction vessels were made of sintered alumina and graphite. Pieces of graphite with a total surface area of 12 cm^2 were utilized as a carbon source. The total weight of aluminium was 12.9 g. For the electrolysis experiments special cell set-up was used. The diagram of this laboratory cell is given in Figure 1. The graphite anode was lowered 2 cm into the melt prior to the start of the experiment, giving a total submerged surface area of 35.5 cm^2. Sodium sulfate was added to the bath as a source of sulfur. Each experiment started with 600-800 ppm of sulfur in the bath. These

values are common in industrial cells depending on the level of contamination of the bath with carbon dust [1].

Samples were taken from the bath every 5 min over a course of 30 min. Following that, samples were taken every 10 min for another 30 min after which the sampling frequency was further reduced to 20 min for 60 min. Total amount of sulfur in the bath was measured by ICP-MS (Inductive Coupled Plasma - Mass Spectrometry).

Furnace off-gases were passed through a process mass spectrometer (MS)-Gaslab300 for qualitative assessment. The gas samples were extracted through a heat-traced capillary at 473 K (200°C). Due to a low internal volume of the sample line and the instrument itself, response times were generally low with 90% of expected response in less than 30 seconds. Electron impact ionization cross sections for detected molecules were taken from the NIST Webbook [22] and are given in Table I. These numbers give the probabilities of ionization, and they provide a basis for qualitative comparison between the molecules (e.g. at equal intensities, COS and SO_2 are approximately equal, CS_2 is higher than COS when the CS_2 to COS factor is 7.970/4.925). The mass spectrometer was the only available gas analyser during the time of the experiments. The gas analysis results were only used to understand the types of gas products formed during the reactions.

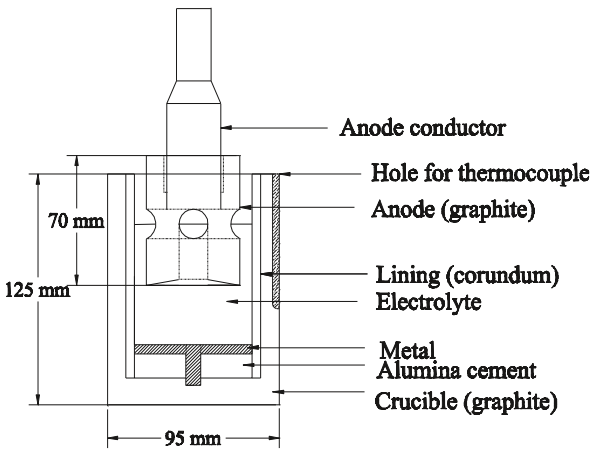

Figure 1. Schematic of the laboratory cell.

Table I. Electron impact ionization cross sections for detected molecules [22].

Constituent	Ionization energy (eV)	Cross section (Å^2)
SO_2	70	4.992
COS	70	4.925
CS_2	70	7.97
CO	70	2.516
CO_2	70	3.521
H_2S	70	4.125
S_2	70	7.927
CS	70	4.548

Results and Discussion

A summary of the results with overall experimental responses are shown in Table II.

Table II. Half-life of sodium sulfate at different conditions.

Carbon	Aluminium	Electrolysis	$T_{1/2}$ (min)
-	-	-	231
+	-	-	116
+	+	-	24
+	+	+	6.9

The half-life was calculated by assuming exponential decay according to the following equation:

$$C_S(t) = C_{S(0)}e^{-kt} \quad (1)$$

where $C_{S(0)}$ is the initial concentration of sulfur at time zero, C_S is the concentration of sulfur remaining after time t, and k is the decay constant. The half-life (the time when sulfur concentration is half of the initial amount) is therefore calculated as:

$$\frac{1}{2}C_{s(0)} = C_{S(0)}e^{-kt} \quad (2)$$

$$t_{1/2} = \frac{\ln 2}{k} \quad (3)$$

where $t_{1/2}$ is the half-life of the sulfur.

Figures 2 and 3 show bath sample and gas analyses under different conditions in the absence of electrolysis. The results (Figure 2 and Table II) indicate that sodium sulfate is relatively stable in cryolite-alumina melts when no electrolysis is taking place and in the absence of other reducing agents, e.g. carbon and aluminium. The total sulfur half-life is equal to 231 min. Gas analysis showed very limited release of gases when no reducing agents were present (Figure 3). The presence of gas signals can be attributed to possible thermal decomposition of Na_2SO_4 even though the Gibbs free energy is positive:

$$Na_2SO_4 = Na_2O + SO_3 \quad (4)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = 381 \text{ kJ/mol}$$

SO_3 is not stable at 1253 K (980 °C) and decomposes into SO_2 and O_2 ($\Delta_rG°$ = -18 kJ/mol). Even though the reaction vessel was made of sintered alumina, the vessel holder was made of graphite. SO_2 gas could have reacted with carbon outside the reaction vessel to produce COS or CS_2 gases:

$$SO_2 + 2C = COS + CO \quad (5)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -171 \text{ kJ/mol}$$

$$2SO_2 + 3C = CS_2 + 2CO_2 \quad (6)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -270 \text{ kJ/mol}$$

H_2S can be formed by hydrolysis due to the presence of moisture in sodium sulfate or the sampling device.

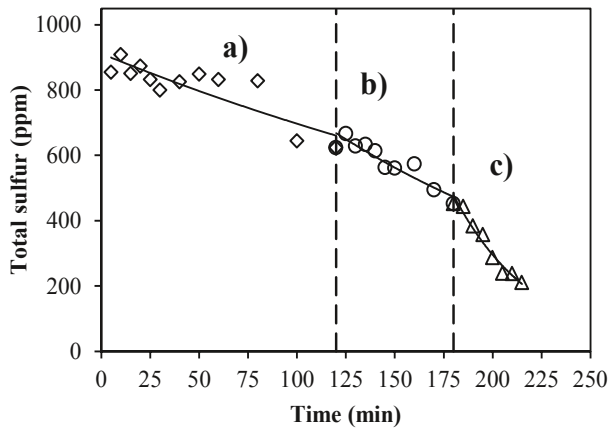

Figure 2. a) Total sulfur concentration as a function of time. Vertical lines mark a change in test conditions: a) cryolite-alumina melt without carbon and aluminium, b) carbon added into the cryolite-alumina melt c) melt contains both carbon and aluminium.

As can be seen from Table I, the half-life of sodium sulfate is reduced by half in the presence of carbon in the bath. The following reactions between sodium sulfate and carbon are possible [7, 23-24]:

$$3Na_2SO_4 + 12C = 3Na_2S + 12CO \quad (7)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -877 \text{ kJ/mol}$$

$$3Na_2SO_4 + 3C = 3Na_2S + 6CO_2 \quad (8)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -581 \text{ kJ/mol}$$

$$3Na_2SO_4 + 2Na_3AlF_6 + 3C = \quad (9)$$
$$12NaF + Al_2O_3 + 3SO_2 + 3CO$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -209 \text{ kJ/mol}$$

$$3Na_2SO_4 + 2Na_3AlF_6 + 3/2C = \quad (10)$$
$$12NaF + 3/2CO_2 + Al_2O_3 + 3SO_2$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -135 \text{ kJ/mol}$$

where $\Delta_rG°$ is the standard Gibbs energy of the reaction. SO_2 formation predicted by Equations 9 and 10 was confirmed by gas analysis (Figure 3b). SO_2 can react with carbon and form CS_2 or COS according to Equations 5 and 6. However, CS_2 can also be formed directly:

$$3Na_2SO_4 + 21/2C + 2AlF_3 = \quad (11)$$
$$3/2CS_2 + Al_2O_3 + 6NaF + 9CO$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -989 \text{ kJ/mol}$$

$$2Na_2SO_4 + 7C = CS_2 + 2Na_2O + 6CO \quad (12)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -89 \text{ kJ/mol}$$

CS_2 can be further oxidized:

$$CS_2 + 3O_2 = CO_2 + 2SO_2 \quad (13)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -918 \text{ kJ/mol}$$

$$CS_2 + CO_2 = 2COS \quad (14)$$
$$\Delta rG° \text{ at } 1253 \text{ K} = -22 \text{ kJ/mol}$$

The reaction in Equation 13 is highly unlikely, however, as experiments were run under inert conditions and the cell was opened only for short periods of time for bath sampling. Besides, O_2 should preferentially react with C to form CO or CO_2. COS may decompose into:

$$2COS = 2CO+S_2 \quad (15)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -8 \text{ kJ/mol}$$

When SO_2 and CO_2/CO are present in equilibrium in the gas phase, the following chemical reaction will take place:

$$SO_2+3CO = COS+2CO_2 \quad (16)$$
$$\Delta rG° \text{ at } 1253 \text{ K} = -72 \text{ kJ/mol}$$

It is difficult to assess the extent to which Equation 16 takes place because the thermodynamics discussed here are at 1253 K (980 °C), whereas the gas temperature in the analyser is 473 K (200 °C). Another shortcoming in the gas analysis is the fact that CO was not recorded. CO measurements were not possible with the utilized MS setup due to signal interference from both CO_2 and N_2.

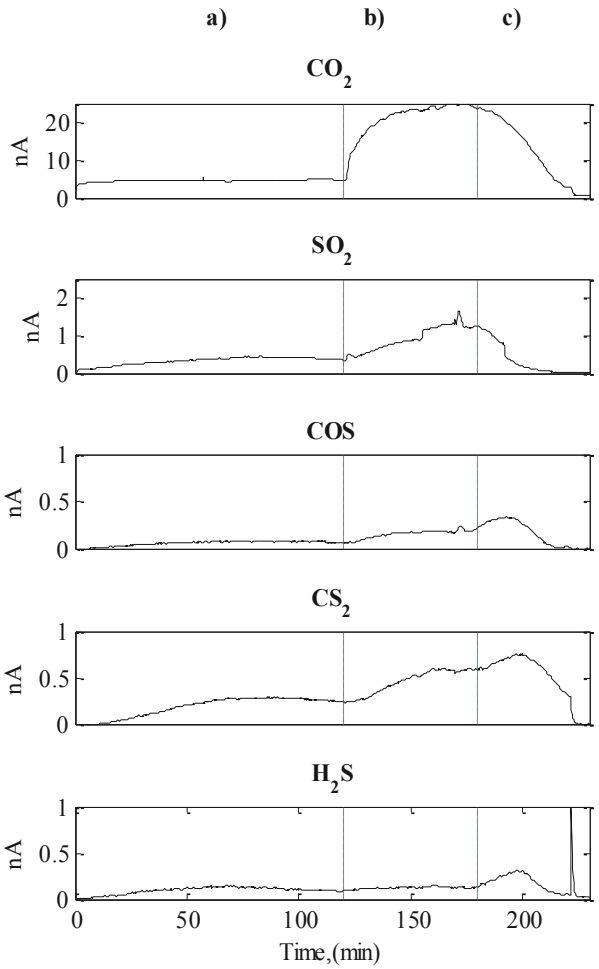

Figure 3. Gas analysis under different conditions. Vertical lines mark a change in test conditions: **a)** cryolite-alumina melt without carbon and aluminium, **b)** carbon added into the cryolite-alumina melt **c)** melt contains both carbon and aluminium.

The half-life of sodium sulfate is reduced to one tenth in the presence of both aluminium and carbon as opposed to when these two entities are absent (Table II). Possible chemical reactions between sodium sulfate and aluminium are [7, 23-24]:

$$3Na_2SO_4 + 8Al = 4Al_2O_3 + 3Na_2S \quad (17)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -3315 \text{ kJ/mol}$$

$$3Na_2SO_4 + 8Al = 3Na_2O + 3Al_2O_3 + Al_2S_3 \quad (18)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -2431 \text{ kJ/mol}$$

$$3Na_2SO_4 + 2AlF_3 + 8Al = 6NaF + 4Al_2O_3 + Al_2S_3 \quad (19)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -3286 \text{ kJ/mol}$$

$$3Na_2SO_4 + 2Na_3AlF_6 + 8Al = 12\ NaF + 4\ Al_2O_3 + Al_2S_3 \quad (20)$$
$$\Delta_rG° \text{ at } 1253 \text{ K} = -3059 \text{ kJ/mol}$$

Ambrova et al. [24] reported formation of Al_2S_3 and Na_2S (identified by XRD) in a solidified bath in the presence of aluminium when a graphite crucible was used. However, none of these reactions fit the rapid release of SO_2 and the fast decay of sulfur concentration from the bath (Figures 2c and 3c).

As conditions become more reducing, a shift from SO_2 to COS and CS_2 is observed (Figure 3c). Increased reducing conditions could also explain the reduction in CO_2 production, which favours more CS_2 and COS formation. However, more extensive gas analysis is required to understand the full picture.

The sharp increase in H_2S at the end of the experiment was observed when the stainless steel rod was immersed all the way to the bottom of the crucible to obtain a large amount of sample from the bath.

A substantial decrease (33-fold in sodium sulfate half-life is observed during electrolysis (Table II). Intensity of COS was observed to be 4.4 times higher than in other experiments [25]. The concentrations of other measured gases were in the same range as in the experiments without electrolysis.

Sulfur concentration as a function of time is given in Figure 4.

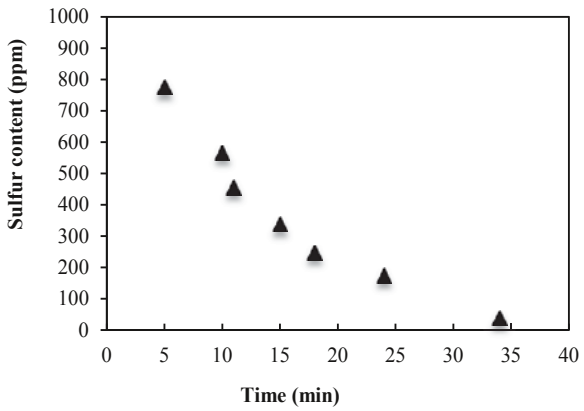

Figure 4. Sulfur concentration as a function of time.

Based on Figure 4, the following relationship for the sulfur decay from 800 ppm can be written:

$$t_{1/2} = \frac{\text{Ln}2}{0.101} = 6.9 \text{ min} \quad (21)$$

where C_s is the concentration of sulfur, t is the time of sulfur addition, and $t_{1/2}$ is the half-life time.

The half-life of sulfur content was found to be 6.9 minutes, which is in accordance with the previous electrolysis tests (5 and 8 minutes) [1].

The fast depletion of sodium sulfate can be attributed to either a chemical or an electrochemical reaction, or possibly both simultaneously. Sodium sulfate can be reduced to sulfides by both carbon and aluminium according to Equations 7 and 8 and Equations 17 to 20.

Both Burnakin et al. [7] and Ambrova et al. [26] reported increased loss of aluminium in the presence of aluminium and carbon. Sulfides can be oxidized electrochemically at the anode according to the following reaction [27]:

$$S^{2-} = S + 2e^- \quad (22)$$

Subsequent reactions resulting in the formation of CS_2 and S_2 were proposed by the same authors [27]:

$$S = S_{(ad)} \quad (23)$$

followed by

$$S_{(ad)} = S_{2(ad)} \quad (24)$$

and/or

$$S_{(ad)} + xC = C_xS_{(ad)} \quad (25)$$

$$2C_xS = CS_{2(ad)} + (2x-1)C \quad (26)$$

Baimakov and Vetyukov [24] also hypothesized that sulfides may be oxidized at the anode to SO_2:

$$S^{2-} + 2e^- = S \quad (27)$$

$$S + 2CO_2 = SO_2 + 2CO \quad (28)$$

Other successive electrochemical reactions were proposed by Hajasova [12] based on electrochemical studies of sodium sulfate:

$$2SO_4^{2-} + 3e^- = S_2O_4^{2-} + 4O^{2-} \quad (29)$$

$$S_2O_4^{2-} + 2e^- = S_2O + 3O^{2-} \quad (30)$$

In our work, further experiments were performed to maintain the sulfur concentration in the bath constant. Sulfur addition was made every 10 minutes. In order to check if the assumption (feeding half of the initial amount of sulfur every 10 min to compensate for the sulfur loss) was accurate, samples were taken 5 and 10 minutes after the addition of sodium sulfate tablets. Samples taken 5 minutes after the addition should give an indication of the maximum concentration after addition, and the 10-minute samples should show the lowest concentration. Analysis of these two samples indicates whether sulfur overfeeding or underfeeding is occurring. Bath samples taken prior to the 5-minute mark after addition contained undissolved tablets and the concentrations obtained from the ICP analyses were thus higher than expected (e.g. 5000 ppm). Sulfur

measurements taken within the first 5 minutes after tablet addition are therefore not representative of the average sulfur concentration in the bath. Further sampling was done every 15 minutes at the same time as the alumina additions.

Figures 5 and 6 show results of experiments where the target levels of sulfur were 1000 ppm and 500 ppm, respectively. It can be concluded that the assumption works reasonably well. If the assumed total time was overestimated, underfeeding would have been observed in Figures 5 and 6. Conversely, overfeeding would have been seen in the case of an underestimated half-life.

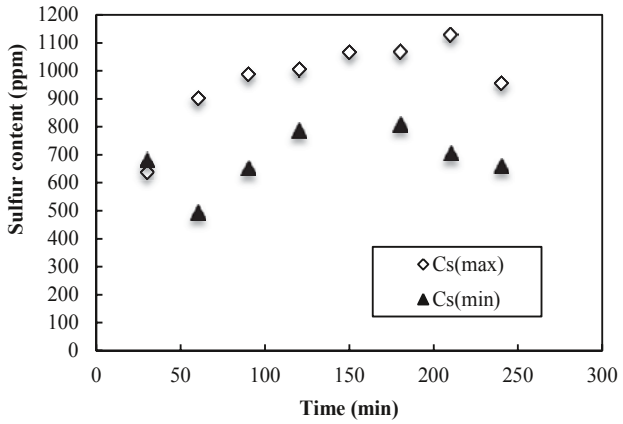

Figure 5. Sulfur concentration in the bath analyzed by ICP analysis (1000 ppm target). C_s(max) are the values sampled at 5 minutes post-addition, and C_s(min) are the values obtained from the 10-minute samples.

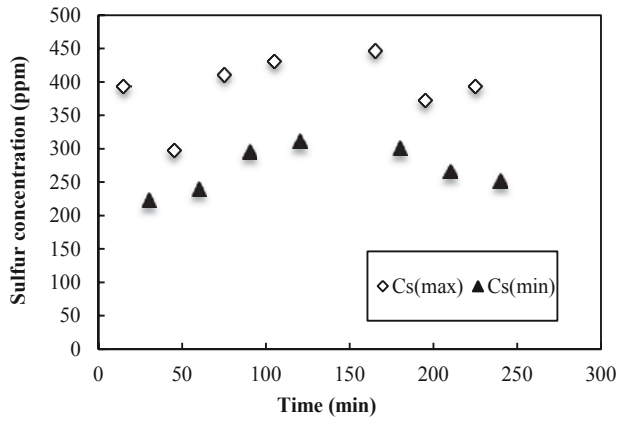

Figure 6. Sulfur concentration in the bath analyzed by ICP analysis (500 ppm target). C_s(max) are the values sampled at 5 minutes post-addition, and C_s(min) are the values obtained from the 10-minute samples.

The effect of sulfur on current efficiency can be studied by either using anodes with different sulfur contents or by adding sodium sulfate directly into the bath and supplementing for its depletion (as utilized here). In order to investigate the sulfur effect on current efficiency, sodium sulfate concentration in the cryolite must remain stable. Industrial cells have a steady supply of sulfur from anodes. Using anodes with different sulfur contents will probably make controlling sulfur concentration easier, but it also has its own disadvantages. High sulfur content in anodes is correlated with other trace impurities [28]. Moreover, anodes with different sulfur levels differ not only by structure, but also in their wetting properties and electrochemical behaviour [29, 30]. In order to isolate the above effects, introducing sulfur directly into the system is an attractive alternative, but both approaches can be helpful in determining the overall effect of sulfur on current efficiency.

Conclusion

Our results indicate that the stability of sodium sulfate in cryolite melts is a function of the presence of carbon ($t_{1/2}$ = 116 min) and carbon-aluminium ($t_{1/2}$ = 29 min). The stability changed dramatically during electrolysis ($t_{1/2}$ = 5-8 min) as compared to $t_{1/2}$ = 231 min in the absence of electrolysis, carbon, or aluminium. It was clear that electrolysis had the strongest influence on the depletion rate. Sulfur removal from the bath increased during electrolysis most likely due to increased stirring and enhanced mass transport as well as gaseous sulfur compounds being produced at the electrodes. Sulfur additions were made after 10 minutes based on the depletion curve and half-life of sodium sulfate in the cryolite-alumina melt during electrolysis. The results are important for obtaining accurate measurements of sulfur concentration in the electrolyte and understanding the value of concentration verification by frequent sampling.

Acknowledgements

The authors would like to acknowledge the financial support from Alcoa Fjarðaál, HRV Engineering, SINTEF and NTNU. Authors are indebted to Dr. Ole Kjos for his assistance with MS Spectrometry. Authors would also like to thank Aksel Alstad for his punctuality with the fabrication of experimental parts.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

POLYVALENT IMPURITIES AND CURRENT EFFICIENCY IN ALUMINIUM CELLS: A MODEL CONCERNING ELECTROCHEMICAL SHORT CIRCUITING

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Keywords: Polyvalent impurities; Current Efficiency; Diffusion; Modelling

Abstract

"Electrochemical short circuiting" describes the loss in current efficiency (CE) due to polyvalent impurity elements that are reduced at the cathode and re-oxidized at the anode. In aluminium cells; the oxidized form of the impurity reacts with dissolved metal near the metal-electrolyte interface, leading to a steeper concentration gradient of dissolved metal. The phenomenon was studied in a model based on the Stefan-Maxwell equations for diffusion in multi component systems and a description of the turbulent diffusion coefficient close to the cathode. The results indicate that the main factors for the slope of CE *vs*. impurity concentration are the CE in absence of impurities and the change in oxidation state during the reduction, while the diffusion coefficient of the impurity is of minor importance. The numerical results could be represented by an analytical expression that fits well with experimental data.

Introduction

It is well known that some impurities present in the Hall-Héroult electrolyte are harmful for the current efficiency (CE). Phosphorus appears to be the worst element per weight, and also the one that has been most extensively studied. The CE loss in laboratory cells has been reported to be between 0.67 and 1.0 percent per 100 ppm phosphorus present in the electrolyte [1-6]. Although the values are often given in percent per 100 ppm or similar; the experimental data generally do not show straight lines, as the slopes decrease with increasing concentration.

The mechanism behind the CE loss appears to be "electrochemical short circuiting". This phenomenon will take place if the impurity element has (at least) two oxidation states and an "electrochemical window" where i) the oxidized state can be reduced by reaction with dissolved aluminium near the cathode, and ii) the reduced state can be oxidized by contact with carbon dioxide or the anode. In this situation, the element may circulate between the cathode and the anode for a long time before it leaves the electrolyte with the gas or with the metal.

The main mechanism for the extra loss in CE is that there will be a reaction zone close to the cathode, where the concentrations of dissolved aluminium and the oxidized state of the impurity element both are zero. Thus, the presence of the impurity element leads to a steeper diffusion gradient for the dissolved aluminium. Principal sketches illustrating the concentration gradients were published by Sterten *et al*. [3].

Although the main principle of electrochemical short circuiting is quite simple and well understood, quantitative modelling of this phenomenon is not straightforward. The challenge is related to the fact that the reduction at the cathode does not take place at the cathode itself (the metal-bath interface), but within the boundary layer close to the cathode. Therefore, the variation in the turbulent diffusion coefficient close to the cathode must be taken into consideration. Furthermore; it must be realized that the transport takes place in a multicomponent system. To the author's knowledge, a quantitative model concerning electrochemical short circuiting has not been published earlier.

Theory

Diffusion in a Multicomponent System

Diffusional transport in multicomponent systems can only be accurately described by the Stefan-Maxwell equations (*e.g*., see Bird *et al.* [7]),

$$\nabla x_i = \sum_{j=1}^{n} \frac{1}{c_{tot} D_{ij}} \cdot \left(x_i J_j - x_j J_i \right) \qquad (1)$$

where x_i is the molar fraction of component i, c_{tot} is the total molar concentration [$molm^{-3}$], D_{ij} (= D_{ji}) is the chemical diffusion coefficient for the binary system i-j [m^2s^{-1}], and J is the molar flux [$molm^{-2}s^{-1}$].

Mass transport depends exclusively on convection in the bulk of the electrolyte. When approaching the cathode, the convection decreases gradually and finally becomes zero at the cathode, at which point the transport depends only on chemical diffusion. Following the treatment in previous papers [8, 9], this can be described by using turbulent diffusion coefficients,

$$D_{tot} = D_c + D_t \qquad (2)$$

where D_{tot} is the total diffusion coefficient, D_c is the chemical diffusion coefficient, and D_t is the turbulent diffusion coefficient. The turbulent diffusion coefficient is zero at the cathode and increases with increasing distance from the cathode (y). As can be justified by the work by Kim *et al*. [10] the following holds close to the cathode,

$$D_t = \alpha \cdot y^3 \qquad (3)$$

where α is a factor [$m^{-1}s^{-1}$] that depends only on the degree of convection (*i.e*, a function of the Reynolds number). By integration of Equation 3 it can be shown that the mass transfer coefficient in the absence of a global flux can be calculated by

$$k_0 = 0.827 \cdot \alpha^{1/3} \cdot D_c^{\,2/3} \qquad (4)$$

The concept described by Equations 2 and 3 is illustrated in Figure 1, which shows the ratio between the chemical diffusion coefficient and the total diffusion coefficient as a function of the distance from the cathode. Close to the cathode the transport takes place solely by diffusion, while convection completely dominates

far from the cathode. Since the flux of a species is constant, while the total diffusion coefficient increases with the distance from the interface; the slope of the concentration gradient decreases with increasing distance (right hand scale in Figure 1).

The thickness of the diffusion layer (δ) can be derived by extrapolating the slope of the concentration curve at zero distance up to the point where the dimensionless concentration is unity (Figure 1). When doing calculations, it is implicitly assumed that there is no convection for $y < \delta$ and "full" convection for $y > \delta$. While the concept is easy to understand; it is also simplistic, and it does not permit detailed studies of transport close to an interface.

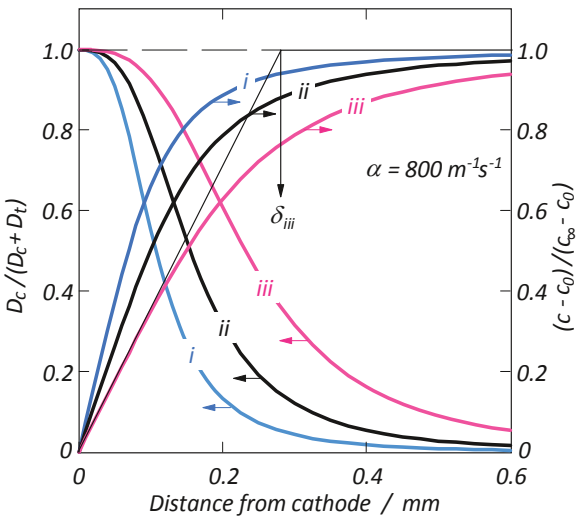

Figure 1. Ratio between the chemical diffusion coefficient and the total diffusion coefficient (left hand scale) and dimensionless concentrations (right hand scale) as a function of the distance from the cathode. i) $D_c = 1 \cdot 10^{-9}$ m^2s^{-1}, ii) $D_c = 3 \cdot 10^{-9}$ m^2s^{-1}, iii) $D_c = 1 \cdot 10^{-8}$ m^2s^{-1}. The thickness of the diffusion layer (δ) is indicated for $D_c = 1 \cdot 10^{-8}$ m^2s^{-1} (see text).

Numerical Solution

Diffusion Equation

To limit the number of components, only the following five substances were taken into consideration; 1) NaF, 2) AlF_3, 3) dissolved aluminium (Al_{diss}), 4) impurity element in its reduced state (i_{red}), and 5) in its oxidized state (i_{ox}). The electrolyte also contains calcium fluoride and alumina, but those substances play "passive" roles inasmuch as they do not take part in reactions at the cathode. For numerical solution in a system with five components, Equation 4 turns into five equations on the form

$$\Delta x_1 = \frac{\Delta y}{\bar{c}_{tot}} \cdot \left(\frac{\bar{x}_1 J_2 - \bar{x}_2 J_1}{D_{12} + \bar{D}_t} + \frac{\bar{x}_1 J_3 - \bar{x}_3 J_1}{D_{13} + \bar{D}_t} + \frac{\bar{x}_1 J_4 - \bar{x}_4 J_1}{D_{14} + \bar{D}_t} + \frac{\bar{x}_1 J_5 - \bar{x}_5 J_1}{D_{15} + \bar{D}_t} \right) \quad (5)$$

where "Δ" represents the difference between neighbouring elements and overlining represents the average of neighbouring elements. The terms within the parentheses in the five equations cancel, which means that the sum of changes in molar fractions always becomes zero.

The calculation domain, *i.e.*, the range between the cathode and 5 mm into the electrolyte, was divided into elements with dimension $\Delta y = 0.0075$ mm. With given molar fluxes and diffusion coefficients, the numerical solution involved using "old" molar fractions to calculate "new" molar fractions by Equation (5), and the latter are again used as "old" molar fractions in a new iteration. The procedure converges after less than ten iterations.

Fluxes and Boundary Conditions

The calculation domain was subdivided into two regions; i) the inner region close to the cathode (inside the reaction zone), where NaF, AlF_3, Al_{diss}, and i_{red} co-exist, and ii) the outer region (outside the reaction zone) where NaF, AlF_3, and i_{ox} co-exist. The following reaction is assumed to take place at the reaction zone,

$$Al_{diss} + \frac{3}{\Delta z} i_{ox} = Al^{3+} + \frac{3}{\Delta z} i_{red} \quad (6)$$

where Δz is the change in oxidation state (*e.g.*, $\Delta z = 2$ if P^{5+} is reduced to P^{3+}). Exactly at the reaction plane, the concentration of Al_{diss} as well as i_{ox} is assumed to be zero.

The net molar flux of NaF (N_1) is zero everywhere. Taking the vector pointing away from the cathode as the positive direction, the molar flux of AlF_3 is constant in every position and given by

$$N_2 = -\frac{i \cdot CE/100}{3F} \quad [mols^{-1}m^{-2}] \quad (7)$$

where i is the current density [Am^{-2}] and F is Faraday's constant [96 485 $Cmol^{-1}$]. The flux of dissolved metal (counted as Al_{diss}) in the inner region is

$$N_3(inner) = \frac{i \cdot (1 - CE/100)}{3F} \quad [mols^{-1}m^{-2}] \quad (8)$$

The flux of Al_{diss} was, of course, set to zero in the outer region since it assumed not to exist outside the reaction plane. For i_{red},

$$N_4(outer) = N_2 \cdot \frac{3}{\Delta z} \quad (9)$$

Finally, for i_{ox},

$$N_5(outer) = -N_4(outer) + N_4(inner) \quad (10)$$

N_4(inner) was set to zero in most cases, which will be discussed below.

The electrolyte 5 mm away from the cathode was assumed to have bulk properties. To start a calculation, the reaction zone was set at a specified distance from the cathode, at which the concentration of Al_{diss} was set to zero. The molar fraction of i_{ox} in the bulk and the CE were adjusted systematically until the concentrations i_{ox} was zero at the reaction point and until the concentration of Al_{diss} at the cathode was equal to the calculated value for the electrolyte composition at the cathode (Equation 11 below).

Input Data. Cases Considered

The concentration of dissolved metal at the cathode was calculated from the equation proposed by Ødegård *et al.* [11],

$$[Al_{diss}] = -0.31759 + 0.01849 \cdot r + 3.44 \cdot 10^{-4} \cdot t - 0.00407[CaF_2] + 9.6 \cdot 10^{-5}[CaF_2]^2 + \frac{0.02285}{3 + [Al_2O_3]^2} \quad (11)$$

where the brackets denote concentration in weight percent, r is the NaF/AlF_3 molar ratio, and t is the temperature in ^oC. The metal solubility was calculated using the NaF/AlF_3 molar ratio computed at the cathode, and also with calcium fluoride and alumina present, although those substances were not taken into account in the diffusion model. The molar fractions were calculated from the total concentration and the density [12], but taking only sodium fluoride and aluminium fluoride into account.

Chemical diffusion coefficients in cryolitic melts are generally unknown. The diffusion coefficient for alumina is about $1.5 \cdot 10^{-9}$ m^2s^{-1} [13], and since the electrolyte species are generally lighter and less voluminous than the dissolved alumina species, it can be expected that the diffusion coefficients are higher than for binary pairs including alumina. The diffusion coefficient for Al_{diss} (which does not exist as such) is relatively high due to the "electronic" nature of dissolved metal. If not otherwise stated, it was assumed that the diffusion coefficient for binary pairs not comprising dissolved metal was $3 \cdot 10^{-9}$ m^2s^{-1}. The diffusion coefficients for binary pairs including dissolved metal was assumed to be $1.5 \cdot 10^{-8}$ m^2s^{-1}. The mentioned diffusion coefficients produced reasonable values for the CE as well as for the cathodic overvoltage, but still; the data used only represents an educated guess.

Three base cases were considered, as shown in Table I. Case I represents a modern cell with high current efficiency. Case II represents the laboratory cell used in much of the work mentioned above [3-6], while Case III represents old cell technology with high molar ratio electrolyte and modest performance in terms of productivity and current efficiency.

Table I. Input data for three cases. The cathodic concentration overvoltage was calculated from the molar ratios in the bulk and at the cathode. CE_0 is the calculated current efficiency in absence of impurities. k_0 is the mass transfer coefficient for AlF_3 (see Equation 4).

	Case I	Case II	Case III
NaF/AlF_3 molar ratio (bulk)	2.2	2.5	2.7
CaF_2 [wt%]	5	5	5
Al_2O_3 [wt%]	3	5	5
Temperature [^oC]	960	980	1000
Current density [Am^{-2}]	9000	8500	7000
$k_0 \cdot 10^5$ [ms^{-1}]	1.27	1.62	2.17
NaF/AlF_3 molar ratio (cathode)	2.70	2.93	2.97
Cath. conc. overvoltage [mV]	85	61	35
CE_0 [%]	95.74	93.00	87.12

Concentration Gradients and Current Efficiency

Concentration Gradients

The concentration gradients for NaF, AlF_3, and Al_{diss} in the three base cases without impurities present are shown in Figure 2. As can be observed, Case III has the steepest gradient of dissolved metal due to the higher mass transfer coefficients in this case, which is the main reason for the relatively low current efficiency.

The boundary layer thickness (defined as $\delta = D/k$) is proportional with $D^{1/3}$ (see Equation 4 and Figure 1), which is reflected in the fact that standard equations for calculating mass transfer coefficients often contains the dimensionless Schmidt number raised to a power of 1/3.

Three examples of concentration gradients with impurities present are shown in Figure 3. Increased concentration of the impurities forces the reaction zone closer to the cathode, giving a steeper concentration gradient of dissolved metal. It should also be noted that although the impurity concentrations shown in Figure 3 are one order of magnitude smaller than the concentration of Al_{diss}, the flux of i_{red} towards the bulk is the same as the flux of Al_{diss}. This is possible because the total diffusion coefficient increases continuously with the distance from the cathode. In the current treatment, the turbulent diffusion coefficient dominates at distances larger than about 0.1 mm from the cathode (Figure 1).

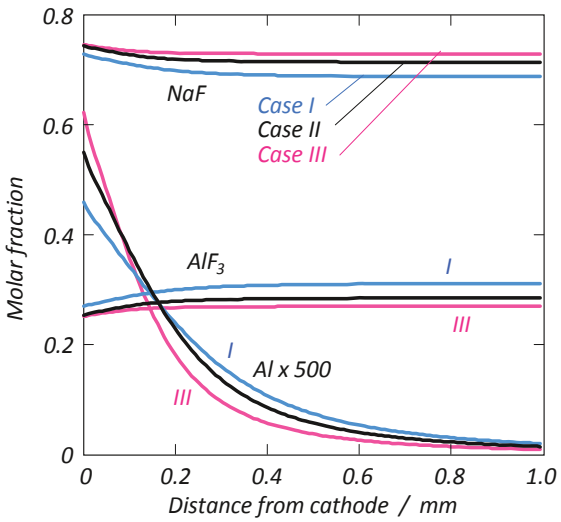

Figure 2. Calculated concentration gradients for NaF, AlF_3, and Al_{diss} for the three base cases.

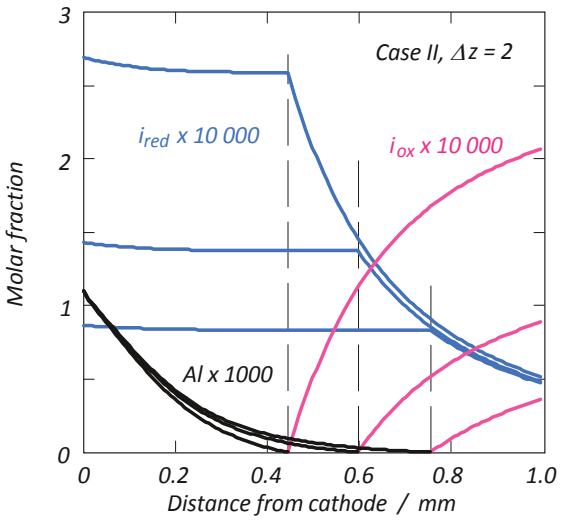

Figure 3. Three sets of concentration gradients for dissolved metal and impurity element in oxidized and reduced form, calculated for Case II with $\Delta z = 2$.

The flux of reduced impurity from the reaction plane, N_4(inner), was set to zero. In that case, the concentration of i_{red} is somewhat higher at the cathode than at the reaction plane, because i_{red} is drifting with the global flow towards the cathode (corresponding to the rate of metal production).

Effect of Chemical Diffusion Coefficient

Impurities have different diffusion coefficients, depending on the molar weight and the way they are complexed in the electrolyte. The effect on the CE of variable diffusion coefficients for the binary pairs including impurity element is illustrated in Figure 4. The effect of the diffusion coefficient is generally small, at least for small and moderate concentrations, which reflects the fact that the transport is dominated by the turbulent part of the total diffusion coefficient.

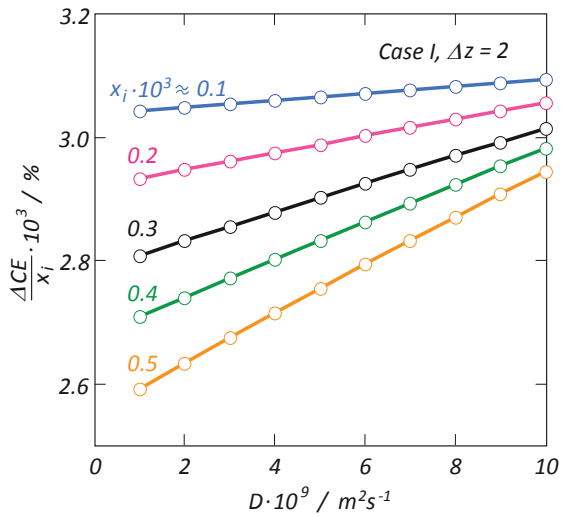

Figure 4. Slope of the CE *vs*. concentration curves as a function of the diffusion coefficient for binary pairs containing impurity elements at different impurity concentration.

Effect of Impurity Concentration

A number of calculations with different impurity concentrations and different change in valence were performed for the three cases shown in Table I. The results are shown in Figure 5. The curves in the figure represent the function

$$CE = CE_0 - \frac{[223 \cdot (100 - CE_0) + 516] \cdot x_i \Delta z}{1 + 105 \cdot x_i \Delta z} \qquad (12)$$

The equation is valid up to $x_i = 2 \cdot 10^{-3}$. The standard deviation between the equation and the numerical data is 0.06 percent CE.

Further Reduction of Impurity at the Cathode

The reduced form of the impurity element may be further reduced at the cathode. Impurities less noble than aluminium prefer to stay in the electrolyte; *i.e.*, a large amount in the electrolyte will be at equilibrium with a small amount in the metal. For impurities more noble than aluminium, the concentration in the metal is governed by mass transfer. The two extreme cases are when a) the impurity is so non-noble that the concentration in the metal becomes very

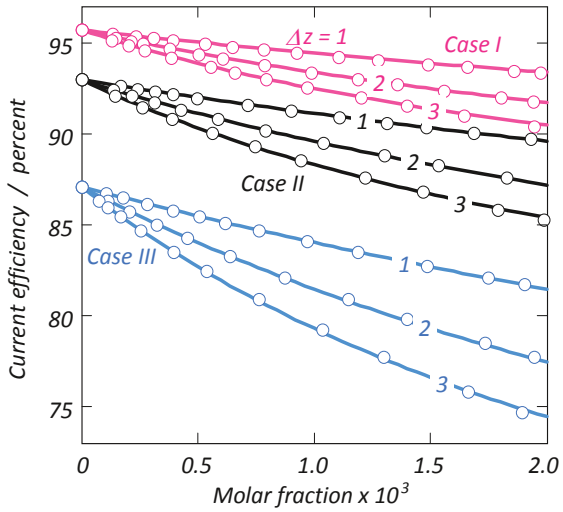

Figure 5. Current efficiency as a function of the molar fraction of impurities at different change in valence. Symbols: Numerical data, curves: Equation 12.

low (*i.e.*, N_4(inner) = 0) and b) the impurity is so noble that the concentration at the cathode drops to zero.

A simulation was performed for the two extreme cases, and the results are shown in Figure 6 and in Table II. The reaction zone was assumed to be at the same distance from the cathode in the two cases. As can be observed; the main difference is found inside the reaction plane, while the concentration gradients in the outer region are less affected. Table II indicates that the flux of i_{red} is very small inside the reaction zone, since the turbulent diffusion coefficient is much smaller close to the cathode.

The results indicate that noble multivalent impurities will have the same influence on the CE as non-noble impurities, provided that the concentrations in the electrolyte are the same. However; noble impurities may have shorter retention time in the electrolyte and generally lower concentrations.

As already mentioned, the transition of a noble impurity element from the electrolyte to the metal is limited by mass transfer at the cathode. Disregarding the global flux; for noble elements that do not have multiple oxidation states, the ratio between the molar fractions in the metal and in the electrolyte becomes

$$\frac{x_i(m)}{x_i(e)} \approx \frac{c_{tot}\, k_0}{N_{Al}} \qquad (13)$$

where c_{tot} is the total molar concentration, k_0 is the mass transfer coefficient at the cathode (calculated from Equation 4), and N_{Al} is the formation rate of aluminium.

The metal concentration was calculated as a function of the electrolyte concentration for a noble element (x_i = 0 at the cathode), and the result is shown in Figure 7.

From the data in Figure 7, it appears that there is generally good agreement between the numerical data and Equation 13, at least when the impurity concentrations are small. In other words, it appears that the reaction with dissolved metal does not affect the transfer of noble impurities between the electrolyte and the metal. This may seem surprising when considering the concentration gradients, but it can be related to the fact that the reaction normally takes place in a region dominated by turbulent diffusion.

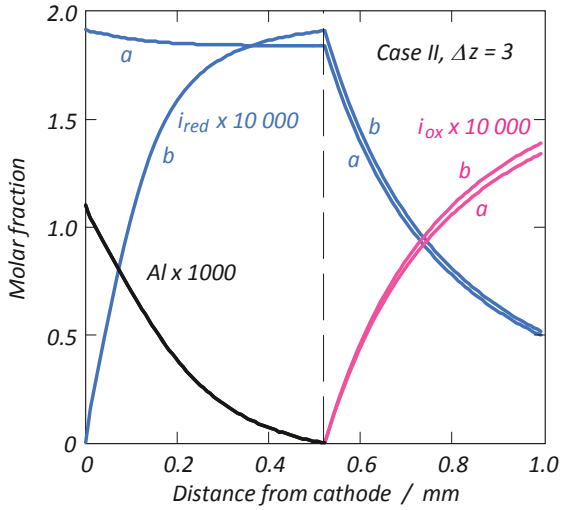

Figure 6. Concentration gradients for dissolved metal and impurity for two cases. Curves a: i_{red} is not reduced into the metal, curves b: reduction of i_{red} is only limited by mass transfer.

Table II. Fluxes, total impurity concentration in the bulk, and current efficiency for the two cases shown in Figure 6.

	a	b
N_3 [molm^{-2}s^{-1}]	$3.415 \cdot 10^{-3}$	$3.545 \cdot 10^{-3}$
N_4 [molm^{-2}s^{-1}]	0	$-1.30 \cdot 10^{-4}$
x_i [1]	$1.837 \cdot 10^{-4}$	$1.907 \cdot 10^{-4}$
CE [%]	92.248	92.247

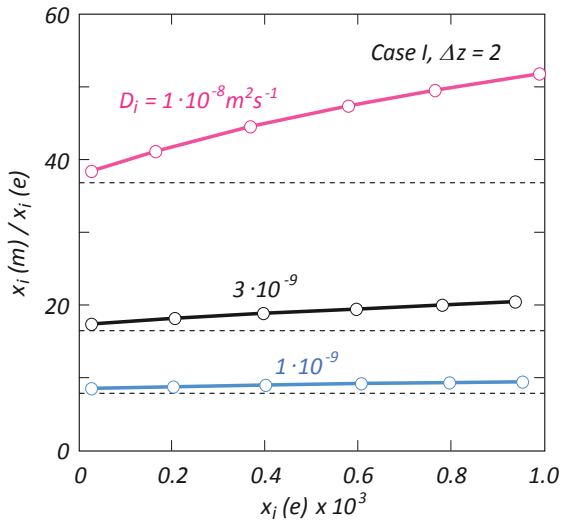

Figure 7. Ratio between noble impurity concentrations in the metal and in the electrolyte as a function of the concentration in the electrolyte, computed with different impurity diffusion coefficients. Symbols and full lines: Numerical data, broken lines: Equation 13 with CE = CE_0.

Discussion. Comparison with Experimental Data

Since the effect of the flux of i_{red} inside the reaction plane is very small (Table II) and the effect of the diffusion coefficient is also quite limited at least for small (industrial) impurity concentrations, it can be suggested that Equation 12 is generally valid. The total molar concentration can be estimated by

$$c_{tot} \approx \frac{\rho}{0.042}\left(\frac{r+1}{r+2}\right) \quad [molm^{-3}] \qquad (14)$$

where ρ is the density of the electrolyte [kgm^{-3}]. The molar fraction x_i in Equation 12 can then be calculated from the corresponding concentration in ppm,

$$x_i \approx ppm_i \cdot \frac{4.2 \cdot 10^{-8}}{M_i} \cdot \left(\frac{r+2}{r+1}\right) \qquad (15)$$

where M_i is the atomic weight of the impurity element. At small impurity concentrations, *i.e.*, when the denominator in Equation 12 is close to 1, we obtain

$$\frac{dCE}{d\,ppm} \approx -4.2 \cdot 10^{-8}\left[223(100 - CE_0) + 516\right] \cdot \frac{\Delta z}{M_i} \cdot \left(\frac{r+2}{r+1}\right) \qquad (16)$$

Equation 16 is presented graphically in Figure 8 together with some of the experimental data available.

It is generally accepted that phosphorus can exist as P^{3+} and P^{5+} in the electrolyte; *i.e.*, $\Delta z = 2$, as was used in Figure 8. The data by Sterten *et al.* [3] extrapolate back to $CE_0 \approx 93$ % at zero concentration of impurities, while the data by Meirbekova *et al.* showed CE_0 about 94 % and the data by Thisted [5] indicate CE_0 = 90.5 %.

The most common oxidation states of sulphur are -2, +2, +4, and +6. In Figure 7 it is assumed that $\Delta z = 2$. Meirbekova's data [14] extrapolate back to CE_0 = 90.9 %.

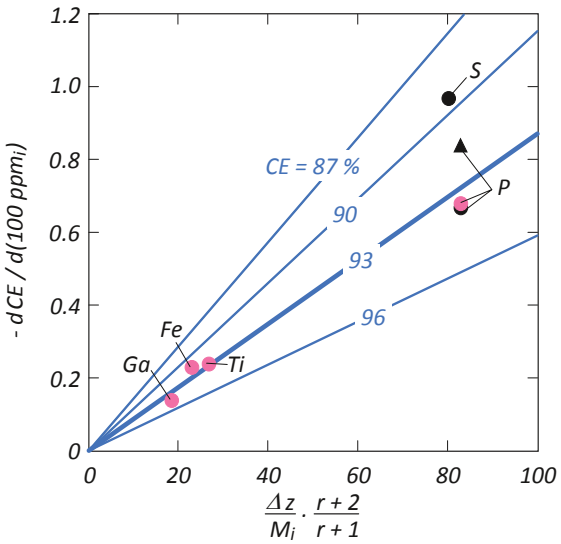

Figure 8. Variation in current efficiency with impurity concentration, plotted for different CEs according to Equation 16. Experimental data: Coloured circles: Sterten *et al.* [3], black circles: Meirbekova *et al.* [6, 14], triangle: Thisted [5].

It is well known that iron has oxidation states of +2 and +3, and $\Delta z = 1$ fits very well with the data by Sterten *et al*. [3]. However, it should be noted that it is possible that the anode potential is too low to oxidize Fe^{2+} to Fe^{3+}, as was discussed by Jentoftsen [16]. The result by Sterten *et al*. [3] can, therefore, be regarded as being controversial.

The oxidation state of gallium is normally +3, but +2 and +1 also exist [18]. $\Delta z = 1$ was used in Figure 7.

Although the impurity elements are often introduced as oxides they react with the electrolyte to form fluorides, *e.g*.,

$$M_2O_3 + AlF_3 = Al_2O_3 + MF_3 \qquad (17)$$

Many fluorides are very volatile. This is the case for vanadium and silicon. Zinc may also be reduced to its metallic form, which has a low boiling point. It is difficult to predict how fast volatile elements disappear from the electrolyte, but there is reason to believe that the adverse effect of V, Si, and Zn on the CE may have been underestimated in the work by Sterten *et al*. [3]. Sulphur also forms volatile compounds, but in the work by Meirbekova [14] the concentration was monitored, and the evaporation loss was compensated by additions of sodium sulphate during the experiments.

It is not straightforward to predict the collective effect on the CE when several impurity elements are present simultaneously, which is the case in industrial cells. The different elements will take part in red-ox reactions of the type

$$i_{1(red)} + i_{2(ox)} = i_{1(ox)} + i_{2(red)} \qquad (18)$$

However, Sterten *et al*. [3] added mixtures of impurities, and the individual effects turned out to be additive within the experimental uncertainty.

Acknowledgement

The present work was partly supported by the project "Sustainable Production and Recycling of Metals" (SUPREME) financed by the Research Council of Norway and Hydro Aluminium, Boliden, Glencore, and Permascand. The author would like to thank Dr. Rauan Meirbekova for inspiration to initiate this study.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SODIUM IN ALUMINUM METAL OF OPERATING PREBAKE CELLS: CONFIRMATION AND NEW FINDINGS

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Keywords: Sodium, current efficiency, cathode overvoltage, turbulence

Abstract

The concentration of sodium impurity in the aluminum metal pool of operating prebake cells has been shown to be an indicator of cell performance; in general, cells operating with higher current efficiency tend to have a higher sodium content in aluminum. The higher sodium content is due to the increase in sodium containing species in the electrolyte near the bath-metal interface as a consequence of reduced hydrodynamic stirring due to cell MHD forces, anode gases and disrupting cell operations. Correlations between the sodium content in aluminum metal and changes in potline amperage, ac-distance, cell operations, bath chemistry, etc. are discussed for individual cell studies and potlines. Correlations are demonstrated between sodium and other alkali/alkaline metals, (lithium, magnesium and calcium) in aluminum in equilibrium with the cryolite electrolyte.

Introduction

The primary cathodic reaction in cryolite-based melts is the reduction of Al^{3+} containing species. The next most favored cathodic reaction is the deposition of sodium which forms an alloy with aluminum, sodium(Al).

$$AlF_3 + 3Na^+ + 3e = Al + 3\ NaF \quad (1)$$

and the main side reaction,

$$Na^+ + e = Na \quad (2)$$

Sodium is produced at the bath-metal interface due to the chemical reaction occurring when reaching thermodynamic equilibrium and sodium partitioning into the two phases based upon equilibrium constants for the system.

$$3NaF + Al = AlF_3 + 3\ Na \quad (3)$$

Since virtually all the current is transported by Na+, whereas Al-containing species are being discharged, there will be concentration gradients of AlF_3 and NaF between the bulk of the electrolyte and the cathode surface. This gives rise to cathodic concentration overvoltage (η_c). It is generally accepted that activation overvoltage is small which means that there is chemical equilibrium at the cathode surface.

Additionally, if lithium, magnesium or calcium fluoride is added to the cryolite based electrolyte, the aluminum will also contain these metals, due to exchange equilibria of the type,

$$MF + 1/3\ Al = M(Al) + 1/3\ AlF3 \qquad M = Na, Li \quad (4)$$

$$MeF_2 + 2/3\ Al = Me(Al) + 2/3\ AlF_3 \qquad Me = Ca, Mg \quad (5)$$

Sodium Content in Non-polarized Equilibrium Cryolite-Aluminum Systems

Tingle et al. [1] first studied in 1981 how the content of sodium in aluminum in equilibrium with NaF – AlF_3 melts varied with time, concentration and temperature. The literature data for sodium content in aluminum in non-polarized equilibrium with cryolite melts is summarized in Figure 1 and was reported in 2001 by Thonstad and co-workers [2].

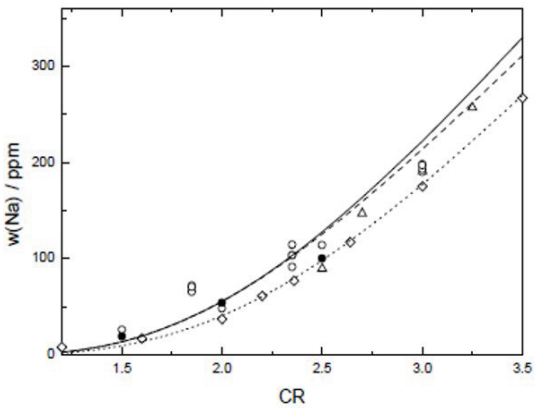

Figure 1. The measured non-polarized equilibrium content of sodium in aluminum, Thonstad [2].

Results

Sodium(Al) Content in Polarized Aluminum-Cryolite Systems

The amount of sodium in the metal extracted from commercial aluminum cells is normally in the order of 90 ppm. This value is somewhat higher than expected from published data concerning the sodium content in aluminum contacting the electrolyte under zero-current conditions.

During electrolysis the content of sodium increases with increasing cathodic current density. This effect is related to the cathodic concentration overvoltage which is mass transfer controlled. It is caused by the fact that the sodium ion is the carrier of current, while aluminum is being deposited. This leads to the formation of a concentration gradient in the boundary layer at the cathode, with enrichment of NaF and depletion of AlF_3 at the interface.

The cathode overvoltage reflects the concentration gradient with respect to the concentration of NaF and AlF_3 at the bath-metal interface. Hence, there should be a relationship between the overvoltage and the sodium content in the aluminum metal in

operating cells. The magnitude of the overvoltage is dependent on the cell design and on the convection pattern in the cell. Modern aluminum cells with good magnetic compensation and low flow rates at the bath-metal interface have a high sodium content, >120 ppm in the aluminum cathode pool. Strong relationships exist between the sodium content in the aluminum pool in cells and the cell current efficiency. Solheim and Sterten [3] estimated that the sodium content at equilibrium doubles for each 75 mV increase in cathodic concentration overvoltage (η_c) at 970 °C.

In 1995-1996, Tabereaux [4] postulated that the sodium concentration in the metal pad, sodium(Al), is directly related to the turbulence at the bath-metal interface and concluded that there is an indirect correlation between high current efficiency and high sodium contents. Tabereaux showed that the average sodium(Al) level in industrial cells depends on the stability of the metal-bath interface, for example:

- Older prebake cell technology with less stable magnetics and Søderberg cells shows lower sodium contents that average about 70 ppm sodium(Al) and lower current efficiencies, < 92%
- Modern prebake cells with well-balanced magnetic fields can average over 100 ppm sodium(Al) and also achieve higher current efficiency, 94-96 %.

Therefore, Tabereaux showed that there is a strong relation between the sodium content of the aluminum and CE%.

Sodium(Al) Content and Current Efficiency in Individual Cells

Normally, it is difficult to correlate sodium(Al) in individual cells to operational parameters because the high fluctuations of the measured sodium(Al) values; they appear to vary widely in a seemingly erratic manner Figure 2. The large variation of sodium in metal samples done within a 24-hour tapping cycle is shown in Figure 3. The average difference between maximum and minimum sodium values within pots is 42 ppm. The few results indicate that there are relationships between the sodium(Al) values and some pot operational parameters: chemistry, temperature and alumina.

The results of a 38-hour study of the variations of the sodium(Al) in three 200 kA AP18 prebake cells are shown in Figure 4.

1) Firstly, no significant change in the sodium(Al) content was observed after tapping metal from the three cells or after the anode change operations.
2) However, the sodium(Al) content in all three individual 200 kA prebake cells was observed to decreased about 4 hours later after the addition of large amounts of AlF_3 to the top ore bins of these cells; the AlF_3 eventually reached the bottom of the ore bins and was discharged with alumina to the bath after about 4 hours.
    - The sodium(Al) content rapidly decreased in all three cells from 100-120 ppm to about 40 ppm due to the immediate decrease in the equilibrium of sodium containing ions at the metal-bath interface as a consequence of the reaction with the additional aluminum ions.
    - The sodium(Al) content slowly return to normal levels over the next four hours.

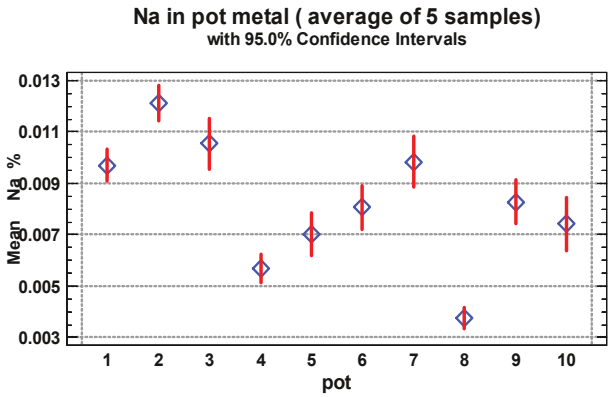

Figure 2. Sodium concentrations in 10 pots in a 200 kA plant.

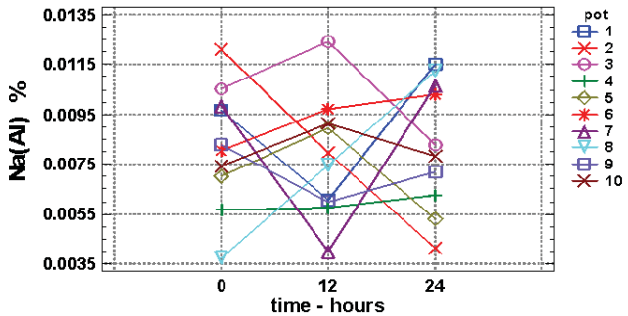

Figure 3. Variation in Na(Al) over a 24-hour period for 10 pots.

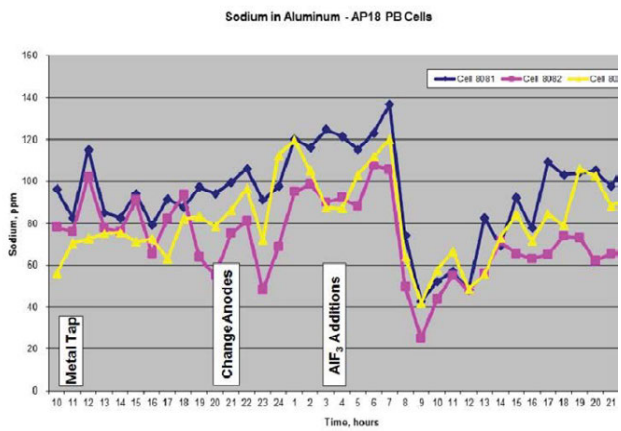

Figure 4. The variations of the sodium(Al) content in three AP18 cells.

Thonstad and co-workers [2] demonstrated in Figure 5 that the sodium(Al) in an individual prebake cell decreased with increased alumina content (~ 5%) in the bath due to 4-hour overfeeding period and increased during underfeeding, (2 - 3%) .

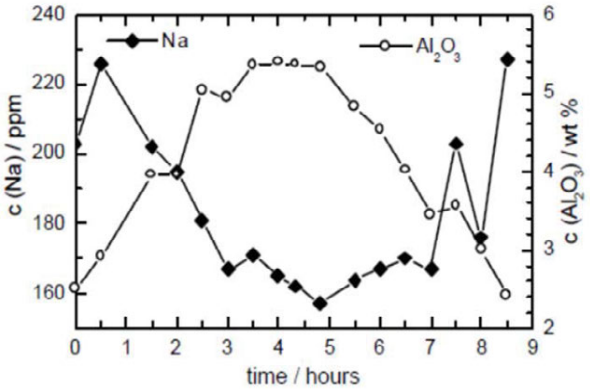

Figure 5. Variations of the sodium(Al) content with alumina in a prebake cell, Thonstad [2].

Average Sodium(Al) Content and Current Efficiency in Potlines

The yearly average for sodium(Al) content over a 10 year period is show in Figure 6 for a 200 kA prebake plant. Both the current efficiency and the sodium(Al) content were very low during the initial year of plant startup. There after the CE% increased to 93% and the sodium(Al) increased to 100 to 120 ppm for five years. Thereafter, the sodium(Al) content decreased to about 90 ppm and the current efficiency fluctuated from 91 to 93%. Current efficiency can be heavily influenced by changes in the metal pad inventory, whereas the sodium(Al) content appears to show a truer gradual decrease in performance during the past 5 years.

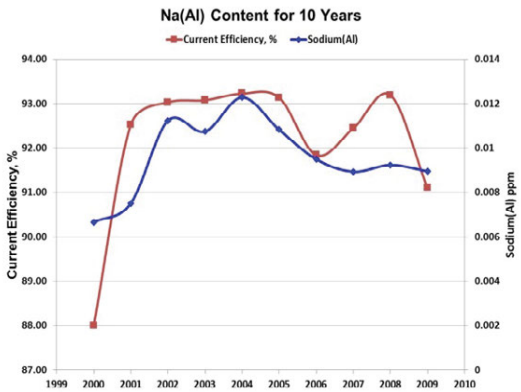

Figure 6. Yearly average sodium(Al) content for a 200 kA prebake smelter over a 10 year period.

Sodium(Al) vs. Current Efficiency Relationships

It has been demonstrated that the average sodium content of all cells in potlines is a good indicator of cell production as showed by current efficiency. Dion and co-workers [5] performed univariate statistical analysis on a potline scale over a five year period to confirm the correlation between the sodium content in aluminum, measured on a monthly basis, and current efficiency. Strong correlations were found for alumina dosage rate and number of point feeder actions. Sodium and current efficiency were found to be strongly correlated with silicon level in aluminum and current intensity and cell resistivity.

In 2009 Lillebuen and co-workers [6] disclosed an excellent linear relationship between the increase in sodium(Al) and individual cell current efficiency average over one year for a potline of 220 kA prebake cells. A regression analysis explained 84% of the variation in the data.

$$\text{Na (ppm)} = -615 + 7.7\ (\text{CE \%}) \quad (6)$$

The linear relationship for the increase in sodium(Al) versus current efficiency is show in Figure 7 for an aluminum smelter with 200 kA prebake cells for a 10 year period. The regression analysis coefficient for the increase in sodium(Al) content versus increase in CE% for the 10 year period was 7.71 sodium(Al) ppm for a one percent increase in current efficiency.

$$\text{Na (ppm)} = -640.6 + 7.9\ (\text{CE \%}) \quad (7)$$

A similar linear relationship is shown in Figure 8 for the increase in sodium(Al) and corresponding increase in the monthly average current efficiency average over a two year period for a potline of 350 kA prebake cells. A regression analysis explained 31% of the data variation in the data.

$$\text{Na (ppm)} = -640.6 + 7.9\ (\text{CE \%}) \quad (8)$$

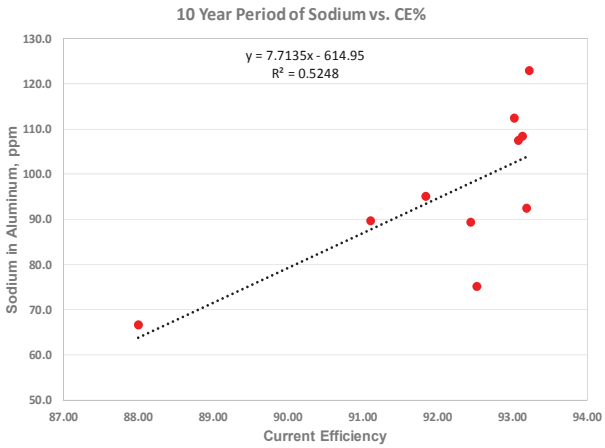

Figure 7. Yearly average sodium(Al) content over a 10 year period.

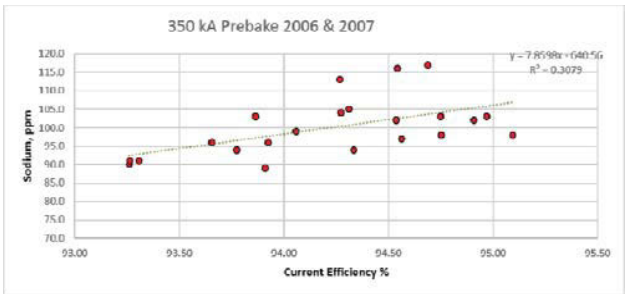

Figure 8. Linear relationship for the increase in sodium(Al) content vs. current efficiency for a potline 350 kA prebake cells.

No similar correlations have been found for older low amperage (100 kA) prebake potlines and Søderberg potlines. Sodium levels are significantly lower in these older lines as well as current efficiency due to the increased turbulence of the aluminum-

electrolyte interface as a consequence of higher MHD forces (earlier potline buss designs); more frequent anode bubbles (higher anode current density); and the significantly larger size of gas bubbles under the Søderberg anodes.

Table I. Regression analysis coefficient for the increase in sodium(Al) per one % increase in CE relationships for aluminum smelters.

Plant	kA	CE% to ppm Na Coefficient
A	205	9.98
B	230	11.09
C	345	12.32
D	157	7.40
E	226	4.70
F	238	0.00
G	169	2.50
Average		6.86

The regression coefficient for the increase in sodium(Al) as a function of the CE% is usually different for different aluminum smelters with different technologies, anode/alumina quality, and work practices as shown in Table I. The coefficient was found to be different for different years for the same smelter, as well as for individual potlines in the same smelter for the same year.

Influence of Calcium and Magnesium on Sodium(Al) Content at a 200 kA Aluminum Smelter.

The linear relationships between sodium(Al) and calcium(Al), lithium(Al) and magnisium(Al) are shown in Figure 9-a, 9-b and 9-c for a potline with 264 cells over a two year period. Peterson and Tabereaux [7] published a regression equation for the contents of sodium, calcium and magnesium in aluminum in equilibrium with cryolite based melts at 970 °C.

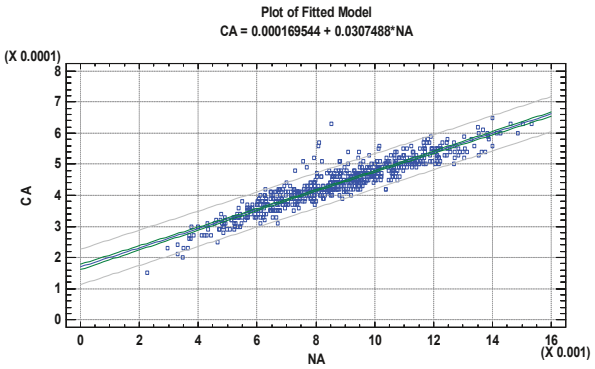

Figure 9-a. Relationship between sodium(Al) and calcium(Al).

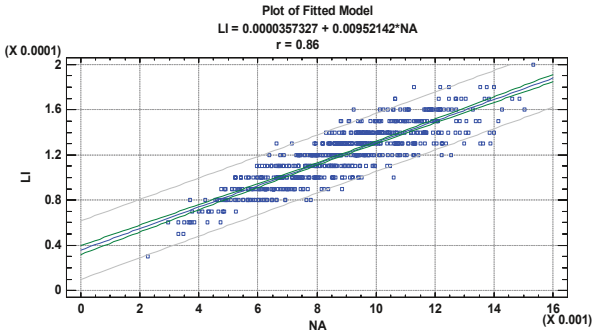

Figure 9-b. Relationship between sodium(Al) and lithium(Al).

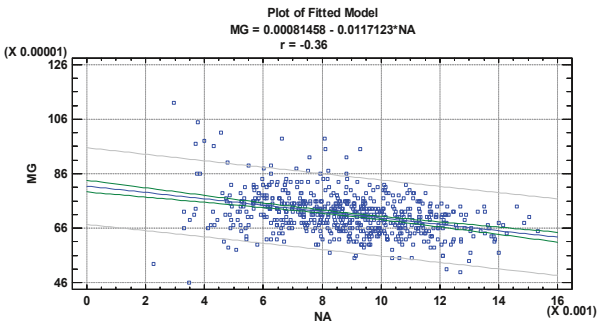

Figure 9-c. Relationship between sodium(Al) and magnesium(Al).

Figure 10 shows how closely sodium(Al) and calcium(Al) are correlated. These results are from weekly metal samples of a single pot in a 200 kA potline.

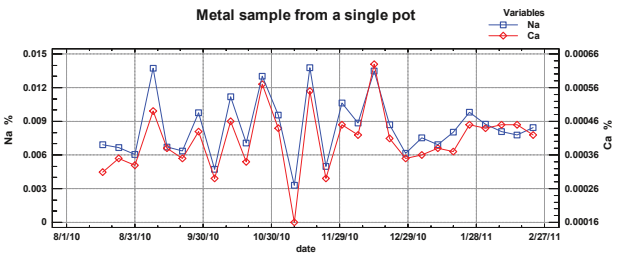

Figure 10. Weekly Na and CA concentrations from a single pot.

Impact of Low Energy Cells and Increase in Pot Line Current on Sodium(Al) and Current Efficiency.

During recent years there has been a major focus on reducing the specific energy consumption for aluminum production by the Hall–Héroult process. Increasing the potline current typically results in squeezing the anode-to-cathode distance (ACD) of all cells due to the need to compensate for the increase in pot resistance at the higher amperage.

Coursol and co-workers [8-9] point out that the key issue when increasing the cell amperage, (and correspondingly reducing cell heat losses and cell voltage) is that a limit is reached in terms of the lowest ACD at which the cell is stable and efficiently operated. Even lower ACDs usually result in a reduction in current efficiency. But ultimately, in some instances depending on electrical power costs, aluminum prices, etc. it is not so much the current efficiency which is the dominate importance, but rather

the specific energy consumption. In general, aluminum plants can afford a small reduction in current efficiency and still maintain or improve the specific energy consumption.

If the ACD is eventually overly "squeezed" and is reduced to the point that the pot instability or bypass electrical shorting current reaches a certain level which prevents a reasonable current efficiency to be obtained, leading to an increase in specific energy consumption.

.In recent years, China has been the world leader for the development of low voltage cell technologies in order to reduce the average voltage as low as possible in the premise of reducing the specific energy consumption, but keeping the same current efficiency as before. According to Liu and Gu [10] at present, application of low voltage technologies has reduced the DC energy consumption to below 12500 kWh/t-Al in industrial potlines. However, as a result of severe reduction in ACD by reducing the cell below 3.8 volt, the current efficiency has reported to have decreased to less than 93%.

Martin et.al. [11] reported in 2012 the results of a stability study, Figure 11, demonstrates that due to its excellent MHD system design that the APXe cell can operate with ACD close to 20 mm without any additional means to stabilize the metal pad (e.g. drained cathode, baffled cell, high metal pad, etc.).

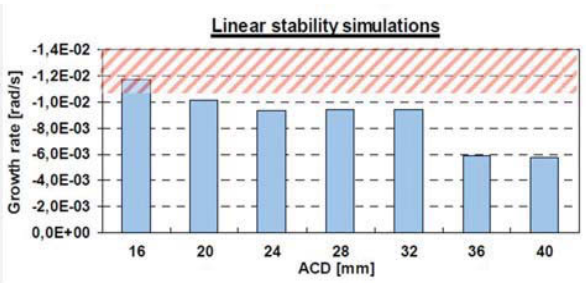

Figure 11. Increase in pot instability versus anode-cathode distance, Martin [11].

Martin also developed the linear relationship shown in Figure 8 for the increase in sodium(Al) and corresponding increase in the monthly current efficiency over a nine month period for a group of APXe prebake cells. A regression analysis explained 70% of the data variation in the data. The lower current efficiency and sodium(Al) values reflect the initial increase in pot instability associated with a squeezed anode-cathode distance until operating parameters and work practices were modified.

Sodium(Al) and CE% decrease with high pot nose due to poor anode current distribution that eventually reduces the ACD causing increased turbulence of the metal-bath interface.

- The reduced ACD may also result in more anode problems like anode spikes and excessive alumina muck that reduce the current efficiency.
- Other anode problems such as excess carbon dust, broken anodes and anode burn-offs contributes to a higher bath resistance and squeezed ACD and increase turbulence of the bath-metal interface that reduces current efficiency.

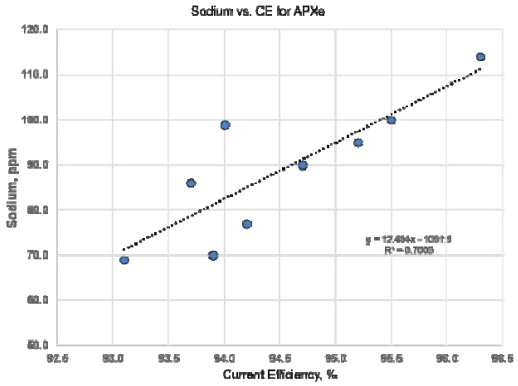

Figure 12. Increase in sodium(Al) content with higher current efficiency for APXe cells, (Martin 12).

Sodium(Al) and CE% values decrease with pot age; cathode current distribution deteriorates with age due to cracks causing increased turbulence of the metal-bath interface, Figure 12.

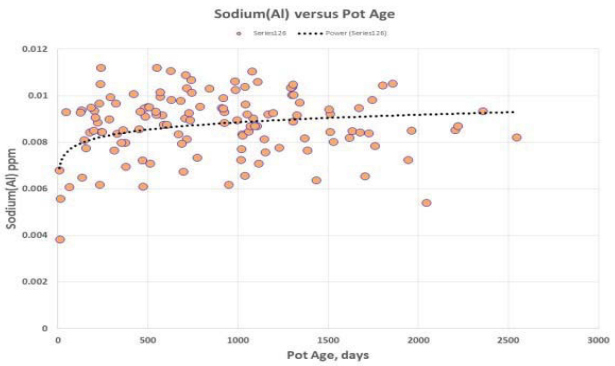

Figure 13. Sodium(Al) versus Pot Age for a Potline of 264 cells.

Sodium(Al) and CE% may eventually decrease with higher potline amperage; high pot resistance ultimately causes the ACD to be squeezed causing increased turbulence of the metal-bath interface.

The variation in sodium(Al) versus pot age is shown in Figure 13. Lower sodium values for the initial 500 days is due to the absorption of sodium into the new carbon cathode lining.

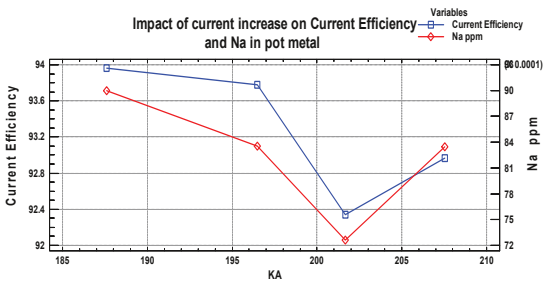

Figure 14. Variations in sodium(Al) and current efficiency with increasing potline current for a 200 kA potline.

Figure 14 shows the impact on sodium(Al) and current efficiency of increasing line current. The initial loss of current efficiency was partially reversed as the plant personnel learnt how to modify the operating parameters and work practices at the higher current.

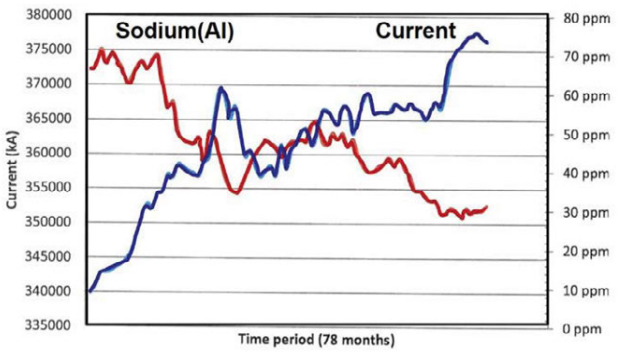

Figure 15. Variations in sodium(Al) with increasing potline current for a 350 kA potline, (Dion 13).

The sodium content of modern 350 kA prebake potline is shown in Figure 15 to decrease from 70 ppm to 30 ppm apparently due to the 35 kA increase in potline amperage over a 3 year time period. Consequently the smelter experienced almost 1.5% decrease in current efficiency during this same period.

Generally aluminum smelters try to avoid this decrease in current efficiency due to squeezing the ACD by decreasing the pot resistance prior to the increase in amperage. This has been achieved mainly as a result of improved cell design (improved MHD, larger anodes, larger anode stubs, etc.) and improved process control strategies that have allowed higher cell currents and smaller ACD.

Conclusions

The results of this study emphasized the importance of the sodium(Al) content as an indicator of cell performance and current efficiency. It is difficult to correlate sodium(Al) in individual cells to operational parameters because of the high fluctuations of the measured sodium(Al) values. Excellent linear relationship were developed for the changes in potline sodium(Al) values and current efficiency; a regression analysis explained 84% of the data variation in the data. In several instances, the sodium content of prebake potlines was shown to decrease significantly, apparently due to increases in potline amperage. However, aluminum smelters try to avoid this decrease in current efficiency due to squeezing the ACD by decreasing the pot resistance prior to the increase in amperage.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE PERFORMANCE OF ALUMINIUM ELECTROLYSIS IN A LOW TEMPERATURE ELECTROLYTE SYSTEM

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Keywords: Aluminium electrolysis, current efficiency, molten KF-AlF_3-NaF, alkali metals, low temperature

Abstract

The electrolyte based on the molten KF-AlF_3-NaF system was investigated for low-temperature aluminium electrolysis in laboratory scale. The current efficiency for aluminium deposition was determined from the amount of aluminium deposited after 4 h at a cathodic current density of 0.85 $A \cdot cm^{-2}$. The cryolite ratio of molar concentrations of alkali fluorides and aluminium fluoride KR = $(N_{KF}+N_{NaF})/N_{AlF3}$, was always retained constant, equal to 1.3 or 1.5. The cryolite-based melts contained 5 and 10 mass% NaF. The temperature was 750 °C for KR=1.3 and 800 °C for KR=1.5. Current efficiencies decreased with increasing content of NaF at cryolite ratios 1.3 and 1.5. The effect of NaF on the cathode process was studied. The contents of sodium and potassium in aluminium based on the KF-NaF-AlF_3 melts were determined as a function of the cathodic current density in the range from 0.5 to 1.2 $A \cdot cm^{-2}$.

Introduction

Industrial production of aluminium is now exclusively based on the Hall-Héroult process. Producing aluminium by this process requires a large amount of energy and produces significant emissions of greenhouse and other harmful gases. In order to reduce harmful waste production and to improve other technological parameters for energy saving, the promising direction is the development of a new electrolyte having high electrical conductivity, high alumina solubility and the possibility to carry out the process at much lower temperature. The electrolyte may be regarded as the heart of the Hall-Héroult process. Most of the changes in electrolyte chemistry throughout the history of the process have aimed at better cell operation by using additives that improve the physicochemical properties of the electrolyte, in order to increase the current efficiency and to reduce the energy consumption of the process [1]. However, the radical decrease of operating temperature is difficult in the frame of the existing technology in which sodium cryolite is a basic component of the electrolyte.

In order to obtain essential decrease of the electrolyte liquidus temperature, the KF-AlF_3-based salts are attractive since the system has a eutectic point at 560 °C. The phase diagram of the KF-AlF_3 system [2, 3] shows that the system has a much wider low temperature liquid range than the NaF-AlF_3 system. Moreover, a KF-AlF_3-based system has higher alumina solubility compared with a NaF-AlF_3-based system. Meanwhile, a KF-AlF_3-based system contributes not only to decreasing the operating temperature but also to the possible application of inert anodes in the Hall-Héroult process [4]. No acceptable inert anode material has yet been found for long-term use in industrial aluminium electrolysers due to the high temperature environment of Al electrolysis cells.

In the KF-AlF_3 system, however, sodium, present as sodium fluoride in the electrolyte, is introduced to the electrolytic cell as an impurity in the alumina feed and can build up over time [5]. So, based on the KF-AlF_3 system, study of the NaF-KF-AlF_3 system is necessary.

Many studies have been performed on the physical and chemical properties and operating parameters in the NaF-KF-AlF_3-Al_2O_3 system. Some alumina solubility measurements have been published by Yan et al. [6] in KF-AlF_3-NaF systems. The liquidus temperature, electrical conductivity and alumina solubility in the KF-NaF-AlF_3 molten mixtures at the [KF]/([KF]+[NaF]) ratio ranging from 0 to 1 and at the fixed values of ([KF]+[NaF])/[AlF_3] ratio equal to 1.3, 1.5 and 1.7 have been measured by Apisarov [7]. Research by Wang et al. [8] suggested that electrolysis was more stable in the KF-NaF-AlF_3 (KF:NaF=1:1) electrolyte with CR=1.5 at 800 °C because high unstable fluctuating voltage (5.96-19.6 V) was observed in cells operated at 700 °C. Voltage fluctuation was explained by the formation of cathode encrustation. According to Zaikov [9, 10], increasing the CR to 1.5 and temperature to 800 °C, as well as modification of the bath composition, allowed for stable aluminium electrolysis in an NaF (12 wt%) - KF (30) - LiF (3) - AlF_3 (55) electrolyte. Sustained, stable operation of low-temperature, potassium cryolite-based aluminium electrolysis in 20 and 100 ampere cells fitted with vertical metal anodes and wetted cathodes was performed by Tkacheva et al. [11].

Sodium is always present as an undesired impurity in aluminium produced by the Hall-Héroult process, but in molten cryolite based electrolytes, dissolved Na must be considered in addition to dissolved Al [12]. Tabereaux [13] found that a high sodium content in aluminium in industrial reduction cells correlates with a high current efficiency during electrolysis. So involving KF as a major component in the KF-AlF_3-NaF system, the metal will also contain potassium and sodium as impurities due to exchange equilibria. The following equilibria are established at the aluminium-electrolyte interface:

$$Al + 3\ KF = 3\ K + AlF_3 \quad (1)$$

$$Al + 3\ NaF = 3\ Na + AlF_3 \quad (2)$$

Most of the previous work was focused on studying the physico-chemical properties and testing the behaviour of inert anodes in the NaF-KF-AlF_3-Al_2O_3 system. The aim of this work is to investigate the NaF-KF-AlF_3-Al_2O_3 system in order to get a quantitative understanding of the potential as a low temperature electrolyte for aluminium electrolysis. The current efficiency was studied in a laboratory cell. The effect on the content of NaF in

the NaF-KF-AlF_3-Al_2O_3 system was studied, involving KF as a major component. The data on the content of potassium and sodium in the KF-AlF_3-NaF based melts are presented.

Experimental

The laboratory cell, which was similar to the cell developed by Solli et al. [14-16], was designed specifically with the aim of obtaining good and reproducible convective conditions, and to give an almost flat metal surface (cathode) during electrolysis. The anode, the cathode, and the electrolyte were contained in a graphite crucible with a cylindrical sintercorundum side lining. The cell was placed in a vertical tube furnace with argon atmosphere and positioned to avoid temperature gradients in the electrolyte. The cathode was aluminium wetted on a stainless steel plate. This ensured a close to flat aluminium surface, and consequently an even current distribution on the cathode surface. The steel plate rested on a layer of alumina on top of a lining made of high purity cast alumina cement. This arrangement ensured negligible transport of aluminium and electrolyte down to the graphite crucible, and a minimum of metal loss due to formation of aluminium carbide. A graphite anode was placed about 1 cm above the steel cathode.

The electrolysis time was 4 hours for each experiment at a cathodic current density of 0.85 $A \cdot cm^{-2}$. The maximum cell voltage was 10 V. The steel plate was weighed before the experiment. After the experiment, the steel plate and aluminium were removed from the cell and the current efficiency was determined from the weight gain. The electrolyte was saturated in alumina at the start of the experiment. The ratio between molar concentrations of the alkali fluorides and the aluminium fluoride, $KR = (N_{KF}+N_{NaF})/N_{AlF3}$, was always kept constant equal to 1.3 or 1.5. The cryolite-based melts contained 0, 5 or 10 wt% NaF, meaning that NaF was replacing KF to keep KR constant. The temperature was 750 °C for KR=1.3 and 800 °C for KR=1.5.

The contents of sodium, potassium in the aluminium pad at different current densities with additive of NaF were determined. Sampling of metal was performed every 1 hour at different current densities during electrolysis. Metal samples were dissolved and analysed by ICP-MS.

Results and Discussion

Current Efficiency

Current efficiencies were obtained at the different contents of NaF at 750 °C for KR=1.3 as shown in Figure 1. The current efficiency decreased with increasing content of NaF. The relationship between cell voltage and time during electrolysis is shown in Figure 2.

The expected benefit of using NaF-KF-AlF_3 mixtures is the combination of the high electrical conductivity of Na-cryolite with the enhanced alumina solubility in K-cryolite melts. However, NaF increases the liquidus temperature in the potassium cryolite system. This fact is especially important when selecting electrolyte composition for aluminium electrolysis. When the cryolite-based melts contained 10 mass% NaF, the liquidus temperature of KF-AlF_3-NaF system at cryolite ratio 1.3 is higher than 750 °C according to Apisarov [7]. Although the electrolysis can still be operated below this calculated liquidus temperature, the process caused high cell voltage fluctuations and changed current distribution, which also interfered with the formation and coalescence of aluminium metal product on the wetted cathode. Some undissolved alumina was found at the anode after the experiment. High unstable voltage fluctuations were observed during electrolysis. When the cryolite-based melts contained 5 wt% NaF, the cell voltage raised during the second half of the process. Since the transport processes at the cathode leads to enrichment of NaF and KF at the cathode (increased KR), it is probable that solid cryolite was formed at the cathode; particularly, in melts rich in NaF, which have higher liquidus temperature. Besides the increased cell voltage, this could also explain that the current efficiency decreased with increasing NaF content, because the metal forms small spheres with larger surface area.

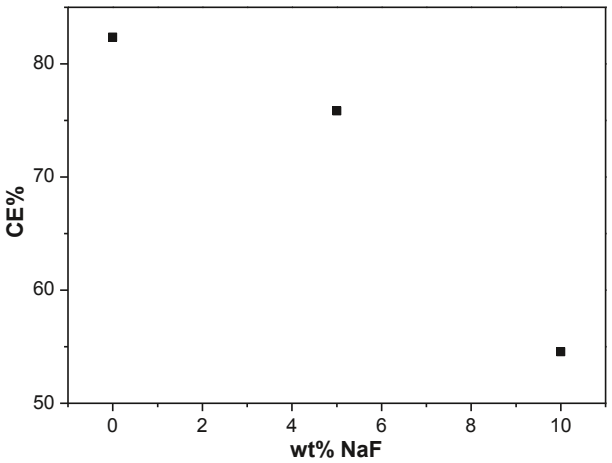


Figure 1. Current efficiency for aluminium deposition as a function of the content of NaF in molten KF-AlF_3-NaF (KR=1.3) - Al_2O_3 (sat) system at 750 °C.

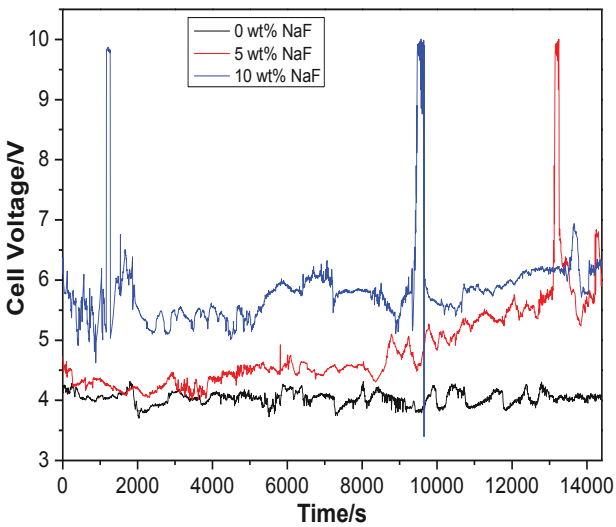


Figure 2. The cell voltages for aluminium deposition as a function of time in molten KF-AlF_3-NaF (KR=1.3) - Al_2O_3 (sat) system at 750 °C as a function the content of NaF during electrolysis.

Current efficiencies were obtained at different contents of NaF at 800 °C for KR=1.5 as shown in Figure 3. The current efficiencies decreased with increasing content of NaF. The relationship

between time and cell voltage during electrolysis is shown in Figure 4.

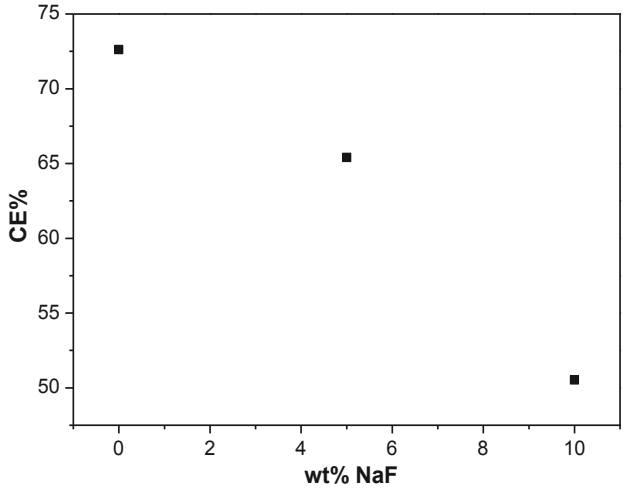


Figure 3. Current efficiency for aluminium deposition as a function of the content of NaF in molten KF-AlF_3-NaF (KR=1.5) - Al_2O_3 (sat) system at 800 °C.

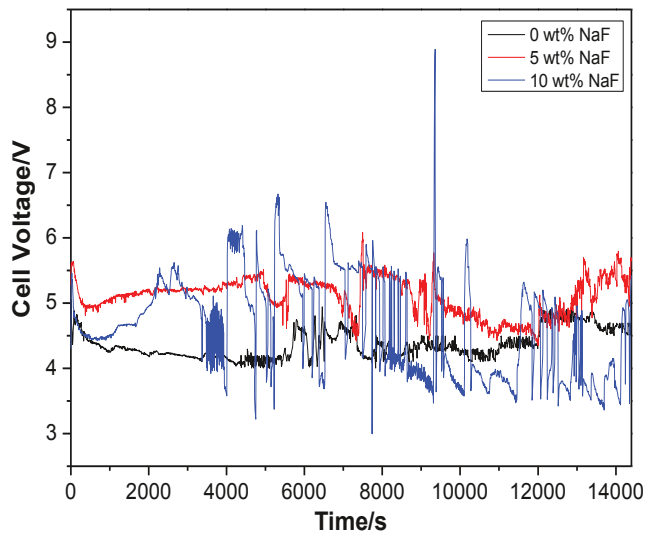


Figure 4. The cell voltages for aluminium deposition as a function of time in molten KF-AlF_3-NaF (KR=1.5) - Al_2O_3 (sat) system at 800 °C for different contents of NaF during electrolysis.

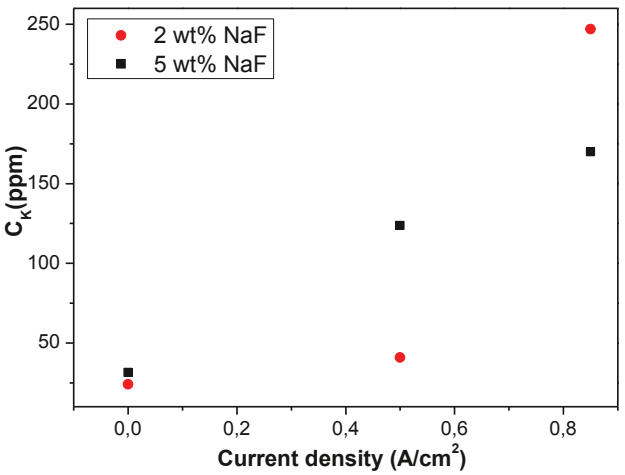
According to Apisarov [7], the liquidus temperature is around 780 and 790 °C respectively when the cryolite-based melts contained 5 mass% and 10 mass% NaF. Large voltage oscillations were observed at 5 and 10 mass% NaF in the melts. This might be because the operation temperature is very close to the liquidus temperature that results in formation of a layer of solid cryolite at the cathode. High cell voltage fluctuations and deterioration of current distribution would inhibit the formation of an adequate coalesced aluminium metal product so that the current efficiency decreased.

It was found that the NaF presence in the potassium cryolite has a negative impact on cell operation, as it caused cell voltage anomalies when experiments were performed. Tkacheva et al. [11] found that an increase in temperature can minimize the amplitude of voltage oscillation. The operating temperature needs to be adjusted to take the NaF content in the electrolyte into account, so the cell voltage anomalies caused by the NaF presence in the potassium cryolite can be avoided by careful control of operating parameters. More experiments are needed to determine the effect of NaF on cell operation in laboratory cells because they will be beneficial for increasing our understanding of processes that cause loss in current efficiency.

The Cathode Process

Sampling of metal was performed every 1 hour at different current densities. After the electrolysis for every current density, a waiting time of 15 min to establish equilibrium before the sampling. The contents of sodium and potassium in polarized aluminium were obtained at different current densities and two different KR with additions of NaF, in shown Figures 5 and 6.

(a) K



(b) Na

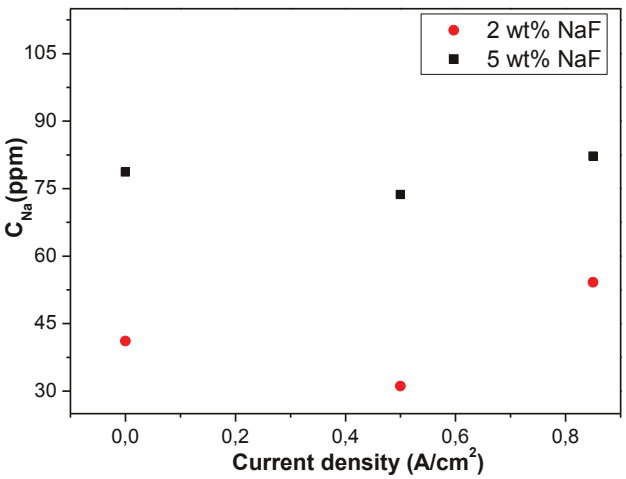
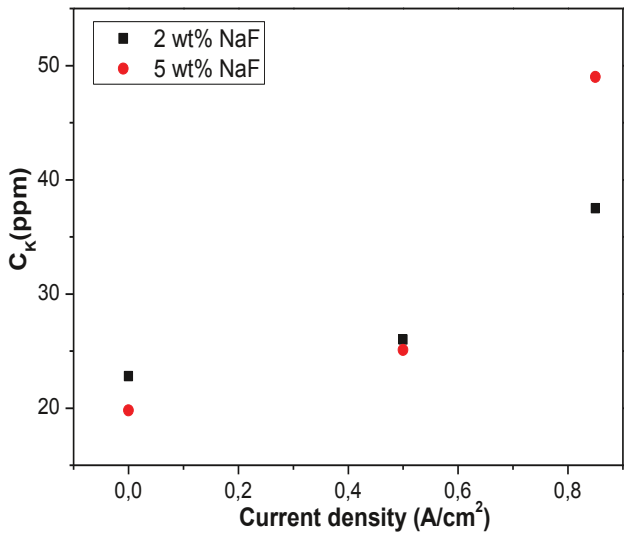


Figure 5. The contents of K and Na in aluminium at different current densities with additions of NaF in molten KF-AlF_3 (KR=1.5) -Al_2O_3 (sat) system at 800 °C.

The rate of chemical and electrochemical reactions taking place at the aluminium/melt interface at the temperatures of electrolysis is high because of the very high exchange current densities [17]. Thus, there is equilibrium between the melt adjacent to the cathode and the metal. The equilibrium content of sodium and potassium in aluminium can be described by the Reactions 1 and 2 in the Introduction.

(a) K



(b) Na

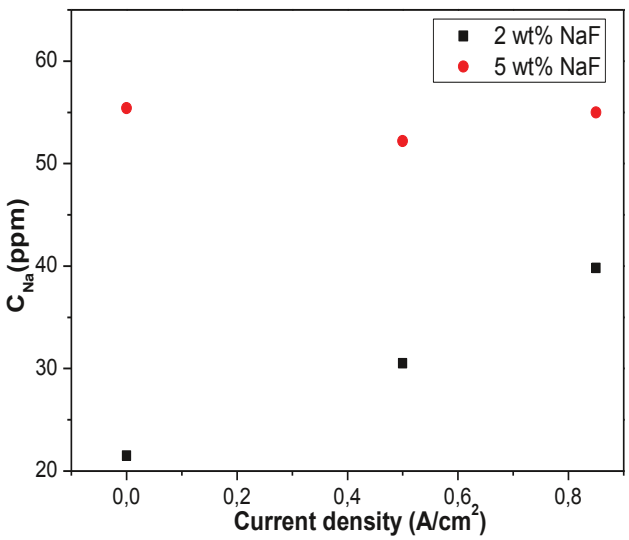


Figure 6. The contents of K and Na in aluminium at different current densities with additions of NaF in molten KF-AlF_3 (KR=1.3) - Al_2O_3 (sat) system at 750 °C.

From Figures 5 and 6, it was found that the concentration of potassium in aluminium increases with increasing current density. As aluminium metal is being produced at the cathode, the local concentration of KF in the near-cathode region can be higher than that in the bulk during electrolysis. Increasing current density leads to the molar ratio KR increases at the metal/melt interface. This change in composition between the bulk of the electrolyte and the layer adjacent to the cathode causes the cathodic concentration overvoltage. With increasing KR the potassium activity increases, and so does the potassium content in the aluminium.

The concentration of sodium in aluminium increases slightly with increasing current density. Although the addition of NaF was rather small, the content of sodium in aluminium is high compared to potassium. This might be explained by the contribution of sodium cations to the transport of electrical charge in the cell because the ionic radius of the latter is indeed slightly bigger, and since K^+ ions are less mobile, the enrichment of NaF at the aluminium/melt interface is more likely.

Concluding Remarks

The current efficiencies achieved in the present work were rather disappointing, considering that the melts used were very acidic and at low temperature. Current efficiencies of about 94 percent are obtained in the same apparatus when using today's standard bath compositions. The reason is probably formation of solid cryolite at the cathode, which may be a serious problem also in an industrial process, at least when using a process where inert anodes are combined with wettable cathodes. The only way to get around this problem seems to be the use of high superheat, which would partly counteract the advantages of using a very acidic bath.

The contents of sodium and potassium in polarized aluminium were obtained at different current densities and at cryolite ratio 1.3 and 1.5 with additions of NaF. The concentration of potassium in aluminium increased with increasing cathodic current density. The concentration of sodium in aluminium increased slightly with increasing cathodic current density.

Acknowledgment

Financial support from Norwegian University of Science and Technology is gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE ROLE OF KEY IMPURITY ELEMENTS ON THE PERFORMANCE OF ALUMINIUM ELECTROLYSIS – CURRENT EFFICIENCY AND METAL QUALITY

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Keywords: Phosphorus, silicon, aluminium electrolysis cells, current efficiency, metal inventory by copper dilution, mass transfer coefficient

Abstract

The behaviour of phosphorus and silicon in industrial aluminium cells was measured in QATALUM cells by analysis of bath and metal after addition of $AlPO_4$, Na_3PO_4 and SiO_2 to the bath. Phosphorus showed a much longer retention time in the bath than silicon. Start-up and anode effects influence the phosphorus and silicon level in the cells. Data trends showed that the amount of phosphorus decreases with increasing temperature. The effect of phosphorus on current efficiency was determined by measuring the metal inventory by copper dilution. The results were compared with current efficiencies estimated from measuring sodium concentration in the produced aluminium. The mass transfer coefficient for silicon was determined to be 3.1 x 10^{-5} m/s. Mass balance for phosphorus was carried out. It was found that massive amounts of phosphorus entering the cells were transported with the off-gases to the dry scrubbers.

Introduction

In the aluminium industry, the current efficiency and metal quality are important parameters. Impurities can either be reduced at the cathode and end up in the metal or take part in cyclic red-ox reactions, thus consuming electric current and reducing the current efficiency [1]. The impurities of interest in this study are phosphorus and silicon. The interest for phosphorus and silicon in aluminium production technology has considerably increased as the cell design has been modified to closed systems with dry scrubbers. The current efficiency for aluminium in modern cells is typically ranging between 90 and 96 %. Further increase in the current efficiency depends on improving a number of factors. One of these factors is the behaviour of impurities in the cells, especially for phosphorus and silicon. For every 100 ppm phosphorous in the electrolyte, the current efficiency is reduced by about 1 % [2]. Moreover, there is a limit on the phosphorus and silicon contents in aluminium, mainly for cast alloys. To evade further problems caused by phosphorus and silicon content, it is essential to improve the understanding of the behaviour of phosphorus and silicon in the electrolysis process.

The main loss in current efficiency is due to the back reaction between dissolved metal and the anode gas as shown in Equation 1 [3]. The other reasons for the decrease in current efficiency are anode effect, electronic conductivity of the bath due to dissolved Na, electrode short circuiting and impurities. The decrease in the current efficiency caused by the impurities can be 2 % in a normal Hall-Héroult electrolysis cell [2].

$$2Al + 3CO_2 = Al_2O_3 + 3CO \quad (1)$$

Primary metal usually has a purity of 99.5 to 99.9 % Al [4]. The rest is impurities introduced to the electrolysis cells. The impurities come with the raw materials, e.g. alumina, carbon anodes and fluorides. Some impurities also come from the cell operation and the production tools. For instance, additional source of iron is from corrosion of tools or anode stubs. Silicon can also come from the refractory lining or silicon carbide side-lining. All the impurities have to a varying degree an effect on the electrolysis process.

Impurities more noble than aluminium are reduced at the cathode and contaminate the produced aluminium metal or they may be re-oxidized at the anode and cause decrease in the current efficiency. Impurities less noble than aluminium will accumulate in the electrolyte where they can react with the components in the bath and thus modify the chemical composition of the melt. Also, they could interact with the anode reaction or lining. Other impurities not so detrimental to the aluminium production process are those which escape from the cell as harmful gases. As such, they do not decrease the current efficiency or lower the metal quality. However, they have negative impact on the environment. The major part of the impurities dissolves in the electrolyte while the minor part leaves the cell as outlet in various ways; in the contaminated aluminium metal, as off gases or adsorbed on carbon dust, alumina or other materials that are skimmed off [5]. However, a significant amount of the impurities is recycled back to the electrolysis process mainly due to the dry scrubber where gaseous and particulate impurities are collected and recycled.

Two effects of phosphorus of high importance are the reduction of the current efficiency and the negative influence on the properties of produced metal. The influence of phosphorus on the total current efficiency loss in industrial cells is difficult to quantify, however, a current efficiency reduction by about 1 % for each 100 ppm of phosphorus is generally accepted [2]. Besides its effect on the current efficiency, phosphorus also lowers corrosion resistance and increases the brittleness of aluminium. The content of phosphorus in aluminium is highly dependent on its concentration in alumina, applied technology and on the operational practices and conditions in a given smelter. In primary aluminium produced in modern smelters equipped with dry scrubbers, the concentration of phosphorus is ranging from a few ppms up to 30 ppm [5-8].

Silicon oxide has lower decomposition potential than alumina. Silicon is nobler than aluminium and tends to be deposited at the cathode electrochemically according to Equation 2, where silicon forms an alloy with aluminium.

$$Si^{4+} + 4e^- = Si \quad (2)$$

Hence, impurities are deposited at the cathode were the limiting current density (i_l) relies on the concentration of the element in the electrolyte (c^o) and on the mass transfer coefficient (k_m) [9].

$$i_l = nFk_m c^o \quad (3)$$

Studies of impurity species in industrial Hall-Héroult cell are accomplished by analysing samples of bath and metal over time after adding of a certain amount of impurity compounds. The concentration of the impurities against time after addition is:

$$c = c_o \exp\left(-\frac{A}{V} kt\right) \quad (4)$$

Thus, the mass transfer coefficient may be determined from Equation 4. Mass transfer coefficients mainly rely on design, technology and operation of industrial electrolysis cells. High convection of bath and metal pad or unstable cell is the main reasons for higher value of mass transfer coefficient.

Some of the silicon impurities are lost together with the exit gas, a process that is believed to be a result of the volatility of SiF_4. SiF_4 is normally formed in this process by an anode reaction or by a reaction in the bath according to Equation 5 [10].

$$4Na_3AlF_6 + 3SiO_2 = 3SiF_4 + 12\,NaF + 2Al_2O_3 \quad (5)$$

For silicon, the maximum concentrations in the metal must be strictly met because of the negative influence on the quality of the final product. Silicon also decreases the current efficiency of the electrolysis process due to the consumption of current which otherwise would be used for aluminium production.

Experimental Set-Up

All measurements were performed at Qatar Aluminium Company (QATALUM). Bath and metal samples were analysed at regular intervals before and after adding a certain amount of Na_3PO_4, $AlPO_4$ and SiO_2. The samples were marked and stored in appropriate containers. Bath samples were analysed by X-ray Fluorescence (XRF). Metal samples were analysed by automated optical emission spectroscopy (OES). Other process variables, as temperature, were studied in connection with phosphorus and silicon additions. Trends in phosphorus and silicon content in produced aluminium were also studied from analytical data.

Also, a certain amount of copper was added and subsequently the copper content in aluminium in the electrolyte cell was analysed before and after the copper was added. Thus, the metal reservoir or inventory could be calculated. Changes in the metal reservoir between two additions of copper can be used together with tapped metal to find the current efficiency of the cell.

The cell type used for the experiments was side-by-side with 30 anodes and 5 feeders. These cells are operated at 300-330 kA. This cell is fed by secondary alumina and the aluminium produced is tapped once a day. Three cells were used for these experiments. These cells were chosen based on specific criteria. The selection criteria of the chosen cells are:

- Reference Cells (Having history in metal inventory)
- Stability (Good in thermal balance and noise)
- Quality (Low impurities content)

Figure 1. A certain amount of Na_3PO_4, $AlPO_4$ and SiO_2 were added in the Qatalum industrial cells.

Figure 2. Bath and metal samples were taken and placed in proper place in able to be analysed.

Figure 3. Bath samples were analysed by X-ray Fluorescence (XRF).

Figure 4. Metal samples were analysed by automated optical emission spectroscopy (OES).

Results and Discussion

Distribution of Phosphorus and Silicon after Addition

Figure 5 and 6 show the concentration of phosphorus in the bath and in the metal as a function of time before and after adding $AlPO_4$ and Na_3PO_4. To compare the behavior of phosphorus to that of metallic impurities, results for silicon are also shown. It can be seen that phosphorus has a much longer retention time in the bath than silicon. Also, it is apparent that for silicon the

decrease in the bath leads to an increase in the metal, the concentration of phosphorus in the bath seems to reduce far more slowly. The increase in the metal is almost immediate after addition and the concentration remains more and less constant during the time of the measurement. A feasible reason is that phosphorus gas is produced at the cathode. Most of it leaves the cathode and is oxidized in the bath or at the anode. This can be repeated several times, prolonging the residence time in the bath and increase the negative effect on current efficiency. This shows that even though phosphorus is a more electropositive element than aluminium, the main part of it is found to remain in the bath. Before phosphorus ends up in the metal, a great part of the current can be lost due to the fact that phosphorus can be ascribed to the following red-ox reaction cycle.

$$P^{5+} + 2\ e^- \rightarrow P^{3+} \quad (6)$$

$$5\ P^{3+} \rightarrow 3\ P^{5+} + 2\ e^- \quad (7)$$

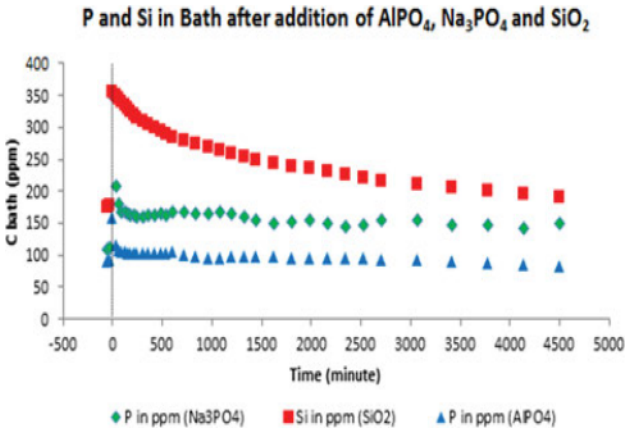

Figure 5. Distribution of phosphorus and silicon in the bath after addition. The dotted line marks the time of addition.

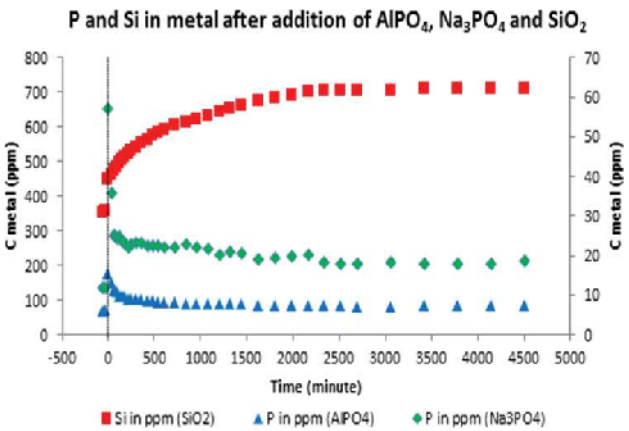

Figure 6. Distribution of phosphorus and silicon in the metal after addition. The dotted line marks the time of addition.

Furthermore, Figure 7 illustrates that the decrease in the silicon concentration in the bath over time is a mass transport controlled deposition of silicon at the aluminium cathode. Nearly all of the added silicon is mainly reduced at the cathode.

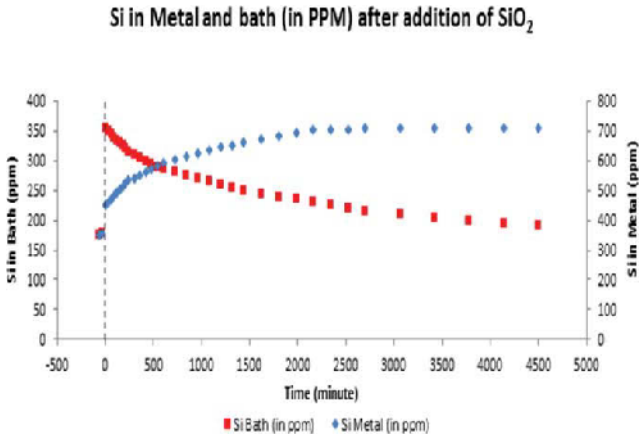

Figure 7. Distribution of silicon in the metal and bath after addition. The addition was made at t = 0 minute.

Phosphorus and Silicon with Bath Temperature

It can be affirmed that the temperature influences the concentration of phosphorus in the metal and the bath. This can be seen from Figure 8 that the phosphorus concentration in the metal for the prebake cells was roughly between 2-9 ppm. Also, it can be seen that the average bath temperature is 952-962 °C. The phosphorus content in the metal shows a striking trend with temperature. A temperature increase leads to a decrease in the concentration of phosphorus in the metal. Figure 8 shows a clear correlation between high bath temperature and low phosphorus concentration in the metal and vice versa.

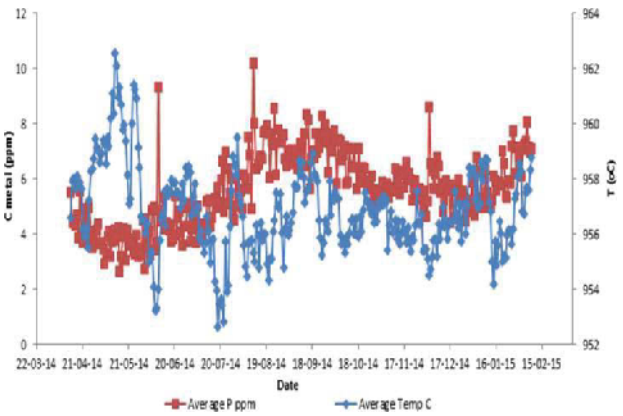

Figure 8. Average values of the phosphorus content in aluminium and bath temperature for QATALUM cells over a period of 1 year.

The high bath temperature causes more phosphorus transfer from the bath to the gas phase. Thus, the phosphorus concentration in the metal is low. Phosphorus evaporation increases with increasing operating temperature. Variations in bath temperature are typically linked with variations in the bath composition. Also, it is linked with changes of the anode-cathode distance (ACD) as well as disturbances such as anode failures or anode effect. These contribute to the low phosphorus concentration in the metal.

It is noticeable that the bath temperature influences the silicon concentration in the metal, see Figure 9. A temperature increase leads to an increase in the concentration of silicon in the metal.

Figure 9 shows a clear correlation between high bath temperature and high silicon concentration in the metal and vice versa. The high bath temperature causes more silicon transfer from the side ledge of the cell to the bath and metal. Therefore, silicon concentration in the produced metal is high. Less side ledge is found with increasing temperature.

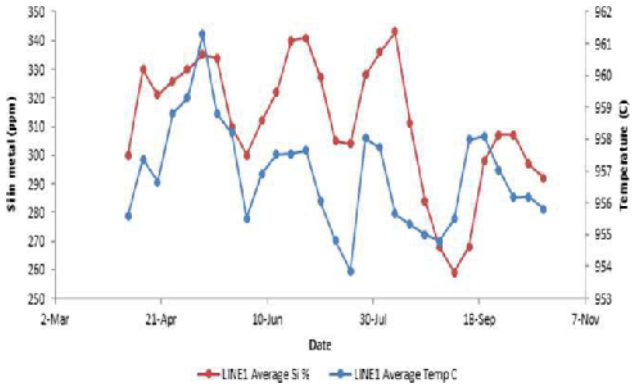

Figure 9. Average values of the silicon concentration in aluminium and bath temperature for QATALUM cells over a period of 8 months.

Silicon Mass Transfer Coefficient

In literature, the mass transfer coefficient is in the range of 10^{-5} – 10^{-6} m/s and it depends on the cell design, technology and operation. Log concentration of silicon in the bath as a function of time after additions of SiO_2 was plotted as shown in Figure 10. The solid line is a linear fit of the data.

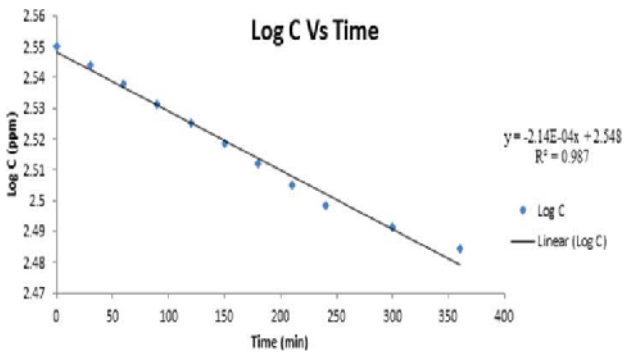

Figure 10. Log concentration of silicon impurity in the bath as a function of time after additions of SiO_2

From Equation 4, Equation 8 is obtained in order to calculate the mass transfer coefficient, k which is approximately found to be 3.1 x 10^{-5} m/s where the area of the cathode is 40 m^2 and the mass of the electrolyte is about 5.8 tonnes. This trend confirms what is known in the literature which is in order of 10^{-5} - 10^{-6} m/s.

$$\log C = \log Co - \frac{A}{V}kt \qquad (8)$$

Phosphorus Mass Balance

For a typical prebake cell at QATALUM, a mass balance concerning phosphorus was derived. The results are summarized in Table I and Table II and briefly sketched in Figure 11. The electrolyte cells are fed with secondary alumina from the dry scrubber. The anodes are covered with a mixture of crushed bath and alumina. Cover materials and solidified bath leave the cells when anodes are changed.

Table I. Input

Material	Phosphorus g/ton Al
Secondary Alumina	19.6
Aluminium Fluoride	1.2
Carbon Anode	3.2
Anode Cover Material	25.7
Soda	0
Calcium Fluoride	0
Total input	49.7

Table II. Output

Material	Phosphorus g/ton Al
Aluminium Metal	6
Carbon Dust	0.6
Excess Tapped Bath	2.8
Mixed Bath Material from Anode Change	20.5
Gas	19.8*
Total output	49.7*

*In order to balance the input and the output, the gas flow has been adjusted accordingly as all the other streams are measured.

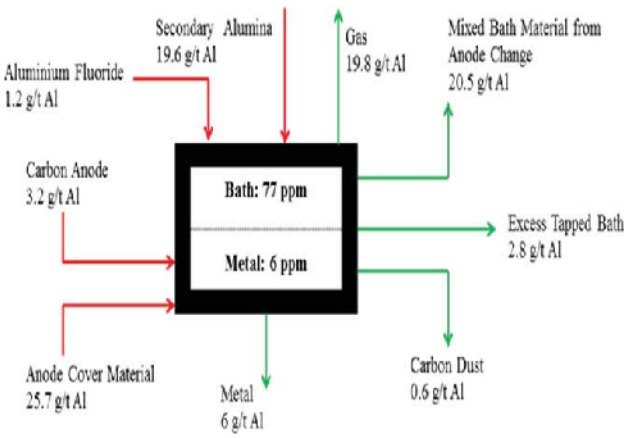

Figure 11. Mass balance for phosphorus. The phosphorus content in the various mass flows for a typical QATALUM cell. The red arrow line marks the inlet. The green arrow line marks the outlet

The difference in phosphorus concentration between primary and secondary alumina depends on the amount of phosphorus escaping from the cells with the process gas. High temperature

leads to high concentration of carbon dust in the bath which results in an increased amount of phosphorus escaping the cell with process gas or with the covering material. Most of the phosphorus in the process gas will be collected in the dry scrubber and recycled to the cells with the secondary alumina. In addition, the phosphorus in the cover material and in the bath removed during anode changing is recycled to the cells as new cover material.

Phosphorus and Current Efficiency

The variation of the sodium content in the produced aluminium can be used to estimate variations in the current efficiency for aluminium. It is a remarkable tool to assist in controlling and analysing the performance of cells. This indicator is frequently utilized in several aluminium smelters to determine and estimate an exact value on a regular basis. It is difficult to properly find out the mass of aluminium produced on a daily basis for an individual cell. The sodium content in the produced aluminium is well linked to the movement at the electrolyte-metal interface. This movement is linked to the current efficiency. Figure 12 shows a solid relationship between the sodium content in the produced aluminium and the current efficiency. This correlation in Figure 12 was obtained from QATALUM industrial cells.

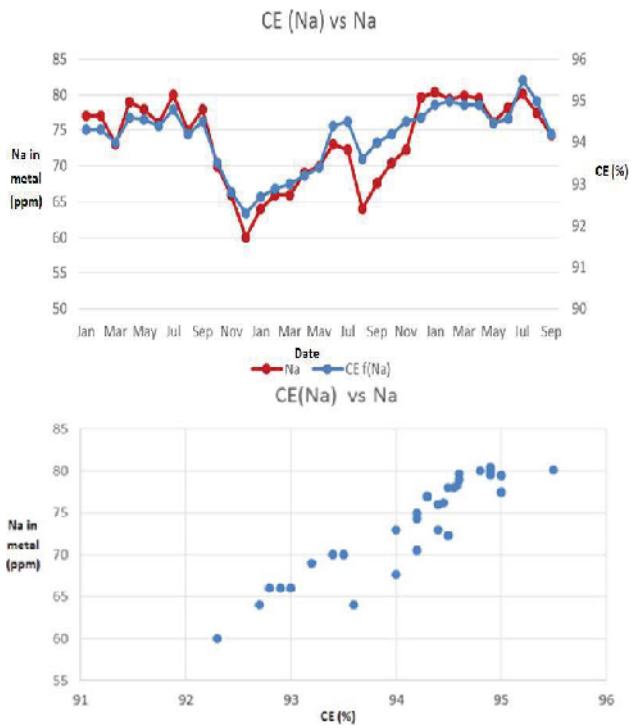

Figure 12. Correlation between sodium concentration in the produced aluminium and the current efficiency

There was an inverse relationship between the phosphorus and the sodium concentration in the metal as shown in Figure 13. A phosphorus concentration increases leads to a decrease in the concentration of sodium in the metal and vice versa. Therefore, the high phosphorus concentration in the metal causes a decrease in the current efficiency of the cell. This trend confirms what is known in the literature and it is the standard in all aluminium smelters.

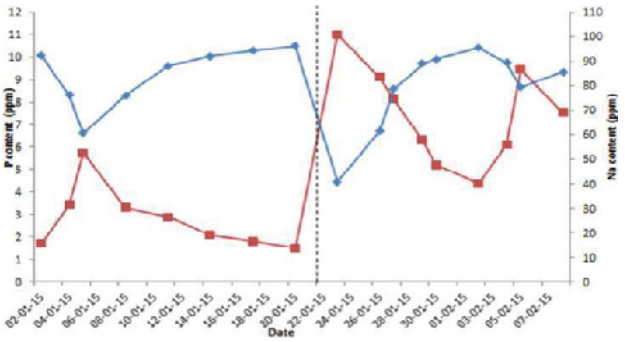

Figure 13. Correlation between phosphorus and sodium content in the metal. The red and blue lines represent the phosphorus and the sodium concentration respectively. The dotted line marks the time of phosphorus compounds addition

Metal Inventory by Copper Dilution

Changes in metal reservoir between two additions of copper can be used together with tapped metal to find the current efficiency of the electrolyte cells. The metal reservoir is calculated using Equation 9.

$$\text{Metal reservoir} = [C_{Cu} \times M_{Cu}] / [1000 \times (C_2 - C_1)] \qquad (9)$$

Figure 14 shows the amount of aluminium metal in the cells is reduced by 600 kg after the phosphorus compounds were added in the cells over a period of two weeks. This indicates that phosphorus has an enormous effect on the production of aluminium due to the huge consumption of current ascribed in the red-ox reaction which otherwise would be used for aluminium production.

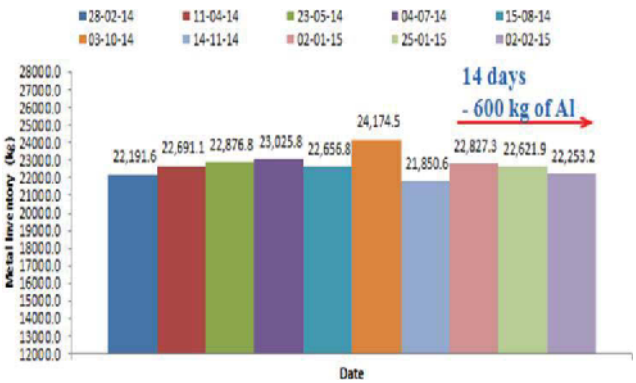

Figure 14. The average values of the metal inventory by copper dilution for the experimental cells at QATALUM over a period of one year. The arrow red line marks the time of phosphorus addition.

Current Efficiency Loss Calculation

There are two approaches to estimate the current efficiency loss. One method to calculate the CE loss is metal inventory approach. At QATALUM, the average metal produced per day is 2400 kg per cell. 600 kg were lost during the two weeks of experiment as shown in Figure 14. 600 kg per 14 days loss is equal to 42.8 kg loss per day. Therefore, for 94.1 CE, tap per day is 2400 kg.

CE loss = [94.1 × 42.8] / 2400 = 1.7 CE loss (10)

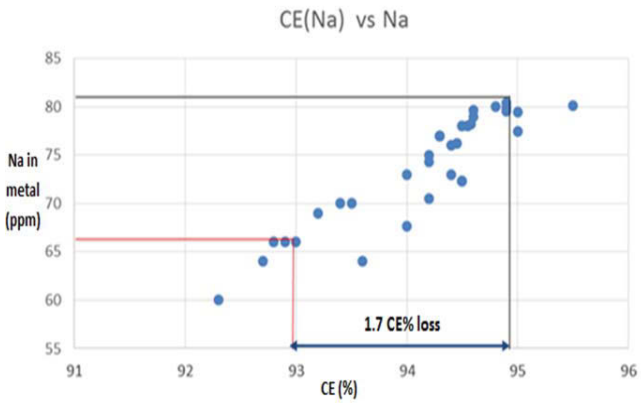

Figure 15. Correlation between Na in the produced aluminium and the current efficiency. The black and red lines mark the Na content before and after the experiment respectively. The blue arrow line marks the total current efficiency loss

The second method to calculate the CE loss is the sodium content in the produced metal approach. The average value of Na in the metal before carrying out the experiment is 81 ppm while the average value of Na after the experiment is 67 ppm. Thus, 14 ppm of Na is lost. It can be noticed from Figure 12 and 15 that the total current efficiency loss during the experiment is 1.7 %. The two approaches demonstrate the same results.

Conclusion

Impurities that can lower the quality of aluminium and cause reduction on current efficiency were added to industrial cells. The measurements showed that the retention time of phosphorus in the bath is much longer than for metallic impurities such as silicon. Also, the data showed that the phosphorus concentration in the produced aluminium decreases with increasing cell temperature while the silicon concentration increased.
Phosphorus behaves differently from other impurities that are nobler than aluminium, e.g. silicon. For metallic impurities nobler than aluminium a mass transport controlled process from bath to metal has been noticed. The mass transfer coefficient for silicon was found to be of the order of 10^{-5} - 10^{-6} m/s.

It was found that a large portion of the phosphorus entering the cells is transported with the off-gases to the dry scrubbers. However, since the secondary alumina is used as feed for the prebake cells, the phosphorus returns and will end up in the produced metal.

Impurity inputs have been linked to cell performance in order to improve Qatalum's process parameters. The following qualitative model for the behaviour of phosphorus impurities in the Hall-Héroult cells is suggested:

- Phosphorus impurities in the raw materials need to be tracked continuously.
- Thermal balance of the cell is considered as the main factor to achieve high level performance.
- Sodium content in the metal should be followed frequently to identify any specific gap regarding cell's performance.

Acknowledgements

The present work was carried out in 2014 and 2015 at Qatar Aluminium Company (QATALUM) and the College of Arts and Sciences in Qatar University. My sincere appreciation goes to QATALUM for giving me the opportunity of carrying out the experiment on impurities addition in the electrolyte cells and is gratefully acknowledged for their financial support.

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Light Metals 2016

ALUMINUM REDUCTION TECHNOLOGY

Fundamentals in Chemistry II

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ALCOA STARPROBE™ – UPDATE IN FURTHER DEVELOPMENT FOR MEASURING CRYOLITE PROPERTIES

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Aluminum reduction, bath chemistry, analysis method, measurement and control, differential thermal analysis

Abstract

Efficient aluminum smelting operation requires routine but necessary measurement and control of individual cell key parameters. Two of the critical measurement and control parameters are cell temperature and bath chemistry. Bath chemistry of interest for the cryolitic bath is cryolite molar ratio and other components such as CaF_2 and Al_2O_3, and sometimes other additives such as LiF and MgF_2. Determination of these properties requires various analytical equipment and rigorous procedures from tedious sampling to analytical preparation. STARprobe™, an Alcoa proprietary instrument specifically developed and designed for use in aluminum smelters, combines tedious time-consuming sampling procedures and all expensive analytical equipment into one simple portable instrument for potroom operators on potroom floor. Now, the STARprobe™, together with Alcoa proprietary Integrated Pot Control system, becomes an integral technology component in Alcoa smelters for an efficient operation. This paper will update our further development in more accurately measuring alumina concentration, and also addition of property measurement including %CaF_2 and %LiF when needed.

Introduction

The efficiency of aluminum smelting cells relies on sophisticated control in maintaining cell's thermal and material balances by regulating resistance, bath chemistry and alumina feed. A good control of the cells is dictated by reliable and accurate measurement of key cell's operating parameters, i.e. cell (bath) temperature and electrolyte chemistry (cryolite ratio and alumina concentration).

Modern cell technology has steadily progressed in the last decades including amperage scale (cell size) and operation practices. Sophisticated pot controls are also incorporated in the modern cells. However, some of the critical information which requires for the control is still primarily replying on the traditionally infrequent manual measurement and sampling/analysis. The aluminum smelting cells – small (as low as 100KA) or big (as high as 600KA) – are very robust in operation, and have one thing in common - material and energy balance states in each cells are different and constantly changing – as referred as "dynamic state": The changes (magnitude/speed) of dynamic state are different from cell to cell even they are at the same amperage potentially due to difference in 1.) Metal pad/bath levels, 2.) Age, feeds, cover, anodes, cleanliness, and others. The changes mean different chemistry (% AlF_3 or ratio), temperature, and superheat. For an efficient operation (also health of the cells), all controls are geared toward to maintain all parameters to targets and variations to minimum. Advanced sensors and novel measurement methods are therefore no longer a nice to have, but rather a necessary first step toward achieving such controls.

The paper published in Light Metals 2011 (1) presented Alcoa's STARProbe™ (shown in Figure 1 as refresher), an apparatus used in potroom floor to measure bath ratio (acidity), alumina, temperature and superheat, overcoming the drawbacks of the traditionally manual measurement of bath temperature and tedious sampling/analysis methods. The apparatus makes it possible to link both material and energy dynamic states for more effective control in productivity and energy efficiency improvement (2).

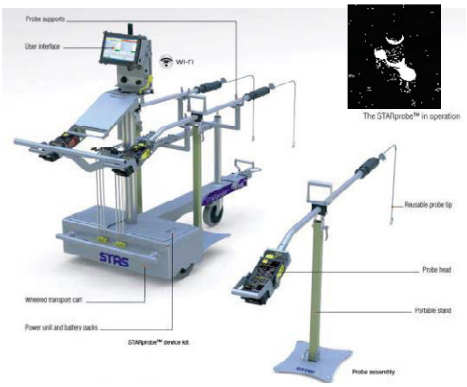


Figure 1 Alcoa STARProbeTM measurement system as manufactured by STAS, Inc (courtesy of STAS, Inc).

Since then, Alcoa has been basically transferred itself in both measurement and control in potroom operation, and at the same time, improvement in the system for robustness to ensure more accurate measurement of key bath properties including more accurate measurement of alumina concentration, additions of measurement features such as %CaF_2 and limited concentration of %LiF were also made. This paper will describe the past progress made in deploying the technology across Alcoa and development of more features for better analysis and therefore more efficient control. A brief update on the STARProbe™ application for active pot control is also provided.

Theoretical Background

Basic theoretical background has been described in the previous paper and will not be repeated here. However, it is worthwhile to re-emphasis the uniqueness and differentiation of Alcoa STARProbe™ from other reported and even commercialized methods which also claim to be capable of measuring cryolitic bath components. It is the adoption of differential thermal analysis (DTA) (3) using two thermocouple system (4): one thermocouple measuring the temperature variation of bath sample (which is going through phase changes/transformation) while the other thermocouple measures the probe body temperature as reference while the whole system is being cooled under various ambient conditions.

When a cryolitic melt is cooled from liquid state to a solid state, it will go through several phase transformation/changes. The two TC system is not only capable of determining the temperatures in the onset of phase transformation (such as freezing point), more importantly, it allows to quantify the phases transformed (such as from liquid to solid) in a certain temperature range based on its amount of heat release. And this amount of heat release may not and cannot be determined by a single temperature system.

For the cryolitic bath, the primary component is cryolite (Na_3AlF_6). As an example, the principle is described below in how to obtain the amount. When a cryolite bath starts to freeze, the Na_3AlF_6 does not freeze out at once, the amount becoming solid depends on not only initial composition (T_{liq}), but also on cooling temperature T (which is also a time dependent).

The total amount Na_3AlF_6 can therefore be obtained by:

$$W_{Na3AlF6} = \int_{Tliq}^{Teut} d(W_{Na3AlF6}) \cdot dT \quad (1)$$

Where T_{liq} and T_{eut} are sample liquidus and eutectic temperatures. Since the heat of fusion (release) of Na_3AlF_6 is constant, the amount of Na_3AlF_6 being frozen out will be proportional to amount of heat released which is then reflected by the magnitude of differential temperature change:

$$\begin{aligned} W_{Na3AlF6} &= \int_{Tliq}^{Teut} d(W_{Na3AlF6}) \cdot dT \\ &= C^* \int_{Tliq}^{Teut} d(f(\Delta H^0_{Na3AlF})) \cdot dT \\ &= C^* \Delta H^0_{Na3AlF6} * \int_{Tliq}^{Teut} d(\Delta T_i) \cdot dT \\ &= C^* \Delta H^0_{Na3AlF6} * \int_{Tliq}^{Teut} d(T_i - T_{ref,i}) \cdot dT \end{aligned} \quad (2)$$

Where $\Delta H^0_{Na3AlF6}$ is entropy (heat of fusion) of pure cryolite (Na_3AlF_6), T_i and $T_{ref,i}$, temperature of sample and reference. The amounts of other components (such as chiolite, $Na_5Al_3F_{14}$) in the cryolitic bath can also be determined in the similar way.

Since the development and deployment of Alcoa STARprobe™ across Alcoa smelters, there have been no basic changes in the probe configuration (design), its component (TCs used) or how it is used for measurement. There are also no changes in how bath temperature, bath liquidus temperature and bath ratio (excess AlF_3 - %XS acidity) is determined by the probe and therefore will not be re-described here in this paper. All changes that occurred have been in the hardware and software improvement of robustness, more user friendly, real time results transfer and active pot control, and addition of new measurement features.

Current STARProbe™ application within Alcoa

Alcoa STARProbe™ has been deployed across Alcoa smelters for active pot control. At the very early stages in 2004, it was primarily used as a pot diagnostic tool when the probe started to take shape and electronics had to be developed, as illustrated in Figure 2. As shown in Figure, it took awhile for it to be truly used as a pot control analysis tool: use as pot diagnostic problem solving tool is easy, to replace the traditional methods and use by operators daily for active pot control is another story. In potroom, bath ratio or acidity and pot temperature are the common phrase – probably only words spoken and referred. There had to be a cultural change and acceptance for transitioning to STARprobe™, a tool like garden hoe or fishing pole from those "sophisticated" and air-conditioned analytical equipment when people expected to wait to know bath chemistry. After it was first proved that the pots (a section of pots) were successfully controlled with STARProbe™ without the use of the traditional analytical methods for a prolonged period of time (as long as half year), plant operations started to accept. It is the change in pot control culture from potroom manager, engineers to potroom supervisors and operators that made the transition to STARprobe™ possible across Alcoa smelters. Now, "Superheat" which measures dynamic balances of energy and materials in each pot, a pot or line healthiness indicator, becomes a fashionable word. STARProbe™ application across Alcoa smelters:

- Started with pot diagnostic tool in 2005
- Started bench scale test in active pot control in 2006
- Smelter wide deployment in 2009
- Peak (both numbers of cells and number of measurements) reached in 2012

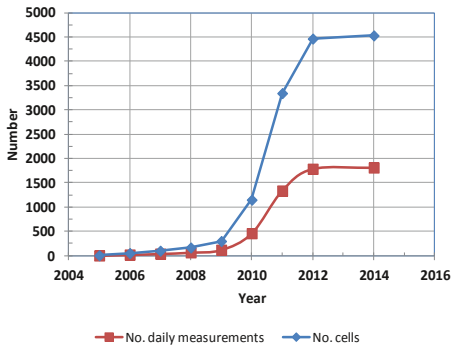


Figure 2 Number of pots and number of daily measurements of STARProbe across Alcoa smelters since 2005 for active pot control.

Improvement in STARProbe™ system

The STARProbe™ is operated by people. Even though a simple step by step procedure is established and required to follow in carrying out a measurement, it is still subjective after all in the real world. The running program has to be smart enough to detect, correct or terminate a test before it goes too far in the process during a measurement. The following improvements have been made to ensure a non-subjective measurement can be made by anyone after given a brief training:

- Proper immersion depth of the probe in molten bath: proper insertion of probe into molten bath of the measuring cell is critical to measure bath temperature and bath properties. Very often operator, either tentative because of afraid of metal contact or not careful, places the probe tip too shallow or sometimes only partially immersed, resulting in a non-representative too low bath temperature and no detectable bath melting temperature. Algorithm was added to best detect such conditions to remind the operator to place he probe properly in molten.
- Proper position of probe in sample hole: Situations are sometimes encountered where operator places the probe too close to (even in contact) endwall or sidewall in the sample hole, resulting improper measurement. During measurement, the program also detects some cases and automatically reminds operator to make corrections.
- Avoiding metal contact: It is not uncommon when aluminum metal (from metal pad) is in contact with probe tip when the probe tip is immersed in molten bath either in getting a stable temperature or in reheating step before dumping out molten bath. In such as cases, program will stop operator and request a probe tip replacement.
- Avoiding molten sample spill from probe tip cup: When operator is lifting the probe tip out of molten bath to cool during measurement, he/she may sometimes incidentally spill molten bath sample out of the probe cup. The program also automatic detects, stops the analysis process and tells operator to repeat the sampling process.
- Detecting improper cooling during cooling stage: for whatever reasons, during the measurement process when the probe tip is out of the molten bath being cooled, probe is back to heating up process even for a short period of time, program has to be modified to add capability to detect and terminate the analysis.
- Probe tip replacement: the probe tip is designed for multiple uses until it reaches to its life. The program automatically tell operator to replace with a new or newly conditioned probe tip to ensure bath property measurement accuracy.

In addition to those improvements mentioned above, some algorithms were also modified to identify and measure key properties of the cooling temperatures in a more robust way. Wireless data/results transfer continues to evolve along with computer/radio wireless technology changes.

Toward more accurate measurement of alumina in bath

Efficient pot operation is all about alumina feed control at desired bathchemistry and temperature conditions. Since alumina is usually controlled within a specific range, its concentration is not usually measured on a routine basis. Alumina concentration can sometimes be out of control either becomes too lean or too rich. Lean conditions can often be a result of blocked feeders or bad alumina ore under which conditions alumina cannot be fed or dissolved into molten bath. Rich alumina conditions can be a result of single or multiple causes such as from leaker feeder, noisy pot or hot pot conditions. Rich alumina conditions significantly reduce electrolyte liquidus temperature as shown in Figure 3, not only causing excess high superheat, but also throwing bath ratio or acidity off balance. Under those conditions, the pot is anything but efficient. Therefore, alumina concentration measurement is necessary for these conditions to be corrected in a timely manner so that pot can be brought back to peak performance.

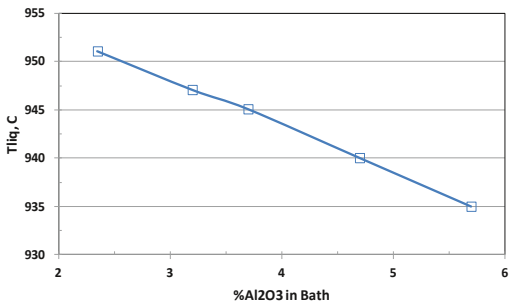


Figure 3 Bath melting temperature or liquidus temperature as impacted with the increase of alumina dissolved in an industrial bath composition as measured with STARprobe™.

Since the release of the STARProbe™, the precision for alumina concentration measurement has been not been able to match with lab analytical equipment such as Leco or XRF due to the alumina precipitation characteristics of low alumina concentration from bath. We continue to manipulate algorithm for improvement in detecting and measuring the alumina temperature response (even overlapped with other bath components) so that alumina in bath especially rich conditions can be measured in a more reasonable manner so that corrective measures can be applied to those problematic pots. If alumina is determined to be within a normal targeted range, no actions in feed control are applied.

Figures 4 and 5 show the comparison of alumina concentration as measured with STARProbe™ and plant LECO or XRF method at two smelter locations. The results as measured by STARprobe™ match the traditional methods fairly well for most cases especially for rich alumina conditions.

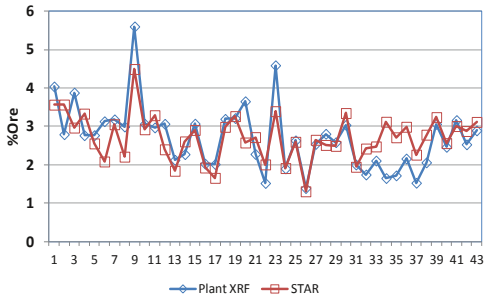


Figure 4 Alumina in bath as measured by STARProbe against plant XRF method of the bath samples taken during STARProbe measurement at one smelter location.

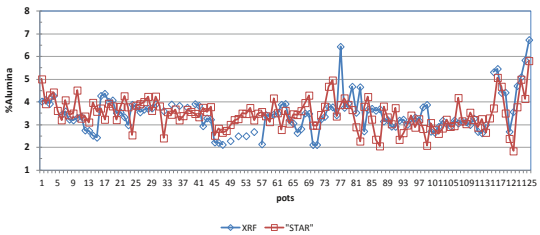


Figure 5 Alumina in bath as measured by STARProbe against plant XRF method of the bath samples taken during STARProbe measurement at another smelter location.

%CaF_2 determination during regular measurement

Calcium fluoride (CaF_2 or spar) level in cryolitic at specific smelters (depending its pot technology and current density) is usually dependent upon the SGA which contains a certain concentration of CaO. Its average level usually does not change much if a specific ore source is used and without varying much amperage. However, the CaF_2 level in individual pot does vary pot from pot because of variation of electrolyte composition and thermal state. Even though not controlled from pot to pot, it is usually managed on the plant average level to a certain range so that the smelter can be operated in a stable and efficient way.

%CaF_2 changes/reduces electrolyte liquidus temperature. Figure 6 shows the bath liquidus temperature as impacted by the variation of %CaF_2. It decreases from 2.4 to 2.8C for each percent increase in %CaF_2 from low ratio (high acidity) to high ratio (low acidity) electrolyte condition. If there is a significant shift in the %CaF_2, its impact on the bath liquidus temperature (together with other physical properties) can no longer be ignored.

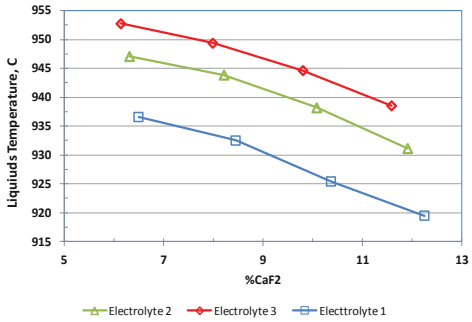


Figure 6 Bath melting temperature or liquidus temperature as impacted with the presence of CaF_2 in bath as measured by STARProbe in three electrolyte compositions.

In a single smelter, CaF_2 can vary pot to pot, Figure 7 shows the %CaF_2 variation as analyzed by XRD, is shows 1.0% variation range.

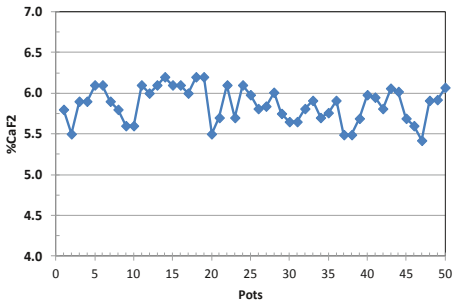


Figure 7 The CaF_2 variation range in a smelter.

The %CaF_2 can also vary significantly smelter from smelter. Figure 8 shows the average of %CaF_2 across most of our smelters from as low as 4% to as high as over 7%. A 3% CaF_2 variation can result as high as over 6C change in bath liquidus temperature which significant impacts bath chemistry (cryolite ratio or acidity) and therefore pot performance.

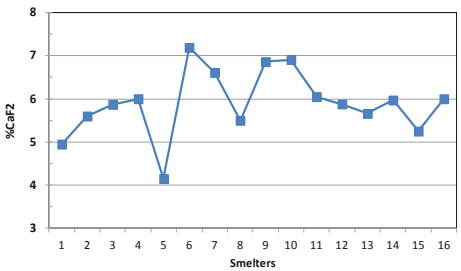
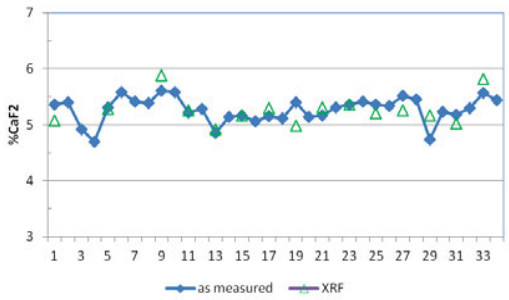


Figure 8 The %CaF_2 average from smelter to smelter.

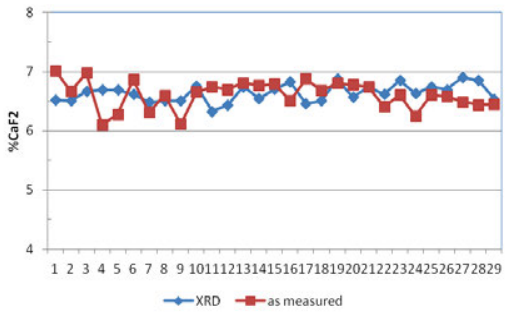
As shown in Figure 8, CaF_2 exists (as one of additives) typically in a range from 4 to 8%. Monitoring %CaF_2 in each smelter has

some practical interest as it will change over time because of ore sources changes.

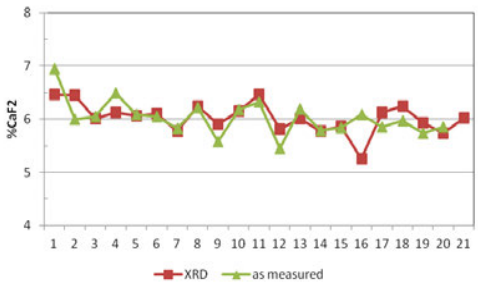
When CaF_2 as a so called "additive" freezes out during sample cooling stage in STARProbeT[M] measurement, it does form solid Ca cryolite ($NaCaAlF_6$) precipitates with release of heat which may be significant enough to impact the shape or bath cooling characteristics. Accordingly, attempt has been made to determine the %CaF_2 in bath during each STARProbe[TM] measurement. Figure 9 (a) to (c) shows the typical results of %CaF_2 in each STARProbe[TM] measurement at 3 smelters. These 3 smelters have a different average CaF_2 level.



(a)



(b)



(c)

Figure 9 %CaF_2 as measured by STARProbe[TM] measurement against plant XRD method: top to bottom: a, b and c smelter locations.

The %CaF_2 as measured by STARprobe[TM] matches with the XRD method is considered to be good enough to allow plant to monitor its average and make adjustment of the spar level necessary.

Addition of measuring LiF modified bathchemistry

LiF is sometimes added as an additive for some bath chemistries in certain smelters. LiF, like NaF, does behave like a base and will increase the basicity of cryolitic bath, and will form solid Li cryolite (Li_3AlF_6) when LiF freezes when cooling temperature reaches a certain temperature.

Figure 10 shows the LiF in reducing bath liquidus temperature: for 1% initial increase of LiF, the bath liquidus temperature can be reduced from 7 to 12C depending on the bath ratio.

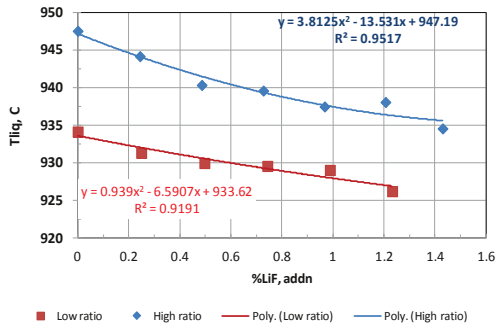


Figure 10 Bath melting temperature or liquidus temperature as impacted with addition of LiF measured with STARProbe[TM].

Attempt has been made to determine %LiF for low LiF containing bath chemistry using STARprobe[TM] as a new feature addition with a certain degree of success. Figures 11 and 12 show the results comparison of %LiF. The STARProbe[TM] was calibrated at ~1.5% LiF level in an operating plant, a year later, a recheck was made to compare STARProbe against XRD, the results still matched well.

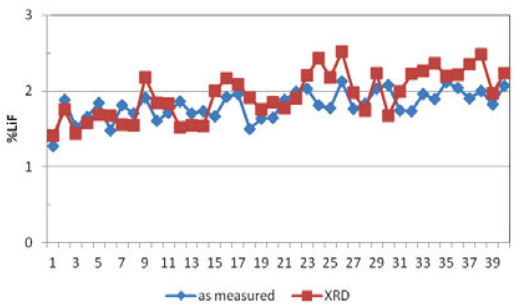


Figure 11 LiF in bath as measured with STARProbe[TM] against XRD method

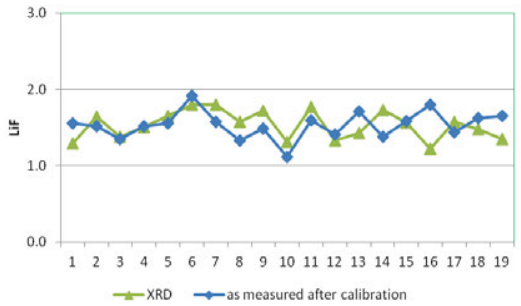


Figure 12 %LiF in bath as measured with STARProbe™ against XRD method

Currently, STARProbe™ is being used to measure %LiF in addition to all other "STAR" information for active pot control. However, it has been limited to low LiF concentration range at this time. Additional development work is needed for high LiF bath chemistries.

Summary

Alcoa STARprobe™ is designed and developed for real time measurement of key cryolitic bath properties, i.e. bath superheat, temperature, alumina and ratio (%AlF_3) for active pot control. The measurement system is based on a sound theoretical principle with a unique engineering design offering an accuracy/precision comparable with traditional laboratory based analytical techniques. The technology has been successfully deployed across Alcoa smelter to together with its own proprietary control package.

The earliest smelter has been using the STARprobe™ for active pot control for over 6 years without disruptions. And now, it has grown into an inseparable technology within Alcoa smelting operations.

Continuous improvements have been made on the user friendliness and robustness of both the hardware and software so that the measurement efficiency and results are no longer subjective to operators. Wireless communication transfers real time measurement results to plant control system while without compromising network security.

Improvement in the determination of alumina concentration, especially at rich conditions, allows for quick and effective actions on those problematic pots so that overall potline performance can be improved or maintained at its peak performance.

Advanced features have been added on the current STARprobe™ including determination of 1.) %CaF_2 which allows smelters to monitor and adjust spar level when necessary, and 2.) %LiF (for LiF containing bath chemistries) to control LiF concentration. The addition of these features further potroom reduces reliance on expensive and yet necessary laboratory equipment.

Acknowledgement

Author wishes to thank Alcoa Global Primary Products for allowing publication of this work. Appreciations go to those who have been involved from development, testing and implementation of the technology across Alcoa smelters.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Analysis and Visualization of Aluminum Reduction Cell Noise Based on Wavelet Transform

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Keywords: Wavelet transformation, reduction cell noise, pseudo resistance, process control

Abstract

The unsteady part of a cell pseudo-resistance signal called noise is closely related to electro-chemical and magneto-hydrodynamics processes, which is consequently reflected in its non-stationary and transient character, particularly in cases of different disturbances. Therefore wavelet transformation has been introduced in order to examine cell noise simultaneously within time-frequency space, thus enabling a deeper view of the process. A special data acquisition system has been installed on the AP18 Pechiney cell, with the capability of conducting continuous measurements of the cell's operating data, performing wavelet analysis and evaluating a result using the data obtained from the plant control system. In this paper the analysis and visualization of cell resistance noise, based on continuous wavelet transformation and wavelet packet transformation is proposed, in order to obtain complementary data for more efficient process control. The research results showed promising potential for using the wavelets for the visualization of dynamics and the detections of different disturbances during the aluminum reduction process.

Introduction

It is difficult to control the aluminum reduction process in real time because corrosive liquid electrolyte, high temperature and the distributive nature of the process prevent the using of any sensors for continuous measurements of the most important process variables like alumina concentration, anode-cathode distance and bath temperature. Therefore all modern real time reduction cell control strategies are based on the continuous measurement of cell pseudo-resistance as the global operational parameter, from which dedicated information's are extracted and used as input data for the main control functions like alumina feed, heat balance and process stability. The unsteady part of the pseudo-resistance signal called noise represents usually only 1% of the total DC signal and is related to physical and chemical processes taking place within the inter-polar space of the electrolysis cell. In case of the process instability the noise pattern is changed and could be used for the identification of a reason for irregular cell operation.
The simple noise-control strategy detect if the amplitude of the noise goes beyond the predefined limits and consequently trigger the alarm or even increases the target cell resistance [4]. More complex systems use advanced methods for noise classification, like special metrics [3], fuzzy logic [7] or neural networks [5].
Taking into account the non-stationary and transient character of pseudo-resistance noise, wavelet transformation could be the perfect method for extracting some complementary information for achieving better process control. Over the last 20 years the wavelet theory has been applied for investigating a multitude of diverse physical phenomena like climate monitoring, seismic signals, rotating machinery, turbulence, medical signals, etc.
In contrast there are fewer publications on the use of wavelets in the aluminum electrolysis process than would be expected [8][9].
In this paper, we focus on applying the wavelets for visualizations of the dynamics and detections of disturbances during the aluminum reduction cell process.

Brief Review of the Wavelet Theory

Wavelet transforms are signal analysis methods which offer simultaneous interpretations of the signal in both time and frequency and thus allows identifications local, transient or intermittent components of the signal. Wavelets are special kinds of functions, with oscillatory behavior, short duration and finite energy, as shown in Figure 1a. The wavelet can be moved to various locations of the signal (Figure 1b) and can be stretched or squeezed (Figure 1c) [1].

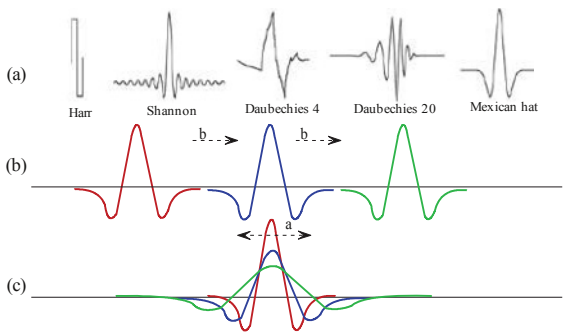

Figure 1: (a) Typical wavelets. (b) Locations. (c) Scale.

Continuous wavelet transform

The continuous wavelet transform (CWT) as a function of scale and position taken on signal *x(t),* is calculated as shown in (1):

$$T(a,b) = \frac{1}{\sqrt{a}} \cdot \int_{-\infty}^{\infty} x(t)\, \psi^*\left(\frac{t-b}{a}\right) dt \qquad (1)$$

Where ψ^* denotes the complex conjugate of the daughter wavelet which is a time-shifted and scaled version of the mother wavelet $\psi(t)$, b is the time-shifting parameter, a is the scaling parameter. Figure 2 illustrates the principle of calculating the wavelet transform represented by the equation (1). For every pair of values *(a,b)*, we obtain wavelet transform coefficients *T(a,b)* which represent how much the scaled wavelet is similar to the function *x(t)* at location *b*.
The total energy contained within a signal x(t) is defined as (2):

$$E = \int_{-\infty}^{\infty} |x(t)|^2\, dt = \|x(t)\|^2 \qquad (2)$$

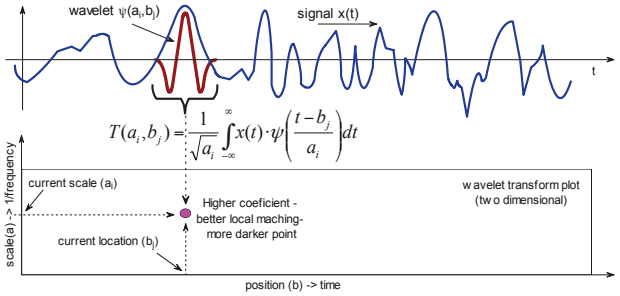

Figure 2: Wavelet transform principle

Taking into the account the energy conservation is the distribution of the signal energy contained at a specific a scale and b location (time) given by the two-dimensional wavelet energy density function, displayed in the graphical plot of which is known as a scalogram (3):

$$E(a,b) = |T(a,b)|^2 \tag{3}$$

The raw wavelet scalograms are not popular for interpreting because of their non-typical forms and the general orientation of engineers towards evaluating physical phenomena by means of Fourier transformation. In order to evaluate the results of the CWT within the context of a time-frequency spectrogram, we may convert the scale-dependent wavelet energy spectrum of the signal to a frequency-dependent wavelet energy spectrum [1]. The wavelet scale is converted to a frequency (more precisely pseudo frequency) by the introduction of a wavelet central frequency, which captures the dominant oscillation of the selected wavelet, as shown in Figure 3.

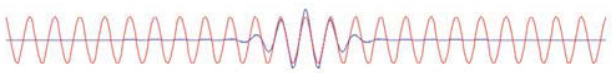

Figure 3: Central frequency of the wavelet

The mapping between scale and pseudo-frequency is done by (4):

$$f_a = \frac{f_c}{a} \tag{4}$$

Where a is the scale and f_c is the wavelet central frequency. The wavelet energy as a function of pseudo frequency and time can be obtained by substituting (4) for a and b=τ using (5):

$$E(f_a,\tau) = \left|\sqrt{\frac{f_a}{f_c}}\cdot\int_{-\infty}^{\infty}x(t)\,\psi^*\left(\frac{f_a}{f_c}(t-\tau)\right)dt\right|^2 \tag{5}$$

Wavelet packet transform

In the Discrete Wavelet Transform (DWT) is an adjustment of the scale and translation parameters carried out in discrete steps. This process is done using the quadrature mirror filter bank structure, illustrated in Figure 4. The original discrete signal is decomposed into vectors of the wavelet coefficients with two-band splitting of the input signal on approximation coefficients cA1 (low frequency) and detail coefficients cD1 (high frequency). These vectors are obtained by filtering and down-sampling at each decomposition level using the finite impulse response low-pass g(n) and high-pass h(n) filters, derived at from the corresponding scaling function and the wavelet function.

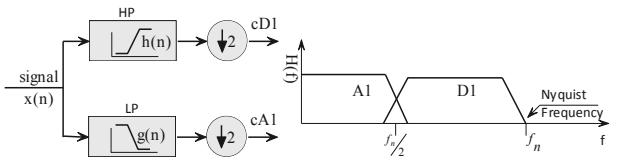

Figure 4: Single level filter bank structure

The wavelet packet analysis is generalization of DWT, where both approximation and detail coefficients can be iteratively split into finer components. The wavelet packet (WP) coefficients of signal x(t) can be calculated as a inner product below (6) [1]:

$$T_{m,n}(k) = \langle x(t), 2^{-\frac{m}{2}}\cdot\psi_n(2^{-m}\cdot t-k)\rangle \tag{6}$$

Where $T_{m,n}(k)$ denotes the n=0,1…2^m-1 set of WP coefficients at the m-th scale parameter, k is the translation parameter and ψ is the wavelet function. The result of this algorithm is presented as the decomposition tree structure with associated frequency bands and time-frequency tiling pattern, illustrated in Figure 5.

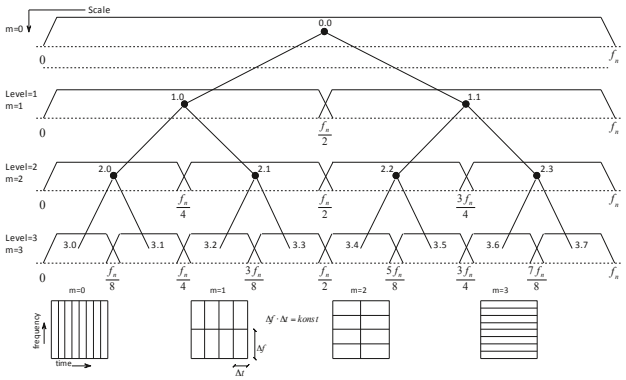

Figure 5: WP decomposition tree with time-frequency tiling

The WPT decomposes a signal into uniform frequency bands of width $f_{bm} = f_n \cdot 2^{-m}$, with coefficients which contain information about the signal's behavior inside the associated frequency bands. The feature extraction methods were used on the wavelet coefficients in order to operate with a reduced quantity of key information, called the feature vectors.
For examining the stationary signal characteristics, the energy feature vector is calculated by (7), which represents the energy stored within the coefficients at each level (frequency band).

$$E_m = \sum_n |T_{m,n}(k)|^2 \tag{7}$$

The total energy of the observed signal is obtained by (8):

$$E_{tot} = \sum_m E_m \tag{8}$$

Shannon entropy of wavelet packet coefficients is proposed for identifying the dynamic characteristics of a signal, calculated as the entropy feature vector (9).

$$S_m = -\frac{E_m}{E_{tot}}\cdot\log\left(\frac{E_m}{E_{tot}}\right) \tag{9}$$

Cell pseudo-resistance noise classification

The cell pseudo-resistance noise classification by frequency is shown in Figure 6. Stable, normally operating cells have background (bubble) noises, Gaussian natures with dominant frequencies of 0.4–2 Hz and amplitudes of less than 30 mV. The reason for bubble noise is due to rapid changes in anodes current densities and localized disturbances in the metal pads from the formations and releases of bubbles from the inter-polar spaces [10]. Unstable cell generate additional noise components with frequency patterns depending on the reasons for instabilities. Cathode noise has a frequency less than 0.1 Hz and occurs due to a standing wave in the metal-bath interface (referred as metal roll or MHD instability). Anode noises have frequencies above 0.1 Hz and occur due to an anode problem like a spike or low-set anode [4]. In addition there are additional temporary technical noises caused from the actuators such as are changes due to the ACD adjustment and alumina feeding.

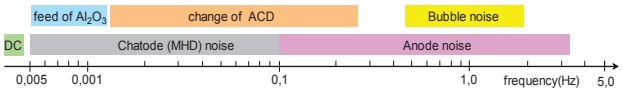

Figure 6: Aluminum reduction cell noise classification

The standard indicator for assessing cell instability is the maximum amplitude of the resistance during a given period (10). In the observed AP18 cell instability is detected if is *wrmi>0.3* and is treated with increasing of the target resistance if is *wrmi>0.8*.

$$\mathrm{wrmi} = \mathrm{MAX}(R_i) - \mathrm{MIN}(R_i) \qquad (10)$$

Wavelet analysis applied to electrolysis cell noise

The wavelet theory was applied mutually with the electrolysis cell monitoring concept using the parameters shown in Table 1, in order to compare the results with data obtained from the potline process control system. For this reason the length of the pseudo resistance input signal is equal to the duration of the pot controller regulation cycle (T_c=300s). The general rule of mother wavelet selection is to choose a mother wavelet having the similarities with the analyzed signal. Using these criteria was selected the *sym8* wavelet.

The CWT spectrogram was used for visualizing the noise simultaneously within the time-frequency space and thus enabled studying of the dynamic and transient phenomena of the reduction cell process. WPT analysis has been used for identifying the global characteristics of a pseudo-resistance noise by calculating the energy and entropy feature vectors for each cell regulation cycle in accordance with the WPT decomposition schema. Thus the obtained feature vectors arranged into frequency bands and chronologically merged as semi spectrograms are suitable for observation of the noise trend, together with the cell regulation data.

The observed frequency band from 0.01Hz to 5Hz was set to capture all individual frequency components of the noise; consequently the sampling of the input signal was fixed at 10Hz. The variable frequency resolution, as a characteristic of the wavelet transformation enables an arbitrary setting of the resolution throughout the whole frequency range. In the case of CWT this was done by calculation of the nonlinear pseudo frequency ramp using an empirical formula (11). The wavelet packet decomposition tree was the main WPT parameter with adequate sub-bands designed to cover all phenomena within the noise classification scheme, shown in the second part of Table 1.

Common parameters		
Length of input signal	T_c	300s
Sampling frequency	f_s	10Hz
Selected wavelet function	ψ	sym8
Analyze frequency range	f_{min} - f_{max}	0,01- 5Hz
Specific CWT parameters		
Number of frequency steps	N_f	256
Step for location	τ	1s
Frequency ramp	$f_a[i] = f_{min} \cdot \left(1 + \frac{i}{N_f} \cdot \left(\left(\frac{f_{max}}{f_{min}}\right)^{\frac{1}{4}} - 1\right)\right)^4$	(11)
Specific WPT parameters – decomposition tree		

Dec Level	Level Bandwidth	Nb. Bands	Frequency Resolution
10	0,000 ÷ 0,010Hz	2	0,005 Hz
9	0,010 ÷ 0,020Hz	1	0,01 Hz
8	0,020 ÷ 0,079Hz	3	0,02 Hz
7	0,079 ÷ 0,157Hz	2	0,04 Hz
6	0,157 ÷ 0,313Hz	2	0,079 Hz
5	0,313 ÷ 0,625Hz	2	0,157 Hz
4	0,625 ÷ 1,250Hz	2	0,313 Hz
3	1,250 ÷ 5,000Hz	6	0,625 Hz
TOTAL	**0,000 ÷ 5,000Hz**	**20**	-

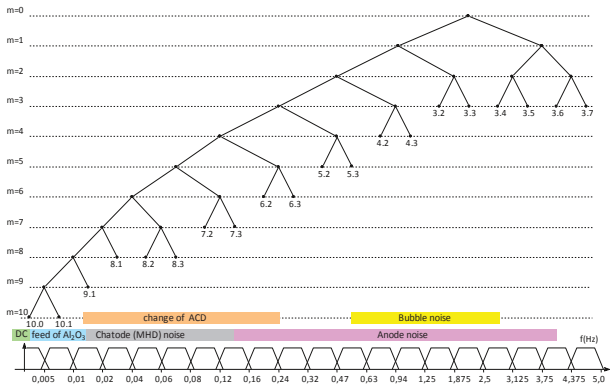

Table 1: Parameters for wavelet analysis

Experimental data acquisition system

A computer data acquisition system was developed and installed on a reduction cell, parallel with the cell control system. National Instruments LabView software and PXI hardware were used for the deployed architecture shown in Figure 7. The installation of measurement system is shown in Figure 8.

For the continuous measurements within harsh smelter environmental conditions the PXI computer system was selected with a PXI-8187 controller, two PXI-4071 7½ digit multimeters (resolution 10μV) and a PXI-6515 digital input card. The measurement system was externally triggered at a sampling rate of f_s=10Hz for synchrony acquisitions of cell voltage U_i, line current signal I_i, feeding rate of Al_2O_3/AlF_3 and orders for anode beam control. After each measurement the calculation of cell pseudo resistance as a basic input variable for wavelet analysis was performed using a well-known equation (12):

$$R_i = \frac{U_i - 1.65}{I_i} \qquad (12)$$

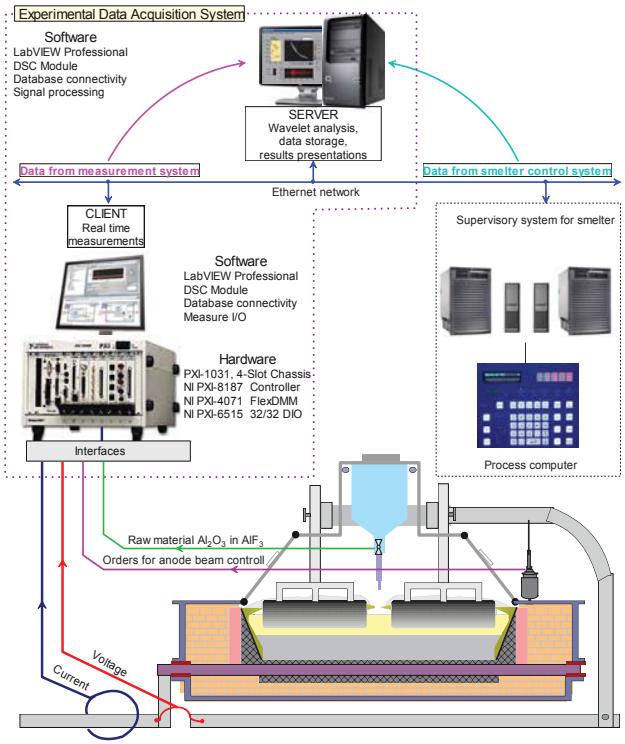

Figure 7:Experimental System's Architecture Overview

The client measurement PXI system and server computer were connected via the ethernet network and exchanged the field measurement data using the LabView shared variable engine. On the server computer ran the LabView application, dedicated to resistance noise analysis using the following main functions:

1) Communication with the distant field measurement system
2) Measurements preprocessing (elimination of DC component)
3) Continuous and wavelet packet transformations
4) Results presentation in graphical forms
5) Database transaction

Figure 8: Installation of measurement system in the potline

Results and discussion

The Pechiney AP18 cell was instrumented using a presented data acquisition system, parallel with a pot control system. During measurements the reduction process ran under regular working conditions without imposing any artificially disturbances.

Continuous wavelet transform

CWT results are presented in the form of a wavelet energy spectrum together with measurement data. Figure 9 shows the CWT spectrogram, which belongs to the stable process (wrmi=0.08). The noise signals were mainly caused by bubble activities, with frequencies between 0.5-1.5Hz.

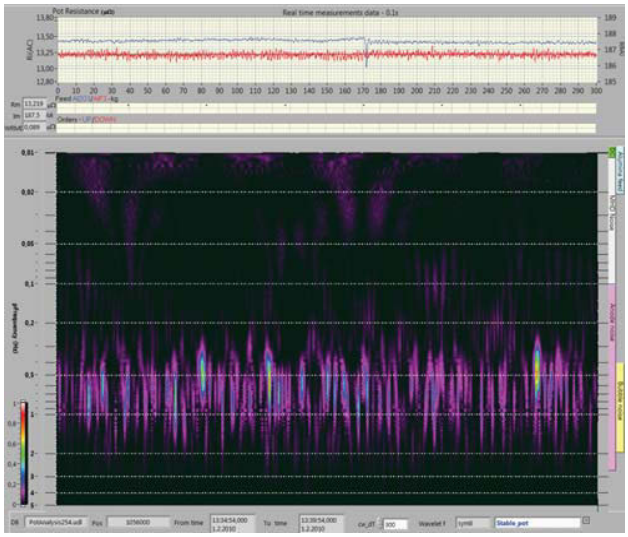

Figure 9: CWT noise spectrogram for stable process

The early stage of instability (wrmi=0.22), caused by an anode problem is shown in Figure 10. The intensities of bubble activities increased due to irregular current distribution and consequently local changes in anode current densities. The noise contained additional frequencies between 0.1-0.2Hz, with the time variable intensities and short-circuit events (spikes).

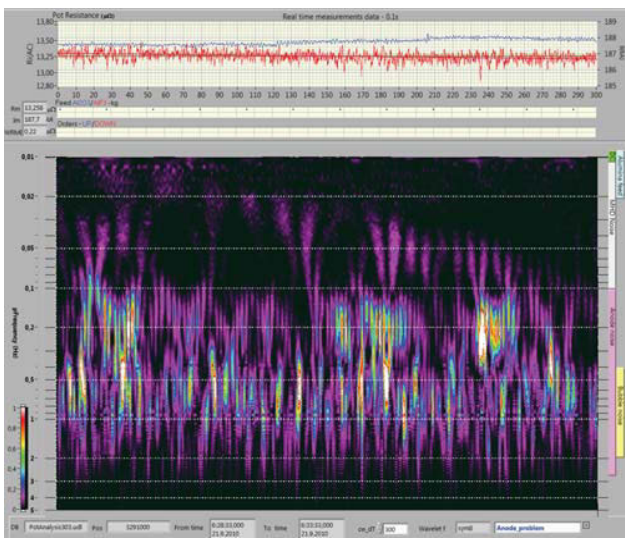

Figure 10: CWT noise spectrogram for anode problem

Figure 11 shows a CWT spectrogram of the anode instability treatment. During the instability period increased bubble activity and numerous short-circuit events were identified. At the 135th second manual anode raising was carried out, which was seen as the momentary increasing of the cell resistance. After the raising of the anode, the cell was stabilized in a very short time.

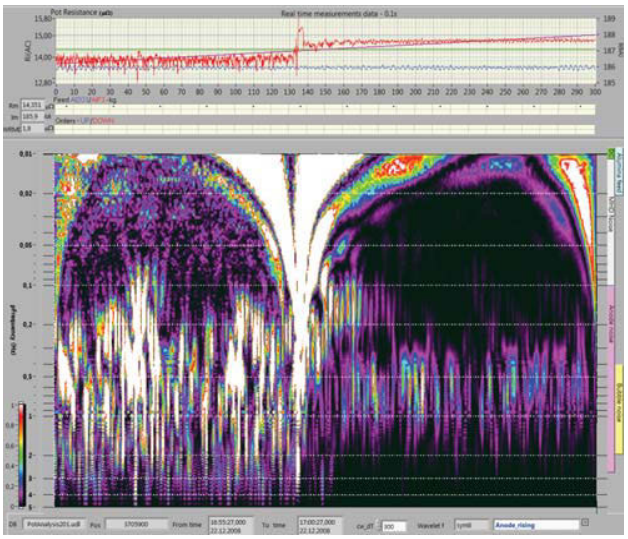

Figure 11: CWT noise spectrogram – instability treatment

A typical metal roll situation with wrmi=0.28 is shown in Figure 12. Dominant bubble noise is superimposed with the temporary oscillations, frequency less than 0.05Hz. Because of liquid fraction fluctuations short-circuit events were detected at the 95th and 130th seconds.

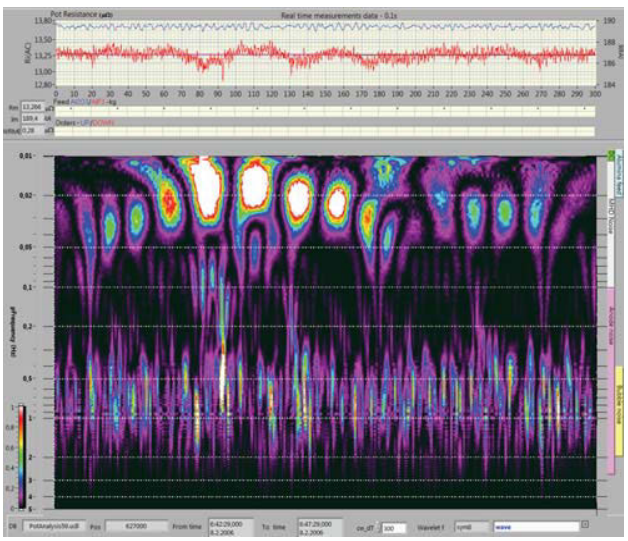

Figure 12: CWT noise spectrogram for metal-roll

Wavelet packet transform

WPT pseudo-resistance noise energy spectrograms for typical isolated process situations are collated in Table 2. For the stable process (a) most of the noise energy was concentrated within a range of 0.4-1.2Hz, with center at 0.6Hz, which matched the characteristic bubble noise pattern. Anode changing (b) disturbed cell stability. The bubble noise increased the energy and spread the frequency pattern. Additionally a MHD noise component may occur because of unbalanced current distribution. The noise pattern corresponding to the anode instability (c) was manifested with increasing intensity of noise within the range 0.1-5Hz. MHD instability (d) was characterized by increasing noise energy within the low frequency range <0.05Hz. The last case (e) shows a response to significant current reduction. The frequency pattern of the bubble noise changed with the shift to lower frequency, as a consequence of lower anodic current density.

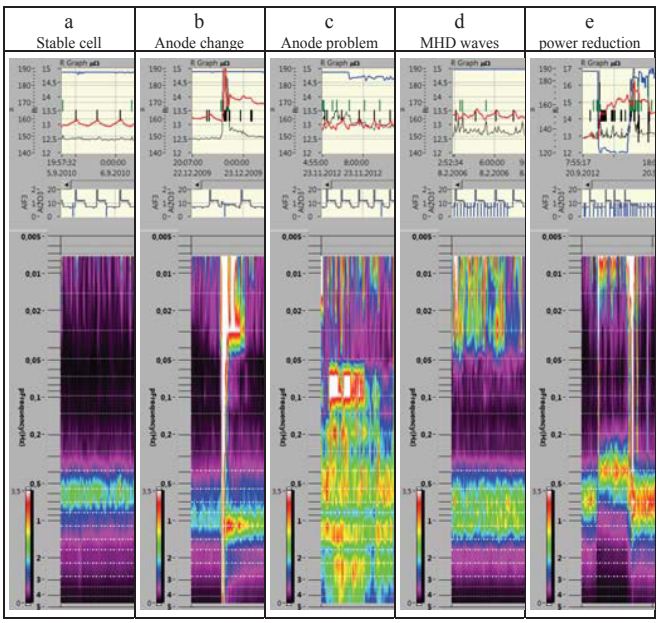

Table 2: WPT spectrograms for typical process situations

For the evaluation of WPT results within the context of a potline control strategy, five days of real time measurements were conducted in order to include all regular cell working operations and eventual disturbances. Figure 13 shows the cell regulation data and working procedure indications obtained from a smelter control system, simultaneous with the WPT energy and entropy semi-spectrograph, which is calculated from the parallel measurement system data, with case markers that are used in comments.

Until the first anode change procedure the process was stable (1). After the first anode changing procedure, a higher level of noise was detected (2). The spectrogram shows increased activities in bubble and the MHD area. After the second anode change procedure the instability level increased. The energy and entropy spectrograms showed increased energy and dynamic within the frequency area of the corresponding anodic problem (3). The anode instability appeared again (4) and was eliminated by manual raising of the low-seated anode (↑). Tapping of the metal (5) caused short transient disturbance only, without any long-term impact on the noise pattern. The last anode change procedure was finished without instability (6).

The present example has demonstrated the capability of the WPT to show the different deviations in a process long before the process control system detects a cell operation irregularity.

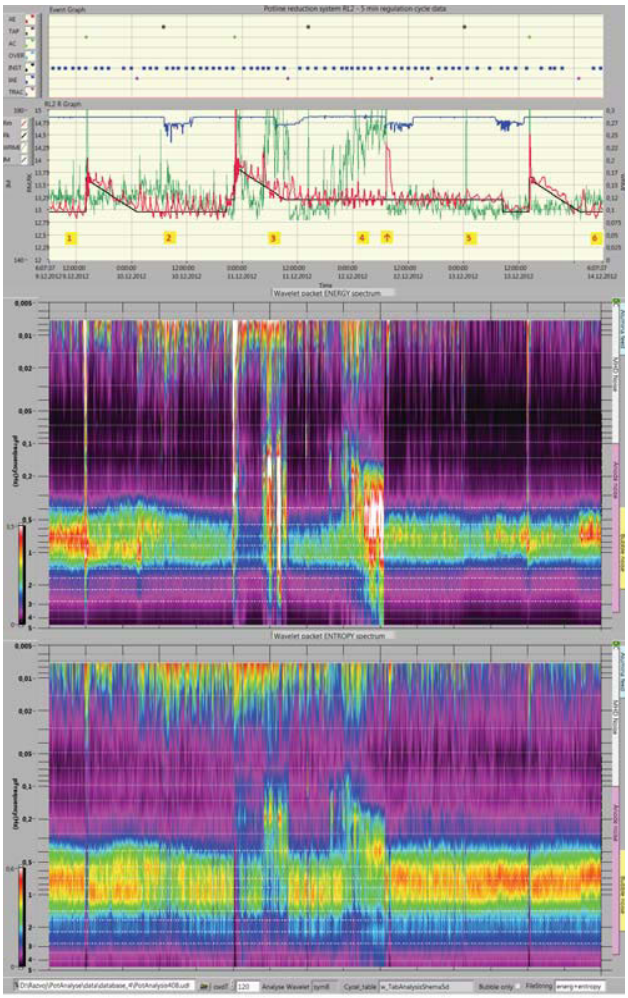

Figure 13: WPT spectrogram and cell control data

Conclusions and future work

An analysis and visualization of the cell resistance noise, based on the wavelet theory was carried out with a specially developed data acquisition system, installed on an AP18 prebaked-anode industrial cell located in the potline C of a Slovenian aluminum producer Talum.

Taking into account the manifestation of process phenomena in the pseudo-resistance noise, a continuous wavelet transformation with nonlinear frequency scale is proposed for visualization of dynamic and transient noise properties, simultaneously with the time-frequency space. The spectrograms obtained with CWT could be the integral part of a real time cell control in order to see certain detail like bubble behavior or transient disturbances.

For the long-term behavior of cell noise, a wavelet packet transformation with a decomposition tree, which covers all phenomena inside cell noise classification, was used to extract energy and entropy feature vectors, which are chronologically displayed as a semi-spectrogram, together with cell regulation cycle data. The presented WPT method is suitable for visualization of the changes in the frequency pattern of a noise in correlation with the process situation as a complementary data.
The results showed a good consistency between the wavelet spectrograms and the expected behavior of the electrolysis cell noise in all observed situations of the process and could be used as the "cell instability footprint".

The application of the wavelets for pseudo-resistance noise analysis demonstrated their great potentials for being integrated into the electrolysis control system as tools for early detection of technological disturbances and a clearer view of the details of process phenomena.

Further theoretical and experimental research is recommended in the following areas:

1) Impact of the anode current distribution on the wavelet spectrum,
2) Developing the methods and metrics for diagnosis of a cell's working conditions using wavelets,
3) Practical implementation within the process control systems.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

STUDY ON EFFECT OF Al-O-C COMPOUND IN ALUMINA CARBOTHERMAL REDUCTION

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Keywords: Alumina carbonthermal reduction, Molecular dynamics simulation, Aluminum, Al-O-C compounds;

Abstract: In this paper, The structures and properties changes for single unit cell of alumina, Al-O-C compounds and aluminium carbide were simulated by first principles method. According to theoretical calculation results, alumina carbothermal reductions with coal and charcoal separately used as reductant were carried out under vacuum. The major equations in carbonthermal reduction reduced to three general equations by reaction temperature; the sequence for each substance which formed in reductions was Al4O4C→Al2OC→Al4C3

Instruction

Aluminum is currently produced industrially via the Hall–Hroult process.The main drawbacks of the electrolytic production are its very high energy consumption (1.35×10^4kWh/t Al)[1], the release of perfluorocarbons, and the high specific CO2-equiv emissions(7.42 kgCO_2-equiv/kg Al)[2]. The greenhouse gas emission by the electrolytic Al production contributes 2.5% to the world anthropogenic CO2-equiv emissions [3]. To find a new method, more and more people put they focus on alumina carbonthermic reduction.

There are two major opinions. Forster[4] proposed in carbothermic reduction Al_4C_3 formed by alumina reacted with carbon at first, then Al_4C_3 reacted with Al_2O_3 formed an intermediate Al_4O_4C, at last Al_4O_4C react with rest Al_4C_3 formed aluminum. Erhard Grunert[5] suggested that aluminum was formed by Al_4C_3 reacted with Al_2O_3, there wasn't any intermediate had been formed.

At present, Aluminum tetraoxycarbide Al_4O_4C and monooxycarbide Al_2OC were discovered during studies of the carbothermal reduction of alumina [6]. To investigate behavior of Al-O-C compound in carbonthermic reduction, thermodynamic calculation and structure simulation were used in this paper.

Method

1 Theoretical calculations

The calculations of the crystal structure and molecular dynamics simulation of α-Al_2O_3、Al_4C_3、Al_4O_4C、Al_2OC were carried out by using the ab initio program of CASTEP based

on quantum mechanics of the pseudo potential plane-wave method in density functional theory (DFT). The geometric crystal structure, energy band, the density of states and other various properties can be predicted accurately by this program. The GGA-PBE functional [7,8], exchange–correlation functional, were adopted in this calculation, which is better than LDA[9,10] functional and the preferred method used in the studies of molecular and materials properties currently. Compared with LDA, all the parameters in PBE are fundamental constants. The exchange part of this function is similar with Becke formula while the correlation part is close to Perdew-Wang function. In addition, this function has a strong physics and reliable numerical results which are frequently used in DFT calculations. Ultra-soft pseudo-potential was used and the spin-polarization of the system was ignored in the calculation with its plane wave 270 eV and the thickness of the two adjacent vacuum layers 1 nm. Whether the energy and charge density distribution of the system is based on convergence in the self-consistent process are the criteria, the accuracy of the energy convergence is better than 2×10^{-5} a.u, and the quality k-point separation in 0.05 $Å^{-1}$.

The *ab initio* dynamics simulation calculations of the optimized α-Al_2O_3、Al_4C_3、Al_4O_4C、Al_2OC crystal structure were carried out at temperatures above 2046℃. It was simulated by NPT system, with the simulation time 1 picoseconds and time step size 1.0 femtoseconds, total 1000 steps. And all the atoms in the studied structures were fully allowed to relax during the optimization procedure. The Nosé thermostat [11] and Andersen barostat[12] were used in the dynamic simulation. The dynamics simulation in CASTEP is based on the velocity Verlet algorithm for integration of the equation of motion. Molecular dynamics in total-energy DFT schemes is implemented in essentially the same way as in conventional forcefield based methods. The main difference is that the energy and atomic forces are derived by solving DFT equations rather than from empirical potentials of interatomic interactions. Electrons are kept on the Born-Oppenheimer surface by means of explicit electronic structure optimization after each MD step. A side effect of this is that evaluation of force and energy from first principles is always the most computationally expensive part of *ab initio* MD. As a result, the efficiency of the MD step itself has no impact on the speed of the calculation[13,14].

Table 1 Lattice parameters of α-Al_2O_3, Al_4C_3,Al_4O_4C,Al_2OC

	Space group	a Å	b Å	c(Å)
α-Al_2O_3	R-3c	5.12846		
Al_4C_3	R-3m	8.55211		
Al_4O_4C	$Cmc2_1$	5.47161		8.81796
Al_2OC	$Pna2_1$	5.65145	6.31885	5.13106

2. Calculation for activity coefficient of Al-Si binary system

Molecular interaction volume model (MIVM) was proposed to describe thermodynamic properties in a multicomponent solution system using its binary interaction parameters. Under some special conditions, this model may be reduced to Flory–Huggins equation, Wilson equation and non-random two liquids (NRTL) equation as well as (MIVM), respectively, and can be verified by Gibbs–Duhem equation, and can express thermodynamic properties of partially miscijble systems. The predicted activities are in good agreement with experimental data of some liquid alloys. The results show that the model is of better predictability and reliability because it has a certain physical basis [15].

Results and Discussion

Using software of Materials Studio simulated the structures of Al_2O_3,Al_4C_3,Al_4O_4C and Al_2OC,then through material structures getting their bond length. Using the same method, the structures of Al_4O_4C and Al_2OC were obtained.

1) Structure of Al_2O_3

Structure of αAl_2O_3 was shown in Figure 1.Bond length of αAl_2O_3 was list in Table 2.

R-3C; a=b=c=0.5196179nm

Fig.1 Structure of αAl_2O_3

Table 2 Bond length of αAl_2O_3

Bond	Bond length(nm)
Al(1)-Al(2)	0.2696
Al(2)-O(3)	0.1879
Al(2)-O(4)	0.1879
Al(2)-O(5)	0.1879
O(3)-O(6)	0.2762

2) Structure of Al_4C_3

Structure of Al_4C_3 was shown in Figure 2.Bond length of Al_4C_3 was list in Table 3.

R3C; a=b=0.3339nm c=2.4996nm

Fig.2 Structure of Al_4C_3

Table 3 Bond length of Al_4C_3

Bond	Bond length(nm)
C(1)-Al(2)	0.3496
Al(2)-C(3)	0.2237
C(3)-Al(4)	0.2098
Al(4)-C(5)	0.1992
Al(4)-C(6)	0.1992
Al(4)-C(7)	0.1992

3) Structure of Al_4O_4C

Structure of Al_4O_4C was shown in Figure 3(a) and after 1Ps Structure of Al_4O_4C was shown in Figure 3(b).Bond length of Al_4O_4C was list in Table 4(a) and after 1Ps bond length of Al_4O_4C was shown in Table 4(b).

(a) (b)

CMC21; a=0.576nm b=0.8537nm c=0.9121nm

Fig.3 Structures of Al4O4C: (a) initial state (b) state after 1PS

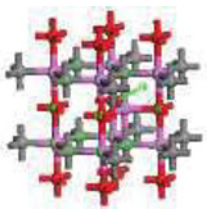
Table 4(a) Bond length of Al_4O_4C

Bond	Bond length(nm)
C(1)-Al(2)	0.2023
Al(2)-O(3)	0.1913
Al(2)-O(8)	0.1913
O(3)-O(8)	0.2893
Al(4)-O(3)	0.1777
Al(4)-O(5)	0.1764
Al(4)-O(9)	0.1781
O(5)-Al(6)	0.1891
O(5)-Al(7)	0.1764
C(1)-Al(6)	0.3387
Al(7)-O(8)	0.1777
Al(7)-O(10)	0.1781

Table 4(b) After 1Ps bond length of Al_4O_4C (2473K,100kPa,1ps)

Bond	Bond length(nm)	Bond	Bond length(nm)
C(Ⅰ)-Al(Ⅱ)	0.1878	C(1)-Al(2)	0.2299
Al(Ⅱ)-O(Ⅲ)	0.4573	Al(2)-O(3)	0.1822
Al(Ⅱ)-O(Ⅷ)	0.3196	Al(2)-O(8)	0.3802
O(Ⅲ)-O(Ⅷ)		O(3)-O(8)	
Al(Ⅳ)-O(Ⅲ)	0.1789	Al(4)-O(3)	0.2077
Al(Ⅳ)-O(Ⅴ)	0.1648	Al(4)-O(5)	0.3120
Al(Ⅳ)-O(Ⅸ)	0.1813	Al(4)-O(9)	0.1694
O(Ⅴ)-Al(Ⅵ)	0.1749	O(5)-Al(6)	0.3989
O(Ⅴ)-Al(Ⅶ)	0.1648	O(5)-Al(7)	0.2302
C(Ⅰ)-Al(Ⅵ)	0.2000	C(1)-Al(6)	0.2973
Al(Ⅶ)-O(Ⅷ)	0.1813	Al(7)-O(8)	0.3989
Al(Ⅶ)-O(Ⅹ)	0.1779	Al(7)-O(10)	0.2063

Structure of Al_2OC and after 2Ps Structure of Al_2OC were both shown in Figure 4.Bond length of Al_2OC was list in Table 5(a) and after 1Ps bond length of Al_2OC was shown in Table 5(b).



$\bar{P}6m2$; a= b=0.317nm c=0.506nm

Fig.4 Structure and after 2Ps Structure of Al_2OC

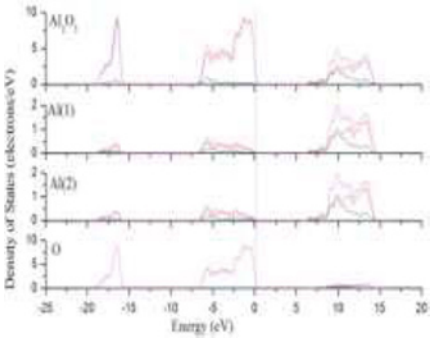
Table 5(a) Bond length of Al_2OC

Bond	Bond length (nm)	Bond	Bond length (nm)
Al(1)-C(2)	0.2010	Al(9)-O(10)	0.2046
C(2)-Al(3)	0.2010	Al(9)-O(11)	0.201
Al(3)-C(4)	0.2010	Al(9)-C(13)	0.2046
Al(3)-O(11)	0.2046	O(10)-Al(14)	0.2046
C(4)-Al(5)	0.2010	O(11)-Al(18)	0.2046
Al(5)-C(6)	0.2010	Al(12)-C(13)	0.2010
Al(5)-O(11)	0.2046	C(13)-Al(14)	0.2010
Al(5)-Al(9)	0.2868	Al(14)-C(15)	0.2010
C(6)-Al(1)	0.2010	C(15)-Al(16)	0.2010
C(4)-Al(7)	0.2010	Al(16)-C(17)	0.2010
C(4)-Al(9)	0.2046	C(17)-Al(18)	0.2010
Al(7)-O(8)	0.2046	Al(18)-C(13)	0.2010
O(8)-Al(9)	0.2010	Al(16)-O(19)	0.2046
O(8)-Al(12)	0.2046	O(19)-Al(1)	0.2046
O(8)-C(13)	0.2868		

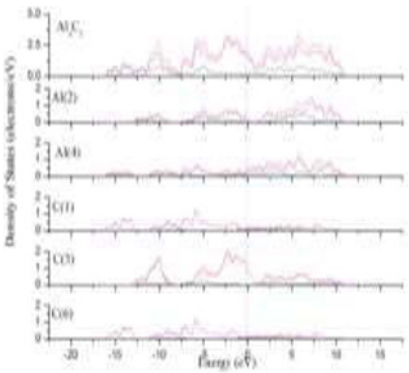
Table 5(b) After 1Ps bond length of Al_2OC(2473K,100kPa,2p)

Bond	Bond length(nm)	Bond	Bond length(nm)
Al(1)-C(2)	0.1941	Al(9)-O(10)	0.2046
C(2)-Al(3)	0.2010	Al(9)-O(11)	0.2057
Al(3)-C(4)	0.1926	Al(9)-C(13)	0.2310
Al(3)-O(11)	0.2046	O(10)-Al(14)	0.2232
C(4)-Al(5)	0.2010	O(11)-Al(18)	0.2232
Al(5)-C(6)	0.2010	Al(12)-C(13)	0.2010
Al(5)-O(11)	0.2046	C(13)-Al(14)	0.2010
Al(5)-Al(9)	0.2868	Al(14)-C(15)	0.2010
C(6)-Al(1)	0.2010	C(15)-Al(16)	0.2010
C(4)-Al(7)	0.2010	Al(16)-C(17)	0.2010
C(4)-Al(9)	0.1884	C(17)-Al(18)	0.2010
Al(7)-O(8)	0.2046	Al(18)-C(13)	0.2010
O(8)-Al(9)	0.2153	Al(16)-O(19)	0.2046
O(8)-Al(12)	0.2232	O(19)-Al(1)	0.2046
O(8)-C(13)	0.3328		

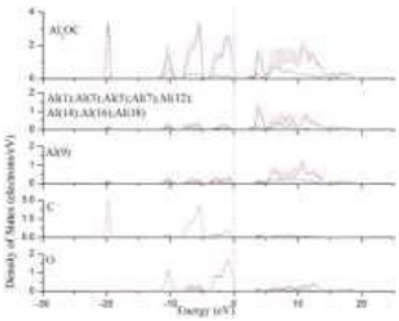
After the structure was optimized, energy band and density of states (DOS) for Al_2O_3, Al_4C_3, Al_2OC, Al_4O_4C, Al_2OC dynamics and Al_4O_4C dynamics were calculated. The results of density of states for those materials were shown in Figure 5.



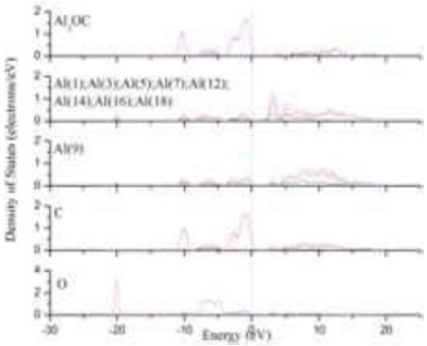
a.Al_2O_3



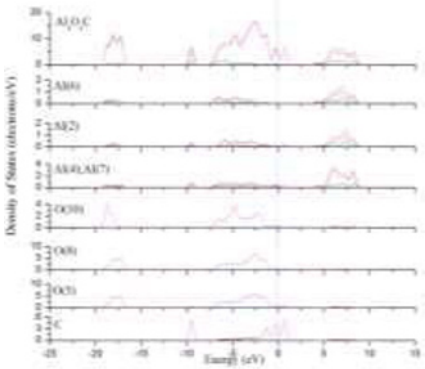
b. Al_4C_3



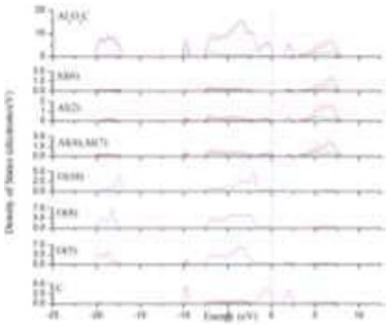
c.Al_2OC



d. Al_2OC dynamics



e.Al_4O_4C



f. Al_4O_4C dynamics

Fig 5 The results of DOS

For calculating results, DOS peaks of Al_2O_3 showed obvious distinguish charac teristic, the peaks of Al in -15~-20eV and 0~7eV were coincided with the peaks of O. It proved that Al and O can form bond obviously.

DOS peaks of Al_4O_4C showed distinguish characteristic, and appeared some broadening than DOS peaks of Al_2O_3.The peaks of Al in -15~-20eV and the peaks of O in 0~10eV compared to DOS peaks of Al_2O_3, the intensity

of Al-O band in Al_4O_4C was lower than Al_2O_3, and there was Al-C band existed in Al_4O_4C in 0~-8eV and -10eV DOS peaks of Al_4O_4C.

DOS peaks of Al_2OC also showed distinguishing characteristic, and appeared some broadening than DOS peaks of Al_4O_4C. But distinguishing characteristic of Al fell in Al_2OC, and appeared a certain extent broadening. From the change of DOS peaks of Al_2OC in -20eV and 0~-10eV, the intensity of Al-O in Al_2OC was weaker than Al_4O_4C.

No distinguishing characteristics appeared in the DOS peaks of Al_4C_3, the energy band of Al_4C_3 was broader than the DOS peaks of Al_2O_3 and moved to direction of low energy. The peaks of Al and C in Al_4C_3 had obviously broadening and character of shifting to lower energy.

It can be seen from before molecular dynamics simulations and after for Al_4O_4C, in 2200K and 100kPa the energy band of Al had some broadening and shifted to direction of lower energy. The DOS peaks of Al and O in -20eV and 0~-8eV at the condition of 2200K and 100kPa, the intensity Al-O had sharply receded. But in same condition the DOS peaks of Al and C showed the intensity of Al-C had some strengthen.

It can be seen from before molecular dynamics simulations and after for Al_2OC, in 2200K and 100kPa the energy band of Al had some broadening and shifted to direction of lower energy. The DOS peaks of Al and O in -20eV and 0~-8eV at the condition of 2200K and 100kPa, the intensity Al-O had sharply receded. But in same condition the DOS peaks of Al and C showed the intensity of Al-C had some strengthen.

Conclusion

The structures and properties changes for single unit cell of alumina, Al-O-C compounds and aluminum carbide were simulated by first principle calculation. Results of molecular dynamics simulations indicated that the mechanism for aluminum formation was Al-O-C compounds reacted with Al4C3, and the sequence for Al-O-C compounds formed in reductions was Al4O4C→Al2OC→Al4C3.

Acknowledgments

This work was supported financially by the Fundamental Research of Yunnan province (Grant NO. 2013FZ033),and National Natural Science Foundation of China (NO.51304095). Corresponding author: Wenlong Jiang; Tel: +86-871-65161583; E-mail: jiangwlkg@qq.com

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE IMPACT OF ALUMINA QUALITY ON CURRENT EFFICIENCY AND ENERGY EFFICIENCY IN ALUMINUM REDUCTION

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Keywords: Alumina Fines, Current Efficiency, Energy Efficiency, Alumina Quality

Abstract

Current efficiency and energy efficiency are the two most important metrics for assessing the overall performance of a given potline. The influence of alumina quality on these parameters is however poorly understood although likely influential. This interplay is considered here with the construction of multiple regression models from daily data of test groups of 60 cells each for a number of important parameters, pertinent alumina properties from detailed materials characterization of judiciously chosen aluminas and relevant weather parameters for approximately 20 months of effective sampling. The concentration of α-Al_2O_3 in the fines is found to be the single most influential parameter affecting current efficiency and energy efficiency. Models of superheat and noise indicate that this is predominantly due to issues associated with alumina dissolution. Energy efficiency is also correlated with gibbsite content of the fines, likely due to the added energy required for the phase transformation from hydroxide to sesquioxide.

Introduction

The need to maintain a protective ledge of frozen cryolite adhered to the cell sidewall necessitates the deliberate shedding of large quantities of heat from a reduction cell in addition to the equally large quantity of heat needed to enable the reduction of alumina. This makes questions of energy efficiency (also termed specific energy consumption a significant concern to the aluminum industry. Closely related, maximizing current efficiency (CE), which references the actual production of metal over the theoretical maximum according to Faraday's Law is the primary metric by which smelters judge process efficiency. Clearly, any energy supplied not reducing alumina to aluminum or stored as heat in the product constitutes an energy loss.

Numerous studies have gone into elucidating the primary influences driving CE losses. A large body of work has demonstrated the role of increasing operating temperature in decreasing CE [1-6]. This is assumed to result from increased metal solubility and therefore susceptibility to re-oxidation. Dissolved alumina concentration has been correlated with CE in some studies [3, 5, 6], but whether positively or negatively is under debate. Additives are found to increase CE, by decreasing aluminum solubility due to lower operating temperatures [4]. Increased superheat has also been linked to CE loss, [4] as has bubble noise [6]. In terms of specific energy consumption, low CE is mathematically correlated to increased energy consumption [7]. Despite all the research, a definitive conclusion on the influence of a number of these parameters is yet to emerge.

Further, relatively little work has been undertaken regarding the impact of alumina quality. As problematic phases (gibbsite, α-Al_2O_3) tend to be concentrated in the fines, the quantity and quality of alumina fines is particularly important. There is increasing evidence that gibbsite in the fines has a considerable influence on HF generation and scrubbing efficiency [8, 9], and even fluoride return to the pots [10]. Therefore, alumina quality is expected to also ultimately influence CE and energy consumption. Over-calcination of the fines, a consequence of longer retention times and potentially better heat transfer to these particles, leads to higher concentrations of the α-Al_2O_3 component of the material. This is expected to decrease dissolution rate and increase sludging, negatively impacting CE. This study details a combination of alumina characterization and multiple regression aimed at determining the role of alumina quality and composition of its fines on CE and energy efficiency.

Potline Configuration and Sampling Methodology

The aluminum smelting process is influenced by a number of interrelated variables; the focus of the current study was to isolate the role of alumina, which requires the nearly impossible task of running a smelter under identical conditions with changing alumina feed as the sole variable. Due to its configuration, the TRIMET Aluminium SE – Essen smelter provided a very good approximation. Here, two separate primary alumina feeds are concurrently supplied to different sections of a single potline and run with almost no mixing of the materials or thermal balances, but for one exception. The treatment of plant-wide evolved fluorides is performed in a common Gas Treatment Centre (GTC) – see the plant overview in Figure 1. Primary alumina pots which are operated in another potline lead thus to an elevated fluoride level in the secondary alumina.

A single potline of 120 pots (10 groups of 12 pots each) was studied. Each end ran different aluminas, therefore daily data was gathered for each pot and averaged over the 60 pots running the same alumina. The potline under study was exclusively fed secondary alumina. The data consists of pot parameters (liquidus temperature, AlF_3 fed, voltage, alumina feed, excess AlF_3 concentration (XsAlF3), superheat, noise, anode effect frequency): weather parameters (relative and absolute humidity, sourced for nearby Düsseldorf); a surface area predictor, ΔBJH(3-300); and a phase descriptor, the weight fraction of either α-Al_2O_3 or gibbsite content of the fines. Averaged alumina properties are used in these models, i.e. pot-to-pot and shipment-to-shipment variations within a given alumina are not yet included. Raw HF generation data has been considered in previous models [10] but does not contribute to models presented in this study.

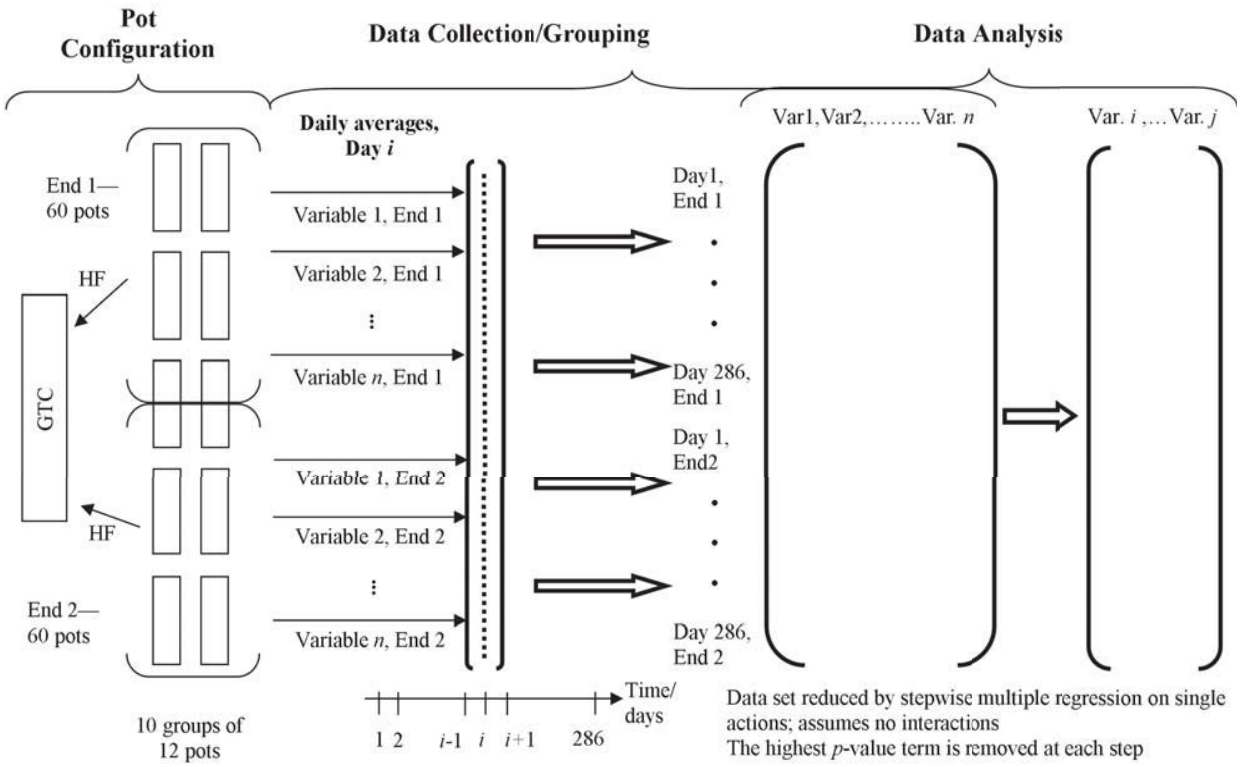

Figure 1: Overview of the potline configuration, data grouping, and regression analyses performed

Table I: Phase quantification (% w/w), as determined by Rietveld refinement, of the bulk and [<45 μm] fractions of four aluminas; n.d. – phase below detection limits.

	Alumina A	Alumina B	Alumina C	Alumina D
Gibbsite	n.d. [6.96]	n.d. [0.58]	n.d. [5.98]	n.d. [n.d.]
γ-Al_2O_3	26.03 [30.38]	23.56 [29.84]	35.54 [28.41]	39.70 [19.74]
γ'-Al_2O_3	38.29 [34.18]	38.74 [34.11]	36.23 [36.19]	37.75 [26.68]
θ-Al_2O_3	33.13 [20.77]	32.24 [21.03]	25.36 [21.65]	15.43 [25.02]
α-Al_2O_3	2.55 [7.71]	5.46 [14.45]	2.86 [7.77]	7.13 [28.56]

Alumina samples of relevance to this work were obtained from harbor silos storing primary-Al_2O_3 as received from the refinery shipments. Four unique aluminas were sampled during two separate 2-3 week long sampling campaigns and were extensively characterized. Plant data obtained spans 2-4 months on both sideof the sampling campaigns leading to approximately 10 months of plant data for each end of the potline associated with only the four aluminas characterized. At 60 pots per end, pot data is therefore effectively averaged over 36000 daily pot readings per variable.

Sample Characterization

The primary analysis technique employed is phase quantification as determined by X-Ray Diffraction (XRD) with subsequent Rietveld refinement. All patterns were collected on a PANalytical Empyrean diffractometer and were recorded such that they were of sufficient quality for Rietveld refinement with at least 10 data points defining each peak above its full width at half maximum. All patterns used the Cu Kα line over angles $10\,° \leq 2\theta \leq 80\,°$. Results for all four aluminas sampled are provided in Table I. Note that these materials exhibit a wide range of gibbsite and α-Al_2O_3 contents, which are concentrated in the fines. As such, models used in this work will examine the fines fraction in determining the role alumina plays on key smelter parameters. This is particularly important as bulk measures alone would suggest there is no gibbsite in any of the materials studied. We find, for the α-Al_2O_3 content at least, that the fines composition should be a good and sensitive proxy for that of the bulk. For example, a bivariate analysis shows a linear correlation with $R^2 = 0.9$ between α-Al_2O_3 in the bulk and its content in the fines. It must also be stated that in the samples collected, gibbsite and α-Al_2O_3 are quite strongly (negatively) correlated. A simple bivariate linear regression fit attains an $R^2 = 0.71$. This is largely as these are expressed as mass percentages, therefore increases in one necessarily decrease the yields of all other phases. This holds as the samples collected do not include materials in which over-calcined alumina is blended with under-calcined material to achieve optimal average properties. Future extensions to this work will be required to consider these more general cases.

Specific Surface Area (SSA) is another important measure of alumina quality. It is associated with fluoride capture capability and consequently recycling to the reduction cells [10]. Pore size and surface area parameters are provided in Table II. SSAs are based on the BET (Brunauer-Emmett-Teller) [11] approach and are typical values reported on Certificates of Analysis. BJH (Barrett-Joyner-Halenda) [12] SSAs, extracted from pore size distributions, are also used. These measures can isolate the surface areas associated with a specific range of pore diameters. Pores smaller than 3 nm do not appear to admit HF on the time-scales

Table II: Specific surface area and pore size characteristics of the four aluminas sampled in this work.

	Alumina A	Alumina B	Alumina C	Alumina D
BET/ m^2 g^{-1}	67.9	65.3	74.1	88.4
BJH(3-300)[a]/ m^2 g^{-1}	56.2	52.4	74.0	71.9
BJH(1.7-300)[b]/ m^2 g^{-1}	66.2	63.7	87.0	94.7
D(4V/A)/ nm	10.36	11.50	8.20	6.54

[a] BJH – calculated SSA, summed over pores of diameter 3 nm ≤ D ≤ 300 nm

[b] BJH – calculated SSA, summed over the conventional range of pore diameters 1.7 nm ≤ D ≤ 300 nm

relevant to dry scrubbing [10, 13]. As such, the commonly utilized BET SSAs (which does not discriminate pore size) does not appear to adequately account for the fluoride carrying capacity of SGAs, whereas an appropriate BJH value (removing the contribution of pores > 3 nm) does.

Regression Modelling

Model Construction

Given the interrelated nature of the variables associated with aluminum smelting, a multivariate regression modelling approach was undertaken in this work. To construct a final model for a given variable, the following methodology was employed:

1. Assemble a multiple linear regression model as a function of all independent variables covering pot, weather and alumina quality parameters. These are termed 'predictors'.
2. Determine the presence of mutual correlation (referred to as collinearity) between predictors which would otherwise lead to unrealistic interpretations of the influence of each predictor. This is deduced by examining the variance inflation factors (VIF), which are found as the terms along the diagonal of the inverse of the r-correlation matrix. Two variables are deemed collinear if their VIFs > 5. In such scenarios, only the predictor having the highest statistical contribution to the final mode is retained (see below).
3. With one of the collinear predictors in each pair removed (if necessary), linear regression is performed and the statistical significance of each predictor ranked by its p-value. High p-values denote low significance. A 95 % confidence level is adopted, i.e. any predictor with $p > 0.05$ is removed on the basis of insignificance. Predictors are removed one at a time, omitting the predictor with the highest p-value only and re-modelling. This proceeds in as many iterations as is necessary to ensure only the most statistically significant core predictors remain, i.e. p-values are all less than 0.05. This is referred to as the 'final model'.

Superheat Models

Alumina dissolution is a common and important source of noise. When sludging/back-feeding cycles occur, bivariate scatter plots of noise vs. alumina feed exhibit a negative correlation. Periods of back-feeding when there is a significant quantity of sludge, and therefore a noisier pot, will require lower alumina feeds and vice versa. When these analyses are performed, only random scatter is seen in general. This is likely a consequence of trends being obscured due to the many interacting parameters – for example,

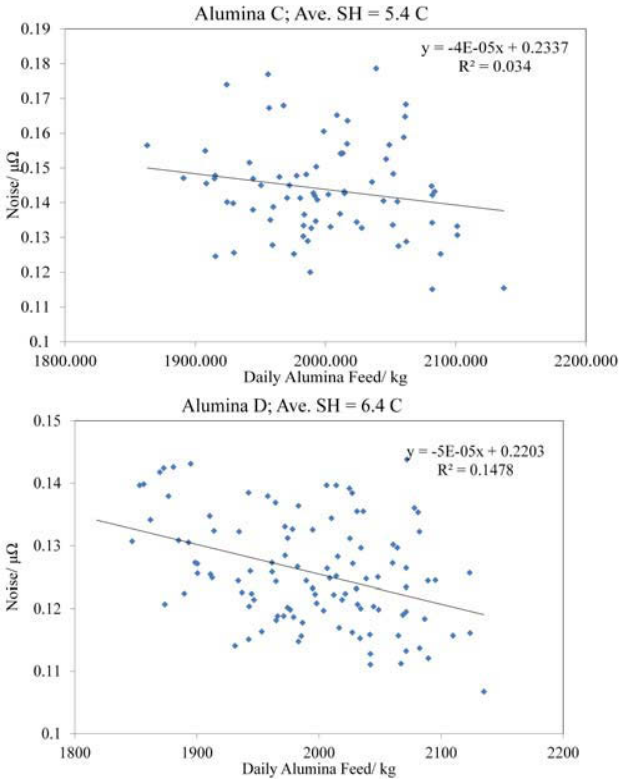

Figure 2: Noise vs. alumina feed for *(top)* low and *(bottom)* high-α aluminas during low superheat periods

additive content impacts alumina dissolution, but also varies superheat, itself also tied to alumina solubility. This highlights the need for multivariate analyses. Together, it also suggests that to effectively model the effects of noise, particularly when assessing the role of α-Al_2O_3, superheat must also be accounted for. To demonstrate this point explicitly, we have explored two short timeframes in which the superheat was particularly low, largely limiting the influence of excess superheat. Bivariate noise-alumina feed plots are provided in Figure 2 for aluminas C and D, which lie at approximately the extremes of the α-Al_2O_3 contents we have examined. The lower α-Al_2O_3 material, C, presents no correlation whereas alumina D shows the expected negative (albeit somewhat weak) correlation. This is in spite of the fact that the average superheat for this period was higher in the alumina D pots, thereby highly suggestive of α-Al_2O_3 driven sludging. These trends are not observed in the other higher superheat regimes.

Table III: Multiple regression final model, using n data points, of superheat (°C).

	Coefficients	*SE*	*P-value*	$\pm t_{0.05,n-1}\frac{\sigma_n}{\sqrt{n}}$
Intercept	150	19	8.9×10^{-14}	38
Liquidus temp.	-0.09	0.01	1.4×10^{-10}	0.02
Voltage	-12	2.00	2.2×10^{-9}	3.9
Al_2O_3 feed	-4.0×10^{-3}	6.0×10^{-4}	4.1×10^{-11}	1.2×10^{-3}
Anode effect	2.2	0.68	1.6×10^{-3}	1.3
Noise	-15	3.1	2.9×10^{-6}	6.0
α-Al_2O_3 fines	-3.4×10^{-2}	6.0×10^{-3}	2.5×10^{-8}	0.01
$XsAlF_3$	0.64	0.08	3.5×10^{-16}	0.15
ΔBJH(3-300)	1.2×10^{-2}	4.6×10^{-3}	8.5×10^{-3}	9.1×10^{-3}
Regression Statistics	Multiple R	*n*	Signif. F	
	0.57	572	1.56×10^{-43}	

A superheat regression model is therefore provided in Table III. Excess AlF_3 is the most statistically, and one of the most physically, significant predictor. Along with the lowest *p*-values, sensitivity analyses examining the change in superheat find $XsAlF_3$ gives one of the largest average changes in superheat (1.7°C) over its measured range of 10.8-13.4%. Larger average changes are found for voltage (2.4°C) and alumina feed (2.0°C) over their measured ranges of 4.32-4.52V and 1717-2230kg/day, respectively. However, these changes are comparable to those driven by $XsAlF_3$, but their lower statistical significance makes these changes more uncertain. The positive correlation with $XsAlF_3$ is potentially explained high fluoride and low bath temperatures driving noise and reducing current efficiency, with the excess energy that ultimately needs to be shed driving up the superheat [14]). Alumina feeds, liquidus, and voltage are relatively well understood. High alumina feeds may draw more superheat to heat the cold alumina, or dissolution issues during low superheat periods prompts higher alumina feeds. The role of liquidus is largely that described for $XsAlF_3$. Voltages are applied when pot temperature, and therefore superheat, are low.

After accounting for the most influential parameters, we still find relatively strong negative correlations linking noise to superheat, and α-Al_2O_3 content of the fines to superheat. The phase correlation is most reasonably due to dissolution. As α-Al_2O_3 dissolves poorly, it leads to increased alumina doses, absorbing more sensible heat in pre-heating the material, i.e. periods where a high α-containing material is being used, superheat will (all other parameters held constant) tend to decrease. Further, the negative correlation with noise indicates it is sludge-driven. When superheat is low and dissolution poor, noise will tend to increase after sludge has accumulated on the cathode.

Consequently, as there are many factors influencing various pot parameters which can be difficult to deduce directly. Removing the mediating role played by superheat can be very revealing. The role of α-Al_2O_3 on noise (see the next section) can be very difficult to determine as high superheat masks the effects, but examining superheat itself can identify a quantifiable link between these two parameters.

Noise Models

A model of noise is provided in Table IV. Clearly, noise is strongly positively tied with anode effect, maybe suggesting a common source. This is consistent with a mechanism in which noise in the pots studied is primarily driven by alumina solubility/availability, where noise is high when alumina availability is low, *i.e.* anode effect frequency is high. It must, however, be noted that noise is responded to by lowering alumina feed, which could also be causing the onset of anode effects. That said, there does not appear to be the expected correlation with alumina feed. There are also strong negative and positive correlations with liquidus and AlF_3 feeds respectively. Both can also be understood in terms of alumina dissolution issues. First, liquidus is lowered by high concentration of additives, which lower alumina solubility [15], increasing sludge-induced noise.

As noted in the previous section, the possible role that α-Al_2O_3 plays may be mitigated if there is sufficient superheat. We find superheat itself is negatively correlated with noise ($p = 6.4 \times 10^{-6}$). Consequently, in periods where superheat is low, noise increases. This is again consistent with a dissolution-based

Table IV: Multiple regression final model, using *n* data points, of pot noise (μΩ).

	Coefficients	*SE*	*P-value*	$\pm t_{0.05,n-1}\frac{\sigma_n}{\sqrt{n}}$
Intercept	1.70	0.17	1.7×10^{-22}	0.33
Liquidus	-1.7×10^{-3}	1.7×10^{-4}	3.7×10^{-21}	3.3×10^{-4}
Superheat	-2.2×10^{-3}	4.9×10^{-4}	6.4×10^{-6}	9.6×10^{-4}
Anode effect	8.5×10^{-2}	7.8×10^{-3}	3.5×10^{-25}	0.015
Abs humid	-6.2×10^{-4}	2.5×10^{-4}	1.2×10^{-2}	4.8×10^{-4}
$XsAlF_3$	3.8×10^{-3}	1.1×10^{-3}	2.8×10^{-4}	2.1×10^{-3}
ΔBJH(3-300)	-1.3×10^{-4}	5.7×10^{-5}	2.7×10^{-2}	1.2×10^{-4}
AlF_3 feed	1.6×10^{-3}	1.9×10^{-4}	9.2×10^{-17}	3.7×10^{-4}
Regression	Multiple R	*n*	Signif. F	
Statistics	0.63	572	1.4×10^{-58}	

mechanism, where sufficiently high superheat aids alumina solubility and thereby prevents noise. While beyond the scope of this work, it may be of interest for future studies to determine whether there is a threshold beyond which there is sufficient superheat such that a correlation with noise is ultimately lost.

Current Efficiency Models

The regression model for CE is given in Table V. Current efficiency is computed through the mass of metal tapped regularly:

$$\overline{CE} = \frac{\overline{m_j}}{m_{theor}}100\% \text{ with } m_{theor} = \frac{M_r(Al)It}{3F} \quad (1)$$

Metal tapped was averaged daily across all pots tapped, with each pot being tapped every 2 days.. In initial surveys of the data, plotting CE as a function of time did not reveal a clear trend and therefore no significant alumina related response. However, the multiple regression model reveals a clearer correlation with alumina quality. In fact, the single most influential parameter associated with CE is the α-Al_2O_3 content of the fines. It is important to recall that α-Al_2O_3 is negatively correlated with gibbsite, and models with this phase find a positive correlation with CE. This raises the possibility that α-Al_2O_3 is a proxy for a true correlation with gibbsite. While low gibbsite levels could increase CE by introducing turbulence and aiding dissolution in the bath, we suspect it is the correlation with α-Al_2O_3 that is genuine. This is in part due to better model quality when using α-Al_2O_3, and that the relatively high gibbsite levels found would arguably lead to volcano effects. The strongest arguments against beneficial effects of gibbsite on CE are presented with the energy efficiency models, however. Further, α-Al_2O_3 is not sufficiently correlated with any other predictor to be spuriously retained. That is, this correlation is very likely real and α-Al_2O_3 is not simply a proxy for another predictor.

Table V: Multiple regression final model, using *n* data points, of current efficiency.

	Coefficients	*SE*	*P-value*	$\pm t_{0.05,n-1}\frac{\sigma_n}{\sqrt{n}}$
Intercept	17	27	0.52	53
Voltage	13	5.8	2.3×10^{-2}	12
Al_2O_3 feed	8.7×10^{-3}	1.9×10^{-3}	4.6×10^{-6}	0.004
Noise	21	9.2	2.6×10^{-2}	18
α-Al_2O_3 fines	-0.10	0.02	1.2×10^{-8}	0.03
ΔBJH(3-300)	3.4×10^{-2}	1.6×10^{-2}	3.0×10^{-2}	0.031
Regression	Multiple R	*n*	Signif. F	
Statistics	0.36	572	9.4×10^{-16}	

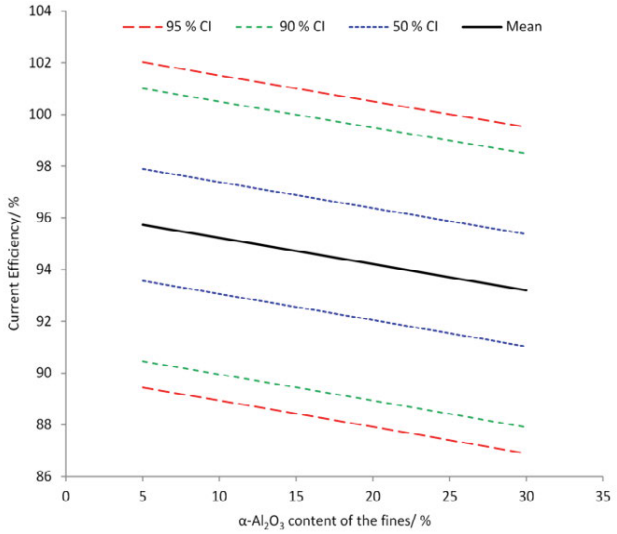

Figure 3: Predicted current efficiency as a function of the α-Al_2O_3 content of the fines; the model of Table V is used, with all other parameters held at their mean values. The 95, 90, and 50 % confidence intervals are provided.

Importantly, the model indicates higher α-alumina will lower CE. The preceding models for superheat and noise are consistent with sludge formation and dissolution issues. It should be noted, however, that while we can demonstrate a statistically significant correlation between α-Al_2O_3 content and CE, the data set is still too small to sufficiently remove the noise required to provide a quantitative tool to positively distinguish the impacts on CE by two different aluminas – see Figure 3. Even at the generous 50 % confidence interval, the differences in CE between two aluminas at the extremes of the samples analyzed cannot yet be distinguished, strictly speaking. Note that we assume normal distributions underlying our confidence intervals leading to upper limits for current efficiency that exceed 100%. A more rigorous analysis should employ a log-normal distribution for example, but this should be sufficient for illustrative purposes here. Our results indicate that future work with much larger data sets could provide some important insights. Future studies should also treat the effects of anode quality and changes in metal and bath inventory, which have been assumed constant in the present models.

Alumina feed is also positively correlated with CE, and quite statistically significantly so ($P = 4.6 \times 10^{-6}$). This is not unexpected as CE is a measure of the mass of metal produced and therefore, necessarily alumina fed. Other parameters kept in the model are only tenuously retained at the 95 % confidence level and will not be discussed due to questions of model reliability for these terms. That said, it is interesting to note that there may be an influence of SSA (and therefore the fluoride balance) as indicated by the retention of the ΔBJH(3-300) predictor.

Energy Efficiency Models

The final model considered is that of energy efficiency, one of the most important smelter operation parameters. Specific energy consumption is found (see the model in Table VI) to depend very heavily on CE; this is unsurprising as energy efficiency/consumption, E_{eff}, is calculated as:

Table VI: Multiple regression final model, using n data points, of energy efficiency.

	Coefficients	*SE*	*P-value*	$\pm t_{0.05,n-1}\frac{\sigma_n}{\sqrt{n}}$
Intercept	-21	11	5.8×10^{-2}	22
Liquidus	3.3×10^{-2}	8.1×10^{-3}	6.2×10^{-5}	0.016
Voltage	3.9	1.2	8.4×10^{-4}	2.3
Abs humid	2.8×10^{-2}	1.3×10^{-2}	2.9×10^{-2}	0.025
Gibbsite-fines	4.9×10^{-2}	1.4×10^{-2}	3.2×10^{-4}	0.027
XsAlF3	-1.5×10^{-1}	5.3×10^{-2}	4.1×10^{-3}	0.11
ΔBJH(3-300)	6.9×10^{-3}	3.2×10^{-3}	3.2×10^{-2}	0.0063
CE	-1.3×10^{-1}	8.0×10^{-3}	3.8×10^{-47}	0.016
Regression	Multiple R	*n*	Signif. F	
Statistics	0.59	572	3.7×10^{-49}	

$$E_{eff} = \frac{24I\sum_i V_i}{\sum_j m_j} \quad (2)$$

where I, V, and m are the amperage, applied voltage and metal mass tapped, respectively. When we employ the definition of CE in equation (1), E_{eff} can therefore be expressed as:

$$E_{eff} = \frac{24I\sum_i \frac{V_i}{n}}{\sum_j \frac{m_j}{n}} = \frac{F\bar{V}}{12M_r(Al)\overline{CE}} = c\,\frac{V}{CE} \quad (3)$$

where F is Faraday's constant, M_r(Al) is the molar mass of aluminum and n is assumed to be both the number of pots voltage is averaged over and the number of pots tapped, which will be valid, at least, in the longer-run averages. Physically, this is also understandable as less metal is produced while the same amount of energy is introduced into the cell by means of voltage and amperage. Importantly, the high significance of CE on E_{eff}, which is in turn determined primarily by α-Al_2O_3 content, implies that E_{eff} is also heavily influenced by α-Al_2O_3. An over-calcined material leads to poor dissolution, therefore to high efficiency losses associated with noise. Note that this causal chain implies α-Al_2O_3 content is positively correlated with specific energy consumption – we will return to this point shortly.

In fact, one can quantify the impact of α-Al_2O_3 on specific energy consumption. On moving from a low α-material in the fines at ~8 % (w/w) to a higher α-Al_2O_3-containing material in the fines at ~28 % (w/w) (see Table I) we would expect a 2% drop in CE based on the α-Al_2O_3 fines predictor (Table V). This CE reduction should yield an increase in energy requirements of around 0.25 kWh/kg(Al).

The predictors liquidus and voltage are also positively correlated with specific energy consumption. High liquidus implies high pot temperatures, as the two are (positively) collinear. This in turn implies eventual increased energy losses by increased heat transfer to the ambient. Additionally, high liquidus implies lower excess AlF_3, increasing metal solubility and lowering CE, eventually raising the energy consumption. Conversely, additional voltage is often specifically employed to heat up pots when temperature becomes too low [14]. Each scenario is associated with either excessive or insufficient heat and demonstrates the importance of maintaining a moderate temperature and an even thermal balance.

Of similar statistical importance (based on p-values) to liquidus and voltage is gibbsite content. Gibbsite and α-Al_2O_3 are negatively collinear, and therefore it could be argued that gibbsite appears as a proxy for α-Al_2O_3. However, there are several reasons to suspect this is not the case.

- The models perform better with gibbsite than α-Al_2O_3; the *p*-values are 3×10^{-4} and 3×10^{-3}, respectively.
- As gibbsite and α-Al_2O_3 are negatively correlated, gibbsite as a proxy would suggest that α-Al_2O_3 should be negatively correlated with specific energy consumption. It has already been discussed above that this is unreasonable, whereas the positive correlation of gibbsite and specific energy consumption is readily explained (see below).
- α-Al_2O_3 already appears in the model, largely through the contribution of CE.

Therefore, the data best supports a statistically significant, genuine, positive correlation between gibbsite and specific energy consumption. Mechanistically, high gibbsite materials contains a significant mass of retained water, requiring a high energy input to remove it through endothermic phase transformations [9]:

$$2Al(OH)_{3(s)} \rightarrow Al_2O_{3(s)} + 3H_3O_{(g)} \quad (4)$$

This in fact requires 260 kJ per mole of gibbsite [16] over and above the 106-130 kJ mol^{-1} needed for alumina dissolution [17], and the energy required to heat the material to operating temperatures. These are clear energy inputs that ultimately constitute a loss.

However, it should be noted that this may not be the only, or even major, contribution of gibbsite content to efficiency losses. At 260 kJ mol^{-1}, an increase in gibbsite content of 6 % in the fines of a material containing 10 % fines demands an additional 0.016 kWh/kg(Al). This is a rather small amount considering the regression model of Table VI suggests an increase of the order of 0.3 kWh/kg(Al). At the 95% confidence level, this predicts a range of 0.13 – 0.45 kWh/kg(Al), an order of magnitude higher than predicted by phase transformation alone, suggesting a missing mechanism. Volcano effects and the subsequent loss of feed control is the likely candidate. This would also drive a negative correlation with CE. This appears to be obscured by the α-Al_2O_3/CE correlation and colinearity between the phase predictors. However, a further implication is that gibbsite levels cannot be such that the positive correlation with CE described in the previous section is real, and is therefore an artefact of a genuine correlation between α-Al_2O_3 and CE. Note, too, this model puts efficiency losses due to gibbsite at a similar magnitude to those associated with α-Al_2O_3 content.

Finally, excess AlF_3 and the ΔBJH(3-300) predictors are both weakly retained. This may indicate that the fluoride cycle plays a non-negligible role, with *p*-values of 10^{-2}-10^{-3}. However, incorporating sensitivity analyses, noting that $XsAlF_3$ varies between 10.813.4 % and ΔBJH(3-300) between -21.6 and -21.6m^2g^{-1}, energy consumption swings of 0.4 and 0.3kWh/kg(Al) respectively are predicted. The fluoride balance therefore likely plays a non-negligible role in the specific energy consumption.

Discussion and Conclusions

The primary metrics of smelter performance and cost are the current efficiency and energy efficiency (specific energy consumption). The current efficiency models presented here implicate α-Al_2O_3 (as measured in the fines) to be the single most influential alumina parameter. High α-Al_2O_3 tends to lower the current efficiency. Models of superheat and noise are consistent with current efficiency losses due to noise driven by dissolution issues.

Given the relationship between energy and current efficiency, this implies that α-Al_2O_3 content also negatively impacts specific energy consumption. Moving from a low α-Al_2O_3 to a material with higher over-calcined content, a current efficiency loss of ~2 % is found, which corresponds to an efficiency loss of ~0.25 kWh/kg(Al). However, energy efficiency models also reveal an important correlation with gibbsite in which its increased content in the fines is associated with increased energy consumption. This is traced, in part, to the additional energy required in the phase transformation of the hydroxide to alumina over and above the energy needed for dissolution and heating to pot temperature. However, these energy losses are only on the order of ~0.02 kWh/kg(Al), whereas the model suggests energy losses of ~0.3 kWh/kg(Al). This is roughly the same size as the energy losses associated with α-Al_2O_3. This indicates a missing mechanism, probably volcano effects, for energy losses due to gibbsite. More importantly, this suggests that alumina quality, in terms of these two phases alone, may cost up to ~0.5 kWh/kg(Al) in additional energy. Thus this study suggests a significant negative impact on specific energy consumption of both over- and under-calcined materials in the alumina fines.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SIDELEDGE FACING METAL IN ALUMINIUM ELECTROLYSIS CELLS: PRELIMINARY MODELLING STUDY OF BATH FILM FORMATION

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Keywords: Sideledge, freezing, melting, model

Abstract

For design and operation of highly efficient aluminium electrolysis cells, knowledge concerning side ledge formation and heat loss through the sides is indispensable. Measurement of ledge profiles and heat fluxes seems to concur with theoretical models only by including a thin film of bath located at the metal-sidewall interface. A transient CFD (Computational Fluid Flow) model including metal circulation is made for these initial studies of how bath film develops. The model includes two liquids (bath and metal) with a bath layer initially sitting at the lowest two cm below the metal, before it proceeds upwards along the side wall by gravitation and surface tension. A parameter study was conducted with surface tension, bath film properties, and metal flow as parameters. It is shown that metal flow has a considerable impact on the film formation rate. The modelled bath film behaviour is in accordance with theoretical considerations.

Introduction

The purpose of this work is to build understanding of the formation of a bath film located at the side walls[1]. The motivation is that the bath film seems necessary for ledge formation, and therefore; proper understanding of this phenomenon might be critical in designing better electrolysis cells and to improve cell operation. A conceptual model for the lower ledge, based on an alumina-saturated bath film is published earlier [1, 2].

A bath film between the metal and the lower sideledge was originally proposed by Haupin [3]. Although Haupin's original explanation for the maintenance of the film appears to be insufficient, the very existence of a film has a large potential of explaining some observed phenomena.

Once a bath film has been established, the flow and its general behaviour can be estimated analytically, as shown elsewhere [4]. The purpose of the current work was to study the initial formation of a bath film in a CFD modelling environment, starting with a given amount of bath residing underneath the molten metal without having a film between the liquid (metal or bath) and the side wall. Also, the effect of metal flow in this context was studied.

Bath Film between Metal and Ledge

The model is based on the volume of fluid approach (VOF) in FLUENT, considering only two phases (bath and metal) so far, for simplicity. It is possible, within the VOF approach, to add a third phase in order to have different properties of the bath film and the bulk of the bath. Bath freezing is not considered in this initial model development.

Two model geometries were built. The first model geometry is shown in Figure 1 and it consists of a sliced electrolysis cell (along the current direction in a side-by-side cell) with 0.8 million mesh elements. A slice model was used to reduce the computational time by reducing the number of cells, while still maintaining sufficient mesh resolution to resolve the flow of bath film and metal. This model is used to study vertical or near vertical metal flows.

The second model, shown in Figure 2 is still a slice model of an electrolysis cell, but the model is wider than the first model to allow investigation of horizontal metal motion along the side wall. To resolve bath film velocities and a thin bath film, inflation layer were added to the cut-cell mesh, as seen in Figure 3. The bath film formation material is initially located at the lowest 2 cm above cathode surface, the metal extends from 2 cm up to 20 cm, and the bulk of the bath film material (representing the bath zone) above 20 cm. The phases are allowed to mix in a transient calculation mode. The mixing process is governed by gravity, surface tension at walls, surface or interfacial tension between fluids, liquid flow, turbulence, and general boundary conditions. The model contains an additional zone (cube in Figure 1) which is used to add a source field in order to set up the metal flow and thereby evaluating the effect of metal flow velocity.

An overview of physical parameters is shown in Table I.

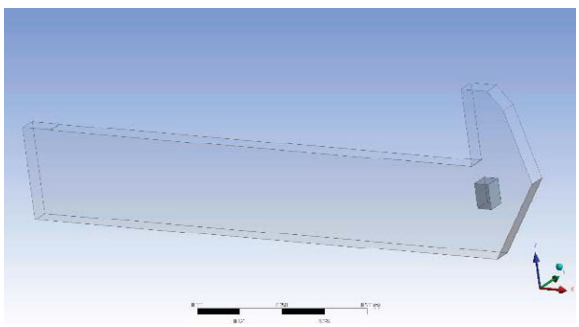

Figure 1. Thin slice model: Model geometry including zone for velocity generation located in the little box at the right hand side of the drawing.

[1] Side wall: interface between cell lining and liquid phase (metal or bath)

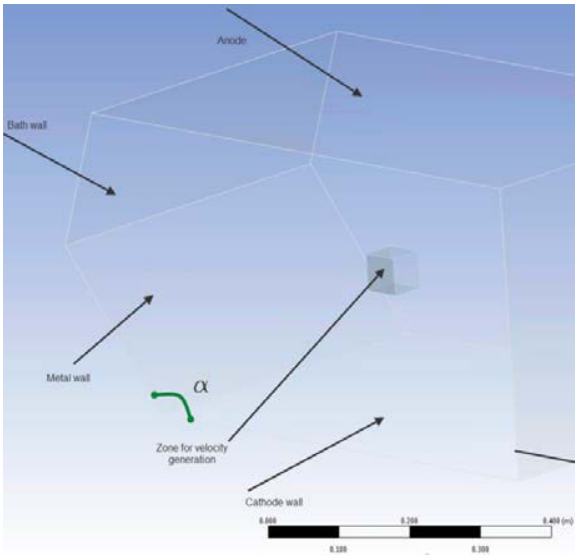

Figure 2. Wide slice model: Case setup with boundary conditions. α is the angle between side wall and cathode top, including zone for velocity generation (small cube).

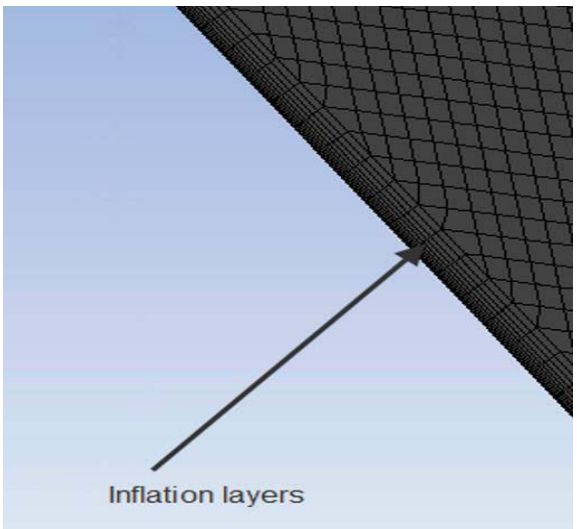

Figure 3. Wide slice model: Generated surface mesh with inflation layers added at the metal wall.

An example of a time series is shown in Figure 4. A similar time series will occur for most parameters, only that the bath film growth velocity will vary. The bath film growth ceases when the cathode surface close to the metal wall is depleted of bath film. Transport of more bath film to this part of the cathode surface is prevented by the surface tension between the metal and the bath film. The bath film trapped on the cathode surface might result in bottom ledge. Bottom ledge is usually located at some distance from the cell wall, as suggested in Figure 4. Beyond the pit, a bottom ledge seems to be under formation.

Table I. Overview used input parameters. The "guesstimate" is the value that was initially assumed to be representative, or the "best" value. This value was kept as standard value as other parameters were changed.

Parameter	Min	Max	"Guesstimate"
Contact angle between wall and liquid [degrees]	10	60	30
Surface tension between bath film and metal [Nm^{-1}]	0.75	10	3
Bath density [kgm^{-3}]	2050	2100	2050
Bath viscosity [$kgm^{-1}s^{-1}$]	0.0015	0.0035	0.0025
Metal density [kgm^{-3}]	2270	2270-	2270
Metal viscosity [$kgm^{-1}s^{-1}$]	0.0012	0.0012	0.0012
Horizontal force density [Nm^{-3}] [1]	0	1000	0
Vertical force density [Nm^{-3}][2]	0	45000	0
Metal wall slopes [degrees][3]	90	130	130

[1] Volume of velocity zone: $1.1 \cdot 10^{-6}$ m^3
[2] Volume of velocity zone: $0.7 \cdot 10^{-6}$ m^3
[3] α as defined as in Figure 2.

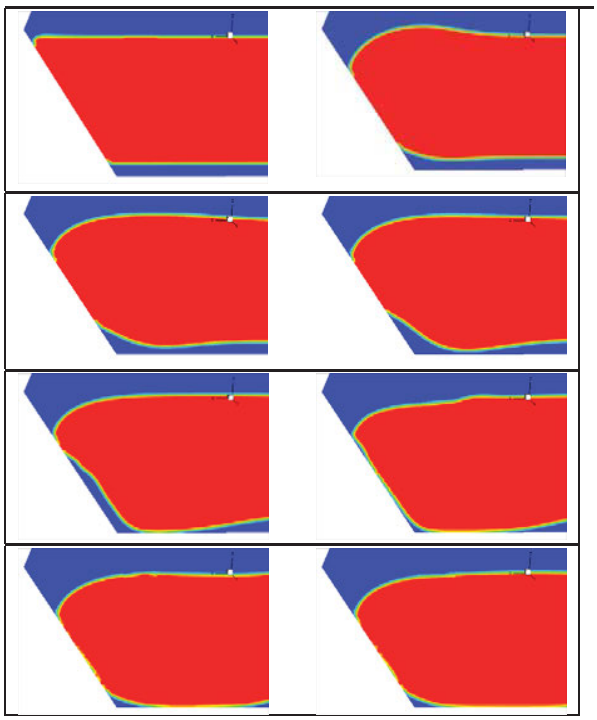

Figure 4. Shape of metal-bath-film region as a function of time with 1 sec time-step. Surface tension 5 N/m, contact angle 30°, bath film density 2050 kg/m^3 and no forced convection (x-velo = 0 N/m^3). The upper left picture is at t = 1 sec, upper right is at t = 2 sec, the picture below upper left is at t = 3 sec and so on.

The effect of the input parameters was analysed by tracking the bath film height along the side wall, as illustrated in Figure 5. A scripted analysis method was developed for comparison of bath film formation rates calculated as functions of tested parameters.

The height above cathode for the obtained point is plotted as a function of time step in the following section.

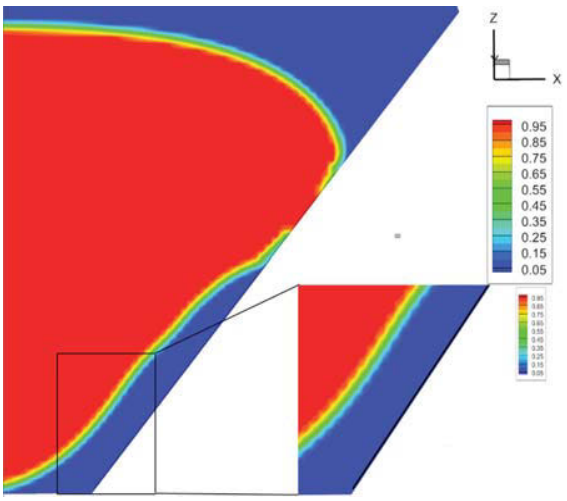

Figure 5. Analysis method based on extracting the bathfilm fraction along a line adjacent to the metal-lining interface.

Results

Parameter study with thin slice model

Before initiating a parameter study, the model mesh was assessed through evaluation of results from two cases with varying mesh density (normal: 0.8 million cells, dense: 2.5 million cells). Difference in mesh is visualised in Figure 6.

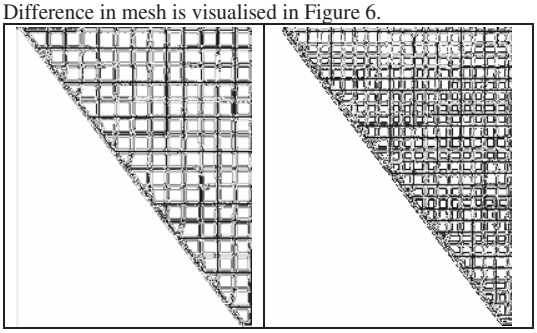

Figure 6. Mesh in extracted region for normal (left) and dense (right) mesh. Extracted region is 54 mm high and 41 mm wide.

Figure 7 shows the bath film maximum height above cathode as a function of time for the case with varying mesh for two different initial bath film heights and for two different mesh densities. The legend provides the bath film height (in cm) and the words "dense" or "normal" for the mesh density. There seems to be some effect of the mesh density, even though the shape is similar. The effect of the initial bath film thickness is dominant and will be further investigated later. For this initial parameter study it is considered sufficient to use the normal mesh density in order to save computational time.

Figure 8 shows the height above cathode of the bath film front as a function of time for varying surface tension between the bath film and the sidewall. The two last digits in the legend provide the contact angle (in °). The contact angle is the angle between the wall and the tangent to the interface at the wall measured from within the bath film. Thus, a contact angle of 10° corresponds to almost full wetting. As would be expected, the bath film transport up along the side wall is faster with increasing wetting.

On top of the metal zone, the bath film also grows downwards along the side wall. The contact angle influences how far the bath film extends down from top of metal zones. This is clearly seen by looking at which height the bath film from bottom meets the bath film from top, as indicated by the black circles in Figure 8.

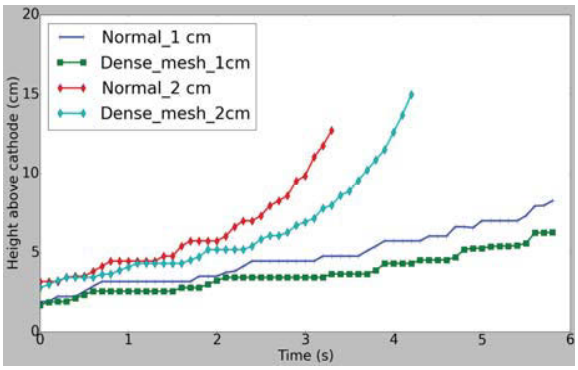

Figure 7. Height above cathode as a function of time for case with varying mesh density. 2050 kg/m^3 is used as film density. Initial bath film thickness was 1 and 2 cm as stated by the legend.

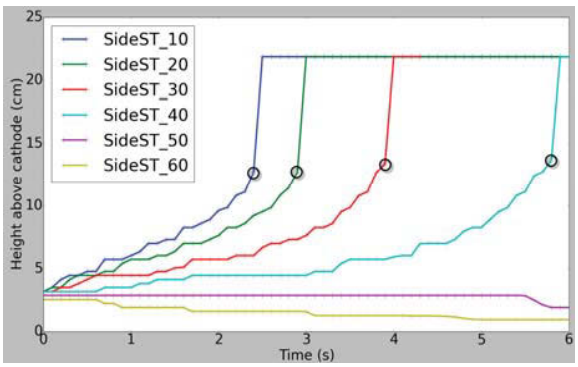

Figure 8. Height above cathode as a function of time for cases with varying contact angle between sidewall and the adjacent liquid (bath film or metal). 2 cm initial bath film thickness, 2050 kg/m^3 bath density and 5 N/m surface tension between bath film and metal. The black circles indicates the last point before the bath film from bottom meets the bath film from top.).

Figure 9 shows the height above cathode as a function of time for cases with varying surface tension between bath film and metal.

The legend provides the surface tension (in N/m) behind the SurfT text.

Figure 10 shows the height above cathode as a function of time for cases with varying bath film density. The legend provides the bath density (in kg/m^3) located at the end of each legend input line. A lower bath density will, as expected, result in faster growth of bath film along the side wall. A step-wise growth-rate of bath film is prominent (See Discussion).

Figure 11 shows the height above cathode of the bath film front as a function of time for varying metal flow velocity. The varying metal velocity was implemented by an imposed force source added to the small acceleration region shown in Figure 1. The legend shows the force density (in N/m^3) behind the "xVelo" text. The velocity is directed from top to bottom of the cell, and as expected; the bath film growth rate will stop at a certain force density. Velocity properties for the applied forces are summarised in Table II.

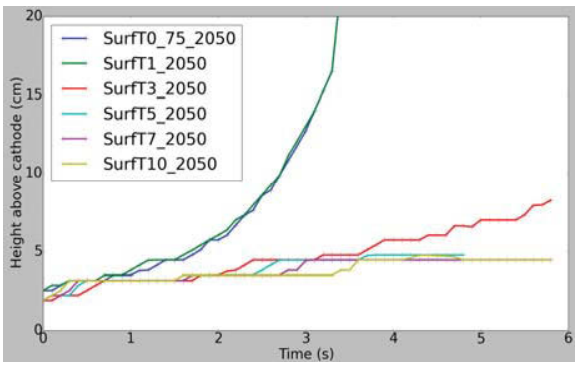

Figure 9**.** Height above cathode as a function of time for case with varying surface tension between bath film and metal.

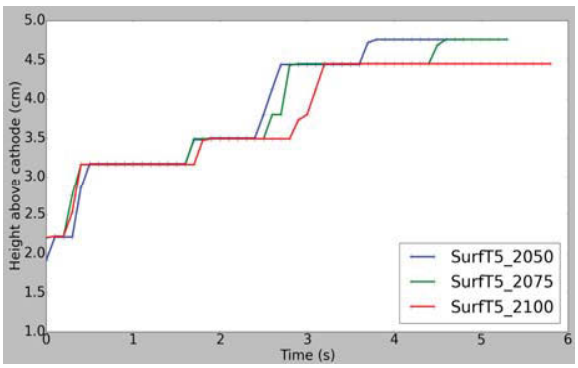

Figure 10**.** Height above cathode as a function of time for case with varying bath film density.

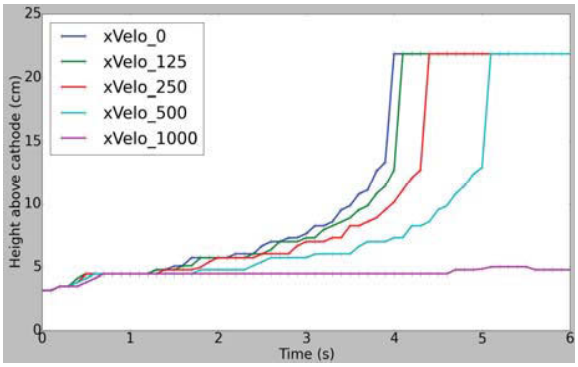

Figure 11**.** Height above cathode as a function of time for case with varying metal flow velocity.

Table II. Average and maximum velocities corresponding to applied force density.

Force density [Nm^{-3}]	125	250	500	1000
Average velocity [ms^{-1}]	0.004	0.005	0.007	0.012
Maximum velocity [ms^{-1}]	0.11	0.13	0.11	0.28

Figure 12 shows the metal – bath-film shape in addition to flow velocity vectors. Figure 13 show the corresponding flow velocity contours for the same case. The figure is produced after the bath film growth ceases for the case with xVelo = 1000 N/m^3 of which the bath film only reaches a height about 5 cm above the cathode top. Apparently, the downward directed metal flow prevents transport of the bath film along the cell side.

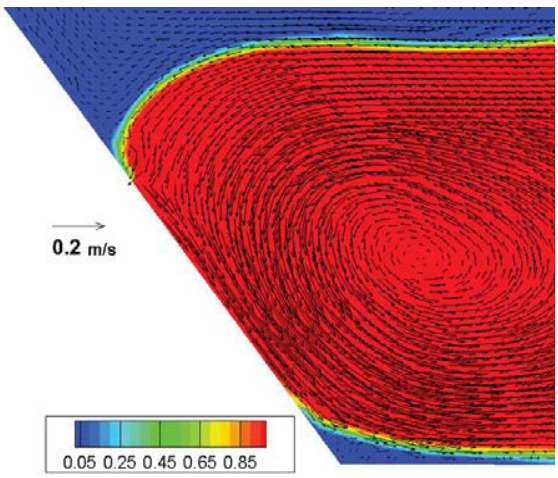

Figure 12**.** Volume fraction contours and flow velocity vectors. Image taken at the stagnation point for bath film transport (xVelo = 1000 N/m^3).

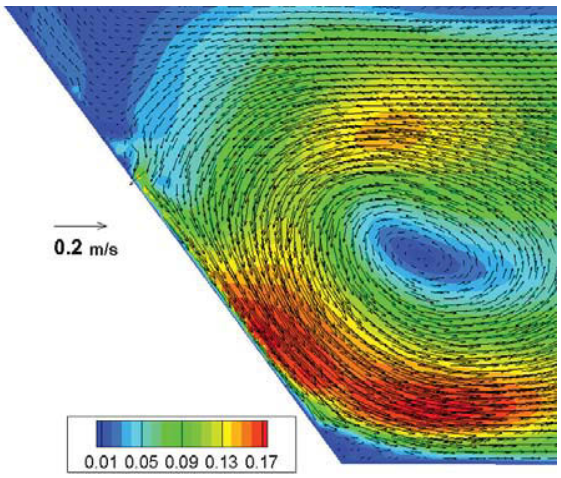

Figure 13. Flow velocity contours and vectors. Image taken at the stagnation point for bath film transport (xVelo = 1000 N/m^3).

Effect of horizontal velocity on local bath film thickness

To evaluate the effect of horizontal metal flow on bath film profile a wider slice model was set up, in which the profile along the cell is visible. The velocity was modified by adding a force in a region located close to the sloped side lining as shown in Figure 3. Figure 14 and Figure 15 show the height of the VOF interface between metal and bath as a function of the y-position, where the y-position is the distance from the end symmetry wall to the bath film - metal interface. The positions of the slices are indicated in Figure 16. For a small horizontal force (500 N/m^3) there is no detectable difference in iso-surface as a function of x-position, as shown in Figure 14. Thus, 500 N/m^3 does not have any effect on bath film formation and thickness. For comparison, the force that results in flow in the vertical direction had a large effect even at 250 N/m^3. A force of 10000 N/m^3 have a clear effect on the local bath film thickness, as the bath film thickness decreases close to the force zone and increases further away from the source, on the other hand, as shown in Figure 15. Thus, the ledge is shifted away for the region with high horizontal metal flow rate. The other parameters used for this investigation were: surface tension 5 N/m, contact angle 30°, and bath film density 2050 kg/m^3.

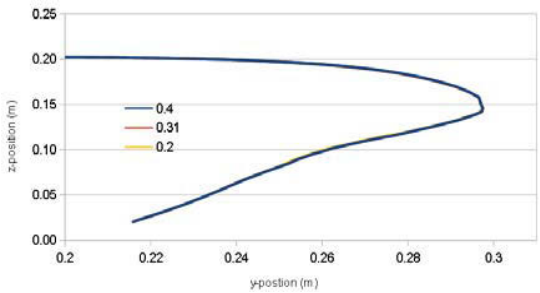

Figure 14. Bath film – metal iso-surface cross-sections with added horizontal velocity (xVelo = 500 N/m^3). The legend shows the distance between the cross-sections and the side symmetry wall as shown in Figure 16.

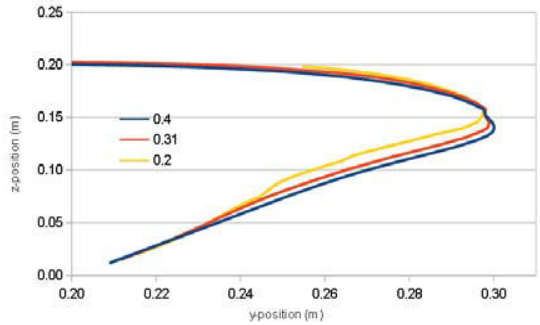

Figure 15. Bath film – metal iso-surface with added horizontal velocity (xVelo = 10000 N/m^3).

Effect of variable metal wall slope

In an attempt to evaluate the effect of the lining slope this parameter was varied in the bath film model, and the velocity of bath film growth up along the lining is evaluated. The results show clearly that the bath film transport rate is dependent on the metal wall slope. In Figure 17 the shape of the phases are shown after 1 second.

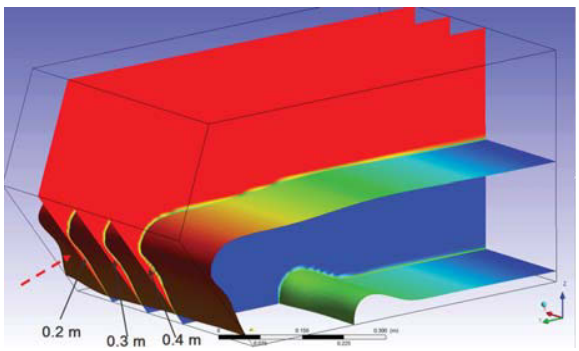

Figure 16. Overview of selected slices. The slices are coloured by volume fraction while the iso-surface is coloured by y-position. The location of the velocity generator is indicated by the dashed arrow at the lower left corner.

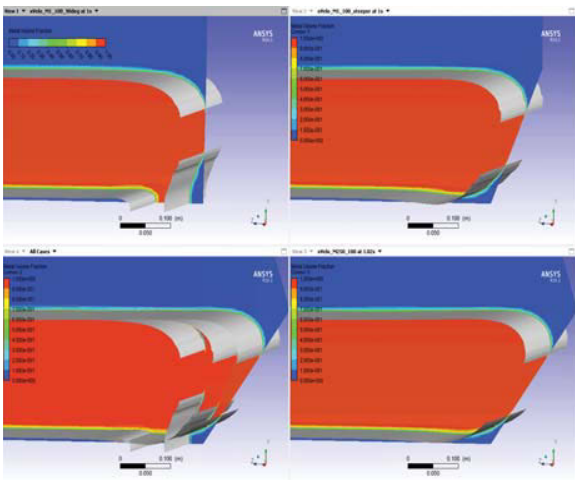

Figure 17. Metal volume fraction contours and metal-bath iso-surfaces after 1 sec real time for various side angles. The lower left figure shows the other three figures superimposed on each other, to allow direct comparison of iso-surfaces.

Discussion

The growth of bath film was modelled transiently from an initial flat position at the cathode top (2 cm thick layer), until it reached the bath by transport along the side wall.
The bath film growth rate was analysed by tracking the bath film front along the side wall. The surface tension between the metal phase and the bath film phase is one of the more important parameters. This is as expected, as a high surface tension will keep the liquids apart from each other. Hence, for making a model with realistic input; it is important to use as realistic values of this parameter as possible.

Another important input parameter is the surface tension between bath film and side wall. At above 60° contact angle no bath film growth is found. This is mainly of theoretical interest, since it seems clear that there will be very good wetting between bath and SiC or between bath and ledge.

Thirdly; the metal obviously is influencing bath film growth. For downward directed metal flow the bath film growth rate is reduced as the metal pushes the bath film downwards. For horizontally directed metal flow the bath film thickness decreases close to the high velocities and increases beyond this.

Having a bath film formation which is dependent on metal flow suggests that metal flow velocities should be minimized to avoid uneven distribution of bath film, which possibly will results in uneven ledge formation. To evaluate this, further studies on solidification and melting of bath film should be performed.

A forth parameter that may affect bath film formation is the geometric design of the cell. A low-sloped side-wall will results in lower bath film transport rate than a more normal slope. A higher slope (steeper wall) will reduce the surface tension force component in the vertical direction. Hence, the bath film formation transport rate will increase. It is assumed that the production of frozen bath is dependent on the transport time along the metal wall, as the transport time is the time available for freezing the bath film material into frozen ledge in the metal zone. If this assumption is true, it is important to optimize the geometrical shape of the cell to achieve a well-distributed and protective ledge. A proper cell wall geometry and construction is also important to have a high enough heat flux through the ledge.

All the plots in Figure 7 to Figure 11 show a step-wise growth rate of bath film along the side wall. This is related to the formation of a thicker bath film in the region just under the bath film front, before the bath film is further advanced upwards along the side wall.

Conclusion

The initial parameter study presented here shows bath film transport along the cell side and the transport rate is heavily dependent on several parameters. The initial results are interesting and we hope to broaden our understanding of side ledge dynamics by further developing the model, including freezing, more realistic metal flow and more realistic cell boundary conditions.

Acknowledgement

The current study was supported by the project "Superior Technology for Energy Efficient Aluminium Production" (STEP), financed by Hydro Aluminium and the Research Council of Norway. Permission to publish the results is gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

PILOT TEST OF ALUMINUM ELECTROLYSIS BY THE $NiFe_2O_4$-M INERT ANODES

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Keywords: $NiFe_2O_4$-M Cermet, Inert Anode, Pilot Test, Aluminum Electrolysis

Abstract

A pilot test of an aluminum electrolysis was completed in the year of 2014 using inert anodes with the composition $NiFe_2O_4$-M (M is metal). The volume of the cell is 110×66×79 cm, the size of inert anode is Φ34×H22 cm, the total number of inert anodes in the cell are six, the current density of the inert anode is about 0.8 A/cm^2, the total electrolysis current fluctuates from 2700 to 4000 A, the electrolysis duration time is 500 hours. The result of the pilot test indicated that the electrolysis process can be run successfully, since above inert anodes have better electric conductivity and corrosive resistance. In addition, none of the six inert anodes cracked during the whole electrolysis process even at the beginning of electrolysis when they were inserted in high temperature electrolyte. The main problem of pilot test is that the content of Fe as impurity is about 1.7% in the product, which is higher than the baseline value of GB/T. In order to remove Fe, related experiments are conducting continually.

Introduction

The Earth Environmental Protection Materials Company of China (EEPMC) investigated a series of inert anodes material candidates applied for aluminum electrolysis from 2004 to 2010 [1-3]. The most promising candidates was $NiFe_2O_4$-M Cermet, Inert Anode, Pilot Test, Aluminum Electrolysis (M is metal) cermet material, because the material demonstrated excellent performance in laboratory testing. It showed increased conductivity and corrosion resistance in the bath of aluminum electrolysis [2, 3]. In the year of 2011, a pilot test of this research was conducted supported by the Yunnan province government of China. The pilot test was finished in 2014. In this paper, the results related to the pilot test are discussed.

Inert anodes, Pilot test cell, and Experimental program

The dimensions of sintered $NiFe_2O_4$-M cermet inert anode are 34 cm in diameter and 20 cm in height, the weight is about 60 kg. Fig.1 indicates a pair of $NiFe_2O_4$-M cermet inert anodes before being inserted in the bath.

Fig.2 is the overhead view of cermet inert anodes distribution in the cell. The bath dimensions are 118×78×65 cm. The first step of the pilot test was roast and start of the cell, the carbon anodes were used as starting anodes. In this step, new built cell was solidified after slowly heating. The last step of the process, bath temperature raised over 1000 ℃, which was high enough to melt 350 kg new electrolyte.

The second step of the pilot test was that carbon anodes were replaced by $NiFe_2O_4$-M cermet inert anodes which must be done one by one in order to avoid the decrease of temperature in the bath. The interval of two replacements was 1.5~2.0 hours.

Fig.1 A pair of $NiFe_2O_4$-M cermet inert anodes which are connected together.

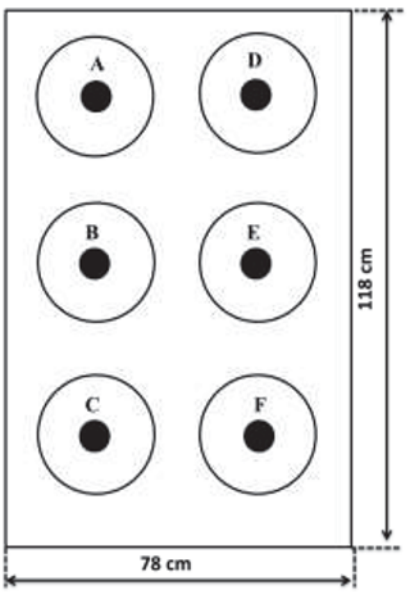

Fig.2 Overhead view of $NiFe_2O_4$-M cermet inert anodes distribution in the cell.

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The third step was aluminum electrolysis using cermet inert anode. The main difference between aluminum electrolysis by cermet inert and carbon as anode is current density of anode in the electrolysis. As for carbon anodes, the current density can be changed from 0.8 to 1.0 A/cm^2 , in the case of $NiFe_2O_4$-M cermet inert anodes, current density must be lower than 0.8 A/cm^2 [4,5]. If this value is overrun, the corrosion of cermet inert anodes will be accelerated. In our pilot test, current density was controlled in the range of 0.64~0.7 A/cm^2, corresponding to general current of 3539 A in the cell. However, if the bath temperature was lower than 970 ℃, the current over than 3539 A was required. In Table I, the time distribution of different stages in pilot test is shown.

Table I Time distribution at different stages in pilot test.

No.	Time (h)	Percent (%)
1. Roast and starting of the cell	120	24
2. Replacing anodes	8	1.6
3. Electrolysis by cermet inert anodes	368	73.6
4. End of cell	4	0.8

Results and discussions

3.1 Sinter results of $NiFe_2O_4$-M cermet inert anode:

The $NiFe_2O_4$-M blocks were sintered by 35 kW vacuum furnace after mold pressing and drying. Fig.3 shows the picture of $NiFe_2O_4$-M cermet inert anode after sintering. The height shrinkage and diameter shrinkage of the inert anodes are about 30% and 20%, respectively, after sintering. The density of $NiFe_2O_4$-M cermet inert anodes is about 5.3 g/cm^3. The $NiFe_2O_4$-M cermet inert anodes possess very good strength and toughness after sintering. If a sintered $NiFe_2O_4$-M cermet inert anode is dropped on cement floor from 1.5 meters height, the inert anode will not break. In addition, sintered $NiFe_2O_4$-M cermet inert anodes may be cut by matching work. The sintered $NiFe_2O_4$-M cermet inert anodes demonstrate excellent electric conductivity, inert anode resistance is in the range of 0~1.5Ω measured from the top to the bottom (different points) by multimeter.

Fig.3 The picture of $NiFe_2O_4$-M cermet inert anode after sintering.

3.2 The better resistance to heat rush:

Before inserting in high temperature bath, the $NiFe_2O_4$-M cermet inert anodes underwent a special heat treatment, did not crack neither when they were inserted in high temperature bath but nor during the electrolyzing process.

3.3 Better electric conductivity:

Fig.4 shows the electric resistances of the bath varied with time in the pilot test aluminum electrolysis. In the figure, the inert anode demonstrated good conductivity in the whole process.

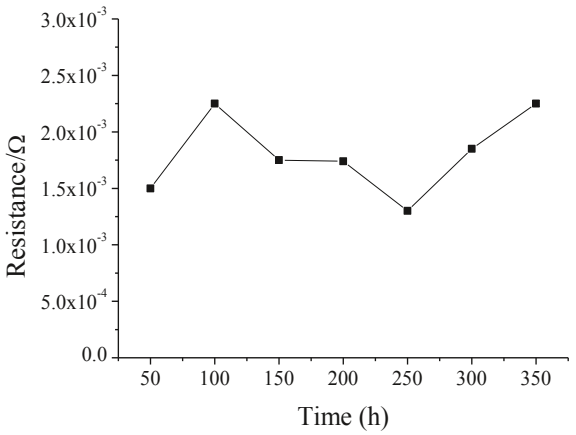

Fig. 4 Bath resistance varied with electrolyzing time in pilot test.

3.4 Corrosive resistance:

Fig.5 is a picture of $NiFe_2O_4$-M cermet inert anode after 100 h aluminum electrolysis on a small scale (1.5 kg) with BR1.25. Table II is the sizes of the small inert anode before and after 100 h electrolysis. According to Fig.5 and Table II, $NiFe_2O_4$-M cermet inert anodes demonstrated good corrosive resistance. However, it is regrettable that we did not measure the data of $NiFe_2O_4$-M cermet inert anodes after pilot test, because the electrolyte in bath froze very quickly after stopping the cell current, these inert anodes can not be raised from the bath.

Fig.5 The picture of $NiFe_2O_4$-M cermet inert anode after 100 h electrolysis (on a small scale).

Table II. Sizes of $NiFe_2O_4$-M cermet inert anode measured before and after 100 h electrolysis.

	Diameter (mm)	Height (mm)
Before electrolysis	25.54	67.54
After 100 h electrolysis	25.48	67.47

3.5 The impurity contents in produced aluminum:

Table III shows concerned impurity contents in the baseline level (GB/T) and electrolytic aluminum in the pilot test.

As shown in Table III, the impurities of Fe and Si in electrolytic aluminum overrun baseline level of GB/T Al 99.00. Fe mainly came from the Fe_2O_3 component of cermet inert anode materials, which was reduced to Fe in electrolytic process, and then went into electrolyzed aluminum. However, Si impurity does not origin from inert anodes, it comes mainly from aluminum silicate fibre used as heat preservation materials on the cell. The silicate fibre dropped into the bath in the pilot test cell during electrolyzing. Because Al_2O_3 is the heat preserving cover material on the industrial scale aluminum electrolysis, the impurity Si can be avoided easily.

Table III. The contents of impurity in electrolyzed Al and concerned standards.

Elements	Pilot test results (%)	GB/T baseline level (%)
Al	97.4811	99.00
Fe	1.3317	0.50
Si	0.9090	0.42
Ti	0.1293	-
Zn	0.0250	0.05
Ga	0.0249	0.05
Cu	0.0240	0.05
Mn	0.0199	-
Ni	0.0189	-

Conclusions

1. The sizes of $NiFe_2O_4$-M cermet inert anode shrank obviously after sintering, the density is about 5.391 g/cm^3, strength and toughness are clearly increased. In addition, the inert anodes do not break after being dropped on cement floor from 1.5 m in height, and the inert anodes may withstand cutting work;
2. Sintered $NiFe_2O_4$-M cermet inert anode performed well in the pilot test aluminum electrolysis, these inert anodes did not crack during heat rush, demonstrating excellent conductivity and corrosive resistance in the whole electrolytic process.
3. It is observed that the impurity of Fe content in electrolytic aluminum product overruns the baseline level of GB/T Al 99.00, Fe came mainly from Fe_2O_3 component of cermet inert anode. In order to overcome this problem, complementary tests continue.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THERMODYNAMIC ANALYSIS AND DYNAMIC SIMULATION ON CARBOTHERMIC CHLORINATION REACTION OF Al_2O UNDER VACUUM

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Keywords: *Ab initio* molecular dynamics; carbothermic-chlorination reaction; thermodynamics; interaction; Al_2O

Abstract

In this research, thermodynamic analysis and CASTEP package of the Material Studio program which is based on density functional theory (DFT) formalism were used to study the carbothermic-chlorination ($AlCl_3$) reaction of Al_2O under vacuum. Thermodynamic calculations indicated that AlCl(g) can be generated by carbothermic-chlorination process at 1760K and 60Pa. The interaction of Al_2O and $AlCl_3$ with C showed that the chemical adsorption of Al_2O and $AlCl_3$ did take place on C(001) crystal plane, at the same time, new chemical bond have been formed between Al atom dissociated from Al_2O and Cl atom dissociated from $AlCl_3$ molecule. The result, after 1ps dynamics simulation, indicated that adsorbed AlCl and CO molecules have been generated in Al_2O-$AlCl_3$-C system, and they would escape from C surface after a longer period of dynamics simulation time. It means that the reaction of Al_2O and $AlCl_3$ with C can be carried out under given constraint condition.

Introduction

Considerable theoretical and experimental efforts have been made in order to provide an insight into the carbothermic reduction of alumina to metallic Al[1-3], owing to the apparent disadvantages of the Hall-Héroult process[4-6]. The direct carbothermal reduction has the potential to reduce energy consumption by up to 38%, capital costs by more than 60%, and decrease CO_2 emissions by up to 30%.[7] Meanwhile, It also reduces overall operating costs by 25-30%, and has no PFCs.[8] In spite of considerable effort, the carbothermic reduction of Al_2O_3 to Al remains to be a formidable technical challenge, because of the high temperature required and aluminum carbide and oxycarbide byproducts formed.[9]

Wang et al.[10] proposed that AlCl(g) would be generated by carbothermal reduction-chlorination of alumina at high temperature zone and disproportion into Al and $AlCl_3$(g) which can be easily separated at lower temperature zone in vacuum furnace. It can be represented by the following overall reactions (1) and (2):

$$Al_2O_3(s)+3C(s)+AlCl_3(g)\xrightarrow{1760K,60Pa}3AlCl(g)+3CO(g) \quad (1)$$

$$3AlCl(g)\xrightarrow{970K,60Pa}2Al(l)+AlCl_3(g) \quad (2)$$

Based on the theory of Wang, Feng et al[11] further pointed out that alumina firstly reacted with carbon to generate gaseous Al_2O, Al, CO in the crucible; Secondly, gaseous AlCl would be formed by chlorination ($AlCl_3$ used as chlorinating agent) of Al_2O, Al gases.

However, the mechanism of chlorination of Al_2O(g) in the presence of carbon, particularly under vacuum atmosphere, has not been studied until now. The purpose of this work is to preliminarily investigate the thermodynamics and kinetics of the formation of AlCl according to the following reaction (3) by vacuum and to further enrich basic theory for the process of aluminum production by carbothermal-chlorination reduction of alumina.

$$Al_2O(g)+C(s)+AlCl_3(g)\xrightarrow{1760K,60Pa}3AlCl(g)+CO(g) \quad (3)$$

Calculation method and model

Thermodynamic calculation

The relationships between standard state Gibbs free energy change estimated in the temperature range of 800.00-2500.00K per 100K (the manual setting value) and temperature for the reaction (3) under ambient pressure (100kPa) were calculated by HSC chemistry 6.0 software with "Gibbs free energy function method[12]" shown from (4) to (10).

$$C_p = A + B\cdot10^{-3}\cdot T + C\cdot10^{5}\cdot T^{-2} + D\cdot10^{-6}\cdot T^{2} \quad (4)$$

Where A, B, C and D are coefficients estimated from experimental data.

$$H_T = H_f(298.15K)+\int_{298.15}^{T}C_p dT+\sum H_{tr} \quad (5)$$

Where H_T is enthalpy of formation at temperature T, H_f(298.15K) is enthalpy of formation at 298.15K, H_{tr} is enthalpy of transformation.

$$S_T = S(298.15K)+\int_{298.15}^{T}\frac{C_p}{T}dT+\sum\frac{H_{tr}}{T_{tr}} \quad (6)$$

Where S_T is entropy of substance at temperature T, S(298.15K) is standard entropy of substance, T_{tr} is temperature of phase change.

$$G = H - TS \quad (7)$$

Where G is Gibbs free energy.

The reaction enthalpy is：

$$\Delta H_r^{\theta} = \sum_B \nu_B H_B^{\theta}(\text{Product}) - \sum_B \nu_B H_B^{\theta}(\text{Reactant}) \quad (8)$$

The reaction entropy is:

$$\Delta S_r^\theta = \sum_B \nu_B S_B^\theta(\text{Product}) - \sum_B \nu_B S_B^\theta(\text{Reactant}) \quad (9)$$

Gibbs energy of reaction:

$$\Delta G_r^\theta = \sum_B \nu_B G_B^\theta(\text{Product}) - \sum_B \nu_B G_B^\theta(\text{Reactant}) \quad (10)$$

Based on the atmospheric pressure calculation, the relationships of the non standard Gibbs free energy change vs temperature under the pressure of 10, 10^2, 10^3 and 10^4Pa have been calculated by using

$$\Delta G_r = \Delta G_r^\theta + RT \ln Q_p \quad (11)$$

Where Q_p is the reaction quotient.

$$Q_p = \frac{(\frac{P_{AlCl(g)}}{100kPa})^3 \times (\frac{P_{CO(g)}}{100kPa})^1}{(\frac{P_{AlCl_3(g)}}{100kPa})^1 \times (\frac{P_{Al_2O(g)}}{100kPa})^1} \quad (12)$$

Theoretical calculation

In this study, the DFT calculations were performed by using the program package Cambridge Sequential Total Energy Package (CASTEP) in the Materials Studio software. Exchange and correlation effects were treated with generalized gradient approximation (GGA) implemented in the Perdew-Burke-Ernzerhof functional (PBE) as exchange-correlation functional. [13-15] Cutoff energy of 300 eV in plane wave basis sets were verified to be sufficient to obtain good numerical accuracy. The ultrasoft pseudopotential method was used.[16]

The C(001) surface was modeled by 8 atomic layers periodic slab. In order to avoid the interaction of periodic configurations, a vacuum layer of 15.0 Å was added to the surface. Optimizations were performed before and after Al_2O and $AlCl_3$ molecules were added to the supercell as the initial structure to calculate the stable structure and the properties. The convergence criterion was as follow: the energy change below 2×10^{-5}eV/atom and displacement change less than 0.0002 Å. The ab initio molecular dynamics (AIMD) [17] simulations were carried out on the optimized C surface in contact with the Al_2O and $AlCl_3$ molecules, using the NPT ensemble at 1760K, 60 Pa, a time step of 1 femtosecond and the simulation time 1 ps. The Andersen barostat[18,19] and Nosé [20] thermostat were used to control pressure and temperature, respectively.

Results and discussion

Results of thermodynamic calculation

The relationships of Gibbs free energy change (△G) versus temperature for the reaction (3) at atmospheric pressure and different vacuum degrees were shown in Fig.1. As can be seen, △G value of reaction (3) decreased with increasing the temperature under a certain pressure and decreasing the system pressure at a constant temperature in the range of 800.00-2500.00K. Thereby, the higher the temperature is, and the lower the system pressure is, the more easily the forward reaction is carried out. Additionally, As a result of a capacity increasing reaction of this process, according to Le Chatelier's principle, decreasing the partial pressure of the product gases reduced the initial reaction temperature at which Gibbs free energy change for reaction (3) equals to zero. The spontaneous reaction temperature T_{start} calculated via the theoretic formula (13) was 1011.37K, 1107.17K, 1223.01K, 1365.94K, 1559.58K under the pressure of 10, 10^2, 10^3, 10^4 and 10^5 Pa, respectively. Therefore, it is concluded that chlorination of $Al_2O(g)$ and C could be performed in conditions of temperature 1760K and the pressure of 60Pa.

$$T_{start} = -\Delta G_r^\theta / R\ln Q_p \quad (13)$$

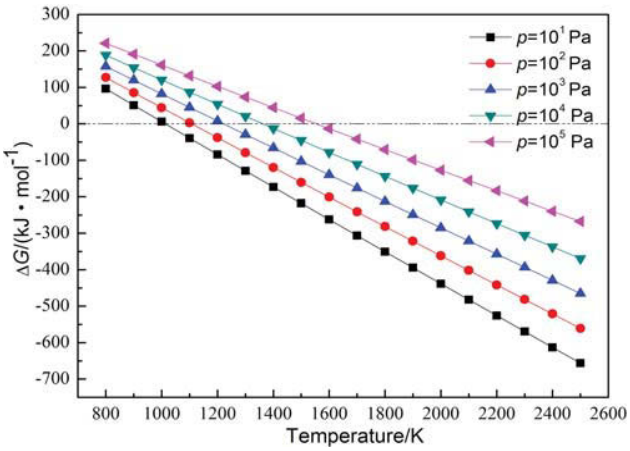
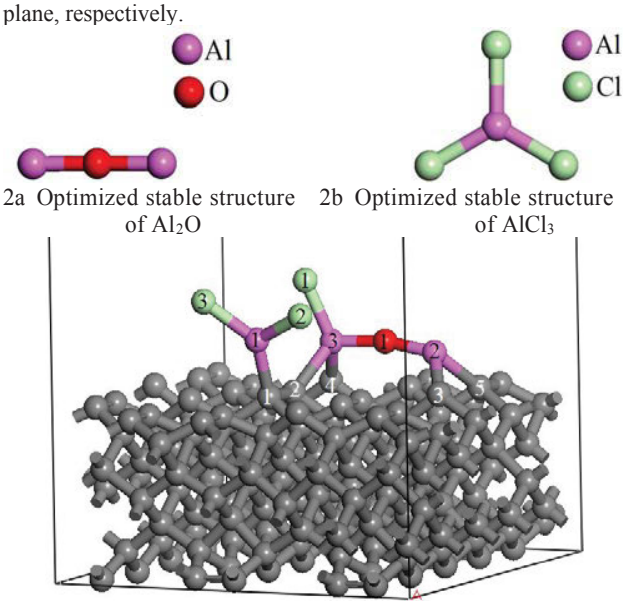


Fig 1. △G of reaction (3) vs temperature at different pressures

The interaction of Al_2O and $AlCl_3$ with C

Fig.2 represented the optimized stable structure of free state Al_2O, $AlCl_3$ and the adsorbed state of Al_2O and $AlCl_3$ on C(001) crystal plane, respectively.



2a Optimized stable structure of Al_2O 2b Optimized stable structure of $AlCl_3$

2c Optimized stable structure of Al_2O and $AlCl_3$ on C(001)

Fig 2. Optimized stable structure of Al_2O, $AlCl_3$ and Al_2O with $AlCl_3$ on C(001)

The Mulliken populations and bond lengths of the free state Al_2O, $AlCl_3$ along with Al_2O and $AlCl_3$ on C(001) optimized were given in table 1, 2, 3.

According to the figures given in table 3, it could be seen that the Mulliken populations of Al and C, and bond lengths were in the range 0.41-0.72, 1.94964-2.16061Å, respectively. Combined with the stable structure of Fig.2a and Fig.2c, it could be concluded that new chemical bond had been formed between Al and C, i.e. the chemical adsorption of Al_2O and $AlCl_3$ did take place on C(001) crystal plane.

Compared with table 1,the Mulliken populations and bond lengths between Al and O in table 3 both slightly increased, which was mainly because of the Al(3)-Cl(1) bond formed. It could be determined that Al_2O still existed in the form of adsorbed state molecule, with no decomposition, by comparing Fig. 2c with Fig. 2a. The bond length of Al(1)-Cl(1) measured was 3.75518Å obviously longer than Al-Cl bond lengths in table 2. Combining the comparison of the structure between Fig.2c and that of Fig.2b, Al(1)-Cl(1) bond could be considered to be dissociation before new Al-Cl bond formed by Al(3) and Cl(1) decomposed. That means $AlCl_3$ adsorbed on substrate C(001) surface had been dissociated in the present of Al_2O.

Table 1. The Mulliken populations and bond lengths of the free state Al_2O

bond	population	bond length(Å)	bond	population	bond length(Å)
Al-O	0.20	1.71412	Al-O	0.20	1.71432

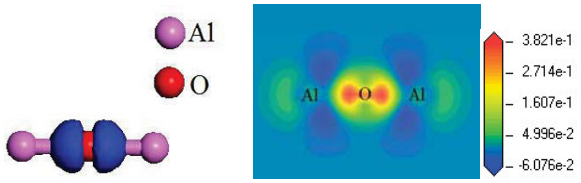
Table 2. The Mulliken populations and bond lengths of the free state $AlCl_3$

bond	population	bond length(Å)	bond	population	bond length(Å)
Al-Cl	0.53	2.07932	Al-Cl	0.53	2.07943
	0.53	2.07870			

Table 3. The Mulliken populations and bond lengths of Al_2O with $AlCl_3$ on C(001)

bond	population	bond length(Å)	bond	population	bond length(Å)
Al(1)-Cl(2)	0.56	2.10714	Al(1)-C(1)	0.60	2.00983
Al(1)-Cl(3)	0.58	2.07812	Al(2)-C(3)	0.72	1.94964
Al(3)-Cl(1)	0.55	2.10892	Al(2)-C(5)	0.41	2.16061
Al(2)-O(1)	0.57	1.72256	Al(3)-C(2)	0.46	2.16061
Al(3)-O(1)	0.45	1.80602	Al(3)-C(4)	0.56	2.04315

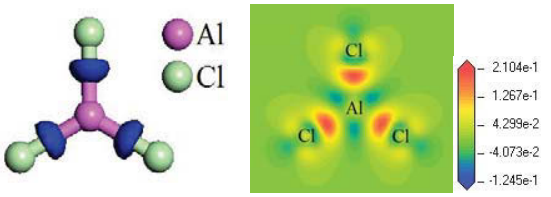
Fig.3 represented charge density difference of the free state Al_2O after optimization, in which Fig.3a was stereogram and Fig.3b was sectional drawing, respectively. It could be intuitively found that electrons mainly gathered between O atom and Al atoms and clustered around O atom from Fig.3.Thereby, the bonds between O and Al were ionic bonds, with a small amount of covalent component.



3a. Stereogram 3b. Sectional drawing

Fig 3. Charge density difference map of the free state Al_2O optimized

Fig.4 showed charge density difference of the optimized free state $AlCl_3$, in which Fig.4a was stereogram and Fig.4b was sectional drawing, separately. According to Fig.4 above, we could obviously see that electrons primarily distributed between Cl atoms and Al atom and were partial to Cl atoms. Thus, the bonds between Cl and Al were covalent bonds.



4a. Stereogram 4b. Sectional drawing

Fig 4. Charge density difference map of the free state $AlCl_3$ optimized

Fig.5 presented charge density difference of optimized stable structure of Al_2O and $AlCl_3$ on C(001) surface. In terms of O(1) and Al(2), Al(3); although Charge density difference between Al and O was slightly different from that of Fig.3, the characteristics of bonds was similar to that of free state Al_2O. To Al(1) and Cl(2), Cl(3); Al(3) and Cl(1); the type of bonds was analogous to that of Al-Cl bonds in free state $AlCl_3$. Concerning Al(1) and C(1); Al(2) and C(3), C(5); Al(3) and C(2), C(4); as can be seen from Fig.5, electrons mainly converged between C and Al atoms and clustered around C atoms. Therefore, the bonds between C and Al were ionic bonds, with a small amount of covalent component. According to the above, new chemical bond had been formed between Al and C, that is to say, the chemical adsorption exists on C(001) crystal plane. This analytical result agreed with that of the Mulliken populations.

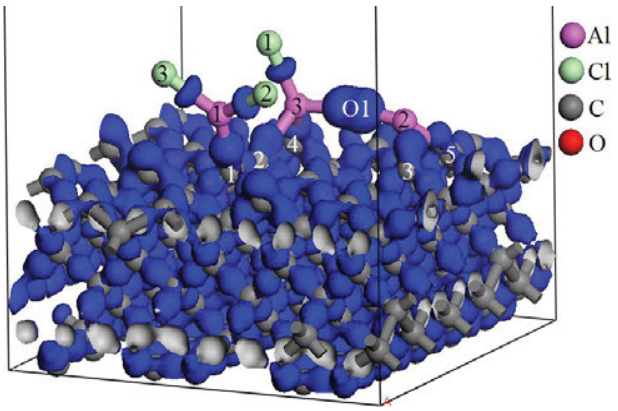


Fig 5. Charge density difference map of Al_2O and $AlCl_3$ on C(001) surface after optimized

Dynamic simulation of Al_2O and $AlCl_3$ with C

Fig.6 showed the structure of Al_2O with $AlCl_3$ and C(001) surface after 1ps dynamics simulation. The model changed obviously by comparing the structure in Fig.2c. The bond length of Al(1)-C(7) was 1.85315Å, C(6)-Cl(2) was 1.89530Å. Al(2)-O(1) and Al(3)-O(1) bonds both dissociated, which was mainly caused by Cl(1) adsorbed on Al(3) and C(6).The $AlCl_3$ molecule had dissociated to absorbed AlCl molecule, but the AlCl molecule did not escape. In addition, it was also observed that there was the formation of the molecular structure of CO, but also not escape (see black circle mark in figure 6). It can be deduced that absorbed Al(2) would react with Cl atom dissociated from one of the other $AlCl_3$ molecule, and Cl(2) also interacted with Al of other Al_2O molecule, generating new absorbed AlCl molecules.

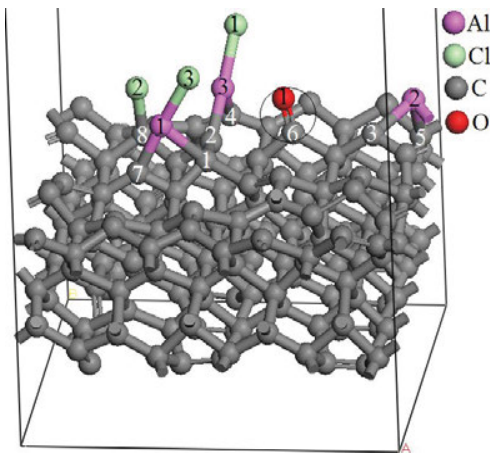


Fig 6. The structure of Al_2O and $AlCl_3$ on C(001) surface after 1ps dynamics simulation

Fig.7 represented the structure and PDOS (partial density of states) of the free state AlCl molecule after optimization. The bond length of the free state AlCl molecule measured was 2.15144Å obviously longer than that of Al(1)-Cl(3) and Al(3)-Cl(1),1.98125Å and 2.12372Å, respectively after 1ps dynamics simulation. Thereby, it could be predicted that the bond lengths of AlCl molecules adsorbed on C(001) surface would increase to 2.15144Å and then escape from the surface after a longer period of dynamics simulation time. From the PDOS of AlCl, including six regions, we noticed that nearby the 0eV, the PDOS was mainly from Al3s orbital, and was mainly contributed by Cl3p state nearby the -2eV region. In the range -4~-3eV and nearby the 6eV, the PDOS was mainly contributed by the Al3p orbital. The PDOS was mainly from Cl3s state between -14 and -13eV.

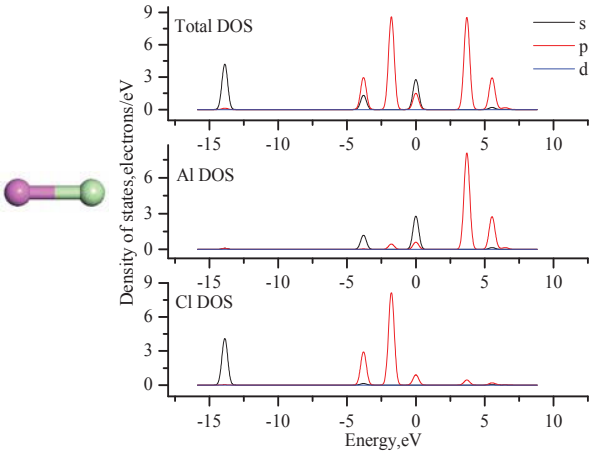


Fig 7. Structure and PDOS of AlCl molecule

Fig.8 showed the PDOS results of Al_2O and $AlCl_3$ on C(001) surface after 1ps dynamics simulation, the PDOS peaks of Cl(1) and Cl(3) atoms in absorbed AlCl molecules in Fig.8 with in Fig.7 roughly the same, therefore, we could further draw conclusion that in conditions of temperature 1760K and under the pressure of 60Pa, the reaction of Al_2O and $AlCl_3$ with C could be carried out through successive steps: In the first step, adsorbed AlCl molecules were formed; these AlCl molecules would escape from C surface and exist in the form of the free state AlCl molecules in the second step. It could be found there was a significant negative shift of the Cl(1) and Cl(3) peaks in Fig.3 at the same time the negative shift of the Cl(3) peaks were more than that of Cl(1) peaks, and that there was an obviously broadened at -5.5~-1eV for Cl(1) and -7~-2eV for Cl(3).

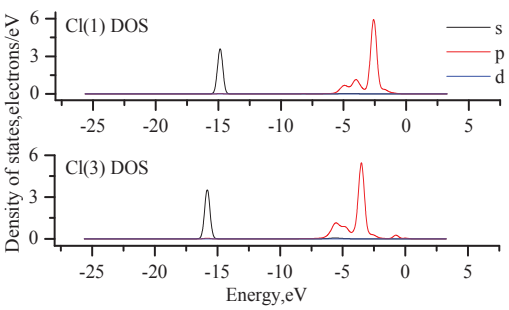


Fig 8. PDOS of Cl(1) and Cl(3) after 1ps dynamics simulation

Fig.9 represented the PDOS of the free state CO molecule after optimization. The bond length of the free CO molecule measured was 1.15090Å shorter than that of C(6)-O(1) bond,1.19648Å after 1ps dynamics simulation, because the CO molecule is interacting with C surface. Thus, it could be predicted that the bond length of CO molecule adsorbed on C(001) surface would decrease to 1.15090Å and then escaped from the surface after more dynamics simulation time. The PDOS of CO, including five regions, showed that at about 0eV, the PDOS was mainly from C2s, 2p orbital, and was primarily contributed by the O2p orbital nearby the -5eV region. The PDOS was from O2p and C2p states nearby the -2.5eV region and 6~7.5eV. The PDOS was mainly from O2s state at about -20eV.

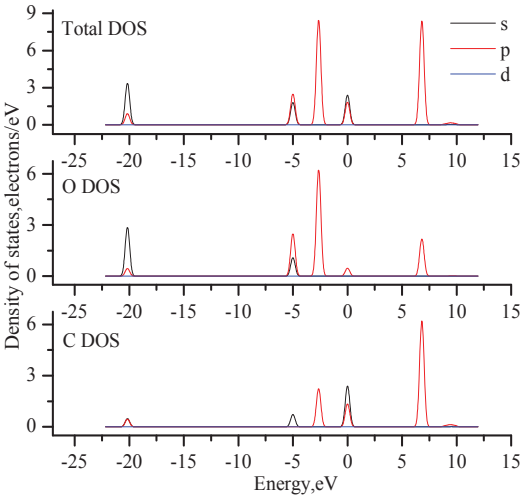


Fig 9. PDOS of the free CO molecule

Fig.10 showed the PDOS of CO molecule absorbed on C(001) surface after 1ps dynamics simulation. The PDOS peaks of CO molecule in Fig.10 roughly the same with the PDOS of free CO molecule in Fig.9, it proved that CO molecule had been generated in Al_2O-$AlCl_3$-C system and would escape from C surface at temperature 1760K and under the pressure of 60Pa after a period of dynamics simulation time. This analysis result well corresponded to that of thermodynamic calculation, indicating the reaction Al_2O and $AlCl_3$ with C could be performed. It could be found there was a significant negative shift of the CO peaks in Fig.10 and there was an obviously broadened at -10~-2eV for O.

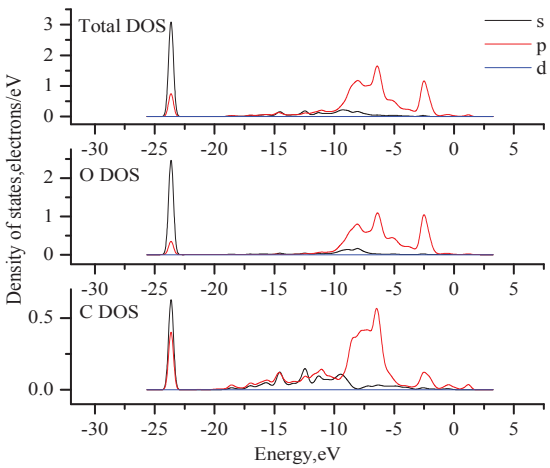


Fig 10. PDOS of CO molecule absorbed on C after 1ps dynamics simulation

Conclusion

The reaction of Al_2O and $AlCl_3$ reacted with C under the condition of 1760K, 60Pa was studied by thermodynamic analysis and CASTEP package of the Material Studio program. The calculation results indicated：

(1) The spontaneous reaction temperature calculated was 1011.37K, 1107.17K, 1223.01K, 1365.94K, 1559.58K under the pressure of 10, 10^2, 10^3, 10^4 and 10^5Pa, respectively. The higher the temperature is, and the lower the system pressure is, the more easily the forward reaction is carried out. The reaction could be performed in conditions of temperature 1760K and under pressure of 60Pa.

(2) $AlCl_3$ adsorbed on substrate C(001) surface dissociated in the presence of Al_2O and then new chemical bond was formed between Al atom in Al_2O and Cl atom from one of Al-Cl bonds in $AlCl_3$ molecule dissociated after optimization. However, Al_2O adsorbed on C(001) crystal surface did not decompose before dynamics simulation.

(3) Adsorbed AlCl and CO molecules was generated in Al_2O-$AlCl_3$-C system, they did not escape after 1ps dynamics simulation, but they would escape from C surface after a longer period of dynamics simulation time, i.e. the reaction (3) could be carried out under given constraint condition, corresponding to the result of thermodynamic calculation.

Acknowledgments

The research leading to these results was supported by funding from the National Natural Science Foundation of China youth fund (No. 51104078), region fund of the National Natural Science Foundation of China (No. 51264023), the Yunnan Province applied basic research fund (No.2010CD022), the Program for Innovative Research Team in University of Ministry of Education of China (No.IRT1250), the National Natural Science Foundation of China-Yunnan United fund (No.U1202271).

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DESIGN, START-UP AND PERFORMANCE OF FOUR GAS TREATMENT CENTERS FOR MA'ADEN ALUMINIUM

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Keywords: Gas Treatment Center (GTC), HF, dry alumina scrubber, hairpin duct, boosted suction.

Abstract

Ma'aden Aluminum Company is a joint venture between Ma'aden and Alcoa that operates the largest vertically integrated aluminum complex in the world. This paper focuses on the four (4) Gas Treatment Centers (GTC) that were supplied by Fives to treat gases from the 740,000 pot exhaust TPY aluminium smelter.
Ma'aden GTCs are among the largest in the world. The paper reviews the special features of these GTCs, which rely on the TGT-RI scrubbing technology including the use of hairpin ducts to cool potline gases, the YPRIOS dual flow system to minimize potroom emissions and the use of HF gas monitoring equipment at the outlet of each filter compartment to improve the tracking of scrubbing performance. It also presents the performance achieved after two years of operation, the minor corrections that were made to improve the reliability of some components and field improvements that could be reused on future projects.

Introduction

In 2010, Fives was awarded the turnkey supply of four (4) large Gas Treatment Centers (GTC) to treat pot exhaust gases from a total of 720 AP-37 pots. Each GTC is composed of eighteen (18) TGT-RI alumina dry scrubber modules and six (6) induced draft fans (ID fans) that are designed to maintain a 2.6 Nm^3/s draft at each pot outlet at all times. Furthermore, each GTC is provided with a boosted suction system (YPRIOS) to double the exhaust flow from 2.6 to 5.2 Nm^3/s extracted at pots during anode change or metal siphoning in order to reduce roof emissions in the potroom. Considering the high summer ambient temperatures (up to 55 °C) and maximum pot gas temperatures (up to 190 °C), pot gas cooling is required to ensure the GTC inlet temperature never exceeds 135 °C. In Ma'aden, this is achieved by combining hairpin ducts (one at each pot) with dilution air. Considering pots under normal draft, pots with boosted flow and the dilution air cooling, each GTC is designed to handle a total flow 2,450,000 Nm^3/h.

The GTCs' fabrication and construction started in 2011 and lasted over two years. Most of the equipment has been fabricated in the GCC region. The TGT-RI filters are made of 960 bags of 7 meters long each. They were pre-assembled at grade on the smelter site (see Figure 1) and then transported in the vertical position to their final location, which was about one kilometer away. Pre-assembly of such large filter units at grade level is very beneficial towards erection and reduces the risks associated with assembly work and welding at a higher elevation. For the last two GTCs, filter bags and cages were also installed in the pre-assembly area to reduce the erection time in the courtyard even further.

The first GTC was started in November 2012 while the fourth GTC was started in December 2013. All 720 pots were in operation by June 2014.

Figure 1: TGT-RI filter modules in pre-assembly area.

Figure 2: Filter module transportation to GTC

Pot Gas Cooling – Choice of Technology and Actual Performance

Fives studied various alternatives for gas cooling [1][2] because using only dilution air was not an option considering the 190 °C maximum pot gas temperature. In that case the total gas volume to be treated and the size of the GTC would have had to be increased by 43% in order to reduce the GTC inlet temperature below 135 °C. Heat exchangers could have been used but would have required installing large air coolers for the water cooling circuit. After evaluating the options, the solution that finally was implemented was to install hairpin ducts at each pot (see Figures 3 and 4) to enhance cooling by air convection. This is the first time this is applied in an aluminium smelter and a solution that is very robust as it requires no maintenance and does not add any motorized equipment or systems. It was designed to increase the overall ductwork convection cooling by 11 °C at maximum pot temperature such that the volume of dilution air needed during the warmest days would be reduced from 43% to 30%. The GTC filtering area and fan capacity were (only) increased by 30% to meet this occasional dilution air need; yet this extra capacity can be beneficial when the air dilution need is low or nil as it reduces the GTC pressure drop if all modules are in operation. The resulting fan power consumption is very low compared to the fan installed power capacity.

Figure 3: View of hairpins and potroom ducts.

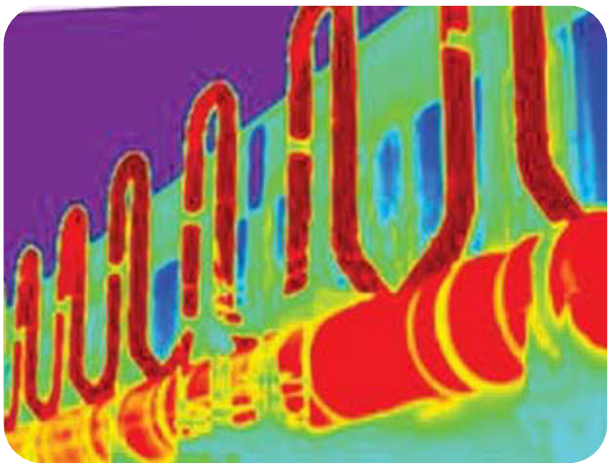

Figure 4: Hairpin ducts temperature overview

After two years of operation, dilution air has been rarely used (less than 1% of the time). This is mostly due to the pot thermal balance which is still exceptionally good, resulting in fewer holes in the pot crust to release heat. Hence, pot gas temperature is presently 25 °C to 30 °C lower than the maximum anticipated value. In the summer days, hairpin ducts and the overall convection losses of the GTC upstream ducts are sufficient to lower the GTC inlet temperature down to 120-130 °C while in the winter months the average inlet temperature is around 100 °C.

Even with the 25-30% lower gas flow at GTC inlet, it was decided to keep in operation all of the 18 filter modules since the velocity in the vertical reactor was still sufficient to entrain the injected alumina (there has been no incident of alumina fall-out into the lower inlet plenum). The lower filtering velocity reduces the filter pressure drop to allow savings in fan power. Furthermore, taking one or two filter modules out of service would require closing the manual reactor damper (as done during periods of filter maintenance) to ensure that there is no leakage of untreated gas through the idle filter units. The motorized filter outlet dampers are not designed for long term shut downs and don't provide a 100% sealing. Therefore, it cannot prevent leakage that increases the GTC fluoride gas emissions – that otherwise would require manual dampers closing[1] fully.

The lower gas volume is then suitable for an operation with four of the six ID fans (compared to the design condition of five fans in operation at N-1). Stopping one extra fan allows for optimal operation without closing excessively the modulating fan inlet

[1] A 1% leak through one idle module corresponds to a leak at the GTC of 0.05%. Considering an inlet concentration of gases fluoride of 350 mg/Nm^3, such leakage would be sufficient to increase by 33% GTC emissions if the 17 operating modules perform with an average outlet emission of 0.05 mg/Nm^3 HF.

dampers. If a lower number of fans had been established in design configuration (four for example), the lower gas volume would not have allowed stopping an extra fan as inlet damper closing would have become too important, thus leading to a poor fan mechanical efficiency. All the benefit of a lower gas volume on fan power consumption would therefore have been lost.

Boosted Suction System – YPRIOS

A boosted suction system has been implemented to ensure low potroom emissions. The objective is to double the flow at a pot outlet during maintenance operation on pots when it is required to remove one or more pot hoods. For such flow increase, the most appropriate solution consists in providing a parallel duct system with booster fans. This technology, YPRIOS, was first implemented at the Alcoa Fjarðaál smelter in Iceland [3] in 2006. An air-actuated diverter valve installed at the pot outlet directs the gas either to the normal ductwork or to the boosted duct system. Booster fans (see Figure 5) are used to generate the extra suction at the pot outlet flange and in order to double the pot flow, the pot static pressure has to be increased by a factor of four over normal suction. Gases drafted by booster fans are redirected into the main ductwork at the GTC inlet. Eight booster fans were supplied for each GTC and up to 14 pots per GTC can be put in boosted suction mode at any given time.

Figure 5: Booster fans shown in the center of the courtyard.

One of the challenges of such system is the reliability of the diverter valves. Their operation in the high magnetic field environment was problematic from the start where in particular that were difficulties at times to move the diverter flap. Also, once in movement, the carbon steel flap would hit the casing with force, causing damage over time to the flap shaft and support. Initially, typical diverter failure could be observed after less than 200 cycles. Fives investigated the problem and tested a number of solutions to improve their reliability. In the end, the 720 diverters were upgraded with a 316 stainless steel flap (SS316 is not affected by the magnetic field) and a new shaft in harder material (35 Cr Mo). The operation is now smooth and very reliable.

The power consumption of the dual flow system is low since booster fans are only in operation when work is being performed on pots. The power consumption of YPRIOS for 14 pots under boosted suction adds approximately 180 kW per GTC and only when pot boosted suction is required. For comparison, to reach the same performance (doubling flow at pots outlet) with a single duct system using air-operated dampers at each pot, the extra power consumption for the GTC ID fans would exceed 1200 kW on a continuous basis, even when boosted suction is not required at all.

GTC Design and Special Features

The Ma'aden GTCs are based on the proven TGT-RI filter modules (960 filter bags) that have been installed in Fjarðaál (Iceland) and Qatalum (Qatar). Because of the large number of modules, Fives selected a double row arrangement with central outlet plenum and ID fans at one end, instead of a single row layout with fans on the side. The fresh alumina distribution uses a central distribution box with individual alumina conveyors to each reactor. The fresh alumina is metered by a volumetric feeder that was not included in Fives' scope. This type of feeder has been used successfully by Alcoa at many of their smelters.

Despite the short distance between filters outlet and the common plenum, a modulating damper and an Annubar flow meter have been included to measure and balance gas flow handled by each filter. At Fives we believe that balancing gas flow and fresh alumina flow at each module is crucial in order to obtain a stable fluoride content in alumina downstream filters. Excessive fluoride content in a single module would lead to high emissions at the module outlet and could be enough to affect the overall GTC performance.

In order to help tracking the GTC performance in terms of HF emissions, Ma'aden decided to install HF monitoring at the outlet of each filter module. It was the first time such system was included in a greenfield GTC system. This monitoring is very helpful for operators, as diagnosing the source of a problem on large GTCs when high HF emissions are detected at stack can be very difficult.

This system allows identifying quickly whether the problem is common to all modules or if it is located on a specific module. It has been useful to rapidly find and address blockages of fresh alumina flow from a glove or other debris that has passed into the alumina handling system. With this monitoring system the reliability of the filters is greatly improved.

Each filter module outlet duct is also provided with a triboelectric dust leak detector, which is necessary to track defective bags in such large GTC with over 17,000 filter bags.

GTC Outlet Duct, ID Fans and Stack

The outlet gas is directed to six (6) DWDI (Double Width Double Inlet) ID fans each, driven by 1600 kW motors. The GTC outlet ducts have been designed in circular sections instead

of rectangular sections, to better resist the vibration that can be induced by flow patterns and acoustic waves. This has been a recurrent issue with large rectangular ducts on past projects.

The ID fans are provided with inlet and outlet silencers and acoustic insulation to reduce fan noise at the stack and around the fans. A site measurement of the sound pressure level around the fans has demonstrated that the guarantee of 85 dB(A) at 1 meter was achieved. The fans are located side by side in an accessible arrangement that makes maintenance easy and that allows removing fan motor and wheel (see Figure 6) with relative ease. They discharge into a common circular duct connected to an 8-meter diameter stack.

Figure 6: GTC 1 exhaust fans

Field Improvements

The overall GTC and ductwork design did not require major site corrective actions other than the upgrade of the diverter valves of the boosted suction system. However, one area of improvement has been to provide better access to the fluorinated air conveyors at the GTC discharge. Even with the double-row configuration, one GTC is nearly 60-meter long and the vertical drop of an air conveyor with 3 degree slope is three meters. This makes it hard to access the ports of the air conveyor at its farthest end. Fives' field representatives along with Ma'aden operators decided to modify the slope of these air conveyors down to one degree, using longer chutes and longer hanger rods. Although conclusive laboratory tests had been made with a lower slope for alumina air conveyors, project specifications rarely provide the opportunity to design with a slope less than three degrees. In Ma'aden, the one-degree fluorinated alumina air conveyors are working well, while making easier the access to the full length of each air conveyor. In the future, using a lower inclination should provide the extra benefit of lowering the GTC support structure and its associated CAPEX.

GTC performance

The GTC performance for the past two years has been excellent. The yearly average HF emission of the four GTCs in 2014 has been around 0.4 mg/Nm^3, with the 2014 summer emissions at an average temperature of 120 - 125 °C being at 0.5 - 0.55 mg/Nm^3. In 2015, the HF summer emissions are slightly higher (0.60 - 0.65 mg/Nm^3), following an increase in fluoride emissions from pots at the end of May. A change in the quality of fresh alumina has triggered more fluoride emission from pots (the fluorination of alumina has then increased from an average of 1.70% -1.75% to an average of 1.80%- 1.85%). The graph in Figure 7 provides GTC 3 HF measurements at stack (in blue), inlet temperature (in red) and the ambient temperature (in green).

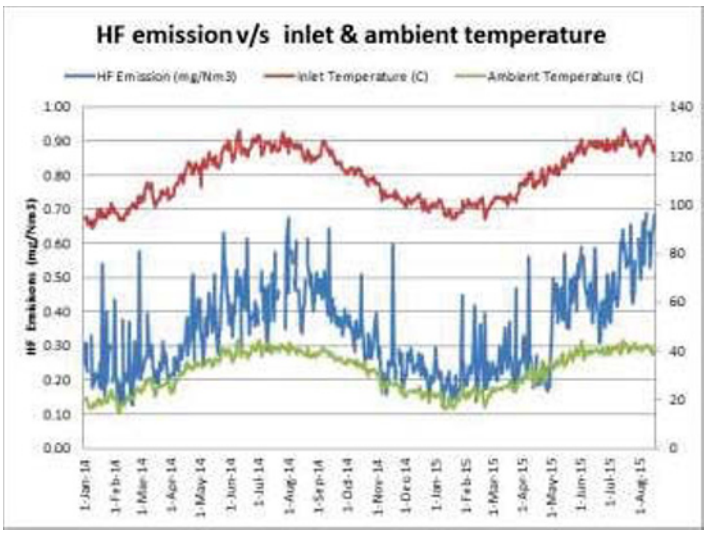

Figure 7: GTC 3 Process Data

The GTC performance is also very good in terms of maintenance and OPEX, considering the following operation parameters:

- The efficient gas cooling with hairpins has resulted in rare usage of dilution air (less than 1 % of the time).
- A low filter pressure drop (after two years of operation, the tubesheet pressure drop is around 1.65 kPa) and less than ten bag failures have been observed (out of 69,120 bags). This performance is partly due to the operation with 18 filter modules at a low filtering velocity (1 m/min).
- The limited use of dilution air and the resulting lower GTC pressure drop allow for the operation of only four ID fans (out of six) all year around. In winter, each fan inlet damper opening is around 60%, against around 70% in summer. The yearly average GTC fan power consumption is about 5300 kW (i.e. only 55% of the 9600 kW installed power).
- Overall, the GTCs have extra capacity to safely handle higher pot gas temperature if pot amperage is further increased.

Conclusion

The Gas Treatment Centers supplied in Ma'aden have demonstrated that low hydrogen fluoride emission can be achieved in hot climate with Fives' TGT-RI technology combined with efficient control of gas inlet temperatures.

The installation of hairpin ducts at each pot is an efficient and robust solution to increase convection gas cooling, hence minimizing the need for dilution air.

The installation of a continuous HF monitoring system at the discharge of each filter module has proven to be an efficient tool to track and quickly correct the cause of sudden HF peak emissions. It is a solution that should be considered in the future for all large GTCs.

It has also been demonstrated that the GTC could be designed with airslide type alumina conveyors having a lower slope than the conventional 3° slope that is part of most specifications. This could provide on future projects some savings in CAPEX.

Acknowledgements

The authors wish to thank Dr. Stephen Lindsay of Alcoa for reviewing this manuscript.

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Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MANAGEMENT AND PERFORMANCE OF THE LARGEST GAS TREATMENT CENTRE AT EMAL POTLINE DURING MAJOR SHUTDOWN OF MAIN EXHAUST FANS

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Keywords: Gas Treatment Centre, GTC, emission, HF, feed rate, emission

Abstract

EMAL commissioned Potline 3 in September 2013, comprising 444 DX+ pots that currently operate at 455 kA. Two large Gas Treatment Centers (GTCs) handle the pot exhaust gas, each one connected to 222 pots and processing 2 million cubic meters of gas every hour. Each GTC is powered by six large Main Exhaust Fans (MEF) through 6.6 kV switchgear, with N-1 design capacity. A major shutdown of three MEFs on the West GTC (5332) was planned on 14 April 2015 to do maintenance work on one section of the 6.6 kV switchgear. This activity lasted for six hours, during which several actions were taken to maintain GTC operation with minimum impact on pot performance and emissions. This paper discusses the actions taken before and during the shutdown, the resulting environmental impact and the lessons learned from this event.

Introduction

In EMAL Potline 3, two GTCs using alumina dry scrubbing technology are located in the courtyard between the two potrooms. The East GTC (5331) and the West GTC (5332) serve half the potline each. In addition, the fumes from the hot anode butt plant are treated by the West GTC (5332), making this GTC slightly larger with two additional compartments[1]. A typical layout of GTC 5332 is shown in Figure 1, showing the ductwork and number of pots connected to each segment.

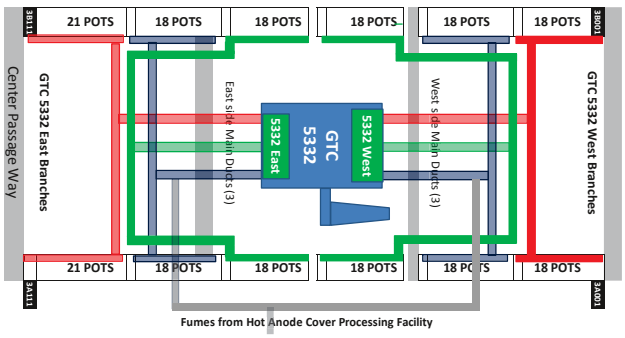

Figure 1. GTC 5332 layout.

Pot duct connections are distributed over six main ducts, three entering each end (half) of a GTC. The ductwork is designed to be extra-long in order to cool the pot exhaust gas below 125 °C by the time it enters the reactor/filter modules. The main ducts at each end merge into the half GTC inlet plenum, which distributes the gases to the reactor/filter modules and three MEFs. A balancing duct between the two halves of the GTC takes care of under pressure balancing during preventive maintenance of any fan.

Each GTC is serviced by one alumina tanker unloading station for primary alumina supply to the primary silo. Reacted alumina is supplied to the pots by two independent Pot Feed Systems for either half of the potline, using hyper-dense phase conveying.

A shutdown was planned on one-half of the 6.6 kV switchgear of GTC 5332 to rectify a chattering noise deficiency. This intervention would force the shutdown of three MEFs. A short test was conducted on 1 April 2015 to assess the impact of shutting down three MEFs and to decide on the operational parameters required during the major shutdown. After completion of the test and getting necessary approvals, three fans were shut down from 09:40 to 15:20 on 14 April 2015.

Data and SCADA screen shots were collected for analysis of the main plant parameters and impact assessment. A report was issued internally [1] explaining the results during the test.

Methodology

The actions taken before and during the reduced load operation at GTC 5332 were the following:

1. A test was conducted on 1 April 2015 by switching off three MEFs connected to Half Board 4531-EMV-002 (MEF #2, #4, and #6) and operating the remaining three MEF in manual mode. The East half of the GTC was operating with MEF #1 & #3 and the West half was operating with MEF #5. This was done to study the impact on potline performance and emissions.
2. A meeting was conducted with representatives from Power Maintenance, GTC Operations, Potline Operation, Reduction Process Control and Reduction Maintenance for discussing the test results and preparing for the planned 8 hour shutdown.
3. During the shutdown on 14 April 2015, the sequence of activity was the following:
   - Low Voltage (LV) changeover was carried out at 07:25 by the Reduction Electrical Maintenance team. All the affected equipment was stopped and restarted during the activity.
   - MEF #2, #4 and #6 were stopped at 09:40.
   - GTC operating conditions were set as shown in Table I, based on the results of the short test conducted on 1 April 2015.
   - The inlet guide vane dampers of MEF #1, #3 and #5 were opened to 100% using manual mode.
   - Inspections were done in the potrooms to observe if any excessive fumes were present.

[1] GTC 5332 has 34 filter compartments, 17 on either side. GTC 5331 has 32 filter compartments, 16 on either side.

Table I: Operating parameters during the planned shutdown.

Parameter	Nominal	During 3 fan operation
Number of Filters	34	34
Number of MEFs – East half	3	2
Inlet Pressure – East half, Pa	1250	860
Number of MEFs – West half	3	1
Inlet Pressure – West half, Pa	1250	680
Inlet damper (fan) open	40%	100%
Bag house differential pressure (DP), Pa	2300 - 2400	1500 – 1900
Pulsing interval, s	As required	
Primary alumina feed rate, t/h	60	50 – 44
Alumina recirculation rate, t/h	40	40 – 35

4. MEF #2, #4 and #6 were restarted at 15:20, upon completing the maintenance activity.
5. After normalizing the GTC, bath temperature measurement was repeated in 10% of the pots (which had already been measured in the morning) to see whether it was adversely affected by lower gas suction from pots.
6. LV changeover was done on the second day, after the shutdown by Reduction Electrical Maintenance team.

Environmental Impacts

Observations during the one hour Test: Operating parameters were set during the one hour test as given in Table II. Since it was a very short duration test, environmental impacts were low - as shown in Figures 2, 3 and 4.

Table II: Changes in operating parameters during one hour test.

Parameter	Nominal	During 3 fan operation
Number of MEFs – East half	3	2
Inlet Pressure – East half, Pa	1250	861
Number of MEFs – West half	3	1
Inlet Pressure – West half, Pa	1250	678
Inlet damper (fan) open	42%	100%
Bag house DP, Pa	2350 - 2475	1300 – 2000
Primary alumina feed rate, t/h	60 - 63	50
Alumina recirculation rate, t/h	40	30

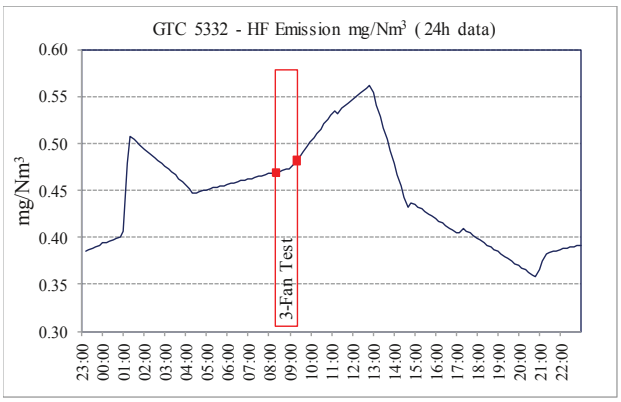

Figure 2. Stack HF emission during 3-Fan Test.

As shown in Figures 2 and 3, a proportional increase and a lagging adverse impact were seen on the HF and SO_2 emissions during the test, though the increase was small. The lagging impact after the test was due to the delay in normalizing primary alumina quantity in the reactors after completing the test. Higher dust emission was observed during the changeover (Figure 4) at the beginning and at the end of the test. This was caused by changes to the bag pulsing interval (Figure 5) to control bag house DP.

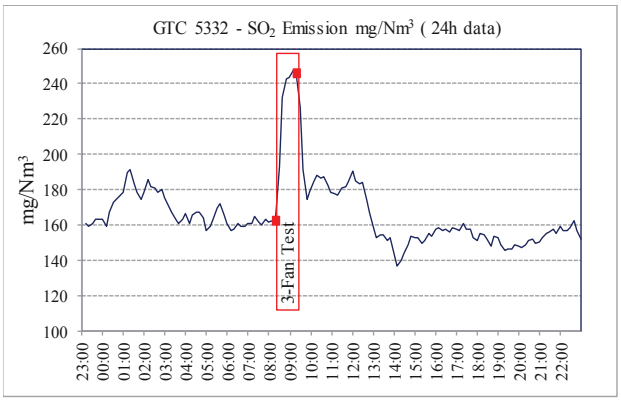

Figure 3. Stack SO_2 emission during 3-Fan Test.

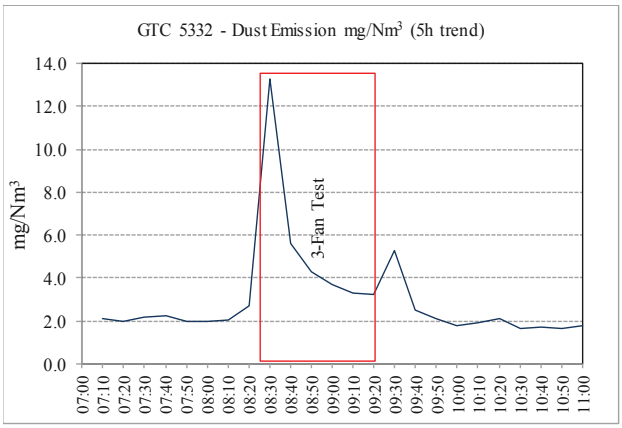

Figure 4. Stack Dust emission during 3-Fan Test.

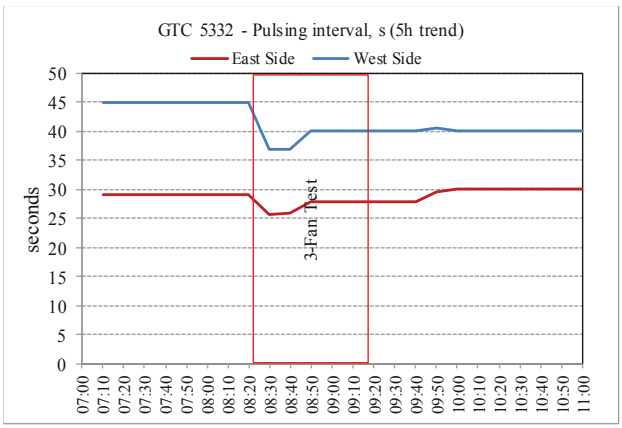

Figure 5. Pulsing interval during the 3-Fan Test.

Plant Operation during the Planned Shutdown: Current loading of the MEFs was kept steady at 40 % to 60 % more than nominal, as shown in Figures 6 and 7. The inlet damper was opened from 40 % to 100 % and maintained for the three operating MEFs.

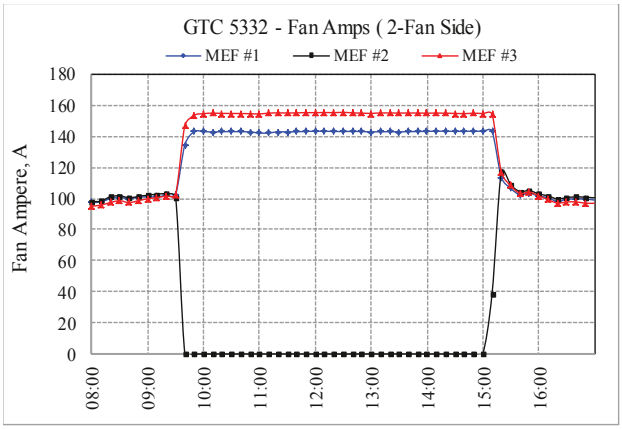

Figure 6. Current loading of MEFs on 2-Fan side (East side).

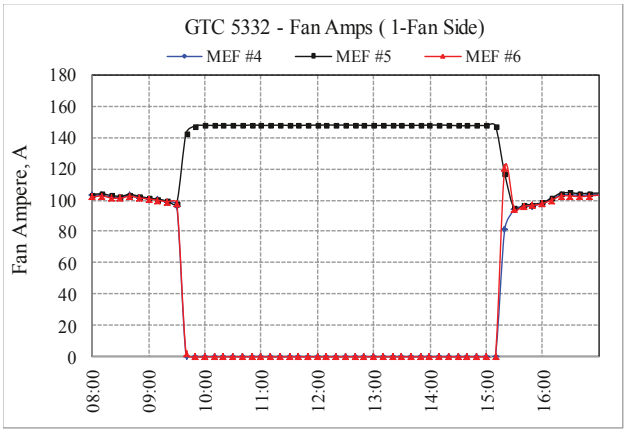

Figure 7. Current loading of MEFs on 1-Fan side (West side).

Inlet pressure was maintained around the target values as shown in Figure 8. West side, with one MEF was operated at lower inlet pressure than the East side. The resultant gas flow, as measured at the inlet duct, is shown in Figure 9. Though there is a difference in gas flow between the two sides, the marginal impact of this on the pot gas flow was considered acceptable.

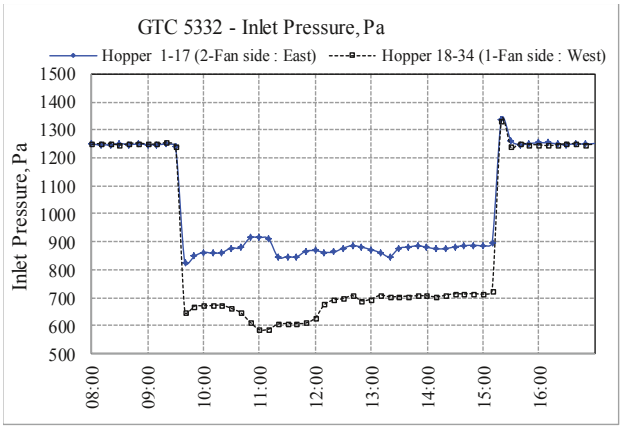

Figure 8. Inlet pressure setting during 3-Fan operation.

When the bag house DP was reduced to a lower than expected level at the start of the shutdown, the pulsing interval was increased. However, noticing the rate of increase in DP, the control target for DP was reduced by adjusting the pulsing interval to maintain smooth operation (see Figure 10 and 11).

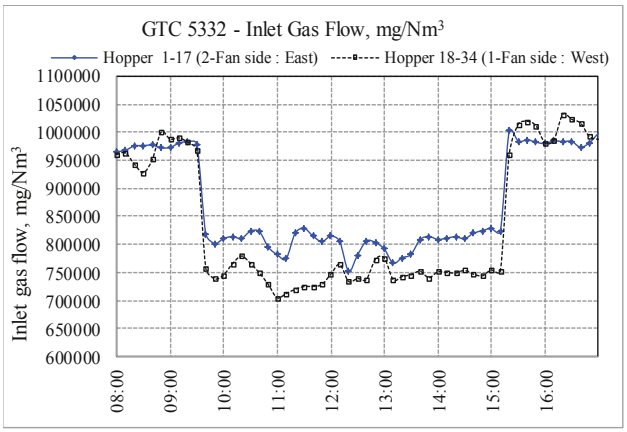

Figure 9. Inlet gas flow during 3-Fan operation.

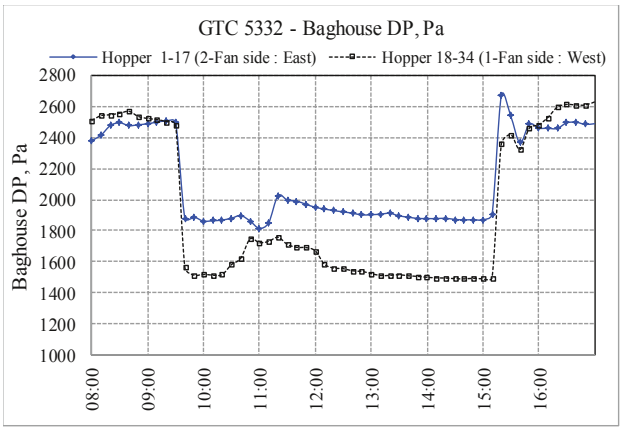

Figure 10. Bag house DP during 3-Fan operation.

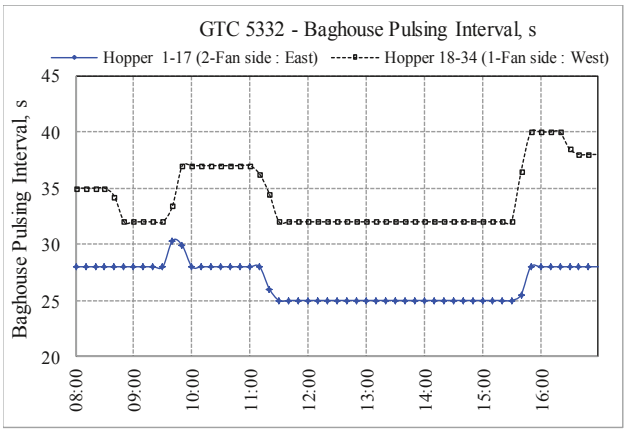

Figure 11. Bag house pulsing interval during 3-Fan operation.

Figure 12 shows the alumina feed rate and recirculation rate trends. The alumina feed rate was reduced to 50 t/h at the beginning of the shift (at 07:00), about two hours before the shutdown. The initial increase to 70 t/h followed by the quick reduction to 30 t/h can be ignored, as both were caused by operator error. The feed rate was further reduced to a minimum 44 t/h about 90 minutes after the shutdown. At the same time, the

alumina recirculation rate was also reduced to 35 t/h. These changes were done based on the DP of the individual filters and the lower fresh alumina feed rate was intended to reduce filter bag pressure drop in order to maximize the pot exhaust rate.

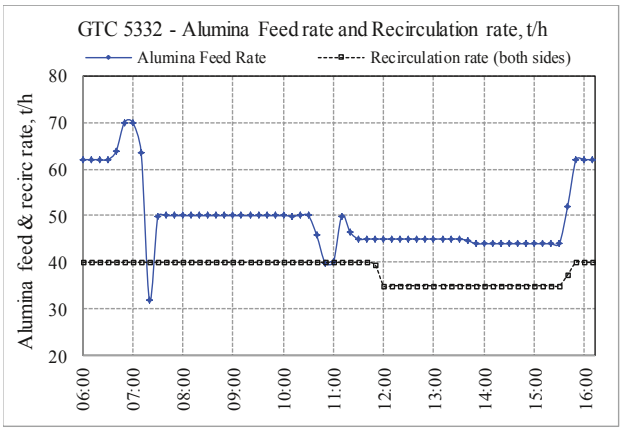

Figure 12. Alumina Feed rate and Recirculation rate control.

Environmental & Process Impacts: Dust, HF and SO_2 emission were monitored using the continuous emission monitors at stack. In addition, data from the continuous potroom roof HF monitors was also reviewed to assess the environmental impact in potroom.

Stack Dust Emission: The dust emissions trend is shown in Figure 13. During the initial changeover, dust emissions increased from normal levels of 2 to 3 mg/Nm^3, up to 10 mg/Nm^3. This increase was due to changes made to the operating parameters. Dust emission returned to normal levels within about 20 minutes. At 11:40, the dust emissions spiked once again, further to changes made to the pulsing interval and alumina feed rate (shown in Figures 11 and 12). These changes were necessary to control the baghouse DP and to sustain the pot gas flow at a minimum required level. It took around one hour to return to normal levels below 5 mg/Nm^3.

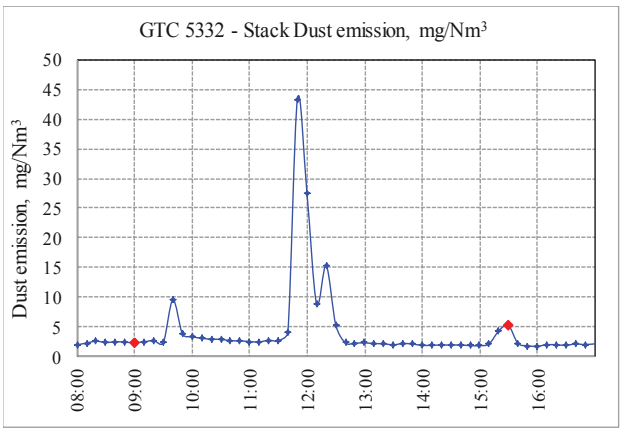

Figure 13. Dust emission trend during 3-Fan operation.

After completing the maintenance activity, while resuming to normal operation, dust emissions increased slightly due to parameter changes during the changeover, similar to that seen during the start of the shutdown. After stabilizing, within 20 minutes, the plant returned to normal dust emission levels below 5 mg/Nm^3.

Stack HF Emission: Based on that seen during the one hour test, an adverse impact on HF emissions was expected during the shutdown. Mainly four factors could cause higher emission; increase in inlet gas HF concentration; reduction in the alumina feed rate; longer shutdown duration and an increase in the inlet gas temperature.

Figure 14 shows the HF emissions trend and Figure 15 shows the corresponding alumina feed rate trend. A summary of the impact is shown in Table III.

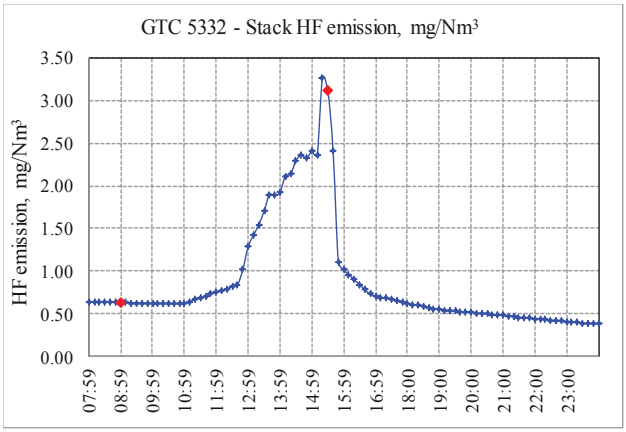

Figure 14. Stack HF emission during and after 3-Fan operation.

Table III: HF emission summary against alumina feed rate and recirculation rate.

Stage	Duration	Feed Rate, t/h	Recirculation rate, t/h	Average HF, mg/Nm^3
I	10:40 – 11:50	45	40	0.65
II	11:50 – 14:00	45	36	1.16
III	14:00 – 15:20	44	35	2.20
IV	15:20 – 17:30	56	38	1.52

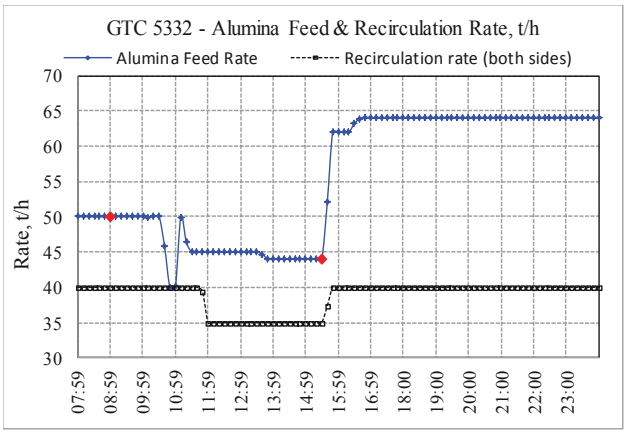

Figure 15. Alumina feed rate and recirculation rate during and after 3-Fan operation.

Figure 16 shows the inlet gas temperature during the shutdown. Due to the cooling of gases in the long ductwork, the inlet gas temperature did not increase with the reduction in gas volume.

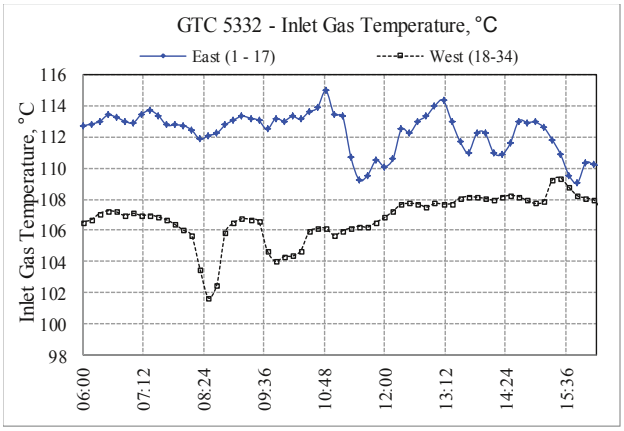

Figure 16. Inlet Gas Temperature during 3-Fan operation.

Discussion on HF emission: For the first two hours after the shutdown of the MEFs, HF emissions did not change. A slow and gradual increase was seen during the next two hours. Since a large quantity of alumina was available in the reactors/filters, HF emissions during the first two hours did not increase much. As the operation continued, alumina in the reactors became more and more saturated with HF, resulting in a progressive reduction in the adsorption rate. This resulted in the steady increase in HF emission beyond two hours.

The surge in HF emissions seen towards the end of 3-Fan operation could not be well explained with the available data. To explain this phenomenon, an earlier study [2] on low alumina feed rate operation that happened on two occasions was referred to and is discussed. The first event was in GTC 5332 on 23 March 2015 (Figure 17) and the second event was in GTC 5331 on 12 August 2015 (Figure 18).

In the first event, the alumina feed rate was reduced to 5 t/h for 150 minutes. Stack HF emissions increased gradually for about two hours and then increased exponentially. Low feed rate operation for longer than two hours had a significant impact on HF emissions causing the exponential increase. In the second event, the feed rate was kept higher than the first case (10 t/h) but for more than three hours (199 minutes). A gradual increase in HF emissions was observed until about 150 minutes. Thereafter the exponential increase in HF emissions continued for more than one hour, reaching as high as 100 mg/Nm^3.

The gradual increase in HF emissions during the first two hours is the combined effect of reduced alumina input and progressive saturation of available alumina recirculated in the reactors. After about four hours, the recirculated alumina became fully saturated and HF adsorption was only done on the fresh alumina feed, which was very low. Comparing these two cases, the duration of low alumina feed (199 minutes versus 150 minutes) was found to have higher impact on HF emission than the change in feed rate (10 t/h versus 5 t/h) itself.

Learning from the one hour test about the impact of alumina saturation on HF emission, attempts were made to maximize the alumina available without jeopardizing plant operation during the planned shutdown.

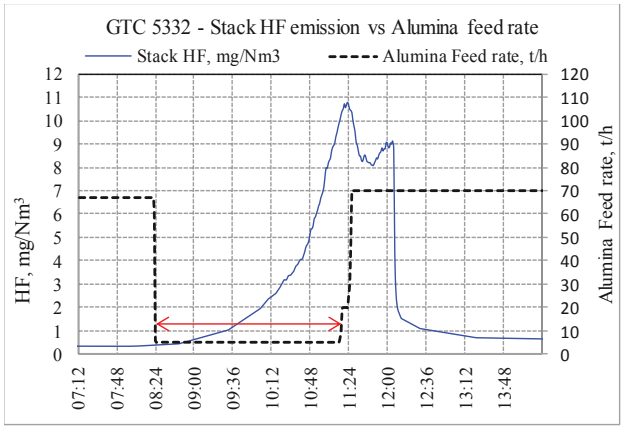

Figure 17. HF trend during low alumina feed operation caused by a maintenance event in GTC 5332 [2].

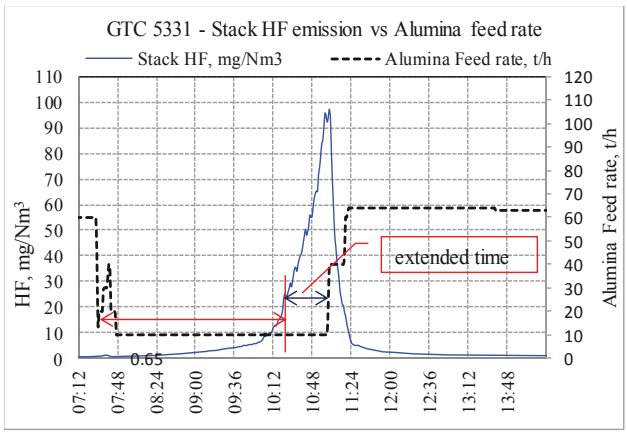

Figure 18. HF trend during low alumina feed operation caused by a maintenance event in GTC 5331 [2].

SO_2 Emissions: SO_2 emissions are monitored by an online monitor located at the stack. During the shutdown, an increase of about 40 % in concentration was seen as shown in Table IV and Figure 19. About 10 % of the increase was due to the reduction in the alumina feed; and about 30 % was simply due to the reduced gas flow, which proportionally increases SO_2 concentration.

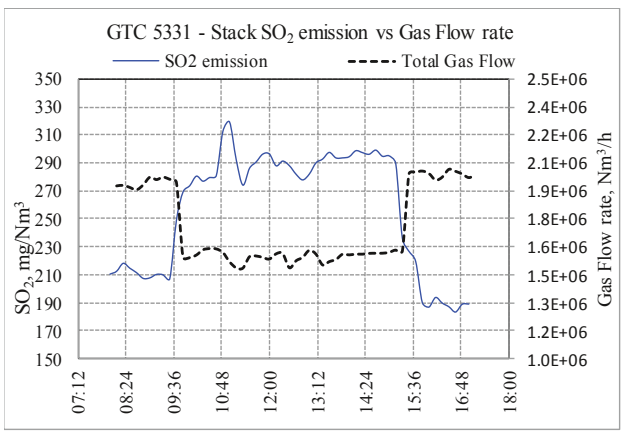

Figure 19. SO_2 emission and total gas flow trend during 3-fan operation.

Table IV. Summary of SO_2 emission during 3-Fan operation.

Stage	SO_2, mg/Nm^3	Gas Flow, Nm^3/h	
		East	West
Normal (07:00 – 09:40)	205	974,195	960,127
Transition (09:40 – 09:50)	249	805,551	737,805
3-Fan (09:50 – 15:20)	289	804,582	744,896
Transition (15:20 – 15:30)	290	824,199	752,254
Normal (15:40 – 16:00)	196	982,063	1,007,816

Potroom Roof HF Emissions: Potroom roof HF monitors are installed across two physical sections in both A-Room and B-Room. GTC 5332 covers pots in physical sections 1, 2, and 3 in A-Room and sections 10, 11 and 12 in B-Room. However, roof monitors are across physical sections 1-2, 3-4, 9-10 and 11-12.

Average roof HF emission trend for sections 1 to 4 and 9 to 12 for 24 hours is shown in Figure 20. There was no noticeable impact on roof HF emissions during the 3-Fan operation. Physical inspection in the potroom also did not show any abnormal fuming from any of the pot shields.

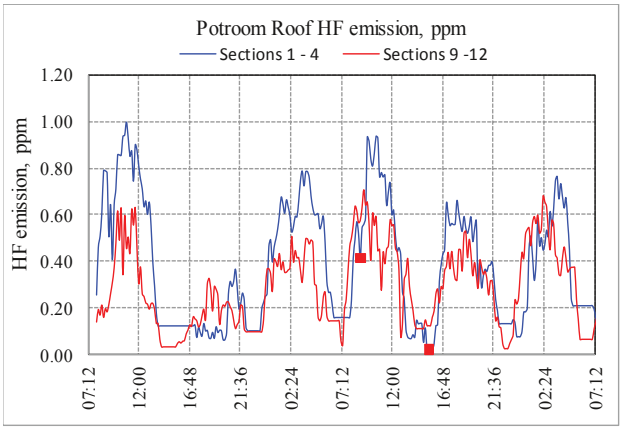

Figure 20. Potrrom Roof HF averaged for A-Room Sections.

Impact on Bath Temperature : Immediately after resuming to normal GTC operation, bath temperature measurements were repeated in 10% of the pots connected to this GTC and compared against the previous measurements. Figure 21 shows the comparison which indicates no measurable impact on thermal control of the pots caused by the lower gas suction.

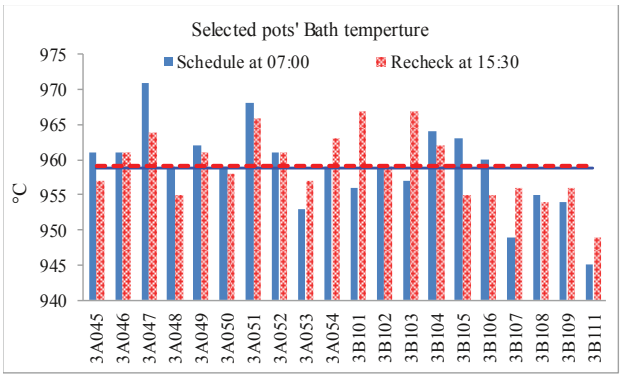

Figure 21. Comparison of selected pots bath temperature before and after 3-fan operation.

Conclusions

Based on the data evaluated, the following can be concluded:

- Environmental impact due to major maintenance activities in a large GTC could be managed by suitable selection of operating parameters.
- During reduced load operation of the GTC, based on the expected duration, selection of gas flow rate, alumina feed rate and alumina recirculation rate play significant role in minimizing environmental impact.
- To minimize the impact on potroom process and roof emissions, a minimum gas flow of 6,500 Nm^3/h per pot, equivalent to about 44,000 Nm^3/t of Al for DX+ pots, (about 75 % of the nominal flow rate) was found effective.
- Maintenance activities that require extremely low alumina feed rate to the GTC should be planned for very short durations or scheduled on multiple occurrences. The duration of extremely low feed rate plays a significant role in HF emissions from the GTC, due to saturation limit of available alumina in the filter hoppers.

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[1] EMAL Reduction Process Control - Internal Test Report "RDGN-PCT-101 Three Fans Operation Test at GTC 5332" dated 8 April 2015

[2] EMAL Reduction Process Control - Internal Report "RGDN-PCT-102_Impact of change in GTC alumina feed rate to Stack HF emission" dated 15 April 2015

Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

COMPACT GTC DESIGN: REDUCING FOOTPRINT AND OVERALL STEEL WEIGHT

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Keywords: GTC, Compact, Footprint

Abstract

Within the primary aluminium smelter, the spatial requirements and capital cost of the GTC remains a major challenge. Due to its large size the CAPEX is predominantly determined by the mass of steel required. A partial solution to reduce footprint and steel weight is to increase the size of the filter modules while subsequently reducing their quantity. Unfortunately this method of footprint reduction is limited by N–1 baghouse flow conditions – the increased filter velocities with one module out of service.

Danieli Corus has addressed this limitation in the new compact GTC design where the low pressure filter cleaning system accommodates longer filter bags. In addition the Danieli Corus compact GTC design is configured with a standby module and exhaust fan thereby avoiding the increased flow associated with N–1 conditions. The new configuration is narrow so it can easily be situated in the courtyard which is often congested with roadways, storage silos and material handling equipment.

There are simplistic advantages to the new configuration which result in a decrease of 15% in the steel weight and overall CAPEX reduction of 10% from traditional Danieli Corus design.

Introduction

The production of primary aluminum is a capital intensive industry with low margins. In the development of projects for new smelters or expansion of existing smelters the capital cost for equipment and the available footprint thereof are under pressure. In this conservative industry proven design dominates the market and there is little tolerance for experimentation. Therefore, new designs should not deviate in basic technology from the proven methods. If the modifications are subtle but lead to substantial benefits, then the design could be very appealing to potential clients.

As the market is CAPEX driven, optimizing the steel weight is of paramount importance since it is the biggest contributor to overall cost. If the physical size of the GTC could be reduced, then a subsequent drop in steel weight is expected.

OPEX could be reduced by regulating the operation of the GTC to suit seasonal changes, especially in very dry arid climates. Very simply, reducing the number of modules online during winter months when a lower actual gas flow needs to be scrubbed. The design should pay close attention to the turn down ratio of the main exhaust fans and baghouse modules during these times.

The development of the new compact design could not diverge too far from proven technologies. Radical changes in bag house design, operation or cooling methods are not acceptable to aluminium manufacturers. Therefore, Danieli Corus started with its standard design and with some subtle modifications optimized it to suit current market needs.

Process criteria for the GTC

The new compact design GTC is configured with square filter modules that are approximately 30% larger in cross section from typical Danieli Corus installations. The new modules house two (2) additional annular rings of filter bags and adapt a filter bag that is 15% longer than conventional. With the additional filter surface area available in each module, the number of modules required for the GTC is reduced. Normal operation sees the new GTC operating continuously with **N** modules at 3–5% higher filter velocities than typical DC design. However, by design the new GTC configuration is **N+1** module as there is a 100% standby unit. Therefore, the continuous filter velocity could be reduced to approximately 90% of traditional figures if the operation philosophy is to operate with all modules (N+1) concurrently. Due to the incorporation of the standby module, the new compact design GTC is more suited towards large potline gas flows. Therefore, the larger the GTC, the more applicable the new compact design by Danieli Corus.

Temperature

A key factor in reducing the GTC footprint is limiting the amount of gas volume it is required to handle. Regardless of the steps taken to optimize the GTC footprint, the majority of the reduction will depend on the chosen method to cool the pot exhaust gases upstream the GTC. Pot gas is cooled because the scrubbing performance at the reactor is influenced by gas temperature. Experience indicates that the removal efficiency of hydrogen fluoride on alumina starts decreasing when temperatures rise above 115°C at the reactor. In cooler climates, GTCs often experience high temperature only during midafternoon of the summer months. But in more arid climates, the gases require some form of constant cooling prior to the GTC to achieve acceptable emission results. There are several proven methods for cooling the gas flow entering the GTC and any could be applied to the new compact design by DC.

Traditional practice cools the hot pot gas by means of dilution with the temperate ambient air. This practice becomes very impractical in extreme climates since the ambient air temperature could reach 50°C during summer months. The GTC may double in size just to accommodate the dilution air volume. If the amount of dilution air is reduced then the GTC is smaller but runs the risk of exceeding the 115°C limit for ideal HF removal. For this reason dilution air is rarely used exclusively on modern installations.

Extended surface technologies are sometimes used to increase the amount of natural convection heat loss through the duct wall, usually where the gases are at peak temperature. Depending on layout and ambient conditions, gas temperatures could experience a 5–10 °C drop. In arid climates, elongating or extending the duct surface is not nearly sufficient as a stand–alone cooling method.

Similarly, the use of gas-liquid heat exchangers can be an efficient method if there is adequate use for the secondary low grade heat

source. Otherwise, fin fan coolers or similar equipment are required to dissipate the low grade heat from the liquid back to the atmosphere.

Water injection is another method of cooling which has been used in limited capacity. The process consists of water mist injection by two phase nozzles – compressed air and water. The hesitancy for water injection as a stand-alone cooling method is the process risks related to the longevity, maintenance and corrosion. However, in conjunction with dilution air the use of water injection is very reasonable. Known as peak shaving, water is only injected when gas temperatures are at their highest - typically 5-10 times per year during midday of midsummer. The remainder of the year, dilution air is used to cool the gas. Peak shaving can reduce the number of filter modules by nearly 20% in some circumstances.

Alumina distribution

In order to achieve optimal HF capture efficiencies, equal distribution of the injected alumina in the reactor is a very important factor. The patented Vertical Radial Injector (VRI) produces the highest contact efficiency between HF and the alumina while creating minimal pressure drop in the system. While developing the new compact GTC, the VRI design and corresponding design parameters were maintained to guarantee that the HF capture efficiency is still the best available.

Baghouse

Downstream of the VRI the dispersed alumina enters the bag house where the clean gas is filtered from the reacted alumina. The Danieli Corus flow distribution baffle ensures a very balanced flow of gases and eliminates peak velocities associated with bag wear and abrasion. A portion of the alumina is separated by gravity from the gas flow which decreases the material loading on the filter bags.

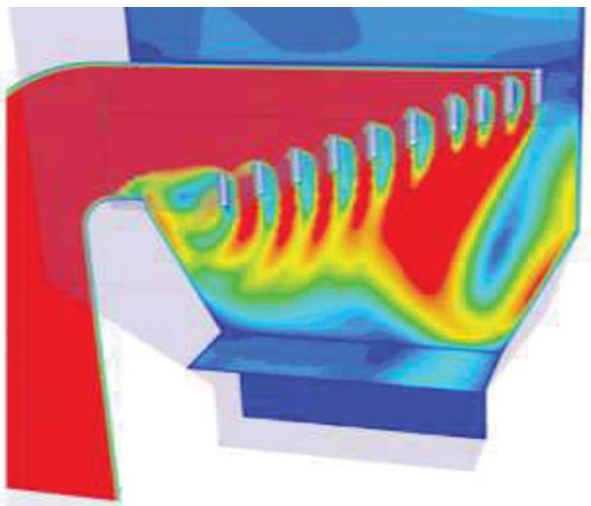

Figure 1 – CFD Hopper Flow Distribution

The baffle system has been optimized using CFD modeling ensuring a prolonged filter bag life and minimal pressure drop. As the module dimensions remain square in the compact design GTC, the flow is balanced within the hopper using the flow distribution baffle – the same technology used in traditional Danieli Corus GTCs.

As previous CFD models have shown, the circular arrangement of filter bags in a square module has the advantage of having geometrical chimneys in each corner as well as a tubular channel in the center. This unique advantage to the Danieli Corus baghouse allows the module to experience higher than typical can velocities without detriment to the bags or pulsing performance. The can velocity refers to the theoretical velocity of the treated gas as it passes upward between the filter bags. The calculated can velocity is actually a theoretical average but in reality there will be areas of higher and lower velocities due to the way the flow is distributed within the module. In the Danieli Corus module, the preferential flow is to the open areas producing a side loading distribution to the filter bags as shown in Figure 2 below.

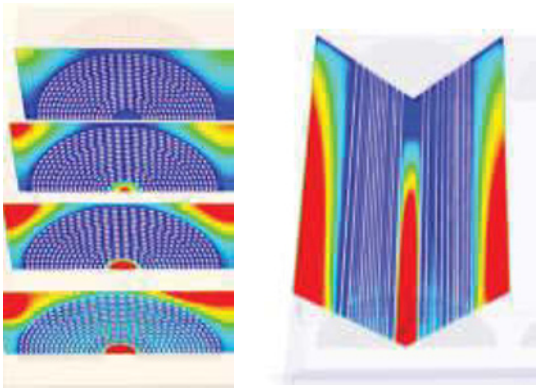

Figure 2 – CFD Gas flow within the module

Experience during filter bag exchange has noted little difference in the failure rate between bags adjacent to the chimneys and those in the middle annular rows. This would suggest that the filter bags are generally equally loaded with alumina particles. These benefits are well known in the industry for providing balanced dust loading, easier bag cleaning and prolonged bag life.

Recycle rate

The added benefit to the high contact efficiency of the VRI reactor is the reduced amount of alumina recycle that is required to achieve high HF removal efficiency. The Danieli Corus design typically requires a recycle rate of only 5:1. As the alumina is highly abrasive, it is desirable to have less alumina circulating within the GTC. Air slides and reactor will require less maintenance and will have a longer lifetime when the recycle rate is low. The means of recycling in the new compact GTC remain unchanged.

Filter Velocity

The filter velocity refers to gas flow per unit area of the filter bags. There is an industrial acceptable range of filter velocity for GTCs commonly between 2.00 – 2.30 cm/s. The high end of the scale is usually associated with one filter module out of operation – meaning the entire flow passes through a reduced filter area hence raising the filter velocity. If maximum filter velocity is exceeded, the GTC footprint could be reduced. But this self–destructive option results in short bag life, erosion problems in the baghouse and unacceptable dust emissions. In the new compact GTC, Danieli Corus has maintained an acceptable filter velocity and added a complete standby module. The addition of the standby module

allows the GTC to operate with lower filter velocities if all modules are operated concurrently.

Footprint

Whether the GTC is to be placed in an existing courtyard or is part of a Greenfield installation, the available space has become very limited. Therefore the new GTC design has adapted to these tighter quarters and become much narrower. The new DC compact design places the modules in a single aligned row offering significant advantages. It provides for a smaller footprint, it makes the configuration of the alumina handling equipment simpler, and lastly, it could provide the option for future expandability by simply adding modules at the discharge end. From a maintenance point of view, the main exhaust fans are easily accessible from the side and are unobstructed by large plenums.

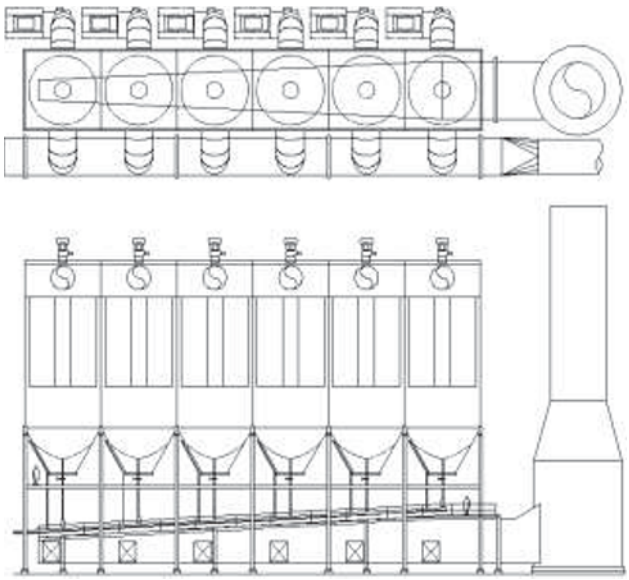

Figure 3 – Compact GTC Plan View and Elevation

The additional annular rings in addition to the longer bag length drastically increase the available filter surface area thus allowing for fewer modules. This small modification in baghouse capacity has a big influence on the steelworks as it requires less platforms and support structures. For example, a six (6) module plot plan decreases by 17% when compared to traditional eight (8) module DC design. The reduction in plot plan increases proportionately with an increase in the gas flow rate. Ie, there are greater savings in plot size as the GTCs size increases.

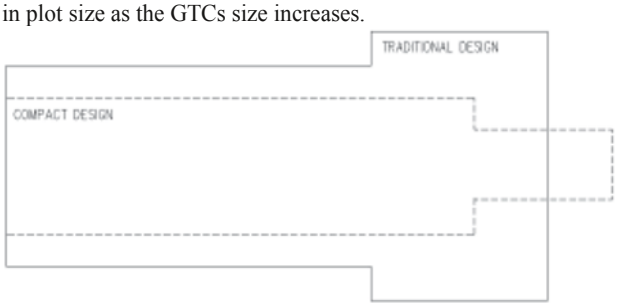

Figure 4 – Footprint Comparison

Mechanical Design Features

In Danieli Corus's conventional design, the gas from each baghouse module was ducted to a common fan inlet plenum where distributed to the fans. The number of fans was usually a balance between power limitations, spatial requirements and investment. For a large GTC, the main exhaust fans were as large as 2500 kW since they included safety factors, contingencies and additional motor capacity. These additions often prevented the fans from operating efficiently during normal operation conditions but were necessary to allow for amperage creep or future expansion. The N+1 fan configuration meant a complete 100% rated standby fan was available but unutilized. The standby fan is a significant investment for a piece of equipment that was only used in case of breakdown of the other units. In general, a small quantity of large fans will have difficulty operating in an efficient range due to the large swings in gas volume throughout the year. This could be avoided if VF drives are incorporated but rarely does this occur.

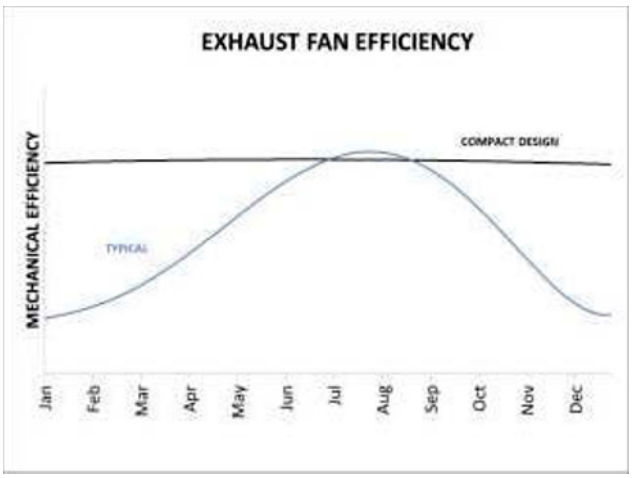

Figure 5 – Fan Efficiency Comparison

In contrast, the Danieli Corus new compact design dedicates a main exhaust fan to each module so the fans are considerably smaller (~800 kW). The design incorporates a complete standby module which would likely remain online but could sit idle if preferred by the owner. Utilization of the main exhaust fans would be 100% when all modules are online, compared to 75-80% in other installations. Since there is a greater quantity of smaller fans, it is much easier to operate the fans within their mechanically efficient range. During winter months when gas volumes are lower, the GTC could place multiple modules on standby to lower OPEX and maintain the fan efficiency. This adds up to a considerable cost savings to the owner since the exhaust fans are a major consumer of power.

Another operational benefit of the new compact design is the guarantee of equal flow distribution among the modules. The large common fan inlet plenum and support structure are eliminated in this configuration allowing for a long straight duct from the module outlet to the fan inlet. This is an ideal location for flow measurement due to the straight length of clean gas and accessibility from the existing hopper platform.

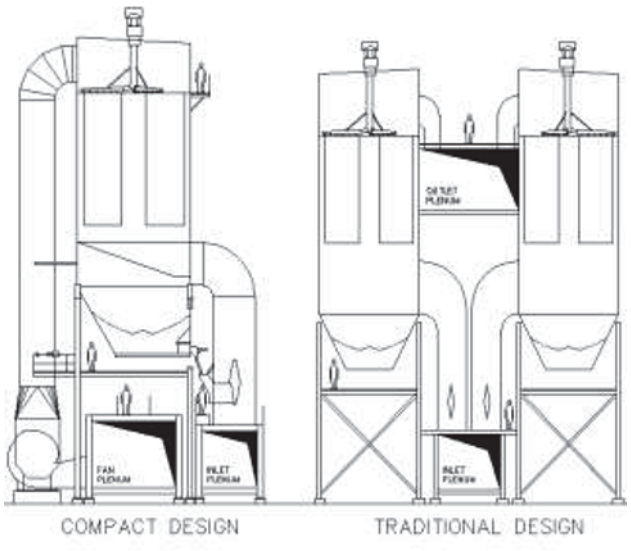

Figure 6 – Elevation Comparison

The simplicity of the compact design is illustrated in the figure above. The traditional design requires a common inlet and outlet plenum as shown in addition to the fan plenum (not shown). The compact design is reduced a common inlet plenum and fan plenum. Overall the complexity of steelwork, ducting and platforms is significantly reduced which reflects in the smaller footprint and CAPEX.

Alumina handling

The new module configuration calls for an optimized and more compact alumina handling system. It is not unusual in old GTCs that the numerous air slides and associated vent piping obstruct some parts of the walking routes across the GTC. In the new module configuration the distribution air slide can be positioned parallel to the baghouses. This fresh alumina supply air slide can act as a distributor, eliminating the elevation required for a conventional distribution box and independent feed air slides. This configuration further eliminates the recycle airslide as the new hopper geometry allows alumina to flow directly to the VRI. Similarly, Danieli Corus has eliminated the transverse airslide which conveyed reacted alumina from each module to a common collection airslide running the length of the GTC. The new arrangement allows the reacted alumina to discharge directly in the collection airslide which is situated directly beneath the modules. Conveniently, the reacted alumina collection airslide and the fan discharge plenum are sloped in the same direction allowing for easy positioning of the access platform. The simplification of the alumina handling equipment also reduces congestion of vent piping and fluidizing air piping associated with the air slides.

Conclusion

A new compact GTC design has been developed utilizing the same reliable technology associated with Danieli Corus. The square modules are slightly larger than previous and an independent exhaust fan is dedicated to each module. The result of new compact design is a 17% decrease in plot plan, a cleaner and simpler alumina handling configuration and a 15% decrease in steel weight. The configuration is much more compact and fits much easier in the narrow courtyards between the pot rooms. The overall savings to the client: 10% CAPEX reduction and reduced OPEX due to lower pressure drop, fewer modules and increased fan efficiency.

The new modular aligned approach is more flexible in design and construction. Each module and fan combination is identical hence the size of the GTC can expand or contract easily. The configuration provides the most efficient turn down ratio possible as modules could be idled as required during the year while maintaining a high mechanical efficiency in the exhaust fans.

Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

TECHNOLOGY FOR REMOVAL OF SULPHUR COMPOUNDS FROM GASES GENERATED DURING ALUMINUM PRODUCTION

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Keywords: wet scrubbers, sulphur, sulphur dioxide, sulphate removal, gas treatment, environment, aluminum reduction

Abstract

RUSAL has been upgrading its aluminium smelters by replacing the scrubbing of the cell off-gases with the "state of the art" alumina dry scrubbers followed by preexisting wet scrubber technology to satisfy today's environmental regulations for fluoride and sulfur emissions. Today, sixty percent of RUSAL's smelters are converted to an alumina dry scrubber followed by a wet scrubber and the others are still using a wet-process that captures the fluorides and sulphur emissions with a liquor of soda.

While the wet scrubber technology meets recovery efficiency in accord with environmental regulations for the recovery of sulfur compounds for all smelters and it recovers fluorides, the sulfur compounds, mainly as sulfates, end up in costly and monitored landfills.

This paper covers RUSAL's development of a technology for producing a commercial grade sodium sulfate slurry, decreasing salt concentrations, delaying the constructing of landfills, and reducing the environmental impact of wet scrubber technology.

Introduction

RUSAL is experiencing an increase of sulphur in anodes due to changes in the quality of petroleum coke used by the aluminum industry. Figure 1, shows historical changes in the weighted average content of sulphur in the Søderberg anode paste produced by RUSAL's smelters from 2003 to 2015. Taking into account the trends in the petroleum coke market [1], the sulphur content is expected to continue increasing as newer, heaver crudes are processed. Moreover, for RUSAL, this means that meeting the environmental standards will require considerable changes to the existing environmental collection and processing facilities used at UC RUSAL's Søderberg smelters.

In the 1960's and 1970's, the Irkutsk branch of the All-Union Aluminum-Magnesium Institute (SibVAMI) developed wet process technology to produce secondary cryolite by regeneration to remove fluorides. The technology, developed for the Søderberg reduction process technology, and later improved [2-5] was implemented in all of the Russian Søderberg smelters. The technology is based on the method of chemical reaction of hydrogen fluoride in the wet scrubber's (WS) soda liquors, (aqueous sodium carbonate and sodium bicarbonate) to form sodium fluoride.

The technology is based on the chemical interactions shown by Reactions (1), (2), and (3). In the course of chemical absorption of hydrogen fluoride, the wet scrubber's sodium carbonate (Na_2CO_3)liquor captures hydrogen fluoride (HF), forming sodium fluoride (NaF) solution (Reactions 1 and 2). By adding sodium aluminate (Na2O*Al2O3) to NaF solution at fluoride regeneration facilities, solid cryolite (Na_3AlF_6) precipitates in a clarifier, and sodium carbonate solution is regenerated and returned to the WS. In addition, concentrated sodium carbonate liquor is added to maintain the required sodium carbonate concentration in the solution.

In addition to regenerated cryolite, carbon dust undergoes flotation in the same facilities. Because of the flotation process, the cryolite mixes with regenerated dehydrated cryolite and is returned to the cells while the flotation tailings are sent to the landfill by hydraulic transport.

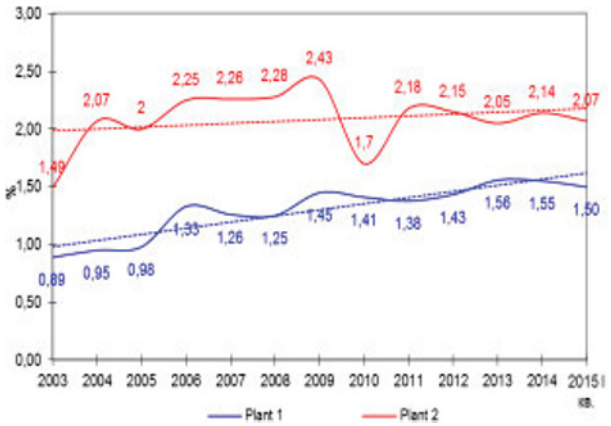

Figure 1: Weighted Average Sulphur Content in Anode Paste, 2003 to 2015

The main cryolite chemical reactions are:

$$\underline{Na_2CO_3}_{dis} + HF_{gas} \rightarrow NaF_{dis} + NaHCO_3 \quad (1)$$

$$NaHCO_3 + HF \rightarrow NaF + H_2O + CO_2\uparrow \quad (2)$$

$$NaF_{dis} + NaHCO_3 + Na_2O{*}Al_2O_3 \rightarrow Na_3AlF_6 + \underline{Na_2CO_3} + H_2O \quad (3)$$

Concurrent reactions:

$$Na_2CO_3 + SO_2 + \tfrac{1}{2}O_2 \rightarrow Na_2SO_4 + CO_2\uparrow \quad (4)$$

$$Na_2CO_3 + CO_2 + H_2O \rightarrow NaHCO_3 \quad (5)$$

Reaction (4) shows that, in addition to hydrogen fluoride, sulphur dioxide is captured from exhaust gases, forming sodium sulfate (Na_2SO_4). Reaction (5) shows the completion of the cycle with the regeneration of sodium bicarbonate in the WS liquor.

The precipitated cryolite, reaction (3) is taken from the clarifier, dried, and returned to the cells while some sodium sulfate remains in solution circulating in the WS's liquors and the liquors are bled off to the landfill along with the carbon dust.

Figure 2 shows sodium sulphate solubility in water as a function of temperature. As shown, the solubility of sodium sulfate is strongly temperature dependent. Furthermore, by controlling the temperature to around zero degrees Centigrade, sodium sulphate precipitates out of solution leading to sludge formation.

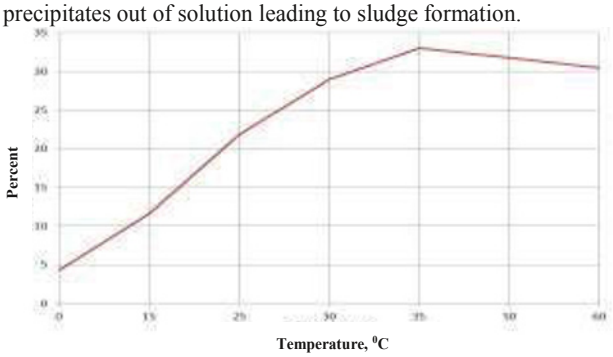

Figure 2: Sodium Sulphate Solubility in Water vs. Temperature

Large number of RUSAL smelters are located in Siberia and operate under severe continental climate conditions, where winter temperatures go down to minus 35 to minus 45 ^oC. Under such conditions, sludge sulfate solidifies and precipitates naturally, forming sulfate salts: primarily NaF * Na_2SO_4; Na_2CO_3 * Na_2SO_4; Na_2SO_4 * 10 H_2O. In the summer, when temperatures go up to plus 30-35 ^oC, sulfate can dissolve into the above semi-aqueous sludge solution resulting in the need to separate the deposited sulfate from the solution. The dusts captured by wet scrubbers along with flotation tailings, carbonaceous-water sludge are used as a cover as the layers buildup in landfill.

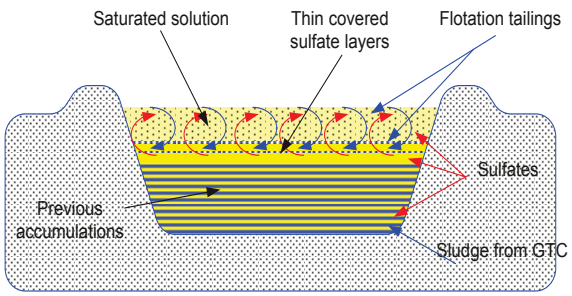

Figure 3: Solid Sludge Accumulation in the Landfill

Usually, sludge ponds with a capacity of one-million cubic meters of sludge and liquor are installed for larger-capacity smelters with an expected service life of 10 years. During this time, the pond is filled with sludge, containing eighty to ninety percent of solids from WS's, carbonaceous flotation tailings and covered solidified sulfate salts. The remaining liquid volume is pumped into a new pond. In the next few years, the newly-constructed pond also gets filled with sludge and, when full, becomes temporarily out of service. It is cleaned from all the deposits. The edges of the first pond are uplifted, and sludge is transferred from the second pond into the first one. After transferring, the first pond is re-cultivated, i.e. the pond surface is covered with clay and a layer of fertile soil, and, then, shrubs are planted. The second pond is re-commissioned after cleaning. Each pond goes through the same life cycle as the first one, so there is a constant need for the construction of new landfills.

Forty percent of RUSAL's Søderberg production capacity still uses wet scrubbing. The OJSC RUSAL Krasnoyarsk Aluminum Smelter (KrAZ) has been converted to dry alumina bag filter gas treatment centers (GTC's)[6] and the wet scrubbers were maintained to capture sulphur dioxide (also known as 'second cleaning stage').

After converting KrAZ to dry scrubbing technology, volatile dust is captured in the bag filters and very little if any ends up in the WS. Consequently, there is insufficient sludge and the quantity of flotation tailings (carbon dust and alumina) is not adequate to cover solidified sulfate salts in the landfill. In winter, the precipitated sulfate salts remain as solids, however in summer when the temperatures rise, and the WS sludge is not sufficient, sulfates dissolve and get into the sludge solution. The salt concentration reaches 120-150 g/l, which leads to deposits in pipelines and on the wet scrubber equipment. In order to prevent deposits, additional clean water is added and diluted WS liquor is discharged to the landfill which leads to higher rate of filling of the landfill and a two to three times reduction in the service life.

Due to high CAPEX for the construction of new ponds (10 to 12 USD/one million cubic meters – this appears to be low; Is it 10-12 USD/m^3) and environmental expenses for the use of new lands, there is a need to use other methods of removing sulphur compounds from the WS's liquors.

GLOBAL EXPERIENCE IN TREATING SULPHUR DIOXIDE

There is a method of capturing SO_2 by seawater and discharging sulfate liquors into the sea. This method requires no reagents to bind sulphur dioxide and it eliminates the need for recycling and collecting the scrubbing water.

There is a method used by Alcoa [7], which involves regeneration of the sorbent that captures sulphur dioxide (regeneration by lime), including producing gypsum, which can be used by the construction industry.

The literature gives other well-known solutions [8], which use different sorbents to capture sulphur dioxide. As a result, the somewhat soluble $CaSO_4$ is immediately formed (Reactions 6 and 7), or, by using an ammonia solution as an intermediary sorbent, calcium sulfate ($CaSO_4$) is formed (Reactions 8,9,10, and 11), and recycled ammonia solution is regenerated by quicklime (Reaction 12). The resulting gypsum can be used by other industries or sent to a landfill.

$$Ca(OH)_2 + SO_2 = CaSO_3\downarrow + H_2O \quad (6)$$

$$CaSO_3 + ½\ O_2 = CaSO_4 \quad (7)$$

$$SO_2 + 2NH_3 + H_2O \rightarrow (NH_4)_2SO_3 \quad (8)$$

$$(NH_4)_2SO_3 + SO_2 + H_2O \rightarrow 2NH_4HSO_3 \quad (9)$$

$$(NH_4)HSO_3 + ½O_2 = (NH4)_2SO4 \quad (10)$$

$$(NH_4)HSO_3 + ½O_2 = NH_4HSO_4 \quad (11)$$

$$(NH_4)_2SO_4 + CaO = CaSO_4 + NH_3 + H_2O \quad (12)$$

UC RUSAL's EXPERIENCE IN TREATING SULPHUR OXIDE

Previously, OJSC RUSAL Bratsk used a technology based on freezing sulfates out of the WS's liquor (Figure 4). Sodium sulfate precipitates from the liquor upon cooling (cooling by means of a cascade of heat exchangers and solidifiers.) Sodium sulfate solidification (crystallization) occurs under further continuous cooling of the liquor in cascade-like stirrers-solidifiers. The temperature in the last solidifier was kept at zero to two degrees Celsius.

As a coolant, a 30-percent solution of calcium chloride (freezing point, minus 31 °C and solution density of 1.240 kg / dm^3) was used. The solution is cooled down to minus 10-12 °C. The coolant circulates in the system in a closed circuit: solidifiers - coolers - solidifiers. For solution cooling, single-stage ammonia coolers with a cooling capacity of 220,000 kcal/h each were used. The boiling point of ammonia in the evaporator is about -25 °C; the temperature of the coolant at the outlet of the evaporator is minus 8-12 °C. The salt slurry, obtained as a result of cooling, is sent to a separator-thickener. Then, it is fed to a centrifuge, where the liquid phase is separated from sodium sulfate crystals. The sodium sulfate with moisture content of 5-8% is fed from centrifuge into a mixer, where it is water-diluted to a concentration of 300-350 g/l. Then, it is loaded into a dedicated tanker for transportation to the customer. The capacity of the unit is up to 10,000 t/slurry of sodium sulfate decahydrate (calculated on the dry basis).

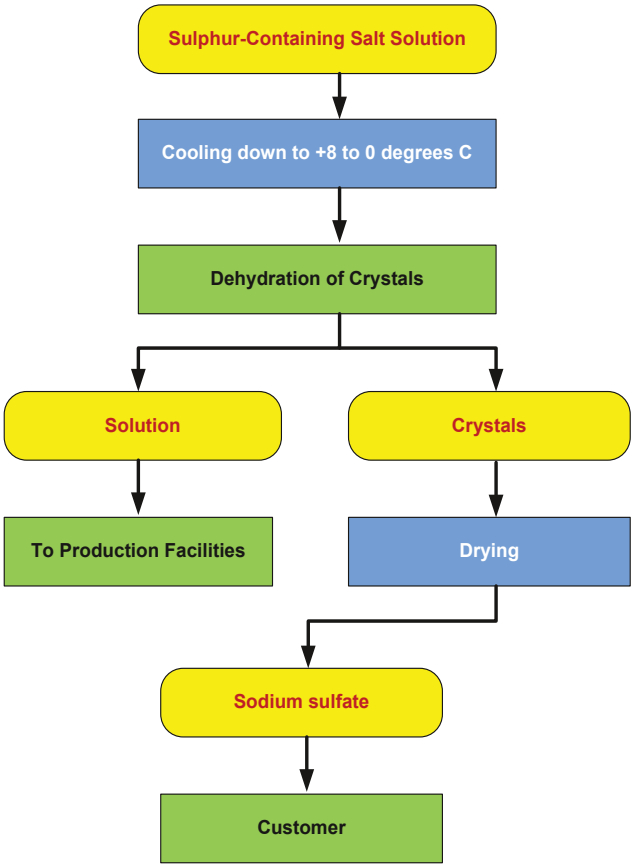

Figure 4: Process Flow Diagram for Sodium Sulfate Freezing-Out

The disadvantage of the unit is low energy efficiency of the coolers, and a limited demand for sodium sulfate decahydrate by customers (it is necessary to calcine the product to obtain a powdery substance). Consequently, a decision was made to temporarily landfill the sulfates. Currently, RUSAL considers upgrading the technology by means of replacing the coolers by a cooling unit that uses natural cooling taking advantage of cold winters [9]. Such an approach would help combine a compact and advanced sulfate removal unit and the energy efficiency due to using the natural conditions of Siberia.

UC RUSAL has developed another technology to evaporate the WS liquor to obtain a concentrated slurry of sodium sulfate decahydrate. This technology has been implemented at OJSC RUSAL Krasnoyarsk Aluminium Smelter. The technology is based on evaporating the WS's liquors by indirect steam in the evaporating tower, obtaining a concentrated solution of sodium sulfate, including sedimentation of the sodium sulfate decahydrate crystals.

The un-saturated salt solution from the WS is fed into the evaporator (Figure 5). Inside the evaporator, the flow of salt solution is mixed with another flow circulating in a closed circuit. (Circulation is performed by means of an axial circulating pump, which is part of the evaporator). The combined solution (starting solution + circulating solution) undergoes boiling. The boiling process is carried out under vacuum, with the help of heat supplied through heat-exchange tubes (the heating steam is condensed in the heating chamber.) When the solution boils, secondary steam is generated, and the solution gets denser. To

improve the heat transfer efficiency (in the course of condensing the heating steam supplied to the heating chamber of the evaporator) and to reduce the heating steam temperature, a condensate is injected into the overheated steam. After boiling, the evaporated solution is drained into a receiving tank that is emptied at regular intervals.

The secondary steam, which is generated in the evaporator in the process of boiling the solution, is fed into a barometric condenser. Inside the condenser, steam condenses (when it gets in contact with the clarified cooling water supplied from the landfill). The non-condensable gases and non-condensable water vapor, which form a gas-vapor mixture, are sent to the vacuum pump which maintains process flow. The heated cooling water and secondary steam condensate are drained by gravity (via barometric tube), and go to the tank with a hydraulic gate (the tank is filled up to the required level) and, then, to the pump. This pump removes the cooling water and condensate from the system.
The resulting product is subject to centrifugation or filtration, including drying for further storage and shipping to the customer.

The advantage of this unit is the ability to use, in the summer, excessive steam produced by the anode paste production facilities. The evaporator aids in:
- Reducing the concentration of sodium sulphate in the WS liquor;
- Extending the service life of the existing landfills;
- Avoiding construction of new landfills;
- Reducing CapEx for the construction and commissioning of new landfills by 1.39 x 10^6 USD per year;
- Improving the efficiency of gas treatment facilities by reducing downtime (for cleaning the wet scrubber);
- Eliminating the use of recycled water for diluting the solutions fed to the WS;
- Ensuring the evaporation of at least 10 m^3/h of the WS's liquors; and
- Ensuring the removal of not less than 2.4 t/h of the concentrated sodium sulfate solution with a concentration of at least 200 g /l.

Figure 5: The Evaporating Tower in OJSC RUSAL Krasnoyarsk

UC RUSAL continuous work in the aforementioned areas and evaluates promising technologies for removing sulfates.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SUSTAINABLE PRACTICES IN SPENT POTLINER - AN INDUSTRIAL ECOLOGY APPROACH

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Keywords: Spent Potliner, Environment, Hazardous Waste

Abstract

In the natural world, the waste from one species is the food for another species. In the same way, industrial resources can be optimized by looking at how ecological systems operate. This paper describes how the principles of "Industrial Ecology" are integrated into a safe, sustainable solution for hazardous spent potliner (SPL) from the aluminum reduction process. All of the SPL can be detoxified at the source and refined into products which retain the valuable chemicals and minerals in an accessible form. These products have genuine value in energy intensive industries and a Life Cycle Analysis shows a net environmental benefit can be achieved. This approach has been proven over 12 years and market demand is presently greater than supply.

Introduction

An ecosystem is a natural system consisting of all plants, animals and microorganisms (biotic factors) in an area functioning together with all the non-living physical (abiotic) factors of the environment (Christopherson 1997) [1]. Central to the ecosystem concept is the idea that living organisms are continually engaged in a set of relationships with every other element constituting the environment in which they exist [2].

More simply, it is both living and non-living things that interact with each other. Crucially, an organism cannot live completely on its own without involving another species of organism.

In natural ecosystems a multitude of symbiotic and synergistic effects are constantly taking place. Life, death, energy consumption and exhaustion and resource utilization are indicators that the system is healthy and performing. In such an ecosystem, the waste of one species is food for other species.

Man can create "Industrial Ecosystems" that are artificial representations of natural ecosystems for exactly the same purpose: to combine and optimize resources used by each species for the synergistic benefits of the entire population. Hence the concept of Industrial Ecology.

Tibbs [3] explains that Industrial Ecology uses the natural environment as a model for solving environmental problems and that *"… the key to creating industrial ecosystems is to re-conceptualize waste as products."*

Fiksel [4] describes the concept of Industrial Ecology as providing *"…a useful systems perspective to support sustainable development while assuring shareholder value creation."* Erkmann [5] explains that Industrial Ecology principles solve much more than just pollution and environmental issues: they consider *"technologies, process economics, the inter-relationships of business, financing, overall government policy and the entire spectrum of issues that are involved in the management of commercial enterprises."*

A particular boost to the emergence of Industrial Ecology came with a 1989 article in Scientific American by Frosch and Gallopoulos [6], who put forward the concept of an industrial ecosystem as a more integrated concept than the traditional industrial model of raw materials being used to make products for sale and waste to be disposed of. They further observed that *"corporate and public attitudes must change to favor the ecosystem approach, and government regulations must become more flexible so as not to unduly hinder recycling and other strategies for waste minimization."*

Defining aluminum processes as part of an Industrial Ecosystem leads to the question: how does Spent Potliner, one of the major waste streams from aluminum metal making, fit into this ecosystem?

The SPL Situation

Primary aluminum smelters are faced with increasing expectations that they will find alternatives to landfilling and/or long-term storage of hazardous waste materials such as SPL. SPL is hazardous because of the presence of cyanide compounds, soluble fluoride and an alarming potential to combine with moisture and generate explosive gases [7]. This material is subject to close regulatory control including the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal [8].

In its raw form, SPL varies in size from fine dust to lumps of up to one metre (Figure 1). It typically presents a wide range in mineral and chemical composition as different materials in the pot lining are mixed together.

Figure 1 Raw Spent Potliner

Over the typical life of a pot (five to eight years), materials such as aluminum metal, calcium, fluorides and sodium infiltrate the cathode lining and cause it to deteriorate. Complex chemical reactions result in the formation of various carbides, nitrides and cyanide within the pot linings. When the linings are removed from the pot, the resulting SPL also contains aluminum metal and sodium metal. These components of SPL readily react with water,

including atmospheric water, to generate explosive gases including hydrogen and methane.

Pawlek [9] noted that about 25kg of SPL results from each tonne of aluminum metal produced and that while in the past most of the SPL has gone to landfill, *"this practice must change if the industry wants to claim a reasonable degree of sustainability and environmentally tolerable emissions."*

There are potential environmental liabilities if leachate from landfills escapes to the environment. For this reason, landfilling of unreacted SPL is prohibited in many regulatory jurisdictions. Godin *et. al.* [10] describe a study aimed at identifying a site remediation option that would minimize overall environmental impacts for a landfill containing 100,000 cubic meters of SPL. The study was based on a comparative lifecycle assessment (LCA) and modelling of contaminant transport in groundwater. The study makes the observation that: a) avoiding environmental impacts by placing the SPL in a secure landfill depended on the assumption that confinement of the waste material would be perfect and b) that if concerns about the quality of the long term confinement are considered, then total destruction of the SPL would be required to avoid adverse environmental impacts.

The 2006 Canadian National Guidelines for Hazardous Waste Landfills makes the observation that "*the contaminating lifespan of some hazardous wastes pose significant challenges to landfill design and operation"*, and further that *"the limited service lives of (engineered) components are an important consideration in the facility design"* [11]. The hazardous chemicals in SPL are stable [12] and if stored under dry conditions have effectively an unlimited contaminating lifespan. Aldrich [13] sets out an approach to predicting long term liability costs in landfill of hazardous wastes, and states *"unless one assumes that a manmade structure will never fail, these safeguards cannot eliminate financial liability for the waste."*

As alternatives to landfill, various methods for treatment of SPL have been well researched and described in the literature [9, 14, 15]. Drawbacks associated with most methods include one or more of the following:

- Not all of the SPL can be processed (e.g. the carbon SPL can be handled but not the refractory SPL or vice versa)
- The SPL brings unwanted hazards (e.g. where the nepheline in the refractory portion of SPL provides an attractive flux for clay brick making but the fluorides may present environmental, health and safety concerns)
- There are residual waste materials without ready disposal options other than landfill.

SPL and the Industrial Ecology Principle

The industrial landscape is rapidly changing with recognition by consumers of the requirement for resource conservation, through movements such as the Circular Economy. The Circular Economy could be thought of as a collection of Industrial Ecosystems. New commercial standards are emerging for the purpose of scrutinizing sustainable manufacturing methods, ensuring "Green" or "Circular" practices. For example the Aluminum Stewardship Initiative (ASI) Standard Version 1 (December 2014) states under Criterion 6.7 for Spent Pot Lining (SPL) [16]:

"The smelter shall maximize recycling of carbon and refractory parts from SPL, and will demonstrate that they continuously review alternative options to land filling of SPL. SPLs shall not be discharged to fresh water or marine environments."

Given the current stocks of SPL around the world, the regulatory consciousness around landfill issues and community concerns, a long term sustainable and cost-effective solution must be found for SPL reduction, re-use or recycling. Aluminum smelters are improving pot lining technology to extend the life of pots [17], but reducing the rate of SPL generation does not solve the present storage and landfill issue. Also the overall volume of SPL will continue to increase as demand for aluminum increases.

If we accept that the aluminum smelting process can be classified as part of an Industrial Ecosystem, then applying Industrial Ecology principles to the situation for SPL leads to the question: "what is the optimal approach to re-use of hazardous SPL?"

This question may be answered with another – "what would happen to such a toxic waste in a natural ecosystem?"

Firstly, the introduction of the waste would cause problems in the ecosystem, perhaps resulting in population reduction as various methods were tried but failed to deal with the waste. Then a natural process of adaptation would lead to an optimum situation as one or another species found optimum ways to process elements of the resource. An example is the way that mangrove plants adapt to deal with salt [18].

The aluminum industry must interact with other species in the industrial ecosystem to survive. This is exemplified by the inability to develop an economical internal recycling loop for SPL in more than 40 years of serious attempts [19]. Working together with other "Industrial Species" as a combined ecosystem could optimize the resource utilization so that many industrial populations may benefit synergistically.

In a different species within our "aluminum industrial ecology", manufacturers of energy intensive products such as cement and clay bricks are faced with increasing energy costs and societal expectation of reduction in carbon dioxide emissions [20, 21]. SPL is rich in particular substances that have beneficial energy saving and carbon dioxide emission reduction properties when used in cement and clay brick manufacture. The key valuable constituents for cement manufacture are:

- carbon as a source of thermal energy
- sodium that acts as a flux, lowering the temperature at which solid materials enter a liquid phase [22, 23, 24]
- fluorine that acts as a mineralizer for cement, accelerating reaction rates and promoting the formation of desired materials [15, 25, 26]
- alumina and silica as useful raw materials.

Fluorine is a widely used and effective mineralizer [25]. Sodium contributes to the clinker melt through its fluxing properties. Sodium also leads to a beneficially more reactive clinker where the clinker raw materials have low to moderate alkali levels. Sodium can be used to balance the important alkali/Sulphur ratio in production circumstances where the clinker plant has excessive Sulphur levels (e.g. from using petroleum coke as fuel).

However, the hazards associated with SPL and the highly variable nature of the material have thwarted realization of these benefits [19, 27]. Simply adding crushed raw SPL to cement kilns can actually degrade cement process performance due to the wide fluctuations in chemical constituents [15]. So the aluminum/cement ecosystem must adapt to make the most of the benefits.

Table I Objectives and Strategies for SPL Processing

OBJECTIVES	STRATEGIES
• 100% beneficial use of SPL with no residue • Low cost of SPL re-processing • Lower energy consumption and greenhouse gas emissions for manufacturing products such as cement and clay bricks • Positive impact on the environment.	1. Secure support of regulatory agencies, environmental groups and community stakeholder groups through effective communication and education in Industrial Ecology principles 2. Develop and commercialise a chemical process to eliminate the explosive gas and cyanide hazards at the smelter, so that only refined products are shipped from the smelter, not hazardous waste materials 3. Use industrial mineral trading and marketing to identify target markets, support the products and promote and distribute the products. Encourage industries to move away from hazardous waste processing fees and handling issues and to see the economic value in SPL-derived products 4. Gain classification of SPL-derived products as safe to transport and use. Thus the refined products are not subject to Basel Convention protocols and meet the by-product requirements of the EU directive on waste [28] 5. Establish and maintain integrity of the SPL solution by ensuring all materials are accounted for and by involving only reputable and trustworthy service providers and end-use customers.

The Design of an SPL Solution

The Industrial Ecology framework is instrumental in formulating the design of an SPL solution and sets the tone for engagement with aluminum smelters, regulatory agencies, community stakeholders and potential markets. SPL can then be viewed as a promising resource to be "mined" for economic and environmental benefit, building wide regulatory and community support.

The objectives and strategies formulated to realize this promise are listed in Table 1.

Design of a process to enable SPL recovery, detoxification and manufacture of valuable products follows from the objectives and strategies outlined in Table 1. A simplified flowsheet is shown in Figure 2.

SPL Preparation involves (a) recovery of material from storage or directly from pots; (b) segregation of aluminum metal, carbon materials and refractory materials; (c) sorting into like material streams and (d) crushing and size classification.

The cyanide and explosion hazards in SPL are eliminated through the **SPL Treatment** process. Neutralization of the reactive compounds is achieved by bringing on the reactions that generate the explosive gases in a controlled environment such that no more gas can be generated. The Treatment Process becomes almost self-sustaining by re-using the gases generated to destroy Cyanide by thermal oxidation i.e. heating the material in the presence of oxygen. No residual materials are produced because there are no other chemical processes or additives.

Mineral products with beneficial fluxing and mineralizing properties are **Manufactured** by refining the detoxified SPL material. Other smelter by-products can be added at this point.

The SPL Industrial Ecosystem

Like ecosystems in the natural world, a solution based on Industrial Ecology principles must fit within a coherent framework that optimizes resources. The SPL solution is based on a multi-party ecosystem (Figure 3) which integrates the physical components of the system with an enabling platform comprising:

- technology to detoxify SPL
- technology to use the products in cement and clay brick manufacture
- regulatory approvals for technology and products
- trading of refined products
- optimized marketing and logistics
- research knowledge to support innovation and ongoing development.

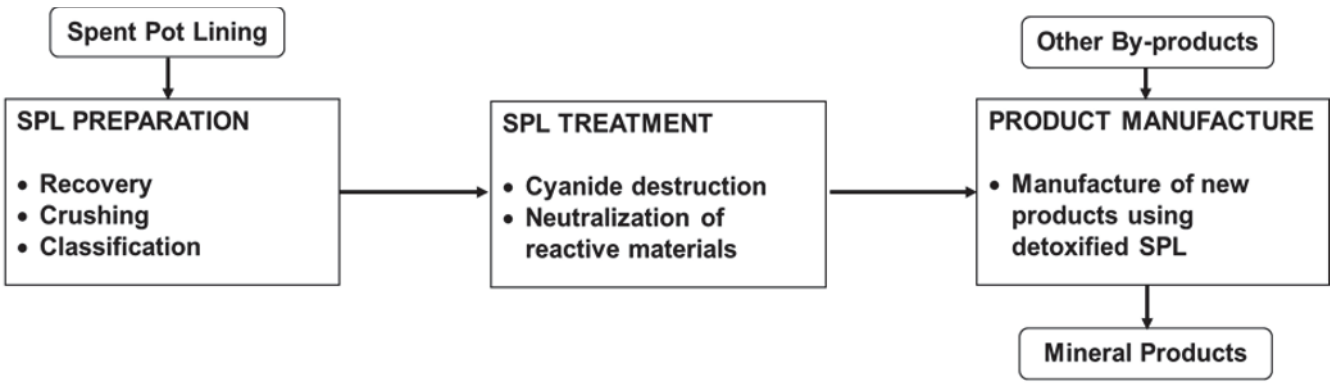

Figure 2 SPL Processing Flow Diagram

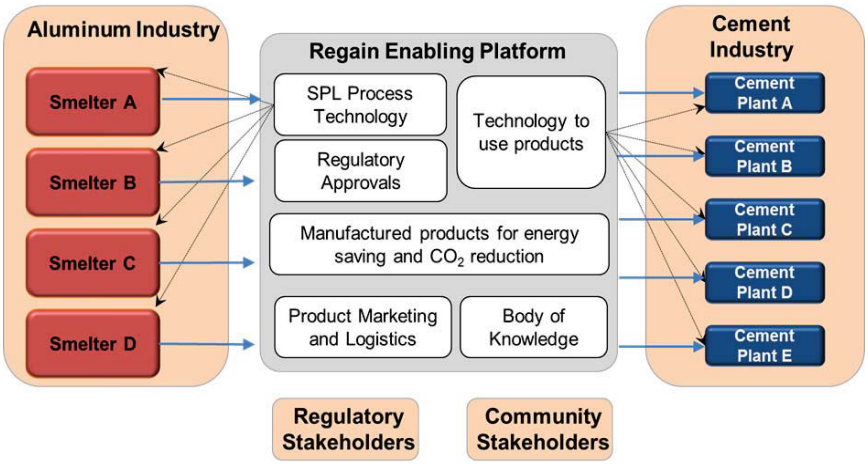

Figure 3 Industrial Ecosystem for SPL

Case Study

The system described above has been developed over a period of fifteen years, providing SPL treatment services for four aluminum smelters in Australia. More than 200,000 tonnes of SPL have been processed through this system and sold to customers as refined products.

The process is typically based on the smelter site near the raw material (SPL). The SPL is detoxified and refined before being shipped outside the smelter boundary. All SPL is used and there are no residual materials. Refined products are used successfully in cement plants and/or clay brick plants in Australia, China, Philippines, Thailand, Morocco, Ecuador, Costa Rica and El Salvador. The refined products have been classified as safe to export and import by governments and/or environmental regulators in all of these countries. The innovation and uniqueness of this SPL detoxification technology has been recognized internationally with patents granted in Australia, Canada, USA, New Zealand, South Africa and UAE.

The success of the solution has been enabled by developing technical expertise on how to maximize the potential value of SPL, designing products tailored for downstream industries, and optimizing the market for the products. Around a quarter of all cement manufacturers can add 2% of quality-controlled products derived from SPL to optimize their production, without seeking rent for handling hazardous waste. With 1 million tonnes of aluminum smelter SPL generated globally each year, and 4 billion tonnes of cement clinker, the cement industry can absorb the generation of SPL by the aluminum industry many times over.

To validate the Industrial Ecology concept, a Lifecycle Assessment (LCA) was conducted on the system of SPL processing, product transport and consumption in a cement plant [29]. The results confirmed net benefits in emissions reduction and energy savings, as shown in Table 2 below.

This approach to dealing with SPL is a mature system. Progressive cost reductions have been realized resulting from the benefits accruing from the technology learning curve, continuous improvement, economies of scale and revenues from sale of refined products (see Figure 4). The costs to a smelter of transforming SPL under this Industrial Ecology approach are now less than the typical liabilities associated with SPL in landfill due to the long term contaminating lifespan and the ongoing landfill remediation requirements [30, 31, 32].

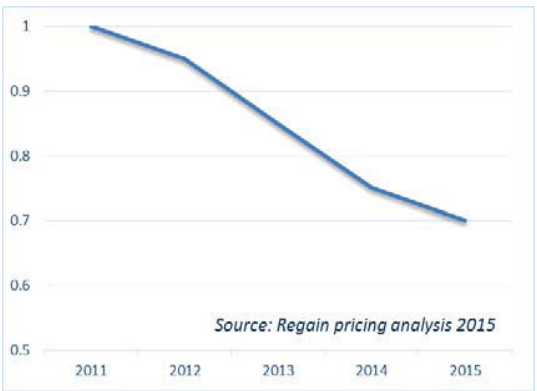

Figure 4 SPL Solution Price Index - Constant 2011 USD Price Level

Table II Summary of Emission and Energy Aspects for One Metric Tonne of SPL

Description	GHG Emission	Thermal Energy	Electrical Energy
SPL Processing	0.2 t CO_2e	1.5 GJ	50 kWh
Savings from SPL-Derived Product Usage	4.2 t CO_2e	17.5 GJ	400 kWh
Net Benefit	**4.0 t CO_2e**	**16.0 GJ**	**350 kWh**

Conclusion

There is a finite resource pool on planet Earth. Understanding natural systems can help us to optimize the value of materials from all possible sources. Regulatory and societal pressures for material stewardship are growing in response to new technologies and our understanding about the impact of industrial progress on the Earth's fragile ecosystem.

While SPL is a tough problem for the aluminum industry, transformation and beneficial re-use of this resource can be achieved safely within an Industrial Ecosystem by:

1. De-toxifying to eliminate the cyanide and explosive gas hazards in the SPL materials, making them safe to transport and re-use in other industries
2. Refining the de-toxified SPL material into quality-controlled products that use the valuable chemicals and minerals to deliver energy savings and CO2 emission reductions in cement and clay brick industries
3. Certifying that the refined products do not require waste regulation (e.g. Basel Convention protocols) because they are no longer a hazardous waste.

A transformation process is described for SPL that has demonstrated success as an innovative model of Industrial Ecology, with net cost benefits for smelters. More than 200,000 tonnes of first-cut and second-cut SPL have been transformed at smelter sites into valuable products which are completely consumed in a developed market. The principles of Industrial Ecology make an environmentally and economically sound argument for industry, regulators and the community to consider SPL as a resource for the future.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE LCL&L PROCESS: A SUSTAINABLE SOLUTION FOR THE TREATMENT AND RECYCLING OF SPENT POTLINING

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Keywords: Spent potlining, LCL&L process, By-product valorization

Abstract

Spent potlining (SPL) is a hazardous waste produced by aluminum smelters. SPL is generated from the internal lining of Al cells, constituted of carbon and refractory bricks and replaced after five to eight years in service. It is classified as a hazardous waste because of its contamination with fluorides and cyanides and its reactivity with water, generating explosive gases. Nowadays, the aluminum industry has made some progress with the SPL issue by recognizing that landfilling is no longer acceptable by most local communities. In 2008, Rio Tinto inaugurated a new plant in Jonquière (Québec) for the treatment of 80 kt of SPL annually, based on the low-caustic leaching and liming process (LCL&L, patent no. US 6,596,252) developed in the early 1990s. This paper describes the advantages of the LCL&L technology, including valorization routes for its by-products and the successful ramp-up of the plant to its nominal capacity reached in 2014.

Introduction

The aluminium smelting process takes place in a steel shell lined with refractory bricks and carbon cathodes. During the operation of the cells, molten fluoride salts and sodium penetrate into the carbon cathode lining and eventually into the alumina refractory lining or firebrick below. Pot failure occurs generally after five to eight years due to the thermo-mechanical stress generated within the pots, which allow attack of the iron collector bars and refractory lining by bath electrolyte or liquid aluminium. During pot shutdown, bath and liquid metal are siphoned off as much as possible. Once cooled, the remaining lining is then broken up and dug out of its steel shell. Iron and large aluminum pieces are sorted and recycled separately. The residual material is called spent potlining (SPL). Figure 1 shows the cross section of a pot.

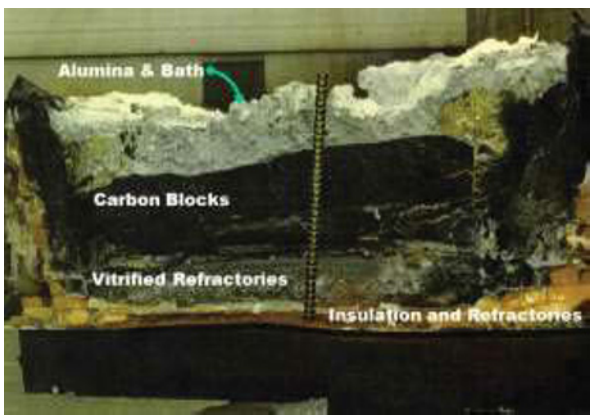

Figure 1. Cross section of an electrolysis pot.

SPL is recognized as hazardous material because it contains significant concentrations of toxic and leachable constituents (cyanides and fluorides). Moreover, in contact with water, the reactive species of SPL, such as residual metallic Al, aluminium carbide and nitrides, have the potential of generating ammonia, hydrogen and methane. Hence, transportation, storage and final disposal of SPL are subject to strict environmental regulations.

Each ton of aluminium produced generates about 22 kg of SPL. Several factors can contribute to the variation in kg of SPL per ton of Al produced or to the variation in chemical composition of SPL. Electrolysis technology, pot operations, achieved lining life, and demolition/relining practices are the major factors. For example, the amount of bath and frozen aluminium that will remain inside the pot and thus in SPL depends on the dismantling procedures of the plant. The fluoride penetration inside the linings depends on the type of materials and the operation lifetime. For Rio Tinto (RT) in Québec, about 20 kt of SPL is generated per year.

Until mid 1980's, RT operated in Jonquière (QC) a plant treating SPL to produce cryolite. That plant was closed because of reduced cryolite consumption in smelters and, since then, RT has accumulated about 600 kt of SPL in safe dedicated storages. Today, less than 500 kt remain stored.
Landfilling of SPL is not an option if the aluminium industry wants to claim an acceptable degree of sustainability. Several options to treat SPL exist and were reported in the literature [1,2]. The biggest challenge for all these options for total recycling of SPL is coming from its heterogeneous composition and its high content in sodium and fluoride. Due to stricter environmental regulations in several places around the world, it becomes more difficult to send SPL without partial or total treatment directly to industries processing hazardous wastes (cement or steel industries).

In the early 1990s, RT has developed at the Arvida Research and Development Centre (ARDC; Jonquière, QC) the hydrometallurgical process called LCL&L (Low Caustic Leaching and Liming), generating inert by-products with high potential for valorization [3,4]. In 2003, after a thorough evaluation of the various available alternatives, RT chose the LCL&L as the most sustainable solution for the treatment of its SPL. In 2008, RT built an 80-kilo ton per year (ktpy) SPL treatment plant in Jonquière (QC) based on this process [5]. This paper is an update of the experience acquired by RT since the plant's start-up [6], including ramp-up capacity, technology challenges that were overcome, and development of by-products valorization routes.

The LCL&L process

The LCL&L process leaches fluorides and cyanides out of spent potlining (SPL) and generates inert by-products that can be valorized (Figure 2).

The treatment is divided in two parts: one dry and one wet sectors. The dry sector includes unloading, handling and storage of SPL containers, SPL grinding (less than 300 microns) by an air swept autogenous mill with air classification and screening, and ground SPL storage. The wet sector consists first of leaching steps in series; the first leach is done with water to extract the water soluble fluorides and most of the cyanide compounds, followed by a low caustic leach to extract the remaining fluorides and cyanides. After filtration, hydrated lime may be added to the inert residues, also named carbonaceous by-product (CBP), to reduce if needed its content in leachable fluorides. Cyanide compounds (such as ferri-ferrocyanide complexes) contained in the leachate are destroyed in pressurized reactors by high temperature hydrolysis (180°C). During cyanides destruction, particulate iron oxide is generated. If necessary, liquor after the flash tanks may need to be filtered to remove this colloidal iron. The resulting liquor is concentrated by evaporation in a multistage evaporation and crystallization circuit, generating caustic liquor at a high concentration compatible, for example, with operation of a Bayer alumina plant, while fluorides are precipitated as sodium fluoride.

NaF is one of the by-products of the LCL&L process. After filtration, the solid sodium fluoride is re-dissolved in water and is reacted with lime to generate inert calcium fluoride pulp and low caustic solution. This caustic solution (causticized liquor) is recycled in the process at leaching, the excess returning to the evaporator to generate a concentrated caustic solution. Thus, the LCL&L process generates three by-products which can be valorized: carbonaceous by-product (CBP), fluoride by-product (FBP) in the form of either NaF or CaF_2, and a concentrated caustic solution (evaporated liquor).

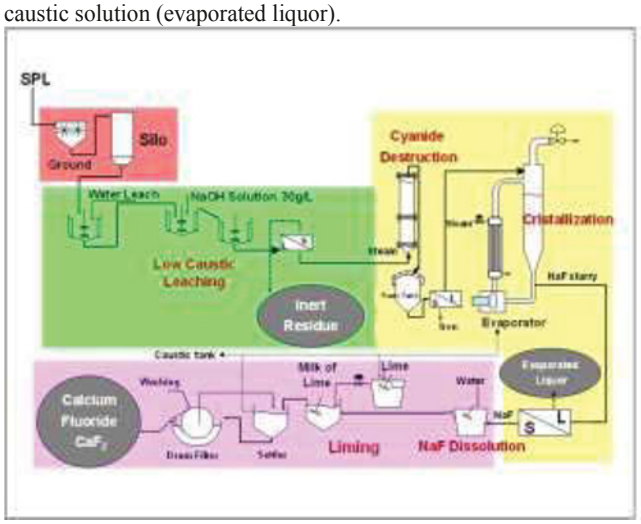

Figure 2. LCL&L process flow diagram.

Technology Package and Analytical Methods

Pilot tests during the process development of the leach step and NaF caustification step were done at a scale of about 25 kg /h SPL at the Centre de Recherche Minérale (CRM). The nominal capacity of the plant built in 2008 is 10.7 t/h SPL, representing a significant up-scaling factor of 450. The basic information required to build the plant was transmitted to the Engineering via the Technology Package, including the Process Flow Diagram with each stream characteristics (flow, temperature, composition). The simulation software Aspen Plus® was used to obtain a detailed steady-state mass and energy balance of the process. All the information (including reaction conversions, SPL composition, compounds solubility) obtained since the start of the plant in mid-2008 was used to continuously improve the reliability of the Aspen model.

For process control, new analytical methods were developed by the ARDC analytical group to measure characteristics of the caustic liquors (fluoride, cyanides, caustic, alumina) or the by-products (pH, leachable fluoride or cyanide contents). Standard methods for measurement of leachable fluoride and cyanide involved delays of 24 hours, which is incompatible with efficient process control. New methods were developed to obtain equivalent measurement in one hour, and thus enabling operators to detect off-spec material generation and adjust production parameters. Development of new analytical methods was also done to determine a detailed composition of SPL (for different electrolysis technologies) and purity of carbonaceous and fluoride by-products.

Ramp-up and Technology challenges

Because of the context of a new technology demonstration, a progressive ramp-up was needed before reaching the nominal plant capacity. The plant was started up in April 2008 and reached its full capacity in 2014 (Figure 3). The first years of operation have been dedicated to major equipment troubleshooting, process adjustments and optimization.

Very early after the plant start-up, heavy scaling of the evaporator-crystallizer unit was experienced. Scale formed on the tube sheet at the exit of the first effect heat exchanger tubes and on the side wall of the evaporating tank, blocking the liquor flow and leading to frequent plant shutdowns. Fundamental causes for this scaling were investigated by the ARDC [7]. Scaling was minimized by modifying the operating parameters and by changing the internal caustic concentration profile within the four different evaporation effects.

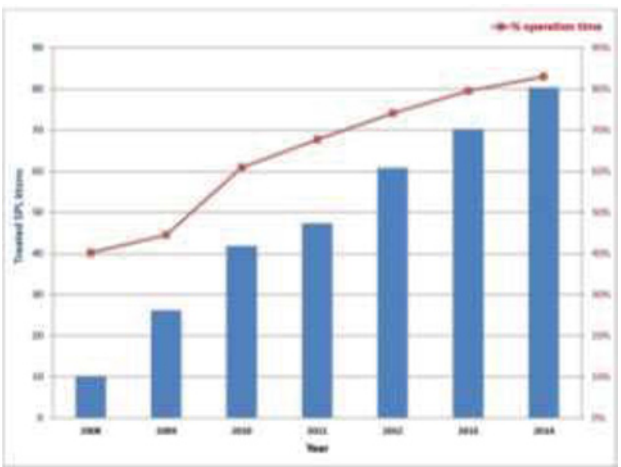

Figure 3. Plant capacity ramp-up.

General control strategy of the wet sector is to keep liquor flow and concentration constant as much as possible. Liquor flow was established to treat the nominal SPL tonnage based on an average SPL composition. The actual SPL tonnage feeding the wet sector is controlled based on the fluoride concentration in solution after caustic leach, to be sure that this liquor is not supersaturated. The plant capacity is then lower if fluoride content in SPL increases, and vice-versa. Caustic concentration at caustic leach is controlled: it has to be high enough to ensure good fluorides extraction, but not too high to avoid fluoride solubility reduction at high caustic concentration which would reduce plant capacity.
Some modifications and simplifications were introduced to the leaching steps since the plant start-up and the initial LCL&L process [5]. Water leach filter, activation and polishing leach steps were by-passed. To improve the separation of solid and liquor, a settling tank was also installed after the leach reactors and before a belt filter. The objective of this decanter was to thicken the slurry out of the caustic leach reactors to reduce the size of the downstream caustic leach filter. Synthetic flocculent is injected to the decanter feed to form flocks and increase solids settling rate. Overflow of the decanter is pumped to the cyanide destruction unit feed tank. R&D was done in the last two years in order to find the best flocculent and to avoid solid accumulation and blockages of equipment in the cyanide destruction sector.

If required by local environment regulations, lime can be added to CBP to reduce its level of leachable fluorides as measured by TCLP procedure. However it must be recognized that lime addition will also increase the pH of CBP to pH ~13 that may be unacceptable depending on local environment rules. If required, it is possible the re-slurry with hot water the limed CBP, and then filter and wash it on a second filter, to remove the caustic generated by the lime and reduce the pH to a value around 12.5. Other additives were also tested and may be used as alternatives to lime addition with respect of low leachable fluoride content (<150 ppm) and low pH (<12.5) of CBP according to Quebec environment regulations.

The plant also faced safety challenges due to the reactivity of SPL with water. The heart of the LCL&L process is the leaching step. Agitated cascade reactors digest the slurry extracting fluorides, cyanides, sodium, alumina, and silica into the caustic liquor, and generating explosive gas. Any intercalated sodium and remaining aluminium metal will react with water and caustic to generate hydrogen:

$$2\ Na + 2\ H_2O \rightarrow 2\ NaOH + H_2 \quad (1)$$

$$2\ Al + 2\ NaOH + 2\ H_2O \rightarrow 2\ NaAlO_2 + 3\ H_2 \quad (2)$$

Aluminium nitrides and carbides will generate ammonia and methane gas respectively:

$$AlN + NaOH + 2\ H_2O \rightarrow NaAlO_2 + NH_4OH \quad (3)$$

$$Al_4C_3 + 4\ NaOH + 4\ H_2O \rightarrow 4\ NaAlO_2 + 3\ CH_4 \quad (4)$$

To avoid accumulation of explosive gazes inside the leaching reactors, dilution air is provided inside the reactors. Continuous measurement of explosive gas on the vent outlet of the first leach reactor is also provided. Freshly ground SPL can also react with moisture to form a mixture of potentially explosive gases. For most of the SPL grinding sector, the risk of explosion has been eliminated by diluting the explosive gas mixture with air to a concentration inferior to 25 % of the lower explosion limit (LEL). However, inside the screw conveyors located underneath the silos, this strategy is not possible because the high air velocities would sweep all the material. Thus, nitrogen is injected downstream the silos to achieve partial or total inerting of the explosive gas mixture in the conveyors.

By-products valorization

As mentioned previously, the plant is producing three main by-products: inert residues also named carbonaceous by-product (CBP), fluoride by-product (FBP) in the form of either NaF or CaF_2, and a concentrated caustic solution (evaporated liquor). In 2014, the plant treated 80 kt SPL and produced 94 kt of CBP, 48 kt of CaF_2 and sent 30 kt of caustic solution (27% NaOH) to the nearby Vaudreuil alumina plant.

Mixed Carbonaceous By-product (CBP)

As mentioned in [2], cement industry and few other industrial pyrometallurgical processes are capable of using the SPL as generated, without any pre-treatment. However, strict restrictions on the sodium and fluoride content in the final product limit the amount of SPL that can be added to these processes. Nowadays, environmental regulations are also stricter and it becomes difficult for these industries to accept unprocessed SPL.

Since 2006, RT has developed an intensive R&D program for valorization alternatives for the CBP. Since the plant treats all of the SPL (both carbon lining and the refractory lining), the CBP contains approximatively 30-40% carbon (on dry basis) and 60-70% of inert materials (mostly SiO_2 and Al_2O_3). Because of its mixed nature, CBP is essentially limited to be used as alternative fuel and raw material for clinker production in the cement industry. Compared to direct SPL valorization in the cement industry, the CBP contains five times less total fluorides and alkalis (sodium), which enables higher dosage into the kiln. The carbon is attractive for its energy content (approx. 13 GJ/t), while the inert material is attractive for its mineral composition (ash) relative to clinker chemistry. Moreover, residual inert fluorides (mostly in the CaF_2 form) in the CBP have the advantage of lowering the energy required to produce clinker due to its fluxing properties. However the mixed nature challenges state of the art clinker production, since minerals are normally fed to the kiln at the cold end while solid fuel is fed to the hot end. Plant scale tests done in 2009 at the Holcim cement facility located in Joliette, Québec, indicated that nearly 20% of the total CBP production could be valorized this way. The use of CBP by a cement plant allows total recovery of the carbon energy value and of the mineral content of CBP, it has no impact on clinker mineralogy and on cement quality, and does not create any significant environmental concern.

Another option identified to valorize mixed CBP is to burn the low sulphur carbon (around 0.1% S) to recover its energy content. The remaining ash could be valorized as raw material for cement plants or in the refractory industry [8].

Since 2013, mixed CBP production has been valorized as low density (1250 kg / m^3) civil engineering construction material at Rio Tinto bauxite residue disposal site. While compacted to its maximum density, CBP has proven to be a very good geotechnical construction material.

CBP valorization by carbon and refractory brick separation

As a promising solution to valorize mixed CBP, carbon separation techniques, such as flotation, were tested in laboratory and in pilot trials. Carbon concentrate grade over 90% C, with carbon recovery up to 95%, and brick concentrate containing less than 5% C were generated. In 2011, more than 150 t of CBP (dry basis) were successfully treated in a pilot unit to produce 50 t of carbon concentrate and 100t of brick concentrate. The carbon concentrate has promising valorization options for the cement and steel industry, but has also an interesting graphitic value. In 2012-2013, 150 t of anodes containing 1% of this carbon concentrate were produced and successfully tested in electrolysis cells at Grande-Baie plant (UGB), Québec.

In the eventuality that the first cut (fraction of SPL above the collector bars, mainly carbon) and second cut (fraction of SPL below the collector bars, mainly refractory brick) would be separated during dismantling of the electrolysis pots, and then treated separately by the LCL&L process, it would be also possible to obtain separate carbon and brick concentrate. An R&D project is on-going at the plant to test this option: a few hundred tonnes of each cut were recently treated separately with success at the plant. The processed brick concentrate contains around 5% C (similar to results obtained by flotation) while the carbon concentrate contains above 80% C. This option seems to be really interesting for valorization of fresh SPL.

Fluoride by-product (FBP) valorization

The valorization of the fluorides extracted from the SPL must take into account characteristics and quality constraints of the local market. In Quebec, because of the limited market for sodium fluoride (NaF), it is better to convert NaF to calcium fluoride (CaF_2), which has also the advantage of being inert. In Jonquière,the conversion to CaF_2 allows the FBP to be used as an alternative raw material in the nearby aluminium fluoride plant, allowing a considerable saving. At its nominal production, the fluoride plant is consuming 90 ktpy of fluorspar (mineral CaF_2) to produce aluminium fluoride, AlF_3 in a two-step process: the first step being the generation of HF by reacting CaF_2 with sulphuric acid while the second step is the production of AlF_3 by reacting anhydrous HF with aluminium hydroxide ($Al(OH)_3$). AlF_3 is then re-used in the electrolysis smelters as an important additive for the bath, allowing to close the loop of the fluorides.

In order to replace fluorspar by LCL&L CaF_2 for AlF_3 production, R&D was done in recent years to improve the quality of FBP to meet the AlF_3 plant requirements. The CaF_2 produced by causticizing NaF with lime in water must contain as little as possible hydroxide and oxide compounds. Indeed, these compounds would later consume sulphuric acid at the fluoride plant, producing unexpected water in the process that must be controlled by addition of oleum. Other impurities in the CaF_2 could also decrease the quality of the produced AlF_3. The other challenge for the CaF_2, in addition to its purity, is the particle size necessary for its introduction at the fluoride plant (around 50µm). As the LCL&L plant is producing CaF_2 paste with small particle size around 5-10 µm and with approximatively 45 wt. % moisture content, the paste must be dried and calcined at high temperature (700-800°C) for sintering/agglomeration.

During the last two years, R&D efforts were focused on both chemistry and transformation of CaF_2. With better control of lime addition and process parameters, side reactions of lime with sodium carbonate or with aluminate and silicate in solution were minimized, avoiding large precipitation of these calcium salts with CaF_2. Pyroprocessing test in rotary kiln also resulted in CaF_2 particles enlargement. The result of these efforts was completed by a pilot trial producing 2000 t of CaF_2 pulp with good purity (87 dry wt.% CaF_2). This pulp was then processed at high temperature in a rotary kiln, and the particle size specification for the dry material target was reached. The final CaF_2 was then introduced at the fluoride plant for several weeks, replacing successfully 25-30% fluorspar feeding with minimal process modifications, and allowing for a full qualification of this valorisation route. Future works will include the engineering of a full scale transformation process (sintering/grinding) in order to close the fluoride loop of the LCL&L process.

Conclusion

The LCL&L process was developed by Rio Tinto to treat spent potlining generated by aluminium electrolysis cells. It is a proven technology with an 80 kt plant in constant operation. The capex and cost per ton of the process is relatively high, however it is a good alternative in situations when transportation costs to cement plants are high or not feasible due to legal compliance requirements.

Meanwhile, R&D was done to develop and implement the valorization routes of the LCL&L by-products. We succeeded to close the loop of the fluorides with the use of CaF_2 by-products at the fluoride plant, making the LCL&L process a sustainable and economical option to efficiently treat SPL. This plant is an interesting option for other aluminium producers in Quebec and in North America for the disposal of SPL. The LCL&L process should also be foreseen as a commercially available technology solution for application in other regions of the world where a significant amount of SPL needs to be managed.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

DEVELOPMENT, PROOF OF CONCEPT AND INDUSTRIAL PILOT OF THE NEW CHAC SCRUBBING TECHNOLOGY: AN INNOVATIVE AND EFFICIENT WAY TO SCRUB SULFUR DIOXIDE

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Keywords: sulfur dioxide, scrubbing, low footprint, recoverable by-product

Abstract

Reducing sulfur dioxide emissions could be a solution to respect different stakeholders' expectations, when considering production growth, raw material changes and/or more stringent environmental regulations.

Owing to the fact that scrubbing is the selected solution, technologies available today require large equipment to handle and recycle reactants as well as multistage treatments for a barely recoverable by-product, thus resulting in high costs.

From 2011 to 2014, Rio Tinto Aluminum division R&D Environmental Group has developed a new patent-pending [1] scrubbing technology named «CHAC», which stands for "Chaux Hydratée Aqua-Catalysée" or Aqua-Catalyzed Hydrated Lime. Tests were conducted at small laboratory scale followed by a larger proof of concept demonstration in a plant environment. The results confirmed the process performance and viability of the new technology. This led to the implementation of a full-scale industrial pilot at the Coke Calciner of the Arvida Smelter, in 2015, to validate the full potential of the CHAC process. The operational and environmental performance was evaluated to confirm the advantages and improvements.

Introduction

The aluminum industry has always generated sulfur dioxide (SO_2), due to the presence of sulfur in raw materials (coke, pitch, etc.) used for anode/cathode manufacturing, as well as during the production of electricity with fossil fuel.

In the last years, there has been a production increase in the aluminum industry and a shortage of carbon raw materials with low sulfur content, which results in a higher total tonnage of SO_2 emitted in local environment. With the latest reports on the impact of SO_2 on human health [2,3], the risk of further increasing the concentration of SO_2 in ambient air remains an extremely important issue.

Technological solutions to decrease SO_2 emissions are limited and among them is the installation of SO_2 scrubbers. Different SO_2 scrubbing technologies are readily applicable but require expensive equipment that also results in high operational and maintenance costs (e.g., energy used and neutralization reactant) [4]. In addition, these technologies produce tonnes of by-products that are difficult to valorize and often end up in landfills.

After reviewing all this information, the issue of SO_2 is not a simple case, when one considers all stakeholders. Since 2006, Rio Tinto Aluminum has had as objective to identify, test and develop a more efficient SO_2 scrubbing technology. The enhanced version of this technology now allows a significant reduction in capital and operating costs compared with other technologies available on the market. Another objective that has been achieved is the reduction of by-product volumes needing to be reused or valorized. In addition, it is also possible, with this updated version, to treat gases having great differences in SO_2 concentrations and lower temperatures (80 °C or more).

The development steps and key parameters of the patent-pending CHAC scrubbing process, from 2011 to date, are reviewed in the present paper.

CHAC scrubbing process development

The development of a new technology or process is not a simple task. It requires know-how and expertise from past experience combined with knowledge in process parameters and performance as well as being conscious of our environmental footprint and having the determination to succeed.

The original idea.

From 2006 to 2010, the R&D work performed on SO_2 scrubbing in Rio Tinto, has allowed delivering a dry SO_2 scrubbing technology based on a batch reactor using sodium bicarbonate as reactant. Fortunately, this technology was abandoned even if the SO_2 scrubbing efficiency was excellent. Indeed, it required high capital and operational costs, the gas temperature was above 135 °C (275 °F), and produced large amounts of by-products to be disposed of in landfills. During the development of that technology, trials were carried out with other reactants including hydrated lime. While the SO_2 scrubbing efficiency was considered acceptable using hydrated lime, efficiency could not be maintained over a long period. At the end of 2010, the Rio Tinto Aluminum R&D team shared preliminary results with colleagues from Rio Tinto, Iron and Titanium (RTIT) division, concerning SO_2 scrubbing. The RTIT division was also working on a similar project. Consequently, these discussions and very preliminary results led to a more structured R&D project, in the Aluminum group, being SO_2 scrubbing using dry hydrated lime as reactant in lower gas temperatures.

Development of a small-scale mobile pilot scrubber and R&D trial results, in 2012 and 2013, using hydrated lime.

At the beginning of 2011, the R&D team decided to use a more "classical" SO_2 scrubber, instead of a batch reactor, to perform tests with hydrated lime reactant. It was also planned to use the low temperature reduction gases sourced directly after the gas treatment centre. In 2011, a hydrated lime supplier (Graymont Inc.) was contacted, and a partnership was initiated to build a small-scale, mobile, pilot scrubber based on a ''classical'' gas treatment centre. The mobile pilot scrubber included a raw reactant silo, injector, reactor, baghouse and a recycling system. All these components were insulated to minimize the temperature drop. This scrubber was sized to fit a truck trailer for optimum versatility and ensure mobility (Figure 1).

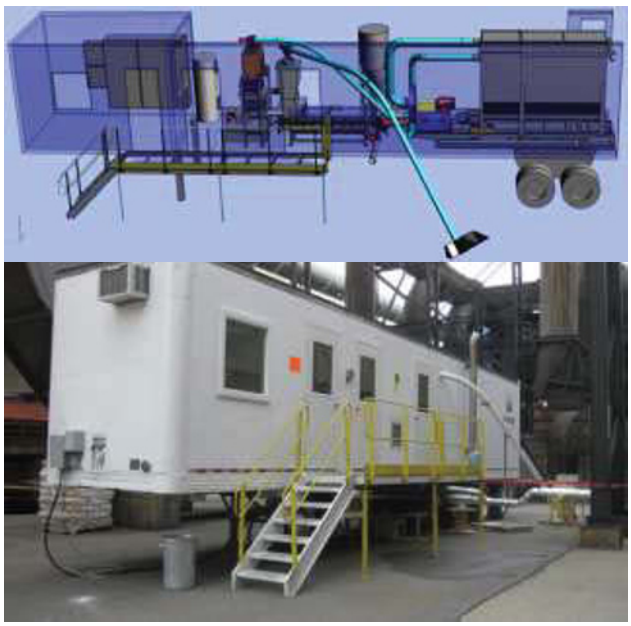


Figure 1. The mobile pilot scrubber

The main operational parameters of the mobile pilot scrubber were:

- raw reactant injection flow rate: 0.1 to 10 kg * $hour^{-1}$
- used reactant recycling flow rate: 60 to 600 kg *$hour^{-1}$
- baghouse: 36 bags with Teflon coating, gas up to 200 °C
- gas flow rate: up to 1200 Nm^3 * $hour^{-1}$
- analyzers: 2 Thermo Scientific models 43i

In 2012, the mobile pilot scrubber was installed at the Grande-Baie Smelter. Many trials were carried out during 2012-2013 on reduction gases located after the gas treatment centre. These gases had a lower temperature and minimal gaseous hydrogen fluoride content. In 2012, trials were also performed with the injection of dry raw and recycled reactants. During these trials, the experimental operational parameters used were:

- raw reactant injection flow rate: 0.4 kg * $hour^{-1}$
- used reactant recycling flow rate: 60 to 600 kg *$hour^{-1}$
- gas flow rate: 900 Am^3 * $hour^{-1}$
- SO_2 concentration in inlet gas: between 100 and 120 ppm
- gas temperature range at the scrubber inlet: 90 to 100 °C
- gas temperature range between reactor and baghouse: 30 to 70 °C

Among the parameters, the filtering bags were coated with raw reactant prior to the trials. The results showed excellent SO_2 scrubbing efficiencies when the gas temperature, in the reactor/baghouse, and the SO_2 concentration, in the gases, were low, and the quantity of reactant (raw + recycling) was very high (Figure 2). In 2012, low temperatures were achieved from dilution but unfortunately the gas temperatures ranged between 90 to 100 °C, at the inlet, and between 60-70 °C, at the outlet.

Accordingly, these results showed that the low gas temperature was a key parameter for SO_2 scrubbing, when using dry hydrated lime as a reactant. More importantly, the amount of recycle reactant can impact the scrubbing efficiency, but an unknown parameter seems to also catalyze the neutralization reaction. Indeed, different scrubbing efficiencies were obtained with gases at 45 °C.

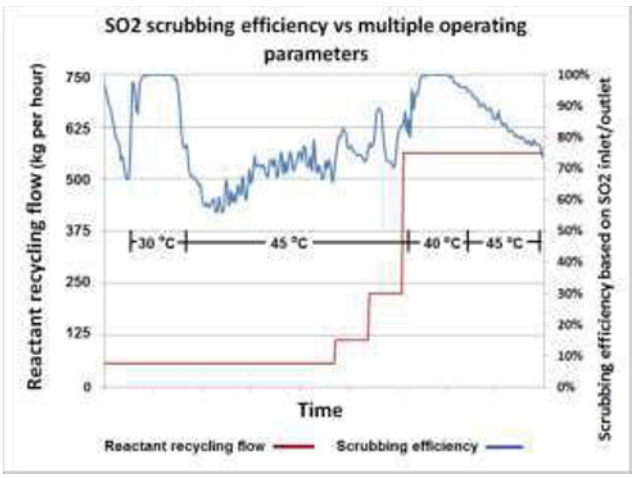


Figure 2. Sulfur dioxide scrubbing efficiency (partial results of the 2012 R&D trials).

In 2013, the mobile pilot scrubber was re-installed at the Grande-Baie Smelter to carry out R&D trials, in July and August, using gases sourced downstream of the potline gas treatment center. Only raw reactant injection was used during these trials.

A heat exchanger was installed on the scrubbing unit to continue trials with lower inlet gas temperatures without using air dilution. The trials were based on injection of dry raw reactant only, with no addition of used reactant. During these trials, the experimental operational parameters were:

- raw reactant injection flow rate: 0.1 to 0.2 kg * $hour^{-1}$
- used reactant recycling flow rate: 0 kg *$hour^{-1}$
- gas flow rate: between 250 and 900 Am^3 * $hour^{-1}$
- SO_2 concentration in inlet gas: between 100 and 120 ppm
- gas temperature range at the scrubber inlet: 35 to 72 °C
- gas temperature range between reactor and baghouse: 15 to 40 °C

A lower gas flow was used in the heat exchanger to obtain the highest temperature drop, thus decreasing the gas speed in the unit and improving the contact time with the reactant. As mentioned, the filtering bags were coated with raw reactant prior to the trials. During these trials, excellent SO_2 scrubbing efficiencies were

achieved with low gas temperature in the reactor/baghouse, and the quantity of raw reactant injected was also low. Another approach to analyze these results is to consider the percentage of relative humidity (RH) of the gases. A gas can contain a maximum quantity of humidity or gaseous water, called saturation vapor density (normally in g/m^3), at a specific temperature. The actual level of humidity of a gas, in gaseous water, is called the actual vapor density (also in g/m^3). The RH can be calculated by Equation 1. When the RH reaches 100 percent, water will condensate when the temperature is lowered; this is called the dew point.

$$H = \frac{\text{Actual vapor density}}{\text{Saturation vapor density}} \times 100\ \% \qquad \text{Eq. 1}$$

The reduction gases, downstream of the potline GTC (Gas Treatment Centre), at 90-100 °C, contain less than 2 percent of total humidity or actual vapor density (less than 12 g/m^3). This represents less than 3 percent of relative humidity (using between 424 and 600 g/m^3 for saturation vapor density of air between 90 and 100 °C). When the gas is cooled down in the heat exchanger, the actual vapor density of the gas remains unchanged, but the saturation vapor density changes as the temperature is lowered. Consequently, by decreasing the gas temperature, the outcome is also an RH increase of the gaseous environment, tending toward the water dew point. The results of the R&D trials performed in 2013, demonstrated that if gas temperature is lowered and/or RH increased, the SO_2 scrubbing efficiency from hydrated lime is improved (Figure 3).

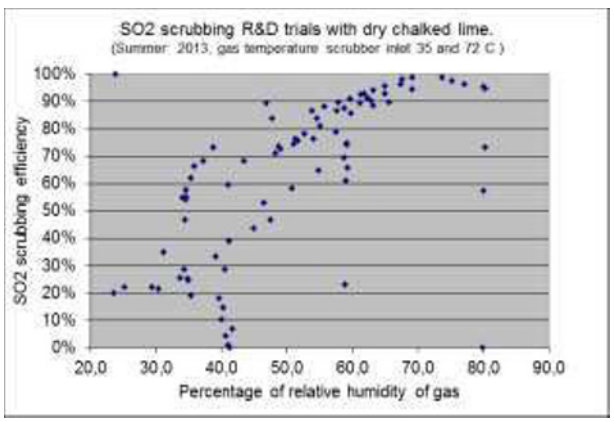


Figure 3. Sulfur dioxide scrubbing efficiency results of the 2013 R&D trials.

In the second half of 2013, a laboratory size experimental set-up was installed at the Arvida Research and Development Center, to carry out further trials with high relative humidity/low temperature gas, using a thin layer of reactant (Figure 4). This set-up was used to confirm SO_2 capture efficiency and neutralization reaction time, at gas temperatures ranging between 24 and 80 °C. The objective was to verify if the neutralization reaction could be improved with a gas containing more gaseous water at a fixed temperature, and if a high RH level was needed at all temperature ranges. The experimental set-up was to use 100 ppm of SO_2 gas, to pass through 1 gram of reactant spread evenly on a 4-cm filter, at different flows, temperatures and relative humidity levels. Results are shown in Figure 5.

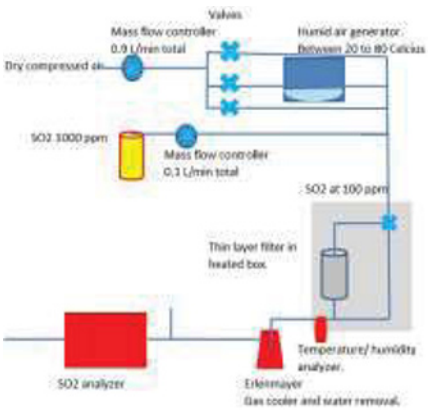


Figure 4. Simplified laboratory experimental set-up.

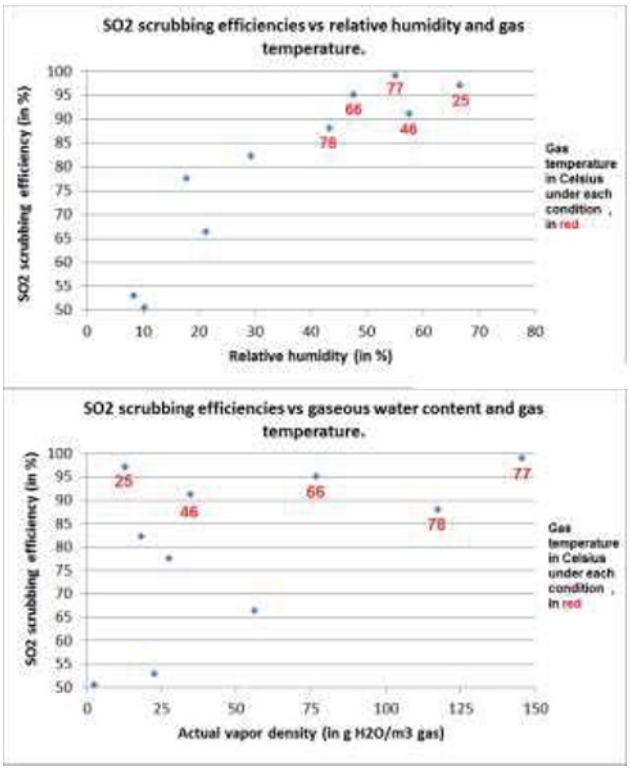


Figure 5. Sulfur dioxide scrubbing efficiency - partial results of the 2013 laboratory scale trials.

The results of these laboratory size trials confirmed that the efficiency of the neutralization reaction of SO_2, by hydrated lime, is improved in low temperature gas with a higher percentage of relative humidity (nearer to water vapor dew point). Similar results were reported in previous technical papers [5, 6]. New information was also found from these latter, for example:

1. when using completely dry air, there is no neutralization reaction of SO_2 with hydrated lime;
2. with higher gas temperature, an efficient neutralization reaction of SO_2 with hydrated lime can be achieved with lower relative humidity level, but with high actual vapor density.

Based on this information, we were able to develop the CHAC scrubbing process. This process involves conditioning the gas to increase its relative humidity/gaseous water content (water

addition in any state and/or decrease temperature of the gas) upstream of the reactant injection point to obtain optimal operating conditions for best scrubbing efficiency. This process allows catalyzing the reaction of SO_2 with the reactant, allowing minimizing the reactant feed rate and the use of few equipment, thus making the scrubbing process affordable and competitive.

Real scale ''proof of concept'' in 2014.

After the trials of 2012 and 2013, Rio Tinto Aluminum decided to complete a ''proof of concept'' on a larger scale. The objective was to confirm that the CHAC scrubbing process was applicable and could reduce SO_2 concentration in a gas, at a near industrial scale. After evaluating all options, the most interesting ''proof of concept'' could be performed using gas coming from one of the two coke calciner kilns, at the Arvida Smelter. This would represent a real industrial scale trial.

The SO_2 scrubbing technology already used for the two kilns is dry injection of sodium bicarbonate, directly in the horizontal duct containing the gas upstream of the baghouse (Figure 6). The two coke calciner kilns have each their own raw reactant silo, reactant injection system, baghouse and fan, but share the same by-product management equipment. The sodium bicarbonate technology has the advantages of being a simple process requiring a relatively low investment to implement, but has some disadvantages such as relatively high operational costs (cost for reactant and its transportation to the site), a yellow fume at the stack (due to the formation of NO_2), carbon dioxide emissions, and in the case of Arvida, one hundred percent of the by-product generated is disposed in a landfill.

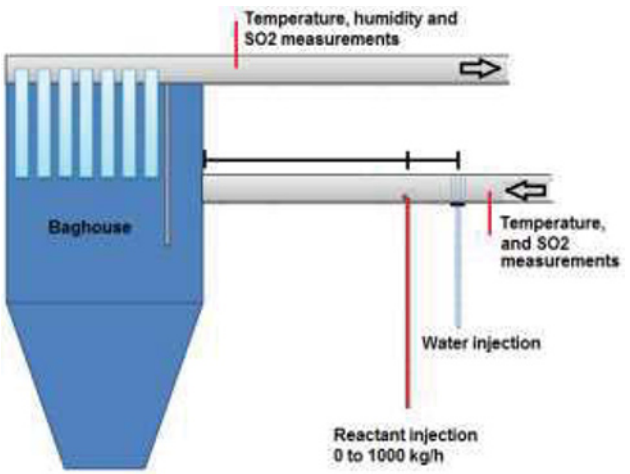


Figure 6. Simplified installation for the ''proof of concept'' of the CHAC scrubbing technology on Coke Calciner Kiln 2.

Completing the ''proof of concept'', at this site, required equipment (silos, injection system, baghouse, etc.) that could operate with different reactants and the addition of a gas conditioning system upstream of the reactant injection location. The decision taken was to rapidly complete the ''proof of concept'' at the Arvida Coke Calciner Kiln 2, from May to November. All actions were taken in this regard. The initial conditions of the process gases, at this location, are 120 000 $Am^3 * h^{-1}$, ranging between 175 and 185 ^oC, containing between 1200 and 2000 ppm of SO_2 and between 1 and 2 g/m^3 of calcined coke particulates. The reactant injection system in place was able to inject between 0 and 1000 kg $* h^{-1}$ of sodium bicarbonate, and the same injection flow was possible with hydrated lime.

The gas conditioning system was designed quickly in order to be able to inject water at 300 PSI, producing a large number of small droplets, to insure maximum vaporization of the water, and a maximum temperature drop of 120 ^oC. The water used was river surface soft water. The system was installed upstream of the reactant injection system. Temperature sensors and SO_2 analyzers were installed upstream of the gas conditioning system and after the baghouse, while a humidity sensor was installed only downstream of the baghouse (Figure 6). As the gases of the two coke calciner kilns end up in the same stack, the plant had already an SO_2 analyzer at the stack.

The ''proof of concept'' trials began September 22, 2014. The experimental plan included different injection flowrates for the reactant and for the water. The CHAC scrubbing process was confirmed ''possible'' during the first two days. SO_2 capture and CHAC scrubbing process were demonstrated when the reactant was injected without a conditioner, as the process gases of the coke calcining contained naturally 8 to 10% of gaseous water (1% RH or actual vapor density of 50 g H_2O/m^3 at 180 ^oC), at that time. When water addition was added upstream of the reactant injection location, SO_2 capture was even improved (Figure 7).

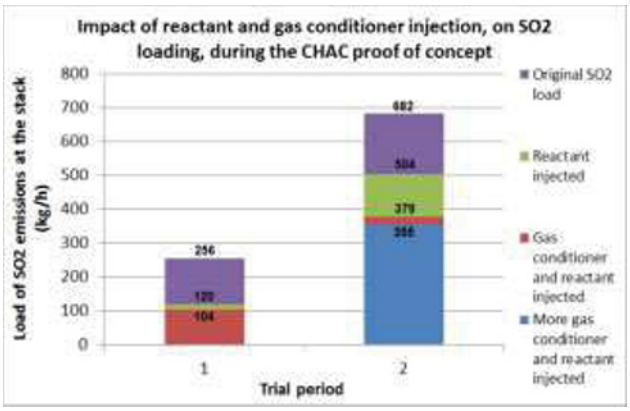


Figure 7. Results of two trials during the CHAC proof of concept, with different operation parameters.

Unfortunately, the "proof of concept" trials were stopped on October 22nd due to damaged nozzles. However, the ''proof of concept'' was nevertheless a success, as five periods of trials confirmed the viability of the CHAC scrubbing process. In 2015, it was therefore decided to move to the next step, which was the installation of an industrial pilot at the Coke Calciner of the Arvida Smelter.

CHAC scrubbing process R&D industrial pilot.

To continue the R&D trials and development of the new CHAC scrubbing technology, the next step was to install a full-scale R&D industrial pilot on both coke calciner kilns of the Arvida Smelter.

One major challenge was to install and fit all of the injection and monitoring equipment on the current existing horizontal ducts and baghouses during the June 2015 maintenance shutdown. The design, selection and installation of the equipment, to demonstrate that innovative scrubbing process, are crucial, as many parameters

would have to be measured and changed, to insure controlled start-up and stabilization, while allowing fine-tuning of the process later. The 2015 R&D industrial pilot should give the critical information and data to confirm the SO_2 scrubbing efficiency achievable in the context of the coke calciner operating conditions.

Conception and installation of the CHAC scrubbing process R&D industrial pilot in 2015.

The critical equipment was designed in order to be ready to purchase, fabricate and deliver on site, between April and beginning of June. The following design and performance criteria were used to design the water and reactant injection systems for the pilot:

- reactant injection on system one, two or both, would be possible from each individual raw reactant silo;
- maximal increase of the gaseous water content of the gas, with maximal temperature decrease of 100 °C;
- minimal length to evaporate a maximal quantity of water injected in an horizontal duct (duct diameter of 1.72 m);
- capability to reduce the water injection flow to a minimal representing 40% of the maximum flow, without significant decrease in droplet quality.

For the baghouse, the existing systems were used without any major modifications. Finally, the monitoring equipment requiring minimal maintenance was installed for temperature, water content, reactant build-up in horizontal duct, and SO_2 concentration.

The gas conditioning system.

As a critical part of the CHAC scrubbing technology, three concepts were designed to address the key design criteria. All these concept results were computed using a computer fluid dynamic (CFD) model. The final selection was based on a system respecting the main design and performance criteria (Figure 8).

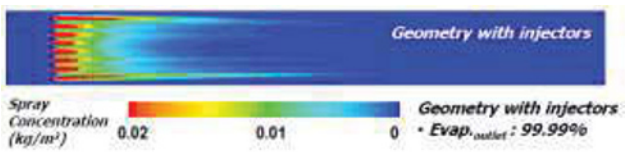


Figure 8. Conceptual CFD study of the selected design for the water injection system.

The complete injection system included injection equipment, as well as pumps, drives, filters, and pressure and flow sensors.

The reactant injection system.

The final concept included to change the location of the injection in the receiving horizontal duct, to maximize distance between the gas conditioning and the reactant injection location. The addition of a three-way valve, on each system, to respect all scenarios of injection requested, was also included. All other equipment such as silos, blowers, and reactant transportation pipes were kept the same. It is important to point out that this equipment is not heated or insulated.

The baghouse, fans and by-product handling equipment.

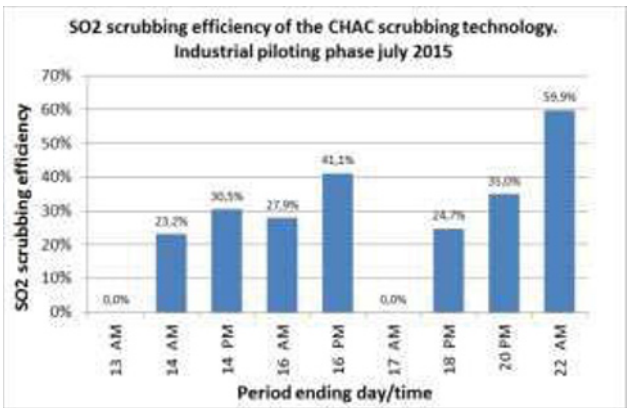
The baghouse operating parameters were changed in order to increase the contact time. These adjustments were limited by the fan capacity. The two baghouses were inspected during the major maintenance shutdown, and problems corrected (leak, declogging system, etc.). For the by-product handling equipment, the new reactant did not request any modification to the equipment or operational parameters.

The monitoring equipment.

As for Coke Calciner Kiln 1, temperature sensors were already in place upstream and downstream of the baghouse, and a humidity sensor was added downstream of the baghouse. As the ''proof of concept'' was performed on Coke Calciner Kiln 2, the humidity and temperature sensors were not changed, but the SO_2 analyzers were changed from extractive mode to in-situ mode. Sensors were installed on the gas duct of both coke calciner kilns, a few meters after the reactant injection location, to detect any build-ups.

Start-up and first results.

Start-up of the industrial pilot of the R&D CHAC scrubbing technology took place on July 6th, after pre-operational verification of the equipment. The piloting started without the injection of reactant and conditioner, to confirm low or no SO_2 scrubbing efficiency of the system. The first addition of reactant and conditioner was performed on July 14th. The pilot trial started with a fixed injection flow of reactant (at 500 kg * $hour^{-1}$) and then the conditioner injection flow was slowly increased. Good results were obtained considering that the maximum conditioner injection flow (''A'' liters * $hour^{-1}$) was not reached. The reactant injection flow rate represents between 1.26 and 1.50 of the stoichiometric ratio (Figure 9).



Period in July 2015	SO_2 load at the inlet	Reactant flow (kg * h^{-1})	Conditioner flow (A is maximum flow, in L * h^{-1})	SO_2 scrubbing efficiency (in %)
13 AM	1,0	0	0 X A	0
14 AM	1,0	500	0 X A	23.2
14 PM	1,1	500	0.23 X A	30.5
16 AM	1,2	500	0.49 X A	27.9
16 PM	1,2	500	0.73 X A	41.2
17 AM	1,1	0	0 X A	0
18 PM	1,2	500	0 X A	24.7
20 PM	1,2	500	0 X A	35.0
22 AM	1,2	500	0.76 X A	59.9

Figure 9. Preliminary results for the CHAC scrubbing process industrial pilot.

At the time of this writing, these were the best results available. The piloting of the CHAC scrubbing technology will continue until the end of 2015, as data found in Figure 9 do not show that the maximum injection flow was not reached for the conditioner and reactant. The objectives are to achieve over 75% SO_2 scrubbing efficiency, using a minimal quantity of reactant and conditioner, with the coke calciner operation conditions, and generating non-hazardous by-products, composed mainly of calcium sulfate, calcium sulfite, calcined coke dust and residual unreacted hydrated lime. The first analyses of the by-product generated showed that calcium sulfate is the main compound created by the neutralization reaction, while other SO_2 scrubbing technologies, using hydrated lime, generate calcium sulfite.

CONCLUSION

In the last years, Rio Tinto Aluminum division has developed the patent-pending [1] CHAC scrubbing process, which is the result of its R&D program on SO_2 scrubbing. This technology represents a very interesting option for SO_2 scrubbing, as this technology can be retrofitted with limited effort at low capital and operational costs.

Work performed to date has confirmed that this scrubbing process is feasible, leading to a full-scale pilot. The R&D CHAC industrial pilot, in Arvida, is providing encouraging first results regarding SO_2 scrubbing efficiency. It is therefore very important to continue the pilot work until the first half of 2016 to identify the highest efficiency achievable, the impact of different operation conditions and the composition of the by-product generated.

Once the pilot is completed, this technology will be available to scrub gases similar to those of the coke calcining process, making the CHAC technology available for retrofit or greenfield in a wide range of industries.

It could also be developed in other industries such as incinerators and coal fired boilers, where coal or petroleum coke is being used.

Acknowledgements

The authors wish to thank the Rio Tinto Arvida Smelter staff for the support provided during the CHAC proof of concept and piloting. Also, thanks to Mmes. Nassima Naas and Sandras de Moor, Messrs. Jean Crépeau and Yann Elghaoui, all from Rio Tinto Aluminum division, for the revision of this document. Finally, special thanks to Ms. Mélanie Minguy and Mr. Joseph Langlais, from the Rio Tinto Engineering and Industrialization department, respectively, for their support and advices.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ALUMINERIE DE BÉCANCOUR CONDITIONING TOWER REPLACEMENT

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Keywords: FTC, Cooling, Conditioning Tower

Abstract

There was an existing cooling tower at Alcoa Aluminerie de Bécancour that had to be replaced by a new improved type to cope with process conditions which had changed since its inception in 1985. The increased volume of Anode Bake Furnace (ABF) fumes, altered pitch burning conditions, anode recipe and the rise in elements such as vanadium and sulfur in the petroleum cokes demanded a larger modern Conditioning Tower design. Danieli Corus (DC) was contracted by the owner's engineer, Hatch Ltd, to design, supply and deliver the new tower to Aluminerie de Bécancour Incorporated (ABI).

During the design and engineering the critical task was to minimize the outage time for tie-in of the new tower to minimize anode production loss and potential fugitive emissions from the ABF. A modular approach was taken in the design whereby the components were fabricated in large sections to minimize the degree of site assembly. Fabrication and pre-assembly of the Conditioning Tower was performed offsite in a manufacturing shop situated in the Republic of China. Inspection and expediting of the components from the shop to the erection site required close cooperation of ABI, Hatch and Danieli Corus BV.

Thermal insulating, alignment and pre-commissioning activities were performed over the winter months leading to tie-in in early spring. A very smooth and safely executed tie-in procedure allowed the anode bake furnace and new Conditioning Tower to be brought online within 48 hours.

Introduction

The Aluminerie de Bécancour Smelter is majestically located along the Quebec shores of the St. Lawrence River and has achieved status in the industrial community of being energy and environmentally conscious. The smelter is fully integrated with a green anode plant, anode bake furnace, reduction potline and finishing facilities all equipped with independent air pollution control facilities. The original anode bake furnace had been installed with dry scrubbing technology to remove pollutant hydrocarbons through fume cooling, gaseous fluorides via adsorbent injection and particulate removal through filtering in a baghouse. In the primary aluminium industry, this dry scrubbing facility is commonly known as a Fume Treatment Center (FTC). Over the course of three decades, the cooling tower of the FTC experienced severe corrosion due to deteriorating equipment continuously operating in a harsh industrial environment. In the same period, the process conditions for which the cooling tower was to service, had changed significantly. The raw ABF fume had an increase in vanadium and sulfur concentrations due to change in recipe and pitch burning practices. As a result, the existing tower was in dire need of replacement with a modern Conditioning Tower to manage the increased fume volume and change in fume composition from the anode bake furnace.

Existing Conditioning Tower

During the initial startup in 1986, the cooling tower was designed to accommodate approximately 220,000 Am^3/hr of fume with a temperature of 140 °C. The tower was fabricated from mild steel, thermally insulated and was essentially an empty vessel designed to cool the fume through evaporation of water. In 1991, the smelter expanded in capacity by 50% with the addition of a new potline and a new (smaller) bake furnace. It was decided to reuse the existing FTC to treat fumes from the new furnace. Filtering capacity and induced draft fans were added to the existing FTC but the cooling tower remained unchanged. Over the years ABI would further ramp up production levels as the demand for the light metal increased; a common practice within primary aluminium industry. Hence, the original cooling tower experienced an increase in operating parameters and was eventually servicing fume volumes of approximately 310,000 Am^3/hr at temperatures of 150°C by the year 2009.

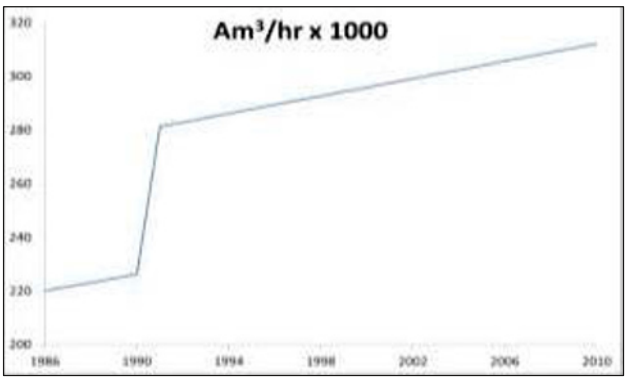

Figure 1 – Fume volume increase from original startup date

The higher gas volume and increased temperature accelerated wear and corrosion within the tower to the point where it was not performing reliably and showed signs of degradation.

Project Approach

The focus of the project execution plan was to modularize equipment to the maximum extent while satisfying all contractual requirements for safety, quality, cost and schedule. This allowed Danieli Corus to have firm control of the fabrication quality and provided an opportunity to shop inspect the tower complete with stairs, hatches and supports. Modifications and adjustments typically associated with fitting supports on site were resolved in the shop prior to shipment. The real advantage was experienced by the erection team since it reduced the logistical challenges and uncertainty typically associated with piece-small construction projects. Having a simpler installation plan increased the team's confidence in the overall erection schedule and ensured that the

Conditioning Tower would be brought online safely and to the owner's satisfaction.

Process Design of Conditioning Tower

The function of the Conditioning Tower is to continuously and consistently maintain an outlet temperature of 105 - 110°C despite the range of inlet flows, temperatures and composition indicated in the tables below.

Condition	Flow Am^3/hr	Temperature °C
Minimum	72,000	110
Normal	360,000	150
Future operation	414,000	160
Maximum heat load	360,000	220

Table 1 – Conditioning tower inlet conditions

The fume flow rate of 360,000 Am^3/hr during maximum heat load is approximately 60% greater than the original design condition of the old cooling tower.

Parameter	Concentration mg/Nm^3
Gaseous fluorides	0 - 50
Total particulates	0 - 500
Condensable Tars	0 - 150
Sulphur Dioxide, SO_2	0 - 400
Sulphur Trioxide, SO_3	0 - 25
Moisture	3 % by vol.

Table 2 – Conditioning tower inlet fume composition

The system achieves this through evaporative cooling of atomized water droplets which are blended into the fume stream via custom spray lances at the top of the tower. These are vital components because they lower the fume temperature to promote the condensation of aromatic hydrocarbons that would otherwise pass through the system in their gaseous state. Once the hydrocarbons condense, they can be removed from the system through inertial separation and filtration on alumina downstream of the Conditioning Tower. As indicated in the graph below, virtually all of the heavy hydrocarbons are condensed when the temperature is reduced to 100°C.

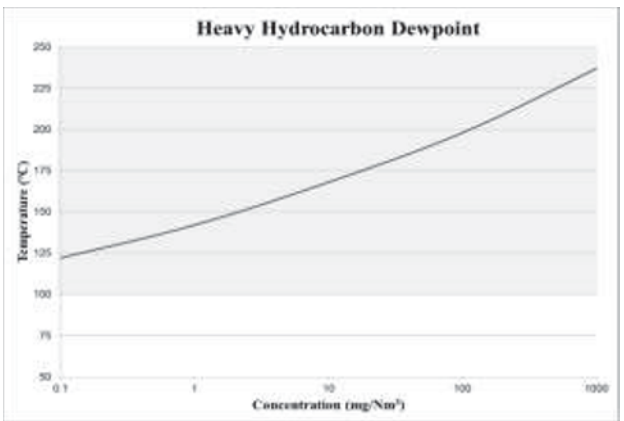

Figure 2 – Condensation Temperature of Heavy Hydrocarbons [1]

The number of lances in operation can be manually varied such that an extremely broad range of water flow can be atomized with high pressure compressed air. The heat load is dependent on the number of fire groups in operation and massive deviations are not expected in short term operation. Therefore, the number of lances in operation remains a manual process to maintain simplicity and economics. The control system monitors and maintains the desired outlet temperature to within +/- 2°C through all cycles of the ABF. Each complete spray lance was fabricated from Hastelloy body and nozzle for a robust and healthy lifespan. Each lance was equipped with a water flow switch which provided online feedback to the operator as to which lances were in operation. The switch provides online confirmation that the lance is receiving water from the valve rack. This is critical because the Conditioning Tower has to provide a balanced spray pattern for optimum performance.

The proprietary spray lances can maintain a water droplet size less than 100 microns over a wide range of fume inlet temperatures with no deterioration of water droplet size. The droplet size is maintained through a constantly regulating supply of compressed air from the valve rack. Danieli Corus prepared equipment specifications and the valve rack was provided by ABI in accordance with the functional requirement. The valve rack was supplied with fully redundant air and water racks, booster pumps, valves and instruments as required.

The Danieli Corus conditioning tower is commonly accepted in the industry as a very efficient and proven means of treating the fumes from the anode bake furnace. The tower was generously sized to allow eight (8) seconds residence time for the water droplet to completely evaporate and ensure it operates with a dry bottom. A fixed inertial separator was located in the coned discharge of the conditioning tower; a component that was not included in the original tower design in 1985. The separator was designed to remove a substantial portion of the condensed tars and particulate which would otherwise short circuit to the inlet plenum like the original tower. This component is included to ease the maintenance requirements associated with cleaning of the inlet plenum and therefore improves system online availability. The particulate separated from the fume collects at the bottom of the Conditioning Tower and discharges through a double dump airlock valve into a collection bin.

Computational Fluid Dynamic Confirmation

As added assurance of the performance, a Computational Fluid Dynamic (CFD) model of the riser duct and Conditioning Tower was performed using Fluent 14.5 software. The objective of the exercise was to confirm dry bottom operation under all inlet conditions and to check that water droplets do not collide with the internal wall. The modeler established boundary conditions, mesh sizes and convergence theories based on significant experience with evaporative cooling simulations. Using a known set of fixed parameters, the model tracked the water droplet size, the fume velocity profile and the thermal fields. Ultimately, the CFD exercise confirmed completion evaporation and no water contact with the internal walls to the satisfaction of all parties.

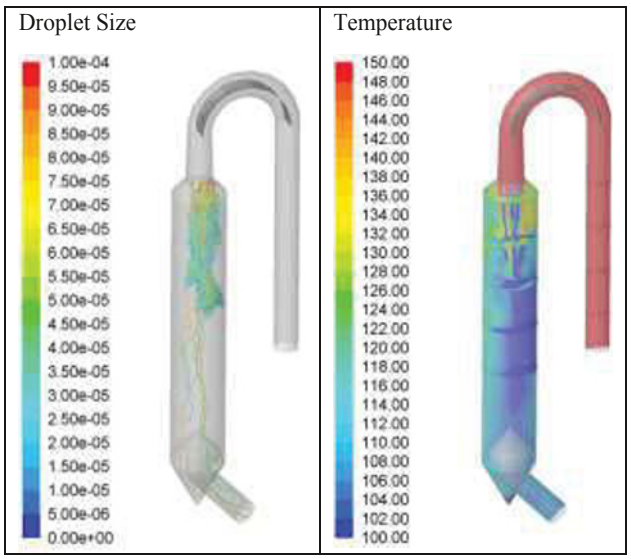

Figure 3 – CFD modelling of Conditioning Tower [2]

Deflagration Analysis

The duct design leading into the Conditioning Tower and the ducting transporting the cooled fume to the alumina injection dry scrubber form an integral part of the FTC. The high temperature of the fume upstream of the Conditioning Tower is sufficient to keep the hydrocarbons in a volatile or gaseous state. Ambient temperatures are much lower so precautions were taken to prevent undesired localized cooling of the ductwork which condenses hydrocarbons on the internal surface. Such carbon depositions are detrimental since:

- carbon accumulation in the ductwork adds resistance to the fume flow and requires additional energy to overcome
- the carbon is a fuel source which adds risk of fire or explosion
- the accumulation is additional dead load which must be accounted for in the structural design

In order to minimize the condensation, the complete system is thermally insulated and lagged including the inlet ducting, Conditioning Tower and outlet ducting to the dry scrubber. In addition to the aromatic hydrocarbons, carbon dust is also present in the solid particulate form. To prevent accumulation of carbon solids, fume velocities within the ductwork were designed to keep the carbon particles airborne and but not high enough to cause wear or unnecessary pressure drop. With all these precautions taken, the risk of fire or explosion is very rare. But due to excursions from normal practice and upset conditions in the anode bake furnace, the risk still exists in the inlet ducting to the Conditioning Tower. Therefore the inlet duct is equipped with deflagration panels which release at a pre-set overpressure. The explosion will still occur, but the free opening of the panels allows the gases to safely expand without causing damage to the equipment. The quantity and location of the deflagration panels are strategically placed and validated by the software MELCOR. This is a computer code that has been used in the nuclear industry to model the progression of accidents in nuclear reactor power plants. Due to the criticality of nuclear explosion, the software was reliably used for the propagation of overpressures in complex systems such as the FTC and applied to the ducting and Conditioning Tower at ABI.

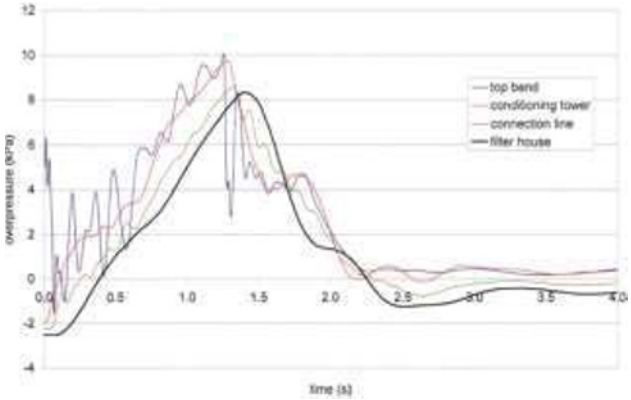

Figure 4 – Deflagration Analysis [3]

Fabrication and Transportation

Due to the critical fabrication tolerance of the Conditioning Tower, it was preferred to construct it as a single unit to eliminate the uncertainty in field alignment and welding. With the team's international experience, it was opted to construct the tower within the manufacturing shop of Changshu Xingda Machinery in the People's Republic of China. The nominal outside dimensions of the tower were 6m diameter and 42m tall and weighed in at over 100 tonnes. It was constructed of mild steel plate ranging from 10mm to 15mm thick and the internal surface was protected with a corrosion resistant coating. The spiral staircase and upper platform were pre-assembled on the tower in the shop to ensure easy fit up with gusset plates and supports. The stair stringers, treads and platform were removed from the tower and loaded into standard containers for standard shipping on container ship. The remaining seven (7) large components including the tower proper, riser duct, lobster-back inlet elbow and outlet ducting required independent transport on the vessel M.V. "Kurt Paul." [4] Over 150 tonnes of steel were loaded on 15 August 2014 and exported from China to the port of Bécancour.

Figure 5 – Sea transport of Conditioning Tower

Upon arrival in port, the components were removed from the shipping vessel by crane and loaded onto truck. Due to the size of the components, a technical method statement and risk assessment for the road transportation was prepared by the freight forwarding company JAS.

Figure 6 – Ground transport of Conditioning Tower

Site Assembly and Erection

As the large duct and vessel components were brought to site, they were strategically placed in the laydown area to allow the site team to prepare them for erection. As the cold Quebec winter was approaching, the major task for the site team was to thermally insulate and lag the components prior to lifting in place. It was a very labour intensive and time consuming assignment but one that was infinitely easier to complete at ground level versus off scaffold or elevated work platforms. Following closely behind the insulation was the installation of the piping and cable tray as the tower lay horizontal at ground level. When DC visited the site in late November for an initial inspection, some minor issues were resolved and a cursory erection philosophy was discussed with the site team. A few weeks later, some hard work and dedication allowed the Conditioning Tower to be erected onto its foundation. Despite the winter conditions, the process of lifting the conditioning tower to securing it on its foundation took less than one hour. A milestone had been completed and the tentative tie-in date was scheduled for last week of March 2015.

In the new year, the site team surveyed the site, measured and pre-assembled the riser duct sections on the ground. Work progressed on insulating and lagging the remaining components and piping was routed from the base of the Conditioning Tower to the valve rack in the new pump room. To be economically conscious, the plan was to reuse some of the existing components from the old valve rack controlling the original cooling tower. But since the old tower was to remain in operation until tie-in, some of the valve rack components were unavailable. Therefore, various components of the valve rack were assembled to ensure one complete water rack with pump and both air racks were completed prior to tie-in. The missing items would be transferred over once the new system was brought online.

The site team fit the Conditioning Tower with a spiral staircase that extended from grade to the spray lance platform 40m above grade. At Bécancour, the spray lances and its access platform were designed inside an enclosure to protect the operators from the extreme winter conditions when maintenance is required. The enclosure was fully insulated all around and was supplied with an HVAC unit to maintain a comfortable work environment at all times. The access door and hoist service doors were rated for winter conditions and installed with door seals to prevent draft. In all, the winterized enclosure ensures a very comfortable and ergonomic work station for operation and maintenance personnel.

Pre-commissioning

To ensure a smooth and efficient tie-in, DC was present three weeks prior to inspect the progress, discuss the tie-in method and perform a set of pre-commissioning activities. The installation at that time exhibited high quality workmanship in regards to alignment of piping & valves, structural welding & fastening, and architectural & painting details. Some minor air/water leakages were noted and corrected but otherwise the system was ready for pre-commissioning.

Figure 7 – Pre-commissioning of Spray Lances

The spray water system at ABI was unique since it utilized the high pressure incoming water supply in conjunction with a variable speed multi-stage pump and a regulating control valve. During pre-commissioning, the challenge was to establish the best combination to reliably provide the desired range of water flows. To achieve this, the pump was set at low speed and the commissioning team manually stepped the control valve through small increments. The flows rates and pressures were tabulated for each valve position. Then the pump speed was increased and the test repeated. Similar trials were performed with different lance configurations and the team compared and analyzed the results for the best combination.

Using recent data for the Bake Furnace operation, the commissioning team agreed to set up the conditioning tower with four (4) spray lances and fixed pump speed of 66%. The complete set of performance charts were handed over for operator reference.

Tie-in

The mandate from ABI was to minimize the tie-in duration since it would require the fume treatment center and anode bake furnace to go offline. In addition, having the furnace unheated for extended periods would allow it to cool. Restarting the chilled furnace could cause disruption in anode quality and spikes in hydrocarbon emissions. Therefore, due to environmental legislation and process risk in the ABF, the tie-in duration was not to exceed forty-eight (48) hours.

As the team prepared for the tie-in, ABI ramped up production on the bake furnace to increase their inventory of baked anodes. During the outage the surplus of baked anodes would be consumed normally within the potroom ensuring that aluminium production was uninterrupted. The site team diligently planned a safe and efficient tie-in schedule which included an extensive list of pre-work to minimize the tasks that had to be performed during shutdown. To ensure minimal downtime and a fluent tie-in, the site

team confirmed all existing components against their final location. All loose material such as duct supports, bracing and expansion joints were accounted for and moved to a location for easy access. Existing insulation and lagging was removed nearby the tie-in locations and where possible some of the existing fasteners were removed from the flanges on the riser duct and inlet plenum. Scaffolding was erected below the existing cooling tower as the entire cone was to be cut out to allow installation of the new ducting. Finally all areas were cleared of potential congestion and the tie-in was to proceed as scheduled on March 24th.

During the tie-in, the existing inlet ducting was removed and the new riser duct with lobster back was installed in single lift. The alignment was near perfect and one crew immediately bolted the bottom flange while a second crew began welding out the top splice connection. Once the load could safely be removed, the crane was no longer required and packed up. The inlet plenum and expansion joint connection was performed at grade with small wheeled crane. The flawless tie-in was performed safely in less than forty hours and the bake furnace was brought back online early morning of March 26th.

Hot Commissioning

It was the end of March but cold wintery conditions were still present during hot commissioning. Shortly after the fires were restarted on the ABF in the early morning, the fume flow was directed to the FTC with all eyes on the performance of the Conditioning Tower. Ironically the excitement subsided because the inlet fume temperatures were very low and the enormous heat sink of the tower absorbed whatever heat came in. As the temperatures slowly rose throughout the morning, to everyone's surprise water began pouring out the bottom of the tower at an alarming rate. The spray water pumps were confirmed off and a check at the spray lance platform confirmed that no water was entering the tower. It was a jolt of comic relief when it was realized that the water was generated from melting ice in the tower. Since the tower was erected in early winter and stood with an open top for almost four months, snow and ice were allowed to accumulate in the bottom.

As the temperature rose throughout the morning, the amount of water from the bottom slowly dwindled to a stop. Nearly half a day later, the fume temperature from the Conditioning Tower reached a point where water injection was required and the control software was trialled. After some fine tuning of the control logic, the FTC was left to operate overnight with the commissioning team on call for the operators. The following morning the tower was inspected and a large amount of dry carbon granulate was discharging from the dump valve – a sign of a very healthy conditioning tower. The addition of the inertial separator in the bottom cone of the tower was new to the operators and they were not expecting that degree of removal.

Once the system was stabilized, the unused spray lances were removed from the process to prevent carbon and soot accumulation in and around the nozzle. The temperature control loop was monitored and tuned as the operators and technical staff became familiar with the operation.

Conclusions

When the project began, the team was challenged with limiting the outage time for tie-in of the new tower to minimize anode production loss and potential fugitive emissions from the ABF. This was achieved through modularization of the large cooling tower and a safe thorough installation and tie-in strategy. In less than forty-eight (48) hours, the team was able to remove the existing equipment and install 40 tonnes of new ducting almost 50 m in length. There was barely a hiccup in the bake furnace operation and fugitive emissions were held to a minimum to the satisfaction of all parties involved. After more than 15,000 working hours there was no reported injury.

Figure 8 – Conditioning Tower alongside existing tower

The conscientious approach to the project kept the costs in check by re-using existing equipment where possible and the site team was able to make progress with the installation throughout the Quebec winter to meet the March startup date. Careful preparation and planning allowed for a smooth transition from the old cooling tower to the new Conditioning Tower measuring almost double in size. The performance stability of the new tower has been recognized by ABI and all parties are very satisfied with the installation.

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TMS (The Minerals, Metals & Materials Society), 2016

Alumina Handling in the Smelter – From Port to Pot

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Keywords: Segregation, Alumina, Storage, Conveying

Abstract

Alumina is the main raw material for primary aluminium production and the smelters rely on continuous, steady streams of alumina to the pots. At the pot, variations in the alumina quality shall be minimized. Fluoride content, particle size distribution, attrition, segregation, powder flowability and impurities all are important parameters that are affected and partly controlled by the alumina handling and treatments including the alumina silo storage, transport, screening and gas treatment /alumina enrichment systems upstream the pots.

The status of alumina handling and research will be reviewed, and new ideas will be presented based on Just-in-time principles, close review of redundancy requirements, storage capacity and how to minimize quality variations all the way from the port to the pot.

Introduction

The flow of alumina from port to pot starts typically with a ship unloading system that fills a large storage silo situated on the docks. After several intermediate transports and process steps, the alumina ends its journey saturated with HF and finally dissolves into the cryolite bath inside the pots.

The requirements of the transport system are many, but we consider the overall parameters to be the most important.

- Availability of alumina must be 100% (i.e. there is always alumina available for the pot)
- Keep alumina quality through the whole line
  - Homogeneity
  - Attrition
- Use as little energy as possible (OPEX/Environment)
- As low as possible installation cost (CAPEX/Environment)
- Recover and distribute fluoride 100% and as evenly as possible

Redundancy, reliability and variations in the alumina quality are the main elements that need to be considered in the evaluation of the alumina handling systems. Several of the design and redundancy requirements for the alumina handling and treatment systems were set in the past and in particular the alumina silo storage requirements will be revisited in order to understand if this can be optimized with modern technology available today.

Storage silos for primary alumina are required especially at the port to receive the large volumes transported by ships (or in some cases by e.g. trains), and have to be sized considering also the reliability of the sourcing and shipping to the storage silo. Additionally silos are needed in the alumina handling and transport system to ensure that alumina are always available at the pot during maintenance and unexpected failures of the alumina handling equipment. It is assumed that for a total abnormal failure situation any quality of alumina whether primary or secondary is acceptable at the pot even if this requires continues AlF3 adjustment during such emergencies. However for normal variations in the flow or for smaller maintenance stop up to 16 hrs, secondary alumina should be available.

Minimizing the sizes and/or eliminating silos obviously reduce the cost of inventory and equipment, and in many cases also the operating cost. In addition the silos typically generate air current segregation that generates much of the undesired alumina fines variations. Without proper exercising of the silos, the quality of the alumina may deteriorate to the degree that it is totally useless for the pots. It is therefore a clear objective of this paper to challenge the present day silo sizing in order to optimize both cost and alumina variations with even improved reliability.

The smelters requirement of alumina

In the pot, alumina is dissolved into a bath of cryolite, and through electrolysis, aluminium is produced. In order to control the electrolysis process the pot requires a narrow window of operating parameters including minimal variations in HF saturation, alumina dissolution properties including water content, particle size, etc. Many of these properties are highly affected by the alumina handling upstream the pot, and directly or indirectly the pot will be affected by the following parameters:

Smelter grade alumina (SGA) is a sandy free flowing alumina made to have the right characteristics for use in the GTC and the pot. In old days, a floury type of alumina was also used, but this now generally not used. Ose and Dyrøy (1) investigated the large variations of SGA. Alumina from different refineries exhibit different properties, though they may still be within the SGA specification. This is well known to any smelter professional.

Fluoride homogeneity. Increasing amperage and improvements in Current Efficiency means pots are operating in a narrower operating mode. Small variations of fluoride content in alumina to pot thus become highly important. The main contributor to fluoride variations is the HF evolution at the pot caused by alumina feeding and anode change.

Particle Size. The typical size range is 50-200 µm, though there is always a certain content of fines typically 8-12%, defined as the portion below 45µm (alternatively the portion below 20 or 10 µm). The fines content is probably one of the most important parameters that affect both the alumina handling and the pot operation.

Surface Area. A large surface area is important for use as adsorbent in the GTC. Typical value is 60-80 m2/g.

Flowability. Flowability is closely linked to particle size. Alumina with a high content of fines will not flow easily.

Fluidization. Much handling equipment is based on fluidization. Alumina is manufactured to allow for easy fluidization. According to the Geldart (2) classification, alumina is class A, (Aeratable). Fluidization is closely linked to particle size. However, alumina with a high content of fines is classified as class C – Cohesive – material and will have poor fluidization properties and low transportability in e.g. airslides.

Erosion. Alumina is a highly erosive material, and the handling equipment must be constructed with this in mind.

Attrition. Attrition (unwanted breakage) of alumina particles is undesirable as it increases the content of fines. There has been a lot of interest for attrition, and it is well known that different aluminas have different attrition index. Attrition (and attrition index) can be measured in the laboratory; however, great care must be taken when trying to determine attrition on site. Due to the easy segregating nature of alumina, relying on measurements of particle size to determine attrition in equipment is difficult. Carefully planned sampling programmes are needed to ensure that the measured "attrition" (i.e. increase in fines content) is not variations due to segregation.

Trace elements. Impurities elements are concentrated in the fine fraction of alumina. This is especially pronounced in enriched alumina. This was shown by Sturm and Wedde (3) with the development of an impurity stripper. Later Gaertner (4) studied pot gas and found that super fines (<1µm) contained mainly quenched bath fumes (NaAlF4) and that micron sized particles contained most impurities.

Segregation. Segregation is not a property, but is often discussed as one. In a broad sense, segregation is separation due to different properties. In alumina handling it is most commonly used to describe the (unwanted) process where fine particles (dust) are separated from the main bulk of alumina. Alumina with a high or varying content of fines is undesirable to both the smelting process and the alumina handling equipment.

Handling operations and their influence on alumina quality

From the port to the pot, there is a long logistics chain with many storage, transport and (one) process steps (GTC) for alumina. Reliable transport and maintaining alumina properties is important when the logistic chain shall be designed, to avoid segregation and increase homogeneity.

The volumes are large: approximately 2 tons of alumina is needed to produce 1 ton aluminium. A smelter producing 500.000 tons of primary aluminium per year will then need 1 million ton alumina, corresponding to transporting more than 110.000 kg per hour.

Dyrøy (5) investigated alumina handling, especially segregation, in several smelters. He clearly showed that most handling processes in the smelter gave a negative quality variation in the form of segregation. He also clearly showed how anode effects can be linked to alumina segregation.

Figure 1 shows homogenising factors for various alumina handling operations. The numbers are from (5). Summary:

- A GTC homogenises alumina. This is due to the long residence time with active recirculation and some agglomeration of fines which mix alumina.
- Transport reduces homogeneity slightly. Material enters and leaves the transport equipment with a time lag that can be controlled.
- Storage units (i.e. silos) reduce homogeneity considerably. This is a result of uncontrollable time lag, segregation in the silos, funnel flow silos and operation practices.

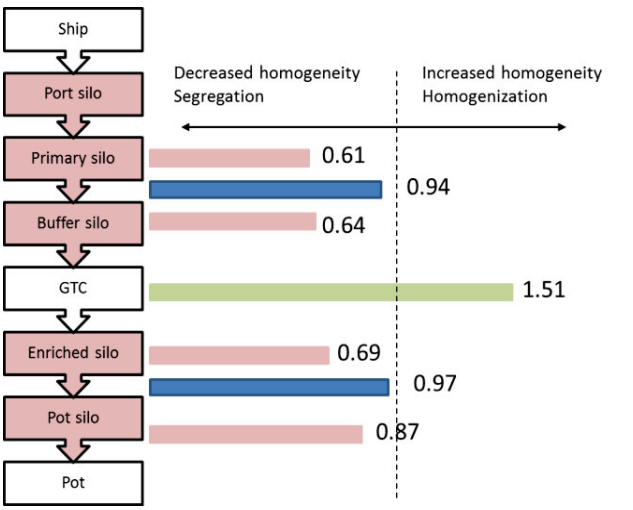

Figure 1 A typical logistic chain for alumina in the smelter and homogenisation factors of various operations in a smelter (numbers from (5)).

Silo operations

The flow pattern in a typical alumina silo is funnel flow. Alumina flows only in a channel, with stagnant material outside. This is a result of the large size and flat bottom silo.

The typical operation pattern of a silo in GTC is to keep the level at 90-95% in order to provide a backup supply of material in case equipment failure. This operation method results in considerable segregation in the silo and large particle size variations when the silos are emptied (high concentration of fines). The operation of GTC and various transport equipment will suffer from this. Such alumina from the secondary silo is high in impurities and fluorides. The pots can suffer from this. Also worth to mention is that secondary alumina kept in a silo can solidify just by air humidity.

The preferred strategy is to empty the silos regularly (exercising) and at least enough to renew the entire cross section surface of the alumina as part of the operation (See Figure 2). Accumulated fines are in this way removed from the silo periodically.

The necessary silo level reduction to renew the surface can be calculated using simple trigonometry. The location of the level transmitter should also be part of the calculation. Typical angle of repose is 35-40° for alumina.

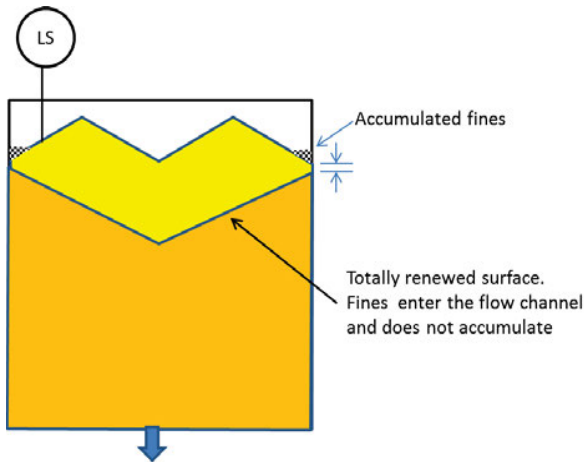

Figure 2. To avoid excessive build-up of fines, silos should be emptied regularly. By renewing the total surface area of alumina (as indicated), fines will not accumulate in the silo.

Power consumption. All transport requires power. Pneumatic conveying is the most power consuming transport process, airlifts require slightly less as the air comes from a blower. Airslides require even less, as the air is produced by a fan. Mechanical transport such as belt transport and bucket elevators requires low power per kg transported. We recommend energy cautious equipment which will be lenient with alumina (Low attrition).

Table: Typical energy ratios of various transport equipments	
Mechanical	1
Airslides	1.3
Hybrids	6.5
Pneumatic /dense	20-35
Pneumatic dilute	N/A

Redundant blowers and fans are generally supplied. If motors fail, a spare motor is kept in store for critical equipment.

Venting

Most equipment in the supply chain can require venting of excess air. This includes airslides, airlifts, pneumatic conveying and bucket elevators. The sizing is the main difference and can be challenging.

The vent volume from airslides is low but constant. Air lifts include more air, but is also fairly constant. Pneumatic conveying systems (e.g. truck unloading systems) put the highest strain on venting systems. The volume flow is highly varying and the dust load high. The design can thus be hard. Use of silo filters is one option, but can result in very large silo filters, and fines accumulation in the silo.

Where vent gas should be routed is often a difficult choice, and the route used is often the shortest possible. Routing to/through a silo is sometimes done, but can give accumulation of fines. If the dust content is acceptable, venting to the main gas duct is often considered best.

Potfeed systems are typically vented to the gas duct. Systems with balancing columns need to be carefully designed to not send fines into the gas duct. To counter this, Alstom's Alfeed system includes a cyclone for the ventilation air from the airslide. The cyclone separates most dust and sends it to the pot, instead of the GTC.

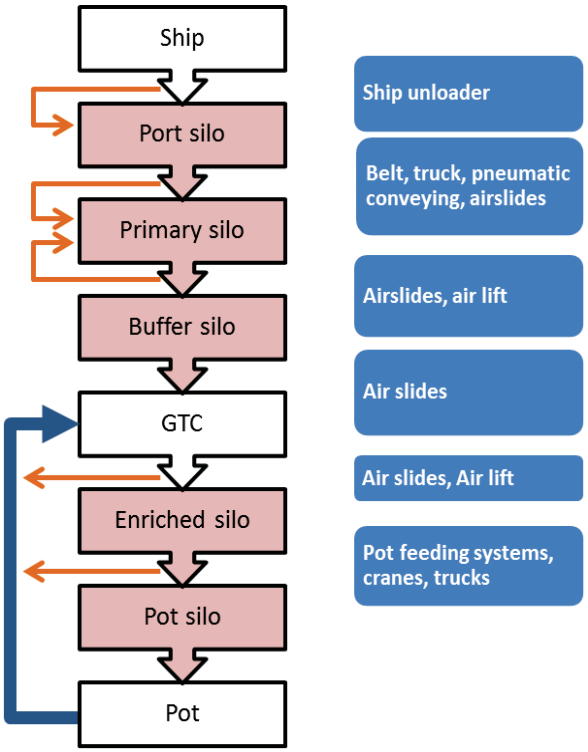

Figure 3 : Overview of handling operations in a smelter. Red areas show storage. The thick blue line indicates gas from pot to GTC. Orange lines indicates ventilation routes. Blue boxes indicate typical transport equipment from one step to the next.

Alumina Operations in the Logistics Chain.

Figure 3 shows the operations in the alumina and typical associated equipment.

Ship loading and unloading. Alumina refineries are often located far from the smelter, and ships are used to transport alumina to the smelter. Behrens (6) studied the quality impact of filling and unloading a ship. He found that loading and unloading homogenises alumina. The likely reason for this is frequent moves of the filling and unloading points to protect the ship. Filling and emptying in just one point can break the ship's hull. As alumina is highly hygroscopic, the hatch covers must be tight to prevent water from entering with resulting solidified alumina. One extra unloader is often used to allow faster unloading of a ship and redundancy in case of breakdown.

Port silo. The port silos should be large enough to be able to empty one ship, although at times, a ship can carry on half full to another smelter. Typical size is therefore 20.000 to 200.000 tons. The silos are large, typically 30-50 meter in diameter and height. Due to large volumes, air current segregation arises and with

funnel flow silos the material out will have size variations. This induced variation cannot be removed unless upstream process and storage units act as homogenisers.

Newer bulk ships are often of Panamax class and carries 60.000 ton (or m3) of alumina. Alumina to smaller and older smelters is often delivered in smaller ships of 20-30.000 ton.

Transport to Primary Silo. Transport from port to primary silo is done in a large variety. Trucks, belt conveyors, air slides and pneumatic conveying are all used.

- Discharge equipment can break down
- Filling system of primary silo
- Utilities such as compressors

- Belt conveyors are used at the some smelters and have large transport capacity and little segregation.
- Trucks give the best redundancy, as a truck is easily replaced.
- Pneumatic conveying has high operation costs and high potential for attrition and segregation at the exit silo.
- Air slides does not transport vertically and are best suited where a natural slope can be used.

Most smelters have a redundant strategy for transporting alumina, most often with trucks and pneumatic truck unloading systems. This would require a minimum available volume in the receiving silo of one truck free storage (30-50 ton).

Primary Silo. The required primary silo size is based on the time allowable for repair. In order to remedy a breakdown in one or more of the preceding systems there need to be some storage close to the GTC, most often 1-3000 ton, corresponding to 24-72+ hours consumption of alumina (Different smelters have different requirements). With appropriate emergency strategies and redundant equipment the long buffer can be challenged. Principally down to the size of one truck.

Transport to and in the GTC. Some GTC include a small buffer silo before feeding to GTC. This transport is normally by airslides and airlifts. The equipment most likely to fail is the airlift.

The buffer silo is mostly found in older GTC's, and was used due to bulky feed arrangements in older GTC's that would have required lifting the primary silo considerably. The size is normally 30-60 ton.

Feed to GTC is using rotary feeders, vibrating screens and airslides. Redundant equipment is installed in case something breaks down. Equipment that fails most often is vibrating screen. Airslides require cleaning at regular intervals.

GTC. GTC's are very efficient in capturing HF from the pot, but the content of HF and dust with fluoride compounds varies during the day and the operation cycle of the pots. To increase the homogeneity of fluoride, the number of re-circulations internally in the GTC is kept as high as possible. In this way alumina saturation is kept high. Varying dust content in raw gas will regardless of this operation result in variation in fluoride content. Storage and segregation in the secondary silo will increase the variation.

In the GTC, primary alumina is used to capture hydrogen fluoride from the gas and the fluorinated alumina is then sent onwards to the cell. The alumina consumption equals the consumption in the cells, and alumina is recycled in loop many times before it leaves the GTC. In this way, HF content of alumina is maximised and with low variations.

Due to the residence time and the recirculation, a GTC homogenises alumina as shown in Figure 3.

In the GTC fines from two distinct different sources are mixed: 1) Fines in the primary alumina and 2) dust from the pot (fumes, bath particles etc). Dust from the pot is the main source of regeneration of HF in dust cake (7). On the other hand, fines from the primary alumina feedstock can be beneficial for HF removal.

Let's consider typical dust content in the raw gas from 400-800 mg/Nm3 and gas volumes typically 10.000 Nm3/hr. For a section of 30 pots, this gives a dust load to the GTC of 120-240 kg/hr from this section.

If we further assume 200 kg/hr/pot alumina with a fines content of 8%, this is a fines load of 480 kg/hr. If there is no agglomeration in the GTC, this would mean an increased in fines content of 25-50%. However, the fines content does typically not increase significantly in a modern GTC, which indicates that agglomeration has taken place.

A GTC includes a number of back up equipment to facilitate repairs and service when necessary.

Transport from GTC to Enriched Silo. Alumina is transported to the enriched alumina silo using airslides and air lifts. Air lifts for enriched alumina often experience scaling of the transport pipe. The pipe is therefore regularly cleaned.

Enriched Silo. The enriched alumina silo (also called secondary or charged) is used to store enriched alumina before it is fed to the pot.

Transport to pot. In old GTC's cranes or vehicles are used to feed the pot silos. Modern smelters use potfeed system (of which there are many suppliers). Older potfeed systems were based on pneumatic conveying, but they often scaled up and due to the pressure pulse of air also gave higher HF emissions. Modern pot feed systems use fluidization to transport alumina. Modern smelters rely totally on the pot feed system. Pots are not designed to be fed by cranes nor are cranes designed with this in mind. A potfeed system must therefore be extremely reliable and require minimal maintenance.

Pot silo. Most smelters technologies are based on pot silos inside the pot. Hydro Aluminium has small wall mounted mass flow silos for each pot.

Proposals for improving the alumina handling system

Theoretically, the ideal handling system would be without storage, just a long constant feeding stream of alumina to the pots. Given the large volumes, the large number of pots, the long distance, 24/365 operation, possibility of equipment failure and more, this is not recommended.

Historical redundancy requirements are not necessarily optimal for today's modern equipment. The main elements to be considered are:

- Reduce number of handling operations
- Reduce number and volume of silos
- Reduce number of satellite storages
- Use mass flow silos where possible
- Use anti-segregation measures where possible
- Storage units should work as mixing units where possible

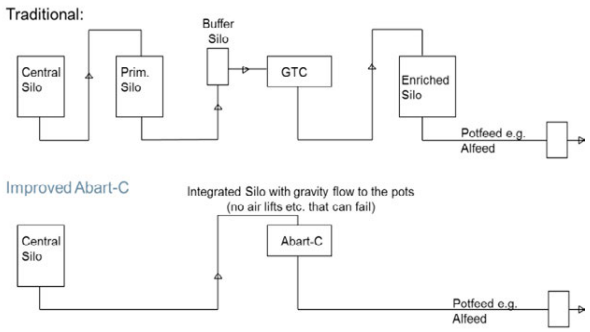

Figure 4 : Traditional alumina handling layout and the proposed Abart-C solution.

Figure 4 shows both the traditional layout and the proposed Abart-C solution. This results in minimal segregation in the handling loop and the number of handling operations where equipment can break down is considerably reduced.

Reduced size of the storage equipment gives also an opportunity to improve bulk solids handling design (e.g. design silos for mass flow).

Figure 5 shows the Alstom Abart-C in more detail and how it integrates the proposed solution. By integrating and lifting a small silo in the structure, gravity flow can be used all the way to the pot avoiding additional lifting operations. The enriched silo is removed, and the potfeed system is fed directly from the filter. A buffer storage is used to smooth variations in alumina flow from the GTC to the potfeed system. The removal of several silos and transport and lifting equipment enhances the reliability of the system.

Alstom's Abart-C concept includes an integrated silo as part of the filter compartment to minimize the space used in the courtyard. Each filter compartment has one silo fed in two points. Each filter compartment has its own feed from the silo. This reduces alumina quality variations. The integrated silo contains 30-48 hours of alumina, depending on the alumina consumption. Less transport and storage operations also increase the reliability of the total system.

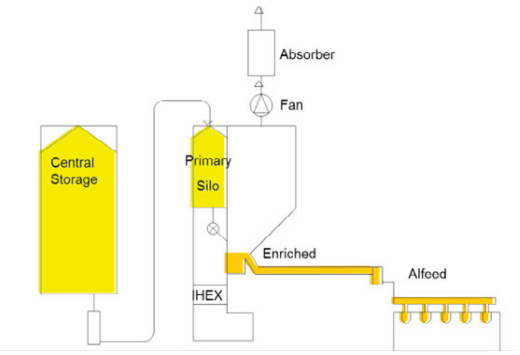

Figure 5 : Abart-C concept feeding directly to potfeed systems.

New developments and old technologies

Mass flow silos.

The (relatively) large size of alumina silos have meant they are not designed as mass flow silos. Typical mass flow silos have very steep hopper walls. With typical alumina volumes, this would mean unreasonably tall silos. Instead alumina silos are often flat bottomed, with fluidization systems to ensure flow, although some silos are built with a cone.

Use of inserts in the silo cone can change a silo flow pattern from funnel flow to mass flow. Inserts can also be used to increase mixing in a silo. Although inserts has been known for 20+ years it has (to our knowledge) not been used in smelters. Use of inverted cones, cone-in-cone and double cone means that steep hopper walls are not needed.

Enriched alumina silo

A large storage silo for enriched alumina is a remnant from the days where pots were fed with cranes etc. It was then necessary to install a large silo that could be fed continuously and emptied occasionally (to the crane).

New smelters are built with pot feed systems to minimize the manpower requirement, and the need for an enriched alumina silo is greatly reduced. We propose that the potfeed system could be fed directly from the GTC.

Pot feeding systems

We advocate to use the potfeeding system as the main storage of enriched alumina. In this way, the homogenisation of alumina in the GTC remains, and depending on the pot feeding technology may also increase further. The available storage volume in potfeed systems is in the order of some hours.

Alstom's Alfeed system (8) includes a cyclone to not send fines back to the GTC but to the pot.

Hydro's pot technology includes a small mass flow silo on the pot room wall and the Hydro pot air slide. A mass flow silo works on the first in – first out principle. There is therefore no accumulation of fines and the alumina quality is kept constant. This silo can be fed with several technologies, including Alstom's Alfeed.

Vertical transport

In old smelters, bucket elevators were frequently used. The last decades air lifts has been the standard lifting equipment in smelters. There has been a considerable development in bucket elevator technology since they were abandoned for air lifts. Rodriguez (9) studied air lifts vs. bucket elevators in his M.Sc. thesis. He found that modern bucket elevators use much less energy than air lifts.

Bucket elevators have many benefits compared with airlift. There is no scaling (secondary alumina), no erosion (primary alumina), they are more forgiving with regards to alumina bulk properties and have reduced power consumption. At Hydro's Årdal smelter, bucket elevators have recently been commissioned.

Conclusions

Alumina handling has been reviewed from port to pot. It has been shown that it is possible to simplify the entire logistic chain of alumina in the smelter.

Air current generated segregation in the silos has been identified as one of the main causes of variation in logistic supply chain. Mixing and recirculation in the GTC improves homogeneity of alumina.

Eliminating and reducing the silos therefore not only reduce the inventory and CAPEX costs, but the simplification also results in more homogeneous alumina to the pot as a result of less segregation.

Furthermore the power consumption is reduced due to less handling operations.

Redundancy requirements have been studied and gravity flow from an integrated silo directly to the pots have been proposed.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Recent Developments in Hyper-Dense Phase Alumina Handling Systems

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Keywords: Alumina handling, fluidized airconveyor.

Abstract

The Hyper-Dense Phase System (HDPS™) was developed in the 1980's by Aluminium Pechiney (AP) to provide its new centre-fed pots with an automatic and cost-effective alumina feeding system. This fluidized conveyor ensures the most stringent technical requirements in terms of alumina quality and reliability are met. Over the years, it has thus contributed to AP Technology's technical leadership, though it is no longer AP specific. Until recently, it was limited to horizontal conveying, which, in some cases, restricted its potential applications. A major R&D program, which spanned over more than 10 years, was initiated to overcome this limitation and has led to the development of the patented HDPS™ siphon. The successful start-up of the siphons installed at the Rio Tinto (RT) Kitimat smelter concluded this R&D phase. This paper will present this development work and the new potential of the HDPS™ system designed by Alesa Technologies.

Introduction

Aluminium Pechiney developed the HPDS™ system in the 1980's in order to provide its new centre-fed pots with an automatic and cost-effective alumina feeding system. The intent was to address the main limitations of the standard of the time, which was based on a discontinuous feeding of the pot hoppers with an overhead crane. In addition to being economically attractive, the continuous-feeding option allows for a safer pot operation – pot hoppers being always full – and a cleaner working environment, with minimal dust emissions.

The Saint-Jean-de-Maurienne AP30 potline was the first smelter equipped with the HDPS™ in 1985. Since then, more than 25 smelters have been equipped, with some in brownfield conditions. In 2007, Alesa Technologies celebrated 100km of installed HDPS™ and has then pursued an ambitious R&D program which aimed at continuously reducing its total cost as well as expanding its potential applications. The following paragraphs briefly present the main concepts behind the system, before presenting the latest development from this R&D program – the patented hyper dense siphon [1].

Principle of Hyper-Dense Phase Alumina Handling Systems

The HDPS™ is based on the ability of some powders with specific characteristics to become "fluidized", a physical state where an ascending flow of gas (usually air) is used to minimize the adherence between particles and thus improve its flowability. The airslide is the most common handling equipment based on the fluidization principle. It consists of two chambers separated by a porous fluidization fabric. Low pressure air is injected in the lower chamber and flows through the fabric to fluidize the upper chamber where the powder is. The airslide is inclined by a few degrees (typically 5-10%) and, with the bed being in a fluid-like state, the material flows in the direction of the incline. Fluidization air is degassed at regular intervals. The two chambers are sized in order to maintain both the product and the air velocities within the acceptable range (a few m/s, depending on the product to handle). In this configuration, the mass flow rate of product needs to be controlled upstream, typically with the help of a mechanical device (such as a rotating valve).

The HDPS™ was designed to overcome the airslide limitations in a potfeed application. Similarly, it has two chambers, the upper one being full of fluidized material, as represented in Figure 1. The fluidization air is collected at the top of the upper channel, creating a layer of air of a few centimeters (the "bubble"). Contrary to the airslide, a hyper-dense phase conveyor is a closed-conduit flow system; in this configuration the driving force for the flow is the product hydrostatic pressure upstream of the conveyor. This means the conveyor can be maintained horizontal, facilitating its integration into the potroom environment and minimizing the investment otherwise induced on the building structure by the airslide slope. In a HDPS™ airconveyor, the material flows at a very low velocity (a few cm/s), ensuring no scaling or attrition occurs. It does not require any mechanical device to control the flow, which is automatically adjusted to the pot consumption.

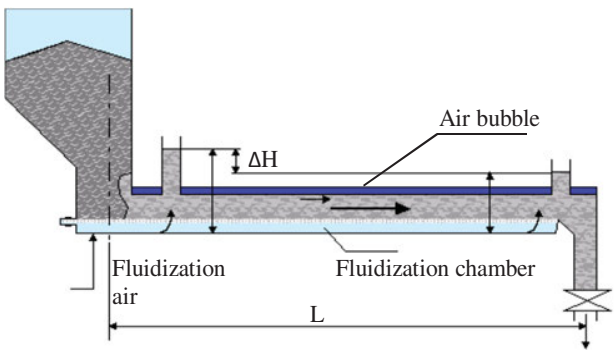

Figure 1 – HDPS™ principle

In industrial conditions, from the silo outlet, the fluorinated alumina circuit consists first of an airconveyor from the silo to the main distribution box and then a second longitudinal airconveyor feeding alumina to each pot through an airslide. The air required to fluidize the material is produced in a centralized way by dedicated fans and degassed into the Gas Treatment Centre (GTC) ductwork through specifically designed degassing columns. These columns aim at reducing the quantity of dust carried along with the fluidization air to the GTC. The overall process is shown in Figure 2.

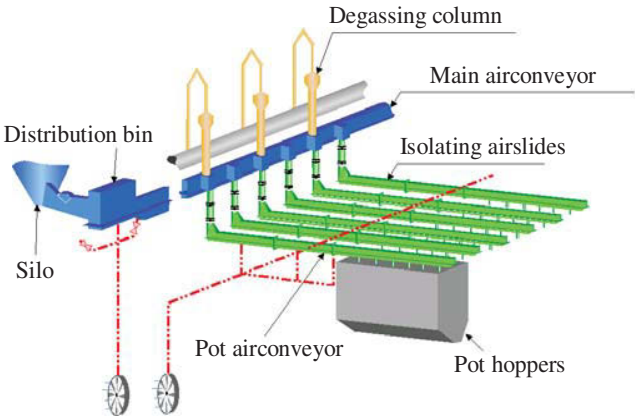

Figure 2 – Fluorinated alumina supply to the pots using a HDPS™ system

Current limitation of the HDPS™ – Siphon objective

For potfeed applications, the HDPS™ system is perfectly adapted to typical greenfield contexts. On the other hand, the historical design was limited to horizontal configurations in straight lines, which prevented its application to more specific layouts where obstacles (building, road, potline…) are present. To be able to address this limitation, Alesa Technologies investigated the design of a system which could couple horizontal and vertical hyper-dense phase fluidized transport sections, a so-called "siphon", as illustrated in Figure 3. Research work at laboratory scale was first conducted in collaboration with the Toulouse University INP-ENSIACET-LGC Unit (France) as part of a PhD. Thesis (G. Turzo, 2013) [2]. It was completed by a pilot testing phase at the Alesa Technologies premises in Gardanne (France).

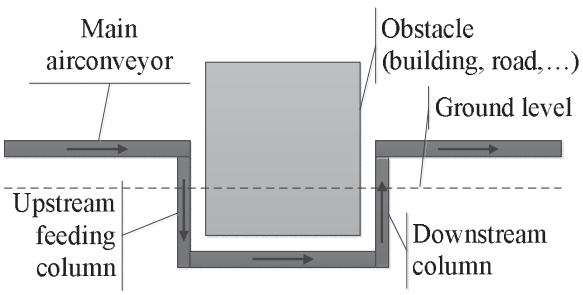

Figure 3 – Hyper Dense Phase Siphon principle

Technical challenges

The main technical challenge associated with this development relates to the mastering of the dense phase fluidized vertical flows which characterize both the upstream and downstream columns. In the upstream vertical column, the material flows downwardly from the top main airconveyor. It is, however, difficult to predict the direction of the fluidization air, which can flow either upwardly or downwardly. It is the relative velocity of the air (U_g) in relation to the particles (U_p) which will condition the local state of the suspension ($U_{sl}=U_g-U_p$; sliding velocity), and thus the capacity of the material to flow without disruption. Earlier experimental work [3] demonstrated that the increase of pressure with depth in the upstream column can result in a local reduction of the air velocity, which in turns leads to a local de-fluidization of the powder (when $U_{sl}<U_{mf}$, where U_{mf}=minimum fluidization velocity) and ultimately a blockage of the flow if the column is too tall. Specific fluidization techniques therefore have to be implemented to overcome this issue.

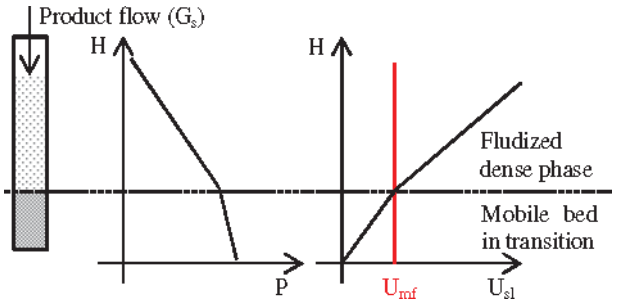

Figure 4 – Typical pressure (P) and sliding velocity (U_{sl}) profiles in a feeding column

In the downstream column, the material flows upwardly; the fluidization air and the powder, therefore, flow co-currently. The air pressure relief along the flow assists the transport of solid material thanks to the particle entrainment in the bubbles' wakes. As has been shown as part of this study, the design of the column fluidization system can stabilize the material flow. It also has to ensure a correct gradient of density between the two columns to force the product to flow in the desired direction.

Laboratory experiments

Laboratory studies were aimed at understanding fluidized dense phase vertical flows. The first part dealt with the hydrodynamic study aimed at determining the relevant physical characteristics from a typical metallurgical alumina. As a second step, laboratory-scale HDPS™ siphons were built and used to determine the key design and process characteristics.

Characterization of the alumina hydrodynamic properties

The characterization of the hydrodynamic properties was conducted with the use of a standard fluidization column (2m. high, 200mm diameter, corresponding to a surface area A_c=314cm^2) fitted with a porous fluidization fabric at the bottom, as represented in Figure 5. The column is equipped with parietal differential pressure transmitters; it is also possible to measure the pressure within the fluidized column at different heights, as well as the fluidization air flow and pressure.

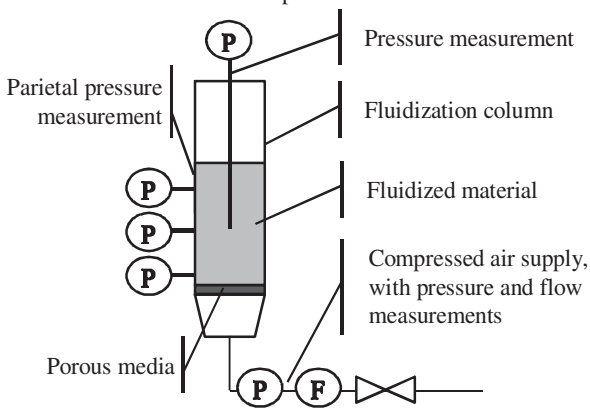

Figure 5 – Fluidization test

For a bed of fluidizable particles, the state of the air / particles suspension will depend of the fluidization velocity, which is the ratio between the fluidization air flow and the fluidization pad surface area:

- For low fluidization velocities, the air will flow through the empty spaces between the particles and the bed therefore remains in a fixed state;
- As velocity is increased, it will reach the minimum fluidization velocity (U_{mf}). At this point, the aerodynamic drag from the air counteracts the gravitational forces and the bed will start to expand. The bed is then in a fluid-like state, and the fluidization is homogeneous.
- As velocity is further increased to the minimum bubbling velocity (U_{mb}), bubbles will appear in the bed of particles and different behavior can be observed, depending of the particle physical characteristics. At some point, the bed will consist of small elements of both gaseous and solid phases that move very rapidly; this is called the turbulent fluidization regime.
- Higher velocities (>U_{ent}) will result in the particles being conveyed along with the air flow; the bed is therefore no longer in a fluidized state.

Figure 6 represents the pressure drop across the bed of particles versus the fluidization velocity, for a given mass of material (m_p= 16kg). It shows that the pressure drop increases continuously until the minimum fluidization velocity is reached, before plateauing. This plateau corresponds to the practical range of velocities for fluidized handling systems applications. Thanks to the Richardson method [4], it is then possible to measure the minimum fluidization velocity U_{mf} – in this case 5 mm.s^{-1}. The minimum bubbling velocity (U_{mb}) is determined by plotting the standard deviation of the temporal evolution of the different parietal pressures versus the fluidization velocity; a step-change can be observed when U_{mb} is achieved (9 mm.s^{-1}), which is due to the development of heterogeneities within the bed. Visually, it also corresponds to the value above which gas pockets can be observed in the fluidized bed.

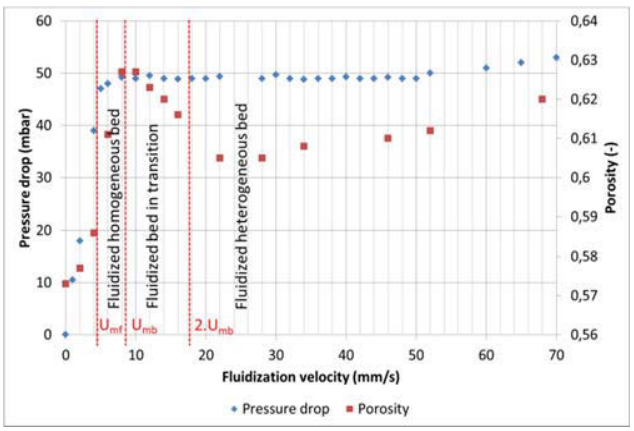

Figure 6 – Pressure drop and porosity evolution across a bed of metallurgical alumina (m_p=16kg)

The global porosity ε_g of the fluidized bed can be evaluated by measuring the bed height (H), knowing that $\varepsilon_g = 1 - \frac{m_p}{\rho_p . A_c . H}$, with ρ_p corresponding to the alumina particles density (2360kg.m^{-3}). It is a measure of the void spaces of the fluidized material $\left(\varepsilon = \frac{Volume\ of\ voids}{Total\ volume\ of\ the\ fluidized\ bed}\right)$ and its evolution is plotted on the same graph. It allows for determining the porosity at the minimum fluidization (ε_{mf}=0.58) and bubbling (ε_{mb}=0.63) velocities. Knowing how porosity, and therefore density, changes with fluidization velocity allows for determining the correct fluidization velocities in each section of the siphon to create a gradient of density and force the flow of material in the right direction.

Characterization of the powder de-fluidization characteristics

De-fluidization of the bed occurs when the flow of fluidization air is stopped (at t_0). The air which is trapped in the aerated bed will then be progressively vented to the atmosphere. Geldart and Wong [5] observed in a two-dimensional transparent mock-up of a fluidization column that two simultaneous phenomena occur during de-fluidization. First is the bed de-aeration; t_b corresponds to the time when all bubbles have escaped. Then the bed height decreases linearly until it reaches its final value at rest. Second is the presence of a rising front of sedimentation at the column's bottom, which starts at t_b and rises at a velocity u_s^* until it reaches the final bed height. Figure 7 represents both phenomena.

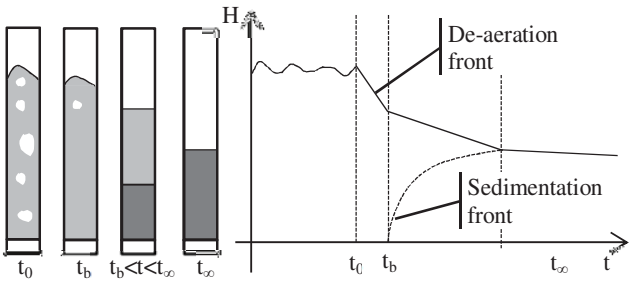

Figure 7 – De-fluidization mechanism: de-aeration and sedimentation fronts progress

Knowing u_s^* is critical since, according to Tavares [3], in fluidized hyper-dense phase downward flows it determines the blocking flow rate (G_s) according to the following equation: $G_s = (1 - \varepsilon_{mf}).\rho_p.u_s^*$; G_s corresponds to the maximum surface mass flow of product that can be transported across the column.

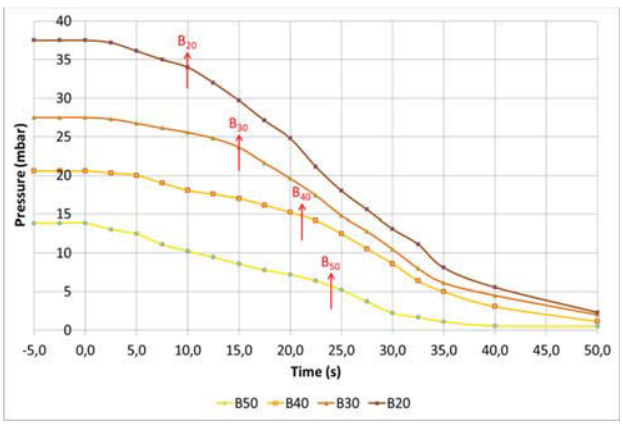

Figure 8 – Local parietal pressures evolution during a de-fluidization test (m_p=16kg; initial fluidization velocity=U_{mb})

As part of this study, a new method was developed to accurately estimate u_s^* while overcoming the limitation of the existing ones (Sharma [6] & Tavares [3]), and in particular their complexity. It is based on the measurement of the parietal pressures of the fluidized column at different heights i (20 / 30 / 40 / 50cm) after

the powder had previously been fluidized at the bubbling velocity (U_{mb}). Figure 8 shows results obtained. It clearly shows break points (B_{xx}), which correspond to the moment when the sedimentation front reaches the different heights *H*. Plotting the temporal gradients $\left(|\Delta P_i|/\Delta t\right)$ versus time allows for determining these different intervals even more accurately. Once they are known, it is then possible to calculate the corresponding sedimentation front velocity $\left(|\Delta H_{xx}|/\Delta t_{xx}\right)$. Interestingly, additional experiments have also shown that u_s^* is independent from the initial fluidization velocity (between U_{mb} and $4.U_{mb}$). It is worth noting that tests results were consistent with the one provided by the two historical methods. Once u_s^* is known, and combined with the other hydrodynamic parameters, it provides for an initial sizing of the upstream column (surface A_c) without any additional aeration, based on a design target flow rate of product to handle ($=G_s.A_c$).

Pilot tests

Different pilot benches were built at Toulouse University in order to optimize the siphon process, including a static one (2m long) and two dynamic ones (2m and 6m long). Each tested siphon is equipped with a feeding device, such as a rotary feeder, itself fed by the downstream column; alumina would therefore flow in a loop in the siphon. Pressure transmitters were installed along the equipment to monitor its operation.

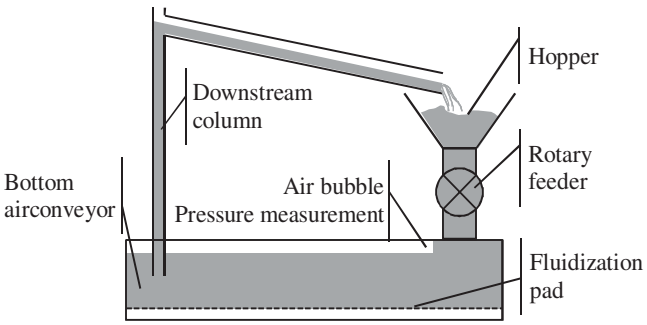

Figure 9 – Principle of the lab-scale siphons

The impact of several parameters was studied on the lab-scale siphons, including the condition of the fluidization of the downstream column and the airconveyor fluidization velocity.

Downstream column fluidization

Figure 10 shows the bottom airconveyor bubble pressure evolution with time (2m pilot, Q_p=1200kg.h^{-1}) for a given fluidization pressure.

Oscillations can be observed, which are directly associated with fluctuations of material flow at the siphon outlet. The phase during which the pressure increases (Phase 1) corresponds to an increase in height in the downstream column, both being directly related. When the material reaches the overflow at the top of the column, the acceleration creates a suction of the suspension in the downstream column, resulting in a bubble pressure reduction (Phase 2). The system is then back in its initial condition, and the two phases alternate.

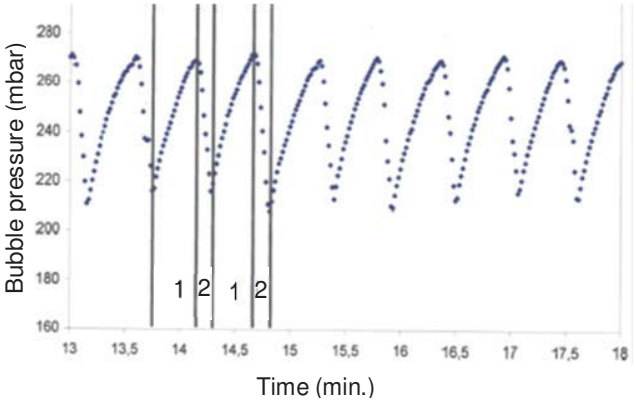

Figure 10 – Bottom airconveyor bubble pressure evolution

In order to dampen these oscillations, different designs of the column fluidization system used in the downstream column were tested (noted A to E). Figure 11 shows the results obtained.

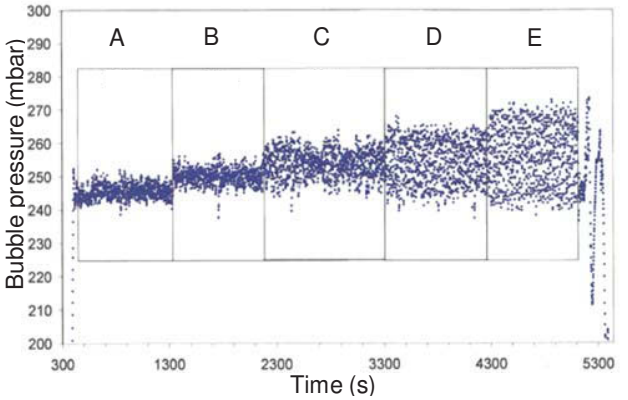

Figure 11 – Bottom airconveyor bubble pressure evolution for different fluidization system designs (downstream column)

This graph clearly shows that the design of the fluidization system has a direct effect on the fluctuation amplitude as well as on the average bubble pressure. The average flow of material is unchanged during all these tests but the impact on the instantaneous flow is significant: for certain designs (D and E), the siphon operates discontinuously (no flow 50% of the time, twice the average flow the remaining 50%); on the other hand, for other designs (A and B) the system is stable and alumina flows continuously. Optimized fluidization systems therefore ensure a stable operation and a reduction of the driving (bubble) pressure.

Airconveyor fluidization velocity

For these tests, the rotary feeder is set at its maximum value and the fluidization velocity of the bottom airconveyor is varied. Figure 12 represents the mass flow rate of product for different fluidization velocities. On this graph, two areas can be distinguished. In Area 1, corresponding to velocities comprised between $1.2.U_{mf}$ and $1.1.U_{mb}$, the fluidization regime is homogeneous and the bed expands with the fluidization velocity; higher velocities result in higher mass flow rates of product. In Area 2, the increase of mass flow rate with the fluidization velocity is more moderate: the fluidization regime is heterogeneous. As already seen, it is characterized by a weak bed expansion.

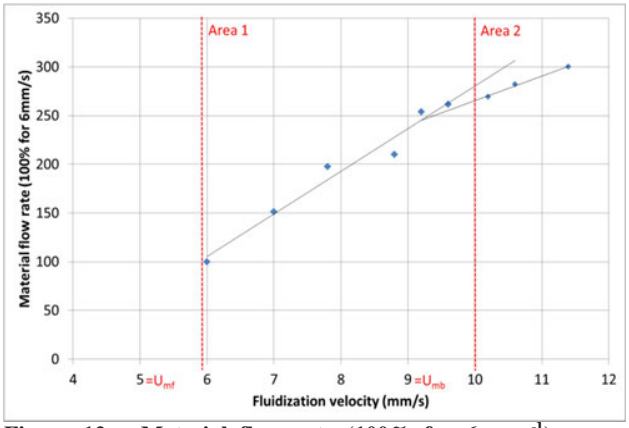

Figure 12 – Material flow rate (100% for 6mm.s^{-1}) versus fluidization velocity

Overall, these tests confirm that the bottom airconveyor fluidization velocity is a flexible parameter that can be used to control the mass flow rate of material if required.

Industrial scale pilot

A fourth industrial-scale prototype, as represented on Figure 13, was built at the Alesa Technologies premises in Gardanne (~8m length, 4.5m height, 12t/h nominal flow rate). This piece of equipment was used to validate the process parameters and detailed design determined on the smaller-scale siphons.

Figure 13 – Picture of the industrial-scale siphon

Figure 14 shows the bubble and top receiver (located at the top of the downstream column) pressures evolution with time. The industrial pilot settings had been determined on the lab-scale siphons (pressures, columns fluidization system design, geometrical characteristics). In this configuration, the siphon is very stable and the flow of material is continuously transported at the design flow rate.

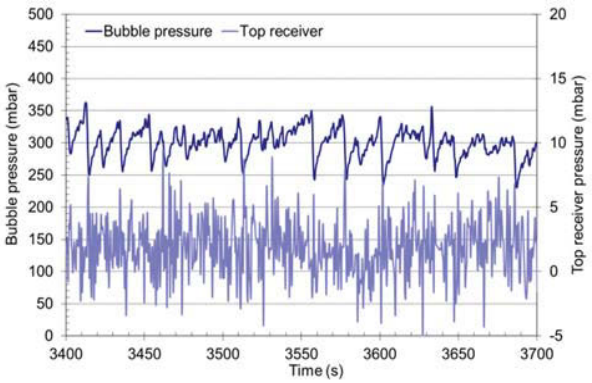

Figure 14 – Bubble and top receiver pressures evolution with time

Having provided satisfactory performance, this configuration served as a basis for the design of the first industrial siphons, which have been installed at the recently started Rio Tinto aluminium smelter in Kitimat (Canada).

Industrial start-up

The Kitimat Modernization Project (KMP) is a multi-billion dollar project for the modernization of the 60 year-old Kitimat smelter. Thanks to the implementation of state-of-the art Aluminium Pechiney (AP) pots, production will be increased by 48% to approximately 420,000t per year. The pots are fed with alumina by a HDPS™ system represented in Figure 15, located for the first time outside the building in order to minimize the potroom size and therefore cost.

Figure 15 – Picture of two main airconveyors installed at the new Kitimat smelter

The KMP smelter has a relatively specific layout, with two GTCs each serving two entire potrooms, crossed by a transversal passageway. These GTCs are located on one side of the passageway and a solution had to be found to feed pots situated on the other side, across the road.

The siphon technology is perfectly adapted to address this sort of issue and a specific potfeed system was designed for KMP. Figure 16 represents an overview of the process flow diagram of the potfeed system installed in Kitimat. It shows the siphon is used to cross below the passageway and feed the remaining 48 pots

located opposite to the GTC. This solution was selected because it was by far the most economically attractive compared to alternatives which would have required additional storage capacities on both sides of the passageway.

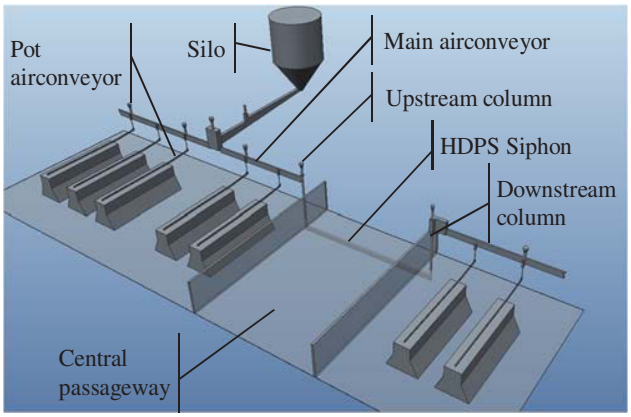

Figure 16 – Overview of the process flow diagram of the potfeed system installed in Kitimat

Since its start-up in May 2015, extensive tests were conducted in order to optimize the siphon operation and validate its performance. During these tests specific focus was paid to the following key performance indicators:

- Nominal flow rate: the siphon was designed to handle a flow rate of 12t/h which corresponds to the nominal pot consumption. With all the pots now operational on the siphon-side, it has been possible to confirm that the system was able to continuously transport this flow in a very stable way, similar to what was observed in the industrial pilot.
- Maximum flow rate: it is critical to ensure pot hoppers can be quickly refilled after a potfeed stoppage (e.g. following a maintenance activity). In Kitimat, the maximum flow rate exceeds 30t/h, well above the minimum that had been required (~20t/h).
- Reliability: the HDPS™ system is well-known for its reliability, and it is therefore critical to not degrade this performance with the siphon introduction. No noticeable issue has been observed during the first few months of operation, giving confidence in its robustness.
- Dust generation: adjustments were made to the initial design in order to ensure no dust would be carried away with the fluidization air through the degassing column and recirculate back to the GTC. Very low dust flow rates were achieved (<2kg/h).

Overall, all of these tests have confirmed the siphon performance was above the pre-defined criteria.

Conclusions

The Hyper Dense Phase System (HDPS™) was developed in the 1980's to provide a cost-effective automatic potfeed system. Since then, it has been successfully implemented in more than 25 smelters. Over the years, this system, which does not require any mechanical device in the material flow, has proved exceptionally reliable. However, it was limited to horizontal straight-line configurations, which prevented its application in some specific cases.

A major development program was initiated by Alesa Technologies with the collaboration of the Toulouse University (France) to design a solution which could couple both horizontal and vertical hyper dense transport sections – the patented HDPS™ siphon. Fundamental research allowed for determining the key process parameters, which were then optimized and validated on different pilots.

The first two industrial siphons have recently been started-up as part of the Kitimat Modernization Project. Since the start-up, different activities have been completed to fine-tune the process and confirm its performance. They have now been operating satisfactorily at their nominal flow rate for weeks, successfully concluding the development phase.

This technology opens many new opportunities for the HDPS™ application, which will prove a valuable solution to simplify layouts and minimize cost, similar to KMP. Other potential prospects include the automatic continuous feeding of end-to-end legacy pot technologies.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE CHALLENGE TO SUPPLY CONSISTENT ALUMINA QUALITY TO ALL POTS ON INCREASINGLY LONGER AND HIGHER CAPACITY POTLINES

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Keywords: DX Technology, Pot feed system.

Abstract

In the last thirty years, the majority of greenfield projects, along with some brownfield upgrades, have switched to using pneumatic conveying technology, typically a form of pneumatic technology. Initially, batch delivery technology supplied consistent alumina quality along the potline. With aerated distributed systems, the assumption is still that all pots would receive consistent alumina quality; and typically this has been the case. However, with high amperage pots combined built in long potlines over the last decade, this assumption may no longer be true as variance in both alumina chemistry and physical properties increases with the distance from the Gas Treatment Centre (GTC).
This paper summarises the detailed investigation during commissioning of the Pot Feed System (PFS) in the world's longest potline, Potline 3 in Emirates Aluminium (EMAL, also known as EGA Al Taweelah), an operating subsidiary of Emirates Global Aluminium (EGA). With a nameplate capacity of 550 ktpa, EMAL Potline 3 uses more alumina than most aluminium smelters and pneumatically conveys the alumina up to a distance of 425 m.

Introduction

Pot Feed Systems (PFS) have evolved from the original batch systems of crude direct shovel from a wheelbarrow, through semi-mechanical breaking the crust "backs" and pouring alumina into the pot open back cavity, to early gantry crane portable hoppers feeding pot hoppers directly, and on to the more continuous or at least semi-continuous aerated distributed systems of the last 25 to 30 years which directly fill the pot hoppers from storage silos.

Throughout this evolutionary process, the key assumption and typical reality is that all pots are fed the same quality of alumina. Any variation in the alumina delivered to one pot compared to another is driven by external factors, such as supply of the material and/or silo segregation or rat holing.

The Growing Challenge

In the early 1990s, typical greenfield pot feed systems were feeding pots of 300 kA in 800 m long potlines of 132 pots from one of two Gas Treatment Centres (GTCs), i.e., 66 pots each per PFS at about 11.5 t/h of alumina for a distance of approximately 190 m along the potroom. In comparison, EMAL Potline 3 is approximately 1 500 m long. The PFS has a conveying distance just short of 400 m along the potroom and faces more than double the alumina demand at 31 t/h.

The added challenge for aerated distributed systems is the limited vertical height, generally less than 10 m, into which systems must be designed to operate. As a consequence of this limited height, in most cases the main transport aerated conveyor is horizontal with no gravity assistance, although Qatalum [1] has employed a slope to aid conveying a distance of 290 m.

To keep in context the challenge required by the PFS for EMAL Potline 3, this single potline has a similar or larger capacity than other multiple potline smelters such as Boyne Smelters, Tomago and Qatalum.

Unusual Commissioning Challenge

Every greenfield potline commissioning presents its own and at times unique challenges. EMAL Potline 3's unique challenge is its approximate 1 500 m length and the five intermediate passageways through which each branch from both the GTCs must pass. These passage ways add additional conveying distance.

The Potline 3 PFS has branches from the GTC which feed 55 to 56 pots each side of the GTC in each room. Each branch includes an approximate 20 m section through a passageway. Due to the distance and limited height constraints, the EMAL Project team chose to employ fluid flow technology as the most cost-effective technology.

During the commissioning of the EMAL Potline 3 PFS, it was discovered that the fines content of the alumina in the pot hoppers was not consistent and that over 90 % of the -45 µm content was lost by the time it reached the last pot on each branch (Figures 1 and 2).

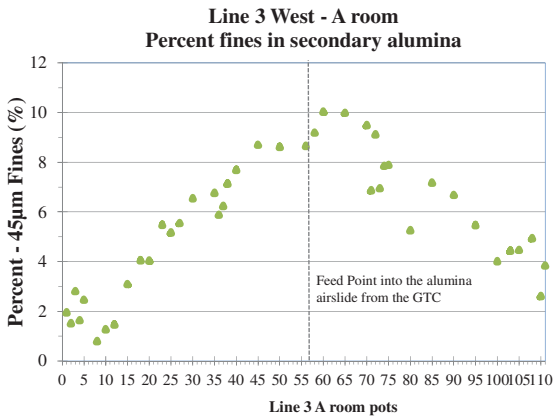

Figure 1. Fines concentration along first half of potroom A EMAL Potline 3.

The fines content declines as the distance from the GTC secondary alumina silo increases. Additionally, further investigation of the chemical analysis revealed fluorination concentration also declined by > 15 %.

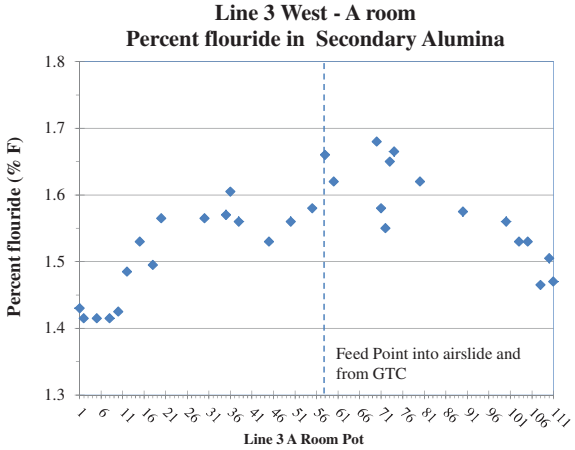

Figure 2. Per cent fluoride concentration along first half of potroom A EMAL Potline 3.

Investigation

As part of the investigation, comparative alumina samples were collected from the significantly shorter PFS in EMAL Phase I (DX Technology at 385 kA), which was commissioned successfully in 2010 using more conventional airslides technology from another supplier with a maximum horizontal conveying distance of 270 to 290 m along the potroom. However, unlike Potline 3, the feed point into the air slide is just three pots from the end of each branch of 45 to 48 pots.

In spite of no concerns or performance issues about this system, the results surprisingly demonstrated a similar trend to that of Potline 3, although the loss was not as significant (Figure 3).

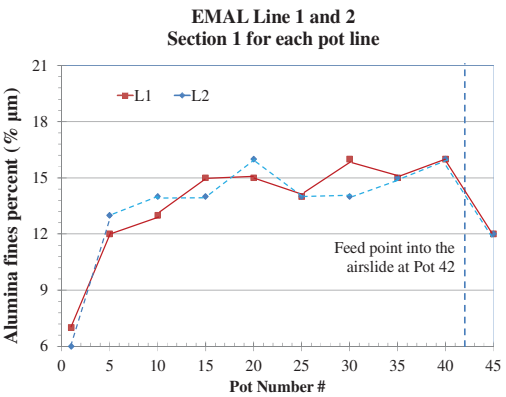

Figure 3. Secondary alumina percent fines concentration along section one of both lines in EMAL Phase I.

Of note is the sudden drop in secondary alumina fines content in only three pots from feed point to the east end of the airslide at Pot 42. Nearly 50 % of the -45 µm content has been lost in a conveying distance of just three pots. While less dramatic, the per cent fluoride concentration of the secondary alumina enters the airslide at 1.8 to 1.9 % and maintains this along the airslide until the last few pots, where it drops to 1.7 %.

Additional comparisons were also undertaken at Dubai Aluminum (DUBAL) Potline 8, comprising 40 pots over two rooms and also using DX Pot technology at 390 kA. The feed points are at pots 16 and 26 for each room respectively. The PFS in Potline 8 is an earlier version of the technology used in EMAL Potline 3, however with a far less challenging distance of a maximum of 15 pots or 110 m including a small passage way. The results are shown in Table I.

Table I. Percent fines and percent fluoride concentration along both rooms of Potline 8 DUBAL.

Pot	3	6	12	15	16	17	18	19
% -45um	11.5	12.6	14.0	14.6	15.0	13.9	14.5	10.7
%F	1.7	1.7	1.8	1.8	1.8	1.8	1.8	1.7

Pot	22	23	24	25	26	29	35	38
% -45um	15.1	19.7	18.7	20.0	20.7	20.5	18.3	16.6
%F	2.0	2.2	1.9	2.0	2.0	2.0	1.9	1.9

Due to operational reasons, samples were not available from the end pots in both rooms. Despite the lack of end pots data, the fines loss was approximately 25 % for all four branches, 5 and 15 pots in length.

The fluorination loss was less significant and was likely masked by the lack of results from the end pots. Additionally, a similar trend was noted along the GTC primary alumina air slide (Table II).

Table II. Primary alumina PSD of an EMAL GTC, start and finish.

Alumina PSD	Start	End
-32 µm	4.2 %	< 0.7 %
-20 µm	2.4 %	0 %

None of the PFSs on any of the four potlines tested were designed with any sampling points. Hence alumina samples were collected from the centre feeder on all pots tested, as this was the only way to have a consistent sampling location across all the potlines tested.

Chemistry Concerns

An alumina size-by-size analysis was undertaken to confirm that the loss of fines was the cause for the reduction of fluorination concentration along the potline. Analysis confirmed that the loss of fines from the initial part of the conveyor was causing a number of interesting impacts to the pots, such as less fluorinated alumina, less impurities and less feeder dumps. Chemical analysis of the secondary alumina by size fraction in Potline 3 is given in Table III.

Table III: Chemical analysis of the secondary alumina by size fraction in Potline 3.

Size Fraction	% F	% Fe	% Si
+75 µm	1.3	0.012	0.013
+45 µm	1.8	0.014	0.016
-45 µm	3.9	0.018	0.019

A 5 % reduction in the average daily number of alumina dumps from the GTC feed point to the end of the PFS branch was also observed. This was assumed to be caused by the physical change in alumina properties (Figure 4).

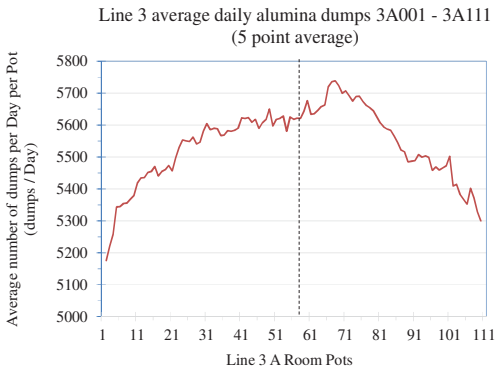

Figure 4. EMAL Potline 3 potroom A, alumina dumps along the potroom.

Root Cause

The Potline 3 fines loss is directly related to the time exposed to the aeration of the system. Typically, the faster the feed rate and/or the further the distance the alumina is transported, the more air is required. With more air per tonne of alumina conveyed, more dust will become suspended in the air. As the material moves along the aerated conveyor, the material slows down due to lower consumption rate for the remaining pots, yet few aerated conveyors change in size.

In older aerated PFSs, the number of pots on one branch is typically 30 to 45 pots – but Potline 3 has 55 to 57 pots per branch. Hence the conveying time (assuming a similar technology) is 50% longer, thus exposing the alumina to twice the aeration time and hence increased loss of fines. Figure 5 shows a modeled example of the residence time for the alumina to get to the end of an aerated PFS Branch (assuming that the starting horizontal velocity is the same for all systems). Additionally if the feed point from the GTC is at some point along the branch but not centralized, then the shorter branch will see even longer exposure to aeration and thus increase the loss of fines over a relatively short distance as seen at DUBAL Potline 8 and EMAL Phase 1.

De-aeration Systems

Most aerated PFSs do not use filtration to separate the fluidising air from the particulate. Rather, one of a number of technologies is employed such as:

- Using the void above the alumina bed in an airslide,
- Expansion chamber (Figure 6) to slow the gas down to a velocity where most of the particulate falls out,
- Mini cyclones (Figure 7).

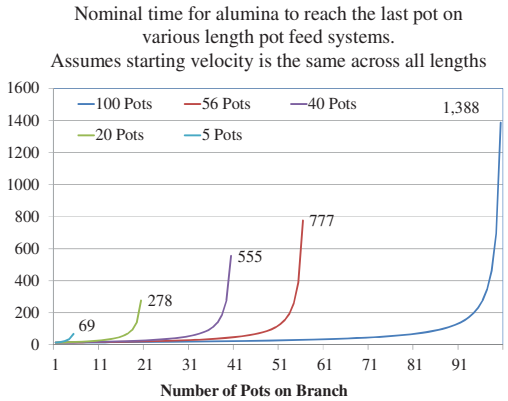

Figure 5. Modeled nominal example of the residence time for the alumina to get to the end of an aerated PFS Branch assuming the alumina velocity.

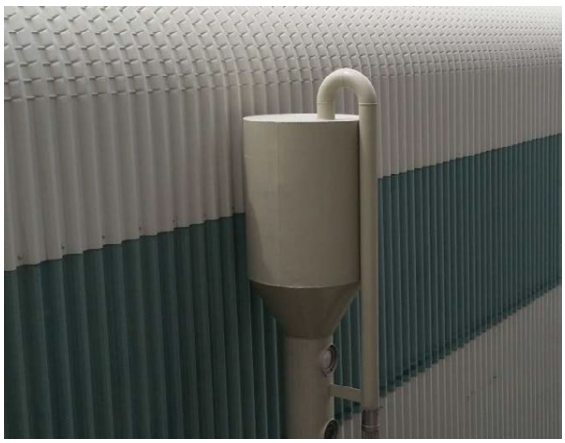

Figure 6. Example of expansion chamber used at EMAL Potline 3.

Figure 7. Example of mini cyclone.

Solution for EMAL Potline 3

The EMAL Project team proposed to increase the diameter of the expansion chamber to reduce the rising gas velocity. Initially a small trial was undertaken, effectively doubling the diameter to 600 mm for the last ten pots. The context for only choosing the last ten pots was the belief that this was where the bulk of the fines loss occurred and hence would have the greatest impact (Figure 8).

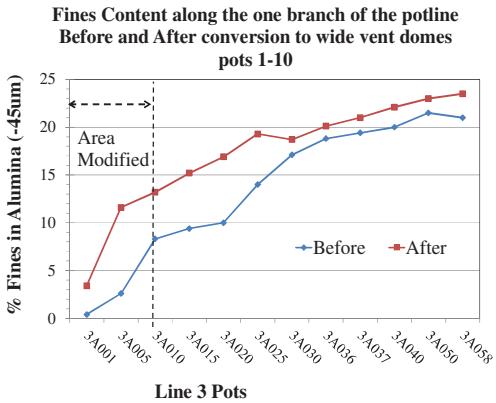

Figure 8. EMAL Potline 3 potroom A per cent fines before and after the small size trial of changing last ten pots only.

This modification achieved an increase in fines content in the last ten pots by 3 %. However, further focus was placed both on the size of the expansion chamber required and the impact of every expansion chamber.

For example, the initial 36 pots to the passage way were initially ignored because the loss of fines to the passageway was less than 3 %. However, with further sampling a simple mass balance was able to determine that the fines loss in the mass flow of the first 36 pots was similar to the middle and last 36 pots groups on each branch, i.e., 200 to 300 kg/h. As such, equal treatment to the expansion chambers needed to be applied along the potline, not just the end.

Additionally, it was decided that the diameter be increased to 800 mm to reduce the rising fluidising gas velocity in the expansion chamber to around 15 % of the initial expansion chambers. Figures 9 and 10 show the results of the modifications.

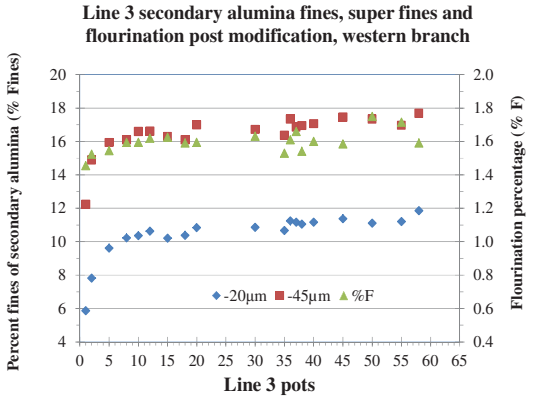

Figure 9. Potline 3 potroom A per cent fines and per cent fluorination content, post-modification.

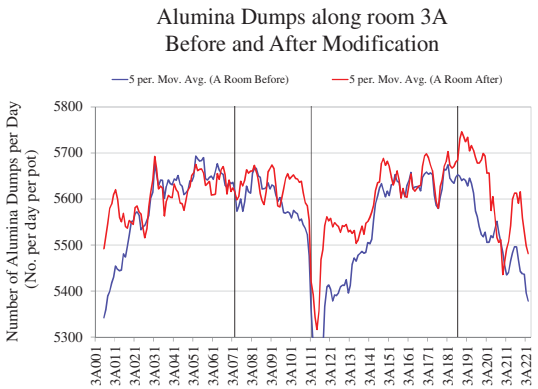

Figure 10. EMAL Potline 3 alumina dumps per pot per day, before and after modification to the expansion chamber.

After the modifications to all expansion chambers were completed, all but the final pots were within 15 % of the initial fines content – with super fines performing to a similar level. Fluorination concentration followed a similar trend.

The sudden reduction in fines for the last pot is an indication that the residence time is still excessive in the aerated conveyor and/or insufficient expansion chamber size. The phenomenon was displayed in the Phase I data as discussed previously and applies to all similar aerated transport systems which have progressive reduction in material flow, yet maintain a similar cross sectional area. Overall the performance is now more successful than the conventional airslides in other pot lines and the primary airslide for the GTC.

Conclusions

Regardless of the PFS technology and/or the distance employed, the loss of alumina fines phenomenon still exists. Historically it was unknown, because it is rare or not common to check for fines loss in PFS unless there is an obvious problem in the pot and/or GTC performance. If alumina is transported to the last pot and fills the hoppers in less than the desired time frame during commissioning, the system is considered successful.

Measures can be implemented to control or limit loss of fines as demonstrated in this paper. However, as the potlines grow longer and amperage rises, supplying consistent quality of alumina to all pots will be an increasingly more difficult challenge and conventional approaches may struggle to succeed.

Alternative options may include

- Using aerated conveyors that decrease in cross sectional area, to reduce the alumina residence time,
- Use of greater than the typical two GTCs per potline, i.e. mini localised GTCs as proposed by some gas treatment vendors and
- Other approaches that reduce the fluidization air/tonne of alumina ratio and thus the transport media that causes the fines loss in the first place.

Both pot feed suppliers and aluminium electrolysis technology providers will need to review and revise traditional alumina quality specifications for PFSs to ensure that even longer and higher amperage potlines in the future are fed consistent quality alumina along the potline.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

DESIGN AND DEMONSTRATION OF AN IMPROVED AUTOMATED POT TAPPING METHOD AND EQUIPMENT

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Keywords: Tapping, automation, bath

Abstract

In the aluminium electrolysis process, metal is retrieved from the cells using a tapping tube connected to a crucible in which vacuum is made to initiate and maintain molted metal flow. Care must be taken not to contaminate the aluminium with electrolyte during the tapping operation. Continuous improvement of cell performance, including reduction of the metal layer depth, increases the challenges related to the tapping operation. Reduction of the metal flow helps reducing the electrolyte quantity that is tapped, however this increases the duration of the operation and poses scheduling problems.

A novel patent pending tapping control method and equipment have been designed and tested in the Alouette smelter, in order to minimize contamination while maintaining or reducing the tapping operation duration. Metal flow control is improved thanks to optimized software. Analysis of the process indicators show that a significant reduction in tapped electrolyte is obtained.

The purpose of this paper is not to describe in detail the design and control algorithm of the control unit, which is proprietary, but rather to describe the results on various performance indicators. Some of these results were unexpected and in first appearance, showing downsides to the improved control. Further analysis shows that these apparent downsides are consequences of improved performance.

Introduction

It is well known that bath carryover from the cells to the casthouse is detrimental to the potroom operations as well as to metal quality and process control. Bath entrainment during the tapping operations encrusts the tapping tubes and crucibles, requiring extensive maintenance and cleaning as well as increased equipment wear. Excess bath an sludge in the casthouse affects the TAC process, furnace quality, skimming and drossing operations, dross quality, metal purity and so on. The purpose of this paper is not to discuss these problems, as these are described very well in the existing literature [1, 3, 4].

One proposed method to reduce bath carryover to the casthouse is to hold the crucibles for a certain period of time [2]. While this reduces the negative impacts of tapped bath to the casthouse, this does not impact the total amount of bath tapped from the cells.

Many factors influence the bath carryover during the tapping operation, including siphon tube positioning, mass flow of aluminium, siphon geometry, metal pad thickness and so on [1, 3, 4].

A video monitoring of the metal stream inside the crucible have been field tested, to improve one existing method where the operators observe the flow through a window in the crucible [6]. However this method is not yet ready for industrial implementation as the results are not satisfactory.

The most promising method to reduce bath carryover is the limitation of the mass flow of tapped aluminium [4] (provided, of course, that proper operating methods are followed). This provides a strong motivation to improve the tapping process, which is still widely a manual operation with little control or monitoring.

One could argue that since bath entrainment is proportional to the metal flow, then the siphon tube will encrust until optimal metal flow is attained and a stable condition would occur. Unfortunately, some bath is always entering the tube when the siphon passes through the bath layer when reaching the bottom of the cell. Pressurizing the crucible is theoretically possible to avoid this, but tapped bath quantity is significantly greater than that coming from this initial tube placement.

Metal flow limitation can be done in many different ways, however care must be taken not to influence the other aspects of the tapping operation.

Description of the proposed method for tapping mass flow control and its benefits will be made further, but let's first begin with an overview of the tapping process.

Tapping operation description

Metal tapping equipment is generally consisting of a crucible with a removable lid on which a siphoning tube is fitted. Vacuum is made inside the crucible using an ejector. The crucible and lid assembly is transported and operated by the means of a lifting beam, itself mounted on a pot tending machine (PTM). The lifting beam is normally equipped with weight measuring means to monitor the metal weight in the crucible. The compressed air for the ejector comes from a compressor located in the PTM.

Fig. 1: Tapping crucible assembly

The tapping operation can be broken down in three (3) different phases (once the siphon tube is inserted in the cell)

Phase 1: Initial vacuum

Phase 1 is the initial vacuum make-up in the crucible in order to lift the aluminium in the tube and initiate the metal flow. This phase must be cleared in the shortest time possible, for many reasons:

- Avoiding siphon tube clogging
- Optimizing operator schedule

The initial vacuum formation is dependent upon many factors, including crucible size, metal level, crucible temperature, siphon length (metal height), ejector performance, compressed air performance and seals performance.

The influence of these factors is summarized in the following table:

Parameter	**Variation**	**Influence on vacuum formation time**
Crucible size	Increase	Increase
Metal level	Increase	Decrease
Crucible temperature	Increase	Decrease
Siphon length (metal height)	Increase	Increase
Ejector performance	Increase	Decrease
Compressed air performance	Increase	Decrease
Seals performance	Increase	Decrease

It is noticeable that the best performance is obtained with a small, hot crucible, with low metal lifting height from the cell, a performing ejector and sealing system with maximal compressed air supply.

In practice, the design of the crucible is fixed and there is no control on the quality of the seals, which is variable from one crucible to another. The ejector performance must thus be maximized and the compressed air supply must be unrestricted and at optimal capacity. This, however, is detrimental to phase 2 performance:

Phase 2: Metal tapping

Phase 2 is the metal tapping phase. The goal is to retrieve an accurate amount of metal, of the highest purity possible, in the least amount of time.

The important parameter is the mass flow of the metal [4], which is dependent upon many factors, including crucible temperature, siphon length and cleanliness (pressure drop along the tube), ejector performance and compressed air supply as well as seals performance.

The basic hypothesis is that if the metal flow is kept below a certain limit (which has to be determined experimentally and will change with operation practices, siphon tube design, metal pad thickness, etc), the entrained bath will be minimized.

The mass flow of metal is controlled by varying the vacuum level in the crucible. No universal value can be used for the vacuum level in order to keep the mass flow constant because the system must compensate for leaks and pressure drop in the siphon tube.

The vacuum level itself is dependent upon the performance of the ejector (a fixed value) and the compressed air supply pressure (which can be controlled). It is therefore possible to change the mass flow of metal by varying the compressed air supply pressure to the ejector.

Having an ejector with fixed performance to optimize the metal tapping phase is difficult, due to variation between crucibles (temperature, quality of the seals, siphon clogging). Moreover, if the ejector performance is optimized for the tapping phase, then the initial vacuum (phase 1) is longer to build. This causes increased operating times and increased probability of having frozen metal in the siphon.

Phase 3: End of tapping

Phase 3 is the end of tapping. The goal is to stop the tapping operation as quickly as possible for the following reasons:

- Minimize the possibility of bath entrainment if the crucible is lifted while metal is still flowing in the tube (thus passing in the bath zone while siphoning)
- Improve the accuracy of the tapped metal weight (minimize weight overshoot)
- Optimizing operator schedule

Typically, the tapping is stopped by closing the compressed air supply valve to the ejector. This requires the operator to anticipate the end of the tapping operation by stopping the air flow before the crucible reaches target weight.

Design criteria for automation success

An ideal automated tapping control system [4] would allow the following:

- Maximize ejector performance and vacuum formation for the initiation of the metal flow (Phase 1)
- Control vacuum level in the crucible to keep metal flow within tight limits (Phase 2)

- Break the vacuum in the crucible as quickly as possible when stopping the siphoning operation (Phase 3)
- Provide optimal performance independently of the variations from one crucible to another

Description of the actual automation system

The system that have been designed, built and tested in real operation conditions is made of a control module that is mounted on the tapping crucible lifting beam.

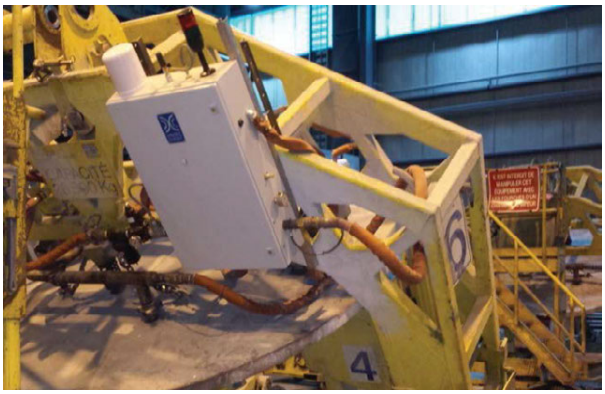

Fig. 2: Control unit prototype mounted on the tapping equipment

A similar system have been tested in another plant using standard components [5]. The present system includes some additional proprietary features.

Construction

The control module is made of a special custom-built compressed air control valve that is mounted in a specific place relative to the ejector position. This valve is controlled using a dedicated, standalone PLC that communicates with the PTM PLC to monitor the weight gain continuously using the PTM load cells. In addition to this, vacuum transducers and pneumatic valves to relief the crucible to atmospheric pressure are provided. Everything is mounted in a sturdy metallic panel for magnetic field protection. Wi-fi communication is provided to monitor the process parameters during the operations.

The control module prototype is mounted in series with the main "on/off" tapping valve located on the lifting beam

.

The prototype unit was built to be easily installed on the existing lifting beam, without any modification except for the electrical connector. A 24V power supply is added for the PLC and valve actuator, and a 4-20 mA communication signal is provided to communicate the load cell data to the PLC. Initiation of the vacuum uses the existing discrete signal.

Functional description

The general operating principle is as follows:
Initial power-up of the control unit allows the PLC to boot and check the quality of the signals and the functioning of the valves. When the unit is ready (requires only a few seconds), the PTM can begin its operations.

Tapping is initiated when the operator sends the starting signal form the PTM remote control. No modification is made to the operating methods, making the control unit transparent from an operator standpoint. As a default, the control valve is 100% open, allowing maximal compressed air flow to the ejector and maximum vacuum generation performance.

The weight of the crucible is continuously monitored and filtered through the control algorithm. The vacuum level is controlled to keep the metal flow constant and close to the operating limit.

When the tapping operation is stopped, the compressed air flow is stopped and valves vent the crucible to the atmosphere to stop the metal flow instantly and minimize the risk of siphoning bath from a depressurized crucible while lifting the crucible from the cell.

A indicator light provides feedback to the operator on the status of the system: continuous green light when the system is ready and the valve is fully open, blinking green light when the valve is restricting the air flow. A red light provides additional (fault) data when the system detects air leaks in the crucible seals or a blocked tube situation.

Plant trials

The control unit was mounted on a lifting beam dedicated for the trials. A complete inspection and tune up of the components was made to ensure optimal performance, particularly of the ejector. All seals were replaced and the tapping tube was new. The unit was used only by a dedicated operator team on specific tapping runs to ensure accurate tracking of the metal in the whole process down to the finished product. The operation procedures were strictly followed. Another trial run was also done with regular tapping equipment, to isolate the effect of the control system over the operation procedures.

Initial observation of the metal surface in the tapping crucible showed immediate, qualitative appreciation of the reduction in entrained bath:

Fig. 3: Comparison of metal surface

The picture on the left shows a crucible with some bath on the metal surface. The picture on the right shows the same crucible with the automated siphoning control system operating.

Results

The Alouette smelter is a producer of primary aluminium. The metal is cast using 3 sow casters. This allow tracking of the metal, directly from the tapping crucible to the final sow. Thus, many performance indicators can be used to assess the quality of the tapping operation.

Results are based on quantitative measurements and qualitative observations. Some results have been modified to protect the confidentiality of production data.

A traditional indicator of the amount of entrained bath is the difference in weight of the empty crucible from one cycle to the other: weight gain between two cycles is proportional to the amount of entrained bath [2, 5, 6]. It is often used because the information is readily available from production data. However it is a poor indicator because the residual weight of a crucible depends upon many factors:

- Only a fraction of the bath adheres to the crucible walls
- This fraction might be different from one crucible to the other depending upon the residence time [2], temperature, wall condition, etc
- An empty crucible contains a significant amount of aluminium, which can be variable from cycle to cycle
- Some residual bath in the crucible can detach and be transferred to the casthouse

In consequence, different indicators have been chosen to evaluate the performance of the control system. The prototype was used in hall D during dayshift. The other halls were operated with standard equipment. The results are presented for halls A, B, C and D for the same period. This allows comparison of the new equipment without interference from other variations coming from casting or potroom equipment or operations.

General finished product quality

A good indicator of the overall performance of the whole plant is the quality of the finished product. Many indicators are already implemented at Alouette to monitor the process. These indicators were used to evaluate the performance.

Bath inclusions

The amount of so-called "bath inclusions", that is, pieces of solidified bath remaining in the sow moulds, is noted by the operators. Significant reduction of bath inclusions have been observed for hall D, although hall B also shows good results:

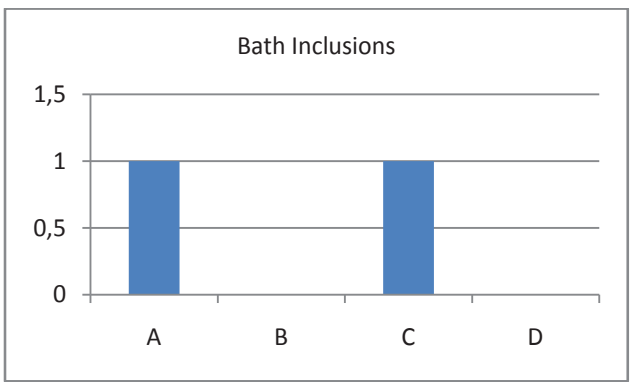

Fig. 4: Relative amount of bath inclusions

Insufficient skimming

"Insufficient skimming" is a tag assigned to a sow showing excessive dross even after the automated skimming process. It is reasonable to say that a reduction of bath carryover would help reduce the amount of dross. Reduced dross means that fewer sows will have remaining dross after the skimming process. Significant reduction of insufficient skimming tags have been observed

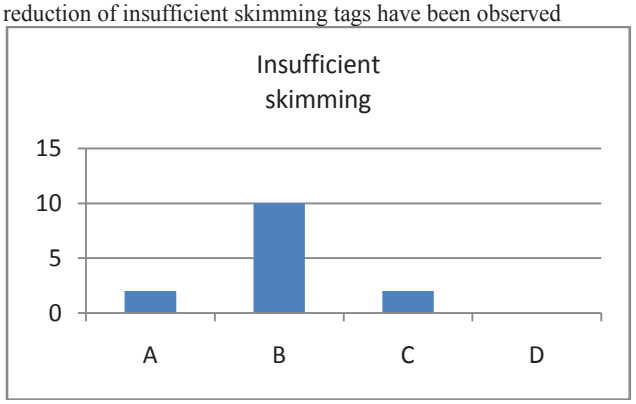

Fig. 5: Insufficient skimming

It is worth mentioning that the automated skimming process does a first skimming pass, then the operator can decide if a second pass is required. An insufficient skimming is assigned after the second skimming pass. It is reasonable to conclude that reduced skiming operations (and accelerated cycle time of the sow caster), as well as reduced melt loss and maintenance requirements on the skimming equipment are a significant benefit of reduced bath entrainment.

Tapped bath (iron content in cells)

Bath quantity in the cell can change depending upon many factors. Bath is added as coverage material for the anodes. Some bath is removed when removing spent anodes and the bulk of removed bath is of course, during the tapping operation.
Unexpectedly, chemical analysis of the metal coming from the controlled tapping operations showed an increase in iron content. This increase was related to an increase of bath level in the cells where the controlled taping operations were made. Thus, bath balance was added to the performance indicators. It shows that these cells need to have more bath removed.

Other performance indicators and benefits

Other than direct finished product quality improvement, many internal improvements are observed, although less quantitative or not directly measured. Some of these improvements can only be evaluated over a long period of time and after all the operations have been converted with the control system. This could be the object of a further paper. However some of the initial observations are summarized hereafter.

Operator acceptance

Operator acceptance is very important when introducing a new technology. The best way to gain operator acceptance is when the process is improved and the operators' task are facilitated at the same time.

In this case, the system is transparent to the operators, that is, no change in the operating procedure is required to operate the system. Moreover, since the Phase 1 of tapping (initial vacuum) is shorter, there is an impression of improved performance from the operators, while reducing the occurrence of blocked siphon tubes.

Tapping tube cleanliness
Since the tapping tube encrusting is decreased, the tubes are cleaned only at the end of the shift. Normally the tubes are cleaned 2-3 times. This reduction represents a significant saving in cleaning machine consumable and maintenance costs.

Crucible cleanliness
Better crucible cleanliness was observed, although not measured due to the reasons mentioned above.

Compressor wear
Another unexpected benefit that was documented by the maintenance team is reduced compressor wear from the reduced overall air consumption of the ejector. The compressed air pressure dropped to approximately 90 psi during tapping with the regulating system. Normally, the pressure drops near 80 psi without tapping regulation due to the excessive demand of the ejector. The reduced pressure drop is an indicator that less air flow is consumed, helping reducing unplanned maintenance issues due to compressor failures.

Conclusion

A tapping regulation system was designed and tested in regular operation conditions. Significant reduction of entrained bath during tapping was observed and documented.
The advantages of tapping regulation (hence reduction of bath entrainment) are well documented [1, 2, 3, 5] and it is not the intent of this paper to enumerate them. Nevertheless, some undocumented benefits have been observed with the prototype unit that are worth mentioning:

- Operator acceptance (improved process quality with equal/reduced workload)
- Retrofittable and failsafe system (should the system fail, it does not restrict the compressed air flow and Thus the tapping process is simply not regulated)
- Cycle time optimization due to minimized Phase 1 and Phase 3 duration
- This technology can also be applied to bath tapping in order to reduce metal entrainment.

More tests are to come to further evaluate the impacts on operations and maintenance.

Acknowledgements

The authors would like to acknowledge the Alouette smelter for the permission to publish the results of the trials. A special thanks to the operator teams for their support and dedication during the trials.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EVOLUTION OF CRUST BREAKER CONTROL FOR DX+ AND DX+ ULTRA TECHNOLOGIES

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Keywords: DX+ cells, crust breakers, Pot Control System, bath sensing

Abstract

Crust breakers equipped with bath sensing modules (BSM) were introduced for DX+ cells since the beginning of their operation in 2010. The modules give signals to pot control system (PCS) when a breaker tip touches liquid bath. This paper describes steps in development and optimization of the break and feed control logics from diagnostic of different kinds of breaker failures to problem treatment procedures. Breaker failures can be linked to different problems both equipment and process. Process problems like bath formation on breakers, narrow breaker holes and alumina accumulation cannot be diagnosed automatically but control logic can help by treating them in a semi-automatic mode. It was noticed that distribution of breaker problems is very dependent on breaker locations. The bath sticking on breakers is most typical for middle breakers located in the hottest zone of the cell. On the other hand, narrow holes and alumina accumulation mostly occur in other breakers. The breaking control logic was modified in order to minimize occurrences of these abnormalities.

Introduction

Crust breaking is one of the most important operations in the aluminium reduction technology. The aim of using crust breakers is to break alumina-cryolite crust which covers the centre channel in order to expose the molten bath for the delivery of alumina. Problems in crust breaking lead to disturbances in alumina feeding, anode effects and a deterioration of cell performance.

Reduction cells are usually equipped with several point breakers actuated by pneumatic cylinders. The cylinders have a constant stroke distance and their position relatively to liquid bath level can be adjusted manually. The cylinder rod is extended by compressed air when control system gives a command to activate the breaker and it is retracted when the command is removed. The command duration should be adjusted in order to ensure full stroke of breaker extension.

It is not mandatory to have the same breaking and feeding rate in each breaker and in time. Breakers can be activated with higher or lower frequency. Pneumatic cylinders consume huge amount of compressed air; some technologies use “alternative” break and feed logic in which breakers are activated every second or third feeding cycle in order to save compressed air.

Abnormalities in crust breaking can be linked to both process and equipment problems. Process factors like varying liquid level and varying bath temperature affect the breaking performance significantly. In cold pots the breaker holes become narrow faster. Variations in bath level require the breaker settings in lowest position in order to reach bath even at minimum bath level. This approach leads to excessive immersion of breakers into bath in case of normal or high bath level, which leads to overheating of breakers, sticking of bath to breaker chisel and, finally, a phenomenon known as “bath build-up” or “elephant foot”.

There are also many mechanical or electrical reasons of breaking problems like low pressure of compressed air, cylinder failures, loose contacts, broken wires, pot controller failures, etc.

Breaking control

Any control system without feedback is a “blind” system. All control actions ideally should be confirmed by proper feedback, otherwise results of the control actions are unknown. Some control systems use signals of breaker activation relays as feedbacks. This approach gives almost nothing, except for detection of some pot controller failures.

The first considerable improvement in the breaker control systems was the introduction of pneumatic cylinders with end position sensors. This kind of feedbacks allows detection of much bigger number of problems like very hard crust, “slow activation” of breaker, air leakage, etc. However such system is not able to prevent problems with overheating of breakers and bath formation due to a constant stroke distance and fixed bottom dwell time [1].

The quality of a feedback signal in a control system is the best if the feedback gives information when the aim of the control action is achieved. The breaking goal is to reach molten bath, so the best feedback would be a bath sensing signal which confirms that the breaker has touched the bath.

Bath sensing

Many different companies like Parker, ECL, Rexroth, Camozzi, etc. have developed their Bath Sensing Modules (BSM) using different approaches.

Technical solution for bath sensing seems to be simple – it only requires to detect a reliable electrical contact between the breaker cylinder rod and the bath when the breaker chisel touches the bath; the voltage between the top of the bath and a reference point on the cathode is used as feedback to the pot control system [2]. The voltage change at bath contact is large enough to detect bath contact near 100 % reliably in spite of working in electrically noisy environment of the pot [3, 4].

Some crust breaker systems for bath sensing use an AC signal generator in the same electrical circuit and detect the bath contact by changes in this AC signal when the contact between the chisel and the bath closes the circuit. This method is independent of the pot voltage but it still requires very good contacts between cylinder and its rod [5, 6].

Another interesting invention is based on changes of electrical capacitance in the circuit cylinder-breaker-bath at the moment of breaker contact with bath. This solution does not require the electrical contact between cylinder and its rod and standard pneumatic cylinders without any modifications can be used in crust breaking systems [7].

Breaking control using BSM experience

DX+ ULTRA pots in DUBAL have two types of bath sensing modules from different suppliers. Both devices have high efficiency of bath sensing (> 97 % since February 2015) as shown in Figure 1. The bath sensing efficiency is defined as percent of correctly given breaker bath sensing feedbacks from total number of breaker activations.

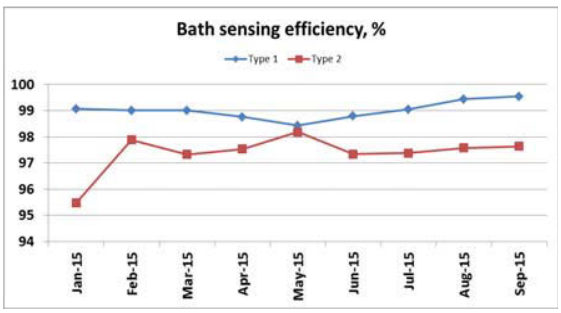

Figure 1. Bath sensing efficiency.

The pots are controlled by PLC based control system developed by DUBAL [8, 9]. The alumina feeding control logic is very effective, which is confirmed by low anode effect frequency (Figure 2). The majority of anode effects occur due to mechanical failures of breaking or feeding equipment.

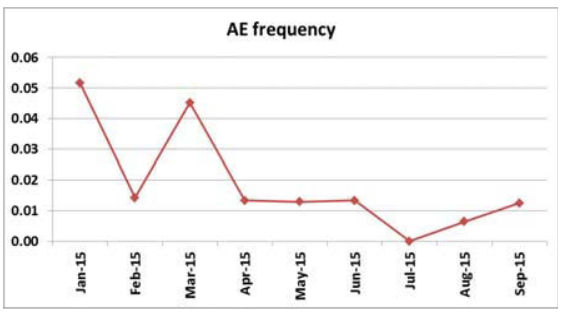

Figure 2. Anode effect frequency pet pot-day.

The breaker control logic is adapted for using of BSM; it monitors the BSM feedback continuously. The logic recognizes the following types of problems:

- "No feedback" – the feedback signal is not detected during the *Maximum Break Time* after breaker activation,
- "Unexpected feedback" – the feedback signal is observed within the *Minimum Break Time* after breaker activation,
- "Stuck feedback" – the feedback signal arrives in the *Break Off Time* after breaker deactivation

The "No feedback" alarm is the most frequently detected event because it can be linked to many process and equipment problems. The "Unexpected feedback" alarm is used as an indicator of incorrect BSM signal and the breaker is activated for the default break time ignoring the feedback. The "Stuck feedback" alarm can be raised in case of the breaker being physically stuck in bath. The alarms are used for audio announcements in potrooms for calling up an operator to check conditions of a certain breaker.

Some specific events on pots were identified as conditions to disable automatically the usage of BSM signals in order to avoid any delays in delivering alumina to the bath. BSM signals are also ignored during events when bath level variation is high.

Breaker problem treatment logic

Using of bath sensing modules in the breaking control reduces significantly the number of abnormalities caused by deep penetration of breakers into molten bath. Also it gives a great advantage in early detection of other breaker related problems. However, problems related to crust breaking can still occur from time to time which requires manual intervention by operators.

Three most frequently faced problems were identified during the development of automatic treatment logic:

- Bath build-up
- Narrow hole
- Alumina accumulation on the crust.

The aim of this logic is to minimize inconsistency in manual intervention by operators. It also helps in recording these issues in the control system. Such problems can not be detected automatically but the control system can help if these problems are identified and entered into the system by the operator. The operator should attend the pot in case of audio announcement or during his routine round, observe conditions of the breaker and inform the control system via a special HMI screen (Figure 3) what kind of problem has been found.

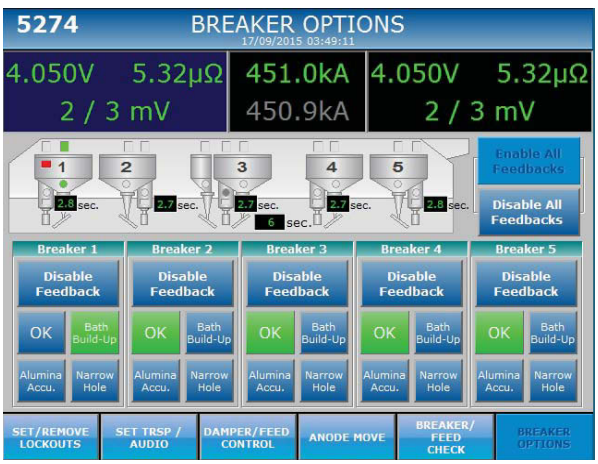

Figure 3. "Breaker treatment options" HMI screen.

The bath build-up problem is caused by overheating of the breaker chisels. It is very frequently observed in locations where breakers are exposed to strong flame. Another reason for overheating is too long breaker residence in the bath when the bath sensing does not work properly. Too hot breaker tip can cause a chain reaction in bath formation: Frozen bath is an electrical insulator, therefore any amount of bath stuck on the breaker impacts bath sensing efficiency, the breaker stays in the bath longer and is heated more, which increases the amount of bath stuck on the breaker and the vicious circle is closed. Sometimes the size of the frozen bath on the breaker can reach 40 - 50 cm (Figure 4).

However there are still some cases that are not solved with breaker treatment logic. In these cases, the build-up continues to grow and may surpass the width of the centre channel or may grow longitudinally along the channel. This prevents the breaker to descend enough to break the crust and prevents alumina to fall into the bath. AlF_3 feeder is also associated with one of the alumina breakers. Large bath build-up on the breakers may also prevent regular AlF_3 feeding, which will cause serious disturbance in bath composition and in pot thermal balance. On the other hand, the AlF_3 feeding itself may be the cause of more frequent bath build-up on this breaker. This is because the feeding increases excess AlF_3 locally, increases local superheat and the conditions of bath cooling on the breakers. In such cases the operator must knock off the build-up manually.

The size of the breaker hole always matches the size of bath formation on the breaker as long as the diameter of the buildup is smaller than the center channel width. In this situation the risk to impede alumina delivery to bath is not too big if the breaker is activated with less frequency. Less activation frequency allows cooling of the breaker and formation the crust below the breaker. Thereafter, during breaking, the crust exerts pressure on the breaker buildup and finally the breaker may be cleared. This decreases the need for manual intervention.

The bath build-up treatment logic can be terminated by the operator manually or is completed automatically in a predefined period of time.

Figure 4. Bath build-up on breakers.

Breaker holes can be narrowed due to very hard crust or during activation of some control logic like anode effect prevention (Near AE logic) or anode effect quenching when anodes are lowered deep into the bath and the bath washes the crust around the holes. It was noticed that sometimes bath sensors give feedback signals when breakers just touch the crust but have not reached the bath; most likely this is due to the crust wetted by the bath. This phenomenon also causes narrowing of the breaking holes.

Narrow breaker holes and other reasons like bad alumina dissolution, strong gas flow through the holes, etc. can make big heaps of alumina around breaker holes.

The treatment of narrow holes and alumina accumulation is similar – decreasing alumina feeding rate and increasing of breaking frequency should help in both cases.

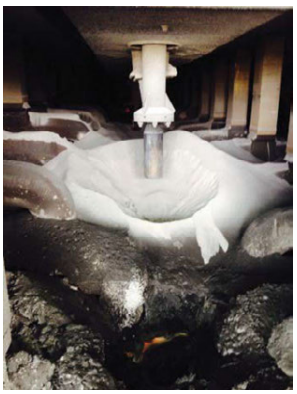

Figure 5. Alumina accumulation around the breaker hole.

The logic for alumina accumulation is slightly different. The breaker pushes some amount of material into bath every breaking cycle in case of alumina accumulation. The normal feeding logic distributes required amount of alumina through other available feeders if a feeder is locked out. This can lead to overfeeding; this is why now the control logic does not feed in that location and does not increase feeding rates in other feeders.

Breaker activity analysis

The analysis of the breaker activity gives clear picture of the location of the crust breaking problems in pots. The bath formation is typical for central breakers where the problem occurred with high frequency (Figure 6 and Table I). End breakers had this issue very rarely. Over five months shown in the graphs, alumina accumulation and narrow holes were more frequent at breaker 4 (Figures 7 and 8 and Tables II and III).

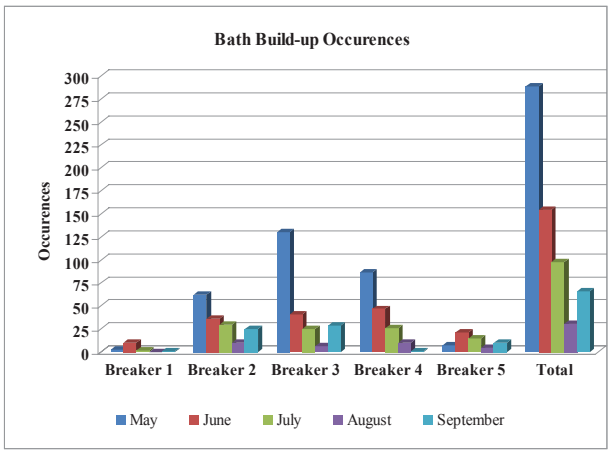

Figure 6. Number of bath build-ups per pot.

Table I. **Number of bath build-ups per pot.**

Month	Breaker 1	Breaker 2	Breaker 3	Breaker 4	Breaker 5	Total
May	3	62	130	87	7	**289**
June	10	36	41	47	21	**155**
July	2	30	25	26	15	**98**
August	0	10	6	10	5	**31**
September	1	25	29	1	10	**66**

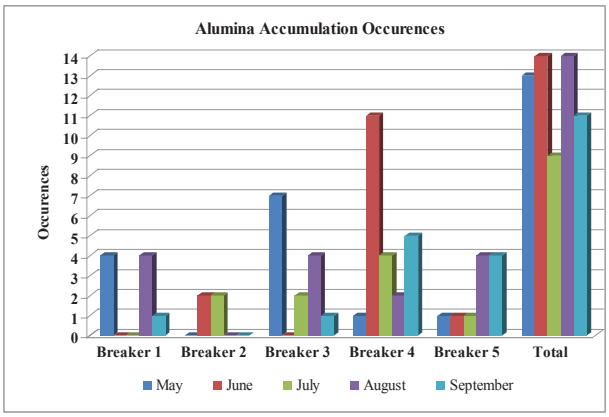

Figure 7. Number of alumina accumulations per pot.

Table II. **Number of alumina accumulations per pot.**

Month	Breaker 1	Breaker 2	Breaker 3	Breaker 4	Breaker 5	Total
May	4	0	7	1	1	**13**
June	0	2	0	11	1	**14**
July	0	2	2	4	1	**9**
August	4	0	4	2	4	**14**
September	1	0	1	5	4	**11**

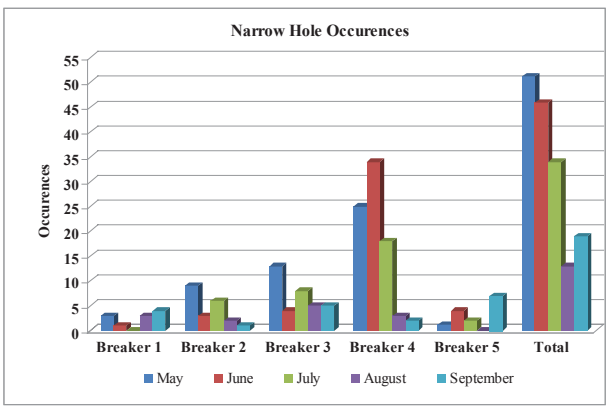

Figure 8. Number of narrow holes per pot.

Table III. **Number of narrow holes per pot.**

Month	Breaker 1	Breaker 2	Breaker 3	Breaker 4	Breaker 5	Total
May	3	9	13	25	1	**51**
June	1	3	4	34	4	**46**
July	0	6	8	18	2	**34**
August	3	2	5	3	0	**13**
September	4	1	5	2	7	**19**

Individual breaker control and results

To solve the problem, the breaking control logic was modified in order to control all breakers using individual parameters. The breaking frequency and the deactivation delays were adjusted to reduce risks of overheating of central breakers. At the same time it was necessary not to compromise breaking efficiency to avoid narrow holes and alumina accumulation issues. The new breaker control logic was introduced in May 2015 and optimized until the end of July 2015.

The results shown in Figures 6 to 8 and Tables I to III are encouraging. The bath build-up occurrences have been reduced by a factor of four and the narrow holes by a factor of three. The alumina accumulation occurrences, which are the least frequent, have been also reduced by 20 %. It is obvious that the three crust breaking conditions discussed in this paper depend on many factors, such as bath composition, bath height control, the state of the crust and alumina properties. Further optimization is being pursued with the goal to eliminate these problems entirely. In the meanwhile, some intervention of the operators will still be required.

Conclusions

Bath sensing is an essential feature of DUBAL's advanced cell technologies. Its bath detection efficiency is very high, > 97 % since February 2015. This prevents many breaker bath build-ups. Remaining rust breaking problems, such as bath build-up, alumina accumulation at breakers and narrow holes in the crust have been greatly reduced with modified control logic since its introduction in May 2015. Further optimization of the crust breaking and feeding logic continues with the goal to eliminate these problems entirely.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SiC IN ELECTROLYSIS POTS: AN UPDATE

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Keywords: SiC manufacture, SiC degradation, SiC corrosion resistance

Abstract

This update reviews recent results of physical and chemical testing of Si_3N_4-bonded SiC sidelining material for aluminium electrolysis pots, as well as the corrosion mechanisms of SiC materials under electrolysis conditions. New test methods have been developed to show how these materials perform in a harsh operating environment.
The attack by low temperature bath was tested on SiC bricks, also with SiAlON and N-SiC as binder phase. Reverse reactive sintering has become an important manufacturing process for Si_3N_4-bonded SiC sidelining material
This review updates [1, 2].

Manufacture

Liu et al. [3] developed self-bonded 95% SiC material, which has a high thermal conductivity and perfect cryolite resistance. α-SiC-bonded SiC or recrystallized SiC is expensive because it requires high α-SiC raw material and a sintering temperature of more than 2000°C. A lower temperature suffices when introducing a special additive. This self-bonded Sicatec 95 produced by LIRR has excellent thermal shock, oxidation and alkali resistance in laboratory tests.
The production of N-SiC was proposed by Danilova et al.[4]. In this process silicon oxide may appear in the end product if the nitrogen flow is not sufficient or if the gas flow diminishes for some reason. The main quality requirement of N-SiC is a long service life. in use as side lining for aluminium reduction cells. The actual content of SiC, Si_3N_4 and SiO_2 may change due to the raw materials and the firing process. A very close control of all technological parameters such as mixing, shaping and firing of N-SiC refractories is essential.
Jones et al. [5, 6] produced Si_3N_4-SiC and β-SiAlON-SiC reaction-bonded silicon carbide refractories for aluminium smelting. They assessed the corrosion resistance under simulated aluminium smelting conditions. The formation of the Si_3N_4 and SiAlON phases were studied by reaction bonding of silicon powders in a nitrogen atmosphere at low temperatures to promote the formation of silicon nitride, followed by a higher heating step to produce β-SiAlON composites of different compositions.
As the costs of the traditional production process for Si_3N_4 – SiC are too high, Sun and co-workers [7, 8] proposed the reverse reaction sintering process (RRS). The properties of this composite reveal that the new sintering method proceeds by the active sintering of neonatal Si_2N_2O or SiO_2 phases. The products have excellent physical and chemical properties due to their dense structure and to in-situ formed oxides or gaseous sub-oxides which are deposited on the boundary layer between Si_3N_4 and SiC particles near the surface. Small pore size is the key factor which limits molten metal or cryolite penetration, and it is apparent that the pore size in the RRS material is noticeably smaller than when made by traditional methods.

SiC in aluminium electrolysis pots

While Grachev [9] and Wu et al. [10] reported the use of Si_3N_4 – SiC composites as sidelining material in aluminium reduction cells, Wang, Skybakmoen and Grande [11] analyzed the chemical degradation of the Si_3N_4 – SiC sidelining. The upper part was subjected to oxidation by the pot gas, where the main corrosion product found was Si_2ON_2. The lower part was infiltrated by $Na_{(g)}$ diffusing from the carbon cathode forming Na_2SiO_3 as the main oxidation product. Over time this changes the thermal conductivity of the sidelining material. But the degradation of the sidelining also depends on the operation conditions of the cell.

SiC degradation of sidelining in aluminium reduction cells

The degradation mechanism of the Si_3N_4 – SiC sidelining of aluminium reduction cells has been examined many times and reviewed in the past [12- 20].
According to Etzion et al. [15], the corrosion results from a combination of attack mechanisms: oxidation of the binder phase (which is thermodynamically more prone to oxidation than the SiC grains) by Na ions which penetrate from the bath, producing SiO_2xNa_2O silicate species. These silicate materials are more prone to oxidation compared to Si_3N_4. CO_2 causes further oxidation of the binder phase. When attacked by corrosive gases like HF and $NaAlF_4$, the oxidized glass releases volatile $SiF_{4(g)}$. CO_2 further oxidizes the remaining binder, and starts another cycle of attack by corrosive gases. It was also observed that βSi_3N_4 in the core of the bricks is more prone to both oxidation and attack by corrosive gases which are found in the reduction cells.
Skybakmoen et al. [19] demonstrated in laboratory scale tests the important role for the chemical stability played by the total oxygen level (mainly as Si_2ON_2, SiO_2), as well as by the phase composition (α-Si_3N_4 and β-Si_3N_4) and microstructure. Apparent porosity, density and the content of pure Si proved to be of less importance regarding chemical oxidation resistance, while an increasing surface area proved to reduce the stability. In the upper part of the cell lining, the degradation proceeds by selective oxidation and by the formation of needle like α-Si_3N_4 by a combination of oxidizing gases (air, CO, CO_2) and HF gas, resulting in the formation of $SiON_2$ and grain detachment. In the electrolyte zone, the lower part of the sidewall, the degradation was characterised by the formation of Na_2SiO_3. The oxidation products filled the pores, so lessening the open porosity, but also lowering thermal conductivity. The nitride binder appears to be the weak part of the material because of chemical kinetics. High oxygen content leads to a more rapid degradation. Micostructure has a significant influence on the chemical stability of the material. SiC with high surface area degrades more severely. Fluoride gases play an important role on the degradation by causing detachment of large SiC grains.
According to Yurkov et al. [20] the upper sidewall region undergoes corrosion when silicon carbide and silicon nitride react

with oxygen from the air and with carbon dioxide from the anode gases as follows:

$$SiC + 2O_2 = SiO_2 + CO_{2(g)} \quad (1)$$

$$SiC + 3CO_2 = SiO_2 + 4CO_{(g)} \quad (2)$$

$$Si_3N_4 + 7\,O_2 = 3\,SiO_2 + 4\,NO_{2(g)} \quad (3)$$

$$Si_3N_4 + 6\,CO_2 = 3SiO_2 + 6\,CO_{(g)} + 4\,N_{2(g)} \quad (4)$$

At the bath surface level, SiC block deterioration occurs according to Proshkin et al. [17] as follows:

$$2SiC + NaAlF_4 + 2O_2 = SiF_4 + NaAlSiO_4 + 4C \quad (5)$$

$$2Si_3N_4 + 3\,NaAlF_4 + 6\,CO_2 = 3SiF_4 + 3NaAlSiO_4 + \\ + 6\,C + 4\,N_2 \quad (6)$$

This corrosion forms a groove at the top of the bath that reduces sidewall block thickness. Therefore a protective, stable, solid sideledge is necessary to prevent severe SiC-block degradation.

Physical and chemical properties

Skybakmoen et al. [21] tested a large number of different commercial nitride-bonded SiC sidelining materials produced worldwide. They summarized the following properties: apparent porosity, density, mineral phase analysis (α-Si_3N_4, β-Si_3N_4, Si_2ON_2, Si, and SiC) chemical analysis, bending strength, hot modulus rupture, cold crushing strength, thermal expansion, thermal conductivity, oxidation resistance, and chemical resistance. Since the first test in 1997, they performed chemical resistance tests on 400 samples from 37 suppliers and found an overall range of resistance: volume loss from 0.07 – 20.65% within an average value of 5.03 ± 3.44%. With respect to chemical resistance the average degree of corrosion is 5 on a scale from 0 – 10 (0 is the best value). Good quality materials are defined as from 0-3 (35%), middle range from 4-7 (42%) and of poor quality range 7-10 (24%). The best protection for the SiC-based sidelining is a stable, frozen sideledge.

Yurkov and co-workers [22] observed that in general SiC and Si_3N_4 are oxidation resistant under temperature conditions of 700-800°C. But in the presence of alkali and fluorine vapours the oxidation reaction is fast, and causes a volume increase due to the reaction of silicon with nitrogen.

While Jiang [23] determined the thermal conductivity of new lining materials, Okayasu [24] determined the mechanical and fatigue strength of silicon ceramics in liquid aluminium and its alloys. The bending strength of Si_3N_4 ceramic in molten aluminium is about 20% lower than obtained at 750°C in air. A significant decrease in fatigue strength occurs due to the iron content in molten aluminium reacting with silicon.

Corrosion resistance tests

Some corrosion resistance tests were developed recently [3, 5, 6, 25-30]. Jones et al. [5, 6] studied the corrosion of SiC bodies in a laboratory scale aluminium electrolysis test, where the samples were attacked by liquid metal, molten bath and corrosive gas. The corrosion resistance of the samples depended on the composition and on the environment during corrosion. The gas phase caused higher corrosion than the bath attack.

Cao et al. [25] also tested SiC bricks for their bath and oxygen resistance. They showed that the corrosion happens according to the reactions cited already by Yurkov et al. [20]. Bricks with a binder content of about 20-22 mass % Si_3N_4 showed good corrosion resistance.

A rotating-CO_2-cryolite test method was developed by Zhao et al. [26] and LIRR [29] in order to simulate the actual working conditions in reduction cells when no sideledge is present. Selective test parameters were found to be 1000°C, holding time 24 hours, CO_2 flow rate 1.0 L/min, rotating speed 45 rpm. The bath composition was 88 wt. % cryolite, 5 wt. % CaF_2 and 7 wt. % Al_2O_3. The corrosion degree was determined by volume loss. Without a protective sideledge the volume loss was found to be about 4.4 %. Especially during anode effects the SiC sidelining is corroded due to the high bath temperature.

Cao [27] proposed also another corrosion resistance method in order to test SiC-Si_3N_4 material. It uses a CO_2-aerated cryolite alumina bath. The CO_2-aeration makes the bath the dynamically corrosive towards the SiC-samples

A potassium cryolite resistance test was proposed by Frolov et al. [28]. The examined SiC-Si_3N_4 sample was found to resist corrosion during 72 hours in potassium cryolite at 800°C. The most corroded part of the sample is the three-phase boundary. Corrosion was found to be five times lower in potassium cryolite at 800°C than in sodium cryolite at 960°C. The oxidation product dissolved in the electrolyte and anode gas CO_2 and SiF_4 together attack the sidewall.

As SiC and Si_3N_4 can react with all components of the bath, gas phases and aluminium, Mikhalev et al. [30] determined the SiC blocks in aluminium electrolysis cells to be chemically unstable. The degree of wear depends on the physical and chemical properties of the block. Samples were kept in air, oxygen, CO_2 and water steam at 500 to 1150°C for 20 to 500 hours, during which the sample weight was measured. In another test, samples were kept in the bath or aluminium at 1000 to 1030°C for 22 to 48 hours. Then the aforementioned methods were combined. In general the quality of SiC blocks can be evaluated according to the three parameters: chemical composition, resistance towards wear and resistance towards oxidation.

Conclusion

Without protection, a sidelining made of silicon nitride-bonded silicon carbide is – under severe test conditions – chemically not stable and will corrode, the binder matrix being the weak component.

The degradation mechanism of sidelinings made of Si_3N_4-SiC has been determined. Oxygen from air and CO_2 from the anodes destroy the upper part of the sidelinings. Also fluoride gases play an important role in the degradation by causing detachment of large SiC grains.

Several corrosion resistance tests have been developed to take into account the corrosive attack by gases together with liquid bath and aluminium.

Acknowledgement

The author gratefully acknowledges with thanks the help of Mr. Bushnell during the preparation of this manuscript.

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Light Metals 2016

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ASSESSING THE ROLE OF SMELTER GRADE ALUMINA POROSITY IN THE HF SCRUBBING MECHANISM

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Keywords: HF, Dry Scrubber, Alumina, Pore Size Distribution

Abstract

The HF scrubbing capacity of SGA and its HF generating potential have been discussed in recent and past publications. Researchers are still in search for the ideal SGAs with optimum surface area and calcination levels that meet potroom and GTC specifications. It has been shown that pore size distribution is a critical parameter for assessing the suitability of SGAs for GTC operations as the bulk of the accessible surface area for HF scrubbing is attributable to internal porosity. Laboratory scale experiments have been carried out to understand the evolution of the bulk alumina pore network as reaction with humid HF gas proceeds. A reduction in reacted alumina BET surface area and total pore volume is observed with time, after a marginal initial increase. In this paper, we discuss findings on the involvement of pores in the mechanism of HF scrubbing and its possible influence on the fluoride capture kinetics.

Introduction

Aluminum smelters all around the world are intensifying efforts to minimize total HF emissions to limits that are challenging to attain. Generation, evolution and scrubbing of HF gas need to be well-understood and treated as parts of a single cycle in order to achieve benchmark limits of 0.2kgF/ton Al [1]. Health, legal and environmental considerations have driven the efforts towards reducing the amounts of PFC emissions, as well as fugitive and stack HF emissions from aluminum smelters. The drive towards higher amperage operations, and the consequently higher fluoride consumption, causes the pot exhaust gases reaching the GTC to be of a higher temperature and fluoride concentration at constant draft. Even with an increased draft, the GTC becomes overloaded, reaching its capacity very quickly, resulting in a reduced gas cleaning efficiency.

Smelter grade alumina (SGA) is the sole adsorbent for removing fluorides from pot exhaust gases. However, it has also been found to be a continuous source of hydrogen, either through moisture or retained hydroxyls, which react with fluoride species in bath to generate HF gas [2]. Over the years, smelters have specified the type of alumina that enables efficient production of aluminum at lower energy costs, lower fluoride consumption, improved dissolution, minimal dusting and minimal total fluoride emissions. Due to the multi-purpose use of alumina in feeding, anode covering and pot gas scrubbing, smelters require SGA with optimum residual hydroxyl content, particle size distribution, specific surface area and pore size distribution. Generally, surface area is a more scrutinized parameter as smelters with GTCs first use the alumina for scrubbing HF gas. From the works of Lamb [3] and Cochran [4] a positive correlation between surface areas of aluminas and adsorption capacity was established. Under-calcined gibbsite in SGA which are the major source of structural hydroxyls, have high specific surface area and very narrow pores [5]. The interplay of specific surface areas, hydroxyl contents and pore size distribution in SGA calls for some critical analyses directed at optimizing the HF adsorption capacity of SGA. In this paper, we analyze the role of SGA porosity in the efforts to return more than 99.5% of total fluoride emission to the reduction cells.

Evolution of Pores in SGA

SGA is produced by the Bayer process which comprises four main sub-processes: digestion, clarification, precipitation and calcination. The pores in alumina are formed predominantly during the calcination process where Bayer gibbsite (aluminium tri-hydroxide, $Al(OH)_3$) is heat-treated in either a rotary kiln or in a fluidized bed calciner up to about 1100°C. The calcination strategy is driven by the requirement to meet a specific surface area target while maintaining a low residual hydroxyl content (Loss on Ignition, LOI). The structural transformations that occur in gibbsite during the initial stages of the thermal dehydroxylation are pseudomorphic, resulting initially in high specific surface area as porosity develops. Further heating causes pore coalescence, growth of capillary pores and pore size expansion which are accompanied by a reduction of the specific surface area [6]. The capillary pores (structural channels) connect the voids and cracks in the calcined agglomerates, forming a complex network of pores and channels which have a bearing on the performance of SGA as a scrubbing medium in dry scrubbers. Figure 1 shows agglomerates of SGA. A sectioned agglomerate is also shown in the SEM image in Figure 2.

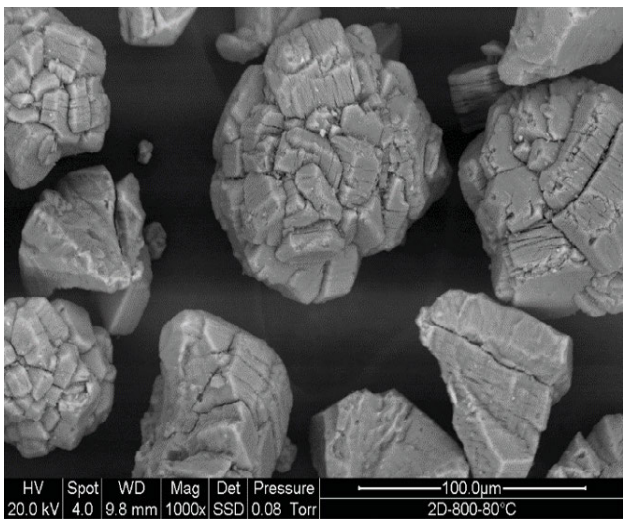

Figure 1: SEM image of agglomerates of Smelter Grade Alumina

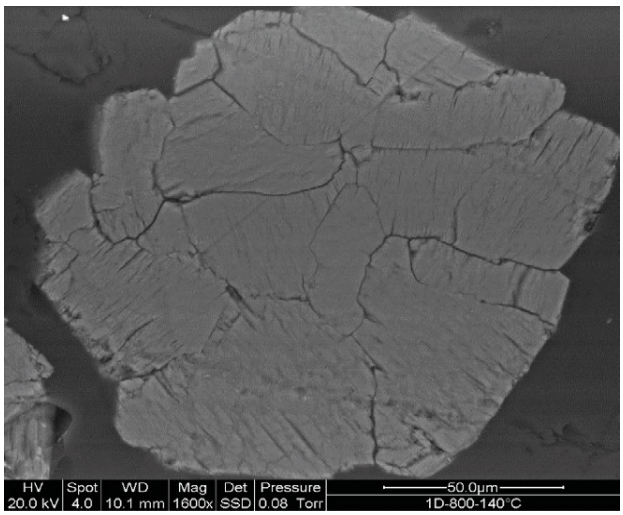

Figure 2: Sectioned grain of SGA showing pores and channels

Nitrogen porosimetry is the most common method for measuring the BET specific surface area and BJH pore size distribution (PSD) for SGA. Alumina specification sheets only report the BET surface area of a consignment. However, it is commonplace to find two different consignments of SGA with similar BET surface areas and yet possess two unique pore size distributions, as fractions of over- and under-calcined alumina can be blended to achieve a target BET surface area. Earlier investigations into the evolution of alumina microstructure by Perander *et al* [5, 7] re-emphasized the importance of pore size distribution in relation to the HF adsorption capacity of SGA. The bulk of the surface area, measured by conventional nitrogen adsorption, is associated with the small pores in the pore network of the alumina. However these smaller pore (1-20nm) are likely to get blocked and rendered inaccessible by the reaction product. PSD plots are shown in Figure 3 for primary unreacted SGA and industrially fluorinated SGA.

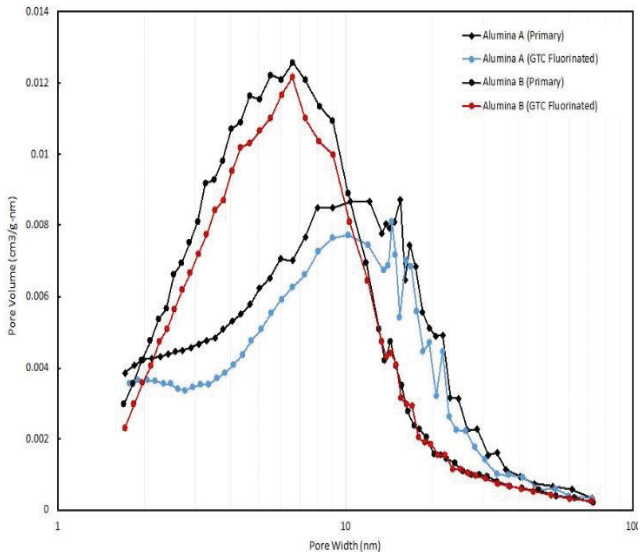

Figure 3: Pore Size Distribution for Primary and Secondary (GTC Fluorinated) SGA [BET SA reduction – Alumina: A 69.0 to 60.7m^2/g. Alumina B: 73.1 to 67.0m^2/g]

Gillespie [8] suggested aluminum hydroxyfluoride ($AlF_x(OH)_{3-x}$) or aluminum oxyfluoride (AlF_2O) as the product formed when humid HF gas reacts with active hydroxyls in the alumina. Considering the reduction in the proportion of small pores (1-20nm) as shown in Figure 3, a study was undertaken to understand the evolution of the pore size distribution of reacted alumina and to assess the influence of the pore geometry on the mechanism and kinetics of HF scrubbing in modern dry scrubbers.

Method

Experimental Setup

A cylindrical stainless steel column (16.5cm long x 1.5cm ID) coated with an HF-resistant fluoropolymer was charged with one gram of primary SGA. Fluidization of the bed of alumina occurred when 320mg/Nm^3 of humidified HF gas was injected at the reactor inlet. A Nafion® humidity exchanger supplied 44gH_2O/kg dry N_2. Porous Teflon retaining frits were installed at reactor inlet and outlet to prevent alumina entrainment in the gas. A constant humid HF gas flow of 500cm^3/min was ensured via high precision mass flow controllers and a reaction temperature of 110°C was maintained by installing the reactor and all associated fittings in an oven. HF concentration was continuously measured in a Teflon gas cell (20cm path length) with a Boreal Laser GasFinder™ 2.0 HF monitor. HF exiting the gas cell was subsequently bubbled through a solution of $Ca(OH)_2$ for disposal.

Breakthrough Curves

In breakthrough experiments, the SGA would usually be reacted with HF until saturation – a point at which the inlet HF concentration equals the outlet HF concentration. The HF concentration versus reaction time curve for alumina saturation is sigmoidal in shape. In this investigation, the alumina was reacted with humid HF for specific time durations of 10, 25, 40, 70, 150 and 208mins (corresponding with breakthrough). The reaction reaches breakthrough when HF concentration at the reactor outlet measures about 8.5mg/Nm^3. The breakthrough threshold of 8.5mg/Nm^3 was chosen as a standard threshold for all experiments based on results from a measurement system analysis of the Laser measurement of HF concentration with variations in experimental conditions. At this value, the signal to noise (S/N) ratio at the instrument detection limit (IDL) far exceeded the statistically accepted value of 3. From an industry point of view, the breakthrough threshold-to-input concentration arrived at is equivalent to detecting the signal at a scrubbing efficiency of ~97.5%. Figure 4 shows overlaid plots of reactor outlet concentration versus time (breakthrough/capture curves). The 'turn off points' labelled on the plot indicate the end of the target dosing times at which a two-way valve was actuated to bypass the reactor, thereby directing HF to the Laser measurement gas cell.

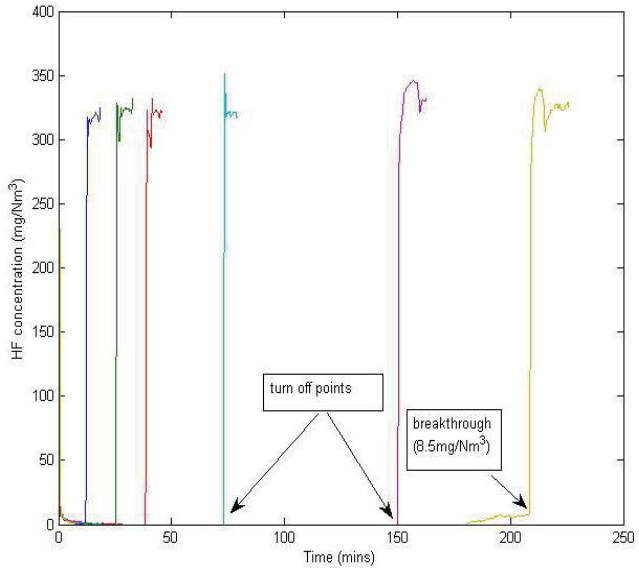

Figure 4: Capture curves (HF concentration vs time). 1g of SGA reacted with humid HF gas for specific durations until the last one reaches breakthrough

Physical and chemical analyses

Surface area and pore size distributions were analyzed with a Micromeritics Tristar 3000 instrument. The Brunauer-Emmet-Teller (BET) and Barrett-Joyner-Halenda (BJH) methods [9] were utilized to extract surface areas and pore size measurements respectively from the N_2 physisorption isotherms. The measurements were conducted under liquid nitrogen temperatures (77K) with nitrogen as the adsorbed gas. Bulk fluoride loading on the fluorinated SGAs were calculated from the reaction times, HF concentration and mass flow rates. These were subsequently confirmed by 19F-NMR measurements.

Results and Discussion

Table 1 below shows the reaction times, the amounts of fluoride captured and the measured BET surface area after reaction.

Table 1: Experimental results of HF loading and surface measurements

Reaction Time (mins)	HF loading mgF/gAl_2O_3 (%)	BET S.A after reaction (m^2/g)	Avg. pore width (nm)
0 (unreacted)	0.0	70.57	9.6
10	0.2	70.92	9.7
25	0.4	71.36	9.7
40	0.6	75.15	9.8
70	1.1	70.77	10.1
150	2.4	62.98	11.0
208 (breakthrough)	3.2	52.62	12.4

Pore size distribution plots for the timed alumina/HF reactions are shown in Figure 5.

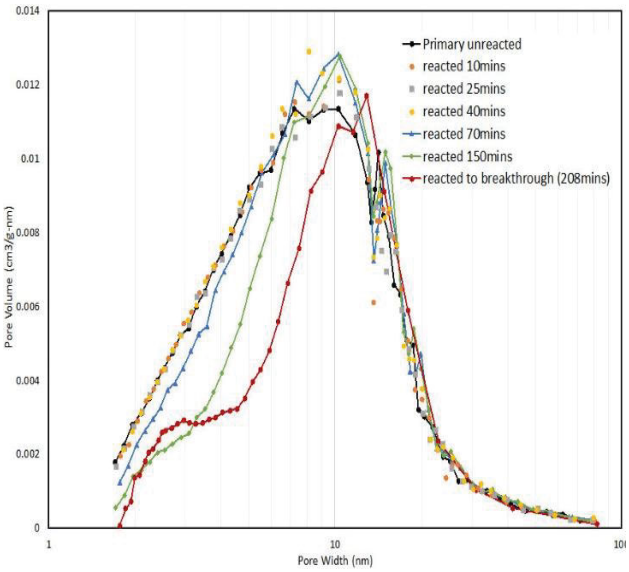

Figure 5: Pore Size Distribution of unreacted and reacted aluminas

There is a general decrease in BET and Langmuir surface areas with increasing reaction time as had been the case in previous experiments by the authors [10, 11]. However, an initial increase in BET SA is observed as reaction proceeds up to the 40-minute reaction time and most likely some minutes beyond. The increase in surface area is likely due to the chemisorbed reaction product, about which little is known in terms of its surface reactivity and structural properties. Errors in calculating the specific surface area may therefore arise as a result of the change in alumina surface chemistry. The subsequent reduction in surface area is the result of gradual pore blocking and coverage of active sites with reaction products. Primary alumina has been reacted with HF to complete saturation in other experiments [10]. The BET surface area after saturation was found to be less than half (~30-35m^2/g) of the starting surface area depending on experimental conditions. From Table 1, average pore width is seen to increase with reaction time. This observation is consistent with smaller pores being 'eliminated' from the distribution. The van der Waals diameter of HF and N_2 molecules are 0.36nm and 0.16nm respectively at STP. During pore size measurements, nitrogen probe molecules would adsorb to the remaining accessible sites on the reacted alumina, most of which is expected to be mesopores (2-50nm) and macropores (>50nm). The population of micropores (<2nm) appears to be significantly reduced after scrubbing as seen from the PSD plots (Figure 5).

Modeling Pore Size Distribution of Fluorinated SGA

The involvement of the SGA pore network in HF scrubbing has been discussed in recent publications [10, 11]. Analyses of the pore size distribution (PSD) of primary alumina reacted with HF until breakthrough and saturated alumina exhibited distinct redistribution of micropores and mesopores in both laboratory and plant studies. McIntosh *et al*. [11] modelled the pore evolution during HF scrubbing by assigning different capture mechanisms/interactions between HF and the different pore size groups within SGA. The model results were at variance with the

widely-published description of the HF capture process by a monolayer model. The monolayer model only fit adsorption in pores above 20nm and also in cases of low fluoride loading on alumina. A more holistic approach towards understanding the role of SGA porosity was to consider monolayer coverage in wide pores, pore length attenuation in the likely-corrugated pores, kinetic inaccessibility of HF molecules due to the effects of diffusion transitions, e.g. gas to Knudsen diffusion in narrow pores and complete plugging of entrances in very small pores (1-3nm). Figure 6 shows the partitioning of the reaction zones according to pore width - namely, pore blocking zone, kinetically inaccessible zone and a mono- or multilayer formation zone.

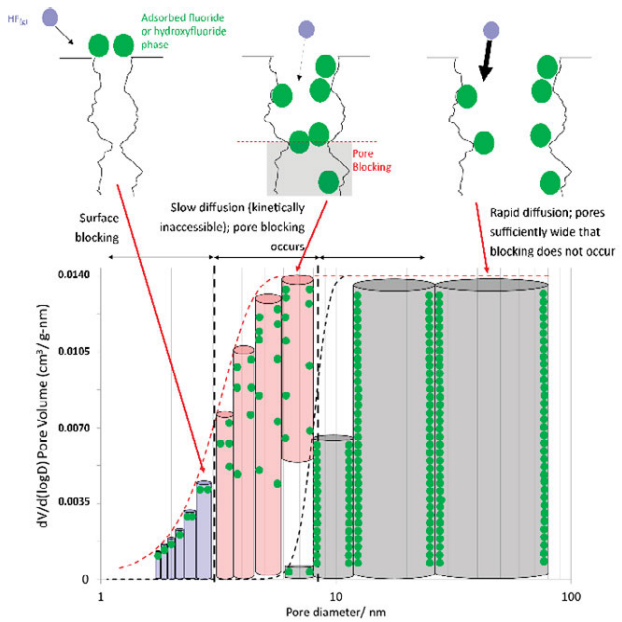

Figure 6: Graphical representation of combined model for pore blocking, pore attenuation and mono/multilayer formation

The model satisfactorily fit the observed reduction in the pore volume of pores between 1.7 – 20nm and also calculated an assumed monolayer thickness of ~0.7nm in agreement with the estimates of Lamb (0.65-0.83nm) [3].

Implications of SGA Porosity on Scrubbing Efficiency in GTCs

There are inherent limitations in BJH method of measuring alumina porosity because the assumptions about the adsorbate-adsorbent interactions become inapplicable below 1.5nm thereby rendering the BJH model unreliable. This notwithstanding, the nature and distribution of pores in SGA can be assessed and its evolution with HF scrubbing can be harnessed to explain the role of porosity in the capture mechanism. Nitrogen adsorption and desorption isotherms support evidence that SGA microstructure is made up of a network of irregularly-shaped and interconnected slits pores. The presence of rod-shaped and ink bottle shapes have also been earlier on inferred [12]. The pores maintain their shape even after saturation with HF in the laboratory reactor (Figure 7). Average pore widths measured for six commercial primary SGAs ranged from 7.5 – 11nm.

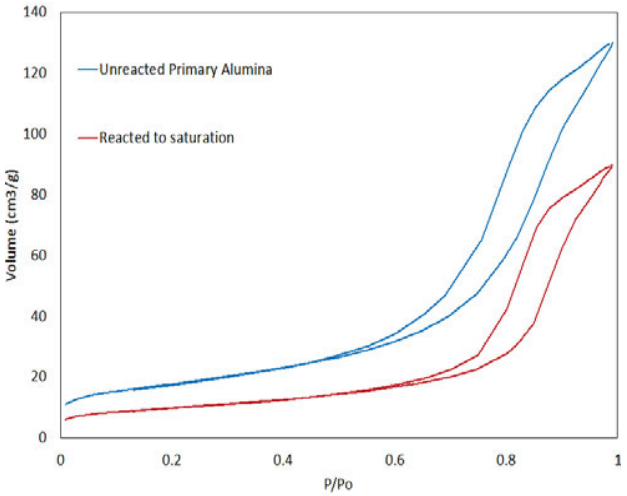

Figure 7: Nitrogen gas adsorption and desorption isotherms for primary and fluoride saturated SGA

Analyses of GTC fluorinated alumina and primary unreacted alumina (Figure 8) shows that the pore size distribution is not significantly altered after scrubbing. Secondary (fluorinated) aluminas unlike the lab fluorinated aluminas include condensed $NaAlF_4$ vapors. Heiberg *et al* [13] reported that these condensed bath fumes were the main source of uncaptured HF from GTC stacks. The mechanism involves humid pot gas reacting with $NaAlF_4$ fines on the filter cake and filter bags to produce about a third of the stack HF emissions.

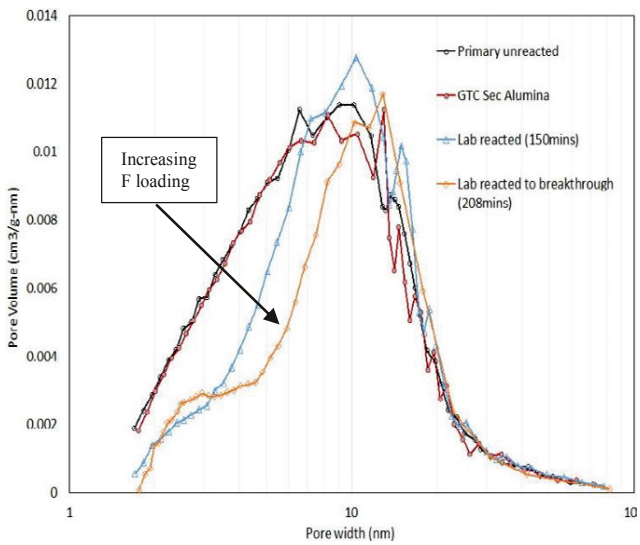

Figure 8: Pore Size Distribution of GTC secondary alumina in comparison with laboratory fluorinated alumina

In the laboratory experiments with humid HF gas, it was generally found that breakthrough capacity for alumina was in excess of 3 - 4% by weight while typical fluoride content of GTC secondary alumina is 1.5-2wt%. SGA can be saturated with HF up to 7wt% depending on the operating conditions [10]. However, the experiments for this publication were designed to assess breakthrough capacity of the unreacted alumina for an equitable comparison to industrial secondary alumina. The lab reacted

aluminas, upon nitrogen adsorption measurements, exhibited a significantly different pore size distribution from the originally unreacted SGA. When pore size distributions of GTC-fluorinated alumina and lab-fluorinated alumina were compared, differences were found in the proportions of micro- and mesopores present after HF capture. The gradual reduction in the proportion of pore volume attributable to pores between 1.7 – 10nm has been observed to positively correlate with increasing HF loading [10, 11]. The commercial aluminas shown in Figure 3 recorded a reduction in BET SA from 69.0 to 60.7m^2/g (12%) for alumina A and reduction from 73.1 to 67.0m^2/g (8%) for alumina B. BET SA of lab fluorinated SGA on the other hand, reduced from 70.6 to 63.0m^2/g (11%) after 150mins of reaction with HF and a further reduction to 52.6m^2/g (25%) at breakthrough (Table 1). Since the major component of the fluoride measured in secondary SGA is contributed by the condensed $NaAlF_4$ fumes, the slight reductions in specific surface area and the accompanying minimal change in pore size distribution suggests under-utilization of the full scrubbing capacity of the primary SGA.

It may be argued that the nature of the lab scale scrubber allows for higher gas/solid contact time for more HF to react with alumina. However, the outcome of this study should kindle an innovative approach to dry scrubber design for aluminum smelters. A combination of knowledge from fluid bed, cascading and multiple-feeder GTC technologies has the potential of realizing the synergistic effect of increasing dry scrubber efficiencies beyond the typical 99.5%, while maximizing the fluoride carrying capacity of the alumina for a compact fluoride recycle loop.

Conclusion

HF scrubbing efficiency in the GTC is dependent on operational, environmental, technological and material selection factors. The procurement of SGA is very important due to its cyclical role in HF scrubbing, aluminum production and HF generation. The pore size distribution of SGA is often overlooked while more attention is given to the specific surface area. Results from ongoing work suggests that the HF capture mechanism and the kinetics are influenced by the nature and distribution of pores in SGA. The evolution of pores was tracked by reacting known amounts of HF with alumina. The evidence suggests that the pore network which holds more than 90% of active reaction sites in alumina, undergoes changes that correlate with increasing HF loading and are consistent with phenomenon such as pore blockage, pore length attenuation and mono- or multilayer coverage. According to pore size distribution model results, the HF capture mechanism involving pores appears to be elemental for both GTC and laboratory fluorinated aluminas. However, industrial secondary aluminas were found to exhibit lesser magnitudes of change in pore size distribution than laboratory reacted alumina. The typical fluoride concentration in GTC secondary alumina is ~1.5 – 2% whereas the breakthrough capacity reached over 3% for lab fluorinated aluminas. These findings present opportunities for gas/solid contact times to be optimized for the purposes of increasing the fluoride loading on secondary SGAs.

Acknowledgement

The authors gratefully acknowledge Hydro Aluminum for permitting data from a related project to be used for this publication. We are also very grateful to David Cotton (LMRC) for the technical support during the experiments.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE COMPETITIVE ADSORPTION OF HF AND SO_2 ON SMELTER GRADE ALUMINA

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Keywords: HF, alumina, SGA, smelter grade alumina, hydrogen fluoride, SO_2, sulfur dioxide, adsorption, porosimetry

Abstract

Smelter grade alumina (SGA) is the preferred agent for recovering HF from aluminum smelter ventilation gas streams since the alumina also serves as feed to the electrolysis cells, thereby allowing return of the recovered fluoride to the process. This alumina-based recovery loop includes sulfur, since SO_2 evolved from cells is in fact adsorbed on the SGA. This paper explores the competitive adsorption/reaction behavior of SO_2 and HF and on smelter grade alumina as well as the impact of increasing HF exposure on the surface area and porosimetry of SGA.

Introduction

Commercial aluminum electrolysis pots evolve HF and SO_2 gases on a continuous basis. The HF gas evolves from the hydrolysis of molten salt bath and atmolite ($NaAlF_4$) vapor.[1] The SO_2 gas evolves from oxidation of COS formed at the anode surface from the 2-3% S content of the anode carbon as it is consumed during electrolysis.[2,3]

Nearly all (>99%) of the evolved fluoride is captured using dry scrubbers that employ smelter grade alumina (SGA) as the active agent. The reacted alumina is then fed back to the electrolysis cells to recover the evolved fluoride to prevent process losses and minimize fugitive emissions.

The reaction of HF on SGA as well as the chemical form of the surface reaction products has been characterized previously. [4,5] In a similar manner, the adsorption of COS, SO_2 on SGA and the influence of temperature and humidity have been investigated in prior art.[3,6-9] Lamb[10] had previously used consecutive adsorptions of SO_2, then HF to show that adsorbed SO_2 could be nearly quantitatively desorbed by subsequent high loadings of HF (>4 wt%). However, very little attention has focused on simultaneous co-adsorption behavior of HF and SO_2, which reflect actual conditions present in commercial aluminum gas treatment centers.

It is interesting to note that reacted SGA at commercial aluminum smelters typically contains ~0.2-0.3 %S and ~1.8-2.2 % F. All of this fluoride and $\geq$99% of this sulfur is from adsorption and reaction of gas phase species (SO_2 and HF) or entrainment of particulate phases from process off-gas. Particulate sulfur phases present in reacted SGA include Na_2SO_3 and $Na_3(SO_4)$.[11] Particulate fluoride phases on reacted ore include Na_3AlF_6, $Na_5Al_3F_{14}$ and $NaAlF_4$. In the case of fluoride, prior work has shown that several amorphous phases are also present, owing to the exhaustive irreversible chemical reaction of gaseous HF with SGA (AlF_3, $AlF_3.H_2O$, $AlF_3.2H_2O$, $AlF_3.3H_2O$ and fluoride that has substituted for terminal hydroxides on the SGA).[5] In other words no physically adsorbed HF exists on the surface of reacted SGA.

This paper employs controlled sequential and simultaneous loadings of HF and SO_2, together with surface area analyses of the reacted SGA to characterize the competitive adsorption behavior of these species as well as the impact on SGA surface area.

Experimental

Individual, sequential and simultaneous adsorptions of HF and SO_2 on SGA were performed under a laboratory hood. The general layout of the adsorption apparatus is illustrated in Figure 1. House air was dried (Drierite®) and passed through tubing to which a temperature controlled HF permeation oven (Kin-Tek Laboratories, Inc.) and permeation tube were employed for the addition of controlled mass flow rates of HF (typically at 200 or 300 ppm). Deliberate SO_2 additions (typically 100 ppm) were from a compressed gas cylinder using a mass flow controller (MFC) to ratio dilution. In this manner, HF and SO_2 concentrations could be maintained at concentrations similar to those observed at commercial dry scrubber inlets.

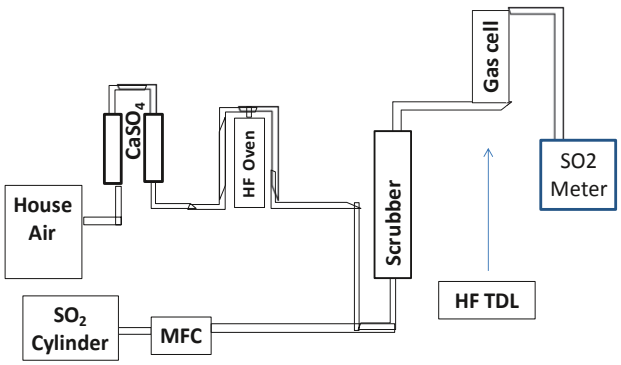

Figure 1. HF and SO_2 adsorption apparatus.

The fluidized bed lab scrubber was operated in a batch manner, where weighed aliquots (typically 5-10 grams) of SGA (various sources) were loading on to a PFA (perfluoroalkoxy) filter supported by a PFA grid in the bottom of a vertically oriented, temperature controlled PFA scrubber body (Savillex, Inc.).

HF and SO_2 concentrations were monitored in real time using an external gas cell with an open-path tunable diode laser HF monitor (Boreal, Inc.) and an internal gas cell based optical SO_2 monitor (Thermo, Inc). While not shown in Figure 1, the apparatus was also valved appropriately to allow measurement of either scrubber inlet or scrubber outlet HF or SO_2 concentrations. The gas stream exiting the SO_2 monitor was bubbled through an aqueous solution of calcium acetate to capture HF gas as CaF_2.

In addition to lab-loaded SGA, alumina from the same source (Alcoa) was loaded using a by-pass stream of off gas from a pilot scale electrolysis cell using a PFA fluid bed reactor. During cell off-gas scrubbing, a 10μ Teflon® filter was placed on the inlet side of the reactor to preclude the entrainment of particulate on the reacted SGA, while allowing vapor phase atmolite to pass.

SGA surface area and micro/mesopore size distributions were characterized from Nitrogen and Argon BET and multi-point adsorption and desorption isotherms using a Micromeritics Tristar II 3020 porosimeter. Sample degassing conditions were 3 hours at 200C under flowing nitrogen. DFT (slit pore models) were employed for fitting the isotherm data to pore size distributions.

Results and Discussion

The impacts of HF and SO_2 adsorption on each other and on the surface area and porosimetry of SGA are addressed in separate sections.

Impact of SO_2 on HF adsorption on SGA

Figure 2 shows temporal "breakthrough" profiles of HF on SGA, as well as the same experiments repeated, where the SGA was pre-loaded to saturation with SO_2 before initiation of SGA exposure to HF.

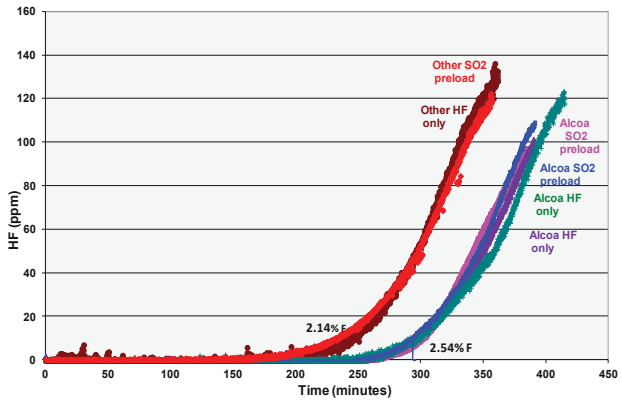

Figure 2. Temporal HF breakthrough behavior for two different SGAs that were pre-loaded to saturation with SO_2.

Identical masses of SGA from two different sources (Alcoa and other), were used to develop the data shown in Figure 2. The "other" SGA exhibits initiation of HF breakthrough, where the scrubber exhaust concentration systematically increases, at 200 minutes, while onset of breakthrough for Alcoa SGA occurs at 260 minutes, owing to the higher surface area of latter (63 vs. 73 m^2/g). For both SGA sources, the temporal breakthrough behavior of the as-is and SO_2 pre-loaded SGAs are identical. This data indicates that SO_2 has no impact whatsoever on the adsorption behavior of HF on SGA. This observation is because SO_2 is physically and reversibly adsorbed on SGA, while HF irreversibly chemically reacts with SGA.

Impact of HF reaction and SO_2 adsorption on SGA

Figure 3 compares temporal SO_2 breakthrough profiles of different aliquots of the same SGA source, where the alumina was pre-treated to four different pre-load conditions (0, 0.3 wt%, 0.6 wt% and 1.2 wt% F).

For each experiment shown in Figure 3, SGA was pre-reacted (using 300 ppm HF) to known fluoride loadings, then the SGA was exposed to SO_2 (100 ppm). The wt% S achieved in the 0 HF case was 0.52% S. This 0 HF experiment was repeated in Figure 3 to show method reproducibility. The maximum wt% S achieved on SGA systematically decreased as a function of wt% F loading, to 0.15% S on SGA that was pre-loaded to 1.2% F.

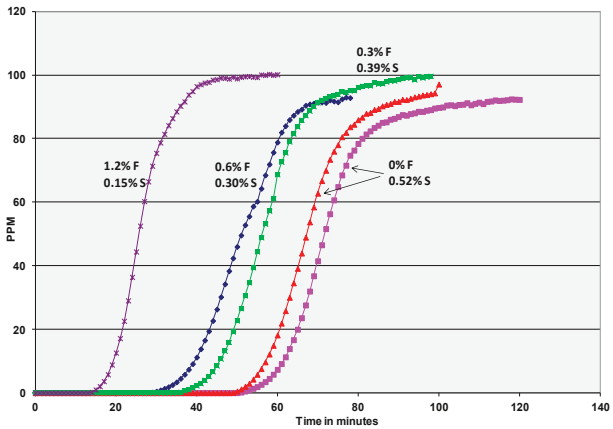

Figure 3. Temporal SO_2 breakthrough behavior for as-is and HF pre-loaded SGA.

This data supports several observations:

1) HF adsorbs at the same SGA surface sites as SO_2, and
2) SO_2 cannot adsorb on SGA surface sites that have reacted with HF.

SO_2 desorption by HF

Figure 4 shows a composite plot of the temporal breakthrough profiles for two separate experiments:

1) The adsorption of SO_2 on neat SGA, and
2) The quantitative desorption of SO_2 and corresponding HF breakthrough profile of SGA that was pre-loaded to saturation with SO_2 prior of HF exposure.

These experiments are all plotted starting at zero time in Figure 4.

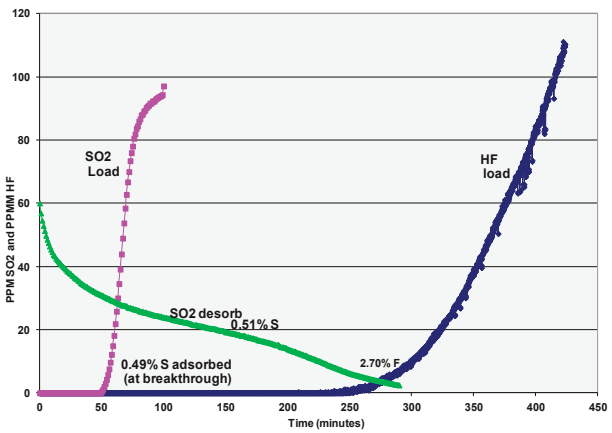

Figure 4. SO_2 load on as-is SGA and desorption due to subsequent HF loading.

The initial SO_2 loading adsorbed 0.49 wt% S on the as-is SGA. Upon exposure to HF (300 ppm) SO_2 desorption started immediately. Integrating the SO_2 monitor response allowed an estimation that 0.51% S had been desorbed from the SGA by the

time HF breakthrough has begun (at 2.7 wt% F load). In other words, adsorbed SO_2 was quantitatively removed from SGA by HF. The slight excess sulfur calculated from the desorption experiment probably results from residual SO_2 in the transfer lines of the test apparatus.

HF and SO_2 co-adsorption on SGA

In an aluminum smelter dry scrubber, SGA is exposed to HF and SO_2 at the same time. A temporal HF (200 ppm) and SO_2 (100 ppm) co-adsorption/reaction profile is shown in Figure 5. Both HF and SO_2 inlet concentrations were invariant over the duration of this test. In addition, the SO_2 only adsorption profile on neat alumina is superimposed on Figure 5 for comparison purposes.

Several observations are evident from the data shown in Figure 5.

1) Both SO_2 and HF adsorb on SGA during initial exposure (0-50 minutes) when there is no competition for surface sites.
2) SO_2 adsorbed on SGA to approximately 0.32 wt% S prior to initiation of SO_2 desorption. This contrasts to an SGA adsorption capacity of 0.49%S in the absence of HF.
3) The maximum concentration of SO_2 in the scrubber exhaust gas exceeded the inlet SO_2 concentration from 50 to 240 minutes of the test period. This is due to the sum of the inlet concentration (100 ppm) plus the desorbed SO_2.
4) The adsorbed SO_2 was quantitatively desorbed from the SGA the the time that HF breakthrough started (240 minutes).

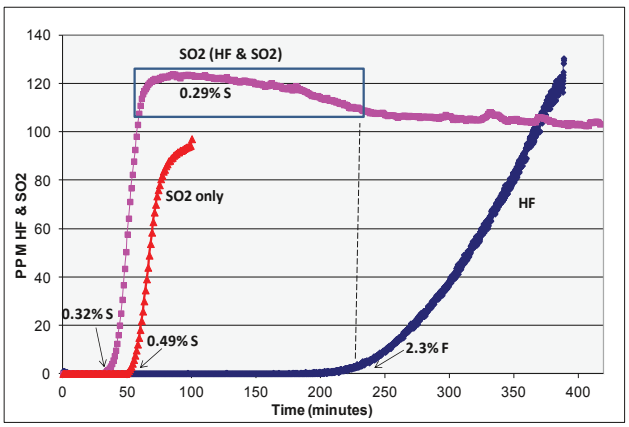

Figure 5. Temporal co-adsorption profile of HF and SO_2 on SGA.

The data shown in Figure 5 indicates that only a minor fraction of the sulfur present in reacted SGA from commercial dry scrubbers exists as physisorbed SO_2. The major fraction of the sulfur present in reacted SGA is in the form of particulate sulfur compounds, as discussed earlier in this paper.

Impact of HF on SGA surface area, pore size distribution, pore area and pore volume

The total adsorption capacity of SGA is proportional to the available surface area. Given the larger size of an SO_2 molecule to that of HF and the corresponding 5-fold difference in relative adsorption capacities at breakthrough, based on wt %S or F shown this report, it is reasonable to assume that SGA pore size may impact the relative adsorption capacities for HF and SO_2.

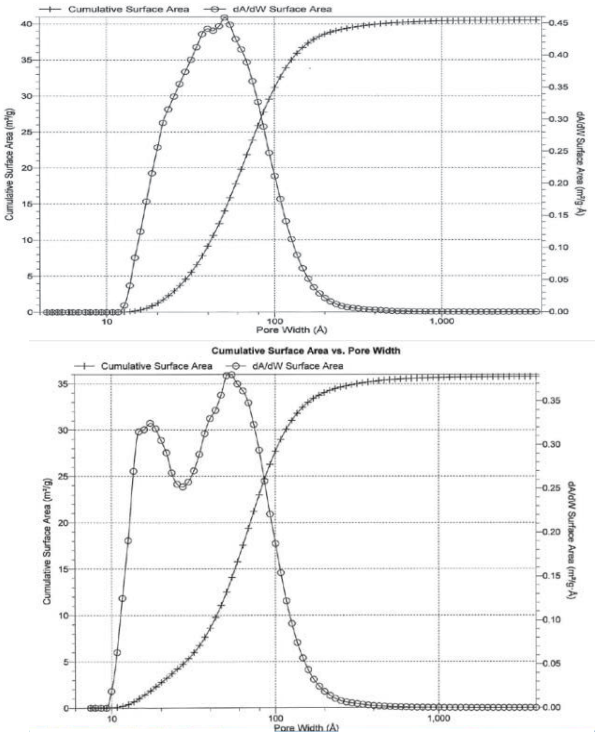

Figure 6. Pore width (angstroms) vs. (upper) cumulative surface area (m^2/g) and (lower) dA/dW surface area (m^2/g-angstrom) for as-is and 2.4%F SGA.

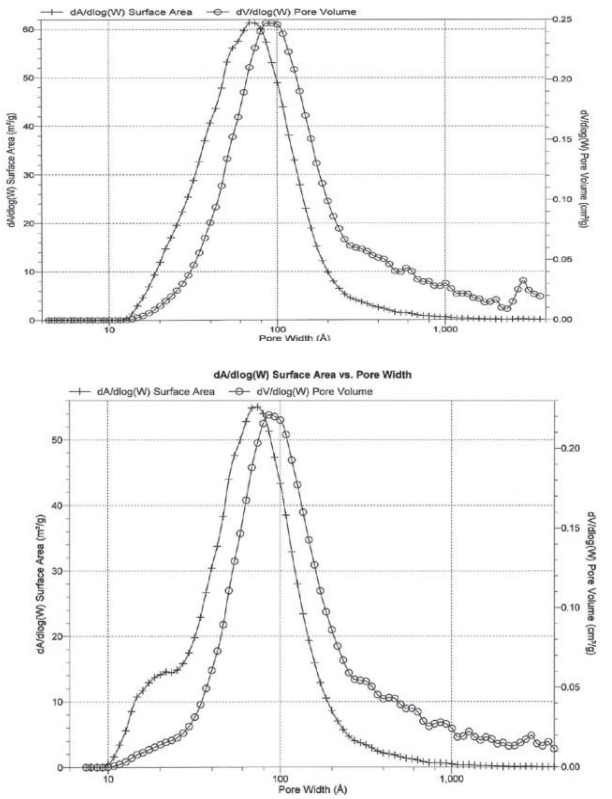

Figure 7. Pore width (angstroms) vs. (upper) dA/dlog W surface area (m^2/g) and (lower) dV/dlogW pore volume (cm^3/g)) for as-is and 2.4%F SGA.

The comparative pore width vs. pore size distributions for as-is SGA and the same SGA after loading to 2.4% F using off-gas from a pilot electrolysis cell are shown in Figure 6. The comparative pore width vs. pore area and pore volume distributions is shown for these same aluminas is shown in Figure 7. As mentioned above, the inlet gas stream to the reactor was passed through a Teflon® filter to preclude entrainment of particulate on the reacted SGA. It is important to note that SGA loaded in this arrangement contains fluoride from both reaction with HF gas and fluoride from physically adsorbed vapor-phase atmolite path passed the inlet filter. In other words, the wt% F loading from HF alone is <2.4%.

The data shown in Figures 6 and 7 support several observations:

1) The SGA pore size and pore volume distribution changed from unimodal to bimodal due to reaction with HF.
2) Upon reaction with HF, SGA size overall surface area and pore volume decreased, while available pore area and volume increased in the micropore domain from 0.9- 20 nm.

Upon closer inspection of the comparative profiles shown in Figures 6 and 7, it is evident that the micropore volume created during HF reaction with SGA is due to closing or narrowing of pore sizes in the 1.3-4 nm domains. This is most likely due to the formation of alumina hydroxyl-fluoride domains on the pore surfaces. This observation is consistent with the known, irreversible formation of several different amorphous aluminum oxy-fluoride and aluminum fluoride hydrate based reaction products on the SGA surface.[5]

Surface area and micro-pore characterizations were performed on as-is SGA and SGA that had been HF loaded to 1, 2, 3 and 4 wt% using a high volume permeation tube as an HF source. The fluoride loading of these SGAs was only from reaction with HF gas. This condition allowed us to exclude process artifacts and evaluate SGA pore size and surface area evolution versus fluoride load in more detail.

Table 1 lists the surface area and porosimetry data for as-is and F loaded SGA, where argon was used as the porosimetry adsorbate gas. It is important to note that Ar adsorbate surface area measurements are typically lower than those observed where N_2 adsorption is employed. For example the N_2 BET surface area measured for this same SGA was 73 m^2/g.

The data in Table 1 supports several observations:

1) The SGA surface area and pore volume decreases with increased wt% F load.
2) The average pore diameter increases with increasing wt% F load.
3) The micropore area decreased by 66% upon fluoride loading to 4 wt%.

Table 1. Surface Area and Porosimetry Data for as-is and F loaded SGA

	BET SA	BJH adsorption	BJH adsorption	BJH adsorption	DFT m^2/g
wt %	Surface area	Surface area	1.7-300 nm	1.7-300 nm	<2.013 nm
% F	m^2/g	m^2/g	Pore volume cm^3/g	Pore Diameter	Micropore area
0	62.13	73	0.19	9.27	2.67
1	60.65	71.5	0.18	9.87	2.62
2	55.73	67.53	0.17	10.3	1.77
3	47.89	58.81	0.17	11.28	1.13
4	42.45	51.4	0.15	12.02	0.91

The data shown in Table 1 indicates that the impact of HF reaction with SGA is most evident in the micropore domain. This observation is likely due the relative impact of two different aspects of the reaction between HF and SGA , as listed below.

1) The initial, most rapid reaction is the substitution of fluoride for outermost surface hydroxyls, converting pore surface Al-OH to Al-F.
2) A secondary, slower reaction is the localized formation of aluminum fluoride hydrate and aluminum fluoride.

As fluoride loading increases, the above two reaction pathways would be followed by exchange of bridging and lattice oxygen by fluoride, as described previously.[5]

It is interesting to note that the smallest change in surface area, pore volume or micropore area observed in Table 1 was evident between the 0% and 1% F SGA loadings. The initial substitution of terminal pore surface hydroxides by fluoride could create an effective increase in accessible micropore surface area at low (<1%) F loadings, given that the effective size of the pendant terminal aluminum hydroxides lining the surface of the micropores would be reduced to that of Al-F. In this scenario, the accessibility of micropore area to reactant or adsorbate gas (HF, N_2 or Ar) would increase initially, at low F loading, then decrease monotonically with increasing F loading, consistent with the data shown in Table 1.

Figure 8 shows a plot of cumulative surface area and derivative surface area (dA/dW) as a function of the pore width for SGA that has been reacted with HF gas from a permeation oven (the same as in Table 1). This 4% loaded SGA exhibits a dramatic loss in surface area over pore widths ranging from 10-50 angstroms, as compared to the as-is SGA porosimetry data shown in Figure 6. As mentioned above, the impact of development of amorphous aluminum fluorides hydrate phases on SGA pore surfaces would be most evident in this pore width domain.

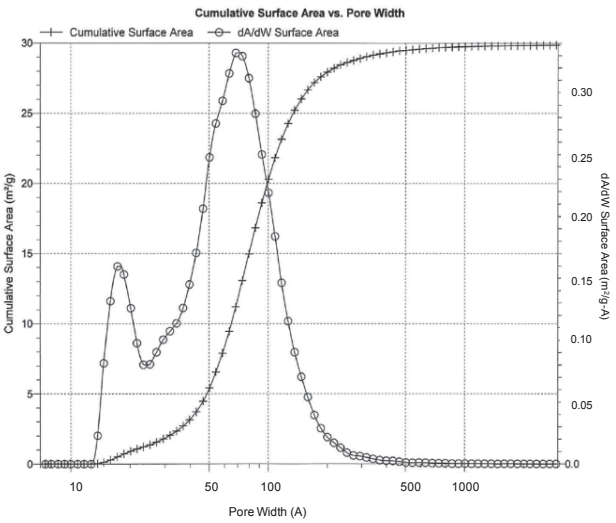

Figure 8. Pore width (angstroms) vs. cumulative surface area (m^2/g) and dA/dW surface area (m^2/g-angstrom) for as 4%F SGA loaded by reaction with HF from a permeation oven.

Conclusion

This work presents a systematic characterization of HF and SO_2 reaction/adsorption on SGA as well as the impact of HF reaction on SGA surface area, pore width, pore volume and micropore area. The key conclusions from this work are:

1) SO_2 is physically, reversibly adsorbed on SGA, while HF irreversibly, chemically reacts with SGA to form aluminum oxyfluoride and aluminum fluoride hydrates.
2) SO_2 has no impact on HF reaction with SGA and is quantitatively desorbed from SGA at high HF loadings (>2% F, when solely from HF).
3) SGA surface area decreases with increased fluoride loading.
4) The decrease in SGA pore area is most evident over the 10-50 angstrom (1-5 nm) pore size range.
5) Upon initial HF loading, accessible SGA pore area may actually increase, owing to the fast initial substitution reaction of aluminum hydroxides lining the pore surface to aluminum fluorides.

Acknowledgements

The authors would like to thank Alcoa's Primary Metals and Environmental Management Teams for encouraging publication of environmental research.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Evaluation of Gas Composition from Laboratory Scale Electrolysis Experiments with Anodes of Different Sulphur Content

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Keywords: Gas characterization, FTIR, anode impurities

Abstract

With the anticipated decrease in available anode raw materials, future anodes are expected to contain higher levels of impurities. Currently the implications of this are being investigated through evaluation of electrochemical performance of anodes and the environmental aspects through gas evolution studies. In this work, four experiments of aluminum smelting have been conducted with anodes of varying sulphur content. The aim of this work was to evaluate the speciation of sulphur in the off-gas. It was found that COS is the main sulphur species in the inert argon furnace atmosphere. The gas composition was further evaluated as a function of set anode voltages in potentiostatic mode. The significance levels of gas composition were evaluated through repeated experiments. As an FTIR multicomponent analyzer was used, less commonly discussed gaseous constituents such as CH_4 and HCl were evaluated with respect to given impurity levels in the anode.

Introduction

With a decrease in the availability of high quality materials for anode production, the anodes produced will show decreasing performance in the physical parameters used as metrics for quality [1]. It is further expected that impurity elements and distribution will be negatively affected. In addition to the physical testing, more recent studies have put emphasis on the electrochemical performance of the anodes, trying to correlate the anodic overpotential with anode composition [2]. Another work by the same author gave evidence of a correlation between anode impurities and anode overvoltage [3].

In future laboratory studies of the electrochemical performance of anodes, a coupling between electrochemical characterization and chemical composition of produced gases has been proposes. In these experiments the sulphur content of the anode has been targeted as an important parameter. The significance level of the sulphur content of the anode has been much discussed: how much change in Sulphur content is required in order to quantify a significant change is sulphur content in the gas phase?

In previous work [4], the potential impact of anode sulphur content on the formation of Perfluorocarbon (PFC) gases was evaluated. In this work, some of these experiments described in [4] have been evaluated for the production of sulphurous gas species rather than the PFC gases. For the high concentrations of CO and CO_2, new gas calibration models were established in order to calculate the sulphur to carbon ratios of the produced gas. Qualitatively, the FTIR spectra were used to evaluate multiple gas species produced, excluding the homonuclear species not providing absorption in the mid-IR range.

Experimental

Laboratory experiments were conducted in an electrolysis setup optimized for anode gas sampling. The anodes were cylindrical with a hollow center. The anode sides were shielded with silicon nitride, extending 0.5 cm below the anode surface. The produced anode gases were fed into the center of the anode by the silicon nitride shielding and a slightly sloping anode surface, and then led out of the furnace through a hollow anode rod. In this way it was possible to recover all produced anode gases with good dynamics. Anode surface area was estimated to be 25 cm^2. A hollow boron nitride rod containing a drop of aluminium metal was used as reference electrode. This setup has previously been described in [4]. The electrolyte used in the laboratory cell was 80% Na_3AlF_6, 12% AlF_3, 5% CaF_2, and 3% Al_2O_3 and the furnace temperature was kept at 960 °C. The anode to cathode distances was kept between 35 and 40 mm.

The experiments conducted are summarized in Table 1.

Table 1. Experimental parameters. S is the baked sulphur content of the anode, Dil is the FTIR dilution flow and ACD the anode to cathode distance.

EXP	Anode	S (%)	Ar (nlpm)	Dil (nlpm)	ACD (mm)
1	A	1.79	0.4		40
2	B	3.29	0.4	0.4	40
3	A	1.79	0.4		35
4	B	3.29	0.4	0.7	35

Due to the high concentration of carbon oxide in the off-gases, various levels of dilution were attempted. The purpose of the dilution was to avoid saturation of carbon oxides while being able to quantify the less abundant gas species.

All experiment were conducted with an inert flow of 0.4 nlpm Ar. Two experiments were conducted with additional dilution of the off-gas as noted above.

The experiments were conducted potentiostatically at varying voltage levels between 3.0 and 4.5 volts. At the end of the experiments, anode effects were induced either amperometrically (EX 1-3) or potentiostatically (EX 4), with the purpose of evaluating any possible difference in off gas composition during the anode effect with respect to sulphur content.

Since the different experiments were conducted with different voltagecurrent profiles and for varying durations, their off-gas concentration versus time plots were not directly comparable. The total concentration of gas constituents produced was estimated by integrating the area under the concentration time curves. For comparison, the current-time curves were integrated and the

charge passed throughout the experiment duration used to estimate total carbon dioxide production under the assumptions of 100 % current efficiency and no losses to Boudouard and/or back reactions.

The gas characterization was conducted with a Protea ProtIR 204M FTIR analyzer. This instrument had a 6.4 meter optical cell and an interferometer resolution of 1 cm^{-1} was applied. The sample line as well as the gas cell was heated to 180 °C. For quantification of gas species, partial least square regression models were built from actual gas constituent spectra. The models were validated for the concentration ranges observed for the experiments conducted.

Results

Interestingly, the only sulphur species found in these experiments were carbonyl sulphide. In order to unequivocally prove the absence of other sulphurous species with the applied gas characterization setup, the calibration spectra of relevant gas species is illustrated in Figure 1. From [5] it was expected to find CS_2 in the samples. Figure 1 shows that the specific absorbance is higher than that of carbonyl sulphide, thus its presence would have been detected if present at comparable concentrations to those of COS. As for sulphur dioxide, this species has lower specific absorbance yet high enough for detection at concentration levels comparable with COS. Sulphur dioxide is however considered a sticky gas and losses to surfaces in the gas extraction channel and analyzer could occur. Sulphur dioxide is considered to be a product of the oxidation of carbonyl sulphide by oxygen gas presence. Hydrogen sulphide has very low specific absorbance thus implying a high quantification level (per cent levels required for detection). The formation of hydrogen sulphide does however require the presence of hydrogen. Even with this criterion fulfilled, its formation would compete with that of methane, hydrochloric and hydrofluoric acids.

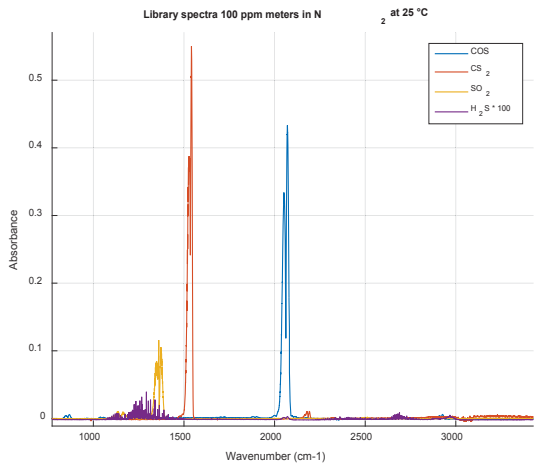

Figure 1. Calibration spectra of relevant gas constituents at 100 ppm meters in N_2 at 25 °C. The H_2S signal is 100 fold amplified for illustrative purposes.

The methane concentration in the off gas was quantified for all experiments. In EX 3 the anode was immersed in the bath for approximately ten minutes before being polarized. As seen from Figure 2, the methane concentration is at its highest when the gas analysis is started upon the anode polarization. This could indicate a non-electrochemical reaction mechanism, which generates up CH_4 at or around the anode, which is then carried out with the electrochemically produced anode gases. For EX 4 it was also found that methane concentration peaked early in the experiment.

As the chlorine content of anode A and B was measured to be 18 and 22 ppm respectively, the hydrochloric acid concentration in the off gas was estimated in order to evaluate the intra-experiment correlation. From Figure 2 it can be seen that the hydrochloric acid concentration profile varies between the experiments. The water content of the anodes has not been determined, but assuming water content between 0.1 and 1 % hydrogen is probably the limiting reagent in the production of hydrofluoric acid, hydrochloric acid and methane. The difference in concentrations for these species is likely due to inhomogeneities in water content of the four 54 mm samples taken from the two anode blocks.

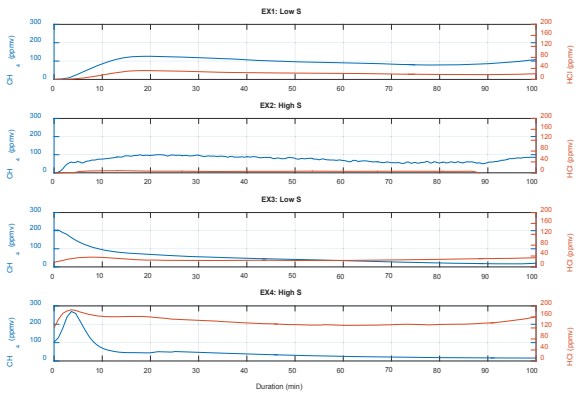

Figure 2. CH_4 and HCl concentration profiles.

For the four experiments conducted, the main gas constituents for carbon and sulphur inventories are carbon monoxide, carbon dioxide and carbonyl sulphide. The concentration profiles for these experiments are plotted along with the current and voltage profiles for the experiments in Figure 3.

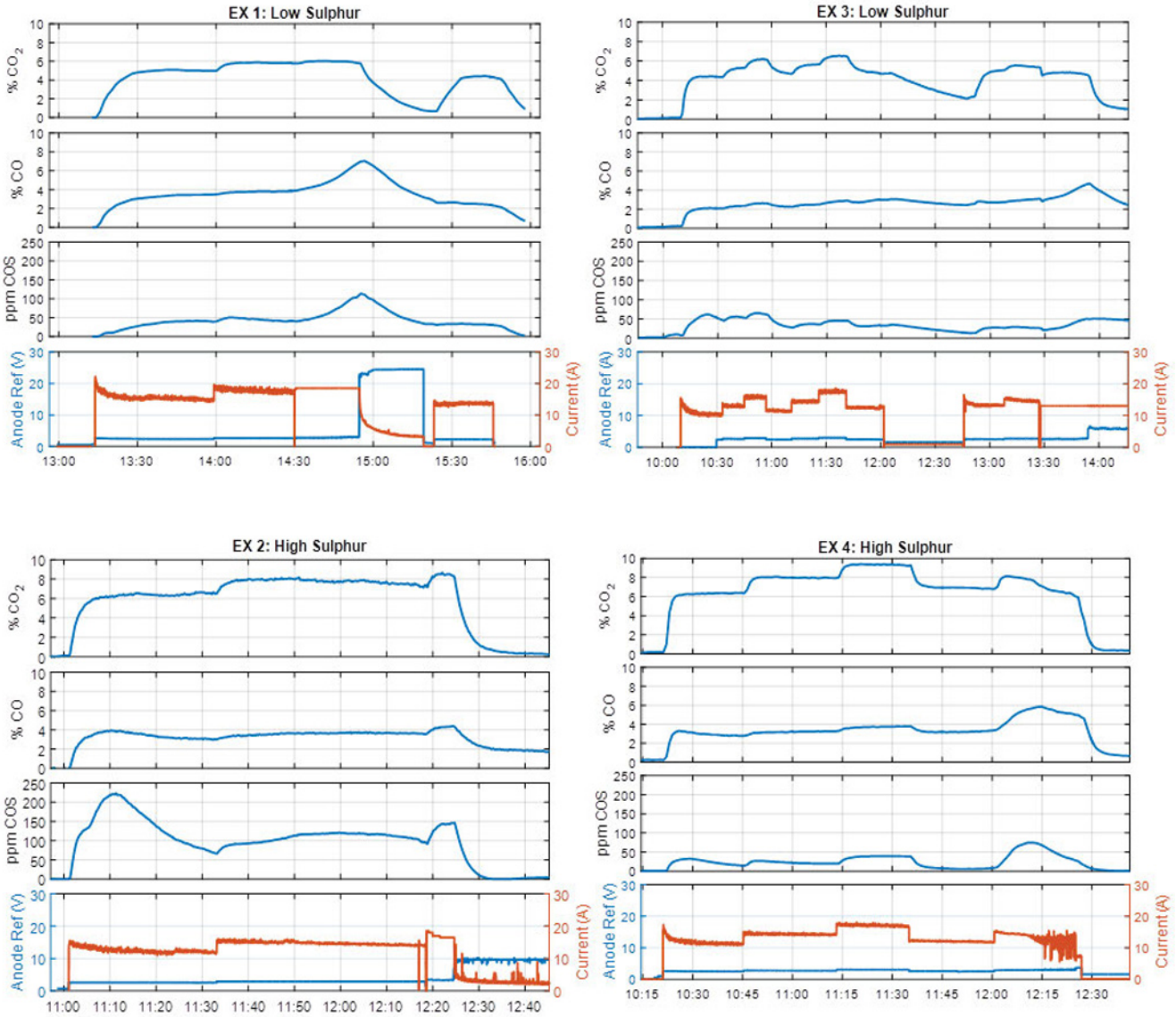

Figure 3. Gas concentration profiles for relevant gas species for the carbon and sulphur inventories. The curves are corrected for the dilution factor given in Table 1.

From the current-voltage curves it can be observed that the current varies significantly between the experiments. This is illustrated in Table 2 where the current and voltage data V is summarized for potentiostatic control at 3.7.

Table 2. Current-voltage characteristics and max. CO_2 concentration at 3.7 V setpoint. Average voltage and current values for the interval operated at 3.7 V given.

EX	Vcell (V)	Vref (V)	CO_2 (%)	Current (A)
1	3.58	2.70	5.9	18.5 A
2	3.57	2.91	8.2	14.6 A
3	3.61	2.67	5.3	13.0 A
4	3.70	2.75	8.1	14.5 A

From Table 2 it is clearly seen that CO_2 production and resulting current varies greatly between the experiments. The large difference in current could not be explained by the 5 mm difference in anode to cathode distance (ACD). The measured difference in ACD (5 mm) was estimated to result in ~ 0.8 A current difference [7] for the experimental conditions in this study.

With the different current voltage applied in the experiment, the total carbon oxide production was compared to a theoretical CO_2 production from the charge passed and assuming 100 % current efficiency and no contribution from Boudouard and back reactions with metal. Total gas production for each experiment was calculated from the initial part of the experiment that was not observably impacted by the anode effect.

Table 3. Gas production ratios. X indicates molar ratio and t theoretical CO_2 production from current passed.

EX	S / C M	X COS	X CO	CO2 / tCO2	CO+CO2/ tCO2
1	0.0012	0.00045	0.40	0.42	0.70
2	0.0030	0.0011	0.33	0.64	0.95
3	0.0015	0.00058	0.32	0.50	0.73
4	0.00054	0.00020	0.30	0.72	1.02

As can be seen from Table 3, the mass ratio of sulphur to carbon does not reflect the content of the anode: the experiments with

high sulphur anodes, EX 2 and EX 4, represent both the highest and the lowest mass ratios. The molar ratio of CO is, with the exception of EX 1 fairly stable. The result for EX 1 is consistent with the low CO_2 production at high current seen in Table 2. The ratio of produced CO_2 to theoretical CO_2 production varies greatly due to the difference in current passed. When calculating ratio of the sum of CO and CO_2 to theoretical CO_2 production, an interesting pattern emerges: the yield in CO and CO_2 produced from current passed is significantly higher for the experiments with high sulphur content. In these experiments, the CO and CO2 yield is very close to the theoretical values.

An interesting feature of the carbonyl sulphide concentration profiles found in all experiments is a marked increase in concentration toward anode effect. This is clearly seen in Figure 4. The same effect is also observed in CO. As the electrochemical formation potential of COS is close to that of CO it can be assumed that the depletion of oxide will have the same impact on COS formation as for CO:

$$Al_2O_3 + C + 3S = 3COS + 2Al \qquad E° = -1.04 \text{ [6]}$$

In the 3.7 V regime, the COS concentration increases to a level that is not sustained whereas CF_4 increases monotonically. This can also be seen in the other experiments for COS and CO in Figure 3. The change from 3.7 V to 18 A setpoint at 14:30 raises the voltage slightly and both COS and CF4 rises monotonically until gas concentrations peak at 14:54. In this period, the anode reference voltage increased from 2.8 to 3.1 V. Half a minute later the voltage reaches its maximum at 22 V.

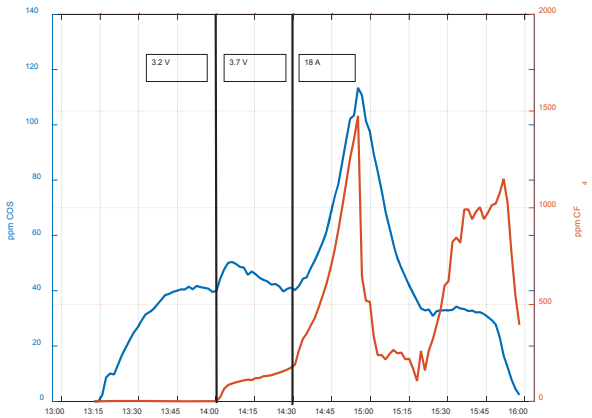

Figure 4. EX 1 COS and CF_4 concentration profiles.

It is likely that during normal electrolysis, enrichment in carbon over sulphur occurs near the anode surface. When exceeding the critical current density of the anode, CO and COS producing reactions become more favorable and the excess sulphur near the anode is consumed.

The correlation between COS and CF_4 is interesting, and it may be speculated that with new quantum cascade lasers [8] available for the mid-IR region, an installation of a gas sensor on the cell duct could function as an early warning sensor for anode effect. How early it would detect the CF_4 onset would depend on the degradation rate of COS to SO2 and also the sensitivity of the laser: the COS increase would have to be discernible from the produced COS from anodes not generating CF_4.

Conclusions

From the current-voltage characteristics of the experiments, it is apparent that the experiments are not amenable to inter-comparison. The variance in current passed is not well understood. Some contribution from Bouduoard and back reactions are expected, but their contributions are not expected to account for the large difference in current. Calculations indicate that the difference in ACD cannot explain the large difference in current. It is possible that shorting through the crucible could vary between the experiments. Measurement of the produced metal could have contributed to the understanding of this problem, however they were not conducted.

The quantitation models built for these experiments appear to be robust. At all dilution levels the models seem to perform well, evaluating from calculation of the residuals. By looking at the gas concentration levels in the various experiments there is no evidence of systematic error in the form of incorrect gas dilution estimations.

The gas composition cannot be corrlated with the sulphur content of the anodes. The sulphur-to-carbon ratios should not be much affected by the currents passed, yet the high sulphur experiments both give the highest and the lowest sulphur to carbon ratios of gas-phase species.

From the concentration profiles of the hydrogenated species, it is concluded that the water content of the samples must vary significantly. It may be speculated that the homogeneity of the sulphur distribution in the 54 mm anode samples is insufficient for the assumption of 1.8 and 3.3 % sulphur in the consumed part of the anode during the experiment. Advanced characterization of anode samples could probably quantify the homogeneity [1].

Further experiments are required in order to gain better experimental control of current-voltage characteristics. Applying identical current-voltage profiles will simplify data inter-comparison of experiments.

Acknowledgement

Hydro Aluminium is acknowledged for providing anode materials of varying sulphur content.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SUSTAINABLE REDUCTION OF ANODE EFFECT AND LOW VOLTAGE PFC EMISSIONS

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Keywords: PFC, perfluorocarbon, tetrafluoromethane, hexafluoroethane, emission monitoring

Abstract

Most aluminum companies have initiated voluntary programs for actively reducing propagating, or above 8 volt, anode effect (AE), perfluorocarbon (PFC) emissions and all modern pre-bake smelters have automated reactive methods for terminating anode effects. The goal of these methods is to reduce cumulative anode effect minutes per cell-day. Any PFCs evolved from aluminum electrolysis pots during periods when pot voltage is below eight volts are considered non-anode effect or low voltage (LV) PFC emissions, consistent with current PFC measurement protocols.

The inclusion of LV-PFC emissions will additively impact aluminum smelter emission inventories in a manner that depends on a number of location-specific factors such as technology type, operating practices, work practices and AE performance and AE kill strategy. The present work outlines a plant-proven approach for systematically and sustainably reducing both AE- and LV-PFC emissions by optimizing alumina feed control.

Introduction

Perfluorocarbons (PFCs) are greenhouse gases with exceptionally long atmospheric lifetimes that are linked to global warming due to their ability to efficiently absorb infrared radiation.[1] Tetrafluoromethane (CF_4) and Hexafluoroethane (C_2F_6), are emitted from aluminum smelters during anode effects (AEs). Aluminum smelters are considered to be one of the largest anthropogenic sources of PFC emissions worldwide.[2]

An anode effect occurs when the alumina (Al_2O_3) concentration in the electrolytic bath drops below ~1.5%, the overall cell voltage rises, and the bath and carbon anodes begin to react.[3] AEs typically start in a localized area of pre-bake electrolysis cells (on one or two anodes). As the AE reaction propagates to additional anodes in the cell, the overall cell voltage rise. When the cell voltage rise above a specific threshold (typically 8V), an anode effect duration counter is initiated in the plant data system to allow tracking of anode effect minutes per cell day.

The United States Environmental Protection Agency (EPA) and the Intergovernmental Panel on Climate Change (IPCC) have provided best practice approaches for measuring and inventorying PFC emissions from aluminum production processes.[4] The most accurate approach, requires periodic on-site PFC measurement during normal potline operation.[4-6] These plant-specific PFC emission coefficients can be used to estimate production normalized plant-specific PFC emissions based upon anode effect minute per pot-day performance.

Most aluminum companies have initiated voluntary programs for actively reducing PFC emissions[4]. All modern pre-bake smelters have implemented automated methods for terminating anode effects. These methods vary by cell technology and operational control strategy; however all have the same goal of minimizing the cumulative anode effect minutes per cell-day at the operating location. These smelter pot feeding and monitoring programs are continually optimized to reduce the AE duration as well as the frequency of these events. These systematic efforts have reduced anode effect related PFC emissions from aluminum smelters by almost 90 percent since 1990.[7]

As previously reported, a proactive focus on anode effect frequency offers a linear route to reducing AE-based PFC emissions. Alternatively, reactive programs that focus on fast anode effect kills to minimize AE duration can reach a point of diminishing returns, given that PFC emission rates (per second of anode effect) are highest during the initial AE onset.[6,8-10]

A focus on reducing anode effect duration via aggressive kill programs can also create an undesirable effect of increasing pot noise, thereby increasing the likelihood of near term operating periods where localized cell anode voltages are above normal (> 4.25 V) but overall pot voltage does not reach 8V. Under these conditions, low voltage (LV) PFC emissions can be generated, however the potroom computer system will not register any anode effect time. This is just one example of a root cause of non anode effect, or low voltage PFC emissions, which have been observed during in-plant studies since 2000.[11]

Any situation or event that de-stabilizes a cell can cause a localized increase in pot voltage and thereby induce PFC emissions from the cell. These include planned work activities or operational controls, such as anode changes, taps, aggressive anode effect kill programs (consisting of large vertical anode displacements), low alumina targets, aggressive underfeeds and low bath targets. Unplanned root causes for LV-PFC emissions can include pot starts, recovery from power outages or fluctuations, interruptions to ore supply or transport lines, improper operation of individual pot feeders and incorrect setting of cell feed control algorithms.[11-16]

The current IPCC protocol for measuring and reporting PFC emissions does not include capable methods for measuring LV PFC emission.[5] Furthermore, regional PFC or CO_{2e} emission permit limits for aluminum smelters were negotiated based on previous knowledge of AE-based PFC performance and do not include provisions for including LV-PFC emissions in a non-punitive manner, assuming they could be capably measured or parametrically estimated.

The impact of including LV-PFC emissions will vary considerably on a location or regional basis, depending on smelter technology as illustrated in Figures 1 and 2. The smelting technologies represented in this portfolio include 15 point-feed prebake (PB) smelters, six Söderberg (S) smelters (both side worked and point feed) and one side worked prebake (SWPB) smelter. [17] The 2010 AE-only PFC emissions for these locations

are shown in Figure 1. The technology type of each smelter is indicated by the letters shown in the X-axis. As expected, PB locations exhibit lower AE-base PFC emissions than non point feed SB and SWPB locations. The order of the plants listed on the x-axis of Figure 1 is identical to those shown in Figure 2.

The impact of estimated NAE-PFC emissions on the total PFC emission profile of each smelting location is shown as a percentage of the sum total (AE + NAE-PFC) for each location, as shown in Figure 2.[17] The plant data shown in Figures 1 and 2 are ranked by percentage of LV-PFC.

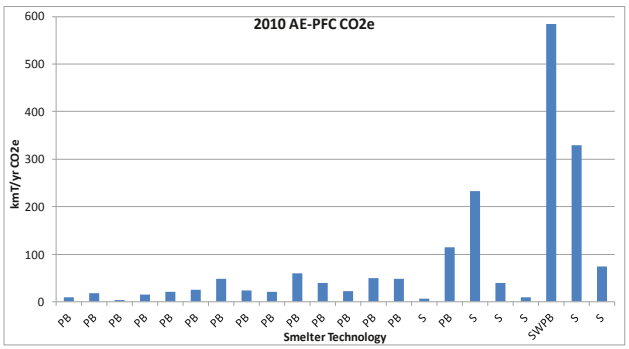

Figure 1. AE-PFC plant emission inventories.

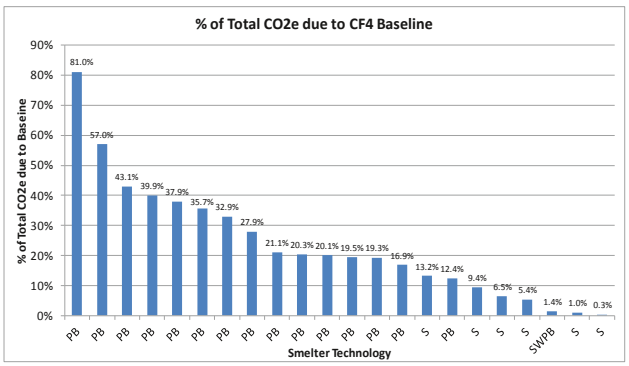

Figure 2. Impact of including LV-PFC on plant emission inventories.

The most obvious take-aways from the data shown in Figure 2 are 1) that all smelters, regardless of technology type or operating capacity, can generate LV-PFC emissions and 2) large pot point feed prebake locations that have most aggressively reduced AE-PFC emissions will likely see the greatest relative increase in their overall PFC emission inventory upon inclusion of LV-PFCs.

If the PFC protocol is revised to include LV-PFC emissions, a key concern will be the impact on local or regional operating permits. These permits may include only single plant or a limited number of plants, and were negotiated prior to any awareness regarding LV-PFC emissions. This issue will need to be addressed to allow non-punitive modification of PFC inventories to include LV-PFC emissions and allow revision of existing operating permits.

As noted above, systematic efforts to reduce AE-based PFC emissions from aluminum electrolysis over the last 25 years yielded significant, sustained reductions in PFC emissions. These efforts need to continue and need to evolve to address both AE and LV-PFC emissions. The present work outlines a plant-proven approach for addressing this goal, to systematically and sustainably reducing both AE and LV PFC emissions.

Experimental

Alcoa has been performing plant-specific PFC emission testing at operating smelters for over 15 years and has the largest data base of real-time monitoring based Tier 3 coefficients in the global aluminum industry.[18] Most of this plant PFC database (and all data discussed herein) employed the use of Fourier transform (FTIR) spectrometers for generating continuous, real-time PFC emission measurements over operating periods ranging from several week to month in duration.

When individual or small pot populations were evaluated, such as during initial designed experiments on feed control modifications, cell off gas was sampled on the "dirty" side, on individual cell ducts or small group collective ducts after the pots but prior to the gas treatment systems. This mode of sampling presents a resource-intensive challenge to assure that capable, representative sampling is maintained throughout the test periods. [6]

When intermediate, moderate or large pot populations were evaluated, PFC measurements were acquired by monitoring collective pot exhaust after the dry scrubber system. This option presents a relatively clean gas stream for sampling, since HF and particulate were removed by the gas treatment system. The sampling procedures employed have been described previously.[6,19]

Normal data analysis practice involved collecting temporal plant AE records from the process control computer and matching these records of AE onset and duration to FTIR based measurements of PFC emissions. In addition, cell operational data collection routines were employed to log individual anode voltages, pot noise, feed status and other cell data that were normally "rolled up" by the data archival system.

In the present study, LV-PFC emission estimates were calculated from FTIR data records by assuming that all observations of non-zero PFC baseline data, when no >8V cell condition were recorded by the plant computer, were due to LV-PFC emissions.

An assumption inherent to all data presented herein is that we assumed that the FTIR limit of quantization (LOQ) is 1 ppb. This is a very conservative assumption given limited corrections to prevent bias or interference from known or unknown gases present in the sampling stream.[6,19]

Results and Discussion

As noted above, Figures 1 and 2 include PFC data from six Söderberg smelters. It is interesting to note that four of these smelters (the 1st, 4th, 5th and 8th plants listed from the right side of the x-axes in Figures 1and 2) exhibited two interesting features:

1) These four plants had AE-based CO_{2e} emission on-par or lower than many of the prebake smelters, and
2) These four plants were among those whose PFC inventories would be least impacted by inclusion of LV-PFCs.

All four of these plants are point-feed Söderberg smelters. Two key operational features that differ from prebake smelters are that:

1) Anode changes do not occur routinely at these locations.
2) Individual electrolysis cells have a single large anode, as opposed to each cell containing an array of individual anodes.

The LV-PFC performance observed at these locations support previous observations that pot instabilities during anode changes are significant contributors to LV-PFC emissions.[15,16,20] In addition, this performance data suggest that the occurrence of sustained LV-PFC conditions is considerably less likely at point–feed Söderberg plants, relative to point-feed prebake smelters.

The use of distributed anodes, universal to all prebake smelters, facilitates the occurrence of localized (at one individual anode), conditions where elevated current density, deficient alumina concentrations or other factors can initiate and sustain localized anode effect conditions that do not propagate to the remaining anodes in the cell, remain undetected by conventional (>8V) methods and thereby create LV-PFC emissions.

Point-feed smelters typically employ alumina feed algorithms that cycle between underfeed and overfeed periods (relative to steady-state production demand) on a periodic basis. This deliberate feed cycling is employed to allow tracking of dissolved alumina concentration using the concomitant change in bath resistance as a process control indicator. Since bath resistance increases with alumina concentration, a common operating philosophy is to run at low alumina targets to minimize pot voltage.

The end of underfeed periods represent recurring transient conditions where localized alumina-deficient conditions could initiate LV-PFC emissions in prebake electrolysis cells, depending on alumina target, bath level and other cell conditions, such as voltage modifiers or anode changes. An example of this behavior is shown in Figure 3 where each spike is at the end of an alumina underfeed period. LV-PFC emission is negligible during low noise, stable operation, as evident in the beginning period of Figure 3. During transient instabilities, such as during an anode change, LV-PFC emissions occur. It is important to note that the average cell voltage was <8V the entire time. When averaged over a large cell population (100-800 cells), these transient events can appear as broad, sustained low level PFC emissions. The optimization of electrolysis cell feed control algorithms present an opportunity to develop deliberate pot control strategies for proactively reducing LV-PFC emissions.

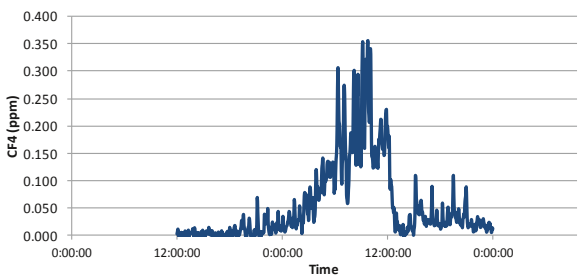

Figure 3. Real-time LV-PFC emission from a single cell.

A number of different alumina feed control algorithms are in use at commercial smelters, worldwide. Fortunately, within a given company, the same feed control algorithms are employed at most or all point feed prebake smelters. This situation allows a detailed, resource-intensive feed algorithm optimization study at one smelter to be broadly applied to other point feed prebake smelters using the same feed control program. The general outline of this strategy is outlined below.

1) Evaluate exiting pot feed parameters and develop a designed experiment (DOX) to quantify the key levers and interaction terms.
2) Perform an in-plant DOX on a small pot population while also continually monitoring real-time PFC emission monitoring.
3) Use data collected during the DOX periods to generate an optimized feed control a model.
4) Implement the refined feed algorithm on a limited pot population.
5) Perform comparative PFC emission measurements on the test and control pot populations.
6) Expand the refined feed algorithm to larger pot populations.

We have employed this strategy to systematically identify and optimize key levers and interaction terms in our existing feed strategy with the express purpose of reducing LV-PFC emission. We have systematically extended these refined algorithms to 1/8, ¼, ½ and full plant scale from 1.5 years ago to present. The key performance indicators from this effort are listed below.

1) 37% reduction in LV-PFC emission.
2) 15-20% reduction in AE-frequency.
3) Lower pot noise.
4) No reduction in current efficiency or metal productivity.
5) Zero CAPEX installation.

Conclusion

The inclusion of NAE-PFC emissions will additively impact aluminum smelter emission inventories in a manner that depends on a number of location-specific factors such as technology type, operating practices, work practices and AE performance. Recurring operations that induce transient destabilizations in point-feed prebake cell performance include anode changes and lean alumina feed control algorithms.

Alumina feed control algorithms can be optimized to sustainably reduce LV-PFC emissions without incurring negative impacts to pot productivity. An additional benefit of this optimization effort is the preventive, concomitant reduction of anode effect frequency.

Given that the same alumina feed control program is typically employed across most or all point-feed prebake smelters within a given company, a resource intensive effort at one plant location can identify key levers and interaction terms common to all point feed prebake smelters using the same feed control program.

Acknowledgements

The authors would like to thank Alcoa's Primary Metals and Environmental Management Teams for encouraging publication of environmental research.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

QCL-BASED PERFLUOROCARBON EMISSION MONITORING

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Keywords: Perfluorocarbon, PFC, tetrafluoromethane, quantum cascade laser, QCL, continuous emission monitor

Abstract

Periodic, short-term, extractive sampling with Fourier transform infrared (FTIR) spectrometers is often employed to measure perfluorocarbon (PFC) emissions at aluminum smelters. FTIRs have been routinely deployed by Alcoa to measure tetrafluoromethane (CF_4) and hexafluoroethane (C_2F_6) gases during 2-4 week sampling campaigns at smelters worldwide. In conjunction with plant process logs, the data collected from these short-duration measurements are used to estimate production-normalized plant-specific PFC emission factors. Quantum cascade lasers (QCL) have recently become available for the mid-infrared monitoring of air pollutants. QCL-based, purpose-designed measurement systems could facilitate long-term in-situ stack measurement of CF_4 at one or a few representative dry scrubber exhaust stacks. This paper discusses paired in-plant measurements by FTIR and QCL-based optical systems employed to develop future QCL measurement platforms and optimize the latter for long term stack PFC monitoring.

Introduction

The aluminum industry is considered the largest anthropogenic emission source of perfluorocarbon (PFC) gases, specifically tetrafluoromethane (CF_4) and hexafluoroethane (C_2F_6). In an effort to reduce its carbon footprint, Alcoa has set environmental goals to monitor and reduce PFC emissions from the aluminum smelting process by coordinating a global program for performance benchmarking that requires periodic in-plant monitoring to document progress towards this systematic reduction in emissions.

By convention, PFC emissions are assumed to occur exclusively when the ore (Al_2O_3) concentration in the electrolytic bath drops below ~1.5%, initiating a reaction between the bath and carbon anode(s) and causing the cell voltage to rise above a trigger threshold (typically 8V). These events are referred to as anode effects (AEs).[1] PFC emissions occurring during AEs are measured and reported using the guidelines outlined by the current IPCC (Intergovernmental Panel for Climate Change) Tier 3 method. Recent reports, however, indicate that PFC generation and emission can also occur during periods of low average cell voltage (<8 V). These so-called low-voltage PFC (LV-PFC) emissions appear to be related to typical potroom activities, such as anode setting and metal tapping, pot starts, power fluctuations and other occasions that may induce a localized increase in anode current density or a localized deficiency in alumina concentration.[2-4] LV-PFCs have been observed at various smelting locations operating with different line amperages or cell technologies.[5]

Fundamentally, because the overall pot voltage remains below 8 V, the current IPCC Tier 3 method does not provide guidelines on how to account for and report LV-PFCs measured during in-plant measurements.[6] Any future revision to the reporting protocol would need to specify the methodology necessary to ensure capable and representative estimates of contributions from LV-PFCs. The development of capable instrumentation for cost effective continuous monitoring of PFCs emissions could be an enabler for facilitating this measurement capability.

To date, a host of commercial technologies have been applied to the measurement of PFC emissions, including process mass spectrometers,[7] Fourier transform infrared spectrometers (FTIR),[8] photoacoustic infrared filter photometers,[9] and tunable diode laser Absorption (TDLA) analyzers.[10] FT-IR spectroscopy is the currently preferred option due to part per billion (ppb) sensitivity, portability, ability to reprocess historical data, multi-component analysis and resistance to magnetic fields. Simultaneously, FT-IRs are generally considered to be higher-cost systems, requiring significant upkeep during sampling campaigns. Few vendors offer process FT-IRs customized to meet the sensitivity requirements for Tier-3 determination of PFC emissions. These often involve use of extractive sampling and long-path gas cells in order to increase method sensitivity, as opposed by measuring the PFC emissions on-line using the exhaust duct diameter of the gas treatment center (GTC), typically 2-6 meters. The typical long-path FTIR gas cell has a significant internal volume which effectively causes a dampening effect of the transient CF_4 concentration measured by the system. Assuming a train encompassing approx. 100 pots, a limit in quantitation (LOQ) of 5-10 ppb is needed in order to accurately measure PFCs downstream of the gas treatment centers (GTC).

Tunable laser absorption spectroscopy (TLAS)-based analyzers represent an attractive route for continuous monitoring of PFCs. TLAS technology has been demonstrated to enable real-time monitoring of HF in potrooms and exhaust stacks.[11-13] Developing a similar analyzer, but tailored to PFCs, would additionally benefit from a lower barrier of entry: TLAS technology is considered mature, offering low-cost solutions with equally low maintenance requirements. Furthermore, exhaust stacks are oftentimes already fitted with flanges for cross-stack gas analyzers, streamlining their installation.

Application of TDLAS analyzers for the detection of CF_4 and C_2F_6 was originally demonstrated in the 1990's.[10,14,15] The analyzers used a Pb-salt mid-infrared source providing low ppb detection levels and with a sufficiently fast sampling rate (few seconds). However, the system used an extractive sampling technique, drawing the gaseous sample from the exhaust stream into a multi-pass gas-cell in order to achieve the sensitivity desired for the application. This, in addition to the need for cryogenic cooling, hindered its adoption as cross-stack sensor for on-line measurements.

Norsk Elektro Optikk A/S (NEO), in collaboration with Corning Incorporated (now Thorlabs Quantum Electronics, Inc.), recently

developed a quantum cascade laser (QCL) mid-infrared source matching the main absorption band of CF_4 and integrated it into a conventional TLAS housing.[16] The LaserGas™ Q sensor uses single-line absorption spectroscopy in concert with wavelength modulation spectroscopy as the detection scheme.[13] Following bench-scale evaluation, the prototype LaserGas™ Q system was deployed at one of our smelters for field testing as a cross-stack CF_4 continuous emission monitor (CEM) in parallel with a conventional extractive FTIR system. Field testing is crucial to the development of any new measurement technology ahead of commercial deployment as it provides direct feedback regarding signal robustness against chemical interferences, instrument drift, changes in sampling conditions (e.g. duct temperature), inclement weather and long-term durability. For these studies, the analyzer's performance with respect to PFC emissions assessed by simultaneous, co-located gas stream monitoring using a conventional FT-IR spectrometer and NEO's prototype analyzer.

The objectives for the QCL-FTIR testing described herein include:

1. In partnership with NEO, develop and evaluate a QCL-based process analyzer for continuous measurement of CF_4. The CEM will be evaluated with respect to (1) its ability to discriminate transient events (AEs) from continuous LV-PFCs, (2) analytical sensitivity equivalent to methodologies currently used to generate Tier 3 slope terms for CF_4 and (3) operation with little-to-no intervention from plant personnel, aside from routine preventative maintenance.

2. To bench-scale calibrate, test and establish analytical figures of merit (e.g. signal linearity, sensitivity and long-term stability) of NEO LaserGas™ Q prototype using certified CF_4 gas standards and perform a preliminary (laboratory) assessment of potential chemical interferents.

3. To directly compare NEO's LaserGas™ Q prototype with a FTIR spectrometer on location at an Alcoa smelter by installing the instrument at the scrubber outlet. The outcome and ultimate metric for success will be a comparison of the estimated contributions from low voltage and total (AE and low voltage) CF_4 emissions.

Experimental

Calibration and measurement parameters:

The LaserGas™ Q analyzer was calibrated using a CF_4 compressed gas cylinder certified to 4.65 ppm (±5 %) using a 735 mm single-pass gas cell. The temperature dependence of the absorption signal was determined by filling the single-pass gas cell with 1000 ppb CF_4 sample generated using an Environics gas mixer and a custom–built heated cell. The sensor response was measured for one concentration at different temperatures allowing for a compensation function to be calculated and programmed into the analyzer's firmware. Field installation of the transmitter and receiver units (Figure 1) was accomplished by securing the components onto existing flanges on the GTC exhaust stack followed by optical alignment across the 2.4 m optical path (the stack diameter). Temperature of the exhaust gas at the sampling location was measured using a thermocouple. The LaserGas™ Q was configured to average the optical signal for 20 seconds and the temperature compensation was fixed to 100°C. The installation did not provide for continuous measurement of gas temperature, therefore it was not possible to feedback real-time temperature fluctuations into the analyzer response.

The FTIR spectrometer (Gasmet, Finland) employed in this comparative study was equipped with a 9.8 m multi-pass gas-cell heated to 120°C (in order to minimize condensation on internal optics) and a TE-cooled mercury-cadmium-telluride detector. The acquired sample spectra were stored as averages of 20 seconds scans collected at 8 cm^{-1} resolution. The sample absorption spectrum was calculated from a 20-second averaged single-beam sample spectrum ratioed to a one-minute averaged single-beam reference spectrum (N_2 gas). The FTIR was calibrated by diluting CF_4 gas from a stock gas cylinder certified as 10.3 ppm (±2 %) with N_2 in order to obtain a curve composed of CF_4 values spanning the range of expected concentrations at the stack. The accuracy of the calibration curve was evaluated by using a separate CF_4 compressed gas cylinder certified to 2.00 ppm (±2%). The estimated LOQ for CF_4 was approximately 10 ppb.

Field test at the Alcoa smelter:

The LaserGas™ Q prototype CEM was mounted on the 2.4 m gas treatment center (GTC) exhaust duct at an Alcoa smelter as shown in Figure 1. A Monel® (Special Metals, Corp) filter probe was installed in close proximity to the transmitter allowing extraction of the GTC exhaust gas to the FTIR spectrometer gas cell using a vacuum pump. This sampling location corresponded to the outlet duct section serving 72 pots.

Figure 1. LaserGas™ Q prototype Laser head and Detector

Results and discussion

Figure 2 an example of the second harmonic signal generated by the analysis for CF_4 concentrations ranging from 0 ppb to 2,580 ppb. The balance gas was N_2. Both C_2F_6 and CH_4 gases display absorption features near the CF_4 band used in this study, as such, the accuracy of the QCL-based system could be affected by the presence of these gases in the process stream.

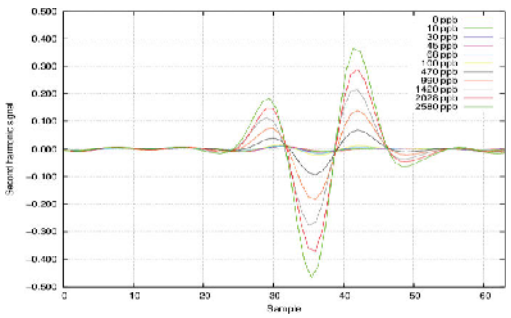

Figure 2. LaserGas™ Q response to different CF_4 concentrations

Presenting the LaserGas™ Q with samples of C_2F_6 and CH_4 demonstrated that it is insensitive towards C_2F_6, however, the current quantification algorithm is impacted by the presence of CH_4. Laboratory trials demonstrated that a 2-5 ppb positive bias in "effective" CF_4 was introduced for each ppm of CH_4 present in the gas sample.

Figure 3 provides a graphical representation of the FTIR response to various CF_4 concentrations in comparison to the response of the QCL-based prototype. Good agreement between the two systems was observed up to values of approx. 7 ppm-m, as demonstrated by the linearity of the trendline. Overall, values measured by the LaserGas™ Q deviated by approximately 6% from those reported by the FTIR spectrometer.

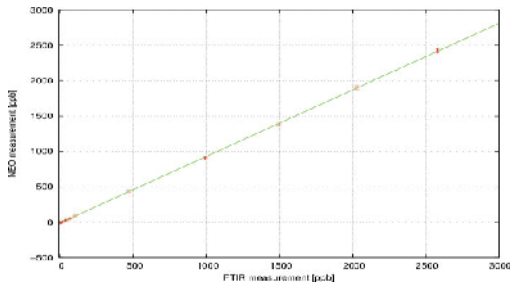

Figure 3. Linearity response of LaserGas™ Q prototype in comparison to the FTIR spectrometer

The dynamic response of the LaserGas™ Q to varying CF_4 levels ranging from 0 ppb (N_2 gas) to 100 ppb is depicted in Figure 4. The analyzer is characterized by fast temporal response to different input values, however it underestimated the CF_4 concentrations by approximately 10 ppb in comparison to the FTIR spectrometer. Figure 4 also indicates that the measurement error at the levels tested ranges from 5 to10 ppb.

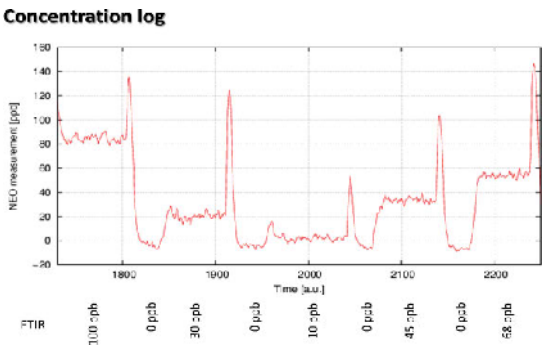

Figure 4. LaserGas™ Q prototype concentration response at low CF_4 levels.

Field testing of the analyzer was performed at a smelter operating two potlines separated into four rooms consisting of 164 pots from Line 1 and 240 pots from Line 2. A sub-section of 72 pots was selected for monitoring during the testing period. Figure 5 shows the CF_4 profile as measured by the FTIR and LaserGas™ Q on September 30, 2015. Overall, the LaserGas™ Q responded well to temporal variations in low voltage CF_4 emissions in the 20-100 ppb range, while a persistent 10 ppb bias was observed between the two instruments.

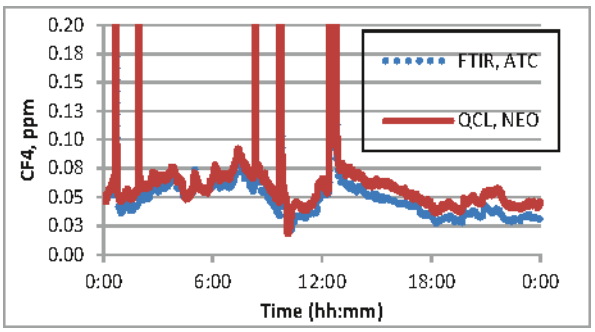

Figure 5. Low voltage PFC emitted at Alcoa smelter

Figure 6 shows seven anode effects (pot voltage >8 V) that occurred on September 30. At first glance it would appear that CF_4 peak concentrations reported by the LaserGas™ Q are slightly lower compared to the FTIR values. However a closer look (Figure 7) shows that the QCL-based CF_4 peaks are also marginally broader. By integrating each individual peak area, a more accurate comparison between the two instruments can be obtained. The peak area values for the LaserGas™ Q and FTIR are within 18 % of each other, a noteworthy achievement considering for the first field evaluation of this system.

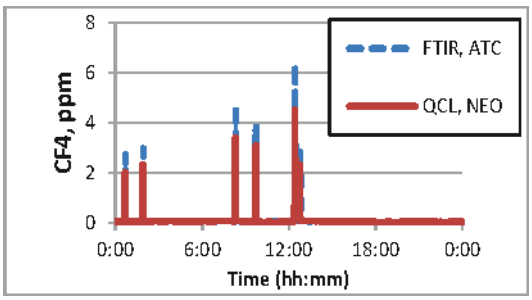

Figure 6. Anode effects (≥8 V) observed at Alcoa smelter

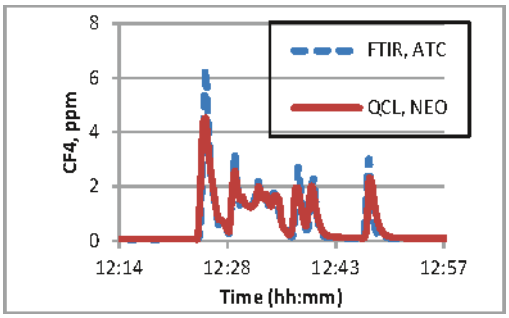

Figure 7. Highlighted temporal profile showing multiple, nearly-concurrent anode effects

Table 1 summarizes CF_4 emissions calculated during the 24-hour period shown in Figure 5 and Figure 6. This table presents values for CF_4 emissions from the seven anode effects (> 8V) *plus* low voltage, the average of total CF_4 for those 24 hours, average of CF_4 related to AEs and average of low voltage CF_4. For the September 30 sampling period, the LaserGas™ Q estimated an average total CF_4 emitted by the potline (AE *plus* low voltage) within 14% (in excess) of the corresponding FTIR value.

In contrast, average AE CF_4 was underestimated by approximately 14% while average low voltage CF_4 was overestimated by 19% with respect to FTIR-based data.

The lower emission values during AEs may be related to a deviation from the expected linear absorption behavior (Beer-Lambert Law) as well as signal saturation at high CF_4 concentrations. Uncertainties associated with independent calibration and low-level sensitivity of both instruments, the absence of real-time temperature compensation for the LaserGas™ Q (spot-checks indicated that gas temperature varies between 85°C and 100°C) and possible interference from CH_4 could have equally contributed to the differences observed for low voltage CF_4 emissions.

Table 1. Calculated AE values from 24 hours period on September 30

	CF4, ppm (FTIR)	CF4, ppm (NEO)
AE(>8v)+ low voltage	311.861	355.333
Average Total	0.077	0.087
Ave AE (>8v)	0.785	0.672
Ave low voltage	0.047	0.056

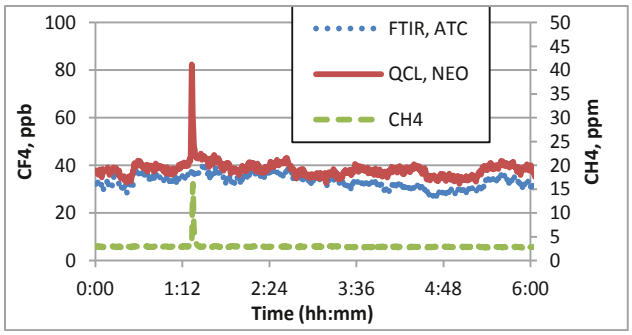

Figure 8. Methane interference in the LaserGas™ Q monitor

Figure 8 serves to confirm the impact of CH_4 as an interferent on CF_4 values reported by the LaserGas™ Q. On average, the CH_4 concentration at the exhaust duct ranged 3 ppm to 4 ppm during the in-plant campaign (measured by the FTIR spectrometer). A sharp increase in CH_4 observed at approximately 1:20am resulted in a false positive detection of CF_4 by the LaserGas™ Q estimating the CF_4 emitted during that transient to be 83 ppb while the FTIR's response remained unaffected (~35 ppb).

Conclusions

A prototype LaserGas™ Q CEM for CF_4 was developed by NEO and comparatively evaluated at an Alcoa aluminum smelting location in conjunction with an FTIR. Following the campaign, a few outstanding items remain to be addresses ahead of commercial deployment. These include (1) reducing interference from CH_4, (2) expanding the linear dynamic range (thereby introducing the ability to more accurately measure AE CF_4 concurrently) and (3) improving sensitivity at low ppb levels. Instrument drift, long-term durability of the laser source and demonstrated low maintenance under plant conditions also remain to be proven and will be addressed in future trials.

The studies summarized here clearly demonstrate the analytical capabilities of the prototype as depicted by the close agreement of the measured CF_4 emissions with the incumbent technique (FTIR spectroscopy). The LaserGas™ Q CEM represents an encouraging step towards providing a solution to the problem of accurate, continuous monitoring low voltage CF_4 emissions at aluminum smelters.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

USING ARTIFICIAL NEURAL NETWORK TO PREDICT LOW VOLTAGE ANODE EFFECT PFCS AT THE DUCT END OF AN ELECTROLYSIS CELL

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Keywords: tetrafluoromethane, perfluorocarbons, GHG emissions, predictive model, neural network, low voltage anode effect

Abstract

Primary aluminum production is generating a significant amount of greenhouse gases. CO_2 is the dominant compound but during anode effects, perfluorocarbons (PFCs) are released as well. Even when no generalized anode effects are present, intermittent emissions of PFC have been reported in small concentrations but the root causes of these emissions are hardly understood.

Measurements were taken at "Aluminerie Alouette" plant using Fourier-transformed Infrared Spectroscopy on individual cells to analyze the evolution of the composition of the gas collected at the duct end. By correlating the variations of the concentration for the emitted gas with cell variables (voltage, intensity, and pseudo-resistivity) and individual anode currents, it was possible to develop a predictive model to quantify the tetrafluoromethane (CF_4) emissions between 10 and 1000 ppb for individual cell emissions.

By analyzing the time history of the resulting data and by applying a post treatment process accordingly, it is possible to reduce the number of false predictions and increase precision of the final results.

Introduction

Primary aluminum production is known to be a major producer of greenhouse gases (GHG). A significant fraction (5 to 10%) [1] of these GHG is composed of perfluorocarbons (PFCs) known to be produced when the alumina concentration in the electrolytic bath reaches a very low concentration (≈1.5% wt.). This situation generally leads the cell to an anode effect (AE) where the cell voltage will rise significantly and the generation of CF_4 and C_2F_6 will become significant.

CF_4 and C_2F_6 have a respective global-warming potential 6500 and 9200 times higher than the CO_2 [2]. Quantification of the PFCs emitted during aluminum production is very important regarding the production of greenhouse gases. Calculations [3] for the total emissions of CF_4 and C_2F_6 are based on the measurement of the cell overvoltage during anode effect and specific emission ratios. These ratios depend on the smelter technology and the operation practices and validation measurements are performed by a third party at least every three years.

In the last decade, many papers were published illustrating the presence of PFC emissions even when the electrolytic cell behavior is not considered as a high voltage anode effect [4-6]. (The cell voltage simply did not rise high enough for the required time to trigger the event). Hence these emissions, characterized as "low voltage anode effect"[1] are not accounted for in the annual calculations performed at the plant. It was shown that the duration of the low voltage anode effect can be from minutes up to hours without necessarily triggering a high voltage anode effect.[4-6]

"Aluminerie Alouette" is currently working on a project to improve the understanding of instantaneous variations within the CF_4 and C_2F_6 rate of production both during high voltage anode effects (HVAE) and low voltage anode effects (LVAE).

This paper focuses on the emissions of CF_4 during operation of an electrolysis cell during low voltage anode effect. The overall methodology (measurements, pre-treatment of data, algorithm and post-treatment) used to develop a predictive model for these emissions is described within the range of 10 to 1000 ppb. Validation of the predictive model is performed and the influence of some independent variables on the predicted CF_4 emissions is briefly discussed.

Acquisition of data and preparation for analysis

Measurements

The cells of interest for this study are prebaked AP40LE cells using point feeders and operating above 390 kA. The cells are equipped with on-line monitoring of individual anode currents and a measurement frequency of 1Hz. Cell current, voltage and pseudo-resistance was measured by the cell control system with the same frequency and all the data was recorded. Only one cell at the time was monitored, therefore there was no dilution other than the gases coming in the cell through the hooding. Inspection of the hooding was frequently performed to make sure it remains in similar condition throughout all the campaign. However, the impact of the changes in dilution caused by manual intervention on the cell (i.e. anode change, or tapping of metal) was not taken into consideration at this point in the study but it could have an effect on the measurements. This effect is even more important for very low concentrations as dilution can decrease the measured concentration of CF_4 below the limit of detection.

Two measurements campaign were performed to collect sufficient data for an accurate model to be developed and validated. The first campaign was performed in winter 2015 and the second one during summer of the same year. During these periods, gas concentrations were measured using a GASMET™ Dx-4000 FTIR (Fourier Transformed InfraRed spectrometer). The sampling probe

[1] In some cases, these emissions are referred to as "Non-anode effect PFC", "background PFC" or "continuous PFC".

was located at the duct end of the electrolysis cell and gas was continuously provided to the analyzer with a volumetric rate of 2.5 liters/minute. The gases were going through a desiccant and an alumina scrubber to remove traces of water and hydrogen fluoride for the protection of the measuring equipment. Concentration measurements were performed continuously with a rate of 10 scans per seconds and the average value measured over five seconds was the corresponding measurement. Using high purity nitrogen, a background spectrum was redefined according to the specifications from the FTIR user manual every 24 hours.

Specific considerations

In this paper, LVAE C_2F_6 emissions were not considered because the concentrations of this component remain inferior to the noise limitation from the analyzer. The noise level with respect to the CF_4 concentration was approximately 5 ppb. By keeping this in mind, and in order to minimize false positive predictions, the lower limit of consideration for the analysis was set to 10 ppb. Henceforth, any measurements below that were considered as "no emissions of PFC". The upper limit of CF_4 concentration for the model was defined arbitrarily as 1000 ppb based on the distribution of data points and the minimum number required for development and validation of the model. The authors acknowledge that the lower limit of the analysis is very close to the limit of detection of the FTIR. However, if misclassification occurs close to this range, the reasonable cause should be the impact of the cell variables and operations on such small concentrations but not the noise from the measuring instrument.

Moreover, data collected during 45 minutes after each anode effects were excluded from the analysis as it represents a recuperation period. Figure 1 illustrates five anode effect recuperation periods. The curves intersect at x=0 once the gas concentration is below the lower limit of detection of 10 ppb. The profile of each curve is very similar to a dilution profile but in some case, unexpected variations could be attributed to residual reactions producing CF_4. Finally, the total time of recuperation is proportional to the anode effect overvoltage. Hence, if more gas is produced during higher overvoltage AE, it will take more time to evacuate it from the cell. For this reason, it is plausible that the CF_4 concentration measured during this period is residual and in no relation with the cell parameter, explaining why this recuperation period was excluded from the analysis.

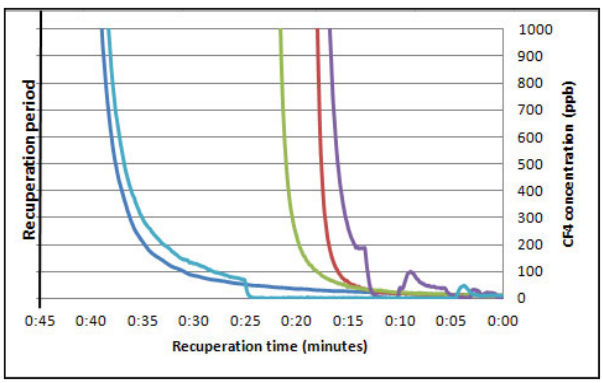

Figure 1: Recuperation time after an anode effect.

Preparation of the data

Before the collected data was considered to be ready for analysis, it was necessary to evaluate the transition time between the moment the CF_4 was produced under the anodes and the time the gas reached the FTIR chamber for analysis. To evaluate this "transportation time" to the detector, anode effects were used as a reference point. It was assumed that the instantaneous increase in voltage will lead to high concentration emissions. Therefore, by measuring the difference between the surge in voltage and the sudden increase in CF_4 it is possible to evaluate the "transportation time" under specific conditions (flow rate, tubing radius, total distance, etc.). If these conditions change, it is required to re-evaluate or re-calculate the "transportation time". In the study, the “transportation time” measured could change from 38 seconds up to 80 seconds. This significant difference was caused by the length of tubing used to transport the gas to the analyzer.

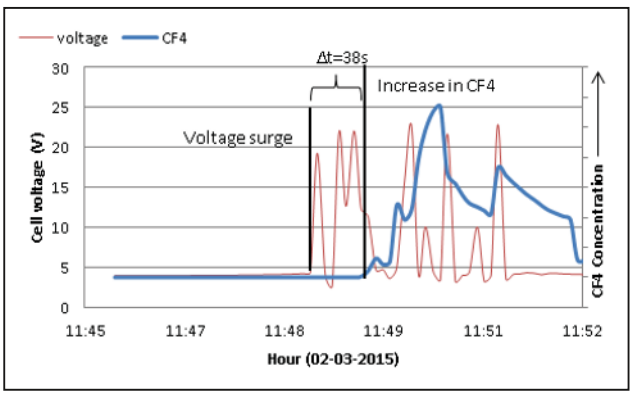

Figure 2: Evaluation of the "transportation time" between gas production and detection by the FTIR

There is necessarily a small error associated with this estimation because of the dilution within the chamber as well as the measurement time itself. However, as LVAE emissions are stable for a significantly longer period than the error, it should not interfere significantly with the results.

Once the emissions were synchronized with the respective cell variables at the time of their production, the data was filtered to eliminate erroneous values. Finally, some variables were normalized for specific purposes.

Distribution of collected data

Approximately 15 days of continuous measurements were acquired using the FTIR. However, for most periods, no LVAE PFC measurements were measured during normal operations. Therefore, most of the measurement periods with LVAE emissions of PFC had to be provoked by changing the normal parameters of the cell to induce a lower alumina concentration in the bath. For this reason, the alumina feeding rate was changed to decrease alumina concentration in the bath for a specific period leading to the generation of PFC. Figure 3 illustrates the typical case where the measured CF_4 concentration was very low (5-10 ppb) during normal feeding rates. However, if the change in feeding rate is high enough (grey zone), then the lower alumina concentration will likely have a direct impact on the CF_4 concentration.

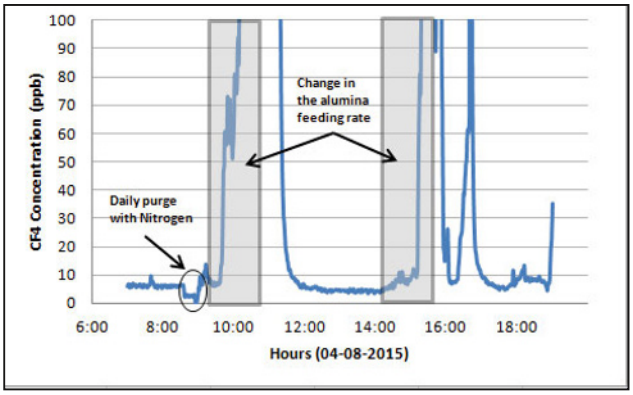

Figure 3: Change in CF_4 concentration with different alumina feeding rate.[2]

During the two sampling periods, more than 11 000 data points were collected with CF_4 concentration within the specified limit for this study (10 ppb to 1000 ppb). This represents LVAE CF_4 measurement spread over 15 days of operations. This data was divided into 13 different populations. These divisions were based on the different periods where CF_4 emissions were measured within the range of interest for a significant period. The duration of each period is different and includes LVAE as well as the absence of emissions. The average duration of these scenarios is approximately 2 hours, however some periods only lasted 30 minutes and others lasted up to 11 hours. These scenarios will be used further in the paper for validation. Although, even if the total number of data points is significant, we can see that the distribution is far from uniform and very concentrated close to the lower limit.

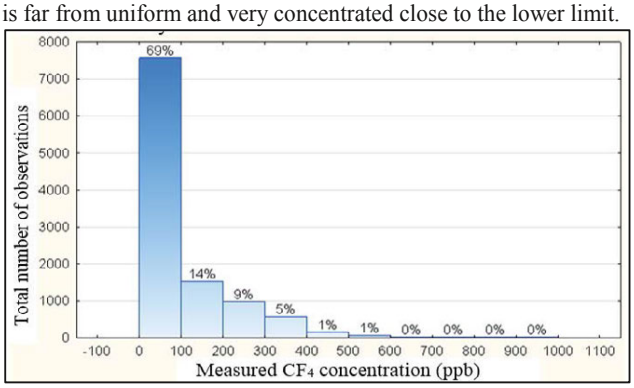

Figure 4: Distribution of the measured CF_4 concentrations (ppb) within the specific limits of this study. (Before separation into two groups)

A non-uniform distribution is not recommended when developing multivariate models. Results will more likely tend to be under predictive regarding the range of the population where the distribution is significantly lower. To have the most efficient model (using most of the data collected), and avoid this problem, the concentration were divided into four categories.

A: 10 ppb to 49.99 ppb
B: 50 ppb to 199.99 ppb
C: 200 ppb to 499.99 ppb
D: 500 ppb to 1000 ppb

Dividing the distribution into such classes will allow us to maximize the potential for good prediction efficiency. However it will require using multiple neural networks model and an algorithm to treat the data accordingly.

Additional measurements with a CF_4 concentration lower than 10 ppb were considered in the analysis to illustrate the behavior of the cell when the PFC emissions are negligible. These data points were used for the development of the first model discussed in the next section.

Presentation of the algorithm

The algorithm has been divided into four steps.[3] It was developed to predict the absence of emissions or in the other case, the concentration of these emissions up to 1000 ppb based on the cell variables. These steps include six different artificial neural networks that use the cell variables to predict a CF_4 concentration at the duct end of the electrolysis cell. The artificial neural networks (ANN) were developed using STATISTICA®. Each individual neural network was developed using uniformly distributed populations to minimize the risk of having over or under prediction.

For validation purpose, the data collected was separated into two groups by using a random sampling algorithm included in the software. Thirty percent of the data was randomly selected to be used for validation and the remaining data has been used to train the artificial neural networks. The four steps of the algorithm are presented and briefly described below:

Step #1: Based on the cell variables, an ANN evaluates if there is presence of LVAE CF_4 emissions? (Yes or No). When developing the model, weights were applied with a ratio of 3:1 (No: Yes) to minimize erroneous predictions of PFC when none are really present.
Step #2: If there are emissions, a second ANN evaluates to which category can they be associated to (A, B, C or D). It was assumed unlikely for the CF_4 concentration to rise abruptly across multiple categories for the short period considered (5 seconds). Hence, a correction algorithm was applied to the results from step #2 to make sure that changes in the categories happen with a reasonable rate.
Step #3: Based on the category defined in step #2, the data is fed through a model respective to its category to define a predicted CF_4 concentration.
Step #4: Results are cross-checked according to time to minimize the effects of wrong predictions. For this paper, a mobile average over approximately 5 minutes was performed based on the current and previous predictions.

Once the four steps are complete, the results provides us with predicted values of LVAE CF_4 emissions that should be similar to the measurements taken with the FTIR. We can anticipate that the first two steps are sufficient to have an idea of the range of the concentration that is expected for specific cell variables. However, the models from the third step give a specific number across this range for calculation purposes.

Every ANN sub-model developed for the purpose of this algorithm used the same 10 independent variables previously described in the

[2] For display purposes, 1 min average are displayed for the CF_4 concentration.

[3] Excluded data from anode effects and recuperation time were removed manually prior to the analysis. However a rule could easily be included in the algorithm itself to perform this operation.

text as "cell variables". The data were collected using the standard control system of the cell or the anode current monitoring system. As described in previous papers from Evans. J et al. [7, 8], individual anode currents were monitored using Hall sensors to follow variations of the magnetic field (gauss) close to anode rod. A model then converts the collected gauss data into current with respect to the cell intensity. However, for the development of the models, only the magnetic field data were considered and not the current values.

The variables used to develop every artificial neural network used in this algorithm are described below:

- Instantaneous stability indicator (nanoOhms): The difference between the maximum and the minimum pseudo-resistance measured for the cell within the last 6 seconds.
- Standard deviation between anodes regarding the magnetic field: For every second, the standard deviation was calculated for individual magnetic field measurements for the entire cell.
- Minimum value of magnetic field: The lowest magnetic field value that has been observed for a single anode in the cell. (During an anode change, this value is equal to zero as one anode is not carrying any current.)
- Range of variation of the magnetic field measurements: For a specific time, this is the difference between the maximum and the minimum magnetic field measured across the entire cell.
- Relative difference between upstream and downstream magnetic field averages: the average magnetic field value is calculated for both halves of the cell and the absolute difference between the two sides is taken into consideration. (This variable will indicate if the current is unbalanced between the anodes)
- Relative difference between tap hole and duct end side magnetic field averages: the average magnetic field value is calculated for both halves of the cell and the absolute difference between the two sides is taken into consideration. (Same as the previous variable but division of the cell was performed across the other axis)
- Average magnetic field measurements: Average values of the measured magnetic field for the entire cell.
- Sum of all magnetic field measurements: Addition of all the measured magnetic fields value for the entire cell.
- Cell voltage (volts): Average value of the cell voltage during the corresponding five second measurement.
- Inhomogeneity factor: Indication of the deviation of the magnetic field for all individual anode with respect to their expected value.

Results and discussion

Validation of the algorithm

As mentioned previously, 30% of the collected data points were taken aside and kept only to validate the entire algorithm. To perform this task, data were fed through the algorithm as illustrated in Figure 5.

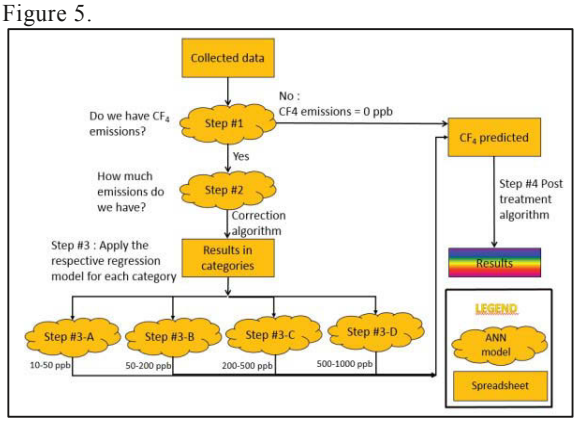

Figure 5: Illustration of the algorithm decision tree

It is important to mention that step #4 (post-treatment algorithm) has little effect if the first two models are not sufficiently precise. Therefore, the efficiency of the prediction is measured after both of these steps as well as for the final results.

As shown in Figure 6, the efficiency of the predictions after the step #1 of the algorithm was 82.5%. This demonstrates that most data set are filtered correctly regarding the production of CF_4 during specific conditions.

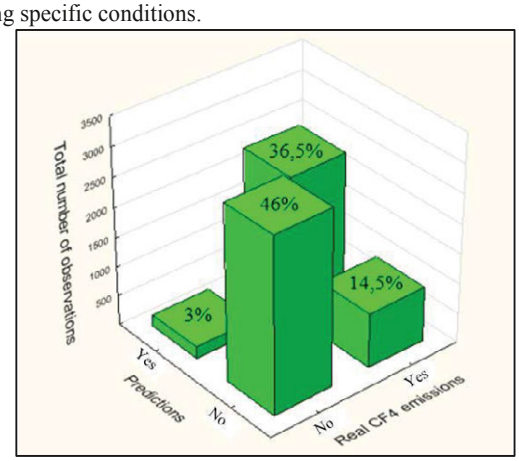

Figure 6: Validation of step #1, efficiency of each prediction

After step #1, data are divided into categories to have an approximate range of concentrations to be expected. Figure 7 illustrates the different characterizations after the correction algorithm has been applied to the results of the second step.

The results show that 71%[4] of the data were classified correctly after two steps in the algorithm. Moreover, we can see that most of the false predictions happened regarding two adjacent

[4] All misclassifications adjacent to a "none" column result from step #1 and are not influenced by the concentration brackets.

categories (26.7%). The effect of this kind of problem should be minimized after steps #3 and #4 have been applied.

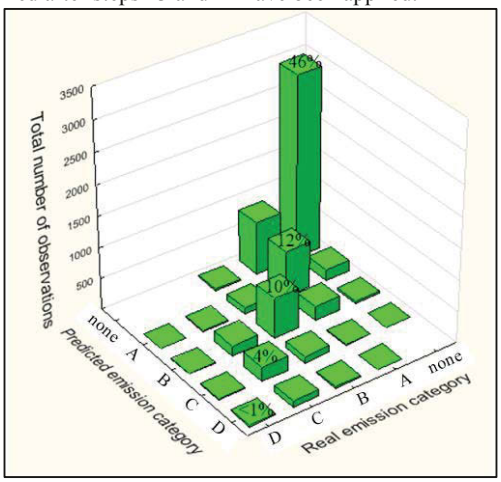

Figure 7: Validation of step #2, efficiency of each prediction (after correction algorithm)

The only difference between steps #3 and #4 is that a correction has been applied on the results. Therefore the validation of the final results was performed only after this correction has been applied.

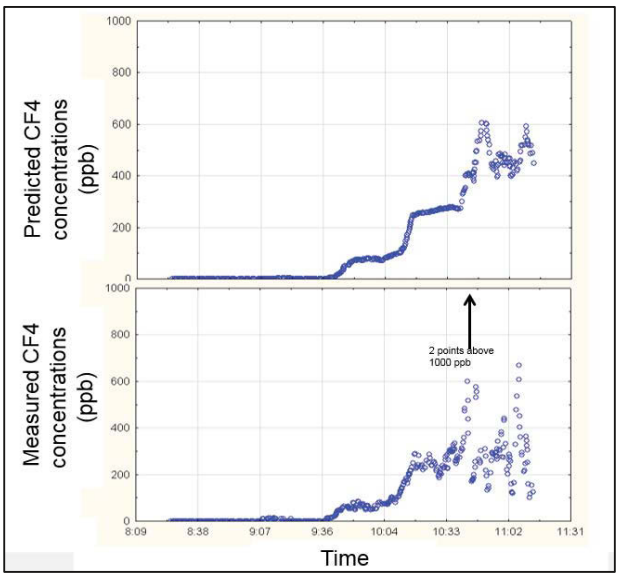

Figure 8: CF_4 emissions behavior for 3 hours of operation during scenario #3. Top) Predictions from the algorithm. Bottom) Measurements from the FTIR

The results from the algorithm appear to be very similar to the real measurements for most of the scenarios with respect to the overall behavior of the PFC emissions as shown in Figure 8. The post-treatment used for this analysis has been chosen for its ability to represent a stable behavior over time. Therefore, it is not ideal when the CF_4 concentration changes more quickly than the mobile average itself. If an explicit need to represent this kind of behavior arises, a different type of post-treatment would need to be performed over the results.

Finally, overall validation was performed by calculating the overall mass of CF_4 emitted during a specific period based on the measurements (scenario 1 through 13) and comparing to the overall mass obtained using the predicted values for the same periods. The mass of CF_4 was calculated using the trapezoid rule for each respective scenario and by multiplying the resulting (ppb*s) by the flow rate at the duct end of the cell and by the CF_4 density for corresponding temperature and pressure.

Table 1 : Validation of different scenarios using the algorithm for predicting LVAE PFC emissions.

Scenario #	1	2	3	4	5	6	7
Mass predicted (g)	2,9	1,3	13,8	4,1	0,8	1,2	7,6
Mass measured (g)	2,7	1,1	10,9	5,2	0,6	1,2	4,7
error (%)	8%	17%	27%	-21%	33%	-2%	62%
Scenario #	8	9	10	11	12	13	Total
Mass predicted (g)	0,3	12,1	0,5	4,4	16,1	0,7	65,8
Mass measured (g)	1,1	14,5	1,0	3,6	11,1	0,6	58,4
error (%)	-70%	-17%	-55%	22%	46%	14%	13%

Validations indicate that the average error in the predictions is approximately 13% with an absolute range going from 2% up to 70 %. Henceforth, the absolute efficiency of the predictions needs to be improved before the model can be used for prediction purposes. It was shown in Figure 4 that very few measurements were within 500 ppb and 1000 ppb. Improvement to the model can be achieved by adding more measurements within this range of concentrations in the learning phase of the neural networks.

It is impossible to compare the efficiency of the presented model with other models as no other quantifying methods regarding LVAE emissions have been previously published. However, if we take into consideration the uncertainties associated with the current models to quantify PFC emissions resulting from high voltage anode effects, the model described in this paper is not too far from achieving a similar goal. In fact, the most precise calculations (level 3) suggested by the International Aluminum Institute (IAI) [2] have an error that can go up to 15% and less accurate calculations (level 1) have an error percentages that can go up to many hundreds of percent.

Finally, the main goal of the approach described in this paper was to offer a better understanding over the effect of some variables describing the electrolytic cell behavior and to have a clearer view about the root cause of low voltage anode effect. The next section clearly indicates that the presented model is sufficient to have quantitative information regarding the effect of different inputs on LVAE PFC emissions.

Sensitivity analysis

The results obtained from the previous algorithm are interesting in terms of prediction efficiency but only a few numbers of cells are equipped with on-line anode current monitoring. Therefore the primary use of this algorithm is to improve the understanding of LVAE by analyzing the influence of the input variables on the predictions. Then, it might be possible to find the root cause of this problem and eventually define a simpler and practical way to quantify these emissions. Because of the complexity of a neural network, combined with the fact that multiple neural networks are used sequentially, a sensitivity analysis seemed to be the most efficient approach to evaluate the

impact of each input on the final results. Most of the information resulting from the sensitivity analysis will be discussed in another publication by the authors. However, some preliminary results are presented and briefly discussed below.

The presented part of the sensitivity analysis investigates the frequency of predicted continuous emission events when variations occur for a selected variable. For each point presented within Figure 9, a total of 6561 different combinations were fed into the model. The respective number of resulting predictions with PFCs is then illustrated to see how the variables affect the predictions over their entire range. These ranges are based on the set of input variables used to develop the models. Values outside of their respective range would be extrapolations and could generate incoherent results.

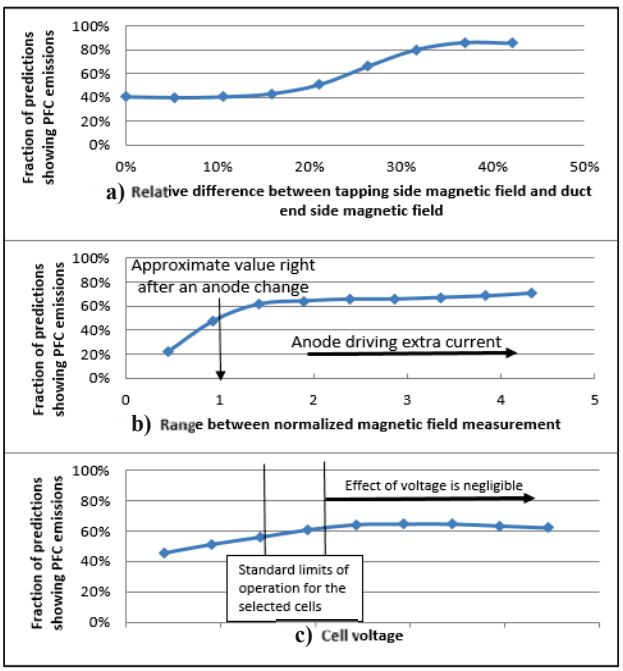

Figure 9: Percentage of "predictions with CF_4 emissions" in comparison to some input variables for the model. a) Effect of the non-uniformity of the current distribution between two sides of the cell. b) Effect of the range between the anodes current distribution. c) Effect of the total cell voltage.

Figure 9a) represents the difference in current between the tapping and the duct end sides of the cell. Values on the x-axis represent the respective deviation in current for all anodes on each side of the cell. A value of 20% on the x-axis indicates that one side is carrying 120% of the normal current and the other is carrying 80% of the normal current. The results clearly indicate that any change in current up to 15% have little effect on LVAE PFC emissions. However, the frequency of those emissions doubles between 15 and 30% and stabilizes afterward. It was shown in a previous paper [8] that a local change in the alumina concentration can be important enough to generate these drifts in anode currents. Therefore, a lower concentration in a localized zone of the cell is in agreement with the presence of PFC emissions.

Figure 9b) investigates the maximum deviation between magnetic fields measured between two anodes in the cell for a specific time (max – min). The figure shows that when all anodes tend to carry equal current (X→0), very low PFC emissions can be expected. On the other hand, the number of cases showing PFC emissions increases with the deviation up to a value of 1.5 and the behavior stabilizes afterward. By assuming that under ideal conditions, the value of this variable should be very close to "1" right after an anode change, these results are in strong agreement with the observations from Jassim et al. [5] where CF_4 emissions were observed after anode setting as long as the current did not reach about 50% of its target value.

Figure 9c) illustrates the cell voltage which has an important effect when quantifying PFC from HVAE. However, in the case of LVAE, the probabilities that the event occurs increases linearly with the cell voltage up to a certain limit, close to the upper limit of normal operation. After this threshold, further increase in voltage does little effect on the probabilities of low voltage anode effect. However, this analysis gives no indication regarding the concentration of CF_4 in these emissions. Hence, it should not be assumed that the total amount of CF_4 emitted remains constant during an event where the cell voltage is higher than this threshold.

Conclusion

In this paper, a successful algorithm to quantify LVAE PFC emissions was presented as well as the steps leading to its development. This algorithm uses different artificial neural networks sequentially and correction algorithms to predict the concentration of LVAE PFC emissions collected at the duct end of an electrolysis cell. The average error of these predictions is approximately 13%, but errors in some scenarios have reached 70%. However, the predictions qualitatively represent the general behavior of the emissions over time with great accuracy.

The main benefit of this algorithm is the improved understanding that it can offer regarding the root cause of low voltage anode effect. The effect of each input variables can be better understood by performing a thorough sensitivity analysis using the developed models and algorithm.

Acknowledgments

The authors deeply appreciate the support from Aluminerie Alouette to publish and present these results.

The helpful recommendations provided by Dr. Jerry Marks for the measurement campaigns are acknowledged and greatly appreciated.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ANODE EFFECT INITIATION DURING ALUMINIUM ELECTROLYSIS IN A TWO-COMPARTMENT LABORATORY CELL

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Keywords: PFC, Anode effect, NAE-PFC, Aluminium electrolysis, GHG, Low voltage AE

Abstract

Most laboratory cells used in the investigation of the alumina reduction process use a single anode. When investigating the initiation of the anode effect an approach with more than one anode might give better results, as the probability of obtaining partial anode effect is higher. Additionally, the design is closer to the industrial, where several anodes are connected in parallel. The system constructed consisted of two anodes in separate electrolyte compartments connected in parallel with a single combined cathode. The results indicate that an anode can go in and out of partial anode effect with little influence on the current, although, kept untreated a full anode effect is likely imminent. The results also show that under certain current and alumina conditions, with only two anodes in parallel, an anode can handle approximately the whole load of a fully passivated anode for a certain time.

Introduction

Anode effects (AEs) have been a part of the aluminium production industry from its infancy. At some point in time a certain AE frequency was wanted as this meant the probability of having sludge in the electrolyte was limited. The big shift in working procedure came together with the invention of the transistor and the rise of the semiconductor industry in the second half of the 21st century. Improved cell control and operation reduced the amount of anode effects, and by extension also the emission of perfluorocarbon (PFC) greenhouse gases CF_4 and C_2F_6. The International Aluminium Institute (IAI) in London reports a decrease in PFC emissions of more than 80 % from 1990 to 2010 [1]. Most of the improvements are the results of a decrease in the traditional AE where cell voltages would rise to levels of 8 V and above [2] with an accompanying erratic current distribution. However, the possibility of producing PFC gases is not governed by the cell voltage, but interfacial electrode potential, and recent investigations show that PFC evolution at normal cell voltages does occur and can be a substantial part of the total [1,3,4,5].

Cell and anode voltage

Most laboratory cells for investigation of aluminium production are a simple electrochemical cell consisting of an anode and a cathode. The cell voltage of such a system would be comprised of the anodic and cathodic potentials, as well as the voltege drops (IR) over the electrolyte, through the anode, the bubble layer (bubble overvoltage) and cathode lining, as described by the following equation (external IR omitted)

$$V_{Cell} = E_A + |E_C| + I_{tot} \cdot (R_A + R_{Bubble} + R_{El.} + R_C) \qquad (1)$$

An industrial cell, on the other hand, have a large number of anodes (20 – 40) connected in parallel. In an ideal case all the parallel segments would be the same with the current shared evenly among them. However, in a real cell there will be as many different systems as there are segments, as discussed by Welch [6]. The anodes are consumed and their resistance will change throughout their lifetime and the number and size of anode slots will influence the bubble resistance. In addition, alumina concentration gradients can develop due to electrolyte mixing or poor feeding. The anode to cathode distance (ACD) may vary due to megnetohydrodynamics (MHD) or work practices.

To maintain the condition that the voltage over each of the parallel circuits is equal, the current can develop non-uniformly over the anodes. By considering the fact that molten aluminium is uniform and has very high electrical conductivity, its potential can be assumed equal for all parallel circuits. Split up on a per anode basis with partial currents the cell voltage becomes:

$$V_{Cell} = E_{A,i} + I_i \cdot (R_{A,i} + R_{Bubble,i} + R_{El.,i}) + |E_C| + I_{tot} \cdot R_C \qquad (2)$$

From this it is understood that current-flow through an anode can vary with anode age, temperature, electrolyte and alumina mixing, and work practises like anode change. Any change of current through one segment will introduce an equal, but opposite change that will be divided over the rest of the segments, or parts of them. The cell voltage, on the other hand, will increase, but only marginally (up to a certain redistribution limit where the critical current density is reached). From this it is evident that a segment carrying low current will have an elevated interfacial electrode potential that can give rise to new reactions. Approximately all of this potential will be located on the anode, as little overpotential is associated with the cathode. In addition the cathode is interconnected by a highly conducting aluminium metal pad and the potential maintained equal for all parallel segments. If the anodic potential rise is of sufficient magnitude co-evolution of fluoride containing gases like CF_4 is possible. This again could give an increased interfacial resistance bringing forth an even larger drop in current and larger redistribution to other segments. Untreated, such an initiation of PFC evolution on one anode can propagate to a full cell anode effect [7].

In this work a two-compartment cell has been constructed in order to extract some of the properties of the parallel circuitry of the large industrial cell.

Parallel circuitry

Properties of parallel anodes

Consider a laboratory cell with a single anode running under constant current conditions. The cell will be described by Equation 1, and given it is properly fed by alumina the interelectrode potential will be of such a magnitude that only production of CO/CO_2 is thermodynamically possible. If the current is decreased, the voltage drop across the different resistances will decrease and the cell voltage will be lowered. This is quite contrary to an industrial cell where the cell voltage only changes marginally with current decrease on a single anode, and in fact to a higher value. Consequently, in an industrial cell this will introduce a redistribution of potential that will mostly be taken up by the anode that saw a lowering of current, perhaps making co-evolution of fluoride containing gases thermodynamically achievable. The laboratory cell circuit, on the other hand, will see little change. The lowering of current will lower the anodic potential slightly, but with just one segment the overall cell voltage will be lowered as the circuit is connected in series and voltage redistribution (to the anode) is not possible.

Another important difference between the single and multi-anode system is the pathway towards an anode effect. In the laboratory cell, when the alumina starts to be depleted, polarization will increase the possibility for other reactions. Evolution of PFC gases will begin to passivate the electrode. The set current has to pass through a smaller and smaller active area, beginning a transition that ultimately ends up in a full anode effect. In the industrial case, when polarisation increases due to limited alumina there will be a corresponding increase in equivalent resistance of that parallel segment, redirecting current elsewhere. Should the alumina concentration improve (as it might with less current) the cell can return to normal. In the meantime, it is likely that the affected anode will produce some PFC.

From the above reasoning, it is understood that normal cell voltage PFC production cannot easily be achieved on a single electrode system. For any redistribution of current to be possible a minimum of two parallel circuits are required. Additionally, the potential needs to be at a level where it is thermodynamically possible to produce PFC. Industrially this is accomplished through the sheer parallelism of all the different resistances in the cell. In the laboratory, this is perhaps easier realised by introducing resistances in series with the anodes, as shown in Figure 1.

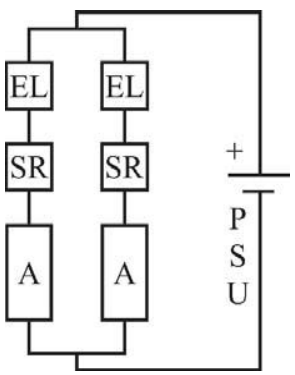

Figure 1. Electric circuit representation of the electrochemical cell. The shunt resistances (0.05 Ω) are used to measure the current of each parallel circuit and the electronic loads are used to bring the total cell voltage to industrial levels. The anodes are depicted as resistances for simplicity.

The shunts are just precise resistances used to measure the current through each of the anodes. The electronic loads (ELs) depict the added resistances that are used to take the cell potential to industrial levels. Regular resistances can be used; however, ELs present better options for control.

All ELs provide a certain resistance to the system, but the way they do so depends on the control mode they are set to. In constant resistance (CR) mode the resistance is set directly. A redistribution of current will give a redistribution of voltage as the voltage drop over the resistance changes. In constant current (CC) mode the maximum current allowed through a segment is set. In this mode the resistance is as low as possible when the current is below its set-point, while it will increase to any possible resistance to keep this current set-point. This makes the mode unfit for anything other than setting a specific current and forcing the rest through the remaining parallel circuit(s).

Constant voltage (CV) mode works by setting a voltage drop, and the load will vary the resistance with the segment current to maintain this voltage drop. In this mode current can to some extent be easily diverted by changing the voltage, however, there will be no automatic redistribution of voltage, from load to anode, with an approaching AE within a segment. With loads of equal magnitude, the setup is no different from just connecting two anodes in parallel, except that the cell voltage will be higher, and more energy dissipated as heat. The extra voltage of the load will not be able to redistribute to the anode, making observation of PFC during normal cell voltages less likely. The CV mode can be improved by constructing a program that changes the set-point from certain system inputs. A possible method would be to have the load change to always keep the total cell voltage at a specific value, making voltage redistribution possible within a segment. This will additionally require very fast response for satisfactory results.

Virtual anodes

A future extension to the system is to increase the parallelism by adding purely electrical elements. When a large industrial cell runs ideally each segment will have an equivalent resistance. Similarly, the series combination of resistance, shunt and anode in the laboratory cell, will also have an equivalent resistance, that will henceforth be denoted the ideal segment resistance, R_S. Each of these electrical segments will act as a virtual anode. Consider a cell having n number of segments, where m of these segments are real live anodes. An equivalent resistance for all the virtual anodes can be created as shown in the following relation:

$$\frac{1}{\mathrm{R}_{eq}} = \frac{1}{R_{S,1}} + \frac{1}{R_{S,2}} + \frac{1}{R_{S,3}} + \cdots + \frac{1}{R_{S,n-m}} \qquad (3)$$

With an increasing number of virtual anodes the equivalent resistance will decrease rapidly, making it necessary with a very stable and precise resistor like an electronic load of good quality. Still, the system will be limited by the short circuit resistance of such a load, but also by the current capabilities of the power supply, since the amperage scales linearly with the number of segments. An example of a circuit diagram with two real anodes and a number of virtual anodes is depicted in Figure 2.

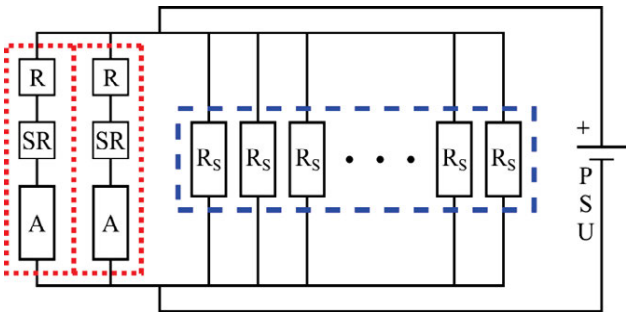

Figure 2. Circuit diagram of two real and a number of virtual anodes connected in parallel. *R* depicts resistances to bring the cell voltage up to industrial levels, *SR* are shunt resistances to measure the current going through the segment, and *A* are the anodes. R_S is the virtual segment resistance.

The system should work in the same manner as the two-segment one described earlier, however it is less likely to go straight to full anode effect when a real segment struggles, since there is a larger number of parallel circuits to redistribute the current across. The virtual anodes will naturally behave ideally at all times and cannot enter a full anode effect. In fact, the likelihood of very high potentials on the real anodes will also decrease with increasing number of virtual anodes, still slightly elevated industrial levels is more than enough to produce PFC.

Methods and Equipment

Experimental cell and electrodes

The experiments were conducted in a regular laboratory furnace with closed atmosphere. The cathode was machined out of a cylindrical block of graphite (Schunk-Tokai). The block had a diameter of 90 mm and an inner height of 110 mm. The two electrolyte compartments were separated by a 5 mm wall. Two steel plates of the same shape (1 mm smaller on all sides) and 5 mm thickness was added to the base of the crucible to improve the cathode aluminium wetting and give a flat and uniform metal pad. A picture of the cathode and electrolyte container is given in Figure 3. Two purely round compartments could have made it possible to easily shield the sides from conducting any current; however, this concept was dropped as it would have made the electrolyte compartments too small to employ reference electrodes or gas collecting tubes.

Figure 3. Graphite two-compartment crucible (cathode) and steel bottom plates.

The electrolyte consisted of cryolite (Aldrich), aluminium fluoride (industrial grade, sublimed at 1090 °C) and alumina (Aldrich). The cryolite ratio (CR, NaF/AlF_3) was set to 2.3 at the beginning of each experiment, but was slightly altered through the experiment with consumption of alumina. The amount of electrolyte in each sub-cell was about 100 cm³ giving a bath height before electrode immersion of approximately 4 cm.

The anodes were made from graphite (Schunk-Tokai, ultrapure) rods (Ø 16 mm) encased in hot-pressed boron nitride (Kennametal) rods (Ø 22 mm). The BN insulation had a 45 ° inward slope towards the bottom of the anodes, making 3 mm of the vertical graphite surface accessible by the electrolyte, as can be seen from Figure 4. In total the initial active surface area was 3.5 cm². The sloping was added to better facilitate gas removal, as purely horizontal surface would create a BN gas trap after the initial graphite layers had been consumed. The anodes were connected electrically by a steel tube (Ø 6 / 8 mm). This gave room inside for an alumina (Ø 5 mm) 5-bore rod encasing both a thermocouple, and a sense wire that connected to a hole in the top of the anode carbon, giving potential and temperature readings as close to the reaction interface as possible. In some experiments, after ~ 5 h of run time electrolyte fumes were observed to penetrate the steel-graphite-BN coupling, dissolving the open-end alumina tube such that the thermocouple shorted with the anode, giving rise to incorrect readings. This has partially been solved by either using a closed end alumina tube or a BN shield. Unfortunately, in the latter case, there is no room for a sense wire so it has to be connected to the current lead, later subtracting the ohmic drop found by impedance measurement.

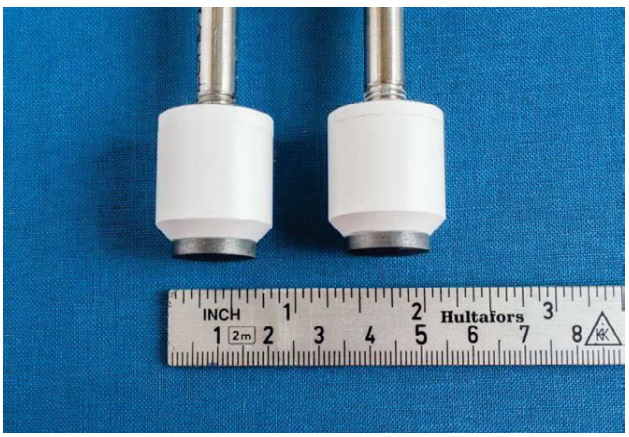

Figure 4. Anode made out of a graphite rod (Ø 16 mm, active height 3 mm) shielded with a 3 mm wall of BN, 5 mm on top. The current is supplied by a hollow steel tube that can be fitted with a sense electrode and/or a thermocouple.

Two aluminium references were used, one for each electrolyte chamber. They were constructed from a milled out BN rod (Ø 5 / 10 mm) of length 10 cm that was threaded onto a steel tube. Aluminium was placed inside, enough to make a pad of ~ 1.5 cm in height and a hole was drilled through the BN rod about 5 mm above the aluminium level. Electrical connection was provided with a tungsten wire (Ø 1 mm) shielded in an alumina tube (Ø 1.5 / 3 mm), except for the bottom 5 cm that would otherwise be in contact with electrolyte and aluminium, which had to be shielded with a BN tube of similar diameter. When doing experiments in alumina saturated melts, the outer BN and steel tube can be

exchanged with an alumina tube for simplicity. A sketch of the complete electrode setup can be viewed in Figure 5.

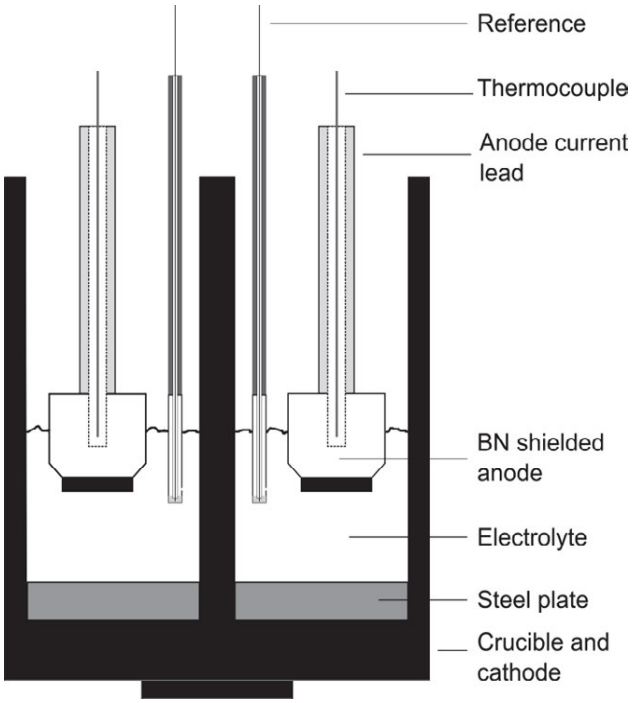

Figure 5. Experimental setup of two-compartment crucible. Each compartment has its own anode, reference and electrolyte.

Electrical equipment and data loggers

The current was supplied to the cell by a power supply (Agilent 6032A) controlled through LabView software. The positive side was split in two parallel sections, each connected through an electronic load (Agilent N3302A in Agilent N3301A chassis) and a shunt resistor (0.05 Ω) before passing through the anode, as showed in Figure 1. All data was logged with a National Instruments CompactRIO 9074 controller with analog voltage modules NI 9229 and NI 9205, as well as thermocouple module NI 9211. The equipment was controlled through LabView and data was sampled at 10 Hz, except the thermocouple module that only updates at 3 – 4 Hz.

Gas analysis

Current voltage characteristics can provide valuable information about cell performance; nonetheless, in order to improve certainty about the reactions occurring at the electrode interface, gas analysis has to be performed to evaluate the gases produced at the anode. Two different mass spectrometers (MS), an FTIR and a CO/CO_2 logger were used in these experiments.

Gas samples were taken from ~ 2 cm above the anode to get reasonable response time (< 1 min) and sensitivity. In the case of the MS that sampled gas at ~ 20 ml/min rate, a thin alumina tube (Ø 2 / 4 mm) was used. The tube would clog from evaporating electrolyte in 1 – 2 hours, requiring it to be cleaned or changed. The FTIR and CO/CO_2 logger were set to sample at ~ 80 ml/min and a wider alumina tube (Ø 4 / 6 mm) was used to extract the gas. The furnace was purged with ~ 100 – 150 ml/min of argon, which was always kept higher than the combined sampling rate of the attached gas equipment, with surplus gas exiting the main exhaust. The main exhaust port could also have been used for gas analysis, which would have eliminated the clogging problem. However, the response time would have been severely decreased and it would have been difficult to distinguish from which anode gas evolution originated from.

Results and Discussion

Figure 6 shows an event observed on a two-compartment cell with a shunt and an electronic load in series with each anode. The load was running in CV mode, the one previously argued to less likely give partial AE, with a potential of 1.4 V. The power supply ran under constant current conditions with a throughput of 6 A, enough to give a current density of 0.85 A/cm² on each anode (relative to initial geometric area). The voltage limit of the power supply was set to 10 V. The temperature set-point was ~ 1000 °C at idle.

At 28730 s anode 2 was abruptly passivated and reduced to carrying less than 10 % of its initial load. On the other hand, anode 1 increased its current load by almost 90 %, and in most cases such a big change at this current density would tend to result in another fully blocked anode. The theoretical alumina concentration at this point was 1.6 wt% and 0 wt% for anodes 1 and 2, respectively, the latter theoretically depleted 50 minutes prior. The FTIR shows the gas analysis on anode 2 with 1281 cm^{-1} being the wavenumber of the strongest absorption line for CF_4. No significant change is visible in the absorbance, possibly because the gas tube was clogging up at about this time. The tube was eventually quickly exchanged just before 29400 s, as can be seen from the cooling of anode 2 at this instance. With the new tube a shift in absorbance is observed; still, it is small and it can just be a new baseline stemming from disturbances introduced by the change.

Anode 1 carried almost the full load of the power supply for ~ 9 minutes before it eventually went on full AE. At this point no further redistribution of current was possible since anode 2 already was fully passivated. The MS shows gas analysis on anode 1 and mass to charge (m/z) ratio 69 is the strongest signal for CF_4/C_2F_6. The signal is not strong, but there is a significant shift just as anode 1 is passivated. After the initial passivation the current stabilized at about 0.12 A/cm² and the anode potential increased to approximately 8.5 V, while the PFC signal has dropped to below the noise level.

In another experiment the electronic load and shunt was replaced with a 1 Ohm resistor. The current was set to ~ 0.73 A/cm² per anode and the temperature set-point was 1000 °C at idle. An event believed to show an anode going in and out of partial anode effect is shown in Figure 7. At time point of 34500 s the theoretical alumina concentration was depleted in chambers 1 and 2 approximately 1.5 h and 0.9 h ago prior to the onset of initial AE. Anode 2 is believed to be under intermittent anode effect in the areas where the observed current is lowered. The current shift is only a mere 10 mA/cm², however it is believed that this change is sufficient for alumina concentration to restore and the anode effect to be extinguished, at least temporarily. The voltage drop related to the extra load on the parallel circuits will in this case fluctuate completely as the current does, giving a slightly decreased voltage drop with decrease of current and a

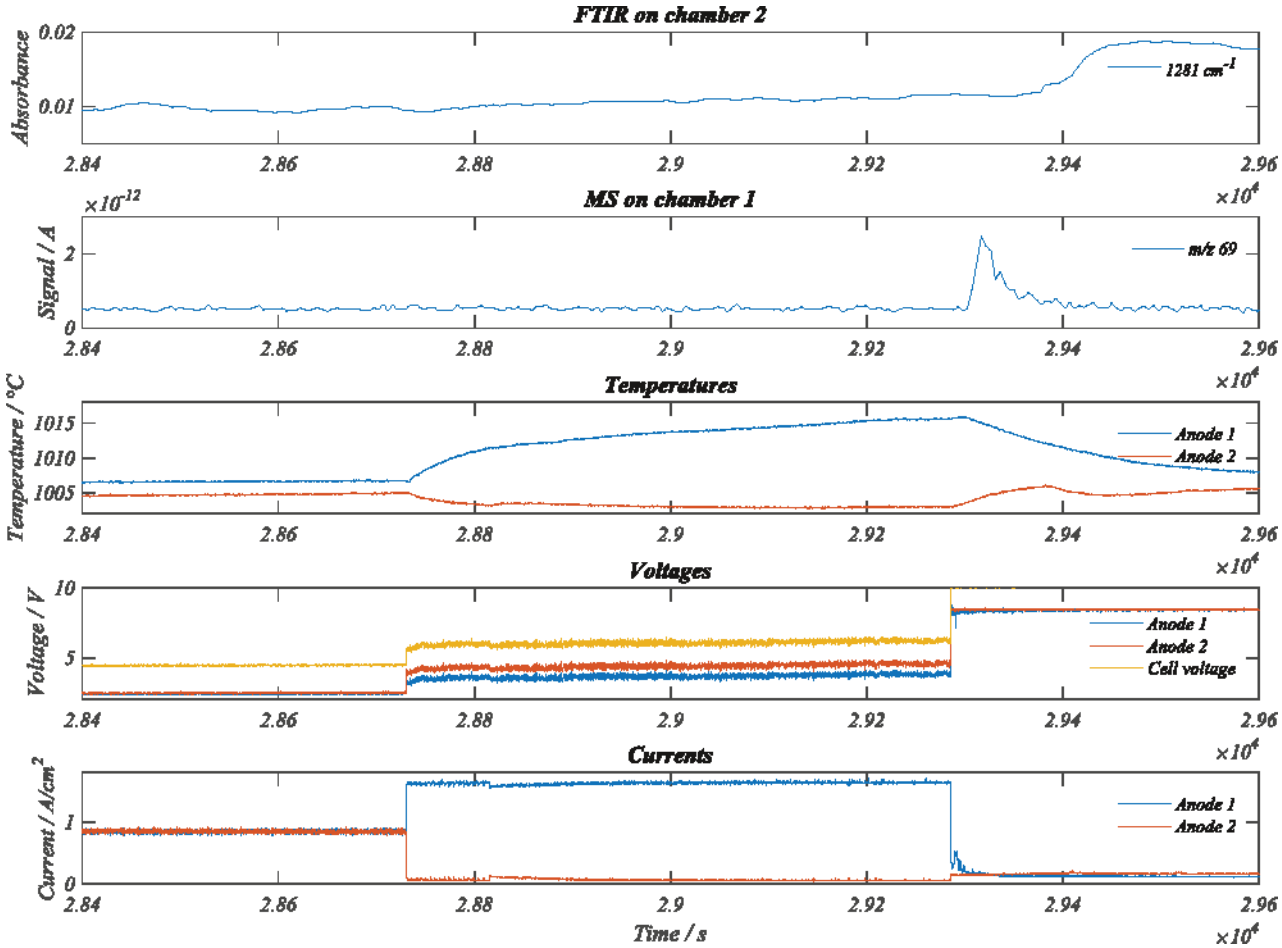

Figure 6. Development of current, voltage, temperature and PFC gas signals, as two anodes make the transition to AE and complete passivation.

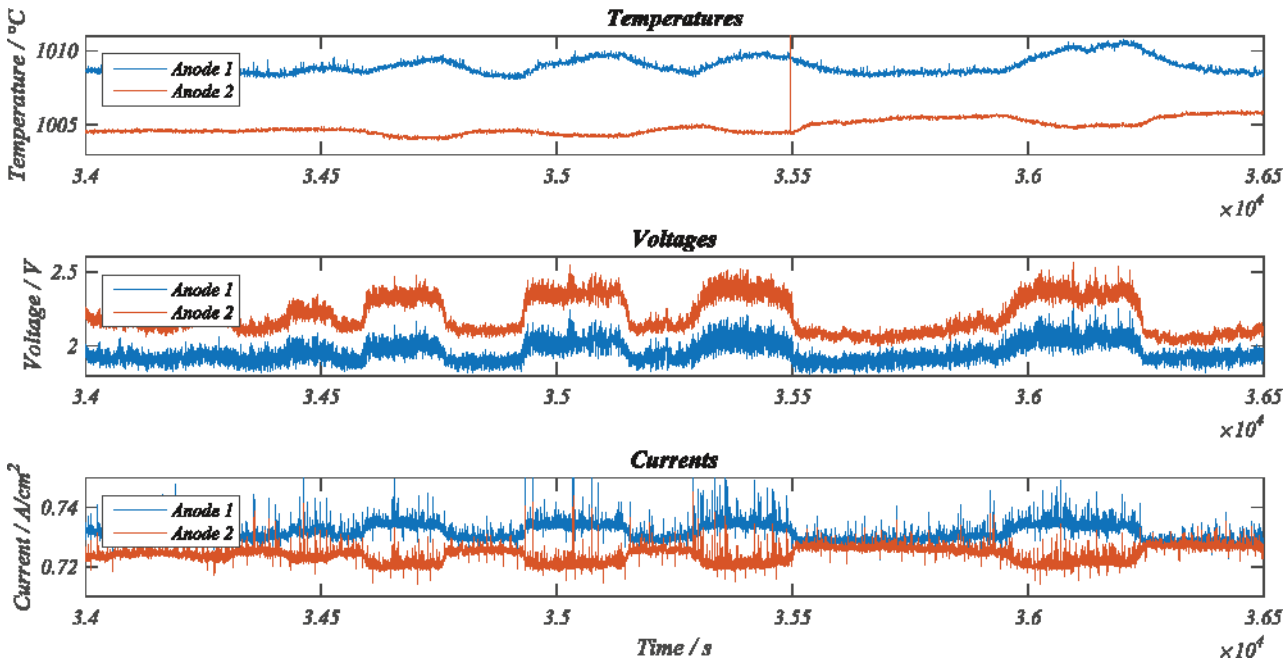

Figure 7. Development of current, voltage and temperature at two anodes connected in parallel. Anode 2 is believed to go in and out of partial anode effect.

corresponding increased voltage drop with increase of current. Thus, by considering segment 1 a slightly higher voltage drop over the load is expected; additionally, the potential of anode 1 is expected to increase with increased current, albeit just marginally. Nonetheless, the potential over anode 1 increased by ~ 100 mV, which is more than what is expected from a small increase in current. The extra shift can be due to degree of parallelism of the system as the total cell voltage is shifted by anode 2's reach for new reactions by increased polarisation. It is expected that a larger degree of parallel circuitry would influence the voltage of non-AE anodes less.

The cell went on full anode effect on both anodes about 150 s after the events shown. Unfortunately gas analysis data are not available to confirm the onset of anode effect.

Conclusions

By employing a laboratory cell with more than one anode it is possible to study how an anode effect on one anode, (either partial or full) affects the rest of the system. Gas analysis can be used to verify the reactions accompanying certain events. Robust laboratory equipment and a well-designed system are needed to observe gas constituents at low concentrations in a relatively low-amperage laboratory cell.

Acknowledgements

The authors acknowledge the financial support received from Norsk Hydro and the Norwegian Research Council.

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ALUMINUM REDUCTION TECHNOLOGY

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ENHANCING PRODUCTION PERFORMANCE BY OPTIMIZATION ALL RESOURCES AT PT INALUM (PERSERO)

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Keywords: Optimization, Improvement, Performance, Inalum

Abstract

PT. Indonesia Asahan Aluminium (Persero) – hereinafter referred to as Inalum is the pioneer of the aluminium smelter in South East Asia region. It was designed by Sumitomo Chemical and started up in February 1982 with the original capacity was 225,000 Tons/year of ingot and 175 kA of amperage.
After transforming to be State Owned Company in the end of 2013 year, Inalum had an ambition to continue the optimizing of production to be 300,000 Tons/year of ingots in 2019 year. For the first step, in 2014, Inalum successfully increased production up to 264,474 Tons/year of ingots (more 17.5% than original capacity) by increasing Line current up to 207 kA.
The technical improvement objectives are increasing of production capacity by new alumina feeding system, new pot design, anode performance improvement and implementation of water turbine efficiency.

Introduction

Inalum transformed to be State Owned Company in December 19, 2013 after founding on January 6, 1976 as Joint Venture Company between the Government of Indonesia with 12 Japanese Investors incorporated in Nippon Asahan Aluminium Co.Ltd.
The smelter has operated by existing design by Sumitomo Chemical with original capacity 225,000 Tpy of ingot with 175 kA of amperage. It consists of 3 pot lines each 170 side-by-side end riser prebaked cells with bar breaker system and supported by Hydroelectric Power Plants with total capacity 603 MW.

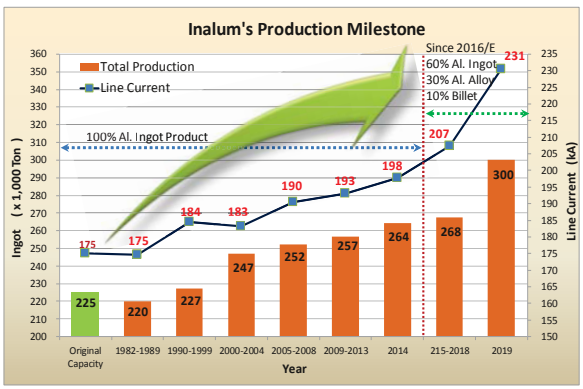

Figure 1. Milestone Production

Since in the end of 2014, Inalum had an ambition to continue the optimizing of production to be 300,000 Tons/year of ingots in 2019 year. For the first step, in 2014, Inalum successfully increased production up to 264,474 Tons/year of ingots by increasing line current up to 207 kA.

Improvement Method

Inalum commit to continuous improvement of the performance and the product quality with safe, stable and environmental friendly operation.

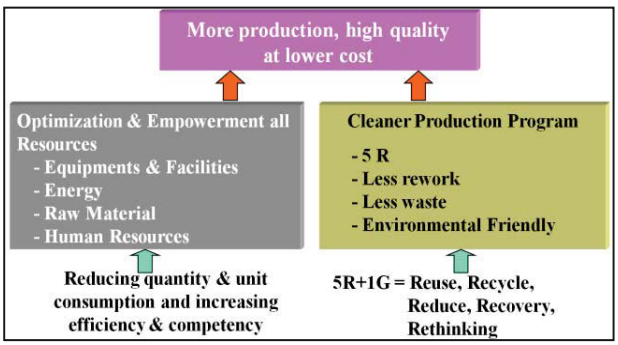

Figure 2. Technical Improvement Method

Summary of Technical Improvement

Table I. The main optimizations which has been applied

Item	Result
Reduction Plant	
a.New Alumina Feeding System	CE : +1.9%
b.Pot Design D5-2	Decrease DC Cons. 32.2 kWh/T.Al
Anode Performance	
a.Anode Quality Improvement	Na <50ppm (< 300 ppm) Butt size over 5 mm: 2~10 % (target 5 %)
b.Enlargement LA3 to be LA4	Line Current up to 215 kA
Power Plant	
Water turbine efficiency	Increasing of power generation 7.9 MW

Inalum Technical Improvement

A. New Alumina Feeding System

Application alumina feeding system has objective to increase productivity by increasing of CE by ca.1% through optimizing of alumina concentration and support the application of higher amperage program.
Basic Philosophy of new alumina feeding system is:

1) Quantity alumina charged for 20 kg/shoot has alumina concentration in bath ± 2.2~3.6% with effect the forming of sludge.

2) Meanwhile, quantity alumina charged for 4 kg/shoot (error frequency ±1 kg/shoot) with more charging frequency than existing equipment has alumina concentration in bath ± 1.5~2.6%, lower than existing with target increasing of pot productivity.

Figure 4. Left (Old Equipment, @20 kg/charge), Right (New Equipment, @4 kg/charge)

The advantages of new alumina feeding utilization as below:
1) Stable condition of operational pot and increase production capacity
2) Environment friendly by lower anode effect frequency
3) Easy maintenance work and reduce maintenance cost

Table 2. The operational result between existing with new alumina slide feeding

Parameter	Unit	Exist	New System	Diff.
Productivity	T/p.y	517.6	528.5	10.9
Current Efficiency	%	90.62	92.49	1.9
DC Consumption	kWh/T.Al	14,198	14,011	(187)
AE Frequency	t/p.day	0.53	0.43	(0.10)
AE Duration	Sec./AE	90.0	71.2	(18.8)
Sludge Formation	cm	7.2	5.7	(1.5)

B. Pot Design Improvement D5-2

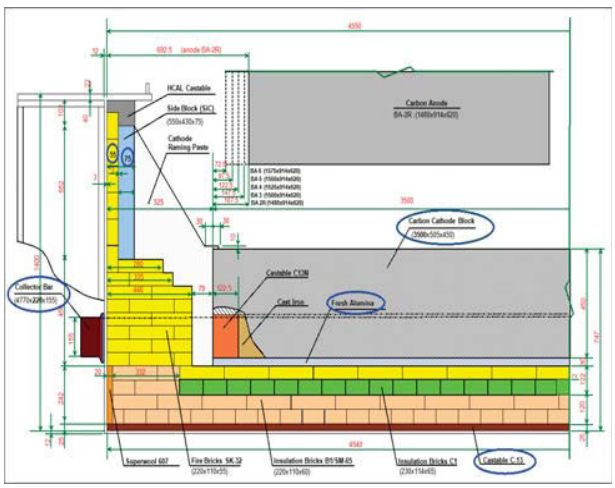

Figure 6. New Pot Design D5-2

One of main factors on pot performance and consequence to metal production cost is pot lining life. The lining life depends on the structure of material and design of pot lining. A number of improving and developing has been done continuously to gain the most suitable and profitable pot design, such as D3-3, D3-4, D3-5, D3-6, D5-0 and the latest is D5-1.
Pot design D5-2 is improved using cathode assembly with longer cathode block (length : 3,500 mm) and bigger collector bar (width : 220 mm), and rand block SiC t: 75 mm with back up brick SK-32 on behind of the rand block; equally combined pot design D3-3 with D5-1 as anticipation to operate at higher amperage up to 210 kA in 2014.

Table 3. Lining construction of (D5-2)

No	Item	Existing		Improvement
		Pot Design D3-3	Pot Design D5-1	Pot Design D5-2
1	Sidewall material	SK-32 back-up brick (55 mm)	Without back-up brick	SK-32 back-up brick (55 mm)
		SiC rand block (75 mm)	SiC rand block (100 mm)	SiC rand block (75 mm)
		Thickness : 130 mm	Thickness : 100 mm	Thickness : 130 mm
2	Bottom Wall Material	Using ramming paste under cathode assembly 30 mm-thick (± 1,200 kg)	Using Alumina under cathode assembly 30 mm-thick (± 1,150 kg)	Using Alumina under cathode assembly 30 mm-thick (± 1,150 kg)
3	Cathode Assembly	Collector bar dimension 4770L x 200W x 155H mm.	Bigger Collector bar dimension 4770L x 220W x 155H mm.	Bigger Collector bar dimension 4770L x 220W x 155H mm.
		Cathode block dimension 3340L x 505W x 450H mm.	Longer Cathode block dimension 3500L x 505W x 450H mm.	Longer Cathode block dimension 3500L x 505W x 450H mm.
			Collector bar in cathode insulated by castable C13N by 92.5 mm	Collector bar in cathode insulated by castable C13N by 92.5 mm

D5-2 pot design improved Cathode Voltage Drop (CVD) by 7.6 mV hence there is opportunity to decrease DC Consumption by 32.2 kWh/T.Al and increase productivity 3.0 kg/pot. day.

C. Anode Quality Improvement

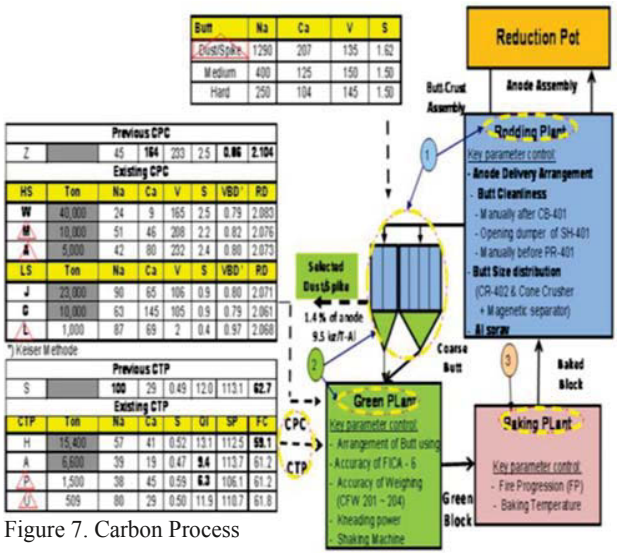

Figure 7. Carbon Process

Inalum has the constraint about lower trend of CTP quality (Low QI, and Meshopase) and vicious cycles of pot deterioration create bad quality of butt (Spike, Soft, dusty).
Improvement target as below:

1. Lower Na in butt (Na < 300 ppm) and Coarser (Size over 5 mm increase 5 %)
2. Higher RRO2 & RRCO2 Anode: 1±0.5 % and lower RDO2 & RDCO2: 0.5 ± 0.2 %.
3. Moderate Thermal Conductivity Anode : 3.5 ± 0.5 W/m°K

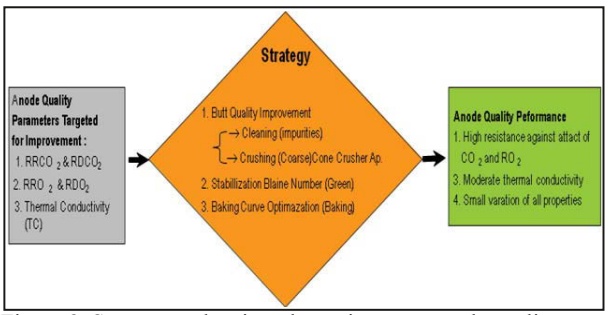

Figure 8. Strategy and action plan to improve anode quality

Result of improvement as below:

1. Improvement of butt cleaning : Na content decrease much < 150 ppm (target < 300 ppm)
2. Utilization of cone crusher: butt size over 5 mm increase 2~10 % (target 5 %)

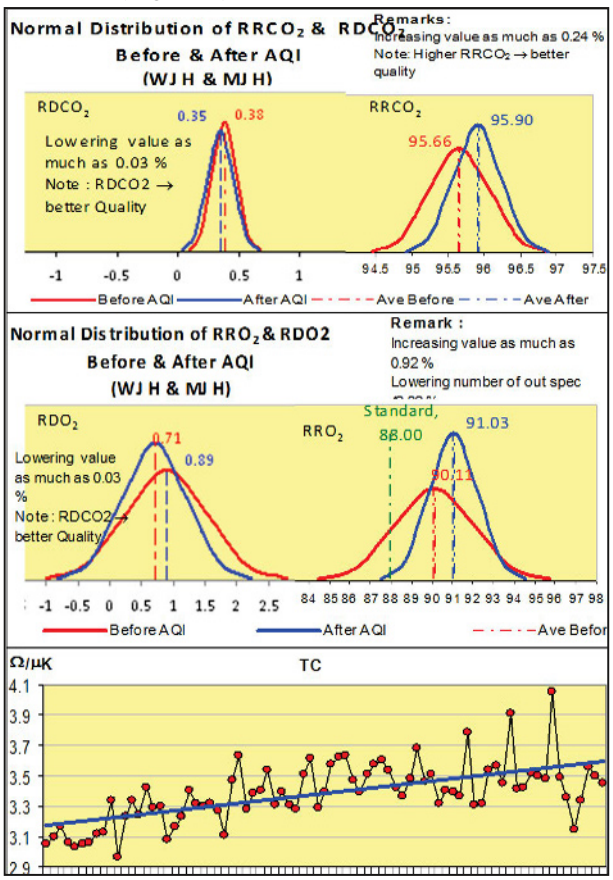

Figure 9. Variance of Air & CO_2 Lower Reactivity & Increase Thermal Conductivity by 3.5 W/m°K

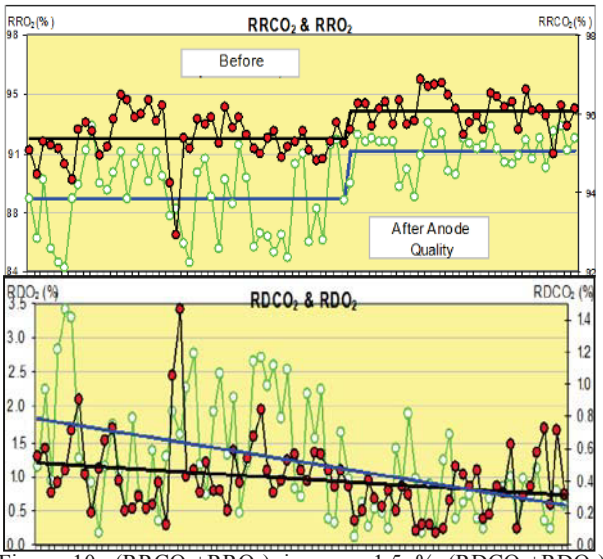

Figure 10. ($RRCO_2$+RRO_2) increase 1.5 % ($RDCO_2$+RDO_2) lower by 0.5 %

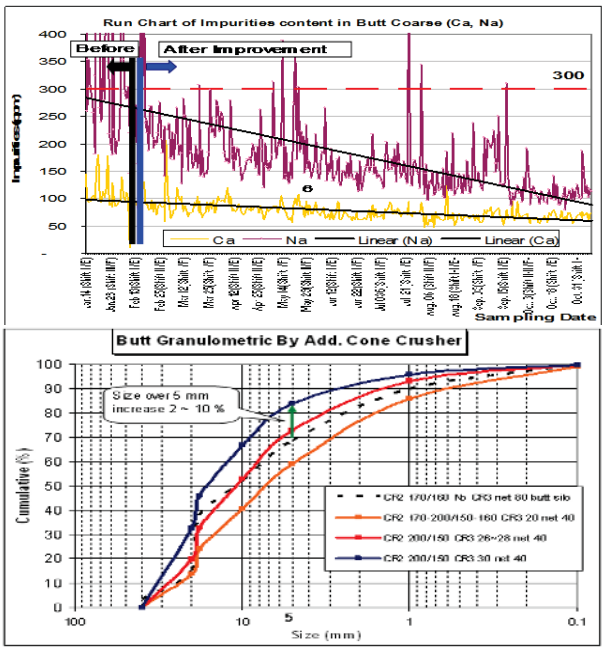

Figure 11. Result of Na Content & Butt Granulometric

D. Anode Size Enlargement

There is an opportunity to increase line current up to 215kA by using LA-4 due to quite similar of anode current density.

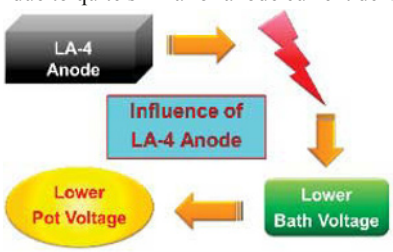

Figure 12. Influence of LA-4 Anode

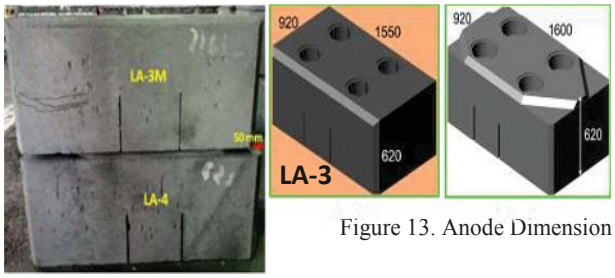

Figure 13. Anode Dimension

Table 5. Test result of LA-4 Anode

No	Item	Unit	193 kA	207	207	215	225
1.	AlF_3	%	8.8	8.8	9.0	9.0	9.0
2.	Bath Temp.	°C	951.7	956.8	954	954	954
3.	Current density	A/cm²	0.728	0.781	0.757	0.786	0.823
4.	ACD	cm	5.4	5.0	4.8	4.6	4.1
5.	Pot voltage	Volt	4.339	4.366	4.339	4.279	4.211
6.	Input power	KW	768	825	821	837	857
7.	Produce Al.	KW	388	411	414	430	450
8.	Lost Power	KW	381	414	407	407	407
Remark			Jul ~ Sep. '14		LA-4 Anode		

E. Implementation of Water Turbine Efficiency

The effective inflows of water to Toba Lake since the operation era has been decreasing from 103.32 m^3/s (average 1963 – 1975) become 91.37 m^3/s (average since 1984). Of course, the electric energy production has been decreasing from its basic capacity which decreases the Aluminum production from its design capacity. For this matter, it is important to re-design the turbine in order to make the design become suitable to the availability of recent water inflow.

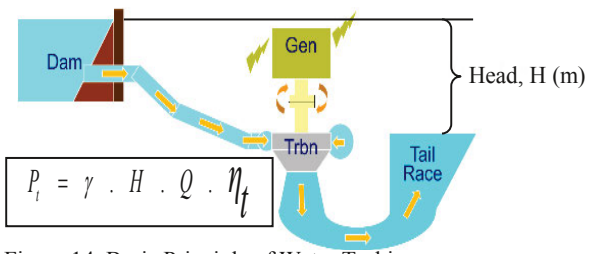

Figure 14. Basic Principle of Water Turbine

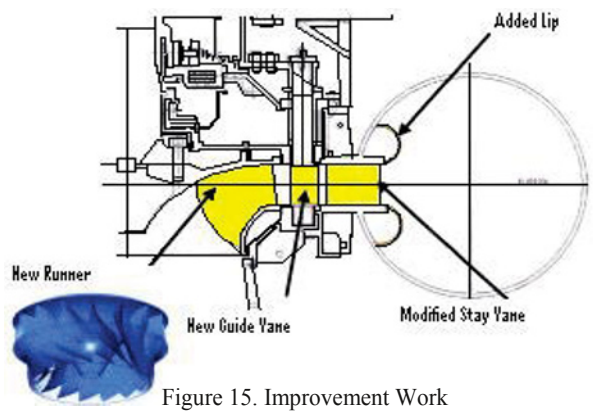

Figure 15. Improvement Work

Result of this program implementation as follows:

1. Turbine can be operated more efficient suitable to the recent availability of water.
2. With the same water discharge, the output energy from turbine will increase.
3. The Aluminum production will increase, following the increasing of the electric production.

Table 5. Result of Turbine Efficiency

Unit	Discharge (m^3/s)	Increase (MW)	Operation (Hour/ Y)	MWH/ Year	Completion Schedule
SGP #1	30	2.45	7,200	17,640	2007
SGP #2	30	2.77	7,200	19,944	2008
SGP #3	30	3.17	7,200	22,824	2011
TNP #2	30	3.94	7,200	28,368	2009
TNP #3	30	3.60	7,200	25,920	2010
TNP #4	30	7.90	7,200	56,880	2012

Conclusion

1. The technical improvement program at Inalum succeeded to be done by achievement of line current 207kA and 264.474 Tpy of production capacity (17.5% than original capacity).
2. Alumina Feeding System has resulted the increasing 1.9% of Current Efficiency which is similar with increasing 10.9 Tons/pot-year of pot productivity and decreasing 187 kWh/Ton-Al of DC Consumption.
3. D5-2 pot design improved Cathode Voltage Drop (CVD) by 7.6 mV hence there is opportunity to decrease DC Consumption by 32.2 kWh/T.Al and increase productivity 3.0 kg/pot. day.
4. Increasing of anode quality has successfully to Na content decrease much < 150 ppm (target < 300 ppm) by improvement of butt cleaning by and increase butt size over 5 mm 2~10 % (target 5 %) by utilizing of cone crusher.
5. Application LA-4 anodes reduce pot voltage 27 mV. At high amperage in the pot, heat loss is controlled 407 KW by reduce anode cover thickness. LA-4 anode is applied until 215 KA.
6. Efficiency of power generation has reached 7.9 MW at TNP#4 or 4 MW over than others by water turbine efficiency.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

NEW ASSISTED GAUGING SYSTEM: ANOTHER STEP TO REDUCE VARIATION ON LOW ACD OPERATION

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Keywords: Automation, PTM, gauging, assisted gauging.

Abstract

Most recent primary aluminium smelting technologies requires a lot of energy (from 12.8 to 14,5 kWh/t Al). Despite the progress made over the past decades, Alouette has always operated in an energy constrain mode ever since its start-up in 1992. Considering the limited amount of energy available and its vision of sustainability, Alouette has been exploring uncharted territories to make the cell technology (Aluminium Pechiney) more 'efficient' by increasing cell productivity while reducing energy consumption. As a result, the smelter has been able to produce more aluminium with its power limitation. Now facing the lower limit of the Anode-Cathode Distance (ACD) of the technology, this has brought additional operational challenges. To overcome this challenging situation, Alouette has developed an assisted anode gauging system on Pot Tending Machines (PTM) coupled with an improved anode setting precision method. This results in an improved cell performance by having a positive impact on Key Performances Indicators (KPI).

Introduction

The production of primary aluminium requires a lot of energy; for Alouette roughly 12.9 kWh/t_{Al}. Considering the fact that having a limited amount of energy from its supplier will constrain further growth, the focus of the company has always focused on "energy efficiency". Then, reducing the voltage in the cells, increasing amperage and developing new practices allowed the company to produce more aluminium with the same power input. But a few years ago, Alouette has faced the lower limit of the anode-cathode distance (ACD) considering the technology and operation practices at this time. Thus, it leads to develop new methods to reduce the specific energy consumption in the potlines even further.

There are many ways to reduce energy consumption in a smelter. One of them consists to minimise the average ACD of the cell. In a collaboration agreement with Rio Tinto (RT), Alouette has implement in early 2013 a new anode gauging method. This method also helped Alouette to debottleneck an operational challenge related to the rate of work and anode consumption rate in the potlines.

With a continuous improvement culture, Alouette wanted to go further with this method by giving the opportunity to the Pot Tending Machine (PTM) operator to set the anode entirely from the cabin, keeping the operator on the floor away from the open pot when the anode is removed. To do that, a fully integrated system was developed and tested, which will be described in further details.

This trend for automation development is not only an Alouette culture, as some other aluminium producer and technology providers have developed embedded systems to improve Health, Safety, Environment and Quality (HSEQ) performance and reduce OPEX [1-4].

Description of the automated system

The developed assisted system takes into consideration some parameters for proper anode setting (see figure 1 and table 1).

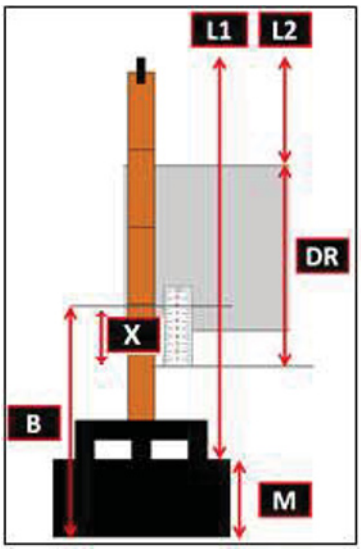

Figure 1: Parameters for the assisted gauging.

B	Distance between the anode bottom and the position line (kept constant).
DR	Distance between the top of the anode beam and the bottom of the ruler (measured for each anode in the potlines).
L1	Distance between laser #1 and the top of the anode.
L2	Distance between laser #2 and the top of the anode beam.
M	Anode height (kept constant).
X	Value of the position mark.

Table 1: Description of the parameters used for anode setting.

Here's the list of the parameters:

- The parameter B is constant for every anode in the potlines. A marking station has been previously introduced at the rodding shop when the method was done manually by the

floor operator. Since the value is constant, it could no longer be relevant to have visual lines on the anodes (except for contingency) as the system use lasers for setting the anode.

- Parameter DR has to be measured for every anode position, as they vary from each anode positions and from pot to pot. One measuring campaign has been done prior to testing. These measurements are stored in a database which can be modified when a pot is refurbished.

- L1 and L2 values are read in real time in the PLC.

- Parameter M is kept constant has the anode height standard deviation is roughly 0.5 mm.

- X is the position mark given from the Rio Tinto's gauging method. This method is based on the calculation of the theorical carbon consumption rate. The calculation uses the carbon consumption rate for different groups of pots with the same characteristics (amperage, anode quality, etc.).

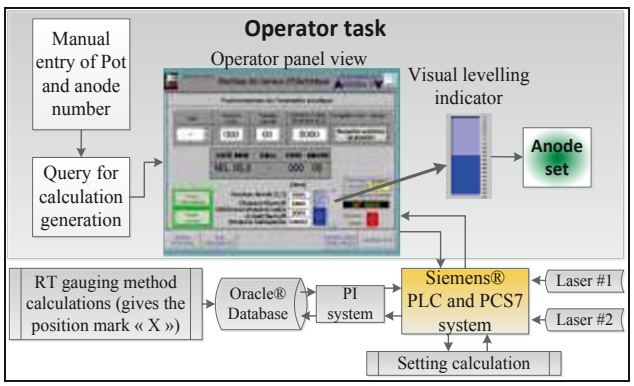

Figure 2: Flowsheet of the information for assisted anode gauging system at Alouette.

Figure 2 describes how the whole system has been built and how the information flows through the different programs. The beginning of the process is done by the operator at the moment when the anode is in the pot and the connector is in place but not fixed. The pot number and anode number are manually typed in the control panel. After that, a query is send manually by the operator to interrogate the PLC. The PLC then transfers this query to different subsystems for the calculation of the anode setting position and to position the anode at its mark.

For the position mark, the Rio Tinto gauging method is executed in an integrated system and the position mark calculated is stored in an Oracle database. The PLC uses a PI request to get the information of the position mark.

That being said, the position is calculated with equation 1, where "X" is the position mark and the other parameters are described in Table 1. The equation is stored in the PLC. Also, without the assisted system developed, the gauging method itself helps Alouette to reduce the anode changing cycle time by a factor 2. This major improvement has been achieved by removing the gauge which was no longer useful has the new gauging method has been introduced.

Setting calculation:

$$(L1 - (B - M)) - (L2 + DR) + X = 0 \quad (1)$$

The PLC will then translate that information in a visual levelling indicator that helps the operator to set the anode properly by fixing the setting position "0" as the target.

Experimental setup and statistical analysis

A trial was performed at Alouette in potline 1 with one PTM equipped with the new system. The PTM equipped with the assisted gauging system was operating only in the test group.

The trail was performed in a 33 pots section. From this section, a group of 31 pots where selected for the analysis and compare to a similar group (30 pots in another section) in terms of location in the potroom, average pot age and design. Only one pot design (AP40LE) has been selected in both groups for statistical analysis because of major differences between the two technologies (AP40LE vs. AP30) in terms of operational KPI results. Two periods were selected for the analysis. The first period was before the test, from April 19th to May 18th (30 days) and the second period was during the test, from May 19th to June 11th 2015 (24 days). Roughly, more than 575 anodes where changed for each period and each group.

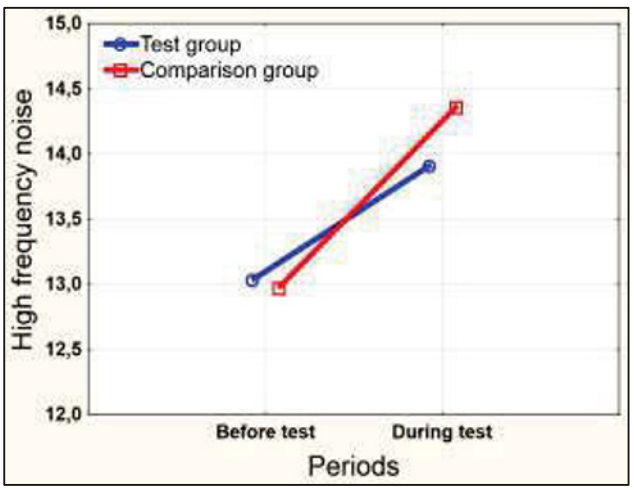

Figure 3: Average values of high frequency noise from both test and comparison groups before and during the test.

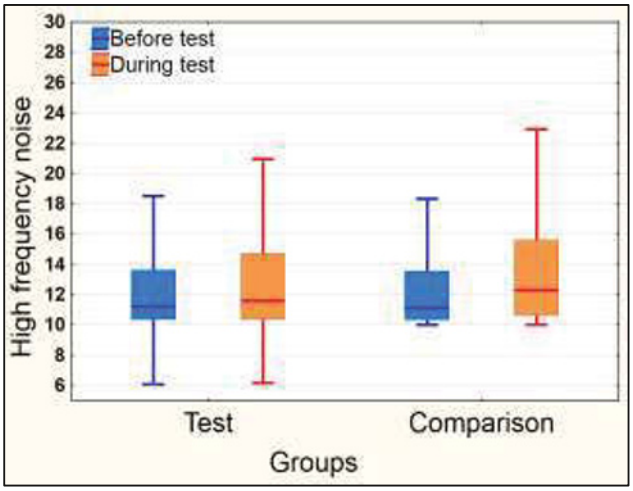

Figure 4: Boxplot comparing high frequency noise from both test and comparison groups before and during the test.

Figure 3 show the results of high frequency noise average for the test and the comparison group. The high frequency noise is commonly known to be a good indicator of the presence of an anodic incident, like spikes. At a low ACD operation, an anode set slightly too low, for example 1 cm, can results in an anodic deviation, or other types of anodic incidents. It is clear from Figure 4 that both groups are statistically different, but the average value in Figure 3 can indicate a smaller increase of the high frequency noise compared to the comparison group.

Also, the anodic effect rate in Figure 5 indicates a very slight (4%) increase compared to the 33% increase of the comparison group. This is in line with a better anode setting in the whole pot, giving a more uniform alumina dissolution from anode to anode as the mV coefficient of variation (CV) for the anodic current distribution is smaller (Table 2).

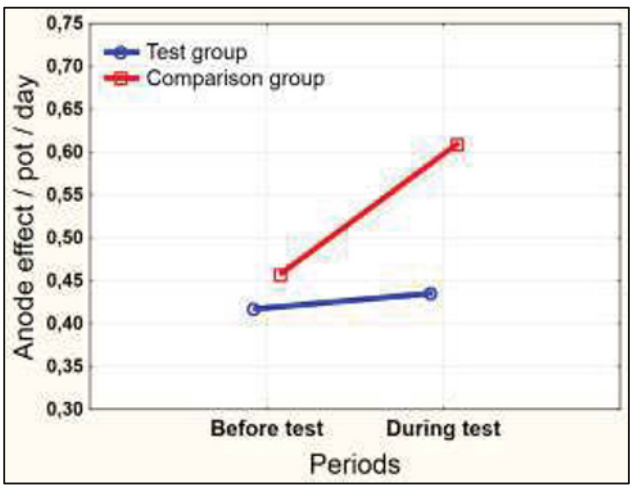

Figure 5: Average anode effect rate from both test and comparison groups before and during the test.

Groups	Periods			
	Before		During	
	CV	σ(CV)	CV	σ(CV)
Test	23,47	8,09	22,53	4,66
Comparison	22,26	9,12	24,81	10,86

Table 2: Coefficient of variation from both test and comparison groups before and during the test.

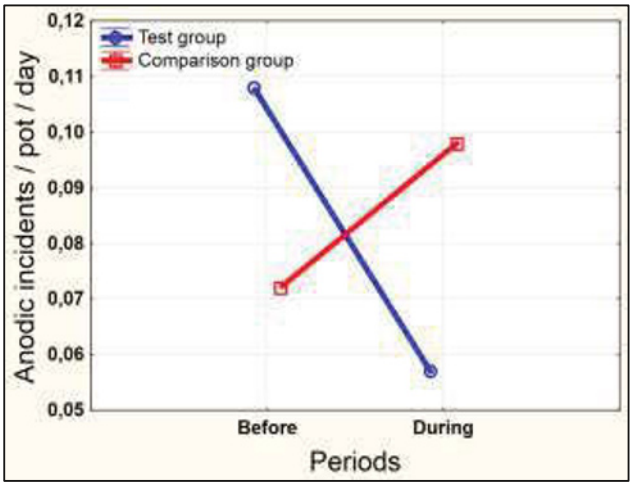

Figure 6: Average anodic incidents from both test and comparison groups before and during the test.

Moreover, anodic incidents were compared by looking at the average anodic incidents per pot per day. The rate decreased by almost a factor of 2 for the test group, while the comparison group increased its rate by 36%. These results indicate a significant improvement of the anode setting.

Conclusions

The anode changing procedure has changed quite a bit in the last three years at Alouette. From a manual gauge to Rio Tinto's gauging method and now with the assisted gauging, many situations have been overcome by those improvements. In the coming years, the project objective is to implement this new method throughout the whole PTM fleet. This will help Alouette achieve its strategic plan by going one step further in HSE improvements and technological development.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

A NOVEL METHOD FOR PROCESSING SODIUM REDUCTION SKIMMING STATION RESIDUE

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Keywords: DX Technology, Sodium removal, TAC

Abstract

The aluminium industry often faces financially conflicting challenges where costs are under increasing pressure but more stringent environmental constraints and customer specifications are imposed. Increasingly, smelters are resorting to treatment of aluminium in crucible technologies to reduce alkali and in particular sodium content in the hot metal prior to delivery to the casthouse. The consequence of this process is a residue or skimming waste that can be costly to process and/or dispose of and typically involves some form of landfill. This paper describes how, faced with a large and growing stockpile of skimming waste, Emirates Aluminium (EMAL, also known as EGA Al Taweelah) used the existing plant to process 100 % of the skimming waste on site with no additional off-site disposal or treatment.

Introduction

In recent decades, continuous development in both greenfield and brownfield aluminium smelting technology has realised ongoing improvement in electrolysis cell current efficiency. As demonstrated by Lukas [1], there is a correlation between electrolysis cell current efficiency and sodium in the aluminum produced in the cell. At the same time, customers are demanding aluminium with tighter specifications on impurities such as sodium.

Increasingly, smelter operators are employing the use of the in crucible alkali treatment process, using aluminium fluoride as the reactant to precipitate out sodium and other alkali metals. At EMAL, this removal of unwanted alkali is undertaken at the Sodium Removal Skimming Station (SRSS) and produces an SRSS residue waste stream.

At 1 300 000 tpa (tonnes per annum) of primary aluminium production, EMAL is one of the world's largest smelters and as such produces an equivalently large amount of SRSS residue, exceeding 4 500 tpa. Additionally, EMAL operates under strict environmental conditions that significantly limit landfilling of process waste streams. Traditional off-site dross treatment process proved costly due to the low metallic content of the SRSS residue and EMAL therefore looked for other options.

Description of the Problem

Within six months of EMAL Phase I achieving full production of approximately 750 000 tpa, EMAL had generated approximately 2 800 t of residue and the volume was growing (see Figure 1) with costly off-site treatment (see Figure 2) deemed the only option.

Figure 1. SRSS residue June 2011.

Unlike normal casthouse dross, EMAL's SRSS residue has lower metallic content at approximately 40 to 45 %, which increases the treatment costs. In 2011, the treatment costs were quoted at 800 US$ per recovered metal tonne. For EMAL Phase I and Phase II combined, the annual cost was estimated to be 3.6 million US$ per year or 2.75 US$ per tonne of aluminium, added to EMAL's operating costs in a tough LME operating environment.

Untreated SRSS residue at EMAL is shown in Figure 3.

Additionally, there is the economic loss of the non-metallic contents 8 % aluminium fluoride, 25 % alumina and the remaining bath material.

EMAL had two unsuccessful trials at processing the residue on the site. The first was to process the material via the Bath Plant Facility (BPF) which caused numerous blockages; and the second was adding directly to the pots, where the volume of material was considered excessive in volume and dust for the teams to handle.

An alternative On-site Process

EMAL's BPF uses autonomous air swept technology with an air flow of approximately 90 000 Nm³/h. Initial trials attempted to process the material with cavity scoop material. However, the flaky metallic pieces were jamming the valves prior to the airlift as well as the vibrating screen.

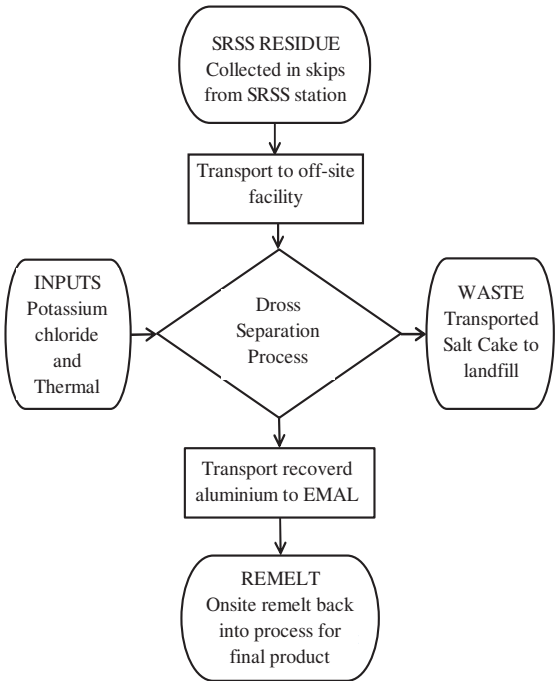

Figure 2. EMAL dross treatment process .

Figure 3. Untreated EMAL SRSS residue.

Typical air swept discharge BPF technology operates by autonomous grinding down to a small enough particle size to enable the particle to be air swept out of the mill. After a period of time, typically 8 h at EMAL, the mill is stopped and the uncrushable residue in the mill is ejected. The uncrushable residue, which typically comprises iron, aluminium and carbon with some bath material, is added back into the cells.

EMAL determined that for the process to be successful, the flaky aluminium pieces need to remain in the mill and be ejected in batch mode from the mill. To do this, the plant would be run exclusively using the skip lift, which feeds the mill directly, and thus avoid mixing with other material. Figure 4 shows a skip containing SRSS residue ready to be fed into the Bath Plant Facility, shown in Figure 5. The bath plant mill circuit is shown in Figures 6a and 6b.

Figure 4. Skip of SRSS residue ready to be fed into the Phase I Bath Plant Facility.

Figure 5. EMAL Phase I Bath Plant Facility.

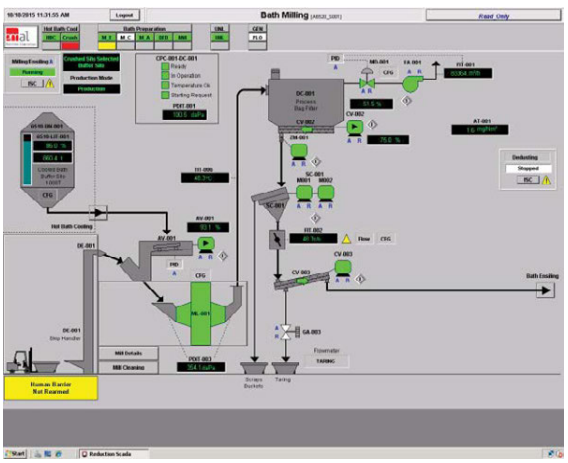

Figure 6a. SCADA screen shot of the bath plant mill circuit.

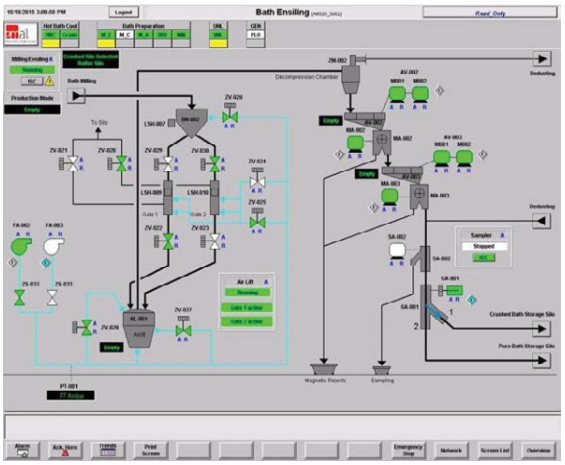

Figure 6b . SCADA screen shot of the bath plant mill circuit.

To prevent the aluminium pieces being carried up with the bath, the airflow rate had to be reduced to the point where the metal remained in the mill. Through a series of trials, it was determined that a gas flow of 40 000 Nm³/h enabled only the non-metallic components to be broken free of the metallic pieces. The mill then ground the non-metallics down to 500 µm where they are small enough to be swept out of the mill.

Mill Residue

The remaining material in the mill is then discharged via the usual process to produce approximately 4 t of "Corn Flake" sized metallic particles of aluminium (see Figures 7 to 9).

Figure 7. Mill residue prior to being ejected.

The non-metallic components pass through to the crushed bath silo which is part of the anode cover circuit.

As approximately 40 to 45 % of the material is metallic, the mill is fed in batches of 20 t of SRSS residue.

Figure 8. Skips of mill residue, metallic aluminium.

Figure 9. The metallic aluminium residue from the mill.

Plant Changes

All changes to the plant were limited to the process control system, SCADA. With the support of the maintenance team, a new operational mode was generated for processing the SRSS material. When starting the plant, the Operator presses SRSS mode, from which the fan speed, input location and destination silo are automatically selected.

The SRSS non-metallic residue has an excess aluminium fluoride content of approximately 8 %, which is lower than the typical 10 to 12 % content for aluminium fluoride. The Bath Plant Facility processes over one hundred times more spent anode cover material than SRSS material per day. Hence the difference has no detectable impact on the pot chemistry control.

Additionally, after 3 years of using this recovery process, there have been no detectable issues in the cells with alpha alumina that may have formed in the dross.

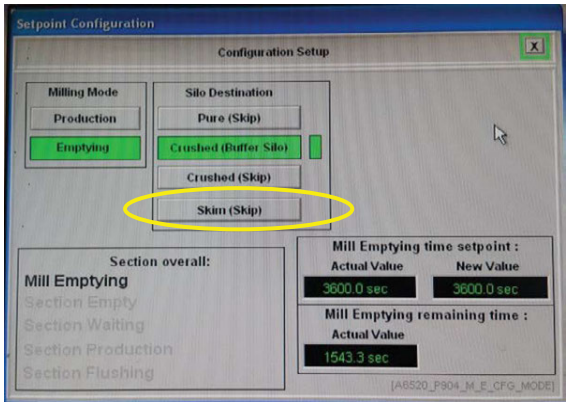

Figure 10. Change to the operators SCADA screen to process SRSS.

Cost Benefits

The advantageous of the onsite treatment process is that the non-metallic components are recovered which is significant as 8 % of the residue is high value aluminium fluoride.

No value has been assigned for the recovery of alumina or bath.

As the figures in Table I confirm, the on-site processing costs are favorable to the annualized value of 2 million US$ per year for EMAL's combined product of Phase I and Phase II.

Table I. Cost comparison in US$, year 2014.

Annualised off-site treatment costs	
Quoted treatment costs	- 1 400 000
Estimated metal recovery	3 100 000
Net benefit for off-site disposal	1 700 000
Annualised on-site treatment costs	
Estimated on-site treatment costs	- 100 000
Estimated recovery costs aluminium	3 400 000
Estimated recovery costs aluminium Fluoride	400 000
Net benefit for on-site disposal	3 700 000

Environmental Benefits

The off-site dross salt treatment process would produce up to 3 100 tpa of landfill waste. It would also require between 250 and 300 extra truck movements on UAE roads per year, adding to UAE road congestion and carbon dioxide emissions.

Other Wastes

The non-high magnesium dross waste streams from the Casthouse could potentially also be processed via the same process. However, at this time EMAL needs to reserve bath plant operating capacity for Phase II as the Phase II Bath Plant uses gravity discharge technology and is not suitable for this process.

Conclusion

Using lateral out-of-the-box thinking and using only plant and human resources available on site, EMAL has found a cost-effective solution to processing sodium skimming residue waste. A saving of 2 million US$ per year, equating to a reduction of 1.5 US$/t Al in site operating costs, eliminated potential landfill products and over 250 off-site truck movements per year.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE 'VIRTUAL BATTERY' – OPERATING AN ALUMINIUM SMELTER WITH FLEXIBLE ENERGY INPUT

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Keywords: Virtual Battery, Shell Heat Exchanger, Flexible Energy Input, Heat Balance Control, Modulation

Abstract

TRIMET Aluminium SE operates four smelters in Western Europe. TRIMET has investigated power modulation and opportunities to provide services to the energy grid for several years. The primary aim has been to stay competitive in a geographic zone with above world-average energy prices as well as having to deal with high volatility in energy prices caused by renewable, non-baseload energy sources.

Recent work has seen the implementation of Shell Heat Exchanger (SHE) technology on 12 pots at TRIMET's plant in Essen, Germany. The heat exchangers are capable of boosting sidewall heat extraction or acting as an insulator depending on active air flow or not. The pots have comfortably maintained heat balance with stable ledge when operated within 150 to 180kA for up to 48h. Greater control of heat loss has also enabled operating voltages to be significantly reduced. By being able to operate significantly above or below the conventional design amperage, the smelter can operate as a 'Virtual Battery' to the grid.

Introduction

Germany is striving to increase the share of electricity supplied by renewable sources. While wind and solar power both have significantly reduced CO_2-emissions, they both have a highly volatile nature. This necessitates conventional power plants to operate in standby modes to balance the grid load, negating much of the possible benefits of renewable energy [1]. Making the energy demand more flexible would be a great benefit given the volatile generation system and allow greater integration of renewable energy production.

Primary aluminium production is, however, a highly energy intensive process that traditionally demands a near-constant energy input. Worldwide about 3% of all generated electricity is used to produce 50 million tpy of aluminium. Germany is slightly below this average with 2% of all generated electricity [2].

TRIMET has modified a test section of 12 reduction pots (from a total of 360) located at their Essen plant with Shell Heat Exchanger technology and an optimized bus bar system. A key objective of this is to enable highly flexible energy input of up to ±25% to better align the plant with the realities of the generation system. So far this has enabled the pots to operate with +20/-13% energy input for up to 48h while staying within the optimum process window.

By enabling variation in power consumption at short notice significantly above or below the usual level, the smelter can act as a 'virtual battery' in the electricity grid, 'returning' capacity to the grid at times of high demand and price and preventing the need for equivalent capacity of conventional energy production that otherwise has to be added.

Shell Heat Exchanger Technology

The Shell Heat Exchanger (SHE) Technology is patented and developed by the Light Metals Research Centre (LMRC) of the University of Auckland.

The purpose of the SHE-technology is to allow dynamic control of the sidewall heat loss to help maintain operable heat balance when there are significant changes in a pot's power input. This is achieved by varying the airflow through the individual exchanger units (controlled via a suction fan) to control the heat transfer rate at the sidewall as per the schematic diagram in Figure 1. Airflow above a base demand allows the sidewall heat loss to be increased, which compensates for power increases (eg. upwards modulation) in the pot. Operating with an associated airflow below the base demand reduces the heat loss thereby creating an insulating system that compensates for power decreases in the pot (eg. downwards modulation) [3, 4]. Figure 2 shows the variation in heat extraction achievable with varying flow rates through the system.

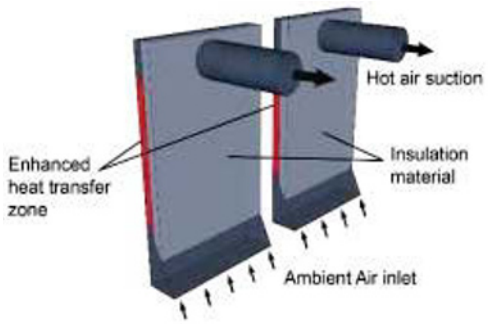


Figure 1 - Schematic of Shell Heat Exchanger (SHE) Concept

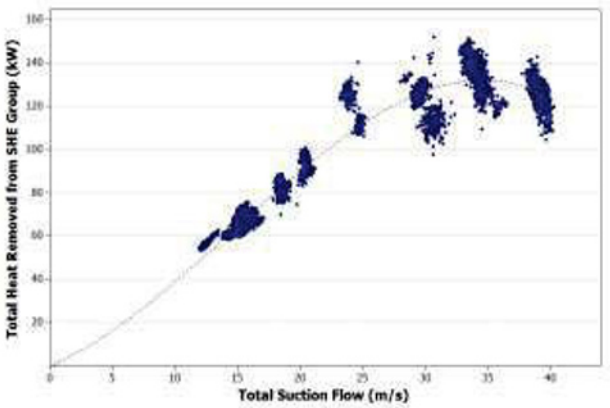
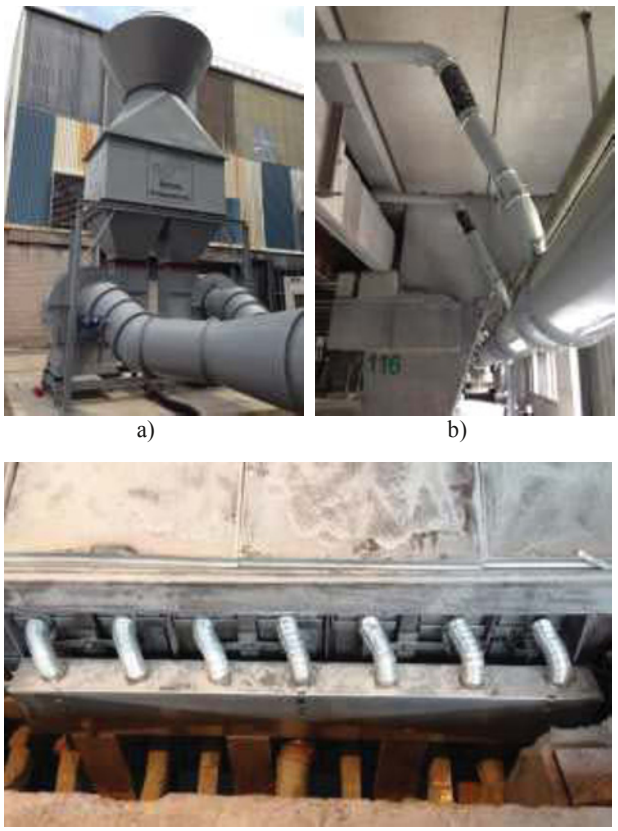


Figure 2- Total heat extraction from the booster group as a function of airflow rate through the SHE system.

Installation and Design

The SHE trial is installed on a mid-line booster group of 12 pots, with capability of increasing current by up to 25kA above the rest of the line. The current of the booster section cannot be reduced independently below the base line current without also reducing current on the rest of the pots in the line, however, which is not possible in a safe manner for more than short periods. For this reason, most trials have involved upwards modulation from the base line current and downwards modulations starting from an elevated state down to the base line current. Some trials have involved reducing the whole line current for up to two days, with additional voltage applied to non-SHE pots.



a) b) c)

Figure 3- SHE trial installation details showing a) Redundant dual-fan system and exhaust stack with silencer, b) branch ducts to pot side in basement, c) SHE units connected to distribution manifold box on pot sidewall

The current 12-pot trial follows an earlier one-pot test design and was installed over 12 days in June 2014. The required ducting and fans were placed ahead of time so that each pot could immediately be connected to the suction flow to prevent overheating. The system comprises of four main ducts and each duct is connected to one side of a pot row down each side of the electrolysis hall equipped with end-to-end pots. The branch ducts are connected to a redundant setup of two fans located on a plinth outside the potroom, with an adjacent control room for fan control and temperature monitoring equipment. The main ducts are connected to three smaller branch ducts per pot-side, which are in turn connected to three manifold boxes mounted above the cathode collector bars adjacent to the shell.

Each manifold box provides suction for 7-8 SHE units, with a total of 44 units per pot. The SHE units are mounted to the wall of a pot shell, one in each inter-cradle space, using a spring loaded clamping system. The ducting and SHE mounting system is completely reversible with no welding or permanent attachment required. The SHE units were installed in-situ while the line was operational, but disconnection of the branch ducts enables the entire pot to be removed to reconstruction with the SHE attached for easier removal later.

Instrumentation consists of 12 thermocouples per pot, at upper- and lower-sidewall positions at the centre and ends of each side, as well as air suction temperature at each pot, and air duct temperatures and flowrates at several locations in the main duct. Two pots have additional instrumentation of 48 thermocouples for more detailed study. Temperatures and airflows are all logged at high frequency (between 1 and 600s) in the monitoring software. High and low temperature alarms are used on all thermocouple measurements to help control sidewall temperatures within target ranges and prevent excess ledge freezing or melting and associated sidewall damage. Currently fan control is manual and alarms alert to when manual fan changes may be required.

The airflow to each manifold of 7-8 SHE units is balanced via damper valves, where the individual SHEs tend to be self-balancing due to pressure drop. The main duct branches are also balanced via damper valves. The main suction flow rate is controlled via a variable speed drive on each fan, with the fan flow rates currently being set manually for each modulation and referenced against pot temperatures.

Heat Balance Control

A key aspect of the SHE system is to allow operation at the existing or as-designed pot heat balance, as well as opening the window for upwards and downwards modulation. Pots equipped with the SHE should be able to run permanently with minimal change in existing procedures or process control even though the energy input is varied significantly. This requires sufficient flexibility in the system to allow both higher heat extraction through increased airflow and also an insulation capability to reduce sidewall heat loss when required. A key stage of commissioning is to find the balance in this heat insulation/extraction range where the pot heat balance can be maintained as prior to SHE installation. This typically requires a moderate suction airflow to compensate for the insulating effects of the SHE.

As the temperatures measured outside the cell are continuously available, suction calibration can be achieved quickly, although some longer term monitoring is required due to the slow speed of changes in ledge profile and heat transfer. Suction calibration also involves flow balancing of the ductwork by manipulation of the damper butterfly valves, both around the main branch ducts and also the flow to each manifold box individually. This can also involve monitoring of temperatures over an extended period of time to ensure the temperature profile is even and stable i.e. the corners especially are neither too hot nor cold.

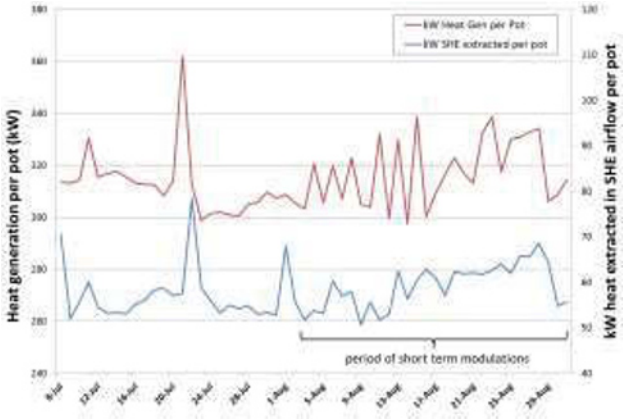


Figure 4 - Pot heat generation and heat extraction from SHE

Balancing of airflow is the main practical difficulty in commissioning, as all dampers are controlled manually. The changes in airflow can be interacting and nonlinear. Once a suitable balance has been found, however, the dampers usually do not need to be adjusted again. As small irregularities in airflow tend to magnify as the airflow is increased it is better to perform flow balancing at a relatively high flowrate. In practice it is difficult to balance the main branch flows to less than 10-20% difference using manual dampers. Otherwise, the airflow distribution has proven to be consistent over one year of modulation trials.

The SHE airflow is currently controlled by regulating the suction fan manually to maintain approximately stable shell temperatures during modulations. This has proven to be easy and stable in operation with only limited adjustment needed. The calculated heat extraction vs pot heat generation is shown in Figure 4, based on measured outlet air temperature and suction rate in the SHE main ducts.

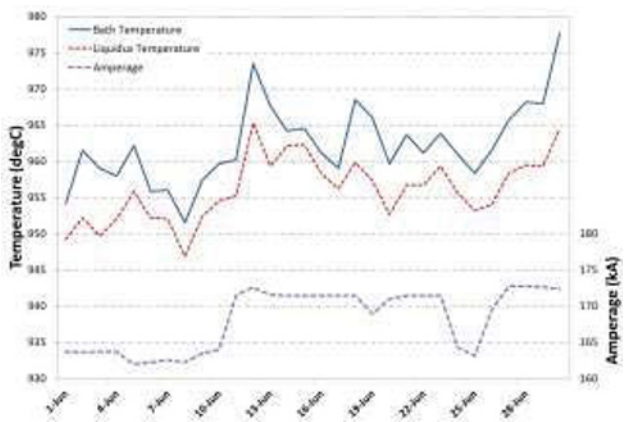


Figure 5 - Long term trial of +10kA for three weeks showing line current, average bath and liquidus temperatures of 12 pots

Further changes in pot heat balance may be required depending on the ultimate goals of power consumption, modulation magnitude and pattern. This may involve long-term changes in the average anode cover level or material mix, including yoke assembly temperature monitoring and auditing of covering practices, changes to average pot exhaust draught rates to the gas treatment center, or bath chemistry changes.

Modulation Trial Design

The initial design of the SHE system was to allow for significant, sustained upwards modulations in line current, using the SHE primarily to increase heat extraction from the sidewall. However, political and economic opportunities have shown that there is potentially more benefit to downwards modulations for power-cutting purposes, on both short and long time scales, which allows the smelter to be used as a 'virtual battery' and release power to the grid for distribution elsewhere in times of peak demand (and cost).

Downward modulations require that line current be significantly reduced from base load, which is not easily possible on a trial group without adversely affecting the entire line. For this reason, most trials were of upwards modulation, or downward modulations from an initially raised base point down to the normal line average. Modulations are tested both with rapid changes over short periods (hours) and sustained changes over long periods (weeks).

The prior single-pot trials had shown stable operation at high heat extraction using both increased booster current and additional voltage for heat generation. The first 12-pot trial consisted of a sustained increase of +10kA for several weeks as shown in Figure 5. While the EPT14 pot design is robust and may cope with this increase without SHE units installed, this system gives much greater flexibility in the timing and method of modulation as well as enabling much larger modulations than without SHE.

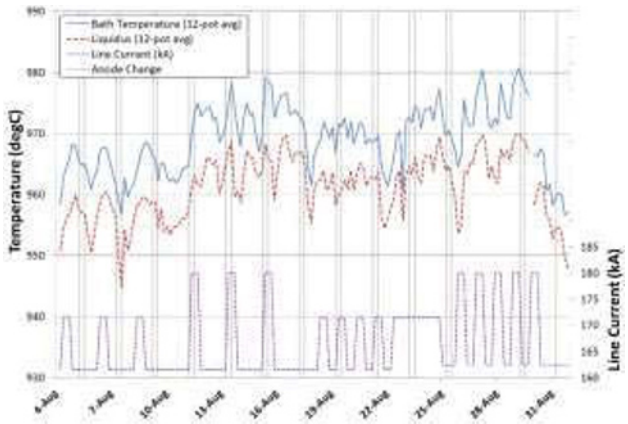


Figure 6 - short term modulations of +10kA and +18kA showing anode change periods and bath and liquidus temperature averages for 12 pots (even side).

The long-term scenario was followed by short-term modulations to simulate rapid response to electricity price spreads e.g. daily/nightly changes. These involved increases of +10kA and +18kA for 8-24hr periods, with normal periods in between to represent periods of regular energy prices.

After this, the pots were allowed to stabilize at +10kA line current as a new 'base' state on the booster group. Short-term downwards modulations of -10kA – which were essentially returning to normal line current – and upwards modulations of +8kA were then tested at short intervals from this new base current. Bath and liquidus temperature responses to modulation are shown in Figure 6 and superheat in Figure 7, also showing the approximate timing of anode changes.

Even larger downwards modulations were then performed by reducing current of the entire line, and adding compensatory voltage to the non-SHE installed pots to maintain heat balance, however, these were limited to periods of maximum 48h to keep the impact on the unmodified rest of the reduction line under control. As a result of this experiment it was found that the pots with compensated voltage required additional heat equal to 0.8 kWh/kg higher energy consumption, while the boosted group with shell heat exchangers was able to keep the same energy consumption as previously. The higher energy consumption was due to both increased cell voltage as well as lower current efficiency, as will be examined later in the paper.

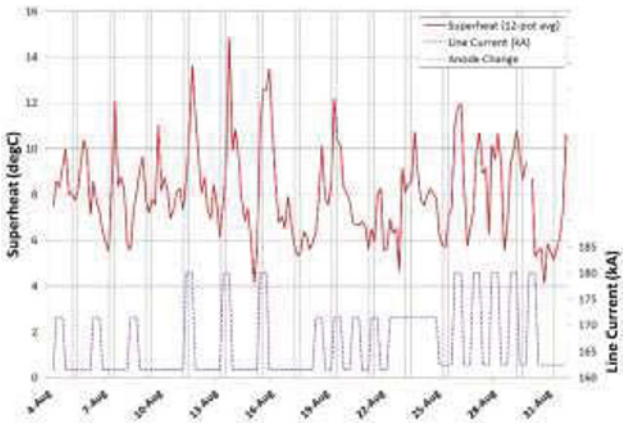


Figure 7 - short term modulations of +10kA and +18kA showing anode change periods and superheat temperature averages for 6 pots (anode change subsection).

Key metrics monitored throughout the trials were the bath temperature, liquidus and superheat as measured using a Heraeus Fibrelab instrument, as well as pot voltage and noise. The current modulation scheme is shown in Figure 8, with the responses in bath and liquidus temperatures shown in Figure 9. Shell temperatures were monitored carefully both to avoid short term hot- or cold-spots and to ensure the long-term average remained steady.

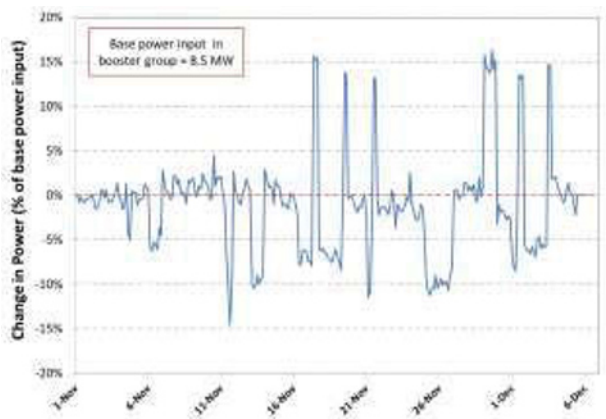


Figure 8 - Repeated downwards and upwards modulations from base line current using corrective voltage on non-SHE pots

Modulation Trial Results

The longer term trials went smoothly, with the pots quickly returning to a steady operating state after 24h at approximately normal temperature, superheat and noise levels. The short term trials, however, were of too fast frequency that the pots were not reaching steady states before successive changes in line current. A key finding is the rapid rise in bath and superheat temperatures at the time of upwards modulation measured by 4-hourly Fibrelab measurement. This is in contrast to the relatively slower fall in temperature following downwards modulation (or removal of upwards modulation/heat generation).

This caused long term increases in bath temperature shown in Figure 6, which also resulted in decreasing measured ledge thickness. This needs to be considered carefully in future, such as by using an 'energy counter' to integrate net energy input changes, which could be compensated using changes in SHE airflow (additional cooling or insulation) i.e. by maintaining increased forced cooling for a period after upwards modulation. Anode changes had large impact on superheat, especially if changed shortly after upwards modulation to reduce the increase in temperature, with a smaller effect if changed before modulation. Anode changes could be leveraged better to reduce heat at desirable times, or by not adding additional anode-setting voltage when superheat is already high.

Pot shell temperatures were well controlled at all times and the surface temperatures responded quickly to changes in suction airflow, although these changes take a long time to propagate to new stable ledge positions. The SHE system does, however, allow for stable operation with thinner than normal ledge thickness due to the forced cooling through the sidewall. Importantly, no damage or increases in metal silicon content were observed, which would indicate sidewall brick erosion.

Bath and superheat temperatures will naturally become higher when operating at high heat input. As sidewall heat loss comprises only approximately 40% of the total heat loss as measured on these pots, it is difficult to entirely control bath temperature via SHE adjustment only. It is estimated that the SHE system can provide around 50% of any additional heat loss/conservation required. A long term strategy will therefore involve careful consideration of factors such as suitable bath composition to maintain alumina dissolution at lower superheats and the overall impact of the entire heat loss of the pot. For example, the next step is to get more control over the top heat loss. Top heat loss can be regulated on the long term through changes in covering practice and cover material properties, but can also be more dynamically controlled through changes in suction rate towards the GTC. However, when varying suction towards the GTC the challenge will be to do so without compromising environmental regulations and therefore this needs to be investigated carefully.

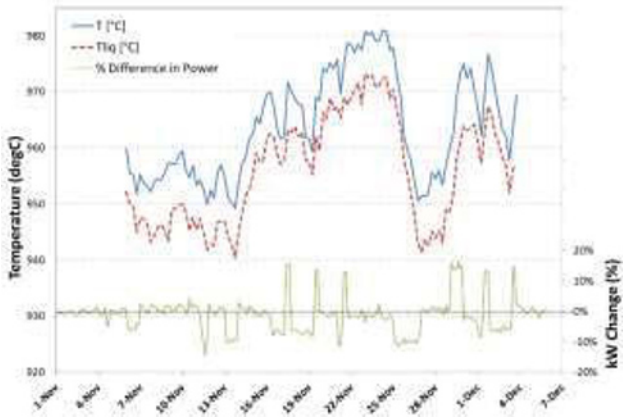


Figure 9 - Bath and liquidus temperature trends over period of upwards and downwards modulations.

Operating Voltage Reduction

In addition to gains in power input flexibility, improvements in energy efficiency have also been made in this trail group when running at the normal operating amperage of 162 kA.

Figure 10 shows the results of an Anode-Cathode Distance (ACD) vs pot stability test carried out on a typical EPT-14 pot at TRIMET Essen. Figure 10 shows that from the normal operating voltage of approximately 4.4V the ACD could be reduced by an equivalent 500mV before significant instability was recorded. This result indicates that the pot is not ACD limited from a stability point of view and that it therefore has potential for significant voltage reductions through further squeezing of the ACD.

From previous experiences it is known, however, that further ACD squeezing in the long term leads to low pot operating temperatures and greater instability caused by ridging. This confirmed that the pots were heat balance limited and that lower heat generation within the pot could not be sustained with the current pot design.

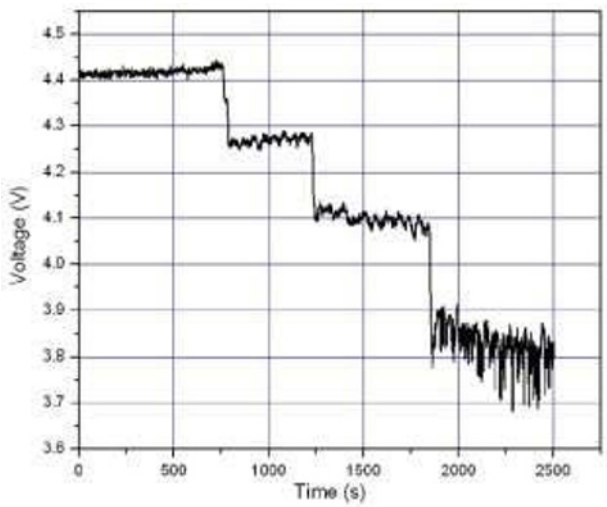


Figure 10 - Impact of ACD reduction on pot stability on a typical EPT-14 pot at TRIMET, Essen.

This heat balance limitation has been partly alleviated through the installation and optimized operation of the Shell Heat Exchangers. Trials were conducted to reduce the ACD (heat generation) whilst concurrently reducing the airflow through the exchangers. Reducing the airflow allowed the heat loss to be controlled to a lower level (operating in a slight insulating mode) which compensated for the lower heat generation in the pot. Initial trials showed that reduction in ACD equivalent to 60mV was easily achieved whilst still providing a long term stable heat balance and operations. This reduction in ACD was also enabled through additional insulation being placed on the shell corners and also having less suction in the exchangers near the corners. These measures have alleviated the significant corner ridging that the cell design is normally prone too. This combined optimization of the heat balance has resulted in more stable operations as ledge and ridge profiles have been able to be controlled. This has resulted in a lower amount of long term pot instability and current efficiencies that are consistently maintained above 93%.

The combination of the reduction in ACD and more stable operation due to optimized heat balance resulted in approximately a 0.25 kWh_{DC}/kg improvement in energy consumption and up to a 0.4% increase in current efficiency in the trial group when compared with the rest of the pots operating in the line at the same amperage (refer to Figure 11 and 12, period April to June 2015).

Overall Performance

Figures 11 and 12 shows the current efficiency and energy consumption of the trial pots in comparison to the rest of the pots in the line since November 2014.

In general it has been found that pots with the SHE Technology are more robust to changes in power input due to modulation. During normal power input operation (162 kA) the trial group maintained current efficiency at a similar level as the rest of the pots in the line (April to date). However, during power modulation it is clear that pots with the SHE maintain current efficiency better than pots without. This is shown in Figure 11 over the November-December modulation trials where it was found that the trial pots were able to maintain current efficiency at up to 1.5% higher over this same power modulation trial period.

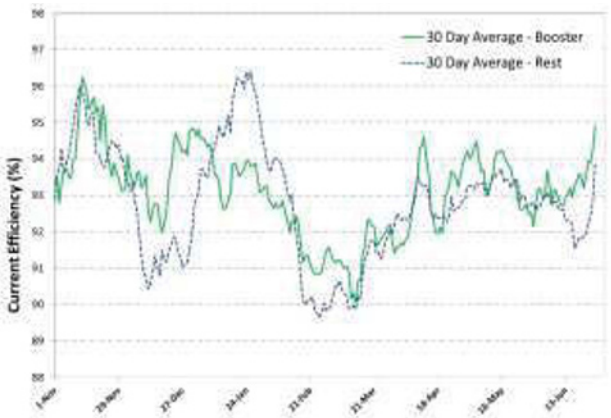


Figure 11 - Trial group current efficiency (%) compared to rest-of-line average over extended trial period

The trial group with Shell Heat Exchanger technology had approximately 1.0 kWh_{DC}/kg lower energy consumption than pots modulated without SHE. However, in normal power input conditions (162 kA during April to June 2015) the benefit of the use of SHE's has been measured at 0.25 kWh_{DC}/kg when compared

to pots without the SHE Technology due to voltage reductions discussed earlier.

It should be noted that from June to August 2015, the significant reduction in energy consumption seen in Figure 12 is due implementation of magnetic compensation in trial group, which has allowed further voltage reductions. This will be discussed in more detail in future publications.

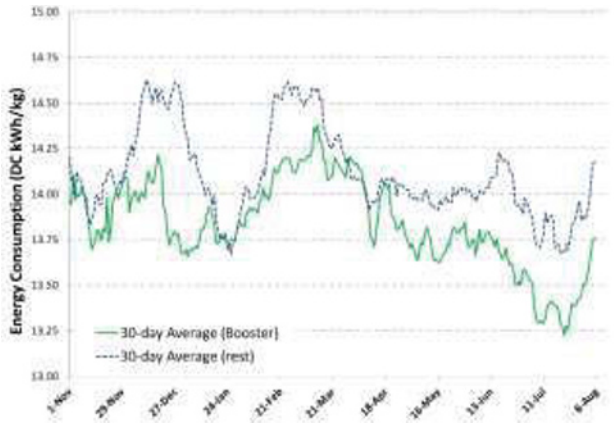


Figure 12 - Trial group energy consumption (DC kWhr/kg Al) compared to rest-of-line average over extended trial period

Also, it should be noted that decreases in current efficiency and increases in energy consumption over the period of January to March 2015 were mainly due to anode quality related issues and not a result of any of the 'virtual battery' trials.

Future Trial Plans

The SHE trial booster group pots have been operated for significant periods of up to 48h stably with +20% and -13% power input compared to base load. The goal for future trials is to achieve a variation in power input of ± 25% for 48 hours.
An advanced 'energy counter' for heat balance control is currently being developed, which will help to achieve this goal. The energy counter is being tested on pots with almost 100 thermocouples each to compare pot shell temperatures all around including dry scrubber flows and SHE air suction rates as well as air temperatures. This should allow better understanding and thus better control over the pot heat balance during energy modulations, especially for larger and longer modulations. This may prevent excessive ledge freezing or melting and promote stability, hence giving the lowest energy consumption and highest current efficiency during flexible operation as well as during traditional, constant operation.

Temperature measurement and bath chemistry control processes will also be investigated further, as these can vary significantly depending on time and location of measurement in the bath, and should be considered in conjunction with any rapid changes in power input.

Further trials also involve newly-installed magnetic compensation at these pots, and further optimization of pot control parameters. It is expected that ACD can be stably lowered significantly further on pots with SHE than pots without, giving even further advantages in energy consumption.

Conclusions

The Shell Heat Exchanger technology tested on 12 pots at TRIMET Essen gave much greater flexibility in energy input to and dissipation from the pots. This opens up a wider operating window for line current while still maintaining stable operation over both short term rapid modulations and long term sustained modulations, which can be highly advantageous given the electricity market in Germany, and particularly the continuing reliance on volatile renewable generation.

The trials performed to date gave good performance at energy inputs -13% to +20% from normal base conditions and provided significant improvements in both current efficiency and energy consumption due to ledge and heat transfer stability improvements. It is planned to push this to ±25% energy flexibility in the near future, although this may require a larger (line-wide) trial to enable significant current reductions without adversely affecting other pots without SHE. This flexibility will give significant ability for the TRIMET Essen plant to act as a 'virtual battery' on the power distribution grid, reducing their power consumption in times of peak demand and price, and reducing the need for further installed back-up power generation capacity in Germany.

A major improvement for future trials will involve a sophisticated energy counter, including existing pot data as well as shell and airflow temperature measurements. This will allow for better long-term control of heat balance and bath temperatures during periods of varying modulations, as trials have shown long term trends due to limited manual control. Modulation changes may also be aligned better with anode changes, as this operation also has a significant and rapid effect on bath and superheat temperature. This necessitates rethinking of standard working procedures if moved beyond the scope of a small test group.

A key outcome of an installed SHE system is that flexible adjustment of pot energy input is no longer counteracted by unstable pot energy balance or increased specific energy consumption. If this 'virtual battery' system is applied to all three TRIMET plants in Germany then the storage capacity energy availability of the grid could increase by 7700 MWh, nearly as much as the largest German pump storage power plant. Almost 40% of this could be achieved solely by installation at Essen.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

UNDERSTANDING THE BASIC REQUIREMENTS OF THE ANODE SET MODIFIER

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Keywords: Anode, Voltage Modifier, Resistance Modifier, Heat Requirement, Specific Heat

Abstract

When spent anodes are replaced with new anodes, cell resistance is changed. A resistance or voltage modifier should be added to the cell's base target and then reduced over the load up period to preserve the anode cathode distance. The initial modifier may need to be increased even more to maintain stability. The size and duration of the modifier should supply extra heat to bring the new anode to steady state operating temperature without allowing or causing a significant heat cycle.

This paper will explore how to determine the minimum initial modifier and consider reasons for additional modifier increase to maintain stability. How to calculate the total kWh required to bring the new anode to steady state operating temperature will be given. The methods of removing the modifier such that the total kWh will be supplied while seeking to avoid significant heat cycling will be discussed.

Introduction

Current Hall-Heroult Cell technology consumes anodes and therefore requires the replacement of spent anodes. Generally the replacement (setting) of anodes is done to one or more on a periodic schedule. In doing so, crust from around the spent anode is broken and goes into the cell. The crust is made up of alumina and bath. If it is not removed, both the bath ratio and feed control will be affected. Removing the spent anode causes the cell resistance to change. Inserting a new anode, which is at ambient temperature, upsets the cells heat balance. In addition, since the new anode is at ambient temperature, bath freezes on the anode preventing it from carrying current immediately after being set and therefore upsets the current distribution until the new anode is fully loaded.[1] The cells resistance does not return to its original value until the anode is fully loaded which, depending on cell technology, may be longer than 24 hours. Given these disturbances, it is not uncommon to see a cell become unstable after the anode set operation.

While it is virtually impossible to eliminate upsetting the cell during and after the anode set operation, it can be minimized by applying the proper modifier (resistance or voltage). The proper modifier must consider the resistance changing over time as well as the kWh required to heat the anode to its steady state operating temperature. ***The anodes average operating temperature when it becomes fully loaded is the temperature we will use as its steady state operating temperature.***

The requirements of the modifier needed to be determined are therefore:

a) The minimum value of the modifier to offset the initial cell resistance change and preserve anode cathode distance
b) The method of its removal during load up to prevent anode cathode distance from being reduced to a point of instability
c) Increasing the initial value and adjusting removal rate or method so that the kWh required to heat the new anode to its steady state temperature is supplied

Method

The Minimum Required Initial Modifier

First consider the required modifier to offset the cells resistance change when one spent anode is removed.

There are several ways to approach an estimate of the minimum value for the initial modifier. Let us start with the simplest approach.

If we consider the electrical circuit in figure 1, the cell target voltage (V_T) is applied across the cell target resistance (R_T) causing a current (I) to flow.

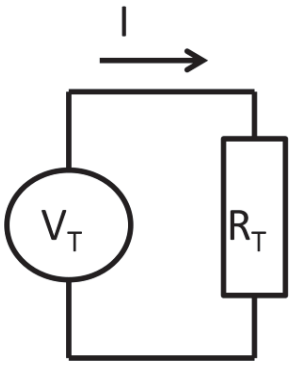

Figure 1

[1] Grjotheim, K., and Barry J. Welch. *Aluminium Smelter Technology*. Dusseldorf: Aluminium-Vertag, 1988, p.238

The relationship between the three factors in a cell is given by:

$$R_T = 1000(V_T - 1.65)/I \qquad (1)$$

The units for resistance, voltage and current are micro ohms, volts and kilo amps respectively. The constant, 1.65 is the back emf.[2]

If we assume the cell resistance (R_T) is the result of N equally resistant anode assemblies (R_A) in parallel, figure 2 would be equivalent to figure 1.

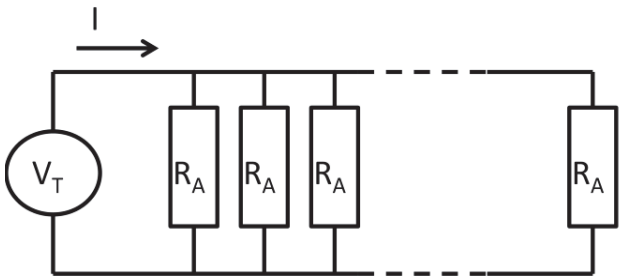

Figure 2

The target resistance is related to the N equal parallel resistances as

$$R_T = \frac{R_A}{N}. \ (2)^3$$

If one anode were removed, the resistance would be $\frac{R_A}{N-1}$ and the ratio of the resistance after removal to before would be,

$$\frac{\frac{R_A}{N-1}}{\frac{R_A}{N}} = \frac{N}{N-1}. \ (3)$$

The new target resistance (R_{NT}) would be,

$$R_{NT} = \frac{N}{N-1}(R_T). \ (4)$$

Since the voltage drop is proportional to the resistance after the back emf is removed, the new target voltage (V_{NT}) would be,

$$V_{NT} = \frac{N}{N-1}(V_T - 1.65) + 1.65. \ (5)$$

The modifier estimate is the difference between the new target and the original target whether you use resistance or voltage.

The previous estimate used the full cell voltage or resistance. Other estimates of the initial modifier may use only a portion of the total resistance but will use the same ratio ($\frac{N}{N-1}$) for the change in the portion used on removal of an anode. The following estimate is an illustration.

[2] Understanding the Hall-Heroult Process for Production of Aluminium, Kai Gejotheim and Halvor Kvande – Editors,p.133

[3] Mark's Standard Handbook for Mechanical Engineers, Eighth Edition, 15-8

The second and more accurate voltage to estimate the minimum modifier is the voltage drop from the anode bus to the metal pad V_B.

This requires a change in the equivalent circuit in figure 2 to that of figure 3. In figure 3, two series resistances (R_{S1}, R_{S2}) are added. The series resistance R_{S1} before the parallel resistances would represent bus, flexes, joints and anode bus. The resistance R_{S2} after the parallel resistances would represent metal pad, cathode drop, flexes, joints and bus. The total resistance (R_{total}), is given by,

$$R_{total} = R_{S1} + \frac{R_B}{N} + R_{S2}. \ (6)$$

The problem with this estimate is that the anode bus to metal pad would have to be measured or estimated on each cell.

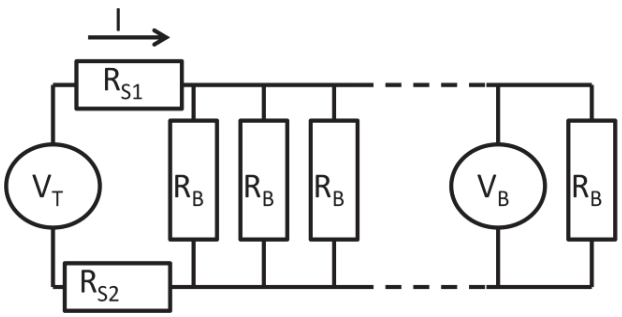

Figure 3

An estimate can be made by estimating the voltage drop on the anode cathode distance (ACD) and adding to that the voltage drop on the anode assembly. If back emf is included in this voltage V_B, the resistance that changes on removal of an anode is,

$$R_{Changing} = \frac{R_B}{N} = \frac{1000(V_B - 1.65)}{I} \mu\Omega. \quad (7)$$

The resistance of the remaining anode assemblies after removal of one anode would be

$$R_{remaining\ anodes} = \frac{N}{N-1} R_B \mu\Omega \ (8)$$

and the resistance modifier would be

$$Resistance\ Modifier = \frac{N}{N-1} R_B - R_B \ \mu\Omega. \ (9)$$

The increased voltage would be given by

$$Increased\ Voltage = \frac{N}{N-1} (V_B - 1.65) + 1.65 \text{ V} \ (10)$$

and the voltage modifier would be

$$Voltage\ Modifier = \frac{N}{N-1} (V_B - 1.65) + 1.65 - V_B \ V. \ (11)$$

For illustration, an estimate can be made by estimating the anode cathode distance (ACD) and knowing that the voltage drop is approximately 0.1V per 0.1 inch. Most cells will have ACD of 1.75- 2.00 inches so the voltage drop on the ACD will be 1.75-

2.00V plus back emf. Add to that the anode assembly voltage drop which will be approximately 0.2V. The total would be 1.95-2.20V plus back emf. For this estimate, V_B would be 3.6- 3.85V. If, for example, the cell had 24 anodes and line current was 200kA, the estimated resistance R_B for the minimum value of V_B would be 9.75 μΩ and would increase to 10.17 μΩ so the modifier would be 0.42 μΩ. The increased voltage would be 3.68V and the voltage modifier would be 0.08V.

Note: In actual practice, an estimate of the voltage on the ACD could be made by using the total cell voltage and subtracting the external anodic voltage, external cathodic voltage and the cathode voltage drop.

Looking back at the first estimate, equation 5, if this same cell had a target voltage of 4.050V, the new target would be

$$V_{NT}=(24/23)(4.050-1.65)+1.65= 4.154V \quad (12)$$

so the modifier would be 0.104V. The first estimate is higher as expected since we included some resistance that does not change when an anode is removed. Since the target voltage is generally set to the minimum value that will allow cell stability and given the upsets in heat balance, current distribution, bath ratio and feed control previously noted, a modifier larger than the minimum required to maintain anode cathode distance may be needed. Since the first estimate is easy to determine, it is the one suggested to begin with as a minimum value.

Determine the Heat Required

To determine the heat required to heat the new anode to steady state temperature, the specific heat capacity (C) of the anode is required. Scientifically, amorphous carbon is free, reactive carbon that does not have any crystalline structure. Prebaked anodes are not amorphous in the strict definition. However, "Contrary to science, in industry "amorphous" refers to very small crystal size rather than complete lack of crystal structure."[4] In the industrial sense, prebaked anodes are amorphous carbon, not graphite, but not truly amorphous either. Amorphous carbons vary widely so the classification does not uniquely determine physical properties. The amorphous form is an assortment of carbon atoms in a non-crystalline, irregular, glassy state, which is essentially graphite but not held in a crystalline macrostructure.[5]

In the United States, the primary feedstock used for making synthetic graphite is calcined petroleum coke and coal tar pitch.[6] The primary difference between the manufacture of baked anodes and synthetic graphite is the baking temperature. The density and porosity of baked anodes and electrode grade graphite are very similar.[7]

One author[8] gave the equation for the specific heat of the free carbon in coal (C) as,

$$C_{free\ C\ in\ Coal} = -0.218 + 3.807 \cdot 10^{-3}T - 1.758 \cdot 10^{-6}T^2 \quad (13)$$

where T is in Kelvin and the units are kJ/kg-K.
The specific heat (C) of carbon graphite[9] is given as,

Specific Heat of Graphite @ T= $2.673+0.002617T-116900/T^2$ (14)

where T is in Kelvin and the units are cal/K-mole. Converting the results of equation (14) to the same units as equation (13), both are displayed in figure 4.

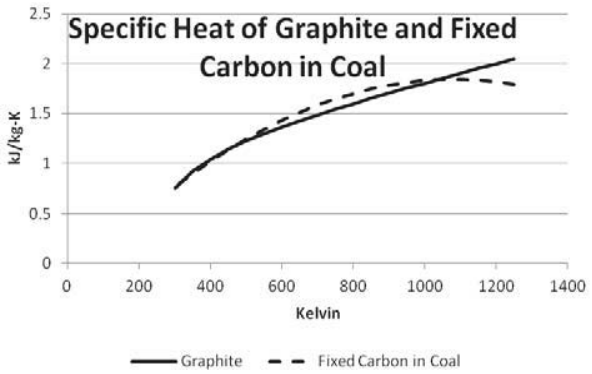

Figure 4

The discussion of the different forms of carbon and the specific heats of those similar to baked anodes is included here to justify using the equation for the specific heat of graphite for the specific heat of prebaked anodes.

Proceed to determine the kWh needed to heat one anode from ambient temperature to steady state operating temperature.

First, transforming equation (14) to the units of kWh/kg-K gives,

Specific Heat of Baked Anode @ $T = 2.59 \cdot 10^{-4} + 2.536 \cdot 10^{-7}T - \frac{11.3274}{T^2}$ (15)

If T is the anode steady state operating temperature and t is ambient (or cold anode temperature), the heat requirement is obtained by integrating the equation for specific heat, evaluating between the limits of t & T. The heat requirement, in kWh/kg, is

$$\text{Heat Requirement} = 2.59\cdot10^{-4}T+1.268\cdot10^{-7}T^2+\frac{11.3274}{T} - \left(2.59\cdot10^{-4}t+1.268\cdot10^{-7}t^2+\frac{11.3274}{t}\right) \quad (16)$$

Example: Assume the line current for the example cell used previously is 200 kA, T=972, t=300 and the anode mass is 1000 kg. The heat requirement is 256 kWh per anode. The average anode operating temperature is determined from the average of

[4] USGS- Mineral Yearbook, 2011,Graphite
[5] *Wikipedia.* August 15, 2001. https://en.wikipedia.org/wiki/Carbon/Allotropes
[6] Albert V. Tamashausky, Director of Technical Services, Ashbury Graphite Mills Inc.,2006, p. 5
[7] Foosnaes, Trygve, et al, *Understanding the Hall-Heroult Process for Production of Aluminium,* Aluminium-Verlag, 1986, p.88
[8] Bartosz LEŚNIAK, Łukasz SŁUPIK, Grzegorz JAKUBINA, Institute for Chemical Processing of Coal, Zabrze, Poland. "The determination of the specific heat capacity of coal based on literature data." *CHEMIK*, 2013
[9] Chemical Engineers' Handbook, Fifth Edition, 3-120

cell temperature, assumed to be 1220K, and the temperature of the top of the anode when it becomes fully loaded, assumed to be 723K. ***The anodes top temperature should be measured for a specific application.***

If one were willing, for simplicity sake to use 972 K and 300 K for anode steady state operating temperature and ambient temperature respectively, the heat requirement would be simply,

$$Heat\ Requirement = 0.256\ A_m \frac{kWh}{anode} \quad (17)$$

where A_m is the anode mass in kg

Removal Rate and Method

If we used the larger modifier, calculated by equation 5, of 0.104V, at 200kA it would take 12.3 hours to supply the required heat.

$$\frac{256\ kWh}{(0.104\ V)(200\ kA)} = \frac{256\ kWh}{20.8 kW} = 12.3\ hours \quad (18)$$

If we removed the modifier linearly it would take 24.6 hours.

$$\frac{2(256\ kWh)}{(0.104\ V)(200\ kA)} = 24.6\ hours \quad (19)$$

See figure 5.

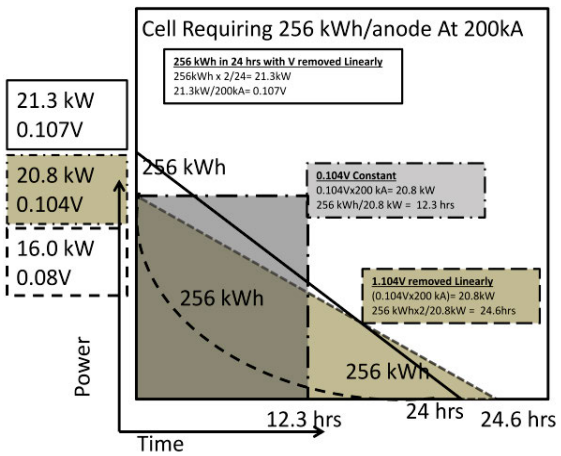

Figure 5

If we look at the minimum modifier for ACD we estimated as 0.08V (16.0 kW) and found for this pot the anode will load to 100% in 24 hours in a non-linear fashion, we could estimate the required modifier to maintain the ACD. The modifier would start at 0.08V and decrease non-linearly to zero as the anode reached full load. The dashed curved line in figure 5 could represent an estimate of the modifier required for preserving ACD. Figure 6 is actual data for a pots anode load up.

Clearly, if we remove the modifier linearly in 24 hours the modifier would need to be larger than our largest estimate of 0.104 volts and not fall below the minimum required for ACD preservation. In our example, it would need to be 21.3 kW/200kA= 0.107V to generates the required heat in 24 hours. If the anode load up time is longer than 24 hours, use that time instead of 24 hours.

In general, if the time for the anode to fully load is Lt, the heat requirement from equation 16 is Hr, the anode mass is A_m and the line current is I, the initial voltage modifier required, to be removed linearly, is given by

$$Voltage\ Modifier = \frac{2A_m H_r}{L_t I} \quad (20)$$

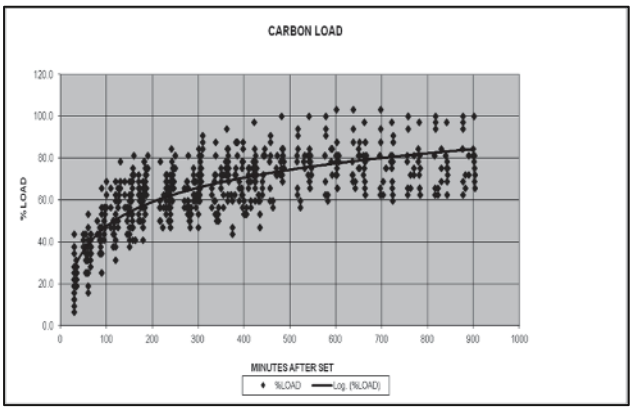

Figure 6

Conclusions and Observations

An acceptable anode set modifier can be determined as follows:

1) Use the cell target voltage to estimate the minimum initial modifier value (equations 4&5)
2) Calculate the kWh required to heat the anode to its steady state operating temperature (anode mass & equation 16)
3) Calculate the initial modifier required to generate the required heat from a linear rate of removal in 24 hours or the actual load up time, whichever is largest (equation 20)

Compare this to the current initial modifier you are using and current kWh generated. If the current modifier or heat generated is smaller than the new modifier and heat required calculated, consider increasing the current modifier to the new values.

If the cell has been running reasonably well with a smaller modifier and/or smaller heat generation than the new values, consider that the current target voltage is likely higher than needed with the new values. Note that in our example the minimum voltage modifier to only preserve the ACD was 0.08 volts but the initial modifier we needed to supply the heat was 0.107V. While that may seem large, it is less than 0.03 inch increase in ACD initially. Furthermore, since we removed it linearly over 24 hours, the average increase in voltage was only 0.053V. If the base target resistance could be reduced by 0.027V, over 48hours, assuming the cell is not set in the following 24 hours, the average voltage would be the same. The advantage now is that the heat input is more "as needed" and less cycling of the side freeze occurs. This will reduce bath ratio cycling and will improve both ratio and feed control.

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Other observations that could be related to the current modifier and removal rate: Frequent anode bridge lowers after set suggest that the removal rate is too fast. Frequent noise control intervention indicates the modifier is too small. Anode effects after set indicates there is too little heat input after set.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

REDUCTION OPERATING EXPERIENCE ON POWER SHADING AT MAADEN

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Keywords: power shading, thermal balance

Abstract

Maaden aluminum smelter in Saudi Arabia commenced operation in 12-12-12. The reduction consists of 2 potlines with 360 cell per line (720 cells) at operating amperage of 376 kA.
On April 30th, power was lost due to a standby rectifier failure during preventive maintenance. This incident lead to operation of Potline 2 at low load (fluctuated from 290 vs 376 kA) for frequent periods (15 time, 4 - 8 hours each) in order to carry out maintenance in the substation without sacrificing potline control.
Ma'aden Rectifiers were assembled using diodes fabricated with a different process (recrystallize structure) that were limiting the rectifiers to operate at a nominal capacity of 105 kA per unit. That makes the cell heat balance control a major challenge for operating at 85 kA below design.
This paper covers the successful controlling of the process during the power limitation.

Introduction

Ma'aden Aluminium has 10 rectifier units rated at 105 kA, 1750VDC servicing two Potlines (PL#1 & PL#2). The rectifier unit function is to convert AC to DC power for supplying the potlines for metal reduction process.
At April 30th, 2015, Rectifier #25 exploded during the process of energizing the unit after planned annual maintenance. The explosion was caused by high short circuit current flow through a grounding cable left connected inside the rectifier container. The ensuing explosion resulted in tripping off Potline #2. No personnel injuries occurred. Catastrophic damages were sustained to rectifier components and structure.
The summarized analysis is listed below:

- In preparation to energize RT#25, Step #5 of the switching order requires the closing of the DC isolating switches;
- A grounding cable was not disconnected from inside the rectifier container as required in Step #6 of the LTV (Lock out Tag out Verification) procedure;
- A Short circuit between the positive & negative buses to ground resulted in a high short circuit current flow;
- An explosion occurred in rectifier container and damaged the equipment
- On RT#25 the Reverse Current Protection & Arc Detection activated;
- Potline #2 Rectifiers tripped to clear the fault;
- RT#22, RT#23 and RT#24 tripped after ~124ms to 127ms while RT#21 tripped after ~169ms;

The Figure 1 (a) and (b) shows part of the damages in the rectifiers.

Figure 1(a) - DC switch damage bottom view

Figure 1(b) - Damage of AC Overvoltage Protection.

Power Modulation

After losing rectifier 25, Potline 2 has experienced operation at low current (290 kA instead of 315 kA) for short periods of time (up to 08:00 h) for maintenance in the substation. The interventions were initially required to change diodes that failed on May 22nd and 23rd, several weeks after the initial incident. Later on the same procedure was applied to replace suspected diodes in Rectifiers 24 and 23 and also to replace all diodes in Rectifier 21.
Ma'aden rectifiers were assembled including diodes fabricated with a different process (recrystallize structure) that were limiting the rectifiers to operate at a nominal capacity of 105 kA per unit. Before the problem on Rectifier 25, only rectifier 22 had all diodes replaced.

On May 22nd, rectifiers 21 and 23 presented diodes failures (one in each unit) that should be replaced as soon as possible. At this time there was not enough time to prepare the potline and the maximum current available was limited to 290 kA on short notice. The duration of intervention was estimated as 3 to 4 hours.

Potline Plan to run at 290 kA

- Add some resistance and/or current to the pots prior to a scheduled shut down (each 0.1 μΩ increases Potline voltage by 13.5 V).
- Add 1.0 μΩ for the period operating at 290 kA (later changed to 1.2 μΩ).
- Stop all AlF3 Feeders
- Stop Forced Cooling Network (FCN)
- Reduce GTC gas flow by 23%
- Tap normally; stop anode change
- Keep additional resistance or current after restoring current to 376 kA.

Calculation of additional resistance

Calculated additional resistance was done to maintain heat in the bath, assuming only power generated at inter-polar distance (ACD). The estimated bath resistance is 4.0 μΩ during normal operation which results in 565.5 kW/pot. To keep this value at 290 kA, total resistance to be added should be 2.7μΩ. Addition of 2.7μΩ would reduce bath heights approximately 10 cm that could represent a serious risk of open circuit, which could cause the potline to shut down. It was then decided to add only 1.0μΩ, considering the expected duration of 4:00h.

Table 1 shows the calculations of additional resistance to different levels of current.

	290 kA	300 kA	310 kA	315 kA	330 kA
Add Resistance - μΩ	2.70	2.28	1.88	1.70	1.19
Pot Voltage- V	4.41	4.37	4.34	4.32	4.28
Pot Volt - V (incl. extenal)	4.46	4.43	4.39	4.38	4.34
Line Voltage - V	1606	1593	1582	1577	1564
Power in Bath -kW	565	565	565	565	565

Table 1 – Additional Resistance Calculation

Rectifier 23 shut down (emergency – 1 diode failure) – duration 3:00 h

22nd May

- 14:53 – Added 0.2 μΩ;
- 15:47 – Added 0.15 μΩ and 2 kA;
- 16:43 – Line Amps to 290 kA, added 0.65 μΩ; (total additional resistance = 1.0 μΩ);
- Stopped FCN, AlF3 Feeders and reduced GTC flow;
- 19:32 – Back to normal amperage (2 kA were kept regard the next day morning stoppage); removed 1.0 μΩ;

Rectifier 21 shut down (emergency – 1 diode failure) – duration 02:25 h

23rd May

- 07:20 – Added 0.3 μΩ;
- 07:33 – Amps to 290 kA Added 0.7 μΩ (total additional resistance = 1.0 μΩ);

Stopped FCN, AlF3 Feeders and reduced GTC flow;

- 10:09 – Current restored to 378 kA; removed 1.0 μΩ;

Figure 2 shows the current modulation for diode repair. The total voltage of the line is presented in Figure 3.

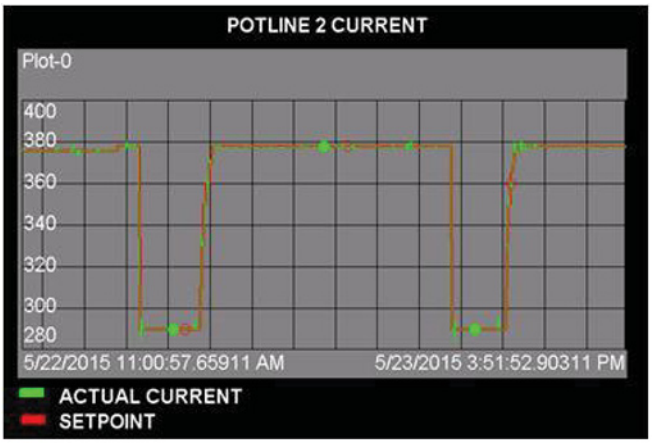

Figure 2 – Line Current for rectifier shutdown on 22nd and 23rd May

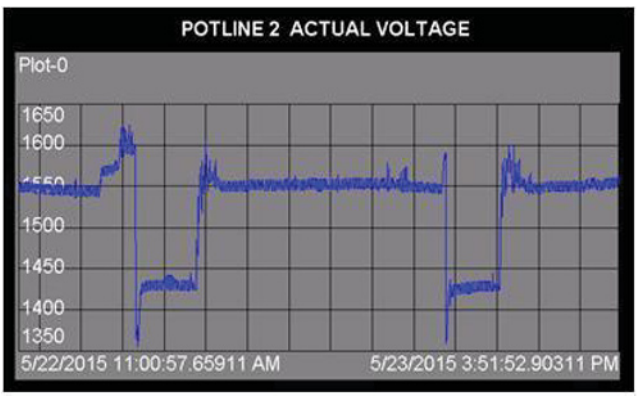

Figure 3 – Line Voltage for rectifier shutdown on 22nd and 23rd May

After these events, it was decided to change suspected diodes mapped by ABB, to avoid future failures as well to allow the rectifiers to run at their nominal capacities. Initially Unit 24 should be stopped to change 8 diodes and then 6 diodes in unit 23. Unit 21 had a high number of suspect diodes, so all the diodes should be replaced. Later on, all diodes would be replaced in units 23 and 24, after receiving the diodes from ABB.

Planed stoppage of Rectifier 24 to replace 8 suspected diodes (duration 02:20 h)

31st May

- 15:22 Added 0.2 μΩ;
- 18:35 Added 1 kA;

1st June

- 05:58 - Current to 295 kA, added more 1.0 μΩ to pot resistance (total of 1.2 μΩ);
  Stopped FCN, AlF3 Feeders and reduced GTC flow;
- 09:08 – Start ramp up to 376 kA, removed 1.2 μΩ;

Planed stoppage of Rectifier 23 to replace 6 suspected diodes (duration 03:10 h)

June 9th

- 18:55 – Added 0.2 μΩ;

June 10th

- 05:58 – Current changed to 300 kA, added more 1.0 μΩ (total 1.2 μΩ);
  Stopped FCN, AlF3 Feeders and reduced GTC flow
- 09:08 – Ramp up to 376 kA;

For Rectifier 21 all the diodes should be replaced. The work was scheduled to be done in 3 days, with stoppages of 8 hours and current at 315 kA (the defective diodes replaced before allowed the rectifiers to run at nominal capacity, slightly higher than before).

1st Stoppage of Rectifier 21 to replace 58 diodes (1/3 of 168 diodes) – duration 08:10

15th June

- Bath tapping table was changed to raise lower limit for bath height;
- Some pots had anode change in advance, but anode change continued during power reduction;
- 18:01 – Added 0.3 μΩ;

16th June

- 05:01 – Current changed to 315 kA added 0.9 μΩ;
- Stopped FCN and reduced GTC flow;
- AlF3 Feeder were kept in operation;
- 13:11 – Ramp up to 376 kA;

During this stoppage more than 50% of diodes were changed, so the following stoppage was planned to complete the work in Rectifier 21, even considering some time extension.

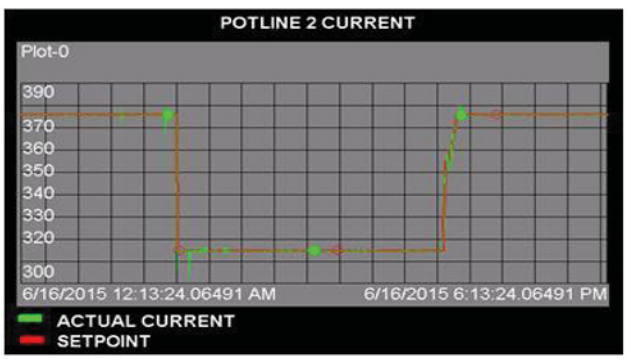

Figure 4 – Line Current for rectifier 21 shutdown on June 16th.

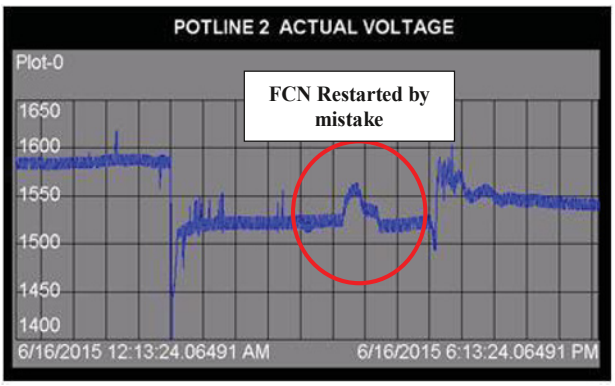

Figure 5 – Line Voltage for rectifier 21 shutdown on June 16th.

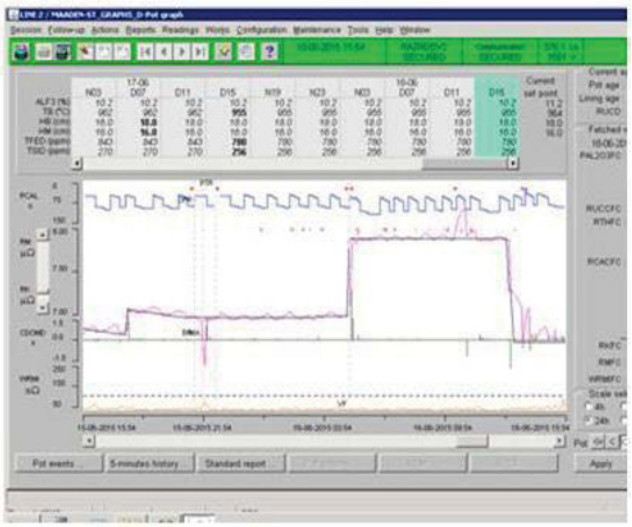

Figure 6 – Pot resistance graph showing the actions taken for rectifier 21 shutdown.

Completion of diodes replacement in Rectifier 21 – duration 08:33 h

17th June

- Bath tapping table was changed to raise lower limit for bath height;
- Some pots had anode change in advance, but anode change continued during power reduction;
- 18:01 Added 0.15 μΩ (smaller value due to high temperature trends);

18th June

- 05:08 Current reduced to 315 kA
- Added more 1.05 μΩ (total 1.2 μΩ;
- Stopped FCN and reduced GTC flow;
- 13:41 Rump up to 376 kA;

Potline process follow up

Below are some running charts Figure 7 ("a" to "e") for key process variables. Each point is the average for 4 hour period (Elementary Period) recorded in ALPSYS database.

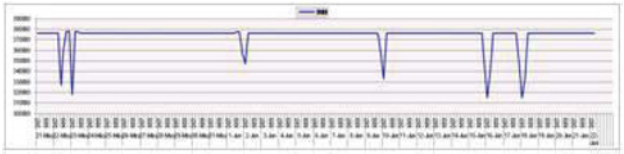

Figure 7(a) - Line Current (A)

Figure 7(b) - Anode Effects/Pot. Day

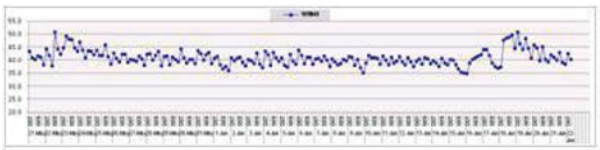

Figure 7(c) - Pot Instability (nΩ)

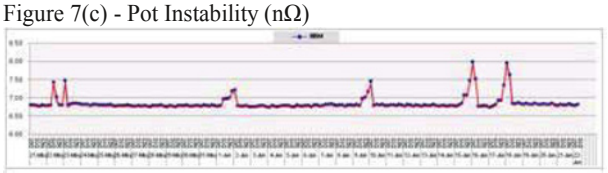

Figure 7(d) - Resistance Average (μΩ)

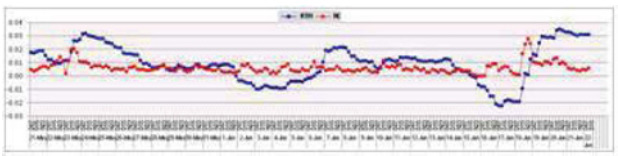

Figure 7(e) - Additional Resistance (μΩ) RTH=thermal; RC Manual Addition

The following variables presented in Figure 8 ("a" to "d") are related to measurements. The termination "D" means latest and "C" means Target. The values represented by round marks and dotted lines can present more variation, because for some elementary periods they can represent on re-checks, taken from small number of pots.

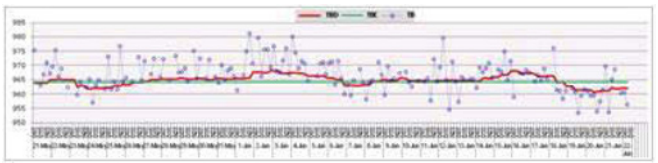

Figure 8(a) - Bath Temperature (°C)

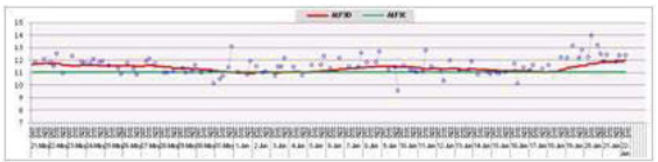

Figure 8(b) - Excess AlF3 (%)

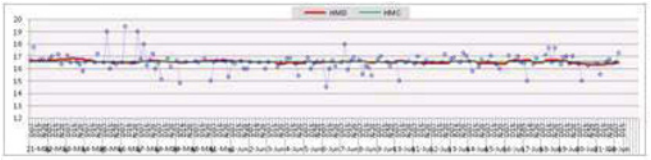

Figure 8(c) - Metal Height (cm)

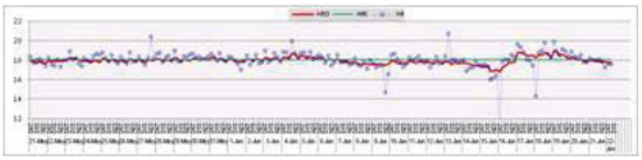

Figure 8(d) - Bath Height (cm)

Conclusions

The changes done for short term current reduction have been effective to keep potline process under control.
Last two longer curtailments (around 8 hours) had a more severe impact on instability and bath temperatures, but nothing that can be considered an out of control situation.
FCN has some strong and quick impact on pot resistance. But the changes observed in resistance seem to be more related to pot shell shape changes and its impact on liquid levels. When FCN was restarted by mistake, potline voltage increased approximately 35 V in a short time. Slope values increased requesting exceptional UFF (Ultra-Fast Feeding) for the pots. The same phenomenon has been observed after restoring power and restarting FCN.
315 kA is a much better condition to operate the Potline, compared to 290 kA, because the total base resistance change required 1.7 μΩ against 2.7. Since we have added 1.2 μΩ without liquid bath addition probably the bath requirement to reach 1.7 μΩ will not be so big. Adding 1.7 μΩ at 315 kA would allow the Potline to run at 315 kA for a medium/long term period of time. Of course, many other process adjustments will be required for a long term operation such cover material and anode size.
If the rectifiers with replaced diodes could work at 110 kA, total 330 kA for the Potline, additional resistance required would be 1.2 μΩ. This is the value that we have experimented for last rectifier's stoppages. 330 kA would allow stable operation for a long period of time without need of adding too much liquid bath in a short period of time.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF CARBON DUST ON THE ELECTRICAL RESISTIVITY OF CRYOLITE BATH

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Keywords: Carbon dust, bath resistivity, ACD squeeze, cell stability

Abstract

For many years anode dusting has been an issue for aluminum producers. Suspended carbon particles enhance the electrical resistivity of the cryolite bath, setting off a chain reaction of adverse effects for the pot. Many studies have documented the negative influence of carbon dust on current efficiency or bath temperature which are indirect effects of contaminated bath. The dependence of bath resistivity on the amount and on the size of carbon particles is poorly understood though. Tube-type cell experiments were performed to determine the electrical resistivity at 980 °C of standard bath mixtures added with carbon dust. The carbon content ranged from 0.06 to 1.01% in weight with mean particle size of 2 µm. A bath resistivity increase of 70% was measured when comparing the bath with the lowest carbon concentration with the one with the highest. From a bath with a carbon content of 0.06% to a one with 0.16%, the change in resistivity was equal to 13%. Thus the difference in bath resistivity between a benchmark bath with 0.03% carbon content and a bath from a dusty pot will be even larger. These results agree with variations of voltage measured during the ACD squeeze of groups of clean and dusty pots.

Introduction

Carbon is one of the impurities present in electrolytic bath along with iron, silicon or sulfur. In contrast to other impurities, carbon does not dissolve in bath. Coarse particles float at the bath surface whereas finer ones are suspended in the electrolyte. Carbon particles have a size ranging from micron to centimeter scale with an average size between 1 and 10 µm [1, 2]. The average carbon concentration in the bath is of the order of 0.05% and varies as a function of bath depth (see Figure 1). It is larger at the bath-metal interface and close to the bath surface where the coarse particles are found. In dusty pots, the average carbon concentration in the bath can reach values as high as 0.4% [3] with severe consequences for the pot performance.

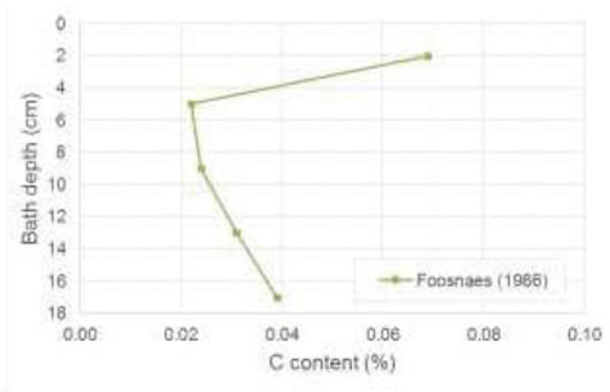

Figure 1. Carbon content as a function of bath depth as measured over 5 prebake pots by Foosnaes [2].

Carbon dust mainly originates from selective burning of the anodes in contact with CO_2 or air. The binder matrix is preferentially burnt, loosening carbon particles into the bath or within the crust/anode cover. The combustion reactions take place at the anode surface or in the anode bulk where the gases are able to permeate. The reaction rates depend on the temperature, surface structure, permeability and reactivity of the anodes constituents [4].

Other sources of carbon dust are identified such as cathode and ramming paste wear. Carbon fines entrained by the fumes and captured by the dry scrubber system return to the pot through secondary alumina feeding. Likewise, recycled anode cover contains carbon particles which contaminate the bath when falling into it.

The cycle of carbon dust in the electrolytic bath is complex and mechanisms of generation of fine particles, combustion rate and accumulation patterns of suspended particles, impact on bath properties are poorly understood. Carbon excess consumption figures give a hint about the total amount of anode carbon which is not used for the electrochemical reactions but fall short on predicting how much, where and for how long carbon dust remains in the bath. Over the years, strategies were developed with the objective of preventing the formation of carbon dust, monitoring the carbon concentration (e.g. in the secondary alumina) and removing carbon dust from the electrolyte.

The primary effect of carbon dust is to increase the electrical resistivity of the bath. Since the cell voltage is prescribed by the process control, a higher dusting level will result in an ACD squeeze. Enhanced bath resistivity is the starting point for a chain reaction of adverse effects. A lower ACD will affect pot stability and current efficiency [5], increasing bath temperature. As a result, the cell performance is reduced and the propensity to dusting is enhanced.

The dependence of bath resistivity on carbon concentration is little documented. Early work by Vetyukov [6] reported significant impact of carbon content in comparison with other bath constituents [7, 8] as shown in Figure 2. However, at levels of carbon content encountered in industrial pots, the increase of bath resistivity found was small i.e. approximately 10% at 1% concentration. It failed to explain the disturbance caused by carbon dust at levels as low as 0.1%. Foosnaes [2] developed a simple model to account for the resistivity increase as the result of a reduction of the conducting cross-section due to suspended particles. The prediction of the model is also represented in Figure 2. The same remark as for [6] applies though, when considering the impact of typically measured carbon dust concentration values.

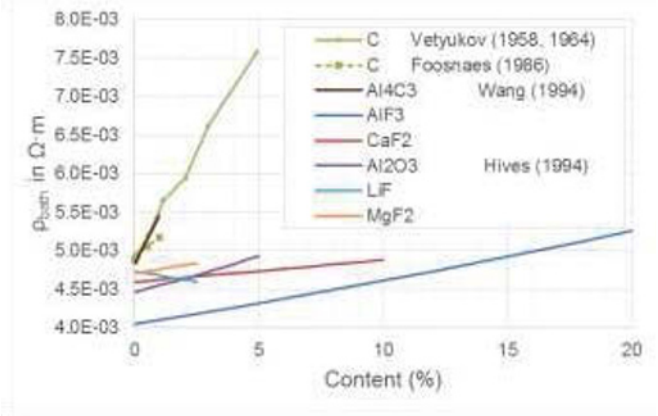

Figure 2. Bath resistivity as a function of its components concentration.

Experimental setup

In order to measure the electrical resistivity of cryolite bath as a function of carbon content, a tube-type cell was used. The cell, also used for the Rapoport cathode test, consists of a graphite crucible with 120 mm outer diameter and 92 mm inner diameter acting as the anode. The bottom of the crucible is insulated electrically by a fused alumina disk. The cathode is made by a 100 mm high graphite cylinder with diameter 30 mm lying on top of the alumina disk. Thus the anode-cathode distance is 31 mm. Solid bath is poured in the crucible and heated up at 980 °C for two hours. The crucible is covered by a graphite cover insulated from the current output by a fused alumina ring. The pressure between the cylinder/crucible and current output/input is ensured by a hydraulic cylinder. The atmosphere of the furnace is controlled through a continuous argon flow. A schematic drawing of the apparatus is shown in Figure 3.

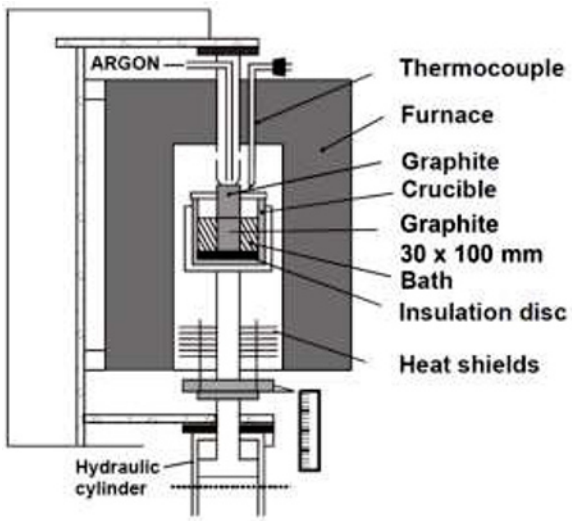

Figure 3. Schematic drawing of the test apparatus.

Bath samples were prepared in another furnace in a graphite crucible according to the following recipe: 80% Na_3AlF_6, 12% AlF_3, 5% CaF_2 and 3% Al_2O_3. The carbon dust was obtained from coke crushed with a vibrating cup mill. The carbon particles have a mean size of 1.9 µm with 90% of volume > 0.5 µm and 10% > 6.3 µm. Three different experiments were performed adding 1%, 0.1% and 0% carbon dust to the bath for a total sample mass of 250 g. The carbon contents of the clean bath and of the bath with 1% were measured separately using the LECO method to assess the sample preparation method and in particular the contamination of the clean bath by the graphite crucible used to melt the ingredients. Three 1 g samples of both baths were analyzed and yielded a mean content of 0.06% ± 0.02% for the clean bath and of 1.01% ± 0.05% for the bath with 1% carbon. Thus the carbon content of the bath samples used in the three experiments was corrected to 1.01%, 0.16% and 0.06%. We assume that carbon dust is uniformly spread in the bath. In industrial pots it is reasonable to admit that the carbon concentration is fairly constant away from the bath surface (see Figure 1) i.e. where most of the current flows. In our experimental setup the bath recirculation might be weaker than in an industrial pot though, causing segregation or clustering of the carbon particles to occur. Thus the measured impact of carbon shall be considered as a lower bound estimate.

Results and discussion

After preheating, voltage (3.87 V) was imposed at fixed power and the current was measured within 30 seconds to avoid changes in bath composition resulting from electrochemical reactions. The current was larger for bath samples with lower carbon content as reported in Table 1. To interpret the current values, we assume the following voltage decomposition:

$$U = C + (R_{bath} + R_{ext}) \cdot I \qquad (1)$$

where C is a constant, R_{bath} is the bath resistance and R_{ext} is the external resistance. The validity of Eq. (1) in the investigated current range was verified by varying the voltage for the bath sample with 0.06% carbon and by fitting the current data linearly (see Figure 5). The constant C = 2.1255 V is obtained as the y intercept value. It shall not be interpreted as the sum of the reversible decomposition voltage and of the overvoltages since the latter are not linear functions of the current. Since R_{ext} is unknown, R_{bath} was determined through modeling of the cell with the clean bath sample assuming a bath resistivity of $4.63 \cdot 10^{-3}$ Ω·m. The current distribution was computed using ANSYS (see Figure 6) yielding first $R_{bath} = U_{bath}/I$ with U_{bath} = 647 mV and second R_{ext} using Eq. (1). R_{ext} was considered to be constant for the two other experiments.

Figure 4. Section of the crucible after two hours of electrolysis for the experiment with the bath sample containing 0.16% carbon

dust. A torus of aluminum around the cylinder bottom is visible as well as the attack of the insulation disk on the sides.

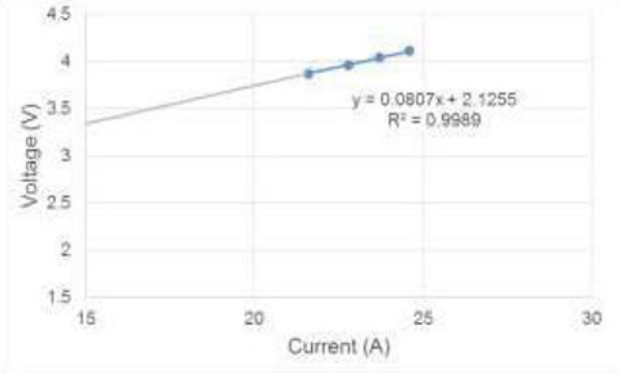

Figure 5. Linear fitting of the voltage as a function of current for the experiment with the bath sample with 0.06% carbon content and determination of the constant C = 2.1255 V.

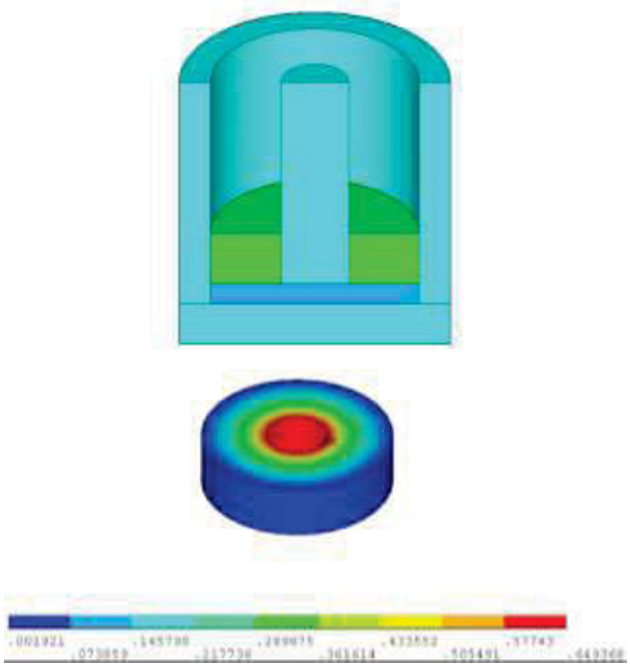

Figure 6. Solid model of the crucible and electric potential in the bath for the experiment with clean bath.

The bath resistance for the three experiments is reported in Table 1 and Figure 7. If we assume that the current distribution is similar for all experiments, R_{bath} is proportional to ρ_{bath}. As a result we conclude that the presence of carbon dust affects the bath resistivity to a larger extent than what has been measured or modeled up to now (Figure 2). At the carbon content of 0.16%, possibly found in a dusty industrial pot, the increase in bath resistivity is equal to 13% with respect to the resistivity of the bath at 0.06% and shall attain 20% with respect to the resistivity of a benchmark bath at 0.03%. The experimental procedure does not allow to measure the resistivity of cleaner bath since a low quantity of carbon is present as a consequence of the bath preparation method.

Table 1. Summary of the three experiments. ρ_{bath} stands for the bath resistivity.

C content	%	1.01	0.16	0.06
I_{init}	A	17.4	20.8	21.8
U	V	3.87	3.87	3.87
R_{bath}	Ω	0.0499	0.0335	0.0297
$\rho_{bath}/\rho_{bath,0.06\%}$	%	168	113	100
R_{ext}	Ω	0.0503	0.0503	0.0503

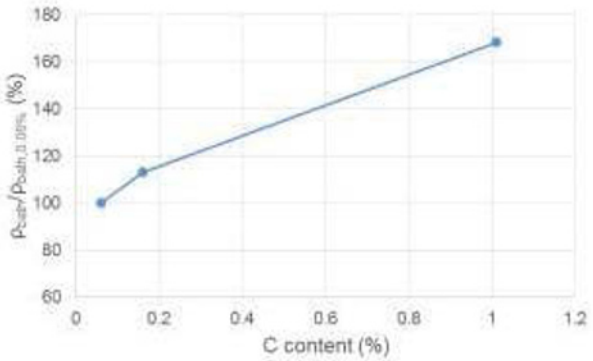

Figure 7. $\rho_{bath}/\rho_{bath,0.06\%}$ versus C content.

Squeeze test of clean and dusty pots

In order to assess the impact of carbon dust on bath resistivity and on the resulting ACD squeeze, a simple method was applied on two groups of pots with different bath carbon content. The beam was moved downwards by a few mm and it was kept at that very position during 10 min until it was moved downwards again. The cell voltage and beam position were recorded as a function of time. The test was stopped when the pot started to oscillate. In order to improve precision and to assess the reproducibility of the measurement, the same operation was repeated with upwards beam displacements. Cell voltage and beam position yield following quantities: 1) the increase of noise level as a function of ACD decrease and 2) the variation of pot voltage per mm of modified ACD. 1) provides information on the stability margin of the pot whereas 2) quantifies the bath electrical resistivity. The test was performed on two groups of pots in one potline: a first group composed of four clean pots and a second group with two dusty pots. Carbon content of bath samples from all 6 pots was determined using LECO method and XRF as control (see Table 2). The samples were taken from the tapping hole at approximately bath mid-height. The average carbon content was equal to 0.07% for the clean pots and 0.11% for the dusty pots. In Figure 8, the voltage of one pot from each group is plotted versus time during the downwards beam movements.

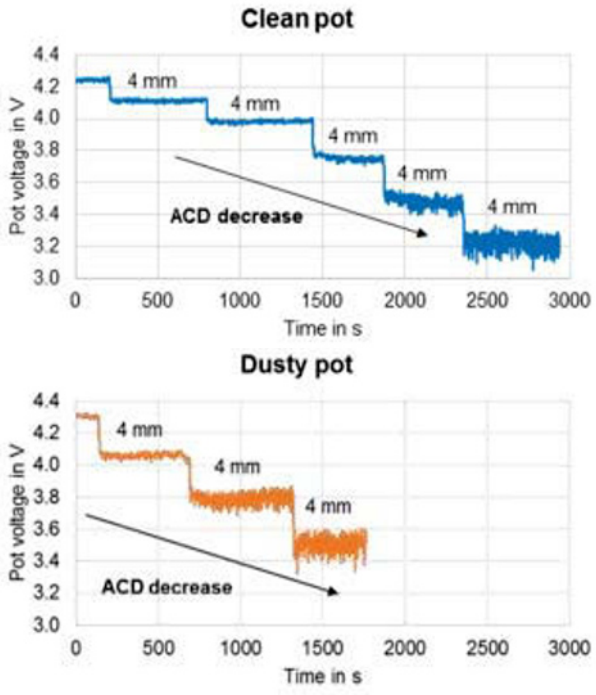

Figure 8: Pot voltage versus time during ACD squeeze in a clean pot and in a dusty pot.

One observes that the increase in noise level occurs earlier for the dusty pot. It is well visible when plotting the noise level (standard deviation of the cell voltage) as a function of the ACD decrease for the two pots (see Figure 9).

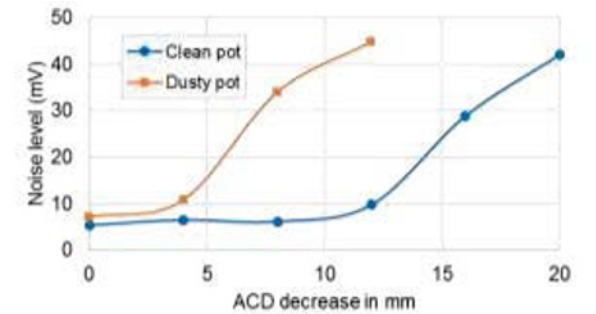

Figure 9: Noise level versus ACD decrease during ACD squeeze in a clean pot and in a dusty pot.

The acceptable noise level limit of 20 mV is exceeded after 6 mm of ACD decrease for the dusty pot and after 14 mm for the clean pot. Since pot design and pot operation are the same, the discrepancy is mainly explained by the fact that the initial ACD levels are not equal (the initial pot voltage values are). This is confirmed by the analysis of the variation of pot voltage per mm of modified ACD which is expected to be proportional to bath resistivity $\Delta U/\Delta ACD = \rho_{bath} \cdot I/S$ with S the total anode surface (see Table 2). Only data points with noise level below 20 mV are considered - at higher noise level, effects related to changing fanning factor and oscillations of the bath-metal interface distort the proportionality relationship - and data points are averaged over each group of pots. From the squeeze test, one obtains that the voltage variation per mm of modified ACD is equal to 47 mV/mm for the clean pots and 61 mV/mm for dusty pots. This represents a 30% larger bath resistivity which can be partly attributed to the presence of carbon dust in the electrolyte. Indeed the voltage variation per mm of modified ACD predicted by the bath resistivity formula [8] using the bath temperature and bath composition data is equal to 42 mV/mm and 41 mV/mm for the clean and dusty pots respectively. Other factors such as poor anode current distribution or bath height might also affect the quantity $\Delta U/\Delta ACD$.

Table 2. Variation of pot voltage per mm of modified ACD for the two groups of pots.

	# pots	ΔU/ΔACD Squeeze test mV/mm	ΔU/ΔACD Bath res. formula mV/mm	Carbon content Leco %
Clean pots	4	47	42	0.07
Dusty pots	2	61	41	0.11

Conclusions

Maintaining a low and steady concentration of carbon dust in the electrolyte bath is a difficult task. It requires careful selection of raw materials and know-how in the manufacturing and rodding of anodes. At operation level, thermally balanced pots, low bath temperature and uniform anode current distribution reduce the risk of dusting. The optimal cost/benefit ratio in terms of raw materials, anode manufacturing optimization and anode cycle time shall be determined. The present work aimed at understanding the root cause for the loss of pot performance due to carbon dust that is the increase of bath resistivity.

Laboratory measurements with a tube-type cell showed that bath resistivity increases significantly with carbon content. A bath resistivity increase of 70% was measured when comparing a bath with 0.06% carbon content with a one with 1.01%. From a bath with a 0.06% carbon content to a one with 0.16%, the change in resistivity was equal to 13%. Thus the difference in bath resistivity between a benchmark bath with a 0.03% carbon content and a bath from a dusty pot will be even larger. Squeezing tests of groups of pots with different bath carbon content confirmed that high levels of dusting affect bath resistivity and consequently the stability margin of the pots.

In this perspective, an objective assessment of the costs generated by carbon dust shall be realized. Under enhanced stress - current creep to increase aluminum production, ACD squeeze to limit specific energy consumption, change in anode recipe - pots shall react very differently depending on carbon dust concentration and stability margin. A comprehensive monitoring of anode quality, carbon content of the bath, butts properties is thus of prime importance.

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ALUMINUM REDUCTION TECHNOLOGY

Process Control in Reduction

SESSION CHAIR
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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

DETECTION OF LOCAL CELL CONDITIONS BASED ON INDIVIDUAL ANODE CURRENT MEASUREMENTS

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Keywords: Aluminum reduction, Individual anode current, Kernel PCA

Abstract

The application of individual anode current measurements in the Hall-Héroult cells has been investigated to aid cell monitoring, in addition to the conventional use of cell voltage measurements. Its advantages are significant, especially in high amperage cells, where information from the voltage signal is heavily diluted. One common use of individual anode current measurements is to identify process faults that can cause a significant change in the local current flow paths between anode busbar and the metal pad. While this detection is simple and direct, there are other consequential changes in cell conditions that impact the current distribution in the vicinity of a problem anode. This paper presents a Moving Window Kernel PCA-based method to extract spatial information from the individual anode current signals by incorporating other known process variables.

Introduction

The measurement of individual anode currents in the Hall-Héroult cells has attracted increasing attention because its capability to reflect changes in cell conditions at the vicinity of the anode has been demonstrated [1-3]. One common use with individual anode current measurement is to continuously monitor the values and trends of the signals, and to infer local cell conditions by matching patterns for known process behaviors, as demonstrated by Jassim et al. [4,5]. While this method is simple and direct, it overlooks the interaction between anodes. This interaction has two levels: (1) the current drawn by one anode affects the cell condition in its vicinity and thus affects the surrounding area through mass and heat transfer, which in turn changes the current on adjacent anodes; (2) since the line current is normally regulated, the individual anode current affects each other via Kirchhoff's Law. Thus, to obtain accurate information from individual anode current measurements, the interaction must be decoupled from the signals. One may argue that this problem can be solved by using the 'resistance' of each anode path by calculating v/i, where v is the cell voltage and i is the individual anode current. However, like other electrochemical reactions, the governing equation in the Hall- Héroult process takes the form of

$$v = a + b\log i + iR \tag{1}$$

where a and b are parameters related to cell conditions, and R is the Ohmic resistance of the anode-cathode path. It then follows that the 'resistance' is actually the impedance of anode paths since it is a function of anode current itself:

$$\frac{v}{i} = \frac{a}{i} + b\frac{\log i}{i} + R \tag{2}$$

Moreover, the impedance also has nonlinear relationships with cell conditions [6]. Therefore, a method that is able to account for the interaction is needed. Some model-based approaches can be found [7,8] whereas the statistical monitoring method which requires less modelling effort is necessary to complement the system.

Principal Component Analysis (PCA) is a classical statistical tool to extract information that explains the majority of the variances in a set of highly correlated data. A number of PCA applications have been proposed for monitoring and fault detections for the Hall-Héroult process. Majid et al. [9,10] use PCA with other multivariate statistical tool for the monitoring and fault diagnosis of the Hall-Héroult process. This approach uses global variables and pinpoints local problems. Individual anode current signals are used with PCA by Batista et al. [11] to predict anode effect. However, PCA is only suitable for data with a linear relationship while the Hall-Héroult process exhibits strong nonlinearity. Therefore, PCA or any other linear multivariate statistical tools may produce inaccurate and misleading results. Kernel Principal Component Analysis (KPCA), on the other hand, is a powerful tool for datasets with nonlinear relationships [12]. It extracts information from the hyper-space of the original set of data using the 'Kernel trick'. In this paper, a Kernel-based PCA with Moving Window (MWKPCA) is proposed to monitor and detect faults in the Hall-Héroult process. The proposed approach is applied to an event of an anode effect caused by a feeder blockage to show its effectiveness in process monitoring and fault detection.

This paper is organized as follows: The KPCA is introduced in the next section, followed by process monitoring and fault detection application of MWKPCA. The results of the MWKPCA-based monitoring are then presented. The paper concludes with discussion of the results and the potential further development of the proposed approach.

Kernel Principal Component Analysis

PCA is a statistical tool that reduces the dimension of the original dataset by solving the eigenvalue problem of its covariance matrix. It produces new linearly independent variables in lower dimensions that are linear combinations of the original variables. KPCA uses a similar approach, however, instead of working on the original dataset, it reduces the dimension of the data set after a nonlinear mapping [12].

Let $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n \in R^m$ be the dataset for training. If the dataset is nonlinearly correlated, it is normally not possible to generate a number of d linearly independent variables where $d < n$. To overcome this, KPCA extends the original variable space into a hyper-dimensional space via a nonlinear mapping $\Phi: R^m \to R^h$. The higher space has arbitrary dimension. The sample covariance of the transformed space can be obtained by

$$C = \frac{1}{n}\sum_{i=1}^{n}(\Phi(\boldsymbol{x}_i) - m_\Phi)\,(\Phi(\boldsymbol{x}_i) - m_\Phi)^T \tag{3}$$

where m_Φ is the sample mean. Denote $\overline{\Phi}(\boldsymbol{x}_i) = \Phi(\boldsymbol{x}_i) - m_\Phi$ and follow the similar procedure in PCA to solve the eigenvalue problem

$$\lambda v = Cv \tag{4}$$

where all solutions $\boldsymbol{v}$ with non-zero λ span $\overline{\Phi}(\mathbf{x}_i)$. It follows that

$$\boldsymbol{v} = \sum_{i=1}^{n} \alpha_i \overline{\Phi}(\boldsymbol{x}_i) \tag{5}$$

and

$$\lambda(\overline{\Phi}(\boldsymbol{x}_i)\boldsymbol{v}) = \overline{\Phi}(\boldsymbol{x}_i)(C\,\boldsymbol{v}). \tag{6}$$

Defining a Kernel matrix

$$K_{ij} = k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \Phi(\boldsymbol{x}_i)\Phi(\boldsymbol{x}_j) \tag{7}$$

and combining Equation (5) and (6), Equation (4) becomes

$$n\lambda\,\boldsymbol{\alpha} = \overline{K}\boldsymbol{\alpha} \tag{8}$$

where

$$\overline{K} = K - KE - EK + EKE \tag{9}$$

with $E_{ij} = 1/n$. It should be noted that $\boldsymbol{\alpha}$ should be scaled to ensure the normality of the principal components $\boldsymbol{v}$. Thus, for a new point $\boldsymbol{x}_{new}$ in the original variable space, its kth projection will be

$$t_{new,k} = \boldsymbol{v}_k\overline{\Phi}(\boldsymbol{x}_{new}) = \sum_{i=1}^{n} \alpha_i^k \overline{k}(\boldsymbol{x}_i, \boldsymbol{x}_{new}) \tag{10}$$

for k=1,2,…,p, where p is the number of principal components in KPCA. The score vector for $\boldsymbol{x}_{new}$ is then

$$\boldsymbol{t}_{new} = [t_{new,1}\ t_{new,2}\ \ldots\ t_{new,p}]^T \tag{11}$$

Process Monitoring with KPCA

Statistical Monitoring Metrics

Similar to its linear counterpart, KPCA also has two metrics [18]. The control limit for Hotelling's T^2 test with a confidence level of $100(1-\alpha)\%$ is

$$T^2_{lim} = \frac{r(n-1)}{n-r}F(r, n-1; \alpha) \tag{12}$$

where $F(p, n-1; \alpha)$ is the F-distribution with r to be the number of retained principal components. The T^2 statistic for each sample in the original variable space can be calculated by

$$T^2 = \boldsymbol{t}^T\Lambda^{-1}\boldsymbol{t} \tag{13}$$

where Λ is the diagonal matrix of significant eigenvalues. Moreover, the control limit for Q statistic SPE (Squared Prediction Error) with a confidence level of $100(1-\alpha)\%$ is

$$\text{SPE}_{lim} = (a/2b)\chi^2_\alpha(2b^2/a) \tag{14}$$

where b and a are the sample mean and variance of the SPE values.

Monitoring for the Hall-Héroult Process with Moving Window

In general, PCA-based monitoring consists of two stages, offline training and online monitoring. In the offline training stage, data during normal operation is collected and used to generate statistical control limits T^2 and SPE, while in the online monitoring stage, T^2 and Q statistics are calculated and compared with the control limits. However, the Hall-Héroult process is, by nature, time varying because its process parameters (anode resistance, ACD, temperature etc.) change with time. This means that the original dataset for offline training may not be suitable for online monitoring as the process proceeds, which could lead to false and missed alarms. Thus, the training dataset needs to be updated regularly to account for the time-varying nature of the process. In this paper, a simplified Moving Window Kernel Principal Analysis (MWKPCA) based on Liu et al. [13] is used to monitor the aluminum reduction process with individual anode current measurements. The procedure of the proposed approach can be summarized as follows:

1. Select the first n data samples as the training set and perform KPCA. This involves the following decisions.
   1.1. The kernel function used in this work is the Gaussian type

$$k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2}{c}\right) \tag{15}$$

   where c is a parameter in the kernel function. In the proposed method, c is determined as suggested by Choi et al. [14].

   1.2. The number of non-zero eigenvalues p is determined by a threshold ε_1 so that p is the smallest value that satisfies

$$\frac{\sum_{i=1}^{p}\lambda_i}{\sum_{i=1}^{n}\lambda_i} \geq 1 - \varepsilon_1 \tag{16}$$

   where ε_1 is chosen as 10^{-4} so that no significant eigenvalues are ignored.

   1.3. The number of eigenvalues to retain r is determined by the similar manner. r is determined as the smallest value that satisfies

$$\frac{\sum_{i=1}^{r}\lambda_i}{\sum_{i=1}^{p}\lambda_i} \geq 1 - \varepsilon_2 \tag{17}$$

   where ε_2 is chosen as 0.05.

2. For data samples $\boldsymbol{x}_{n+1}$ to $\boldsymbol{x}_{n+s}$, T^2 and Q statistics are calculated to check the number of points that exceed the control limits. If an unacceptable number of points

exceeds the control limits, an alarm is raised and the signals are check for any possible faults.

3. If the monitoring metrics are acceptable, the s points are included in the training set, discarding the first s points. The training set is then $\boldsymbol{x}_{s+1}$ to $\boldsymbol{x}_{n+s}$ and the procedure from step 1 is repeated with this new training dataset.

In this work, the MWKPCA algorithm is applied to the dataset that contains anode current measurements feed rate, cell voltage and beam movement. If at one point the monitoring metrics are out of bounds, the MWKPCA algorithm is then applied to every anode current measurement with the relevant process data to find the localized problematic areas. The following section shows an example of using MWKPCA in the Hall-Héroult process.

Results and Discussions

Fault Detection

In this example, operation data from an industrial cell leading to an anode effect were recorded and used with PCA, KPCA and MWKPCA to illustrate the monitoring performance of different algorithms. The thresholds for the monitoring statistics have a significance level of 95%.

A period of approximately 300 minutes of operation is used in this illustration. The unusual voltage signal was detected by the cell controller at t = 290mins, which was approximately 50 minutes before the anode effect was detected by the cell controller and the blocked feeder was observed by the operator. During this time, individual anode current signals, anode and beam movement, cell voltage and feed rate were recorded and used for the algorithms. In the Kernel-based PCA algorithms, the size of the training set is 120 minutes. Figure 1 shows the T^2 and Q statistics for the dataset considered using PCA. As can be seen, it is not obvious that the monitoring statistics are out of the control limits. Moreover, there are some points that could lead to potential false alarms, especially in the SPE plot. This is because the fact that PCA assumes linear relationship between variables in the dataset, thus it cannot handle the nonlinearity in the process.

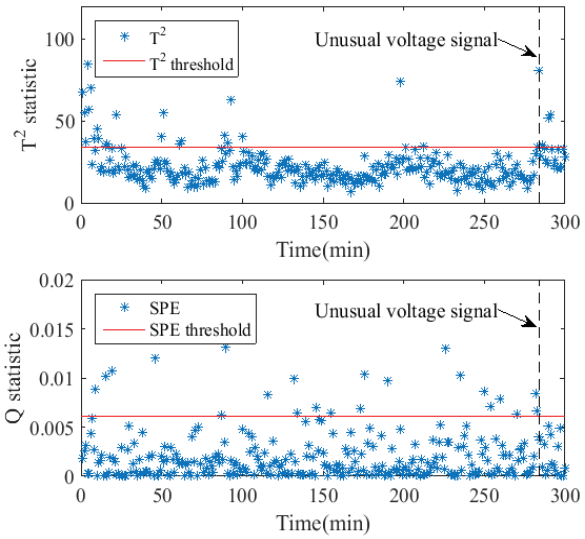

Figure 1 T^2 and Q statistics using PCA

Figure 2 depicts the T^2 and Q statistics for the dataset using KPCA. It is obvious that on both plots, the statistics dramatically increase towards the end, indicating a possible process fault. However, it can be seen that the statistics keep increasing after the KPCA training, creating a significant number of false alarms. This is due to the time-varying nature of the process. Since the process parameters keep changing, the original training set may not be suitable for the data samples after a period of time.

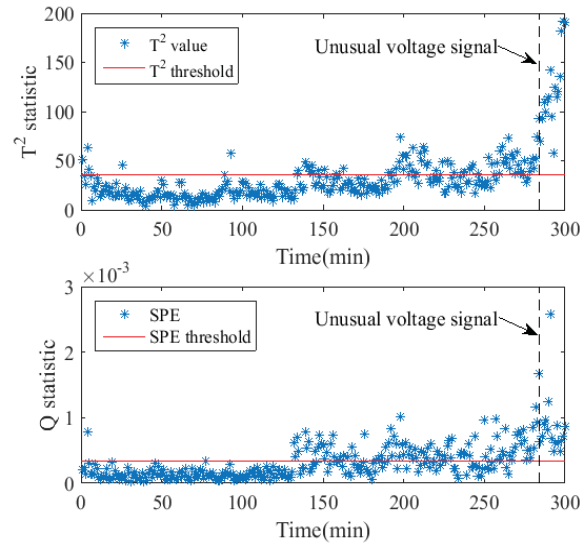

Figure 2 T^2 and Q statistics using KPCA

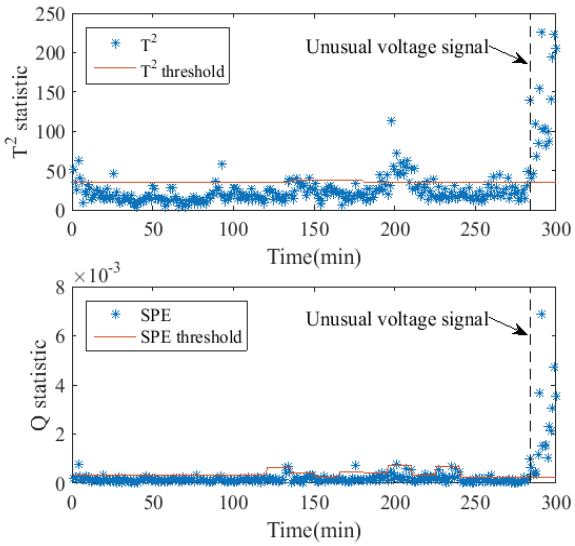

Figure 3 T^2 and Q statistics using MWKPCA

Figure 3 shows the best monitoring performance is achieved using MWKPCA. It can be seen that there are significant increases on the monitoring metrics when the unusual voltage signal is detected, and before that, the false alarm rate is relatively low.

Fault Isolation

When the monitoring statistics are out of the control limits, indicating that a possible fault is detected on the cell level, the proposed approach then applies MWKPCA on every anode to locate the problematic area. It uses the dataset when the monitoring statistics are still in control as the training set, and each anode is examined with T^2 and Q statistics. In the end, a visualization of results is produced for analysis.

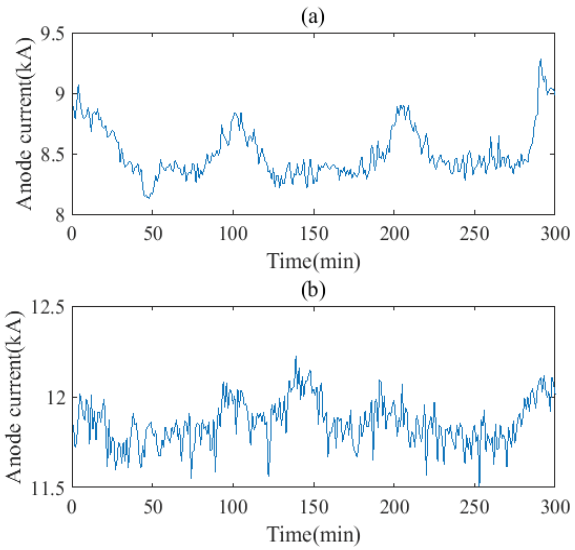

Figure 4 (a) Anode current signals for the anode close to the blocked feeder; (b) Anode current signals for the anode far away from the blocked feeder

As stated in the previous section, it is difficult to identify the localized process faults by observing the individual anode current signal itself. This is illustrated in Figure 4. It shows the current signals of respective anodes that are close to and far away from the blocked feeder. Both signals show an increase when the feeder was blocked, while the reasons behind the increases are different. For the one close to the blocked feeder, it may be caused directly by the blocked feeder, but for the one that is far away from the said feeder, it may just be showing the consequential changes caused by the blocked feeder. Similarly, anode current signals on other anodes do not exhibit distinguishable behaviors that can be of help in identifying problematic anodes. Pinpointing the problematic area using anode current measurements is therefore not straightforward. This is where MWKPCA can be of great assistance. Figure 5 and Figure 6 show the monitoring metrics for the respective anodes after the algorithm has found a possible fault in the last section. It is evident that the T^2 statistic for the anode close to the blocked feeder has more points that are out of control limit and they are significantly further away from that limit compared to that of the anode away from the problematic feeder. This demonstrates that MWKPCA can deal with the interaction between anodes and thus find the anode with fault directly associated with it.

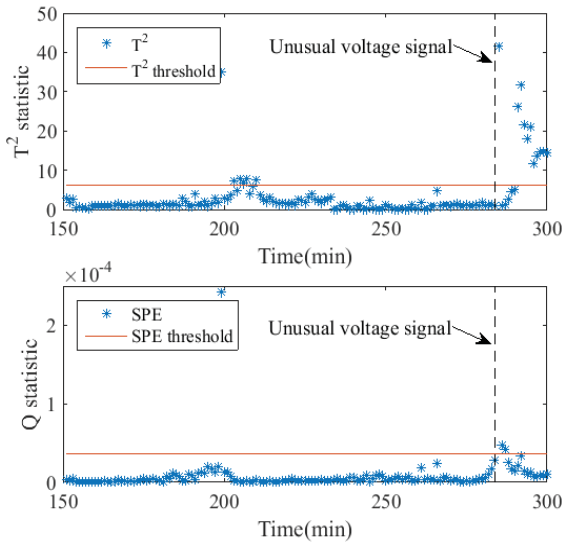

Figure 5 T^2 and Q statistics for the anode close to the blocked feeder

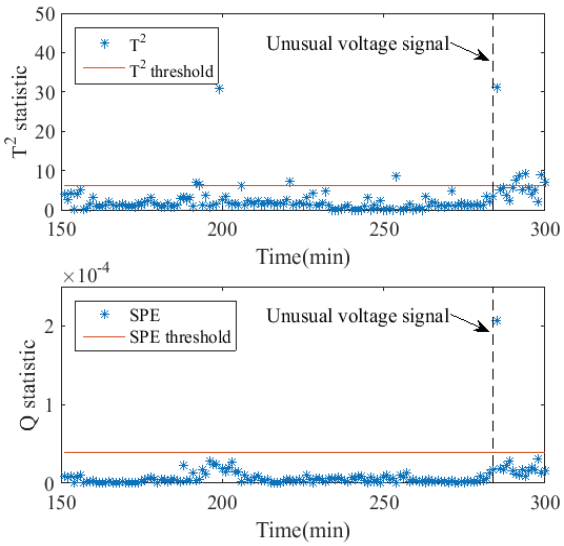

Figure 6 T^2 and Q statistics for the anode away from the blocked feeder

To visualize the results for troubleshooting, a heat map is created based on the number of out-of-limit points on the monitoring metrics plot and their values. Figure 7 shows a heat map on Red-Yellow-Green scale with red representing the highest value, and each block representing an anode and its vicinity. As can be seen, most of the red areas are close to the blocked feeder. The result shows the effectiveness of MWKPCA in identifying and clustering the problematic anodes to diagnose the fault. While such method can be used on the discretization based on individual anodes, the discretization can be changed if the computational cost becomes a concern, or when there is a sensor failure. As an example, two anodes in the upper right corner are treated together due to a sensor problem. By changing the discretization, instrumental error can be eliminated effectively to avoid the possibility false alarm.

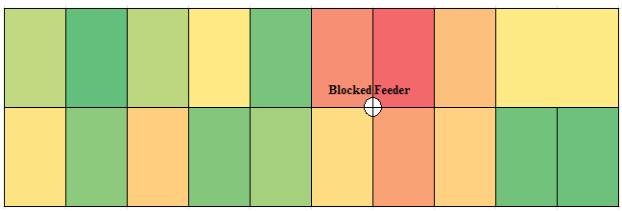

Figure 7 Heat map showing the problematic anodes in the cell

Conclusion and Further Development

This paper proposes a method based on Moving Window Kernel Principal Component Analysis (MWKPCA) for the process monitoring, fault detection and isolation with individual anode current measurements. By extending the original variable space into a hyperdimesional one, in which the linearly independent variables can be extracted, the proposed method overcomes the shortcomings in direct uses of individual anode current signals due to their two levels of interaction and the nonlinear nature of the process. It has been demonstrated that the monitoring performance of MWKPCA is superior to that of PCA and KPCA. In addition, a visualization of the cell condition can be produced based on the results from MWKPCA using individual anode current signals. This has great potential in the operation of the Hall-Héroult cells and can be complementary to some of the existing model-based process monitoring approaches.

This work can be improved by incorporating the dynamic version of PCA so that the dynamic features of the process can be captured as the process is not only time-varying, but also dynamic due to the dissolution of the fed alumina. In this way, the algorithm would be more robust to noises and sensitive to faults.

Acknowledgement

The authors wish to acknowledge the technical and financial support from Emirates Global Aluminum, Jebel Ali Operations.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

DYNAMIC RESPONSE OF CRYOLITIC BATH AND INFLUENCE ON CELL HEAT AND MASS BALANCE WITH LARGE SCALE POTLINE POWER SHIFTS

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Keywords: Smelter heat balance shift, heat transfer, bath chemistry

Abstract

An appropriate energy balance is critical to achieve high energy efficiency in aluminium electrolysis. However, to satisfy the electricity market, and the market demand for aluminium, a wider operational energy input window is now also required. Unfortunately this would require large scale adjustment of input power and heat dissipation, and is not possible with existing smelter technologies in which heat dissipations are not able to be adjusted widely or regulated. The associated wider cell operational window would also need to be matched to flexible energy supply without causing alumina concentration problems. Side ledge control is also a challenge under such a flexible energy input scenario, and the bath chemistry will be perturbed by dynamic ledge behaviour in conventional cryolitic baths. Thus, observations of the bath behaviour and heat transfer rates in the cell system are necessary in order to understand the limits of long term power adjustment. In this paper an analogue laboratory experiment is proposed, to model the industrial cell system and allow prediction of the effect of large scale power shifts.

Introduction

In aluminium electrolysis, the reduction cell is designed based on the principle of heat balance. Achievement of a good energy balance sets the maximum energy efficiency since almost half of the energy exits the cell as heat dissipated to the ambient in all cell designs available. Beyond the economic consideration, a more important reason to keep heat balance is to maintain control of bath temperature, composition and ACD, in order to achieve high current efficiency. Therefore, good understanding and maintenance of heat balance is a necessity for aluminium smelting process.

Theoretically, heat balance in a smelting cell can be explained by the conservation EQ. (1) [1]:

$$(V_{decomposition} + V_{IR} + \eta_{anode} + \eta_{cathode} - V_{REACT}) \times I = \int_A h(T_b - T_i)\, dA_{ledge} + \int_A h(T_b - T_i)\, dA_{crust} + \int_A h(T_b - T_i)\, dA_{metal} \quad (1)$$

$V_{decomposition}$ is the decomposition potential for the alumina electrolysis reaction; V_{IR} is the ohmic voltage drop in the molten bath; $\boldsymbol{\eta}$ is the over-voltage at each of the anode and cathode; V_{REACT} is the voltage equivalent of the enthalpy required for the total chemical reaction, including alumina dissolution, back reaction of aluminium, heat-up of anodes and steel assembly, and minor reactions, I is the amperage in this zone of the cell. h is the local heat transfer coefficient at interfaces i between molten bath and sidewalls, top crust and aluminium, which varies widely throughout the cell; $T_b - T_i$ is the temperature difference between the bath temperature T_b and the solidified interface temperature T_i; A is the interface area.

The left-hand side of EQ. (1) is defined as surplus heat generation, *surplus* Q_{GEN}, which is available in the cell between melt/solid interfaces and the molten bath. Clearly a major portion of this surplus heat is generated in the bath just below the anodes due to the low electrical conductivity of bath, which causes 75% voltage drop [2].

The right-hand side of EQ. (1) is defined as Q_{DIST}, which is the heat distributed from molten bath and metal to the interfaces. At steady state, Q_{DIST} is equal to the heat extraction (Q_{EXT}) through the cell external boundaries to the environment. Therefore the balance between *surplus* Q_{GEN} and Q_{EXT} is critical because these two variables determine the above described heat balance.

Smelter Heat Balance Shift Technology

For a flexible operation window, the key challenge in the maintenance of heat balance is that the surplus heat generation in the cell will change substantially with amperage adjustment, requiring a matching change in the cell heat dissipation to the ambient. Global electricity and aluminium markets are driving a need to change smelter input conditions [3] to achieve different production rates and also lower energy consumption per tonne of aluminium.

In this paper a new technology is proposed to achieve flexible potline operation, and is defined as "Smelter Heat Balance Shift".This technology aims to shift each cell and the potline from one steady state to another with minimum disturbance to cell operation, as shown in EQ. (2).

State I: $Surplus\ Q^I_{GEN} = Q^I_{DIST} = Q^I_{EXT}$

Amperage Shift ⇩ ⇩ Heat Extraction Shift

State Π: $Surplus\ Q^{\Pi}_{GEN} = Q^{\Pi}_{DIST} = Q^{\Pi}_{EXT}$ (2)

When amperage is shifted, surplus heat generation will change according to the left-hand side of EQ. (1). Bath temperature and superheat will vary according to the imbalance between *surplus* Q_{GEN} and Q_{DIST}. In order to maintain the heat balance and minimize the change in bath temperature and superheat, total heat distribution needs an adjustment in a manner complimentary to the surplus heat generation change. In the new technology this change is realized by a heat extraction shift through the side wall. In this way, the overall heat balance adjustment or 'shift' can be

achieved. It is recognized that the cell sidewall lining itself will still undergo some dynamics while the new surplus Q_{GEN}/Q_{EXT} balance is re-established. These dynamics and their consequences need to be studied.

The "Smelter Heat Balance Shift" technology is proposed to meet three aluminium industry demands:

- Power modulation due to short term electricity availability/price
- Longer term (e.g. seasonal or supply driven) electricity availability in grid
- Flexible aluminium production as a response to supply and demand for metal being out of balance.

Power Modulation

Power modulation isn't a new concept in aluminium electrolysis. In response to variable electricity price over a day and/or week, or to other peaks in demand, small line amperage adjustments have been carried out in some smelters [4-6]. According to these results, only minor disturbances to cell thermal balance appeared in the short term, if proper control strategies were applied. However the build-up of side ledge and particularly bottom ridge limit efficient control after intermittent modulation in the long term [7]. So a better solution is required to make line amperage modulation more feasible in smelters, especially if more than 5% modulation of amperage is required.

Electricity Availability in the Grid

Global electricity grids are connected in networks across different areas/countries, and renewable energy forms an increasing portion of the total electricity generation, including hydroelectric power. However, some of these electricity resources have potential risks of failed supply, which can become a serious constraint to aluminium smelters on the network. For example, to combat a hydroelectricity shortage in dry seasons, a four-month power-saving project was carried out in a Norwegian smelter to save electricity for the national grid [8]. In this work, the cells suffered many operation difficulties during their continued operation at 11% reduced power input (from 135kA to 120 kA), such as increased side ledge thickness, serious alumina dissolution problems and a high frequency of anode effects. In this type of situation, if a smelter was able to control the input power flexibly without losing its cell heat balance, e.g. turn up/turn down the line amperage to satisfy the capacity of grid to supply electricity, the dilemma of limited and varying electricity grid capacity would be manageable both for the network and the smelters. In addition, electricity price fluctuation would be able to be damped, having a beneficial impact on aluminium production cost.

Flexible Aluminium Production

According to market analysis [9], the aluminium price is undergoing a depressed period at present, while only one year ago prices were much higher. To meet the unstable market demand for aluminium, some smelters consider shutting down potlines when the market suffers price depression such as at present. However the overall operating cost then becomes even worse because it takes much more time and economic efforts to restart a potline.

Smelter heat balance shift technology will combat aluminium market dynamics through providing a flexible electrolysis window for each potline technology, so that these smelters can keep operating at reduced load instead of shutting down whole potlines. Through this mechanism, aluminium production rate can be adjusted to satisfy the market demand – increasing the production when aluminium price is high and decreasing when the market is over-supplied.

Heat Extraction System

Light Metals Research Centre has developed the Shell Heat Exchanger (SHE) technology to control the heat extraction of smelting cells as required [10]. By varying the SHE coverage area on the cell and/or air flowrates through the SHE, the heat removal can be increased or decreased for cooling or insulating purposes. Internal design of the SHE gives high sensitivity of heat transfer coefficient (HTC) as flowrate is varied. In steady-state experiments [11], the technology has been demonstrated to insulate the cell efficiently at low flowrate due to the sensitivity of HTC.

In the Smelter Heat Balance Shift technology, SHE regulates heat extraction to offset the input amperage turn-up/turndown. For example, the SHE in insulation mode can compensate the lower surplus heat generation at lower amperages. The degree of turn-up and turn-down of $Q^{\Pi}{}_{DIST}$ in EQ. (2) can be increased further if the top heat dissipation of the cell is also able to be regulated through adjustment of the cell draught (suction) into the potline ducts. This has been investigated previously and has been shown to be possible as long as hooding efficiency and standards are maintained [12].

Simplification and Computation of Cell Dynamic Heat Balance

Even though many studies have been performed, it is still a challenge to predict the dynamic behaviour of operating industrial cells due to their complexity and the multiple sources of variability which are present. This will also be the case for the present heat balance shift technology where both amperage and heat extraction are proposed to be adjusted simultaneously.

Dynamic Heat Balance in a Smelting Cell

Heat balance disturbances are inevitable in a smelting cell to sustain continuous electrolysis due to batchwise operations such as anode replacement, metal tapping, alumina feeding/dissolving, and anode covering and dressing. These dynamics result in alumina-based sludge and solidified ridge, and side ledge movement causing bath composition and temperature changes. These interactions have been documented in many previous investigations and can cause the imbalances in the bath. Sensing and correction, or elimination of sustained energy imbalances are two ways to minimize these dynamics.

However it is evident that uncontrolled variation in ledge profile is still one of the active constraints in modern smelting cell operations. Cells can become almost inoperable in some cases where side ledge becomes too thin to protect the sidewall, or too thick for normal cell operation (anode change difficulty or cathode ledging and voltage instability are examples) [1].

Potential imbalances in all cells develop between the surplus Q_{GEN}, and the heat distribution from the molten bath and metal phases to the cell lining, Q_{DIST}. These lead to rapid changes in the molten bath and metal temperatures, and the bath superheat.

Imbalances between Q_{DIST} and the heat extracted from the cell boundaries, Q_{EXT} are then created and these are long lasting because of the thermal inertia of the solid cell materials and the tendency of the molten bath to freeze and remelt on the walls and into the top crust of the cell.

Formulations of these balances have been established based on the above described dynamic thermal behaviours in a smelting cell [13]. In the heat balance shift technology, these imbalances are considered in advance, and corrected using pre-planned changes in Q_{EXT} on the external walls of the cell, and at the top of the cell via the duct extraction system.

Imbalances between *Surplus* Q_{GEN} and Q_{DIST} are usually caused by small mismatches between energy input to the cell and heat leaving the liquid bath and metal. These are perturbations from the steady state. However, an imbalance due to a change in amperage I is much larger (EQ. (1)), and requires immediate correction using a heat balance shift through changed heat extraction, Q_{EXT} as in (EQ. (2)).

Q_{EXT} is the new degree of freedom enabling the heat balance shift technology through the shell heat exchangers (SHE) [11]. Consequently, Q_{EXT}, and then Q_{DIST} are tuned to match up with the new surplus Q_{GEN} in EQ. (2). Dynamic behaviour will still occur as the walls re-equilibrate and solidification/melting of the side ledge and crust occur. But these dynamics should not be sustained for a long period if the heat extraction can be matched to the surplus heat generation.

The dynamic imbalances of *Surplus* $Q_{GEN} - Q_{DIST}$ and $Q_{DIST} - Q_{EXT}$ require a deeper experimental analysis to support the heat balance shift technology.

Analogue Laboratory Experiment

Since no previous work has been done on bath temperature and composition under heat balance shift conditions, an analogue laboratory system has been designed. In this analogue, the adjustment of heat available in the bath - *surplus* Q_{GEN}, can be achieved by changing the power input to the furnace. The heat extracted from the bath, Q_{EXT}, can be varied independently using an analogue heat exchanger, allowing the heat distribution from the molten bath to be reduced through reduction in the power input (amperage in the industrial cell). Thus the heat balance shift in EQ. (2) can be achieved.

Experimental Design

The design and methodology of power control in the furnace is achieved by a Loop Command Controller, which is able to control a furnace at temperature/power modes. In temperature mode, the controller is maintaining the furnace temperature at set value by adjusting the power input to the system. In power mode, the input power to the furnace interior is constant regardless of the temperature fluctuation in the system. Thus, the power input can be adjusted at different levels to realize the power turn-up/turn down (mimicking industrial amperage I adjustment).

The heat extraction is adjusted using a carbon 'cold finger' heat exchanger immersed in molten bath (mimicking the Shell Heat Exchanger), which is redesigned based on a previous work [14]. Figure 1 shows the detailed construction sketch of the 'cold finger' heat exchanger.

The 'cold finger' heat exchanger mainly consists of two parts: Inlet – stainless tube and internal carbon cylinder (OD 22mm× H75 mm); Outlet – stainless pipe and external carbon cylinder (OD/ID 50/26 mm × H 110 mm). Nitrogen enters the inlet stainless tube and flows along the path illustrated by arrows. When nitrogen passes by the concentric annulus (2.0mm wide ×75 mm height) between internal/external carbon cylinders, heat is transferred from the bath to the nitrogen flow through the wall of the cold finger cylinder. Finally nitrogen exits the cold finger to the outlet stainless pipe. Heat extraction can be controlled by controlling nitrogen flowrates.

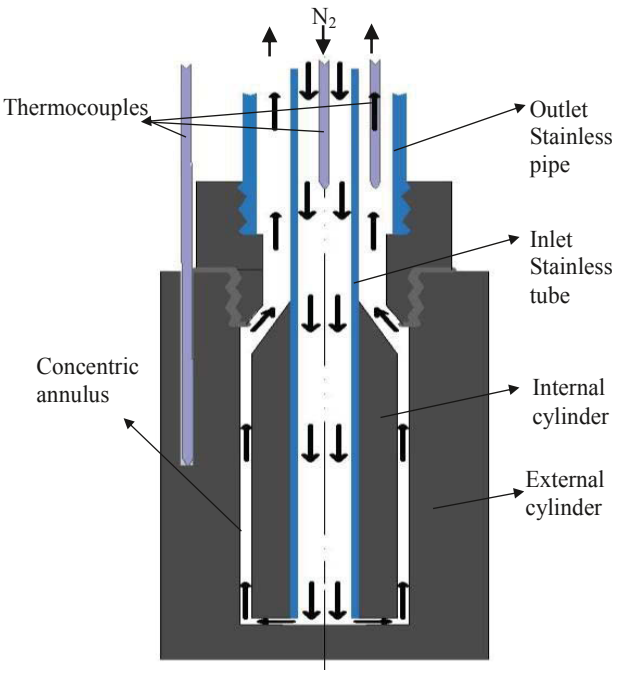
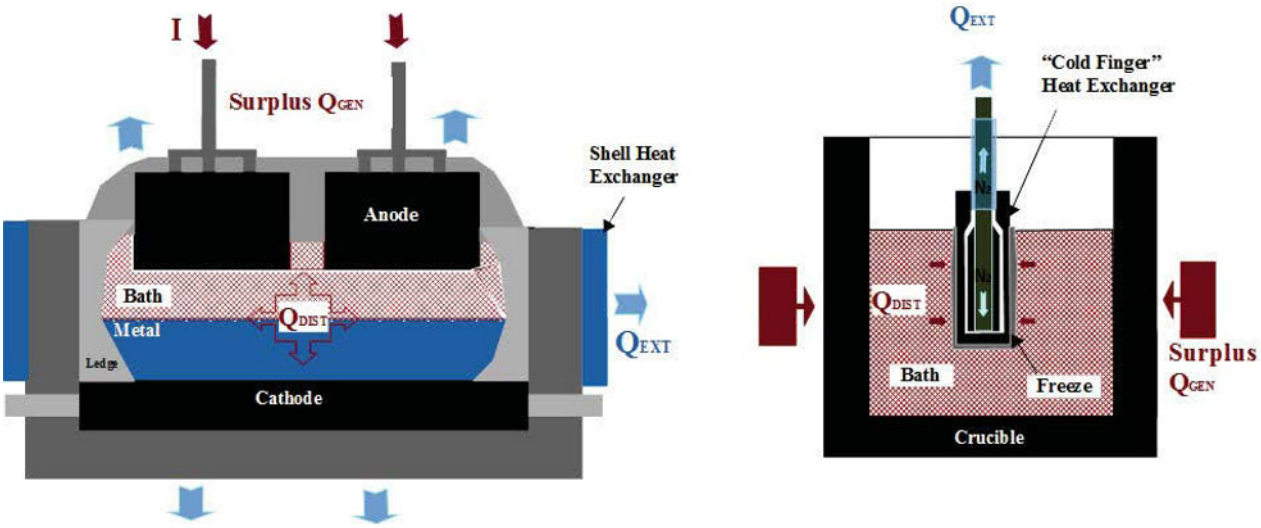


Figure 1 Construction details for 'cold finger' heat exchanger body (not to scale)

With combined adjustment of furnace power input and heat extraction, the bath freeze layer on the surface of the cold finger will change to equalize Q_{DIST} and Q_{EXT}. This is shown schematically in Figure 2(b). Thermocouples (not shown in the diagram) are used to measure the temperature in the wall of the containing crucible, in the wall of the cold finger, in the cold finger nitrogen flow inlet and outlet, and at various heights and radiuses in the molten bath allowing both the transient and steady-state conditions to be monitored. Two stirrers (not shown in the diagram) are placed on opposite sides of the cold finger to realize uniform thermal profile and bath mixing. The thickness of the freezing layer will be observed at the end of each run after a new steady state, as described in EQ. (2) is achieved. The cold finger will be removed from the molten bath to make these freeze observations.



(a) Industrial cell heat balance (b) Analogue experimental system

Figure 2 Simplification of smelter heat balance shift system to an analogue laboratory experiment.

The Simplified Heat Balance Analogue

The essential simplifications from the industrial case are illustrated in Figure 2. The key features of the analogue system in Figure 2(b) are:

i) The relationship between surplus Q_{GEN}, Q_{DIST} and Q_{EXT} can be controlled and/or monitored independently.

ii) To exclude thermal disturbances to the heat balance shift, smelter operations related to alumina feeding, electrochemical reaction, anodes, liquid metal etc are eliminated from the experiment.

Thus heat balance shift descibed in EQ. (1) and (2) can be observed with just the variation of furnace power input and cold finger heat extraction. The simplified balance in the furnace is:

$$Surplus\ Q_{GEN} = Q_{Furnace} - Q_{DISS} - Q_{EXT}$$

$$Q_{DIST} = H_{bath/freeze} \times A_{freeze} \times (T_{bath} - T_L)$$

$$Q_{EXT} = \dot{V} \cdot C_{p,N_2} \cdot (T_{outlet} - T_{inlet})$$

$Q_{Furnace}$ is the controlled and measured power supply of the furnace to the crucible; Q_{DISS} is the heat dissipating fr the crucible; Q_{EXT} is set by nitrogen flowrate $\dot{V}$, heat capacity and inlet/outlet temperatures T_{inlet}/T_{outlet} of the nitrogen, and is initially zero when no nitrogen flows through the "cold finger" heat exchanger. Q_{DIST} is the heat distributed to the cold finger.

The newly designed analogue allows the measurement/control of $Q_{Furnace} - Q_{DISS}$ through the Loop Command Control system. Before the "cold finger" heat exchanger is switched on, the furnace temperature will be maintained at a set value by the controller. Time averaged power supply then balances the rate of heat dissipated from the crucible and bath to the ambient, e.g. $Q_{Furnace} = Q_{DISS}$. Then the controller is switched to power control mode, the power input to the crucible will be kept constant at the same level as Q_{DISS}.

In the experiment, bath temperature variation will be controlled in the range +/-25 °C around 960 °C, so the differential heat radiation from the actual furnace to the environment will be very small across the range of operating conditions. Therefore Q_{DISS} is constant (assumption will be tested at different bath temperatures.)

When the "cold finger" heat exchanger is turned on by introducing nitrogen, a heat imbalance between Q_{DIST} and Q_{EXT} will be immediately set up at the freeze/melt interface on the cold finger. Bath freezing on the cold finger commences until the heat balance between Q_{DIST} and Q_{EXT} is achieved.

In the interim, the imbalance causes sensible enthalpy change of the bath in the crucible and of the cold finger, as well as latent enthalpy release as the freeze layer is formed. The dynamics of the heat balance around the cold finger interface can be expressed simply as:

$$Q_{EXT} - (Q_{DIST}(t) + \sum_i H_i(t)) = 0 \qquad (3)$$

Where H_i is the Enthalpy of system component *i*. These enthalpy changes will be closely monitored during the dynamic part of each experiment.

Heat Balance Shift in the Analogue

Once a new equilibrium point is established for $Q_{Furnace}$, Q_{DIST} and Q_{EXT} with the cold finger operating, the input power ($Q_{Furnace}$) may be varied through the Loop Command Controller. Q_{EXT} may be varied in concert with $Q_{Furnace}$. The subsequent response of the bath is a temperature/superheat change and a fraction of the bath undergoing freezing. The changed Q_{DIST} caused by this thermal perturbation will create an imbalance with Q_{EXT} at the freeze/melt interface, leading to further freezing/melting dynamic behaviours.

Therefore the heat balance in EQ. (3) will be disturbed by the input power adjustment, as shown in EQ. (4):

$$Q_{Furnace}(set) - Q_{DISS} = Q_{EXT}(set) - (Q_{DIST}(t) + \sum_i H_i(t)) \quad (4)$$

The equilibrium will be restored by adjustment of the right-hand side of EQ. (4) to match the power input change. The input furnace power, $Q_{Furnace}$ and heat extracted by cold finger, Q_{EXT} can be tuned with different set values flexibly.

For example, in the case of power turn-down, the reduced surplus heat generation (the left-hand side of EQ. (4)) will disturb the steady state. Less heat is distributed to the cold finger interface (analogue of the solidified ledge/crust boundary around the side wall/anode/cathode in an industrial cell).

More molten bath then freezes at the interface according to EQ. (3), to maintain the dynamic energy balance. In an industrial cell built-up of freeze will eventually cause operating problems, as described earlier.

According to EQ. (4), an option to restore heat balance is to introduce an equivalent reduction in Q_{EXT} by reducing the nitrogen gas flowrate. Thus a match between surplus heat generation (*surplus* Q_{GEN}) and heat extraction (Q_{EXT}), connected by Q_{DIST}, may maintain the heat balance, without massive bath freezing. The freeze layer thickness is expected eventually to be at a similar level to the base case, when a new steady state is achieved.

This adjustment of surplus Q_{GEN}, Q_{DIST} and Q_{EXT} will take some time in the actual cell due to the higher heat capacity. However, in the analogue experiment, less time will be required to shift the heat balance due to the smaller dimensions of the cell. Thus an efficient experimental detection of the bath response during the heat balance shift process in EQ. (2) may be achieved, by changing the surplus Q_{GEN} and Q_{EXT}. However the starting bath composition may also affect the dynamics and the final minimum bath temperature sustained.

Future Plan

The heat balance shift from the base state to new steady states will be detected in the experiments described here, including turn-up/turndown of the input power/heat extraction by different degrees.

Bath dynamics and heat transfer rate will be observed and compared with modeling of industrial cells [2], to assess the bath chemistry changes, and also the degree of freezing which will occur due to various heat balance shifts. The bath response in complicated bath chemistries (with Li, K, Mg fluorides) to the heat balance shift is also going to be detected.

The aim is to find the optimal strategy for managing the heat balance shift in terms of its impact on a potline of smelting cells.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SIMULATIONS ON THE BATH CHEMISTRY VARIABLES USING NEURAL NETWORKS

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Keywords: Bath Chemistry Modeling, Simulation, Single Layer Feedforward Neural Networks

Abstract

In this work we present a single layer neural network based model for bath chemistry variables in the aluminum smelting process. This model is designed to be simulated with real data as if it worked online in parallel with the process. The model is built using a very fast machine learning algorithm, the Extreme Learning Machines, which provides excellent results in regression problems in a very short time. Also we applied statistical analysis for data collection, preprocessing and filtering and for validation we performed several simulations to attest the neural model's capability to respond to new data. A comparison of this model against linear and traditional nonlinear structures is performed to show how single layer neural networks can be applied on the bath chemistry modeling.

Introduction

Modeling in the Hall-Heroult Process has been largely applied in Aluminum Smelter over the last decades [1-4]. As a powerful tool for design and optimization of aluminum electrolysis cells, cell modeling provides a number of benefits ranging from understanding process phenomenological behaviors to performing simulations without compromising real cells. Usually CFD (Computer Fluid Dynamics) models are used [5], in order to represent properly all the components of an Aluminum Reduction Cell, including the liquid bath, magnetic flux among others. Expensive modeling software solutions such as ANSYS® as well as powerful computer hardware are required to afford such modeling tasks, however, good results are obtained. On the other hand, data driven modeling can also produce good results with a much lower computational consumption [6,7].

From the fact that every industry maintains large production databases, one can consider applying these data into a model structure that is capable to reproduce the process dynamics. These models are usually built empirically but can also be developed analytically. The works from Branco [8], Frost & Karri [9], Fontes [10] and Souza [11] have proven how mathematical modeling can be useful in Aluminum Cell Optimization.

A previous version of this work [12] was based in Artificial Neural Networks, which is a Nonlinear Modeling Tool for Complex Process, whereby a Multivariable Model on the bath chemistry, a key sub-process in Aluminum Smelting, was developed. The modeling principle is fairly simple and straightforward, it needs only to select data and find a model structure capable of delivering good results. However, it has been proved that careful data selection and model design such as the choice of model structure, number of delays, etc., has a significant impact on the model's accuracy.
This work presents an extended version of the MIMO (Multiple Input Multiple Output) model presented in [12], now with a simpler Neural Architecture including new variables and applied in a simulation environment. We also performed statistical analysis to find out how good this new model is against some traditional structures such as ARX (Auto Regressive with eXogenous inputs) and NARX (Nonlinear ARX) models, as well as other complex BP (Back Propagation) neural networks architectures. To examine model's accuracy we also performed statistical analysis on the results obtained through model simulations.

Current Bath Chemistry Challenges

It is known that bath chemistry has a strong impact on current efficiency, which is a key variable that shows how productive a cell is [13]. Bath chemistry control algorithm is performed on cells in order to keep the mass balance stable and ensure alumina solubility. One great challenge is to keep temperature and aluminum fluoride stable. Typically their variations are higher than desired and there is always a need for improving or designing better control algorithms [14]. One additional difficulty regarding the measurements of bath temperature is that it is usually measured once a day or a week, and the bath composition is measured at an even lower frequency via laboratory analysis [15]. This leaves large gaps between measurements, which makes it challenging to design an effective control algorithm for bath chemistry. Furthermore, a computer model could allow investigation of some phenomena and evaluate control systems.

Simulation Environment

Figure 1 illustrates how the simulation environment is laid out.

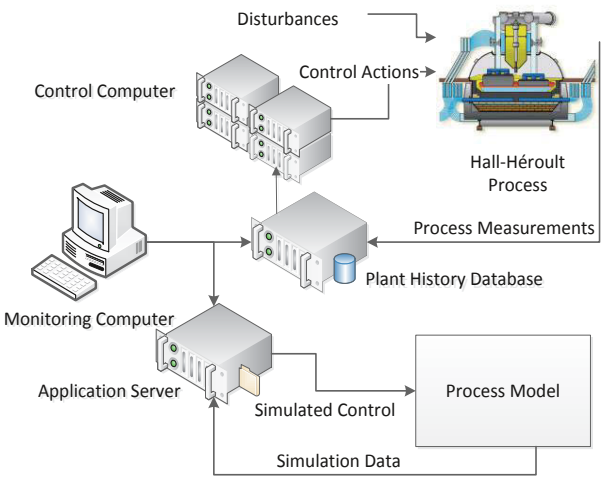

Figure 1. Simulation Environment for Hall-Héroult Process

A plant history database collects real time data from the Hall-Heroult Process, from which the computer control takes control actions. Parallel to the smelting process, another computer runs the process model built with the same data collected in the plant

database. The advantage is that the application server hosting the process model can be simulated to check or investigate any process behavior, including those which usually compromise the production. From a monitoring computer, a user could visualize at the same time the process being run and the model.

Data-Driven Models

A process model functions like a block composed of inputs, outputs and disturbances [6]:

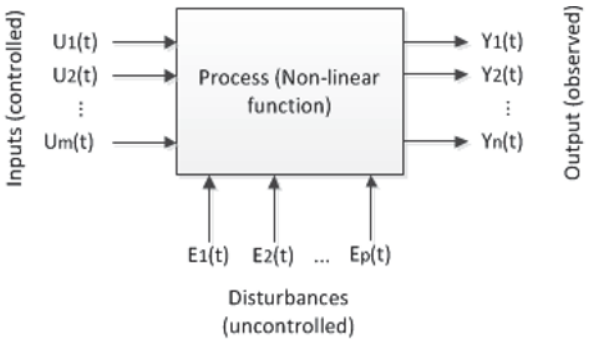

Figure 2. Representation of a model block and concerning signals

Mathematically this block can be represented by the following equation:

$$Y(t) = F(U(t), E(t), t) \quad (1)$$

Where the function F can be any function that produces the appropriate model outputs according to model inputs. This block diagram in this general form includes the disturbances. The structure of this block for industrial process is usually non-linear, however, linear representations are preferred for simplicity and their easy-to-analyze mathematical behavior.
Bringing this approach to a data-driven modeling, the collected real data from observed variables Y(t) and controlled variables U(t) can build a matrix of regressors $\vartheta(t)$:

$$\vartheta(t) = \left[Y(t) \dots Y(t - n_y) U(t) \dots U(t - n_u)\right] \quad (2)$$

The objective of the data-driven modeling is to find the coefficients of the function F by applying system identification algorithms. Considering, for example a linear ARMAX (ARX with Moving Average) model:

$$Y(t) = A_1 . Y(t-1) + \cdots + A_{n_y} . Y(t - n_y) + B_1 . U(t-1) + \cdots + B_{n_u} . U(t - n_u) + E(t) \quad (3)$$

Combining the matrix of regressors in equation 3, we can apply the Least Squares Method to find the coefficients A_i and B_i, which are matrices:

$$Y(t) = \left[A_1\ A_2 \dots A_{n_y}\ B_1\ B_2 \dots B_{n_u}\right]^T \vartheta(t-1) + E(t) \quad (4)$$

By making $\theta = \left[A_1\ A_2 \dots A_{n_y}\ B_1\ B_2 \dots B_{n_u}\right]^T$, X(t)= $\vartheta(t-1)$ and the equation 4 becomes:

$$Y(t) = X(t)\theta + E(t) \quad (5)$$

Which is solved with the Least Squares method [6]:

$$\theta = [X(t)^T X(t)]^{-1} X(t)^T Y(t) \quad (6)$$

This method is a very basic one for system identification, however it serves as a basis for understanding how data-driven models are built.

Single Layer Neural Networks

Neural Networks are well known for their ability to learn from data [16]. For nonlinear regression task, which is usually a supervised learning, artificial neural networks have proven to be excellent choice in modelling complex nonlinear processes with a good and accurate quality regarding generalization [7]. In this sense we highlight the single hidden layer neural networks (SLFNs) which have been extensively preferred for approximating nonlinear mappings, besides being simpler than multiple hidden layer networks (MLFNs) and therefore require less memory [17]. The following figure shows how a SFLN looks like:

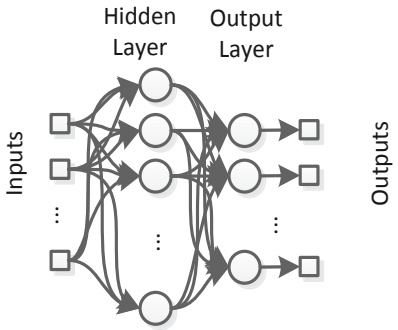

Figure 3. Schematic of a SLFN structure

Data flow from inputs to outputs passing through small processing units called neurons, hence the data are modified by neural connections, and then finally at the output, when these neural connections are properly adjusted, the desired values are produced. An SLFN can be described by the following equation:

$$Y(X) = \sum_{i=1}^{N} \beta_i g(W_i X + b_i) \quad (7)$$

Where Y is the SLFN output, X is the input, N is the number of hidden nodes, β is the output weight matrix g(x) is the activation function, W is the input weight matrix and b is the bias vector. The weight matrices W and β and the bias vector b represent the neural network knowledge and capacity to map the inputs X to the outputs Y. These matrices are found through training algorithms.

Extreme Learning Machines

Neural networks traditional training algorithms such as Back Propagation (BP) or Levenberg-Marquardt (LM) are based on a gradient descendent method, which finds an optimal solution by iterating over an objective function. LM is considered an optimization over the BP, since it is based on a Gauss Newton algorithm which speeds up the gradient descendent iterative process because it finds in the first epochs very good weights. However, for some nonlinear mappings, the training time using BP or even LM can be very long.

In pursuit of a faster learning algorithm Huang et al [18] focused on SLFNs learning by applying incremental constructive hidden layers by varying the number of hidden nodes. Moreover, Huang and Chen [17] have shown that the output layer weights could be trained in similar way as Support Vector Machines and this special training algorithm has been named Extreme Learning Machines (ELM). In comparison with BP and LM, ELM provides a significant reduction in training time, without compromising generalization. In addition, it is simpler since the parameter settings include the number of hidden nodes and activation function only. The ELM algorithm calculates the output weight matrix as follows.

$$\beta = H^{T}(HH^{T})^{-1}T \tag{8}$$

$$H_{nxN} = \begin{bmatrix} g(W_1X_1 + b_1) & \cdots & g(W_NX_1 + b_N) \\ \vdots & \ldots & \vdots \\ g(W_1X_n + b_1) & \ldots & g(W_NX_n + b_N) \end{bmatrix} \tag{9}$$

Where β is the output weight matrix, H is the hidden layer output matrix and T is the target or desired output matrix. The choice of input weights W and bias b is random, although sometimes it may cause the hidden layer output matrix to be not full column rank which can simply make the regression task impossible. This shortcoming has not occurred in the multivariable estimation addressed in this work and the results are really impressive.

Variable and Data Selection

Data selection is a crucial step in the design of data-driven development models [7]. One should pay attention to relevant variables, according to the current temperature, and other factors like sampling time. In the case of this work, bath chemistry variables are usually measured once a day, so every cycle of measurement is assumed as being the sampling time. Bath chemistry operation concentrates on monitoring mass and thermal balance, so the key variables are bath temperature, aluminum fluoride, calcium fluoride and alumina concentrations, so they were output (observed) variables.

Regarding input variables, correlation analysis has been performed in order to detect which of the original 100 variables present in the process database have influences with each of the output variables, and up to which delay point. The Pearson correlation index was applied to that end. This correlation index (between -1 and 1) indicates a measure of how much a variable influence other variable:

- A value near -1 indicates an inverse correlation (one variable goes in the opposite direction of the other);
- A value near 1 indicates direct correlation (both variables tend to go in the same direction);
- And an index near 0 indicates no correlation between the variables.

Usually absolute values higher than 0.5 denote significant influence, however one may consider in some cases a lower threshold such as 0.3. Table I shows the correlation indexes of the chosen input variables to each output and up to which delay the correlation index was greater than 0.3 for at least one variable.

Table I. Selected Variables for Bath Chemistry Model with delays

Variable	Unit	Corr. w/ TMP	Corr. w/ ALF	Corr w/ CAF	Corr w/ ALM	Delays
Cell Voltage	V	-0.477	0.44	-0.151	0.19	2
Cell Resistance	μΩ	-0.52	0.541	-0.19	0.08	1
Metal Level	cm	-0.19	0.23	0.021	0.34	1
Alumina Feed	Kg	-0.677	0.412	0.049	0.51	3
Bath Temperature (TMP)	ºC	1	-0.88	0.17	0.21	2
Aluminium Fluoride percentage (ALF)	%wt	-0.621	1	-0.209	-0.14	2
Calcium Fluorite percentage (CAF)	%wt	0.21	-0.233	1	-0.07	2
Alumina Percentage (ALM)	%wt	0.13	-0.26	-0.03	1	3
Aluminium Fluoride Feed	Kg	0.589	-0.522	0.197	-0.18	3
Calcium Fluoride Feed	Kg	-0.235	0.192	-0.534	-0.02	2

Finally the structure of the model considered 10 variables (including output variables fed back), and we initially applied 3 delays for each variable, resulting in 30 inputs.

The data was collected from an aluminum industry database over a one-year period. After applying 3-sigma rules [7], total database has been reduced from 23118 to 9205 records.

Model Development

Having collected data, one should choose a structure to build the model. There is no rule of thumb that might determine how the parameters could be structured, for example, using SLFNs one may vary the number of hidden nodes, the activation functions among other parameters. In order to have better results, one can apply multiple experiments and select the structure that presented the best result. A good metric to use is the Mean Square Error (MSE):

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{P} e_k(i)^2 \tag{10}$$

where *N* is the size of the dataset, *P* is the number of the model's output variables, $e_k(i)$ is the error between the predicted output and the actual value for the variable *k* at the instant *i*. Acceptable MSE errors for each of the output variables are listed in table II.

Table II. MSE Criteria for each model variable

Variable	MSE Error
Alumina Concentration	0,5%
Aluminum Fluoride	1,5%
Bath Temperature	5ºC
Calcium Fluoride	1%

Using the fast training algorithm Extreme Learning Machines for SFLNs, it is possible to perform multiple experiments without consuming too much time or computing resources. The methodology for building the model was defined in the following way:

1. Define the errors listed in the table II for each variable as the satisfaction criteria;
2. Define a number of SFLNs with activation functions sigmoid and hyperbolic tangent and varying the number of hidden units from 10 until 2000;
3. Given the random weights initialization, each neural network created was run at least 100 times, in order to ensure the weights to cover a wide area in the search hyperspace;
4. Get the best MSE errors of each network;
5. If no SFLN achieved the satisfaction criteria, go back to step 3 and increase the number of hidden units; otherwise select the SFLN that provided the best MSE and stop the experiment.

This experiment has been carried out on a common Dell® laptop with Windows 10 operating system, 6 GB RAM and 2.8 4 cores Intel Core i5 processor. The algorithm was implemented using Matlab ® language, but using the free environment Octave.
After training, the following configuration of the SFLN was found:

Table III. MSE training dataset results

Parameter	**Value**
Hidden Neurons	213
Activation Function	Sigmoid
Normalized overall MSE	0,003191
MSE for ALF	1,55 %
MSE for TMP	7,13 ºC
MSE for CAF	0,23 %
MSE for ALM	0,89 %

From the total number of records, 20% were taken out for testing the model accuracy, the following table shows the MSE performance for test datasets.

Table IV. MSE training dataset results

Parameter	**Value**
Normalized overall MSE	0,003258
MSE for ALF	1,62 %
MSE for TMP	6,71 ºC
MSE for CAF	0,28 %
MSE for ALM	0,89 %

Figures 4 to 7 shows a comparison of the model results versus test results.

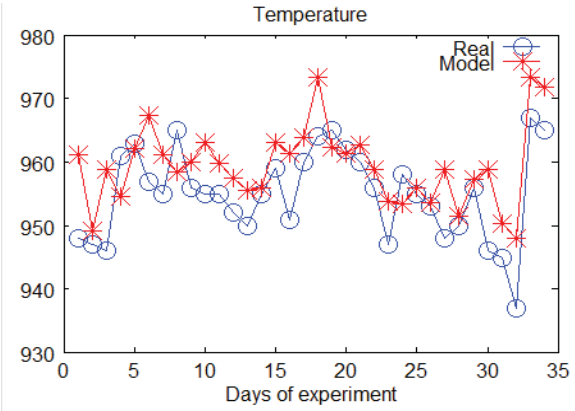

Figure 4. Time series plot for Bath Temperature

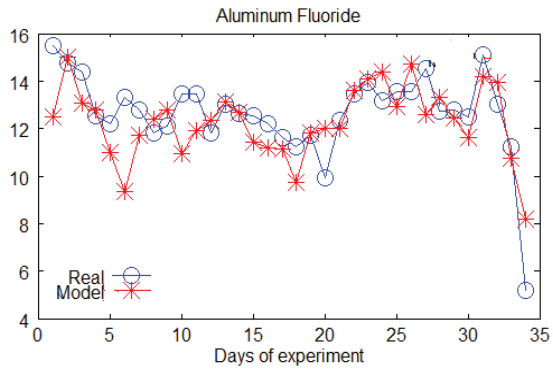

Figure 5. Time series plot for Aluminum Fluoride

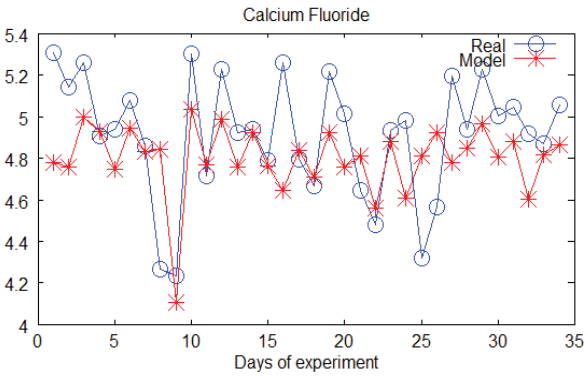

Figure 6. Time series plot for Calcium Fluoride

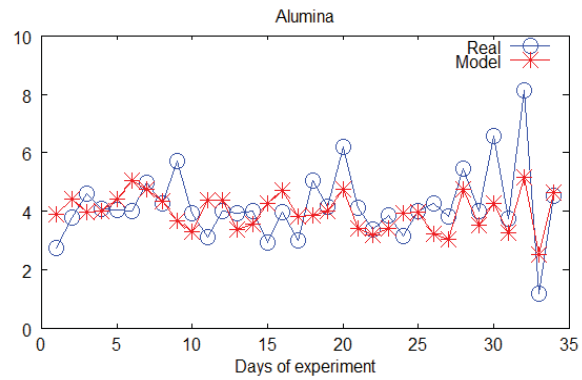

Figure 7. Time series plot for Calcium Fluoride

Comparison with Traditional Models

As an alternative to MSE, we also considered the error histogram and statistical measurements such as kurtosis, skewness and standard deviation and compared the performance of the model against other traditional models. The standard deviation means how "regular" is the error, for example a lower value of standard deviation denotes a smaller average error. The kurtosis shows how concentrated the errors are in the histogram, for example a higher value of kurtosis means a lower variability of the error. It is desirable for the kurtosis measurement to be high, so the predictions of the model would be more reliable. Finally, the skewness shows how asymmetric is an error histogram. Values at both extremes (positive or negative) mean that the error probability density has a longer tail, the error is more concentrated on one specific region, and thus it is desirable for the skewness to be closer to 0. The error histograms for all variables and models are shown in figure 8 and table V shows the statistical measurements for all models built considering the variable Temperature, which shows greater values.

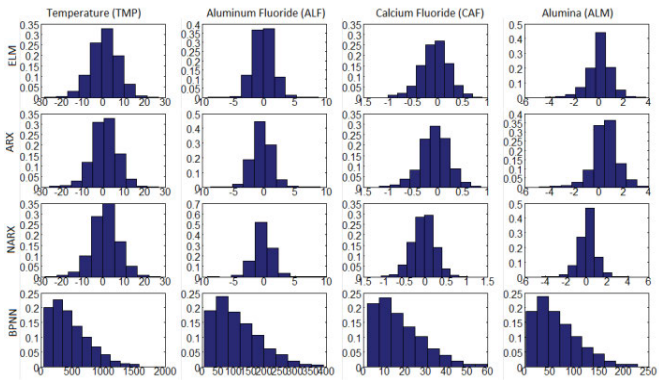

Figure 8. Error Histograms for all variables (Temperature – TMP, Aluminum Fluoride – ALF, Calcium Fluoride – CAF, Alumina – ALM) and the models studied (ELM – Extreme Learning Machines, ARX – Autoregressive with eXogenous inputs, NARX – Polynomial NARX, BPNN – Backpropagation Neural Network)

Table V. Statistical Analysis of the errors in the Temperature

Model Structure	Extreme Learning Machines (ELM)	BP Neural Network (BPNN)	Linear ARX (ARX)	Nonlinear Polynomial NARX (Poly2)
Standard Deviation	6.6902	304.398	6.2981	6.57212
Kurtosis	1.5458	0.52325	1.2055	1.1541
Skewness	-0.14391	0.96372	-0.2918	-0.18371

Closed Loop Simulations

Finally, to examine the model's ability to work on a production environment, we performed a number of tests. In the first case we simulated bath temperature control with high aluminum fluoride additions. A given pot will receive much aluminum fluoride than needed. This pot's life is about 4.5 years, and after 10 control cycles we simulated an extra addition of 40 kg of aluminum fluoride, to check its response.

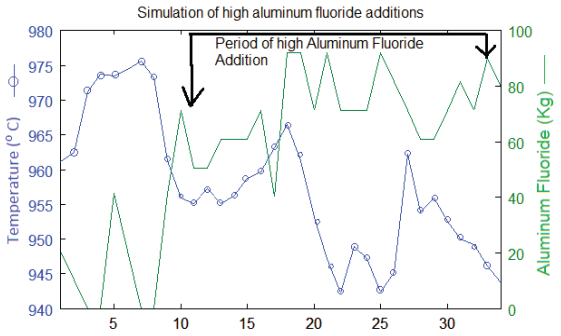

Figure 9. Simulation of high aluminum fluoride additions

From the 10th point in the chart, the temperature started to drop, until considerably fall below 945 ºC after receiving a high amount of Aluminum Fluoride more than 90 kg, or 300 kg after 4 cycles.
Secondly we induced a low aluminum fluoride addition from a given point in time and checked how a virtual pot would react to it regarding aluminum fluoride variable. This virtual pot's life is around 5 years, and from the 10th control cycle we simulated the situation as if the pot received only one third of fluoride.

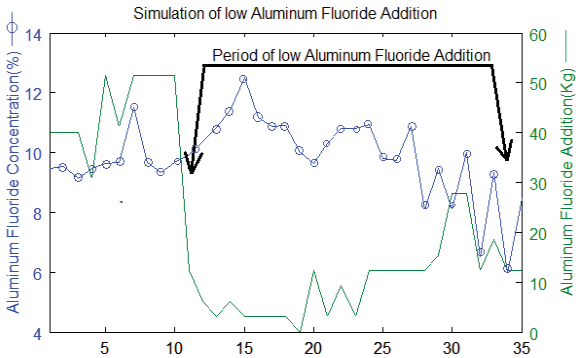

Figure 10. Simulation of low aluminum fluoride addition.

When the period of low fluoride addition starts, the fluoride concentration continues to grow because of high additions that occurred before (more than 200 kg after 5 cycles), but it started to drop gradually after a few cycles.
Finally we put inline a fuzzy control system, previously developed by Pereira [19] to control the virtual pot. Given the situation shown in table VI, the pot's control was passed to the fuzzy system. The two main control variables are shown in the time series chart of Figure 11.

Table VI. Situation on a given pot before the application of fuzzy control

Variable	Value
Pot life	758 days
Temperature	954 °C
Aluminum Fluoride %	13.7 %
Temperature Variation	+1 °C
Aluminum Fluoride Variation	-0.54 %
Cumulative Aluminum Fluoride	31,2 kg
Action that fuzzy control would take	**Add 18.5 kg**
Action that was taken	**Add 10.4 kg**

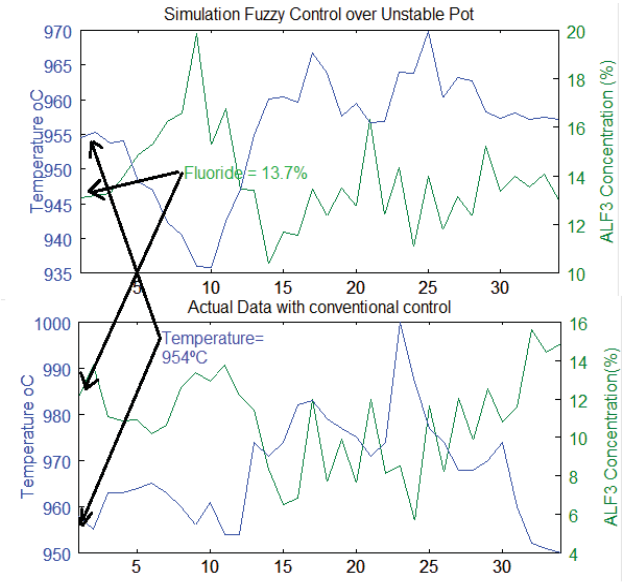

Figure 11. Comparison between the fuzzy control simulation (upper plot) and the actual data under conventional control (lower plot).

According to the simulation, although a fuzzy control of the virtual plot would initially drive temperature below 940 degrees, the pot would be stabilizing after a few control cycles, whereas the conventional control kept the pot out of control for a few cycles, but eventually caused the fluoride concentration to drop and the temperature to rise.

Conclusions

In this work we examined the power of data-driven modeling for the Hall-Heroult process, using simple models and tools that could run on cheap desktop computers. The results show good accuracy and appear to be in conformity with corresponding literature; however some adjustments may still be required. A model would be a suitable tool that might be useful for many tasks, such as testing control algorithms before putting it into production or investigating some unusual event. It simple structure allows easy implantation on an industrial environment, where it can actually work as a virtual cell laboratory. Another impressive point is that the time required for building such models is very little, since it combines purely linear algebra with artificial intelligence techniques, which drives faster convergence. This allows also easy re-adaptation of the models to a new environment.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

TECHNOLOGY RESEARCH ON DECREASING THE ALUMINUM SURFACE WAVES AND REDUCING THE CATHODE VOLTAGE DROP IN ALUMINUM ELECTROLYSIS CELL

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Keywords: Aluminium electrolysis, Aluminum surface waves, Cathode voltage drop, Horizontal current, Higher conductivity cathode steel bar.

Abstract

The influence of the materials of cathode carbon block and cathode steel bar, cathode steel bar size and steel bar assembly way on the horizontal current of aluminum liquid and cathode voltage drop was studied using simulation calculation and industrial cell experiment. It is aimed to obtain the technical parameters for decreasing the longitudinal fluctuation of aluminum liquid and reducing the cathode voltage drop, increase the stability of electrolytic cell and realize the cells operated at low voltage and high efficiency. The results of simulation study and industrial cell tests show that it can effectively reduce the liquid aluminum surface waves and the cathode voltage drop through comprehensive optimization of cathode carbon block and steel bar materials, reasonable design of cathode steel bar size and optimizing steel bar assembly ways. The results of 350 kA test cell indicate that the cell voltages are about 3.82-3.85 mV at the current density of 0.73 A/cm^2 and the cell voltages fluctuation range is about 6-8 mV. Starting after 180 days, the cathode voltage drop is about 190-220 mV. It realized high current efficiency of production under the condition of low voltage, and reduced the energy consumption for aluminum electrolysis production.

Introduction

For more than 100 years, the Hall-Héroult process is the main industrial process for the production of primary aluminum. A direct current is applied and passes through the anode, bath, molten aluminum pad, cathode blocks and flows horizontally through cathode steel bars to the risers to the next cell. In this process, the interaction of magnetic field with the current produces an electric magnetic field which causes the fluctuation of the molten fluid. This will increase the cell voltage and the energy consumption. Nowadays the capacity of the cell has been extended from 300 to 600 kA, stable and efficient operation of the electrolysis process has become more and more important topics [1-4]. Especially for the aluminum electrolytic cells in China, where most of them are operating with lower ACD, the need to improve the metal/bath interface stability become more pressing [5-7].

The metal pad waves have been studied by several authors [8-11]. Feng et al. [12] invented a new type of cathode structure for aluminum reduction cells to reduce the velocity of liquid aluminum and weaken the surface oscillation of liquid aluminum. Das et al. [13] studied the inclined cathode to reduce the horizontal component of the current density. Zhou et al. [14] studied a new collector bar design to reduce the horizontal current in the metal pad.

In this work, a numerical model based on the multiphase flow fluid feature is built to simulate and calculate the influence of the materials of both cathode carbon block and cathode steel bar, cathode steel bar size and steel bar assembly design on the horizontal currents in the metal pad and cathode voltage drop. This paper provides a new technology for the large series of electrolytic cells that are operated at low voltage and high efficiency.

Mathematical Model and Experimental Setup

A finite element model of half a cathode block was built to simulate the current distribution for various cases. The pertinent details of the model can be found in our early work [2]. All the simulations are based on a 350 kA cell which includes carbon block, collector bars and binder materials. Based on a base case model, other cases were used to study the impact of other factors, such as cathode block materials and steel bar design and so on.

In one of the smelters in China, operating at 350 kA, ten test cells were selected to demonstrate the results of the new designs. The new designs developed in this study are called “new type steady flow and energy saving technology”. Five of 350 kA traditional cells were selected as reference cells to compare the performances. The aluminum cell test for energy balance and its calculation method was based on the Industry Standards of the People’s Republic of China [15]. The test results used in this study are the average values of 180 days of operation after starting the test.

Results and Discussion

Decreasing the Aluminum Surface Waves

Figure 1 shows the vector plot of current distribution in the molten aluminum in the 350 kA cell. The current flows out of the cell through the cell sides. The electrical resistance is small in the path of current flows near the end of cathode steel bar. The current automatically chooses the path with the least resistance where it obliquely flows into the molten aluminum and carbon cathode. As a result, it generates a horizontal currents component along the short axis of the cell. The electromagnetic force generated by the interaction of the vertical magnetic field with the horizontal currents causes metal circulation which indirectly induce waves.

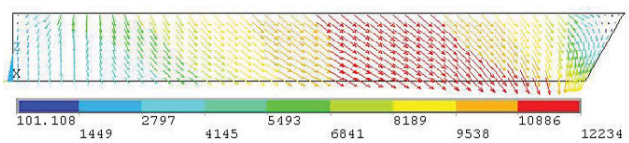

Figure 1. Vector plot of current distribution in the molten aluminum (A/m^2).

Figure 2 shows the schematic diagram of molten metal waves (h) and its effects on the effective ACD of the cell during cell operation. If the effective ACD ($L=L_0-0.5h$) is maximized (i.e., h=0), it can greatly reduce the cell voltage drop by lowering the ACD.

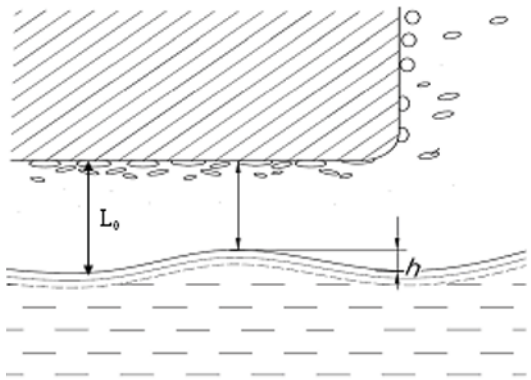

Figure 2. Schematic diagram of the fluctuation of molten melt effects on the effective ACD.

The resistivity of cathode block plays a major role on the level of horizontal currents hence impacting the molten aluminum waves amplitude. In addition, the cathode block resistivity is directly related to the cathode voltage drop. Figure 3 shows the simulated results of horizontal currents for cathode blocks with different electrical resistivities.

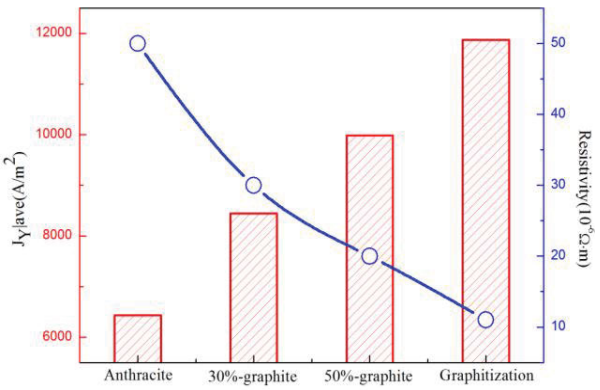

Figure 3. Horizontal current density vs. cathode block resistivity.

As shown in Figure 3, the horizontal current density is about 11867 A/m^2 when the resistivity is 1.1×10^{-5} Ω·m for graphitized cathode block. On the other hand, the horizontal current density is about 6432 A/m^2 when the resistivity is 5.0×10^{-5} Ω·m for anthracitic cathode block. It means that lower electrical resistivity leads to higher horizontal current density hence higher molten metal waves. The higher horizontal current density will cause the faster flow velocity of molten metal which may lead the cathode blocks to be eroded more rapidly [16]. When designing the cell, and considering the cell stability, one should choose the optimum electrical resistivity of the cathode carbon block that will avoid very high horizontal current density but still reduce cathode voltage drop.

Figure 4 shows the horizontal current density vs. cathode steel bar resistivity. It demonstrates that the horizontal current density is greatly increased when increasing the resistivity of cathode steel bar. This means that an appropriate horizontal current density could be obtained by the optimization of steel bar resistivity. For low resistivity steel bar (S4) of the latest development and application, the resistivity decreased by more than 40% at the room temperature and by about 30%, at 600°C (from 0.73×10^{-5} Ω·m to 0.52×10^{-5} Ω·m,) when compared with the traditional steel bar material (S1). Application results show that the horizontal current density could reduce by 28% for the cell with the new steel bars (S4).

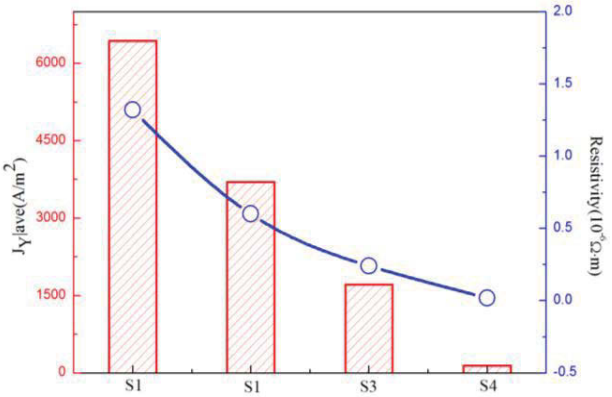

Figure 4. Horizontal current density vs. cathode steel bar resistivity.

Table I presents the simulated horizontal current densities of various cathode steel bar cross sections. The horizontal current density decreases with higher cross sectional area of the cathode steel bar (from 6607 A/m^2 for 160 mm high × 50 mm wide, to 4050 A/m^2 for 200 mm high × 100 mm wide). Increasing the cross sectional area of steel bar is a simple and effective way to reduce the horizontal current density and cathode voltage drop.

Table I. Results of horizontal current densities of various cathode steel bars sizes.

No.	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
H× W (mm)	160× 50	200× 50	240× 50	200× 65	200× 80	200× 90	200× 100
$J_Y\|max$ (A/m^2)	10805	10328	10189	9800	8600	8040	7380
$J_Y\|ave$ (A/m^2)	6607	6322	6236	5440	4710	4360	4050

Based on the results in Figure 4 and Table I, the optimum case would be to use the low resistivity cathode steel bar material with highest cross sectional area.

Another important parameter is the cathode block / steel bar connection method. It has a great influence on the horizontal current density. The phosphorus pig iron casting and a paste binder are the current methods which are widely applied in the smelters in China. Compared with the paste binder, the phosphorus pig iron casting is a relatively better choice at present although it may introduce higher horizontal current densities.

From the simulation results above, it is shown that the way of reducing the horizontal current and getting a more even current distribution include an appropriate adjustment of the cathode resistance, the cathode steel bar resistance and the cathode / steel bar connection method. The vector plot of current distribution for the optimized case is showed in Figure 5. Comparing Figure 1 and

Figure 5, it is clear that the direction of the current distribution vector is more vertically downwards in the optimized case.

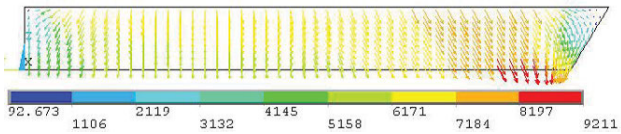

Figure 5. Vector plot of current distribution for the optimized case (A/m^2).

Reducing the Cathode Voltage Drop

The cathode voltage drop comprises three parts: the voltage drop of cathode block, the voltage drop of cathode block / steel bar connection and the voltage drop of steel bar. Each part account for about a third. Over the cell life, the cathode voltage drop will be increased primarly caused by the deformations and/or corrosions of the steel bar for the cells.

Figure 6 shows the cathode voltage drop vs. cathode blocks resistivity. It shows that the cathode voltage drop is increased when increasing cathode blocks resistivity. The resistivity of graphitized cathode block is the smallest (1.1×10^{-5} Ω·m) and the cathode voltage drop is also the lowest, about 215 mV. Following comprehensive study of the effects of lower resistivity cathode blocks on the horizontal current density, it is found that very low resistivity cathode block may not be favorable for achieving the goal of lower energy saving. Based on the current research results, that the optimum cathode block electrical resistivity would be around 2.5×10^{-5} Ω·m, which is equivalent to a cathode voltage drop of less than 300 mV.

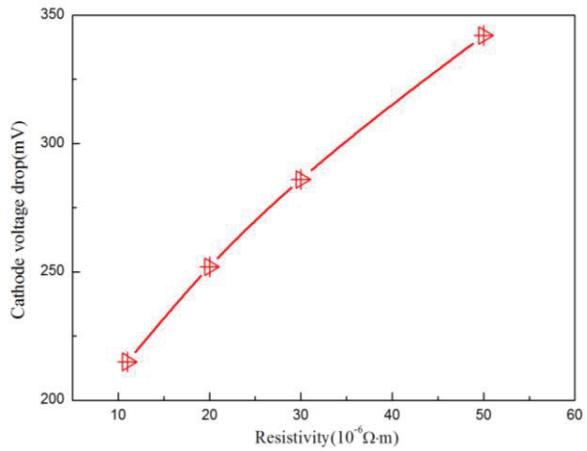

Figure 6. Cathode voltage drop vs. cathode blocks resistivity.

Figure 7 shows the cathode voltage drop vs. cathode steel bar resistivity. The cathode voltage drop increased when increasing cathode steel bar resistivity. However, experience and plant data indicates that the cathode voltage drop could be very low level when using low resistivity cathode steel bar during early life of the cell. However, the cathode voltage drop is observed to increase over the cell life. This could be due to the fact that the steel bar is getting attacked by the carbon material or the melt which result in decreasing the electrical conductivity [17].

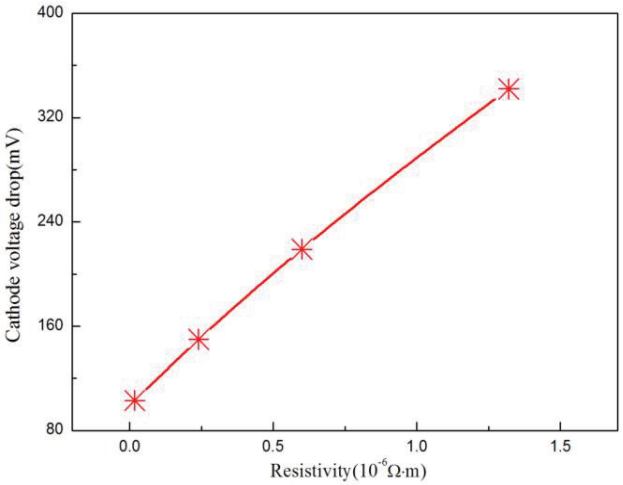

Figure 7. Cathode voltage drop vs. cathode steel bar resistivity.

Furthermore, the effects of cathode steel bar cross sectional area were also simulated. The results indicate that increasing the cross sectional area of the steel bar results in decreasing the cathode voltage drop. Considering minimizing horizontal current density, it is recommended to increase the steel bar width rather than the height.

The phosphorus pig iron casting will cause lower cathode voltage drop. The main reason is the volume of the high phosphorus cast iron expands when its temperature increases during the process [18]. Compared with the paste connection, the phosphorus pig iron connection causes the cathode voltage drop to be reduced by about 40-50 mV.

Results of testing the new technology

Based on the simulation results and overall consideration of various factors, we optimized the selection of materials of cathode block and cathode steel bar, optimized the cathode steel bar cross section and optimized the method of cathode block / steel bar connection. The objective is to even out the current distribution and reduce the cathode voltage drop. The new technology is called "New type steady flow and energy saving cell". The optimized solutions are shown in Table II.

Table II. Optimized solutions for 350 kA pilot cells.

No.	Cathode block	Cathode steel bar	Cathode steel bar size (mm)	Connection method
Case 1	30% graphite	Traditional	180×70	Paste
Case 2	50% graphite	Low resistivity	200×90	Paste
Case 3	50% graphite	Low resistivity	200×90	High phosphorus pig iron casting

The optimized solutions were applied on 350 kA pilot cells in a smelter in China, with nominal current density of 0.73 A/cm^2. Five cells were selected to implement the solution of each case, where case 1 is base case representing the common cathode design in the smelter.

The results of testing the new technology are shown in Table III (data are averaged for a period of 180 days of operation).

As shown in Table III, the cell voltage is 3890 mV and 3838 mV and the fluctuation range is about 6-8 mV for the cells using the new technology. Compared with the results of base case, the cathode voltage drop is reduced from 318 mV to about 190-220 mV and the current efficiency is increased from 91.4% to about 92.9%. The lower flucuations of the molten aluminum waves allowed the cell to be operated at a lower ACD (4.3-4.4 cm) hence decrease the interpolar voltage drop for the cell using the new technology (cases 2 & 3). The power consumption for cases 2 & 3 are 12478 kWh/t-Al and 12325 kWh/tAl recpectively, which is lower by 781 kWh/tAl and 934 kWh/tAl respectively compared to the base case 1 (13259 kWh/tAl).

Table III. Results of 350 kA pilot cells.

Cell type	Case1	Case2	Case3
Thickness of side ledge (cm)	8.1	15	17.5
ACD(cm)	4.6	4.4	4.3
Liquid aluminum height(cm)	29.5	23.5	23
Melt height(cm)	19.5	19	18.5
Cathode voltage drop (mV)	318	250-270	190-220
Interpolar voltage drop(mV)	3154.9	3009.4	3014.4
Cell voltage (mV)	4066.7	3890.0	3838
Cell voltage fluctuation range(mV)	12-15	6-8	6-8
Current efficiency	91.4%	92.9%	92.8%
DC(kWh/t-Al)	13259	12478	12325

Conclusions

(1) The influence materials selection for cathode carbon block and cathode steel bar, cathode steel bar cross section and cathode block / steel bar connection method on the horizontal current density in the molten metal and cathode voltage drop was studied using finite element simulation and testing in industrial cells.
(2) Based on the simulation results and overall consideration of various factors, the optimum solutions for decreasing the aluminum surface waves and reducing the cathode voltage drop were identified.
(3) Based on the simulation results of the horizontal current density and the cathode voltage drop, the optimized solutions were applied on 350 kA pilot cells. The results of the test cells confirmed the improved performance using the optimized solutions.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

HALL-HEROULT CELL SIMULATOR: A TOOL FOR THE OPERATION AND PROCESS CONTROL

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Keywords: Cell simulator, Process control, Process optimization, Energy saving

Abstract

There is a wide range of complex chemical and physical phenomena in the Hall-Héroult process that can be described by mathematical tools. Using the MatLab-Simulink software, a mathematical model has been developed to solve the dynamic status of an aluminium reduction cell. Simulink provides a very powerful graphical user interface for building sub-models as block diagrams. The cell simulator determines various process interactions. The key operating parameters such as alumina concentration, bath temperature, ledge thickness, cell voltage and many others are computed as a function of time. Raw materials and process variations effects are predicted. The model may aid at improving operating strategies that can be implemented in the process control. Applications such as the variation of ledge thickness, specific energy, bath level, AlF_3 emissions are presented.

Introduction

Despite the fact that the Hall-Heroult process is based on a simple theory, there is a wide range of complex physical-chemical phenomena that must be taken into consideration when modeling the overall process. Several dynamic cell simulators have been published with different objectives from increasing the understanding of the process to development and analysis of process control [10 – 15]. The objective of this work is to develop a mathematical model of a Hall-Heroult reduction cell as well as develop a better understanding of the reduction process itself. The effects of variations in operational parameters on the dynamical behavior of an aluminum reduction cell are predicted. Various coupling exists between the complex physiochemical phenomena. Compromises have to be made between simplicity and accuracy of the model. The most relevant aspects of the Hall-Heroult process are preserved. The cell simulator is structured in three main modules:

- a material balance model
- a cell voltage model
- a thermal balance model

The three modules are then combined into one overall cell model which can be used for improving control strategies, cell operation as well as developing a predictive tool for the process itself. The model is not limited in the period of prediction. Operations such as anode change, tapping, alumina and aluminum fluoride feeding can be defined easily by the user. A period of 26 days can be simulated in less than 15 minutes. The simulation of one hour operation needs about 1 second computer time. The model can therefore be used online to help at process control strategies.

The cell simulator is based on MatLab-Simulink software. It integrates:

- computation visualization programming of problems which are expressed in familiar mathematical notation
- a graphical user interface (GUI) for building models as block diagrams, allowing you to draw models as you would with pencil and paper.
- a tool for simulating dynamics systems
- block diagram windows, in which models are created and edited by mouse driven command.
- interactive graphical environment simplifying the modeling process, eliminating the need to formulate differential equations
- open modeling system to add new algorithms such as smoothing of cell resistance or anode effect detection

Aluminum Production Rate

The mass conservation law for a system with chemical reaction is:

$$\text{Input} + \text{Generation} = \text{Output} + \text{Accumulation} \quad (1)$$

The aluminium production is given by Faraday's law [15]:

$$m = \frac{1}{eN_A} \cdot \frac{QM}{n} = \frac{1}{F} \cdot \frac{QM}{n} \quad (2)$$

m: Mass of the substance produced at the electrode [g]
M: Molar mass of the substance [g/mol]
Q: Electric charge passed through the solution [C]
n: Number of free electrons per atom of the substance
e: The elementary charge carried by a single electron
N_A: Avogadro'c constant [atoms/mol]
F: Faraday' constant [A s/mol]

Taking the current efficiency into account, the production rate can be expressed as follows [15]:

$$\frac{dm_{Al}}{dt} = 9.322 \cdot 10^{-8} \cdot I \cdot \eta \quad (3)$$

Where I is the cell current in amperes and η the current efficiency.

Carbon Consumption Rate

Pearson and Waddington assumed that the loss in current effciency is only due to the internal recombinaison of aluminum and carbon dioxide therefore [4]:

$$2\eta Al_2O_3 + 3C = 4\eta Al + 3(2\eta\text{-}1)\ CO2 + 6(1\text{-}\eta)\ CO \quad (4)$$

From the rate of aluminum produced and the electrolysis equation, the theoretical carbon consumption rate can be expressed as [15]:

$$\frac{dm_C}{dt} = 3.11213 \cdot 10^{-8} \cdot I \quad (5)$$

Correspondently, the production rate of carbon monoxide and carbon dioxide can be expressed by [15]:

$$\frac{dm_{CO}}{dt} = 1.45154 \cdot 10^{-7} \cdot I \cdot (1 - \eta) \quad (6)$$

$$\frac{dm_{CO_2}}{dt} = 1.140324 \cdot 10^{-7} \cdot I \cdot (2\eta - 1) \quad (7)$$

There are various additional loss mechanisms that contribute to the overall consumption, such as air burning and detachment of anode carbon particles from the blocks. The anode consumption rate is thus considerably higher in practice than expressed by equation (5).

Alumina Consumption Rate

The alumina consumption is governed by the principal electrolysis equations. As for carbon, the alumina consumption can be expressed from the rate of aluminum produced and the modified electrolysis equations [15]:

$$\frac{dm_{Al_2O_3}}{dt} = 1.76126 \cdot 10^{-7} \cdot I \cdot \eta \quad (8)$$

The theoretical consumption is about 1.89 kg of pure alumina per kilogram of aluminum produced. In practice, the alumina is not pure, so the total amount of alumina that is required for the reduction process is thus higher than the theoretical consumption. Up to 1% of the alumina mass is natural impurities and about 2-3% of the mass is moisture. The main impurity is sodium oxide, while calcium, silicon, iron and titanium oxides are also present. Furthermore, when using dry scrubbers for removing fluorides from the effluent pot gases and bag filters for the particulates, the alumina is used as a gas absorbent before it is fed to the cell. A part of the feed material is thus recycled fluoride which exists in the alumina bulk material mainly as hydrogen fluoride. When using dry scrubbers, the fluoride content can range from 1.0 to 1.5% of the mass, depending on the absorbability of alumina used. This means that only about 94-95% of the raw feed material is in fact pure alumina so the actual bulk consumption of alumina can be 5-6% higher than expressed by the above equation.

Cell Emissions

For a cell with prebaked anodes and normalized to 1 metric ton of aluminum, the value is around 20 kg of hydrogen fluoride (HF), 20 kg of particulate fluorides (i.e., in $NaAlF_4$) and 10 kg of sulfur dioxide (SO_2).

There are a number of other chemical species that make up the emissive gases, such as the greenhouse gases tetrafluoromethane (CF_4) and hexafluoroethane (C_2F6) as well as the other constituents of the particulate material. Since the mass ratio of these chemicals to the total amount of emissive substances is very small compared to HF, $NaAlF_4$ and SO_2, they will be neglected. The fine particulate fluorides and gaseous hydrogen fluoride are of greatest concern in relation to the material balance, since fluorides are lost from the electrolyte. These are present in the anode gases as a result of:

- Gaseous fluoride evolution
- Particulate emission
- Entrainment of particulate materials

The subsequent sections concentrate on these.

Hydrogen Fluoride Evolution

The sources of the hydrogen H can be from the anode carbon or from moisture present in the air or in the alumina feed. Part of the hydrogen reacts with cryolite to produce hydrogen fluoride according to equation [4]:

$$2Na_3AlF_6\ (l) + 3H_2\ (g) \rightarrow 2Al\ (l) + 6NaF\ (l) + 6HF\ (g) \quad (9)$$

The most favored reaction that water may undergo with the electrolyte constituents is that it may react with aluminum fluoride either in the electrolyte or in the emissions to produce hydrogen fluoride. The reaction of the dissolved water with the electrolyte may be described in simplified form by [4]:

$$2AlF_3\ (s) + 3H_2O\ (g) \rightarrow Al_2O_3\ (s) + 6HF\ (g) \quad (10)$$

According to Haupin [1], the total rate of the gaseous fluoride evolution in kilograms per metric ton of aluminum, can be estimated with the equation:

$$F_G = 1.4537 \cdot 10^7 \cdot e^{(0.78127R_b^2 - 3.1733 \cdot R_b - 8444/T_b)} \cdot \left(\frac{469 - 191 \cdot R_b}{P_b \cdot \%CE}\right) \cdot \left(\frac{C_{H_2O}}{37.44} + \frac{C_H}{21.5}\right)^{0.5} \cdot \left(\frac{C_{Al_2O_3}}{C_{Al_2O_3}^{max}}\right)^{-0.462} \quad (11)$$

Rb : Bath ratio (C_{NaF} /C_{AlF_3})
Pb : Barometric pressure [kPa]
%CE : Current efficiency [%]
Tb : Bath temperature [°K]
C_{Al2O3} : Concentration of alumina [wt %]
C^{max}_{Al2O3}: Maximum solubility of alumina [wt %]
CH_2O : Conc. of H_2O in alumina feed mat. [wt %]
CH : Conc. of hydrogen in the anode [wt %]

Haupin suggests that the average concentration of moisture in the alumina feed material is approximately 2.8% and the concentration of hydrogen in the carbon anode 0.093%.

The production of electrolyte constituents due to cell fluoride emissions can then be approximated as follows in kilograms per metric ton of aluminum produced:

$$\Delta mAl = 0.011552\ F_G \quad (12)$$

$$\Delta mAl_2O_3 = 0.8726\ F_G \quad (13)$$

$$\Delta mAlF_3 = -1.473\ F_G \quad (14)$$

Vaporization

The $NaAlF_4$ is the most volatile species existing above cryolite-alumina melts. The vapor exists both as the monomer $NaAlF_4$ and as $Na_2Al_2F_8$, from the reaction of sodium fluoride with aluminum fluoride, according to:

$$NaF + AlF_3 \rightarrow NaAlF_4 \quad (15)$$

The rate of loss of vapors is dependent on the saturation vapor pressure. The total vapor pressure PT in [kPa] wich is equal to the sum of the partial pressures of the individual gases can be expressed as a function of bath composition and temperature by equation [1]:

$$\log(P_T) = B - A/T_b \quad (16)$$

Where:

$$A = 7101.6 + 3069.7R_b - 635.77R_b^2 + \frac{764.5 \cdot C_{Al_2O_3}}{1 + 1.0817 \cdot C_{Al_2O_3}} + 13.2C_{CaF_2}$$

$$B = 7.0174 + 0.6844R_b - 0.08464R_b^3 + \frac{1.1385 \cdot C_{Al_2O_3}}{1 + 3.2029 \cdot C_{Al_2O_3}} + 0.0068C_{CaF_2}$$

Assuming that the transpiring gases are saturated with vapor, the rate of fluoride evolution in kilograms per metric ton of aluminum, due to particulate emissions resulting from volatilization given by Haupin [1], can be approximated by:

$$F_{VP} = \frac{2040}{\eta} \cdot \frac{P_m + 2P_d}{P_b} \quad (17)$$

Where P_b is the barometric pressure and P_m and P_d are the partial pressure of the monomer and dimer respectively, given by:

$$P_m = \frac{\sqrt{1+4K \cdot P_T} - 1}{2K} \quad [kPa] \quad (18)$$

$$P_d = P_T - P_m \quad [kPa] \quad (19)$$

Where (K) is the dimensionless equilibrium constant for dimerization, given by:

$$K = \exp\left(\frac{-21414}{T_b} + 15.6\right) \quad (20)$$

Finally one can express the consumption of electrolyte constituents due to particulate in kilograms per metric ton of aluminum as:

$$\Delta m_{NaF} = -0.5525\ F_{VP} \quad (21)$$

$$\Delta m_{AlF3} = -0.105\ F_{VP} \quad (22)$$

Entrainment of Particulate Materials

The mechanism when liquid droplets or solid particulates are entrapped in a flowing gas is known as entrainment. Haupin developed an empirical model for estimating the rate of fluoride evolution due to entrainment [1]:

$$F_E = \frac{1}{\%CE} \cdot (-17030 + 29800 \cdot R_b - 13000 \cdot R_b^2 + 67 \cdot C_{Al_2O_3} - 173 \cdot \tau - 0.3896 \cdot \tau^2 + 141.6R_b \cdot \tau) \quad (23)$$

Where:

$$\tau = T_b - 1243$$

The total fluoride emission will thus become:

$$F_T = F_G + F_{VP} + F_E \quad (24)$$

From the fluoride loss and by assuming that the concentration of the slowly consumed constituents remains constant as before, the consumption of electrolyte constituents due to entrainment can be approximated by proportional calculations as follows, in kilograms per metric ton of aluminum.

$$\Delta mAlF_3 = -1.473\ CAlF_3\ F_E\ f_F^{-1} \quad (25)$$

$$\Delta mNaF = -2.210\ CNaF\ F_E\ f_F^{-1} \quad (26)$$

$$\Delta mAl_2O_3 = -C_{Al2O3}\ \Delta M\ f_F^{-1} \quad (27)$$

Where f_F is the fraction of the electrolyte that is made up of fluoride compound and ΔM is the total mass change in the fluoride compound. Assuming that alumina is the only constituent that is not a fluoride compound,

$$f_F = (100 - C_{Al2O3}) \quad (28)$$

Neutralization

One of the key factors in order to obtain optimal aluminium production efficiency is to maintain the bath composition as stable as possible around specified target values. Usually the concentration of AlF_3 is kept between 10% to 13%. There are two basic oxides entering the cell together with alumina: Na_2O and CaO. They neutralize AlF_3 in the bath according to equations [4]:

$$3Na_2O + 4AlF_3 \rightarrow 2Na_3AlF_6 + Al_2O_3 \quad (29)$$

$$3CaO + 2AlF_3 \rightarrow 3CaF_2 + Al_2O_3 \quad (30)$$

The first reaction produces neutral bath and the second reaction forms CaF_2. The aluminium fluoride must be regularly added into the cell to readjust bath chemistry back to the targets. The AlF_3 addition rate is well predictable and depends on the alumina composition and scrubber effciency. Another factor influencing the AlF_3 demand is the sodium content of the cathodes. Depending on the cathode status, the AlF_3 feed rate may be below the prediction because part of the sodium provided by the alumina is absorbed into the cathode in the early months of the cathode life.

Alumina Feeding

The alumina feed algorithm is based on the process control algorithm and depends on the technology. It is normaly based on a demand feed strategy analyzing the voltage or calculated bath resistance curve from the voltage. Figure 1 shows a typical dependence of the voltage as function of the alumina concentration. The anode effect detection is also integrated in the alumina control model and takes place when the alumina reaches a critical low value.

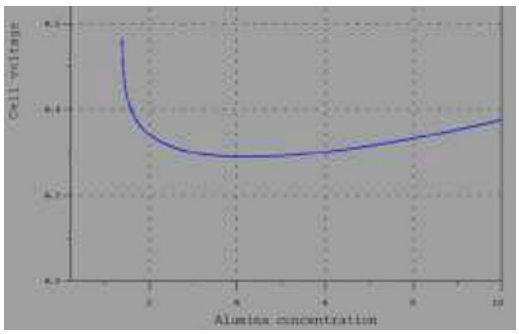

Figure 1: Cell voltage versus alumina concentration.

Alumina Dissolution

The alumina dissolution is a process by itself. Let us just mention that two specific time constant are used in the model associated to fast and slow alumina dissolution after alumina feed.

Corrective Fluoride additions

Fluoride is required for correcting the electrolyte bath ratio. Another obvious reason for corrective additions is to maintain sufficient electrolyte volume. To increase the ratio, sodium fluoride is added to the electrolyte in the form of sodium carbonate

(Na_2CO_3), which reacts with aluminum fluoride to form sodium fluoride (NaF), according to the reaction [4]:

$$3Na_2CO_3 + 2AlF_3 \rightarrow 6NaF + Al_2O_3 + 3CO_2 \quad (31)$$

Sodium fluoride also enters the cell as the natural occurring impurity, sodium oxide Na_2O in the feed material, which reacts with aluminum fluoride to form sodium fluoride [4]:

$$3Na_2O + 2AlF_3 \rightarrow 6NaF + Al_2O_3 \quad (32)$$

Adding aluminum fluoride also reduces sodium deposition, since excess AlF_3 reacts with sodium to form cryolite, according to the reaction [4]:

$$AlF_3 + 3NaF \rightarrow Na_3AlF_6 \quad (33)$$

The physical properties of the electrolyte are strongly influenced by its composition and temperature.

Density

Several empirical equations have been published for calculating the density of the molten electrolyte. The empirical relationship given by Haupin is considered [2]:

$$\rho_{EL} = 10^5/\left(\frac{C_{Na_3AlF_6}}{3.305-0.000937\cdot T_b} + \frac{C_{AlF_3}}{1.987-0.000319\cdot T_b+0.094\cdot C_{AlF_3}}\right. \quad (34)$$
$$\left. + \frac{C_{CaF_2}}{3.177-0.000391\cdot T_b+0.0005\cdot C_{CaF_2}^2} + \frac{C_{Al_2O_3}}{1.449+0.0128\cdot C_{Al_2O_3}}\right.$$
$$\left. + \frac{C_{MgF_2}}{3.392-0.000524\cdot T_b-0.01407\cdot C_{MgF_2}} + \frac{C_{Lif}}{2.358-0.00049\cdot T_b}\right)$$

Where T_b is the bath temperature in Kelvin degrees. From the above equation the theoretical density of the electrolyte at 960 °C was found to be 2113 kg/m^3. For comparison, the density of molten aluminum is close to 2305 kg/m^3.

Viscosity

The viscosity of the electrolyte influences several hydrodynamic processes in the cell: movement of metal droplets in the electrolyte, dissolution and sedimentation of alumina particles, release of the gas bubbles from the anode surface. A simplified equation for the Viscosity of the Electrolyte is given by Grjotheim and Welch [3]:

$$\mu_{EL} = 11.557 - 9.158\cdot 10^{-3}\cdot(T_b - 273) \quad (35)$$
$$-1.587\cdot 10^{-3}\cdot C_{AlF_3}$$
$$+\left(-2.049\cdot 10^{-3} + 1.853\cdot 10^{-5}\cdot(T_b - 1273)\right)$$
$$\cdot C_{AlF_3}^2 - 2.168\cdot 10^{-3}\cdot C_{Al_2O_3}$$
$$+\left(5.925\cdot 10^{-3} - 1.938\cdot 10^{-5}\cdot(T_b - 1273)\right)\cdot C_{Al_2O_3}^2 \ [mPa\cdot s]$$

Alumina Solubility

The maximum amount of alumina that can be dissolved in the electrolyte depends on both the electrolyte composition and temperature. The maximum alumina solubility can be expressed by the following equation, given by Skybakmoen [4]:

$$C_{Al_2O_3}^{max} = A\cdot\left(\frac{T_b - 273}{1000}\right)^B \quad (36)$$

Where:

$$A = 11.9 - 0.062\cdot C_{AlF_3} - 0.0031\cdot C_{AlF_3}^2 - 0.20\cdot C_{CaF_2}$$
$$-0.50\cdot C_{LiF} - 0.30\cdot C_{MgF_2} + \frac{42\cdot C_{AlF_3}\cdot C_{LiF}}{2000 + C_{AlF_3}\cdot C_{LiF}}$$
$$B = 4.8 - 0.048\cdot C_{AlF_3} + \frac{2.2\cdot C_{LiF}^1.5}{10 + C_{LiF} + 0.001\cdot C_{AlF_3}^3}$$

When the concentration of alumina reaches the maximum alumina solubility, the electrolyte becomes saturated with alumina, and all excess alumina sinks to the bottom of the cavity, where it can form sludge. From the above equation and typical values related to the bath composition, the theoretical maximum alumina solubility of the electrolyte at 960 °C was found to be approximately 8 wt%.

Aluminium Solubility

The tendency for metal to dissolve in the electrolyte is the primary cause for the lowering cell current efficiency. The lower the metal solubility, the higher resulting current efficiency. Based on correlation of measurements, the aluminum solubility can be expressed by the following equation [3]:

$$\log(C_{Al}^{max}) = 1.8251 - \frac{0.2959}{R_{bath}} \quad (37)$$
$$-\frac{3429}{T_b} - \frac{0.0339\cdot C_{Al_2O_3}}{C_{Al_2O_3}^{max}} - 0.0249\cdot C_{LiF} - 0.0241\cdot C_{MgF_2}$$
$$-0.0381\cdot C_{CaF_2}$$

From the above equation and typical values related to the bath composition, the theoretical maximum aluminum solubility of the electrolyte at 960°C was found to be approximately 0.034 wt%.

Current Effciency

The internal recombination of aluminum and carbon dioxide is the main cause of the difference between the actual quantity of aluminum produced and the theoretical quantity stated by Faraday's law of electrolysis. The current efficiency is affected by a number of variables, but the solubility of the metal in the bath and the mass transfer conditions within the electrolyte play the most important roles. The current efficiency can be expressed by the following relationship (Lillebuen model) [5]:

$$\%CE = 100\cdot\left(1 - \frac{r_m}{r_{Al}}\right) = 100\cdot\left(1 - \frac{3F\cdot r_m}{I}\right)(\%) \quad (38)$$

Where r_m/r_{Al} is the ratio of the metal transfer through the bulk of the electrolyte and the theoretical rate of aluminum production, F is the Faradays constant and I the cell current.

Predicting the current efficiency from the above equation has limitations due to the assumptions made and due to secondary effects such as metal velocity variations. The equation shows that the current effciency is a function of bath composition, bath temperature, anode to cathode distance and velocity. Corrrelations to real cells results led us to combine the above equation to another important work performed concerning the mathematical modeling of the current efficiency by P.A. Solli. The details of the work can be found in References [6-8]. The authors are considering an average of both theories.

Energy Considerations

The aluminum reduction process requires energy to keep the electrolyte at the reaction temperature and to produce aluminum, which is furnished as electric energy. The authors cannot develop the equations in the frame of this paper but would like to summarize the important elements imbedded in the model:

Enthalpy of reaction for the primary reaction [9]:

$$\Delta H_{react} = \frac{1}{2} \cdot \Delta H_f(Al_2O3) - \frac{3}{4}(2 - \frac{1}{\eta}) \cdot \Delta H_f CO_2 - \frac{3}{2}(\frac{1}{\eta} - 1) \cdot \Delta H_f(CO) \quad (39)$$

$$\Delta H_f(Al_2O_3) = 1692.44 - 0.011235 \cdot (T_b 1100) \quad (40)$$

$$\Delta H_f(CO_2) = 394.84 + 0.002095 \cdot (T_b - 1100) \quad (41)$$

$$\Delta H_f(CO) = 112.59 + 0.00642 \cdot (T_b - 1100) \quad (42)$$

Table I. Enthalpy of reaction for the secondary reactions [9]

Chemical reaction	Enthalpy of reaction [kJ/mol]
$Na_3AlF_6(l) + 3/2H_2(g) \rightarrow Al(l) + 3NaF(l) + 3HF(g)$	$\Delta H_{react} = -793.26 + 0.1258 \cdot T_b$
$2AlF_3(s) + 3H_2O(g) \rightarrow Al_2O_3(s) + 6HF(g)$	$\Delta H_{react} = -324.03 + 0.07591 \cdot T_b$
$NaF + AlF_3 \rightarrow NaAlF4$	$\Delta H_{react} = -895.21 + 0.93468 \cdot T_b$
$S + 2CO_2 \rightarrow SO_2 + 2CO$	$\Delta H_{react} = -210.9 + 0.00748 \cdot T_b$
$3Na_2CO_3 + 2AlF_3 \rightarrow 6NaF + Al_2O_3 + 3CO_2$	$\Delta H_{react} = -24.784 + 0.14984 \cdot T_b$
$AlF_3 + 3NaF \rightarrow Na_3AlF_6$	$\Delta H_{react} = -169.3 + 0.09972 \cdot T_b$
$3Na_2O + 2AlF_3 \rightarrow 6NaF + Al_2O_3$	$\Delta H_{react} = -844.58 - 0.18601 \cdot T_b$

Enthalpy of heating the reactants and bath additives as well as enthalpy of alumina dissolution must be considered [9]:

$$H_{\gamma \rightarrow \alpha} = (\Delta H_f(\alpha) - \Delta H(\gamma)) \cdot c_\gamma(\%) \quad [KJ/mol] \quad (43)$$

$\Delta H_f(\alpha)$: Heat of formation for alumina in alpha phase (-1675.69 kJ/mol)

$\Delta H(\gamma)$: Heat of formation for alumina in gamma phase (-1656.86 kJ/mol)

$c_\gamma(\%)$: Concentration of gamma phase alumina (wt%)

The enthalpies of formation of the main substances involved in the overall aluminum reduction process can be found in the litterature for various temperatures. The theoretical net energy requirements of the reduction process during 24 hours operation of a cell operating at 94% current effciency and 200 kA nominal current are given in table II:

Table II. Theoretical net energy requirements of the reduction process during 24 hours operation of a cell operating at 94% current efficiency and 200 KA nominal current

Reactions	Energy [kJ]
Primary reaction	$3.127 \cdot 10^7$
Evolution reaction	$3.921 \cdot 10^4$
Vaporization reaction	$-3.756 \cdot 10^4$
Heating alumina	$3.049 \cdot 10^5$
Alumina dissolution	$1.143 \cdot 10^6$
Total energy	$3.632 \cdot 10^7$

Finally all types of events such as metal tapping, beam rizing, current increase and feedings can be scheduled. The alumina feeding is of course of prime importance as it is used to control the bath voltage. It can be easily modified in the simulator to reflect the exact algorythms used for any particular smelter. Same remarks apply for the aluminium fluoride, calcium fluoride or any desired addition.

Cell voltage model

The cell voltage is the most important operating parameter. It depends strongly on the alumina concentration in the electrolyte, the bath temperature and the anode to cathode distance (beam movements). Analytic expressions can be found in literature for each component of the cell voltage. The relation with alumina concentration is shown in figure 1.

Thermal balance model

Thermodynamic tell us how much heat is produced by the process [4]:

$$Q = I\,(U_{cell} - 1.65\,\eta - 0.48) \quad (44)$$

U_{cell} : Cell voltage [V]
I : Current in the line [kA]
η : Current efficiency [1]
Q : Cell heat loss [kW]

Conduction, convection, radiation and heat capacitance tell us what happens to the heat. The ledge freezing and melting processes play a key role on the cell heat balance. It changes the bath chemistry which impacts on the electrical conductivity of the bath, itself impacting on the voltage. The software was designed to run much faster than real time, it was therefore important to simplify some partial differencal equations to achieve fast answers still with the adequate accuracy. The total heat loss is devided into shell bottom, long side top, long side bath, long side metal, ends, collector bars, stubs and top heat losses.

Application

The cell is described by a set of over 300 parameters. This defines the cell geometry, the operating procedures and materials properties. Each parameter can be modified to analyse its impact on the process. Figure 2 shows a set of output from the model. After 8 hours one anode is changed, after 22 hours tapping is imposed to the cell. The analysis can be performed over a full anode cycle and beyond (more than 30 days). The feeding cycles are based on a "demand feed" algorithm and is triggered by the slope of the bath electrical resistance curve. A current fluctuation is imposed to the model (not compulsory) to simulate the rectifiers' fluctuations. As all formulas are somehow impacted by the current, it helps at validating the smoothing algorithms needed to control the process fluctuations.

Figure 2 shows 24 hours plot of the beam displacements, feeding rate in % of nominal feed and cell voltage.

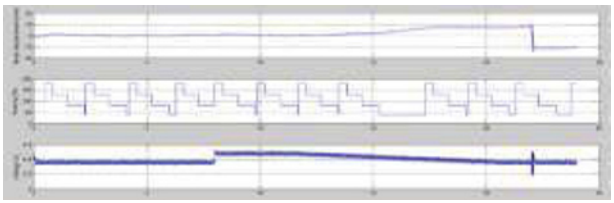

Figure 2. 24 hours plot of the beam displacements, feeding rate in % of nominal feed, cell voltage.

Figure 3 shows the bath and metal temperature and ledge at metal and bath level.

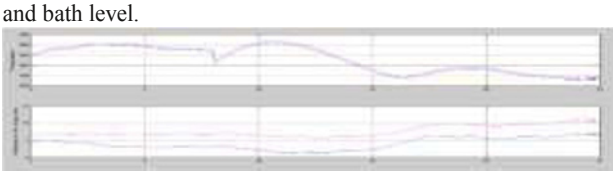

Figure 3. 24 hours plot of the bath and metal temperature and ledge at metal and bath level.

Figure 4 shows the bath resistance, current efficiency and energy consumption for the analysed period. The high frequency signals result from the noise level imposed on the current.

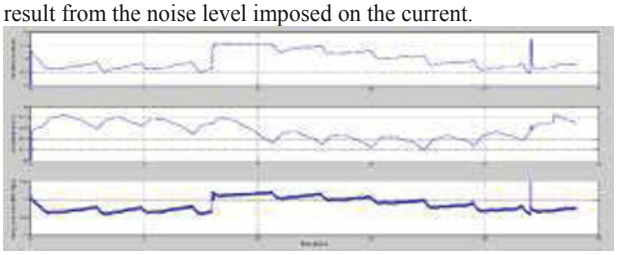

Figure 4. 24 hours plot of the bath resistance, current efficiency and energy consumption

There are many more plots analysing the energy balance, the bath chemistry and any desired parameter as function of time. Predictions were validated with measurements and plant data.

Conclusions

A sophisticated cell simulator has been developed using MATLAB-SIMULINK software. The software is fully open and can be easily modified to implement each plant feedings strategies or specificities. It highlights how heat losses, current efficiency and other key operating parameters fluctuate during the day. It can be used to optimize the process.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

STUDIES ON ANODE PRE-HEATING USING INDIVIDUAL ANODE SIGNALS IN HALL-HÉROULT REDUCTION CELLS

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Keywords: online anode current monitoring, anode pre-heat, anode change

Abstract

The proportional increase in modern cells' amperage and anode size while aiming for low energy input has resulted in slowing the process of anode current pick-up rate. Changes in cell thermal and electrical balance due to anode replacement (while a thick layer of frozen electrolyte is formed under a new anode) causes a slower anode current pick-up rate which may require 4 days to reach the target current. Variations in cell conditions following anode replacement result in increasing cell magnetohydrodynamic instability and prolonged presence of low superheat zones. This increases the possibility of forming local abnormalities, such as anode spikes. As part of a continuous enhancement programme, the latest developed individual anode current monitoring system in DUBAL (EGA Jebel Ali) was used to study the benefits of pre-heating anodes up to 500 °C. The work includes studying the impact of anode pre-heating on current pick-up rate while conducting operations such as anode dressing.

Introduction

Because of the continuous anode consumption process in an aluminium reduction cell, an anode replacement activity is carried out in the cell every 1 to 3 days, resulting in frequent disturbances to cell magnetohydrodynamic (MHD) stability [1]. Formation of a quenched frozen insulating layer of molten electrolyte under newly set anodes hinders current flow and diverges toward anodes which operate at lower ohmic resistance [2, 3]. The presence of a low superheat zone in the vicinity of newly set anodes lowers the efficiency of electrolyte mixing and alumina dissolution [4, 5]. Accordingly, the temporal changes in local cell conditions increase cell MHD instability and result in reduced current efficiency [6].

The objective of this study was to evaluate the potential benefits of anode pre-heat on accelerating the rate of current pick-up of newly set anodes, which could result in minimising the period of cell operation under non-uniform current distribution. The ultimate set-up of an anode pre-heat station would be interconnected with a novel heat exchanger developed by Aljasmi et al [7], which demonstrates the vision of Emirates Global Aluminium (EGA) to maintain its status as one of the world's leading providers of high cell energy efficiency and low emission technology.

An earlier study by Ali et al [8] showed that the newly set cold anode requires a minimum of 4 to 5 days to reach full target current draw. Zoric et al [6] estimated a longer period, probably due to different operational practise and cell design parameters.

The higher initial temperature of a pre-heated anode should lower the local thermal deficit in the zone of the replaced anode, leading to a lower heat extraction from the molten electrolyte and less entropic cooling effect than desired for improved cell performance during anode replacement [9]. Furthermore, this would provide a better superheat condition to improve alumina solubility during anode setting.

Limited research is published that discusses the possible improvement in process efficiency when using pre-heated anodes, although in a work performed by Fortini et al [10] anodes were electrically pre-heated and set in 4 test cells to a temperature range of 484 °C to 512 °C. They observed a 0.5 % to 1.0 % increase in average current efficiency after a period of 60 days of analysis, although further verification of this effect is desirable. Many factors are likely to affect any improvements in current efficiency associated with pre-heated anodes, but there is little information available in the literature. Furthermore, pre-heated anode temperature profiles during replacement have not been reported previously. In this work, the continuous current pick-up rate for newly set cold and pre-heated anodes has been studied and the impact of anode dressing activity by using anode cover recycled material (ACRM) has been systematically investigated in order to better evaluate any benefits that might be derived from anode pre-heating.

Experiment set-up

Two cells (with four point feeders) fitted with DUBAL's latest individual anode current monitoring system were selected for the experiment. The first cell was designated to receive cold anodes whereas the second received pre-heated anodes. The gas pre-heat station was designed to pre-heat anodes to a target temperature of 470 °C to500 °C to avoid initiation of any anode air burn reaction, with the ability to pre-heat pairs of anodes at the same time, as illustrated in Figure 1.

Figure 1. Capability of pre-heating pairs of anodes in the in-house designed pre-heat station.

The pre-heat station in Figure 2b was constructed of a steel cradle lined with a layer thermal insulation bricks to minimise heat loss and protect the steel structure against internal thermal stress. To avoid localised heating spots or air burn while pre-heating the anodes, the throughput flame from the gas pre-heater was placed at a safe distance to eliminate any direct contact with the anode block as shown in Figure 2a.

Figure 2. Pre-heat station a) Location of the flame. b) Outer steel cradle structure of pre-heat station.

During pre-heat, the anode temperature was continuously monitored by a k-type thermocouple which was inserted at a depth of 25 mm from the upper anode surface as shown in Figure 3.

Figure 3. Top view for the location of monitoring thermocouple in pre-heated anode.

Due to logistics constraints, the anode pre-heat facility was constructed outside the potroom and hence the pre-heated anodes experienced a reduction in temperature during transfer to the cell. The average initial temperature upon setting was thus measured as 300 °C with ±11 % variation.
A total of 8 anodes were pre-heated and their location, shaded in grey in Figure 4, was deliberately selected from different stall positions to minimise the individualistic impact of anode stall location on the current pick-up curve.

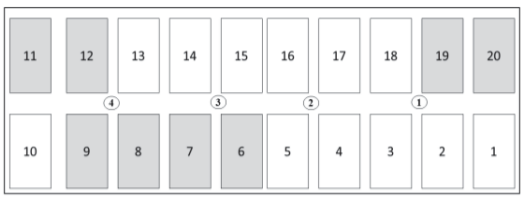

Figure 4. Location of pre-heated anodes (coloured in grey).

Pre-heated anode setting procedure

A similar amount of anode set adder was applied when setting pre-heated and cold anodes through adjusting the heat input to supply the additional energy required to heat the new anodes. This involved adjusting the cell pseudo-resistance target through increasing anode cathode distance (ACD) to compensate for the increased current density in the remaining anodes and avoid metal pad instability [13]. However, due to the higher initial temperature of the pre-heated anodes, their anode set point adder was slightly modified as shown in Figure 5. Pre-heated anodes will have lower deficit in electrolyte superheat, which will allow faster dissolving process of quenched frozen layer and thereby facilitate relatively quicker anode current pick-up rate. Nevertheless, during the initial six hours of setting, the energy input remained the same for both cold and pre-heated anodes.

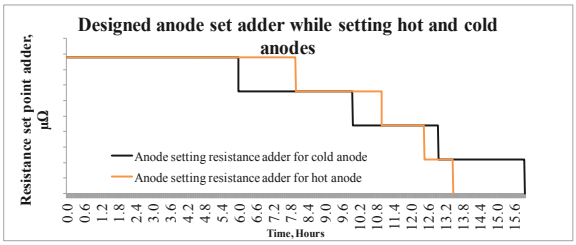

Figure 5. Designed resistance set point adder for pre-heated and cold anodes.

The higher initial temperature of pre-heated anodes allowed for setting them at a slightly lower position, without inducing any thermal cracks. Furthermore, pre-heated anodes were dressed immediately upon setting to avoid air burn, while cold anodes were dressed after 8 hours once the crust layer around anodes had formed. The effect of anode dressing time was, however, investigated for both cold and pre-heated anodes.

Results and discussion

Analysis of current pick-up curves for pre-heated and cold anodes was divided into two time frames. The first involved analysis of the initial 6 hours upon setting, where both test and control cells received the same amount of additional anode set point adder. During this time-frame, disturbances to cell MHD stability caused by other operational activities such as metal tapping were also avoided since the operational procedure in the smelter sets a minimum of 6 to 8 hours' time gap between anode setting and any operational activities. The second set of analysis studied the current pick-up curves of both cells after 24 hours of operation.
Pre-heated anodes experienced a faster evolution in current, by 42 % on average, in the initial 6 hours of operation as shown in Figure 6. The higher initial anode temperature and lower extracted energy from the molten electrolyte intuitively provided a better local superheat and electrolyte mixing condition in the test cell.

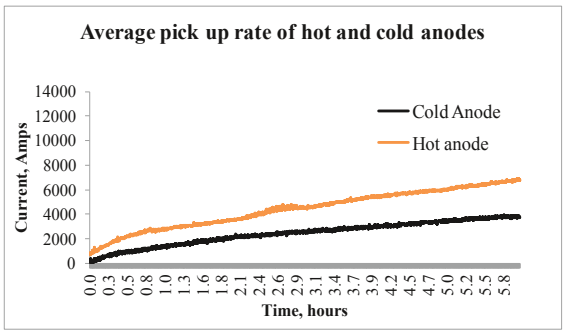

Figure 6. Average anode current curve for pre-heated and cold anodes during first 6 hours of setting. Dressing times are: within first hour for pre-heated anodes and at around 8 hours for cold anodes, after anode change

Analysis of individual anode currents showed that pre-heated anodes always have a higher current draw, as shown in Figure 7.

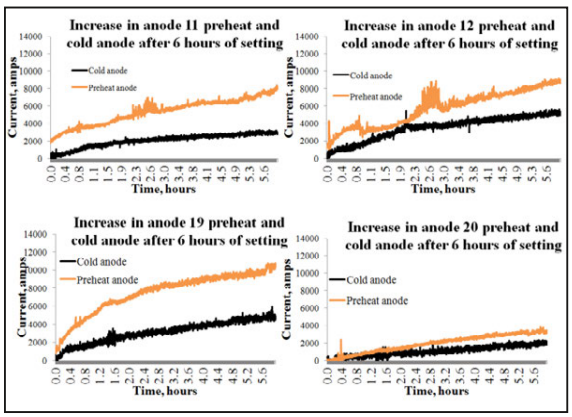

Figure 7. Individual anode current curves for pre-heated and cold anodes during first 6 hours of setting. Dressing times as indicated in Figure 6.

However, individual anode current curves in pre-heated cells showed a higher spread in current and a 10 % excess over cold anodes after 6 hours of replacement as shown in Figure 8.

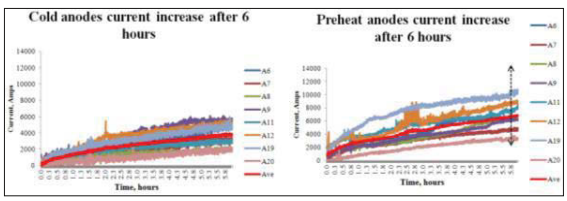

Figure 8. Individual anode current increase in pre-heat and cold anodes after 6 hours of replacement. Dressing times as indicated in Figure 6.

The higher spread in the pre-heated anode current curves is attributed to the non-uniform cooling rate of anodes while transporting from the pre-heat station to the test cell which resulted in a ±11 % initial anode surface temperature variation, as illustrated in Figure 9.

The generally observed reduction in anode surface temperature (shown in Figure 9) during the first hour after setting was associated with anode dressing activity that cools the surface of the pre-heated anodes when the room temperature ACRM is applied.

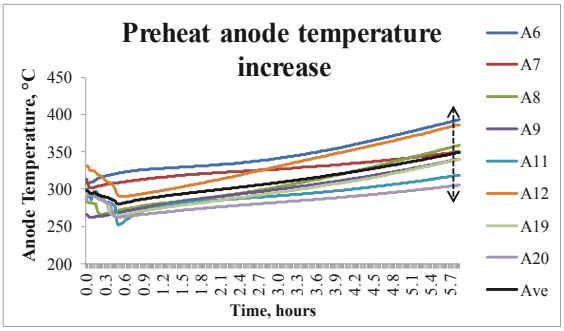

Figure 9. Variation in pre-heated anode temperature measured at 25 mm depth of the anode surface for different stall locations. Dressing time: within first hour after anode change

The current rise after 24 hours of operation remains higher in pre-heated anodes, as shown in Figure 10a. The accelerated current rise allows pre-heated anodes to reach 70 % of current load after 9 hours of setting. while cold anodes reach the same level after 15 hours. The differences in average current profile between pre-heated and cold anodes are plotted in Figure 10b. This shows that the dominant variance in current occurred in the initial 14 hours of setting, which is primarily controlled by rate of freeze dissolution [8], and appears to be faster in pre-heated anodes.

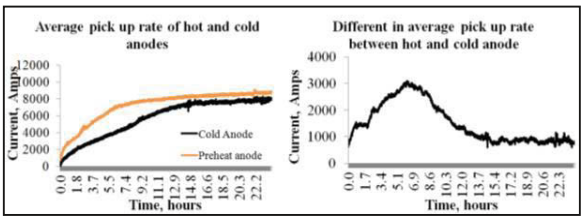

Figure 10. Current increase in pre-heated and cold anodes a) Average current increase. b) Difference in average current increase.

In terms of the cyclic anode current pattern which has been characterised by Jassim et al [11] as being associated with the co-evolution of fluorocarbon species CF_4, none of the pre-heated anodes showed a cyclic anode current, although this appeared in the cold anode (as shown in Figure 11). The combined effect of alumina depletion at the end of the search window and low superheat, result in the formation of an alumina concentration gradient that induces the potential of the cold anode to increase beyond the enabling limits for perfluorocarbons species co-evolution, thereby lowering anode current toward the end of the search window.

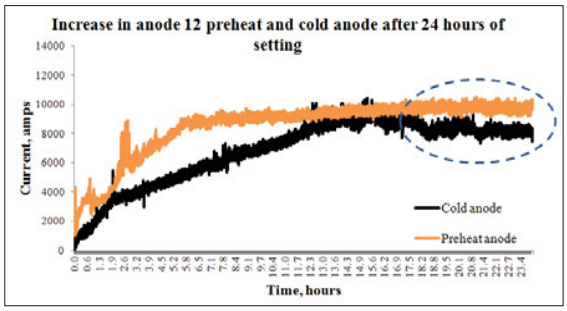

Figure 11. Cyclic anode current in cold anode 12. Impact of anode dressing on cold anode

The impact of the anode dressing activity was studied for the cold anodes dressed after 8 hours of anode setting activity, during which around 650 kg of ACRM is applied. Dressing appears to have an insignificant impact on the rate of anode current pick-up as shown in Figure 12 (with the exception of the corner anode 20).

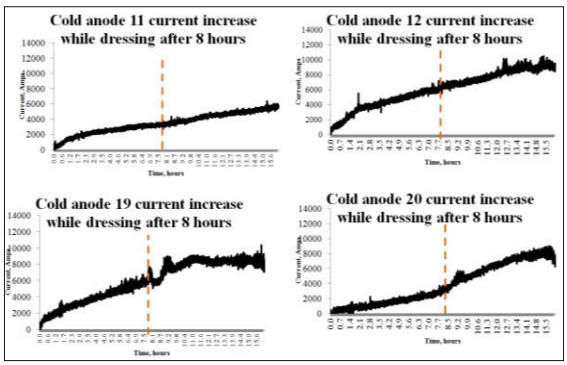

Figure 12. Cold anodes current while dressing crushed bath material after 8 hours of setting.

Figures 9 and 13 show a slight reduction in the pre-heated anode surface temperature upon dressing. However, the impact on current appears negligible as anodes showed normal anode current dynamics upon dressing, except for anode 19 where the change in current occurred prior to anode dressing.

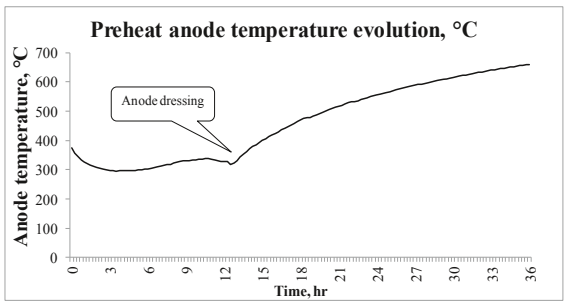

Figure 13. Change in pre-heated anode temperature while dressing after 13 hours of setting in previous work- temperature measured at 25 mm depth from anode surface.

A 2D thermal model to simulate the study's changes in anode surface temperature upon dressing was developed by Arkhipov [12] and a similar drop in temperature was estimated as illustrated in Figure 14a. Dressing will have an impact on the rate of heat loss above the anodes and will affect the bulk temperature profiles of the set anodes, so its effect on current pick-up rates needs to be explored further.

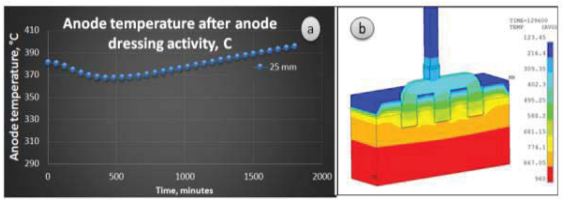

Figure 14. Thermal 2D-model for change in surface anode block temperature after dressing a) Simulation of change in anode temperature due to dressing activity b) Anode temperature distribution.

Analysis of pre-heated and cold anodes current pick-up rate while dressing at the same time

An analysis of cold and pre-heated anode current pick-up curves was conducted while applying the dressing material (ACRM): a) 8 hours after anode setting, or b) within the first hour after anode setting.

The average current profiles while dressing immediately or dressing after 8 hours are shown in Figures 15a and 15b respectively. It is seen that the pre-heated anodes have a higher current pick-up rate in both scenarios.

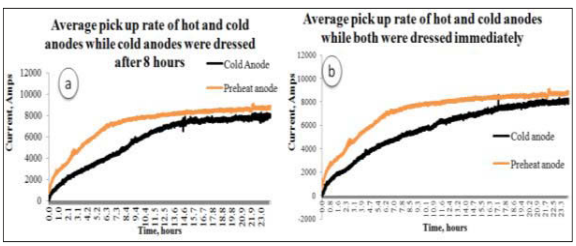

Figure 15. Comparison of pre-heated and cold anodes current curve while a) Dressing cold anodes 8 hours after anode setting. b) Dressing both anodes within first hour after anode setting.

The difference in current pick-up rate between pre-heated and cold anodes when dressing cold anodes after 8 hours (Figure 16a) or dressing immediately (Figure 16b) has shown a comparable profile, whilst the predominant variance in current evolution occurred in the initial 14 hours of operation – which is controlled by different heat transfer and side as well as bottom freeze melting process. Furthermore, pre-heating anodes allows for setting them at a slightly lower reference position, without inducing any thermal cracks, which assisted in accelerating the anode current pick-up rate.

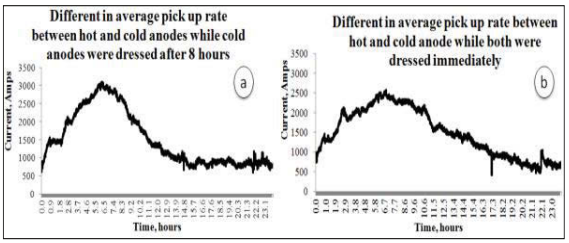

Figure 16. Difference in average current curve of pre-heated and cold anodes while a) Dressing cold anodes after 8 hours; b) Dressing pre-heated and cold anodes after setting immediately.

Analysis of individual anode current after dressing immediately showed that pre-heated anodes have either a higher or similar current draw. Two examples are shown in Figure 17, where in the case of anode stall 6 (left), the current pick-up rate of both pre-heated and cold anodes becomes similar 12 hours after the anodes setting, while in the case of anode 19 (right), the current pick-up rate of the pre-heated anode is clearly higher than cold anodes. The phenomenon observed in the case of anode 6 could be attributed to some other local spatial changes in cell condition.

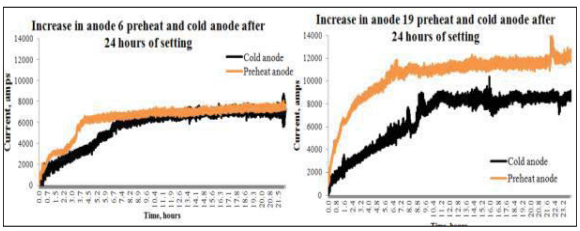

Figure 17. Individual pre-heated and cold anodes current while dressing immediately under identical anode set point energy adder.

Conclusion

Studies have shown that in cells operating at lower energy input and with larger anode size, the reduction in current draw upon anode setting results in an extended period of cell operation under non-uniform current density because of bath freeze around the cold set anodes. Pre-heating the anodes has shown the possibility of accelerating anode current pick-up rate under normal cell conditions, which could result in minimising the period of operating at high MHD instability induced by the anode change activity, thereby improving process efficiency in theory. The reduction in heat from the molten electrolyte when using pre-heated anodes will lower the local heat deficit and enhance alumina mixing and dissolution rates. Any benefits derived from improved cell performance should not be negated by increased energy consumption for anode pre-heating, so the use of waste heat from the cell needs to be considered. The Hall-Héroult reduction cell is considered a highly heat dissipative system with around 50 % of energy used in aluminium electrowinning being wasted as radiated heat to the environment. An industrial application of pre-heated anodes could thus become feasible through connecting the anode pre-heat station to a heat recovery system as developed by Aljasmi et al [7]. This system recovers dissipated heat from the cell side walls and the vented hot off gases from the duct, resulting in improved overall cell energy efficiency. Locating the anode pre-heat station near the cell would also avoid the unnecessary loss in anode heat during transportation, providing further energy efficiency gains. The capital and operating costs of such as system still need to be assessed.

Acknowledgment

The authors offer their sincere gratitude to EGA and the University of New South Wales for their financial and technical support in developing the anode current system through a collaborative research project. Appreciation particularly goes to the EGA pot repair crew for their enormous support while developing and executing the pre-heat station.

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35 Years of Contributions to Cast Shop Research and Development – Honoring Prof. Dr. Wolfgang Schneider

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Keywords: Filtration, Grain refinement, Solidification, DC Casting

Abstract

Casting processes of aluminum and especially DC casting processes exist since less than one century. Although this area has a short history the findings in science of casting and the related solidification process of aluminum are key assets for the development of the Al industry and its related fast growth.
In his now 35 years lasting industrially based research work Prof. Wolfgang Schneider always focused on the link between sound theoretical understanding and industrial application of scientific findings. This led to various significant contributions to our industry, which is also documented by a number of patents he holds. His development activities span from filtration processes, grain refinement and modification to shape and DC casting processes itself. He published more than 150 papers and received numerous awards for his work such as Light Metals Award in 1990 and 1995 or the Madhu Nilmani Award in 2001.
Besides his research he was Professor for Metallurgy at the Technical University of Berlin and gave lectures at RWTH in Aachen. He was also active in related professional networks such as Verein Deutscher Gießereifachleute VDG, Germany, Deutsche Gesellschaft für Materialkunde DGM, Germany, and The Minerals, Metals and Materials Society TMS, U.S.A, where he was the first European President in 2012.

Introduction

Within the area of aluminum production the invention of DC casting in the early 1930ies represented a major step regarding the application of aluminum in construction, aerospace, transportation and packaging. Taking this into account tells us that DC casting and also the related process chain before and after this step represents a rather young part of industrial progress and scientific development.

Looking on the 35 years of contribution of Prof. Wolfgang Schneider in this area with this in mind his work on the scientific and development side shows a quite impressive impact. His own and those together with co-workers published contributions always considered a global view on the DC casting process and the resulting products. Therefore, research driven by him or a related team did not only concentrate on one specific scientific topic, but spread the research focus along the relevant part of the process chain.
Key or outstanding contributions of these varying groups can be found in the area of filtration, grain refinement and the casting process itself. The purpose of this paper is to briefly review these and to highlight those results which significantly impacted related research.
Although Prof. Schneider started his scientific career in the 80ies with work on grain refinement principles this review will more or less follow the process chain and therefore start with the work towards melt treatment and filtration, which concentrated as well on the related mechanism as on interfering forces and finally led to the development of an optimized filter concept. The contributions to research on grain refinement and DC casting will then be discussed in the following sections organized along the process chain.

Filtration

His and related co-workers activities within this area was driven by the need to achieve suitable material quality for demanding applications such as lithographic sheet or very thin foil for liquid packaging. Based on extensive material characterization the appropriate removal of non-metallic inclusions was identified as key step to establish a reliable process. Starting in 1996 an experimental program was carried out under his lead to evaluating on a fundamental basis the performance of in-line melt treatment and more specifically filtration concepts with regard to efficiency and reliability. Publications within JOM [1] or Light Metals [2-7] together with partners from Foseco covered a comparison of the filtration efficiency of different filter types as is demonstrated in Figure 1 on the basis of various casting campaigns monitored by LiMCA measurements to detect the amount of non-metallic impurities before and after the tested filters.

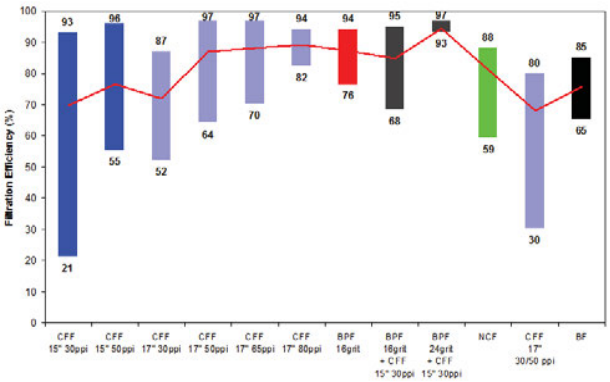

Figure 1 Filtration Efficiency of various filter types and combinations, taken from [6]

Within this work lasting for about half a decade major findings were achieved which also influenced and initiated work of other groups regarding these aspects.
The most important of these are listed in the following section

- "Bridging" of inclusion agglomerates within filter pores of fine filters enhance performance
- Reliability of achieving satisfactory filtration efficiency is strongly improving with finer filter pores

As mentioned above these results were based on an extensive campaign with LiMCA measurements before and after the in line units and the comparison of the results of these parallel acquisitions.

Including further disturbing effects such as heavy stirring of the furnace or the addition of grain refiner before the filter revealed even more interesting phenomena as can be seen in Figure 2 representing typical LiMCA curves monitored during an observed casting trial [8].

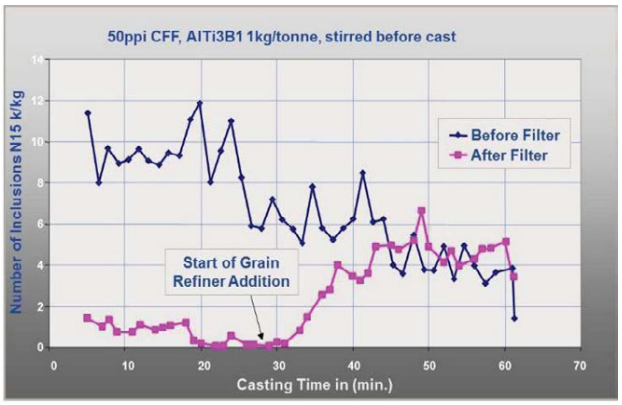

Figure 2 3/1 TiBAl rod fed partway through cast - "high" inclusion load, taken from [9]

The repetition of the trials under the above modified different conditions led to the achievement of the following results [9-11]:

- Ceramic foam filters can achieve high levels of filtration efficiencies even with high inclusion loading in the absence of grain refiner addition before filter
- Addition of grain refiner significantly reduces filtration efficiency in case of high inclusion loading conditions. Impact
- It was assumed that the reason for the performance loss of the filters was the prevention of the building of the above mentioned “bridges” especially in finer pore CFF due to the presence of grain refiner particles

Confirmation of the headlines of these findings, which were achieved for low alloyed material, even for high Mg alloy material was published a few years later by a different group of researchers [12].
Finally, all these results converged in the development of a new filtration system, which combined a low footprint, high filtration efficiency and cost savings by multi-use of CFF [13-14]. This concept (see Fig. 3) represented a combination of a CFF, adjusted grain refiner addition location and a small bed filter as safety belt

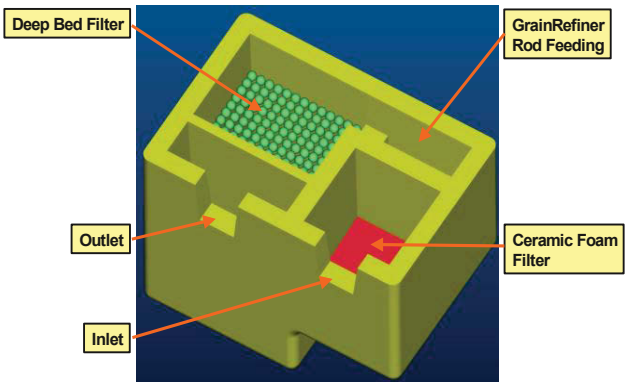

Figure 3 Typical layout of the XC Filter showing grain refiner addition between the two filtration stages, taken from [13]

concerning the impact of the grain refiner quality on global inclusion loading.
Performance data of a pilot scale version of the concept (XC Filter) confirmed its capabilities as can be seen in the LiMCA data evaluation for consecutive casts in Figure 4.

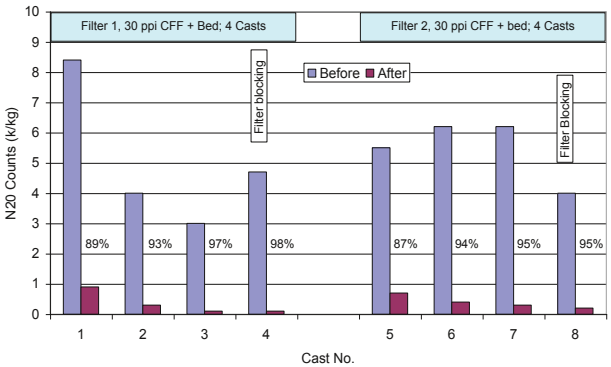

Figure 4 LiMCA N20 results for several casting trials using the XC Filter, taken from [13]

Key results achieved at that time were [14]:

- Improved filtration efficiency over standard CFFs can be achieved through the use of double filtration for example with the XC Filter.
- The XC filter permits the addition of the grain refiner after the CFF, which can lead to significant improvements in the performance of the CFF. In this configuration the detrimental effect of Al-Ti-B and Al-Ti-C grain refiners on CFF performance can be avoided.
- Multiple use of CFFs in the XC Filter can lead to reduced filtration costs despite the inclusion of a second filtration stage.

Besides the more applied work he also initiated a scientific program on a more fundamental level where the focus of the investigation was the internal flow within the filter material [15-16]. The work addressed geometrical factors and hydrodynamics in the internal cellular regime and how they impact on the flow field and related residence times of particles. Aim of this work was to achieving a better understanding of the mechanism and developing filter concepts with improved capabilities.

Grain Refinement

Research on grain refinement represented the area where Prof. Schneider began his scientific work. This started with experiments to study fundamental principles behind the process and the poisoning influence of specific alloying elements on the efficiency of different Titanium based grain refiners [17-19]. Later he developed new refiners together with partners from university and suppliers in order to minimize the limiting impact of those elements on the refinement performance [20-24].

Exemplary for this work should be a view on the comparison of AlTiB and AlTiC based refiners regarding their behavior under varying casting conditions [22]. Here, the focus of the work was to identify the main differences and possible advantages of the two alternatives with regard to performance and efficiency. Performing a set of casting trials using an AA8079 alloy as standard with the addition of the different grain refiners under defined conditions the resulting grain size was measured as can be seen in Figure 5.

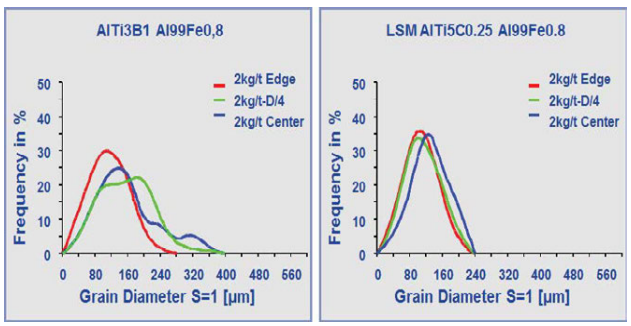

Figure 5 Distribution of grain size at edge, d/4 and center of AA8079 billets refined with 2kg/t of Al-3Ti-1B and Al-5Ti-0.25C, taken from [22]

In addition to that program Alcoa Cold Finger-Tests were used to evaluate the efficiency of the different types with regard to alloys having higher amounts of Zr, which is known to hinder refinement when using TiB_2 based material. Figure 6 shows the result of such a test for two different master alloys.

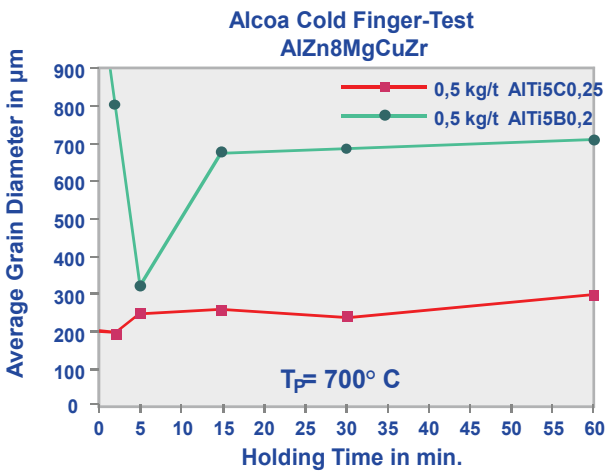

Figure 6 Influence of Zr on the grain refinement with AlTiB and AlTiC master alloys, taken from [27]

Based on these results it was revealed that:

- AlTiC grain refiners can be used in place of AlTiB in most wrought alloys though to achieve best results the casting temperature should be kept as low as possible.
- AlTiC offers advantages in alloys containing Zr; Cr which can otherwise poison AlTiB refiners. Conversely, Mn-bearing alloys e.g. AA3004 appear more difficult to refine with AlTiC.

Parallel to these investigations to find and evaluate new refiners he continued to work on the expansion of knowledge concerning standard TiB_2 master alloys. Besides the research on appropriate TiB ratios [25], he and his coworkers also looked on the impact of different casting parameters on the effectiveness of grain refinement. A brief summary of the underlying fundamentals and consequences represents a publication of him together with other experts in the field in 2008 on the ICAA11 conference [26].

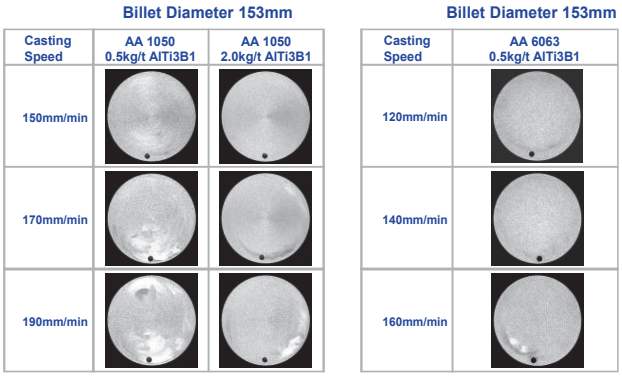

Figure 7 Influence of casting speed on grain refinement with AlTiB master alloys, taken from [28]

A nice overview on all this work was given in two contributions together with Douglas Granger [27, 28] where they presented experimental evidence how parameters like casting speed, pouring temperature and even metal distribution in the liquid pool within the mold affect the refinement and consequently the final microstructure within solidified ingots. Figure 7 displays such results for the impact of chosen casting speed on the final structure. Related to the increase of the casting speed the almost uniform grain structure achieved under reference conditions develops deviating features as can be seen in the cross sections for the higher casting speed cases shown in the lower part of the figure. A closer look at the casting conditions revealed a higher probability of the appearance of such deviation in the area opposite to the pouring gate of the mold which is about the position of the black spot in the different slices. Further research on the basis of numerical modelling of the casting process itself could show the relationship between the flow conditions within the mushy zone in that area and the development of feathery crystals opposite to the pouring gate of the level pour mold.

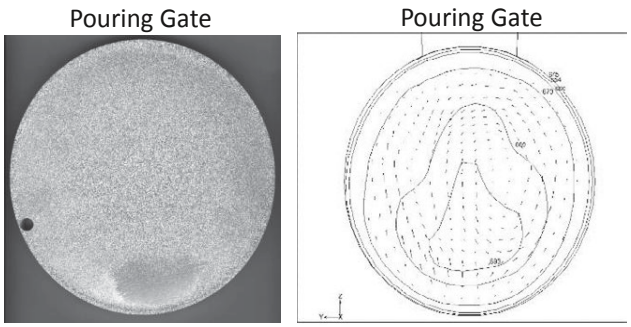

Figure 8 Feathery crystal growth opposite to the pouring gate and modelled related temperature and velocities at a horizontal cross section referring to the structure, taken from [28]

Figure 8 shows a comparison of a final microstructure slice of a cast extrusion ingot with a velocity pattern in a corresponding 3-dimensional coupled fluid flow and solidification model at a horizontal cross section within the solidifying ingot shortly below the bottom of the pouring gate.
Based on these first observations even more detailed investigations carried out within the related scientific group finally revealed the impact of shear force generated by the individual flow pattern on feathery grain growth [29].

DC Casting

In comparison with the activities discussed in the previous paragraphs it can be stated that the research against the DC Casting process represented the main focus area of Prof. Schneider's scientific career. He worked as well on fundamental understanding as specific technical applications of the DC casting process for extrusion and rolling ingots.
In the beginning he studied solidification conditions close to the mold surface and the related shell zone behavior [30], which then led to the development of new types of mold for rolling ingot and extrusion billet casting applying adjusted solutions with regard to the target goal [31-33]. Key feature within both was a hot top solution, while for the extrusion ingot additionally a sophisticated cooling/lubrication system was introduced.
In the following he, together with colleagues within the European based industry, studied the impact of casting conditions on starting cracks during the DC casting of extrusion ingots. Final findings of this research work supported by the "Continuous Casting" committee of the German Society for Material Science and Technology, DGM were presented by Einar Jensen, a long-standing research partner and friend and him at Light Metals 1990 [34, 35].
Within this program the group investigated the influence of:

- starting conditions (starting rate, cooling water volume, inlet temperature, filling time and grain refiner addition rate),
- starting block shape and material,
- liquid sump development in transient casting phase,
- temperatures in ingot butt and starting block

by means of a dedicated trial campaign and accompanying numerical modelling.
This work delivered a comprehensive understanding of the underlying mechanism for generating and consequently for avoiding starting cracks.

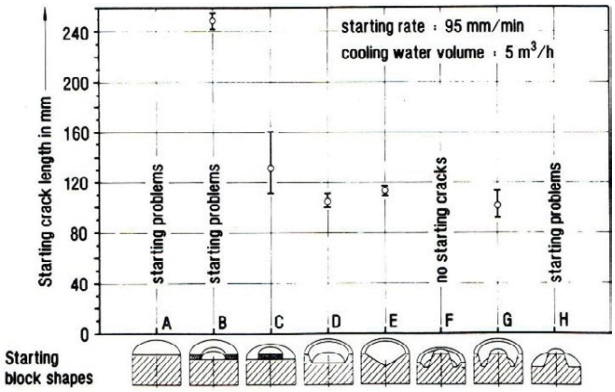

Figure 9 Influence of starting block shape on starting crack length, taken from [34]

Testing a variety of different shapes most favorable designs were identified (see Figure 9) and then the conditions of these solutions analyzed in more detail to achieve relevant criteria guaranteeing prevention of starting cracks.
This analysis included experimental evaluation of the sump depth development for the different starter blocks and casting conditions as well as numerical simulation of this transient phase of DC casting.
Figure 10 shows such a development for the shape F which represents the design avoiding any crack formation as documented in Figure 9. The main criterion for the avoidance was found to be the limitation of the liquid sump depth to lowest possible values during the starting phase.

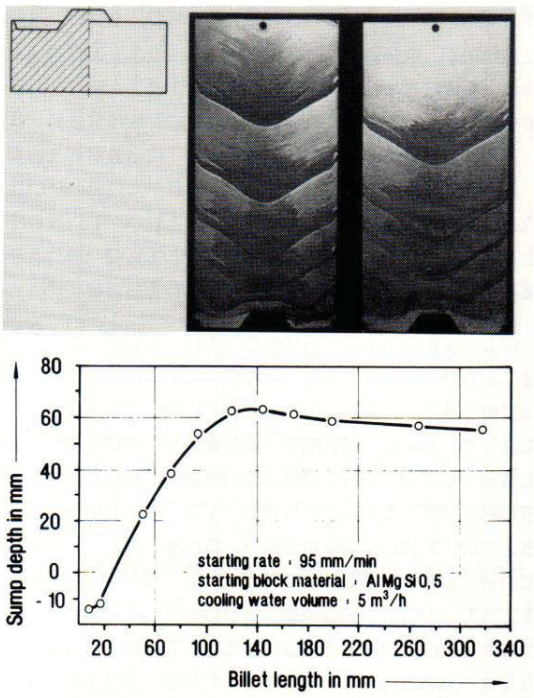

Figure 10 Sump depth formation in the ingot butt using starting block shape F, taken from [34]

Supported by the simulation results following conclusions regarding cracking risk reduction were drawn [35]:

- Changing starting block material from Steel/Cast-Iron to Al or Cu

- Reducing initial casting rate
- Changing starting block shape to type F geometry

At that time this was considered a significant contribution to Cast Shop related research and Prof. Schneider and colleagues were awarded with the Light Metals Award 1990.

Figure 11 Al-4Cu-1Sn-0.5Bi billet cast with the modified AirsolVeil® mold with graphite ring insert. Casting speed was 95 mm/min and casting lubricant 2 ml/min, taken from [38]

In the following he steered the continuing further development of the extrusion ingot casting technology AirSolVeil® which resulted in significant improvements regarding casting control and consequently billet surface quality [36] and later to even better performance in case of operational problems and easier maintenance [37].
Latest stage of development under his lead represented a version including a non-porous graphite ring insert as mold surface below the gas/lubricant feeding system which proved to enable this technology to cast specialty alloys not possible with the standard version at that time [38].
Figure 11 shows an ingot section of an Al-4Cu-1Sn-0.5Bi alloy trial. Based on the experimental verification work it was shown that due to this modification the lubricant consumption could be significantly decreased in case of casting standard alloys compared to the standard mold set up.

Concerning rolling ingot DC casting research a corresponding story can be told. Together with colleagues he first studied the impact of casting parameters and alloys on the development of butt curl and swell of rolling ingots [39] using pilot scale casting campaigns where the time dependent butt curl was measured with displacement transducers during the transient starting phase.
On the basis of these results he started another collaborative attempt to construct an improved rolling ingot starting block design in order to achieve lower butt curl and swell values allowing for less scrap due to butt sawing and finally safer casting conditions [40, 41].

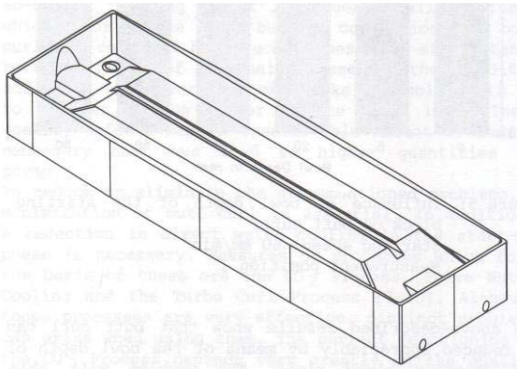

Figure 12 Optimized starting block including a middle cone, taken from [40]

Figure 12 shows the final shape of the improved starter block design which consisted of a defined depth bowl shaped block in combination with an inner rectangular cone. In the pilot scale experimental program, the supporting numerical temperature modelling in combination with measurements in the butt and mold and as well in the following full scale casting trials the improved performance was demonstrated.

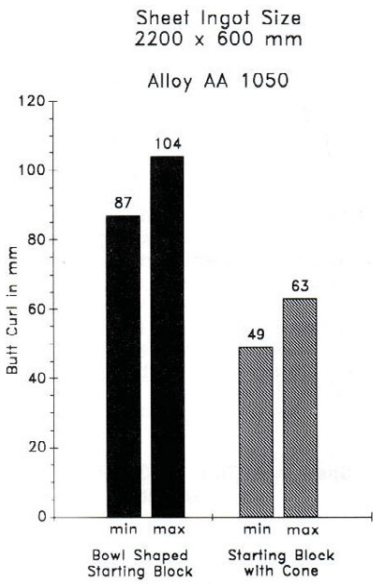

Figure 13 Butt curl of sheet ingots cast with bowl shaped starting block and with a center cone, taken from [41]

Figure 13 summarizes the significant step forward showing a comparison of minimum and maximum butt curl measured for standard and optimized mold design for a full scale 2200mm x 600mm ingot size. A reduction to almost 50% of the standard value could be achieved.
Being published in Light Metals this again was considered as a remarkable contribution and the Light Metals Award 1995 was presented to him and his co-authors.

Reviewing the impact of Prof. Schneider's contribution to DC casting research would not be complete if one would not mention his early recognition of the importance of numerical simulations of the related physical phenomena to increase as well fundamental understanding as implementation of sophisticated and pioneering solutions within the area of DC casting.

Studies on impact of melt flow on temperature and solidification in extrusion ingot DC casting [42] as well as numerical modelling based launder design for multi strand DC casting extrusion ingot units [43] stand for this visionary research approach he pursued in this area while his and his co-workers activities towards the starting phase of rolling ingot DC casting [44-46] including modelling of mold filling phase and thermo-mechanical development during cast start including butt curl and ingot shape change clearly demonstrate that for the rolling ingot area.

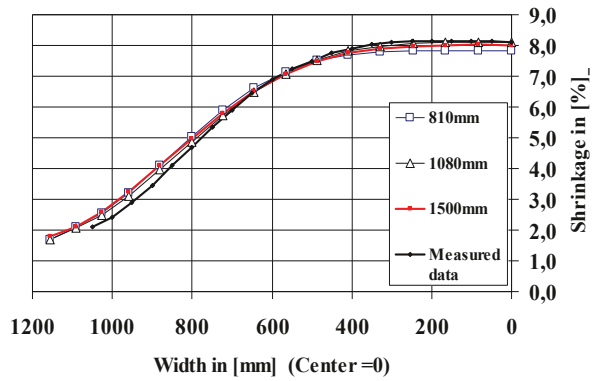

Figure 14 Calculated and measured ingot shrinkage for a 2200x600mm ingot cast at 60mm/min, taken from [46]

Figure 14 should stand as an example how these sophisticated models are capable to predict the ingot behavior and can be used to design molds and adjust casting parameters.

Already in 2002 he presented a summary on the achievements regarding research and development on DC casting in the past, the current state at this time and a personal outlook into the future where further improvements or step changes regarding this technology would be necessary [47].

Concluding remarks

Although this paper represents only a very brief review of the contributions of Prof. Schneider to the research and development of the DC casting process chain and also totally excludes his work on shape casting and foundry alloy research it is clearly visible that he significantly influenced the progress within this area during his entire research.

In addition to direct scientific work his involvement in committees and leading positions of various research and material science related organizations together with being Professor for Metallurgy at Technical University of Berlin and giving lectures at RWTH in Aachen might be considered even more important in the long term perspective. Steering activities in the Continuous Casting Committee of the German Society of Material Science, membership in different committees of the German Foundrymen Society and the Committee Light Metals of Deutsche Gesellschaft für Bergbau, Metallurgie, Rohstoff- und Umwelttechnik GDMB and last but not least his intensive involvement in the Minerals, Metals & Materials Society TMS which culminated in being the first European President of this organization in 2012.

His retirement at the end of 2014 clearly represented a loss for the entire research community of the DC casting process.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF LIQUID METAL DISTRIBUTION ON THE FLOW FIELD AND MACROSEGREGATION DURING DIRECT CHILL CASTING OF ALUMINUM ALLOY 7050

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Keywords: Direct Chill Casting, Macrosegregation, Alloy 7050

Abstract

A fully transient 2-D axisymmetric numerical model is used to investigate the effects of liquid metal distribution on the flow field and macrosegregation during DC casting. The model is applied to 500 mm diameter billets of aluminum alloy 7050 cast at 30, 60, and 90 mm/min. The predicted flow fields and composition profiles are compared for three metal feeding systems; one containing only a submerged nozzle, another with the addition of a rigid combo bag, and a final with the addition of a flow diffuser plate. The results from the numerical model show that the flow field and macrosegregation are strongly influenced by the nozzle exit velocity, which is primarily a function of the nozzle to ingot area ratio. For area ratios below 5%, jetting of the nozzle flow down the centerline is predicted, resulting in higher levels of macrosegregation. For area ratios above 5%, jetting of the nozzle flow is not predicted, but oscillations in the flow field at higher casting speeds are. The addition of either a low permeability combo bag or a flow diffuser plate is shown to prevent jetting of the nozzle flow down the centerline and to stabilize the flow field.

Introduction

Direct Chill (DC) casting is the primary casting method for wrought aluminum alloys. A major irreversible effect that occurs in DC casting is macrosegregation, defined as the spatial variation of alloying elements on the length scale of the ingot. [1] Macrosegregation impairs the quality of the ingot through nonuniform mechanical properties, caused by inhomogeneous solid solution strengthening and secondary phase fractions. Unlike microsegregation, macrosegregation cannot be alleviated by homogenization, so it is important to control during the casting process. Modifying the design of the liquid metal distribution system has been experimentally shown to improve the centerline segregation by up to 15%. [2] However, the effects of the liquid metal distribution on the flow field mechanics are still not fully understood. In this study, a continuum mixture model for the conservation of mass, momentum, energy, and species for the multicomponent solidification of alloy 7050 during DC casting is used. The effects of casting speed, nozzle diameter, and distribution design on the flow field and macrosegregation are investigated.

Numerical Model

Model Description

The conservation equation for mass (1) and momentum (2, 3) are taken from the axisymmetric continuum mixture model of Vreeman et al. [3] The source terms in the momentum equations include the thermosolutal buoyancy, the retardation of flow through the rigid mushy zone due to dendritic solidification, and the effect of free-floating solid in the slurry zone. The momentum equations are grid using the SIMPLER algorithm [4] on a staggered grid with upwind differencing for the advection terms.

$$\frac{\partial}{\partial t}(\rho)+\nabla\bullet\left(\rho\vec{V}\right)=0 \tag{1}$$

$$\begin{aligned}&\frac{\partial}{\partial t}(\rho u)+\nabla\bullet\left(\rho\vec{V}u\right)=\nabla\bullet\left(\mu_l\frac{\rho}{\rho_l}\nabla u\right)+\nabla\bullet\left(\mu_l u\nabla\left(\frac{\rho}{\rho_l}\right)\right)\\&-\nabla\bullet\left(\mu_l\frac{\rho f_s}{\rho_l}\nabla u_s\right)-\nabla\bullet\left(\mu_l u_s\nabla\left(\frac{\rho f_s}{\rho_l}\right)\right)+\nabla\bullet\left(\mu_l u_s\nabla g_s\right)\\&+\nabla\bullet\left(\mu_s g_s\nabla u_s\right)-\nabla\bullet\left[\left(\frac{\rho f_s}{f_l}\right)\left(\vec{V}-\vec{V}_s\right)\left(u-u_s\right)\right]+\rho\mathrm{B}_z-\frac{\partial P}{\partial z}\end{aligned} \tag{2}$$

$$\begin{aligned}&\frac{\partial}{\partial t}(\rho v)+\nabla\bullet\left(\rho\vec{V}v\right)=\nabla\bullet\left(\mu_l\frac{\rho}{\rho_l}\nabla v\right)-\mu_l\frac{\rho}{\rho_l}\frac{v}{r^2}\\&+\nabla\bullet\left(\mu_l v\nabla\left(\frac{\rho}{\rho_l}\right)\right)-\nabla\bullet\left(\mu_l\frac{\rho f_s}{\rho_l}\nabla v\right)+\mu_l\frac{\rho f_s}{\rho_l}\frac{v}{r^2}\\&-\nabla\bullet\left(\mu_l v\nabla\left(\frac{\rho f_s}{\rho_l}\right)\right)+\nabla\bullet\left(\mu_l v\nabla g_s\right)+\nabla\bullet\left(\bar{\mu}_s g_s\nabla v\right)\\&-\bar{\mu}_s g_s\frac{v}{r^2}-\frac{\partial P}{\partial r}\end{aligned} \tag{3}$$

Solid is assumed to nucleate as free-floating solid grains, forming a slurry of particles suspended in the liquid metal. As the particles grow and interact, the volume fraction solid, g_s , increases to a critical value (the packing fraction, $g_{s,crit}$), at which point a connected permeable structure is formed. The packing fraction is assumed to be 30% in this model, according to previous observation. [5, 6] In the slurry region, ($g_s < g_{s,crit}$), the effect of free-floating solid is included in the mixture equations. The relative vertical velocity of the solid and liquid in the slurry region is found by Stokes law,

$$\vec{V}_s-\vec{V}_l=\frac{\left(1-g_s\right)}{18\mu_m}\left(\rho_s-\rho_l\right)d^2\vec{g} \tag{4}$$

where d is the average particle diameter and μ_m is the mixture viscosity. This approximation treats the equiaxed dendrites as uniform solid spheres of a specified diameter.

In the rigid mushy zone ($g_S > g_{s,crit}$), the free-floating solid grains pack into a connected permeable structure, where there is no solid motion ($u_S = v_S = 0$), and the terms accounting for the transport of solid are removed from equations (2) and (3). To represent the liquid drag in the rigid mushy zone, the terms $\frac{\mu\rho}{K\rho_l}u$ and $\frac{\mu\rho}{K\rho_s}v$ are added to equations (2) and (3), respectively. The resulting momentum equations in the rigid mushy zone for axial and radial directions are:

$$\frac{\partial}{\partial t}(\rho u)+\nabla\bullet\left(\rho\vec{V}u\right)=\nabla\bullet\left(\mu_l\frac{\rho}{\rho_l}\nabla u\right)-\frac{\mu_l\rho}{K\rho_l}(u-u_s)-\rho B_z-\frac{\partial P}{\partial z} \quad (5)$$

$$\frac{\partial}{\partial t}(\rho v)+\nabla\bullet\left(\rho\vec{V}v\right)=\nabla\bullet\left(\mu_l\frac{\rho}{\rho_l}\nabla v\right)-\frac{\mu_l\rho}{K\rho_l}v-\mu_l\frac{\rho v}{\rho_l r^2}-\frac{\partial P}{\partial r} \quad (6)$$

where K is the permeability, modeled using the Blake-Kozeny equation, $K=\frac{\lambda_2^2}{180}\frac{(1-g_s)^3}{g_s^2}$, which is a function of the secondary dendrite arm spacing (λ_2) and the volume fraction solid (g_S).

Thermosolutal buoyancy is included in the mixture equations as a source term in the axial direction, ρB_z. The expanded form of the buoyancy source term is expressed in equation (7), as a function of the differences in temperature and composition for both the solid and liquid phases.

$$\begin{aligned}\rho B_z = {} & g_s(\rho_s-\rho_l)g-g_s\rho_s g\left[\beta_{T,s}(T-T_0)+\beta_{S,s}\left(C_s^i-C_0^i\right)\right]\\ & -g_l\rho_l g\left[\beta_{T,l}(T-T_0)+\beta_{S,l}\left(C_l^i-C_0^i\right)\right]\end{aligned} \quad (7)$$

The energy equation is based on the mixture temperature:

$$\begin{aligned}&\frac{\partial}{\partial t}\left(\rho c_p T\right)+\nabla\bullet\left(\rho c_p T\vec{V}\right)=\nabla\bullet(k\nabla T)-\frac{\partial}{\partial t}\left(\rho f_l L_f\right)\\ &-\nabla\bullet\left(\rho f_l L_f\vec{V}\right)-\nabla\bullet\left(\rho f_s\left(\left(c_{pl}-c_{ps}\right)T+L_f\right)\left(\vec{V}-V_s\right)\right)\end{aligned} \quad (8)$$

The species conservation is provided for N-1 elements of an N component alloy:

$$\begin{aligned}&\frac{\partial}{\partial t}\left(\rho C^i\right)+\nabla\bullet\left(\rho f_l\vec{V}C^i\right)=\nabla\bullet\left(\rho f_l D\nabla C^i\right)\\ &+\nabla\bullet\left(\rho D\nabla\left(C_l^i-C^i\right)\right)-\nabla\bullet\left(\rho f_s\vec{V}\left(C_l^i-C^i\right)\right)\end{aligned} \quad (9)$$

where *i* is the component tracked using the mixture composition, $C^i=f_sC_s^i+f_lC_l^i$.

Aluminum alloy 7050 is assumed to have compositions in the middle of the ranges according to ASTM B247. In this study, only Zn, Cu, and Mg are tracked based on their influence on buoyancy and tendency to segregate. To link the thermodynamic phase diagram to the energy equation, the liquidus temperature is used,

$$T=933.15+0.63-174.16C_l^{Zn}-271.55C_l^{Cu}-494.68C_l^{Mg} \quad (10)$$

where the first term is the melting temperature of pure Al and the second term is a correction for the alloying elements not tracked in this study. The partition coefficients ($k = C_s/C_l$) for the alloying elements were found to be $k^{Zn}=0.39$, $k^{Cu}=0.09$, and $k^{Mg}=0.29$. The liquidus temperature and partition coefficients were found from Thermo-Calc™ using the TCAL1 database. [7]

In this model, level-rule solidification is assumed. Depending on the initial composition of the metal, the metal can undergo either primary solidification, where most of the macrosegregation occurs, or primary solidification followed by a single secondary reaction if the mixture composition is enriched enough to undergo an isothermal eutectic reaction at 755 K. The relevant thermophysical properties for aluminum alloy 7050 are provided in Fezi et. al. [7]

Domain and Boundary Condition

The simulation domain extends from the top of the bottom block to the top of the mold. Liquid metal is added through a submerged nozzle as a boundary condition at the top of the domain. Two nozzle diameters, 20 and 40 mm, are simulated at the top-center of the domain by setting the viscosity in the control volumes of the nozzle wall to an arbitrarily high value, 10^{15}kg/ms. The nozzle is submerged 20 mm into the liquid metal and has a wall thickness of 5 mm.

During casting, the velocity of the nozzle flow is set to feed the casting speed and shrinkage in the ingot frame of reference. In order to grow the numerical domain, the top row of control volumes is expanded incrementally each time step until twice its initial height, and then is split in half. A schematic of this procedure is shown in Figure 1. The transport properties are linearly interpolated for the split control volumes.

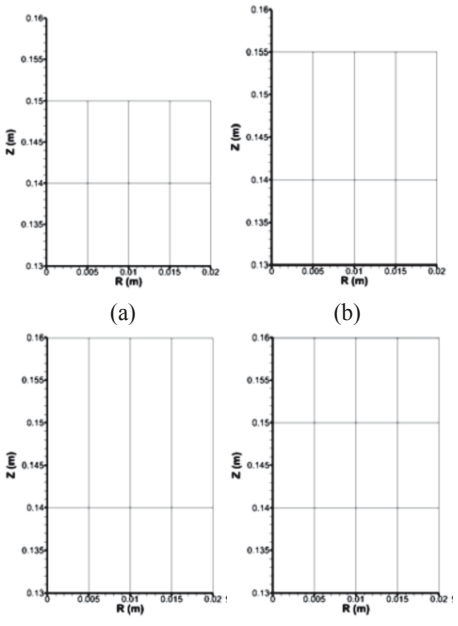


Figure 1. Schematic of the control volume expansion and splitting technique used to grow the domain: (a) start of expansion, (b) during expansion, (c) just before splitting, and (d) after splitting

Initially, the liquid metal sits at a height of 10 cm in the mold, where it is held for 60 sec in order to develop a solid shell along the bottom block and near the mold wall. The thermal boundary conditions and dimensions of the mold are described in Fezi et al. [7] After the specified hold time, the casting speed is stepped to 50% and then increased with time to 100% of the steady state value. As the domain grows, thermal boundary conditions for the water jet and falling water, taken from Weckman and Niessan [8], are applied to the region below the mold.

The combo bag and flow diffuser plate each are added to the domain by setting the permeability on the designated momentum control volume faces to 8.2 X 10^{-10} m^2 and 1.0 X 10^{-15} m^2, respectively. A schematic of the numerical domain with the addition of a combo bag and flow diffuser plate are shown with their dimensions in Figure 2.

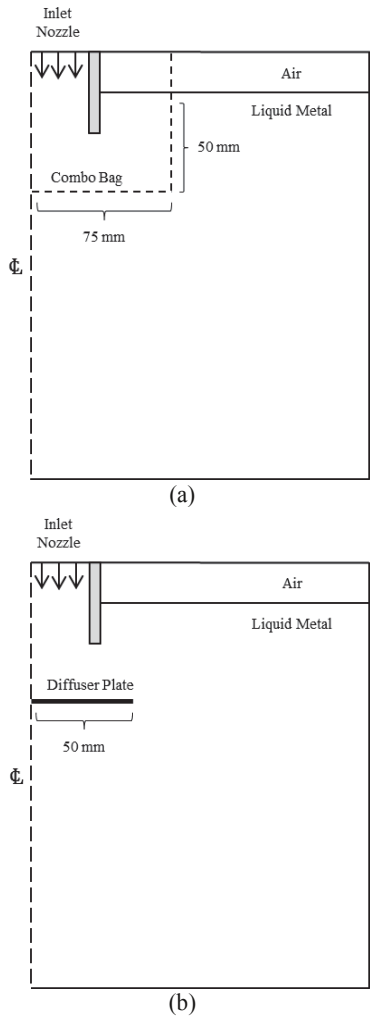


Figure 2. Schematic of the numerical domain for the (a) combo bag and (b) flow diffuser plate.

Results and Discussion

The DC casting model described is used to investigate the effects of liquid metal distribution on the flow field and macrosegregation during DC casting. A parametric study was performed focused specifically on the effects of nozzle diameter and distribution design on the flow field mechanics and level of macrosegregation. The standard operating conditions for DC casting through a submerged nozzle are provided in Table 2.

Table I. Operating Conditions for the Base Case

Ingot Diameter	500 mm
Casting Speed	30 mm/min
Nozzle Diameter	40 mm
Distribution System	Nozzle Only

During DC casting through a submerged nozzle, there are two competing flow fields in the ingot, the thermosolutal buoyancy and the forced nozzle flow. The momentum source for the thermosolutal buoyancy is the high rate of heat extraction from the mold and water jets on the outer surface of the ingot, which is primarily dependent on the casting speed. The momentum source for the nozzle flow is primarily dependent on the nozzle to ingot area ratio. Nozzle diameters of 40 and 20 mm were chosen based on comparing the nozzle to ingot area ratio to the resulting nozzle exit velocity, shown in Figure 3. The results from this comparison show that the nozzle exit velocity is very sensitive to the nozzle to ingot area ratio below values of 0.1.

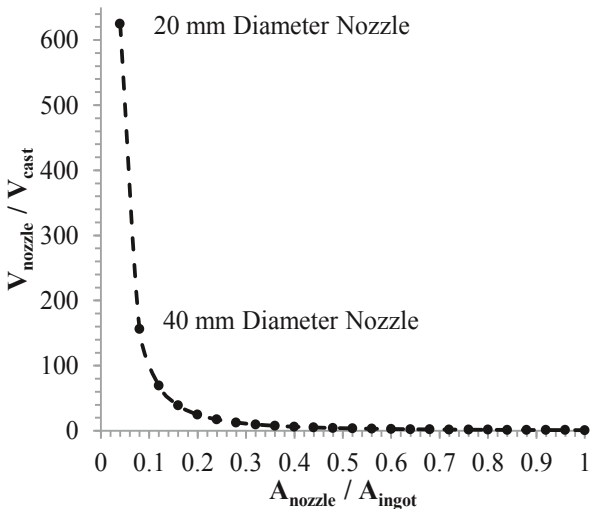


Figure 3. Plot of the normalized velocity as a function of the nozzle to ingot area ratio.

For the base case, the maximum mass flow rate caused by the forced convection is -0.0476 kg/s, while the maximum mass flow rate caused by the thermal buoyancy effects is 0.356 kg/s. The order of magnitude difference between the strengths of the two momentum sources indicates that thermal buoyancy effects dominate the flow field. As liquid metal enters mold through the submerged nozzle it becomes entrained by the thermosolutal buoyancy towards the outer surface of the ingot. Due to the slow casting speed, the profile of the sump at the centerline is flat, so there no significant horizontal component to the shrinkage driven flow to pull enriched liquid away from the centerline, as seen in the close-up view of the rigid mushy zone in Figure 4. The result is a region of positive centerline segregation that extends to ~5 mm of the ingot radius.

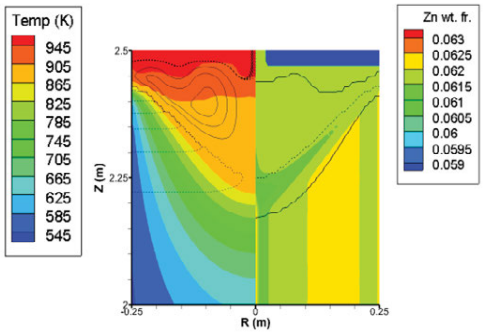
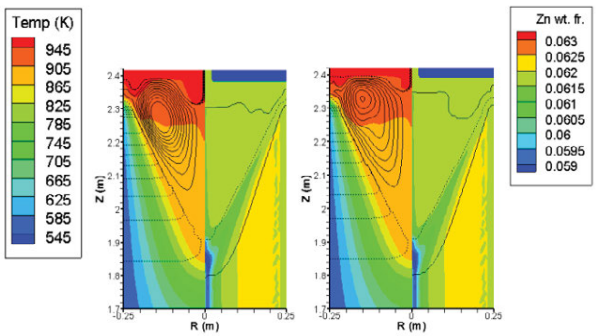


Figure 4. Close up of steady state rigid mushy zone for the standard operating case. The temperature and flow fields are shown on the left and the mixture composition, the solidus and liquidus lines (solid), and the packing interface (dashed) are shown on the right. The solid streamlines show the clockwise flow with $0.1 < \rho\Psi < 1$ and $\Delta\rho\Psi = 0.1$ kg/s and dotted streamline show the clockwise shrinkage driven flow with $-0.005 < \rho\Psi < -0.000025$ and $\Delta\rho\Psi = 0.0005$ kg/s.

The sump depth for the 40 mm diameter submerged nozzle cases increases directly with casting speed from 24 cm cast at 30 mm/min, to 59 cm at 60 mm/min, to 79 cm at 90 mm/min. The increase in sump depth indicates that the temperature gradients, and thus, the magnitude of the thermosolutal buoyancy flow cell is greater at higher casting speeds. Because the magnitude of the nozzle flow also increases with casting speed, competition between the two momentum sources prevents entrainment of the nozzle flow, destabilizing the flow field at higher casting speeds. The flow field no longer has a steady state solution and oscillates between two phases; the nozzle flow penetrating the slurry region, and the nozzle flow becoming entrained by the thermosolutal buoyancy. The oscillations in the flow field are shown in Figure 5 for a 60 mm/min casting speed.



(a) (b)

Figure 5. Close up of the rigid mushy zone, showing the oscillations in the flow field for a submerged nozzle cast at 60 mm/min: (a) nozzle flow penetrates into slurry region and (b) nozzle flow entrained by thermosolutal buoyancy.

The effect of the oscillating flow field on macrosegregation is a slight banding of the composition field along the outer radius of the ingot in the axial direction, which increases in magnitude with casting speed. The oscillations do not have a significant effect on the composition field along the centerline because the flow does not penetrate that far. As the solidification front becomes more vertical with increasing casting speed, transport of the solute-enriched liquid in the rigid mushy zone by the shrinkage driven flow has a greater horizontal component, resulting in a transition from positive centerline segregation at 30 mm/min, to negative centerline segregation at 60 and 90 mm/min. The segregation becomes more negative with casting speed, as shown in Figure 6.

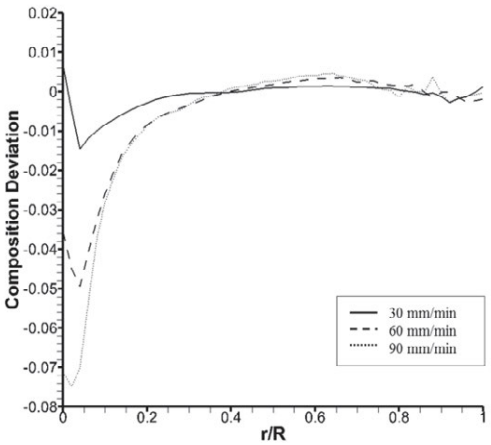


Figure 6. Normalized radial Zn segregation profile at 1.6 m from the bottom block, at V_{cast} = 30, 60 and 90 mm/min.

The nozzle to ingot area ratio decreases from 6.4% for the 40 mm diameter nozzle to 1.6% for the 20 mm diameter nozzle. This change results in a 400% increase in nozzle exit velocity for the 20 mm diameter nozzle case. Due to this significant increase in nozzle exit velocity, the nozzle flow jets down the centerline of the ingot resulting in a cliff shaped sump, as seen in Figure 7. The effect of a jetting nozzle down the centerline has been shown to increase the level of macrosegregation in the ingot, and cause banding of the composition field in the axial direction, so it is generally beneficial to avoid such an effect. [9,10] Currently, the nature of the solid packing model used to define the demarkation between the slurry and rigid mushy region cannot resolve the interaction of the nozzle jet and thermosolutal bouyancy at the top of the sump cliff. While the predicted composition field resembles reported results, its nature here is a numerical artificact, not a true model of the physical phenomenon.

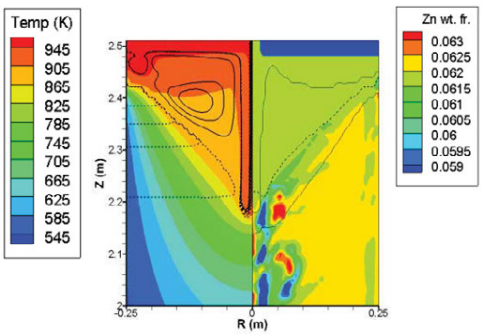
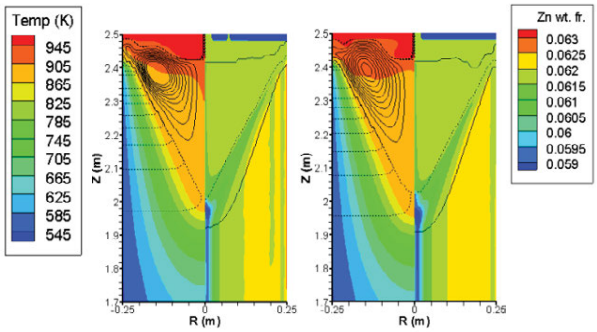


Figure 7. Close up of the rigid mushy cast at 30 mm/min, showing the effect of a jetting nozzle flow at the centerline.

In order to prevent jetting of the nozzle flow down the centerline or oscillations in the flow field, combo bags and flow diffuser plates are used. The purpose of the combo bag is to distribute the flow uniformly at the top of the liquid metal pool by damping the nozzle flow. The purpose of the flow diffuser plate is to redirect

the nozzle flow horizontally to the outer surface of the ingot. In both systems, competition between the nozzle flow and thermosolutal buoyancy is reduced resulting in the entrainment of the nozzle flow by the thermosolutal buoyancy flow cell.

For the 40 mm diameter nozzle configuration, both the combo bag and flow diffuser plate stabilize the flow field for both the 60 and 90 mm/min casting speeds. The stabilized flow fields are shown in Figure 8 for the 60 mm/min casting speed.



(a) (b)

Figure 8. Close up of the rigid mushy zone cast at 60 mm/min for the (a) combo bag and (b) diffuser plate

Although the oscillations (and related banding near the outer radius) are removed from the flow field, the net effect on macrosegregation is relatively small as shown in Figure 9 for the 60 mm/min casting speed.

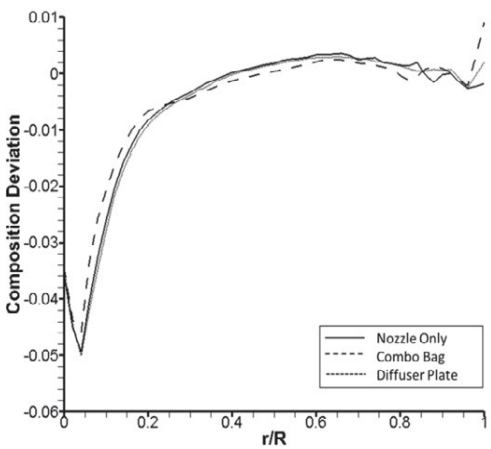
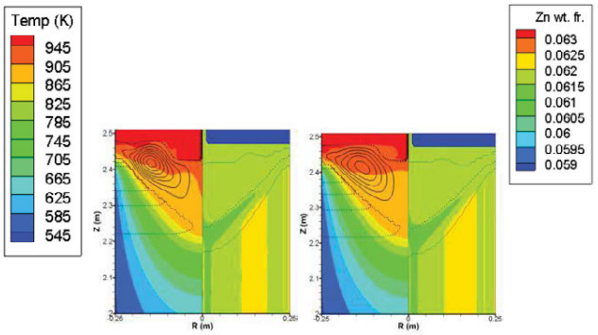


Figure 9. Normalized radial Zn segregation profile at 1.6 m from the bottom block, comparing different distribution systems for a 40 mm diameter nozzle cast at 60 mm/min.

For the 20 mm diameter nozzle, both the combo bag and flow diffuser plate prevent jetting of the nozzle flow down the centerline, as shown in Figure 10.



(a) (b)

Figure 10. Effect of the (a) combo bag and (b) diffuser plate for the 20 mm diater nozzle cast at 30 mm/min. Comparison to Fig. 7 shows the elimination in these cases of the centerline jetting.

A negligible change in macrosegregation cast at 30 mm/min is predicted for the 20 mm diameter nozzle cases compared to the base case (Figure 11).

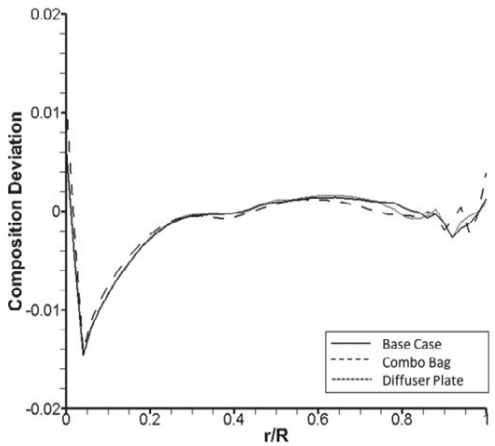


Figure 11. Normalized radial Zn segregation profile at 1.6 m from the bottom block, comparing different distribution systems for a 20 mm diameter nozzle cast at 30 mm/min.

Conclusions

A fully transient numerical model for DC casting is used to investigate the effects of liquid metal distribution on the flow field and macrosegregation of aluminum alloy 7050. The results from the numerical model show that the flow field and macrosegregation are strongly influenced by the nozzle exit velocity, which is strong function of the nozzle to ingot area ratio for values below 0.1. For the 40 mm nozzle, entrainment of the nozzle flow by the thermosolutal buoyancy is predicted at low casting speed. As casting speed increases, competition between the nozzle flow and thermosolutal buoyancy destabilizes the flow field and oscillations in the flow field develop. For the 20 mm nozzle, jetting of the nozzle flow down the centerline is predicted. The addition of either a combo bag or a flow diffuser plate is shown to prevent jetting of the nozzle flow down the centerline and to stabilize the flow field. For a given casting speed, changes in the predicted segregation profile are negligible for cases where jetting of the nozzle flow into the rigid mushy zone does not occur.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ALUMINUM BILLETS D.C. CASTING: LEVEL-POUR VS. FALL-POUR: A TECHNO-HISTORICAL APPROACH

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Keywords: Aluminium, Billets, D.C. Casting, hot-top, dip-tube, float

Abstract

Hot-top billet casting technology paved the road to excellence in many primary and secondary aluminum alloys billets cast houses during the last forty years. But the conventional dip-tube and float system kept running many others that couldn't afford the investments associated with hot-top. This paper intends to shed some light on the future of this peaceful coexistence between hot-top and spout and float billet casting systems.

Introduction

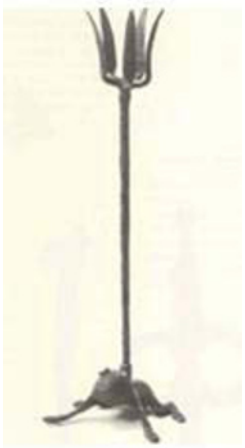


Fig. 1

The oldest known casting in existence, a copper cast frog (Fig.1) found in Mesopotamia, was cast about 3200 b. C.[1].

The English word **cast**, meaning "to form in a mold" and "that which is cast" is from late 15c and comes from uncertain Scandinavian sources [2]. An earlier meaning from mid-13c was "a throw, an act of throwing" and this sense carried an idea of "the form a thing takes after it has been thrown", which led to the meaning of forming the metal cast in a mold.

The Spanish and French translations of cast come from the latin *fondere*, pouring a liquid. *Cast House* is f*onderie* in France and *atelier de coulèe* in Québec, but *pouring* is *coulèe* in both countries. Comparing Mexico and Spain the situation is similar with *fundición* and *colada*. Conclusion: *casting* in America is also *colada* or *coulèe* and in Europe f*undición* or f*onderie*. It´s only words.

This linguistic introduction supports that words are important in technology and some scholars hold that some ideas are better explained in one language than in another. It could be added that translation is something like interpreting and English *hot-top* is not the same as French "*coulèe en charge*" , though both have the same meaning but the way of thinking in every language and the origin of the idea have an influence in the vocabulary. There are also "false friends" like English *foundry* and Spanish *fundir*. We'll come back to this point later.

The invention of Direct Chill casting is attributed to Jungham in the 1930's in Germany [3] who used it to cast bronze billets but it was soon adopted by the aluminum industry everywhere. This new technology, as usual, had pros and cons in comparison with the old permanent mold casting that had been used since the start of extrusion at the end of 19c.

The extrusion demands and internal innovation in 1980's led to much more sophisticated casting systems than the Jungham's, mainly by the integration of other operations as batching, degassing, filtering that had made casting aluminum billets a full automated process. In parallel with this integration, a new billet casting technology was made available, the so called hot-top, that had an extraordinary success since then.

But as far as metal flow rate control was concerned, the conventional old float device still competes with the hot-top, in spite of so many different features among both technologies, one of which is the price.

The above referred book by Grandfield, Eskin and Bainbridge on D.C. Casting states in page 255: *Open molds using floats and spouts for metal level control are used in some older plants, but the current standard technology is gas-pressurized hot-top molds*.

"Current standard" could be associated with "best standard" but this has never been demonstrated. This paper does a brief comparison of hot-top with spout and float technologies, aiming at finding some development lines to improve the older and cheaper system.

Description

Hot-top and dip-tube/float systems are both conventional today so in the rest of the paper the names *level-pour* and *fall-pour* respectively will be used for each one respectively.

In the earliest D.C. casting systems (see Fig. 2) a ladle was used to supply metal to each billet [4]:

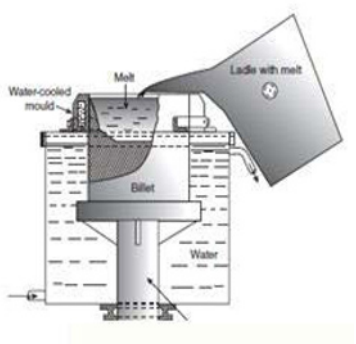


Fig. 2 Early fall-pour system

Later a trough and a distributor pan were used to connect the furnace with the molds. The furnace was emptied through its spout by free fall on the trough which led to the metal distributor

on the molds. The furnace-trough fall was later on avoided with a mechanical articulation of the furnace spout and the trough. But there were two metal levels, one in the trough and another in the mold [5], i.e. metal supply from the distributor was made by a fall (See Fig. 3).

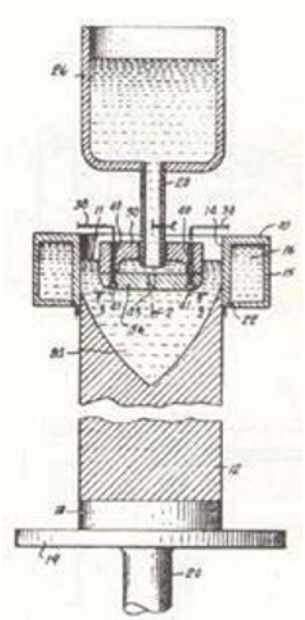


Fig. 3 Alcan fall-pour system 1970

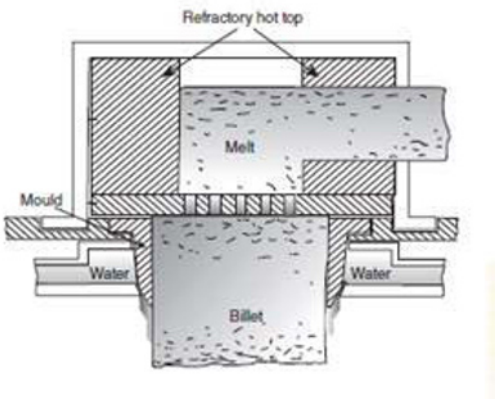


Fig. 4 Early level-pour system

Finally hot-top avoided the fall between de distributor and the mold (See Fig. 4) and there was only one metal level in the trough [6].

The name hot-top soon became popular although it didn't reflect its real technological change. *Hot top* might have meant for the first marketing campaigns the fact that on the hot-top molds there was a higher amount of metal than on the fall-pour molds. *Hot-top* was kept later on with some brands that referred to improvements of the original.

But the real change provided by hot-top had less to do with temperature than with metal falls. The important new feature of the hot-top was that the metal level in the trough was the same as the level in the mold, i.e., there was not any metal fall between them.

The French name for the new system was *coulèe en charge* that explained better the metalfluid-dynamical changes in the *hot-top*, but *hot-top* was and it is still the name kept by salesmen and users.

Metal-fall consequences

Metal fall has some important practical consequences in **turbulence** and **metal height** in the mold.

Turbulence: Spouts and dip-tubes are not synonymous. A spout is *"a tube or lip projecting from a container, through which liquid can be poured* ". To dip is *"to put something quickly into a liquid and take it out again"*. Dip-tube are supposed to be immersed in the liquid metal to guarantee a metal supply below the metal surface to avoid oxide films being entrained into the billet. It was like that until 1970s, see Fig. 3, with heavy floats, even loaded at the bottom with an steel plate serving as shutter of the dip-tube.

But once the full refractory float was implemented, the low density of the advanced refractories used for contact with molten aluminum, prevented the immersion of the float and the tip of the dip-tube doesn't dip into the molten metal, enough to guarantee subsurface supply of metal to the mold. The result is that in most of fall-pour systems, metal supply to the mold is turbulent.

This issue of turbulences became a concern of fall-pour casters, mainly when billets were extruded in thin walled shapes, but market might have adapted requirements and capabilities automatically, because no definite innovation to eliminate these turbulences was carried on.

On. top of that, the measurement of oxide films entrained inside the billet cannot be done accurately and comparison of this characteristic in both types of billets has to be done with extrusion trials, what is also cumbersome and not very reliable.

This is an unresolved matter, but paradoxically it can be solved with simple modifications of the float to ensure it sinks enough in the liquid pool to guarantee subsurface supply of metal.

Metal height in the mold: Another weak point for the ancient fall-pour casting system is the need to control the metal level in the mold. The shorter the metal-mold contact the thinner the shell zone will be, but the risk of cold shuts will increase. So fall-pour systems used to make 10mm thick shell zone before hot-top appearance, which design avoided shell-zone formation.

This was the main competitive advantage claimed by former hot-top marketing campaigns, not the cleanliness issue that was more difficult to proof. Thickness of the shell zone started then to be controlled in the fall-pour systems and it was soon admitted the disadvantage of fall versus level pour, because average shell zone was bigger with fall-pour than with level-pour, where it was neglectible.

This drawback was soon fixed by simply adding a more insulating material to the mold, first graphite, then calcium silicate, etc.

Economical comparison

1. Safety

Molten aluminum explosions during the cast are by far the worst risk associated to vertical D.C. It is well known that these explosions are strongly related to the coating of metallic structures that can be in contact with molten metal, in the case of a spillage.

Another safety issue related to casting are metal splashes at the start of the cast, when the tooling is cold and may have

moisture. Splashes are a minor problem compared to explosions, but they represent indeed an important safety concern of molten metal handling, one of the first safety issues in any aluminum casting operation.

Explosions and splashes belong to different categories of safety issues and the comparison must be done separately.

Explosions.

Trying to make an economic assessment of the explosion risks associated with two casting systems must start with statistics and probability.

But a reference to a real case seemed preferable to start this paragraph because explosions are a very odd event and one could tend to neglect its importance. The case selected was the only duly registered molten aluminum explosion occurred in Spain during a D.C. cast. It was in 1962, in the Avilés smelter of ENDASA and produced three deaths. Fig 6 pictures need no comments.

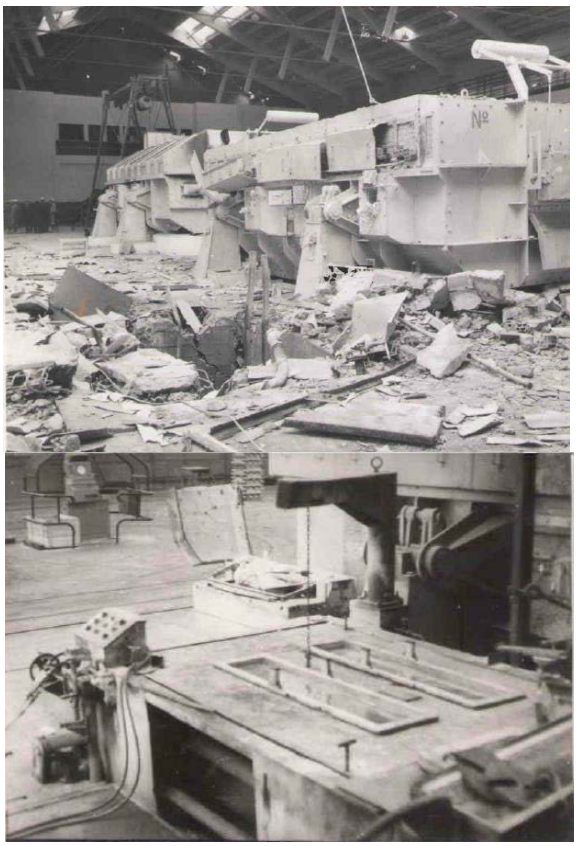


Fig. 6. Explosion in a D.C. casting pit

What is the probability of an explosion with loss of human life during the cast? It can range between 1/100.000 and 1/100.000.000, the high numbers coming from the casting engineers and the low from the sales specialists. A thorough study of this question is out of this paper's scope but the previous estimations show that an accurate answer will be impossible to find.

It is pertinent in this point to ask another question: How many explosions with loss of human life during the cast have occurred in the last 50 years? The author does not know any statistics where the relationship between different casting systems and explosions in D.C. castings could be studied. These events, like any other with human life loss, have many socio-economic implications and are not dealt by companies with transparency, even internally, not to mention the reluctance to share information between them.

This is not a question of company size, nor geographic area, most of companies do research of these fatalities but don't feel confident in sharing their conclusions. This is an unfinished issue, not only for aluminum companies but also for any kind of national or international gathering that include owners or operators of a DC casting machine. And overall it is an unfinished issue for every institutions that represents companies and/or professionals involved in the operation of DC casting units.

Without statistical data about explosions it is difficult to go ahead in the comparison of both casting systems, and if it is not possible to compare the highest risk, the rest of comparisons are irrelevant on trying to find out a whole economic assessment of both technologies.

To explore this question it is suggested to read the *Report on the 1986 Challenger Shuttle Explosion* written by the Nobel Prize Richard Feynman. This report started with the same question of probability that starts this paragraph. And Feynman's conclusion is about confusing reality and public relations, because there is no way to know the probability of such an explosion. In D.C. casting the conclusion is similar, so as far as explosions is concerned, fall-pour and level-pour are equivalent.

Splashes

Splashes lack also of an accurate quantification, what makes comparison something subjective. But it goes without saying that fall-pour operation has more unsafe working conditions than level-pour, as far as metal splashes during the start-up is concerned. Level pour casting system has allowed a higher degree of automation, that cannot be achieved with the fall-pour because of the necessary manipulation of floats at the start of the cast, what is usually performed by operators standing on the casting table during the start, as fifty years ago.

2. Investment

It is clear enough that level pour is much more expensive.

3. Quality

The chances of oxides entrained into the billet are much higher in fall-pour. The influence of this quality feature depends of the thickness of the extrusions to be made.

4. Operation

Changing sizes and alloys is much easier with fall-pour casting equipment but automation is much easier in level-pour.

Conclusions/Improvements in fall-pour system

The comparison of both billet casting systems has given two lines of development of the *fall-pour* system to make it sustainable. One is related to quality and the other to safety.

A new float

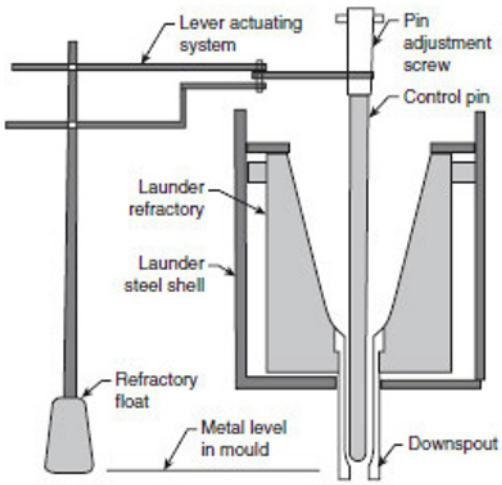
Elimination of turbulences at the metal discharge on the float requires a new float with minor modifications, removing some immersed material of the float, so it sinks more and its shutter

surface is 4-5 mm below metal level, enough to avoid turbulences. The author performed a new float trial ten years ago, that failed but gave clues for a new trial that was never done for several circumstances. But it can be said that changing the float design to guarantee a metal feeding without turbulences is something that any casting operator could do.

An automated steady-eddy

The steady-eddy was one of the first solutions used in DC casting to fight turbulences, when adjusting the metal flow rate was manual. It is easier to control the flow through the dip tube inserting a rod in the upper hole than putting a plug in the lower hole. This was the steady-eddy principle.

But the number of billets increased and controlling manually each billet was unaffordable, and the float did the job by flotation, i. e., by itself. Steady-eddy is a standard technology in most of sheet ingot casting systems. See Fig. 7.



With more than one hundred billets per cast, a steady-eddy approach has many peculiarities that haven't been solved with sheet ingot casting, where the number of pieces involved are below ten. The investment required for this development is much bigger than for the new float, but the potential benefits are proportional to that investment.

Fall-pour has other features that mean a competitive advantage, as the independence of the casting table and the metal distributor and the open mold. These features justify the current smaller investment and better operation costs for fall-pour.

An automated steady-eddy for billets could make fall-pour overtake level-pour, because it would even up safety and quality, with less investment and lower operation costs.

One must believe in a miracle to make it happen but this steady-eddy for billets is not a fantasy when one has a look at the 3711 patents on continuous casting registered from 1960 to 1980 [9].

Epilogue

Casting aluminum billets is today an 80 years old technology which during the last 40 has maintained two alternatives in operation to control the metal flow rate to the mold: hot-top and dip-tube and float. Among the many explanations given to this duality the most common was that each one is suitable for its own extrusion market niche. But the reluctance to change and the lack of necessity could be other reasons.

History proofs that most of inventions, in any field, are generally improvements of previous ones, and new inventions are often superseded by revamped old ones. It could happen this with the level-pour casting system for billets, that forty year after its invention, could be put aside by a revamped fall-pour, much safer than its predecessors and more efficient than its competitors.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

HOT TEARING IN DC CASTING INGOT OF 7XXX ALUMINUM ALLOYS

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Keywords: Stress-strain curve, Semi-solid state, Hot tearing

Abstract

The initiation mechanism of hot tearing in direct chill casting ingot of 7XXX aluminum alloy was investigated. In 7XXX aluminum alloys (arranged 7055 aluminum alloy), the temperature range between solidus (463℃) and 583℃ was classified into brittle range on mechanical properties and deformation behaviors in semi-solid state. The brittle range of 7XXX aluminum alloys was larger than that of other general aluminum alloys. In 7XXX aluminum alloy, the healing of the crack which initiated from shrinkage cavity was difficult to occur especially in the center of the direct chill casting ingot which was the last solidification part. Therefore, it was considered that the hot tearing in direct chill casting ingot of 7XXX aluminum alloys was easy to initiate the crack in the center of the ingot.

Introduction

Ingots produced via direct chill (DC) casting of aluminum alloys are, in general, prone to hot tearing. Many studies have focused on determining the mechanism that governs hot tearing [1-3]. Experimental results indicated that the susceptibility to, and morphology of, hot tearing vary with the chemical composition of these alloys. For example, 1XXX series alloys are almost completely resistant to hot tearing. In contrast, the high-strength 7XXX and 2XXX series are highly susceptible to hot tearing. Hot tearing occurs in the center of the ingots of these alloys and the cracks are referred to as “center cracks” or “star cracks”. On the other hand, hot tearing of the 5XXX and 3XXX series occurs primarily on the surface of the ingot and hence these cracks are referred to as “surface cracks”. In recent years, the computer-aided engineering (CAE) has been used to evaluate hot tearing [4-6]. However, the relation between the chemical composition and morphology of hot tearing remains unclear. The mechanical properties in semi-solid state of pure aluminum and Al-4.5mass%Cu have been evaluated in previous studies [7-9]. In this study, the mechanical properties of 7XXX series aluminum alloys are evaluated; these properties are then compared with those of the semi-solid aluminum and Al-4.5mass%Cu, in order to determine the correlation between the chemical composition and the morphology of hot tearing.

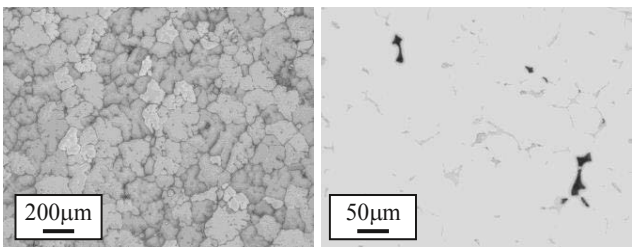
Experimental procedure

Materials and specimen preparation

A 7XXX series alloy (arranged 7055), which has the same Mg and Zn content as 7055 aluminum alloy, was selected as the sample material. Billets (diameter: 330 mm) of the arranged 7055 were fabricated via DC casting. The chemical composition of each billet is summarized in Table I. For use in the tensile test, plate-type specimens with parallel sections having dimensions of 4 mm (thickness) × 3 mm (width) × 10 mm (length) were fabricated from constant domains of this billet; Figure 1 shows the dendritic microstructures of the specimens. In addition, scanning electron microscopy combined with energy dispersive spectroscopy revealed that Al, Zn, Cu, and Mg containing compounds formed in the grain boundaries and inter-dendrite arms. Figure 2 shows the result of the differential thermal analysis (DTA) of the arranged 7055 specimens. The solidus and liquidus occur at 463℃ and 635℃, respectively, and a peak occurs at temperatures ranging from the solidus to 475℃. Figure 3 shows the phase change that occurs during the solidification of arranged 7055, as determined by JMatPro V8.0 (Practical software for material properties, Sente Software Ltd.). The solidus and crystallization temperature of α-Al, 450.0℃ and 631.2℃, respectively, are similar to those determined via the DTA. In addition, the eutectic reaction of α-Al and Mg_2(Zn, Cu, Al) at 468.0℃ confirms that the compounds in the grain boundaries and inter-dendrite arms are Mg_2(Zn, Cu, Al). Therefore, the peak observed at temperatures ranging from the solidus to 475℃ (as determined from the DTA) resulted from the eutectic reaction of the α-Al and Mg_2(Zn, Cu, Al).

Table I. Chemical composition (mass%)

Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Zr	Al
0.06	0.09	2.34	<0.01	1.70	<0.01	8.88	0.04	0.12	Bal.



(a) Low magnification (Anodized and observed with polarized light)

(b) High magnification (as polished)

Figure 1. Microstructures of ingot.

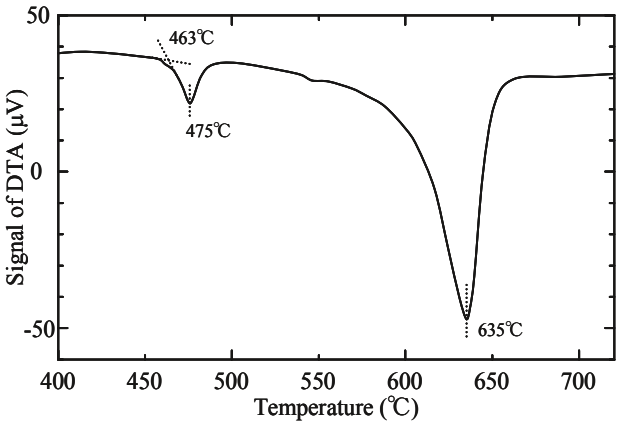


Figure 2. Result of differential thermal analysis of ingot.

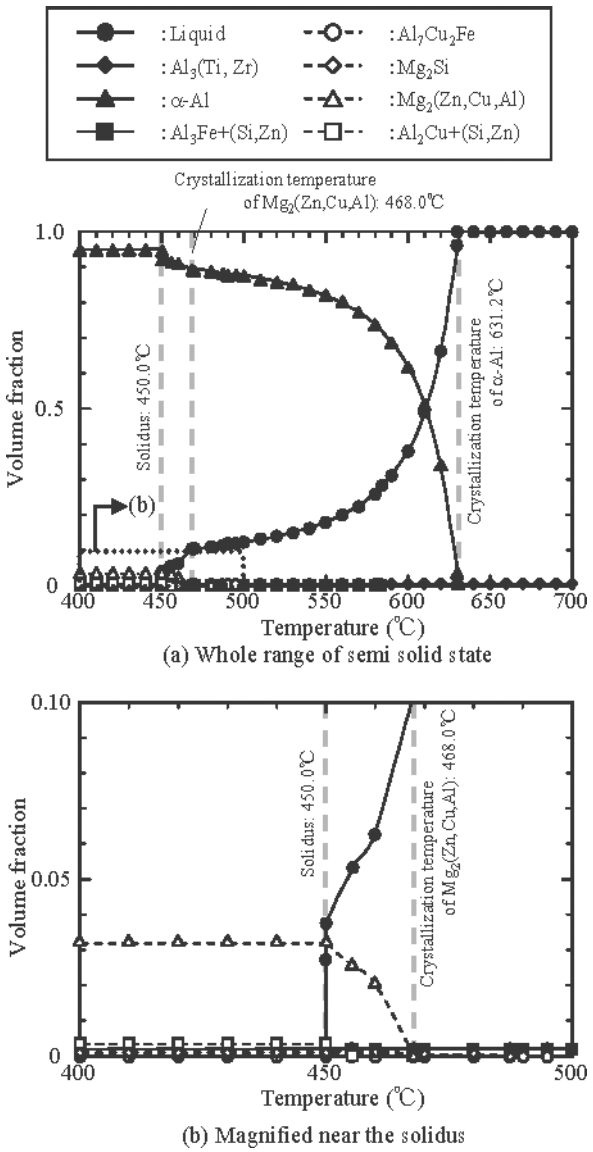


Figure3 Calculation result of phase changing during solidification in arranged 7055 by JMatPro V8.0.

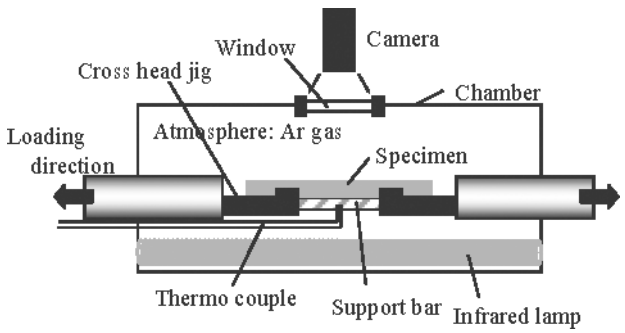


Figure 4. Schematic drawing of tensile test machine.

Tensile test in the semi-solid state

A schematic of the instrument used in the tensile test is shown in Figure 4. The details of the tensile test have been provided in previous studies [7-9].

Results

Tensile test and in situ observation

Figure 5 shows the tensile properties and the classification of deformation behaviors of arranged 7055; the corresponding data, which were reported in previous studies of pure aluminum and Al-4.5mass%Cu, are also shown in the figure. As the figure shows, arranged 7055 has a tensile strength of >15 MPa in the solid state, i.e., at temperatures lower than the solidus (463℃) and undergoes significant elongation, without fracturing. The elongation decreased significantly at temperatures ranging from the solid state to the solidus. In addition, the specimen exhibited brittle fracture at the solidus (Figure 6), and did not become elongated in the semi-solid state (i.e., at temperatures higher than the solidus). The tensile strength decreased significantly at temperatures ranging from the solidus to 475℃ and was only 1.3 MPa at 475℃ ; the tensile strength decreased more slowly at temperatures ranging from 475℃–597℃. Moreover, the sample underwent healing at temperatures higher than 583℃ and hence did not break after crack initiation, because the liquid healed the cracks in this temperature range (Figure 7). The tensile load could not be measured at 606℃, as there was no clear boundary between the solid and the liquid at this temperature. Furthermore, after crack initiation (Figure 8), the specimen deformed to a sandglass shape and then became a slurry.

Fracture surface

The fracture surfaces formed at the solidus (463℃), 468℃, 475℃ , and 483℃ are shown in Figure 9. The morphology of the fracture surface changes at temperatures ranging from 468℃–475℃. In addition, Al, Zn, Cu, and Mg containing compounds (i.e., $Mg_2(Zn, Cu, Al)$) that stuck to the fracture surfaces at both the solidus (463℃) and 468℃, were not observed at 475℃ and 483℃. The fracture surfaces formed at 475℃ and 483℃ were also more rounded than those formed at 463℃ and 468℃. Furthermore, an Al, Fe, and Cu containing compound (i.e., Al_7Cu_2Fe) was observed on the fracture surfaces formed at 475℃ and 483℃; Al_7Cu_2Fe crystallized at temperatures higher than 475℃ (Figure3).

Discussion

Mechanical properties

・The tensile strength decreased significantly, modestly, and could not be measured at temperatures of 463℃–475℃,475℃–597℃, and over 603℃, respectively.

・The eutectic reaction that occurred at temperatures ranging from 463℃–475℃ was identified via the DTA.

・The fracture surfaces formed at temperatures ranging from 468℃–475℃ had a more rounded morphology than those formed at 463℃. Moreover, at temperatures higher than 475℃, Mg_2(Zn,

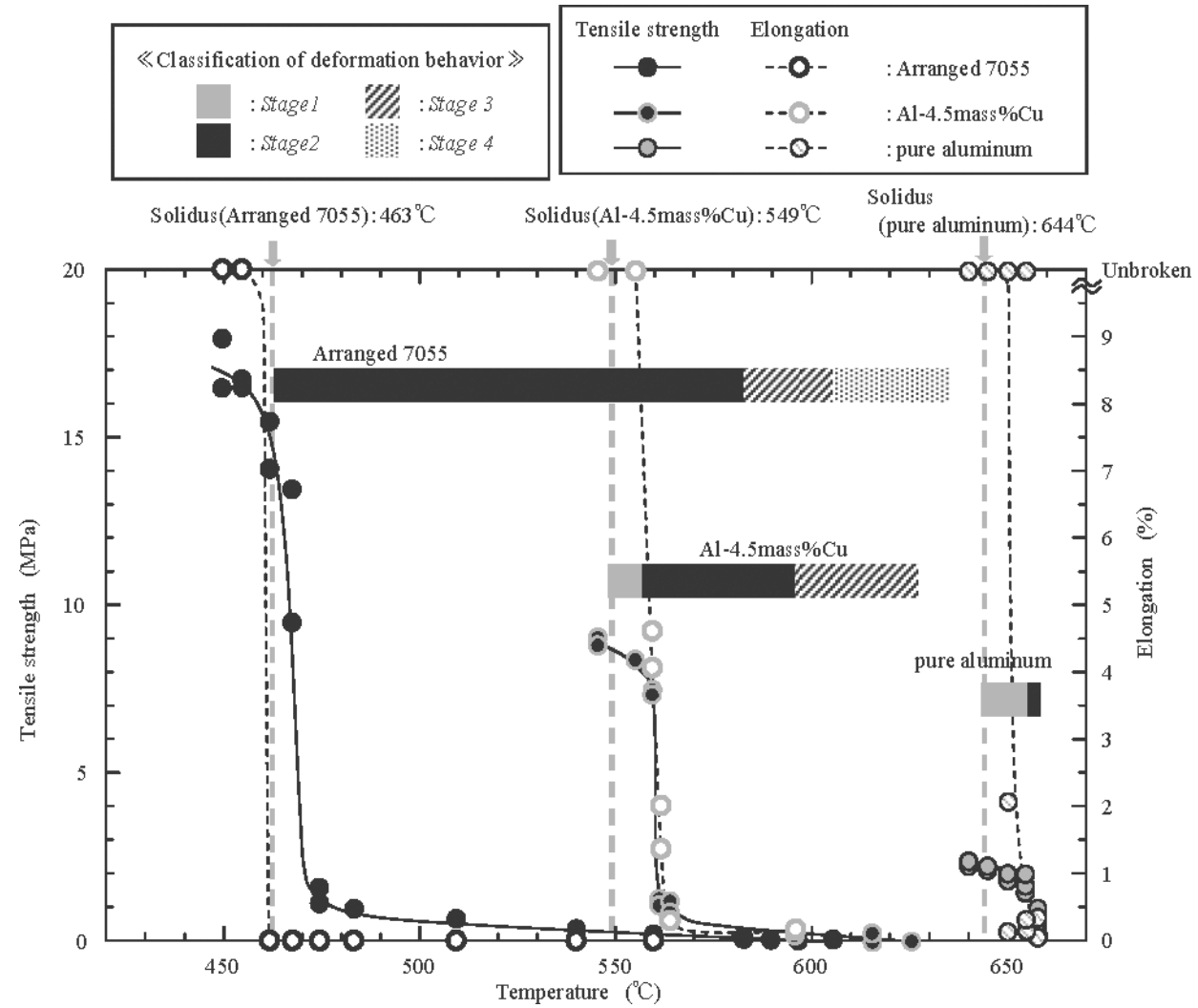
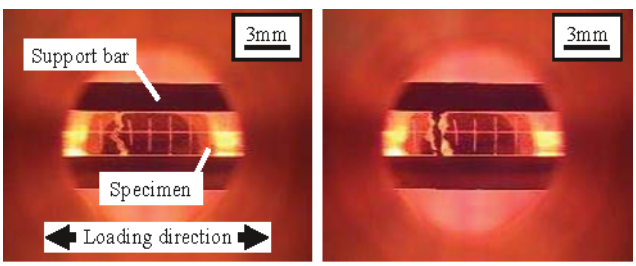
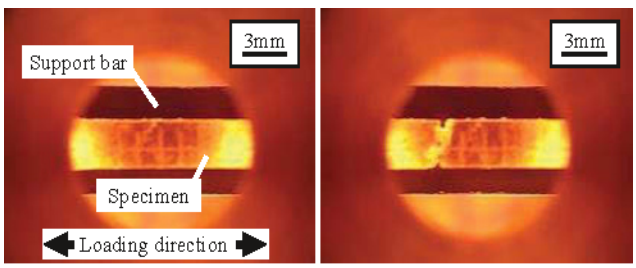


Figure 5. Change in the mechanical properties and deformation behaviors of arranged 7055, Al-4.5mass%Cu and pure aluminum. (Classification of deformation behaviors was explain in Table II.)

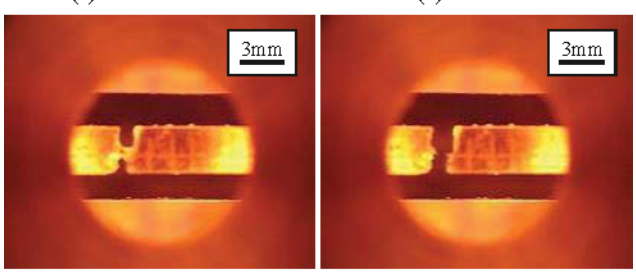


(a) Crack initiation (b) After 15s

Figure 6. Deformation behavior of arranged 7055 tested in solidus(463℃).



(a) Crack initiation (b) After 20s



(c) After 30s (d) After 40s

Figure 7. Deformation behavior of arranged 7055 tested in 583℃.

Cu, Al) did not form on the fracture surfaces, despite being present at lower temperatures.

These results indicate that the significant decrease in tensile strength at temperatures ranging from 463℃–475℃ stems from the melting of Mg_2(Zn, Cu, Al) at the grain boundaries. The sticking that occurs on the fracture surface of semi-solid Al-4.5mass%Cu, and which constitutes remnants of solid-solid contact, does not occur at 475℃. Therefore, at 475℃, the fracture surface is covered mainly by liquid and the solid-solid contact ceases. Moreover, smoothing of the solid grains, and the accompanying decrease in the mechanical resistance to deformation, results in a corresponding decrease in the tensile strength at temperatures ranging from 475℃–603℃.

At temperatures near the solidus, the tensile strength of solid-state arranged 7055 was higher than that of other aluminum alloys, such as Al-4.5mass%Cu. Therefore, if the solid grain is not deformed, then arranged 7055 undergoes brittle fracture owing to the stress concentration at the grain boundary, stemming from partial melting of the compound at the boundary. The compound observed in the fracture surface at 463℃ and 468℃ corresponds to the trace of brittle fracture of the compound in the grain boundary. At temperatures higher than 475℃, the grain boundary is covered by the liquid, and solid-solid contact ceases. As such, the solid grain did not deform in this temperature range.

Classification of the deformation behavior

The deformation behavior of arranged 7055 is classified into four stages, as shown in Table II [8]. As the table shows, the elongation at the solidus is significantly lower than at other temperatures; cracks heal at temperatures higher than 583℃ and the specimen becomes a slurry at 603℃ . Therefore, temperature ranges of 463 ℃–583 ℃ , 583 ℃–608 ℃ , and 608 ℃–635 ℃ (liquidus) were classified as Stage2, Stage3, and Stage4 (slurry), respectively. Stage1 was, however, not confirmed for the semi-solid state of arranged 7055.

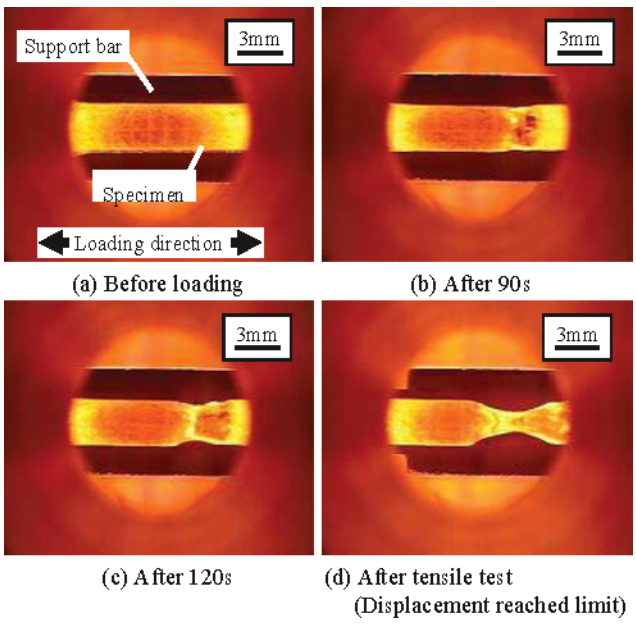


Figure 8. Deformation behavior of arranged 7055 tested in 606℃.

The hot tearing of arranged 7055

The schematic in Figure 10 shows the phase change that occurs during the solidification of arranged 7055. As the figure shows, α-Al crystallizes at the liquidus but constitutes only a low volume fraction and hence, floats on the liquid in Stage4. Contact between the solid grains (α-Al) is initiated in Stage3 (temperatures lower than 603 ℃), with increasing volume fraction of the solid. However, the surface of the solid grain (grain boundary) is still covered by liquid; this corresponds to the mushy state. In Stage2 (temperatures lower than 583℃), the high volume fraction of solid restricts the flow of the liquid in the grain boundary. The liquid exists only in the form of a film in the boundary and cracks are not healed in this stage. JMatPro V8.0 was used to determine the crystallization of Al_7Cu_2Fe, which was observed in the fracture surface at temperatures higher than 475℃. The eutectic reaction occurs at 475℃ and results in a significant decrease in the volume fraction of the liquid; the specimen becomes fully solidified at the solidus. In Stage3 and Stage4, cracks initiated by solidification shrinkage or an external force, are healed by the liquid. Therefore, crack propagation is hindered in this temperature range. However, cracks are not healed in Stage2 and hence crack initiation and propagation occur readily in this temperature range.

The schematic in Figure 11 shows the solidification that occurs during DC casting of 7XXX and other series of aluminum alloys, such as Al-4.5mass%Cu. The sump of 7XXX series aluminum alloy is deeper than that of the other series, because the temperature range of the semi-solid state in the former is wider than that of the latter; the magnitude of the temperature range of Stage2 is also larger in the case of the 7XXX series. In fact, cracks initiated in Stage2 of this series, are not easily healed by the liquid. Liquid flow to the cracks is possible in Stage3, but the cracks are separated by large distances. Crack propagation occurs readily in the sump, because there is no Stage1 (in the sump), and the cracks are located at large distances from the ductile solid

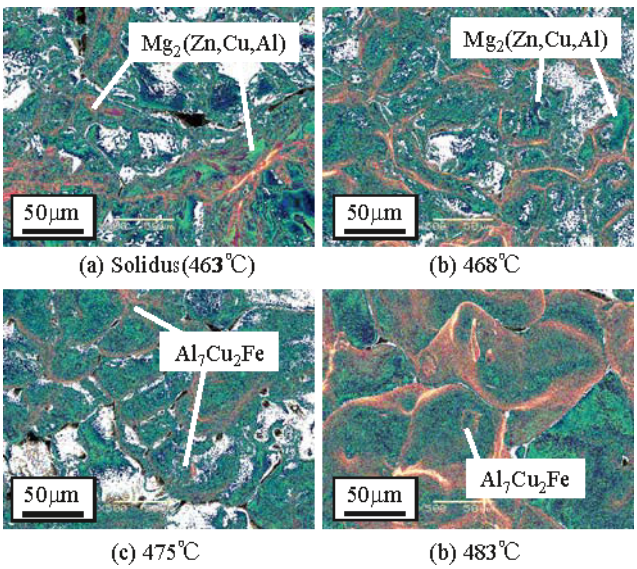


Figure 9. SEM micrographs of fracture surfaces on arranged 7055 in semi-solid state.

state. Moreover, the distance between Stage2 and Stage1 or Stage3 in other series of aluminum alloys is lower than that of the 7XXX series; crack initiation and propagation are therefore more difficult in the other alloys than in the 7XXX series. Furthermore, shrinkage of the ingot during the final stage of solidification leads to a stress concentration. The shrinkage cavity, which becomes the crack initiation site, is more easily formed in the center of the ingot than in other regions; the center of the ingot solidifies last. Therefore, hot tearing occurs in the center of the ingot of the 7XXX series aluminum alloy.

The occurrence of hot tearing is influenced by the casting method and casting design. However, the magnitude of the temperature range of Stage2, where crack initiation and propagation occur readily, is common to all, but independent of the, casting methods and design. This magnitude can therefore be used to evaluate the susceptibility of the ingot to hot tearing.

Table II. Classifications of deformation behavior in semi-solid state

Low Temperature ⇕ *High Temperature*	*Stage1*	Large uniform deformation and high tensile strength are obtained.
	Stage2	Embrittlement with liquid film is occurred.
	Stage3	Healing behavior of crack by liquid phase flow is occurred.
	Stage4	Surly zone. Specimen could not keep original shape.

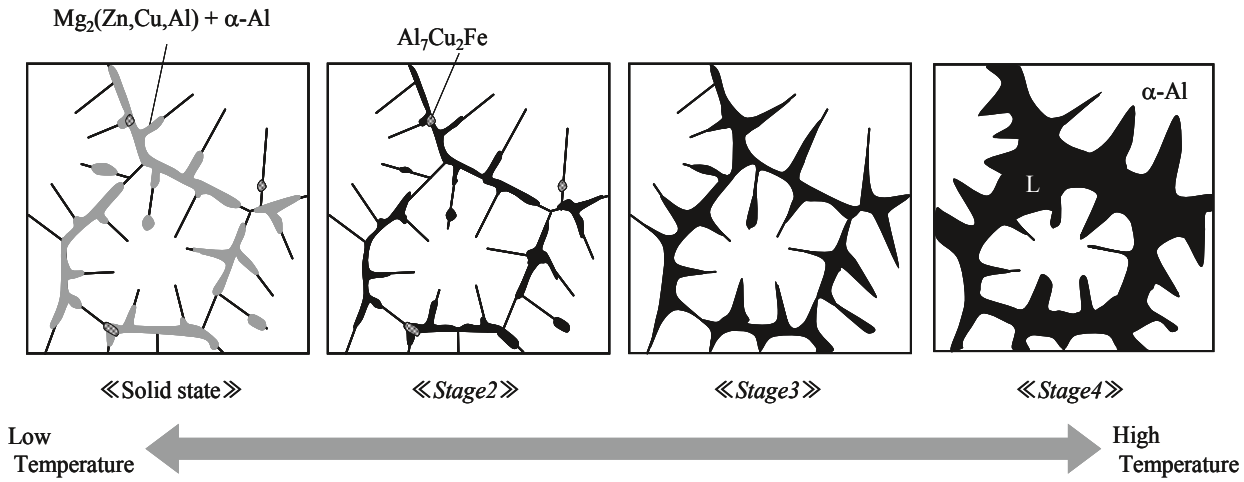
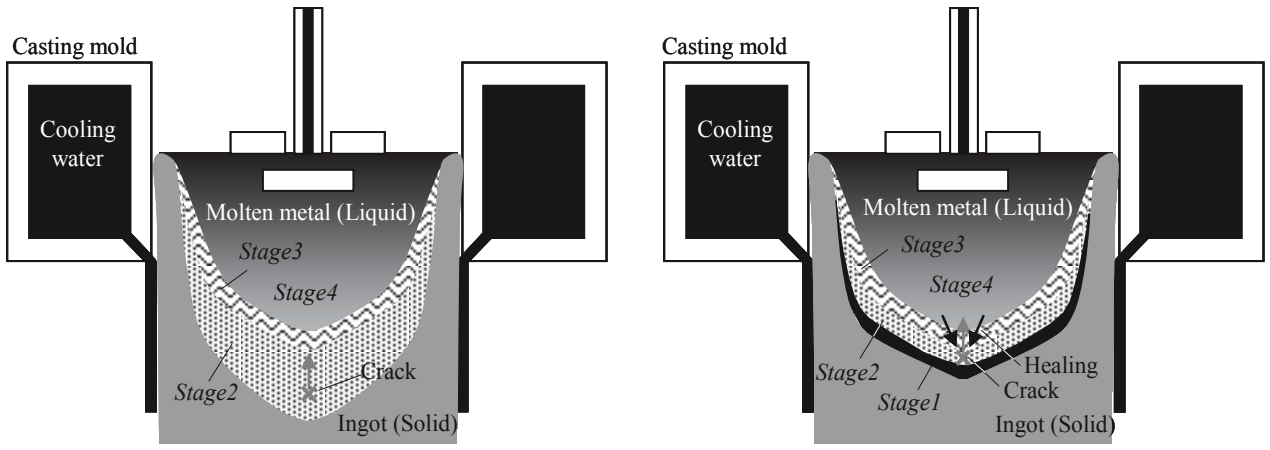


Figure 10. Schematic drawings of solidification of arranged 7055.



(a) 7XXX series aluminum alloys (b) Other series aluminum alloys

Figure 11. Schematic drawings of solidifications of 7XXX series aluminum alloys and other series aluminum alloys in DC casting.

Conclusion

An alloy which has the same Mg and Zn content as the 7055 aluminum alloy, exhibited brittle behavior and existed as a semi-solid at temperatures ranging from the solidus to 583℃. This temperature range was wider than those of the other aluminum alloys (pure aluminum and Al-4.5mass%Cu). Therefore, in the case of the 7XXX series, the magnitude of this temperature range in the sump is larger than those of the other series of alloys. Compared to the other alloys, crack initiation and propagation occurred more readily in the 7XXX series; this resulted from the greater difference between the range of temperatures at which brittle behavior was exhibited and crack healing occurred (or ductility was generated). During DC casting, the center of the ingot solidified last, and stress concentrations generated by the shrinkage led to the formation of a shrinkage cavity. Therefore, hot tearing occurred in the center of the ingot of the 7XXX series aluminum alloy.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Initial Development of Micro-Shrinkage Crack During Early Stages of Direct Chill Casting of Al-4.5% Cu Alloy

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Keywords: Direct Chill Casting, Thermal Convection, Interdendritic strain, Macrosegregation, Micro-Shrinkage Crack, Aluminum alloys.

Abstract

An investigation has been carried out under controlled laboratory conditions designed to approximate the conditions encountered during the early stages of direct chill of Al-4:5% Cu aluminum alloy ingots to study the formation of micro-shrinkage crack. Also, a couple models have been established at the author's organization to compute convection patterns, thermal fields, interdendritic thermo-metallurgical strain and macrosegregation distributions. Comparisons between the available microstructure taken from the samples and model predictions of convection streams to verify the mechanism of the random sedimentation of free crystals and therefore the micro-shrinkage crack formation were performed where good agreements were obtained. The effects of convection streams of various melt superheats at different locations in the aluminum ingots on the formation micro-shrinkage crack have been studied and analyzed. The mechanism of micro-shrinkage crack formation with different convection streams patterns and superheats were also discussed.

I- INTRODUCTION

Micro-shrinkage crack together with micro-porosity have always been recognized as major defects in casting of aluminum alloys.[1,2] The micro-shrinkage crack being typical of semi-continuous casting whereas the micro-porosity is dominant in shape castings. As explained in the lecture, these two defects interconnected and their surfaces are intergranular profile exhibiting a smooth surface with a few small spikes.[1-3]

The formation of micro-shrinkage crack is associated with two different mechanisms,[4] namely the lack of interdendritic melt flow to feed the solidification shrinkage, and a too large thermally induced deformation caused by the non-uniform cooling contraction of the casting. The formation mechanisms of this crack have been subject of numerous studies, which are very well summarized in the recent review of Eskin et. al.[5] From this summary of numerous experimental and simulations studies on this topic, it appears that this crack nucleate with the formation of random network of equiaxed grains associated with the sedimentation of free crystals controlled by convection stream patterns.[6-8] As these crystals grow in incoherent and coherent mushy regions, the mushy solid globally contracts, this can compensate solidification shrinkage. In the coherent mushy region where it is under global expansion, the dilatation of the mushy solid adds to the solidification shrinkage. This may induce micro-shrinkage crack if liquid feeding is in-sufficient. However, the thermal strains impose significant deformations to the partially coherent solid in the last stages of the coherent mushy region. This will be localized in the coherent mushy region where the mushy zone is most vulnerable to tensile strains, i. e. at grain boundaries.

In the practices of semi-continuous direct chill (DC) casting process, variables such as fluid flow, thermal, strain, and strain-rate combinations will generate unfavorable flow patterns and spawn abnormally high thermal stresses that result in the formation in several types of imperfections within the cast component. Defects including loss of dimensional stability and hot tearing, micro-shrinkage cracks are three of the main quality concerns related to thermo mechanical behavior with unfavorable sedimentation of free crystals associated with unfavorable convection stream patterns. Over the past few decades, a number of models have been developed[9-12] to predict the fluid flow patterns and the generation of stresses and strains within the casting, which in turn provides an assessment of when and how these defects can arise. One of the

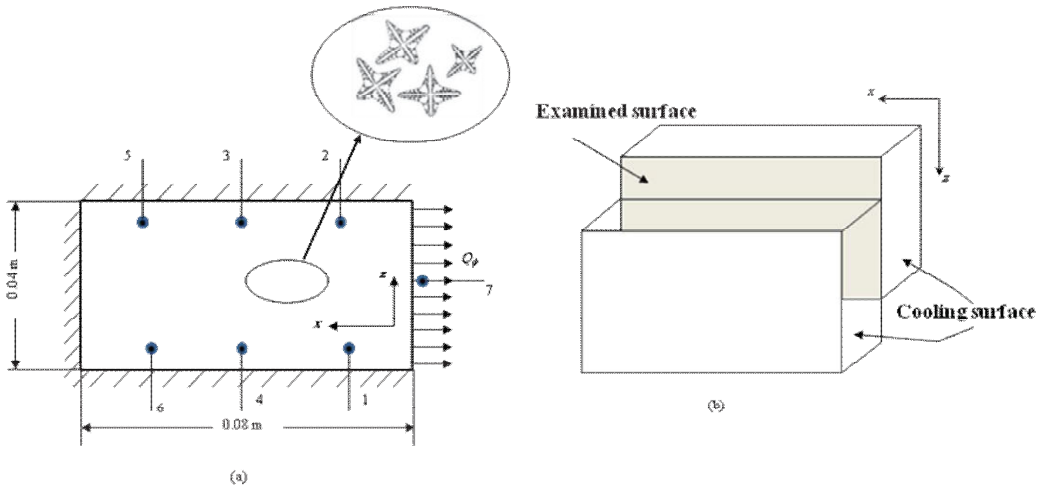

Fig. 1: Schematic illustration of a) experimental and physical system and b) longitudinal cross section for micro-examinations

main findings has been the relative importance of processing parameters.[9,13,14] Convection streams that affects micro-shrinkage crack formation due to the formation of random network of equiaxed grains with large misoriented angle. Casting speed is believed to be the most important parameter that affects micro-shrinkage crack defects[10,14,15], while pouring temperature and cooling water flow rate are of reduced importance.[11] The physical effect of increased casting speed is both an overall increase in the solidification rate, and a proportional increase in the thickness of the mushy zone region which also affects the convection stream patterns significantly.

In this contribution, therefore, the laboratory experiments and model of El-Bealy[16] is further extended in order first, to account for the non-uniform temperature and convection stream fields in the liquid and different mushy regions with various melt superheats and second, to study the effects of these fields and their interactions on the formation of micro-shrinkage crack during the early stages of direct chill of Al-4:5% Cu aluminum alloy ingots. After presenting this work in sections II, III, a few experimental and theoretical results are given in section IV, while section V briefly general discussion based on the findings of this contribution.

II- EXPRIMENATL WORK

A commercial Al-0.4.5%Cu alloy was used in this work and its chemical analysis is tabulated in Ref. [16]. Three different experimental groups have per-formed to study the coupled effect of natural convection streams and interdendritic strain on the micro-shrinkage crack formation of Al-4.5%Cu aluminum alloy solidified in horizontal rectangular parallelepiped cavity at different superheats as shown in Fig. 1(a).

In order to measure the temperature distributions, six thermocouples were inserted into alloy melt and another thermocouple was inserted at steel chill at 2 mm from the chill surface as seen in Fig. 1(a). Temperature singles from thermocouples were acquired automatically with multi-channels digital thermometer.

The solidified ingots were sectioned according to Fig. 1(b) and their surfaces were examined metallo-graphically and grain size of equiaxed structure was measured. The description of the experimental work steps and measurements were explained and discussed in details in Ref. [16].

III- MATHEMATICAL MODEL

The mathematical model which has been developed from the first principles to couple the general frame-work with governing equations of continuity, momentum by using Navier stoke equations,[16] heat transfer presented by Poirier co workers,[17] inter-dendritic thermo-metallurgical strain and macro-segregation proposed by El-Bealy[19] as well as solute redistribution approach under no diffusion in the dendritic solid derived by El-Bealy.[16] Most of these equations are chosen for simplicity and could be changed without introducing fundamentals changes to the model. Only the main references are provided here, and the reader is referred to the original

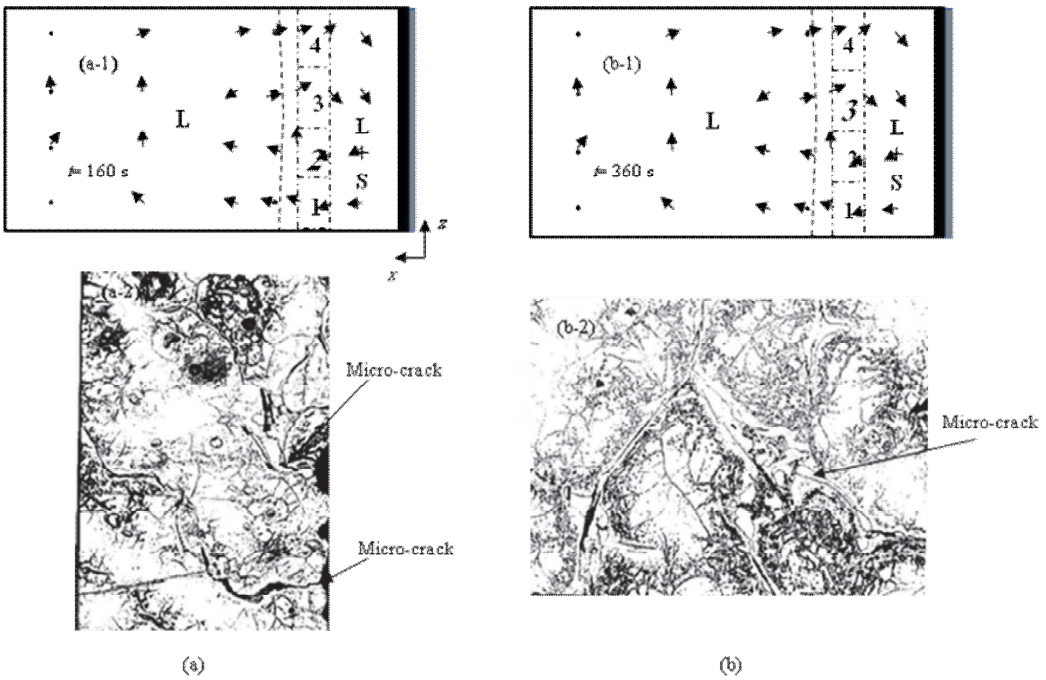

Fig. 2 : Schematic illustration of a-1) and b-1) thermal convection patterns and micrographs of a-2) surface and b-2) interior areas of Al-4.5%Cu with superheat 20 °C, magnification 400 times.

references for the details of the model derivations, assumptions, nomenclature and properties of aluminum alloy. The model equations together with supplementary relations presented therein are again summarized in Ref. [16].

IV- EXPERIMENTAL & PREDICTED RESULTS ANALYSIS

The thermal convection patterns are used often as criterion for the formation of random network of grains and therefore, the micro-shrinkage crack formation during different stages of dendritic solidification.[3,6,8] The results of three predicted and experimental results of convection streams patterns and micrographs of surface and interior areas are presented here in Figs. 2(a,b), 3(a,b) and 3(a,b) for 20, 50 and 75 °C superheats , respectively.

A- Superheat 20 °C

The results shown in Figs. 2(a-1) and (b-1) illustrate that the mushy zone forms with irregular liquidus isotherm and thin solid shell which appears as black straight area close to ingot surface for superheat 20 °C. The convection stream consists of a single clockwise rotating convection cell. These streams in the liquid zone are mostly upward, particularly in the hotter regions in left hand side of ingot.

The micrographs 2(a-2) and 2(b-2) exhibit equiaxed structure where the micro-shrinkage cracks propagate through boundaries of equiaxed crystals with length values from 47 to 92 µm. The surfaces of these cracks exhibit an intergranular profile exhibiting smooth surfaces with small spikes resulting from local plastic deformation accommodated by local redistribution of the liquid. Also, these micrographs show that these cracks locate at grain boundaries but the configuration of these boundaries is associated with nucleation and growth of grains. These cracks form mostly in random network of grain areas which results from a random sedimentation of free crystals associated with convection streams shown in Figs. 2(a-1) and 2(b-1).[6] The random network of grains may result in large misorientation values of grains (>15 degree) particularly at the ingot surface as illustrated in micrographs 2(a-2) and 2(b-2).[6,7]

Another interesting phenomenon arises from the micrographs examination where two criteria can be expected to detect this percolation of grain boundaries and therefore, the propagation of micro-shrinkage crack.[3] The first is that the grains are coalesced and the interfaces are dry as shown in Fig. 2(a-2) whereas the second one is that the grains can be in mechanical contact, but still separated by micrometric liquid films as revealed in Fig. 2(b-2). These two percolation criteria are called percolation by contact and percolation by coalescence or percolation by bridging, respectively. Therefore, the morphology and severity of micro-shrinkage crack vary also based on percolation criterion.[3]

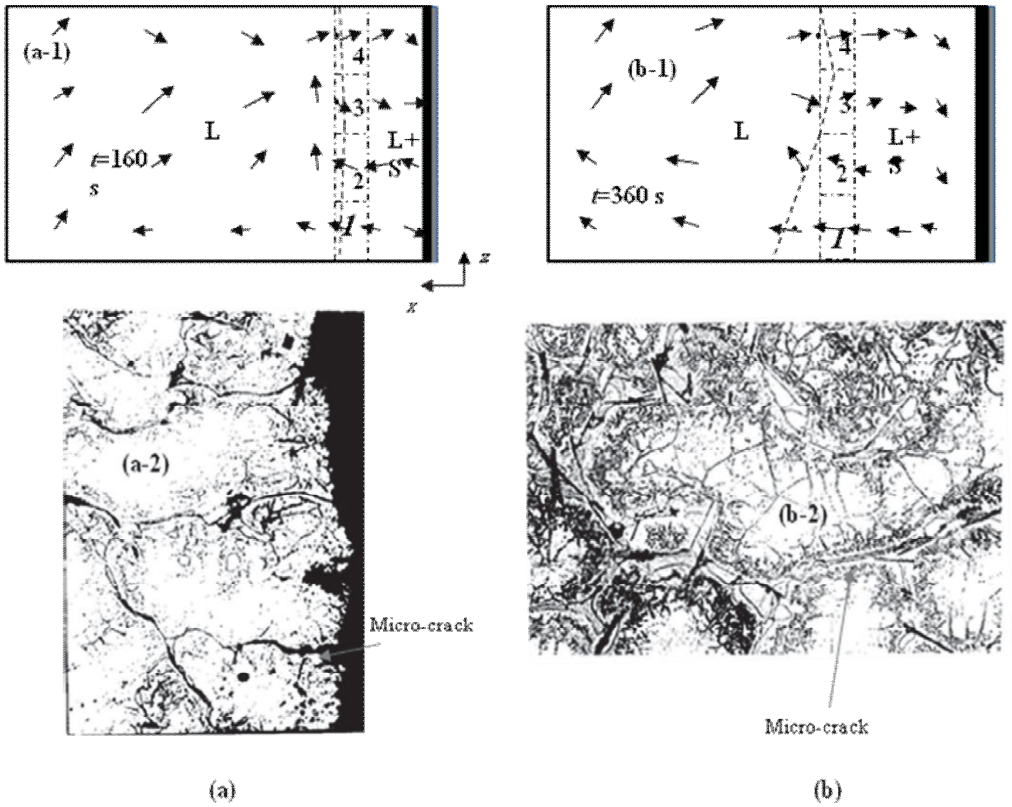

Fig. 3 : Schematic illustration of a-1) and b-1) thermal convection patterns and micrographs of a-2) surface and b-2) inner areas of Al-4.5%Cu with superheat 50 °C, magnification 400 times.

B- Superheat 50 °C

Simulations metallographic results of experiment with superheat 50 °C are shown in Figs 3(a,b). The convection streams patterns shown in Figs. 3(a-1) and 3(b-1) follow the same trend of previous case. These patterns show an interesting feature of liquids isotherms where these isotherms are not straight and bottom parts always cooler than the upper parts. This observation becomes clear with increasing the time as illustrated in Fig, 3(b-1). This observation results from the effect of the convection streams direction where these streams transfer the heat from bottom regions into upper regions close to liquidus isotherm as revealed in Figs. 3(a-1) and 3(b-1).[16,20] Because of a high solidification rate of the crystals sedimentation, a significant amount of latent heat is released in the lower portion controlling the extended of liquidus isotherm in this portion. Another interesting phenomenon arises from these figures where the solidus isotherm straight, thin and its thickness 2.1-3.9 mm for different times. This isothermal straightness results from a wide mushy zone associated with long solidification interval approximately (100 °C) and also is due to small quantities of convection streams.[16]

Micrographs 3(a-2) and 3(b-2) reveal the same trend of dendritic equiaxed structure formation and its crystals are finer than the previous case. Also, it is observed that the decreasing in the surface grain size in the direction of *z* becomes noticeably. This results from increasing in the number of crystals sedimentation in the bulk liquid close to the chill. This is due to small convection streams and increasing in the crystals weight growing on the chill surface. The morphology of micro-shrinkage cracks exhibits that the cracks form also at grain boundaries in random network of grains. This results from a random sedimentation of free crystals associated with convection streams. These micrographs reveal also that the surface cracks form with no any segregates of rich copper at grain boundaries as shown in Fig. 3(a-2) whereas the interior cracks appear with heavy copper segregation as observed sediment on the equiaxed boundaries as revealed in Fig. 3(b-2).

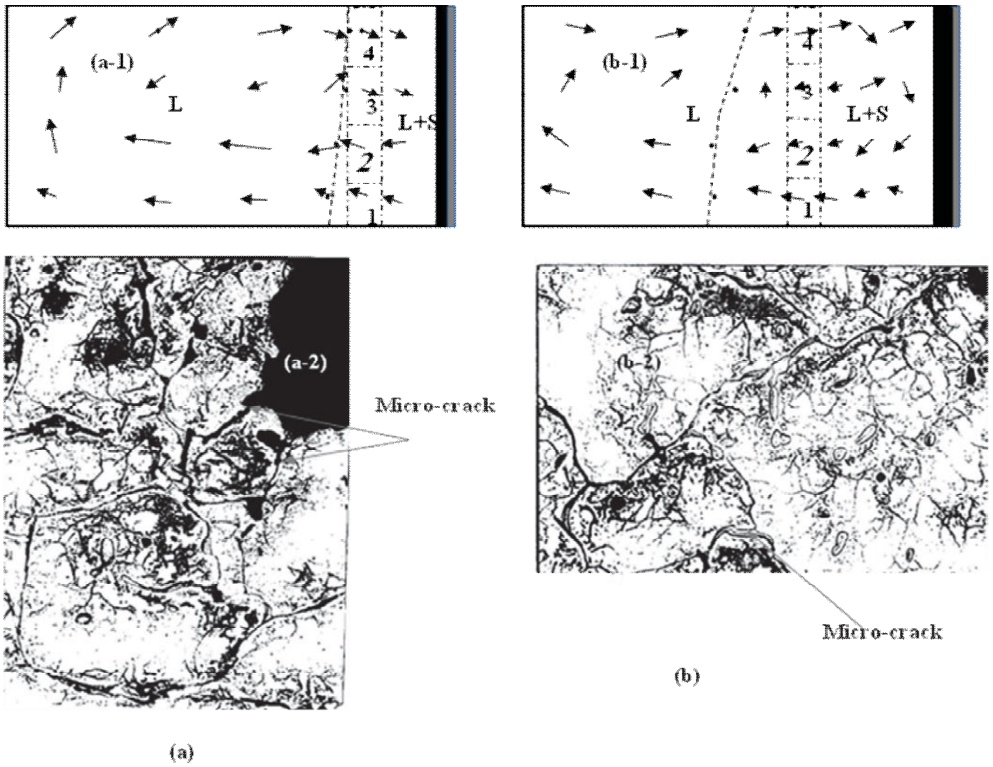

Fig. 4 : Schematic illustration of a-1) and b-1) thermal convection patterns and micrographs of a-2) surface and b-2) inner areas of Al-4.5%Cu with superheat 75 °C, magnification 400 times.

B- Superheat 75 °C

Figures 4(a,b) depict the convection patterns at different times and micrographs of surface and interior areas of superheat 75 °C. The convection streams shown in Figs. 4(a-1) and 4(b-1) indicate the same trend of convection streams directions with non straight liquidus and straight solidus isotherms. The magnitudes of the surface and interior streams are higher than the above cases where the high superheat increases the temperature gradient and therefore increases the difference in densities between the different areas. Also, it is interesting to note that the direction of convection streams of the interior areas at and close to the ingot bottom is one directional parallel to ingot bottom and is similar to the previous cases.

The microstructure and the micro-shrinkage crack morphologies shown in Figs. 4(a-2) and 4(2-b) at different times show the same trend as the above cases. This microstructure reveals the same dendritic equiaxed structure with finer grain size than the above two cases and the same cracking morphology as the previous cases for surface and interior areas. However, the direction of sedimentation of equiaxed crystals in the interior areas is in the direction parallel to ingot bottom as shown in micrographs 3(b-2) and 4(b-2). This results from the nature direction and magnitude of convection streams of bottom interior areas as illustrated in Figs. 3(a-1) and 4(a-1). Also, these may create slight differences of sedimentation of equiaxed crystals which affect the micro-shrinkage crack formation associated with different misorientation values of grains.[6,7]

V- GENERAL DISCUSSION

The findings of this study indicate unambiguously that the formation of micro-shrinkage cracks resulted mostly from the mechanisms of nucleation, convection streams, sedimentation of free crystals and their interactions.[8,16,20] These cracks form due to the random network of equaxied grains associated with random sedimentation of free crystals floating with different convection stream patterns. Therefore, it is appropriate to analysis the present experimental and predictions results where this analysis combined with knowledge of previous work of the micro-shrinkage crack formation may lead to a consistent mechanism for the formation of this crack in a direct chill casting process of Aluminum alloys.[6,7,16]

As solidification starts, the velocities of convection streams reduce observably and become insufficient to affect the floating crystals movements or their sedimentation. Therefore, the mushy zone is approximately convectively stable and the change in grain density of different areas becomes small.[40-43] This changing reduces rapidly during solidification due to a continuous increasing of mushy

permeability. Also, the random network of grains forms in the different final morphologies based on its location in the ingot. For this reason, the grain density increases generally from bottom to top areas where the direction of convection streams of top part help heavy density free crystals to sediment by various densities in the surface areas.

As these crystals grow, two criteria can be expected to detect this percolation of grain boundaries and therefore, the propagation of micro-shrinkage crack.[21] The first is misorientation values of grains whereas the second type of grains contact. Vernéde and Rappaz[6] as well as Rappaz et. al,[7] concluded that the possibility of this crack formation increases with large angle of misoriented grains (>15 degree) which forms a high angle grain boundaries as shown in micrographs 2(a-2) to 4(a-2). However, Vernéde et. al[21] pointed out that The grains contact can be in mechanical contact, but still separated by micro-metric liquid films as shown in micrographs 2(b-2) to 4(b-2) whereas the second is that the grains are coalesced and the interfaces are dry as revealed in micrographs 2(a-2) to 4(a-2). These two percolation criteria are called percolation by contact and percolation by coalescence or percolation by bridging, respectively. Therefore, the morphology and severity of micro-shrinkage crack vary also based on percolation criterion.[53]

VI- CONCLUSIONS

Micro-shrinkage crack formation during dendritic solidification of Al-4.5%Cu in ingot casting to simulate the same conditions occurred in direct chill casting process has been studied both experimentally and mathematically. The micro-shrinkage crack formed during dendritic solidification have been analyzed. The key findings of this study are summarized below as follows;

1- The heterogeneity in the cast structure is due to a formation of the random sedimentation of free crystals associated with different convection stream patterns.
2- The micro-shrinkage cracks form in different ingot locations through grain boundaries where the crack morphologies are different in surface compared to interior areas.
3- The surface crack morphology always is dry contact whereas its morphology in the interior areas is separated by micrometric liquid films of copper segregates.
4- The reason behind micro-shrinkage crack formation is formation of random network of grains with high angle grain boundaries associated with large misoriented grains. This is associated with random sedimentation mechanism of free crystals.

ACKNOWLEDGEMENT

The author is also especially grateful for financial support of Companies' Chair of the Swedish Iron Masters Association, Stockholm, Sweden.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Successful Implementation of a New Rolling Slab Casting Technology, AFM, within Hydro

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Keywords: Casthouse, Rolling Slab Casting, Geometry, Scalping

Abstract

Reduced scrap generation in the value chain is one of the most powerful ways of saving money and energy in the production of rolled products due to less re-melting and metal loss. For the total value chain, geometry deviations are one of the most important reason for scrapped material. Hydro and Hycast has for a long time being working on optimizing the rolling slab geometry to minimize both cut off material in casthouse and scalped off material in the rolling plant. Two decades ago Hydro implemented the flexible mould technology to eliminate the butt swell of rolling slabs.

This work describes the implementation of the newest development within rolling slab casting, the Adjustable Flexible Mould (AFM) technology, within Hydro. The resulting geometries, shell zones thickness and the reduction of scrapped material in the total value chain are discussed and quantified by some examples.

Introduction

Rolling slab geometry is an important quality parameter. Due to the rolling process the rolling faces of the slab need to be absolutely flat prior to rolling. This is obtained by precision scalping of the surface. For many applications the outer surface with slow solidification (shell zone) needs to be removed prior to rolling to avoid unwanted surface defects. All geometrical defects leads to higher scalping scrap. Most rolling mills therefore specify an acceptance limit with regards to geometry. In Europe many uses the EN 487 standard that describes some tolerances for rolling slabs. It is a common assumption that these tolerances will tougher in the future. These limits will determine the butt cut-off length in the casthouse and will also limit the casting speed window than may be used when using conventional fixed moulds.

Elastical deformation of moulds during the initial stages of casting is not a new idea. Alcan patented a flexible mould as early as 1974/75, [Veilette 1974], [Rodenchuk 1975]. In 1976 R. R. Lawrence (Alcan) published a paper on flexible moulds, [Lawrence 1976]. He lists up the following benefits with flexible moulds: 1) Reduce of eliminate the need to saw cut butts, 2) Reduce scalping, 3) Reduce aluminium melt loss due to reduction of scrap generated from an ingot, 4) Reduce overall equipment inventory as several different alloys can be cast with the same mould, 5) Increase effective capacities in casting furnaces, swarf furnaces, soaking pits, storage areas, scalping operations, saws, etc., 6) Reduce conveyor damage, 7) Permit higher casting speeds without forming a large ingot butt.

The Hycast flexible mould technology has been in regular production since 1996 and is described in [Håkonsen 2012].

Adjustable mould technologies with fixed mould opening have been used in the Aluminium industry for several decades, see e.g. [Takagi 1993]. The main challenge for these technologies is the overall geometry of the rolling slabs, especially for large adjustment ranges, resulting in excessive scalping.

In this work the implementation of the new adjustable version of the flexible mould technology, Hycast ® AFM (Adjustable Flexible Mould), is described. Figure 1 shows the AFM pilot installed in February 2014 in Hydro Aluminium Høyanger, and the commissioning team after the first cast.

Figure 1. The AFM pilot and the commissioning crew after the first cast in February 2014.

There are several reasons for implementing the Hycast ® AFM technology in Hydro: 1) automate the start-up and end-of cast, 2) eliminate the usage of ceramic fibers as hot-top material on the moulds, 3) be able to produce all soft alloys.

Theory

When using moulds with a fixed geometry the rolling slab will be thicker in the butt end. This phenomenon is commonly called “butt swell”. The basic mechanisms behind the formation of the butt swell is well understood and is related to the inward bending of the rolling faces during steady state, see [Drezet 1995], [Fjær 1997]. This deformation is compensated for by making the moulds convex (thicker on the middle of the rolling faces). During the initial stages of casting the steady state thermal field is not established and the pull-in is close to zero. During the first ~2 slab

thicknesses of casting the steady state thermal field is established and pull-in is gradually increased to the steady state level. The pull-in is varying across the rolling face of the slab and is largest on the middle (thickness reduction of ~5-9%), and smallest (~1.8%) at the corners.

C. E. Weaver (Alcan) was the first to point out that butt swell was related to the development of the thermal field during start-up and inevitable when using moulds with fixed geometry, see [Weaver 1976]. He was also the first to point out that the pull-in was approximately proportional to the casting speed and to the square of the slab thickness. In other words will the optimal mould shape be dependent on the casting speed.

The steady state "optimal" mould opening can be estimated using the equation given in [Håkonsen 1997], see equation (1).

$$D_M = (1+\varepsilon_0)D + C_{Alloy}\Theta V D^2 \qquad (1)$$
$$\Theta = \exp\left[-1{,}9(X_A)^{1{,}9}\right] - \exp\left[-1{,}9\left(\frac{W}{D} - X_A\right)^{1{,}9}\right]$$

D_M is the optimal mould opening, ε_0 the total thermal contraction during casting ($=\beta(T_{coh}-T_0)\approx 0{,}018$), D the nominal thickness of the ingot, C_{alloy} is a alloy dependent constant ($\approx 130 s/m^2$), V is the casting speed, X_A is the dimensionless distance from one short side ($=x/D$) and W is the nominal width of the ingot. Θ Is named the pull-in function and may be regarded as the dimensionless mould opening.

Equation (1) is a further development of the model proposed by Weaver. An observation worth mentioning is that the butt swell length and the thickness difference between bottom and steady state are both proportional to the casting speed and the square of the slab thickness. This thickness difference between the bottom and steady state is given by (1). The length of the butt swell is given by the casting length until steady state thermal field is established. It may be shown that this casting length also is proportional to the casting speed and the square of the slab thickness.

In Figure 2 the optimal (dimensionless) mould shape for different width to thickness ratios are shown.

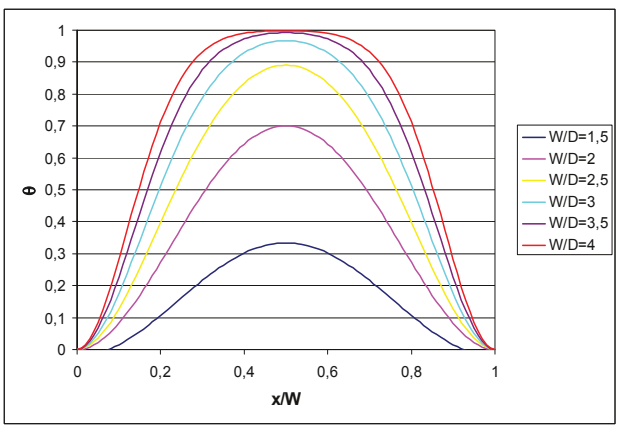

Figure 2. The dimensionless mould opening for varying width to thickness ratios.

During the start-up of cast, there is no pull-in and the mould should therefore be rectangular at this stage. As the thermal field develops and the pull-in phenomenon starts to evolve the mould should gradually be opened to give rectangular ingots from start. At a certain casting length, the thermal field will stabilize and the pull-in will be constant from this point and the rest of the cast, assuming constant casting parameters. This means that the mould then should be kept at a constant mould opening from this point.

Technology

The current understanding of the butt swell formation tells us that butt swell can not be avoided when using moulds with fixed geometry. We also know that moulds with fixed geometry have to be designed to give optimal slab geometry in steady state, for one fixed casting speed. In addition we know that the size (thickness difference) and length of the butt swell is proportional to the casting speed. Therefore most casthouses choose a low design casting speed to limit the need for butt cut-off.

Hydro applied in 1999 for a patent on flexible moulds where the stiffness of the rolling face mould wall is varied/designed to give a close to optimal shape when deformed, [Steen 1999]. By this it is possible to make flexible moulds with close to perfect steady state geometry for any width to thickness ratio.

Based on this patented technology Hycast and Hydro developed a flexible mould to eliminate butt swell and to give the casthouse freedom of choosing different casting speeds for different alloys in the same mould set. 3D modeling tools are utilized in this design process, see Figure 3 for an example of calculated deformation. The figure shows an example of a mechanical deformation of the mould wall corresponding to maximum flexing during casting. Von Mises stresses are shown in the figure since yield in the mould wall material will give a lasting deformation, and thus Von Mises stress is a parameter of interest.

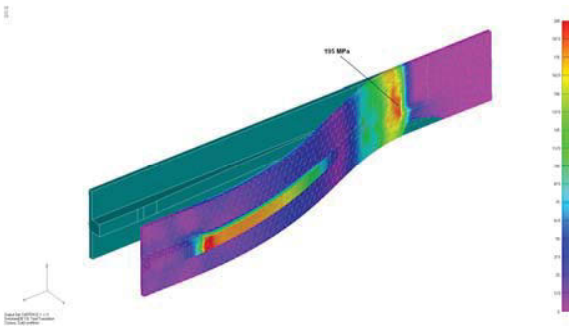

Figure 3. 3D simulation (ANSYS) of a deformed long side of a flexible mould. Equivalent tensile stress (Von Mises) is shown with different colours (0-200MPa scale).

The deformation of the mould walls is done by one or one or two flexing actuators and a system to transfer the actuator movement to the moulds – the flexing system. The flexing movement is accurately controlled by a PLC system and a pre-programmed flexing curve. This curve may be dimension and alloy dependent

allowing different casting speed, and thus flexing, for different alloys

In the technology described in this work the flexible mould technology is further developed. In this technology, the end-walls (short sides) are movable to allow for casting different ingot widths in the same mould set. By doing this fewer mould sets will be needed in the casthouse, giving more efficient maintenance, logistics and reduced investment cost. The mould table has fully automated stepless width adjustment, with no manual work. This includes automatic adjustment of the blocking of water in the area close to corners. The standard width adjustment range is 350mm. Figure 4 shows a 3D model of the AFM technology.

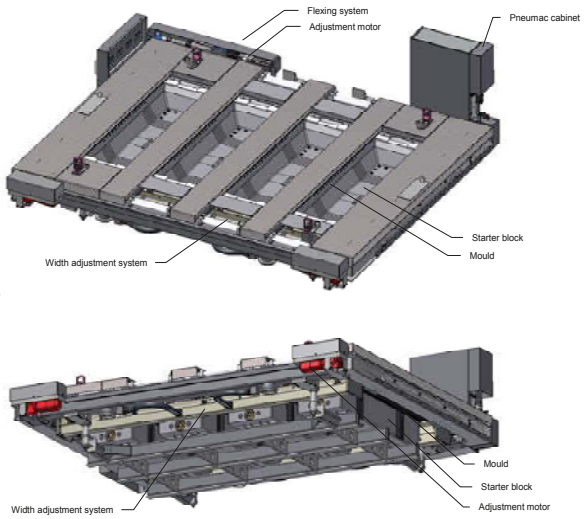

Figure 4. 3D model of an Hycast ® AFM (Adjustable Flexible Mould) casting table.

The starter blocks are adjusted using spacers with defined thickness. Typically, this is done in 50mm steps but may be done in finer steps dependent on the spacer thickness used.

The moulds are equipped with a PLC-controlled continuous lubrication system. The AFM technology is patented [Næss 2008].

To be able to control the metal level in each mould a Hycast ® LCS (Level Control System) is used. A stopper/nozzle concept is used with state-of the art lasers and actuators for accurate metal level control. Figure 5 shows a 3D model of such a system (T-version).

Results

The results presented in this work is based on a Pilot installed in Hydro aluminium Høyanger in February 2014. This was a tree-slab system consisting of a Level Control System and a 600x1600-1950mm casting table. During the first nine months of 2015 the total production with the AFM pilot was just approximately 13.000 tons of rolling slabs. Recorded scrap rates for some alloy groups are shown in Table 1. The definition of pit recovery is; ton of good rolling slabs from the casting process relative to total tonnage cast. The total scrap rate will in addition include cut-off, drained metal and scrap due to chemistry deviations or other reasons than casting.

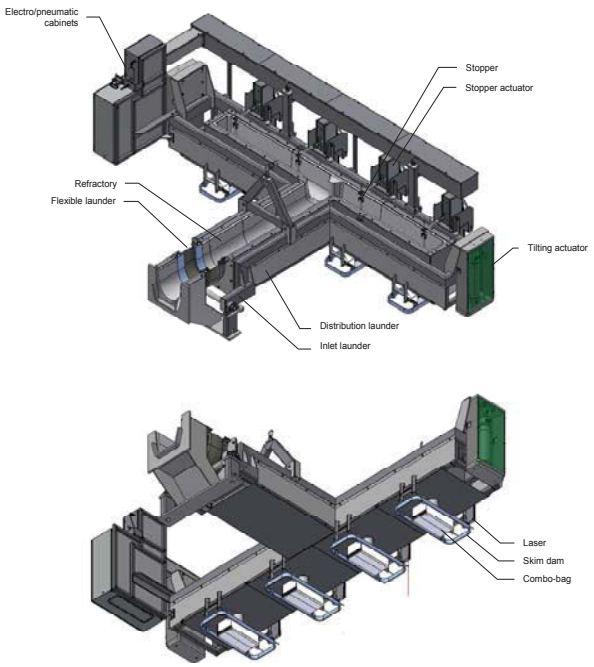

Figure 5. 3D model of a T-version Hycast ® LCS.

Table 1. Pit recovery versus alloy group.

Alloy group	Pit recovery (%)
1xxx	94,1
5xxx	96,0
8xxx	96,0

The lubrication oil consumption has gradually been reduced from the implementation, from ~30ml/ton to ~8ml/ton. Analysis of the cooling water shows that only 20% of the added oil ends up in the cooling water. The rest of the oil is then most likely burned off or attached to the rolling slab surface after cast. Typical water consumption during casting is 3.8m^3/ton. 8ml/ton will then result in an oil concentration of only 4ppm in the cooling water.

The bottom and top cut-off is typically 170mm and 120mm respectively. Some customer specify longer cut-off lengths in their standard specification but these cut-off lengths should be sufficient for the AFM technology. The typical cut-off for 9m casting lengths are then 4-5%.

Hydro is supplying all major Rolling Mills in Europe, with a variety of surface critical products. The qualifying processes in the rolling industry are quite extensive and time consuming. In total 24 alloys have been shipped. In total 12 Alloys have been qualified. To qualify one alloy typically several trial orders are required. Table 2 is an overview of the trial orders supplied from installation.

Shell zone thickness were measured using light microscope in the middle of the rolling face. The measured shell zones varied between 1.5-8mm. Table 3 is a summary of the measurements for different alloy groups.

Table 2. Trial orders produced on the AFM pilot.

Alloy/Application	# of trials orders	Success-rate (%)
1200/Litho	5	100
1200/8xxx Foil(Tetrapack, Pharma, etc)	9	100
5005 (Anodize)	3	100
6xxx (Automotive)	2	100

Table 3. Measured shell zones of AFM ingots.

Alloy group	Typical shell zone (mm)	Maximum shell zone (mm)
1xxx	6	6
5xxx	5	8
8xxx	5	7

The geometry tolerance of the slabs is within ±2mm in flatness and ±3mm in thickness from start to end. Due to this excellent geometry and shell zone values, one rolling client have reduced scalping with 1% (relative to total volume of ingot) in their 600mm thick slabs when production were shifted to AFM. This corresponds to ~4$/ton in saved re-melt cost of scalping chips.

The technology has proven to be very easy to operate. The total turn-over time to change dimension (width) is 10 minutes.

Discussion

The Hycast ® AFM implementation has met the performance goals. The start and stop of cast is hands-off. The geometry is within the design criteria values of ±2mm flatness from start. The cut-off lengths are reduced to what we would expect for flexible moulds. The ceramic fiber hot-top is no longer needed and the technology is validated for all relevant soft alloys; 1xxx, 3xxx, 5xxx, 6xxx and 8xxx.

Further implementation of AFM is now being carried out in the Hydro casthouses. Level Control System is simultaneously being implemented.

The typical reduction in scrap rate compared to conventional fixed moulds is estimated to 5%, see [Håkonsen 2012], which corresponds to 10$/ton saved re-melt cost in the casthouse. Here a re-melt cost of 200$/ton cut-off is assumed. The overall improved geometry of the rolling slabs, in combination with thin shell zones may give additional savings due to reduced scalping. Using the 1% reduced scalping compared to fixed moulds mentioned above this corresponds to an additional saving of 4$/ton. A re-melt cost of 400$/ton scalping chips is assumed. This adds up to 14$/ton in reduced re-melt cost in the complete value chain. In addition there is a potential by reduced investment cost, and handling/maintenance cost, compared to the alternative of having several fixed mould sets to cover the different widths.

Conclusions

A new casting technology for rolling slabs is developed and implemented in operation. The new technology is based on the Hycast ® Flexible Mould technology but is in addition adjustable in width.

A pilot has been in operation since February 2014 and is validated for all relevant alloys. The geometry of the slabs are within ±2mm (flatness). Hydro is now in process of implementing this new technology in the Norwegian primary rolling slab casthouses.

Acknowledgement

The authors would like to acknowledge all the skilled people that during the last decade have been involved in the development of the AFM technology. Harald Næss Jr., Ulf Bersås, Rune Fjærvik, Tore Strandheim, Daniel Sjølseth, the Hydro Reference Centre crew and many more.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

UNCERTAINTY PROPAGATION IN NUMERICAL MODELING OF DIRECT CHILL CASTING

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Keywords: Uncertainty Quantification, Direct Chill Casting, Macrosegregation

Abstract

Complex solidification processes are often simulated to gain insight into physical phenomena that cannot be experimentally observed. Validation of these models is done through direct comparison to scarce experimental data, in which agreement can be misinterpreted due to uncertainties from the model and experimental measurements. A fully transient numerical model of the direct chill casting process of Al-4.5 wt.% Cu is used to examine the propagation of input uncertainty to outputs of interest. This model solves equations for momentum, temperature, and species conservation. The effect of material property input uncertainties are examined. Probability density functions are calculated based on these input uncertainties for metrics that characterize the ingot macrosegregation and sump depth.

Introduction

The most common processing method for wrought aluminum alloys is direct chill (DC) casting, a schematic of which is shown in Figure 1. Because of its complex nature, numerical simulations are often used to improve understanding of the dominant physical phenomena and the behavior of ingots during the process [1,2]. Such numerical predictions are affected by model inputs and thermal boundary conditions that are obtained experimentally, such as material properties. In this study, a continuum mixture model for the conservation of mass, energy, species, and momentum is used to examine the effect of input uncertainty on the uncertainty of select outputs of interest. The transport model is coupled with a multicomponent solidification model to simulate the production of Al-4.5 wt.% Cu billets.

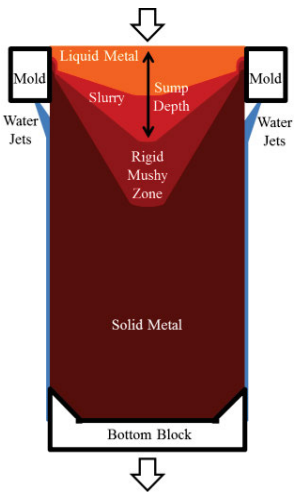


Figure 1. Schematic of the DC casting process showing each region of the mushy zone.

The uncertainty in the inputs to this solidification process model, like the material property values, may significantly alter the numerical predictions [3]. Quantification of model output uncertainty will help identify which inputs would be most profitable to improve. This kind of analysis is not often performed for solidification process models. The Prism Uncertainty Quantification (PUQ) framework is used to propagate material property input uncertainty through the DC casting model [4]. The outputs of interest from the DC casting model are the macrosegregation level and steady state sump depth.

Numerical Model

DC Casting Model Description

The model used for this study has been described previously [5], in which the governing continuum mixture equations for momentum and species were taken from Vreeman et al. [2,6], so only a brief description of the model will be given here. The conservation equations are discretized using the finite volume method and solved using the SIMPLER algorithm for the flow field on a staggered grid [7]. The mass and momentum equations in axisymmetric coordinates for liquid and free-floating solid are:

$$\frac{\partial}{\partial t}(\rho)+\nabla\bullet\left(\rho\vec{V}\right)=0 \tag{1}$$

$$\begin{aligned}&\frac{\partial}{\partial t}(\rho u)+\nabla\bullet\left(\rho\vec{V}u\right)=\nabla\bullet\left(\mu_l\frac{\rho}{\rho_l}\nabla u\right)+\nabla\bullet\left(\mu_l u\nabla\left(\frac{\rho}{\rho_l}\right)\right)\\&-\nabla\bullet\left(\mu_l\frac{\rho f_s}{\rho_l}\nabla u_s\right)-\nabla\bullet\left(\mu_l u_s\nabla\left(\frac{\rho f_s}{\rho_l}\right)\right)+\nabla\bullet\left(\mu_l u_s\nabla g_s\right)\\&+\nabla\bullet\left(\mu_s g_s\nabla u_s\right)-\nabla\bullet\left[\left(\frac{\rho f_s}{f_l}\right)\left(\vec{V}-\vec{V}_s\right)\left(u-u_s\right)\right]+\rho\mathrm{B}_z-\frac{\partial P}{\partial z}\end{aligned} \tag{2}$$

$$\begin{aligned}&\frac{\partial}{\partial t}(\rho v)+\nabla\bullet\left(\rho\vec{V}v\right)=\nabla\bullet\left(\mu_l\frac{\rho}{\rho_l}\nabla v\right)-\mu_l\frac{\rho}{\rho_l}\frac{v}{r^2}\\&+\nabla\bullet\left(\mu_l v\nabla\left(\frac{\rho}{\rho_l}\right)\right)-\nabla\bullet\left(\mu_l\frac{\rho f_s}{\rho_l}\nabla v\right)+\mu_l\frac{\rho f_s}{\rho_l}\frac{v}{r^2}\\&-\nabla\bullet\left(\mu_l v\nabla\left(\frac{\rho f_s}{\rho_l}\right)\right)+\nabla\bullet\left(\mu_l v\nabla g_s\right)+\nabla\bullet\left(\bar{\mu}_s g_s\nabla v\right)\\&-\bar{\mu}_s g_s\frac{v}{r^2}-\frac{\partial P}{\partial r}\end{aligned} \tag{3}$$

The momentum equations account for the motion of free-floating solid and relate the solid and liquid velocities through Stokes law:

$$\vec{V}_s - \vec{V}_l = \frac{(1-g_s)}{18\mu_m}(\rho_s - \rho_l)d^2\vec{g}\,. \quad (4)$$

This equation treats the equiaxed dendrites as uniform solid spheres with an average particle diameter (d) and mixture viscosity (μ_m). The fluid flow is predominantly driven by thermal and solutal buoyancy forces which are given in equation (5).

$$\begin{aligned} \rho B_z &= g_s(\rho_s - \rho_l)g \\ &- g_s\rho_s g\left[\beta_{T,s}(T-T_0) + \beta_{S,s}\left(C_s^i - C_0^i\right)\right] \\ &- g_l\rho_l g\left[\beta_{T,l}(T-T_0) + \beta_{S,l}\left(C_l^i - C_0^i\right)\right] \end{aligned} \quad (5)$$

The solid initially forms as equiaxed particles that are free to move with the liquid until a critical fraction solid, $g_{s,crit}$, is reached. Once $g_{s,crit}$ is reached in a given control volume and one neighbor is a rigid solid, the free-floating solid packs and becomes part of the rigid mushy zone. This critical solid fraction has been observed to be below 30% [2,8] and for this study a value of 15% is used. The flow in the rigid mushy zone is damped by the coarse dendritic solid structure and the permeability (K) is modeled using the Blake-Kozeny model, which is a function of the secondary arm spacing (λ_2) and the volume fraction solid (g_s). When the mushy zone is rigid ($g_s > g_{s,crit}$), the terms $\frac{\mu\rho}{K\rho_l}u$ and $\frac{\mu\rho}{K\rho_s}v$ are added to equations (2) and (3), respectively, to represent the drag due to the packed dendrites.

The species and energy conservation equations are given by:

$$\begin{aligned} &\frac{\partial}{\partial t}\left(\rho C^i\right) + \nabla\bullet\left(\rho f_l \vec{V} C^i\right) = \nabla\bullet\left(\rho f_l D\nabla C^i\right) \\ &+ \nabla\bullet\left(\rho D\nabla\left(C_l^i - C^i\right)\right) - \nabla\bullet\left(\rho f_s \vec{V}\left(C_l^i - C^i\right)\right) \end{aligned} \quad (6)$$

and

$$\begin{aligned} &\frac{\partial}{\partial t}\left(\rho c_p T\right) + \nabla\bullet\left(\rho c_p T\vec{V}\right) = \nabla\bullet\left(k\nabla T\right) - \frac{\partial}{\partial t}\left(\rho f_l L_f\right) \\ &- \nabla\bullet\left(\rho f_l L_f \vec{V}\right) - \nabla\bullet\left(\rho f_s\left(\left(c_{pl} - c_{ps}\right)T + L_f\right)\left(\vec{V} - V_s\right)\right) \end{aligned} \quad (7)$$

The solidification of the binary alloy follows the lever law and the liquidus and solidus lines are assumed to be linear. The thermophysical properties for this alloy are shown in Table I.

Table I. Thermophysical properties of Al-4.5 wt.% Cu.

Liquid Density (ρ_l) [kg/m3]	2460
Solid Density (ρ_s) [kg/m3]	2750
Specific Heat (C_p) [J/kg K]	1006
Latent Heat (L_f) [J/kg]	$3.9\text{x}10^5$
Thermal Conductivity (k) [W/m K]	137.5
Liquid Viscosity (ν_l) [kg/m s]	0.0023
Eutectic Temperature (T_{eut}) [K]	821
Liquid Thermal Expansion ($\beta_{T,l}$) [1/K]	1.17x10-4
Solid Thermal Expansion ($\beta_{T,s}$) [1/K]	2.25x10-5
Liquid Solutal Expansion ($\beta_{S,l}$) [1/K]	-0.73
Solid Solutal Expansion ($\beta_{S,s}$) [1/K]	-0.87
Partition Coefficient (k_p)	0.17

Initially the mold is filled with quiescent molten metal. The mold consists of a hot top, a graphite ring, and an aluminum mold, shown schematically in Figure 2 along with the dimensions and various thermal boundary conditions.

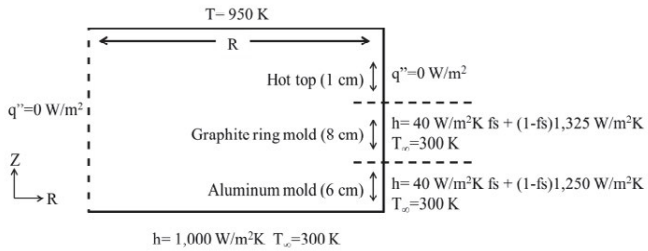


Figure 2. Schematic of the mold showing dimensions and thermal boundary conditions of the mold.

The heat transfer coefficients in the mold are functions of the fraction solid of the control volume next to the surface, accounting for the effect of solidification shrinkage. Below the mold, the ingot is actively cooled by free falling water. The boundary conditions for the water are taken from Weckman and Niessan [9]. Initially the entire domain is set to the superheated pouring temperature of 950 K (677°C), corresponding to a superheat of 31.8 K, and the initial composition. To develop a solid shell near the mold wall before moving the bottom block, the liquid metal is initially held in the mold for one minute. After this holding period, the casting speed is set to 30 mm/min, which is 50% of the steady state value, and linearly increased to steady state. The ramp rate is set so that, when the steady state casting speed is reached, the axial length has grown by the radius of the ingot. The grid spacing in the axial direction is 10 mm and 7.5 mm in the radial direction.

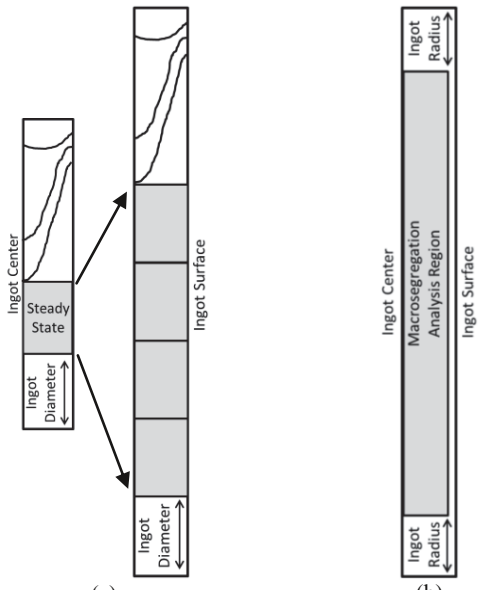


Figure 3. Schematic of the simulation domain showing (a) how the ingot was lengthened and (b) the area of the ingot analyzed for macrosegregation.

The process continues using the steady state casting speed until 1000 mm of solid ingot is solidified, after which the casting speed is ramped down at the same rate it was ramped up. Once the

casting speed reaches 30 mm/min, it is then set to zero to allow the remaining liquid to solidify. The steady state region of the casting, where the casting speed, sump profile, and composition field, are axially constant, was determined to be the portion of the ingot extending from an ingot diameter from the bottom block to when the casting speed began ramping down (1000 m). This region was duplicated 3 times to extend the length of the final billet, shown schematically in Figure 3(a). The top, bottom, and surface of the final ingot are "sectioned" before analyzing the macrosegregation level as shown in Figure 3(b). The length sectioned from the top and bottom corresponds to the ingot radius and 50 mm are removed from the surface.

Uncertainty Quantification Model Descriptions

This study uses the publicly available PRISM Uncertainty Quantification (PUQ) framework to perform the uncertainty quantification. Details of this software can be found elsewhere [4], so only a brief description will be provided here. PUQ calculates a PDF for each output quantity of interest, given the known uncertainty of select inputs. The outputs of interest are sampled from the full numerical model to generate a response surface, which acts as a surrogate to running the full process model. In PUQ, the computationally efficient Smolyak sparse grid algorithm[10–13] is used to determine simulation conditions of the runs required of the numerical model to generate an adequate surrogate model. Different levels of the Smolyak algorithm, which correspond to different orders of the polynomial functions, can be used to develop the surrogate model. A level one Smolyak, which uses a surrogate model linear in the input variables, varies each uncertain input independently and requires the least number of cases from the full process model. Level two and three Smolyak calculation result in quadratic and cubic surrogate models respectively, and considers the effect of varying multiple inputs at a time. Even higher levels are possible, but require progressively more full process model cases to be, raising the computational cost.

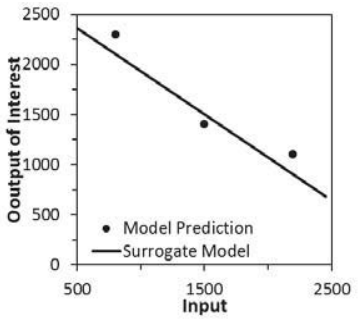
The fit of the response surface to the sampled process model outputs is quantified by using the root mean square error (RMSE). Each uncertain input parameter is randomly sampled 10,000 times and the surrogate model is used to calculate the output PDFs, which are characterized by their normal standard deviation. These output PDFs give the probability that the output has any value in its range.

The relative sensitivities of the outputs to changes in the input parameters are calculated using the sampled points from the model and the Elementary Effects Method (EEM) [14,15]. The EEM calculates two sensitivity measures to determine whether the inputs have negligible, linear, or nonlinear effects on the outputs. A distribution of elementary effects is calculated for each input parameter by varying them independently with a constant step change across all levels of the other parameters. The definition of an elementary effect (EE) for given inputs, x_j, and outputs, y_j, is shown in equation 8, in which *(xmax – xmin)* is 6.15σ, the inputs' standard deviation.

$$d(x_j) = EE_j = \frac{\Delta y_j}{\dfrac{\Delta x_j}{(x_{\max} - x_{\min})}} \qquad (8)$$

The mean sensitivity, μ^*, is the estimate of the mean of the distribution of the absolute value of the elementary effects: $\mu^* = \frac{1}{r}\sum_{j=1}^{r}\left|d(x_j)\right|$. For μ^* values near zero, the input has a small or negligible effect on the output; values much greater indicate the input plays a significant role in determining the output. An example data set, where the mean input is 1,500 with a standard deviation of 225, is shown in Figure 4 with the corresponding elementary effects calculations.

Input	Output
807	2295
1500	1406
2193	1102



$$EE_1 = \frac{(2295-1406)}{\left(\dfrac{(1500-807)}{(2193-807)}\right)} = 1778$$

$$EE_2 = \frac{(1406-1102)}{\left(\dfrac{(2193-1500)}{(2193-807)}\right)} = 608$$

$$\mu^* = \frac{1}{2}(|1778|+|608|) = 1193$$

Figure 4. Plot of an example surrogate model fit to model predictions for a level 1 Smolyak run with the corresponding model results and elementary effects calculations.

The second parameter, σ^*, is a measure of the standard deviation of the elementary effects. A high value of σ^* (a wide distribution), indicates the effects are strongly influenced by other inputs. A low value of σ^* indicates a linear dependence on each input with little interdependence. A level one Smolyak is capable of assessing which uncertain inputs have the most impact on the output of interest, even without a low RMSE fit of the surrogate model.

Results and Discussion

The DC casting model described above has been used to study the effects of uncertain material property values on the macrosegregation and sump formation of Al-4.5 wt.% Cu. The steady state casting velocity and ingot diameter for this study is the 60 mm/min and 500 mm respectively. The probability distributions for the material property values were assumed to be Gaussian with the nominal reported values as the means and a specified standard deviation (σ). Two uncertainty levels, in which 95% (2σ) of the distributions were within 5% or 10% of the mean, were studied (Table II). A level one Smolyak run was performed on both cases to examine which properties affect the outputs of interest the most. Higher Smolyak level runs, with more fidelity in the response surface at the expense of more calculations per input parameter, were then performed using this smaller set of uncertain inputs. To avoid the solid and liquid densities from having overlapping uncertainty distributions, the liquid density is set based on ρ_s and $\Delta\rho$ ($\rho_l - \rho_s$).

Table II. Uncertainty of material property inputs.

Property	Mean (μ)	Case A ($2\sigma = \pm 10\%\mu$)	Case B ($2\sigma = \pm 5\%\mu$)
ρ_s [kg/m^3]	2,750.0	275	137.5
$\Delta\rho$ [kg/m^3]	-290.0	29	14.5
k [W/m K]	137.5	13.75	6.875
C_p [J/kg K]	1,006.0	100.6	50.3
L_f [J/kg]	390,000.0	39,000	19,500
$\beta_{T,s}$ [1/K]	$2.25\text{x}10^{-5}$	$2.25\text{x}10^{-6}$	$1.125\text{x}10^{-6}$
$\beta_{T,l}$ [1/K]	$1.17\text{x}10^{-4}$	$1.175\text{x}10^{-5}$	$5.85\text{x}10^{-6}$
$\beta_{S,s}$	-0.87	0.087	0.0435
$\beta_{S,l}$	-0.73	0.073	0.0365

The outputs of interest for cases A and B are the macrosegregation leve and the sump depth (*SD*). The former is described using the normalized Weibull deviation (*W*) described by Fezi et al. [16]. This metric is found by plotting the composition field as a continuous distribution function (CDF) and fitting a Weibull distribution to it. The CDF is calculated by ranking the composition field and corresponding control volume sizes (*V*) from lowest to greatest. Then the CDF value for each unique composition, C_j, is calculated as: $CDF(C_j) = \sum_{i=1}^{i=j} V_i \Big/ \sum_{i=1}^{i=n} V_i$. A Weibull distribution is then fit to the constructed CDF and the standard deviation is calculated to quantify the amount of macrosegregation in the ingot. The *SD* was calculated as the distance from the top of the mold to the point where the control volume at the center of the ingot became packed. A level one Smolyak was run for cases A and B to compare the sensitivities of the outputs of interest to the changes in material property values and their uncertainty level. The composition field, sump profile, and flow field in the steady state region of the ingot for the mean values in Table 2 are shown in Figure 5. The sensitivity results for Cases A and B are shown in Figures 6 and 7, respectively, in which the height of each bar is μ^* and the error bars are $\pm 2\sigma^*$.

The flow in the slurry (liquid and moving solid) region is driven by thermal and solutal buoyancy. This motion has only a small effect on the mixture composition field because the solid and liquid move almost entirely together and so very little segregation occurs. Solid nucleates in the slurry and is depleted in solute. This solid is transported by the buoyancy driven flow cell in the slurry. As solidification progresses, the viscosity of the slurry increases by an order of magnitude until the point at which the equiaxed dendrites attached to the rigid mushy zone at locations different from where they nucleated. The only relative motion of solid and liquid occurs due to settling of the copper-poor particles, which does contribute to macrosegregation. However, most of the redistribution of solute occurs in the rigid, and very low permeability mushy zone, where shrinkage driven flow is the dominant transport mechanism. The free-floating solid particles that attach to the rigid mushy zone are coarse and form an interconnected network that severely damps the buoyancy driven flow. The shrinkage driven flow acts perpendicular to the advancing solidification front, transporting the enriched interdendritic liquid deep in the mushy zone from the center of the ingot towards the surface due to the "V" shaped steady state sump profile.

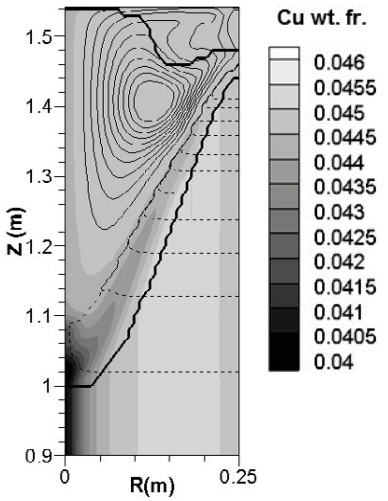


Figure 5. Contour plot of the Cu composition field, sump profile, and flow field during steady state. The bold solid lines indicate the extent of the mushy zone. The other lines flow streamlines. The solid lines are clockwise-rotating flow cell in the slurry with $0.1 < \rho\Psi < 1$ and $\Delta\rho\Psi=0.1$ kg/s. The dotted streamlines show the counter clockwise rotating flow in the rigid mushy zone with $0.005 < \rho\Psi < -0.000005$ and $\Delta\rho\Psi=0.0004995$ kg/s.

The transport phenomena described above form in the steady-state region of the ingot a solute-depleted area at the center of the ingot and a mid-radius that is slightly enriched. The depleted subsurface region and enriched surface is caused by another effect. The surface of the ingot is the first region to freeze and form the rigid mushy zone. Associated shrinkage driven flow transports the solute rich liquid from the subsurface region to the surface, forming the enriched surface and depleted subsurface. At the center of the casting, the enriched liquid is lost to the mid-radius due to shrinkage and is replaced by liquid from the slurry which is closer to the nominal composition. This forms the depleted region at the center and the slightly enriched region at the mid-radius.

For *SD* and *W* the relative sensitivities to the inputs are similar for Case A and B. The sump depth was most affected by k, ρ_s, and L_f. The Stefan number ($St = C_p(T_{liq} - T_{sol})/L_f$) varies between 0.23 and 0.16 based on the uncertainty of the inputs, showing that the latent heat release over the freezing range is much more than the sensible heat removal necessary to drop the temperature from T_{liq} to T_{sol}. Consequently, the overall heat removal during solidification is controlled by latent hear (L_f) and so it should have a larger influence on *SD*. The release and advection of latent heat are also dictated by the density, which contributes to the importance of L_f in determining *SD*. Because the formation of the sump is a conduction-dominated heat transfer problem, the thermal conductivity determines the rate at which heat is conducted radially from the sump to the mold and water, which explains the high sensitivity of *SD* on k.

For the macrosegregation metric, *W*, the most influential material properties are $\Delta\rho$, C_p, k, and ρ_s for both levels of uncertainty. The solid solutal and thermal buoyancy terms had a negligible effect on *W* in both Case A and B, while the liquid thermal and solutal expansion coefficients ($\beta_{T,l}$ and $\beta_{S,l}$) had the same mean effect as L_f, but with less variability. This result suggests that the buoyancy driven flow in the slurry is primarily dictated by the liquid buoyancy terms. The most influential material property on *W* is

$\Delta\rho$, which is a large factor (along with the solidification rate) in determining the strength of the shrinkage driven flow. The solidification rate depends on how quickly heat can be removed from the mushy zone and how much latent heat is released during solidification. Therefore, C_p, k, and ρ_s (which make up the thermal diffusivity) also have a significant impact on W.

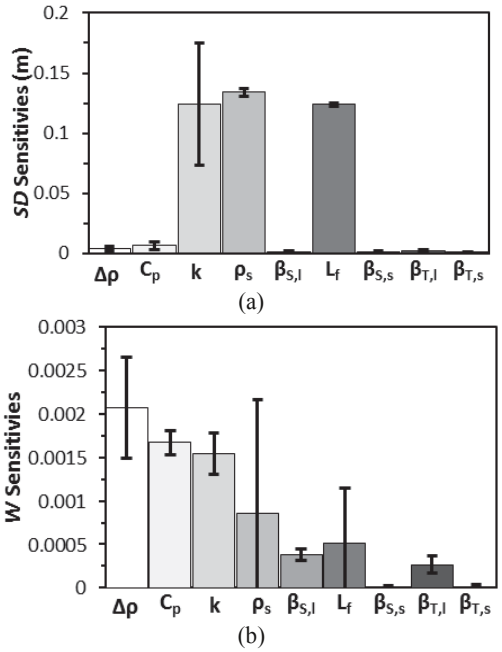


Figure 6. Plots of the sensitivities of the output quantities of interest to the uncertain inputs for Case A: (a) sump depth (b) macrosegregation level.

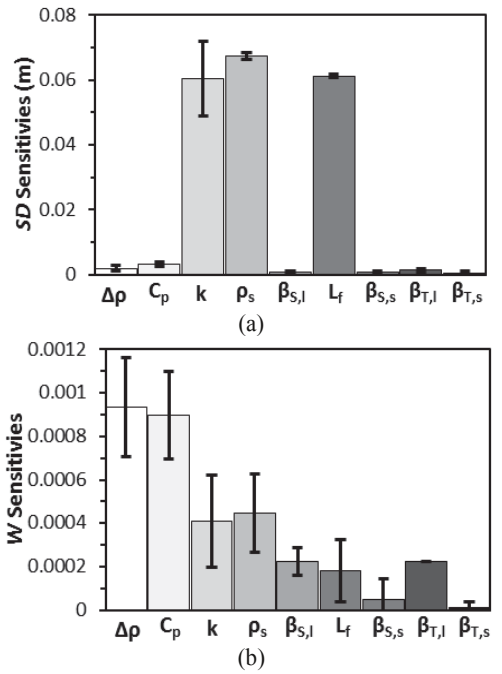


Figure 7. Plots of the sensitivities of the output quantities of interest to the uncertain inputs for Case B: (a) sump depth (b) macrosegregation level.

The resulting probability distributions for each output of interest are shown in Figure 7. The mean predicted values of *SD* and *W* are 0.469 m and 0.009, respectively, for Case A and 0.466 m and 0.009 for Case B. The smaller input uncertainty in Case B caused more certain outputs while the mean value was not altered. In order to predict the SD within 14% of the mean ($2\sigma/\mu$), k, ρ_s, and L_f of the Al-4.5 wt.% Cu alloy must be known better than 10% of their measured values since *SD* is most sensitive to them. For the prediction of W to be known within 10% of the mean value, $\Delta\rho$, C_p, k, and ρ_s must be known with more than 10% certainty.

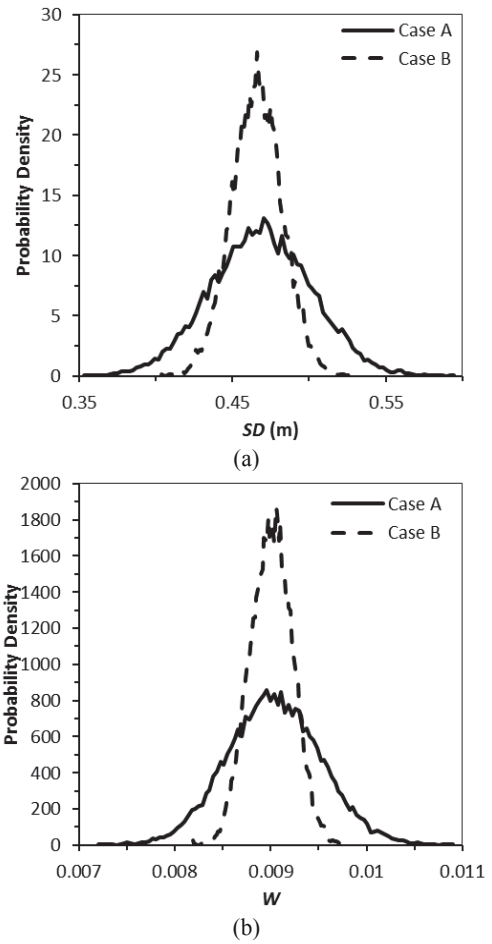


Figure 8. Probability distribution functions of the model outputs for Cases A and B: (a) sump depth (b) macrosegregation level.

Summary

The PRISM Uncertainty Quantification (PUQ) framework has been used to quantify the uncertainty of the sump depth and macrosegregation of an Al-4.5 wt. % alloy predicted by a fully transient numerical model for DC casting. Two different uncertainty levels of the material properties were examined. The prediction of the sump depth was mostly affected by the thermal conductivity, solid density, and latent heat release. The macrosegregation level was primarily affected by the density difference between the solid and liquid, thermal conductivity, specific heat, and solid density. In order to reduce the output uncertainty to reasonable levels (less than 10%), the most sensitive material properties must be known within 5% of the mean values.

Acknowledgements

The authors thank Martin Hunt and Alejandro Strachan of Purdue with their invaluable assistance with PUQ.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE STUDY CONDITIONS OCCURRENCE OF HOT TEARING IN THE BILLETS ALLOY EN AW6060 PRODUCED WITH THE PROCESS OF DIRECT CHILL CASTING

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Keywords: Hot Tearing, Direct Chill Casting, Alloy EN AW6060

Abstract

During *Direct Chill* casting of aluminum alloys technology *Wagstaff AirSlip*, the primary and secondary cooling causes strong thermal gradients, which leads to the uneven thermal contraction in different sections of the billet. The consequences are manifested appearance of casting defects such as hot tearing and residual stresses. In order to reduce the occurrence of these defects were studied different effects on the formation of hot tearing: changes of technological parameters, chemical composition and microstructure of the alloy EN AW 6060 according to the length and cross section, and the conditions of occurrence and progression of cracks describes the model the ratio of light and heavy metals in the billet. Additional testing of as - cast, cracked and homogenized sample billets in multiple directions cross-section detects that the rapid cooling of the surface of the billet affect the development of fields measured hardness.

Introduction

During production AlMgSi billets by the procedure Wagstaff AirSlip quality tree should meet high requirements. In cases where the disorder occurs in the management process, the appearance of surface or internal defect involves different paths of fluid flow, solidification, microstructure development, stresses due to uneven heat transfer in the system billet - molds. The disorder some of the critical variables (casting speed, temperature and level of melt, flow and water quality, etc.), pulls other disorders, which can build up and make the process unstable. The high speed casting, high temperature melt or uneven distribution of water across the surface of billet are some of the disorders that cause "hot casting conditions" and increase the tendency to develop the appearance of hot tearing in the billets.

2. Experimental details

This paper investigates the conditions under which comes to the appearance of hot tearing in two charges: 815 605 and 822 723, and their manifestation in the mechanical properties of billets diameter of 178 mm and 203 EN AA6060 alloy produced in the foundry company "Aluminij" d.d Mostar. Research can be described by establishing relationships: macro / micro alloy structure - process parameters - alloy composition - hot tears [1].

2.1 The billet diameter 178 mm

In the first phase of the study, data was collected share of Foundry Defects and separate results for billet diameter of 178 mm, which had the largest share of errors with cracks. We studied the state of the surface of billet, microstructure, chemical composition, technological parameters of the process and mechanical property of hardness.

2.1.1 Material

The subject of research is batch 37.39 t, which was 21 billet with internal cracks alloy EN AW6060. Ultrasonic inspection showed that the cracks formed in the final phase of the casting. From rejected billetss is highlighting one billet for testing.

2.1.2 Results

The main goal of the study is to identify the initial micro cracks rejected in the billet. The billet is cut into plate-like sample 19 of thickness 20 - 25 mm of which is selected 7 samples: one sample at the start of casting, the three samples in the stable phase of casting and three samples at the final stage of casting. For samples No. 18 and 19 (Figure 1a) in the last stages of the casting is visible macro cracks.

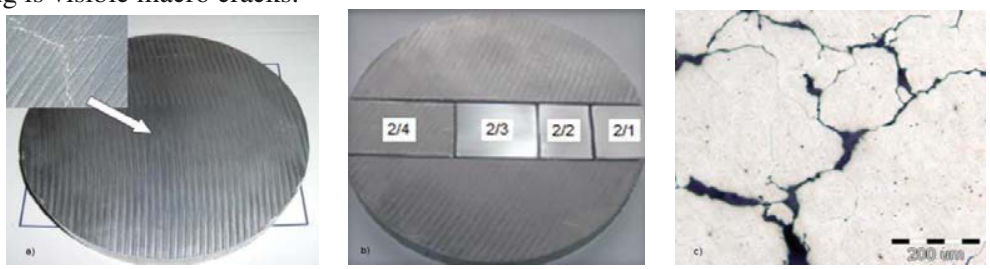

Figure 1. Macro cracks (2×) a), individual samples b) and micro-cracks (100×) c) of billet

In the second phase, samples of mark 2, 7, 12, 16, 17, 18 and 19 are cut out individual samples for metallographic observation in the direction of solidification which is included a central part, for the area and the surface of of billet (Figure 1b). In Figure 1c sample 19/3 shows that developed open microcracks at grain boundaries.

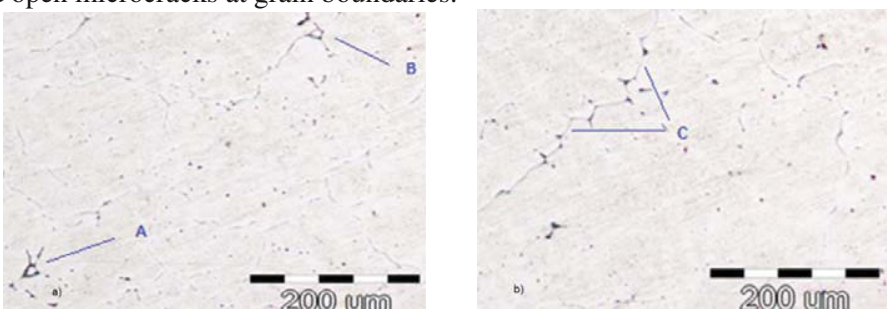

Figure 2. Initial micro - cracks on a sample number 17/3 100×

Figure 2 you can see a variety of alloys developed microcracks sample 17/3 formed by two mechanisms: (1) nucleation of voids resulting from a series of small holes (Detail A and B figure 2a) and (2) cracks formed by separating along the border intermediate particles (Detail C Figure 2b).

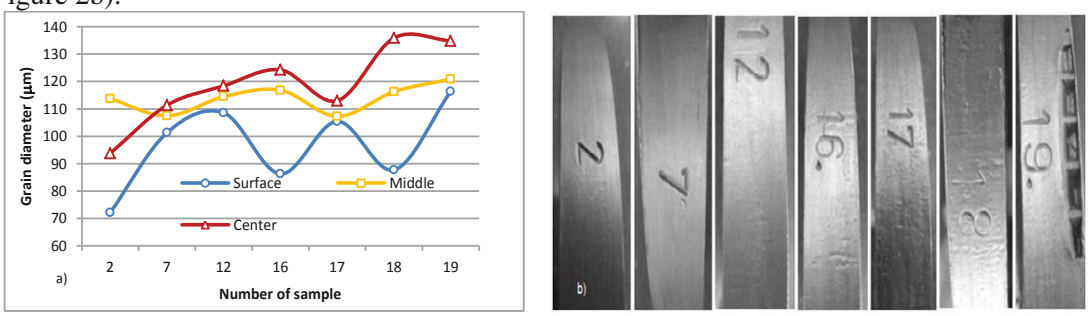

Figure 3. Changes in grain size a) and surface conditionb) of billet alloy EN AW 6060

Analysis of changes in particle size is performed in relation to the cross section and along the length of billet using quantitative image analysis (Leica QWin) (Figure 3b) indicates coarser

grain in the center. Condition sample surface of the billets shows that samples 2, 7 and 12 have very fine surface (Figure 3b), a sample of 17 which appeared microcracs further, visible coarse texture and liquation, and therefore changes in stable conditions as characteristics hot casting conditions. Analysis of the chemical composition of the melt (Table 1) indicates low content of Fe, which plays an important role in controlling the occurrence of hot tearing when with increasing Fe develops stronger 3D network intermetallic phase AlFeSi [2].

Table I - Chemical composition of molten alloy EN AW 6060 batch 815605

Chemical composition (%)		Si	Fe	Cu	Zn	Mg	Mn	Ti	Cr
Required	Min.	0,41	-	-	-	0,38	-	0,007	-
	Max.	0,45	0,18	0,01	0,015	0,42	0,01	0,01	0,01
Achieved	Average	0,435	**0,0704**	0,0002	0,0119	0,0396	0,0013	0,0079	0,0006

Analysis of the chemical composition of the sample after casting (Figure 1.a,b) in the length and cross-section in Figure 4 shows the contents of a key element of iron and boron. Segregation and reduced iron content are present along the billet and per section (Figure 4a). Although the content of Fe in the central parts of the of billet prior to the formation of cracks increases, it is enough to form a strong network of intermetallic phases AlFeSi. Also, boron added to the master alloys (Master alloys) AlTi5B1 segregate along of billet. The particles for nucleation in the melt are hexagonal tiles TiB2 coated with a thin layer of adsorbed Al3Ti. Therefore, the B content of the primary mechanism of nucleation, while the content of Ti in excess means [3]. This is confirmed by the grain size of the sample 12/3 to 19/3 (Figure 3a) where a drop of B content in the sample 16/3 leads to poor grain nucleation (Figure 4b).

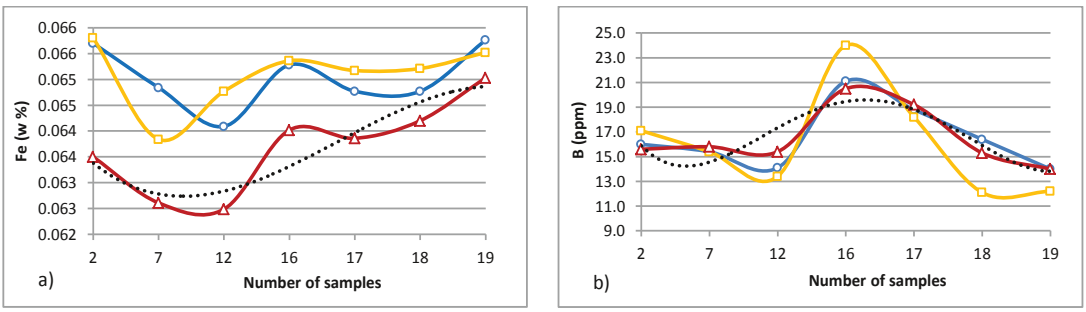

Figure 4.The content of iron a) and boron b) alloy EN AW 6060 to billet batch 815605

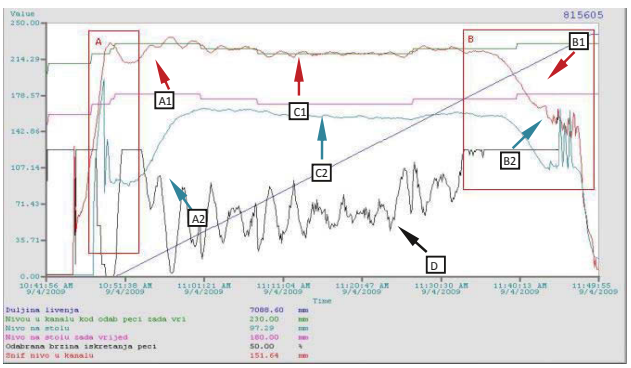

Figure 5. The level of melt in the channel and table casting batch 815605

Measuring technological casting parameters was performed *AutoCast* existing system to control and manage the facility foundry company "Aluminium". Studied parameters controlling the process of casting showed no deviation from the set value unless the parameter which regulates the level of the melt (Figure 5) in the channel (red curve) and on the casting table (blue curve). Deviations are visible at the beginning of the casting (details A1 and A2) as

at the end of casting (details B1 and B2), and somewhat less in the working phase (details C1 and C2) when carried out periodically tilting furnaces (detail D) and maintains sufficient the level of melt at both measurement sites. These disorders affect the poor efficacy AlT5B1 alloy and grain nucleation. Subsequently studied mechanical property of hardness billet alloy EN AW6060 which has presented an assessment of thermo-mechanical regime the procedure of casting on the samples with relatively stable conditions of cooling and solidification (Figure 6a). The influence of residual stress is reduced to a minimum value (multiple cutting samples) as well as natural aging examining hardness ~ 48 months after casting (hardness for alloy EN AW 6060 stabilizes after 10 000 hours).

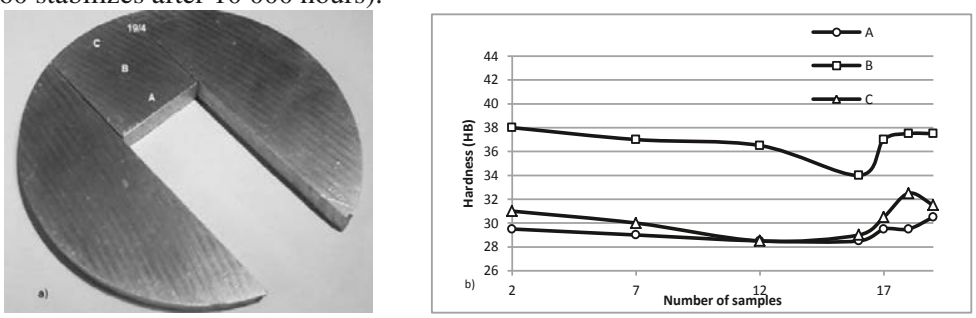

Figure 6. The individual sample results 19/4 a) and of hardness over the length billet b)

Results of measuring of hardness samples at measuring points A, B and C according to the method *Brinell* are shown in Figure 6b. We notice the steady decline in of hardness until the appearance of the initial failure of the 17 where the hardness increased which indicates a change thermomechanical regime of casting. Looking at a stable equilibrium measurement point B, after conditions that lead to cracks, hardness of the sample stabilizes.

2.2 The billet diameter 203 mm

Research alloys EN AW 6060 are further expanded on the 822,723 batch testing in multiple transverse directions billet, larger diameter billet, but have been reduced to two critical phases of casting: the beginning and end of casting.

2.2.1. Material

Subject of the research is the batch of 43,55 t from which they were cast billets with a diameter Φ 203 mm alloy EN AW6060. For testing were allocated three billets: 1) log with internal cracks, 2) As cast billet and 3) homogenized billet. Cylindrical specimens height of 60 mm (Figure 7 a) cut off from the initial and final stages of casting: head and foot billet, designed for mechanical hardness test (Figure 7b). For other tests are cut out a cylindrical sample in the central zone mark 7 and samples stable zone solidification in the form of circular clips mark 1 - 6 (Figure 7c).

Figure 7. Sample of billet a) to test of hardness b) and chemical composition c)

2.2.2 Results

On the marked samples from the image 7b measured hardness of the center to the surface of the billet on the 18 directions. The measured field of hardness of the head (Figure 8a) cracked billet (mark C - H) show less variation of hardness in the cross section of the center to the interior of the billet except the maximum expressed on the surface. But at the foot billet

cracked hardness of the center is larger (Figure 8b), which is in contradiction with the foot properly As cast billet (Figure 8c) which indicates a high residual stresses [4].

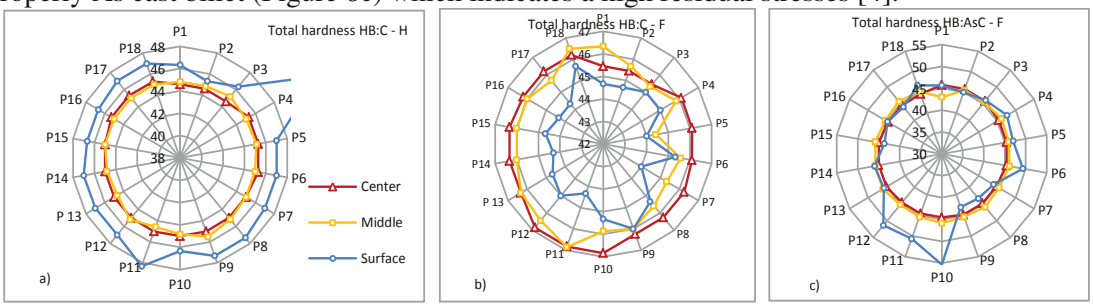

Figure 8. The measured field of hardness in the cross-section of the foot and the head of cracked billet a,b) and foot of As cast billet c)

Analysis of the chemical composition of the melt (Table 2) show reduced content of Ti and Mn.The chemical composition of the two classes billet: with internal cracks and As cast billet, shown i figure 9 with marks: CH and CF (head and foot cracked billet), ASC-H; and ASC-F (head and feet As cast billet). Increased content of Fe on the samples 1 - 7 foot (curve CF, ASC-F, Figure 9a) cracked billet will increase the thermal conductivity alloys hotter center to the cold surface of the billet and increase the stress caused by solidification. Also, a greater sensitivity to cracks due to reduced content of B shows the head of both billets (Figure 9b).

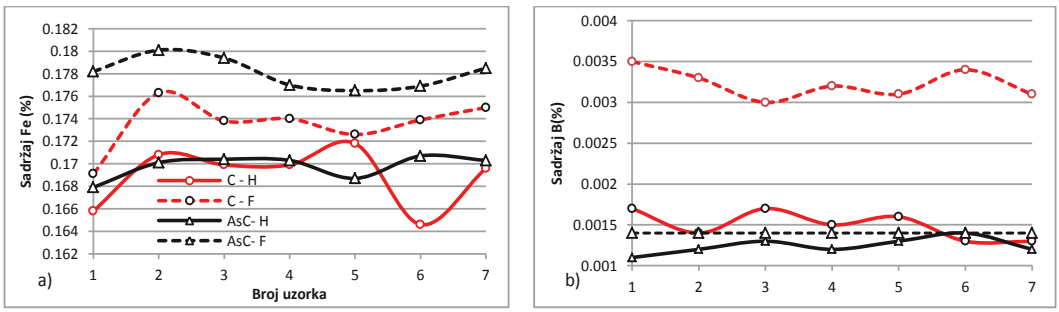

Figure 9. Iron content) and boron b) heads and foot cracked and As cast billet

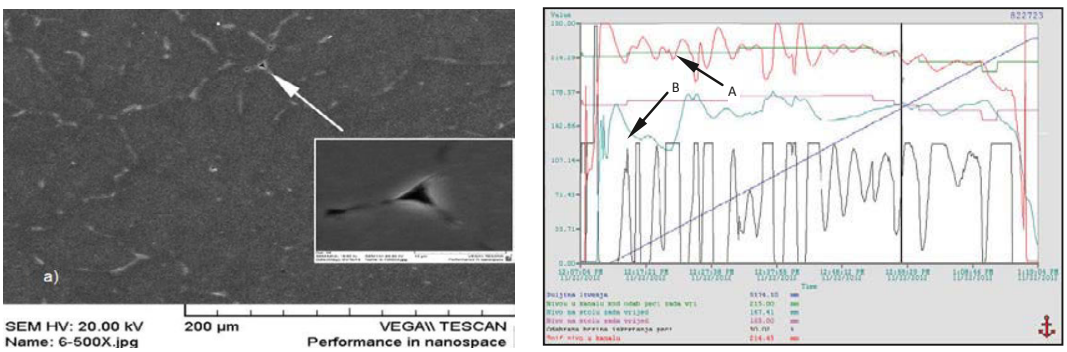

Figure 10. SEM micrographs foot of bilet with crack a) and level of melt batches 822,723 b)

Testing by SEM microstructural characteristics (Tescan, 2008, Vega TS) exhibits microcracking in the central sample foot of bilet with internal cracks (Figure 10 a), and confirmed by EDX method (Al 99.01%, atomic ratios Mg:Si =10: 1. Measuring the parameters of the casting process shows that the SNIF level of melt in the channel (Figure 10b, detail A) and the level of on the casting table (Figure 10b detail B) lower throughout the casting, which is a disorder similar to that of bilet diameter 178mm.

3. Impact of the ratio of heavy and light metal sensitivity occurrence of internal cracks

In order to reduce thermally induced casting defects, it is necessary to act preventively in the preparation phase melt. That is followed by the content of light and heavy metal along the length of bilet batch 815 605 Significant segregation of heavy (or light) metal (Figure 11a,b) up to 600 cm are evidence that in some locations, dendrite branches that break away, fresh melt can be replaced and cracks heal, and the absence of segregation of heavy metals (Figure 11a) at the end of casting (650 -700 cm) favors fully opening cracks. This established coefficient ratio ($K_{AV} = \Sigma\ w_{(Hm)} / w_{(Lm)}$) between the content of light and heavy metals where w (Hm) and w(Lm) by weight of heavy and light metals.

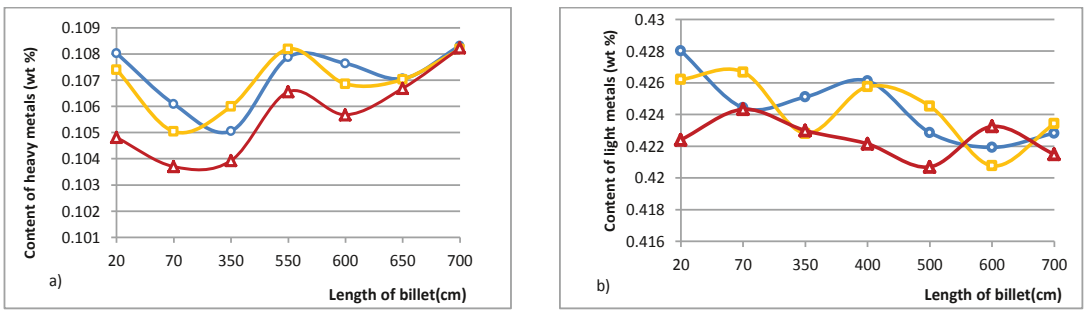

Figure 11. Heavy a) and light metals b) the length of billet batch 815605

Model function adopted border average ratio of heavy and light metals $k_{AV} = 0.25$ as a measure of the midpoint of the sensitivity of occurrence of internal cracks that in addition to the valid parameters of casting, provide a mechanism for increasing strength in relation to the increase in stress caused by contraction of the alloy. Verification of the model value for k_{AV} samples in the central part of the of billet for batch 822723 is shown in Table 3, indicates that samples of the head and foot cracked of billet found in this sensitive area, with the exception of samples at the feet of as cast billet.

Table II - The calculated values of the coefficient k_{AV}

Batch 822723	Cracked billet		As cast billet	
	Head	Foot	Head	Foot
k_{AV}	0,235	0,237	0,237	0,224

4. Conclusion

The paper focuses on the results that may suggest possible mechanisms and the causes of hot cracks alloy EN AW6060, and the parameters of the structure, chemical composition and mechanical properties describe the conditions that determine the stability of the campaign casting billets. Repeatability critical phases of casting can sometimes be absent in the process control system that balances the hot and cold casting conditions. At some point in time they grow to the level of response when critical casting parameters in time will fail to achieve their inter balance, which indicates the need for preventive action in preparation melt.

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Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

WEIBULL ANALYSIS FOR THE REPEATIBILITY OF DIE CASTINGS MADE BY AN Al-Mg-Si-Mn ALLOY

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Keywords: Aluminium alloys; Casting; Mechanical property; Microstructure; Weibull analysis

Abstract

The Weibull statistical model with three parameters has been used to assess the repeatability of the high pressure die-castings made by an Al-Mg-Si-Mn alloy. It was found that the round samples had a maximum Weibull modulus, indicating the best repeatability. The machined samples exhibited the second best of Weibull modulus. Among the square samples, the 2 mm thick samples had the lowest and the 5 mm thick samples had the highest Weibull modulus, indicating that the repeatability was worsened for the castings with thinner wall thickness. The microstructural uniformity and porosity levels are critical factors in determining the repeatability of high pressure die-castings.

Introduction

In the manufacturing of thin-wall components by high pressure die-casting (HPDC) process, one of the most important characteristics is that the melt is filled into the die cavity with turbulent flow during casting [1,2]. This results in a high probability of forming defects, in particular porosities in individual die-casting. More importantly, the defect distribution in the die-castings is very random and irregular, which results in a large uncertainty in the mechanical properties and poor repeatability between different types of die-castings and even in the same location of a casting made in different manufacturing cycles [3]. In reality, a relatively large design margin is usually given in order to compensate for the detrimental effect caused by internal defects in die-castings. However, this significantly increases the casting weight, which is one of the major concerns in the specific structural applications for the automotive and aerospace industry. Therefore, the repeatability of castings becomes greatly important in the manufacturing of thin-wall castings.

In the present study, the tensile properties were measured for a set of standard samples that were directly made by HPDC process and a set of samples that were obtained from machining samples from various locations of a larger automotive die-casting. The distributions of the yield strength, ultimate tensile strength (UTS) and elongation of the samples were analysed by the average value with standard deviations and by the Weibull statistical model with three parameters. The correlations between the tensile properties and the microstructural features, porosity levels and fracture morphology were investigated for the different types of samples.

Experimental

The Al-Mg-Si-Mn alloy was supplied by an industrial supplier with the measured composition of 4.90-5.11wt.%Mg, 1.89-2.05wt.%Si, 0.56-0.61wt.%Mn, 0.16-0.18Ti, 0.08- 0.1wt.%Fe, 10-12 ppm Be and Al as balance. During experiments, the alloy was melted at 710 ^{o}C. Two sets of die were used for the present study. One was to cast six ASTM standard samples with three ϕ6.35 mm round samples and three square samples with cross sections of 6.35×2mm, 6.35×3mm and 6.35×5mm, respectively [4,5]. Another die casting was an automotive component with a typical wall thickness of 2.8mm and the overall size of casting is 300×45×40 mm. 20 pieces of trial castings were selected for making tensile samples. Five standard square samples with a cross section of 6.35×2.8mm were machined from each trial casting. All parameters had been optimized for the HPDC machine during casting trials.

The tensile tests were conducted following ASTM standard B557, using an Instron 5500 Universal Electromechanical Testing System equipped with Bluehill software. All tensile tests were performed at ambient temperature. The gauge length of the extensometer was 25 mm and the ramp rate for extension was 2 mm/min. The microstructure of each alloy was examined using a Zeiss optical microscope with quantitative metallography, and a Zeiss SUPRA 35VP scanning electron microscope (SEM).

Results and Discussion

1. Mechanical properties

The measured tensile properties of the Al-Mg-Si-Mn alloy are shown in Fig. 1 for 99 of φ6.35 mm round samples, 40 of square samples for each thickness and 100 machined samples from 20 pieces of trial castings with two standard deviations (also termed as two-sigma). The average of the round samples was 146.6 MPa for yield strength, 302.7 MPa for UTS and 16.5% for elongation. Majority of the measured data were within the square defined by the two-sigma. This confirms the good consistence of tensile properties. Meanwhile, for square samples, the average value was 7.9% of elongation, 174.8 MPa of yield strength and 295.6 MPa of UTS for 2mm thick square samples, and the average value was 10.2% of elongation, 164.2 MPa of yield strength and 306.0MPa of UTS for 3mm thick square samples, and the average value was 14.9% of elongation, 165.7 MPa of yield strength and 314.9MPa of UTS for 5mm thick square samples. Furthermore, the machined samples were in square with two as-cast surfaces and two machined surfaces. The results in Fig. 1 showed that the measured data were almost overlapping for the different locations, although slight differences could be observed. The total average was 9.44% of elongation, 151.3 MPa of yield strength and 271.4 MPa of UTS. The averages of the measured data at each location were quite close the total average because of the small standard deviations.

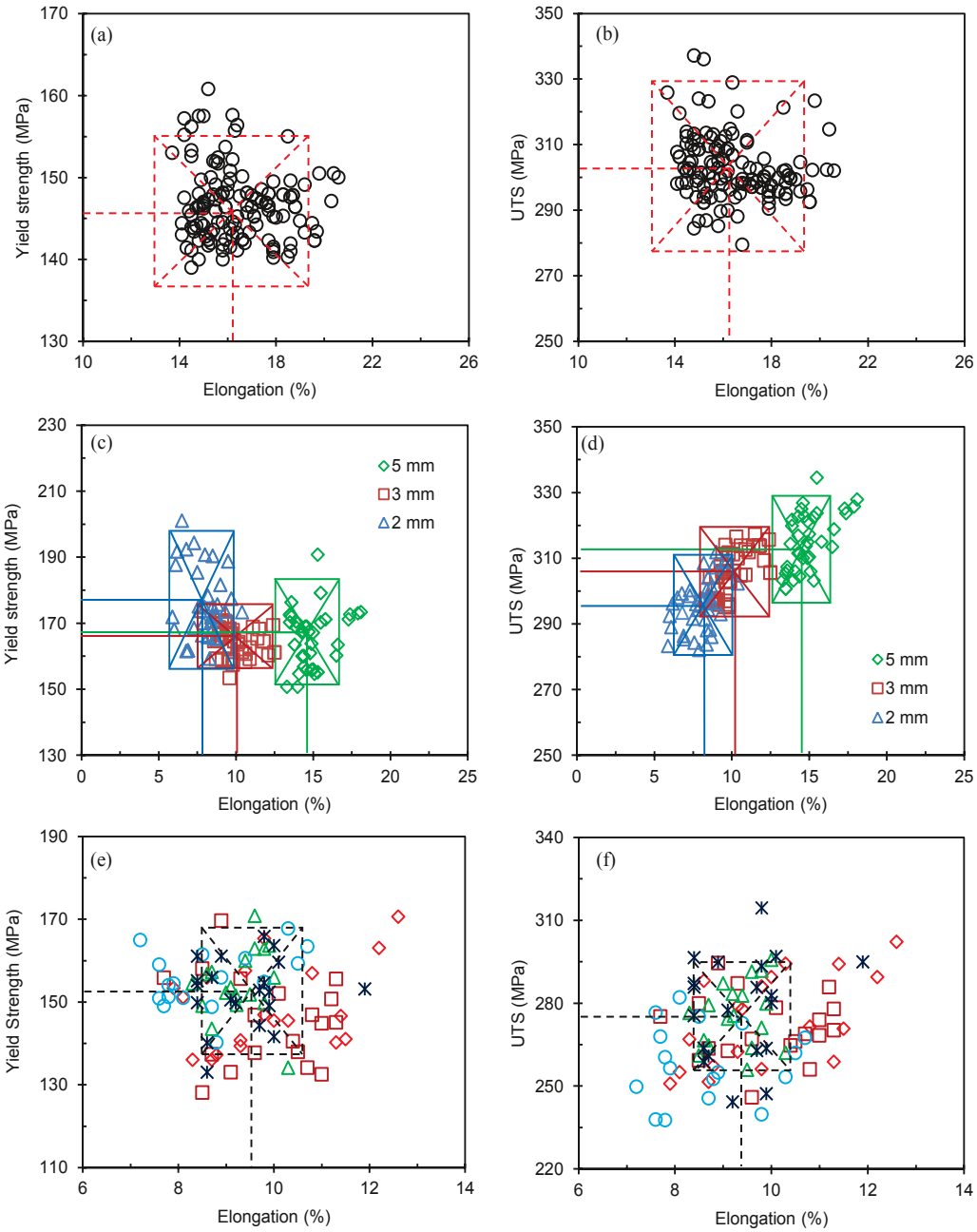

Figure 1. The (a, c, e) yield strength and the (b, d, f) ultimate tensile strength (UTS) as a function of the elongation of (a, b) the standard φ6.35mm round samples, (c, d) the standard flat tensile samples with a cross section of 6.35×2mm, 6.35×3mm, 6.35×5mm, (e, f) the flat tensile samples cut from the as-cast castings with a cross section of 6.35×2.8mm. The squares represent the value of two times standard deviation of the tested samples and the centre of squares is the average value of tested data.

Table 1. Calculated parameters in Weibull statistics analysis of the yield strength, ultimate tensile strength and elongation of the tensile samples with as-cast surface and with machined surface.

Property	Sample	m	σ_u	σ_0	R^2
Yield strength	ϕ6.35mm, as-cast surface	4.43	131	26.78	0.9899
	5×6.35 mm, as-cast surface	2.54	145	23.36	0.9612
	3×6.35 mm, as-cast surface	2.78	145	18.67	0.9747
	2×6.35 mm, as-cast surface	1.65	158	14.82	0.9755
	2.8×6.35 mm, machined surface	3.92	117	37.71	0.9885
Ultimate tensile strength (UTS)	ϕ6.35mm, as-cast surface	4.17	260	43.82	0.9759
	5×6.35 mm, as-cast surface	2.81	297	19.17	0.9728
	3×6.35 mm, as-cast surface	2.71	293	17.67	0.9425
	2×6.35 mm, as-cast surface	2.11	279	14.86	0.9787
	2.8×6.35 mm, machined surface	3.34	223	54.26	0.9924
Elongation	ϕ6.35mm, as-cast surface	4.81	13.2	11.36	0.9913
	5×6.35 mm, as-cast surface	2.40	4.5	3.88	0.9837
	3×6.35 mm, as-cast surface	1.89	8.5	3.84	0.9904
	2×6.35 mm, as-cast surface	1.32	3.7	2.68	0.9671
	2.8×6.35 mm, machined surface	3.50	7.4	5.48	0.9814

The results showed that the thinner samples exhibited lower elongation and lower UTS, but higher yield strength for square samples. In comparison of the tensile properties obtained by the round and the square tensile samples, the yield strength, UTS and elongation were all higher for the round samples. In particular, the average elongation of the round samples was doubled over that of the 2mm thick square samples, although the yield strength was higher and the UTS was lower for the 2mm thick square samples. On the other hand, it is also seen that an overlapping among the measured data for the different types of samples. The overlapping had different levels, but over 35% of the measured strength was seen at similar levels, which indicates that the overlapping of the data is significant. Overall, it is seen that the application of average value with standard deviations is capable of assessing the tensile properties, but the overlapping is significant. It is thus very rough to differentiate the data scatter and error distribution in the trial castings.

2. Weibull analysis

In order to assess the repeatability using Weibull analysis, the data in Fig. 1 was reorganised in ascending order for the calculated Pf value, which is calculated using Bernard method [6] for the failure probability as $P_f = (i\text{-}0.3)/(n+0.4)$, where i is the ranked position of the specimen strength (or strain) in that set of castings and n is the total number of specimens. During modelling, the representative Weibull plots were fitted with

$$ln\left[ln\left(\frac{1}{1-P_f}\right)\right] = m[ln(\sigma - \sigma_u) - ln\sigma_0] \quad (1)$$

where σ is the variable being measured (yield strength, UTS or elongation to fracture); σ_u is the threshold parameter, i.e. the characteristic stress (or strain) below which no specimen is expected to fail; σ_0 is the scaling parameter, also called the characteristic stress (or strain) at which the probability of failure is 63.2%; m is the shape parameter, also referred to as the Weibull modulus [7].

The best-fit line was obtained by the maximum R-square value. After finding the maximum R-square value, it was immediately used to evaluate the applicability of the Weibull distribution for the measured results by utilising a goodness-of-fit. For assessing the goodness-of-fit, the critical values of R^2 with α=0.05 ($R^2_{0.05}$) was calculated according to the formula

$$R^2_{0.05} = 1.0637 - \frac{0.4174}{n^{0.3}} \quad (2)$$

For the as-cast round samples, the sample number was 99, the calculated $R^2_{0.05}$ was 0.9585. The principle of assessing the measured data is that the Weibull distribution is not applicable if the R^2 of the linear regression from the Weibull probability plot is less than the value of R^2 with α=0.05. This means that the R^2 from the regression equation needs to be larger than the $R^2_{0.05}$, in order to satisfy the Weibull distribution. By comparing the R^2= 0.9585 from the equation (2), the R^2 value was 0.9759 for the best-fit line for the as-cast round samples. The higher R^2 value means that the measured data can be well described by Weibull distribution. By using the same methodology, the statistic results were obtained for different parameters with different types of samples. Table 1 lists the characteristic parameters in Weibull distribution of the yield strength, UTS and elongation of the tensile samples with as-cast surface and with machined surface.

The R-square data in Table 1 confirmed that the tensile test data can be adequately modelled by three-parameter Weibull statistical analysis. For the yield strength, the as-cast round samples had a maximum Weibull modulus, which indicates that the as-cast round samples exhibited a best reliability, repeatability and reproducibility. Meanwhile, the machined samples exhibited the second best of Weibull modulus, which means that the machined samples have better repeatability than the as-cast square samples. Among which square samples the 2 mm thick samples had the lowest and the 5 mm thick samples had the highest Weibull modulus. This indicates that the repeatability was worsened for

the castings with thinner wall thickness. The square samples showed a lower value of Weibull modulus than the machined samples and the as-cast round samples. On the other hand, as described above, the parameters σ_0 and σ_u are preferred to be higher in the castings because no sample is expected to fail below σ_u and the probability of failure is 63.2% at σ0. However, it was not observed the highest σ_u for the yield strength, UTS in the as-cast round samples and only the σ_u for elongation is the highest value. On the other hand, although slight inconsistent in the parameters for elongation, the as-cast samples provided the higher σ_u. This means the samples with as-cast surface provided better strength than the machined samples. When reviewing the data of σ0, the results showed that the as-cast round samples had the highest value and the machined samples had the second best results. This is consistent with the data of Weibull modulus (m). Overall, comparing the as-cast round samples and the as-cast square samples, the as-cast round samples showed good repeatability and reliability. It was expected that the machined samples would show worse ductility than the as-cast samples because of the removal of casting surface. However, good performance as described by the Weibull parameters was shown through the different types of samples. The repeatability and reliability of the machined samples was reasonably good, although the strength under which the samples were not failed was slightly lower than the directly cast samples. It is important to note that the values in the Weibull parameters for the as-cast square samples, in particular the 3 mm thick square samples were relatively close to those obtained from the machined samples.

3. Microstructure

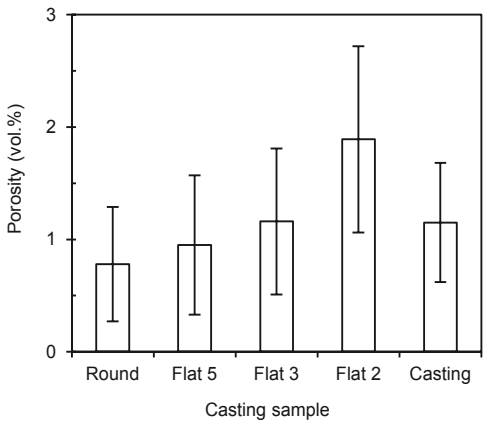

Figure 2. The porosity levels of the tensile samples obtained from the different shapes and surface conditions. 'Round' represents the ϕ6.35mm sample; 'Flat 5' represents the 5×6.35mm sample; 'Flat 3' represents the 3×6.35mm sample; 'Flat 2' represents the 2×6.35mm sample; 'Casting' represents the 2.8×6.35mm sample machined from the trial castings.

The porosity levels were quantitatively measured using optical microscope on the as-polished metallographic surfaces of the different type samples, which were also selected from the castings made under the same batch as the fractured samples. The results are shown in Fig. 2. The average porosity levels were from 0.8 vol.% to 1.9 vol.%. The as-cast round samples had the least amount of porosities and the as-cast 2 mm thick square samples had the highest amount of porosities. The porosity levels in the trial castings were similar with that in the 3 mm thick square samples; both of them were higher than that in the as-cast round samples. The results of porosity levels are consistent to the parameters obtained from Weibull analysis, in which the samples with less porosity provided the increased Weibull modulus, threshold parameter and scaling parameter. Therefore, the improvement of repeatability, the increase of the failed strength and the failure probability of 63.2% were achieved.

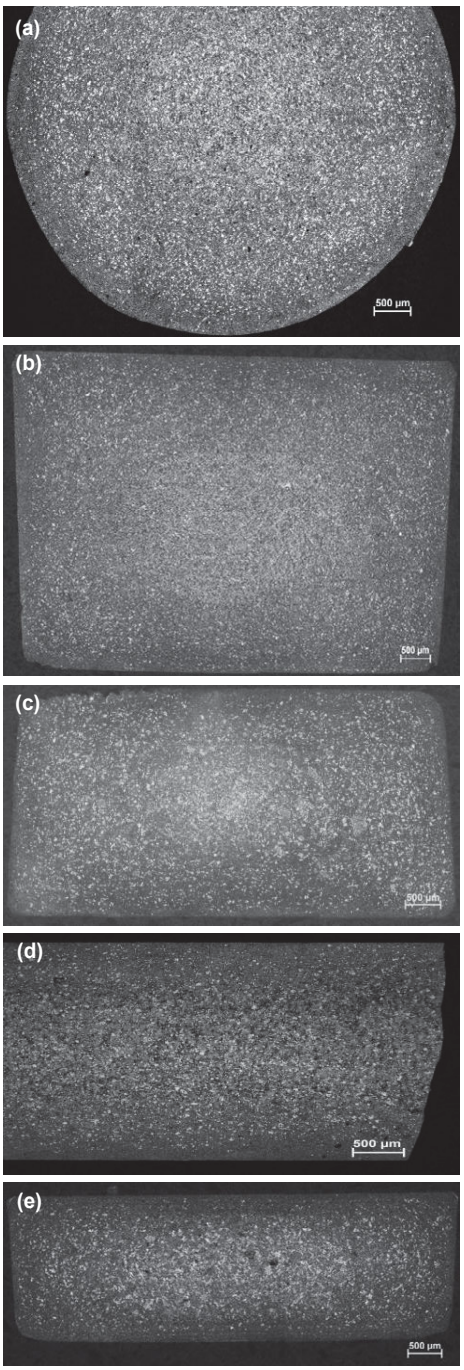

Figure 3. Optical micrographs showing the microstructure of the die-cast tensile samples on a cross section of (a) φ6.35 mm, (b) 5×6.35 mm, (c) 3×6.35 mm, (d) 2.8×6.35 mm with machined surface, and (e) 2×6.35 mm.

Generally, the microstructure plays key roles in determining the materials tensile properties. Therefore, the as-cast microstructures were examined for the different types of samples. The similar results were observed in different casting samples. Fig. 3 shows the microstructure on a cross section of the die-cast tensile samples. As seen, the microstructure showed an apparent skin layer with less primary α-Al phase. The thick samples exhibited severe segregations. However, the detail microstructures were very similar in the skin region and the central region of different samples. The mixture of coarse and fine primary α-Al grains was observed, although the centre had coarser primary α-Al grains and slightly increased defect levels. From these results, the microstructural uniformity and defect levels are the fundamentals to result in the difference in the tensile properties and in the Weibull parameters obtained in Weibull statistical analysis.

Summary

The repeatability and reliability of castings made by die-cast Al-Mg-Si-Mn alloy can be modelled by three-parameter Weibull analysis because of the high R-square value in the linear fitting of experimental data. The round samples had a maximum Weibull modulus for the modelled yield strength, thus showed the best repeatability. The machined samples exhibited the second best of Weibull modulus, which indicates that the repeatability of machined samples is better than the square samples. Among the square samples, the 2 mm thick samples had the lowest and the 5 mm thick samples had the highest Weibull modulus, indicating that the repeatability was worsened for the castings with thinner wall thickness. The microstructural uniformity and porosity levels are critical factors in determining the repeatability of high pressure die-castings. The less segregation in the microstructure can uniform the stress distribution and the less porosity in the casting can reduce the sources for brittle fracture. These can improve the repeatability in casting production.

Acknowledgement

The authors acknowledge the Engineering and Physical Sciences Research Council (EPSRC), Technology Strategy Board (TSB) and Jaguar Land Rover (JLR) in United Kingdom for financial support.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THERMO-MECHANICAL PROPERTIES OF WROUGHT ALUMINIUM ALLOYS PRODUCED FROM SCRAP MIXING

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Keywords: Scrap, Recycling, Aluminium Alloys, JMatPro
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Abstract

A significant demand in the use of aluminium alloys, most especially in the automobile industries, has been recorded over the past few decades. This is due to light-weight properties of aluminium alloys which when considered in many industries has a significant increase on the fuel efficiency economy of automobiles. The sustainable production of aluminium from recycling of scraps helps in better energy efficiency, cost effectiveness and reducing carbon footprints. Scraps for recycling can easily be obtained through the post-consumers and post-manufacturers. 100% compatibility of the feedstock from these scraps has been a major challenge in the automotive industry. It is very difficult to get alloys which have compatibility of 100% substitution rate. This study compares and predicts using JMatPro the thermo-physical and mechanical properties of these alloys after mixing and the aftermath effects on their service life

Introduction

Aluminium is a soft, reactive and non-magnetic ductile metal which exists in the earth crust in a combined form known as bauxite due to its high chemical reactivity. Excellent properties like ductility, malleability, lightweight, corrosion resistance etc. makes it one of the most sought after metallic alloys in the world and it is widely used in the aerospace, transportation, packaging, and construction industries etc. It is obvious that present development of industry is coupled with necessity for application of lightweight materials with beneficial mechanical and tribological characteristics that can fulfil technical and technological requirements. One of the best suited material with characteristics that can provide most of the modern demands is aluminium and its alloys [1-2].

There has been a steady increase in the demand for the use of aluminium alloys in automobiles over the past few decades and the availability of old and new scraps has made recycling a viable option for the production of aluminium alloys. The advantages of using lighter metals e.g. aluminium, over steel is enormous. Key areas where these advantages are obvious are in energy savings and efficiency as related to the fuel economy; cost effectiveness, carbon footprint reduction etc. which can all be summarized under the caption called Sustainability. Today, more than 50% of automobile parts come from recycled materials and studies have also shown that the recyclability is enhanced with increased content of light metals (aluminium and magnesium). Aluminum does not lose any of its important properties or damage the metal's structure it can then be recycled indefinitely. In the automobile industry, aluminium alloys of various grades have already gained wide acceptance in applications like engine blocks, transmission casings, radiators etc. Selecting purer scrap grades will help in obtaining aluminium alloys with increased castability and machinability, improved performance and properties (toughness and ductility). This makes it obvious and imperative that recycling saves natural resources and reduces the environmental impact of the solid waste posed by discarding what can be a valuable resource [3-6].

Recycling is a secondary process in aluminum production (melting from aluminum ingots, scraps, etc.) which offers significant energy savings and environmental benefits over primary aluminum production since the former consumes only five percent of the energy used in the latter process. In other words; recycling saves energy, reduces pollution and preserves environmental condition, preserves natural resources, and saves the space that is used for waste disposal. Higher compositional adaptability in combining various grades of clean scrap and dross are efficiently addressed. While today's recycling metals markets also include ferrous metals like iron and steel, and nonferrous metals like copper and brass, aluminum recycling is the engine of recycling economics. The growth in aluminum usage in transportation applications, relative decline in aluminum beverage can recycling, and increasing reliance of the fabrication industry on secondary aluminum have combined to create new needs in both the materials design and processing space. To economically utilize these scrap streams, new approaches to developing acceptable materials processes to control properties suitable for an expanded range of applications are needed. Aluminum remains the most economically attractive material from which to make aircraft and space vehicles, and new construction proceeds at a prodigious rate [7-10].

Of recent, large volumes of wrought aluminium alloys have been from aluminium scraps. But due to the variation in the elemental composition of these gathered scraps, mixing to obtain an alloy of great quality has been a great challenge from the metallurgical standpoint. In recent times, more research both in academic and industrial projects have been dedicated to the complete understanding of compositional tolerance in wrought aluminum alloys. Adequate quality and quantity of scrap which correlates with the chemical composition of the intended alloy must be in check hence, careful pre-screening, pre-sorting and separation of different alloys must be done before mixing and melting takes place [11].

For melting of aluminium scrap the main emphasis is on meeting the ideal ecological and economic conditions in the best possible way [12]. Scraps for recycling can easily be obtained through the post-consumers and post-manufacturers. Scraps from post-consumers contain paints, oil, water etc. Most of these impurities exist with the molten alloy in the melting furnaces and these

causes oxidization of aluminium alloys. The energy consumed in the production process due to the presence of impurities is very high while the production yield of the finished products is always very low in this regard. 100% compatibility of the feedstock from these scraps has been a major challenge in the automotive industry. It is very difficult to get alloys which have compatibility of 100% substitution rate [6].

The major challenges of these srap alloys prior to mixing are;

a) To automate and optimize pre-sorting, shredding, and separation technologies, and make them broadly available is a herculean task.
b) To identify more useful by-products to handle elemental residual unsuitable for reuse in recycled metal is another issue.
c) To broaden the number of available aluminum alloys whose specifications will readily directly accept recycled metal and will perform well in high-quality, value added products.
d) To be able to achieve the prescribed compositional tolerance or proper chemical composition when obtaining wrought aluminium alloys from scraps.

Scrap segregation is critical because of the highly alloyed product mix. Managing scrap containing high concentrations of zinc, copper and magnesium into an existing primary location greatly increases the risk of cross contamination from poor scrap control procedures [13].

To convince wrought aluminium alloy users of its improved mechanical properties achieved through recycling, proper organization of scrap yard of different material streams each with proper and well-controlled chemical composition is of paramount importance. And to achieve the desired alloy composition, a software package dedicated mainly to calculating the proper combination of material streams is highly essential. Figuring out how to achieve standard composition of pre-melting mixture of material streams and better mechanical properties of required alloys whereby the production cost of the entire processing chain can be minimized will suggest solutions to challenges faced by scrap mixing [14].

It is very difficult to get alloys which have compatibility of 100% substitution rate. This study compares and predicts using JMatPro the thermo-physical and mechanical properties of these alloys after mixing and the aftermath effects on their service life

Discussion

In this study, aluminium wrought alloys 6111 and 5182 which are among the most important sheet alloys in the body sheet and body structure of the new aluminium-intensive vehicles are explored. Table I shows the minimum and maximum concentrations of these alloys [6, 15].

In this case, the compatibility of moving from one alloy to another was analyzed. There was a careful review in the entire range of element to determine if they are 100% compatible and go directly from one alloy to a new desired alloy but major problem is if the stamping skeleton scrap is to purposely go into an alternative (desired) alloy or if the alloy can just be mixed. Using alloy 6111 as a feed material for making 5182, the main limiting elements are Si and Cu while the limiting element in the case of 5182 going into is Mg.

No single alloy can go back into another alloy at a 100% addition rate therefore it is critical that all returning alloys be properly identified and segregated. Charging the incorrect scrap to a melting furnace can lead to off-analysis conditions and can become a costly mistake for the melt shop. Another potential problem for the melt shop is having mixed scrap loads with more than one alloy present being returned from scrap yard. Unless further advanced sorting is performed a mixed alloy scrap is guaranteed. As has been previously shown, none of the alloys are compatible at a 100% substitution rate. However, we may choose to use lesser amounts of one alloy as a feedstock for another and make up the balance of the furnace charge with the parent alloy or with a prime based charge [6].

Even though these alloys are not 100% compatible, understanding the resultant properties of the alloy as result of the mixture of one scrap to obtain a specific alloy would be of great importance. This will set precedence and better guidelines in future research as regards scrap mixing. JMatPro [16]; a metal thermodynamic software was using in the prediction analysis. It was used in determining the mechanical and thermo-physical properties of these alloys.

Fig. 1 and 2 shows the mechanical strength of alloys 6111 and 5182→6111. The tensile strength of 6111 as the temperature increases is 250 MPa which is lower than the strength obtained when 5182 is used as feedstock to get 6111. The tensile strength of the new alloy is around 370 MPa affirming that there is a big difference in properties of both alloys. The T5-T6 strength (heat treatment) as a function of time for both alloys matches perfectly. The strength decreases as the time increases. Accordingly, the solidification rate in a metallurgical system can be estimated by the enthalpy (H) and heat capacity (C_p), and how these thermodynamic properties reflect the system thermal state and heat energy requirements. Fig. 3–6 accounts for the total enthalpy and specific heat of alloys 6111 and 5182→6111. Fig. 3 and fig. 4 shows that the enthalpies of both alloys increases with temperature and follows the same curve. There is a slight change in the enthalpy curves around 600^{o}C and the enthalpy of both alloys is slightly above 1000 J/g. For the specific heat curves, there is a phase change at certain temperatures. For 6111 (fig. 5), the specific heat was constant up to 500^{o}C (1.0 J/gK); a slight change around 550^{o}C (2.0 J/gK) and a big jump around 630^{o}C (16 J/gK). The jump may be as a result of latent heat. Fig. 6 shows the total specific heat for alloy 5182→6111. Phase changes occurred around 450^{o}C (4.3 J/gK) and 630^{o}C (12 J/gK).

Table I: Concentrations of Selected Aluminium Alloys [6, 15]

Alloy	Si		Fe		Cu		Mn		Mg		Cr		Zn		Ti	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
6111	0.7	1.1		0.4	0.5	0.9	0.15	0.45	0.5	1		0.1		0.15		0.1
5182		0.2		0.35		0.15	0.2	0.5	4	5		0.1		0.3		0.1
6111 → 5182	1.1*		0.4		0.9*		0.45		1		0.1		0.15		0.1	
5182 → 6111	0.2		0.35		0.15		0.5		5*		0.1		0.3		0.1	

* Limiting Elements

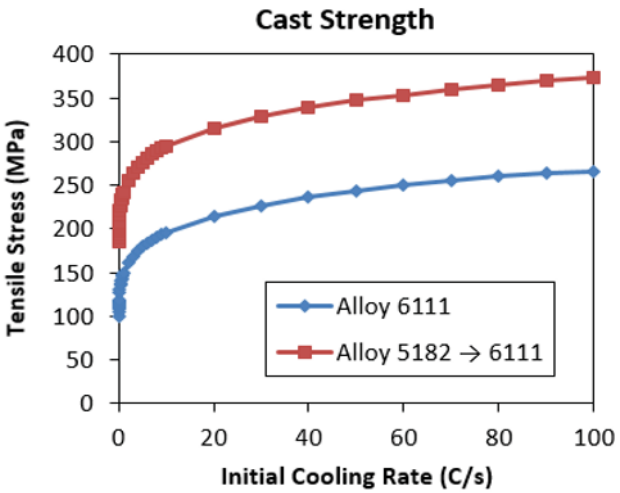

Fig. 1: Mechanical Strength vs Temperature of Alloys 6111 and 5182→6111

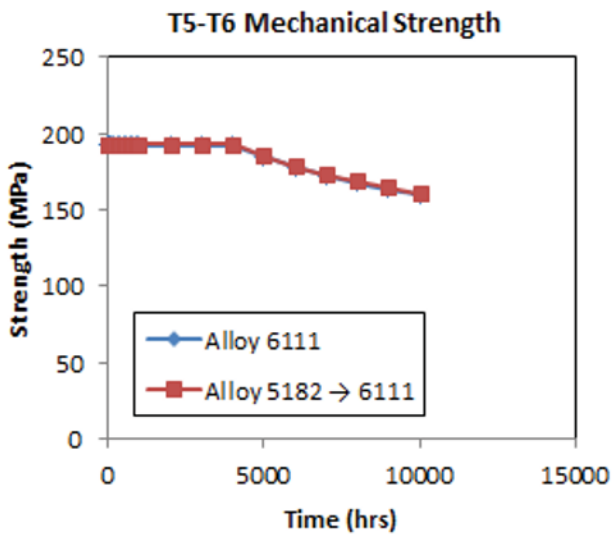

Fig. 2: Mechanical Strength vs Time of Alloys 6111 and 5182→6111

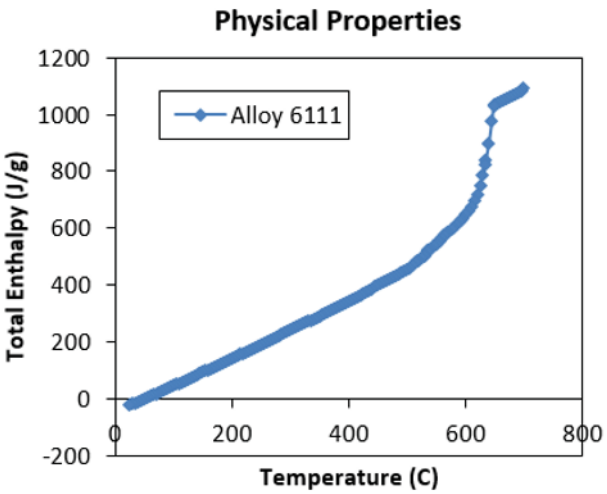

Fig. 3: Total Enthalpy of Alloy 6111

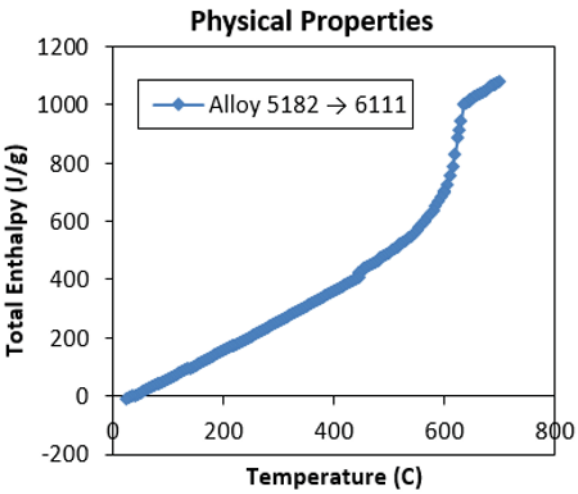

Fig. 4: Total Enthalpy of Alloy 5182→6111

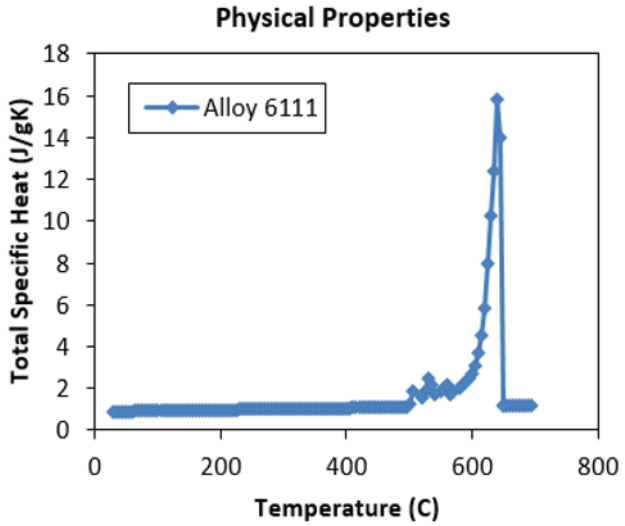

Fig. 5: Total Specific Heat of Alloy 6111

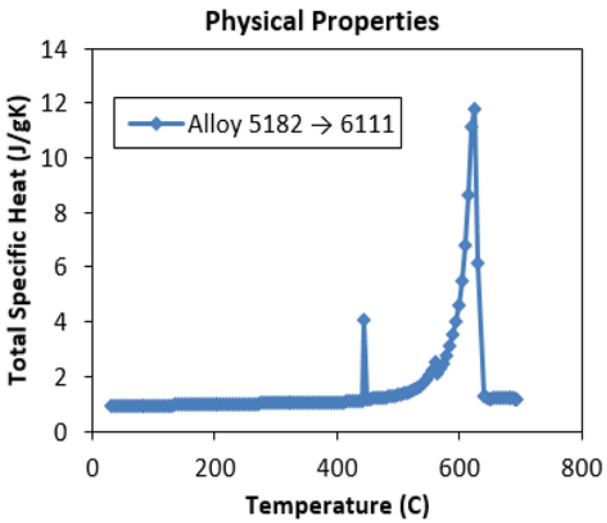

Fig. 6: Total Specific Heat of Alloy 5182→6111

The measured mechanical strength, enthalpy and specific heat results of alloys 5182 and 6111→5182 are plotted as shown in Fig. 7–12. Fig. 7 compares the tensile strength of alloy 5182 and 6111→5182. Alloy 5182 has higher tensile strength while 6111→5182 has lower strength reiterating the incompatibility in this feedstock. According to fig. 8, the T5-T6 strength (heat treatment) as a function of time for alloy 6111→5182 is greater than alloy 5182 but both strength decreases as the time increases. The strength decreases as the time increases. The enthalpy and specific heat of both alloys are also considered. The result in fig. 9 and fig. 10 shows that the enthalpy increases with an increase in temperature and both curve follows the same pattern. There is a slight change in the enthalpy curves around 600°C and the enthalpy of both alloys is slightly above 1000 J/g. For the specific heat curves (fig. 11 and 12), there is a phase change at certain temperatures. For 5182, the specific heat was constant up to 450°C (1.0 J/gK) followed by a phase change at this temperature and then a conspicuous phase change at around 630°C (12 J/gK). The jump may be as a result of latent heat. Fig. 12 shows the total specific heat for alloy 6111→5182. Phase changes occurred around 550°C (2.0 J/gK) and 630°C (16 J/gK).

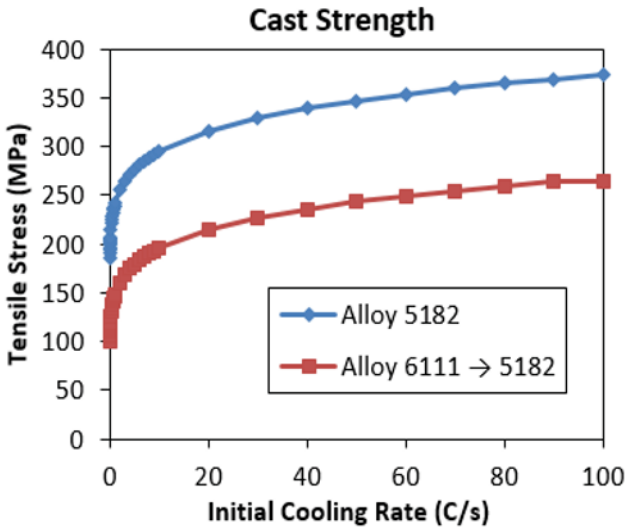

Fig. 7: Mechanical Strength vs Temperature of Alloys 5182 and 6111→5182

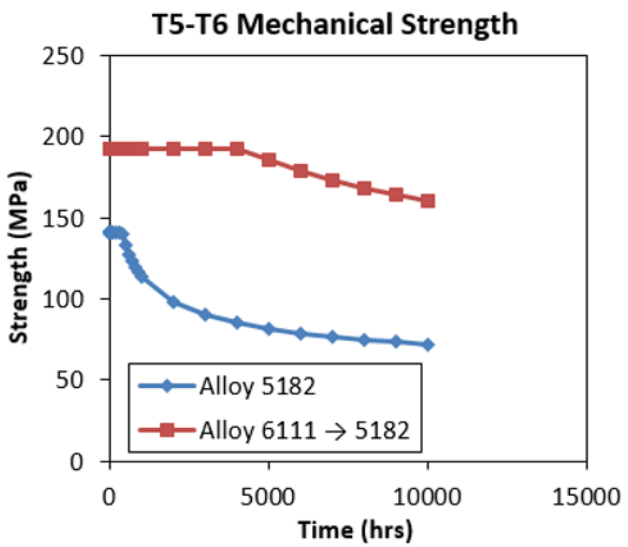

Fig. 8: Mechanical Strength vs Time of Alloys 5182 and 6111→5182

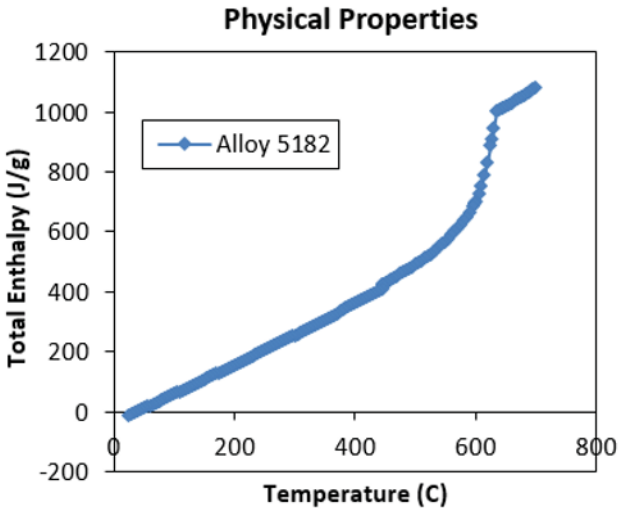

Fig. 9: Total Enthalpy of Alloy 5182

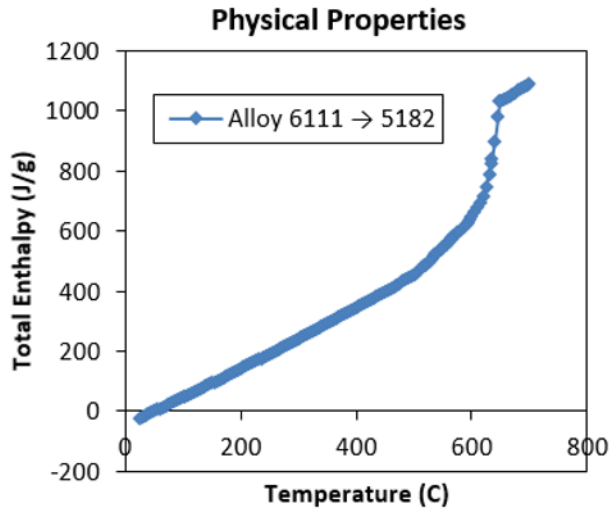

Fig. 10: Total Enthalpy of Alloy 6111→5182

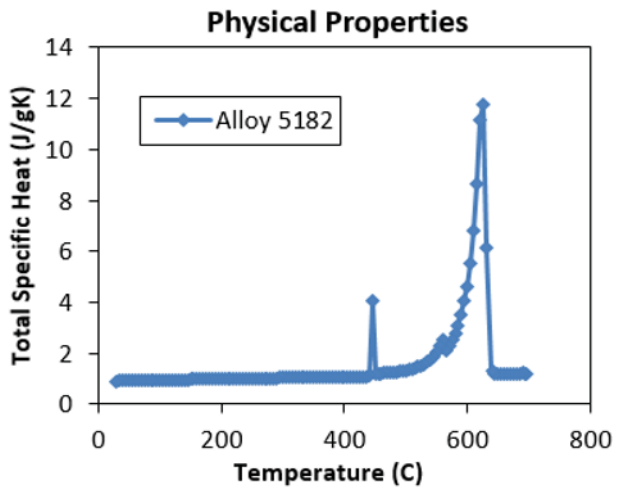

Fig. 11: Total Specific Heat of Alloy 5182

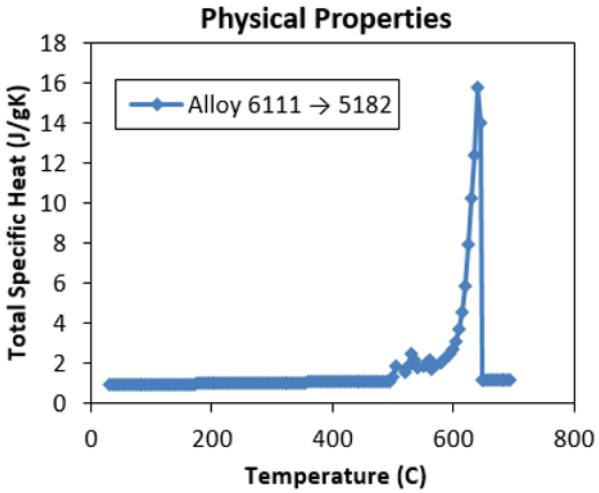

Fig. 12: Total Specific Heat of Alloy 6111→5182

Conclusion

Scrap handling and segregation of alloys will be critical to the success of recycling of these alloys into useful products. The results obtained from the JMatPro predictions shows that feeding an alloy of different composition to obtain another alloy of desired composition is very difficult to obtain. Although the software is prone to some errors, the results obtained are still close to the ideal solutions and an experiment might be a better fit in this case. Conclusively, sourcing alloy 5182 to get alloy 6111 looks more feasible since it contains only one limiting element compared to sourcing alloy 6111 to obtain alloy 5182 which has more limiting element. Demagging will be a better way out in obtaining good results for alloy 5182→6111.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

HISTORY AND DEVELOPMENT OF SLAG AND DROSS PRESSING

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Abstract

The hot dross pressing technology was first invented for the lead industry in 1896. This early patent was later followed by the hot pressing of zinc slag. The first commercially marketed pressing systems using this 1896 patent as a reference for aluminum, were available in the early 1980's and manufactured by Pechiney in France and Showa Aluminum in Japan. In the mid 1980's Anaconda Aluminum also developed a dross press system internally that was used in several of its operations. The first commercially successful dross presses were developed and marketed by Altek International in the early 1990's. Since that time there have been many commercial manufactures and home made versions of the dross press found around the world. Some have been successful performing the process and some have not been successful. Dross pressing is the most widely used system for hot dross processing used through out the aluminum industry today. The original developing industries of lead and zinc do not widely use this process to recover liquid metals from their slags.

Review of Aluminum Dross Generation

Dross is an oxidation product of the aluminum melting process. When it is mechanically removed from the furnace with a skim hoe, molten aluminum is naturally contained in the dross. The concentration of aluminum in dross can vary significantly with melting process, the skimming method and tools. These techniques have improved over the past 40 years producing a higher quality material for subsequent processing.

There are many types of aluminum dross: white dross, black dross, foundry dross, induction furnace dross and primary aluminum dross just to name a few. Each dross acts differently in the pressing process.

Lead and Lemons

There were two technical breakthroughs that occurred in 1896 that were the dawn of todays technology. First, Mr. W. H. Howard of Pueblo, Colorado was a visionary when it came to processing slag. The device he invented used a flat plate to contact the lead slag in a cylindrical container. The device put pressure on the lead slag mass and let the molten lead flow back into the lead kettle. The patent covered a simple mechanical system and an early hydraulic system to provide the squeeze pressure.

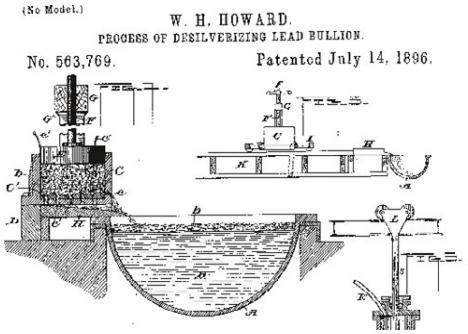
(No Model.)
W. H. HOWARD.
PROCESS OF DESILVERIZING LEAD BULLION.
No. 563,769. Patented July 14, 1896.

Figure 1 - The 1896 Lead Bullion

The next big advance, the significance of which no one really knew at the time, was invented by Mr. J. T. White of New York, New York. He developed a mechanical advantage device that used a pressing head and container with holes to separate a liquid that was contained in a solid. The lemon squeezer was born. These two ideas were the foundation of all the patents that were to follow.

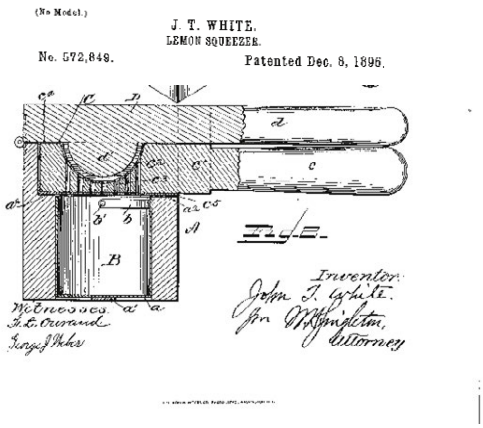
(No Model.)
J. T. WHITE.
LEMON SQUEEZER.
No. 572,848. Patented Dec. 8, 1896.

Figure 2 -The 1896 Lemon Squeezer

The Basic Process

The process of hot slag/dross pressing is based on the observation that when a solid matrix, that has a trapped liquid within it, is placed under pressure, the liquid will separate from the solid and flow to the area of highest concentrated pressure (the head and pan surfaces, aluminum escaping thru the drain hole locations). The basic press system consists of a press head and skim container/sow mold set, where compression forces are exerted either by gravity, a mechanical advantage or a hydraulic unit. All

of these systems have been used successfully over the past 120 years of the press development.

Systems developed over the past 10 years have focused more around the ancillary equipment such has safety doors, spray coating systems and pollution control units. All gadgets that are really nonessential to the basic process, but can make operation simpler, safer and appear environmentally more acceptable.

The Development of the Modern Press

There were three very important patents that lead to today's successful aluminum dross press.

The first was invented by Mr. G. P. Ross and Mr. J. L. Bedortha of National Steel Corporation in 1977 for the separation of molten zinc from zinc slag. This was the first patent in slag pressing that really took into consideration the importance of matching the compression head contours to the collection pan contours. This was another breakthrough idea for understanding the type of pressure forces required to get the liquid metal separation at low enough values to be commercially practical. It was the understanding of this idea that made the next two patents possible.

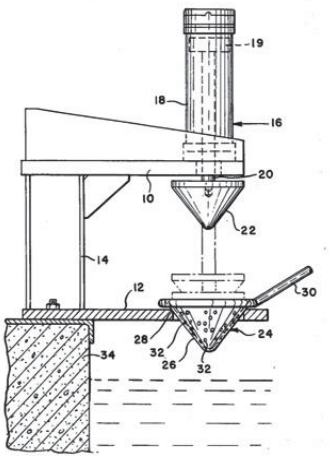

Figure 3 - 1977 Molten Metal from Dross

The second was invented by Mr. D. J. Roth, Mr. L. R. Culler and Mr. R. D. Heffner of the Anaconda Company in 1983 for the separation of molten aluminum from aluminum dross. This was the first patent in this area of aluminum dross pressing but others soon followed and used cylinders to contain the dross and flat plates to press the material. The pressures required to drive out the aluminum from the dross with these type of flat heads and cylindrical containers were extraordinary and made the units far too expensive and commercially unacceptable. Presses manufactured using the Anaconda couture head and pan technology were placed at several Anaconda plants. The shortcoming of this technology and the reason for the relatively short internal company use was due to deformation and cracking of the fabricated steel pans and the difficulty this caused in removal of the cold dross from these pans. The most important part of this development was the understanding of the aluminum plating process that occurred at the right pressures due to the matching of the contours of the head and container designs. This was the first major step in increasing the recovery of aluminum from dross from the 20% range that was typical in the early 80's; to the 50 - 60% range that is typical in today's pressed dross.

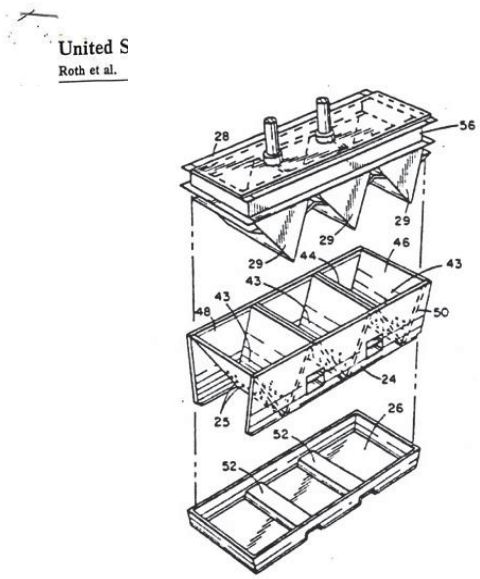

Figure 4 – 1983 Metal Recovery from Dross

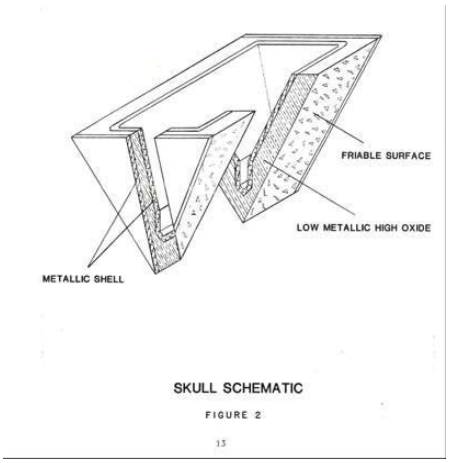

Figure 5 – Plating Action under Pressure

The third major step was the invention by Mr. D. J. Roth of the Altek International Company in 1997. This patent talked of the use of ribs on the head that matched the contours of the pan. The genesis of this breakthrough idea came from old tests from the 1980's that were done in the Anaconda development period and the Remetal Rotary Cooler Systems. The Remetal use of cast steel skim pans was very unique at the time. By going away from the fabricated pans, the dross pressing process could now be commercially acceptable from the viewpoint of a long service life and a moderate sales price. The first press frames and hydraulic systems were modified scrap paper compactors.

They were inexpensive but did the job. Today's standard commercial versions use the same pan and head concepts of the original designs but are built like tanks to last in the severe cast house environments. Fork truck and cast house operators seem to know only one speed, full ahead!

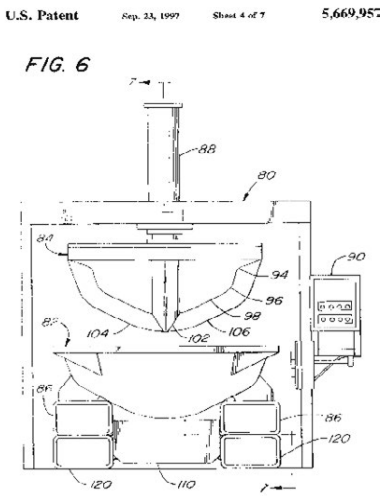

Figure 6 – 1997 Dross Compression Apparatus

There have now been more than 25 US and international patents in this technology area.

Dead Weight Pressing

The use of the press head only started in the Anaconda Company in the early 1980's. All the early test and designs used this strategy. This "dead" head technique has been used by several other companies since the 1980's. The heavily weighted "dead" head would cool the material but it was not able to get the plating action that was so important to the high recovery values achievable from hydraulically pressed dross. However, for generators of small amounts of dross and when shallow dross pans can be used, this is a viable less expensive solution over the use of a hydraulic press. The overall recovery numbers will not be as good as a standard dross press

Figure 7 - "Dead Head" Pressing

The Key to a Successful Dross Press

Dross press technology is really not about the press but about good skim pan and head designs. High grade steel castings, which allow for optimal pressing configurations have made the process what it is today. The skim pot has holes to allow for the maximum natural drain and pressed drain. Good casting designs also promote dross cooling and newer designs optimize skull sizing for downstream processing.

The process starts when dross is skimmed into the skim pot set. The most important free metal drain starts here. When projecting the process total drain the rule of thumb is that 2/3 of the total drain will come from the free drain while skimming and 1/3 will come from the pressing action. After skimming, the skim pot sets are brought to the press by fork truck, where the press head is pressed into the dross. This pressure forces additional molten aluminum out of the dross and into the sow mold. The more important action however is the coalescing the fine particles of liquid aluminum into large solid plates of aluminum within the dross matrix. This action is the key behind increased recoveries and cooling of the pressed dross. The mode of the press cylinder cycle is very important to get the separation of the aluminum from the oxides for down stream processing. The plates of aluminum formed by the high head pressures recover at a much higher melt recovery than the typical – ¼" (6mm) aluminum particles that you see in non-pressed dross. A properly pressed dross will have 70 - 80% of the solid aluminum above + 6 mm (1/4"). A non-pressed or poorly pressed dross will have 50 – 60% under 6 mm (1/4"). The size of these aluminum particles plays a major role in their overall recovery.

Figure 8 - Pressed Dross

The in-house drain metal recovery will range from 0 -10% on black dross and salt cake where cooling and metal coalescing is more important than draining. The free drain can run 15 - 40% on white dross alloys. Secondary recovery ranges the entire spectrum depending on the type of dross or salt cake pressed.

Although the hot dross pressing process is by far the most acceptable and best method commercially put forward to date for cooling and increasing the overall recovery of dross, it is not useful for all materials.

Cold skimmed dross will not press and this practice is the preferred skimming method for some companies. Wildly thermiting dross will press but with poor results and potentially significant damage to the press and pans. Pressed thermiting dross typically continues to burn and fume after pressing.

Future Dross Presses

Those of us who work in this area of dross recycling have to continually ask ourselves how we can make things better. What could we do to be even more effective in dross pressing and handling, saving energy units, aluminum units and minimizing our industries landfill and CO_2 generation.

GPS - Global Solutions is working on a press system to minimize the amount of metal pulled out of the furnaces with the dross during the skimming process. The goal is to cut the amount removed in half from the current averages. Imagine cutting the plants dross generation in half by the use of a unique in furnace dross press / skimming machine. A significant amount of the dross removed from the furnace is aluminum that should have remained in the furnace. The development of this process has the opportunity to significantly reduce melt losses, energy consumption from remelting metallic concentrates and cause a significant contribution to real landfill and CO_2 reductions in the area where dross is processed and handled.

Summary

There is an industry need to keep moving forward in its dross processing technology development. The first step is keeping as much aluminum as possible inside the melting furnace. The next step is keeping as much recycled aluminum in-house for use in the facility that generates it and producing dross pressed skulls that are easily recyclable by non-thermal means. The dross press developed over the past 120 years has added significantly to where we are today. We have continuously moved forward over the past 50 years and we still have the momentum for change and improvement. Our industry understands the value of the aluminum and the oxide products and the value of efforts to preserving both.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

TESTING PPE FOR MOLTEN ALUMINUM SPLASH RESISTANCE

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Keywords: Safety, PPE, Molten Aluminum, Vinex, Zirpro Wool

Abstract

Understanding the performance of aluminum cast house PPE (Personal Protective Equipment) is vitally important for employee safety. The ASTM F955 standard describes the testing of flat sections of PPE fabric and is the basis of this testing. This paper will show that wash cycles affect the performance of some articles of PPE. Vinex fabric was tested as new, washed and dried numerous times with and without fabric softener. Vinex that was heavily washed performed better than new Vinex. The performance of damp Vinex and dry Vinex over a damp undershirt was also tested. A test was developed to evaluate curved fabrics such as the fingers of gloves. The fabrics tested include Vinex, Zirpro wool and cotton.

Introduction

The performance of primary and secondary aluminum cast house Personal Protective Equipment (PPE) must be understood. This can be accomplished by testing PPE as described by ASTM F955-07. If a change in PPE is needed it should be tested before being implemented. Even the addition of high visibility components to secondary PPE should be evaluated.

Secondary PPE is clothing designed for continuous everyday wear in locations where there may be occasional exposure to molten aluminum splash, radiant heat and flame. It is designed so that it does not continue to burn after exposure to or after removal of an ignition source. It may not prevent all burns, however it should reduce the number and severity.

Primary PPE is worn over secondary PPE as additional protection for activities where exposure to molten splash, flame and radiant heat is likely. Some of the activities where primary PPE is worn include, melting, alloying, sampling, transferring and casting.

For additional information on primary and secondary aluminum PPE consult The Aluminum Association, "Guidelines for Handling Molten Aluminum", Third Edition, July 2002.

Test Description and Modifications

Testing was based on the ASTM F955-07 standard. The test method specifies that 1 kg of 1100 series aluminum be used. The tests performed used approximately 0.75 kg of molten AA5182. Using AA5182 better represents the actual conditions in the test cast house. Less molten aluminum was used to reduce the severity of the test so differences in material performance would be easier to detect. Note that the freezing range of AA5182 is wider than for AA1100. Additional tests were made with AA3104. The fabric tested was Vinex FR-9B (85% Vinal and 15% polynosic rayon).

ASTM F955-07 only addresses the testing of flat panels of material or fabric. The slug calorimeters specified in the testing standard must be made of electrical grade copper, weigh 18± 1g and measure 40 mm in diameter and 1.6 mm thick. There are some articles of PPE that are constructed of more than one component and are not flat. Several of the flat slug calorimeters fabricated for the testing were formed so that they would approximate the curvature of a human finger (see Figure 1).

Figure 1. Curved copper slug calorimeter for glove testing. A Type J (Fe/Constantan) thermocouple is installed.

When flat pieces of material are tested they are placed on an inclined refractory board that has two imbedded flat calorimeters (see Figure 3). For glove testing, Kaowool refractory blanket was packed behind each calorimeter in the glove to ensure the calorimeter was pressed against the material to be exposed to molten aluminum. The top calorimeter in each glove was flat and the bottom calorimeter curved (see Figure 2). The top cross (see Figure 2) is the target for first contact with molten aluminum.

Figure 2. Zirpro Wool glove installed in the test apparatus. The top cross (+) shows the location of the top flat calorimeter and the bottom cross shows the location of the curved finger calorimeter.

Test Equipment and Utilities

The equipment and utilities used are listed below:

- Pyradia Electric Oven – For pre-heating crucibles
- Natural gas fired (Fire Assay) furnace – For melting and holding of aluminum
- Graphite crucibles
- Test Apparatus (Figure 3) and Data Acquisition Electronics (property of SECAT Inc.) - For molten splash testing
- Crucible carrying and pouring tool, tongs, ladles
- Primary and secondary PPE for molten aluminum
- Video and photographic equipment.
- Fire extinguisher
- Natural gas, compressed air and 110v.

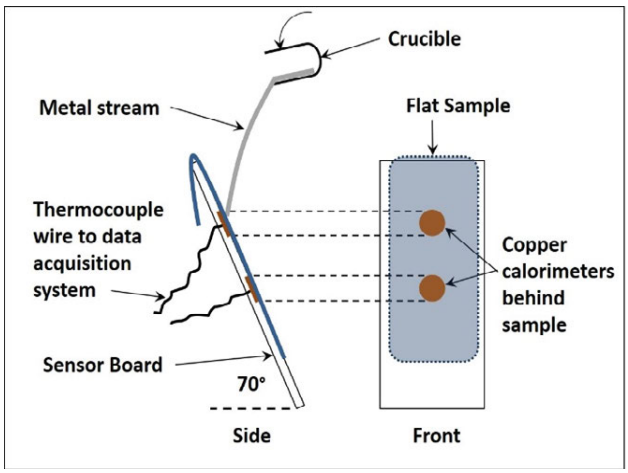

Figure 3. Test apparatus schematic for flat samples. After a sample was attached to the sensor board a measured amount of molten aluminum, at the prescribed temperature, was poured at a controlled rate onto the sample. Any heat that passed into the calorimeters was measured and collected by the data acquisition system. The layout of the test equipment was carefully considered to ensure safe and smooth work flow to reduce the chance of molten aluminum spill or injury.

Sample Preparation

Several sets of Vinex secondary PPE were washed and dried both with and without fabric softener. Testing brand new fabric does not represent the actual conditions of PPE in the cast house. PPE is only new once. Tests were performed on Vinex 200 g/m^2 (6 oz/yd^2) shirts and Vinex 280 g/m^2 (8.5 oz/yd^2) pants as new, with 1 wash, 15 washes and 30 washes. The units of g/m^2 are the weight of the fabric per unit area. Half of the Vinex samples were washed and dried with fabric softeners. One used Vinex shirt was soaked in tap water to test the effect of damp fabric on performance. Additionally, a cotton undershirt was soaked in water and then placed under a used dry Vinex shirt for testing. Damp fabric was tested at the request of employees and was intended to simulate perspiration soaked clothing.

Results

Following the ASTM F955-07 method provides heat transfer and temperature data as well as a visual assessment of each sample following the test. The heat data measured from the calorimeters is compared to a Stoll Curve. The calculated Stoll curve is a predictor of a second degree (blistering skin) burn. If the heat measured by either the top or bottom calorimeter is above the Stoll response curve then a second degree burn or worse is likely.

Vinex Fabric Test

A plot of the top and bottom calorimeter along with the Stoll curve (see Figure 4) shows the performance of new 280 g/m^2 (8.5 oz/yd^2) Vinex. The heat, with a peak of 28.9 J/cm^2, measured by the top calorimeter far exceeds the Stoll response curve and a burn is likely to occur. The visual inspection of the sample showed that there was significant thermally caused shrinkage, break-open and ignition of the New Fabric sample (see Figure 6).

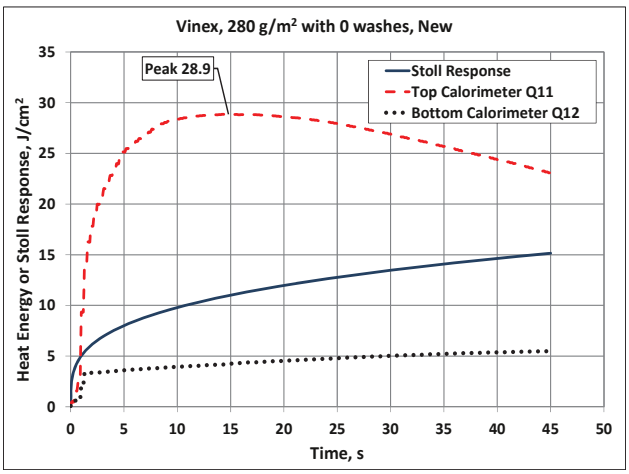

Figure 4. Stoll response curve along with top and bottom calorimeter data for new and unwashed Vinex 280 g/m^2 (8.5 oz/yd^2) material.

Test results (see Figure 5) of 280 g/m^2 (8.5 oz/yd^2) Vinex, washed 30 times, show it performed better than new Vinex. The heat transfer peaked at 17 J/cm^2. Compare these results to those of new Vinex (see Figure 4). Because heat measured by the top calorimeter was above the Stoll response curve for a period of time a burn would still have been likely. Note that there was less heat for less time passing through the fabric that was washed 30 times.

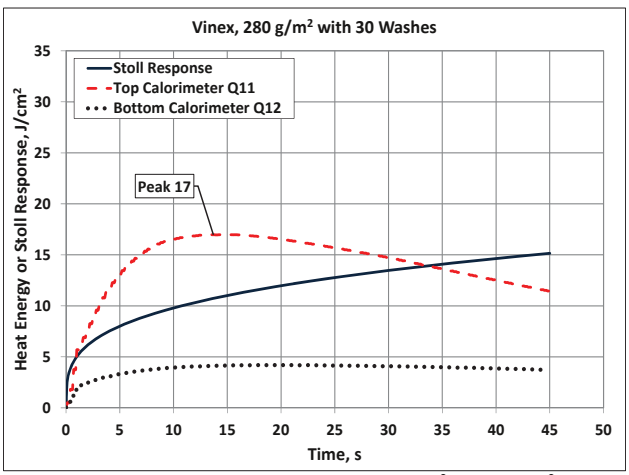

Figure 5. The Stoll response curve for 280 g/m^2 (8.5 oz/yd^2) Vinex that was washed 30 times.

As the number of wash cycles increased the damage to the Vinex was reduced. A progression of washing from 0 to 30 cycles (see Figure 6) shows the effect of washing. The use of fabric softeners did not have any measurable effect on performance.

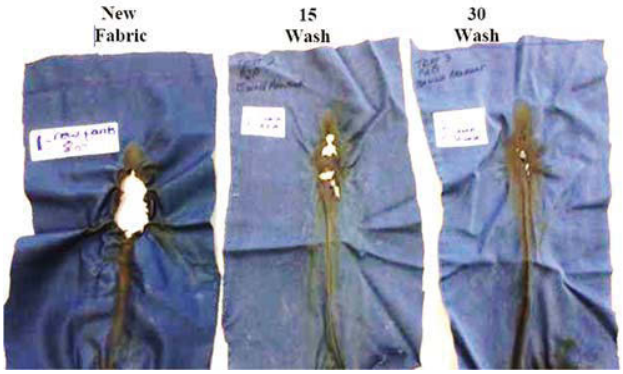

Figure 6. Progression of wash cycles (0, 15, 30) and visual damage to 280 g/m^2 (8.5 oz/yd^2) Vinex material.

The Effect of Washing Vinex FR-9B on Thermal Shrinkage, Break-open, Adhesion and Ignition

Additional washing generally reduces thermal shrinkage, break-open, adhesion and ignition (see Figure 7). Thermal shrinkage and break open were the factors that reduced the most. Review of test video showed that heavily washed samples (30 washes) had little or no ignition compared to brand new Vinex (zero washes).

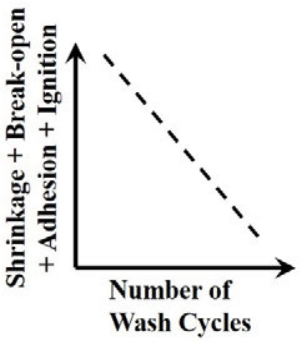

Figure 7. The relationship between Vinex FR9B Washing and Thermal Shrinkage + Break-open + Adhesion + Ignition.

Uniform Changes and High Visibility Components

The addition of high visibility and reflective components must be tested before being implemented (see Figure 8).

Figure 8. Blue Vinex 200 g/m^2 (6 oz/yd^2) shirt with an orange Vinex stripe with an added synthetic reflective stripe.

Two layers of Vinex (blue and orange) topped by a reflective stripe provided more protection than one layer of 200 g/m^2 (6 oz/yd^2) Vinex. Vertical seams and stripes are better than horizontal so that molten aluminum will shed off of the material.

Vinex Weave Evaluation

As part of the investigation into why heavy washing positively influenced the performance of Vinex, several fabric samples were examined under a microscope. At 100x the samples of new Vinex and Vinex washed 30 times appear similar (see Figure 9).

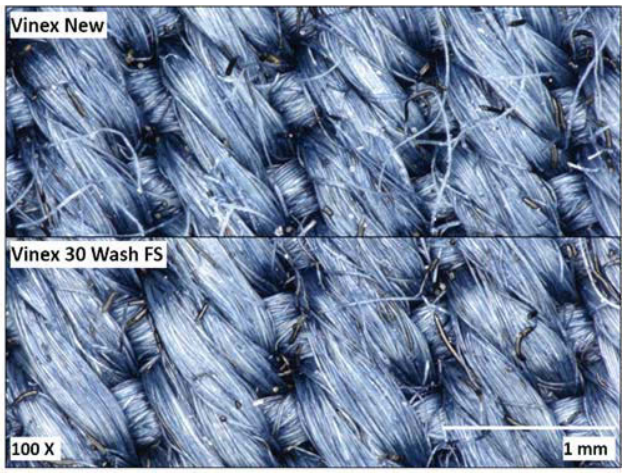

Figure 9. Comparison of Vinex as New and with 30 washes.

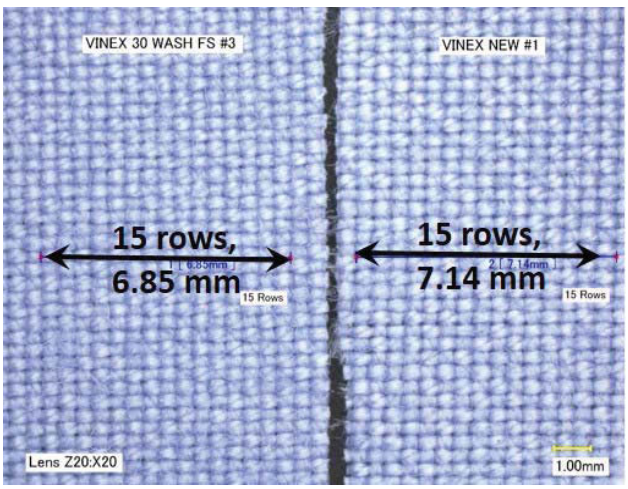

Figure 10. Weave comparison of Vinex washed 30 times (left) and new Vinex. Approximately 4% shrinkage after 30 washes.

Samples were then examined at 20x and the weave dimensions measured and compared. With progressive washings the weave of the fabric becomes tighter. Vinex shrinks approximately 4% after 30 washings (see Figure 10). A tighter weave may make it more difficult for molten aluminum to adhere to the fabric. If molten aluminum can not adhere to the fabric it will shed off. When molten aluminum quickly sheds from PPE then less heat passes through to the wearer.

Results showed that the heavier 280 g/m^2 (8.5 oz/yd^2) Vinex performs better than the lighter weight 200 g/m^2 (6 oz/yd^2) material.

Glove Testing

Understanding the performance of hand protection, or any other item of PPE should be given the same attention as shirts, pants and coats. For glove testing a curved calorimeter was designed and paced in a finger of each glove tested.

Zirpro wool gloves, used as primary PPE, are constructed of 882 g/m^2 (26 oz/yd^2) orange Zirpro wool, a Kevlar jersey palm and a 509 g/m^2 (15 oz/yd^2) wool liner. The response curve for Zirpro wool gloves (see Figure 11) shows that the heat into both the top and bottom calorimeters did not cross above the Stoll curve so a burn is unlikely. Video footage of the test showed that molten aluminum quickly shed off of the orange wool leaving only some discoloration. Examination of the glove liner shows no metal penetration or other effect (see Figure 12). The heat transfer peaked at 4.3 J/cm^2 at 45 seconds. Zirpro gloves performed better than all other gloves tested.

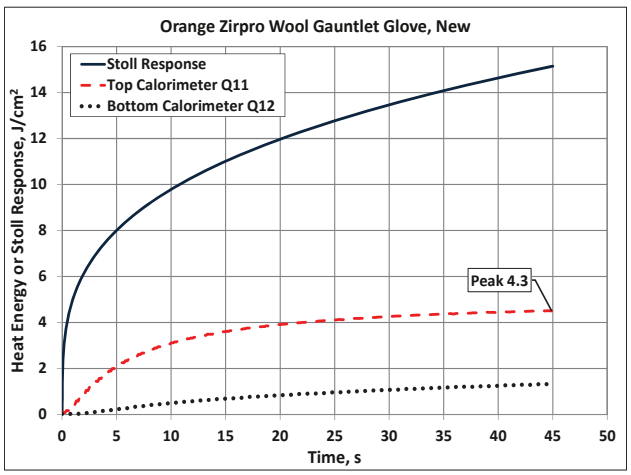

Figure 11. Stoll response curve for orange Zirpro wool gloves.

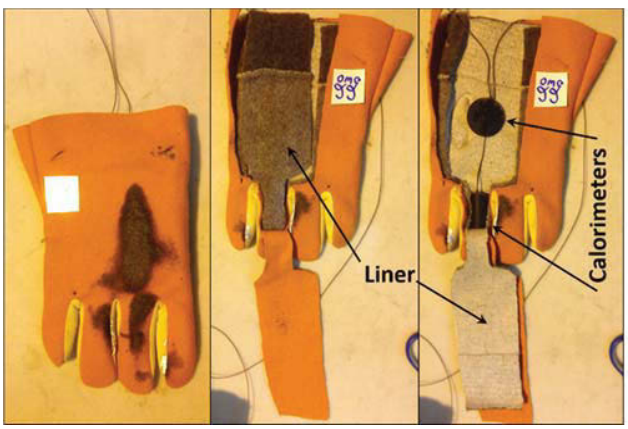

Figure 12. Visual inspection of Orange Zirpro wool glove after testing. No evidence of metal penetration.

A Stoll response curve graph for an 850g (30 oz) cotton "Hot Mill Glove" (see Figure 13) shows its performance. This glove is used as both primary and secondary PPE depending on the process and risk. Note that the top calorimeter crossed the Stoll curve at 2.5 seconds into the test. After 2.5 seconds the heat level became great enough to begin causing a skin burn. If the wearer were to remove the gloves quickly a burn may be avoided. Some aluminum adhered to the Hot Mill Glove (see Figure 14). The adhering aluminum allowed heat to continue to transmit through the glove and into the calorimeter. The glove continued to smolder past the 45 seconds of the test. The heat transfer peaked at 16.9 J/cm^2.

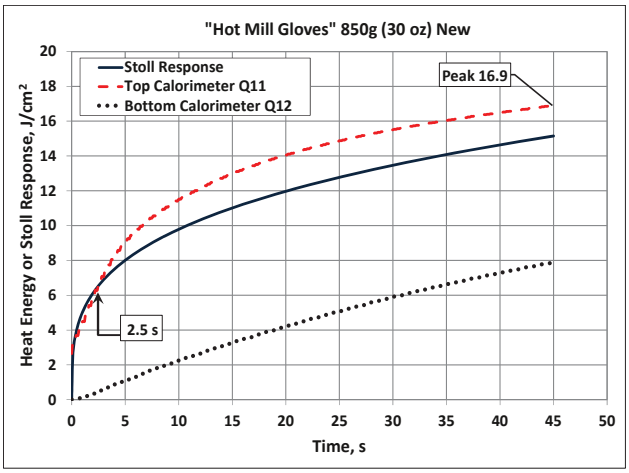

Figure 13. Stoll response curve for an 850g Hot Mill Glove.

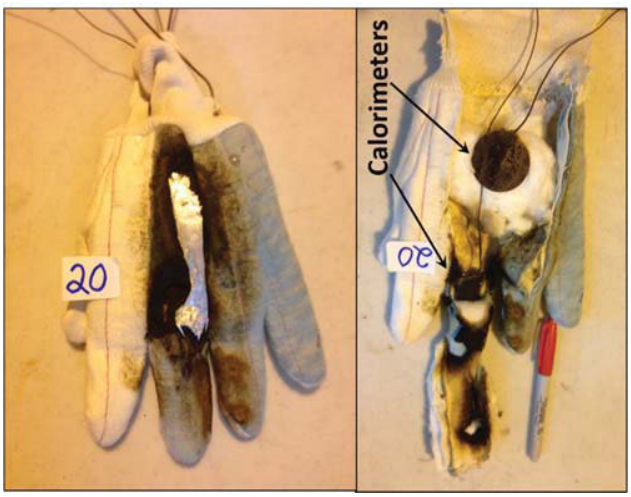

Figure 14. Cotton Hot Mill Glove. Left photo panel shows adhering aluminum. Right photo panel shows the interior of the glove after the test.

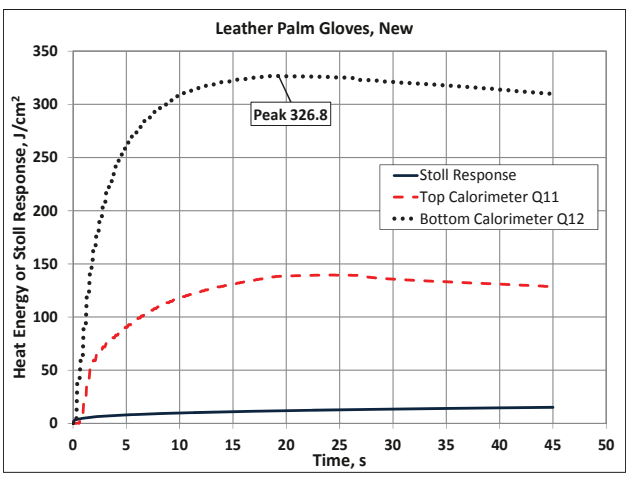

Figure 15. Both the top and bottom calorimeter far exceeded the Stoll response curve for Leather Palm Gloves.

A leather palm glove was also tested. The glove includes both cotton and leather components. It performed poorly as shown by the Stoll response curve graph (see Figure 15). The heat measured by the bottom calorimeter exceeded that of the top calorimeter. The bottom calorimeter peaked at 326.8 J/cm^2. The glove sustained severe damage and would provide no protection from molten aluminum and burns would be certain (see Figure 16). Leather palm gloves must not be used as molten aluminum PPE.

Figure 16. Remains of a leather palm glove.

Damp Fabric Testing

Employees questioned the effectiveness of damp Vinex PPE. A used 200 g/m^2 (6 oz/yd^2) Vinex shirt was soaked in ordinary tap water so the effect of damp clothing on PPE performance could be investigated. Tap water was used to simulate perspiration. The sample was weighed before and after it was soaked in water. The sample was then quickly fastened to the test apparatus and the test initiated. Visual observation and review of the test video showed that the molten aluminum quickly shed off of the sample. The top and bottom calorimeters stayed well below the Stoll Response curve after about 3 seconds (see Figure 17). It is believed a steam barrier developed between the Vinex sample and the molten aluminum. This appeared to protect the sample from damage in this test (see Figure 18). The heat transfer peaked at 6.1 J/cm^2. A burn would not be likely.

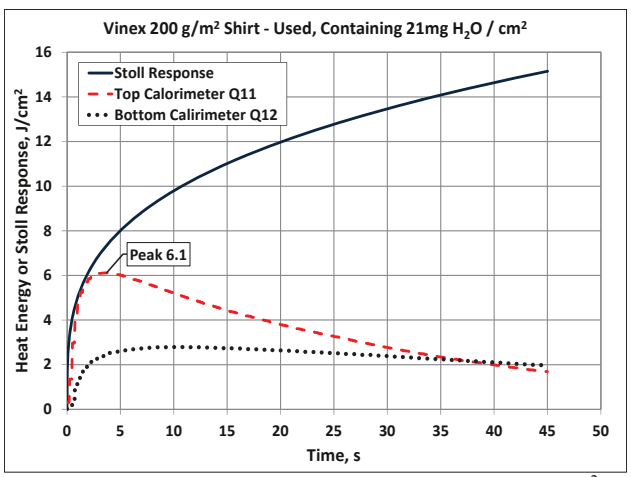

Figure 17. Stoll Response curve graph for a Vinex 200 g/m^2 (6 oz/yd^2) shirt containing approximately 21 mg H_2O / cm^2. Tap water was used to dampen the material.

Figure 18. Front and back of Vinex 200 g/m^2 (6 oz/yd^2) material after testing. There was little shrinkage and no metal penetration.

An additional test configuration where a damp cotton T-shirt was placed under dry 200 g/m^2 (6 oz/yd^2) Vinex also showed promising results (see figure 19). Both the top and bottom calorimeter readings were below the Stoll Response curve. The heat transfer peaked at 8.8 J/cm^2. A burn would not likely with this configuration.

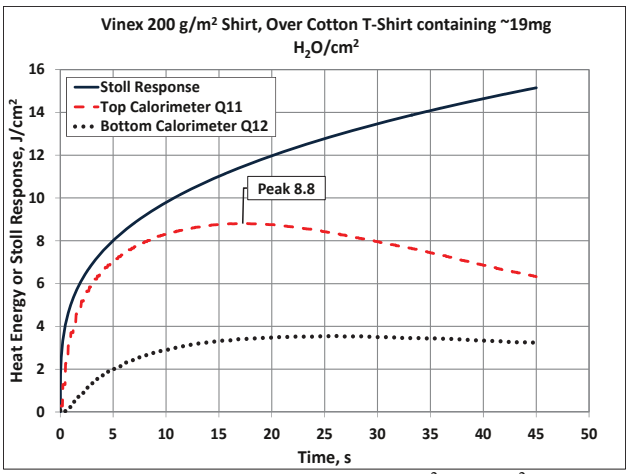

Figure 19. Response curves for Vinex 200 g/m^2 (6 oz/yd^2) shirt over a damp cotton T-shirt.

Summary

Each aluminum cast house must understand the capability of their PPE as it ages. Molten metal splash resistance can be evaluated using the test described in ASTM F955-07. Any changes to PPE must be evaluated before implementation. PPE made from 200 g/m^2 (6 oz/yd^2) and 280 g/m^2 (8.5 oz/yd^2) Vinex FR-9B performs better after many washings. Vinex performs the worst when brand new and never washed. The weave dimensions of Vinex samples were measured and it was determined that Vinex shrinks approximately 4% from 0 to 30 washes. The tighter weave from shrinkage may make it more difficult for molten aluminum to adhere to the fabric, thus allowing the aluminum to shed more quickly. The heavy use of fabric softener did not have a measurable effect on Vinex performance. This is the opposite of what was

understood from the Vinex uniform supplier. The user must follow the washing instructions provided by the uniform supplier or fabric manufacturer. Gloves can be tested using the apparatus and data collection system described in ASTM F955-07. For gloves a curved calorimeter can be placed in one finger and is the bottom calorimeter of the test. A flat calorimeter is used in the upper portion of the glove. Zirpro wool provides excellent protection from molten aluminum splash. Testing found that a damp Vinex shirt or a damp cotton undershirt under a dry Vinex shirt reduced the chance of burn.

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Grain refinement of self-hardening aluminum alloys

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Keywords: Al-based alloy, self-hardening, grain refinement, morphology, mechanical properties.

Abstract

Grain refinement offers several benefits in aluminum casting applications. It can be performed by inoculation, representing one of the mostly used method at industrial level and is based on the addition of nucleating agents. By using grain refinement method important benefits can be reached, for both cast and wrought aluminum alloys: a fine distribution of the second phases, improved castability, reduction of shrinkage porosity, higher mechanical properties, as well superior fatigue life.
In the present paper investigation of the effect of Al-5Ti-1B grain refiner on the microstructure and on the mechanical properties of self-hardening aluminum alloy will be presented and discussed. The appropriate use of grain refiner contributes to improve the mechanical properties, particularly the strength and the ductility of the alloy.

Introduction

Aluminum alloys, produced through traditional or innovative technologies, play an important role in specific engineering applications, thanks to their outstanding properties, like high strength to weight ratio compared to ferrous alloys, good corrosion resistance, etc. [1-3].
The mechanical strength of aluminum alloys can be amplified reducing the grain size, according to the theory and to the Hall-Petch law, through the addition of grain refiners. The use of grain refiners (GR) allow to increase the mechanical properties (strength and fatigue resistance), reducing the grain size leading to an equiaxed microstructure, preventing the development of hot cracking and reducing the evolution of porosity (microporosity and shrinkage) [4].
The goal of the present study is the evaluation of the effect of Al-5Ti-1B grain refiners on the microstructure and on the mechanical resistance of the Al self-hardening alloys. This class of alloys have specific properties: after a period of 7 to 10 days, they are subjected to a natural ageing, reaching high mechanical performances at room temperature without any heat treatment [5].

Grain Refiners

Initially, the grain refiners (GR) was realized by the addition of titanium to the melt [6], producing the modification of the microstructure and the reduction of the grain size as well as higher castability of the aluminum alloy. Without the introduction of titanium, the microstructure shows a dendritic morphology, where the intermetallic phase and the porosities are concentrated at the grain boundaries. With an equiaxed microstructure, both the porosity and the pore size, are reduced: the shrinkage porosities have a tendency to disappear, and consequently higher mechanical properties are achieved. The use of titanium, as one of the first grain refiners, can be explain due to its capacity to increase the melting point of aluminum to 665°C, as reported in Figure 1.

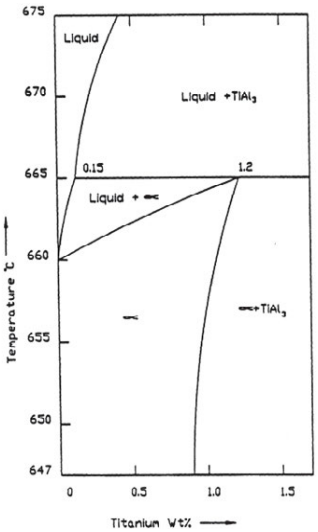

Figure 1. Phase diagram Al-Ti [7].

This variation affects the nucleation and the growth of the aluminum grains according to [8]. As reported in Figure 2, when the crystals of titanium aluminide compound $TiAl_3$ are in contact with the melt, the liquid close to the crystals becomes enriched in titanium and starts to solidify at a temperature above the melting point of aluminum. The nucleation starts at the surface of aluminide particles and after this the growth of α-Al crystals occur around the surface of $TiAl_3$. When Ti is completely consumed, the growth of the crystals stops. The solidification continues with the cooling of the metal and the formation of dendrites. Figure 3 reports the cooling curve of the non-refined and of the Ti-refined alloy: by the addition of Ti grain-refiners the nucleation temperature T_N is modified and is situated above the growth temperature T_G.

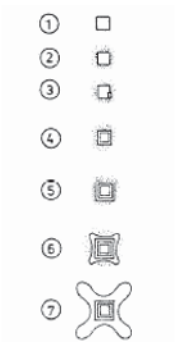

Figure 2. $TiAl_3$ nucleates aluminum grain [8].

The dissolution of $TiAl_3$ at normal casting temperature, makes disappear the effect of GR and the time depends on the titanium content, the metal temperature, the type of alloy and the $TiAl_3$ particle size. Rods have small particles and dissolve in 10-15 minutes while waffles need more time for the same process (30-40 minutes). The amount of titanium has to be superior than 0,15%wt at 665°C.

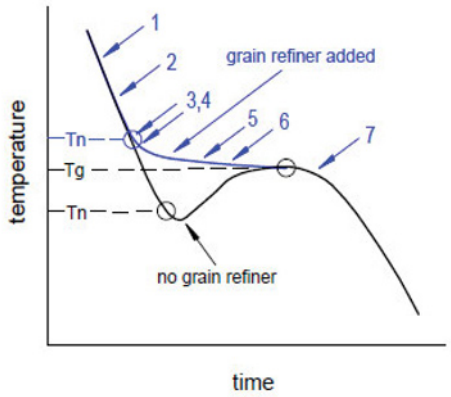

Figure 3. Cooling curves during solidification.[8]

From the 1970s was introduced boron, in combination with titanium and this allows reaching a superior grain refinement. The main reason of this effect is related to the solubility of $TiAl_3$ crystals which are very soluble in aluminum, while TiB_2 reveals a very low solubility in the liquid aluminum. This characteristic induces the formation of many potential nuclei. As reported in Figure 4 boron is a better grain refiner than titanium. AlB_2 are the best nucleus allowing to produce the smallest grain size, but it can dissolve in aluminum and it reacts with titanium and strontium which are usually present as modifier.

To obtain a grain refiner with the characteristics of AlB_2, avoiding some difficulty presented earlier, the development of a mix of crystals of $(Al,Ti)B_2$ (one of the main used grain refiners) has been realized. Today, GR are made with the introduction of boron (10-20 ppm) in the form of rod Al-5Ti-1B and Al-3Ti-1B. Lower dissolved Ti levels provide better grain refinement and in some alloys improved resistance to hot cracking [8].

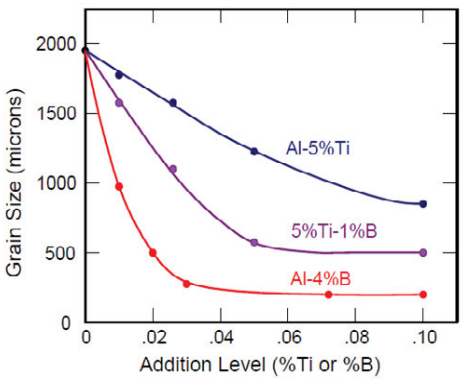

Figure 4. Grain refining of A356 alloy by three master alloys [9].

Another grain refiner commonly used is Al-Ti-C. In the same way, the particles of GR promote the nucleation of new grains leading to an equiaxed microstructure. Addition of Ti and B in the liquid alloy in the form of powder tablets is frequently applied. Anyway, the control and the management of this type of GR is difficult because it takes more time to dissolve; additionally it could have a low efficiency. The melt bath has to be maintained at a temperature lower than 750°C, although the GR effect will be reduced. Grain refiners can be used for both cast and wrought alloys. Good properties and good surface finishing properties are acquired. The main inoculant are Al-Ti, Al-Ti-B, Al-Ti-C, Al-B and Al-Sr-B, inducing the development of fine equiaxed grains. The efficiency of GR can be reduced by the presence of Si, Zr, V, while other elements like Mn, Cr, Zr can positively affect the refinement of the grain size. Depending on the alloying elements, which are present in the material, the effect of the GR can change. The main commercially available GR are listed in Table 1 [10].

Table 1: Commercially available grain refining alloys

Alloy type	Composition
Al-Ti	Al-10Ti, Al-6Ti
Al-Ti-B	Al-3Ti-1B, Al-5Ti-1B, Al-5Ti-0.6B, Al-3Ti-0.2B, Al-5Ti-1B, Al-5Ti-0.2B, Al-10Ti-0.4B, Al-1.6Ti-1.4B (TiBloy for hypoeutectic Al-Si alloys), Al-1.2Ti-0.5B (Hydloy), Al-3Ti-3B, Al-1Ti-3B
Al-B	Al-10B, Al-5B, Al-3B, Al-10Sr-2B, Strobloyä, (Al-10Sr-1.6Ti-1.4B)
Al-Ti-C	Al-6Ti-0.02C, Al-3Ti-0.15C
Al-Sc	Al-1Sc, Al-2Sc
Mg-Zr	AM cast (Mg-25Zr), Zirmax (Mg-33.3Zr)

An inappropriate quantity of grain refiners increases the porosity. Reduction of grain size can be obtained also by thermomechanical treatments during the solidification of the liquid metal and by recrystallization of the deformed material.

Heterogeneous nucleation

The GR particles act as nucleation site for the heterogeneous nucleation of aluminum grains, that needs a lower critical free energy. In this way it is activated the nucleation of new grains and the final microstructure presents grains with finest size. The grain refinement is related to the heterogeneous nucleation that presents a lower barrier of free energy that the homogeneous one according to the equations 1 and 2:

$$\Delta G^{*}\text{homogeneous} = \frac{16\pi\gamma_{sL}^{3}}{3\Delta G_{V}^{2}} \quad (1),$$

$$\Delta G^{*}\text{heterogeneous} = \frac{16\pi\gamma_{sL}^{3}}{3\Delta G_{V}^{2}} f(\theta) \quad (2),$$

where γ_{sL} is the interface surface energy of a solid–liquid interface in [J/m^2], ΔGv is the solidification driving force, $f(\theta)$ is function of the contact angle θ on the substrate where there is the nucleation. $f(\theta)$ is always ≤ 1, so the critical free energy is always less than or equal to that for homogeneous nucleation, in heterogeneous one. The nucleation rate increases if the contact angle θ tend to zero [11], as reported in Figures 5 and 6.

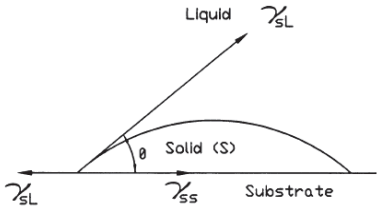

Figure 5. Formation of a cap of solid (s) on a substrate, contact angle and surface tension forces [12].

There are many theories about which grain refiner is better and more effective [8]. However, the chemical composition of each alloys determines the efficiency of the GR and the identification of the most suitable GR for new alloys require particular investigation.

Self-hardening aluminum alloy

Aluminum alloys can be strengthened through thermal treatment or by the cold working process. Heat treated alloys show a uniform distribution of small precipitates, allowing to achieve high strength. The ability of the precipitates to inhibit the dislocations movement is associated to higher mechanical properties. The strength is determined by the size and distribution of the precipitates and by the coherency of the precipitates with the matrix. Nowadays the tendency is to find alternative methods to the heat treatments, for the aluminum alloys, reducing as much as possible, the environmental impact with an evident economic returns.
One of the new class of aluminum alloys that follow this goal is the self-hardening aluminum alloy (main alloying elements are zinc, silicon and magnesium).

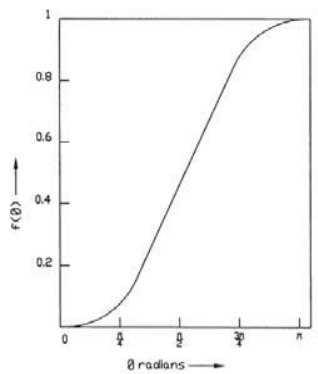

Figure 6. Variation of $f(\theta)$ with θ where $f(\theta)$ is equal to $(2 - 3\cos\theta + \cos^3\theta)/4$ [12].

The main feature of these alloys is their capacity to reach good mechanical performances, without the application of any heat treatment, reducing the cost of the process and the risk of deformation [13,14]. In about 7-10 days after casting, during which alloys are stored at room temperature, the alloy reach an interesting mechanical behavior. About 50% of the mechanical strength is already reach one day after the casting.
These alloys have been developed recently and their properties and potentiality are still the goal of research. The goal of this paper is to investigate on the improvement of the mechanical properties, of these self-hardening alloys, using different kind of grain refiners.

Experimental

Table 2 reports the chemical composition of the self-hardening alloys produced by die-casting technique (Teksid Aluminum Srl Carmagnola, Torino-Italy) and investigated in the present paper. The main difference between the two alloys is related to the concentration of Mg.

Table 2. Chemical composition (wt.%) of the two alloys produced

	Si	Fe	Cu	Mn	**Mg**	Zn	Ti	Al
AlZn10Si8Mg	7,5-9,5	0,30	0,10	0,15	**0,2-0,5**	9,0-10,5	0,15	bal.
AlZn10Si8Mg1	7,5-9,5	0,30	0,10	0,15	**1,0**	9,0-10,5	0,15	bal.

Al-5Ti-1B master alloy, has been chosen as grain refiner and a content of 500 ppm has been used to refine the self-hardening alloys. The mould used for the new casting is reported in Figure 7. Preheating of the die has been performed at 250°C and it was monitored with a thermocouple to guarantee the complete filling of the mould and furthermore to reduce the risk of gas porosity development. About 600 g of alloy is employed to completely fill up the mould.

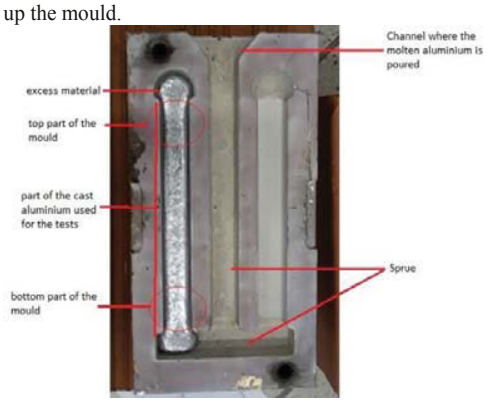

Figure 7: Mould and the casting geometry.

The casting temperature has been set at 720°C. In order to evaluate the homogeneity of the microstructure and the porosity distribution, samples for the microstructural analysis, have been extracted from different zone of the casting (top and bottom) as indicated in figure 8. This because elements like titanium, that present higher density, can concentrate at the bottom part of the mould during the gravity casting [11]. The castings performed are reported in Table 3.

Table 3. Composition of the castings executed

Number of the casting	Composition
1	AlZn10Si8Mg
2	AlZn10Si8Mg + 500ppm of TiB
3	AlZn10Si8Mg1
4	AlZn10Si8Mg1 + 500ppm of TiB

The microstructural characterization has been performed using an optical microscope, (OM, MeF4 Reichart-Jung) and Scanning Electron Microscopy (SEM, Leo 1450VP) equipped with Energy X-rays Dispersive Spectroscopy unit (EDS, Oxford microprobe) used for compositional analysis. The mechanical properties have

been investigated by hardness measurements and Charpy impact test.

Results and discussion

The microstructure of the investigated alloys consists, as illustrated in Figure 8, in α-Al phase, eutectic Al-Si phase, Mg-Si intermetallic and intermetallic phases containing Fe, Mn and Cu. Probably, the quantity of zinc is higher in the α phase than the value shown by the EDS analysis, but due to its high solubility in aluminum, it is difficult to detect.
Figure 9 shows the different microstructures of the self-hardening alloys AlZn10Si8Mg and AlZn10Si8Mg1 with and without grain refiners. Some differences, concerning the microstructure and the eutectic phase have been detected. For a deeper analysis the samples were chemically etched with Keller reagent, to make the grains visible and to allow the measurement of their size. The microstructures obtained after the chemical etching are reported in Figure 10.

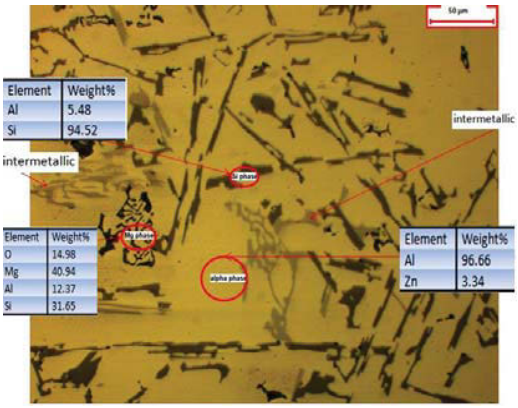

Figure 8: OM microstructure of the self-hardening aluminium alloy AlZn10Si8Mg (200x) and EDS analysis results.

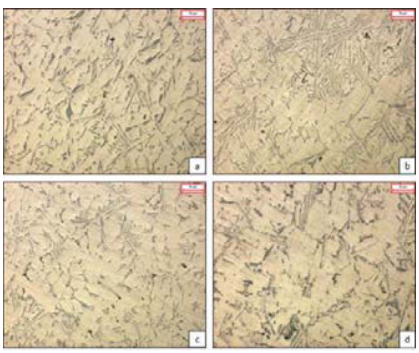
Figure 9. Microstructure of self-hardening alloys with and without grain refinement: a) AlZn10Si8Mg without grain refiner, b) AlZn10Si8Mg with 500ppm of TiB, c) AlZn10Si8Mg1 without grain refiner d) AlZn10Si8Mg1 with 500ppm of TiB, (200X).

By the addition of grain refiner, in the case of the AlZn10Si8Mg alloy, the grain dimension has been reduced of one order of magnitude, while on the contrary, for the AlZn10Si8Mg1 alloy, comparable grains dimensions have been obtained, with and without the grain refiner. A higher Mg content, determine an increased volume fraction of Mg-based precipitates, which can act as a barrier for the grain boundaries movement, consequently a refined microstructure can be achieved. The average grains area values are reported in Figure 11.
Figure 11 a) shows that more than the 75% of the grains in the AlZn10Si8Mg alloy have an average area bigger than 1μm , while in the same alloy, following the addition of TiB, grains present an average dimension of about 0.1-0.2 μm , Figure 11 b). For AlZn10Si8Mg1, the distribution of the grain dimension is very similar for both the cases, with and without grain refiner, Figure 11 c) and Figure11 d). The majority of the grains have an area between 0.1 and 0.2 μm .

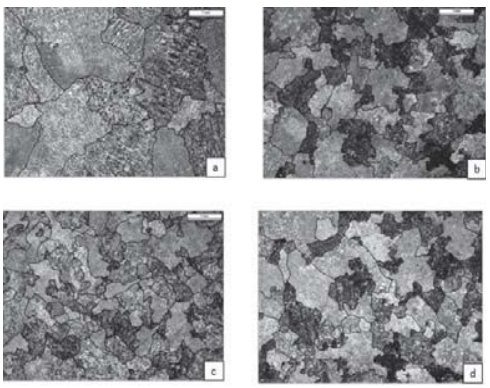
Figure 10. Microstructure of self-hardening alloys chemically attacked: a) AlZn10Si8Mg without grain refiner, b) AlZn10Si8Mg with 500ppm of TiB, c) AlZn10Si8Mg1 without grain refiner d) AlZn10Si8Mg1 with 500ppm of TiB, (200X).

Figure 12 reports the energy absorbed by the material during a Charpy impact test, carried out at room temperature (using a pendulum of 50J and un-notched samples of dimension 55mm x 10mm x 10mm). For each case, a minimum of 4 samples have been tested. When the quantity of magnesium increases, the impact energy of the aluminium decreases, as a consequence of the larger intermetallic particles development. The values obtained for the AlZn10Si8Mg alloy are in a good agreement with the results reported in [14, 15].
For the AlZn10Si8Mg1 alloy, the higher value of impact resistance in the presence of grain refiner, is partially due to the casting conditions which are more evident than the addition of TiB.
The AlZn10Si8Mg presents the highest values of absorbed energy, while the presence of grain refiner and a concentration of 1% of Mg decrease the impact resistance values.
The hardness measurements have been made on polished samples applying a load 5 kg for 15 seconds for each measurement and were performed on each samples a minimum of 10 indentations. As reported before, these self-hardening alloys reach their highest mechanical properties after a storage period between 7 and 10 days, at room temperature, during which they are subjected to a natural ageing. Figure 13 reports the hardness measurement results: the hardness value is higher for the alloy enriched in magnesium. However, the AlZn10Si8Mg alloy has a hardness similar to other kind of AlSi alloys (between 100HV and 110HV), the AlZn10Si8Mg1 alloy has a hardness around 120HV. This can be caused by a higher amount of Mg_2Si hardening-precipitates that increase the global hardness of the alloy. In this case, the difference of hardness between the samples extracted from the top of the

casting and the samples coming from the bottom has been investigated and the results are perfectly comparable.

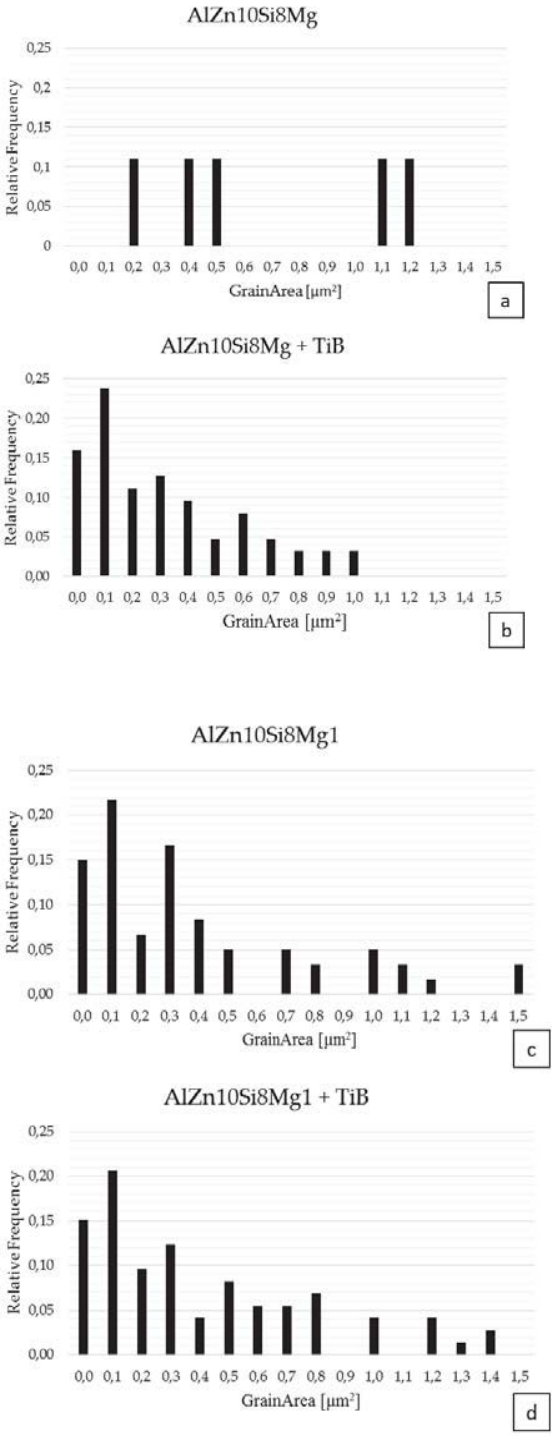

Figure 11. Relative frequency of the grain area measured: a) AlZn10Si8Mg without grain refiner, b) AlZn10Si8Mg with 500ppm of TiB, c) AlZn10Si8Mg1 without grain refiner d) AlZn10Si8Mg1 with 500ppm of TiB.

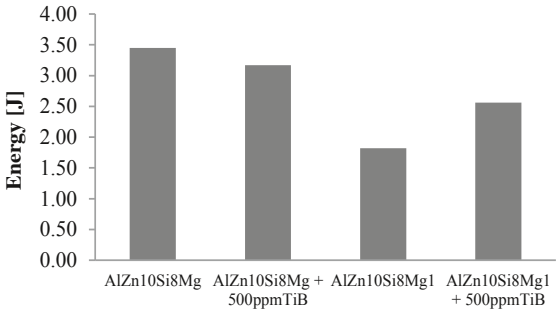

Figure 12: Impact energy for the different self-hardening alloys with and without grain refiner tested.

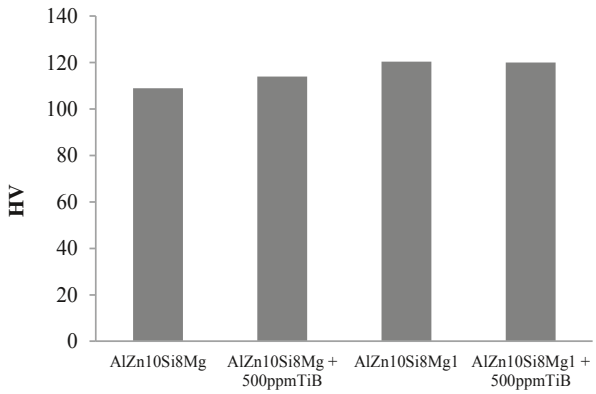

Figure 13: Vickers Hardness values for the different self-hardening alloys with and without grain refiner.

As can be observed in the graph of Figure 14, the hardness results for the samples with TiB grain refiner extracted from the top of the mould, are very similar to the values obtained without the addition of grain refiners, because titanium is heavier than the aluminium and it remains in the bottom part of the mould during the casting process.

Therefore, different densities value can determine a heterogeneous final microstructure.
With the addition of TiB, a slight increase of the average hardness value, has been obtained, especially for the AlZn10Si8Mg alloy. The effect is not so evident in the case of AlZn10Si8Mg1.

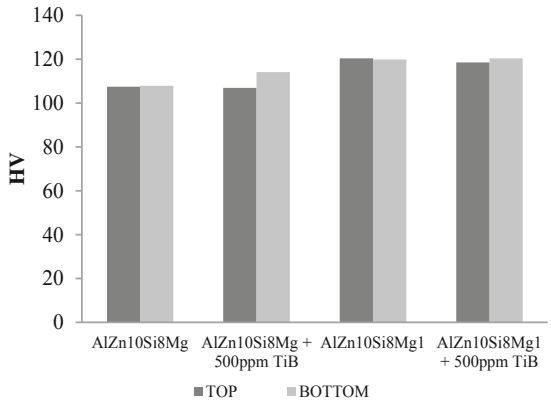

Figure 14. Vickers Hardness values for the different self-hardening alloys with and without grain refiner, in different region of the cast.

Conclusions

In this paper the effect of Al-5Ti-1B grain refiner on the microstructure and on the mechanical properties of two self-hardening aluminum alloys was investigated.
Based on the results obtained up to now the following conclusions can be drawn:

- Al-5Ti-1B grain refiner allows reaching a refined-microstructure for the AlZn10Si8Mg alloy, while on the contrary, in the case of the AlZn10Si8Mg1 alloy, the same average grain size was achieved, with or without the grain refiner addition;
- Probably high concentrations of Mg in the alloy activate the heterogeneous nucleation, creating smaller grains;
- Higher values of impact energy was obtained for the AlZn10Si8Mg alloy, both with or without grain refiner, with respect to the AlZn10Si8Mg1 alloy in the same condition;
- As a consequence of a higher volume fraction of Mg_2Si hardening precipitates, the AlZn10Si8Mg1 alloy, in both conditions, with and without the addition of Al-Ti-B grain refiner, reveals superior hardness values than the AlZn10Si8Mg alloy, with or without Al-Ti-B.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MODIFICATION OF EUTECTIC Si AND REFINEMENT OF EUTECTIC GRAIN IN Al-Si-Mg ALLOYS BY CrB_2 AND Sr ADDITION

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Keywords: Al-Si alloy; Refinement of eutectic grain; Modification of eutectic Si; Casting defect.

Abstract

The addition of CrB_2 and Sr into Al-10Si-0.3Mg (wt.%) based alloys not only modify the eutectic Si, but also refine the eutectic grains. CrB_2 particles were found to nucleate the eutectic grains, despite the AlP particles being partly removed or depleted by the interaction between Sr and AlP. Furthermore, P level was found to be one of the dominant factors that affect the nucleation of eutectic grains. CrB_2 addition with a low P level results in a refined eutectic grain size, despite less AlP particles being present to nucleate the eutectic grains. Porosity was found to be mainly located at the boundaries of eutectic grains, highlighting the importance of the refinement of eutectic grains to reduce the size and optimize the distribution of porosity. This investigation demonstrates a novel approach to refine eutectic grains and modify eutectic Si simultaneously into Al-10Si-0.3Mg (wt.%) based alloys, and thereby suggests possible industrial applications.

Introduction

Hypoeutectic Al-Si foundry alloys normally contain between 5 and 12 wt. % Si, together with additional Mg, Cu and Fe / Mn elements, to improve or control the mechanical properties [1, 2]. Modifying eutectic Si structure from a flake-like morphology to a fine fibrous morphology has been widely investigated. Different modifiers (e.g. Na [3-5], Sr [5-9], Eu [10]) have been used. Possible modification mechanisms (i.e. impurity-induced twinning (IIT) growth mechanism [5] and twin plane re-entrant edge (TPRE) growth mechanism [11, 12] as well as poisoning of the TPRE [13]) have been proposed. However, the interaction between the modifiers and the primary potent nuclei (i.e. AlP) for the eutectic colonies was also noted. Such interaction can reduce the nucleation frequency [14, 15] and thereby result in a coarse eutectic grain size. The eutectic grain size in Sr-modified alloys has been reported to be mainly in a range from 3 to 4 mm (in diameter) [16-18]. In Na-modified alloys, the eutectic grain size has been reported to be in a range from 5 to 25 mm [19], indicating that Na may have a stronger interaction with AlP than Sr.

Modifying eutectic Si has also been reported to cause a redistribution of pores and an increase of casting porosity [20]. In contrast, refining the eutectic grain size could optimize the distribution and reduce the size of pores. Due to the interaction between the modifiers and the primary potent nuclei (i.e. AlP) for the eutectic colonies, other available nucleation sites, instead of AlP, are required in order to nucleate eutectic grains. A combined addition of Sr and AlB has been reported [21] to increase the eutectic nucleation frequency and thereby decrease the eutectic grain size. Furthermore, the Si morphology is still well modified even at 500 ppm B levels. A combined addition of Sr and CrB_2 in Al-10 wt.% Si alloy has also been reported [22] to decreases the size of eutectic grains by increasing the number of nucleation sites for the eutectic grains. A well-modified eutectic Si and a refined eutectic grain size as well as an accompanying reduced level of porosity were obtained.

In the present paper, different CrB_2 addition level into Al-10Si-0.3Mg (wt.%) based alloys modified by Sr were investigated using multiple microstructure characterization techniques, with a special aim to elucidate the role(s) of CrB_2 during the refinement of eutectic grains.

Experimental material and procedures

A series of Al-10Si-0.3Mg based alloys (wt. % is used throughout the paper unless otherwise noted) with various CrB_2 and / or Sr additions were prepared using commercial purity (CP) Al (2N8, 99.8), Al-13Si master alloy, Al-25Mg master alloy, and other additions of Al-3B master alloy, Al-2Ti-1B master alloy, Al-10Sr master alloy, and Al-2Cr-1B master alloy, respectively. The chemical contents of these Al-10Si-0.3Mg based alloys were determined by spark analysis, as given in Table 1. The Ti concentrations reported in Table 1 arise from the Al-2Ti-1B master alloy additions and the base impurity level of the CP Al ingot. No excess Ti (i.e. Al-Ti master alloy) was added.

Table 1. Measured compositions of Al-Si-Mg alloys with CrB_2 and / or Sr additions using spark analysis. (wt.%)

No.	Si	Mg	Ti	Sr	Cr	Al
A	10.25	0.2383	0.0671	0.0004	0.0051	Balance
B	10.02	0.1823	0.0320	0.0203	0.0053	Balance
C	10.19	0.1782	0.0262	0.0204	0.0528	Balance

Note: Unrefined unmodified alloy (A) is Al-10Si-0.3Mg based alloy with the addition of Al-3B master alloy, and is used as a reference. Sr-modified alloy (B) was modified with Sr. CrB_2-refined Sr-modified alloy (C) was alloyed with the Al-2Cr-1B master alloy and modified with Sr.

In order to elucidate the individual effect of different CrB_2 addition, a series of Al-10Si-0.3Mg based alloys without the addition of AlB and Al-Ti-B, but with the additions of 0.05, 0.1 and 0.2 CrB_2 were also prepared using the same procedure.

Experimental alloys were melted in an electric resistance furnace and the temperature of the melt was maintained at 720 °C. Each alloy was produced in a single batch. Rotary degassing using Ar gas (about 15 mins) was performed prior to casting. After degassing, reduced pressure tests and subsequent density measurements were performed to make sure the gas content (H_2)

in the Al melts and the corresponding density index was acceptable for casting.
Samples for spark analysis were cast for subsequent chemical composition measurements (Table 1). Samples for mechanical properties tests were taken from the Diez samples. At least three tests for one condition were performed. Samples for microstructural characterisation were taken from the feeder parts of Diez samples. The specimens for OM investigation were mechanically ground, polished and then etched using a mixture of 60 ml water, 10 g sodium hydroxide and 5 g potassium ferricyanide (Modified Murakami reagent). The same samples but without etching, were further investigated using scanning electron microscopy (SEM) and electron probe microanalyzer (EPMA). Element mappings were performed on a JEOL JXA-8500F EPMA equipped with wavelength dispersive X-ray spectrometers (WDS) to elucidate the distribution of Mg element, which segregates to eutectic grain boundaries, and thereby can be used to determine the eutectic grain size.

Results and discussions

Figure 1 shows the eutectic grains observed by OM of unrefined unmodified alloy (Figure 1a), Sr-modified alloy (Figure 1b), and CrB_2-refined Sr-modified alloy (Figure 1c), respectively. Prior to Sr modification, the eutectic grains (Figure 1a) are too small to be clearly determined by OM. Therefore, the eutectic grain size was not evaluated. After Sr modification, the eutectic grains (Figures 1b, 1c) can be clearly determined by OM.

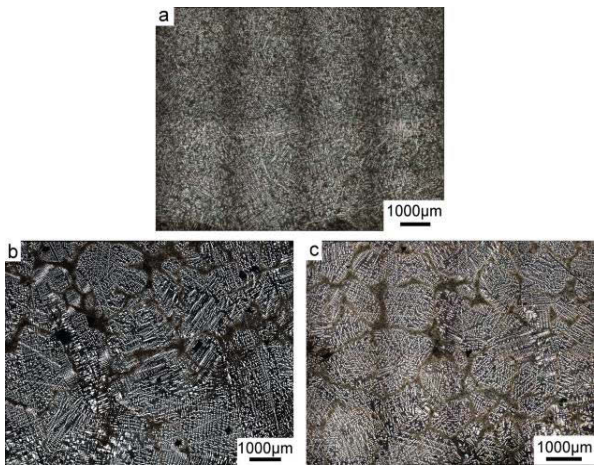

Figure 1: Eutectic grains observed by OM of unrefined unmodified alloy (a), Sr-modified alloy (b), and CrB_2-refined Sr-modified alloy (c), respectively.

The measured eutectic grains size for Sr-modified alloy is about 2015 ± 162 µm, which is smaller than 3 to 4 mm (in diameter) in other previous reported Sr-modified alloys [16, 18]. The measured eutectic grains size for CrB_2-refined Sr-modified alloy is 1538 ± 341 µm. Clearly, a more refined eutectic grain size was obtained with the addition of Al-2Cr-1B master alloy.

In order to elucidate the individual effect of different CrB_2 addition, a series of Al-10Si-0.3Mg based alloys without the addition of AlB and Al-Ti-B, but with the additions of 0.05, 0.1 and 0.2 CrB_2, were also investigated. Figure 2 shows the eutectic grains observed by OM of Sr-modified low-P alloys without (Figure 2a) and with the addition of 0.05 (Figure 2b), 0.1 (Figure 2c) and 0.2 (Figure 2d) CrB_2. Clearly, the addition of Al-2Cr-1B master alloy leads to a more refined eutectic grain size, although no significant change of the secondary dendrite arm space was observed. It should be noted that more than $0.1CrB_2$ addition (e.g. $0.2CrB_2$) results into an increased eutectic grain size (Figure 2a).

It should be noted that the eutectic grain, instead of eutectic cell, is used here as eutectic grains can become unstable and form cells themselves. Different types of eutectic cells can be present within the eutectic grain. After modification, most eutectic grains are approximately spherical [23], while eutectic cells appear to be cellular or dendritic depending on the coupled eutectic growth. Solute elements (i.e. Mg, Fe) can be rejected to not only the eutectic cell boundaries but also eutectic grain boundaries. However, during the last stages of solidification, such type of solute segregation will eventually decorate the eutectic grain boundaries. It has been reported that Sr-modified eutectic grains are large approximately spherical grains and grow with a high interface velocity [23].

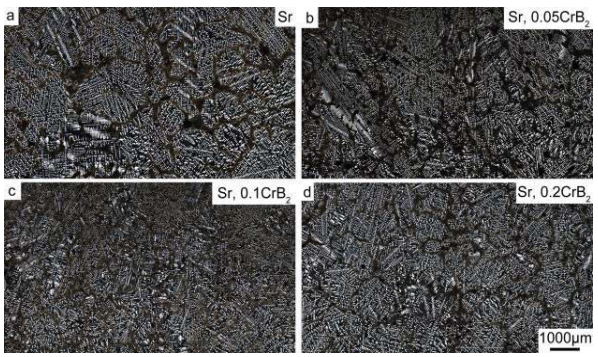

Figure 2: Eutectic grains observed by OM of Sr-modified alloys without (a) and with the additions of 0.05 (b), 0.1 (c) and 0.2 (d) CrB_2.

Figure 3 shows an SEM image (Figure 3a) and the element mappings of Al (Figure 3b), Si (Figure 3c), Fe (Figure 3d), and Mg (Figure 3e) in unrefined unmodified alloy, respectively. Here, it should be noted that π-$Al_8Si_6Mg_3Fe$ phase enriched in Mg and Fe forms during the last stages of solidification, decorating the eutectic grain boundaries. The dark regions observed in Figure 1 and Figure 2 are actually from the etched π-$Al_8Si_6Mg_3Fe$ phase located along the eutectic grain boundaries. Therefore, through the Mg mapping (Figure 3e), it is possible to elucidate the eutectic grain boundaries. From the Mg mapping (Figure 3e), the eutectic grain size was determined to be about 200 µm, as marked with red circles. A similar technique was also performed for other alloys. However, it should be noted that determining eutectic grain boundaries is still not reliable because of the low Mg content (< 0.3 wt.%) and thereby the low volume fraction of π-$Al_8Si_6Mg_3Fe$ phase as well as the limited evaluated regions.

Figure 4 shows an SEM image (Figure 4a) and the element mappings of Al (Figure 4b), Si (Figure 4c), Fe (Figure 4d), Mg (Figure 4e) and Sr (Figure 4f) in CrB_2-refined Sr-modified alloy, respectively. From the Mg mapping (Figure 4e), the eutectic grain size appears to be in the order of 650 µm, as marked with red circles, indicating that the presence of Sr has removed or depleted some of the possible nucleation sites of eutectic grains (most likely AlP). As mentioned above, care needs to be exercised in interpreting the measured eutectic grain sizes, even using the EPMA technique, since it is possible that they may not represent

the eutectic grain sizes but rather just part of the eutectic grains (i.e. cell sizes), due to the limited measurement area.

In terms of eutectic grain size, Sr addition increases the eutectic grain size, as shown in Figure 1b, which can be mainly attributed to the fact that Sr depletes or removes AlP. The presence of AlP particles in CP Al alloys (at a level of about 15 ppm, Table 1) is believed to play an important role during the nucleation of eutectic grains [6, 14, 15]. Although the eutectic grains are too small to be easily seen using OM, EPMA Mg mapping shows that the eutectic grain size is about 200 µm (Figure 3e), indicating that AlP can nucleate eutectic grains in the absence of Sr. However, in the presence of Sr, the eutectic nucleation density is reduced and thereby the number of formed eutectic grains is decreased. It should be noted here that the eutectic Si appear to be modified well in the present investigation. Therefore, the decreased eutectic grain size in CrB_2-refined Sr-modified low-P alloy can be attributed to the fact that more CrB_2 particles, despite less AlP, can nucleate the eutectic grains. Although the competing nucleation potency between CrB_2 and AlP is a subject for further investigation, it appears that CrB_2, in contrast to AlP, does not affect the modification of eutectic Si caused by Sr.

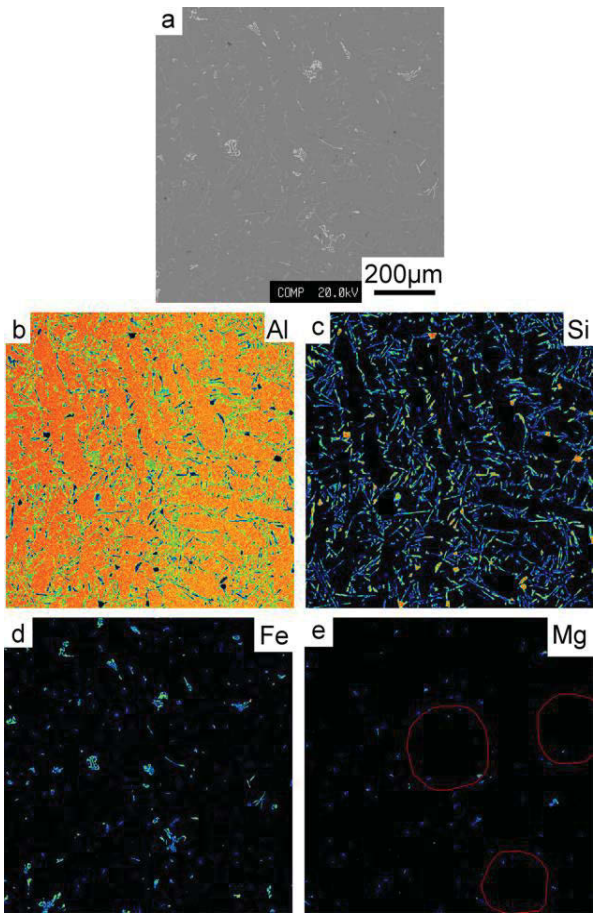

Figure 3: SEM image (a), and element mapping of Al (b), Si (c), Fe (d), and Mg (e) in unrefined unmodified alloy, respectively.

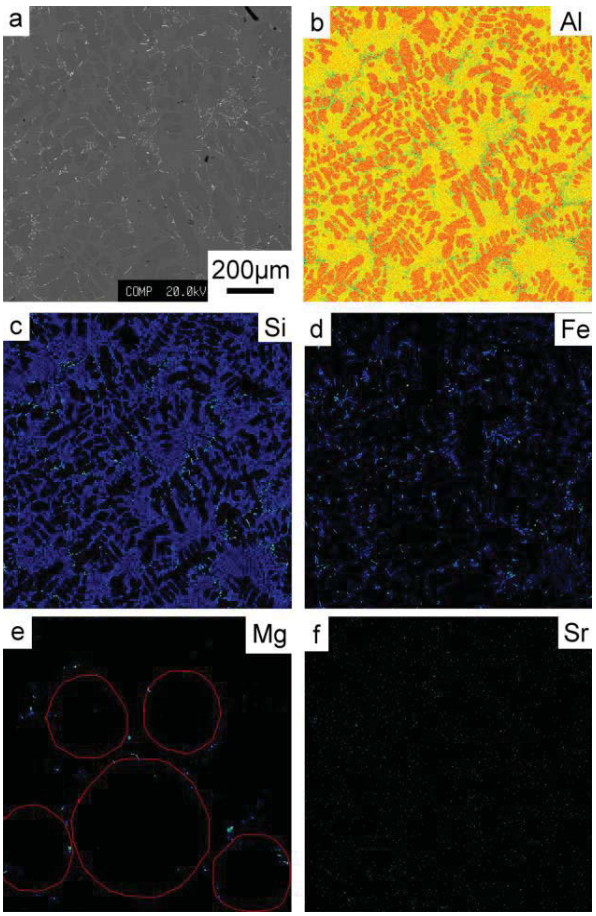

Figure 4: SEM image (a), and element mapping of Al (b), Si (c), Fe (d), Mg (e) and Sr (f) in CrB_2-refined Sr-modified alloy.

Conclusions

1. Chemical etching (Modified Murakami reagent) and EPMA permits a reliable determination of eutectic grain size in Al-Si-Mg based alloys.
2. CrB_2 particles can nucleate the eutectic grains, despite the AlP particles being partly removed or depleted by the interaction between Sr and AlP.
3. A combination of CrB_2 and Sr additions results in a simultaneous modification of eutectic Si and a refinement of eutectic grain sizes.
4. This investigation demonstrates a novel approach to refine eutectic grains and modify eutectic Si simultaneously, and thereby suggests possible industrial applications.

Acknowledgements

J.H. Li gratefully acknowledges the financial support from the Major International (Regional) Joint Research Project (No. 51420105005) from China.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MODIFICATION OF MACROSEGREGATION PATTERNS IN ROLLING SLAB INGOTS BY BULK GRAIN MIGRATION

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Keywords: DC Casting, Macrosegregation, Stirring, Mixing

Abstract

In recent years the metallurgical community has debated on the influence of floating grains on macrosegregation in Direct-Chill cast rolling-slab ingots. The majority of research to date has been primarily focused on the origin and trajectory of these grains and how to incorporate them into numerical simulations. We propose a novel mixing technique to promote the resuspension of these grains. In order to test this approach, pairs of ingots were cast using the standard DC and mixing techniques. Non-grain refined P1050 ingots were cast in order to illustrate the change in microstructure, while Al4.5Cu ingots were cast to investigate the potential change in macrosegregation. The mixing trials indicate the potential to significantly modify microstructure and macrosegregation in rolling-slab ingots.

Introduction

Macrosegregation continues to be one of the major, irreversible defects present in direct-chill (DC) cast aluminum ingots. This significant variation in chemical composition throughout the cross section of an ingot typically results in the variation of thermal and mechanical properties, which lead to deleterious effects downstream in final products[1,2].

Modern macrosegregation theory was pioneered by M.C. Flemings in the 1960's-1970's. Since that time, significant efforts have been devoted to the development of computer models for simulation of macrosegregation in castings, fewer efforts have been dedicated to experimental studies, and even less have been aimed at modifying existing casting structures.

The basis for all macrosegregation theory is the relative movement of liquid and solid phases during solidification[3,4]. This relative movement enables the microscopic partitioning of solute between liquid and solid phases (microsegregation) to be translated to differences in chemical composition on the macroscopic scale (macrosegregation). There are different variations of this relative movement that are characteristic of different parts of the transition region (sump) of a casting (See 3 for a full treatment). In this study we will focus uniquely on the movement and sedimentation of free-moving grains.

We know that commercially cast alloys tend to solidify as dendrites, forming an overall equiaxed structure in the casting due to the addition of exogenous nucleation sites (grain refiner) during the casting process. In the slurry region (between the liquidus and coherency isotherms in the phase transition region) these equiaxed grains are free to move and can travel short or long distances depending on the magnitude of the surrounding convective currents. In the mushy zone (between the coherency isotherm and non-equilibrium solidus) however, these dendrites interlock to form a continuous, porous network and thus have a fixed position in the ingot. From this point on they are considered immobile and that their only movement is in the casting direction at the specified casting speed.

Movement and sedimentation/growth of solid grains in the slurry zone is often considered as the primary mechanism of centerline segregation in DC cast ingots [5,6]. As the sump angle exceeds the static angle of repose of the dendrites in the slurry region they fall to the bottom of the sump, or center of the ingot. Assuming solid transport is continuous from many regions of the ingot to the center, the fraction solid is greater than equilibrium conditions predict. As the primary solid in hypoeutectic aluminum alloys is always depleted of solute, the resultant accumulation of free-moving grains contributes to negative (depleted) segregation in the center of ingots. While the existence of these grains was debated for many years, primarily due to the difficulty of their observance in post-mortem analysis, the industry has largely acknowledged their existence and effects on macrosegregation[3].

Much of the research to date on sedimenting grains has been focused on their origin, trajectory, and composition. While invaluable to those working on computational models; there has been little work done on *mitigating* the deleterious effects of sedimenting grains in large castings. Since grain sedimentation is inherited from grain refinement and the geometry of an ingot sump, this attitude of "alea iacta est" is understandable. It is generally accepted that macrosegregation is magnified in larger castings, and in those cast at higher speeds. With an aim to increase productivity in the metals industry, it is requisite to not only understand macrosegregation[7], but also understand how to *m*itigate it. This paper aims to be a first approach at modifying macrosegregation by inhibiting the bulk accumulation of sedimenting grains in the center of industrial rolling slab ingots

In the chemical processing industry, sediment buildup in reacting vessels is avoided in order to ensure stable reaction kinetics and prevent blockage. In order to maintain suspension of the particulate jet mixers are often used. The volumetric flow rate of such jets is typically limited by the

pump, which often become cost-prohibitive for large vessels. In order to utilize as small a pump as possible, while maximizing flow rate, educting nozzles are often used (see Figure 1).

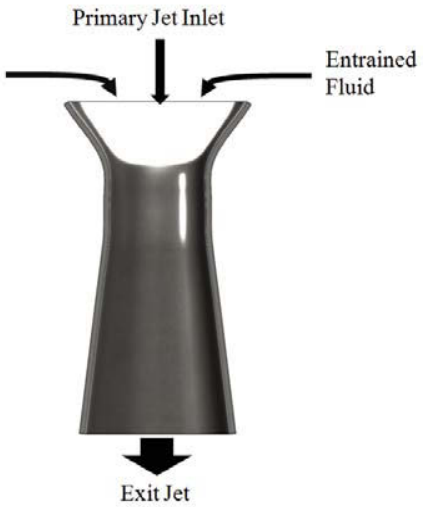

Figure 1: Educting Nozzle Schematic

Educting nozzles entrain the surrounding fluid due to the venturi housing mounted downstream of the entrance (primary) nozzle. As the working fluid passes through the narrower "neck" region, it develops a low pressure region capable of entraining additional surrounding fluid and generating a jet downstream of the housing with a volumetric flow rate up to five times that of the primary jet.

The purpose of this research was to investigate the possibility of using a turbulent jet to promote the bulk migration of sedimenting grains from the bottom of the sump. As a first step in this research, this paper presents the results obtained after sectioning and analyzing industrial ingots of P1050 and Al4.5Cu cast with an educting nozzle.

Experimental Apparatus and Procedure

For this experiment, charges were loaded into a commercial gas burner furnace. The melt was degassed and filtered according to commercial standards. The Al4.5Cu ingot was inoculated with 25ppm TiB commercial grain refiner, while the P1050 ingot was not grain refined. The metal was cast using a bi-level pour method through a fused silica downspout; no combination bag was used for these trials. A typical "coffin type" bottom block was used along with a 1750x600mm WAGSTAFF LHC™ open top mold. The steady state casting speed was 65 mm/min. The casting speed and metal level were varied during the first 500mm of the cast before reaching steady state values. At 600mm of cast length an eductor nozzle machined from 4140 steel was suspended by inconel rods beneath the downspout. The top of the nozzle was placed 25mm below the downspout exit to prevent the submersion of an oxide from the surface, and to ensure proper metal flow into the nozzle. The nozzle was removed from the ingot at the end of the cast before final solidification had occurred.

The cold ingots were cross sectioned at cast lengths of 1200mm, 1800mm, and 2200mm. Each cross section was divided into quadrants and a series of 45 samples were removed using a core drill for metallographic analysis. The opposite quadrant was marked with a set of 455 points and was analyzed for composition using an Olympus Delta Professional Alloy Plus XRF analyzer (see Figure 2).

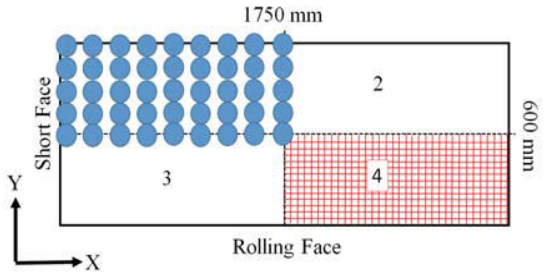

Figure 2: Location of Samples taken for Metallographic and XRF Analysis

The metallographic samples were etched with a dilute hydrofluoric acid etchant and analyzed for grain size and dendrite arm spacing using an optical microscope according to the line-intercept method[8].

During the course of the experiments particles could be observed impacting the meniscus on the surface of the molten pool during casting. In an attempt to capture these particles, a quench mold was fabricated following the design of Martinez and Flemings[9]. A set of 12.5mm copper plates were clamped over a gasket made of 2mm graphite felt, forming an opening 2x25mm in cross section. One end of the cavity was sealed and attached to a vacuum generator. Cellophane wrapping was placed over the mouth of the opening in order to maintain the vacuum. To take a sample, the mold was quickly dipped at the surface of the molten pool causing the cellophane to melt immediately and draw in a thin sample of quenched alloy.

Results

Figure 3A is a surface plot representing one quadrant of a standard cast Al4.5Cu alloy. For more discussion of the various attributes of this pattern the reader is directed to [7]. Figures 3B-D represent the macrosegregation pattern as measured for the educting nozzle cast ingots at cast lengths of 1200,1800, and 2200 mm respectively. The pattern in each of these zones appears to be fairly similar in its distortion of the standard profile. There is a significant reduction in the size of the depleted centerline region, as well as an accumulation of enriched material along the short face. While overall the pattern is far from homogenous, it does not exhibit any large zone of inconsistent composition that could not be removed through a reasonable homogenization time.

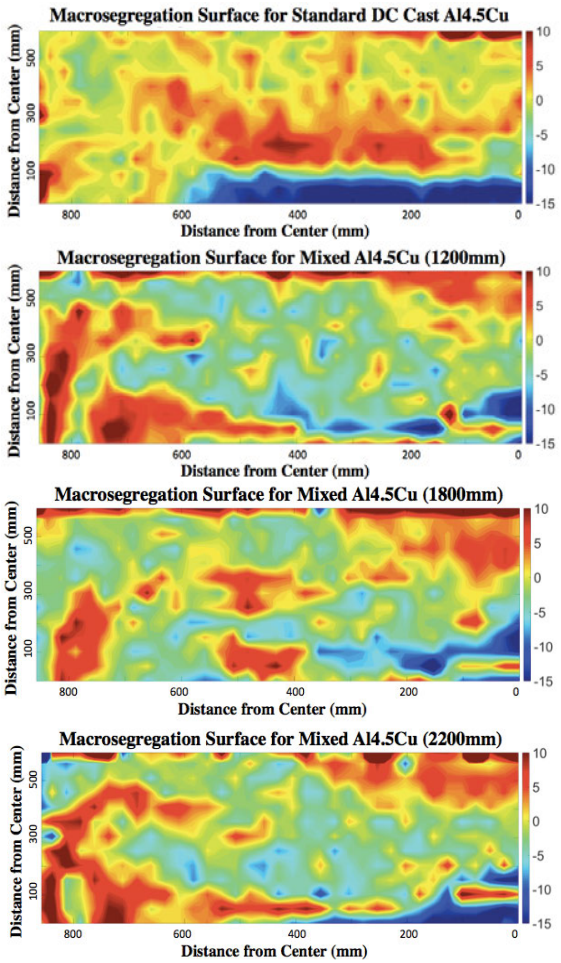

Figure 3: A,B,C,D From top to bottom, are macrosegregation plots based on XRF analysis of one quadrant of an ingot cross section. (A) Standard DC Cast with Combination Bag. (B) Educting Nozzle cast at 1200mm (C)1800mm (D)2200mm. Al4.5Cu Alloy

Figure 4A is a representation of one quadrant of the standard grain size profile in a DC cast ingot. The grain size varies from around 50 um at the short face where solidification is most rapid, to over 250um near the center where the solidification times are longest. Figures 4B-D represent similar profiles but for the educting nozzle ingot at cast lengths of 1200,1800, and 2200mm. All scale bars are identical for ease of comparison. While in each of the stirred ingots we can see the trend towards slower cooling (and larger grain size) near the center, the range of grain sizes is greatly reduced. For each of the slices the majority of the grains are near 50 um in size, with the largest barely reaching 150um in diameter.

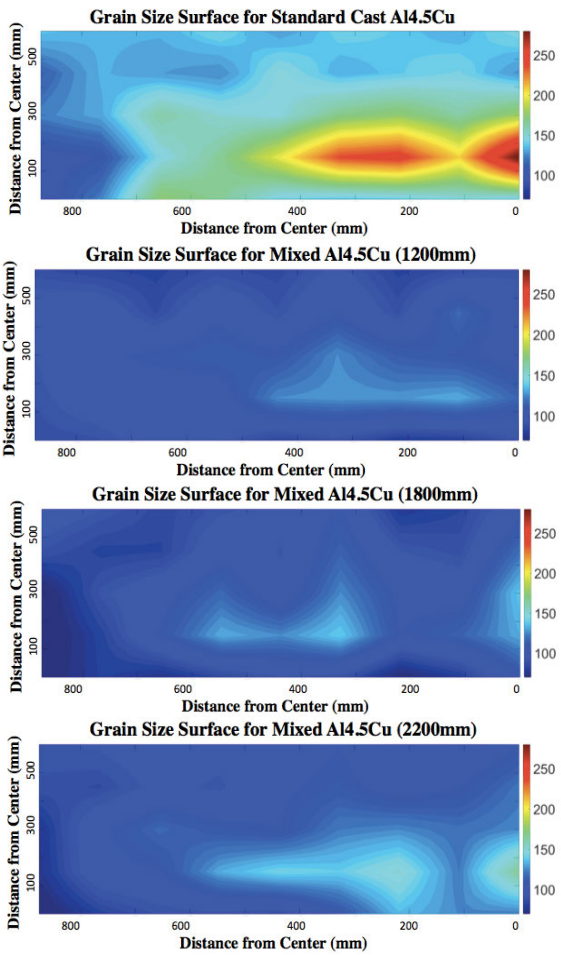

Figure 4: A,B,C,D From top to bottom , are grain size plots of one quadrant of an ingot cross section. (A) Standard DC Cast with Combination Bag. (B) Educting nozzle cast at 1200mm (C)1800mm (D)2200mm. Al4.5Cu Alloy

Figure 5A is a representation of one quadrant of the standard secondary arm spacing profile in a DC cast ingot. We can see small dendrite arms of approximately 30um surrounding the periphery of the casting, and growing to approximately 55um near the center. Figures 5B-D represent profiles taken from the educting nozzle cast ingot at cast lengths of 1200, 1800, and 2200mm. For each of the mixed profiles we can see an overall increase in dendrite arm spacing. While the smallest dendrite arms are still on the order of 30um apart, the largest arms are now over 70um apart near the center of the ingot. The largest dendrite arm spacings are on par with the average grain size for these mixed castings, limiting our ability to distinguish grains or dendrite arm spacing as anticipated for very coarse grains. Such potential crosstalk between our data sets however does not affect the observed trends.

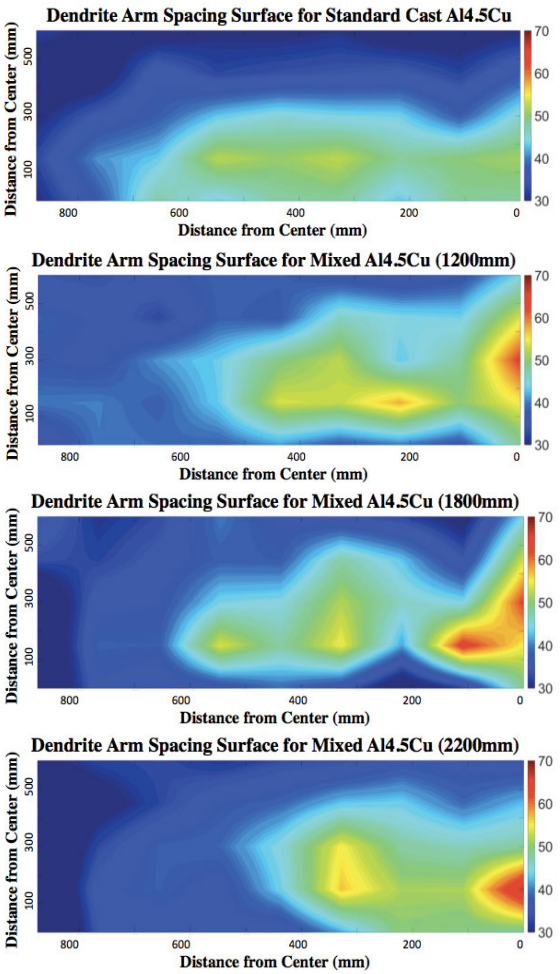

Figure 5:A,B,C,D From top to bottom, are dendrite arm spacing plots of one quadrant of an ingot cross section. (A) Standard DC Cast with Combination Bag. (B) Educting nozzle cast at 1200mm (C)1800mm (D)2200mm. Al4.5Cu Alloy.

The grains observed impinging on the meniscus during casting were readily observed after sectioning and etching of the thin samples obtained by the quench mold. Figure 6 is a representative sample of the microstructure obtained from the quench mold. Large globular grains were found to be surrounded by large numbers of fine crystals. Performing a line scan of these grains found them to be approximately 1% Cu as compared to the furnace composition of 4.5% Cu. It is presumed that the larger globular crystals were transported via forced convection from the center of the ingot. Their morphology evolved via a fragmentation route as described by Martinez[9]. When captured by the quench mold their morphology was frozen in place thereby preserving their microstructure. While there was found to be a distribution of globular grain sizes , the average size was found to be 52um. In comparing this information to that from Figure 4 we can see that this represents the smallest final grain size present in the ingot. This seems to indicate that while these grains are mobile, they do continue to grow in size.

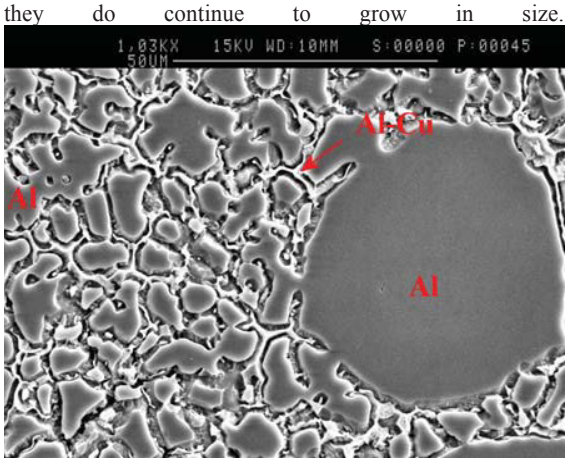

Figure 6: SEM Image of the grains captured by the quench mold during the casting process

P1050 is a nominally pure alloy that contains 10 ppm silicon and 50ppm iron. Due to the low levels of alloying elements in these castings, macrosegregation profiles would not be able to be measured within the statistical noise of the XRF detector. By using a non-grain refined alloy however we can easily see the microstructural changes caused by the addition of the educting nozzle. Figure 7 is a cross section of a P1050 ingot that has been polished and etched with a dilute hydrofluoric acid etchant. We can see on the outer periphery a small chill zone of small, regular crystals. Adjacent to the chill zone is a large region of twinned-columnar grains (TCG). In the center of the casting is a large zone of equiaxed crystals. These three regions exemplify the typical ingot microstructure. Figure 8 is a cross section of a P1050 ingot that has been processed using an educting nozzle. We can still observe the small chill zone surrounding the periphery of the casting, but the large TCG zone is completely absent. Instead it has been replaced by a fine equiaxed structure that extends nearly to the center of the ingot. At the ingot center there is a small columnar zone growing from either side of the equiaxed zone and then impinge on one another at the plane of thermal convergence.

Figure 7: Polished and etched cross section of a P1050 ingot illustrating Chill, Columnar, and Equiaxed zones.

Figure 8: Polished and etched cross section of a P1050 ingot processed with an educting nozzle.

Conclusion

We have proposed a novel method to help modify the existing macrosegregation patterns present in rolling slab ingots. By introducing intense forced convection along the centerline of the ingot using an educting nozzle we have been able to observe the motion of sedimenting grains in the molten pool of an ingot. By re-suspending these grains during the solidification process we have been able to show the potential to modify existing macrosegregation patterns in rolling slab ingots. In addition to the changes in compositional homogeneity, significant microstructural changes are also observable. While these results are a promising first step, they also illustrate the need for additional study. The bulk migration of grains from the bottom of the sump is a useful tool, but if they continue to settle preferentially in another location in the ingot, the advantage could be lost. Additional study is required to determine the velocity, and form of the jet which will best evenly distribute these grains and generate superior microstructures.

Acknowledgement

A special thanks to the NOVELIS Solatens team for the use of their facility as well their support and assistance throughout the course of this study.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF HIGH INTENSITY ULTRASONIC TREATMENT ON THE MICROSTRUCTURE, CORROSION AND MECHANICAL BEHAVIOUR OF AC7A ALUMINUM ALLOY

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Keywords: Ultrasonic treatment, Aluminum alloy, Microstructure, Corrosion, Mechanical

Abstract

Ultrasonic treatment is one of the most efficient and cost effective processes that are developing in today's technology. It improves mechanical properties, corrosion resistance and homogeneity of the microstructure. In this study, the effect of casting conditions and ultrasonic treatment on the microstructure, hardness and corrosion behaviour of the Al-4.5%Mg alloy was investigated. Alloy ingots were melted in resistance furnace at 750C, and then cast into metallic molds. Pouring temperatures were 750C (superheated) and 639C (liquidus temperature). One sample was poured at 655C and left to solidify in ultrasonic field down to 615C within 2-3 minutes. The microstructure showed a completely coarse dendritic morphology for casting at 750C, and less dendritic morphology with finer grain size when cast at 639C. The finest and most globular aluminum grains and intermetallic particles were obtained by ultrasonic treatment during solidification. Higher hardness values and corrosion resistance were recorded for ultrasonic treated sample in comparison to cast one.

Introduction

The commercial importance of aluminum-silicon alloys is based on their high fluidity and low shrinkage in casting, brazing and welding applications, in addition to their high specific strength, low thermal expansion and good corrosion resistance. These alloys are increasingly used in automotive applications to reduce vehicle weight and improve fuel economy. They are used for manufacturing several important components, such as pistons, engine blocks, and others. The microstructure of the commercial hypoeutectic alloys usually contains α-Al, Al-Si eutectic and Fe-intermetallic phases and other minor phases [1].

The aluminium 4.5% magnesium alloy is known for exceptional performance in extreme environments: this alloy is highly resistant to attack by both seawater and industrial chemical environments. It also retains exceptional strength after welding. It has the highest strength of the non-heat treatable aluminum alloys [2].

It is well accepted that the microstructure is one of the major factors that define the mechanical properties of aluminum alloys. For general casting applications, refining is used to achieve equiaxed and fine grains will increase fluidity, reduce hot cracking, and reduce microporosity. Non-dendritic structures will be obtained if solidification produces a well grain-refined structure [3]. In shape casting, the advantages include better distribution of porosity and second phase particles, improved surface finish and machinability, greater fatigue strength and better pressure tightness. In ingot and DC (Direct Chill) casting, it greatly facilitates the DC casting by reducing tendency of ingot cracking and provides improved mechanical properties and several other beneficial effects [4].

The ultrasonic treatment showed potential effects in refining the primary silicon and Fe-intermetallic phases in hypo- and hyper-eutectic Al-Si alloys, as well [5-6]. This effect superseded that of phosphorous addition (50 ppm) in both Si-particle refining and its effect on wear resistance [7]. It was also found that the use of ultrasonic in the production of billets for thixocasting is beneficial, since the UST produces a much slower-growing aluminum grains during soaking treatment, with a uniquely very small amount of entrapped eutectic, giving a merit of better formability in the semisolid state [8]. The ultrasonically-treated thixocast parts showed a highly uniform nondendritic microstructure with higher mechanical properties than the untreated billets [9-10].

The corrosion resistance of aluminum-magnesium alloys is related to the formation of an oxide (passive) film, which naturally develops on the alloy surface under normal atmospheric conditions [11-12]. However, the presence of magnesium may decrease the corrosion resistance as a result of oxide formation. The oxide film formed on the aluminum alloy surface is non-uniform, thin and non-coherent. Therefore, it imparts a certain level of protection under normal conditions. When exposed to environments containing halide ions, of which the chloride (Cl^-) is the most frequently encountered in service, the oxide film breaks down at specific points leading to the formation of pits on the aluminum surface. This type of corrosion is known as pitting corrosion [13-14]. The structural characteristics of the film formed on the aluminum surface, and the intensity of corrosion attack are influenced by chemical composition of the exposed alloys, the presence and distribution of micro-defects (vacancies, voids, etc.) as well as macro-defects (inclusion, second phase particles - their size and shape) and electrolyte composition (e.g. pH, halide concentration, temperature, fluid velocity). In general, pitting on the aluminum alloy surface takes place in the presence of an electrolyte within a 4.5–8.5 pH range and tends to increase with an increase in temperature, concentration of aggressive ions and stagnation of the electrolyte [15].

Owing to the advantages of ultrasonic treatment, in this work a systematic study has been carried out to determine the effect of

ultrasonic vibrations on the microstructure and the corrosion and mechanical behaviour of aluminum-magnesium AC7A alloy.

Experimental work

The chemical composition of the base alloy, Al-4.5 wt% Mg, used in the present work is shown in Table. 1. For investigating the effect of pouring temperature, the alloy ingots were melted in an electrical resistance furnace where the temperature was kept at 750C for a holding time of one hour. No fluxes were added to the ingots. After removing the crucibles out of the furnace, a digital thermometer was used to detect the melt temperature in the crucibles just before pouring. Pouring was done in a metallic mold of dimensions (φ25 mm x 120 mm). Pouring was done at different temperatures relative to the liquidus temperature of the alloy as shown in Table 2.

Table I. Chemical Composition of the investigated alloys.

Al-4.5%Mg	Si	Mg	Cu	Fe	Zn	Ti	Mn
	0.08	3.65	0.02	0.7	0.03	0.00	0.47

For investigating the effect of ultrasonic treatment, molten alloys were poured into the crucible that is connected to the ultrasonic vibrating system as shown in Figure 1, then the ultrasonic vibrations were turned on and continued during solidification. The ultrasonic vibrations were transferred to the molten alloys contained in the crucible through a vertical steel rib. The ultrasonic system consists of 2000 watts – generator, a 19.5 kHz transducer, steel wave guides and a stainless steel crucible. The steel crucible was coated with boron nitride to avoid melt soldering with the crucible. For the alloy Al-4.5%Mg, the molten alloy was poured at 660C into the steel crucible horn. The ultrasonic vibrations continued from 660C down to 615C, within 2-3 minutes. These are typically liquid metal treatments followed by solidification under the ultrasonic field. No grain refiner or Si-modifying additions were used with both alloys.

Table II. Different pouring temperatures used in the current study.

Alloy	Pouring Temperature C
Al-4.5%Mg	750C
	639C

* Liquidus temperature.

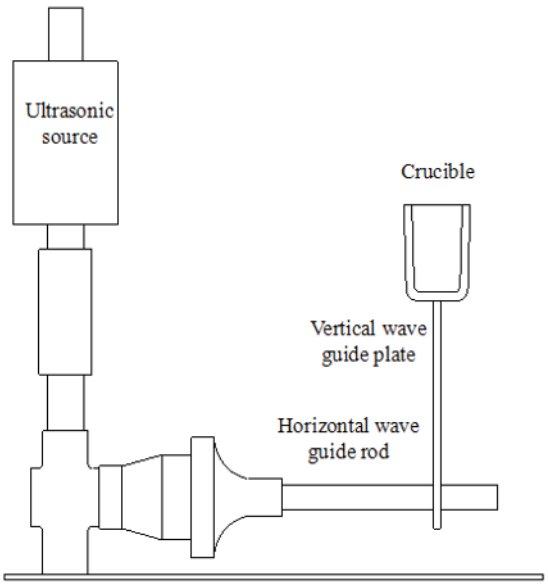

Figure 1. Apparatus used in ultrasonic treatment.

The specimens were prepared for microstructure investigation by the standard techniques of grinding and polishing, followed by etching using a solution of 0.5% HF. The grain size and microstructure of alloys were investigated by Zeiss optical microscopy. The grain size was measured using the linear intercept procedure. Vickers hardness test was carried out at a load of 50 N, and the results were compared for the different pouring and treatment conditions. Electrochemical measurements were carried out in 3.5% NaCl solution using a conventional three-electrode electrochemical glass cell with a platinum counter electrode, a saturated calomel electrode (SCE) as reference and the working electrode (the tested material). All the potentials were measured relative to the SCE. Corrosion rates and pitting potential measurements were used to evaluate the electrochemical behaviour of the samples.

The pitting behaviour of samples was characterised, with the initial potential set at -1000 mV and scan rate 0.2 mV/s. The scan direction was set to reverse from anodic to the cathodic direction. When the current starts to increase, potential for pitting E_{pit} is determined, whereas when the current is reversed, the potential of repassivation E_{rep} is obtained.

Results and discussion

Figure 2 (a-c) shows the microstructure of the AC7A alloys. It is clearly seen that Fig. 2 (a) shows the smallest grain size in comparison to the AC7A 750 ^{o}C alloy (fig. 2b) without the application of UST. However both samples represented dendritic structure. The application of ultrasonic treatment to the AC7A alloy has greatly enhanced the microstructure as shown in Fig. 2 (c). Ultrasonic treatment has yielded much finer, non dendritic, globular grains. The aluminum grains have become highly equiaxed and round when the ultrasonic treatment is applied to the AC7A alloy for 2-3 minutes.

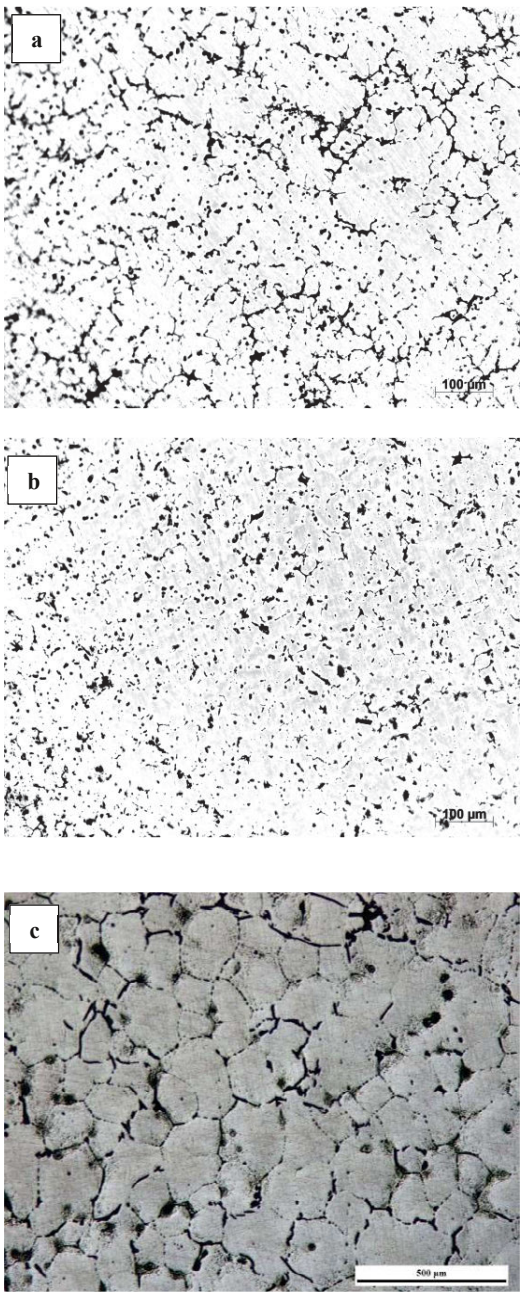

Figure 2. Microstructure of AC7A alloy: (a) without UST poured at 639C. (b) without UST poured at 750C. (c) with the application of UST at 660-615C.

Table 3 indicates that the HV of the as-cast AC7A alloy poured at 639C is higher than 750C pouring temperature sample, without the application of UST. On the other hand, the HV of the AC7A alloy with UST is greater than in the case without UST. Pouring at 639C (just above liquidus) and 750C showed dendritic structure versus fine equiaxed structure (Figure 2(c)) and this may explain the effect of microstructure on hardness.

Table III. The hardness average values

AC7ASample	639C	750C	UST
HV	70±3	67±2	75±3

The reason why higher pouring temperature (above liquidus temperature) produced coarser Al grain sizes is because, higher pouring temperature produces slower cooling rate; leading to longer time in the liquid state; resulting in lower nucleation rate and longer growth time and this eventually leads to larger grain sizes [16].

The refining effect of UST when applied to both alloys can be described according to two main theories:

1. Activation of inclusion particles in the melt, where they become nucleation sites upon solidification [17].
2. Formation of solid crystals at some temperatures above the liquidus temperature, which act as nucleation sites upon solidification [1].

According to the first theory the pronounced effect of ultrasonic vibrations when applied in the liquid state seems to be the activation of inclusion particles in the melt. Among the physical effects accompanying the propagation of high-power ultrasonic oscillations in a fluid (a melt), cavitation, i.e. the formation of bubbles, almost empty or filled with gases dissolved in the melt is most important. These bubbles are formed by tensile stresses created by the sound wave at the half-period of rarefaction (negative pressure) and continue to grow by inertia for some time. The intense cavitation activates impurities suspended in the melt leading to the formation of a large number of nuclei, which act as nucleation sites during solidification, resulting in the formation of fine non-dendritic grains.

From an electrochemical point of view, it is known that aluminum is nobler than magnesium, which means that magnesium is more vulnerable to corrosion when being subjected to a violent medium. However the process of samples preparation may have significant effect in their behavior. Figure 3 represents the Tafel's curves of the three samples; the estimated values of the corrosion rates indicate that 655-6150C UST sample has the lowest corrosion rate (36 µm/year) in comparison to 130 and 158 µm/year for AC7A 750C and 639C respectively. Also, the 655-615C UST showed more repassivation and less pitting potential (figure is not shown here) than 639C, however, the 750C showed no passivation behavior at all.

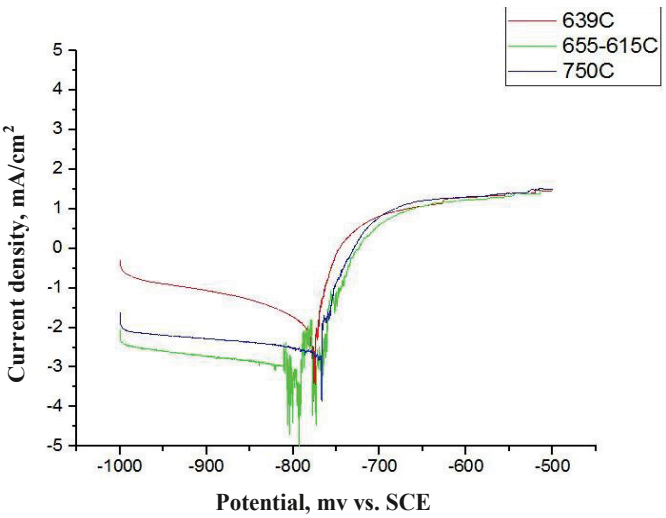

Figure 3. Tafel's curves of AC7A 639C, 750C and UST samples in 3.5% NaCl

The increase in the corrosion resistance might be due the fact that ultrasonic treatment had refined the microstructure, fine grain size structure, whereas the other two samples have the usual dendritic structure. These results agree with [18-19] in their researches in which the UST samples reflected higher corrosion resistance and higher resistance to pitting formation.

Conclusions

This study was done to investigate the effect of pouring temperature and ultrasonic treatment of the AC7A alloy on the microstructure, corrosion and mechanical behaviour. From the obtained results the following points can be drawn:

1. The pouring temperature has a pronounced effect on α - Al grain size, where lower pouring temperature (above liquidus temperature) produced finer Al grain sizes and less developed dentritic structure than higher pouring temperatures, and has yielded the higher hardness.
2. The application of ultrasonic treatment to the AC7A alloy has enhanced the microstructure reflecting higher hardness values of the alloy
3. All samples have been subjected to corrosion attack, a significant improvement was noticed for the ultrasonically treated sample while for pouring temperature samples (without UST) had a negative effect.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MECHANISM OF ZIRCONIUM POISIONING EFFECT ON TiB_2 INOCULATION IN ALUMINIUM ALLOYS

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Keywords: Grain refinement, Al alloys, Heterogeneous nucleation, Solidification.

Abstract

It was found that the efficiency of TiB_2 inoculation to achieve grain refinement was decreased dramatically by Zr addition. In the present work, the mechanism of so-called "poisoning" effect was investigated by advanced analytical microscopy, including Cs-corrected high resolution STEM. For the first time, adsorption of Zr atoms was observed in a mono-layer on {0001} surface of TiB_2. Significantly, atomic resolution STEM Z-contrast imaging revealed that the brightness of atomic columns in the monolayer varied periodically when viewing in $[10\bar{1}0]$ direction, indicating that the adsorbed Zr atoms in the layer were ordered in 2D. Deviation by 8 degrees from the well define orientation relationship, $(0001)_{TiB2}$ 8deg ~ $(111)_{Al}$, and $[11\bar{2}0]_{TiB2} \parallel [0\bar{1}1]_{Al}$, was observed. The effect of Zr atom adsorption at the interface on mechanism of heterogeneous nucleation was discussed.

Introduction

As the most widely used grain refiner for Al alloys, Al–Ti–B master alloy provides the inoculating TiB_2 particles which promote heterogeneous nucleation during solidification, resulting in the formation of fine, uniform and equiaxed grains. Since the introduction of Al–Ti–B-based grain refiners over 60 years ago [1], extensive research work has been carried out to understand the mechanism of grain refinement [2–4]. Pioneering TEM work was done by Schumacher and Greer [5, 6] who used a metallic-glass technique. They claimed that Al_3Ti coating phase was essential for TiB_2 to be potent. Recently, a direct observation of such grain refining TiB_2 particles in solidified Al samples has been conducted using state-of-the-art electron microscopy [7]. It was revealed that there existed only a Ti-rich monolayer on the $\{0001\}_{TiB2}$ surface. It was the adsorption of Ti atoms that played an important role in promoting the potency of TiB_2 particles for the heterogeneous nucleation of Al.

However, the grain refining efficiency of Al-Ti-B is dramatically reduced by some alloying elements, such as Zr, Si and Cr [3]. This deteriorating effect is so-called "poisoning". When Ti and Zr are added together in Al melt, the grain refining effect is less than that which occurs with either individual addition, a phenomenon termed "co-poisoning" [8]. Considerable work has been carried out to eliminate or reduce this deleterious effect. One approach is to increase the addition level of Al-5Ti-1B master alloy [9, 10], and another is to develop alternative grain refiners, such as Al-Ti-C [11-13]. In this work, the mechanism of the so-called "poisoning" effect caused by Zr addition was investigated. By using Cs-corrected high resolution STEM, the TiB_2 particles collected from both Zr-free and Zr-containing Al melts were subjected to extensive examinations which were focused on the nucleating surfaces in order to establish the atomic configuration across the interface. For the first time, Zr atom adsorption at TiB2 surfaces was experimentally revealed. The effect of the Zr adsorption on heterogeneous mechanism is discussed.

Experimental

Commercial purity Al (>99.86 wt%) was used as the base material with Zr concentration being 0.058, 0.088, 0.168, 0.206, 0.31 and 0.45 (in wt.%). The Al-Zr melt was being held at 800°C for 4 hours followed by addition of 0.1wt% commercial Al-5Ti-1B grain refiner. After 4 more hours holding, the melt was cast into TP-1 mould [14]. Quantitative metallography was carried out using a Zeiss optical microscope. In order to facilitate direct examination of TiB_2 particles, melt filtration was used to collect the particles from melt. In the process, Al-Zr melt containing 0.2 wt% Al-5Ti-1B grain refiner was transferred into the crucible in the pressure chamber. Ar was introduced to force the melt to flow through the porous filter attached at the bottom. TiB_2 particles were thus collected above the filter together with the remaining melt. The solidified material above the filter, which contained the locally concentrated TiB_2 particles, was subjected to examinations. To prepare thin foils for TEM/STEM examinations, 3 mm diameter discs from the filtered residue material were ground to a thickness less than 70 μm followed by ion beam thinning. TEM analyses were conducted on a Jeol 2100F microscope operated at an accelerating voltage of 200 kV. Atomic resolution STEM Z-contrast (HAADF) imaging was carried out on Cs-corrected FEI Titan 80-200 instrument.

Results and discussion

Assessment of Zr poisoning effect on TiB_2 inoculation

Fig. 1 shows the macrostructure of the as-solidified TP-1 samples of pure Al containing Al-5Ti-1B and Zr. It is seen that, a coarse and columnar grain structure was resulted when no grain refiner was added (Fig 1a). As expected, the addition of 0.1 wt% Al-5Ti-1B grain refiner resulted in a fine and fully equiaxed grain structure (Fig 1b). However, the grain structure was back to coarse and columnar when 580ppm (0.058 wt%) of Zr was added to the TiB_2-containing Al sample (Fig.1c), indicating that the effect of the grain refiner was reduced dramatically. The optical micrographs in Fig.2a show the detailed microstructure of the refined equiaxed grain structure, being about 175 μm in size, compared to the coarse columnar grain structure for Zr-containing sample (Fig.2b). Experiments also showed that, with further increasing the Zr concentration, the size (width) of the columnar grains became coarser, being 621±98 μm at 0.45wt% Zr, the maximum content of Zr added in this study, for instance.

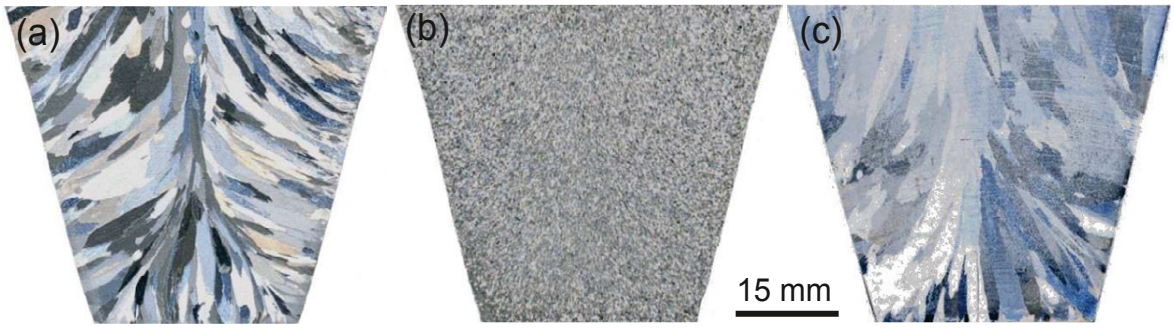

Fig.1 Macrographs showing grain structures on vertical-sections of Al solidified in TP-1 mould with (a) with no addition of, (b) with addition of 0.1wt.% Al-5Ti-1B grain refiner, and (c) with both additions of 0.1wt% such grain refiner and 0.048wt% of Zr.

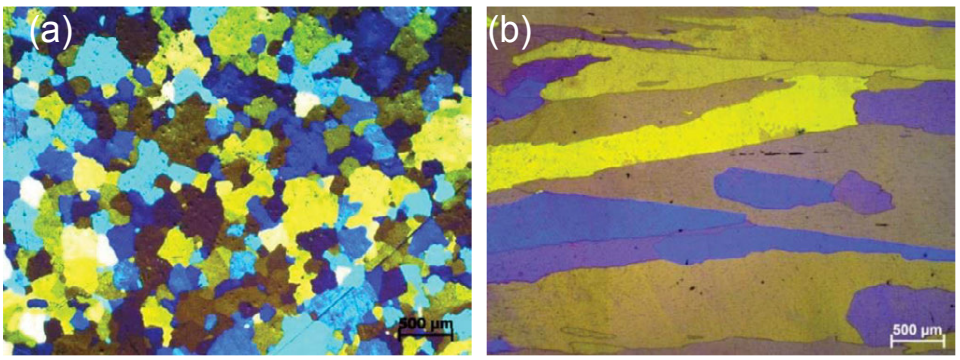

Fig.2 Optical micrographs showing microstructure of TP-1 sample of (a) Zr-free Al containing 0.1wt% Al-5Ti-1B grain refiner and (b) Al containing both 0.1wt% Al-5Ti-1B and 0.058 wt% Zr.

Morphology and size of TiB_2 particles

SEM and TEM of TiB_2 particles, which were collected by pressurised melt filtration of Al melt containing 0.2wt% Al-5Ti-1B grain refiner, was carried out focusing their {0001} termination surfaces. The TEM image in Fig.3a, with the incident electron beam being almost perpendicular to their <0001> direction, shows two such TiB_2 particles, indicating its faceted {0001} basal planes. Being viewed along its [0001] direction in Fig 3b, the boride particle displays a hexagonal shaped morphology. The average size of the boride particles collected from the melt was 0.2-1.5 µm. The selected area electron diffraction (SAED) pattern and the corresponding index in Fig.3 confirm that the TiB_2 particle was also faceted with its $\{10\bar{1}0\}$ prismatic planes. Heterogeneous nucleation of Al is believed to occur on the {0001} planes of the borides [9].

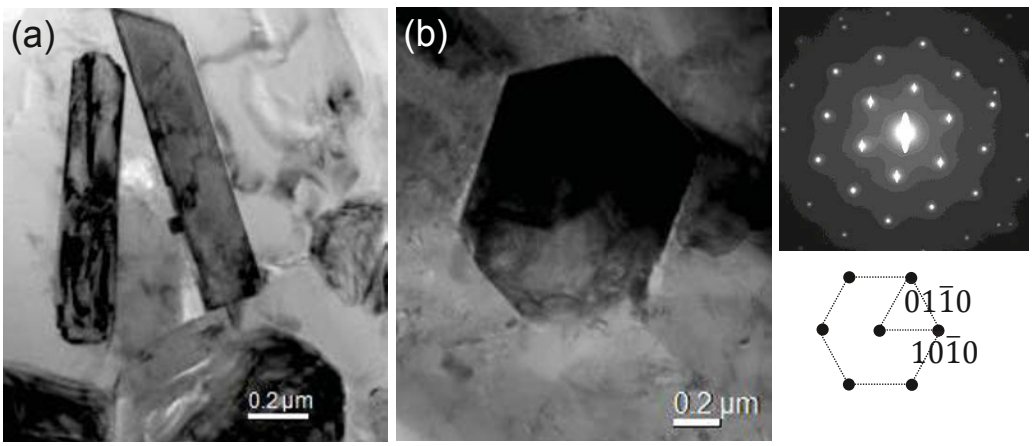

Fig.3 TEM bright field images showing morphology of TiB_2 particles being viewed along (a) $[11\bar{2}0]$ and [0001] direction respectively, with the [0001] SAED pattern and its indexing.

Adsorption of Zr atoms on TiB_2 surface

In order to understand the Zr poisoning mechanism, TiB_2 particles collected from both the Zr-free and Zr-containing Al samples have been extensively examined high resolution TEM/STEM, in particular the aberration-corrected STEM. Adsorption or segregation, if any, of alloying atoms or impurities at the interface is expected to be revealed by the atomic resolution STEM Z-contrast HAADF imaging which provides both structural and chemical information at the same time. It was shown that the morphology of the TiB_2 particles collected in Zr-free Al sample was the same as those collected in the Zr-containing sample. However, STEM Z-contrast images clearly show the adsorption of heavier (brighter) atoms on the (0001) surface of TiB_2, as shown in Fig.4. It is seen from Fig.4a that, being viewed in $[11\bar{2}0]_{TiB2}$ direction, an atomic mono-layer on the (0001) surface of TiB_2 shows a brighter contrast that the Ti columns of TiB_2 (please refer to the schematic in Fig.4a). High resolution EDS analysis confirmed the existence of Zr atoms at the surface, confirming that these atom columns with a brighter contrast in the monolayer contained Zr atoms. It is interesting to note that the adsorption of Zr atoms occurred only in one atomic layer. More significantly, by tilting the specimen for 30 degrees from $[11\bar{2}0]$ to $[10\bar{1}0]$, the brightness of the mono-layer shows a periodical variation, with two darker atomic columns between every brighter atomic column, as shown in Fig.4b and the corresponding schematic. The periodical variation in the brightness in the HAADF image indicated that the adsorption of Zr atoms in the layer was ordered in 2D. It was likely that the adsorbed Zr atoms were in the form of 2D intermetallic, similar to the adsorption of Ti atoms at TiB_2 surface to form 2D Al_3Ti layer when no Zr was present [7]. In this study, however, TEM/STEM failed to find any atom adsorption layer rather than Zr adsorption on the (0001) surface of TiB_2 particles. The disappearance of the previously existing Ti adsorption layer and substitution by Zr atoms must have caused a significant decrease in the potency of TiB_2 as heterogeneous nucleation substrates, resulting in the elimination of grain refinement observed in the Zr-containing Al samples.

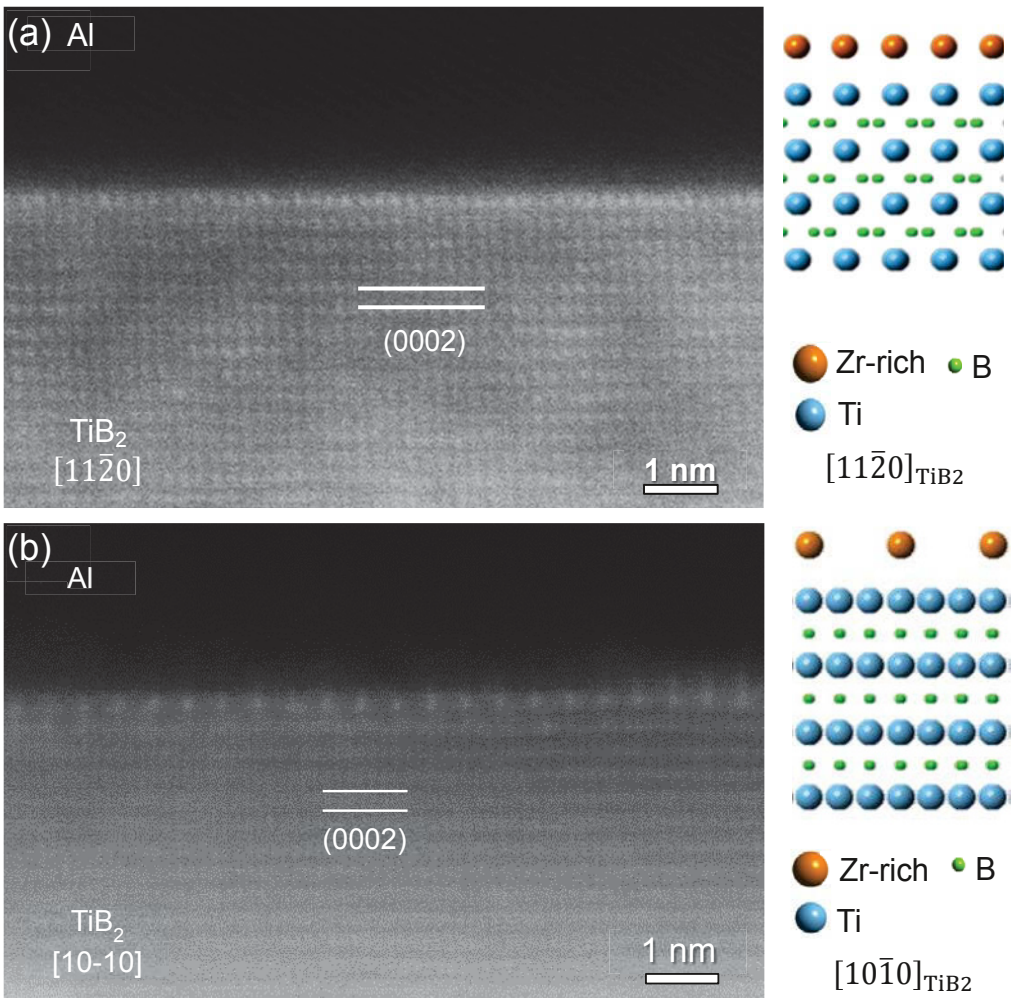

Fig. 4 High resolution STEM Z-contrast (HAADF) images showing adsorption of Zr atoms in a mono-layer on the (0002) surface of TiB_2 particle which is viewed in its (a) $[11\bar{2}0]$ and (b) $[10\bar{1}0]$ direction respectively. The corresponding schematics are showing the possible atomic arrangement at the interface for the two directions, respectively.

A thermodynamic analysis [9] was carried out and it was concluded that Zr would substitute for Ti in the boride to form a layer of ZrB_2 on the TiB_2 particle surface; this would be accompanied by a lattice parameter expansion of the boride. Bunn *et al* [15] also assumed that the dramatic decrease in potency of TiB_2 was attributed to the interference of both the Al_3Ti phase coating on TiB_2 and the TiB_2 particles. They claimed that Zr atoms would preferentially substituted in Al_3Ti coating phase, and that the most probable reason for the destruction of potency of TiB_2 was Zr substitution into the boride phase to form (Ti, Zr)B_2 phase, resulting in an increase of lattice misfit. Indeed, it was found in this work that Zr atoms interfered the 2D Ti adsorption layer as determined recently [7], rather than a 3D Al_3Ti phase. This resulted in the disappearance of the 2D Ti adsorption layer and a Zr adsorption monolayer was present instead. The dramatic decrease of the potency of TiB_2 is attributed to the disappearance of the 2D Ti adsorption layer, which caused by the preferential adsorption of Zr atoms. According to the results in this study, it is unlikely that (Ti, Zr)B_2 phase forms at the TiB_2 surface since Zr atoms were adsorbed in a monolayer only.

Orientation relationship between TiB_2 and Al

It was showed that the majority of TiB_2 particles were not in any defined orientation relationship (OR) with the adjacent Al, indicating that the majority of the boride particles did not nucleate Al during solidification. This is consistent with the conclusion derived from the free growth model that only 0.1–1% of the added TiB_2 particles are active for heterogeneous nucleation [16]. However, those few ones that did nucleate Al might be observed with sufficient TiB_2 particles being examined. An example of such TiB_2 particle in a Zr-free specimen was given previously [7], and the well-defined OR was $(0001)[11\bar{2}0]_{TiB2} \parallel (111)[0\bar{1}1]_{Al}$. The observation of the much expected OR in a solidified Al-TiB_2 sample provides the evidence to verify that TiB_2 particle nucleates Al on its basal planes.

With sufficient boride particles from the Zr-containing Al melt being examined, those that were in specific OR with Al were also observed in the study. Fig.5 shows the interface between such a TiB_2 particle and the adjacent Al, where the electron incident beam is parallel to both the $[11\bar{2}0]_{TiB2}$ and $[0\bar{1}1]_{Al}$ direction. However, unlike the OR found in the Zr-free sample, (0001) TiB_2 and (111) Al planes are not parallel to each other here in the Zr-containing Al sample. Measurement through both SAED and fast Fourier transformation (FFT) patterns confirmed a deviation of 8° between the crystal planes, as show in the SEAD pattern and the corresponding index in Fig.5. Therefore, the OR was $(0001)_{TiB2}$ 8 deg ~ $(111)_{Al}$; $[11\bar{2}0]_{TiB2} \parallel [0\bar{1}1]_{Al}$, with Al crystal being rotated 8 degrees against TiB_2. This deviation could be attributed to the Zr adsorption since the adjustment in OR is essential in order to accommodate the increase of the misfit at the interface and thus the interfacial energy.

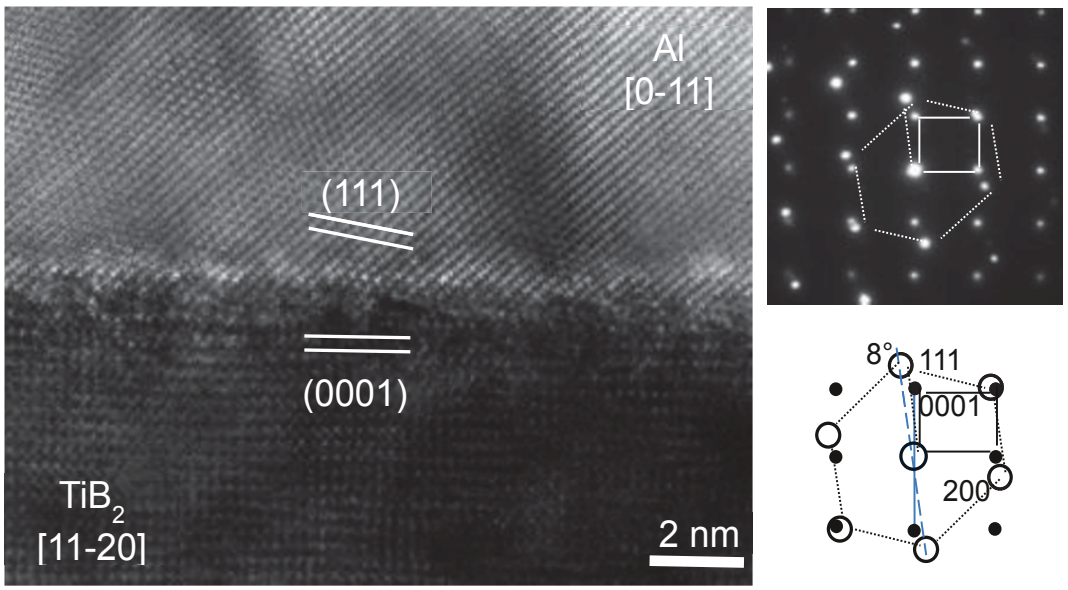

Fig. 5 High resolution TEM image showing the Al/TiB_2 interface when the incident electron beam is parallel to both $[11\bar{2}0]_{TiB2}$ and $[0\bar{1}1]_{Al}$ direction with SAED pattern taken from both TiB_2 particle and the adjacent Al and the indexed corresponding pattern.

Summary

For the first time, high resolution TEM/STEM revealed that Zr atoms were adsorbed on the {0001} surfaces of TiB_2 particles, resulting in disappearance of the previously existing Ti adsorption layer. The substitution of the Ti-rich layer by the adsorbed Zr atoms on the TiB_2 surfaces was believed to cause a decrease in the potency of the TiB_2 particles for heterogeneous nucleation of Al. This in turn resulted in the dramatic destruction of grain refinement efficiency of the TiB_2 grain refiner. In

addition, the change in the potency of the TiB_2 substrates resulted in a deviation in the orientation relationship between TiB_2 and Al from the well-defined one found in Zr-free Al samples, in order to accommodate the increase of the misfit at the interface and thus the interfacial energy. For instance, the OR observed in the Zr-containing Al sample was $(0001)_{TiB2}$ 8 deg. ~ $(111)_{Al}$; $[11\bar{2}0]_{TiB2} \parallel [0\bar{1}1]_{Al}$, with Al being rotated 8° relative to TiB_2.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Study of Manganese Dissolution in Aluminum Melts

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Keywords: Manganese briquette, Dissolution rate, Intermetallic phases, Heat transportation, Freeze-on layer.

Abstract

Problems with the yield during Mn alloy additions and the occurrence of undissolved Mn rich phases in as cast material containing Ti is the background to the present study of the dissolution of Mn compacts in aluminum melts.
The dissolution rate of Mn in liquid aluminum, have been studied in pure Al and in Al-0.12%Ti melts. It was found that Ti additions to the melt decreased the dissolution rate of Mn compacts. It was also shown that the intermediate phases formed at the interface between Mn and liquid aluminum was different after small Ti additions. Moreover, an undissolved Mn briquette, found after casting in a furnace, was examined and the conditions for this to happen have been discussed. The discussion was based on calculations of heat balances during the initial dissolution steps and studies of the transformations occurring within the briquette.

Introduction

There are different methods of adding alloying elements into the aluminum melt in a cast house. The cold addition practice of the compacted additives in the shape of a briquette or compacts is common. The dissolution process of the briquettes occurs in three steps of incubation, exothermic reaction and dispersion of additives [1]. During the exothermic reaction, intermetallic compounds are formed between the aluminum component of the compact and manganese particles and subsequently cause the swell of the compact forming a porous structure, which allows the aluminum melt to penetrate the solid Mn by a capillary action. Subsequently, the briquette is disintegrated into discrete particles and disperses in the melt to continue the dissolution process. [1,2,3,4].
A dissolution performance is favorable when additives dissolve within a short time otherwise it results in an inappropriate dissolution [2].
Poor dissolution behavior of manganese in aluminum alloys, as alloys A and B in Table I, is a common problem in cast houses producing alloys of AA3003 type with relatively high Mn content and it is normally shown as a limited yield of the added Mn. In alloys containing Ti it also has been found that Mn rich phases can occur in rolled sheets of the material, and as this may be coupled to poor dissolution, it is of interest to study the effect of Ti on the Mn dissolution rate. It is also of interest in this context to study the manganese dissolution mechanism in a more general approach.

Table I. Chemial composition of studied alloys

Alloy	Si	Fe	Cu	Mn	Ti	Mg
A	0.08	0.22	0.35	1.5	0.15	0.008
B	0.1	0.2	0.85	1.7	0.02	0.02
C	0.06	0.08	0.01		0.12	0.01
D	0.05	0.07	0.21		0.12	0.01

Experimental Procedure

Three different experimental investigations were performed and are outlined in the following, additional to that a post mortem analyses of particles in flat rolled product was performed.

Analyses of Dissolution Rates

To study the dissolution process of manganese at different temperatures in molten aluminum, two series of sampling tests with pure Al at two different temperatures 770 °C and 730 °C respectively, were carried out. Also the effect of Ti additions was studied in a series of tests with Al-0.12%Ti at 770 °C. For the first two series of tests, 500 g of pure Al and for the final one 500 g of Al-0.12%Ti, alloy C in Table I, were put in a crucible and held in a furnace at the target temperature, which was monitored by a thermocouple (K type). When the melt reached to the desired temperature, the crucible was taken out and an amount of Mn tablet (90%Mn10%Al) was added to give a concentration of 1.5% Mn, and then the crucible was returned to the furnace. Subsequently, samples were cast in cylindrical molds of 25 mm diameter with an insulated periphery after 1, 2, 3, 5, 7 and 9 minutes, respectively. Stirring was conducted for 5 seconds before each sampling test. During the sampling tests, the melt temperatures fell and the measured temperatures of the last samples were 730 °C and 700 °C for experiments with furnace temperatures of 770 °C and 730 °C respectively, which gave average temperatures of 750 °C and 715 °C for the two series of tests. Also a similar procedure was repeated for Al-0.12%Ti at an average temperature of 750 °C. All collected samples were thereafter analyzed using OES technique to obtain the Mn concentrations. Some samples were also studied in LOM, SEM and analyzed with EDX.

Analyses of Dissolution Mechanism

To follow the dissolution mechanism in detail a 50 gr melt of both pure Al and an alloy with a composition of 0.12% Ti and 0.21% Cu, alloy D in Table I, were melted in cylindrical crucibles with a diameter of 35 mm giving a melt depth of 20 mm in a furnace with a temperature of 760 °C. When the melt temperature was stabilized, pieces of pure Mn flakes were added to the melt, and after 8 minutes the crucibles were taken out of the furnace and cooled at ambient temperatures. Then, the samples were cut to

find the positions where the Mn flakes were located and metallographically prepared and etched for about 15 seconds in a solution of 1% NaOH at 60 °C to reveal the phases formed around the flakes.

Analyses of Undissolved Mn Tablets

A possibility to examine the reaction processes between aluminum and manganese in a briquette at an industrial scale occurred when an undissolved briquette (90%Mn10% Al) was raked out of a cast house tilting furnace after a casting. The furnace was charged with alloy B in Table I. The briquette was cut off through the center and studied by both macroscopic and microscopic methods.

Results

Effect of Composition and Temperature on Dissolution Rate

The results from three series of dissolution tests with different temperatures and compositions are shown in figure 1. One trial series was made for each of the parameters, corresponding to respective curve, and each measurement point was the average of three OES sparks on each sample. As the deviations between the sparks was less than 0.1%, this can be regarded as the accuracy of the average composition shown as point in the diagram. As seen in the two series of tests with pure Al, the experiments at 750 °C clearly give a higher dissolution rate than the tests at a temperature of 715 °C, which quantitatively shows the effect of temperature on dissolution rate. It can also be seen in the last made sample (9 minutes), that the Mn content reached a level of 1.85%, which is below the saturation limit at this temperature. In the microstructure observation of the 9 minute sample at 750 °C, figure 2, some large particles were found and analyzed with EDX which showed a composition corresponding to Al_6Mn. Furthermore, the SEM investigation also revealed that some undissolved Mn particles were attached to oxide films. In the sample made at 9 minutes at 715 °C, a few similar Al_6Mn particles with smaller sizes and to a smaller extent could be seen. In the sample from 750 °C larger amounts of eutectic structure were observed, compared to the sample from 715 °C. All the microstructure observations support the results of the diagram showing large variations in dissolution rate. It is important to notice that during sampling the melt level decreases and at the end less than half of the original melt remains. This means that the melt concentration will be higher than the 1.5%, which should have been the maximum concentration for a constant melt volume. In dissolution tests performed with larger melt volumes, as described in the ASTM TP-2 standard, the curve shape normally is parabolic with an asymptotic approach towards the targeted concentration as was obtained in [3,7]. In the present case the shape of the curves indicate that it is only the pure Al at 750 °C that is approaching an equilibrium with a parabolic shape at the end, and the other curves are at an earlier stage of dissolution due to slower dissolution rate. This experimental approach simulates what happens if the dissolution rate is to slow and undissolved Mn remains in the furnace.
The dissolution rate for the melt containing 0,12% Ti, alloy C, is significantly decreased compared to the pure Al melt at the same temperature, 750 °C. It can therefore be concluded that the Mn dissolution rate is substantially lowered by the addition of the 0.12% Ti.

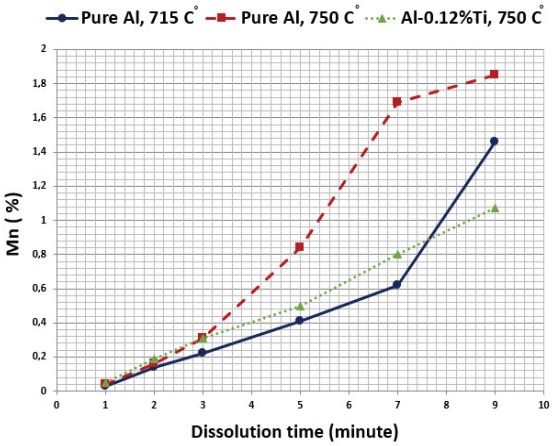


Figure 1. Mn content in the melt during dissolution as a function of time after addition of a Mn table.

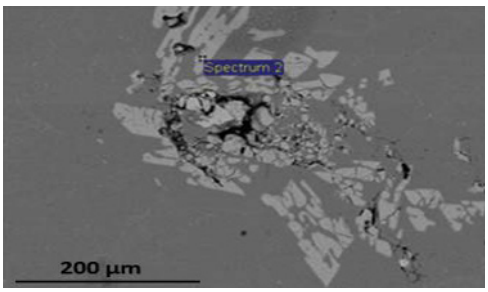


Figure 2. SEM image of precipitated particle in pure Al from sample taken after 9 minutes at 750 °C. The corresponding EDX spectrum showed 85%Al and 15% Mn in atomic percent.

Dissolution Mechanism of Pure Mn Flakes

Figure 3, shows intermediate phases formed around the Mn flakes after reactions in pure Al for 8 minutes. Three different phases are clearly visible at the interface between Al liquid and Mn. EDX spectra show that the Al/Mn atomic ratio of the inner phase, close to the light Mn, the middle phase and the outer phase are about 50/50, 67/33 and 75/25, respectively.
In figure 4 the corresponding structure from the dissolution of Mn flakes in alloy D containing 0.12% Ti and 0.21% Cu is shown. Here only two thin layers have grown between the liquid Al and the Mn flake. The EDX analysis shows that the inner phase, close to the light Mn phase, has an Al/Mn atomic ratio of 67/33, while the ratio of the outer phase is 75/25.
Comparison of the intermediate phases formed in alloy D with pure Al shows that the innermost phase with the Al/Mn ratio of 50/50 was not formed in this alloy. The total thickness of the intermetallic layers is on average 30-50 µm in pure Al, compared to 20 µm in the alloy containing Ti. These observations indicate that the kinetics of the formation of intermetallic phases in the Al-Mn system has been affected by the addition of Ti.
Further studies of Mn dissolution in the Ti containing alloy D, figure 5a, shows that some phase is growing out into the melt from the outer intermetallic compound. The EDX spectra show that this phase has an Al/Mn atomic ratio of 82/18, but it also

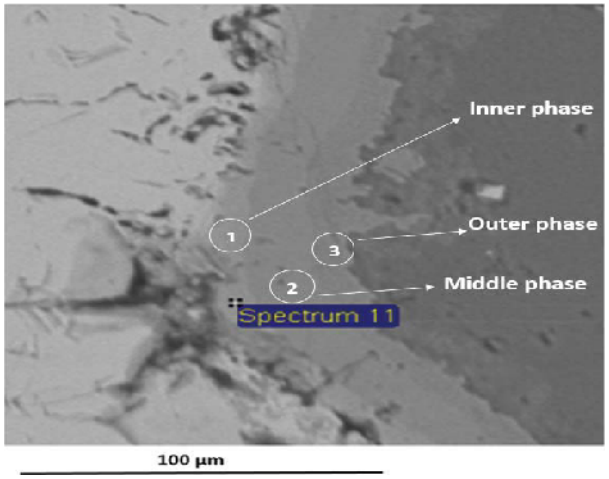


Figure 3. SEM picture of three layers formed during Mn flake dissolution in pure Al melt. Light area at the left side is Mn flake.

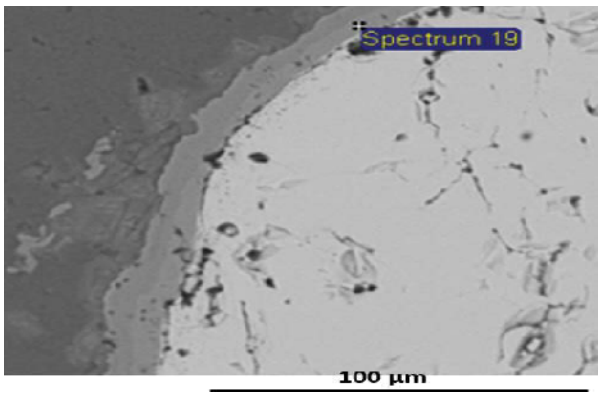


Figure 4. SEM picture of two layers formed during Mn flake dissolution in Al-0.12%Ti-0.21%Cu, alloy D. Light area at the right side is Mn flake.

contains 0.5% Fe. Away from the interface smaller particles can be observed inside the Al matrix, figure 5b. The atomic Al/Mn ratio is also here about 82/18 but these particles contain 0.5% Ti.

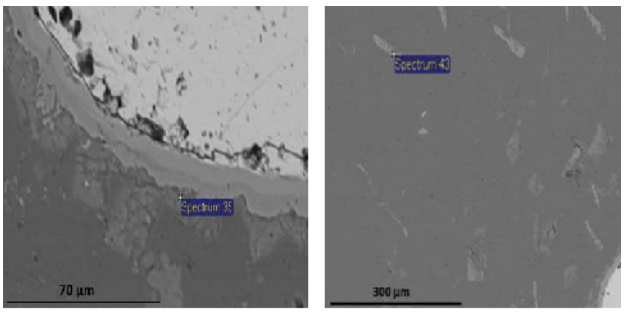


Figure 5. SEM pictures of Mn dissolution in Al-0.12%Ti-0.21%Cu, alloy D: (a) phases close to outer phase layer. Light area at the top is Mn flake, (b) precipitated particles in the Al matrix.

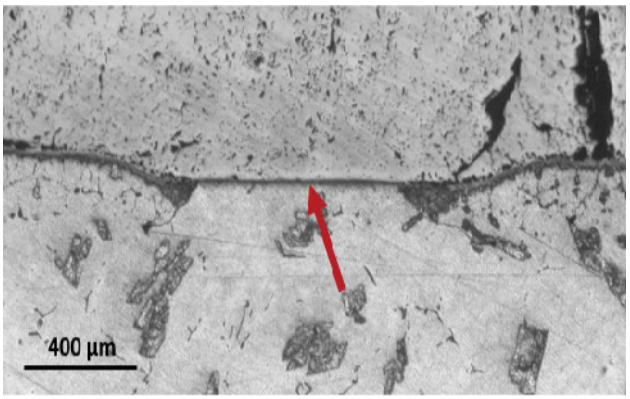
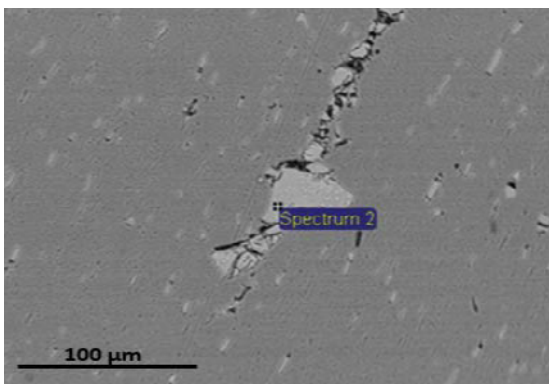


Figure 6. LOM picture of reaction surface during dissolution of alloy D at spot where oxide layer has eliminated reactions. Top Mn side and bottom Al side. Flat part of interface is covered by oxide showing position of original surface. Arrow shows oxide film.

At some spots along the Mn flake surfaces oxide remain after the dissolution experiments, thus revealing the position of the original surface. Such a place is shown in figure 6, where the flat part is the original surface, which is covered by oxide. At the sides of the spot the reaction has proceeded into the Mn (upwards in figure) and it can be estimated that about 90 µm of Mn flake has been dissolved during the 8 minutes.

Analysis of Rolled Sheet

A post mortem study of a rolled sheet of alloy A was performed, and in the center of the sheet large particles of the type shown in figure 7 were revealed. These particles were analyzed by EDX and the content in weight percent was determined as following: 72% Al, 20.3% Mn, 2.8% Ti and 1.7 % Cu, with the atomic ratio Al/Mn of 88/12. Smaller particles, shown as light dots in figure 7, were found distributed in the matrix elongated in the rolling direction. EDX spectra of those particles showed the atomic ratio of 83/17 for Al/(Mn,Fe), corresponding to the ratio for the $Al_6(Mn,Fe)$ phase , which is normally precipitated in this type of alloy, but the larger particles are clearly of another phase.



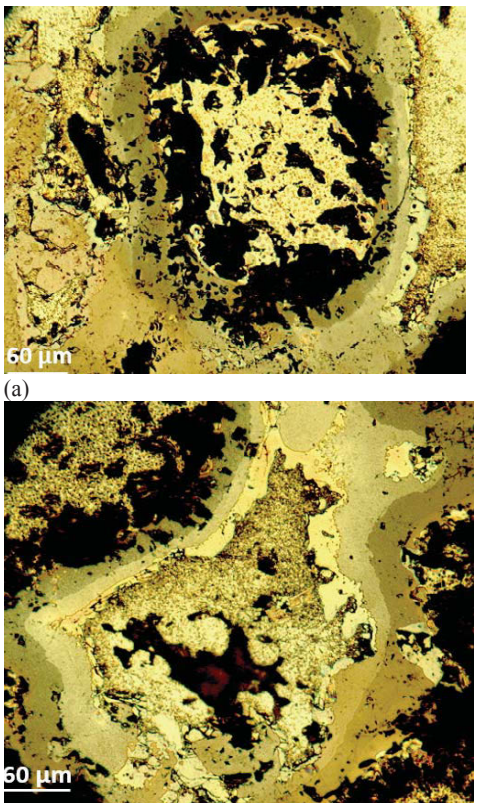
Figur 7. SEM image of large particle in the center of rolled sheet of alloy A together with smaller particles, which were scattered in the rolling direction of the matrix.

Undissolved Mn Tablet after Casting

Three undissolved Mn briquettes containing 10% Al, in shapes of tablets with diameter and height of 80 mm and 30 mm, respectively have been sintered together and found and raked out of a holding furnace after casting. The outer appearance is shown

(a) (b)

Figure 8: (a) Three Mn briquettes were sintered together and found undissolved in the furnace. (b) Transverse cut through the undissolved Mn briquettes. The arrows show the places where aluminum has penetrated into the Mn block.



(a)

(b)

Figure 9. Microstructure from the inside of the Mn briquette at the reaction sites: (a) partially dissolved Mn surrounded by intermediate phases (b) aluminum isolated by intermediate phases.

in figure 8a. The surface of the compact reveals many holes, which penetrate through the solid shell into the briquette. A macro view of a transverse cut through the briquette, figure 8b, shows how these holes and channels proceed into the material from the lower right corner of the cut section. One can see that the aluminum melt has penetrated into the briquette in different places, e.g. in the left bottom corner where it continued as a bent curve pattern and also as a band of thin streaks at the right side. Figures 9a and b show the microstructure from inside of the compact at the reaction sites. It can be seen that intermediate phases have grown between Mn and aluminum. Figure 9a shows an Mn particle isolated by intermediate phases, which is typical for what could be seen throughout the structure. It can also be seen on the right side that some aluminum has remained unreacted, but in the lower side a reaction between Mn and aluminum has occurred and formed intermediate phases filling out a large area. In some areas also aluminium has been isolated by the intermediate phases, which form an irregular circle of interconnecting phases, as in figure 9b.

Briquette Dissolution Model

Once the briquette is immersed into the melt, the low temperature of the briquette chills down the surrounding melt and therefore can create a freeze-on layer around the briquette [1, 7]. Formation of this solid shell is detrimental to the dissolution performance, as it aids the briquette to retain its rigidity within the melt. Thus, causing a delay of the start of the dissolution process, as firstly the solid shell has to be remelted [7]. The heat transport into the briquette from the melt occurs through conduction, and the relatively low thermal conductivity of Mn results in poor heat transportation and thus the following boundary conditions can be assumed; at start the temperature at the interface of the briquette is equal to the melt temperature, and the inside temperature of the briquette is equal to room temperature. By using these boundary conditions and balancing the heat flux into the solid with the latent heat and heat capacity of a growing freeze on layer, the general law of thermal conduction gives the following equation for computation of the shell thickness as a function of temperature and time [5].

$$y_l(t) = \frac{2}{\sqrt{\pi}} \frac{T_i - T_o}{\rho_1 \left(-\Delta H + C_p^1 \Delta T\right)} \sqrt{k \ \rho_2 C_p^2} \ \sqrt{t} \qquad (1)$$

Where T_i and T_o are the melt temperature and briquette temperature at the time zero, respectively, ΔH is the latent heat of aluminum (- 398 kJ/kg) [5], ρ_1 and ρ_2 are densities of the aluminum and briquette respectively, k is thermal conductivity of the briquette, C_p^1 and C_p^2 the heat capacities of the melt and briquette, respectively. The density of the aluminum melt was assumed to be 2385 kg m^{-3} [6] and the heat capacity 1.25 kJ/kg K at 660 °C [5]. The density of manganese was taken to be 7400 kgm^{-3}, the heat capacity 0,486 kJ/kg K and the thermal conductivity 7.8 J/m s K [6]. The primary temperature of the briquette was assumed to be 24 Cº. Thereafter, required times to grow the layers into three different possible thicknesses of 5 mm, 7 mm and 10 mm, respectively at different melt temperatures were calculated using equation (1) and the results are displayed in figure 10. It can be seen that a thickness of 6-7 mm is reached after less than 5 seconds and that at higher temperatures the growth rate of the solid shell is lowered. However, the calculation is only valid for the melt temperature of 660 °C, equal to the melting point. At the two higher temperatures no consideration have been taken to the heat flux from the melt, which should be subtracted from the flux going into the briquette to obtain the right

balance. At 800 °C the temperature away from the surface is 140 °C above the melting point and assuming a boundary layer of 5 mm the flux can be estimated by the expression:

$$\frac{dq}{dt} = Ak\,\frac{dT}{dy} \quad (2)$$

Using that estimation it is found that the flux is of the same order of magnitude as the flux into the briquette, indicating that alloy additions at this temperature and higher can be done without forming a freeze-on layer. A more accurate calculation should be done by numerical methods to handle the transients of the heat flux at the surface to obtain better data, but this simple estimation shows that a freeze-on layer may form under realistic conditions, and temperatures below 800 °C.

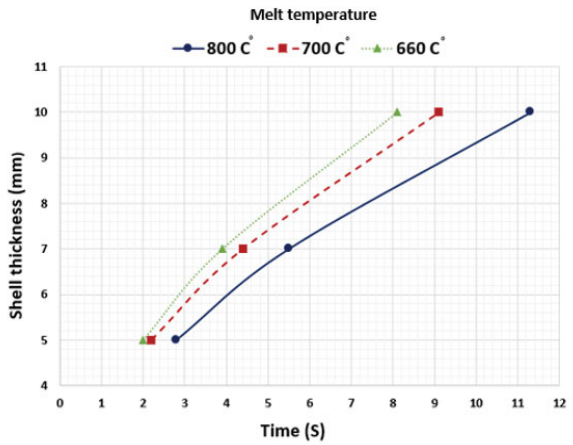


Figure 10. Solid shell growth at different temperatures.

Discussion

Dissolution Process of Mn in Liquid Aluminum

The obtained results from the dissolution rate investigation, figure 1, reveal that it is only at the end of the test at 750 C°, with pure Al, that the composition approaches full dissolution as the curve is levelling out. The shapes of the curves is a result of that melt is removed and the concentration is increased both due to decreased amount of melt and increased Mn content from the dissolution. A constant amount of melt would have given parabolic curves approching the nominal composition (1.5%). In the top curve, 750 C°, with pure Al, a composition above the nominal is reached and during cool down the melt becomes locally supersaturated, and Al_6Mn particles are precipitated as a primary phase and not only in the eutectic reaction, figure 2. The measured composition is below the saturation limit and also slightly below the eutectic composition, which means that local supersaturation must have occurred, and it was also observed undissolved floating Mn particles in the sample. This has similarities with the large particles in the sheet material, figure 7, which also has been precipitated primary in contrast to the smaller eutectic particles in the matrix. In that case it was also slightly different composition of the large particles compared to the eutectic phase, showing that it is a primary precipitation, but the melt must have been locally supersaturated with regard to Mn for this to happen.

Figure 1, reveals in a quantitative way strong effects on the dissolution rate both from temperature and small Ti additions. The effect of temperature is well known and a result of higher driving force and diffusion rate at higher temperature, which has been discussed before [2,7]. However, the observation of that small Ti additions lower the dissolution rate of Mn is not known before. The dissolution of alloying elements takes place through the formation of several intermetallic compounds. In the present work, different intermetallic phases at the interface between the Mn and aluminum liquid have been found. In figure 6, on both sides of the flat oxide covered interface, it can be seen how the layer with intermetallic phases have been moving into the Mn. It can be concluded that it is not the thickness of the layer of intermetallic phases that increase but it is rather the layer in itself that moves into the Mn during the dissolution. The dissolution rate is controlled by the diffusion through these phases. The results indicate that the intermetallic phases have preferentially grown towards the liquid Al while the βMn-phase grows into αMn-phase, figure 3 and 4, which is different from what recently has been observed for dissolution of Fe in aluminum [7,8], showing strong growth of intermetallic phase into the solid Fe.

The results from the dissolution process show that the addition of Ti influences the kinetics. In the test with pure Al three distinct phases could be observed at the interface, figure 3, while after a small Ti addition only two phases could be revealed, figure 4. The EDX data do not exactly fit to compositions in the phase diagram [9], but by assuming a slight overestimation of the Al content in the measurements the Al/Mn ratios 50/50, 67/33 and 75/25 corresponds to the phases βMn-phase, γ2 (Al_8Mn_5) and $Al_{11}Mn_4$, respectively. In the pure Al melt all three phases appeared and the total thickness of the layer was 30-50 μm, while after Ti addition only the two outer phases, i.e. γ2 (Al_8Mn_5) and $Al_{11}Mn_4$ were observed, and the thickness was about 20 μm.

There are also phases found growing out into the melt and floating in the melt, figure 5, and their Al/Mn ratio of 82/18 indicate that it is the μ-phase. Most likely these phases have been formed during cool down of the samples. This atomic ratio was also measured in the phases found in rolled sheet, figure 7, also containing Ti and Cu, and as the floating particles in the dissolution tests, figure 5b, contained some Ti it is likely that the μ-phase is stabilized by Ti additions. The Ti additions might therefore increase the risk for forming such primary phases during casting.

The reason why the Ti additions decrease the dissolution rate and make the Mn-rich βMn-phase to disappear is unclear. According to figure 6 the dissolution process proceeds with the intermediate phases moving into the Mn. The likely process is thus that first when Mn comes in contact with the liquid the intermediate phases forms and grows to thicknesses giving a steady state flux through the total layer. The next stage is that a constant flow through the layers dissolves the Mn while the intermediate phases move with the dissolution front. If one of the phases disappears, as in this case at Ti additions, no change in the total flux should occur, as the flux through each phase must be the same as through the whole layer at steady state. The diffusion problem is very complicated and both thermodynamic and diffusion data are lacking for a meaningful analysis, but a ternary phase diagram, in [10] at 1000 °C, shows that Ti destabilizes the βMn-phase, which is supported by the disappearance of that phase here.

A possible explanation to the Ti effect on the dissolution rate can possibly be attributed to a very small value of the titanium diffusion coefficient in liquid aluminum [7]. At steady state aluminium have to be transported through the interface layers at the same speed as Mn moving in the other direction and if Ti interact with the aluminium flux at the liquid side the total flux may be lowered.

Undissolved Briquettes

Theoretically, the dissolution process is described by the following sequence. When the briquette reaches the melting temperature of the aluminum, the aluminum ingredients melt and start to react with the Mn particles inside the briquette. Expansion occurs by forming voids within the compact allowing more liquid aluminum to penetrate and contribute to the collapsing of the compact. Subsequently, the disintegrated particles are exposed to the main melt where they can be dissolved easily. Therefore, a quick disintegration and liquid penetration will facilitates the dissolution performance.

The investigation of the undissolved briquettes gives an idea from practical point of view of what can happen and what can cause the limited yield of the dissolution process, which is sometimes observed. Analysis of the undissolved briquettes (90%Mn10%Al), figures 8b, shows that aluminum succeeds to peneterate the biquette only in limited areas, and the briquette has not been broken into smaller pieces. If the Mn particles are not sufficiently disintegrated within a short time, a reaction between the Mn and aluminum results in the formation of a number of intermediate phases at the interfaces instead, figures 9a and b. Ultimately, only solid phases will remain and hold together the Mn particles, as can be seen in the lower part of figure 9a. The growing intermediate phases can also isolate aluminum areas and thus prevent the peneteration of the aluminum into the regions between the Mn particles, as seen in figure 9b. In the observed case, collapse of the briqutte has not occurred fast enough, i.e. before the intermediate phases has grown to such an extent that not enough liquid remains to split the briquette.

Depending on the temperature when the briquettes are immersed into the melt, more or less freeze-on layer will form around them [7], and if the temperature is too low such layers may be too stable to remelt fast enough to expose the briquette to the aluminum melt. In such cases the internal sintering, by growth of intermediate phases, can proceed and hold the tablets together. Figure 10, from heat balance calculations shows that such an scenario is possible and that the melt temperature strongly influences the occurance of the freeze-on layers. In the case studied here three briquettes have been sintered together, which will give a stronger freezing effect than if they had been better separated.

Other factors that may cause an inappropriate dissolution is uneven distribution of the aluminium ingredients inside the briquette giving areas with a content of less than 10% Al. Therefore, insufficient penetration can occur in these areas with a small amount of aluminum and the reaction can not occur properly in the whole briquette. Conclusively, briquettes with a higher aluminum may be more safe to give control over the dissolution process [2].

Conclusions

From studies of the dissolution process of Mn in liquid Al the following conclusions can be made:

-The dissolution rate of Mn decreased after addition of Ti.

-The formation of intermetallic compounds at the interface between the liquid Al and Mn was different after addition of a small amount of Ti.

-Mn should be added before Ti during alloying.

From studies of the heat balance during additions of an Mn briquette the following conclusions can be made:

-A freeze-on layer can form around the briquette if the melt temperature is below 800 °C.

-A too low fraction of Al in the briquette can result in undissolved Mn.

-It is important that the single briquettes are separated from each other at addition to the melt.

Acknowledgments

The cast-house employees at Kubikenborg Aluminium AB are acknowledged for providing materials, facilitating experimental tests, and hosting fruitful discussions.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ULTRASONIC GRAIN REFINING OF CONTINUOUS CAST ALUMINUM: MICROSTRUCTURE AND PROPERTIES

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Keywords: Ultrasonics, Grain refining, Aluminum, Cavitation, Acoustic streaming

Abstract

Southwire Company located in Carrollton, Georgia U.S.A. has developed ultrasonic grain refining technology to be used on the Southwire Continuous Rod (SCR) process. Grain refining of aluminum occurs when the crystal size of the solid phases is reduced using chemical, physical, or mechanical means. The ultrasonic grain refining process uses power ultrasound to give rise to non-linear effects such as cavitation, acoustic streaming, and radiation pressure. These non-linear effects are then used to nucleate new grains and break up dendrites during the solidification process of an alloy. The first section of this paper discusses the mechanism of grain formation and refinement using acoustic irradiation. Then the results from the lab trials will be presented. Following this, the benefits of ultrasonic grain refining will be reviewed. Finally the test data from initial scale-up of the ultrasonic grain refining technology at the Southwire Kentucky Plant will be summarized and discussed.

Introduction

The objective of the Southwire ultrasonic grain refining process is to grain refine aluminum and its alloys without the use of external particulates that are generally used in the conventional grain refining process.

The Southwire process of ultrasonic grain refining continues to be optimized for the SCR wheel and belt continuous cast system. The SCR system is a wheel and belt casting system, which consists of a water cooled copper casting wheel and a flexible steel band. The casting wheel has a groove in the outer periphery and the flexible steel band goes approximately half way around the casting ring to enclose the casting groove. Therefore the casting groove in the wheel and the flexible steel band that encloses the casting groove form the mold cavity. A tundish, pouring spout and a metering device, mounted on the casting machine, deliver molten aluminum into the casting groove as the wheel rotates creating a cast bar. The cast bar is then guided into a series of roll stands that hot rolls the cast bar into rod. The rod is then quenched and coiled into rod. Throughout the process, extensive data has been collected on the electrical conductivity and mechanical properties of rod made using the ultrasonic grain refiner on the SCR process. Data was then obtained from the aluminum wire that was drawn for various electrical applications.

Grain refining is a solidification processing step by which the crystal size of the solid phase is reduced by chemical, physical or mechanical means [1]. When casting aluminum, it is necessary to control the dendritic grain growth within the cast. A coarse dendritic grain structure is more susceptible to cracking. Grain refining results in a finer grain structure that impacts the castability of the cast metal and minimizes hot tearing during the continuous casting process. The most conventional method of grain refining in the aluminum industry is using a combination of titanium and boron (TiBor). TiBor is a chemical nucleant that results in the formation of an equiaxed grain structure in the solidified aluminum [2]. There are large costs associated with this method. In addition these additives can negatively impact the mechanical and electrical properties as well as the machinability of the cast metal.

Mechanisms of Grain Formation and Grain Refining

Grain refining of aluminum is important to the cast metal's physical properties and formability. The process in which aluminum is grain refined can be broken into two stages: the nucleation stage and the growth stage. The nucleation stage occurs when the molten aluminum is undercooled below the liquidus temperature thus forming solid particles that can grow within the liquid called nuclei. The growth stage occurs when the nuclei begin to develop into dendrites by solidifying the molten aluminum around it into a unidirectional pattern.

The pattern in which the aluminum solidifies is very important. When the aluminum solidifies in a unidirectional pattern, the dendrites form a columnar grain structure which is more susceptible to cracking and causes a non-uniform distribution of the secondary phases. Figure 1 below is an example of a columnar grain structure of pure aluminum (1350 EC) that was cast with no grain refining additions.

Figure 1. 1350 EC cast bar sample cast with no grain refiner.

When the aluminum solidifies with a uniform growth of dendrites in all directions; an equiaxed grain structure is formed within the casting. Castings with an equiaxed grain structure can be cast continuously and have excellent formability.

During the solidification of commercial metals and alloys, heterogeneous nucleation occurs. Solid particles usually exist in the melt. Any solid particles having a wetting angle, θ, smaller than 180° can serve as nucleation site of the solid phase to be formed and thus substantially reduce the size of the embryos that become

stable nuclei. Figure 2 illustrates the use of a foreign solid substrate for the nucleation of a solid phase from a melt. A cluster, or an embryo, of much small size can have a radius larger than the critical size required for growth. In order to achieve an equiaxed grain structure there has to be a substantial formation of these nuclei within the melt.

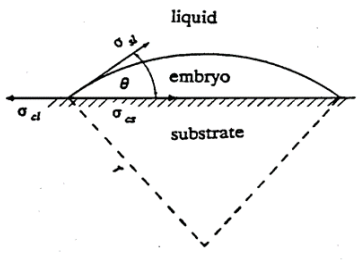


Figure 2. Schematic illustration of heterogeneous nucleation [2].

The nuclei formed during solidification grow into solid grains known as dendrites. The dendrites can also be broken into multiple small fragments by stirring or by other methods such as electromagnetic fields and ultrasonic irradiation. The dendritic fragments thus formed can grow into new grains and result in the formation of small grains; thus creating an equiaxed grain structure [2].

For aluminum alloys, the most widely used grain refiners are master alloys containing Titanium boride (TiBor). However it is still very challenging to obtain an equiaxed grain structure on pure aluminum even with additions of grain refiners as shown in Figure 3 below.

Figure 3. 1350 EC cast bar sample cast using TiBor grain refiners.

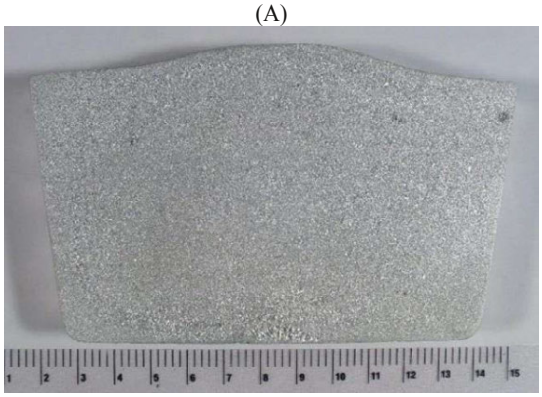
When using the ultrasonic grain refining process, no additives are needed to create nuclei within the melt. Once the nuclei begin to solidify into dendritic form, the dendrites are broken into smaller fragments by ultrasonic irradiation. These smaller fragments can then form new smaller grains thus creating an equiaxed grain structure.

Experimental Data of Ultrasonic Grain Refining: Lab Scale

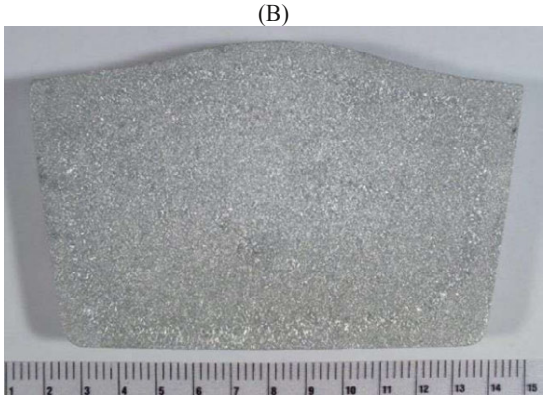
Ultrasonic grain refinement of molten metals has been evaluated and studied since the early 1950's. Through these studies, it was discovered that high intensity ultrasonic vibrations during solidification create sites for heterogeneous nucleation to occur [3, 4]. This in turn produces a finer grain structure. However, large production quantities could not be produced using ultrasonics. The work carried out by Southwire has demonstrated that industrial scale production of fine grained aluminum rod can be produced using the SCR process.

Before proceeding with a discussion of the benefits of the ultrasonic grain refining process at an industrial level; it is appropriate to summarize the work that was carried out in the lab. In order to replicate the SCR process on a lab scale, a section of a copper continuous casting wheel was used as a mold. Pure aluminum directly from a smelter was used in all the experiments to ensure that no residual chemical grain refiners were present. The metal was heated to 700°C and then poured into the copper mold as ultrasonic vibrations were applied to the melt. Results show that each aluminum alloy that was cast using the ultrasonic process had an equiaxed grain structure as shown in Figure 4 A-D.

(A)



(B)



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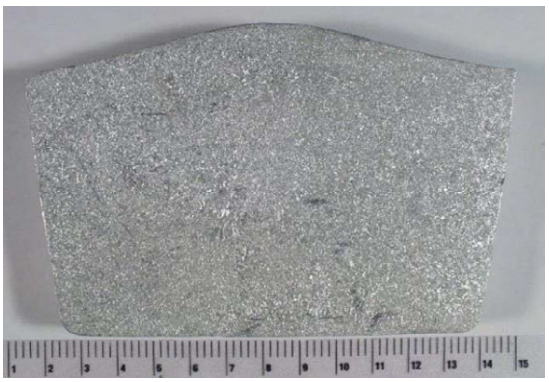

(D)

Figure 4. Aluminum alloy samples cast using ultrasonics: (A) 1350, (B) 5154, (C) 6201, (D) AlZr

Figure 4(A) shows the equiaxed grains formed in AA 1350 alloy, which is a commercially pure aluminum metal. TiBor grain refining has been not very effective, as the grain size shown in Figure 3 is still columnar and very large, similar to that shown in Figure 2.Ultrasonic grain refining significantly eliminates columnar grains and yields equiaxed grains.

Figure 4(B) shows grain size in AA5154 alloy, which is an aluminum magnesium based cast alloy. The equiaxed grains formed in this alloy are much smaller than those in the commercially pure AA 1350 alloy. This is in agreement with experimental observation that grain refining alloys is comparatively then grain refining pure aluminum metal.

Extremely small equiaxed grains are also produced in AA 6201 alloy. The grain size in this alloy is larger than that in AA5152 but much smaller than that in A1350. This experimental results suggests that grain size under the influence of acoustic vibration is still inversely proportional to the solidification interval of the alloy, $\Delta T=T_L-T_S$, where T_L is the liquidus temperature of the alloy and T_S is the solidus temperature of the alloy [4, 5].

The AlZr alloy is an aluminum cable alloy, in which Zirconium (Zr) is added into the commercially pure aluminum metal to slow down recrystallization at elevated temperatures. In addition, our data indicates that grain size in Figure 4 (D) is smaller than that shown in Figure 4(A). We have tentatively concluded that the addition of Zr in aluminum tends to reduce grain size, especially when the alloy is cast using ultrasonic grain refining. This result is also in agreement with past work. [4, 5].

Comparing to the columnar grains obtained using TiBor addition shown in Figure 3, the grains in 4 commercial alloys shown in Figure 4 are all equiaxed. These equiaxed grains are in the range of 200 to 800 microns, much smaller than that obtained using TiBor as chemical grain refiners. These results indicate that ultrasonic vibration can be effective in eliminating columnar grains in a relative large copper mold under continuous casting conditions.

The mechanisms of grain refining using high-intensity ultrasonic vibration are still not fully understood. The application of high-intensity ultrasonic vibration in the molten metal creates some non-linear effects such as ultrasonically induced cavitation and acoustic streaming. Both are effective in reducing grain size by fragmentation in a small castings. These fragments can then grow into small grains in a copper mold during continuous casting process. Thus, the experimental results shown in Figure 4 could be partially due to the effects of acoustic streaming and cavitations.

It has to be pointed out that most of the work on ultrasonic grain refining of metals and alloys was performed on samples smaller than one kilogram. Results shown in Figures 2-4 were obtained in a section of a full size copper mold for a continuous production of aluminum cables.

Benefits of Ultrasonic Grain Refining

Results from the trials concluded that by using only ultrasonics to grain refine aluminum, an equiaxed grain structure can be obtained without the use of external particulates being added to the melt. Based on the previous work as well as the current work aluminum castings with an equiaxed grain structure have good ductility and can easily be machined and formed into desired dimensions. In addition, the operational cost is significantly lower using the ultrasonic grain refining process when compared to using TiBor additives. Another benefit with the elimination of additives to the melt is the potential for a reduction in the inclusion levels within the casting.

In specific applications such as electrical conductor aluminum alloys like 1350 EC; there is a potential for an increase in conductivity and mechanical properties due to the elimination of TiBor inclusions within the cast.

Initial Scale-up to Southwire Continuous Rod Process

The SCR process was developed in 1964 as demand was increasing for electrical conductor aluminum wire. This process was then adapted to produce not only electrical grade aluminum conductors but mechanical aluminum alloys as well. Recently Southwire has had many technology breakthroughs in aluminum processing such as Southwire's Ultra-D™ degasser and Southwire's ultrasonic grain refining process [6, 7].

The ultrasonic grain refining process was tested on one of Southwire's largest aluminum rod lines that primarily casts electrical grade aluminum rod. When testing the ultrasonic grain refiner, cast bar samples were collected and etched. A comparative analysis of the cast bar showed that the grain structure produced using the ultrasonic grain refiner was comparable to those cast with TiBor additives. Comparable microstructures of the hot rolled aluminum rod were also observed.

At the rod mill, the rod is tested for tensile, elongation, and conductivity before it is sent to the downstream process to be drawn into final dimensions. A comparative analysis was completed on the rod properties between rod that was cast using ultrasonic grain refining process and rod cast using conventional TiBor grain refiners. Table I shows the results of rod processed using the ultrasonic grain refiner. Table II shows the results of rod processed using TiBor grain refiners.

Table I. Rod processed with ultrasonic grain refining.
(Averages of 100 coils)

Ultrasonic Grain Refined (.375" diameter) 1350 EC		
Test Completed	**Ranges**	**Averages**
Tensile [a] (KSI)	12.3 – 16.0	14.4
Elongation [b]	14.0 – 25.0	19.6
Conductivity [c]	61.5 – 62.4	61.9

a: 1000 lbs. per sq. in.; b: % of Elongation; c: Reported as % IACS

Table II. Rod processed with TiBor grain refiners.
(Averages of 100 coils)

TiBor Grain Refined (.375" diameter) 1350 EC		
Test Completed	**Ranges**	**Averages**
Tensile [a] (KSI)	13.6 – 15.9	14.9
Elongation [b]	15.0 – 23.0	18.2
Conductivity [c]	61.6 – 62.1	61.8

a: 1000 lbs. per sq. in.; b: % of Elongation; c: Reported as % IACS

Once the rod passes the quality check to ensure the specifications are met; it is then sent to the drawing mill. During the testing of the ultrasonic grain refining process data was collected on coils that were drawn to 0.174" for use in overhead conductors. There are two separate conductors that this wire could be used for: aluminum conductor steel supported (ACSS) or aluminum conductor steel reinforced (ACSR). The major difference between the two processes of making the conductors is that the ACSS aluminum wire is annealed after stranding. A comparative analysis was completed on aluminum drawn wire that was processed with the ultrasonic grain refining process and aluminum wire that was processed using conventional TiBor grain refiners. All specifications as outlined in the ASTM standards for 1350 electrical conductor wire were met on the drawn samples. There was a marginal improvement in the non-annealed drawn samples tensile strength and conductivity; however no definitive conclusions are being made at this time.

Conclusions

Southwire's ultrasonic grain refining process is still being optimized and tested on its aluminum rod mills. It has been proven that an equiaxed grain structure can be obtained on pure aluminum and aluminum alloys using the ultrasonic grain refining process. In the case of the .375" diameter rod; comparable properties were achieved without the use of external particulates. In the case of drawn aluminum wire comparable results were achieved; however in the non-annealed samples there was a slightly higher tensile and conductivity of the wire processed using the ultrasonic grain refining process.

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Aluminium Casting Furnace Energy Efficiency: Recent Improvements in Rio Tinto Aluminium Casthouses

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Keywords: Energy Efficiency, Casthouse, Furnace, Sustainability, Energy Performance Indicator

Abstract

In 2005, Rio Tinto Aluminium started an ambitious program to reduce casting furnace energy consumption in its casthouses. A major reduction step was reached in 2008 by implementing this energy efficiency program in all casthouses, focusing more specifically on the reduction of natural gas consumption (m^3 of gas) and specific energy consumption (MJ/t of cast metal). In order to further reduce energy consumption, a new strategy was recently developed and is described in this paper.

The main tool of this strategy is a standardized energy performance indicator. This indicator takes into account not only intrinsic furnace operating parameters such as burner efficiency and pressure control but also some specific production characteristics (casthouse layout, temperature of the potroom metal, mass of remelt and mass of alloying ingredients). This indicator allows to adequately compare energy performance between casthouses and to identify best practices and potential improvements. Some industrial applications of this tool are also presented.

Introduction

Over the last two decades, Rio Tinto Aluminium group has developed and implemented strategies in order to optimize energy efficiency of its casthouse furnaces and minimize their energy consumption. From 2005 to 2010, a global casthouse furnace energy efficiency program [1] was implemented. The key points of this program were to (see Figure 1):

- set up a Steering Committee, whose mandate is to promote energy efficiency and provide a challenging energy consumption target per plant;
- set up a network of key resources from plants and R&D in order to develop, document and implement best practices for furnace energy efficiency.
  Furnace energy efficiency is intimately linked to optimized pressure [2-3] (airtightness of furnace, positive pressure target, and adequate physical control system) and burner [3-6] (maximal power used, air/gas ratio control, and pilot flame) control.

As a result, a 20% decrease of energy consumption was obtained while maintaining the same level of melting.

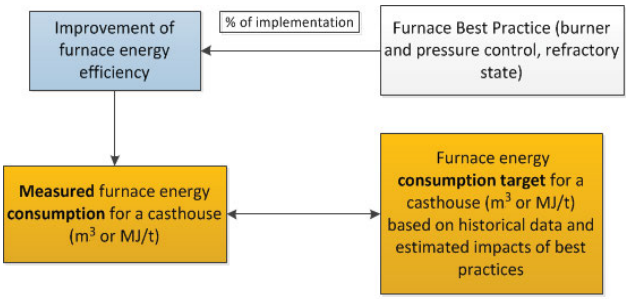


Figure 1. Energy consumption follow-up approach used from 2005 to 2010.

New energy performance approach proposed

Several challenges remained to be addressed to keep the focus on energy efficiency optimization. First, the state of mind and the momentum have to be maintained despite the decreasing price of natural gas. Second, the implementation of best practices has to be finalized and adapted to the configuration of each casting center. Finally, the analysis and follow-up of the energy efficiency of the casting furnaces had to be performed at a higher level and a more global analysis needed to be developed.

Indeed, optimal energy consumption performance, for a casting center, can only be obtained with energy efficient furnaces and effective management of various production inputs such as the mass of solid to be melted, maintenance requirement, or batch scheduling. Furthermore, even with optimal management of the production inputs, energy consumption could vary drastically, from a period to another, on a same furnace, due to variations of these production inputs i.e., the remelt performed or the scheduled production. Other parameters influence the energy consumption of each casting center; they are related to their layout such as the number of furnaces, type of heating (electric or combustion), etc.

To adequately analyze the energy consumption results obtained in casting centers, and detect if some improvements should and could be performed, consumption targets need to be calculated during consecutive periods of analysis:

- considering that furnaces best practices are fully implemented,
- taking into account the layout of the casting center,
- considering the production parameters of the analysis period,
- assuming that the management of the production parameters was optimal.

An energy performance indicator (EP) can then be calculated by comparing the real energy consumption to the energy consumption target that could be achieved. Figure 2

schematically illustrates the elements to consider for the calculation of the energy performance indicator.

$$EP(\%) = \frac{Energy\ Consumption\ Target}{Real\ Energy\ Consumption} x100$$

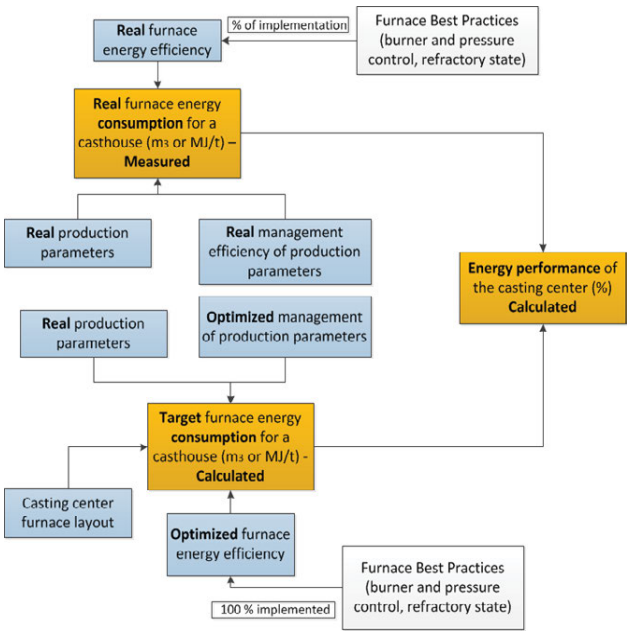


Figure 2. Energy Performance (EP) calculation approach used since 2011.

Examples of problematic when using only the furnace energy consumption

In order to illustrate the limitation of the approach based only on the measurement of the energy consumed to optimize the energy efficiency of a casting center, a fictive case is presented hereafter. Four different casting centers, having the same production volume of 15 kt per month, are considered. Their global characteristics are presented in Table I while, their monthly specific energy consumption are illustrated in Figure 3.

Table I. Global characteristics of each casting center

Casting Center	Monthly production (t)	Heating method	Usual percentage of remelt per batch (%)
1	15 000	Electrical elements	< 5%
2		Gas burner	20-30 %
3		Gas burner	20-30 %
4		Gas burner	5-15 %

When analyzing these monthly results, some questions may be raised:

- What should be the energy consumption target for each of these casting centers? Are they on target? Are there any feasible energy consumption reductions possible, especially for casting centers nos. 2 and 3?
- The specific energy consumption of casting center no. 3 was reduced by 20% from May to June. Is this reduction due to higher furnace efficiency (improvement of best practices), management optimization or to a variation of the production parameters?
- What are the reasons for the gradual 10% increase in specific energy consumption observed for casting center no. 2, during the last four months? Is it due to deterioration of the furnace performance or to the production input management?

Such questions can be answered using the energy performance indicator approach and are discussed later in the paper.

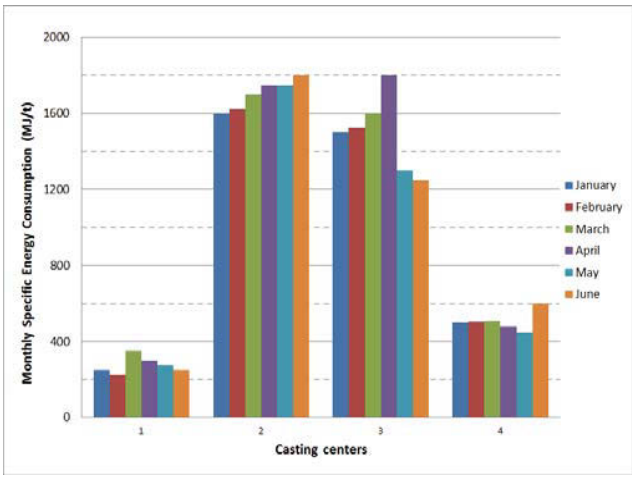


Figure 3. Monthly energy specific consumption (MJ/t) measured for four casting centers

Production and configuration parameters influencing the furnace energy consumption

Prior to describing and using the new energy performance indicator that was developed for the Rio Tinto casthouse furnaces, production parameters and configurations influencing furnace energy consumption must be fully understood.

Production parameters

Percentage of remelt added per batch. Typically, the temperature of the liquid metal delivered from the potroom is much higher than the temperature needed to perform the alloying preparation and the holding period before casting. This overheat allows melting a given amount of solid metal without any addition of energy from the heating system. This amount of solid metal is an 'energy-free' remelt for the batch. The percentage of free remelt varies generally between 2 and 10%, depending on the electrolytic metal temperature, heel characteristics (mass and temperature) and the temperature needed for the addition of alloying ingredients.

As an example, a batch having the following parameters:

- no heel,
- potroom metal delivered at 850 °C,
- temperature needed for the addition of alloying ingredients at 760 °C

would have a free remelt of around 8.5%.

The energy needed during the melting/heating phase, for a 60 to 100-tonne capacity furnace, using cold air burners, can vary typically from:

- 0 m^3 of natural gas for a batch with a remelt addition below the 'free remelt' to
- 1000 to 1500 m^3 of natural gas for batches around 20% of remelt to
- 7000 to 10,000 m^3 of natural gas for batches above 90% of remelt (typical for a recycling furnace).

Theoretically, specific energy consumption targets could be as low as 50-100 MJ/t, for batches with low remelt, and up to 2,500-3,000 MJ/t, for batches having 90% of remelt and more.

Remelt distribution between batches. The average mass of remelt added per batch will have an impact on the energy consumption of the furnaces. However, this impact will highly depend on the way that the remelt is distributed between each batch. The more uniform is the distribution, the lower the global energy consumption will be, for the same amount of remelt performed. This is schematically presented in Figure 4.

Two production scenarios are considered, when keeping the same furnace, and the same intrinsic furnace energy efficiency. Both scenarios consider two successive batches having the same input of hot metal at 850 °C, same temperature target of alloying addition of 760 °C, no heel, and 15% of remelt to be performed. In the first scenario, the remelt is added to the first batch only (Figure 4a). In the second scenario, the remelt is equally distributed between the two batches (Figure 4b). The energy consumption difference during the remelt period, for both scenarios, can be deduced as follows:

- the percentage of free remelt is 8.5%,
- Scenario no.1: for the first batch, energy has to be provided to remelt 6.5% of solid, whereas for the second batch, the liquid metal inside the furnace will be too hot,
- Scenario no 2: no energy input is required for both batches during the remelt period.

Therefore, if the first scenario is used, more energy will be required.

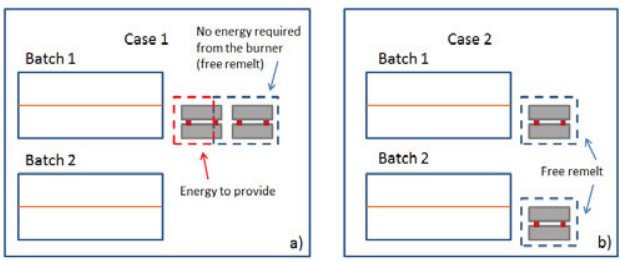


Figure 4. Scenarios considered a) remelt is introduced in the first batch only; b) remelt is equally distributed between both batches.

Temperature of the potroom metal. As mentioned, temperature of the molten metal delivered from the potroom to the casting center has a direct impact on the percentage of "free remelt". Indeed, a 10 °C loss of hot metal temperature delivered to the furnace typically reduces the free remelt by 1%. This temperature loss can be due to several causes:

- metal flow between potroom and the casting center (waiting time between the end of the tapping run and the transfer of the hot metal into the furnace),
- insulation of the potroom crucibles,
- cleaning frequency of potroom crucibles (colder crucibles after this operation),
- alkalis and bath pretreatment inside the crucibles,
- transfer method from the crucibles to the furnaces (influences the furnace heat loss due to door opening).

Alloying addition. The type, concentration (pure or master alloys), format (powder, tablets) and mass of the required alloying ingredients, as well as their addition method and schedule will impact the metal temperature target needed for their addition and the temperature loss that will occur:

- minimal temperature required for adequate dissolution of manganese and chrome.
- increase of temperature during pure addition of iron, copper or manganese.
- decrease of temperature during addition of silicium, magnesium, zinc, and chrome.
- for alloying ingredients blended with aluminium, additional heat loss has to be considered for the heating and melting of aluminium.

This will have a direct impact on the energy consumption target of the product to be cast.

Net production performed. Energy used during a batch is the summation of the energies utilized throughout the:

- melting period to reach the temperature needed for the alloying phase,
- preparation operations (alloying, fluxing, skimming),
- cast,
- holding periods between all these operations in order to maintain the furnace temperature.

The longer the holding temperature periods are, the higher is the holding energy consumption and the lower is the production. These holding periods may result, for example, from issues in product scheduling, hot metal delivery delay or casting unit availability.

Figure 5 illustrates the weekly specific energy consumption in function of the weekly production rate for two periods of production measured for the same furnaces. For the first period (marked in black), the best practices for furnace efficiency were not implemented, whereas they were implemented for the second period (marked in red). It can be noted that:

- for both periods, a specific energy consumption increase of around 35 to 40% was measured, when the production was reduced by 20 to 25% (see also [7]). This increase is higher before the implementation of best practices, as the heat loss during the holding periods was not minimized (more air infiltration);
- in this case, the variation of the production rate had a higher impact on the specific energy consumption than the implementation of the best practices.

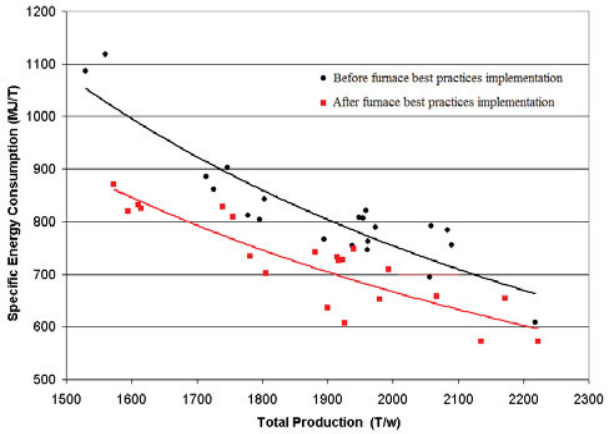


Figure 5. Impact of implementation of best practices and production rate on weekly specific energy consumption.

Configuration parameters

Number of furnaces in operation. For each analysis period, the number of furnaces in operation has to be known to adequately quantify the global duration of holding periods. For the same tonnage of production, the greater the number of furnaces used (while keeping the same furnace capacity), the greater the energy consumption will be during the holding periods.

Type of energy input: gas/oil burner vs. electric heating elements. A major distinction exists between electric and combustion furnaces. Indeed, the latter have to evacuate the combustion gas via a chimney, which leads to two impacts:

- combustion gas has a restraint residence time inside the furnace. At full power, with cold air burners, the maximal heat transfer to the charge and refractory is around 50-55%. The rest of the energy is evacuated by the chimney;
- during holding periods at low power, the pressure target is usually not reached due to the natural draft of the chimney, and cold air infiltration occurs through the door gaps.

In an electric furnace, either is there no chimney, or this latter is totally closed during all the periods that do not require gas evacuation (furnace cleaning, alloying, fluxing or skimming). Therefore, the heat transfer is optimal during heating or holding periods, and cold air infiltration remains minimal. For these reasons, electric furnaces are energetically very efficient for production at low remelt levels. For these types of productions, the consumption target for electric furnaces should be twice to three times lower than that of the combustion furnaces. However, the use of electric furnaces remains limited for remelt due to their low heating rate [8].

Casting temperature target. From a casting center to another, the temperature target used to cast the same product can vary in function of trough configuration (total length, design, insulation, preheating, metal flowrate) or of the equipment used for the inline metal treatment (degasser, filtration, etc.). Indeed, each configuration will have its own heat loss that will need to be compensated.

The type of product cast will also have an impact on the casting temperature target.

Energy Performance (EP) Indicator

Calculation of the energy consumption target

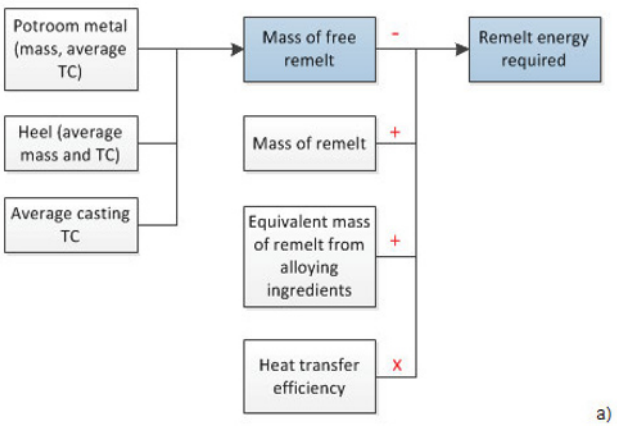
As mentioned previously, the calculation of the energy performance indicator requires calculating an energy consumption target. The method used is schematized in Figure 6.

First, the mass of free remelt (1) that should have been obtained during the period is calculated taking into account the inlets of hot metal (average mass and temperature), heel (average mass and temperature) and average casting temperature. For this calculation, the mass of remelt is considered uniformly distributed between the batches.

The energy required to remelt the excess of solid metal and to heat, remelt or dissolve alloying ingredients (2) can then be calculated taking into account the heat transfer efficiency obtained when best practices are implemented.

The necessary heating time (3) to deliver the remelt energy required is then deduced. The remaining time is considered being for temperature holding (4). The energy consumption target (5) during this holding period is then deduced using the lowest energy consumption (6) obtained during holding periods for furnaces where best practices are implemented.

Finally, the total energy consumption target can be deduced in summing the remelt energy required and the holding energy required. The energy performance indicator is then obtained by dividing this target by the real furnace energy consumption.



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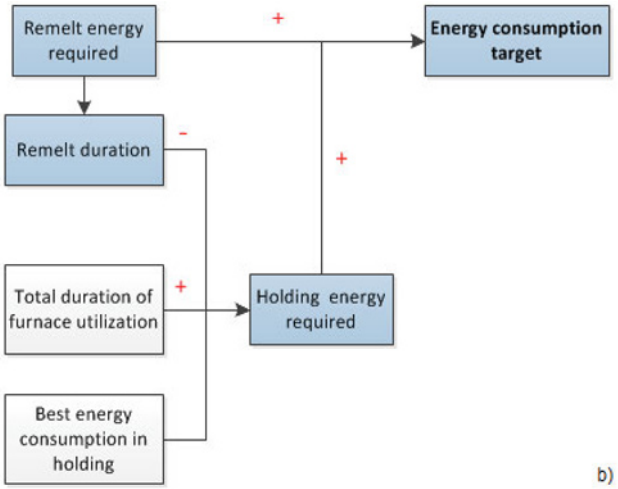

Figure 6a) Flowchart for the calculation of the free remelt and the remelt energy required 6b) Flowchart for the holding energy required and the total energy target.

Discussion

The energy performance indicator allows detecting potential variations of performance for a given casting center. These variations are only due to variations of furnace energy efficiency or management efficiency of production parameters. Indeed, the impacts of production parameters are already considered in the calculation.

Furthermore, as the configuration parameters specific to each casting center are also included in the energy performance indicator calculation, all casting center results can be now compared on the same scale (0-100%). Their performance can also be compared to a global energy performance target defined by the Steering Committee. The discrepancies between this global target and the results obtained, by each casting center, provide the percentage of improvement that could be obtained in each casting center.

To translate these potential improvements in terms of energy consumption savings, the real energy consumption has to be considered.

Performance diagnostic

To illustrate the benefits of using this indicator, the monthly specific energy consumption results (Figure 2), obtained for the four casting centers, are analyzed here after together with their monthly energy performance (Figure 7). The global energy performance target desired is set at 85%.

The following conclusions can be drawn:

- Casting center no. 1 has the lowest specific energy consumption but also the lowest energy performance. The low specific energy consumption obtained results from the type of furnace used (electrical) and the low level of remelt performed. The low energy performance results from a poor implementation of best practices. Therefore, even if the specific energy consumption is the lowest, there are still improvement opportunities.
- Casting center no. 2, on the contrary, presents the best energy performance, close to the targeted value of 85% and the higher specific energy consumption. This results from the fact that for this casting center,
    - the best practices were well implemented,
    - the management of the production parameters was adequate,
    - the percentage of remelt was relatively high.

  In addition, even if the specific energy consumption has increased over the last four months, no diminution of the energy performance is noted. The energy consumption increase is a consequence of production input variation and is not caused by a deterioration of the furnace energy efficiency. An increase of the remelt performed could explain this increase.
- Similarly, the energy consumption decrease noted for casting center no. 3, during the last two months, is not a result of an energy performance improvement, the performance indicator remaining stable. The consumption decrease should also be due to a change in the production inputs.
- On the other hand, the energy consumption increase observed, for the last month, for casting center no. 4, is linked to a decrease of the energy performance indicator. A furnace control or production parameter management problematic has occurred in this case and should be investigated.

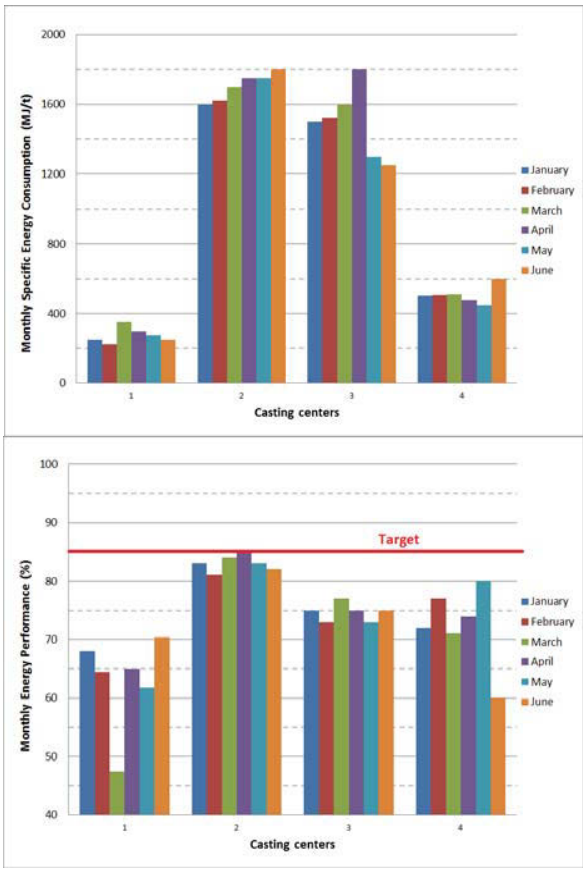

Figure 7. Monthly evolution of specific energy consumption and energy performance indicators for the four casting centers.

Potentials savings

The joint analysis of the specific energy consumption and the energy performance indicator also allows quantifying the potential savings that may be obtained.

Table II illustrates this fact, while assuming that all casting centers have the same monthly production of 15,000 tonnes, the same equivalent energy cost at 0.34 $/m^3 [9] and a performance target fixed at 85 %. The followings conclusions can be drawn:

- Casting center no. 1 presents the worst performance results (average of 63%), but potential savings are only 40% of those available in casting center no. 3. Indeed, casting center no. 3 has an average energy performance 12% higher than that of casting center no. 1, but its monthly average specific energy consumption is six times higher due to its type of furnaces and its remelt percentage. The annual potential savings would be of 270 k$, for casting center no. 3, and 106 k$, for casting center no. 1.
- Casting center no. 2 has the highest specific energy consumption, its potential for savings is however relatively low due to its higher energy performance.
- Casting center no. 4 presents a relatively low potential of improvement during the first five months, but offers a significant opportunity for the last month. Rapid action should be taken to restore good practices.

Table II. Calculation of potential monthly savings per casting center.

	Average consumption (MJ/t)	Average performance (%)	Potential savings (GJ/month)	Potential savings (k$/month)
CC 1	275	62.8	914	8.9
CC 2	1700	83.0	511	5.0
CC 3	1496	74.7	2319	22.5
CC 4	508	72.3	964	9.4

Results obtained in Rio Tinto aluminium casting centers

Since 2011, this indicator has been calculated and followed by the Steering Committee and by plant key resources. It has allowed quantifying energy reduction and potentials for savings, in order to prioritize improvement activities and to promote energy performance for plants having the worst results.

Moreover, energy consumption targets, specific to each casting center, were quantified in function of their production plan.

Actions taken between 2011 and 2015 have allowed improving the global energy performance of the Rio Tinto Aluminium casting centers by 10%, and reduce the specific energy consumption by 15%.

Conclusion

A new energy performance indicator for the casting center furnaces was developed and successfully implemented in Rio Tinto Aluminium casting centers. This indicator allowed comparing the furnace energy consumption to a targeted consumption for each casting center, taking into consideration the production parameters and the casting center configuration.

This indicator allowed to effectively follow-up the energy consumption performance for each casting center furnace, providing adequate consumption targets, and identifying opportunities for reduction and prioritizing actions to be taken.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

CASE STUDY ON ROUND-TOP FIRE RATES

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Keywords: Energy, Furnace, Melter

Abstract

Energy makes up a major portion of the controllable costs involved in melting aluminum. While technology such as regenerative burners and multi-chamber furnaces improve energy efficiency, these technologies are expensive and difficult to cost justify given current U.S. natural gas prices. Low cost methods to improve energy efficiency without high capital outlays include process optimization, better maintenance, and furnace settings. The burners on many furnaces are over-fired and poorly controlled. This case study shows the results from a trial made with different fire rates and control schemes on a round-top aluminum melting furnace.

Introduction

Round-top melters are circular batch furnaces that are charged by removing the roof and filling the furnace with a loose mound of cold metal charge. Two to four burners are located around the perimeter of the furnace[1]. After charging with cold material, the roof is replaced and the burners are turned on. Typically, the burners will fire at high fire until the metal melts into a flat bath and the molten aluminum reaches the desired temperature. The molten aluminum then transfers to a holding furnace for further processing and to the casting process.

For the furnace in this trial, the burners were controlled by the flue temperature so that they ran at high fire for the majority of the melt cycle until the flue temperature reached the set point. Energy efficiency was thought to be satisfactory, but after comparing performance with external benchmarks it was realized that improvement was possible. Trials were run consisting of modifying the fire rate and the control schemes on the furnace over several months. The performance was analyzed using weekly data since shorter time periods such as batch data showed too much variation.

Fire Rate Optimization

Studies in literature[2,3,4] suggest that reducing the fire rate to an optimum level provides energy savings. Many furnaces are over-fired under the belief that a higher fire rate increases melt rate. However, heat transfer is limited based on the exposed metal surface. Fire rates over the optimum level results in excessive heat loss in the flue gases without substantially increasing the melt rate. High gas flow rates also reduce the residence time of products of combustion within the furnace. An optimum fire rate can be found to obtain the designed melt rate and improve energy efficiency.

The previous computer models and studies recommend a firing density of 208 kW/m^2 [66,000 BTU/hr/ft^2]. Firing density is the burner fire rate (kW or BTU/hr) divided by the area of the surface of the molten bath within the main chamber (square meter or square feet).

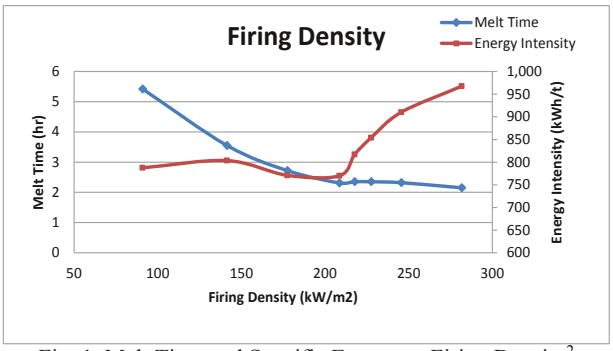

Fig. 1. Melt Time and Specific Energy vs Firing Density[2]

However these studies were based on a rectangular reverberatory melting furnace using mostly a continuous process and assuming a flat bath.

Round-top melters are batch furnaces with a higher surface area due to the loose mound of material. This surface area has been modeled as a cone[1], but the reality is that the surface area is complex and varied with different charge types. While a cone compared to a flat bath only increases the surface area by 20% for these furnaces, a higher firing density appeared to be necessary based on previous experience. A goal of 315 kW/m^2 [100,000 BTU/hr/ft^2] was used which is 67% higher than a continuous furnace. The exact area varies depending on charge types and charging practices. A higher firing density based on the flat bath area would most likely be appropriate for all batch furnaces.

The starting fire density for the trial furnace was 394 kW/m^2 [125,000 BTU/hr/ft^2] based on the molten bath surface area. It can be noted that flames were occasionally seen going up the flue and door maintenance was high.

The first trial consisted of lowering the high fire so that the firing density was lowered to 347 kW/m^2 [110,000 BTU/hr/ft^2]. No change was seen in the melt rate, but an energy savings of 12.8% was obtained.

A second step to 315 kW/m^2 [100,000 BTU/hr/ft^2] firing density was instituted. This again showed no change in melt rate with a total energy savings of 15.2% from the baseline or an increase of 2.4% improvement over the higher firing density.

Further decreases might have been possible. However, it was felt that a high rate was needed at the beginning of the melt cycle when there was a high surface area and a high delta T (difference in temperature) between the combustion gases and

the cold surfaces. The decision was made to investigate a different burner control scheme.

Control Scheme

At the beginning of the melt cycle, the metal surface area is high. Later in the cycle, the solid metal slumps down and eventually reaches a flat bath. Metal surface area decreases over time.

Since the roof is removed during the charging process, the refractory furnace walls and refractory or fiber roof material loses much of their stored heat. At the beginning of the melt cycle, heat transfer is to both the metal surface and into the walls and roof. As the melt cycle continues, the walls and roof temperature increases and heat transfer into these surfaces decrease given the lower ΔT between the flame and the walls and roof.

Over the melt cycle, the temperature differential between the flame temperature and the bath surface lowers. The formation of thicker metal oxide layers (dross) also reduces the heat transfer into the surface over time as this dross insulates the surface.

All of these factors mean that a lower fire rate is appropriate later in the melt cycle. This can be done in one of three ways:

1) The fire rate can be programmed to step-down over time. This can be done step-wise or a gradual ramp.
2) Program logic can be set to drop the fire rate a certain percentage when the flue temperature reaches a setpoint.
3) Cascade logic can be used. At the beginning of the melt cycle the flue temperature will be low and bath temperature controls the burner. Over time the flue (or roof) temperature will rise. Cascade logic lowers the fire rate as the flue reaches a certain setpoint.

For the first two options, the burner controls must be capable of this type of programming. Many of the older controls still prevalent in industry cannot adjust the fire rate in this manner. For the third option, this cascade logic is widespread in industry. However, the flue (or roof) setpoint is often set high so the fire rate only begins to lower late in the cycle. A lower setpoint can be appropriate.

In this case, the furnace did not have a bath thermocouple until a molten bath existed. The flue temperature controlled the burners during the majority of the cycle. The furnace had newer burner controls that could be programmed using simple programming changes. Option 1 was used with fire rates ramping over time. An estimated melt time is calculated based on the weight of the charge. Fire rate started at high fire. After a portion of the melt time, the fire rate decreased and ramped down until later in the melt cycle. The melt cycle finished at this lower firing density (Fig. 2). This change resulted in 3.4% improvement over the fixed fire rate or a total savings of 18.6% since the beginning of these trials.

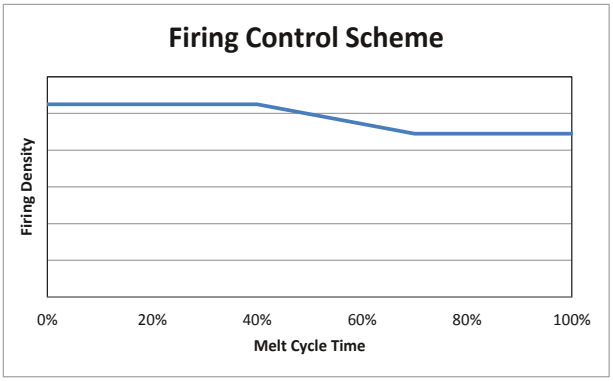

Fig. 2. Firing Control Scheme

Issues

The biggest issue for a project like this was getting over the historical notion that a high fire rate is needed. Containing the trials to one furnace helped overcome this issue. It is to be noted that the management of the casthouse was aware of the change while the operators were not notified. As in many tests, a blind test is best so that users don't blame any problems on the change. If there was a problem such as a lower melt rate, management could make the decision to modify the testing procedure.

As with any project, time is always an issue. There are higher priorities than energy in a casthouse. This time issue caused the test to continue longer than necessary (a year). However the extra time meant more data was collected so we could be confident in the change.

Variation in production caused an issue in interpreting results. A simple Energy Intensity (BTU/lb or kWh/t) couldn't be used since production needs varied each week. Production variation alone greatly affects Energy Intensity. In this case, weekly energy used was plotted against production (Fig. 3). The goal is to reduce the amount of energy needed for the same production. The trend line moves down with better energy performance.

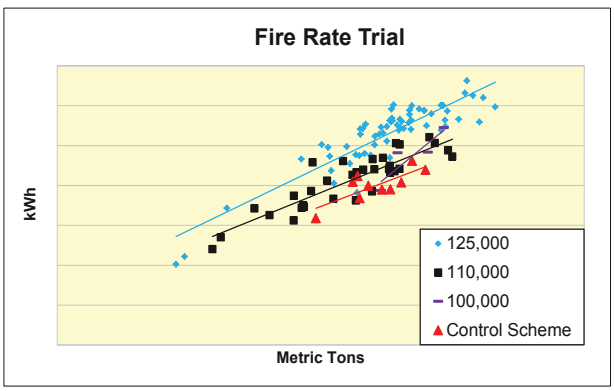

Fig. 3. Energy Use vs Production per Trial

Conclusions

Low cost methods such as burner fire rate and modified burner control schemes can provide substantial savings in a melting furnace. Round-top furnaces require a higher firing density at

the beginning of the cycle compared to continuous furnaces due to the high surface area of the metal and cold furnace surfaces. The firing density should be lowered as the surface area decreases and walls heat up. In this case study, savings of 18.6% was achieved (Fig. 4) without spending money on capital. Further savings could be possible with additional trials to optimize the firing control scheme on this furnace.

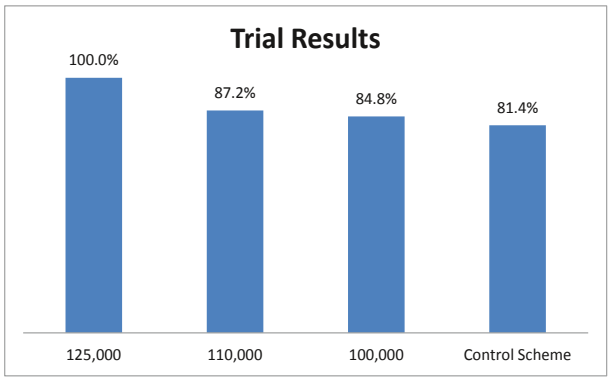

Fig. 4. Trial Results

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INCREASING HOLDING FURNACE CAPACITY FROM 30 TO BE 40 TONS MOLTEN ALUMINIUM THROUGH MODIFICATION OF LINING DESIGN

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Keywords: Optimization, Improvement, Holding Furnace, Inalum

Abstract

PT. Indonesia Asahan Aluminium (Persero) – hereinafter referred to as Inalum - is the pioneer of the Aluminium Smelter in the South East Asia region. It was designed by Sumitomo Chemical and started up in February 1982. The original capacity was 225,000 Tpy of ingot and 175 kA of amperage reduction cell and 30 Tons of Holding Furnace (HF) capacity in the casting house.
Since operation commenced in 1982, only HF-706 had been modified to increase capacity to 36 Tons in 2007. Inalum has now successfully increased furnace capacity to be 40 Tons at HF-704 in October 2014. Improvement of HF-704 has been done through optimized lining design by new materials, an improved heat-up period, and optimized reconstruction time.
The improvement is done to support the increasing of production up to 300,000 Tpy in 2019 followed by diversification of Aluminium product (100% of Aluminium Ingot to be 60% of Aluminium Ingot, 30% of Ingot Alloy and 10% of Billet).

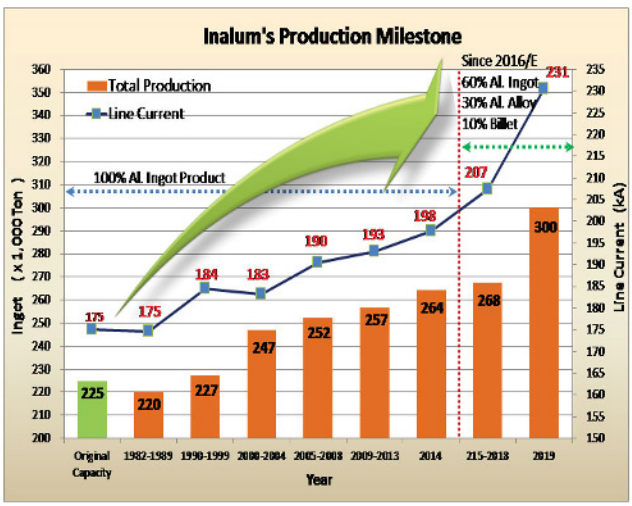

Figure 1. Production Milestone

Introduction

As with all furnaces, HF also needs maintenance work, namely "partial and whole repair". Based on collected maintenance data, the partial repair of the furnace is normally conducted after 3~4 years or after 80,000~110,000 ton capacity. Meanwhile, total refractory replacement is normally conducted after 3~4 partial repairs (10~15 years).
HF lining design for the original capacity of 30 Tons has 5 layers. In the recent reline, there was a change to modify the hearth design to be 4 layers in order to increase its capacity up to 40 Tons for covering more metal production in the future.

Table 1. Thickness of HF Hearth Layer

Layer	Unit	HF Existing (30 Tons)	HF Improvement (40 Tons)
1	mm	55	58
2		120	114
3		65	65
4		65	230
5		230	0
Total		**535**	**467** (-68 than existing)

The heat-up process for installation of refractory material especially monolithic (castable) is very important to be done before furnace is operated by objective to evaporate the water content in refractory material. The heat-up must be done by increasing of temperature step by step per hour which is depend on refractory type and thickness of installation.
By decreasing of HF hearth layer, optimization of the heat-up period has been done by decreasing the heat-up period from 53 to 19 days with the existing standard of the heat-up rate and final temperature. The time saved of up to 34 days will increase the productivity of furnace (Fig. 2).

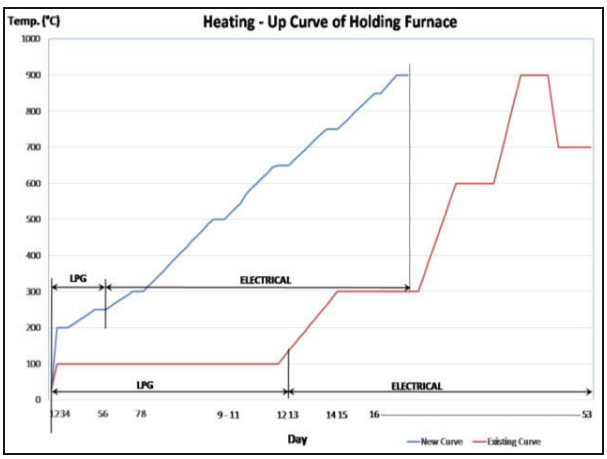

Figure 2. Curve of Heat-Up Temperature – Comparison between Existing with New

Improvement Method

Inalum commits to continuous improvement of the performance and the product quality with safe, stable and environmental friendly operation (Fig. 3).

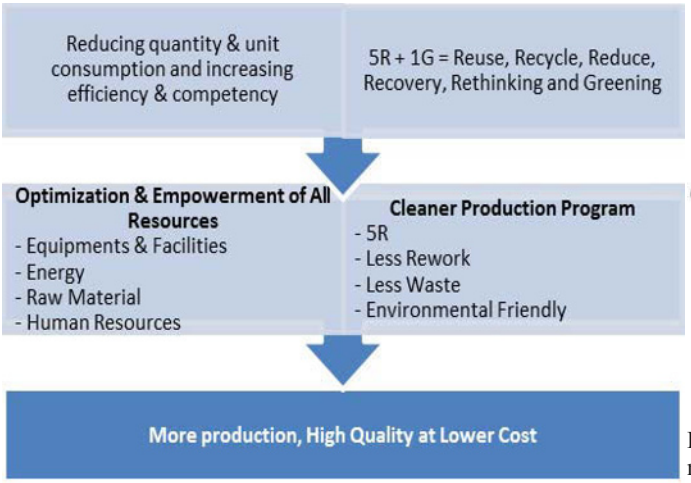

Figure 3. Technical Improvement Method

Lining Design Of Holding Furnace

The improvement objective was to increase HF capacity from 30 to be 40 Tons at normal capacity (an increase of 33.3%). The modification of HF has been done especially on the bottom lining through decreasing the thickness as much as 68 mm by eliminating the 5th layer (Insulating Brick CWR-8-3) and using castable material instead of insulating brick.

Table 2. Lining Material of HF Layers for Bottom

Layer	Ref. Old HF (30 Tons)	Improvement HF-704 (40 Tons)
1	Insulating concrete	Insulating Castable IC-11 HS
2	Insulating Brick B-1	Insulating Brick B-2
3	Insulating Brick B-2	Castable C-HAS
4	Castable Hisicon HC-AL	Castable Calde Cast Gibram AL-4
5	Insulating brick CWR 8-3	-

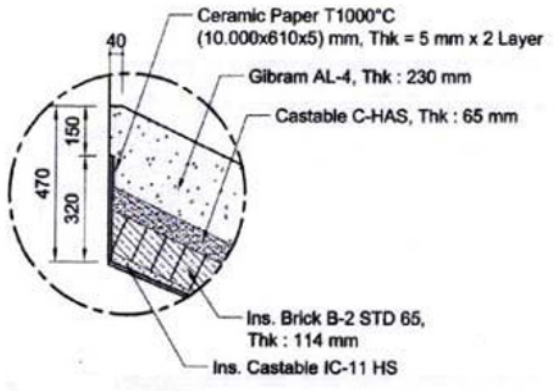

Figure 4. New Bottom Lining Design

Meanwhile, the number of layers in the sidewall lining is kept equal to the existing design (4 layers). The only change was the use of different brick in the 4th layer by Castable C-16 instead of Chamotte Special Brick SK-32.

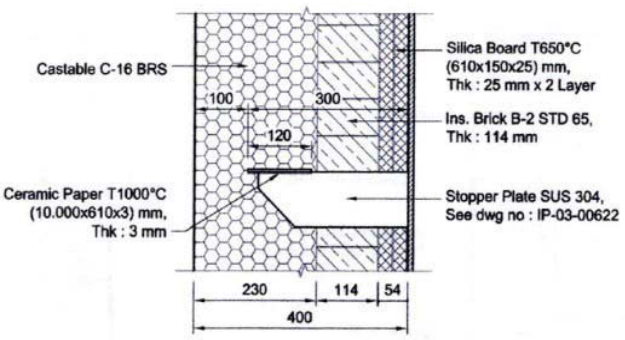

Figure 5. Sidewall Lining Design

Design of the roof brick with 6 layers is kept to the existing materials and thicknesses which have been used since 2004 with good reliability.

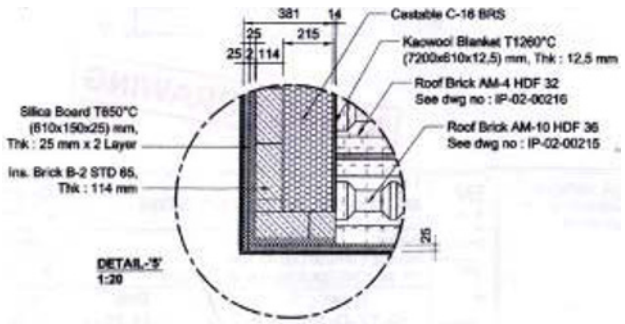

Figure 6. Roof Lining Design

Reconstruction Time

Reconstruction time of HF was optimized for installation new refractories at bottom, sidewall and roof lining from 55 to be 30 days. It was successful to be done by optimizing of working time.

Table 3. Schedule of Installation New Refractory

No	Installation New Refractory	Day	Day Installation 1~10	11~20	21~30	31~40	41~50	51~60
1	Old Schedule	**55**						
	- Bottom	16	■	■				
	- Sidewall	23		■	■	■		
	- Roof	16					■	■
2	Optimizing	**30**						
	- Bottom	15	■	■				
	- Sidewall	15		■	■			
	- Roof	12			■			

Application Test of Holding Furnace

A. Megger Check

A megger check (megger is name of equipment) by a Mega Ohm Tester is done to ensure that there is no leakage current during the operation of the HF moreover it is a new design which use castable refractory material in lining design with high moisture content.

In the existing HF, a megger check is always done one week after the completion of the installation. It is done during the first part of

the heat-up cycle when heat is provided by LPG *(liquid petroleum gas)* on HF-704. The standard value for the megger check of greater than 0.1 mega ohm was reached by the 4th day of the heat-up period. The result of measurement shown that there was no current leakage by measurement result was higher than 0.1 mega ohm (Table 4).

Table 4. Result of Megger Check

Measurement		Unit	Stan-dard	HF 704	Ref. Old HF
				4th Day (Heat-up)	After 2 Months
Heater 1	U1	MΩ	> 0.1	1.67	1.2
	V1			6.30	2.6
	W1			3.2	6.1
Heater 2	U2			6.17	0.6
	V2			2.96	1.9
	W2			1.41	4.1

B. Heat-Up Process

1. Monitoring of Heat-Up Process

Curve of heat-up based on new standard (heat-up period: 19 days) which will save 34 operational days (existing heat-up period: 53 days).

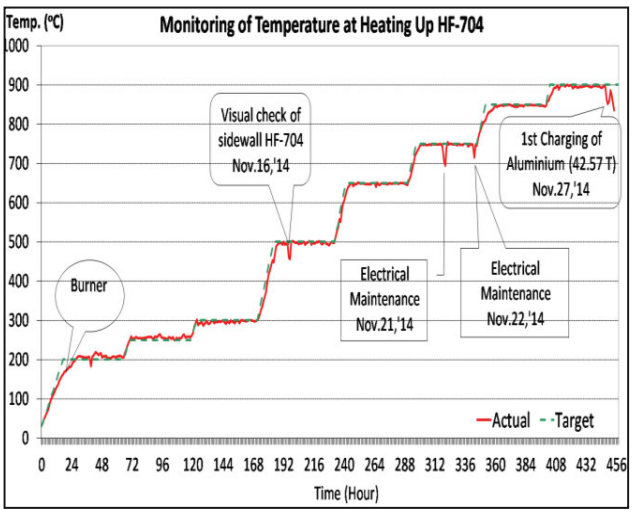

Figure 7. Heat-Up Curve

On the other hand, utilization of castable material which has high water content on the bottom lining at HF-704 will increase consumption of LPG during the heat-up process. But it will be compensated by the reduction of the total dry-out cycle for reduced down-time.

2. Measurement of Temperature at Outside Furnace

The results of measurements have shown that the outer temperature at the bottom, sidewall, and roof lining of HF-704 was lower than the reference furnace (HF-706). The data was measurement result in equivalent time after reconstruction. The most significant improvement occurred at the bottom lining by decreasing up to 32.4°C compared to the existing HF. This condition has been caused by the improved value of thermal conductivity on the bottom lining which was 22.4% higher than the reference HF (the 4th layer of HF-704: 2.08kcal/mh°C which is compared by the 5th layer of HF-706: 1.7kcal/mh°C).

Table 5. Result of Temperature Measurement at Outside Furnace

No	Measurement Point	Average of Temperature (°C)		
		HF 704	Ref. HF-706	Diff.
1	SIDEWALL-1	88.2	88.7	-0.5
2	SIDEWALL-2	93.1	93.8	-0.7
3	FRONT WALL (Charging Door)	147.0	165.3	-18.3
4	BACK WALL (Skimming Door)	122.3	140.3	-18.0
5	ROOF	78.1	79.0	-0.9
6	BOTTOM	91.1	123.5	-32.4

Normal Operation

A. Capacity of HF-704

Through decreasing the thickness of refractory material on the bottom lining of HF-704 as much as 68 mm, normal capacity increased by 40 Tons (increase 33.3%).
Based on an actual test of metal charging on Nov.27,'14, HF-704 can accept molten Aluminium as much as 40 Tons (instead of maximum of reference capacity of 34 Tons).

B. Electrical Power Consumption

Measurement of electrical power consumption is done by Power Meter at the control panel of DS HF-704.

Table 6. Electrical Power Consumption

Item	Unit	HF 704	Ref. Old HF	Diff.	Remark
Temp. Casting	°C	766	760	6	Start
Temp. Casting	°C	698	708	-10	End
Power Consumption	kWh/hr	177.36	129.77	47.6 (36.7%)	
Charging of Al	Ton	319.9	-	-	
Ingot	Ton	311.85	-	-	
Power Consumption	kWh	1317.8	-	-	Avg. Heater "on" = 7.43 hours/day
	kWh/ T.Al	12.6	6.6	6.0 (90.9%)	

Electrical power consumption of HF-704 is higher 6.0 kWh/T.Al (1.9 times) than the old HF. Based on measurement result, to increase Aluminium temperature as much as 10°C, HF-704 need 1.9 hours which is compared the old HF, only 1.1 hours. Besides that, heater of HF-704 is often in switch on position than the reference HF.

C. Silicon Content

Higher silicon content of castable material with low cement content especially the Gibram block material (upper layer at bottom lining) does not give a negative effect for quality of molten aluminium during casting process.

Table 7. Monitoring of Si Content in Ingot Product

No. Sample	Aluminium Charging (Ton)	Aluminium Ingot (Ton)	Si Content (%)
1	42.57	40.58	0.041
2	40.93	40.79	0.039
3	36.19	35.64	0.036
4	39.15	38.50	0.037
5	40.49	39.88	0.038
6	40.55	40.23	0.036
7	39.96	39.54	0.038
8	40.06	36.70	0.042
9	37.66	37.41	0.039
10	33.16	32.67	0.040
11	37.32	36.58	0.039
12	39.08	38.84	0.042
13	38.41	38.10	0.041
14	42.10	40.10	0.038
15	35.67	35.40	0.040
Average Si Content (%)			**0.039**
Standard Deviation			**0.002**

Next Steps

To support the increasing of production of up to 300,000 Tpy in 2019, as the first phase, improvement of HF will be done on HF 707~710. Improvement of HF-707~710 is wholly to support project diversification of Aluminium product (from 100% of Aluminium Ingot to be 60% of Aluminium Ingot, 30% of Ingot Alloy and 10% of Billet).

Generally, the properties of refractory materials for future design are nearly the same as the new improvement at HF-704, except the Insulation Brick B-2 at existing furnace will be changed to be calcium silicate board which has high resistance to carbon monoxide with objective to increase reliability of lining from melting by carbon monoxide.

Table 8. New Material Reconstruction for HF Bottom Lining

Layer	Ref. Old HF (30 Tons)	Improvement HF-704 (40 Tons)	Future Design HF (40 Tons)
1	Insulating concrete	Insulating Castable IC-11 HS	1260°C C.F. Paper
2	Insulating Brick B-1	Insulating Brick B-2	Calcium Silicate Board
3	Insulating Brick B-2	Castable C-HAS	Matricast 2500AC
4	Castable Hisicon HC-AL	Castable Calde Cast Gibram AL-4	Matriflo 84ACX
5	Insulating brick CWR 8-3	-	

With the future HF design with new refractory material, the heat-up process is estimated to be done in 14 days, which is 5 days less than HF-704 (heat-up: 19 days). Overall, this will be 39 days less than the existing HF (heat-up: 53 days).

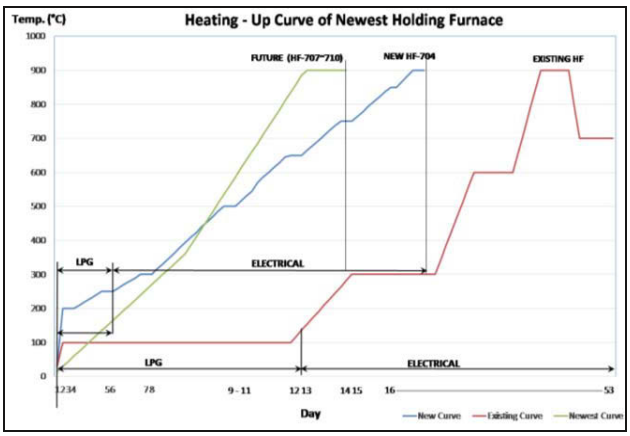

Figure 8. Existing, New (HF-704) and Estimation of Newest Holding Furnace Curve (HF-707 ~ 710) of Heat-Up Temp.

Meanwhile, maintenance work is estimated to be able to less with 4~5 times partial repairs until a total reline is required (instead of 3~4 partial repairs).

Conclusion

1. Improvement of HF-704 has been successful to optimize the heat-up period from 53 to be 19 days with the existing standard of heating temperature. The time saved was as much as 34 days which will increase productivity of the furnace.
2. Through further planning for HF-707~710, the heat-up period is estimated to be optimized to only 14 days. This condition will increase time saved from the heat-up process and is expected to be followed by increasing productivity of the furnace.
3. Improvement of HF-704 has been successful to decrease the bottom lining as much as 68 mm and using castable material instead of insulation brick which resulted in increasing HF capacity from 30 to be 40 Tons (33.3 %).
3. This improvement will become the blue print to improve the others holding furnaces to support the increasing production capacity to be 300,000 Tpy in 2019 through enhancing the existing resources at smelting plant.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

FURNACE MODELING FOR EFFICIENT COMBUSTION GAS CIRCULATION

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Keywords: Aluminum reduction, cast house, furnace modeling, exhaust gas, circulation

Abstract

In the conventional open flame aluminum casting furnace, a large proportion of the heat generated by the burners is wasted. Proper circulation of the flame's combustion gases could increase the fraction of heat transferred to the molten metal. In this work, we develop models for the aluminum holding furnace with an aim of optimizing the exhaust gas dynamics, for a more efficient casting process. Using the Finite Volume Method (FVM), fully coupled combustion, heat transfer and fluids dynamics models of the furnace are developed, on the commercial code StarCCM+. The exhaust and burner locations of a typical furnace design are varied to examine the effect of combustion gasses recirculation on the furnace's first law efficiency. In all models, the non-premixed flame (heat source) is simulated using the Eddy Break Up (EBU), 3-step reaction mechanism. The participating media radiation models are adapted for calculating the radiation heat transfer. Turbulence is modeled using the standard K-epsilon approach. Overall, our results highlight ventilation design considerations for the casting furnace.

Introduction

The enormous energy consumption of primary aluminum production has long been a global concern. Despite recent progress in aluminum recycling -- which demands only 5% of the energy required for primary aluminum production energy-- the industry still ranks as one of the largest energy consuming sector worldwide. Evidence that climate change is a result of human activities has bolstered an increased push for improved energy management within the industry.

Energy utilization in this industry is spread across the various aluminum production processes and several energy saving strategies can be drawn for each process. However, a high impact process for energy management is the holding/casting process which is common to both primary and secondary aluminum production. In this process, molten aluminum is held at a specific temperature required to attain an adequate quality for further extrusion or forging processes [1]. To do this, a reverberatory furnace is utilized which is typically operated at a low thermal conversion efficiency of about 12% [2, 3].

The first step in any energy management program is to understand the energy source and dissipation channels. In the aluminum casting process, the gas (or oil) fired reverberatory furnaces are most widely used due to their generally higher thermal-efficiency [4, 5] -- in comparison to the resistance and induction heated furnaces. The gas-fired holding furnaces, unlike its melting counterpart, usually do not directly impinge flames on the metal [1]. Therefore, heat transfer from the flame to the metal, is predominantly controlled by radiation.

Initiatives to maximize casting/holding furnace productivity are mostly driven by economics, or more specifically, the cost per ton of produced aluminum. This topic has been approached from the operational culture [6], dross management [7], furnace selection [1] and, energy savings [8, 9, 10] perspectives. Considering that the primary aluminum production industry is very energy-intensive, every little increase in energy efficiency could translate to significant financial savings in a global scale. Hence, it comes as no surprise that the majority of efforts in improving cast house productivity has been directed at minimizing energy utilization. Some of the known energy savings options are refractory lining optimization [11], exhaust gas energy recuperation [12, 13, 14], use immersion heating [9, 12], and burner design optimization [15, 16]. However, an energy saving strategy inherent in appropriate furnace gases circulation has been long ignored. Heat transferred to the metal is not only by direct radiation or convection from the flame. The hot combustion gases within the furnace also play a role in increasing the metal's heat content. Depending on the burners' orientation and the exhaust duct's location, the hot gases could exit the furnace with a minimal circulation within the furnace. This could lead to: (1) high temperature exhaust gas which signifies low thermal conversion efficiency and; (2) hot spots/regions which could lead to a higher dross formation [7, 1]. As such, maximizing this resident time and spread within the furnace, using ventilation strategies can be advantageous to the industry [4].

Due to their allowance for inexpensive parametric studies, mathematical models have proven to be very useful in assessing heat transfer and ventilation systems for the aluminum smelting industry [17]. Unsurprisingly, many models have been built to assess furnace performance and test cast house energy saving strategies [8, 10, 15, 18, 19, 20]. These models have succeeded in showing the energy saving potential of immersion heaters [9], maximizing oxy-fuel burners' benefits [19] and, studying the effect of burner injection angle [18] and furnace roof height [10] on the overall thermal efficiency. However, to our knowledge, none of such modeling efforts have demonstrated the potential of applied mathematics in optimizing the furnace's hot gas circulation – while many experimental studies in this field have been encountered [21, 8].

The main goal of the present study is to depict the capability of modeling tools in aiding furnace design from an exhaust gas circulation perspective. To achieve this, coupled Combustion-CFD-Heat transfer models of five different furnace ventilation configurations are developed. Using these modes, the energy efficiencies of the furnace configurations are computed and compared. Calculated fluid dynamics streamlines are then used to explain the variation in energy efficiency between the furnaces.

Mathematical Modeling

To demonstrate the effect of different ventilation strategies on the furnace efficiency, coupled combustion-CFD-heat transfer models were developed for three furnace configurations on the commercial code Star CCM+. All modeled furnaces have a 40 tons aluminum capacity with natural gas-fired non-premixed burners. The all have an identical measurement of 4.5 x 3.3 x 1.7 m (LxBxh). The benchmark furnace configuration adapted is that of a typical casting furnace presented in reference [9] (See Figure 1).

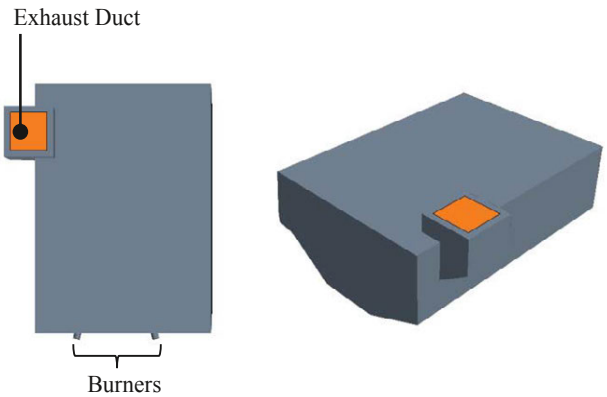

Figure 1: Benchmark furnace configuration: Configuration (a)

The typical furnace shown in Figure 1 consists of two burners with the exhaust port located at the adjacent end of the burners. One problem with this configuration is that the momentum of the flame is directed towards the exhaust duct hence, no significant circulation of exhaust gas is experienced -- the combustion gases have a minimum resident time.

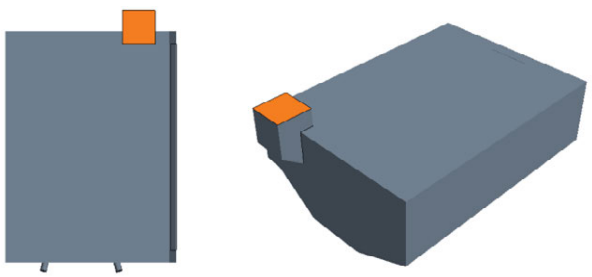

Figure 2: Furnace configuration with exhaust duct located at the opposite end of the flame direction: Configuration (b)

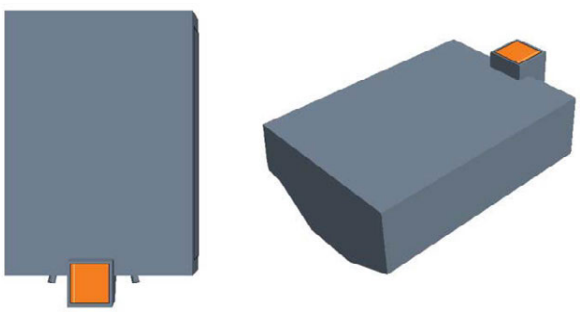

Figure 3: Furnace configuration with exhaust duct located at the same end of the burners: Configuration (c)

In the furnace configuration of Figure 2, the location of the exhaust duct is moved to the opposite transverse end of the furnace. This is expected to increase the combustion gases' travel distance and improve circulation within the furnace. The configuration shown in Figure 3 attempts to further increase the gases' resident time by placing the exhaust duct at the same end as the burners. As the pressure outlet is located at the end where the burners are located, it is expected that the combustion gasses would be forced to travel the full girth of the furnace before exiting via the exhaust duct.

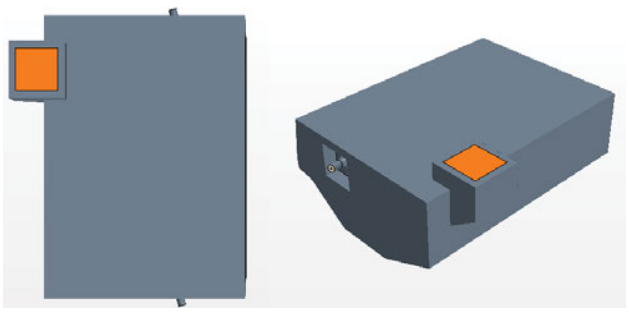

Figure 4: Furnace configurations with burners located at the longitudinal ends of the furnace: Configuration (d)

Figure 4 shows a configuration with the burners located at opposite longitudinal ends of the furnace while Figure 5 presents a three burner furnace. It is important to note that regardless of the increased number of burners, Configuration (e) has an equal amount of heat input as the previous configurations.

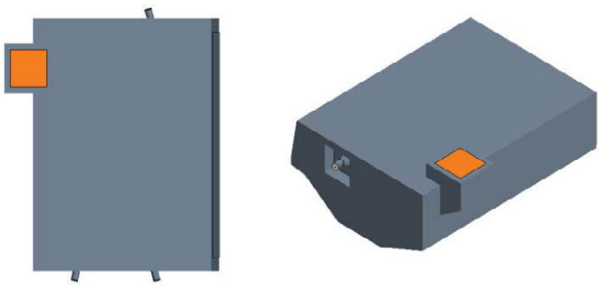

Figure 5: Furnace configuration with three burners: Configuration (e)

Each furnace configuration is modeled with the three domains: combustion gases, metal pad and refractory wall. The combustion gasses region is the region where combustion occurs and the hot gases are dissipated. The metal pad is connected to this region using a baffle interface which permits heat transfer between the combustion gas region and the metal pad but not mass transfer. The interface between the combustion gases and the furnace walls also permits heat transfer only. A clearer depiction of each region can be found in our previous publication [9].

The specific mathematical models and approaches adapted are available in our previous publication [9]. Polyhedral meshes with

a base size of 0.05m are adapted for the mesh continuum. Mesh sensitivity analysis and validations are also presented in the reference [9].

Each furnace is modeled with total fuel (Methane) and air flow rates of 0.0238kg/s and 0.4138kg/s respectively. This is equivalent to a heat source of about 1300 kW. Fuel and air are supplied at ambient conditions. Temporal density, specific heat, thermal conductivity and viscosity of air and methane used can be found in reference [22].

Results

The flow velocity-streamline profile of Configuration (a) shown in Figure 6 portrays minimum recirculation of the combustion product gases within the furnace. Even though the left hand side (LHS) burner is tilted towards the right hand side (RHS) of the furnace, a combination of the pressure difference between the inner and outer portion of the furnace and the interaction between the two flames, convey the combustion gases from the LHS burner directly to the exhaust duct. The flame from the RHS burner interacts with the LHS flame which forces the combustion gases to diffuse away from the exhaust duct. However, this effects a minimal combustion gasses recirculation according to the temperature profile shown in Figure 6 where the LHS of the furnace has a relatively higher temperature gradient.

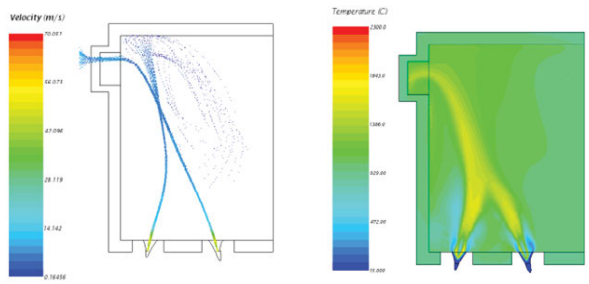

Figure 6: Velocity and temperature profile of Configuration (a)

In Configuration (b), where the exhaust duct of Configuration (a) is moved to the RHS of the furnace, the dynamics of the flame's combustion fluid is altered as shown in Figure 7. The combustion gases of the LHS burner are forced to recirculate upon interaction with the RHS flame. Even though the inclination of the RHS burner is towards the LHS of the furnace, the pressure difference generated by the exhaust duct tilts the flame towards the RHS. This favors an increased spread of the combustion gases within the furnace, with a slight bias towards the LHS of the furnace. This is evident from the temperature profile that indicates the more uniform temperature distribution within this furnace.

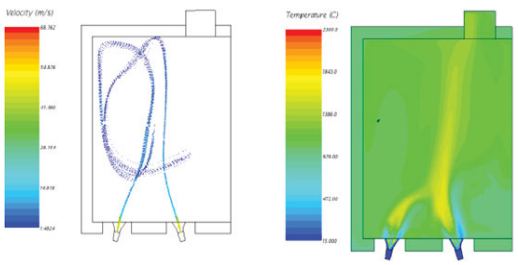

Figure 7: Velocity and temperature profile of Configuration (b)

Figure 8 shows that placing the exhaust port in between the two burners, as demonstrated in Configuration (c), further improve the dynamics of the combustion gases within the furnace. From the flow velocity profile in Figure 8, a full circulation of both burners' combustion gases is observed. This is expected since the burners' momentum is directed away from the exhaust port so the gases would have to reverse in order to exit the furnace. Nonetheless, the temperature profile of Configuration (c) indicated a lower temperature contour. This observation can be justified by watching the combustion completeness along the diffusion flame in the furnace. Due to the closeness of the exhaust duct to the burners, some of the hot gases exit the furnace immediately after leaving the burner with incomplete combustion. As the furnace is fired by non-premixed flames, some un-burnt fuel and air also exit the furnace immediately, as an evidence, the flue gases showed a volume concentration of methane of 1.16×10^{-3} kmol/m^3 versus 1.25×10^{-5} kmol/m^3 in the configuration (b). As such, the heat energy circulated within the furnace is reduced and temperature within the furnace is not uniform.

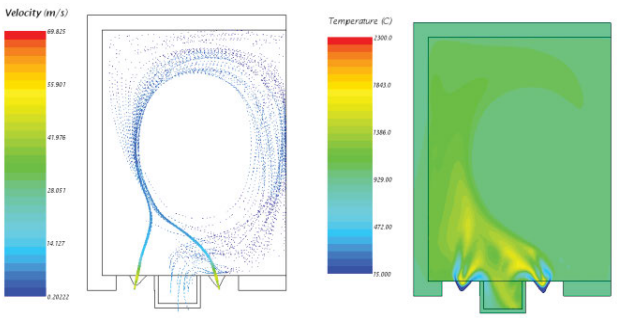

Figure 8: Velocity and temperature profile of Configuration (c)

The velocity profile of Figure 9 shows that moving the LHS burner of Configuration (a) to a position adjacent to the RHS burner (see Configuration (d)) causes a minimum interaction between the flames from the two burners. The combustion gases from the bottom burner (according to the orientation of Figure 9) are conveyed directly to the exhaust duct while the gasses from the top burner are driven away from the duct then reversed by the pressure difference to exit the furnace. The temperature profile of Figure 9 suggests a temperature profile with improved uniformity in comparison to Configuration (a).

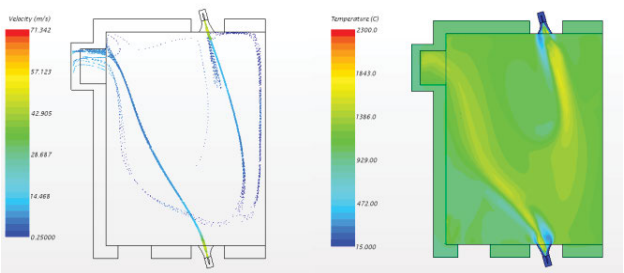

Figure 9: Velocity and temperature profile of Configuration (d)

Including another burner to Configuration (a), as demonstrated in Configuration (d), evidently increases the interaction between the flames within the furnace (see Figure 10). This improved interaction coupled with the extended reach of the flame results in an increased temperature distribution uniformity which is depicted in the temperature profile of Figure 10. Since the three-burner

furnace of Configuration (e) is supplied with the same amount of fuel as the two-burner furnaces, the observed increased uniformity in the furnace's temperature can be attributed to solely to the improved distribution of heat within the furnace.

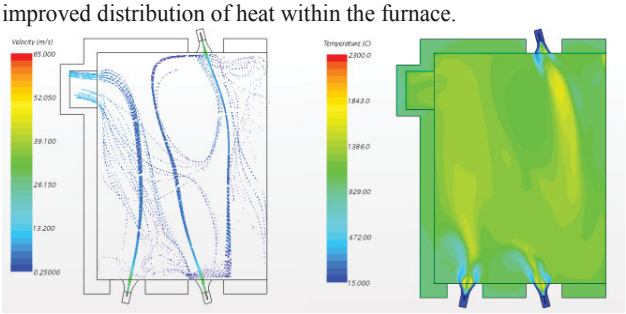

Figure 10: Velocity and temperature profile of Configuration (e)

A breakdown of the heat transfer for each furnace configuration is presented in Table 1. From our model's calculation, the reference furnace (Configuration (a)) has a first law efficiency (based on the amount of heat transferred to the metal) of 13.4%. Configuration (b) has a slightly higher thermal efficiency of 15.6% due to the improved combustion gas circulation it offers. The thermal efficiency of Configuration (c) is about half that of Configuration (a). The relatively low quantity of heat transfer to the metal, in the Configuration (c) furnace, can be attributed to the wasted un-burnt fuel via the exhaust duct which is in close proximity to the burners. Regardless of the improved heat distribution in Configuration (d) and Configuration (e) visible from Figure 9 and Figure 10, the thermal efficiencies of these furnaces are lower than the reference furnace. This is due to the increased heat losses via the doors of the furnace. Figure 9 and Figure 10 show that the top furnaces are inclined at an angle that causes the flame to impinge on the furnace's door. As the furnace door in modeled as a wall thinner and having less thermal resistance, regardless of the improved heat distribution, most of the heat from that burner is lost via this channel.

Table 1: Heat transfer breakdown of all furnace configurations

	(a)	(b)	(c)	(d)	(e)
Exhaust Heat (KW)	940.78	977.66	890.21	964.99	950.57
Metal Pad Heat Flow (KW)	174.24	202.15	76.79	94.03	114.74
Wall Temperature (C)	1054	1075	904	1028	923
Metal Pad Heat transfer Eff.	13.4%	15.6%	5.9%	7.2%	8.8%

Table 1 also highlights an increase of metal heat transfer efficiency with furnace wall temperature. A higher temperature wall signifies a lower thermal gradient between the flame and the wall which favors an increased heat transfer from the flame to the metal. Also, as the temperature of the wall is higher than that of the metal, heat is re-radiated from the wall to the metal. Hence, a higher wall temperature allows for an increased heat transfer from the wall to the metal. This is a key furnace feature which is adequately captured by our model.

Conclusion

CFD models are used to examine the effect of varying the ventilation configuration of a furnace on its heat distribution and thermal efficiency. Five different furnace configurations were considered. Our models suggest that the orientation and location of the burners relative to that of the exhaust duct affects the heat distribution and thermal efficiency of the furnace. However, an improved combustion gases circulation does not necessarily translate to an increase in thermal efficiency. This was highlighted by a configuration which improved combustion gases' recirculation but also portrayed an increased CH_4 concentrations in the exhaust flow and hence a reduced fuel efficiency. Metal pad heat transfer efficiency was depicted to increase with the furnace's wall temperature.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

FURNACE PRESSURE CONTROL TECHNOLOGY FOR FUEL EFFICIENCY

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Keywords: Furnaces, Energy Efficiency, Carbon Emissions, Greenhouse Gases, Furnace pressure, HFT

Abstract

This paper covers the results of the first trials of an invention called the Heat Flow Transmitter. A pilot project on a casthouse furnace, using a technology package that includes the Heat Flow Transmitter (HFT) and an air curtain, succeeded in reducing the fuel cost by 33%, and is expected to provide a reduction of carbon emissions of 3,200 tons/year for the casthouse.

Introduction

With fuel efficiency, there is no single solution that will solve everything. To obtain a fuel efficient furnace, all components must function efficiently and all operations must be under control. A hot super furnace waiting with no load is not fuel efficient, neither is a well insulated furnace burning high amounts of excess air, nor is a furnace with a negative atmosphere pressure inducing cold air infiltration, and so on. A single item cannot guarantee good performance but a single defective item will kill the fuel efficiency. Today's technology provides all means to track furnace consumption and give warnings when a system fails and consumption goes up. After setting up a fuel efficient tracking program, we can identify the recurring weak points and correct them. One of the most important recurring problems is furnace pressure control and it is the subject of this paper.

Potential for Savings

Our survey on fuel consumption, covering 50 liquid aluminum furnaces from 7 different casthouses, concluded that the cost of excess fuel consumed varied from $33,000 to $320,000 per furnace per year. In most furnaces surveyed, energy was wasted in the form of cold air infiltrating the furnace, being heated and going out through the stack. It is generally agreed that furnace pressure control is determinant in limiting cold air infiltration and is an essential component to obtain fuel efficiency.

Fig 1 Furnace Pressure Control Damper

Present Furnace Pressure Control Technology

Furnace pressure is normally controlled by a mechanical damper and/or an air curtain mounted in the stack. The damper opening set-point modulates in function of a process value signal from a transmitter recording furnace pressure. Alternately, this function can also include other variables, such as the burner firing power. The problem with the existing technology is the very small 0,005 kPa (0,02 in.wc) set-point required to control the damper, since reading tends to drift and requires constant recalibration. The pressure value being measured can also be disrupted by random noise from the dynamic pressure of the burner flame.

Fig. 2 Casthouse environment

Measuring furnace pressure is like weighing a feather in the dusty and turbulent environment of a casthouse. Nonetheless up to now, it was the best available system on the market.

Cost of Air Infiltration

As previously mentioned, all aspects of the furnace equipments need to be well tuned in order to maintain fuel efficiency. Furnace assessment by an outside source can reveal deficiencies that have gone unnoticed for years such as the pressure transmitter and its calibration. Next time you are in the casthouse, you can easily perform this simple test: On the control panel, locate the furnace atmosphere pressure reading; this value comes from the furnace pressure transmitter, most likely a Rosemount or N.A. Mfg. Epic instrument. This value is the process value that controls the furnace pressure control damper. When the burners are off and the doors open, it should read zero, which is often not the case. I have often seen it as being off by 0,07in.wc. (0,017KPa) or more. Let's calculate the impact of such a pressure offset on fuel consumption.

Fig.3 Typical openings leading to air infiltration

If the zero reading on the instrument is offset by +0,07 in.wc. the actual furnace pressure will be -0,05 in.wc (0,012 KPa)instead of the desired +0,02 in.wc. This negative pressure will draw air into the furnace at a velocity of 15 ft/sec. Assuming the total opening area in the furnace walls including pour spout, door seals, etc. totals 1 ft², then 54,000 scfh (1500Nm³/h) of cold air will be drawn into the furnace. The cooling effect of the cold air will consume 2,835,000 Btu/h (2.7 GJ/h) at a cost of 85 litres/h of oil or (720,000 litres/year). This calculation (presented below) is a rough approximation; however it gives an idea of the magnitude and cost of air infiltration and the need to properly maintain furnace pressure.

The new HFT Furnace Pressure Control Technology

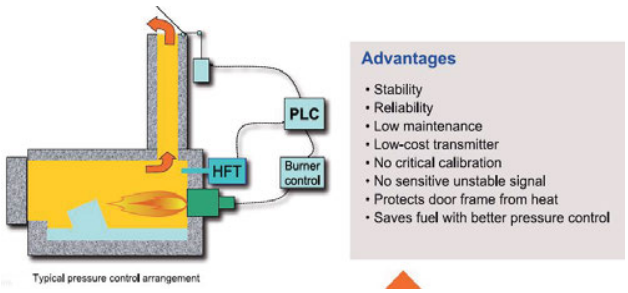


Fig.4 HFT Arrangement Schematic

The Heat Flow Transmitter (HFT) is a new invention and is capable of supplying a control signal that replaces the traditional furnace atmosphere pressure gauge. The Hatch HFT will supply a reliable detection of cold air entering the furnace or of the amount of hot gas escaping the furnace. This information can then be used in the algorithm driving the furnace pressure control system.

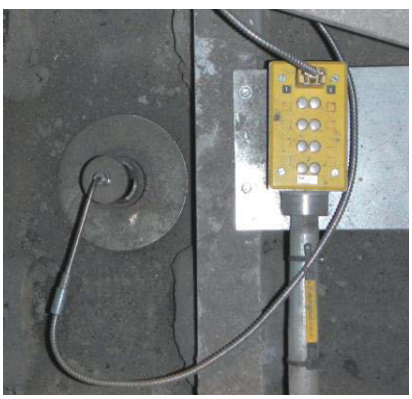


Fig. 5 HFT mounted in furnace wall

HFT Pilot Project

A pilot project to evaluate the HFT technology was undertaken on a primary aluminum furnace. The 82T tilting furnace has 2 (12MMBtu/h) cold air burners and is equipped with mass-flow combustion control. The HFT transmitters were installed in the same receptacles previously used for the furnace pressure transmitters, and the HFT transmitter signal connected to the PLC control. A special algorithm was used for the interpretation of the Process Value from the HFT, combining it to the firing rate signal of the burners in order to calculate the Control value that drives the pressure control damper. Overall, this addition required only a small modification to the furnace hardware.

Fig 6 Air curtain piping

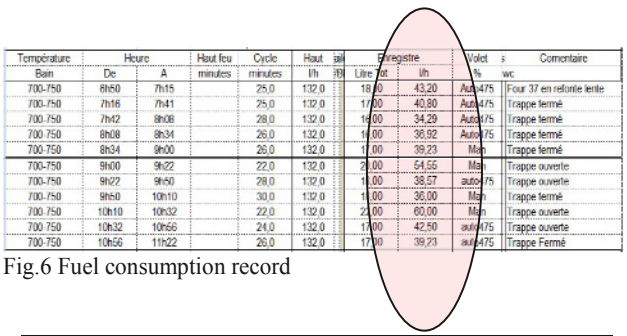
Addition of an Air Curtain

During the energy audit, a second problem was detected on the furnace. Under certain conditions, it was not possible to maintain positive furnace pressure. When the furnace was running on low fire (heating only with pilots), the stack damper in the closed position did not seal tightly enough to prevent leakage of the flue damper which caused negative furnace pressure and resulted in cold air infiltration. This was corrected by adding an air curtain, installed in the stack breach, before the pressure control damper. The new air curtain uses the air available from the combustion air blower. This blower is not fully used when the burners are on low fire or shut off.

Testing the HFT

During start-up, some tests were performed to observe the effects of the new HFT equipment. Among those, simulate a defect in the furnace enclosure sealing and record the behaviour of the HFT control. The furnace is equipped with a charge well consisting of a side opening with mechanised cover. This was used to test the efficiency of the new pressure control. The burners were set in holding mode, maintaining metal at a steady temperature. We have recorded the fuel consumption during periods of time where the furnace operated with the charging trap cover closed; then opened the charging trap cover and measured the amount of excess fuel consumed. The HFT was then activated and consumption recorded again. The test was then repeated in random sequences. The results confirmed that the HFT detected the leak and corrected the stack damper position to compensate for the defect, thus maintaining best practice consumption.
Fig.6 shows fuel consumption during the test and Fig.7 the analysis of this data. One can see that when charge door is shut

closed, best practice consumption is maintained, with or without the HFT, this was to be expected. When the trap door is open without the HFT system activated consumption increases by about 50% and when the HFT system is activated, the consumption reverts to best practice. This indicates that the pressure loss from the open trap door is detected by the HFT, and the stack damper corrected accordingly.



Température Bain	Heure De	Heure A	Haut feu minutes	Cycle minutes	Haut lb/h	Enregistre Litre tot	Enregistre lb/h	Volet %	Commentaire wc
700-750	6h50	7h15		25,0	132,0	18,00	43,20	Auto475	Four 37 en refonte lente
700-750	7h16	7h41		25,0	132,0	17,00	40,80	Auto475	Trappe fermé
700-750	7h42	8h08		28,0	132,0	16,00	34,29	Auto475	Trappe fermé
700-750	8h08	8h34		26,0	132,0	16,00	36,92	Auto475	Trappe fermé
700-750	8h34	9h00		26,0	132,0	17,00	39,23	Man	Trappe fermé
700-750	9h00	9h22		22,0	132,0	21,00	54,55	Man	Trappe ouverte
700-750	9h22	9h50		28,0	132,0	15,00	38,57	auto475	Trappe ouverte
700-750	9h50	10h10		30,0	132,0	15,00	36,00	Man	Trappe fermé
700-750	10h10	10h32		22,0	132,0	23,00	60,00	Man	Trappe ouverte
700-750	10h32	10h56		24,0	132,0	17,00	42,50	auto475	Trappe ouverte
700-750	10h56	11h22		26,0	132,0	17,00	39,23	auto475	Trappe Fermé

Fig.6 Fuel consumption record

Furnace fuel consumption in Litres of oil per hour		
	No HFT	With HFT
Charge trap door shut	39, 36	39
Charge trap door open	54, 60	38, 42

Fig.7 Table

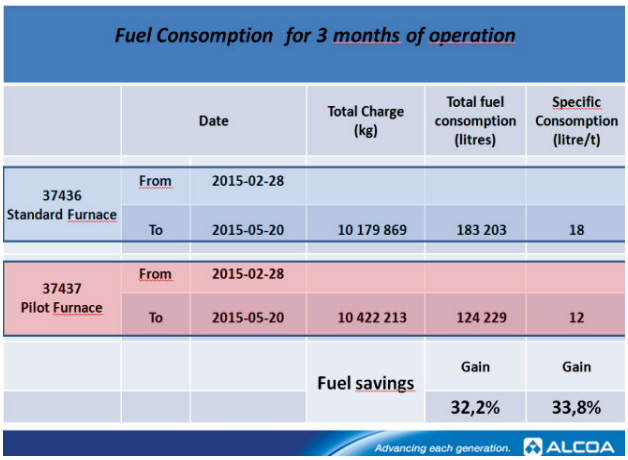
Testing the Air Curtain

Performance of the air curtain was also investigated. Again burners were set in holding mode, maintaining metal at a steady temperature. As burners were cycling on and off we measured consumption and recorded burner cycling on a chart (Fig.8). On the left side of the chart, the air curtain was off and on the right side the curtain was on. It becomes clear that with the air curtain activated the furnace conserves its heat for a longer period before requiring the burner, resulting in a reduction in consumption from 55 litres per hour (without air curtain) to 35 litres per hour (with air curtain) or an improvement of 36%.

Fig.8 (Burner firing) vs (time)

Consumption during Production

The furnace was handed over to production and consumption evaluated. Since these furnaces work in pairs and are connected to a common casting machine it is easy to compare performance between this pilot furnace and the standard. After confirming a similar amount of metal re-melted and of metal cast for the two furnaces, we compared their respective fuel consumption. The result exceeded our expectations.



Fuel Consomption for 3 months of operation

		Date	Total Charge (kg)	Total fuel consomption (litres)	Specific Consomption (litre/t)
37436 Standard Furnace	From	2015-02-28			
	To	2015-05-20	10 179 869	183 203	18
37437 Pilot Furnace	From	2015-02-28			
	To	2015-05-20	10 422 213	124 229	12
			Fuel savings	Gain	Gain
				32,2%	33,8%

Advancing each generation. ALCOA

Fig.9 Fuel Consumption for 3 Months of Operation

Casthouse records indicated that under the same operating conditions over a three month period, the average fuel consumption of the pilot furnace was reduced by 33%. Needless to say this is now being rolled out on the other casthouse furnaces. The fuel consumption for this casthouse was expected to be 4,000,000 litres of oil for 2016, therefore a saving of 33% will reduce the consumption for this casthouse by 1,200,000 litres/year The expected reduction in fuel will also impact GHG emissions by reducing CO2 emissions by 3,200 tons/year

Conclusion

"If you are not part of the solution you are the problem". Global warming is the responsibility of us all, and Carbon emission reduction by itself should justify the business case for fuel efficiency projects.

Calculations

(1) Assuming loss coefficient of a cracked furnace door or other furnace opening as (100%) then a negative furnace pressure of -0.05 (in.wc.) will induce cold air at a velocity of 15 ft/sec. From $P = \rho \times (V/1097)^2$ [2]
- 1097 x(0,05in.wc/0,075lb/ft³)½ =900 ft/min. = 15 ft/sec. (4,5m/s)

(2) Air leaking into a 1 ft² wall opening at a velocity of 15 ft/sec results in 54,000 ft³/h (1500 Nm³/h) of air infiltrating the furnace
- 15 ft/sec. x 3600sec/h x 1 ft² = 54,000 ft³/h

(3) A cold air infiltration of 54,000 ft³/h heated to 1460 °F with 50% combustion efficiency will cost 2,800,000 Btu/h

- 54,000 ft³/h x 1400 °F x 0,075 lb/ft3 x 0,25 Btu/lb / 50% = 2,835,000 Btu/h (3GJ)

(4) Heat content of oil

- 17,855 Btu/lb =33,500Btu/l= 10,000kcal/l

(5) Fuel consumed to produce 2,835,000 Btu/h (3,0 GJ)

- *2,835,000 Btu/h / 33,500Btu/l x 8640 h/year = 720,000 l/h*

(6) Saving 1,200,000 litres /year of oil saves 3,264 ton /y of CO^2 [1]

- 1,200,000liters x (0,85 Kg/Liter) x (3,2 kg CO2/Kg oil) = 3,264,000 Kg

(7) Heat content of Natural gas:

19700 btu/lb 1,000 btu/ft3 11,000kcal/l

References

[1] North American Combustion Handbook, (North American Mfg. Co. Cleveland, Ohio)

[2] Robert Jorgensen editor, (Fan Engineering Buffalo Forge Company. Buffalo, New York)

Light Metals 2016
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TMS (The Minerals, Metals & Materials Society), 2016

CALCULATED ALUMINUM OXIDATION RATES DURING ROTARY FURNACE MELTING THROUGH FLUE GAS ANALYSIS – PART TWO

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Keywords: Aluminum, Oxidation, Melt Loss, Combustion, Flue Gas, Furnace, Oxygen, Analysis

Abstract

Continuous flue gas composition measurements (CO, CO2, O2, H2) were made for four rotary aluminum melt furnace heats, with air/oxy/gas combustion. In many cases, during rotary furnace aluminum melting, the furnace atmosphere becomes increasingly reducing (higher CO and H2, lower O2 and CO2) as the heat progresses. This suggests that aluminum is being oxidized.

For these four heats, melting aluminum chips, the measured flue gas CO, CO2, O2 and H2 compositions were utilized, along with C, H and O balance equations, to calculate instantaneous aluminum oxidation rates throughout the heat.

Calculated total aluminum oxidation loss ranged from 0.5% to 1.4%. Instantaneous oxidation rates are significantly higher at the end of each heat, corresponding to highest temperature just before tapping. Oxidation rate correlates most strongly with measured flue gas temperature, and does not seem to correlate with % O2 participation or % excess oxidizer from the burner settings.

Introduction

Flue gas analysis, consisting of measuring CO, CO2, O2 and H2 concentrations in the flue gases, can be utilized to improve combustion system performance in fuel-fired melting or heating furnaces. For example, with an empty furnace or under non-reactive furnace load conditions, flue gas concentration measurements can allow one to calibrate and/or confirm the combustion control system settings, to achieve the desired combustion atmosphere (oxidizing, reducing or neutral). Flue gas analysis can also allow one to calculate furnace infiltration air flow rate, and guide in adjusting the flue damper or furnace pressure control system to minimize infiltration air. Flue gas analysis can also help to diagnose and correct problems with combustion control equipment. Examples of the application of flue gas analysis to achieve these goals, under empty furnace or non-reactive load conditions, have been summarized in a prior publication [1].

Flue gas analysis can also be utilized for a fuel-fired melt furnace, where the load has chemical interaction with the furnace gases (reactive load). One example is for fuel-fired rotary furnace aluminum melting.

Flue gas analysis has been performed on several rotary aluminum melting furnaces. Flue gas CO, CO2, O2 and H2 compositions, along with flue gas temperature, are recorded throughout the batch melting cycle. It has often been observed that even though a reducing burner setting is employed (example lambda 0.95 or 5% reducing), in many cases, as the heat progresses, the furnace atmosphere becomes increasingly reducing (increased CO and H2, reduced O2 and CO2). Since the burner settings have not changed, these increasingly reducing conditions indicate that aluminum oxidation must be occurring, acting as a "sink" for O2, as the heat progresses. This effect is typically more pronounced when melting dross or light gauge scrap, than with heavier scrap, which suggests that aluminum oxidation rates are higher for higher surface area material types.

It is possible to utilize the measured flue gas concentrations, along with measured combustion natural gas and O2 flow rate, to calculate the instantaneous aluminum oxidization rate (kg/min) throughout the melt cycle. It can be useful to accurately identify the actual amount of aluminum oxidized during melting, since the tapped vs charged weight-based definition of 'melt loss' includes other apparent losses such as Al trapped in the dross/slag, any oxides originally contained in the charge material, and any other moisture or impurities contained in the charge material. It is interesting to observe how calculated aluminum oxidation rate changes throughout the heat.

In this paper, 4 rotary aluminum melt furnace heats, melting aluminum chips, are analyzed, and calculated aluminum oxidation rates are presented and discussed.

Definitions and Combustion Equations: for air/oxy/gas combustion

G = natural gas flow rate NM3/hr (SCFH). G =1 as basis.

Natural gas is assumed to be 100% CH4.

A = total air/gas ratio (combustion air plus infiltration air).
Ac = Combustion air/gas ratio; Ai = Infiltration air/gas ratio
A = Ac + Ai
X = oxygen/gas ratio (from pure i.e. purchased O2)

Perfect (stoichiometric) combustion equation:

$$1\ CH4 + 2\ O2 \rightarrow 1\ CO2 + 2\ H2O$$

Note that for complete CH4 combustion, O2 is distributed equally between C and H.

The term Lambda (L) has been developed to quantify the total amount of combustion O2 supplied:

Lambda = 1.0: stoichiometric combustion (neutral)
Lambda > 1.0: oxidizing conditions
Lambda < 1.0: reducing conditions

For air/oxy/gas combustion, Oxygen Participation (P) expresses the amount of combustion O2 provided by pure (purchased) O2 versus air. Air is assumed to be 21% O2 and 79% N2.

Relating the oxy/gas ratio X and the total air/gas ratio A to lambda (L) and O2 participation (P):

P = X / (0.21A + X) and L = (0.21A + X) / 2
Or, X = 2PL and A = (2L – X) / 0.21

For example, for air/gas combustion X = 0 and P = 0. For 100% oxy/gas combustion, P = 1 and A = 0 with respect to combustion air supplied by blower (Ac). For air/oxy/gas combustion, P will be between 0 and 1.

During "empty furnace" or non-reactive load conditions, the following relations apply:

Oxidizing Conditions, Non-Reactive Load:

$$1\ CH4 + X\ O2 + A\ (0.21O2 + 0.79N2) \rightarrow 1\ CO2 + 2\ H2O + 0.79A\ N2 + (0.21A + X - 2)\ O2$$

Reducing Conditions, Non-Reactive Load:

Incomplete products of combustion will include CO and H2 in addition to CO2 and H2O:

$$1\ CH4 + X\ O2 + A\ (0.21O2 + 0.79N2) \rightarrow d\ CO2 + m\ CO + w\ H2O + h\ H2 + 0.79A\ N2 + 0\ O2$$

For analyzing reducing conditions, it can be useful to assume an equal split of O2 between C and H, or alternately, one could utilize the water-gas shift reaction equilibrium relations to calculate the split of CO and H2 as a function of gas temperature.

From these relations, one can generate equations to express dry flue gas concentrations (CO, CO2, H2, O2, N2). Flue gas compositions are reported as dry basis (H2O removed), as water is removed from the gas sample before going to the analyzers.

Calculating Al Oxidation Rate from Flue Gas Analysis

During aluminum melting, a general equation to cover either oxidizing or reducing conditions is:

$$1\ CH4 + X\ O2 + A\ (0.21O2 + 0.79N2) \rightarrow d\ CO2 + m\ CO + w\ H2O + h\ H2 + 0.79A\ N2 + z\ Al2O3 + y\ O2 + a\ H2$$

A distinction is made between h which indicates the gaseous H2 evolving into the furnace gases, and a which indicates the H2 (H) dissolving into the (molten) aluminum. However, based on well-known values for maximum H solubility in molten aluminum, it can be shown that a is negligible compared to h, and a is assumed to equal zero for the purpose of these calculations.

From the above equation, mass balances for C, H and O are:

C: d + m = 1
H: w + h = 2 (assumes CH4 is the only source for H)
O: X + 0.21A = d + (m/2) + (w/2) + (3z/2) + y

Dry flue gas volume = d + m + h + 0.79A + y = 1 + h + 0.79A + y

Measured flue gas concentrations are (dry basis):

[CO] = m / (1 + h + 0.79A + y)
[CO2] = d / (1 + h + 0.79A + y)
[H2] = h / (1 + h + 0.79A + y)
[O2] = y / (1 + h + 0.79A + y)

And, by difference:

[N2] = 1 – [CO] – [CO2] – [H2] – [O2] = 0.79A / (1+h+0.79A+y)

To solve for the 7 unknowns (A, d, m, w, h, z, y), we have 7 equations (4 measured flue gas concentrations H2, O2, CO2, CO, plus 3 element balance equations C, H and O). The oxy/gas ratio X is known, since both are measured by accurate flow meter. We treat A as an unknown, since infiltration air is unknown, and the combustion air flow measurement is typically less accurate than gas or O2.

With these 7 equations and 7 unknowns, we can solve for all terms, and then aluminum oxidation rate is determined by multiplying z times the natural gas flow rate.

Data and Observations

Flue gas dry CO, CO2, O2 and H2 concentrations were measured along with flue gas temperature at 10 second intervals throughout four batch rotary furnace heats, melting aluminum machining chips. Prior to charge, the aluminum chips had been centrifuged but otherwise not dried through any post-processing such as a delacquering kiln. There were two charges of chips per heat.

Due to build up on the flue thermocouple, the indicated flue gas temperatures were judged to be about 100-200 C too low. While not accurate on an absolute basis, the measured flue gas temperatures are judged to be reliable on a relative basis.

Due to some additional refractory wear around the door seal, there was a larger air gap between the furnace door and barrel, compared to 2014 conditions when a prior study had been performed [2]. Consequently, there was a higher level of infiltration air than in the prior study [2]. This is summarized in the V-curve shown in Figure 1 – note the higher measured flue gas [O2] concentrations for the empty furnace, for 2015 vs 2014.

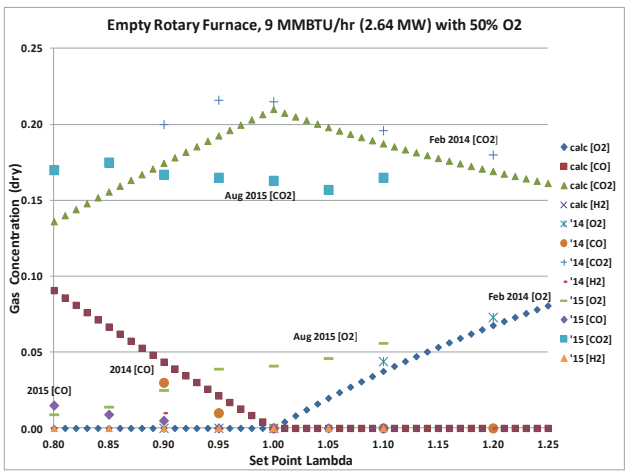

Figure 1. Measured vs Calculated Flue Gas Compositions for Hot Empty Rotary Furnace, 2014 vs 2015.

There were times when both O2 and CO were measured in the flue gas. This indicated that some of the infiltration air must have been rising directly up the flue (not all pulled into the furnace). Consequently, the combined oxidizing/reducing equation shown above (includes O2, H2 and CO species) was utilized in calculations.

In Table 1, an empty furnace calibration of the calculation procedure is shown. For the 4 burner settings utilized, with a hot empty furnace, the calculated fraction of total O2 reacting with Al is essentially zero, to within less than +/- 1%. (In other words, essentially 100% of the O2 is accounted for). This suggests that these measurements and calculation method should be reasonably accurate, for determining instantaneous aluminum oxidation rates during melting.

Table 1. Empty Furnace Measurements and Calculations

Heat #	MW	O2 %	Lambda	[O2] %	[CO2] %	calc Al oxid kg/hr	calc frac of total O2 rctng with Al
1	2.00	50%	0.95	7.1	13.3	2.1	0.0028
3	2.00	50%	0.85	5.9	13.6	-3.1	-0.0043
2	2.00	80%	0.95	12.6	14.8	-8.1	-0.0097
8	2.64	80%	0.85	6.3	21.0	-7.1	-0.0087

Figure 2 shows measured flue gas CO, CO2, H2 and O2 concentration along with flue gas temperature, throughout heat #3. Heat #3 is shown as a first example (2 MW or 7 MMBTU/hr, 50% O2, and Lambda = 0.85). In the early stages after charge #1, [O2] slowly increases, this indicates that the remaining oil in the chips is being burned away (indeed, a small orange flame was sometimes observed during this time). During this time period, [CO2] is slowly decreasing, and the small initial amount of [CO] and [H2] fades. Eventually all flue gas concentrations level off, at this point the calculation of aluminum oxidation rate was begun. Prior to this time, aluminum oxidation rate was assumed zero, as the oil in the chips was the "sink" for O2, not aluminum (relatively cold temperatures, chips still solid).

The same effect is seen, for a shorter time period, as the second charge is heated. As the heat progresses, temperature is steadily increasing, at an essentially constant rate. Near the end of the heat, there is sharp drop in [O2] and [CO2], while [CO] and [H2] rise sharply. During this same time, the flue temperature rises at a steeper rate. This indicates that aluminum oxidation rate has sharply increased, during this time period, corresponding to roughly the last ~10 minutes of the heat. This is commonly referred to as the "breakdown" phase, as the heated aluminum is converted from solid to liquid, and no longer "climbs the walls" during rotation. The furnace operators are observant, well trained and experienced, they know that at this point the furnace is ready to tap – if heating is continued excessively, melt loss will be further increased.

This same general progression in flue gas composition and temperature was observed for all 4 heats. Heat #3 is shown here as a first example. Note, we consistently observed "baseline" H2 concentration of -1%, this was assumed to be a calibration offset and assigned a value of zero for calculation purposes.

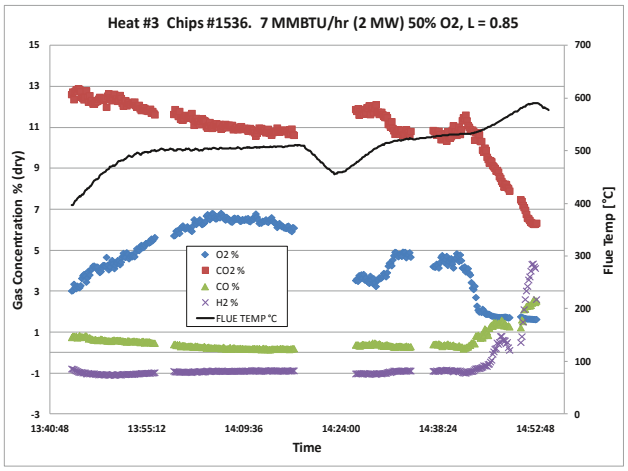

Figure 2. Flue Gas Composition and Temperature for Heat #3

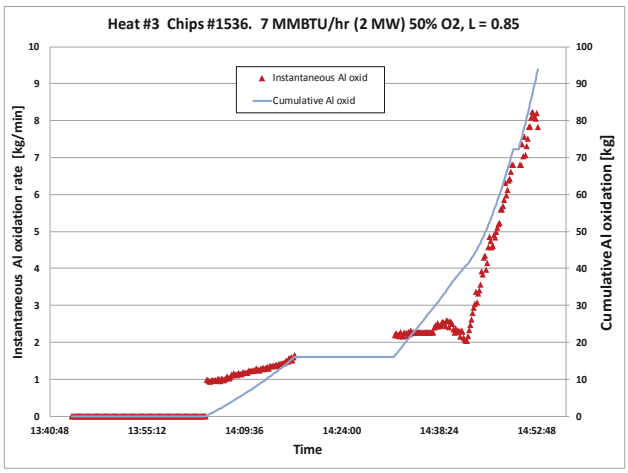

Figure 3. Calculated Aluminum Oxidation for Heat #3

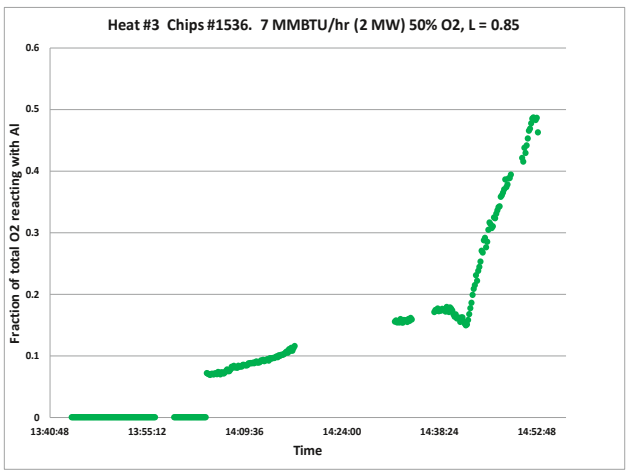

Figure 4. Calculated Fraction of Total O2 Reacting with Al, for Heat #3

Calculations and Results

Figure 3 shows calculated instantaneous aluminum oxidation rate (kg/min) and cumulative total kg of aluminum oxidized, throughout heat #3. Cumulative total kg oxidized is calculated by summing each 10 second segment. Calculated instantaneous Al oxidation rate climbs slowly and steadily, as charge 1 and charge 2 are heated. Then there is a sharp increase in aluminum oxidation rate, for the final ~10 minutes of the heat. Figure 3 shows that most of the total (cumulative) aluminum oxidation for the heat occurs in this final ~10 minutes (breakdown phase).

Figure 4 shows the corresponding calculated fraction of total O2 reacting with aluminum, for heat #3. The shape of Figure 4 follows the shape of Figure 3. At the end of heat #3, nearly 50% of total O2 supplied to the furnace (from oxygen, combustion air and infiltration air, total) is shown to be reacting with aluminum!

Heat #3 showed the highest instantaneous oxidation rates, and highest total cumulative aluminum oxidation, and also exhibited the highest temperatures. All 4 heats showed the same general pattern as shown in Figures 2, 3 and 4 for heat #3.

As a second example, Figures 5, 6 and 7 show flue gas compositions, temperature, and calculated aluminum oxidation for heat #8 (8.5 MMBTU/hr or 2.64 MW, 80% O2, lambda = 0.85).

In both heats shown, one can see where the operator briefly opened the door near the end of each heat, to look at the load. And in heat #3, one can see two brief time periods where flue gas sampling was interrupted, so that we could un-clog the sample probe.

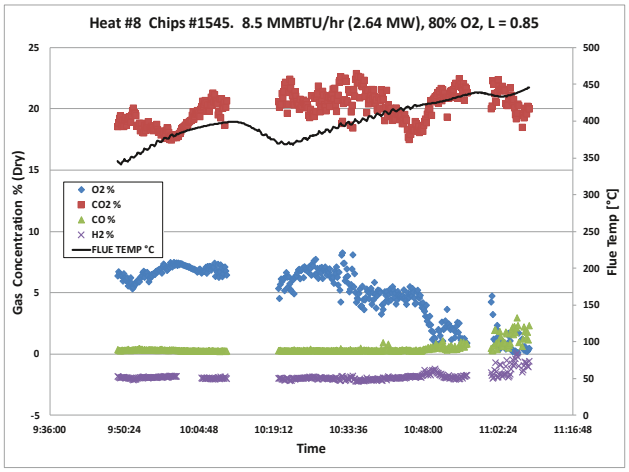

Figure 5. Flue Gas Composition and Temperature for Heat #8

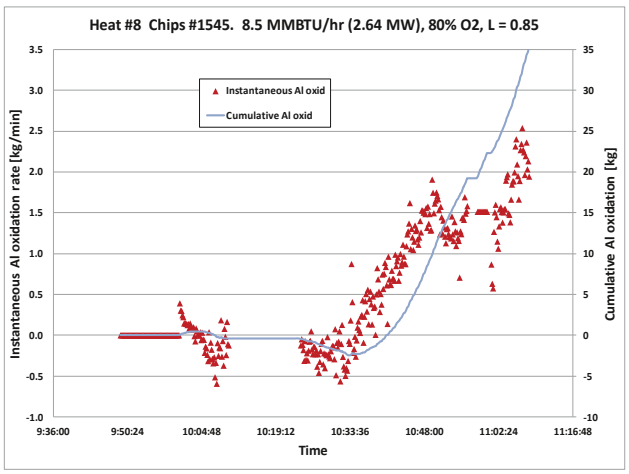

Figure 6. Calculated Aluminum Oxidation for Heat #8

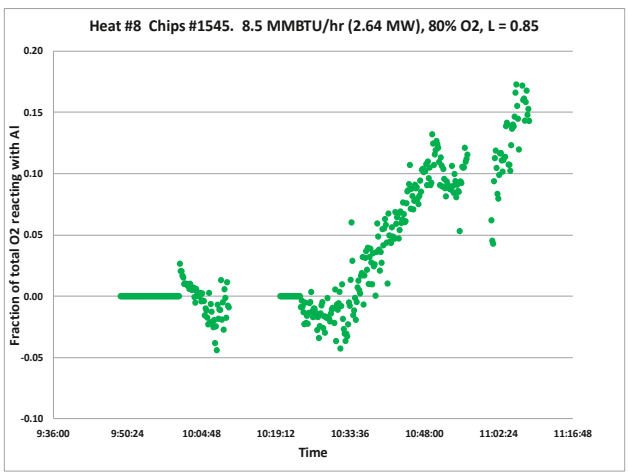

Figure 7. Calculated Fraction of Total O2 Reacting with Al, for Heat #8

Figure 8 shows calculated instantaneous aluminum oxidation rate as a function of flue gas temperature for all 4 heats, superimposed on the same graph.

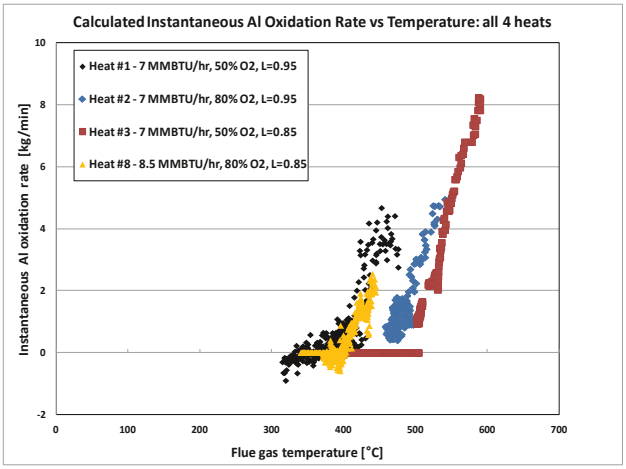

Figure 8. Aluminum Oxidation Rate vs Temperature, all 4 Heats

It is interesting to see in Figure 8 that all 4 heats exhibit a very similar pattern. Calculated oxidation rate rises slowly and steadily early in the heat, then sharply increases for the last ~10 minutes, prior to tapping. Calculated oxidation rate seems to correlate very clearly with temperature, and final oxidation rate for each heat is the highest, corresponding to the final (tap) temperature.

For two of the heats (#1 and #8), early in the heat some slightly negative aluminum oxidation rate values were calculated. These were judged to result from "data scatter" and experimental error, and the contribution to cumulative Al total oxidation was very small, just a few kg, contributing a small % to the total. And, again, due to buildup on the flue thermocouple, indicated temperatures were judged to be reading 100-200 C too low. Flue temperature data cannot be considered accurate on an absolute basis, but is judged to provide a fairly accurate relative indication.

Table 2 provides a summary of the 4 heats analyzed.

Table 2. Summary of Results for all 4 Heats

Heat			Calculated	Calculated		peak kg/min	peak frac	peak end
#	MW	kg chrgd	kg Al oxid	% Al oxid	minutes	Al oxidtn	O2 rctng w Al	T deg C
1	2.00	6,947	40.6	0.59%	77	4.67	0.30	470
3	2.00	6,819	93.9	1.38%	69	8.24	0.49	590
2	2.00	7,263	58.6	0.81%	80	4.95	0.31	541
8	2.64	7,099	35.3	0.50%	79	2.54	0.17	445

It is interesting to observe that for Heat #3, during the final 3 minutes of heating, the cumulative calculated aluminum melt loss increases from 1.06% to 1.38% (additional loss of 21.6 kg aluminum). This final 3 minutes of heating, after the operator had quickly cracked open the door to observe the load, is easy to visualize in Figures 2, 3, 4 and 8. For this heat, the operator would have preferred to tap the heat shortly after cracking the door open for this quick observation, but we continued heating for an additional 2 minutes, for the purposes of data collection, and this contributed to additional aluminum oxidation.

Conclusions

As demonstrated with these 4 chip melting heats, continuous flue gas analysis can be utilized during rotary furnace aluminum melting to indicate and calculate both the timing and magnitude of aluminum oxidation throughout the heat. Cumulative total kg of aluminum oxidized can also be calculated by summation.

These data and calculations show that instantaneous aluminum oxidation rate increases sharply near the end of the heat, corresponding to roughly the final 10 minutes before tap ("breakdown" period or main solid to liquid transition). Most of the cumulative aluminum oxidation for the heat occurs during this final time period. This illustrates the importance of avoiding overheating, any over-extended heating time can significantly increase aluminum oxidation losses.

Aluminum oxidation rate appears to correlate consistently with flue temperature. Aluminum oxidation rates are highest at the end of each heat, corresponding to highest temperature.

For the 4 heats analyzed, aluminum oxidation rate does not appear to correlate with burner settings, either % O2 (50% vs 80%) or % excess oxidizer (lambda 0.85 vs 0.95).

For these 4 heats, total aluminum oxidation loss was calculated to be between 0.5% and 1.4%. The % of total O2 (from purchased O2, combustion air and infiltration air) that reacts with Al is shown to be small for most of the heat, but increases sharply at the end of each heat, increasing to a maximum of 49% at the end of heat #3, corresponding to the highest temperatures observed.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ON THE CAST HOUSE EXERGY MANAGEMENT

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Keywords: exergy analysis, holding, melting, furnace, heat transfer.

Abstract

The energy efficiency of aluminum casting furnaces has been widely reported. However, the conventional energy efficiency analyses are based on the first law of thermodynamics which do not shed adequate light on the processes' degradation of energy. This just gives a general idea of the furnace's performance with no reference to possible improvement strategies. In this study, we apply exergy analyses on the aluminum holding and melting furnaces to identify the location and causes of energy degradation. The exergy analyses which are based on a real life furnace conditions highlight the possible locations for technology improvement in a typical cast house. With this established, methods of minimizing the cast house's exergy losses are assessed.

Introduction

The aluminum industry is well-known for its enormous energy consumption and high emission release. The push to reverse this inclination is now even more persistent with the recent climate change actions. This has led to an increased energy management scrutiny on every process along the aluminum production line.

A vital production process of both the primary and secondary aluminum is the heat treatment carried out in a holding and melting furnace [1, 2]. Although the melting and holding processes are typically carried out in a single furnace, they inherently constitute different procedures. The holding process is used to keep molten aluminum at a certain temperature required for the casting process, while the melting furnace, as the name implies, is used to melt metal. This means that during the melting process, the metal which acts as the heat sink is introduced into the system at a lower temperature.

Energy management of both processes requires methodologies which involves thermodynamic analysis. Most efforts in this line have relied on the use of energy balances which is based only on the first law of thermodynamics [3, 4, 5]. However, simple energy balances do not explicate on the energy degradation occurring in the process, nor does it quantify the usefulness of the heat content in the various energy streams [6]. For instance, based on only the first law, one could wrongfully attribute the available work in 1J of a heat stream at 100˚C to be equal to that of a 1J heat stream at 1000˚C. Similarly, the environment is considered an infinite heat reservoir, but as insufficient temperature difference exist between the environment and a device in its thermal ground state, this energy cannot be used to drive any prime mover.

The exergy (or availability) analysis, which is based on the first and second laws of thermodynamics, overcomes the shortcomings of the energy balance approach. Its use specifies the site of energy degradation which could lead to the development of strategies for an improved process or technology [6]. Furthermore, this approach allows for the quality of a heat stream to be adequately quantified as work availability relative to the datum temperature rather than a mere unit of energy. As such, the exergy approach, or "*exergy management*", has a greater chance of resulting in a significantly reduced depletion rate in natural resources and hence, reducing emission rates of pollutants. The potential of this methodology in the aluminum industry has been demonstrated by Kazakov and Lipin [7] who used exergy analysis to optimize bauxite digestion heat flow diagrams for greater heat stream regeneration.

In the present study, the first and second laws of thermodynamics are used to deduce adequate methodologies for exergy management in the aluminum cast house. Subsequently, these formulations are used to perform exergy analyses on typical holding and melting furnaces used in aluminum smelters. From our analysis, we estimate the magnitude of exergy losses in the furnaces and identify their causes. Upon establishing these, strategies for improving exergy efficiency are considered.

Theory

The first step of any thermodynamics analysis is the definition of the entity being analyzed [8]. In this case, the furnaces being examined can be considered as a simple open system (control volume) shown in Figure 1.

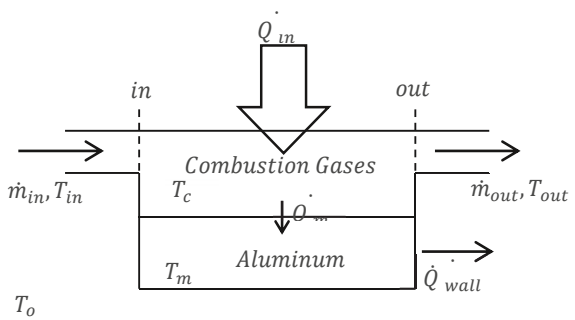

Figure 1: Representation of the furnace as a control volume.

Neglecting the kinetic and potential energy terms, and irreversibility, the first law interpretation of this system is:

$$\dot{Q}_{in} = (\dot{m}_{in}.h_{in} - \dot{m}_{out}.h_{out}) + \dot{W} + \dot{Q}_m + \dot{Q}_{wall} \qquad (1)$$

Where $\dot{Q}_{in}$, $\dot{Q}_m$ *and* $\dot{Q}_{wall}$ are the total inlet heat rate, aluminum absorption heat rate and rate of heat transfer through the furnace's walls respectively. The normal smelting furnace is a simple thermal machine hence, energy is transferred in and out of the system only via heat transfer. Hence, the work rate, $\dot{W}$ is typically nil leaving just the enthalpy terms on the RHS of

Equation 1. Also, from the fundamental conservation of mass principle, at steady state, the inlet mass $\dot{m}_{in}$ must be equal to the outflow of mass, $\dot{m}_{out}$. This reduces equation 1 to the reduced form of the system's energy balance equation:

$$\dot{Q}_{in} = \underbrace{\dot{m}_{in}(h_{in} - h_{out})}_{mass\ flow\ energy} + \underbrace{\dot{Q}_m + \dot{Q}_{wall}}_{Heat\ energy} \quad (2)$$

To the cast house, the heat quantity which is of principle importance is $\dot{Q}_m$ and hence, the term which the system's efficiency is based upon i.e.

$$\eta_{energy} = \dot{Q}_m / \dot{Q}_{in} \quad (3)$$

However, this is not a fair depiction of the furnace's performance as the maximum fraction of heat available to be transferred from the heat source to the metal, which requires the complete reversibility of a Carnot Engine, is less than $\dot{Q}_{in}$. This is where the second law comes into the analysis.

From the first and second laws, the maximum available work $\dot{W}_{rev}$ of the Carnot cycle is [9]:

$$\dot{W}_{rev} = \dot{Q} - \dot{Q}_{en} \quad (4)$$

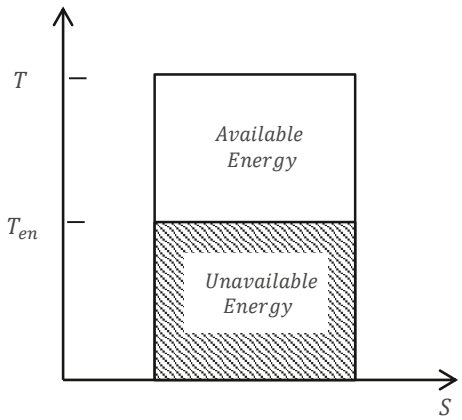

Figure 2: T-S diagram for the Carnot Cycle

Looking at this from the T-S diagram shown in Figure 2, based on the area under the total heat and available heat curves, it can be said that:

$$\frac{\dot{Q}}{T} = \frac{\dot{Q}_{en}}{T_{en}} = \Delta S \quad (5)$$

Where, T_{en} and $\dot{Q}_{en}$ are the temperature and heat transfer to the surrounding environment. Hence,

$$\dot{W}_{rev} = \dot{Q}(1 - \frac{T}{T_{en}}) \quad (6)$$

It is important to note that in the case of the furnace, this work term is synonymous to the heat transferred to the metal. So, even if the furnace attained thermodynamic perfection, the maximum amount of heat that could be transferred to the metal is$\dot{W}_{rev}$. As such, the degree of perfection or exergetic efficiency of the heat transfer between the furnace's heat source and the metal is:

$$\eta_{exergy} = \frac{\dot{Q}_m}{\dot{Q}_{in}(1 - \frac{T_m}{T_c})} = = \frac{\eta_{energy}}{\eta_{maxc}} \quad (7)$$

Where, η_{maxc} is the maximum energetic efficiency of the furnace.

From equation 7, it can be observed that the maximum heat energy available to be transferred to the metal (exergy or availability) is:

$$\dot{Q}_{in}(1 - \frac{T_m}{T_c}) \quad (8)$$

In a similar fashion, the availability of energy for the furnace walls is:

$$\dot{Q}_{in}(1 - \frac{T_{wall}}{T_c}) \quad (9)$$

However, we have only covered the heat terms in equation 2. This is the portion of the supplied energy that is exchange within the furnace. For a reversible flow of gases through the furnaces, the mass flow energy is:

$$\dot{Q}_{mf} = \dot{m}_{in}(h_{in} - h_{out}) \quad (10)$$

In the smelter, the process, like every other real process, is actually irreversible. Equation 10 for a real process can be represented as:

$$\dot{m}_{in}T_{in}(s_{out} - s_{in}) - \dot{Q}_{mf} = T_{in}s_{gen} > 0 \quad (11)$$

Where, $T_{in}s_{gen}$ is the irreversibility of the combustion gases flow, and s_{in} and s_{out} are the specific entropy conditions of the combustion gases at the inlet and outlet of the furnaces. To minimize this irreversibility, the source must be the minimum possible temperature which is the ambient value T_o. If this is the case, the second law for the reversible process shows that:

$$\dot{m}_{in}T_{in}(s_{out} - s_{in}) - \frac{\dot{Q}_0{}^{rev}}{T_o} = \dot{Q}_{mf}{}^{rev} \quad (12)$$

Making $\dot{Q}_{rev}$ in Equation 12 the subject of the formulae and substituting into the first law equation of a reversible process:

$$\dot{Q}_{mf}{}^{rev} = -\dot{Q}_0{}^{rev} + (h_{out} - h_{in}) \quad (13)$$

We obtain:

$$\dot{Q}_{mf}{}^{rev}\left(1 - \frac{T_0}{T_{in}}\right) = \dot{m}_{in}T_0(s_{out} - s_{in}) - \dot{m}_{in}(h_{out} - h_{in}) \quad (14)$$

Equation 14 represents the maximum reversible energy or availability from the flow of combustion gasses. The less irreversibility this flow experiences, the greater the amount of energy that can be extracted from this source.

In the next section, we would be using these equations to calculate the availability of the metal, furnace wall and the flow of the combustion gases for the holding and melting furnaces. Exergy efficiencies are then calculated to depict the degree of perfection of the various energy transfer streams. Based on this analysis and prevailing equations suggestions are made on strategies for improving the furnace's overall exergy efficiency.

Exergy Analysis

Taking the holding furnace presented in reference [10] into consideration, the following operating conditions are adapted for illustration purposes:

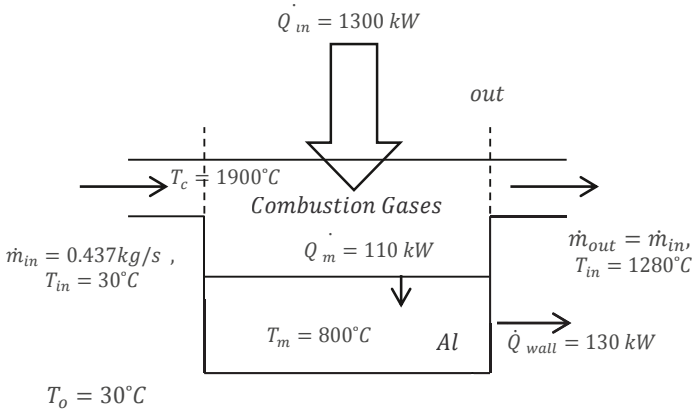

Figure 3: Operating conditions of a typical real life aluminum holding furnace

In this case, the inlet and outlet temperatures are the mass-weighted average values of the combustion gasses' constituent elements. It is important to note that these are steady state parameters obtained under the assumption that the burner runs continuously at full fire. For a complete analysis, temporal furnace parameters would have to be adapted.

From Equation 8, the exergy (maximum available energy) of the metal during the holding process is:

$$1300 * \left(1 - \left(\frac{1073}{2173}\right)\right) = 658\ kW$$

Comparing this to the exergy available for the metal at the time of introduction into the melting furnace ($40^{\circ}C$) – $1100\ kW$ – it is easy to see why the energy efficiency of the melting process is typically higher than that of the holding process.

In the analyzed furnace, only $110\ kW$ (16.7%) of the exergy is used up by the metal. The remaining portion of the "*maximum available energy*" is the exergy losses within the furnace. Based on our CFD about 90% of these exergy is lost in the radiation process while 10% is due to exergy losses in the convection heat transfer process. To minimize the radiation irreversibility, one has to minimize the radiation space resistances between the flame and the metal. Radiation space resistance is inversely proportion to the view factor which is dependent on the emissivity of the flame and the metal [11]. Consequently, about 498kW of the metal's available exergy can be recovered by optimizing the emissivity of the metal and the flame. Little can be done to control the flame's emissivity however; the metal's emissivity is dictated by dross formation which, to some extent, can be controlled. Dross which has a very low emissivity of about 0.1 is usually formed when molted aluminum is oxidized. If this oxidization is reduced, the radiation exergy losses are also reduced. Similarly the furnace's convection heat transfer exergy losses (about $54\ kW$) can be minimized by increasing the convective heat transfer coefficient between the combustion gases and the metal. Increment in the convective heat transfer coefficient can be attained by increasing the velocity of turbulence of the combustion gases running over the metal surface. Strategies for attaining this by optimizing burner type and orientations are discussed in reference [12]. Another strategy to improve radiation heat transfer between the flame and the metal is to increase the flame spread in order to maximize the heat source's surface area. It is also important to note that the thermal resistivity of the dross also serves as a resistance to conduction heat transfer between momentary stagnated gases and the metal.

Minimizing exergy losses through the furnace's walls (estimated to be 544 kW from Equation 9) is also desirable. Heat flow through the walls of the furnace is irreversible hence; exergy expended to transfer heat via conduction through the walls of the furnace ends up being lost. However, if this exergy is not allowed to be expended in this way, it can be diverted as available exergy for processes which can put it to better use. Minimizing the heat flow through the walls of the furnace can be achieved by modification of the refractory. However, this approach is limited by the formation of freeze plane from within the hard working lining into the backup insulation which increases the chance of refractory failure [5]. From the theoretical heat transfer perspective presented in Equation 9, another approach could be to maximize the surface temperature of the inner refractory lining in order to reduce the temperature gradient between the flame and the walls which in turn minimizes the heat flux. To achieve this, ideas of coating the inner walls with highly reflective metallic materials comes to mind. However, this might practicable in a real life furnace especially in furnaces which utilizes magneto-hydrodynamics (MHD) stirrers. From a transient perspective, the radiation and convection heat transfer can also be enhanced by increasing the exhaust gas resident time via ventilation strategies.

Assuming a reversible heat transfer to the metal and zero heat losses to the furnace walls, the exergy of the outlet stream is about $516\ kW$ (applying Equation 6 to the exhaust heat stream with T_o as the temperature of the environment). The exergy available in this exhaust stream can be used up in several ways. In the cast house, two of the most widely adapted approaches for this are combustion air and scrap metal preheating. Combustion air preheating involves the use of a heat exchanger to transfer energy from the hot exhaust gases to the combustion processes' supply air. This increases the flame temperature [13] which, based on Equation 8, increases the exergy available for the metal. However, based on Equation 9, this also increases the exergy availability for the wall losses.

Increasing the temperature of the scrap metal has a similar two-sided effect. Although, it reduces the energy requirement from the heat source, as the flame temperature does not change in the process, it also has an effect of reducing the exergy available for the metal – from Equation 8, if T_m increases, available exergy reduces. However, this can be compensated for by reducing the overall heat into the system in order to account for the savings in terms of fuel consumption.

Table 1 summarizes the analysis for the exergy analysis for the holding furnace. Based on the preceding discussion, the exergy losses at each portion of the furnace are quantified, causes of these losses are highlighted and strategies for avoiding such losses are suggested.

Table 1: Exergy losses of the analyzed holding furnace

Location of Exergy Losses	Exergy Losses (kW)	Exergy Management Strategies
Metal Radiation	493	Dross minimization; flame spread maximization
Metal Convection	54	Improvement of flame dynamics; maximization of hot gasses' resident time
Furnace Walls	544	Insulation and thermal conditioning of wall materials
Exhaust Stream (with furnace wall losses minimized)	516	Exhaust gas recuperation

Walls of recent holding furnaces are usually well insulated. Hence, in practice, the wall losses are most often below the values stated in Table 1. This evident from the analyzed furnace in which only about 24% of the exergy available to be transferred through the furnace's wall is actually lost to the furnace walls. Similarly, exhaust gas recirculation is also already a well-established technology in the aluminum production industry. As such, the greatest portion of the losses that occur in modern holding furnaces is due irreversibility in the flame-metal radiation heat transfer. Dross management, flame spread maximization and the maximization of the combustion gases resident time are some of the suggested strategies to minimize these losses.

On the other hand, utilizing the fire tube furnace design presented in reference [10] totally eliminates the reliance of the flame-metal heat transfer on radiation. Consequently, exergy losses due to irreversibility of the radiation process are avoided. Rather, the heat conduction exergy losses would have to be managed by minimizing the thermal resistance of the fire tubes. Higher energy efficiencies recorded for this design suggests a reduction in exergy losses

Conclusion

This article provides a methodology for exergy management of the aluminum reverberatory furnace. Fundamental thermodynamics principles are applied on the furnace to deduce exergy equations. These equations are applied on the aluminum holding and melting furnaces to identify the location and causes of exergy degradation. We find that the radiation heat transfer between the flame and the metal accounts for exergy losses which are most often neglected. To minimize this energy loss, it is recommended that either the fire tube furnace design is used or dross management strategies are formulated.

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DESIGN OF SQUARE INDUCTION COILS FOR THE ELECTROMAGNETIC PRIMING OF CERAMIC FOAM FILTERS

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Keywords: Induction, Coil, Aluminum, Ceramic Foam Filters, Priming

Abstract

Ceramic Foam Filter (CFF) ElectroMagnetic (EM) priming systems with square-shaped solenoidal coils are in the process of full-scale industrialization. The demands on the EM fields used in such a system are substantially different from the requirements in the existing EM pumping systems used for liquid metals. The same fundamental physics are applied, *i.e.* Lorentz forces induced by time varying EM fields; however, in the present case the forces are used to overcome the metal surface energy and force metal into the cold air filled pores of the CFF instead of producing bulk metal movement.

The present study discusses the design methodology for square shaped solenoidal coils applied to EM priming using a combination of analytical models, experimental measurements and Finite Element Modeling (FEM) with COMSOL® Multiphysics. The EM force field at different induction coil designs (4 [10.16] and 9 [22.86] inches [cm]) is estimated using FEM and validated against experimental results.

Introduction

The impact of the EM field generated by different induction coils inducing electromotive forces in liquid aluminum has been investigated in earlier publications by the present authors [1, 2], and is also the topic of this study. A reduction of the required metal head to successfully prime commercial filters has been discovered, and is now the subject of an international patent application [3]. The observed effect on the required metal height is shown in

Figure 1.

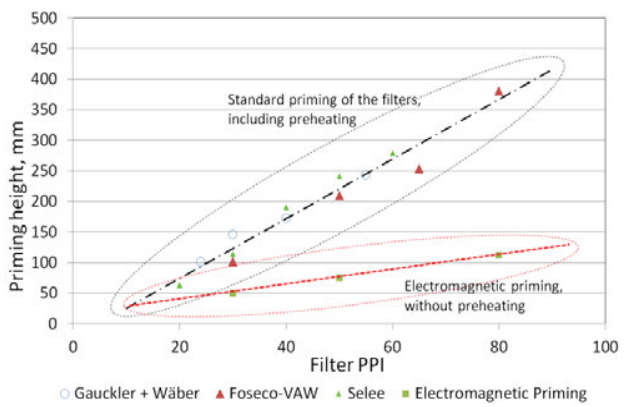

Figure 1: Priming height versus filter pores per inch (ppi) for filters from different producers [4-6] compared to earlier electromagnetic priming trials using different 4" round filters and a flux density of 0.2 T [1, 7].

The focus of the present paper is on the induction coil itself, the priming of 3 stacked filters and how the shape of the magnetic field alters the priming of the most commonly applied physical filtration system, i.e. Ceramic Foam Filters (CFF) [8]. The applied frequency was 50 Hz and the variable field strength was obtained by the application of alternating currents of up to 1250 A RMS maximum. The main goal of the present study has been to compare the behavior of square and round double coils, validating the use of a 9'' (inch) square double coil for industrial trials.

Theory

To design an induction coil with a required magnetic flux density B_z, the inductance L can be applied as a useful parameter. The inductance L is defined as the flux linkages per unit current and is directly related to the magnetic flux density B_z, the area of the coil A and the applied current I, as shown in (Eq. 1):

$$L = \frac{N\Phi}{I} = \frac{B_z N A}{I} \tag{1}$$

where L is the inductance [H], Φ the total flux [Wb], I the current [A], N the number of turns [unitless], A the total area of the cross section of the induction coil [m^2] and B_z the magnetic flux density in the z-direction [T]. Only the z-component of the magnetic flux needs to be considered for symmetrical induction coils, as the sum of the r-components is zero after integration around the periphery. The computation of area A varies with geometry, *e.g.* round vs. square, which has a directly proportional effect on the inductance L. A good correlation between different coil shapes and the diameter can be obtained by using the equivalent diameter for the area A of the considered coil shape.

It is possible to calculate the exact inductance L for theoretical 'current sheet' cylindrical induction coils using the equation of Lorenz from 1879 [9]. This equation involves the solution of the double elliptical integrals $K(k)$ and $E(k)$ of the first and second kinds:

$$L_{Lorenz} = \mu_r \mu_0 \frac{8N^2 r^3}{3l_c^2}\left[\left(\frac{2k^2-1}{k^3}\right)E(k) + \left(\frac{1-k^2}{k^3}\right)K(k) - 1\right],$$
$$k = \sqrt{\frac{4r^2}{4r^2+l_c^2}} \tag{2}$$

Lorenz's equation was reformulated by Nagaoka in 1908 [10], as shown in Eq. (3), using a short coil correction factor K which he tabulated to 6 decimals. The short coil correction factor K is

dimensionless and varies between 1 and 0 with an increasing coil diameter to length ratio.

$$L_1 = K\frac{\mu_0\mu_r\pi N^2 r^2}{l_c} \quad (3)$$

where l_c is the length of the coil [m], $\mu_0 = 4\pi * 10^{-7}$ the permeability of free space [H/m], $\mu_r = 1$ the relative magnetic permeability for air [unitless], and K the short coil correction factor or Nagaoka coefficient [unitless], which considers accounting for the geometry of the coil.

In 1928 Wheeler presented an empirical formula for the inductance of a single layer short coil [11]. Wheeler's equation was subsequently reformulated by Knight [12] to estimate Nagaoka's short coil correction factor K to approximately 3 decimal precision for coils of 'typical' induction coil dimensions:

$$K = \frac{1}{1+0.4502\left(\frac{D_c+\delta_c}{l_c}\right)} \quad (4)$$

where D_c is the inner diameter of the coil conductors, and δ_c the electromagnetic penetration depth into the coil defined as:

$$\delta_c = \sqrt{\frac{2\rho}{\omega\mu_o\mu_r}} = \sqrt{\frac{\rho}{\pi f\mu_o\mu_r}} \quad (5)$$

where ρ is the resistivity of the copper conductor (1.7241 x 10^{-8} Ω·m for 100% IACS copper [13]), and f the frequency [Hz].

The magnetic field of a single layer induction coil can be found by equating Eq. (1) and Eq. (3) and substituting Eq. (4):

$$B_z = \frac{1}{1+0.4502\left(\frac{D_c+\delta_c}{l_c}\right)}\frac{\mu_0\mu_r NI}{l_c} \quad (6)$$

In 1982 Wheeler published a letter [14] where he presented new and more accurate empirical equations designed for use with both square and round single layer coils:

$$L_2 = \mu_0 N^2 r\left(\frac{4}{\pi}\right)\left(\ln\left(1+\frac{\pi r}{l_c}\right)+\frac{31.496}{3.64+2\frac{l_c}{r}+0.51\left(\frac{l_c}{r}\right)^2}\right) \quad (7)$$

where L_2 is the inductance of a single layer coil [μH], N the number of turns, r the radius measured on the wire centerline [m], and l_c the length measured top to bottom of the coil.

Eq. (2) through Eq. (6) can be used to estimate the average z-component of magnetic fields of single layer coils with good precision. The equations can, however, not be applied with precision to multi-layer coils. For multi-layer coils the magnetic flux density in the z-direction B_z can be found by first estimating or measuring the inductance L (where the coil already physically exists), and then applying Eq. (1). Another empirical equation of Wheeler [11] can be used to estimate the inductance L for a multi-layer coil, with good precision, prior to constructing the coil. The equation in question has been transformed from imperial to SI units with dimensions in [m] as follows:

$$L_3 = \frac{31.496\, r^2 N^2}{6r+9l_c+10c} \quad (8)$$

where L_3 is the inductance of a multi-layer coil [μH], N the number of turns, r the average radius [m], l_c the length measured from the top to the bottom of the coil, and c the thickness of the coil [m] as shown in Figure 2.

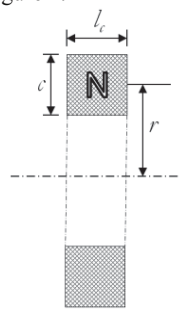

Figure 2: Model of a multi-layer induction coil with N turns, radius r, length $\mathbf{l_c}$, and thickness c. [11]

To prime a CFF with a conducting liquid metal, such as aluminum, a physical force in the form of the metallostatic pressure, has to be overcome. Earlier studies indicated that the Lorentz forces, induced by the electromagnetic flux density B_z, can help to prime non-preheated CFFs [2, 3, 7, 15]. In this case the Lorentz forces act as a driving force, as the axial variation of the magnetic field of real induction coils generates a curl [16], and thus produces electromagnetic or Magneto-Hydro-Dynamic (MHD) mixing. This mixing effect is well known and is discussed in several studies related to metal cleaning [17, 18].

The Lorentz forces F_L [N/m^3] are proportional to the square of the magnetic flux density B_z [19], and can be calculated as the cross product of the induced current density and the magnetic flux, e.g. from Eq. (6):

$$F_L = J \times B_z \quad (9)$$

where J is the induced current density [A/m^2].

The priming of the filters is induced by the EM flux and the curl in the Lorentz forces as shown Figure 3.

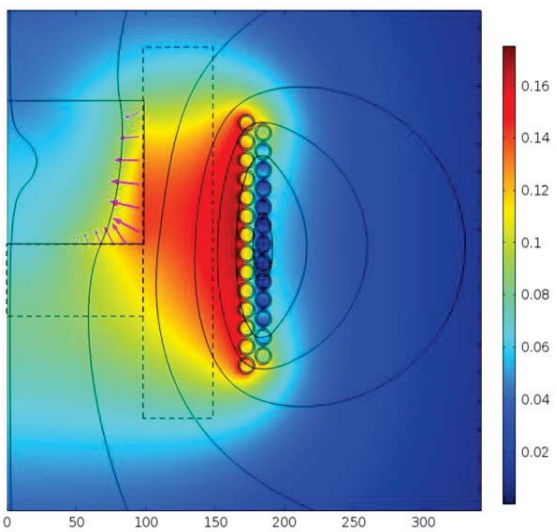

Figure 3: Real 9'' coil geometry, showing an axisymmetric COMSOL® 5.0 simulation of the magnetic flux density [T]. The arrows represent the relative magnitude of the time averaged Lorentz forces inside the liquid metal above the filter prior to priming (shown as a black box). The dotted

lines represent the Monalite® crucible and the filter element below the metal.

The design equations described in the present theory section are only applicable to an empty 'air-core' coil, *i.e.* a coil prior to priming. After priming, the space over, under and within the CFF's are filled with metal. This metal alters both the electrical resistance and the reactance of the system in a complicated way described elsewhere [20]. In practical terms the reduction of reactive impedance is greater than the increase in effective coil resistance, and for a fixed applied voltage the current and magnetic flux density increase as the CFFs become primed. Hence, the air-core case represents a minimum and therefore a reasonable design case.

Experimental

Experimental data were collected using two different round coils, *i.e.* a short single coil and a short double coil, shown in Figure 4 a) and b). In addition two square 9'' short coils were used, *i.e.* one single coil and one double coil, shown in Figure 5 a) and b) respectively.

The 4'' round coils, shown in Figure 4, have been designed to allow for approx. 1 kA at maximal cooling rates. The highest estimated magnetic flux densities vary from coil to coil and are in the center of coil a) ~0.13 and b) ~0.23. The multi-layer coils were connected by a jumper (the 'hoop' rising up in the picture), which allowed the current to flow between the two coils in series without disturbing the magnetic field inside the coil, as shown in Figure 4 b). Cooling water was provided in parallel, while current flowed in series.

The 9'' coil, as shown in Figure 5, was designed to allow for 2 kA, generating an estimated magnetic field of 0.2 T in the center of the coil. The inner coil prior to being insulated is shown in Figure 5 a) and after coupling both coils to create one double coil in Figure 5 b). The coils were connected by a jumper, visible in Figure 5 b). The square shape of the coil generated strong electromagnetic curls in the Lorentz forces at the corners of the coil, as well as strong 100 Hz electro-mechanical vibrations due to the 50 Hz alternating current (passing through zero twice each cycle). This required powerful restraints to maintain the physical stability of the coil, as indicated in the photograph shown in Figure 5 b).

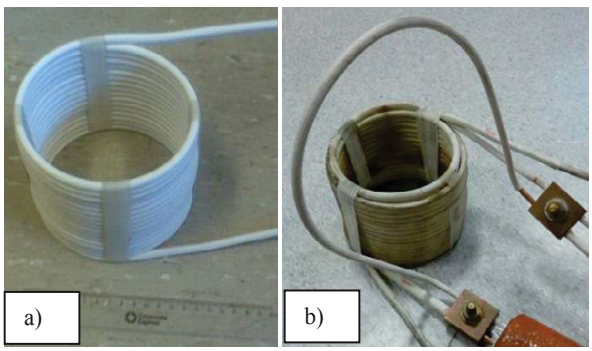

Figure 4: The two round 4'' coils used, a) the single coil with 16 turns and b) the double short coil with 31 turns.

The liquid metal priming (Al alloy 3003) trial casting temperature was aimed to be 700°C. No pre-heating was applied to the 3 stacked 50 ppi CFF, and they remained at room temperature until contact with the liquid metal. The coil was electrified with 1250 A generating an EM flux of 0.12 T (empty coil) as the metal was poured into the crucible. In total, a metal head of approx. 100 mm was poured into the crucible and a meniscus was formed instantaneously, as observed and discussed during earlier trials [2].

Two different experimental power supplies have been used, *i.e.* a step down transformer (45 kW, 230/30 V, 196/1500 A) for the small round coils, and two adjustable welding transformers (each 80 kW, 44/89 V, 500/1800 A), wired in parallel for the 9'' coils. The measurements for the magnetic fields were executed using a Hall Effect gauss meter model 6010 supplied by Pacific Scientific OECO, F.W. Bell®. The axial probe was standardized using an axial standard of 0.05 T ± 0.3 mT. The coil current was measured using a Fluke® inductive current probe i1000S with a resolution of 1 ampere and ±1% precision. The real (W) and reactive (VAR) power were measured using a Fluke® 43B Power Quality Analyzer with a resolution of 100 W. The experimentally derived coil inductances could then be calculated using the measured reactive power. The resulting experimental data were used to validate the FEM simulation results without 'fitting'.

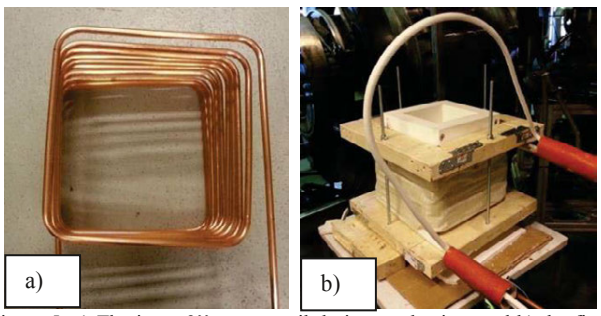

Figure 5: a) The inner 9'' square coil during production, and b) the final 9'' coil setup, insulated with fiberglass sleeves and the Monalite® crucible for liquid metal experiments. The coupling between the coils, by using a jumper, can be seen in b).

Results and Discussion

The validation of the present method to design simple induction coils of various shapes can be accomplished by comparison of the predicted analytical results of the modeled flux densities with the measured flux densities and by comparison of the calculated inductance and measured inductance. In the present study magnetic flux density of the designed coils was first estimated, then calculated by application of the software COMSOL® 5, and finally compared to experimental measurements of real coils. The data from the inductance measurements and the results of the priming trials are presented in Table I. Liquid metal priming of 3 stacked 50 ppi CFF filters, installed within the 9'' setup, was successfully conducted, as shown in Figure 6.

The round coils were built and tested by electromagnetic priming of liquid aluminum with CFFs, and the effect of the EM flux density (induced by the coils designed to generate the strongest EM flux) was shown in earlier investigations by the present authors for 4'' [7] and 9'' systems [2].

Table I: Summary of the experimental coil dimensions, measured and theoretical inductance and flux density values. All inductance estimates are for a frequency of 50 Hz.

			4'' (10.16 cm) coil		9'' (22.86 cm) coil	
			Single short	Double short	Single	Double
			Round		Square	
	Unit	Abr.	1	2	3	4
Inner Dia. Coil	[mm]	r	63	66	169,5	172,5
Total Height	[mm]	l_c	111	111	182	182
Coil Thickness	[mm]	c	6,5	14	13	26
Turns		N	16	31	14	27
Short coil cor.		K	0,660	0,650	0,543	0,538
Example			At 956 A	At 1 kA	at 1,250 kA	
Measured	[µH]	L	27,6	104	58	240
Lorenz (Eq.3)	[µH]	L_1	25,4	96,1	66,2	254
Single Coil (Eq.6)	[µH]	L_2	24,8	94,1	66,4	255
Multilayer (Eq.8)	[µH]	L_3	23,6	85,3	63,6	233
Calc. Flux D.	[T]	B_z	0,123	0,228	0,066	0,125
Meas. Flux D.	[T]		0,121	0,23	0,062	0,127

In the present experimental results the electromagnetic field, induced by a double square coil, was able to prime a stack of 3 cold 9'' 50 ppi CFF after a total of four minutes (see Table II). During the required 4 minutes, the EM field induced both MHD mixing within the liquid metal and heating. Solidification and freezing at the filter metal interface was avoided as the heat of the metal preheated the filter.

Table II: Summary of the events during the priming of the stacked filters.

Time	Comment
~60 seconds after pouring on the filter	Metal level dropped by approx. 50 mm; metal was refilled to a head of 100 mm.
~90 seconds after second filling	Metal level dropped another 50 mm; metal was refilled to a head of 100 mm.
~30 seconds after the 3third filling	The metal level dropped rapidly. Remaining metal was poured into the Monalite® crucible keeping the level above the filter.

The metal primed the stack of filters in three stages, *i.e.* each individual filter in sequence. The required times together with the observations are shown in Table II. A total of four minutes was required to prime three stacked cold 50 ppi CFFs, after which the coil was switched off.

After the filter reached the required temperature, the metal can penetrate into the filter, *i.e.* when the initial solidified aluminum 'skin' had been melted. The electrical conductivity of solid aluminum is approx. 2.5 times larger than for liquid aluminum at the same temperature. This focuses the induced current into any frozen metal inducing preferential heating. This induced heat, together with heat convected from the liquid metal due to MHD, rapidly melts any solidified material and avoids permanent blockage of the filter. The actual priming is accomplished by the velocity head created by the bulk MHD flow induced by the Lorentz forces, driving the metal into the filter. The curl of the magnetic field within the filter continues to generate MHD after the initial phase of priming, and assists in the removal of trapped gases. Mechanical vibrations generated by the alternating current are believed to produce the excellent wetting, which has been observed between the CFF and the metal. The change of the direction of the current produces a 'zero' point in the Lorentz forces twice in each cycle (*e.g.* 2x50 Hz), inducing mechanical vibrations in the liquid metal. The direction of the Lorentz force remains constant regardless of the current polarity, as both the magnetic field and the current reverse directions together. The final result of priming is shown in Figure 6 b) and c).

The 3D model designed with COMSOL® 5 gives an estimation of the resulting magnetic flux density of a square coil, but more computational resources and time is required to design a proper model. The advantages of such a 3D model are given by the possibility to visualize the corner effects, which can't be estimated by the axisymmetric models. However, one of the main disadvantages of 3D modeling is that the resolution of the mesh is typically reduced due to lack of computer memory. This is a particularly significant issue when applying high frequencies [21]. The total error obtained for the axisymmetric model is 2.65 % and for the 3D model showing minimal overall improvement 2.25 %. The presently obtained results are plotted together with an estimated flux density, shown as a black dotted line, in Figure 8. The calculations are based on the application of the empirical approach of Wheeler using the inductance L_3, from Eq. (8) and Eq. (6) to calculate B_z.

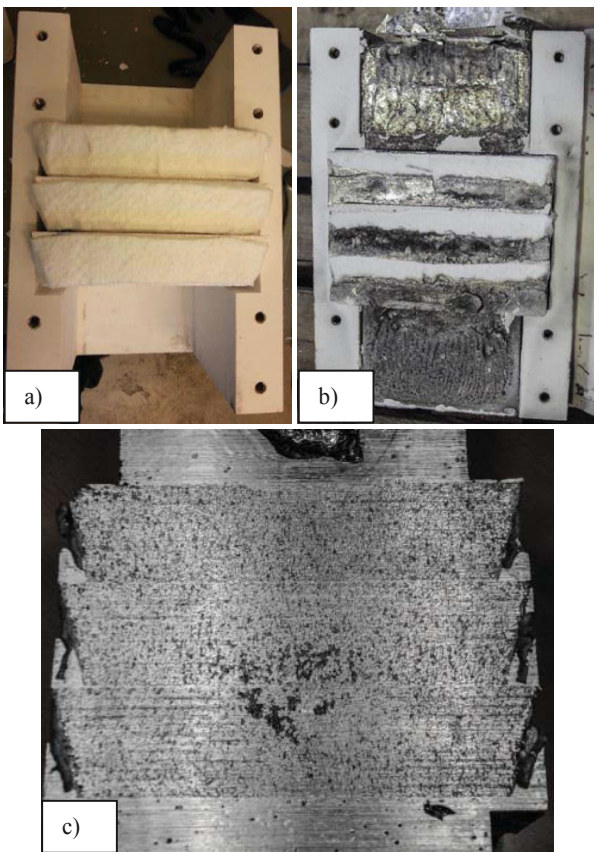

Figure 6: a) The open Monalite® crucible showing the three stacked 9'' 50 ppi CFFs prior to the liquid metal priming experiments. b) After the liquid metal priming experiment the completely filled filters with metal above and below are shown. The sealing, visible as the white foam around each

filter, has worked properly and bypassing of metal has been prevented. c) A cut through the center of the sample reveals fully priming of the three stacked 50 ppi CFF. The total filter thickness was 150 mm. No ceramic filter surfaces are visible, affirming the total priming of the filter.

The coil geometry and the resulting magnetic flux density of the 9" (22.86 cm) square coil and the round coil is visualized in Figure 7 a) and b) respectively. The Figures 7 a) and b) were visualized by applying COMSOL® 5 using a 3D model of the square and round geometries.

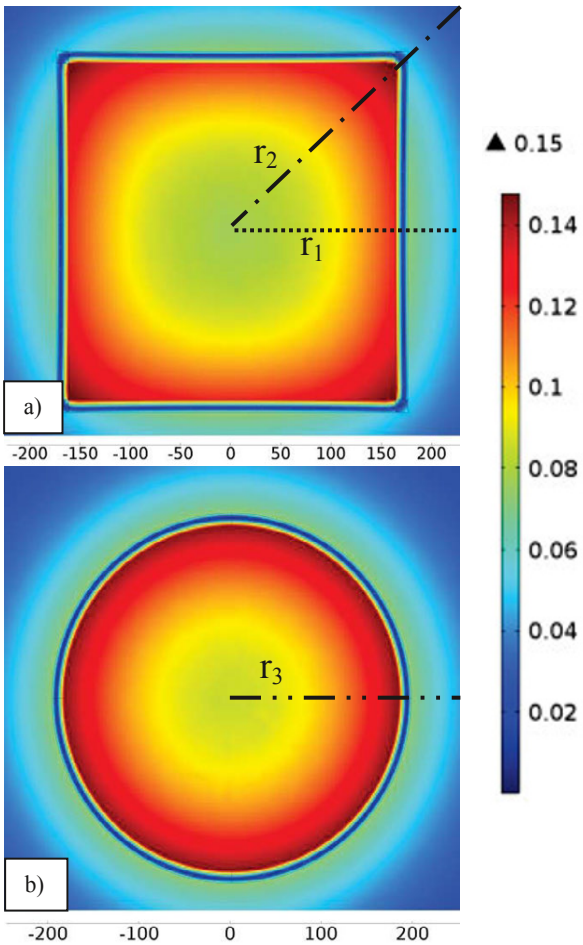

Figure 7: COMSOL® 5 models of the a) designed 9" (22.86 cm) square coil, and b) an equivalent diameter 9" round coil having the same total area. Both models have been meshed suitable for low frequency (spacing of less than δ_c). The applied current is equal to the measured current during the real experiments, *i.e.* 1250 A RMS. The maximal magnetic flux density is calculated to be 0.144 T in both models. The dashed lines show the different radiis r_1, r_2 and r_3 where measurements have been taken.

The COMSOL® 5 calculations and the measured data of the flux density are compared in Figure 8. As can be seen from the figure, the calculated values of the 9'' square 3D model at the centerline r_1, the diagonal r_2, and the 9'' round 3D model at r_3 are plotted together with the measured data from square coil no. 4 along the diagonal and along the centerline. The round coil 3D model with the equivalent diameter of the square coil gives the lowest flux density, whereas the square on the perpendicular 'centerline' gives the highest values. The real measurements fall in between these two sets of values.

The geometry has proven to have a strong effect on the resulting magnetic field. In the 3D model, the variation in the flux density, as a result of the coil shape, can be seen along the diagonal, whereas the 2D axisymmetric solution cannot provide this information. Hence, 3D modelling was necessary to understand the corner effects, and provide an accurate estimate of the flux density in this area. The comparisons of the measurements and the calculations from the 3D-model are also shown in Figure 8.

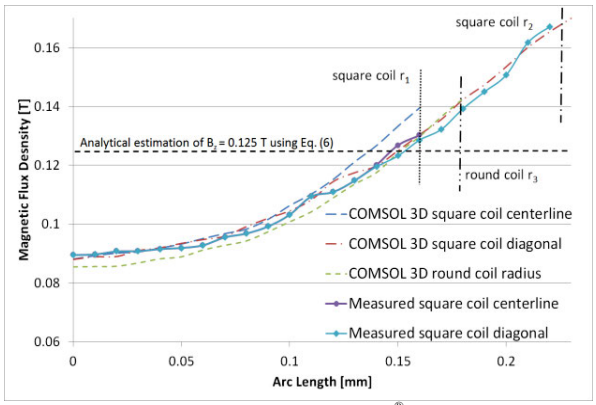

Figure 8: Plotted data from the 3D COMSOL® 5 models compared to measured data from square coil no. 4, while applying 1250 A. The measurements were taken every 10 mm from the center to the side, where the square coil has a short 'radius', shown as r_1 (measured perpendicular to the sidewall), and a long diagonal side, shown as r_2. The radius of the equivalent round coil is shown as r_3.

Conclusions

A simple method was shown capable of designing round, square, single and multi-layer solenoidal coils capable of generating a target magnetic flux density. The method allows the fitting of an induction coil to the limits of the available power transformer, and/or the specification of a new transformer, to achieve a target average flux density for a given coil dimension. Modern FEM software, such as Comsol® 5, allows the solution of simple, fast and accurate 2D axisymmetric models for round and symmetric coils. It also allows the design of 3D models, which can deliver a greater spatial resolution of the physics for any required coil geometry. The presently generated analytical solutions give the smallest error, whereas the FEM allows for the visualization of the resulting fields. These fields have been compared to high precision manual measurements of real coils.

The priming of a stack of three commercial 9'' 50 ppi CFF by liquid aluminum (3003 alloy), *i.e.* a 150 mm thick filter or 300% of the normally applied thickness of a 50 mm filter has been demonstrated by application of electromagnetic fields (0.12T) while only using 50% of the normal industrially applied metal head [4, 5].

Investigation of the metal samples after the experimental trial showed excellent results regarding priming and wetting of the filters at all levels. Metal was observed in all regions of the filter setup, and the alumina wool around each filter was not wetted by aluminum excluding bypassing. The EM field forces the metal towards the corners of each filter, ensuring the complete priming of the total filter area.

Future Work

The 9” square coils designed with the present theory will be up-scaled for the priming of 23” and 26’’ commercial CFF systems using induction coils. The future trials will also include fine filters, up to 80 ppi CFF, and from 50 to 150 mm thickness. These experiments are in the process of being executed in collaboration with an industrial partner.

Acknowledgement

The authors also wish to express their gratitude to Nigel Clear from Pyrotek® EMP™ Technologies and Dr. Neil Keegan from Pyrotek® Metallurgical Services Group, Kurt Sandaunet and Roar Jensen from SINTEF Materials and Chemistry, Shahid Akhtar from Hydro Aluminium in Karmøy and Trygve Schaunche from NTNU for their support, good discussions and help and to thank the Institute for Material Technology (IMT) at NTNU, Trondheim in Norway for using the casting laboratory equipment.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ASSESSMENT OF ACTIVE FILTERS FOR HIGH QUALITY ALUMINIUM CAST PRODUCTS

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Keywords: LiMCA, Filtration, Inclusions, Efficiency, Molten aluminium

Abstract

The quality requirements of aluminium products are constantly increasing. It is critical that ingots and billets have the least amount of inclusions in the as-cast product. Filtration is therefore applied to all critical products, ceramic foam filtration (CFF) being the most frequently used technique. In an attempt to increase inclusion capture by CFF the influence of the filter surface chemistry on the filtration efficiency of aluminium alloys has been evaluated. Lab scale filters have been produced with coatings of different chemical composition. The filters have been characterized prior to usage and lab scale casts have been made on a pilot unit in conjunction with two LiMCA for inclusion removal efficiency determination. The paper will present the results from the characterization done and the observations will be discussed.

Introduction

Melt cleanliness is essential for high quality aluminum production. The main cleanliness criteria are dissolved hydrogen (potentially resulting in porosity) and inclusions. Inclusions have different origins, e.g. from scrap melting, oxidation of the melt surface, and erosion of the refractory material. In aluminum production, oxides are the most common inclusions found in the melt (Al_2O_3, MgO, $MgAl_2O_4$, $3Al_2O_3$ $2SiO_2$, SiO_2). In thin rolled products (e.g. aluminum cans and foil) inclusions can create pinholes, while in thick rolled products (used e.g. in transportation), inclusions may restrict fatigue life. The continuous improvement of melt cleanliness is essential to meet the ever-increasing quality requirements of today's final products. The industry relies on ceramic foam filters (CFF) and deep bed filters (DBF) to obtain the inclusion quality required for their end products.

The efficiency of CFF has been shown to vary widely, the filter grade being a major factor [1]. Experiments to understand the inclusion capture in CFF showed that inclusions concentrate in the entry side of the filter. Bridges form in recessed passages, favouring inclusion agglomeration [1, 2]. However CFF are sensitive to inclusion releases. It has been shown both in unalloyed [3] and highly alloyed melts [1] that the addition of the refiner upstream of the CFF lead to significant inclusion releases from high grade CFF at high inclusion loads. CFF are also sensitive to hydrodynamic pressures that also negatively impact the inclusion removal efficiency [4].

There is only limited information on the influence of the filter surface chemistry on the filtration efficiency of liquid aluminum although reactive filters have been considered to enhance inclusion adhesion on CFF. Zhou et al. [5] use the term "active coating" for an enamel coating that can capture nonmetallic inclusions and dissolve Al_2O_3 during the casting of aluminum. They find an increase in the elongation at fracture of aluminum samples filtered with enamel-coated filters compared to aluminum samples filtered with uncoated filters [5]. Filter surfaces made of NaBr are tested by Oosumi et al. [6] and Luyten et al. [7] for the filtration of small particles and intermetallic particles. Syvertsen et al. compare the filtration of Al_2O_3 and SiC foam filters, and find better filtration efficiency for the SiC filters [8]. A possible explanation for this difference could be the better wetting of the SiC.

Active filtration has been assessed during steel casting. In this high temperature range inclusions were sintering on the filter surface [9]. Based on this approach Voigt et al. investigate the filtration of aluminum when using filters with different surface chemistries (Al_2O_3, $MgAl_2O_4$, 3 Al_2O_3 2 SiO_2 and TiO_2). From scanning microscopic observations, they find that Al_2O_3 and $MgAl_2O_4$ exhibit better filtration behavior than the other two materials [10].

Although scanning electron microscopy may indicate trends the filtration efficiency of filters is most often characterized by LiMCA (Liquid Metal Cleanliness Analyzer) and/or PoDFA (Porous Disk Filtration Apparatus). LiMCA is considered the reference technology for quantifying inclusions in molten aluminum [11, 12]. Based on the principle of the Coulter Counter, LiMCA facilitates the quantification of inclusion number and size in molten aluminum. PoDFA is a method for the metallographic observation of inclusions in a concentrated filter cake [13].

In order to contribute to the understanding of the impact of surface chemistry on aluminum filtration, the efficiency of filters with different surface compositions was assessed. The surface compositions are Al_2O_3 (alumina), $MgAl_2O_4$ (spinel), $3.Al_2O_3.2SiO_2$ (mullite), and TiO_2 (rutile). Experimental casts were performed under controlled test conditions, including a particle seeding process. The casts were monitored by LiMCA, and the spent filters were fully characterized by metallographic methods. The paper reports the first results from this work.

Materials and methods

Filtration pilot line

The filtration pilot line used for the trials is represented in Fig. 1 and consists of a 750 kg resistance tilting furnace, a launder system, a filter box, and a ladle to collect the filtered material.

The AlSi7Mg0.3 ingots were melted in the furnace at 1023 K (750°C). A rotary degassing treatment was applied with a mix of Ar/Cl_2, and the metal was left to settle.

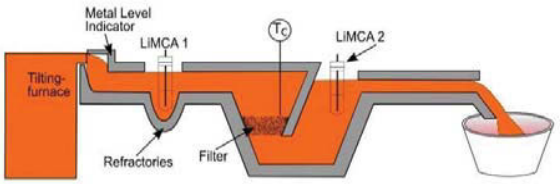


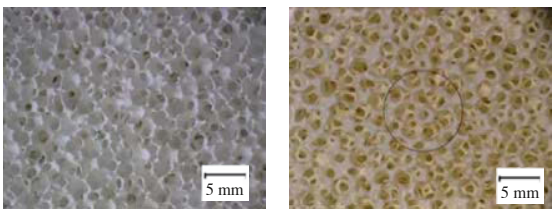
Figure 1: Filtration pilot line

The filter used for the trial was preheated up to 923 K (650°C) with a hot air burner. Once the temperatures were reached, the molten metal was seeded with non-metallic particles and mechanically stirred just before the cast was started. The cast duration was

typically 60 min, allowing enough LiMCA data points to be collected. After the cast, the metal was solidified in the filter and the tile was removed and cut in order to perform metallographic observations of the inclusion distribution inside the filter.

Ceramic foam filters

In this study, 30 ppi (pores per inch) alumina filters were used as starting material. Each tile had a pyramidal shape with a cross section of 90 mm x 90 mm on the small side (melt outlet) and a thickness of 50 mm. The initial alumina skeletons were coated using the modified replica technique [14]. This process produces filters with comparable pore size and varying surface properties, and a more rounded morphology of pores and struts in the coated filters. This is illustrated in figure 2 which shows a typical industrial filter (high alumina) and a coated filter from the replica method. In the present paper, the following surface chemistries were considered: Al_2O_3 (alumina), $MgAl_2O_4$ (spinel), 3 $Al_2O_3 \cdot 2\ SiO_2$ (mullite), and TiO_2 (titanium dioxide). The filters were installed in the filtration box and an expandable gasket was used to prevent filter flotation or metal leaks on the lateral sides. The metal flow rate in the filter section (empty box) was in the range of 3 to 7 mm/sec.



Industrial filter Coated filter

Figure 2: Picture of an industrial and a coated filter

Molten metal characterization

In order to assess the inclusion content of the melt, two LiMCA II devices (ABB Inc., Quebec, Canada) were installed on the casting line, respectively before and after the filter. The LiMCA measurements are expressed in k/kg (thousands of inclusions per kg of molten aluminum). N20 indexes represent the total amount of inclusions larger than 20 μm, expressed in k/kg. Similarly, N20-30 represents the number of inclusions in the size range 20-30 μm. These measurements also enabled the determination of the filtration efficiency E of the filters.

$$E = \frac{N20_O - N20_i}{N20_O} \times 100 \quad (1)$$

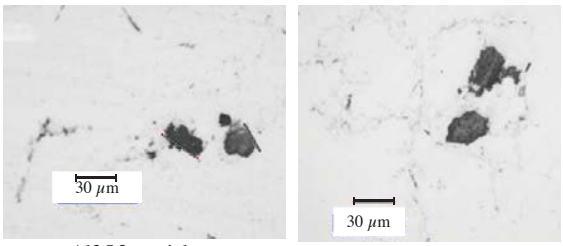
where $N20_0$ and $N20_i$ are respectively, the average number of inclusions larger than 20 μm before and after the filter [1]. PoDFA sampling was also performed.

Particle seeding

In order to guarantee similar inclusion contents in each trial, it was decided to seed the metal before casting by using composite ingots. The choice of the non-metallic particles in the ingots was based on two specific criteria:

- The nature of the non-metallic particles in the ingots should be similar to the inclusions in industrial products (chemistry, density etc.)
- The inclusion size should be in the range of 20-100 μm (to allow quantification by the LiMCA and by metallography).

The following particle types were chosen: alumina (Al_2O_3), spinel ($MgAl_2O_4$), mullite ($3Al_2O_3 \cdot 2SiO_2$), and silicon carbide (SiC). Figure 3 shows examples of cross sections of the seeding ingots.



Al2O3 particles SiC particles

Figure 3: Examples of micrographs of the seeding ingots

Filters characterization

Scanning electron microscopy (SEM) was used to characterize the functionalized filters before and after the filtration trials.

The geometric characteristics of the filter tiles chosen in the present article are relatively close to each other for the different functionalized coatings [15].

In the case of the spent filters, a more detailed investigation of the filtration behavior was carried out analyzing several samples taken at the inlet, in the middle, and at the outlet of the filters. The inclusions found were imaged with a suitable magnification and analyzed with EDX. Every non-metallic particle larger than 5 μm was counted as one inclusion.

Results

Before performing the trials with the experimental coated filters it was decided to run a blank test to assess the potential impact of the pilot set-up on the inclusion density. In this configuration the filtration box was left empty, without CFF. Figure 4 plots the LiMCA N20 indexes respectively before and after the filtration box. There was a technical issue with the LiMCA used after the filter box which did not allow to measure data points between 10 and 35 min into the cast. The curves show similar indexes in terms of inclusion content.

This trial validates the fact that the casting pilot set-up does not, by itself, result in inclusion capture. Based on this result, the reported inclusion removal are attributed to the tested filter natures.

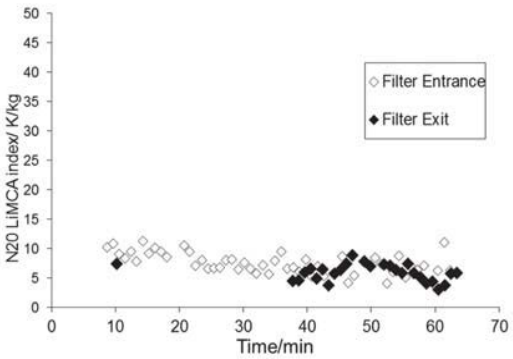


Figure 4: Blank test – Set up validation

The subsequent filtration trials were carried out in order to study the effect of the functionalized filter coatings on the filtration efficiency.
The inclusion content at the furnace exit, assessed by LiMCA, is reported in Figure 5. The incoming metal quality mainly ranges between 10 and 25 k/kg throughout the duration of the cast. The curves show that the filtration pilot is able to reproduce inclusion contents in the range of those measured at the outlet of industrial casting furnaces. Similarly the trends are those observed during industrial casts: progressive reduction of the inclusion number at the beginning of the cast (due to particles settling in the furnace) followed by a more stable period, and finally an increase in inclusions content at the end of the cast (associated to the release of the settled inclusions).

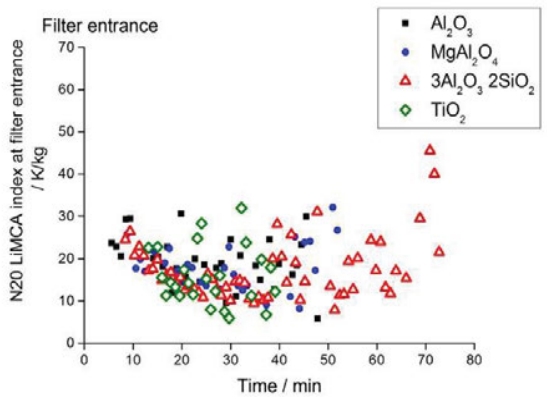


Figure 5: LiMCA results at the filter entry

The LiMCA inclusion index at the filter exit is reported in Figure 6. The trend shows an inclusion count reduction at the start of the cast followed by a quite stable inclusion content during the cast.

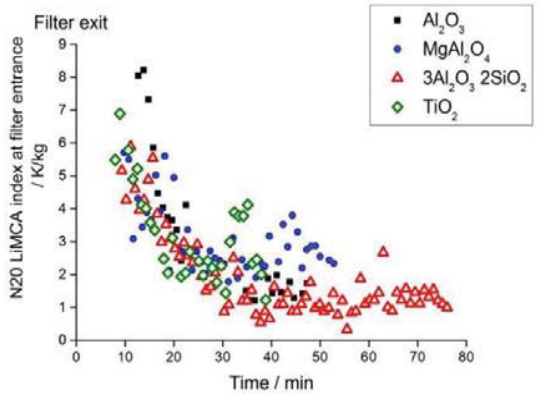


Figure 6: LiMCA results at the filter exit

Figure 7 shows the analysis of LiMCA efficiency by size classes of 10 μm width. For this analysis the mean LiMCA data through the cast are used. The graph shows high filtration efficiencies for particle size ranging between 20 and 80 μm, with quite similar figures for all tested coatings. However, as larger inclusions sizes are concerned, mullite and rutile coatings exhibited lower efficiencies than alumina and spinel coatings. This behavior can be seen in figure 6 and was already observed in a previous work [12].

As already mentioned the experimental set-up allows to recover the in-situ solidified CFF and hence the study of the spatial distribution of inclusions in the ceramic tile.

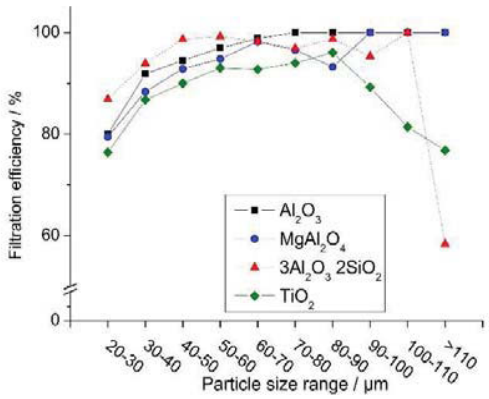


Figure 7: LiMCA efficiency by size range

Figure 8 shows a typical solidified spent filter after removal from its seat. The tiles have been cut lengthwise and transversally in order to perform specific metallographic observations. The individual inclusions were identified and counted in a scanning electron microscope. These observations of the filters confirmed they were primed on the full section, in all tests conditions.

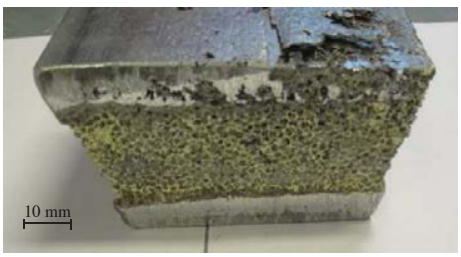


Figure 8: Solidified filter after removal at the end of the cast

Discussion

Industrial filters are typically 17 to 26" in side dimension (430 to 660 mm). However assessing the efficiency of experimental filters is done on a dedicated set up convenient for an initial research stage (90 mm on the outlet side). The experimental set up has been shown to be convenient for the downsizing of production conditions: no measurable impact of the filtration system itself (blank test), filtration velocity in the range of the filter recommendations, inclusion counts comparable with those of a casting furnace exit, cast duration similar to a (short) industrial cast.
The experimental procedure required the seeding of the melt in selected inclusions. This process resulted in the introduction of other inclusions in addition to the voluntarily added particles, in particular oxide films (figure 9). Indeed the Ar/Cl_2 furnace treatment was done before particle seeding in order not to float the added particles. Oxides pre-existing in the seeding ingots and oxides generated by the manual stirring of the furnace are not removed and contribute to the filtration efficiency.

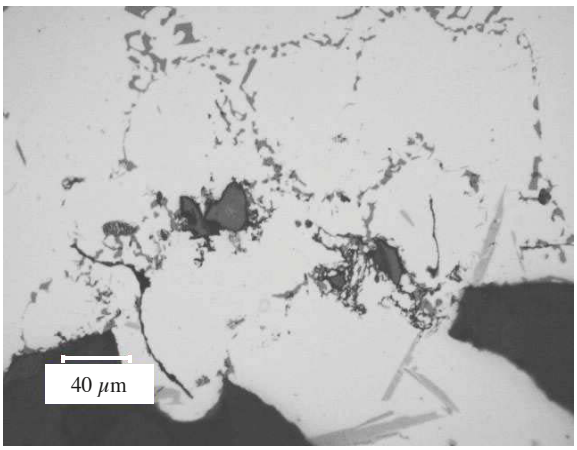


Figure 9: Typical PoDFA sample (furnace exit). Voluntarily added particles and oxides generated during the process

The spent and cut filters were observed in first instance to check the integrity of the coatings which is a few tens of μm thick. This is a prerequisite for the filter usage. In addition released parts of the coating may have impacted the filtration result. It is shown in figure 10 that all the coatings tested were not damaged during the casts. In addition, a larger scale observation did not reveal filter or coating parts removed for the filter struts. It can be assumed that the filter coating production is adequate for the aluminum casting processes (preheating and casting).

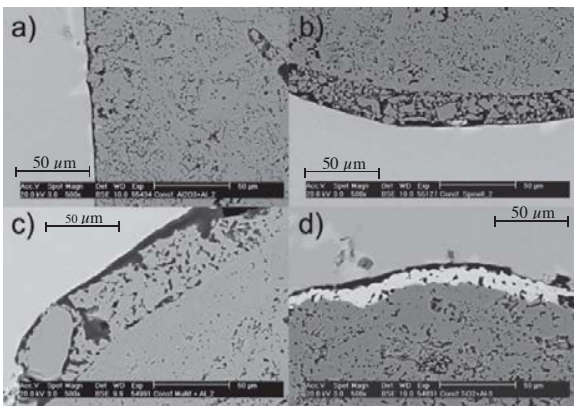


Figure 10: SEM micrograph of the alumina skeletons with different coatings after the casting trials
a) Al_2O_3, b) $MgAl_2O_4$, c) 3 Al_2O_3 2SiO_2, and d) TiO_2

The measured LiMCA efficiencies are high as compared to industrial data for 30 ppi CFF [1]. This observation is valid for all filters. It is likely related to the fact there was no refiner used in these trials. As already mentioned oxide skins were observed in the PoDFA samples taken at the filter exit. It is known that oxide films play a significant role in building bridges inside the CFF filters which collect inclusions. In industrial production most often a refiner is used which is known to destroy the internal bridges and reduce the efficiency of CFF [1, 3].
In order to understand the mechanisms involved in the coated filters the inclusions were methodically counted and analyzed in sections of the spent filters. The number of inclusion captured is reported in figure 11 as function of the filter surface chemistry. The overall mean LiMCA efficiency is included in the same graph. It is shown that both indicators give the same trends for the different filter natures.

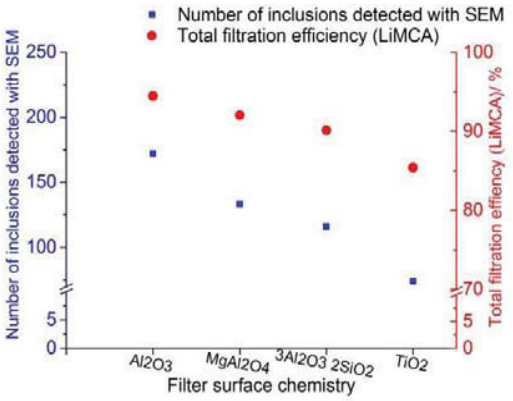


Figure 11: Number of inclusions counted (SEM) and mean filtration efficiency (LiMCA) as function of the coating chemistry

In addition, figure 12 details the inclusion natures observed with the 4 tested coatings. The figure suggests all inclusion families are removed regardless of the coating nature. This observation differs from what has been seen when active filters were tested for steel production [9]. It can likely be explained by the difference in relative temperatures during the filtration process. Illustrating for alumina filtration in steel production: the relative temperature (melt temperature / inclusion melting point) in steel production is about 0.8, which can explain enhanced capture by sintering when an oxide substrate is used. In aluminum production the relative temperature is less than 0.4 so that sintering processes will not occur. Hydrodynamic phenomena will be more important in aluminum filtration while chemical (reactive interfacial phenomena) may play a larger role in steel filtration.

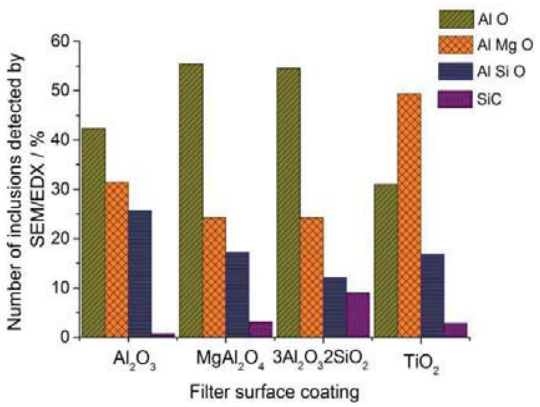


Figure 12: Proportion of inclusions of a given nature captured as function of the coating nature

Conclusions

A rigorous lab-scale methodology was used to evaluate the behavior and the efficiency of ceramic foam filters for use in aluminum alloy production. Lab scale casts were performed which lasted for about 1h and during which LiMCA measurements and PoDFA sampling were done. Filters with four different surfaces were evaluated with this methodology (alumina, mullite, spinel, and rutile). It has been seen that the filter coatings were not damaged during the preheating and casting processes.

The filtration efficiency was determined by LiMCA. The resulting LiMCA efficiencies were elevated and quite similar for all coatings tested. The absence of grain refiner during the trials likely contributed to the high efficiencies measured. The distribution of inclusions in the filter suggest a limited selectivity in inclusion removal. This confirms the importance of the hydrodynamic phenomena in ceramic foam filtration.

Further work is needed to understand the interaction between the particles and the filter, in particular, the impact of the morphology and chemical interaction between the filter and the inclusion.

Acknowledgments

The authors would like to thank the German Research Foundation (DFG) for supporting these investigations through the Collaborative Research Centre 920 “Multi-Functional Filters for Metal Melt Filtration – A Contribution towards Zero Defect Materials”, Subprojects A02, S03, A03 and A06. The authors also would like to acknowledge the support of G. Schmidt and J. Hubálková (S01).

The contribution of Alexis Janot, Robert Rey-Flandrin and Yves Pouyet from Constellium C-TEC is gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Numerical Simulation of Degassing Phenomena in Direct Chilled Casting Process under External Static Magnetic Field on Flow Pattern in Slab Mold

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Keywords: Degassing; numerical simulation; direct chilled casting (DCC); Static magnetic field; slab mold.

Abstract

For good quality billet, metal degasing is important consideration in billet casting. In-line degasser processes help to remove 61 to 66% of the dissolved gas. An improved billet quality can increase extrusion speed. A model of degassing delineates the most important key factors for optimization of product design for high quality, productivity and cost savings using computer simulations. The present work focuses on aluminum direct chilled casting process with gas injected in the mold through a submerged entry nozzle. The effects of gas into the melt and a direct casting magnetic field on the mold fluid flow are studied. Considering the effect of complex flow, injection and heat transfer on the gas bubble size the particles model reveals to be an efficient method to study the degassing phenomenon. However, the model treats the gaseous phase as bubbles with multiple sizes providing a new approach to simulate multiphase flow in continuous casting.

1. Introduction

Conitnuous casting processes for both aluminium alloys and steel were developed several decades ago to produce shapes for subsequent semi-fabrication processes such as extrusion or rolling. As-cast product shapes include billets, thin slabs, strips (thickness between 1 and 12 mm for both aluminium alloys and steel), and rounds/extrusion billets (100- to 500-mm diameter for both aluminum alloys and steel). In recent decades, a dramatic growth of this primary metal processing technology has been realized in both aluminum industries and steel, owing to a substantial increase in yield, energy savings, and productivity over static casting. However, the technological advancement has taken distinctly different routes for these two metal industries. Over the years, the casting procedures for aluminum alloy and steel products have developed distinctive features in terms of casting practices, machinery, and process and quality control methodologies.

In the conventional continuous (or strand) casting of steel, shown in Fig. 1(a), liquid steel flows from the bottom of a ladle into a small intermediate vessel known as the tundish. It leaves the tundish bottom through a submerged nozzle, according to the position of a stopper-rod or slide-gate flow control system. The liquid flow is directed into the mold (usually 700 to 1200 mm in length), and freezes a thin shell against the water-cooled copper walls. At steady state, the solid shell exiting the mold forms a stable strand, which has adequate mechanical strength to support the liquid metal core (typically 5 to 30 m in depth, depending on the casting speed and thickness). The direct chill (DC) casting process for aluminum alloys is shown schematically in Fig.1(b). In contrast to the continuous casting process for steel, DC casting is only semi-continuous; as the strand is withdrawn vertically for a short length (10 m) until the process must be stopped and restarted when the cast ingot reaches the bottom of the casting pit. Thus, considerable attention must focus on the initial start-up stage, when defects are most likely to be initiated. To start the process, a bottom block is partially inserted into an open rectangular mold (usually 100 to 150 mm in length). Superheated liquid aluminum flows through a launder, down the nozzle spout, through a distribution bag, and into the mold, generating a magneto-hydrodynamic environment. Once the molten metal fills the bottom block to a prescribed height, the bottom block and cast ingot are lowered into a casting pit.

This paper is undertaken to study the utilization of a magnetic field in cast-house providing an ingenious and efficient appliance for an adequate and contactless flow control in the mold, which facilitates substantial capabilities to improve the quality of the aluminium and to increase the productivity of the process. The external magnetic field is of high capability to monitor the flow of liquid aluminium in the mold cavity by damping, accelerating, levitating, and stirring.

The principle of an electromagnetic brake technology usage in cast-house industry employing a static magnetic field aligned perpendicular to the main flow direction is used. It relies on the interaction between the electrically conducting melt and the applied magnetic field resulting in a retarding force to slow down the mold flow and to damp substantial velocity variations. A uniform reduction of the melt flow specifically in the vicinity of the jet emerging from the submerged entry nozzle (SEN) is the major objective of the flow control because unsteady flows at high velocities are assumed to engender an entrapment of gas (CO_2 bubbles) or non-metallic inclusions damaging therefore the metal purity extremely. When a magneto-hydrodynamic (MHD) flow is studied in the presence of an external magnetic field, we essentially confront two boundary conditions: (a)-electrically insulating wall and, (b)-electrically conducting wall. When an electrically conducting viscous incompressible fluid flows through ducts and a uniform steady magnetic field acts perpendicular to the channel walls, the distribution of electric currents describe significant variations. For electrically insulating wall, the electric currents make closure inside the channel and are reinforced in the vicinity of the walls. The walls normal and parallel to the direction of the external magnetic field are named Hartmann wall and side wall, respectively. For

an electrically conducting wall, the electric currents are able to pass through the interface between liquid and solid shell. Clearly, the boundary condition is very essential when tackling with the MHD flow. To study the electromagnetic effect in a continuous casting aluminium slab mold, the correct and detailed expressions of the boundary conditions are required. The magnetic field is applied perpendicular to the wide face; the cyan solid line presents the induced electric currents. A layer of re-solidified flux prevails in the region between the liquid metal pool and the water-cooling mold. The electrically insulating wall is assumed on account of the small electrical conductivity of the re-solidified flux. Therefore, the electrically insulating wall boundary condition was carried out by most of numerical simulations in the past. However, in real application the electrical conductivity of the solidified metal shell has to be considered, which is larger compared to that of the molten metal.

Although various electromagnetic device designs have already been adopted for industrial application since more than 20 years [1], the effects of a magnetic field on such highly turbulent and complex flows is a complicated phenomenon and has not been fully understood until now. Contrary to the usual expectations, static magnetic fields may even destabilize liquid metal flows. Similar observations have been found in convection experiments in liquid metals, where it could be demonstrated that a weak DC magnetic field can enhance the convective heat transfer [2]. First, direct observations of the destabilizing effects on the velocity field by an applied magnetic field were reported by Zhang et al. [3], who considered the application of a horizontal magnetic field on a bubbly-driven flow inside a cylindrical liquid metal column. For a certain parameter range, the magnetic field gives rise to the development of distinctive transient flow pattern with increased turbulent perturbations. In spite of physical models, computational fluid dynamics (CFD) is now used widely in industries, for instance, the effects of a magnetic field on the melt metal flow in the continuous casting mold was addressed by many numerical studies considering various magnetic field configurations or examining the influence of the variations of different casting parameters on the magnetic field effects exerted by a given assembly. There is no doubt that a magnetic affects the jet char-acteristics and the entire flow pattern in the mold significantly. Most of the authors rely on a k-ε turbulence model and report a suppression of the mean flow and the turbulent fluctuations as well. The convective heat transfer is usually decreased and uncontrollable deflections of the free surface can be reduced. Cukierski et al. [4] studied the configuration of a local electromagnetic device and documented remarkable effects of the magnetic field on the jet angle, the impingement depth, the recirculation zones and the surface velocity. Kim et al. [5] employed a revised variant of a low Reynolds-number *k-ε* turbulence model to simulate the mold flow under the influence of a local magnetic field. Wang et al. [6] obtained a lowering of the turbulent kinetic energy at the SEN outports and a more uniform flow pattern in the mold if a local magnetic field is applied. Similar findings were presented for local magnetic field in [7], magnetic fields of the ruler type [8] or in a flow control mold [9].

The efficiency of braking the velocity at the free surface or the penetration depth of inclusions was suggested to depend on the position and intensity of the magnetic field. A comparison between local and ruler electromagnetic brake was drawn by Harada et al. [10] showing a higher braking efficiency provided by a level magnetic field. These numerical results based on Large Eddy Simulations (LES) are supported by accompanying mercury model experiments. It is interesting to note that a sensitive dependence of the surface velocity on the nozzle conditions was observed in case of a local magnetic field. A related study based on LES was performed by Takatani [11] who described the general effect of the local magnetic field as a suppression of the turbulent flow and a tendency of the flow pattern to become two-dimensional. Furthermore, he found that the electromagnetic brake may accelerate the fluctuations and the meniscus flow in some cases, in particular for an improper selection of the nozzle discharge angle.

Gas injection then is necessary to prevent SEN from clogging and to reduce the quantity of inclusions in molten aluminium by attaching. But with gas injection, the gas phase may often cause the entrainment of mold slag into the liquid pool resulting in defects. In aluminium continuous casting process, the bubbly flow regime is the most desirable operational regime because the small discrete bubbles provide the greatest collection efficiency for inclusions, and also minimize the generation of unwanted turbulence within the casting mold. In principle, interactions of bubbles exert a great impact on the flow pattern in a bubbly flow system. Three basic flow regimes were observed by Evans et al [12] during the experimental trials in the down-flowing column for the range of gas and liquid velocities. These were (i) bubbly, (ii) churn-turbulent flow, and (iii) annular flow. Flow regimes are closely dependent on the gas void fraction. Wallis et al.[13] defined the critical void fraction, 0.27, which is the void fraction at zero gas drift flux where bubble coalescence commences and churn-turbulent flow begins. It is in reasonable agreement with the value obtained by Evans et al.[14] for turbulent liquid flow in a down-flowing air-water system.

Some few theoretical papers are devoted to the magnetic field impact on the shape of a single bubble rising in a liquid metal. Unfortunately, respective experimental results are not available because of the still missing availability of measuring techniques to provide a reliable reconstruction of the surface of a gas bubble inside an opaque liquid metal. Shin et al. [15] considered an incompressible gas bubble in an axisymmetric straining flow under the influence of a uniform magnetic field. They predicted an elongation of the bubble along the direction of the magnetic field. This bubble elongation increases monotonously as the magnetic interaction parameter increases. The phenomenon of bubble elongation in field direction was also reported by Shibasaki et al. [16] for a bubble rising inside a steady magnetic field parallel to the gravity force. Takatani [17] studied two configurations of a bubble ascending in either a vertical or a horizontal magnetic field. In case of a longitudinal field the bubble contour is transformed to a bullet-like shape, which reduces the flow resistance, but leaves the terminal velocity almost unaffected compared to

the situation without magnetic field. In contrast, the shape of the bubble becomes flat if a transverse field is applied. The resulting bubble velocity is supposed to be lower than that without magnetic field. Experimental investigations on rigid spheres revealed an increase of the drag coefficient proportional to the square root of the interaction parameter, Galfgat et al. [18] and Eckert et al. [19] investigated the influence of a transverse magnetic field on the velocity slip ratio in a channel flow. The linear dependence of the drag coefficient on the magnetic field strength would imply a continuous decrease of the slip ratio with rising magnetic flux. However, the slip ratio was found to be reduced only at moderate values of the magnetic field strength, but increases again at higher values of the field intensity. This tendency is caused by the braking effect on the liquid metal flow, which is proportional to B^2 and becomes dominant at high field intensities therefore.

Experiments with single bubbles in stagnant liquid metal pools have demonstrated that an imposed DC magnetic field modifies the drag coefficient. A vertical magnetic field damps the horizontal components of the bubble velocity. This effect forces the bubble into a straighter path and reduces the apparent drag force. Similar observations were made by Mori et al. [20] in a transverse magnetic field. An elimination of the zig-zag motion of the bubble leads to a higher final velocity. By contrast, the final velocity of bubbles moving along a rectilinear way decreases with increasing field strength. Another study was focused on turbulent dispersion of gas l
duct flow which were initially injected
The application of a transverse magneti
anisotropic distribution of the void fraction over the duct cross-section with a significantly higher dispersion coefficient found for the direction parallel to the field lines. This finding indicates that the damping of turbulent fluctuations is much more pronounced in the direction parallel to the magnetic field than in the perpendicular direction. That means that the bubble dispersion is determined by the existence of quasi-two-dimensional fluctuations with a vorticity along the magnetic field direction being well-known for MHD turbulence. The rise of gas bubbles drives a flow inside the liquid metal and acts as a source of turbulence. Gherson et al. [21] investigated a mercury pipe flow with dispersed nitrogen bubbles. At large magnetic fields they found regions with liquid turbulent fluctuations higher than in the case without magnetic field. The authors explain this observation by a magnetically- induced redistribution of the void fraction with the formation of large but unstable bubbles. The higher frequency of bubble break-up phenomena engenders an additional turbulence production. Recently, Zhang et al. [22] presented an experimental investigation with respect to the effects of magnetic field on a bubble plume in a cylindrical liquid metal column. Measurements of the liquid velocity shave shown that a transverse magnetic field might lead to a destabilization of the global flow resulting in transient, oscillating flow structures with predominant frequencies. This conclusion seems to be contrary to usual expectations, because the Lorentz force is often regarded to cause a deceleration of the mean flow and a damping of turbulent fluctuations. The overall objective of this work focuses on studying the effects of magnetic field and gas injection on the metallic flow in a continuous casting slab mold numerically. Numerical simulations will be performed by means COMSOL 5.0 package. This work is carried out to provide a flexible numerical model which is able to deal with complex industrial metallic flow under the influence of magnetic field and to obtain better understanding of aluminium flow control in the continuous casting process. Next, in Section 2 the mathematical model for a multiphase fluid flow system submitted to a magnetic field during continuous casting is presented. In Section 3, we discuss the physical model used in this present study. Then, we demonstrate the efficiency of our turbulent numerical model and its utilization in COMSOL 5.0 package to conduct the numerical simulations in Section 4. Finally, Section 5 concludes this paper.

Fig.1: **Schem**

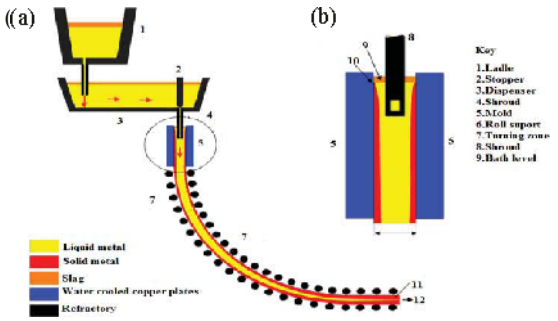

Fig.1: schematic of (a) the continuous casting process for steel slabs anbillets and (b) the DC casting process for aluminum sheet ingots.

2. Mathematical model for multiphase fluid flow in a direct chilled casting slab mold under an external magnetic field

In a multiphase fluid flow (liquid aluminium and gas bubbles), the flow pattern in a bounded flow is dependent on the void fraction of gas phase, slip velocity, interfacial shear, etc. In general, the flow regime is characterized as: bubbly flows, slug flow, churn flow and annular flow. In bubbly flow regime, numerous bubbles are distinguishable as the gas is dispersed in the form of discrete bubbles in the continuous liquid phase, whereby the bubbles may vary in both size and shape. In slug flow regime, coalescence of gas bubbles expand and form larger bubbles as a result of increasing void fraction of gas, which are similar in size to the pipe diameter and have a characteristic geometrical shape similar to a bullet with a hemispherical nose and a blunt tail. In churn flow regime, the structure of the flow becomes unstable with the fluid traveling up and down in an oscillatory way. Once the interfacial shear of the high velocity gas on the liquid film becomes dominant over the gravity force, the liquid is expelled from the center of the pipe and flows as a thin film on the wall. In addition, liquid may be entrained in the gas core as small droplets.

In the process of aluminium continuous casting, generally, CO_2 is introduced in Submerged Entry Nozzle (SEN) to prevent nozzle ports from clogging. We expect the gas phase injected in the liquid behaves as discrete CO_2 bubbles. To simulate gas bubbles expansion and migration within the metallic flow, the interphase momentum transfer

models are introduced. The interphase momentum, $M_{liq/gas}$, occurs due to interfacial forces acting on liquid aluminium phase (liq.) by the interaction with gas phase (gas) and vice versa. The interfacial forces between two phases are equal and opposite, therefore the net interfacial forces sum to zero. The detailed mathematical models are given in the following section.

2.1 Hydrodynamics equations

In our simulations, we apply the Euler-Euler approach considering both the liquid aluminim and the gaseous phase with certain volume fractions with the volume fractions of liquid (λ_{liq}), and gas (λ_{gas}), respectively. On the supposition that any mass exchange between the phases (liquid and gas) can be neglected, the system of equations of continuity and momentum is given as follows [17]:

$$\frac{\partial(\lambda_i \rho_i)}{\partial t} + \nabla \cdot \left(\lambda_i \rho_i \vec{U}_k\right) = 0 \quad (1)$$

$$\frac{\partial\left(\lambda_i \rho_i \vec{U}_i\right)}{\partial t} + \nabla \cdot \left(\lambda_i \rho_i \vec{U}_i \vec{U}_i\right) = -\nabla p - \nabla \cdot \left(\lambda_i \tau_i\right) + \lambda_i \rho_i \vec{g} + \vec{F}_{\text{int},i} + \vec{F}_{mag,i} \quad (2)$$

Where the subscript i = liq represents the liquid aluminium phase and i =gas is the gas phase, respectively. The terms on the right hand side of Eq. (2) represent the pressure gradient, the turbulent viscous stress, the buoyancy force, the interfacial forces $\vec{F}_{\text{int},i}$ and the electromagnetic force $\vec{F}_{mag,i}$.

In multiphase fluid flow, the summation of the net interfacial forces is equal to zero. The interfacial forces $\vec{F}_{\text{int}/liq}$ in Eq. (2) can be written as follows,

$$\vec{F}_{\text{int}/liq} = -\vec{F}_{\text{int}/gas} \quad (3)$$

Where the term on the right hand side of Eq. (3) represents the drag force, lift force, the wall lubrication force etc.

This model has been extended for the situation of a liquid-bubbles mixture flow. Sato et al. [25] proposed a model that takes into account the turbulence induced by the motion of dispersed gas bubbles. They assumed that the effective viscosity of the liquid aluminium phase is composed of laminar viscosity, the turbulent eddy viscosity and the bubble-induced turbulence contributions. However, Deen et al. [26] have found that an addition of bubbles-induced viscosity does not significantly decrease the accuracy of the numerical results. Considering the expansion of gas bubbles through the liquid aluminium and the effects of bubbles-induced turbulence, Viollet et al. [27] included additional source terms into the turbulence models.

3. Physical model

In aluminium continuous casting process, gas (CO_2) is injected to prevent submerged entry nozzle (SEN) from clogging and to reduce the quantity of inclusions in molten aluminium by attaching.

Therefore, an applicable multiphase model is required when studying the electromagnetic effects on continuous casting slab mold. The validation of our turbulent model on multiphase fluid flow has been performed successfully. In this paper, the effects of gas injection and the electromagnetic field is studied. A poly-dispersed particle model has been developed on the precondition that the gas phase is regarded as gas bubbles. The mathematical model has been introduced in the following Section. Numerical simulations are performed by using COMSOL 5.0. Fig.2 shows the full scale computational domain. At the inlet flow rate of the liquid aluminium is 0.1 l/s and the flow rate of gas is given by 450cm^3/min. In the poly-dispersed particle model, gas bubbles can be classified by size in which the diameters of the gas bubbles vary from 0.1 to 10 mm. The inlet velocity of the molten aluminium was calculated by the flow rate. Standard equations for flow in pipe were used to describe this velocity profile.

The turbulent kinetic energy and the dissipation rate at the inlet are estimated by using the semi-empirical relations [35]. The flow rate of gas was calculated by the effective injection area and the volume ratio of molten aluminium to gas. At the meniscus, a degassing boundary condition was specified, at which only the bubbles were allowed to escape. A zero-shear condition is specified for molten aluminium without considering the effect of the mold flux. Normal gradients of all variables are set to zero. To opening pressure and direction is specified. Once the flow direction is into domain, the pressure value is taken to be the total pressure based on the normal component of velocity. While it is leaving the domain, it is taken to be relative static pressure. At the outlet, the relative static pressure is set to zero. A no-slip condition for liquid aluminium, free-slip condition for gas bubbles and automatic-near-wall function is used for all walls. The magnetic field is static inhomogeneous and obtained from, which is implementation into COMSOL 5.0 as a three-dimensional (3D) external magnetic field. For an electrically insulating boundary, the gradient of the electrical potential is set to zero at the wall. For an electrically conducting boundary, the electric current is conserved through the interface between fluids domain and the solid domain.

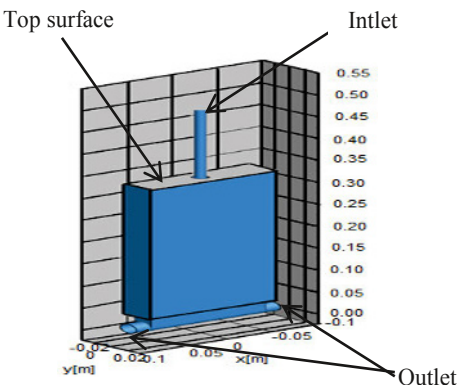

Fig.2: Physical domain used in this study.

4. Numerical results

4.1. Model-based poly-dispersed particle

In industrial application, the gas injected into the mold distributes non-uniformly with variation of the gas bubbles size. Therefore, further numerical prediction will be performed with the same model. As described previously, gas bubbles discharged from the submerged entry nozzle ports will separate from the jets. Large gas bubbles drift up rapidly and most of them migrate in the upper recirculation zone. In addition, small bubbles float up slowly due to the influence of drag force and wake. Smaller bubbles can be subsequently carried by flowing downstream and are easily entrapped into the lower recirculation zone. The distribution of dissolved gas varies with wall boundary conditions as shown in Fig. 3. It is clear that the gas plume is remarkably modified under the action of the electromagnetic field. For a case where only the buoyancy force reigns as in Fig. 3-(a), gas bubbles widespread in the upper recirculation zone. In comparison to the case with magnetic field, the gas plume was more restricted to the SEN area under the influence of magnetic field in Figs. 3. In particular, gas bubbles are mainly concentrated in the vicinity of the SEN till they escape from the top surface for the case of electrically conducting wall (Fig. 3-(c)).

The turbulent kinetic energy is one of the most important variables in gas/liquid flow, which is the ratio of root-means-square of the turbulent velocity fluctuations and the average velocity. The magnetic field also influences the turbulence. Fig.4 depicts the time-averaged distributions of turbulent kinetic energy in the mid-plane across the wide face. For a case of electrically insulating wall, the turbulent kinetic energy is slightly decreased in comparison to the case without magnetic field in Figs. 4-(b)-(c). On the other hand, the turbulent kinetic energy is significantly reduced in case of an electrically conducting wall such as in Fig.4-(c). Vortex cores for the different boundary conditions are shown in Fig.4. The eddies in the mold cavity are oriented parallel to the direction of the external magnetic field. Due to the Joule dissipation, some of the eddies are reduced (Fig.5-(b)-(c)) compared to the case without magnetic field.

5. Conclusion

In an aluminium continuous casting process, generally, CO2 gas is injected in the mold through the submerged entry nozzle (SEN). In this study, the effects of gas injection and magnetic field on the fluid flow in the mold were studied using the poly-dispersed particle model.

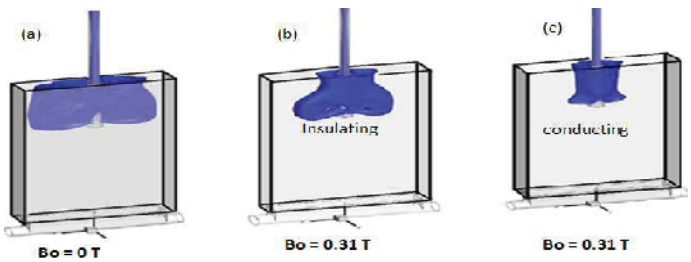

Fig.3: Three-dimensional representation of gas fraction over time varying in relation with electrical wall boundary conditions.

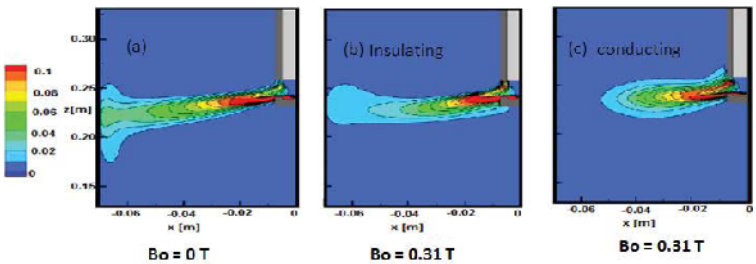

Fig.4: Time-averaged contours of turbulent kinetic energy in the mid-plane.

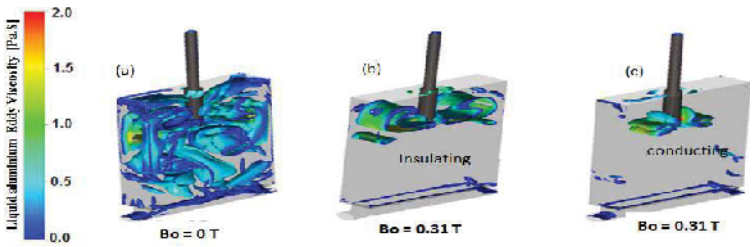

Fig.5: Vortex cores for the different boundary conditions.

Simulation results show that the distribution of the gas phase is dependent on the bubble diameter. Considering the effect of complex multiphase fluid flow (liquid aluminium/ gas bubbles), injection method, heat transfer, etc. to the

bubble size, the poly-dispersed particle model treats the gas phase as gas bubbles with different bubble diameters, thus providing a new way to simulate the two-phase flow in the continuous casting process. The overall results show that the gas injection and the magnetic field play an important role on the flow pattern. Moreover, to optimize the flow pattern in the industrial casting process, several crucial factors must be taken into account simultaneously. The Simulation results show that the distributions of gas phase depend on the bubble diameter by using the poly-dispersed particle model. To deal with the bubble flow with different bubble diameters, the poly-dispersed particle model gives rise to more accurate predictions compared to the mono-dispersed particle model. The flow pattern in the mold is sensitive to the strength of a static magnetic field and the wall conductance ratio. An improperly employed magnetic field may cause severe side-effects, such as additional large eddies or a deepening of the penetration depth. Considering the solidified shell in the mold cavity, the electrical wall boundary condition is of importance to be taken into account in order to get more accurate predictions.

Acknowledgements

The authors would like to thank Masdar Institute of Science and Technology and Emirates Global Aluminum for support.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE PROBLEM OF CAVITIES IN OPEN MOLD CONVEYOR REMELT INGOTS

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Keywords: Shrinkage cavities, porosity, remelt ingot

Abstract

Open mold conveyor casting machines for 8-25kg ingot produce millions of tonnes of remelt ingot every year. However, these ingots suffer from cavity formation. These cavities represent a safety risk during remelting of the ingots due to the possibility of water molten metal explosions if the cavities contain water. The formation mechanisms of these cavities and the effect of alloy, machine design and operational parameters are reviewed. Alternative means of controlling cavities are examined and assessed. The need for a new approach to the problem is highlighted.

Introduction

Millions of tonnes of aluminium remelt ingots are produced every year globally. These ingots can be in the form of T-shapes, large sows or smaller ingots ranging from 8-25kg. Purity grades of aluminium e.g. P1020 99.85% Al and aluminium silicon alloys are commonly produced in these formats.

Remelt ingots can be produced by a variety of casting processes (see Grandfield 2009[1]), the most popular being open mold conveyor casting machines. Conveyor casting involves pouring aluminium into cast iron or steel molds, mounted on a conveyor usually by a wheel-type system at the start of the conveyor. As the moulds move along the conveyor the ingots cool and solidify. The line is run at maximum speed such that the ingots are fully solid when they reach the end of the conveyor and are discharged from the molds for further cooling, stacking and strapping into bundles. In the case of primary smelter machines, the molds are generally water-cooled in order to achieve a high production rate of 20 t/hr or greater.

The ingots produced on these conveyors are a type of shape casting however, they do not have any kind of feeding system like most shape castings. Consequently they often have shrinkage cavities in them. These cavities present a potential safety problem. If during transport and storage these ingots come in contact with water, the water can find its way into the cavities in the ingots. If these ingots are then charged into a molten bath of metal a water/molten metal explosion can occur. Producers take steps to educate customers about this risk and recommend the use of ingot drying and preheating to avoid this possibility. Charging to a dry furnace hearth is also recommended.

Despite these efforts, incidents still occur within the industry resulting in injury and equipment damage. Small foundries which often require the manual charging of individual Al-Si alloy ingots are particularly at risk because the operator is close to the melt when an incident occurs.

Alternatives to conveyor casting include continuous casting processes such as vertical and horizontal direct chill casting or Properzi type wheel and belt casting. In these processes, liquid is available to feed the porosity and cavity free product can be produced. However, they can also suffer from centerline cracking and other forms of cavity formation. In the case of wheel and belt casting the liquid pool is deep and sometimes centerline porosity can form depending on the water spray configuration. HDC casting can also have porosity forming in the top surface in the form of worm holes due to gas evolution in the mold from casting lubricant [2]. These continuous casting processes also generally have higher operating costs than conveyor casting machines.

A goal for the industry is to produce cavity free conveyor ingot. This paper reviews the known factors which influence the cavity formation in open mold conveyor ingots, presents various approaches which are used or have been investigated in the industry and discusses the prospects for achieving this goal.

Cavity formation mechanism

The formation of porosity in the form of small pores or large cavities is a well understood phenomena in castings. The basic mechanism of porosity formation is determined by the increase in density from liquid to solid (reducing volume) during solidification resulting in shrinkage (Figure 1). In the case of most shape castings some form of feeder is provided to deliver liquid during solidification to fill the reduced volume. However, no feeder is available for conveyor cast ingots. The shrinkage can manifest as either a drop in the level of metal in the mold, shrinkage pipes or internal cavities depending on the heat flow characteristics.

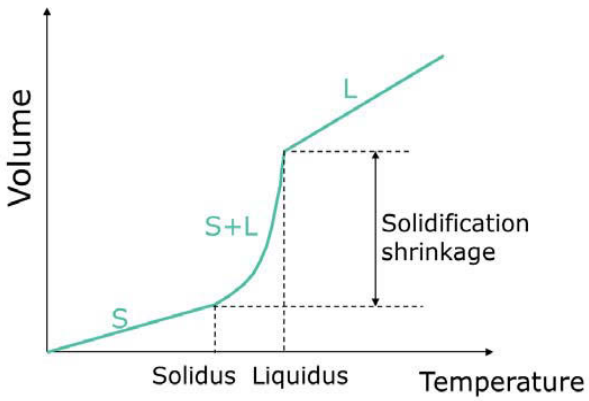

Figure 1 Change in volume going from liquid to solid.

The trapezoid shape of remelt ingots encourages a central cavity or centrally located network of smaller pores (Figure 3). If the ingot is solely air cooled, a large central cavity tends to form.

Some of the 23kg ingots cast are apparently free of porosity or internal cavities (Figure 4) but not all ingots are cavity free.

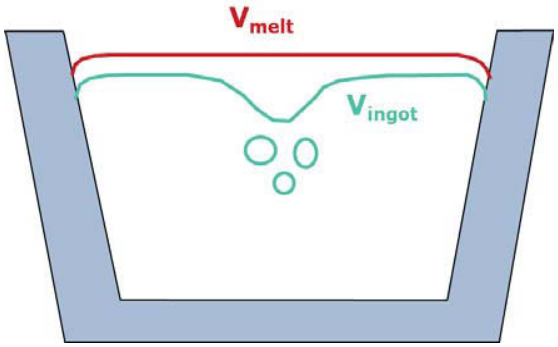

Figure 2 Schematic of porosity formation in remelt ingots.

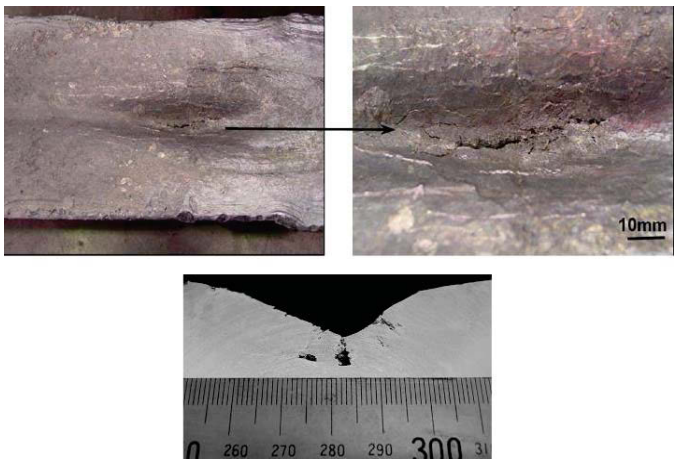

Figure 3 Typical cavities in 23kg P1020 99.85% Al ingot.

Figure 4 Section through 99.85% Al 23kg ingot showing central shrinkage but no internal voidage.

Part of the problem with the trapezoid ingots is that heat is flowing to the walls and the base of the mold. Consequently, the solidification front progresses from the walls inward and the base upward and the top surface begins to solidify at an early stage. If the whole top surface were to solidify at the final stage then this would eliminate the tendency for a central cavity. Thus, the aspect ratio of width to height also effects the tendency to form a cavity.

A low profile wide ingot has a smaller proportion of heat flow from sides and the solidification front teds to progress from the bottom up.

The columnar grain structure and solidification behavior often seen with P1020 purity ingots (Figure 4) also encourages shrinkage formation. AlSi alloys generally solidify with an equiaxed grain structure (Figure 5). If equiaxed grains form in the liquid pool they may settle to the bottom and essentially increase the progression of the solidification front from the bottom up (Figure 6). On the other hand, if these equiaxed grains bridge across the top surface, as they often do with silicon alloys, then large sub-surface voids can form (Figure 7). Sometimes the top surface collapses leaving an open void (Figure 8). The exact behavior is alloy dependent (Figure 8).

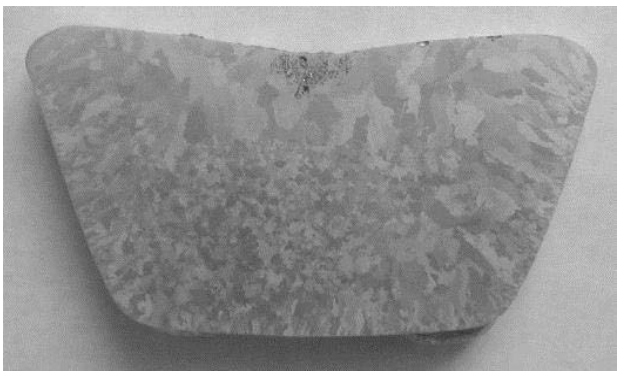

Figure 5 Equiaxed grain structure in Al-Si alloy ingot.

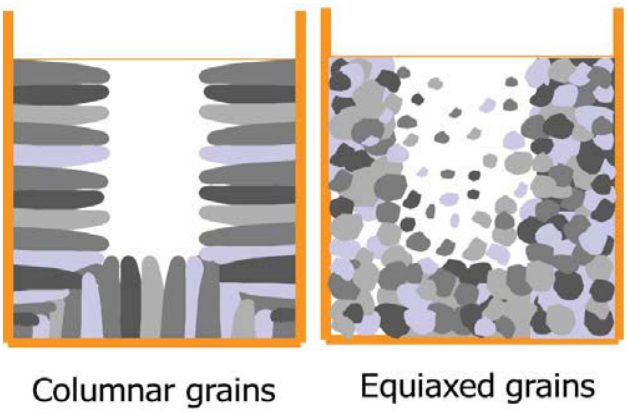

Figure 6 Schematic of columnar and equiaxed grain solidification.

Figure 7 Sub surface cavity in Al-7% silicon alloy 12kg ingot.

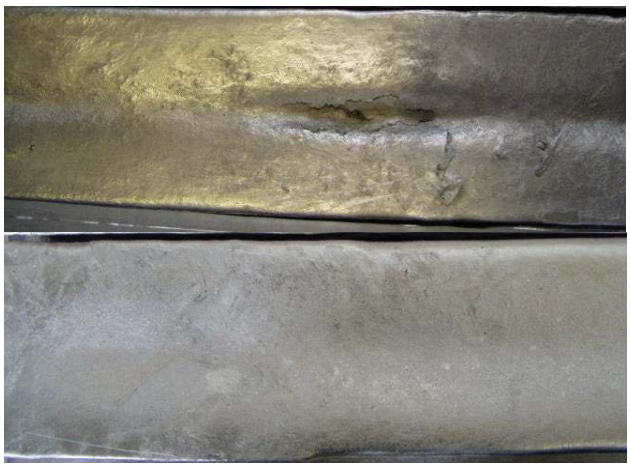

Figure 8 Cavities in ADC12 alloy ingot top and A313 alloy bottom cast under the same conditions.

Low profile ingots

The problem of cavities in large sow ingots has been tackled [3] by improving mold design and producing low profile sows i.e. sows with a low aspect ratio but even these shapes do not guarantee completely cavity free sows This approach also works with small ingots. Some producers in the industry do have wide low profile conveyor ingots.

The lower aspect ratio reduces the proportion of cooling coming from the walls.The solidification front thus tends to proceed from bottom up. However, there is a reduction in caster productivity as the wider pitch of these ingots reduces productivity per line length. Productivity, can be calculated from the following

$$P = \frac{3.6WL}{I_s t_s}$$

Where P is tonnes per hour, W ingot weight in kg, L line length in m, I_s, ingot spacing (pitch) in m, and t_s ingot solidification time in seconds [4]. The 3.6 is to account for the mixed units.

Hot topping

Hot topping is used in many casting processes. The intent is to keep the top of the ingot liquid until the last phase of solidification to prevent cavities. This approach has been used quite successfully in the case of zinc jumbo remelt ingot casting. However, the high latent heat of Al precludes hot topping. Trials on small format AlSi ingots (Figure 11) showed that crystals form on the top surface very soon after filling and it is very difficult to get sufficient heat into the ingots to melt these crystals and prevent a solid crust forming. These crystals seem to be held on the top surface despite their higher density than the melt.

Hot topping also results in extra energy cost and a productivity drop as the solidification time is increased. Similarly, insulating the top surface appears to give no benefit (Figure 9).

Figure 9 Insulating top surface ingot to encourage solidification from bottom to top.

Melt Temperature

Melt temperature may also effect shrinkage cavity formation in conveyor ingots but does not seem to have been studied carefully. In general it appears lower casting temperatures tend to reduce cavity formation.

Directional heat flow bottom to top

There are several ways one may imagine to encourage the solidification front to progress from bottom to top rather than in from the sides. Little success has been achieved with refractory coating of the mold walls (Figure 10). It is difficult to engineer the mold wall materials to be completely refractory in a practical way.

Lowering the water level on the conveyor so that only the base of the mold is water cooled and the walls are only air cooled is another approach that has been tested but with little benefit.

All these approaches tend to increase the overall solidification time and reduce line productivity as a result.

Figure 10 Attempt to reduce heat flow through the mold walls using a refractory mold coating.

Figure 11 Attempt at hot topping of small 12kg foundry ingot.

Top water sprays

Some success has been obtained by the use of top water sprays to control cavities within the industry. In general those plants using a fine mist spray seem to achieve better results. Stronger water sprays cause the top surface to freeze over before the ingot is fully solid thus causing a sub surface cavity (Figure 12). Water sprays can also cause staining and marks on the ingot surface. Air cooling of the top surface has also been tried and sometimes leads to improvement but results have been mixed.

Figure 12 Sub-surface cavity in 99.85% Al alloy 23kg ingot formed by strong top water sprays.

Skimming and stirring

There is some anecdotal evidence that ingot skimming seems to improve cavity characteristics. This may well be that during skimming the stirring of the melt at an early stage during ingot solidification generates seed crystal which settle into the bottom of the ingot and effectively increase the bottom up solidification. Deliberate mechanical stirring would achieve a similar effect, however there is an increased chance of dross formation and also the possibility of introducing contaminants into the ingot from the refractory stirrers.

Discussion

As can be seen from the above, there is not currently an entirely satisfactory solution to the problem of controlling cavities in remelt ingots. The use of low profile ingots is quite effective but results in a productivity loss per line length. One clear pathway is then to work on maximizing productivity for low profile ingots.

Other methods such as hot topping or top water sprays are either not effective, are unreliable or have other downsides. Reducing wall heat flow increases solidification time and reduces productivity. Top water sprays in particular may make the problem worse by encouraging sub-surface cavity formation.

Mathematical modelling helps understand what's happening with different alloys (Figure 13) and to assist in designing improved molds (Figure 14) with shorter solidification times and better solidification front progression [5]. However, although this approach can lead to improvement it does not give a comprehensive solution.

Continuous casting produces higher quality ingots with reduced chance of cavities but these processes are more expensive than chain conveyors.

A new technology is needed to control cavities in remelt ingots. A general approach would be to investigate ways to encourage unidirectional heat flow to the base of the mold.

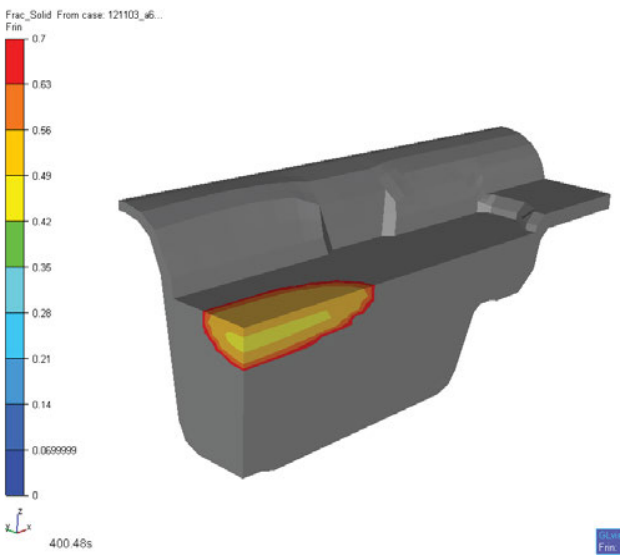

Figure 13 Alsim model prediction of shrinkage cavity formation in Al-Si alloy 23 kg ingot showing tendency for sub-surface closed porosity.

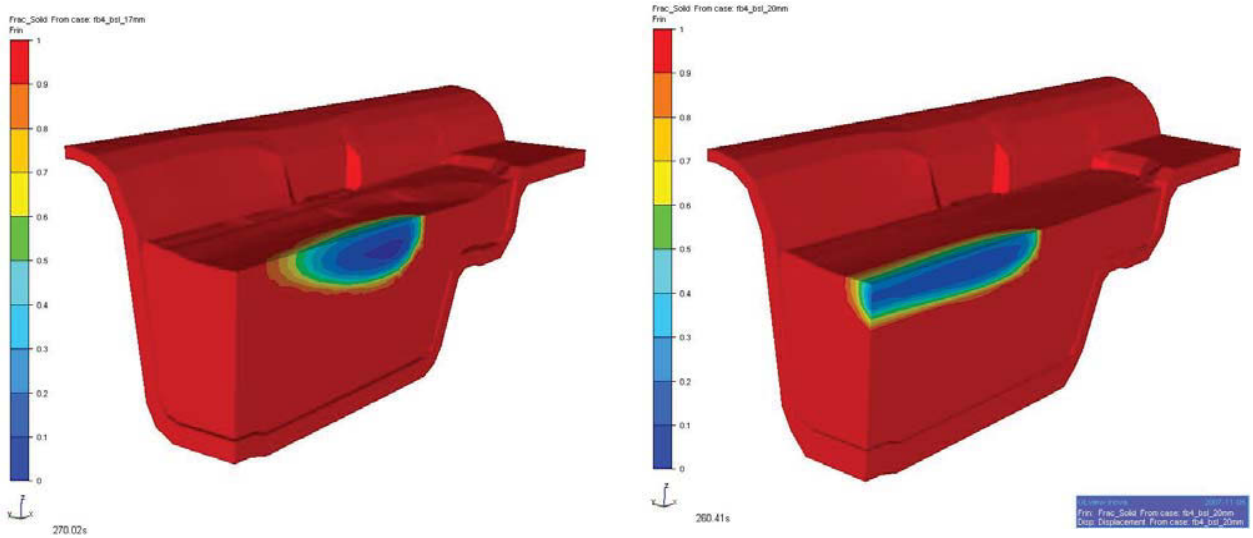

Figure 14 Mathematical model simulation using Alsim software of two different 23kg ingot mold designs showing different cavity formation tendencies. Blue area is last liquid to solidify.

Acknowledgements

Many thanks to those I have collaborated with over the years on ingot casting technology including CSIRO/CAST, odt Engineering, IFE, Elkem, Hydro, Rio Tinto Alcan, Portland, Mozal, Hillside & Alba.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THEORY AND PRACTICAL APPLICATION OF ULTRASONIC DEGASSING AT J. W. ALUMINUM

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Keywords: Ultrasonic Degassing, Inclusions, Twin-Roll Casting.

Abstract

Following the initial molten metal quality studies with the MetalVision MV 20/20 Inclusion Analyzer[1,2,3,4], presented previously, the need for more robust molten metal treatment equipment and methods was identified. Elimination of fluxing gas and improved filtration techniques were identified as opportunities. New degassing and cleaning technologies such as ultrasonic degassing were tested. This paper presents the results obtained during the initial trials of the ultrasonic SCR Ultra-D™ Degasser[5]. Some unexpected and surprising results led to further investigation (theory and confirmation tests), which then led to modifications in equipment and further process improvement.

Introduction

Continuous twin roll casting presents special challenges with regards to long casting campaigns with low molten metal flow rates. Typical twin roll casters require close tolerances with regard to molten metal level and temperature[6]. Interventions such as replacing filters, cleaning the degasser box, scraping the launders and cleaning the holding furnace therefore need to be kept to a minimum, yet running too long without cleaning also puts product quality at risk.

"Tribal" knowledge built up from experience over years presents the solution of "we have always done it this way." Challenging the status quo needs a data driven, scientific approach. This include the use of systematic techniques such as lean and six sigma, the fundamentals of science, and most importantly solid and reliable measurement and data gathering in order to make informed decisions.

In the last few years J. W. Aluminum has challenged the status quo at their plants by the systematic implementation of the above mentioned techniques. The initial phase concentrated on procedures, methods and immediate incremental gains. The second phase concentrates on longer term targets, such as introducing new technologies that results in a step change.

This paper describes introducing new ground breaking technologies such as the SCR Ultra-D™ Degasser.

Process Description

The predominant molten metal configuration at J. W. Aluminum consists of a three chamber melter (pump well, charge well and main hearth), followed by a holder and then molten metal treatment using a rotor degasser and two filters in series. Casters are twin roll continuous, producing cast coils to be rolled to final gauge. Casting is 24/7 with maximum utilization of equipment in order to satisfy customer demand.

Challenges with 24/7 Continuous Production.

Batch processes, such as DC casting, provide ideal opportunities for cleaning the molten metal supply chain. Melters and holders cycle between empty and full. During the empty stage, the furnaces can be cleaned. On the casting side, launders cycle between empty and full, while filter bowls can be drained and cleaned. Degassers can be drossed and cleaned between casts.

A previous paper[6] described the cleaning cycle developed by J. W. Aluminum using the MetalVision MV 20/20. Moving from this incremental to a step change improvement with a 24/7 casting operation presents a challenge. A survey of molten metal equipment suppliers reveals a strong bias towards DC or batch casting. Often equipment designed for batch casting is awkwardly re-engineered to provide sub-optimal solutions for continuous casting. Complexity generally results in operator dissatisfaction and higher risks of caster shutdowns. The SCR Ultra-D™ Degasser was developed by Southwire for their own continuous rod mill casting operations.

Results from the Ultra-D™ Degasser trial

Results in the trough upstream of the degasser box

The first trial was conducted at the Mount Holly caster 11, where a long trough before the caster allowed for testing before the degasser box. The metal flowrate was approximately 1400 kg/hr which exceeded the recommended flow per Ultra-D™ Degasser probe of up to 1134 kg/hr. The first trial consisted of a single probe inserted into a widened and deepened section of the trough. The probe was run with pure Argon (no flux gas) at 560 liters/hr. The MV 20/20 inclusion analyzer was inserted after the ultrasonic degasser. Alscan™ and PoDFA measurements were taken before and after ultrasonic degassing. Figure 1 shows the MV 20/20 inclusion measurements during this first trial and Table I the hydrogen removal results.

The PoDFA samples taken after the ultrasonic degasser probe did not flow well and only pulled about 0.2 kg. The frits separated during removal of the samples from the crucible. This turned out to be a consistent problem with all the PoDFA samples taken after the ultrasonic degasser. (The separated samples were sent for analysis in order to determine the reason for this consistent result.) PoDFA results obtained are shown in Figure 2.

The first trial using 1 probe was followed by a trial using 2 probes. The results obtained were similar to those found using a single probe. After these two trials in the trough upstream of the rotor degasser, it was decided to replace the rotor flux injector in the degasser box with two probes ultrasonic degassing, and compare the rotor results with the ultrasonic degassing results.

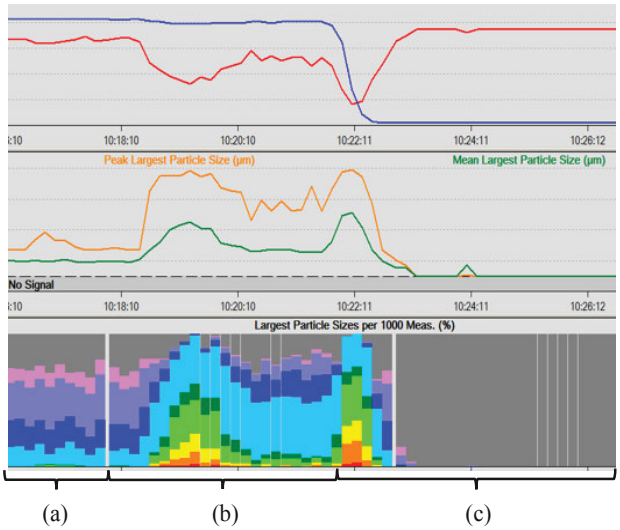

Figure 1: MV 20/20 inclusion measurements in the trough below the single ultrasonic probe. (a) is the baseline without running the probe. (b) is disturbing the trough during the insertion and adjustments and (c) is the response once the probe was activated (note the initial large particles and then loss of signal).

The MV 20/20 measurements after the degasser box and two fine pore filters are shown in Figure 3. Note the same loss of signal after the ultrasonic probes are started. After 2 hours running the probes, cast lines appeared on the cast sheet and after 2.5 hours the filters backed up and the trial had to be aborted. PoDFA results are shown in Figure 4 and the hydrogen removal drop and efficiencies are shown in Table II.

Table I: Alscan™ Hydrogen Measurements across the single probe ultrasonic degasser

Flowrate = 1400 kg./hr per probe.	Difference
Temperature drop (°C)	1.7
Alscan™ Hydrogen Reading drop (mL/100g)	0.252
% Efficiency	64.1%

Results with two ultrasonic probes in the degasser box

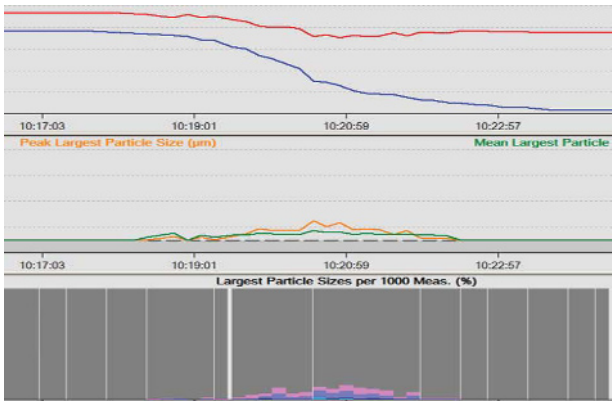

Figure 3: MV 20/20 inclusion measurements before and during the start of two ultrasonic probes in the degasser box.

Table II: Alscan™ Hydrogen Measurements across the degasser box, as measured before the box and close to the caster head-box using 2 ultrasonic probes (final hydrogen before casting).

Flowrate = 1550 lb./hr per probe.	Rotor	2 probe ultrasonic
Temperature drop (°C)	63	62
Alscan™ Hydrogen Reading drop (mL/100g)	0.34	0.33
% Efficiency	81%	82.5%

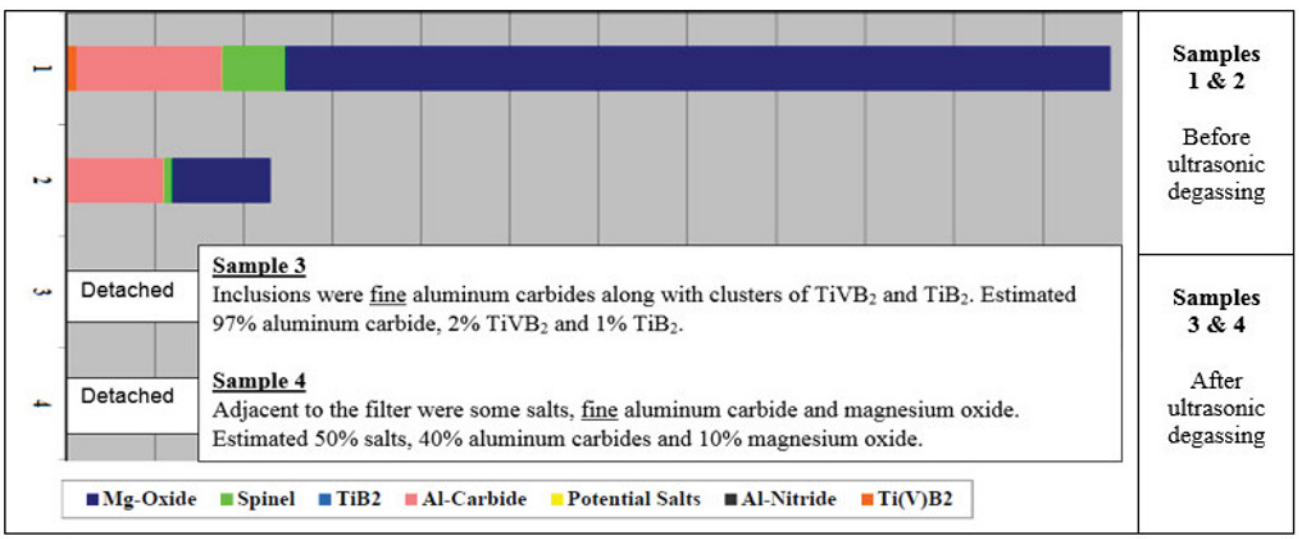

Figure 2: PoDFA results before and after the single probe ultrasonic degassing (flowrate 1400 kg./hr). Samples taken upstream in the trough close to the holder taphole and after the ultrasonic degasser.

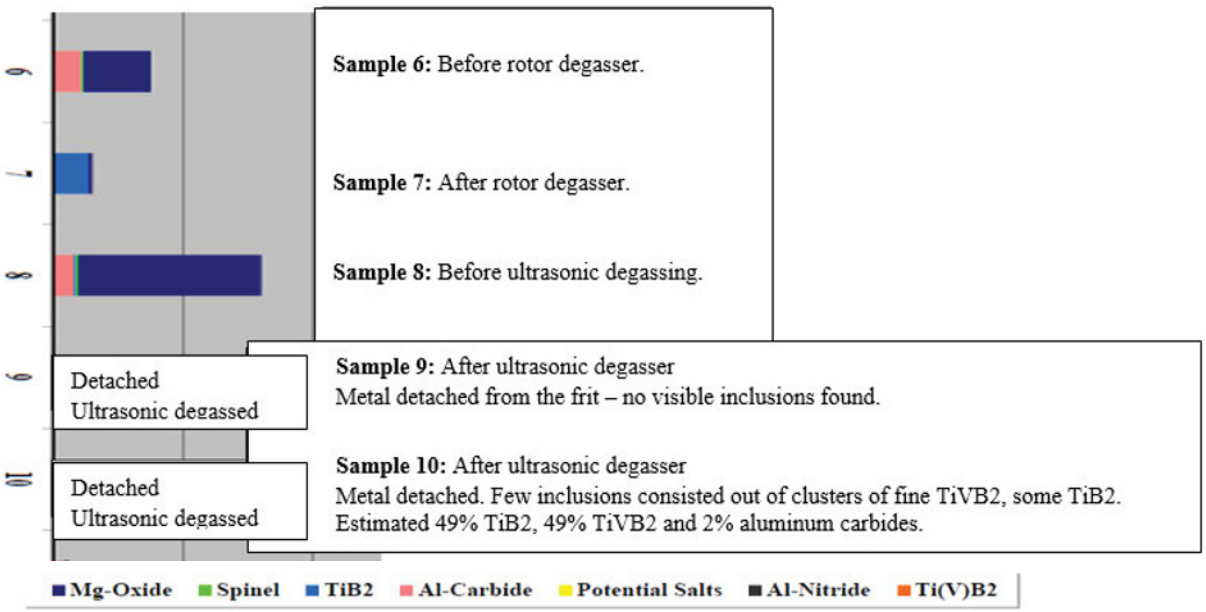

Figure 4: PoDFA results before and after the rotor and ultrasonic degasser (flowrate per probe was 700 kg/hr)

The short cast coil produced were followed through the production line and the final coil quality noted in finishing at the tension leveler. A picture of the cast lines at finishing is shown in figure 5.

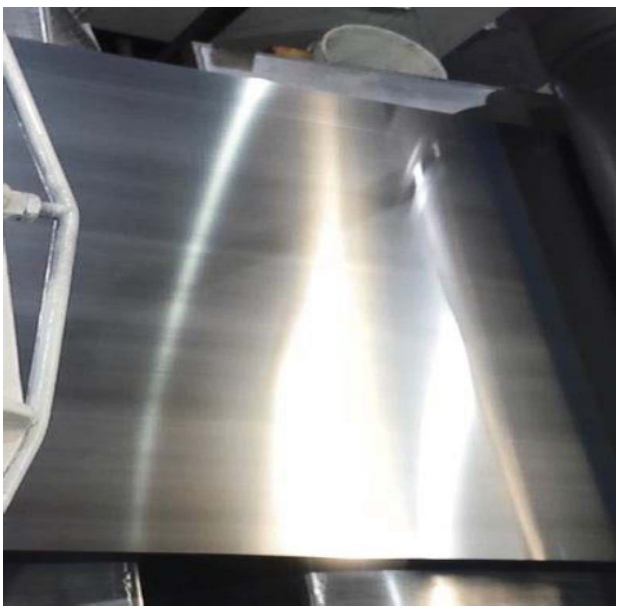

Figure 5: Cast lines as they appeared in the final product made with the ultrasonic degasser.

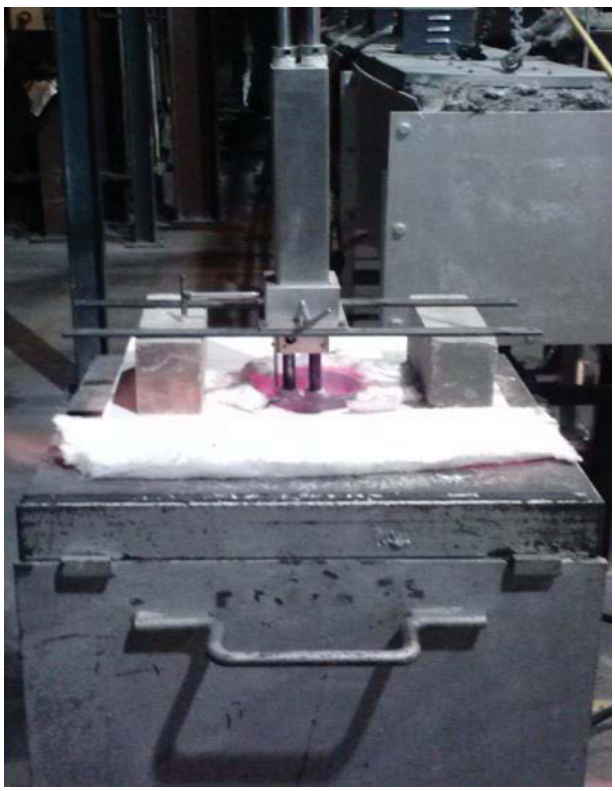

Figure 6: Crucible remelt and MetalVision inclusion testing of the ultrasonic degassed product.

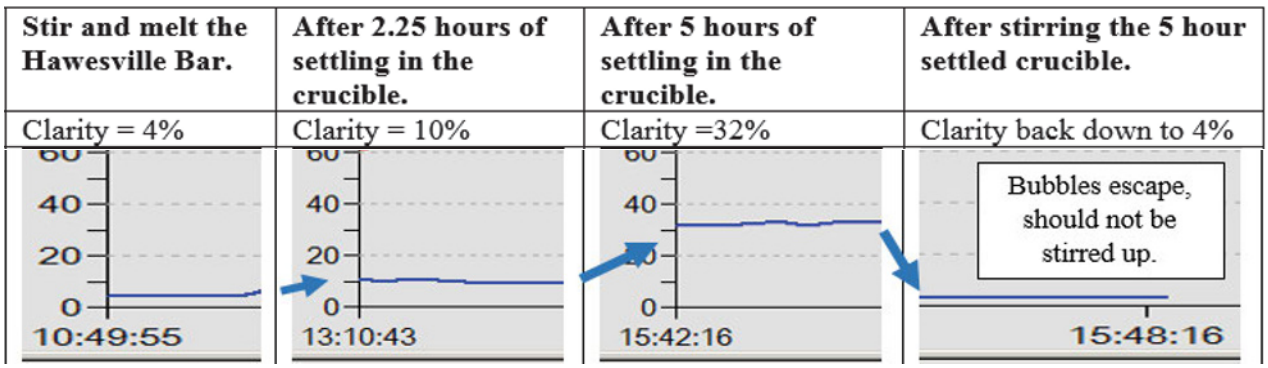

Figure 7: MetalVision MV 20/20 attenuation results of the crucible re-melting and settling of product cast using the ultrasonic degasser.

Crucible Remelt Trials

The MV 20/20 attenuation results showed a consistent drop and loss of signal once the ultrasonic degasser probes were switched on. In order to determine if this was due to fine bubbles generated by the ultrasonic degasser, products cast with the ultrasonic degasser were re-melted in a crucible and tested using the MV 20/20 (Figure 6). A typical attenuation (blue line or clarity of the metal) result obtained during settling is shown in Figure 7.

It is clear that a very slow process of settling occurs, as evidenced by the recovery of the blue line. It is also evident that the settling that occurred is reversed by stirring the sample again.

Discussion

The hydrogen measurements indicated that the ultrasonic degasser performed as good as or slightly better than the rotor degasser with regards to hydrogen removal. The behavior with regards to inclusion removal needs further explanation.

The MV 20/20 inclusion analyzer features an attenuation measurement where the loss of signal received back from the reflector is measured[1,2,4]. This measurement effectively quantifies all inclusions and gas bubbles present in the melt, even inclusions or gas bubbles smaller than 20 microns, the cut-off for inclusion identification. In this study, the MV 20/20 probe reflector distance was set at 76 mm.

The response of the MV 20/20 system, when the ultrasonic degassing is started, shows a drop in attenuation and release of greater than 20 micron particles, until the signal drops so low that all measurements fail. The loss of signal and drop in attenuation when the ultrasonic degasser probes were initiated therefore indicated an ultrasonic "cleaning" effect (large particles "shaken" loose), followed by either the presence of gas bubbles or inclusions finer than 20 microns. In order to determine if this loss was due to bubbles or inclusions, crucible remelt tests were performed on the cast product. After 5 hours there were a recovery in attenuation of approximately 32%

Five hours represents 18000 seconds and using the work done by Shu et.al[7], the size of the remaining inclusions causing the loss in attenuation has to be less than 3 microns, or bubbles has to be less than 1.5 microns to still be within the 76 mm distance between the MV 20/20 probes and the reflector after 5 hours (Figure 8). If the loss of signal was due to bubbles, stirring the melt would not have re-introduced these bubbles as they would have escaped the melt and no change in the attenuation is to be expected.

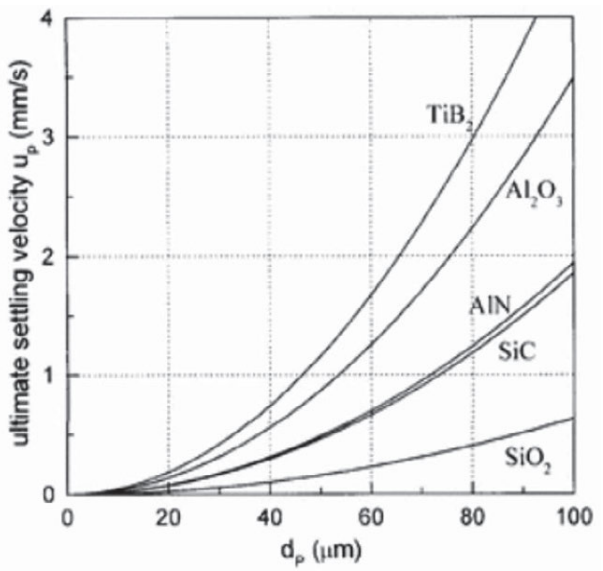

Figure 8: Ultimate settling velocity of spherical particles in molten aluminum according to Shu et. al[7].

Since the attenuation dropped back to pre-settling numbers, it is safe to assume that the loss of signal was due to inclusions in the range larger than 3 – 20 microns that have settled. The fact that the signal only partially recovered within 5 hours indicated that there were still substantial amounts of inclusions less than 3 microns present in the melt.

A survey of literature on ultrasonic interaction with particles indicates that ultrasonic milling is used for reducing particle size[8,9]. The behavior of the settled melt after stirring proves the occurrence of ultrasonic milling of inclusions introduced into the ultrasonic power field. The microscopic analysis of the PoDFA samples taken after ultrasonic degassing right at the separation of the frit and metal, also indicated that the inclusions present were very fine.

The presence of very fine particles explains the PoDFA and ceramic foam filtration behavior. Abundant very fine inclusions will blunt the PoDFA frit (filter) and stop the flow. A thin layer

of these fine inclusions could result in frit separation from the metal. Ceramic foam filtration is a surface attachment process based on non-wetting[10]. Pulverization leads to larger surface area (Figure 9 example).

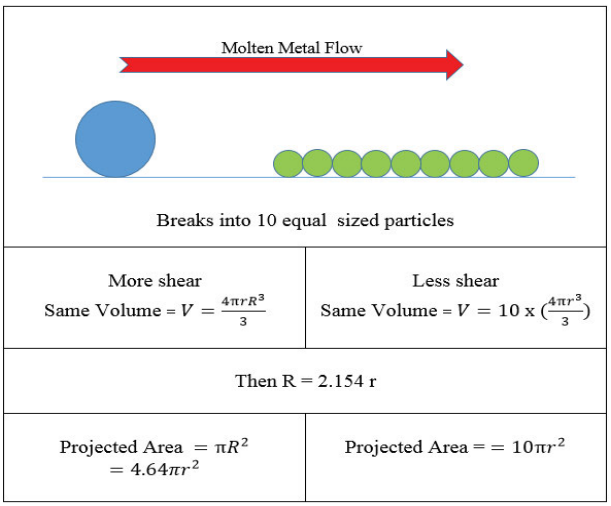

Figure 9: An example of how pulverization (ultrasonic milling) increases inclusion surface area, enhance bonding to the ceramic foam filter and reduces the tendency to release inclusions.

Large inclusions has a smaller attachment area and a larger shear due to their protrusion into the passing molten metal stream. This make them more prone to be swept out of the ceramic foam filter. Therefore large inclusions provide the filter with a "self-cleaning" effect through releases. Even though smaller inclusions are less likely to be captured within a ceramic foam filter[10] (Figure 10), once they are captured they are held tight, because of their large area to volume ratio, larger attachment area, lower protrusion into the molten metal stream and resultant stronger bonding to the filter. Over time, these fine inclusions constricts flow and therefore have an increased tendency to plug the filters compared to larger inclusions.

The initial "shaking loose" or cleaning effect of the ultrasonic probes combined with the ultrasonic milling therefore explains why the ceramic foam filters plugged prematurely. Subsequent trials indicated that filter "life' could be extended by making modifications to the equipment and procedures.

Since prior work[10] (Figure 10) clearly indicated low capture probability for very small particles, a large portion of the fine inclusions do pass through the filters. In the twin-roll caster tip, the velocity of the metal is greatly reduced as the metal spreads thinly along the wide tip. Under these conditions, fine particles have enhanced conditions to settle and/or attach to the refractory tip, and once captured, are held tight, just as is the case with the filters. It is therefore only a matter of time before enough of these fines are attached to the tip to cause flow in the tip to be changed. It is this flow alterations that leads to low flow in certain areas and high flow in other areas. Depending on the flow velocity, the kissing point therefore moves closer to the tip in low flow and further away in high flow areas, resulting in bands of different shade on the cast sheet.

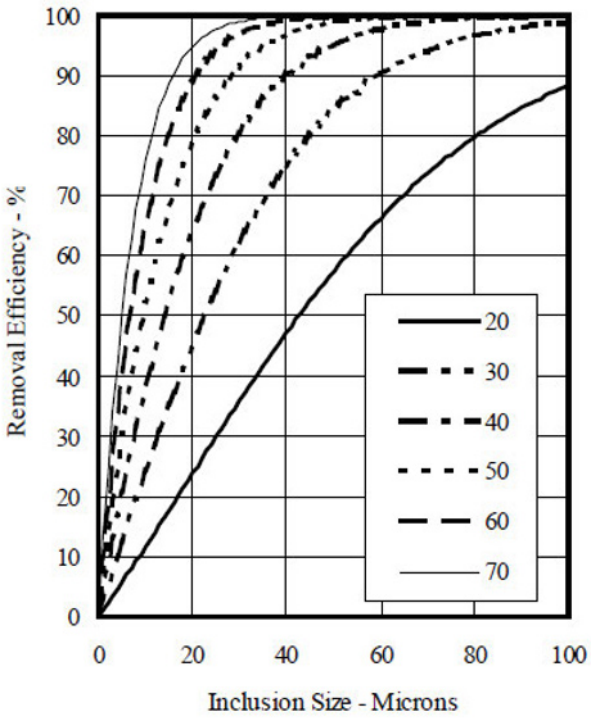

Figure 10: Filtration efficiency as reported by Barbis et.al[10]. and based on the Grandfield model.

Conclusion

Ultrasonic degassing as applied with the Ultra-D™ Degasser performs as good as or even better than rotary degassing while at the same time eliminating the need for a fluxing bi-gas.

The application of ultrasonic degassing in twin-roll casting did reveal some side effects. These side effects resulted from ultrasonic milling, which led to altered ceramic foam filter behavior and flow modification in the caster tip. The flow modification in the tip led to cast bands developing on the sheet.

The initial feasibility trials described in this paper were followed by optimization trials. During these optimization trials, the disadvantages of ultrasonic milling were significantly reduced, to the extent that cast bands were eliminated and the ceramic foam filter life significantly increased. Benefits of ultrasonic milling were also identified, such as improved grain refining, and the consequential partial offset of the Ultra-D™ Degasser probe costs with the reduction in grain refiner addition. These optimization trials will be covered in the next paper, as some of the process changes made are currently subject to a patent application process.

Acknowledgements

The authors would like to acknowledge the metal cleanliness team, Randy Proffitt, Randy Gibson and Kenny Light as well as the operators at caster 11 and the management of J. W. Aluminum for their support.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

TiB_2 PARTICLE DETECTION IN LIQUID ALUMINUM VIA LASER INDUCED BREAKDOWN SPECTROSCOPY

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Keywords: Metal Cleanliness, LIBS, Inclusions, Liquid Aluminum

Abstract

Because aluminum alloy castings are becoming commonplace for critical applications in the automotive and aerospace industries, tight control over the cleanliness of the melt (mitigation of solid particle inclusions) and microstructure must be achieved. In order to control cleanliness, it must first be well defined and measured. Very few techniques exist in industry that can quantitatively measure inclusion levels in-situ. Laser-induced breakdown spectroscopy (LIBS) is presented as a promising technique to quantify solid particles, desired or undesired, in aluminum melts. By performing LIBS with subsequent statistical analysis on liquid Al with varying amounts of TiB_2 particles, calibration curves for B and Ti were generated.

Introduction

There is a plethora of laboratory and foundry floor techniques to assess inclusion content in aluminum and its alloys. They include methods ranging from traditional optical metallography to filtration and ultrasound with each method having its own pros and cons [1-6]. Other than LiMCA (Liquid Metal Cleanliness Analyzer), a coulter counter technique, and Metalvision, an ultrasonic technique, there are no other methods of measuring inclusion concentration in–situ [7]. One of the main drawbacks to these techniques is their inability to discern particles with different chemistries.

Laser-induced breakdown spectroscopy (LIBS) has been of great interest to the metals processing field as tool for bulk chemistry measurement. Similar to conventional OES, LIBS uses a short laser pulse to form a micro-plasma from a sample of metal. The plasma light is processed with a spectrograph to determine the concentration of elements present. The volume ablated in this process is quite small (10^{-8} to 10^{-5} cm^3 depending on laser parameters and substrate material) and allows for many measurements to be taken without compromising the bulk material [8]. LIBS requires a direct line of sight to the substrate for analysis, making it attractive for use in extreme environments. LIBS systems have been adapted for high temperature applications and have been used to measure composition of molten metals [9]. The fundamentals of LIBS can found in several books and review papers [10-12].

In this work, the premise has been that LIBS can be used as a means of detecting inclusions and second phase particles in molten metals. If a particle is present in the sampling volume, the spectra will reveal its presence and chemistry. Such work has been performed on solid steel [13, 14] and aluminum [15, 16]. Studies with Al_2O_3, AlB_2, and SiC particles in liquid Al have been recently performed by the authors [16-18].

Experimental

Sample Preparation

99.99% pure aluminum (Belmont Metals) was melted in an electric resistance furnace to a temperature of 800±1°C. Al-Ti-B master alloy rods (Milward Alloys) were used to introduce boride particles into the melt. Base metal and rod additions were weighed beforehand to ensure each charge had a total weight of 2.5 kg. The melt was gently agitated immediately after master alloy addition. Composition of the base aluminum is compiled in Table I. Weights and compositions of each trial are compiled in Table II.

Elemental composition of the metal was measured by Spark OES (SpectroMaxX). The maximum level of boron detectable via OES was 0.026 wt.%. To estimate the actual amount of boron in the metal, the measured amount of Ti was multiplied by the B/Ti ratio of the master alloy.

Table I: Composition of base aluminum used in experiments

Element	Si	Fe	Ga	Others	Al
wt.%	0.00079	0.00092	0.0007	Trace	99.995

Table II: Weights, compositions, and elemental ratio of each experiment.

Base Al (kg)	**Master Alloy (kg)**	**Ti/B Ratio**	**Composition (wt.%, From OES)**	
			Ti	**B**
2.40	Al-3Ti-1B (0.10)	3	0.092	0.030
2.30	Al-3Ti-1B (0.20)	3	0.151	0.050
2.22	Al-3Ti-1B (0.30)	3	0.319	0.162
2.37	Al-2.6Ti-2.2B (0.10)	1.18	0.064	0.054
2.38	Al-5Ti-1B (0.10)	5	0.205	0.041

LIBS Measurements

LIBS trials were performed using a ceramic probe developed by Energy Research Company (Plainfield, NJ) [19]. The probe was submerged 5-7cm below the melt surface. Purified nitrogen gas from a liquid source was used for coolant and bubbling flow. The immersed probe is shown in Figure 1.

The laser apparatus consisted of a Q-switched, 20 Hz Nd:YAG laser, operated at 1064 nm with a 50 mJ pulse energy (Quantel Laser). Emitted plasma light was collected via a fiber-optic cable and fed into an Echelle-type spectrometer (LLA Instruments).

10-20 test measurements were taken to account for any transient signals. 500 successive laser shots were then fired one at a time

into the melt. Spectra data was gathered with ESAWIN software and analyzed via Microsoft Excel. Measurement frequency was approximately 1 Hz.

Figure 1: LIBS Probe Immersed in the furnace [16].

A small amount of metal was also drawn from the crucible and cast in an iron, cylindrical mold. Cast cylinders had a 7.5 cm height and 4 cm diameter. The bottom 1.5 cm of each cylinder was then sectioned off for metallographic analysis.

Microanalysis

Samples were mounted in diallyl phthalate studs, ground, and polished with diamond suspension on a semi-automatic polisher. Back-scatter SEM/EDS analysis was employed to determine particle composition. ASPEX analysis software was used to determine diameter, volume fraction, of the inclusions. Particles were identified based on contrast and EDS count. Particles with both bright contrast (high video level) and high Ti content (> 3% after subtracting background Al signal) were classified as TiB_2.

Results and Discussion

LIBS Measurements

Boron's highest intensity peaks in emission spectroscopy occur within the 200-300 nm range. Boron's emission line at 249.772 nm was analyzed for these experiments. Titanium produces a large numberof lines within the 300-400 nm range. Its peak at 399.8636 nm was used to analyze the effect of Ti concentration on the melt. To analyze individual particles, a weaker but more sensitive peak at 323.4513 nm was used. To account for the variability in signal between individual measurements, boron and titanium peak intensities were normalized by the intensity of the aluminum peak at 308.852 nm [20].

As seen in Figure 2 and 3, elemental signal noticeably increases with solute content. Data points represent the averaged signal over 500 laser pulses. There is a good linear agreement for the ranges of Ti and B desired in grain refinement. Depending on the alloy, typical target chemistries range from 0.2-0.5 wt.% Ti and 0.01-0.03 wt.% B [21]. The solubility of boron in liquid aluminum is extremely low (less than 0.05% at 750-800°C) [22]. This is lower than the typical limits of detection of LIBS for solute elements in aluminum [23]. Therefore, the assumption can be made that any time a boron line is detected, a boron containing particle was present in the sampling volume.

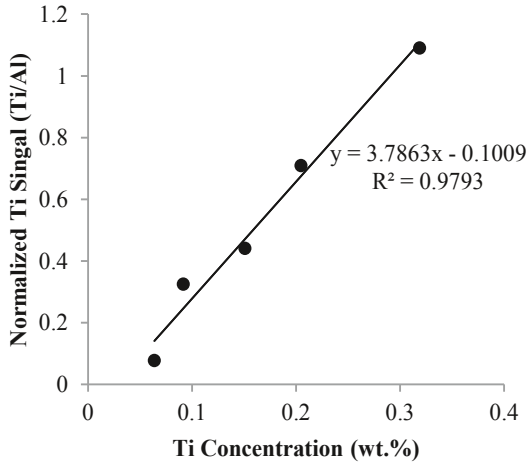

Figure 2: Calibration curve of average Ti signal vs. concentration.

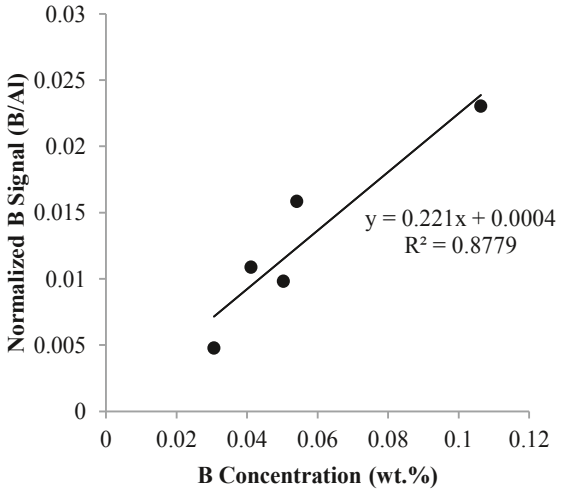

Figure 3: Calibration curve of average C signal vs. concentration.

Two-tailed analysis of variance (ANOVA) was applied to the LIBS datasets. Statistics on each element, compiled in Table III, show that with 95% confidence there is no statistically significant overlap between the data.

Table III: ANOVA on Ti and B measurements

Element	Calculated F	p-Value	Critical F
B	33.268	4.52×10^{-27}	2.375
Ti	286.417	7.40×10^{-203}	2.375

The Nalimov test was used to separate baseline signal from that of a particle hit. For boron, the solubility is extremely low in liquid aluminum and below the sensing capabilities of the spectrometer. Therefore, any substantial signal above the background can be attributed to a boride particle. For titanium, there is an appreciable amount dissolved within the matrix. In this case, the baseline is the signal coming from dissolved titanium. By using the Nalimov test on the Ti and B datasets, it was found that for approximately 10% of the time, a spike in B intensity did not coincide with a spike in Ti intensity. This indicates that the test requires some optimization, or perhaps a more conditional analysis approach is required to separate out background elemental signal from signal corresponding to a particle hit. Application of the Nalimov test to Ti and B signals yielded linear relationships between average particle hit intensities and volume fraction determined by microanalysis (Figure 4 and 5). Even though there is a linear relationship between particle hit intensity and volume fraction, using overall Ti and B signal from Figure 2 and 3 can be used to determine when grain refining elements are at their optimal levels.

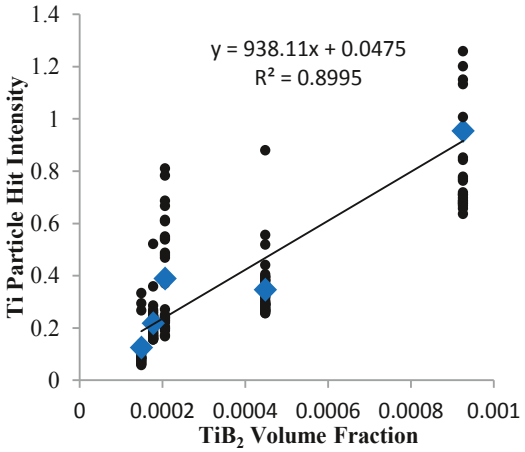

Figure 4: Ti particle hits from LIBS as a function of TiB_2 volume fraction. Diamonds correspond to average values.

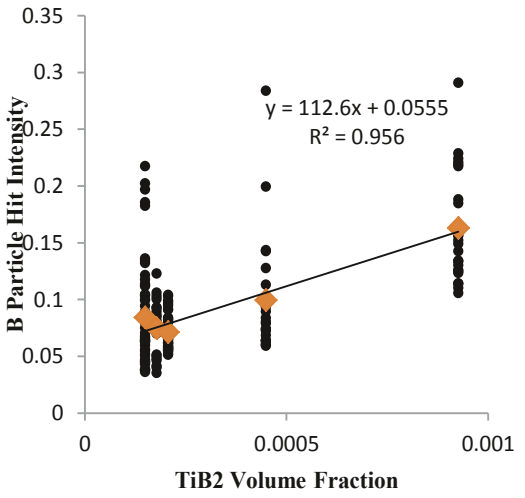

Figure 5: B particle hits from LIBS as a function of TiB_2 volume fraction. Diamonds correspond to average values.

Microanalysis

Irrespective of volume fraction, most TiB_2 particles were less than 20 µm and were narrowly distributed. Little to no AlB_2 or $TiAl_3$ particles were observed during microanalysis. TiB_2 was seen as bright particles in the pure Al matrix. Irrespective of volume fraction, over 90% of particles were less than 10 µm and were narrowly distributed (Figure 6). It should be noted that LiMCA, with a detection limit of 20 µm, would in principle be unable to detect such particles. The fact that they were able to be detected by LIBS is significant.

Both the number and volume fraction of TiB_2 particles are linearly proportional to the concentration of Ti and B. However, deviations occurred when the ratio of Ti/B was too skewed. It is known that, for most alloys, the optimal ratio is approximately 2.22 wt.% Ti to 1 wt.% B for optimal grain refinement [21]. If the ratio is too high, the resultant volume fraction of particles will decrease due to the lack of boron in the matrix. This can be seen in Figure 7, where the data point for Al-0.205Ti-0.041B (Ti/B ratio of 5) significantly deviates from the linear relationship.

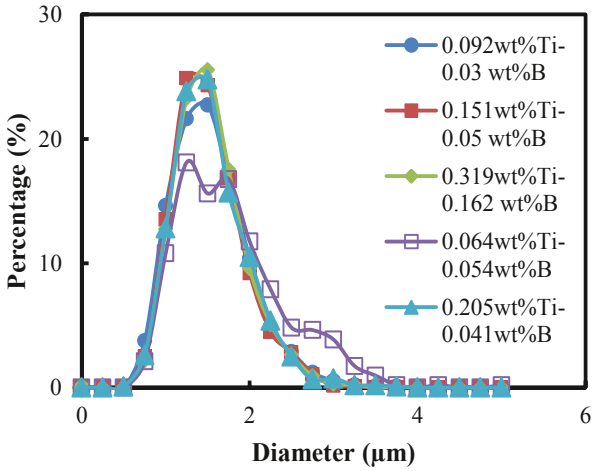

Figure 6: TiB_2 particle size distribution.

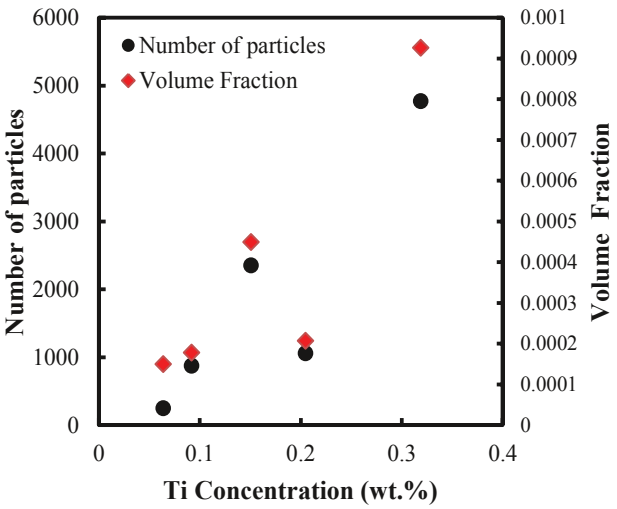

Figure 7: TiB_2 particle count and volume fraction as a function of Ti concentration.

Conclusions and Future Work

Pure aluminum with varying amounts of Al-Ti-B master alloys were analyzed using in-situ LIBS. It was found that over the course of many laser measurements, TiB_2 particles could be detected by monitoring Ti and B signal. Calibration curves were developed for Ti and B in pure Al within the compositional ranges for grain refinement, allowing LIBS to be use as a possible tool for monitoring grain refining elements. Nearly all particles that were observed in microanalysis were less than 6 μm in size, well below the 20 μm sensing limit of LiMCA. The Nalimov test was used to determine elemental signal that was not from the bulk chemistry, but from particles. In the case of separating out Ti signal from a particle hit from that in the bulk metal, the Nalimov test performs a reasonable job. However, its identification of false negative requires a more conditional analysis.

Due to the small size of the melt, there was enough turbulence to enable random sampling with the laser. However, this is not always the case in larger melts. Further tests with larger melt sizes need to be conducted.

Acknowledgements

We kindly thank Dave Smith and Wenwu Shi (Foseco) for metallography and image analysis as well as the member consortium of WPI's Advanced Casting Research Center for funding this ongoing project.

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Inline melt treatment for low to medium metal flow rates

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Keywords: Casthouse, De-gassing, Inline melt treatment

Abstract

The Aluminum casthouse industry is striving to reduce operating cost, emissions and energy consumption. During the last decades technologies have been launched to able drain-free de-gassers with no use of Chlorine or Chlorine containing salt.

Ten years ago, Hycast launched the I-60 SIR unit with a unique siphoning concept, and bottom mounted rotors, allowing for long residual time of the Argon gas. This unit works very well for all flow rates with only two rotors. However, for low flow rates, the unit over-performs and the purpose of this work was to develop a one-rotor siphon inert reactor for low and medium flow rates up to 25 tons per hour. In this work, the basic design and the metallurgical performance of the new SIR I-25 are presented. The removal efficiency of both Hydrogen and inclusions are well within the requirements in the industry.

Introduction

In the late 1980's a box-type de-gasser was developed at the R&D centre of Hydro Aluminium located at Sunndalsøra, [Myrbostad 1986]. This was the start of the HI-technology, see Figure 1. Several versions of this reactor were made, from one and up to four rotors for different flow rates of metal. To be able to commercialize this new technology the spin-off company Hycast was established in 1990.

Figure 1. The first Hycast de-gasser launched in 1990.

In September 2001 Hycast launched the second generation inline melt treatment technology the I-60 SIR, see Figure 2 and [Mæland 2002]. This was a innovative technology with two bottom mounted rotors and no liquid metal in the reactor between casts. The metal was elevated using siphoning (under-pressure) to ensure good removal efficiency and to be able to empty the reactor completely after casting. The reactor was designed for treating metal flow rates up to 60 tones per hour but is successfully used up to 75 tones per hour today. The unit is also used for flow rates down to ~5 tones per hour today but is over-performing at these conditions. To further reduce the investment cost, operation cost and footprint needed Hycast decided to develop a one-rotor version for low and medium flow rates. This technology, named I-25 SIR, is described in this paper.

Figure 2. The I-60 SIR unit launched in 2001.

Theory

Assuming 1. order kinetics in the removal efficiency the removal in each reactor chamber might be described as follows; see e.g. [Rosenqvist 1983].

$$C = C^0 \exp(-kAt) \quad (1)$$

This implies that the removal efficiency is independent of the inlet level of the impurity. If e.g. the removal efficiency of each chamber is E_1 the removal efficiency of n chamber in series will be:

$$E_n = 1-(1-E_1)^n \quad (2)$$

For example, if the removal efficiency of each chamber is 0.6 (60%) the total efficiency of two chambers in series will be $1-(1-0.6)^2 = 0.84$ (84%). In [Håkonsen 2004] and [Steen 2010] the removal efficiency of inclusions and hydrogen in the I-60 SIR unit was published. Using 2x75Nl Argon per ton Aluminium the typical inclusion removal efficiencies were 0.98 for inclusions larger than 40µm, and 0.70 for Hydrogen inlet level of 0.4ml/100gAl. Simply applying (2) to these number one should for a one-chamber reactor then expect removal efficiencies of 0.86 and 0.45 for inclusions (larger than 40µm) and Hydrogen (inlet level of 0.4ml/100gAl) respectively.

Technology

The following design criteria were set up prior to the development:

- Drain free unit
- Inert inside reactor
- Reduced footprint compared to I-60 SIR
- Metal flow rates between 5 – 25t/h
- It should be possible to by-pass the unit
- The investment cost should be significantly lower than I-60 SIR
- The unit should be able to handle 7xxx alloys
- Possible to preheat if unit is by-passed
- Should be possible to use Reactor Replacement Program (spare reactor at Hycast), as for I-60 SIR

It was decided early in the development program to go for a one-rotor and one reactor chamber concept. We assumed that this would give sufficient removal efficiency and significant lower operating cost and foot-print compared to the existing two-rotor I-60 SIR. The design of the prototype is shown in Figure 3 and 4.

The design is based on the existing I-60 SIR with the same rotor, metal elevation height, rotor bearing package, under pressure and dust filter system. The inlet and outlet are slightly modified and the launder is design for bypassing capabilities. The reaction chamber is identical to the two chambers in I-60 SIR but is rotated 90°. Since the inlet and outlet is in the same chamber a turbulent flow inside the chamber is a prerequisite to ensure good mixing and no short-cuts of metal. Water model experiments and the experience with the I-60 SIR indicates complete mixing conditions in the chambers.

Other design features is that the chassis and reactor are combined in the same unit for the I-25 SIR. The unit is design as a stand-alone unit with integrated control system.

Figure 3. The new I-25 SIR, prototype testing at the Hydro Reference Centre.

Figure 4. The new I-25 SIR, dust filter, gas and control cabinet toward camera.

Figure 5 shows the inside of the I-25 SIR unit, with one reactor chamber and one rotor.

Figure 5. The new I-25 SIR, dust filter, gas and control cabinet toward camera.

Results

A prototype was made and tested at the Hydro Reference Centre, a full scale R&D casthouse located at Sunndalsøra. Figure 3 and 4 shows the prototype during testing.

The metal flow rate varied between 7.4 and 17 tones Aluminium per hour, and the gas amount was kept at 26Nl/min or 60Nl/min.

The inclusion content were measured by using two LiMCA II simultaneously, one up-stream and one down-stream the I-25 SIR unit. Figure 6 and 7 shows the results.

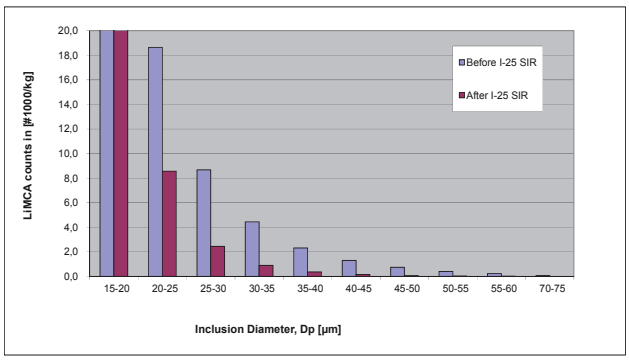

Figure 6. LiMCA measurements of inclusion size distribution up- and down-stream the I-25 SIR unit.

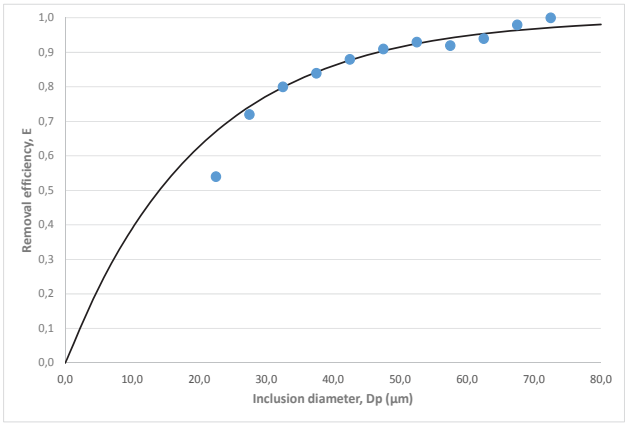

Figure 7. Measured removal efficiency of inclusions versus inclusion diameter (blue circles).

The Hydrogen content up- and down-stream the unit were measured by using an AlSCAN apparatus. In these trials, the inlet Hydrogen was relatively low and varied between 0.17 to 0.22ml/100gAl. The Hydrogen removal results are shown in Table 1, Figure 8 and 9. As can be seen from Figure 8 there is a significant effect of relative Argon flow rate on the removal efficiency. Figure 9 is a comparison between previous published results for I-60 SIR, see [Steen 2010], and the results from this work.

Table 1. Hydrogen measurements in this work

Metal flow rate (t/hr)	Ar flow (Nl/min)	Ar flow (Nl/t)	Hin (ml/100gAl)	Hout (ml/100gAl)	E
12,7	26	123	0,19	0,11	0,42
14,9	26	105	0,17	0,1	0,41
7,4	60	486	0,22	0,081	0,63
12,7	60	283	0,18	0,089	0,51
14,9	60	242	0,17	0,089	0,48
17	60	212	0,17	0,093	0,45

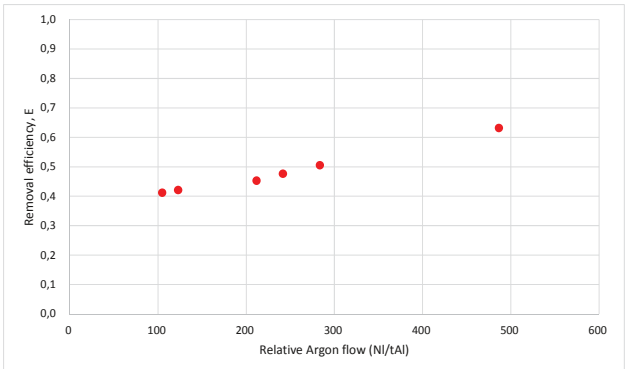

Figure 8. Measured removal efficiency of Hydrogen versus relative Argon flow.

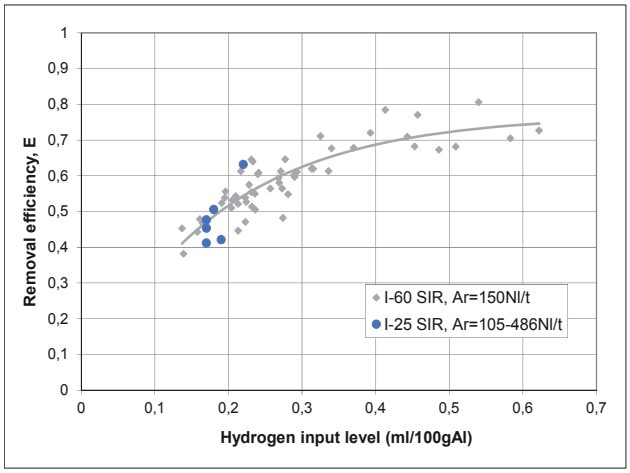

Figure 9. The measured removal efficiencies for I-25 SIR combined with previous published results for I-60 SIR.

Discussion

Two conditions indicates that the removal efficiency actually may be much higher for the one-chamber situation for Hydrogen than stated in the Theory section. We know that the removal of Hydrogen do not follow 1. order kinetics. The removal efficiency is higher for high inlet levels compared to low inlet values, see e.g. [Steen 2010]. This will give a relatively higher removal in the first chamber since the inlet level here is higher. Secondly, the removal efficiency of inclusion and Hydrogen is highly dependent on the Argon flow relative to the metal flow. There is a physical limit on how much Argon you can use in a Hydro-rotor design. However, this is independent on metal flow rates. This means that for low metal flow rates we may use relatively more Argon per rotor, giving increased removal efficiencies. This explains the relative high removal efficiencies we found in this work.

The disturbance of LiMCA measurements down-stream de-gassers by micro-bubbles is well known; see e.g. [Håkonsen 2004]. This disturbance is probably the reason for the relative low measured removal efficiency for 22.5µm inclusion size shown in Figure 7. In these trials, however the micro-bubble disturbance is lower that earlier trials with I-60 SIR. This is most probably due to the relative low metal flow rate and thus fewer micro-bubbles carried over to the launder due to the low velocity of the metal in the reactor.

The results in this work supports that the assumption of complete mixing in the reactor is correct.

Conclusions

A new inline melt-refining unit, I-25 SIR, for low to medium metal flow rates is developed. The unit has been verified at the Hydro Reference Centre and has demonstrated excellent removal of inclusions and Hydrogen. The I-25 SIR is based on the principles of the well proven I-60 SIR. Due to the simple design and the one-rotor concept, the unit will be relatively low in investment and operating cost and has a small footprint. The unit may be by-passed in the launder system and.

Acknowledgement

The authors would like to acknowledge all the skilled people that have been involved in the development and verification of the new I-25 SIR unit. This include the Hydro Reference Centre crew and our colleagues at Hycast.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF SOAKING TREATMENT ON THE MICROSTRUCTURE AND WEAR BEHAVIOR OF THE ULTRASONIC MELT-TREATED B390 HYPEREUTECTIC Al-Si ALLOY

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Keywords: Soaking, Hypereutectic, Al-Si, Wear, Microstructure, Ultrasonic

Abstract

The hypereutectic Al-Si alloys constitute an important family of alloys because of their excellent wear resistance and low thermal expansion. The current work aimed at studying the effect of soaking treatment on the microstructure and wear behavior of the ultrasonic melt-treated B-390 hypereutectic Al-Si alloy. The results showed that the ultrasonic treatment resulted in uniform distribution and refinement of the primary silicon particles. The soaking treatment resulted in coarsening of the primary silicon particles, with larger particles obtained at longer soaking times. It was also observed that particle coalescence is the primary ripening mechanism for the primary Si, especially in the ultrasonic-treated samples. The wear weight loss changed with the soaking time, with the ultrasonic-treated samples exhibiting more consistent trend with little variations of weight loss at different soaking times. The optimum particle size and distribution is obtained after 5 min. soaking of the conventionally cast samples, and 10 min. for the ultrasonic-treated samples. The worn surface of these conditions is characterized by fine and shallow wear scratches with little particle detachments. Soaking at high temperature leads to reducing the volume fraction of the primary silicon, which consequently, caused high weight losses during wear test.

Introduction

Hypereutectic Al-Si alloys containing from 14 to 25 % Si exhibit excellent wear resistance, high hardness, and low thermal expansion. Because of these properties, these alloys find applications in some critical components such as internal combustion engine parts, cylinder bodies of compressors and pumps, brake systems, etc.

The prominent properties of these alloys depend mainly on the morphology and size of the primary Si particles and the eutectic Si. In conventional casting solidified at a moderate cooling rate, the primary silicon crystallizes in the form of hexagonal plates joined together at the center into star-shaped particles. These coarse platelet particles have deleterious effect on the mechanical properties of the alloys [1]. These microstructural features can be controlled by the addition of phosphorus as a refiner for the primary Si and modifiers such as Sr or Na for the eutectic Si and/or by controlling cooling rate encountered during solidification. However, the addition of both agents, i.e., refiners and modifiers, leads to counteracting effects because of their chemical incompatibility, which leads to the formation of strontium and sodium phosphides, with evident weakening of both additions.

Therefore, the current melt treatment technology of the hypereutectic Al-Si alloys does not allow the coexistence of modified eutectic with primary Si phase having fineness and uniformity of distribution. Therefore, other techniques are used for primary Si refining in the hypereutectic Al-Si cast alloys, which has no interaction with the eutectic Si modification. The ultrasonic treatment (UST) was proven to be an efficient method for this purpose.

The ultrasonic treatment showed potential effects in refining the primary silicon and Fe-intermetallic phases in hypo- and hyper-eutectic Al-Si alloys, as well [2,3]. This effect superseded that of phosphorous addition (50 ppm) in both Si-particle refining and its effect on wear resistance [4]. It was also found that the use of ultrasonic in the production of billets for thixocasting is beneficial, since the UST produces a much slower-growing aluminum grains during soaking treatment, with a uniquely very small amount of entrapped eutectic, giving a merit of better formability in the semisolid state [5]. The ultrasonically-treated thixocast parts showed a highly uniform nondendritic microstructure with higher mechanical properties than the untreated billets [6,7]. In addition, the advantages of ultrasonic treatment in preparing ingots of hyper-eutectic Al-Si alloys with improved deformation in the semi-solid state was previously reported [8,9].

The hypereutectic Al-Si alloys are usually cast by low-pressure casting, high-pressure die casting, and permanent mold casting methods. Due to the high latent heat and the consequent long solidification time which results in die wear, segregation and excessive growth of primary silicon particles, and unfavourable shrinkage behavior, thixoforming was recently considered to be a viable alternative in the production of these alloys [10]. Thixoforming provide high-integrity die-cast type components, with reduced cycle time, reduced enthalpy, improved casting integrity and properties and reduced solidification shrinkage. In addition, the die filling process can be controlled to eliminate porosity [11].

Based on this, the objectives of the current study are to study the effect of soaking treatment, which is the first step in thixocasting, on the microstructure and wear behavior of ultrasonic-treated and untreated ingots of the B390 hypereutectic Al-Si alloy. These objectives were achieved through wear testing and metallographic studies on optical microscope.

Experimental

The hypereutectic Al-Si alloy B390 was used in the current study. This alloy contains ~17 pct Si, 4.5 pct Cu, 0.55 pct Mg,<1.5 pct Zn,<1.3 pct Fe, and lower levels of Mn, Ni, and Sn. Ingots of this alloy were melted in a resistance furnace set at 750 ^{o}C, and then allowed to cool down to pouring temperature, which is 680^{o}C. The molten alloy was poured into the crucible mold for ultrasonic treatment (UST).

The UST of the solidifying alloy was carried out using the ultrasonic system shown in Figure 1. Ultrasonic vibrations were

transferred through a horizontal guide rod and vertical guide plate to the mold, which is made of stainless steel in the form of crucible. The vibrations are, thus, transferred to the solidifying alloy through the bottom and walls of the crucible horn. The ultrasonic system provides a power of 2 KW, frequency of 19.5 kHz, and vibration amplitude of about 25 to 30 µm at the bottom of crucible. The capacity of the crucible is about 1.3 kg liquid aluminum. The mold (crucible) temperature was kept at room temperature during all experiments.

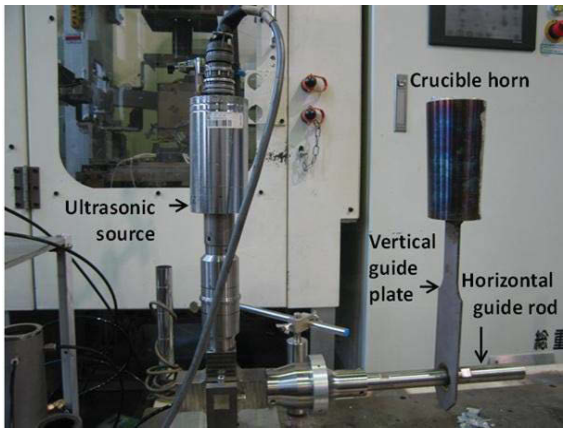


Figure 1. The ultrasonic treatment system.

The ultrasonic vibrations were turned on before pouring and continued during solidification. The effect of ultrasonic melt treatment on the size and morphology of primary Si particles in the commercial alloy B390 with and without using phosphorus refining was studied previously [4].

Treated and untreated ingots were cut, so that several pieces were taken from the bottom of the ingot. The alloy pieces were soaked in 2 Kg salt bath: consisting of a mixture of 40% barium chloride ($BaCl_2$), 30% potassium chloride (KCl), 15% sodium chloride (NaCl) and 15% calcium chloride ($CaCl_2$). The salt bath temperature was kept at 610 ± 5°C. Different soaking times were investigated, namely: 0, 1, 3, 5, and 10 minutes. Three pieces of the same alloy condition were soaked at the same time.

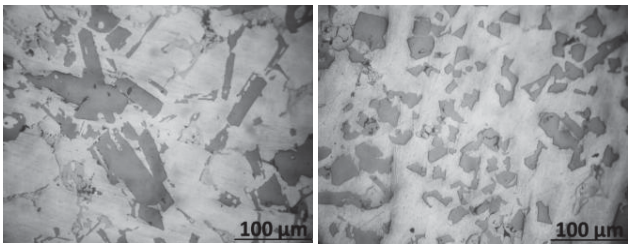
The temperature interval for the development of the primary Si crystals is very wide [12] and for this reason the soaking temperature was raised above the eutectic temperature in the current study to provide better conditions for the morphological modification and ripening of the primary Si particles.

The microstructure of the soaked pieces was studied using optical microscopy. In addition, the wear properties were determined. The wear test was carried out to measure weight loss of the samples using a pin on ring wear tester. The specimen was loaded against a steel ring by a dead weight loading system. The steel ring was of diameter 40 mm × 10 mm as a counterpart made of hardened and tempered steel (HRC 65 ± 5). The pressure used was 0.3 bar, the sliding velocity was 150 RPM and the sliding time was 30 min. The weight loss, Δ, was the difference between the sample weight before test and after test.

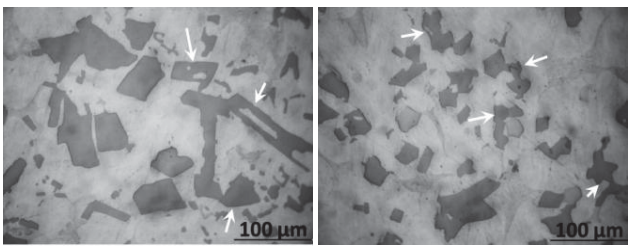
Results and Discussion

Microstructure of the soaked samples

The microstructures of the soaked samples are shown in Figures 2 to 5. The main focus of these figures is the size and distribution of the primary silicon particles, which have the dominant effect on wear and tribological properties of the hypereutectic Al-Si alloys [4]. Figure 2 shows the microstructure of the B390 alloy in the as-cast condition, without and with UST. It can be seen from these micrographs that the ultrasonic treatment resulted in uniform distribution and refinement of the primary silicon particles (see Figure 2(b)) as compared to the sample which was conventionally cast without ultrasonic treatment (Figure 2(a)).

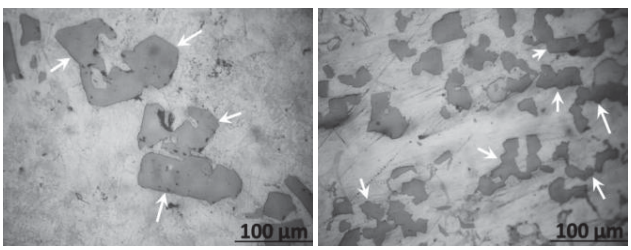


(a) (b)
Figure 2. Microstructure of the B390 alloy in the as-cast condition: (a) without UST, and (b) with UST.



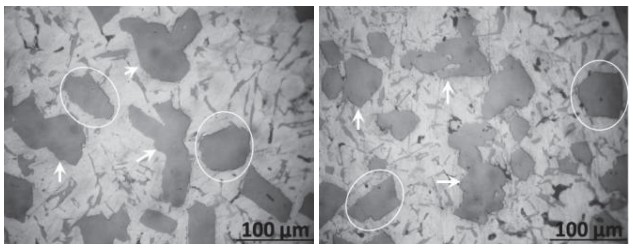
(a) (b)
Figure 3. Microstructure of the B390 alloy after soaking for 1 minute: (a) without UST, and (b) with UST.

The soaking treatments at 610 ± 5°C resulted in coarsening the primary silicon particles, with the general trend: the longer the soaking treatment the larger the Si particles. It can also be seen that soaking has a more evident effect on the samples which were cast without ultrasonic treatment. This can be seen by comparing Figures 2(a), 3(a), 4(a), and 5(a) with Figures 2(b), 3(b), 4(b), and 5(b), respectively.



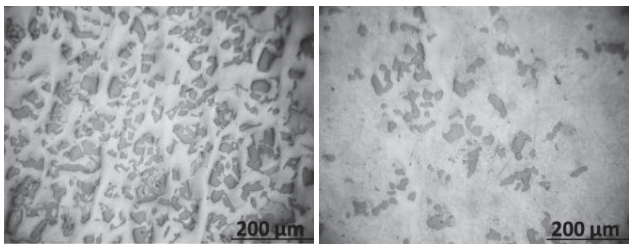
(a) (b)
Figure 4. Microstructure of the B390 alloy after soaking for 5 minutes: (a) without UST, and (b) with UST.

When the soaking temperature exceeded the 615°C *(i.e.,* overheating), the primary silicon particles melt back into the liquid phase, and when quenched in water after soaking, they exhibited much finer particles sizes, and extremely uneven distribution (congregation), with a total decrease of the primary silicon volume fraction. Figure 6 shows the microstructure of the as-cast and the overheated sample. The overheated samples showed higher weight loss during wear test, as can be seen in the flowing discussion of wear and weight loss.



(a) (b)

Figure 5. Microstructure of the B390 alloy after soaking for 10 minutes: (a) without UST, and (b) with UST.



(a) (b)

Figure 6. Wide field microstructure of the B390 alloy cast with ultrasonic treatment: (a) as-cast condition, and (b) after soaking for 3 minutes at 625 ± 5°C.

The size and morphology of the primary silicon particles after soaking depend mainly on the original as-cast structure which was controlled to different extents by the ultrasonic treatment, and the time of the soaking treatment. Although particle sizes increase continuously with soaking time, the difference between the two groups of ultrasonically-treated and that without ultrasonic treatment (No UST) continues, so that the grain sizes of the former case remain smaller than the latter at all soaking times. This observation is in agreement with a previous work on the soaking of alloy A356 [5].

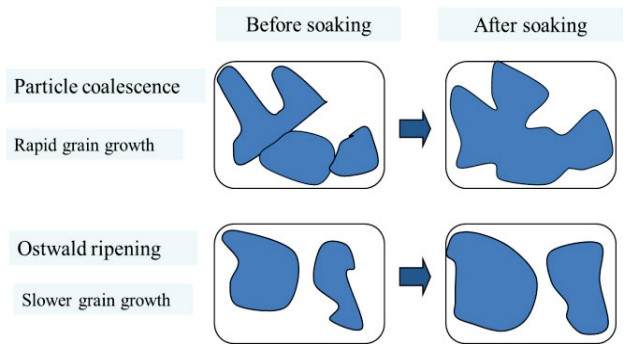


Figure 7. Schematic presentations of the mechanisms of primary silicon growth during soaking treatment.

During the soaking/ripening treatment, the primary silicon particles grow in size and acquire more curved surfaces as can be seen from Figures 2 to 6. The driving force for surface curving is the thermodynamic tendency of minimizing the surface energy of the particle. The growth mechanisms proposed for the ripening in the semi-solid state are the grain coalescence and the Ostwald ripening [13,14].Figure 7 shows schematics of these mechanisms. The Ostwald ripening is the natural growth of individual particles, and it usually takes place at slow rates. In the other hand, the particle coalescence takes place if two or more neighboring grains have low angle solid-solid grain boundaries or if the grains form coherent twins within the solid skeleton in the semi-solid state. The probability of having such neighboring grains in Al-7 wt % Si is 7%, and that tendency for the primary silicon particles in Al-Si alloys was not reported in the literature. However, the current results (see the arrowed particle agglomerates in Figures 3 to 5) gave evidence that the particle coalescence is the primary ripening mechanism for the primary Si, more specifically in the ultrasonic-treated samples. The particle coalescence mechanism is usually used to rationalize the rapid or unexpected grain growth during the soaking experiments.

The particle growth during soaking experiments is studied using the LSW (Lifshitz-Slyozov and Wanger) relationship [13, 14]:

$$d_t^3 = d_0^3 + Kt \quad (1)$$

where d_t is the particle size after soaking time t, d_0 is the starting particle size and K is the growth rate constant and is usually expressed as $\mu m^3/s$. High growth rate constants indicate dominance of particle coalescence, while low growth rates indicate dominance of Ostwald ripening.

It is also believed that, in the current study, the individual Si particles underwent Ostwald ripening, since large individual particles were seen after soaking for longer times (see circled particles in Figure 5). Therefore, it can be concluded that the operating growth mechanisms of the primary Si particles in the current experiments include simultaneous agglomeration/ coalescence and Ostwald ripening.

Wear behavior of the soaked samples

The wear behavior of the as-cast and soaked specimens was assessed using a pin on ring wear tester, which induces abrasive wear conditions during the test. The wear tests were carried out at fixed conditions, *i.e.*, 0.3 bar pressure, 150 RPM sliding velocity and 30 min sliding time. The results are presented as weight losses in milligrams and are presented in Figure 8. The 3-dimentional macrographs of the worn surfaces are shown in Figures 9 to 12.

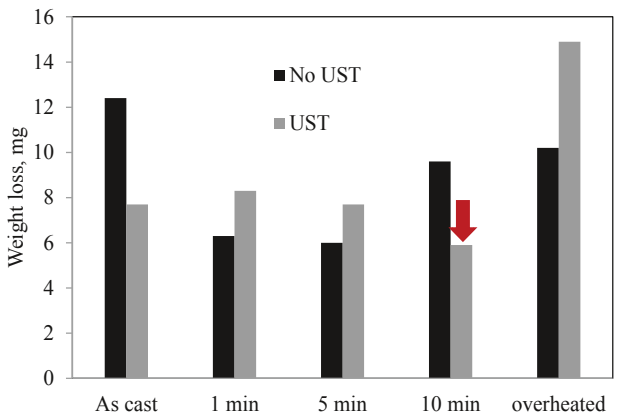
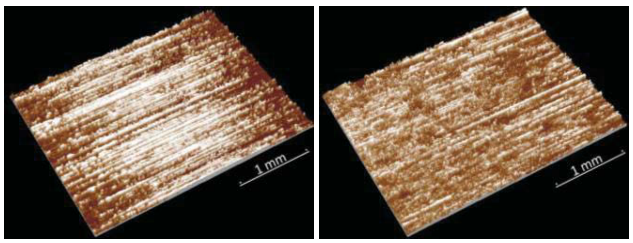


Figure 8. Weight loss of the B390 soaked samples, cast without and with UST. Soaking times are shown in minutes. Overheated samples were soaked at 625 ± 5°C for 3 minutes.

In the as-cast condition, the ultrasonic treated samples showed lower weight loss (7.7 mg) than the conventionally cast sample (12.4 mg). This is mainly because of the refining effect of ultrasonic treatment which brought about finer primary Si particles and enhanced their distribution (See Figure 2 for

comparison). The worn surfaces of Figure 9 show an evident difference. The ultrasonic-treated sample with the lower weight loss exhibited much finer and shallower scratches (Figure 9(b)) than the conventionally cast sample without UST (Figure 9(a)).

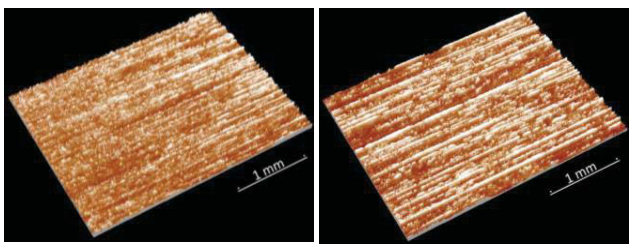


(a) (b)

Figure 9. Worn surface of the B390 alloy in the as-cast condition: (a) without UST, and (b) with UST.

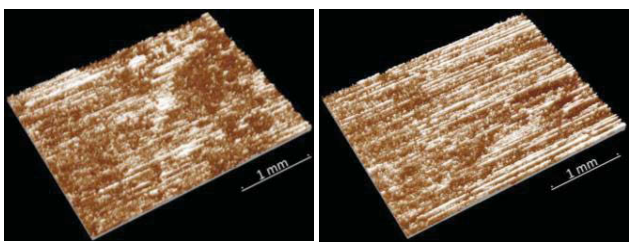
It can be seen from Figure 8 that the weight loss changes with soaking time. It can also be observed that the wear behavior of the ultrasonic-treated samples exhibited a consistent trend with little variations of weight loss with the soaking time, *i.e.*, 7.7, 8.3, 7.7 and 5.9 mg weight losses for 0, 1, 5 and 10 min. soaking times, respectively. The conventionally cast samples (*i.e.*, without UST) showed large variations of weight loss with soaking time.

It is well explained in the literature that the small and well dispersed silicon particles improve the wear resistance of the alloy by avoiding the massive detachment of these particles. Wear resistance of the alloy is improved with the finer primary silicon particles distributed in the Al matrix [15,16].



(a) (b)

Figure 10. Worn surface of the B390 alloy soaked for 5 minutes: (a) without UST, and (b) with UST.

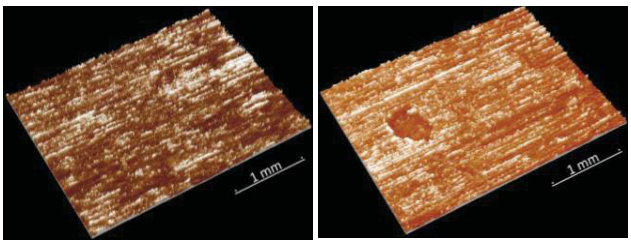


(a) (b)

Figure 11. Worn surface of the B390 alloy soaked for 10 minutes: (a) without UST, and (b) with UST.

It is believed that the wear behavior is optimum (corresponding to the minimum weight loss) at certain Si particle size range and distribution, and it deteriorates (corresponding to weight loss increase) below and above this range. Figure 8 gives an insight about this behavior, *e.g.*, the optimum particle size and distribution is obtained after 5 min. soaking of the conventionally cast samples, and after 10 min. for the ultrasonic-treated samples. Figures 10(a) and 11(b) show the worn surface macrographs of these optimum conditions, respectively. The common feature of these surfaces is the fine and shallow wear scratches with little particle detachments. The counterpart samples for these optimum conditions showed either deep wear scratches, as can be seen in Figure 10(b); or excessive detachments, as can be seen in Figure 11(a).

It worth mentioning here that the minimum weight loss of the investigated samples, which is 5.9 mg (arrowed in Figure 8), was observed in the ultrasonic-treated sample after 10 minutes soaking.



(a) (b)

Figure 12. Worn surface of the B390 alloy in the overheated soaking condition: (a) without UST, and (b) with UST.

The wear behavior of the samples soaked at overheating condition showed very high weigh losses compared with the other soaking conditions. These losses were 10.2 and 14.9 mg for the no-UST and UST conditions, respectively, as can be seen from Figure 8. The main reason for this is that the volume fraction of the primary silicon particles is lower than the other conditions since the silicon particles dissolved in the liquid phase because of overheating. The worn surfaces of Figure 12 showed massive detachments which can explain the high weight loss. It can be concluded here that soaking at high temperature leads to high weight losses and improper wear properties.

Conclusions

The effect of soaking treatment on the microstructure and wear behavior of the ultrasonic melt-treated B-390 hypereutectic Al-Si alloy was studied in this paper. The results showed that:

- The ultrasonic treatment resulted in uniform distribution and refinement of the primary silicon particles.
- The soaking treatment resulted in coarsening the primary silicon particles, with larger particles obtained at longer soaking times.
- Particle coalescence is the primary ripening mechanism for the primary Si, especially in the ultrasonic-treated samples.
- The wear weight loss changed with the soaking time, with the ultrasonic-treated samples exhibiting a more consistent trend with little variations of weight loss at different soaking times.
- The optimum particle size and distribution is obtained after 5 min. soaking of the conventionally cast samples, and after 10 min. for the ultrasonic-treated samples. The worn surface of these conditions is characterized by fine and shallow wear scratches with little particle detachments.
- Soaking at high temperature leads to reducing the volume fraction of the primary silicon, which consequently, caused high weight losses during wear test.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INFLUENCE OF OXIDATION ON CONTACT ANGLE BETWEEN LIQUID ALUMINUM AND Al_2O_3

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Keywords: Contact angle, Aluminum, Al_2O_3, Oxidation

Abstract

Aluminum can be easily oxidized at high temperature so that it is difficult to accurately measure the contact angle between the liquid aluminum and Al_2O_3 substrate. In the current study, the measurement of the contact angle between liquid aluminum and Al_2O_3 was performed under Ar gas purging and vacuum conditions. The effect of oxidation on the contact angle between liquid aluminum and Al_2O_3 was investigated. It was observed that aluminum samples in both Ar gas and vacuum condition were oxidized. With further increasing of the temperature, the oxide film gradually reacted with liquid Al and became thinner. The disruption of the thin oxide film caused the mutation of contact angle. The contact angle after mutation was approximately 130^o for both cases. The temperature of the turning point was lower under vacuum condition than under Ar gas purging. Under vacuum condition, the final stable contact angle was approximately 63^o above 1275 ^{o}C.

Introduction

The wettability between liquid and solid is usually related to the physical and chemical interaction between them. It's one of the important surface features of material. The wettability between Al and ceramic material plays an important role in diverse applications, such as the manufacture of metal-ceramic composite material, melting and casting process, corrosion, wear protection, coating and so on [1-3]. The contact angle between liquid Al and other solid material is usually used to evaluate the wettability. The wettability is one of the basic parameter of whether and how a new composite material can be prepared, and it also shows the adhesion property. Pure Al is an active metal, it can strongly combine with oxygen for its high reducibility. Under room temperature, a dense oxide film will be easily formed on the surface of Al. Under high temperature, especially when the Al becomes liquid phase, the oxidation of Al is serious [4, 5]. The oxidation of Al becomes the most important fact that affects the measurement of contact angle between liquid Al and Al_2O_3 substrate [6-9]. Shen [10] et al. measured the contact angle between liquid Al and Al_2O_3 in both a conventional and an improved sessile-drop method. The oxidation of Al occurred in both cases, and had different influence on the contact angles. Only when the temperature was over a critical value, the results would be free of the influence of the Al surface oxidation. Bao [11] et al. pointed out that even in vacuum condition above 1000 ^{o}C, the oxidation of Al could not be prevented. It still took a long time to remove the oxidation film, and the time needed at 1000 ^{o}C was amazingly longer than 12 hours.

The contact angle is formed by the interaction among three interfacial tensions, i.e., $\gamma_{gas\text{-}liquid}$, $\gamma_{gas\text{-}solid}$ and $\gamma_{solid\text{-}liquid}$. When the Al is melted, the surface of the liquid Al may be coved by a solid oxide film. The interfacial tension, $\gamma_{gas\text{-}liquid}$ is not the real value as is expected, and the morphology of the liquid Al may deviate from the spherical shape. Thus it's difficult to measure the contact angle accurately [6, 12]. Figure 1 shows the contact angle reported by several researchers [10, 11, 13-16], the oxidation of Al is one of main reasons for the large scatter of results. The lower temperature, especially in the range of 600 ^{o}C to 1000 ^{o}C, the more scattered of the results. Previous researchers made their great efforts to reduce the oxidation of Al, while the oxidation of Al still could not be prevented. How this phenomenon acting on the measurement of contact angle was not fully illustrated. In the current study, the measurement of contact angle between liquid Al and Al_2O_3 substrate was conducted. Based on the several common experiment conditions, the effect of oxidization of Al under high temperature on the contact angle measurement was studied to identify the accurate value.

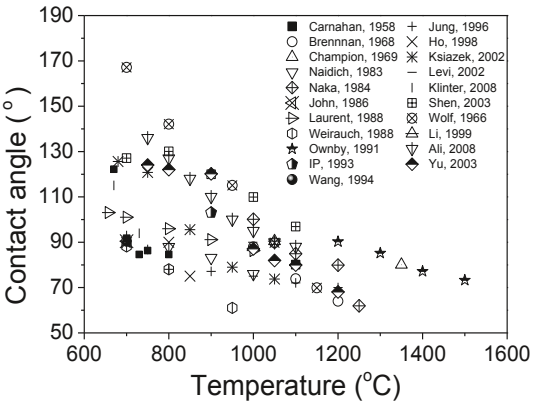
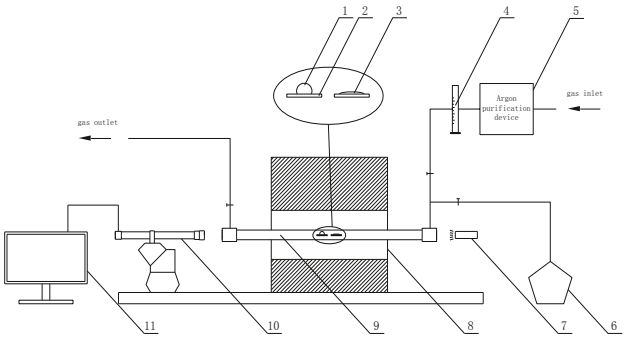


Figure 1. Contact angle reported by several researchers

Experiment Setup and Procedure

The sessile drop method was employed to measure the contact angle between liquid Al and Al_2O_3 substrate. The purity of Al was 99.999%, each sample for the experiment was about 0.056 g. The Al_2O_3 substrate was flat and smooth. The Al sample was cleaned with ethyl alcohol by ultrasonic cleaning and placed on the Al_2O_3 substrate. Both of them were placed in the constant temperature zone of the horizontal resistance furnace. Ar gas was introduced into the furnace or vacuum condition was used in this experiment. The heating rate was 5 ^{o}C/min. Both sides of the reaction tube were sealed by quartz plate. The illuminant passed through the tube from one side, and the shadow images of the samples were captured by a CCD camera on the other side. The experiment time and temperature were also recorded with each frame of the image. Figure 2 shows the connection schematic diagram.



1-sample; 2-Al_2O_3 substrate; 3-Ti powder; 4-gas flowmeter; 5-argon purification device; 6-vacuum pump; 7-halogen light; 8-resistance furnace; 9-reaction tube; 10-CCD camera; 11-computer

Figure 2. Connection schematic diagram of the experiment equipment

Contact Angle Measurement under Ar Gas Purging

Ar gas deoxidized by Ti powder

In this experiment, the Ar gas was first purified by an Ar purification device, the moisture was absorbed and the oxygen content in the gas was purified to less than 0.2 ppm, then the gas was further purified by Ti powder added in the reaction tube to remove the residual oxygen, a common method used in many experiments [11, 17]. Figure 3(a) was the initial morphology of the sample, the melting point of which ought to be 660 ^oC. In this experiment, the pure Al started to melt at 663 ^oC, with the temperature rising, the Al shrinked at the corner and kept the shape as Figure 3(b) for a long time. Since the sample was not a spherical drop, the contact angle was not measured in this period.

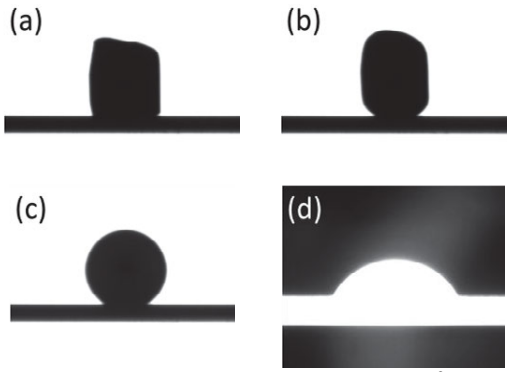


Figure 3. (a) Initial morphology of pure Al (25 ^oC) (b) Morphology of the sample at certain temperature above the melting point (692 ^oC) (c) Morphology of the sample when it just forms a spherical liquid drop (1086 ^oC) (d) Morphology of the sample at the final temperature (1512 ^oC) in the experiment added with Ti powder

When the temperature elevated to a higher value, the shape of Al sample changed again, and the initial contact angle of 148.0^o was formed at 1086 ^oC, shown in Figure 3(c). From this time on, the contact angle continuously decreased with temperature rising. However, there was a sudden drop of contact angle from 138^o to 128^o around 1150 ^oC. At the end of the experiment, the contact angle was 60.8^o. The variation of contact angle and temperature with time were shown in Figure 4.

In the cooling stage, the tube was still inputted with Ar gas until it reached the room temperature. Figure 5 showed the sample after experiment, and the gas flow direction was presented. The Al sample was dark without any metal luster and the surface of the sample was wrinkled. Besides, it was surrounded by several small particles with different color. Detected by EDS, the composition of particles were conformed to be Al_2O_3.

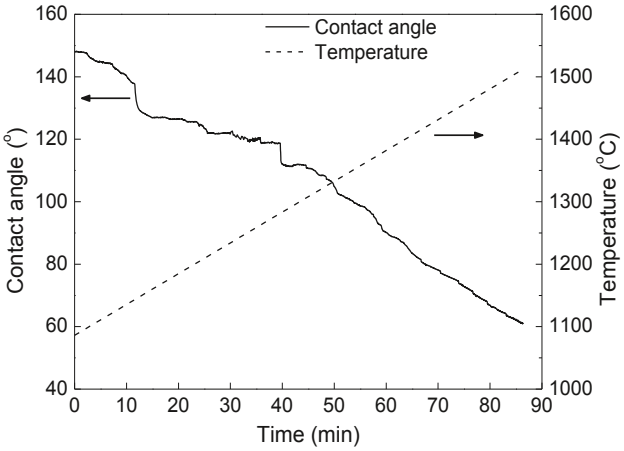


Figure 4. The variation of the contact angle with temperature rising in the experiment added with Ti powder

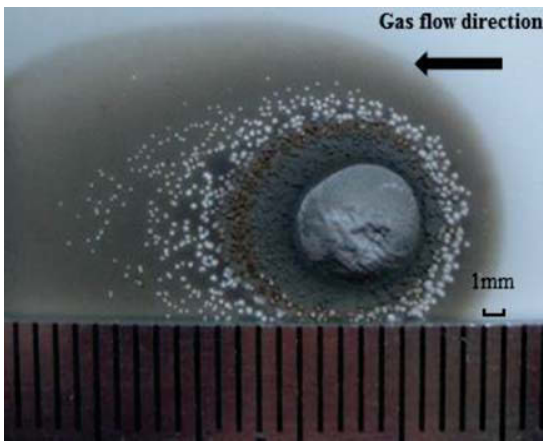


Figure 5. Morphology of the sample after experiment conducted in Ar gas deoxidized by Ti powder

Ar gas added with H_2

As a reducing gas, H_2 was widely used to prevent the oxidation of metal [18, 19]. In the Al related experiment, the H_2 was also employed [6, 13, 20]. In this experiment, the Ti powder was replaced by adding 3% H_2 into the Ar gas to compare the effect of the two methods. The morphology of the sample at different stage was shown in Figure 6. The Al sample started to form a spherical drop at a relative higher temperature of 1344 ^oC compared with part (1), and some crystal whisker were formed on the surface of the liquid drop. With temperature rising further, the amount of the crystal whisker increased first, then decreased, and finally disappeared. The contact angle seemed to decrease in the overall process. Since the surface of the drop was obviously not liquid phase in most of the time, so the exact contact angle was not measured. Figure 7 showed the sample after experiment. The morphology of the sample was similar to the previous one, and the small particles on the substrate showed the same composition.

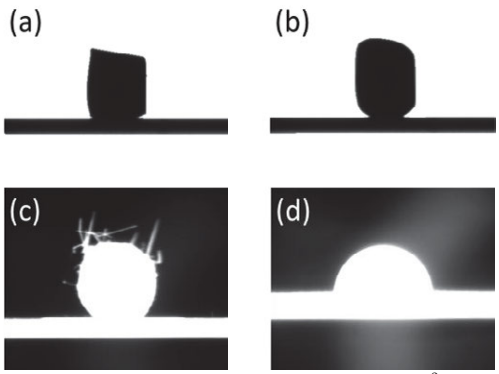


Figure 6. (a) Initial morphology of pure Al (25 ^{o}C) (b) Morphology of the sample at certain temperature above the melting point (676 ^{o}C) (c) Morphology of the sample when it forms crystal whisker (1427 ^{o}C) (d) Morphology of the sample when the crystal whisker disappears (1533 ^{o}C) in the experiment added with H_2

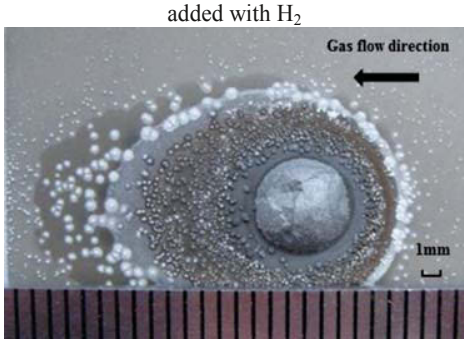


Figure 7. Morphology of the sample after the experiment conducted in Ar gas added with H_2

Contact Angle Measurement in Vacuum

In this experiment, the outlet of the reaction tube was sealed, while the inlet was connected to a vacuum pump. The initial vacuum degree was controlled to two levels, that was 1.0×10^{-4} Pa and 1.0×10^{-3} Pa before the temperature rising. The contact angle decreased with temperature rising in both experiments, the results were shown in Figure 8. The temperature and contact angle at some feature points were listed in Table I. Figure 9 showed the sample morphology at the condition of the initial vacuum degree of 1×10^{-4} Pa.

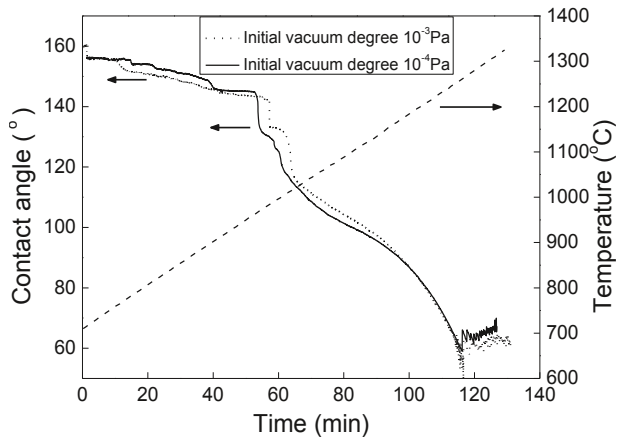


Figure 8. The variation of the contact angle under the vacuum condition with the time and temperature increasing

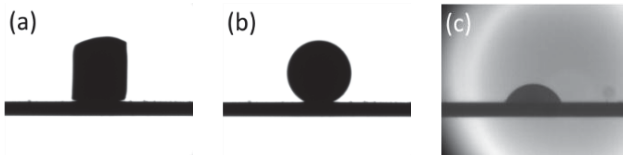


Figure 9. (a) Initial morphology of pure Al (25 ^{o}C) (b) Morphology of the sample when Al melts and forms a stable spherical liquid drop (715 ^{o}C) (c) Morphology of the sample at the final temperature (1275 ^{o}C) in the experiment started with the initial vacuum degree of 1×10^{-4} Pa

Table I. Temperature and contact angle between Al and Al_2O_3 under different feature points

Initial vacuum degree (Pa)		1.0×10^{-4}	1.0×10^{-3}
Thermal vacuum degree (Pa)		8.1×10^{-4}~ 2.0×10^{-3}	1.4×10^{-3}~ 7.6×10^{-3}
Initial contact angle	Temperature (^{o}C)	715	156.1
	Contact angle (^{o})	710	159.4
Contact angle mutating	Temperature (^{o}C)	964	144→130
	Contact angle (^{o})	979	142→132
Stable contact angle	Temperature (^{o}C)	1275	64
	Contact angle (^{o})	1278	64

After the measurement, the vacuum pump kept working until the temperature cooled down. The morphology of the two samples were nearly the same, **Figure 10** showed the sample in the former experiment. There was metal luster on the surface of the sample, and the net structure covered the whole surface. According to EDS result, there was only little oxygen on the whole surface, and no composition difference, which meant that the net structure should be the solidification structure. By the way, no particles were observed surrounding the Al sample.

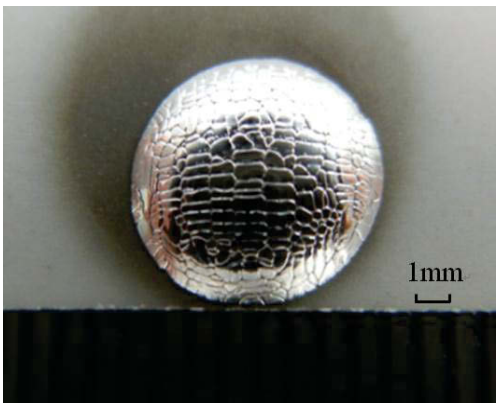


Figure 10. Morphology of the sample after the experiment with vacuum condition

The results in vacuum condition were relatively better. However, there was still a obvious contact angle turning point. In order to investigate the phenomenon, another six experiments were conducted. The temperature was elevated to the designed value, and then cooled down immediately. The morphology of the samples and contact angle of the six designed point were shown in Figure 11 and Figure 12. The contact angles before the turning point didn't fit the contact curve very well, while those after that showed the consistency. The morphology of samples after experiment were shown in Figure 13.

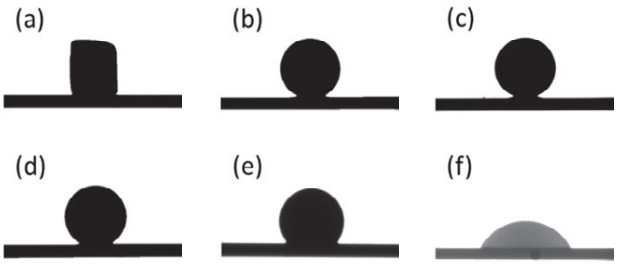


Figure 11. Morphology of the samples (a) before melting (663 ^{o}C) (b) after melting (720 ^{o}C) (c) at the middle stable stage (788 ^{o}C) (d) before turning point (896 ^{o}C) (e) after turning point (1007 ^{o}C) (f) at the final stable stage (1259 ^{o}C)

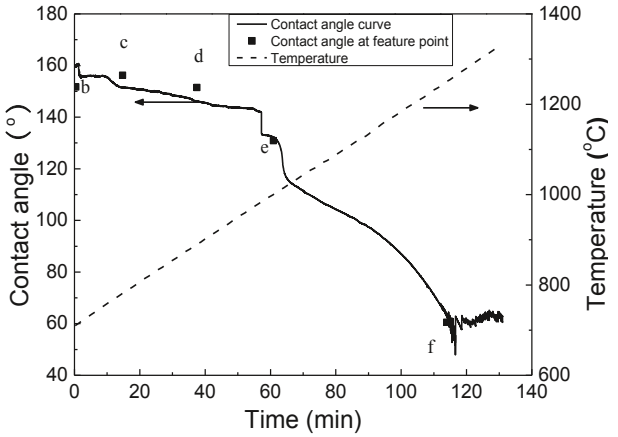


Figure 12. Contact angle of different feature points

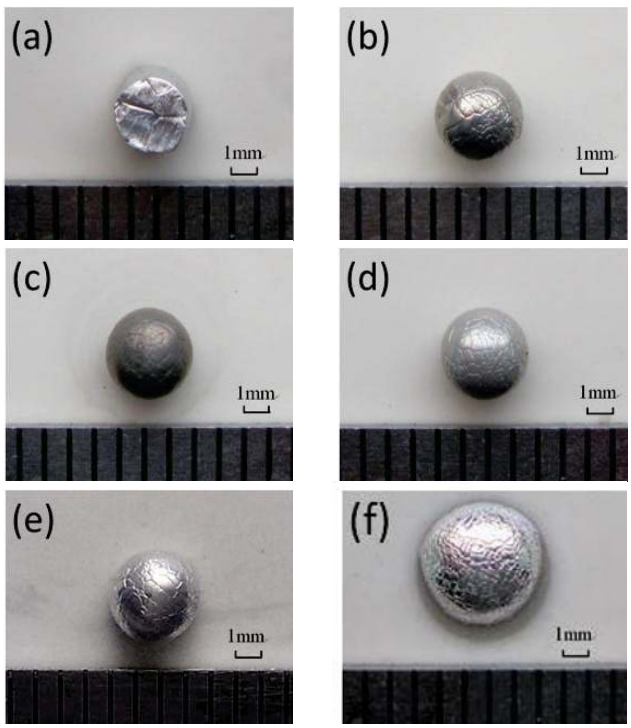


Figure 13. Morphology of the samples after cooling at the point (a) before melting (b) after melting (c) of middle stable stage (d) before turning point (e) after turning point (f) of final stable stage

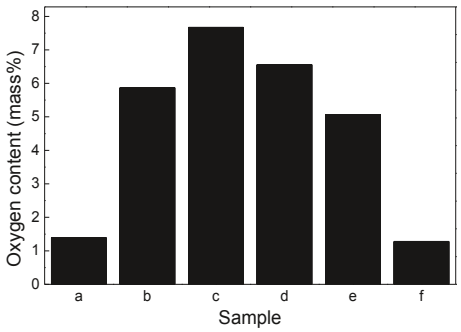


Figure 14. Oxygen content on the surface of each sample after cooling

After the experiment, the surface of the six samples were analyzed by EDS, the oxygen content was shown in Figure 14. The oxygen content increased first and then decreased later, but all the values were not so high. The EDS result was in consistence with the shininess of the samples in Figure 13.

Effect of Oxidation on Contact Angle

Ar gas deoxidized by Ti powder

Although the Ar gas was purified by Ar purification device and deoxidized by Ti powder, but the trace oxygen in the gas could still cause the oxidization of Al, and a thick oxide layer covered the Al surface. Affected by this solid Al_2O_3 layer, the sample didn't form a spherical liquid drop when reached its melting point and kept the same morphology for a long time.

$$4Al(l)+Al_2O_3(s)=3Al_2O(g),\ \Delta G^{\theta}=1180020-479.55T \qquad (1)$$

When the temperature was elevated further, the liquid Al reacted with Al_2O_3 layer according to reaction 1 [4, 13, 21]. The oxide film was consumed and the morphology of the sample changed gradually. At 1086 ^{o}C, a spherical drop was formed and the contact angle was 148.0^{o}. However, the surface of the drop was still covered with thin oxide film. Until the temperature reached 1150 ^{o}C, the film was broken. The sudden disruption of the film caused the quick spreading of liquid Al and the sudden decrease of contact angle in a short time [10]. After that, the real interface between liquid Al and Al_2O_3 substrate was achieved, the contact angle was 128^{o}. The further rising of temperature accelerated the evaporation of the liquid phase. Part of the Al vapor was oxidized by the trace oxygen in the gas and stick on the substrate. The decrease of the contact angle might be caused by the following aspects:

1) Reaction wetting between Al and Al_2O_3 [20].
2) The decrease of surface tension for the temperature rising [22].
3) The decrease of drop volume for the evaporation of liquid Al.

In the cooling process, the temperature decreased. The Al sample was oxidized again, and a new solid oxide layer was formed.

Ar gas added with H_2

Compared with the above experiment, the oxidization of the Al was much obvious, the thickness of the oxide layer was thicker. The influence of the H_2 gas on this experiment could be concluded into two aspects.

1) H_2 could not prevent the oxidation of Al

Some related reaction in the experiment could be expressed as reaction 2 to 4. If the oxidation of Al could be prevented by H_2, the Gibbs free energy of reaction 4 under the experiment temperature should be negative. However, it was not satisfied.

$$4Al(l)+3O_2(g)=2Al_2O_3(s),\ \Delta G^{\theta}=-3361440+642.66T \quad (2)$$

$$2H_2(g)+O_2(g)=2H_2O(g),\ \Delta G^{\theta}=-499150+114.22T \quad (3)$$

$$Al_2O_3(s)+3H_2(g)=2Al(l)+3H_2O(g),\Delta G^{\theta}=931995-150T \quad (4)$$

2) The oxygen and moisture in the H_2 gas could accelerate the oxidation

In this experiment, the H_2 gas was not properly purified. Not only the little O_2 but also the moisture in the gas could react with the Al, causing the serious oxidation of Al. The oxidation of Al by moisture could be express by the opposition direction of reaction 4 [4]. When the oxide film was disruption, the oxidation of Al vapor and the consumption went on simultaneously. The oxidation product prefer to grow on the residual oxide shell and form the dendrite, which had some common point with the reference [4]. When the consumption speed was much faster than the oxidation speed at higher temperature, the oxide shell and the dendrite began to disappear and finally formed a spherical drop.

As is stated above, the added H_2 gas could not provide a reducing atmosphere for liquid Al, instead it provided an oxidizing atmosphere. However, if the moisture could be effectively removed before inputting the furnace, the H_2 could work as a deoxidizing agent under a proper high temperature. This might be the reason for no obvious oxidization of Al in the reference [6]. In view of the disadvantage of H_2 gas, the solid deoxidizing agent such as Ti powder or Mg powder may perform better in this experiment.

Vacuum condition

The vacuum condition was a widely used method to prevent the oxidation of sample, the effect was related to the vacuum degree to some extent. According to the variation trend of the contact angle, especially the turning point, the Al was still oxidized in the vacuum condition. Figure 14 indicated that Al was gradually oxidized by the trace oxygen when the temperature rose, then the oxide film was gradually removed by the reaction between Al and Al_2O_3. In addition, the trace oxygen could be consumed by these reactions, and an oxygen-free gas environment was formed. Thus the Al was not oxidized during the cooling stage. Once the oxide film was disruption, the contact angle mutated and started to decrease fast. The higher vacuum degree contained less oxygen, thus thinner oxide layer and was also beneficial to reaction 1. Figure 15 shows the relationship between equilibrium partial pressure of Al_2O and temperature [23]. If the reaction tube was only filled with Al_2O gas, the thermal vacuum degree could be treated as the equilibrium partial pressure of Al_2O. The turning temperatures for reaction 1 were listed in Table II. The lower turning temperature corresponding to the higher thermal vacuum degree, which was in consistence with experiment results.

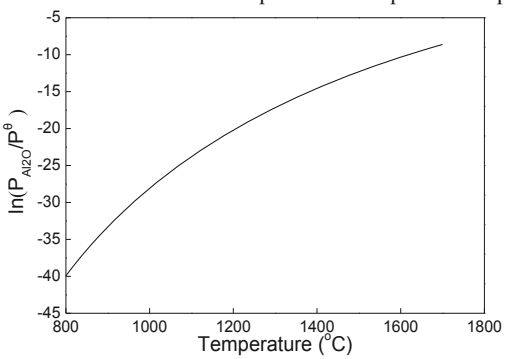


Figure 15. The relationship between Al_2O equilibrium partial pressure and temperature

Table II. Calculated temperature by different thermal vacuum according to reaction 1

Initial vacuum degree (Pa)	Thermal vacuum degree (Pa)	Calculated theoretical temperature (°C)	Measured temperature of turning point (°C)
1×10^{-4}	8.1×10^{-4}~ 2.0×10^{-3}	976~1007	964
1×10^{-3}	1.4×10^{-3}~ 7.6×10^{-3}	995~1055	979

Figure 16 showed the variation of liquid volume and contact angle with time. Before the turning point, the evaporation of Al was prevented by the oxide film. The expand of liquid volume was caused by the elevated temperature. After the turning point, the residual oxide film gradually disappeared, and the evaporation of Al became intensively. However, the former process needed enough to complete. Thus the fast decrease of liquid volume was delayed compared with the turning point of contact angle. Since the volume of liquid Al continuously decreased, the final stable contact angle ought to be the receding contact angle.

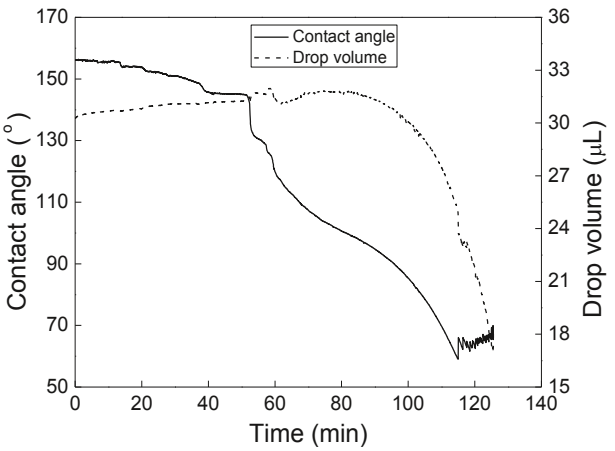


Figure 16. The variation of contact angle and droplet volume with time increasing

Conclusions

(1) In this study, the Al sample was oxidized in both Ar gas and vacuum condition and a oxide film was generated on the surface of the sample. The Al was seriously oxidized in Ar gas for the relative higher oxygen content.

(2) Under the experiment temperature, the added H_2 gas could not prevent the oxidation of Al. Instead, the Al could be oxidized by both the oxygen and H_2O existed in the H_2 gas.

(3) In both Ar gas or in vacuum condition, the contact angle decreased with temperature rising. During this process, there was a reaction between liquid Al and solid Al_2O_3, which lead to the removal of the oxide film and the mutation of contact angle.

(4) In both Ar gas and vacuum condition, there was a turning point of contact angle because of the disruption of oxide film.

The contact angle was about 130° after mutation. The temperature corresponding to the turning point was lower in vacuum than in Ar gas. Besides, the contact angle kept stable when the temperature was higher than 1275 °C in vacuum, the final stable contact angle was about 63°.

Acknowledgements

The authors are grateful for support from the National Science Foundation China (Grant No. 51274034 and No. 51404019), State Key Laboratory of Advanced Metallurgy, Beijing Key Laboratory of Green Recycling and Extraction of Metals (GREM), the Laboratory of Green Process Metallurgy and Modeling (GPM^2) and the High Quality steel Consortium (HQSC) at the School of Metallurgical and Ecological Engineering at University of Science and Technology Beijing (USTB), China. Correspondence author: Lifeng Zhang, E-mail:zhanglifeng@ustb.edu.cn.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

OPTIMIZATION OF THE ULTRASONIC PROCESSING IN A MELT FLOW

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Keywords: cavitation, light-metal alloys, ultrasonic melt processing

Abstract

Ultrasonic cavitation treatment of melt significantly improves the downstream properties and quality of conventional and advanced metallic materials. However, the transfer of this technology to treating large melt volumes has been hindered by a lack of fundamental knowledge, allowing for the ultrasonic processing in the melt flow. In this study, we present the results of experimental validation of an advanced numerical model applied to the acoustic cavitation treatment of liquid aluminum during continuous flow [1]. This was achieved by using a calibrated high-temperature cavitometer. The acoustic spectrum was analyzed at various points across the launder while acoustic pressures were calculated at the frequencies of interest.

Introduction

It is recognized that ultrasonic treatment of metal alloys is a powerful, environment friendly, and cost effective process [2-4]. Liquid melt is subjected to high-intensity acoustic waves, resulting in important non-linear effects in liquids, i.e., cavitation and acoustic streaming. The dynamic behaviour of the cavitation bubbles has been proven effective and promising in degassing and structure refining of metallic melts as shown and reviewed in [4]. Cavitation is the formation, growth, oscillation, collapse, and implosion of bubbles in liquids [5]. In the vicinity of collapsing bubbles, extreme temperatures (>10000 K) [6], pressures (>400 MPa) [6, 7], and cooling rates (>10^{11} K/s) [8] occur. To enable cavitation, a sufficient amount of acoustic energy should be introduced in the melt to set up a pressure variation that initiates bubbles formation. Typically a peak-to-peak amplitude of about 10 μm at 20 kHz, corresponding to acoustic pressures of greater than 0.5 MPa, is sufficient for cavitation inception in liquid aluminum [4, 9].

Many studies have been recently published on the effect of ultrasonic vibration on modifying and controlling the solidification structure of aluminum and magnesium alloys [9-11]. However, the fundamental mechanisms are still under scrutiny due to the experimental challenges [4, 12-13]. The prime benefit of this technology is that the ultrasonic vibrations degas the melt and refine the solidification structure of many alloys, therefore improving the quality of the end products. Cast components with refined and uniform grain structure have many advantages including significant improvement of product quality, processability, and mechanical properties [3].

A key feature of most advanced large scale casting processes is that they are performed in continuous or semi-continuous mode. Thus, to transfer and establish this promising technology into an industrial process routine, laboratory scale experiments should change emphasis from conventional static batch treatment to processing in continuous liquid flow. This brings up an important point of choosing appropriate parameters for evaluation of cavitation intensity, and thus of determining the spread of the cavitation activity in a moving volume flow. The main hypothesis is that the liquid processing occurs by passing unit volumes of the treated liquid through the active region where cavitation bubbles are formed and collapse, with acoustic and secondary flows acting as mixing and transporting means. Therefore, the dimensions and geometry of cavitation zone and the flow pattern of the melt play an important role in improving treatment efficiency. Ideally, they should be adjusted in such a way that all the metal passes through the cavitation zone. To the best of our knowledge, experimental validation and proper discussion on these issues are lacking in the scientific literature and further research is essential to reveal the conditions of a more controllable and efficient ultrasonic processing regime in different alloying systems and in large moving volumes of liquid melts, paving the way to extensive industrial use.

This study moves towards that direction as it focuses on the cavitation processing efficiency, obtained by measuring acoustic pressures in a continuous flow. A specially modified launder with baffles at specific points is used. Experiments in this part of the study are performed only in water, as water and aluminum share similar fluid properties and results obtained are easily scalable [14,15]. The effect of baffles separation on the pressure field is also determined. The intensity of the cavitation zone and the extent of the cavitation region in the upstream and downstream regions of the moving flow are directly measured using a novel high-temperature calibrated cavitometer [16].

Methodology

In the current study, characterization of the cavitation intensity and the corresponding acoustic pressure fields in water was conducted using an experimental set-up schematically shown in Figure 1. Ultrasonic excitation was achieved by a 500 W piezoelectric generator (Sonics/UK) which oscillates in a nominal fundamental frequency at 20 kHz. Vibrations were introduced into the liquid media by vertically immersing a stepped titanium sonotrode with 20-mm tip diameter to a depth of approximately 20 mm into the liquid volume. Peak-to-peak vibration amplitudes at the tip of the sonotrode were varied from 0.25 μm to 30 μm. The sonotrode was placed in the middle of the 1000 mm in length launder and equidistant from both baffles. Some of the key material properties for water are listed in Table 1.

The launder is 1000 mm along y and 90 mm along x. The bottom section is a half cylinder of radius 45 mm. The liquid is filled up to a height of 50 mm from the bottom of the launder. The baffles have a thickness 8 mm. The lower opening of the upstream baffle (baffle 1) is 20 mm, and the height of the downstream baffle

(baffle 2) is 40 mm. Two cases were considered in this study with: i) moving flow at 5.5 liters per minute (cm^3/min) and ii) stationary liquid (0 cm^3/min).

The distance between the baffles is fixed at 148 mm tailored to the sound wavelength of 74 mm at 20 kHz for water. To investigate the effect of the baffles on cavitation intensity, the measurements of acoustic emissions were taken at several points as shown in Fig. 1 (indicated by the capital letters A, B, C). Acoustic pressure is also measured at 3 points in the domain A (below the sonotrode and inside the cavitation zone), B (in the upstream of the flow) and C (in the downstream of the flow), with and without moving flow across the launder. To achieve reproducible and comparable data, each test was carried out at least 3 times.

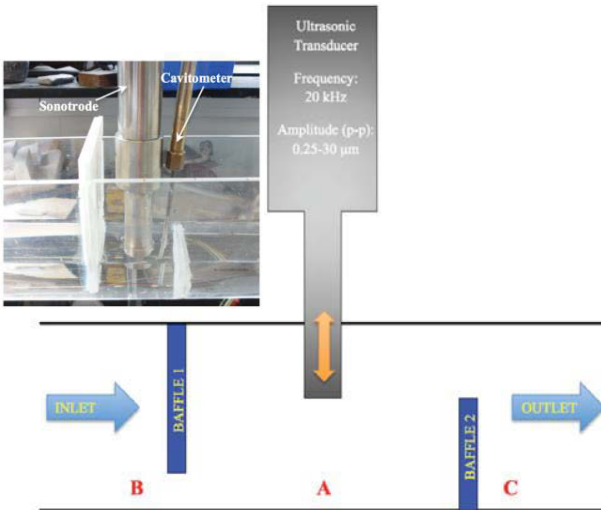

Figure 1: Schematic overview of the test arrangement with the sonotrode placed in the middle of the launder and the baffles at equal distance (7.5 cm) from the sonotrode in the upstream (Baffle 1) and downstream (Baffle 2) of the moving flow. Measurements performed with a calibrated cavitometer (see inset in Fig. 1) placed in the points indicated by the letters A (inside the cavitation zone), B (upstream), C (downstream).

Features of the acoustic emissions were captured and converted into meaningful acoustic pressures using an advanced calibrated cavitometer ICA-3HT (BSUIR/Belorussia) equipped with a 4-mm diameter tungsten probe, with a spatial resolution of 50 ± 10 mm and a bandwidth of up to 10 MHz. The cavitometer was specifically designed to measure cavitation activity in high temperature melts and in high power ultrasonic fields, i.e. in molten metals, although it can measure equally well cavitation activity in low temperature liquids. A full account of the cavitometer design and performance can be found in [16].

Table 1: Material properties of water.

Material Property	Water (20 °C)
Sound speed (m/s)	1482
Density (kg/m^3)	1000
Dynamic viscosity (mPa s)	1.004

The frequency spectrum was acquired by an external digital oscilloscope device, Picoscope, attached to the cavitometer. The Picoscope allowed real-time signal monitoring of the cavitometer sensor's data and ultrasonic parameters. The raw voltage signal is transformed to the frequency values via a Fast Fourier Transform. A number of 30 signal averages of the acquired signal were typically taken using a resolution bandwidth of 500 Hz. The time for this signal acquisition was approximately 30 x 2 ms (time gate) = 60 ms. A total of 1000 wave patterns were analyzed in each of the measurement points from Fig. 1.

Results

The local cavitation phenomena in the launder for both cases can be explained based on the spectral characteristics of acoustic emissions. RMS (root mean square) acoustic pressures are shown in Figures 2 and 3 for the driving frequency 20 kHz and at an acoustic frequency of 1 MHz (associated with acoustic pressures exerted from the stable cavitation behavior of tiny in size bubbles, i.e. in the range of 3 μm, or from the transient activity of cavitation bubbles of any size) at the points of interest (A, B, C) for stationary liquid as well as for the flow.

In Figure 2, acoustic pressures in the range of the driving frequency for both of the studied cases are shown for the three different positions along the launder. In the case where the liquid volume is stationary and at low, pre-cavitation amplitudes of vibration, there is a significantly higher range of pressures in the upstream region (point B) compared with the downstream (point C). After a period of rapid increase up to the cavitation threshold a significant drop in measured acoustic pressures is observed due to the cavitation cloud shielding effect, which limits the penetration of the driving acoustic field into the liquid. Shielding is the formation of cluster of bubbles just in front of the tip of the sonotrode distracting the propagation of acoustic emissions from the incident source and the cavitation bubbles [17]. In the downstream region, the values increase and remain rather steady at higher amplitudes reaching the pressure about 50% lower than the maximum value. The reason that acoustic pressure is significantly lower and remains fairly steady in the downstream region is possibly due to the blockage of the liquid pathway from baffle 2 where only a small portion of incident or secondary acoustic emissions escape from the upper part of the baffle reaching the cavitometer probe, as the opening is about 1 cm above the surface of the sonotrode tip. Finally in the area below the sonotrode, and inside the cavitation zone, acoustic pressures are one order of magnitude higher than the corresponding ones in the upstream and downstream, demonstrating the importance of the liquid volume to passing through the cavitation zone for an efficient processing.

When the moving volume is considered results are slightly different especially for the two regions near the baffles as shown in Fig. 2b. In the upstream region (point B), acoustic pressures are gradually increasing with amplitude although this time the increment lasts until 15 μm displacements before a significant drop occurs. In the downstream region (point C), a significant increment of the acoustic pressures is observed until a prominent pressure peak at 7.5 μm displacement is reached. Cavitation intensity is then decreased with pressure values being similar or higher than that in the upstream region, depending on the amplitude of oscillations. Inside the cavitation zone (point A) pressures are similar to the stationary case, staying mainly unaffected from the flow regime.

When the flow is introduced, the initially measured downstream acoustic pressure gets significantly higher, about 40%, reaching

similar pressure values to that of the upstream region. The main hypothesis is that since the flow is much slower than the speed of sound it should not influence the acoustic propagation. Therefore, the only reasonable explanation is that the transport of cavitating bubbles towards the downstream region causes the change in acoustic pressure pattern. Bubbles migrate with flow towards the downstream end and due to their non-linear behavior or subsequent collapse, contribute further to the acoustic pressure at 20 kHz (superimposed to the main/incident signal from the source). In addition, the shielding effect is less pronounced at the moderate amplitudes as the bubbly cloud is dispersed by the flow.

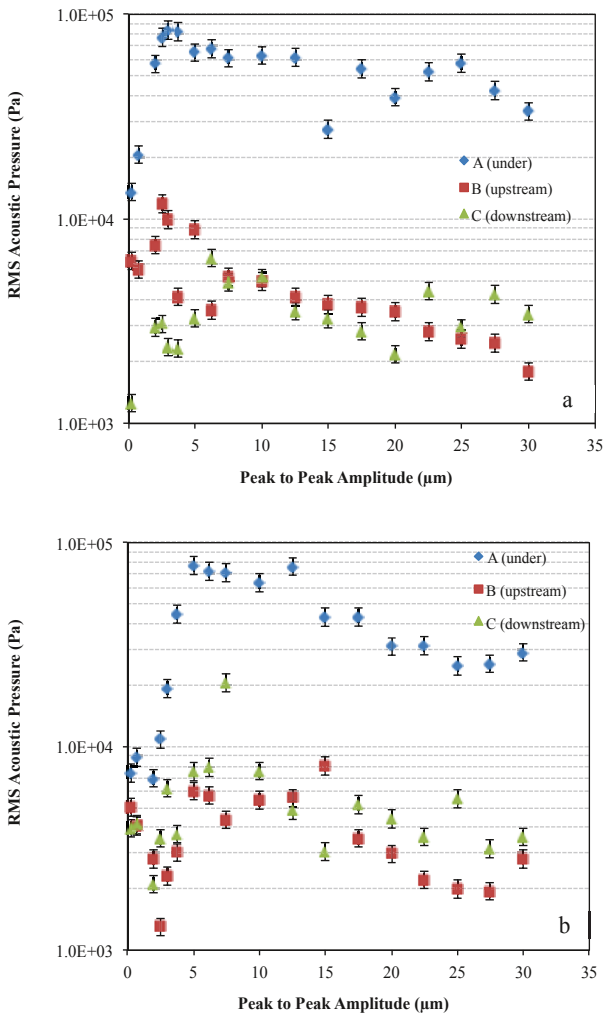

Figure 2: Variation in RMS acoustic pressures of the driving frequency (20 kHz) as a function of the tip amplitude for three different regions across the experimental launder. Measurements were taken for a) stationary flow and b) moving flow.

Results in Figure 3 indicate that acoustic emissions from cavitation bubbles at 1 MHz and thus the cavitation regime in general are not affected by the fluid flow. It is clear, for all the studied regions, that cavitation intensity from bubbles behavior follows the same pressure pattern regardless the flow regime. It is interesting that the upstream pressure is actually higher that the downstream, which makes the processing time of the unit volume longer and effective even before the liquid reaches the cavitation zone. This also implies that fluid flow is not necessary increases the efficiency of bubbles' transport downstream as bubbles can be also migrated with the acoustic streamers induced by the sonication process. For the migration of larger in size cavitation bubbles (larger than 150 µm as the resonance size of bubbles at 20 kHz is 160 µm), fluid flow maybe necessary in order for these bubbles to reach areas behind baffle 2. However, for tiny cavitation bubbles in the size of a few microns, which are registered with the acoustic emissions at 1 MHz, fluid flow does not play any significant role.

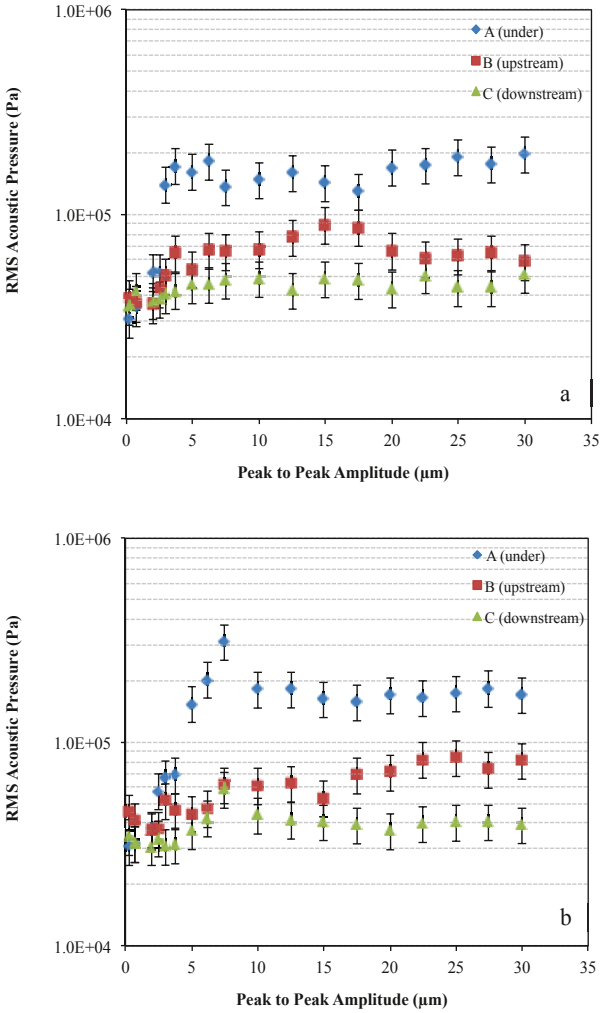

Figure 3: Variation in RMS acoustic pressures at 1 MHz frequency (associated with transient cavitation bubble behaviour and emissions from stable oscillating cavitation bubbles in the range of 3 µm) as a function of the tip amplitude for three different regions across the experimental launder. Measurements were taken for a) stationary flow and b) moving flow.

Conclusions

The interaction between the cavitation zone and the moving flow is an important component in ultrasonic melt processing. In this study, cavitation acoustic pressures generated along a specially modified with baffles launder were measured with the use of a calibrated cavitometer. Baffles were used to slow down the flow

passing through the cavitation zone and thus to increase the residence time of the moving flow into the cavitation zone for better treatment. Results showed that flow can enhance the pressure regime at low frequencies associated with larger in size cavitation bubbles along the studied regions and especially in the downstream region where without the flow the absence of high acoustic pressures was noticeable. When acoustic pressures at 1 MHz, mainly associated with the stable cavitation emissions from smaller in size bubbles, are considered, the acoustic pressure pattern stayed mainly unaffected implying that the transportation of such a small in size bubbles can be easily achieved with the acoustic streamers generated from the acoustic source. The higher emissions from cavitation bubbles observed upstream suggest that the zone of active cavitation processing extends beyond the cavitation zone itself and can be managed by the flow management. The findings of this study along with the used technique are needed for eventual industrial implementation and scale up of ultrasonic processing technologies. A better control of the acoustic pressure fields and cavitation development holds the key for the optimization of casting processes.

Acknowledgments

This work is performed within the Ultramelt Project supported by the EPSRC Grants EP/K005804/1 and EP/K00588X/1.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

A COMPARISON OF COLD AND HOT PoDFA PROCEDURE FOR PARTICLE MONITORING IN LIQUID ALUMINIUM

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Keywords: Inclusions, PoDFA, Melt Quality

Abstract

PoDFA, widely used in Aluminium cast houses to monitor the concentration of non-metallic inclusions, takes a two-step approach. An initial filtration step increases the particle concentration to a level suitable for the subsequent metallographic investigation of the solidified filter cake, giving information on the amount and nature of the inclusions present. The filtration step is generally done on-site using the PoDFA equipment, but equipment availability or operational difficulties may make it necessary to apply the "Cold PoDFA" procedure. Here the liquid metal is solidified on-site and remelted offline before filtration.

Trials with simultaneous Hot PoDFA and Cold PoDFA sampling methods were conducted in different alloy groups to evaluate a) the effect of the Cold PoDFA process on the particle concentration, b) possible changes in the particle nature due to reactions occuring during remelting and c) the influence of the sample cooling rate and the remelting procedure on the PoDFA results.

Introduction

The product quality of flat Aluminium products is largely influenced by the concentration of non-metallic inclusions transferred with the liquid Aluminium into the product. In particular, products with thin final gauges like 6 µm foil for aseptic applications or products with high surface quality requirements like lithographic sheet require extremely low concentration and a complete absence of large non-metallic inclusions.

Cast houses for high quality rolling ingots therefore employ a sophisticated melt treatment process chain with several steps like salt- or gas fluxing, degassing and/or mechanical filtration shown schematically in Figure 1 [1]. Continuous monitoring of the efficiency of the applied processing steps is applied to ensure a consistent product quality. Different technologies are available on the market to detect particles in liquid Aluminium [2], [3]. The more sophisticated are capable of monitoring the melt quality online such as the technologies based on ultrasound or the LiMCA (Liquid Metal Cleanliness Analyser) based on the coulter-counter principle. Nevertheless, the melt quality monitoring principle that is applied most in the Aluminium industry is the PoDFA (Porous Disc Filtration Apparatus) process due to its simplicity and robustness.

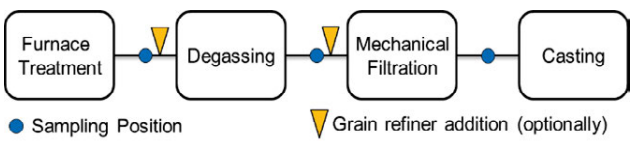

Figure 1. Melt treatment processing route and sampling position

The PoDFA principle applies a two-step approach. Pre-concentration of the non-metallic inclusions by mechanical filtration and a metallographic investigation of the solidified filter cake [3]. The pre-concentration step is generally performed on-site by sampling melt from the process step of interest as shown in Figure 2. This metal is then poured into a crucible and filtered through an Aluminiumoxide filter by applying negative pressure in case of PoDFA-f or overpressure in case of Prefil [4]. Prefil additionally utilizes the filtered metal weight vs. the filtration time to judge the melt quality. In the second step, the solidified filter cake is cut, polished and qualitatively and quantitatively analysed using an optical microscope. This procedure will be called within this study "Hot PoDFA".

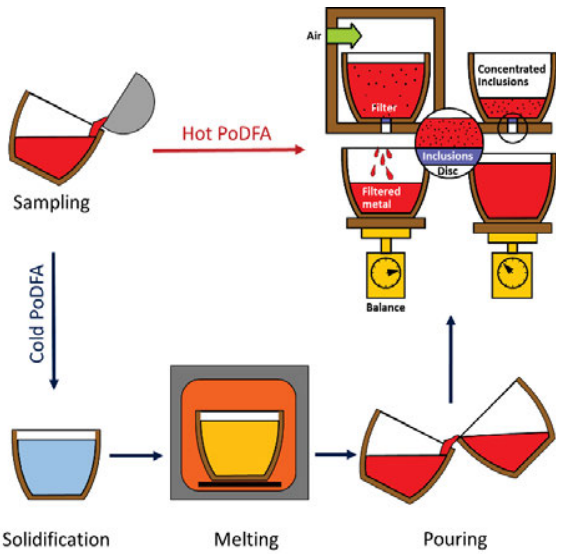

Figure 2. Pre-concentration step of PoDFA: The upper process chain describes the standard PoDFA process ("Hot PoDFA"), while the lower process chain describes the "Cold PoDFA" procedure with additional solidification, melting and pouring steps

To perform Hot PoDFA it is necessary to have a PoDFA unit ready and the crucibles pre-heated at the point and at the time of interest. In situations these requirements are not fulfilled the Cold PoDFA procedure is regularly applied. Here, the liquid metal sample is solidified in a suitable container at the casting line and later on remelted and poured for the PoDFA filtration step, which might happen offside the casting line.

The Cold PoDFA process applies three additional process steps compared to the Hot PoDFA procedure: solidification of the melt, remelting of the solid sample and pouring from the remelting

container into the PoDFA crucible. Figure 2 illustrates the difference in the processing steps between Hot PoDFA and Cold PoDFA. The additional process steps of the Cold PoDFA procedure implies the risk to change the particle concentration in the metallographically analysed sample and therefore the reported PoDFA particle concentration. Some examples of processes that might occur during the additional process steps of Cold PoDFA are discussed in the following points:

- Particle generation
  Liquid Aluminium reacts due to its ignoble character with the ambient atmosphere and generates Al_2O_3 films. Reaction promoted by pouring of the liquid aluminium, when a fresh surface is continuously provided [5]. In the case of a relevant concentration of Magnesium the oxidation is further enhanced and other reaction products like $MgAl_2O_3$ or MgO might be generated [6].

- Particle removal
  Particles might be moved to the dross layer by the progressing solidification front during the solidification step or even evaporate in case of salt residues [7]. Denser particles might settle and attach to the crucible wall [8].

- Change of the particle nature
  Independent of the total particle level it is feasible that the type, shape and structure of the particles is changed during the Cold PoDFA processing. Al_4C_3 particles may corrode in contact with the ambient atmosphere to Al_2O_3 particles [9], [10], [11]. MgO particles may further react in liquid Aluminium to form $MgAl_2O_3$ at elevated temperatures [12].

The reported study was performed to investigate the effect of the Cold PoDFA procedure on the reported PoDFA value regarding:

a) Total particle concentration

b) Influence of the solidification rate

c) Influence of type of furnace and cover gas used in the cold PoDFA remelting step.

Procedure

The evaluation of the impact of the Cold PoDFA procedure on the reported particle content was performed by parallel Hot PoDFA and Cold PoDFA sampling at Hydro Aluminium production sites. In total 112 PoDFA samples (44 Hot PoDFA; 68 Cold PoDFA) have been analysed, representing 15 different combinations of alloy, material source, and sampling position. Table 1 shows an overview of the experimental variations.

Three different alloys have been included in the current study: a low alloyed AA1110 with a max. Mg level of 0.25%, the high alloyed AA5083 (AlMg4.5Mn0.7) and the extrusion ingot alloy AA6060 (AlMgSi) with an Mg level around 0.5%.

The second variation within the current study has been the metal sources used to produce the monitored charges. 100% primary liquid metal from the electrolysis (+ alloying material) have been utilized in all AA6060 charges and in one of the AA5083 charges, while 100% scrap with a constant ratio of pre- and postconsumer scrap have been used in the rest of the AA5083 charges. The low alloyed AA1110 charges were based on the standard production mix of primary liquid metal and a mix of internal and external scrap.

For all three alloys standard Hot PoDFA and standard Cold PoDFA monitoring was performed. In this study the standard Cold PoDFA procedure (Var. 0 in Table 1) consisted of solidification in an A8 clay-graphite crucible, remelting of the solid sample in the same crucible in a resistance heated furnace and manual pouring into a reusable PoDFA crucible. The pre-concentration step was performed for all samples using an overpressure of 0.8 bar and Prefil filters.

Table I. Overview of experimental conditions

Alloy	**Metal source**	**Sampling positions**	**Hot PoDFA**	**Var. 0:** (crucible solidification + electric melting + atmosphere)	**Var. 1:** (**mould solidification** + elec. melting + atmosphere)	**Var. 2:** (**mould solidification** + **inductive melting**+r atmosphere)	**Var. 3:** (**mould solidification** + **inductive melting**+ **cover gas**)
1110	Mix of primary metal and scrap (pre- and postconsumer)	4	X	X	X	X	X
5083	100% primary metal **or** 100% scrap (pre- + post-consumer)	3	X	X			
6060	Primary metal	3	X	X			

The AA1110 charges were used for the evaluation of the impact of the solidification and remelting step on the Cold PoDFA results. In addition to the standard Cold PoDFA procedure three variants of the Cold PoDFA process have been investigated (Table 1):

- Variant 1: The liquid metal samples taken from the process chain was solidified in an ingot mould to increase the solidification rate as compared to the standard practice using the clay-graphite crucible. One section of the ingot was further processed according the standard Cold PoDFA procedure described above.

- Variant 2: A second section of the Ingot was remelted in an induction furnace to study the influence of the remelting furnace technology on the PoDFA results.

- Variant 3: The last part of the ingot was then also remelted in the induction furnace but Argon cover gas was used during the remelting process until pouring of the melt into the PoDFA crucible.

Results and Discussion

The results of the current study will be reported in two sections:

1. Standard Cold PoDFA compared to Hot PoDFA
   Within this section all three alloys will be discussed by comparing the total particle concentration and the particle type mix of both standard procedures. The impact of the alloy and the sampling position along the process chain will be discussed.

2. Impact of solidification and remelting on Cold PoDFA
   Here only charges of the alloy AA1110 will be evaluated regarding the impact of the solidification rate and the remelting procedure on the total reported particle concentration and the particle type mix.

Standard Cold PoDFA compared to Hot PoDFA

Parallel Cold and Hot PoDFA sampling was conducted along the process chain in several charges each of the three chosen alloys as shown in figure 1. Figure 3 shows the total particle concentration reported by Hot PoDFA and Cold PoDFA (Var. 0) for one charge of the alloy AA5083 (AlMg4.5Mn0.7). Here the sampling was performed at three positions of the launder system between casting furnace and casting pit. "Furnace Exit" refers to the launder section upstream the first inline melt treatment process step, "After degasser" to the section after a degassing unit and before the grain refiner addition point, while "after Filtration" refers to the last launder section ahead of the casting pit.

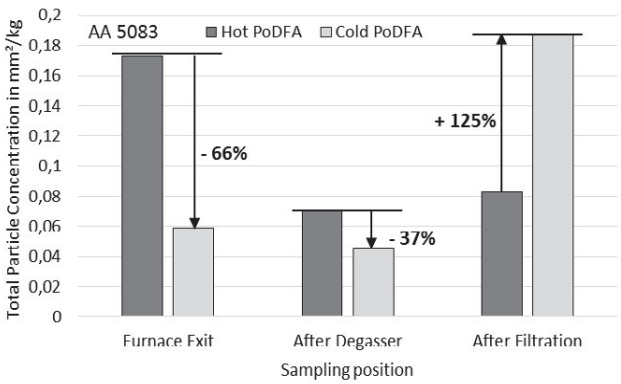

Figure 3. Total particle concentration reported by Hot PoDFA and standard Cold PoDFA procedure* in an AA5083 charge
(* solidification in clay-graphite crucible and remelting in a resistant heated furnace)

The Hot PoDFA results of the AA5083 charge in figure 3 indicated a reduction of the particle concentration by the inline melt treatment process chain. The opposite was evident within this charge according the Cold PoDFA data due to lower reported concentration at Furnace Exit (-66%) and After Degasser (-37%) and higher reported concentration after the mechanical filtration step (+125%). The large differences might be considered as a first hint regarding the comparability of the two methods. However due to the large inherent uncertainty of the PoDFA evaluation a statistical analyses was necessary to reliably identify the impact of the Cold PoDFA procedure on the reported PoDFA results.

Figure 4 shows therefore the average difference of the reported particle concentration of the Cold PoDFA and Hot PoDFA method for samples taken along the inline process chain for the three alloys.

A lower total particle concentration was reported by Cold PoDFA at "Furnace Exit" in all three alloys, while for "After Filtration" and "After Degasser + Grain Refiner" a similar (AA6060) or significantly higher concentration (AA1110 + AA6060) was reported. In the intermidiate sampling position "After Degasser" the behavior seemed to vary depending on the alloy with higher reported Cold PoDFA data for the low alloyed AA1110 charges than the high alloyed AA5083.

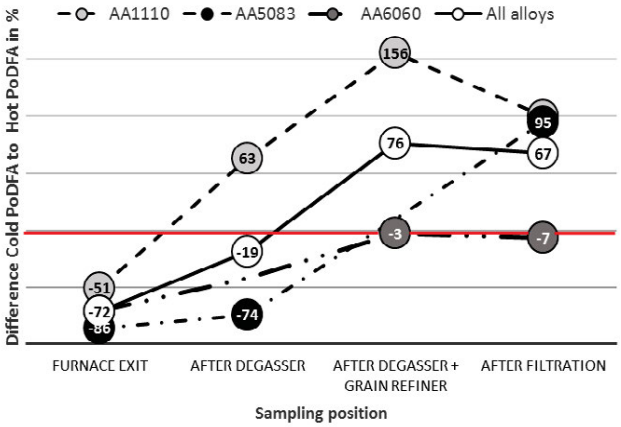

Figure 4. Difference of the total particle concentration reported by standard Cold PoDFA procedure compared to Hot PoDFA (red line indicates identical total concentration in Hot and Cold PoDFA)

The evaluation of the total particle concentration in figure 4 neglects the influence of the alloy on the particle types present in the melt. Changes to the type of particles could explain the variations shown inthe Cold PoDFA analyses for the three alloys and sampling positions. Figure 5 and 6 show the particle fraction for the three alloys at the positions "Furnace Exit" and "After Filtration". The influence of the cold PoDFA procedure on the total particle concentration was quite consistent except for the results of the AA6060 alloy "After Filtration", which will be discussed later.

Figure 5 shows for each of the three alloys tested the fractions of the six main types of non-metallic inclusions found in the melt at the position "Furnace Exit" during this study: Aluminiumoxide (Al_2O_3), Titaniumdiboride (TiB_2), Aluminiumcarbide (Al_4C_3), Magnesiumoxide (MgO), Magnesium- Aluminium-Spinel ($MgAl_2O_4$) and salt. In some cases minimal amounts of other particle types were detected but neglected for this evaluation.

A good agreement regarding the particle type mix reported by Hot PoDFA and Cold PoDFA was evident for the alloy AA1110 although the average total particle concentration reported by Cold PoDFA was roughly 50% less than by the Hot PoDFA procedure. Tentatively an disproportionately high reduction of Al_4C_3 was evident as was an relative increase of Al_2O_3. The corrosion of Al_4C_3 to Al_2O_3 by reaction with humidity as described by C. Simensen [9] and in the PoDFA catalogue [13] could be an explanation for these changes.

The progression of the Carbide corrosion seems to proceed slowly in the low alloy charges of the alloy AA1110 and AA6060 but appears to be quite fast in the charges of the Magnesium containing alloy AA5083. Here no Al_4C_3 particles were present in the Cold PoDFA samples, while a significant fraction was present in Hot PoDFA.

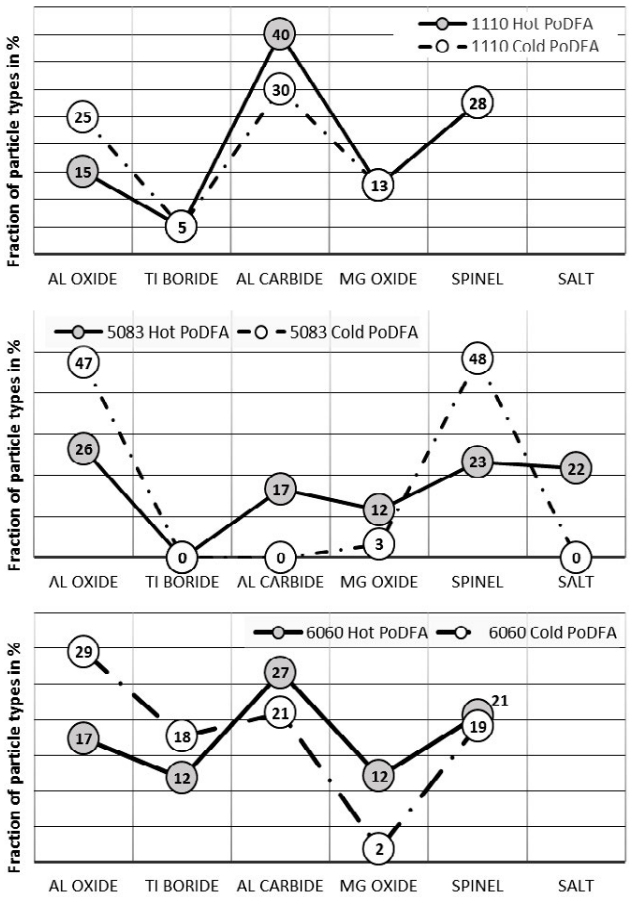

Figure 5. Fraction of main particle types present at position "Furnace Exit" for various alloy groups

A similar result was observed in the concentration of salt, which was detected to a significant amount in the Hot PoDFA but was absent in the Cold PoDFA of the alloy AA5083. Movement of the salt into the dross layer and evaporation during the additional solidification and remelting step of Cold PoDFA may have occured. Salt deposition on the crucible wall during remelting was observed during Cold PoDFA processing, which supports this explanation.

In addition to the lower concentration of Carbides and Salt in the Cold PoDFA samples, the alloys AA5083 and AA6060 also showed a disproportionally high reduction of MgO and a relative increase of Al_2O_3 and Spinel. The later may have resulted from a reaction of MgO with the liquid Aluminium forming Spinel as described by Impey [12].

The comparison of the particle type fractions at the position "After Filtration" shown in Figure 6 showed a similar change of the particle type mix in the Cold PoDFA results. Salt residues present in the Hot PoDFA samples of AA5083 were fully absent in the parallel Cold PoDFA samples. The Al_4C_3 level was significantly reduced in all three alloys and being completely eliminated in case of the more reactive AA5083 charges. Again the amount of MgO was significantly lower for the lower Mg containing alloys AA1110 and AA6060 while the fraction of the reaction products Al_2O_3 and Spinel increased. Generally these reactions appear to occur predominantly at the position "After Filtration" for the alloys AA1110 and AA6060 but further upstream for the more reactive alloy AA5083.

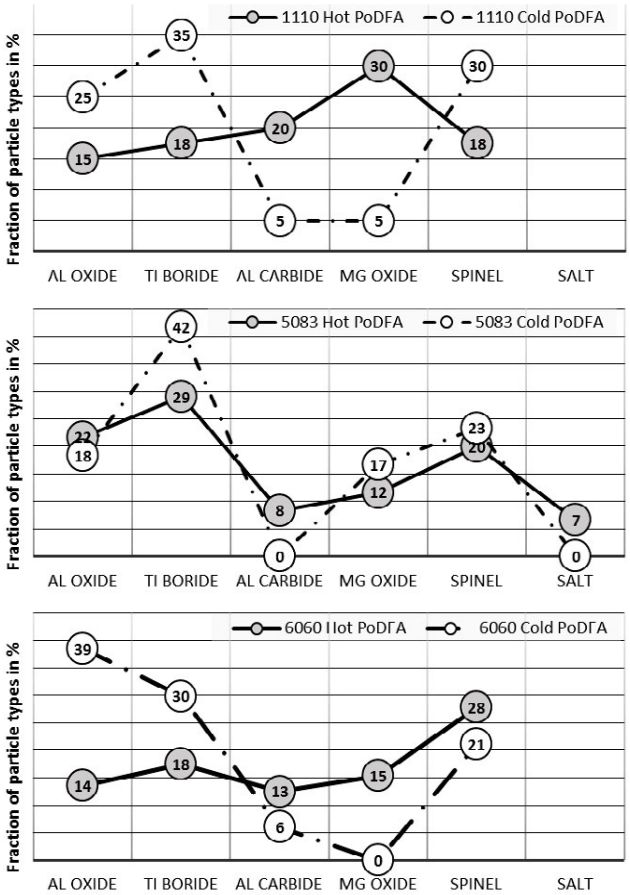

Figure 6. Fraction of main particle types present at position "After Filtration" for various alloy groups

The total particle concentration "After Filtration" for the alloys AA1110 and AA5083 obtained using cold PoDFA was double the concentration of the Hot PoDFA result (see figure 4). Both these alloys showed a large increase in the fraction of TiB_2 particles in the "After Filtration" Cold PoDFA samples (Figures 6), which might be the main reason for the absolute increase of the particle concentration. Agglomeration of the TiB_2 particles with Al_2O_3 films as reported for spent ceramic foam filters [14] could contribute to this effect. The larger agglomerate might be better captured by the PoDFA filter than the small individual TiB_2 grain refiner particles.

The AA6060 charges showed contrary to the AA1110 and AA5083 charges a similar total particle concentration reported by Hot PoDFA and Cold PoDFA (Figure 4). Some of the AA6060 charges

have been produced without grain refiner addition ahead of the monitoring positions and figure 7 shows therefore the difference of the total particle concentration along the inline process chain reported by Hot and Cold PoDFA for AA6060 charges with grain refiner (added before degassing) and charges with complete absence of grain refiner. The two processing conditions showed a significantly different impact onto the correlation of Hot PoDFA and Cold PoDFA, which indicated that the grain refiner had an impact on at least one of the two PoDFA procedures.

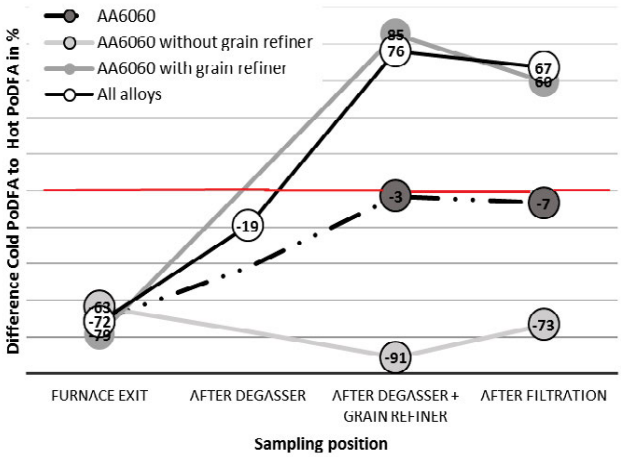

Figure 7. Difference of the total particle concentration reported by standard Cold PoDFA procedure compared to Hot PoDFA in AA6060 charges with and without grain refiner addition

Impact of solidification and remelting on Cold PoDFA Process

For the alloy AA1110 four different variants of the Cold PoDFA procedures have been performed as described above. The solidification rate was increased in Var. 1-3 by using a cold metal mould instead of a clay-graphite crucible. The remelting procedure then involved splitting the faster solidified sample into three pieces and remelting these either in a resistance-heated furnace or in an induction furnace with and without an Argon cover gas.

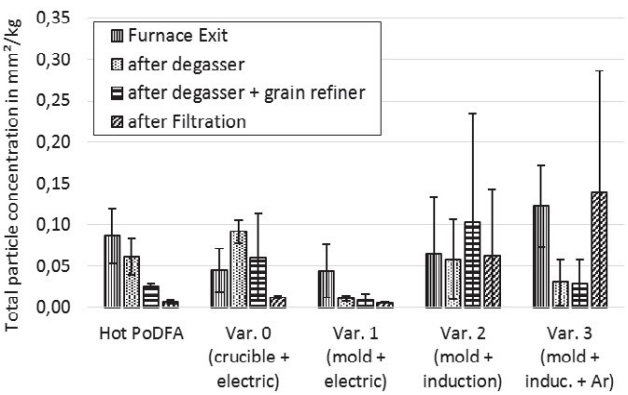

Figure 8. Average total particle concentration for AA1110 charges reported by Hot PoDFA and different Cold PoDFA procedures (variants described in detail in "Procedure" section)

The discussion regarding the impact of the Cold PoDFA process chain on the reported particle concentration will be divided between the total particle concentration and the particle type mix. Figure 8 shows for the Hot PoDFA samples and the four Cold PoDFA variants, the average total particle concentration including standard variation at the four sampling positions.

The Hot PoDFA results showed a typical reduction of inclusion concentration in the melt along the inline process chain. According to these results a stable end quality of the final product was achieved despite variations in metal source and furnace practices. The standard Cold PoDFA procedure reported a similar total particle concentration but significant different inclusion concentrations at the various monitoring position. The statistical variance was also significantly higher in this case. The variance could be reduced by generating a higher solidification rate by using the metal mould in Var. 1 but a significantly lower total particle concentration was reported. Variants Var. 2+3 using the induction furnace for the remelting step showed both large difference in total particle concentration compared to Hot PoDFA and significantly higher variances.

The fraction of the main particle types measured for the 1110 alloy and the various PoDFA procedures are shown for the position "Furnace Exit" in figure 9 and the position "After Filtration" in figure 10.

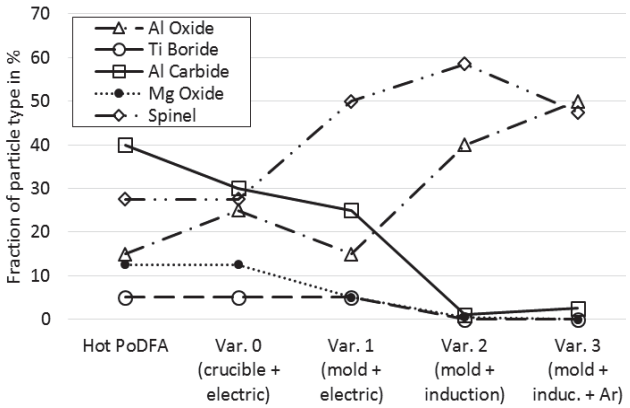

Figure 9. Fraction of main particle types found at position "**Furnace Exit**" for the 1110 alloy charges depending on PoDFA procedure

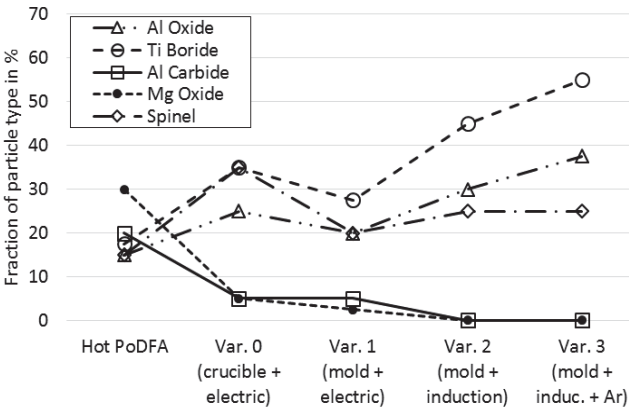

Figure 10. Fraction of main particle types present at position "**After Filtration**" for the 1110 alloy charges depending on PoDFA procedure

In the case of the 1110 alloy, the standard Cold PoDFA procedure showed, as previously discussed, a reasonable agreement with the Hot PoDFA procedure at the upstream position "Furnace Exit". This is, to lower extent, also valid for the Var. 1 with faster solidification of the sample and remelting in a resistance heated furnace. The alternative remelting procedures using induction heating however, changed the particle type mix significantly. Compared to Hot PoDFA, Al_4C_3 and MgO were almost completely absent in these samples, while the fractions of Al_2O_3 and Spinel were increased significantly. It can be assumed that the induced convection during remelting in the induction furnace favoured the reaction of Al_4C_3 to Al_2O_3 and MgO to Spinel by exposure of these particles to the ambient air. The use of Argon as cover gas (variant 3) did not appear to mitigate these reaction processes.

The change of the particle type mix: less Al_4C_3 and MgO and larger fractions of Al_2O_3 and Spinel, was also evident at the downstream position "After Filtration" (figure 9) for all four Cold PoDFA variants. This change was more pronounced for remelting in an induction furnace. Additionally all AA1110 Cold PoDFA samples showed an increased fraction of TiB_2, which supported the assumption of an agglomeration of grain refiner and oxide films as discussed in the previous chapter.

Summary and Conclusion

The impact of the additional solidification and remelting step in the Cold PoDFA procedure on the reported total particle concentration and the reported particle type mix was evaluated by parallel Hot PoDFA and Cold PoDFA sampling. The sampling was performed along the inline process chain in charges of three alloys with a Magnesium content between 0.25 % and 4.5%. Four different Cold PoDFA processing routes were investigated. The results of this study are summarized in the following points:

- All Cold PoDFA procedures evaluated seem to propagate reaction between the particles present in the melt, the ambient atmosphere and the liquid Aluminium. The concentration of Salt residues, Al_4C_3, and MgO was reduced in the Cold PoDFA samples, while the fraction of the reaction products Spinel and Al_2O_3 was increased

- The completeness of the particle reactions seem to be dependent on the Cold PoDFA procedure used. Faster solidification of the sample slowed the transformation reactions, while remelting in an induction furnace resulted in a full progression of the chemical reactions.

- Presence of TiB_2 grain refiner in the melt seemed to have an impact onto either Hot PoDFA or Cold PoDFA methods. More effective capture of TiB_2 and Al_2O_3 agglomerates on the PoDFA filter is believed to be responsible for this increase.

- The total particle concentration reported by the cold PoDFA procedure differed significantly from the Hot PoDFA procedure with generally lower upstream levels and depending on the grain refiner concentration higher downstream levels.

The current study considered only selected alloys and processing parameters but it can be concluded that Hot PoDFA and Cold PoDFA are not directly comparable. The reactions that appear to occur during the Cold PoDFA process have to be considered when judgements about the melt quality based on Cold PoDFA results are to be made

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ASSESSMENT OF SETTLING BEHAVIOR OF PARTICLES WITH DIFFERENT SHAPE FACTORS BY LiMCA DATA ANALYSIS

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Keywords: Inclusions, Settling, Al_2O_3, Shape factor, LiMCA

Abstract

Non-metallic inclusions have a crucial impact on the quality of aluminum products and their behavior in aluminum melts needs to be understood in order to control them. Oftentimes LiMCA (Liquid Metal Cleanliness Analyzer) is used for the quantification of non-metallic inclusions in the melt. However, since LiMCA provides data with the assumption that all particles are spherical, it is currently impossible to obtain information about the particle shape during an online detection. In this work, aluminum oxide particles were added into an aluminum melt and the inclusions were detected by LiMCA in a demo-scale crucible furnace to observe how particles with different shapes settle. The effect of blocky alumina particles versus thin alumina films on the experimental results was assessed. Analytical modellings were also performed to support the experimental findings. First results on the impact of the shape factor on settling will be presented in this paper.

Introduction

Aluminum is extensively used in industrial applications such as transportation, packaging, automotive, aerospace, electronics… The inclusion content has been for a long time a major quality criterion for many products [1]. And it has continued gaining in importance with an increase of inclusion-critical applications such as products with low final thickness and high surface quality.

Inclusion characterization is performed by off-line techniques such as PoDFA, Prefil and K-Mold or on-line methods such as LiMCA and ultrasound. Inclusion characterization must be fast since the production process is continuous and that makes the on-line methods preferred [2].

LiMCA is one of the most common online inclusion detection methods; it is based on the electric sensing zone principle. LiMCA detects the inclusion by measuring the change in electrical resistance between electrodes which is caused by a particle passing through the orifice of a glass tube immersed in liquid aluminum. The voltage peaks are then used for defining the inclusion size and number [3, 4]. LiMCA does not provide information on the shape of the detected particle since a spherical shape is considered in the calculation of equivalent size of the inclusion. As this kind of information is of relevance for aluminum producers, different approaches already exist to extract more information about the characteristics of detected particles from the LiMCA data.

At McGill University, a new technology was applied to better discriminate the particles by using Digital Signal Processing (DSP) technology. In this method, all single events during a LiMCA measurement were characterized with regard to parameters like start slope, end slope and time to maximum voltage to extract more information about the properties of the detected particles. The parameters were found to describe the voltage peaks of LiMCA and they were used to classify the peaks according to their characteristics. Finally, a DSP-based LiMCA method was carried out to characterize the measurement peaks during the on-line operation [5]. This methodology is however not used in the industrial LiMCA.

In another study [1], Martin et al. observed more than one settling trend after stirring the melt and three different settling rates were studied. Continuous improvement of the melt quality was observed in each time zone during settling. However, to reach the same decrease in inclusion concentration in a later stage of the settling curve, the melt required always more time than in the previous time zone. It was concluded that those differences might be caused by different inclusion densities and sizes. The different settling trends suggest that the inclusions settle differently and cannot be easily described by a single settling exponential curve.

Figure 1 presents the LiMCA N20 values at the exit of a holding furnace after the melt was deliberately stirred by gas fluxing. This industrial test was shown in the work of Badowski et al. [6]. After some further evaluation of the data, it was observed that the settling of particles occurred in two different trends as indicated in Figure 1. This situation raised the question whether different settling trends of particles with different chemical and physical properties can be extracted by analysis of LiMCA data.

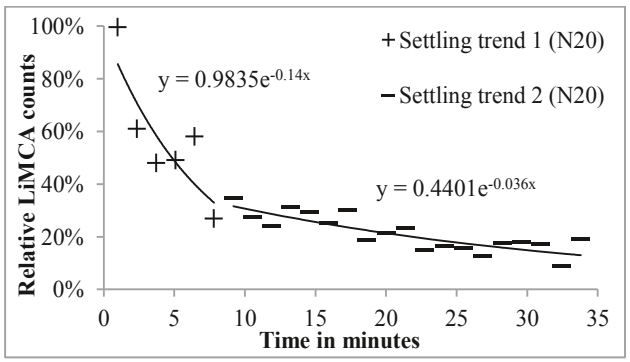

Figure 1: LiMCA settling curve at the furnace exit during casting after intermediate stirring by gas fluxing in the furnace

The target of the present paper is to contribute to the studies on settling behavior of particles. For this, LiMCA trials were performed in molten aluminum before and after addition of particles into the melt in order to observe different settling trends of inclusions with different shapes. The obtained results were evaluated by LiMCA data analysis taking inclusion density, size and shape into consideration.

Experimental Procedure

In the present work, a number of demo-scale scale experiments were performed in different conditions. A LiMCA II unit was used for the detection of inclusions in a test setup consisting of a resistance heated crucible furnace and 120 kg molten metal based on 99.8 % aluminum ingots. The temperature was between 720 and 730 °C during the experiments.

The melt was manually stirred before the particle monitoring started in order to observe the settling of the particles. In selected trials, DURALCAN W6S.14A composite material was added to increase the inclusion concentration in a defined way in order to allow for a better monitoring of particles. The composite material contains 14% ± 2% Al_2O_3 particles in an aluminium matrix with an average size of 17µm (Figure 2).

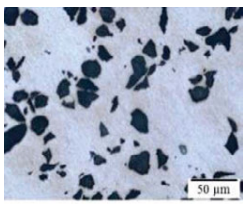

Figure 2: Microscopic picture of alumina particles in MMC

The different parts of the composite material were examined under a light microscope and the dimensions of the particles were measured by image analysis. The composite material consists of particles with the av. length of 18 µm and the av. width of 11 µm. However, the maximum length of the particles is measured as 64 µm. The data was used to compare the size of the charged particles with the base inclusions in the melt.

In addition to LiMCA, PoDFA (Porous Disc Filtration Apparatus) was used to analyze the melt cleanliness. The PoDFA samples were taken from the stirred and settled melt to characterize the particle types under a light microscope. The PoDFA sampling and particle addition procedure during the trials are shown in Table I.

Table I: Particle addition and PoDFA sampling during the trials

Test	Reference Melt	1	2	3	4
Al_2O_3 presence	-	✓	✓	✓	✓
PoDFA samples	P1 (settled)	-	-	P2 (stirred)	P3 (stirred)

Results

The settling curves of the small scale trials, starting with different inclusion concentrations, will be discussed by using their exponential coefficients as they represent the rate of settling [1].

Figure 3 shows the selected LiMCA settling curves before and after particle addition. The first LiMCA settling curve in Figure 3 corresponds to the initial melt before particle addition. The associated PoDFA sample P1 presents mainly Al_2O_3 films and γ-Al_2O_3 as expected (Figure 4). The high inclusion content before MMC material addition (typically 30 k/kg after melt stirring) is mainly attributed to the presence of oxide films, likely due to manual transfer of the melt that was required by the set up used. However, a melt treatment afterwards was not performed because the presence of sufficient inclusions was required for a statistically valid study of settling behavior. After particle addition, the inclusion concentration rapidly increases and reaches about ≈50k/kg after stirring. Sample P2 shows a very high amount of blocky Al_2O_3 particles due to charging of DURALCAN MMC.

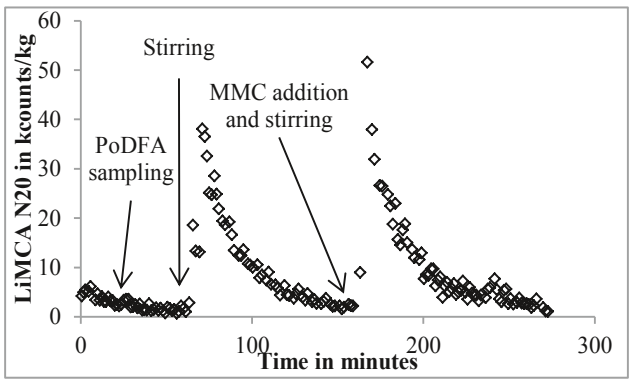

Figure 3: LiMCA run chart showing two tests with PoDFA results before (P1) and after (P2 and P3) particle addition

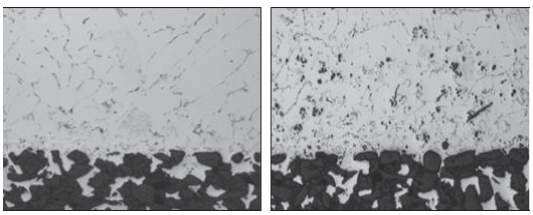

Figure 4: PoDFA samples taken from the surface before (P1) and after (P2) MMC addition

LiMCA settling curves will be investigated separately in terms of k values of the trend lines, where the time "0" corresponds to the stirring of the melt.

Figure 5 represents the settling of the inclusions after stirring (t=0) in the reference melt with an exponential coefficient of -0.030 (min^{-1}). This trend can be assumed as the base inclusion settling trend of the melt.

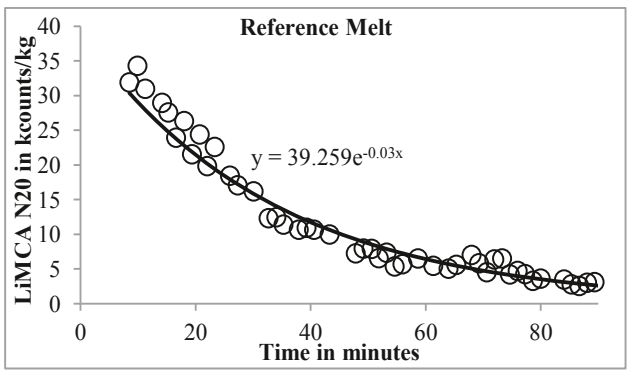

Figure 5: LiMCA settling curve after stirring of the melt with base inclusion content (without particle addition)

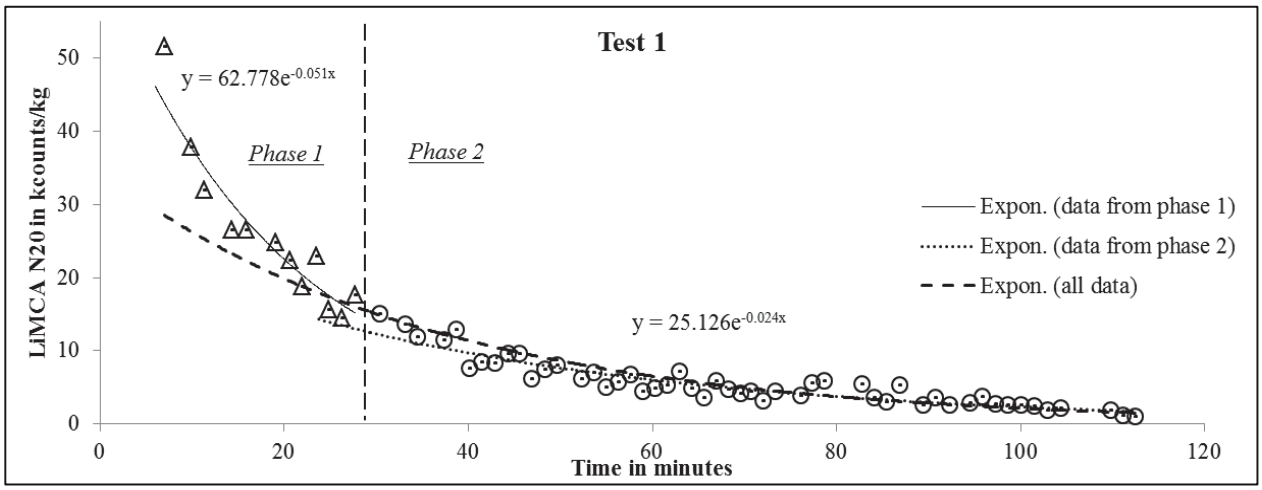

Figure 6: Comparison of the different regressions on data from a settling trial

After particle addition, the nature of the observed trend line changes (Figure 6). A single exponential trend line (dashed line) no longer represents the entire trend as precisely as in the reference melt. It is obvious that the settling behavior is described more accurately by using two exponential functions with a transition in the range of 15-20 kcounts (N20). From now on, the two parts of the curve will be named phase 1 and phase 2 respectively. The separate curves show an exponential coefficient of -0.051 for the first and -0.024 for the second phase.

While the exponential coefficient of the phase 2 in Figure 6 is quite similar to that of the base inclusion settling trend in the reference melt (-0.030), the coefficient of phase 1 differs significantly.

After observing those independent trends in settling, more tests were analyzed in phase 1 and phase 2 in order to obtain a better understanding of the settling behavior. In test 2, only phase 2 is available due to technical issues of the LiMCA unit during the start of the settling (Figure 7). However, the exponential coefficient of the curve (-0.036) is close to the reference melt and indicates that the exponential coefficient of this settling phase seems to be characteristic for the base inclusions in phase 2 in the utilized setup.

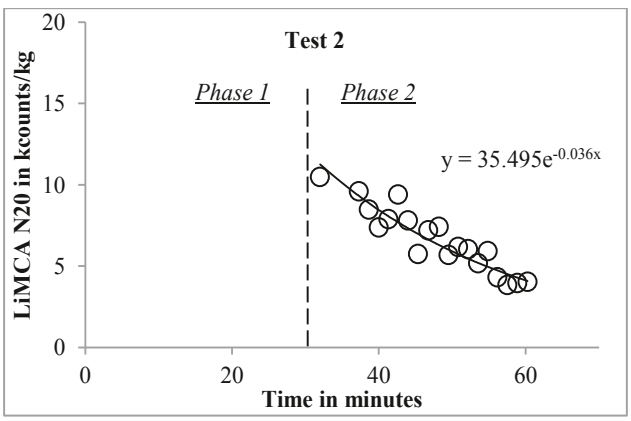

Figure 7: LiMCA settling curve after stirring the melt with added alumina particles

Moreover, two settling trials were evaluated in tests 3 and 4. In test 3 the settling was interrupted by PoDFA sampling so that phase 2 of the curve is not available.

The exponential coefficient of the settling curve in the Figure 8 is -0.055 and therefore comparable to the first test (-0.051). Accordingly, this range of exponential coefficients seems to be characteristic for phase 1 with specific particles in the utilized setup.

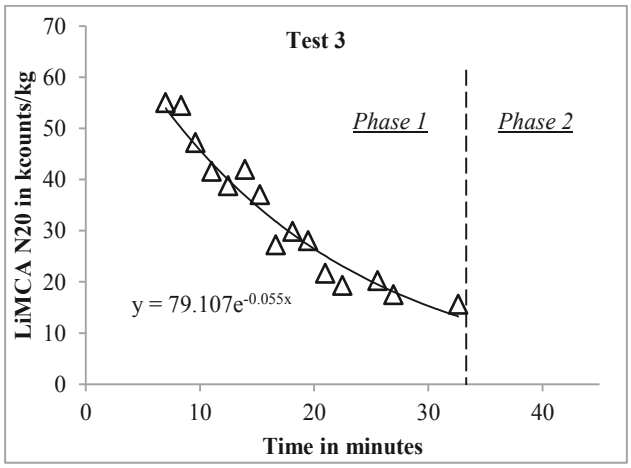

Figure 8: LiMCA settling curves of test 3 after stirring the melt with added alumina particles

Figure 9 also shows two settling phases after stirring. The settling in the phase 1 has the rate of -0.088 which is quite higher than the rate in tests 1 and 3. This value might be caused by the limited number of measurement points or a lower starting inclusion concentration. However, it is still obvious that the particles in phase 1 represent another settling trend. On the other hand, the second phase of the curve has a settling rate of -0.036 which is very similar to the results in phase 2 of the other tests.

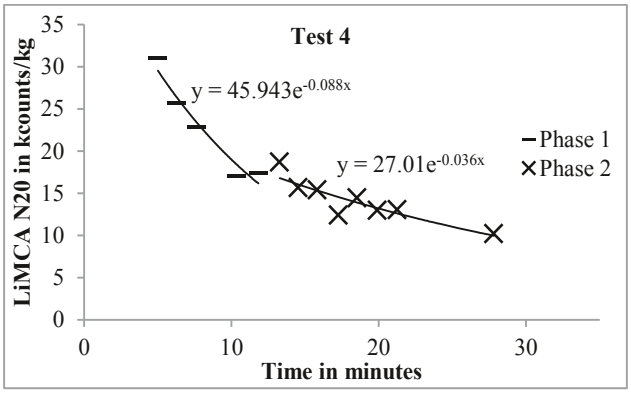

Figure 9: LiMCA settling curves of test 4 after stirring the melt with added alumina particles

The obtained results show that the difference in these coefficients might be a result of a different settling behavior of the initial and the subsequently added particles. Nevertheless, the different settling trends of the particles with different properties can be observed by LiMCA data and the major effect causing this variation will be studied in the discussion part.

Discussion

The settling of a melt with alumina particle additions has been studied by LiMCA measurements. It has been observed that the settling curves were different and two separate exponential curves were needed to describe the settling behavior when alumina particles were added. With alumina particle addition, the settling is apparently driven by the added particles (k typically close to -0.05 min^{-1}). After a period of time, the settling behavior of both melts (with and without particles) becomes similar (k typically 0.03 min^{-1}) due to the settling of the added particles (Table 2).

Table 2: The exponential coefficients -k (min^{-1}) of the settling curves in each test condition

N20 data	Phase 1	Phase 2
Reference melt	0.030	0.030
Test 1	0.051	0.024
Test 2	n.a.	0.034
Test 3	0.055	n.a.
Test 4	0.088	0.036

As stated before, PoDFA evaluations showed that the initial melt contained mainly alumina thin films before the addition of composite material. Therefore, the settling trends after particle addition can be interpreted by assuming that LiMCA detects both the settling of the thin alumina films and the added alumina particles in the melt.

The observed settling curves with two exponential regressions after stirring in test 1 to 4 were seen only after the addition of particles. This different behavior might be caused by the different properties of the added particles in comparison to the base inclusions in the initial melt. From the results it can be concluded that the first phase of the curve in the test 2 presents likely the settling of the added particles and the second phase the base particles, which were already present in the initial melt.

The particle properties, which might have caused the different settling trends, can be density, size and shape factors of the particles. In the following, they will be taken into consideration with regard to the interpretation of the results.

Density

Due to the buoyancy and gravity forces (Eq. 1 and 2), the density has an important influence on the settling velocity of a particle and the exponential coefficient of the settling curve accordingly.

$$F_G = m_s \cdot g = \rho_{particle} \cdot V_{particle} \cdot g \quad (1)$$

$$F_v = m_f \cdot g = \rho_{fluid} \cdot V_{particle} \cdot g \quad (2)$$

with
- g the gravitational acceleration (m/s^2)
- ρ_p the mass density of the particle (kg/m^3)
- ρ_f the mass density of the fluid (kg/m^3)
- $V_{particle}$ the volume of the particle (m^3)

However, the detailed PoDFA analysis showed that the melt contained mainly α-Al_2O_3, γ-Al_2O_3 and Al_2O_3 films with approx. the same density and therefore the density does not seem to be the major cause for the difference in the settling velocity.

Size

The size of a particle has a direct influence on its settling velocity due to the gravitational force (Eq. 1). The settling curves of all tests were compared by studying the settling curves in different LiMCA size ranges, namely 20-40 and 40-60 µm, in order to investigate the influence of the particle size (Table III).

In reference melt and the test 1, there is virtually no difference between the different size ranges with regard to their exponential coefficients. This shows that the inclusions settle without distinctive dependence on the particle size. However, in tests 2 to 4 the size ranges do not show similar settling rates anymore. This change might be caused by insufficient detected particle concentration in the size rage 40-60 µm. The very low inclusion concentration in larger size ranges does not allow representative results and it is hard to observe a clear settling trend in case of insufficient number of detected inclusions.

Table III: Rate of settling (-k) for each test in size ranges of 20-40 and 40-60 µm

Test	Phase	-k (20-40) (min)	-k (40-60)
Reference		0.030	0,032
Test 1	1	0.048	0,051
	2	0.021	0,023
Test 2	1	n.a.	n.a.
	2	0.034	0.045
Test 3	1	n.a.	n.a.
	2	0.052	0.072
Test 4	1	0.082	0.118
	2	0.037	0.058

The difference of particle size distribution between added and base inclusions influences the characteristics of the settling curves. The normalized size distribution of the particles measured by LiMCA is shown in the Figure 10 for the five test conditions. The first measurement point after stirring for each test was normalized by dividing of the respective size ranges by the number of particles in the N20-40 class to observe the ratio between classes. The histograms can be compared to each other by the means of the given exponential coefficients.

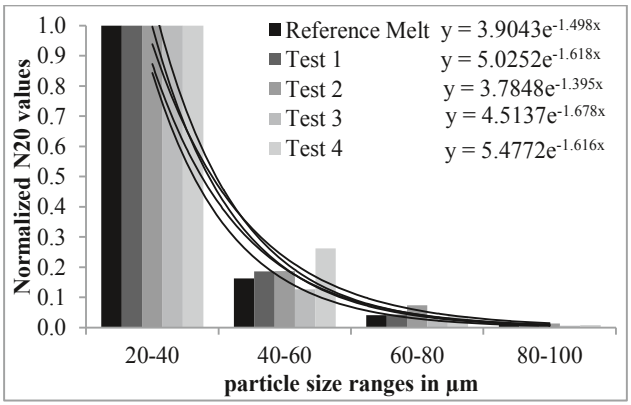

Figure 10: Size distribution of the first detected point after stirring in each test

The results do not show a significant difference in terms of size distribution between the tests before and after the addition of particles. Furthermore, the particles in the added MMC material are in a comparable size range. This might be the reason why no clear size effect on settling was observed in the results.

These results suggest that the particle size range is not the major factor affecting the settling trends in this particular case.

Shape Factor

The influence of the shape factor on the settling of inclusions can be described via the drag coefficient (C_D) of the particles. This coefficient changes the drag force (Eq. 3) and determines the particle movement during settling in a fluid.

$$F_D = ½\ \rho_{fluid} \cdot \upsilon^2_{particle} \cdot C_D \cdot A \qquad (3)$$

with C_D the drag coefficient of the settling particle
υ the terminal settling velocity of a particle (m/s) (vertically downwards if $\rho_{particle} > \rho_{fluid}$)
A the surface area of the particle

The settling of the inclusions is analytically investigated in this work based on the PoDFA observation that shows that the major fraction of the initial inclusions consisted of thin Al_2O_3 films and that α-Al_2O_3 particles were added. Although they have the same density, their settling velocities will be different due to their shapes. A microscopic picture of the added particles is shown in Figure 2 and the shapes of the particles seem blocky and spherical-like. They will have a smaller C_D value in comparison with thin and thick films in the melt, which results in higher settling velocities.

The terminal settling velocity of a spherical particle can be expressed by Stoke's equation (Eq. 4) under laminar flow [7]. The added particles in this study have the aspect ratio of 1.6-1.7) and therefore, the terminal velocity of added particles is approximated with the assumption of spherical particles. The terminal velocity (Eq. 5) of a thin film is calculated by an approximation of thin disc geometry with the aspect ratio of 0.1 [8].

$$v_P = \frac{(\rho_P - \rho_F) \cdot d^2 \cdot g}{18 \cdot \mu} \qquad (4)$$

$$v_P = \frac{(\rho_P - \rho_F) \cdot \pi \cdot d \cdot D \cdot g}{32 \cdot \mu} \qquad (5)$$

with D the thickness of the disc
d the diameter of the particle
µ the viscosity of the fluid

The following diagram shows the theoretical settling velocities of thin discs and spherical particles calculated by using the equations 4 and 5. Here, the spherical particles with the same density of thin discs reach approximately 5.5 times larger settling velocities. Since this value is quite high, this might be the major effect for the observation of the two settling trends after stirring during the performed trials. However, these values should only be seen as indicators, since Badowski et al. [6] already showed that the applicability of Stoke's law for melting furnaces is limited and that the overall melt velocity pattern needs to be taken into account for the calculation of proper particle settling velocities.

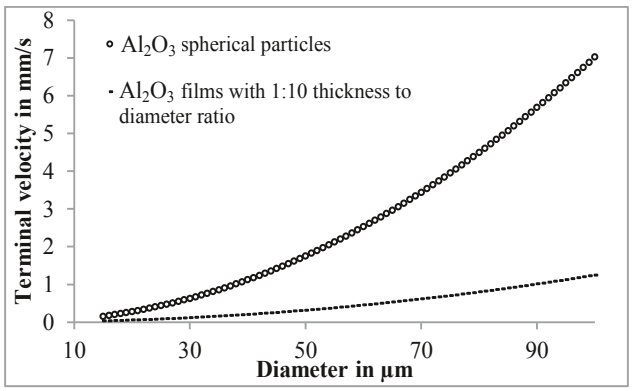

Figure 11: Analytical settling velocities for disc-shaped and spherical Al_2O_3 particles

The calculations above suggest that the drag coefficient and therefore the shape factor might have a dominating influence on the settling velocities in this particular case since all particle types have the same chemical composition and a similar size range.

Conclusions

The settling of Al_2O_3 films and particles was investigated in a demo-scale crucible furnace based on LiMCA data. The analysis and the results are summed up as follows:

- Large amounts of oxide films were observed in the initial melt by PoDFA samples and their settling could be described by a single exponential function.

- After addition of a MMC material that contained spherical-like α-Al_2O_3 particles, two different settling trends were observed by LiMCA after stirring the melt.
- In the present setup, the transition of the two settling curves of films and particles was in the range of 15-20 kcounts (LiMCA N20 value). But, this value depends on the inclusion concentration of the melt.
- The reason of different settling trends due to particle addition were investigated by analytical calculations, in which density, shape and size of the particles were taken into consideration
- The shape factor of an inclusion particle was shown to be having a major influence on the particle settling in this particular case.
- This study has shown that, in specific configurations, information on particle characteristics can be retrieved from LiMCA curves when known inclusions are present. However the approach can hardly be considered for the analysis of industrial melts where a mix of inclusions is most often present and the LiMCA curve will depend on the mix of all inclusion characteristics

Acknowledgments

The research leading to these results has been carried out within the framework of the AMAP (Advanced Metals and Processes) research cluster at RWTH Aachen University, Germany.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

Modeling of Inclusion Behavior in an Aluminum Induction Furnace

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Keywords: Induction Melting, Aluminum Bath, Inclusion, Metal Quality, Simulation

Abstract

Crucible induction furnaces are widely used in the aluminum industry, for scrap remelting, metal treatments and casting. The operation principle results in an intense circulation within the furnace, raising a specific question with regard to inclusion dynamics within the melt, reflected by LiMCA measurements at the furnace exit. In an effort to understand the involved phenomena, a hydrodynamic model of an induction furnace was built and complemented by an inclusion module that takes into account the transport of inclusions and the interaction of inclusions with other inclusions (aggregation) or with the crucible walls. A numerical inclusion distribution has been developed that reflects the characteristics of the inclusions present in the melt. The model and results of its application are presented in this paper.

Introduction

Electromagnetic forces are widely used in molten metal production, including aluminum cast houses. These forces can be very powerful and provide means for non-contact actions on the very corrosive molten aluminum. Electromagnetic forces are used for induction melting of aluminum, in particular for scrap melting [1]. In this application, a molten heel is often used and the turbulence generated by the electromagnetic field enhances the immersion of the solid in the liquid metal. This, together with the absence of gas burners, reduces the oxidation of the scrap prior to remelting and after the solid has been molten.

Electromagnetic forces are frequently used to stir aluminum furnaces. Different technologies exist: side stirrers can be installed on most furnaces while bottom stirrers are limited to tilting furnaces. Stirrers can provide several benefits, including the thermal homogenization of the metal bath, an enhanced melt rate, significant energy savings, an improved alloying efficiency, and even also a reduction in dross generation [2]. Transport ladles can also be stirred by the same process, avoiding thermal gradient and reducing dross generation [3].

Other applications of electromagnetic forces are considered for molten aluminum filtration. Lorentz forces are studied for the separation of non-metallic inclusions in a molten aluminum flow by electromagnetophoresys [4, 5]. It has also been discovered that priming of ceramic foam filters may be significantly enhanced when Lorentz forces are applied [6]. This allows finer or thicker filters to be used for aluminum production and hence an increased metal quality to be expected. Electromagnetic forces are also used to lift the metal: electromagnetic pumps and rising launders [7]. The main application of induction in aluminum casthouses is in melting furnaces. Coreless induction furnaces are attractive in particular for the flexibility they offer for alloy production and the resulting low metal loss. It is known by experience that process parameters widely impact the hydrodynamics in the furnace: frequency, power, filling ratio in particular have a significant impact.

A model has been developed to describe the hydrodynamics of molten aluminum in a coreless induction furnace [8]. This model was validated based on the comparison of the free surface deformation, both at lab scale [8] and on an industrial furnace [9]. Given the fact that the inclusion quality is a critical parameter for metal quality, there is a strong desire to understand inclusion behavior in these specific furnaces where induction forces are applied.

Only limited work has been done to understand and predict the inclusion behavior in a casting furnace and this work has been focused on gas fired furnaces [10]. The nature of the metal flow in an induction furnace is very different from that of a gas fired furnace as the melt is continuously stirred. The impact is illustrated in Figure 1, where the relative amount of inclusions exiting a furnace is reported for several casts done with an induction furnace (A) or a gas fired furnace (B). The drastic difference in inclusion behavior is most often explained by the stirring provided by the induction force; with very limited further understanding of the physical phenomena acting on the inclusions.

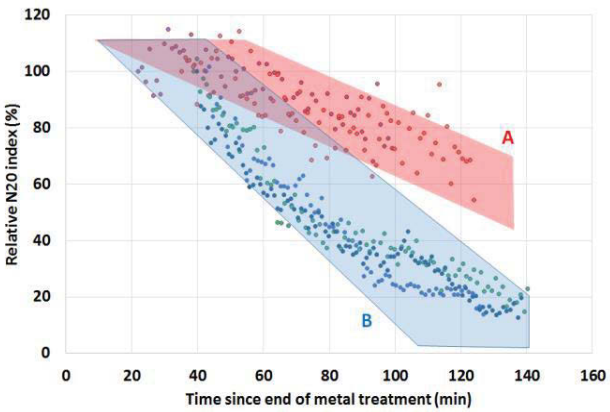

Figure 1: Relative LiMCA N20 index versus time measured during casts from an induction furnace (A) and a gas fired furnace (B), after an Ar-Cl2 metal treatment.

It is the aim of this work to describe the impact of the hydrodynamics on inclusions in an induction furnace, based on the numerical description of the physics involved. If the simulation of a population of inclusions in gas-stirred ladle of liquid steel or aluminum has received great attention by the

metallurgists [11, 12, 13], very few have focused on inclusion behavior in an induction furnace. Some experimental and numerical investigations such as [14, 15] have explored the inclusion behavior in a liquid metal bath subjected to an alternating magnetic field and note a general consensus over the important role of inclusion separation due to EM migration within the EM skin depth. However the numerical modeling has been mostly restricted to a Lagrangian technique where some inclusions were tracked within the bath [16, 17]. While the smaller inclusions were found to follow the bath flow structures and thus take a long time settling or migrating towards the wall, the larger inclusions quickly exited these streamlines and migrated towards the interfaces due to the reigning EM influence.

Physical data

Furnace

In this work, the numerical model was applied to simulate a series of experiments performed in an industrial scale induction furnace using a 5 t molten metal bath of aluminum alloy Al-2xxx series.

Properties

The density, dynamic viscosity, electrical conductivity and surface tension of the liquid of aluminum alloy Al-2xxx series at 750°C were taken equal to 2416 $kg.m^{-3}$, $1.04x10^{-3}$ Pa.s, $3.38x10^{6}$ $\Omega^{-1}.m^{-1}$ and 0.795 $N.m^{-1}$ respectively [18, 19].

Particles

Typically, when the transport of inclusions in liquid aluminum is modelled, the characteristics of non-metallic and undissolved inclusions (type, density, size, shape) originate from measurements by LiMCA and PoDFA techniques. The calculation does not use the full distribution of inclusions, but many couples of samples density / inclusion size [10].

Our approach is slightly different. Initially, we also used measurements by LiMCA and PoDFA techniques to determine the type of inclusions in the furnaces and their characteristics. Then, a database was developed, describing for each type of inclusion, several characteristics such as: density, minimum, mean and maximum diameters, diameter distribution, shape and a frequency index of observation in PoDFA samples. Knowledge of inclusion shape is important since it allows choosing the best approach to calculate drag and lift coefficients for different particles. An extract of this database is shown in Figure 2 for $MgAl_2O_4$ inclusions (spinel).

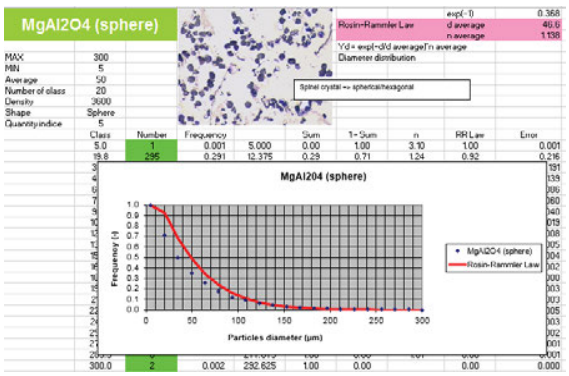

Figure 2: Characteristics of spinel particles.

From the diameter distribution based on observations in metal and the frequency index for each type of inclusion, a numerical particle size distribution can be reconstructed, which correspond to distributions measured by LiMCA. The current paper is illustrating the approach for the transport of $MgAl_2O_4$ inclusions in the melt. The inclusion's size distribution may be approximated using a log-normal distribution.

Numerical Model

According to the symmetry of the induction furnace and its operating conditions a transient 2D axisymmetric model was developed based on the ANSYS Fluent Software, while many separate User Defined Functions (UDF) were hooked in to simulate specific phenomena such as electromagnetic induction, population balance, aggregation, particle/liquid drift velocity etc. Since the solid particle weight content remains very small (much lower than 50 ppm) the motion of the inclusion population does not affect the turbulent liquid flow at a macroscopic level, and therefore hydrodynamics and inclusion behavior can be decoupled. As a consequence the modeling is divided into two steps: the simulation of the turbulent hydrodynamic Al bath, followed by the behavior of an inclusion population in the bath.

Magneto-Hydro Dynamic model

The MHD model employed in this work considers, in a single framework (i.e. the same computational mesh), the induction phenomena, the turbulent flow of the liquid metal inside the crucible, the deformation of the free surface of the metal bath and the presence of an oxide layer covering the metal bath. Note that the temperature in the stirred metal bath is considered homogeneous and thus the energy equation is not solved. This model was presented in detail [8] and only the main aspects of the model are summarized below.

The electromagnetic problem is solved using the harmonic A-V formulation, with an imposed current density in the coil which is described as a hollow cylinder with a rectangular cross section. Using the hypothesis of axisymmetry, the electric current density j and the vector potential A have only one non zero component, along θ. The transport equation for A_θ is solved over the whole domain, including a part of atmosphere large enough to enable the magnetic lines to close freely (chosen here as ten times the coil size in both directions), with boundary conditions $A_\theta=0$ on its outer surface and $\partial A_\theta/\partial r=0$ on its axis. A magnetic yoke, present in the industrial scale furnace, has been modelled as a hole in the mesh, and the condition $B_t=0$ was imposed, which would be correct for a perfectly magnetic material surrounding the whole coil (as the model is axisymmetric).

As far as hydrodynamic phenomena are concerned, standard models are used to describe the fluid flow (Reynolds-Averaged Navier-Stokes equations, with the electromagnetic force F=jxB as a source term), the turbulent friction (k-ω SST model for turbulence) and the deformation of the free surface using a Volume Of Fluid method with a geometric reconstruction scheme. The oxide layer which covers the free surface of the metal bath is considered homogeneous and behaving like a pseudo deforming wall. It impacts both the surface tension σ of the metal and the friction of the metal on the interface. Volumetric forces are included in the model to represent the surface tension, and the value of σ has been defined from experimental data of the literature [19] to account for the effect of the oxide layer. The

friction between the metal and the oxide layer is described using a volumetric force proportional to the fluid velocity, since the oxide is supposed fixed:

$$\mathbf{F}_{vol} = \boldsymbol{\tau}_{oxide} A = -C_f A(\mathbf{V_m}.\mathbf{t})\mathbf{t} \quad (1)$$

where C_f is a friction coefficient, A is the interfacial area density and t is the unit vector tangent to the interface. During the implementation phase, it was demonstrated that high values of C_f (above 1500 $kg.m^{-2}.s^{-1}$) are required for the oxide layer to be assimilated to a wall and that for higher values of C_f the flow solution is not sensitive to the precise value of C_f.

At each time step of the simulation, the induction model first calculates the distribution of the Lorentz forces, which is then transferred to the flow and free surface calculations. The newly determined free surface shape is finally provided back to the induction model to continue the next time step, which ensures a strong coupling between the electromagnetics and hydrodynamics calculations.

Population Balance Model (PBM)

The MHD simulation [8] provides results in the form of velocity, turbulence and Lorentz force fields which are then used while modeling the transient inclusion behavior in the bath. With the aim of modeling both transport of inclusions within the Al bath and mesoscopic interactions between inclusions (namely aggregation mechanisms) a PBM has been adopted rather than a Lagrangian method. As presented by Kumar and Ramkrishna [20], population balance is a powerful way of synthesizing the behavior of a population of discrete particles from the behavior of single particles in their local environment. The behavior of the inclusion population, defined by a distribution function of particle size (N_i is the number of inclusions of class *i* –where the volume of the particle is between v_i and $v_{i+1,}$ per m^3 of liquid metal), is described by the following population balance equation.

$$\forall i, \quad \frac{\partial N_i}{\partial t} + \frac{\partial}{\partial x_j}(N_i u_{p,ij}) = H_i \quad (2)$$

In equation 2, the macroscopic transport phenomenon of inclusions is given by the left member whereas the inclusion aggregation is modeled by the right member. At the inclusion scale, the application of the fundamental dynamic equation leads to the expression of the drift velocity i.e. the relative velocity between inclusions (of size *i*) and liquid metal:

$$\vec{u}_{p,i} - \vec{u}_f = \vec{u}_{sed,i} + \vec{u}_{em,i} = \frac{dp_i^2}{18\mu_f}(\rho_p - \rho_f)\vec{g} - \frac{dp_i^2}{24\mu_f}\vec{F}_{em} \quad (3)$$

$\vec{u}_{sed,i}$, $\vec{u}_{em,i}$ and $\vec{F}_{em}$ respectively represent the Stokes velocity, the electromagnetic drift velocity of class i and the Lorentz forces. At this stage of development of the numerical model, the turbulent diffusive contribution was not taken into account in the macroscopic transport. However the PBM takes bath turbulence into account while calculating the aggregation collision kernel.

Since the solution of the coupled set of partial differential equations (2) is a tricky mathematical problem, a time splitting technique is used:

$$\forall i, \quad \frac{\partial N_i}{\partial t} + \frac{\partial}{\partial x_j}(N_i(u_{f,j} + u_{sed,ij} + u_{em,ij})) = 0 \quad (4)$$

$$\forall i, \quad \frac{\partial N_i}{\partial t} = H_i = B_i - D_i \quad (5)$$

The first step (Equation 4) is to solve the transport of the inclusion population. While caution is required when implementing these drift velocities within ANSYS Fluent, it nevertheless allowed a faster calculation as the velocity terms are completely implicit and are treated directly by ANSYS Fluent. In the current work, the bath free surface and the crucible walls are treated as sinks and therefore inclusions are considered as *captured* at these boundaries, either due to the settling or the EM migration flux. There is no release of inclusions once inclusions are captured at the wall.

The second step (Equation 5) consists in solving the Population Balance equation in each control volume of the domain. This step deals with the aggregation of the inclusions. Following the collision/aggregation mechanism, H_i is simply the number of inclusions which have died D_i subtracted from those which have taken birth B_i in that particular class. Using Kumar's Cell average method [21], the aggregation term is further described in terms of a collision kernel and a redistribution operator. Furthermore while describing the collision kernel, we considered turbulence [22] and gravitational collision terms. Finally, Figure 3 summarizes the general flow of the Population Balance Model.

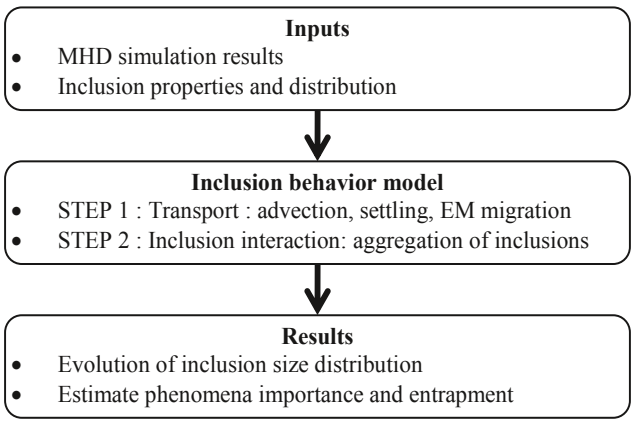

Figure 3: General layout of the Population Balance Model

Results and Discussion

MHD results

In this paper, two extreme operating conditions that correspond to very different numerical cases, useful in validating the model on an industrial scale were investigated experimentally and simulated. The inductor works in both cases at full power. In the first case, the furnace is full with molten metal (Figure 4a) while in the second case the furnace is only filled to 2/3 of its capacity (Figure 4b). Experimentally, we observe strong fluctuations of the metal surface with full furnace configuration (E1), especially near the crucible wall. The surface of metal is more stable in the second case.

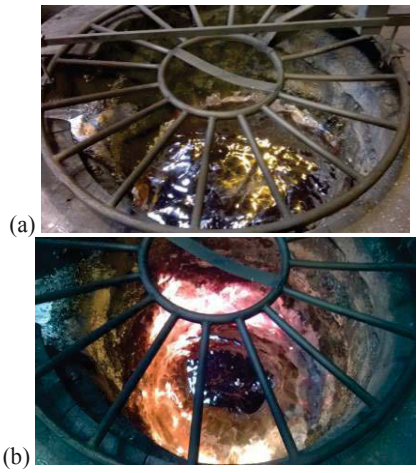

Figure 4: Free surface of industrial induction furnace. (a) Large fluctuations (E1), (b) small fluctuations

Figure 5 presents numerical results with the temporal evolution (over a time interval of 6 s) of the dome profiles for the two experimental sets (E1 & E2). While for E1, large fluctuations are recorded along with an irregular profile deformation (especially near the crucible wall), for E2, a smooth deformation profile is obtained with a much smaller relative fluctuation amplitude.

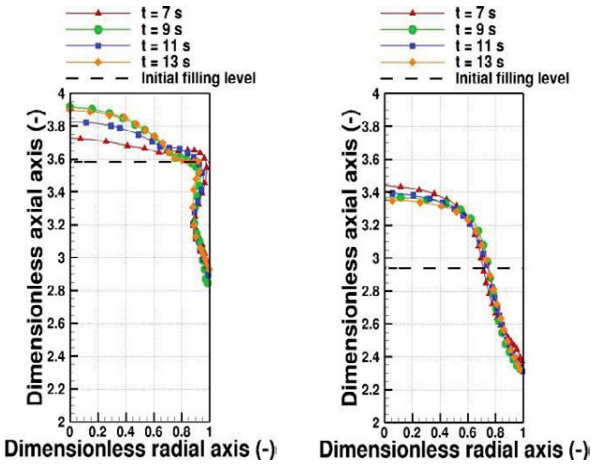

Figure 5: Temporal evolution of domes profiles – E1 (left) & E2 (right)

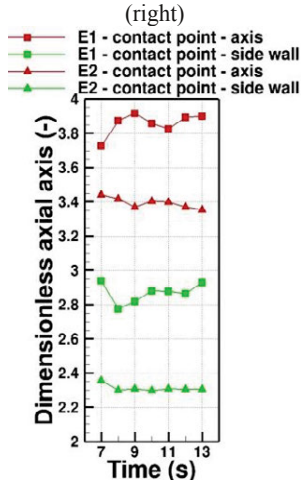

Figure 6: Temporal evolution of dome contact points – E1 & E2.

Figure 6 presents the contact points on the crucible axis or side wall, extracted from these dome profiles. This figure confirms the stability of the metal surface in the second case. 3. Since the model is axisymmetric, it imposes a symmetry to the problem, which is probably not given in the reality: indeed, the real perturbations observed in Figure 4 reveal a less symmetric appearance.

Moreover we also investigated an experimental case (E3) which corresponds to the holding mode operation with a fully filled furnace at only 50 % of the maximum power. Figure 7 illustrates the melt velocity distribution in the bath along with the flow structures in the form of two recirculation zones. Using this bath hydrodynamics, the next section describes the modeling of the inclusion behavior in the holding mode.

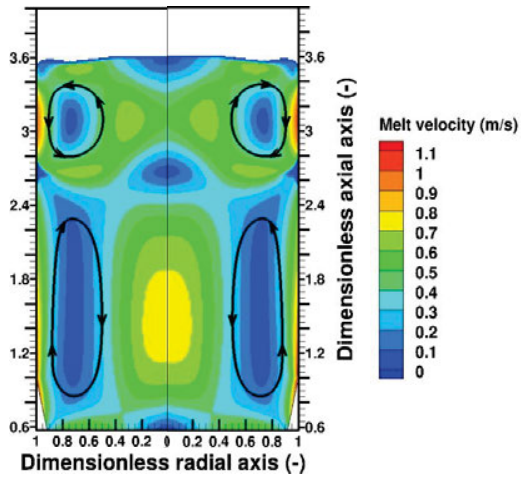

Figure 7: Bath hydrodynamics: melt velocity contours – E3.

Population Balance Model Results

According to the inclusion analysis presented earlier we have used a log-normal law to describe the initial particle size distribution with a total concentration equal to 5.2×10^{-3} kg/m^3 corresponding to 2 ppm of total mass content, considering a high population of $MgAl_2O_4$ inclusions.

The PBM allows investigation of the evolution of the particle size distribution for this inclusion population, with respect to time, as reported in Figure 8.

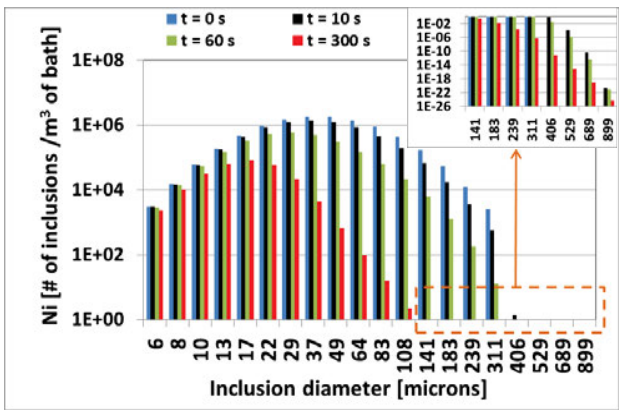

Figure 8: Temporal evolution of inclusion population distribution.

It clearly highlights a strong reduction of the population within the bath after few minutes of treatment regardless of the inclusion size. The inclusion removal appears to be more effective for large sized inclusions. For instance we notice a twentyfold decrease of the density number for 22 µm inclusions after 5 minutes while 141 µm inclusions seem to have been totally separated. A closer look at these results (see inset zoom within Figure 8) reveals that during an intermediate period of time, large particle (>300 µm) are generated into the bath before their elimination. It can be readily explained by the agglomeration of smaller size particles to produce large agglomerates.

The striking drop of the total mass of these inclusions in the liquid bath (shown in Figure 9) can be explained by the relative role of the two removal mechanisms i.e. sedimentation and electromagnetic separation. Only 4% of the removed particles were due to the sedimentation at the crucible bottom (Figure 10). It appears that the Lorentz force is high enough for transporting inclusions toward the crucible wall. The numerical model illustrates the transport of inclusions from the bulk flow toward the crucible wall but the behavior of the inclusions impacting the wall is not modelled so far and should be investigated (aggregation, cementation…).

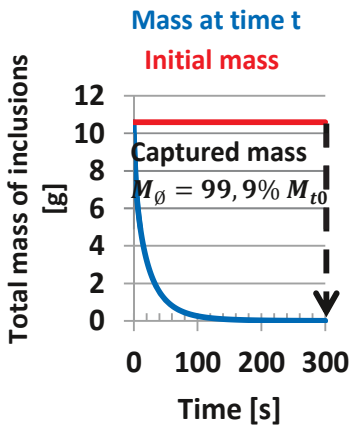

Figure 9: Evolution of total mass of inclusions in the bath.

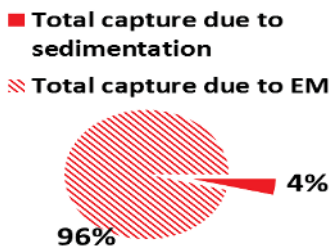

Figure 10: Relative capture share due to Settling and EM migration.

The force of the electromagnetic field is primarily confined in a limited region adjacent to the wall (called electromagnetic skin). Hence the transport of the inclusion population toward the wall combines two mechanisms, as is clearly illustrated in Figure 11 for two sizes of particles and at three different processing times.

The inclusions are dragged by the liquid metal following the two recirculation loops as seen in Figure 7. While the inclusions are approaching the wall and enter the electromagnetic skin they are dragged by the EM drift velocity. An important part of the population is then collected at the side wall. The larger the inclusion size, the stronger the effect of the EM capture. Indeed the square size dependence of the settling and electromagnetic drift velocities (see Equation 3) explains the sensitivity on the particle size resulting in the fact that larger particles are more easily transported by electromagnetic force and sedimentation than the smaller ones.

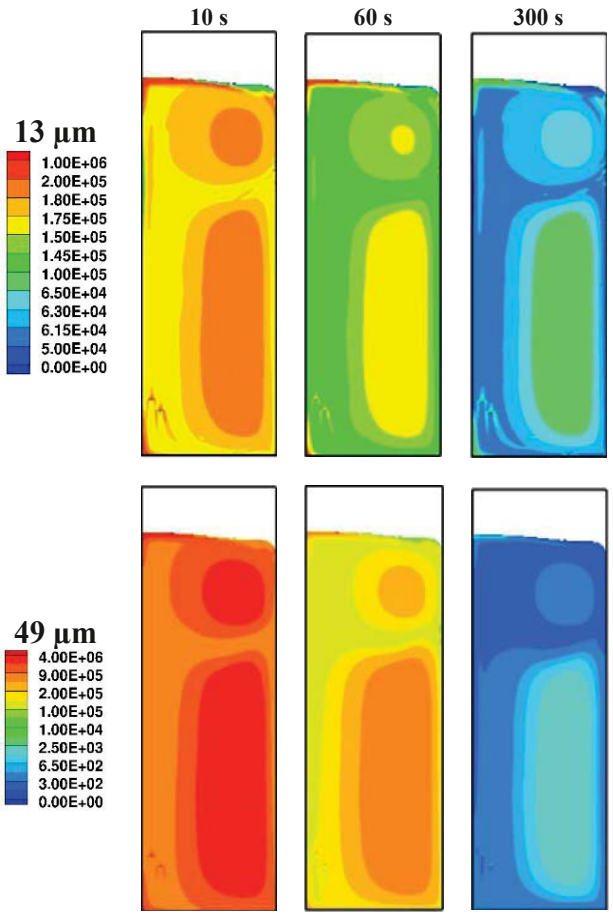

Figure 11: Temporal evolution of density [# of inclusions per m^3 of melt] distribution contours for inclusion size = 13 µm and 49 µm.

Conclusions and perspectives

A numerical tool was developed to calculate the hydrodynamics in an induction furnace and further applied to an industrial scale furnace. This model reproduces the results which are experimentally observed on the industrial furnace (5 tons) when varying the filling rate or power. A population balance model was also developed in order to model transport of inclusions with the following phenomena: settling, electromagnetic migration, aggregation, particle/liquid drift velocity... A model particle size distribution was built, based on several experimental campaigns during the pouring by LiMCA and PoDFA techniques.

Using both Magneto-Hydrodynamic and Population Balance models and these measurements, the behavior of a spinel inclusion population, in an industrial induction furnace operated in holding mode, was modelled. The model shows that spinel type inclusions

(heavier than aluminum) are quickly captured by the furnace walls, mainly thanks to electromagnetic migration. However the time scale seems very short which might be due to the assumption of inclusion capture at the wall (100% capture, no release). Accordingly a perspective to this study is to develop the understanding of inclusion adherence on the crucible wall.

The experimental validation of the inclusion part of the model is quite challenging. Inclusion measurements can hardly be done in an induction furnace in operation as it is sometimes possible in gas fired furnaces [23]. As a result the quality is measured in the metal stream exiting the induction furnace. The behavior of the inclusions present in the furnace during casting has not been considered in this work and might be a topic of further studies.

Acknowledgements

This research has been supported by the French National Research Agency within the framework of the ANR project PRINCIPIA, (2010-RMNP-0007).

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Light Metals 2016

ELECTRODE TECHNOLOGY

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Electrode Materials and Characterization

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

CHARACTERIZATION OF CARBON ANODE MATERIALS BY IMAGE ANALYSIS

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Keywords: Wetting, microscopy, carbon anodes, characterization of carbon materials, calcined petroleum coke, coal tar pitch

Abstract

Carbon anodes are made of calcined petroleum coke, recycled carbon materials (butts and anodes), and coal tar pitch. The pitch and coke properties strongly influence the anode quality. High-quality anodes help reduce the energy consumption and environmental emissions during the electrolysis. Methods have been developed to characterize the structural aspects of green and baked anodes, coke-pitch and biocoke-pitch samples from the sessile-drop wettability tests as well as the solid particles of various pitches. These methods are based on the analysis of the images obtained by the optical and scanning electron microscopy techniques. Such data help determine the possible impact on anode quality. Biocokes could serve as alternate raw materials for the partial replacement of petroleum coke, and biocoke samples were prepared from wood by calcination at high temperatures. The wetting of biocokes by pitch was compared with that of petroleum coke. This article presents the results of the characterization work.

Introduction

Carbon anodes used in aluminum industry consist of different carbon materials such as petroleum coke, recycled carbon materials (butts and anodes), and coal tar pitch. These exhibit different structures and forms in anodes due to different mixing conditions and heat treatments that they are subjected to during anode fabrication. The structure of coke and the interface between coke and pitch formed in the anode during baking are some of the important factors which influence the quality of baked anodes.

Many researchers have worked on the characterization of carbon material surfaces using different techniques [1-8]. Optical microscopy and scanning electron microscopy (SEM) were used to characterize the structure of different metallurgical cokes using various etchants [1,4]. It was seen that the carbon textural constituents observed by microscopy contribute to the coke strength [1]. It was also reported that the textural composition of carbonized pitch studied using SEM can successfully explain the mechanical properties of the corresponding anode [3]. Hays et al. [2] fabricated baked carbon electrodes using a petroleum coke and four different coal tar pitches as binder. They used SEM to differentiate coke, coke/pitch interface, and carbonized pitch in carbon anodes. They used the laboratory baked carbon anodes which were fabricated from coke particles with a narrow range of size distribution. However, they did not study the commercial carbon anodes containing highly fine coke particles.

The object of the present study is to investigate a number of cokes and carbon materials with the particle size distribution used in anode production by means of microscopic characterization methods (optical microscopy and scanning electron microscopy). An attempt was made to compare the differences in the textures of green and baked anodes as well as pure pitch or pitch/wood mixtures before and after carbonization. The pitch/wood mixture was heat treated to produce carbonized pitch/biocoke.

In addition, wetting tests were performed by measuring the dynamic contact angles of pitch on coke beds using the sessile-drop method. The wettability of pitch on biocoke and petroleum coke particles were investigated from SEM micrographs of pitch-biocoke and pitch-petroleum coke sessile-drops; and these were correlated with the wettability test results. The applications of optical microscopy and SEM are presented for different carbon materials with the aim of identifying suitable techniques.

Experimental

Calcination of biocoke

The biocoke samples received from Boisaco Inc. were calcined to 1200°C in a TGA (thermogravimetric analyzer) with induction heating at the University of Quebec at Chicoutimi (UQAC). A heating rate of 40 °C/h and a holding time of 15 min at the maximum temperature were used, which are typical values for green petroleum coke calcination. The TGA with induction heating was chosen because high heating rates required for coke calcination can be achieved in such a system. The calcination was carried out under pure nitrogen atmosphere to prevent the oxidation of carbon materials.

Wetting tests

The dynamic contact angles between pitch and biocoke as well as pitch and petroleum coke were measured using a sessile-drop experimental system. This system consists of a tube furnace (Thermolyne 21100), an Inconel tube with a pitch injection system, a graphite crucible for the coke sample, a digital video camera (B/W) (APPRO, model KC), and a secondary rotary vacuum pump (GE, Precision Vacuum Pump, Model D25). Biocoke and petroleum coke particles were ground, and particles less than 125 µm (-125 µm) were used for the wetting test [9]. The particles were compacted in the sample crucible to have a smooth bed surface. A drop of pitch was formed on the coke substrate by keeping the temperature constant at 170 °C and controlling the pressure difference between the metal injection chamber and the tube furnace. The experiments were conducted under protective nitrogen atmosphere. The FTA 32 software was used to measure the contact angle from the dynamic images captured during the wetting experiments. The dynamic contact-angle data were used to assess the wettability. The coke-pitch sessile-drops were examined with an optical microscope and an SEM.

Production of green and baked anodes

Anodes were produced in the laboratory using an anode recipe similar to a typical industrial recipe. The coke and pitch were mixed at a certain temperature, and the anode paste was vibro-compacted to obtain green anodes. Then, these were baked under the conditions typically used in industry.

Investigation with the Optical Microscope and the SEM

Coal-tar pitch, green anode, baked anode, calcined biocoke-pitch, coke-pitch samples from the sessile-drop wettability tests, and carbonized pitch/biocoke mixtures were characterized using optical microscopy and SEM.

The samples were mounted into the epoxy liquid resin and polished by a Struers polisher to create a scratch free surface. The textual morphology was investigated using a Nikon Eclipse ME600 optical microscope (Nikon Inc., Melville, NY), and the images were analyzed with Optical Image Analyzer (Clemex JS-2000). Then, the epoxy resin of polished block samples was removed to have the polished carbon samples which were subsequently mounted onto the surface of an aluminum block using an electrically conducting paste and sputter-coated with a palladium/gold layer to have electrically conductive surface. The samples were scanned using a Jeol SEM (JSM 6480LV).

Results and discussion

Green and baked anodes

Optical microscopy images illustrating the appearance of green and baked anode surfaces at low magnification are shown in Figures 1a and 1b. Due to the different hardnesses of coke and pitch in green anodes, it is difficult to obtain a sufficiently even surface (Figure 1a). This makes the focusing on all the elements of the surface difficult at the same time while taking images with the optical microscope. It is possible to observe coke and pitch seperately. The coke particles in green anodes as well as the small particles in pitch exhibit optical anisotropy. Even at this low magnification (500x), differences in the surface morphology of green and baked anodes can be detected (Figures 1a and 1b). The surface of a baked anode (Figure 1b) shows that the pores are created in pitch during baking due to volatile release. Anisotropic cokes are present in a baked anode, and both isotropic (I in Figure 1b), and anisotropic (A in Figure 1b) structures occur in pitch after baking to high temperature.

Due to the limit of optical microscopy, the structure of pitch in green anodes can not be clearly observed with this technique. The uneven and porous nature of anode surfaces necessitates their examination using SEM (Figure 1c). The particles of pitch in green anodes can be visually identified in a SEM micrograph (Figure 1c) even after polishing the sample surface. Spherical particles of pitch were observed in the baked anode sample (Figure 1d), but there was no indication of the presence of an anisotropic structure that was seen by the optical microscopy (Figure 1b). Anisotropic or isotropic structures of coke in both green and baked anodes could not be seen clearly by SEM for such polished surfaces. It is possible that the high magnification and small field of view of SEM micrographs made it difficult to see the anisotropic surfaces.

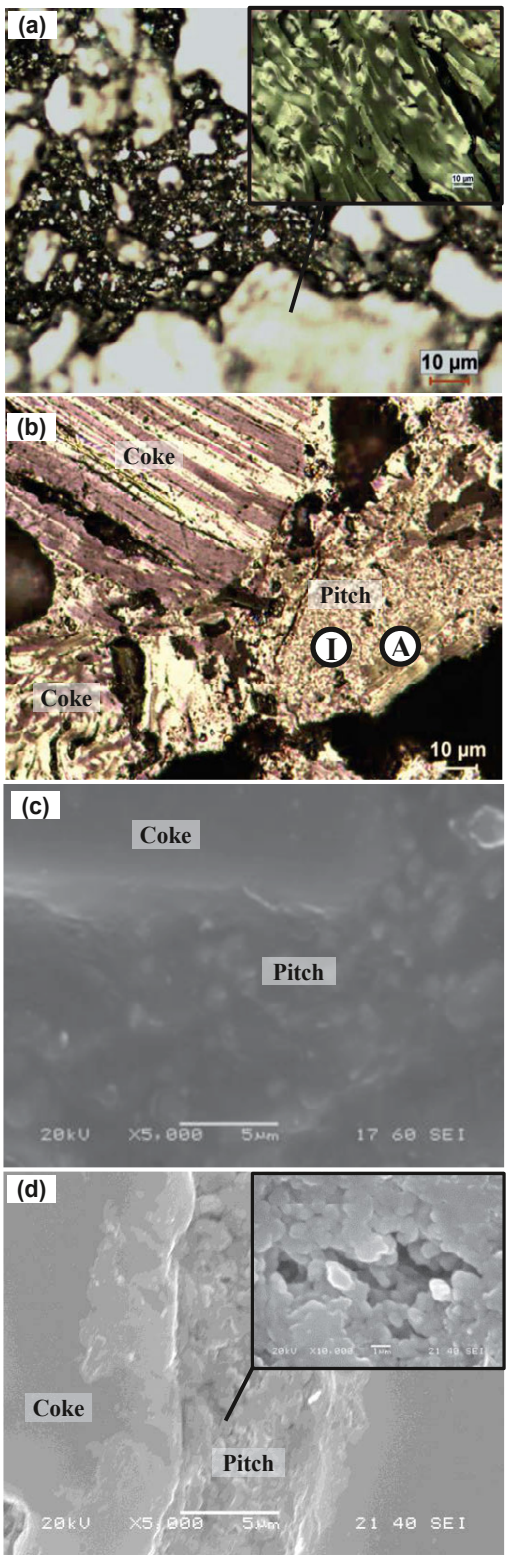

Figure 1 Optical microscopy images of (a) green anode, (b) baked anode (I : isotropic; A: anisotropic); and SEM images of (c) green anode, (d) baked anode

It was reported that atomic oxygen and chromic acid solutions can be used as etchants to help obtain images of metallurgical coke and carbon structures [4]. As carbonized pitch is more reactive to

oxidizing agents compared to calcined petroleum coke, selective oxidation of pitch surface can be done by the use of etchants. The change in the surface texture of carbonized pitch by etching helps differentiate pitch from the coke particles. The etching method provides more structural information on the coke particles and pitch/carbonized pitch in both green and baked anodes. This will be further studied. In addition, micro-crack formation is not observed between coke and pitch in anode samples even after baking, which indicates the good bonding between them for the samples studied. Furthermore, interaction zones at the interface of calcined coke and carbonized pitch show possible reaction. The possibility of the presence of reaction products has been assumed based on the difference in texture and the brightness of the surface at the interface compared to those of the coke surface.

Pitch and pitch/wood mixture before and after carbonization

The pitch samples were prepared in epoxy resin and polished, and then their optical microscopy images were taken in such a way that the background appeared dark. Figure 2a shows the solid particles in pitch. Then, the image was binarized as shown in Figure 2b where the solid particles appear white and the rest of pitch appears black. The solid particles in the binary images are separated with clear boundaries. The particles sizes (maximum length) were determined using the image analysis software of Clemex. Figure 2c shows the comparison of the solid particle size distribution in two different pitches using the image analysis technique. This technique may help relate the pitch properties to different anode properties.

The SEM images in Figures 3a and 3b illustrate the appearance of the solid particles of pitch. The polishing process has removed the softer components of pitch preferentially leaving the harder solid particles (see Figures 3a and 3b). The micrographs show that spherical solid particles of less than 1 µm in diameter are present in pitch. It was reported that the mesophase spheres are 10-20 µm in size [2]. Thus, the SEM analysis results indicate that the pitch in this study does not contain mesophase. An SEM image illustrating the appearance of agglomerated submicron particles is shown in Figure 3b. These particles appear to be carbon black according to their characteristics [2].

The carbonized pitch was obtained by baking the pure coal tar pitch using the same heating rate used to bake the anode. The polished surface of carbonized pitch has been examined by both optical microscope and SEM (Figure 4 a -c).

It can be observed from Figure 4a that the appearance of the carbonized pitch exhibit similar surface textures found in metallurgical cokes [2]. Hays et al. [2] mentioned that the texture in carbon particles can be characterized based on lamellar, intermediate, and granular components. The carbonized pitch in the current work contains only two components, namely, lamellar (position L) and granular (position G). The granular component is present in greater proportion than the lamellar one. The appearance of the two components in the carbonized pitch is clearly visible from the SEM images in Figures 4 b and c, which shows the details of the polished carbonized pitch surface.

The nature of lamellar components on the polished surface can be easily identified when the pure pitch is baked as seen in SEM images (Figure 4b), but not in pitch-coke of anodes. The granular components in carbonized pure pitch appear as small pits with small spheres. Micron-size particles are visible in the granular area as shown in Figure 4c. Corresponding views visible in baked anode surfaces are given in Figure 1d as small spheres. The appearance of pitch in baked anodes is quite different from that of carbonized pure pitch in SEM images. However, the optical microscopy images of pitch in baked anodes and carbonized pitch obtained from pure pitch show a similar structure (see Figures 1b and 4a).

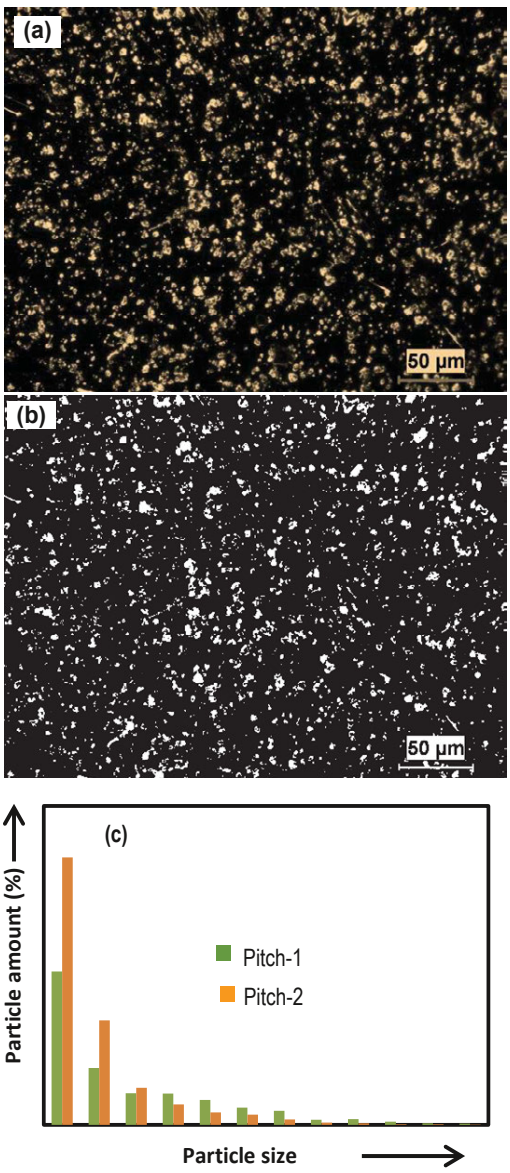

Figure 2 (a) Optical microscopy image, (b) Binary image, (c) Solid particle size distribution in two different pitches

This might be due to the high magnification of SEM which focuses on a small region in an image which cannot contain all the structural components. The presence of a relatively small amount of pitch-coke in baked anodes makes the observation of all its structural components difficult using image analysis techniques, which might be another reason for the difference between optical microscopy and SEM analysis.

Pitch/wood mixtures were carbonized under similar conditions as the anode baking process to produce carbonized pitch/biocoke. Optical microscopy and SEM images illustrating the appearance of carbonized pitch/biocoke are shown in Figure 5. The carbonized pitch/biocoke in Figure 5a consists of lamellar structure close to the pore surface where the volatiles are passing and granular structure close to the matrix of carbonized biocoke. The matrix of biocoke shows cell walls that are usually present in wood. This phenomenon indicates that the presence of cell walls stops the expansion of anisotropic structure during carbonization. The pitch can easily enter the cell lumen of biocoke and mix with small biocoke particles present in the cell lumen, which helps form the granular component of pitch during heating. The appearance of the cell wall of the biocoke in the SEM image of Figure 5b is indicative of the structural unit present, namely, flat component (Position "F" in Figure 5b) [3].

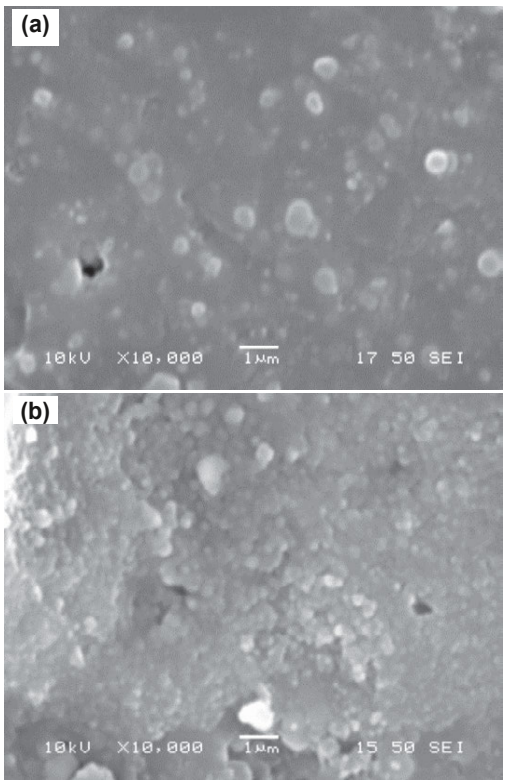

Figure 3 SEM images of solid components of coal tar pitch: (a) solid particles, (b) carbon black particles

Pitch-biocoke and pitch-petroleum coke sessile-drops

The information on the wettability of coke by pitch is useful in understanding the interaction mechanism between coke and pitch and exploiting the possibility of using biocoke as a raw material for anodes. During this study, the dynamic contact angle (wettability) of a biocoke/pitch sample was measured, and the results were compared with those of a calcined petroleum coke/pitch sample. Figure 6 presents the dynamic contact angle of coke/pitch system as a function of time for a biocoke and a petroleum coke. As it can be seen in this figure, the contact angles of biocoke were lower than those of petroleum coke at all times. This shows that the biocoke was wetted better than the petroleum coke by pitch used in this study. Most of the pitch penetrated into the biocoke within 400 s while there is still pitch present on the petroleum coke surface even at 1500 s.

The differences in coke structure and particle shape can cause differences in the wettability behavior. In order to investigate the influence of biocoke structure on pitch/biocoke interactions, the relevant microstructures of pitch/biocoke interfaces were meticulously examined. The interface of pitch and coke was studied by vertically cutting (see Figure 7) the sessile-drops obtained from the contact angle tests and analyzing them with the optical microscope and SEM (Figure 8).

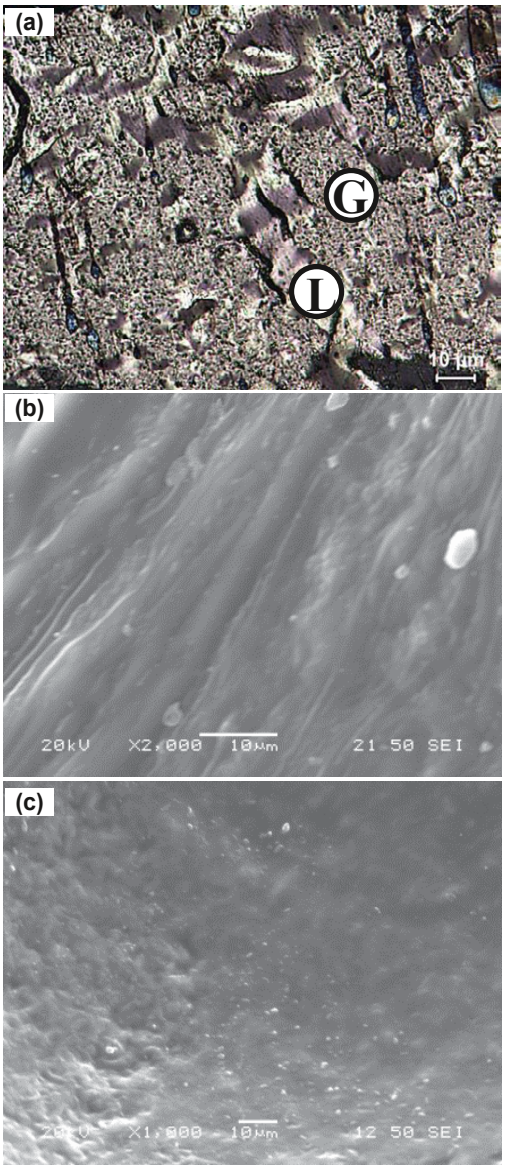

Figure 4 Optical microscope micrograph showing (a) general view (L: lamellar structure, G: granular structure); and SEM images showing (b) lamellar component, (c) granular component in carbonized pure pitch

The biocoke and petroleum coke particles as well as pitch can be easily identified from both the optical microscopy (Figures 8a and 8b) and SEM images (Figures 8c and 8d). The appearance in the interface shows that biocoke and petroleum coke have different

particle shapes (Figures 8a and b), and the structural factors play an important role in wettability during the penetration of pitch.

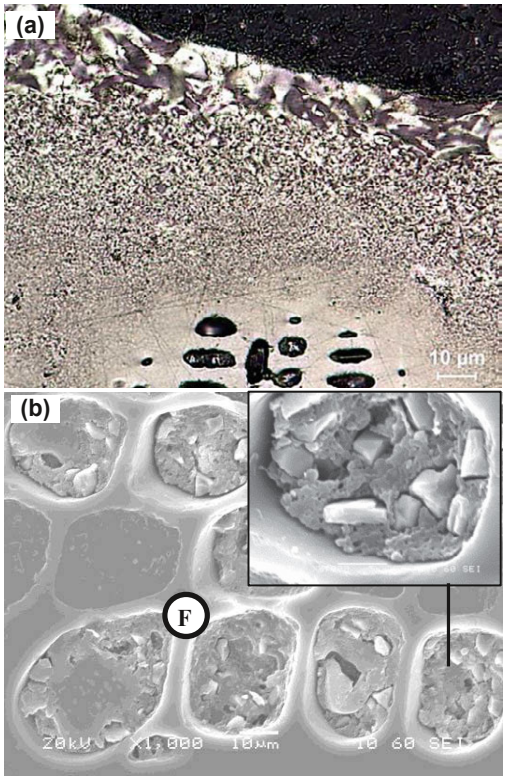

Figure 5 (a) Optical microscopy and (b) SEM images of carbonized pitch/biocoke (F = flat component)

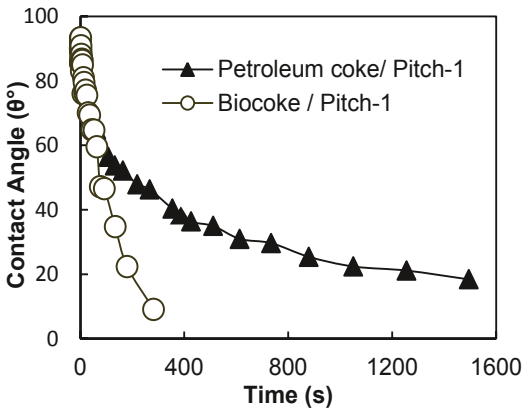

Figure 6 Comparison of the dynamic contact angles of pitch on petroleum coke and biocoke beds

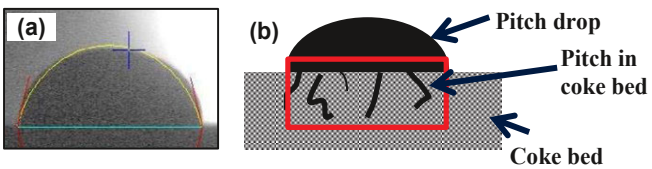

Figure 7 (a) Image of a sessile-drop, (b) the schematic representation of a sample taken from the sessile-drop cross-section to study the coke-pitch interface

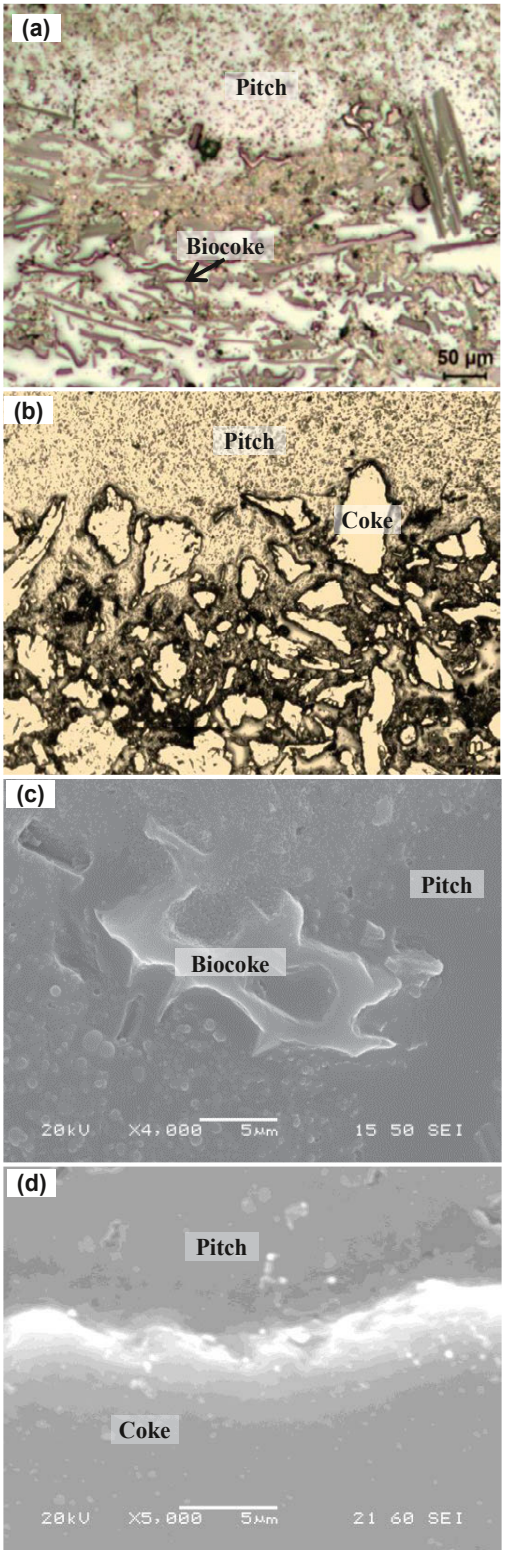

Figure 8 Optical microscopy images of (a) pitch/biocoke and (b) pitch/petroleum coke; SEM images of (c) pitch/biocoke and (d) pitch/petroleum coke

It can be seen from Figure 8 that biocokes are more anisotropic than isotropic, and their porosities are higher than those of petroleum coke. The porous structure of biocoke contributes to higher wettability by pitch where pitch component can penetrate into the pores as shown in Figure 8c. Figure 8d shows a section of the petroleum coke-pitch interface. The difference in brightness close to the interface of pitch and biocoke suggests the possible presence of reaction products. Figure 8d does not show the presence of pores which are apparent in Figure 8c.

Conclusions

The optical microscope and SEM can be used to characterize the structure of different carbon materials such as green and baked anodes, coke-pitch and biocoke-pitch samples from the sessile-drop wettability tests as well as pure pitches and pitch/wood mixtures before and after carbonization. The two techniques have their own advantages for different carbon materials, and they can be used as complementary techniques for a better characterization of a given sample. Furthermore, their combination contributes to the understanding of the internal structure within carbon materials and at the interface between pitch and biocoke or petroleum coke, consequently giving an insight into the mechanism of wettability.

Acknowledgements

The technical and financial support of Aluminerie Alouette Inc. as well as the financial support of the National Science and Engineering Research Council of Canada (NSERC), Développement économique Sept-Îles, the University of Quebec at Chicoutimi (UQAC), and the Foundation of the University of Quebec at Chicoutimi (FUQAC) are greatly appreciated. The authors would like to thank the Fonds de recherche du Québec nature et technologies by the intermediary of the Aluminum research centre – REGAL for their contribution for student participation in this conference.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ELECTROCHEMICAL REACTIVITY AND WETTING PROPERTIES OF ANODES MADE FROM ANISOTROPIC AND ISOTROPIC COKES

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Keywords: Carbon Anodes, Anisotropic Coke, Isotropic Coke, Electrochemical Reactivity, Wetting Properties

Abstract

High quality anodes require stable raw materials. As the quality of the coke available for production of anodes is changing, future production will rely on cokes that are more isotropic with higher impurity levels than traditional raw materials. The purpose of this work was to improve the understanding of the electrochemical reactivity of different cokes. Anode:electrolyte wetting is a key factor towards determining electrochemical reactivity, which in turn is affected by anode properties, type of coke and polarization. Pilot anodes were fabricated from single source cokes; one anisotropic coke low in impurities, and one isotropic coke with significantly higher impurity levels. Electrochemical characterisation included chronopotentiometry, cyclic voltammetry and impedance spectroscopy. Wetting properties were studied on both unpolarised and polarised samples by a dedicated wetting apparatus, and indirectly by computed tomography (CT) images of frozen electrolyte films.

Introduction

The quality of anode grade coke has changed in the past decade due to shortage in supply of traditional anisotropic sponge coke, and cokes that were previously not regarded as anode grade coke are now accepted by the aluminium industry. These cokes are blended with more traditional anisotropic coke [1, 2].

An improved knowledge on how isotropic cokes influence the electrochemical behaviour of anodes is needed and has been investigated in [3, 4, 5]. The overpotential is a source of increased energy consumption. Equation 1 summarises the measured anodic potential, where E^{rev} is the reversible potential (1.187 V w.r.t. Al/Al^{3+} [6]), η_c is the concentration overpotential, and in a saturated melt this overpotential may be considered negligible. η'_r is the reaction overpotential which will depend on the anode material [4]. η_h is overpotential caused by hyperpolarisation due to bubbles on the surface. The last term in Equation 1 describes the increased ohmic resistance at the anode caused by bubble build-up: δR_s denotes the ohmic resistance due to bubbles blocking the anode surface and R'_s is the ohmic resistance with no bubbles screening the anode. Due to the vertical anode setup used, η_h, $I \cdot \delta R_s$ and $I \cdot R'_s$ are negligible. Thorne *et al.* found a slight reduction in reaction overpotential when using isotropic coke compared to anisotropic sponge coke in anodes [4].

$$E_{measured} = E^{rev} + \eta_c + \eta'_r + \eta_h + I(R'_s + \delta R_s) \quad (1)$$

Electrical impedance spectroscopy (EIS) can be used in order to investigate the capacitance of anodes, which is generally assumed to represent the wetting of the anode surface by the electrolyte [7]. An increase in capacitance of about 45 % was seen on a polarised anode sample compared with a fresh anode sample, and this capacitance increase was attributed to higher surface area exposed to electrolyte. Thonstad [7] measured a double layer capacitance range for pyrolitic graphite between 45-75 $\mu F/cm^2$ and a range for baked carbon between 150-600 $\mu F/cm^2$ using a horizontal anode setup. In this work, capacitance is used in order to determine the charge per anode area for the different anode materials used. Thorne *et al.* [4] found a slightly higher capacitance for an isotropic coke anode compared with an anisotropic coke anode, denoted as Anode 4 and Anode 1, respectively.

The wetting properties have been studied for various

electrolyte compositions and also for polarised *vs.* unpolarised samples [6, 8]. Solheim *et al.* [8] described how wetting properties change after polarisation of an anode surface compared with an unpolarised surface, using a specially constructed wetting apparatus for the immersion and emersion of a carbon anode cup. It was observed that the receding wetting angle during emersion changed when a crucible containing a cryolitic electrolyte was pulled away from an anode sample, before and after polarisation. CT imaging can also be used to investigate the frozen electrolyte image on an anode sample and is hence anticipated to reflect the wetting properties on the anode:electrolyte interface [9]. The sessile drop method was used in order to investigate the anode:electrolyte interface on an unpolarised anode surface by Thorne *et al.* [5]. It was found that for both graphite and the anisotropic coke materials, the anode:electrolyte wetting angles were of non-wetting character, i.e. θ>90°. The isotropic anode materials showed a wetted character between anode:electrolyte, i.e. θ<90°. The sessile drop method is an equilibrium measurement, where a very small electrolyte sample is placed on the anode surface. There are some sources of error with this method, including the small electrolyte sample on the very small anode surface area, the fact that the electrolyte:anode interface is stagnant, sensitivity to surface inhomogeneities and that no polarisation is applied.

In this work, the electrochemical reactivity of anodes towards a cryolitic electrolyte was investigated by reaction overpotential measurements and cyclic voltammetry. The wetting properties of the anode towards the electrolyte were investigated using three different methods. By capacitance measurements from EIS, by use of a dedicated wetting apparatus where the effect of polarisation was investigated, as explained in [8, 10] and finally, by CT imaging investigating the effect of the bathfilm thickness when pulling the anode out of the electrolyte with and without current on. The purpose of the work was to investigate correlations between electrochemically active surface area and the geometric surface area, as well as the electrochemical reactivity.

Experimental

Pilot anodes (Ø = 130 mm, h = 180 mm) were specially designed and made at Hydro Aluminium in Årdal for this experimental work. The anodes were produced from single source industrial grade cokes, one of anisotropic character: Anode A, and the other of purely isotropic character: Anode B. The same industrial grade pitch was used for both pilot anodes. Coke aggregate were produced to a common sieve curve from ball milled coke (fines <63 μm), 0-1 mm and 1-2 mm fractions. Fractions of <2 mm coke particles were chosen in order to maintain a representative exposed surface area during small scale electrochemical tests. For clarity, Anodes A and B in this work are identical to Anode A and B in Sommerseth *et al.* [9], also Anode A = Anode 1 and Anode B = Anode 4 in [3], [4] and [5], however, new parallels samples have been made for all electrochemical tests. Ultrapure graphite (CMG grade) from Svensk Specialgrafit AB was used as reference carbon material.

During electrochemical tests an experimental setup and anode assembly similar to that described in Figure 1 by Thorne *et al.* [3] was used. The vertical anode assembly gave a well-defined, geometrical surface area of 1.52 cm^2 and negligible bubble noise. The surface area was not corrected for consumption of the anodes, however, the electrolysis time was kept to a minimum in order to avoid changes of the geometrical anode surface area. The electrolyte was a cryolite melt with a $NaF:AlF_3$ molar ratio of 2.3, corresponding to 9.8 wt% excess of aluminium fluoride, AlF_3. The cryolite was from Sigma-Aldrich (>97 %) and the AlF_3 was sublimed in-house for purification. The electrolyte was saturated in γ-Al_2O_3 (9.4 wt%) from Merck.

Electrochemical reactivity of anode A and B was determined by slow sweep-rate cyclic voltammetry (CV) at 0.1 V/s from open circuit potential (OCP) to 2.5 V and chronopotentiometry (CP) at 1.0 A/cm^2. Capacitance was measured by EIS at 1.5 V (non-IR corrected). Measurements were performed using a Zahner IM6 with built in EIS module and 20 A booster (PP201, from Zahner-Elektrik). Ohmic resistance was determined by taking the high frequency Z_{Re} intercept from the Nyquist plot obtained by EIS at OCP and this resistance was used to IR correct all electrochemical tests. An aluminium reference electrode was used and a new Al reference electrode was produced per duplicate run. Capacitance was extracted from Nyquist plots (EIS at 1.5 V) by three different methods: 1) modelling by the equivalent circuit LR(CR) and extracting the double layer capacitance (C_{dl}) directly, 2) modelling by the equivalent circuit LR(Q(R(LR))) and calculating the effective capacitance from Q according to Equation 13.10 in [11] and 3) extracting C_{dl} directly by extracting the

inductance directly at the highest frequency (100 000 Hz) and using the high frequency range of the Nyquist plot to determine the capacitance as described in [7].

Rod shaped anodes (Ø = 9.7 mm, h = 5 mm) were polarised for 1500 s at 1.0 A/cm^2 and electrolyte removed by soaking in saturated $AlCl_3$ solution until all remaining electrolyte on the surface was removed. The horisontal surface area was then investigated in a confocal microscope (Infinitefocus from Alicona 3D) in order to determine the surface roughness of the anode surface after polarisation. Based on data from the image analysis, the ratio of true area (including all voids and pores) to projected (geometrical) area of the horisontal surface was calculated. This true area over projected area was then used to normalise the capacitance for true surface area including all voids and pores.

The dynamic wetting angle for the anode:electrolyte interface was determined before and after polarisation at 0.7 A/cm^2 for 10 s, using the method described in [8] and [10]. This experimental setup was used with a slight modification: the anode cup samples in this work had an outer diameter of 30 mm and an inner diameter of 22 mm. The advancing wetting angles are reported here (during immersion of the anode sample into the electrolyte) due to more noisy recordings during the emersion of the sample.

The wetting properties of the anode:electrolyte interface for Anode A and Anode B were also investigated using CT imaging. Two test pieces of each Anodes A and B, respectively, were electrolysed for 2000 s, then one was pulled out of the furnace with current still on (hot-pulled) and the other was left in the electrolyte for 5 mins after current had been turned off and then pulled out (non-hot-pulled). The thickness of the bathfilm left on the sample surface was determined using the ImageJ software (from U. S. National Institutes of Health) where carbon and electrolyte were distinguished from each other by setting a threshold to the limit value of the density of the two different matters. Contour images of the bathfilm thickness were also produced. Methodology of CT imaging is thoroughly described in [9] and will not be described further here.

Results and Discussion

Figure 1 shows the IR corrected potentials corresponding to 0.6 A/cm^2, 0.8 A/cm^2 and 1.0 A/cm^2 extracted from CV curves and CP results at 1.0 A/cm^2. The

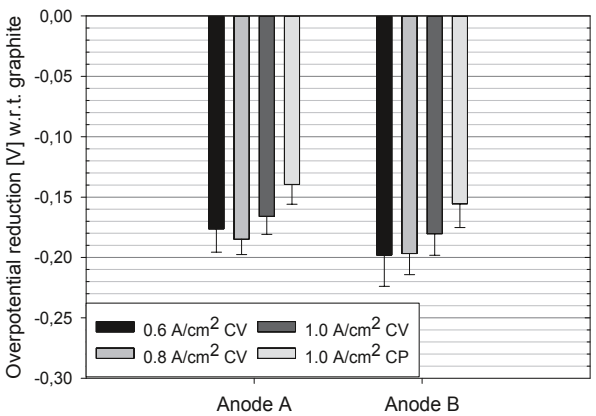


Figure 1: Overpotential reduction for Anode A and and Anode B w.r.t. graphite reference at three different current densities (0.6, 0.8 and 1.0 A/cm^2) obtained by CV curves and by CP at 1.0 A/cm^2.

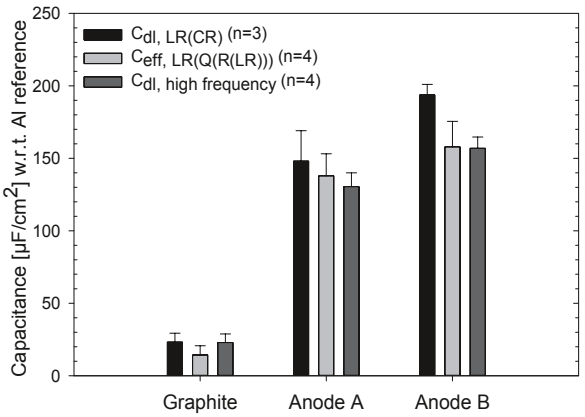


Figure 2: Capacitance for Graphite, Anodes A and Anode B obtained by running electrical impedance spectroscopy at 1.5 V (non-IR corrected).

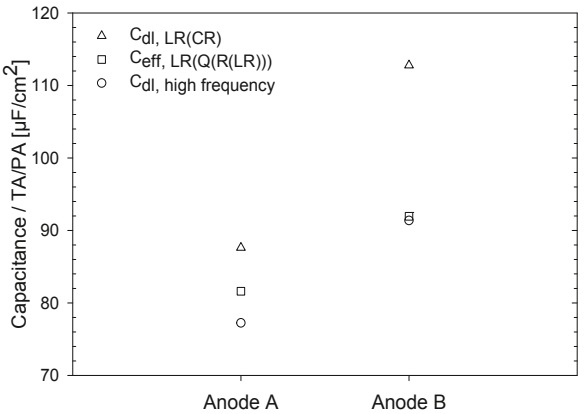


Figure 3: Surface area normalised double layer capacitance for Anode A and Anode B.

potential response was recorded with respect to Al reference and normalised with respect to graphite in order to compensate for some shifts in potential between duplicate runs with new reference electrodes. In general it can be seen that the purely isotropic coke based anode (Anode B), has a slightly lower reaction overpotential (η'_r) of about 15-20 mV, compared with Anode A, at all current densities and independent of measuring technique. This is in accordance with results found and reported elsewhere [4].

Figure 2 shows the capacitance for Graphite, Anode A and Anode B, extracted using three different methods, as described previously. For all three methods it is clear that Graphite has a very low capacitance at around 20 μF/cm^2 due to its non-porous and almost polished-like surface. The industrial anodes show higher capacitance where Anode A have values between 130-150 μF/cm^2 and Anode B have values between 155-195 μF/cm^2, depending on how the capacitance was extracted from the raw data. The Graphite values are somewhat lower than what Thonstad described [7], but this is probably due to the graphite materials used in the present work being non-identical to Thonstad's, as well as different instruments recording the spectras. Also, the industrial coke anodes are within the lower range to that Thonstad found for his baked carbon even though these materials are certainly not identical.

Confocal microscopy was used to determine the real surface area of the anode surfaces including all voids and pores, and this value was reported as true area over projected geometrical area (TA/PA). It was found that Anode B had a slightly higher TA/PA than Anode A. These area values were then used to determine if the higher capacitance for Anode B compared with Anode A, was merely due to increased surface area of Anode B. The capacitance was normalised in terms of TA/PA as seen in Figure 3. When normalising the capacitance values for surface area, capacitance values for Anode A and Anode B are still not equal. This indicates that there is a real capacitance difference between the two materials. It was also shown in [9] that pores had to be very large and of a concave form in order to be filled with electrolyte, with the conclusion that a surface area including all pores and voids is not necessarily equal to the electrochemical active surface area. Hence, when normalising capacitance values using TA/PA, the area effect may be over-estimated.

Figure 4 shows measured weight corrected for theoretical weight (m_m-m_t) when taking buoyancy into consideration and calculated wetting angle for Graphite, Anode A and Anode B using the immersion-emersion technique as described in [8], [10]. Before polarisation, the wetting difference between Anode A and Anode B is not so pronounced while Graphite shows a higher wetting angle. After polarisation, the wetting changes noticeably. For all anodes tested the wetting towards the electrolyte increases, but the shift is most prominent for Anode B, changing from a wetting angle of about 140° to nearly 90°. This change is highly relevant for the electrochemical reactivity of the anode, and is contrary with the finding in [8] where no difference between wetting angle before and after polarisation during immersion was found. Better wetting between anode and electrolyte probably means better anode performance in the potrooms.

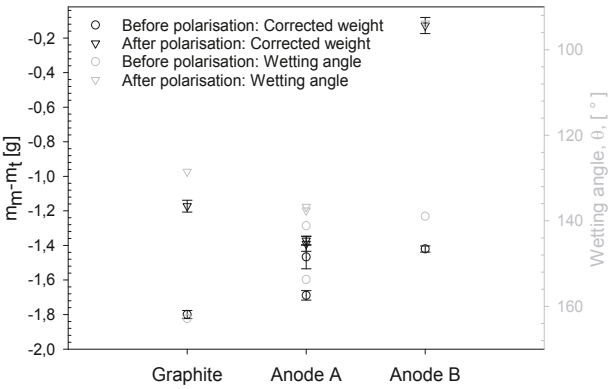


Figure 4: Corrected weight (black) and wetting angle (grey) before (circles) and after (triangle) polarisation.

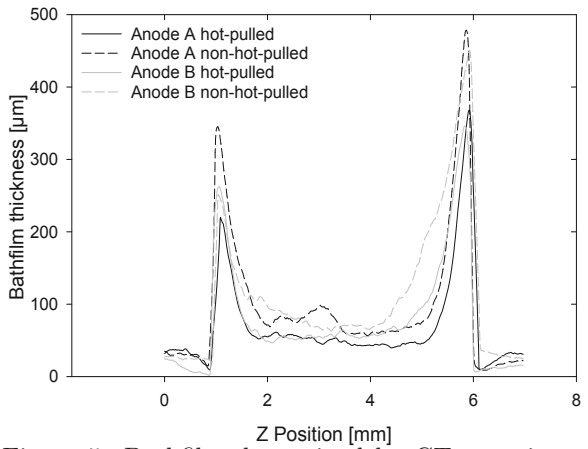
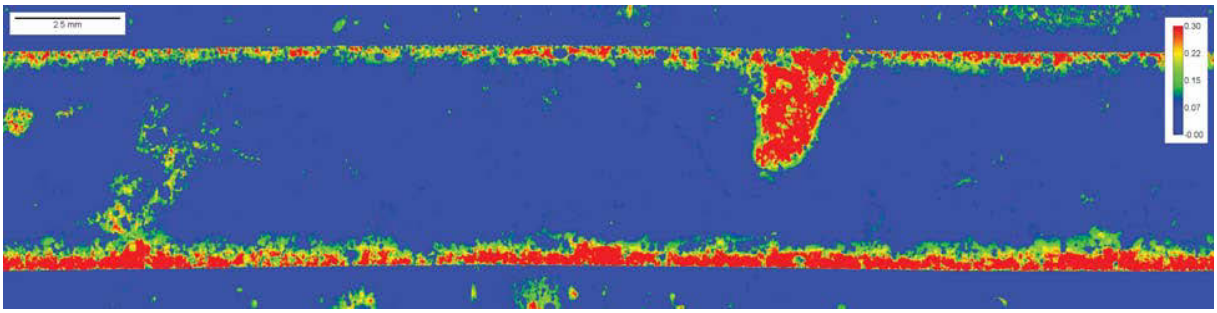
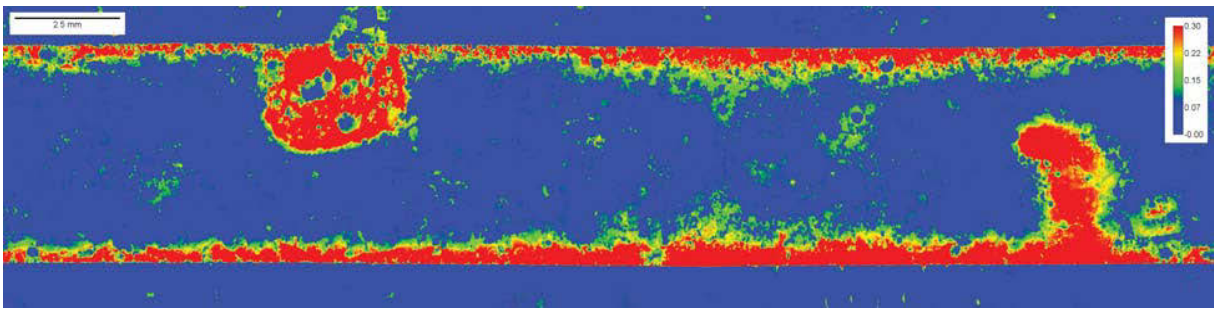


Figure 5: Bathfilm determined by CT scanning of hot-pulled and non-hot-pulled samples.



(a)



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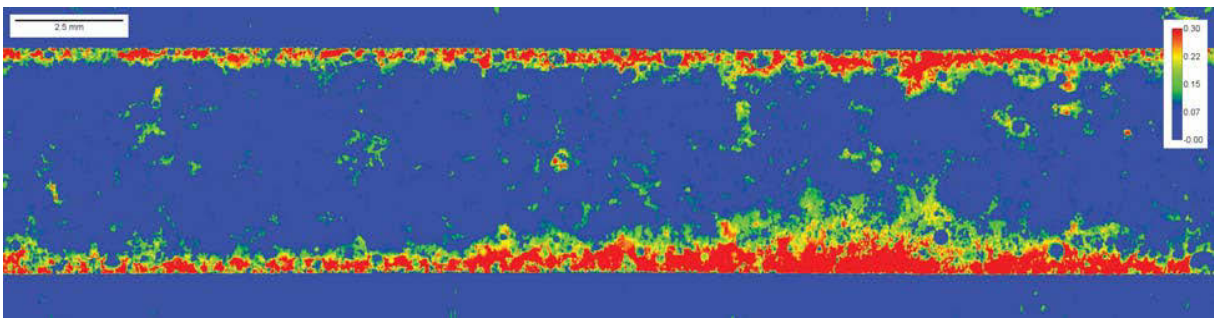

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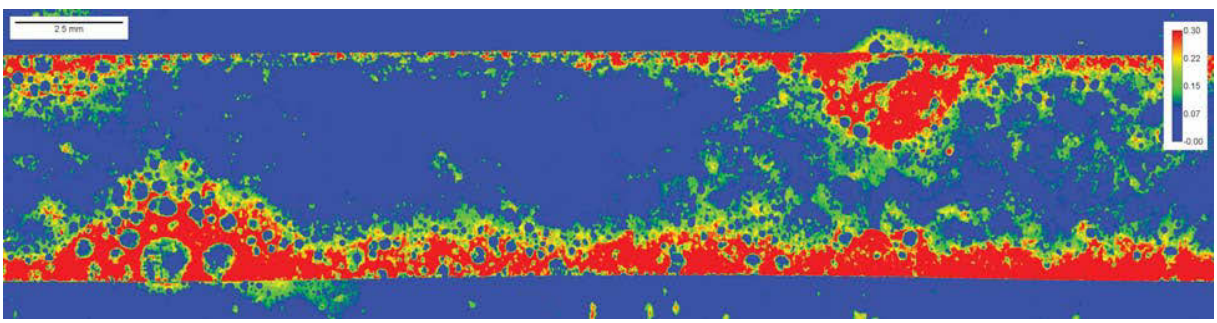

(d)

Figure 6: Contour plots of bathfilm obtained by CT. (a) Anode A hot-pulled, (b) Anode A non-hot-pulled, (c) Anode B hot-pulled and (d) Anode B non-hot-pulled.

Wetting behaviour of anode:electrolyte was also investigated by polarising the anodes at 1.0 A/cm^2, then either hot-pulling or not-hot-pulling. Then the anodes were investigated using CT to obtain the bathfilm thickness on the anode surfaces. Figure 5 shows the bathfilm thickness for Anode A and Anode B, both hot-pulled and non-hot-pulled. Anode B has a slightly thicker bathfilm both during hot-pulling and during non-hot-pulling compared with Anode A, supporting better wetting between Anode B and electrolyte after polarisation compared with Anode A. An edge effect is seen where the bathfilm is thicker towards both ends of the vertical anode sample. The thickness is largest at the bottom (to the right in Z position), as expected due to gravitational forces. Figure 6 shows corresponding contour images of the bathfilm thickness. Blue areas have thinner bathfilm, while red areas have thicker bathfilm. For Anode A, the large red areas are due to electrolyte around "bubble" coke, but on the bubble coke itself, the bathfilm is thin as was also found in [9]. During hot-pulling CO_2 gas bubbles will keep some of the electrolyte off the anode surface, and this is a likely explanation to why the hot-pulled anodes have a thinner bathfilm than the non-hot-pulled anodes. This work also supports the beneficial effects of hot-pulling anodes while changing anodes in the potroom as hotpulling will require less cleaning of butts.

Conclusion

Isotropic coke anodes showed both higher electrochemical reactivity in terms of lower overpotential and better wetting properties towards the electrolyte compared with an anode produced from anisotropic coke. The wetted surface area is not directly related to the geometric surface area, and thus wetting properties are material dependent.

Acknowledgement

This work was funded by The Norwegian Research Council and Hydro Aluminium AS through the research project Hal Ultra Performance. A great thank you is sent to Aksel Alstad, Cristian Torres, Ole Tore Buset and Julian Tolchard.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

STUDY OF THE WETTING OF COKE BY DIFFERENT PITCHES

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Keywords: Wetting, carbon anodes, calcined petroleum coke, coal tar pitch

Abstract

Prebaked anodes are made of dry aggregates (coke, recycled butts and rejected green/baked anodes) and the binder pitch. During the mixing process, the wettability of coke by pitch influences the anode paste behavior. Coke particles (-125 +100 μm) were prepared from the same source using two different procedures, and the wettability of cokes by five pitches with different properties was studied using a sessile-drop system at 170ºC. The contact angle results show that the wettability of coke by all pitches follows a similar trend for both cases, but the actual contact angles are different. The cokes and the pitch-coke interfaces were also investigated by the optical and scanning-electron microscopy (SEM) techniques. The image analysis results indicated that the coke particle size distribution depends on the coke preparation, which seems to influence its wettability.

Introduction

Prebaked carbon anodes consist of approximately 65% petroleum coke, 20% recycled anodes and butts (about 85% dry aggregates), and 15 % coal tar pitch (binder). They are used in the electrolytic alumina reduction process and thus are consumed during the electrolysis. Good binding between dry aggregates and pitch results in dense anodes and thereby greatly affects the final anode properties [1]. Wettability of coke by pitch is one of the important parameters which determines the quality of binding between coke and pitch [2].

Wetting can be physical due to intermolecular interactions known as adhesive and cohesive forces between solid (coke) and liquid (pitch) or can be chemical due to reactions at the solid-liquid interface [3, 4]. The interaction between coke and pitch depends on the properties of both pitch (i.e., chemical composition, surface tension, contact angle, viscosity, QI, mesophase content, etc.) and coke (i.e., particle size, structure, texture, chemical functional groups on the surface, porosity, etc.) [3-7]. For this reason, the modification of pitch and/or coke may alter the coke wettability by pitch.

The degree of wettability of a solid by a liquid can be described in terms of the angle formed between them when they are brought into a contact. The contact angle is a measure of the ability of a liquid to spread on the solid surface and penetrate through it if the solid is porous [2, 5, 6, 8-10]. It is reported in the literature that the sessile-drop technique, which monitors the change in contact angle with time under an inert gas atmosphere, can be used successfully for wettability studies [2, 5, 6, 8-10].

Different authors studied the structures of coke and/or pitch using SEM and optical microscopy techniques. They found that the variations in composition as well as physical properties of coke and pitch are accompanied by differences in their microstructures [5, 6, 8-12]. It was reported that coke particles were ground and sieved to have particles of -125 +100 μm size for the wetting tests [8]. However, there does not seem to be any published work on the effect of coke particle size distribution, caused by different coke preparation techniques, on the wettability of coke by pitch.

The objective of this study is to investigate the wetting of coke by different pitches. The work also involves the SEM characterization of the shape and

size of coke particles. Furthermore, the pitch-coke interfaces were investigated using optical microscopy. The wettability tests were carried out using cokes which have particles in the same size range, but prepared using two different procedures. These procedures affected the particle size distribution. The wettability results for two cokes (from the same source, but prepared differently) with five pitches were compared. In addition, the results were correlated with the coke structures as well as the characteristics of pitch-coke interfaces obtained from the sessile-drops after the tests.

Methodology

Materials used

Five coal-tar pitches with different properties and one petroleum coke were used in this study. Table 1 summarizes the properties of the pitches as well as the coke. The calcined coke particles were sieved with 100 μm and 125 μm sieves. Two different preparation procedures were used as follows: 1) the coke particles were crushed and then sieved to get particles in the desired size range (marked as the crushed sample), 2) fine particles from the coke were sieved also with 100 μm and 125 μm sieves (marked as the original sample). Both series of samples were sieved for 2 minutes with the 125 μm sieve to remove the larger particles and for 4 minutes with the 100 μm sieve to remove the smaller particles.

Sessile-drop system

The wettability of coke by pitch was determined using a sessile-drop system at 170℃, which is a typical mixing temperature used in industry. The sessile-drop system consists of a tube furnace, a pitch injection system, a graphite crucible, a digital video camera, and a vacuum pump (see Figure 1). The coke sample is placed in the graphite crucible and compacted in order to have a smooth coke bed surface. The injection chamber holds the solid pitch sample. This chamber has a small hole which is placed just above the coke substrate during the experiment. The air in the system is first purged with nitrogen, and then the experiments were conducted under nitrogen atmosphere. In order to decrease the oxygen and humidity content of nitrogen, the gas is passed through a number of traps before it enters the system. There are two entry lines for nitrogen. The main line is directly connected to the furnace tube for maintaining the inert atmosphere inside the tube. The other line that connects the injection chamber to the inert gas supply carries the nitrogen gas necessary for slightly pressurizing this chamber in order to force the molten pitch droplet out onto the surface of the coke substrate. A video of the drop is captured, and the images are saved in a computer. To measure the contact angle, the FTA 32 software is used.

Table 1: Properties of the coal tar pitches and the calcined petroleum coke

Material	Atomic percentages (wt%)				S.P.	Q.I.	T.I.	Beta resin	C.V.
	C	O	N	S					
Pitch-1	96.62	1.89	1.21	0.29	118.4	3.4	25.9	22.5	58.9
Pitch-2	96.93	1.74	1.07	0.25	119.6	6.9	29.1	22.2	59.1
Pitch-3	97.98	1.05	0.70	0.26	118.0	10.8	33.0	22.2	61.2
Pitch-4	98.01	1.25	0.53	0.21	121.5	7.5	29.6	22.1	59.9
Pitch-5	97.61	1.18	0.94	0.27	119.4	5.1	28.0	22.9	59.6
Coke	95.89	2.53	0.60	0.98	-	-	-	-	-

S.P. :Softening point (°C)
Q.I. :Quinolone insoluble (wt %)
T.I. :Toluene insoluble (wt %)
Beta resin :T.I. – Q.I (wt %)
C.V. :Coking value (wt %)

Each experiment was repeated twice, and the contact angle was taken as the average of these two results. After the experiments, the pitch-coke drops were cut vertically and studied with an optical microscope.

Scanning electron microscope

The cokes prepared with two different procedures were investigated using SEM. The original coke particles and the crushed samples were uniformly dispersed in a conductive tape on a specimen plate

and vacuum-dried for one day at room temperature prior to SEM analysis. The SEM analysis was done using JEOL-JSM-6480LV with secondary electron scattering, a voltage of 20 kV and a WD (Work Distance) of 19 mm.

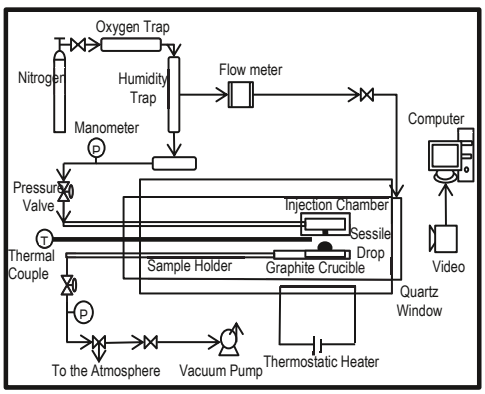

Figure 1 : Schematic diagram of sessile drop experimental set-up at UQAC

Optical microscope

The optical microscope allows the examination of a larger surface area compared to that of the SEM technique. The solidified pitch-coke sessile-drops were sectioned vertically and polished to obtain smooth and flat surfaces containing the interface between coke particles and pitch. These surfaces were investigated using the Nikon Eclipse ME600P optical microscope and analyzed with the image analysis software Clemex Vision 4.0. This allows the visualization of pitch penetration through the coke particles as well as the interface of pitch and coke in the drop samples

Results and discussion

Wettability of coke by five different pitches: Effect of particle preparation

Figure 2 and 3 show the evolution of contact angles with time for five different pitches on two cokes (original and crushed), respectively. The contact angles for both pitch/coke systems decreased with the increasing time. The results also showed that the contact angles were different for the two cokes although their wettability by five pitches followed a similar trend; that is, Pitch-1 was the best-wetting and Pitch-5 was the least-wetting. The other pitches also followed the same order. The complete penetration time into the bed for the pitches (reaching zero contact angle) was significantly shorter for the original coke than the crushed coke. Some of the pitches did not penetrate completely into the crushed coke bed even after 1200 s as shown in Figure 3. However, the original coke absorbed all pitches within 300 s as shown in Figure 2.

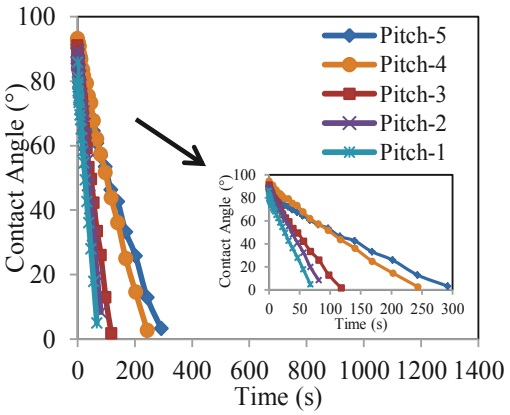

Figure 2 : Dynamic contact angles of five different pitches on original coke

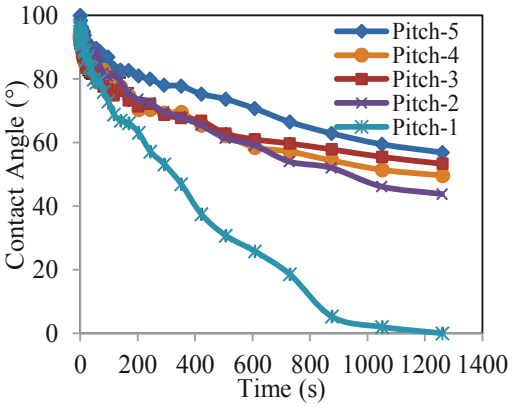

Figure 3 : Dynamic contact angles of five different pitches on crushed coke

The contact angles measured at the initial time and at 60 s are shown in Figure 4 and 5, respectively, for all pitches as well as for the original and the crushed cokes. The initial time was set at 5 s. The initial contact angles of all pitches on the original coke are slightly lower compared to those of the crushed coke (Figure 4). The initial contact angles for the same coke are similar with different pitches. However, the contact angles measured at 60s for the original coke and the crushed coke are significantly different. The contact angles are much greater with the crushed coke showing that it is wetted less by all the pitches compared to the original coke.

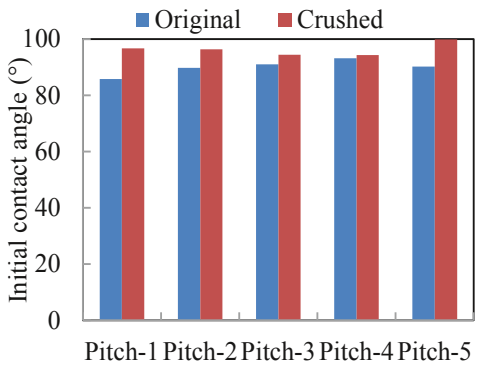

Figure 4 : Initial contact angles of five different pitches on coke

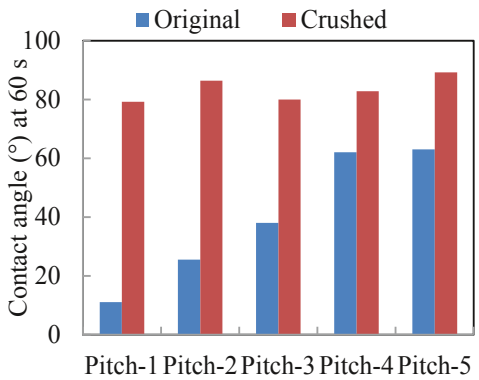

Figure 5 : Contact angle at 60s of five different pitches on coke

In order to understand the difference in wettability of two coke beds, optical microscopy and image analysis of the cokes were carried out. The interfaces of pitch and coke in sessile-drops were also investigated using optical microscopy to understand the wetting mechanism. SEM is used to study the coke structure.

SEM analysis

SEM is a useful tool to visualize the shape and the size of coke particles. It was found that there is no notable difference in particle shape and surface texture between original and crushed coke particles within the same size range (see Figure 6 and 7). The SEM images of original coke in Figure 6 show that the particle sizes are in the range of -125 +100 μm. However, the crushed coke particles seem to have finer particles than those in the original coke particles even though they were both prepared using the same sieving procedure. Thus, crushing caused the generation of finer particles. The finer particles fill the space between bigger particles. The mixture of coarse and fine particles of the crushed coke results in a more compact coke bed than that of the original coke, and this reduces the pitch penetration into the coke bed.

Optical microscopy analysis

Figures 8 and 9 present the optical microscopy images of the pitch 3-coke interfaces in the sessile-drops for both cokes. Coke, pitch, and solid particles in the pitch can be easily seen on these images. Figure 8 shows that pitch penetrated into the original coke bed homogenously. Moreover, the solid particles of pitch are distributed homogenously in pitch around the coke particles. The coke particles are totally embedded by the pitch. The crushed coke bed however displays completely different characteristics. Different components of pitch seem to form layers during wetting as well as during the penetration of pitch (see Figure 9). Some of the solid particles of pitch were blocked by the finer crushed coke particles which prevent them entering into the coke bed.

Consequently, a layer of solid particles was formed on the surface of the crushed coke bed. This layer acts as a filter and slows down the pitch penetration into the coke bed. This explains the lower wettability of crushed coke particles by the same pitches compared to the wettability of original coke particles. It can also explain the differences in contact angles observed at initial times and 60s. It takes some time for the filter-like layer to form. Until the formation of this layer, the angles are closer for both cokes. Later, the difference becomes more pronounced.

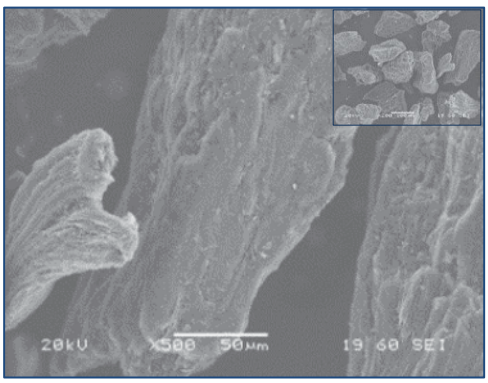

Figure 6 : SEM image of original coke

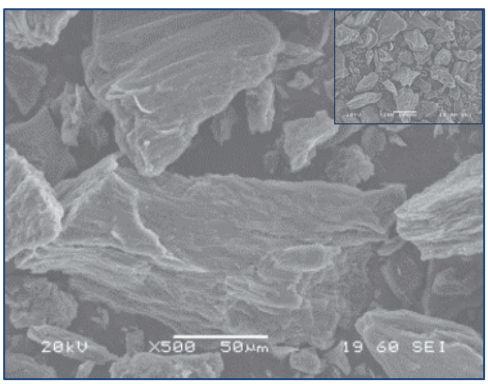

Figure 7 : SEM image of crushed coke

Conclusions

The dynamic contact angle results showed that the wettability of two coke samples by the same pitches is significantly different. These cokes were from the same source, but they were prepared differently. Therefore, the only difference between them was the amount of finer particles.

The SEM results indicated that the two cokes contained different amounts of finer particles. There are a lot of finer particles formed during crushing, and these filled the spaces between larger particles resulting in a better compaction of the coke bed. Thus, the pitch penetration was delayed. The presence of finer particles in coke seems to have a strong influence on the penetration of pitch into the coke.

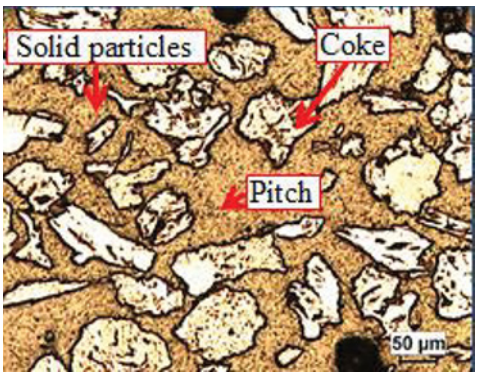

Figure 8 : Optical microscopy image of Pitch-3 with original coke

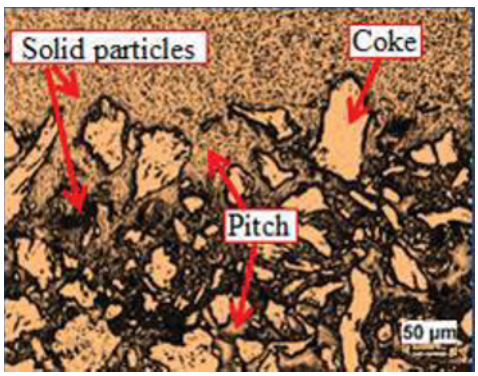

Figure 9 : Optical microscopy image of Pitch-3 with crushed coke

The optical microscopy analysis showed that the solid particles in pitch are distributed differently at the pitch-coke interface depending on the coke particle size distribution. When the particle size is more or less uniform, pitch solid particles are distributed uniformly. If there is a wide coke particle size distribution, pitch solids can form a filter-like layer at the interface which slows down the pitch penetration into the coke bed. This, in turn, decreases the wettability. The complete mechanism of pitch/coke interactions is complex, and further work involving the chemical analysis of coke and pitch is needed to better understand this mechanism.

Acknowledgements

The technical and financial support of Aluminerie Alouette Inc. (AAI) as well as the financial support of the Natural Sciences and Engineering Research Council of Canada (NSERC), Développement économique Sept-Îles, the University of Québec at Chicoutimi (UQAC), and the Foundation of the University of Québec at Chicoutimi (FUQAC) are greatly appreciated. The authors would like to thank the Fonds de recherche du Québec nature et technologies by the intermediary of the Aluminum research centre – REGAL for their contribution for student participation in this conference.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

QUANTIFICATION OF SODIUM PRESENT IN DRY AGGREGATES AND ANODES

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Keywords: Carbon anodes, anode quality, sodium content, electrophoresis, specific sodium-ion electrode

Abstract

In aluminum industry, it is important to determine the concentration of contaminants present in anode raw materials as rapidly as possible in order to adjust the anode recipe. Sodium, which is an impurity coming largely from anode butts, significantly influences the anode reactivity, and an increase in its concentration increases the anode consumption. A simple and inexpensive method was developed to quantify the sodium content in dry aggregates and anodes without grinding the samples. The method is based on potentiometric principles using a sodium-ion specific electrode. A sample can be analyzed easily within thirty minutes. In order to prevent the rapid degradation of the specific electrode due to experimental conditions, sodium is extracted from samples by electrophoresis prior to the test. The comparison shows that the measured sodium concentrations obtained are similar to the results determined by other test methods.

Introduction

Primary aluminum is produced by the electrolysis of alumina (Al_2O_3) dissolved in cryolite (Na_3AlF_6) using the Hall-Héroult process. The reduction reaction takes place at the cathode to form the molten aluminum. During this process, oxygen is removed in the form of carbon dioxide (CO_2) on the anode side. The carbon required for the production of CO_2 is supplied by the carbon anodes used in the electrolytic process. Thus, the anode is consumed and has to be replaced at regular intervals [1-3].

A green anode is made from anode paste which consists of about 85% dry aggregate (fresh petroleum coke, butts, green and baked rejects) and 15% binder (coal tar pitch). The rejected green and baked anodes and butts are crushed and added to the dry aggregate, and they constitute around 20 % of the anode paste [2-4].

The theoretical value for the consumption of carbon for the electrolytic production of aluminum is 0.334 kg C/kg Al [3], but due to various losses (reaction with air and CO_2) and current efficiency, the actual consumption is over 0.4 kg C/kg Al [3]. The rates of reaction of air and CO_2 with the anode are influenced by the presence of different impurities in the anode. Sodium is one of the impurities, which catalyzes the anode reactivity [1]. The major source of sodium is butts. The anodes are submerged into the electrolytic bath, which contains cryolite (Na_3AlF_6). During the electrolysis, they absorb sodium. When the anode is consumed up to a certain height, they (called butts) are taken out of the electrolytic bath, cleaned and crushed and used in anode production. Even though they are cleaned externally, a significant amount of sodium still stays in the butt particles. The presence of butt particles in the anode recipe influences the percentage of sodium in anodes and, in turn, the reactivity of the anode [2-4].

The quality of anodes has a direct impact on the cost of metal, energy consumption, and environmental emissions. One of the key goals for industry is to improve/maintain anode quality in spite of the variations in the quality of the raw materials. Availability of tools for monitoring the impurity levels will contribute significantly to achieve this goal [2-4].

There are different standard methods to quantify sodium. The American Society for Testing and Materials (ASTM) developed different test methods using Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES), Atomic Absorption Spectrometry (AAS), Flame Emission Spectroscopy (FES), and X-ray Fluorescence Spectroscopy (XRF) [5]. Batista and Da Silveira [1] characterized samples of coke, pitch, and butt powder by XRF and AAS. Many researchers employed ICP-AES and AAS for the determination of various elements in lubricants [6] and in soil [7]. The XRF method has been utilized in a number of cases [8-10] for the estimation of sodium. Carter et al. [11] presented a review on the application of AAS in the industrial analysis of metals, chemicals, and advanced materials. An update of this review was published by Gibson et al [12].

A number of researchers have used zinc uranyl acetate for the estimation of sodium in the presence of interfering ions such as phosphate [13-15]. The concentrations of sodium in different samples were measured by Emission Flame Photometry [16-19].

In aluminum industry, the estimation of sodium is usually done by different methods such as XRF and AAS, which require intensive sample preparation, highly skilled personnel, and costly reagents. The methods are usually time-consuming. Thus, there is a need for simple methods for fast and efficient estimation of sodium. In this context, researchers have developed potentiometric methods for the estimation of sodium using sodium-ion specific electrode (SISE) [20-23]. SISE is usually suitable for a system which does not contain any particulate materials. In the presence of particulate materials, its life reduces to a great extent.

The aim of this work is to develop a simple, quick, and inexpensive method to estimate reliably the quantity of sodium present in anode raw materials using SISE in the presence of particulate materials.

Materials and methods

Materials

Industrial anode raw materials (cokes and butts) were used to test the method. The samples contained different concentrations of sodium. Ionic strength adjuster (ISA) and standard solutions of sodium were acquired from Thermo Fisher Scientific Inc.

Methodology

Potentiometry is an electrochemical method used for estimating the concentrations of chemical species present in a solution. In this method, a working electrode (WE) and a reference electrode (RE) are dipped into the solution and the potential difference between the two electrodes are measured at a constant current. Usually, for most of the potentiometric methods of analysis, the value of current is set to zero by using high impedance associated with the reference electrode. Zero current helps minimize the variation in the half-cell potential of the reference electrode during the analysis. The potential difference (ΔE) between WE and RE varies with change in the concentration of ions in the solution. [24-26].

SISE measures sodium in a solution potentiometrically. The concentration of sodium ion (C_{Na^+}) in a solution can be correlated with ΔE using the equation (1) [24-26].

$$\Delta E = \Delta L - \beta \frac{RT}{F} ln C_{Na^+} \quad (1)$$

where

ΔL : constants of the system
β : membrane efficiency
R : ideal gas constant (8.314 $J \cdot K^{-1} \cdot mol^{-1}$)
T : temperature in Kelvin (K)
F : Faraday's constant (96485 Coulombs (C))

In this work, a SISE (ROSS Sodium Ion Selective Electrode Thermo Scientific, 8611BNWP) was employed. This probe contains both WE and RE.

The SISE is a probe made of porous glass membranes containing specific chemical compounds that facilitate selective migration of sodium in the electrode (see Figure 1). The membrane usually does not allow the passage of ions larger than the sodium ion. The ions smaller than sodium can interfere with the response of the electrode; however, these interferences are usually low compared to the response of sodium. The probe glass is fragile in certain mediums and deforms with time. The glass membrane of SISE degrades faster in the presence of particulate impurities, which can block the pores with time. Their presence can reduce the response of the SISE [24-26].

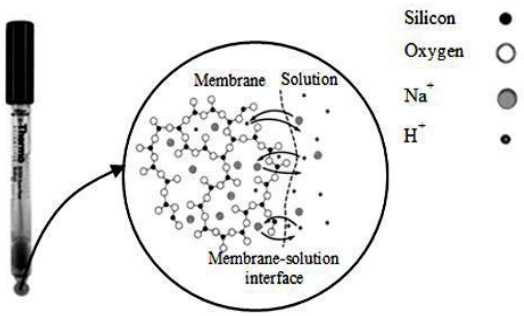

Figure 1. Composition of a specific electrode membrane [26, 27].

Analysis of sodium in fresh coke or butts using SISE is challenging due to two major reasons. First, the sodium ions do not leave the surface of the samples easily in aqueous medium. Second, the particulate sample can degrade the SISE within a short time. To solve these two issues, electrophoresis technique was employed (a) to extract the sodium ions from the samples and (b) to measure the extracted sodium using SISE in the absence of particles.

Researchers have used electrophoresis technique to separate different ions based on their ionic mobilities through a medium by the application of direct current. Rengan et al. [28] developed an electrophoresis method to separate different ions. Electrophoresis was used by Janos [29] to separate the lanthanide and the actinide series elements.

Usually, agar gel is utilised as the medium in electrophoresis. A small well is created inside the agar gel and filled with the sample. The gel along with the sample is placed in a conducting solution and direct current is passed through the system using two electrodes (anode and cathode) by applying a potential difference. The passage of current causes the positive ions to move towards the cathode. Due to the resistance offered by the agar gel and the size of the ions, different positive ions move towards the cathode at different speeds for a particular time. The ionic force of the conducting solution and the amount of applied current have to be sufficient in order to have a good separation [24-26].

Thus, in this work, the quantity of sodium in fresh coke and butt particles were estimated in two steps. In the first step, electrophoresis was used to transfer the sodium in the samples to the conducting solution outside the agar gel. This ensured that the coke or butt particles stayed inside the agar gel, whereas the sodium ions moved to a solution containing no particles. Then, in the second step, SISE was used to estimate the sodium transferred to the solution.

The efficiency of sodium separation depends on the applied potential, ionic strength and nature of the conducting solution, and the time of passage of current.

ISA was chosen as the conducting solution, as recommended by the supplier of the SISE, to eliminate interferences in the measurement.

For a specific value of current passing through a solution of known ionic strength, the time of passage of current should be such that most of the sodium ions are extracted from the sample. Based on Faraday's law of electrolysis, it can be stated that 1 Faraday (or 96485 Coulombs (C)) of electricity enables the transfer of 1 g equivalent of sodium (23 g). For a current of "I" (A) passed for a period of "t" (s), the electric charge "Q" (C) can be expressed as shown by Equation (2):

$$Q = It \quad (2)$$

If the amount of sodium in a given sample is "m", then the charge of "Q_{Na}" (C) required to transfer the sodium is given by Equation (3):

$$Q_{Na} = \frac{m \times 96485}{23} \quad (3)$$

Thus, the time "t_{Na}" (s) required to pass the current can be expressed by Equation (4):

$$t_{Na} = \frac{Q_{Na}}{I} \quad (4)$$

Experimental

Preliminary work to identify the role of electrophoresis in the movement of ions

In two identical electrophoresis set-ups, $NiCl_2$ solutions of same concentration and volume were placed into the wells inside the agar gel. Then, the systems were filled with ISA. Dimethylglyoxime (DMG) solution was placed outside the agar gel. DMG produces pink color with Ni salts. Since sodium cannot produce any colored salt, Ni was used to study the effect of electrophoresis on the ionic movement. In one of the set-ups, a potential difference was applied between the two electrodes whereas no potential was applied in the other one. After some time, a pink color was observed in the set-up where the current was present. No pink color was detected in the second set-up.

The concentration of Ni in the well is higher than that on the outside of the gel. It is possible that mass diffusion can take place due to the concentration gradient. If mass diffusion plays an important role in the transfer of Ni^{+2}, then pink color should be observed in both set-ups. The absence of pink color when there is no current clearly indicates that the passage of current forced the Ni^{+2} ions to move towards the exterior of the gel. Thus, there was no ion movement due to the concentration difference.

Electrophoresis set-up for the extraction of sodium

The extraction of sodium from the raw materials was carried out with electrophoresis. Figure 2 shows a schematic representation of the electrophoresis set-up. It comprises of two copper electrodes and an agar gel block in a glass container.

A measured quantity of sample in 1 ml of ISA/distilled water (10% v/v) was placed in the well of the agar gel. A 25 ml conducting solution containing 0.1 % v/v ISA/distilled water was then put in the glass container, and a potential difference of 17.5 V was applied between the two electrodes for a certain period of time. After the experiment, the flow of current was turned off and the solution outside the agar gel was transferred to another beaker.

Measurement of sodium concentration in solution using SISE

The SISE was dipped in the solution containing the sodium ions (see Figure 3). The potential difference between the WE and RE of the SISE was recorded with respect to time.

Results and discussion

During the electrophoresis step, at the voltage of 17.5 V, it was observed that the current flowing through the system was in the order of 0.002 A. Part of the butt sample used for sodium analysis was analyzed by XRF, and it was found to contain around 1600 ppm of sodium. Since butt particles contain much higher quantity of sodium compared to that of fresh coke, the sodium percentage present in butt sample was used to calculate the minimum time required to transfer the sodium ions by electrophoresis.

Due to the small size of the well in the agar gel, it was possible to add 0.01 g of sample into the well. 0.01 g of a 1600 ppm sodium sample contains

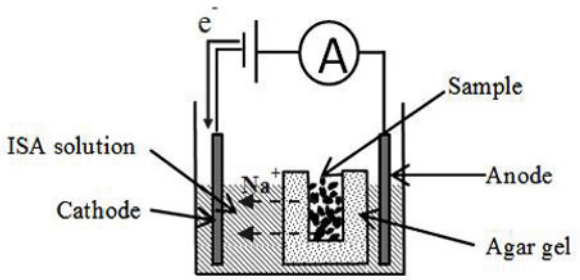

Figure 2. Schematic representation of the electrophoresis method.

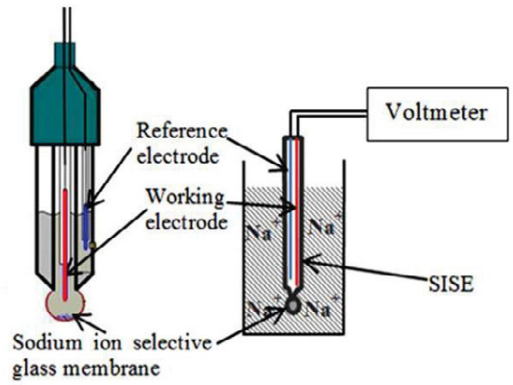

Figure 3. Schematic representation of the potentiometric method.

(0.01 g sample)(1600 g Na/ 10^6 g sample) (1/23 g /g.mol) = 6.956 x 10^{-7} g.mol of Na. As Na has a valency of 1, the amount of sodium in mole equivalent will be 6.956 x 10^{-7} g.mol equivalent. Thus, the electrical charge required to transfer the sodium ions will be (6.956 x 10^{-7} g.mol)(96485 C/g.mol) = 0.067 C. Thus, for a current of 0.002 A, time required for the transfer of sodium will be (0.067 C)/ (0.002 A) = 33.56 s. = 0.56 min. This calculation shows that the transfer of sodium could be completed in less than 1 minute. However, in a real system, there are other ions as well as loss of electricity (in the form of heat dissipation) due to the resistance of the agar gel. Hence, the time required for the passage of current was determined experimentally. The required time should be more than the theoretically required time which was calculated above.

An experiment was performed to determine the time required to transfer the sodium ions from a given sample. A measured quantity of solution with a known ppm of sodium was added in the well of the agar gel. The voltage of the solution outside the gel was measured periodically using SISE. Figure 4 shows the response of the SISE at different time intervals. The voltage increased significantly with increasing time up to 6 min and a maximum value was obtained at around 8 min. The voltage is proportional to the sodium concentration, and the atteinment of a maximum shows that all the sodium is transferred at this time. Thus, the time used for electrophoresis for further experiments was chosen as 8 min.

A calibration curve (see Figure 5) was prepared for the estimation of sodium in samples containing unknown concentrations of sodium. Solutions of different known concentrations (1, 5, 10, 25, and 40 ppm) of sodium were prepared using the purchased

standard solutions and ISA. 1 ml of the solutions of different concentrations was added to the well of agar gel. After adding measured quantity of the ISA solution, the electrophoresis was performed for 8 min. Then, the solution outside the gel was analyzed using SISE. The voltage response was plotted against the concentration of the sodium solution used for calibration.

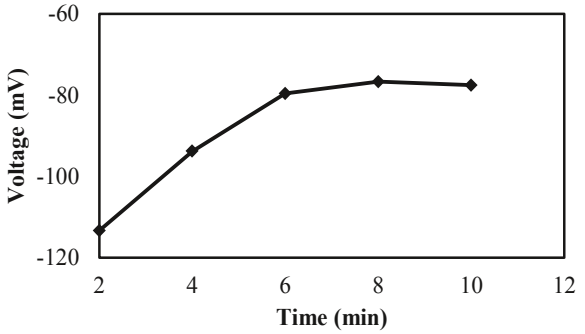

Figure 4. Variation of the response of SISE with time.

The calibration curve shows that the voltage increased linearly with increasing sodium concentration in the range of 1 to 40 ppm. As stated earlier, a butt sample was found to contain around 1600 ppm of sodium. The calibration curve was prepared so that it would be possible to measure the sodium concentrations of butt samples up to 4000 ppm which is much higher than that measured for butt sample used in this study.

A butt sample of 0.01 g was placed in a 1 ml solution inside the well of the agar gel. If 0.01 g of sample containing 4000 ppm of Na is analyzed, then this sample contains $40x10^{-6}$ g of Na. In the 1 ml solution, this amount of sodium is equivalent to $40x10^{-6}$ g of Na $\times$ 10^6 ppm.g^{-1} or 40 ppm of Na. Therefore, a calibration curve up to 40 ppm is capable of measuring butt samples up to 4000 ppm.

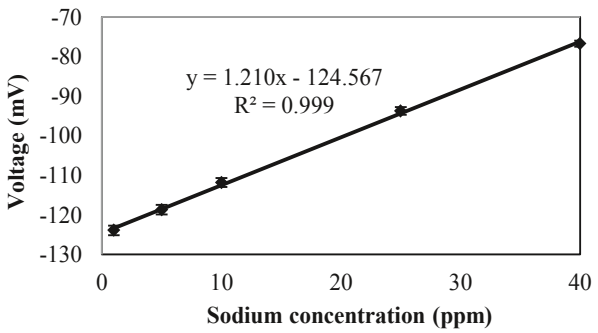

Figure 5. Calibration curve obtained from sodium standards.

For a sample of unknown sodium concentration, the ppm of sodium was calculated using the calibration curve based on the measured voltage response. If the calculated ppm is x, then the concentration of sodium in the carbon sample is 100x ppm.

The calibration curve was later used to estimate the concentrations of sodium in two coke samples and one butt sample. The results of these measurements are presented in Table 1. The samples analyzed had a particle size of less than 1 mm. The results show that the measured values are similar to the known values. The known sodium concentrations for the two coke samples are from the supplier data for the whole shipment. It is possible that due to non-homogeneity of the raw materials, the measured values did not match exactly with the known concentrations for these samples. The sodium concentration of butt was found to match well with the XRF measurement of the same sample. The developed method can give a good estimation of sodium concentration. The time requirement for the estimation of sodium using this method is around 30 min.

Table 1. Comparison of known and measured values of sodium concentration in different samples.

Samples	Sodium concentration analysis (ppm)	
	Known	Measured
Coke A	106 (supplier data)	68
Coke B	60 (supplier data)	47
Butt A	1576 (measured by XRF)	1593

Further study will be carried out to improve the manipulation and fabrication of the delicate agar gel. Later, tests will be done using samples of dry aggregates and anodes.

Conclusions

A simple and effective method was developed to determine sodium concentration in coke and butt samples using SISE. Extraction of sodium by electrophoresis prior to the potentiometric measurement using SISE helped prevent the degradation of the electrode. The comparison shows that the measured sodium concentrations are similar to the concentrations determined by other test methods.

Acknowledgements

The technical and financial supports of Aluminerie Alouette Inc. (AAI) as well as the financial support of the Natural Sciences and Engineering Research Council of Canada (NSERC), Développement économique Sept-Îles, the University of Québec at Chicoutimi (UQAC) and the Foundation of the University of Québec at Chicoutimi (FUQAC) are greatly appreciated. The authors would like to thank the Fonds de recherche du Québec nature et technologies by the intermediary of the Aluminum research centre – REGAL for their contribution for student participation in this conference.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

INTERFACIAL BOUNDARY BETWEEN CARBON ANODES AND MOLTEN SALT ELECTROLYTE

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Keywords: carbon anodes, aluminium electrolysis, double layer capacitance

Abstract

The properties of calcined petroleum cokes (CPC), used in the anodes in the Hall-Héroult process, are important for the electrochemical behaviour of the anodes. CPCs from different sources vary not only in impurity level but also in microstructure. The trend of available CPCs for anode production indicates an increase in impurities as well as introduction of more isotropic structure. Hence, a better understanding of the behaviour of these materials during electrolysis is essential for optimizing performance. In previous studies, different anode cokes were characterized with respect to anode overpotential and the gas evolution during polarization. Present work focuses on investigation of the interfacial contact between the anode and the molten salt electrolyte. The interfacial contact is important for the electrochemically active area of the porous electrodes, which is different from the geometric area. The focus is on the assessment of electroactive area by measurements of the double layer capacitance, which may be performed during electrolysis. This is a great advantage as the active surface area is known to depend on the polarization. Double layer capacitance is compared for anodes made from very different carbons, and correlated to the overpotential.

Introduction

Prebaked carbon anodes used in electrolysis of aluminium have a significant influence on the economy of the process. Despite relatively high current efficiency of industrial electrolysis reaching more than 90 % [1], the process is very energy-consuming, and the cell voltage – approx. 4 V. The total contribution in cell voltage related to the anode is approx. 1.9 V [2]. This value consists of anode electrochemical overpotential (≈ 1.12 V), resistance due to gas bubbles evolution (≈ 0.48 V) and electronic resistance of anode material (≈ 0.33 V) [2]. Therefore reaching the lowest possible voltage drop influences the overall process economy positively. Other factors which impact the costs of process are an excessive consumption of anodes by dusting and CO_2/air- burn as well as unexpected failures. The problem of producing optimal quality anode becomes more complex due to use of multiplicity of input source materials (CPC and pitch) with different physical and chemical properties.

As the operational overpotential related to anodes during electrolysis is a very important factor it has been the subject of several studies in the past *i.a.* [3–10]. Due to differences in experimental setups, the general difficulty in proper estimation of iR drop in a cell and differences in reference electrodes the absolute values obtained in various papers are not consistent. However one general observation recurs – electrodes made of graphite always exhibit higher overpotential than carbon materials. It is still not clear how the total overpotential of various anode materials is affected by the actual differences in electrochemical reactivity, and differences in electrochemically active surface area, and how both of these relate to surface chemistry and impurities. Since in all studies the geometrical surface area of the working electrode is applied for the calculation of the current density (CD), differences in the real current density may occur due to differences in the electroactive surface area. Considering the strong dependence of the anode potential on the current density (especially in the range of higher CD), the influence of apparent to real surface area ratio on the potential values can be substantial.

It is generally agreed that the double layer capacitance correlates with the electroactive surface area. It is typically measured by electrochemical impedance spectroscopy, which allows for assessment of the active surface area during electrolysis. The latter makes it a very powerful tool, as surface wetting properties are known to depend on the polarization, and the active area might also change during the experiments. The capacitance of the double layer of carbon electrodes in cryolitic melts in laboratory scale measurements has been investigated in many previous studies. In case of graphite, values of C_{dl} ranging from 14-40 $\mu F \cdot cm^{-2}$ at 0.03 $A \cdot cm^{-2}$ [11] to 47-97 $\mu F \cdot cm^{-2}$ at 0.05 $A \cdot cm^{-2}$ [12] were obtained For baked carbon anodes reported values of C_{dl} are in average 104 $\mu F \cdot cm^{-2}$ at 0.05 $A \cdot cm^{-2}$ [12], 110-130 $\mu F \cdot cm^{-2}$ at 1.2 V [13], an average of 140 $\mu F \cdot cm^{-2}$ at 1.2 V [14], an average of 350 $\mu F \cdot cm^{-2}$ in the range 0-0.15 $A \cdot cm^{-2}$ [15]. Thus, there is a large scatter of measured values of double layer capacitances in the literature. The main reason can be an inhomogeneity of the microstructure of measured materials and pretreatment of the electrode surface. This effect is clearly visible in aqueous electrolytes [16]. Also, little is currently known on how impurities and surface chemistry would influence the electrochemical reactivity and electrolyte wetting. Another reason lies in the different frequencies used for the calculation of the double layer capacitance. Still, even though the absolute values of C_{dl} are different in the various works their relative values should be correct.

The aim of this work is to improve the methodology of double layer capacitance measurements by electrochemical impedance spectroscopy. Measurements of the double layer capacitances were obtained for anodes made from very different carbon materials (graphite and cokes), and changes over short periods of time were monitored. The double layer capacitance was also obtained at different polarization conditions and correlated to measured overpotential for each sample.

Materials and Methods

The electrochemical experiments were performed in a closed furnace with an inert atmosphere maintained by passing argon with 100 ml·min^{-1} flow rate, as described previously [10]. As electrolyte, a cryolite melt with CR = 2.3 composed of synthetic Na_3AlF_6 (≥97 %, Sigma-Aldrich), 9.8 wt% excess of AlF_3 (industrial grade, in-house sublimed) and 9.4 wt% of γ-Al_2O_3 (≥99.5 %, Merck) was used. The temperature of the electrolyte during experiments was maintained at 980 ºC. Tested samples were made from 4 different single source calcined petrol coke (CPC) with a grain size limited to 0-2 mm (the fractions were 2-1, 1-0 and mill product) and correspond to samples used in previous works [9,10], with the same labels: anode 1 and anode 4. These anodes showed the differences in terms of overpotential and have completely different microstructure – anode 1, 2 – anisotropic, anode 3, 4 – isotropic. Furthermore, the cokes differ in terms of the microstructure, with a higher fraction of edge planes at the surface of anode 4 (80 % vs. 60 %), and also a higher content of acidic oxygen surface groups in the coke of anode 4. For comparison purposes also ultrapure graphite was tested or measured. The graphite has a lower fraction of edge planes in the surface (40 %) and a corresponding higher fraction of basal planes. The working electrodes with different geometrical area (Figure 1) were shielded by boron nitride (BN) parts to expose only the vertical surface in order to minimize effect of gas bubbles evolution accompanying the anodic reactions. The design of this type of electrode and its damping effect on voltage noise during polarization was presented previously [10]. In present work only two different dimensions of the anodes were used – 0.5 and 0.75 cm of height. As a reference electrode an Al pseudo-reference electrode was used. It consisted of a closed bottom alumina tube where the ~1.6 g of pure aluminium was placed. The tube had a hole above the aluminium melt level to introduce the electrolyte. As an electrical contact to aluminium pool a tungsten wire shielded by alumina tube was used. The graphite crucible with shielded bottom by alumina disc acted as a cathode.

Electrochemical measurements were performed using potentiostat with power booster (IM6e and PP201 by Zahner-Elektrik). The measurement procedure consisted of five steps: impedance spectroscopy at open circuit potential (OCP) every 5 min for 30 min after immersion electrode to the electrolyte, polarization at constant current density of 1 A·cm^{-2} by a period of 3.5 min followed by EIS at OCP every 5 min for 30 min, EIS at different potentials and finally EIS in psudo-galvanostatic mode (voltage perturbation with maintaining constant current density of 1 A·cm^{-2}). EIS measurements were carried out at frequency range 500-10 kHz with 20 mV_{pp} AC amplitude except EIS at 1 A·cm^{-2} that, due to higher noise, were performed at frequency range 150-10 kHz with 50 mV_{pp} AC amplitude. The double layer capacitance of electrodes was calculated using frequencies range (150-50 kHz) assuming simple equivalent circuit of resistance, inductance and capacitance in series, see Figure 2. Fitting of spectra was performed by ZView software (Scribner Associates).

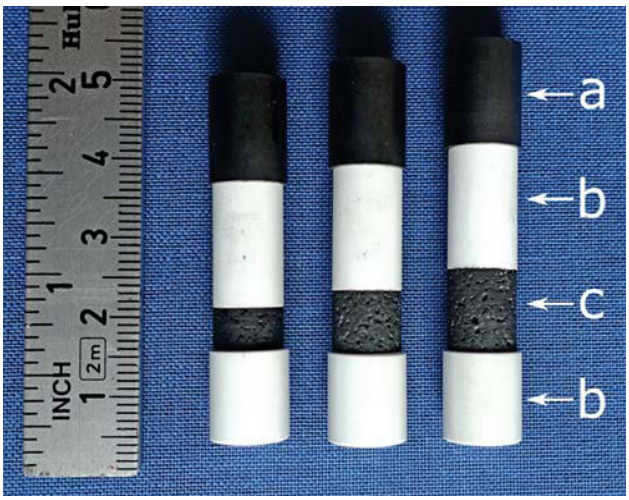

Figure 1. Electrodes used in electrochemical measurements; a – graphite connector, b – boron nitride shields, c – carbon sample

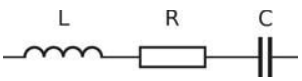

Figure 2. Equivalent electrical circuit used for fitting impedance spectra.

The inductance (L), resistance (R) and capacitance (C) represent the inductance of leads to the cell, resistance of electrolyte between working and reference electrodes (including lead to the working electrode) and capacitance of working electrode.
Presented results in first part of this work (except EIS at 1 A·cm^{-2}) were performed on one batch of samples in order to keep better consistency. One repetition on the similar batch of samples was made to check repeatability of measurements.

Results and Discussion

A typical Nyquist plot of EIS measurement for investigated samples is shown in Figure 3a), and the corresponding Bode plot is shown in Figure 3b). The capacitance values were determined from the equivalent circuit shown in Figure 2. Using such a simple equivalent circuit was based on the assumption that for sufficiently high frequencies no faradaic reactions occur (or occur at negligible rate), such that the only contributions to the imaginary impedance are the double layer capacitance and the inductance from the external leads. The vertical part of curve on the Nyquist plot demonstrates only small change of real part of impedance over this frequency range which indicates no faradaic charge transfer. Therefore other elements in the electrical circuit model which represent charge transfer, diffusion, adsorption etc. can be omitted. Values at the highest frequencies were used to calculate inductance which value was fixed during fitting of spectra between frequencies from 150 to 50 kHz. Calculation of inductance at the highest frequency was performed to minimize contribution from the capacitance on the total impedance.

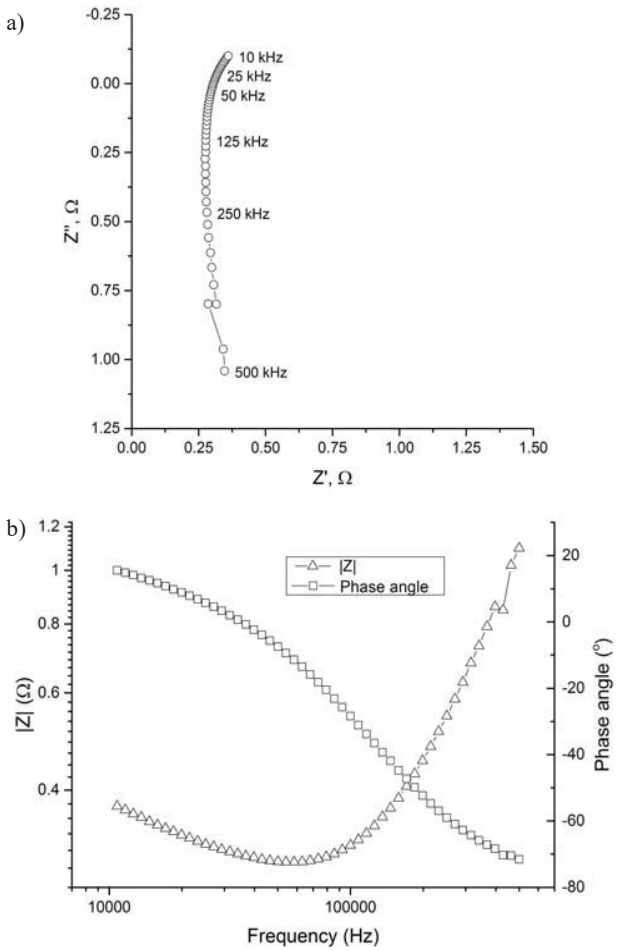

Figure 3. Nyquist (a) and Bode (b) plots measured at open circuit potential.

After immersion of the anodes in the electrolyte they exhibited relatively low capacitance – the highest for the graphite sample, slightly lower for anode 1 and much lower for anode 4 (Figure 4). The values of capacitance should correspond to the wetting properties – higher value indicates a larger interfacial contact surface area. Therefore, without any current applied, the "fresh surface" of the graphite was better wetted by electrolyte than the tested anodes. The lower values of capacitance for the coke anodes could be related to the fact that these have significantly higher porosities/lower density compared to the graphite, and in particular anode 4 has a high fraction of large pores [17,18], and therefore a lower geometric surface area, provided that the electrolyte is not intruding the pores. Another difference to graphite is the fact that the coke anodes were machined to make proper size electrode. Thus machining could change their surfaces providing worse wettability, especially in case of isotropic anode 4 which is characterized by much higher hardness. There is also visible in Figure 4 a constant increase of the capacitance for all the anodes, hence there is some conditioning of the surface upon immersion in the electrolyte causing an improved interfacial contact over time even at zero-current conditions [13].

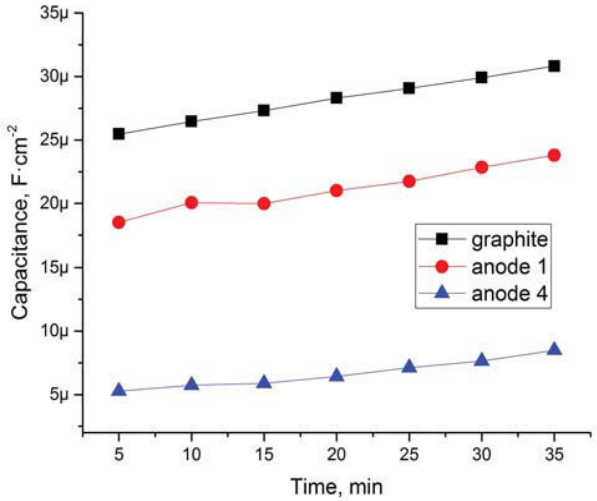

Figure 4. Anodes capacitances changes during time at open circuit potential after immersion to the electrolyte.

Plots on the chart in Figure 5 depict potentials changes of the electrodes during polarization at constant current density. The values of potentials were corrected by IR drop across the electrolyte (R was read out from impedance spectra). The graphite sample is characterized by the highest overpotential what is in accordance with all previous works. The coke anodes have a lower potential (about 100 mV) than graphite, but more interesting is the profile of these curves. In case of graphite and anode 1, the level of the potential is relatively stable during the polarization time. However, in the case of anode 4, the potential decreases during polarization. This indicates that the anode undergoes some kind of a slow conditioning process such that the active area of the electrode increases, causing decreasing effective current density, and hence the declining potential.

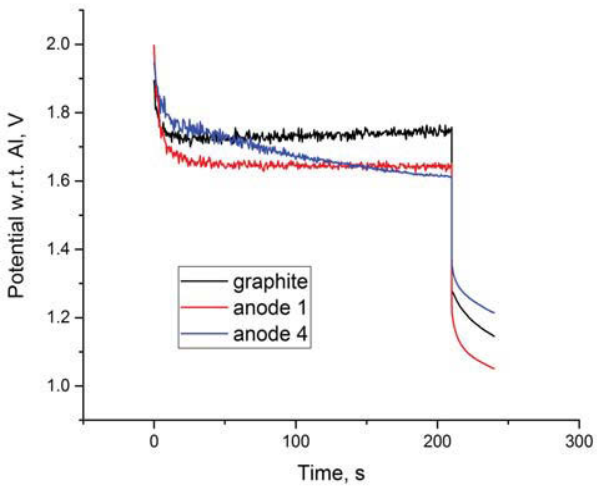

Figure 5. Potential curves during polarization at constant current density of 1 A·cm^{-2}.

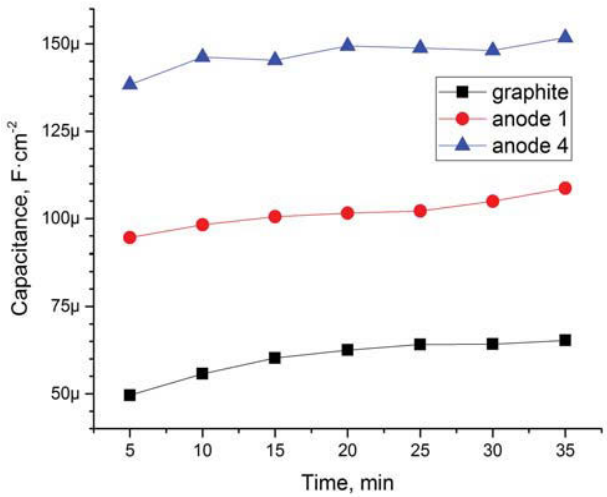

Figure 6. Anodes capacitances changes during time at open circuit potential after polarization by 3.5 minutes at 1 A·cm^{-2}.

After polarization the values of the capacitance increased significantly with respect to the values before polarization. For the graphite sample from around 31 to 50 μF·cm^{-2}, for anode 1 from 24 to 94 μF·cm^{-2} and for anode 4 from 9 to 138 μF·cm^{-2} which gives an increase by a factor of approximately 2, 4 and 16 respectively. Such large changes of the capacitance before and after polarization are likely to be caused by change of wetting properties and developing of electrode surface during oxidation. The exact reasons for the huge change of anode 4 are not known, but compared to anode 1, the coke used for anode 4 is more isotropic, a higher fraction of reactive surface sites, more impurities, and is also differing with respect to the surface oxygen groups.

Figure 7 shows the capacitance calculated at different potentials with respect to Al pseudo reference electrode. Because of current limitations of the main potentiostat measurements were performed up to 1.5 V vs. Al which corresponded to 139, 128 and 49 mA·cm^{-2} DC for anode 1, anode 4 and graphite respectively. As is visible from Figure 6, there is decrease of capacitance with increasing potential of the electrode reaching minimum at about potential 1.4 V and slight increase going to higher potentials. Similar trend can be observed in work of Kisza *et al.* [11] for pyrolytic graphite considering the same current densities. This pattern is distinct for anode 4, less for anode 1, and not visible for graphite anode, most likely due to much smaller current densities at these potentials

In figure 8 the double layer capacitance obtained at different potentials is plotted together with the corresponding current density. It is evident that in the region of potentials where the capacitance is decreasing, the current density is very small. At small current densities the reaction of carbon oxidizing to CO is favoured [19]. Prior to the evolution of CO, adsorption of CO on the surface is frequently assumed. Therefore a reduced active surface area can be an effect of adsorbed CO on the surface of the electrode. Another explanation of this phenomenon can be coverage of the electrode surface by adsorbed oxyanions. When the potential is higher, the rate of electrochemical reactions at the interface accelerates as is indicated by the rapid increase of current (Figure 8).

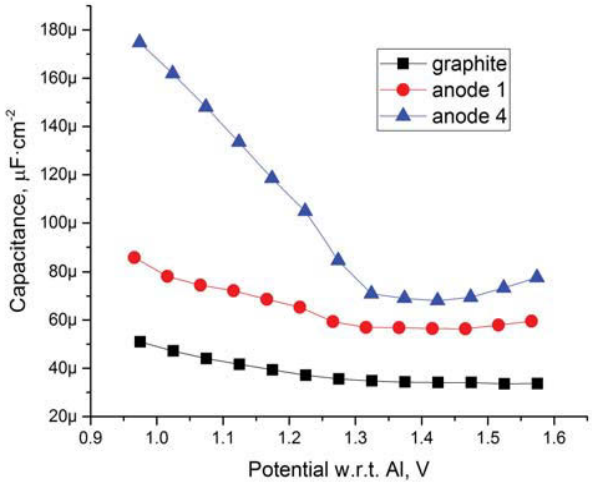

Figure 7. Capacitances of anodes at different potentials.

In this region the double layer capacitance stabilizes. Faster generation of CO_2 establishes certain coverage of the anode by adsorbed species.

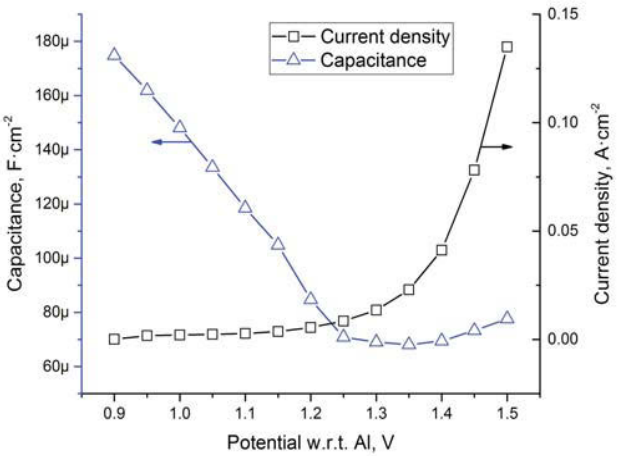

Figure 8. Anode capacitance at different potentials with corresponding current densities.

In order to perform EIS measurements at 1 A·cm^{-2} DC current density range a power booster in pseudo galvanostatic mode was used.

Several materials were evaluated in the previous work [10]. On figure 9 the calculated values of capacitance were plotted versus potentials of these materials during constant current polarization at 1 A·cm^{-2}. There is a certain correlation between the active surface area and the values obtained for the overpotentials. The higher capacitance and hence higher active surface area the lower potential were observed during constant current polarization. However, the differences in the potentials of anode 2 and 3 are not so significant while differences in C_{dl} are high. It means that the different mictrostrucure and the surface chemistry of cokes can also influence the overpotential.

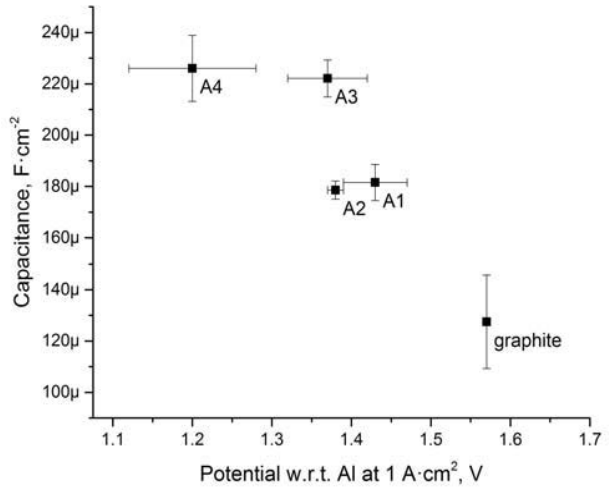

Figure 9. Anodes capacitances measured at current density 1 $A \cdot cm^{-2}$ and their potentials at this current density taken from previous work [10]. A1 – anode 1; A2 – anode 2; A3 – anode 3; A4 – anode 4.

It was found that the measured C_{dl} correlates well with the permeability of the tested materials (Figure 10). This is reasonable considering that permeability to some extent is a measure of the macroporosity of the materials.
Because current density usually has been related to the geometrical area of the electrode, which can be very different from the active surface area for various materials, the local rates of the electrochemical reactions may be very different, and thus the measured overpotentials will differ.

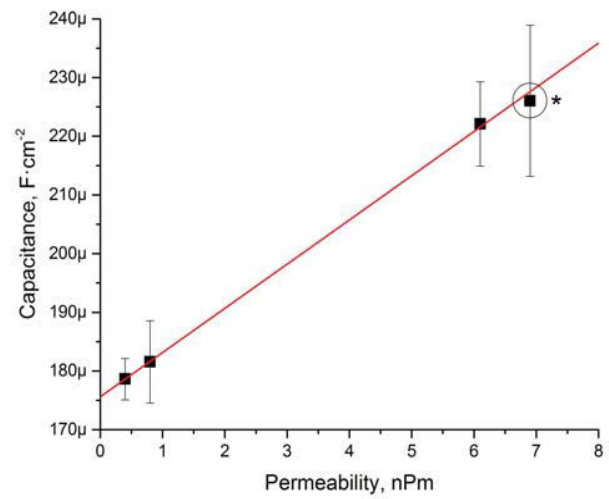

Figure 10. Permeability of anode materials and measured anodes capacitances measured at current density 1 $A \cdot cm^{-2}$. (* value of permeability 6.9 nPm may be even higher due to limitation of measuring apparatus)

Jarek and Thonstad [13] made a correction of current densities using active surface area from the double layer capacitance. After this correction, the curves on Tafel plots for different carbon materials moved closer to each other. It is thus very important when one try compare different anode materials in terms of their electrochemical properties to identify also the differences in the wetting properties. The cause of differences in wetting properties is still not well known, and this area certainly deserves further investigations.

Conclusions

Performed measurements showed substantial increase of double layer capacitance after polarization of anode. Different values of capacitances for tested anode materials were found which correlated, to a certain extent, with observed polarization potentials. Therefore the differences in overpotential can be partly explained by different active surface areas of the anodes, which influences the real current density during polarization. A strong dependence between the double layer capacitance and permeability of anode materials was observed.

Acknowledgements

This work was financed by Hydro Aluminium and the Research Council of Norway. Thanks are due to the NTNU workshop where fabrication of experimental parts was required. Additionally, the contributions of Egil Skybakmoen and others at SINTEF are gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MEASUREMENT OF THE ELECTRIC CURRENT DISTRIBUTION IN AN ANODE

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Keywords: Carbon anodes, anode quality control, electrical current distribution, electrical resistance distribution

Abstract

Carbon anode, which carries the electricity and provides the carbon necessary for the electrochemical reaction in the electrolytic cell, is an essential component of the primary aluminum production. Energy efficiency and environmental impact are still the key issues in modern aluminum smelters; thus, it is important to control the quality of anodes which strongly influences the energy consumption and the greenhouse gas emissions.

As the uniform current distribution appears to be one of the indicators of the anode quality, an experimental system has been developed to measure the current distribution in an anode. This article describes the measurement system used for laboratory anodes and presents the experimental results. The results are also compared with the electrical resistance distribution measurements in the anodes, and the comparison shows good agreement.

Introduction

The production of primary aluminum is carried out via the Hall-Heroult electrochemical process. During the electrolysis, high amperage is used to transform the alumina into aluminum. This current is supplied to the electrolytic cell through carbon anode blocks which are manufactured from dry aggregates (composed of calcined petroleum coke, recycled butts, and rejected green and baked anodes). The aggregates, after mixing with coal tar pitch, are pressed or vibrocompacted to form green anodes which are then baked in large furnaces to produce baked anodes [1, 2].

The carbon anodes must have low electrical resistivity to have good energy efficiency in the cell. At the end of the baking cycle, a core sample from approximately 1.5% of the anodes produced is subjected to quality control tests. These include physical, electrical, mechanical, and chemical tests to ensure that the good quality anodes are installed in the electrolytic cells [3].

Several studies have been reported in the literature showing the direct impact of the electric current distribution on the operation and performance of the electrolytic cells. The flow of electric current controls the magneto-hydrodynamic (MHD) forces in the cell [4-6]. The high intensity electrical currents in the electrolytic cell generate high magnetic fields that lead to the onset of motion within the molten aluminum layer [7]. The movement of the ions affects the distribution of current in the cell. These flows are of MHD nature, and they are of concern to the cell stability. It was also reported that the measurement of individual anode currents can help detect the anode effects earlier [7]. Figure 1 shows a typical electric current distribution in an electrolytic cell.

Only a few studies have been published on the measurement of current distribution in anodes. Kuang, and Thonstad measured the potential distribution in the electrolytic bath by introducing probes connected to the position sensors in the cell [5] and correlated the voltage distribution to the current distribution in the cell. They measured the current distribution along the sides of the anodes in commercial Søderberg aluminium cells. Urata and Evans [7] measured the magnetic field in an anode with sensors placed near the anode stubs. Later, they converted the measured magnetic field into current by using a proportionality constant. This method is commonly used in industrial smelters for the measurement of the electric current distribution.

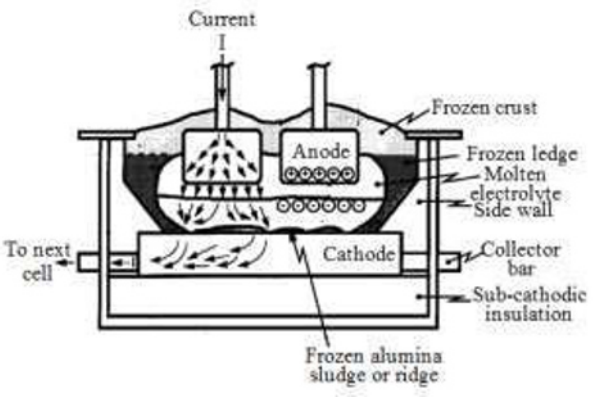

Figure 1: Current flow in an aluminum smelting cell [6]

Aluminum smelters follow the ASTM D6120-97 standard to determine the electrical resistivity of baked anodes using small cylindrical cores. These cores are taken from the top of the anode which does not come into contact with the electrolytic bath. This method does not take into account the non-homogeneity of the anodes. Smelters, therefore, need methods that can enable them to evaluate the resistivity of the entire anode in order to identify the defective anodes as early as possible in the production line.

A number of researchers worked on the measurement of electrical resistivity (or conductivity) of the entire anode, the evaluation of contact resistances, and the estimation of voltage drop at the stub-cast iron-anode interfaces. Seger [8] has developed a multiple contact system to measure the electrical resistivity of baked anodes.

Chollier-Brym et al. [9], based on the work of Seger, have developed a device called the Instantaneous Measurement of Electrical Resistance of Anode (MIREA). This equipment was used to measure the voltage drop along the height of the anode. The performance of the device was compared with the resistivities of the cores taken from different positions in the anode. The first

version of MIREA was used to measure the electrical resistivities of anodes at predetermined positions on the current path. However, measurements were relatively slow due to the inflatable metallic bags used to apply the current in the stub holes. Subsequently, Leonard et al. [10] improved the MIREA by changing the inflatable metallic bags with metallic brushes which decreased the measurement time. They carried out six measurement campaigns using numerous baked anodes in different plants around the world. The results led to a better understanding of the variation of the resistivity in the anodes. The results also revealed the problems of homogeneity in anodes and possible issues related to the compactors. At the end of the measurement campaigns, it was concluded that MIREA could be used as a tool to compare the anodes in different plants through the measurements of the electrical resistivity. This method is applicable to only baked anodes, needs calibration for each plant and it can't detect the defects outside the current path

Kocaefe et al. [11] have developed a tool (SERMA) for measuring the resistivity distribution in both green and baked anodes with the objective of identifying defective green anodes before they are baked and rectifying possible problems in the plant so that better quality green anodes are produced. The resistivity measurements agree well with the X-ray tomography results showing that the high resistivity regions correspond to the defective regions of the anode.

There are also reports on the inverse methods based on electrical or magnetic impedance tomography for the measurement of voltage or current distribution in different materials [12-15]. These methods utilize the experimental results as input to numerical models to obtain the distribution of voltage and current at different points of the anode. However, in most electrolytic cell modelling studies, the anodes are assumed to be homogeneous and the effect of anode non-homogeneity on cell current distribution is not taken in to account [18-20].

The goal of this study was to build a set-up to measure the current distribution in an anode.

Theory

Electrical conductivity (σ) is a fundamental material property that quantifies the ability of the material to carry a current when placed in an electrical field. Electrical resistivity (ρ), the inverse of conductivity, is the resistance (R) of a unit volume as measured across one dimension. Electrical resistivity has the unit of Ω.m, while that of the electrical conductivity is $S.m^{-1}$ ($\Omega^{-1}.m^{-1}$). With resistivities around 50-70 μΩ.m, a baked carbon anode is an electric conductor, and the current generated by an electric field is based on electronic conduction.

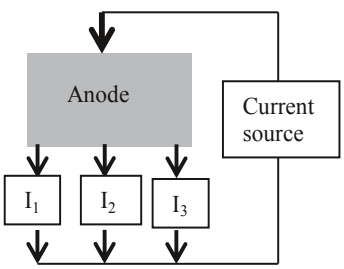

Figure 2: Current distribution measurement in an anode

Current measurement

In an electrical circuit, the current (I) is measured usually by an ammeter. The principle for the measurement of a current distribution in an anode can be represented schematically as shown in Figure 2.

Even though it is possible to directly use the ammeter for current measurements, sometimes it is more practical to determine the current indirectly by measuring the voltage drop across a resistance placed in the circuit. If the value of the resistance (R) is known, then current (I) can be calculated from the measured voltage (V) using Ohm's law (equation 1):

$$I = \frac{V}{R} \tag{1}$$

The schematic of measurement of current in an anode using this method is shown in Figure 3.

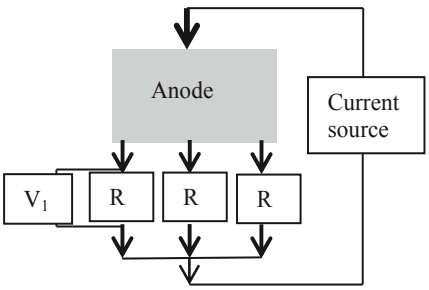

Figure 3. Determination of current by measuring voltage V across a resistance R

An anode can be represented as a combination of number of resistances. Figure 4 shows a simplified representation of an anode as a combination of three parallel resistances. Current I is applied to the anode at point 1 and leaves at points 3, 4, and 5 as

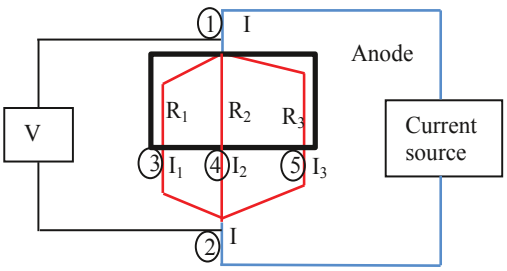

Figure 4. Representation of a combination of resistances in an anode

I_1, I_2, and I_3, respectively. The sum of these three currents should be equal to I at point 2. The resistances of the current paths 1-3-2, 1-4-2, and 1-5-2 are R_1, R_2, and R_3, respectively. The potential difference between the points 1 and 2 is V.
As given by Ohm's law:

$$V = I_1 R_1 = I_2 R_2 = I_3 R_3 \tag{2}$$

Thus,

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \qquad \frac{I_3}{I_2} = \frac{R_2}{R_3} \tag{3}$$

Therefore, the ratio of currents between two points is equal to the inverse ratio of the resistances at those points. If the resistances are constant, then the ratio of currents will be constant for any I.

The resistances at different points of the anode should remain constant with the passage of current if the heating (Joule effect) is not significant. Thus, the total current applied to the anode should not influence the current distribution profile.

As indicated by Equation (2), if it is assumed that $R_2>R_1$, then $I_2<I_1$ in order to maintain the voltage drop V constant. Consequently, the value of current will be lower at the point where the resistance is higher compared to that of a point with a lower resistance. Accordingly, the resistivity distribution will have the opposite trend of the electric current distribution.

Electrical resistivity measurement

In general, the electrical resistivity measurement methods can be broadly classified as either 2-point or 4-point probe methods. In the 2-point probe method (Figure 5(a)), two electrodes are used to apply an electrical current (I), and the same electrodes are used to measure the corresponding drop in voltage (V) across the specimen. Electrodes must be in intimate contact with the specimen to induce a current within that specimen. Metallic electrodes can be surface-mounted using conductive gels and pastes. If a direct current (DC) with a value I is applied to the specimen, and (V) is the potential drop, then the resistivity (or conductivity) can be calculated using equation 1:

$$\rho = \frac{1}{\sigma} = \frac{V}{I}\frac{Wh}{L} \tag{4}$$

where L, W, and h correspond to electrode spacing, specimen width, and specimen height, respectively. Although 2-point DC probe measurements are easy to perform, they are not suitable for the measurements on small samples since the contact resistances at the electrode-specimen interface influences significantly the resistance of the specimen.

The 4-point probe method (also called Kelvin method) is the preferred approach for measuring the resistivity (or conductivity) of conducting materials which have low values of resistivity [16]. This method uses four independent electrodes along the length of a specimen (Figure 5(b)). The two outer electrodes are used to supply an electric current (DC or AC) into the medium while the two inner electrodes are used for measuring the electrical potential developed over the length L. Again, if it is assumed that the cross-sectional area of the specimen is W*h, then Equation 4 is still valid. The effects of contact resistances are minimized in the 4-point probe method because the electrodes used for passing the current and those for measuring the voltage drop are different. The measurement of electrical resistivity is more accurate when compared with the 2-point probe method.

Applied force

The contact resistances have a parasitic influence on the electrical current measurements. According to the literature [17], the contact resistance varies inversely with the contact pressure which is itself proportional to the applied force. Figure 6 shows the variation of a contact resistance with applied force. It can be seen from this figure (at the asymptotic part of the curve) that there is no significant variation in the contact resistance above a certain threshold pressure. Therefore, it is important to measure the threshold pressure to minimize the effect of contact resistance.

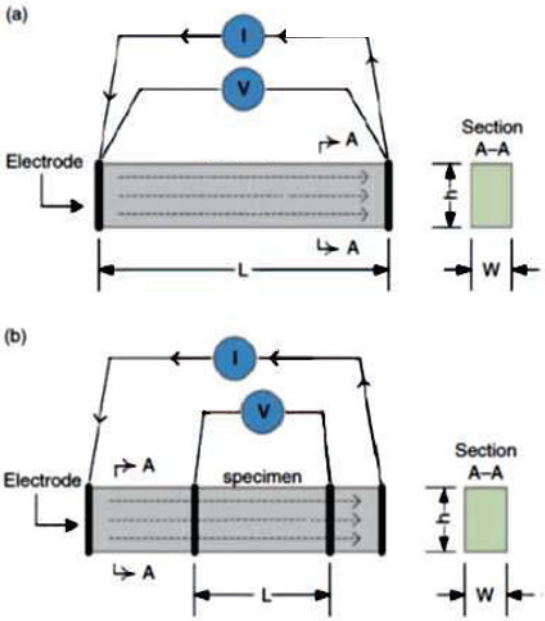

Figure 5: Resistivity (or conductivity) measurement based on the (a) 2-point and (b) 4-point probe methods [16]

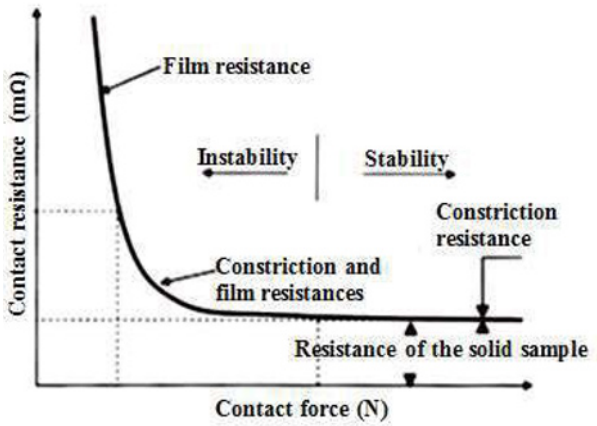

Figure 6: Evolution of the electrical contact resistance as a function of contact pressure [22]

During this study, Kelvin (or 4-point probe) method was used to measure the contact resistances.

Results and Discussion

Measurement system

The principle of the measurement requires the application of current to the anode between the upper and lower surfaces and the measurement of the current at different points. The current is applied in the middle of the anode top surface. The bottom surface is fitted with a number of contacts placed in the form of a grid. Each point in the grid is connected to a resistance of known value. A schematic representation and a picture of the experimental system are given in Figures 3 and 7, respectively.

Figure 7: Experimental measurement system

According to the literature [22], the contact resistance gradually decreases as the contact pressure increases until it reaches an asymptotic region. In this region, the variation in resistance is not significant despite the increase in contact pressure. The results in Figure 7 show the same tendency and are in agreement with those found in literature.

The results in Figure 8 were obtained by measuring contact resistances at four points. Due to the surface roughness of anodes, there may be a slight variation in contact resistances for some of the other points. However, this is not expected to be significant.

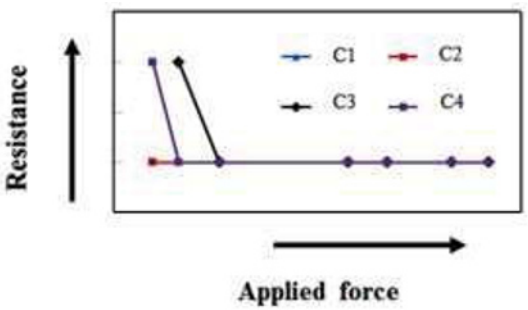

Figure 8: Variation of contact resistances with applied force

Current distribution in the anode

Total currents of 8 A and 20 A were applied to the top surface of the anode, and the current distributions were measured as described above. The applied force was greater than the threshold force determined previously. Figure 9 shows the measured contour plots of the current distributions for the two cases. Figure 9 indicates that the two current distributions are similar. Ideally, the total current should not influence the current distribution. The slight differences observed might be due to the difficulty in placing all the contacts at the same places during the two measurements and surface asperities.

In both cases (Figure 9 (a) and 9 (b)), the top left corners have the highest current and the bottom right corners have the lowest current. This might be due to the structure of the anode. The porous regions have a low density and a high electrical resistivity. Higher current preferentially passes from the regions of low electrical resistivity.

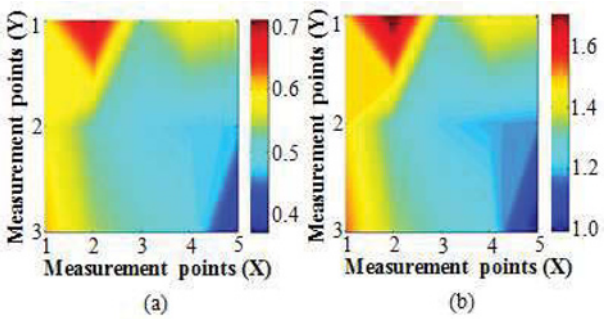

Figure 9: Current distributions for different total input currents of (a) 8 A and (b) 20 A

Electrical resistivity distribution in the anode

Resistivity distribution measurements were carried out for the test anode in order to determine the high and low resistivity regions. The measurement system and the methodology have been previously explained by Kocaefe et al [11]. A schematic representation of the specific electrical resistivity measurement apparatus (SERMA) is shown in Figure 10. A pair of plates containing a certain number of flexible contacts at exactly the same positions to feed current from one anode surface and to receive from the opposite anode surface were used for the measurement. These contacts were connected to a power supply and made a complete circuit. Close to the current contacts, the voltage contacts were installed on both plates to be able to measure the voltage drop at a given position. The current and voltage contacts were distributed over the plate covering the entire anode surface. At multiple points, a current was passed through the anode block from one surface to the opposite surface, and the corresponding voltage drop was measured. The electrical resistivities were calculated across the anode at all the measurement positions.

Figure 11 shows the resistivity distribution measured in the same direction as the one used for the current measurement. There was a difference between the positions of the contacts and the grid sizes of the current and resistivity measurements. During the measurement of the current distribution, the current is applied from the middle of the top surface. On the other hand, resistivity measurements require the feeding of current through a large number of points on the top surface corresponding to each point of voltage measurement at the bottom surface. Therefore, certain discrepancy is expected between these two measurements.

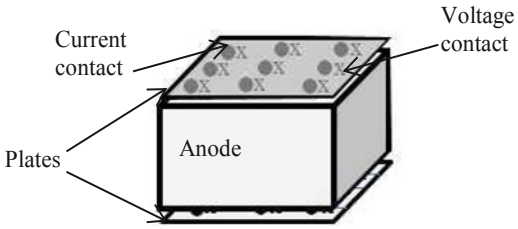

Figure 10: Schematic representation of the specific electrical resistivity measurement apparatus (SERMA) [12]

Normally, the current (Figure 9) corresponding to a low resistivity point (Figure 11) should be high. Thus, a blue region in the resistivity map should correspond to a yellow or red region in the current map, and a red or yellow region in the resistivity map should correspond to a blue region in the current distribution map. The results demonstrate that this relation holds. This shows that the electrical current distribution system gives reasonable results.

As mentioned previously, usually the non-homogeneity of anodes is not taken into account in cell models. The next step is to study the effect of non-homogeneity of anodes on the current distribution. Measurement of the electrical resistivity distribution using SERMA or the optical density distribution by X-ray tomography can help understand the non-homogeneity of an anode. Understanding this effect will permit to better resolve anode non-homogeneity issues and help modelling studies represent the anodes more realistically.

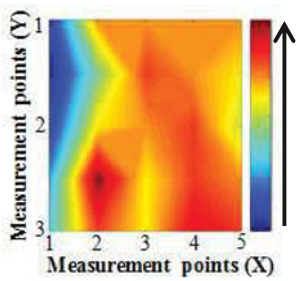

Figure 11: The resistivity distribution measured using SERMA

Conclusions

This project focuses on a thorough understanding of the current distribution in a carbon anode. An experimental measurement system was developed to measure the current distribution in the anode. Preliminary results show that the current distribution obtained is in good agreement with the electrical resistivity distribution measured using the equipment developed by the carbon group of the University of Quebec at Chicoutimi (UQAC). Within the range of experimental error, the current distribution was found to be independent of the applied current and inversely correlated with the resistivity distribution as expected.

The results of this study might help more realistic representation of the current distribution in electrolytic cell models for aluminum production.

Acknowledgements

The technical and financial support of Aluminerie Alouette Inc. as well as the financial support of the National Science and Engineering Research Council of Canada (NSERC), Développement Économique Sept-Îles, the University of Quebec At Chicoutimi (UQAC), and the Foundation of the University of Quebec At Chicoutimi (FUQAC) are greatly appreciated. The authors would like to thank the Fonds de recherche du Québec nature et technologies by the intermediary of the Aluminum research centre – REGAL for their contribution for student participation in this conference.

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CATHODE WEAR IN ELECTROWINNING OF ALUMINUM INVESTIGATED BY A LABORATORY TEST CELL

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Keywords: Aluminum electrolysis, cathode wear, limiting current density, mass transport

Abstract

The lifetime of an aluminum electrolysis cell is mainly determined by the cathode wear, especially for high amperage cells utilizing graphitized carbon cathodes. The cathode wear mechanisms are, however, complex and still subject to debate. Laboratory tests using inverted cell configurations have previously been used to study commercial cathode materials. Here we report on results obtained in the last 5 years for three different commercial cathode materials. The current density applied in the tests was in the range from 0 to 2 A/cm^2 and the rotation speed of the cathode was from 0 to 125 rpm, corresponding to surface velocity 0-19.6 cm/s. The wear mechanisms with respect to current density, limiting current of Al_4C_3 formation and mass transport are discussed. In addition, similarities and differences between wear observed in the laboratory and in industry cells are discussed.

Introduction

The combination of high thermal conductivity, high electrical conductivity and low sodium expansion, compared to traditional anthracitic, semi-graphitic and graphitic carbon materials, make graphitized carbon cathodes the state-of-the-art cathode material in Hall-Héroult aluminum electrolysis cells. The retrofit of old cells to graphite cathodes is also advantageous when smelters gradually raise the cell amperage, and with it the current density. Despite the excellent properties, the graphitized cathode material experiences higher local wear rate compared to the traditional carbon materials. The observed wear rate by industry is: graphitized > graphitic > anthracitic carbon [1].

Cathode wear has become one of the decisive factors which determine the lifetime of the electrolysis cell, especially for the high amperage cell utilizing graphitized carbon materials. The typical observed wear rates are in the range 2-6 cm/year, although parts of the cell see less wear. The uneven wear of the cathode usually gives a cross-section profile resembling a "W" shape or sometimes a "WW" shape [1-4]. It is usually the deepest wear that determines the cell life. To reduce the cathode wear and prolong the lifetime of the cell, it is important to understand the reasons for the uneven wear.

One of the main reasons for the carbon material loss is believed to be the formation, dissolution and transport of aluminum carbide (Al_4C_3) away from the surface [1]. In principle, Al_4C_3 could be formed through both chemical routes described by Reaction 1 and 2 and/or an electrochemical route according to Reaction 3. The chemical formation of Al_4C_3 according to Reaction) and 2 is thermodynamically favorable at the electrolysis temperature [5]. Reaction 1 is however kinetically hindered due to the passive alumina skin present on the aluminum surface as it is likely that the oxygen potential is not low enough in the cell to avoid the oxide skin. Reaction 2 relies on a significant amount of intercalated sodium, which was also argued to be unlikely since highly graphitized carbon has been observed to have highest wear rate but at the same time have minimal Na uptake [1, 6].

$$4Al(l) + 3C(s) \rightarrow Al_4C_3(s) \quad (1)$$
$$\Delta G(960\ ℃) = -147.6\ kJ/mol$$

$$4Na_3AlF_6(l) + 12Na(in\ C) \rightarrow Al_4C_3(s) + 24NaF(l) \quad (2)$$
$$\Delta G(960\ ℃) = -217.4\ kJ/mol$$

$$4Al^{3+} + 3C(s) + 12e^- = Al_4C_3(diss) \quad (3)$$

A tremendous amount of research works have been conducted over the last decades to study the wear mechanisms as well as to establish a "standard accelerated" test method for ranking of commercial carbon cathode materials in terms of wear resistance [7-13]. Although the underlying mechanisms are still under debate, it is generally accepted that the formation of Al_4C_3 is electrochemical in nature. The chemical reactions however, cannot be completely ruled out when fluoride melt is present and/or the Na activity is high.

The Al_4C_3 formed can dissolve and be transported through the metal and bath before being oxidized by the anode gases or at the anode surface. "Fresh" bath should be available in order to continue the wear process. Though the formation of Al_4C_3 is likely to be electrochemical in nature, the wear rate is not proportional to the current density in all current range. Gudbrandsen *et* al.[14] found that the limiting current (i_L) was 0.1 A/cm^2 in their cathodic dissolution experiments at CR equal 1.8 at 960 °C. According to Reaction 3, the limiting current for Al_4C_3 formation can be expressed as Equation 4

$$i_L = 12 \times F \times c_{Al_4C_{3,sat}} \times k \quad (4)$$

where F is Faraday constant equal to 96485 A·s/mol, $c_{Al4C3,\ sat}$ is the saturation of Al_4C_3 in bulk (mol/m^3), and k is the mass transfer constant in (m/s).

Equation 4 demonstrates that the amount of Al_4C_3 formed by the electrochemical reaction is determined by the solubility of Al_4C_3 in bulk and the mass transfer coefficient. The solubility of carbon in aluminum is 105 ppm counted as Al_4C_3 at 960 °C. The solubility in bath can be two orders higher. Ødegård revealed that Al_4C_3 had maximum solubility around 2 wt% at CR=1.8 at 1020 °C and varied with bath composition [15]. In a typical industry bath composition the solubility is about 1 wt%. Empirical expressions for the solubility and mass transfer coefficient for a vertical rotating cylinder electrode are presented in a later section.

The wear rate results reported here are based on an "inverted cell" configuration originally designed by Rafiei *et* al. [16]. The test cell has been utilized by the authors since 2010 and several modifications has been made [11-13]. This "accelerated" test cell was initially designed to explore the possibility to rank commercial carbon cathode materials at standard operating conditions. Recently the method has been used to investigate the cathode wear dependency on several parameters, e.g. current densities, hydrodynamics and rotation speeds, in order to better understand the wear mechanisms.

Experimental

Materials

Three types of commercial carbon cathode materials were used in the wear test, namely high density graphitized (A), graphitized (B) and anthracitic (C). The physical properties of these materials, including density, porosity and electrical resistivity, have been published previously [13]. Cathode A has the highest density and lowest porosity. Cathode B has the highest porosity. The electrical resistivity of cathode A and B are comparable, while cathode C has three times higher electrical resistivity.

Laboratory Test Cell - "Inverted Cell"

A schematic drawing of the "inverted cell" with a vertical rotating cathode is shown in Figure 1. Carbon samples were cathodic polarized during the test, while the crucible was positive charged and acted as anode.

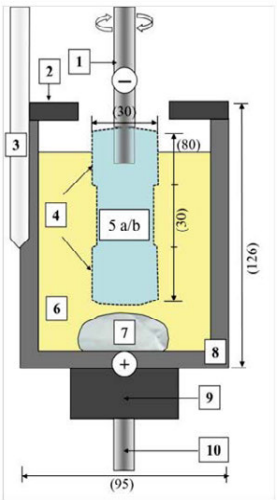

Figure 1. Schematic drawing of the experimental set-up. Dimensions in mm are in parenthesis. 1) rotating cathode connecting rod, 2) sintered alumina lid, 3) thermal couple, 4) Si_3N_4 lining covering both ends, 5 a/b) carbon sample (cathode), a/b represents sample with/without slots, 6) electrolyte, 7) aluminum metal (100 g), 8) graphite crucible (anode) , 9) graphite support, 10) anode lead.

A few high density graphitized samples had slots on the top part of the sample surface, which was used to study the influence of hydrodynamics on the cathode wear. The results have been discussed elsewhere [8]. For a slotted sample, the measurements in this report were taken from the parts without slots.

The initial electrolyte composition is shown in Table I with the cryolite ratio (CR) equal to 2.14. The vertical electrode rotation speeds used were 0, 50, 80 100 and 125 rpm at constant current density (CD = 1 A/cm^2) to study the influence of mass transfer on the cathode wear. A speed of 50 rpm corresponds to a surface velocity of 0.08 m/s, which is a typical linear speed of the bath and metal pad in industrial cells. In addition, the cathodic current density was set to 0, 0.2, 0.5, 1, 1.5 and 2 A/cm^2 at constant electrode rotation speed of 50 rpm. All the tests reported here were conducted for 24 hours at 960 °C. The numbers of samples for each test are summarized in Table II and Table III. Two tests without aluminum present initially were also performed for high density graphitized samples. The detailed test set up and test procedures were given elsewhere [11].

Table I. Electrolyte or bath composition, unit in wt%.

Cryolite/Na_3AlF_6	74.8
Excess AlF_3	12.0
Al_2O_3	8.2
CaF_2	5.0

Table II. Wear tests (number of experiments) at different rotation speeds and constant current density (CD = 1 A/cm^2).

Rotation speed [rpm]		0	50	80	100	125
Graphitized	A	1	7			2
	A-S*	1	4	1	1	1
	B		5			3
	B-S*		1			1
Anthracitic	C		3			3
	C-S*					1

*) S denotes cathode samples with slots.

Table III. Wear tests (number of experiments) at different current densities with constant rotation speed 50 [rpm].

Current density [A/cm^2]		0	0.2	0.5	1	1.5	2
Graphitized	A				7		
	A-S*	2	1	1	4	2	1
	B				5		
	B-S*				1		
Anthracitic	C				3		

*) S denotes cathode samples with slots

After each test, the adhered bath at the cathode surface was carefully removed. Special care was taken to avoid loss of carbon material. The average diameters before and after each test were measured by a caliper and recorded. The wear rate was calculated as the half of this diameter change and normalized to yearly rate. The carbon consumptions were also calculated based on the volume loss and material density by Equation 5

$$c_{loss} = \pi \times h \times (r_{before}^2 - r_{after}^2) \times \rho_c \quad (5)$$

where h is the height of the sample, r_{before} and r_{after} is the average diameter before and after the wear test, respectively, and ρ_c is the density of the sample.

Theory of Limiting Current and Mass Transfer

According to Equation 4, the limiting current for producing Al_4C_3, the main source of carbon materials loss, is proportional to the solubility and mass transfer coefficient. The solubility has a

maximum at CR=1.8 and varies on bath composition. An empirical equation was derived by Ødegård for CR>1.8 as expressed in Equation 6 [15]

$$\log(c_{Al_4C_3}) = 1.041 - 0.3949 \times CR - 1.274 \times 10^{-3} \times (t - 1020) - 0.3028 \times \left(\frac{c_{Al_2O_3}}{c_{Al_2O_3(sat)}}\right) - 2.45 \times 10^{-2} \times c_{CaF_2} \quad (6)$$

where t is temperature [°C], $c_{Al2O3(sat)}$ is the saturation concentration. The concentrations in this equation are in wt%.

In the "inverted" test cell, the rotating cylinder electrode (RCE) was enclosed within a concentric cell. In this case, the critical Reynolds number (Re) for laminar flow is around 200 corresponding to rotation speeds of less than 10 rpm [17, 18]. Dimensionless group correlation has been described as Equation 7

$$Sh = 0.0791 \times Re^{0.7} \times Sc^{0.356} \quad (7)$$
$$Sh = k \times R_i / D$$
$$Re = R_i \times U / \nu$$
$$Sc = \nu / D$$
$$\nu = \eta / \rho$$

where *Sh* denotes Sherwood number, *Sc* denotes Schmidt number, k is mass transfer coefficient, *Ri* is radii of electrode (m), D is diffusion coefficient (m^2/s), U is peripheral velocity (m/s), ν is kinetic viscosity (m^2/s), η is viscosity (Pa·s) and ρ is density (kg/m^3). Rearranging Equation 7, the mass transfer coefficient k can be described by Equation 8

$$k = 0.0791 \times U^{0.7} \times (R_i)^{-0.3} \times \nu^{-0.344} \times D^{0.644} \quad (8)$$

Equations 4 and 6-8 demonstrate that the limiting current depends on the bath composition, physical properties of bath and rotating speeds.

Results

The appearance of the tested cathode materials before and after 24 hours electrolysis at 1 A/cm current density and rotation speed of 125 rpm is shown in Figure 2. The anthracitic sample C showed more uneven wear pattern.

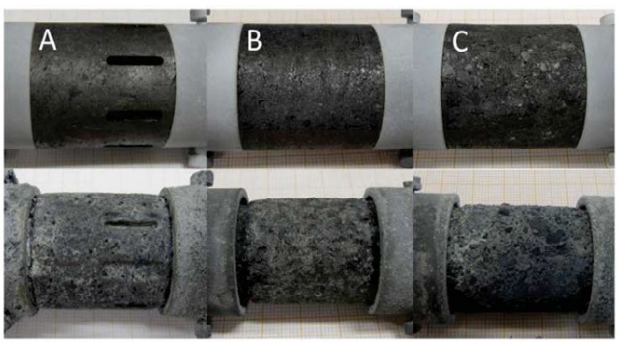

Figure 2. Appearance of tested cathode materials before (top) and after (bottom) 24 hours electrolysis at 1 A/cm current density and 125 rpm. A) high density graphitized, with slots B) graphitized and C) anthracitic.

The average wear rate as a function of rotation speeds for the three commercial cathode materials is shown in Figure 3. The tests were carried out with CR=1 A/cm^2 for 24 hours and the dimensional changes were recalculated to cm/year.

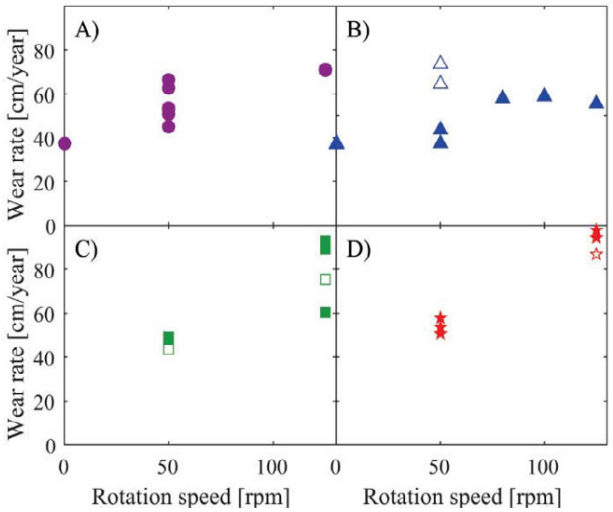

Figure 2. Average wear rates v.s. rotation speeds recalculated to cm/year for three types of cathode materials. A) high density graphitized, B) high density graphitized with slots, open triangle markers represent tests without initial aluminum present in the cell, C) graphitized. Open square and star markers represent the cylinder samples with slots.

The wear rate at stationary bath (no rotation) is significant. This indicates that the natural convection is sufficient to circulate the bath for the formed Al_4C_3 to be dissolved and transported away from the cathode surface and the saturated bath is replaced by "fresh" bath to retain the wear process. With increasing rotation speeds, the forced convection enhances the mass transfer, so that the wear rate increases correspondingly. This could be observed in most of the tests. The rough sample surface after tests made it, however, challenging to obtain accurate measurements and reproducibility is to some degree dependent on subjective interpretation by the person doing the measurements. Consequently, some scatter may be anticipated in spite of equal experimental conditions. The uncertainties due to the measurements have been evaluated. The standard deviations were between 2-6 cm/year after normalization. To avoid confusions in figures, error bars are only shown in Figure 4. The values are representative.

The relationship between wear rate and total cathodic current density for the high density graphitized cathode at constant rotating speed of 50 rpm is shown in Figure 4.

The results clearly demonstrate that there was no wear without polarization, which proves that the cathode wear or the formation of Al_4C_3 is electrochemical in nature. The wear rate is, however, not proportional to the current density. There seems to be a plateau for current densities between 0.2 A/cm^2 (the lowest current applied in the tests) and up to about 1 A/cm^2. The constant wear rate region has the characteristic features of a limiting current, which is in agreement with Gudbrandsen *et* al. [14]. In this case, the limiting current is below 0.2 A/cm^2. The wear rate

increased again in the high current density region, which implied another wear mechanism may need to be taken into consideration.

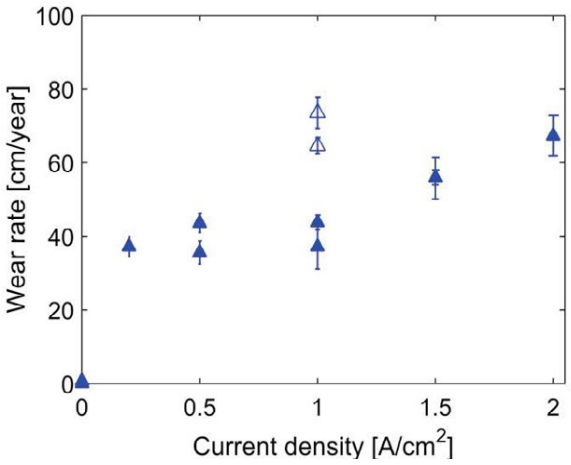

Figure 3. Wear rate as a function of current density for high graphitized carbon material at constant rotation speed of 50 rpm. Open triangle markers represent the tests without adding initial aluminum metal.

Discussion

The general assumption in the calculations provided here is that the wear rate was determined only by the formation, dissolution and transportation of Al_4C_3. Formation of Al_4C_3 was assumed to be the only source for carbon material loss. In this case, according to Equations 4 and 6-8, the limiting current density was proportional to the solubility of Al_4C_3 in the bath and the mass transfer coefficient. To derive the limiting current analytically, the bath composition, solubility and physical properties of the bath are required. Several useful empirical relations are available through previous research [19]. The bath used in the present tests, specified in Table 1, has a viscosity of 2.86×10^{-3} Pa·s and a density of 2056 kg/m^3, which give a kinematic viscosity of 1.39×10^{-6} m/s^2. The solubility limit of Al_2O_3 in this melt was 8.17 wt% and the corresponding solubility of Al_4C_3 derived by Equation 6 was 0.70 wt%. If the cylinder electrode rotated at a speed of 50 rpm, the mass transfer coefficient estimated from Equation 8 would be 5.88×10^{-6} m/s, which results in a limiting current density of 0.068 A/cm^2. The diffusion coefficient in this calculation was chosen to be 9.0×10^{-10} m^2/s. The calculated theoretical limiting current density was much lower than the current densities applied in the reported wear tests.

The amount of electrochemical formed Al_4C_3 could be calculated by Faraday's law according Reaction 9

$$m_{Al_4C_3} = \left(\frac{Q}{F}\right) \times \left(\frac{M_{Al_4C_3}}{z}\right) \qquad (9)$$
$$Q = i \times A \times t$$

where Q is the total charge, M_{Al4C3} is the molar mass, i is the current density, A is the area, t is the time of electrolysis and z=12 according to Reaction 4.

Figure 5 shows the carbon consumption as a function of current density for high density graphitized carbon material at constant rotation speed of 50 rpm after 24 hours electrolysis. Open triangle markers represent the tests without adding initial aluminum metal. Dashed lines are the calculated carbon consumption at limiting current i_L = 0.068 A/cm^2 as derived above.

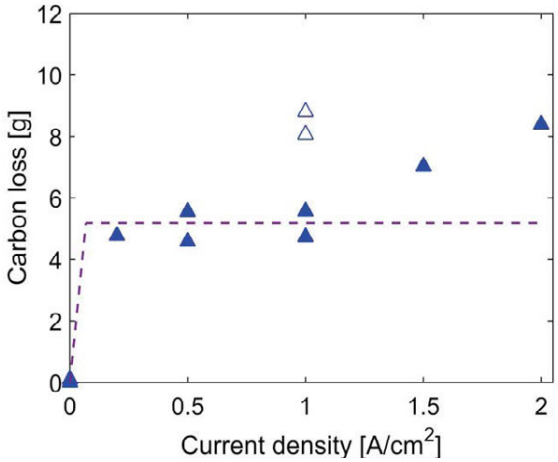

Figure 4. Carbon loss as a function of current density for high density graphitized carbon material at constant rotation speed of 50 rpm. Open triangle markers represent the tests without adding initial aluminum metal. Dashed lines are the calculated carbon consumption at i_L = 0.068 A/cm^2.

There is no doubt that the carbon loss is associated with the current, since no current resulted in no wear. The theoretical carbon consumption is only proportional to the current density until the limiting current is reached and then remained constant with further increased total cathodic current. The calculated carbon consumption fits well the experimental data until the cathodic current density exceeded 1 A/cm^2. The current used to produce Al_4C_3 is limited by the solubility of Al_4C_3 and mass transfer and does not rely on the applied total cathodic current. The limiting current density for this particular bath and rotating speed was close to the estimated value of 0.068 A/cm^2. That is also in agreement with industrial observations: When the local current density was reduced by half, the wear rate did not decrease correspondingly. However, for the areas that the current were cut completely, the wear stopped.

The carbon consumption increases again at the current density region above 1 A/cm^2. The reason for this behavior is not fully understood. The trend of the data in Figure 6 is, however, similar to the Na expansion as a function current density [6]. Na expansion is known to be related to the Na activity in the bath. This similarity implies that the increased Na activity at high current densities may cause increased carbon consumption. The chemical formation Al_4C_3 according Reaction 2 depends on the Na activity. At low current density region, the Na activity is low and the amount of Al_4C_3 formed by this reaction may be limited. In the high current density region, however, when Na activity is high, the amount of Al_4C_3 formed by this reaction may become significant, observed as increased carbon consumption.

The limiting current density depends on the mass transfer in the melt. For a static bath with no stirring or at low rotating speed, the circulation of the bath is dominated by natural convection. The natural convection in the bath is induced by the density difference

between the bath in the vicinity of cathode surface and in bulk, as well as the temperature gradient in the bath. With polarization, the bath in the cathodic diffusion layer becomes more basic compared to the bulk due to increased NaF activities [20]. The corresponding density is also higher for the bath in the diffusion layer. Natural convection is likely with respect to the particular test cell set up, so that the carbon loss even in the static bath is quite significant. Increasing rotating speed over a certain level will eventually change the convection from natural convection control to forced convection control. The forced convection enhances the mass transfer and the corresponding limiting current for Al_4C_3 formation, which leads to increased material loss.

Figure 6 shows the carbon consumption as a function of rotation speed for different types of carbon materials. The current density was kept at 1 A/cm^2 for all tests. The calculated carbon consumptions are also displayed in the Figure. Region 1 and 2 in the Figure represents the natural convection region and the forced convection region, respectively. The calculations imply that the transformation from natural to forced convection control occurs at a rotating speed slightly below 50 rpm. These theoretical calculations do not consider the different properties of carbon materials and therefore should be the same for all materials.

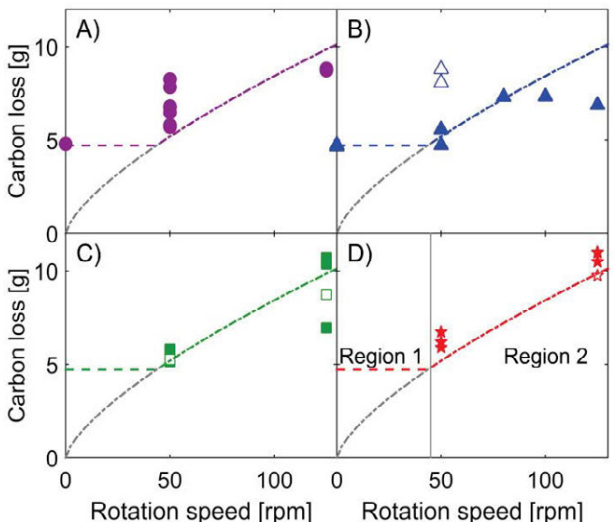

Figure 6. Carbon consumption as a function of rotation speed for A) high density graphitized, B) high density graphitized with slots, open triangle makers represent test without adding initial aluminum, C) graphitized, D) anthracitic. Open square and star markers represent the cylinder samples with slots. For a slotted sample, the measurements were taken from the parts without slot. Dashed and dash-dotted lines are calculated carbon consumptions. Region 1 and 2 represents natural convection region and forced convection region respectively.

The actual wear rates were expected to be higher than the theoretical calculations due to the formation of anode gas bubbles, which may give additional agitation to the bath and therefore faster mass transfer. The test results, despite some scattering, follow the trend of the theoretical calculations, which implies that the wear process is mass transfer controlled. This test method is, however, quite demanding in its practical execution. Especially at high rotating speeds, where the alignment of the vertical electrode is difficult and may cause vibrations that introduces additional uncertainties.

Dependence of Limiting Current on Bath Composition

The solubility of Al_2O_3, as well as the viscosity and density of the bath varies with bath composition, which leads to different Al_4C_3 solubility and mass transfer coefficient. The limiting current density and the amount of carbon loss due to Al_4C_3 formation will then change correspondingly. Figure 7 illustrates these changes as a function of the CR ratio in the range 1.8 and 4, i.e. from acidic bath to basic bath.

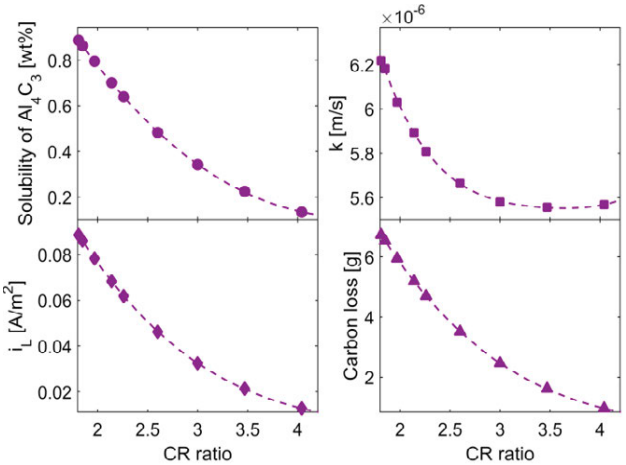

Figure 7. Solubility of Al_4C_3, mass transfer coefficient, k, limiting current density (i_L), and carbon loss as a function of bath CR ratio. Caron loss is calculated based on 24 hours electrolysis in the present "inverted cell". Markers represent the calculated points and the dashed lines are polynomial fitting curves.

The calculations demonstrate that increasing the acidity of the bath will increase both Al_4C_3 solubility and mass transfer coefficient (for mass transfer coefficient valid from CR less than around 3.5), and therefore increase the limiting current density and carbon loss. The carbon loss dependency on CR is significant. For example, increasing the excess AlF_3 from 5 wt% to 10 wt% will increase the carbon loss by 14 %. This calculation is based on the wear test set up, where the carbon cathode is exposed directly to the bath and without any aluminum protection layer.

In fact, when the smelters increase the cell amperage gradually, many other improvements are also introduced to the cell to optimize the operation. For example, the amount of excess AlF_3 is increased for improving current efficiency. Despite some advantages, increased bath acidity will increase the solubility of Al_4C_3, which may contribute to increased carbon loss as discussed previously.

Cathode Wear in Laboratory Test Cell and in Industrial Cells

The cathode wear process in the laboratory and the industry cells is similar, which is mainly due to the formation, dissolution and transport of Al_4C_3. The measured wear in the test cell is, however, more than 10 times higher than in the industry with respect to the graphitized carbon cathode, even in a stationary bath. The main difference between the two cells is that in the test cell, the cathode is exposed to fresh bath, whereas in the industrial cell, the cathode

is to a much larger degree protected by the aluminum pad. The Al_4C_3 solubility is round 0.70 wt% and 0.01 wt% in the bath in the test cell and liquid aluminum, respectively. According to the observed wear rates in the industrial cells, the aluminum pad is, however, not likely the main source for dissolving and transporting the formed Al_3C_4 in the wear process. Sludge, rich in bath and alumina, has been reported present on the carbon cathode surface, most likely moving with metal pad. The bath in sludge has been found to be less acidic than the bulk, and more or less neutral. It has been also suggested that the cathode is covered by a thin film of bath with similar composition as the sludge. If neutral sludge were the main source for dissolution and transport of Al_4C_3, the carbon loss would be around 45 % lower than that in the test cell environment.

Both the test results and theoretical calculations have demonstrated that the cathode wear is strongly dependent on the local bath chemistry and mass transfer condition. The uneven wear along the cathode surface in industrial cells is actually expected due to the dynamic local environment. The local current density is also important in terms of magneto-hydrodynamic driven (MHD) flow which is related to mass transfer; and Na activity which is related to chemical formation of Al_4C_3 by Reaction 2.

Conclusion

One of the main reasons for the cathode material loss is the formation, dissolution and transport of Al_4C_3. The formation of Al_4C_3 is electrochemical in nature. The wear process is mass transfer controlled. The material loss is proportional to the supplied total cathodic current density only up to a limiting current determined by Al_4C_3 solubility and mass transfer. The mass transfer coefficient for the test cell running at 1 A/cm^2 and a rotating speed of 50 rpm is close to 5.88×10^{-6} m/s, and the limiting current in this case is around 0.068 A/cm^2, which is less than 10 % of the total supplied current. It is the limiting current that determines the amount of material loss. The rotating speed and bath chemistry have a great impact on this. At high current densities, when the Na activity is high, the chemical formation of Al_4C_3 by Reaction 2 cannot be neglected, which might be a plausible reason for the increasing amount of carbon loss at current densities above 1 A/cm^2.

Acknowledgement

The present work was carried out in the project “Durable Materials in Primary Aluminium Production” (DuraMat), financed by the Research of Norway, Hydro Primary Metal Technology, Sør-Norge Aluminium, and Elkem Carbon. Permission to publish the results is gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

COPPER BARS FOR THE HALL-HÉROULT PROCESS

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Keywords: Copper, Collector bars, Productivity increase, Energy saving, Magneto-hydrodynamic, Cell design

Abstract

The use of copper inserts in steel collector bars has proven to be an efficient way of improving the Hall-Héroult process. It helps at decreasing the specific energy consumption and improving the cell productivity. A further step is presented for which only copper is used inside the carbon cathode. The feasibility was modeled, tested at laboratory scale and measured in a real cell. The issues, advantages and potential of using 100% copper bars as current collectors inside the carbon blocks are discussed. Modeling results are validated by measurements.

Introduction

Copper has been proposed for many decades for improving the Hall-Héroult process [1]. A number of patent publications have proposed different approaches for minimizing the voltage drop from the liquid metal to the end of the collector bars. The idea is to use a high conductive material as a complement to the existing steel collector. A number of patents disclose solutions using copper inserts inside steel collector bars. Reference [2] splits the cathode and the collector bars in sections. Reference [3] uses high conductive materials inside the collector bars. References [4, 5] present the use of copper in combination with steel collector bars. Today there are a number of smelters using copper inserts in the collector bars but none report about the actual results as it can give a real competitive advantage.

On the one hand copper helps at decreasing the cathode voltage drop (CVD) and on the other hand it helps at improving the cell magneto-hydrodynamic (MHD) state. If copper has not made a breakthrough up until recently, it must be mainly linked to the misunderstanding on the global thermal-magneto-hydrodynamic effects. Indeed, copper must be designed to decrease the CVD but also to realize the right heat extraction through the collector bars, the right isotherms in the cathode and the right current density in the liquid metal. Therefore the full thermal-electrical and magneto-hydrodynamic effects must be analysed. A correctly designed cathode leads to significant advantages [6]. In order to achieve the desired current distribution in the liquid metal, collector bars must be electrically insulated on some areas. The insulated areas not only depend on the collector bars parameters such as height, width, length but also on the cathode dimensions and properties. The other important parameter is the pressure between collector bar and carbon block at operating temperature. The combination of all factors makes it impossible to design a solution without referring to modeling work. One way of decreasing the complexity of using copper inserts with their various designs is to consider only copper inside the cathode [7]. Copper is chemically stable with carbon and its melting point is 1085°C. Copper inserts are usually imbeded in the steel bars which are further implemented in the carbon cathode by casting cast iron around them. This requires the productionof liquid cast iron and i pre-heating the cathode before pouring the cast iron. Furthermore, at the end of the cell life, it becomes very difficult to recover the valuable copper as it is fused together with the steel bar. The authors have studied a solution that should remove the costly operation of casting in the steel bars using cast iron, and that would allow recovery of at least e95% of the copper. So how could we remove the need for cast iron and stop the fusion of copper with the steel bar?

The concept is patented [7] and consists in using only copper inside the cathode and couple the copper bar to a steel bar outside the cathode. One can not use copper over the full length as the temperature would be too high at the end of the bar. It would not be possible to connect the copper to the aluminium flexes and heat losses would be too high.

This paper presents the approach to validate the concept. There were many open questions and following had to be answered:

- what is the contact resistance between copper and carbon in operation conditions?
- will copper melt inside the carbon block if the liquid aluminium in the cell will reach 1100 °C?
- where and how should the copper bar connect to the steel bar?
- what is the CVD when using a copper bar?

In order to answer these questions tests were performed with small samples in a laboratory at room temperature and up to 900°C. A test was performed in a real electrolysis cell with in situ measurements. All experiments were supported by modeling work to predict the electrical potential and temperature fields.

Laboratory tests

Figure 1 shows a sample of graphite cathode that was used to evaluate the electrical contact resistance between copper and carbon as function of the temperature, pressure and surface conditions.

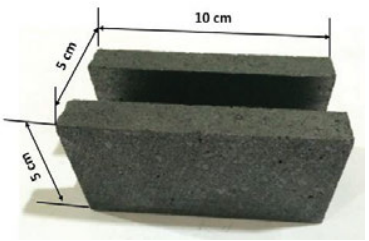

Figure 1: Graphite cathode sample

To be as close as possible to the situation existing in the cell, the

sample was heated at 900°C before performing measurements. Figure 2 shows the box used to heat the sample at 900°C. The sample was covered with coke granules to create a reducing atmosphere.

Figure 2: Box at 900°C containing the sample

Figure 3 shows one carbon sample together with a copper element at 800°C. A current of 10 amperes was injected at one end leading to a current density of 1 A/cm^2 between the copper element and the carbon sample. The voltage was measured simultaneously. The comparaison of measurement to modeling results allows to determine the electrical contact resistance.

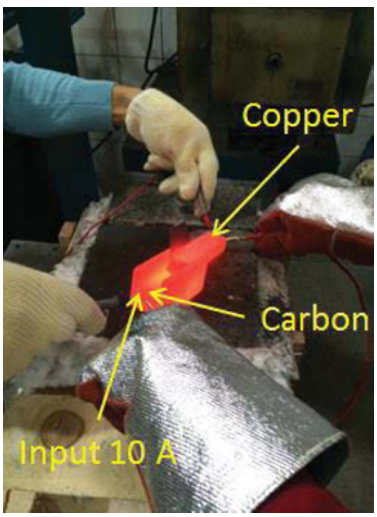

Figure 3: Carbon sample and copper at 800°C

A second experiment consisted in registering the voltage over time while heating the sample from room temperature to 900°C. Figure 4 shows where the electrical potential was recorded during the heating process at four different locations (A, B, C, D).

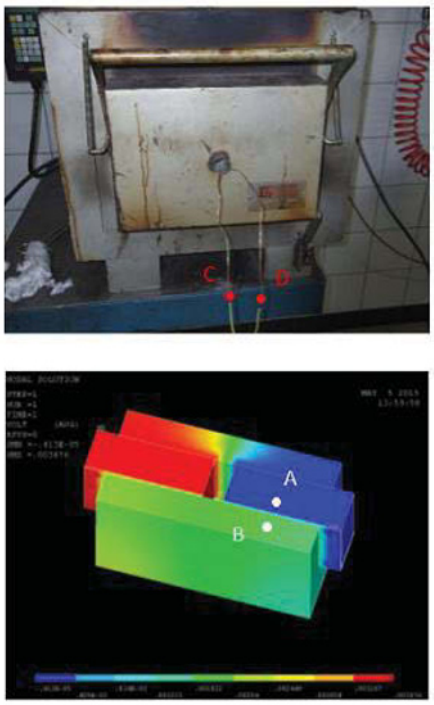

Figure 4: Measurement locations of the electrical potential

Figure 5 shows the voltage between points A and B as a function of temperature. The voltage is composed of the voltage due to the carbon and copper elements and to the contact over-voltage. The decrease in voltage is due to the change of pressure as copper expands more than carbon.

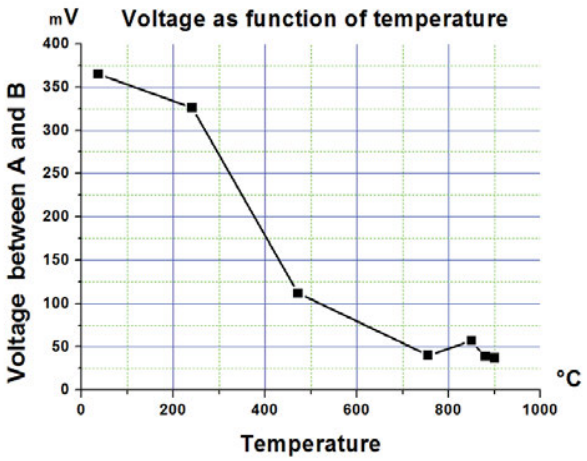

Figure 5: Voltage over the sample as a function of temperature

In order to separate the impact of temperature from the impact of pressure, measurements were performed at room temperature while varying the pressure in a hydrostatic press as shown in Figure 6.

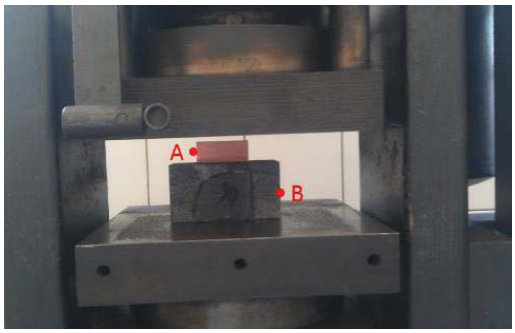

Figure 6: Contact voltage measurements as a function of pressure at room temperature

Figure 7 shows the voltage between point A and B (copper, carbon and contact resistance) as a function of pressure at room temperature. As the electrical contact resistance is very sensitive to the status of the surface, two type of surface treatments were considered for the test. Obviously increasing pressure decreases the electrical contact resistance. The surface "treatment" plays also an important role.

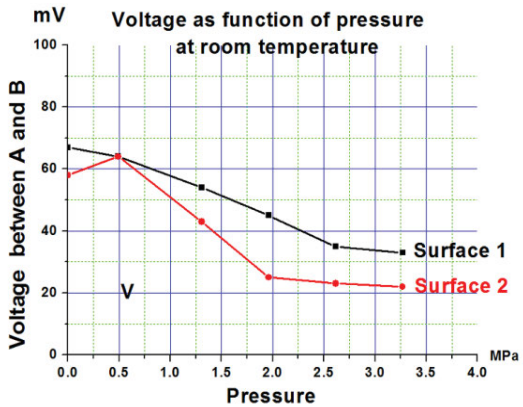

Figure 7: Voltage between A and B as a function of pressure for two type of copper surface treatment

By using these measurements and numerical simulation the electrical contact resistance between copper and carbon as function of temperature and pressure can be determined. The good news is that it corresponds to less than 30 mV in standard operation conditions (about 850°C, 3MPa, 1 A/cm2) if the surface conditions are realized.

Before going to an industrial test, numerical simulation was performed to determine the copper bars dimensions, the steel bar dimensions, the expected CVD, temperature and heat loss.

Numerical simulation

First a 3D slice model of a cell (Figure 8) including two collector bars and one anode was used. In a second stage, when the heat balance was satisfied, a full 3D cell model was used to determine the metal velocity field, metal deformation and cell magneto-hydrodynamic stability.

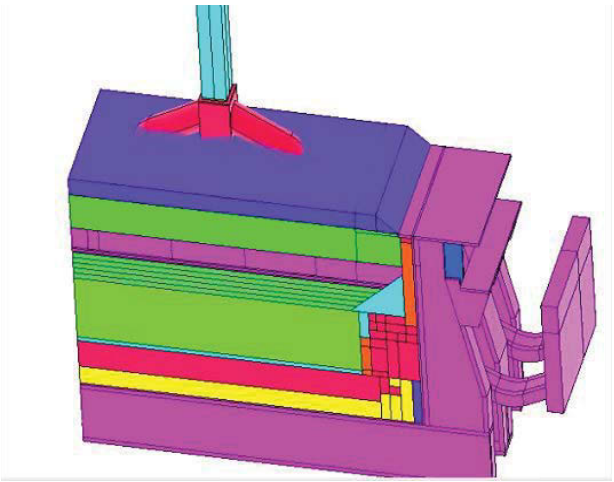

Figure 8: 3D-slice, two collector bars, one anode.

By using a 70 mm X 70 mm copper bar inside the cell connected to 45 cm of steel bar at the end very interesting results came out assuming 20 mV contact voltage drop.

The voltage from liquid metal to the aluminium flexes (end of the steel bar) was found to be 168 mV without any contact voltage. It is equal to 347 mV with the standard steel bars. Assuming 30 mV contact voltage between copper and carbon and zero contact between copper and steel this represents a potential saving of 149 mV. Figure 9 shows the calculated electrical potential. It is interesting to see that the equipotential line is horizontal. This means that the current is vertical in the liquid metal.

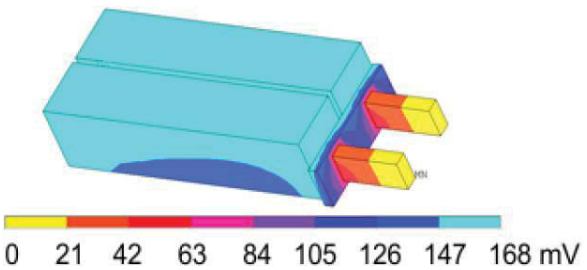

Figure 9: Calculated CVD liquid metal to collector bar (V) (no electrical contact resistance)

One fear was that the copper might reach its melting temperature which is 1085°C. In this respect the collector bar temperature was calculated assuming that the liquid metal in the cell is 965 °C and then 1100°C at steady state. Under such conditions the copper bars will never reach the melting point as the temperature gradient is typically 15 °C between the cathode surface and the collector bar. This value depends on the properties of the cathode block. Figure 10 shows the expected temperature under operation conditions when the liquid metal is 965°C. It can be observed that the steel part is necessary to decrease the temperature from 800 °C to 200°C at the end of the bar while the copper bar (thin part) varies from 950°C to 800°C from the center of the cell to the steel bar.

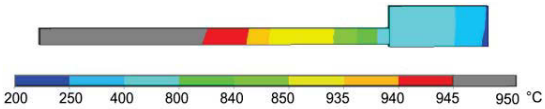

Figure 10: Temperature of the copper bar assuming 965°C liquid metal in the cell

Another important factor is the thermal expansion. Copper expands typically 3 to 4 times more than carbon. Therefore, thermo-mechanical stresses develop during the startup phase of the cell that could damage the cathode. Figure 11 shows how the cathode and copper bar expand when the temperature is changed from room temperature to the operating temperature. The length is increased by 24 mm towards the outside of the cathode and 5 mm towards the inside.

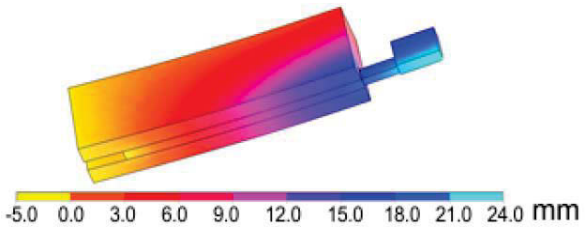

Figure 11: Cathode and copper bar thermal expansion from room temperature to operating temperature

Industrial test

Based on the laboratory and modeling results a test was realized in an industrial electrolysis cell. Figure 12 shows the copper bar connected to the steel end.

Figure 12: Copper bar coupled to a steel bar

It was decided to use a threaded connection to connect the copper bar to the steel bar, as shown in Figure 13.

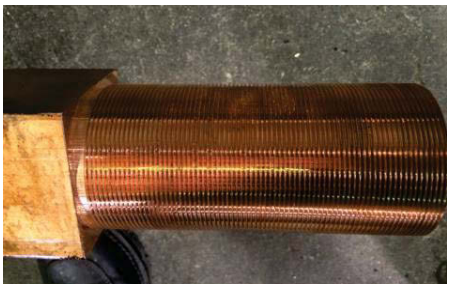

Figure 13: Copper thread

In order to avoid any risk of cracking the carbon cathode due to mechanical stresses generated by the thermal expansion of the copper bar, a gap filled with ramming paste was used between the copper bar and the carbon cathode. The copper bar was insulated electrically over some distance at the end of the block by an insulating cloth as shown in Figure 14.

Figure 14: Block, ramming paste and electrical insulation

The bars were equipped with electrical potential and temperature sensors at two locations as shown in Figure 15.

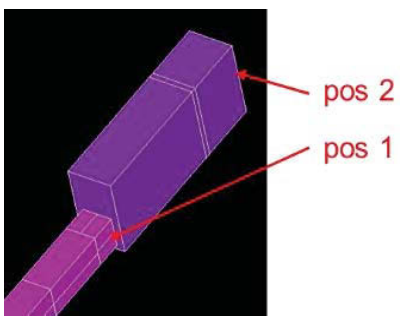

Figure 15: Voltage and temperature measurements

It was of high interest to record the temperature and the voltage at these locations as it would indicate any risk of high

temperature for the copper bar and tell us about the electrical contact resistance between the copper and the steel bar of the threaded connection. Both could be compared to the modeling results. In fact, very soon after the cell startup, a higher temperature was measured at position 1 on the copper bar. Figure 16 shows the copper temperature for two bars over a one month period including the startup period. The maximum temperature at the end of the copper bar during the startup phase was 900 °C. It reached stability after 10 days at 810 °C. When the temperature reached 900°C, the bath temperature in the cell was close to 1100°C. One thermocouple was lost after 9 days. Modeling predicted 800 °C which is in line with the measurements. However very soon the higher measured temperature was attributed to heat produced at the threaded contact Cu-Steel. This was confirmed by voltage measurements from the end of the copper bar at position 1 to the end of the steel bar at position 2.

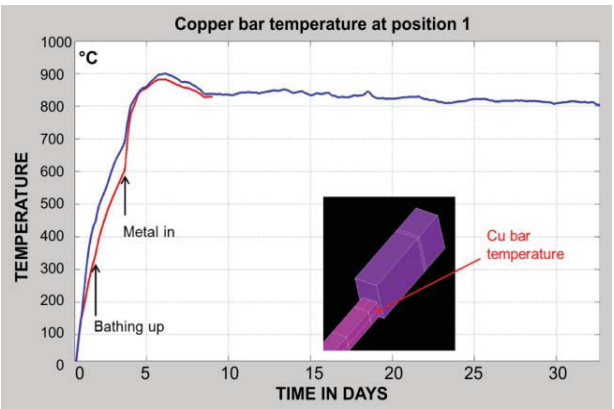

Figure 16: Temperature at the end of the copper bar during the first 30 days

The electrical potential was recorded every second and is shown for the first 147 days in Figure 17. A drastic increase of the voltage can be observed for the two measured bars after 40 days. The initial voltage was already higher than calculated from the numerical model.

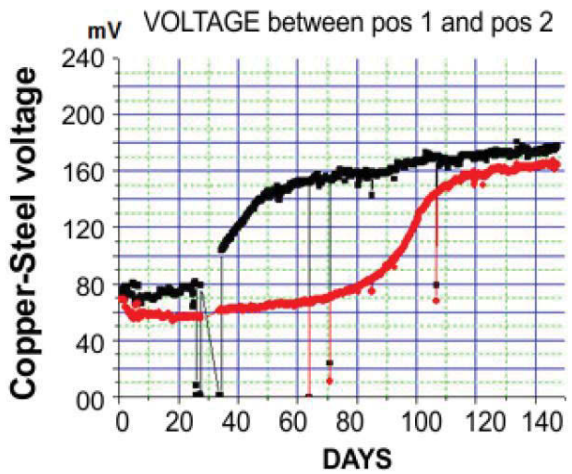

Figure 17: Voltage drop from position 1 to 2 for two coll. bars

The targeted electrical resistance from the end of the copper bar to the end of the steel bar was 10 μΩ. Figure 18 shows the evolution of the electrical resistance which started around 40 μΩ and reached up to 220 μΩ for one bar after 147 days. It confirms that the threaded connection cannot be used or was not realized as it should. In other words the threaded connection did not work as expected, most likely due to copper oxidation or other type of corrosion at the copper surface inside the steel bar! This affects the CVD and is not acceptable. As a consequence, a new type of connection was designed.

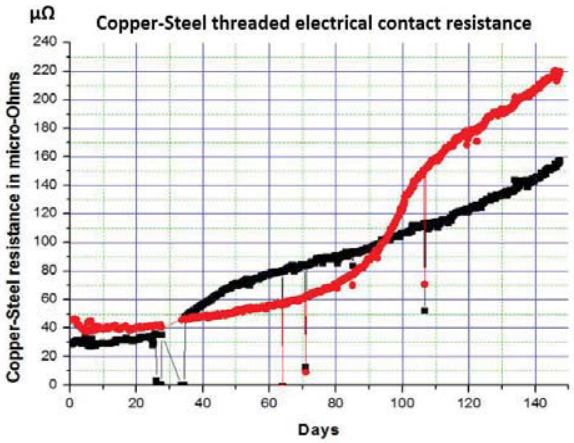

Figure 18: Cu-Steel electrical resistance

A second parameter affected the CVD. Indeed, the use of ramming paste is not optimal for the electrical contact between the copper and the cathode. Soon after startup, the CVD measurements led to an average electrical resistance of 120 μΩ. This is much higher than the numerical model results of 28 μΩ. In fact it is twice more resistive than the existing conventional steel bar showing 60 μΩ. Obviously this design is not recommended. However the experiment showed that copper can be considered as a candidate for extracting the current from the cathode.

Next steps

Based on these results, a new cathode design was realized using two copper bars directly imbedded in the cathode and connected to a steel bar as shown in Figure 19.

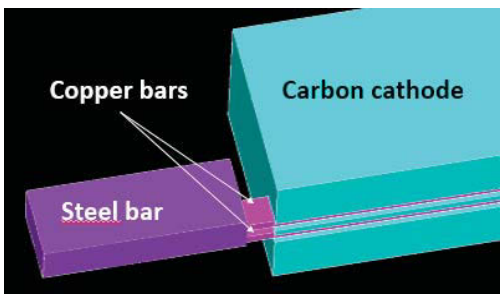

Figure 19: Copper bars imbedded in carbon cathode

Using appropriate dimensions and contacts, this solution predicts a CVD of 190 mV which is more than 150 mV lower than the solution using steel bars. At the same time modeling predicts that

the current density at the cathode surface is very smooth ranging from 0.7 A/cm^2 to maximum 1 A/cm^2 as shown in Figure 20.

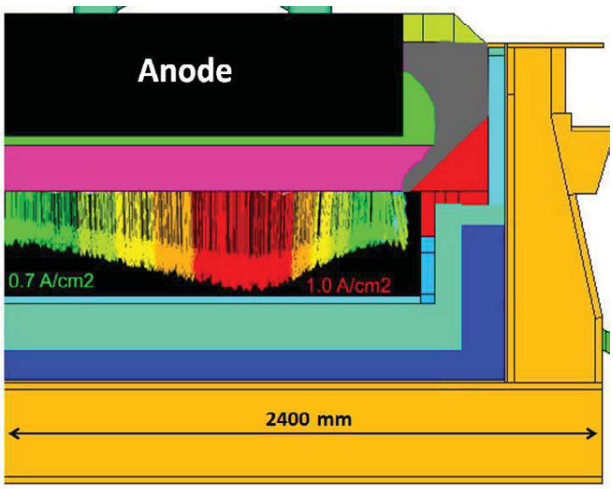

Figure 20: Current density at cathode surface

Another benefit of using copper bars is the impact on the magneto-hydrodynamic cell stability. Figure 21 shows the current density inside the liquid metal. There is quasi no horizontal current flowing towards the cathode edge. In other words the current is almost vertical and this helps at decreasing the ACD while keeping a stable cell.

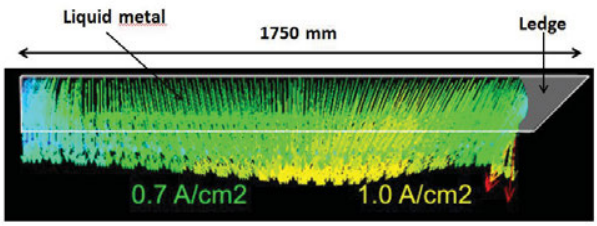

Figure 21: Current density inside the liquid metal

Therefore a new industrial test needs to be performed using this new cathode design.

Conclusions

A new concept has been patented and is presently implemented in a cell. The first industrial test failed due to the poor electrical contact of the threaded connection between copper and steel. The use of ramming paste as interface between the copper and the cathode was most likely also a weakness. The positive result is that after 150 days the copper is still conducting current and the decrease of intensity is related to the threaded connection. Laboratory tests permitted to identify important parameters such as the pressure between copper and the carbon cathode. Modeling work confirmed possibilities of using copper bars inside the cathode leading to very low CVD and improved MHD.

The measurements showed that:

- The contact voltage between copper and carbon must be lower than 30 mV at 850°C with 1 A/cm2 and about 3MPa pressure
- A threaded connection betweem copper and steel is most likely not a solution
- Ramming paste interface between the copper bar and the carbon cathode is not very favorable
- Copper is not affected in the cathode even when the temperature is reacing 1100 °C in the liquid metal

As comment let us mention that there are numerous potential advantages when using copper bars:

- No need to preheat the cathode
- No need of cast iron
- Very easy to recycle 95% of the copper
- Very low CVD
- Improved cell MHD stability

This must lead to:

- Lower cell specific energy consumption
- Longer cell life (less electro-erosion)
- Deeper cell cavity (low collector bars)
- Lighter cathode (cheaper)

Acknowledgments

Many thanks go to R&D Carbon management who helped us by allocating the necessary facilities for the laboratory measurements.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MODELING GRAVITY WAVE IN 3D WITH OPENFOAM IN AN ALUMINUM REDUCTION CELL WITH REGULAR AND IRREGULAR CATHODE SURFACES

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Keywords: Modeling, irregular cathode, cell stability, gravity wave, OpenFoam

Abstract

In recent years extensive modeling work has been done to assess if the usage of an irregular cathode surface does or does not increases the MHD cell stability.

So far, 2 types of studies have been carried out: full 3D steady-state analysis and 2D shallow layer dynamic analysis. Cell stability being a transient phenomenon, steady-state results are not providing any direct answer to the question.

2D shallow layer dynamic analysis can directly answer the question, but unfortunately, irregular cathode surface introduces a third dimension to the flow, so this type of 2D analysis is not the best suited to analyze this type of 3D flow problem.

The current work presents a new way to analyze the problem and answer the question. Lateral gravity waves have been simulated in a 3D cell slice model using VOF formulation in OpenFoam. Results obtained for cells using regular and irregular cathode surfaces are compared.

Introduction

Irregular cathode surface technology is still the subject of research in China where it is still quite popular. The most recent Chinese paper known to the authors on the subject was published in Metallurgical and Materials Transaction B in 2014 [1].

That paper presents a very detailed 3D model based on ANSYS and CFX solvers. Four steady-state solutions are presented with and without irregular cathode and with and without considering the effect of the gas release under the anodes.

In addition to these four 3D steady-state solutions, two 2D transient solutions were presented with and without irregular cathode that model only the gas release.

When comparing the two 2D transient solutions, no major difference in the global deformation or evolution of the bath/metal interface can be identified (see Figure 16 of [1]).

When comparing the two 3D solution of the conventional cathode with and without the effect of the gas release, a major difference in the local deformation of the bath/metal interface in the small channel between anodes can be identified (comparing Figure 14 a) with Figure 28 a) of [1]).

This local effect is far less intense in Figure 28 b) when compared with Figure 28 a) for the irregular cathode which seems to contradict what is observed in Figure 16.

In any cases, as presented in last year review [2], it is the opinion of the authors that a transient cell stability analysis is required to make any prediction of the impact of irregular cathode surface technology on the cell stability as bath/metal interface wave dynamic is a time dependant phenomenon.

As demonstrated in [2], MHD-Valdis is the perfect tool to model and hence study MHD driven cell stability in the case of regular flat cathode surface as the model only represents and solves the key physics required, nothing more, which make MHD-Valdis an extremely efficient model.

One of the key simplification in MHD-Valdis solver is the use of the 2D shallow layer CFD model to solve the bath and metal flow. Yet, as Figure 12 b) of [1] clearly demonstrates, for cells using irregular cathode technology, this simplification is no longer valid as the flow is now fully 3D in nature.

MHD-Valdis also does not consider the impact of gas bubble release on the bath flow. This was demonstrated to be quite a valid simplification as it is clear that the dynamic of the gas bubble release under the anodes, which strongly affects the global bath resistance, is decoupled from the cell stability problem. Otherwise no cell stability model developed up to now would be valid.

In [2] and in previous studies presented before that [3,4,5], MHD-Valdis could not clearly show the impact of irregular cathode technology on cell stability, maybe because of the 2D flow structure simplification.

Yet it is not by mistake that results of a full 3D transient MHD driven cell stability analysis was not presented in [1]. Even nowadays, CPU resources are still too sparse and too expensive for such an analysis to be carried out without a huge R&D budget.

The need for a third modeling approach

If reducing the metal flow to a shallow layer representation is not a justified simplification in the case of irregular cathode surface and if solving a full 3D transient cell stability problem is still not practical, clearly there is a need to find a new modeling approach.

There are two parts to the MHD driven cell stability issue: the energy source part coming from the presence of the variable Lorentz force in the metal pad as explained by Urata and Davidson among others [6,7] and the energy dissipation part coming from viscous damping.

Clearly the aim of irregular cathode surface technology is to affect this second energy dissipation part by increasing the viscous damping in the metal pad.

Yet for a bath/metal interface wave to move around, not only the metal must be displaced but also the bath. Furthermore, since the ACD layer thickness is much less than the metal pad thickness, the required bath flow velocity needs to be much greater than the required metal flow velocity.

So clearly, the viscous damping in the bath is very important and must be considered in the analysis which will be the case in a 3D analysis of the damping rate of a gravity bath/metal interface wave in an aluminium reduction cell.

This approach reduces the difficulty of a study of the viscous damping effect of irregular cathode surface which is perfectly valid as this is the key effect that needs to be investigated.

Furthermore, if we choose to study a lateral gravitational wave, the geometry of the problem can be reduced to a cell side slice as presented in Figure 1.

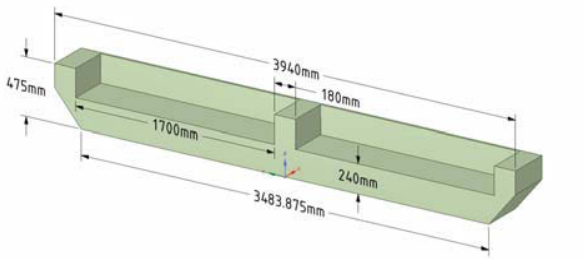

Figure 1: Geometry of the cell side slice model

OpenFoam and the free interface wave

Even reduced both in terms of physic and geometry, we are still left with a quite difficult question to solve namely a transient 3D multiphases (three in this case: metal, bath and air) flow. Very few codes are able to cope with this problem, the open source code OpenFoam being one of them.

OpenFoam has quickly become a very popular code in many fields such as marine applications due to its free surface modeling capabilities [8]. Its free surface capabilities are comparable to other VOF solvers like CFX. A direct comparison between experimental, OpenFoam and CFX results for a free surface study are presented in [9].

The free interface wave between a gas and a liquid or between two non-miscible liquids in a closed rectangular container has been extensively studied experimentally, as can be seen per example in Figure 2, a reproduction of Figure 16 in [10]. As explained in [11], it is very difficult to measure experimentally in a reproducible manner the damping rate of such a gravity driven, viscous damping wave problem.

Recently, OpenFoam has been quite successfully used to model this type of free surface wave topic, per example [12] is a Ph.D. thesis on the subject.

The problem of modeling the damping of a gravity bath-metal interface wave in an aluminium reduction cell is a very similar question with the extra difficulty that there are immersed anodes in the top liquids and that it will be very difficult to get physical measurements for model validation.

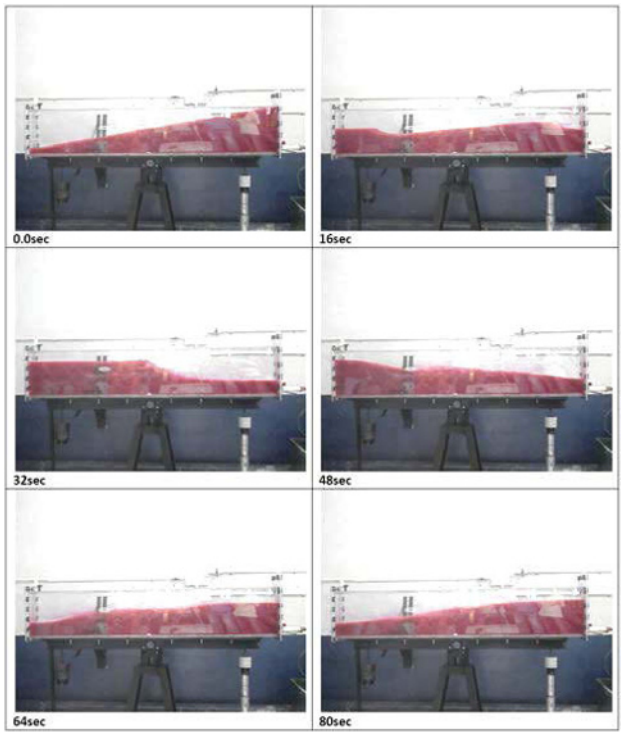

Figure 2: Experimental results for a free interface wave motion between two liquids in a closed container (Figure 16 in [10])

Base case model setup

The geometry of the base case model, with regular flat cathode surface was already presented in Figure 1. The model depth extends from a front frictionless symmetry plane located at half the anode width to the back frictionless symmetry plane located at half width of the small channel between two anodes. The length of the model is typical of a cell cavity width minus a 10 cm uniform ledge thickness in both ends: 3.94 m. The height of the model is enough cavity depth to leave room for 20 cm of metal pad thickness, 20 cm of bath thickness and 7.5 cm of air on top.

The model mesh is presented in Figure 3. The mesh is fine enough to resolve fairly well the boundary layer problem close to the solid surfaces (cathode, ledge and anodes). It is constituted of hexagonal finite volumes of approximately uniform size. The mesh also perfectly aligned with the initial bath-metal position in order to have a perfectly smooth initial position of that bath-metal interface.

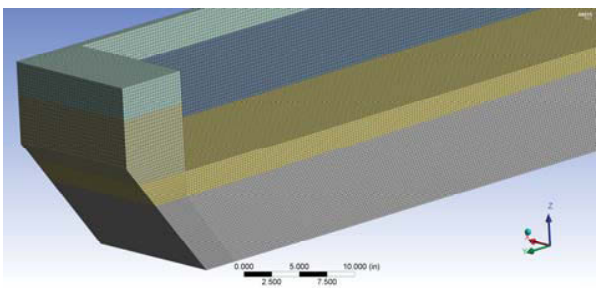

Figure 3: Mesh of the cell side slice model

The model contains 1,180,980 hex finite volumes with an orthogonal quality of 0.77. It uses a k-ω SST (shear stress transport) turbulence model because of its demonstrated capability to well predict drag [13].

The bath and metal properties utilized where obtained using Peter Entner's AlWeb application [14]. A quite standard bath composition has been selected, see Figure 4.

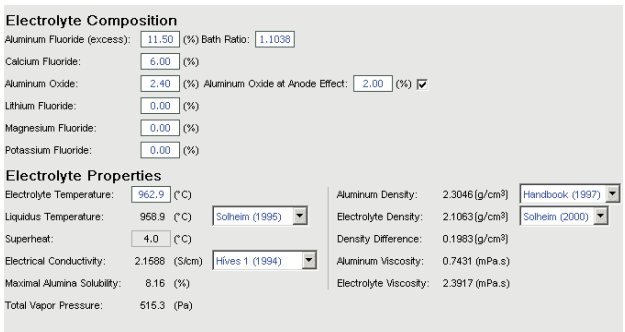

Figure 4: Bath and metal properties from AlWeb

The transient evolution is starting from a resting position having a sloped bath metal interface of -2 cm on the left side to +2 cm on the right side as shown in Figure 5.

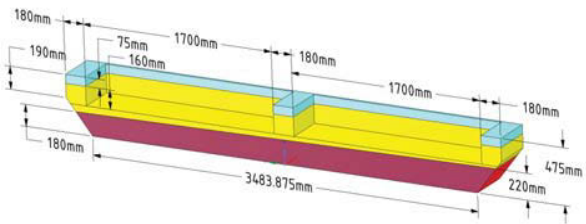

Figure 5: Initial bath-metal interface position

The transient evolution of the system from that starting point is calculated using an explicit solver available in OpenFoam 2.3.0 [15], the multiphase Euler solver using a maximum courant number of 0.05 and a maximum time step of 0.002 seconds.

The transient evolution of the system was calculated for a total of 60 seconds which is more than 1 total period of the lateral wave oscillation. The calculations were performed using a Dell 28 cores Xeon ES-2697 V3 computer having 128 GB of RAM at its disposal. That computer took about 30 CPU hours to solve that problem using all 28 cores.

Base case model solution

Figure 6 is showing the position of the bath-metal interface position every 15 seconds. That gravity lateral wave happens to die almost completely in a single period.

The maximum velocity is reached a little before the 15 sec. mark. Figure 7 is showing the velocity field of the front plane. The solver assumed continuity of the velocities at the interfaces so the solution shows that the bath flow drags the top layer of the metal so the flow reversal is occurring in the metal pad and not at the bath-metal interface. The maximum bath velocity is about 3 cm/s.

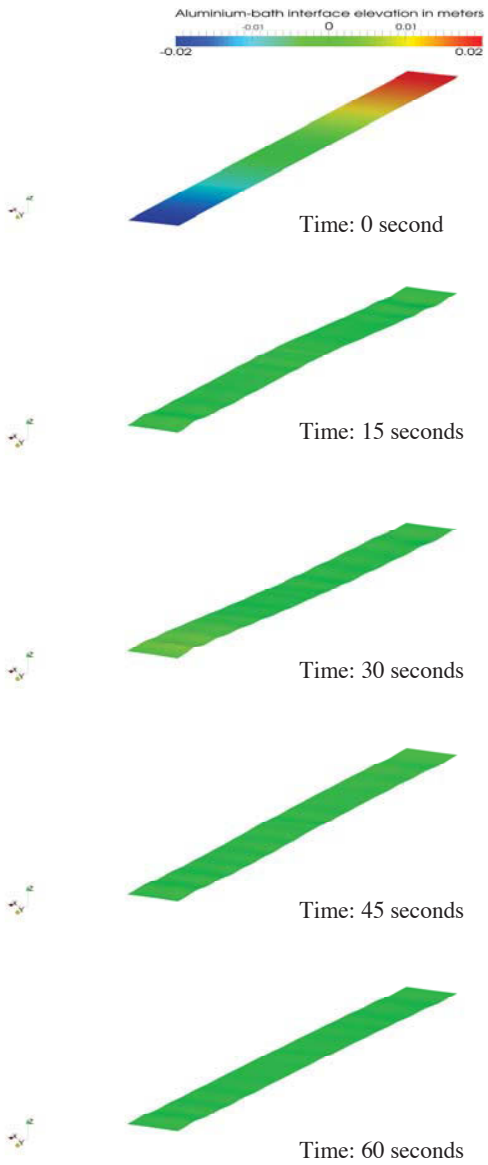

Figure 6: Position of the bath-metal interface every 15 seconds from 0 to 60 seconds

Figure 8 is showing the velocity field after 60 seconds, indicating that the wave has been already almost completely damped down. Figure 9 illustrates the turbulent viscosity after 15 seconds. Since the laminar viscosity of the metal is 3.224e-7 m^2/s and the maximum turbulent viscosity 4.66e-4 m^2/s, the maximum turbulent viscosity is 1447 time the laminar viscosity.

Irregular cathode surface case model setup

The geometry of the irregular cathode surface case model is presented in Figure 10. The geometry of the cathode surface has been changed when compared to the base case model. But the mass of metal, the mass of bath and the 4 cm ACD have remained the same.

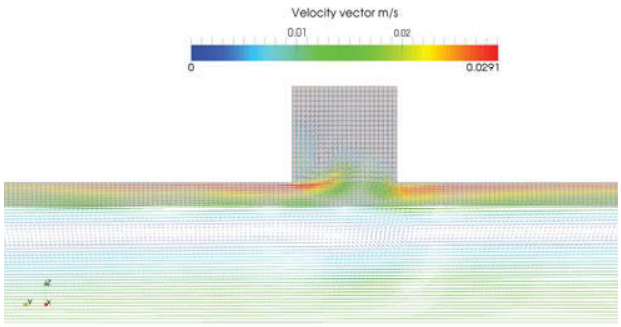

Figure 7: Velocity field after 15 seconds (bath region has gray background)

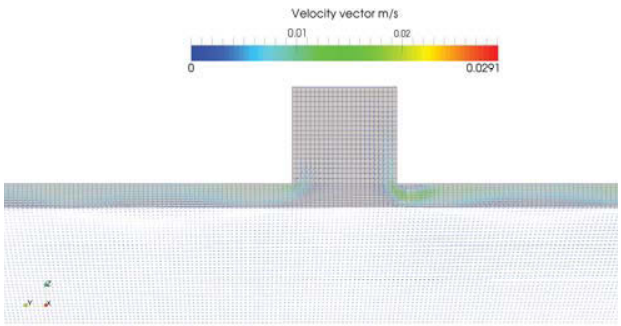

Figure 8: Velocity field after 60 seconds (bath region has a gray background)

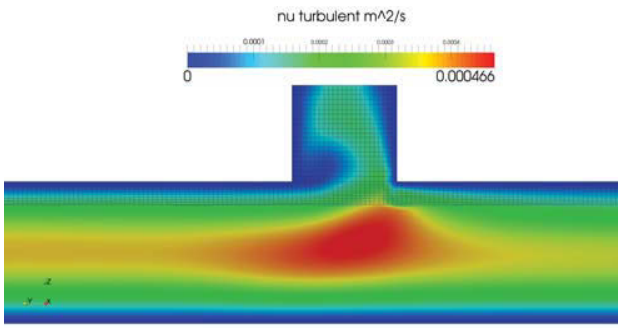

Figure 9: Turbulent viscosity after 15 seconds (bath mesh is visible)

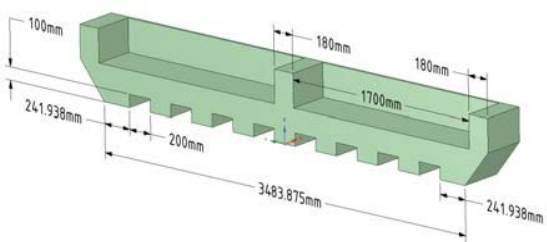

Figure 10: Geometry of the cell side slice model with irregular cathode surface

The model mesh presented in Figure 11 contains 1,152,016 hex finite volumes. Figure 12 is showing the initial bath-metal interface position. Figures 13 and 14 illustrate the velocity and the turbulent viscosity after 15 seconds. Figure 15 is showing the position of the bath-metal interface position every 15 seconds.

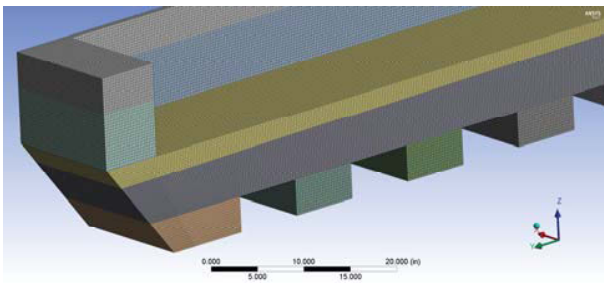

Figure 11: Mesh of the cell side slice model with irregular cathode surface

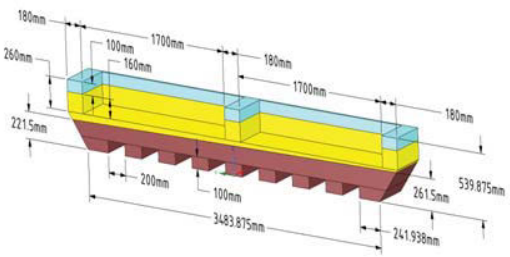

Figure 12: Initial bath-metal interface position

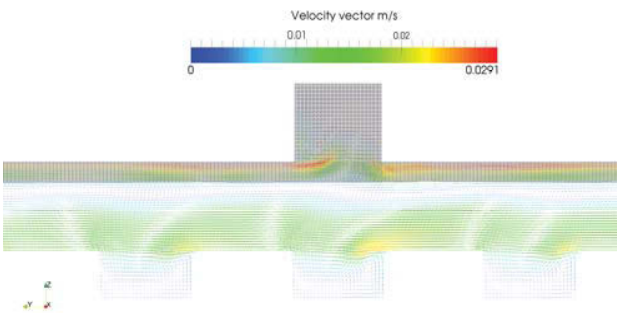

Figure 13: Velocity field after 15 seconds (bath region has a gray background)

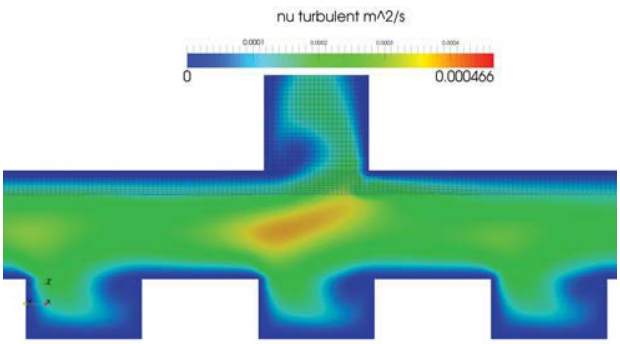

Figure 14: Turbulent viscosity after 15 seconds (bath mesh is visible)

Due to the presence of the flow obstacles, the flow in the metal pad is now quite different. Notice that flow around obstacles has been extensively studied and successfully modeled using OpenFoam [16]. Notice also that the mesh density used in [16] makes the one used in this study looking somewhat coarse!

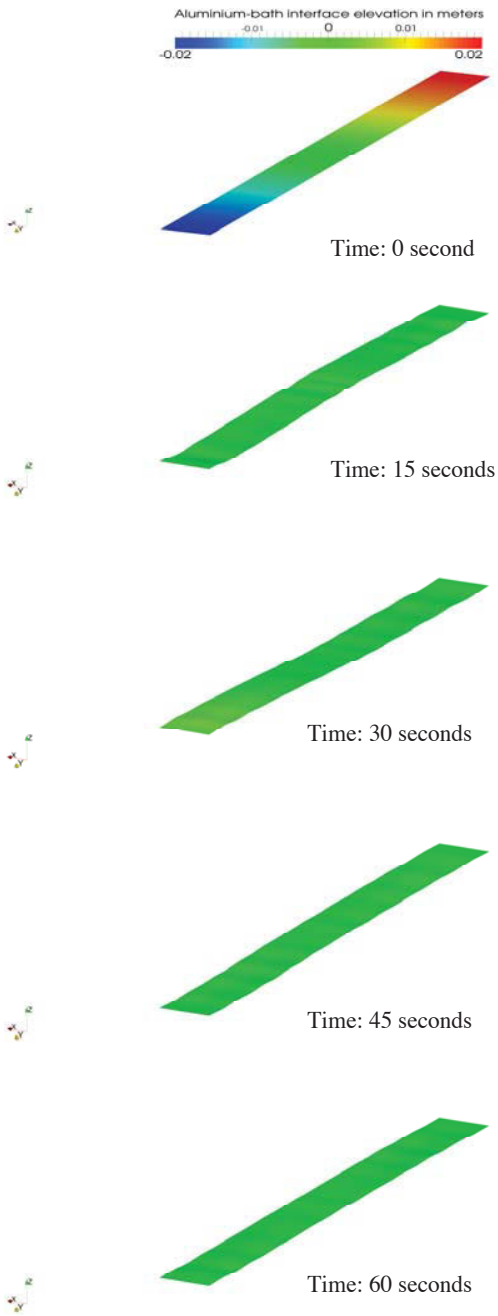

Figure 15: Position of the bath-metal interface every 15 seconds from 0 to 60 seconds

Comparison of the damping rate

The comparison between Figure 6 and Figure 15 interface positions reveals very little difference. Figure 16 is more useful for that, for it shows the transient evolution of the vertical position of the front left corner of the interface for the two cases.

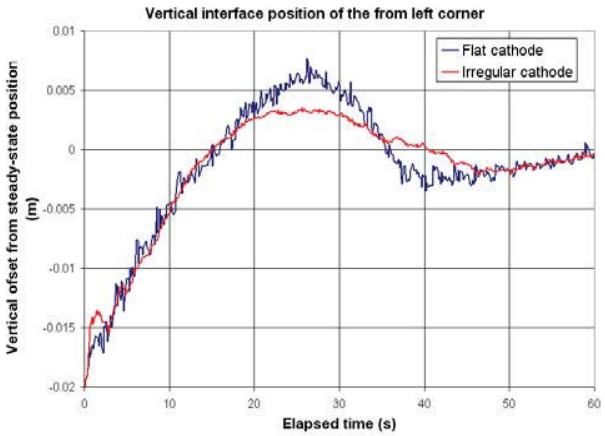

Figure 16: Evolution of the interface front left corner

There is definitively less overshoot in the case of the irregular cathode and also less secondary ripples on the interface so clearly the obstacles are somewhat performing as intended [17].

Yet this observation is not in contradiction with what was previously published in [2,3,4,5] in general and in Figure 7 of [5] in particular. The damping effect of the irregular cathode surface technology is not very important so many other changes to the cell design can have more impact on the cell stability.

Future work

The geometry of the cell side slice model is coming from the cell design presented in Figure 17 produced using Peter Entner CellVolt application [18]. That cell geometry was inspired from the GY420 420 kA cell design presented in [19]. Since that cell design has 48 anodes, modeling a longitudinal gravitational wave in a half cell model using the same mesh refinement used in that study would require a model more than 24 times bigger. Even with a linear increased of the required CPU time, solving such a half cell slice model would require about 750 CPU hours which is about 1 month of CPU time on the computer used in this study.

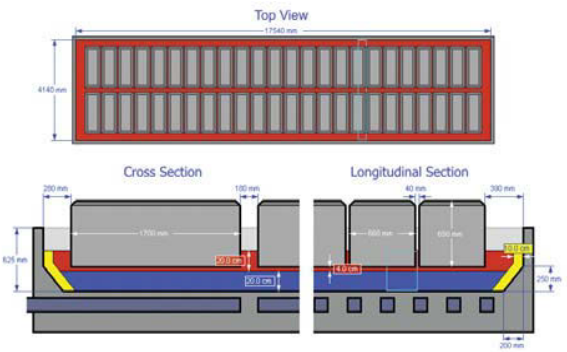

Figure 17: Sketch of the GY420 cell design that inspired the cell side slice model geometry

A model optimization study might reveal that a coarser mesh and a bigger time step could be used without losing much accuracy, so performing such a model optimization would be important. Yet, it is probable that a bigger computer than the Dell 28 cores Xeon ES-2697 V3 computer used in this study would be required in

order to obtained a practical turn around time to solve a transient 3D full cell gravitational wave VOF OpenFoam model.

Adding the MHD physic to an even bigger 3D full cell OpenFoam model is also quite possible to do. OpenFoam has already been successfully used to solve MHD flows [20,21].

Conclusions

Lateral gravity wave can be successfully simulated in a 3D cell side slice model using VOF formulation in OpenFoam.

Solving for just 60 seconds of transient evolution using a Dell 28 cores Xeon ES-2697 V3 computer took about 30 CPU hours.

Comparing regular flat cathode case model results with the irregular cathode surface case model results revealed that there is definitively less overshoot in the case of the irregular cathode so clearly there is somewhat more damping in that second case.

Yet this observation is not in contradiction with what was previously published using MHD-Valdis 2D shallow layer model as this new study confirms that the extra damping effect of the irregular cathode surface technology is not that significant. Many other changes to the cell design can have more impact on the cell stability.

A bigger computer than the Dell 28 cores Xeon ES-2697 V3 computer used in the present study would be required in order to obtain a practical turn around time to solve a transient 3D half cell VOF model to study a longitudinal gravitational wave.

Adding the MHD physic to an even bigger 3D full cell OpenFoam model is also quite possible to do. OpenFoam has already been successfully used to solve MHD flows.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

POROUS CARBON ANODES FOR THE SUPPLY OF METHANE DURING ELECTROWINNING OF ALUMINIUM

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Keywords: Aluminium electrolysis, Porous anodes, Methane

Abstract

One of the major downsides of the current aluminium production process is the high amount of CO_2 emission. One alternative is to replace the consumable carbon anodes with inert anodes so that oxygen evolves instead of CO_2 and PFC emissions. However, so far a sufficiently inert anode has not been found. Another option is to utilize natural gas through porous anodes. This will decrease CO_2 emission remarkably and also eliminate PFC emissions and anode effect. The porous anode could be made of carbon or it can be inert. However, the as-mentioned problem still exists regarding porous inert anodes. Therefore, at the moment porous carbon anodes seem to be the best practical option. In this study, porous anodes made of different grades of graphite were used for electrolysis experiments. Also, off-gas analysis was performed to get an insight of the ongoing reactions. Our results show that for some types of graphite anodes, methane participates effectively in the anodic reaction.

Introduction

In the current industrial aluminium production process (Hall-Héroult process) consumable carbon anodes react with the oxygen-containing ions in the electrolyte at the anode/electrolyte interface. As a result, a large amount of CO_2 as the main gaseous product is emitted. The following reaction can be considered as the overall reaction of the process [1]:

$$Al_2O_3\ (diss.) + 3/2C\ (s) = 2Al\ (l) + 3/2CO_2\ (g) \quad (1)$$

This large amount of generated CO_2 is one of the major weaknesses of aluminium production process; since CO_2 is a greenhouse gas. To solve this problem an inert anode can be used which changes the anodic reaction to oxygen evolution. But a suitable material as an inert anode in the current aluminium electrolysis process must fulfil several requirements [2]. There have been only laboratory and bench scales tests to try inert anodes so far [3]. In conclusion, a prospective industrial inert anode still seems to be unreachable; at least in the near future.
Another possibility is to supply a reducing gas (*e.g.* CH_4) to the anode/electrolyte interface through a porous anode. Then, the gas participates in the anodic reaction and the overall reaction changes from (1) to the one shown below:

$$Al_2O_3\ (diss.) + 3/4CH_4\ (g) = 2Al\ (l) + 3/4CO_2\ (g) + 3/2H_2O\ (g) \quad (2)$$

For supplying the gas a porous anode must be used. The porous anode could be made of carbon or an inert material. If it is made of an inert material, then according to the stoichiometry of reaction (2) the amount of emitted CO_2 can be decreased to half. However, as mentioned earlier a sufficiently inert anode has not been found so far. Therefore, a porous anode made of carbon seems to be a more practical choice at the present time. If the anode is made of carbon, either of the two reactions can occur at the anode. The theoretical cell voltage of reaction (2) is 1.1 V while for reaction (1) it is equal to 1.2 V at 1233 K (960°C) [4]. Therefore, thermodynamically reaction (2) is slightly more favourable. This small difference in theoretical cell voltage is due to depolarization effect of methane [4, 5]. Besides, utilizing the reducing gas, *e.g.* CH_4, results in reduced CO_2 emission. The degree of CO_2 emission reduction depends on which of these two reactions dominates as the anodic reaction.
The idea of utilizing a reducing gas for the anodic reaction of aluminium electrolysis has been tested before. Injecting of methane to graphite anodes in two series of experiments resulted in lowering the polarization voltage by 0.3-0.4 V [6]. In another study [7], gas electrodes made of graphite (50% porosity) were flushed with methane. But the anode was clogged by soot. When H_2 and CO were used some depolarization was observed; though the carbon consumption increased and the anodes disintegrated. Also anodes made of magnetite were used. They showed higher stability but eventually disintegrated after long time [7]. In a similar study porous graphite anodes showed depolarization when methane and H_2 were used. It was mentioned that due to high temperature of the process, methane decomposition occurs considerably and methane can be considered electrochemically equivalent to hydrogen. Carbon monoxide reacted only in the presence of catalysts and showed much less reactivity. Considerable fluoride losses from electrolyte occurred when hydrogen-containing fuels were used [8]. Lately, a study from New Zealand, have reported use of nickel alloy hydrogen diffusion anodes tested in a potassium-based electrolyte for aluminium production. Although a noticeable depolarization was observed, the metallic anode showed relatively low stability [9]. There are also a few patents in this field. One is a non-consumable gas anode based on the type used for Solid Oxide Fuel Cells (SOFC). Although, this anode is not suitable for the current aluminium electrolysis process, and could be utilized in a modified Hall-Héroult process [10]. The other patent is an anode made of porous graphite or a carbon based material. The anode has the roles of both conducting electricity and conveying and distributing the reducing gas [4].
We have been working on this concept of reducing gas-supplied anodes for aluminium electrolysis using porous anodes; both inert (*e.g.* SnO_2) and graphitic; where methane and H_2 were chosen as reducing gases and a modified electrolyte at 850°C was used [5, 11, 12] Considerable depolarization was detected when using SnO_2 anodes. Flushing methane through graphite anodes also showed a tiny depolarization effect [5]. Due to probable dissolution of SnO_2 anode in the electrolyte, in the present study

we have focused on graphite anodes using an electrolyte similar to what is used in industry for production of aluminium.

Experimental

The electrolyte composition was 6 wt. % AlF_3 (Noralf, Boliden Odda AS) and 5 wt. % CaF_2 (Merck, > 97 %), 4.5 wt. % anhydrous γ-Al_2O_3 (Merck, > 98 %) and remaining Na_3AlF_6 (natural cryolite, Greenland). The cryolite ratio was 2.5 and it was saturated with alumina. Figure 1 illustrates the schematic of the experimental set-up. A graphite crucible contained the electrolyte. The walls of the crucible were lined with alumina and its bottom served as the cathode. A hollow steel tube screwed to the porous carbon anode was used as current collector. The anode and cathode were positioned horizontally in respect to each other. A molybdenum wire (Norsk Spesialmetall, 99.9 pct) wire (1 mmØ) was used as the current collector. The wire was passed through an alumina tube and both were placed into another alumina tube. The outer alumina tube contained aluminium at the bottom. There was a small hole near the bottom of the outer alumina tube where electrolyte could enter the reference electrode. The whole system served as an Al^{3+}/Al reference electrode. All of the potentials were measured versus this aluminium reference electrode. The crucible containing the bath was dried in air at 120°C overnight. Furnace and bath were also dried at 200°C in N_2 for a few hours before heating up the furnace to the working temperature at 970°C. The furnace was continuously flushed with N_2. The inlet gas composition for the anode was controlled using mass flow controllers (Bronkhorst) and the inlet gas pressure was measured. The whole electrochemical cell was placed in a vertical tube furnace heated by resistance wires, and connected to a temperature controller. The gases were flushed into the porous anode with the gas flow equal to 20 ml min^{-1}.

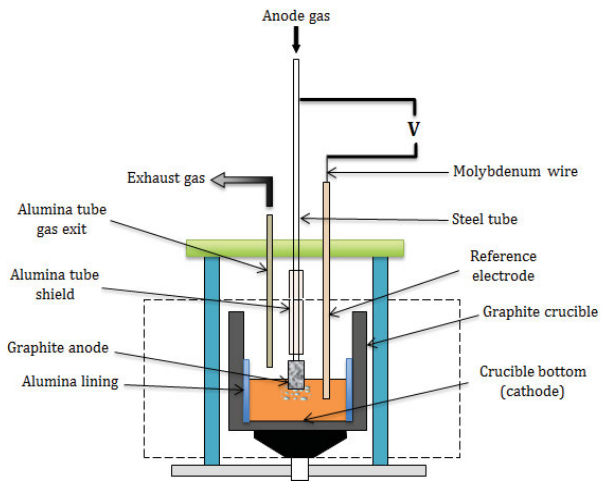

Figure 1: Schematic of the electrolysis cell.

Galvanostatic electrolysis experiments were performed using porous anodes made of three different graphite grades (Tokai Carbon Group). Some of the properties of graphite grades used for preparing anodes are given in Table I. Each experiment was started by applying a constant current of 2.3 A to the cell - corresponding to an apparent current density of 0.4 Acm^{-2} – while N_2 was passed through the porous anode for the first 45 min of the electrolysis time and afterwards changing the gas to CH_4 and continuation of the electrolysis for 4 h (total time: 285 min). This procedure enabled us to detect if there is any depolarization upon introduction of the CH_4 to the anode. Also another series of electrolysis experiments were carried out without using methane for comparison and in order to have a better insight of the process and for comparison. In this series, only nitrogen was flushed into the anode; while the rest of the experimental conditions were unchanged. The weight of the anodes was measured before and after each experiment to check the consumption of the anodes and it was compared with the theoretical values.
The graphite anodes were studied by Scanning Electron Microscope (SEM, Hitachi S-3400 N).

Table I: Typical properties of different graphite grades[13]

Grade name	Specific gravity (g/cm^3)	Porosity (%)	Grain size (µm)	Grade
G347	1.85	12	11	Isotropic
G140	1.7	20	1000	Moulded
KWPSY	1.6	20	2000[1]	Extruded

[1]Maximum grain size

Results and Discussion

Table II summarizes the consumption of the porous graphite anodes in electrolysis experiments. The theoretical consumption based on reaction (1), *i.e.* assuming that CH_4 does not take part in the anodic reaction, is 1.2 g.
As it can be seen, the isotropic graphite (G347) behaved quite differently compared to two other grades. There was a significant change in weight loss of the anode during electrolysis when CH_4 was supplied through G347 compared when N_2 was supplied through the anode. The moulded (G140) and extruded (KWPSY) grades were also consumed slightly less, when CH_4 was flushed into the anode but the difference is less significant compared to G347. It seems when isotropic grade was used, CH_4 participated in the anodic reaction remarkably, while for the two other grades anode consumption is almost equal to the theoretical value. For the extruded grade anode (KWPSY) with and without CH_4, the difference in the weight loss is not very small. But this is due to the fact that this anode showed a higher consumption when N_2 was used, compared with two others. The reason for this is not clear at the moment.
Generally, the amount of anode consumption is expected to be higher than the theoretical value. In industry the carbon consumption is around 110-120 % of the theoretical value. This is due to the unwanted consumptions of anode such as CO_2 burn (Boudouard reaction) especially when the gas penetrates into the pores of the anode, air-burn, and dusting [1]. Unlike the industrial anodes, the anodes used in this study were porous. Higher porosity enhances the Boudouard reaction even more effectively [1]. So, higher consumption of porous anodes was expected in these cases.
However, it is seen here that for all the graphite grades when CH_4 was applied, the weight loss was less than theoretical consumption. Hence, it is likely that CH_4 was involved in the anodic reaction - at least to some extent - for all graphite grades. This conjecture sounds more probable when we consider the probable thermal cracking of CH_4 resulting in precipitation of carbon in the porous structure of the anode. Methane becomes unstable in terms of its elements from 530°C. However, the reaction kinetics is slow. The equilibrium constant for cracking of methane is around 87 at T = 970°C [14]. Therefore, this reaction is most likely to happen in our experiments, at least to some

extent. Precipitated carbon can add to the final weight of the anodes. One experiment was performed to check this, where all the conditions were unchanged, except that no current was passed. So, the sole factor influencing the anode weight was the amount of carbon precipitation. It was found that around 0.2 g carbon was precipitated in 4 h (the same time CH_4 was flushed into anodes during electrolysis experiments). Consequently, the amount of graphite consumption was probably even less than the values reported in Table II. This means that methane has been involved in the anodic reaction even more.

Table II: The consumption of graphite anodes when supplied with only N_2 and when supplied with $N_2 + CH_4$ during electrolysis in cryolite-based electrolyte at 970°C for 285 min. The theoretical consumption is 1.2 g.

Graphite type	Anode gas	Weight loss (g)	Consump. (%)
Isotropic (G347)	$N_2 + CH_4$	0.67	56
	N_2	1.28	107
Moulded (G140)	$N_2 + CH_4$	1.15	96
	N_2	1.24	103
Extruded (KWPSY)	$N_2 + CH_4$	1.16	97
	N_2	1.4	117

Apart from the electrochemical reaction and cracking reaction, there exists another factor which might have changed the anode weights. This is the electrolyte which enters the porous structure of the graphite during electrolysis and can add to the final weight of the anode. However, this seems to be negligible since in all the experiments when the electrolysis was finished the anode was pulled out of the bath and was flushed with N_2 for 1 hour. This caused the remained electrolyte, if some, to be pushed out of the anode. SEM studies of the graphite anodes also confirmed that the graphites structure was essentially electrolyte free after the experiments.

Figure 2, shows the pressure measured before the anode, upon introduction of gas (N_2) to the anode for different graphite grades.

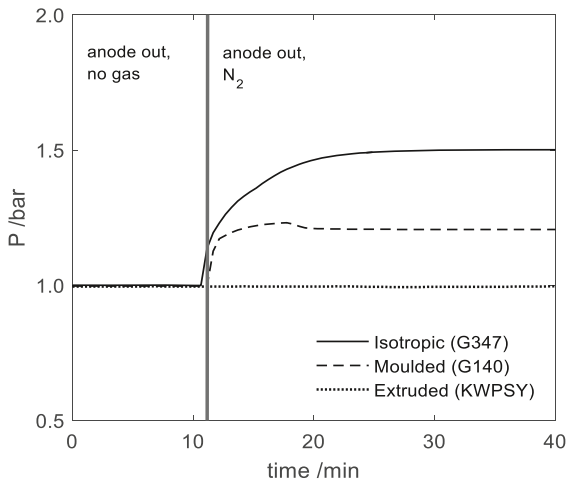

Figure 2: Measured pressure before the anode, upon introduction of gas (N_2) to the graphite anode for different graphite grades; Solid line: Isotropic (G347), dotted line: Moulded (G140) and dashed line: Extruded (KWPSY).

As can be seen, different graphite grades behaved differently when the gas (N_2) was introduced into the anode. This was before submerging the anode into the salt bath and the anode was placed 2 cm above the melt. The isotropic grade showed the largest pressure increase, around 0.5 bars, the moulded grade showed less increase, around 0.25 bars; while the extruded grade did not show any pressure increase. This difference can be attributed to the large difference in grain/pore size of these graphite grades. The pores in the extruded grade were large enough to avoid any resistance for the flow of the gas.

Figure 3 shows the cell potential and pressure changes during aluminium electrolysis using isotropic grade (G347) anode for two different experiments; Figure 3 (a) when only N_2 was supplied through the anode and Figure 3 (b) when N_2 was supplied through the anode 45 min prior to introducing CH_4. In the second case the electrolysis continued for 4 more hours.

As can be seen, when only nitrogen was supplied to the anode, the potential increased gradually during the electrolysis, but the pressure remained constant. This is due to the consumption of graphite resulting in decrease of surface area and consequently increased current density. This is in agreement with previous studies demonstrating an increase in the anodic overvoltage when current density is increased [15].

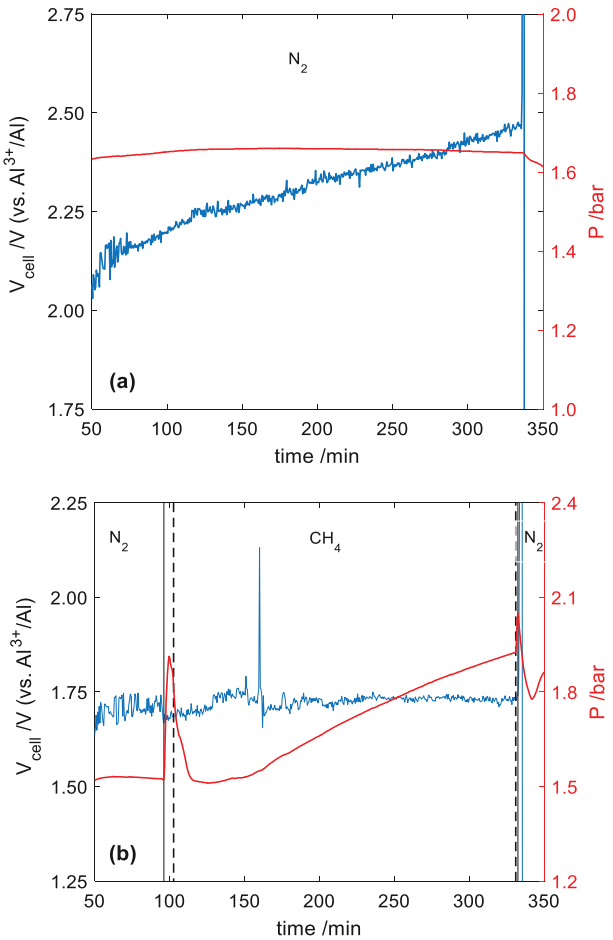

Figure 3: Potential and pressure changes during aluminium electrolysis for two experiments where in (a), only N_2 and in (b) $N_2 + CH_4$ were flushed into the porous anodes. Anode was made of isotropic graphite (G347). I=0.4 A.cm^{-2}, T=970°C.

A rough calculation demonstrates that the change in the surface area was noticeable. The surface area prior to electrolysis was 6.28 cm^2 and the approximate value for the final surface area was: $A_f \approx 5$ cm^2. The final surface area was calculated based on the geometrical surface of the anode after electrolysis. Assuming the change in surface area and a constant current the apparent current density increased from 0.37 $A.cm^{-2}$ to 0.46 $A.cm^{-2}$ during electrolysis.

However, when CH_4 was introduced to the anode the potential became stable and it remained almost constant throughout the electrolysis period. The vertical lines in Figure 3 (b) represent the time when the gas anode was shifted from nitrogen (dashed line) to methane (solid line). There was an abrupt increase and decrease in pressure which is because of supply of both gases during the shifting time. After that as can be seen, introduction of methane to the anode caused a gradual pressure increase which lasted as long as methane was flushed; *i.e.* until the end of electrolysis. This is due to carbon precipitation from the cracking reaction of methane. If the electrolysis had continued for a longer time, it might have led to clogging; as observed earlier in another study [7].

The change in cell voltage agrees with the weight loss data (Table II). It is clear that methane was significantly involved in the anodic reaction, so the potential was stable and did not increase. However, the problem of clogging might prevent long term electrolysis. A suitable anode design could possibly prevent the probable and undesirable clogging.

There is a small difference in the cell potential in the beginning between these two experiments, although the anodes were similar and both experiments were started by supplying N_2 to the anode. The reason is not clear.

The potential and pressure changes for the two other graphite grades (G140 and KWPSY) are illustrated in Figure 4; both when there was only N_2 Figure 4 (a), and when CH_4 was also used as the gas anode, Figure 4 (b) and (c). As can be seen, when N_2 was introduced through the anode the potential showed the gradual increase during electrolysis as also observed for isotropic grade Figure 4 (a); which was due to increased current density.

In contrast to the isotropic grade (G347), Figure 3 (b), providing the anode made of the other two graphite grades with methane, did not change the potential behaviour, Figure 4 (b) and (c). The potential behaviours were similar to the experiments where only N_2 was flushed to the anode, Figure 4 (a). So, it suggests that the contribution of the methane in the anodic reaction was not significant in these cases and this is in agreement with results from the weight change results. Moreover, by comparing the cell voltage during electrolysis between the isotropic grade and the two other grades, it is revealed that the potential fluctuations were much less pronounced. The observed fluctuations in cell voltage can be due to bubble formation when introducing the gas though the anode. It is assumed that smaller grain pore size due to smaller grain size will result in smaller bubble sizes which finally will result in weaker fluctuations in the cell voltage.

Pressure changes are also shown for the experiments were CH_4 was used as reducing gas, Figure 4. It is clear that in the case of moulded grade, Figure 4 (b), similar to isotropic grade (G347), carbon precipitation led to pressure build up; although to a lower degree. However, the extruded grade, Figure 4 (c), did not show any pressure build up. This can be attributed to the larger grain/pore size of this grade, as mentioned before. The pressure changes during electrolysis for these grades are consistent with those observed upon introduction of gas to the anode, Figure 2. In the experiments were only N_2 was used no pressure build-up was observed; since there was no carbon precipitation from the gas.

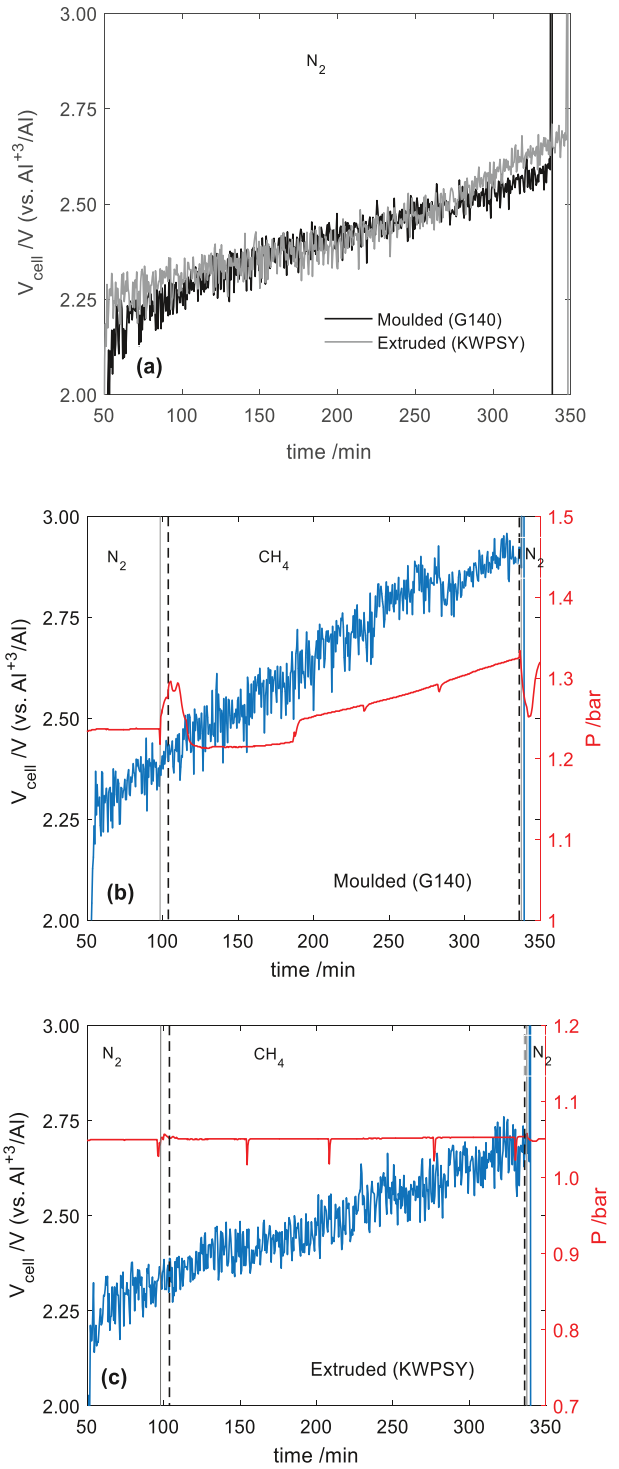

Figure 4: Potential and pressure changes during aluminium electrolysis where only N_2 (a), or N_2 + CH_4 were flushed into the porous anodes made of moulded (b), and extruded (c) graphite grades. I=0.4 $A.cm^{-2}$, T=970°C.

Figure 5 shows micrographs of fracture surfaces of the extruded (KWPSY) and isotropic (G347) graphite grades after electrolysis in cryolite-based electrolyte for 285 min at 970°C; i = 2.3 A. For the extruded grade Figure 5 (a) and (b)) it is clear that there was

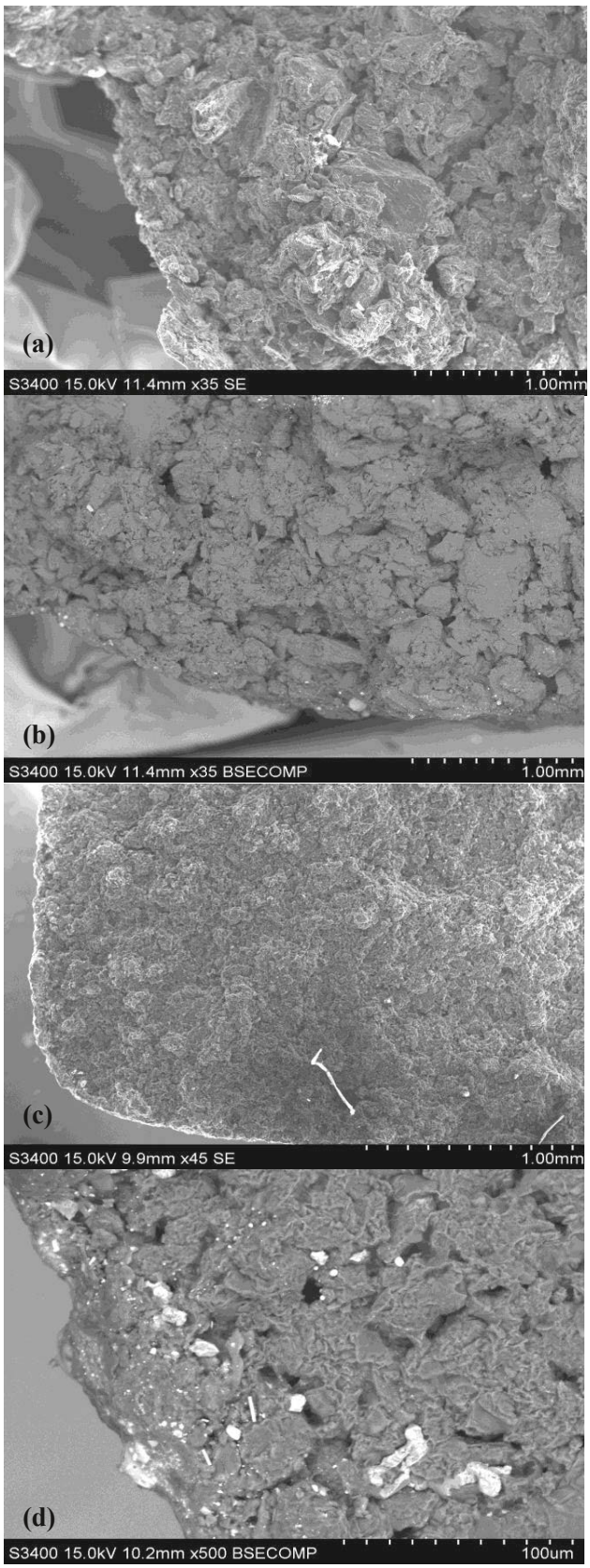

Figure 5: SEM images of porous anodes. (a) SE and (b) BS image (extruded grade). (c) SE image, (d) BS image (isotropic grade).

no electrolyte left inside as little elemental contrast is observed in the back scattered electron micrograph after electrolysis and consecutive N_2 flushing. This grade is quite coarse as can be seen in the images.

Figure 5 (c) and (d) show fracture surface areas from and isotropic (G347) graphite grade. This grade is much finer. There are small amounts of electrolyte left inside the structure very close to outer surface. Chemical analysis confirmed that the second phase is electrolyte. This might be due to smaller size of pores and finer structure of graphite which hinders the exit of electrolyte from the anode. Nevertheless, this amount seems to be negligible regarding the weight change of the anode.

Conclusion

In conclusion, the isotropic graphite grade (G347) showed a better performance as a gas anode compared to the moulded and extruded grades. From the weight change results and potential behaviour it is clear that methane participates in the anodic reaction in a large extent when isotropic grade was used compared to the two other grades of graphite. The fact that isotropic grade behaved differently from the two others is mainly due to the much finer structure. It seems the finer grain/pore of the isotropic grade provided a better gas distribution and the three-phase boundary was well established. Further studies including off-gas analysis which can confirm our findings is undergoing.

Acknowledgment

Financial support is gratefully acknowledged from the Research Council of Norway, GASSMAKS program, and grant number 224985.

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Light Metals 2016
Edited by: Edward Williams
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CREEP BEHAVIOR AND CHANGE OF POROUS STRUCTURE OF GRAPHITE CATHODE MATERIAL IN $NaF-AlF_3-Al_2O_3$ MELT UNDER EXTERNAL PRESSURE

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Keywords: Creep behavior, Porous structure, Graphite cathode, Aluminum reduction

Abstract

Investigation of creep behavior of graphite cathode material during aluminum electrolysis in $NaF-AlF_3-Al_2O_3$ melt were carried out under external 4 MPa pressure with modified rapoport method in a laboratory cell at 965 ^{0}C. The effects of cathode current density (i=0 A/cm^2, 0.25 A/cm^2, 0.50 A/cm^2 and 1.0 A/cm^2), cryolite ratio (CR=2.34, 3,0 and 4.0) of the cryolitic melts have been investigated. The results demonstrate that increasing current density or CR value can yield enlarged creep strain. The porous structures of the cathode samples were characterized using image analysis, in which the porosity increased with more long and narrow pores appearing after 150 min of electrolysis. The information acquired from this work can be valuable in materials selection and performance improvement of graphite cathodes..

Introduction

Cathode blocks in Hall-Heroult cell are vital components for maintaining cell operation safety and efficiency. However, they work in rigorous condition, and the material properties may deteriorate due to sodium/bath penetration, electrochemical wear, thermal stress as well as external pressure from the cell shell [1, 2].

Creep deformation of cathode materials can play an important role in how the cathode deterioration develops. Hop [3] has studied in detail the creep phenomenon of carbon cathode materials during electrolysis process. Picard [4] studied ambient creep behavior of three industrial cathodes, and found that graphitized material has the highest specific creep strain amplitude at room temperature, followed by graphitic and then semi-graphitic material. Zolochevsky [5, 6] established a constitutive model based on creep experimental data of cylindrical cathode samples to describe the relationship between creep strain, external pressure and time during tests. Xue [7] found that introduction of TiB_2 can increase the creep resistance of cathodes significantly, which would facilitate extension of cathode service life.

In the cathode materials, porosity is an important factor that affects sodium and bath penetration into the cathode body during electrolysis [8, 9]. The pores inside the cathode materials also influence the creep behavior of the cathode materials. Chen [10] studied the porous structure change of graphite material during high-temperature creep deformation, and found that the total porosity of cathodes decreased in the first stage of creep deformation and then increased in the second creep stage, accompanied by the disappearance of some initial pores and the emergence of new pores.

It is known that increasing the cathode current density or the cryolite ratio (CR) of cryolitic melts enhances sodium penetration, which results in higher sodium expansion of the cathode during electrolysis. However, the influence of the current density and CR on the creep deformation and porous structures of the cathode materials under external pressure still remain unclear.

In the present paper, creep behavior of high-purity graphite cathode material with varying cathode current density and cryolite ratio has been experimentally studied. Burgers model was introduced to fit the measured data to quantitatively describe the creep behavior of cathode materials. Additionally, the porous structures of the cathode sample before and after electrolysis were characterized using image analysis in association with the creep behaviors observed.

Experimental Procedure

Tests of High-Temperature Creep

Figure 1 is the schematic drawing of the experimental apparatus for high-temperature creep tests. The high-purity graphite cathode samples used in these tests were cylindrical (Φ25×50 mm), and provided by Shanghai Carbon. The cathode sample was set vertically and its top end was connected to the graphite loading support rod. An alumina guide rod (Φ10 mm) was placed through a hole inside the graphite loading support rod along the axial direction. Through this alumina rod, the signals of creep strain were transferred. A loading frame was mounted on the top of the loading support rod, which provided the required pressure on the cathode sample during experiments. Constant pressure was generated by a hydraulic workstation, by accounting for the weight of the loading frame, the total pressure applied on the cathode sample was 4 MPa. This pressure was maintained for 150 min until the end of the creep test. During the test, the axial creep strain of the cathode sample was measured using a laser vertical displacement (LVD) transducer (range 10 mm, resolution 1 μm, Micro-Epsilon), which was connected to computer for automated data recording at 60 s intervals.

Temperature of the furnace was maintained at 965 °C for 30 min before electrolysis to homogenize the temperature distribution and to finish thermal expansion of the cathode sample. Before application of external pressure, the sample was allowed to expand freely for 60 min after the current was supplied. Thereafter constant pressure was applied via the loading frame. Argon gas was used to protect the cathode sample against oxidation during the whole experiment process.

The cryolite melt was contained in a graphite crucible. For each test, 260 g electrolyte (pre-melted and crushed) was required and 45 mm of the sample height was immersed in the melt. The composition of the electrolyte was 87 wt% NaF/AlF_3, 8 wt%

Al_2O_3 and 5 wt% CaF_2, with cryolite ratios of 4.0, 3.0 and 2.34, respectively. A constant current was generated by a MPS302DC power supply, where the cathode current densities of 0 A/cm^2, 0.25 A/cm^2 and 0.50 A/cm^2 and 1.0 A/cm^2, were used respectively.

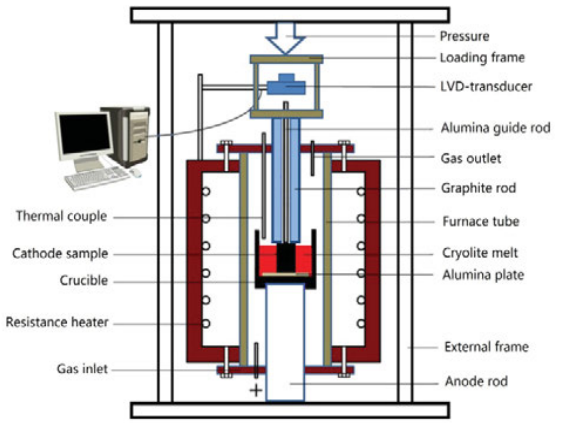


Figure 1. Experimental apparatus for in-situ measurements of the cathode creep strain during aluminum electrolysis

Characterization of Porous Structure

Cathode samples after electrolysis were sliced to form 1 cm thick cylindrical specimens and sealed in a mixture of epoxy and curing agent to prevent air contact, then carefully ground and polished to eliminate all visible scratches on the cross-section surface. Images at 40× magnification of a Φ3 mm area were taken from the central area of each specimen surface, with a standard polarizing microscope (Leica, DMRX), and photographed with Canon 500D camera. These colored images were converted to binary images using Image-Pro-Plus software (a public domain, Java-based image processing program developed at the National Institute of Health), thereafter porous structure parameters of specimens were calculated with the same software. More details of the image analysis method can be found in our previous publication [9, 10].

Results and Discussion

Effect of Current Density and Cryolite Ratio

Creep strain (R) is defined by equation (1):

$$R = \frac{h_t - h_0}{h_0} \times 100\% \quad (1)$$

Where h_t represents the sample height at testing time t; h_0 is height of the sample before loading pressure.

Figure 2 and Figure 3 are the creep curves for the primary and secondary (stationary) creep stages of the cathode material. The primary creep stage is known as the decelerating creep stage, and in this stage, the initial strain rate is relatively high but reduces over time. When the strain rate reaches a minimum and becomes stable, the creep deformation enters the stationary stage, the material continues to deform at this rate for a long time. In our case, the trend of all the creep tests was to slowly level out in the second stage, and would require a much longer time (beyond the time limit available in lab) to see a transition into the tertiary creep stage.

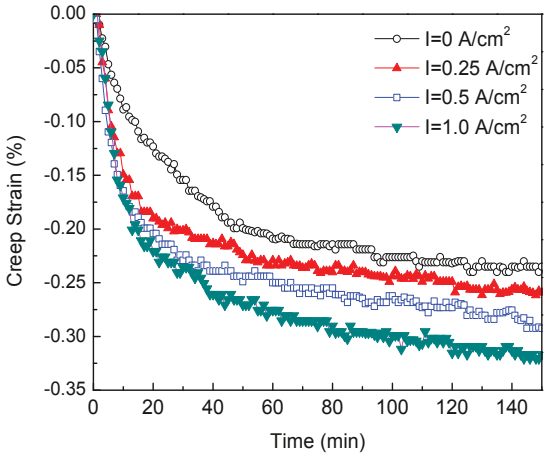
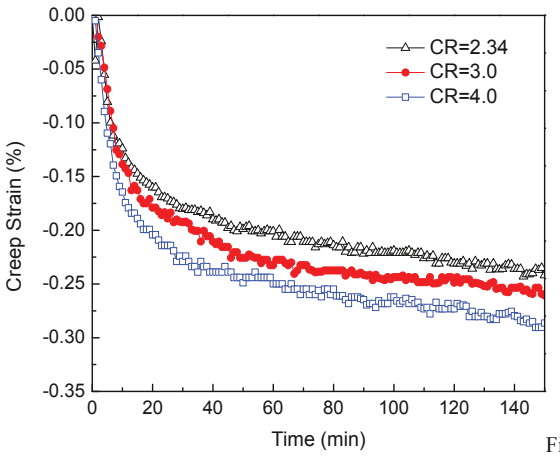


Figure 2. Creep curves of the graphite cathodes during electrolysis with various current densities (CR=4.0, external pressure=4.0 MPa, at 965 ^{o}C)



Fi

gure 3. Creep curves of the cathodes during electrolysis with various CR during aluminum electrolysis (Current density=0.50 A/cm^2, external pressure=4.0 MPa, at 965 ^{o}C)

The influence of cathode current density on the creep behavior of graphite material in NaF-AlF_3-Al_2O_3 melt is shown in Figure 2. The primary creep stage is shortened from about 40 minutes (with no current applied) to less than 20 minutes when current is applied. Thereafter the curves follow a similar pattern of enlarged creep deformation. Moreover, the creep strain (R) at the same testing time is clearly increased with the current density. This may relate to formation of metallic Na on the cathode surface, which usually increases with current density during electrolysis.

Figure 3 shows the creep curves of cathode materials during electrolysis with various CR, and the results indicate that CR does have a definite effect on creep deformation. The creep strain increased with higher CR. The influence of CR on creep behavior of graphite cathodes can be explained by the deterioration of the cathode structure caused by sodium intercalation. Increasing current density and CR facilitate sodium absorption by the cathode and more C-Na intercalation compounds are produced

during aluminum electrolysis. Therefore, sodium atoms would lubricate the slip and cleavage movement of graphite basal planes after they have intercalated into the graphite interlayer, as Guilliat [11] proposed. Such action could help relieve internal stress by making the material deformation easier. In this way, the creep resistance of cathode material can be reduced and this reduction is enhanced with higher CR due to the increased amount of sodium absorption.

The results above suggest that cell service life may be prolonged by using lower CR so that the graphite material maintains a stronger creep resistance. While the selection of cathode current density is a more complicated design matter because of the trade-off of other factors, increasing the cathode current density should be carefully considered for its possible long term influence on the cathode life.

Burgers Model and Measured Creep Curves

Figure 4 shows Burgers model as combination of Maxwell model and Kelvin model in series, which can be applied to describe the creep deformation of carbonaceous cathode materials at primary stage and stationary stage [7]. In our case, this model was introduced to fit the creep curves measured in the electrolysis experiments with various values of CR or current density. The constitutive equation of Burgers model is given in Equation (2).

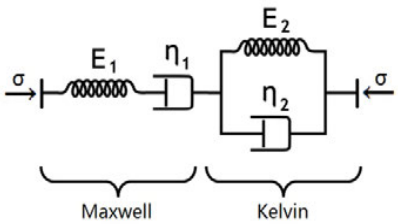


Figure 4. Structure schematic of Burgers model

$$R = \frac{\sigma}{E_1} + \frac{\sigma \cdot t}{\eta_1} + \frac{\sigma}{E_2}[1 - \exp(-\frac{E_2}{\eta_2} \cdot t)] \qquad (2)$$

Where R is the cathode creep strain; σ is pressure applied on the cathode; t represents function duration of pressure; E_1 is delayed modulus of elasticity in Kelvin model; η_1 is viscosity coefficient in Kelvin model; E_2 is Maxwell instantaneous modulus of elasticity; η_2 is Maxwell viscosity coefficient.

Selected curves fitted against the measured creep data are presented in Figure 5. The modeling results could be used to assess creep resistance properties of graphite cathode material in aluminum electrolysis under different electrolysis conditions.

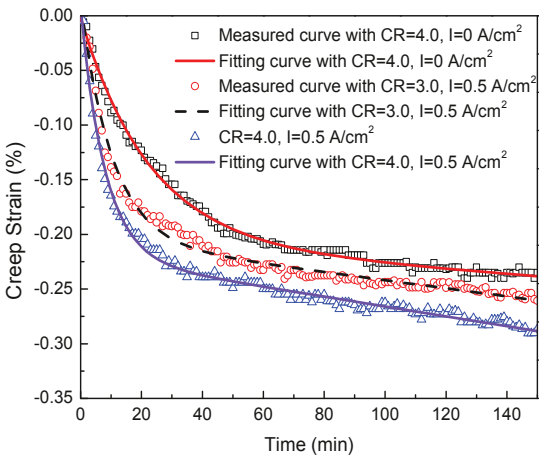


Figure 5. Selected fitted curves vs. measured creep curves for graphite cathode samples in aluminum electrolysis at 965 ^{0}C

Table 1 lists the parameters derived from the fitted curves of Burgers model that are in good agreement with the measured creep strain data, showing a high correlation. As the current density increases from 0 to 1.0 A/cm^2, Burgers model parameters E_1, E_2, η_1, η_2 have all decreased. This corresponds to the deterioration in creep resistance of graphite cathode material. However, this deterioration can be reduced to some extent by decreasing CR values from 4.0 to 2.3. It is believed that an appropriate combination of material parameters, as obtained by Burgers model, may contribute to composition optimization for better cathode creep resistance.

Table I. Burgers Model Parameters Based on Measured Creep Curves

Electrolysis conditions	E_1/(MPa)	η_1/(MPa·min)	E_2/(MPa)	η_2/(MPa·min)	Correlation
CR=4.0; i=0 A/cm^2	2209.96	19.34	20312.32	442.51	0.968
CR=4.0; i=0.25 A/cm^2	668.09	17.92	9713.62	146.00	0.991
CR=4.0; i=0.50 A/cm^2	241.01	17.45	8331.58	138.93	0.973
CR=4.0; i=1.0 A/cm^2	162.80	15.00	7207.61	126.46	0.986
CR=3.0; i=0.50 A/cm^2	290.06	17.68	8729.10	141.56	0.983
CR=2.34; i=0.50 A/cm^2	683.76	18.02	10362.76	150.26	0.976

Characterization of Porous Structure

Figure 6 shows the binary images [10] acquired by image-analysis of the porous specimens taken from 5 cathode samples: the original cathode sample (HS-1), and the samples after creep tests at 0 A/cm^2 (HS-2), 0.20 A/cm^2 (HS-3), 0.50 A/cm^2 (HS-4) and 1.0 A/cm^2 (HS-5), respectively. In these binary images, the black areas represent pores and the white areas represent carbonaceous substrate. It can be clearly seen that after creep deformation under external pressure for 150 minutes, remarkable changes have occurred to pores inside the cathodes. The number of pores and the value of porosity have both evidently increased, and while pores in the original cathode sample are uniform and neat, the

pores inside the cathode samples after creep tests appear more irregular.

Porous structure parameters calculated from the binary images (in Figure 6) are listed in Table II - where the definition of these parameters can be found elsewhere [10]. It can be seen that the porosity of cathode samples increases after creep tests. The increase in the aspect ratio and fractional dimension indicate that most pores after creep deformation have become longer and narrower, and bigger in size, and the edge of pores has become more complex and irregular. In general, the results demonstrate a strong influence of current density on the change of cathode porous structure in association with the creep deformation.

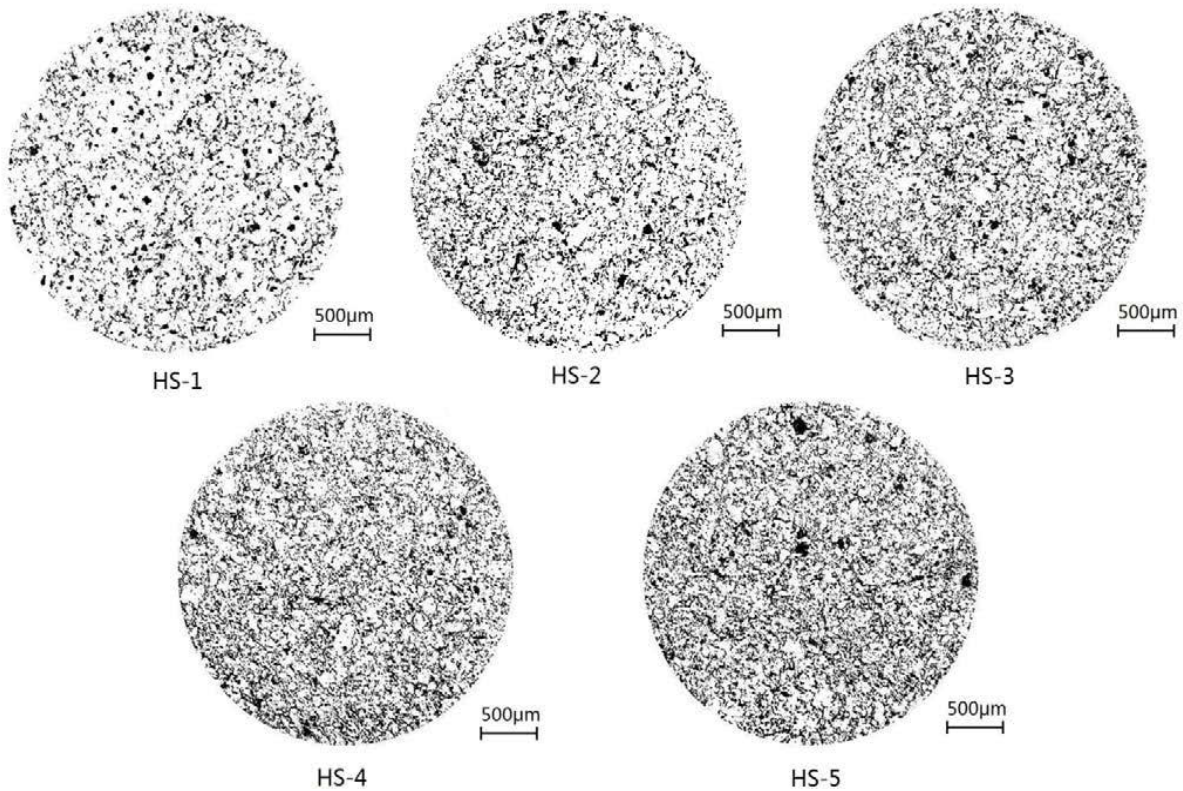


Figure 6. Binary images for 5 cathode samples before and after creep tests, HS-1: Original cathode sample; HS-2: cathode sample after creep test with current density=0 A/cm^2; HS-3: after creep test with current density=0.25 A/cm^2; HS-4: after creep test with current density=0.50 A/cm^2; HS-5: after creep test with current density=1.0 A/cm^2.

Table II. Calculated Porous Structure Parameters of 5 Samples in Figure 6

Sample	Porosity (%)	Aspect ratio	Fractional dimension	Mean diameter (μm)	Coordination number	Connectivity (%)
HS-1	13.50	2.13	1.108	22.13	1.116	5.21
HS-2	17.06	2.17	1.132	25.43	1.129	12.12
HS-3	19.12	2.20	1.165	25.95	1.145	13.29
HS-4	20.02	2.21	1.172	26.16	1.154	15.35
HS-5	21.16	2.23	1.186	26.53	1.156	16.40

The porous structure evolution of the graphite cathode during creep deformation under electrolysis condition should not be considered merely as a physical process, where the impacts of sodium and electrolyte penetration into the cathode body should take into account. The change in volume and profile of the pores could be due to Na-C intercalation and re-crystallization of the infiltrated salts within grain boundaries of the carbonaceous material in the cathodes, resulting in additional internal stress. It seems that the quantitative information acquired by image analysis is sensitive enough to reflect the variation in the porous structure of the graphite due to creep deformation with respect to current density. Additional operating parameters will be investigated in future work.

Conclusions

1. Creep deformation of graphite cathode materials under external pressure during aluminum electrolysis in NaF-AlF_3-Al_2O_3 melt

increases with increased current density and cryolite ratio.

2. Burgers model is found to fit well with the measured creep data, from which the model parameters E_1, E_2, η_1, η_2 can be derived to enable determination of the creep resistance of the graphite cathode material for potential improvement in cell service life.

3. Quantitative information on evolution of the porous structure inside the graphite cathode material during various stages of creep deformation can be acquired using image analysis, and shows the the pores after creep deformation to become longer and narrower, as well as larger in size..

Acknowledgement

Financial support from National Natural Science Foundation of China (No. 51434005) is gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

UNEVEN CATHODE WEAR IN ALUMINIUM REDUCTION CELLS

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Keywords: Cathode wear, Electrochemical, Aluminium carbide, Bath film

Abstract

The service life of aluminium reduction cells is limited by cathode wear. A typical wear pattern has been reported as a single or double "W" profile. The uneven cathode wear appears to be controlled by multiple mechanisms such as non-uniform current distribution and carbide dissolution. According to our analyses, formation of aluminium carbide that dissolves into a bath film located on the cathode surface is the dominating mechanism. A dynamic model was developed in order to calculate the local carbide formation as well as cathode wear rate. The present model utilizes a simplified geometry where the metal flow, which is driven by the Lorentz force, results in shear stress and motion of the bath film. The simulations demonstrate how the local current distribution and the mass transfer of aluminium carbide from the cathode surface and into the bath film result in uneven cathode wear profiles.

Introduction

The cathode wear has been attributed to physical abrasion and chemical corrosion. A typical wear pattern was reported to be a single or double "W" profile [1], where the lowest point determines the life time of the aluminium reduction cells. Liao and Øye [2] concluded that chemical and electrochemical corrosion is a stronger wear process than physical abrasion. Many researchers [3-12] have concluded that electrochemical formation of aluminium carbide is probably dominant in the cathode wear mechanism. The presence of an electric field enhanced the formation rate of aluminium carbide. Novak et al. [3-5] revealed that the aluminium carbide grows from the bottom of the layer where the carbon is available. The aluminium carbide layer was observed covering the cathode surface [14] and the pores inside the cathode [9, 14].

Considering the presence of the sludge at the bottom of the aluminium reduction cells, a bath film [14] is assumed to be located between the molten metal and cathode. The aluminium carbide layer is then to be removed by dissolving into the bath film. However, the aluminium carbide layer in the pores may not be easily removed due to its slower mass transfer process. The present authors' group developed a dynamic model of aluminium carbide formation [15] on the cathode surface as a function of physical properties and thickness of the aluminium carbide layer, thickness of bath film as well as the voltage drop over the bath film and aluminium carbide layer. The model delivered a local dynamic aluminium carbide formation rate, but without taking the removal of aluminium carbide into account.

Einarsrud et. al [13] predicted the cathode wear profile by using a magnetohydrodynamic (MHD) model in a simplified representative cell. In this model, the cathode wear rate was assumed to be proportional to the mass transfer of carbon containing species from cathode into molten metal. The mass transfer coefficient is derived on basis of the local rate of strain [16] in the molten aluminium close to the cathode. However, the bath film between the molten metal and cathode was not taken into account.

In this study we demonstrate the influence of bath film on cathode wear by modelling the flow of the bath film. The mass transfer of aluminium carbide into the film is discussed. The cathode wear rate is predicted by proportionally associating it with the local aluminium carbide dissolution rate.

Model of fluid flow

Bulk of molten aluminium and bath

The bulk of fluid flow is driven by the magnetic hydro dynamic force in the aluminium reduction cells. The electric and magnetic fields are governed by the Maxwell equations [17] and constitutive closure relations. The electrical current density $\boldsymbol{j}$ is given by the continuum form of Ohm's law.

$$\boldsymbol{j} = \sigma\left(\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B}\right) \tag{1}$$

where $\boldsymbol{E}$ is the electric field, $\boldsymbol{B}$ is the magnetic field, $\boldsymbol{u}$ is the velocity of the fluid, and σ is the corresponding electrical conductivity.
The electric field can be expressed by the electrical potential ϕ,

$$\boldsymbol{E} = -\nabla\phi \tag{2}$$

The conservation of electric charge is in this case governed by

$$\nabla \cdot \boldsymbol{j} = 0 \tag{3}$$

Combining equations (1)-(3), the transport equations of electrical potential is,

$$\nabla \cdot \left(\sigma \nabla \phi\right) = \nabla \cdot \left(\sigma \boldsymbol{u} \times \boldsymbol{B}\right) \tag{4}$$

The multiphase volume of fluid (VOF) model is applied to the mass conservation of metal and bath phases,

$$\frac{\partial}{\partial t}\left(\alpha_k \rho_k\right) + \nabla \cdot \left(\alpha_k \rho_k \boldsymbol{u}\right) = 0 \tag{5}$$

where α_k is the local volume fraction of the *k*-th phase.
The momentum of the fluid flow is described by the Navier-Stokes equations,

$$\frac{\partial}{\partial t}\left(\rho_k u_i\right) + \frac{\partial}{\partial x_j}\left(\rho_k u_i u_j\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu \frac{\partial u_i}{\partial x_j}\right) + \boldsymbol{F}_{Li} \tag{6}$$

where, $\boldsymbol{F}_{Li}$ is the Lorentz force acting on the neutral fluid in which the current is passing through. The Lorentz force is given by,

$$\boldsymbol{f}_L = \boldsymbol{j} \times \boldsymbol{B} \tag{7}$$

The magnetic field $\boldsymbol{B}$ is a superposition of an externally imposed field and an internally generated field, reading,

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} + \boldsymbol{B}_{ext} \tag{8}$$

where, $\boldsymbol{A}$ is the internally generated magnetic vector potential which is calculated by solving the following Poisson equation,

$$\nabla^2 \boldsymbol{A} = -\mu_0 \boldsymbol{j} \tag{9}$$

The above equations close the model of bulk fluid flow in the present model, which is inherited from the previous work [13].

Modelling of bath film

The flow of the bath film is driven by the shear force of molten aluminium. It is not realistic to model the flow of bath film by a VOF model, because it is often too thin comparing the scale of the computational cell.

The solution in the current model is to assign a virtual bath film in the first layer of grid on the cathode surface. Figure 1gives a 1-D sketch of a virtual bath film in an aluminium reduction cell. The h_c and h_f are the height of the cell and the local bath film respectively.

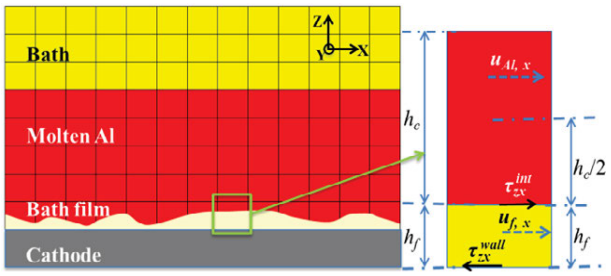


Figure 1. The virtual bath film in the first layer of grid above the cathode surface

The continuity equation of the bath film is expressed in term of its thickness,

$$\frac{\partial h_f}{\partial t}+\nabla\cdot(h_f\boldsymbol{u}_f)=S \tag{10}$$

where S is the source term of the bath film that comes from the releasing of bath by dissolving bottom sludge. The focus of the current model is to establish the governing model of the bath film and demonstrate its importance on mass transfer. Therefore, the bath film is simulated as a closed system with source term S of zero.

The shear force between the molten metal and bath film at $z=h_f$ is given by,

$$\tau_{zx}^{\text{int}}=\mu_{Al}\frac{u_{Al,x}-u_{\text{int},x}}{h_c/2}=\mu_{Al}\frac{u_{Al,x}-u_{f,x}^{\text{int}}}{h_c/2} \tag{11}$$

While the shear force on the carbon cathode from the bath film is,

$$\tau_{zx}^{wall}=\mu_f\frac{u_{f,x}^{\text{int}}}{h_f} \tag{12}$$

The flow in the bath film is assumed to be laminar since it is very thin, has a low velocity and a correspondingly small Reynolds number. Thus we derive,

$$u_{f,x}^{\text{int}}=2u_{f,x} \tag{13}$$

To extend those above equations into three dimensions, the governing momentum equations of the bath film in x-coordinate is,

$$\rho_f V_f\left(\frac{\partial u_{f,x}}{\partial t}+u_{f,x}\frac{\partial u_{f,x}}{\partial x}+u_{f,y}\frac{\partial u_{f,x}}{\partial y}+u_{f,z}\frac{\partial u_{f,x}}{\partial z}\right)=-\frac{\partial p_x}{\partial x}V_f+ \tag{14}$$
$$(\Delta\tau_{xx}\mathrm{A}_x+\Delta\tau_{yx}\mathrm{A}_y+\Delta\tau_{zx}\mathrm{A}_z)+\rho_f g_x V_f$$

where, V_f is the volume of the bath film in an arbitrary grid cell; A_x, A_y, A_z are the surface areas of the cell.

The bath film is assumed to be parallel to the cathode surface, thus its velocity normal to the cathode surface is zero. Therefore, the viscous shear stress and their velocity gradients in x and y directions are ignored in the present model. By substituting equations (11)-(13) into equation (14), it can be simplified as,

$$\rho_f\frac{\partial u_{f,x}}{\partial t}=\left(\mu_{Al}\frac{u_{Al,x}-2u_{f,x}}{h_c/2}-\mu_f\frac{2u_{f,x}}{h_f}\right)\frac{\mathrm{A}_z}{V_f}+\left(\rho_f-\rho_{Al}\right)g_x \tag{15}$$

Similarly, the momentum equation of the bath film in y-direction is given by,

$$\rho_f\frac{\partial u_{f,y}}{\partial t}=\left(\mu_{Al}\frac{u_{Al,y}-2u_{f,y}}{h_c/2}-\mu_f\frac{2u_{f,y}}{h_f}\right)\frac{\mathrm{A}_z}{V_f}+\left(\rho_f-\rho_{Al}\right)g_y \tag{16}$$

The system of bath film is closed by using equations (11), (15), and (16) combining the velocity of the molten aluminium derived from the model of bulk metal and bath.

Simulation parameters

The simulations are run in the simplified electrolysis cell which has the same geometry as in Einarsrud et al. [13]. The geometry is a cylinder shape cell shown in Figure 2. The thickness of the bath film is set to be 1 mm initially.

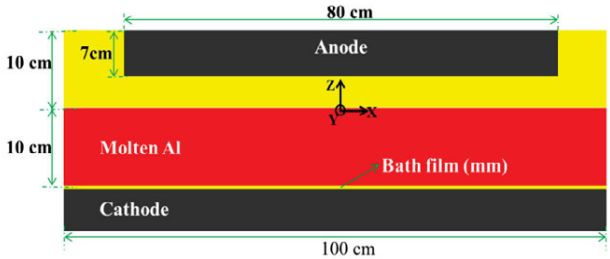


Figure 2 Geometry of electrolysis cell.

The simulations are performed in ANSYS Fluent R15.0 by using a 3D transient solver. The VOF model is applied to the multiphase bulk of fluid containing molten metal and bulk of bath. The electrical potential, current density, magnetic field, and bath film thickness and velocity are described by User defined Scalars (UDS).

Turbulence is modelled by the realizable k-ϵ model with standard wall functions. All boundaries are treated as no-slip walls for momentum. A constant reference potential is given on the top surface of the anode. The cathode is given a specified electrical current density. Two types of current distributions are investigated in this study, which are shown in Figure 3. The non-uniform current distribution is proposed by Eq. 17, based on considerations of sideledge effects and location of current collector bar in the cathode block [13,18]. All other surfaces are assumed to be non-conducting.

$$j(\mathrm{r})=a\cdot\sin\left(2\pi|r|\right)\cdot\left(r_{\max}-\sqrt{x^2+y^2}\right) \tag{17}$$

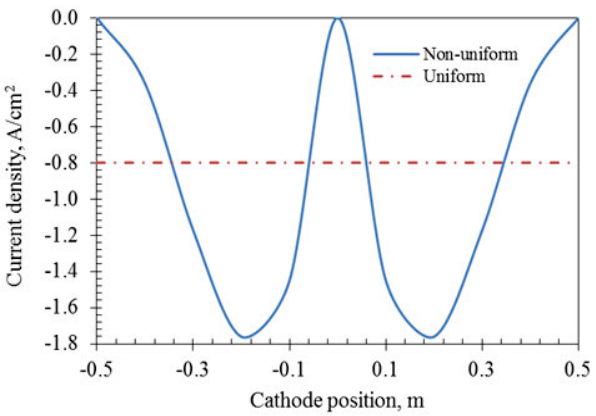


Figure 3 Current density distributions on the cathode surface.

Exact (Dirichlet) boundary conditions are specified for the z-component of the magnetic vector potential (the principal component), while academic boundary conditions are used for the remaining components [13, 18]. All the properties of materials are taken from previous work [13, 18-19].

Mass transfer of cathode wear

The aluminium carbide formation rate is definitely proportional to the electrochemical cathode wear rate locally. At the current state of model development we simply assume that the aluminium carbide formation rate is locally in equilibrium to its removal rate. Thus, the local cathode wear rate is anticipated to be proportional to the aluminium carbide removal rate which is a result of carbide dissolution into the bath film. The dissolved aluminium carbide in the film partly transfers into the molten aluminium and is partly taken out from the bath film through the sideledge film in the cells. In this manner, the simulation of the bath film was done by using a closed system without including the outflow of the bath film. Thus, we focus on the mass transfer of aluminium carbide and demonstrate the importance of the presence of the bath film. Figure 4 shows the mass transfer of aluminium carbide into the molten metal through the bath film.

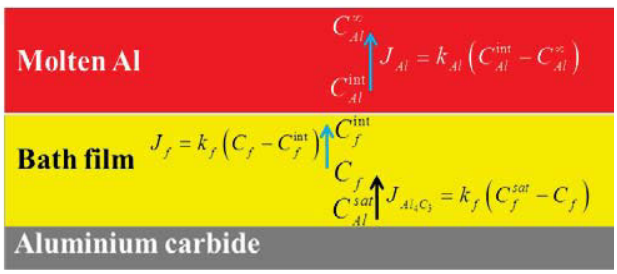


Figure 4. Mass transfer of aluminium carbide.

Due to the low velocity in the thin bath film, diffusion dominates the mass transfer of the aluminium carbide in the vertical direction in the bath film. The concentration of aluminium carbide in the film, C_f is assumed to be equal to the concentration on the bath film side (C_f^{int}) of the metal-film interface. The flux of carbide from the bath film and into the interface with the molten Al is:

$$J_f = k_f\left(C_f - C_f^{int}\right) \tag{18}$$

where, $k_f = \dfrac{D_{Al_4C_3,f}}{h_f/2}$ (19)

On the molten metal side of the interface with the bath film, the flux of aluminium carbide is given by,

$$J_{Al} = k_{Al}\left(C_{Al}^{int} - C_{Al}^{\infty}\right) \tag{20}$$

where, k_{Al} is the mass transfer coefficient that proposed by Davies [20],

$$k_{Al} = 0.075 \cdot Sc^{-2/3} u_{\tau} \tag{21}$$

Sc is the Schmidt number and u_{τ} is the friction velocity of aluminium which is expressed by the shear stress on the metal at the interface and density of molten aluminium,

$$u_{\tau} = \sqrt{\tau^{int}/\rho_{Al}} \tag{22}$$

and, C_{Al}^{∞} is the concentration of aluminium carbide in the bulk of the molten aluminium. Considering the general carbon wear rate and mass transfer between metal and bath, C_{Al}^{∞} was set equal to 50 % of saturation. .; C_{Al}^{int} is the concentration of aluminium carbide on the metal side of the interface, which is in equilibrium with C_f^{int} on the bath film side of the interface.

$$C_{Al}^{int} = \eta \cdot C_f^{int} \tag{23}$$

where, $\eta = 0.0049$ is calculated from the equilibrium of saturated concentration of aluminium carbide in bath and metal, which are 1 wt% [21] and 0.066 kg/m^3 [19] at 960 °C respectively.

The flux of aluminium carbide from the bath film is calculated by closure of equations (18)-(23), reading,

$$J_f = J_{Al} = \frac{k_f k_{Al}}{k_{Al}\eta + k_f}\left(\eta C_f - C_{Al}^{\infty}\right) \tag{24}$$

The aluminium carbide removal is controlled by diffusion from the cathode surface.

$$J_{Al_4C_3} = k_f\left(C_f^{sat} - C_f\right) \tag{25}$$

While the C_f in the bath film is controlled by diffusion and convection due to the flowing bath film,

$$\frac{\partial C_f}{\partial t} + \nabla\cdot(\boldsymbol{u}_f C_f) = \nabla\cdot(D_{Al_4C_3}\nabla C_f) + S_f \tag{26}$$

where, S_f is the source term of the aluminium carbide concentration, which comes from the dilution by refreshing bath film, the flux of carbide into the molten metal and the dissolution flux from the solid aluminium carbide layer on the cathode surface.

Results and discussions

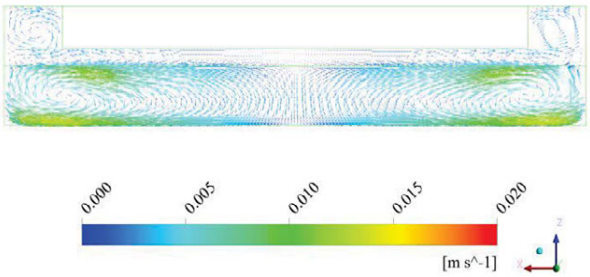
Flow of bulk fluid and bath film

The flow of the fluid is driven by the magnetohydrodynamic force caused by the electrical and magnetic fields. Thus the current passing through the reduction cells would have a large impact on the fluid flow for a given and representative magnetic field. Figure 5 shows the flow pattern of molten aluminium and bulk of bath at the profile of x=0 (see Figure 2). The changes of the predicted bath film thickness with time are shown in Figure 6.

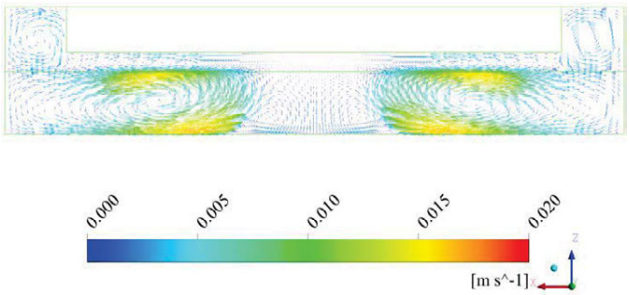
When a uniform current density is predefined on the cathode surface, the flow of the molten aluminium pushes the bath film towards the periphery of the cell. The film is built up quickly at the beginning and approaches steady state after 200 s. Nevertheless, the bath film is squeezed towards the centre of the cell when a non-uniform current distribution is assumed. The two peaks at the center position in the bath film are a result of the flow of molten metal in z-direction.

Actually, the flow of bulk fluid is controlled by both the electrical current distribution and the magnetic field distribution. However, in this case the magnetic field is induced by changing the electrical field, while the external magnetic field is not included in the present model. In this way, the electrical current density becomes the only parameter that controls the flow of the bulk fluid. Therefore, the behaviour of the bath film can deviate from the present results when a strong external magnetic field is introduced.

According to the change of bath film thickness with time shown in Figure 6, all the numbers for calculating mass transfer and cathode wear rate are taken from 300 s when the bath film has reached a pseudo-steady state.

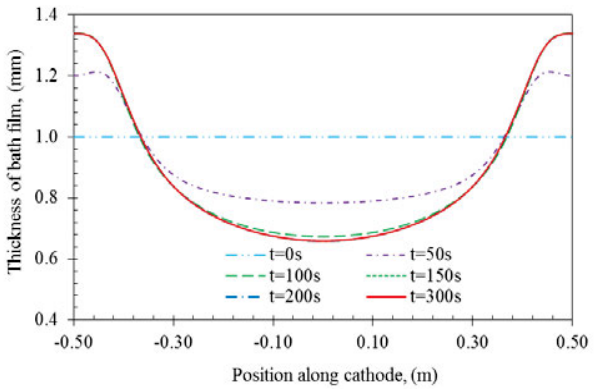


(a). Uniform current distribution.

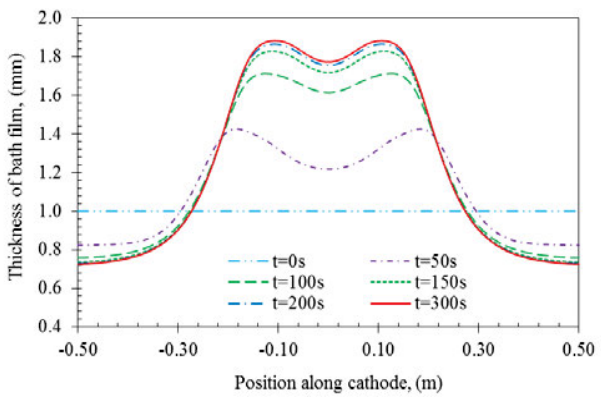


(b). Non-uniform current distribution.

Figure 5 Flow pattern of molten aluminium and bulk of bath.



(a). Uniform current distribution.

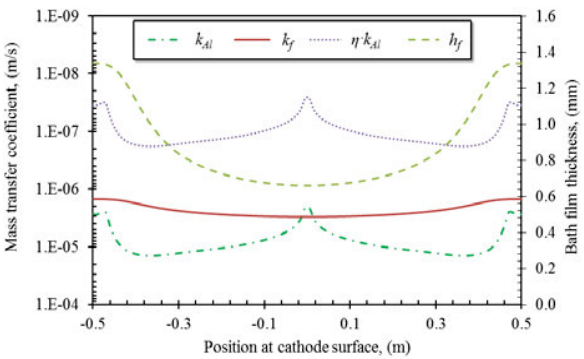


(b). Non-uniform current distribution.

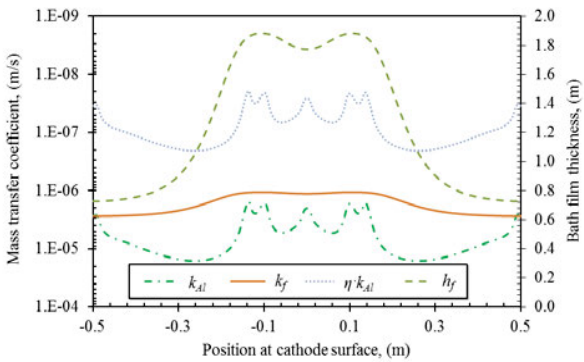
Figure 6 Change of bath film thickness with time.

Mass transfer of aluminium carbide

Figure 7 shows the mass transfer coefficient of aluminium carbide in bath and molten metal as well as the bath film thickness (t=300s). As seen Figure 7, $\eta \cdot k_{Al}$ is two orders of magnitude smaller than k_f . Recalling equation (24), the resistivity of aluminium carbide transport from bath film into metal would mainly locate it in the metal side of the interface. When the aluminium carbide is saturated in the bath film, the aluminium carbide removal rate would be independent of the bath film thickness. Figure 8 shows the cathode wear profiles dominated by mass transfer from bath film into molten aluminium. The results appear to be similar to that concluded by Einarsrud et. al [13], in which C_f and C_{Al}^{∞} are assumed as the saturated concentration and zero (half of saturated concentration in this study) respectively.



(a). Uniform current distribution.



(b). Non-uniform current distribution.

Figure 7 Mass transfer coefficient of aluminium carbide.

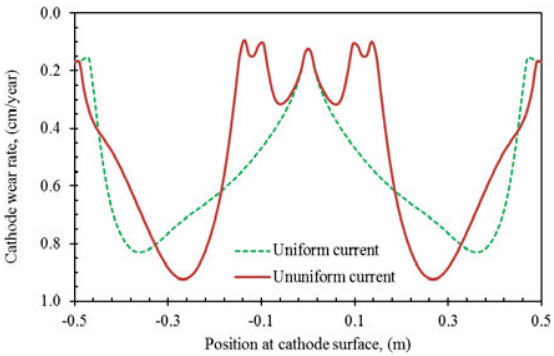
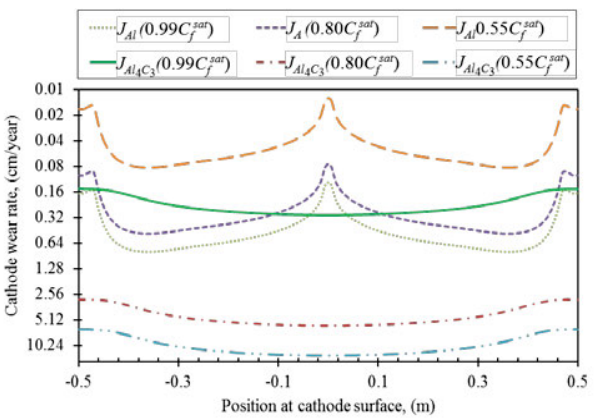


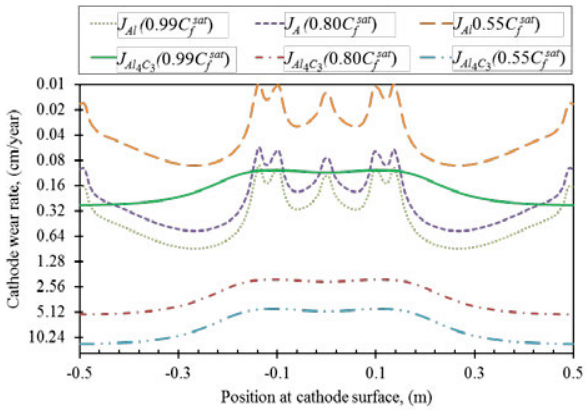
Figure 8 Cathode wear rate dominated by mass transfer from bath film into molten aluminium.

However, the aluminium carbide concentration is probably not saturated throughout the entire bath film, but is a function of diffusion and advection in Equation (26). The cathode wear profiles in Figure 8 are actually a limiting case where the bath film is saturated with aluminium carbide and carbide can only escape by mass transfer into the molten metal flowing above the cathode. On the other hand, when the aluminium carbide concentration is less than half of its saturated concentration in the bath film, as seen in Equation (24), it would yield no mass transfer from the bath film directly into the molten metal. The cathode wear rate is thus controlled by the mass transfer from the solid aluminium carbide layer into bath film, as seen in Equation (25). And the dissolved aluminium carbide will be brought out of the bath film by the drainage of the bath film. The realistic average cathode wear rate was reported as 2 cm/year; and the maximum cathode wear rate is around 5 cm/year. Nevertheless, the average values which was computed due to mass transfer into the molten metal is 0.58 cm/year and 0.54 cm/year, for uniform and non-uniform current density distribution (Figure 8) ,respectively. Those average values are based on the assumption that the aluminium carbide is saturated in the bath film. The contribution of the mass transfer into the molten metal is not large enough to account for the realistic cathode wear rate. Therefore, the outflow of bath film (film drainage) contributes largely to remove the dissolved aluminium carbide away from the cathode.

Figure 9 shows the influence of the aluminium carbide concentration in the bath film on the cathode wear rate. J_{Al} indicates the cathode wear rate due to mass transfer from bath film into molten metal; while $J_{Al_4C_3}$ indicates the cathode wear rate by dissolving aluminium carbide from the cathode surface. The concentration of aluminium carbide in the bath film is specified in the brackets. The aluminium carbide dissolution rate is definitely proportional to the cathode wear rate. When the aluminium carbide concentration is close to its saturated value in the bath film, the mass flux $J_{Al_4C_3}$ and J_{Al} are similar, but the aluminium carbide dissolution rate is still too low to account for the realistic cathode wear rate. In the case of uniform current density, the cathode wear rate by aluminium carbide dissolution reaches realistic values when $C_f = 0.8C_f^{sat}$. The contribution of the mass flux of aluminium carbide into the molten metal is shown by $J_{Al}(0.8C_f^{sat})$ and the rest of the dissolved aluminium carbide is taken out of the bath film by the outflow (film drainage). Similar conclusions can also be drawn in the case of non-uniform current density.



(a). Uniform current distribution.



(b). Non-uniform current distribution.

Figure 9 Influence of aluminium carbide concentration in the bath film on the cathode wear rate.

The outflow of bath from the bath film was not simulated in this study. The aluminium carbide concentration in the bath film probably varies along the radial direction in the reduction cell. We next discuss and estimate the influence of aluminium carbide diffusion and advection in the bath film along the radial direction.

The fresh bath is released into the bath film from the melting of sludge which contains 40 wt% Al_2O_3 [22]. By assuming that the bath flowing out of the bath film is saturated with aluminium carbide, the relations of sludge dissolution rate, cathode wear rate and bath refreshing rate are shown in Figure 10.

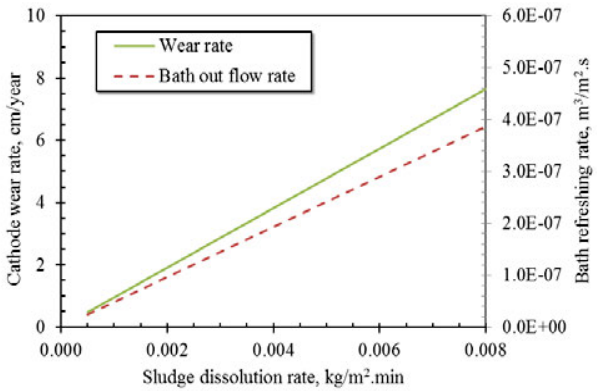


Figure 10 Relations of sludge dissolution rate, cathode wear rate and bath refreshing rate.

The outflow mass rate (R_f) of bath from the bath film is the integral of the mass flow of fresh bath being released on the entire bottom of the aluminium reduction cell. Though the simulations in this study are done with a closed bath film without out flow and source, the velocity of the bath film towards the periphery of the cell is estimated as,

$$u_f^* = \frac{R_f}{2\pi r_c h_f} \tag{27}$$

By the difference of realistic average cathode wear rates (2 cm/year) and those shown in Figure 8, the out flow of bath from the periphery of the reduction cell is calculated to be around 2.68×10^{-5} m/s and 5.05×10^{-5} m/s. The Peclet number is given by,

$$Pe = \frac{u_f^* \cdot h_f}{D_{Al_4C_3}} \quad (28)$$

The orders of magnitude of u_f^*, h_f, $D_{Al_4C_3}$ are 10^{-5}, 10^{-3} and 10^{-9} respectively. Thus the Peclet number in the bath film is around 10, which indicates that the advection of aluminium carbide is not negligible in Equation (26).

In the future work, the mass transfer will be coupled with the fluid flow of the bath film. This is expected to give more realistic and quantitative cathode wear profiles. Furthermore, the external magnetic field would largely impact on the flow pattern of bulk fluid and, accordingly, on the bath film.

Conclusions

In this study, we focus on the presence of a bath film. A preliminary model of the bath film flow is established to approximate the realistic cathode wear mechanism. The influence of the bath film on the cathode wear rate is discussed.

(1) In the case of uniform current density on the cathode surface, the bath film is tending to be pushed towards the periphery of the aluminium reduction cells, which yield a thicker bath film close to the periphery. The bath film is built up in the center of the cell when a "W" shape current density is distributed on the cathode surface.
(2) At the interface between the bath film and the molten aluminium, the mass transfer of aluminium carbide is dominated by the mass transfer on the metal side. The mass flux into the molten aluminium is not enough to account for the realistic cathode wear rate, even when the molten metal side mass flux is on its maximum, which is in the case that the bath film is saturated with aluminium carbide.
(3) The dissolution rate of aluminium carbide from the cathode surface is close to realistic cathode wear rate as the aluminium carbide concentration in the bath film is averagely close to $0.8C_f^{sat}$. This is only possible with a significant loss of bath film due to outflow (film drainage). The outflow of the bath film contributes largely to the cathode wear.
(4) The aluminium carbide concentration probably varies in the bath film. The advection plays an important role in the concentration distribution.
(5) The external magnetic field would greatly impact on the flow pattern of bulk fluid and bath film. In order to predict the cathode wear profile more realistically in the future, it is important to couple the mass transfer processes and the flow of the bath film.

Acknowledgement

The present work was carried out in the project Durable Materials in Primary Aluminium Production (DuraMat), financed by the research Council of Norway, Hydro Aluminium, Sør-Norge Aluminium (Søral), and Elkem Carbon. Permission to publish the results is gratefully acknowledged.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF CATHODE COLLECTOR COPPER INSERTS ON THE HALL-HÉROULT CELL MHD STABILITY

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Abstract

A composite cathode design with copper inserts could reduce the voltage drop and the horizontal currents in the liquid aluminium, which are responsible for the MHD wave instability and the intense metal circulation. A model incorporating full 3d busbar configuration coupled to the electric current in the liquid is used to study the MHD response when the copper inserts are included in the cathode construction. The results confirm the reduction of electric current component in liquid metal along the collector bar direction, however show the equally important horizontal current in the metal pad in the long direction of the cell due to the current distribution over the outgoing collectors. The time dependent MHD solution is applied to industrial cells incorporating the composite cathode, variable bottom and the full neighbor cell network effect. Examples of two types of cells show the possibility for different scenarios: either decreasing or increasing the MHD stability depending on the overall cell design and operating conditions.

1. Introduction

Industrial aluminium production cells are highly optimised for magnetic field and electric current distribution in order to avoid MHD instabilities, increase the energy savings and to maintain a stable electrolytic process during normal operation. A composite cathode design [1-7] where steel collector bars are supplemented by copper inserts, has been previously described as beneficial to reduce the cathode voltage drop and the horizontal currents in the liquid aluminium, which are known to be responsible for the MHD instability development. However, the conclusion about reduced horizontal current in liquid metal is valid only for the component along the collector bar direction as demonstrated in previous numerical solution examples [1,2,7]. The current distribution over all the outgoing collectors is equally important in order to reduce the horizontal current in the metal pad in the long direction of the cell, which in turn could affect the MHD stability. The linear stability theory, which is often used to estimate the MHD stability, assumes a uniform current distribution over the cathode collectors, neglecting the full bus bar network effect on the development of the longitudinal currents and the associated MHD waves.

A model incorporating full 3d busbar configuration and fully coupled electric currents in the liquid metal and bath is used for a systematic study of the MHD response [8]. The code has been extensively tested, see for example [9,10] and in other cases. The model computes time-dependent currents, voltages, magnetic fields, bath-metal interface shape and turbulent magnetically driven flow in the bath and liquid metal. The electric current in the bus bar network is computed at all times following the wave development on the interface between liquid layers. The dome shaped time average deformation of the metal surface is projected to the bottom of the anode block to account for the anode bottom burn-out effect for a constant ACD, except in the case of an initial disturbance, such as an anode change operation or an imposed gravity wave of a specified mode. The electric current distribution in the liquid zone is computed from all the bus bar network connections to the anodes and cathode collectors, coupled with the cell interior details like the ledge profile, bottom shape, collector connections, electrolyte channels and the electrochemical voltage drop. The temperatures of the bus bars are adjusted according to the computed Joule heating and losses to the air. The magnetic field is optionally recomputed at all times in order to follow the full magneto-hydrodynamic wave and flow development. The detailed representation of the cell steel elements adds non-linearity to the overall magnetic field properties. The MHD response of different commercial cells is quite variable depending on their design (side- to-side, end-to-end, Soderberg, etc.), the line amperage, operational practices (tapping, anode change, bottom erosion), and many other factors.

In the present paper a dynamically coupled mathematical software is applied to two particular cells when the copper inserts of given position, length, cross section and variable electrical conductivity are included in the cathode construction. The two types of cells analyzed show the possibility for two different scenarios: either decreasing or increasing the MHD stability depending on the overall cell design and operating conditions. An attempt is made to analyze the governing physical field distributions to pinpoint the source for the instability.

2. Modelling results for 180 kA cell

Figure 1 is a model representation of the TRIMET 180 kA cell potline [9] showing a zoomed view to all the conductors that are part of the simulation. The conductors are colored by their computed currents and the dark zone in the test cell is due to the steel construction elements included for the magnetic field calculation. The individual collector bars are easier to visualize at the neighbor cells included in the full solution. Copper inserts are included as additional bars electrically connected to the normal steel collectors. The size (length and cross section), start/end positions and the electrical conductivity can be prescribed for the steel and the copper bars independently, permitting to optimize their properties.

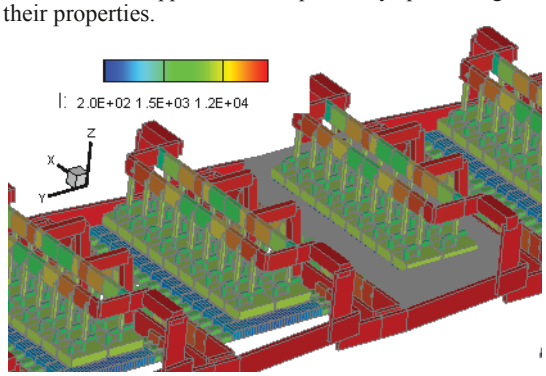

Figure 1. Busbar configuration for 180 kA Trimet cell.

The present simulations used a fixed insert positioning on top of the steel bar. The copper insert has 4 by 10 cm cross section, starts at 20 cm from the cell central axis and extends to the cell side wall position. The electrical conductivity of the steel bar was adjusted to the temperature and the copper insert has similarly adjusted conductivity to the local temperature. There is also a contact resistance between the collector and the carbon, which is modeled as a uniformly distributed ohmic resistance prescribed in the input material data file.

In the case of the 180 kA cells without the Cu inserts the typical cathode voltage drop CVD was computed in the range 0.24 – 0.26 V, depending of the collector position along the cell. The computed CVD value is within the range for typical cathode voltage drop in commercial cells [2,11], however this information is variable due to the cell aging and various design factors. Note, the voltage drops are unequal for different collector numbers due to the distribution of the outgoing electrical currents in the collector bars as shown in the Figure 2. When the Cu inserts were included, the CVD dropped to 0.18 – 0.20 V, about 60 mV less for this type of cell modification. The presence of the inserts makes the distribution of the outgoing currents a little steeper in comparison to the standard case.

The respective horizontal currents in the liquid metal are significantly reduced when using the inserts, as can be seen comparing the Figures 3 and 4. The MHD stability calculation involves an initial sloshing wave perturbation of a small amplitude, a=5 mm in this case, prescribed as a user input. This results in the current redistribution in the whole electrical network including the liquid metal shown in the Figure 5 at time = 30 s after the initial perturbation. The corresponding case with the copper inserts is illustrated in the Figure 6, showing a rather moderate increase of the longitudinal electric current density.

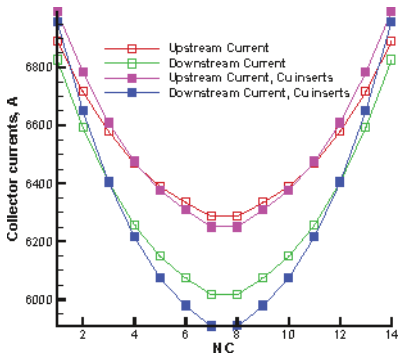

Figure 2. The collector currents exiting the cell compared for the effect of the copper collector inserts for the 180 kA cell.

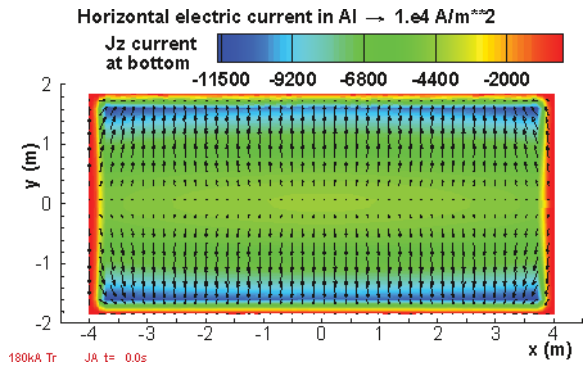

Figure 3. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 180 kA cell without the inserts.

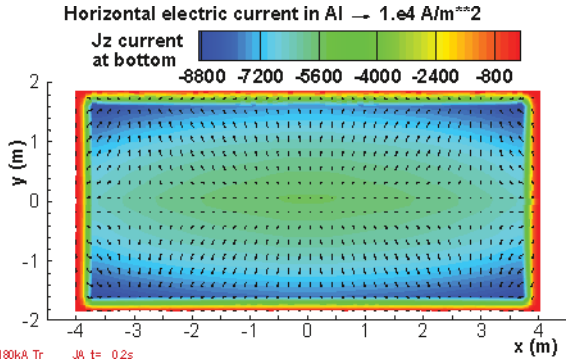

Figure 4. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 180 kA cell with the Cu inserts.

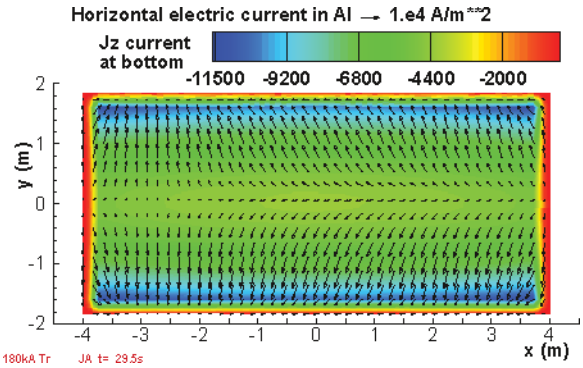

Figure 5. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 180 kA cell without the Cu inserts after 30 s of the perturbation.

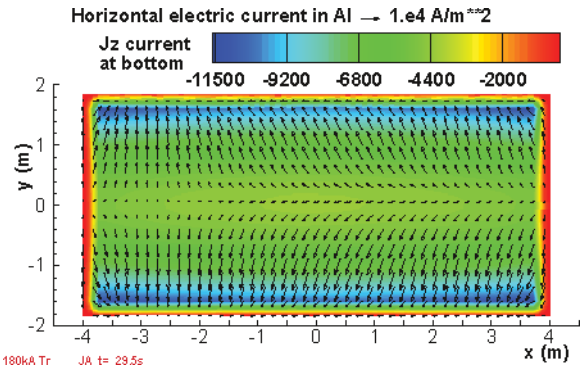

Figure 6. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 180 kA cell with the Cu inserts after 30 s of the perturbation.

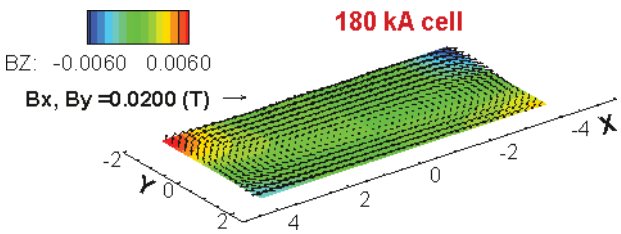

Figure 7. The horizontal (arrows) and vertical magnetic field (color levels) distribution for the 180 kA cell without the Cu inserts.

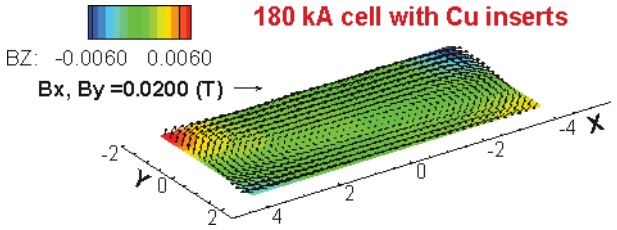

Figure 8. The horizontal (arrows) and vertical magnetic field (color levels) distribution for the 180 kA cell with the Cu inserts.

The respective magnetic field distribution in the middle of the metal layer is practically the same for both cases, the Figures 7 and 8. The dome shaped interface deformation is changed very little with the addition of the copper inserts as the Figures 9 and 10 show. There is more change for the horizontal velocity in the liquid metal after the Cu inserts are included. The typical magnitudes of the horizontal circulation reduces by about half to approximately 0.07 m/s if compared to the standard case where the velocity magnitude is about 0.15 m/s (Figures 12 and 11, respectively). This leads to a 'no surprise' faster damping of the MHD wave perturbation, meaning an enhanced stability of this cell after the copper insert inclusion, Figure 13. The reduced horizontal current density both in the transversal (y) and the longitudinal (x) directions are apparently responsible for the improved MHD stability of the cell. An attempt to gain more energy savings by reducing the ACD by 2 mm was equally successful, leaving this particular cell in the MHD stable condition (Figure 13). This dynamic simulation indicates a potential benefit when the copper inserts are included in the 180 kA cell construction.

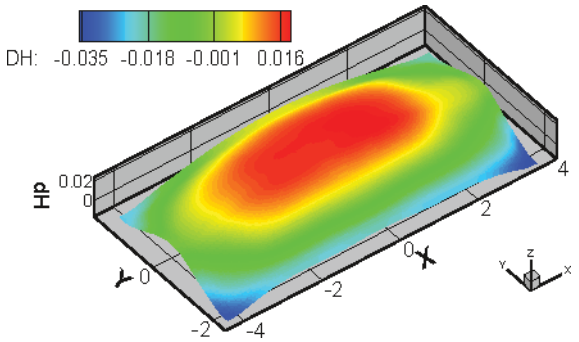

Figure 9. The interface between liquid metal and electrolyte (color levels) distribution for the 180 kA cell without the Cu inserts.

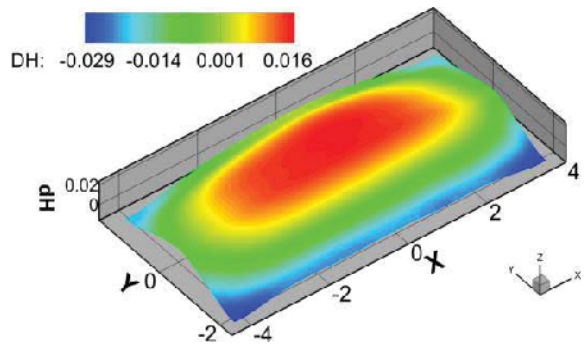

Figure 10. The interface between liquid metal and electrolyte (color levels) distribution for the 180 kA cell with the Cu inserts.

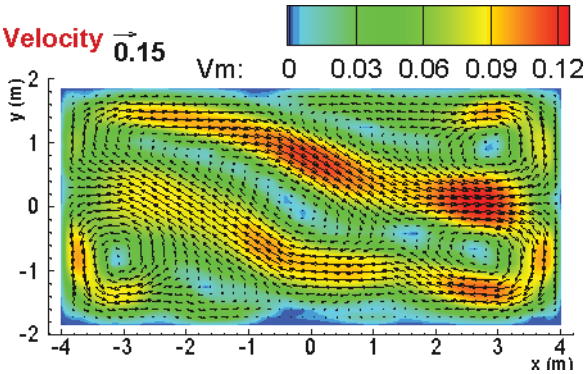

Figure 11. The horizontal velocities in liquid metal for the 180 kA cell without the Cu inserts.

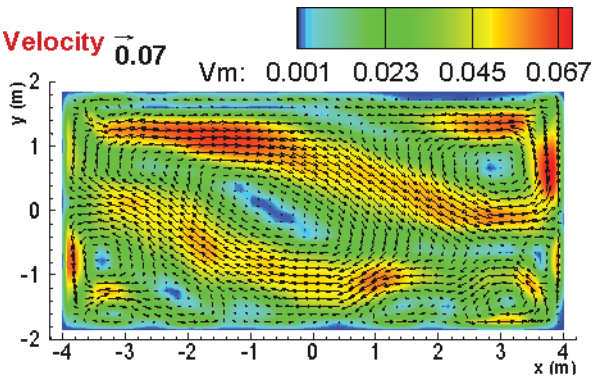

Figure 12. The horizontal velocities in liquid metal for the 180 kA cell with the Cu inserts.

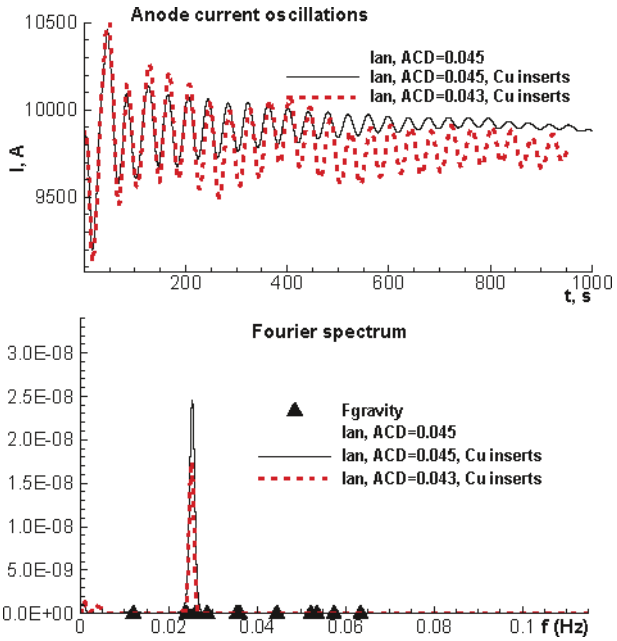

Figure 13. The effect of the collector bar Cu inserts for the 180 kA cell after 5mm sloshing (1,0) perturbation, ACD=0.045m, H_{al}=0.17m, and comparison to the case of reduced ACD= 0.043m.

2. Modelling results for 500 kA cell

The same modeling procedure was repeated for a 500 kA cell, known to be stable to MHD waves according to our previous simulation results [8,12]. Figure 14 shows a rather more complex arrangement of the bus bar network for this type of cells, consisting of 6 risers and 13 electrically separated cathode sections to distribute the collector currents evenly and achieve 49.50% upstream, 50.5% downstream cathodic current split. The collector current distribution at initial equal ACD interface position is shown in Figure 15. For the larger cell the collector currents are less uniformly distributed and the effect of the cathodic sections on the current distribution is clearly visible. Still, the inclusion of the similar copper inserts as in the previous example of the smaller cell leads to little change in the current distribution along the collectors, but a significant reduction in the horizontal current in the liquid metal at the stage when the ACD is unperturbed: Figures 17 and 18. However, when a gravity wave sloshing perturbation of mode (1,0) and amplitude 0.005m is applied, the current distribution change can be felt even for the collector bar distribution as seen comparing the Figures 15 and 16. This leads to a significant growth of the longitudinal current component Jx in the liquid metal, which in the case of the copper inserts present gives even more pronounced Jx magnitude increase, Figures 19 and 20.

In the case of the noncommercial concept 500 kA cells without the Cu inserts the typical cathode voltage drop CVD was computed in the range 0.44 – 0.46 V, depending of the collector position along the cell. When the Cu inserts were included, the CVD dropped to 0.35 – 0.37 V, about 90 mV less for this type of cell modification. The inserts make the distribution of the outgoing currents more variable in comparison to the standard case.

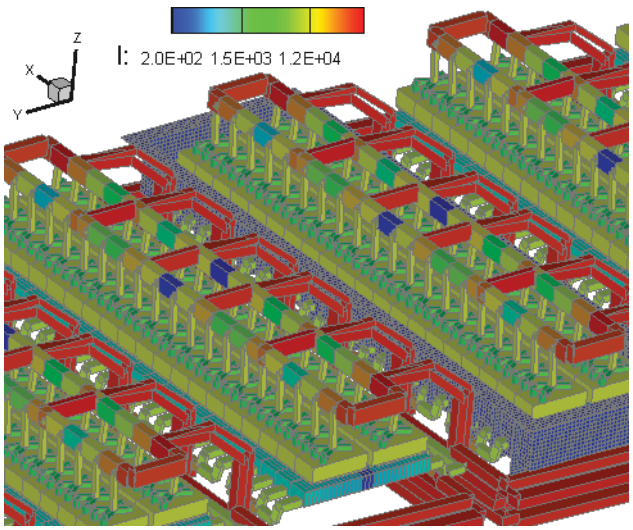

Figure 14. Busbar configuration for 500 kA cell.

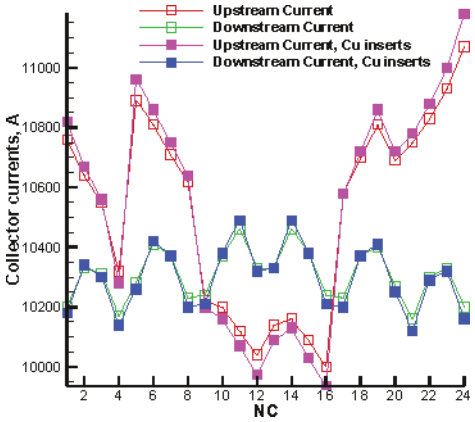

Figure 15. The collector currents exiting the cell compared for the effect of the copper collector inserts for the 500 kA cell for the uniform ACD= 0.045m.

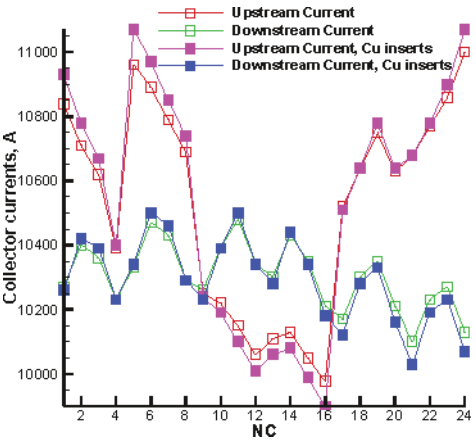

Figure 16. The collector currents exiting the cell compared for the effect of the copper collector inserts for the 500 kA cell for the deformed interface after 30 seconds when the 5 mm sloshing wave perturbation is applied.

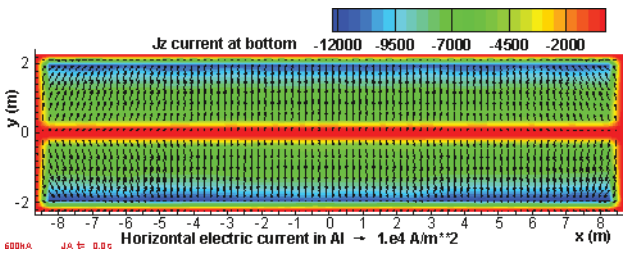

Figure 17. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 500 kA cell without the inserts.

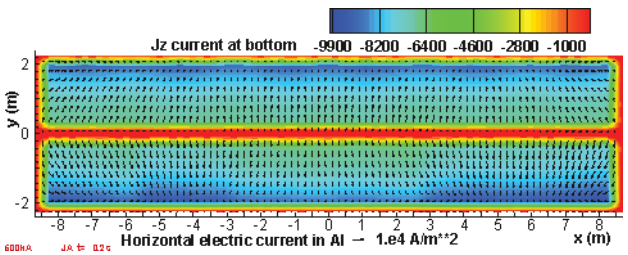

Figure 18. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 500 kA cell with the inserts.

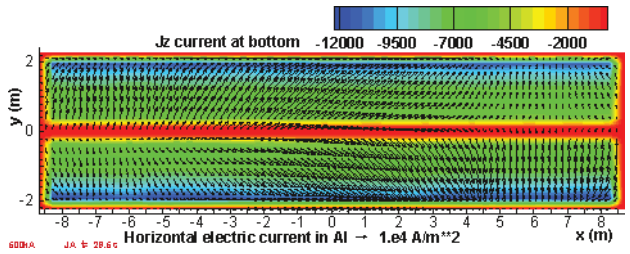

Figure 19. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 500 kA cell without the Cu inserts after 30 s of the perturbation.

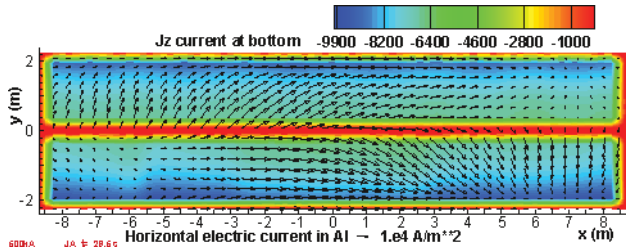

Figure 20. The horizontal (arrows) and vertical bottom current (color levels) distribution for the 500 kA cell with the Cu inserts after 30 s of the perturbation.

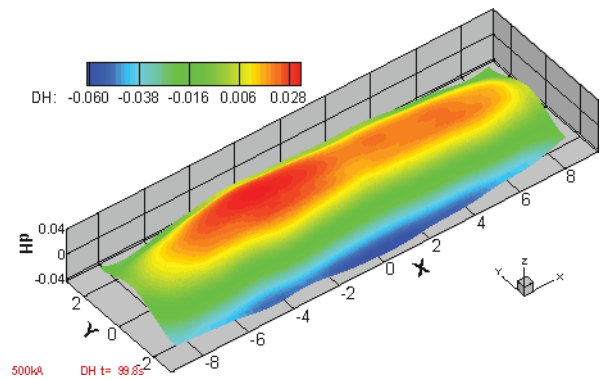

Figure 21. The initial interface between liquid metal and electrolyte (color levels) distribution for the 500 kA cell with Cu inserts.

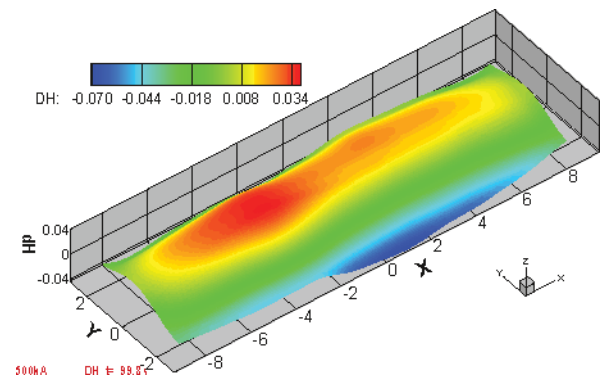

Figure 22. The interface at 100 s after the perturbation is applied for the 500 kA cell with Cu inserts.

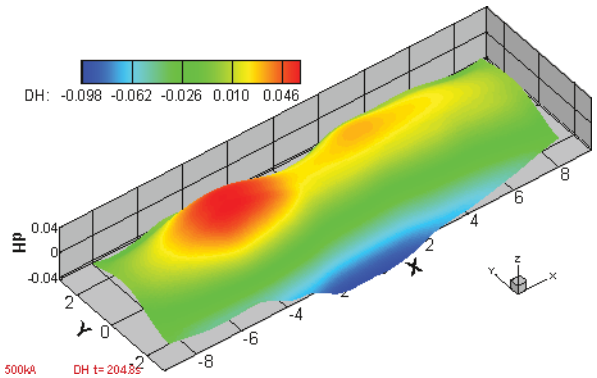

Figure 23. The interface before short-circuiting at 205 s after the perturbation is applied for the 500 kA cell with Cu inserts.

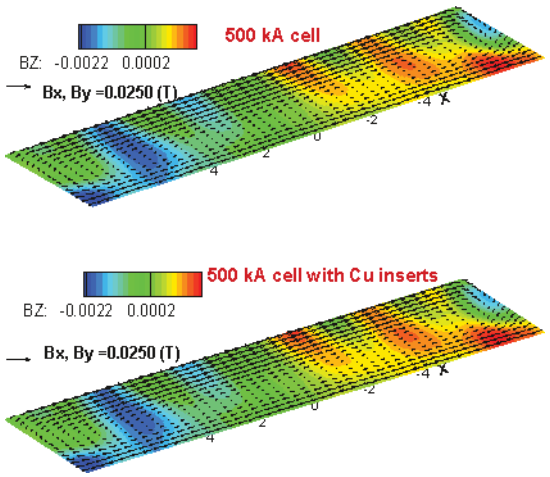

Figure 24. The comparison of magnetic field distribution for the 500 kA cell with and without the Cu inserts: horizontal (arrows) and vertical magnetic field (color levels).

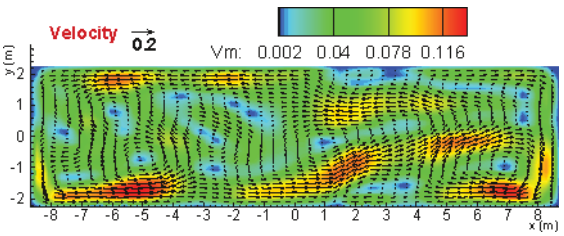

Figure 25. The horizontal velocities 30 s after perturbation in liquid metal for the 500 kA cell without the Cu inserts.

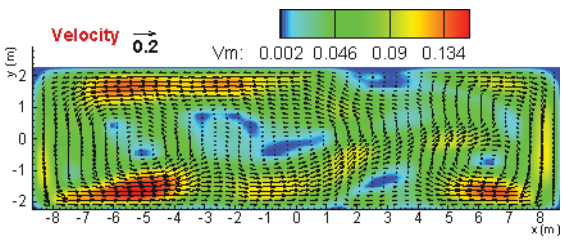

Figure 26. The horizontal velocities 30 s after perturbation in liquid metal for the 500 kA cell with the Cu inserts.

The rise in the perturbed horizontal current density in the liquid metal is detrimental for the MHD stability, giving the growth of the dome shaped interface in a sloshing mode illustrated in the Figures 21, 22 and 23 for consecutive time intervals of about 100 s. The magnetic field calculated with the effects of these internal and external current perturbations (in the presence of the ferromagnetic construction elements) shows a rather small change, Figure 24. Hence we conclude that the electric current longitudinal component Jx growth is primarily responsible for the MHD instability growth (Figure 27) and larger horizontal velocities (Figures 25 and 26) when the perturbation is applied. A further increase of the ACD by at least 3 mm to the value ACD=0.048m is required to restore the cell to stable operation. This means that in the case of a larger, more complex cell the cathode inserts did not function as intended. The energy savings due to the lower cathode voltage drop are completely lost due to the required increase of the ACD.

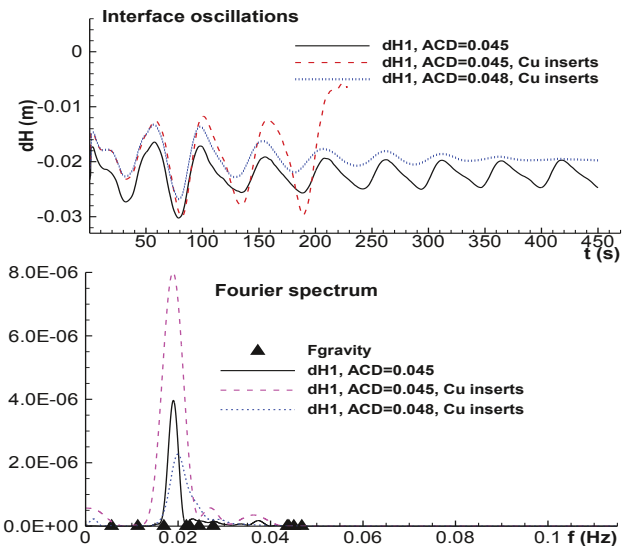

Figure 27. The effect of the collector bar Cu inserts for the 500 kA cell after 5mm sloshing (1,0) perturbation, ACD=0.045 m, H_{al}=0.20 m, and comparison to the case of increased ACD= 0.048 m.

Conclusions

The MHD model for full cell stability analysis is applied to the Trimet 180 kA and experimental 500 kA cells to compare the results of including copper inserts of similar design in the collectors. The case of the larger, more complex cell demonstrates that a simple repeat of the same recipe of the insert design could lead to no gain in the energy savings and a reduced MHD stability of the cell. Further studies will be needed to find a particular design of the collector with inserts for an economically beneficial solution.

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Light Metals 2016

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ANODE BAKING FURNACE FLUEWALL DESIGN EVOLUTION: A RETURN OF EXPERIENCE OF LATEST BAFFLELESS TECHNOLOGY IMPLEMENTATION

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Keywords: anode baking furnace, fluewall design

Abstract

In the early 2000's, a project to redefine fluewall design was started at the LRF, St Jean de Maurienne, France. The goal of this project was to improve the performance of the fluewall in term of safety, flow and heat distribution, pressure drop and thermo mechanical behavior. The idea came to remove the baffles and, with the help of finite element modelling, a long and difficult development process started with several generations of design that were tested in various smelters across the world. Some pitfalls were found on the road like uneven heat distribution, too fast degassing or fluewall pinching. However, those problems were progressively addressed and baffle less furnaces erected at Tomago in 2008 and Qatalum in 2009 are showing now, after several years of operation, that initial goals are being reached.

Baffleless design development origin

Back in the 90's, the idea came to challenge the typical anode baking furnace fluewall design as it was assumed that it was close to its full potential [1] .
The question was raised: if the fluewall design was to be reinvented, what the perfect fluewall would look like?
First, of course, the design should be better in terms of safety. Then, the fluewall should last longer. Pressure drop would be reduced in order to reduce heat sink to optimize energy consumption. Temperature would also be as homogeneous as possible in order to reduce anode baking deviation. Brick quantity would be reduced as much as possible to reduce fluewall cost and energy consumption.

Common standard fluewalls concerns

Main risk in an anode baking furnace is an explosion. One cause that can lead to an explosion is having a blocked fluewall. Standard design can be blocked if some packing coke is flowing through any gaps or cracks in the fluewall and blocking the bottom baffle as shown in figure 1. A by-pass is often included as a mitigation at the baffle top in the design but it allows only a little flow (cf. Figure 1).

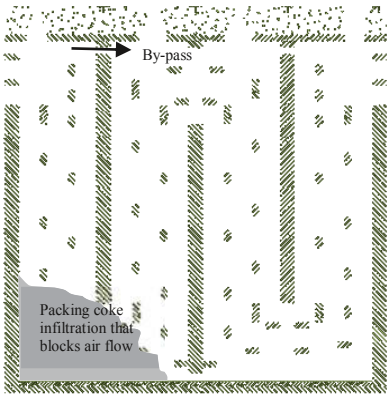

Figure 1: Packing coke infiltration blocking the flow at bottom baffle of a standard ABF fluewall

Fluewall lifetime expectancy is not so easy to explain and to compare between smelters because it depends on many factors external to fluewall design itself as brick quality, operations and maintenance standards [2] and the sodium content of the anodes being baked – the quality of the butt cleaning process. However, it has also been highlighted that fluewall design itself is a main contributor to fluewall lifetime expectancy: tie bricks number and positioning, brick thickness (cf. figure 2) are known to influence fluewall robustness. Fluewall design can also impact negatively on fluewall lifetime expectancy by inducing thermomechanical stresses. In fact, in a standard fluewall design, the baffle acts like a screen to the heat exchange, which generates a temperature gradient between two consecutive peephole shafts. This gradient leads to a thermomechanical constraint across the refractory (Figure 3 represents shearing strengths concentrated around the mechanical singularities which are the baffles) and ends with fluewall deformation like pinching and bowing. However, impact of design on thermomechanical constraints and fluewall lifetime expectancy is not correlated and measured accurately yet.

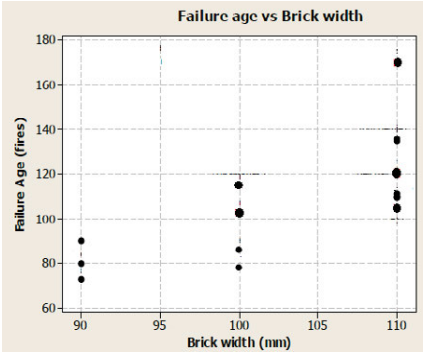

Figure 2 – Brick thickness impact on fluewall lifetime expectancy

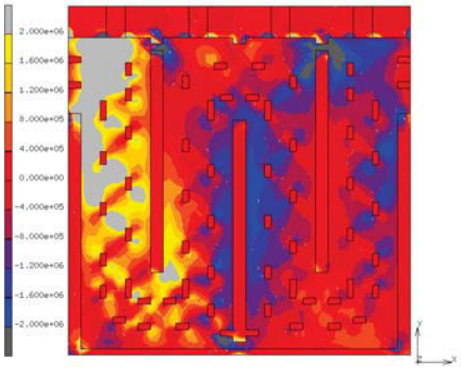

Figure 3 – Shearing stress (MPa) in a standard fluewall over preheating, calculation through finite elements modelling

High fluewall pressure drop leads to air infiltrations and exfiltrations, nomatter how good fluewall tightness may be. This high pressure drop means, at the bottom line, energy loss. Based on experience and modelling, an increase of 5% of air infiltration and exfiltration costs 160 MJ/tba. Figure 4 shows thermal losses with an infrared camera of a top furnace. Figure 5 shows typical consequences of high pressure drop combined with poor sealing.

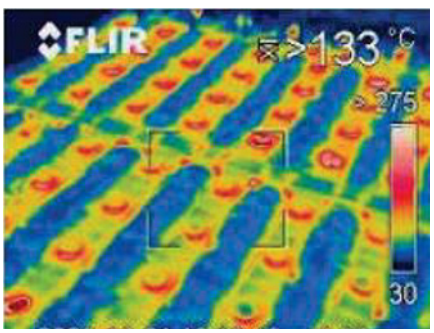

Figure 4 – Anode baking furnace heat losses

Figure 5 – Unsatisfactory top furnace tightness

Baking temperature deviation is also linked to the fluewall design and it should be as low as possible. In figure 6, we can observe examples of anode baking cartographies that reflect hot spots induced by gas injectors positioned in peephole shaft separated by the baffles.

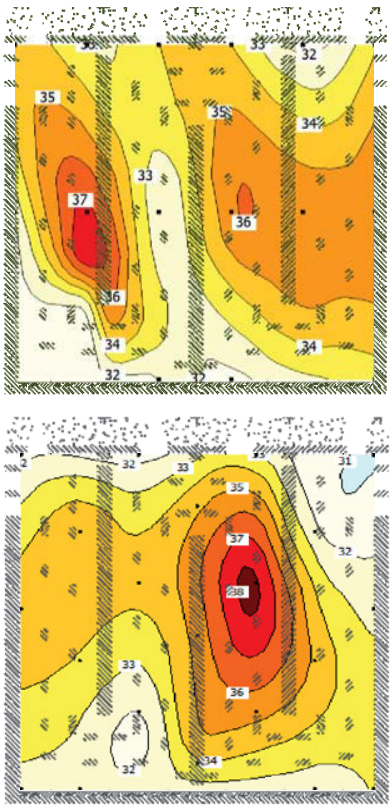

Figure 6 – Examples of baking level (Lc) in a pit

Baffleless design concept vs. standard fluewall

The different solutions included in the baffleless design [3], applied to tackle the common standard fluewalls concerns, will be detailed now.

From a safety point of view, the baffleless design cannot be blocked (cf. figure 7). On the top of that, a packing coke infiltration incident will not impact seriously the anode baking level of the pit whereas a blocked standard fluewall will end with pit re-baking.

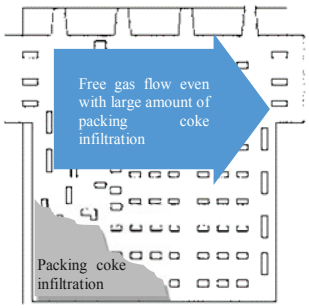

Figure 7: Baffleless fluewall design with blockage proof positive safety design

From a mechanical point of view, the baffleless design concept allows an even distribution of the tie bricks along the surface of the fluewall. In figure 8, the density of tie bricks per surface unit is represented. The surface of the fluewall has been marked out in 100 squares and for each square the ratio tie brick modules over square surface has been calculated. Baffleless design shows a considerably more homogeneous profile than the standard design and, most important, no area with a zero density. On the top of that, average density is lower for baffleless design (0.10 tie brick/surface unit vs. 0.14), that means that with baffleless design, a more homogeneous structure can be achieved with less bricks (-5%).

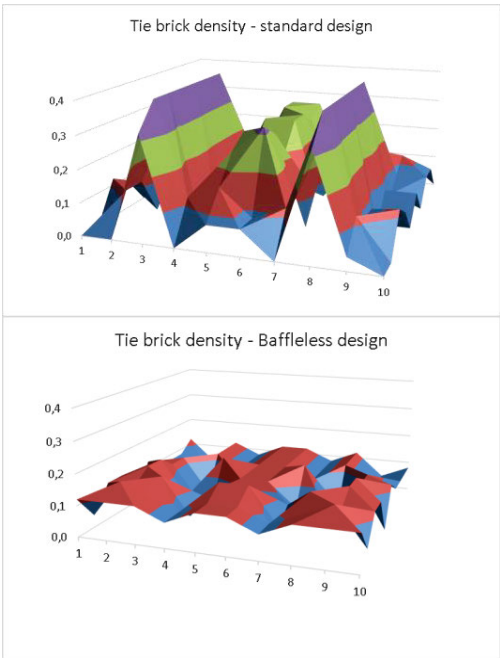

Figure 8 – Figure represents tie brick density per square (horizontal plane represents fluewall's surface and has been divided in 10x10 squares)

One striking point concerning baffleless design compared to standard design is the gas speed. For the same mass flow, the average gas velocity is around 5 times lower (cf. Figure 9).

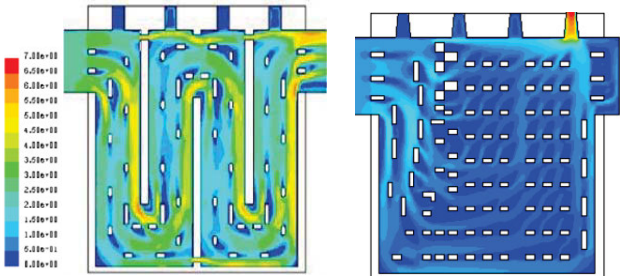

Figure 9 – Gas velocity in different fluewall designs (m.s^{-1}) through finite element modelling (first baffleless design generation)

The pressure drop in a pipe is described in (1):

$$\Delta P = R_{point\ loss}.\lambda_{surface\ loss}.\frac{L}{D}.\rho.v^2 \quad (1)$$

Assuming that:

- surface losses factor and air mass density are equivalent in both standard and baffleless designs,
- gas velocity in a baffleless design is 1/5th of a standard design for the same air flow,

it is possible to write the following equality:

$$\Delta P_{Baffleless} \approx \frac{1}{100}.\frac{R_{Baffleless}}{R_{Standard}}.\Delta P_{Standard} \quad (2)$$

Interpretation of (2) gives the following critical indication: baffleless design is much less constrained by hydraulic resistance than standard design. In other words, in a standard design, tie bricks must be carefully positioned trying not to increase the hydraulic resistance whereas in a baffleless fluewall there is much more freedom to position tie bricks in respect of other criteria like mechanical homogeneity whilst targeting a desired fluewall pressure drop.

From a baking homogeneity perspective, a baffleless fluewall is an homogenous combustion chamber. There are no baffles that act like "radiation screens" and create hot spots below the gas injectors. Temperature during forced heating is more evenly distributed (cf. Figure 10).

Figure 10 – Gas temperature in different designs through finite elements modelling (first baffleless design generation)

The baffleless technology demonstration took several years and some pitfalls were found on the road. The main difficulties were encountered in Lynemouth smelter and Tomago, for different reasons. The design had four major evolutions:

- The first generation (from first trials, asymmetric design)

- The second generation (including Lynemouth and Tomago BO1, symmetric design)
- The third generation (Qatalum)

Lynemouth return of experience

The first full scale implementation of baffleless design was achieved in 2006 at Lynemouth smelter during BF2 refurbishment. Early after start-up, a major concern appeared with the design. One of the baffleless design advantage was pushed too far: the fluewall pressure drop was too low. The negative pressure at the first section was in a range of [-80;-40 Pa]. The insufficient negative pressure ended with pitch volatile matters degassing directly out of the pit, creating yellow deposits (Benzo (a) pyrene) on the packing coke and fumes in the building, cf. Figure 11.

Figure 11 – Evidence of anode degassing through the pit at Lynemouth BF2

The problem was managed applying the '"alternative preheating regulation" (APR), a specific technique from RTA [4], that allowed to increase the draft whilst managing the pitch front advance (cf. Figure 12).

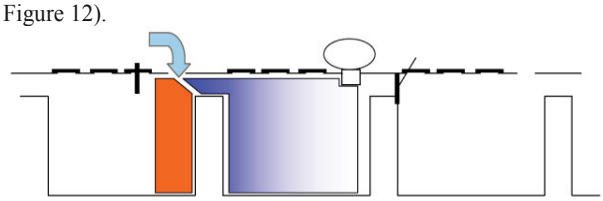

Figure 12 – APR technique

After this experience, the design criteria for the baffleless fluewall were changed, setting the pressure drop above a minimum value to ensure volatile matters extraction.

Tomago return of experience

The following full scale implementation of baffleless design was done at Tomago Bake Oven 1 during refurbishment in 2008. Tomago needed to increase the anode height and refurbishment was an opportunity to increase the pit width by 30 mm. Taking into account the fast fire cycle (21h) and the already narrow fluewall design (460 mm), it would have been unsafe to keep a standard fluewall design because of the risk to have a dramatic increase of the pressure drop into the fluewall line and probable combustion problems. On top of that, Tomago had been facing fires in BO1 in the early 2000's, it was known that furnace capacity had reached its limit. Then it was decided to install narrow baffleless fluewall (430 mm width). The design was adjusted to get the right pressure drop, and, from a process point of view, those fluewalls are running smoothly since then.

The concern observed at Tomago about one year after start-up, was a mechanical weakness at the first peephole, at the upstream gas injector position (cf. Figure 13). Some pinching were noticed. The decision was taken to strengthen this area with additional tie bricks. 8 years after start-up, fluewalls are reaching 95 round of fire lifetime expectation, which is comparable to the performance of the old design. Although this fluewall life is well below the benchmark, one has to take into account the narrowness of fluewalls and brick (90mm), and the very short fire cycle, which negatively affect the fluewall lifetime in this case.

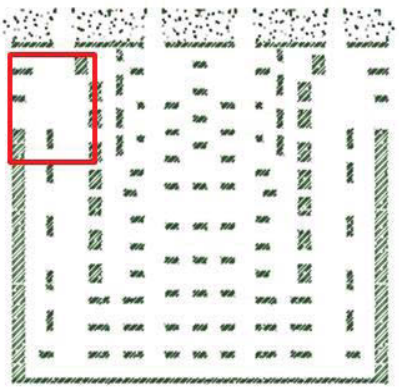

Figure 13 – Weak area is highlighted (Second generation baffleless design)

Then came Qatalum in 2010. The first project with baffleless technology outside of RTA.

Qatalum return of experience

Qatalum has two baking furnaces, of identical section design. One furnace has 66 sections (4 fires), the other has 50 sections (3 fires). The sections comprise 8 pits, baking 144 anodes. Dry-out commenced on 1st January 2010. The furnaces have been operated at 32, 28 and 26 hours FAC. The fire configuration is relatively standard with the exception of having a second cooling ramp. There are three pre-heat sections, three heating ramps, three in heat recovery and three in cooling. On each furnace there is one refractory window comprising 6 sections, the other gaps between the fires being 4 sections. Each furnace has a refractory maintenance platform in the refractory zone. Top-sealing is performed before each fire. Qatalum has prioritized having a high focus on preventative maintenance on refractory – which is paying off in terms of refractory life and performance. As of 1st September 2015, the furnaces are 100 and 101 cycles old. Qatalum has not started a flue wall replacement program as yet, having only replaced a few walls for the purpose of practice and commissioning of tools and equipment. Start of routine fluewall replacement was initially planned in 2016 but, after RTA expert visit and the outstanding shape of the ABF, it was decided to postpone the main replacement program by one year (at least). For 2016, Qatalum has only a contingency for 50 wall replacements if and when it should be necessary. Figure 14 shows a comparison between Qatalum ABF and a similar ABF with standard fluewall design at exactly the same age (90 fire cycles). Qatalum ABF is showing no elevation of fluewalls and headwalls.

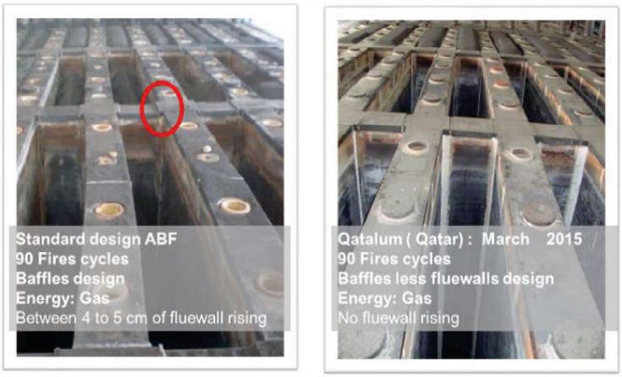

Figure 14 – Anode baking furnace ageing comparison

During the initial operation of the furnaces difficulties were experienced in getting the vertical heat distribution correct in the pre-heat phase. The pitch-front was spread over two sections from where it occurred at the bottom of the flue wall to where it was in the higher parts. The adjustments were made to the gas nozzles - making them of larger diameter, the gas-pulse timings (longer pulses) and the ratio of gas going to the upstream and downstream nozzles. These actions resulted in a much more even pitch front, the entire pitch front being within the space of one section. No refractory modifications were necessary. The result is a benchmark baking level homogeneity shown in figure 15 (Average Lc 35.09 Å, Lc standard deviation 1.29 Å).

The Lc values for all routine core samples over one year are shown in figure 16. Samples are taken in pit 5 top, middle and bottom layers (same anode positions within the layers always: bottom anode #6, middle anode #3 , top anode #2 counting in the direction of the fire). The fact that the bottom anode #6 is slightly lower in Lc is also due to it being the anode next to a headwall. This show very consistent Lc values – each layer having almost the same average Lc, and the standard deviations also as good as identical, despite fire cycle and peak fire changes over the year.

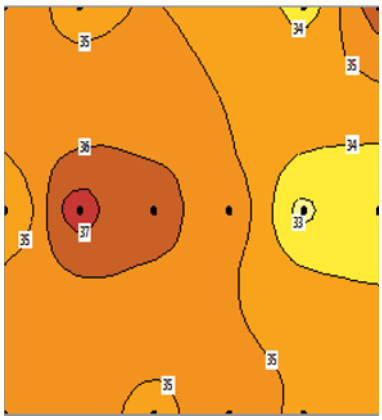

Figure 15 – Pit baking profile at Qatalum ABF

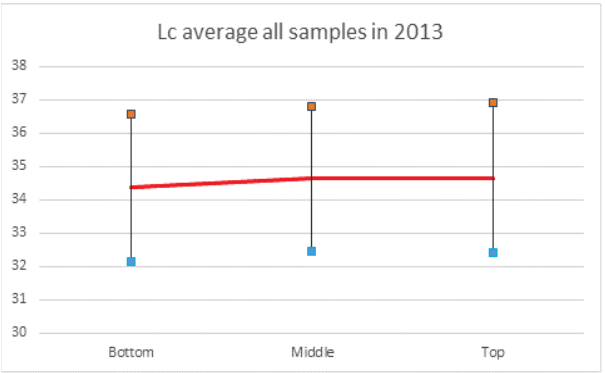

Figure 16 – ABF baking trend at Qatalum

The quality of the anode baking has been excellent with reduction experiencing extremely low levels of spikes and deformations (typically 0.025%). Pot dusting has not been a concern. Specific energy consumptions (gas alone) in 2015 is 2.5 GJ/ton, which is consistent with the high baking level (Lc 35 Å) and the furnace dimensions. Actually, all other things being equal, the baffleless design should lead to reduction of gas energy consumption by at least 0.10 GJ/t compared to a standard design if it is taken advantage of the lower baking level deviation to lower the baking average target.

Conclusion

After a long development, baffleless design is mature. The design advantages have been explored and optimized with the result that both Qatalum ABFs became AP technology benchmarks.

Based on the return of experience, the following improvements are demonstrated with the baffleless design:

- a blockage proof design
- a 5% lighter fluewall with about 25 fire cycles extra-lifetime expectancy
- a better baking homogeneity (pit σLc < 1.5 Å) for better anode robustness on pots
- a low-cost solution to reduce fluewall width in anode height creeping projects
- a reduced capex for a greenfield ABF (through lighter and possibly narrower fluewalls)
- an opportunity to reduce gas consumption

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EFFECT OF HEATING RATE DURING BAKING ON THE PROPERTIES OF CARBON ANODES USED IN ALUMINUM INDUSTRY

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Keywords: Carbon anodes, anode baking, heating rate, soaking time

Abstract

Carbon anodes are called the heart of aluminum electrolysis cells and contribute to about 15% of total production cost of aluminum. In case of poor quality, this contribution can reach 25%. The final quality of the carbon anodes and their behavior in the electrolysis cells depend on the conditions during the baking process. In any case, a low-quality green anode cannot be improved during baking. However, the quality of green anodes can deteriorate if the parameters of the baking process are not properly adjusted.

The objective of this work is to study the effect of introducing a plateau during baking on the properties of carbon anodes. The samples were characterized before and after baking by measuring their density and electrical resistivity which are important in defining the final quality of the anode samples. The experimental data were also analysed using the artificial neural network model (ANN). The article will present the results of this study.

Introduction

Carbon anodes are used in Hall-Heroult cells to produce aluminum. Calcined petroleum coke, butts and rejected anodes are crushed to obtain a desired particle size distribution. These are then mixed with coal tar pitch to obtain a carbonaceous paste. The paste is then compacted and baked to produce baked anodes with appropriate properties such as high density, low electrical resistivity, and low air and CO_2 reactivities. Increasing anode quality helps decrease energy consumption, production cost as well as greenhouse emissions.

Baking is by far the costliest stage in the manufacturing of anodes [1]. The final quality of the carbon anodes and their behavior in the electrolysis cells depend on the conditions of the baking process.

A few researchers investigated the effect of baking parameters on the anode quality. Belitskus [2] studied the effect of the final anode baking temperature on the quality of anodes and found that it has a major effect on anode electrical resistivity and excess anode consumption during electrolysis. The same parameter was also studied by Fischer et al. [3]. They found that low final baking temperatures or short soaking times lead to selective air and CO_2 burning of the anodes in the electrolytic cells. This problem can be detected by an increase in the standard deviations of the reactivities. They recommended the utilization of baking temperatures in the range of 1050°C to1200°C to optimize the CO_2 and air reactivities and the thermal conductivity of anodes. They also pointed out the importance of heating rate on anodes. Their results showed that the presence of high anode temperature gradients significantly increases the standard deviations of electrical resistivity and flexural strength; hence, this results in nonhomogeneous anode properties. They recommended that the maximum heating rate should not exceed 10 to 14°C/h between 200-600°C to control the devolatilization of tar components from pitch.

Severo et al. [4] studied the influence of the final baking temperature on the energy consumption in the baking furnace. They found that increasing the baking temperature by 50˚C increases the energy consumption by about 0.2 GJ/t of anodes.

Amrani et al. [5] showed that crack formation increases at high heating rates. Kocaefe et al. [6] reported that the baked anode properties such as air permeability as well as air and CO_2 reactivities are improved if the anodes are baked at low heating rates.

Although a low heating rate leads to an increase in baking time, several researchers have strongly recommended the utilization of such a rate to obtain high quality anodes. Keller et al. [7] mentioned that minimizing energy consumption should not be the goal of the baking process. It should be the improvement of the anode quality. They also suggested that a good control of the burning of pitch volatiles can minimize the energy consumption and the emission of greenhouse gases during the baking process.

In this study, the effect of the presence of a plateau at different temperatures during baking on the quality of anodes is investigated. Figure 1 shows qualitatively an example of a plateau during anode baking.

Due to large variations in the standard variations in the data analysed, the experimental data were also analysed using an artificial neural network (ANN) model.

ANN is a tool, often used in the field of artificial intelligence, to find the relation between a set of independent parameters and an output parameter where no straight forward mathematical relationship exists. Researchers have described feedforward artificial neural

network with sigmoidal transfer function and backpropagation training as a universal function approximator [8]. A multi-layered feed forward ANN model was developed.

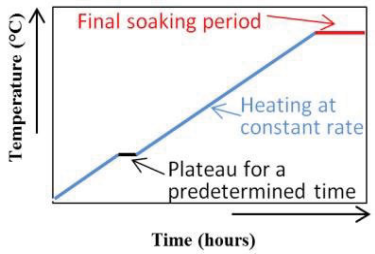

Figure 1 : Figure showing the plateau and the soaking time

Methodology

Four sets of cylindrical samples (130 mm in length and 50 mm in diameter) were taken from the top and the bottom parts of two different industrial anodes (Anode 1 and Anode 2). They were baked at a heating rate similar to those typically used in industry in a laboratory baking furnace at the University of Quebec at Chicoutimi (UQAC).

A plateau for a predetermined time period was tested at different temperatures (200ºC<T1<T3<T5<T7<600ºC) using different samples. This temperature range coincides with the devolatilization range of nearly all pitch volatiles [3]. Also, all anode samples were maintained at the maximum baking temperature for eight hours (final soaking time). Four samples from each anode were baked for each condition studied. The densities and electrical resistivities of the samples were measured before and after baking. Table 1 presents the specifications of the two industrial anodes used. The percentages of pitch in both anodes were nearly the same.

Table 1 : Specification of the industrial anodes

	Anode 1	Anode 2
Type of coke	Coke 1 55% Coke 2 45%	Coke 3 low S 30% Coke 4 high S 70%
Type of pitch	Pitch 1	Pitch 1 90% Pitch 2 10%

Figure 2 shows the experimental system. The anode samples were placed in packing coke, and nitrogen was passed through the system to prevent their oxidation. The furnace was programmed to have the desired heating rate. A thermocouple was placed in the packing coke, and the temperature was monitored during the experiment using a data acquisition system. After baking, the anode samples were cooled down to the room temperature under nitrogen atmosphere.

Measurement and representation of anode properties

The bulk density and the electric resistivity of the anodes were measured before and after baking using the ASTM standards [9, 10]. The bulk density was measured according to ASTM D5502-00 (2005). The apparent density of the cylindrical core samples of carbon anodes can be determined with this standard. The electrical resistivity was measured according the ASTM D6120-97 (2007). This test method is used for the determination of the electrical resistivity of cylindrical anode samples at room temperature.

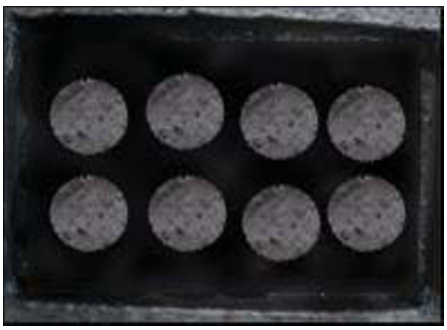

Figure 2: Placement of samples in the laboratory baking furnace

In this work, the densities of green and baked anode samples are expressed in dimensionless form using equation 1:

$$DD = (GAD \text{ or } BAD)/AVD \quad (1)$$

where DD is the dimensionless density, GAD is the green anode density, BAD is the baked anode density, and AVD is the the average of the experimental values of GAD and BAD of all samples.

The green anode resistivities are expressed in dimensionless form using equation 2:

$$DGR = GAR/AVGR \quad (2)$$

where DGR is the dimensionless green anode resistivity, GAR is the green anode resistivity, and AVGR is the average of the experimental values of GAR of all samples.

Similarly, the baked anode resistivities are expressed in dimensionless form using equation 3:

$$DBR = BAR/AVBR \quad (3)$$

where DBR is the dimensionless baked anode resistivity, BAR is the baked anode resistivity, and AVBR is the average of the experimental values of BAR of all samples.

Results and Discussion

Eight samples (four from each anode) were baked under the same baking condition to account for the non-homogeneity of the industrial anodes in each case.

Figures 3 and 4 show the average dimensionless densities and resistivities of the green anode samples, respectively. These samples were later baked to study the impact of

adding a plateau during the devolatilization period while the anodes are baked. The dimensionless values were calculated as explained above (Equations (1) and (2)). "No plateau" implies that no plateau was present when these samples were later baked. "Plateau Ti" implies that the baking temperature was maintained constant at this temperature for a predetermined time during the baking of these samples.

The standard deviations of density and resistivity of the green anode samples show that there was a significant variation in the properties of the green anodes used for the analysis. The variations in the density can be explained in terms of the non-homogeneity of the anodes with respect to the nature and distribution of particles, binder pitch and pores/cracks. To compare the effect of having a plateau at different temperatures, it is ideally necessary to use identical green anode samples with the same density and electrical resistivity. However, this is not possible when industrial anodes are used. Thus, as explained previously, a number of samples were used to test each condition.

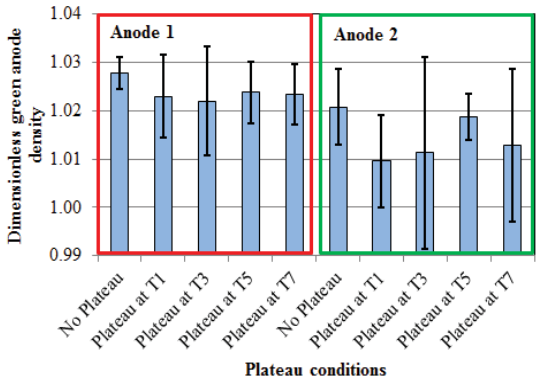

Figure 3: The average dimensionless densities and the standard deviations of the green anode samples

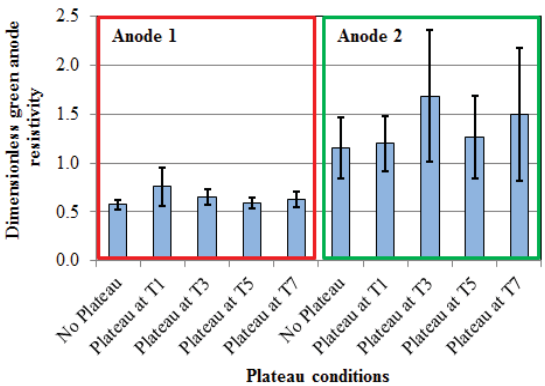

Figure 4: The average dimensionless electrical resistivities and the standard deviations of the green anode samples

As the initial values of the densities and resistivities of the green anode samples were different, the ratio of BAD to GAD for the density and the ratio of BAR to GAR were used to understand the effect of plateau temperature on the anode properties. The ratio shows how much a property is improved. Since a high BAD is required, a high BAD/GAD is desirable. On the other hand, a low BAR is desirable. Thus, a low BAR/GAR value indicates good quality anode.

Figures 5 and 6 present the ratios of BAD/GAD and BAR/GAR for samples from anodes 1 and 2. The objective is to identify the plateau which will give high BAD/GAD and low BAR/GAR ratios. It can be seen from these figures that this objective is reached for both anodes, around T7. This suggests that, by introducing a plateau during the devolatilization period, it might be possible to produce baked anodes with higher density and lower electrical resistivity than those baked without a plateau.

The standard deviations in Figures 5 and 6 show that the variations in BAD/GAD and BAR/GAR depend on the variations in green anode properties. As described earlier, it is difficult to find identical green anode samples. Thus, the artificial neural network (ANN) [11] was used to predict the density and resistivity of baked anode samples with identical GAD and GAR.

In this work, the experimental data for both anodes were used to train the ANN model. The model was validated with a set of experimental data which was not used for training. The output of the model was in good agreement with the experimental results.

This model was used to predict the BAD and BAR corresponding to the top and bottom parts of Anode 1 and Anode 2. ANN is based on pattern identification and recognition. The predicted properties should be in the range for which experimental data are available (T1 to T7). Thus, the BAD and BAR were predicted for a particular value of GAD and GAR corresponding to plateau temperatures of T1, T2, T3, T4, T5, T6, and T7. The chosen plateau temperatures are the averages of the experimental ones on either side of the temperature used for prediction in ANN as given below:

$$T2 = (T1+T3)/2,\ T4 = (T3+T5)/2,\ T6 = (T5+T7)/2 \qquad (4)$$

The output parameters BAD and BAR for different anode positions (top and bottom) are predicted based on the pattern identified from the experimental data set during the training phase of the ANN model.

Figures 7 and 8 show the predicted BAD and BAR corresponding to bottom and top parts of Anode 1, respectively, at different plateau temperatures. The predicted BAD and BAR at different plateau temperatures at the bottom and top of Anode 2, respectively, are presented in Figures 9 and 10.

It can be observed from Figures 7-10 that the predicted BAD and BAR varied at the top and bottom positions of the two anodes for the same GAD and GAR. The predictions of the model varied based on the pattern identified in the training data set. The figures also showed that at around temperature T6, the predicted BAD is the maximum and BAR is the minimum. The predictions of ANN can be improved if more experimental data is available.

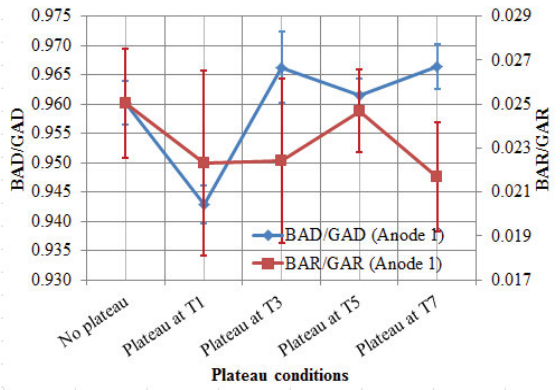

Figure 5: Ratios of BAD/GAD and BAR/GAR for Anode 1 for different baking conditions (without plateau and with plateau at different temperatures)

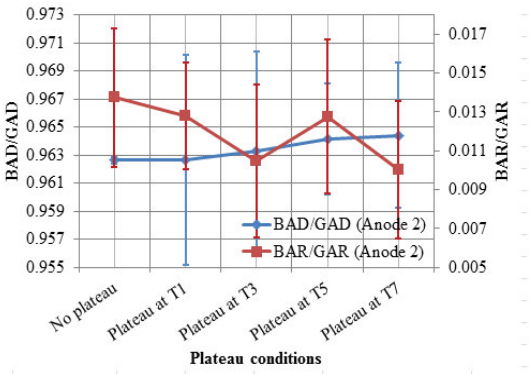

Figure 6: Ratios of BAD/GAD and BAR/GAR for Anode 2 for different baking conditions (without plateau and with plateau at different temperatures)

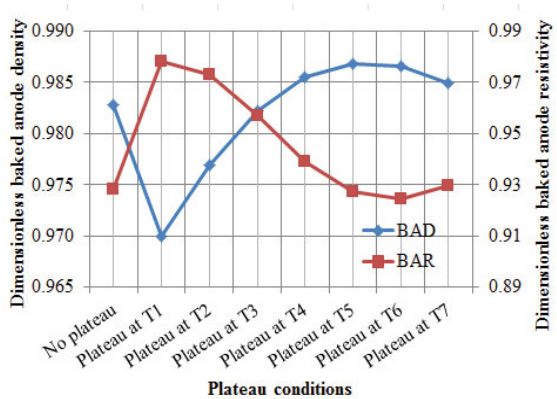

Figure 7: Predicted BAD and BAR for samples from the bottom part of Anode 1 for constant GAD and GAR

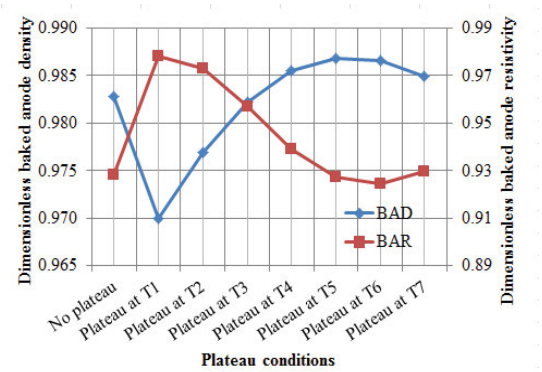

Figure 8: Predicted BAD and BAR for samples from the top part of Anode 1 for constant GAD and GAR

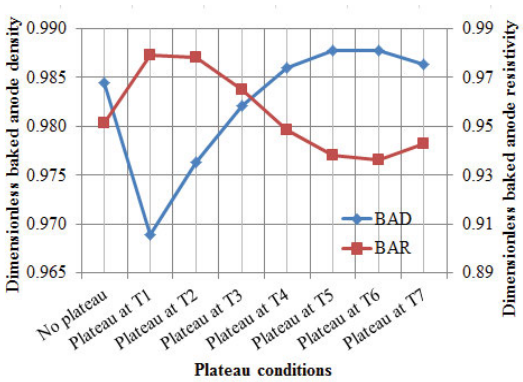

Figure 9: Predicted BAD and BAR for samples from the bottom part of Anode 2 for constant GAD and GAR

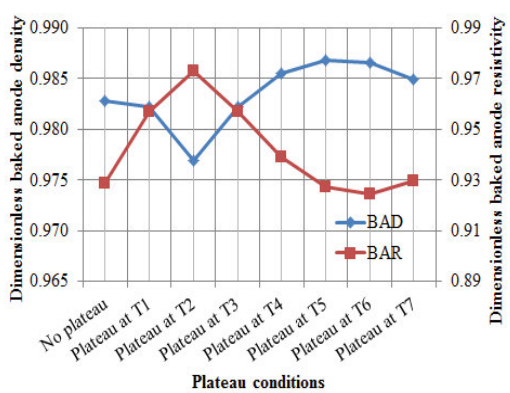

Figure 10: Predicted BAD and BAR for samples from the top part of Anode 2 for constant GAD and GAR

It may be noted that no experiment was performed at the plateau temperature of T6 exactly. The ANN model and the experimental data analyses showed that plateau temperatures of T6 and T7, respectively, might improve the

baked anode density and resistivity. These two temperatures are very close. The ANN and the experimental data analyses identified similar plateau temperatures. It is possible that adding a plateau in this temperature range would seem to help control the rate of volatiles release from pitch during baking consequently improve the baked anode quality.

Conclusions

Anode quality is crucial for aluminum smelters and anode baking is an important step in producing good quality anodes. The baking process influences the anode quality as well as the production and operation costs. In this work, the effect of adding a plateau at a certain temperature during the devolatilization period was studied. The ratios obtained from the experimental data for BAD/GAD and BAR/GAR as well as the results from the artificial neural network analysis showed that introducing such a plateau could improve the baked anode quality.

Similar plateau temperatures were obtained from both analyses. The study indicated that a good quality baked anode with high density and low electrical resistivity can be produced if the plateau is set in the range of T6 and T7 for a predetermined time.

Acknowledgements

The technical and financial contributions of Aluminerie Alouette Inc. and the financial support from the Natural Sciences and Engineering Research Council of Canada (NSERC), the Economic Development Sept-îles, the University of Quebec at Chicoutimi (UQAC), and the Foundation of the University of Quebec at Chicoutimi (FUQAC) are greatly appreciated. The authors would like to thank the Fonds de recherche du Québec nature et technologies by the intermediary of the Aluminum research centre – REGAL for their contribution for student participation in this conference.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

EMPIRICAL MODELING OF THE BAKING FURNACE TO PREDICT BAKED ANODE PROPERTIES

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Keywords: Anode baking, Anode properties, Principal Component Analysis (PCA), Partial Least Squares (PLS)

Abstract

A soft-sensor model developed from historical carbon plant data and multivariate statistical methods was proposed in past work to obtain quick predictions of individual anode properties right after baking for quality control purposes. It could only be used for anodes baked at the coldest and hottest positions within the furnace due to the core sampling method used at the partner plant and the way the data is collected. To complement the soft-sensor, this work proposes a strategy to account for the thermal history of anodes baked at any position. It is shown that combining categorical variables for pit and baking positions and routinely available firing equipment data is sufficient to predict the temperature history of anodes baked in different positions, which is typically not measured during normal operation. The model will be integrated to the soft-sensor in the future to account for the impact of baking position on anode properties.

Introduction

Anodes are made from three types of raw materials: petroleum coke, coal tar pitch and anode butts. These are two by-products of the petroleum refineries and metallurgical coke production, respectively. In spite of the rising prices for these two materials, their quality is not following the same trend [1]. Therefore, frequent supplier changes are made to reduce purchase costs and blending of materials is performed to achieve desired targets on some material properties. This situation introduces variability in baked anode properties.

The anode quality control strategy currently used in the industry based on core sampling and lab testing is insufficient to prevent setting faulty anodes in reduction cells. A very small proportion of the anodes manufactured are typically cored and analyzed in the laboratory. In addition, the delay to receive core characterization results is too long to take timely corrective actions when necessary. Hence, alternative and more rapid methods are needed for anode quality control. Some papers have reported the use of a phenomenological model to predict anode properties often focusing on only one or few properties [2]. Empirical models have also been developed to predict net carbon consumption in the cells using some raw materials or anode properties [1,3].

An empirical soft-sensor model was recently proposed by Lauzon-Gauthier et al. [4] to predict anode core properties right after the baking cycle. It uses all the available plant data (raw materials properties, paste plant, baking furnace operation and anode core properties). Good predictions were obtained for most of the measured properties, but only for anodes baked at two specific positions within the furnace: the hottest and the coldest. This is because core data was available only at those two baking positions (core sampling strategy implemented at the specific plant).

The objective of this work is to determine how to incorporate the impact of baking position in the soft-sensor model such that predictions can be made for anodes baked at any position. The impact of baking on anode quality is discussed in [5,6]. Anodes are stacked vertically in a pit and it is well known they may be submitted to a different thermal history depending on their location in that pit. Anode temperature is not routinely measured during baking due to operational issues and high costs. Several phenomenological models of the baking furnace have been proposed to predict anode temperature [7-9], however these require long computing time. This work proposes to use routinely available baking furnace operating data and binary variables to represent anode baking position in order to predict anode temperature during the baking cycle. Anode temperatures measured during special pit surveys are used to calibrate and validate the proposed empirical modelling approach.

This paper first describes the sources of the data used to build the models. The baking process is briefly explained and more details about anode position within the furnace are provided. The latent variables methods used to build the empirical models, namely Principal Component Analysis (PCA) and Partial Least Squares (PLS), are overviewed. Finally, the modelling results are presented and discussed, and some conclusions are drawn.

Methodology

Baking process and available data

Anodes are baked in one of the 2 open-top ring furnaces each containing 34 sections of 6 pits. The anodes are stacked vertically in the pits. A series of equipment bridges are moved from one section to the next at each fire cycle time (~22-24 hours). The hot gases flow through the flues to bake the anodes. Figure 1 illustrates the configuration of the furnace.

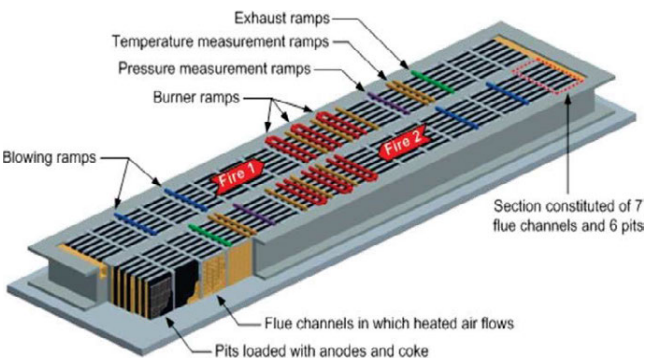

Figure 1. Baking furnace configuration [7].

The fire is controlled to meet a specific heat-up rate (maximum ~15°C/h depending on anode formulation and raw material

properties) and to reach a desired soaking temperature (above ~1100°C), which is maintained during a minimum period of time. These parameters are very critical for some anode properties especially electrical resistivity, apparent and real density, Lc and mechanical properties. Flue depression and burners opening are monitored and controlled to meet a flue gas temperature target and achieve the above mentioned specifications [10].

Part of the data used to build the models were collected during pit surveys conducted between 2011 and 2014 at the Alcoa Deschambault smelter in Quebec, Canada (ADQ). During the surveys thermocouples were installed in 9 anodes located within 1 or 2 pits in the same section of the furnace. Their position is identified by a red dot in Figure 2. At the end of the baking cycle, the temperature profile of each anode is available for analysis. These 9 anodes are also cored and characterized in the laboratory.

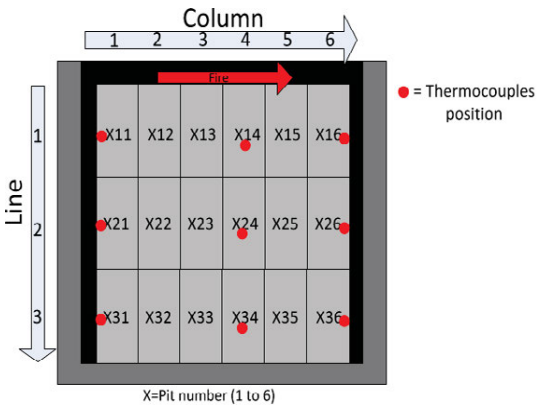

Figure 2. Illustration of a baking furnace pit with thermocouples locations during a survey. Anodes are identified by the pit number (X in the figure), followed by their line and column numbers.

In addition to pit survey data (available occasionally), the baking process data routinely collected at the plant during every baking cycle were also retrieved from a data historian. The data available in real-time include flue gas temperatures on both sides of a pit, flue pressure at different locations, packing coke temperature measured in pits 1 and 3, and natural gas flow rates.

Data organization

Binary variables were used to identify the anode position (line, column) in one of the two furnaces and the 6 pits selected for the pit surveys. Table 1 shows how the binary variables were assigned for a few combinations of furnace, pit and anode positions. The anodes are uniquely identified using four numbers, for example 1-135 for furnace #1 - pit #1, line #3 and column #5.

Table 1: Example of how position is defined within the model

Anode	**Furnace**	**Pit**				**Line**			**Column**			
Furnace-position	1=1 or 2=0	1	2	…	6	1	2	3	1	2	…	6
1-135	1	1	0	0	0	0	0	1	0	0	1	0
1-223	1	0	1	0	0	0	1	0	0	0	0	0
2-331	0	0	0	0	0	0	0	1	1	0	0	0
1-514	1	0	0	1	0	1	0	0	0	0	0	0

The overall data structure is shown in Figure 3. It is divided in two data matrices **X** and **Y**. Each row in these matrices correspond to the conditions under which the 79 anodes tested in the available pit surveys have been baked. The anode position (binary variables) and the average of the flue gas temperature trajectories measured in both flue channels around the pit where the anode was baked are stored as a long row vector in **X**. The corresponding anode temperature trajectory measured during a pit survey is stored as a row vector in **Y**. A total of 9 binary variables were used for anode position, and the flue gas and anode temperature trajectories were measured at 200 and 300 sampling points, respectively. Hence, the dimensions of both data matrices are **X** (79×209) and **Y** (79×300).

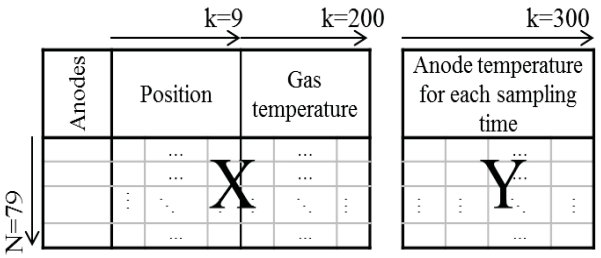

Figure 1. Data structure available empirical modelling.

Two models were built from this data to achieve to different objectives. Principal Component Analysis (PCA) applied first to the **Y** data to better understand the sources of variations in the anode temperature trajectories. Then, Partial Least Squares (PLS) regression was applied to **X** and **Y** to show that anode temperatures can be predicted routinely available data (anode positions and flue gas temperatures). Both PCA and PLS are briefly described next. These techniques were selected because they efficiently handle large and highly collinear sets of data as is the case in this study. Note that prior to applying these methods, the data was mean-centered and scaled to unit variance.

Principal Component Analysis (PCA)

PCA is a basic latent variable method which approximates a high dimensional dataset **X** (N×K) by a much lower dimensional subspace **T** (N×A), A<<<K (i.e. a plane or hyperplane) capturing most of the variance in the data. It performs the following bilinear decomposition of the variance-covariance matrix of **X**:

$$\mathbf{X} = \mathbf{T}\mathbf{P}^{T} + \mathbf{E} \quad (1)$$

Where $\mathbf{T} = [\mathbf{t}_1\ \mathbf{t}_2 \dots \mathbf{t}_A]$ (N×A) is the lower dimensional subspace formed by the A orthogonal score vectors (**t**'s) and $\mathbf{P} = [\mathbf{p}_1\ \mathbf{p}_2 \dots \mathbf{p}_A]$ (K×A) contains the corresponding orthonormal loading vectors (**p**'s) defining the orientation of the subspace. In fact, each score vector $\mathbf{t}_i$ is defined as a linear combination of the columns of **X** (i.e. $\mathbf{t}_i = \mathbf{X}\,\mathbf{p}_i$) and the weights of those linear combinations are contained in the corresponding loading vector $\mathbf{p}_i$. **E** contains the model residuals. The components (pairs of **t**'s and **p**'s) are ordered in such a way that the first (t_1) explains the greatest amount of variations in the data, whereas the second explains the greatest amount of variance orthogonal to the first, and so on. The number of principal components is typically selected by a cross-validation procedure. Projection of the data into the lower dimensional score space **T** allow to visualize the clustering pattern of the data. The loading vectors **P** are used to interpret what

combinations variables explain such a clustering pattern. For more details on PCA the reader is referred to [11].

Partial Least Squares (PLS)

PLS is a regression method based on the same latent variable concept as PCA. It finds a small number of orthogonal latent variables **T** that simultaneously explain the variations in **X** and **Y** while building a relationship between them. The PLS model is expressed as follows:

$$\mathbf{X} = \mathbf{T}\mathbf{P}^{T} + \mathbf{E} \quad (2)$$

$$\mathbf{Y} = \mathbf{T}\mathbf{Q}^{T} + \mathbf{F} \quad (3)$$

$$\mathbf{T} = \mathbf{X}\mathbf{W}^{*} \quad (4)$$

Where the score vectors **T** are those linear combinations of **X** that are the most predictive of **Y**. The weight vector **W*** contains the weights of each X-variable in those linear combinations. **P** and **Q** are the loading vectors of the **X** and **Y** spaces, respectively, and the model residuals are stored in **E** and **F**. The number of orthogonal components are selected to maximize predictive ability of **Y**. Again, the scores T are used to visualize the data and W* and Q allow interpretation of the relationships between the variables in **X** and **Y**. Further details on PLS can be found in [12].

Results and discussion

In this section, the results of applying PCA on the anode temperature trajectories (**Y**) are presented first, followed by the analysis of the PLS regression model built for predicting the anode temperature trajectories (**Y**) using the variables in **X** as predictors. Finally, the prediction performance of the PLS models will be assessed against temperature measured during pit surveys.

Application of PCA to anode temperature trajectories

Three PCA components were found statistically significant by a cross-validation procedure. The percent variance of **Y** explained (R^2) and predicted by the model in cross-validation (Q^2) are provided in Table 2 for each component. A total of 97.8% of the variance in the anode temperature trajectories are captured by the PCA model using only 3 components. In the very similar value of the cumulative Q^2 suggests no sign of overfitting the data.

Table 2. Explained and predicted variance of Y by the PCA model

Components	R^2 (%)	R^2_{cum} (%)	Q^2 (%)	Q^2_{cum} (%)
[1]	84.5	84.5	84.1	84.1
[2]	9.5	94.0	9.6	93.7
[3]	3.8	97.8	3.8	97.5

The score plot for the first two components is shown in Figure 4. Note that t_Y is used for the PCA scores to distinguish the PLS scores discussed in the next section. The observations are colored according to the month and year of the pit survey. Each dot corresponds to the thermal history of one anode (one row in **Y**). Anodes (dots) falling near one another in the score plot are have a similar thermal history. The anodes baked in the different pit surveys show a similar spread along the $t_{Y,1}$ and $t_{Y,2}$ axes, except for those baked in the pit survey of November 2012 (blue dots), which cluster separately in the right superior quadrant. The loading values are now used to interpret the variations captured by the first two (most dominant) components.

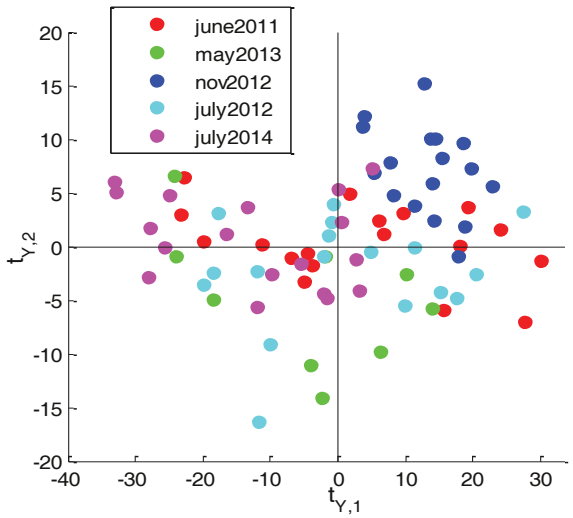

Figure 2. Score plot $t_{Y,1}$-$t_{Y,2}$ of the PCA model applied on anode temperatures (**Y**). Colors according to date (pit survey).

The loadings values of the first two PCA model components ($p_{Y,1}$ and $p_{Y,2}$) are presented in Figure 5. A red vertical line was added to separate the pre-heating and the burning phases in the baking cycle, when temperature measurements were available.

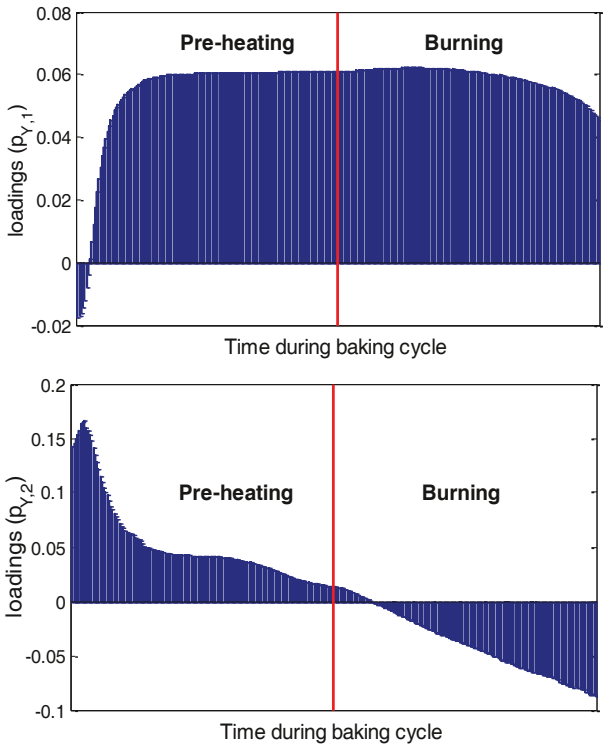

Figure 3. Loading plot for the first two PCA model components.

The absolute value of a loading is proportional to the importance of the corresponding variable (anode temperature at a certain time point) in a given component. Its sign indicates whether the correlation between the variables is positive or negative. For example, the loading values in the first component ($p_{Y,1}$) are positive for most the of cycle except early in the pre-heating phase. This suggests that the first component captures systematic

variations in the anode temperature trajectories occurring over most of the baking cycle. Hence, the anodes projecting in the positive $t_{Y,1}$ region (Figure 4) have been warmer for most of the cycle compared to those falling in the negative $t_{Y,1}$ region. Therefore, the first component is likely related with variations in the total energy input to the anode either due to higher temperature set-points or to the position where the anode was baked. Indeed, the score values have been recolored according to the position (column and line) in the pit where the anodes were baked (Figure 6). It seems that anodes baked in column 1 were warmer than those baked in column 6. This may be explained by the way the burner bridges are moved on top of the section (see figure 2). The anodes in column 1 are heated first but the fire reaches those in column 6 at a later stage. Considering that cycle time is the same for all anodes, those baked in column 6 may have been submitted to a different thermal history. The impact of the line is not as clear as for columns.

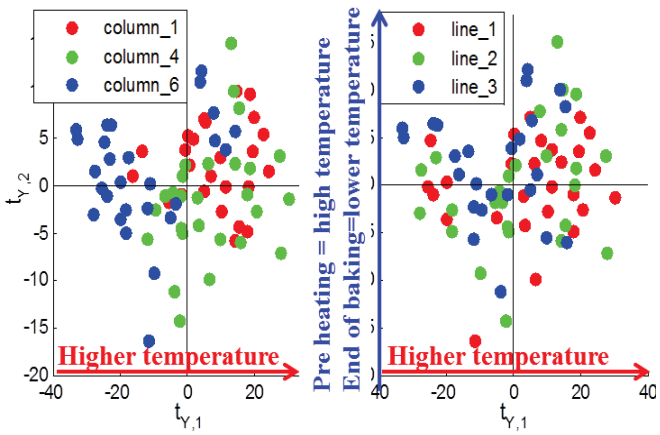

Figure 6. Score plot $t_{Y,1}$-$t_{Y,2}$ of the PCA model applied on anode temperatures (**Y**). Colors according to column (left) and line (right) within a pit.

The loadings of the second component ($p_{Y,2}$) in Figure 5 rather shows a contrast between the pre-heating and burning phases because the loadings values in the two periods have a different sign. Anodes having positive $t_{Y,2}$ values were warmer earlier during the pre-heating and colder towards the end of the burning phase compared with those falling in the negative $t_{Y,2}$ region which are characterized by the opposite behavior.

The overlap observed in Figure 6 between groups of anodes not baked in the same position is probably explained by the use of different operating conditions of the baking furnace from one pit survey to another. This will be investigated using the PLS model.

PLS model for predicting anode temperature trajectories

Although 8 PLS components were found significant by cross-validation (cumulative R^2 and Q^2 of 83.3% and 76.3%, respectively), only the statistics for the first three components are presented in Table 3. Only those will be interpreted in this section.

Table 3. Explained and predicted variance of **Y** by the PLS model

Components	R^2 (%)	R^2_{cum} (%)	Q^2 (%)	Q^2_{cum} (%)
[1]	29.7	29.7	27.1	27.1
[2]	7.0	36.7	4.7	31.8
[3]	32.4	69.1	18.8	50.6

Components 1 and 3 clearly explained most the variance in the anode temperatures (**Y**) and so, the analysis of this model will focus on these 2 components. The PLS model $t_{X,1}$-$t_{X,3}$ score plot and the corresponding loading weights ($w^*_{X,1}$ and $w^*_{X,3}$) are presented in Figure 7 and Figure 8, respectively.

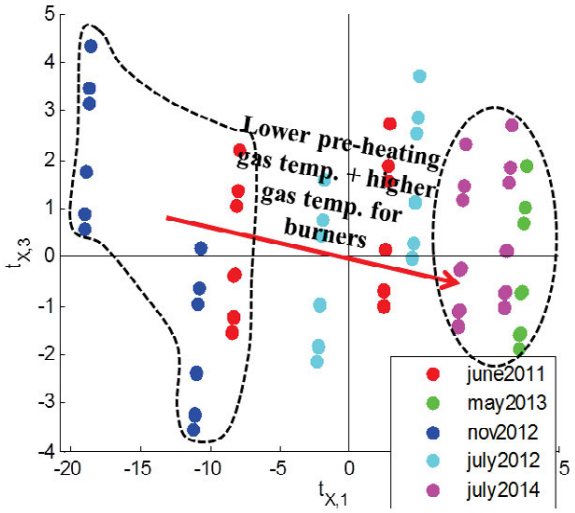

Figure 7. PLS model scores $t_{X,1}$-$t_{X,3}$. Colors according to date (pit survey)

The first component mainly explains the impact of changes in the baking furnace operation because the anodes cluster according to pit survey along the first component $t_{X,1}$. The third component shows intra-pit variability (e.g. position within the pit).

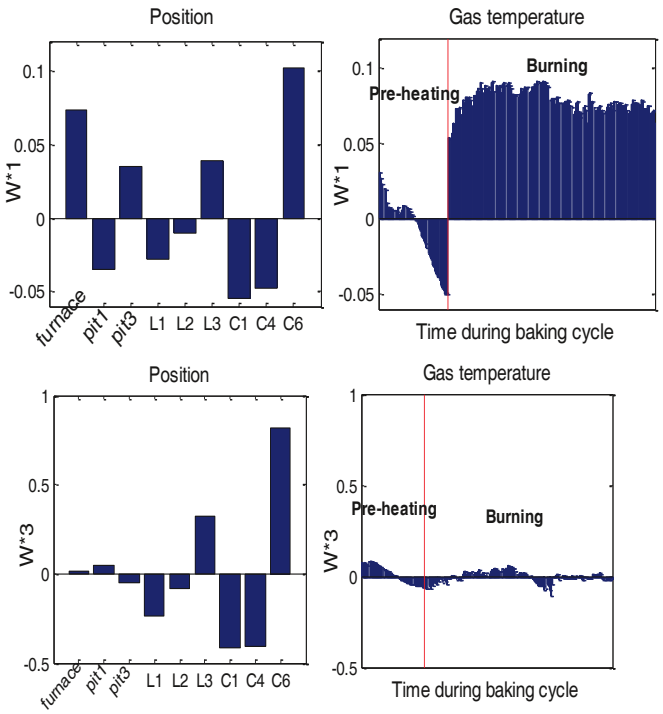

Figure 8. PLS model weights. Top row: first component ($w^*_{X,1}$). Bottom row: 3rd component ($w^*_{X,3}$). The weights for the position and flue gas temperature blocks appear in separate plots.

The PLS weights of the flue gas temperatures in the first component ($w^*_{X,1}$) reveal that the X-space weights are negative at

the end of the pre-heating, but all high and positive when the burners are used. Thus, anodes falling in the high positive $t_{X,1}$ regions (e.g. pit surveys of May 2013 and July 2014) have been baked under lower gas temperatures during preheating, but higher temperatures when the burners were set on the section. However, the temperature of theses anodes were found to be lower in this score region (by looking at the data itself). Hence anode temperature show the opposite trend as the flue gas temperatures. This may seem contradictory, but when gases are cold during pre-heating the anodes are baked to a lesser extent and when burners are set on the section, more heat is necessary for reaching the desired soaking temperature. These differences in anode temperature can also be due to different furnace configuration (pit and flues width), parameters not taken in account in this model.

The third component mainly explains the effects of baking position as shown by the $w^*_{X,3}$ weights (Figure 8), which are much more important for the position block compared to the gas temperature block. The PLS score plot was colored according to positions and shown in Figure 9. This clearly confirms the difference of the anodes baked in column 6, which were colder during the cycle (high $t_{X,3}$ associated with colder anodes, confirmed by looking at the data). Some subtle differences can also be observed among the baking lines. Those anodes baked in line 3 (high $t_{X,3}$ values) were also cooler.

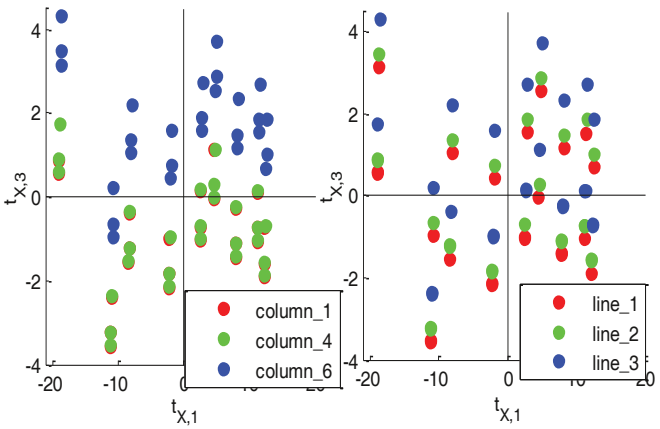

Figure 9. PLS model scores $t_{X,1}$-$t_{X,3}$. Colors according to column (left) and line (right) within a pit.

Predictive performance of the PLS model

Prediction results of the anode temperature trajectories are presented in Figure 10 for three different pit surveys. The measured temperatures are also presented in the figure. Anodes baked in position 11, 24 and 36 (line and column number, refer to Figure 2) within pit 1 and pit 3 were selected in order to span a wide range of anode thermal history. The measure anode temperature is shown by a solid line and a dashed line is used for predictions.

Very good prediction results were obtained. The distinction between hotter and cooler anodes is clear. The predictions follow the same trends as those measured even if a gap is present with the measured profiles for some anodes.

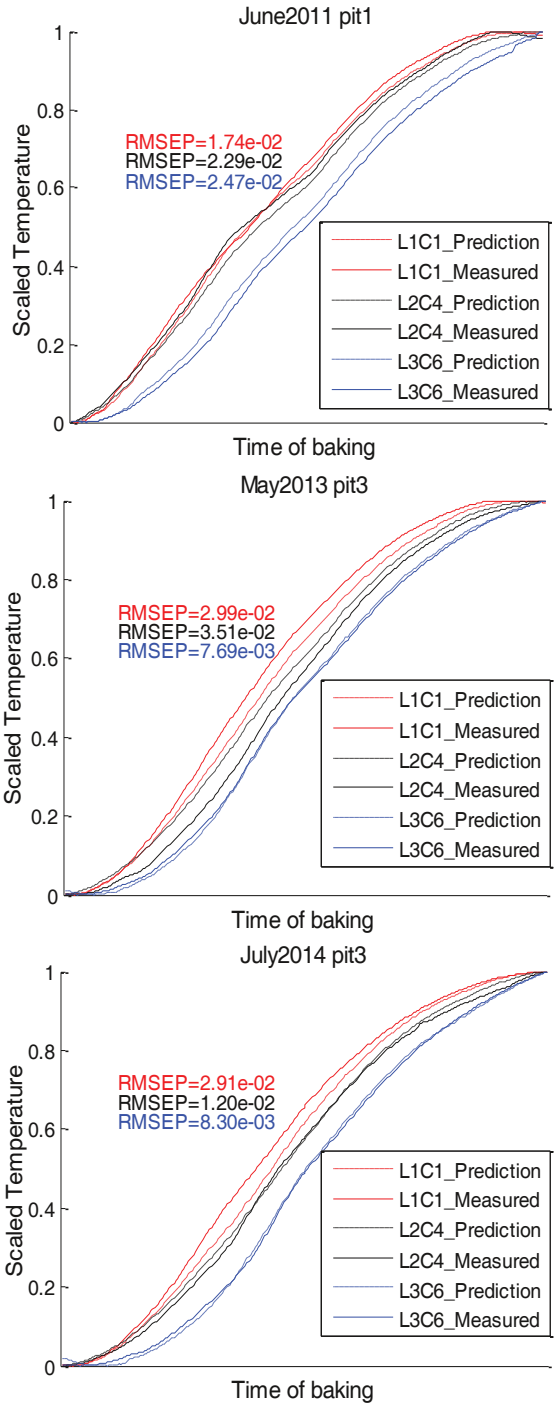

Figure 10. Measured anode temperature trajectories in three different pit surveys and their prediction using the PLS model. In the legends, L and C refer to baking line and column, and RMSEP is the root mean square error in prediction.

Conclusions

This work has shown that anode temperature trajectories during the baking cycle in open-top ring furnaces can be predicted with good accuracy by using flue gas temperature trajectories and binary variables for describing the anode position within a pit. This information will be incorporated into a previously published

soft-sensor model [4] in order to predict the properties of anodes baked in different position within the furnace.

The proposed methodology involved data collected during 5 pit surveys conducted at the ADQ smelter between 2011 and 2014 and the routinely measured baking furnace data. Latent variable methods (i.e. PCA and PLS regression) were used for data analysis and model building.

Validation of the modeling results is planned in the future. This will involve collecting core samples from anodes baked in a variety of positions and characterizing them. The predictions of the new soft-sensor model (incorporating the impact of baking position) will be compared against those.

Acknowledgement

The authors would like to acknowledge the financial support and collaboration of Alcoa Inc. A part of the research presented in this work is financed by the National Science and Engineering Research Council of Canada (NSERC) by the intermediary of the aluminum Research Centre – REGAL.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

IN SITU INVESTIGATION OF THE BEHAVIOR OF ANODE ASSEMBLIES

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Keywords: Anode Assembly, Anode Voltage Drop, Current Distribution

Abstract

In aluminium production, the performance of the anode assembly is a great concern which is mainly associated with its energy consumption, aging, fabrication and cleaning process. Therefore, a better understanding of its *in situ* behavior is required to efficiently optimize the process using an improved design. In this work, electrical and thermal data have been measured on anode assemblies using a specific *in situ* instrumentation during a complete anode cycle at Aluminerie Alouette Inc. With the help of a numerical model developed on the ANSYS Workbench platform, the evolution of the electrical resistance and current distribution along the anode assembly components has been determined over a complete anode cycle. Considering these information, an innovative prototype has been designed and gave promising results at the laboratory scale. The next step is to confirm the performance of this new design in a real cell.

Introduction

The aluminum production requires a considerable amount of electrical energy to supply the necessary electrons to the process. Fig. 1 shows the electrical resistance decomposition between the stem and the cathode of an aluminium reduction cell. From these sections, the anode assembly itself represents an electrical loss of about 300 mV which accumulate over the potline leading to important power consumption.

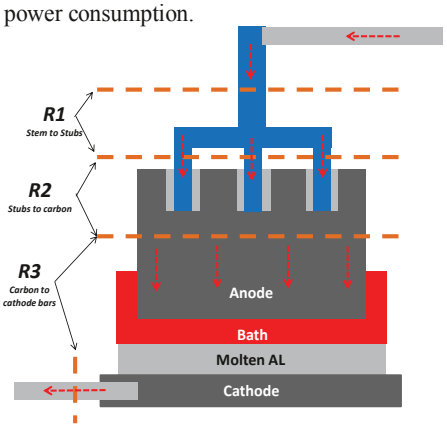

Fig. 1 – Schematic representation of the electrical resistance distribution in the anode assembly

Another important aspect is to minimise the metal pad instability which is partially obtain by a uniform current distribution among the anode assemblies. This means that the electrical resistance difference between each anode position in a cell has to be minimal.

So, the purpose of this study is to provide a better understanding of the evolution of the electrical resistance distribution on each section of the anode assembly (Fig.1) over a complete anode cycle. More specifically, these results would lead to a better understanding about the impact of the electrical resistance distribution on the current variation circulating in the anode. Finally, those relations could lead to the determination of key elements to reduce the electrical resistance and improve the MHD stability via the optimisation of the design of the anode assembly.

Previous work

In 1976, using *in situ* measurements, Peterson [1] has estimated that 7% of the total voltage drop is due to the anode assembly, out of which 25% attributed to the stub to carbon (STC) connections.

In 1984, Brooks and Bullough [2] demonstrated that the electrical contact resistance (ECR) is a function of temperature and contact pressure. This dependence is then quantified in 1985 by Rhedey and Castonguay [3] for a carbon/steel interface showing low ECR above 2.5 MPa.

In 2007, Wilkening and Côté [4] used a small scale set-up to measure the STC resistance at isothermal condition to conclude that in the lower temperature range up to about 600°C, the ECR increase with thickness of the cast iron sleeves due to the higher shrinkage of the thicker cast iron layers.

Wilkening and Côté [4] did also measurements of the stub-anode voltage drop in operational cells. Fig. 2 shows a typical voltage drop evolution between the stub/thimble in function of the temperature and the anode current.

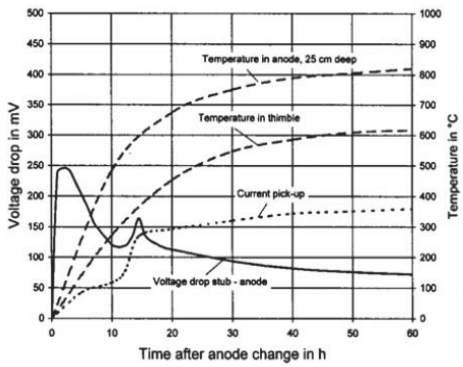

Fig. 2 – Stub to carbon voltage drop as function of the current pickup and temperature

It can be seen that the current circulating in the anode takes about 15 hours to achieve most of his current due essentially to the melting of the frozen bath at the base of the new anode. Also, the

temperature increase in the stub hole region leads to an increase of the contact pressure at the cast iron to carbon interface due to the stub expansion which decreases the ECR and thereby, allows possible increases of the current pickup.

In 2015, using high frequency measurements of individual anode current loads, Steinar Kolås et al. [5] study the current variation over the anode lifetime. Fig. 3 shows the average current pickup for the 30 anode positions as a function of the anode age.

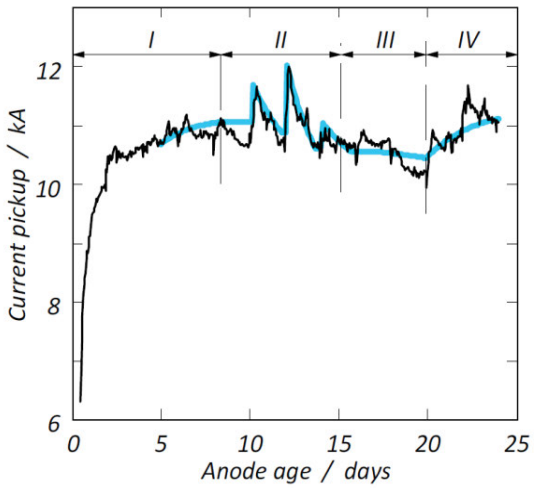

Fig. 3 – Current pickup in function of the anode age

They suggest that:

- The sharp current increase during the first 16 hours represents the re-melting period;
- The current increase after 16 hours is probably related to the decreasing ACD and electrical resistance at the cast iron to carbon interface;
- Large current increase at 10, 12 and 14 days are caused by newly set of resistive neighbouring anodes leading to current redistribution;
- From 15 to 20 days, the active area of the anode being smaller (air burn and widening of the slots), the anode is more resistive and will consume more rapidly leading to a slightly ACD increase (the beam being lowered at constant speed) and then less current in the anode;

Experimentation

The measurements were performed at Aluminerie Alouette Inc. on 4 anode assemblies (including two anodes each) during their full cycle. Two anode assemblies were tested at the same time in different cell at the same position.

Approach

The approach consists of recording in-plant voltage drop between each section of interest to evaluate their influence on the electrical performance of the anode assembly.

As described on Fig 1, the monitored voltage drop corresponds to the electrical resistance:

- between the stem and the stubs (R_1): Variable material resistivity;
- between the stubs and the carbon (R_2): Variable ECR, material resistivity and carbon cross section area;
- between the carbon and the cathode (R_3): Variable material resistivity, carbon cross section area and height, anode-cathode distance (ACD), bubble resistance and anodic overvoltage.

Instrumentation

Besides the voltage drop, the resistance between two monitored points depends also on the current passing through these points. This specific current I [A] can be determined using the equation:

$$I = \frac{\Delta V}{R} = \frac{\Delta V}{\left(\frac{\rho(T) * L}{A_{Cross\ Section}}\right)} \qquad \text{Eq. 1}$$

where V is the voltage drop between the two selected points, $\rho(T)$ is the material resistivity at a given temperature [Ω m], L is the length between the voltage probes [m], $A_{Cross\ Section}$ is the cross section area perpendicular to the electric field lines [m^2].

To accurately evaluate the electrical resistance between each section of the anode assembly as well as the temperature at each section, the position of each voltage probe/thermocouple has to be strategically determined. Fig. 4 shows the instrumentation used to capture these voltage drops. This experimental approach is justified in the following lines.

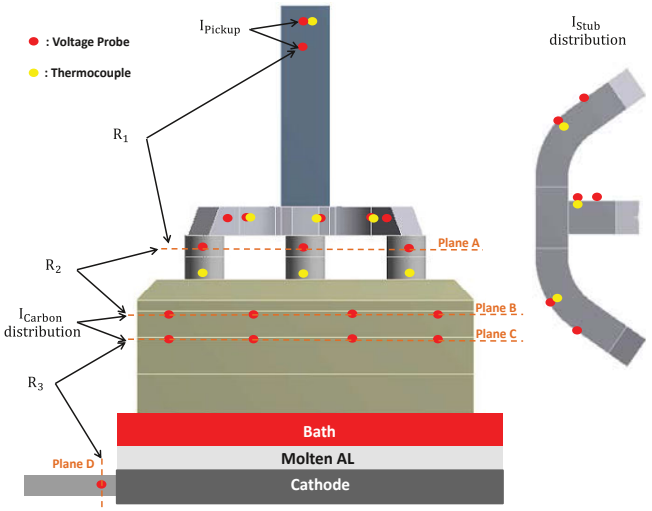

Fig. 4 – Instrumentation representation

I_{Pickup}

To evaluate the current passing through the anode assembly (called I_{Pickup}), two voltage probes (ΔV at distance L from each other) and one thermocouple on the stem (of known $A_{Cross\ Section}$ and $\rho(T)$) are required to determine each parameters of Eq.1.

Thereby, the current pickup is the amperage passing through plane A-B-C-D (Fig.4) if the electric field lines are perpendicular to the planes. Then, the location of the measured plane (*i.e.* plane that includes the voltage probe) has to be in a zone where the electric

field lines are perpendicular to the measured plane to allow accurate computation of the electrical resistance.

Plane A
The total current passing through the stubs corresponds to I_{Pickup}. However, the current passing through each stub isn't the same since the resistance from each corresponding path (specific stub to cathode) is different. Then, by using one voltage probe per stub, the average voltage drop between each stub and the stem gives the resistance between plane A and B (Fig 4) since the total current passing through is known.

Plane B
The same principle applies in the carbon block since the current density isn't the same along the block. The variation of the current density mainly comes from the variation of the electrical resistance between the base of the carbon block and the cathode including local variable ACD/bath movement, bubble electrical resistance and anode overvoltage but also, from the electrical resistance at the cast iron to carbon interface which differs for each stub hole. Using multiple voltage probes along plane B, the average voltage drop between plane A and B allows a better interpretation of the electrical resistance.

Also, the plane B has to be distant to the stub hole because there is an important variation of the current density in the bottom part of the stub hole where the main part of the current goes from the connector to the carbon block. Using a numerical model of the anode assembly with the proper material properties, it is possible to estimate the variation of the current density in the carbon block (Fig. 5). In this way, the plane B can be locate in the carbon block at a position where the measurements are less sensitive to these variation of current density.

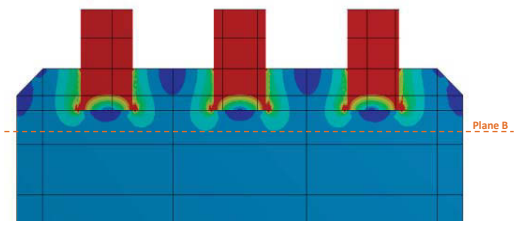

Fig. 5 – Current density variation in the carbon block

Plane D
The resistance between the carbon block and the cathode can be estimated by inserting voltage probes on each cathode bars below the selected anode assembly. Assuming that the current pickup is the same as the current circulating in the cathode bars below the anode assembly, the electrical resistance can be evaluated using the average voltage drop between plane B and D.

Current distribution in the anode assembly

The current distribution in the stubs and the carbon block can lead to a better understanding of the electrical behavior of the anode assembly via correlations with the measured voltage drops of R_1 to R_3.

Stubs current distribution
Using the monitored voltage drop, the determination of the current passing through a specific stub as well as the current pickup requires knowing the electrical resistance. Then, two voltage probes and one thermocouple on each leg of the hexapod (of known $A_{Cross\ Section}$ and $\rho(T)$) allow to determine the current passing through each stub (Fig.4). It should be noted that measuring the current directly from the stubs could lead to a bad estimation considering the high variation of the temperature gradient along the stubs coming from the crust growing.

Carbon block current distribution
For the carbon block, each voltage probe in plane B as a corresponding probe located 100 mm below plane B. All these corresponding probes define the plane C. The electrical resistance between plane B and C can then be estimated from the average voltage drop between the two planes assuming that the electric potential lines are vertically aligned and the carbon electrical resistivity is a constant.

Considering the important current fluctuations troughs the anodes, the entire electrical measurements have to be taken at the same time to extract an accurate and stable electrical resistance value.

Results and interpretations

Electrical resistance decomposition

Fig. 6 shows the electrical resistance as a function of anode age and the current pickup between the stem and the base of the carbon blocs and between the base of the carbon blocs and the cathode bars:

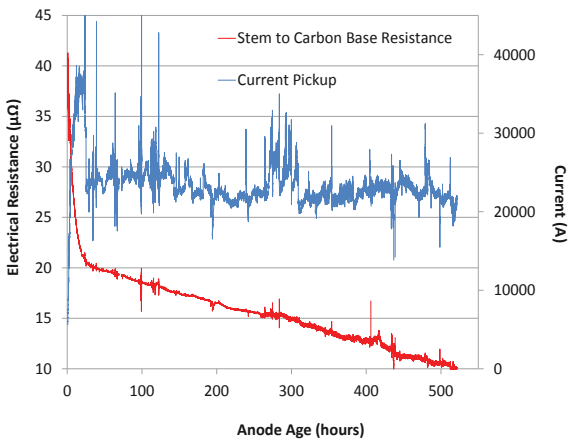

Fig. 6 – Total electrical resistance and current pickup as a function of the anode age

It can be seen that:

- After 20 hours, the electrical resistance between the stem and the base of the carbon blocs goes from 20 to 10 µΩ representing 13 to 6% of the total electrical resistance;
- Compared to the noisy current pickup (cause by the bath movement, bubble and anodic overvoltage), considering the scale of the electrical resistance axis, the stable electrical resistance shows the efficiency of the approach and instrumentation to accurately determine the resistance from the measured voltage drop in the anode assembly.

To reduce the electrical resistance through the design of the anode assembly, it is necessary to identify their behavior during the anode cycle.

Fig. 7 shows the anode assembly electrical resistance decomposition as a function of the anode age.

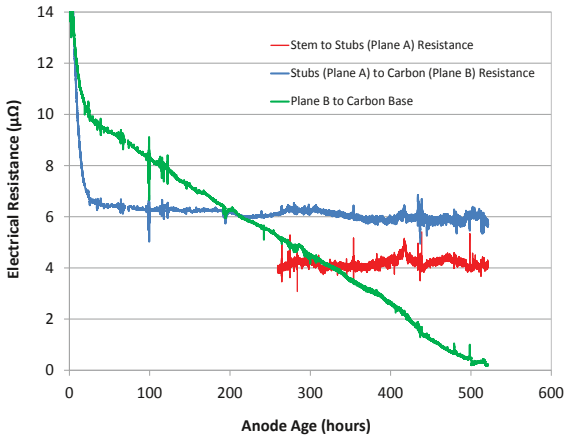

Fig. 7 – Decomposition of the electrical resistance of the anode assembly as function of the anode age

The linear reduction of the electrical resistance of the anode assembly comes mainly from the carbon bloc consumption which represents 41 to 0 % of the total resistance from 24 to 510 hours (From Plane B to Carbon Base).

Also, the electrical resistance between the stem and the stubs represent approximately 25 % and varies following the electrical resistivity coming from the temperature evolution.

The electrical resistance including the anode connection (Plane A to B) drops rapidly till 24 hours to a value of 6.5 µΩ approximately and represent 33 to 47 % of the total resistance afterward. Using a thermoelectrical model including thermal boundary conditions from measurements, a resistance of 7 µΩ between plane A and B correspond approximately to an electrical contact resistance (ECR) of 0.25 Ω*mm^2 for the given configuration. Note that the ECR is assumed to be constant on the entire cast iron-carbon interface. By reducing further the ECR to 0.125 Ω*mm^2, the electrical resistance is reduced by only 0.3 µΩ suggesting that, with a non-deteriorated anode assembly, the ECR is mainly optimal. For a specific geometry and boundary conditions, the electrical resistance begins to increase rapidly over 0.8 Ω*mm^2 corresponding to an electrical resistance of 8.53 µΩ between plan A and B.

ECR influence on the current pickup

It is well known that the resistance between the stubs and the carbon bloc is strongly influence by the ECR at the cast iron-carbon interface. This ECR is mainly directed by the dominant radial thermal dilatation of the stubs.

Then, the current increase during the first hours comes mainly from the decreasing ECR of the anode connectors [5] and the re-melting period of the bath. Those two elements lead to important electrical resistance reduction on a short period. Fig. 8-9 show the influence of ECR reduction on the current pickup compare to the re-melting of the bath.

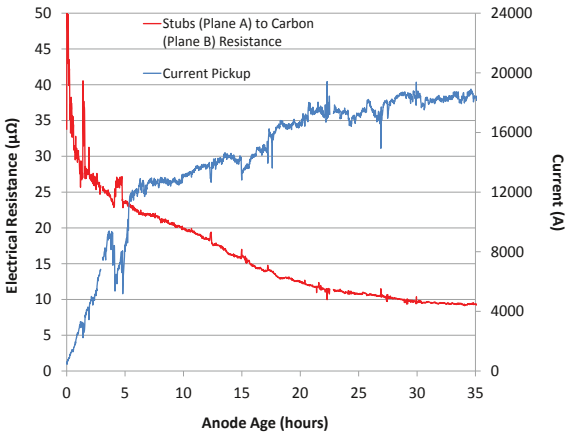

Fig. 8 – R_2 and current pickup as a function of the anode age

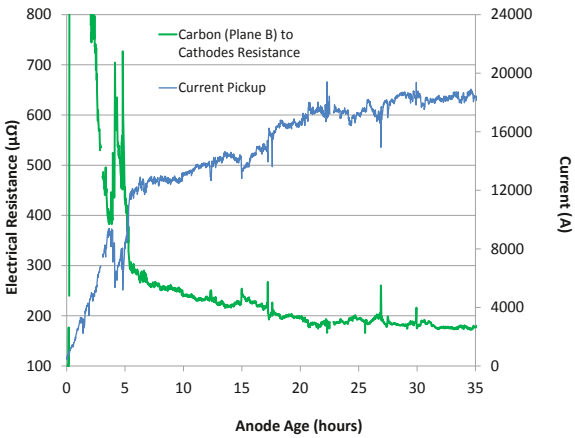

Fig. 9 – R_3 and current pickup as a function of the anode age

It can be seen that:

- R_2 reach near its lower resistance after approximately 30 hours corresponding to a reduction from 50 to 10 µΩ;
- The current pickup gets 60 % of his current after 6 hours (from 1800 to 300 µΩ) and 75% after 20 hours (from 300 to 200 µΩ) which coincide with the electrical resistance variations of R_3 and then correspond to the re-melting period;
- The current pick-up variation is influence essentially by the re-melting of the bath till 15 hours base on variation correlation between the current pickup and R_3;
- Between 6 and 15 hours, the electrical resistance variation coming from R_2 (7 µΩ) seems negligible compare to R_3 (100 µΩ). Note that, in that case, the stub diameter was not reduced or deteriorated which minimise the electrical resistance variation at early age for R_2.

Then, to allow the current to circulate in the anode assembly has fast has possible, preheating the anodes leading to a faster re-melting period is possibly a better option instead of optimising the ECR at low temperature via connector design modifications

leading potentially to carbon cracking afterward (at higher temperature).

In Situ behavior of the anode connection

As state previously, the ECR is function of the thermomechanical behavior of the stubs. Previous works have shown that key indicators based on variations between the thermal expansion coefficient and the electrical changes can help to determine the connector behavior and interface electrical performance [6].

Fig. 10 shows R_2 as function of temperature.

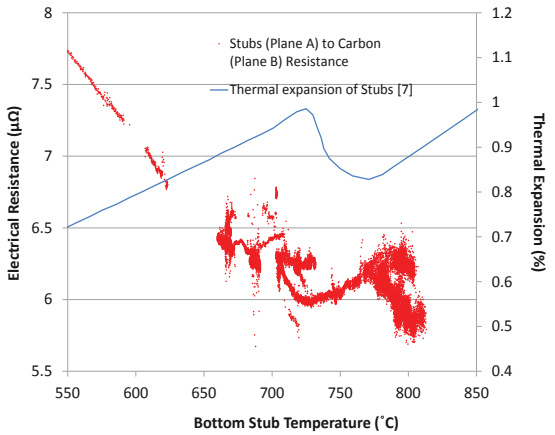

Fig. 10 – R_2 and thermal expansion of stubs as function of the temperature

These results show that, for this specific anode connection configuration, a temperature of 650 ˚C approximately is required to generate acceptable contact pressures.

Also, the shrinkage of the steel (between 725 and 770 ˚C) coming from the effect of the α-γ phase transition cause an increase of the electrical resistance of approximately 0.4 µΩ in that case. Then, considering the thermal expansion of the steel, the temperature of the anode connection region should avoid to be around 770 ˚C and less than 650 ˚C to obtain a minimal ECR (assuming no carbon cracking).

To efficiently quantify the economic impact of those behaviors, it is necessary to determine the duration and the current pickup corresponding to those temperature ranges. Fig. 11 shows the R_2 (from Fig.10) as function of the anode age with the critical temperature range.

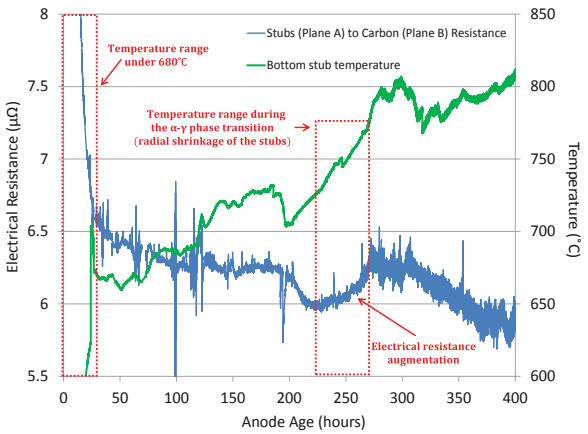

Fig. 11 – R_2 and thermal expansion of stubs as function of the temperature

From 0 to 30 hours, even if the electrical resistance is important, the reduced current pickup lowers the energy consumption. However, a stub diameter reduction requires a higher temperature to reach an acceptable ECR leading to a higher electrical resistance at a higher current pickup which means a lot more energy consumption.

The stubs shrinkage leads to an increase of 0.4 µΩ on a period of 40 hours at full current pick up. This period takes place in the middle of the anode life which means that it can be on a larger period considering the high variability in the temperature in the connector region. Also, based on previous work [7], because of the nonlinear ECR relation with contact pressure [3], the lower is the contact pressure before the phase change, the higher the increase of the electrical resistance at the phase change will be. That element may be critical regarding the energy consumption with reduced stub diameter.

Finally, the anode connection optimisation requires to:

- Minimise the electrical impact of the α-γ phase transition of the steel;
- Achieve a low ECR before the major current pickup which his approximately 24 hours;
- Minimise the electrical impact of the geometry degradation over the cycles;
- Maximise the nominal contact area by maintaining an ECR under 0.8 Ω*mm^2.

Current distributions

The sum of the measurements from each leg of the hexapod is in good agreement with the measured current pickup from the stem. This shows that it is possible to accurately quantify the current passing through each anode/stub and thereby a better understanding of the electrical behavior of the anode assembly.
Fig. 12 shows a typical current distribution in each stub one anode as a function of the anode age.

In Fig. 12, it can be seen that, after 24 hours, the current difference between the stubs stays mostly between 31 to 34 % showing a uniform current distribution between the three stubs.

Also, in general, the current offset between stubs stays relatively stable on a short period of time even if there's important resistance fluctuations along the anode (variable bubble resistance, bath movement or localise ACD). This uniformity can be explained by two different facts. Firstly, the electrical resistance of the carbon bloc represent a very small percentage of the resistance under the anode (4% at midlife approximately, 160 VS 6 μΩ). Secondly, the electrical resistance of 160 μΩ under the anode varies by a small percentage **along the anode base**.

Then, the exceptional high offset before 24 hours is possibly caused by the remaining solid bath at the shell side under the anode. This leads to a very large variation of the electrical resistance along the anode base. Once the re-melt complete, the current distribution becomes uniform along the stubs.

It can be seen in Fig 13 that abrupt change occurs in the current distribution along the stubs when the current pick up is unstable (hour 75,100 and 115). This behavior may also come from the very large variation of the electrical resistance along the anode. Considering the fluctuation rate, this large variation may come from large bubble formations, unstable ACD or anodic overvoltage.

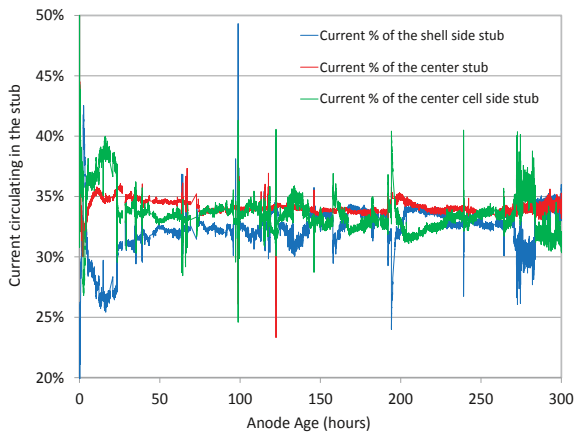

Fig. 12 – Current distribution in each stub as a function of the anode age (0 to 300 hours)

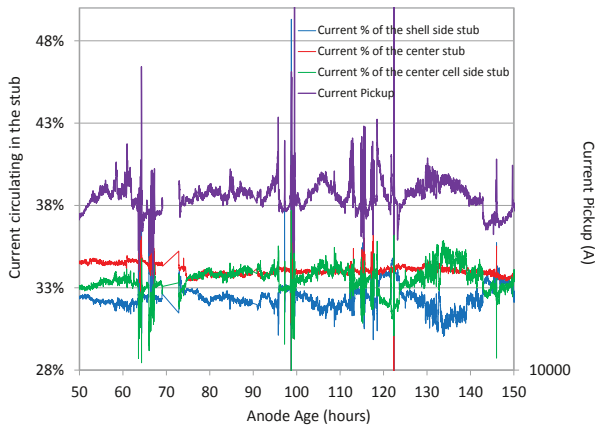

Fig. 13 – Current distribution in each stub as a function of the anode age (50 to 150 hours)

The three stubs representing a parallel circuit, an augmentation of the current pickup will lead to a superior current augmentation in the less resistive path (compare to other stub current). Then, the source of the path resistance variation can be determined by correlations between the known behaviors, measurements values (electrical and thermal) and variation rates.

Conclusion

Using a specific *in situ* instrumentation of an anode assembly, it has been possible to determine the electrical resistance decomposition over the anode life.

Also, correlations have been made between the thermomechanical behavior of the anode connection and the *in situ* electrical measurements to identify key elements for the design optimisation of an anode assembly.

Finally, in plant measurements of the current distribution can lead to a better understanding of the electrical behavior of the anode assembly in response of different phenomena.

Acknowledgments

The author acknowledge the financial support of the *Fonds québécois de la recherche sur la nature et les technologies* trough the Aluminum Center – REGAL, Nature Sciences and Engineering Research Council of Canada (NSERC) and particularly our industrial partners Aluminerie Alouette Inc. Special thanks also to the industrial partner team in this project, namely Mr. Guy Ladouceur, Mr. Marc Gagnon, Mr. Charles-Luc Lagacé and Mr. François Laflamme for their professional advice and experimental help.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

LOW RESISTANCE ANODE ASSEMBLY USING STEEL STUBHOLE CONDUCTORS ACROSS THE CAST IRON TO CARBON INTERFACE

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Keywords: Anode Voltage Drop, Contact Resistance, Stub to Carbon, Cast Iron, Power Consumption

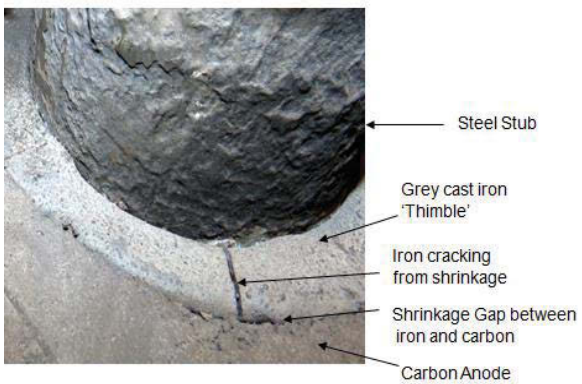
Abstract

Anode assemblies suffer a significant electrical contact resistance across the cast iron to carbon interface that is inversely dependent on contact pressure and area. Industry efforts have incrementally reduced this electrical resistance by increasing stub diameter, changing iron chemistry and by improving the stubhole shape.
The additional use of multiple steel conductors to bridge across the cast iron to carbon interface provides a means to further reduce the electrical resistance. The function of the conductors is independent of the iron to carbon contact pressure, the stub temperature, iron chemistry, and the stubhole shape. The steel conductors are tightly driven into the carbon anode at one end, with the other end bonded into the cast iron.
This paper includes in-pot performance testing results which demonstrate the reduced resistance when using stubhole conductors.

Introduction

The electrical resistance (contact resistance) across the cast iron to carbon interface in prebaked anode assemblies is a significant cause of wasted energy consumption in aluminum electrolysis. The casting process and resulting iron connection are shown below.

Cast iron rodding of stubs to carbon anode.



Shrinkage gap and cracks of cast iron connection.

Causes of Electrical Contact Resistance

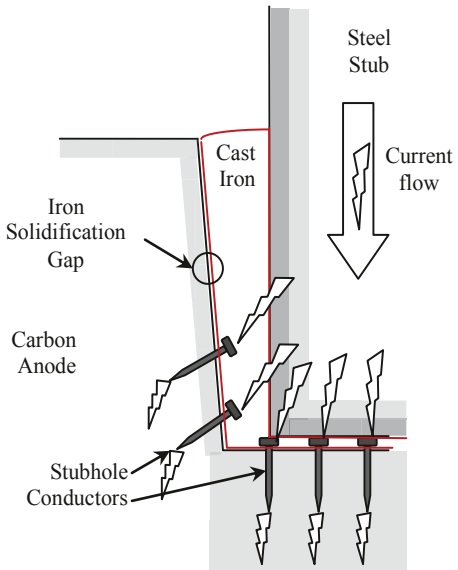
Material characteristics, operating conditions and physical designs contribute to the causes of high electrical contact resistance of the cast iron to carbon connection, including:

- **Thermal expansion characteristics** – the higher coefficient of thermal expansion [1][2] of the steel versus anode carbon enables an increasing contact pressure and lower contact resistance as the steel stub heats up.
- **Cast Iron Chemistry** – % CE impacts solidification shrinkage and phase transition expansion [3][4].
- **Stub Diameter** – the greater the contact area the lower the electrical resistance, subject to contact pressure, with deteriorating stub diameter increasing the resistance [5][6].
- **Contact pressure** – contact resistance decreases with increasing contact pressure [7].
- **Cast iron and stub preheat temperature** - these temperatures influence the cast iron crystalline structure during rodding, and the thermal expansion possible during operation [3].
- **Cast iron thickness** – cast iron thickness impacts the solidification shrinkage gap which impacts the contact pressure from subsequent thermal expansion of the stub [8].
- **Stubhole and Flute shape** – these shapes impact both contact pressure and contact area at the cast iron to carbon interface [3][6].
- **Rod Yoke rigidity and toe-in of stubs** – Yoke thermal expansion affects contact pressure on the outer stubs, while residual yoke & stub deformation (toe-in) causes variation in the cast iron thickness around stubs [9].
- **Stub surface texture and cleanliness** – these factors impact contact resistance and electrical contact area [10].

General findings lean toward contact pressure as the predominant factor of electrical contact resistance of the cast iron to carbon connection, whether that contact pressure is caused by thermal expansion of the stub and iron, by cast iron phase expansion, by stubhole shape, or by yoke expansion. However, contact pressure has limited benefits where the rate of reduction of contact resistance with the increase of contact pressure decreases significantly when contact pressure exceeds 1 Mpa, and no significant change of contact resistance occurs solely due to contact pressures above 10 Mpa [6].

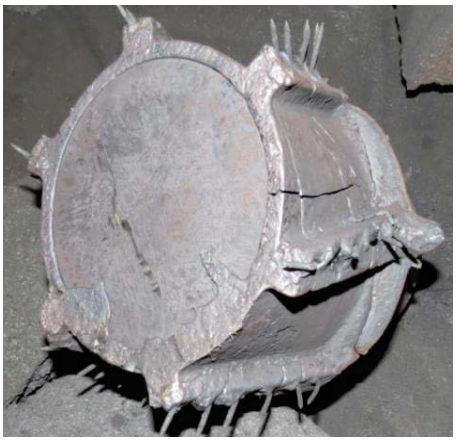
Stubhole Conductors

Research by Hatch into improvements of the cast iron to carbon connection have led to the method of embedding multiple thin carbon steel conductors into the inner carbon surface of anode stubholes, with the exposed end of the conductors then encapsulated by the cast iron. This method overcomes many of the causes of electrical contact resistance as listed above. A cross section depiction of current flow is shown below.



The result is multiple conductors bridging the shrinkage gap between the frozen cast iron and the carbon. The conductors may be located in any combination of bottom or sides of the stubhole, as shown below.

Example patterns of the cast-in conductors are shown below.



The stubhole conductors consist of small hardened carbon steel spikes (nails) which are driven into the anode carbon with a pneumatic nailer. The embedded thin conductors have a very high localized contact pressure at the yield point of the carbon. However, the total contact force against the carbon by each conductor is limited by the conductors' small contact area, and by the porosity of the carbon which enables it to compress locally without promoting long cracks away from the conductor.

The cast iron freezes and shrinks around the exposed end of each conductor exhibiting both high contact pressure and metal to metal contact similar to the stub to cast iron connection.

The total electrical resistance of the combined conductors is dependent on the total contact area and conductor cross section. A minimum quantity of conductors is required to reduce risk of melting the conductors from resistance heating, while the maximum quantity is based on diminishing benefit and cycle time requirements for installation.

The conductors are bonded into the cast iron 'thimble' and following the anode cycle they are stripped from the anode with the thimble. Depending on conductor spacing additional thimble tumble cleaning may be required to remove carbon prior to remelting the iron & conductors.

Performance Tests

Previous research (circa 2006) by Aluminerie Alouette Inc. (Alouette) in Sept Iles, Quebec found cast iron-carbon resistance of approximately 10-12 micro-ohms[3, Fig.34]. More recent changes to the ALOUETTE stubhole shape has further reduced this resistance. ALOUETTE agreed to conduct in-pot tests of the Stubhole Conductors from April-August 2015.

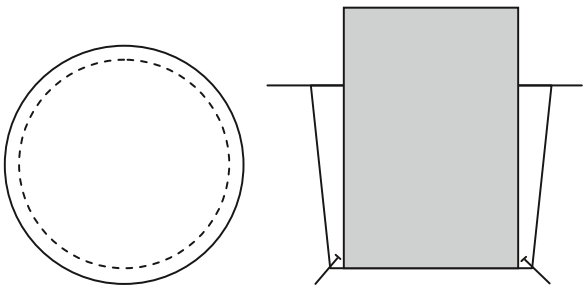
Alouette performed two in-pot tests as follows:

1. Test 1: Three pairs of anodes B133-135 (one anode of each pair with test conductors, the other anode without

conductors as reference). All anodes had manually cleaned stubholes to thoroughly remove all packing coke remaining from the baking process. Each pair was installed in a separate reduction cell. Voltage drop was collected by datalogger for each anode assembly through the stem and for each stub.

2. Test 2: Three pairs of Anodes in cells B136-138, same as Test one, however all anodes had partially cleaned stubholes with a 'typical' amount of packing coke remaining in the stubhole from the normal mechanized cleaning process.

The pattern of the stubhole conductors in the test anode included sixty conductors installed at approximately 45 degrees in the bottom perimeter corner of the stubhole. The pattern is depicted in the illustration below.



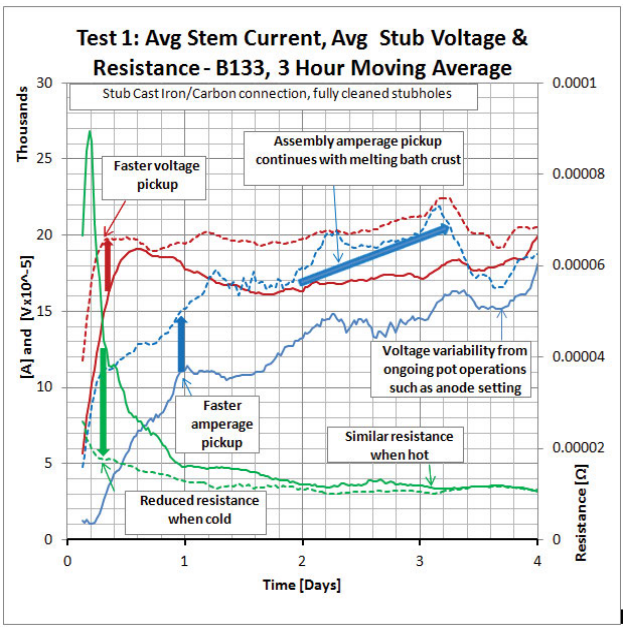
Alouette Stubhole Conductor installation pattern.

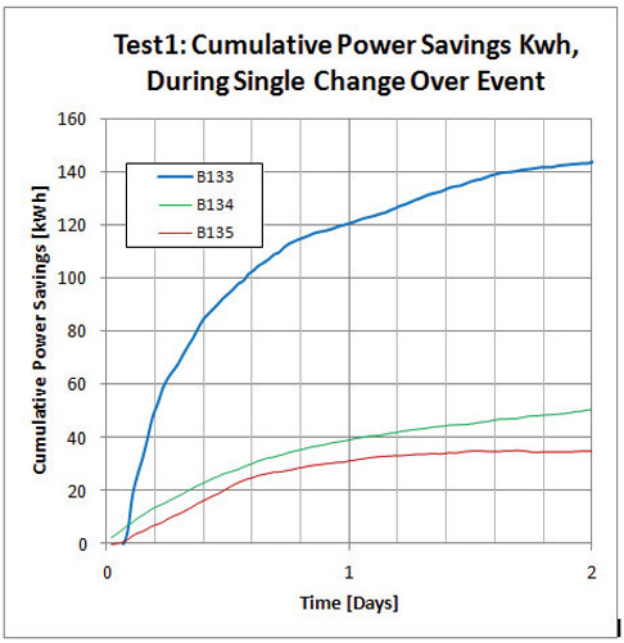
Test Anodes voltage measurement leads and wiring setup.

Condition of conductors after full anode cycle and butt removal.

The collected data was cleaned to remove data from failed voltage leads, which occurred between 4~6 days after the start of the 25 day anode cycle. The collected data was analyzed to determine the resistance of each stubhole, the assembly amperage and voltage pickup, and to estimate potential power savings for the anodes with conductors versus those with fully or partially cleaned stubholes.

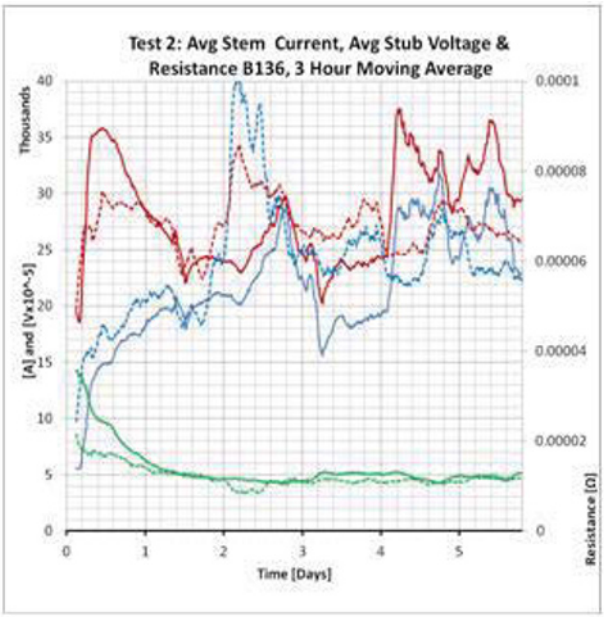
The typical comparison between one pair of test anode (dotted line) and reference anode (solid line) in one reduction cell is provided in the following graph.



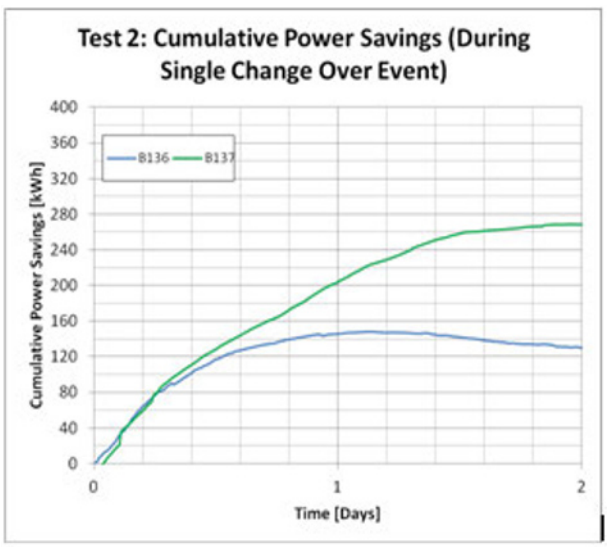


The results of Test 1, for three pairs of anodes with clean stubholes, provided energy savings of between 35-142 Kwh per anode change event, primarily occuring over the first two days of operation. This potential savings can be extrapolated annually to the number of anode changes per smelter per year.

Test 2 included anodes with partially cleaned stubholes which caused the reference resistance to be higher, both at the start of the cycle and for a longer duration into the cycle. Data collection was incomplete due to failed wire leads after 5-6 days. The analyzed data is provided in the following graph.



The average stubhole resistance of both Test 1 anodes with clean stubholes was approximately 10.0 micro-ohms by day 5 of the anode cycle. The Test 2 anodes with partially cleaned stubholes had average resistance at day 5 of approximately 12.5 micro-ohm for the reference anode (without conductors) and approximately 10.5 micro-ohms for the test anode (with conductors).
The difference in resistance between the Test 1 and Test 2 anodes may be the previously mentioned causes; however the main difference during preparation was the remaining coke in the stubholes from the mechanized cleaning process.



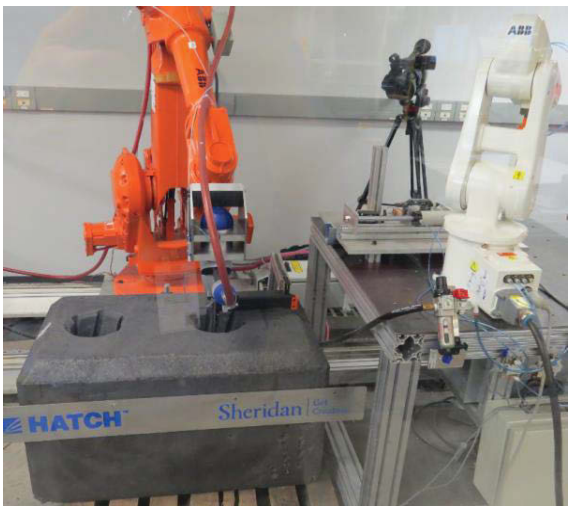
The predicted power savings per anode change event is 125-250 Kwh when the stubholes are only partially cleaned of packing coke.

Two additional observations of the comparative test data included:

1. A reduced short term variation of voltage drop between stubs in the same anode assembly, as measured after the initial heat up period (during days 1-4). The stub to stub voltage variation reduced from approximately 33% (without conductors), to approximately 16% (with conductors), The stub to stub voltage drop variation is transient and will normally increase as the stubs and the rod yoke heat up. The conductors appear to reduce the impact of material thermal expansion on the stub to stub voltage variation.
2. A reduced short term fluctuation of anode assembly voltage drop (measured in ten minute increments over 3 hour average) from approximately >30% without conductors, reduced to <10% with the conductors. The anode assembly voltage fluctuation appears to be 'calmer' when using the conductors.

Method of Applying the Stubhole Conductors

The stubhole conductors have been developed from hardened T-nails, with a modified pneumatic nailer to reach into the stubhole (shown below). The nailer can be used manually, or robotically with automated reloading. The use of Stubhole Conductors has been patented by Hatch.



Summary and discussion

The Stubhole Conductors have demonstrated three potential benefits for smelter operations:

1. To reduce the electrical resistance across the cast iron to carbon connection, and reduce its associated power consumption. In optimal conditions this reduction is mainly during the early anode and stub heat up period. The improvement is more pronounced and for longer duration in the case that the normal cast iron to carbon electrical resistance is relatively high, whether due to low contact area, low contact pressure, worn stub diameters, presence of packing coke in the stubholes, high current density, etc.
2. To reduce voltage drop variation between stubs in an anode assembly. This improves current distribution through the anode, and may reduce the occurrence of hot stubs and associated air burn of the carbon.
3. To reduce the short term fluctuation of voltage drop through an anode assembly. This may provide a 'calmer' pot operation.

The benefits of improved current distribution may also occur during new pot startup with electrical preheating, when all of the anodes start out cold. In this situation the reduced and more consistent electrical resistance may then improve current distribution across the cathode blocks.

Further comparative resistance versus temperature testing is planned in a controlled furnace with micro-ohm resistance measurement, in order to remove the variation from normal cell operations and to remove the test wiring from the hostile environment of a reduction cell.

Further in-pot testing with pot noise analysis may determine if the reduction in anode assembly voltage fluctuation is statistically significant.

Acknowledgement

The authors wish to express their thanks to Aluminerie Alouette Inc. for participating with the in-pot tests.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

UPGRADE OF THE FIRING AND CONTROL SYSTEM AT EGYPTALUM FOR DUAL FUEL FIRING

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Keywords: Anode Baking Furnace, Firing and Control System, Dual Fuel Firing

Abstract

Based on the recent upgrade of the existing anode baking furnace no.1 at Egyptalum, Egypt, the paper will demonstrate the improvements that have been achieved by introducing a new and flexible firing and control system on the baking furnace.

The focus was set on a dual fuel firing capability and strategy. The firing system is able to operate either on heavy fuel oil or natural gas on the same system. This feature provides maximum flexibility in the daily challenges of the primary energy market to carbon operations. The paper covers the most relevant parameters of the baking process such as combustion performance, temperature regime, baking level, anode quality and consistency, fuel efficiency and the resulting operation flexibility.

Introduction

Egyptalum is a primary smelter located in Nag Hammadi, Egypt (see Fig.1). The first Soederberg potline was put into operation in 1975. From this starting point, the Smelter continuously increased its capacity and finally started the first prebake Potline no. 6 in 1997. The actual aluminium output is around 320.000 t/a . The carbon plant was started in 2000 after the first prebake potline was sucessfully in operation. It consists of a complete paste plant and 1 open type Anode Baking Furnace, having a baking output capacity of 160.000 t/a. Up to March 2015 it operated on heavy fuel oil. In the near future the source for heavy fuel oil in Egypt will become volatile. So it was a strong demand to secure the anode production by looking for alternative energy sources. Finally a natural gas supply could be secured. And to avoid any further bottle necs in the future, it was decided to install a dual fuel firing technology, which either can operate on natural gas or fuel oil.

Fig. 1 Egyptalum

Details of the baking furnace no. 1

The baking furnace no.1 consists of 66 sections, each with 9 flues and 8 pits. The firing system operates with 4 fire groups on heavy fuel oil. Fig. 2 allows a side view on the anode baking furnace.

Fig. 2 Anode Baking Furnace no. 1 at Egyptalum

Dual Fuel Firing – the Challenge and the Strategy

Egyptalum needs to change the energy source from heavy fuel oil to natural gas. In parallel they want to keep as much as possible from the existing firing system. So the challenges of the development and implementation of a dual fuel firing system are:

- Check if the existing firing ramps are in a healthy condition
- Check if there is any space available to implement a second energy distribution system and a second set of burner equipment onto the existing burner ramp.
- How can a transition period be realized without or with minimum loss of production
- How can a safe and reliable operation be realized with 2 different types of fuel on board
- The existing control system is only able to handle fuel oil. A new control system has to be implemented which is able to handle dual fuel applications.

An expert team was sent to Egyptalum to check the current situation. All disciplines had to be examined starting from mechanical structures, oil burner system, pump feeding system, electrical installation and controls. After one week of intensive examination, a strategy was developed which made the challenge feasible. Figure 3 shows the situation of the burner ramps before the conversion.

Fig. 3 heavy fuel oil burner ramp at Egyptalum

It can be seen from the picture above (Fig. 3) that the existing burner ramp is already full of equipment, necessary to serve the fuel oil operation. So the idea was to implement a second level which can accommodate the whole gas system parallel to the fuel oil circuit.

After examination of the equipment it became obvious, that the transition from fuel oil to natural gas will entail a certain loss of production. To minimize this loss of production, the level for the gas distribution was fully pre-fabricated and assembled. The first out of the four fires was stopped for a period of 4 days and converted to dual fuel firing. Also the control panels were completely fabricated and tested in advance, ready to be "switched-on". Figure 4 shows the prefabricated gas distribution pipes including the control valves, ready for the transition work.

Fig. 4 natural gas circuits and control panels fully preassembled

After fire stop, three (3) burner ramps were immediately transported into the working area and modified in parallel. Fig. 5 and 6 give some impressions of the working situation.

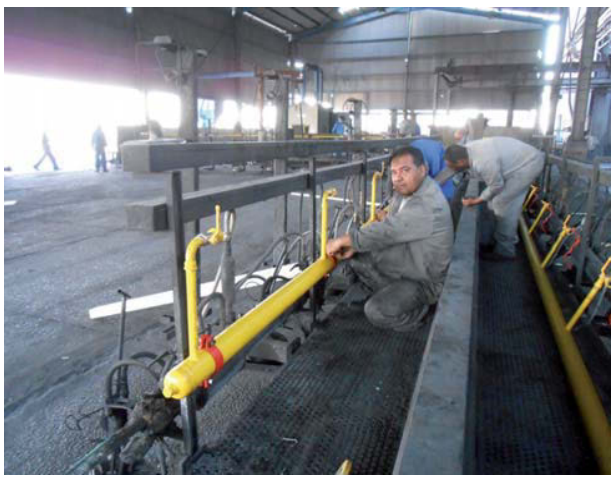

Fig. 5 transition of 3 burner ramps in parallel

Fig. 6 3 burner ramps ready to go

Finally the converted ramps were put back onto the furnace and the fire was restarted in the following sequence:

- Restart fire 1 on heavy fuel oil for one week
- Change from oil to natural gas on fire 1
- Fire 2-4 stay on heavy fuel oil
- Repeat procedure for Fires 2,3 and 4

Fig.7 shows the fire after restart on the furnace on heavy fuel oil, now capable of either operating on fuel gas or heavy fuel oil.

Fig. 7 dual fuel fire restart on heavy fuel oil

During this week, intensive training of operators was executed to prepare the production team for the next step namely the operation on fuel gas. The fuel gas system is a high efficient pulse burner system, operating at 1,5 bar gas pressure at a pulse length of 450 msec. It contains multiple safety interlocks and safety circuits for reliable and safe operation on fuel gas. Fig. 8 and 9 illustrate the operation on fuel gas after 1 week.

Fig. 8 fire operation on fuel gas

Fig. 9 fire operation on fuel gas, oil burners on stand-by

Upgrade of the process control system

A new and advanced process control system was installed which accommodates the parameters needed for fuel oil firing as well as for natural gas firing. It also includes features for a smooth change over from either use of fuel gas or fuel oil. This can be even selected fire by fire. Figure 10 shows the dual fuel burner parameters activated in the central control system.

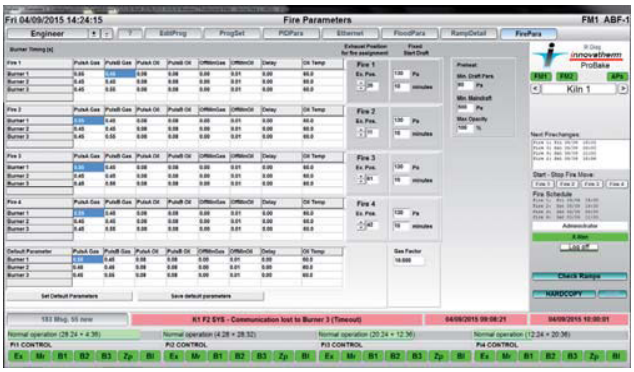

Fig. 10 Dual fuel burner parameters

In parallel new advanced control modules were implemented to gain a better combustion efficiency, namely

- The flooding index module
- The pitch burn module

The control modules ensure the optimum utilization of the available draft to combust the pitch and the fuel gas or the fuel oil. With the on-line calculation of the Flooding Index the actual oxygen levels in the firing area of the furnace can be evaluated, particularly at the front burner ramp. Any lack of oxygen is automatically prevented by the Flooding Index Module by a dynamic limitation of the burner capacity, especially during fuel gas firing. Figure 11 illustrates the fuel efficiency and the calculated flooding index for each flue.

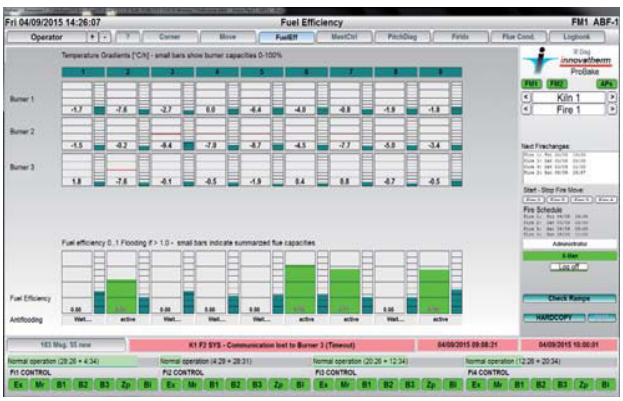

Fig. 11 Flooding Index and Fuel efficiency

The Pitch Combustion requires a specific recognition of the oxygen levels during the pitch burn phase. The advanced Pitch Burn Module as shown in Figure 12 incorporates the opacity reading in the collection pipe of the exhaust ramp and the temperature gradients in the preheat sections. As a result of these observations the pitch burn module forces the pitch combustion by dynamically increasing the draught (volume) in the individual flues and/or reduction of either the fuel oil or the fuel gas input on the dual fuel burner ramps.

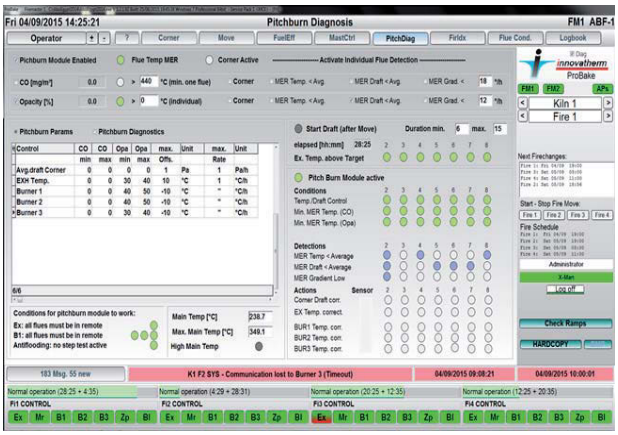

Fig. 12 Pitch burn optimisation

Figure 13 shows the installation of the opacity meter at the existing exhaust ramps.

Fig. 13 Opacity meters installed at Egyptalum

Achieved Results

Egyptalum is now able to react dynamically on shortages of either fuel oil or fuel gas. Also change of fuel energy suppliers available on the market can be studied and taken into consideration. Each fire can be individually operated on gas or oil. Within one hour or during a fire move the transition from one energy source to the other can be executed and vice versa. This provides maximum flexibility and reliability in production and prevents the risk of interruptions.

Additionally the combustion efficiency increased significantly by introducing the short pulse, high pressure fuel gas technology accompanied by a new process control system. As a matter of fact also the anode quality parameters show a positive and robust trend in consistency. Fig. 14 shows the achieved results of the anode

parameters apparent density, specific resistance, CO2 reactivity and Air reactivity as follows:

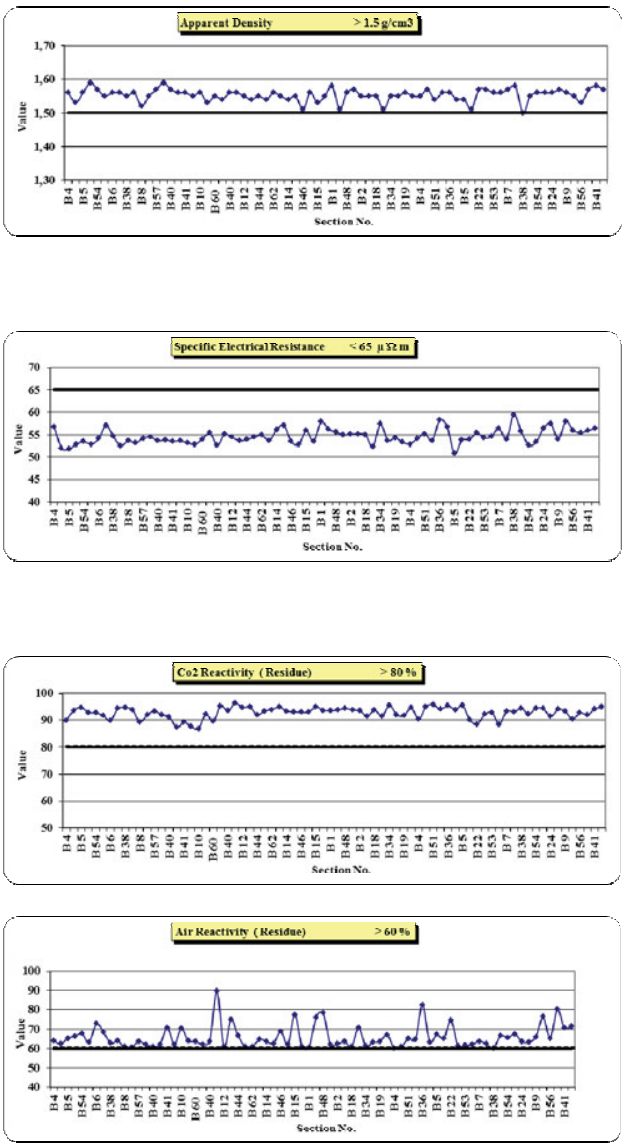

Fig. 14 Anode Quality analysis data

In August 2015, Egyptalum produced 14150 t of baked anodes, which is a monthly production record. Taking this into account, the production can increase up to 169.000 t / a. Figure 15 shows the dedicated firing curve leading to this optimized process.

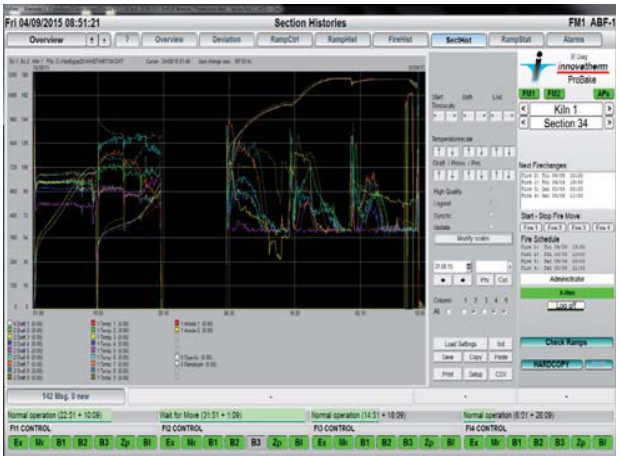

Fig. 15 Specific Firing Curve optimized in the central control system

The final top temperature could be lowered to 1160 °C, still providing an anode temperature of 1085 °C (see Figure 15). The maximum heat-up rate of the anode was measured at 14 °C/h. Finally the specific energy consumption dropped significantly. In August 2015 the amount of natural gas used was at 874.870 m^3. This leads to a greatly improved specific energy consumption which is now in a range of 2,4 GJ/t.

Conclusions

This paper discussed the challenges and strategies for an implementation of a dual fuel firing. The results demonstrate very impressively how the targets were reached and showed further improvements that have been achieved.

Dual fuel firing will provide more flexibility in production to withstand future challenges in energy supplies.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

MIREA: AN ON-LINE QUALITY CONTROL EQUIPMENT INTEGRATION IN AN OPERATIONAL CONTEXT

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Keywords: Carbon resistivity, Anode resistance, Anode voltage drop, On-line measurement

Abstract

Electrical resistivity is one of the key performance parameters of prebaked anodes. Nowadays, the desire to lower anodic incidents and energy consumption can be attained with a state-of-the-art equipment, which is able to accurately measure the electrical resistance of baked anodes on a production line. Anode/carbon plant improvements will be achieved by a better understanding of the impact of the physical properties of the anode in the cell and the fundamental causes of anode defects. This paper presents an overview of the milestones and challenges of implementing such a device into a mature carbon plant and ensuring acceptance from the production (or operation) team. This innovative device will provide opportunities for smelters to improve their cost-effectiveness by reducing process variation and maximizing anode performance in the cell.

Introduction

Fives ECL has introduced on the market a new online tool called MIREA. The acronym is derived from a French translation: "Instantaneous Measurement of Anodic Electrical Resistance". This online tool is used to measure the quality of baked anodes, which are used in pot lines for the production of primary aluminium. The system is based on a non-destructive measurement of the anode electrical resistance by mimicking the current distribution of an anode in service. It is derived from the R&D device developed by Rio Tinto [1] and used in numerous measurement campaigns carried out worldwide under the supervision of Rio Tinto/AP Technology™ experts since 2012. This industrial MIREA is the result of a technological development partnership between Rio Tinto, Fives ECL and Alouette Smelter (Sept-Îles, Québec).

With the use of the industrial MIREA device, all anodes can be measured in real time to provide not only an electrical resistance estimate of the anode population, but the exact resistance and dispersion of the population itself. This device can therefore help the carbon plant of a primary aluminium smelter to control more effectively its anode production process. It also helps the smelter identify and reject anodes having internal deficiencies, which can lead to premature failure of the anodes in use or excessive energy consumption. Globally, the use of the online MIREA helps reduce carbon waste and increasing pot performance.

This technology is now under commissioning at Alouette, which is the largest primary aluminium smelter in the Americas having an annual capacity of 600 000 metric tons of metal. The smelter has a well-established R&D philosophy promoting innovative technologies such as the MIREA. Furthermore, it is also well-known that Alouette is very demanding in terms of anode quality. This smelter has already implemented various control systems to reject a fraction of its baked anode population. These control systems are based on visual inspection of the baked anode surface according to a predefined set of parameters such as the presence of longitudinal cracks, stub hole cracks, agglomerated packing coke, missing anode parts, etc.

However, it has been demonstrated in previous works, using the R&D MIREA device [2], that visual observation alone can result in the misclassification of individual anode in terms of their electrical performance. These categorization errors are costly since unacceptable anodes are still reaching the pot lines and fair anodes are wrongfully discarded from the production. It must also be noted that visual observation is not very discriminating. At best, an anode population can be divided into three categories such as normal, fair-mediocre and unacceptable anodes. This is not the case with the MIREA device where an individual ranking of an anode, in the overall population, is possible [3]. This ranking can be done using indicators such as the MIREA global voltage drop of the anode, the voltage drop residuals against the voltage drop pattern of a reference anode, the combination of a multivariate analysis with the Mahalanobis distance [4], or other means.

Furthermore, Alouette, like other smelters in the world, relies on the traditional core sampling process to estimate the resistivity of its anode population. This evaluation of the electrical performance of an anode population is based on weekly resistivity averages taken on a limited number of anode core samples. Few anodes are cored weekly since this operation is time-consuming and destructive. Thus, this cumbersome method cannot be used as an online method to reject a fraction of the baked anode population.

Core sampling also provides limited information on the spatial variability within an anode. For example, cracks in a precise anode section could remain undetected due to the absence of sampling in that specific region. This problematic was observed in a previous MIREA campaign highlighting resistance heterogeneity problems such as highly resistive tops from two anode plants producing a similar anode format [2]. This gap of apparent resistivity between the two plants was estimated at ~17 μΩm in the top anode section. This resistivity difference had never been detected before due to the absence of core samples in this specific region.

Furthermore, resistivity measurements are typically based on the ISO standard 11713 [5]. With this standard, the current pattern in the core sample is almost ideal (top bottom current distribution). Chollier et al. [1] showed that coring samples underestimated the

resistivity in the upper part of the anode since the standard cannot take into account the specific current distribution near the stub hole vicinity. Indeed, the current pattern for an anode in service is mostly horizontal in the stub hole region.

Consequently, anode coring methods cannot provide an accurate assessment of the electrical performance of an anode population due to the insufficient number of samples taken every week and the lack of spatial resolution. Furthermore, resistivity results are usually obtained weeks after the introduction of the anode in the pot. Therefore, technical teams in smelters cannot take sound decisions or immediate corrective actions. These shortcomings emphasize the needs for implementation of devices such as the industrial MIREA device in carbon plants.

This paper outlines the important milestones in the industrial MIREA development from its initial conception, integration in the Alouette process chain (Figure 1), start-up and the next steps for its final appropriation by the Alouette operation team. This paper also includes a brief description of the MIREA measurement principles and an illustration of typical MIREA results. It must be noted that another paper [3] offered a first glimpse of the capacity of this online tool by summarizing the resistance measurements of more than 10 000 anodes produced between June and August 2015, at the Alouette Smelter.

Figure 1. Industrial MIREA device on the baked anode conveyor with its machine protection system.

Measurement principles

The measurement principles of the MIREA device can be found in several publications [1-3]. Briefly, the MIREA device replicates the anodic current distribution using stub hole devices imitating the standard anodic rodding. These stub hole devices establish an electrical connection to the anode, thereby allowing the passage of an electric current. The devices were designed to adapt to irregular stub hole shapes such as holes with missing flutes or ellipsoid geometries. The electric circuit is completed with a carpet in contact with the anode bottom surface. A voltmeter is used to establish a voltage drop map of the anode side surfaces.

Voltage drops are measured between reference points (H_0) located on the top of the anode and at predefined positions on the anode side surfaces. These positions follow a predefined template consisting of four columns of measures (C_1 to C_4) with seven position levels (H_1 to H_7), as illustrated in Figure 2. A total of 28 "mV" measures are taken for each anode. This measurement template is considered a good trade-off between the operating mode of the device, measurement cadence and anode representativeness.

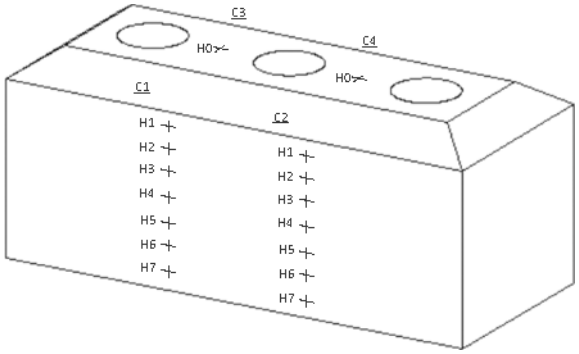

Figure 2. MIREA measurement position template.

The millivolt measurements of the anode surface can be converted into apparent resistivity values [1]. This data transformation provides a physical grasp of the MIREA to standard resistivity units and can allow comparison between different anode formats, as presented in another paper summarizing six MIREA campaigns [2]. Numerical simulations, modeling the electrical potential drop on the anode surface, are used for this unit conversion. These simulations are based on the anode blueprint, use a constant resistivity value and match the current intensity target of the MIREA. A simulation of a generic anode format is illustrated in Figure 3.

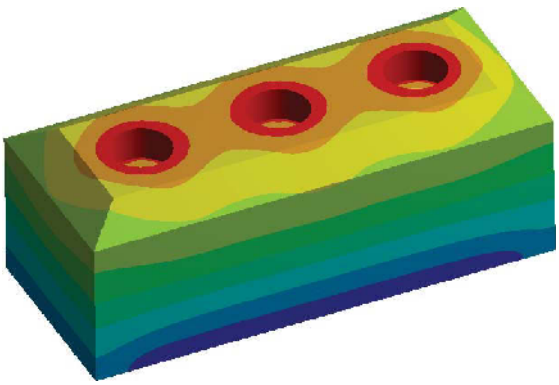

Figure 3. Numerical simulation of a generic anode format showing the voltage drop on the anode surface.

To perform a resistivity conversion, it is assumed that the resistivity of an anode slice section, $\rho(S_i)$, is proportional to the slice voltage drop, $\Delta V(S_i)$, measured between the slice boundaries H_{i-1} and H_i (Figure 2). The proportionality factor of this relation is the ratio between the resistivity of the simulation (ρ_s) and the predicted voltage drop of S_i, $\Delta V_s(S_i)$:

$$\rho(S_i) = \frac{\rho_s}{\Delta V_s(S_i)} \Delta V(S_i)$$

$\rho(S_i)$	Apparent resistivity of section S_i (μΩm);
ρ_s	Resistivity of the simulation (μΩm);
$\Delta V(S_i)$	Voltage drop of S_i (mV);
$\Delta V_s(S_i)$	Predicted voltage drop of S_i (mV).

It must be noted that this assumption is valid if the current distribution in real life anode is similar to that of the simulated case. However, some biases are always present between the real and simulated current distribution due to the heterogeneity of the real life anode. Therefore, this document continually refers to an apparent resistivity value, when presenting the MIREA resistivity data.

Partnership description

During the development of the industrial MIREA device, Rio Tinto, Fives ECL and Alouette had continuous interactions to ensure a smooth progress in the development of this online technology. Rio Tinto coordinated the interactions between Fives ECL and Alouette. They also assumed the technical lead in the data treatment and data analysis. Fives ECL developed the industrial MIREA based on the R&D version of the MIREA and on the Alouette requirements. Fives ECL also offered support for the installation, start-up, and maintenance of the MIREA as well as training support. Alouette followed the evolution of the project at every stage to ensure the proper development of the technology according to its needs. Alouette handled the assembly of the MIREA, the start-up, operation and maintenance of the device. They also assumed the lead in the realization of several tests to validate the performance of the online device. The technological development team of Alouette led the project on the smelter side. The team was supported by the Alouette production, operation, maintenance, automation, and HSE groups.

Major project milestones

The MIREA industrial development follows the management project flowchart presented in Figure 4. This plan includes seven different stages with major milestones such as the collection of the voice of the customer, installation of the online device in Alouette and initial start-up of the device. The project will end with the final gate review where the project will be delivered to the Alouette operation team to allow its final appropriation.

As of this writing (September 2015), the project is at the proof of concept stage (Analyse phase), where final works are completed before integrating a rejection algorithm and conducting the first real time rejection test to recycle unacceptable anodes. The proof of concept stage is further described in the next section of this paper.

Voice of the customer (Define/Measure stage)

At the beginning of the project, a basic data list was produced to include the major requirements for the project. First, the industrial version of the MIREA needed to reproduce the concept of the R&D device while keeping a similar level of precision and repeatability.

As for the location of the device in the carbon plant, Alouette wished to integrate the industrial MIREA device on their baked anode conveyor, servicing two anode baking furnaces, on Production Line no.2. Consequently, every baked anode produced by the two baking furnaces needed to pass through the MIREA device. This position allowed sampling of half the total baked anode production of the carbon plant. The other half of the baked anode production was provided by two other anode baking furnaces on Production Line no. 1, where control systems with visual inspection would remain. Anodes rejected, based on the MIREA results, needed to be redirected to a nearby alternative conveyor to be withdrawn from the production.

In order to measure every anode on the second production line, the industrial MIREA needed to be designed to reliably achieve one anode measurement per minute. The measurement sequence includes the following chronological steps: position the baked anode into the device, introduce current devices into the anode stub holes, measure the anode voltage drop, withdraw current devices from stub holes, and remove anode from the MIREA. At the time of this writing, with top production rates, the MIREA is able to complete the measurement sequences in time intervals varying between 60 to 66 seconds. This cycle time measurement was achieved for more than 15 % of the population measured between June and August 2015. At other times, this cadency was not required since the production rate was lower. It must be noted that this measurement cadence can be further improved.

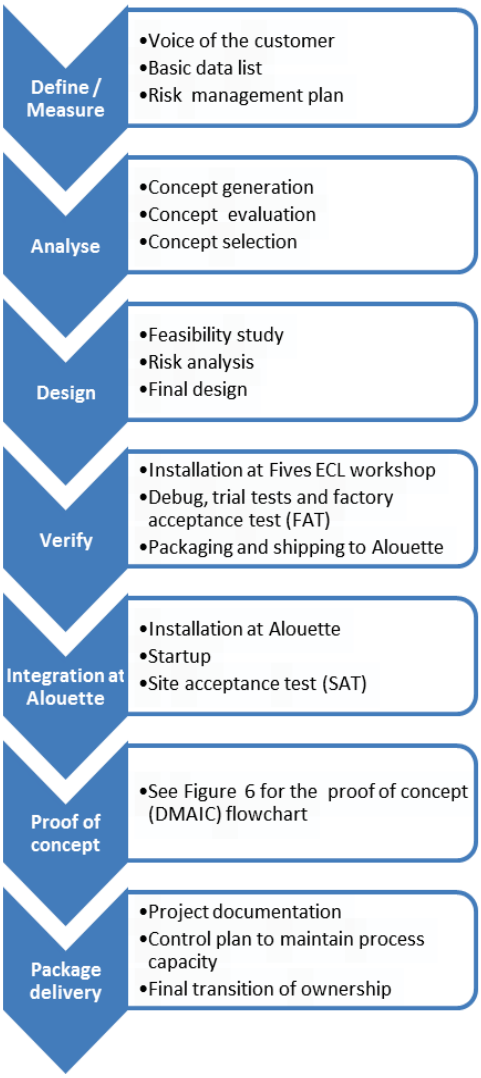

Figure 4. Global project management flowchart.

Since the MIREA was located at the exit of the anode baking furnaces, it was designed to handle baked anodes up to 400 °C.

During start-up and the proof of concept stage, it was found that the MIREA was exposed to anode temperatures varying between 25 and 250 °C, at most. This temperature interval is much lower than the maximal intended limit of the device and provides an important security margin.

As the R&D MIREA device only measured anodes at room temperature, corrections needed to be included in the automation of the industrial MIREA to correct the bias associated with varying anode temperatures. The reference baseline could be set at room temperature, to allow comparison of resistivity measurements with core samples, or set to pot temperature, to provide an estimate of the anodic voltage drop in service.

It should be noted that the MIREA device was completely automated and designed to require limited maintenance. Anodes with excessive irregular dimensions could be automatically bypassed and rejected, if needed. For example, this situation could arise when a thick layer of packing coke (more than 1 cm) is agglomerated to the anode surface.

MIREA assembly (Integration stage)

The installation of the industrial MIREA at Alouette followed a defined method to minimize the disturbance on the baked anode production. The construction followed the normal weekly maintenance shutdown of the plant (8-hour shutdown/week). It required a total of seven shutdowns. The design of MIREA assembly facilitated the installation since the device was designed with independent plug and play modules. One module could be assembled during a shutdown without needing to install the second module during the same shutdown or in the following one. Figure 5 shows the installation of the device frame as an example. Between shutdowns, baked anodes were passing through the MIREA setup without any production disturbances.

Figure 5. Frame assembly steps for the MIREA.

The installation steps included the modification of the baked anode conveyor at the MIREA location and the integration of the different MIREA modules (frame, upper and lower servicing modules, measurement module, alimentation module and control/acquisition box). Work on automation was also required to establish proper communication for the manipulation of the baked anodes between the baked anode conveyor, MIREA device and the alternative conveyor used for rejection purposes. Prior to starting the equipment, protectors compliant with international standards and local regulations were set up in order to ensure safe operation.

MIREA start-up (Integration stage)

Alouette dedicated a Lean Six Sigma black belt team for the start-up of the device. The team was composed of members from the technological development, operation, maintenance and automation teams. Fives ECL was present to provide support for the start-up and offered training for the operation and maintenance teams. Rio Tinto was also present to provide support during the first data analysis.

Alouette and Fives ECL automation personnel insured that the MIREA was in symbioses with the process chain. This required adjusting the automation control scheme since some bugs and problems, often experienced with prototypes, appeared during the initial start-up. These deficiencies were corrected and the start-up was rapidly completed allowing moving to the next stage of the project, which is the proof of concept.

Project stage: proof of concept

The proof of concept plan was designed to assess the capacity of the industrial MIREA as an online measurement system to evaluate the electrical performance of the baked anodes produced by Alouette. Two aspects were covered with this plan: evaluation of the resolution the MIREA and evaluation of the concordance between the MIREA results and the performance of the anodes in the pots. The plan was divided into 5 phases, as illustrated in Figure 6. It describes the goal, required tests and expected results of each phase. This proof of concept plan was partly inspired by the Lean Six Sigma DMAIC improvement process methodology and the ASTM standard guide for measurement system analysis [6].

Measurement system analysis

The tests identified in Figure 6 were required to assess the performance of the MIREA as a measurement system. The following properties were evaluated: the degree of repeatability, stability (variation of the master reference anode batch with time), consistency (change in repeatability with time) and resolution (ability to discriminate between different anodes) of the device.

A reference anode batch (20 anodes) was assembled by Alouette to create the master reference anode batch required to assess the performance of the device. The lot included anodes considered as excellent, fair and unacceptable after visual observation of the anode surface. This reference batch was used several times during the different proof of concept phases. When required, the reference anodes were manually introduced on the baked anode conveyor with a forklift, measured by the MIREA and collected back with a forklift from the alternative conveyor. They were then stored for the next usage.

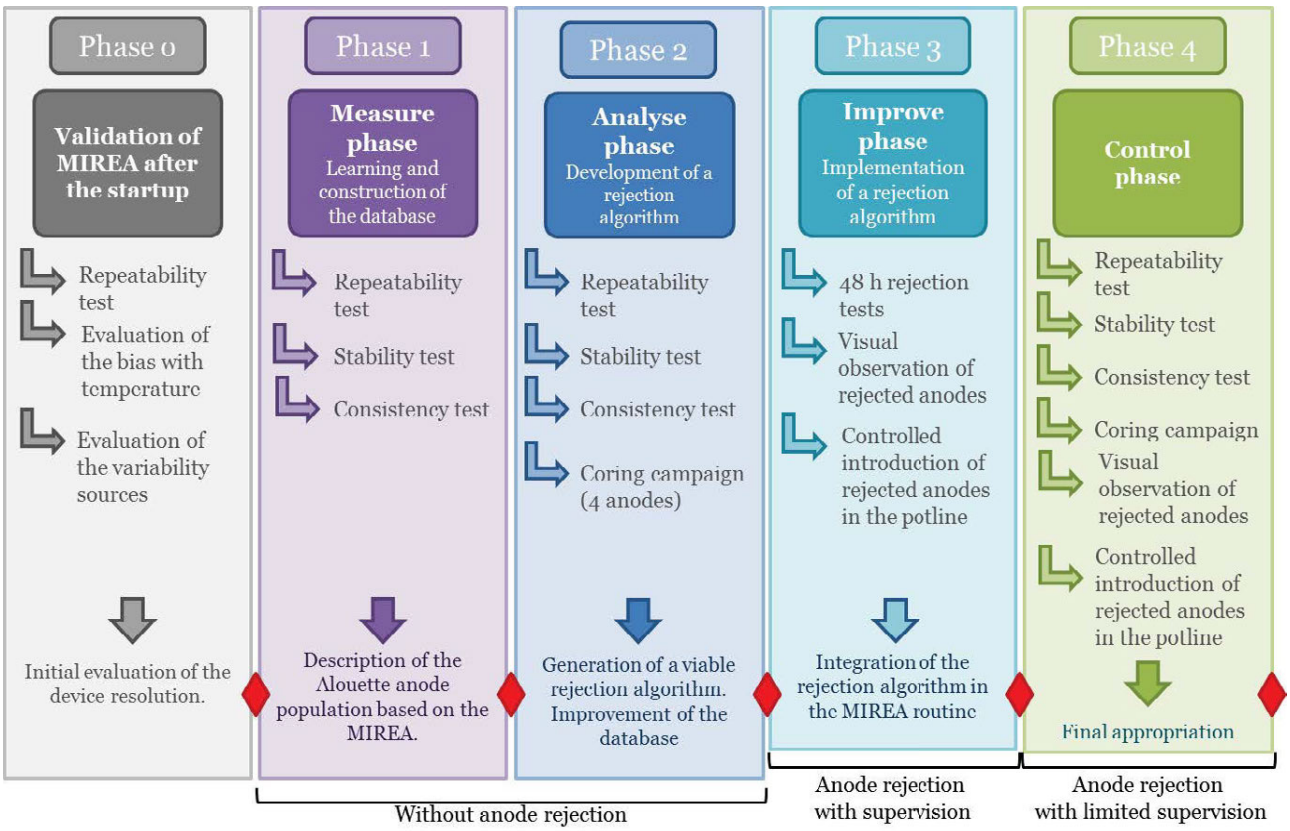

Figure 6. Proof of concept flowchart to assess the capacity of the industrial MIREA as an online measurement system.

Validation phase to the Analysis phase (Phases 0 to 2)

Validation of the MIREA (Phase 0) was performed just after the start-up of the MIREA device to ensure the conformity of the instrument after being sent from the Fives ECL warehouse to Alouette. The completion of this phase was led by the Alouette technical team. This phase was also used as a first initiation for the Alouette operation team to the MIREA device.

In phases 1 and 2, the operation team took partial ownership of the MIREA in order to insure continuous measurement of the baked anode production. This appropriation allowed to measure, in real time, more than 16 000 anodes during phases 1 and 2. Meanwhile, Alouette continued to reject anodes using the visual inspection systems on both process lines.

This extensive number of MIREA measures was used to build a MIREA database to characterize the Alouette population. During Phase 2, this database was also used to develop parameters for the rejection algorithm. This important source of information allowed evaluating the potential impact of the rejection parameters on the baked anode rejection rate.

Furthermore, during this phase, four anodes were cored for a qualitative comparison with the MIREA results. These anodes were initially measured with the MIREA device and discarded from the production to perform a resistivity coring campaign. Four cores were collected from each anode and were located near each MIREA measurement columns. This test showed a good correspondence between the real time MIREA measures and the coring campaign results [3].

Next phases for the full implementation (Phases 3 to 4)

At the time of this writing, Alouette was planning to start the third phase of the proof of concept. It consists of rejecting anodes based solely on the MIREA measurement for 48 hours with the proposed rejection parameters issued during Phase 2. This test will be carried out with a certain level of supervision to ensure proper production during this phase. Thus, the control systems, based on visual inspections of the baked anode surface, will be deactivated on Production Line no. 2 during this test.

Rejected anodes during this 48-hour test will be visually inspected and cored to qualitatively evaluate the efficiency of the MIREA. In addition, some of the rejected anodes will be introduced in pots to assess the relationship between the performance of the anode in service and its MIREA resistance characteristics. Using this knowledge, the rejection algorithm will be reviewed and improved, if required. The success of this phase will lead to the final control phase where the MIREA will continue to reject anodes in solo with minimal supervision. A quality control audit will be planned to insure maintaining the performance observed during the previous phase. The project will end after this control phase, and the complete ownership of the MIREA will be transferred to the Alouette operation team.

Illustration of typical MIREA results

Figure 7 shows an example of typical MIREA results illustrating the discrimination ability of this device to identify unacceptable baked anodes from the production. It shows the apparent resistivity fingerprints collected by the MIREA device for an

arbitrary selection of anodes considered as “mediocre” (red) and “fair” (blue). The “average” fingerprint of the population (more than 10 000 anodes) is also included for comparison purposes (black).

These fingerprints present the apparent resistivity according to different sliced sections of an anode (S_i). Each point represents an average of the measurements at the specified anode position level (H_i) for the four measurement columns. This average treatment hides some variability between columns; however, it allows a simple visual representation of the anode general behavior, which can then be compared to other anodes. More complex comparison techniques can also be used [3]. The results presented in Figure 7 are normalized to a temperature of 25 °C.

As it can be seen in Figure 7, “mediocre” anodes deviate from the “fair” anodes and the averaged fingerprint of the population. These deviations indicate the presence of anode defects at the corresponding position level. It must be noted that the presence of a very low resistivity value, such as the one highlighted in Figure 7, is not associated to a good resistivity value, but to an anodic defect, which disrupts the current intensity in this layer. The current distribution of the actual anode is not equivalent to the simulated case for this section. It is believed, in this case, that there is a preferred pathway that does not pass in the vicinity of the four “mV” probes on Slice S_2, thereby reducing the current intensity in this specific region. However, the presence of these deviations in an anode, from normality, clearly highlights them as problematic anodes.

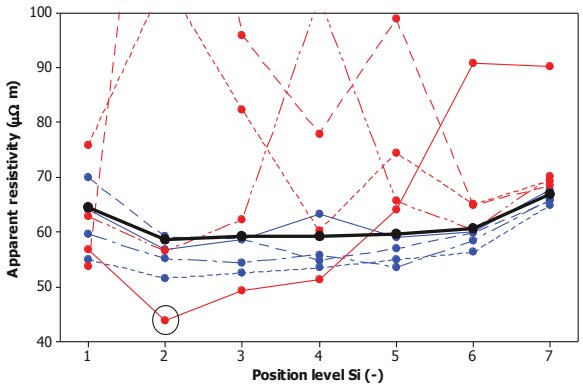

Figure 7. Anode apparent resistivity profile for a selection of individual anodes. In blue: fair anodes; in red: mediocre anodes; in black: population average.

Conclusions

This paper outlines the important milestones of the industrial MIREA development from its initial conception, its integration in Alouette process chain, the start-up and the initial stages of the proof of concept. The MIREA is now operated in real time in an industrial environment. At the time of this writing, some major milestones still remain such as the commissioning of the rejection algorithm and the final ownership by Alouette operation team. However, preliminary results confirm the efficiency of the MIREA to reject unacceptable anodes in real time [3]. Furthermore, it must be noted that the Alouette smelter is on track to fully characterize the electrical performance of their anode population. This achievement represents a major breakthrough in baked anode quality control.

Acknowledgements

The authors wish to acknowledge the numerous contributors of the Rio Tinto Reduction R&D team for their ingenuity and tenacity, which resulted in the development of the MIREA device. Finally, the precious help of Ms. Sandras de Moor in reviewing this paper is greatly appreciated.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

JOURNEY TOWARDS WORLD-CLASS OPERATIONAL EFFECTIVENESS AT DUBAL (EGA JEBEL ALI) PASTE PLANT

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Keywords: Paste plant, Production capacity, Total effective equipment performance, Availability, Loading effectiveness

Abstract

Over the last decade, metal production at DUBAL (also known as EGA Jebel Ali) has increased by 358 000 tonnes, through expansion and capacity creep, resulting in the plant's anode requirement exceeding the Paste Plant capacity. This increase was supported by anode procurement at premium from the global market. To handle these challenges, DUBAL Paste Plant embarked on a journey towards best-in-class operational effectiveness where paste plant throughput was increased and the conventional maintenance systems were revolutionized. The Plant throughput was increased by 9 % above the design capacity and an improvement of 19 % in Total Effective Equipment Performance (TEEP) was achieved, while continuously improving anode quality. As a result, anode procurement was reduced by more than 85 000 anodes per year, reducing dependency of the smelter on purchased anodes and resulting in significant financial benefits.

Introduction

Since its inception in 1979 with a production capacity of 150 000 tonnes per year, EGA Jebel Ali Operations ("DUBAL") has grown into one of the largest single site smelters with a production capacity of 1 039 000 tonnes per year at the end of 2014. Currently, DUBAL has seven potlines with 1 573 cells operating at amperages ranging from 203 kA to 455 kA.

The increase in metal production resulted in the annual demand for baked anodes increasing from 115 000 to 465 000 anodes. The size of anodes has also been increased in stages to cope with higher line current. Presently, DUBAL Carbon Plant can produce baked anodes with dimensions of 1585 × 805 × 645 mm (3 stub holes) which was initially 1130 × 795 × 635 mm (2 stub holes).

Expansions in DUBAL's aluminium production capacity and continuous capacity creep through amperage increase over the past ten years (Figure 1) resulted in the anode demand of the smelter exceeding the Paste Plant capacity by up to 139 000 anodes in reference to the paste plant production capacity in 2004 (Figure 2). These anodes were purchased from the international market at a premium.

With the installation of new potlines and in-house technology developments, anode dimensions and types have also increased. Thus, the Paste Plant has to produce multiple sizes of anodes to cater for four out of the six different cell technologies at DUBAL (anodes for DUBAL's latest cell technologies, DX and DX+, cannot be produced in house due to dimensional constraints).

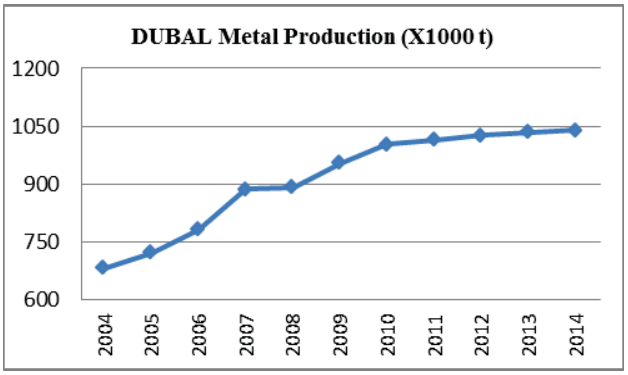

Figure 1. DUBAL Metal production over past 10 years.

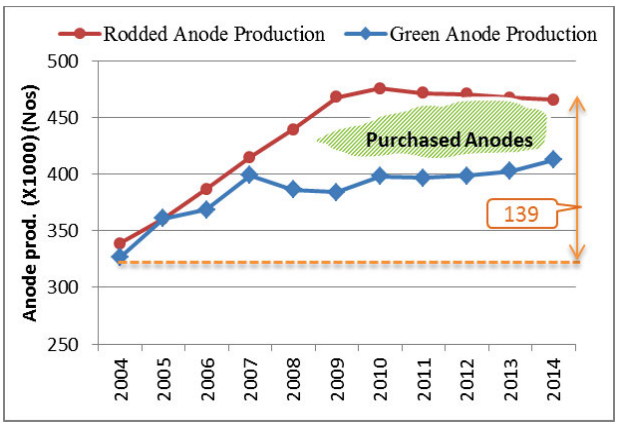

Figure 2. Dubal rodded anode and green anode production.

To reduce anode procurement and provide the required anode size and types to the smelter, anode production had to be increase by increasing the plant throughput and improving plant efficiency. This paper describes the initiatives undertaken at DUBAL to increase the throughput and achieve world-class paste plant efficiency over past 10 years using 2004 performance as the baseline.

DUBAL Paste Plant

The first Paste Plant at DUBAL was installed in 1979 with a designed capacity of 33 t/h. In 1996, a new mixing line was added to bring the capacity of the plant to 66 t/h with an upgrade of the common screening and crushing system to accommodate both lines. Figure 3 indicates the various upgrades and modifications undertaken at DUBAL paste plant to meet the smelter operational needs. Currently, DUBAL Paste Plant operates at 72 t/h, with a conventional three-fraction anode recipe.

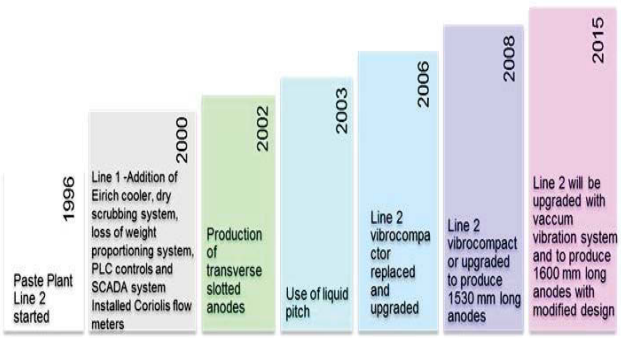

Figure 3. Paste plant major modifications / upgrades.

Paste Plant Throughput Increase from 66 t/h to 72 t/h

In late 2004, to absorb the raw material deterioration and explore the capability of the existing plant, DUBAL embarked on an anode quality improvement programme, with R&D Carbon using Dynamic Pitch Optimisation (DPO)[1]. The anode recipe was modified and the following adjustments were carried out to maintain anode quality:

- Mixing temperature was increased by 10º C.
- Mixer parameters (power and speed) were adjusted to maintain consistent mixing energy.
- Fines Blaine Index was increased from 2500 to 4000

The trial helped to reduce the pitch content in anodes by 0.7 % without compromising the anode quality; and in addition increased throughput by 6 % (i.e. 66 t/h to 70 t/h) over the nominal level. Further increase of the throughput at that stage was limited by the capacity of fines and medium fractions proportioning units and compactor cycle time.

In 2007, the capacity of the proportioning units was increased, paving the way for further throughput increase. The compactor cycle time was optimised and anode size increase was increased, thereby enabling another 2 t/h throughput increase. The throughput of the Paste Plant was therefore increased by 9% beyond the design limit, resulting in 28 600 tonnes of additional good paste production per year. Figure 4 shows the throughput of the paste Plant and equivalent additional paste production from 2004 to 2014.

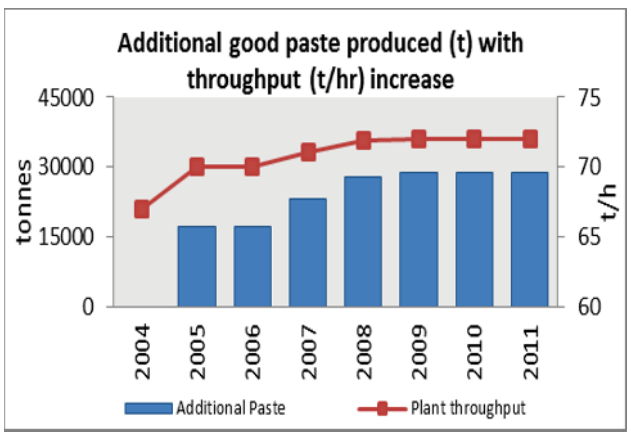

Figure 4. Additional paste production with increase of throughput.

Increasing Total Effective Equipment Performance (TEEP)

Key Terminologies

Total Effective Equipment Performance (or TEEP) and Overall Equipment Effectiveness (or OEE) are two closely related measurements that report the overall utilisation of facilities, time and material for manufacturing operations. As the calculation methods for these indicators may vary from one plant to another, it is important to explain the method of calculation and the terminology used in this article. At a global level, OEE and TEEP can be defined as follows:

- OEE = Good products output time / Scheduled time
- TEEP = Good products output time / Calendar time

Figure 5 below shows the break-up of the calculation of these indicators.

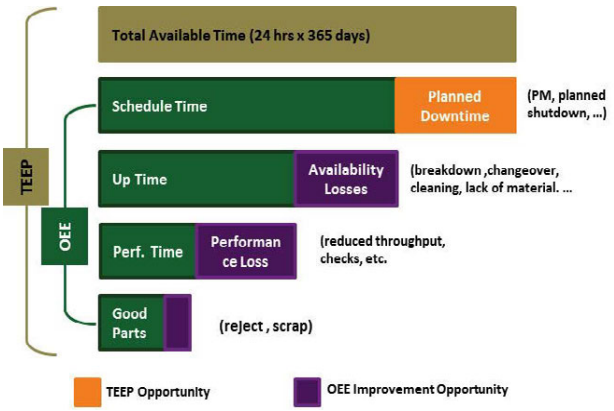

Figure 5. OEE and TEEP calculation description.

Overall Equipment Effectiveness is a combination of three metrics - Availability, performance and quality.
OEE = Performance x Availability x Quality [2] where:

- Availability is the percentage of scheduled time which the plant is ready to operate. In a Paste Plant, the availability is generally an indicator of frequency and duration of equipment breakdowns.
- Performance is a comparison between the actual production speed and the design speed. Typically, in a Paste Plant, it indicates operating at lower throughput than the design throughput due to equipment performance or idle time.
- Quality represents the percentage of good quality products out of the total products manufactured.

TEEP adds a fourth metric generally known as "Loading Effectiveness", which is the percentage of the calendar time that the plant is scheduled to operate. In other words TEEP = OEE x Loading Effectiveness.

When a plant is required to operate continuously 24 hours, 7 days a week to meet the demand, TEEP is a more appropriate indicator as it reports the overall utilisation of the facility. Therefore, DUBAL uses TEEP as an indicator of its Paste Plant efficiency.

Key Initiatives

To increase the anode production and optimise the anode procurement, DUBAL implemented various initiatives over a period of ten years to improve the TEEP of the Paste Plant by addressing the contributing components.

The key initiatives were:

- Increasing plant availability,
- Reducing rejection rate,
- Increasing the Loading Effectiveness.

The "**Performance**" component of TEEP was not addressed since both paste plants were operating at their design capacities.

Increasing Plant Availability

Availability is a vital factor in improving TEEP. The DUBAL Paste Plant has successfully improved and maintained its availability at 90%, and achieved a record performance of 92% availability in 2014 (Figure 6). To attain and sustain high plant availability, several maintenance improvement programmes have been implemented over the period. Some of the most significant are discussed below.

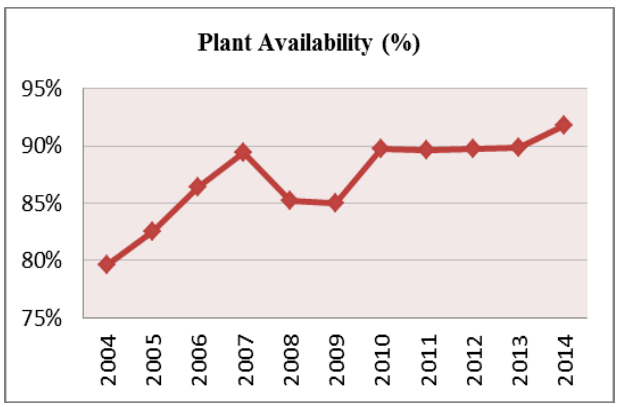

Figure 6. Plant availability.

Breakdown Analysis

The Paste Plant maintenance and operation teams developed and established the process of routine breakdown analysis using process improvement tools like Pareto and Cause and effect (6M analysis) illustrated by the example given in Figure 7.

The findings of such analysis are discussed during routine communication forums:

- **Daily Meetings** between operation supervisor, maintenance supervisors and planner: to analyze breakdowns of the past 24 hours, review actions taken by the shift technicians and identify additional corrective and preventive actions.
- **Weekly review meetings** between operation and maintenance superintendents and supervisors: to review the progress of actions from the daily meeting and identify opportunities for efficiency improvement.

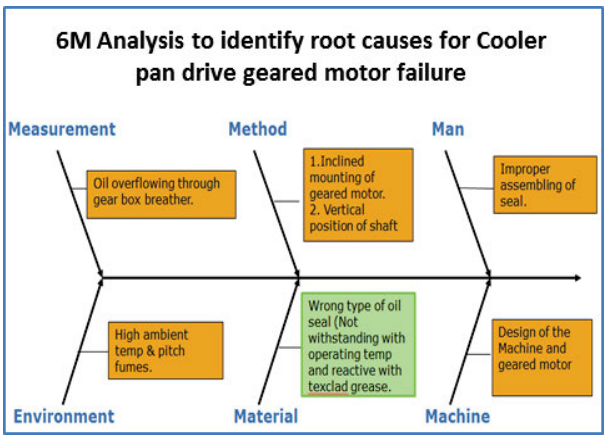

Figure 7. Breakdown analysis using 6M tool.

Another benefit of the routine meetings was reinforcement of team work and communication between operation and maintenance, which was the fundamental for achieving high performance.

Reliability Improvement Programme

To improve the availability of critical Paste Plant equipment such as paste kneader, paste cooler, anode compactor, etc., DUBAL initiated extensive reliability improvement programmes on these machines. The programmes incorporated all aspects of equipment maintainability. The following were some of the key aspects:

- Real time equipment performance monitoring,
- Root Cause Failure Analysis (RCFA),
- Optimisation of maintenance frequencies and review of checklists,
- Implementing condition monitoring and condition based maintenance,
- Using visual maintenance aids and early system failure detection tools,
- Improving the availability of maintenance manuals, maintenance procedures, drawings and critical spare parts.

Suggestion Scheme

DUBAL has been running a very successful suggestion scheme for more than three decades. Through this scheme, the operators propose improvements or solutions to existing problems. After review and approval by area management, the idea is implemented and the operator is rewarded.

Several recurrent breakdowns and equipment design issues have been solved through suggestions, contributing to the improvement of the plant availability. Some examples of such suggestions were:

- Modification of the holding plate of shaking table support pins with a saving of 48 hours breakdowns per year,
- Installation of guide plates on anode turn table at exit of compactor which resulted in 60 hours breakdown savings yearly.

Reducing Total Rejection

The total rejection in a Paste Plant has two components: block rejection and loose paste rejection. These two components are distinguished in two categories:

- Start-up rejection which occurs during production start,
- Steady state rejection which occurs during continuous production.

It was observed through statistical data analysis that if the plant operation is stable, the percentage of Steady state rejection is relatively small compared to start-up rejection. Thus, the focus at DUBAL was to reduce start-up rejection.

Reducing start-up rejection mainly involved reducing the frequency of breakdowns and optimising the production start and stop sequences. The improvement of plant availability described above has helped to improve start-up rejection, but two main initiatives described below have contributed the most in achieving the total rejection of around 2 %.(as shown in Figure 8).

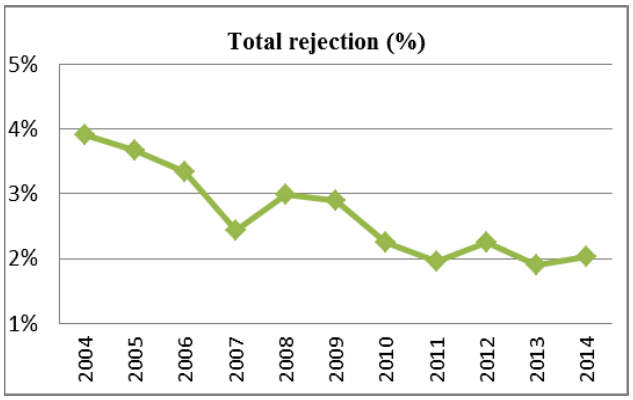

Figure 8. Total paste and anode rejection.

Reduced Duration from Start to First Good Quality Anode from 20 Minutes to 14 Minutes

This was achieved through detailed analysis of different start-up parameters to determine a target rejection time after plant start-up (Figure 9) and optimisation of start-up parameters (kneader load ramp-up, temperature setting and adjustments, start-up pitch percentage).

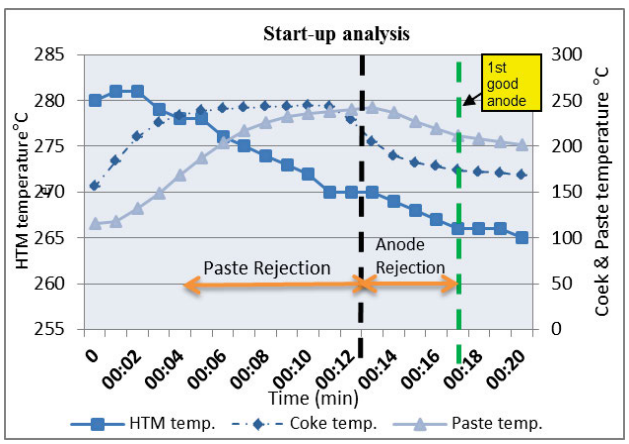

Figure 9. Plant start–up analysis.

Implementation of Compactor Simulation Programme for Start-up

Following any major issue, or after a prolonged stoppage of the anode compactor, it is often difficult in local or maintenance mode to be sure that all interlocks and sequences of the compactor are ready for operation in auto mode. A significant amount of paste was rejected frequently whenever the compactor sequence failed to run while the paste was ready for production. The programme developed in-house allows the operator to simulate several compactor cycles without material. After implementation of new program, the production starts only when the compactor has been tested and ready to operate, reducing paste rejection while adjusting the compactor startup.

Increasing Loading Effectiveness

The only means to increase loading effectiveness, was to reduce the duration of total annual planned maintenance shutdowns. Reducing planned maintenance shutdown time without prerequisite study and supporting actions can lead to reduced availability in the long-term. In the first few years from 1979 onwards, the Paste Plant was operated at 28 t/h, five days a week with three weeks of operation and one week of maintenance. In 1990, with the increase in anode demand, the Paste Plant started operating 24-hour, seven days per week.

DUBAL paste plant started to treat the OEM recommended maintenance frequencies and equipment replacement rates as recommendations only, and challenged it by conducting controlled trials wherever possible. As a result, DUBAL paste plant has gradually reduced the total numbers of planned shutdowns per year from 50 to 25 days. (Figure10).

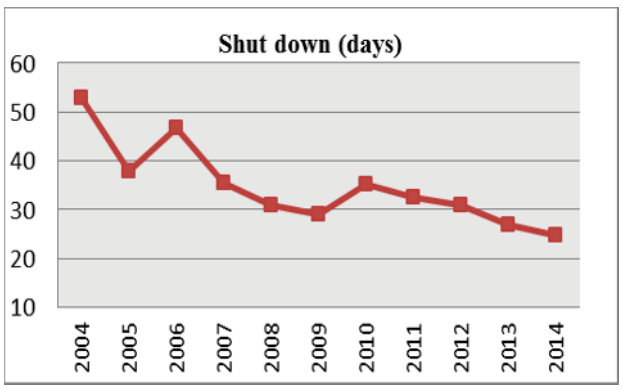

Figure 10. Total shutdown days per year.

To support this change without affecting the plant availability, several initiatives have been implemented such as:

- Opportunity repair and maintenance,
- Condition monitoring of crucial equipment,
- Reduction of mould change frequency and duration through effective production planning and reducing the number of anode types in coordination with potrooms,
- Daily equipment inspection by assigned technicians.

The initiative of “opportunity repair” had the highest impact in TEEP improvement in the initial years. As a percentage of planned maintenance activities were being performed during

"opportunities", the DUBAL Paste Plant was able to reduce the duration and frequency of planned maintenance shutdowns. The basics principles of this type of maintenance are:

- Maintaining a list of maintenance activities based on routine planned maintenance, breakdown analyses and observations from daily inspections, to be performed whenever there is an opportunity (breakdowns, unplanned stoppage etc.),
- Assigning the activities to shift maintenance technicians with required spare parts readily available,
- Whenever an opportunity arises, the technician (based on manning availability) coordinates with the operation supervisor to perform the maintenance activity.

In recent, live condition monitoring has ensured effective planning, resulting in optimum shutdowns. All these initiatives have resulted in the loading effectiveness of the Paste Plant increasing from 86 to 93% (Figure 11).

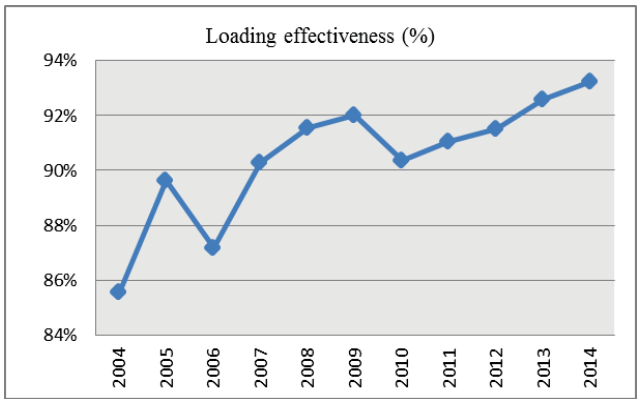

Figure 11. Plant loading effectiveness.

Results and Discussion

Throughput Increase

While conducting DPO, the DUBAL Paste Plant throughput increased by 6 % from 66 t/h to 70 t/h and was further increased by 3 % from 70 t/h to 72 t/h in 2007. This resulted in an overall annual increase of 28 600 tonnes of good paste production.

TEEP Achievement

DUBAL implemented various initiatives stated in this paper to improve the TEEP of the Paste Plant by 19 % from 65 % to 84 % (Figure 12), which has increased the good paste production by 115 600 tonnes per year compared to that in 2004 and can be considered as world benchmark performance. The increased efficiency through various components is shown in Figure 13.

To summarize, the DUBAL Paste Plant has through, optimisation of capacity utilisation, increased plant availability and reduced rejection rate. This has increased its production by more than 85 000 anodes (Figure 14) per year, reduced the equivalent dependency of the smelter towards purchased anodes with a significant financial benefit due to the cost differential between DUBAL and purchased anodes.

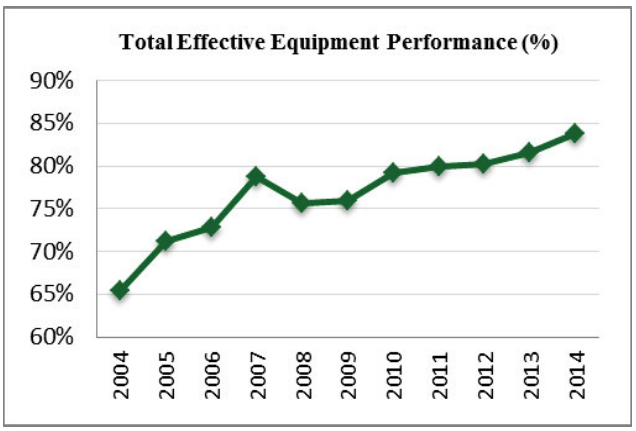

Figure 12. Paste Plant TEEP %.

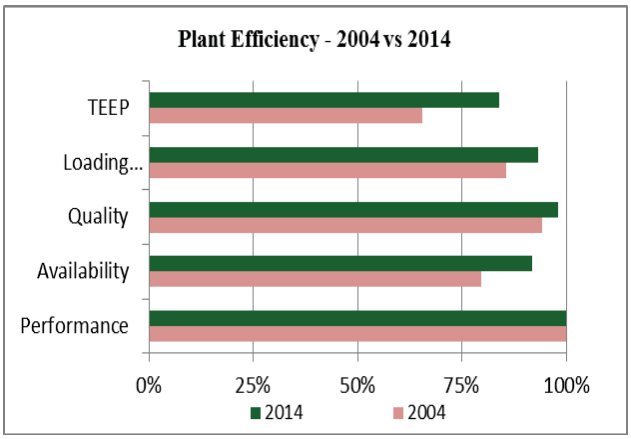

Figure 13. TEEP and its components 2004 vs 2014.

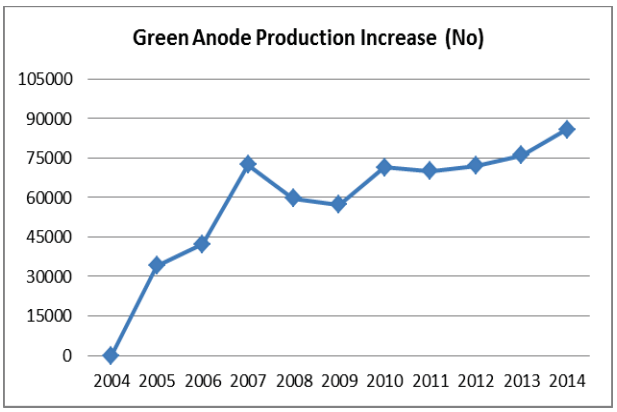

Figure 14. Green anode production increase.

The other noteworthy point of this capacity creep journey is the fact that neither the operational safety nor anode quality has been impacted negatively. On the contrary, the quality of anode has been gradually improved [4] while the plant has achieved a record of 18 years without lost time injury.

Conclusions

The DUBAL Paste Plant has succeeded in achieving world-class efficiency through the use of well-established problem solving and efficiency improvement tools, extensively supported with process observations, data and facts. Use of concepts like

opportunity repair and condition based maintenance have challenged the historical approach of accepting mediocre plant availabilities as the standard. The availability and easy access to long-term and detailed breakdown records, production data and maintenance documentation were the fundamental to this efficiency improvement drive.

Higher loading effectiveness was achieved by challenging the OEM recommended equipment replacement and maintenance frequencies by conducting controlled trials wherever possible. The exercise has yielded fewer planned maintenance shutdowns, reduced maintenance cost and more stable plant operation, since the plant would have longer durations of uninterrupted production.

Finally, the significance of nurturing a culture of continuous improvement within any organisation cannot be over stated. DUBAL's award winning Suggestion Scheme has shown that even the most complex plant and equipment related problems may be solved with surprising simplicity and creativity if people who are closest to the shop floor are encouraged to participate in finding the solutions.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

THE STARTUP AND THE OPERATION PERFORMANCE OF THE TWIN GREEN ANODE PLANT AT MAADEN ALUMINIUM SMELTER IN SAUDI ARABIA

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Keywords: Green anode plant, Paste plant, Ma'aden, Alcoa, Fives, Intensive Mixing, Vibrocompactor, Dry mix preparation, Safety

Abstract

MA'ADEN, Saudi Arabia's government owned mineral resources company, and ALCOA, the world's leading producer of Aluminium, signed a JV (Joint Venture) agreement in December 2009 to develop the largest integrated Aluminium complex in Saudi Arabia on Arabian Gulf coast at Ras Al-Khair industrial zone.

Aluminium smelter capacity of MA'ADEN is 740,000 mtpy of hot metal. With local bauxite mining capacity of 4,000,000 mtpy, alumina refinery capacity of 2,000,000 mtpy and rolling mill capacity 380,000 mtpy, MA'ADEN is the world largest integrated Aluminium complex.

The carbon anode manufacturing of this complex consists of two 40 tph green anode plants using Rhodax dry mix preparation and intensive mixing cascade technologies and one liquid pitch terminal with 2x6,000 tonnes storage tanks.

This paper covers the design concept of the green anode plants and its operation performance since its inception in 2011 with the focus being on the commissioning and start-up management as well as the technical challenges encountered.

Introduction

FIVES was awarded by MA'ADEN Aluminium Company (MA) the turn key supply of several industrial facilities such as a green anode plant (GAP), a carbon recycling unit, a bath treatment plant (BTP) and a liquid pitch terminal (LPT) at MA industrial port.

The 740,000 mtpy pot lines required approximately 450,000 mtpy of green anode production which corresponds to 80 tph green anode plant capacity.

The highest single line green anode plant capacity is 60 tph at Qatalum. Today, there is no available technology allowing a capacity as high as 80 tph with one single line. It was, therefore, decided to go with two parallel production lines of 40 tph capacity each. Moreover, producing anodes with two parallel lines brings more operation and maintenance flexibility, consequently, lowers the risk of anode shortage during unscheduled stops.

The green anode plant is featured with Fives proven technology which is already considered as a reference for capacity range from 35 tph (Alba, Sohar, Mahan) [1] and up to 60 tph (Qatalum) [2]. It includes:

- The Rhodax® process for dry material preparation,
- The IMC® process (Intensive Mixing Cascade) for the paste preparation lines,
- The Xelios vibrocompactor for the anode forming [3],
- The Amelios software for the green anode plant performance monitoring [4]

Ma'aden Green Anode Plant Design Concept

The green anode plant (Figures 1 and 2) includes one carbon recycling unit, the coke and scrap storage silos, one coke and scrap feeding tower, one paste plant building with two twin green anode production lines followed by two twin cooling tunnels, all the HTM and Pitch facilities and the substation.

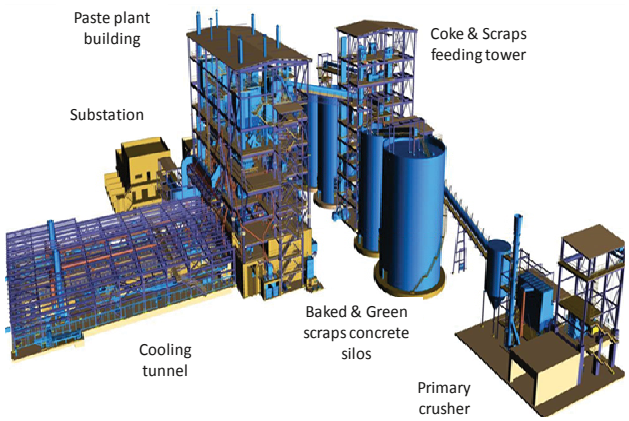


Figure 1. Ma'aden Green Anode Plant Birdview

Some key figures of the GAP construction are:

- Civil works = 10,000 m^3 concrete,
- Structural steel = 3,200 tons
- Mechanical equipment = 750 tons & 9,200 lm of piping
- E & I = 260,000 lm of cables

Figure 2. MA'ADEN Green Anode Plant

The raw material storage consists of:

- Two 600 t fresh calcined coke silos,
- One 1800 t baked scrap silo,
- One 500 t green scrap silo,
- And one 300 t pitch daily storage tank.

The 40 tph carbon recycling unit consists of one primary crushing unit with a MR15 jaw crusher fed by green or baked scraps followed by a secondary crushing with a RC1200 roll crusher producing a final product with a maximum top particle size of 80 mm. It also includes an eddy's current separator to remove aluminium scraps and thus prevent from disturbing downstream comminution process.

One single dry material proportioning line is used to mix different cokes as well as green and baked scrap fractions before sending them to two parallel green anode production lines.

Each green anode production line consists of one dry aggregate preparation section (Figure 3) based on the Rhodax® process including:

- one variable gap and speed Rhodax® crusher RX600 (Figure 4)
- two variable speed TSV 1600 dynamic classifiers
- one 2,6 m x 6,3 m variable speed and fully air swept ball mill

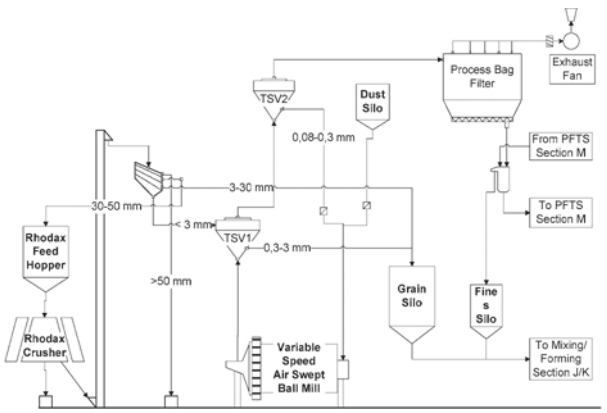


Figure 3. Dry Aggregate Preparation Line

The dry aggregate final product consists of two fractions only – 80% coarse particles (0.3 to 30 mm) and 20% fines (70% < 32 µm ⇔ 6,000 BLAINE) with a ratio Grains/Sands (+300 µm / [30-300 µm]) above 5. The fractions are stored in two buffer silos equipped with loss-in-weight dosimeters.

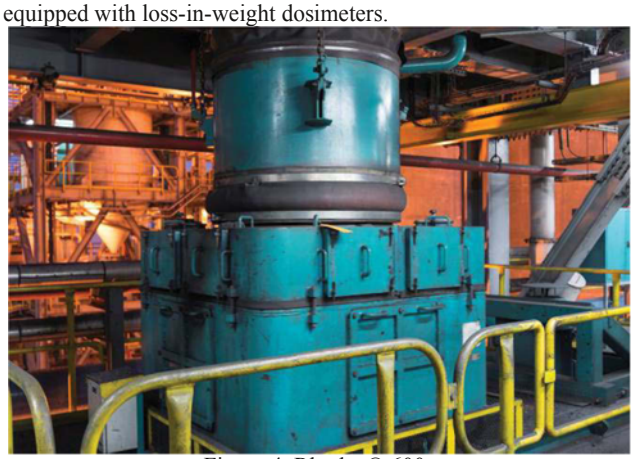


Figure 4. Rhodax® 600

The weighed dry mix is then preheated in a preheating unit with four hollow shaft screws which are 800 mm in diameter by 6,5 m in length. The preheated dry mix is then fed along with liquid pitch into the IMC® (Intensive Mixing Cascade is shown in Figure 5).

The mixed and cooled paste is then fed into a weighing hopper and dumped into a transfer hopper to feed two Xelios vibrocompactor moulds alternatively. The vibrocompactors are equipped with counter pressure and a vacuum unit and produce up to 60 anodes per hour, each anode weighing 1060 kg.

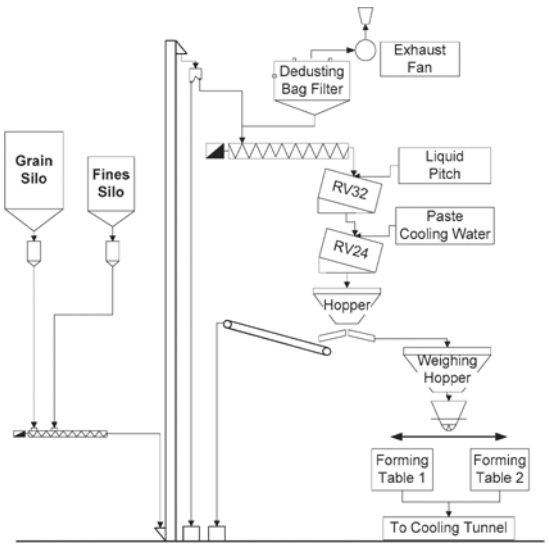


Figure 5. Paste Mixing and Anode Forming Line

The formed green anode blocks are then cooled down below 70°C by passing them through a water spraying cooling tunnel in less than 75 min of residence time.

Aligned with world class standards, the stack emission limits are stringent. Several dust collection systems have been designed to maintain an optimum working environment and to limit dust stack emissions to less than 10 mg/Nm^3. A pitch fume treatment system using a dry scrubber has been designed to limit the condensed tar emission levels to less than 5 mg/Nm^3.

Table I compares the size of key pieces of equipment required for a 40 tph and a 60 tph capacity modern green anode plant.

Table I. Equipment Size vs GAP Capacity

Key Pieces of Equipment	**GAP Capacity 40 tph**	**GAP Capacity 60 tph**
Rhodax® crusher	RX600 25 – 40 tph (1)	RX1100 40 – 80 tph (1)
TSV	TBF & HF1600	TBF & HF1800
Ball Mill	Diam. 2.6 m	Diam. 3.0 m
Preheating screw	Quatro 8.65 Four 800 mm x 6.5 m screws	Quatro 8.90 Four 800 mm x 9.0 m screws
IMC®	1 RV32 boosted 1 RV24 boosted	2 RV32 boosted
Xelios Vibrocompactor	2 tables	2 tables
Cooling tunnel	75 min	75 min
Pitch fume treatment system	Dry Scrubber only	Dry Scrubber + RTO

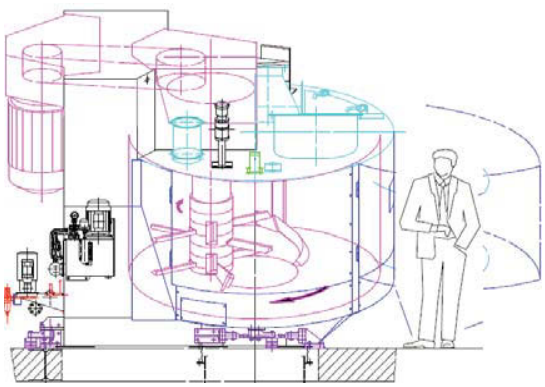
(1) Paste equivalent throughput

All key pieces of process equipment, Rhodax® crusher, TSV dynamic classifiers, Ball Mill, IMC® preheating screws and mixers, and Xelios formers selected for this application have been installed and operating in other Fives plants with similar capacities. Therefore, the expected plant process performance was known with a high degree of confidence.

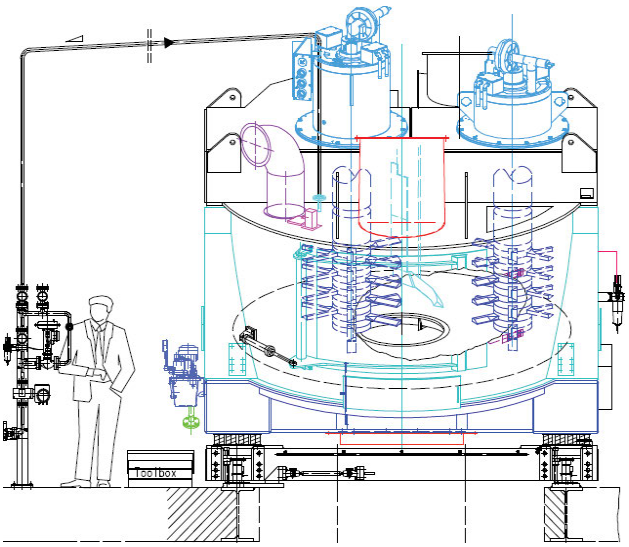
For IMC®, the mixing principle remains unchanged and due to the "intermediate" capacity range of each line (40 tph), it consists of an original combination of:

- one bi-rotor and continuous RV32 mixer with 410 kW installed power distributed over two variable speed high torque motors,
- and one single rotor RV24 mixer-cooler equipped with one 250 kW motor

as shown in Figures 6a and 6b



(a)



(b)

Figure 6. (a) EIRICH RV24 remixer-cooler,
(b) EIRICH RV32 mixer *(by courtesy of EIRICH)*

Xelios Vibrocompactor model, shown in Figure 7, benefits from the return on experience from the highly demanding Qatalum site (60 tph single line). Several mechanical improvements such as new design of suspension rod swivel joint, cover lifting and counter pressure bellows were implemented to this vibrocompactor and have shown a significantly improved reliability since production start-up [7].

Figure 7. Xelios Vibrocompactor

Furthermore, Amelios software (Figure 8) was implemented just before plant start-up to monitor the process and equipment, and to estimate the quality of the anodes produced from ramp-up to contractual performance tests. This system also guaranteed MA'ADEN to have a full snapshot of the green anode plant at the final handover of ownership.

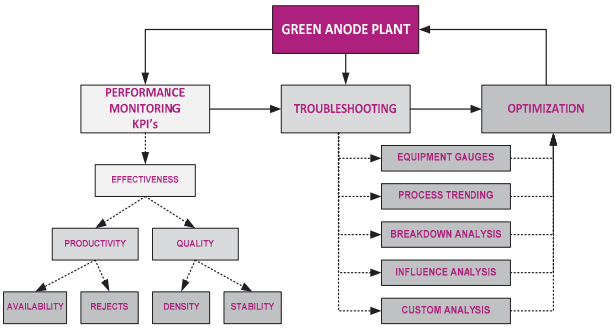


Figure 8. Amelios – System work flow

Commissioning & Start-up

A key factor of success was the organization implemented among all parties, which are:

- FIVES commissioning team,
- MA operations and maintenance team,
- ALCOA Commissioning and start-up team (C&S).

The biggest challenge of this twin GAP commissioning was to supply the required anode quantity with the required anode quality to feed the reduction ramp-up and to prevent excessive anode production that could not have been handled in the anode storage facilities. It was therefore decided to commission line 2 after line 1 instead of commissioning the two lines simultaneously.

Line 1 produced its first anode on the 17th of August 2012 and successfully completed the performance tests on the 29th of January 2013.

Line 2 was commissioned just after Line 1 with a first anode produced on the 8th of April 2013 and the final performance tests on the two lines together were completed on the 3rd of July 2013.

Another major challenge was to actually build the first smelter complex in Saudi Arabia with a fresh work force with no experience in the aluminium Industry and gradually building a knowledge base. A lot of training was required to bring the team members up to speed with the technology and the carbon anode manufacturing aspects.

Health and Safety management during the commissioning

As a common strongest shared value, the Health and Safety has been the top priority during the commissioning both for MA and ALCOA.

Commissioning, by nature is a transition stage between construction activities and the regular operation, with strong interfaces between construction, pre-commissioning testing and operations, including all the other sectors of the smelter.

FIVES has the ownership of its systems and implemented the safety procedures to manage safely these activities and interactions with all the parties.

On the entire FIVES Project scope, from the construction to the commissioning phase, **a total 6,429,779 h without a lost time injury** have been achieved including **4,361,093 h** dedicated for the green anode plants only.

As shown in Figure 9, during the entire project, the period covered by commissioning the recorded incident typology classification was as follows (for all GAP, BTP and LPT):

- 62 recorded incidents during the all project on all activities,
- 26 of them occurred while commissioning activities had started,
- 9 incidents related to commissioning activities.

The 26 incidents that occurred during commissioning had the typology shown in Figure 9.

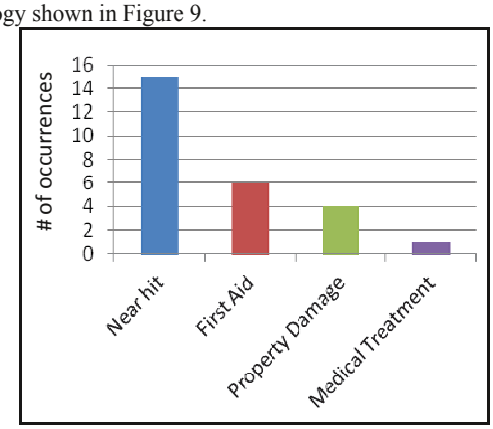


Figure 9. Incident Typology

Commissioning and Start-up Lines 1 and 2

The technical feedback from Line 1 commissioning was collected and implemented on line 2 resulting in the successful performance testing of line 2 in less than 3 months, to be compared to the 6 months for the successful commissioning of line 1.

To reach this, the feedback collection was organized through an operational quality driven system, collecting on a daily basis:

- Feedback Record Sheets (FRS) used to collect and record all improvements identified during the project: 136 FRS established in all domains as shown in Figure 10.

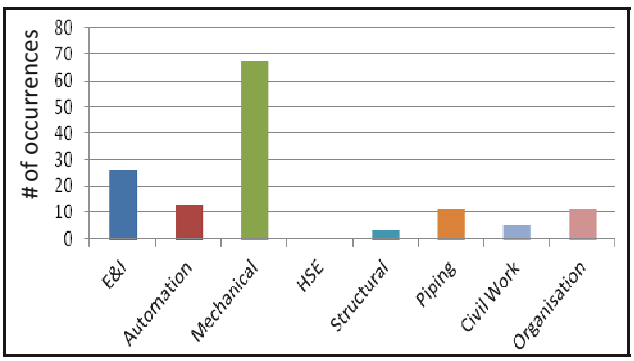


Figure 10. FRS Typology

- Non conformances (NC) used when a task, an equipment or a process function does not reach or does not comply with engineering requirement (Figure 11).

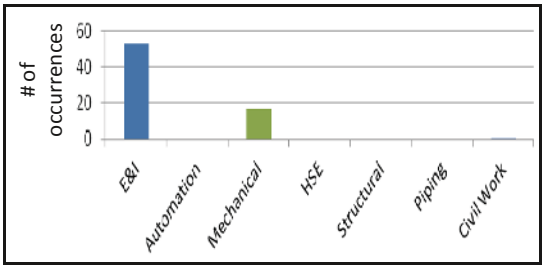


Figure 11. NC Typology

- A root cause analysis (RCA) was systematically done on all events with a high occurrence or gravity rate, all RCA were implemented commonly between Fives and MA.

Commissioning progress and follow-up with Amelios:

The use of Amelios during the plant's ramp-up allowed the commissioning teams to decline the contractual performance requirements on a KPIs daily basis follow-up.

During the commissioning, the teams could also progressively rely on the data provided by the Keops supplied, Carbon –MES to provide the production reports and the plant performance indicator.

Availability rates, anode quality, process deviations and breakdowns were followed-up on a daily basis, constituting a strong decision making support to identify the priorities to implement during the commissioning phase.

Green Anode Plant Operation and Performance

The first anode was produced in green mill line 1 in August 2012. Several technical challenges came with production. After reasonable time of troubleshooting and tuning, these challenges have been overcome. Line 2 started production in April 2013, with many fine tunings and lessons learned from line 1 start-up.

The full production demand was only reached in March 2014, 720 pots were fully operational by end of June 2014. This explains the lower green anode plant uptime (Figure 12) and green anode block production (Figure 13) before June 2014.

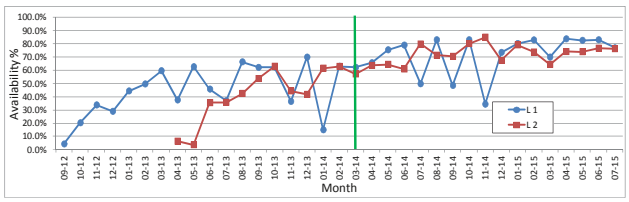


Figure 12. Green Anode Plant Uptime since Start-up

Since April 2014, the average green anode plant uptime and good green anode block production remain steady respectively above 80% and 30,000 anodes per month.

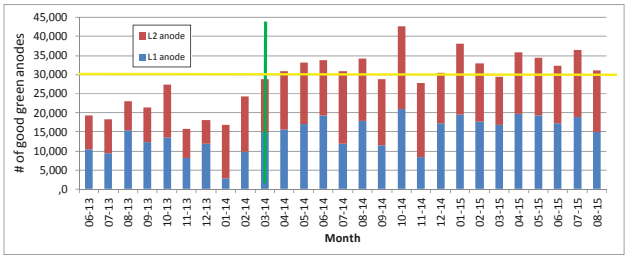


Figure 13. Good Anode Production since Start-up

After a period of ramp-up and fine tuning, the green anode density was stabilized between 1.64 and 1.65 g/cm^3 and the baked anode density was between 1.58 and 1.60 g/cm^3 as shown in Figure 14.

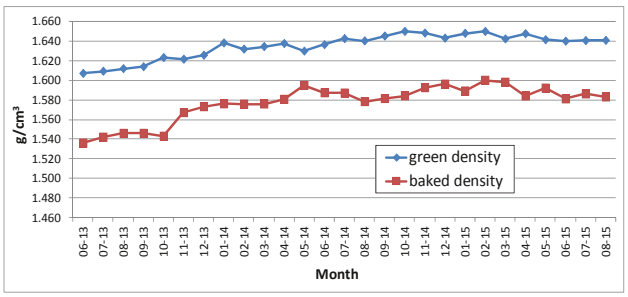


Figure 14. Green and Baked Anode Densities

Simultaneously, pitch content was continuously optimized by MA'ADEN operation team and reduced from more than 14% down to 12.5% as shown in Figure 15 representing significant savings.

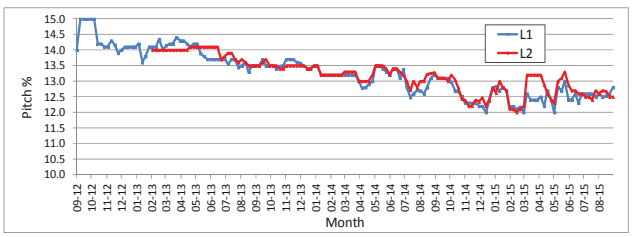


Figure 15. Pitch Ratio in Anode Recipe

Increasing the butt content in the recipe was an important enabler to achieve the lower pitch % in the recipe and to increase the baked anode density. The increase in butt dosage, as shown in figure 16, was gradual as the baked scrap volume became available and the carbon recycling was more reliable.

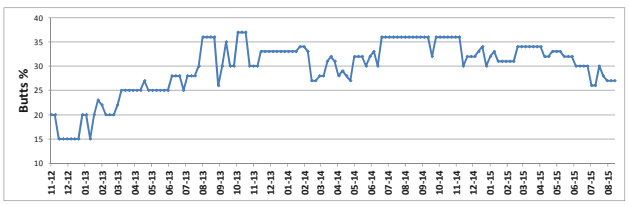


Figure 16. Butts Dosage in Anode Recipe

The rhodax crusher system made possible to operate at a level of up to 36% baked scrap with a stable anode production level of 40 tph. Considering the premature wear associated with the increase in baked scrap flow, the business case justified the increase in butt

dosage for reducing the carbon inventory and minimizing production unit costs.

Table II. Green and Baked Anode Quality Results
(GAD: Green Anode Density; BAD: Baked anode Density)

		MA'ADEN 2015		MA'ADEN 2013	
Properties	Units	Mean	Standard deviation	Mean	Standard deviation
GAD	kg/dm3	1.645	0.003	1.635	0.008
BAD	kg/dm3	1.59	0.007	1.55	0.023
Electrical Resistivity	mWcm	5324	215	5434	346
CO2 Reactivity Residue	%	92.4	2.3	91.1	8.9
Air Reactivity Residue	%	77.0	4.9	74.4	9.2

Overall the anode quality has improved drastically from start-up in 2013 until August 2015. Table II shows a summary of the anode properties and their standard deviations. Many properties show substantial improvement with a significant reduction in the standard deviation between 2013 and 2015. Meanwhile, multiple process improvements such as baking practices in the baking furnace and anode butt cleaning process in the rod shop, have played a major role in decreasing the sodium content and the electrical resistivity of anodes.

Anode performance in the pots has proven to be extremely good, with only few reported anode incidents. Based on this stability, the set cycle was increased from 25.5 days to 27 days in late 2014. The Net Carbon Consumption for the technology was expected to be at 410-415 kg C/ MT Al produced, while the 2015 results are at an average of 406 kg C/MT Al after 8 months.

All together, these green anode plant good performance figures are in-line with what can be expected from a proven technology combined with good operation and maintenance practices.

Conclusions

The two MA'ADEN 40 tph green anode plants are based on a proven technology. This technology contributes to the good production performance currently achieved.

A step by step ramp-up, continuous equipment and process performance monitoring as well as combined efforts of MA'ADEN operation, ALCOA and Fives commissioning teams made this project a success and a reference in term of Health and Safety. This contributed to the success of the MA'ADEN smelter project.

Beyond the many challenges faced during this project, the quality of the anodes produced have contributed to the successful and complete ramp up of all the pot complement at MA'ADEN Aluminium and is now considered to be part of the best anode plant in the world.

Acknowledgement

The Authors wish to thank MA'ADEN for the permission to publish this paper.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

SIMULATION-BASED DECISION SUPPORT IN CATHODE RE-LINING FACILITY DESIGN

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Keywords: Cathode, Re-lining, Modeling, Simulation

Abstract

Within Bechtel Mining & Metals (Bechtel), the Aluminium Center of Excellence tests and validates design alternatives and operational variants to optimize equipment and facility design. In a recent study for a cathode re-lining facility for Ma'aden Aluminium (Ma'aden), a discrete event model (DEM) was developed to demonstrate the achievable potshell throughput by matching planned resources and the expected operating schedule.

The initial number of stalls and stations in the cathode re-lining facility layout was based on static calculations. The DEM provided the tools required to optimize the design resulting in capital expense (CAPEX) savings. Also, the need for extra potshells, spent pot lining (SPL) scrap containers and mobile equipment container delivery units was addressed. The model covered the full cycle of re-lining a potshell: from stopping the cell, cathode cooling time in the potroom, delivery of the potshell to the re-lining facility, undergoing all the steps of cooling, de-lining, shell repair, re-lining and return to potroom. Various operational patterns were tested; including the timing of cell shutdown, cathode cooling location (potroom or lining facility), potshell movement by transfer gantry and cranes, shift durations, operating schedule and other variables.

Introduction

The operation of the Ma'aden Aluminium Smelter began in 2012. The construction of the cathode re-lining facility and the SPL processing facility was deferred during the initial smelter construction until re-lining of the aging 720 smelter potshells with new cathodes was required.

Bechtel was engaged by Ma'aden to prepare a front-end engineering and design (FEED) study for the Ma'aden cathode re-lining facility and SPL processing facility. The design of the cathode re-lining facility had to take into account the de-lining, repairing and there-lining of approximately 250 potshells a year and the SPL processing facility had to be designed to process approximately 20,000 tonnes of SPL material per year.

It was essential to design the re-lining process for 50 weeks of operation per year, five days per week and two twelve hour shifts per day. One twelve hour shift per week needed to be reserved for maintenance.

It was necessary to construct the new re-lining facility between the existing cathode sealing shop and the smelter center passageway (used to transport the potshells). The previously designated construction area for the cathode re-lining facility was fixed in size and location during the initial construction of the smelter. The facility design required effective use of the available space.

The SPL facility design was carried out to such that it can operate on a similar schedule as the re-lining facility but on the day shift only. This design allows the SPL 1st and 2nd cut materials to be processed separately during alternating two twelve hour shifts per week.

In determining the design and configuration of the new facilities, several key questions arose including:

- How many de-lining pits and lining stations would be required?
- Should potshell repair be done in-house or done by an outside contractor?
- How many potshell repair stations would be required?
- How many potshell cooling stations would be required?

The typical method used to answer these questions is usually to create a static, sequential schedule with estimated time to accomplish different operations within an assumed operating and maintenance schedule.

For the first cathode re-lining cycle in a new smelter all the cells require re-lining within a short time period. Therefore, the cathode re-lining facility needs to be capable of sustaining this peak initial re-lining period production rate. Figure 1 shows the proposed layout provided by Ma'aden. The initial scenario included the following work positions:

- 5 - Cooling stations
- 2 - Transfer pits
- 2 - De-lining pits
- 3 - Cathode shell repair stations
- 6 - Lining stations including one at floor level called the cathode ready station.

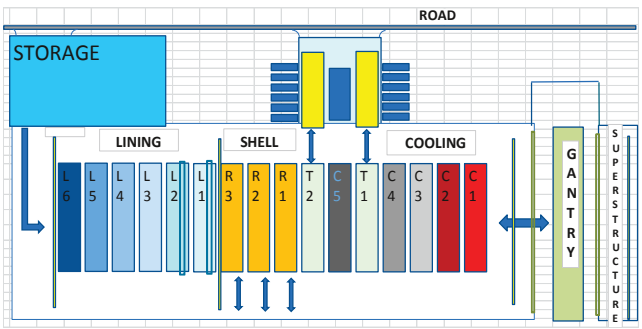

Figure 1 - Proposed Layout

A new DEM was developed to validate the initial scenario. The DEM is an ideal platform to simulate dynamically the cathode re-lining process, incorporating events occurring in the plant such as equipment breakdowns and delays, different crew efficiencies and shift durations. The DEM provided flexibility for the user to explore and rapidly test different numbers of lining stations, de-lining pits, operating schedule variants, cathode cooling time, maintenance shift requirements, breakdown frequency, crew efficiencies and other multiple iterations.

While developing the model, Ma'aden provided the tentative cycle time for each of the operations, developed the operating strategy, work schedules and maintenance philosophy, established typical frequency of breakdown, time to repair etc. While these process and operating parameters were being developed by Ma'aden, the model developer set up the logic block flows and began building and testing the model.

The model created an opportunity to review the re-lining operation in terms of cooling stations, lining stations, de-lining pits, shell repair stations, transfer pit, equipment and tools. This is an optimized approach to the design the cathode re-lining facility. The model review included two main considerations: lean manufacturing and capital cost effectiveness.

The purpose of the DEM was to optimize the facility configuration and to answer basic process design questions such as the following:

- Can the proposed layout and equipment reliably achieve the cathode re-lining of five (5) cells a week on a 5 day week, based on a 24 hour operation per day?
- What would be the required number of work stations for the various re-lining operations?
- What would be the potshell turnaround time?
- What would be the number of in-circuit potshells required?

The model development also served to help Ma'aden to validate their preliminary work design for the future cathode re-lining operations. Indeed, to assemble the model, the full process logic was required including detailed input tables. The task of defining all the operating parameters puts the operator in a very good position to confidently prepare the future work organization.

Model Development

Modeling Approach

DEM was chosen as a natural fit for the cathode re-lining facility. The process was modeled in FlexSim, a commercially available modeling and simulation tool (from FlexSim Software Product Inc.). FlexSim is a simulation tool that models, simulates, predicts, and visualizes systems in a wide variety of logistical applications. FlexSim was developed to optimize present and planned processes, identify and decrease waste, reduce CAPEX, and increase revenue. Material flows were composed of FlexSim library modules, e.g. network nodes, routes, queues, transporters. Operational logic was captured by Ma'aden and Bechtel. Particular schedules were set in timetables. Data pre-processing, data input to model, extraction of simulation results and then subsequent post-processing were all performed in MS Excel.

Application of standard tools made the model verification process fast and clear. Model parameters (e.g., available routes, assigned routes, distances, allowed traveling speeds, time of travel) were based on Ma'aden's actual plant. Model validation was achieved by vigorous comparison of historical plant data and design data with simulation results over various time periods (e.g. day shift versus night shift, shift change, weekday versus weekend performance).

Aluminium Potline Model

As part of the design and construction of the smelter project, the Ma'aden potline facility was previously mapped with buildings, cranes, vehicles, anode pallets, liquid metal transporters and anode haulers.

Figure **2** shows the cathode re-lining area as captured in the model; with colored blocks positioned correctly on the layout to mark the main buildings and crane routes assigned to specific tasks.

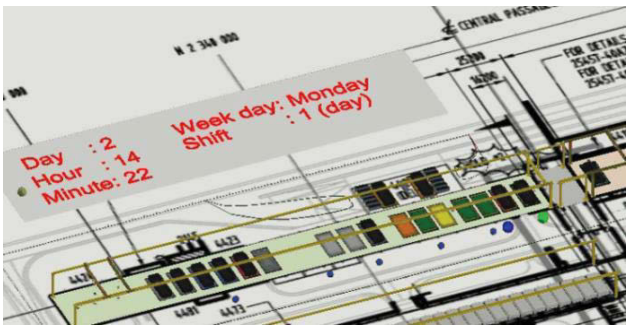

Figure 2 - Re-lining Area

The model was loaded with input tables: including operating schedules, cycle times per task, maintenance activities, crew efficiencies, equipment mean time between failures (MTBF), mean time to repair (MTTR), equipment speeds and other relevant operating parameters.

Table 1 is an example of an Excel input table. Parameters can be updated at any time using the applicable information to generate multiple iterations for analysis.

Table 1 - Excel Input Table (example)

Positions	Quantity		Basic Scenario	Data Source
	min	max		
Cooling	3	5	5	Ma'aden
Transfer Pit	2	2	2	Ma'aden
Delining Pits	2	2	2	Ma'aden
Lining Pits	4	6	6	Ma'aden
Shell Repair	1	3	3	Ma'aden

To produce the model, the logic flow of each operation was defined and summarized in a block diagram. Each block diagram was uploaded into the model. An example of a block diagram is shown in Figure 3.

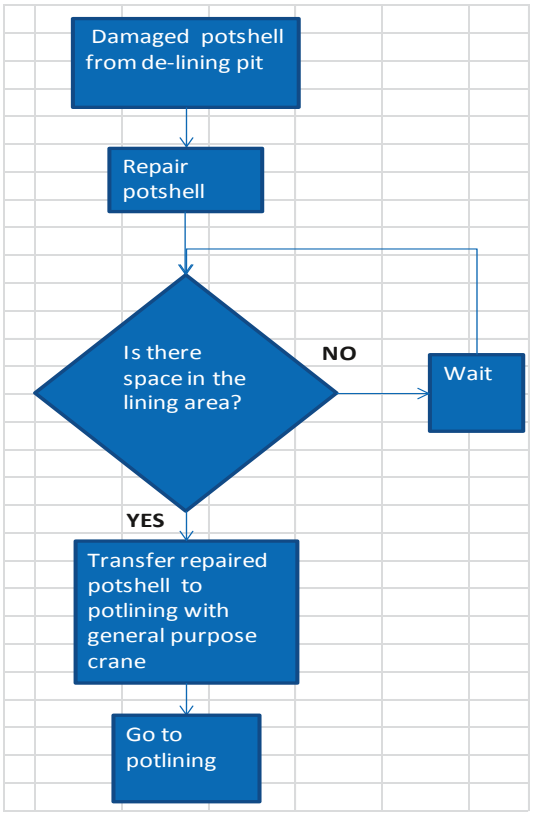

Figure 3 – Potshell Transfer Block Diagram (example)

Model Runs

The operations were modeled for a one year period. The first runs were performed to verify and validate the initial Ma'aden scenario, which would serve as the base case.

In order to achieve an accurate analysis of the results, several iterations were performed and graphs were produced to analyze the results, and determine opportunities to optimize the layout. Obviously, one of the critical targets was to achieve the re-lining of 5 potshells per week. Figure **4** shows the time remaining to complete each potshell at the end of the week. With 6 lining stations in use, but the 6th one indicating re-lining is incomplete, implies an occurrence when a potshell was not completed on time and was not ready to be transferred to the potroom.

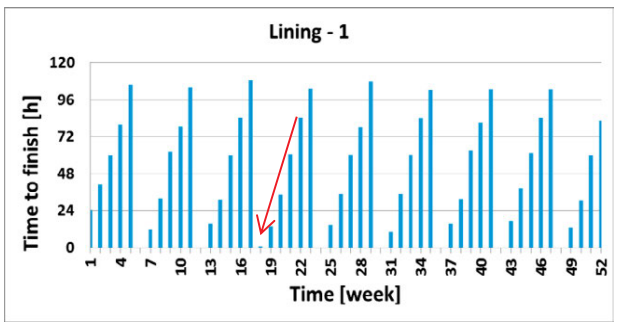

Figure 4 - Time Remaining to Complete a Cathode

Other results were also important: overtime required to complete the tasks as shown on Figure 5 such as cell turnaround time (time from cell out of service to cell back into service), equipment utilization, total time for a potshell in circulation, lining station utilization, shell repair station utilization, etc.

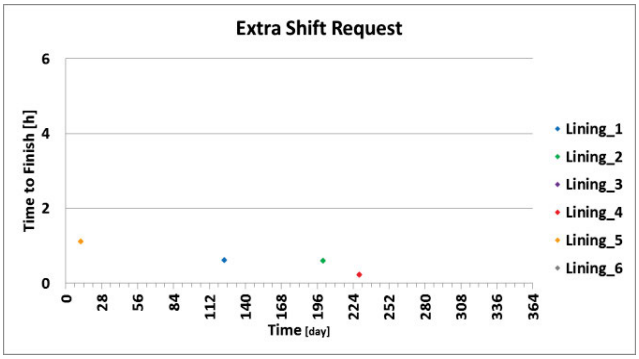

Figure 5 - Overtime Required to Complete a Cathode

Lean Approach Opportunities

The analysis of the charts provided input for potential increased utilization opportunities. As an example, Figure 6 shows the availability of the cooling stations in time. For ease of the representation only part of the year is shown.

Number of Cooling Stations

It appears that all cooling stations from Figure 6 have some excess availability. Removing a cooling station might be possible but it is not necessarily advantageous as these stations are inexpensive and provide flexibility to accommodate different cooling times between seasons.

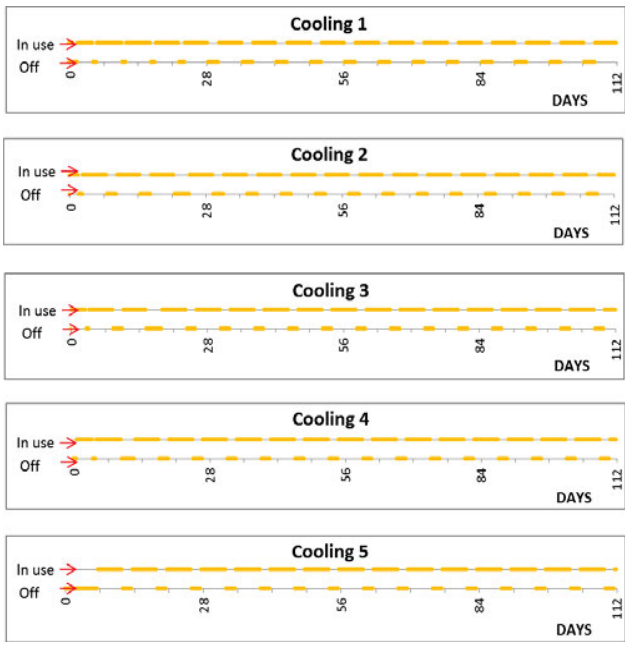

Figure 6 - Availability of the Cooling Stations (partial)

Cathode Lining Stations

As shown on Figure **7**, all lining stations would be occupied. However, when the initial suggested cathode re-lining time was compared to the benchmark re-lining time from other smelters, it appeared that it is possible to reduce the cathode lining time and evaluate the possibility of removing one lining station.

Also it should be noted that the 6th station would be used only during the first re-lining peak period and would be redundant during the following re-lining campaigns.

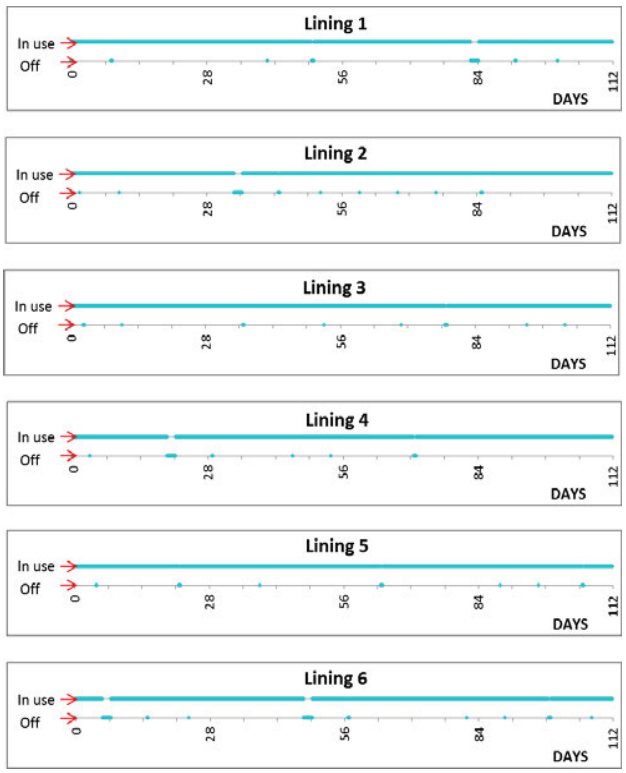

Figure 7 - Lining Station Availability

In the initial 6 cathode lining stations scenario, there was also a lining station, called "cathode ready", located at floor level for the final cathode inspection and installation of the cathode stuffing boxes. It was demonstrated with the model that a combination of avoiding double handling (lean), reducing cathode lining time and completion of the final activities in the 5 cathode lining stations allowed the removal of the cathode ready station or 6th lining station and to maintain still a targeted 7 days cell turnaround time. The weekly operation was shown to be optimized by having adequate hours of operation in the week, an adequate number of cathode lining and cooling stations, and strategic maintenance time planning.

Optimizing Maintenance Shift

The re-lining facility would use equipment purchased during the original construction of the smelter, including equipment such as cathode ramming machines, construction cranes of various sizes, and cathode transfer crane. The results from the model demonstrated a low utilization rate of the existing equipment as shown in Figure **8**.

In the initial base case scenario, all operations were stopped for 12 hours for maintenance. Due to quantity of equipment and the high equipment availability, an opportunity to reduce the downtime from 12 to 6 hours and doing the preventive maintenance on spare equipment was demonstrated as feasible.

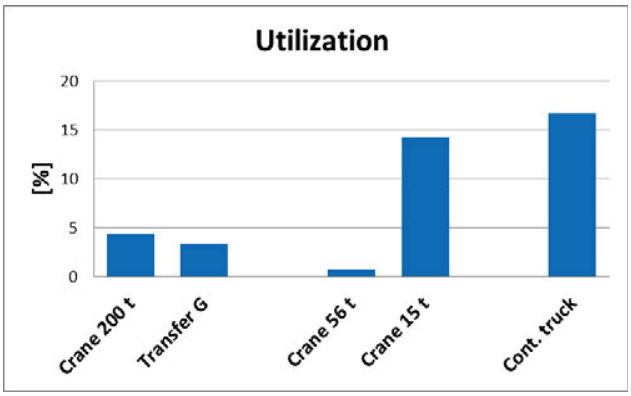

Figure 8 - Utilization

Number of Transfer Pits and De-lining Pits

Transfer trolleys located in pits would only be used a few minutes a day; however, the trolley was planned to remain with the potshell in the de-lining pit. De-lining pits were demonstrated to be fully occupied as shown on Figure 9. Therefore, there was no opportunity to reduce number of transfer pits without introducing an additional transfer or switching system.

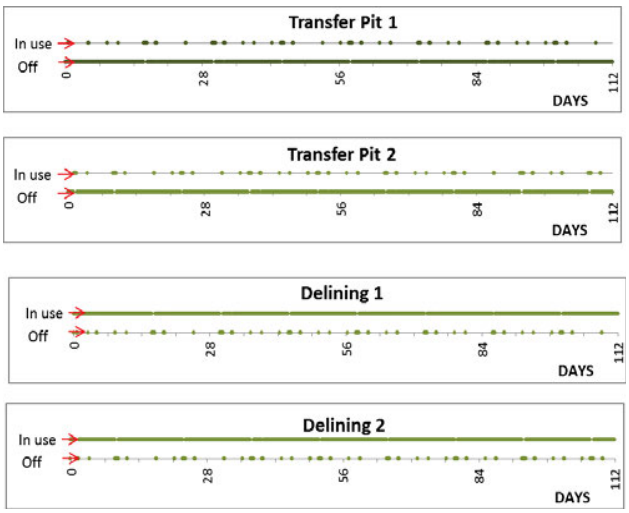

Figure 9 - Transfer and De-lining Pits

Number of Potshell Repair Stations

The potshell repair time has been determined to be 60 h for major repairs, with a 5 % variation due to crew efficiency. On that basis, the potshell repair stations have been demonstrated as being fully occupied, according to Figure **10**. In the event of

catastrophic potshell damage, or repairs requiring greater than 60 h, it has been concluded that the potshell would be sent off-site to a contractor rather than being repaired in the potshell repair stations. During the re-lining peak period, there could be an opportunity to operate with 2 potshell repair stations and use external resources for certain special selected repairs. Based on the plant experience developed in the 1st re-lining campaign, the model could be expanded to analyze a modified repair operation.

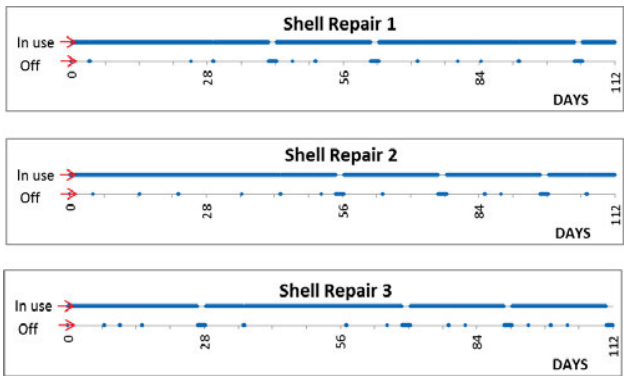

Figure 10 - Shell Repair

Results

The model permitted the quick analysis of multiple iterations of improvements and step by step changes. The graphic validation of various design approaches and operational options gave confidence to the team that the improvements identified during the analyses would be attained. Overall, millions of items of data generated by the model were drawn into simple graphs for ease of understanding. Analyzing the model results with experienced people provided an efficient way to get multiple combinations of parameters assessed. Some scenarios, when analyzed further, were eventually dropped while others were accepted as promising results under potential future conditions. While most runs included some constant features like the number of transfer pits and de-lining pits, other parameters were adjusted keeping in mind that some results were economically essential: i.e. cell turnaround time, minimal overtime, minimum 5 cathodes re-lined per week etc. Although field trials of adopted crew efficiency would be required for confirmation, the results demonstrated that re-lining 5 potshells a week was achievable with a total of 5 cathode lining stations.

Another area where efficiency gains were envisioned was the number of cooling stations. Many model runs showed only minor use of the 5th cooling station. The reason was identified as the idle state of the cathode lining facility during the two day weekend compared to the potline schedule. Although it would have been possible to assume an odd schedule permitting seven days a week operation and avoid the 5th cooling station, it was preferable to Ma'aden to keep the two consecutive weekend days off.

A common practice in the aluminium smelter industry has been to minimize the potshell turnaround time for de-lining and re-lining to minimize any production loss. Many DEM runs were done to achieve the minimum turnaround time. The average downtime was monitored as well as the standard deviation. Indeed, although a low average downtime was desirable, a high standard deviation indicated a high incidence of cathode re-lining process failures and the resulting downtime. This high failure rate and downtime would necessitate a recurring overtime requirement, delayed cell start-ups and delayed operations which would have been disruptive and inefficient to the whole potline organization. Moreover, the DEM led to significant advantages and confidence in terms of operability and scheduling.

The four best scenarios for comparison, made with different number of positions and different number of potshells, the need to re-line 5 cathodes a week were selected. Considering the smelter location in a hot climate, it was decided to target a potshell turnaround time of less than 7 days including in-situ cooling. Table 2 indicates that the best scenario would be run number 4.1 (blue column) which would have the minimum occurrences of a cell being out of service for more than 7 days. Indeed, the model predicted only one occurrence of 7.4 days in the first year, but this was during the initialization of the model before reaching a steady state. The conditions of run 4.1 were clearly the most effective scenario.

Table 2 Best Scenario

	Run 3.1	Run 4.1	Run 5.1	Run 6.1
Lining hours	84	84	84	84
Number of shells	13	13	12	14
Number of Lining spaces	5	5	5	5
Number of cooling spaces	4	5	4	5
Number of hours Maintenance shift	6	6	6	6
Number of Cell off Exceedances > 7days	50+	1	51	51
Result of maximum cell off production day	7.49	7.43	7.48	7.45

From the dynamic modeling point of view, the adopted DEM approach fulfilled its goals in terms of validating the best scenario by applying lean manufacturing principles and cost effectiveness.

Conclusions

A basic toolset, containing all the major equipment of a generalized de-lining and re-lining facility has been presented. This new DEM is now available to support both the study and the design work at the Bechtel Center of Excellence. During recent years, modeling toolsets have been systematically developed in many smelter areas. Now, boundary condition modules and control logic modules are available in commercial software for easy model development to suit most smelter customer needs.

Benefits through flexible models, pre-developed with the parameters used in most plants today and combined with the ability to quickly adjust to a user who is willing to explore innovations, offers potential cost savings and an increased level of confidence.

The DEM has proven to be an efficient tool for the layout and operation optimization of a de-lining and re-lining facility from the removal of a failing potshell to the installation of a re-lined potshell into the potline stall. This optimization includes recommendations for the most effective number of stalls or stations in the de-lining and re-lining facility, the most efficient

operating and maintenance work schedule and crew efficiencies, analysis of equipment availabilities and breakdowns, and other relevant variables. Moreover, the DEM would allow the establishment of early operational and maintenance work organization and procedures which would increase the confidence for an efficient and reliable facility start-up and operation.

Trademarks

FlexSim is registered trademark of Flexsim Software Products, Inc.

Acknowledgements

The authors thank *Bechtel* and *Ma'aden* for permitting the publication of this study.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

TWIN-ROLL CASTING OF CARBON FIBER-REINFORCED AND GLASS FIBER-REINFORCED ALUMINUM STRIPS

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Keywords: Twin-roll casting, fiber-reinforcement, aluminum, metal matrix composites

Abstract

The modern requirements for lightweight products force one to make constructional materials with brand new properties. One such group of materials that meets these demands are fiber-reinforced metal-matrix composites. A combination of light and good deformable aluminum with a strong fiber allows obtaining high strength, ductility and corrosion resistance in a single product. In the present article the application of energy-efficient technology of twin-roll casting for production of flat aluminum-matrix composites reinforced with carbon and glass fiber is observed. The corresponding experimental study on the manufacturing of thin strips of pure aluminum reinforced with carbon and glass fiber rovings was carried out using a 370 mm roll diameter labor twin-roll caster. The obtained flat products of 3…5 mm thickness were analyzed by means of mechanical tests and electron microscopy. The results have shown a great potential of the twin-roll casting for production of high-strength fiber-reinforced aluminum strips.

Introduction

Aluminum-matrix composites allow one to obtain an outstanding combination of properties that cannot be reached using conventional construction materials. The combination of strong and stiff fibers with ductile, corrosion-, heat- and chemical resistant matrix material, opens wide the horizons for the application of such compounds. First of all, they are important in modern lightweight construction, civil engineering and structural design. Areas of application of aluminum alloys are defined by the requirements of corrosion resistance, lightness and functionality of structural shapes [1]. The two latter properties can be extended by the reinforcement of aluminum with materials having a much higher Young's modulus, as well as greater tensile strength than aluminum. At the same time, the reinforcing material must not eliminate one of the main advantages of aluminum alloys: their low specific weight. From this point of view, it is more preferable to use materials in the form of continuous fibers that allows reaching of their maximal strength and stiffness. The combination of fibers and conventional constructional materials results in a considerably higher effect on mechanical characteristics of the latter than the use of common techniques such as grain refinement, solid solution-, strain- and precipitation hardening [2]. The different methods for production of the fiber-reinforced metal-matrix-composites have developed in the last decades. However, the modern scientific literature has not paid enough attention to the fabrication of such composites by means of twin-roll casting.

Fiber-reinforced metal matrix composites

The most widely used constructional materials in the form of fibers are carbon, glass and aramid. These fiber-reinforced polymers are finding application in almost every field of machine building and construction [3]. Nevertheless, their main disadvantages are low ductility, crash- and heat resistance. The optimal combination of reinforcing fibers and metallic matrix material creates the best properties in the resulting compound. In this case, a good bond between the matrix and fiber and the relation between their strength and Young's modulus play a decisive role [4]. From these relations, the optimal content of the fibers and matrix material in the cross section of the composite can be determined. In some cases, the calculated optimal reinforcement conditions cannot be reached [2]. The applied manufacturing method is one of the primary restricting factors in this case. The methods can be divided into four groups: liquid metal processes, solid state processes, deposition processes and in-situ processes [5]. The liquid metal processes are mostly related to the casting processes in which the reinforcement fibers are joined to a molten metallic matrix with a subsequent solidification of the latter. A well-known process for the production of single-piece metal-matrix composites is die casting [6]. Die casting consists of placement of the fiber in the form and the subsequent filling of the form with a molten metal under pressure. In [7], The application of different aluminum alloys for production of continuous fiber-reinforced composites by means of investment casting process is shown Ceramic fibers were used as the reinforcing material. The results showed a good bonding between the fibers and aluminum matrix. The technology of squeeze casting is also often used for manufacturing of discontinuous metal-matrix composites. In the works [4, 8, 9], examples of production of composites on the basis of aluminum and magnesium alloys reinforced with carbon fibers are shown. In the works [10, 11], the liquid silicon infiltration process for production of such metal-matrix composites is described. The experiments on squeeze casting showed a very good infiltration of the melt in the bundles of fiber due to the high pressure used during the procedure. The disadvantage of the process is a relatively long contact time between fiber and melt. This provokes the formation of undesirable brittle carbides on the interface between the matrix and the reinforcements. Another negative effect from such interaction is a degradation and erosion of the reinforcing filaments. As a result, the initial cross-section of the fiber bundle collapses and does not correspond to the expected loading. One of the processes used to manufacture continuous hybrids of fiber and metal matrix is shown in [12]. The composite in form of a wire with a carbon fiber core and aluminum coating was produced by means of a liquid metal pressure infiltration process. Here it was shown that a short contact time between molten aluminum and carbon fiber creates a strong bonding without a thick layer of carbides at the interface between the materials. Although the fibers were not coated with a protective layer, the maximum tensile strength of the composite could be reached at infiltration times less than 0.5…1 s. When this time was exceeded, brittle carbides Al4C3 grew in the aluminum matrix from the surface of the fibers. This significantly deteriorated the bonding strength and the properties of the composite as a whole. By application of glass fibers as reinforcing

material for the aluminum, the main known reaction between the two materials is the Al reduction of SiO_2 with formation of Al_2O_3 [13]. The diffusion between glass and aluminum first occurs in the range of temperatures that are higher than used for the aluminum casting [14]. The reinforcement of aluminum matrix with short glass fibers by means of compocasting [15] is also well known. The main advantage of the glass fiber is the much lower price in comparison with carbon fibers.

The optimal conditions for formation of fiber-reinforced metal-matrix-composites can be summarized as follows: short contact time with the melt and correspondingly fast solidification and high pressure in the bonding zone to ensure the infiltration of the melt between the filaments. These conditions of metal-fiber composite formation can be fulfilled by use of the twin-roll casting process. One of the distinguishing characteristics of this method is an extremely high melt solidification speed. This allows avoidance of long contact times of fibers with aggressive molten metal. The process is well-known for the production of thin strips of different metals directly from the melt [16]. Due to exclusion of intermediate heating operations on the way from the melt to the finished product, the process can be characterized as economical, ecological and resource-saving in comparison with a common thin strip production. Additionally, a high solidification speed and instant melt mixing are advantages for the manufacture of strips of high-alloyed alloys [17]. A nontrivial use of the twin-roll casting technology was shown in [18] for production of thin aluminum-steel bimetallic strips. The solid state steel strip was fed between the two rotating casting rolls together with an aluminum melt and exited the caster in the form of a two-layer strip with a high bonding strength between the layers. Analogously the manufacture of aluminum strips reinforced with a steel mesh by means of twin-roll casting is shown by Haga et al. [19]. A similar method can be used to produce fiber-reinforced strips with an aluminum matrix. The use of twin-roll casting is described in [20-23] as possible and promising for the production of thin metal-matrix composites. During the twin-roll casting, the aluminum strips can be uni-directionally reinforced with single rovings or with multiaxial fiber fabrics.

The aim of the present work is a further investigation of twin-roll casting of fiber-reinforced metal-matrix composites and the characterization of their properties.

Experimental procedure

For the investigation of the formation process of thin fiber-reinforced metal-matrix strips, the experiments are performed on a twin-roll casting machine. The reinforced strips are produced using a laboratory twin-roll caster belonging to the Chair of Materials Science of the University of Paderborn [24] (See Figure 1). The laboratory caster has two rolls, each with a diameter of 370 mm and a 200 mm barrel length. The thickness of the rolls' sleeves of the steel X38CrMoV5-3 is 15 mm. The rolls are cooled with water at a temperature of 13 °C over their entire inner surfaces. The casting rolls are driven with an 11 kW electric motor. The twin-roll casting machine is equipped with a conductive furnace for the melt preparation under a protective gas atmosphere. The furnace is positioned directly above the roll gap. This configuration provides the direct melt feeding into the strip forming zone through the tundish using the vertical twin-roll casting scheme.

The melt nozzle is equipped with a device for positioning of the reinforcement fibers in the middle of the strip forming zone and correspondingly in the core layers of the finished strip after its solidification. For the fixation of the reinforcement fibers, and to apply their back tension, a support roll having a contact with one of the casting rolls is used.

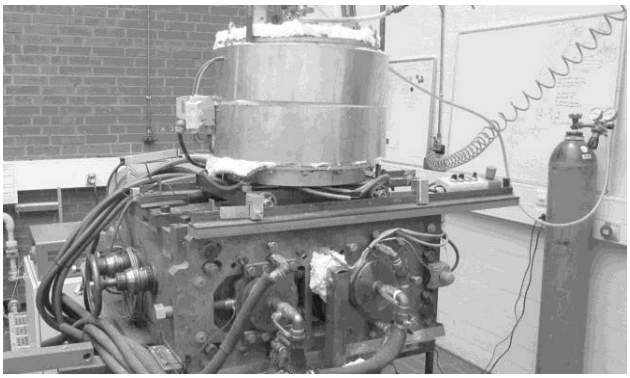

Figure 1. The twin-roll casting machine used for the experiments on manufacturing of metal-matrix composites.

In the first series of experiments the 4 mm thick single-roving fiber-reinforced metal-matrix composite strips are manufactured by means of twin-roll casting. Commercially pure aluminum EN AW-1070 is used as the matrix material and two types of reinforcing fibers in the form of single threads were implemented. The first fiber is a continuous glass fiber of E-glass Silane 2400 tex. The cross section of roving is 0.945 mm² and consists of approximately 1500 single filaments with a middle diameter of 25 µm. Another type of reinforcement fiber used is a HT carbon fiber PYROFIL® TRH50 60M 3200 tex. The area of the roving's cross-section is 1.8 mm² and consists of 60,000 single filaments of 6 µm diameter. The properties of the materials used in the study are shown in Table I.

Table I. The properties of materials used for the experiments

Material	Glass roving	Carbon roving	Aluminum EN AW-1070
Number of the filaments in the roving	1500	60000	-
Filament diameter, µm	25	6	-
Density, kg/m³	2540	1810	2700
Tensile strength, MPa	3952	4830	70
Young's modulus, GPa	73	250	70
Elongation at fracture, %	3.5	1.9	35

The rovings are positioned symmetrically and parallel to the longitudinal axis of the strip at an equal distance of about 50 mm from the strip center and the strip's edges. This arrangement minimizes the influence of the unevenness in the strip's formation caused by cooling through the side weirs. The fibers are fed into the roll gap prior to the melt feeding and are then stabilized during the twin-roll casting in the core layer of the strip. The main twin-roll casting parameters used for the experiments are given in Table II. As a result of the performed experiment, an aluminum strip with continuous fiber-reinforcement is manufactured. The strip's length is 4 meters; it has no visible casting defects and has the characteristic imprint of the rolls' relief on the surface which corresponds to a plastic deformation of the matrix during the strip formation. The specimens for mechanical testing and metallographic analysis are sampled from the end section of the strip. For a comparative analysis, the samples of reinforced and non-reinforced aluminum matrix are subjected to mechanical testing.

Table II. The main parameters of twin-roll casting used for the manufacturing of fiber-reinforced metal-matrix composites

Process parameter	Value
Casting temperature, °C	661
Casting speed, m/min	3.5
Setback, mm	70
Strip reduction in twin-roll caster, %	40
Coolant flow rate, l/min	115
Coolant temperature, °C	13

The uniaxial tensile tests are performed using small dog-bone specimens with a cross-section of 3 mm x 4 mm. The specimens are sampled from the strips by means of electro-discharge machining, with consideration that the rovings of glass and carbon fiber lay in the middle of the specimens' tailored section. In order to remove surface unevenness, the samples are polished down to a grit size of 15 μm prior to the actual experiments. The uniaxial tensile tests are conducted on a MTS 858 table top system. The loading direction of all specimens coincides with the initial twin-roll casting direction.

The microstructure of the interaction area between the fibers and the matrix material is investigated by means of electron microscopy. The cross-section specimens of the reinforced strips are ground, polished down to 5 μm and subsequently subjected to ultrasonic cleaning to remove the small particles of aluminum matrix from the filaments. A Zeiss Ultra Plus scanning electron microscope (SEM) equipped with secondary electron (SE), back-scatter (BSE) and in-lens detectors is used for the analysis.

In the second series of twin-roll casting experiments on the reinforcing of commercially pure aluminum matrix, the continuous multi-roving fabrics of glass and carbon fibers are applied as reinforcing material. The glass filaments are used in form of continuous biaxial fabric made of E-glass Silane 390 g/m² twill with a thickness of 0.429 mm. The width of the clutch is 100 mm. The used carbon fiber fabric is unidirectional Tairyfil® TC33 400 tex with a width of 45 mm. For every type of named reinforcement, a single experiment is performed using the twin roll casting parameters given in Table II. In both experiments, the reinforcing clutches are positioned in the middle of the strip and placed in its core layer with the help of a feeding device in the melt nozzle. Analogous to the first series of experiments, strips with a high quality of matrix that undergo the plastic deformation in the caster are manufactured. The specimens of the strips' end sections are taken for further analysis.

Results and discussion

As can be seen in Figure 2, both rovings of glass and carbon fibers lie in the core layer of the twin-roll cast strip. However, both filaments bundles have lost their round form during the strip production. This may be caused by plastic deformation of the aluminum matrix subsequent to its solidification in the caster. It can be also seen that the filaments are tightly surrounded by matrix material without any visible pores or defects.

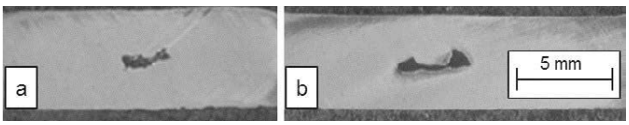

Figure 2. The cross-sections of reinforced sections of the twin-roll cast composite strip: a – with a glass roving, b – with a carbon roving.

There are no traces from the rovings detected on the strip's surface. During the tensile tests, the different behavior of the reinforced specimens is observed. In some cases the rovings are sheared-off in the specimen by the clamping jaws during the fastening in the tensile testing machine (See Figure 3a).Conversely, the application of low fastening forces causes the specimens slide out from the clamp during the test. This is observed in the specimens reinforced with a strong carbon fiber. The fastening force is stepwise increased until the optimal value, approximately 3 bars, is found at which the maximal tensile loading of specimen is reached without slippage and cutting of fibers. Another feature observed during the tension of some specimens reinforced with either glass or carbon rovings is that the bonding between most of the filaments and aluminum matrix is unsatisfying and the specimens slide out from the aluminum matrix under the tensile loading (See Figure 3b). Nevertheless, most specimens fracture in their middle evenly with the reinforcing fibers (See Figure 3c).

The results of tensile tests of reinforced and non-reinforced samples of the strips are shown on the diagram in the Figure 4.

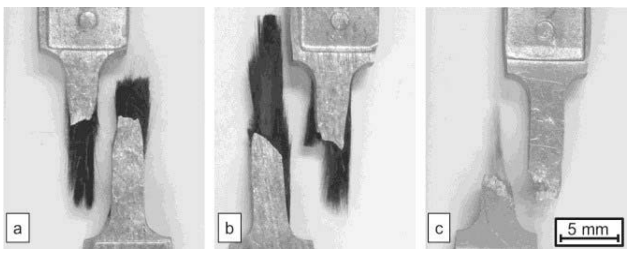

Figure 3. The samples of fiber-reinforced strips after the tensile testing: a – with sheared-off carbon fibers, b – with slid out carbon fibers, c – with an even fracture of glass fiber and aluminum matrix.

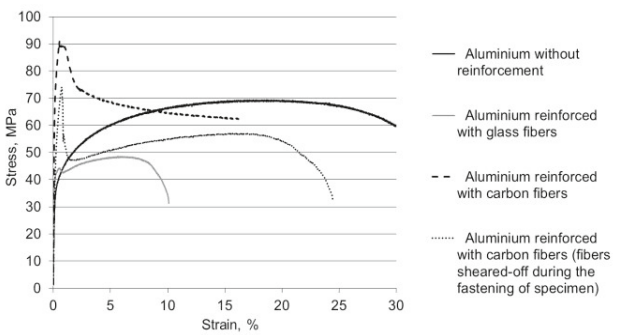

Figure 4. Stress-strain curves of the twin-roll cast metal-matrix composites under the uni-axial tensile loading.

As can be seen from the diagram, the reinforcement of aluminum strip with carbon fibers significantly improves the yield strength of the composites. The two types of stress-strain charts are shown in the diagram (See Figure 4) for the carbon-fiber reinforced specimens. In the specimen corresponding to the lower curve, the carbon roving is partially sheared-off with clamping, so in this case the maximum load cannot be reached. This is a typical case when the fiber material fractures before the complete hardening of the metallic matrix occurs as described in [2]. It is obvious that the residual cross-section of the metallic matrix, excluding the roving, cannot resist loadings at the same level as the solid aluminum samples of the same size. On the contrary, in the upper chart, the maximum stress of 91 MPa is reached by both the

reinforced filament and the hardened matrix. It can be clearly seen that directly after the fracture of the fibers, the composite loses its loading capacity. The poor results of the glass-reinforced specimens can be reasoned by an inconsistency of the E-glass fiber to the thermal and deformational conditions of the twin-roll casting. The strength and elasticity of glass fibers have very similar values to aluminum values. Additionally, they may be deteriorated due to the applied deformation, as well as the formation of thermal stresses in the strip forming zone.
The potential of improvement of the composite properties can be seen in the bonding conditions between the fibers and matrix material. In Figure 5, the SEM images of the reinforced area of the composite strips are presented. The images, made by use of SE detector, show the partial infiltration of the aluminum between the filaments in the core layers of rovings. No pores or delaminations between the roving and aluminum matrix are observed.

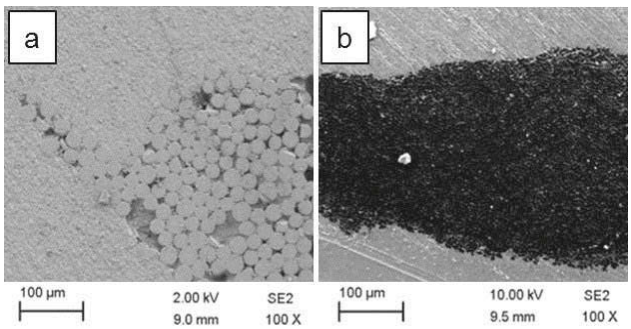

Figure 5. SEM images of the reinforced sections of the twin-roll cast aluminum-matrix composite: a – with glass fibers, b – with carbon fibers.

For a more detailed investigation, the analysis of the interaction zone is carried out in the scanning electron microscope at magnitudes of 1000x. The representation of the interface between the aluminum matrix and glass fibers, made using SE, BSE and in-lens detectors, is shown in Figure 6. As can be seen in Figure 6a, the fibers of the roving's external layer are completely wrapped by aluminum matrix from the outside. Aluminum protrusions into the deeper lying layers of fibers are observed. The BSE and in-lens images in Figures 6b and 6c show that the spaces between the fibers are filled with an aluminum matrix.
Figure 7 shows the interface zone between the carbon-fiber roving and aluminum matrix of the strip. Although the carbon fibers are thinner than the glass fibers, the protrusions of aluminum into several outer layers of filaments are also observed. The analysis by means of BSE and in-lens detectors (See Figures 7b and 7c) demonstrates the presence of aluminum between the carbon fibers in the near-surface zone of the bundle.
Further investigation of the interaction between the carbon fibers and aluminum was performed at SEM by magnitude of 2000x using SE and BSE detectors (See Figure 8). Due to the difference in the density of aluminum and graphite, the presence of aluminum in the carbon bundle, as well as the presence of carbon particles in the aluminum matrix, can be clearly seen in the BSE image (See Figure 8b). In the transition zone between two materials, some areas can be seen that potentially correspond to carbide formation. Such an effect is not observed on the whole interface between aluminum and fibers and has an irregular character. The small impurities in the aluminum are detected in the near-surface zone of the matrix. However, they should not be characterized as carbides due to their size and location at some distance from the interaction surface. On the whole, the interface line between the fibers and aluminum is sharp and restricts the formation of any transition phases. The investigation of bonding area between the glass fiber and aluminum by means of BSD analysis is impeded due to their similar density.

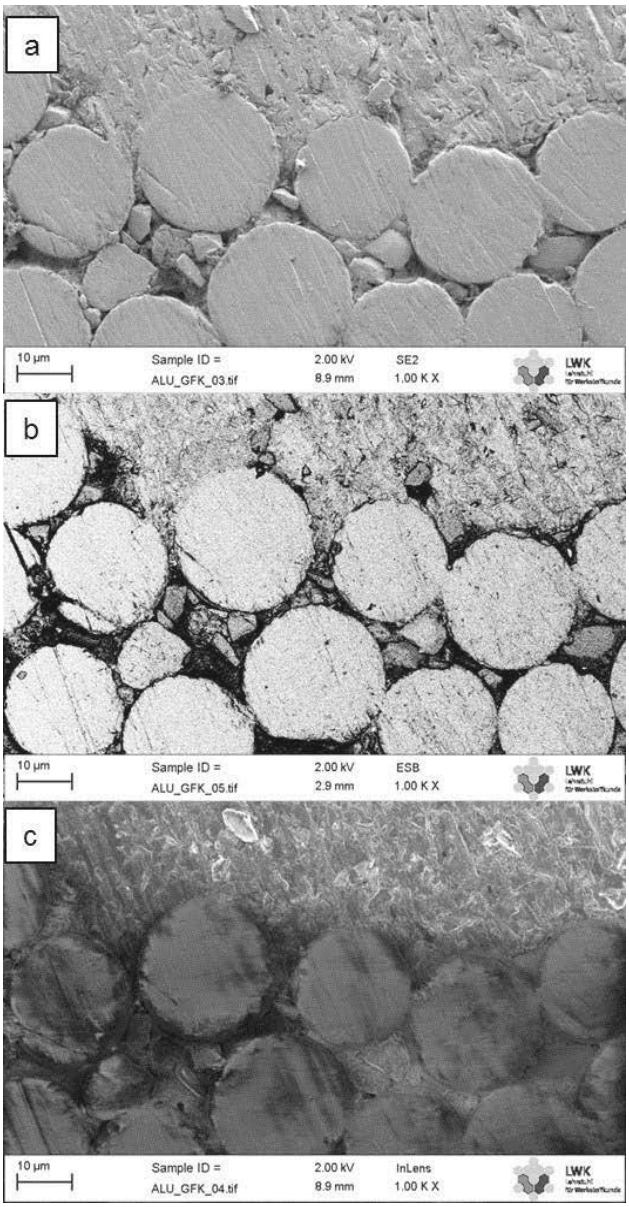

Figure 6. SEM images of the interface zone between aluminum and reinforcing glass fiber in the twin-roll cast composite strip: a – SE image, b – BSE image, c – in-lens image.

Thus, the conditions for manufacturing of single-roving fiber-reinforced strips by means of twin-roll casting are seen to be optimal. The second series of the experiments is dedicated to proving the possibility of production of aluminum strips reinforced with multi-roving fiber fabrics. As a result of the experimental procedure, strips with continuous reinforcement are produced. The pictures demonstrating the cross-sections of the twin-roll cast strips with inlaid glass- as well as carbon-fiber reinforcement are shown in Figure 9.

The fiber clutches are deposited in the core layer of the strip without any strong deviations in the position over their width and length. The strips with carbon-fiber reinforcement show a good bonding strength between the clutch and aluminum matrix. On the contrary, the glass-fiber reinforced strips show the distinctive inclination to a debonding of the matrix from the clutch when subjected to a minimal load. During the observation of delaminated areas, no fractures in the glass rovings, both in the longitudinal and transversal directions, are detected. Poor bonding in the glass-fiber reinforced specimens can serve a reason to explain the low strength properties of this type of composite. As can be seen in Figure 10, the glass rovings of longitudinal direction slide out from the matrix material at the compound fracture and do not use their full strength.

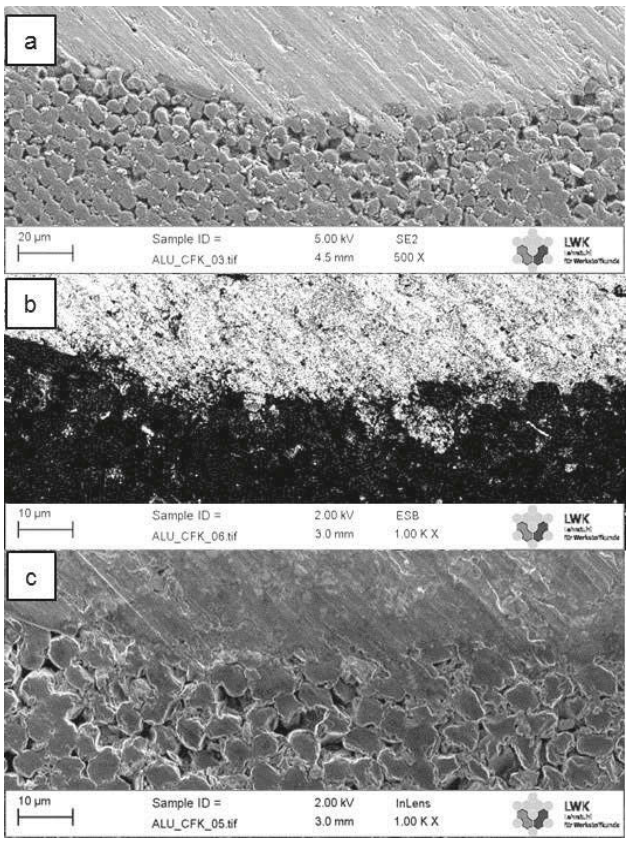

Figure 7. SEM images of the interface zone between aluminum and reinforcing carbon fiber in the twin-roll cast composite strip: a – SE image, b – BSE image, c – in-lens image.

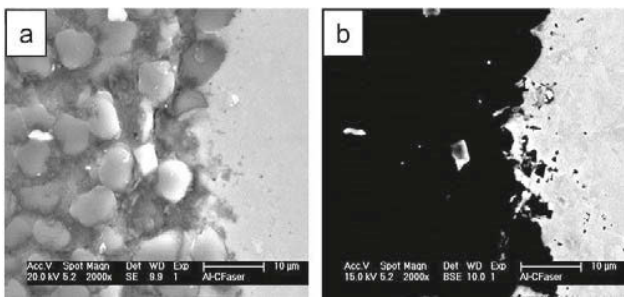

Figure 8. SEM images of the interface between aluminum and reinforcing carbon fiber in the twin-roll cast composite strip at a 2000x magnitude: a – SE image, b – BSE image.

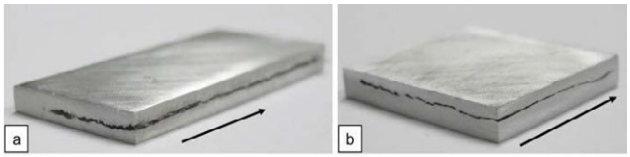

Figure 9. Samples of the twin-roll cast strips reinforced with glass fiber fabric (a) and carbon fiber fabric (b). The arrows show the twin-roll casting direction.

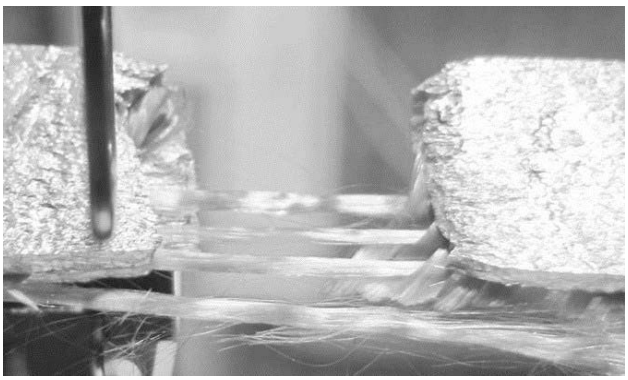

Figure 10. The sample of the aluminum-matrix strip reinforced with biaxial glass fiber fabric fractured during tensile testing.

The theoretical value of maximum strength of the reinforced composites can be calculated using the rule of mixture [2]:

$$\sigma_{fm} = (\eta)\sigma_f \frac{V_f}{V_{fm}} + \sigma_m(1 - \frac{V_f}{V_{fm}}), \quad (1)$$

where η – reduction factor depending on the form of fibers (η = 0.5…1), σ – tensile strength, $\frac{V_f}{V_{fm}}$ – contingent of the fibers in the whole composite cross-section; indexes f concerning the fibers, m – metal matrix and fm – the composite.

Using the values from Table I and the data from the charts in Figure 4, the comparative analysis of the theoretical and real reinforcement level by means of Formula 1 is performed. The calculation shows that the potential of reinforcement for the glass fiber-aluminum composite strips is only used for 13% from the maximum theoretical value. This value for the carbon-reinforced strips equals 12%. At the same time, the values of specific strength for the strips with glass-fiber reinforcements are lower than those for the unreinforced ones. On the contrary, the reinforcement with carbon fibers increases the specific strength by 40% compared to the solid aluminum strip.

The further improvement of the compound's properties can be reached by means of several arrangements. The first is the application of other fiber materials with a higher load and thermal stability. First of all, consider the E-glass fibers, which show poor resistance to deformation as well as to high processing temperatures above 650 °C, characteristic for the twin-roll casting. As can be seen in the Figure 11, some of the glass fibers are broken due to the loadings experienced during the composite production. The broken fibers do not sustain the tensions in the strip; however, they do increase its weight.

Another method is the use of clutches with optimized form. In this case the fastening of the filaments in the metallic matrix can be provided by melt protrusions between the gaps in the clutch during the twin-roll casting.

Improvement in the direct bonding between the fibers and matrix can be realized by changing of the twin-roll casting parameters. In this way, the twin-roll casting with higher casting temperatures, higher deformation strain as well as longer infiltration time can be analyzed. As can be seen in Table II, the casting temperature was only 1 °C higher than the melting point of pure aluminum. That enabled the rapid solidification of the melt but could suppress the melt infiltration process in the fiber bundle.

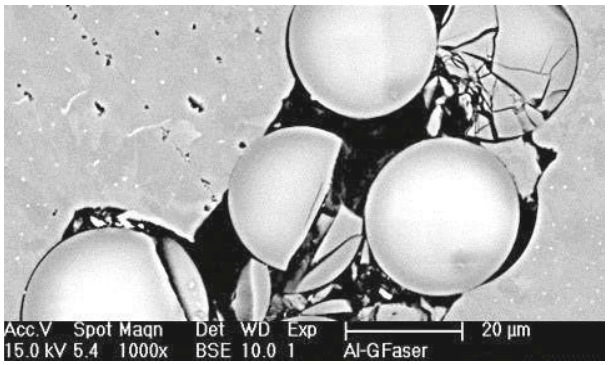

Figure 11. SEM image of reinforcing glass fibers crushed during the manufacture of the composite strip.

Conclusions and outlook

The main conditions for the production of metal-matrix composites reinforced with glass or carbon fibers are brief contact time and high pressure between the molten metal of matrix and fibers. The fiber-reinforced aluminum-matrix composites were manufactured by means of twin-roll casting. The 4 mm strips of pure aluminum with single-roving reinforcement of E-glass and HT-carbon filaments were produced using a labor twin-roll caster. The specimens of the strips with reinforcements were subjected to uniaxial tensile testing and were analyzed by means of SEM. The results showed an increase of the specific strength in the carbon-fiber reinforced strips by 40% compared to solid aluminum twin-roll cast strips. The formation of transition phases on the interface between the carbon filaments and aluminum was minimal. The reinforcement of aluminum matrix with E-glass fibers has a negative effect on the mechanical properties of strip due to the fracture of the glass fibers and poor bonding between the fiber and matrix. Application of the twin-roll casting for manufacturing of multi-roving fiber-reinforced aluminum strips is shown. The estimation of optimal conditions of strip formation will be the investigation of future works.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

PRODUCTIVITY IMPROVEMENTS IN INDUSTRIAL TWIN-ROLL CASTING BY HEAT LOSS ANALYSIS ALONG THE PROCESS CHAIN

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Keywords: Twin-Roll Casting (TRC), Aluminum production, Heat balance model, AA3003

Abstract

The main limitation in industrial twin-roll casting of aluminum with respect to productivity is heat removal from the melt in order to ensure complete solidification before the material leaves the roll gap. For this purpose a heat loss analysis was performed along the process chain starting at the melting furnace. By reducing heat losses in the launder system, filter box, and the ceramic nozzle it was shown that melt temperature can be decreased considerably at all stages. This leads to savings of energy and CO_2 emissions. Furthermore, lower headbox temperatures reduce the heat to be diverted in the cooling system. This advantage can be directly converted into an increased strip speed. For example for alloy AA3003, a reduction in headbox temperature by 12 °C did increase the productivity by up to 9% in a full campaign. The findings are now consequently used in production.

Introduction

Twin-roll casting (TRC) is an industrial standard process in the aluminum industry for the continuous production of strip in one process step combining solidification and deformation. Omitting process steps which are necessary in the direct chill casting route like ingot casting, heating and hot rolling leads to a shortened process chain with lower investment costs, space requirements, energy input, and emissions. Especially when the process is fed with liquid primary metal like in case of the Hydro Aluminum Karmøy plant, the process is highly energy efficient. The main energy input into the whole processing chain is mainly covered by the superheat of the primary aluminum melt. On the backside, the TRC process is commonly regarded as being limited to relatively low alloyed products in the 1xxx, 3xxx, 5xxx, and 8xxx alloy classes and medium surface qualities.

In twin-roll casting, liquid metal is supplied to the gap formed by two internally water-cooled rolls. In the roll gap, the material is solidified and formed within a very confined space and time interval. Solidification has to occur uniformly and must be finalized at a certain position, named solidification length or sump depth, before the exit to control the process and to achieve homogeneous material properties and high strip quality. The rate-limiting factor in twin-roll casting of aluminum is the dissipation of latent and caloric heat from the melt and the formed strip. Here especially the heat flux through the steel shell on the water-cooled cores proves to be limiting [1]. Even though strong and complex interactions of thermal aspects with mechanical and quality aspects are present in the process [2], the control of process temperatures is of high importance for a stable and quality-oriented production.

Therefore in this work, the heat losses along the process chain are reviewed with a special focus on the losses occurring after the headbox as these lead to a lateral temperature variation in the process. Based on previously described simulations of heat losses in the ceramic tip [3], melt temperature in the headbox is minimized for a full scale trial campaign producing AA3003. The reduced heat flow into the roll gap is utilized by increasing the strip speed while keeping solidification conditions stable. A simple heat balance model is used for prediction of the possible productivity increase. The lowered headbox temperature allows to decrease melt temperature upstream of the headbox furthermore, increasing the total energy efficiency of the process.

Process Temperatures in Twin-Roll Casting

The set-up of an industrial twin-roll casthouse is typically composed of a melting furnace, a holding or casting furnace, a filter unit combined with degassers, and a tundish or a headbox and a ceramic tip as shown in Fig. 1. Typically at all these stages, the melt temperature is monitored and controlled, although continuous temperature measurements in the tip are unusual due to difficulties in correct positioning of the thermocouples without disturbing the melt flow. Furthermore, the surface temperature of the rolls as well as the strip temperature after the roll exit are measured.

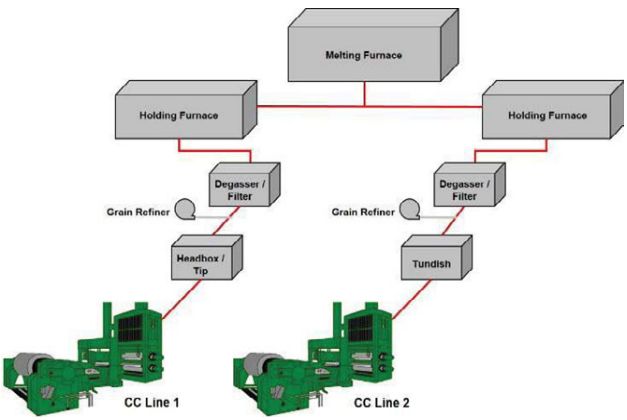


Fig.1: Setup of an industrial twin-roll casthouse for the continuous production of aluminum strip.

The heat losses in this process chain have to be categorized into two sections. Any losses between the holding furnace and the headbox are energy losses with limited consequences on quality and nearly no influence on productivity. Heat losses in the tip are more critical as they cause a lateral temperature distribution in the process, which means different solidification and deformation conditions over casting width. Hence these losses significantly influence both, quality and productivity of the process [3].

Heat Losses between Holding Furnace and Headbox

Nevertheless, heat losses between holding furnace and headbox should not be disregarded. The higher the heat losses upstream of the headbox, the higher the furnace temperature set-point has to be chosen. A higher furnace temperature leads to increased use of heating energy combined with extensive emissions of carbon dioxide or equivalents. Furthermore, higher melt temperature leads to increased solubility of hydrogen in the melt [4], enhanced dross formation, and enhanced wear of launder and furnace linings.
High heat losses in the launder are mainly caused by damaged or insufficient backside insulation of the launder. Furthermore, launder section transitions and corners in the launder system lead to distinct localized temperature losses as shown in Fig. 2.

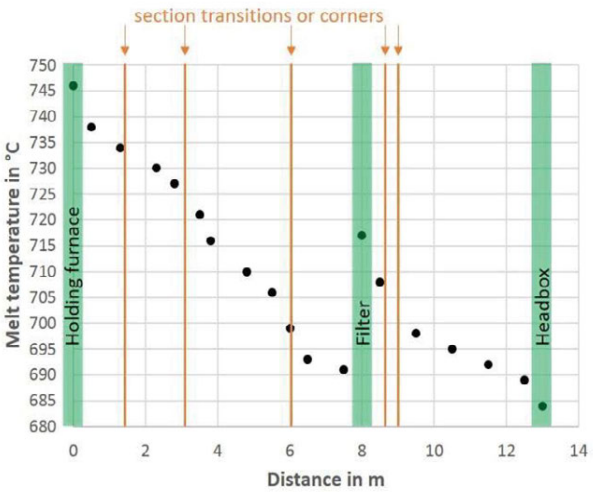


Fig.2: Evolution of melt temperature along the process chain between holding furnace and headbox.

Heat Losses after the Headbox

The last position before the solidification and deformation zone in the roll gap, where a correct and reliable temperature measurement is easily possible lies within the headbox. Following the headbox the melt flow is spread in lateral direction within the ceramic nozzle, commonly called tip. Inevitably this leads to a distribution of melt temperature across the cast width with the coldest spots at the edges. A strong focus has to be put on the lateral variation of temperatures in the tip, on the rolls and on the strip.

Local variation in melt temperature leads to different solidification and deformation conditions which influence productivity and product quality in manifold ways. Diverse quality issues like e.g. heat lines arise from inhomogeneous heat transfer [5-6]. Transversal strip profile is also influenced by local variation of solidification and rolling length. Finally also productivity can be limited by severe temperature gradients across the casting width. To avoid pre-solidifications in the tip, the headbox temperature has to be adjusted to a level sufficiently high to compensate for temperature losses in the tip even on the longest streamline without reaching the alloy's liquidus temperature. For this reason the design of the tip is a very critical point in the heat management in the twin-roll casting process being frequently studied [e.g. 3, 7-8]

Determination of Headbox Temperature and Strip Speed

Based on fluid dynamics simulations of heat losses in the tip shown in Fig. 3 and described in detail in [3], a thorough analysis of recorded process data, and the respective liquidus temperatures of the alloys produced by twin-roll casting, minimum applicable headbox temperatures can be determined.

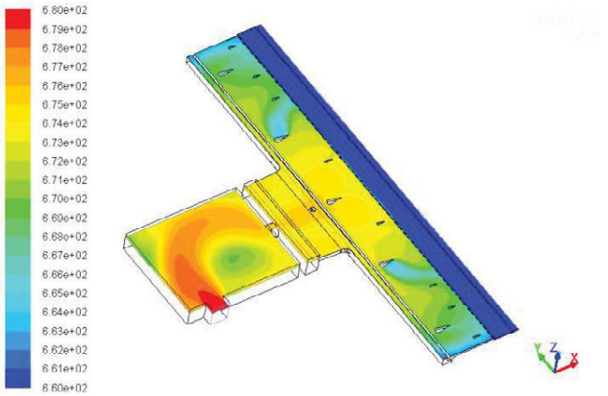
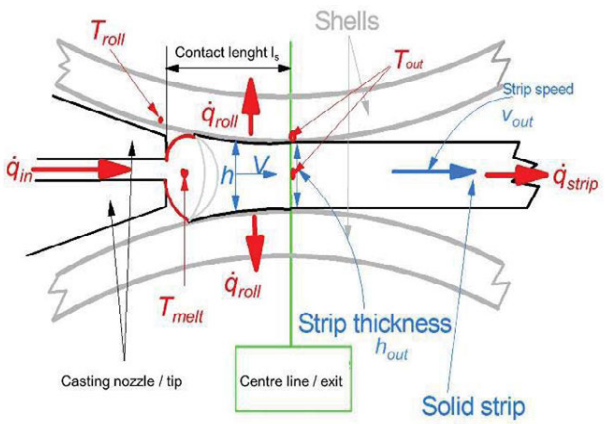


Fig. 3: Temperature field in the metal distribution system consisting of headbox and tip for AA1050 [3].

The applicable strip speed in twin-roll casting as a measure of productivity depends on numerous process parameter like e.g. metal level, setback and melt temperature as well as on material properties of the alloy to be cast and the roll shell. To efficiently run the process, the solidification and rolling lengths have be kept in a certain balance. Typically this balance is discussed in terms of the sump depth in literature and the influencing parameters are well known [e.g. 9].



$$v_{out} = a \cdot \frac{\sqrt{\lambda_{roll} \cdot l_s \cdot v_{roll} \cdot c_{p_roll} \cdot \rho_{roll}} \cdot [T_{out} - T_{roll}]}{\rho_{Al} \cdot h_{out} \cdot [c_{s_Al} + c_{p_Al} \cdot (T_{melt} - T_{out})]}$$

Fig.4: Heat balance in the roll gap during twin-roll casting and derived equation for strip speed v_{out}. For details see [3].

Among them is also melt temperature in the headbox. A lower headbox temperature reduces the sump depth, which can be balanced by increasing roll speed and hence enhancing productivity

of the process, while the conditions of solidification and therewith strip the quality and material properties remain unchanged. Although many sophisticated models of the twin-roll casting process have been developed [e.g. 10-12], for a simple case such as to estimate the potential strip speed increase by lowered melt temperature a simple heat balance model, as shown in Fig.4 and described in detail in [3], is sufficient.

Casting Trial

A full scale casting trial covering a campaign of several days production on a third generation Pechiney 3CM JumboCaster was performed producing the alloy AA3003 in 5 mm as-cast thickness and 1570 mm strip width with the goal of reducing the headbox temperature to a minimum as previously determined. This shall lead to reduced use of energy upstream of the caster as well as to higher productivity. Further benefits as discussed above like decreased wear of furnace and launder linings are to be expected in long-term but cannot be proven in this single trial campaign.

The minimum headbox temperature was determined considering the maximum thermal loss between headbox and tip exit as determined by a simulation as shown in Fig. 3 for AA1050 of 12 K. Accounting for the typically lower strip speed in production of AA3003 and the lower liquidus temperature of 652 °C this results in a value of 677 °C compared to the previous average of 689 °C. Applying the heat balance model displayed in Fig. 4 the reduced headbox temperature is expected to reduce the heat flow into the roll gap $\dot{q}_{in}$ in such way that strip speed v_{out} can be increased by 3.4 %.

For the start of the full-scale trial campaign, the headbox temperature was set to 679 °C as a first step (see Fig. 5). Other process parameter like cooling water flow and temperature, setback, and metal level remained on standard levels. With this setup an average productivity of 2.10 metric tons per hour was reached for 13 full coils. In the following, the melt temperature in the headbox was further decreased to 677.6 °C in average for another 31 coils at an average productivity of 2.14 t/h. The recorded values of headbox temperature and productivity are shown in Fig. 5 in comparison to a previous campaign of production of the same alloy in same width on the same continuous casting machine. In the previous campaign the only relevant differences in process parameter beside the intended variation in headbox temperature were a reduced roll shell diameter and a slightly reduced setback in the trial campaign.

The achieved improvements in productivity in comparison to the previous campaign with standard parameter accounted for 6.5 % in the first step and 8.6 % in total when the full temperature decrease to 677.6 °C was reached. This outperforms the expectations as predicted by the heat balance model somewhat. According to operational experience, the shell diameter difference should enable an additional 1.4 % increase in productivity in the trial campaign compared to the previous campaign. Furthermore during the trial as well as in the previous campaign an increased roll cooling capacity was available at the continuous casting machine, which might explain the even higher step in productivity than expected.

Furthermore, as a consequence of the decreased headbox temperature, the temperature in the holding furnace could be reduced by 10 °C leading to reduced energy consumption and CO_2 emissions. Additionally the before mentioned quality aspects like reduced hydrogen solubility and dross formation as well as reduced wear of launder and furnace linings can be expected in long-term from these operational improvements.

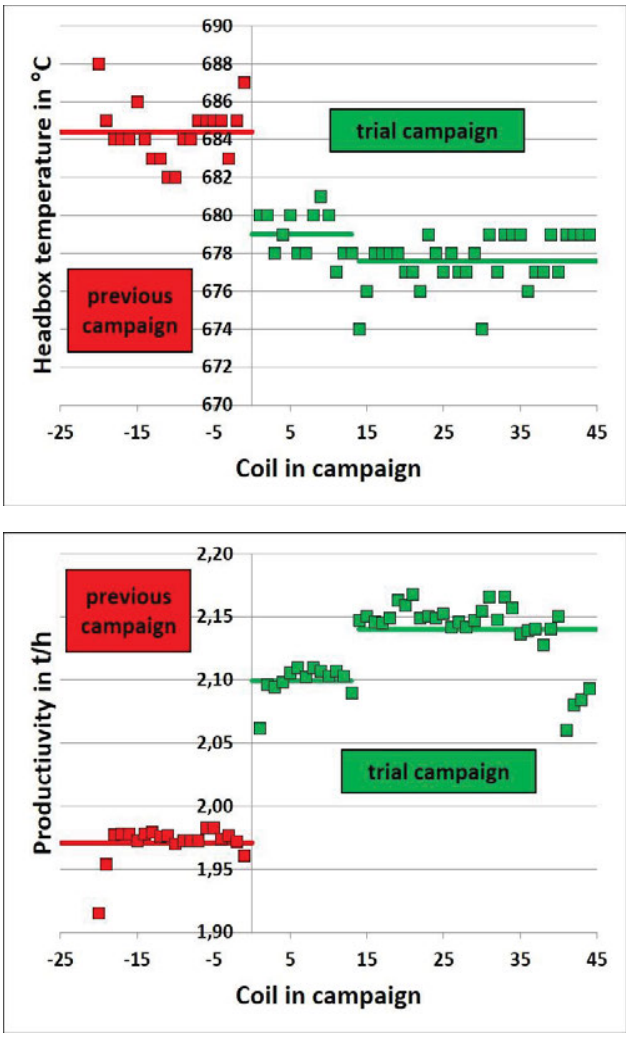


Fig.5: Recorded headbox temperature and productivity during the AA3003 full scale casting trial compared to a previous campaign under nearly identical conditions.

Summary

Based on fluid dynamics simulations of heat losses within the ceramic tip, the minimum applicable headbox temperature for the production of AA3003 in 1570 mm cast width was determined. This led to a decrease of headbox temperature by 12 °C compared to the previously used average, which in turn could be transferred in higher roll speed and hence increased productivity of the twin-roll casting process. For estimation of the possible increase in strip and roll speed a simple heat balance model was used. Based on these predictions a full scale industrial trial campaign was run with a two-step decrease in headbox temperature down to the predicted value of 677 °C. Due to other beneficial effects like a small shell diameter and a high roll cooling capacity, the predicted productivity increase of 3.4% was clearly exceeded and an 8.6% improvement was achieved. The new process parameter set leads to identical sump depth and hence solidification and rolling conditions are

expected to remain unchanged. This is indirectly proven by strip quality and material properties which did not change in a noticeable extent. During the trial campaign 44 full coils were produced without any serious quality issue and without any scrap production. Beyond the considerable improvement of productivity at stable quality, the decreased headbox temperature allowed to reduce the melt temperature in the holding furnace by 10 °C leading to additional energy savings in the process and expectedly reduced wear of furnace and launder linings. The presented findings are now consequently used in production and were transferred to other alloys as well.

Acknowledgement

It is a pleasure to thank Mr. Jakob Kvalevåg, Mr. Øystein Pedersen as well as the staff of Hydro Aluminium Rolled Products AS, Karmøy for support and close collaboration in joint process development and in execution of plant trials.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

DEVELOPMENT AND NUMERICAL SIMULATION OF A COMPOUND BELT CASTING PROCESS

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Keywords: Compound Belt Casting, Clad strip, Aluminium Alloy, Numerical Simulation

Abstract

A new belt casting process has been developed to cast aluminum-aluminum alloy compound strips. The first strip is cast by a conventional glass fiber belt caster. At a defined distance from the location of the primary solidification, a second casting unit is placed. The local upper surface temperature of the substrate is essential for the forming of a sound metallurgical compound. Numerical simulation models of an existing single strip belt casting process were set up and extended to describe the new double strip casting process. The influence of process parameters (cooling conditions, casting speed) and constructive modifications on the crucial surface temperature of the substrate were investigated. Temperatures measured in the casting strip and in the cooling plates were used to calibrate the simulations. Casting parameters for the new compound casting process as well as the appropriate position of the second casting unit were derived from the simulation results.

Introduction

Strip-shaped aluminium clad materials are used in different industrial applications for example in structural or decorative elements in the automotive or aviation industry or as heat exchanger material in the industrial engineering [1, 2]. Advantages are manifold resulting from the combination of different aluminium alloys with various mechanical, physical or chemical properties forming readily a composite [3, 4]. These clad materials are usually produced by a mixture of hot and cold rolling processes in several production steps [5, 6]. Constantly increasing demands on clad materials make it necessary to exchange the classic compound alloys to high-strength, high-performance alloys. But with high-strength alloys the roll bonding process reaches its limits as the cold roll process is a type of solid-phase welding which occurs during the plastic deformation of the join partners [7, 8, 9]. Therefore, an alternative process route for the production of clad material is the compound casting of different layers in a single process step, forming a metallurgical compound [10, 11]. This leads to a simplified production process, reduces production costs and enables new combinations of aluminium alloys [12].
The aim of this work is the development of a compound casting process based on the twin belt casting process, where the strip is transported by glass fibre belts. The local upper surface temperature of the substrate strip is of great importance to achieve a cohesive bonding layer between two aluminium alloys. Even small temperature variations from the narrow process window can lead to considerable reductions in compound quality. Therefore, the influences of casting parameters on the substrate temperature need to be quantified to determine the local upper surface temperature along the cooling section. A numerical simulation model of the compound belt casting process was set up to study the variation of process parameters (temperature and amount of cooling water, casting speed, heat transfer from the casting strip to the copper cooling plates) and to indicate the maximum temperatures at the compound interface during casting. The simulation was calibrated by various temperature measurements in a single strip casting machine used in production for several years.

Compound Belt Casting Process

The process is shown schematically in Figure 1. The principle of operation is to cast the molten metal into the cavity between two water-cooled copper plates via a casting nozzle [13]. Two glass fiber belts are circulating between the casting strip and the copper cooling plates to transport the strip continuously with a defined casting velocity. In the cavity the strip alloy solidifies and the delivered heat is transferred through the belts to the water-cooled copper plates on both sides. After solidification and leaving the upper copper cooling plate the moving substrate strip reaches the compound casting region where clad alloy is casted onto the solidified strip through a vertical slot. At suitable thermal condition a metallurgical compound forms shortly after the clad alloy gets in contact with the substrate strip. In the further process the compound strip cools down and is cut at to pre-chosen lengths.

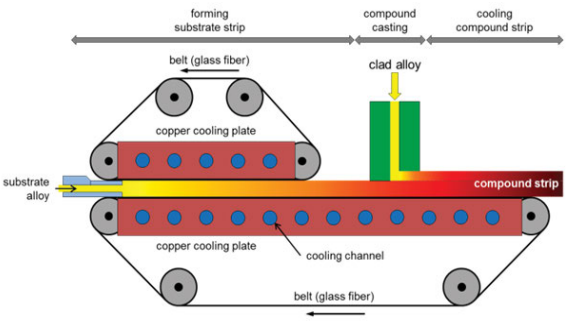
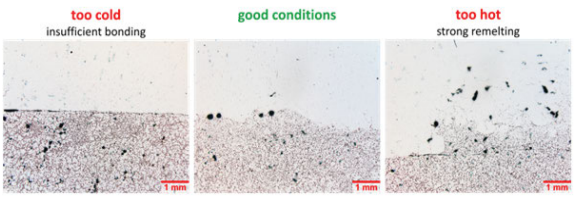


Figure 1. Layout of the compound belt casting process.

Initial Experiments

For initial experiments a small-scale compound casting test plant was designed and built to produce clad plates in a quasi-continuous casting process. Details are given in [14]. With these clad plates the possibility and the necessary metallurgical and thermal conditions to form a metallurgical compound of different aluminium alloys can be elaborated. As mentioned before, the

process window to cast a high quality compound is quite narrow. Figure 2 shows the microstructure at the interface of clad plates cast at different casting parameters. If the temperature in the bonding zone reaches a certain temperature, the surface of the substrate strip will partly remelt forming a metallurgical bond between these two layers (Figure 2b). If the temperature at this interface is too low, no cohesive bonding occurs (Figure 2a). Higher temperatures result in an unwanted mixing of the two materials due to strong remelting of the substrate alloy (Figure 2c). The thermal conditions at the interface and therefore the compound quality are influenced by the thickness ratio of the layers, the melt compositions, the temperature of the clad alloy and mainly by the substrate strip temperature [15].



(a) (b) (c)

Figure 2. Microsections of the compound layer (below: substrate alloy / above: clad alloy).

Numerical Simulation of the Compound Belt Casting Process

A 3D macroscopic finite element model of the new compound strip belt casting process was set up to analyse constructive modifications and process parameters. The numerical simulations were carried out using the commercial simulation software FLOW-3D, Version 10.1 (Flow Science Inc., Santa Fe, New Mexico, USA). The movement of the liquid melt in the mold cavity is modelled by classical flow equations (conservation of mass and momentum) considering the temperature development in the liquid and solidified melt as well as in the whole casting device (conservation of energy). Due to a relatively low casting speed and therefore a low velocity of the moving strip, the fluid flow of the liquid metal was assumed to be laminar. The movement of the solidified casting strip was modelled by translating the temperature field in casting direction at each calculation time step according to the casting speed.

The heat transfer coefficients at the strip/belt and belt/copper plate interfaces, used to characterize the heat removal rate from the casted strip, is a critical input value to the simulation model which cannot be directly measured [16]. The heat transfer coefficients incorporate the thermal resistance at the copper plate and belt interface, the glass fiber belt, the coating layer on the belt and the air gap between the belt and the solidifying strip which has a strong insulating effect [17].

The modelling of the process was simplified in several ways. The gap formation due to shrinkage and gravitation is not taken into account directly as the heat transfer is the same at the top and bottom of the cast. Another simplification was to eliminate the glass fiber belts and to neglect their movement relatively to the liquid melt. This hardly affects the thermal conditions of the strip or the copper cooling plates because liquid melt is limited to just a few hundred millimetres from the nozzle inlet [17]. But to achieve a correct heat removal rate from the strip the belts were considered thermally by heat transfer coefficients at the newly created strip/copper plate interface.

Figure 3 shows the meshed simulation model. The blue mesh covers the casting box which is linked to the casting nozzle (green mesh) to deliver fresh melt as smooth as possible. The pink mesh discretizes the area of the copper cooling plates and the yellow one the compound casting unit. The mesh is refined at the beginning of the cooling plates and beneath the compound casting unit where the highest solidification rates occur. Due to its axial symmetry only one half of the belt caster was simulated to reduce calculation time.

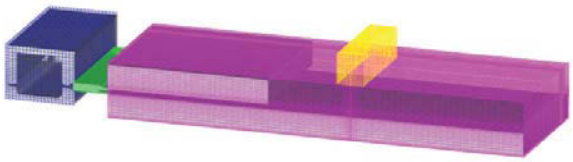


Figure 3. Meshed simulation model.

Simulation Model Calibration with Measured Data

Various temperature measurements in the casting strip and in the copper cooling plate of an existing single strip belt casting machine were performed to calibrate the simulation model and to define a realistic temperature dependent heat transfer coefficient at the strip/copper plate interface (Figure 4). The coefficient is relatively low even at temperatures at which the melt is in liquid phase due to the strong insulating effect of the glass fiber belt. The strip shrinkage during solidification leads to a strong decrease of the heat transfer coefficient.

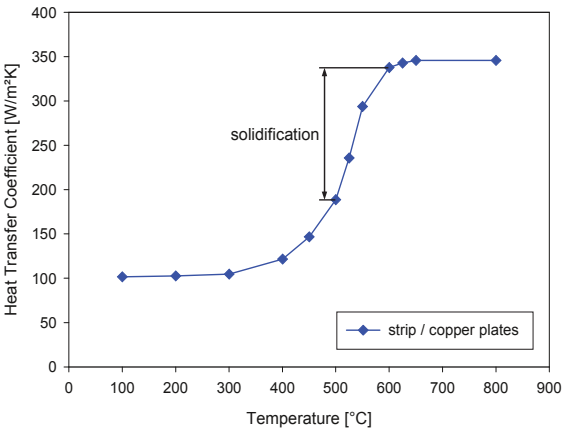


Figure 4. Temperature dependent heat transfer coefficient at the strip/copper plate interface.

For the measurement of the strip temperature along the entire casting length, Type-K thermocouples with a diameter of 1.5 mm and a length of several meters were carried along with the casting strip. The thermocouple was inserted through a geometrically modified casting nozzle and pulled into the liquid melt between the moving belts. As soon as the strip solidified after a certain distance from the casting nozzle, the thermocouple was entrained by the strip and moves at the same speed. The feed of the thermocouple along a stationary scale was recorded on video to link the measured temperature data with the exact position of the

thermocouple in the strip. Afterwards, the casted strip was cut to check the measuring position of the thermocouple in thickness direction [18]. Figure 5 shows measured and calculated temperature profiles in cast aluminium strips with different compositions; they are in noticeable good agreement over the whole casting length. For both alloys the same simulation parameters and heat transfer coefficients were used. The calibration with strip temperatures is essential for the compound casting model.

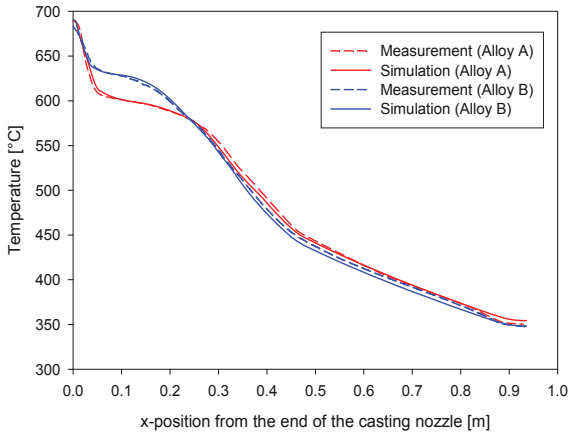


Figure 5. Measured and calculated temperature profiles of substrate strips.

In addition the temperature distribution of the lower water-cooled copper plate was measured at 32 temperature measurement points just below the surface. The points are at center axis, at the quarter width and at the margins of the copper plate. The highest temperatures were measured in the center of the plate with decreasing temperature towards the margins. Figure 6 illustrates the measured and calculated steady state temperature profiles with a maximum discrepancy of approximately 8 °C at the inlet area of the casting nozzle at center axis.

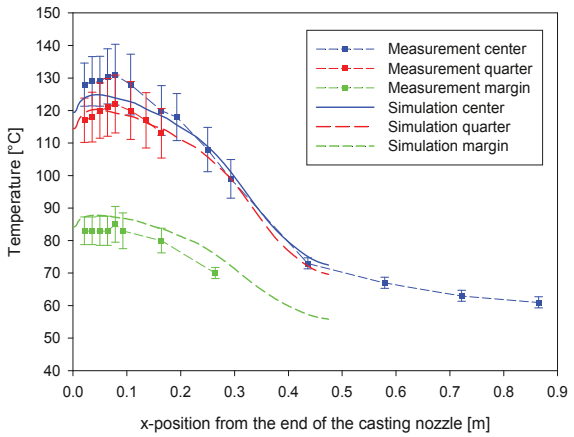


Figure 6. Measured and calculated temperature distribution in the lower copper cooling plate.

Results of numerical variations of casting parameters during single strip casting

To define the main process parameters on the substrate temperature at given strip dimensions the following process parameters were varied:

- Casting speed:
  (0.4 m/min; 0.5 m/min; 0.6 m/min)
- The heat transfer at the strip/copper plate interface:
  (± 10 %; ± 20 %)
- The amount of cooling water in the copper cooling plates:
  800 W/m²K; 10,000 W/m²K (expressed in a heat transfer coefficient between copper plate and cooling water)
- Cooling water temperature:
  (15 °C; 25 °C; 35 °C).

The casting speed as well as the variation of heat transfer at the interface have the strongest influence on the substrate strip temperature among all examined parameters within their simulated ranges. Figure 7 shows the computed strip temperature distribution for three different casting speeds at steady state conditions. The temperature difference between the lowest and highest casting speed can reach up to 100 °C, making the casting speed the most sensitive casting parameter. Also the variation of the heat transfer coefficient of the strip/copper plate interface between -20 % and +20 % has a significant influence on the strip temperature and effects temperature differences up to 80 °C. Possible causes of variable heat transfer coefficient can be an irregular application of the belt coating layer, bad belt guiding or large shrinkage of the solidifying strip.

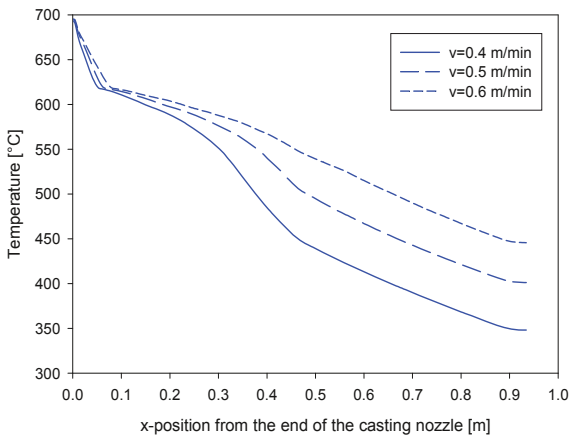


Figure 7. Simulated variation of the casting speed during single strip casting.

Different strip temperature profiles at varying cooling water temperatures and flow quantities at steady state are illustrated in Figure 8. The heat transfer coefficient of 800 W/m²K is equal to a low cooling performance with a flow rate of 1 l/min at a cooling channel diameter of 0.01 m and 10,000 W/m²K means a high one with 11 l/min respectively. The small temperature differences show that the amount of cooling water and even less the temperature of the cooling water for cooling the copper plates have only little influence on the strip temperature.

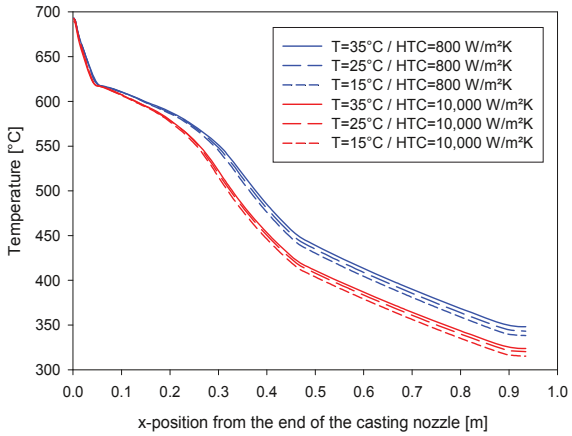


Figure 8. Simulated variation of the amount of cooling water and the cooling water temperature during single strip casting.

Results of the numerical compound belt casting process

Figure 9 and Figure 10 show the simulation model with computed temperature distribution and solid fraction at steady state during compound casting. The substrate alloy leaves the casting nozzle at a temperature of approximately 700 °C. Between the copper plates the substrate strip starts solidifying and cools down according to the cooling effects of the copper cooling plates. After leaving the upper cooling plate the strip keeps cooling, moving towards the compound casting unit. Through a vertical slot the clad alloy with a temperature of 700 °C is cast onto the substrate strip and heats up the substrate strip again. As the heat transfer between the two layers is assumed to be almost ideal the temperature differences balance out very fast. The clad alloy which gets in direct contact with the substrate strip solidifies immediately due to the lower temperature of the substrate strip as can be seen in Figure 10. At the end of the lower copper cooling plate the compound strip is completely solidified for further processing.

Figure 11 shows the temperature profile at the upper surface of the substrate strip. The left vertical line indicates the end of the upper cooling plate; at the vicinity of the compound casting unit the temperature raises again. The change amounts 120 °C for 0.4 m/min, 76 °C for 0.5 m/min and 50 °C for 0.6 m/min. The maximum temperature at the position of the right vertical line is crucial to obtain a sound compound strip. If the heat flux into the substrate strip surface is high enough, the surface will partly remelt forming a compound. The maximum temperature at the position of the compound casting unit depends directly on the substrate strip surface temperature before compound casting.

Figure 9. Temperature distribution during compound casting at steady state.

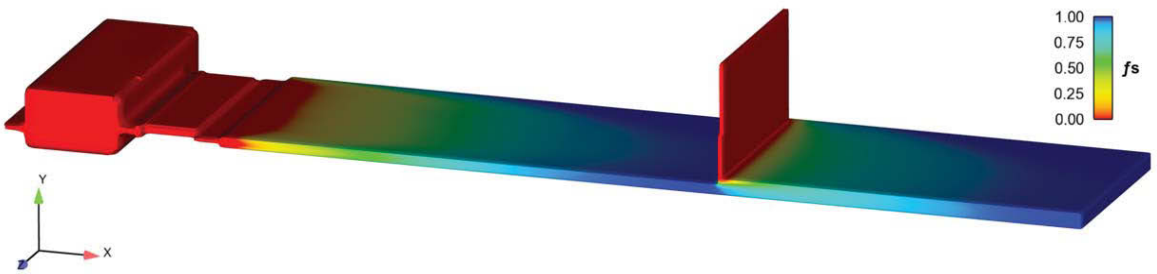


Figure 10. Solid fraction during compound casting at steady state.

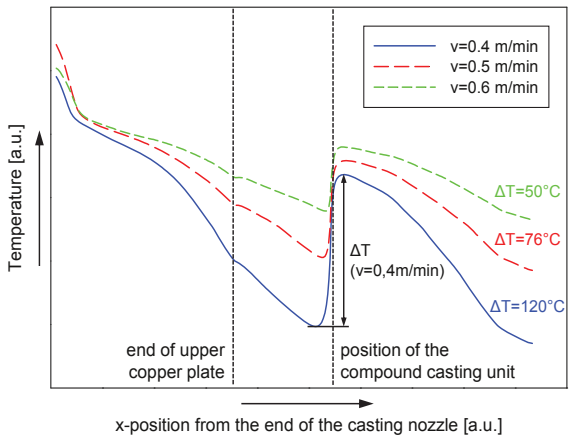


Figure 11. Temperature of the upper surface of the substrate strip during compound casting.

Conclusions

- Necessary metallurgical and thermal conditions to form a compound between different aluminium alloys were initially investigated with a small-scale compound casting test plant.
- A 3D numerical simulation model of a full-scale compound belt casting process was developed. Casting conditions are predicted to control the substrate strip temperature at the compound casting unit position. To ensure a high compound quality, the (maximum) temperatures at the bonding interface were determined. Furthermore, the thermal effects of geometrical variations of the compound strip e.g. the width, thickness or the thickness ratio of substrate and clad alloy can be investigated.
- Temperature measurements in the casting strip and in the lower copper cooling plate of an existing single strip belt casting machine were used for calibration.
- Among the examined variations of casting parameters the casting speed is the most influencing and sensitive parameter on the substrate strip temperature. The effect of variable cooling water temperatures in the copper cooling plates is negligible.

Acknowledgement

This work is financially supported by the Austrian Research Promotion Agency (FFG), Vienna, within the project no. 838821.

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CRYSTALLOGRAPHIC TEXTURE DEVELOPMENT OF AS-CAST 3105 ALLOY PRODUCED BY St/Cu SHELL PAIR

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Keywords: copper shell, twin roll casting, texture

Abstract

Altering operational casting parameters has limited effect on increasing productivity of twin roll cast (TRC) alloys. Further improvement can be accomplished by enhanced heat transfer between shell material and solidifying metal. In that respect copper alloys are commercially utilized in twin roll casting technology. However, it can be easily predicted that microstructure and texture development would be different than those produced by conventional steel shells. Present study investigates crystallographic texture development of as-cast 3105 alloy produced by St/Cu shell pair. Pole figures were employed to characterize the dominant texture. Complimentary studies were performed by macro and microstructural investigations.

Introduction

Physical and mechanical properties of the caster rolls have substantial impact on heat extraction from the liquid metal and attaining required geometrical features (crown) of the as-cast sheet. Caster roll is composed of two components, namely core and shell. While core provides structural integrity and stability of the system with its design features, basic function of the shell is to extract heat from the liquid metal introduced in between the roll gap and then transfer it to cooling media circulating at the interface between core and shell in cooling groves. Shell is assembled to core with shrink fitting operation and as its diameter reduces to a critical one with repeated turning and grinding operations after each casting campaigns, it is replaced with the new one. Producer companies offer different grades of caster shells varying in chemical compositions and properties. However the base material for conventional caster shells has not changed for years, which is steel. Heat conductivity of the shell cannot be readily changed without taking other requirements into account. A perfect compromise between mechanical and thermal properties is essential for achieving its lowest consumption rate and highest possible conductivity. Due to the priorities of the companies, this balance can be changed in favor of one or another. Preference is either to obtain remarkable increase in productivity figures with optimized consumption rate or putting up with traditional productivity figures with much longer shell life.

Its high thermal conductivity makes copper alloys a preferred candidate for caster shell material. Low mechanical properties of copper is increased with alloying elements, such as Beryllium and Cobalt. Deterioration in thermal conductivity with alloying element is unavoidable but it is still much higher than that of steel shells.

More heat flux from liquid metal to shell in the course of solidification process by employing copper shell is expected to result in different microstructural constituents and features to be generated. TRC is a casting method having one of the highest solidification rate in industrial scale production method of commercial aluminum alloys. Use of copper shell brings it to another scale in the solidification but with some altered material characteristics. Present study investigates the crystallographic texture development of a commercial aluminum alloy of 3105 cast by employing copper-steel shell pair. They are represented with the pole figures and analysis were conducted on both surfaces. Despite of using only one copper shell, average increase in productivity figure (ton/h/m) was between 50-60%. This is an optimized productivity figure to obtain ultimate quality for the whole quality spectrum of end products. Along with the findings on crystallographic texture development on both surfaces, microstructural constituents, grain size, intermetallic particle size/distribution, hardness profile through the thickness were investigated for both surfaces exposed to different solidification paths. Results were compared with those obtained from standard production of the same alloy by using steel/steel pair.

Experimental

Commercial grade 3105 aluminum alloy was produced in industrial scale twin roll caster by using Steel/Copper shell pair (St/Cu). Reference materials compared with in characterization methods were samples produced out of Steel/Steel shell pair (St/St). The convention to identify the samples is; “Sample A” designates the samples produced out of St/Cu shell pair, “Sample B” was used to designate sample of St/St shell pair. All the samples are representative of the real scale coils and related production parameters. The thickness of Sample A and Sample B are 6.5mm and 6 mm, respectively. Microstructural investigations were carried out on polished and polished+electrochemically etched samples to reveal distribution of intermetallic particles and grain structure. Optical microscope (Zeiss AxioTech 100) and SEM (JEOL 5600) were utilized for microstructural characterization. Through thickness hardness profile of the samples were determined with microhardness measurements by using 100 g load for 10 sec. As-cast samples were exposed to recrystallization anneal at the temperature of 480°C for 5 hours to observe recrystallization behavior of the samples at the outermost skin. Variation in grain structure through the thickness was reported. Complementary studies were also carried out to understand the saturation of alloying elements in the vicinity of the free surface.

Macroscopic texture analysis of the samples were performed by using PANalytical X’Pert Pro operated at 45 kV and 40 mA on slightly polished surfaces of the specimens. Results were evaluated from Al(111), Al(200) and Al(220) pole figures.

Results and Discussion

Figure 1 gives the grain structures across the strip thickness in the casting direction of both samples. Figure 1b shows a classical TRC as cast grain structure where size, shape and orientations of the grains changes symmetrically from both surfaces towards to middle. The outermost layers are composed of very fine grains and long fine oriented grains takes place in subsurface regions whereas quasi equiaxed grains become common in the middle of the strip. However, as can be seen in Figure 1a, this symmetrical structure changed when St/Cu shell pair was used.

Shifting of centre line segregation towards to steel side and changes in grain structure morphology at outermost layer contacting with copper side are the indicators of this observation. This finding can be attributed to the higher heat transfer capability of copper shell than that of steel shell.

Figure 2 shows grain structures and intermetallic particle distributions of outermost layers of Sample A. As it can clearly be seen in Figure 2, copper side exhibits larger and equiaxed grains as well as less and finer intermetallics when compared to those of steel side. This observation can be explained by higher solidification rate on copper side than that of steel side. Higher solidification rate inhibits precipitation of intermetallics leading to a more supersaturated matrix. Owing to this fact, there are less obstacles hindering grain growth which results in larger grains. BSE SEM images of outermost layers of Sample A which shows differences between intermetallic size and distribution are given in Figure 3.

Microstructural investigations of outermost layers of annealed sample revealed that steel side completed recrystallization and even passed to coarsening phase while the recrystallization has just started on copper side (Figure 4). This partial recrystallization is a consequence of precipitation of dispersoids from a highly supersaturated matrix which is known as Zener Drag phenomena [1]. The size and distribution of intermetallic particles have a fundamental role not only in the recrystallization kinetics of the alloy, but also in controlling its microstructure evolution during downstream processes. Fine dispersoids in nanoscale formed during annealing exert a dragging force on subgrain and grains which retards recrystallization to higher temperatures [2].

Figure 4 shows the grain structures of Sample A after recrystallization annealing performed at as-cast thickness. Grain growth at the outermost skin after annealing is a typical phenomenon for TRC materials, regardless of alloy composition. Whilst sheet surface in contact with steel side support this typical observation, grains size of copper side after annealing contradicts with it. There is almost no gradient in grain size from the surface to the center plane in as-cast material and they are almost in cellular geometry, contrary to well-known historical data of "very fast" solidification rate. However, repeated castings by employing Cu shell in industrial scale productions consistently confirmed this observation.

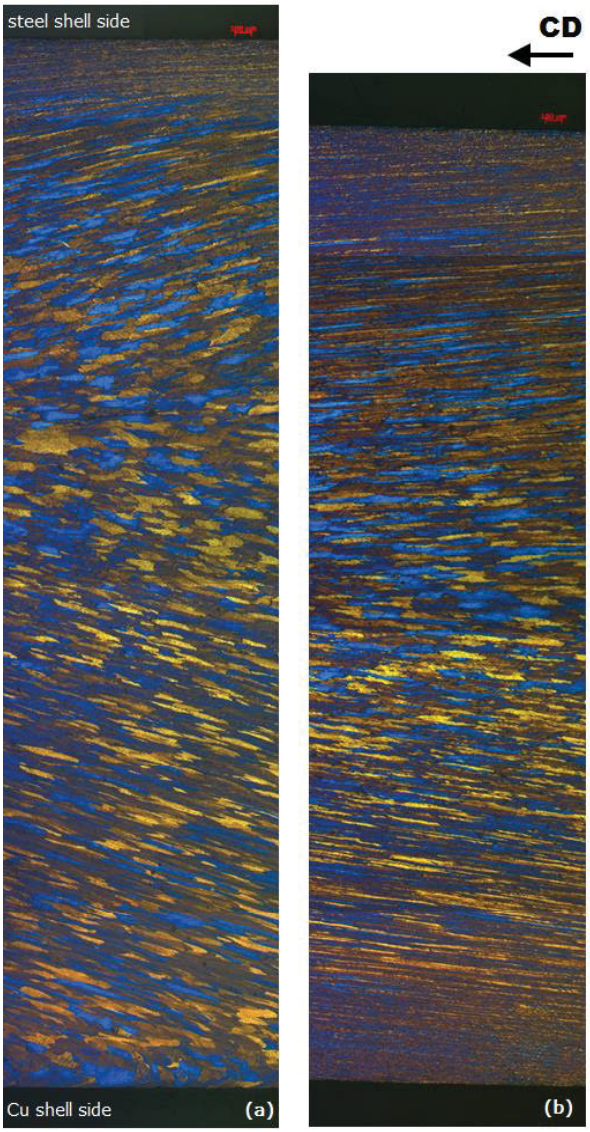

Figure 1. As-cast grain structure- cross section. (a) Sample A (b) Sample B

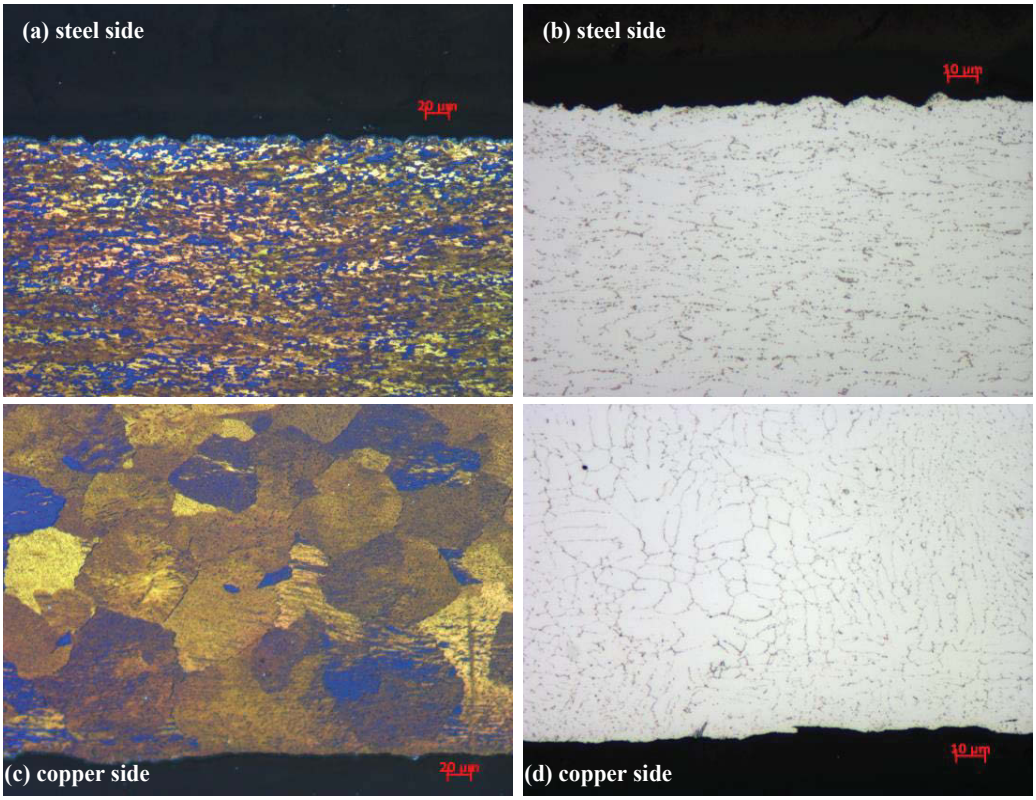

Figure 2. Grain structures and intermetallic particle distributions of outermost layers of Sample A- (cross section-transverse to casting direction)

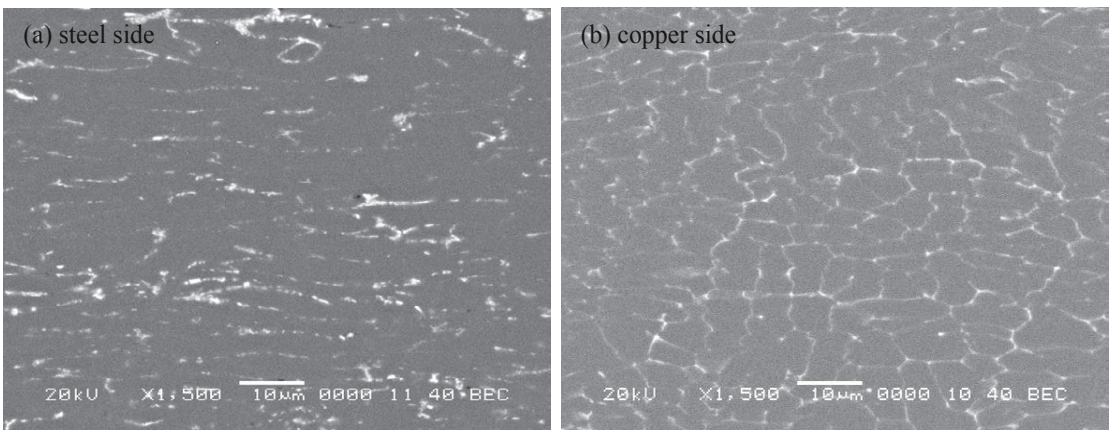

Figure 3. BSE images of the outermost layers of Sample A.

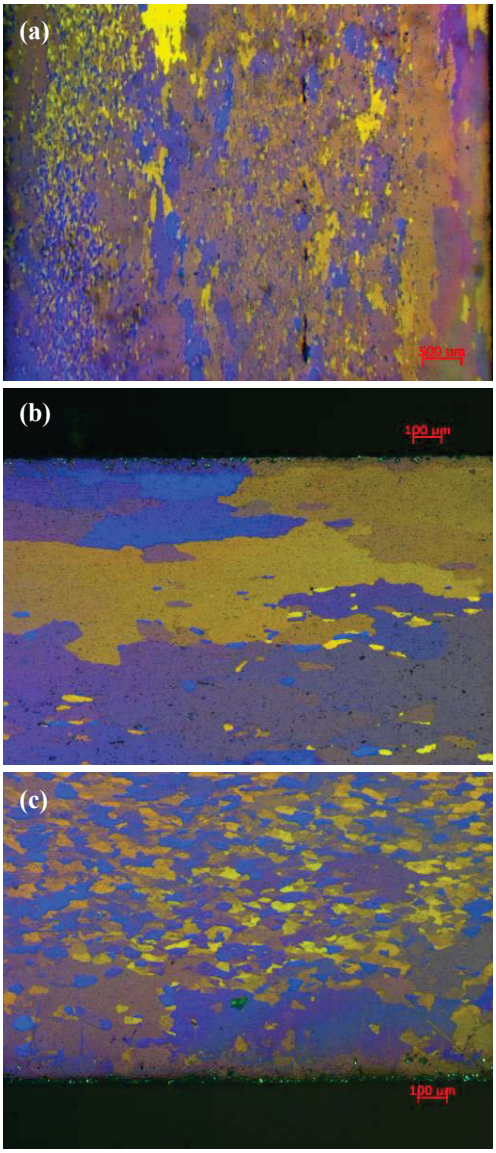

Figure 4. Grain structure after annealing of Sample A-transverse to casting direction (a) whole cross section (b) steel side (c) copper side

To endorse above explanations electrical conductivity measurements were conducted on as-cast and annealed samples since it was not possible to observe fine dispersoids by optical microscope and SEM. As it is well known, electrical conductivity decreases with increasing supersaturation of matrix [3-5]. For as-cast condition, the minimum conductivity value was measured on copper side of Sample A which support that it has the highest supersaturation level. Maximum conductivity difference between as-cast and annealed condition was found on copper side of Sample A, which indicates higher potential of copper side for precipitation. Measured conductivity values were given in Table I.

Mechanical properties and results of hardness measurements performed on both surfaces of as-cast samples were summarized in Table II and Table III, respectively. Sample A exhibited higher yield and tensile strength and hardness values with respect to Sample B. Hardness measurements also revealed that there was a remarkable difference between the surfaces of the Sample A. Hardness of copper side is higher than that of steel side as expected. This can also clearly be seen on through thickness hardness profiles of samples which is given in Figure 5. Results are congruent with microstructural findings: Higher hardness values on copper side is due to enhanced cooling rate leading to a higher supersaturation.

Table I. Electrical conductiRevity values of samples

Sample		Electrical Conductivity (MS/m)	
		As-cast	Annealed
A	top surface steel side	24.1	28.6
	bottom surface copper side	23.3	29.8
B	top surface	24.6	28.9
	bottom surface	24.7	28.8

Table II. Mechanical properties of the as-cast samples

	Yield Strength, MPa	Tensile Strength, MPa	Elongation (A100) %
Sample A	133	165	9,6
Sample B	108	143	10,3

Table III. Brinell hardness values of as-cast samples

Sample		Hardness (HB)
A	top surface steel side	50
	bottom surface copper side	59
B	top surface	45
	bottom surface	44

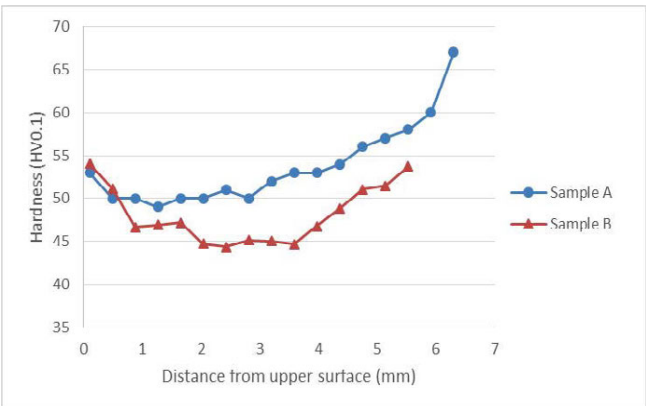

Figure 5. Hardness profiles through the thickness of as-cast samples

Samples were slightly polished to remove topographical features imprinted by the shell surface. Crystallographic texture of the surfaces were analyzed with Al(111) and Al(200) pole figures. Both surface of sample A clearly show the presence of two dominant texture, they are [100] <*uvw*> and [111] <*uvw*>. These results are in accord with the findings of Hunt, Meredith and Gras [6]. However, remarkable difference is that the intensity of [100] <*uvw*> shear texture is higher on the surface that is in contact with steel shell than that of surface contacting with copper shell. This is also confirmed with the Al(200) pole figures of the same surfaces. The pole figures obtained from both surface of sample B show strong [100] <*uvw*> texture that is similar to observation on steel side of sample A. Solidification is not the only physical transformation at the roll gap. Fraction of solid at the gap is determined by the casting parameters. Upon solidification is established, solid fraction in the roll gap increases in casting direction and solidified metal begins to be deformed until reaching the aimed casting thickness. During this particular course of casting, some of the deformation mechanisms, similar to those of cold rolling, are operative within limited thickness of the as-cast sheet surface. The deformation exerted by the rolls creates a deformation gradient barely reaching to the center plane of the sheet. However, shear deformation between the shell and free surface of the sheet prevails starting from neutral point of deformation to the exit side of the rolls. Thus, detecting a shear texture component on the surface of TRC material is reasonable and can be attributed to the deformation component of the casting process. Regardless of type of caster shell, same plausible explanation can be proposed. However, significantly higher intensity of [100] <*uvw*> texture on steel side and identical observations reported for steel/steel pair should be interpreted by coupling with some features of solidified structure. They are; differences in degree of supersaturation of some elements due to heat extraction on both surfaces, size and distribution of microstructural constituents formed.

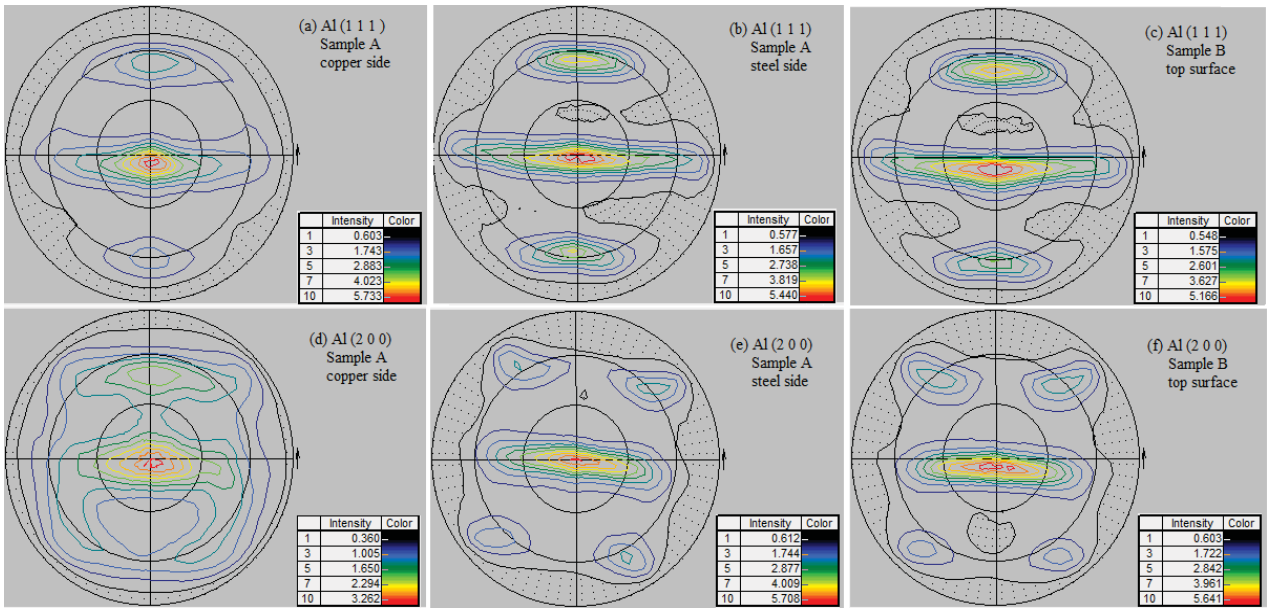

Figure 6. (1 1 1) and (2 0 0) pole figures of samples.

Conclusions

The results of this study can be summarized as follows:

CLS was found to be shifted towards steel side due to higher heat extraction capability of copper shell.

Microstructural constituents such as intermetallics and grain structure are quite different between surfaces contacting with each shells. Whilst grains of copper side are equiaxed and larger, steel side exhibits a featureless zone with quite fine grains. This can be attributed to difference in cooling rates dictated by thermal conductivities of shell materials.

As verified by electrical conductivity measurements, metal solidifying on copper side exhibits higher solute level with respect to steel side. Thus, hardness of copper side was found remarkably higher than that of steel side.

Enrichment of solute elements on copper side is thought to exert a dragging force (Zener Drag) on grain boundry movements which retards recrystallization to higher temperatures during annealing.

It is clear that not only due to the plane strain deformation exerted by the rolls but also shear deformation between the roll and the sheet determines the type of dominant texture which is in favor of shear on steel side in this particular case.

Acknowledgement

The Authors would like to thank Mr. Nihat Yılmaz, technician of Assan Aluminum Laboratory, for his valuable help in sample preparation for metallographic investigations.

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Light Metals 2016
Edited by: Edward Williams
TMS (The Minerals, Metals & Materials Society), 2016

ANNEALING CURVE OF 3105 ALLOY PRODUCED BY TWIN ROLL AND BELT CASTING METHOD

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Keyword: 3105 alloy, Twin roll casting, twin belt casting, annealing curve, work hardening curve

Abstract

Twin roll and twin belt casting method are widely used in the aluminum industry for the production of 3105 alloy for painted products (mainly architectural products). Work hardening curve and final annealing data of 3105 alloy produced from the two casting methods are presented in the paper.

Recrystallization annealing was carried out at intermediate cold rolling gauges to study the effect of cold rolling reduction on the recrystallized grain size of 3105 alloy produced by both casting methods. The different casting method results in a different work hardening rate and final annealing response of 3105 alloy.

1. Introduction

The two basic strip casting methods in aluminum production are twin roll casting method (TRC) and the twin belt casting method (TBC).
In the TRC method liquid metal is fed through a ceramic nozzle into a converging cavity set by two internally cooled rotating rolls.

In the TBC method the liquid metal is feed through a ceramic nozzle into a near parallel cavity comprised of internally water cooled counter rotating belts.
In the TBC method, in- line mill stands at the caster exit are installed in order to hot roll the slab in coiling gauge around 1 to 2 mm (see Fig 1)..

In TBC method, the solidification completes over a longer distance (long sump depth) relative to the TRC method (short sump depth)(Fig 2). The cooling rate of TBC method is less than 100 Ks^{-1}, when the cooling rate of TRC method is around 1000 Ks^{-1}.

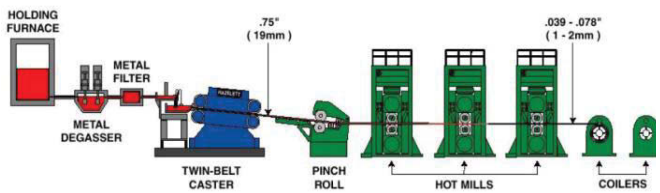


Fig 1: Hazellet twin belt caster configuration with in line 3 stand hot mill at the caster exit [1].

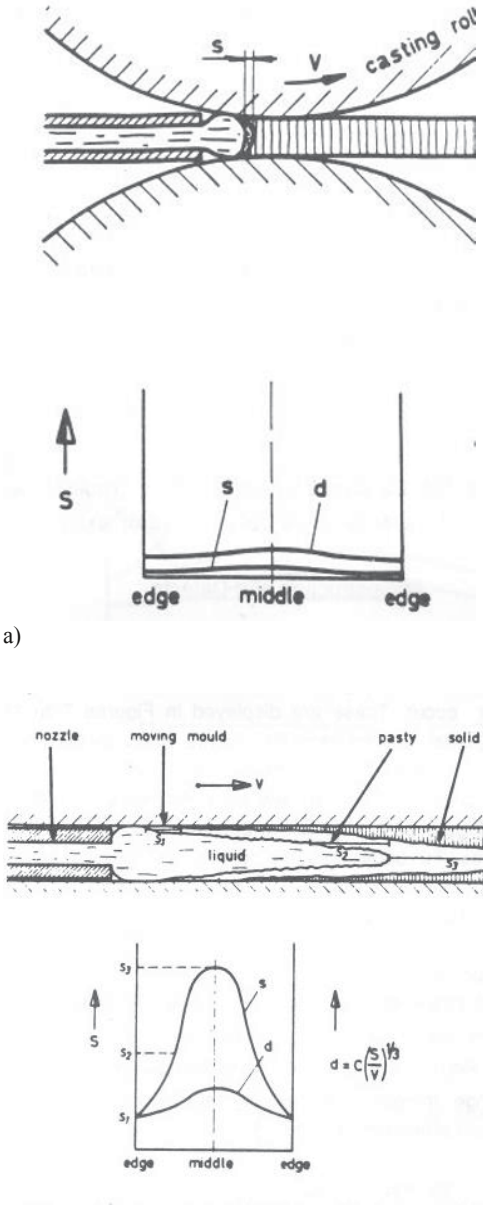


Fig 2: a) short sump depth of TRC and b) long sump depth of TBC method [2]

2. Experimental procedures

Paint stock coil of 3105 alloy were cast in a Hazellet Twin Belt Caster (TBC). 3105 TBC coils incoming gauge in the cold mill is 2,5 mm. The coils were further cold rolled at final gauge of 0,4 mm without intermediate annealing heat treatment.

Coils of 3105 alloy were cast at Fata Hunter twin roll caster at exit gauge of 5,5 mm. The 3105 alloy cast coils were further cold rolled at final gauge of 0,4 mm without intermediate annealing heat treatment.

Full width as rolled samples collected at the initial, intermediate and final cold rolling gauges.

Tensile testing of the as cold rolled samples was carried out in Instrom 5582 tensile test machine. Tensile test specimen axis is parallel to the rolling direction

Laboratory annealing test for fully recrystallization was carried out in all the cold rolled samples in Nabertherm furnace with constant heating rate of 20 ^{0}C per hours and soak time at peak temperature of 3 hours.
Laboratory annealing test at different annealing temperatures were carried out at final gauge of 0,4 mm for both 3105 TRC and TBC method in order to determine the temper annealing curve.

The lab fully annealed samples surface was etched using Tucker's reagent (mixture of acids) to reveal the surface grain size. The surface average grain size was determined under the light optical stereoscope using the linear intercept method. Large length lines draw randomly on the surface grain structure and count the number of intersection of the line and the grain boundaries per line length.

The grain structure of annealed material was examined under the light optical microscope after metallography preparation of cross sections parallel to the rolling direction and electropolishing with Barker's solution.

Metallography preparation steps includes 3 grinding step with SiC papers and finally two polishing step with polishing media of 3µm diamond suspension and colloidal silica.

The as rolled samples conductivity was determine using SIGMATEST D2.068 electrical conductivity measurement device using a frequency of 120 kHz and straight probe with a temperature sensor.

3. Work hardening curves of 3105 alloy produced by TRC and TBC method

Figures 3 and 4 illustrates the work hardening curves (yield strength and tensile strength values as a function of rolling strain) for the material produced by TRC and TBC method. .

At initial low rolling strains the work hardening curve of 3105 alloy TRC method shows around 40 MPa higher Yield and Ultimate tensile strength values relative to the work hardening curve of 3105 alloy produced by TBC method.
At higher rolling strains values, the strength difference of the work hardening curves is reduced significantly (difference around 10 MPa).

Therefore we can conclude that 3105 alloy produced by TRC method shows a higher work hardening rate, especially at low rolling strains values, relative to the 3105 alloy produced by TBC method.

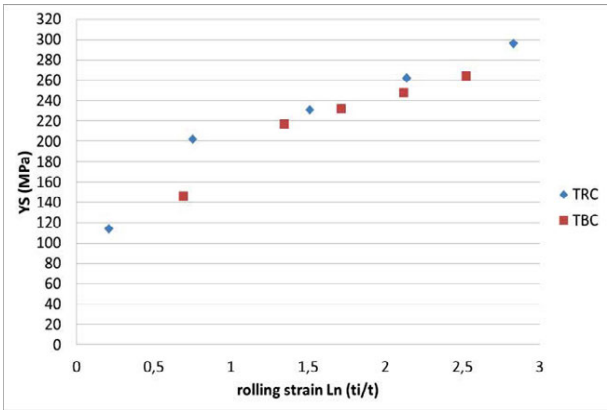


Fig 3: Diagram of Yield strength versus the rolling strain for alloy 3105 produced by TRC and TBC method.

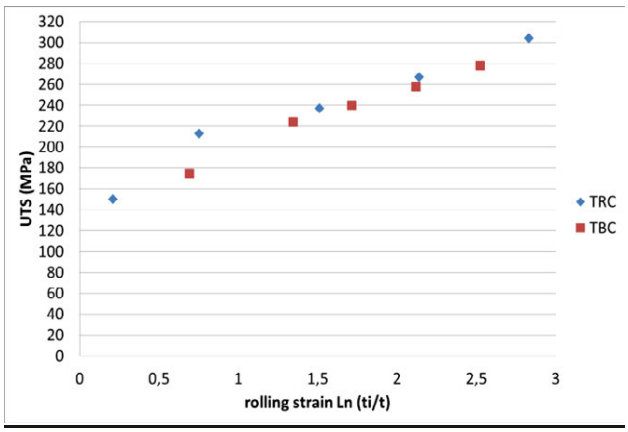


Fig 4: Diagram of tensile strength versus the cold rolling strain for alloy 3105 produced by TRC and TBC method.

4. Annealing Curves of 3105 Alloy produced by TRC and TBC method

Figure 5 and **6** shows the annealing curves of 3105 alloy at the final gauge of 0,4 mm for both material produced by TRC and TBC method.
The annealing curve of 3105 alloy produced by TRC method is displaced at higher strength values relative to the 3105 alloy produced by TBC method.

Recrystallization start temperature could be considered as the annealing temperature where the annealing curve slope becomes very steep (rapid drop of mechanical properties).
According to the annealing curves, alloy 3105 produced by TRC start to recrystallize at temperature of 320 ^{0}C.
The 3105 alloys TBC start to recrystallize at temperature of 300 ^{0}C.
The above findings suggests that the alloy 3105 produced by TRC method is more resistance to the recovery and recrystallization softening mechanisms compare to 3105 alloy produced by TBC method.

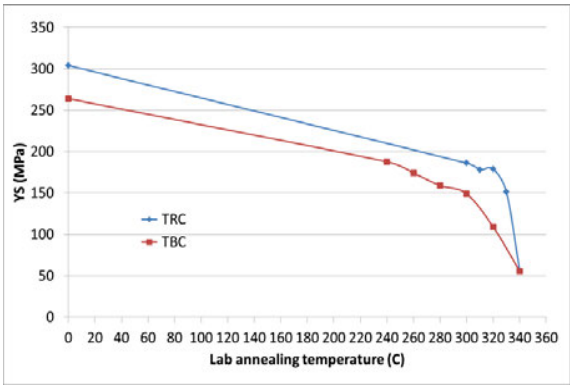


Fig 5: Diagram of Yield strength versus lab annealing temperature for alloy 3105 produced by TRC and TBC method.

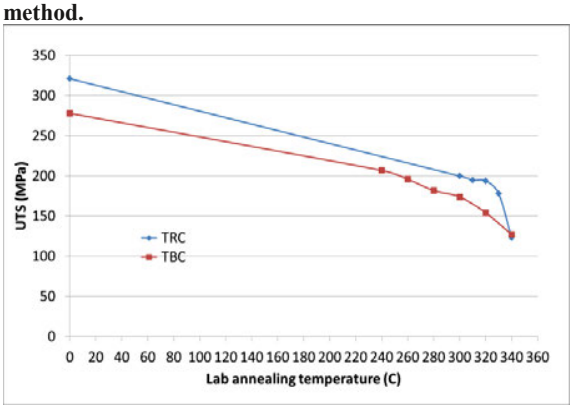


Fig 6: Diagram of tensile strength versus lab annealing temperature for alloy 3105 produced by TRC and TBC method.

5. Surface grain size of full anneal 3105 alloy at intermediate and final cold rolling gauges

Fig 7 diagram shows the average surface grain size of fully recrystallized lab annealed samples as a function of the rolling strain. The alloy 3105 produced by TBC method has an increase of the average grain size by increasing the rolling strain (initial Grain Size of 250 μm increase up to 600 μm).
In the case of 3105 TRC, the increase of the rolling strain results in reduction of the average surface grains size (initial grain size of 2.500 μm reduced down to 500 μm).

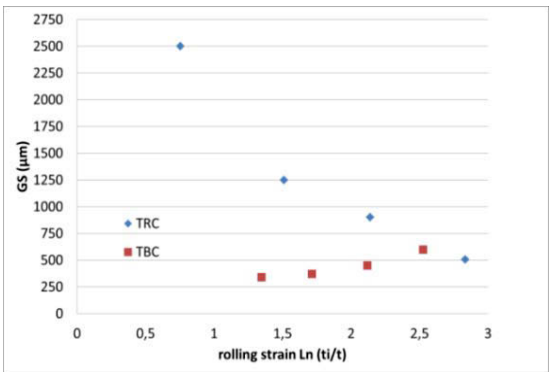


Fig 7: Diagram of average grain size versus cold rolling strain for alloy 3105 produced by TRC and TBC method

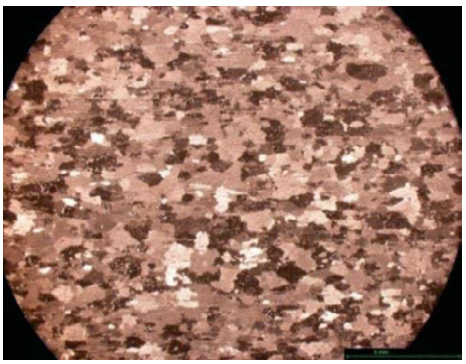


Fig 8: **Stereoscope image of full anneal surface Grains. Gauge :1,3 mm. 3105 TBC.**

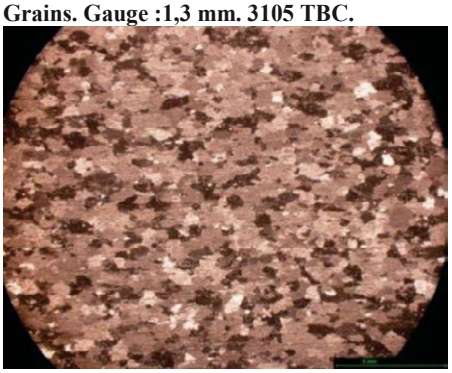
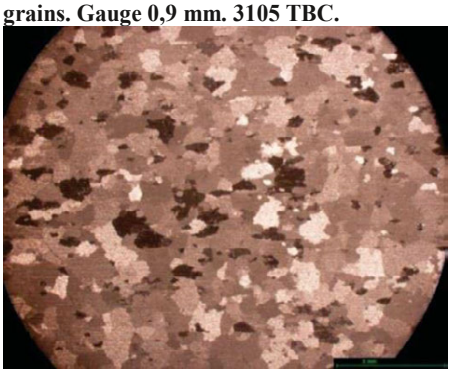


Fig 9: Stereoscope image of full anneal surface grains. Gauge 0,9 mm. 3105 TBC.



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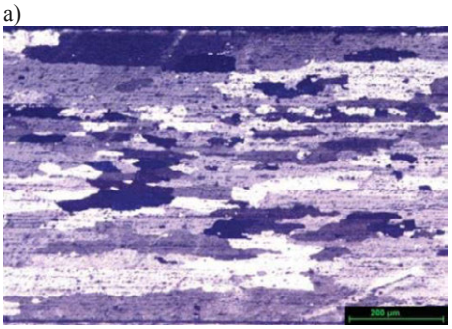

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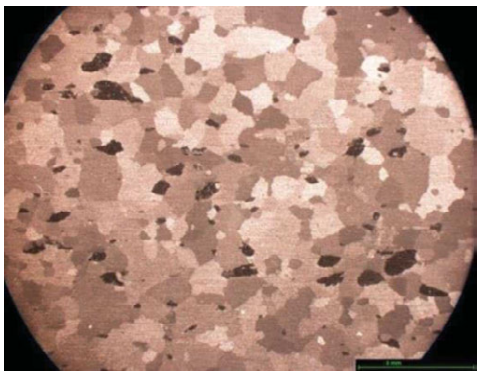
Fig 10: a) stereoscope image of full anneal sample surface grains b) optical microscope image of full anneal grains in cross section. Gauge 0,6 mm.3105 TBC



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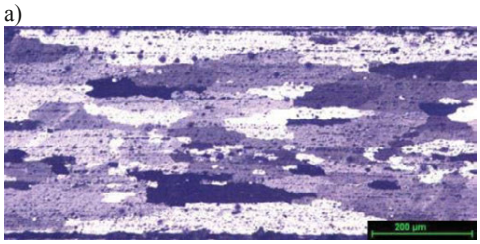


b)

Fig 11: a) stereoscope image of full anneal sample surface grains b) optical microscope image of full anneal grains in cross section Gauge 0,4 mm.3105 TBC.

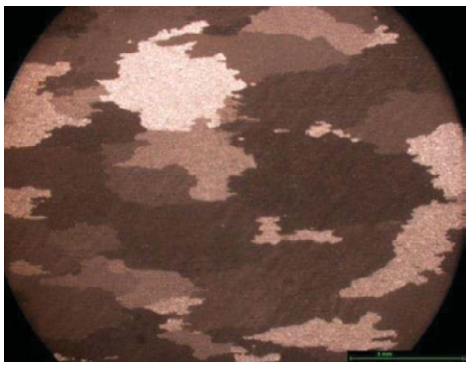


Figure 12: Stereoscope image full anneal surface Grains. Gauge :3,2 mm. 3105 TRC.

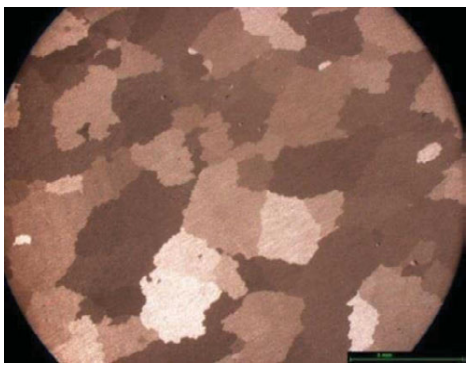
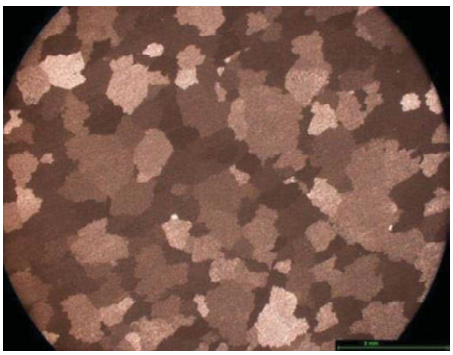


Fig 12:Stereoscope image of full anneal surface Grains. Gauge :1,7 mm. 3105 TRC.



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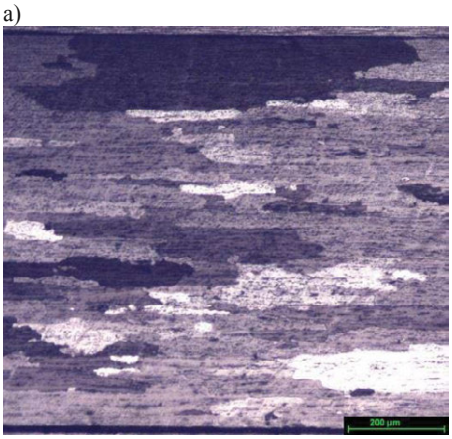

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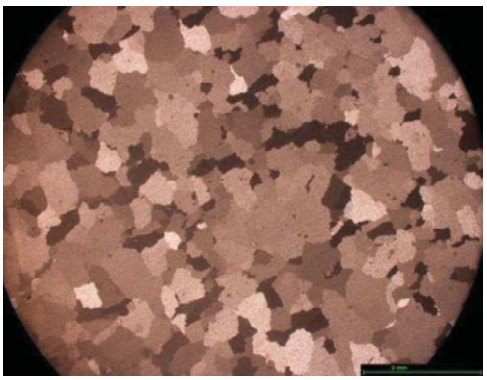
Fig 13: a) stereoscope image of full anneal sample Surface grains b) optical microscope image of full anneal grains in cross section Gauge :0,8 mm.3105 TRC.



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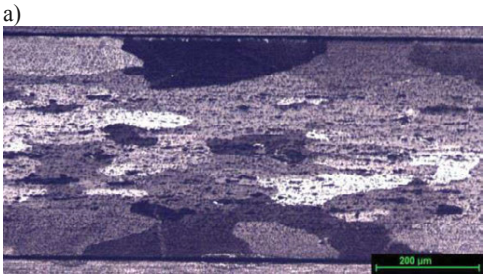

b)

Fig 14: a) stereoscope image of full anneal sample surface grains b) optical microscope image of full anneal grains in cross section.Gauge:0.4 mm. 3105 TRC.

6. Conductivity measurements at intermediate cold rolling gauges and final gauge

Fig 15 graph shows the conductivity values of 3105 alloy produced by TBC method at each cold rolling gauge.
Fig 16 graph shows the conductivity values of 3105 alloy produced by TRC method at each cold rolling gauge.
The alloy 3105 produced by TRC method exhibits a significantly lower conductivity values (23,86MS/m) relative to the 3105 alloy produced by TBC method (25,95).

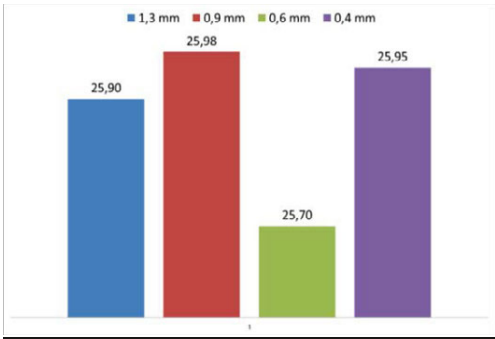

Figure 15: Conductivity (MS/m) values of 3105 TBC method sheet at various cold rolling gauge.

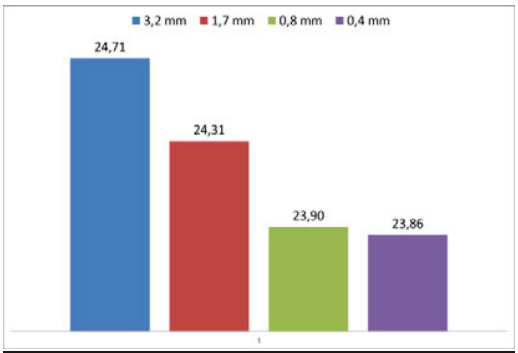

Figure 16: Conductivity (MS/m) values of 3105 TRC method sheet at various cold rolling gauge.

7. Discussion and final conclusions

The significantly lower conductivity values measured in all the cold rolling gauges of 3105 TRC material compare with the 3105 TBC methods indicates a higher amount of alloying element dissolve in the aluminum matrix solid solution.

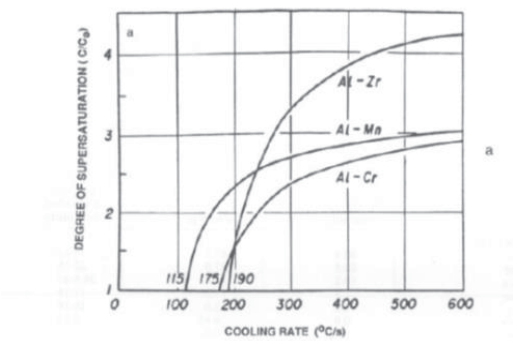

Fig 17: **Degree of supersaturation versus the cooling rate for the binary systems Al-Zr, Al-Mn and Al-Cr [2]**

According to **Fig 17**, the supersaturation degree initially increase rapidly with cooling rate but eventually plateaus, further increase in cooling rate brings marginally increase in the degree of superstation [3].
Therefore the higher cooling rates of TRC method relative to TBC method cause a significantly higher degree of alloying element supersaturation.

The higher degree of supersaturation of alloying elements in the case of 3105 alloy produced by TRC method relative to the TBC method has the following effects in the thermomechanical process of the material:

1. **High work hardening rate** especially at the initial rolling strains (40 MPa higher strength value of 3105 alloy produced by TRC method relative to 3105 alloy produced by TBC method)

2. **Displacement of the final annealing curve at the higher strength levels**. The higher amount of alloying element into the solid solution reduces mechanical properties loss due to the recovery and increase the recrystallization temperature. Recrystallization start at 320 ^{0}C for 3105 TRC material and 300 ^{0}C for 3105 TBC material.

The recrystallized surface grain size of the full recrystallized 3105 TRC method is reduced by the increase of rolling strain.
In the case of 3105 TBC alloy, the recrystallized surface grain size increase by increasing the rolling strain.
Such recrystallization could be considered "anomalous" because is not following the typical recrystallization law which states that grain size depends primarily on the amount of deformation, being smaller for large amounts of deformation [4].

According to E.Nes, the "anomalous" recrystallization behavior observed in alloy 3105 produced by TBC method is caused by a surface layer crystallographic texture. When the hot rolled band material is unrecrystallized, a strong shearing texture component ({001}<110>) is developed on the material surface [5].

Acknowledgments.

The authors would like to express their acknowledgment to Mr. Bogos Giakoupian ELVAL SA Depute General Director for his support and the approval to present the paper's data.

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Light Metals 2016
Edited by: Edward Williams
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EFFECT OF AS-CAST STRIP THICKNESS AND REDUCTION PRIOR TO SOFT ANNEALING ON THE FORMABILITY OF TWIN-ROLL CAST 5754 SHEETS

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Keywords: 5754 aluminum alloy, forming limit curves, twin roll casting

Abstract

5XXX aluminum alloys are preferred in automotive applications owing to their high strength, good formability and corrosion resistance. However, these alloys are mostly produced by direct chill casting and there is a limited number of attempt for twin-roll cast (TRC) 5754. With the motivation of the fact that microstructure and properties of sheets produced via TRC are highly influenced by casting parameters and cooling gradient originated throughout the strip thickness, this study investigates the effects of as-cast strip thickness and amount of cold rolling ratio prior to soft annealing on the microstructural evolution as well as mechanical properties of 1 mm thick TRC 5754 sheets. Microstructural characterizations were conducted by optical and SEM examinations on as-cast samples and after intermediate annealing as well as soft annealing performed at final gauge. Mechanical properties of final products processed by different routes were determined by tensile tests and forming limit diagrams.

Introduction

With development of new environmental legislations triggered by global warming, reducing the emissions of vehicles has been gaining much attention. One way of providing the required emissions is to reduce the overall weight of the vehicles. At this point, owing to their superior specific strength, good formability, corrosion resistance as well as recyclability, use of aluminum alloys instead of steels is preferred to build lightweight vehicles [1-5]. However, deformation characteristics of aluminum alloys are different than those of steels and thus successful forming of aluminum alloys in stamping forming is challenging since fracture occurs before necking instability due to their low ductility [2]. On the other hand, this drawback of aluminum can be eliminated by introducing suitable alloying elements. Among the aluminum alloys, high Mg containing 5xxx series possesses enhanced strength and reasonable formability together.

High Mg containing 5xxx wrought aluminum alloys are mainly produced by direct chill casting followed by repetitive hot rolling and if needed cold rolling [6]. Direct chill casting is an expensive technique since numerous thermo-mechanical steps such as scalping, homogenization annealing as well as hot rolling are required [3]. On the other hand, twin roll casting, where liquid metal transforms directly to 3-7 mm thick strip with the combination of casting and hot rolling, can reduce greenhouse emissions and the manufacturing cost of high Mg containing 5xxx aluminum alloys through reduced capital, energy consumption, operating costs and scrap rate when compared with direct chill casting [3,5-8]. However, since the number of thermo-mechanical steps are less than those in direct chill casting due to lower as-cast strip thickness, it is challenging to obtain homogeneous microstructure and good formability [5].

The motivation of this study is to increase the formability of twin roll cast 1 mm thick 5754-O through changing as-cast strip thickness and amount of reduction prior to soft annealing. The effects of as-cast strip thickness and amount of cold rolling ratio prior to soft annealing on the microstructural as well as mechanical properties of 1 mm thick twin roll cast 5754 sheets were investigated.

Experimental Studies

In this study, 5754 aluminum alloy was cast at strip thicknesses of 5.5 mm and 7.0 mm utilizing industrial scale Novelis twin roll caster. Table I lists the chemical compositions of the samples used in this study.

Table I. Chemical compositions of the alloys.

	Element, wt. %					
As-cast strip thickness, mm	Si	Fe	Mn	Mg	Cr	Ti
7.0	0.11	0.17	0.06	2.93	0.07	0.03
5.5	0.09	0,16	0.08	2.81	0.07	0.02

Samples were processed to achieve 1 mm O temper by the combinations of cold rolling and annealing. In order to determine the effect of reduction amount prior to soft annealing, intermediate annealing was conducted on 4.0 mm and 2.5 mm thick samples at 350°C for 6h in a batch type industrial scale furnace. After intermediate annealing, samples were cold rolled to 1 mm and soft annealed at 350°C for 4h to achieve O temper. Table II summarizes different downstream processes conducted in this study.

Table II. Downstream processes followed in this study.

Process number	As-cast strip thickness, mm	Intermediate annealing	Soft annealing
1	7	4.0 mm, 350°C -6h	1.0 mm, 350°C -4h
2		2.5 mm, 350°C -6h	
3	5.5	4.0 mm, 350°C -6h	
4		2.5 mm, 350°C -6h	

Microstructural and mechanical characterization of the samples were conducted. Microstructural characterization was carried out by optical microscope examinations on the cross sections of as-cast and annealed samples after preparing the samples according to standard metallographic methods. Mechanical properties of the samples were determined by hardness measurements, tensile tests and forming limit curves (FLCs). Hardness measurements were conducted on polished cross sections of as-cast samples utilizing a micro hardness tester with a Vickers indenter under an indentation load of 10 g. Tensile tests were performed on as-cast strips and 1.0 mm soft annealed samples at room temperature on

Zwick tensile test machine at a nominal cross speed of 10 mm/min with a 50 mm extensometer. In order to determine plastic anisotropy ratios and strain hardening exponent, longitudinal and transverse strain gauges were utilized during tensile tests of soft annealed 1.0 mm thick rectangular dog-bone samples prepared in 0°, 45° as well as 90° to the rolling direction. Plastic anisotropy ratios and strain hardening exponent values were measured at plastic strains between 4 and 6 %. Results of tensile tests were evaluated by averaging the results of 5 successive tensile tests.

Forming limit curves (FLCs) were determined based on Nakajima model. As shown in Figure 1, blanks with six different geometries (radius) were employed for FLCs to obtain different strain paths. In order to calculate the strain values, during forming tests, 10 pictures of samples in a second were captured. Strains calculated from 5 different sections taken on the last pictures just before failure were used for FLCs. Forming studies were performed three times for each geometry. Therefore, each point in the FLCs were the average of 15 measurements.

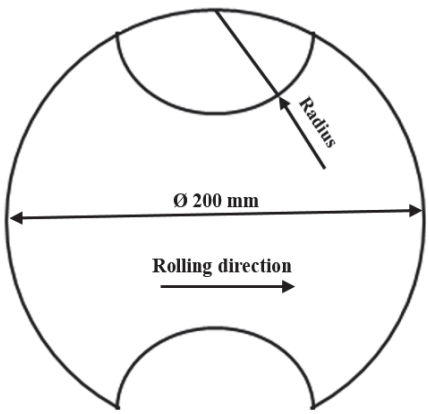


Figure 1. Blank geometries used in forming tests.
Radius: 0 mm, 35 mm, 45 mm, 50 mm, 55 mm, 85 mm.

Results and Discussion

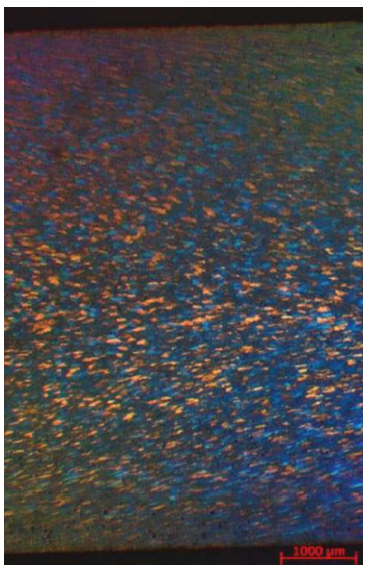
Mechanical properties of as-cast strips are given in Table III. Tensile tests results have revealed that as-cast strip thickness has no significant influence on mechanical properties.

Table III. Mechanical properties of as-cast strips.

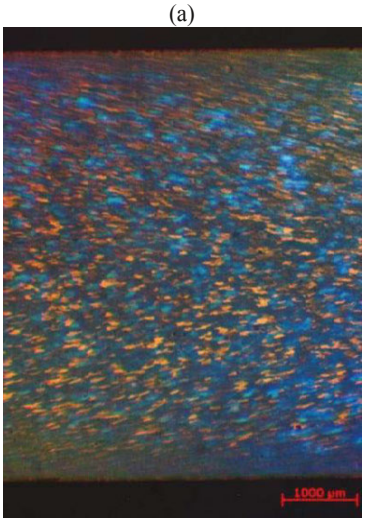
As-cast strip thickness, mm	Yield strength, MPa	Tensile strength, MPa	Elongation A50, %	Hardness, $HV_{0.01}$
7.0	157	226	9.5	79±8.8
5.5	162	228	9.9	77±12.5

Typical etched cross sectional optical microscope images of 7.0 mm and 5.5 mm thick as-cast strips are shown in Figure 2. As expected, grain sizes of as-cast strips increase gradually from outermost surfaces to mid-plane due to gradient in solidification rate. As can be seen in the cross sectional images of etched samples, 7.0 mm thick as-cast strip exhibits slightly finer grain structure when compared to that of 5.5 mm thick as-cast strip.

Figure 3 shows typical etched microstructures of all processes after intermediate annealing. Microstructures of samples have revealed that intermediate annealing at 350°C for 6h is sufficient for fully recrystallization at 4.0 mm and 2.5 mm. However, after intermediate annealing conducted at 4.0 mm and 2.5 mm, strips produced from 7.0 mm thick as-cast strips show finer grains when compared to those of produced from 5.5 mm thick as-cast strips. This result can be attributed to higher amount of cold rolling prior to intermediate annealing as well as smaller grain structure of 7.0 mm thick as-cast strip. At the same time, for a constant as-cast strip thickness, after intermediate annealing, 2.5 mm thick samples exhibit finer grains when compared to those of 4.0 mm thick samples. This observation can also be attributed to increased amount of reduction for 2.5 mm thick samples at a constant as-cast strip thickness.



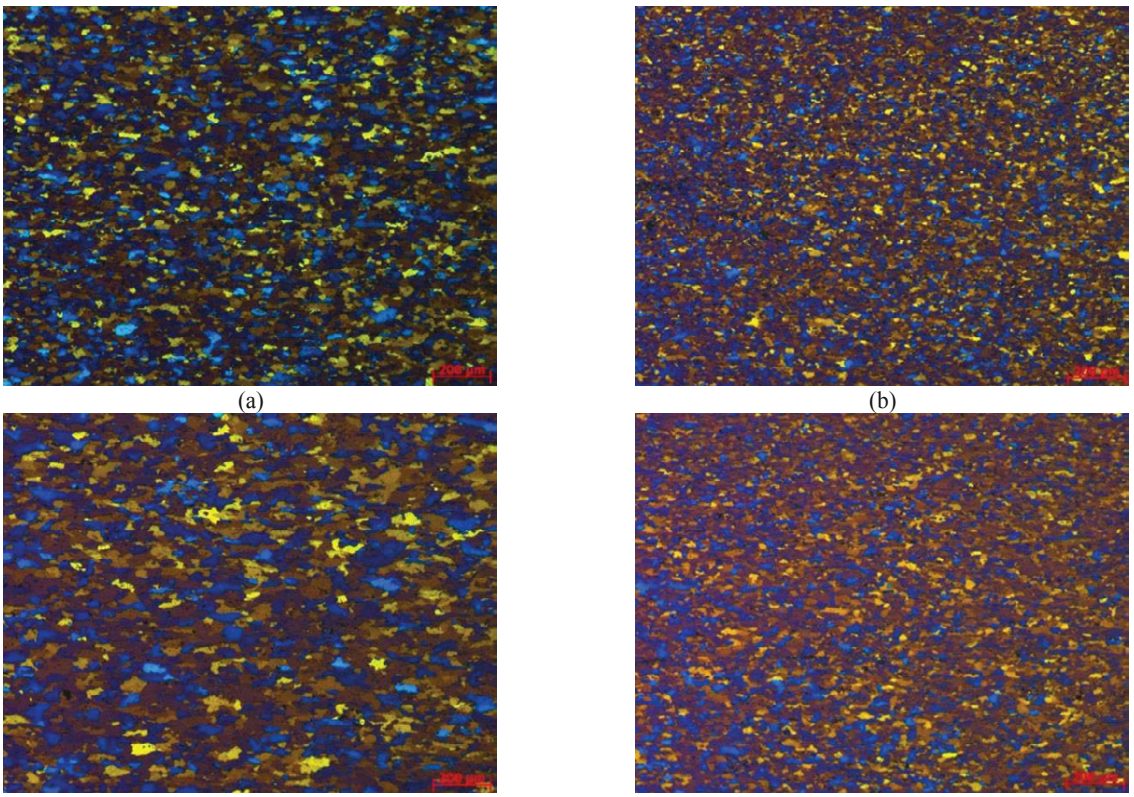
(a)



(b)

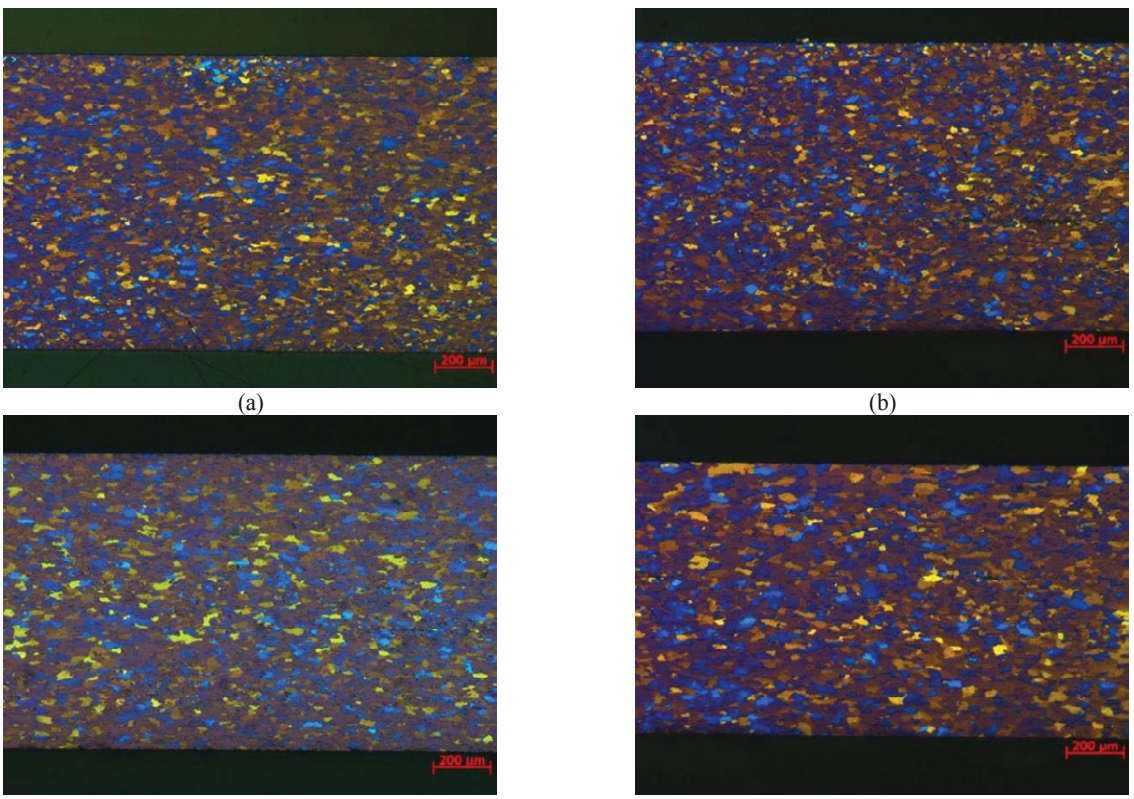
Figure 2. Typical etched cross sectional optical microscope images of (a) 7.0 mm thick and (b) 5.5 mm thick as-cast strips.

Figure 4 represents etched microstructures of 1 mm thick samples after soft annealing. Microstructural investigations revealed that for the same intermediate annealing thickness, increased as-cast strip thickness led to finer grains.



(a) (b) (c) (d)

Figure 3. Etched microstructures of samples after intermediate annealing (a) process 1, (b) process 2, (c) process 3 and (d) process 4.



(a) (b) (c) (d)

Figure 4. Etched microstructures of 1 mm thick samples after soft annealing (a) process 1, (b) process 2, (c) process 3 and (d) process 4.

Table IV. Tensile test results with r and n values.

Process number	Direction	Yield strength, MPa	Tensile strength, MPa	Uniform elongation A50, %	Total elongation A50, %	r	n	$R_{vertical}$	R_{planar}
1	0	101±0.8	210±2.5	15.3±0.71	17.9±0.66	0.85	0.27	0.7350	0.3300
	45	98±1.1	205±3.1	17.4±1.84	20.7±0.94	0.57	0.29		
	90	101±1.6	206±2.7	16.3±1.91	19.5±0.92	0.95	0.30		
2	0	108±0.9	224±1.0	16.0±1.27	18.8±1.22	0.89	0.27	0.7825	0.2650
	45	105±1.7	215±1.1	18.0±1.24	20.1±1.31	0.65	0.28		
	90	107±1.0	216±2.4	16.2±1.32	19.1±0.91	0.94	0.28		
3	0	100±1.0	211±1.0	15.4±1.74	17.8±1.23	0.95	0.28	0.7200	0.3800
	45	96±1.4	202±0.9	16.6±1.23	19.6±0.93	0.53	0.29		
	90	98±0.9	205±0.5	15.5±1.27	18.1±0.89	0.87	0.28		
4	0	99±0.9	214±1.6	15.7±1.05	18.4±0.45	0.98	0.30	0.7725	0.4250
	45	95±1.1	206±1.3	16.7±1.70	20.3±1.86	0.56	0.30		
	90	98±1.0	209±1.1	16.1±0.86	18.0±0.82	0.99	0.30		

Tensile test results of 1 mm thick O temper samples are shown in Table IV with plastic anisotropy ratios (r) and strain hardening exponent (n).

In general, samples possess maximum strength and total elongation values at 0° and 45° to rolling direction, respectively. This trend is comparable with those reported by Jain et al [9].

For samples produced from the same as-cast strip, the r-values of samples taken parallel and perpendicular to rolling direction are much higher than those of taken at 45° to rolling direction. This observation indicates that samples experience higher thinning at 45° when compared with those of other directions. Among the other directions, in general, samples taken perpendicular to rolling direction possess slightly higher r-values. This trend in r-values are in good agreement with those reported by Jain et al [9] and Abedrabbo et al [10], who studied forming performance of 5754 Al alloy. It is interesting to note that samples produced from 5.5 mm thick as-cast strip show slightly higher r-values at 0° and 90° directions where they exhibit lower r-values at 45° direction as compared to those of samples produced from 7.0 mm thick as-cast strip. However, when vertical ($R_{vertical}$) and planar (R_{planar}) anisotropy coefficients were calculated, it was found that samples produced from 7.0 mm thick as-cast strip have relatively higher vertical and lower planar anisotropy coefficients indicating that increased as-cast strip thickness provides resistance to thinning and earing during forming.

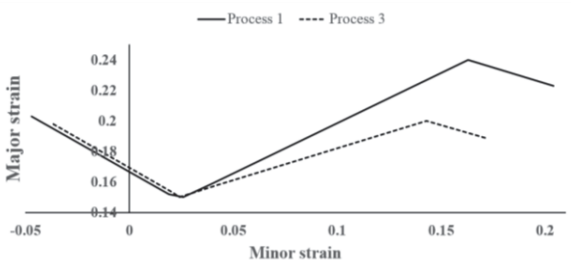


Figure 5. Forming limit curves of 1 mm thick samples produced by processes 1 and 3.

Figure 5 shows FLCs of 1 mm thick samples produced from processes 1 and 3. Along with thick gauge casting, relatively higher deformation (43%) applied prior to intermediate annealing, in the former process, appears to have an influence on the maximum strain before onset of failure at biaxial deformation (ε_{major}, ε_{minor}>0) side of FLCs.

Conclusions

The results of this study can be summarized as follows:

Increased strip thickness in casting leads to finer grain structure without any significant change in mechanical properties of as-cast strips.

Intermediate annealing conducted at 350°C for 6h is sufficient for completing recrystallization of all the samples exposed to the prior deformation ranging from 27-64% from as-cast thickness. Due to higher amount of cold rolling and finer grain structure at as-cast strip thickness, samples produced from 7.0 mm thick strip exhibit finer grains after intermediate annealing. For a constant as-cast strip thickness, increased amount of reduction before intermediate annealing results in fine grained structure after intermediate annealing, regardless of casting thickness.

For the same intermediate annealing thickness, increased as- cast strip thickness leads to finer grains in soft annealed 1 mm thick samples.

Plastic anisotropy values of samples taken 45° to the rolling direction are the lowest when compared to those of other directions indicating that samples experience higher levels of thinning at 45°. Increased as-cast strip thickness results in slightly lower r-values at 0° and 90° directions while higher r-values at 45° direction. However, increased as-cast strip thickness results in higher vertical and lower planar anisotropy coefficients which are favorable for thinning and earing, respectively.

Higher deformation applied prior to intermediate annealing in thick gauge cast material results in higher fracture strains that can be accommodated on biaxial deformation mode.

Acknowledgements

Authors wish to thank Mr. Nihat Yılmaz, technician of Assan Alüminyum Laboratory, for his valuable help with metallographic studies.

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